

NEET

OBJECTIVE PHYSICS



DC Pandey

Volume **01**

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DC Pandey

[B.Tech, M.Tech, Pantnagar, ID 15722]



ARIHANT PRAKASHAN (Series), MEERUT



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Preface

Medical offers the most exciting and fulfilling of careers. As a doctor you can find satisfaction from curing other persons. Although the number of medical colleges imparting quality education and training has significantly increased after independence in the country but due to the simultaneous increase in the number of serious aspirants, the competition is no longer easy for a seat in a prestigious medical college today.

For success, you require an objective approach to your study in the test subjects. This does not mean you 'prepare' yourself for just 'objective questions'. Objective Approach means more than that. It could be defined as that approach through which a student is able to master the concepts of the subject and also the skills required to tackle the questions asked in different formats in entrances such as NEET (National Eligibility cum Entrance Test). These two-volume books, Objective Physics (Vol.1 & 2) are borne out of my experience of teaching physics to medical aspirants, fill the needs of such books in the market.

The plan of the presentation of the subject matter in the books is as follows

- The whole chapter has been divided under logical topic heads to cover the whole syllabi of NEET developing the concepts in an easy going manner, taking the help of suitable examples.
- Important points of the topics have been highlighted in the text under notes, some extra points regarding the topics have been given in Notes to enrich the students.
- The **Solved Examples** given with different concepts of the chapter make the students learn the basic problem solving skills in Physics. It has been ensured that given examples cover all aspects of a concepts comprehensively.
- **Check Point** Exercises given in between the text of all chapter help the readers to remain linked with the text given as they provide them an opportunity to assess themselves while studying the text.
- Exercises at the end of the chapters have been divided into three parts:
 - Part A- 'Taking it together'** has Objective Questions of the concerned chapter. The special point of this exercise is, all the questions have been arranged according to level of difficulty, providing students a systematics practice.
 - Part B- 'Medical Entrance Special Format Question'** this section covers all special type of questions, other than simple MCQs, generally asked in NEET & other Medical Entrances. Here Assertion-Reason, Statement based and Matching Type Questions have been given.
 - Part C- 'Medical Entrances' Gallery'** covering all questions asked in last 11 years' (2021-2011) in NEET & other Medical Entrances.
- The answers / solutions to all the questions given in different exercises have been provided.

At the end I would like to say that suggestions from the respected Teachers & Students for the further improvement of the book will be welcomed open heartedly.

DC Pandey

DEDICATION

This book is dedicated to my honourable grandfather

(Late) Sh. Pitamber Pandey

a Kumaoni poet, and a resident of village Dhaura (Almora), Uttarakhand

Syllabus

UNIT I Physical World and Measurement

Physics Scope and excitement, nature of physical laws, Physics, technology and society. Need for measurement, Units of measurement, systems of units, SI units, fundamental and derived units. Length, mass and time measurements, accuracy and precision of measuring instruments, errors in measurement, significant figures. Dimensions of physical quantities, dimensional analysis and its applications.

UNIT II Kinematics

Frame of reference, Motion in a straight line, Position-time graph, speed and velocity. Uniform and non-uniform motion, average speed and instantaneous velocity. Uniformly accelerated motion, velocity-time and position-time graphs, for uniformly accelerated motion (graphical treatment). Elementary concepts of differentiation and integration for describing motion. Scalar and vector quantities: Position and displacement vectors, general vectors and notation, equality of vectors, multiplication of vectors by a real number, addition and subtraction of vectors. Relative velocity. Unit vectors. Resolution of a vector in a plane-rectangular components. Scalar and Vector products of Vectors. Motion in a plane. Cases of uniform velocity and uniform acceleration, Projectile motion. Uniform circular motion.

UNIT III Laws of Motion

Intuitive concept of force. Inertia, Newton's first law of motion; momentum and Newton's second law of motion, impulse, Newton's third law of motion. Law of conservation of linear momentum and its applications. Equilibrium of concurrent forces. Static and Kinetic friction, laws of friction, rolling friction, lubrication. Dynamics of uniform circular motion. Centripetal force, examples of circular motion (vehicle on level circular road, vehicle on banked road).

UNIT IV Work, Energy and Power

Work done by a constant force and variable force, kinetic energy, work-energy theorem, power. Notion of potential energy, potential energy of a spring, conservative forces, conservation of mechanical energy (kinetic and potential energies), non-conservative forces, motion in a vertical circle, elastic and inelastic collisions in one and two dimensions.

UNIT V Motion of System of Particles and Rigid Body

Centre of mass of a two-particles system, momentum conservation and centre of mass motion. Centre of mass of a rigid body, centre of mass of uniform rod. Moment of a force, torque, angular momentum, conservation of angular momentum with some examples. Equilibrium of rigid bodies, rigid body rotation and equation of rotational motion, comparison of linear and rotational motions, moment of inertia, radius of gyration. Values of MI for

simple geometrical objects (no derivation). Statement of parallel and perpendicular axes theorems and their applications.

UNIT VI Gravitation

Kepler's laws of planetary motion. The universal law of gravitation. Acceleration due to gravity and its variation with altitude and depth. Gravitational potential energy, gravitational potential. Escape velocity, orbital velocity of a satellite. Geostationary satellites.

UNIT VII Properties of Bulk Matter

Elastic behavior, Stress-strain relationship. Hooke's law, Young's modulus, bulk modulus, shear, modulus of rigidity, poisson's ratio; elastic energy. Viscosity, Stokes' law, terminal velocity, Reynold's number, streamline and turbulent flow. Critical velocity, Bernoulli's theorem and its applications. Surface energy and surface tension, angle of contact, excess of pressure, application of surface tension ideas to drops, bubbles and capillary rise. Heat, temperature, thermal expansion, thermal expansion of solids, liquids and gases. Anomalous expansion. Specific heat capacity, C_p , C_v - calorimetry; change of state - latent heat. Heat transfer- conduction and thermal conductivity, convection and radiation. Qualitative ideas of black body radiation, Wein's displacement law and green house effect. Newton's law of cooling and Stefan's law.

UNIT VIII Thermodynamics

Thermal equilibrium and definition of temperature (zeroth law of Thermodynamics). Heat, work and internal energy. First law of thermodynamics, Isothermal and adiabatic processes. Second law of the thermodynamics, Reversible and irreversible processes. Heat engines and refrigerators.

UNIT IX Behaviour of Perfect Gas and Kinetic Theory

Equation of state of a perfect gas, work done on compressing a gas. Kinetic theory of gases, Assumptions, concept of pressure. Kinetic energy and temperature, degrees of freedom, law of equipartition of energy (statement only) and application to specific heat capacities of gases, concept of mean free path.

UNIT X Oscillations and Waves

Periodic motion-period, frequency, displacement as a function of time. Periodic functions. Simple harmonic motion(SHM) and its equation, phase, oscillations of a spring-restoring force and force constant, energy in SHM-kinetic and potential energies, simple pendulum-derivation of expression for its time period, free, forced and damped oscillations (qualitative ideas only), resonance. Wave motion. Longitudinal and transverse waves, speed of wave motion. Displacement relation for a progressive wave. Principle of superposition of waves, reflection of waves, standing waves in strings and organ pipes, fundamental mode and harmonics. Beats, Doppler effect.

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CHAPTER 0

Basic Mathematics

TRIGONOMETRY

It is one of the important branch of mathematics which deal with relations of sides and angles of triangle and also with the relevant functions of any angle.

Consider a ray OA . If this ray revolves about its end point O in anti-clockwise direction and takes position OB , then we say that the angle $\angle AOB$ has been generated as shown in the following figure.

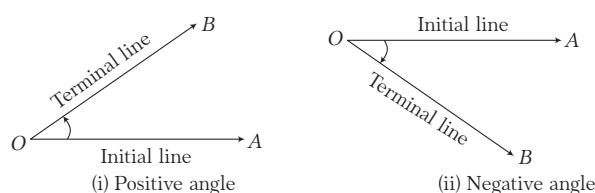


Fig. 0.1

Or simply say that angle is a measure of an amount of revolution of a given ray about its initial point.

The angle is positive (or negative), if the initial line revolves in anti-clockwise (or clockwise) direction to get the terminal line.

System of measurement of angles

- (i) **Sexagesimal system** In this system, a right angle is divided into 90 equal parts, called degree.

Thus, 1 right angle = 90° (degrees)

$$1^\circ = 60' \text{ (minutes)}$$

$$1' = 60'' \text{ (seconds)}$$

Inside

- 1 Trigonometry
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- (ii) **Circular system** In this system, the unit of measurement is radian.

$$\theta = \frac{l}{r} \text{ radian or rad}$$

$$1 \text{ right angle} = \frac{\pi}{2} \text{ rad}$$

$$1 \text{ straight angle} = \pi \text{ rad}$$

$$\text{and } 1 \text{ complete angle} = 2\pi \text{ rad}$$

One radian (denoted by 1^c) is the measure of an angle subtended at the centre of a circle by an arc of length equal to the radius of the circle.

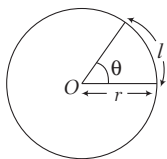


Fig. 0.2 Circular system

$$1 \text{ rad } (1^c) = \frac{180^\circ}{\pi} \approx 57^\circ 17' 45'' \approx 57.3^\circ$$

Note (i) $\pi = \frac{22}{7} = 3.14$ (ii) $360^\circ = 2\pi \text{ radian}$, $\frac{\pi}{2} \text{ radian} = 90^\circ$

Example 0.1 Find the radian measures corresponding to the following degree measures.

- (i) 75° (ii) $-37^\circ 30'$ (iii) $5^\circ 37' 30''$

Sol. We have, $1^c = \frac{180^\circ}{\pi} \Rightarrow 1^\circ = \left(\frac{\pi}{180}\right)^c$

$$(i) 75^\circ = \left(75 \times \frac{\pi}{180}\right)^c = \left(\frac{5\pi}{12}\right)^c$$

$$(ii) -37^\circ 30' = -\left(37\frac{1}{2}\right)^\circ = -\left(\frac{75}{2}\right)^\circ = -\left(\frac{75}{2} \times \frac{\pi}{180}\right)^c = -\left(\frac{5\pi}{24}\right)^c$$

$$(iii) 5^\circ 37' 30'' = \left(5\frac{5}{8}\right)^\circ = \left(\frac{45}{8}\right)^\circ = \left(\frac{45}{8} \times \frac{\pi}{180}\right)^c = \left(\frac{\pi}{32}\right)^c$$

Example 0.2 Find the degree measures corresponding to the following radian measures.

- (i) $\left(\frac{2\pi}{15}\right)^c$ (ii) $\left(\frac{\pi}{8}\right)^c$ (iii) $(-2)^c$

Sol. We have,

$$(i) \left(\frac{2\pi}{15}\right)^c = \left(\frac{2\pi}{15} \times \frac{180}{\pi}\right)^\circ = 24^\circ$$

$$(ii) \left(\frac{\pi}{8}\right)^c = \left(\frac{\pi}{8} \times \frac{180}{\pi}\right)^\circ = \left(\frac{45}{2}\right)^\circ = \left(22\frac{1}{2}\right)^\circ = 22^\circ \left(\frac{1}{2} \times 60\right)' = 22^\circ 30'$$

$$(iii) (-2)^c = \left(-2 \times \frac{180}{\pi}\right)^\circ = -\left(2 \times \frac{180}{22} \times 7\right)^\circ = -\left(114\frac{6}{11}\right)^\circ = -114^\circ \left(\frac{6}{11} \times 60\right)' = -114^\circ \left(32\frac{8}{11}\right)' = -114^\circ 32' \left(\frac{8}{11} \times 60\right)'' \approx -[114^\circ 32' 44'']$$

Trigonometrical ratios (or T-ratios)

Consider the two fixed lines $X'OX$ and YOY' intersecting each other at right angle at point O as shown in the following figure

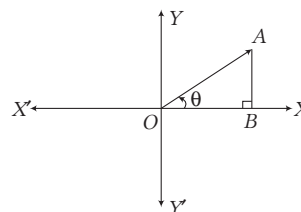


Fig. 0.3 Trigonometrical ratios

Then,

- (i) Intersection point O is called origin.
- (ii) $X'OX$ and YOY' are called X -axis and Y -axis, respectively.
- (iii) The portion XOY , YOX' , $X'OY'$ and $Y'OX$ are known as I, II, III and IV quadrants, respectively.

Now, consider that the revolving line OA has traced out an angle θ in anti-clockwise direction (in I quadrant).

From point A , draw $AB \perp OX$ which results a right angled $\triangle ABO$, where AB = perpendicular, OA = hypotenuse and OB = base.

The three sides of right angled triangle are related to each other through side having different ratios, called trigonometrical ratios or T -ratios, which are given as

$$(i) \sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{AB}{OA} \quad (\text{From Fig. 0.3})$$

$$(ii) \cos \theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{OB}{OA}$$

$$(iii) \tan \theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{AB}{OB}$$

$$(iv) \cot \theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{OB}{AB}$$

$$(v) \sec \theta = \frac{\text{Hypotenuse}}{\text{Base}} = \frac{OA}{OB}$$

$$(vi) \operatorname{cosec} \theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}} = \frac{OA}{AB}$$

Fundamental of T -ratios or trigonometric functions

For any acute angle say θ ($< 90^\circ$), the functions are given as

$$(i) \operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad (ii) \sec \theta = \frac{1}{\cos \theta}$$

$$(iii) \cot \theta = \frac{1}{\tan \theta}$$

$$(iv) \sin^2 \theta + \cos^2 \theta = 1$$

$$(v) 1 + \tan^2 \theta = \sec^2 \theta$$

$$(vi) 1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Signs of trigonometric ratios or T -ratios in various quadrants

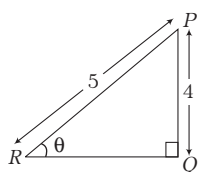
II only sin and cosec are + ve	I All + ve
III only tan and cot are + ve	IV only cos and sec are + ve

Fig. 0.4 Sign of T -ratios

- (i) In I quadrant, all T -ratios are positive.
- (ii) In II quadrant, $\sin \theta$ is + ve, $\cos \theta$ and $\tan \theta$ are - ve.
- (iii) In III quadrant, $\tan \theta$ is + ve, $\sin \theta$ and $\cos \theta$ are - ve.
- (iv) In IV quadrant, $\cos \theta$ is + ve, $\sin \theta$ and $\tan \theta$ are - ve.

Example 0.3 If $\sin \theta = \frac{4}{5}$, where θ lies in the first quadrant, then find all the other T -ratios.

Sol. Let ΔPQR be right angled triangle, right angled at Q .



$$\sin \theta = \frac{4}{5} \quad (\text{Given})$$

$$\therefore PR = 5 \text{ and } PQ = 4$$

On applying Pythagoras theorem in ΔPQR , we have

$$(PR)^2 = (PQ)^2 + (QR)^2$$

$$\Rightarrow (5)^2 = (4)^2 + (QR)^2$$

$$\Rightarrow QR = \sqrt{25 - 16} = \sqrt{9} = 3$$

[Taking positive value of square root]

$$\text{Now, } \cos \theta = \frac{QR}{PR} = \frac{3}{5},$$

$$\tan \theta = \frac{PQ}{QR} = \frac{4}{3},$$

$$\cot \theta = \frac{QR}{PQ} = \frac{3}{4},$$

$$\sec \theta = \frac{PR}{QR} = \frac{5}{3}$$

$$\text{and } \operatorname{cosec} \theta = \frac{PR}{PQ} = \frac{5}{4}$$

T -ratios of allied angles

In trigonometry two angles are said to be allied angles when their sum or difference is a multiple of 90° .

The T -ratios of the following allied angles are as

- (i) When angle (say) θ is negative, then
 - (a) $\sin(-\theta) = -\sin \theta$
 - (b) $\cos(-\theta) = \cos \theta$
 - (c) $\tan(-\theta) = -\tan \theta$
- (ii) When angle θ is less than 90° (i.e., lies in I quadrant), then
 - (a) $\sin(90^\circ - \theta) = \cos \theta$
 - (b) $\cos(90^\circ - \theta) = \sin \theta$
 - (c) $\tan(90^\circ - \theta) = \cot \theta$
- (iii) When angle θ lies between 90° and 180° (i.e., lies in II quadrant), then
 - (a) • $\sin(90^\circ + \theta) = \cos \theta$
 - $\cos(90^\circ + \theta) = -\sin \theta$
 - $\tan(90^\circ + \theta) = -\cot \theta$
 - (b) • $\sin(180^\circ - \theta) = \sin \theta$
 - $\cos(180^\circ - \theta) = -\cos \theta$
 - $\tan(180^\circ - \theta) = -\tan \theta$

- (iv) When angle θ lies between 180° and 270° (i.e., lies in III quadrant), then

- (a) • $\sin(180^\circ + \theta) = -\sin \theta$
- $\cos(180^\circ + \theta) = -\cos \theta$
- $\tan(180^\circ + \theta) = \tan \theta$
- (b) • $\sin(270^\circ - \theta) = -\cos \theta$
- $\cos(270^\circ - \theta) = -\sin \theta$
- $\tan(270^\circ - \theta) = \cot \theta$

- (v) When angle θ lies between 270° and 360° (i.e., lies in IV quadrant), then

- (a) • $\sin(270^\circ + \theta) = -\cos \theta$
- $\cos(270^\circ + \theta) = \sin \theta$
- $\tan(270^\circ + \theta) = -\cot \theta$
- (b) • $\sin(360^\circ - \theta) = -\sin \theta$
- $\cos(360^\circ - \theta) = \cos \theta$
- $\tan(360^\circ - \theta) = -\tan \theta$

Values of T -ratios of some standard angles

Angle (θ)	0°	30° $\left(=\frac{\pi}{6}\right)$	45° $\left(=\frac{\pi}{4}\right)$	60° $\left(=\frac{\pi}{3}\right)$	90° $\left(=\frac{\pi}{2}\right)$	120° $\left(=\frac{2\pi}{3}\right)$	135° $\left(=\frac{3\pi}{4}\right)$	150° $\left(=\frac{5\pi}{6}\right)$	180° $(=\pi)$
sin θ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
tan θ	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

Example 0.4 Find the value of

(i) $\sin(-45^\circ)$

(ii) $\tan 225^\circ$

(iii) $\cos 300^\circ$

(iv) $\sec 120^\circ$

Sol. (i) $\sin(-45^\circ) = -\sin 45^\circ$ [$\because \sin(-\theta) = -\sin \theta$]
 $= -\frac{1}{\sqrt{2}}$ [$\because \sin 45^\circ = \frac{1}{\sqrt{2}}$]
 (ii) $\tan 225^\circ = \tan(270^\circ - 45^\circ) = \cot 45^\circ$
 $= 1$ [$\because \tan(270^\circ - \theta) = \cot \theta$]
 $[\because \cot 45^\circ = 1]$
 (iii) $\cos 300^\circ = \cos(270^\circ + 30^\circ) = \sin 30^\circ$
 $= \frac{1}{2}$ [$\because \cos(270^\circ + \theta) = \sin \theta$]
 $[\because \sin 30^\circ = \frac{1}{2}]$
 (iv) $\sec 120^\circ = \sec(180^\circ - 60^\circ) = -\sec 60^\circ$
 $= -2$ [$\because \sec(180^\circ - \theta) = -\sec \theta$]

Some important formulae of trigonometry

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$
- $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1$
 $= 1 - 2 \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$

- $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$

- $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$
- $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$
- $\cos(A + B) + \cos(A - B) = 2 \cos A \cos B$
- $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$

- $\sin C + \sin D = 2 \sin \left(\frac{C + D}{2} \right) \cdot \cos \left(\frac{C - D}{2} \right)$

- $\sin C - \sin D = 2 \cos \left(\frac{C + D}{2} \right) \cdot \sin \left(\frac{C - D}{2} \right)$

- $\cos C + \cos D = 2 \cos \left(\frac{C + D}{2} \right) \cdot \cos \left(\frac{C - D}{2} \right)$

- $\cos C - \cos D = -2 \sin \left(\frac{C + D}{2} \right) \cdot \sin \left(\frac{C - D}{2} \right)$

Example 0.5 Find the value of (i) $\sin 15^\circ$ (ii) $\tan 75^\circ$

Sol. (i) We have,

$$\begin{aligned} \sin 15^\circ &= \sin(45^\circ - 30^\circ) \\ &= \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \\ &[\because \sin(A - B) = \sin A \cos B - \cos A \sin B] \\ &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &(\because \sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}, \cos 30^\circ = \frac{\sqrt{3}}{2} \\ &\text{and } \sin 30^\circ = \frac{1}{2}) \\ &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \end{aligned}$$

(ii) We have, $\tan 75^\circ = \tan(45^\circ + 30^\circ)$

$$\begin{aligned} &= \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \cdot \tan 30^\circ} \left(\because \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} \right) \\ &= \frac{1 + \frac{1}{\sqrt{3}}}{1 - 1 \cdot \frac{1}{\sqrt{3}}} \left(\because \tan 45^\circ = 1 \text{ and } \tan 30^\circ = \frac{1}{\sqrt{3}} \right) \\ &= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \end{aligned}$$

CHECK POINT 0.1

1. Find the radian measures corresponding to the following degree measures.

- (i) 25° (ii) $-47^\circ 30'$
(iii) $39^\circ 22' 30''$

Ans. (i) $\frac{5\pi}{36}$ (ii) $-\frac{19\pi}{72}$ (iii) $\frac{7\pi}{32}$

2. Find the degree measures corresponding to the following radian measures.

- (i) $\left(\frac{18\pi}{5}\right)^\circ$ (ii) $(-3)^\circ$
(iii) $\left(-\frac{5\pi}{6}\right)^\circ$ (iv) $\left(\frac{9\pi}{5}\right)^\circ$

Ans. (i) 648° (ii) $-171^\circ 49' 5''$ (iii) -150°
(iv) 324°

3. Find $\sin \theta$ and $\tan \theta$, if $\cos \theta = -\frac{12}{13}$ and θ lies in the third quadrant.

Ans. $\sin \theta = -\frac{5}{13}$ and $\tan \theta = \frac{5}{12}$

4. Find the values of other five T -ratios, if $\tan \theta = -\frac{3}{4}$ and θ lies in II quadrant.

Ans. $\sin \theta = \frac{3}{5}$, $\cos \theta = -\frac{4}{5}$, $\operatorname{cosec} \theta = \frac{5}{3}$, $\sec \theta = -\frac{5}{4}$ and $\cot \theta = -\frac{4}{3}$

5. Find the values of the following T -ratios

- (i) $\operatorname{cosec} 315^\circ$ (ii) $\cos 210^\circ$
(iii) $\sin(-330^\circ)$

Ans. (i) $-\sqrt{2}$ (ii) $-\frac{\sqrt{3}}{2}$ (iii) $\frac{1}{2}$

6. Find the value of

- (i) $\sec 165^\circ$ (ii) $\cot 105^\circ$

Ans. (i) $(\sqrt{2} - \sqrt{6})$ (ii) $\frac{1 - \sqrt{3}}{1 + \sqrt{3}}$

CALCULUS (DIFFERENTIATION)

Differentiation in calculus, is the process of finding the derivative. The derivative is the instantaneous rate of change of a function with respect to one of its variable. This is equivalent to finding the slope of the tangent line to the function at a point.

Variable

A quantity, which can take different values, is called a variable quantity. A variable is usually represented by x , y , z etc.

Constant

A quantity, whose value remains unchanged during mathematical operations, is called a constant quantity. The integers, fractions such as π , e , etc are all constant.

Function

A quantity y is called a function of a variable x , if corresponding to any given value of x , there exists a single definite value of y . The phrase ' y is function of x ' is represented as $y = f(x)$.

e.g., Consider that y is a function of the variable x which is given by $y = 4x^2 + 3x + 7$ and

$$y = \sin x + e^x.$$

Here, we will treat x as independent variable and y as dependent variable, *i.e.*, the value y depends on x . If we change the value of x , then y will change.

Physical meaning of $\frac{dy}{dx}$

- (i) The ratio of small change in the function y and the variable x is called the average rate of change of y w.r.t. x .

e.g. If a body covers a small distance Δs in small time Δt , then average velocity of the body, $v_{av} = \frac{\Delta s}{\Delta t}$

- (ii) The limiting value of $\frac{\Delta y}{\Delta x}$, when $\Delta x \rightarrow 0$, *i.e.*,

$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$, is called the instantaneous rate of change of y w.r.t. x .

Thus, the differentiation of a function w.r.t. a variable implies the instantaneous rate of change of the function w.r.t. that variable.

e.g., Instantaneous velocity of the body,

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

Theorems of differentiation

- (i) If c is constant, then $\frac{d}{dx}(c) = 0$

- (ii) If $y = cv$, where c is a constant and v is a function of x , then $\frac{dy}{dx} = \frac{d}{dx}(c \cdot v) = c \frac{dv}{dx}$

(iii) If $y = x^n$, where n is a real number, then

$$\frac{dy}{dx} = \frac{d}{dx} (x^n) = nx^{n-1}$$

(iv) If $y = u \pm v \pm w$, where u , v and w are functions of

$$x, \text{ then } \frac{dy}{dx} = \frac{d}{dx} (u \pm v \pm w) = \frac{du}{dx} \pm \frac{dv}{dx} \pm \frac{dw}{dx}$$

(v) **Product rule** If $y = u \cdot v$, where u and v are

$$\text{functions of } x, \text{ then } \frac{dy}{dx} = \frac{d}{dx} (u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$$

(vi) **Division rule** If $y = \frac{u}{v}$, where u and v are functions

$$\text{of } x, \text{ then } \frac{dy}{dx} = \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Example 0.6 Differentiate the following functions

(i) $y = x^{-3}$

(ii) $6x^5 + 4x^3 - 3x^2 + 2x - 7$

(iii) $y = (x+2)(x^2+1)$

(iv) $y = \frac{(x^3+4)}{(x+1)}$

Sol. (i) We have, $y = x^{-3}$

On differentiating both sides w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= -3x^{-3-1} \quad \left(\because \frac{d}{dx} x^n = nx^{n-1} \right) \\ &= -3x^{-4} = \frac{-3}{x^4} \end{aligned}$$

(ii) Let $y = 6x^5 + 4x^3 - 3x^2 + 2x - 7$

On differentiating both sides w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= 6 \frac{d}{dx} x^5 + 4 \frac{d}{dx} x^3 - 3 \frac{d}{dx} x^2 + 2 \frac{d}{dx} x - \frac{d}{dx} 7 \\ &\quad \left(\because \frac{d}{dx} (c \cdot v) = c \cdot \frac{dv}{dx} \right) \\ \Rightarrow \frac{dy}{dx} &= 6 \cdot 5x^{5-1} + 4 \cdot 3x^{3-1} - 3 \cdot 2x^{2-1} + 2 \cdot x^{1-1} - 0 \\ &\quad \left(\because \frac{d}{dx} x^n = nx^{n-1} \text{ and } \frac{d}{dx} c = 0 \right) \end{aligned}$$

$$\Rightarrow \frac{dy}{dx} = 30x^4 + 12x^2 - 6x + 2 \times 1 \quad (\because x^{1-1} = x^0 = 1)$$

$$\Rightarrow \frac{dy}{dx} = 30x^4 + 12x^2 - 6x + 2$$

(iii) We have, $y = (x+2)(x^2+1)$

On differentiating both sides w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= (x+2) \frac{d}{dx} (x^2+1) + (x^2+1) \frac{d}{dx} (x+2) \\ &\quad \text{(By product rule)} \\ &= (x+2)(2x+0) + (x^2+1)(1+0) \\ &= 2x(x+2) + x^2 + 1 = 2x^2 + 4x + x^2 + 1 = 3x^2 + 4x + 1 \end{aligned}$$

(iv) We have, $y = \frac{x^3+4}{x+1}$

On differentiating both sides w.r.t. x , we get

$$\frac{dy}{dx} = \frac{(x+1) \frac{d}{dx} (x^3+4) - (x^3+4) \frac{d}{dx} (x+1)}{(x+1)^2}$$

(By division rule)

$$\begin{aligned} \frac{dy}{dx} &= \frac{(x+1)(3x^2+0) - (x^3+4)(1+0)}{(x+1)^2} \\ &= \frac{3x^2(x+1) - x^3 - 4}{(x+1)^2} = \frac{3x^3 + 3x^2 - x^3 - 4}{(x+1)^2} \\ &= \frac{2x^3 + 3x^2 - 4}{(x+1)^2} \end{aligned}$$

Formulae for differential coefficient of trigonometric, logarithmic and exponential function

- $\frac{d}{dx} (\sin x) = \cos x$
- $\frac{d}{dx} (\cos x) = -\sin x$
- $\frac{d}{dx} (\tan x) = \sec^2 x$
- $\frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x$
- $\frac{d}{dx} (\sec x) = \sec x \cdot \tan x$
- $\frac{d}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$
- $\frac{d}{dx} (\log x) = \frac{1}{x}$
- $\frac{d}{dx} (e^x) = e^x$

Example 0.7 Differentiate the following functions

(i) $y = \sin x + e^x$

(ii) $y = 3x^2 + \log x + 4e^x + 5$

(iii) $y = e^x \cdot \tan x$

Sol. (i) We have, $y = \sin x + e^x$

On differentiating both sides w.r.t. x , we get

$$\frac{dy}{dx} = \frac{d}{dx} (\sin x + e^x) = \cos x + e^x$$

(ii) We have, $y = 3x^2 + \log x + 4e^x + 5$

On differentiating both sides w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} (3x^2 + \log x + 4e^x + 5) \\ &= 3 \frac{d}{dx} x^2 + \frac{d}{dx} (\log x) + 4 \frac{d}{dx} e^x + \frac{d}{dx} 5 \\ &= 3 \cdot 2x^{2-1} + \frac{1}{x} + 4e^x + 0 = 6x + \frac{1}{x} + 4e^x \end{aligned}$$

(iii) We have, $y = e^x \cdot \tan x$

On differentiating both sides w.r.t. x , we get

$$\begin{aligned} \frac{dy}{dx} &= e^x \cdot \frac{d}{dx} \tan x + \tan x \cdot \frac{d}{dx} e^x \quad \text{(By product rule)} \\ &= e^x \cdot \sec^2 x + \tan x \cdot e^x = e^x (\sec^2 x + \tan x) \end{aligned}$$

Chain rule

This rule is applied, when the given function is the function of function, i.e., a function is in the form of $f[g(x)]$.

$$\therefore \frac{d}{dx} f[g(x)] = f'[g(x)] \cdot g'(x)$$

Example 0.8 Differentiate $\sin(x^2 + 5)$ w.r.t. x .

Sol. Let $y = \sin(x^2 + 5)$

On differentiating both sides w.r.t. 'x', we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \sin(x^2 + 5) = \cos(x^2 + 5) \cdot \frac{d}{dx} (x^2 + 5) \\ &= \cos(x^2 + 5) \cdot (2x + 0) = 2x \cos(x^2 + 5) \end{aligned}$$

Maxima and minima

Let $y = f(x)$, where y is a function of x .

For y to be minimum or maximum, put $\frac{dy}{dx} = 0$, then

find x .

If $\frac{d^2y}{dx^2} < 0$, y is maximum and if $\frac{d^2y}{dx^2} > 0$, y is minimum.

Note Most of time, it is known from physical situation whether the quantity is a maximum or minimum; therefore, there is no need to check maximum or minimum with the help of $\frac{d^2y}{dx^2}$.

Example 0.9 Divide a number 1000 in two parts such that their product is maximum.

Sol. Let the two parts be x and $(1000 - x)$.

$$\therefore \text{Their product, } P = x(1000 - x) \Rightarrow P = 1000x - x^2$$

On differentiating both sides w.r.t. x , we get

$$\frac{dP}{dx} = 1000 - 2x \quad \dots(i)$$

For P to be maximum or minimum,

$$\frac{dP}{dx} = 0 \Rightarrow 1000 - 2x = 0 \Rightarrow x = 500$$

On differentiating both sides of Eq. (i) w.r.t. x , we get

$$\frac{d^2P}{dx^2} = -2 < 0$$

$\therefore P$ is maximum at $x = 500$

On dividing equally, the two parts are (500, 500).

Remembering points

- All the problems of maxima/minima cannot be solved by the above methods e.g., $y = x^2$, y is maximum when x is maximum.
- If $y = \sin x$, by simple observation, y is maximum if $\sin x$ is maximum, i.e., $\sin x = 1$. The value of sine or cosine functions lies between -1 and $+1$.
- The product of the two parts is the maximum when the parts are equal.

Applications of differentiation in physics

(i) If the displacement is a function of time t , then to find the velocity, differentiate s w.r.t. t .

$$s = f(t), v = \frac{ds}{dt}$$

(ii) If the velocity is a function of time t , then to find acceleration, differentiate v w.r.t. t

$$\begin{aligned} v &= f(t), a = \frac{dv}{dt} \\ &= \frac{d}{dt} \left(\frac{ds}{dt} \right) = \frac{d^2s}{dt^2} \end{aligned}$$

(iii) If the velocity is a function of displacement s , then to find the acceleration, differentiate v w.r.t. t and use the expression, $a = v \frac{dv}{ds}$

(iv) Consider the motion along the X-axis

- If $v > 0$, s is increasing, then the particle is moving along the positive X-axis.
- If $v < 0$, s is decreasing, then the particle is moving along the negative X-axis.
- If $a > 0$, $v > 0$, then speed is increasing along the positive X-axis.
- If $a > 0$, $v < 0$, then speed is decreasing along the negative X-axis.
- If $a < 0$, $v > 0$, then the speed is decreasing along the positive X-axis.
- If $a < 0$, $v < 0$, then the speed is increasing along the negative X-axis.

Note

(i) If v and a have same sign, then the speed is increasing.

(ii) If v and a have opposite sign, then the speed is decreasing.

Example 0.10 The displacement of a particle as a function of time t is given by $s = \alpha + \beta t + \gamma t^2 + \delta t^4$, where α , β , γ and δ are constants. Find the ratio of the initial velocity to the initial acceleration.

Sol. First find the velocity and acceleration in terms of time t , then use $t = 0$ to find the initial values.

$$s = \alpha + \beta t + \gamma t^2 + \delta t^4 \quad (\text{Given})$$

On differentiating both sides w.r.t. t , we get

$$\frac{ds}{dt} = \beta + 2\gamma t + 4\delta t^3$$

$$\Rightarrow v = \beta + 2\gamma t + 4\delta t^3 \quad \left(\because \frac{ds}{dt} = v \right) \quad \dots(i)$$

On differentiating both sides of Eq. (i), w.r.t. t , we get

$$\frac{dv}{dt} = 2\gamma + 4 \cdot 3\delta t^2$$

$$\Rightarrow a = \frac{dv}{dt} = 2\gamma + 12\delta t^2 \quad \left(\because \frac{dv}{dt} = a \right)$$

At $t = 0$, $v = \beta$ and $a = 2\gamma$

$$\therefore \frac{\text{Initial velocity}}{\text{Initial acceleration}} = \frac{\beta}{2\gamma}$$

Example 0.11 The position of a particle moving along the X -axis varies with time t as $x = 6t - t^2 + 4$. Find the time interval during which the particle is moving along the positive x -direction.

CHECK POINT 0.2

1. Differentiate the following function

(i) $y = 3x^4 + 2\frac{1}{x^2} + \log x$ (ii) $y = (x^2 + 1)(x + 2)$

(iii) $y = \frac{3x^2}{x+1}$ (iv) $y = \sin x$

(v) $y = \tan(x^2 + 3x + 1)$

(vi) $y = 3 \cos x + 7e^x + \log(x^2 + 1)$

Ans. (i) $12x^3 - 4\frac{1}{x^3} + \frac{1}{x}$ (ii) $3x^2 + 4x + 1$

(iii) $\frac{3x^2 + 6x}{(x+1)^2}$ (iv) $\cos x$

Sol. Given, $x = 6t - t^2 + 4$

$$v = \frac{dx}{dt} = \frac{d}{dt}(6t - t^2 + 4) = 6 - 2t + 0 = 2(3 - t)$$

At $t < 3$, $v > 0$, then the particle is moving along the positive x -direction.

At $t > 3$, $v < 0$, then particle is moving along the negative x -direction.

At $t = 3$, $v = 0$

For time-interval $t = 0$ to $t = 3$, the particle is moving along the positive x -direction.

(v) $(2x + 3) \sec^2(x^2 + 3x + 1)$ (vi) $-3 \sin x + 7e^x + \frac{2x}{x^2 + 1}$

2. A particle is moving with velocity $v = t^3 - 6t^2 + 4$, where v is in m/s and t is in seconds. At what time will the velocity be maximum/minimum and what is it equal to?

Ans. $v_{\max} = 4$ m/s at $t = 0$ s and $v_{\min} = -28$ m/s at $t = 4$ s

3. If the time and displacement of particle along the positive X -axis are related as $t = (x^2 - 1)^{\frac{1}{2}}$, then find the acceleration in terms of x .

Ans. $\frac{1}{x^3}$

INTEGRATION

It means summation. It is the process of finding the function, whose derivative is given. In other word, integration is the reverse process of differentiation. Its symbol is $\int$.

Consider a function $f(x)$, whose derivative w.r.t. x is equal to $f'(x)$, then $f(x) + C$ is called integration of $f'(x)$, where C is called constant of integration. Symbolically, it is written as $\int f'(x)dx = f(x) + C$

Here, $f'(x) dx$ is called element of integration and $\int$ is called indefinite integral.

Some basic formulae of integration

$$\bullet \int x^n dx = \frac{x^{n+1}}{n+1} + C; n \neq -1 \quad \left(\because \frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n \right)$$

$$\bullet \int 1 dx = x + C \quad \left(\because \frac{d}{dx} x = 1 \right)$$

$$\bullet \int \sin x dx = -\cos x + C \quad \left(\because \frac{d}{dx} (-\cos x) = \sin x \right)$$

$$\bullet \int \cos x dx = \sin x + C \quad \left(\because \frac{d}{dx} (\sin x) = \cos x \right)$$

$$\bullet \int \frac{1}{x} dx = \log_e x + C \quad \left(\because \frac{d}{dx} \log_e x = \frac{1}{x} \right)$$

$$\bullet \int e^x dx = e^x + C \quad \left(\because \frac{d}{dx} e^x = e^x \right)$$

Example 0.12 Evaluate the following integrals.

(i) $\int \left(e^x + \frac{1}{x} + 2x^2 + 3 \right) dx$

(ii) $\int \left(\cos x + 3x^{1/2} + \frac{3}{x} + \frac{4}{x^2} \right) dx$

Sol. (i) Let $I = \int \left(e^x + \frac{1}{x} + 2x^2 + 3 \right) dx$

$$= \int e^x dx + \int \frac{1}{x} dx + 2 \int x^2 dx + 3 \int 1 dx$$

$$= e^x + \log_e x + 2 \cdot \frac{x^{2+1}}{2+1} + 3x + C$$

$$= e^x + \log_e x + \frac{2}{3}x^3 + 3x + C$$

$$\begin{aligned} \text{(ii) Let } I &= \int \left(\cos x + 3x^{1/2} + \frac{3}{x} + \frac{4}{x^2} \right) dx \\ &= \int \cos x \, dx + 3 \int x^{1/2} dx + 3 \int \frac{1}{x} dx + 4 \int \frac{1}{x^2} dx \\ &= \sin x + 3 \cdot \frac{x^{(1/2)+1}}{(1/2)+1} + 3 \log_e x + 4 \frac{x^{-2+1}}{-2+1} + C \\ &= \sin x + 2x^{3/2} + 3 \log_e x - \frac{4}{x} + C \end{aligned}$$

Other important formula of integration

$$I = \int f'(ax+b) \, dx = \frac{f(ax+b)}{\left[\frac{d}{dx}(ax+b) \right]}$$

Example 0.13 Evaluate the following

$$\text{(i) } \int (2x+1)^3 dx \qquad \text{(ii) } \int \left(\frac{1}{a-x} \right) dx$$

$$\text{(iii) } \int (x^2 + 3x + 4)^4 dx \qquad \text{(iv) } \int \sin(2x^2) \, dx$$

$$\begin{aligned} \text{Sol. (i) Let } I &= \int (2x+1)^3 dx = \frac{(2x+1)^{3+1}}{(3+1) \left[\frac{d}{dx}(2x+1) \right]} + C \\ &= \frac{(2x+1)^4}{4 \cdot 2} + C = \frac{(2x+1)^4}{8} + C \end{aligned}$$

$$\begin{aligned} \text{(ii) Let } I &= \int \left(\frac{1}{a-x} \right) dx = \frac{\log(a-x)}{\frac{d}{dx}(a-x)} + C \\ &= \frac{\log(a-x)}{-1} + C = -\log(a-x) + C \end{aligned}$$

$$\begin{aligned} \text{(iii) Let } I &= \int (x^2 + 3x + 4)^4 dx \\ &= \frac{(x^2 + 3x + 4)^{4+1}}{(4+1) \frac{d}{dx}(x^2 + 3x + 4)} + C \\ &= \frac{(x^2 + 3x + 4)^5}{5(2x+3)} + C \end{aligned}$$

$$\begin{aligned} \text{(iv) Let } I &= \int \sin(2x^2) \, dx = \frac{-\cos(2x^2)}{\frac{d}{dx}(2x^2)} + C \\ &= \frac{-\cos(2x^2)}{4x} + C \end{aligned}$$

Definite integral

When a function is integrated between a lower limit and an upper limit, it is called a definite integral.

A definite integral has definite value.

If $\frac{d}{dx} f(x) = f'(x)$, then $\int_a^b f'(x) \, dx$ is called definite integral,

where a and b are called lower and upper limit, respectively of variable x .

After carrying out integration, the result is evaluated between upper and lower limits as shown below

$$\int_a^b f'(x) \, dx = [f(x)]_a^b = f(b) - f(a)$$

Example 0.14 Evaluate the following

$$\text{(i) } \int_0^2 (4x^3 + 2x^2 + 2x + 1) \, dx$$

$$\text{(ii) } \int_0^{\pi/4} (\sin x + \cos x) \, dx$$

$$\text{(iii) } \int_2^4 \frac{dx}{x}$$

$$\text{Sol. (i) Let } I = \int_0^2 (4x^3 + 2x^2 + 2x + 1) \, dx$$

$$\begin{aligned} &= 4 \int_0^2 x^3 \, dx + 2 \int_0^2 x^2 \, dx + 2 \int_0^2 x \, dx + \int_0^2 1 \, dx \\ &= 4 \left[\frac{x^4}{4} \right]_0^2 + 2 \left[\frac{x^3}{3} \right]_0^2 + 2 \left[\frac{x^2}{2} \right]_0^2 + [x]_0^2 \\ &= 4 \left(\frac{2^4 - 0^4}{4} \right) + 2 \left(\frac{2^3 - 0^3}{3} \right) + 2 \left(\frac{2^2 - 0^2}{2} \right) + (2 - 0) \end{aligned}$$

$$= 16 + \frac{2}{3} \times 8 + 4 + 2$$

$$I = 22 + \frac{16}{3} = \frac{66+16}{3} = \frac{82}{3}$$

$$\text{(ii) Let } I = \int_0^{\pi/4} (\sin x + \cos x) \, dx$$

$$= \int_0^{\pi/4} \sin x \, dx + \int_0^{\pi/4} \cos x \, dx$$

$$= [-\cos x]_0^{\pi/4} + [\sin x]_0^{\pi/4}$$

$$= -\cos \frac{\pi}{4} + \cos 0 + \sin \frac{\pi}{4} - \sin 0$$

$$= -\frac{1}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}} - 0 = 1$$

$$\text{(iii) Let } I = \int_2^4 \frac{dx}{x} = [\log_e x]_2^4 = \log_e 4 - \log_e 2$$

$$= \log_e \left(\frac{4}{2} \right) = \log_e 2 \quad \left[\because \log_e a - \log_e b = \log_e \left(\frac{a}{b} \right) \right]$$

Example 0.15 Evaluate the following

$$(i) \int_1^2 e^{(x+4)} dx \quad (ii) \int_0^{\pi/4} \cos(2x^2 + x) dx$$

$$(iii) \int_1^2 \frac{dx}{(3x+4)} \quad (iv) \int_0^4 (3x^2 + 4x + 5) dx$$

Sol. (i) Let $I = \int_1^2 e^{(x+4)} dx = \left| \frac{e^{x+4}}{\frac{d}{dx}(x+4)} \right|_1^2$

$$= \left| \frac{e^{x+4}}{1} \right|_1^2 = e^{2+4} - e^{1+4} = e^6 - e^5 = e^5(e-1)$$

(ii) Let $I = \int_0^{\pi/4} \cos(2x^2 + x) dx$

$$= \left| \frac{\sin(2x^2 + x)}{\frac{d}{dx}(2x^2 + x)} \right|_0^{\pi/4} = \left| \frac{\sin(2x^2 + x)}{4x + 1} \right|_0^{\pi/4}$$

$$= \frac{\sin\left(2 \cdot \frac{\pi^2}{16} + \frac{\pi}{4}\right)}{4 \cdot \frac{\pi}{4} + 1} - \frac{\sin(2 \times 0^2 + 0)}{4 \times 0 + 1}$$

$$= \frac{\sin\left(\frac{\pi^2 + 2\pi}{8}\right)}{(\pi + 1)} - \frac{\sin 0}{1}$$

$$= \frac{\sin\left(\frac{\pi^2 + 2\pi}{8}\right)}{\pi + 1} - 0 = \left(\frac{1}{\pi + 1}\right) \sin\left(\frac{\pi^2 + 2\pi}{8}\right)$$

(iii) Let $I = \int_1^2 \frac{dx}{(3x+4)} = \left| \frac{\log(3x+4)}{\frac{d}{dx}(3x+4)} \right|_1^2$

$$= \left| \frac{\log(3x+4)}{3} \right|_1^2$$

$$= \frac{\log(3 \times 2 + 4) - \log(3 \times 1 + 4)}{3}$$

$$= \frac{1}{3} [\log 10 - \log 7] = \frac{1}{3} \log\left(\frac{10}{7}\right)$$

(iv) Let $I = \int_0^4 (3x^2 + 4x + 5) dx$

$$= \left| 3 \frac{x^{2+1}}{2+1} + \frac{4x^{1+1}}{1+1} + 5x \right|_0^4$$

$$= \left| 3 \frac{x^3}{3} + 4 \frac{x^2}{2} + 5x \right|_0^4$$

$$= |x^3 + 2x^2 + 5x|_0^4$$

$$= 4^3 + 2 \times 4^2 + 5 \times 4 - 0$$

$$= 64 + 32 + 20 = 64 + 52 = 116$$

Application of integration in physics

We know that, $v = \frac{ds}{dt}$, $a = \frac{dv}{dt}$ or $v \frac{dv}{ds}$

If the displacement is given and we have to find the velocity and acceleration, then we use differentiation.

But if the acceleration is given and we have to find the velocity and displacement, then we use integration.

Example 0.16 A particle is moving under constant acceleration $a = 3t + 4t^2$. If the position and velocity of the particle at start, i.e., $t = 0$ are x_0 and v_0 , respectively, then find the displacement and velocity as a function of time t .

Sol.

O	A
$t = 0$	$t = t$
$x = x_0$	$x = x$
$v = v_0$	$v = v$

Given, $a = 3t + 4t^2$

$$\frac{dv}{dt} = 3t + 4t^2 \quad \left(\because a = \frac{dv}{dt} \right)$$

$$\Rightarrow \int_{v_0}^v dv = \int_0^t (3t + 4t^2) dt \quad (\text{Integrating both sides})$$

$$\Rightarrow |v|_{v_0}^v = \left| 3 \frac{t^2}{2} + 4 \frac{t^3}{3} \right|_0^t$$

$$\Rightarrow (v - v_0) = \frac{3}{2}(t^2 - 0) + \frac{4}{3}(t^3 - 0)$$

$$\Rightarrow v = v_0 + \frac{3}{2}t^2 + \frac{4}{3}t^3 \quad \dots(i)$$

$$\Rightarrow \frac{dx}{dt} = v_0 + \frac{3}{2}t^2 + \frac{4}{3}t^3 \quad \left(\because v = \frac{dx}{dt} \right)$$

$$\Rightarrow \int_{x_0}^x dx = \int_0^t \left(v_0 + \frac{3}{2}t^2 + \frac{4}{3}t^3 \right) dt$$

$$\Rightarrow x - x_0 = \left| v_0 t + \frac{3}{2} \cdot \frac{t^3}{3} + \frac{4}{3} \cdot \frac{t^4}{4} \right|_0^t = v_0 t + \frac{1}{2}t^3 + \frac{1}{3}t^4 - 0$$

$$\Rightarrow x - x_0 = v_0 t + \frac{1}{2}t^3 + \frac{1}{3}t^4 \Rightarrow x = x_0 + v_0 t + \frac{1}{2}t^3 + \frac{1}{3}t^4$$

Average value

If the velocity is variable and depends on time t , then find the average value of velocity ($\bar{v}$) for time interval $t = t_1$ to $t = t_2$.

$$\text{Let } v = f(t); \bar{v} = \frac{\int_{t_1}^{t_2} v dt}{\int_{t_1}^{t_2} dt}$$

To find the average value of square of velocity

$$\bar{v}^2 = \frac{\int_{t_1}^{t_2} v^2 dt}{\int_{t_1}^{t_2} dt} \Rightarrow \bar{v}^4 = \frac{\int_{t_1}^{t_2} v^4 dt}{\int_{t_1}^{t_2} dt}$$

If velocity is a function of displacement, $v = f(x)$, for average of v from $x = x_1$ to $x = x_2$

$$\Rightarrow \bar{v} = \frac{\int_{x_1}^{x_2} v dt}{\int_{x_1}^{x_2} dt}$$

The above procedure can be applied to find the average value of any quantity; like acceleration, force, etc.

Example 0.17 The velocity of a particle is given by $v = v_0 \sin \omega t$, where v_0 is constant and $\omega = 2\pi / T$. Find the average velocity in time interval $t = 0$ to $t = T/2$.

Sol. We have, $v = v_0 \sin \omega t$

$$\begin{aligned} \text{Average velocity, } \bar{v} &= \frac{\int_0^{T/2} v_0 \sin(\omega t) dt}{\int_0^{T/2} dt} = \frac{v_0 [-\cos(\omega t)]_0^{T/2}}{\left| t \right|_0^{T/2}} \\ &= \frac{v_0 \left[-\cos\left(\omega \frac{T}{2}\right) + \cos 0 \right]}{\left(\frac{T}{2} - 0\right)} \\ &= \frac{\frac{v_0}{T} \left[-\cos\left(\frac{2\pi}{T} \cdot \frac{T}{2}\right) + 1 \right]}{\frac{T}{2}} \quad \left(\because \omega = \frac{2\pi}{T} \right) \\ &= \frac{2v_0 T}{2\pi} \frac{(1 - \cos \pi)}{T} = \frac{v_0}{\pi} [1 - (-1)] \\ &= \frac{v_0}{\pi} (1 + 1) = \frac{2v_0}{\pi} \quad (\because \cos \pi = \cos 180^\circ = -1) \end{aligned}$$

CHECK POINT 0.3

1. Evaluate

(i) $\int \left(\sin x + \frac{1}{x} + 2\frac{1}{x^2} + 3x^3 \right) dx$

(ii) $\int (3\cos x + e^x + 4x^2 + x + 5) dx$

Ans. (i) $-\cos x + \log x - \frac{2}{x} + \frac{3}{4}x^4 + C$

(ii) $3\sin x + e^x + \frac{4}{3}x^3 + \frac{x^2}{2} + 5x + C$

2. Evaluate the following

(i) $\int (2x^2 + 4x + 1)^2 dx$ (ii) $\int \frac{1}{x^2 + 2} dx$

(iii) $\int \cos(x + 2) dx$

Ans. (i) $\frac{1}{12} \frac{(2x^2 + 4x + 1)^3}{(x + 1)} + C$ (ii) $\frac{1}{2x} \log(x^2 + 2) + C$

(iii) $\sin(x + 2) + C$

3. Evaluate

(i) $\int_1^3 \left(4x + \frac{1}{x} + 1 \right) dx$ (ii) $\int_0^{\pi/4} (\sin x - \cos x) dx$

Ans. (i) $18 + \log 3$

(ii) $1 - \sqrt{2}$

4. Evaluate the following

(i) $\int_0^2 \frac{dx}{(x^2 + 4x + 1)}$

(ii) $\int_1^3 (4x^3 + 3x^2 + 2x + 1) dx$

Ans. (i) $\frac{1}{8} \log 13$ (ii) 116

5. A particle is moving in a straight line under acceleration $a = kt$, where k is a constant. Find the velocity in term of t , if the motion starts from rest.

Ans. $\frac{kt^2}{2}$

6. A particle is moving in a straight line such that its velocity varies as $v = v_0 e^{-\lambda t}$, where λ is a constant. Find the average velocity during the time interval in which the velocity decrease from v_0 to $\frac{v_0}{2}$.

Ans. $\frac{v_0}{\log_e(2)}$

GRAPHS

It is defined as pictorial representation showing the relation between variable quantities, typically two variables, each measured along one of a pair of axes at right angles.

- (i) If a graph is concave up (curved upward), the slope is increasing.

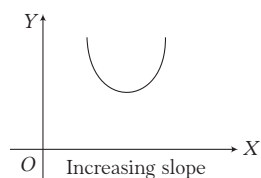


Fig. 0.5

- (ii) If a graph is concave downward (curved downward), the slope is decreasing.

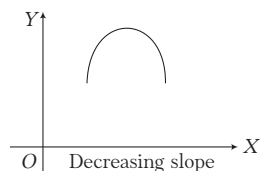


Fig. 0.6

- (iii) If the graph is a straight line, the slope is constant.

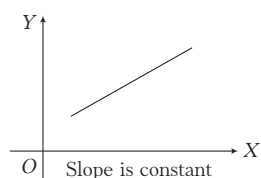


Fig. 0.7

- (iv) The general equation of a straight line is of the form $y = mx + c$ where, m is the slope of line, $m = \tan \theta$ and c is the intercept on the Y-axis.

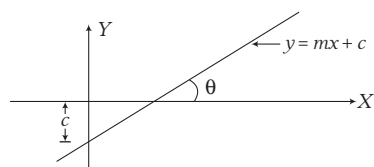


Fig. 0.8

- (a) If m is +ve, i.e., $0^\circ < \theta < 90^\circ$, then lines will be of the type,

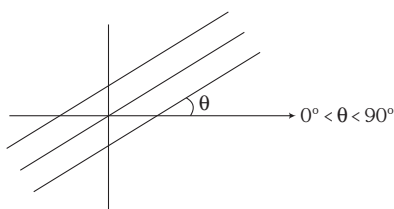


Fig. 0.9

- (b) If m is -ve, i.e., $90^\circ < \theta < 180^\circ$, then the lines will be of the type,

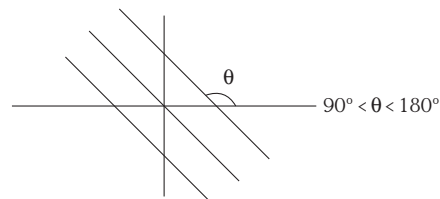


Fig. 0.10

- (c) If c is +ve, then the lines will cut the Y-axis above the origin.

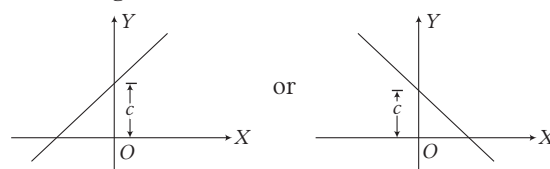


Fig. 0.11

- (d) If c is -ve, then the lines will cut the Y-axis below the origin.

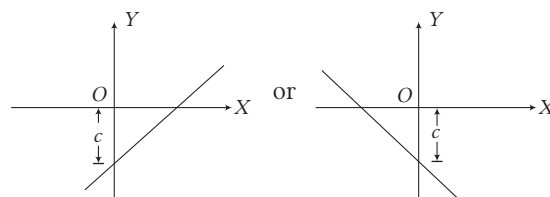


Fig. 0.12

- (e) If $c = 0$, then the lines will pass through the origin.

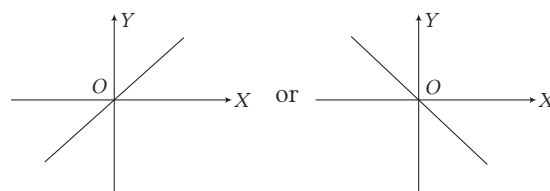


Fig. 0.13

- (v) **Parabola** Some standard forms of parabola are as follows

- (a) $y^2 = kx$, a parabola passing through the origin and opens rightward.

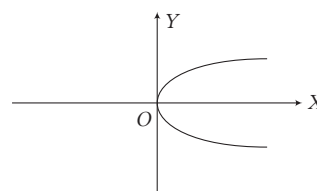


Fig. 0.14

- (b) $y^2 = -kx$, a parabola passing through the origin and opens leftward.

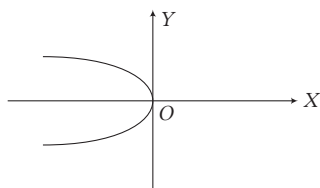


Fig. 0.15

- (c) $x^2 = ky$, a parabola passing through the origin and opens upward.

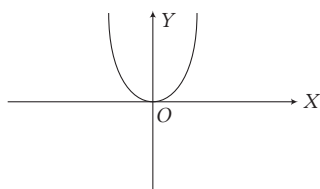


Fig. 0.16

- (d) $x^2 = -ky$, a parabola passing through the origin and opens downward.

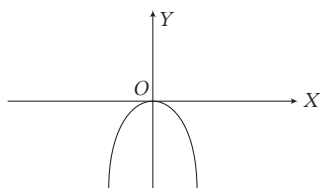


Fig. 0.17

- (vi) **Exponential graph** The most popular graph based on exponential (e) are

- (a) $y = e^{-x}$

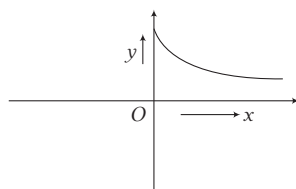


Fig. 0.18

- (b) $y = 1 - e^{-x}$

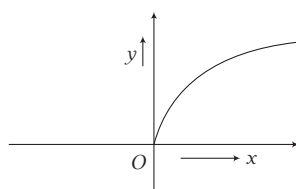


Fig. 0.19

- (vii) **Circle** If equation of circle is $x^2 + y^2 = a^2$, where centre of circle $\equiv (0, 0)$ and radius of circle $= a$

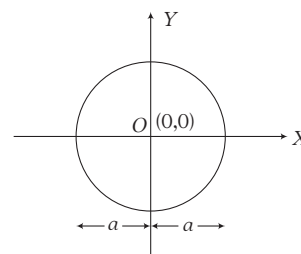


Fig. 0.20

- (viii) **Ellipse** Equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$), where, coefficient of $x^2 \neq$ coefficient of y^2 .

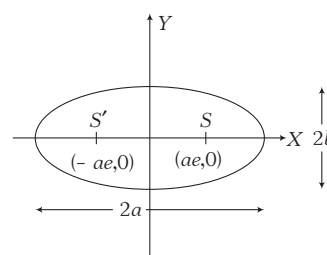


Fig. 0.21

Here, $2a$ is major axis and $2b$ is minor axis.

$$\text{Eccentricity, } e = \sqrt{1 - \frac{b^2}{a^2}}$$

For ellipse, $e < 1$

Focus, $S \equiv (ae, 0)$, $S' \equiv (-ae, 0)$

Area of ellipse $= \pi ab$

- (ix) **Hyperbola** Equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

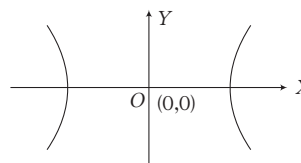


Fig. 0.22

- (x) **Rectangular Hyperbola** If $x \propto \frac{1}{y}$ or $xy = \text{constant}$, then

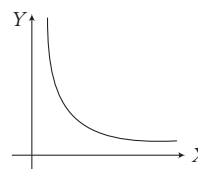


Fig. 0.23

Sketches of some standard curves

Equation of curve	Diagram/Sketch
1. (a) Straight lines $x = a$ and $x = -a$, where $a > 0$	
(b) Straight lines $y = b$ and $y = -b$, where $b > 0$	
2. Straight lines $y = x$ and $y = -x$	
3. Straight lines $\frac{x}{a} + \frac{y}{b} = 1$, $a \neq b$ $x + y = a$, $a = b$	
4. Modulus function $y = x $ $y = \begin{cases} x, & \text{for } x \geq 0 \\ -x, & \text{for } x < 0 \end{cases}$	
5. Circle $x^2 + y^2 = a^2$ Centre $= (0, 0)$ Radius $= a$	

Equation of curve	Diagram/Sketch
6. (a) Parabola $y^2 = 4ax$ and $y^2 = -4ax$ Vertex , $O = (0, 0)$ Focus , $S = (a, 0)$ and $(-a, 0)$ Length of latusrectum $= 4a$	
(b) Parabola $x^2 = 4ay$ and $x^2 = -4ay$ Vertex , $O = (0, 0)$ Focus , $S = (0, a)$ and $(0, -a)$ Length of latusrectum $= 4a$	
7. (a) Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, when $a > b$ Vertices $= (\pm a, 0)$ Centre , $O = (0, 0)$	
(b) Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, when $a < b$ Vertices $= (0, \pm b)$ Centre , $O = (0, 0)$	
8. (a) Sine function $y = \sin x$	
(b) Cosine function $y = \cos x$	

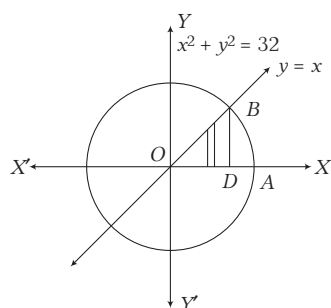
Example 0.18 Find the area of the region in the first quadrant enclosed by the X-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$.

Sol. We have, circle

$$x^2 + y^2 = 32 \quad \dots(i)$$

and line $y = x \quad \dots(ii)$

It is clear from the figure that region for finding area is OABO.



On putting the value of y from Eq. (ii) in Eq. (i), we get

$$x^2 + x^2 = 32$$

$$\Rightarrow 2x^2 = 32 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$$

From Eq. (ii), we get $y = \pm 4$

Thus, line and circle intersect each other at two points $(4, 4)$ and $(-4, -4)$. So, coordinates of $A(4\sqrt{2}, 0)$, and $B(4, 4)$ in I quadrant.

Now, area of OABO = Area of ODBO + Area of DABD

$$\begin{aligned} &= \int_0^{4\sqrt{2}} y \, dx + \int_4^{4\sqrt{2}} y \, dx \\ &= \int_0^4 x \, dx + \int_4^{4\sqrt{2}} \sqrt{32 - x^2} \, dx \quad [\text{From Eq. (i) and Eq. (ii)}] \\ &= \left[\frac{x^2}{2} \right]_0^4 + \int_4^{4\sqrt{2}} \sqrt{32 - x^2} \, dx \\ &= \frac{4^2 - 0^2}{2} + \left[\frac{x}{2} \sqrt{32 - x^2} + \frac{32}{2} \sin^{-1} \left(\frac{x}{4\sqrt{2}} \right) \right]_4^{4\sqrt{2}} \\ &= \frac{16}{2} + \left[\frac{4\sqrt{2}}{2} \sqrt{32 - 32} - \frac{4}{2} \sqrt{32 - 16} + \frac{32}{2} \sin^{-1} \left(\frac{4\sqrt{2}}{4\sqrt{2}} \right) - \frac{32}{2} \sin^{-1} \left(\frac{4}{4\sqrt{2}} \right) \right] \\ &= 8 + 0 - 2 \times 4 + 16 \sin^{-1}(1) - 16 \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \\ &= 8 - 8 + 16 \left(\frac{\pi}{2} \right) - 16 \left(\frac{\pi}{4} \right) \\ &= 8\pi - 4\pi = 4\pi \text{ sq unit} \end{aligned}$$

Hence, the required area of region is 4π sq units.

Example 0.19 Find the area of region bounded by the curve $y^2 = 4x$ and the line $x = 4$.

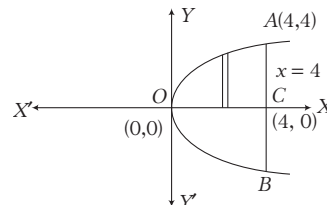
Sol. Given curve is a parabola, $y^2 = 4x \quad \dots(i)$

Which is of the form $y^2 = 4aX$ having vertex $(0, 0)$

and line, $x = 4 \quad \dots(ii)$

Then, the region for which we have to find area is OACBO.

Also, region OACO is symmetrical about X-axis.



On putting the value of x from Eq. (ii) in Eq. (i), we get

$$y^2 = 4(4) = 16 \Rightarrow y = \pm 4$$

$\therefore$ Area of bounded region OACBO = 2(Area of region OACO)

($\because$ Parabola is symmetrical about X-axis)

$$\begin{aligned} &= 2 \int_0^4 y(\text{parabola}) \, dx = 2 \int_0^4 2 \cdot x^{1/2} \, dx \quad [\text{From Eq. (i)}] \\ &= 4 \int_0^4 x^{1/2} \, dx = 4 \left[\frac{x^{3/2}}{3/2} \right]_0^4 = 4 \cdot \frac{2}{3} [x^{3/2}]_0^4 \\ &= \frac{8}{3} [4^{3/2} - 0] = \frac{8}{3} \times 4\sqrt{4} = \frac{8}{3} \times 4 \times 2 = \frac{64}{3} \text{ sq units} \end{aligned}$$

Hence, the required area is $\frac{64}{3}$ sq units.

Example 0.20 Find the area of the region bounded by the line $y = 3x + 2$, the X-axis and the ordinates $x = -1$ and $x = 1$.

Sol. Given lines are $y = 3x + 2 \quad \dots(i)$

$y = 0$ [on X-axis] $\dots(ii)$

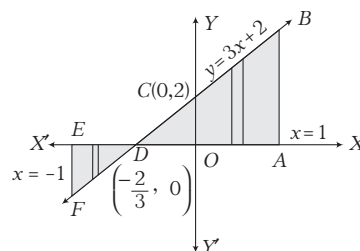
$x = -1 \quad \dots(iii)$

and $x = 1 \quad \dots(iv)$

Now, table for $y = 3x + 2$

x	0	$-\frac{2}{3}$
y	2	0

The given region bounded by $y = 3x + 2$, X-axis and the ordinates $x = -1$ and $x = 1$ is represented by shaded region.



∴ Required area

$$\begin{aligned}
 &= \text{Area of region EFDE} + \text{Area of region ABDA} \\
 &= \left| \int_{-1}^{-2/3} y_1 dx \right| + \int_{-2/3}^1 y_2 dx \\
 &= \left| \int_{-1}^{-2/3} (3x+2) dx \right| + \int_{-2/3}^1 (3x+2) dx \\
 &= \left| \left[\frac{3x^2}{2} + 2x \right]_{-1}^{-2/3} \right| + \left[\frac{3x^2}{2} + 2x \right]_{-2/3}^1 \\
 &= \left| \left[\frac{3}{2} \left(\frac{4}{9} \right) - \frac{4}{3} \right] - \left(\frac{3}{2} - 2 \right) \right| + \left[\left(\frac{3}{2} + 2 \right) - \left\{ \frac{3}{2} \left(\frac{4}{9} \right) - \frac{4}{3} \right\} \right] \\
 &= \left| \left[\frac{2}{3} - \frac{4}{3} + \frac{1}{2} \right] \right| + \left[\frac{7}{2} - \frac{2}{3} + \frac{4}{3} \right] \\
 &= \left| \left[\frac{4-8+3}{6} \right] \right| + \left[\frac{21-4+8}{6} \right] \\
 &= \left| \left[-\frac{1}{6} \right] \right| + \frac{25}{6} = \frac{1}{6} + \frac{25}{6} = \frac{26}{6} \\
 &= \frac{13}{3} \text{ sq units}
 \end{aligned}$$

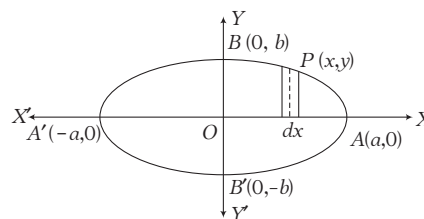
Hence, the required area is $\frac{13}{3}$ sq units.

Example 0.21 Find the area of the region bounded by the

$$\text{ellipse } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Sol. We have, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$... (i)

Since, power of x and y both are even in the equation of the curve. So, it is symmetrical about the axes as shown in figure.



Area enclosed by the ellipse = 4 [Area enclosed by the ellipse and coordinate axes in 1st quadrant]

$$\begin{aligned}
 \Rightarrow A &= 4 \int_0^a y dx = 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx \quad [\text{From Eq. (i)}] \\
 &= \frac{4b}{a} \left[\frac{1}{2} x \sqrt{a^2 - x^2} + \frac{1}{2} a^2 \sin^{-1} \frac{x}{a} \right]_0^a \\
 &= \frac{4b}{a} \left\{ 0 + \frac{1}{2} a^2 \sin^{-1} (1) \right\} = \frac{4b}{a} \times \frac{1}{2} a^2 \left(\frac{\pi}{2} \right) = \pi ab \text{ sq units}
 \end{aligned}$$

CHECK POINT 0.4

1. Find the area of region bounded by the curve $y = x^3$, $y = x + 6$ and $x = 0$.

Ans. 10 sq units

2. Find the area of the region included between the parabola $y = \frac{3x^2}{4}$ and the line $3x - 2y + 12 = 0$.

Ans. 27 sq units

3. For the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$, evaluate the area of the region under the curve and above the X -axis.

Ans. 3π sq units

4. Find the area of the region bounded by $y = \sqrt{x}$ and $y = x$.

Ans. $\frac{1}{6}$ sq units

Units, Dimensions & Error Analysis

In this chapter, we will discuss about units and dimensions of different physical quantities and errors that occur in measurement.

PHYSICAL QUANTITIES AND UNITS

All the quantities which are used to describe the laws of physics are called **physical quantities**, *e.g.* length, mass, volume, etc.

To express the measurement of a physical quantity, we need to know two things as given below

- (i) The unit in which the quantity is measured.
- (ii) The numerical value or the magnitude of the quantity.

i.e. The number of times that unit is contained in the given physical quantity
 $= nu$

$$\therefore n \propto \frac{1}{u} \Rightarrow nu = \text{constant}$$

Here, n = numerical value of the physical quantity and u = size of unit.

We may also write it as

$$n_1 u_1 = n_2 u_2$$

where, n_1 and n_2 are values of the physical quantity in two different units u_1 and u_2 .

The standard amount of a physical quantity chosen to measure the physical quantity of same kind is called a **physical unit**.

The essential requirements of physical unit are given below

- (i) It should be of suitable size.
- (ii) It should be easily accessible.
- (iii) It should not vary with time.
- (iv) It should be easily reproducible.
- (v) It should not depend on physical conditions like pressure, volume, etc.

Inside

1 Physical quantities and units

- System of units

2 Dimensions of physical quantities

- Applications of dimensional analysis
- Defects or limitations of dimensional analysis

3 Significant figures

- Rules to determine significant figures
- Mathematical operations of significant figures
- Rounding off
- Order of magnitude

4 Error in measurement

- Expression of errors
- Combination of errors

System of units

A complete set of units which is used to measure all kinds of fundamental and derived quantities is called a system of units.

Some of the commonly used systems of units are as follows

- (i) **CGS system** In this system, the units of length, mass and time are centimetre (cm), gram (g) and second (s), respectively. The unit of force in this system is dyne and that of work or energy is erg.
- (ii) **FPS system** In this system, the units of length, mass and time are foot (ft), pound (lb) and second (s), respectively. The unit of force in this system is poundal.
- (iii) **MKS system** In this system, the units of length, mass and time are metre (m), kilogram (kg) and second (s), respectively. The unit of force in this system is newton (N) and that of work or energy is joule (J).
- (iv) **International system (SI)** This system of units helps in revolutionary changes over the MKS system and is known as **rationalised MKS system**. It is helpful to obtain all the physical quantities in physics.

Note

- (i) The FPS system is not a metric system. This system is not in much use these days.
- (ii) The drawback of CGS system is that many of the derived units on this system are inconveniently small.

Fundamental quantities and fundamental units

Those physical quantities which are independent of each other and not defined in terms of other physical quantities, are called **fundamental quantities** or **base quantities**. The units of these quantities are called **fundamental** or **base units**.

Derived quantities and derived units

The quantities which can be expressed in terms of the fundamental quantities are called **derived quantities**. The units of these quantities are called **derived units**.

e.g. Unit of speed = ms^{-1} can be derived from fundamental units, *i.e.* unit of length and time as

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{\text{m}}{\text{s}} = \text{ms}^{-1}$$

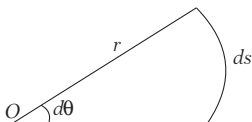
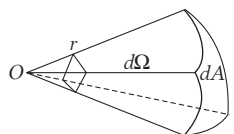
Supplementary quantities and supplementary units

Other than fundamental and derived quantities, there are two more quantities called as **supplementary quantities**. The units of these quantities are known as **supplementary units**.

Fundamental quantities and their SI units

Base quantity	SI units	
	Name/ Symbol	Definition
Length	Metre (m)	It is defined by taking the fixed numerical value of the speed of light in vacuum c to be 299792458 when expressed in the unit ms^{-1} , where the second is defined in terms of the caesium frequency $\Delta\nu_{\text{Cs}}$.
Mass	Kilogram (kg)	It is defined by taking the fixed numerical value of the Planck constant h to be $6.62607015 \times 10^{-34}$ when expressed in the unit Js, which is equal to $\text{kg}\cdot\text{m}^2\text{s}^{-1}$, where the metre and the second are defined in terms of c and $\Delta\nu_{\text{Cs}}$.
Time	Second (s)	It is defined by taking the fixed numerical value of the caesium frequency $\Delta\nu_{\text{Cs}}$, the unperturbed ground state hyperfine transition frequency of the caesium-133 atom, to be 9192631770 when expressed in the unit Hz, which is equal to s^{-1} .
Electric current	Ampere (A)	It is defined by taking the fixed numerical value of the elementary charge e to be $1.602176634 \times 10^{-19}$ when expressed in the unit C.
Thermodynamic temperature	Kelvin (K)	It is defined by taking the fixed numerical value of the Boltzmann constant k to be 1.380649×10^{-23} when expressed in the unit JK^{-1} , which is equal to $\text{kg}\cdot\text{m}^2\text{s}^{-2}\text{K}^{-1}$, where the kilogram, metre and second are defined in terms of h , c and $\Delta\nu_{\text{Cs}}$.
Amount of substance	Mole (mol)	One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities. This number is the fixed numerical value of the Avogadro constant N_A , when expressed in the unit mol^{-1} and is called the Avogadro number. The amount of substance, symbol n , of a system is a measure of the number of specified elementary entities. An elementary entity may be an atom, a molecule, an ion, an electron, any other particle or specified group of particles.
Luminous intensity	Candela (cd)	It is defined by taking the fixed numerical value of the luminous intensity of monochromatic radiation of frequency 540×10^{12} Hz, K_{cd} , to be 683 when expressed in the unit lm W^{-1} , which is equal to cd sr W^{-1} , or $\text{cd sr kg}^{-1}\text{m}^{-2}\text{s}^3$, where the kilogram, metre and second are defined in terms of h , c and $\Delta\nu_{\text{Cs}}$.

Supplementary quantities and their SI units

Supplementary quantity	SI units	
	Name/Symbol	Definition
Plane angle	Radian (rad)	One radian is the angle subtended at the centre by an arc equal in length to the radius of the circle.  <i>i.e.</i> $d\theta = \frac{ds}{r}$
Solid angle	Steradian (sr)	One steradian is the solid angle subtended at the centre of a sphere, by that surface of the sphere, which is equal in area, to the square of radius of the sphere.  <i>i.e.</i> $d\Omega = \frac{dA}{r^2}$

Some other units

(Not contained in SI units)

Length

- (i) 1 micron = $1\mu\text{m} = 10^{-6}\text{ m}$
- (ii) 1 nanometre = $1\text{ nm} = 10^{-9}\text{ m}$
- (iii) 1 Angstrom = $1\text{ \AA} = 10^{-10}\text{ m} = 10^{-8}\text{ cm} = 10^{-4}\mu\text{m}$
- (iv) 1 fermi = $1\text{ fm} = 10^{-15}\text{ m}$
- (v) 1 astronomical unit = $1\text{ AU} = 1.496 \times 10^{11}\text{ m}$
- (vi) 1 light year = $1\text{ ly} = 9.467 \times 10^{15}\text{ m}$
- (vii) 1 parsec = $3.08 \times 10^{16}\text{ m} = 3.26\text{ ly} = 206267\text{ AU}$

Mass

- (i) 1 quintal = 100 kg
- (ii) 1 tonne or 1 metric ton = 1000 kg = 10 quintal
- (iii) 1 megagram = 10^3 kg
- (iv) 1 gigagram = 10^6 kg
- (v) 1 teragram = 10^9 kg
- (vi) 1 slug = 14.57 kg
- (vii) 1 pound = 1 lb = 0.4536 kg

Time

- (i) 1 millisecond = 10^{-3} s
- (ii) 1 microsecond = 10^{-6} s
- (iii) 1 shake = 10^{-8} s
- (iv) 1 nanosecond = 10^{-9} s
- (v) 1 picosecond = 10^{-12} s
- (vi) 1 hour = 60 min = 3600 s
- (vii) 1 day = 24 hours = 86400 s
- (viii) 1 year = 365 days = $3.15 \times 10^7\text{ s}$
- (ix) 1 century = 100 years

Common SI Prefixes and Symbols for Multiples and Sub-multiples

Multiple			Sub-multiple		
Factor	Prefix	Symbol	Factor	Prefix	Symbol
10^{18}	Exa	E	10^{-18}	atto	a
10^{15}	Peta	P	10^{-15}	femto	f
10^{12}	Tera	T	10^{-12}	pico	p
10^9	Giga	G	10^{-9}	nano	n
10^6	Mega	M	10^{-6}	micro	μ
10^3	Kilo	k	10^{-3}	milli	m
10^2	Hecto	h	10^{-2}	centi	c
10^1	Deca	da	10^{-1}	deci	d

Example 1.1 The acceleration due to gravity is 9.8 ms^{-2} . Give its value in ft s^{-2} .

Sol. As, $1\text{ m} = 3.28\text{ ft}$

$$\begin{aligned}\therefore 9.8\text{ ms}^{-2} &= 9.8 \times 3.28\text{ ft s}^{-2} \\ &= 32.14\text{ ft s}^{-2} \\ &\approx 32\text{ ft s}^{-2}\end{aligned}$$

Example 1.2 The value of gravitational constant G in MKS system is $6.67 \times 10^{-11}\text{ N-m}^2\text{ kg}^{-2}$. What will be its value in CGS system?

Sol. Given, $G = 6.67 \times 10^{-11}\text{ N-m}^2\text{ kg}^{-2}$
 $= 6.67 \times 10^{-11}(\text{kg-ms}^{-2})\text{ m}^2\text{ kg}^{-2}$
 $= 6.67 \times 10^{-11}(\text{m}^3)(\text{s}^{-2})(\text{kg}^{-1})$
 $= 6.67 \times 10^{-11}(10^2\text{ cm})^3(\text{s})^{-2}(10^3\text{ g})^{-1}$
 $= 6.67 \times 10^{-8}\text{ cm}^3\text{ g}^{-1}\text{ s}^{-2}$
 $= 6.67 \times 10^{-8}\text{ dyne-cm}^2\text{ g}^{-2}$

Example 1.3 The wavelength of a light is of the order of $6400\text{ \AA}$. Express this in micron and metre.

Sol. As, $1\text{ \AA} = 10^{-10}\text{ m}$
 $\therefore$ Wavelength of light = $6400\text{ \AA}$
 $= 6400 \times 10^{-10}\text{ m} = 6.4 \times 10^{-7}\text{ m}$

Also, 1 micron = 10^{-6} metre

$$\therefore \text{Wavelength of light (in micron)} = \frac{6.4 \times 10^{-7}}{10^{-6}} \text{ micron} \\ = 0.64 \text{ micron}$$

Example 1.4 How many microns are there in 1 light year?

Sol. 1 ly = 9.46×10^{15} m

$$\begin{aligned} \text{As, } 1 \text{ m} &= 10^6 \text{ micron} \\ \therefore 1 \text{ ly} &= 9.46 \times 10^{15} \times 10^6 \text{ micron} \\ &= 9.46 \times 10^{21} \text{ micron} \\ &\approx 10^{22} \text{ micron (approx.)} \end{aligned}$$

Example 1.5 How many microseconds are there in 10 minutes?

Sol. As, 1 second = 10^6 microseconds

$$\begin{aligned} 10 \text{ minutes} &= 10 \times 60 \text{ seconds} \\ &= 10 \times 60 \times 10^6 \\ &= 6 \times 10^8 \text{ microseconds} \end{aligned}$$

Example 1.6 Calculate the angle of

- (i) 1° (degree)
 - (ii) $1'$ (minute of arc or arc minute) and
 - (iii) $1''$ (second of arc or arc second)
- in radians. (Use $360^\circ = 2\pi$ rad, $1^\circ = 60'$ and $1' = 60''$)

Sol. (i) $1^\circ = \frac{2\pi}{360} \text{ rad} = \frac{\pi}{180} \text{ rad}$

$$= \frac{22}{7 \times 180} \text{ rad} = 1.746 \times 10^{-2} \text{ rad}$$

(ii) $1 \text{ arc min} = 1' = \frac{1^\circ}{60} = \frac{1}{60} \times \frac{\pi}{180} \text{ rad} = 2.91 \times 10^{-4} \text{ rad}$

(iii) $1 \text{ arc second} = 1'' = \frac{1'}{60} = \frac{1^\circ}{60 \times 60} = \frac{1}{60 \times 60} \times \frac{\pi}{180} \text{ rad}$

$$= 4.85 \times 10^{-6} \text{ rad}$$

DIMENSIONS OF PHYSICAL QUANTITIES

The dimensions of a physical quantity are the powers (or exponents) to which the fundamental quantities must be raised to represent that quantity completely.

e.g. Density = $\frac{\text{Mass}}{\text{Volume}} = \frac{\text{Mass}}{(\text{Length})^3}$

or Density = (Mass) (Length) $^{-3}$... (i)

Thus, the dimensions of density are 1 in mass and -3 in length. The dimensions of all other fundamental quantities are zero.

Dimensional representation of physical quantities

For convenience, the fundamental quantities are represented by one letter symbols. The dependence of all other physical quantities on these base quantities can be expressed in terms of their dimensions.

Thus, the seven dimensions of physical quantities are represented as follows

[M]	for mass
[L]	for length
[T]	for time
[A]	for electric current
[K] or [θ]	for thermodynamic temperature
[cd]	for luminous intensity
[mol]	for amount of substance

The physical quantity that is expressed in terms of the base quantities is enclosed in square brackets.

Thus, from Eq. (i), dimensions of density can be represented as $[ML^{-3}]$.

Dimensional formula and dimensional equation

The expression of a physical quantity in terms of its dimensions is called its **dimensional formula**. e.g. Dimensional formula for density is $[ML^{-3}T^0]$, the dimensional formula of force is $[MLT^{-2}]$ and that for acceleration is $[M^0LT^{-2}]$.

An equation which contains a physical quantity on one side and its dimensional formula on the other side, is called the **dimensional equation** of that quantity.

Dimensional equations for a few physical quantities are given below

$$\begin{aligned} \text{Speed } [v] &= [M^0LT^{-1}] \\ \text{Area } [A] &= [M^0L^2T^0] \\ \text{Force } [F] &= [MLT^{-2}], \text{ etc.} \end{aligned}$$

The physical quantities having same derived units have same dimensions.

Dimensional formulae of some physical quantities

The table given below gives the dimensional formulae and SI units of some physical quantities frequently used in physics

S. No.	Physical quantity	SI units	Dimensional formula
1.	Velocity = displacement/time	m/s	$[M^0 L T^{-1}]$
2.	Acceleration = velocity/time	m/s^2	$[M^0 L T^{-2}]$
3.	Force = mass $\times$ acceleration	$kg\cdot m/s^2 = \text{newton or N}$	$[M L T^{-2}]$
4.	Work = force $\times$ displacement	$kg\cdot m^2/s^2 = \text{N}\cdot m = \text{joule or J}$	$[M L^2 T^{-2}]$
5.	Energy	Joule or J	$[M L^2 T^{-2}]$
6.	Torque = force $\times$ perpendicular distance	N-m	$[M L^2 T^{-2}]$
7.	Power = work/time	J/s or watt	$[M L^2 T^{-3}]$
8.	Momentum = mass $\times$ velocity	$kg\cdot m/s$	$[M L T^{-1}]$
9.	Impulse = force $\times$ time	N-s	$[M L T^{-1}]$
10.	Angle = arc/radius	radian or rad	$[M^0 L^0 T^0]$
11.	Strain = $\frac{\Delta L}{L}$ or $\frac{\Delta V}{V}$	No units	$[M^0 L^0 T^0]$
12.	Stress = force/area	N/m^2	$[M L^{-1} T^{-2}]$
13.	Pressure = force/area	N/m^2	$[M L^{-1} T^{-2}]$
14.	Modulus of elasticity = stress/strain	N/m^2	$[M L^{-1} T^{-2}]$
15.	Frequency = 1/time period	per second or hertz (Hz)	$[M^0 L^0 T^{-1}]$
16.	Angular velocity = angle/time	rad/s	$[M^0 L^0 T^{-1}]$
17.	Moment of inertia = (mass) $\times$ (distance) ²	$kg\cdot m^2$	$[M L^2 T^0]$
18.	Surface tension = force/length	N/m	$[M L^0 T^{-2}]$
19.	Gravitational constant = $\frac{\text{force} \times (\text{distance})^2}{(\text{mass})^2}$	$N\cdot m^2/kg^2$	$[M^{-1} L^3 T^{-2}]$
20.	Angular momentum	$kg\cdot m^2/s$	$[M L^2 T^{-1}]$
21.	Coefficient of viscosity	$N\cdot s/m^2$	$[M L^{-1} T^{-1}]$
22.	Planck's constant	J-s	$[M L^2 T^{-1}]$
23.	Specific heat (s)	J/kg-K	$[M^0 L^2 T^{-2} \theta^{-1}]$
24.	Coefficient of thermal conductivity (K)	watt/m-K	$[M L T^{-3} \theta^{-1}]$
25.	Gas constant (R)	J/mol-K	$[M L^2 T^{-2} \theta^{-1} \text{mol}^{-1}]$
26.	Boltzmann constant (k)	J/K	$[M L^2 T^{-2} \theta^{-1}]$
27.	Wien's constant (b)	m-K	$[M^0 L T^0 \theta]$
28.	Stefan's constant (σ)	$\text{watt}/m^2\cdot K^4$	$[M L^0 T^{-3} \theta^{-4}]$
29.	Electric charge	C	$[M^0 L^0 T A]$
30.	Electric intensity	N/C	$[M L T^{-3} A^{-1}]$
31.	Electric potential	volt (V)	$[M L^2 T^{-3} A^{-1}]$
32.	Capacitance	farad (F)	$[M^{-1} L^{-2} T^4 A^2]$
33.	Permittivity of free space	$C^2 N^{-1} m^{-2}$	$[M^{-1} L^{-3} T^4 A^2]$

S. No.	Physical quantity	SI units	Dimensional formula
34.	Electric dipole moment	C-m	$[M^0 L T A]$
35.	Resistance	Ohm	$[ML^2 T^{-3} A^{-2}]$
36.	Magnetic field	tesla (T) or weber/m ² (Wb/m ²)	$[ML^0 T^{-2} A^{-1}]$
37.	Coefficient of self-induction	henry (H)	$[ML^2 T^{-2} A^{-2}]$
38.	Magnetic flux	Wb (weber)	$[ML^2 T^{-2} A^{-1}]$
39.	Permeability of free space	Hm ⁻¹	$[MLT^{-2} A^{-2}]$
40.	Magnetic moment	Am ²	$[M^0 L^2 T^0 A]$

Quantities having same dimensions

S. No.	Quantities	Dimensions
1.	Strain, refractive index, relative density, angle, solid angle, phase, distance gradient, relative permeability, relative permittivity, angle of contact, Reynolds number, coefficient of friction, mechanical equivalent of heat and electric susceptibility.	$[M^0 L^0 T^0]$
2.	Mass and inertia	$[M^1 L^0 T^0]$
3.	Momentum and impulse	$[M^1 L^1 T^{-1}]$
4.	Thrust, force, weight, tension and energy gradient.	$[M^1 L^1 T^{-2}]$
5.	Pressure, stress, Young's modulus, bulk modulus, shear modulus, modulus of rigidity and energy density.	$[M^1 L^{-1} T^{-2}]$
6.	Angular momentum and Planck's constant (h).	$[M^1 L^2 T^{-1}]$
7.	Acceleration, acceleration due to gravity and gravitational field intensity.	$[M^0 L T^{-2}]$
8.	Surface tension, free surface energy (energy per unit area), force gradient and spring constant.	$[M^1 L^0 T^{-2}]$
9.	Latent heat and gravitational potential.	$[M^0 L^2 T^{-2}]$
10.	Thermal capacity, Boltzmann constant and entropy.	$[ML^2 T^{-2} \theta^{-1}]$
11.	Work, torque, internal energy, potential energy, kinetic energy, moment of force, (q^2/C), (LI^2), (qV), (V^2C), (I^2Rt), $\left(\frac{V^2}{R}t\right)$, (VIt), (pV), (RT), (mL) and ($mc\Delta T$).	$[M^1 L^2 T^{-2}]$
12.	Frequency, angular frequency, angular velocity, velocity gradient, radioactivity, $\left(\frac{R}{L}\right)$, $\left(\frac{1}{RC}\right)$ and $\left(\frac{1}{\sqrt{LC}}\right)$.	$[M^0 L^0 T^{-1}]$
13.	$\left(\frac{l}{g}\right)^{1/2}$, $\left(\frac{m}{k}\right)^{1/2}$, $\left(\frac{R}{g}\right)^{1/2}$, $\left(\frac{L}{R}\right)$, (RC), ($\sqrt{LC}$) and time.	$[M^0 L^0 T^1]$
14.	Power (VI), (I^2R) and (V^2R).	$[ML^2 T^{-3}]$

Example 1.7 Find the dimensional formulae of

- (i) coefficient of viscosity, η (ii) charge, q
 (iii) potential, V (iv) capacitance, C and
 (v) resistance, R

Some of the equations containing above quantities are

$$F = -\eta A \left(\frac{\Delta v}{\Delta l} \right), \quad q = It, \quad U = VIt,$$

$$q = CV \quad \text{and} \quad V = IR$$

where, A is the area, v is the velocity, l is the length, I is the electric current, t is the time and U is the energy.

Sol.

$$(i) \quad \eta = -\frac{F}{A} \frac{\Delta l}{\Delta v}$$

$$\therefore [\eta] = \frac{[F][l]}{[A][v]} = \frac{[MLT^{-2}][L]}{[L^2][LT^{-1}]} = [ML^{-1}T^{-1}]$$

$$(ii) \quad q = It$$

$$\therefore [q] = [I][t] = [AT]$$

$$\begin{aligned}
 \text{(iii)} \quad U &= VIt \\
 \therefore V &= \frac{U}{It} \\
 \text{or } [V] &= \frac{[U]}{[I][t]} = \frac{[ML^2T^{-2}]}{[A][T]} = [ML^2T^{-3}A^{-1}] \\
 \text{(iv)} \quad q &= CV \\
 \therefore C &= \frac{q}{V} \\
 \text{or } [C] &= \frac{[q]}{[V]} = \frac{[AT]}{[ML^2T^{-3}A^{-1}]} = [M^{-1}L^{-2}T^4A^2] \\
 \text{(v)} \quad V &= IR \\
 \therefore R &= \frac{V}{I} \\
 \text{or } [R] &= \frac{[V]}{[I]} = \frac{[ML^2T^{-3}A^{-1}]}{[A]} = [ML^2T^{-3}A^{-2}]
 \end{aligned}$$

Example 1.8 If C and R denote capacitance and resistance, then find the dimensions of CR .

Sol. The capacitance of a conductor is defined as the ratio of the charge given to the rise in the potential of the conductor,

$$\begin{aligned}
 C &= \frac{q}{V} = \frac{q^2}{W} \quad \left(\because V = \frac{W}{q} \right) \\
 C &= \frac{\text{ampere}^2 \cdot \text{s}^2}{\text{kg} \cdot \text{metre}^2 / \text{s}^2}
 \end{aligned}$$

Hence, dimensions of C are $[M^{-1}L^{-2}T^4A^2]$.

From Ohm's law, $V = iR$, therefore dimensions of resistance,

$$\begin{aligned}
 R &= \frac{V}{i} = \frac{\text{Volt}}{\text{Ampere}} \\
 &= \text{kg} \cdot \text{metre}^2 \cdot \text{s}^{-3} \cdot \text{ampere}^{-2}
 \end{aligned}$$

Dimensions of $R = [ML^2T^{-3}A^{-2}]$

$$\begin{aligned}
 \therefore \text{Dimensions of } RC &= [ML^2T^{-3}A^{-2}][M^{-1}L^{-2}T^4A^2] \\
 &= [M^0L^0T^1A^0]
 \end{aligned}$$

Example 1.9 Which amongst the following quantities is (are) dimensionless?

$$\text{(i) } \frac{\text{Work}}{\text{Energy}} \quad \text{(ii) } \sin \theta \quad \text{(iii) } \frac{\text{Momentum}}{\text{Time}}$$

Sol. (i) Since, work and energy both have the same dimensions $[ML^2T^{-2}]$, therefore their ratio is a dimensionless quantity.

(ii) $\sin \theta$, here θ represents an angle. An angle is the ratio of two lengths, i.e. arc length and radius. Therefore, θ is dimensionless, hence $\sin \theta$ is dimensionless.

$$\text{(iii) } \left[\frac{\text{Momentum}}{\text{Time}} \right] = \left[\frac{MLT^{-1}}{T} \right] = [MLT^{-2}]$$

Hence, the given ratio is not dimensionless.

Example 1.10 In the formula $x = 3yz^2$, x and y have dimensions of capacitance and magnetic induction respectively, then find the dimensions of y .

Sol. Given, $x = 3yz^2$

$$\begin{aligned}
 \Rightarrow y &= \frac{x}{3z^2} = \frac{\text{Capacitance}}{(\text{Magnetic induction})^2} \\
 [y] &= \frac{[M^{-1}L^{-2}T^4A^2]}{[MT^{-2}A^{-1}]^2} \\
 &= [M^{-3}L^{-2}T^8A^4]
 \end{aligned}$$

Applications of dimensional analysis

The method of studying a physical phenomenon on the basis of dimensions is called dimensional analysis.

The three main uses of a dimensional analysis are described in detail in the following section

1. Checking the dimensional consistency of equation

Every physical equation should be dimensionally balanced. This is called the **principle of homogeneity**. This principle states that, the dimensions of each term on both sides of an equation must be the same. On this basis, we can judge whether a given equation is correct or not. But a dimensionally correct equation may or may not be physically correct.

e.g. In the physical expression $s = ut + \frac{1}{2}at^2$, the

dimensions of s , ut and $\frac{1}{2}at^2$ all are same.

Note The physical quantities separated by the symbols $+$, $-$, $=$, $>$, $<$ etc., should have the same dimension.

Example 1.11 Show that the expression of the time period T of a simple pendulum of length l given by $T = 2\pi \sqrt{l/g}$ is dimensionally correct.

Sol. Given, $T = 2\pi \sqrt{\frac{l}{g}}$

$$\text{Dimensionally, } [T] = \sqrt{\frac{[L]}{[LT^{-2}]}} = [T]$$

As in the above equation, the dimensions of both sides are same. Therefore, the given expression is dimensionally correct.

Example 1.12 Check the correctness of the relation

$s = ut + \frac{1}{2}at^2$, where u is initial velocity, a is the acceleration, t is the time and s is the displacement.

Sol. Writing the dimensions of either side of the given relation.

$$\text{LHS} = s = \text{displacement} = [\text{M}^0\text{LT}^0]$$

$$\text{RHS} = ut = \text{velocity} \times \text{time} = [\text{M}^0\text{LT}^{-1}] [\text{T}] = [\text{M}^0\text{LT}^0]$$

$$\text{and } \frac{1}{2}at^2 = (\text{acceleration}) \times (\text{time})^2 \\ = [\text{M}^0\text{LT}^{-2}] [\text{T}]^2 = [\text{M}^0\text{LT}^0]$$

As LHS = RHS, so the relation is dimensionally correct.

Example 1.13 Write the dimensions of a and b in the relation,

$$P = \frac{b - x^2}{at}$$

where, P is power, x the distance and t the time.

Sol. The given equation can be written as $Pat = b - x^2$

$$\text{Now, } [Pat] = [b] = [x^2] \text{ or } [b] = [x^2] = [\text{M}^0\text{L}^2\text{T}^0]$$

$$\text{and } [a] = \frac{[x^2]}{[Pt]} = \frac{[\text{L}^2]}{[\text{ML}^2\text{T}^{-3}] [\text{T}]} = [\text{M}^{-1}\text{L}^0\text{T}^2]$$

Example 1.14 The velocity v of a particle depends upon the time t according to the equation $v = a + bt + \frac{c}{d + t}$. Write the dimensions of a , b , c and d .

Sol. From principle of homogeneity,

$$[a] = [v] \text{ or } [a] = [\text{LT}^{-1}]$$

$$[bt] = [v]$$

$$\text{or } [b] = \frac{[v]}{[t]} = \frac{[\text{LT}^{-1}]}{[\text{T}]} \text{ or } [b] = [\text{LT}^{-2}]$$

$$\text{Similarly, } [d] = [t] = [\text{T}]$$

$$\text{Further, } \frac{[c]}{[d + t]} = [v] \text{ or } [c] = [v][d + t]$$

$$\text{or } [c] = [\text{LT}^{-1}] [\text{T}]$$

$$\text{or } [c] = [\text{L}]$$

$$\therefore \text{Dimensions of } a = [\text{LT}^{-1}]$$

$$\text{Dimensions of } b = [\text{LT}^{-2}]$$

$$\text{Dimension of } c = [\text{L}]$$

$$\text{Dimension of } d = [\text{T}]$$

Example 1.15 The following equation gives a relation between the mass m_1 , kept on a surface of area A and the pressure p exerted on this area

$$p = \frac{(m_1 + m_2)x}{A}$$

What must be the dimensions of the quantities x and m_2 ?

Sol. Since, all the terms of a mathematical equation should have the same dimensions.

$$\text{Therefore, } [p] = \left[\frac{(m_1 + m_2)x}{A} \right] \quad \dots(i)$$

Only the quantities having same dimensions and nature can be added to each other.

Here, m_2 is added to mass m_1 .

$$\text{Hence, } [m_2] = [m_1] = [\text{M}]$$

Also, the quantity obtained by the addition of m_1 and m_2 would have the same dimensions as that of mass.

$$\therefore [m_1 + m_2] = [\text{M}]$$

Now, going back to Eq. (i),

$$[p] = \frac{[m_1 + m_2][x]}{[A]}$$

$$\Rightarrow [\text{ML}^{-1}\text{T}^{-2}] = \frac{[\text{M}][x]}{[\text{L}^2]}$$

$$\Rightarrow [\text{ML}^{-1}\text{T}^{-2}] = [\text{ML}^{-2}][x]$$

$$\Rightarrow \frac{[\text{ML}^{-1}\text{T}^{-2}]}{[\text{ML}^{-2}]} = [x] \Rightarrow [x] = [\text{LT}^{-2}]$$

Hence, the quantity x represents acceleration. In this example, it is the acceleration due to gravity g . $(m_1 + m_2)g$ represents the weight exerted by two masses m_1, m_2 on the area A .

2. To convert a physical quantity from one system of units to other system of units

This is based on the fact that the product of the numerical value (n) and its corresponding unit (u) is a constant, i.e.

$$n(u) = \text{constant} \text{ or } n_1[u_1] = n_2[u_2]$$

Suppose the dimensions of a physical quantity are a in mass, b in length and c in time. If the fundamental units in one system are M_1, L_1 and T_1 and in the other system are M_2, L_2 and T_2 , respectively. Then, we can write

$$n_1 [M_1^a L_1^b T_1^c] = n_2 [M_2^a L_2^b T_2^c] \quad \dots(i)$$

$$n_2 = n_1 \frac{u_1}{u_2} = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

Here, n_1 and n_2 are the numerical values in two system of units, respectively. Using Eq. (i), we can convert the numerical value of a physical quantity from one system of units into the other system.

Example 1.16 Find the value of 100 J on a system which has 20 cm, 250 g and half minute as fundamental units of length, mass and time.

Sol. The dimensional formula of work is $[\text{ML}^2\text{T}^{-2}]$

To convert a physical quantity from one system of units to other system of units, we use the following formula

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

$$n_2 = 100 \left[\frac{1 \text{ kg}}{250 \text{ g}} \right]^1 \left[\frac{1 \text{ m}}{20 \text{ cm}} \right]^2 \left[\frac{1 \text{ s}}{0.5 \text{ min}} \right]^{-2}$$

$$= 100 \left[\frac{1000 \text{ g}}{250 \text{ g}} \right]^1 \left[\frac{100 \text{ cm}}{20 \text{ cm}} \right]^2 \left[\frac{1 \text{ s}}{30 \text{ s}} \right]^{-2}$$

$$= 100 \times 4 \times 25 \times 30 = 9 \times 10^6 \text{ new units}$$

Example 1.17 The value of gravitational constant is $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$ in SI units. Convert it into CGS system of units.

Sol. The dimensional formula of G is $[M^{-1}L^3T^{-2}]$.

To convert a physical quantity from one system of units to other system of units, we use the following formula

$$n_1[M_1^{-1}L_1^3T_1^{-2}] = n_2[M_2^{-1}L_2^3T_2^{-2}]$$

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^{-1} \left[\frac{L_1}{L_2} \right]^3 \left[\frac{T_1}{T_2} \right]^{-2}$$

$$= 6.67 \times 10^{-11} \left[\frac{1 \text{ kg}}{10^{-3} \text{ kg}} \right]^{-1} \left[\frac{1 \text{ m}}{10^{-2} \text{ m}} \right]^3 \left[\frac{1 \text{ s}}{1 \text{ s}} \right]^{-2}$$

$$\text{or } n_2 = 6.67 \times 10^{-8}$$

Thus, value of G in CGS system of units is $6.67 \times 10^{-8} \text{ dyne cm}^2 / \text{g}^2$.

3. Deducing relation between the physical quantities

If we know the factors on which a given physical quantity depends, we can find a formula relating to those factors.

Example 1.18 The frequency (f) of a stretched string depends upon the tension F (dimensions of force), length l of the string and the mass per unit length μ of string. Derive the formula for frequency.

Sol. Suppose, the frequency f depends on the tension raised to the power a , length raised to the power b and mass per unit length raised to the power c .

$$\text{Then, } f \propto (F)^a (l)^b (\mu)^c$$

$$\text{or } f = k (F)^a (l)^b (\mu)^c \quad \dots(i)$$

Here, k is a dimensionless constant of proportionality.

$$\text{Thus, } [f] = [F]^a [l]^b [\mu]^c$$

$$\text{or } [M^0L^0T^{-1}] = [MLT^{-2}]^a [L]^b [ML^{-1}]^c$$

$$\text{or } [M^0L^0T^{-1}] = [M^{a+c}L^{a+b-c}T^{-2a}]$$

For dimensional balance, the dimensions on both sides should be same.

$$\text{Thus, } a + c = 0 \quad \dots(ii)$$

$$a + b - c = 0 \quad \dots(iii)$$

$$\text{and } -2a = -1 \quad \dots(iv)$$

Solving these three equations, we get

$$a = \frac{1}{2}, \quad c = -\frac{1}{2}$$

$$\text{and } b = -1$$

Substituting these values in Eq. (i), we get

$$f = k(F)^{1/2}(l)^{-1}(\mu)^{-1/2} \text{ or } f = \frac{k}{l} \sqrt{\frac{F}{\mu}}$$

Experimentally, the value of k is found to be $\frac{1}{2}$.

$$\text{Hence, } f = \frac{1}{2l} \sqrt{\frac{F}{\mu}}$$

Example 1.19 The centripetal force F acting on a particle moving uniformly in a circle may depend upon mass (m), velocity (v) and radius (r) of the circle. Derive the formula for F using the method of dimensions.

Sol. Let $F = k (m)^x (v)^y (r)^z \quad \dots(i)$

Here, k is a dimensionless constant of proportionality.

Writing the dimensions of RHS and LHS in Eq. (i), we have

$$[MLT^{-2}] = [M]^x [LT^{-1}]^y [L]^z$$

$$= [M^x L^{y+z} T^{-y}]$$

Equating the powers of M , L and T on both sides, we have

$$x = 1, \quad y = 2 \text{ and } y + z = 1$$

$$\text{or } z = 1 - y = -1$$

Putting the values in Eq. (i), we get

$$F = kmv^2 r^{-1} = k \frac{mv^2}{r}$$

$$F = \frac{mv^2}{r} \quad (\text{where, } k = 1)$$

Defects or limitations of dimensional analysis

The method of dimensional analysis has the following limitations

- The value of dimensionless constant involved in a formula cannot be deduced from this method.
- By this method, the equation containing trigonometrical, exponential and logarithmic terms cannot be analysed.
- This method does not work when physical quantity depends on more than three variables because we only have three equations by equalising the power of M , L and T .
- If dimensions are given, physical quantity may not be unique. *e.g.* Work, energy and torque all have the same dimensional formula $[ML^2T^{-2}]$.
- It gives no information whether a physical quantity is a scalar or a vector.

CHECK POINT 7.1

- In the SI system, the unit of temperature is
 - degree centigrade
 - kelvin
 - degree celsius
 - degree Fahrenheit
- Which amongst the following is not equal to watt?
 - joule/second
 - ampere $\times$ volt
 - (ampere)² $\times$ ohm
 - ampere/volt
- Joule $\times$ second is the unit of
 - energy
 - momentum
 - angular momentum
 - power
- Which amongst the following pairs has the same units?
 - Wavelength and Rydberg constant
 - Relative velocity and relative density
 - Thermal capacity and Boltzmann constant
 - Time period and acceleration gradient
- Density of liquid in CGS system is 0.625 g cm^{-3} . What is its magnitude in SI system?
 - 0.625
 - 0.0625
 - 0.00625
 - 625
- Dimensions of surface tension are
 - $[M^2L^2T^{-2}]$
 - $[M^2LT^{-2}]$
 - $[MT^{-2}]$
 - $[MLT^{-2}]$
- The dimensions of impulse are equal to that of
 - force
 - linear momentum
 - pressure
 - angular momentum
- Which of the following does not possess the same dimensions as that of pressure?
 - Stress
 - Bulk modulus
 - Thrust
 - Energy density
- What is the dimensional formula of gravitational constant?
 - $[ML^2T^{-2}]$
 - $[ML^{-1}T^{-1}]$
 - $[M^{-1}L^3T^{-2}]$
 - None of these
- Which one of the following have same dimensions?
 - Torque and force
 - Potential energy and force
 - Torque and potential energy
 - Planck's constant and linear momentum
- The force F on a sphere of radius a moving in a medium with velocity v is given by $F = 6\pi \eta a v$. The dimensions of η are
 - $[ML^{-3}]$
 - $[MLT^{-2}]$
 - $[MT^{-1}]$
 - $[ML^{-1}T^{-1}]$
- The dimensional representation of specific resistance in terms of charge Q is
 - $[ML^3T^{-1}Q^{-2}]$
 - $[ML^2T^{-2}Q^2]$
 - $[MLT^{-2}Q^{-1}]$
 - $[ML^2T^{-2}Q^{-1}]$
- The dimensional formula for Planck's constant and angular momentum is
 - $[ML^2T^{-2}]$ and $[MLT^{-1}]$
 - $[ML^2T^{-1}]$ and $[ML^2T^{-1}]$
 - $[ML^3T^{-1}]$ and $[ML^2T^{-2}]$
 - $[MLT^{-1}]$ and $[MLT^{-2}]$
- The dimensions of $\frac{1}{2} \epsilon_0 E^2$ (ϵ_0 is the permittivity of the space and E is electric field), are
 - $[ML^2T^{-1}]$
 - $[ML^{-1}T^{-2}]$
 - $[ML^2T^{-2}]$
 - $[MLT^{-1}]$
- The units of length, velocity and force are doubled. Which of the following is the correct change in the other units?
 - Unit of time is doubled
 - Unit of mass is doubled
 - Unit of momentum is doubled
 - Unit of energy is doubled
- Given that $y = a \cos \left(\frac{t}{p} - qx \right)$, where t represents time and x represents distance; which amongst the following statements which is(are) true?
 - The unit of x is same as that of q
 - The unit of x is same as that of p
 - The unit of t is same as that of q
 - The unit of t is same as that of p
- The dimensions of $\frac{a}{b}$ in the equation $p = \frac{a - t^2}{bx}$, where p is pressure, x is distance and t is time, are
 - $[M^2LT^{-3}]$
 - $[MT^{-2}]$
 - $[LT^{-3}]$
 - $[ML^3T^{-1}]$
- The equation of a wave is given by $y = a \sin \omega \left(\frac{x}{v} - k \right)$ where, ω is angular velocity and v is the linear velocity. The dimensions of k will be
 - $[T^{-2}]$
 - $[T^{-1}]$
 - $[T]$
 - $[LT]$
- If 'muscle times speed equals power', then what is the ratio of the SI unit and the CGS unit of muscle?
 - 10^5
 - 10^3
 - 10^7
 - 10^{-5}
- If p represents radiation pressure, c represents speed of light and Q represents radiation energy striking a unit area per second, then for what values of non-zero integers x , y and z , $p^x Q^y c^z$ is dimensionless?
 - $x = 1, y = 1, z = -1$
 - $x = 1, y = -1, z = 1$
 - $x = -1, y = 1, z = 1$
 - $x = 1, y = 1, z = 1$
- Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon the velocity v of the water, its density ρ and the acceleration due to gravity g . Then, m is directly proportional to
 - v^3
 - v^4
 - v^5
 - v^6

SIGNIFICANT FIGURES

The significant figures are normally those digits in a measured quantity which are known reliable or about which we have confidence in our measurement plus one additional digit that is uncertain.

e.g. If length of some object is 185.2 cm, then it has four significant figures. The digits 1, 8 and 5 are reliable and digit 2 is uncertain.

Note Significant figures indicate the precision of the measurement which depends on the least count of the measuring instrument.

Rules to determine significant figures

For determining number of significant figures, we use the following rules

Rule 1 All non-zero digits are significant, *e.g.* $x = 2567$ has four significant figures.

Rule 2 The zeros appearing between two non-zero digits are significant, no matter where the decimal point is, if at all, *e.g.* 6.028 has 4 significant figures.

Rule 3 If the number is less than 1, the zero(s) on the right of decimal point but to the left of first non-zero digit are not significant.

e.g. 0.0042 has two significant digits.

Rule 4 The terminal or trailing zero(s) in a number without a decimal point are not significant. Thus, 426 m = 42600 cm = 426000 mm has three significant figures.

Rule 5 In a number with decimal, zeros to the right of last non-zero digit are significant.

e.g. 4.600 or 0.002300 have four significant figures each.

Point of confusion and its remedy

Suppose we change the units of a physical quantity, then we will write

$$2.30 \text{ m} = 230 \text{ cm} = 2300 \text{ mm} = 0.00230 \text{ km}$$

When we are considering 2300 mm, then from Rule-4, we would conclude erroneously that the number has two significant figures, while in fact it has three significant figures and a mere change of units cannot change the number of significant figures.

To remove such ambiguities in determining the number of significant figures, apply following rules

Rule 6 The power of 10 is irrelevant to the determination of significant figures. *e.g.* In the measurements,

$$\begin{aligned} 2.30 \text{ m} &= 2.30 \times 10^2 \text{ cm} = 2.30 \times 10^3 \text{ mm} \\ &= 2.30 \times 10^{-3} \text{ km} \end{aligned}$$

The significant figures are three in each measurement, because all zeros appearing in the base number in the scientific notation (in the power of 10) are not significant.

Rule 7 A choice of change of different units does not change the number of significant digits or figures in a measurement.

e.g. The length 7.03 cm has three significant figures. But in different units, the same value can be written as 0.0703 m or 70.3 mm. All these measurements have the same number of significant figures (digits 7, 0 and 3) namely three.

Rule 8 The exact numbers appearing in the mathematical formulae of various physical quantities have infinite number of significant figures. *e.g.* Perimeter of a square is given by $4 \times \text{side}$. Here, 4 is an exact number and has infinite number of significant figures.

$\therefore$ It can be written as 4.0, 4.00, 4.0000 as per the requirement.

Some significant figures of measured values given in the table below

Measured values	Number of significant figures	Rule
12376	5	1
6024.7	5	2
0.071	2	3
410	2	4
2.40	3	5
1.6×10^{10}	2	6
$\ln 2$ ($l + b$), digit 2	Infinite	8

Example 1.20 How many significant figures are there in the following measured values?

- (i) 227.2 g (ii) 3600 g
(iii) 0.00602 g (iv) 2.50×10^{10} g

Sol. (i) 227.2 g has all the non-zero digits. Hence, it has four significant figures.

(ii) According to rule number 4, trailing zeros are not significant. Hence, 3600 g has 2 significant figures.

(iii) According to the rule number 3, the zeros on the right of decimal point but to the left of first non-zero digit are not significant. Hence, 0.00602 g has 3 significant figures.

(iv) According to the rule number 6, it has 3 significant figures.

Mathematical operations of significant figures

The result of a mathematical operation involving measured values of quantities cannot be more accurate than the measured value themselves.

So, certain rules have to be followed while doing mathematical operations with significant figures, so that precision in final result is consistent with the precision of the original measured values.

Addition or subtraction

Suppose in the measured values to be added or subtracted, the least number of significant digits after the decimal is n . Then, in the sum or difference also, the number of significant digits after the decimal should be n .

e.g. $1.2 + 3.45 + 6.789 = 11.439 \approx 11.4$

Here, the least number of significant digits after the decimal is one. Hence, the result will be 11.4 (when rounded off to smallest number of decimal places).

Similarly, *e.g.* $12.63 - 10.2 = 2.43 \approx 2.4$

Example 1.21 Add 6.75×10^3 cm to 4.52×10^2 cm with regard to significant figures.

Sol. Let $a = 6.75 \times 10^3$ cm, $b = 4.52 \times 10^2$ cm
 $= 0.452 \times 10^3$ cm $= 0.45 \times 10^3$ cm
 (upto 2 places of decimal)

$\therefore$ Addition of significant figures

$$a + b = (6.75 \times 10^3 + 0.45 \times 10^3) \text{ cm} = 7.20 \times 10^3 \text{ cm}$$

Example 1.22 Two sticks of lengths 12.132 cm and 10.2 cm are placed end to end. Find their total length with due regard to significant figures.

Sol. Length of first stick = 12.132 cm (5 significant figures)
 Length of second stick = 10.2 cm (3 significant figures)
 $\therefore$ Total length of two sticks = $12.132 + 10.2 = 22.332$
 The answer should be rounded off with least number of significant digits after the decimal.
 $\therefore$ Total length of two sticks will be 22.3 cm.

Multiplication or division

Suppose in the measured values to be multiplied or divided, the least number of significant digits be n , then in the product or quotient, the number of significant digits should also be n .

e.g. $1.2 \times 36.72 = 44.064 \approx 44$

The least number of significant digits in the measured values are two. Hence, the result when rounded off to two significant digits become 44. Therefore, the answer is 44.

Similarly, *e.g.* $\frac{1100}{10.2} = 107.8431373 \approx 110$

As 1100 has minimum number of significant figures (*i.e.* 2), therefore the result should also contain only two significant digits. Hence, the result when rounded off to two significant digits becomes 110.

Example 1.23 The voltage across a lamp is 6.32 V when the current passing through it is 3.4 A. Find the power consumed upto appropriate significant figures.

Sol. Voltage across a lamp, $V = 6.32$ V (3 significant figures)
 Current flowing through lamp, $I = 3.4$ A (2 significant figures)
 $\therefore$ Power consumed, $P = VI = (6.32)(3.4) = 21.488$ W
 Answer should have minimum number of significant figures. Here, the minimum number of significant figures is 2.
 $\therefore$ Power consumed = 21 W

Example 1.24 A thin wire has a length of 21.7 cm and radius 0.46 mm. Calculate the volume of the wire upto correct significant figures.

Sol. Given, $l = 21.7$ cm, $r = 0.46$ mm $= 0.046$ cm
 Volume of wire, $V = \pi r^2 l = \frac{22}{7} (0.046)^2 (21.7)$
 $= 0.1443 \text{ cm}^3 \approx 0.14 \text{ cm}^3$

Example 1.25 The time taken by a pendulum to complete 25 vibrations is 88.0 s. Find the time period of the pendulum in seconds upto appropriate significant figures.

Sol. Time period of oscillation = $\frac{\text{Total time taken}}{\text{Number of oscillations}}$
 $= \frac{88.0}{25} \text{ s} = 3.52 \text{ s}$

Out of the two quantities given in the data, 25 is exact, hence has infinite significant figures. Therefore, the answer should be reported to three significant figures, *i.e.* 3.52 s.

Example 1.26 5.74 g of substance occupies 1.2 cm^3 . Express its density by keeping the significant figures in view.

Sol. Here, mass, $m = 5.74$ g, volume, $V = 1.2 \text{ cm}^3$
 As density, $\rho = \frac{\text{mass}}{\text{volume}} = \frac{5.74 \text{ g}}{1.2 \text{ cm}^3} = 4.783 \text{ g cm}^{-3}$

As mass has 3 significant digits and volume has 2 significant digits, therefore as per rule, density will have only two significant digits, rounding off, we get $\rho = 4.8 \text{ g cm}^{-3}$.

Rounding off

The process of omitting the non-significant digits and retaining only the desired number of significant digits, incorporating the required modifications to the last significant digit is called rounding off the number. In physics, calculation is a vital part and during that we shall reduce the number to the required extent and that is why there is a need to round off numbers.

Like mathematical operations of significant figures, rounding off numbers also follow certain rules.

ERROR IN MEASUREMENT

We use different kinds of instruments for measuring various quantities. However, these measurements always has a degree of uncertainty related to it. This uncertainty is called as error in the measurement. Thus, the difference between the measured value and the true value of a quantity is known as the error of measurement.

$$\therefore \text{Error} = \text{True value} - \text{Measured value}$$

Errors may arise from different sources and are usually classified as follows

1. Systematic errors

These are the errors whose causes are known to us. They can be either positive or negative.

One of the common source of systematic errors is as follows

Instrumental errors

These errors are due to imperfect design or erroneous manufacture or misuse of the measuring instrument.

These are of following types

- (i) **Zero error** If the zero mark of vernier scale does not coincide with the zero mark of the main scale, the instrument is said to have zero error. A metre scale having worn off zero mark also has zero error.
- (ii) **Least count or permissible error** This error is due to the limitation imposed by the least count of the measuring instrument. It is an uncertainty associated with the resolution of the measuring instrument.

Note Least Count (LC) of

$$(i) \text{ Vernier callipers} = \frac{\text{Value of 1 MSD}}{\text{Number of divisions}}$$

$$\text{or LC} = 1 \text{ MSD} - 1 \text{ VSD on vernier scale}$$

$$(ii) \text{ Screw gauge} = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

- (iii) **Constant error** The errors which affect each observation by the same amount are called constant errors. Such errors are due to faulty calibration of the scale of the measuring instrument.
- (iv) **Backlash error** Backlash error occurs in screw gauge, when we try to rotate the screw very fast to measure a reading. Due to this, there is some slipping between the different screws instead of the rotation, which gives an incorrect reading. To avoid this we should rotate the screw slowly in only one direction.

Causes of systematic errors

Few causes of systematic errors are as follows

- (i) Instrumental errors may be due to erroneous instruments. These errors can be reduced by using more accurate instruments and applying zero correction, when required.
- (ii) Sometimes errors arise on account of ignoring certain facts. *e.g.* In measuring time period of simple pendulum, error may creep because no consideration is taken of air resistance. These errors can be reduced by applying proper corrections to the formula used.
- (iii) Change in temperature, pressure, humidity, etc., may also sometimes cause errors in the result. Relevant corrections can be made to minimise their effects.

2. Random errors

The errors which occur irregularly and at random, in magnitude and direction are called random errors. The causes of random errors are not known. Hence, it is not possible to remove them completely. These errors may arise due to a variety of reasons.

e.g. The reading of a sensitive beam balance may change by the vibrations caused in the building, due to persons moving in the laboratory or vehicles running nearby. The random error can be minimised by repeating the observation a large number of times and taking the arithmetic mean of all the observations. The mean value would be very close to the most accurate reading.

Example 1.29 In a vernier callipers, 1 main scale reading is 1 mm and 9th main scale division coincide with 10th vernier scale. Find the least count of vernier.

Sol. Given, 1 main scale reading or division (MSD)

$$= 1 \text{ mm}$$

$$9 \text{ MSD} = 10 \text{ VSD}$$

$$\Rightarrow 1 \text{ VSD} = \frac{9}{10} \text{ MSD} = \frac{9}{10} \times 1 = \frac{9}{10} \text{ mm}$$

$$\therefore \text{LC} = 1 - \frac{9}{10} = \frac{1}{10} \text{ mm or } 0.1 \text{ mm}$$

Expression of errors

Errors can be expressed in following way

- (i) **Absolute error** The difference between the true value and the measured value of a quantity is called an absolute error. Usually the mean value a_m is taken as the true value. So, if

$$a_m = \frac{a_1 + a_2 + \dots + a_n}{n} = \frac{1}{n} \sum_{i=1}^n a_i$$

Then by definition, absolute errors in the measured values of the quantity are

$$\begin{aligned}\Delta a_1 &= a_1 - a_m \\ \Delta a_2 &= a_2 - a_m \\ &\vdots \\ \Delta a_n &= a_n - a_m\end{aligned}$$

Absolute error may be positive or negative.

Mean absolute error It is the arithmetic mean of the magnitudes of absolute errors. Thus,

$$\Delta a_{\text{mean}} = \frac{|\Delta a_1| + |\Delta a_2| + \dots + |\Delta a_n|}{n} = \frac{1}{n} \sum_{i=1}^n |\Delta a_i|$$

Thus final result of measurement can be written as

$$a = a_m \pm \Delta a_{\text{mean}}$$

This implies that value of a is likely to lie between $a_m + \Delta a_{\text{mean}}$ and $a_m - \Delta a_{\text{mean}}$.

- (ii) **Relative or fractional error** The ratio of mean absolute error to the mean value of the quantity measured is called relative or fractional error.

Thus,
$$\text{Relative error} = \frac{\Delta a_{\text{mean}}}{a_m}$$

- (iii) **Percentage error** When the relative error is expressed in percent, it is called percentage error. It is denoted by δa .

Thus,
$$\delta a = \frac{\Delta a_{\text{mean}}}{a_m} \times 100\%$$

Example 1.30 The length of a rod as measured in an experiment is found to be 2.48 m, 2.46 m, 2.49 m, 2.49 m and 2.46 m. Find the average length, the absolute error in each observation and the percentage error.

Sol. Average length = Arithmetic mean of the measured values

$$x_{\text{mean}} = \frac{2.48 + 2.46 + 2.49 + 2.49 + 2.46}{5} = \frac{12.38}{5} = 2.476 \text{ m}$$

$\therefore$ True value, $x_{\text{mean}} = 2.48 \text{ m}$

Absolute errors in various measurements,

$$\begin{aligned}|\Delta x_1| &= |x_1 - x_{\text{mean}}| = |2.48 - 2.48| = 0.00 \text{ m} \\ |\Delta x_2| &= |2.46 - 2.48| = 0.02 \text{ m} \\ |\Delta x_3| &= |2.49 - 2.48| = 0.01 \text{ m} \\ |\Delta x_4| &= |2.49 - 2.48| = 0.01 \text{ m} \\ |\Delta x_5| &= |2.46 - 2.48| = 0.02 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Mean absolute error} &= \frac{|\Delta x_1| + |\Delta x_2| + |\Delta x_3| + |\Delta x_4| + |\Delta x_5|}{5} \\ &= \frac{(0.00 + 0.02 + 0.01 + 0.01 + 0.02)}{5} = \frac{0.06}{5} = 0.012\end{aligned}$$

$$\Delta x_{\text{mean}} = 0.01 \text{ m}$$

Thus, $x = 2.48 \pm 0.01 \text{ m}$

$$\begin{aligned}\text{Percentage error, } \delta x &= \frac{\Delta x_{\text{mean}}}{x} \times 100 \\ &= \frac{0.01}{2.48} \times 100 = 0.40\%\end{aligned}$$

Example 1.31 The diameter of a wire as measured by a screw gauge was found to be 2.620 cm, 2.625 cm, 2.630 cm, 2.628 cm and 2.626 cm. Calculate

- mean value of diameter,
- absolute error in each measurement,
- mean absolute error,
- fractional error,
- percentage error and
- express the result in terms of percentage error.

Sol. (i) Mean value of diameter,

$$\begin{aligned}a_m &= \frac{2.620 + 2.625 + 2.630 + 2.628 + 2.626}{5} \\ &= 2.6258 \text{ cm} = 2.626 \text{ cm}\end{aligned}$$

(rounding off to three decimal places)

- (ii) Taking a_m as the true value, the absolute errors in different observations are

$$\begin{aligned}\Delta a_1 &= 2.620 - 2.626 = -0.006 \text{ cm} \\ \Delta a_2 &= 2.625 - 2.626 = -0.001 \text{ cm} \\ \Delta a_3 &= 2.630 - 2.626 = +0.004 \text{ cm} \\ \Delta a_4 &= 2.628 - 2.626 = +0.002 \text{ cm} \\ \Delta a_5 &= 2.626 - 2.626 = 0.000 \text{ cm}\end{aligned}$$

- (iii) Mean absolute error,

$$\begin{aligned}\Delta a_{\text{mean}} &= \frac{|\Delta a_1| + |\Delta a_2| + |\Delta a_3| + |\Delta a_4| + |\Delta a_5|}{5} \\ &= \frac{0.006 + 0.001 + 0.004 + 0.002 + 0.000}{5} \\ &= 0.0026 = 0.003\end{aligned}$$

(rounding off to three decimal places)

$$(iv) \text{ Fractional error} = \pm \frac{\Delta a_{\text{mean}}}{a_m} = \pm \frac{0.003}{2.626} = \pm 0.001$$

$$(v) \text{ Percentage error} = \pm 0.001 \times 100 = \pm 0.1\%$$

$$(vi) \text{ Diameter of wire can be written as} \\ d = 2.626 \text{ cm} \pm 0.1\%$$

Example 1.32 The refractive index (n) of glass is found to have the values 1.49, 1.50, 1.52, 1.54 and 1.48. Calculate

- the mean value of refractive index,
- absolute error in each measurement,
- mean absolute error,
- fractional error and
- percentage error.

Sol. (i) Mean value of refractive index,

$$\begin{aligned}n_m &= \frac{1.49 + 1.50 + 1.52 + 1.54 + 1.48}{5} \\ &= 1.506 = 1.51\end{aligned}$$

(rounded off to two decimal places)

- (ii) Taking n_m as the true value, the absolute errors in different observations are

$$\Delta n_1 = 1.49 - 1.51 = -0.02$$

$$\Delta n_2 = 1.50 - 1.51 = -0.01$$

$$\Delta n_3 = 1.52 - 1.51 = +0.01$$

$$\Delta n_4 = 1.54 - 1.51 = +0.03$$

$$\Delta n_5 = 1.48 - 1.51 = -0.03$$

- (iii) Mean absolute error,

$$\begin{aligned}\Delta n_{\text{mean}} &= \frac{|\Delta n_1| + |\Delta n_2| + |\Delta n_3| + |\Delta n_4| + |\Delta n_5|}{5} \\ &= \frac{0.02 + 0.01 + 0.01 + 0.03 + 0.03}{5} = 0.02\end{aligned}$$

- (iv) Fractional error = $\frac{\pm \Delta n_{\text{mean}}}{n_m} = \frac{\pm 0.02}{1.51} = \pm 0.0132$

- (v) Percentage error = $(\pm 0.0132 \times 100) = \pm 1.32\%$

Combination of errors

Most of our experiments involves the measurement of various physical quantities. We then put these measurements in appropriate formula, to calculate the required quantity. Therefore, we must know how the errors in all the measurements combine and appear in the final quantity.

1. Error in sum or difference

Let $x = a \pm b$

Further, let Δa be the absolute error in the measurement of a , Δb be the absolute error in the measurement of b and Δx be the absolute error in the measurement of x .

$$\begin{aligned}\text{Then, } x + \Delta x &= (a \pm \Delta a) \pm (b \pm \Delta b) \\ &= (a \pm b) \pm (\pm \Delta a \pm \Delta b) \\ &= x \pm (\pm \Delta a \pm \Delta b) \text{ or } \Delta x = \pm \Delta a \pm \Delta b\end{aligned}$$

The four possible values of Δx are $(\Delta a - \Delta b)$, $(\Delta a + \Delta b)$, $(-\Delta a - \Delta b)$ and $(-\Delta a + \Delta b)$. Therefore, the maximum absolute error in x is

$$\Delta x = \pm(\Delta a + \Delta b)$$

i.e. The maximum absolute error in sum or difference of two quantities is equal to sum of the absolute errors in the individual quantities.

Example 1.33 The volumes of two bodies are measured to be $V_1 = (10.2 \pm 0.02) \text{ cm}^3$ and $V_2 = (6.4 \pm 0.01) \text{ cm}^3$.

Calculate the sum and difference in volumes with error limits.

Sol. Given, $V_1 = (10.2 \pm 0.02) \text{ cm}^3$ and $V_2 = (6.4 \pm 0.01) \text{ cm}^3$

$$\Delta V = \pm(\Delta V_1 + \Delta V_2) = \pm(0.02 + 0.01) \text{ cm}^3 = \pm 0.03 \text{ cm}^3$$

$$V_1 + V_2 = (10.2 + 6.4) \text{ cm}^3 = 16.6 \text{ cm}^3$$

$$\text{and } V_1 - V_2 = (10.2 - 6.4) \text{ cm}^3 = 3.8 \text{ cm}^3$$

$$\text{Hence, sum of volumes} = (16.6 \pm 0.03) \text{ cm}^3$$

$$\text{and difference of volumes} = (3.8 \pm 0.03) \text{ cm}^3$$

2. Error in product

Let $x = ab$

$$\text{Then, } (x \pm \Delta x) = (a \pm \Delta a)(b \pm \Delta b)$$

$$\text{or } x \left(1 \pm \frac{\Delta x}{x}\right) = ab \left(1 \pm \frac{\Delta a}{a}\right) \left(1 \pm \frac{\Delta b}{b}\right)$$

$$\text{or } 1 \pm \frac{\Delta x}{x} = 1 \pm \frac{\Delta b}{b} \pm \frac{\Delta a}{a} \pm \frac{\Delta a}{a} \cdot \frac{\Delta b}{b} \quad (\because x = ab)$$

$$\text{or } \pm \frac{\Delta x}{x} = \pm \frac{\Delta a}{a} \pm \frac{\Delta b}{b} \pm \frac{\Delta a}{a} \cdot \frac{\Delta b}{b}$$

Here, $\frac{\Delta a}{a} \cdot \frac{\Delta b}{b}$ is a very small quantity, so can be neglected.

$$\text{Hence, } \pm \frac{\Delta x}{x} = \pm \frac{\Delta a}{a} \pm \frac{\Delta b}{b}$$

$$\begin{aligned}\text{Possible values of } \frac{\Delta x}{x} &\text{ are } \left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right), \left(\frac{\Delta a}{a} - \frac{\Delta b}{b}\right), \\ &\left(-\frac{\Delta a}{a} + \frac{\Delta b}{b}\right) \text{ and } \left(-\frac{\Delta a}{a} - \frac{\Delta b}{b}\right).\end{aligned}$$

Hence, maximum possible value of

$$\frac{\Delta x}{x} = \pm \left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right)$$

Therefore, maximum fractional error in product of two (or more) quantities is equal to sum of fractional errors in the individual quantities.

3. Error in division

$$\text{Let } x = \frac{a}{b}$$

$$\text{Then, } x \pm \Delta x = \frac{a \pm \Delta a}{b \pm \Delta b}$$

$$\text{or } x \left(1 \pm \frac{\Delta x}{x}\right) = \frac{a \left(1 \pm \frac{\Delta a}{a}\right)}{b \left(1 \pm \frac{\Delta b}{b}\right)}$$

$$\text{or } \left(1 \pm \frac{\Delta x}{x}\right) = \left(1 \pm \frac{\Delta a}{a}\right) \left(1 \pm \frac{\Delta b}{b}\right)^{-1} \quad \left(\because x = \frac{a}{b}\right)$$

As $\frac{\Delta b}{b} \ll 1$, so expanding binomially, we get

$$\left(1 \pm \frac{\Delta x}{x}\right) = \left(1 \pm \frac{\Delta a}{a}\right) \left(1 \mp \frac{\Delta b}{b}\right)$$

$$\text{or } 1 \pm \frac{\Delta x}{x} = 1 \pm \frac{\Delta a}{a} \mp \frac{\Delta b}{b} \pm \frac{\Delta a}{a} \cdot \frac{\Delta b}{b}$$

Here, $\frac{\Delta a}{a} \cdot \frac{\Delta b}{b}$ is a very small quantity, so can be neglected.

Hence, $\pm \frac{\Delta x}{x} = \pm \frac{\Delta a}{a} \mp \frac{\Delta b}{b}$

Possible values of $\frac{\Delta x}{x}$ are $\left(\frac{\Delta a}{a} - \frac{\Delta b}{b}\right), \left(\frac{\Delta a}{a} + \frac{\Delta b}{b}\right), \left(-\frac{\Delta a}{a} - \frac{\Delta b}{b}\right)$ and $\left(-\frac{\Delta a}{a} + \frac{\Delta b}{b}\right)$. Therefore, the maximum value of

$$\frac{\Delta x}{x} = \pm \left(\frac{\Delta a}{a} + \frac{\Delta b}{b} \right)$$

Therefore, the maximum value of fractional error in division of two (or more) quantities is equal to the sum of fractional errors in the individual quantities.

Example 1.34 Calculate focal length of a spherical mirror from the following observations. Object distance $u = (50.1 \pm 0.5)$ cm and image distance $v = (20.1 \pm 0.2)$ cm.

Sol. Formula for focal length of a spherical mirror,

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \quad \dots(i)$$

or $f = \frac{uv}{u+v} = \frac{(50.1)(20.1)}{(50.1+20.1)} = 14.3$ cm

On differentiating Eq. (i), we get

$$\frac{\Delta f}{f^2} = \frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}$$

or $\Delta f = \frac{f^2}{u^2} \times \Delta u + \frac{f^2}{v^2} \Delta v$

$$= \left(\frac{14.3}{50.1}\right)^2 \times 0.5 + \left(\frac{14.3}{20.1}\right)^2 \times 0.2$$

$$= 0.0407 + 0.1012$$

$$= \pm 0.1419 \text{ cm} \approx \pm 0.1 \text{ cm}$$

$\therefore f = (14.3 \pm 0.1)$ cm

4. Error in quantity raised to some power

Let $x = \frac{a^n}{b^m}$

Then, $\ln(x) = n \ln(a) - m \ln(b)$

Differentiating both sides, we get

$$\frac{dx}{x} = n \frac{da}{a} - m \frac{db}{b}$$

In terms of fractional error, we may write

$$\pm \frac{\Delta x}{x} = \pm n \frac{\Delta a}{a} \mp m \frac{\Delta b}{b}$$

Therefore, maximum value of

$$\frac{\Delta x}{x} = \pm \left(n \frac{\Delta a}{a} + m \frac{\Delta b}{b} \right)$$

Example 1.35 The radius of sphere is measured to be (2.1 ± 0.5) cm. Calculate its surface area with error limits.

Sol. Surface area, $S = 4\pi r^2 = (4) \left(\frac{22}{7} \right) (2.1)^2$

$$= 55.44 = 55.4 \text{ cm}^2$$

Further, $\frac{\Delta S}{S} = 2 \frac{\Delta r}{r}$ or $\Delta S = 2 \left(\frac{\Delta r}{r} \right) (S)$

$$= \frac{2 \times 0.5 \times 55.4}{2.1} = 26.38 = 26.4 \text{ cm}^2$$

$\therefore S = (55.4 \pm 26.4) \text{ cm}^2$

Example 1.36 The mass and density of a solid sphere are measured to be (12.4 ± 0.1) kg and $(4.6 \pm 0.2) \text{ kg m}^{-3}$. Calculate the volume of the sphere with error limits.

Sol. Here, $m \pm \Delta m = (12.4 \pm 0.1)$ kg

and $\rho \pm \Delta \rho = (4.6 \pm 0.2) \text{ kg m}^{-3}$

Volume, $V = \frac{m}{\rho} = \frac{12.4}{4.6} = 2.69 \text{ m}^3 = 2.7 \text{ m}^3$

(rounding off to one decimal place)

Now, $\frac{\Delta V}{V} = \pm \left(\frac{\Delta m}{m} + \frac{\Delta \rho}{\rho} \right)$

or $\Delta V = \pm \left(\frac{\Delta m}{m} + \frac{\Delta \rho}{\rho} \right) \times V$

$$= \pm \left(\frac{0.1}{12.4} + \frac{0.2}{4.6} \right) \times 2.7 = \pm 0.14$$

$\therefore V \pm \Delta V = (2.7 \pm 0.14) \text{ m}^3$

Example 1.37 A thin copper wire of length L increase in length by 2% when heated from T_1 to T_2 . If a copper cube having side 10 L is heated from T_1 to T_2 , what will be the percentage change in

- area of one face of the cube and
- volume of the cube ?

Sol. (i) Area, $A = 10 L \times 10 L = 100 L^2$

Percentage change in area

$$= \frac{\Delta A}{A} \times 100 = 2 \times \frac{\Delta L}{L} \times 100$$

$$= 2 \times 2\% = 4\%$$

(ii) Volume, $V = 10 L \times 10 L \times 10 L = 1000 L^3$

Percentage change in volume

$$= \frac{\Delta V}{V} \times 100 = 3 \frac{\Delta L}{L} \times 100 = 3 \times 2\% = 6\%$$

Example 1.38 Calculate percentage error in determination of time period of a pendulum

$$T = 2\pi \sqrt{\frac{l}{g}}$$

where, l and g are measured with $\pm 1\%$ and $\pm 2\%$ errors.

Sol. Percentage error in time period,

$$\begin{aligned}\frac{\Delta T}{T} \times 100 &= \pm \left(\frac{1}{2} \times \frac{\Delta l}{l} \times 100 + \frac{1}{2} \times \frac{\Delta g}{g} \times 100 \right) \\ &= \pm \left(\frac{1}{2} \times 1\% + \frac{1}{2} \times 2\% \right) = \pm 1.5\%\end{aligned}$$

Example 1.39 Find the relative error in Z , if $Z = \frac{A^4 B^{1/3}}{CD^{3/2}}$ and the percentage error in the measurements of A, B, C and D are 4%, 2%, 3% and 1%, respectively.

Sol. $\therefore \frac{\Delta Z}{Z} = 4 \left(\frac{\Delta A}{A} \right) + \frac{1}{3} \left(\frac{\Delta B}{B} \right) + \frac{\Delta C}{C} + \frac{3}{2} \left(\frac{\Delta D}{D} \right)$

Given, $\frac{\Delta A}{A} \times 100 = 4\%$

$$\frac{\Delta B}{B} \times 100 = 2\%$$

$$\frac{\Delta C}{C} \times 100 = 3\%$$

and $\frac{\Delta D}{D} \times 100 = 1\%$

$$\begin{aligned}\therefore \frac{\Delta Z}{Z} \times 100 &= (4 \times 4\%) + \left(\frac{1}{3} \times 2\% \right) + 3\% + \left(\frac{3}{2} \times 1\% \right) \\ &= \left(16 + \frac{2}{3} + 3 + \frac{3}{2} \right)\% \\ &= 21.16\%\end{aligned}$$

The percentage error in the measurement of Z is 21.16%.

Therefore, the relative error in Z is 0.2116.

CHECK POINT 1.3

- A spherometer has 100 equal divisions marked along the periphery of its disc and one full rotation of the disc advances on the main scale by 0.01 cm. The least count of this system is
(a) 10^{-2} cm (b) 10^{-4} cm
(c) 10^{-5} cm (d) 10^{-1} cm
- Three measurements are made as 18.425 cm, 7.21 cm and 5.0 cm. The mean of measurements should be written as
(a) 10.212 cm (b) 10.21 cm (c) 10.22 cm (d) 10.2 cm
- If error in measuring diameter of a circle is 4%, the error in measuring radius of the circle would be
(a) 2% (b) 8%
(c) 4% (d) 1%
- The length of a rod is (11.05 ± 0.2) cm. What is the net length of the system of rods, when these two rods are joined side by side?
(a) (22.1 ± 0.05) cm (b) (22.1 ± 0.1) cm
(c) (22.10 ± 0.05) cm (d) (22.10 ± 0.4) cm
- A body travels uniformly a distance of (13.8 ± 0.2) m in a time (4.0 ± 0.3) s. The velocity of the body within error limit is
(a) $(3.45 \pm 0.2) \text{ ms}^{-1}$ (b) $(3.45 \pm 0.3) \text{ ms}^{-1}$
(c) $(3.45 \pm 0.4) \text{ ms}^{-1}$ (d) $(3.45 \pm 0.5) \text{ ms}^{-1}$
- A cuboid has volume $V = l \times 2l \times 3l$, where l is the length of one side. If the relative percentage error in the measurement of l is 1%, then the relative percentage error in measurement of V is
(a) 18% (b) 6%
(c) 3% (d) 1%
- A force F is applied on a square plate of side L . If the percentage error in the determination of L is 2% and that in F is 4%. What is the permissible error in pressure?
(a) 8% (b) 6%
(c) 4% (d) 2%
- The heat generated in a wire depends directly on the resistance, current and time. If the error in measuring the above are 1%, 2% and 1%, respectively. The maximum error in measuring the heat is
(a) 8% (b) 6% (c) 18% (d) 12%
- If the error in the measurement of momentum of a particle is (+ 100%), then the error in the measurement of kinetic energy is
(a) 100% (b) 200% (c) 300% (d) 400%
- The radius of a ball is (5.2 ± 0.2) cm. The percentage error in the volume of the ball is (approximately)
(a) 11% (b) 4% (c) 7% (d) 9%
- The values of two resistors are $(5.0 \pm 0.2) \text{ k}\Omega$ and $(10.0 \pm 0.1) \text{ k}\Omega$. What is the percentage error in the equivalent resistance when they are connected in parallel?
(a) 2% (b) 5% (c) 7% (d) 3%

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1 If dimensions of A and B are different, then which of the following operation is valid?
 (a) $\frac{A}{B}$ (b) $e^{-A/B}$ (c) $A - B$ (d) $A + B$
- 2 The diameter of a wire is measured to be 0.0250×10^{-4} m. The number of significant figures in the measurement is
 (a) five (b) four (c) three (d) nine
- 3 Dimensional formula for electromotive force is same as that for
 (a) potential (b) current (c) force (d) energy
4. The number of significant figures in 0.06900 is [NCERT Exemplar]
 (a) 5 (b) 4
 (c) 2 (d) 3
- 5 The sum of the numbers 436.32, 227.2 and 0.301 in appropriate significant figures is [NCERT Exemplar]
 (a) 663.821 (b) 664
 (c) 663.8 (d) 663.82
- 6 The dimensional formula for magnetic flux is
 (a) $[ML^2T^{-2}A^{-1}]$ (b) $[ML^3T^{-2}A^{-2}]$
 (c) $[M^0L^{-2}T^{-2}A^{-2}]$ (d) $[ML^2T^{-1}A^2]$
- 7 If the dimensions of a physical quantity are given by $[M^a L^b T^c]$, then the physical quantity will be
 (a) force, if $a = 0, b = -1, c = -2$
 (b) pressure, if $a = 1, b = -1, c = -2$
 (c) velocity, if $a = 1, b = 0, c = -1$
 (d) acceleration, if $a = 1, b = 1, c = -2$
- 8 What is the units of $k = \frac{1}{4\pi\epsilon_0}$?
 (a) $C^2N^{-1}m^{-2}$ (b) $N\cdot m^2C^{-2}$
 (c) $N\cdot m^2C^2$ (d) Unitless
- 9 The radius of a circle is 2.12 m. Its area according to the rule of significant figures is
 (a) 14.1124 m^2 (b) 14.112 m^2
 (c) 14.11 m^2 (d) 14.1 m^2
- 10 If the value of resistance is $10.845\ \Omega$ and the value of current is 3.23 A, the value of potential with significant numbers would be
 (a) 35.0 V (b) 3.50 V
 (c) 35.029 V (d) 35.030 V
- 11 The position of the particle moving along Y-axis is given as $y = At^2 - Bt^3$, where y is measured in metre and t in second. Then, the dimensions of B are
 (a) $[LT^{-2}]$ (b) $[LT^{-1}]$ (c) $[LT^{-3}]$ (d) $[MLT^{-2}]$
- 12 Out of the following four dimensional quantities, which one qualifies to be called a dimensional constant?
 (a) Acceleration due to gravity
 (b) Surface tension of water
 (c) Weight of a standard kilogram mass
 (d) The velocity of light in vacuum
- 13 If the random error in the arithmetic means of 100 observations is x , then the random error in the arithmetic mean of 400 observations would be
 (a) $4x$ (b) $\frac{1}{4}x$ (c) $2x$ (d) $\frac{1}{2}x$
- 14 The damping force on an oscillator is directly proportional to the velocity. The unit of the constant of proportionality is
 (a) $kg\text{ ms}^{-1}$ (b) $kg\text{ ms}^{-2}$ (c) kgs^{-1} (d) kgs
- 15 The square root of the product of inductance and capacitance has the dimensions of
 (a) length (b) time
 (c) mass (d) no dimension
- 16 The frequency of vibration of string is given by $f = \frac{p}{2l} \left[\frac{F}{m} \right]^{1/2}$. Here, p is number of segments in the string and l is the length. The dimensional formula for m will be
 (a) $[M^0LT^{-1}]$ (b) $[ML^0T^{-1}]$
 (c) $[ML^{-1}T^0]$ (d) $[M^0L^0T^0]$
- 17 The numbers 2.745 and 2.735 on rounding off to 3 significant figures will give [NCERT Exemplar]
 (a) 2.75 and 2.74 (b) 2.74 and 2.73
 (c) 2.75 and 2.73 (d) 2.74 and 2.74
- 18 The mass and volume of a body are 4.237 g and 2.5 cm^3 , respectively. The density of the material of the body in correct significant figures is [NCERT Exemplar]
 (a) 1.6048 g cm^{-3} (b) 1.69 g cm^{-3}
 (c) 1.7 g cm^{-3} (d) 1.695 g cm^{-3}

- 19** Which of the following measurement is most precise? [NCERT Exemplar]
 (a) 5.00 mm (b) 5.00 cm
 (c) 5.00 m (d) 5.00 km
- 20** The mean length of an object is 5 cm. Which of the following measurements is most accurate? [NCERT Exemplar]
 (a) 4.9 cm (b) 4.805 cm
 (c) 5.25 cm (d) 5.4 cm
- 21** If the energy (E), velocity (v) and force (F) be taken as fundamental quantities, then the dimensions of mass will be
 (a) $[Fv^{-2}]$ (b) $[Fv^{-1}]$
 (c) $[Ev^{-2}]$ (d) $[Ev^{-1}]$
- 22** The dimensional formula for molar thermal capacity is same as that of
 (a) gas constant (b) specific heat
 (c) Boltzmann's constant (d) Stefan's constant
- 23** The displacement of an oscillating particle is given by $y = A \sin(Bx + Ct + D)$. The dimensional formula for $(ABCD)$ is
 (a) $[M^0L^{-1}T^0]$ (b) $[M^0L^0T^{-1}]$ (c) $[M^0L^{-1}T^{-1}]$ (d) $[M^0L^0T^0]$
- 24** A force F is given by $F = at + bt^2$, where t is time. The dimensions of a and b are
 (a) $[MLT^{-3}]$ and $[MLT^{-4}]$ (b) $[MLT^{-4}]$ and $[MLT^{-3}]$
 (c) $[MLT^{-1}]$ and $[MLT^{-2}]$ (d) $[MLT^{-2}]$ and $[MLT^0]$
- 25** The length, breadth and thickness of a block are given by $l = 12$ cm, $b = 6$ cm and $t = 2.45$ cm, respectively. The volume of the block according to the idea of significant figures should be
 (a) $1 \times 10^2 \text{ cm}^3$ (b) $1.76 \times 10^2 \text{ cm}^3$
 (c) $1.764 \times 10^2 \text{ cm}^3$ (d) None of these
- 26** A physical quantity Q is calculated according to the expression

$$Q = \frac{A^3 B^3}{C\sqrt{D}}$$
 If percentage errors in A, B, C, D are 2%, 1%, 3% and 4%, respectively. What is the percentage error in Q ?
 (a) $\pm 8\%$ (b) $\pm 10\%$ (c) $\pm 14\%$ (d) $\pm 12\%$
- 27** With usual notation, the following equation said to give the distance covered in the n th second, i.e.

$$s_n = u + a \frac{(2n-1)}{2}$$
 is
 (a) only numerically correct
 (b) only dimensionally correct
 (c) Both dimensionally and numerically correct
 (d) Neither numerically nor dimensionally correct
- 28** The length and breadth of a rectangular sheet are 16.2 cm and 10.1 cm, respectively. The area of the sheet in appropriate significant figures and error is [NCERT Exemplar]
 (a) $164 \pm 3 \text{ cm}^2$ (b) $163.62 \pm 2.6 \text{ cm}^2$
 (c) $163.6 \pm 2.6 \text{ cm}^2$ (d) $163.62 \pm 3 \text{ cm}^2$
- 29** Which of the following pairs of physical quantities does not have same dimensional formula? [NCERT Exemplar]
 (a) Work and torque
 (b) Angular momentum and Planck's constant
 (c) Tension and surface tension
 (d) Impulse and linear momentum
- 30** Measure of two quantities along with the precision of respective measuring instrument is
 $A = 2.5 \text{ ms}^{-1} \pm 0.5 \text{ ms}^{-1}$, $B = 0.10 \text{ s} \pm 0.01 \text{ s}$. The value of AB will be [NCERT Exemplar]
 (a) $(0.25 \pm 0.08) \text{ m}$ (b) $(0.25 \pm 0.5) \text{ m}$
 (c) $(0.25 \pm 0.05) \text{ m}$ (d) $(0.25 \pm 0.135) \text{ m}$
- 31** Young's modulus of steel is $1.9 \times 10^{11} \text{ Nm}^{-2}$. When expressed in CGS units of dyne/cm^2 , it will be equal to ($1 \text{ N} = 10^5 \text{ dyne}$, $1 \text{ m}^2 = 10^4 \text{ cm}^2$) [NCERT Exemplar]
 (a) 1.9×10^{10} (b) 1.9×10^{11} (c) 1.9×10^{12} (d) 1.9×10^{13}
- 32** If voltage $V = (100 \pm 5) \text{ V}$ and current $I = (10 \pm 0.2) \text{ A}$, the percentage error in resistance R is
 (a) 5.2% (b) 25% (c) 7% (d) 10%
- 33** A wire has a mass $(0.3 \pm 0.003) \text{ g}$, radius $(0.5 \pm 0.005) \text{ cm}$ and length $(0.6 \pm 0.006) \text{ cm}$. The maximum percentage error in the measurement of its density is
 (a) 1 (b) 2 (c) 3 (d) 4
- 34** If $x = 10.0 \pm 0.1$ and $y = 10.0 \pm 0.1$, then $2x - 2y$ is equals to
 (a) (0.0 ± 0.1) (b) zero (c) (0.0 ± 0.4) (d) (20 ± 0.2)
- 35** Dimensions of Ohm are same as
 (a) $\frac{h}{e}$ (b) $\frac{h^2}{e}$ (c) $\frac{h}{e^2}$ (d) $\frac{h^2}{e^2}$
 (where, h is Planck's constant and e is charge)
- 36** The equation of state of some gases can be expressed as

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT$$
 where, p is the pressure, V is the volume, T is the absolute temperature and a, b and R are constants. The dimensions of a are
 (a) $[ML^5T^{-2}]$ (b) $[ML^{-1}T^{-2}]$ (c) $[L^3]$ (d) $[L^6]$

- 37** Using mass (M), length (L), time (T) and current (A) as fundamental quantities, the dimensions of permeability are

(a) $[M^{-1}LT^{-2}A]$ (b) $[ML^2T^{-2}A^{-1}]$
(c) $[MLT^{-2}A^{-2}]$ (d) $[MLT^{-1}A^{-1}]$

- 38** In a system of units, the units of mass, length and time are 1 quintal, 1 km and 1 h, respectively. In this system, 1 N force will be equal to

(a) 1 new unit (b) 129.6 new unit
(c) 427.6 new unit (d) 60 new unit

- 39** Given that $\int \frac{dx}{\sqrt{2ax - x^2}} = a^n \sin^{-1} \left[\frac{x - a}{a} \right]$

where, $a = \text{constant}$. Using dimensional analysis, the value of n is

(a) 1 (b) zero
(c) -1 (d) None of these

- 40** The magnetic force on a point charge is

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$$

Here, q = electric charge,

$\mathbf{v}$ = velocity of point charge

and $\mathbf{B}$ = magnetic field.

The dimensions of $\mathbf{B}$ are

(a) $[MLT^{-1}A]$ (b) $[M^2LT^{-2}A^{-1}]$
(c) $[MT^{-2}A^{-1}]$ (d) None of these

- 41** A quantity is given by $X = \frac{\epsilon_0 lV}{t}$, where V is the potential difference and l is the length. Then, X has dimensional formula same as that of

(a) resistance (b) charge
(c) voltage (d) current

- 42** The length of a strip measured with a metre rod is 10.0 cm. Its width measured with a vernier callipers is 1.00 cm. The least count of the metre rod is 0.1 cm and that of vernier callipers is 0.01 cm. What will be error in its area?

(a) $\pm 13\%$ (b) $\pm 7\%$ (c) $\pm 4\%$ (d) $\pm 2\%$

- 43** The length of cylinder is measured with a metre rod having least count 0.1 cm. Its diameter is measured with vernier callipers having least count 0.01 cm.

Given that length is 5.0 cm and radius is 2.0 cm. The percentage error in the calculated value of the volume will be

(a) 1.5% (b) 2.5% (c) 3.5% (d) 4%

- 44** You measure two quantities as $A = 1.0 \text{ m} \pm 0.2 \text{ m}$, $B = 2.0 \text{ m} \pm 0.2 \text{ m}$. We should report correct value for $\sqrt{AB}$ as [NCERT Exemplar]

(a) $1.4 \text{ m} \pm 0.4 \text{ m}$ (b) $1.41 \text{ m} \pm 0.15 \text{ m}$
(c) $1.4 \text{ m} \pm 0.3 \text{ m}$ (d) $1.4 \text{ m} \pm 0.2 \text{ m}$

- 45** If E = energy, G = gravitational constant, I = impulse and M = mass, then dimensions of $\frac{GIM^2}{E^2}$ are same as

that of

(a) time (b) mass (c) length (d) force

- 46** The relation $p = \frac{\alpha}{\beta} e^{\frac{-\alpha Z}{k\theta}}$, where p is pressure, Z is distance, k is Boltzmann constant and θ is temperature. The dimensional formula of β will be

(a) $[M^0L^2T^0]$ (b) $[ML^2T]$ (c) $[ML^0T^{-1}]$ (d) $[M^0L^2T^{-1}]$

- 47** If E , M , L and G denote energy, mass, angular momentum and gravitational constant respectively, then the quantity (E^2L^2/M^5G^2) has the dimensions of

(a) angle (b) length (c) mass (d) energy

- 48** If the energy $E = G^p h^q c^r$, where G is the universal gravitational constant, h is the Planck's constant and c is the velocity of light, then the values of p , q and r are respectively

(a) $-1/2$, $1/2$ and $5/2$ (b) $1/2$, $-1/2$ and $-5/2$
(c) $-1/2$, $1/2$ and $3/2$ (d) $1/2$, $1/2$ and $-3/2$

- 49** A gas bubble formed from an explosion under water oscillates with a period T proportional to $p^a d^b E^c$, where p is the pressure, d is the density of water and E is the total energy of explosion. The values of a , b and c are

(a) $a = 1$, $b = 1$, $c = 2$ (b) $a = 1$, $b = 2$, $c = 1$
(c) $a = \frac{5}{6}$, $b = \frac{1}{2}$, $c = \frac{1}{3}$ (d) $a = -\frac{5}{6}$, $b = \frac{1}{2}$, $c = \frac{1}{3}$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) If both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.
(c) If Assertion is true, but Reason is false.
(d) If Assertion is false, but Reason is true.

1 Assertion When we change the unit of measurement of a quantity, its numerical value changes.

Reason Smaller the unit of measurement, smaller is its numerical value.

2 Assertion The error in the measurement of radius of the sphere is 0.3%. The permissible error in its surface area is 0.6%.

Reason The permissible error is calculated by the formula

$$\frac{\Delta A}{A} = 4 \frac{\Delta r}{r}$$

3 Assertion If $x = \frac{a^n}{b^m}$, then $\frac{\Delta x}{x} = n \left(\frac{\pm \Delta a}{a} \right) - m \left(\frac{\pm \Delta b}{b} \right)$

The change in a or b , i.e. Δa or Δb may be comparable to a and b .

Reason The above relation is valid when $\Delta a \ll a$ and $\Delta b \ll b$.

4 Assertion Dimensional formula of the given quantity

$$\sqrt{\frac{\text{Magnetic dipole moment} \times \text{magnetic induction}}{\text{Moment of inertia}}} \text{ is } [M^0 L^0 T^{-1}].$$

Reason The given dimension is that of frequency.

5 Assertion $\sqrt{\frac{\text{Modulus of elasticity}}{\text{Density}}}$ has the unit ms^{-1} .

Reason Acceleration has the dimensions of $\frac{1}{(\sqrt{\epsilon_0 \mu_0}) t}$.

Statement based questions

1 Which of the following statement(s) is/are incorrect?

- Method of dimensions cannot be used for deriving formulae containing trigonometrical ratios.
- The light year and wavelength consist of dimensions of length.
- Both light year and wavelength represent time.
- Pressure has the dimensions of energy density.

2 Which of the following statement(s) is/are incorrect?

- Systematic errors and random errors fall in the same group of errors.
- Both systematic and random errors are based on the cause of error.
- Absolute error cannot be negative.
- Absolute error is the difference between the real value and the measured value of a physical quantity.

3 Which of the following statement(s) is/are incorrect?

- Dimensional formula of thermal conductivity (K) is $[M^1 L^1 T^{-3} K^{-1}]$.
- Dimensional formula of potential (V) is $[M^{-1} L^2 T^{-3} A^{-1}]$.
- Dimensional formula of permeability of free space (μ_0) is $[M^1 L^1 T^{-2} A^{-2}]$.
- Dimensional formula of RC is $[M^0 L^0 T^1]$.

4 Which of the following statement(s) is/are correct?

- Out of two measurements $l = 0.7 \text{ m}$ and $l = 0.70 \text{ m}$, the second one is more accurate.
 - In every measurement, the last digit is not accurately known.
- Only I
 - Only II
 - Both I and II
 - None of these

5 Which of the following statement(s) is/are correct?

- A screw gauge having a smaller value of pitch has greater accuracy.
 - The least count of screw gauge is directly proportional to the number of divisions on circular scale.
- Only I
 - Only II
 - Both I and II
 - None of these

Match the columns

1 Match the following columns.

Column I	Column II
(A) R/L	(p) Time
(B) CR	(q) Frequency
(C) E/B	(r) Speed
(D) $\sqrt{\epsilon_0 \mu_0}$	(s) None

Codes

- | | | | | | | | |
|-------|---|---|---|-------|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) p | r | q | s | (b) q | p | r | s |
| (c) s | q | p | r | (d) p | r | s | q |

2 Match the following columns.

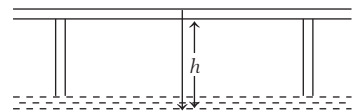
Column I (Physical quantity)	Column II (Dimensions)
(A) $GM_e M_s$	(p) $[M^2 L^2 T^{-3}]$
(B) $3kT/M$	(q) $[L^2 T^{-2}]$
(C) $F^2/q^2 B^2$	(r) $[L^2 T^{-1} \theta^{-1}]$
(D) GM_e/R_e	(s) None

Codes

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) r | r | p | q |
| (b) p | p | s | r |
| (c) s | q | r | s |
| (d) s | r | q | q |

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1** A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale.
The pitch of the screw gauge is [NEET 2020]
(a) 0.25 mm (b) 0.5 mm (c) 1.0 mm (d) 0.01 mm
- 2** Taking into account of the significant figures, what is the value of $9.99 \text{ m} - 0.0099 \text{ m}$? [NEET 2020]
(a) 9.98 m (b) 9.980 m
(c) 9.9 m (d) 9.9801 m
- 3** Dimensions of stress are [NEET 2020]
(a) $[\text{ML}^2\text{T}^{-2}]$ (b) $[\text{ML}^0\text{T}^{-2}]$
(c) $[\text{ML}^{-1}\text{T}^{-2}]$ (d) $[\text{MLT}^{-2}]$
- 4** The angle of $1'$ (minute of arc) in radian is nearly equal to [NEET 2020]
(a) $2.91 \times 10^{-4} \text{ rad}$ (b) $4.85 \times 10^{-4} \text{ rad}$
(c) $4.80 \times 10^{-6} \text{ rad}$ (d) $1.75 \times 10^{-2} \text{ rad}$
- 5** Time intervals measured by a clock give the following readings 1.25 s, 1.24 s, 1.27 s, 1.21 s and 1.28 s.
What is the percentage relative error of the observations? [NEET 2020]
(a) 2% (b) 4% (c) 16% (d) 1.6%
- 6** The SI unit of thermal conductivity is [NEET 2019]
(a) $\text{J m}^{-1}\text{K}^{-1}$ (b) W m K^{-1}
(c) $\text{W m}^{-1}\text{K}^{-1}$ (d) J m K^{-1}
- 7** In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4%, respectively. Then, the maximum percentage of error in the measurement X , where $X = \frac{A^2 B^{1/2}}{C^{1/3} D^3}$ will be [NEET 2019]
(a) 16% (b) -10% (c) 10% (d) $\left(\frac{3}{13}\right)\%$
- 8** The main scale of a vernier callipers has n divisions/cm. n divisions of the vernier scale coincide with $(n - 1)$ divisions of main scale. The least count of the vernier callipers is [NEET 2019]
(a) $\frac{1}{(n+1)(n-1)} \text{ cm}$ (b) $\frac{1}{n} \text{ cm}$
(c) $\frac{1}{n^2} \text{ cm}$ (d) $\frac{1}{n(n+1)} \text{ cm}$
- 9** Calculate the mean percentage error in five observations, 80.0, 80.5, 81.0, 81.5 and 82. [AIIMS 2019]
(a) 0.74% (b) 1.74% (c) 0.38% (d) 1.38%
- 10** If mass $[M]$, distance $[L]$ and time $[T]$ are fundamental quantities, then find the dimensions of torque. [JIPMER 2019]
(a) $[\text{ML}^2\text{T}^{-2}]$ (b) $[\text{MLT}^{-2}]$
(c) $[\text{MLT}]$ (d) $[\text{ML}^2\text{T}]$
- 11** What is the dimensions of energy in terms of linear momentum $[p]$, area $[A]$ and time $[T]$? [JIPMER 2019]
(a) $[p^1 A^1 T^1]$ (b) $[p^2 A^2 T^{-1}]$
(c) $[p^1 A^{1/2} T^{-1}]$ (d) $[p^{1/2} A^{1/2} T^{-1}]$
- 12** A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of -0.004 cm , the correct diameter of the ball is [NEET 2018]
(a) 0.053 cm (b) 0.525 cm (c) 0.521 cm (d) 0.529 cm
- 13** In an experiment to measure the height of a bridge by dropping stone into water underneath. If the error in measurement of time is 0.2s at the end of 4s, then the error in estimation of height of bridge will be (neglect the water resistance, i.e. thrust) [AIIMS 2018]

(a) $\pm 19.68 \text{ m}$ (b) $\pm 17.22 \text{ m}$
(c) $\pm 7.84 \text{ m}$ (d) $\pm 12.22 \text{ m}$
- 14** $R = (65 \pm 1) \Omega$, $l = (5 \pm 0.1) \text{ mm}$ and $d = (10 \pm 0.5) \text{ mm}$. Find error in calculation of resistivity. [JIPMER 2018]
(a) 21% (b) 13% (c) 16% (d) 41%
- 15** Dimensions of force are [JIPMER 2018]
(a) $[\text{M}^2\text{L}^1\text{T}^{-1}]$ (b) $[\text{ML}^1\text{T}^{-2}]$
(c) $[\text{M}^2\text{L}^{-1}\text{T}^{-2}]$ (d) $[\text{ML}^1\text{T}^{-1}]$
- 16** A physical quantity of the dimensions of length that can be formed out of c , G and $\frac{e^2}{4\pi\epsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge] [NEET 2017]
(a) $\frac{1}{c^2} \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$ (b) $c^2 \left[G \frac{e^2}{4\pi\epsilon_0} \right]^{1/2}$
(c) $\frac{1}{c^2} \left[\frac{e^2}{G 4\pi\epsilon_0} \right]^{1/2}$ (d) $\frac{1}{c} G \frac{e^2}{4\pi\epsilon_0}$

- 17** Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimensions of length? [NEET 2016]

(a) $\frac{\sqrt{hG}}{c^{3/2}}$ (b) $\frac{\sqrt{hG}}{c^{5/2}}$
 (c) $\sqrt{\frac{hc}{G}}$ (d) $\sqrt{\frac{Gc}{h^{3/2}}}$

- 18** If energy (E), velocity (v) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be [CBSE AIPMT 2015]

(a) $[Ev^{-2}T^{-1}]$ (b) $[Ev^{-1}T^{-2}]$
 (c) $[Ev^{-2}T^{-2}]$ (d) $[Ev^{-2}v^{-1}T^{-3}]$

- 19** In terms of basic units of mass (M), length (L), time (T) and charge (Q), the dimensions of magnetic permeability of vacuum (μ_0) would be [AIIMS 2015]

(a) $[MLQ^{-2}]$ (b) $[LT^{-1}Q^{-1}]$
 (c) $[ML^2T^{-1}Q^{-2}]$ (d) $[LTQ^{-1}]$

- 20** The dimensional formula for electric flux is [AIIMS 2015]

(a) $[ML^3T^{-1}T^{-3}]$ (b) $[M^2L^2I^{-1}T^{-2}]$
 (c) $[ML^3T^{-1}T^{-3}]$ (d) $[ML^{-3}I^{-1}T^{-3}]$

- 21** In terms of time t and distance x , the force F is given by $F = A \sin Ct + B \cos Dx$, then dimensions of $\frac{A}{B}$ and

$\frac{C}{D}$ are given by [UK PMT 2015]

(a) $[M^0L^0T^0]$, $[M^0LT^{-1}]$ (b) $[MLT^{-2}]$, $[M^0L^0T^{-1}]$
 (c) $[MLT^{-2}]$, $[M^0L^{-1}T^0]$ (d) $[M^0LT^{-1}]$, $[M^0L^0T^0]$

- 22** The wrong unit conversion among the following is

(a) 1 angstrom = 10^{-10} m (b) 1 fermi = 10^{-15} m [Kerala CEE 2015]
 (c) 1 light year = 9.46×10^{15} m
 (d) 1 astronomical unit = 1.496×10^{-11} m
 (e) 1 parsec = 3.08×10^{16} m

- 23** The mass of the liquid flowing per second per unit area of cross-section of the tube is proportional to (pressure difference across the ends) n and (average velocity) m of the liquid. Which one of the following relation is correct? [CG PMT 2015]

(a) $m = n$ (b) $m = -n$ (c) $m^2 = n$ (d) $m = -n^2$

- 24** The ratio of the dimensions of Planck's constant and that of moment of inertia has the dimensions of

(a) angular momentum (b) time [KCET 2015]
 (c) velocity (d) frequency

- 25** Match the column I with column II and mark the correct option from the codes given below.

[Guj. CET 2015]

Column I	Column II
(A) Electrical resistance	1. $[ML^3T^{-3}A^{-2}]$
(B) Electrical potential	2. $[ML^2T^{-3}A^{-2}]$
(C) Specific resistance	3. $[ML^2T^{-3}A^{-1}]$
(D) Specific conductance	4. None

Codes

(a) A-2, B-3, C-1, D-4 (b) A-2, B-4, C-3, D-1
 (c) A-1, B-2, C-4, D-3 (d) A-1, B-3, C-2, D-4

- 26** The three physical quantities x , y and z have units $g\text{ cm}^2\text{ s}^{-5}$, $g\text{ s}^{-1}$ and cm s^{-2} , respectively. The relation between x , y and z is [AFMC 2015]

(a) $x = yz^2$ (b) $x = y^2z$
 (c) $y^2 = xz$ (d) $z = x^2y$

- 27** If force (F), velocity (v) and time (T) are taken as fundamental units, then the dimensions of mass are [CBSE AIPMT 2014]

(a) $[FvT^{-1}]$ (b) $[FvT^{-2}]$
 (c) $[Fv^{-1}T^{-1}]$ (d) $[Fv^{-1}T]$

- 28** The dimensional formula for Reynold's number is [MHT CET 2014]

(a) $[L^0M^0T^0]$ (b) $[LMT]$
 (c) $[L^{-1}MT]$ (d) $[LMT^{-1}]$

- 29** The relation between force F and density d is

$F = \frac{x}{\sqrt{d}}$. The dimensions of x are [MHT CET 2014]

(a) $[L^{-1/2}M^{3/2}T^{-2}]$ (b) $[L^{-1/2}M^{1/2}T^{-2}]$
 (c) $[L^{-1}M^{3/2}T^{-2}]$ (d) $[L^{-1}M^{1/2}T^{-2}]$

- 30** If the absolute errors in two physical quantities A and B are a and b respectively, then the absolute error in the value of $A - B$ are [EAMCET 2014]

(a) $b - a$ (b) $a \neq b$
 (c) $a + b$ (d) $a - b$

- 31** If the unit of force is kN, the length is 1 km and time 100 s, then what will be the unit of mass? [KCET 2014]

(a) 1000 kg (b) 1 kg
 (c) 10000 kg (d) 100 kg

- 32** If n denotes a positive integer, h the Planck's constant, q the charge and B the magnetic field, then

the quantity $\left[\frac{nh}{2\pi qB} \right]$ has the dimension of [WB JEE 2014]

(a) area (b) length
 (c) speed (d) acceleration

- 33** In an experiment, four quantities a , b , c and d are measured with percentage error 1%, 2%, 3% and 4%, respectively. Quantity P is calculated as follows

$$P = \frac{a^3 b^2}{cd} \%, \text{ error in } P \text{ is} \quad [\text{NEET 2013}]$$

- (a) 14% (b) 10%
(c) 7% (d) 4%
- 34** The density of glass is 2.8 g/cc in CGS system. The value of density in SI unit is [Kerala CEE 2013]
- (a) 2.8×10^{-3} (b) 2.8×10^{-2}
(c) 2.8×10^2 (d) 2.8×10^3
(e) 2.8×10^6

- 35.** In the equation $\left(\frac{1}{p\beta}\right) = \frac{y}{k_B T}$, where p is the pressure,

y is the distance, k_B is Boltzmann constant and T is the temperature. Dimensions of β are [EAMCET 2013]

- (a) $[M^{-1} L^1 T^2]$ (b) $[M^0 L^2 T^0]$ (c) $[M^1 L^{-1} T^{-2}]$ (d) $[M^0 L^0 T^0]$
- 36** A physical quantity X is defined by the formula

$$X = \frac{IFv^2}{WL^3}$$

where, I is moment of inertia, F is force, v is velocity, W is work and L is length, the dimensions of X are [MP PMT 2013]

- (a) $[MLT^{-2}]$ (b) $[MT^{-2}]$ (c) $[ML^2T^{-3}]$ (d) $[LT^{-1}]$
- 37** A physical quantity X is given by

$$X = \frac{2k^3 l^2}{m\sqrt{n}}$$

The percentage error in the measurements of k , l , m and n are 1%, 2%, 3% and 4%, respectively. The value of X is uncertain by [AMU 2012]

- (a) 8% (b) 10%
(c) 12% (d) None of these

- 38** The quantities A and B are related by the relation $m = A/B$, where m is the linear density and A is the force. The dimensions of B are of [BCECE 2012]

- (a) pressure (b) latent heat
(c) work (d) None of these

- 39** A physical quantity is given by $X = [M^a L^b T^c]$. The percentage error in measurement of M , L and T are α , β and γ , respectively. Then, the maximum % error in the quantity X is [AFMC 2012]

- (a) $a\alpha + b\beta + c\gamma$ (b) $a\alpha + b\beta - c\gamma$
(c) $\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma}$ (d) None of these

- 40** The dimensions of $(\mu_0 \epsilon_0)^{-1/2}$ are [CBSE AIPMT 2011]

- (a) $[L^{-1}T]$ (b) $[LT^{-1}]$
(c) $[L^{-1/2}T^{1/2}]$ (d) $[L^{1/2}T^{-1/2}]$

- 41** If $p = \frac{RT}{V-b} e^{-\alpha V/RT}$, then dimensional formula of α is [UP CPMT 2011]

- (a) p (b) R
(c) T (d) V

- 42** Velocity v is given by $v = at^2 + bt + c$, where t is time. What are the dimensions of a , b and c , respectively?

- (a) $[LT^{-3}]$, $[LT^{-2}]$ and $[LT^{-1}]$ [UP CPMT 2011]
(b) $[LT^{-1}]$, $[LT^{-2}]$ and $[LT^{-3}]$
(c) $[LT^{-2}]$, $[LT^{-3}]$ and $[LT^{-1}]$
(d) $[LT^{-1}]$, $[LT^{-3}]$ and $[LT^{-2}]$

- 43** From the dimensional consideration, which of the following equations is correct? [Haryana PMT 2011]

- (a) $T = 2\pi\sqrt{\frac{R^3}{GM}}$ (b) $T = 2\pi\sqrt{\frac{GM}{R^3}}$
(c) $T = 2\pi\sqrt{\frac{GM}{R^2}}$ (d) $T = 2\pi\sqrt{\frac{R^2}{GM}}$

ANSWERS

- **CHECK POINT 1.1**

1. (b)	2. (d)	3. (c)	4. (c)	5. (d)	6. (c)	7. (b)	8. (c)	9. (c)	10. (c)
11. (d)	12. (a)	13. (b)	14. (b)	15. (c)	16. (d)	17. (b)	18. (c)	19. (a)	20. (b)
21. (d)									

- **CHECK POINT 1.2**

1. (b)	2. (c)	3. (b)	4. (c)	5. (c)	6. (d)	7. (a)	8. (b)	9. (d)	10. (a)
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- **CHECK POINT 1.3**

1. (b)	2. (d)	3. (c)	4. (d)	5. (d)	6. (c)	7. (a)	8. (b)	9. (c)	10. (a)
11. (d)									

(A) Taking it together

1. (a)	2. (c)	3. (a)	4. (b)	5. (c)	6. (a)	7. (b)	8. (b)	9. (d)	10. (a)
11. (c)	12. (d)	13. (b)	14. (c)	15. (b)	16. (c)	17. (d)	18. (c)	19. (a)	20. (a)
21. (c)	22. (a)	23. (b)	24. (a)	25. (b)	26. (c)	27. (c)	28. (a)	29. (c)	30. (a)
31. (c)	32. (c)	33. (d)	34. (c)	35. (c)	36. (a)	37. (c)	38. (b)	39. (b)	40. (c)
41. (d)	42. (d)	43. (b)	44. (d)	45. (a)	46. (a)	47. (d)	48. (a)	49. (d)	

(B) Medical entrance special format questions

- **Assertion and reason**

1. (c) 2. (c) 3. (d) 4. (a) 5. (b)

- **Statement based questions**

1. (c) 2. (c) 3. (b) 4. (c) 5. (a)

- **Match the columns**

1. (b) 2. (d)

(C) Medical entrances' gallery

1. (b)	2. (a)	3. (c)	4. (a)	5. (d)	6. (c)	7. (a)	8. (c)	9. (a)	10. (a)
11. (c)	12. (d)	13. (c)	14. (b)	15. (b)	16. (a)	17. (a)	18. (c)	19. (a)	20. (a)
21. (a)	22. (d)	23. (b)	24. (d)	25. (a)	26. (a)	27. (d)	28. (a)	29. (a)	30. (c)
31. (c)	32. (a)	33. (a)	34. (d)	35. (b)	36. (b)	37. (c)	38. (b)	39. (a)	40. (b)
41. (a)	42. (a)	43. (a)							

Hints & Explanations

• CHECK POINT 1.1

3 (c) Since, $(mvr) = n \cdot \frac{h}{2\pi}$ and $E = h\nu$

So, unit of h = joule second = angular momentum

5 (d) We know that, $\text{density} = \frac{\text{mass}}{\text{volume}}$

In CGS system, $d = 0.625 \text{ g cm}^{-3}$

In SI system, $d = \frac{0.625 \times 10^{-3} \text{ kg}}{10^{-6} \text{ m}^3} = 625 \text{ kg m}^{-3}$

7 (b) Impulse = Change in linear momentum.

12 (a) Since $R = \frac{\rho l}{A}$, where ρ is specific resistance.

$$\therefore [\rho] = \left[\frac{RA}{l} \right], R = \frac{V}{i}, V = \frac{W}{Q}$$

$$[\rho] = [\text{ML}^3\text{T}^{-1}\text{Q}^{-2}]$$

15 (c) Since, units of length, velocity and force are doubled.

$$\text{Hence, } [m] = \frac{[\text{force}][\text{time}]}{[\text{velocity}]}, [\text{time}] = \frac{[\text{length}]}{[\text{velocity}]}$$

Hence, unit of mass and time remains same.
Momentum is doubled.

17 (b) Given, $p = \frac{a - t^2}{bx}$, where p is the pressure and t is the time

$$[pbx] = [a] = [t^2]$$

$$\text{Hence, } [b] = \frac{[t^2]}{[px]}$$

$$\text{Dimensions of } \frac{a}{b} = [px] = [\text{MT}^{-2}]$$

18 (c) $\therefore \frac{x}{v}$ have the same dimension as of k .

$$\therefore \text{Dimensional formula of } [k] = \frac{[x]}{[v]} = \frac{[L]}{[\text{LT}^{-1}]} = [T]$$

19 (a) Muscle $\times$ Speed = Power

$$\text{or Muscle} = \frac{\text{Power}}{\text{Speed}} = \frac{\text{Work}}{\text{Time} \times \text{Speed}} = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{T}][\text{LT}^{-1}]} = [\text{MLT}^{-2}]$$

$$= \text{Mass} \times \text{Acceleration} = \text{Force}$$

$$\text{Now, } \frac{\text{SI unit of force}}{\text{CGS unit of force}} = \frac{\text{kg} \times \text{m} \times \text{s}^{-2}}{\text{g} \times \text{cm} \times \text{s}^{-2}} = 10^3 \times 10^2 = 10^5$$

20 (b) Since, $p^x Q^y c^z$ is dimensionless. Therefore,

$$[\text{ML}^{-1}\text{T}^{-2}]^x [\text{MT}^{-3}]^y [\text{LT}^{-1}]^z = [\text{M}^0\text{L}^0\text{T}^0]$$

Only option (b) satisfies this expression.

$$\text{So, } x = 1, y = -1, z = 1$$

21 (d) $m \propto v^a p^b g^c$. Writing the dimensions on both sides,

$$[M] = [\text{LT}^{-1}]^a [\text{ML}^{-3}]^b [\text{LT}^{-2}]^c$$

$$[M] = [\text{M}^b \text{L}^{a-3b+c} \text{T}^{-a-2c}]$$

$$\therefore b = 1$$

$$a - 3b + c = 0$$

$$-a - 2c = 0$$

Solving these, we get

$$a = 6$$

$$\text{Hence, } m \propto v^6$$

• CHECK POINT 1.2

3 (b) Only in 20.2, all zeros are significant because in this, zero lies between two non-zero digits.

6 (d) Out of 107.88 and 0.610, least number of significant digits is 3, so the product must contain 3 significant digits. So, the right answer is 65.8.

8 (b) Given, $R = 0.16 \text{ mm}$

$$\text{Hence, } A = \pi R^2 = \frac{22}{7} \times (0.16)^2 = 0.080457$$

Since, radius has two significant figures, so answer will also have two significant figures.

Hence, option (b) is correct.

9 (d) As 2.54 has least number of significant digits which is 3, so the result must have 3 digits as significant.

Hence, the correct result is 38.4.

10 (a) The product must have two significant digit as both the figures being multiplied have two significant digits, i.e. 25×10^{-14} . So, the order of magnitude of result is -14 .

• CHECK POINT 1.3

1 (b) Given, number of divisions on circular scale = 100

$$\text{Pitch} = 0.01 \text{ cm}$$

$$\therefore \text{Least count (LC)} = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}} = \frac{0.01}{100} = 10^{-4} \text{ cm}$$

3 (c) Since, $D \propto r$

where, D = diameter of a circle
and r = radius of the circle.

Therefore, there will be no change in error, it will remain 4% for radius also.

6 (c) Volume of cuboid,

$$V = l \times 2l \times 3l = 6l^3$$

$$\therefore \frac{\Delta V}{V} \times 100 = 3 \left(\frac{\Delta l}{l} \right) = 3\% \quad \left(\because \frac{\Delta l}{l} \times 100 = 1\% \right)$$

7 (a) $p = \frac{F}{A} = \frac{F}{L^2} = FL^{-2}$

% error in pressure = (% error in F) + 2(% error in L)
 $= (4\%) + 2(2\%) = 8\%$

8 (b) We know that, $H = i^2 R t$

$\therefore$ % error in $H = 2(\text{% error in } i) + (\text{% error in } R) + (\text{% error in } t)$
 $= 2(2\%) + 1\% + 1\% = 6\%$

9 (c) We know that, $K = \frac{p^2}{2m}$

The error in measurement of momentum is + 100%.

Therefore, the actual momentum with error, $p' = p + p = 2p$

$\therefore$ Kinetic energy with error,

$$K' = \frac{(p')^2}{2m} = \frac{(2p)^2}{2m}$$

or $K' = \frac{4p^2}{2m}$

or $K' = 4K$

So, percentage change in kinetic energy,

$$\text{KE} = \left(\frac{K' - K}{K} \right) \times 100 = \left(\frac{4K - K}{K} \right) \times 100$$

$$= \left(\frac{4 - 1}{1} \right) \times 100 = 300\%$$

10 (a) Radius of ball = 5.2 cm

Volume, $V = \frac{4}{3} \pi R^3$

$$\frac{\Delta V}{V} = 3 \left(\frac{\Delta R}{R} \right)$$

$$\left(\frac{\Delta V}{V} \right) \times 100 = 3 \left(\frac{0.2}{5.2} \right) \times 100 \approx 11\%$$

(A) Taking it together

6 (a) Magnetic flux, $[\phi] = [BS] = [MT^{-2}A^{-1}]$ $[L^2] = [ML^2T^{-2}A^{-1}]$

11 (c) As, $[y] = [B][T^3] \Rightarrow [L] = [B][T^3] \Rightarrow [B] = [LT^{-3}]$

13 (b) Since, error is measured for 400 observations instead of 100 observations. So, error will reduce by 1/4 factor.

14 (c) Given, damping force $\propto$ velocity

$$F \propto v \Rightarrow F = kv$$

$$\Rightarrow k = \frac{F}{v}$$

$$\text{Unit of } k = \frac{\text{Unit of } F}{\text{Unit of } v} = \frac{\text{kgms}^{-2}}{\text{ms}^{-1}} = \text{kg s}^{-1}$$

15 (b) $f = \frac{1}{2\pi\sqrt{LC}}$ or $\sqrt{LC} = \frac{1}{2\pi f} = \frac{T}{2\pi}$ ($\because T = 1/f$)

Thus, $\sqrt{LC}$ has the dimensions of time.

16 (c) Given, $f = \frac{p}{2l} \left[\frac{F}{m} \right]^{1/2}$

Squaring the equation on either side, we have

$$f^2 = \frac{p^2}{4l^2} \left(\frac{F}{m} \right) \Rightarrow m = \frac{p^2 F}{4l^2 f^2}$$

$$[m] = \frac{[MLT^{-2}]}{[L^2][T^{-1}]^2} = [ML^{-1}T^0]$$

17 (d) Rounding off 2.745 to 3 significant figures, it would be 2.74. Rounding off 2.735 to 3 significant figures, it would be 2.74.

18 (c) In this question, density should be reported to two significant figures.

$$\text{Density} = \frac{4.237\text{g}}{2.5\text{ cm}^3} = 1.6948\text{ g cm}^{-3}$$

On rounding off the number, we get

$$\text{Density} = 1.7\text{ g cm}^{-3}$$

19 (a) All given measurements are correct upto two decimal places. As here 5.00 mm has the smallest unit and the error in 5.00 mm is least (commonly taken as 0.01 mm, if not specified), hence 5.00 mm is most precise.

Note In solving these type of questions, we should be careful about units although their magnitude is same.

20 (a) Given, length, $l = 5\text{ cm}$

Now, checking the errors with each options one by one, we get

$$\Delta l_1 = 5 - 4.9 = 0.1\text{ cm}$$

$$\Delta l_2 = 5 - 4.805 = 0.195\text{ cm}$$

$$\Delta l_3 = 5.25 - 5 = 0.25\text{ cm}$$

$$\Delta l_4 = 5.4 - 5 = 0.4\text{ cm}$$

Error Δl_1 is least.

Hence, 4.9 cm is most accurate.

23 (b) Displacement,

$$y = A \sin(Bx + Ct + D)$$

$$A = y = [L]$$

As each term inside the brackets is dimensionless, so

$$B = \frac{1}{x} = [L^{-1}]$$

$$C = \frac{1}{t} = [T^{-1}]$$

and D is dimensionless.

$$\therefore [ABCD] = [L][L^{-1}][T^{-1}] = [M^0L^0T^{-1}]$$

25 (b) Using the relation for volume,

$$V = \text{Length} \times \text{Breadth} \times \text{Thickness}$$

$$= 12 \times 6 \times 2.45 = 176.4\text{ cm}^3$$

$$V = 1.764 \times 10^2\text{ cm}^3$$

The minimum number of significant figure is 3. Hence, the volume will contain only 3 significant figures. Therefore, $V \approx 1.76 \times 10^2\text{ cm}^3$.

26 (c) Given, $Q = \frac{A^3 B^3}{C\sqrt{D}}$

$$\frac{\Delta Q}{Q} = 3 \left(\frac{\Delta A}{A} \right) + 3 \left(\frac{\Delta B}{B} \right) + \left(\frac{\Delta C}{C} \right) + \frac{1}{2} \left(\frac{\Delta D}{D} \right)$$

$$\begin{aligned}\text{Here, } \frac{\Delta A}{A} \times 100 &= 2\%, \frac{\Delta B}{B} \times 100 = 1\%, \\ \frac{\Delta C}{C} \times 100 &= 3\%, \frac{\Delta D}{D} \times 100 = 4\% \\ \therefore \frac{\Delta Q}{Q} \times 100 &= (3 \times 2\%) + (3 \times 1\%) + (3\%) + \frac{1}{2} \times (4\%) = \pm 14\%\end{aligned}$$

28 (a) Given, length, $l = (16.2 \pm 0.1) \text{ cm}$

Breadth, $b = (10.1 \pm 0.1) \text{ cm}$

Area, $A = l \times b = (16.2 \text{ cm}) \times (10.1 \text{ cm}) = 163.62 \text{ cm}^2$

Rounding off to three significant digits, area, $A = 164 \text{ cm}^2$

$$\therefore \frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta b}{b} = \frac{0.1}{16.2} + \frac{0.1}{10.1} = \frac{1.01 + 1.62}{16.2 \times 10.1} = \frac{2.63}{163.62}$$

$$\Rightarrow \Delta A = A \times \frac{2.63}{163.62} = 163.62 \times \frac{2.63}{163.62} = 2.63 \text{ cm}^2$$

$\Delta A = 3 \text{ cm}^2$ (By rounding off to one significant figure)

$\therefore$ Area, $A = A \pm \Delta A = (164 \pm 3) \text{ cm}^2$

29 (c) (a) Work = Force $\times$ Distance = $[\text{MLT}^{-2}][\text{L}] = [\text{ML}^2\text{T}^{-2}]$

Torque = Force $\times$ Distance = $[\text{ML}^2\text{T}^{-2}]$

(b) Angular momentum = $mvr = [\text{M}][\text{LT}^{-1}][\text{L}] = [\text{ML}^2\text{T}^{-1}]$

$$\text{Planck's constant} = \frac{E}{\nu} = \frac{[\text{ML}^2\text{T}^{-2}]}{[\text{T}^{-1}]} = [\text{ML}^2\text{T}^{-1}]$$

(c) Tension = Force = $[\text{MLT}^{-2}]$

$$\text{Surface tension} = \frac{\text{Force}}{\text{Length}} = \frac{[\text{MLT}^{-2}]}{[\text{L}]} = [\text{ML}^0\text{T}^{-2}]$$

(d) Impulse = Force $\times$ Time = $[\text{MLT}^{-2}][\text{T}] = [\text{MLT}^{-1}]$

Momentum = Mass $\times$ Velocity = $[\text{M}][\text{LT}^{-1}] = [\text{MLT}^{-1}]$

Note One should not be confused with the similar form tension in both the physical quantities-surface tension and tension. **Dimensional formula** for both of them is not same.

30 (a) Given, $A = 2.5 \text{ ms}^{-1} \pm 0.5 \text{ ms}^{-1}$, $B = 0.10 \text{ s} \pm 0.01 \text{ s}$

$$x = AB = (2.5)(0.10) = 0.25 \text{ m}$$

$$\frac{\Delta x}{x} = \frac{\Delta A}{A} + \frac{\Delta B}{B} = \frac{0.5}{2.5} + \frac{0.01}{0.10} = \frac{0.05 + 0.025}{0.25} = \frac{0.075}{0.25}$$

$\Delta x = 0.075 = 0.08 \text{ m}$ (rounding off to one significant figure)

$$AB = (0.25 \pm 0.08) \text{ m}$$

31 (c) Given, Young's modulus, $Y = 1.9 \times 10^{11} \text{ Nm}^{-2}$

$$1 \text{ N} = 10^5 \text{ dyne}$$

Hence, $Y = 19 \times 10^{11} \times 10^5 \text{ dyne/m}^2$

We know that, $1 \text{ m} = 100 \text{ cm}$

$$\begin{aligned}\therefore Y &= 19 \times 10^{11} \times 10^5 \text{ dyne}/(100)^2 \text{ cm}^2 \\ &= 1.9 \times 10^{16-4} \text{ dyne/cm}^2 \\ Y &= 1.9 \times 10^{12} \text{ dyne/cm}^2\end{aligned}$$

32 (c) Given, voltage, $V = (100 \pm 5) \text{ V}$

Current, $I = (10 \pm 0.2) \text{ A}$

From Ohm's law, $V = IR \Rightarrow$ Resistance, $R = \frac{V}{I}$

Maximum percentage error in resistance,

$$\begin{aligned}\left(\frac{\Delta R}{R} \times 100\right) &= \left(\frac{\Delta V}{V} \times 100\right) + \left(\frac{\Delta I}{I} \times 100\right) \\ &= \left(\frac{5}{100} \times 100\right) + \left(\frac{0.2}{10} \times 100\right) = 5 + 2 = 7\%\end{aligned}$$

33 (d) Density, $\rho = \frac{m}{\pi r^2 L}$

$$\therefore \frac{\Delta \rho}{\rho} \times 100 = \left[\frac{\Delta m}{m} + \frac{2\Delta r}{r} + \frac{\Delta L}{L} \right] \times 100$$

After substituting the values, we get the maximum percentage error in density = 4%.

35 (c) $\therefore$ 'Ohm' is the unit of resistance and having dimension = $[\text{ML}^2\text{T}^{-3}\text{A}^{-2}]$

From option (c),

$$\Rightarrow \frac{[h]}{[e]^2} = \frac{[\text{ML}^2\text{T}^{-1}]}{[\text{AT}]^2} = [\text{ML}^2\text{T}^{-3}\text{A}^{-2}] = \text{Dimensions of resistance } (\Omega)$$

37 (c) $B = \mu NI \Rightarrow \mu = \frac{B}{NI}$

As, $Bqv = F$

$$\Rightarrow B = \frac{F}{qv}$$

So, $\mu = \frac{F}{qvNI}$, where N is the number of turns per unit length

$$[\mu] = \frac{[\text{MLT}^{-2}]}{[\text{AT}][\text{A}][\text{L}^{-1}]} = [\text{MLT}^{-2}\text{A}^{-2}]$$

38 (b) $[\text{Force}] = [\text{MLT}^{-2}]$

$$\therefore 1\text{N} = \left(\frac{1}{100}\right) \left(\frac{1}{1000}\right) (3600)^2 = 129.6 \text{ units}$$

40 (c) Magnetic force, $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$ or $F = qvB \sin \theta$

$$\therefore [B] = \left[\frac{F}{qv} \right] = \frac{[\text{MLT}^{-2}]}{[\text{AT}][\text{LT}^{-1}]} = [\text{MT}^{-2}\text{A}^{-1}]$$

42 (d) Area of strip = lb

$$\begin{aligned}\therefore \left(\frac{\Delta A}{A}\right) \times 100 &= \left(\frac{\Delta l}{l}\right) \times 100 + \left(\frac{\Delta b}{b}\right) \times 100 \\ &= \frac{0.1}{10} \times 100 + \frac{0.01}{1} \times 100 = \pm 2\%\end{aligned}$$

43 (b) Volume of cylinder, $V = \pi r^2 L$, $r = \left(\frac{D}{2}\right)$

$$\begin{aligned}\therefore \left(\frac{\Delta V}{V}\right) \times 100 &= 2 \left(\frac{\Delta D}{D}\right) \times 100 + \left(\frac{\Delta L}{L}\right) \times 100 \\ &= 2 \left(\frac{0.01}{4.0}\right) \times 100 + \left(\frac{0.1}{5}\right) \times 100 = 2.5\%\end{aligned}$$

44 (d) Given, $A = 1.0 \text{ m} \pm 0.2 \text{ m}$, $B = 2.0 \text{ m} \pm 0.2 \text{ m}$

Let $Y = \sqrt{AB} = \sqrt{(1.0)(2.0)} = 1.414 \text{ m}$

Rounding off to two significant digit $Y = 1.4 \text{ m}$

$$\therefore \frac{\Delta Y}{Y} = \frac{1}{2} \left[\frac{\Delta A}{A} + \frac{\Delta B}{B} \right] = \frac{1}{2} \left[\frac{0.2}{1.0} + \frac{0.2}{2.0} \right] = \frac{0.6}{2 \times 2.0}$$

$$\Rightarrow \Delta Y = \frac{0.6Y}{2 \times 2.0} = \frac{0.6 \times 1.4}{2 \times 2.0} = 0.212$$

Rounding off to one significant digit, $\Delta Y = 0.2 \text{ m}$

Thus, correct value for $\sqrt{AB} = Y + \Delta Y = 1.4 \text{ m} \pm 0.2 \text{ m}$

- 46** (a) In the given equation, $\frac{\alpha Z}{k\theta}$ should be dimensionless.

$$\therefore [\alpha] = \left[\frac{k\theta}{Z} \right] \Rightarrow [\alpha] = \frac{[\text{ML}^2\text{T}^{-2}\text{K}^{-1}][\text{K}]}{[\text{L}]} = [\text{MLT}^{-2}]$$

$$\text{and } p = \frac{\alpha}{\beta} \Rightarrow [\beta] = \left[\frac{\alpha}{p} \right] = \frac{[\text{MLT}^{-2}]}{[\text{ML}^{-1}\text{T}^{-2}]} = [\text{M}^0\text{L}^2\text{T}^0]$$

- 47** (d) The dimensions of $E = [\text{ML}^2\text{T}^{-2}]$

Dimension of $M = [\text{M}]$

Dimensions of $L = [\text{ML}^2\text{T}^{-1}]$

Dimensions of $G = [\text{M}^{-1}\text{L}^3\text{T}^{-2}]$

$$\therefore \text{Dimensions of } \left[\frac{E^2 L^2}{M^5 G^2} \right] = \frac{[\text{ML}^2\text{T}^{-2}]^2 [\text{ML}^2\text{T}^{-1}]^2}{[\text{M}]^5 [\text{M}^{-1}\text{L}^3\text{T}^{-2}]^2} = [\text{ML}^2\text{T}^{-2}] = \text{Energy}$$

- 49** (d) Given, $T \propto p^a d^b E^c$

We have, $[\text{M}^0\text{L}^0\text{T}] = k [\text{ML}^{-1}\text{T}^{-2}]^a [\text{ML}^{-3}]^b [\text{ML}^2\text{T}^{-2}]^c$

where, k is a constant.

$$\Rightarrow [\text{M}^0\text{L}^0\text{T}] = k [\text{M}^{a+b+c} \text{L}^{-a-3b+2c} \text{T}^{-2a-2c}]$$

On comparing powers of M , we have

$$0 = a + b + c \quad \dots(i)$$

On comparing powers of L , we have

$$0 = -a - 3b + 2c \quad \dots(ii)$$

On comparing powers of T , we have

$$1 = -2a - 2c \quad \dots(iii)$$

On solving Eqs. (i), (ii) and (iii), we have

$$a = -\frac{5}{6}, b = \frac{1}{2}, c = \frac{1}{3}$$

(B) Medical entrance special format questions

• Assertion and reason

- 2** (c) $A = 4\pi r^2$

$$\frac{\Delta A}{A} \times 100 = 2 \frac{\Delta r}{r} \times 100 = 2(0.3)\% = 0.6\%$$

- 5** (b) $\sqrt{\frac{\text{Modulus of elasticity}}{\text{Density}}} = \sqrt{\frac{[\text{ML}^{-1}\text{T}^{-2}]}{[\text{ML}^{-3}\text{T}^0]}} = \text{m/s}$

• Statement based questions

- 1** (c) Both light year and wavelength has dimensions of length.

- 2** (c) Absolute error may be negative or positive.

- 3** (b) Dimensional formula of potential $[V] = [\text{ML}^2\text{T}^{-3}\text{A}^{-1}]$

- 5** (a) The least count of screw gauge is

$$\text{LC} = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

• Match the columns

- 1** (b) $A \rightarrow q$; $B \rightarrow p$; $C \rightarrow r$; $D \rightarrow s$

$$(A) [R]/[L] = \frac{[\text{ML}^2\text{T}^{-3}\text{A}^{-2}]}{[\text{ML}^2\text{T}^{-2}\text{A}^{-2}]} = [\text{T}]^{-1}$$

This is the dimensions of frequency.

(A) $\rightarrow$ (q)

$$(B) [CR] = [\text{M}^{-1}\text{L}^{-2}\text{T}^4\text{A}^2] [\text{ML}^2\text{T}^{-3}\text{A}^{-2}] = [\text{T}]$$

This matches with the dimensions of time given in the column.

(B) $\rightarrow$ (p)

$$(C) [E]/[B] = \frac{[\text{MLT}^{-3}\text{A}^{-1}]}{[\text{MT}^{-2}\text{A}^{-1}]} = [\text{LT}^{-1}]$$

This is the dimensions of speed.

(C) $\rightarrow$ (r)

$$(D) [\epsilon_0 \mu_0]^{1/2} = \{[\text{M}^{-1}\text{L}^{-3}\text{T}^4\text{A}^2] [\text{MLT}^{-2}\text{A}^{-2}]\}^{1/2} = [\text{L}^{-2}\text{T}^2]^{1/2} = [\text{L}^{-1}\text{T}]$$

This does not matches with the dimensions given in the column.

(D) $\rightarrow$ (s)

(C) Medical entrances' gallery

- 1** (b) Given, least count = 0.01 mm

Number of divisions on circular scale = 50

$$\begin{aligned} \text{Pitch of the screw gauge} &= \text{Least count} \times \text{Number of divisions on circular scale} \\ &= 0.01 \times 50 = 0.5 \text{ mm} \end{aligned}$$

- 2** (a) The difference between 9.99 m and 0.0099 m is
 $= 9.99 - 0.0099 = 9.9801 \text{ m}$

Taking significant figures into account, as both the values has two significant figures after decimal.

So, their difference will also have two significant figures after decimal, i.e. 9.98 m.

- 3** (c) $\therefore \text{Stress} = \frac{\text{Force}}{\text{Area}}$

$$\therefore \text{Dimensions of stress} = \frac{[\text{MLT}^{-2}]}{[\text{L}^2]} = [\text{ML}^{-1}\text{T}^{-2}]$$

- 4** (a) 1 minute = $\frac{1}{60}$ degree = $\frac{1}{60} \times \frac{\pi}{180} = 2.91 \times 10^{-4} \text{ rad}$

- 5** (d) Mean time interval,

$$\bar{T} = \frac{1.25 + 1.24 + 1.27 + 1.21 + 1.28}{5} = \frac{6.25}{5} = 1.25 \text{ s}$$

Mean absolute error,

$$\Delta \bar{T} = \frac{|\Delta T_1| + |\Delta T_2| + |\Delta T_3| + |\Delta T_4| + |\Delta T_5|}{5}$$

$$\Rightarrow = \frac{|1.25 - 1.25| + |1.24 - 1.25| + |1.27 - 1.25| + |1.21 - 1.25| + |1.28 - 1.25|}{5}$$

$$\Rightarrow = \frac{0 + 0.01 + 0.02 + 0.04 + 0.03}{5} = \frac{0.1}{5} = 0.02 \text{ s}$$

$$\therefore \text{Percentage relative error} = \frac{\Delta \bar{T}}{\bar{T}} \times 100 = \frac{0.02}{1.25} \times 100 = 1.6\%$$

6 (c) The SI unit of thermal conductivity is $\text{Wm}^{-1}\text{K}^{-1}$.

7 (a) Given, $X = \frac{A^2 B^{1/2}}{C^{1/3} D^3}$

The percentage error in X is given by

$$\frac{\Delta X}{X} \times 100 = 2 \left(\frac{\Delta A}{A} \right) \times 100 + \frac{1}{2} \left(\frac{\Delta B}{B} \right) \times 100 + \frac{1}{3} \left(\frac{\Delta C}{C} \right) \times 100 + 3 \left(\frac{\Delta D}{D} \right) \times 100 \dots (i)$$

$$\text{Given, } \frac{\Delta A}{A} \times 100 = 1\%, \frac{\Delta B}{B} \times 100 = 2\%,$$

$$\frac{\Delta C}{C} \times 100 = 3\%, \frac{\Delta D}{D} \times 100 = 4\%$$

Substituting these values in Eq. (i), we get

$$\frac{\Delta X}{X} \times 100 = 2(1\%) + \frac{1}{2}(2\%) + \frac{1}{3}(3\%) + 3(4\%) = 2\% + 1\% + 1\% + 12\% = 16\%$$

Thus, maximum % error in X is 16%.

8 (c) As it is given that, n divisions of vernier scale coincide with $(n-1)$ divisions of main scale, i.e.

$$n(\text{VSD}) = (n-1)\text{MSD}$$

$$\Rightarrow 1 \text{ VSD} = \frac{(n-1)}{n} \text{MSD} \dots (i)$$

The least count is the difference between one Main Scale Division (MSD) and one Vernier Scale Division (VSD).

$\therefore$ Least Count (LC) = 1 MSD - 1 VSD

$$= 1 \text{ MSD} - \frac{(n-1)}{n} \text{MSD} \quad [\text{From Eq. (i)}]$$

$$= \left(1 - \frac{(n-1)}{n} \right) \text{MSD} = \frac{1}{n} \text{MSD}$$

$$\text{Given, } 1 \text{ MSD} = \frac{1}{n} \text{ cm}$$

$$\Rightarrow \text{LC} = \frac{1}{n} \times \frac{1}{n} \text{ cm} = \frac{1}{n^2} \text{ cm}$$

9 (a) Mean of the five observations,

$$\mu = \frac{80.0 + 80.5 + 81.0 + 81.5 + 82}{5} = \frac{405.0}{5} = 81$$

$$\therefore \text{Mean error} = \frac{\left[|80 - \mu| + |80.5 - \mu| + |81.0 - \mu| + |81.5 - \mu| + |82 - \mu| \right]}{5}$$

$$= \frac{\left[|80 - 81| + |80.5 - 81| + |81.0 - 81| + |81.5 - 81| + |82 - 81| \right]}{5}$$

$$= \frac{1 + 0.5 + 0 + 0.5 + 1}{5} = \frac{3}{5} = 0.6$$

$$\therefore \text{Mean \% error} = \frac{0.6}{81} \times 100\% = 0.74\%$$

10 (a) Dimensions of torque,

$$\begin{aligned} \tau &= [\mathbf{F} \times \mathbf{r}] = [\text{MLT}^{-2}] [\text{L}] \\ &= [\text{ML}^2\text{T}^{-2}] \end{aligned}$$

11 (c) Dimensions of energy in terms of linear momentum (p), area (A) and time (T), is related to

$$E = p^a A^b T^c \dots (i)$$

Writing dimensional formula of both sides, we get

$$[\text{ML}^2\text{T}^{-2}] = [\text{MLT}^{-1}]^a [\text{L}^2]^b [\text{T}]^c$$

$$[\text{ML}^2\text{T}^{-2}] = [\text{M}^a \text{L}^{a+2b} \text{T}^{-a+c}]$$

Comparing the exponents,

$$a = 1, a + 2b = 2$$

$$2b = 1$$

$$\Rightarrow b = \frac{1}{2}$$

$$-a + c = -2$$

$$\Rightarrow -1 + c = -2$$

$$\Rightarrow c = -1$$

$\therefore$ From Eq. (i), we have

$$\Rightarrow E = [pA^{1/2}T^{-1}]$$

12 (d) Given, least count of screw gauge,

$$\text{LC} = 0.001 \text{ cm}$$

$$\text{Main Scale Reading (MSR)} = 5 \text{ mm} = 0.5 \text{ cm}$$

Number of coinciding divisions on the circular scale, i.e.

$$\text{Vernier Scale Reading (VSR)} = 25$$

$$\text{Here, zero error} = -0.004 \text{ cm}$$

Final reading obtained from the screw gauge

$$= \text{MSR} + \text{VSR} \times \text{LC} - \text{zero error}$$

$$= 0.5 + 25 \times 0.001 - (-0.004)$$

$$= 0.5 + 0.025 + 0.004$$

$$= 0.5 + 0.029 = 0.529 \text{ cm}$$

Thus, the diameter of the ball is 0.529 cm.

13 (c) We know that, $s = ut + \frac{1}{2}at^2$

$$\text{or } h = (0)t + \frac{1}{2} \times 9.8 \times (4)^2 \quad \left(\because a = g \text{ and } u = 0 \right)$$

$$= 78.4 \text{ m}$$

$$\text{Given, } \Delta t = 0.2 \text{ s, } t = 4 \text{ s}$$

$$\text{Now, for error, } \frac{\Delta h}{h} = \pm 2 \left(\frac{\Delta t}{t} \right) = \pm 2 \left(\frac{0.2}{4} \right) = \pm 0.1$$

$$\text{or } \Delta h = \pm 0.1 \times h = \pm 0.1 \times 78.4 = \pm 7.84 \text{ m}$$

14 (b) Given, $R = 65 \Omega$, $\Delta R = 1 \Omega$,

$$l = 5 \times 10^{-3} \text{ m, } \Delta l = 0.1 \times 10^{-3} \text{ m,}$$

$$d = 10 \times 10^{-3} \text{ m and } \Delta d = 0.5 \times 10^{-3} \text{ m}$$

$$\therefore \text{ Resistivity, } \rho = \frac{RA}{l} \text{ or } \rho = \frac{R\pi (d/2)^2}{l} = \frac{\pi R d^2}{4l}$$

$$\therefore \frac{\Delta \rho}{\rho} = \frac{\Delta R}{R} + 2 \frac{\Delta d}{d} + \frac{\Delta l}{l}$$

$$\Rightarrow \frac{\Delta \rho}{\rho} = \frac{1}{65} + 2 \left(\frac{0.5 \times 10^{-3}}{10 \times 10^{-3}} \right) + \frac{0.1 \times 10^{-3}}{5 \times 10^{-3}}$$

$$\Rightarrow \frac{\Delta \rho}{\rho} = 0.0153 + 0.1 + 0.02 \Rightarrow \frac{\Delta \rho}{\rho} \approx 0.1353$$

So, error in calculation of resistivity is $13.5\% \approx 13\%$.

15 (b) We know that, $F = ma$

$$\therefore [F] = [M][a] = \frac{[M][L]}{T^2} = [M^1 L^1 T^{-2}]$$

16 (a) As, force, $F = \frac{e^2}{4\pi\epsilon_0 r^2} \Rightarrow \frac{e^2}{4\pi\epsilon_0} = r^2 \cdot F$

Putting dimensions of r and F , we get

$$\Rightarrow \left[\frac{e^2}{4\pi\epsilon_0} \right] = [ML^3 T^{-2}] \quad \dots(i)$$

$$\text{Also, force, } F = \frac{Gm^2}{r^2} \Rightarrow [G] = [M^{-1} L^3 T^{-2}] \quad \dots(ii)$$

$$\text{and } \left[\frac{1}{c^2} \right] = \frac{1}{[L^2 T^{-2}]} = [L^{-2} T^2] \quad \dots(iii)$$

Now, checking optionwise

$$= \frac{1}{c^2} \left(\frac{Ge^2}{4\pi\epsilon_0} \right)^{1/2} = [L^{-2} T^2] [L^6 T^{-4}]^{1/2} = [L]$$

17 (a) In forms of h , c and G , length can be expressed as

$$L = (h)^a (c)^b (G)^c$$

Writing dimensions on both sides, we get

$$[M^0 L T^0] = [M L^2 T^{-1}]^a [L T^{-1}]^b [M^{-1} L^3 T^{-2}]^c$$

$$= M^{a-c} L^{2a+b+3c} T^{-a-b-2c}$$

On comparing powers of M , L and T on both sides, we get

$$a - c = 0, 2a + b + 3c = 1$$

$$\text{and } -a - b - 2c = 0$$

On solving, we get

$$a = c = \frac{1}{2} \text{ and } b = -\frac{3}{2}$$

$$\therefore \text{Dimensions of length, } L = (h)^{1/2} (c)^{-3/2} (G)^{1/2} = \frac{\sqrt{hG}}{c^{3/2}}$$

18 (c) We know that, surface tension (S) = $\frac{\text{Force } [F]}{\text{Length } [L]}$

$$\text{So, } [S] = \frac{[MLT^{-2}]}{[L]} = [ML^0 T^{-2}]$$

$$\text{Energy } (E) = \text{Force} \times \text{Displacement} \Rightarrow [E] = [ML^2 T^{-2}]$$

$$\text{Velocity } (v) = \frac{\text{Displacement}}{\text{Time}} \Rightarrow [v] = [LT^{-1}]$$

$$\text{As, } S \propto E^a v^b T^c$$

where, a , b and c are constants.

From the principle of homogeneity,

$$[\text{LHS}] = [\text{RHS}]$$

$$\Rightarrow [ML^0 T^{-2}] = [ML^2 T^{-2}]^a [LT^{-1}]^b [T]^c$$

$$\Rightarrow [ML^0 T^{-2}] = [M^a L^{2a+b} T^{-2a-b+c}]$$

Equating the power on both sides, we get

$$a = 1, 2a + b = 0 \Rightarrow b = -2$$

$$\text{and } -2a - b + c = -2$$

$$\Rightarrow c = (2a + b) - 2 = 0 - 2 = -2$$

$$\text{So, } [S] = [E v^{-2} T^{-2}]$$

19 (a) The force per unit length experienced due to two wires in which current is flowing in the same direction is given by

$$\frac{dF}{dl} = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d} \Rightarrow \frac{[M L T^{-2}]}{[L]} = [\mu_0] \frac{[A^2]}{[L]}$$

$$\Rightarrow \frac{[MLT^{-2}]}{[L]} = [\mu_0] \left[\frac{Q^2}{T^2 L} \right] \Rightarrow [\mu_0] = [MLQ^{-2}]$$

20 (a) As electric flux is given by, $\phi_E = EdS = \frac{F}{q} dS$

$$\therefore \text{Dimensions of } \phi_E = \left[\frac{MLT^{-2}}{IT} \right] [L^2] = [ML^3 T^{-1} T^{-3}]$$

21 (a) Given, $F = A \sin Ct + B \cos Dx$...(i)

where, t = time and x = distance

As, we know that trigonometric ratios are dimensionless. This implies

$$\sin Ct = \text{dimensionless and } \cos Dx = \text{dimensionless}$$

$$\text{Also, } [C] = \left[\frac{1}{t} \right] = [T^{-1}] \text{ and } [D] = \left[\frac{1}{x} \right] = [L^{-1}]$$

As, Eq. (i) represents the force. So, A and B both have the dimensions as that of force. So, A/B is dimensionless, i.e. $[M^0 L^0 T^0]$.

$$\text{While } \left[\frac{C}{D} \right] = \left[\frac{T^{-1}}{L^{-1}} \right] = [M^0 L T^{-1}]$$

22 (d) Option (d) is wrong because 1 astronomical unit
 $= 1.5 \times 10^{11} \text{m}$

23 (b) According to the question, we have

$$\frac{m}{t \cdot A} \propto p^n \cdot v^m \text{ or } m/t \cdot A = k p^n v^m$$

where, k is proportionality constant.

Using principle of homogeneity, we get

$$[ML^{-2} T^{-1}] = k [ML^{-1} T^{-2}]^n [LT^{-1}]^m$$

$$\text{or } [ML^{-2} T^{-1}] = k [M]^n [L]^{-n+m} [T]^{-2n-m}$$

Equating both sides, we find $n = -m$ or $m = -n$

24 (d) We know that, energy of an emitted particle,

$$E = h\nu \Rightarrow h = \frac{E}{\nu}$$

$$\text{Planck's constant, } [h] = \frac{[ML^2 T^{-2}]}{[T^{-1}]} = [ML^2 T^{-1}] \quad \dots(i)$$

$$\text{and moment of inertia, } I = mr^2 \Rightarrow [I] = [ML^2] \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{[h]}{[I]} = \left[\frac{ML^2 T^{-1}}{ML^2} \right] = [T^{-1}] = \frac{1}{T}$$

$$\text{i.e. } \frac{[h]}{[I]} = [T^{-1}] = \text{Dimensions of frequency of a particle}$$

25 (a) The dimensions of electrical resistance,

$$R = \frac{V}{I} = \frac{\left(\frac{W}{q} \right)}{I} = \frac{W}{It} = \frac{W}{I^2 t} = [ML^2 T^{-2} T^{-1} A^{-2}] = [ML^2 T^{-3} A^{-2}]$$

Then, (A) $\rightarrow$ (2)

The dimensions of electrical potential,

$$V = \frac{W}{q} = \frac{W}{It} = [\text{ML}^2\text{T}^{-2}\text{A}^{-1}\text{T}^{-1}] = [\text{ML}^2\text{T}^{-3}\text{A}^{-1}]$$

Then, (B) $\rightarrow$ (3)

The dimensions of specific resistance,

$$\rho = R \frac{A}{l} = [\text{ML}^2\text{T}^{-3}\text{A}^{-2}] [\text{L}] = [\text{ML}^3\text{T}^{-3}\text{A}^{-2}]$$

Thus, (C) $\rightarrow$ (1)

And the dimensions of specific conductance,

$$\sigma = \frac{1}{\rho} = \frac{1}{[\text{ML}^3\text{T}^{-3}\text{A}^{-2}]} = [\text{M}^{-1}\text{L}^{-3}\text{T}^3\text{A}^2]$$

= not given in column

Thus, (D) $\rightarrow$ (4)

26 (a) Given, $x = \text{g cm}^2 \text{ s}^{-5} = [\text{ML}^2\text{T}^{-5}]$

$$y = \text{g s}^{-1} = [\text{ML}^0\text{T}^{-1}]$$

and $z = \text{cm s}^{-2} = [\text{M}^0\text{L}\text{T}^{-2}]$

Now, $z^2 = [\text{M}^0\text{L}^2\text{T}^{-4}]$

and $yz^2 = [\text{ML}^0\text{T}^{-1}][\text{M}^0\text{L}^2\text{T}^{-4}] = [\text{ML}^2\text{T}^{-5}] = x$

i.e. $x = yz^2$

27 (d) We know that, $F = ma \Rightarrow F = \frac{mv}{t} \Rightarrow m = \frac{Ft}{v}$

$$\therefore \text{Dimensions of } [m] = \frac{[F][T]}{[v]} = [\text{Fv}^{-1}\text{T}]$$

28 (a) Reynold's number describes the ratio of inertial force per unit area to viscous force per unit area for a flowing fluid.

Thus, Reynold's number is the ratio of two physical quantity of same dimension which cancel out each other. Hence,

Reynold's number is dimensionless $[\text{M}^0\text{L}^0\text{T}^0]$ quantity.

29 (a) Substituting dimensions, $[\text{MLT}^{-2}] = \frac{x}{\sqrt{[\text{ML}^{-3}]}}$

$$\Rightarrow x = [\text{M}^{3/2}\text{L}^{-1/2}\text{T}^{-2}]$$

30 (c) The absolute error in the value $A - B$ will be $a + b$.

31 (c) Let $x \text{ kg}$ be unit of mass, then

$$f = (x \text{ kg})(1000 \text{ m})(100 \text{ s})^{-2}$$

$$\Rightarrow 1000 = x \times 1000 \times \frac{1}{10000}$$

$$\therefore x = 10000 \text{ kg}$$

$$= 10^4 \text{ kg}$$

32 (a) The quantity is given as $\left[\frac{nh}{2\pi qB} \right]$, where n and 2π are

dimensionless quantities.

$$\therefore mvr = \frac{nh}{2\pi} \quad \dots(i)$$

Also using, $Bqr = mv$

$$\Rightarrow Bq = mv/r \quad \dots(ii)$$

Using Eqs. (i) and (ii), we get

$$\left[\frac{nh}{2\pi qB} \right] = \frac{mvr}{mv/r} = [r^2]$$

As, r has the dimension of length, thus the given quantity has the dimension of area.

33 (a) Here, $P = \frac{a^3b^2}{cd}$

$$\begin{aligned} \therefore \frac{\Delta P}{P} \times 100 &= \left(\frac{3\Delta a}{a} + \frac{2\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d} \right) \times 100 \\ &= 3 \frac{\Delta a}{a} \times 100 + 2 \frac{\Delta b}{b} \times 100 + \frac{\Delta c}{c} \times 100 + \frac{\Delta d}{d} \times 100 \\ &= 3 \times 1\% + 2 \times 2\% + 3\% + 4\% \\ &= 3\% + 4\% + 3\% + 4\% = 14\% \end{aligned}$$

34 (d) Given, density of glass in CGS system = 2.8 g/cc
 $= 2.8 \text{ g/cm}^3$

$$\begin{aligned} \text{Value of density in SI system} &= \frac{2.8 \times 10^{-3} \text{ kg}}{10^{-6} \text{ m}^3} \\ &= 2.8 \times 10^3 \text{ kgm}^{-3} \end{aligned}$$

35 (b) Given equation, $\frac{1}{p\beta} = \frac{y}{k_B T}$

where, p = pressure, y = distance,

k_B = Boltzmann constant and T = temperature.

$$\begin{aligned} \text{Dimensions of } [\beta] &= \frac{[\text{Dimensions of } k_B] [\text{Dimensions of } T]}{[\text{Dimensions of } p] [\text{Dimensions of } y]} \\ &= \frac{[\text{ML}^2\text{T}^{-2}\text{K}^{-1}][\text{K}]}{[\text{ML}^{-1}\text{T}^{-2}][\text{L}]} \\ &= [\text{M}^0\text{L}^2\text{T}^0] \end{aligned}$$

36 (b) Dimensions of $[X] = \frac{\text{Dimensions of } [IFv^2]}{\text{Dimensions of } [WL^3]}$

$$\begin{aligned} &= \frac{[\text{ML}^2] [\text{MLT}^{-2}] [\text{LT}^{-1}]^2}{[\text{ML}^2\text{T}^{-2}] [\text{L}^3]} \\ &= \frac{[\text{M}^2\text{L}^5\text{T}^{-4}]}{[\text{ML}^5\text{T}^{-2}]} = [\text{MT}^{-2}] \end{aligned}$$

37 (c) Here, $X = \frac{2k^3l^2}{m\sqrt{n}}$

Percentage error in X

$$\begin{aligned} &= \frac{3 \times \Delta k}{k} \times 100 + 2 \times \frac{\Delta l}{l} \times 100 + \frac{\Delta m}{m} \times 100 + \frac{1}{2} \times \frac{\Delta n}{n} \times 100 \\ &= \left(3 \times 1\% + 2 \times 2\% + 3\% \times 1 + 4\% \times \frac{1}{2} \right) = 12\% \end{aligned}$$

38 (b) Given, $m = \frac{A}{B}$

$$\therefore \text{Dimensions of } B = [B] = \frac{[A]}{[m]}$$

Here, A = force $= [\text{MLT}^{-2}]$

and m = linear density = mass per unit length $= \frac{[M]}{[L]}$

$$\therefore [B] = \frac{[MLT^{-2}]}{[ML^{-1}]} = [L^2T^{-2}]$$

These are same dimensions as that of latent heat.

39 (a) Given, $X = [M^a L^b T^c]$

Maximum % error in $X = a\alpha + b\beta + c\gamma$

40 (b) $\because (\mu_0 \epsilon_0)^{-1/2} = \frac{1}{(\mu_0 \epsilon_0)^{1/2}} = \text{speed of light}$

$$\Rightarrow [c] = \frac{[L]}{[T]} = [LT^{-1}]$$

41 (a) Given, $p = \frac{RT}{V-b} e^{-\alpha V/RT}$

So, $\frac{\alpha V}{RT}$ is dimensionless.

$$\text{Hence, } [\alpha] = \left[\frac{RT}{V} \right] = \frac{[ML^2T^{-2}\theta^{-1}][\theta]}{[L^3]} = [ML^{-1}T^{-2}]$$

This is the dimensional formula of pressure (p).

42 (a) Dimensions of velocity are $[v] = [L][T^{-1}]$

So, dimensions of $[at^2] = [LT^{-1}]$

$$\Rightarrow [a][T^2] = [LT^{-1}]$$

$$[a] = [LT^{-3}]$$

Dimensions of $[bt] = [LT^{-1}] \Rightarrow [b][T] = [LT^{-1}]$

$$\Rightarrow [b] = [LT^{-2}]$$

Dimensions of $[c] = [LT^{-1}]$

43 (a) Taking, $T = 2\pi\sqrt{\frac{R^3}{GM}}$

Substituting the dimensions, LHS = $T = [T]$

$$\text{RHS} = 2\pi\sqrt{\frac{R^3}{GM}} = \sqrt{\frac{[L]^3}{[M^{-1}L^3T^{-2}][M]}} = \sqrt{[T]^2} = [T]$$

Thus, LHS = RHS for $T = 2\pi\sqrt{\frac{R^3}{GM}}$.

CHAPTER 02

Vectors

In physics, we study a large number of physical quantities. These physical quantities can either have magnitude or magnitude and direction both. On this basis, we have broadly categorised physical quantities into two categories : scalars and vectors. In this chapter, we will study about the vector quantities and their operations in detail.

SCALAR AND VECTOR QUANTITIES

Scalar quantities

A physical quantity which can be described completely by its magnitude only and does not require a direction is called scalar quantity. Addition, subtraction, multiplication or division of scalar quantities can be done according to the general rules of algebra. Mass, volume, density, etc., are few examples of scalar quantities.

Vector quantities

A physical quantity which has both magnitude and particular direction and obeys the triangle law of vector addition or equivalently the parallelogram law of vector addition is called a vector quantity. Displacement, velocity, acceleration, etc., are few examples of vector quantities.

Note The physical quantity current has both magnitude and direction but it is still a scalar as it disobeys the laws of vector algebra.

General points regarding vectors

General points regarding vectors are as follows

Vector notation Usually a vector is represented by a bold capital letter with an arrow (or without arrow) over it, as $\vec{A}$, $\vec{B}$, $\vec{C}$ or simply **A**, **B**, **C**.

The magnitude of a vector **A** is represented by A or $|\mathbf{A}|$.

Graphical representation of a vector Graphically a vector is represented by an arrow drawn to a chosen scale, parallel to the direction of the vector. The length of the arrow represents the magnitude and the tip of the arrow (arrow-head) represents the direction.

Suppose that a car *A* is running with a velocity of 10 m/s towards east; and another car *B* is running with a velocity of 20 m/s towards north-east.

Inside

1 Scalar and vector quantities

- General points regarding vectors
- Types of vectors
- Multiplication and division of vectors by scalars

2 Addition of two vectors

3 Subtraction of two vectors

- Resolution of vectors

4 Product of two vectors

- Scalar product of two vectors
- Vector product of two vectors

These velocities can be represented by vectors drawn in Fig. 2.1.

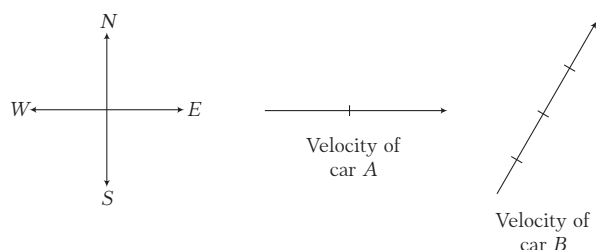


Fig. 2.1

To draw these vectors, it has been assumed that 1 cm length represents a velocity of 5 m/s. The velocity vector of car A is an arrow of length 2 cm and its tip is directed towards east, while that of B is an arrow of length 4 cm with its tip directed towards north-east.

Angle between two vectors (θ) Angle between two vectors is the smaller of two angles between the vectors when they are placed tail to tail.

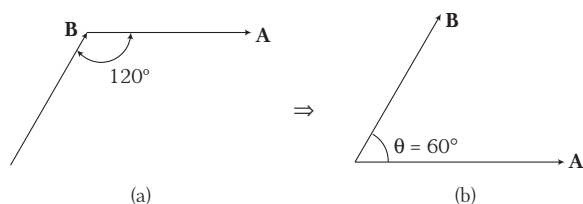


Fig. 2.2

For example, in Fig. 2.2, angle between **A** and **B** is 60° not 120° . Because in Fig. (a), their tails are not together while in Fig. (b), they are drawn correctly.

If a vector is displaced parallel to itself, it does not change.



Fig. 2.3

$\therefore$

$$\mathbf{B} = \mathbf{A}$$

If a vector is rotated by an angle ($\theta \neq 2n\pi, n = 1, 2, 3, \dots$), the vector is changed.

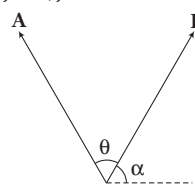


Fig. 2.4

$\therefore$

$$\mathbf{B} \neq \mathbf{A}$$

Types of vectors

Different types of vectors are given below

- (i) **Polar vectors** These are the vectors which have a initial starting point or a point of application. *e.g.* Displacement, force, etc.
- (ii) **Axial vectors** These are the vectors which represent rotational effect and act along the axis of rotation in accordance with right hand screw rule or right hand thumb rule. *e.g.* Angular velocity, angular acceleration, torque, etc.

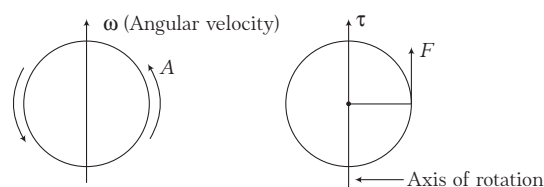


Fig. 2.5 Axial vectors

- (iii) **Equal vectors** These are the vectors which have equal magnitude and same direction. In Fig. 2.6, **A** and **B** are equal vectors. *i.e.*, $\mathbf{A} = \mathbf{B}$

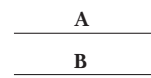


Fig. 2.6 Equal vectors

- (iv) **Parallel vectors** These are the vectors which have same direction but their magnitude may be equal or different. The angle between two parallel vectors is always 0° .

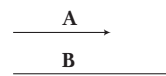


Fig. 2.7 Parallel vectors

- (v) **Anti-parallel vectors** These are the vectors which have opposite direction but their magnitude may be equal or different. The angle between two anti-parallel vectors is always 180° .

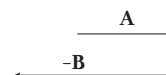


Fig. 2.8 Anti-parallel vectors

- (vi) **Collinear vectors** These are the vectors which lie along the same line. Angle between them can be 0° or 180° . *e.g.*

- | | |
|--|--|
| (a) $\longleftrightarrow$ ($\theta = 0^\circ$) | (b) $\longrightarrow$ ($\theta = 0^\circ$) |
| (c) $\longleftrightarrow$ ($\theta = 180^\circ$) | (d) $\longrightarrow$ ($\theta = 180^\circ$) |

(vii) **Zero or null vector** A vector having zero magnitude is known as zero vector. Its direction is not specified and hence is arbitrary. It is represented by 0. *e.g.* Displacement, velocity and acceleration of a particle at rest or acceleration of a particle moving with uniform velocity.

(viii) **Unit vector** A vector whose magnitude is one unit and points in a particular direction, is called unit vector. It is represented by $\hat{\mathbf{A}}$ ($\mathbf{A}$ cap or $\mathbf{A}$ hat). The unit vector along the direction of $\mathbf{A}$ is

$$\hat{\mathbf{A}} = \frac{\text{Vector}}{\text{Magnitude of the vector}} = \frac{\mathbf{A}}{|\mathbf{A}|}$$

(Here, $\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}$)

$$\therefore \mathbf{A} = \hat{\mathbf{A}} |\mathbf{A}|$$

$$\text{where, } |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

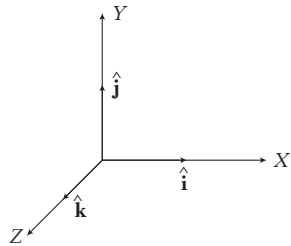


Fig. 2.9

In cartesian coordinates $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are the unit vectors along X-axis, Y-axis and Z-axis, respectively.

$$\text{where, } |\hat{\mathbf{i}}| = |\hat{\mathbf{j}}| = |\hat{\mathbf{k}}| = 1$$

Note A unit vector is used to specify the direction of a vector.

(ix) **Coplanar vectors** These are the vectors which always lie in the same plane.

Note Two vectors are always coplanar vectors.

(x) **Negative or opposite vector** If the direction of a vector is reversed, the sign of the vector get reversed. It is called negative vector of the original vector.

In Fig. 2.10, $\mathbf{B}$ is negative vector of $\mathbf{A}$, i.e. $\mathbf{A} = -\mathbf{B}$.



Fig. 2.10

Example 2.1 If vectors $\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ and $\hat{\mathbf{i}} - 3\hat{\mathbf{j}} - a\hat{\mathbf{k}}$ are equal vectors, then find the value of a .

Sol. Two vectors are said to be equal, if their magnitude is equal and direction is same.

$$\text{i.e. } \mathbf{A} = \mathbf{B}$$

$$\text{Here, } \mathbf{A} = \hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}} \text{ and } \mathbf{B} = \hat{\mathbf{i}} - 3\hat{\mathbf{j}} - a\hat{\mathbf{k}}$$

$$\Rightarrow \hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}} = \hat{\mathbf{i}} - 3\hat{\mathbf{j}} - a\hat{\mathbf{k}}$$

Comparing on both sides, we get

$$a = -5$$

Example 2.2 Check whether the vector $\left(\frac{\hat{\mathbf{i}}}{\sqrt{2}} + \frac{\hat{\mathbf{j}}}{\sqrt{2}}\right)$ is a unit vector or not.

Sol. A unit vector is a vector with magnitude equals to 1.

The magnitude of given vector is

$$\left| \frac{\hat{\mathbf{i}}}{\sqrt{2}} + \frac{\hat{\mathbf{j}}}{\sqrt{2}} \right| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$$

As magnitude of given vector is 1.

$\therefore$ It is a unit vector.

Example 2.3 Find the unit vector of $4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

Sol. Let the given vector be $\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}$

$$= 4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

$$\therefore A_x = 4, A_y = -3, A_z = 1$$

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(4)^2 + (-3)^2 + (1)^2} = \sqrt{26}$$

$$\therefore \text{Unit vector, } \hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{4\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}}{\sqrt{26}}$$

Example 2.4 If $\mathbf{A} = 4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $\mathbf{B} = 24\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$, find the vector having the same magnitude as $\mathbf{B}$ and parallel to $\mathbf{A}$.

Sol. Magnitude of $\mathbf{B}$, $|\mathbf{B}| = \sqrt{24^2 + 7^2} = 25$

$$\text{Magnitude of } \mathbf{A}, |\mathbf{A}| = \sqrt{4^2 + 3^2} = 5$$

$$\text{Unit vector } \mathbf{A}, \hat{\mathbf{A}} = \frac{4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}}{5}$$

$$\text{Required vector, } \mathbf{r} = 25 \frac{(4\hat{\mathbf{i}} + 3\hat{\mathbf{j}})}{5} = 20\hat{\mathbf{i}} + 15\hat{\mathbf{j}}$$

Multiplication and division of vectors by scalars

The product of a vector $\mathbf{A}$ and a scalar m gives a vector $m\mathbf{A}$ whose magnitude is m times the magnitude of $\mathbf{A}$ and which is in the direction or opposite to $\mathbf{A}$ accordingly, if the scalar m is positive or negative. Thus, $m(\mathbf{A}) = m\mathbf{A}$. Further, if m and n are two scalars, then

$$(m+n)\mathbf{A} = m\mathbf{A} + n\mathbf{A} \text{ and } m(n\mathbf{A}) = n(m\mathbf{A}) = (mn)\mathbf{A}$$

The division of vector $\mathbf{A}$ by a non-zero scalar m is defined as the product of $\mathbf{A}$ and $\frac{1}{m}$.

CHECK POINT 2.1

1. Which of the following is not the vector quantity?

- (a) Torque (b) Displacement
(c) Dipole moment (d) Electric flux

2. Surface area is

- (a) scalar
(b) vector
(c) Both (a) and (b)
(d) Neither scalar nor vector

3. I. Pressure II. Temperature

III. Momentum IV. Work

Which of the following physical quantities are scalars ?

- (a) I and II (b) I, II and III
(c) I, II and IV (d) II and III

4. A vector multiplied by the number 0, results into

- (a) 0 (b) $\mathbf{A}$
(c) $\mathbf{0}$ (d) $\hat{\mathbf{A}}$

5. Find the vector that must be added to the vector

$\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $3\hat{\mathbf{i}} + 6\hat{\mathbf{j}} - 7\hat{\mathbf{k}}$, so that the resultant vector is a unit vector along the Y-axis.

- (a) $-4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ (b) $-4\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$
(c) $4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ (d) $-4\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$

6. The magnitude of $\hat{\mathbf{i}} + \hat{\mathbf{j}}$ is

- (a) 2 (b) 0
(c) $\sqrt{2}$ (d) 4

7. The direction of unit vector along $\hat{\mathbf{i}} + \hat{\mathbf{j}}$ is

- (a) $\hat{\mathbf{k}}$ (b) $\hat{\mathbf{i}} + \hat{\mathbf{j}}$ (c) $\frac{\hat{\mathbf{i}} + \hat{\mathbf{j}}}{\sqrt{2}}$ (d) $\frac{\hat{\mathbf{i}} + \hat{\mathbf{j}}}{2}$

8. What happens, when we multiply a vector by (-2) ?

- (a) Direction reverses and unit changes
(b) Direction reverses and magnitude is doubled
(c) Direction remains unchanged and unit changes
(d) None of the above

ADDITION OF TWO VECTORS

Vectors cannot be added by simple laws of algebra, which are applicable to scalars. To add two vectors, we must follow certain laws. These laws are described below

1. Triangle law of addition of two vectors

It states that, if two vectors acting on a particle at the same time are represented with magnitude and direction by the two sides of a triangle taken in same order, then their sum or resultant is represented in magnitude and direction by the third side of the triangle taken in opposite order.

As is evident from the figure that the resultant $\mathbf{R}$ is the same irrespective of the order in which the vectors $\mathbf{A}$ and $\mathbf{B}$ are taken.

Thus,

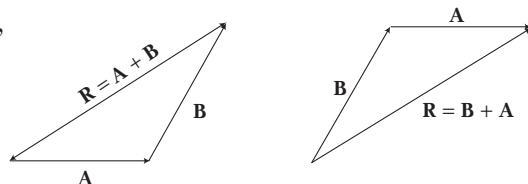


Fig. 2.11

$$\mathbf{R} = \mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

This is the geometrical method of vector addition.

2. Parallelogram law of vector addition

It states that, if two vectors acting on a particle at the same time can be represented with magnitude and direction by the two adjacent sides of a parallelogram drawn from a point, then their resultant vector is represented in magnitude and direction by the diagonal of the parallelogram drawn from the same point.

(i) **Magnitude of resultant vector** Let $\mathbf{R}$ be the resultant of two vectors $\mathbf{A}$ and $\mathbf{B}$. According to parallelogram law of vector addition, the resultant $\mathbf{R}$ is the diagonal of the parallelogram of which $\mathbf{A}$ and $\mathbf{B}$ are the adjacent sides as shown in figure below.

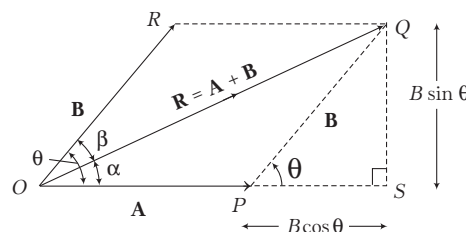


Fig. 2.12

Magnitude of $\mathbf{R}$ is given by

$$\mathbf{R} = \sqrt{A^2 + B^2 + 2AB \cos \theta} \quad \dots(i)$$

Here, θ = angle between $\mathbf{A}$ and $\mathbf{B}$.

Eq. (i) is also known as law of cosines.

Also, $\frac{R}{\sin \theta} = \frac{A}{\sin \beta} = \frac{B}{\sin \alpha}$ is known as law of sines.

Special cases

- Resultant of two vectors will be maximum when they are parallel, i.e. angle between them is zero.
or $R_{\max} = A + B$
- Resultant of two vectors will be minimum when they are anti-parallel, i.e. angle between them is 180° .
or $R_{\min} = A - B$
- Resultant of two vectors of unequal magnitude can never be zero.

(ii) **Direction of resultant vector** Let θ be the angle between **A** and **B**, then

$$|\mathbf{A} + \mathbf{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

If R makes an angle α and β with **A** and **B** respectively, then

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}$$

$$\tan \beta = \frac{A \sin \theta}{B + A \cos \theta}$$

Polygon law of vector addition for more than two vectors

It states that, if n number of vectors acting on a particle at the same time are represented in magnitude and direction by various sides of an open polygon taken in the same order, then their resultant is represented in magnitude and direction by the closing side of the polygon taken in opposite order.

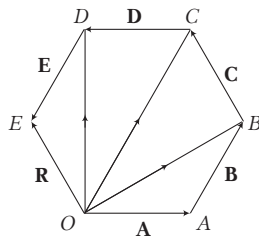


Fig. 2.13

Thus, in the figure,

$$\mathbf{OE} = \mathbf{OA} + \mathbf{AB} + \mathbf{BC} + \mathbf{CD} + \mathbf{DE}$$

$\therefore$

$$\mathbf{R} = \mathbf{A} + \mathbf{B} + \mathbf{C} + \mathbf{D} + \mathbf{E}$$

Note

- Resultant of two vectors is always located in their common plane.
- Vector addition is commutative, i.e. $A + B = B + A$
- Vector addition is associative
i.e. $A + (B + C) = (A + B) + C$
- If vectors are of unequal magnitude, then minimum three coplanar vectors are required for zero resultant.

Example 2.5 Two forces whose magnitude are in the ratio 3 : 5 give a resultant of 28 N. If the angle of their inclination is 60° , find the magnitude of each force.

Sol. Let A and B be the two forces, then $A = 3x$, $B = 5x$

$$\therefore R = 28 \text{ N and } \theta = 60^\circ$$

$$\text{Now, } R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

$$\therefore 28 = \sqrt{(3x)^2 + (5x)^2 + 2(3x)(5x) \cos 60^\circ}$$

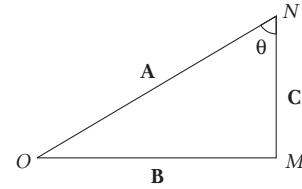
$$28 = \sqrt{9x^2 + 25x^2 + 15x^2} = 7x \text{ or } x = 4$$

$$\therefore A = 3 \times 4 = 12 \text{ N}$$

$$\text{and } B = 5 \times 4 = 20 \text{ N}$$

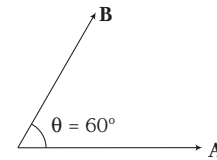
Example 2.6 If $\mathbf{A} = \mathbf{B} + \mathbf{C}$ have scalar magnitudes of 5, 4, 3 units respectively, then find the angle between **A** and **C**.

Sol. Here, triangle OMN is given with vectors **A**, **B** and **C** are its adjacent sides.

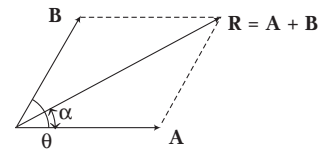


$$\text{As, } \cos \theta = \frac{MN}{ON} \Rightarrow \theta = \cos^{-1} \left(\frac{|C|}{|A|} \right) = \cos^{-1} \left(\frac{3}{5} \right)$$

Example 2.7 Find the sum of vectors **A** and **B** as shown in the figure, also find the direction of sum vector. Given, $A = 4$ unit and $B = 3$ unit.



Sol. According to the question, we draw the following figure



Resultant of the vectors **A** and **B**,

$$\begin{aligned} R &= \sqrt{A^2 + B^2 + 2AB \cos \theta} \\ &= \sqrt{16 + 9 + 2 \times 4 \times 3 \cos 60^\circ} \\ &= \sqrt{37} \text{ unit} \end{aligned}$$

$\therefore$ Direction of the sum vector,

$$\begin{aligned} \tan \alpha &= \frac{B \sin \theta}{A + B \cos \theta} \\ &= \frac{3 \sin 60^\circ}{4 + 3 \cos 60^\circ} = 0.472 \end{aligned}$$

$$\therefore \alpha = \tan^{-1}(0.472) = 25.3^\circ$$

Thus, resultant of $|\mathbf{A}|$ and $|\mathbf{B}|$ is $\sqrt{37}$ unit at angle 25.3° from $\mathbf{A}$ in the direction shown in figure.

SUBTRACTION OF TWO VECTORS

Negative of a vector say $-\mathbf{A}$ is a vector of the same magnitude as vector $\mathbf{A}$ but pointing in a direction opposite to that of $\mathbf{A}$.

Thus, $\mathbf{A} - \mathbf{B}$ can be written as $\mathbf{A} + (-\mathbf{B})$ or $\mathbf{A} - \mathbf{B}$ is the vector addition of $\mathbf{A}$ and $-\mathbf{B}$.



Fig. 2.14

Suppose angle between two vectors $\mathbf{A}$ and $\mathbf{B}$ is θ . Then, angle between $\mathbf{A}$ and $-\mathbf{B}$ will be $180^\circ - \theta$ as shown in Fig. 2.15 (b).

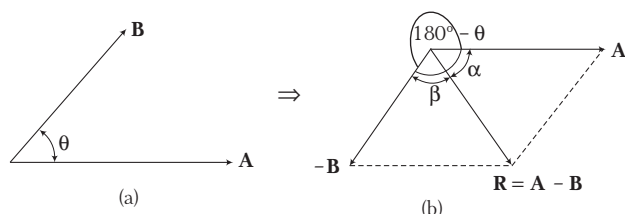


Fig. 2.15

Magnitude of resultant vector $\mathbf{R} = \mathbf{A} - \mathbf{B}$ will be thus given by

$$|\mathbf{R}| = |\mathbf{A} - \mathbf{B}| = \sqrt{A^2 + B^2 + 2AB \cos(180^\circ - \theta)}$$

$$\text{or } |\mathbf{R}| = \sqrt{A^2 + B^2 - 2AB \cos \theta} \quad \dots(i)$$

For direction of resultant vector $\mathbf{R}$, we will either calculate angle α or β , where

$$\tan \alpha = \frac{B \sin(180^\circ - \theta)}{A + B \cos(180^\circ - \theta)} = \frac{B \sin \theta}{A - B \cos \theta} \quad \dots(ii)$$

$$\text{or } \tan \beta = \frac{A \sin(180^\circ - \theta)}{B + A \cos(180^\circ - \theta)} = \frac{A \sin \theta}{B - A \cos \theta} \quad \dots(iii)$$

Special cases

- If two vectors have equal magnitudes, i.e. $|\mathbf{A}| = |\mathbf{B}| = a$ and θ is the angle between them, then

$$|\mathbf{A} - \mathbf{B}| = \sqrt{a^2 + a^2 - 2a^2 \cos \theta} = 2a \sin\left(\frac{\theta}{2}\right)$$

$$\text{If } \theta = 60^\circ, \text{ then } 2a \sin\left(\frac{\theta}{2}\right) = a$$

$$\text{i.e. } |\mathbf{A} - \mathbf{B}| = |\mathbf{A}| = |\mathbf{B}| = a \text{ at } \theta = 60^\circ$$

- If two vectors are such that their sum and difference have equal magnitude, then angle between the given vectors $\theta = 90^\circ$.

$$\text{i.e. } |\mathbf{A} + \mathbf{B}| = |\mathbf{A} - \mathbf{B}|$$

$$\text{then } \cos \theta = 0 \text{ or } \theta = 90^\circ$$

- If $\mathbf{A} + \mathbf{B} = \mathbf{A} - \mathbf{B}$

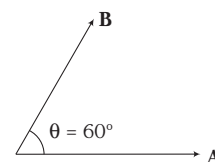
$$\text{then } \mathbf{B} = \mathbf{0} \text{ (a null vector)}$$

Note

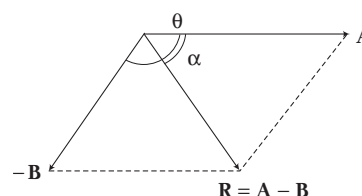
(i) The vector subtraction does not follow commutative law, i.e. $\mathbf{A} - \mathbf{B} \neq \mathbf{B} - \mathbf{A}$.

(ii) The vector subtraction does not follow associative law, i.e. $\mathbf{A} - (\mathbf{B} - \mathbf{C}) \neq (\mathbf{A} - \mathbf{B}) - \mathbf{C}$

Example 2.8 Find the subtraction of vector $\mathbf{A}$ and $\mathbf{B}$ as shown in the figure, also find the direction of subtraction vector. Given, $A = 4$ unit and $B = 3$ unit.



Sol. According to the question, we draw the following figure.



Magnitude of resultant of the vectors $\mathbf{A}$ and $\mathbf{B}$,

$$\begin{aligned} R &= \sqrt{A^2 + B^2 - 2AB \cos \theta} \\ &= \sqrt{16 + 9 - 2 \times 4 \times 3 \cos 60^\circ} \\ &= \sqrt{13} \text{ unit} \end{aligned}$$

and direction is

$$\begin{aligned} \tan \alpha &= \frac{B \sin \theta}{A - B \cos \theta} \\ &= \frac{3 \sin 60^\circ}{4 - 3 \cos 60^\circ} = 1.04 \end{aligned}$$

$$\therefore \alpha = \tan^{-1}(1.04) = 46.1^\circ$$

Thus, $\mathbf{A} - \mathbf{B}$ is $\sqrt{13}$ unit at 46.1° from $\mathbf{A}$ in the direction shown in figure.

Example 2.9 Obtain the magnitude of $2\mathbf{A} - 3\mathbf{B}$, if

$$\mathbf{A} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}} \text{ and } \mathbf{B} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}.$$

Sol. The magnitude of $2\mathbf{A} - 3\mathbf{B}$ is

$$\begin{aligned} 2\mathbf{A} - 3\mathbf{B} &= 2(\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}) - 3(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}) \\ &= -4\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 7\hat{\mathbf{k}} \\ \therefore \text{Magnitude of } 2\mathbf{A} - 3\mathbf{B} &= \sqrt{(-4)^2 + (5)^2 + (-7)^2} \\ &= \sqrt{16 + 25 + 49} \\ &= \sqrt{90} \end{aligned}$$

Example 2.10 Two vectors $\mathbf{P}$ and $\mathbf{Q}$ have equal magnitudes.

If the magnitude of $(\mathbf{P} + \mathbf{Q})$ is 'k' times the magnitude of $(\mathbf{P} - \mathbf{Q})$, then calculate the angle between $\mathbf{P}$ and $\mathbf{Q}$.

Sol. Given, $|\mathbf{P}| = |\mathbf{Q}|$

$$\text{or } P = Q \quad \dots(i)$$

Let magnitude of $(\mathbf{P} + \mathbf{Q})$ is R and for $(\mathbf{P} - \mathbf{Q})$ is R'

$$\text{Now, } \mathbf{R} = \mathbf{P} + \mathbf{Q}$$

$$\text{and } R^2 = P^2 + Q^2 + 2PQ \cos \theta$$

$$R^2 = 2P^2 + 2P^2 \cos \theta \quad \dots(ii)$$

$$\text{Again, } \mathbf{R}' = \mathbf{P} - \mathbf{Q}$$

$$(R')^2 = P^2 + Q^2 - 2PQ \cos \theta$$

$$(R')^2 = 2P^2 - 2P^2 \cos \theta \quad \dots(iii)$$

$$\text{Given, } R = kR' \text{ or } \left(\frac{R}{R'}\right)^2 = k^2$$

Dividing Eq. (ii) by Eq. (iii), we get

$$\frac{k^2}{1} = \frac{1 + \cos \theta}{1 - \cos \theta}$$

$$\begin{aligned} \text{or } \frac{k^2 - 1}{k^2 + 1} &= \frac{(1 + \cos \theta) - (1 - \cos \theta)}{(1 + \cos \theta) + (1 - \cos \theta)} \\ &= \frac{2 \cos \theta}{2} = \cos \theta \end{aligned}$$

$$\therefore \cos \theta = \left(\frac{k^2 - 1}{k^2 + 1}\right) \text{ or } \theta = \cos^{-1} \left(\frac{k^2 - 1}{k^2 + 1}\right)$$

Resolution of vectors

The resolution of a vector is opposite to vector addition. If a vector is resolved into two vectors whose combined effect is the same as that of the given vector, then the resolved vectors are called the components of the given vector.

Resolution of vectors into rectangular components

When a vector is splitted into components which are at right angle to each other, then the components are called rectangular or orthogonal components of that vector.

Refer Fig. (a)

We have resolved a two dimensional vector (in XY -plane) $\mathbf{R}$ in mutually perpendicular directions x and y .

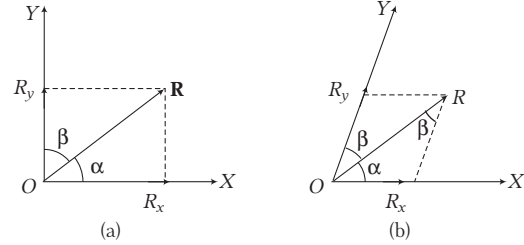


Fig. 2.16

Component along X -axis $= R_x = R \cos \alpha$ or $R \sin \beta$

Component along Y -axis $= R_y = R \cos \beta$ or $R \sin \alpha$

If $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ be the unit vectors along X and Y -axes

respectively, we can write, $\mathbf{R} = R_x \hat{\mathbf{i}} + R_y \hat{\mathbf{j}}$

Refer Fig. (b)

Vector $\mathbf{R}$ has been resolved in two axes such that x and y not perpendicular to each other. Applying sine law in the triangle shown, we have

$$\frac{R}{\sin [180^\circ - (\alpha + \beta)]} = \frac{R_x}{\sin \beta} = \frac{R_y}{\sin \alpha}$$

$$\text{or } R_x = \frac{R \sin \beta}{\sin (\alpha + \beta)} \text{ and } R_y = \frac{R \sin \alpha}{\sin (\alpha + \beta)}$$

If $\alpha + \beta = 90^\circ$, $R_x = R \sin \beta$ and $R_y = R \sin \alpha$

$$\text{Also, } \tan \alpha = \frac{R_y}{R_x} \text{ or } \alpha = \tan^{-1} \left(\frac{R_y}{R_x}\right)$$

Rectangular components of a vector in three dimensional space

Let R_x, R_y and R_z are the components of resultant vector $\mathbf{R}$ in X, Y and Z -axes respectively, and $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are unit vectors along these directions. Then, a vector $\mathbf{R}$ and its magnitude can be written as

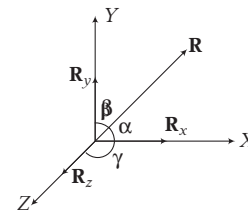


Fig. 2.17

$$\mathbf{R} = R_x \hat{\mathbf{i}} + R_y \hat{\mathbf{j}} + R_z \hat{\mathbf{k}} \text{ or } \mathbf{R} = R_x \hat{\mathbf{i}} + R_y \hat{\mathbf{j}} + R_z \hat{\mathbf{k}}$$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2}$$

This vector $\mathbf{R}$ makes an angle of

$$\alpha = \cos^{-1} \left(\frac{R_x}{R}\right) \text{ with } X\text{-axis}$$

$$\beta = \cos^{-1} \left(\frac{R_y}{R} \right) \text{ with } Y\text{-axis}$$

and $\gamma = \cos^{-1} \left(\frac{R_z}{R} \right) \text{ with } Z\text{-axis}$

Note

(i) A vector can be resolved into maximum infinite number of components.

For example, $10\hat{i} = \hat{i} + \hat{i} + \hat{i} \dots 10 \text{ times}$

$$\text{or } = \frac{\hat{i}}{2} + \frac{\hat{i}}{2} + \frac{\hat{i}}{2} \dots 20 \text{ times}$$

(ii) A vector is independent of the orientation of axes but the components of that vector depends upon the orientation of axes.

(iii) The component of a vector along its perpendicular direction is always zero.

Example 2.11 Find the angle that the vector $\mathbf{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ makes with Y -axis.

Sol. According to the resolution of the vector,

$$\cos \theta = \frac{A_y}{A} = \frac{3}{\sqrt{(2)^2 + (3)^2 + (-1)^2}} = \frac{3}{\sqrt{14}}$$

$$\therefore \theta = \cos^{-1} \left(\frac{3}{\sqrt{14}} \right)$$

Example 2.12 A vector is given by $\mathbf{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$. Find the magnitude of $\mathbf{A}$, unit vector along $\mathbf{A}$ and angles made by $\mathbf{A}$ with coordinate axes.

Sol. We have, magnitude, $|\mathbf{A}| = A = \sqrt{A_x^2 + A_y^2 + A_z^2}$

$$= \sqrt{(3)^2 + (4)^2 + (5)^2} = 5\sqrt{2}$$

$$\text{Unit vector, } \hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{3\hat{i} + 4\hat{j} + 5\hat{k}}{5\sqrt{2}}$$

Angles made by $\mathbf{A}$ with coordinate axes,

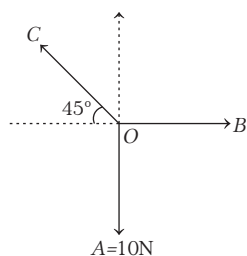
$$\cos \alpha = \frac{A_x}{|\mathbf{A}|} = \frac{3}{5\sqrt{2}} \Rightarrow \alpha = \cos^{-1} \left(\frac{3}{5\sqrt{2}} \right)$$

$$\cos \beta = \frac{A_y}{|\mathbf{A}|} = \frac{4}{5\sqrt{2}} \Rightarrow \beta = \cos^{-1} \left(\frac{4}{5\sqrt{2}} \right)$$

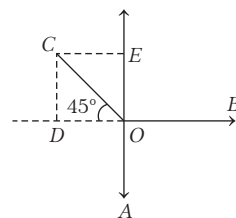
$$\cos \gamma = \frac{A_z}{|\mathbf{A}|} = \frac{5}{5\sqrt{2}} \Rightarrow \gamma = \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}$$

Example 2.13 Find the magnitude of vectors $\mathbf{OB}$ and $\mathbf{OC}$.

If sum of three vectors gives a value equals to 0 as shown in figure below.



Sol. Consider the figure shown below.



Resolve $\mathbf{OC}$ into two rectangular components,

$$OD = OC \cos 45^\circ \text{ and } OE = OC \sin 45^\circ$$

To obtain zero resultant,

$$OE = OA \text{ or } OC \sin 45^\circ = 10 \text{ N}$$

$$\Rightarrow OC \times \frac{1}{\sqrt{2}} = 10 \text{ N}$$

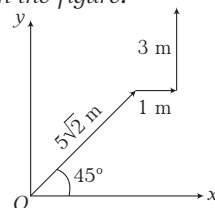
$$|OC| = 10\sqrt{2} \text{ N and } OD = OB$$

$$\Rightarrow OC \cos 45^\circ = OB$$

$$\Rightarrow OB = 10\sqrt{2} \times \frac{1}{\sqrt{2}} = 10 \text{ N}$$

Thus, the magnitude of $\mathbf{OB}$ and $\mathbf{OC}$ is $10\sqrt{2} \text{ N}$ and 10 N .

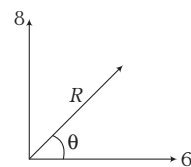
Example 2.14 Find the resultant and direction of three vectors as shown in the figure.



Sol. From the figure,

$$\text{On } X\text{-axis, } x = 5\sqrt{2} \cos 45^\circ + 1 = 5 + 1 = 6 \text{ m}$$

$$\text{On } Y\text{-axis, } y = 5\sqrt{2} \sin 45^\circ + 3 = 5 + 3 = 8 \text{ m}$$



$\therefore$ Magnitude of resultant of given vectors,

$$R = \sqrt{x^2 + y^2} \\ = \sqrt{(6)^2 + (8)^2} = 10 \text{ m}$$

$\therefore$ Direction of resultant vectors,

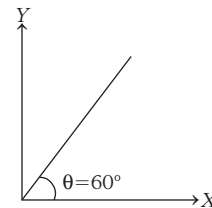
$$\tan \theta = \frac{y}{x} = \frac{8}{6} = \frac{4}{3}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{4}{3} \right)$$

Thus, resultant vector makes an angle of $\tan^{-1} \left(\frac{4}{3} \right)$ with X -axis.

CHECK POINT 2.2

- For the resultant of two vectors to be maximum, what must be the angle between them?
(a) 0° (b) 60°
(c) 90° (d) 180°
- Minimum number of vectors of unequal magnitude whose vector sum can equal to zero is
(a) two (b) three
(c) four (d) Any
- Two vectors having magnitudes 8 and 10 can have maximum and minimum value of magnitude of their resultant as
(a) 12, 6 (b) 10, 3
(c) 18, 2 (d) None of these
- Given that, $\mathbf{P} + \mathbf{Q} + \mathbf{R} = 0$. Which of the following statement is true?
(a) $|\mathbf{P}| + |\mathbf{Q}| = |\mathbf{R}|$ (b) $|\mathbf{P} + \mathbf{Q}| = |\mathbf{R}|$
(c) $|\mathbf{P}| - |\mathbf{Q}| = |\mathbf{R}|$ (d) $|\mathbf{P} - \mathbf{Q}| = |\mathbf{R}|$
- $(\mathbf{P} + \mathbf{Q})$ is a unit vector along X -axis. If $\mathbf{P} = \hat{i} - \hat{j} + \hat{k}$, then what is the value of $\mathbf{Q}$?
(a) $\hat{i} + \hat{j} - \hat{k}$ (b) $\hat{j} - \hat{k}$ (c) $\hat{i} + \hat{j} + \hat{k}$ (d) $\hat{j} + \hat{k}$
- $\mathbf{A} = 2\hat{i} + \hat{j}$, $\mathbf{B} = 3\hat{j} - \hat{k}$ and $\mathbf{C} = 6\hat{i} - 2\hat{k}$.
Value of $\mathbf{A} - 2\mathbf{B} + 3\mathbf{C}$ would be
(a) $20\hat{i} + 5\hat{j} + 4\hat{k}$ (b) $20\hat{i} - 5\hat{j} - 4\hat{k}$
(c) $4\hat{i} + 5\hat{j} + 20\hat{k}$ (d) $5\hat{i} + 4\hat{j} + 10\hat{k}$
- At what angle should the two vectors $2P$ and $\sqrt{2}P$ act, so that the resultant force is $P\sqrt{10}$?
(a) 45° (b) 60°
(c) 90° (d) 120°
- Three vectors each of magnitude A are acting at a point such that angle between any two consecutive vectors in same plane is 60° . The magnitude of their resultant is
(a) $2A$ (b) $\sqrt{2}A$
(c) $\sqrt{3}A$ (d) $\sqrt{6}A$
- If $\mathbf{P} + \mathbf{Q} = \mathbf{P} - \mathbf{Q}$, then
(a) $\mathbf{P} = 0$ (b) $\mathbf{Q} = 0$
(c) $\mathbf{P} = 1$ (d) $|\mathbf{Q}| = 1$
- Resultant of two vectors $\mathbf{A}$ and $\mathbf{B}$ is given by $\mathbf{R} = \{\mathbf{A} - \mathbf{B}\}$, angle between $\mathbf{A}$ and $\mathbf{B}$ will be
(a) 90° (b) 180°
(c) 0° (d) None of these
- If $|\mathbf{A}| = 2$ and $|\mathbf{B}| = 4$ and angle between them is 60° , then $|\mathbf{A} - \mathbf{B}|$
(a) $\sqrt{13}$ (b) $3\sqrt{3}$
(c) $\sqrt{3}$ (d) $2\sqrt{3}$
- A vector inclined at an angle θ to the horizontal as shown in figure below. If its component along X -axis is 50 N, then its magnitude in y -direction is



- (a) 50 N (b) 72 N (c) 64 N (d) 87 N

PRODUCT OF TWO VECTORS

The multiplication of two vector quantities cannot be done by simple algebraic method. The product of two vectors may be a scalar as well as a vector.

If the product of two vectors is a scalar quantity, then it is called scalar product (or dot product); if the product is a vector quantity, then it is called vector product (or cross product).

Scalar product of two vectors

The scalar product (or dot product) of two vectors is defined as the product of their magnitude with cosine of the angle between them.

Thus, if there are two vectors $\mathbf{A}$ and $\mathbf{B}$ having angle θ between them, then their scalar product is written as

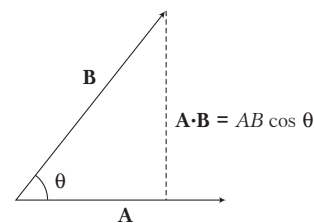


Fig. 2.18

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta \quad (\text{A scalar quantity})$$

e.g. work done (W) = $\mathbf{F} \cdot \mathbf{s}$ and power (P) = $\mathbf{F} \cdot \mathbf{v}$

Note (i) The scalar or dot product of two vectors $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \cdot \mathbf{B}$ and is read as **A dot B**.

(ii) Dot product is always a scalar, which is positive, if angle between the vectors is acute (i.e. $\theta < 90^\circ$) and negative, if angle between them is obtuse (i.e. $90^\circ < \theta < 180^\circ$).

Important points regarding dot product

The following points should be remembered regarding the dot product

- (i) $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$ (i.e. dot product is commutative)
- (ii) $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$ (i.e. dot product is distributive)
- (iii) $\mathbf{A} \cdot \mathbf{A} = A^2$ (also called self-dot product)
- (iv) $\mathbf{A} \cdot \mathbf{B} = A(B \cos \theta) = A$ (component of $\mathbf{B}$ along $\mathbf{A}$)
or $\mathbf{A} \cdot \mathbf{B} = B(A \cos \theta) = B$ (component of $\mathbf{A}$ along $\mathbf{B}$)
- (v) $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = (1)(1) \cos 0^\circ = 1$
- (vi) $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{k}} = \hat{\mathbf{i}} \cdot \hat{\mathbf{k}} = (1)(1) \cos 90^\circ = 0$
- (vii) $(a_1\hat{\mathbf{i}} + b_1\hat{\mathbf{j}} + c_1\hat{\mathbf{k}}) \cdot (a_2\hat{\mathbf{i}} + b_2\hat{\mathbf{j}} + c_2\hat{\mathbf{k}})$
 $= a_1a_2 + b_1b_2 + c_1c_2$
- (viii) $\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{AB}$ (cosine of angle between $\mathbf{A}$ and $\mathbf{B}$)
- (ix) Two vectors are perpendicular (i.e. $\theta = 90^\circ$), if their dot product is zero.
- (x) Dot product of two vectors will be maximum when vectors are parallel (i.e. $\theta = 0$) $(\mathbf{A} \cdot \mathbf{B})_{\max} = AB$

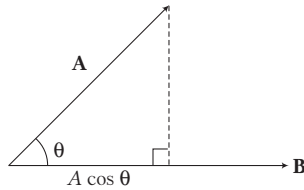
Projection of A along B (Components of dot product)

Fig. 2.19

- (i) **In scalar form** : Projection or scalar component of $\mathbf{A}$ along $\mathbf{B}$

$$= A \cos \theta = A \times \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{\mathbf{A} \cdot \mathbf{B}}{B} = \mathbf{A} \cdot \hat{\mathbf{B}}$$

- (ii) **In vector form** : Projection or vector component of $\mathbf{A}$ along $\mathbf{B}$

$$= (A \cos \theta) \hat{\mathbf{B}} = \left(A \times \frac{\mathbf{A} \cdot \mathbf{B}}{AB} \right) \hat{\mathbf{B}}$$

$$= \frac{\mathbf{A} \cdot \mathbf{B}}{B} \cdot \hat{\mathbf{B}} = (\mathbf{A} \cdot \hat{\mathbf{B}}) \hat{\mathbf{B}}$$

Example 2.15. Find the projection of $\mathbf{A} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ on $\mathbf{B} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

Sol. Projection of $\mathbf{A}$ on $\mathbf{B} = A \cos \theta$ (where, θ = angle between $\mathbf{A}$ and $\mathbf{B}$)

$$= \frac{\mathbf{A} \cdot \mathbf{B}}{B} = (2\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}) \cdot \frac{(\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}})}{\sqrt{(1)^2 + (-2)^2 + (1)^2}}$$

$$= \frac{2 + 2 + 1}{\sqrt{6}} = \frac{5}{\sqrt{6}}$$

Example 2.16 Prove that the vectors $\mathbf{A} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{B} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ are mutually perpendicular.

Sol. $\mathbf{A} \cdot \mathbf{B} = (2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}) \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$
 $= (2)(1) + (-3)(1) + (1)(1)$
 $= 0 = AB \cos \theta$ ($\because \mathbf{A} \cdot \mathbf{B} = AB \cos \theta$)
 $\therefore \cos \theta = 0$ (As $A \neq 0, B \neq 0$)
or $\theta = 90^\circ$ ($\because \cos 90^\circ = 0$)
or the vectors $\mathbf{A}$ and $\mathbf{B}$ are mutually perpendicular.

Example 2.17 Find the angle between two vectors

$$\mathbf{A} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}} \text{ and } \mathbf{B} = \hat{\mathbf{i}} - \hat{\mathbf{k}}$$

Sol. $A = |\mathbf{A}| = \sqrt{(2)^2 + (1)^2 + (-1)^2} = \sqrt{6}$
 $B = |\mathbf{B}| = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$
 $\mathbf{A} \cdot \mathbf{B} = (2\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) \cdot (\hat{\mathbf{i}} - \hat{\mathbf{k}}) = (2)(1) + (-1)(-1) = 3$
Now, $\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{3}{\sqrt{6} \cdot \sqrt{2}} = \frac{3}{\sqrt{12}} = \frac{\sqrt{3}}{2}$
 $\therefore \theta = 30^\circ$

Example 2.18 Find the component of vector $\mathbf{A} + \mathbf{B}$ along (i) X-axis (ii) and $\mathbf{C}$.

Given, $\mathbf{A} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}}$, $\mathbf{B} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{k}}$ and $\mathbf{C} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$.

Sol. $\mathbf{A} + \mathbf{B} = (\hat{\mathbf{i}} - 2\hat{\mathbf{j}}) + (2\hat{\mathbf{i}} + 3\hat{\mathbf{k}}) = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$

(i) Component of $\mathbf{A} + \mathbf{B}$ along X-axis is 3.

(ii) Component of $\mathbf{A} + \mathbf{B} = \mathbf{R}$ (say) along $\mathbf{C}$ is

$$\mathbf{R} \cdot \mathbf{C} = RC \cos \theta$$

$$\therefore R \cos \theta = \frac{\mathbf{R} \cdot \mathbf{C}}{C} = \frac{(3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}})}{\sqrt{(1)^2 + (1)^2}} = \frac{3 - 2}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Example 2.19 Find the (i) scalar component and (ii) vector component of $\mathbf{A} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ on $\mathbf{B} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$.

Sol. (i) Scalar component of $\mathbf{A}$ along $\mathbf{B}$ is

$$A \cos \theta = \mathbf{A} \cdot \hat{\mathbf{B}} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 5\hat{\mathbf{k}}) \cdot \frac{(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})}{\sqrt{3}}$$

$$= \frac{3 + 4 + 5}{\sqrt{3}} = \frac{12}{\sqrt{3}} = 4\sqrt{3}$$

(ii) Vector component of $\mathbf{A}$ along $\mathbf{B}$ is

$$(A \cos \theta) \hat{\mathbf{B}} = (\mathbf{A} \cdot \hat{\mathbf{B}}) \hat{\mathbf{B}} = (4\sqrt{3}) \frac{(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})}{\sqrt{3}} = 4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$$

Vector product of two vectors

The vector product or cross product of two vectors is defined as a vector having magnitude equal to the product of their magnitudes with the sine of angle between them, and its direction is perpendicular to the plane containing both the vectors according to right hand screw rule.

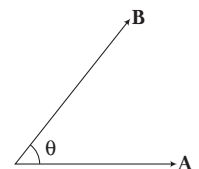


Fig. 2.20

Thus, if $\mathbf{A}$ and $\mathbf{B}$ are two vectors, then their vector product, *i.e.* $\mathbf{A} \times \mathbf{B}$ gives a vector $\mathbf{C}$ and is defined by $\mathbf{C} = \mathbf{A} \times \mathbf{B} = AB \sin \theta \hat{\mathbf{n}}$.

where, $\hat{\mathbf{n}}$ is a unit vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$.

The direction of $\mathbf{C}$ (or of $\hat{\mathbf{n}}$) is determined by right hand screw rule and right hand thumb rule.

- (i) **Right Hand Screw Rule** Rotate a right handed screw from first vector ($\mathbf{A}$) towards second vector ($\mathbf{B}$). The direction in which right handed screw moves gives the direction of vector ($\mathbf{C}$) as shown in Fig. 2.21.

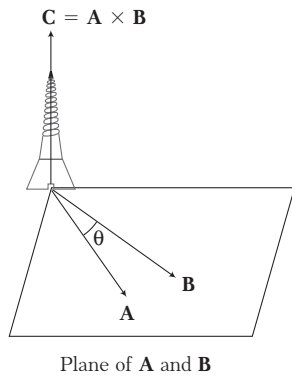


Fig. 2.21

The direction of $\mathbf{C}$ (or of $\hat{\mathbf{n}}$) is perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$; and its sense is decided by right hand screw rule.

- (ii) **Right Hand Thumb Rule** If the fingers of the right hand be curled in the direction in which vector $\mathbf{A}$ must be turned through the smaller included angle θ to coincide with the direction of vector $\mathbf{B}$, the thumb points in the direction of $\mathbf{C}$ as shown in Fig. 2.22.

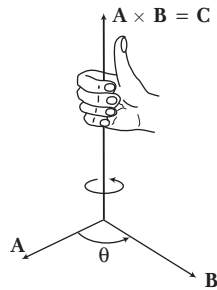


Fig. 2.22

Important points regarding vector product

- (i) $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$
 (ii) The magnitude of cross product of two parallel vectors is zero, as $|\mathbf{A} \times \mathbf{B}| = AB \sin \theta$ and $\theta = 0^\circ$ for two parallel vectors. Thus,

$$\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} \times \hat{\mathbf{k}} = 0$$

- (iii) If two vectors are perpendicular to each other, we have $\theta = 90^\circ$, *i.e.* $\sin \theta = 1$. So that, $\mathbf{A} \times \mathbf{B} = AB \hat{\mathbf{n}}$. These vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A} \times \mathbf{B}$ thus form a right handed system of mutually perpendicular vectors. It follows at once from the above that in case of the orthogonal triad of unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ (each perpendicular to each other)

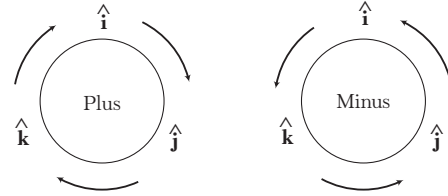


Fig. 2.23

$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \quad \hat{\mathbf{j}} \times \hat{\mathbf{i}} = -\hat{\mathbf{k}}$$

$$\hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{j}} = -\hat{\mathbf{i}}$$

and

$$\hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}}, \quad \hat{\mathbf{i}} \times \hat{\mathbf{k}} = -\hat{\mathbf{j}}$$

- (iv) $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$

- (v) A vector product can be expressed in terms of rectangular components of the two vectors and put in the determinant form as may be seen from the following

$$\text{Let } \mathbf{A} = a_1 \hat{\mathbf{i}} + b_1 \hat{\mathbf{j}} + c_1 \hat{\mathbf{k}}$$

$$\text{and } \mathbf{B} = a_2 \hat{\mathbf{i}} + b_2 \hat{\mathbf{j}} + c_2 \hat{\mathbf{k}}$$

Putting it in determinant form, we have

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

It may be noted that the scalar components of the first vector $\mathbf{A}$ occupy the middle row of the determinant.

- (vi) A unit vector ($\hat{\mathbf{n}}$) perpendicular to $\mathbf{A}$ as well as $\mathbf{B}$ is given by $\hat{\mathbf{n}} = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|}$

- (vii) If $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are coplanar, then $[\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})] = 0$

- (viii) Angle between $(\mathbf{A} + \mathbf{B})$ and $(\mathbf{A} \times \mathbf{B})$ is 90° .

- (ix) Two vectors can be shown parallel to each other, if

- The coefficient of $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ of both the vectors bear a constant ratio. For example, a vector $\mathbf{A} = a_1 \hat{\mathbf{i}} + b_1 \hat{\mathbf{j}} + c_1 \hat{\mathbf{k}}$ is parallel to another vector $\mathbf{B} = a_2 \hat{\mathbf{i}} + b_2 \hat{\mathbf{j}} + c_2 \hat{\mathbf{k}}$, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

- The cross product of both the vectors is zero. For instance $\mathbf{A}$ and $\mathbf{B}$ are parallel to each other, if

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

- (x) The area of triangle bounded by vectors $\mathbf{A}$ and $\mathbf{B}$ is $\frac{1}{2} |\mathbf{A} \times \mathbf{B}|$.

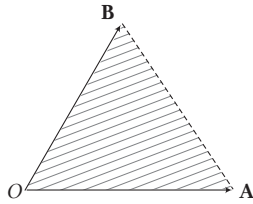


Fig. 2.24

Area of triangle ABC If position vector of A is $\mathbf{a}$, position vector of B is $\mathbf{b}$ and position vector of C is $\mathbf{c}$, then

$$\text{Area of triangle ABC} = \frac{1}{2} |\mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}|$$

- (xi) Area of parallelogram shown in figure is

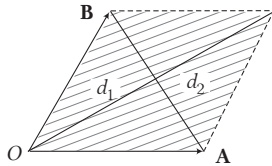


Fig. 2.25

$$= |\mathbf{A} \times \mathbf{B}| = \frac{1}{2} |d_1 \times d_2|$$

where, d_1 and d_2 are diagonals.

Example 2.20 If $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} \neq 0$ with $\mathbf{a} \neq -\mathbf{c}$, then show that $\mathbf{a} + \mathbf{c} = k\mathbf{b}$, where k is scalar.

Sol. Given, $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c}$
 $\mathbf{a} \times \mathbf{b} = -\mathbf{c} \times \mathbf{b}$

$$\therefore \mathbf{a} \times \mathbf{b} + \mathbf{c} \times \mathbf{b} = 0$$

$$(\mathbf{a} + \mathbf{c}) \times \mathbf{b} = 0$$

Given, $\mathbf{a} \times \mathbf{b} \neq 0$, $\mathbf{b} \times \mathbf{c} \neq 0$, $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ are non-zero vectors.

$$(\mathbf{a} + \mathbf{c}) \neq 0$$

Hence, $\mathbf{a} + \mathbf{c}$ is parallel to $\mathbf{b}$.

$$\therefore \mathbf{a} + \mathbf{c} = k\mathbf{b} \quad (\text{where, } k \text{ is scalar})$$

Example 2.21 Prove that, $|\mathbf{a} \times \mathbf{b}|^2 = a^2 b^2 - (\mathbf{a} \cdot \mathbf{b})^2$

Sol. Let $|\mathbf{a}| = a$, $|\mathbf{b}| = b$

and θ be the angle between them.

$$\therefore |\mathbf{a} \times \mathbf{b}|^2 = (ab \sin \theta)^2 = a^2 b^2 \sin^2 \theta$$

$$= a^2 b^2 (1 - \cos^2 \theta) = a^2 b^2 - (ab \cos \theta)^2$$

$$= a^2 b^2 - (\mathbf{a} \cdot \mathbf{b})^2 = \text{RHS}$$

Example 2.22 Show that the vector $\mathbf{A} = \hat{i} - \hat{j} + 2\hat{k}$ is parallel to a vector $\mathbf{B} = 3\hat{i} - 3\hat{j} + 6\hat{k}$.

Sol. A vector $\mathbf{A}$ is parallel to an another vector $\mathbf{B}$, if it can be written as

$$\mathbf{A} = m\mathbf{B}$$

$$\text{Here, } \mathbf{A} = (\hat{i} - \hat{j} + 2\hat{k}) = \frac{1}{3} (3\hat{i} - 3\hat{j} + 6\hat{k})$$

$$(\because \mathbf{B} = 3\hat{i} - 3\hat{j} + 6\hat{k})$$

$$\therefore \mathbf{A} = \frac{1}{3} \mathbf{B}$$

This implies that $\mathbf{A}$ is parallel to $\mathbf{B}$ and magnitude of $\mathbf{A}$ is $1/3$ times the magnitude of $\mathbf{B}$.

Example 2.23 Find a unit vector perpendicular to

$$\mathbf{A} = 2\hat{i} + 3\hat{j} + \hat{k} \text{ and } \mathbf{B} = \hat{i} - \hat{j} + \hat{k} \text{ both.}$$

Sol. Given, $\mathbf{A} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\mathbf{B} = \hat{i} - \hat{j} + \hat{k}$

Now, $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ is a vector, perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.

Hence, a unit vector $\hat{n}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$. It can be written as

$$\hat{n} = \frac{\mathbf{C}}{C} = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|}$$

$$\text{Here, } \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & -1 & 1 \end{vmatrix}$$

$$= \hat{i}(3+1) + \hat{j}(1-2) + \hat{k}(-2-3)$$

$$= 4\hat{i} - \hat{j} - 5\hat{k}$$

$$\text{Further, } |\mathbf{A} \times \mathbf{B}| = \sqrt{(4)^2 + (-1)^2 + (-5)^2} = \sqrt{42}$$

$$\therefore \text{The desired unit vector is } \hat{n} = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|}$$

$$\text{or } \hat{n} = \frac{1}{\sqrt{42}} (4\hat{i} - \hat{j} - 5\hat{k})$$

Example 2.24 Let $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ be unit vectors. Suppose that $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \cdot \mathbf{C} = 0$ and that the angle between $\mathbf{B}$ and $\mathbf{C}$ is $\pi/6$, then prove that, $\mathbf{A} = \pm 2 (\mathbf{B} \times \mathbf{C})$

Sol. Since, $\mathbf{A} \cdot \mathbf{B} = 0$, $\mathbf{A} \cdot \mathbf{C} = 0$

Hence, $(\mathbf{B} + \mathbf{C}) \cdot \mathbf{A} = 0$

So, $\mathbf{A}$ is perpendicular to $(\mathbf{B} + \mathbf{C})$ and $\mathbf{A}$ is a unit vector perpendicular to the plane of vectors $\mathbf{B}$ and $\mathbf{C}$.

$$\mathbf{A} = \frac{\mathbf{B} \times \mathbf{C}}{|\mathbf{B} \times \mathbf{C}|}$$

where, $|\mathbf{B} \times \mathbf{C}| = |\mathbf{B}| |\mathbf{C}| \sin \theta$

$$= |\mathbf{B}| |\mathbf{C}| \sin \frac{\pi}{6}$$

$$= 1 \times 1 \times \frac{1}{2} = \frac{1}{2}$$

$$\therefore \mathbf{A} = \frac{\mathbf{B} \times \mathbf{C}}{|\mathbf{B} \times \mathbf{C}|} = \pm 2 (\mathbf{B} \times \mathbf{C})$$

$$\left(\because \theta = \frac{\pi}{6} \right)$$

Example 2.25 If $\mathbf{a} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}} - 4\hat{\mathbf{k}}$, $\mathbf{b} = 6\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, then find the area of a triangle whose adjacent sides are determined by $\mathbf{a}$ and $\mathbf{b}$.

Sol. Cross product of vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & 1 & -4 \\ 6 & 5 & -2 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(-2 + 20) - \hat{\mathbf{j}}(-6 + 24) + \hat{\mathbf{k}}(15 - 6) = 18\hat{\mathbf{i}} - 18\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$$

Magnitude of $\mathbf{a}$ and $\mathbf{b}$,

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{(18)^2 + (-18)^2 + (9)^2} = \sqrt{729} = 27$$

$$\therefore \text{Area of } \Delta = \frac{1}{2} |\mathbf{a} \times \mathbf{b}| = \frac{27}{2} \\ = 13.5 \text{ sq. units}$$

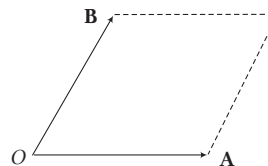
Example 2.26 If the diagonals of a parallelogram are $2\hat{\mathbf{i}}$ and $2\hat{\mathbf{j}}$, then find its area.

Sol. Let $\mathbf{A} = 2\hat{\mathbf{i}}$ and $\mathbf{B} = 2\hat{\mathbf{j}}$

$$\begin{aligned} \text{Area of parallelogram} &= \frac{1}{2} |\mathbf{A} \times \mathbf{B}| = \frac{1}{2} |2\hat{\mathbf{i}} \times 2\hat{\mathbf{j}}| \\ &= \frac{1}{2} |4(\hat{\mathbf{i}} \times \hat{\mathbf{j}})| = \frac{1}{2} |4\hat{\mathbf{k}}| \quad (\because \hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}) \\ &= 2 \text{ sq. units} \end{aligned}$$

Example 2.27 The adjacent sides of a parallelogram is given by two vectors $\mathbf{A}$ and $\mathbf{B}$, where $\mathbf{A} = 5\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ and $\mathbf{B} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$. Calculate the area of parallelogram.

Sol. Here, $\mathbf{A}$ and $\mathbf{B}$ represents the adjacent sides of a parallelogram.



$$\mathbf{A} = 5\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$$

$$\mathbf{B} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$$

$$\text{Area of parallelogram} = |\mathbf{A} \times \mathbf{B}|$$

$$\begin{aligned} \therefore \mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 5 & -4 & 3 \\ 3 & -2 & -1 \end{vmatrix} \\ &= \hat{\mathbf{i}}(4 + 6) - \hat{\mathbf{j}}(-5 - 9) + \hat{\mathbf{k}}(-10 + 12) \\ &= 10\hat{\mathbf{i}} + 14\hat{\mathbf{j}} + 2\hat{\mathbf{k}} \\ \Rightarrow |\mathbf{A} \times \mathbf{B}| &= \sqrt{(10)^2 + (14)^2 + (2)^2} \\ &= \sqrt{300} = 10\sqrt{3} \text{ sq. units} \end{aligned}$$

CHECK POINT 2.3

1. The modulus of the vector product of two vectors is $\frac{1}{\sqrt{3}}$ times their scalar product. The angle between vectors is

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

2. What is the dot product of two vectors having magnitude of 3 and 5; and the angle between them is 60° ?

- (a) 5.2 (b) 7.5
(c) 8.4 (d) 8.6

3. The vector projection of a vector $3\hat{\mathbf{i}} + 4\hat{\mathbf{k}}$ on Y-axis is

- (a) 5 (b) 4
(c) 3 (d) zero

4. Three vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ satisfy the relation $\mathbf{A} \cdot \mathbf{B} = 0$ and $\mathbf{A} \cdot \mathbf{C} = 0$. Then, the vector $\mathbf{A}$ is parallel to

- (a) $\mathbf{B}$ (b) $\mathbf{C}$
(c) $\mathbf{B} \cdot \mathbf{C}$ (d) $\mathbf{B} \times \mathbf{C}$

5. The condition $(\mathbf{a} \cdot \mathbf{b})^2 = a^2 b^2$ is satisfied when

- (a) $\mathbf{a}$ is parallel to $\mathbf{b}$ (b) $\mathbf{a} \neq \mathbf{b}$
(c) $\mathbf{a} \cdot \mathbf{b} = 1$ (d) $\mathbf{a} \perp \mathbf{b}$

6. When $\mathbf{A} \cdot \mathbf{B} = -|\mathbf{A}| |\mathbf{B}|$, then

- (a) $\mathbf{A}$ and $\mathbf{B}$ are perpendicular to each other
(b) $\mathbf{A}$ and $\mathbf{B}$ acts in the same direction
(c) $\mathbf{A}$ and $\mathbf{B}$ acts in the opposite direction
(d) $\mathbf{A}$ and $\mathbf{B}$ can act in any direction

7. Given $|\mathbf{A}| = 2$, $|\mathbf{B}| = 5$ and $|\mathbf{A} \times \mathbf{B}| = 8$. If angle between $\mathbf{A}$ and $\mathbf{B}$ is acute, then $\mathbf{A} \cdot \mathbf{B}$ is

- (a) 6 (b) 3 (c) 4 (d) 7

8. If $|\mathbf{A} \times \mathbf{B}| = \sqrt{3} \mathbf{A} \cdot \mathbf{B}$, then the value of $|\mathbf{A} + \mathbf{B}|$ is

- (a) $\left(A^2 + B^2 + \frac{AB}{\sqrt{3}}\right)^{1/2}$
(b) $A + B$
(c) $(A^2 + B^2 + \sqrt{3} AB)^{1/2}$
(d) $(A^2 + B^2 + AB)^{1/2}$

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1 Out of the following quantities, which is scalar?
(a) Displacement (b) Momentum
(c) Potential energy (d) Torque
- 2 The vector quantity among the following is
(a) mass (b) time
(c) distance (d) displacement
- 3 A vector is added to an equal and opposite vector of similar nature, forms a
(a) unit vector (b) position vector
(c) null vector (d) displacement vector
- 4 The component of a vector along any other direction is
(a) always less than its magnitude
(b) always greater than its magnitude
(c) always equal to its magnitude
(d) None of the above
- 5 Which of the following is a unit vector?
(a) $\hat{i} + \hat{j}$ (b) $\cos \theta \hat{i} - \sin \theta \hat{j}$
(c) $\sin \theta \hat{i} + 2 \cos \theta \hat{j}$ (d) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j})$
- 6 A vector $\mathbf{P} = 3\hat{i} - 2\hat{j} + a\hat{k}$ is perpendicular to the vector $\mathbf{Q} = 2\hat{i} + \hat{j} - \hat{k}$. The value of a is
(a) 2 (b) 1
(c) 4 (d) 3
- 7 If $\mathbf{A} = \hat{i} + \hat{j} + \hat{k}$ and $\mathbf{B} = -\hat{i} - \hat{j} - \hat{k}$, then what is the angle made by $(\mathbf{A} - \mathbf{B})$ with $\mathbf{A}$?
(a) 0° (b) 180°
(c) 90° (d) 60°
- 8 The angle between the two vectors $-2\hat{i} + 3\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$ is
(a) 45° (b) 90°
(c) 30° (d) 60°
- 9 Component of the vector $\mathbf{A} = 2\hat{i} + 3\hat{j}$ along the vector $\mathbf{B} = (\hat{i} + \hat{j})$ is
(a) $\frac{5}{\sqrt{2}}$ (b) $4\sqrt{2}$
(c) $\frac{\sqrt{2}}{3}$ (d) None of these
- 10 Vector $\mathbf{P} = 6\hat{i} + 4\sqrt{2}\hat{j} + 4\sqrt{2}\hat{k}$ makes an angle with Z-axis which is equal to
(a) $\cos^{-1}\left(\frac{\sqrt{2}}{5}\right)$ (b) $\cos^{-1}(2\sqrt{2})$
(c) $\cos^{-1}\left(\frac{2\sqrt{2}}{5}\right)$ (d) None of these
- 11 Which of the following is the unit vector perpendicular to $\mathbf{A}$ and $\mathbf{B}$?
(a) $\frac{\hat{\mathbf{A}} \times \hat{\mathbf{B}}}{AB \sin \theta}$ (b) $\frac{\hat{\mathbf{A}} \times \hat{\mathbf{B}}}{AB \cos \theta}$ (c) $\frac{\mathbf{A} \times \mathbf{B}}{AB \sin \theta}$ (d) $\frac{\mathbf{A} \times \mathbf{B}}{AB \cos \theta}$
- 12 Condition under which vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are parallel is
(a) $\mathbf{a} \perp \mathbf{b}$ (b) $|\mathbf{a}| = |\mathbf{b}|$
(c) $\mathbf{a} \neq \mathbf{b}$ (d) $\mathbf{a}$ is parallel to $\mathbf{b}$
- 13 The forces, which meet at one point but their lines of action do not lie on one plane, are called
(a) non-coplanar non-concurrent forces
(b) non-coplanar concurrent forces
(c) coplanar concurrent forces
(d) coplanar non-concurrent forces
- 14 Consider a vector $\mathbf{F} = 4\hat{i} - 3\hat{j}$. Another vector perpendicular to $\mathbf{F}$ is
(a) $4\hat{i} + 3\hat{j}$ (b) $6\hat{i}$ (c) $7\hat{k}$ (d) $3\hat{i} - 4\hat{j}$
- 15 If the angle between two non-zero vectors $\mathbf{A}$ and $\mathbf{B}$ is 120° , its resultant $\mathbf{C}$ will be
(a) $C = |\mathbf{A} - \mathbf{B}|$ (b) $C < |\mathbf{A} - \mathbf{B}|$
(c) $C > |\mathbf{A} - \mathbf{B}|$ (d) $C = |\mathbf{A} + \mathbf{B}|$
- 16 Two vectors $\mathbf{A}$ and $\mathbf{B}$ inclined at angle θ have a resultant $\mathbf{R}$ which makes an angle ϕ with $\mathbf{A}$. If the directions of $\mathbf{A}$ and $\mathbf{B}$ are interchanged, then the resultant will have the same
(a) magnitude
(b) direction
(c) magnitude as well as direction
(d) None of the above
- 17 Which of the following is correct?
(a) $|\mathbf{a} - \mathbf{b}| = |\mathbf{a}| - |\mathbf{b}|$ (b) $|\mathbf{a} - \mathbf{b}| \leq |\mathbf{a}| - |\mathbf{b}|$
(c) $|\mathbf{a} - \mathbf{b}| \geq |\mathbf{a}| - |\mathbf{b}|$ (d) $|\mathbf{a} - \mathbf{b}| > |\mathbf{a}| - |\mathbf{b}|$

18 If $\mathbf{A} = \mathbf{B}$, then which of the following is not correct?

- (a) $\hat{\mathbf{A}} = \hat{\mathbf{B}}$ (b) $|\mathbf{A}| = |\mathbf{B}|$
(c) $\mathbf{AB} = \mathbf{BA}$ (d) $\mathbf{A} + \mathbf{B} = \hat{\mathbf{A}} + \hat{\mathbf{B}}$

19 The angle between vectors $(\mathbf{A} \times \mathbf{B})$ and $(\mathbf{B} \times \mathbf{A})$ is

- (a) zero (b) π
(c) $\pi/4$ (d) $\pi/2$

20 Unit vector parallel to the resultant of vector $8\hat{\mathbf{i}}$ and $8\hat{\mathbf{j}}$ will be

- (a) $(24\hat{\mathbf{i}} + 5\hat{\mathbf{j}})/13$ (b) $(12\hat{\mathbf{i}} + 5\hat{\mathbf{j}})/13$
(c) $(6\hat{\mathbf{i}} + 5\hat{\mathbf{j}})/13$ (d) None of these

21 The area of the parallelogram represented by the vectors $\mathbf{A} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $\mathbf{B} = \hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ is

- (a) 14 units (b) 7.5 units
(c) 10 units (d) 5 units

22 A vector is represented by $3\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$. Its length in XY-plane is

- (a) 2 (b) $\sqrt{14}$
(c) $\sqrt{10}$ (d) $\sqrt{5}$

23 What is the angle between $(\mathbf{P} + \mathbf{Q})$ and $(\mathbf{P} \times \mathbf{Q})$?

- (a) Zero (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) π

24 If three vectors along coordinate axes represent the adjacent sides of a cube of length b , then the unit vector along its diagonal passing through the origin will be

- (a) $\frac{\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}}{\sqrt{2}}$ (b) $\frac{\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}}{\sqrt{36}}$ (c) $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ (d) $\frac{\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}}{\sqrt{3}}$

25 The angle between $\mathbf{A} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\mathbf{B} = \hat{\mathbf{i}} - \hat{\mathbf{j}}$ is

[NCERT Exemplar]

- (a) 45° (b) 90° (c) -45° (d) 180°

26 Resultant of which of the following may be equal to zero?

- (a) 10 N, 10 N, 10 N (b) 10 N, 10 N, 25 N
(c) 10 N, 10 N, 35 N (d) None of these

27 A vector perpendicular to both the vectors $2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ and X-axis is

- (a) $\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ (b) $\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$ (c) $5\hat{\mathbf{j}} + \hat{\mathbf{k}}$ (d) $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$

28 The resultant of $\mathbf{A}$ and $\mathbf{B}$ makes an angle α with $\mathbf{A}$ and β with $\mathbf{B}$, then

- (a) $\alpha < \beta$ (b) $\alpha > \beta$, if $A < B$
(c) $\alpha < \beta$, if $A = B$ (d) $\alpha < \beta$, if $A < B$

29 The vector that must be added to the vectors $\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ and $6\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 7\hat{\mathbf{k}}$, so that the resultant vector is a unit vector along the Y-axis is

- (a) $4\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ (b) $-7\hat{\mathbf{i}} + 4\hat{\mathbf{k}}$
(c) $3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ (d) null vector

30 If a unit vector is represent by $0.5\hat{\mathbf{i}} + 0.8\hat{\mathbf{j}} + c\hat{\mathbf{k}}$, then the value of c is

- (a) 1 (b) $\sqrt{0.11}$
(c) $\sqrt{0.01}$ (d) $\sqrt{0.39}$

31 $\mathbf{A}$ and $\mathbf{B}$ are two vectors given by $\mathbf{A} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $\mathbf{B} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$. The magnitude of the component of $\mathbf{A}$ along $\mathbf{B}$ is

- (a) $\frac{5}{\sqrt{2}}$ (b) $\frac{3}{\sqrt{2}}$ (c) $\frac{8}{\sqrt{5}}$ (d) $\frac{5}{\sqrt{13}}$

32 If $\mathbf{F}_1$ and $\mathbf{F}_2$ are two vectors of equal magnitudes F such that $|\mathbf{F}_1 \cdot \mathbf{F}_2| = |\mathbf{F}_1 \times \mathbf{F}_2|$, then $|\mathbf{F}_1 + \mathbf{F}_2|$ equals to

- (a) $\sqrt{(2 + \sqrt{2})} F$ (b) $2F$
(c) $F\sqrt{2}$ (d) None of these

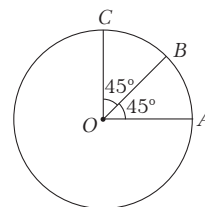
33 The angle between the vectors $\mathbf{A}$ and $\mathbf{B}$ is θ . The value of the triple product $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{A})$ is

- (a) A^2B (b) zero (c) $A^2B \sin \theta$ (d) $A^2B \cos \theta$

34 If $\hat{\mathbf{A}}$ is a unit vector in a given direction, then the value of $\hat{\mathbf{A}} \cdot \frac{d\hat{\mathbf{A}}}{dt}$ is

- (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) 2

35 Find the resultant of three vectors $\mathbf{OA}$, $\mathbf{OB}$ and $\mathbf{OC}$ shown in the following figure. (Radius of the circle is R)



- (a) $2R$ (b) $R(1 + \sqrt{2})$
(c) $R\sqrt{2}$ (d) $R(\sqrt{2} - 1)$

36 If $\mathbf{A} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ and $\mathbf{B} = 7\hat{\mathbf{i}} + 24\hat{\mathbf{j}}$, the vector having the same magnitude as $\mathbf{B}$ and parallel to $\mathbf{A}$ is

- (a) $5\hat{\mathbf{i}} + 20\hat{\mathbf{j}}$ (b) $15\hat{\mathbf{i}} + 10\hat{\mathbf{j}}$
(c) $20\hat{\mathbf{i}} + 15\hat{\mathbf{j}}$ (d) $15\hat{\mathbf{i}} + 20\hat{\mathbf{j}}$

37 Let $\mathbf{A} = \hat{\mathbf{i}}A \cos \theta + \hat{\mathbf{j}}A \sin \theta$ be any vector. Another vector $\mathbf{B}$ which is perpendicular to $\mathbf{A}$ can be expressed as

- (a) $\hat{\mathbf{i}}B \cos \theta - \hat{\mathbf{j}}B \sin \theta$ (b) $\hat{\mathbf{i}}B \sin \theta - \hat{\mathbf{j}}B \cos \theta$
(c) $\hat{\mathbf{i}}B \cos \theta + \hat{\mathbf{j}}B \sin \theta$ (d) $\hat{\mathbf{i}}B \sin \theta + \hat{\mathbf{j}}B \cos \theta$

38 If two vectors $2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $-4\hat{\mathbf{i}} - 6\hat{\mathbf{j}} - \lambda\hat{\mathbf{k}}$ are parallel to each other, then value of λ is

- (a) 0 (b) 2 (c) 3 (d) 4

- 39** If $\mathbf{P} + \mathbf{Q} = \mathbf{R}$ and $|\mathbf{P}| = |\mathbf{Q}| = \sqrt{3}$ and $|\mathbf{R}| = 3$, then the angle between $\mathbf{P}$ and $\mathbf{Q}$ is
 (a) $\pi/4$ (b) $\pi/6$
 (c) $\pi/3$ (d) $\pi/2$
- 40** The angles which the vector $\mathbf{A} = 3\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ makes with the coordinate axes are
 (a) $\cos^{-1}\left(\frac{3}{7}\right)$, $\cos^{-1}\left(\frac{6}{7}\right)$ and $\cos^{-1}\left(\frac{2}{7}\right)$
 (b) $\cos^{-1}\left(\frac{4}{7}\right)$, $\cos^{-1}\left(\frac{5}{7}\right)$ and $\cos^{-1}\left(\frac{3}{7}\right)$
 (c) $\cos^{-1}\left(\frac{3}{7}\right)$, $\cos^{-1}\left(\frac{4}{7}\right)$ and $\cos^{-1}\left(\frac{1}{7}\right)$
 (d) None of the above
- 41** Resultant of two vectors of equal magnitude A is
 (a) $\sqrt{3}A$ at 0° (b) $\sqrt{2}A$ at 90°
 (c) $2A$ at 120° (d) A at 180°
- 42** The component of a vector $\mathbf{r}$ along X -axis will have maximum value, if [NCERT Exemplar]
 (a) $\mathbf{r}$ is along positive Y -axis
 (b) $\mathbf{r}$ is along positive X -axis
 (c) $\mathbf{r}$ makes an angle of 45° with the X -axis
 (d) $\mathbf{r}$ is along negative Y -axis
- 43** If $\mathbf{A} \cdot \mathbf{B} = 0$ and $\mathbf{A} \times \mathbf{B} = 1$, then A and B are
 (a) perpendicular unit vectors
 (b) parallel unit vector
 (c) parallel
 (d) anti-parallel
- 44** If $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$, then $\mathbf{a} \times \mathbf{b}$ is equal to
 (a) $\mathbf{b} \times \mathbf{c}$ (b) $\mathbf{c} \times \mathbf{b}$
 (c) $\mathbf{a} \times \mathbf{c}$ (d) None of these
- 45** If $\mathbf{A} = 4\hat{\mathbf{i}} - 3\hat{\mathbf{j}}$ and $\mathbf{B} = 6\hat{\mathbf{i}} + 8\hat{\mathbf{j}}$, then magnitude and direction of $\mathbf{A} + \mathbf{B}$ with X -axis will be
 (a) $5, \tan^{-1}(3/4)$ (b) $5\sqrt{5}, \tan^{-1}(1/2)$
 (c) $10, \tan^{-1}(5)$ (d) $25, \tan^{-1}(3/4)$
- 46** The direction cosines of vector $(\mathbf{A} - \mathbf{B})$, if $\mathbf{A} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\mathbf{B} = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$ are
 (a) $0, \frac{1}{\sqrt{5}}, \frac{-2}{\sqrt{5}}$ (b) $0, \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}$
 (c) $0, 0, \frac{1}{\sqrt{5}}$ (d) None of these
- 47** Two vectors $\mathbf{A}$ and $\mathbf{B}$ are such that $\mathbf{A} + \mathbf{B} = \mathbf{C}$ and $\mathbf{A}^2 + \mathbf{B}^2 = \mathbf{C}^2$. If θ is the angle between $\mathbf{A}$ and $\mathbf{B}$, then the value of θ is
 (a) π (b) $\frac{2\pi}{3}$
 (c) 0 (d) $\frac{\pi}{2}$
- 48** If vectors $\mathbf{A}$ and $\mathbf{B}$ have an angle θ between them, then value of $|\hat{\mathbf{A}} - \hat{\mathbf{B}}|$ will be
 (a) $2 \cos \frac{\theta}{2}$ (b) $2 \tan \frac{\theta}{2}$
 (c) $2 \sin \frac{\theta}{2}$ (d) None of these
- 49** Given $\mathbf{A} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ and $\mathbf{B} = 6\hat{\mathbf{i}} + 8\hat{\mathbf{j}}$, then which of the following option is correct?
 (a) $\mathbf{A} \times \mathbf{B} = 50$ (b) $\frac{|\mathbf{A}|}{|\mathbf{B}|} = \frac{1}{2}$
 (c) $|\mathbf{A}| = 15$ (d) $\mathbf{A} \cdot |\mathbf{B}| = 48$
- 50** The resultant of two vectors $3\mathbf{P}$ and $2\mathbf{P}$ is R . If the first vector is doubled, then the resultant is also doubled. The angle between the two vectors is
 (a) 60° (b) 120°
 (c) 90° (d) 180°
- 51** The sum of two vectors $\mathbf{A}$ and $\mathbf{B}$ is at right angles to their difference. Then, the correct relation is
 (a) $A = B$ (b) $A = 2B$
 (c) $B = 2A$ (d) None of these
- 52** If the sum of two unit vectors is a unit vector, then magnitude of difference in two unit vectors is
 (a) $\sqrt{2}$ (b) $\sqrt{3}$ (c) $1/\sqrt{2}$ (d) $\sqrt{5}$
- 53** What is the angle between $\mathbf{P}$ and the cross product of $(\mathbf{P} + \mathbf{Q})$ and $(\mathbf{P} - \mathbf{Q})$?
 (a) 90° (b) $\tan^{-1}(P/Q)$
 (c) $\tan^{-1}(Q/P)$ (d) 0°
- 54** A vector having magnitude 30 unit makes equal angles with each of X, Y and Z -axes. The components of vector along each of X, Y and Z -axes are
 (a) $10\sqrt{3}$ unit (b) $20\sqrt{3}$ unit
 (c) $15\sqrt{3}$ unit (d) 10 unit
- 55** The resultant of two forces, one double the other in magnitude, is perpendicular to the smaller of the two forces. The angle between the two forces is
 (a) 120° (b) 135° (c) 90° (d) 150°
- 56** What is the angle between P and the resultant of $(\mathbf{P} + \mathbf{Q})$ and $(\mathbf{P} - \mathbf{Q})$?
 (a) Zero (b) $\tan^{-1}(P/Q)$
 (c) $\tan^{-1}(Q/P)$ (d) $\tan^{-1}(P - Q)/(P + Q)$
- 57** There are N coplanar vectors each of magnitude V . Each vector is inclined to the preceding vector at angle $\frac{2\pi}{N}$. What is the magnitude of their resultant?
 (a) $\frac{V}{N}$ (b) V
 (c) Zero (d) $\frac{N}{V}$

- 58 At what angle must the two forces $(x + y)$ and $(x - y)$ act, so that the resultant may be $\sqrt{x^2 + y^2}$?

- (a) $\cos^{-1} \left[-\frac{x^2 + y^2}{2(x^2 - y^2)} \right]$
 (b) $\cos^{-1} \left[-\frac{2(x^2 - y^2)}{x^2 + y^2} \right]$
 (c) $\cos^{-1} \left[-\frac{(x^2 + y^2)}{(x^2 - y^2)} \right]$
 (d) $\cos^{-1} \left[-\frac{(x^2 - y^2)}{(x^2 + y^2)} \right]$

- 59 If $\mathbf{a}$ and $\mathbf{b}$ are two vectors, then the value of $(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b})$ is

- (a) $2(\mathbf{b} \times \mathbf{a})$ (b) $-2(\mathbf{b} \times \mathbf{a})$
 (c) $\mathbf{b} \times \mathbf{a}$ (d) $\mathbf{b} \times \mathbf{a}$

- 60 Given that $\mathbf{A} + \mathbf{B} = \mathbf{C}$ and that $\mathbf{C}$ is perpendicular to $\mathbf{A}$. Further, if $|\mathbf{A}| = |\mathbf{C}|$, then what is the angle between $\mathbf{A}$ and $\mathbf{B}$?

- (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
 (c) $\frac{3\pi}{4}$ (d) π

- 61 The resultant of vectors $\mathbf{A}$ and $\mathbf{B}$ is $\mathbf{R}_1$. On reversing the direction of vector $\mathbf{B}$, the resultant becomes $\mathbf{R}_2$. What is the value of $R_1^2 + R_2^2$?

- (a) $A^2 + B^2$ (b) $A^2 - B^2$
 (c) $2(A^2 + B^2)$ (d) $2(A^2 - B^2)$

- 62 Unit vector perpendicular to vector $\mathbf{A} = -3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$ and $\mathbf{B} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 6\hat{\mathbf{k}}$ both is

- (a) $\frac{3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}}{\sqrt{13}}$ (b) $\frac{3\hat{\mathbf{k}} - 2\hat{\mathbf{j}}}{\sqrt{13}}$
 (c) $\frac{-\hat{\mathbf{j}} + 2\hat{\mathbf{k}}}{\sqrt{13}}$ (d) $\frac{\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}}{\sqrt{13}}$

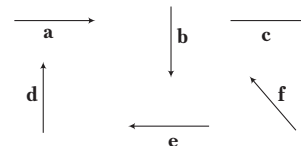
- 63 The velocity of a particle is $\mathbf{v} = 6\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$. The component of the velocity parallel to vector $\mathbf{a} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ in vector form is

- (a) $6\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ (b) $2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$
 (c) $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ (d) $6\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

- 64 If $a\hat{\mathbf{i}} + b\hat{\mathbf{j}}$ is a unit vector and it is perpendicular to $\hat{\mathbf{i}} + \hat{\mathbf{j}}$, then value of a and b is

- (a) 1, 0 (b) -2, 0
 (c) 0.5, -0.5 (d) None of these

- 65 Six vectors have magnitude and direction as indicated in the figure. Which of the following expression is true?

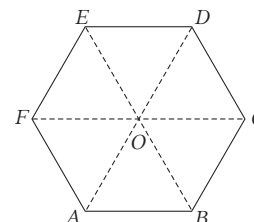


- (a) $\mathbf{b} + \mathbf{e} = \mathbf{f}$ (b) $\mathbf{b} + \mathbf{c} = \mathbf{f}$
 (c) $\mathbf{d} + \mathbf{c} = \mathbf{f}$ (d) $\mathbf{d} + \mathbf{e} = \mathbf{f}$

- 66 The sum of the magnitudes of two forces acting at a point is 18 and the magnitude of their resultant is 12. If the resultant is at 90° with the force of smaller magnitude, what are the magnitudes of forces?
 (a) 12, 6 (b) 14, 4 (c) 5, 13 (d) 10, 8

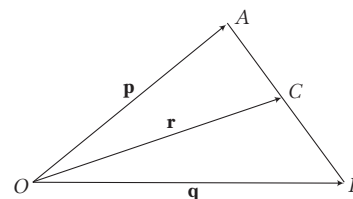
- 67 The resultant of two vectors $\mathbf{P}$ and $\mathbf{Q}$ is $\mathbf{R}$. If $\mathbf{Q}$ is doubled, the new resultant is perpendicular to $\mathbf{P}$. Then, $\mathbf{R}$ equal to
 (a) $\mathbf{P}$ (b) $(\mathbf{P} + \mathbf{Q})$ (c) $\mathbf{Q}$ (d) $(\mathbf{P} - \mathbf{Q})$

- 68 In the figure shown, $ABCDEF$ is a regular hexagon. What is the value of $\mathbf{AB} + \mathbf{AC} + \mathbf{AD} + \mathbf{AE} + \mathbf{AF}$?



- (a) $\mathbf{AO}$ (b) $2\mathbf{AO}$
 (c) $4\mathbf{AO}$ (d) $6\mathbf{AO}$

- 69 Figure shows three vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$, where C is the mid-point of AB . Then, which of the following relation is correct?



- (a) $\mathbf{p} + \mathbf{q} = 2\mathbf{r}$ (b) $\mathbf{p} + \mathbf{q} = \mathbf{r}$ (c) $\mathbf{p} - \mathbf{q} = 2\mathbf{r}$ (d) $\mathbf{p} - \mathbf{q} = \mathbf{r}$

- 70 A vector $\mathbf{a}$ is turned without a change in its length through a small angles $d\theta$. The value of $|\Delta \mathbf{a}|$ and Δa are, respectively.

- (a) 0, $a d\theta$ (b) $a \cdot d\theta$, 0
 (c) 0, 0 (d) None of these

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) These questions consists of two statements each printed as Assertion and Reason. While answering these question you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is true but Reason is false.
- (d) If Assertion is false but Reason is true.

1 Assertion Angle between $\hat{i} + \hat{j}$ and $\hat{i}$ is 45° .

Reason $\hat{i} + \hat{j}$ is equally inclined to both $\hat{i}$ and $\hat{j}$ and the angle between $\hat{i}$ and $\hat{j}$ is 90° .

2 Assertion $(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B})$ is always positive.

Reason This is positive if $|\mathbf{A}| > |\mathbf{B}|$.

3 Assertion $\mathbf{A} \times \mathbf{B}$ is perpendicular to both $\mathbf{A} + \mathbf{B}$ as well as $\mathbf{A} - \mathbf{B}$.

Reason $\mathbf{A} + \mathbf{B}$ as well as $\mathbf{A} - \mathbf{B}$ lie in the plane containing $\mathbf{A}$ and $\mathbf{B}$ while $\mathbf{A} \times \mathbf{B}$ lies perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$.

4 Assertion $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{A})$ is $-A^2 B^2 \sin^2 \theta$.

Here θ is the angle between $\mathbf{A}$ and $\mathbf{B}$.

Reason $(\mathbf{A} \times \mathbf{B})$ and $(\mathbf{B} \times \mathbf{A})$ are two anti-parallel vectors provided $\mathbf{A}$ and $\mathbf{B}$ are neither parallel nor anti-parallel.

5 Assertion If $|\mathbf{A}| = |\mathbf{B}|$, then $(\mathbf{A} + \mathbf{B})$, $(\mathbf{A} - \mathbf{B})$ and $(\mathbf{A} \times \mathbf{B})$ are three mutually perpendicular vectors.

Reason Dot product of a null vector with any other vector is always zero.

Statement based questions

1 Which of the following statement is true?

- (a) When the coordinate axes are translated, the component of a vector in a plane changes.
- (b) When the coordinate axes are rotated through some angle, components of the vector change but the vector's magnitude remains constant.
- (c) Sum of $\mathbf{a}$ and $\mathbf{b}$ is $\mathbf{R}$. If the magnitude of $\mathbf{a}$ alone is increased angle between $\mathbf{b}$ and $\mathbf{R}$ decreases.
- (d) The cross product of $3\hat{i}$ and $4\hat{j}$ is 12.

2 Which one of the following statement is true?

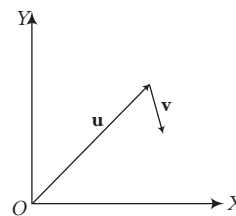
[NCERT Exemplar]

- (a) A scalar quantity is the one that is conserved in a process.
- (b) A scalar quantity is the one that can never take negative values.
- (c) A scalar quantity is the one that does not vary from one point to another in space.
- (d) A scalar quantity has the same value for observers with different orientation of the axes.

3 Figure shows the orientation of two vectors $\mathbf{u}$ and $\mathbf{v}$ in the XY -plane.

If $\mathbf{u} = a\hat{i} + b\hat{j}$ and $\mathbf{v} = p\hat{i} + q\hat{j}$, then which of the following statement is correct?

[NCERT Exemplar]



- (a) a and p are positive while b and q are negative.
- (b) a , p and b are positive while q is negative.
- (c) a , q and b are positive while p is negative.
- (d) a , b , p and q are all positive.

4 Two unit vectors when added give a unit vector. Then, choose the correct statement.

- (a) Magnitude of their difference is $\sqrt{3}$.
- (b) Magnitude of their difference is 1.
- (c) Angle between the vectors is 90° .
- (d) Angle between the sum and the difference of the two vectors is 180° .

5 I. Displacing a vector parallel to itself leaves the vector unchanged.

II. Three equal vectors cannot add upto zero.

Which of the following statement(s) is/are correct?

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

6 Unit vector

I. has dimensions and a unit.

II. when multiplied by a scalar quantity, it results a scalar.

Which of the following statement(s) is/are incorrect?

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

Match the columns

- 1** Vector **A** is pointing eastwards and vector **B** northwards. If $|\mathbf{A}| = |\mathbf{B}|$, then match the following two columns and mark the correct option from the codes given below.

Column I	Column II
(A) $(\mathbf{A} + \mathbf{B})$	(p) North-east
(B) $(\mathbf{A} - \mathbf{B})$	(q) Vertically upwards
(C) $(\mathbf{A} \times \mathbf{B})$	(r) Vertically downwards
(D) $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{A} \times \mathbf{B})$	(s) None

Codes

A	B	C	D	A	B	C	D
(a) p	r	s	q	(b) p	s	q	s
(c) q	r	p	s	(d) q	p	s	r

- 2** A vector has a magnitude x . If it is rotated by an angle θ , then magnitude of change in vector is nx . Match the following two columns and mark the correct option from the codes given below.

Column I	Column II
(A) $\theta = 60^\circ$	(p) $n = \sqrt{3}$
(B) $\theta = 90^\circ$	(q) $n = 1$
(C) $\theta = 120^\circ$	(r) $n = \sqrt{2}$
(D) $\theta = 180^\circ$	(s) $n = 2$

Codes

A	B	C	D
(a) p	r	q	s
(b) r	s	p	q
(c) r	q	s	p
(d) q	r	p	s

- 3** Two vectors **A** and **B** have equal magnitude x . Angle between them is 60° . Then, match the following two columns and mark the correct option from the codes given below.

Column I	Column II
(A) $ \mathbf{A} + \mathbf{B} $	(p) $\frac{\sqrt{3}}{2}x^2$
(B) $ \mathbf{A} - \mathbf{B} $	(q) x
(C) $\mathbf{A} \cdot \mathbf{B}$	(r) $\sqrt{3}x$
(D) $ \mathbf{A} \times \mathbf{B} $	(s) None

Codes

A	B	C	D
(a) p	s	q	r
(b) p	r	q	s
(c) r	q	s	p
(d) r	s	p	q

(C) Medical entrances' gallery

Collection of questions asked in NEET and various medical entrance exams

- 1** If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is [NEET 2016]
(a) 90° (b) 45° (c) 180° (d) 0°
- 2** If a vector $2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 8\hat{\mathbf{k}}$ is perpendicular to the vector $4\hat{\mathbf{j}} - 4\hat{\mathbf{i}} + \alpha\hat{\mathbf{k}}$, then the value of α is [Manipal 2015]
(a) $1/2$ (b) -1 (c) $-1/2$ (d) 1
- 3** The angle θ between the vector $\mathbf{p} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and unit vector along X-axis is [MHT CET 2014]
(a) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$
(c) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (d) $\cos^{-1}\left(\frac{1}{2}\right)$
- 4** Consider three vectors $\mathbf{A} = \hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, $\mathbf{B} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{C} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$. A vector $\mathbf{X}$ of the form $\alpha\mathbf{A} + \beta\mathbf{B}$ (α and β are numbers) is perpendicular to $\mathbf{C}$. The ratio of α and β is [EAMCET 2014]
(a) $1 : 1$ (b) $2 : 1$
(c) $-1 : 1$ (d) $3 : 1$
- 5** Two equal vectors have a resultant equal to either of the two. The angle between them is [UK PMT 2014]
(a) 90° (b) 60°
(c) 120° (d) 0°
- 6** Which of the following not a vector quantity? [KCET 2014]
(a) Weight
(b) Nuclear spin
(c) Momentum
(d) Potential energy
- 7** The scalar product of two vectors $\mathbf{A} = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$ and $\mathbf{B} = -\hat{\mathbf{j}} + \hat{\mathbf{k}}$ is given by [J&K CET 2013]
(a) $\mathbf{A} \cdot \mathbf{B} = 3$ (b) $\mathbf{A} \cdot \mathbf{B} = 4$
(c) $\mathbf{A} \cdot \mathbf{B} = -4$ (d) $\mathbf{A} \cdot \mathbf{B} = -3$
- 8** If $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \times \mathbf{B}$, then angle between $\mathbf{A}$ and $\mathbf{B}$ is [AMFC 2012]
(a) 45° (b) 30°
(c) 60° (d) 90°

- 9 If a vector **A** having a magnitude of 8 is added to a vector **B** which lies along *X*-axis, then the resultant of two vectors lies along *Y*-axis and has magnitude twice that of **B**. The magnitude of **B** is [JCECE 2012]

(a) $\frac{6}{\sqrt{5}}$ (b) $\frac{12}{\sqrt{5}}$ (c) $\frac{16}{\sqrt{5}}$ (d) $\frac{8}{\sqrt{5}}$

- 10 The value of λ for which two vectors $\mathbf{a} = 5\hat{\mathbf{i}} + \lambda\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\mathbf{b} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$ are perpendicular to each other is [WB JEE 2011]

(a) 2 (b) -2
(c) 3 (d) -3

ANSWERS

• CHECK POINT 2.1

1. (d) 2. (a) 3. (c) 4. (c) 5. (a) 6. (c) 7. (c) 8. (b)

• CHECK POINT 2.2

1. (a) 2. (b) 3. (c) 4. (b) 5. (b) 6. (b) 7. (a) 8. (a) 9. (b) 10. (b)
11. (d) 12. (d)

• CHECK POINT 2.3

1. (a) 2. (b) 3. (d) 4. (d) 5. (a) 6. (c) 7. (a) 8. (d)

(A) Taking it together

1. (c)	2. (d)	3. (c)	4. (a)	5. (b)	6. (c)	7. (a)	8. (b)	9. (a)	10. (c)
11. (c)	12. (d)	13. (b)	14. (c)	15. (c)	16. (a)	17. (c)	18. (d)	19. (b)	20. (d)
21. (d)	22. (c)	23. (b)	24. (d)	25. (b)	26. (a)	27. (c)	28. (b)	29. (b)	30. (b)
31. (c)	32. (a)	33. (b)	34. (a)	35. (b)	36. (d)	37. (b)	38. (b)	39. (c)	40. (a)
41. (b)	42. (b)	43. (a)	44. (a)	45. (b)	46. (a)	47. (d)	48. (c)	49. (b)	50. (b)
51. (a)	52. (b)	53. (a)	54. (a)	55. (a)	56. (a)	57. (c)	58. (a)	59. (a)	60. (c)
61. (c)	62. (a)	63. (b)	64. (d)	65. (d)	66. (c)	67. (c)	68. (d)	69. (a)	70. (b)

(B) Medical entrance special format questions

• Assertion and reason

1. (b) 2. (d) 3. (a) 4. (b) 5. (b)

• Statement based questions

1. (b) 2. (d) 3. (b) 4. (a) 5. (a) 6. (c)

• Match the columns

1. (b) 2. (d) 3. (c)

(C) Medical entrances' gallery

1. (a) 2. (c) 3. (a) 4. (a) 5. (c) 6. (d) 7. (d) 8. (a) 9. (d) 10. (c)

Hints & Explanations

• CHECK POINT 2.1

- 1 (d) Electric flux is a scalar quantity.
- 2 (a) Surface area has magnitude only. Therefore, it is a scalar quantity.
- 3 (c) Pressure, temperature and work are scalar quantities while momentum is a vector quantity.
- 4 (c) When a vector is multiplied by zero, it results into zero vector, *i.e.* a vector having zero magnitude. It is written as **0**.

$$5 (a) (\hat{i} - 3\hat{j} + 2\hat{k}) + (3\hat{i} + 6\hat{j} - 7\hat{k}) + \mathbf{r} = \hat{j} \\ \Rightarrow \mathbf{r} = -4\hat{i} - 2\hat{j} + 5\hat{k}$$

- 6 (c) Let the given vector be **A**, where

$$\mathbf{A} = \hat{i} + \hat{j} \\ \therefore |\mathbf{A}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$$

- 7 (c) Let $\mathbf{A} = \hat{i} + \hat{j}$

$$|\mathbf{A}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2} \\ \therefore \text{Unit vector along } \mathbf{A} \text{ will be } \hat{\mathbf{A}} \text{ is given by}$$

$$\hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

- 8 (b) When a vector is multiplied by a negative scalar number (*i.e.* -2), then its magnitude gets doubled and direction gets reversed.

• CHECK POINT 2.2

- 1 (a) Resultant of two vectors will be maximum when they are parallel, *i.e.* angle between them is zero.
- 2 (b) Minimum three vectors of unequal magnitude are required to make vector sum equal to zero.
- 3 (c) Maximum resultant, $R_{\max} = |\mathbf{A} + \mathbf{B}| = 18$
Minimum resultant, $R_{\min} = |\mathbf{A} - \mathbf{B}| = 2$
- 4 (b) If $\mathbf{P} + \mathbf{Q} + \mathbf{R} = 0$, then $|\mathbf{P} + \mathbf{Q}| = |\mathbf{R}|$.

$$5 (b) \text{ Given, } \mathbf{P} = \hat{i} - \hat{j} + \hat{k}$$

$$\text{and } \mathbf{P} + \mathbf{Q} = \hat{i} \Rightarrow \mathbf{Q} = \hat{i} - \hat{i} + \hat{j} - \hat{k} = \hat{j} - \hat{k}$$

$$6 (b) \text{ Given, } \mathbf{A} = 2\hat{i} + \hat{j}, \mathbf{B} = 3\hat{j} - \hat{k} \text{ and } \mathbf{C} = 6\hat{i} - 2\hat{k}$$

$$\therefore \mathbf{A} - 2\mathbf{B} + 3\mathbf{C} = (2\hat{i} + \hat{j}) - 2(3\hat{j} - \hat{k}) + 3(6\hat{i} - 2\hat{k}) \\ = 2\hat{i} + \hat{j} - 6\hat{j} + 2\hat{k} + 18\hat{i} - 6\hat{k} \\ = 20\hat{i} - 5\hat{j} - 4\hat{k}$$

$$7 (a) P\sqrt{10} = \sqrt{4P^2 + 2P^2 + 4\sqrt{2}P^2 \cos \theta}$$

$$\therefore \theta = 45^\circ$$

$$8 (a) |\mathbf{A} + \mathbf{C}| = \sqrt{A^2 + A^2 + 2A^2 \cos 120^\circ} \\ = A$$

$$\text{Now, } |\mathbf{A} + \mathbf{B} + \mathbf{C}| = A + A = 2A$$

- 9 (b) If $\mathbf{P} + \mathbf{Q} = \mathbf{P} - \mathbf{Q}$, then $\mathbf{Q} = 0$ (a null vector).

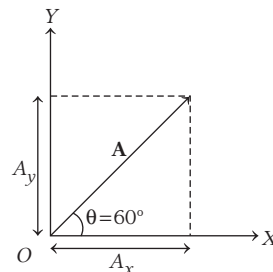
- 10 (b) Resultant of two vectors **A** and **B**,

$$R = \{|\mathbf{A}| - |\mathbf{B}|\}$$

$$\text{then } \theta = 180^\circ$$

$$11 (d) |\mathbf{A} - \mathbf{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta} \\ = \sqrt{4 + 16 - 2 \times 2 \times 4 \times \frac{1}{2}} \\ = \sqrt{12} = 2\sqrt{3}$$

- 12 (d) Let the vector be **A** as shown below.



$$\tan \theta = \frac{A_y}{A_x} \text{ or } A_y = A_x \tan \theta$$

$$A_y = 50 \tan 60^\circ = 50\sqrt{3} = 86.6 \text{ N} \\ \approx 87 \text{ N}$$

• CHECK POINT 2.3

$$1 (a) AB \sin \theta = \frac{1}{\sqrt{3}} AB \cos \theta$$

$$\therefore \tan \theta = \frac{1}{\sqrt{3}} \text{ or } \theta = 30^\circ = \frac{\pi}{6}$$

- 2 (b) Let the two vectors be **A** and **B**.

$$\text{We know, } \mathbf{A} \cdot \mathbf{B} = AB \cos \theta \\ = (3)(5) \cos 60^\circ = 7.5$$

- 3 (d) As the multiple of $\hat{j}$ in the given vector is zero, therefore this vector lies in *XZ*-plane and projection of this vector on *Y*-axis is zero.

- 4 (d) Given, $\mathbf{A} \cdot \mathbf{B} = 0$

$$\therefore \mathbf{A} \perp \mathbf{B}$$

$$\mathbf{A} \cdot \mathbf{C} = 0$$

$$\therefore \mathbf{A} \perp \mathbf{C}$$

As, $\mathbf{B} \times \mathbf{C}$ is $\perp$ to both **B** and **C**.

So, $\mathbf{B} \times \mathbf{C}$ is parallel to **A**.

5 (a) $(\mathbf{a} \cdot \mathbf{b})^2 = a^2 b^2 \cos^2 \theta = a^2 b^2$ (Given)

$$\therefore \theta = 0^\circ$$

$\Rightarrow \mathbf{a}$ is parallel to $\mathbf{b}$.

6 (c) We have, $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$

Here, $\mathbf{A} \cdot \mathbf{B} = -AB$, i.e. $\cos \theta = -1$

$$\therefore \theta = 180^\circ$$

So, $\mathbf{A}$ and $\mathbf{B}$ acts in the opposite direction.

7 (a) We have, $\mathbf{A} \times \mathbf{B} = AB \sin \theta$

$$8 = (2)(5) \sin \theta$$

$$\therefore \sin \theta = 4/5 \text{ or } \cos \theta = 3/5$$

$$\therefore \mathbf{A} \cdot \mathbf{B} = AB \cos \theta$$

$$= (5)(2)(3/5) = 6$$

8 (d) $AB \sin \theta = \sqrt{3} AB \cos \theta$ or $\tan \theta = \sqrt{3}$

$$\therefore \theta = 60^\circ$$

$$\text{Now, } |\mathbf{A} + \mathbf{B}| = \sqrt{A^2 + B^2 + 2AB \cos 60^\circ} \\ = \sqrt{A^2 + B^2 + AB}$$

(A) Taking it together

1 (c) Potential energy is a scalar quantity.

2 (d) Displacement has both direction as well as magnitude. So, it is a vector quantity.

3 (c) When a vector is added to another vector of similar nature having equal magnitude and opposite sign, it forms a null vector.

4 (a) The component of a vector is always less than its magnitude along any other direction.

5 (b) Unit vector has a magnitude equal to 1.

$$\text{Here, } \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$$

$\therefore$ Option (b) is the correct answer.

6 (c) Since, vector $\mathbf{P}$ is perpendicular to the vector $\mathbf{Q}$.

$$\therefore \mathbf{P} \cdot \mathbf{Q} = 0 \Rightarrow (3\hat{i} - 2\hat{j} + a\hat{k}) \cdot (2\hat{i} + \hat{j} - \hat{k}) = 0$$

$$\Rightarrow 6 - 2 - a = 0 \Rightarrow a = 4$$

7 (a) $\mathbf{A} - \mathbf{B} = (\hat{i} + \hat{j} + \hat{k}) - (-\hat{i} - \hat{j} - \hat{k})$

$$= 2\hat{i} + 2\hat{j} + 2\hat{k} = 2\mathbf{A}$$

i.e. $\mathbf{A} - \mathbf{B}$ and $\mathbf{A}$ are parallel.

Hence, $\theta = 0^\circ$

8 (b) Let $\mathbf{A} = -2\hat{i} + 3\hat{j} + \hat{k}$

$$\mathbf{B} = \hat{i} + 2\hat{j} - 4\hat{k}$$

$$\therefore \mathbf{A} \cdot \mathbf{B} = (-2\hat{i} + 3\hat{j} + \hat{k}) \cdot (\hat{i} + 2\hat{j} - 4\hat{k})$$

$$= -2 + 6 - 4 = 0$$

Since, dot product of two vectors is zero, so vectors will be perpendicular (i.e. $\theta = 90^\circ$) to each other.

9 (a) Component of $\mathbf{A}$ along $\mathbf{B} = A \cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{B} = \frac{2+3}{\sqrt{1+1}} = \frac{5}{\sqrt{2}}$

10 (c) $\mathbf{P} = 6\hat{i} + 4\sqrt{2}\hat{j} + 4\sqrt{2}\hat{k}$

Let $\mathbf{P}$ makes angle γ with Z -axis.

$$\therefore \gamma = \cos^{-1} \left(\frac{P_z}{P} \right)$$

$$\text{Here, } P_z = 4\sqrt{2}$$

$$\text{and } P = \sqrt{(6)^2 + (4\sqrt{2})^2 + (4\sqrt{2})^2} = 10$$

$$\therefore \gamma = \cos^{-1} \left(\frac{4\sqrt{2}}{10} \right) = \cos^{-1} \left(\frac{2\sqrt{2}}{5} \right)$$

11 (c) Unit vector perpendicular to $\mathbf{A}$ and $\mathbf{B}$,

$$\hat{\mathbf{n}} = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} = \frac{\mathbf{A} \times \mathbf{B}}{AB \sin \theta}$$

12 (d) Their cross product should be zero.

$$(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = 0$$

$$\therefore 2(\mathbf{a} \times \mathbf{b}) = 0$$

So, $\mathbf{a}$ is parallel to $\mathbf{b}$.

14 (c) $\mathbf{F}$ lies in XY -plane. Hence, $7\hat{\mathbf{k}}$, perpendicular to XY -plane is also perpendicular to $\mathbf{F}$.

15 (c) Resultant of two vectors lies between $|\mathbf{A} + \mathbf{B}|$ and $|\mathbf{A} - \mathbf{B}|$.

$$\text{i.e. } |A - B| < C < |A + B|$$

16 (a) Only the magnitude will remain same.

17 (c) $(\mathbf{a} - \mathbf{b})$ is nothing but addition of $\mathbf{a}$ and $-\mathbf{b}$.

$$|\mathbf{a}| = |\mathbf{a} - \mathbf{b} + \mathbf{b}| \leq |\mathbf{a} - \mathbf{b}| + |\mathbf{b}|$$

$$\text{or } |\mathbf{a}| - |\mathbf{b}| \leq |\mathbf{a} - \mathbf{b}|$$

Hence, option (c) is correct.

19 (b) The angle between vectors $(\mathbf{A} \times \mathbf{B})$ and $(\mathbf{B} \times \mathbf{A})$ is 180° or π .

20 (d) Resultant of two given vectors will be

$$\sqrt{(8)^2 + (8)^2} = 8\sqrt{2}$$

$$\therefore \text{Unit vector parallel to the given vectors will be, } \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

21 (d) Area = $|\mathbf{A} \times \mathbf{B}|$

$$\text{Here, } \mathbf{A} \times \mathbf{B} = (2\hat{i} + 3\hat{j}) \times (\hat{i} + 4\hat{j})$$

$$= 8\hat{\mathbf{k}} - 3\hat{\mathbf{k}} = 5\hat{\mathbf{k}}$$

$$\therefore \text{Area} = |\mathbf{A} \times \mathbf{B}| = 5 \text{ units}$$

22 (c) In XY -plane, vector is $3\hat{i} + \hat{j}$

$$\therefore \text{Length in } XY\text{-plane} = \sqrt{9+1} = \sqrt{10}$$

23 (b) $\mathbf{P} \times \mathbf{Q}$ is perpendicular to the plane formed by $\mathbf{P}$ and $\mathbf{Q}$.

$\mathbf{P} + \mathbf{Q}$ lies in this plane. Hence, $\mathbf{P} + \mathbf{Q}$ is perpendicular to $\mathbf{P} \times \mathbf{Q}$.

24 (d) Diagonal vector, $\mathbf{A} = b\hat{i} + b\hat{j} + b\hat{k}$

$$\text{or } A = \sqrt{b^2 + b^2 + b^2} = \sqrt{3}b$$

$$\therefore \mathbf{A} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$$

- 25** (b) Given, $\mathbf{A} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$ and $\mathbf{B} = \hat{\mathbf{i}} - \hat{\mathbf{j}}$

We know that, $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta$

$$\Rightarrow (\hat{\mathbf{i}} + \hat{\mathbf{j}}) \cdot (\hat{\mathbf{i}} - \hat{\mathbf{j}}) = (\sqrt{1+1})(\sqrt{1+1}) \cos \theta$$

where, θ is the angle between $\mathbf{A}$ and $\mathbf{B}$.

$$\Rightarrow \cos \theta = \frac{1-0+0-1}{\sqrt{2}\sqrt{2}} = \frac{0}{2} = 0 \Rightarrow \theta = 90^\circ$$

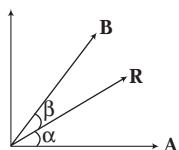
- 26** (a) If three vectors form a triangle, then the resultant of these vectors will be zero when the sum of two smaller sides of triangle is greater than its third side. This is only possible in option (a).

- 27** (c) Cross product of two perpendicular vector should be zero.

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -1 & 5 \\ 1 & 0 & 0 \end{vmatrix} = -(0-5)\hat{\mathbf{j}} + (0)\hat{\mathbf{k}} = 5\hat{\mathbf{j}} + \hat{\mathbf{k}}$$

The vector perpendicular to both vectors is $5\hat{\mathbf{j}} + \hat{\mathbf{k}}$.

- 28** (b) Consider the figure,



Resultant is inclined towards a vector having greater magnitude.

- 29** (b) $\hat{\mathbf{j}} = (\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) + (6\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 7\hat{\mathbf{k}}) + \mathbf{C}$

$$\text{Hence, } \mathbf{C} = -7\hat{\mathbf{i}} + 4\hat{\mathbf{k}}$$

- 30** (b) For a unit vector, $\sqrt{(0.5)^2 + (0.8)^2 + c^2} = 1$

On solving, we get $c = \sqrt{0.11}$

- 31** (c) Component of $\mathbf{A}$ along $\mathbf{B} = \frac{\mathbf{A} \cdot \mathbf{B}}{B} = \frac{4+12}{\sqrt{(2)^2 + (4)^2}} = \frac{16}{2\sqrt{5}} = \frac{8}{\sqrt{5}}$

- 32** (a) $F \cdot F \cos \theta = F \cdot F \sin \theta$ or $\tan \theta = 1$ or $\theta = 45^\circ$

$$|\mathbf{F}_1 + \mathbf{F}_2| = \sqrt{F^2 + F^2 + 2F \cdot F \cos 45^\circ} = (\sqrt{2} + \sqrt{2})F$$

- 33** (b) $\mathbf{B} \times \mathbf{A}$ is perpendicular to $\mathbf{A}$. Hence, $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{A})$ will be zero.

- 34** (a) Since, $\hat{\mathbf{A}}$ is a unit vector in a given direction. It should be a constant unit vector.

$$\text{or } \hat{\mathbf{A}} \cdot \frac{d\hat{\mathbf{A}}}{dt} = 0$$

- 35** (b) $R_{\text{net}} = R + \sqrt{R^2 + R^2} = R + \sqrt{2}R = R(\sqrt{2} + 1)$

- 36** (d) $|\mathbf{A}| = \sqrt{3^2 + 4^2} = 5$

$$\text{Desired vector, } \mathbf{r} = |\mathbf{B}| \hat{\mathbf{A}} = (\sqrt{7^2 + 24^2}) \times \frac{3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}}{5} = 15\hat{\mathbf{i}} + 20\hat{\mathbf{j}}$$

- 37** (b) Dot product of two perpendicular vectors should be zero.

$$\therefore \mathbf{B} = \hat{\mathbf{i}}B \sin \theta - \hat{\mathbf{j}}B \cos \theta$$

$$[\therefore (\hat{\mathbf{i}}A \cos \theta + \hat{\mathbf{j}}A \sin \theta) \cdot (\hat{\mathbf{i}}B \sin \theta - \hat{\mathbf{j}}B \cos \theta) = 0]$$

- 38** (b) The coefficients of $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ should be in constant ratio.

$$\text{or } \frac{2}{-4} = \frac{3}{-6} = \frac{1}{-\lambda} \text{ or } \lambda = 2$$

- 39** (c) Resultant, $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$

$$3 = \sqrt{(\sqrt{3})^2 + (\sqrt{3})^2 + 2(\sqrt{3})(\sqrt{3}) \cos \theta}$$

Squaring both sides in the above equation, we get

$$9 = 3 + 3 + 6 \cos \theta$$

$$\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ \text{ or } \frac{\pi}{3}$$

- 40** (a) Magnitude of vector, $|\mathbf{A}| = \sqrt{(3)^2 + (6)^2 + (2)^2} = 7$

$$\alpha = \cos^{-1}\left(\frac{3}{7}\right), \beta = \cos^{-1}\left(\frac{6}{7}\right), \gamma = \cos^{-1}\left(\frac{2}{7}\right)$$

- 41** (b) Resultant of two vectors $A_R = 2A \cos \frac{\theta}{2} = 2A \cos 45^\circ = \sqrt{2}A$, at $\theta = 90^\circ$.

- 42** (b) Let $\mathbf{r}$ makes an angle θ with positive X -axis, so component of $\mathbf{r}$ along X -axis,

$$\begin{aligned} r_X &= |\mathbf{r}| \cos \theta \\ (r_X)_{\text{maximum}} &= |\mathbf{r}| (\cos \theta)_{\text{maximum}} \\ &= |\mathbf{r}| \cos 0^\circ (\because \cos \theta \text{ is maximum, if } \theta = 0^\circ) \\ &= |\mathbf{r}| \end{aligned}$$

As $\theta = 0^\circ$

i.e. $\mathbf{r}$ is along positive X -axis.

- 43** (a) Given, $\mathbf{A} \cdot \mathbf{B} = 0$

$$\Rightarrow \mathbf{A} \perp \mathbf{B} \Rightarrow \mathbf{A} \times \mathbf{B} = 1$$

$$AB \sin \theta = 1 \Rightarrow AB \sin 90^\circ = 1$$

$$\text{or } AB = 1 \Rightarrow A = 1 \text{ and } B = 1$$

So, A and B are perpendicular unit vectors.

- 44** (a) We have, $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$

$$\therefore \mathbf{a} + \mathbf{c} = -\mathbf{b}$$

$$\text{or } (\mathbf{a} + \mathbf{c}) \times \mathbf{b} = -\mathbf{b} \times \mathbf{b} = 0$$

$$\Rightarrow (\mathbf{a} \times \mathbf{b}) + (\mathbf{c} \times \mathbf{b}) = 0$$

$$\text{Hence, } \mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c}$$

- 45** (b) $\mathbf{A} + \mathbf{B} = 10\hat{\mathbf{i}} + 5\hat{\mathbf{j}}$

$$\therefore |\mathbf{A} + \mathbf{B}| = \sqrt{100 + 25} = 5\sqrt{5}$$

Angle of $\mathbf{A} + \mathbf{B}$ with X -axis,

$$\theta = \tan^{-1}\left(\frac{5}{10}\right) = \tan^{-1}\left(\frac{1}{2}\right)$$

- 46** (a) We have, $\mathbf{A} - \mathbf{B} = \hat{\mathbf{j}} - 2\hat{\mathbf{k}} = \mathbf{C}$

$$C = \sqrt{1+4} = \sqrt{5}$$

$$\cos \alpha = \frac{0}{\sqrt{5}} = 0, \cos \beta = \frac{1}{\sqrt{5}} \text{ and } \cos \gamma = \frac{-2}{\sqrt{5}}$$

- 47** (d) Here, $\mathbf{A} + \mathbf{B} = \mathbf{C}$ and $\mathbf{A}^2 + \mathbf{B}^2 = \mathbf{C}^2$

$$\therefore (\mathbf{A} + \mathbf{B})^2 = (\mathbf{C})^2$$

$$|\mathbf{A}|^2 + |\mathbf{B}|^2 + 2|\mathbf{A}||\mathbf{B}| \cos \theta = |\mathbf{C}|^2$$

$$|\mathbf{C}|^2 + 2|\mathbf{A}||\mathbf{B}|\cos\theta = |\mathbf{C}|^2$$

$$\therefore 2|\mathbf{A}||\mathbf{B}|\cos\theta = 0$$

$$\Rightarrow \cos\theta = 0$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

$$48 \text{ (c) } |\hat{\mathbf{A}} - \hat{\mathbf{B}}| = \sqrt{1 + 1 - 2 \times 1 \times 1 \times \cos\theta} = 2\sin\frac{\theta}{2}$$

Angle between $\hat{\mathbf{A}}$ and $\hat{\mathbf{B}}$ is also θ , but their magnitudes are 1.

$$49 \text{ (b) } |\mathbf{A}| = \sqrt{9 + 16} = 5 \text{ and } |\mathbf{B}| = \sqrt{36 + 64} = 10$$

$$\Rightarrow \frac{|\mathbf{A}|}{|\mathbf{B}|} = \frac{1}{2}$$

$$50 \text{ (b) Resultant, } R = \sqrt{9\mathbf{P}^2 + 4\mathbf{P}^2 + 12\mathbf{P}^2 \cos\theta} \quad \dots(i)$$

$$2R = \sqrt{36\mathbf{P}^2 + 4\mathbf{P}^2 + 24\mathbf{P}^2 \cos\theta} \quad \dots(ii)$$

Equating Eqs. (i) and (ii), we get

$$\cos\theta = -\frac{1}{2}$$

$$\therefore \theta = 120^\circ$$

$$51 \text{ (a) } \mathbf{R} \cdot \mathbf{S} = (\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B})$$

$$= \mathbf{A} \cdot \mathbf{A} - \mathbf{A} \cdot \mathbf{B} + \mathbf{B} \cdot \mathbf{A} - \mathbf{B} \cdot \mathbf{B}$$

$$\text{or } 0 = A^2 - B^2 \quad (\because \mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A})$$

$$\therefore A^2 = B^2 \text{ or } A = B$$

52 (b) Sum of two unit vectors is a unit vector, means angle between those two unit vectors is 120° .

$$\therefore \text{Difference, } |\mathbf{S}| = \sqrt{1 + 1 - 2 \times 1 \times 1 \times \cos 120^\circ} = \sqrt{3}$$

53 (a) Cross product of $(\mathbf{P} + \mathbf{Q})$ and $(\mathbf{P} - \mathbf{Q})$ is perpendicular to the plane formed by $(\mathbf{P} + \mathbf{Q})$ and $(\mathbf{P} - \mathbf{Q})$ or $\mathbf{P}$ (or $\mathbf{Q}$).

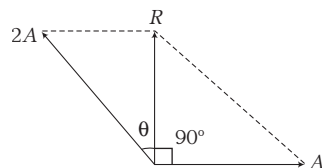
$$54 \text{ (a) We have, } A_x = A_y = A_z$$

$$\text{Resultant, } A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$= \sqrt{3} A_x$$

$$\therefore A_x = \frac{A}{\sqrt{3}} = \frac{30}{\sqrt{3}} = 10\sqrt{3} \text{ unit}$$

$$55 \text{ (a) We have, } \tan 90^\circ = \frac{2A \sin\theta}{A + 2A \cos\theta} = \infty$$



$$\therefore A + 2A \cos\theta = 0$$

$$\text{or } \cos\theta = -\frac{1}{2} \text{ or } \theta = 120^\circ$$

56 (a) Resultant of $(\mathbf{P} + \mathbf{Q})$ and $(\mathbf{P} - \mathbf{Q})$ is $\mathbf{P} + \mathbf{Q} + \mathbf{P} - \mathbf{Q}$ or $2\mathbf{P}$ which is parallel to $\mathbf{P}$.

So, angle between $\mathbf{P}$ and $2\mathbf{P}$ will be zero.

57 (c) Since, each of N coplanar vector is inclined at $\frac{2\pi}{N}$ to the preceding vector. Hence, they will form a closed polygon. Therefore, their resultant must be equals to zero.

58 (a) Let the two forces be A and B .

$$\text{Resultant, } R^2 = A^2 + B^2 + 2AB \cos\theta \quad \dots(i)$$

$$\text{Substituting, } A = (x + y), B = (x - y)$$

$$\text{and } R = \sqrt{x^2 + y^2}$$

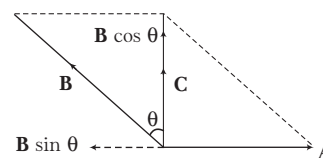
On substituting these values in Eq. (i), we get

$$\theta = \cos^{-1} \left[-\frac{(x^2 + y^2)}{2(x^2 - y^2)} \right]$$

$$59 \text{ (a) } (\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = \mathbf{a} \times \mathbf{a} - \mathbf{a} \times \mathbf{b} + \mathbf{b} \times \mathbf{a} - \mathbf{b} \times \mathbf{b} \\ = 2(\mathbf{b} \times \mathbf{a})$$

$$60 \text{ (c) From the figure, } \mathbf{B} \cos\theta = \mathbf{C} \quad \dots(i)$$

$$\text{and } \mathbf{B} \sin\theta = \mathbf{A} \quad \dots(ii)$$



$$\text{Given, } |\mathbf{A}| = |\mathbf{C}|$$

$$\therefore |\mathbf{B} \cos\theta| = |\mathbf{B} \sin\theta| \text{ or } \theta = 45^\circ$$

$$\text{Hence, angle between } \mathbf{A} \text{ and } \mathbf{B} \text{ is } 135^\circ \text{ or } \frac{3\pi}{4}.$$

$$61. \text{ (c) We have, } R_1^2 = A^2 + B^2 + 2AB \cos\theta$$

$$\text{and } R_2^2 = A^2 + B^2 - 2AB \cos\theta$$

$$\therefore R_1^2 + R_2^2 = 2(A^2 + B^2)$$

$$62 \text{ (a) Unit vector is given by } \hat{\mathbf{n}} = \frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|}$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -3 & -2 & -3 \\ 2 & 4 & 6 \end{vmatrix}$$

$$= \hat{\mathbf{i}}(-12 + 12) - \hat{\mathbf{j}}(-18 + 6) + \hat{\mathbf{k}}(-12 + 4) = 12\hat{\mathbf{j}} - 8\hat{\mathbf{k}}$$

$$|\mathbf{A} \times \mathbf{B}| = \sqrt{(12)^2 + (-8)^2} = \sqrt{208} = 4\sqrt{13}$$

$$\therefore \hat{\mathbf{n}} = \frac{12\hat{\mathbf{j}} - 8\hat{\mathbf{k}}}{4\sqrt{13}} = \frac{3\hat{\mathbf{j}} - 2\hat{\mathbf{k}}}{\sqrt{13}}$$

63 (b) Magnitude of component of $\mathbf{v}$ along $\mathbf{a}$

$$= \frac{\mathbf{v} \cdot \mathbf{a}}{a} = \frac{6 + 2 - 2}{\sqrt{3}} = 2\sqrt{3}$$

$$\text{Now, } \hat{\mathbf{a}} = \frac{\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}}{\sqrt{3}}$$

$\therefore$ Component of velocity in vector form

$$= 2\sqrt{3}\hat{\mathbf{a}} = (2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$$

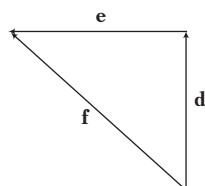
64 (d) $(a\hat{i} + b\hat{j}) \cdot (\hat{i} + \hat{j}) = 0$ or $a + b = 0$... (i)

Further, $\sqrt{a^2 + b^2} = 1$ or $a^2 + b^2 = 1$... (ii)

Solving Eqs. (i) and (ii), we get

$$a = \pm \frac{1}{\sqrt{2}} \quad \text{and} \quad b = \pm \frac{1}{\sqrt{2}}$$

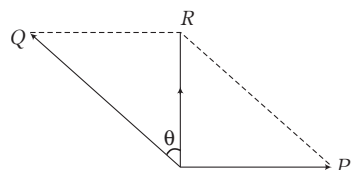
65 (d) On arranging vectors,



$$\mathbf{f} = \mathbf{d} + \mathbf{e}$$

66 (c) We have, $P + Q = 18$... (i)

and $R = 12$



$$Q \sin \theta = P \quad \text{and} \quad Q \cos \theta = R$$

Squaring and adding, we get

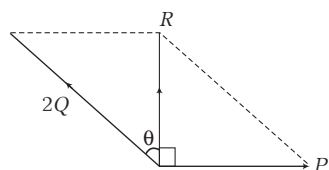
$$P^2 + R^2 = Q^2$$

or $Q^2 - P^2 = R^2 = 144$... (ii)

Solving Eqs. (i) and (ii), we get

$$P = 5 \quad \text{and} \quad Q = 13.$$

67 (c) We have, $2Q \sin \theta = P$, as $R \perp P$



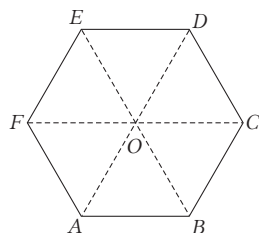
$$\text{Now, } R = \sqrt{P^2 + Q^2 + 2PQ \cos(90^\circ + \theta)}$$

$$= \sqrt{P^2 + Q^2 - 2PQ \sin \theta}$$

$$= \sqrt{(2Q \sin \theta)^2 + Q^2 - 2(2Q \sin \theta)Q \sin \theta}$$

$$= \sqrt{4Q^2 \sin^2 \theta + Q^2 - 4Q^2 \sin^2 \theta} = Q$$

68 (d) $\mathbf{AB} + \mathbf{AC} + \mathbf{AD} + \mathbf{AE} + \mathbf{AF}$



$$= \mathbf{AB} + (\mathbf{AB} + \mathbf{BC}) + (\mathbf{AB} + \mathbf{BC} + \mathbf{CD}) + (\mathbf{AB} + \mathbf{BC} + \mathbf{CD} + \mathbf{DE}) + \mathbf{AF}$$

$$= 4\mathbf{AB} + 3\mathbf{BC} + 2\mathbf{CD} + \mathbf{DE} + \mathbf{AF}$$

$$(\because \mathbf{AB} = -\mathbf{DE}, \mathbf{BC} = -\mathbf{EF}, \mathbf{AF} = \mathbf{CD})$$

$$= 3(\mathbf{AB} + \mathbf{BC} + \mathbf{CD}) = 3\mathbf{AD} = 6\mathbf{AO}$$

69 (a) From the figure, $\mathbf{OB} + \mathbf{BC} = \mathbf{r}$

$$\text{or} \quad \mathbf{q} + \mathbf{BC} = \mathbf{r} \quad \dots (i)$$

$$\mathbf{OA} + \mathbf{AC} = \mathbf{r}$$

$$\text{or} \quad \mathbf{p} + \mathbf{AC} = \mathbf{r} \quad \dots (ii)$$

Adding these two equations, we get,

$$\mathbf{p} + \mathbf{q} = 2\mathbf{r}, \text{ as } \mathbf{AC} \text{ and } \mathbf{BC} \text{ are equal and opposite vectors.}$$

70 (b) $|\Delta a| = \sqrt{a^2 + a^2 - 2 \cdot a \cdot a \cdot \cos(d\theta)} = 2a \sin\left(\frac{d\theta}{2}\right)$

For small angles $\sin \frac{d\theta}{2} \approx \frac{d\theta}{2}$

$$\therefore |\Delta a| = 2a \times \frac{d\theta}{2} = a \cdot d\theta$$

Δa means change in magnitude of vector,

$$\text{i.e.} \quad a - a = 0 \Rightarrow \Delta a = 0$$

(B) Medical entrance special format question

• Assertion and reason

1 (b) $\hat{i}$ and $\hat{j}$ are mutually perpendicular to each other and angle between $(\hat{i} + \hat{j})$ and $\hat{i}$ is 45° .

2 (d) Value of $(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B})$ can be positive or negative.

3 (a) $(\mathbf{A} \times \mathbf{B})$ is perpendicular to $(\mathbf{A} + \mathbf{B})$ and $(\mathbf{A} - \mathbf{B})$ because $(\mathbf{A} \times \mathbf{B})$ lies perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$.

4 (b) Angle between $\mathbf{A} \times \mathbf{B}$ and $\mathbf{B} \times \mathbf{A}$ is 180° .

$$\therefore (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{A}) = (AB \sin \theta)(AB \sin \theta)(\cos 180^\circ) = -A^2 B^2 \sin^2 \theta$$

5 (b) $(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = A^2 - B^2$

This is the dot product that possesses zero value, if $A = B$.

Therefore, they are perpendicular. Further, $(\mathbf{A} \times \mathbf{B})$ is perpendicular to both $(\mathbf{A} + \mathbf{B})$ and $(\mathbf{A} - \mathbf{B})$.

• Statement based questions

1 (b) When the coordinate axes are rotated through some angle, only the components of the vector changes, whereas magnitude remains constant.

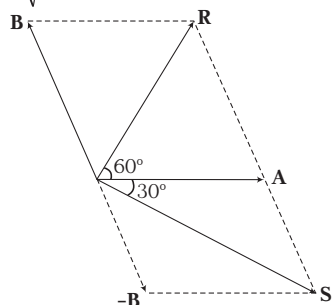
2 (d) A scalar quantity is independent of direction, hence has the same value for observers with different orientation of the axes.

3 (b) Clearly, from the diagram, $\mathbf{u} = a\hat{i} + b\hat{j}$

As $\mathbf{u}$ is in the first quadrant and located upward, hence its both components a and b will be positive.

For $\mathbf{v} = p\hat{i} + q\hat{j}$, as it is in positive x -direction and located downward, hence its x -component p will be positive and y -component q will be negative.

4 (a) Sum, $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$



We have, $1 = \sqrt{1 + 1 + 2 \cos \theta}$ or $\cos \theta = -\frac{1}{2}$

$\therefore \theta = 120^\circ$

Difference, $S = \sqrt{A^2 + B^2 - 2AB \cos \theta}$
 $= \sqrt{1 + 1 - 2 \times 1 \times 1 \times \left(-\frac{1}{2}\right)} = \sqrt{3}$

- 5 (a) If a vector is displaced parallel to itself, it does not change.

Thus, statement (I) is correct.

Three equal vectors can add up to zero when each of them is inclined at an angle of 120° with each other.

Thus, statement (II) is incorrect.

- 6 (c) A unit vector is a vector of unit magnitude and points in a particular direction.

If we multiply a unit vector, say $\hat{n}$ by a scalar λ , then the result is a vector $= \lambda \hat{n}$.

So, both statements I and II are incorrect.

• Match the columns

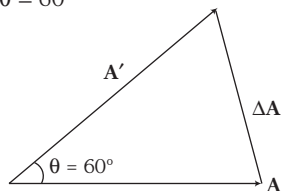
- 1 (b) Using right hand rule, direction of $(\mathbf{A} \times \mathbf{B})$ will be vertically upwards and $(\mathbf{A} + \mathbf{B})$ will point towards north-east.

The $(\mathbf{A} - \mathbf{B})$ will point towards south-east and

$(\mathbf{A} \times \mathbf{B}) \times (\mathbf{A} \times \mathbf{B})$ is a null vector having arbitrary direction.

Hence, $A \rightarrow p, B \rightarrow s, C \rightarrow q, D \rightarrow s$.

- 2 (d) (A) When $\theta = 60^\circ$

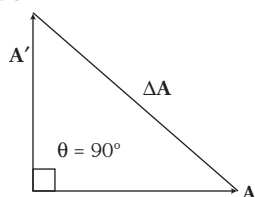


$\Rightarrow \mathbf{A} = \mathbf{A}'$

$\therefore$ Change in $|\mathbf{A}|$, $\Delta A = \sqrt{x^2 + x^2 - 2 \cdot x \cdot x \cdot \cos 60^\circ} = x$

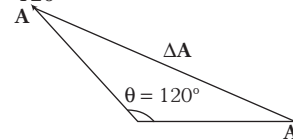
$\Rightarrow n = 1$

- (B) When $\theta = 90^\circ$



$\therefore$ Change in $|\mathbf{A}|$, $\Delta A = \sqrt{x^2 + x^2} = \sqrt{2}x \Rightarrow n = \sqrt{2}$

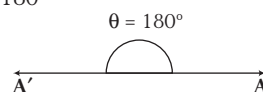
- (C) When $\theta = 120^\circ$



$\therefore$ Change in $|\mathbf{A}|$, $\Delta A = \sqrt{x^2 + x^2 \times x^2} = \sqrt{3}x$

$n = \sqrt{3}$

- (D) When $\theta = 180^\circ$



$\therefore$ Change in $|\mathbf{A}|$, $\Delta A = \sqrt{x^2 + x^2 + 2x^2} = 2x$

$\Rightarrow n = 2$

Hence, $A \rightarrow q, B \rightarrow r, C \rightarrow p, D \rightarrow s$.

3 (c) (A) $|\mathbf{A} + \mathbf{B}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$

Here, $|\mathbf{A}| = |\mathbf{B}| = x$ and $\theta = 60^\circ$

$\therefore |\mathbf{A} + \mathbf{B}| = \sqrt{3}x$

(B) $|\mathbf{A} - \mathbf{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta} = x$

(C) $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta = \frac{x^2}{2}$

(D) $|\mathbf{A} \times \mathbf{B}| = AB \sin \theta = \frac{\sqrt{3}}{2} x^2$

Hence, $A \rightarrow r, B \rightarrow q, C \rightarrow s, D \rightarrow p$.

(C) Medical entrances' gallery

- 1 (a) Suppose two vectors are $\mathbf{P}$ and $\mathbf{Q}$.

It is given that

$$|\mathbf{P} + \mathbf{Q}| = |\mathbf{P} - \mathbf{Q}|$$

Let angle between $\mathbf{P}$ and $\mathbf{Q}$ is ϕ .

$\therefore P^2 + Q^2 + 2PQ \cos \phi = P^2 + Q^2 - 2PQ \cos \phi$

$\Rightarrow 4PQ \cos \phi = 0$

$\Rightarrow \cos \phi = 0$

$(\because P, Q \neq 0)$

$\Rightarrow \phi = \frac{\pi}{2} = 90^\circ$

- 2 (c) Dot product of these two vectors should be equal to zero as they are perpendicular to each other.

$\therefore \mathbf{A} \cdot \mathbf{B} = -8 + 12 + 8\alpha = 0$

$8\alpha = -4 \Rightarrow \alpha = -\frac{1}{2}$

- 3 (a) The angle between, $\mathbf{p} = \hat{i} + \hat{j} + \hat{k}$

and X -axis, $\mathbf{x} = \hat{i}$ is given by

$$\cos \theta = \frac{\mathbf{p} \cdot \mathbf{x}}{|\mathbf{p}| |\mathbf{x}|} = \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i})}{\sqrt{1^2 + 1^2 + 1^2} \cdot \sqrt{1^2}} = \frac{1}{\sqrt{3}}$$

$\therefore \theta = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$

- 4 (a) Vector $\mathbf{X}$ of the form $\alpha\mathbf{A} + \beta\mathbf{B}$.

$$\begin{aligned}\mathbf{X} &= \alpha\mathbf{A} + \beta\mathbf{B} \\ &= \alpha(\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}) + \beta(\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}) \\ \mathbf{X} &= \hat{\mathbf{i}}(\alpha + \beta) + \hat{\mathbf{j}}(\alpha - \beta) + \hat{\mathbf{k}}(-2\alpha + \beta)\end{aligned}$$

A vector $\mathbf{X}$ is perpendicular to $\mathbf{C}$, i.e. $\mathbf{X} \cdot \mathbf{C} = 0$

$$[\hat{\mathbf{i}}(\alpha + \beta) + \hat{\mathbf{j}}(\alpha - \beta) + \hat{\mathbf{k}}(-2\alpha + \beta)] \cdot [2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}] = 0$$

$$\Rightarrow 2(\alpha + \beta) - 3(\alpha - \beta) + 4(-2\alpha + \beta) = 0$$

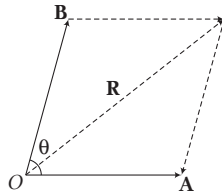
$$\Rightarrow 2\alpha + 2\beta - 3\alpha + 3\beta - 8\alpha + 4\beta = 0$$

$$\Rightarrow -9\alpha + 9\beta = 0$$

$$\text{or } \alpha = \beta \Rightarrow \frac{\alpha}{\beta} = \frac{1}{1}$$

$$\text{or } \alpha : \beta = 1 : 1$$

- 5 (c) Let two vectors are $\mathbf{A}$ and $\mathbf{B}$, inclined at an angle θ .



Resultant of the two vectors $\mathbf{A}$ and $\mathbf{B}$,

$$|\mathbf{R}| = \sqrt{|\mathbf{A}|^2 + |\mathbf{B}|^2 + 2|\mathbf{A}||\mathbf{B}|\cos\theta} \quad \dots(i)$$

Let $|\mathbf{A}| = |\mathbf{B}| = a$

According to the question, $|\mathbf{R}| = a$

From Eq. (i), we get

$$a = \sqrt{a^2 + a^2 + 2aa\cos\theta}$$

$$\Rightarrow a^2 = a^2 + a^2 + 2a^2\cos\theta$$

$$\Rightarrow 2a^2\cos\theta = -a^2$$

$$\Rightarrow \cos\theta = -1/2$$

$$\Rightarrow \theta = 120^\circ$$

- 6 (d) Potential energy is not a vector quantity.

- 7 (d) Given, $\mathbf{A} = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$ and $\mathbf{B} = -\hat{\mathbf{j}} + \hat{\mathbf{k}}$

Scalar product, $\mathbf{A} \cdot \mathbf{B} = (2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}) \cdot (-\hat{\mathbf{j}} + \hat{\mathbf{k}})$

Using $\hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = 1, \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = 1, \hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = 1$

We have, $\mathbf{A} \cdot \mathbf{B} = -2 - 1 = -3$

- 8 (a) Given, $\mathbf{A} \cdot \mathbf{B} = \mathbf{A} \times \mathbf{B}$

$$AB\cos\theta = AB\sin\theta$$

$$\tan\theta = 1$$

$$\theta = 45^\circ$$

- 9 (d) According to given condition,

$$\mathbf{A} + \mathbf{B}\hat{\mathbf{i}} = \mathbf{R}\hat{\mathbf{j}} \quad (\text{where, } \mathbf{R} \text{ is the resultant vector})$$

$$\text{Also, } \mathbf{R} = 2\mathbf{B}$$

$$\therefore \mathbf{A} + \mathbf{B}\hat{\mathbf{i}} = 2\mathbf{B}\hat{\mathbf{j}} \quad \text{or } \mathbf{A} = 2\mathbf{B}\hat{\mathbf{j}} - \mathbf{B}\hat{\mathbf{i}}$$

$$\therefore A^2 = 4B^2 + B^2 = 5B^2$$

$$\text{Here, } A = 8$$

$$\therefore 64 = 5B^2$$

$$\Rightarrow B = \sqrt{\frac{64}{5}} = \frac{8}{\sqrt{5}}$$

- 10 (c) For two vectors $\mathbf{a}$ and $\mathbf{b}$ to be perpendicular, $\mathbf{a} \cdot \mathbf{b} = 0$

$$\text{Thus, } (5\hat{\mathbf{i}} + \lambda\hat{\mathbf{j}} + \hat{\mathbf{k}}) \cdot (\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}) = 0$$

$$5(\hat{\mathbf{i}} \cdot \hat{\mathbf{i}}) - 2\lambda(\hat{\mathbf{j}} \cdot \hat{\mathbf{j}}) + 1(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}) = 0$$

$$\Rightarrow 5 - 2\lambda + 1 = 0$$

$$\Rightarrow 6 - 2\lambda = 0 \Rightarrow \lambda = 3$$

CHAPTER 03

Motion in One Dimension

Motion is defined as the change in position of an object with time. When the object moves along a single axis, the motion is known as **one dimensional motion** or **rectilinear motion** and such a motion is along a straight line only, which may be horizontal or vertical. In this chapter, we shall learn about motion using the concepts of velocity and acceleration along with the basic physics of one dimensional motion.

Frame of reference

A system of coordinate axes which defines the position of a particle or an event in two or three dimensional space along with a clock constitutes a frame of reference. The simplest frame of reference is the cartesian system of coordinates, in which the position of the particle (P) is specified by its three coordinates x , y and z .

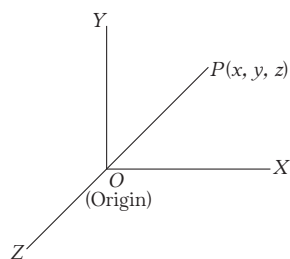


Fig. 3.1 Frame of reference for position of a particle P

Types of frame of reference

Frame of references are of two types

- (i) **Inertial frame of reference** It is a frame of reference, where Newton's law holds good. *e.g.* An object will remain at rest or in uniform motion unless acted by an external force. An inertial frame of reference is either at rest or moving with a constant velocity.
- (ii) **Non-inertial frame of reference** An accelerating frame of reference is called a non-inertial frame of reference. In this reference frame, Newton's law will not hold true.

Inside

- 1 Rest and motion
 - Some basic terms related to motion
- 2 Kinematic equations for uniformly accelerated motion
- 3 Motion under gravity
 - Equations for motion under gravity
- 4 Non-uniformly accelerated motion
- 5 Graphical representation of motion
- 6 Relative velocity
 - Different cases of relative velocity
 - Examples of relative motion

REST AND MOTION

If the position of an object does not change w.r.t. its surrounding with the passage of time, it is said to be at rest. *e.g.* Book lying on the table, a person sitting on a chair, etc. And if the position of an object is continuously changing w.r.t. its surrounding, then it is said to be in the state of motion. *e.g.* The walking man, crawling insects, water flowing down a dam, etc.

Rest and motion are relative terms

Rest and motion are always relative but never absolute. It means an object, can be at rest for an observer but the same object can be in motion when observed by other observer. *e.g.* A person sitting in his house is at rest w.r.t. earth but is in motion w.r.t. moon.

Classification of motion

On the basis of the number of coordinates required to specify the position of an object, the motion of the object can be classified as

One dimensional motion

The motion of an object is considered as one dimensional, if only one coordinate is needed to specify the position of the object.

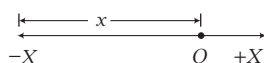


Fig. 3.2 One dimensional motion

In one dimensional motion, the object moves along a straight line. *e.g.* A boy running on a straight line, motion of freely falling body, etc.

Two dimensional motion

The motion of an object is considered as two dimensional, if two coordinates are needed to specify the position of the object.

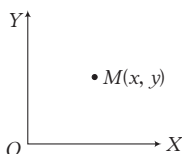


Fig. 3.3 Two dimensional motion

In two dimensional motion, the object moves in a plane. *e.g.* Motion of billiard ball.

Three dimensional motion

The motion of an object is considered as three dimensional, if all the three coordinates are needed to specify the position of the object.

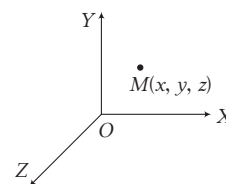


Fig. 3.4 Three dimensional motion

In three dimensional motion, the object moves in a space. *e.g.* Butterfly flying in garden, the motion of water molecules, etc.

Some basic terms related to motion

1. Point object

An object is considered as point object, if the size of the object is much smaller than the distance, it moves in a reasonable duration of time. *e.g.* Earth can be considered as a point object during its revolution around the sun because it covers much larger distance than its own size.

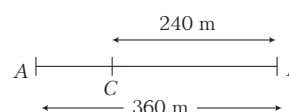
2. Distance and displacement

Distance

The length of the path covered by the object in a given time interval, is known as its distance. It is a scalar quantity. The unit of distance is metre in SI or MKS and centimetre in CGS. Its dimensional formula is $[M^0 L T^0]$.

Example 3.1 A scooter is moving along a straight line AB covers a distance of 360 m in 24 s and returns back from B to C and covers 240 m in 18 s. Find the total distance travelled by the scooter.

Sol. From the above question, we draw the following figure



Hence, to find out the total path distance, it does not matter how much time is taken by a scooter to reach at B and the time taken to return at C.

$$\therefore \text{Total distance} = AB + BC = 360 + 240 = 600 \text{ m}$$

Example 3.2 A wheel completes 2000 revolutions to cover the 9.5 km distance. Find the diameter of the wheel.

Sol. Given, number of revolutions, $n = 2000$

$$\text{Distance, } x = 9.5 \text{ km} = 9.5 \times 10^3 \text{ m} = 9500 \text{ m}$$

$\therefore$ Distance covered in n revolutions is equal to the circumference of the wheel,

$$x = n \cdot 2\pi r \Rightarrow x = n \cdot \pi D \quad (\because \text{Diameter} = 2 \times \text{Radius})$$

$$9500 = 2000 \times \pi \times D$$

$$\Rightarrow \text{Diameter, } D = \frac{9500}{2000 \times 3.14} = 1.5 \text{ m}$$

Displacement

It is the shortest distance between the initial and final position of the moving object.

If x_1 and x_2 are the initial and final positions of an object, respectively. Then, displacement of the object is given by

$$\Delta x = x_2 - x_1$$

- If $x_2 > x_1$, then Δx is positive.
- If $x_1 > x_2$, then Δx is negative.
- If $x_1 = x_2$, then Δx is zero.

i.e. The displacement of an object in motion can be positive, negative or zero while distance can never be negative or zero.

Displacement has both magnitude and direction. The unit of displacement is metre in SI or MKS and centimetre in CGS. Its dimensional formula is $[M^0 L T^0]$.

To understand distance and displacement clearly, let us consider the following example

Suppose a person (moving body) moves from A to B (4 km) towards east and from B to C (3 km) due north as shown in figure, then the distance travelled by the person is

$$AB + BC = 4 \text{ km} + 3 \text{ km} = 7 \text{ km}$$

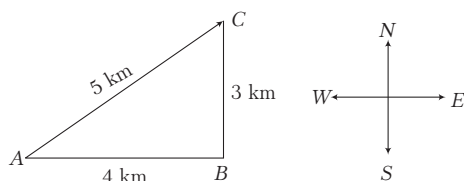


Fig. 3.5

But the displacement of the person is AC which can be calculated by Pythagoras' theorem,

$$\begin{aligned} \text{i.e. } AC^2 &= AB^2 + BC^2 \\ &= (4)^2 + (3)^2 = 16 + 9 = 25 \end{aligned}$$

$$\Rightarrow AC = 5 \text{ km}$$

Distance versus Displacement

- The magnitude of displacement may or may not be equal to the distance traversed by an object.
- The magnitude of the displacement for a course of motion may be zero but the corresponding distance can never be zero.
- If a particle moves in a straight line without change in direction, the magnitude of displacement is equal to the distance travelled otherwise displacement is always less than distance. Thus,

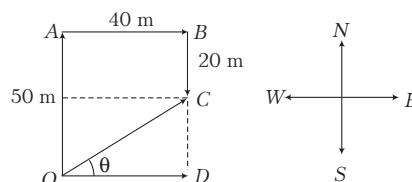
$$|\text{Displacement}| \leq \text{Distance}$$

- Distance depends on the path while displacement is independent of the path but depends only on initial and final positions.

Example 3.3 A man starts from his home and walks 50 m towards north, then he turns towards east and walks 40 m and then reaches to his office after moving 20 m towards south.

- What is the total distance covered by the man from his home to office?
- What is his displacement from his home to office?

Sol. Let O represents the position of home, then according to the question, the man moves from O to A (50 m) towards north, then from A to B (40 m) towards east and from B to C (20 m) towards south as shown in figure.



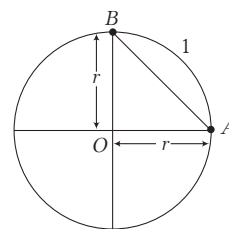
- Total distance travelled by the man is
 $OA + AB + BC = 50 + 40 + 20 = 110 \text{ m}$
- Displacement of the person is OC, which can be calculated by Pythagoras theorem, i.e.

$$\begin{aligned} OC^2 &= OD^2 + CD^2 = (40)^2 + (30)^2 \\ &= 1600 + 900 = 2500 \\ \Rightarrow OC &= 50 \text{ m} \end{aligned}$$

Example 3.4 An object covers $(1/4)$ th of the circular path. What will be the ratio of the distance and displacement of the object?

Sol. Distance covered by object = $1/4$ th of the circular path
 $= AB$ through path (1)

$$= 1/4 \text{th of circumference of circular path} = \frac{2\pi r}{4} = \frac{\pi r}{2}$$



Displacement = Shortest distance between initial position (A) and final position (B)

$$\begin{aligned} AB &= \sqrt{OA^2 + OB^2} = \sqrt{r^2 + r^2} = r\sqrt{2} \\ \therefore \frac{\text{Distance}}{\text{Displacement}} &= \frac{\pi r/2}{r\sqrt{2}} = \frac{\pi}{2\sqrt{2}} \end{aligned}$$

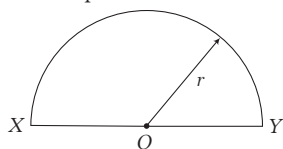
Example 3.5 Displacement of a person moving from X to Y along a semicircular path of radius r is 200 m. What is the distance travelled by him?

Sol. Given, displacement = 200 m

Distance travelled by the person from X to Y is equal to the circumference of the semicircular path,

$$x = \frac{\pi D}{2} \Rightarrow x = \pi r \quad (\because \text{Diameter} = 2 \times \text{Radius}) \dots(i)$$

According to the question, the shortest distance between the final position Y and initial position X is $XY = 2r$.



$\therefore$ The displacement traversed by the person is $2r$.

$$200 = 2r \Rightarrow r = 100 \text{ m}$$

Putting $r = 100 \text{ m}$ in Eq. (i), we get

$$\text{Distance, } x = \pi \times 100 = 3.14 \times 100 \Rightarrow x = 314 \text{ m}$$

Example 3.6 An athlete completes one round of a circular track of diameter 200 m in 40 s . What will be the distance covered and the displacement at the end of $2 \text{ min } 20 \text{ s}$?

Sol. Diameter of circular track, $D = 200 \text{ m}$

Circumference of circular track

$$\begin{aligned} &= 2\pi r = \pi \times (D) \\ &= \frac{22}{7} \times 200 = \frac{4400}{7} \text{ m} \end{aligned}$$

Time taken by athlete for completing one round = 40 s

In 40 s , distance covered by athlete

$$= \frac{4400}{7} \text{ m}$$

$\therefore$ Distance covered by athlete in 2 min and 20 s ($= 140 \text{ s}$)

$$= \frac{4400}{7} \times \frac{140}{40} = 2200 \text{ m}$$

As the athlete returns to the initial point A in 40 s , so his displacement = 0

In 40 s , the number of round, around the track = 1

$\therefore$ In 140 s , the number of rounds around the track

$$= \frac{140}{40} = 3 \frac{1}{2}$$

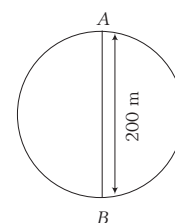
For each complete round, the displacement is 0 .

$\therefore$ For 3 complete rounds, the displacement will be 0 .

Hence, the final displacement will be due to $1/2$ round.

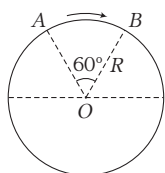
Thus, his displacement = diameter of circular track = 200 m

$\therefore$ Displacement after $2 \text{ min } 20 \text{ s} = 200 \text{ m}$



CHECK POINT 3.1

- Which of the following is a one-dimensional motion?
 - Landing of an aircraft
 - Earth revolving around the sun
 - Motion of wheels of moving train
 - Train running on a straight track
- A person moves towards east for 3 m , then towards north for 4 m and then moves towards for 5 m . What is his distance now from the starting point?
 - 12 m
 - 5 m
 - 10 m
 - 20 m
- A particle moves in a circle of radius R from A to B as shown in figure. The distance covered by the object is



- $\frac{\pi R}{3}$
 - $\frac{\pi R}{2}$
 - $\frac{\pi R}{4}$
 - πR
- A wheel of radius 1 m rolls forward half a revolution on a horizontal ground. The magnitude of displacement of the point of the wheel initially in contact with the ground is
 - 2π
 - $\sqrt{2}\pi$
 - $\sqrt{\pi^2 + 4}$
 - π

- The three initial and final position of a man on the X -axis are given as
 - $(-8 \text{ m}, 7 \text{ m})$
 - $(7 \text{ m}, -3 \text{ m})$
 - $(-7 \text{ m}, 3 \text{ m})$
 Which pair gives the negative displacement?
 - (i)
 - (ii)
 - (iii)
 - (i) and (iii)
- The numerical ratio of displacement to the distance covered is always
 - less than one
 - equal to one
 - equal to or less than one
 - equal to or greater than one
- A particle moves along a circular path of radius R . The distance and displacement of a particle after one complete revolution is
 - $0, 2\pi r$
 - $2\pi r, 0$
 - $0, \pi r$
 - $\pi r, 0$
- A particle starts from the origin, goes along X -axis to the point $(20 \text{ m}, 0)$ and then returns along the same line to the point $(-20 \text{ m}, 0)$. The distance and displacement of the particle during the trip are
 - $40 \text{ m}, 0$
 - $40 \text{ m}, 20 \text{ m}$
 - $40 \text{ m}, -20 \text{ m}$
 - $60 \text{ m}, -20 \text{ m}$

3. Speed

The time rate of distance travelled by an object in any direction is called speed of the object.

$$\text{Speed } (v) = \frac{\text{Distance travelled}}{\text{Time taken}}$$

It is a scalar quantity.

The unit of speed in SI or MKS system is ms^{-1} and in CGS system is cms^{-1} . Its dimensional formula is $[\text{M}^0 \text{L} \text{T}^{-1}]$.

For a moving body, speed is always positive and can never be negative or zero.

Average speed

The ratio of the total distance travelled by the object to the total time taken is called average speed of the object.

$$\text{i.e. Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

Average speed of particles in different cases

Case I. If a particle travels distance $s_1, s_2, s_3, \dots$, etc., with speeds $v_1, v_2, v_3, \dots$, etc., in same direction, then the distance travelled = $s_1 + s_2 + s_3 + \dots$

$$\text{Total time taken} = \frac{s_1}{v_1} + \frac{s_2}{v_2} + \frac{s_3}{v_3} + \dots$$

$$\text{Average speed, } v_{\text{av}} = \frac{s_1 + s_2 + \dots}{\left(\frac{s_1}{v_1} + \frac{s_2}{v_2} + \dots\right)}$$

If $s_1 = s_2 = s$, i.e. the body covers equal distances with different speeds, then

$$v_{\text{av}} = \frac{2s}{s\left(\frac{1}{v_1} + \frac{1}{v_2}\right)}$$

$$v_{\text{av}} = \frac{2v_1 v_2}{v_1 + v_2}$$

Case II. If a particle travels with speeds $v_1, v_2, v_3, \dots$, etc., during time intervals $t_1, t_2, t_3, \dots$, etc., then total distance travelled, $s = v_1 t_1 + v_2 t_2 + v_3 t_3 + \dots$

$$\text{Total time taken} = t_1 + t_2 + t_3 + \dots$$

$$\text{So, average speed, } v_{\text{av}} = \frac{v_1 t_1 + v_2 t_2 + v_3 t_3 + \dots}{t_1 + t_2 + t_3 + \dots}$$

Case III. If $t_1 = t_2 = t_3 = \dots = t_n$, then we have

$$v_{\text{av}} = \frac{(v_1 + v_2 + \dots + v_n)t}{nt}$$

$$v_{\text{av}} = \frac{v_1 + v_2 + \dots + v_n}{n}$$

Instantaneous speed

The speed of a particle at any instant of time is known as its instantaneous speed.

$$\text{Instantaneous speed} = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

where, s represents distance.

Example 3.7 Abdul while driving to school, computes the average speed for his trip to be 20 kmh^{-1} . On his return trip along the same route, there is less traffic and the average speed is 40 kmh^{-1} . What is the average speed for Abdul's trip?

Sol. Let t_1 and t_2 be the time taken by Abdul to go to school and come back from the school, respectively. Let s be its distance covered in one way, then

$$t_1 = \frac{s}{20} \text{ h and } t_2 = \frac{s}{40} \text{ h}$$

$$\text{Total time taken} = t_1 + t_2 = \frac{s}{20} + \frac{s}{40} = \frac{3s}{40}$$

$$\text{Total distance covered} = 2s$$

$$\therefore \text{Average speed} = \frac{2s}{\frac{3s}{40}} \times 40 = \frac{80}{3} = 26.67 \text{ kmh}^{-1}$$

Example 3.8 A car covers the first half of the distance between two places at a speed of 40 kmh^{-1} and second half at 60 kmh^{-1} . Calculate the average speed of the car.

Sol. Given, speed in first half, $v_1 = 40 \text{ kmh}^{-1}$

Speed in second half, $v_2 = 60 \text{ kmh}^{-1}$

$\therefore$ Car covers equal distance with different speeds.

$\therefore$ Average speed of car,

$$v_{\text{av}} = \frac{2v_1 v_2}{v_1 + v_2}$$

$$\Rightarrow v_{\text{av}} = \frac{2(40)(60)}{40 + 60} = 48 \text{ kmh}^{-1}$$

Example 3.9 A car moves from X to Y with a uniform speed v_u and returns to X with a uniform speed v_d . Find average speed for this round trip.

Sol. We know that, average speed = $\frac{\text{distance travelled}}{\text{time taken}}$

Let t_1 and t_2 be times taken by the car to go from X to Y and then from Y to X, respectively.

$$\text{Then, } t_1 + t_2 = \frac{XY}{v_u} + \frac{XY}{v_d} = XY \left(\frac{v_u + v_d}{v_u v_d} \right)$$

$$\text{Total distance travelled} = XY + XY = 2XY$$

Therefore, average speed of the car for this round trip is

$$v_{\text{av}} = \frac{2XY}{XY \left(\frac{v_u + v_d}{v_u v_d} \right)}$$

$$\text{or } v_{\text{av}} = \frac{2v_u v_d}{v_u + v_d}$$

Example 3.10 A particle travelled half the distance with a speed v_0 . The remaining part of the distance was covered with speed v_1 for half the time and with speed v_2 for the other half of the time. Find the average speed of the particle.

Sol. If s is the total distance travelled by the particle, then

$$\frac{s}{2} = v_0 t_1 \Rightarrow t_1 = \frac{s}{2v_0}$$

If t is the time taken by particle to travel remaining distance

$$s/2, \text{ then } \frac{s}{2} = \frac{v_1 t}{2} + \frac{v_2 t}{2} = (v_1 + v_2) \frac{t}{2} \text{ or } t = \frac{s}{(v_1 + v_2)}$$

$$\text{Average speed} = \frac{s}{t_1 + t} = \frac{s}{\frac{s}{2v_0} + \frac{s}{(v_1 + v_2)}} = \frac{2v_0(v_1 + v_2)}{v_1 + v_2 + 2v_0}$$

4. Velocity

The rate of change of position or displacement of an object with time is called the velocity of that object.

i.e.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

It is a vector quantity. The unit of velocity in SI or MKS system is ms^{-1} and in CGS system is cms^{-1} . Its dimensional formula is $[M^0 L T^{-1}]$. In 1-D motion, the velocity of an object is taken to be positive, if the object is moving towards the right of the origin and is taken to be negative, if the object is moving towards the left of the origin.

Average velocity

The ratio of the total displacement to the total time taken is called average velocity.

$$\text{Average velocity} = \frac{\text{Total displacement } (\Delta x)}{\text{Total time } (\Delta t)}$$

If velocity of the object changes at a uniform rate, then

$$\text{Average velocity} = \frac{\text{Initial velocity} + \text{Final velocity}}{2}$$

Note For a given time interval, average velocity has single value while average speed can have many values depending on path followed.

Instantaneous velocity

The velocity of a particle at any instant of time is known as its instantaneous velocity.

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Note The magnitude of average velocity may be less than or equal to the average speed for a particular motion. But the magnitude of instantaneous velocity is always equal to the instantaneous speed for a particular instant.

Uniform and non-uniform velocity

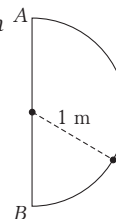
An object is said to have uniform velocity, if the magnitude and direction of its velocity remains constant. This is only possible when the object moves along a straight line without reversing its direction.

However, an object is said to have non-uniform velocity, if either magnitude or direction of velocity change w.r.t. time.

Velocity versus Speed

- Velocity of an object can be changed by changing the object's speed or direction of motion or both.
- For an object in a time interval (t); $|\text{Velocity}| \leq \text{Speed}$
i.e. The magnitude of velocity of an object is always equal to or less than its speed.
- If a body is moving in a straight line, then the magnitude of its speed and velocity will be equal.
- Average velocity could be zero or positive or negative but average speed is always positive for a moving body.

Example 3.11 In one second, a particle goes from point A to point B moving in a semicircular path as shown in figure. Find the magnitude of average velocity.



Sol. Given, $t = 1$ s and radius, $R = 1.0$ m

The shortest distance between the final position (B) and initial position A is AB, which is the displacement of the particle.

$$\therefore \text{Total displacement } (AB) = 2R = 2 \times 1.0 = 2 \text{ m}$$

$$\text{Thus, average velocity, } v_{av} = \frac{\text{Total displacement}}{\text{Total time taken}} = \frac{AB}{\Delta t} = \frac{2.0}{1.0} \text{ ms}^{-1} = 2 \text{ ms}^{-1}$$

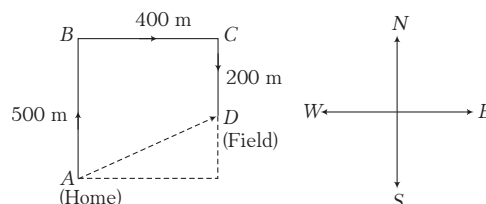
Example 3.12 A farmer has to go 500 m due north, 400 m due east and 200 m due south to reach his field. If he takes 20 min to reach the field,

- what distance he has to walk to reach the field?
- what is the displacement from his house to the field?
- what is the average speed of farmer during the walk?
- what is the average velocity of farmer during the walk?

Sol. (i) Distance = $AB + BC + CD$

$$= (500 + 400 + 200) = 1100 \text{ m}$$

$$\begin{aligned} \text{(ii) Displacement} &= AD = \sqrt{(AB - CD)^2 + BC^2} \\ &= \sqrt{(500 - 200)^2 + (400)^2} = 500 \text{ m} \end{aligned}$$

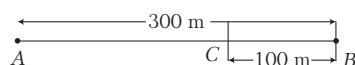


- (iii) Average speed = $\frac{\text{Total distance}}{\text{Total time}} = \frac{1100}{20} = 55 \text{ m/min}$
- (iv) Average velocity = $\frac{AD}{t} = \frac{500}{20} = 25 \text{ m/min (along AD)}$

Example 3.13 Joseph jogs from one end A to the other end B of a straight 300 m road in 2 min 50 s and then turns around and jogs 100 m back to point C in another 1 min. What are Joseph's average speeds and velocities in jogging (i) from A to B and (ii) from A to C?

Sol. Given, distance covered from A to B = 300 m

$$\text{Time} = 2 \text{ min } 50 \text{ s} = (2 \times 60) + 50 \text{ s} = 170 \text{ s}$$



- (i) Average speed = $\frac{\text{Distance covered}}{\text{Time}} = \frac{300}{170} = 1.76 \text{ ms}^{-1}$
- Average velocity = $\frac{\text{Displacement along AB}}{\text{Time}} = \frac{300}{170} = 1.76 \text{ ms}^{-1} \text{ along AB}$
- (ii) When Joseph turns around from B to C towards A, then
- Average speed = $\frac{\text{Distance covered}}{\text{Time}} = \frac{400}{230} = 1.74 \text{ ms}^{-1}$
- Average velocity = $\frac{\text{Displacement AC}}{\text{Time}} = \frac{200}{230} \text{ along AC} = 0.87 \text{ ms}^{-1} \text{ along AC}$

CHECK POINT 3.2

- A car has to cover the distance 60 km. If half of the total time, it travels with speed 80 kmh^{-1} and in rest half time, its speed becomes 40 kmh^{-1} , the average speed of car will be
(a) 60 kmh^{-1} (b) 80 kmh^{-1} (c) 120 kmh^{-1} (d) 180 kmh^{-1}
- During the first 18 min of a 60 min trip, a car has an average speed of 11 m min^{-1} . What should be the average speed for remaining 42 min, so that car is having an average speed of 21 m min^{-1} for the entire trip?
(a) 25.3 m min^{-1} (b) 29.2 m min^{-1}
(c) 31 m min^{-1} (d) 35.6 m min^{-1}
- A man walks on a straight road from his home to a market 2.5 km away with a speed of 5 kmh^{-1} . Finding the market closed, he instantly turns and walks back home with a speed of 7.5 kmh^{-1} . The average speed of the man over the interval of time 0 to 40 min is equal to
(a) 5 kmh^{-1} (b) $\frac{25}{4} \text{ kmh}^{-1}$
(c) $\frac{30}{4} \text{ kmh}^{-1}$ (d) $\frac{45}{8} \text{ kmh}^{-1}$
- A particle is constrained to move on a straight line path. It returns to the starting point after 10 s. The total distance covered by the particle during this time is 30 m. Which of the following statements about the motion of the particle is true?
(a) Displacement of the particle is zero
(b) Average speed of the particle is 3 ms^{-1}
(c) Displacement of the particle is 30 m
(d) Both (a) and (b)
- A 150 m long train is moving with a uniform velocity of 45 kmh^{-1} . The time taken by the train to cross a bridge of length 850 m is
(a) 56 s (b) 68 s (c) 80 s (d) 92 s
- An insect crawls a distance of 4 m along north in 10 s and then a distance of 3 m along east in 5 s. The average velocity of the insect is
(a) $\frac{7}{15} \text{ ms}^{-1}$ (b) $\frac{1}{5} \text{ ms}^{-1}$
(c) $\frac{1}{3} \text{ ms}^{-1}$ (d) $\frac{4}{5} \text{ ms}^{-1}$

Example 3.14 The position of an object moving along X-axis is given by $x = 3t - 4t^2 + t^3$, where x is in metres and t in seconds. Find the position of the object at the following values of t : (i) 2 s, (ii) 4 s, (iii) What is the object's displacement between $t = 0 \text{ s}$ and $t = 4 \text{ s}$?; and (iv) What is its average velocity for the time interval from $t = 2 \text{ s}$ to $t = 4 \text{ s}$?

Sol. Using $x = 3t - 4t^2 + t^3$ with SI units ... (i)

- (i) Substituting $t = 2 \text{ s}$ in Eq. (i), we get

$$x_2 = 3(2) - 4(2)^2 + (2)^3 = -2 \text{ m}$$

Thus, the position of the object at $t = 2 \text{ s}$ is $x_2 = -2 \text{ m}$.

- (ii) Substituting $t = 4 \text{ s}$ in Eq. (i), we get

$$x_4 = 3(4) - 4(4)^2 + (4)^3 = 12 - 64 + 64 = 12 \text{ m}$$

Thus, the position of the object at $t = 4 \text{ s}$ is $x_4 = 12 \text{ m}$.

- (iii) The displacement of the object between $t = 0 \text{ s}$ and $t = 4 \text{ s}$ can be calculated as follows:

The position of the object at $t = 0 \text{ s}$ is $x = 0$.

$$\Delta x = \text{Final position} - \text{Initial position} = x_4 - x_0 = 12 \text{ m} - 0 \text{ m} = 12 \text{ m}$$

Hence, the displacement between $t = 0 \text{ s}$ and $t = 4 \text{ s}$ is 12 m.

- (iv) The displacement of the object from $t = 2 \text{ s}$ to $t = 4 \text{ s}$ is

$$\Delta x = x_4 - x_2 = 12 \text{ m} - (-2 \text{ m}) = 14 \text{ m}$$

The time interval from $t = 2 \text{ s}$ to $t = 4 \text{ s}$ is

$$\Delta t = 4 \text{ s} - 2 \text{ s} = 2 \text{ s}$$

The average velocity of the object from $t = 2 \text{ s}$ to $t = 4 \text{ s}$ is given by the relation

$$v_{av} = \frac{\Delta x}{\Delta t} = \frac{14 \text{ m}}{2 \text{ s}} = 7 \text{ ms}^{-1}$$

7. A particle traversed $(3/4)$ th of the circle of radius R in time t . The magnitude of the average velocity of the particle in this time interval is

- (a) $\frac{\pi R}{t}$ (b) $\frac{3\pi R}{2t}$
(c) $\frac{R\sqrt{2}}{t}$ (d) $\frac{R}{\sqrt{2}t}$

8. A boy is running over a circular track with uniform speed of 10ms^{-1} . What is the average velocity for movement of boy along semicircle (in ms^{-1})?

- (a) $\frac{10}{\pi}$ (b) $\frac{40}{\pi}$
(c) 10 (d) $\frac{20}{\pi}$

5. Acceleration

The time rate of change of velocity of a body is called acceleration.

$$\therefore \text{Acceleration} = \frac{\text{Change in velocity } (\Delta v)}{\text{Time interval } (\Delta t)}$$

If in a given time interval, the velocity of a body changes from u to v , then acceleration a is expressed as

$$a = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{Time interval}} = \frac{v - u}{t}$$

It is a vector quantity. Its SI unit is ms^{-2} and CGS unit is cms^{-2} .

Its dimensional formula is $[M^0 L T^{-2}]$. Its direction is same as that of change in velocity (not of the velocity).

There are three possible ways by which change in velocity may occur

- When only direction of velocity changes**, then acceleration is perpendicular to velocity.
e.g. Uniform circular motion.
- When only magnitude of velocity changes**, then acceleration is parallel or anti-parallel to velocity.
e.g. Motion under gravity.
- When both magnitude and direction of velocity changes**, then acceleration has two components : one is perpendicular to velocity and another is parallel or anti-parallel to velocity.
e.g. Projectile motion.

Retardation

When the velocity of a body increases with time, acceleration is positive and when the velocity of a body decreases with time (*i.e.* $u > v$), then acceleration becomes negative. This negative acceleration is also called **deceleration** or **retardation**. In other words, retardation opposes the motion of body.

Average acceleration

When an object is moving with a variable acceleration, then the average acceleration of the object for the given motion

is defined as the ratio of the total change in velocity of the object to the total time taken.

$$\text{Average acceleration} = \frac{\text{Total change in velocity}}{\text{Total time taken}}$$

Note The average acceleration can be positive or negative depending upon the sign of change of velocity. It is zero, if the change in velocity of the object in the given interval of time is zero.

Instantaneous acceleration

The acceleration of an object at a given instant of time or at a given point during the motion, is called its instantaneous acceleration. *i.e.*

$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt} = \frac{d^2 \mathbf{s}}{dt^2}$$

Key points regarding acceleration

Following are the important points for motion of an object having some acceleration

- A body falling down from a height or a body rolling down on a smooth inclined plane has uniform acceleration.
- If a car travelling along a straight road, increases its speed by unequal amounts in equal intervals of time, then the car is said to be moving with non-uniform acceleration.
- The acceleration is created by accelerator of the vehicles and the applications of breaks give the uniform deceleration to the vehicles. However, the acceleration produced in spring block system is non-uniform acceleration.
- If a particle is accelerated for time t_1 with acceleration a_1 and for time t_2 with acceleration a_2 , then average acceleration is

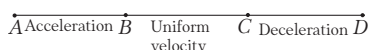
$$a_{av} = \frac{a_1 t_1 + a_2 t_2}{t_1 + t_2}$$

- (v) Acceleration can also be written as

$$a = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \left(\frac{dv}{dx} \right)$$

Example 3.15 A car starts from rest, attains a velocity of 18 km h^{-1} with an acceleration of 0.5 ms^{-2} , travels 4 km with this uniform velocity and then comes to halt with a uniform deceleration of 0.4 ms^{-2} . Calculate the total time of travel of the car.

Sol. Let the car be accelerated from A to B, it moves with uniform velocity from B to C of 4 km distance and then moves with uniform deceleration of 0.4 ms^{-2} from C to D as shown below.



For motion of car from A to B, $a = 0.5 \text{ ms}^{-2}$

$$u = 0 \text{ and } v = 18 \text{ km h}^{-1} \\ = 18 \times \frac{5}{18} \text{ ms}^{-1} = 5 \text{ ms}^{-1}$$

$$\text{Time, } t_1 = \frac{v - u}{a} \quad \dots(i)$$

Substituting given values of v , u and a for A to B motion, we get

$$t_1 = \frac{5 - 0}{0.5} = 10 \text{ s} \quad \dots(ii)$$

For motion of car from B to C,

$$s = 4 \text{ km} = 4000 \text{ m}$$

and

$$v = 5 \text{ ms}^{-1}$$

$$t_2 = \frac{\text{distance}}{\text{velocity}} \\ = \frac{4000}{5} = 800 \text{ s} \quad \dots(iii)$$

For motion of car from C to D, $v = 0$, $u = 5 \text{ ms}^{-1}$

and $a = -0.4 \text{ ms}^{-2}$ (negative sign shows deceleration)

$$\text{Time taken, } t_3 = \frac{v - u}{a} = \frac{0 - 5}{-0.4} \\ = \frac{-5}{-0.4} = 12.5 \text{ s} \quad \dots(iv)$$

Total time taken, $T = t_1 + t_2 + t_3$

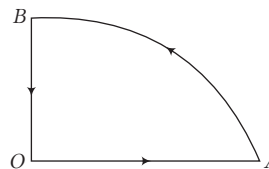
Substituting values of t_1 , t_2 and t_3 from Eqs. (ii), (iii) and (iv) respectively, we get

$$T = (10 + 800 + 12.5) \text{ s} = 822.5$$

Thus, total time of travel of the car is 822.5 s.

CHECK POINT 3.3

- Acceleration of a particle changes when
 - direction of velocity changes
 - magnitude of velocity changes
 - Both (a) and (b)
 - speed changes
- If a particle moves with an acceleration, then which of the following can remain constant?
 - Both speed and velocity
 - Neither speed nor velocity
 - Only the velocity
 - Only the speed
- The average velocity of a body moving with uniform acceleration travelling a distance of 3.06 m is 0.34 ms^{-1} . If the change in velocity of the body is 0.18 ms^{-1} , then during this time, its uniform acceleration is
 - 0.01 ms^{-2}
 - 0.02 ms^{-2}
 - 0.03 ms^{-2}
 - 0.04 ms^{-2}
- A car travelling with a velocity of 80 km/h slowed down to 44 km/h in 15 s. The retardation is
 - 0.67 ms^{-2}
 - 1 ms^{-2}
 - 1.25 ms^{-2}
 - 1.5 ms^{-2}
- An object is moving along the path OABO with constant speed, then
 - the acceleration of the object while moving along the path OABO is zero
 - the acceleration of the object along the path OA and BO is zero
 - there must be some acceleration along the path AB
 - Both (b) and (c)



KINEMATIC EQUATIONS FOR UNIFORMLY ACCELERATED MOTION

When a body is moving along a straight line with uniform acceleration, then its motion is called **uniformly accelerated motion**. For this motion, we can establish the relation between velocity, acceleration and the distance travelled by the body in a particular time interval by a set of equations. These equations are known as **kinematic equations** or **equations of motion**.

The three equations of motion on a straight line are

$$(i) v = u + at \quad (ii) s = ut + \frac{1}{2} at^2$$

$$(iii) v^2 - u^2 = 2as$$

where, u is the initial velocity of the body, a is the uniform acceleration of the body, v is the final velocity of

the body after t second and s is the distance travelled in this time.

- Distance travelled by a body in n th second,

$$s_n = u + \frac{1}{2} a (2n - 1)$$

Note

- (i) If initial position of a particle is r_0 , then position at time t can be written as

$$r = r_0 + s = r_0 + ut + \frac{1}{2} at^2$$

- (ii) **Stopping distance** When brakes are applied to a moving vehicle, the distance it travels before stopping is called stopping distance. It is an important factor for road safety. It is given by

$$s = \frac{u^2}{2a}$$

where, u is initial velocity and a is the retardation produced by brakes.

Key points regarding kinematic equations

Following are the important points in case of one dimensional motion with constant acceleration

- If the motion starts from rest, then initial velocity is taken as zero, i.e. $u = 0$.
- If the object comes to rest after the motion, then final velocity is taken as zero, i.e. $v = 0$.
- If velocity of moving object increases with time, then acceleration is taken as positive and if velocity decreases with time, acceleration is taken as negative.
- If velocity and acceleration both have same sign like $v > 0$; $a > 0$ or $v < 0$; $a < 0$, then object is speeding up. Similarly, if velocity and acceleration both have opposite sign like $v < 0$; $a > 0$ or $v > 0$; $a < 0$. Then, the object is speeding down.
- For motion of an object along a straight line, normally we take vertically upward direction positive (and downward negative) and horizontally rightwards positive (or leftwards negative). Sign convention for (a) motion in vertical direction (b) motion in horizontal direction is shown in figure.

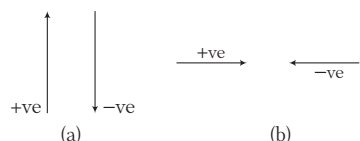


Fig. 3.6 Sign convention for vertical and horizontal direction

Example 3.16 Two cars start off a race with velocities 2 ms^{-1} and 4 ms^{-1} travel in straight line with uniform accelerations 2 ms^{-2} and 1 ms^{-2} , respectively. What is the length of the path, if they reach the final point at the same time?

Sol. Let both cars reach at same position in same time t , then

$$\text{from } s = ut + \frac{1}{2} at^2$$

$$\text{For 1st car, } s = 4(t) + \frac{1}{2} (1) t^2 = 4t + \frac{t^2}{2} \quad \dots(i)$$

$$\text{For 2nd car, } s = 2(t) + \frac{1}{2} (2) t^2 = 2t + t^2$$

Equating above equations, we get

$$4t + \frac{t^2}{2} = 2t + t^2 \Rightarrow t = 4 \text{ s}$$

Substituting the value of t in Eq. (i), we get

$$s = 4(4) + \frac{1}{2} (1)(4)^2 = 16 + 8 = 24 \text{ m}$$

Example 3.17 A car was moving at a rate of 18 kmh^{-1} .

When the brakes were applied, it comes to rest at a distance of 100 m . Calculate the retardation produced by the brakes.

Sol. Given, $v = 0$, $u = 18 \text{ kmh}^{-1} = 5 \text{ ms}^{-1}$, $s = 100 \text{ m}$

Using the equation of motion,

$$v^2 - u^2 = 2as \quad \dots(i)$$

$$\Rightarrow -u^2 = 2as \quad (\because v = 0)$$

$$\Rightarrow a = -\frac{u^2}{2s}$$

$$\Rightarrow a = \frac{-5 \times 5}{2 \times 100} = -\frac{1}{8} = -0.125 \text{ ms}^{-2}$$

So, the retardation produced by the brakes is 0.125 ms^{-2} .

Example 3.18 Two car travelling towards each other on a straight road at velocity 10 ms^{-1} and 12 ms^{-1} , respectively. When they are 150 m apart, both the drivers apply their brakes and each car decelerates at 2 ms^{-2} until it stops. How far apart will they be when both of them come to rest?

Sol. Let x_1 and x_2 be the distance travelled by the car before they stop under deceleration.

From third equation of motion,

$$v^2 = u^2 + 2as$$

$$\Rightarrow 0 = (10)^2 - 2 \times 2x_1 \Rightarrow x_1 = 25 \text{ m}$$

$$\text{and } 0 = (12)^2 - 2 \times 2x_2 \Rightarrow x_2 = 36 \text{ m}$$

Total distance covered by the two cars

$$= x_1 + x_2 = 25 + 36 = 61 \text{ m}$$

Distance between the two cars when they stop

$$= 150 - 61 = 89 \text{ m}$$

Example 3.19 A train travelling at 20 kmh^{-1} is approaching a platform. A bird is sitting on a pole on the platform. When the train is at a distance of 2 km from pole, brakes are applied which produce a uniform deceleration in it. At that instant, the bird flies towards the train at 60 kmh^{-1} and after touching the nearest point on the train flies back to the pole and then flies towards the train and continues repeating itself. Calculate how much distance the bird covers before the train stops?

Sol. For retardation of train, $v^2 = u^2 + 2as$

$$\Rightarrow 0 = (20)^2 + 2(a)(2)$$

$$\Rightarrow a = -100 \text{ kmh}^{-2}$$

Time required to stop the train, $v = u + at$

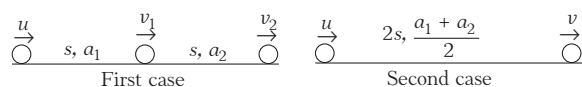
$$\Rightarrow 0 = 20 - 100t \Rightarrow t = \frac{1}{5} \text{ h}$$

For bird, speed = $\frac{\text{distance}}{\text{time}}$

$$\Rightarrow s_B = v_B \times t = 60 \times \frac{1}{5} = 12 \text{ km}$$

Example 3.20 A particle starts with an initial velocity and passes successively over the two halves of a given distance with accelerations a_1 and a_2 , respectively. Show that the final velocity is the same as if the whole distance is covered with a uniform acceleration $\frac{(a_1 + a_2)}{2}$.

Sol. In the first case,



$$v_1^2 = u^2 + 2a_1s \quad \dots(i)$$

$$v_2^2 = v_1^2 + 2a_2s \quad \dots(ii)$$

Adding Eqs. (i) and (ii), we get

$$v_2^2 = u^2 + 2\left(\frac{a_1 + a_2}{2}\right)(2s) \quad \dots(iii)$$

In the second case,

$$v^2 = u^2 + 2\left(\frac{a_1 + a_2}{2}\right)(2s) \quad \dots(iv)$$

From Eqs. (iii) and (iv), we can see that

$$v_2 = v$$

Example 3.21 In a car race, car A takes a time t less than car B at the finish point and passes the finishing point with speed v more than that of the car B. Assuming that both the cars start from rest and travel with constant acceleration a_1 and a_2 , respectively. Show that $v = \sqrt{a_1 a_2} t$.

Sol. Let A takes t_1 second, then according to the given problem B will take $(t_1 + t)$ seconds. Further, let v_1 be the velocity of B at finishing point, then velocity of A will be $(v_1 + v)$. Writing equations of motion for A and B,

$$v_1 + v = a_1 t_1 \quad \dots(i)$$

$$v_1 = a_2 (t_1 + t) \quad \dots(ii)$$

From these two equations, we get

$$v = (a_1 - a_2) t_1 - a_2 t \quad \dots(iii)$$

Total distance travelled by both the cars is equal.

$$\text{or } s_A = s_B$$

$$\text{or } \frac{1}{2} a_1 t_1^2 = \frac{1}{2} a_2 (t_1 + t)^2$$

or

$$t_1 = \frac{\sqrt{a_2} t}{\sqrt{a_1} - \sqrt{a_2}}$$

Substituting this value of t_1 in Eq. (iii), we get

$$v = (\sqrt{a_1 a_2}) t$$

Example 3.22 A body starting from rest has an acceleration of 4 ms^{-2} . Calculate distance travelled by it in 5th second.

Sol. Given, $u = 0$, $a = 4 \text{ ms}^{-2}$

Distance travelled by the body in 5th second is

$$s_n = u + \frac{1}{2} a(2n - 1)$$

$$s_5 = 0 + \frac{1}{2} \times 4(2 \times 5 - 1)$$

$$= \frac{1}{2} \times 4(9) = \frac{36}{2} = 18 \text{ m}$$

Example 3.23 A particle starts from rest and moves under constant acceleration in a straight line. Find the ratio of displacement (i) in successive second and (ii) in successive time interval t_0 .

Sol. (i) Displacement in 1 s or 1st second,

$$s_1 = ut + \frac{1}{2} at^2 = 0 + \frac{1}{2} a(1)^2 = \frac{1}{2} a$$

$$\text{or } s_1 = u + \frac{1}{2} a(2t - 1) = 0 + \frac{1}{2} a(2 \times 1 - 1) = \frac{1}{2} a$$

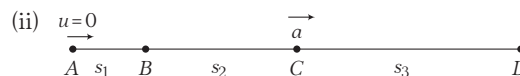
Displacement in the 2nd second,

$$s_2 = u + \frac{1}{2} a(2t - 1) = 0 + \frac{1}{2} a(2 \times 2 - 1) = \frac{3}{2} a$$

Displacement in the 3rd second,

$$s_3 = 0 + \frac{1}{2} a(2 \times 3 - 1) = \frac{5}{2} a$$

$$s_1 : s_2 : s_3 : \dots = \frac{1}{2} a : \frac{3}{2} a : \frac{5}{2} a : \dots = 1 : 3 : 5 : \dots$$



A to B: Displacement in the first t_0 second,

$$s = ut + \frac{1}{2} at^2 \Rightarrow s_1 = 0 + \frac{1}{2} at_0^2 = \frac{1}{2} at_0^2$$

A to C, $t = t_0 + t_0 = 2t_0$

$$s_1 + s_2 = 0 + \frac{1}{2} a(t_0 + t_0)^2 = 2 at_0^2$$

Displacement in the next t_0 second,

$$s_2 = \frac{3}{2} at_0^2$$

A to D, $t = t_0 + t_0 + t_0 = 3t_0$

$$s_1 + s_2 + s_3 = 0 + \frac{1}{2} a(t_0 + t_0 + t_0)^2 = \frac{9}{2} at_0^2$$

Displacement in the next t_0 second,

$$s_3 = \frac{5}{2} at_0^2 \Rightarrow s_1 : s_2 : s_3 = 1 : 3 : 5$$

CHECK POINT 3.4

- Velocity of a body moving along a straight line with uniform acceleration (a) reduces by $\frac{3}{4}$ of its initial velocity in time t_0 . The total time of motion of the body till its velocity becomes zero is
 (a) $\frac{4}{3}t_0$ (b) $\frac{3}{2}t_0$
 (c) $\frac{5}{3}t_0$ (d) $\frac{8}{3}t_0$
- The displacement of a body in 8 s starting from rest with an acceleration of 20 cm s^{-2} is
 (a) 64 m (b) 64 cm
 (c) 640 cm (d) 0.064 m
- The motion of a particle is described by the equation $v = at$. The distance travelled by the particle in the first 4 s is
 (a) $4a$ (b) $12a$
 (c) $6a$ (d) $8a$
- A particle starts with a velocity of 2 ms^{-1} and moves in a straight line with a retardation of 0.1 ms^{-2} . The first time at which the particle is 15 m from the starting point is
 (a) 10 s (b) 20 s
 (c) 30 s (d) 40 s
- A particle starts from rest, accelerates at 2 ms^{-2} for 10 s and then moves with constant speed of 20 ms^{-1} for 30 s and then decelerates at 4 ms^{-2} till it stops after next 5 s. What is the distance travelled by it?
 (a) 750 m (b) 800 m (c) 700 m (d) 850 m
- A body is moving with uniform velocity of 8 ms^{-1} . When the body just crossed another body, the second one starts and moves with uniform acceleration of 4 ms^{-2} . The time after which two bodies meet, will be
 (a) 2 s (b) 4 s
 (c) 6 s (d) 8 s
- Two bodies A and B start from rest from the same point with a uniform acceleration of 2 ms^{-2} . If B starts one second later, then the two bodies are separated at the end of the next second, by
 (a) 1 m (b) 2 m
 (c) 3 m (d) 4 m
- A train accelerating uniformly from rest attains a maximum speed of 40 ms^{-1} in 20 s. It travels at this speed for 20 s and is brought to rest by uniform retardation in further 40 s. What is the average velocity during this period?
 (a) $(80/3) \text{ ms}^{-1}$ (b) 40 ms^{-1}
 (c) 25 ms^{-1} (d) 30 ms^{-1}
- A particle starts from rest and traverses a distance l with uniform acceleration, then moves uniformly over a further distance $2l$ and finally comes to rest after moving a further distance $3l$ under uniform retardation. Assuming entire motion to be rectilinear motion, the ratio of average speed over the journey to the maximum speed on its ways is
 (a) $1/5$ (b) $2/5$ (c) $3/5$ (d) $4/5$
- A body travelling with uniform acceleration crosses two points A and B with velocities 20 ms^{-1} and 30 ms^{-1} , respectively. The speed of the body at mid-point of A and B is
 (a) 25 ms^{-1} (b) 25.5 ms^{-1}
 (c) 24 ms^{-1} (d) $10\sqrt{6} \text{ ms}^{-1}$
- If a body starts from rest and travels 120 cm in the 6th second, then what is the acceleration?
 (a) 0.20 ms^{-2} (b) 0.027 ms^{-2}
 (c) 0.218 ms^{-2} (d) 0.03 ms^{-2}

MOTION UNDER GRAVITY

The objects falling towards the earth under the influence of gravitational force alone, are called freely falling objects and such fall is called **free fall**.

Whenever an object falls towards the earth, an acceleration is involved, this acceleration is due to the earth's gravitational pull and is called **acceleration due to gravity**. The value of acceleration due to gravity near the earth surface is 9.8 ms^{-2} . It is independent of the mass of freely falling objects and is denoted by g .

Though the value of g is independent of freely falling mass, a feather falls much slowly than a coin when released from a height. This is due to the resistance offered by air to the falling mass. If both the bodies were released at the same time in vacuum (no air resistance), they would reach the earth surface within the same duration of time.

Equations for motion under gravity

When the objects fall under the influence of gravity, its motion is uniformly accelerated motion. Hence, equations of motion are applicable in this case also.

Equation for motion under gravity are given below

(i) If particle is thrown vertically upwards

In this case, applicable kinematics relations are

$$v = u - gt \quad \dots(i)$$

$$h = ut - \frac{1}{2}gt^2 \quad \dots(ii)$$

$$v^2 = u^2 - 2gh \quad \dots(iii)$$

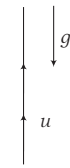


Fig. 3.7

Here, h is the vertical height of the particle in upward direction.

In this case, acceleration due to gravity is taken as negative.

At maximum height (say h), $v = 0$
 ($\because$ at maximum height, the particle stops moving upwards that means its velocity becomes zero)

$\therefore$ From the Eq. (i), $u = gt$

or $t = \frac{u}{g}$, which is called **time of ascent**.

For motion under gravity, for the same distance, the time taken to go up is same as time taken to fall down.

$\therefore$ Time of ascent = Time of descent

Total flight time, $T = 2 \times$ Time of ascent or descent

$$\boxed{\text{Total flight time (T)} = \frac{2u}{g}}$$

From Eqs. (ii) and (iii), we get

$$h = \frac{1}{2} gt^2$$

$$\text{and } u^2 = 2gh \text{ or } h = \frac{u^2}{2g}$$

(ii) **If particle is thrown vertically downward with some velocity from some height.**



Fig. 3.8

In this case, $v = u + gt$... (i)

$$h = ut + \frac{1}{2} gt^2 \quad \dots (ii)$$

$$v^2 = u^2 + 2gh \quad \dots (iii)$$

Here, h is the vertical height of particle in downward direction.

In this case, acceleration due to gravity is taken as positive.

(iii) **If a particle is dropped from some height.**

In this case, initial velocity is taken zero ($u = 0$), so equations of motion are

$$v = gt \quad \dots (i)$$

$$h = \frac{1}{2} gt^2 \quad \dots (ii)$$

$$v^2 = 2gh \quad \dots (iii) \quad \text{Fig. 3.9}$$

For fast calculation in objective problems, remember the following results

(a) Maximum height attained by a particle, thrown upwards from ground,

$$h = \frac{u^2}{2g}$$

(b) Velocity of particle at the time of striking the ground when released ($u = 0$) from a height h is

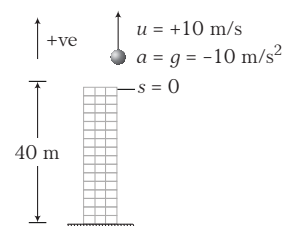
$$v = \sqrt{2gh}$$

(c) In above point (b), time of collision of particle with ground,

$$t = \sqrt{\frac{2h}{g}}$$

Example 3.24 A ball is thrown upwards from the top of a tower 40 m high with a velocity of 10 m/s. Find the time when it strikes the ground. (Take, $g = 10 \text{ m/s}^2$)

Sol. According to the question, the condition is as shown below.



Given, $u = +10 \text{ m/s}$, $a = -10 \text{ m/s}^2$ and $s = -40 \text{ m}$ (at the point, where stone strikes the ground)

Substituting in $s = ut + \frac{1}{2} at^2$, we get

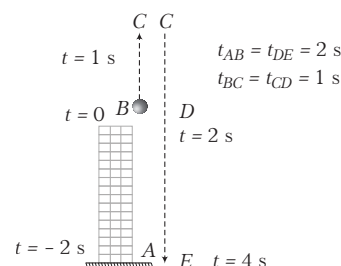
$$-40 = 10t - 5t^2 \text{ or } 5t^2 - 10t - 40 = 0$$

$$\text{or } t^2 - 2t - 8 = 0$$

Solving this, we have

$$t = 4 \text{ s and } -2 \text{ s. Taking the positive value } t = 4 \text{ s.}$$

Note The significance of $t = -2 \text{ s}$ can be understood by following figure



Example 3.25 A rocket is fired vertically up from the ground with a resultant vertical acceleration of 10 ms^{-2} . The fuel is finished in 1 min and it continues to move up.

(i) What is the maximum height reached?

(ii) After finishing fuel, calculate the time for which it continues its upwards motion. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. (i) The distance travelled by the rocket during burning of fuel (1 minute = 60 s) in which resultant acceleration is vertically upwards and is 10 ms^{-2} will be $h_1 = ut + \frac{1}{2}gt^2$

$$= 0 \times 60 + (1/2) \times 10 \times 60^2 = 18000 \text{ m} = 18 \text{ km}$$

and velocity acquired by it will be

$$v = u + at = 0 + 10 \times 60 = 600 \text{ ms}^{-1}$$

Now, after 1 min, the rocket moves vertically up with initial velocity of 600 ms^{-1} and acceleration due to gravity opposes its motion.

So, it will go to a height h_2 from this point, till its velocity becomes zero such that

$$v^2 - u^2 = -2gh \Rightarrow 0 - (600)^2 = -2gh_2 \quad (g = 10 \text{ ms}^{-2})$$

$$\text{or } h_2 = 18000 \text{ m} = 18 \text{ km}$$

So, the maximum height reached by the rocket from the ground, $H = h_1 + h_2 = 18 + 18 = 36 \text{ km}$

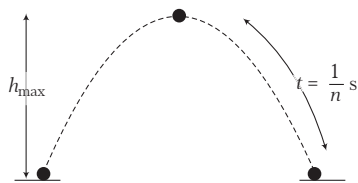
(ii) As after burning of fuel, the initial velocity attained will be 600 ms^{-1} and gravity opposes the motion of rocket, so from first equation of motion time taken by it till its velocity $v = 0$ is given as,

$$0 = 600 - gt \Rightarrow t = 60 \text{ s}$$

Example 3.26 A juggler throws balls into air. He throws one ball whenever the previous one is at its highest point. How high does the balls rise, if he throws n balls each second? Acceleration due to gravity is g .

Sol. Juggler throws n balls in one second, so time interval

between two consecutive throws is $t = \frac{1}{n} \text{ s}$



Each ball takes $\frac{1}{n} \text{ s}$ to reach maximum height.

$$\text{So, } h_{\max} = \frac{1}{2} \times gt^2 = \frac{1}{2} \times g \left(\frac{1}{n} \right)^2 \Rightarrow h_{\max} = \frac{g}{2n^2}$$

Example 3.27 From an elevated point A, a stone is projected vertically upwards. When the stone reaches a distance h below A, its velocity is double of what it was at a height h above A.

Show that the greatest height attained by the stone is $\frac{5}{3}h$.

Sol. Let u be the velocity with which the stone is projected vertically upwards.

$$\text{Given, } v_{-h} = 2v_h \text{ or } (v_{-h})^2 = 4v_h^2$$

According to the kinematic equation,

$$v_h^2 = u^2 - 2gh$$

$$\text{and } v_{-h}^2 = u^2 - 2g(-h)$$

$$\Rightarrow u^2 - 2g(-h) = 4(u^2 - 2gh)$$

$$\therefore u^2 = \frac{10gh}{3}$$

$$\text{Now, maximum height, } h_{\max} = \frac{u^2}{2g} = \frac{5h}{3}$$

Example 3.28 A ball is thrown vertically upwards with a velocity of 20 ms^{-1} from the top of a multistorey building. The height of the point from where the ball is thrown is 25 m from the ground. How long it will take before the ball hits the ground? (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Let us take the positive Y-axis in the vertically upward direction with zero at the ground.

$$\text{Now, } v_0 = +20 \text{ ms}^{-1}, a = -g = -10 \text{ ms}^{-2}, v = 0 \text{ ms}^{-1}$$

The total time taken can also be calculated by noting the coordinates of initial and final positions of the ball with respect to the origin chosen and using equation

$$y = y_0 + v_0t + \frac{1}{2}at^2$$

Now, $y_0 = 25 \text{ m}$, $y = 0 \text{ m}$,

$$v_0 = 20 \text{ ms}^{-1}, a = -10 \text{ ms}^{-2}$$

$$\therefore 0 = 25 + 20t + \left(\frac{1}{2} \right) (-10)t^2$$

$$\Rightarrow 5t^2 - 20t - 25 = 0$$

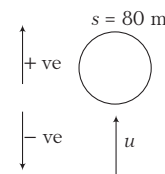
Solving this quadratic equation for t , we get

$$t = 5 \text{ s}$$

Example 3.29 A ball is thrown upwards from the ground with an initial speed u . The ball is at a height of 80 m at two times, for the time interval of 6 s . Find the value of u .

Sol. Here, $a = g = -10 \text{ ms}^{-2}$ and $s = 80 \text{ m}$

Substituting the values in $s = ut + \frac{1}{2}at^2$,



$$\text{we get } 80 = ut - 5t^2$$

$$\text{or } 5t^2 - ut + 80 = 0$$

$$\text{or } t = \frac{u + \sqrt{u^2 - 1600}}{10}$$

$$\text{and } t = \frac{u - \sqrt{u^2 - 1600}}{10}$$

Now, it is given that

$$\frac{u + \sqrt{u^2 - 1600}}{10} - \frac{u - \sqrt{u^2 - 1600}}{10} = 6$$

$$\Rightarrow \frac{\sqrt{u^2 - 1600}}{5} = 6$$

$$\Rightarrow \sqrt{u^2 - 1600} = 30$$

$$\Rightarrow u^2 - 1600 = 900$$

$$\therefore u^2 = 2500$$

$$\Rightarrow u = \pm 50 \text{ ms}^{-1}$$

Ignoring the negative sign, we get

$$u = 50 \text{ ms}^{-1}$$

Example 3.30 A particle is thrown vertically upwards from the surface of the earth. Let T_P be the time taken by the particle to travel from a point P above the earth to its highest point and back to the point P .

Similarly, let T_Q be the time taken by the particle to travel from another point Q above the earth to its highest point and back to the same point Q . If the distance between the points P and Q is H , find the expression for acceleration due to gravity in terms of T_P , T_Q and H .

Sol. Time taken by the particle to travel from point P back to point P ,

$$T_P = T_{PR} + T_{RP}$$

Here, $T_{PR} = T_{RP}$, then $T_P = 2T_{PR}$

Using second equation of motion,

$$(H + h) = \frac{1}{2} g T_{PR}^2$$

$$\Rightarrow T_{PR} = \sqrt{2 \frac{(H + h)}{g}}$$

$$\Rightarrow T_P = 2 \sqrt{2 \frac{(H + h)}{g}}$$

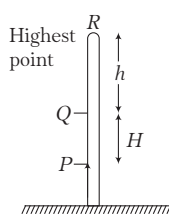
Then, similarly time taken by the particle to travel from point Q back to point Q ,

$$T_Q = 2 \sqrt{2 \frac{h}{g}}$$

$$T_P^2 = \frac{8(h + H)}{g}$$

and $T_Q^2 = \frac{8h}{g}$

$$\Rightarrow T_P^2 = T_Q^2 + \frac{8H}{g} \Rightarrow g = \frac{8H}{T_P^2 - T_Q^2}$$



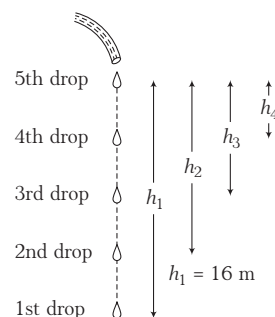
Example 3.31 From the top of a building, 16 m high water drops are falling at equal intervals of time such that when the first drop reaches the ground, the fifth drop just starts. Find the distance between the successive drops at that instant.

Sol. Let the interval of time be t_0 .

First drop is released at $t = 0$, second drop at $t = t_0$, third drop at $t = 2t_0$, fourth drop at $t = 3t_0$, fifth drop at $t = 4t_0$.

Therefore, first drop has fallen for time $4t_0$, second drop for

$3t_0$, third drop for $2t_0$, fourth drop for t_0 when fifth drop is about to fall. The location of drops are as shown in the figure.



For 1st drop,

$$h_1 = \frac{1}{2} g (4t_0)^2, \quad 16 = \frac{1}{2} g \times 16t_0^2$$

$$\frac{1}{2} g t_0^2 = 1 \text{ m}$$

For 2nd drop, $h_2 = \frac{1}{2} g (3t_0)^2 = 9 \text{ m}$

For 3rd drop, $h_3 = \frac{1}{2} g (2t_0)^2 = 4 \text{ m}$

For 4th drop, $h_4 = \frac{1}{2} g t_0^2 = 1 \text{ m}$

For 5th drop, $h_5 = 0$

Separation between drops

1st and 2nd : $h_1 - h_2 = 7 \text{ m}$

2nd and 3rd : $h_2 - h_3 = 5 \text{ m}$

3rd and 4th : $h_3 - h_4 = 3 \text{ m}$

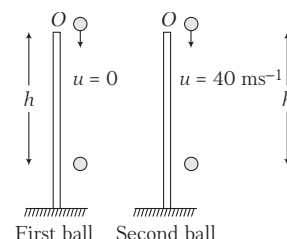
4th and 5th : $h_4 - h_5 = 1 \text{ m}$

Note If the 1st drop is at the ground and the 5th drop is about to fall, the time for which the first drop has fallen $(5 - 1)t_0 = 4t_0$, where t_0 is the regular interval of time.

Example 3.32 A ball is dropped from the top of a tower.

After 2s another ball is thrown vertically downwards with a speed of 40 ms^{-1} . After how much time and at what distance below the top of tower the balls meet?

Sol. Let the balls meet at distance h below the top of tower at t second after dropping of first ball. The second ball takes time $(t - 2)$ seconds.



For first ball,

$$h = \frac{1}{2} g t^2 \quad \dots(i)$$

$$\text{For second ball, } h = 40(t - 2) + \frac{1}{2} g (t - 2)^2 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$40(t-2) + \frac{1}{2}g(t-2)^2 = \frac{1}{2}gt^2$$

$$40(t-2) = \frac{1}{2}g[t^2 - (t-2)^2]$$

$$40(t-2) = \frac{1}{2} \times 10(2t-2) \times 2$$

$$4t - 8 = 2t - 2 \Rightarrow t = 3 \text{ s}$$

Distance below the top of tower, where the balls meet,

$$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 3^2 = 45 \text{ m}$$

NON-UNIFORMLY ACCELERATED MOTION

When acceleration of particle is not constant, motion is known as non-uniformly accelerated motion. Then, basic equations of velocity and acceleration can be written as

$$(i) \mathbf{v} = \frac{d\mathbf{s}}{dt} \text{ or sometimes } \mathbf{v} = \frac{d\mathbf{r}}{dt}$$

$$(ii) \mathbf{a} = \frac{d\mathbf{v}}{dt}$$

$$(iii) d\mathbf{s} = \mathbf{v} dt$$

$$(iv) d\mathbf{v} = \mathbf{a} dt$$

For one dimensional motion, above relations can be written as under

$$(i) v = \frac{ds}{dt} \quad (ii) a = \frac{dv}{dt} = v \frac{dv}{ds}$$

$$(iii) ds = v dt \text{ and } (iv) dv = a dt \quad \text{or} \quad v dv = a ds$$

Such problems can be solved either by differentiation or integration (with some boundary conditions).

$$s-t \longrightarrow v-t \longrightarrow a-t \quad (\text{Differentiation})$$

$$a-t \longrightarrow v-t \longrightarrow s-t \quad (\text{Integration with boundary conditions})$$

Note

- (i) By boundary condition, we mean that velocity or displacement at some time (usually at $t = 0$) should be known to us. Otherwise we cannot find constant of integration.
- (ii) Equation $\mathbf{a} = \mathbf{v} dv/ds$ or $\mathbf{v} dv = \mathbf{a} ds$ is useful when acceleration-displacement equation is known and velocity-displacement equation is required.

Example 3.33 Velocity-time equation of a particle moving in a straight line is

$$v = (10 + 2t + 3t^2) \text{ (SI units)}$$

Find

- (i) displacement of particle from the mean position at time $t = 1 \text{ s}$, if it is given that displacement is 20 m at time $t = 0$.
- (ii) and acceleration-time equation.

Sol. (i) The given equation can be written as

$$v = \frac{ds}{dt} = (10 + 2t + 3t^2) \quad \dots(i)$$

$$\text{or} \quad ds = (10 + 2t + 3t^2) dt$$

$$\text{or} \quad \int_{20}^s ds = \int_0^t (10 + 2t + 3t^2) dt$$

$$\text{or} \quad s - 20 = [10t + t^2 + t^3]_0^t$$

$$\text{or} \quad s = 20 + 12 = 32 \text{ m}$$

(ii) Acceleration-time equation can be obtained by differentiating Eq. (i) w.r.t. time. Thus,

$$a = \frac{dv}{dt} = \frac{d}{dt} (10 + 2t + 3t^2)$$

$$\text{or} \quad a = 2 + 6t$$

Example 3.34 Displacement-time equation of a particle moving along X-axis is

$$x = 20 + t^3 - 12t \text{ (SI units)}$$

- (i) Find position and velocity of particle at time $t = 0$.
- (ii) State whether the motion is uniformly accelerated or not.
- (iii) Find position of particle when velocity of particle is zero.

Sol. (i) Given, displacement-time equation of a particle

$$x = 20 + t^3 - 12t \quad \dots(i)$$

$$\text{At } t = 0, \quad x = 20 + 0 - 0 = 20 \text{ m}$$

Velocity of particle at time t can be obtained by differentiating Eq. (i) w.r.t. time, i.e.

$$v = \frac{dx}{dt} = 3t^2 - 12 \quad \dots(ii)$$

$$\text{At } t = 0, \quad v = 0 - 12 = -12 \text{ m/s}$$

(ii) Differentiating Eq. (ii) w.r.t. time t , we get the acceleration,

$$a = \frac{dv}{dt} = 6t$$

As acceleration is a function of time, the motion is non-uniformly accelerated motion.

(iii) Substituting $v = 0$ in Eq. (ii), we get

$$0 = 3t^2 - 12$$

Positive value of t comes out to be 2 s from this equation. Substituting $t = 2 \text{ s}$ in Eq. (i), we get

$$x = 20 + (2)^3 - 12(2) \text{ or } x = 4 \text{ m}$$

Example 3.35 The velocity of particle moving in the positive direction of X-axis varies as $v = \alpha\sqrt{x}$, where α is a positive constant. Assuming that at moment $t = 0$, the particle was located at the point $x = 0$.

Find

- (i) the time dependence of the velocity of the particle.
- (ii) the mean velocity of the particle averaged over the time that the particle takes to cover first s metres of the path.

Sol. (i) Given, the velocity of the particle moving in the positive direction of X-axis,

$$v = \alpha\sqrt{x}$$

$$\Rightarrow \frac{dx}{dt} = \alpha\sqrt{x}$$

$$\text{By integrating } \int_0^x \frac{dx}{\sqrt{x}} = \alpha \int_0^t dt$$

$$\Rightarrow \frac{x^{-1/2+1}}{-(1/2)+1} = \alpha t$$

$$\Rightarrow x = \frac{\alpha^2 t^2}{4}$$

Time dependence of the velocity of the particle,

$$v = \frac{dx}{dt} = \frac{2\alpha^2 t}{4} = \frac{1}{2} \alpha^2 t$$

$$(ii) \text{ Distance, } x = \frac{\alpha^2 t^2}{4}; \text{ for } s \text{ distance } s = \frac{\alpha^2 t^2}{4}$$

$$\text{Time taken to cover first } s \text{ distance, } t = \sqrt{\frac{4s}{\alpha^2}}$$

The mean velocity of the particle,

$$v_{av} = \frac{s}{t} = \frac{s}{\sqrt{4s/\alpha^2}} = \frac{\alpha\sqrt{s}}{2}$$

CHECK POINT 3.5

- If a stone is thrown up with a velocity of 9.8 ms^{-1} , then how much time will it take to come back?
(a) 1 s (b) 2 s
(c) 3 s (d) 4 s
- If a ball is thrown vertically upwards with speed u , the distance covered during the last t second of its ascent is
(a) $ut - (gt^2/2)$ (b) $(u + g)t$
(c) ut (d) $gt^2/2$
- A person throws balls into air after every second. The next ball is thrown when the velocity of the first ball is zero. How high do the ball rise above his hand?
(a) 2 m (b) 5 m (c) 8 m (d) 10 m
- A particle is thrown vertically upwards. Its velocity at half of the height is 10 ms^{-1} . Then, the maximum height attained by it is (Take, $g = 10 \text{ ms}^{-2}$)
(a) 16 m (b) 10 m
(c) 20 m (d) 40 m
- When a ball is thrown up vertically with velocity v_0 , it reaches a maximum height of h . If one wishes to triple the maximum height, then the ball should be thrown with velocity,
(a) $\sqrt{3} v_0$ (b) $3 v_0$ (c) $9 v_0$ (d) $3/2 v_0$
- A stone thrown upward with a speed u from the top of the tower reaches the ground with a speed $3u$. The height of the tower is
(a) $3u^2/g$ (b) $4u^2/g$
(c) $6u^2/g$ (d) $9u^2/g$
- A body thrown vertically up from the ground passes the height of 10.2 m twice in an interval of 10 s. What was its initial velocity?
(a) 52 ms^{-1} (b) 61 ms^{-1}
(c) 45 ms^{-1} (d) 26 ms^{-1}
- A body is projected with a velocity u . It passes through a certain point above the ground after t_1 second. The time interval after which the body passes through the same point during the return journey is
(a) $\left(\frac{u}{g} - t_1\right)$ (b) $2\left(\frac{u}{g} - t_1\right)$
(c) $\left(\frac{u}{g} - t_1\right)$ (d) $\left(\frac{u^2}{g^2} - t_1\right)$
- A body is thrown vertically upwards from the top A of tower. It reaches the ground in t_1 second. If it is thrown vertically downwards from A with the same speed, it reaches the ground in t_2 second. If it is allowed to fall freely from A , then the time it takes to reach the ground is given by
(a) $t = \frac{t_1 + t_2}{2}$ (b) $t = \frac{t_1 - t_2}{2}$
(c) $t = \sqrt{t_1 t_2}$ (d) $t = \sqrt{\frac{t_1}{t_2}}$
- A man in a balloon rising vertically with an acceleration of 4.9 ms^{-2} releases a ball 2 s after the balloon is let go from the ground. The greatest height above the ground reached by the ball is (Take, $g = 9.8 \text{ ms}^{-2}$)
(a) 14.7 m (b) 19.6 m
(c) 9.8 m (d) 24.5 m
- A stone falls freely under gravity. The total distance covered by it in the last second of its journey equals the distance covered by it in first 3 s of its motion. The time for which stone remains in air, is
(a) 5 s (b) 12 s
(c) 15 s (d) 8 s
- A body falls from a height $h = 200 \text{ m}$. The ratio of distance travelled in each 2 s, during $t = 0$ to $t = 6 \text{ s}$ of the journey is
(a) 1 : 4 : 9 (b) 1 : 2 : 4
(c) 1 : 3 : 5 (d) 1 : 2 : 3
- A ball is released from height h and another from $2h$. The ratio of time taken by the two balls to reach the ground is
(a) $1 : \sqrt{2}$ (b) $\sqrt{2} : 1$
(c) 2 : 1 (d) 1 : 2
- A particle is dropped under gravity from rest from a height h ($g = 9.8 \text{ ms}^{-2}$) and it travels a distance $9h/25$ in the last second, the height h is
(a) 100 m (b) 1225 m
(c) 145 m (d) 167.5 m
- A body dropped from the top of a tower covers a distance $7x$ in the last second of its journey, where x is the distance covered in first second. How much time does it take to reach the ground?
(a) 3 s (b) 4 s (c) 5 s (d) 6 s
- The displacement (in metre) of a particle moving along X -axis is given by $x = 18t + 5t^2$. The average acceleration during the interval $t_1 = 2 \text{ s}$ and $t_2 = 4 \text{ s}$ is
(a) 13 ms^{-2} (b) 10 ms^{-2} (c) 27 ms^{-2} (d) 37 ms^{-2}

17. The displacement of a particle moving in a straight line is described by the relation $s = 6 + 12t - 2t^2$. Here, s is in metre and t is in second. The distance covered by particle in first 5 s is

(a) 20 m (b) 32 m
(c) 24 m (d) 26 m

18. The displacement of a particle moving in a straight line depends on time as $x = \alpha t^3 + \beta t^2 + \gamma t + \delta$.

The ratio of initial acceleration to its initial velocity depends on

(a) α and γ only (b) β and γ only
(c) α and β only (d) α only

19. The acceleration of a particle is increasing linearly with time t as bt . The particle starts from the origin with an initial velocity v_0 . The distance travelled by the particle in time t will be

(a) $v_0 t + \frac{1}{6} bt^3$ (b) $v_0 t + \frac{1}{3} bt^3$
(c) $v_0 t + \frac{1}{3} bt^2$ (d) $v_0 t + \frac{1}{2} bt^2$

20. The acceleration a (in ms^{-2}), of a particle is given by $a = 3t^2 + 2t + 2$, where t is the time. If the particle starts out with a velocity $v = 2 \text{ ms}^{-1}$ at $t = 0$, then the velocity at the end of 2 s is

(a) 12 ms^{-1} (b) 14 ms^{-1}
(c) 16 ms^{-1} (d) 18 ms^{-1}

21. A particle is moving such that $s = t^3 - 6t^2 + 18t + 9$, where s is in metre and t is in second. The minimum velocity attained by the particle is

(a) 29 ms^{-1} (b) 5 ms^{-1}
(c) 6 ms^{-1} (d) 12 ms^{-1}

GRAPHICAL REPRESENTATION OF MOTION

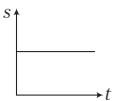
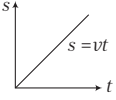
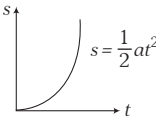
Motion of a point or body or a particle in all aspects can be shown with the help of the graph, such as displacement-time graph and velocity-time graph, etc.

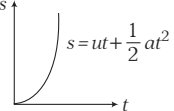
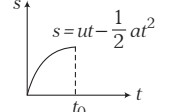
Displacement-time and velocity-time graphs for one dimensional motion are shown in tabular forms.

1. Displacement-time graph

- Displacement-time graph gives instantaneous value of displacement at any instant.
- The slope of tangent drawn to the graph at any instant of time gives the instantaneous velocity at that instant.
- The s - t graph cannot make sharp turns.

Different cases of displacement-time graph

Different cases	s - t graph	Main features of graph
At rest		Slope = $v = 0$
Uniform motion		Slope = constant, $v = \text{constant}$ or $a = 0$
Uniformly accelerated motion with $u = 0$, $s = 0$ at $t = 0$		$u = 0$, i.e. slope of s - t graph at $t = 0$ should be zero.

Different cases	s - t graph	Main features of graph
Uniformly accelerated motion with $u \neq 0$ but $s = 0$ at $t = 0$		Slope of s - t graph gradually goes on increasing.
Uniformly retarded motion		θ is decreasing, so v is decreasing, a is negative.

2. Velocity-time graph

- Velocity-time graph gives the instantaneous value of velocity at any instant.
- The slope of tangent drawn on graph gives instantaneous acceleration.
- Area under v - t graph with time axis gives the value of displacement covered in given time.
- The v - t curve cannot take sharp turns.

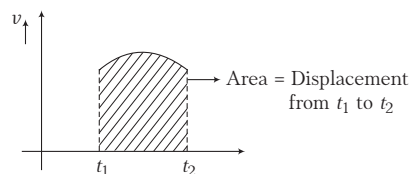
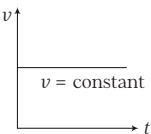
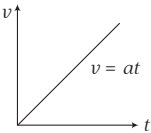
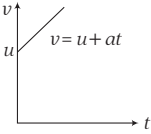
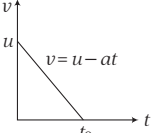
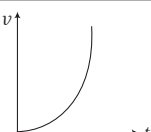
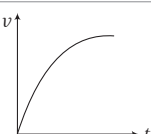


Fig. 3.10

Different cases in velocity-time graph

Different cases	v-t graph	Main features of graph
Uniform motion		(i) $\theta = 0^\circ$ (ii) $v = \text{constant}$ (iii) Slope of v-t graph = $a = 0$
Uniformly accelerated motion with $u = 0$ and $s = 0$ at $t = 0$		Slope of v-t graph is constant, so $a = \text{constant}$, $u = 0$, i.e. $v = 0$ at $t = 0$
Uniformly accelerated motion with $u \neq 0$ but $s = 0$ at $t = 0$		Positive constant acceleration because θ is constant and $< 90^\circ$ but the initial velocity of the particle is positive.
Uniformly retarded motion		Slope of v-t graph = $-a$ (retardation)
Non-uniformly accelerated motion		Slope of v-t graph increases with time. θ is increasing, so acceleration is increasing.
Non-uniformly decelerating motion		θ is decreasing, so acceleration is decreasing.

Note

- (i) Slope of s-t or v-t graphs can never be infinite at any point because infinite slope of s-t graphs means infinite velocity. Similarly, infinite slope of v-t graph means infinite acceleration. Hence, the following graphs are not possible.

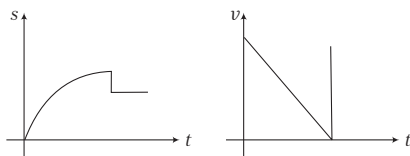


Fig. 3.11

- (ii) At one time, two values of velocity or displacement are not possible. Hence, the following graphs are not acceptable.

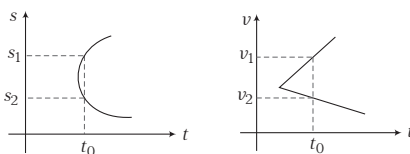


Fig. 3.12

3. Acceleration-time graph

The area of the a-t graph between time t_1 to t_2 gives the change in velocity.

$$\text{As, } a = \frac{dv}{dt}$$

$$\Rightarrow dv = a dt$$

$$\Rightarrow \int_{v_1}^{v_2} dv = \int_{t_1}^{t_2} a dt$$

$$v_2 - v_1 = \int_{t_1}^{t_2} a dt$$

Change in velocity = Area of the a-t graph

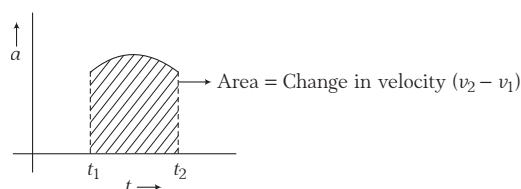
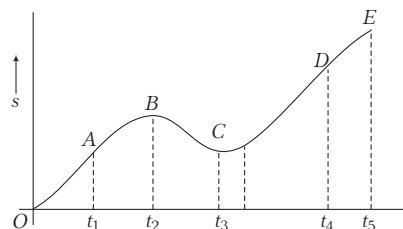
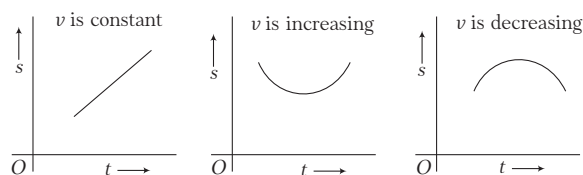


Fig. 3.13 Acceleration-time graph

Example 3.36 A particle is moving along the X-axis and its position-time graph is shown. Determine the sign of acceleration.



Sol. By observing the s-t graph, we can determine the sign of acceleration. Recall, if the graph is concave upwards, the slope is increasing; if it is concave downwards, the slope is decreasing; and if the graph is straight line, the slope is constant.



OA : Slope is increasing, v is increasing and a is positive.

AB : Slope is constant, v is constant and $a = 0$.

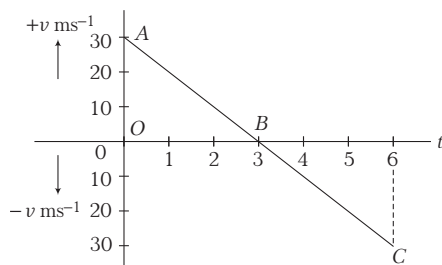
BC : Slope is decreasing, v is decreasing and a is negative.

CD : Slope is increasing, v is increasing and a is positive.

DE : Slope is constant, v is constant and $a = 0$.

Example 3.37 With the help of the given velocity-time graph, find the

- (i) displacement in first three seconds and
(ii) acceleration



Sol. (i) Displacement in first three seconds = Area of triangle OAB

$$= \frac{1}{2} (OB) \times (OA) = \frac{1}{2} (3) \times (30) = +45 \text{ m}$$

(ii) Acceleration = Slope of v - t graph

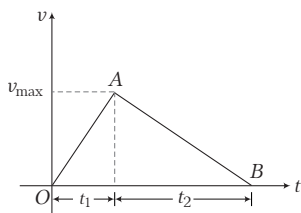
As, v - t graph is a straight line. So, consider the slope of line AB .

$$\therefore \text{Slope of line } AB = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - 30}{3} = -10 \text{ ms}^{-2}$$

So, the acceleration is negative.

Example 3.38 A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β to come to rest. If the total time elapsed is t second evaluate (i) the maximum velocity reached and (ii) the total distance travelled.

Sol. (i) Let the car accelerates for time t_1 and decelerates for time t_2 . Then, $t = t_1 + t_2$... (i)
and corresponding velocity-time graph will be as shown in figure.



From the graph,

$$\alpha = \text{slope of line } OA = \frac{v_{\max}}{t_1}$$

$$\text{or } t_1 = \frac{v_{\max}}{\alpha} \quad \dots \text{(ii)}$$

$$\text{and } \beta = -\text{slope of line } AB = \frac{v_{\max}}{t_2}$$

$$\text{or } t_2 = \frac{v_{\max}}{\beta} \quad \dots \text{(iii)}$$

From Eqs. (i), (ii) and (iii), we get

$$\frac{v_{\max}}{\alpha} + \frac{v_{\max}}{\beta} = t$$

$$\text{or } v_{\max} \left(\frac{\alpha + \beta}{\alpha\beta} \right) = t \quad \text{or } v_{\max} = \frac{\alpha\beta t}{\alpha + \beta}$$

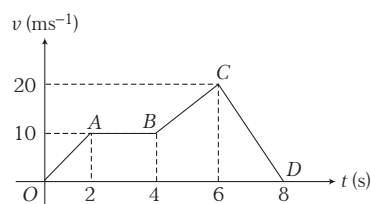
(ii) Total distance = Displacement = Area under v - t graph

$$= \frac{1}{2} \times t \times v_{\max}$$

$$= \frac{1}{2} \times t \times \frac{\alpha\beta t}{\alpha + \beta}$$

$$\text{or } \text{Distance} = \frac{1}{2} \left(\frac{\alpha\beta t^2}{\alpha + \beta} \right)$$

Example 3.39 Velocity-time graph of a particle moving in a straight line is shown in figure.



Plot the corresponding displacement-time graph of the particle, if at time $t = 0$, displacement $s = 0$.

Sol. Displacement = Area under velocity-time graph

$$\text{Hence, } s_{OA} = \frac{1}{2} \times 2 \times 10 = 10 \text{ m}$$

$$s_{AB} = 2 \times 10 = 20 \text{ m}$$

$$\text{or } s_{OAB} = 10 + 20 = 30 \text{ m}$$

$$s_{BC} = \frac{1}{2} \times 2(10 + 20) = 30 \text{ m}$$

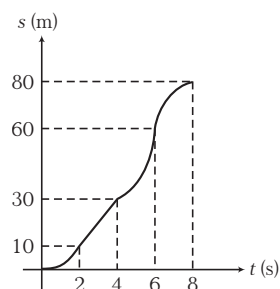
$$\text{or } s_{OABC} = 30 + 30 = 60 \text{ m}$$

$$\text{and } s_{CD} = \frac{1}{2} \times 2 \times 20 = 20 \text{ m}$$

$$\text{or } s_{OABCD} = 60 + 20 = 80 \text{ m}$$

Between 0 s to 2 s and 4 s to 6 s, motion is accelerated, hence displacement-time graph is a parabola. Between 2 s to 4 s, motion is uniform, so displacement-time graph will be a straight line. Between 6 s to 8 s, motion is decelerated, hence displacement-time graph is again a parabola but inverted in shape.

At the end of 8 s velocity is zero, therefore slope of displacement-time graph should be zero. The corresponding graph is shown in figure.



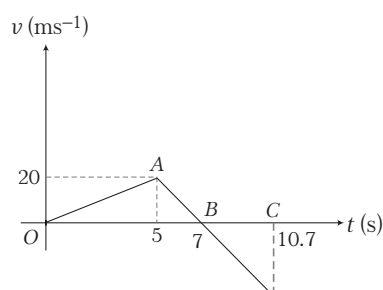
Example 3.40 A rocket is fired vertically upwards with a net acceleration of 4 ms^{-2} and initial velocity zero. After 5 s, its fuel is finished and it decelerates with g . At the highest point, its velocity becomes zero. Then, it accelerates downwards with acceleration g and return back to ground. Plot velocity-time and displacement-time graphs for the complete journey. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. In the graphs, $v_A = at_{OA} = (4)(5) = 20 \text{ ms}^{-1}$

$$\Rightarrow \quad v_B = 0 = v_A - gt_{AB}$$

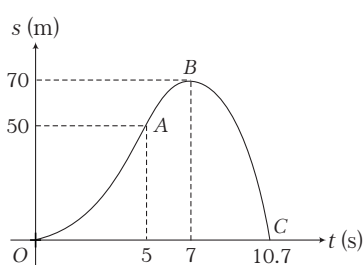
$$t_{AB} = \frac{v_A}{g} = \frac{20}{10} = 2 \text{ s}$$

$$\therefore \quad t_{OAB} = (5 + 2) \text{ s} = 7 \text{ s}$$



Now, s_{OAB} = area under v - t graph between 0 to 7 s

$$= \frac{1}{2} (7) (20) = 70 \text{ m}$$



Now, $|s_{OAB}| = |s_{BC}| = \frac{1}{2} gt_{BC}^2$

$$\therefore \quad 70 = \frac{1}{2} (10) t_{BC}^2$$

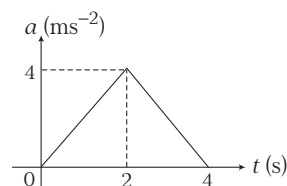
$$\Rightarrow \quad t_{BC} = \sqrt{14} = 3.7 \text{ s}$$

$$\therefore \quad t_{OABC} = 7 + 3.7 = 10.7 \text{ s}$$

Also, s_{OA} = area under v - t graph between OA

$$= \frac{1}{2} (5) (20) = 50 \text{ m}$$

Example 3.41 Acceleration-time graph of a particle moving in a straight line is shown in figure. Velocity of particle at time $t = 0$ is 2 ms^{-1} . Find velocity at the end of fourth second.

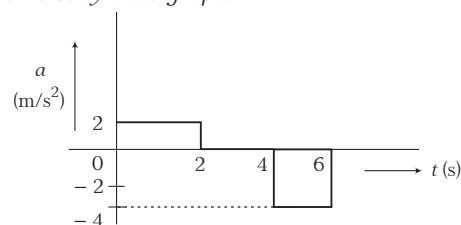


Sol. According to acceleration time-graph, $dv = a dt$
or change in velocity = area under a - t graph

Hence, $v_f - v_i = \frac{1}{2} (4) (4) = 8 \text{ ms}^{-1}$

$$\therefore \quad v_f = v_i + 8 = (2 + 8) \text{ ms}^{-1} = 10 \text{ ms}^{-1}$$

Example 3.42 The acceleration versus time graph of a particle moving along a straight line is shown in the figure. Draw the respective velocity-time graph.



Assume at $t = 0$, $v = 0$.

Sol. From $t = 0$ to $t = 2 \text{ s}$, $a = +2 \text{ m/s}^2 \Rightarrow v = at = 2t$

or v - t graph is a straight line passing through origin with slope 2 m/s^2 .

At the end of 2 s, $v = 2 \times 2 = 4 \text{ m/s}$

From $t = 2$ to 4 s, $a = 0$.

Hence, $v = 4 \text{ m/s}$ will remain constant.

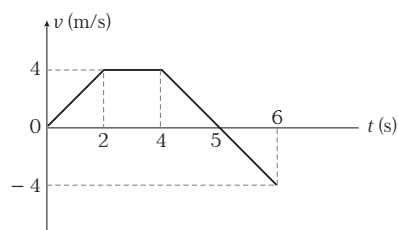
From $t = 4$ to 6 s, $a = -4 \text{ m/s}^2$.

Hence, $v = u - at = 4 - 4t$

$v = 0$ at $t = 1 \text{ s}$ or at 5 s from origin.

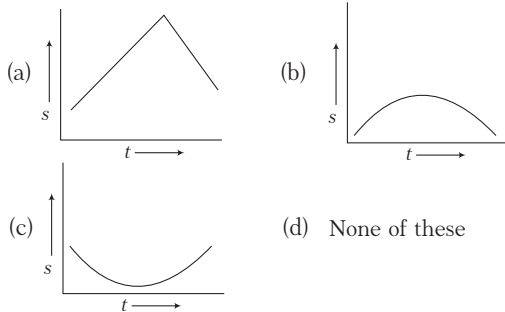
At the end of 6 s (or $t = 2 \text{ s}$) $v = -4 \text{ m/s}$.

Corresponding v - t graph as shown below.

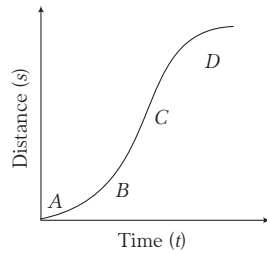


CHECK POINT 3.6

1. Which of the following graph represents the uniform motion?

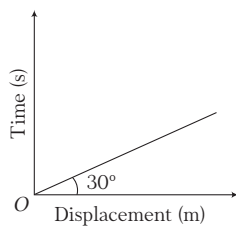


2. A particle shows distance-time curve as given in this figure. The maximum instantaneous velocity of the particle is around the point



- (a) A (b) B (c) C (d) D

3. From the displacement-time graph, find out the velocity of a moving body.

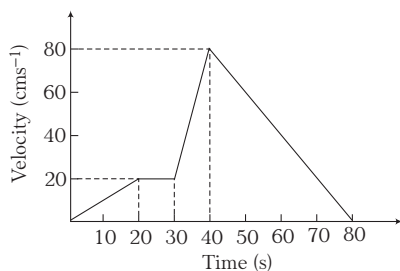


- (a) $\frac{1}{\sqrt{3}} \text{ ms}^{-1}$ (b) 3 ms^{-1} (c) $\sqrt{3} \text{ ms}^{-1}$ (d) $\frac{1}{3} \text{ ms}^{-1}$

4. The distance-time graph of a particle at time t makes angle 45° with the time axis. After one second, it makes angle 60° with the time axis. What is the average acceleration of the particle?

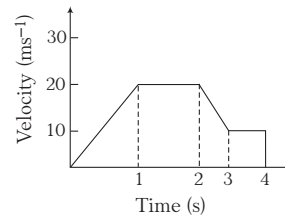
- (a) $\sqrt{3} - 1$ (b) $\sqrt{3} + 1$ (c) $\sqrt{3}$ (d) 1

5. The v - t graph of a moving object is shown in the figure. The maximum acceleration is



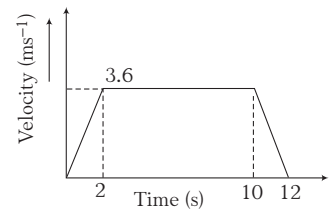
- (a) 1 cms^{-2} (b) 2 cms^{-2} (c) 3 cms^{-2} (d) 6 cms^{-2}

6. The variation of velocity of a particle with time moving along a straight line is illustrated in the adjoining figure. The distance travelled by the particle in 4 s is



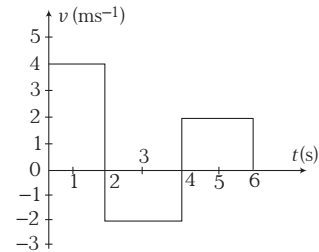
- (a) 60 m (b) 55 m (c) 25 m (d) 30 m

7. A lift is going up. The variation in the speed of the lift is as given in the graph. What is the height to which the lift takes the passengers?



- (a) 36 m
(b) 28.8 m
(c) 36.0 m
(d) Cannot be calculated from the above graph

8. The velocity-time graph of a body moving in a straight line is shown in the figure. The displacement and distance travelled by the body in 6 s are respectively

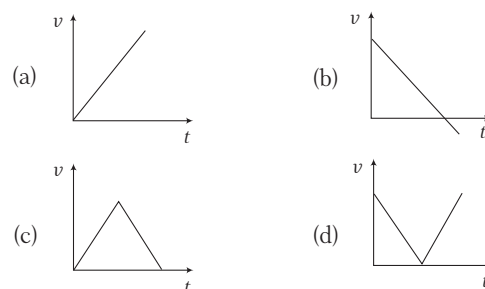


- (a) 8 m, 16 m (b) 16 m, 32 m
(c) 16 m, 16 m (d) 8 m, 18 m

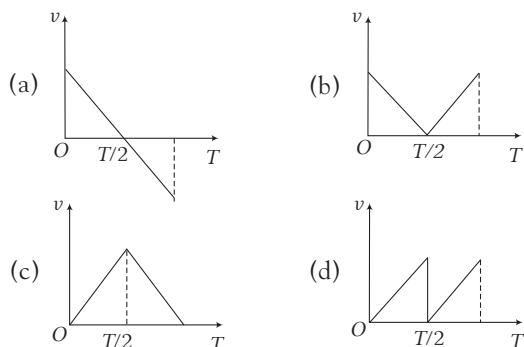
9. The x - t equation is given as $x = 2t + 1$. The corresponding v - t graph is

- (a) a straight line passing through origin
(b) a straight line not passing through origin
(c) a parabola
(d) None of the above

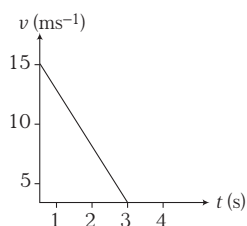
10. Which of the following graphs correctly represents velocity-time relationship for a particle released from rest to fall freely under gravity?



11. A particle projected vertically upwards returns to the ground in time T . Which graph represents the correct variation of velocity (v) against time (t)?

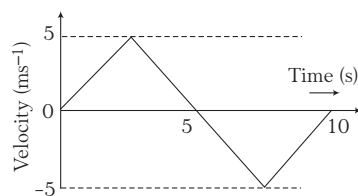


12. The velocity-time graph is shown in the figure, for a particle. The acceleration of particle is



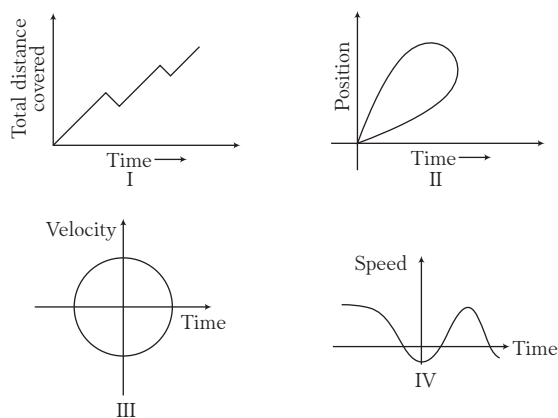
- (a) 225 ms^{-2} (b) 5 ms^{-2} (c) -5 ms^{-2} (d) -3 ms^{-2}

13. The v - t plot of a moving object is shown in the figure. The average velocity of the object during the first 10 s is



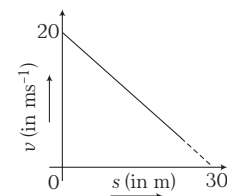
- (a) zero (b) 25 ms^{-1} (c) 5 ms^{-1} (d) 2 ms^{-1}

14. Which of the following graphs cannot possibly represent one dimensional motion of a particle?



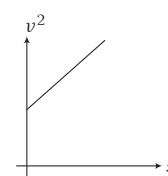
- (a) I and II (b) II and III (c) II and IV (d) All of these

15. If the velocity v of a particle moving along a straight line decreases linearly with its displacement s from 20 ms^{-1} to a value approaching zero at $s = 30 \text{ m}$, then acceleration of the particle at $v = 10 \text{ ms}^{-1}$ is



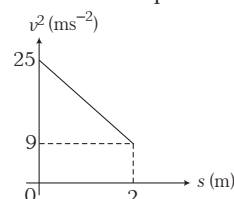
- (a) $\frac{2}{3} \text{ ms}^{-2}$ (b) $-\frac{2}{3} \text{ ms}^{-2}$
(c) $\frac{20}{3} \text{ ms}^{-2}$ (d) $-\frac{20}{3} \text{ ms}^{-2}$

16. v^2 versus s graph of a particle moving in a straight line is shown in the figure below. From the graph some conclusions are drawn. State which amongst the following statement(s) is wrong?



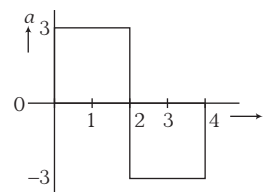
- (a) The given graph shows a uniformly accelerated motion.
(b) Initial velocity of particle is zero.
(c) Corresponding s - t graph will be a parabola.
(d) None of the above

17. A graph between the square of the velocity of a particle and the distance s moved by the particle is shown in the figure below. The acceleration of the particle is

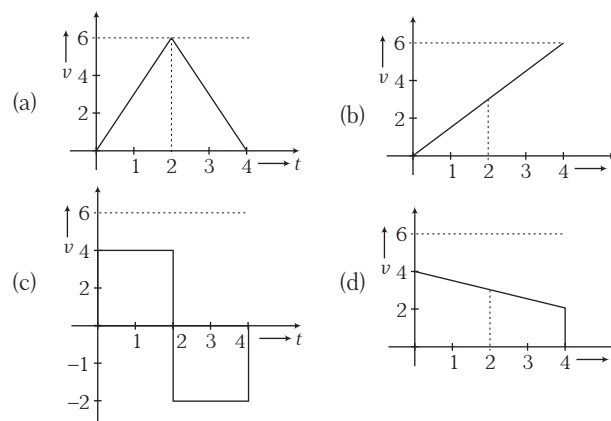


- (a) -8 ms^{-2} (b) -4 ms^{-2} (c) -16 ms^{-2} (d) None of these

18. A particle starts from rest at $t = 0$ and undergoes an acceleration a in ms^{-2} with time t in second which is as shown?



- Which one of the following plot represents velocity v (in ms^{-1}) versus time t (in s)?



RELATIVE VELOCITY

The term relative is frequently used for comparison of displacement, velocity and acceleration of the two objects. The time rate of change of relative position of one object with respect to another is called **relative velocity**.

Let two objects A and B are moving along the +ve direction on X -axis. At time t , their displacement from the origin be x_A and x_B .

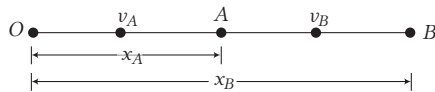


Fig. 3.14

$\therefore$ Their velocities are $\mathbf{v}_A = \frac{d\mathbf{x}_A}{dt}$ and $\mathbf{v}_B = \frac{d\mathbf{x}_B}{dt}$

The displacement of B relative to A ,

$$\mathbf{x}_{BA} = \mathbf{x}_B - \mathbf{x}_A$$

Rate of change of relative displacement w.r.t. time is

$$\frac{d(\mathbf{x}_{BA})}{dt} = \frac{d}{dt}(\mathbf{x}_B - \mathbf{x}_A) \Rightarrow \frac{d\mathbf{x}_{BA}}{dt} = \frac{d\mathbf{x}_B}{dt} - \frac{d\mathbf{x}_A}{dt}$$

$$\therefore \boxed{\mathbf{v}_{BA} = \mathbf{v}_B - \mathbf{v}_A}$$

Similarly, relative acceleration of A with respect to B is

$$\boxed{\mathbf{a}_{AB} = \mathbf{a}_A - \mathbf{a}_B}$$

If it is a one dimensional motion, we can treat the vectors as scalars just by assigning the positive sign to one direction and negative to the other. So, in case of a one dimensional motion, the above equations can be written as

$$v_{AB} = v_A - v_B$$

and

$$a_{AB} = a_A - a_B$$

Further, we can see that $\mathbf{v}_{AB} = -\mathbf{v}_{BA}$ or $\mathbf{a}_{BA} = -\mathbf{a}_{AB}$

Different cases of relative velocity

Case I When the two objects move with equal velocities, i.e. $v_A = v_B$ or $v_B - v_A = 0$. It means, the two objects stay at constant distance apart during the whole journey.

In this case, the position-time graph of two objects are parallel straight lines.

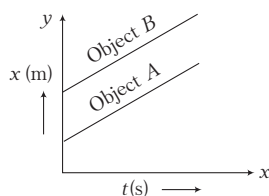


Fig. 3.15

Case II If both objects A and B move along parallel straight lines in the opposite direction, then relative velocity of B w.r.t. A is given as

$$\mathbf{v}_{BA} = \mathbf{v}_B - (-\mathbf{v}_A) = \mathbf{v}_B + \mathbf{v}_A$$

and the relative velocity of A w.r.t. B is given by

$$\mathbf{v}_{AB} = \mathbf{v}_A - (-\mathbf{v}_B) = \mathbf{v}_A + \mathbf{v}_B$$

Case III If $v_A < v_B$, $v_A - v_B$ is negative.

Then, $x - x_0 < 0$

$$\Rightarrow x < x_0$$

where, x_0 = initial displacement of object A w.r.t. B

and x = displacement of object A w.r.t. B at time t .

i.e. $(x - x_0)$ is negative. It means the separation between the two objects will go on decreasing and two objects will meet and object B will overtake object A at this time. In this case, relative velocity of A w.r.t. B , i.e.

$$\begin{aligned} v_{AB} &= v_A - v_B \\ &= -v_{BA} \end{aligned}$$

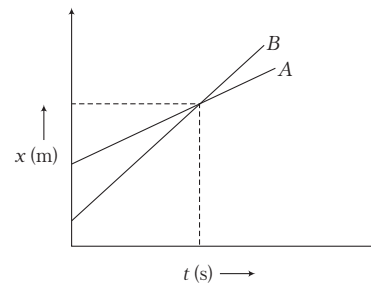


Fig. 3.16

Case IV If $v_A > v_B$, $v_A - v_B$ is positive.

Then, $x - x_0 > 0$, i.e. $(x - x_0)$ is positive.

It means the separation between the two objects will go on increasing with time, i.e. the separation $(x - x_0)$ between them will increase by an amount $(v_A - v_B)$ after each unit of time.

Therefore, their position-time graphs will open out gradually as shown below.

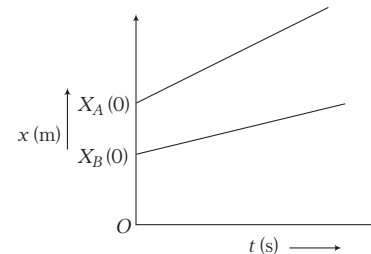


Fig. 3.17

Example 3.43 Seeta is moving due east with a velocity of 1 ms^{-1} and Geeta is moving due west with a velocity of 2 ms^{-1} . What is the velocity of Seeta with respect to Geeta?

Sol. It is a one dimensional motion. So, let us choose the east direction as positive and the west as negative.
Now, given that

$$v_S = \text{velocity of Seeta} = 1 \text{ ms}^{-1}$$

$$\text{and } v_G = \text{velocity of Geeta} = -2 \text{ ms}^{-1}$$

$$\text{Thus, } v_{SG} = \text{velocity of Seeta with respect to Geeta} \\ = v_S - v_G = 1 - (-2) = 3 \text{ ms}^{-1}$$

Hence, velocity of Seeta with respect to Geeta is 3 m/s due east.

Example 3.44 A man A moves due east with velocity 6 ms^{-1} and another man B moves in $N-30^\circ E$ with 6 ms^{-1} . Find the velocity of B w.r.t. A.

Sol. Given, $\mathbf{v}_A = 6\hat{\mathbf{i}}$,
 $\mathbf{v}_B = v_B \cos 60^\circ \hat{\mathbf{i}} + v_B \sin 60^\circ \hat{\mathbf{j}}$
 $= 6\left(\frac{1}{2}\right)\hat{\mathbf{i}} + 6\left(\frac{\sqrt{3}}{2}\right)\hat{\mathbf{j}} = 3\hat{\mathbf{i}} + 3\sqrt{3}\hat{\mathbf{j}}$

To find the velocity,

$$\mathbf{v}_{BA} = \mathbf{v}_B - \mathbf{v}_A = (3\hat{\mathbf{i}} + 3\sqrt{3}\hat{\mathbf{j}}) - 6\hat{\mathbf{i}} \\ = -3\hat{\mathbf{i}} + 3\sqrt{3}\hat{\mathbf{j}}$$

$$|\mathbf{v}_{BA}| = \sqrt{(-3)^2 + (3\sqrt{3})^2} \\ = \sqrt{9 + 27} = \sqrt{36} = 6 \text{ ms}^{-1}$$

Here, $\hat{\mathbf{i}}$ is $-ve$ and $\hat{\mathbf{j}}$ is $+ve$. So, second quadrant is possible.

Direction,

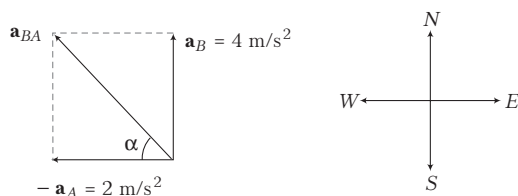
$$\tan \alpha = \frac{\text{coefficient of } \hat{\mathbf{j}}}{\text{coefficient of } \hat{\mathbf{i}}} = \frac{3\sqrt{3}}{-3} = -\sqrt{3}$$

$$\Rightarrow \alpha = -60^\circ$$

Example 3.45 Car A has an acceleration of 2 m/s^2 due east and car B, 4 m/s^2 due north. What is the acceleration of car B with respect to car A?

Sol. It is a two dimensional motion, therefore

$$\mathbf{a}_{BA} = \text{acceleration of car B with respect to car A} \\ = \mathbf{a}_B - \mathbf{a}_A$$



Here, $\mathbf{a}_B$ = acceleration of car B = 4 m/s^2 (due north)

and $\mathbf{a}_A$ = acceleration of car A = 2 m/s^2 (due east)

$$|\mathbf{a}_{BA}| = \sqrt{(4)^2 + (2)^2} \\ = 2\sqrt{5} \text{ m/s}^2$$

$$\text{and } \alpha = \tan^{-1}\left(\frac{4}{2}\right) = \tan^{-1}(2)$$

Thus, $\mathbf{a}_{BA}$ is $2\sqrt{5} \text{ m/s}^2$ at an angle of $\alpha = \tan^{-1}(2)$ from west towards north.

Example 3.46 A police van moving on a highway with a speed of 30 kmh^{-1} fires a bullet at a thief car which is speeding away in the same direction with a speed of 190 kmh^{-1} . If the muzzle speed of the bullet is 150 ms^{-1} , find speed of the bullet with respect to the thief's car.

Sol. Let v_b is velocity of bullet, v_p is velocity of police van and v_t is velocity of thief's car.

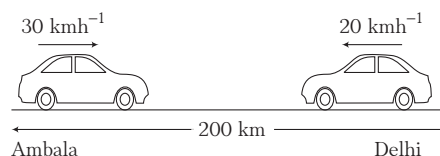
Then, speed of the bullet with respect to the thief's car,

$$v_{bp} = v_b - v_p \\ v_b = v_{bp} + v_p = 150 \times \frac{18}{5} \text{ kmh}^{-1} + 30 \text{ kmh}^{-1} = 570 \text{ kmh}^{-1} \\ v_{bt} = v_b - v_t = 570 \text{ kmh}^{-1} - 190 \text{ kmh}^{-1} = 380 \text{ kmh}^{-1}$$

Example 3.47 Delhi is at a distance of 200 km from Ambala. Car A set out from Ambala at a speed of 30 kmh^{-1} and car B set out at the same time from Delhi at a speed of 20 kmh^{-1} . When will they meet each other? What is the distance of the meeting point from Ambala?

Sol. Relative velocity, $\mathbf{v}_{AB} = \mathbf{v}_A - \mathbf{v}_B$

$$= 30 - (-20) = 50 \text{ kmh}^{-1}$$



They will meet after time

$$t = \frac{s}{v_{AB}} = \frac{200}{50} = 4 \text{ h}$$

Distance from Ambala, where they will meet,

$$x = 30 \times 4 = 120 \text{ km}$$

Examples of relative motion

1. Relative velocity of rain

Consider a man walking east with velocity $\mathbf{v}_m$ represented by OA . Let the rain be falling vertically downwards with velocity $\mathbf{v}_r$, represented by OB . To find the relative velocity of rain w.r.t. man (i.e. $\mathbf{v}_{rm}$) being the man at rest by imposing a velocity $-\mathbf{v}_m$ on man and apply this velocity on rain also.

Now, the relative velocity of rain w.r.t. man will be the resultant velocity of $\mathbf{v}_r$ ($= OB$) and $-\mathbf{v}_m$ ($= OA$), which will be represented by diagonal OC of rectangle $OACB$.

$$\therefore \mathbf{v}_{rm} = \sqrt{v_r^2 + v_m^2 + 2v_r v_m \cos 90^\circ} = \sqrt{v_r^2 + v_m^2}$$

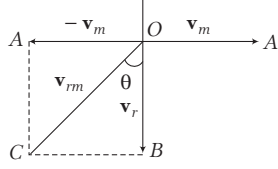


Fig. 3.18 Relative velocity of rain

If θ is the angle which $\mathbf{v}_{rm}$ makes with the vertical direction, then

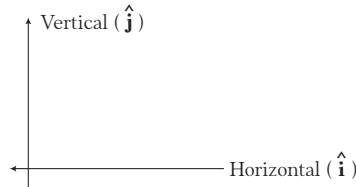
$$\tan \theta = \frac{BC}{OB} = \frac{v_m}{v_r} \quad \text{or} \quad \theta = \tan^{-1} \left(\frac{v_m}{v_r} \right)$$

Here, angle θ is from vertical towards west and is written as θ , west of vertical.

Note In the above problem, if the man wants to protect himself from the rain, he should hold his umbrella in the direction of relative velocity of rain w.r.t. man, i.e. the umbrella should be held making an angle θ from west of vertical.

Example 3.48 To a man walking at the rate of 3 km/h, the rain appears to fall vertically. When he increases his speed to 6 km/h, it appears to meet him at an angle of 45° with vertical. Find the speed of rain.

Sol. Let $\hat{i}$ and $\hat{j}$ be the unit vectors in horizontal and vertical directions, respectively.



Let velocity of rain,

$$\mathbf{v}_r = a\hat{i} + b\hat{j} \quad \dots(i)$$

Then, speed of rain will be

$$|\mathbf{v}_r| = \sqrt{a^2 + b^2} \quad \dots(ii)$$

In the first case, $\mathbf{v}_m = \text{velocity of man} = 3\hat{i}$

$$\therefore \mathbf{v}_{rm} = \mathbf{v}_r - \mathbf{v} = (a - 3)\hat{i} + b\hat{j}$$

It seems to be in vertical direction. Hence,

$$a - 3 = 0 \quad \text{or} \quad a = 3$$

In the second case $\mathbf{v}_{rm} = 6\hat{i}$

$$\therefore \mathbf{v}_{rm} = (a - 6)\hat{i} + b\hat{j} = -3\hat{i} + b\hat{j}$$

This seems to be at 45° with vertical. Hence, $|b| = 3$

Therefore, from Eq. (ii), speed of rain is

$$|\mathbf{v}_r| = \sqrt{(3)^2 + (3)^2} = 3\sqrt{2} \text{ km/h}$$

2. Crossing the river

To cross the river over shortest distance, i.e. to cross the river straight, the man should swim upstream making an angle θ with OB such that, OB gives the direction of resultant velocity ($\mathbf{v}_{mR}$) of velocity of swimmer and velocity of river water as shown in figure.

Let us consider

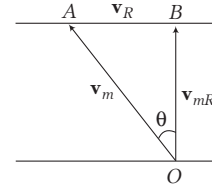


Fig. 3.19 Crossing the river

$AB = v_R$ (velocity of river water),

$OA = v_m$ (velocity of man in still river water)

and $OB = v_{mR}$ (relative velocity of man w.r.t. river).

$$\text{i.e.} \quad v_{mR} = \sqrt{v_m^2 - v_R^2}$$

$$\text{In } \triangle OAB, \quad \sin \theta = \frac{v_R}{v_m}$$

where, θ is the angle made by man with shortest distance OB ,

$$\tan \theta = \frac{v_R}{v_{mR}} = \frac{v_R}{\sqrt{v_m^2 - v_R^2}}$$

(i) **Time taken to cross the river** If d be the width of the river, then time taken cross to the river given by

$$t_1 = \frac{d}{v_{mR}} = \frac{d}{\sqrt{v_m^2 - v_R^2}}$$

(ii) **To cross the river in possible shortest time** The man should go along OA . Now, the swimmer will be going along OB , which is the direction of resultant velocity of $\mathbf{v}_m$ and $\mathbf{v}_R$.

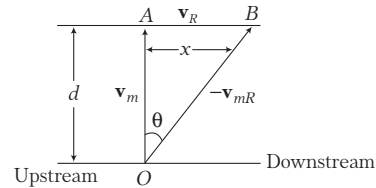


Fig. 3.20 Crossing the river in possible shortest time

$$\text{In } \triangle OAB, \quad \tan \theta = \frac{AB}{OA} = \frac{v_R}{v_m} \quad \text{and} \quad v_{mR} = \sqrt{v_m^2 + v_R^2}$$

Time of crossing the river,

$$t = \frac{d}{v_m} = \frac{OB}{v_{mR}} \Rightarrow t = \frac{\sqrt{x^2 + d^2}}{\sqrt{v_m^2 + v_R^2}}$$

The boat will be heading the point B instead of point A . If $AB = x$, then $\tan \theta = \frac{v_R}{v_m} = \frac{x}{d} \Rightarrow x = \frac{dv_R}{v_m}$

In this case, the man will reach the opposite bank at a distance AB downstream.

Drift

It is defined as the displacement of man in the direction of river flow as shown below.

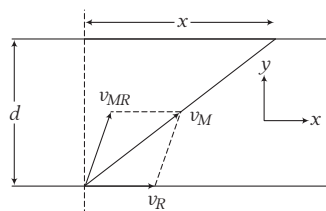


Fig. 3.21

It is simply the displacement along X -axis. During the period, the man crosses the river. $(v_{MR} \cos \theta + v_R)$ is the components of velocity of man in the direction of river flow and this component of velocity is responsible for drift along the river flow. If drift is x , then

$$x = (v_{MR} \cos \theta + v_R) \times \frac{d}{v_{MR} \sin \theta}$$

Note If $v_R > v_{MR}$, then it is not possible to have zero drift. In this case, the minimum drift corresponding to shortest possible path is non-zero and the condition for minimum drift can be proved to be $\cos \theta = -\frac{v_{MR}}{v_R}$ or $\sin \phi = \frac{v_{MR}}{v_R}$ for minimum but non-zero drift.

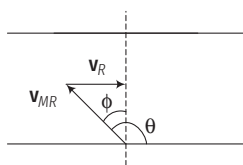


Fig. 3.22

Example 3.49 A man crosses a river in a boat. If he cross the river in minimum time, he takes 10 min with a drift 120 m. If he crosses the river taking shortest path, he takes 12.5 min, find

- width of the river,
- velocity of the boat with respect to water
- and speed of the current.

Sol. Let v_r = velocity of river,

v_{br} = velocity of boat in still water

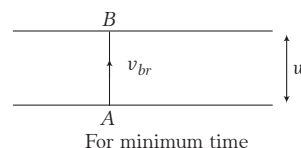
and w = width of river.

Given, $t_{\min} = 10 \text{ min}$ or $\frac{w}{v_{br}} = 10$... (i)

Drift in this case will be

$$x = v_r t$$

$\therefore 120 = 10 v_r$... (ii)



Shortest path is taken when v_b is along AB . In this case,

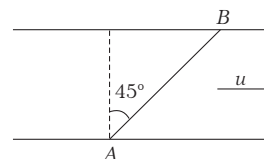
$$v_b = \sqrt{v_{br}^2 - v_r^2}$$

Now, $12.5 = \frac{w}{v_b} = \frac{w}{\sqrt{v_{br}^2 - v_r^2}}$... (iii)

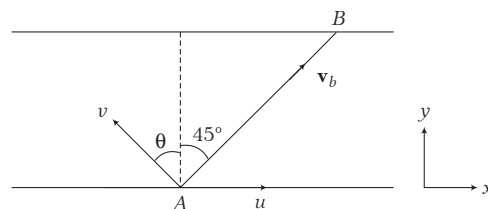
Solving these three equations, we get

$$v_{br} = 20 \text{ m/min, } v_r = 12 \text{ m/min and } w = 200 \text{ m}$$

Example 3.50 A man wants to reach point B on the opposite bank of a river flowing at a speed u as shown in figure. What minimum speed relative to water should the man have, so that he can reach point B ? In which direction should he swim?



Sol. Let v be the speed of boatman in still water.



Resultant of v and u should be along AB . Components of v_b (absolute velocity of boatman) along x and y -directions are

$$v_x = u - v \sin \theta \quad \text{and} \quad v_y = v \cos \theta$$

Further, $\tan 45^\circ = \frac{v_y}{v_x}$

or $1 = \frac{v \cos \theta}{u - v \sin \theta}$

$$\therefore v = \frac{u}{\sin \theta + \cos \theta} = \frac{u}{\sqrt{2} \sin (\theta + 45^\circ)}$$

v is minimum, at

$$\theta + 45^\circ = 90^\circ$$

or $\theta = 45^\circ$

and $v_{\min} = \frac{u}{\sqrt{2}}$

Example 3.51 A man can row a boat with 4 km/h in still water. If he is crossing a river, where the current is 2 km/h.

- In what direction will his boat be headed, if he wants to reach a point on the other bank, directly opposite to starting point?
- If width of the river is 4 km, how long will the man take to cross the river, with the condition in part (i)?
- In what direction should he head the boat, if he wants to cross the river in shortest time and what is this minimum time?
- How long will it take him to row 2 km up the stream and then back to his starting point?

Sol. (i) Given, $v_{br} = 4$ km/h and $v_r = 2$ km/h

$$\begin{aligned}\therefore \theta &= \sin^{-1} \left(\frac{v_r}{v_{br}} \right) \\ &= \sin^{-1} \left(\frac{2}{4} \right) \\ &= \sin^{-1} \left(\frac{1}{2} \right) = 30^\circ\end{aligned}$$

Hence, to reach the point directly opposite to starting point he should head the boat at an angle of 30° with AB or $90^\circ + 30^\circ = 120^\circ$ with the river flow.

- (ii) Time taken by the boatman to cross the river,

w = width of river = 4 km

$v_{br} = 4$ km/h

and $\theta = 30^\circ$

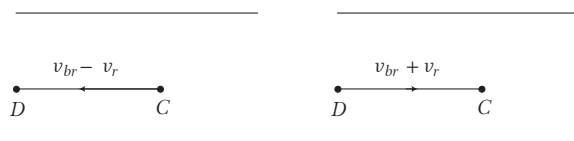
$$\therefore t = \frac{4}{4 \cos 30^\circ} = \frac{2}{\sqrt{3}} \text{ h}$$

- (iii) For shortest time $\theta = 0^\circ$

$$\text{and } t_{\min} = \frac{w}{v_{br} \cos 0^\circ} = \frac{4}{4} = 1 \text{ h}$$

Hence, he should head his boat perpendicular to the river current for crossing the river in shortest time and this shortest time is 1 h.

- (iv) $t = t_{CD} + t_{DC}$

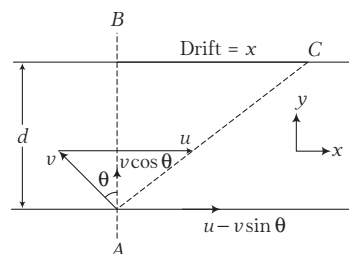


$$\begin{aligned}\text{or } t &= \frac{CD}{v_{br} - v_r} + \frac{DC}{v_{br} + v_r} \\ &= \frac{2}{4 - 2} + \frac{2}{4 + 2} = 1 + \frac{1}{3} = \frac{4}{3} \text{ h}\end{aligned}$$

Example 3.52 A boat moves relative to water with a velocity v is n times less than the river flow u . At what angle to the stream direction must the boat move to minimise drifting?

Sol. In this problem, one thing should be carefully noted that the velocity of boat is less than the river flow velocity. Hence, boat cannot reach the point directly opposite to its starting point, i.e. drift can never be zero.

Suppose boat starts at an angle θ from the normal direction up stream as shown.



Component of velocity of boat along the river,

$$v_x = u - v \sin \theta$$

and velocity perpendicular to the river,

$$v_y = v \cos \theta$$

Time taken to cross the river is

$$t = \frac{d}{v_y} = \frac{d}{v \cos \theta}$$

$$\text{Drift } x = (v_x) t = (u - v \sin \theta) \frac{d}{v \cos \theta}$$

$$= \frac{ud}{v} \sec \theta - d \tan \theta$$

The drift x is minimum when $\frac{dx}{d\theta} = 0$

$$\text{or } \left(\frac{ud}{v} \right) (\sec \theta \cdot \tan \theta) - d \sec^2 \theta = 0$$

$$\frac{u}{v} \sin \theta = 1 \Rightarrow \sin \theta = \frac{v}{u}$$

So, for minimum drift, the boat must move at an angle

$$\theta = \sin^{-1} \left(\frac{v}{u} \right) = \sin^{-1} \left(\frac{1}{n} \right)$$

from normal direction.

3. Minimum distance between two bodies in motion

When two bodies are in motion, the questions like, the minimum distance between them or the time when one body overtakes the other can be solved easily by the principle of relative motion. In these type of problems one body is assumed to be at rest and the relative motion of the other body is considered.

By assuming so two body problem is converted into one body problem and the solution becomes easy.

Following example will illustrate the statement.

Example 3.53 Car A and car B start moving simultaneously in the same direction along the line joining them. Car A moves with a constant acceleration $a = 4 \text{ ms}^{-2}$, while car B moves with a constant velocity $v = 1 \text{ ms}^{-1}$. At time $t = 0$, car A is 10 m behind car B. Find the time when car A overtakes car B.

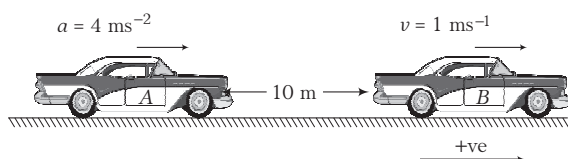
Sol. Given, $u_A = 0$, $u_B = 1 \text{ ms}^{-1}$, $a_A = 4 \text{ ms}^{-2}$ and $a_B = 0$

Assuming car B to be at rest, we get

$$u_{AB} = u_A - u_B = 0 - 1 = -1 \text{ ms}^{-1}$$

$$a_{AB} = a_A - a_B = 4 - 0 = 4 \text{ ms}^{-2}$$

Now, the problem can be assumed in simplified form as follows

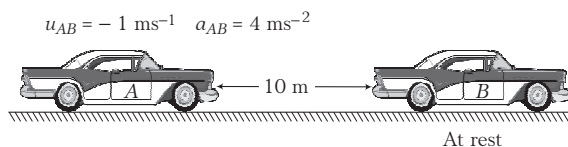


Substituting the proper values in equation

$$s = ut + \frac{1}{2}at^2, \text{ we get}$$

$$10 = -t + \frac{1}{2}(4)(t^2)$$

$$\text{or } 2t^2 - t - 10 = 0$$



$$\text{or } t = \frac{1 \pm \sqrt{1+80}}{4} = \frac{1 \pm \sqrt{81}}{4} = \frac{1 \pm 9}{4}$$

$$\text{or } t = 2.5 \text{ s and } -2 \text{ s}$$

Ignoring the negative value, the desired time is 2.5 s.

Note The above problem can also be solved without using the concept of relative motion as follows

At the time, when A overtakes B, $s_A = s_B + 10$

$$\therefore \frac{1}{2} \times 4 \times t^2 = 1 \times t + 10$$

$$\text{or } 2t^2 - t - 10 = 0$$

which on solving gives $t = 2.5 \text{ s}$ and -2 s , the same as we found above.

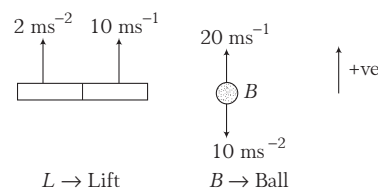
As per my opinion, this approach (by taking absolute values) is more suitable in case of two body problem in one dimensional motion. Let us see one more example in support of it.

Example 3.54 An open lift is moving upwards with velocity 10 ms^{-1} . It has an upward acceleration of 2 ms^{-2} . A ball is projected upwards with velocity 20 ms^{-1} relative to ground. Find

- time when ball again meets the lift,
 - displacement of lift and ball at that instant
 - and distance travelled by the ball upto that instant.
- (Take, $g = 10 \text{ ms}^{-2}$)

Sol. (i) At the time, when ball again meets the lift, $s_L = s_B$

$$\therefore 10t + \frac{1}{2} \times 2 \times t^2 = 20t - \frac{1}{2} \times 10t^2$$



Here,

$L \rightarrow$ Lift

$B \rightarrow$ Ball

Solving this equation, we get

$$t = 0 \quad \text{and} \quad t = \frac{5}{3} \text{ s}$$

$\therefore$ Ball will again meet the lift after $\frac{5}{3} \text{ s}$.

$$\begin{aligned} \text{(ii) At this instant, } s_L &= s_B = 10 \times \frac{5}{3} + \frac{1}{2} \times 2 \times \left(\frac{5}{3}\right)^2 \\ &= \frac{175}{9} \text{ m} = 19.4 \text{ m} \end{aligned}$$

(iii) For the ball $u \uparrow$ $a \downarrow$. Therefore, we will first find t_0 , the time when its velocity becomes zero.

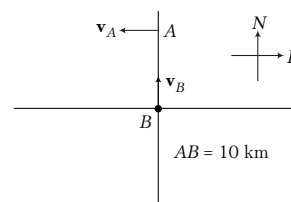
$$t_0 = \left| \frac{u}{a} \right| = \frac{20}{10} = 2 \text{ s}$$

As $t \left(= \frac{5}{3} \text{ s} \right) < t_0$, distance and displacement are equal

$$\text{or } d = 19.4 \text{ m}$$

Example 3.55 Two ships A and B are 10 km apart on a line running south to north. Ship A farther north is streaming west at 20 kmh^{-1} and ship B is streaming north at 20 kmh^{-1} . What is their distance of closest approach and how long do they take to reach it?

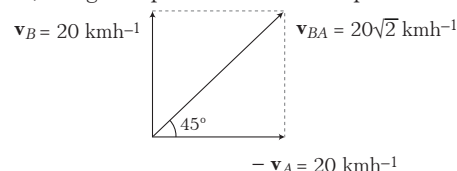
Sol. Ships A and B are moving with same speed 20 kmh^{-1} in the directions as shown in figure. It is a two dimensional, two body problem with zero acceleration. Let us find $\mathbf{v}_{BA}$.



$$\mathbf{v}_{BA} = \mathbf{v}_B - \mathbf{v}_A$$

$$\text{Here, } |\mathbf{v}_{BA}| = \sqrt{(20)^2 + (20)^2} = 20\sqrt{2} \text{ kmh}^{-1}$$

i.e. v_{BA} is $20\sqrt{2} \text{ kmh}^{-1}$ at an angle of 45° from east towards north. Thus, the given problem can be simplified as

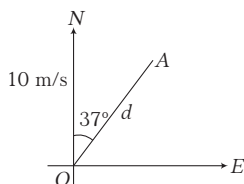


A is at rest and B is moving with $\mathbf{v}_{BA}$ in the direction as shown in figure. Therefore, the minimum distance between the two is

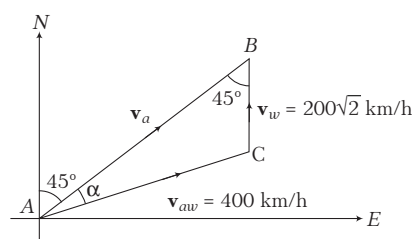
$$t = \frac{BC}{|\mathbf{v}_{BA}|} = \frac{5\sqrt{2}}{20\sqrt{2}} \quad (\because BC = AC = 5\sqrt{2} \text{ km})$$

$$= \frac{1}{4} \text{ h} = 15 \text{ min}$$

Example 3.56 An aeroplane has to go from a point O to another point A , at distance d due 37° east of north. A wind is blowing due north at a speed of 10 ms^{-1} . Find the air speed of the plane is v , (i) the direction in which the pilot should head the plane to reach the point A (ii) and the time taken by the plane to go from O to A .


$$\sin^{-1}\left(\frac{6}{12}\right) \text{ east of line } OA$$
$$t = \frac{d}{10 \cos 37^\circ + v \cos \theta} = \frac{d}{8 + v \cos \theta}$$

Sol. Given, $v_w = 200\sqrt{2}$ km/h, $v_{aw} = 400$ km/h and $\mathbf{v}_a$ should be along AB or in north-east direction. Thus, the direction of $\mathbf{v}_{aw}$ should be such as the resultant of $\mathbf{v}_w$ and $\mathbf{v}_{aw}$ is along AB or in north-east direction.


$$\frac{AC}{\sin 45^\circ} = \frac{BC}{\sin \alpha} \text{ or } \sin \alpha = \left(\frac{BC}{AC} \right) \sin 45^\circ = \left(\frac{200\sqrt{2}}{400} \right) \frac{1}{\sqrt{2}} = \frac{1}{2}$$

$$\frac{|\mathbf{v}_a|}{\sin (180^\circ - 45^\circ - 30^\circ)} = \frac{400}{\sin 45^\circ}$$

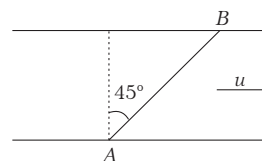
$$\begin{aligned} \text{or } |\mathbf{v}_a| &= \frac{\sin 105^\circ}{\sin 45^\circ} \times (400) \text{ km/h} \\ &= \left(\frac{\cos 15^\circ}{\sin 45^\circ} \right) (400) \text{ km/h} \\ &= \left(\frac{0.9659}{0.707} \right) (400) \text{ km/h} \\ &= 546.47 \text{ km/h} \end{aligned}$$

$$t = \frac{AB}{|\mathbf{v}_a|} = \frac{1000}{546.47} \text{ h}$$

$$t = 1.83 \text{ h}$$

CHECK POINT 3.7

- A 100 m long train crosses a man travelling at 5 kmh^{-1} , in opposite direction in 7.2 s, then the velocity of train is
(a) 40 ms^{-1} (b) 25 ms^{-1} (c) 20 ms^{-1} (d) 45 ms^{-1}
- Two parallel rail tracks run north-south. Train A moves north with a speed of 54 kmh^{-1} and train B moves south with a speed of 90 kmh^{-1} . Find the relative velocity of B w.r.t. A.
(a) 40 ms^{-1} N to S (b) 10 ms^{-1} N to S
(c) 10 ms^{-1} S to N (d) 40 ms^{-1} S to N
- Two bodies are held separated by 9.8 m vertically one above the other. They are released simultaneously to fall freely under gravity. After 2 s, the relative distance between them is
(a) 4.9 m (b) 19.6 m (c) 9.8 m (d) 39.2 m
- A particle (A) moves due north at 3 kmh^{-1} and another particle (B) moves due west at 4 kmh^{-1} . The relative velocity of A with respect to B is (Take, $\tan 37^\circ = 3/4$)
(a) 5 kmh^{-1} , 37° north of east
(b) 5 kmh^{-1} , 37° east of north
(c) $5\sqrt{2} \text{ kmh}^{-1}$, 53° east of north
(d) $5\sqrt{2} \text{ kmh}^{-1}$, 53° north of east
- A man standing on a road has to hold his umbrella at 30° with the vertical to keep the rain away. He throws the umbrella and starts running at 10 kmh^{-1} . He finds that raindrops are hitting his head vertically. What is the speed of rain with respect to ground?
(a) $10\sqrt{3} \text{ kmh}^{-1}$ (b) 20 kmh^{-1}
(c) $\frac{20}{\sqrt{3}} \text{ kmh}^{-1}$ (d) $\frac{10}{\sqrt{3}} \text{ kmh}^{-1}$
- A stationary man observes that the rain is falling vertically downward. When he starts running with a velocity of 12 kmh^{-1} , he observes that the rain is falling at an angle 60° with the vertical. The actual velocity of rain is
(a) $12\sqrt{3} \text{ kmh}^{-1}$ (b) $6\sqrt{3} \text{ kmh}^{-1}$
(c) $4\sqrt{3} \text{ kmh}^{-1}$ (d) $2\sqrt{3} \text{ kmh}^{-1}$
- A boy is running on the plane road with velocity v with a long hollow tube in his hand. The water is falling vertically downwards with velocity u . At what angle to the vertical, he must inclined the tube, so that the water drops enter it without touching its sides?
(a) $\tan^{-1} \left(\frac{v}{u} \right)$ (b) $\sin^{-1} \left(\frac{v}{u} \right)$
(c) $\tan^{-1} \left(\frac{u}{v} \right)$ (d) $\cos^{-1} \left(\frac{v}{u} \right)$
- The speed of boat is 5 kmh^{-1} in still water. It crosses a river of width 1 km along the shortest possible path in 15 min. Then, velocity of river will be
(a) 4.5 kmh^{-1} (b) 4 kmh^{-1}
(c) 1.5 kmh^{-1} (d) 1 kmh^{-1}
- A ship X moving due north with speed v observes that another ship Y is moving due west with same speed v . The actual velocity of Y is
(a) $\sqrt{2}v$ towards south-west (b) $\sqrt{2}v$ towards north-west
(c) $\sqrt{2}v$ towards south-east (d) v towards north-east
- A river is flowing from west to east at a speed of 8 m per min. A man on the south bank of the river, capable of swimming at 20 m/min in still water, wants to swim across the river in the shortest time. He should swim in a direction,
(a) due north (b) 30° east of north
(c) 30° west of north (d) 60° east of north
- The rowing speed of a man relative to water is 5 kmh^{-1} and the speed of water flow is 3 kmh^{-1} . At what angle to the river flow should he head, if he wants to reach a point on the other bank, directly opposite to starting point?
(a) 127° (b) 143° (c) 120° (d) 150°
- A man wants to reach point B on the opposite bank of a river flowing at a speed as shown in figure. With what minimum speed and in which direction should the man swim relative to water, so that he can reach point B?



- (a) u , 45° north-west (b) u , 45° north-east
(c) $\frac{u}{\sqrt{2}}$, 45° north-west (d) $\frac{u}{\sqrt{2}}$, 45° north-east

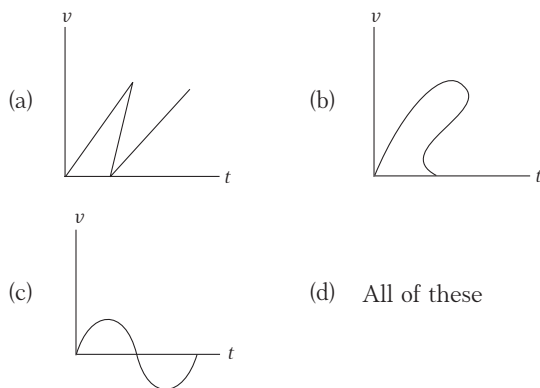
- Two trains, each 50 m long moving parallel towards each other at speeds 10 ms^{-1} and 15 ms^{-1} respectively, at what time will they pass each other?
(a) 8 s (b) 4 s
(c) 2 s (d) 6 s
- A ball is dropped from the top of a building 100 m high. At the same instant, another ball is thrown upwards with a velocity of 40 ms^{-1} from the bottom of the building. The two balls will meet after
(a) 5 s (b) 25 s
(c) 2 s (d) 3 s

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

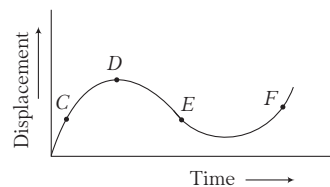
- 1 A boy walks to his school at a distance of 6 km with constant speed of 2.5 kmh^{-1} and walks back with a constant speed of 4 kmh^{-1} . His average speed for round trip expressed (in kmh^{-1}), is
(a) $24/13$ (b) $40/13$ (c) 3 (d) $1/2$
- 2 The distance travelled by a particle starting from rest and moving with an acceleration $\frac{4}{3} \text{ ms}^{-2}$, in the third second is
(a) $\frac{10}{3} \text{ m}$ (b) $\frac{19}{3} \text{ m}$ (c) 6 m (d) 4 m
- 3 A vehicle travels half the distance l with speed v_1 and the other half with speed v_2 , then its average speed is
[NCERT Exemplar]
(a) $\frac{v_1 + v_2}{2}$ (b) $\frac{2v_1 + v_2}{v_1 + v_2}$ (c) $\frac{2v_1v_2}{v_1 + v_2}$ (d) $\frac{l(v_1 + v_2)}{v_1v_2}$
- 4 Which of the following speed-time (v - t) graphs is physically not possible?



- 5 A particle moves in a straight line with a constant acceleration. It changes its velocity from 10 ms^{-1} to 20 ms^{-1} while passing through a distance 135 m in t second. The value of t (in second) is
(a) 12 (b) 9 (c) 10 (d) 1.8
- 6 A man walks on a straight road from his home to a market 2.5 km away with a speed of 5 kmh^{-1} . Finding the market closed, he instantly turns and walks back home with a speed of 7.5 kmh^{-1} . The average speed of the man over the interval of time 0 to 50 min is equal to

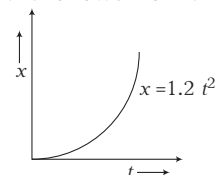
- (a) 5 kmh^{-1} (b) $\frac{25}{4} \text{ kmh}^{-1}$
- (c) $\frac{30}{5} \text{ kmh}^{-1}$ (d) $\frac{45}{8} \text{ kmh}^{-1}$

- 7 A car moving with a velocity of 10 ms^{-1} can be stopped by the application of a constant force F in a distance of 20 m. If the velocity of the car is 30 ms^{-1} , it can be stopped by this force in
(a) $\frac{20}{3} \text{ m}$ (b) 20 m (c) 60 m (d) 180 m
- 8 A body A starts from rest with an acceleration a_1 . After 2 seconds, another body B starts from rest with an acceleration a_2 . If they travel equal distance in the 5th second, after the start of A , then the ratio $a_1 : a_2$ is equal to
(a) 5 : 9 (b) 5 : 7 (c) 9 : 5 (d) 9 : 7
- 9 The displacement-time graph of a moving particle is shown below. The instantaneous velocity of the particle is negative at the point



- (a) E (b) F (c) C (d) D

- 10 Figure given shows the distance-time graph of the motion of a car. It follows from the graph that the car is



- (a) at rest
 - (b) in uniform motion
 - (c) in non-uniform accelerated motion
 - (d) uniformly accelerated motion
- 11 The velocity of a body depends on time according to the equation $v = \frac{t^2}{10} + 20$. The body is undergoing
(a) uniform acceleration (b) uniform retardation
(c) non-uniform acceleration (d) zero acceleration

- 12** The velocity v of a particle as a function of its position (x) is expressed as $v = \sqrt{c_1 - c_2 x}$, where c_1 and c_2 are positive constants. The acceleration of the particle is

(a) c_2 (b) $-\frac{c_2}{2}$
(c) $c_1 - c_2$ (d) $\frac{c_1 + c_2}{2}$

- 13** A person walks up a stalled escalator in 90 s. When he is just standing on the same moving escalator, then he is carried for 60 s. The time it would take him to walk up the moving escalator will be

(a) 27 s (b) 50 s
(c) 18 s (d) 36 s

- 14** Particle A is moving along X -axis. At time $t = 0$, it has velocity of 10 ms^{-1} and acceleration -4 ms^{-2} . Particle B has velocity of 20 ms^{-1} and acceleration -2 ms^{-2} . Initially both the particles are at origin. At time $t = 2 \text{ s}$, distance between the two particles is

(a) 24 m (b) 36 m (c) 20 m (d) 42 m

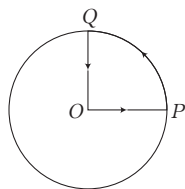
- 15** The displacement of a body along X -axis depends on time as $\sqrt{x} = t + 1$. Then, the velocity of body

(a) increases with time (b) decreases with time
(c) independent of time (d) None of these

- 16** A car starts moving along a line, first with acceleration $a = 5 \text{ ms}^{-2}$ starting from rest, then uniformly and finally decelerating at the same rate a and comes to rest. The total time of motion is 25 s. The average speed during the time is 20 ms^{-1} . How long does particle move uniformly?

(a) 10 s (b) 12 s
(c) 20 s (d) 15 s

- 17** A cyclist starts from the centre O of a circular park of radius one kilometre, reaches the edge P of the park, then cycles along the circumference and returns to the centre along QO as shown in the figure. If the round trip takes ten minutes, the net displacement and average speed of the cyclist (in metre and kilometre per hour) is



(a) 0, 1 (b) $\frac{\pi + 4}{2}$, 0
(c) $21.4, \frac{\pi + 4}{2}$ (d) 0, 21.4

- 18** A body starts from rest with uniform acceleration a . The acceleration of the body as function of time t is given by the equation $a = pt$, where p is a constant, then the displacement of the particle in the time interval $t = 0$ to $t = t_1$ will be

(a) $\frac{1}{2} pt_1^3$ (b) $\frac{1}{3} pt_1^2$
(c) $\frac{1}{2} pt_1^2$ (d) $\frac{1}{6} pt_1^3$

- 19** A particle moves along a straight line OX . At a time t (in seconds), the distance $x = 40 + 12t - t^3$. How long would the particle travel before coming to rest?

(a) 24 m (b) 40 m (c) 56 m (d) 16 m

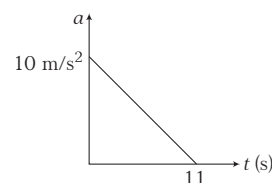
- 20** A bullet emerges from a barrel of length 1.2 m with a speed of 640 ms^{-1} . Assuming constant acceleration, the approximate time that it spends in the barrel after the gun is fired, is

(a) 4 ms (b) 40 ms (c) 400 μs (d) 1 s

- 21** The ratios of the distance traversed, in successive intervals of time by a body, falling from rest, are

(a) 1:3:5:7:9:... (b) 2:4:6:8:10:...
(c) 1:4:7:10:13:... (d) None of these

- 22** A particle starts from rest. Its acceleration (a) versus time (t) graph as shown in the figure. The maximum speed of the particle will be



(a) 110 ms^{-1} (b) 55 ms^{-1}
(c) 550 ms^{-1} (d) 660 ms^{-1}

- 23** A ball is dropped onto the floor from a height of 10 m. It rebounds to a height of 5 m. If the ball was in contact with the floor for 0.01 s, what was its average acceleration during contact?

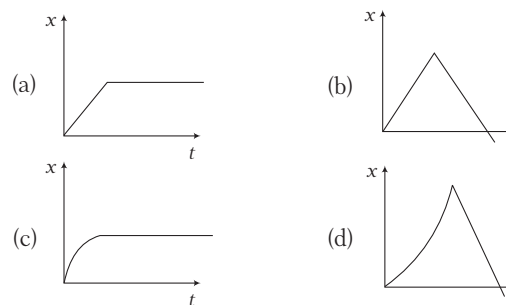
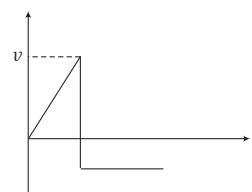
(Take, $g = 10 \text{ ms}^{-2}$)

(a) 2414 ms^{-2} (b) 1735 ms^{-2}
(c) 3120 ms^{-2} (d) 4105 ms^{-2}

- 24** Two boys are standing at the ends A and B of a ground, where $AB = a$. The boy at B starts running in a direction perpendicular to AB with velocity v_1 . The boy at A starts running simultaneously with velocity v and catches the other boy in a time t , where t is

(a) $a / \sqrt{v^2 + v_1^2}$ (b) $\sqrt{a^2 / (v^2 - v_1^2)}$
(c) $a / (v - v_1)$ (d) $a / (v + v_1)$

- 25** A boggy of uniformly moving train is suddenly detached from train and stops after covering some distance. Then, which amongst the following option is correct about the relation between the distance covered by the boggy and distance covered by the train in the same time?
- Both will be equal
 - First will be half of second
 - First will be $1/4$ of second
 - No definite ratio
- 26** A body moves for a total of nine second starting from rest with uniform acceleration and then with uniform retardation, which is twice the value of acceleration and then stops. The duration of uniform acceleration is
- 3 s
 - 4.5 s
 - 5 s
 - 6 s
- 27** The displacement x of a particle varies with time t as $x = ae^{-\alpha t} + be^{\beta t}$, where a, b, α and β are positive constants. The velocity of the particle will
- go on decreasing with time
 - be independent of α and β
 - drop to zero when $\alpha = \beta$
 - go on increasing with time
- 28** A stone is allowed to fall freely from rest. The ratio of the time taken to fall through the first metre and the second metre distance is
- $\sqrt{2} - 1$
 - $\sqrt{2} + 1$
 - $\sqrt{2}$
 - None of these
- 29** Amongst the following equation of motion, which represents uniformly accelerated motion?
- $x = \sqrt{\frac{t+a}{b}}$
 - $x = \frac{t+a}{b}$
 - $t = \sqrt{\frac{x+a}{b}}$
 - $x = \sqrt{t+a}$
- 30** A point initially at rest moves along X-axis. Its acceleration varies with time as $a = (6t + 5) \text{ ms}^{-2}$. If it starts from origin, then the distance covered in 2 s is
- 20 m
 - 18 m
 - 16 m
 - 25 m
- 31** A particle moves a distance x in time t according to equation $x = (t + 5)^{-1}$. The acceleration of particle is proportional to
- (velocity) $^{3/2}$
 - (distance) 2
 - (distance) $^{-2}$
 - (velocity) $^{2/3}$
- 32** A particle moves along a straight line. Its position at any instant is given by $x = 32t - \frac{8t^3}{4}$, where x is in metre and t is in second. Find the acceleration of the particle at the instant when particle is at rest.
- -16 ms^{-2}
 - -27.6 ms^{-2}
 - 32 ms^{-2}
 - 16 ms^{-2}
- 33** Two particles P and Q simultaneously start moving from point A with velocities 15 ms^{-1} and 20 ms^{-1} respectively. The two particles move with accelerations equal in magnitude but opposite in direction. When P overtakes Q at B , then its velocity is 30 ms^{-1} . The velocity of Q at point B will be
- 30 ms^{-1}
 - 5 ms^{-1}
 - 20 ms^{-1}
 - 15 ms^{-1}
- 34** A man is 45 m behind the bus when the bus start accelerating from rest with acceleration 2.5 ms^{-2} . With what minimum velocity should the man start running to catch the bus?
- 12 ms^{-1}
 - 14 ms^{-1}
 - 15 ms^{-1}
 - 16 ms^{-1}
- 35** A point moves in a straight line, so that its displacement x at time t is given by $x^2 = t^2 + 1$. Its acceleration is
- $1/x$
 - $1/x^3$
 - $-1/x^2$
 - $-1/x^3$
- 36** A point moves with uniform acceleration and v_1, v_2 and v_3 denote the average velocities in the three successive intervals of time t_1, t_2 and t_3 . Which of the following relations is correct?
- $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_2 + t_3)$
 - $(v_1 - v_2) : (v_2 - v_3) = (t_1 + t_2) : (t_2 + t_3)$
 - $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_1 - t_3)$
 - $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_2 - t_3)$
- 37** The velocity-time graph for a particle moving along X-axis is shown in the figure. The corresponding displacement-time graph is correctly shown by



- 38** The vertical height of point P above the ground is twice that of Q . A particle is projected downward with a speed of 5 ms^{-1} from P and at the same time, another particle is projected upward with the same

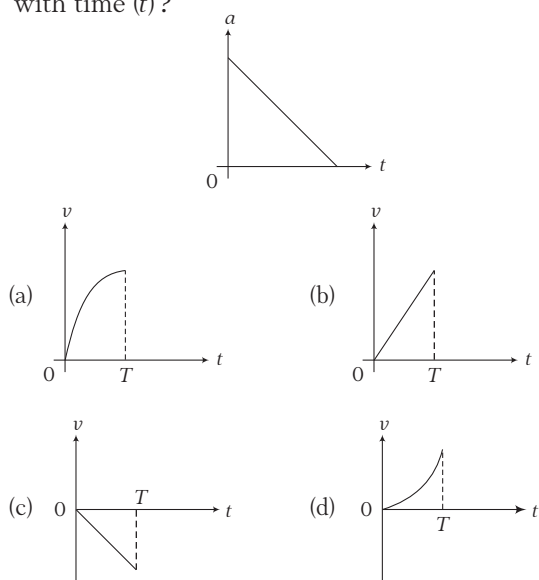
speed from Q . Both particles reach the ground simultaneously, then

- (a) $PQ = 30$ m
 (b) time of flight of stones = 3 s
 (c) Both (a) and (b) are correct
 (d) Both (a) and (b) are wrong

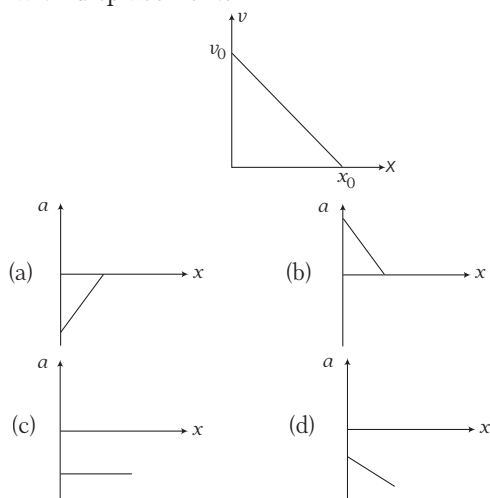
39 A body falling from a high Minaret travels 40 m in the last 2 seconds of its fall to ground. Height of Minaret in metre is (Take, $g = 10 \text{ ms}^{-2}$)

- (a) 60 (b) 45 (c) 80 (d) 50

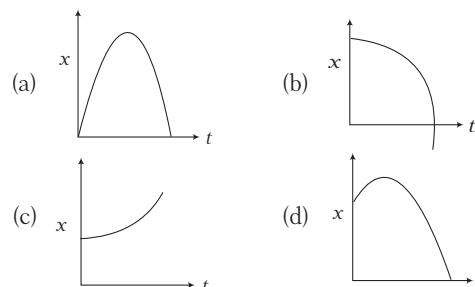
40 The acceleration-time (a - t) graph for a particle moving along a straight line starting from rest is shown in figure. Which of the following graph is the best representation of variation of its velocity (v) with time (t)?



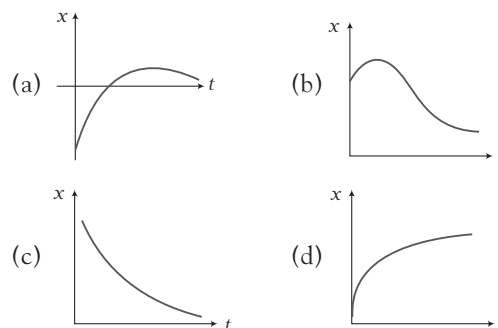
41 The given graph shows the variation of velocity with displacement. Which one of the graphs given below correctly represents the variation of acceleration with displacement?



42 The displacement x of a particle in a straight line motion is given by $x = 1 - t - t^2$. The correct representation of the motion is



43 Among the four graph shown in the figure, there is only one graph for which average velocity over the time interval $(0, T)$ can vanish for a suitably chosen T . Which one is it? [NCERT Exemplar]



44 A lift is coming from 8th floor and is just about to reach 4th floor. Taking ground floor as origin and positive direction upwards for all quantities, which one of the following is correct? [NCERT Exemplar]

- (a) $x < 0, v < 0, a > 0$ (b) $x > 0, v < 0, a < 0$
 (c) $x > 0, v < 0, a > 0$ (d) $x > 0, v > 0, a < 0$

45 The displacement of a particle is given by $x = (t - 2)^2$, where x is in metre and t in second. The distance covered by the particle in first 4 seconds is [NCERT Exemplar]

- (a) 4 m (b) 8 m (c) 12 m (d) 16 m

46 A body falls freely from the top of a tower. It covers 36% of the total height in the last second before striking the ground level. The height of the tower is

- (a) 50 m (b) 75 m
 (c) 100 m (d) 125 m

47 A particle moving along X -axis has acceleration f , at time t , given $f = f_0 \left(1 - \frac{t}{T}\right)$, where f_0 and T are constants. The particle at $t = 0$ has zero velocity. When $f = 0$, the particle's velocity (v_x) is

- (a) $\frac{1}{2} f_0 T$ (b) $f_0 T$ (c) $\frac{1}{2} f_0 T^2$ (d) $f_0 T^{-2}$

- 48** An elevator car whose floor to ceiling distance is 2.7 m starts ascending with a constant acceleration of 1.2 ms^{-2} . 2 s after the start, a bolt falls from the ceiling of the car. The free fall time of the bolt is (Take, $g = 9.8 \text{ ms}^{-2}$)

- (a) $\sqrt{\frac{2.7}{9.8}} \text{ s}$ (b) $\sqrt{\frac{5.4}{9.8}} \text{ s}$
 (c) $\sqrt{\frac{5.4}{8.6}} \text{ s}$ (d) $\sqrt{\frac{5.4}{11}} \text{ s}$

- 49** A parachutist after bailing out falls 50 m without friction when parachute opens, it decelerates at 2 ms^{-2} . He reaches the ground with speed of 3 m/s. At what height did he bail out?

- (a) 293 m (b) 111 m
 (c) 91 m (d) 182 m

- 50** Two cars A and B are travelling in the same direction with velocities v_1 and v_2 ($v_1 > v_2$). When the car A is at a distance d ahead of the car B, the driver of the car A applied the brake producing a uniform retardation a . There will be no collision when

- (a) $d < \frac{(v_1 - v_2)^2}{2a}$
 (b) $d < \frac{v_1^2 - v_2^2}{2a}$
 (c) $d > \frac{(v_1 - v_2)^2}{2a}$
 (d) $d > \frac{v_1^2 - v_2^2}{2a}$

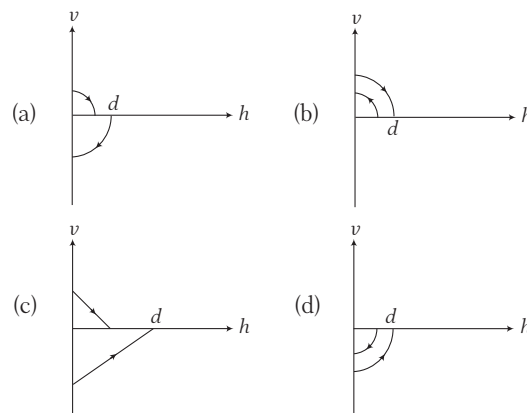
- 51** Water drops fall at regular intervals from a tap which is 5 m above the ground. The third drop is leaving the tap at the instant, the first drop touches the ground. How far above the ground is the second drop at that instant?

- (a) 2.50 m (b) 3.75 m (c) 4.00 m (d) 1.25 m

- 52** A body is thrown vertically up with a velocity u . It passes three points A, B and C in its upward journey with velocities $\frac{u}{2}$, $\frac{u}{3}$ and $\frac{u}{4}$, respectively. The ratio of the separations between points A and B and between B and C, i.e. $\frac{AB}{BC}$ is

- (a) 1 (b) 2
 (c) $\frac{10}{7}$ (d) $\frac{20}{7}$

- 53** A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up vertically to a height $d/2$. Neglecting subsequent motion and air resistance, its velocity v varies with the height h above the ground can be plotted as



- 54** The position x of a particle with respect to time t along X-axis is given by $x = 9t^2 - t^3$, where x is in metres and t in second. What will be the position of this particle when it achieves maximum speed along the positive x -direction?

- (a) 24 m (b) 32 m (c) 54 m (d) 81 m

- 55** A car A moves along north with velocity 30 km/h and another car B moves along east with velocity 40 km/h. The relative velocity of A with respect to B is

- (a) 50 km/h north-east
 (b) 50 km/h north-west
 (c) 50 km/h at angle $\tan^{-1}(3/4)$ north of west
 (d) 50 km/h at angle $\tan^{-1}(4/3)$ west of north

- 56** Rain is falling vertically downward with velocity 4 m/s. A man is moving horizontally with velocity 3 m/s, the velocity of rain with respect to man is

- (a) 5 m/s at an angle $\tan^{-1}(4/3)$ with horizontal
 (b) 5 m/s at an angle $\tan^{-1}(3/4)$ with vertical
 (c) 5 m/s at an angle $\tan^{-1}(4/3)$ with vertical
 (d) Both (a) and (b)

- 57** A ship is travelling due east at a speed of 15 km/h. Find the speed of a boat heading 30° east of north, if it always appears due north from the ship.

- (a) 30 km/h (b) $\frac{15\sqrt{3}}{2}$ km/h (c) $10\sqrt{3}$ km/h (d) 20 km/h

- 58** A man takes 3 h to cover a certain distance along the flow of river and takes 6 h to cover the same distance opposite to the flow of river. In how much time, he will cross this distance in still water?

- (a) 3.5 h (b) 4 h (c) 4.5 h (d) 5 h

- 59** A river 500 m wide is flowing at a rate of 4 m/s. A boat is sailing at a velocity of 10 m/s with respect to the water in a direction perpendicular to the river. The time taken by the boat to reach the opposite bank is

- (a) 30 s (b) 40 s
 (c) 50 s (d) 60 s

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is correct but Reason is incorrect.
- (d) If Assertion is incorrect but Reason is correct.

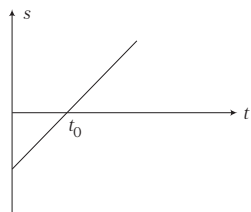
1 Assertion A body is momentarily at rest at the instant, it reverses the direction.

Reason A body cannot have acceleration, if its velocity is zero at a given instant of time.

2 Assertion The v - t graph perpendicular to time axis is not possible in practice.

Reason Infinite acceleration cannot be realised in practice.

3 Assertion In the s - t diagram as shown in figure, the body starts moving in positive direction but not from $s = 0$.



Reason At $t = t_0$, velocity of body changes its direction of motion.

4 Assertion If acceleration of a particle moving in a straight line varies as $a \propto t^n$, then $s \propto t^{n+2}$.

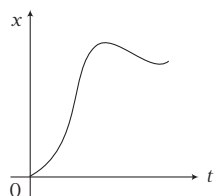
Reason If a - t graph is a straight line, then s - t graph may be a parabola.

5 Assertion A lift is ascending with decreasing speed means acceleration of lift is downwards.

Reason A body always moves in the direction of its acceleration.

Statement based questions

1 The displacement (x)-time (t) graph of a particle is shown in figure. Then, which of the following statement is correct?



- (a) Particle starts with zero velocity and variable acceleration.
- (b) Particle starts with non-zero velocity and variable acceleration.
- (c) Particle starts with zero velocity and uniform acceleration.
- (d) Particle starts with non-zero velocity and uniform acceleration.

2 A particle moves along X -axis as

$$x = 4(t - 2) + a(t - 2)^2$$

Which of the following statement is true ?

- (a) The initial velocity of particle is 4.
- (b) The acceleration of particle is $2a$.
- (c) The particle is at origin at $t = 0$.
- (d) None of the above

3 In one dimensional motion, instantaneous speed v satisfies $0 \leq v < v_0$. Then, which of the following statement is true? [NCERT Exemplar]

- (a) The displacement in time T must always take non-negative values.
- (b) The displacement x in time T satisfies $-v_0 T < x < v_0 T$.
- (c) The acceleration is always a non-negative number.
- (d) The motion has no turning points.

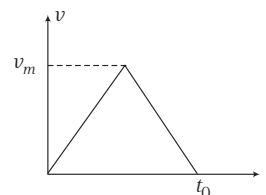
4 I. If a particle is thrown upwards, then distance travelled in last second of upward journey is independent of the velocity of projection.

II. In last second, distance travelled is 4.9 m. (Take, $g = 9.8 \text{ ms}^{-2}$)

Which amongst the statement(s) is/are correct?

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

5 I. In the v - t diagram as shown in figure, average velocity between the interval $t = 0$ and $t = t_0$ is independent of t_0 .



II. Average velocity in the given interval is $\frac{1}{2} v_m$.

Which amongst the statement(s) is/are correct?

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

Match the columns

- 1 Match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Constant positive acceleration	(p) Speed may increase
(B) Constant negative acceleration	(q) Speed may decrease
(C) Constant displacement	(r) Speed is zero
(D) Constant slope of a - t graph	(s) Speed must increase
	(t) Speed must decrease

Codes

A	B	C	D	A	B	C	D
(a) q,r	s	p	t	(b) p,q	p,q	r	p,q
(c) p	s,r	s	q	(d) q	p	t	s,q

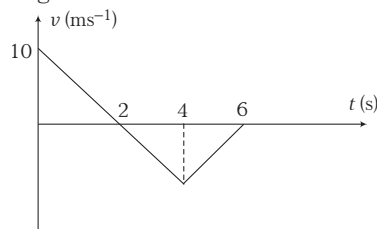
- 2 In the s - t equation ($s = 10 + 20t - 5t^2$), match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Distance travelled in 3 s	(p) -20 units
(B) Initial acceleration	(q) 15 units
(C) Velocity at 4 s	(r) 25 units
	(s) -10 units

Codes

A	B	C	A	B	C
(a) p	q	s	(b) r	q	s
(c) r	s	p	(d) q	s	r

- 3 For the velocity-time graph as shown in the figure, in a time interval from $t = 0$ to $t = 6$ s, match the following columns and mark the correct option from the codes given below.



(C) Medical entrances' gallery

Collection of questions asked in NEET & Various Medical Entrance Exams

- 1 A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is (Take, $g = 10 \text{ m/s}^2$)

[NEET 2020]

(a) 340 m (b) 320 m (c) 300 m (d) 360 m

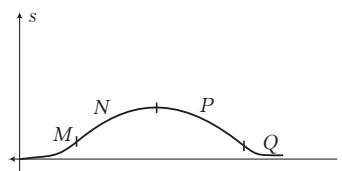
- 2 A person sitting in the ground floor of a building notices through the window of height 1.5 m, a ball dropped from the roof of the building crosses the

Column I	Column II
(A) Change in velocity	(p) $-5/3$ SI unit
(B) Average acceleration	(q) -20 SI unit
(C) Total displacement	(r) -10 SI unit
(D) Acceleration at $t = 3$ s	(s) -5 SI unit

Codes

A	B	C	D
(a) p	r	q	s
(b) p	r	s	q
(c) r	p	r	s
(d) q	p	s	r

- 4 Let us call a motion, A when velocity is positive and increasing, A^{-1} when velocity is negative and increasing. R when velocity is positive and decreasing and R^{-1} when velocity is negative and decreasing. Now, match the following two columns for the given s - t graph and mark the correct option from the codes given below.



Column I	Column II
(A) M	(p) A^{-1}
(B) N	(q) R^{-1}
(C) P	(r) A
(D) Q	(s) R

Codes

A	B	C	D	A	B	C	D
(a) p	r	q	s	(b) r	s	p	q
(c) s	p	r	q	(d) q	p	s	r

window in 0.1 s. What is the velocity of the ball when it is at the topmost point of the window?

(Take, $g = 10 \text{ m/s}^2$)

[NEET 2020]

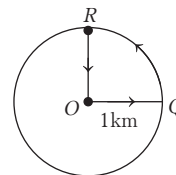
(a) 15.5 m/s (b) 14.5 m/s (c) 4.5 m/s (d) 20 m/s

- 3 A person travelling in a straight line moves with a constant velocity v_1 for certain distance x and with a constant velocity v_2 for next equal distance. The average velocity v is given by the relation

[NEET 2019]

- (a) $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}$ (b) $\frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$
 (c) $\frac{v}{2} = \frac{v_1 + v_2}{2}$ (d) $v = \sqrt{v_1 v_2}$
- 4** The speed of a swimmer in still water is 20 ms^{-1} . The speed of river water is 10 ms^{-1} and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path the angle at which he should make his strokes w.r.t. north is given by [NEET 2019]
 (a) 0° (b) 60° , west
 (c) 45° , west (d) 30° , west
- 5** Find the average velocity when a particle complete the circle of radius 1m in 10 s. [JIPMER 2019]
 (a) 2 ms^{-1} (b) 3.14 ms^{-1}
 (c) 6.28 ms^{-1} (d) Zero
- 6** Speed of a particle at 3rd and 8th second are 20 ms^{-1} and zero respectively, then average acceleration between 3rd and 8th second will be [JIPMER 2019]
 (a) 3 ms^{-2} (b) 4 ms^{-2}
 (c) 5 ms^{-2} (d) 6 ms^{-2}
- 7** A toy car with charge q moves on a frictionless horizontal plane surface under the influence of a uniform electric field $\mathbf{E}$. Due to the force $q\mathbf{E}$, its velocity increases from 0 to 6 ms^{-1} in one second duration. At that instant, the direction of the field is reversed. The car continues to move for two more seconds under the influence of this field. The average velocity and the average speed of the toy car between 0 to 3 s are respectively [NEET 2018]
 (a) 1 ms^{-1} , 3.5 ms^{-1}
 (b) 1 ms^{-1} , 3 ms^{-1}
 (c) 2 ms^{-1} , 4 ms^{-1}
 (d) 1.5 ms^{-1} , 3 ms^{-1}
- 8 Assertion** A body is momentarily at rest at the instant, if it reverse the direction.
Reason A body cannot have acceleration, if its velocity is zero at a given instant of time. [AIIMS 2018]
 (a) Assertion and Reason both are correct and Reason is the correct explanation of Assertion.
 (b) Assertion and Reason both are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Assertion is incorrect but Reason is correct.
- 9** Velocity is given by $v = 4t(1 - 2t)$, then find the value of time at which velocity is maximum. [AIIMS 2018]
 (a) 0.25 s (b) 1 s
 (c) 0.45 s (d) 4 s

- 10** A runner starts from O and goes to O following path $OQRO$ in 1 h. What is net displacement and average speed? [JIPMER 2018]



- (a) 0, 3.57 kmh^{-1} (b) 0, 0 kmh^{-1}
 (c) 0, 2.57 kmh^{-1} (d) 0, 1 kmh^{-1}
- 11** A ball is thrown upwards with a speed u from a height h above the ground. The time taken by the ball to hit the ground is [JIPMER 2018]
 (a) $\sqrt{2h/g}$ (b) $\sqrt{8h/g}$ (c) $\frac{\sqrt{u^2 + 2gh}}{g}$ (d) $\frac{u}{g} + \sqrt{\frac{2h}{g}}$
- 12** Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1 . On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2 . The time taken by her to walk up on the moving escalator will be [NEET 2017]
 (a) $\frac{t_1 + t_2}{2}$ (b) $\frac{t_1 t_2}{t_2 - t_1}$ (c) $\frac{t_1 t_2}{t_2 + t_1}$ (d) $t_1 - t_2$
- 13** What will be the a versus x graph for the following graph? [AIIMS 2017]
-
- (a) (b) (c) (d)
- 14** Which of the following statements is true for a car moving on the road? [Manipal 2017]
 (a) With respect to the frame of reference attached to the ground, the car is at rest.
 (b) With respect to the frame of reference attached to the person sitting in the car, the car is at rest.
 (c) With respect to the frame of reference attached to the person outside the car, the car is at rest.
 (d) None of the above

- 15** If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1s and 2s is [NEET 2016]

(a) $3A + 7B$ (b) $\frac{3}{2}A + \frac{7}{3}B$
 (c) $\frac{A}{2} + \frac{B}{3}$ (d) $\frac{3}{2}A + 4B$

- 16** A particle of unit mass undergoes one dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$, where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x , is given by [CBSE AIPMT 2015]

(a) $-2n\beta^2 x^{-2n-1}$ (b) $-2n\beta^2 x^{-4n-1}$
 (c) $-2\beta^2 x^{-2n+1}$ (d) $-2n\beta^2 x^{-4n+1}$

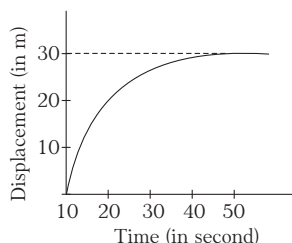
- 17** A ball is dropped from a bridge 122.5 m above a river. After the ball has been falling for 2s, a second ball is thrown straight down after it. What must be the initial velocity of the second ball, so that both ball hit the water at the same time? [AIIMS 2015]
- (a) 40 ms^{-1} (b) 55.5 ms^{-1} (c) 26.1 ms^{-1} (d) 9.6 ms^{-1}

- 18** A ball is thrown vertically upwards from the ground with a speed of 25.2 ms^{-1} . How long does it take to reach its highest point and how high does it rise? (Take, $g = 9.8 \text{ ms}^{-2}$) [UK PMT 2015]

(a) 2.75 s, 3.24 m (b) 25.7 s, 34.2 m
 (c) 2.57 s, 32.4 m (d) 27.5 s, 3.42 m

- 19** A vehicle moving with a constant acceleration from A to B in a straight line AB , has velocities u and v at A and B , respectively. C is the mid-point of AB . If time taken to travel from A to C is twice the time taken to travel from C to B , then the velocity of the vehicle v at B is [EAMCET 2015]
- (a) $5u$ (b) $6u$ (c) $7u$ (d) $8u$

- 20** The displacement of a particle as a function of time is shown in figure. It indicates that [Kerala CEE 2015]

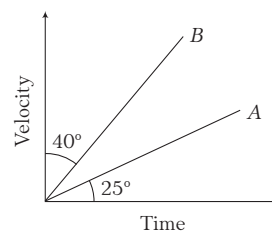


- (a) the velocity of the particle is constant throughout
 (b) the acceleration of the particle is constant throughout
 (c) the particle starts with a constant velocity and is accelerated
 (d) the motion is retarded and finally the particle stops
 (e) None of the above

- 21** A car starts from rest and accelerates uniformly to a speed of 180 kmh^{-1} in 10 s. The distance covered by the car in the time interval is [Manipal 2015]

(a) 200 m (b) 300 m
 (c) 500 m (d) 250 m

- 22** The velocity-time graph for two bodies A and B are shown in figure. Then, the acceleration of A and B are in the ratio [KCET 2015]



(a) $\sin 25^\circ$ to $\sin 50^\circ$ (b) $\tan 25^\circ$ to $\tan 40^\circ$
 (c) $\cos 25^\circ$ to $\cos 50^\circ$ (d) $\tan 25^\circ$ to $\tan 50^\circ$

- 23** A ball thrown vertically upwards after reaching a maximum height h returns to the starting point after a time of 10 s. Its displacement after 5 s is [Kerala CEE 2014]

(a) h (b) $2h$ (c) $10h$ (d) $20h$
 (e) $5h$

- 24** A police jeep is chasing with velocity of 45 kmh^{-1} , a thief in another jeep moving with velocity 153 kmh^{-1} . Police fires a bullet with muzzle velocity of 180 ms^{-1} . The velocity with which it will strike the car of the thief is [EAMCET 2014]
- (a) 150 ms^{-1} (b) 27 ms^{-1} (c) 450 ms^{-1} (d) 250 ms^{-1}

- 25** A particle moves with constant acceleration along a straight line starting from rest. The percentage increase in its displacement during the 4th second compared to that in the 3rd second is [WB JEE 2014]
- (a) 33% (b) 40% (c) 66% (d) 77%

- 26** A car covers the first half of the distance between the two places at 40 kmh^{-1} and another half at 60 kmh^{-1} . The average speed of the car is [UK PMT 2014]
- (a) 40 kmh^{-1} (b) 48 kmh^{-1} (c) 50 kmh^{-1} (d) 60 kmh^{-1}

- 27** A particle starts moving from rest with uniform acceleration. It travels a distance x in first 2 s and distance y in the next 2 s. Then, [EAMCET 2014]
- (a) $y = 3x$ (b) $y = 4x$ (c) $y = x$ (d) $y = 2x$

- 28** At time $t = 0$, two bodies A and B are at the same point. A moves with constant velocity v and B starts from rest and moves with constant acceleration. Relative velocity of B w.r.t. A when the bodies meet each other is [EAMCET 2014]

(a) $\frac{v}{2}$ (b) $\frac{v}{3}$ (c) v (d) $2v$

- 29** A car moves from A to B with a speed of 30 kmh^{-1} and from B to A with a speed of 20 kmh^{-1} . What is the average speed of the car? [KCET 2014]

(a) 25 kmh^{-1} (b) 24 kmh^{-1}
(c) 50 kmh^{-1} (d) 10 kmh^{-1}

- 30** A body starts from rest and moves with constant acceleration for t second. It travels a distance x_1 in first half of time and x_2 in next half of time, then [KCET 2014]

(a) $x_2 = x_1$ (b) $x_2 = 2x_1$ (c) $x_2 = 3x_1$ (d) $x_2 = 4x_1$

- 31** The acceleration of a moving body is found from the [Kerala CEE 2014]

(a) area under velocity-time graph
(b) area under displacement-time graph
(c) slope of distance-time graph
(d) slope of velocity-time graph
(e) None of the above

- 32** A stone falls freely under gravity. It covers distances h_1 , h_2 and h_3 in the first 5 s, the next 5 s and the next 5 s, respectively.

The relation between h_1 , h_2 and h_3 is [NEET 2013]

(a) $h_1 = 2h_2 = 3h_3$ (b) $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$
(c) $h_2 = 3h_1$ and $h_3 = 3h_2$ (d) $h_1 = h_2 = h_3$

- 33** The motion of a particle in straight line is an example of

(a) constant velocity motion [J&K CET 2013]
(b) uniformly accelerated motion
(c) non-uniformly accelerated motion
(d) zero velocity motion

- 34** The velocity-time graph of particle comes out to be a non-linear curve. The motion is [J&K CET 2013]

(a) uniform velocity motion
(b) uniformly accelerated motion
(c) non-uniform accelerated motion
(d) Nothing can be said about the motion

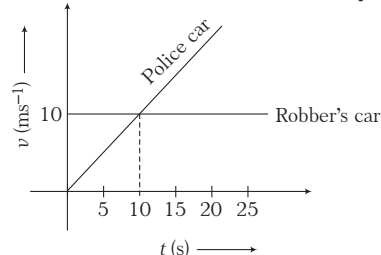
- 35** A body is thrown vertically upward from a point A 125 m above the ground. It goes up to a maximum height of 250 m above the ground and passes through A on its downward journey. The velocity of the body when it is at a height of 70 m above the ground is (Take, $g = 10 \text{ ms}^{-2}$) [EAMCET 2013]

(a) 50 ms^{-1} (b) 60 ms^{-1} (c) 80 ms^{-1} (d) 20 ms^{-1}

- 36** A person reaches a point directly opposite on the other bank of a river. The velocity of the water in the river is 4 ms^{-1} and the velocity of the person in still water is 5 ms^{-1} . If the width of the river is 84.6 m, time taken to cross the river (in seconds) is [EAMCET 2013]

(a) 28.2 (b) 9.4
(c) 2 (d) 84.6

- 37** The velocity-time graph of robber's car and a chasing police car are shown in the following graph. Police car crosses the robber's car in time [UP CPMT 2013]



(a) 10 s after it starts (b) 15 s after it starts
(c) 20 s after it starts (d) Never crosses

- 38** Initial speed of an α -particle inside a tube of length 4m is 1 kms^{-1} , if it is accelerated in the tube and comes out with a speed of 9 kms^{-1} , then the time for which the particle remains inside the tube is [BCECE 2013]

(a) $8 \times 10^{-3} \text{ s}$ (b) $8 \times 10^{-4} \text{ s}$
(c) $80 \times 10^{-3} \text{ s}$ (d) $800 \times 10^{-3} \text{ s}$

- 39** The motion of a particle along a straight line is described by equation [CBSE AIPMT 2012]

$$x = 8 + 12t - t^3$$

where, x is in metre and t in second. The retardation of the particle when its velocity becomes zero, is

(a) 24 ms^{-2} (b) zero (c) 6 ms^{-2} (d) 12 ms^{-2}

- 40** A particle moves along with X-axis. The position x of particle with respect to time t from origin given by $x = b_0 + b_1t + b_2t^2$. The acceleration of particle is [AIIMS 2012]

(a) b_0 (b) b_1 (c) b_2 (d) $2b_2$

- 41** A body X is projected upwards with a velocity of 98 ms^{-1} , after 4 s, a second body Y is also projected upwards with the same initial velocity. Two bodies will meet after [BCECE 2012]

(a) 8 s (b) 10 s (c) 12 s (d) 14 s

- 42** A scooter starts from rest have an acceleration of 1 ms^{-2} while a car 150 m behind it starts from rest with an acceleration of 2 ms^{-2} . After how much time, the car catches up with the scooter? [BHU 2012]

(a) $\sqrt{700} \text{ s}$ (b) $\sqrt{300} \text{ s}$
(c) $\sqrt{150} \text{ s}$ (d) None of these

- 43** Let $\mathbf{r}_1(t) = 3t\hat{i} + 4t^2\hat{j}$
and $\mathbf{r}_2(t) = 4t^2\hat{i} + 3t\hat{j}$

represent the positions of particles 1 and 2, respectively, as function of time t , $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ are in metre and t in second. The relative speed of the two particles at the instant $t = 1\text{s}$, will be [AMU 2012]

(a) 1 m/s (b) $3\sqrt{2} \text{ m/s}$ (c) $5\sqrt{2} \text{ m/s}$ (d) $7\sqrt{2} \text{ m/s}$

ANSWERS

- **CHECK POINT 3.1**

1. (d)	2. (a)	3. (a)	4. (c)	5. (b)	6. (c)	7. (b)	8. (d)
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- **CHECK POINT 3.2**

1. (q)	2. (q)	3. (d)	4. (d)	5. (c)	6. (c)	7. (c)	8. (d)
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- **CHECK POINT 3.3**

1. (c)	2. (d)	3. (b)	4. (a)	5. (d)
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- **CHECK POINT 3.4**

1. (a)	2. (c)	3. (d)	4. (a)	5. (a)	6. (b)	7. (c)	8. (c)	9. (c)	10. (b)
11. (c)									

- **CHECK POINT 3.5**

1. (b)	2. (d)	3. (b)	4. (b)	5. (a)	6. (b)	7. (a)	8. (b)	9. (c)	10. (a)
11. (a)	12. (c)	13. (a)	14. (b)	15. (b)	16. (b)	17. (d)	18. (b)	19. (a)	20. (d)
21. (c)									

- **CHECK POINT 3.6**

1. (d)	2. (c)	3. (c)	4. (a)	5. (d)	6. (b)	7. (c)	8. (a)	9. (b)	10. (a)
11. (a)	12. (c)	13. (a)	14. (d)	15. (d)	16. (b)	17. (b)	18. (a)		

- **CHECK POINT 3.7**

1. (d)	2. (a)	3. (c)	4. (b)	5. (b)	6. (c)	7. (a)	8. (d)	9. (b)	10. (a)
11. (a)	12. (c)	13. (b)	14. (b)						

(A) Taking it together

1. (b)	2. (a)	3. (c)	4. (d)	5. (b)	6. (c)	7. (d)	8. (a)	9. (a)	10. (d)
11. (c)	12. (b)	13. (d)	14. (a)	15. (a)	16. (d)	17. (d)	18. (d)	19. (c)	20. (b)
21. (a)	22. (b)	23. (a)	24. (b)	25. (b)	26. (d)	27. (d)	28. (b)	29. (c)	30. (b)
31. (a)	32. (b)	33. (b)	34. (c)	35. (b)	36. (b)	37. (d)	38. (c)	39. (b)	40. (a)
41. (a)	42. (b)	43. (b)	44. (a)	45. (b)	46. (d)	47. (a)	48. (d)	49. (a)	50. (c)
51. (b)	52. (d)	53. (a)	54. (c)	55. (c)	56. (d)	57. (a)	58. (b)	59. (c)	

(B) Medical entrance special format questions

- **Assertion and reason**

1. (c) 2. (a) 3. (c) 4. (b) 5. (c)

- **Statement based questions**

1. (a) 2. (b) 3. (b) 4. (c) 5. (c)

- Match the columns

1. (b) 2. (c) 3. (c) 4. (b)

(C) Medical entrances' gallery

1. (c)	2. (b)	3. (b)	4. (d)	5. (d)	6. (b)	7. (b)	8. (c)	9. (a)	10. (a)
11. (c)	12. (c)	13. (c)	14. (b)	15. (b)	16. (b)	17. (c)	18. (c)	19. (c)	20. (d)
21. (d)	22. (d)	23. (a)	24. (a)	25. (b)	26. (b)	27. (a)	28. (c)	29. (b)	30. (c)
31. (d)	32. (b)	33. (b)	34. (c)	35. (b)	36. (a)	37. (c)	38. (b)	39. (d)	40. (d)
41. (c)	42. (b)	43. (c)							

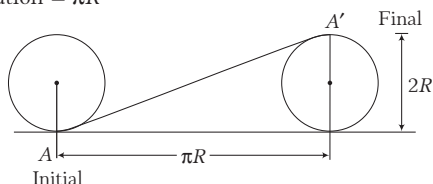
Hints & Explanations

• CHECK POINT 3.1

2 (a) Distance from starting point = $3 + 4 + 5 = 12$ m

3 (a) Distance = Length $AB = 2\pi R \times \frac{60}{360} = \frac{\pi R}{3}$

4 (c) Horizontal distance covered by the wheel in half revolution = πR



So, the displacement of the point which was initially in contact with ground = $AA' = \sqrt{(\pi R)^2 + (2R)^2}$

$$= R\sqrt{\pi^2 + 4} = \sqrt{\pi^2 + 4} \quad (\text{As, } R = 1 \text{ m})$$

5 (b)

(i) The displacement of the man from A to E is $\Delta x = x_2 - x_1 = 7 \text{ m} - (-8 \text{ m}) = +15 \text{ m}$ directed in the positive x-direction.

(ii) The displacement of the man from E to C is $\Delta x = -3 \text{ m} - (7 \text{ m}) = -10 \text{ m}$ directed in the negative x-direction.

(iii) The displacement of the man from B to D is $\Delta x = 3 \text{ m} - (-7 \text{ m}) = +10 \text{ m}$ directed in the positive x-direction.

6 (c) Since, displacement is always less than or equal to distance but never greater than distance. Hence, numerical ratio of displacement to the distance covered is always equal to or less than one.

• CHECK POINT 3.2

1 (a) $v_{av} = \frac{\text{Total distance}}{\text{Total time}} = \frac{80t + 40t}{2t} = 60 \text{ kmh}^{-1}$

2 (a) $21 = \frac{11 \times 18 + 42 \times v}{60}$
 $\therefore v = 25.3 \text{ m min}^{-1}$

3 (d) Man walks from his home to market with a speed of 5 kmh^{-1} . Distance = 2.5 km and time = $\frac{2.5}{5} = \frac{1}{2} \text{ h} = 30 \text{ min}$ and he returns back with speed of 7.5 km/h in rest time, i.e. 10 min .

$$\text{Distance} = 7.5 \times \frac{10}{60} = 1.25 \text{ km}$$

$$\begin{aligned} \text{So, average speed} &= \frac{\text{total distance}}{\text{total time}} \\ &= \frac{(2.5 + 1.25) \text{ km}}{(40/60) \text{ h}} = \frac{45}{8} \text{ kmh}^{-1} \end{aligned}$$

4 (d) Displacement of the particle will be zero because it comes back to its starting point.

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{30 \text{ m}}{10 \text{ s}} = 3 \text{ ms}^{-1}$$

5 (c) $t = \frac{d}{v} = \frac{150 + 850}{45 \times \frac{5}{18}} = 80 \text{ s}$

6 (c) $v_{av} = \frac{\text{Total displacement}}{\text{Total time}} = \frac{\sqrt{(4)^2 + (3)^2}}{10 + 5} = \frac{1}{3} \text{ ms}^{-1}$

7 (c) $v_{av} = \frac{\text{Displacement}}{\text{Time}} = \frac{\sqrt{2}R}{t}$

8 (d) $v_{av} = \frac{\text{Displacement}}{\text{Time}} = \frac{2R}{(\pi R/v)} = \frac{2v}{\pi} = \frac{20}{\pi} \text{ ms}^{-1}$

• CHECK POINT 3.3

1 (c) Acceleration is a vector quantity. So, it changes when either magnitude or direction of velocity changes or both changes.

3 (b) Time = $\frac{\text{Distance}}{\text{Average velocity}} = \frac{3.06}{0.34} = 9 \text{ s}$

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time}} = \frac{0.18}{9} = 0.02 \text{ m/s}^2$$

4 (a) $a = \frac{\text{Change in velocity}}{\text{Time taken}} = \frac{v_f - v_i}{t}$
 $= \frac{\left(44 \times \frac{5}{18}\right) - \left(80 \times \frac{5}{18}\right)}{15} = -0.67 \text{ m/s}^2$

Negative sign represents the retardation.

5 (d) For path OA and BO, the magnitude of velocity (speed) and direction is constant, hence acceleration is zero.

For path AB, since this path is a curve, so the direction of the velocity changes every moment but the magnitude of velocity (speed) remains constant.

Since, the direction of velocity is changing, i.e. there must be some acceleration along the path AB.

• CHECK POINT 3.4

1 (a) $u - \frac{3u}{4} = u - at_0$
 $a = \frac{3u}{4t_0}$ or $\frac{u}{a} = \frac{4}{3} t_0$

$$\text{Now, } 0 = u - at \text{ or } t = \frac{u}{a} = \frac{4}{3} t_0$$

2 (c) $s = \frac{1}{2} at^2 = \frac{1}{2} \times (0.2) (64) = 6.4 \text{ m} = 640 \text{ cm}$

3 (d) $s = \frac{1}{2} at^2 \Rightarrow s = \frac{1}{2} \times a \times (4)^2 \Rightarrow s = 8a$

$$4 \text{ (a) } 15 = 2t - \frac{1}{2} \times (0.1)t^2 \text{ or } t = 10 \text{ s}$$

$$\begin{aligned} 5 \text{ (a) } s &= s_1 + s_2 + s_3 \\ &= \frac{1}{2} \times 2 \times (10)^2 + (20)(30) + \frac{1}{2} \times 4 \times (5)^2 \\ &= 100 + 600 + 50 = 750 \text{ m} \end{aligned}$$

6 (b) Displacements of both should be equal.

$$\text{or } 8t = \frac{1}{2} \times 4 \times t^2 \text{ or } t = 4 \text{ s}$$

$$7 \text{ (c) } s_B = \frac{1}{2} \times 2 \times (1)^2 = 1 \text{ m}$$

$$s_A = \frac{1}{2} \times 2 \times (2)^2 = 4 \text{ m}$$

$$\therefore s_A - s_B = 3 \text{ m}$$

$$8 \text{ (c) } 40 = (20) a_1 \Rightarrow a_1 = 2 \text{ ms}^{-2}$$

$$\text{Further } 40 = (40) a_2$$

$$\therefore a_2 = 1 \text{ ms}^{-2}$$

Therefore, acceleration is 2 ms^{-2} and retardation is 1 ms^{-2} .

$$\text{Now, } s_1 = \frac{1}{2} a_1 t_1^2 = \frac{1}{2} \times 2 \times (20)^2 = 400 \text{ m}$$

$$s_2 = v_{\max} t_2 = 40 \times 20 = 800 \text{ m}$$

$$s_3 = \frac{v_{\max}^2}{2a_2} = \frac{(40)^2}{2 \times 1} = 800 \text{ m}$$

$$\begin{aligned} \text{Now, average velocity} &= \frac{\text{total displacement}}{\text{total time}} \\ &= \frac{400 + 800 + 800}{20 + 20 + 40} = 25 \text{ ms}^{-1} \end{aligned}$$

9 (c) Let v_m be the maximum speed,

$$v_m = a_1 t_1 = \sqrt{2a_1 l} \quad (\because u = 0)$$

$$t_2 = \frac{2l}{v_m} \text{ and } v_m = a_2 t_3 = \sqrt{2a_2(3l)}$$

$$\text{Now, average speed, } v_{\text{av}} = \frac{l + 2l + 3l}{t_1 + t_2 + t_3}$$

$$\begin{aligned} v_{\text{av}} &= \frac{6l}{(v_m/a_1) + (2l/v_m) + (v_m/a_2)} \\ &= \frac{6l}{\left(\frac{v_m}{v_m^2/2l}\right) + \left(\frac{2l}{v_m}\right) + \left(\frac{v_m}{v_m^2/6l}\right)} = \frac{6l}{(10l/v_m)} = \frac{3v_m}{5} \end{aligned}$$

$$\Rightarrow \frac{v_{\text{av}}}{v_m} = \frac{3}{5}$$

$$10 \text{ (b) } (30)^2 = (20)^2 + 2a(2s) \text{ or } 2as = 250$$

$$\text{Now, } v^2 = u^2 + 2as \Rightarrow v^2 = (20)^2 + 250 \Rightarrow v^2 = 650$$

$$\therefore v = 25.5 \text{ ms}^{-1}$$

$$11 \text{ (c) } s_n = u + \frac{a}{2}(2n-1) \Rightarrow 1.2 = 0 + \frac{a}{2}(2 \times 6 - 1)$$

$$\Rightarrow a = \frac{1.2 \times 2}{11} = 0.218 \text{ ms}^{-2}$$

• CHECK POINT 3.5

$$1 \text{ (b) } t = \frac{2u}{g} \Rightarrow t = \frac{2 \times 9.8}{10} = 1.96 \approx 2 \text{ s}$$

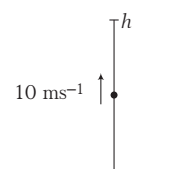
2 (d) Let t second of upward journey = first t second of downward journey (with zero initial velocity).

$$\therefore \text{Desired distance} = \frac{1}{2} gt^2$$

3 (b) Time taken to reach maximum height is 1 s.

$$\text{Height} = \text{free fall distance in } 1\text{s} = \frac{1}{2} gt^2 = 5 \text{ m}$$

$$4 \text{ (b) } h = \frac{(10)^2}{2g} = 5 \text{ m}$$



$$\therefore \text{Total height} = 2h = 10 \text{ m}$$

$$5 \text{ (a) } H_{\max} \propto v^2$$

$$\Rightarrow v \propto \sqrt{H_{\max}}$$

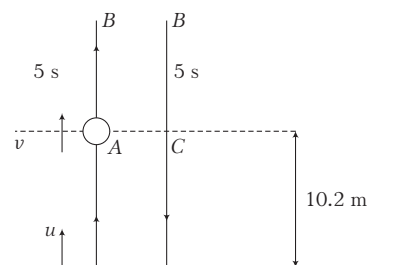
i.e. To triple the maximum height, ball should be thrown with velocity $\sqrt{3} v_0$.

$$6 \text{ (b) } v^2 = u^2 + 2gh$$

$$\Rightarrow (3u)^2 = (-u)^2 + 2gh$$

$$\Rightarrow h = \frac{4u^2}{g}$$

$$7 \text{ (a) } t_{ABC} = 10 \text{ s} \Rightarrow t_{AB} = 5 \text{ s}$$

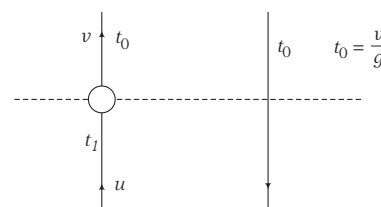


At B, velocity becomes zero. Hence at A, velocity should be 50 ms^{-1} .

$$\text{Now, } (50)^2 = (u^2) - 2 \times 10 \times 10.2$$

$$\therefore u = 52 \text{ ms}^{-1}$$

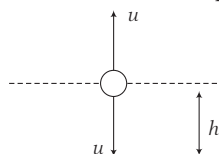
8 (b) Velocity of particle of this instant will be $v = (u - gt)$



Now, the desired time interval will be $\frac{2v}{g}$.

or
$$\frac{2(u - gt_1)}{g} = 2\left(\frac{u}{g} - t_1\right)$$

9 (c) Taking downward direction as the positive direction.



$$+h = -ut_1 + \frac{1}{2}gt_1^2 \quad \dots(i)$$

$$+h = ut_2 + \frac{1}{2}gt_2^2 \quad \dots(ii)$$

Multiplying Eq. (i) by t_2 and Eq. (ii) by t_1 and adding, we get

$$h(t_1 + t_2) = \frac{1}{2}gt_1t_2(t_1 + t_2) \quad \text{or} \quad h = \frac{1}{2}gt_1t_2$$

For free fall from rest, $h = \frac{1}{2}gt^2$

$$\therefore t^2 = t_1t_2 \Rightarrow t = \sqrt{t_1t_2}$$

10 (a) After 2 s, velocity, $v = 4.9 \times 2 = 9.8$

and
$$h = \frac{1}{2} \times 4.9 \times (2)^2 = 9.8 \text{ m}$$

Greatest height, $h_{\max} = h + \frac{v^2}{2g} = 14.7 \text{ m}$

11 (a) Here, $u = 0$

$$S_3 = 0 + \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 9 = 45$$

$$S_{\text{th}} = u + (2t - 1)g/2$$

$$\Rightarrow S_{\text{th}} = 0 + 5(2t - 1)$$

$$45 = 5(2t - 1)$$

$$\Rightarrow 2t - 1 = 9 \Rightarrow t = 5 \text{ s}$$

12 (c) $h_1 = \frac{1}{2} \times g \times (2)^2 = 2g$

$$h_2 = \frac{1}{2} \times g \times (4)^2 - 2g = 6g$$

$$h_3 = \frac{1}{2} \times g \times (6)^2 - 8g = 10g$$

$$\therefore h_1 : h_2 : h_3 = 1 : 3 : 5$$

13 (a) $t = \sqrt{\frac{2h}{g}}$ or $t \propto \sqrt{h} \Rightarrow \frac{t_1}{t_2} = \sqrt{\frac{h}{2h}} = 1 : \sqrt{2}$

14 (b) Let h be the distance covered in t second, $h = \frac{1}{2}gt^2 \quad \dots(i)$

Distance covered in t th second $= \frac{1}{2}g(2t - 1)$

$$\Rightarrow \frac{9h}{25} = \frac{g}{2}(2t - 1) \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$h = 122.5 \text{ m}$$

15 (b) In first second distance travelled, $x = \frac{1}{2} \times g \times t^2 = 5 \text{ m}$

and similarly, in last second distance travelled, $7x = 35 \text{ m}$

Now, $s_t = \left(u + at - \frac{1}{2}a\right)$

$$\Rightarrow 35 = 0 + 10 \times t - \frac{1}{2} \times 10$$

$$\therefore t = 4 \text{ s}$$

16 (b) $x = 18t + 5t^2$

$$v = \frac{dx}{dt} = \frac{d}{dt}(18t + 5t^2) = 18 + 10t$$

$$\therefore v = 10t + 18$$

At $t_1 = 2 \text{ s}$, $v_1 = 10(2) + 18 = 38 \text{ m/s}$

At $t_2 = 4 \text{ s}$, $v_2 = 10(4) + 18 = 58 \text{ m/s}$

$$\therefore a = \frac{v_2 - v_1}{t} = \frac{20}{2} = 10 \text{ m/s}^2$$

17 (d) $v = \frac{ds}{dt} = 12 - 4t$

Comparing with $v = u + at$, $u = 12 \text{ ms}^{-1}$ and $a = -4 \text{ ms}^{-2}$

Velocity will become zero at time $t_0 \Rightarrow 0 = 12 - 4t_0$ or $t_0 = 3 \text{ s}$.

Since, the given time $t = 5 \text{ s}$ is greater than $t_0 = 3 \text{ s}$

distance > | displacement |

$$\begin{aligned} \text{Distance, } d &= |s_0 - t_0| + |s_t - t_0| = \frac{u^2}{2|a|} + \frac{1}{2}|a|(t - t_0)^2 \\ &= \frac{(12)^2}{8} + \frac{1}{2} \times 4 \times (2)^2 = 26 \text{ m} \end{aligned}$$

18 (b) $v = \frac{dx}{dt} = \frac{d}{dt}(\alpha t^3 + \beta t^2 + \gamma t + \delta)$

$$v = 3\alpha t^2 + 2\beta t + \gamma; v_{t=0} = v_i = \gamma$$

$$a = 6\alpha t + 2\beta; a_{t=0} = a_i = 2\beta$$

$$\therefore \frac{v_i}{a_i} = \frac{\gamma}{2\beta}$$

19 (a) $a = bt$

$$\therefore \int_{v_0}^v dv = \int_0^t a dt = \int_0^t bt \cdot dt \Rightarrow v = v_0 + \frac{bt^2}{2}$$

Further integrating the above equation w.r.t. time, we get

$$s = v_0t + \frac{bt^3}{6}$$

20 (d) $dv = a dt$

$$\therefore \int_2^v dv = \int_0^2 (3t^2 + 2t + 2) dt = [t^3 + t^2 + 2t]_0^2$$

or $v = 18 \text{ ms}^{-1}$

21 (c) $s = t^3 - 6t^2 + 18t + 9 \Rightarrow v = \frac{ds}{dt} = 3t^2 - 12t + 18$

v is minimum or maximum at time t , which can be calculated as

$$a = \frac{dv}{dt} = 6t - 12 = 0$$

$$\Rightarrow t = 2 \text{ s}$$

At $t = 2$ s,

$$\frac{d^2v}{dt^2} = 6 > 0, \text{ i.e. } v \text{ is minimum at } t = 2 \text{ s}$$

$$v_{\min} = 3(2)^2 - 12(2) + 18 = 6 \text{ m/s}$$

• CHECK POINT 3.6

1 (d) Uniform motion means uniform velocity or constant slope of s - t graph.

3 (c) Slope of the s - t graph is velocity, $v = \tan \theta$

But it is valid only when angle is measured with time axis.

So, for the given graph, angle from time axis
 $= 90^\circ - 30^\circ = 60^\circ$

$$\text{Now, } v = \tan 60^\circ = \sqrt{3} \text{ ms}^{-1}$$

4 (a) $v_i = \text{slope of } s\text{-}t \text{ graph} = \tan 45^\circ = 1 \text{ ms}^{-1}$

$$v_f = \text{slope of } s\text{-}t \text{ graph} = \tan 60^\circ = \sqrt{3} \text{ ms}^{-1}$$

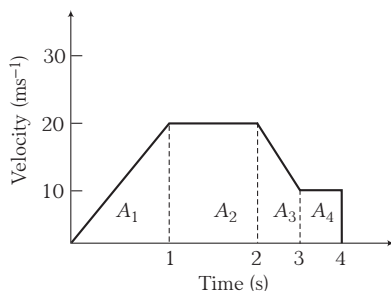
$$\text{Now, } a_{\text{av}} = \frac{v_f - v_i}{\Delta t} = \frac{\sqrt{3} - 1}{1} = (\sqrt{3} - 1) \text{ units}$$

5 (d) Maximum acceleration means maximum change in velocity in minimum time interval.

In time interval, $t = 30$ s to $t = 40$ s

$$\therefore a = \frac{\Delta v}{\Delta t} = \frac{80 - 20}{40 - 30} = \frac{60}{10} = 6 \text{ cms}^{-2}$$

6 (b) Distance = Area under v - t graph = $A_1 + A_2 + A_3 + A_4$

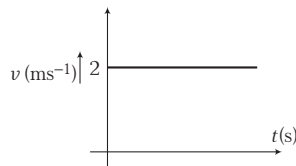


$$= \frac{1}{2} \times 1 \times 20 + (20 \times 1) + \frac{1}{2} (20 + 10) \times 1 + (10 \times 1)$$

$$= 10 + 20 + 15 + 10 = 55 \text{ m}$$

7 (c) Distance = Area of trapezium = $\frac{1}{2} \times 3.6 \times (12 + 8) = 36 \text{ m}$

9 (b) $v = \frac{dx}{dt} = 2 \text{ ms}^{-1} = \text{constant}$



10 (a) Velocity will continuously increase (starting from rest).

11 (a) Velocity first decreases in upwards direction, then increases in downward direction.

12 (c) $a = \text{Slope of } v\text{-}t \text{ graph} = -\frac{15}{3} = -5 \text{ ms}^{-2}$

13 (a) Since, total displacement is zero, hence average velocity is also zero.

14 (d) I is not possible because total distance covered by a particle cannot decrease with time. II is not possible because at a particular time, position cannot have two values. III is not possible because at a particular time, velocity cannot have two values. IV is not possible because speed can never be negative.

$$15 (d) a = v \left(\frac{dv}{ds} \right) = (10) \left(-\frac{20}{30} \right) = -\frac{20}{3} \text{ ms}^{-2}$$

16 (b) For uniformly accelerated motion, $v^2 = u^2 + 2as$, i.e. v^2 versus s graph is a straight line with intercept u^2 and slope $2a$. Since, intercept is non-zero, initial velocity is non-zero.

$$17 (b) v^2 = u^2 + 2as, \text{ slope} = 2a = -\frac{16}{2} = -8 \text{ ms}^{-2}$$

$$\text{or } a = -4 \text{ ms}^{-2}$$

18 (a) Taking the motion from 0 to 2 s,

$$u = 0, a = 3 \text{ ms}^{-2}, t = 2 \text{ s}, v = ?$$

$$v = u + at = 0 + 3 \times 2 = 6 \text{ ms}^{-1}$$

Taking the motion from 2s to 4s,

$$v = 6 + (-3)(2) = 0 \text{ ms}^{-1}$$

• CHECK POINT 3.7

$$1 (d) 7.2 = \frac{100}{(v + 5)(5/18)} \text{ or } v = 45 \text{ ms}^{-1}$$

2 (a) Let positive direction of motion be from south to north.

$$\text{Given, } v_A = +54 \text{ kmh}^{-1} = 15 \text{ ms}^{-1},$$

$$v_B = -90 \text{ kmh}^{-1} = -25 \text{ ms}^{-1}$$

$\therefore$ The relative velocity of B w.r.t. A ,

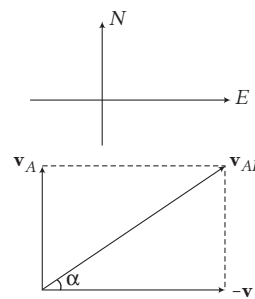
$$v_{BA} = v_B - v_A = -25 - 15 = -40 \text{ ms}^{-1}$$

i.e. The train B appears to A to move with a speed of 40 ms^{-1} from north to south.

3 (c) Both the particles will fall same distance in same time interval.

So, the relative separation will remain unchanged.

4 (b) $v_{AB} = v_A - v_B$



$$v_{AB} = \sqrt{v_A^2 + v_B^2} = 5 \text{ kmh}^{-1}$$

$$\alpha = \tan^{-1} \left(\frac{v_A}{v_B} \right) = \tan^{-1} \left(\frac{3}{4} \right)$$

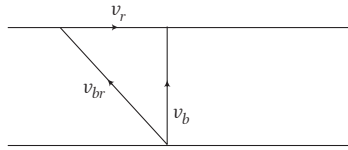
$$= 37^\circ \text{ east of north}$$

- 5 (b) Velocity of rain is at 30° in vertical direction. So, its horizontal component is $v_R \sin 30^\circ = \frac{v_R}{2}$. When man starts walking with 10 kmh^{-1} rain appears vertical. So, horizontal component $\frac{v_R}{2}$ is balanced by his speed of 10 kmh^{-1} . Thus,
- $$\frac{v_R}{2} = 10 \text{ or } v_R = 20 \text{ kmh}^{-1}$$

6 (c) $\tan 60^\circ = \frac{v_H}{v_V}$ or $\sqrt{3} = \frac{12}{v_V}$
 $\therefore v_V = 4\sqrt{3} \text{ kmh}^{-1}$

7 (a) $\tan \theta = \frac{v_H}{v_V} = \frac{v}{u}$
 or $\theta = \tan^{-1} \left(\frac{v}{u} \right)$

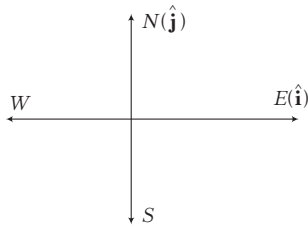
8 (d) $v_b = \frac{1}{1/4} = 4 \text{ kmh}^{-1}$



$$v_{br} = 5 \text{ kmh}^{-1}$$

$$\therefore v_r = v_{br} - v_b = 1 \text{ kmh}^{-1}$$

9 (b) $\mathbf{v}_x = -v \hat{j}$



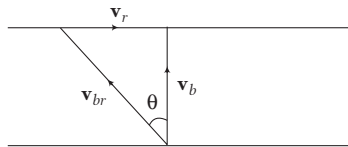
$$\therefore \mathbf{v}_Y = \mathbf{v}_{YX} + \mathbf{v}_X = -v \hat{i} + v \hat{j}$$

$$|\mathbf{v}_Y| = \sqrt{2} v$$

Direction of $\mathbf{v}_Y$ is north-west.

- 10 (a) For shortest time, one should swim at right angles to river current.

11 (a) $\sin \theta = \frac{v_r}{v_{br}} = \frac{3}{5}$

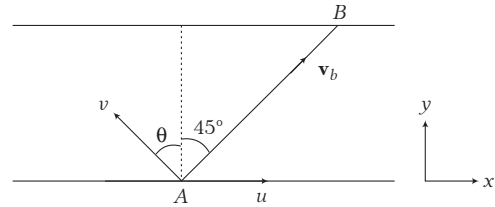


$$\therefore \theta = 37^\circ$$

The required angle is therefore

$$90^\circ + \theta = 90^\circ + 37^\circ = 127^\circ$$

- 12 (c) Let v be the speed of boatman in still water. Resultant of v and u should be along AB . Components of $\mathbf{v}_b$ (absolute velocity of boatman) along x and y -directions are,



$$v_x = u - v \sin \theta \text{ and } v_y = v \cos \theta$$

Further, $\tan 45^\circ = \frac{v_y}{v_x}$ or $1 = \frac{v \cos \theta}{u - v \sin \theta}$

$$\therefore v = \frac{u}{\sin \theta + \cos \theta} = \frac{u}{\sqrt{2} \sin(\theta + 45^\circ)}$$

v is minimum at,

$$\theta + 45^\circ = 90^\circ \text{ or } \theta = 45^\circ \text{ and } v_{\min} = \frac{u}{\sqrt{2}}$$

13 (b) $v_r = 10 + 15 = 25 \text{ ms}^{-1}$

where, v_r is relative velocity.

$$\therefore t = \frac{50 + 50}{25} = 4 \text{ s}$$

- 14 (b) Relative acceleration = 0, relative velocity is 40 ms^{-1} and relative separation is 100 m .

$$\therefore t = \frac{100}{40} = 2.5 \text{ s}$$

(A) Taking it together

1 (b) Average speed of a boy = $\frac{2v_1v_2}{v_1 + v_2} = \frac{2 \times 2.5 \times 4}{2.5 + 4}$
 $= \frac{20}{6.5} = \frac{40}{13} \text{ km/h}$

2 (a) $s_n = u + \frac{a}{2}(2n - 1)$

$$\Rightarrow s_3 = 0 + \frac{4/3}{2}(2 \times 3 - 1) \Rightarrow s_3 = \frac{10}{3} \text{ m}$$

3 (c) Time taken to travel first half distance, $t_1 = \frac{l/2}{v_1} = \frac{l}{2v_1}$

Time taken to travel second half distance, $t_2 = \frac{l}{2v_2}$

$$\text{Total time} = t_1 + t_2 = \frac{l}{2v_1} + \frac{l}{2v_2} = \frac{l}{2} \left(\frac{1}{v_1} + \frac{1}{v_2} \right)$$

We know that, v_{av} = Average speed

$$= \frac{\text{total distance}}{\text{total time}} = \frac{l}{\frac{l}{2} \left(\frac{1}{v_1} + \frac{1}{v_2} \right)} = \frac{2v_1v_2}{v_1 + v_2}$$

- 4 (d) In all the given graphs, we have more than one value of speed corresponding to a single time. However, the graph must have a unique value of speed corresponding to a single time. Thus, all the three graphs are not possible.

5 (b) From third equation of motion, $v^2 = u^2 + 2as$

$$\Rightarrow a = \frac{v^2 - u^2}{2s} = \frac{(20)^2 - (10)^2}{2 \times 135} = \frac{300}{270} = \frac{10}{9} \text{ ms}^{-2}$$

From first equation of motion,

$$v = u + at \Rightarrow t = \frac{v - u}{a} = \frac{20 - 10}{10/9} = \frac{10}{10/9} = 9 \text{ s}$$

- 6 (c) Man walks from his home to market with a speed of 5 kmh^{-1} . Distance = 2.5 km and time

$$= \frac{d}{v} = \frac{2.5}{5} = \frac{1}{2} = 30 \text{ min}$$

and he returns back with speed of 7.5 kmh^{-1} in rest a time of 20 min.

$$\text{Distance} = 7.5 \times \frac{20}{60} = 2.5 \text{ km}$$

$$\text{So, average speed} = \frac{\text{total distance}}{\text{total time}} = \frac{(2.5 + 2.5)}{(50/60) \text{ h}} = \frac{30}{5} \text{ km/h}$$

- 7 (d) $s \propto u^2$. If u becomes 3 times, then s will become 9 times, i.e. $9 \times 20 = 180 \text{ m}$

- 8 (a) Distance travelled by body A in 5th second and distance travelled by body B in 3rd second of its motion are equal.

$$0 + \frac{a_1}{2} (2 \times 5 - 1) = 0 + \frac{a_2}{2} (2 \times 3 - 1)$$

$$9a_1 = 5a_2 \Rightarrow \frac{a_1}{a_2} = \frac{5}{9}$$

- 9 (a) $\therefore$ Slope of displacement-time graph at the point E is negative. Thus, at the point E, the instantaneous velocity of the particle is negative.

- 10 (d) Since $x = 1.2t^2$, comparing it with equation of motion $x = \frac{1}{2}at^2$.

Thus, the motion is uniformly accelerated.

- 11 (c) It is given that, $v = \frac{t^2}{10} + 20$

Differentiating both sides w.r.t. time, we get

$$\frac{dv}{dt} = \frac{2t}{10} + 0 = \frac{t}{5}$$

$$\therefore \text{Acceleration, } a = \frac{dv}{dt} = \frac{t}{5} \neq \text{constant}$$

Hence, it is a case of non-uniform acceleration.

- 12 (b) Given, $v = \sqrt{c_1 - c_2x}$

$$\text{We know that, } a = v \frac{dv}{dx}$$

$$\Rightarrow a = \sqrt{c_1 - c_2x} \cdot \frac{d}{dx} 9(\sqrt{c_1 - c_2x})$$

$$= \frac{1}{2} (-c_2) = -\frac{c_2}{2}$$

- 13 (d) $v_1 = \frac{s}{90}, v_2 = \frac{s}{60}$

$$\text{Now, } t = \frac{s}{v_1 + v_2} = \frac{s}{\frac{s}{90} + \frac{s}{60}}$$

$$= \frac{90 \times 60}{90 + 60} = 36 \text{ s}$$

- 14 (a) At $t = 2 \text{ s}$; $x_A = 10 \times 2 - \frac{1}{2} \times 4 \times (2)^2 = 12 \text{ m}$

$$\text{and } x_B = 20 \times 2 - \frac{1}{2} \times 2 \times (2)^2 = 36 \text{ m}$$

$\therefore$ Distance between A and B at that instant is 24 m.

- 15 (a) Given, $\sqrt{x} = t + 1$

Squaring both sides, we get

$$x = (t + 1)^2 = t^2 + 2t + 1$$

Differentiating it w.r.t. time t , we get

$$\frac{dx}{dt} = 2t + 2$$

$$\text{Velocity, } v = \frac{dx}{dt} = 2t + 2$$

- 16 (d) Average velocity = $\frac{\text{Total displacement}}{\text{Total time}}$

$$\begin{array}{c} \xrightarrow{t} \quad \xleftarrow{25-2t} \quad \xrightarrow{t} \\ \text{---} \end{array}$$

$$v = at = 5t$$

$$\text{or } 20 = \frac{\frac{1}{2} \times a \times t^2 + (at)(25-2t) + \frac{1}{2} \times a \times t^2}{25}$$

Solving this equation with $a = 5 \text{ ms}^{-2}$, we get

$$t = 5 \text{ s}$$

Thus, the particle moved uniformly for $(25 - 2t)$ or 15 s.

- 17 (d) Net displacement = 0 and total distance = $OP + PQ + QO$

$$= 1 + \frac{2\pi \times 1}{4} + 1 = \frac{14.28}{4} \text{ km}$$

$$\text{Average speed} = \frac{14.28}{4 \times 10 / 60} = \frac{6 \times 14.28}{4} = 21.42 \text{ km/h}$$

- 18 (d) Integrate twice to convert a - t equation into s - t equation.

$$\therefore a = \frac{v}{t} \Rightarrow v = at$$

$$\Rightarrow \int dv = \int a \cdot dt \Rightarrow v = \int pt \cdot dt = \frac{pt^2}{2}$$

$$\therefore s = v \times t \Rightarrow ds = v dt$$

$$\Rightarrow \int ds = \int_0^t v dt \Rightarrow s = \frac{1}{6} pt^3$$

- 19 (c) Distance travelled by the particle is $x = 40 + 12t - t^3$

We know that, speed is defined as the rate of change of

$$\text{distance, i.e. } v = \frac{dx}{dt}$$

$$\therefore v = \frac{d}{dt} (40 + 12t - t^3) = 0 + 12 - 3t^2$$

But final velocity, $v = 0$

$$\therefore 12 - 3t^2 = 0 \Rightarrow t^2 = \frac{12}{3} = 4$$

$$\Rightarrow t = 2 \text{ s}$$

Hence, distance travelled by the particle before coming to rest is given by

$$x = 40 + 12(2) - (2)^3 = 40 + 24 - 8 = 64 - 8 = 56 \text{ m}$$

- 20 (b) Given, $s = 1.2 \text{ m}$, $v = 640 \text{ ms}^{-1}$,

$$\begin{aligned} a &= ?; u = 0; t = ? \\ 2as &= v^2 - u^2 \\ \Rightarrow 2a \times 1.2 &= 640 \times 640 \\ \Rightarrow a &= \frac{8 \times 64 \times 100}{3} \\ v &= u + at \\ \Rightarrow t &= \frac{v}{a} = \frac{640 \times 3}{8 \times 64 \times 100} = 37.5 \times 10^{-3} \text{ s} \approx 40 \text{ ms} \end{aligned}$$

- 21 (a) Here, $u = 0$, $a = g$

Distance travelled in n th second is given by

$$\begin{aligned} D_n &= u + \frac{a}{2}(2n - 1) \\ \therefore D_n &\propto (2n - 1) \\ \therefore D_1 : D_2 : D_3 : D_4 : D_5 : \dots &= 1 : 3 : 5 : 7 : 9 : \dots \end{aligned}$$

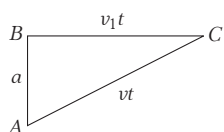
- 22 (b) The area under acceleration-time graph gives change in velocity.

As acceleration is zero at the end of 11 s,

$$\text{i.e. } v_{\max} = \text{Area of } \triangle OAB = \frac{1}{2} \times 11 \times 10 = 55 \text{ ms}^{-1}$$

- 23 (a) $a_{\text{av}} = \frac{v_f - v_i}{\Delta t}$ (As they are in opposite direction)
- $$\begin{aligned} &= \frac{\sqrt{2gh_f} + \sqrt{2gh_i}}{\Delta t} = \frac{\sqrt{2 \times 10 \times 5} + \sqrt{2 \times 10 \times 10}}{0.01} \\ &= 2414.21 \approx 2414 \text{ ms}^{-2} \end{aligned}$$

- 24 (b) Let two boys meet at point C after time 't' from the starting. Then, $AC = vt$, $BC = v_1 t$ (see figure)



$$(AC)^2 = (AB)^2 + (BC)^2 \Rightarrow v^2 t^2 = a^2 + v_1^2 t^2$$

By solving, we get

$$t = \sqrt{\frac{a^2}{v^2 - v_1^2}}$$

- 25 (b) Let a be the retardation of boggy, then distance covered by it be s_b . If u is the initial velocity of boggy after detaching from train (i.e. uniform speed of train)

$$\begin{aligned} v^2 &= u^2 + 2as \Rightarrow 0 = u^2 - 2as \\ \Rightarrow s_b &= \frac{u^2}{2a} \end{aligned}$$

Time taken by boggy to stop

$$v = u + at \Rightarrow 0 = u - at \Rightarrow t = \frac{u}{a}$$

In this time t , distance travelled by train, $s_t = ut = \frac{u^2}{a}$

$$\text{Hence, ratio } \frac{s_b}{s_t} = \frac{1}{2}.$$

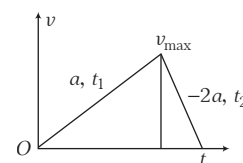
- 26 (d) Let acceleration is a and retardation is $-2a$.

Then, for accelerated motion,

$$t_1 = \frac{v}{a} \quad \dots(i)$$

For retarding motion,

$$t_2 = \frac{v}{2a} \quad \dots(ii)$$



$$\text{Given, } t_1 + t_2 = 9 \Rightarrow \frac{v}{a} + \frac{v}{2a} = 9 \Rightarrow \frac{3v}{2a} = 9 \Rightarrow \frac{v}{a} = 6$$

Hence, duration of acceleration, $t_1 = \frac{v}{a} = 6 \text{ s}$.

- 27 (d) Given, $x = ae^{-\alpha t} + be^{\beta t}$

$$\text{So, velocity, } v = \frac{dx}{dt} = -a\alpha e^{-\alpha t} + b\beta e^{\beta t}$$

$$a = \frac{dv}{dt} = a\alpha^2 e^{-\alpha t} + b\beta^2 e^{\beta t} = +ve \text{ all time.}$$

$\therefore v$ will go on increasing.

- 28 (b) $1 = \frac{1}{2} g t_1^2$ or $t_1 = \sqrt{\frac{2}{g}}$

$$2 = \frac{1}{2} g t^2 \text{ or } t = \sqrt{\frac{4}{g}}$$

$$\text{But } t_2 = t - t_1 = \sqrt{\frac{4}{g}} - \sqrt{\frac{2}{g}}$$

$$\begin{aligned} \therefore \frac{t_1}{t_2} &= \frac{\sqrt{2/g}}{\sqrt{4/g} - \sqrt{2/g}} = \frac{\sqrt{2}}{2 - \sqrt{2}} \\ &= \frac{\sqrt{2}(2 + \sqrt{2})}{2} = (\sqrt{2} + 1) \end{aligned}$$

- 29 (c) $t = \sqrt{\frac{x+a}{b}}$ or $(x+a) = bt^2$ or $x = -a + bt^2$

Comparing this equation with general equation of uniformly accelerated motion, $s = s_i + ut + \frac{1}{2} at^2$

we see that $s_i = -a$, $u = 0$ and acceleration $= 2b$.

- 30 (b) Given, acceleration, $a = 6t + 5$

$$\therefore a = \frac{dv}{dt} = 6t + 5, dv = (6t + 5)dt$$

Integrating it, we have

$$\int_0^v dv = \int_0^t (6t + 5)dt$$

$v = 3t^2 + 5t + C$, where C is a constant of integration.

When $t = 0$, $v = 0$, so $C = 0$

$$\therefore v = \frac{ds}{dt} = 3t^2 + 5t \text{ or } ds = (3t^2 + 5t)dt$$

Integrating it within the conditions of motion, i.e. as t changes from 0 to 2s, s changes from 0 to s , we have

$$\int_0^s ds = \int_0^2 (3t^2 + 5t)dt$$

$$\therefore s = \left[t^3 + \frac{5}{2} t^2 \right]_0^2 = 8 + 10 = 18 \text{ m}$$

31 (a) Given, $x = \frac{1}{t+5}$

Differentiating both sides w.r.t. t , we get

$$\frac{dx}{dt} = -\frac{1}{(t+5)^2} \times 1$$

$$\therefore \text{Velocity, } v = \frac{dx}{dt} = -\frac{1}{(t+5)^2} \quad \dots(i)$$

Again, differentiating both sides w.r.t. t , we get

$$\frac{dv}{dt} = -\frac{1 \times 2}{(t+5)^3} (1) = -\frac{2}{(t+5)^3} \quad \dots(ii)$$

Now, from Eqs. (i) and (ii), we get

$$\frac{dv}{dt} = -2v \times \frac{1}{(t+5)} = (-2v)(\sqrt{v}) = -2(v)^{3/2}$$

$$\therefore \text{Acceleration, } a = \frac{dv}{dt} = -2v^{3/2} \Rightarrow a \propto (\text{velocity})^{3/2}$$

32 (b) $v = \frac{dx}{dt} = 32 - 6t^2$, $v = 0$ at $t = 2.30$ s

$$a = \frac{dv}{dt} = -12t$$

At 2.30 s, $a = -27.6 \text{ ms}^{-2}$

33 (b) For P, $30 = 15 + at$ or $at = 15 \text{ ms}^{-1}$

For Q, $v = 20 - at$ or $v = 20 - 15 = 5 \text{ ms}^{-1}$

34 (c) Let the man will catch the bus after t second. So, he will cover distance ut .

Similarly, distance travelled by the bus will be $\frac{1}{2}at^2$.

For the given condition,

$$ut = 45 + \frac{1}{2}at^2 = 45 + 1.25t^2 \quad (\text{As, } a = 2.5 \text{ ms}^{-2})$$

$$\Rightarrow u = \frac{45}{t} + 1.25t$$

To find the minimum value of u

$$\frac{du}{dt} = 0 \Rightarrow \frac{du}{dt} = \frac{-45}{t^2} + 1.25 = 0$$

So, we get $t = 6$ s, then $u = \frac{45}{6} + (1.25 \times 6) = 15 \text{ ms}^{-1}$

35 (b) $x^2 = t^2 + 1$ or $2x \cdot \frac{dx}{dt} = 2t$ or $x \cdot \frac{dx}{dt} = t$

$$\therefore v = \frac{dx}{dt} = \frac{t}{x} = \frac{t}{\sqrt{t^2 + 1}}$$

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = \frac{\sqrt{t^2 + 1} - \frac{t^2}{\sqrt{t^2 + 1}}}{(t^2 + 1)} = \frac{1}{(t^2 + 1)^{3/2}} = \frac{1}{x^3}$$

36 (b) Average velocity in uniformly accelerated motion is given by

$$v_{av} = \frac{s}{t} = \frac{ut + \frac{1}{2}at^2}{t} = u + \frac{1}{2}at$$

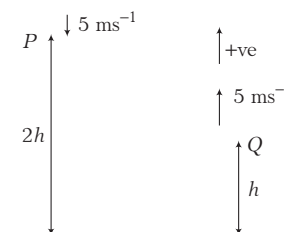
Now, $v_1 = u + \frac{1}{2}at_1$, $v_2 = (u + at_1) + \frac{1}{2}at_2$

and $v_3 = u + a(t_1 + t_2) + \frac{1}{2}at_3$

$$\therefore (v_1 - v_2) : (v_2 - v_3) = (t_1 + t_2) : (t_2 + t_3)$$

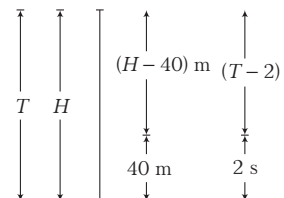
37 (d) Motion is first uniformly accelerated in positive direction, then it is uniform in negative direction.

38 (c) $h = -5t + 5t^2$ and $2h = 5t + 5t^2$



From these two equations, we get $t = 3$ s and $h =$ distance between P and Q = 30 m.

39 (b) Let height of Minaret be H and body take time T to fall from top to bottom.



$$H = \frac{1}{2}gT^2 \quad \dots(i)$$

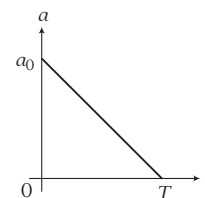
In last 2 s, body travels distance of 40 m, so in $(T-2)$ second, distance travelled = $(H-40)$ m

$$\Rightarrow (H-40) = \frac{1}{2}g(T-2)^2 \quad \dots(ii)$$

By solving Eqs. (i) and (ii), we get

$$T = 3 \text{ s and } H = 45 \text{ m}$$

40 (a) Given acceleration-time graph is shown in figure.



Let $a(t=0) = a_0 =$ constant

From the graph, we have

$$\frac{t}{T} + \frac{a}{a_0} = 1$$

$$\Rightarrow a_0t + aT = a_0T \quad \dots(i)$$

Also, we know that,

$$a = \frac{dv}{dt} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{a_0T - a_0t}{T} = \frac{dv}{dt}$$

Integrating both sides, we get

$$\int_0^t (a_0 T - a_0 t) dt = T \int_0^v dv$$

$$\Rightarrow a_0 T t - \frac{a_0 t^2}{2} = T v \Rightarrow v = a_0 t - \frac{a_0}{2T} t^2$$

Therefore, v - t graph must be parabolic.

Now, at $t = 0, v = 0$

$$t = T, v = \frac{a_0 T}{2} = \text{constant}$$

These conditions are satisfied by graph (a).

- 41** (a) Given line has positive intercept but negative slope. So, its equation can be written as

$$v = -mx + v_0 \left(\text{where, } m = \tan \theta = \frac{v_0}{x_0} \right) \quad \dots(i)$$

By differentiating with respect to time, we get

$$\frac{dv}{dt} = -m \frac{dx}{dt} = -mv$$

Now, substituting the value of v from Eq. (i), we get

$$\frac{dv}{dt} = -m(-mx + v_0) = m^2 x - mv_0$$

$$\therefore a = m^2 x - mv_0$$

i.e. The graph between a and x should have positive slope but negative intercept on a -axis. So, graph (a) is correct.

- 42** (b) $\frac{dx}{dt} = v = -1 - 2t$

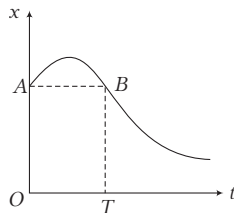
Comparing with $v = u + at$, we get

$$u = -1 \text{ ms}^{-1} \text{ and } a = -2 \text{ ms}^{-2}.$$

At $t = 0, x = 1 \text{ m}$. Then, u and a both are negative. Hence, x -coordinate of particle will go on decreasing.

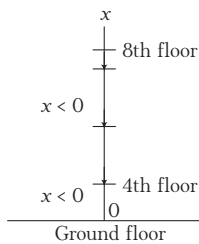
- 43** (b) In graph (b) for one value of displacement, there are two different points of time. Hence, for one time, the average velocity is positive and for other time is equivalent negative.

As there are opposite velocities in the interval 0 to T , hence average velocity can vanish in (b). This can be seen in the figure alongside. Here, $OA = BT$ (same displacement) for two different points of time.



- 44** (a) As the lift is coming in downward directions, displacement will be negative. We have to see whether the motion is accelerating or retarding. We know that due to downward motion, displacement will be negative. When the lift reaches 4th floor and is about to stop, hence motion is retarding in nature, hence $x < 0$; $a > 0$.

As displacement is in negative direction, velocity will also be negative, i.e. $v < 0$.



- 45** (b) Given, $x = (t - 2)^2$

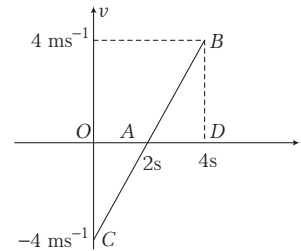
$$\text{Velocity, } v = \frac{dx}{dt} = \frac{d}{dt} (t - 2)^2 = 2(t - 2) \text{ ms}^{-1}$$

$$\text{Acceleration, } a = \frac{dv}{dt} = \frac{d}{dt} [2(t - 2)] = 2(1 - 0) = 2 \text{ ms}^{-2}$$

$$\text{When } t = 0; v = -4 \text{ ms}^{-1}, t = 2 \text{ s}; v = 0 \text{ ms}^{-1}$$

$$t = 4 \text{ s}; v = 4 \text{ ms}^{-1}$$

v - t graph is shown in figure below.



$$\begin{aligned} \text{Distance travelled} &= \text{Area under the } v\text{-}t \text{ graph} \\ &= |\text{Area } OAC| + \text{Area } ABD \\ &= \left| \frac{4 \times 2}{2} \right| + \frac{1}{2} \times 2 \times 4 = 8 \text{ m} \end{aligned}$$

- 46** (d) Let height of tower is h and body takes t time to reach to ground when it fall freely.

$$\therefore h = \frac{1}{2} g t^2 \quad \dots(i)$$

In last second, i.e. t th second body travels $= 0.36 h$

$$\begin{aligned} \text{It means in rest of the time, i.e. in } (t-1) \text{ second, it travels} \\ = h - 0.36 h = 0.64 h \end{aligned}$$

Now, applying equation of motion for $(t-1)$ second

$$0.64 h = \frac{1}{2} g (t-1)^2 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$t = 5 \text{ s and } h = 125 \text{ m}$$

- 47** (a) Acceleration, $f = f_0 \left(1 - \frac{t}{T} \right)$

$$\frac{dv}{dt} = f_0 \cdot \left(1 - \frac{t}{T} \right) \quad \left(\because f = \frac{dv}{dt} \right)$$

$$\Rightarrow dv = f_0 \cdot \left(1 - \frac{t}{T} \right) dt \Rightarrow \int dv = \int f_0 \left(1 - \frac{t}{T} \right) dt$$

$$v = f_0 t - f_0 \frac{t^2}{2T} + C \quad \dots(i)$$

where, C is constant.

When $t = 0, v = 0$ thus from Eq. (i), $C = 0$

$$v = f_0 t - \frac{f_0}{2T} t^2 \quad \dots(ii)$$

$$\text{As, } f = f_0 \left(1 - \frac{t}{T} \right)$$

When $f = 0$

$$\therefore f_0 \left(1 - \frac{t}{T} \right) = 0; f_0 \neq 0 \Rightarrow t = T$$

Putting $t = T$ in Eq. (ii), we get

$$v = f_0 T - \frac{f_0}{T} \cdot \frac{T^2}{2} = \frac{f_0 T}{2}$$

48 (d) Relative to lift,

$$u_r = 0, a_r = (9.8 + 1.2) = 11 \text{ ms}^{-2}$$

$$s_r = \frac{1}{2} a_r t^2 \Rightarrow 2.7 = \frac{1}{2} \times 11 \times t^2 \Rightarrow t = \sqrt{\frac{5.4}{11}} \text{ s}$$

49 (a) After bailing out from point A, parachutist falls freely under gravity. The velocity acquired by it will be v .

$$\text{From } v^2 = u^2 + 2as = 0 + 2 \times 9.8 \times 50 = 980$$

$$(\text{As, } u = 0, a = 9.8 \text{ ms}^{-2}, s = 50 \text{ m})$$

At point B, parachute opens and it moves with retardation of 2 ms^{-2} and reach at ground (point C with velocity of 3 ms^{-1}).

For the part BC by applying the equation $v^2 = u^2 + 2as$

$$v = 3 \text{ ms}^{-1}, u = \sqrt{980} \text{ ms}^{-1}, a = -2 \text{ ms}^{-2}, s = h$$

$$\Rightarrow (3)^2 = (\sqrt{980})^2 + 2 \times (-2) \times h$$

$$\Rightarrow 9 = 980 - 4h$$

$$\Rightarrow h = \frac{980 - 9}{4} = \frac{971}{4} = 242.7 \approx 243 \text{ m}$$

$$\text{So, the total height by which parachutist bails out} \\ = 50 + 243 = 293 \text{ m}$$

50 (c) Initial relative velocity $= v_1 - v_2$, final relative velocity $= 0$

$$\text{From } v^2 = u^2 - 2as \Rightarrow 0 = (v_1 - v_2)^2 - 2 \times a \times s$$

$$\Rightarrow s = \frac{(v_1 - v_2)^2}{2a}$$

If the distance between two cars is s , then collision will take place. To avoid collision $d > s$

$$\therefore d > \frac{(v_1 - v_2)^2}{2a}$$

where, d = actual initial distance between two cars.

51 (b) Time taken by first drop to reach the ground, $t = \sqrt{\frac{2h}{g}}$

$$\Rightarrow t = \sqrt{\frac{2 \times 5}{10}} = 1 \text{ s}$$

As the water drops fall at regular intervals from a tap, therefore time difference between any two drops $= \frac{1}{2} \text{ s}$.

In this given time, distance of second drop from the tap

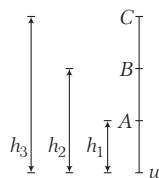
$$= \frac{1}{2} g \left(\frac{1}{2} \right)^2 = \frac{10}{8} = 1.25 \text{ m}$$

Its distance from the ground $= 5 - 1.25 = 3.75 \text{ m}$

$$\textbf{52} \text{ (d) } A \Rightarrow \frac{u^2}{4} - u^2 = -2gh_1$$

$$B \Rightarrow \frac{u^2}{9} - u^2 = -2gh_2$$

$$C \Rightarrow \frac{u^2}{16} - u^2 = -2gh_3$$



$$\therefore AB = \frac{u^2}{2g} \left(\frac{8}{9} - \frac{3}{4} \right) = \frac{u^2}{2g} \cdot \frac{5}{36}$$

$$BC = \frac{u^2}{2g} \left(\frac{15}{16} - \frac{8}{9} \right) = \frac{u^2}{2g} \cdot \frac{7}{144}$$

$$\therefore \frac{AB}{BC} = \frac{5}{36} \times \frac{144}{7} = \frac{20}{7}$$

53 (a) For the given condition, initial height $h = d$ and velocity of the ball is zero. When the ball moves downward, its velocity increases and it will be maximum, when the ball hits the ground and just after the collision, it becomes half and in opposite direction. As the ball moves upwards, its velocity again decreases and becomes zero at height $d/2$. This explanation match with graph (a).

54 (c) The position x of a particle w.r.t. time t along X-axis,

$$x = 9t^2 - t^3 \quad \dots(i)$$

Differentiating Eq. (i) w.r.t. time, we get speed, i.e.

$$v = \frac{dx}{dt} = \frac{d}{dt} (9t^2 - t^3)$$

$$\Rightarrow v = 18t - 3t^2 \quad \dots(ii)$$

Again, differentiating Eq. (ii) w.r.t. time, we get acceleration, i.e.

$$a = \frac{dv}{dt} = \frac{d}{dt} (18t - 3t^2)$$

$$\Rightarrow a = 18 - 6t \quad \dots(iii)$$

Now, when speed of particle is maximum, its acceleration is zero.

$$a = 0$$

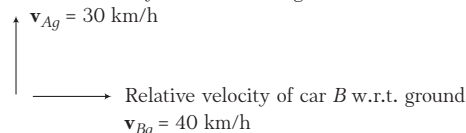
$$\Rightarrow 18 - 6t = 0$$

$$\Rightarrow t = 3 \text{ s}$$

Putting in Eq. (i), we obtain position of particle at that time

$$x = 9(3)^2 - (3)^3 = 9(9) - 27 \\ = 81 - 27 = 54 \text{ m}$$

55 (c) Relative velocity of car A w.r.t. ground



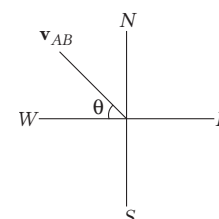
$\therefore$ Relative velocity of A w.r.t. B,

$$\mathbf{v}_{AB} = \mathbf{v}_{Ag} - \mathbf{v}_{Bg} = \mathbf{v}_{Ag} + (-\mathbf{v}_{Bg})$$

$$|\mathbf{v}_{AB}| = \sqrt{(30)^2 + (40)^2} = 50 \text{ km/h}$$

$$\therefore \tan \theta = \frac{30}{40} = \frac{3}{4}$$

$$\Rightarrow \theta = \tan^{-1}(3/4) \text{ N of west}$$

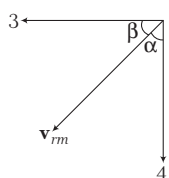


56 (d) Relative velocity of man w.r.t. ground $\mathbf{v}_{mg} = 3 \text{ m/s}$

Relative velocity of rain w.r.t. ground, $\mathbf{v}_{rg} = 4 \text{ m/s}$

Relative velocity of rain w.r.t. man, $\mathbf{v}_{rm} = ?$

$$\mathbf{v}_{rm} = \mathbf{v}_{rg} - \mathbf{v}_{mg} = \mathbf{v}_{rg} + (-\mathbf{v}_{mg})$$

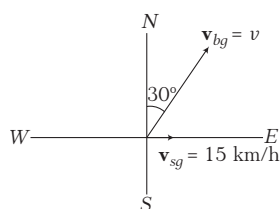


$$v_{rm} = 5 \text{ m/s}$$

$$\tan \alpha = \frac{3}{4}, \quad \tan \beta = \frac{4}{3}$$

$\mathbf{v}_{rm} = 5 \text{ m/s}$ at $\tan^{-1}\left(\frac{3}{4}\right)$ with vertical or at $\tan^{-1}\left(\frac{4}{3}\right)$ with horizontal.

57 (a)



where, $\mathbf{v}_{bg}$ is the velocity of boat w.r.t. ground and $\mathbf{v}_{sg}$ is the velocity of ship w.r.t. ground.

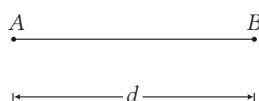
Relative velocity of boat w.r.t. ship $\mathbf{v}_{bs}$ is along north.

$\mathbf{v}_{bs}$ is resultant of $\mathbf{v}_{bg}$ and $\mathbf{v}_{sg}$ in opposite direction.

$$v \cos 30^\circ = v_{bs}$$

$$v \sin 30^\circ = 15 \Rightarrow v = 30 \text{ km/h}$$

58 (b)



$u : v_{rg}$ (velocity of river w.r.t. ground)

$v : v_{mr}$ (velocity of man w.r.t. river)

$$\text{Along the flow, } v_{mg} = v + u, \quad 3 = \frac{d}{v + u} \quad \dots(i)$$

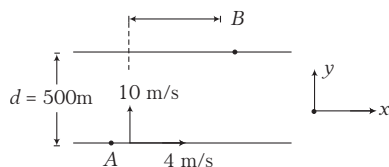
where, v_{mg} is velocity of man w.r.t. ground.

Opposite to the flow,

$$v_{mg} = v - u, \quad 6 = \frac{d}{v - u} \quad \dots(ii)$$

$$\Rightarrow t_0 = \frac{d}{v} = 4 \text{ h}$$

59 (c)



y : displacement = velocity $\times$ time

$$500 = 10t$$

$$\Rightarrow t = 50 \text{ s}$$

(B) Medical entrance special format questions

• Assertion and reason

1 (c) When a particle is released from rest, $v = 0$ but $a \neq 0$.

2 (a) $a = \frac{dv}{dt}$ = slope of v - t graph.

Perpendicular to t -axis, slope = ∞ , therefore $a = \infty$

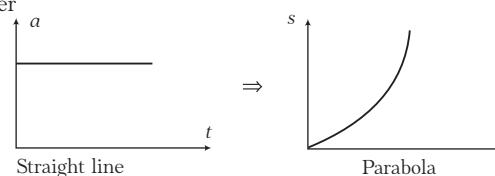
3 (c) Slope of s - t graph = Velocity = Positive

At $t = 0$, $s \neq 0$. Further at $t = t_0$: $s = 0$, $v \neq 0$.

Thus, v is constant.

4 (b) By differentiating a - t equation two times, we will get s - t equation.

Further



5 (c) Ascending means velocity is in upward direction. Speed is decreasing. It means acceleration is downwards.

Further, the body moves in the direction of velocity.

• Statement based questions

1 (a) From the given x - t graph, it is clear that

$$\text{at } t = 0, \quad \frac{dx}{dt} = 0$$

$$\therefore \text{Velocity, } v = \frac{dx}{dt} = 0, \text{ at } t = 0$$

As time passes $\frac{dx}{dt}$ increases and then decreases.

$\therefore$ Velocity and hence, acceleration changes.

2 (b) $x = 4(t - 2) + a(t - 2)^2$

$$\text{At } t = 0, \quad x = -8 + 4a = 4a - 8$$

$$v = \frac{dx}{dt} = 4 + 2a(t - 2)$$

$$\text{At } t = 0, \quad v = 4 - 4a = 4(1 - a)$$

$$\text{But acceleration, } a = \frac{d^2x}{dt^2} = 2a$$

3 (b) For maximum and minimum displacement, we have to keep in mind the magnitude and direction of maximum velocity.

As maximum velocity in positive direction is v_0 maximum velocity in opposite direction is also v_0 .

Maximum displacement in one direction = $v_0 T$

Maximum displacement in opposite directions = $-v_0 T$

Hence, $-v_0 T < x < v_0 T$.

4 (c) Distance travelled in last second of upward journey is independent of velocity of projection.

$$\text{Distance travelled} = \frac{1}{2} g t^2 = \frac{1}{2} \times 9.8 \times (1)^2 = 4.9 \text{ m}$$

$$\begin{aligned} 5 \text{ (c) Average velocity} &= \frac{\text{Displacement}}{\text{Time}} \\ &= \frac{\text{Area of } v-t \text{ graph}}{\text{Time}} \\ &= \frac{(1/2)v_m t_0}{t_0} = \frac{1}{2}v_m \end{aligned}$$

• Match the columns

- 1 (b) With constant positive acceleration, speed will increase when velocity is positive, speed will decrease, if velocity is negative.

Similarly, with constant negative acceleration, speed will increase, if velocity is negative and speed will decrease, if velocity is positive.

Hence, A → p,q, B → p,q, C → r, D → p,q.

- 2 (c) At $t = 3$ s, $s = 10 + 20(3) - 5(3)^2 = 25$ unit

$$v = \frac{ds}{dt} = 20 - 10t$$

At $t = 4$ s, $v = 20 - 10(4) = -20$ unit

$$a = \frac{dv}{dt} = -10 \text{ units}$$

Hence, A → r, B → s, C → p.

- 3 (c) $v_i = +10 \text{ ms}^{-1}$ and $v_f = 0$

$$\therefore \Delta v = v_f - v_i = -10 \text{ ms}^{-1}$$

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{-10}{6} = -\frac{5}{3} \text{ ms}^{-2}$$

Total displacement = Area under $v-t$ graph (with sign)

$$= \frac{1}{2} \times 10 \times 2 - \frac{1}{2} \times 2 \times 10 - \frac{1}{2} \times 2 \times 10 = -10 \text{ m}$$

and acceleration = slope of $v-t$ graph

$$= -\frac{10}{2} = -5 \text{ ms}^{-2}$$

Hence, A → r, B → p, C → r, D → s.

- 4 (b) For M , slope of $s-t$ graph is positive and increasing. Therefore, velocity of the particle is positive and increasing. Hence, it is A type motion. Similarly, N , P and Q can be observed from the slope.

Hence, A → r, B → s, C → p, D → q.

(C) Medical entrances' gallery

- 1 (c) Given, $u = 20 \text{ m/s}$, $v = 80 \text{ m/s}$ and $h = ?$

From kinematic equation of motion, $v^2 = u^2 + 2gh$

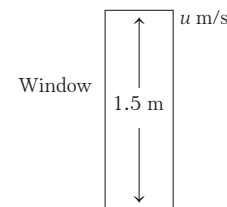
$$\begin{aligned} \Rightarrow h &= \frac{v^2 - u^2}{2g} = \frac{(80)^2 - (20)^2}{2 \times 10} \quad (\because \text{Given, } g = 10 \text{ m/s}^2) \\ &= 300 \text{ m} \end{aligned}$$

- 2 (b) According to question, time taken by the ball to cross the window,

$$t = 0.1 \text{ s}, h = 1.5 \text{ m}$$

If u be the velocity at the top most point of the window, then from equation of motion,

$$\begin{aligned} h &= ut + \frac{1}{2}gt^2 \\ \Rightarrow 1.5 &= u \times 0.1 + \frac{1}{2} \times 10 \times (0.1)^2 \\ \Rightarrow 1.5 &= 0.1u + 0.05 \\ \Rightarrow u &= \frac{1.5 - 0.05}{0.1} = \frac{1.45}{0.1} = 14.5 \text{ m/s} \end{aligned}$$



- 3 (b) For distance x , the person moves with constant velocity v_1 and for another x distance, he moves with constant velocity of v_2 , then

Total distance travelled by the person,

$$D = x + x = 2x$$

Total time taken to cover that distance,

$$T = t_1 + t_2 = \frac{x}{v_1} + \frac{x}{v_2} \quad \left(\because t = \frac{\text{Distance}}{\text{Velocity}} \right)$$

Average velocity,

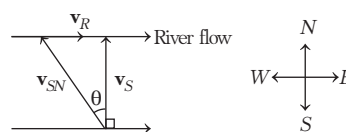
$$v_{av} = \frac{\text{Total distance}}{\text{Total time}} = \frac{D}{T}$$

$$v = \frac{2x}{\frac{x}{v_1} + \frac{x}{v_2}} = \frac{2}{\frac{1}{v_1} + \frac{1}{v_2}} \quad (\because v_{av} = v)$$

$$\Rightarrow \frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$$

- 4 (d) Given, speed of river, $v_R = 10 \text{ ms}^{-1}$

Speed of swimmer in still water, $v_{SN} = 20 \text{ ms}^{-1}$



For the shortest path to cross the river, he should swim at an angle $(90^\circ + \theta)$ with the stream flow. From the above figure,

$$\mathbf{v}_{SN} = \mathbf{v}_R + \mathbf{v}_S$$

So, angle θ is given by

$$\sin \theta = \left| \frac{\mathbf{v}_R}{\mathbf{v}_{SN}} \right| = \frac{10}{20} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$

As the river is flowing in east direction, so he should swim towards west.

- 5 (d) When a particle completes one revolution in circular motion, then average displacement travelled by particle is zero.

Hence, average velocity

$$= \frac{\text{average displacement}}{\Delta t} = \frac{0}{\Delta t} = 0$$

- 6 (b) Time interval between 8th and 3rd second,

$$\Delta t = 8 - 3 = 5 \text{ s, i.e. } \Delta t = 5 \text{ s}$$

Change in velocity,

$$\Delta v = 20 - 0 = 20 \text{ ms}^{-1}$$

$$\therefore \text{Average acceleration} = \frac{\Delta v}{\Delta t} = \frac{20}{5} = 4 \text{ ms}^{-2}$$

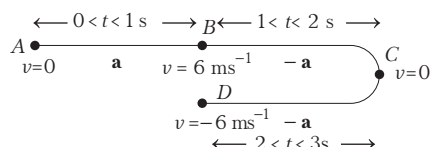
7 (b) According to the question,

For the time duration $0 < t < 1$ s,

the velocity increase from 0 to 6 ms^{-1} .

As the direction of field has been reversed, for $1 < t < 2$ s, the velocity firstly decreases from 6 ms^{-1} to 0.

Then, for $2 < t < 3$ s; as the field strength is same; the magnitude of acceleration would be same, but velocity increases from 0 to -6 ms^{-1} .



Acceleration of the car,

$$|a| = \left| \frac{v-u}{t} \right| = \frac{6-0}{1} = 6 \text{ ms}^{-2}$$

The displacement of the particle is given as

$$s = ut + \frac{1}{2}at^2$$

For $t = 0$ to $t = 1$ s,

$$u = 0, a = +6 \text{ ms}^{-2} \Rightarrow s_1 = 0 + \frac{1}{2} \times 6 \times (1)^2 = 3 \text{ m}$$

For $t = 1$ s to $t = 2$ s,

$$u = 6 \text{ ms}^{-1}, a = -6 \text{ ms}^{-2}$$

$$\Rightarrow s_2 = 6 \times 1 - \frac{1}{2} \times 6 \times (1)^2 = 6 - 3 = 3 \text{ m}$$

For $t = 2$ s to $t = 3$ s,

$$u = 0, a = -6 \text{ ms}^{-2} \Rightarrow s_3 = 0 - \frac{1}{2} \times 6 \times (1)^2 = -3 \text{ m}$$

$\therefore$ Net displacement,

$$s = s_1 + s_2 + s_3 = 3 \text{ m} + 3 \text{ m} - 3 \text{ m} = 3 \text{ m}$$

Hence, average velocity

$$= \frac{\text{net displacement}}{\text{total time}} = \frac{3}{3} = 1 \text{ ms}^{-1}$$

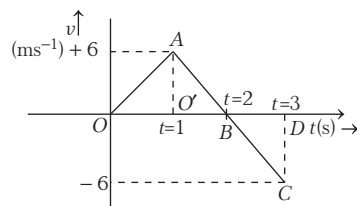
Total distance travelled, $d = 9 \text{ m}$

Hence, average speed

$$= \frac{\text{total distance}}{\text{total time}} = \frac{9}{3} = 3 \text{ ms}^{-1}$$

Alternate Method

Given condition can be represented through graph also as shown below.



$\therefore$ Displacement in three seconds

= Area under the graph

= Area of $\triangle OAO'$ + Area of $\triangle AO'B$ - Area of $\triangle BCD$

$$= \frac{1}{2} \times 1 \times 6 + \frac{1}{2} \times 1 \times 6 - \frac{1}{2} \times 6 \times 1 = 3 \text{ m}$$

$$\therefore \text{Average velocity} = \frac{3}{3} = 1 \text{ ms}^{-1}$$

Total distance travelled, $d = 9 \text{ m}$

$$\therefore \text{Average speed} = \frac{9}{3} = 3 \text{ ms}^{-1}$$

8 (c) When a particle is released from rest position under gravity, then $v = 0$, but $a \neq 0$.

Thus, Assertion is correct but Reason is incorrect.

9 (a) To find value of time at which velocity is maximum, taking differentiation of v with respect to time

$$\frac{dv}{dt} = 0$$

Given, $v = 4t(1-2t)$

$$v = 4t - 8t^2 \Rightarrow \frac{d}{dt}(4t - 8t^2) = 0$$

$$\Rightarrow 4 - 16t = 0 \Rightarrow t = \frac{1}{4} \text{ s} = 0.25 \text{ s}$$

Again, taking differentiation, we get

$$\Rightarrow \frac{d^2v}{dt^2} = -16 < 0$$

So, at $t = 0.25$ s, velocity is maximum.

10 (a) $\therefore$ Runner starts from O and goes to O following the path $OQRO$, so net displacement is zero.

$$\begin{aligned} \therefore \text{Average speed} &= \frac{\text{Total distance}}{\text{Total time}} = \frac{OQ + QR + RO}{\text{Total time}} \\ &= \frac{1 \text{ km} + (2\pi r)(1/4) + 1 \text{ km}}{1 \text{ h}} \\ &= 2 + \frac{\pi}{2} = 3.57 \text{ kmh}^{-1} \end{aligned}$$

11 (c) Time taken to reach the highest point from the height h is obtained from equation, $v = u - gt$

$$\therefore 0 = u - gt \text{ or } t = \frac{u}{g}$$

($\therefore$ At highest point, final velocity of ball = 0)

Height attained above h is obtained from $v^2 - u^2 = 2(-g)h_1$

$$\text{or } 0 - u^2 = 2(-g)h_1 \text{ or } h_1 = \frac{u^2}{2g}$$

$$\text{Total height, } h_2 = \left(h_1 + h = \frac{u^2}{2g} + h \right)$$

Time taken to hit the ground is obtained from

$$h_2 = ut + \frac{1}{2}at^2 \text{ or } \frac{u^2}{2g} + h = 0 + \frac{1}{2}gt^2$$

$$\therefore \text{Total time taken, } t = \sqrt{\frac{(u^2 + 2gh)}{g}}$$

12 (c) Speed of walking, $v_1 = \frac{h}{t_1}$

Speed of escalator, $v_2 = \frac{h}{t_2}$

Time taken when she walks over moving escalator,

$$t = \frac{h}{v_1 + v_2}$$

$$\Rightarrow \frac{1}{t} = \frac{v_1}{h} + \frac{v_2}{h} = \frac{1}{t_1} + \frac{1}{t_2}$$

$$\Rightarrow t = \frac{t_1 t_2}{t_1 + t_2}$$

13 (c) For the given v - x graph,

Slope = $-v_0 / x_0$

and intercept = v_0

From the general equation of straight line, *i.e.*

$$y = mx + c$$

where, m is slope and c is the intercept.

The equation of motion of the v - x graph can be given as

$$v = -\frac{v_0 x}{x_0} + v_0 \quad \dots (i)$$

(-ve sign signifies that the slope is decreasing)

On differentiating Eq. (i) with respect to t , we get

$$\frac{dv}{dt} = -\frac{v_0}{x_0} \frac{dx}{dt} + 0 \quad \text{or} \quad a = -\frac{v_0}{x_0} v \quad \dots (ii)$$

Putting the value of v from Eq. (i) in Eq. (ii), we get

$$a = -\frac{v_0}{x_0} \left[-\frac{v_0}{x_0} x + v_0 \right] = \frac{v_0^2}{x_0^2} x - \frac{v_0^2}{x_0}$$

This equation shows that at $x = 0$,

$$a = -\frac{v_0^2}{x_0} = \text{constant and at } a = 0, x = x_0.$$

So, option (c) is correct.

14 (b) For a car in motion, if we describe this event w.r.t. a frame of reference attached to the person sitting inside the car, the car will appear to be at rest as the person inside the car (*i.e.* observer) is also moving with same velocity and in the same direction as car.

15 (b) Velocity of the particle is given as $v = At + Bt^2$

where, A and B are constants.

$$\Rightarrow \frac{dx}{dt} = At + Bt^2 \quad \left(\because v = \frac{dx}{dt} \right)$$

$$\Rightarrow dx = (At + Bt^2) dt$$

Integrating both sides, we get

$$\int_{x_1}^{x_2} dx = \int_1^2 (At + Bt^2) dt$$

$$\Rightarrow \Delta x = x_2 - x_1 = A \int_1^2 t dt + B \int_1^2 t^2 dt$$

$$= A \left[\frac{t^2}{2} \right]_1^2 + B \left[\frac{t^3}{3} \right]_1^2$$

$$= \frac{A}{2} (2^2 - 1^2) + \frac{B}{3} (2^3 - 1^3)$$

$\therefore$ Distance travelled by the particle between 1s and 2s is

$$\Delta x = \frac{A}{2} \times (3) + \frac{B}{3} \times (7) = \frac{3A}{2} + \frac{7B}{3}$$

16 (b) Given, $v = \beta x^{-2n}$

$$a = \frac{dv}{dt} = \frac{dx}{dt} \cdot \frac{dv}{dx}$$

$$\Rightarrow a = v \frac{dv}{dx} = (\beta x^{-2n})(-2n\beta x^{-2n-1})$$

$$\Rightarrow a = -2n\beta^2 x^{-4n-1}$$

17 (c) Let the ball hit water in t second.

$$\text{For first ball, } s = ut + \frac{1}{2} at^2$$

$$122.5 = 0 + \frac{1}{2} \times 9.8 \times t^2 = 4.9 t^2$$

$$\Rightarrow t = \sqrt{\frac{122.5}{4.9}} = \sqrt{25} = 5 \text{ s}$$

$$\text{For second ball, } 122.5 = u(5-2) + \frac{1}{2} \times 9.8 \times (5-2)^2$$

$$= 3u + 44.1$$

$$\Rightarrow 3u = 122.5 - 44.1$$

$$3u = 78.4 \Rightarrow u = 26.1 \text{ ms}^{-1}$$

18 (c) Given, upward velocity of ball = 25.2 ms^{-1}

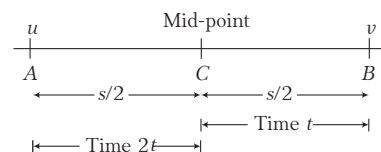
Height attained by the ball,

$$H = \frac{u^2}{2g} = \frac{25.2 \times 25.2}{2 \times 9.8} = 32.4 \text{ m}$$

Now, time taken by the ball to attain 32.4 m is

$$t = \frac{u}{g} = \frac{25.2}{9.8} = 2.57 \text{ s}$$

19 (c) According to question, the given condition can be depicted as



For motion from A to C,

$$\frac{s}{2} = 2ut + \frac{1}{2} a \times (2t)^2 = 2ut + 2at^2 \quad \dots (i)$$

Also, for motion from C to B,

$$\frac{s}{2} = v' t + \frac{1}{2} at^2 = (u + 2at)t + \frac{1}{2} at^2$$

(Let initial velocity be $v' = u + 2at$)

$$= ut + 2at^2 + \frac{1}{2} at^2$$

Now, from Eq. (i), we get

$$2ut + 2at^2 = ut + 2at^2 + \frac{1}{2} at^2 \Rightarrow u = \frac{at}{2}$$

For overall motion,

$$v = u + 3at = \frac{at}{2} + 3at = \frac{7at}{2} = 7u \quad \left(\because u = \frac{at}{2} \right)$$

20 (d) At first, the slope is decreasing, therefore the motion is retarded. Finally, the displacement becomes constant, thus the particle stops.

- 21 (d) Given, $u = 0$, $v = 180 \text{ kmh}^{-1} = 50 \text{ ms}^{-1}$

$$\text{Time taken, } t = 10 \text{ s, } a = \frac{v - u}{t} = \frac{50}{10} = 5 \text{ ms}^{-2}$$

Distance covered by the car,

$$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 5 \times (10)^2 \\ = \frac{500}{2} = 250 \text{ m}$$

- 22 (d) The slope of velocity-time graph gives acceleration,

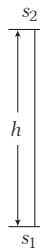
$$\text{i.e. } \frac{a_A}{a_B} = \frac{\tan \theta_1}{\tan \theta_2}$$

Here, θ_1 is the angle made by line A with time axis and θ_2 is the angle made by line B with time axis

$$\Rightarrow \frac{a_A}{a_B} = \frac{\tan 25^\circ}{\tan 50^\circ}$$

- 23 (a) The position of an object is always expressed w.r.t. some reference point. If the initial position of an object w.r.t. a reference point is s_1 and after sometime, it changes to s_2 , then the magnitude of the displacement of the object is $s_2 - s_1$. Since, the given ball returns to its starting point in 10 s. So, in 5s, it reaches the highest point which is at s_2 .

Given, s_2 is at a height h w.r.t. s_1 . Thus, the displacement after 5s will be $s_2 - s_1 = h$.



- 24 (a) Let v_{PG} = velocity of police w.r.t. ground,
 v_{TG} = velocity of thief w.r.t. ground,
 v_{TP} = velocity of thief w.r.t. police $= v_{TG} - v_{PG}$
 $= \left(\frac{153 - 45}{18} \right) \times 5 = 30 \text{ ms}^{-1}$

$$v_{BC} = \text{velocity of bullet w.r.t. car} \\ = 180 - 30 = 150 \text{ ms}^{-1}$$

- 25 (b) We know that, $s_{nth} = u + \frac{1}{2}a(2n - 1)$

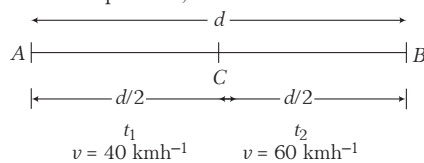
$$s_{3rd} = 0 + \frac{1}{2}a(2 \times 3 - 1) = \frac{5}{2}a \quad (\text{For } n = 3\text{s})$$

$$s_{4rd} = 0 + \frac{1}{2}a(2 \times 4 - 1) = \frac{7}{2}a \quad (\text{For } n = 4\text{s})$$

$$\text{So, the percentage increase} = \frac{s_{4th} - s_{3rd}}{s_{3rd}} \times 100$$

$$= \frac{\frac{7}{2}a - \frac{5}{2}a}{\frac{5}{2}a} \times 100 = \frac{\frac{2}{2}a}{\frac{5}{2}a} \times 100 = \frac{2}{5} \times 100 = 2 \times 20 = 40\%$$

- 26 (b) As from the question,



Average speed of the car, v_{av}

$$= \frac{\text{Total distance travelled by the body } (d)}{\text{Total time taken } (t_1 + t_2)} \quad \dots(i)$$

So, now time taken in 1st half of the distance,

$$t_1 = \text{Time} = \frac{\text{Distance}}{\text{Velocity}} = \frac{d}{2 \times 40}$$

$$\Rightarrow t_1 = \frac{d}{80} \quad (\because \text{Distance} = \frac{d}{2}, \text{velocity} = 40 \text{ kmh}^{-1})$$

$$\text{Time taken for 2nd half, } t_2 = \frac{d}{2 \times \text{Velocity}}$$

$$\text{So, } t_2 = \frac{d}{2 \times 60} = \frac{d}{120}$$

$$(\because \text{distance} = \frac{d}{2}, \text{velocity} = 60 \text{ kmh}^{-1})$$

$$\text{Now, average velocity, } v_{av} = \frac{d}{\frac{d}{80} + \frac{d}{120}} \quad [\text{From Eq. (i)}] \\ = \frac{d \times 80 \times 120}{200d} = 48 \text{ kmh}^{-1}$$

- 27 (a) The distance covered in 2 s, $x = \frac{1}{2}a(2)^2$ ($\because u = 0$)
 $= 2a$

The distance covered in next 2 s,

$$y = \frac{1}{2}a(4)^2 - \frac{1}{2}a(2)^2 = 6a$$

$$\text{Now, } x/y = \frac{2a}{6a} = \frac{1}{3} \Rightarrow y = 3x$$

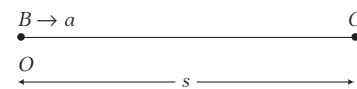
- 28 (c) $A \rightarrow v_A$ (constant velocity)

$B \rightarrow a$ (constant acceleration)

C be the fixed point at which both A and B meet in time t .

Now, $s = v_A t$ and $v_B = at$

$$A \rightarrow v_A = v$$



$$s = \frac{1}{2}at^2 = v_A t$$

$$\Rightarrow a = \frac{2v_A}{t} \quad \dots(i)$$

$$v_B = u + at$$

$$(\because v = 0)$$

$$\Rightarrow v_B = at = 2v_A$$

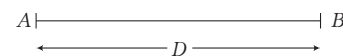
$$[\text{From Eq. (i)}]$$

$$v_{BA} = \text{velocity of } B \text{ w.r.t. } A$$

$$\Rightarrow v_{BA} = v_B - v_A = 2v_A - v_A = v_A$$

$$\Rightarrow v_{BA} = v \quad (\because v_A = v)$$

- 29 (b) v_{AB} = Speed of car from A to B

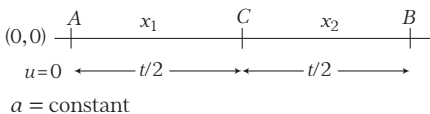


$$v_{AB} = 30 \text{ kmh}^{-1}$$

v_{BA} = Speed of car from B to A

$$v_{BA} = 20 \text{ kmh}^{-1}$$

$$v_{av} = \frac{2v_{AB} v_{BA}}{v_{AB} + v_{BA}} = 24 \text{ kmh}^{-1}$$

30 (c) 

Let A be the original point.

From equation of motion,

$$x = ut + \frac{1}{2}at^2 = \frac{1}{2}at^2 \quad (\because u = 0)$$

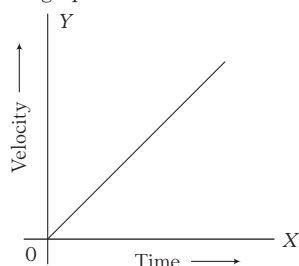
For point C, $x_1 = \frac{1}{2}a(t/2)^2 \Rightarrow x_1 = \frac{1}{8}at^2$

Distance travelled by the body in t second is

$$x_1 + x_2 = \frac{1}{2}a(t)^2 = 4x_1$$

$$x_2 = 4x_1 - x_1 = 3x_1$$

31 (d) The acceleration of a moving body is found from the slope of velocity-time graph.



$$a = \frac{dv}{dt} = \text{Slope of } v-t \text{ curve}$$

32 (b) The distance h_1 covers by stone,

$$h_1 = \frac{1}{2}gt^2 \times \frac{1}{2}10 \times (5)^2 = 125$$

The distance h_2 covers by stone,

$$h_2 = \frac{1}{2} \times 10 \times (10)^2 - \frac{1}{2} \times 10 \times (5)^2 = 375$$

The distance h_3 covers by stone,

$$h_3 = \frac{1}{2} \times 10 \times (15)^2 - \frac{1}{2} \times 10 \times (10)^2 = 625$$

The relation between h_1 , h_2 and h_3 is $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$

33 (b) When velocity of a body changes by equal amount in equal intervals of time, then the body is said to have uniform acceleration, this holds true for straight line motion.

34 (c) Velocity-time graph gives the instantaneous value of acceleration at any instant. For non-uniformly accelerated motion, $v-t$ graph is non-linear.

35 (b) The initial velocity of the body, $u^2 = 2gh$

$$\Rightarrow u = \sqrt{2gh} = \sqrt{2 \times 10 \times 125} = \sqrt{2500} = 50 \text{ ms}^{-1}$$

The final velocity from equation of motion,

$$v^2 = u^2 + 2gh$$

$$v = \sqrt{(50)^2 + 2 \times 10 \times 55}$$

$$v = \sqrt{2500 + 1100}$$

$$= \sqrt{3600} = 60 \text{ ms}^{-1}$$

36 (a) Given, velocity of water, $v_w = 4 \text{ ms}^{-1}$

Velocity of person, $v_p = 5 \text{ ms}^{-1}$, width of the river = 84.6 m

We know that, time taken, $t = \frac{s}{\sqrt{v_p^2 - v_w^2}}$

$$\Rightarrow t = \frac{84.6}{\sqrt{25 - 16}}$$

$$\Rightarrow t = \frac{84.6}{3} \approx 28.2 \text{ s}$$

37 (c) From graph, velocity of robber's car = 10 ms^{-1}

Let police car crosses it after t second.

Distance travelled by robber's car = $10t$ m

Police car is moving with a constant acceleration of 1 ms^{-2} as it attains a velocity of 10 ms^{-1} in 10 s after starting from rest.

$$\text{Distance travelled in } t \text{ second} = \frac{1}{2} \cdot a \cdot t^2 = \frac{1}{2}t^2$$

When the police car crosses the robber's car, distance travelled by the both cars should be same from the starting point of chase.

$$\therefore \frac{1}{2}t^2 = 10t \Rightarrow t = 20 \text{ s}$$

38 (b) Given, initial speed, $u = 1 \text{ kms}^{-1} = 1000 \text{ ms}^{-1}$

Final speed, $v = 9 \text{ kms}^{-1} = 9000 \text{ ms}^{-1}$

By using the relation, $v^2 = u^2 + 2as$

$$(9000)^2 = (1000)^2 + 2 \times a \times 4 \Rightarrow a = 10^7 \text{ ms}^{-2}$$

$\therefore$ The time for which the particle remains in the tube

$$v = u + at$$

$$\Rightarrow t = \frac{v - u}{a} = \frac{9000 - 1000}{10^7} = 8 \times 10^{-4} \text{ s}$$

39 (d) Given, $x = 8 + 12t - t^3$

We know that, $v = \frac{dx}{dt}$ and $a = \frac{dv}{dt}$

$$\text{So, } v = 12 - 3t^2$$

$$\text{and } t = 2 \text{ s,}$$

$$v = 0$$

$$\text{and } a = -12 \text{ ms}^{-2}$$

So, retardation of the particle = 12 ms^{-2} .

40 (d) Distance, $x = b_0 + b_1t + b_2t^2$

$$\text{Velocity, } v = \left(\frac{dx}{dt} \right) = b_1 + 2b_2t$$

$$\text{Acceleration, } a = \frac{d^2x}{dt^2} = 2b_2$$

41 (c) Let t second be the time of flight of the first body after meeting, then $(t - 4)$ second will be the time of flight of the second body.

As the initial velocity at which the bodies A and B projected are same and also the position of meeting will be also same.

So, $h_x = h_y$

$$\therefore 98t - \frac{1}{2}gt^2 = 98(t-4) - \frac{1}{2}g(t-4)^2$$

On solving, $t = 12$ s

- 42** (b) Let after time t , car catches the scooter and the distance travelled by scooter in time t ,

$$x = \frac{1}{2} \times (1) \times t^2 = \frac{t^2}{2} \quad \dots(i)$$

The distance travelled by car in time t

$$x + 150 = \frac{1}{2} \times 2 \times t^2 = t^2 \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$t = \sqrt{300} \text{ s}$$

- 43** (c) Given, $\mathbf{r}_1(t) = 3t\hat{\mathbf{i}} + 4t^2\hat{\mathbf{j}}$

$$\therefore \frac{d\mathbf{r}_1}{dt} = 3\hat{\mathbf{i}} + 8t\hat{\mathbf{j}}$$

At $t = 1$ s,

$$\mathbf{v}_1 = \frac{d\mathbf{r}_1}{dt} = 3\hat{\mathbf{i}} + 8\hat{\mathbf{j}}$$

Again, $\mathbf{r}_2(t) = 4t^2\hat{\mathbf{i}} + 3t\hat{\mathbf{j}}$

$$\Rightarrow \frac{d\mathbf{r}_2}{dt} = 8t\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$$

At $t = 1$ s,

$$\mathbf{v}_2 = \frac{d\mathbf{r}_2}{dt} = 8\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$$

Relative speed, $\mathbf{v}_{\text{rel}} = \mathbf{v}_2 - \mathbf{v}_1$

$$= (8\hat{\mathbf{i}} + 3\hat{\mathbf{j}}) - (3\hat{\mathbf{i}} + 8\hat{\mathbf{j}})$$

$$= 5\hat{\mathbf{i}} - 5\hat{\mathbf{j}}$$

$$\therefore |\mathbf{v}_{\text{rel}}| = \sqrt{(5)^2 + (-5)^2}$$

$$= 5\sqrt{2} \text{ m/s}$$

CHAPTER 04

Motion in a Plane and Projectile Motion

In this chapter, we will study motion of those particles which do not move in a straight line rather, they move in a plane. In our daily life, examples of motion in a plane are motion of a football, circular motion, motion of a stone fired from a slingshot (gulel), etc.

MOTION IN A PLANE

(TWO DIMENSIONAL MOTION)

Motion of an object is called two dimensional motion when two of the coordinates (x - y , y - z or z - x) from the three coordinates (x , y , z) are required to specify the change in position of the object in space, with respect to time. In two dimensional motion, the object moves in XY -plane, YZ -plane or ZX -plane, therefore it is called motion in a plane. *e.g.* Projectile motion, circular motion, etc.

When an object moves in a plane or in two dimensional motion, different physical quantities related with it changes with time. *e.g.* Position vector, displacement vector, velocity vector and acceleration vector. In XY -plane, these physical quantities are studied in terms of their x , y -components.

Position vector

A vector that extends from a reference point to the point at which particle is located is called position vector.

Let $\mathbf{r}$ be the position vector of a particle P located in a plane with reference to the origin O in XY -plane as shown in the figure.

$$\mathbf{OP} = \mathbf{OA} + \mathbf{OB}$$

Position vector, $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$

Direction of this position vector $\mathbf{r}$ is given by the angle θ with X -axis, where

$$\tan\theta = \left(\frac{y}{x}\right)$$

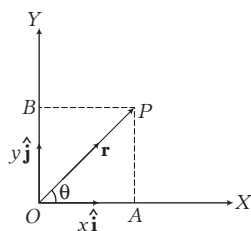


Fig. 4.1 Representation of position vector

Inside

1 Motion in a plane (Two dimensional motion)

- Position vector
- Displacement vector
- Velocity vector
- Acceleration vector

2 Projectile

- Projectile motion
- Projectile projected obliquely on the surface of the earth
- Projectile fired at an angle with the vertical

3 Projectile from a point above the ground

- Projectile projected from a tower
- Shooting a freely falling target

$$\Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

In three dimensions, the position vector is represented as

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

Example 4.1 A particle moves in a plane such that its coordinates changes with time as $x = at$ and $y = bt$, where a and b are constants. Find the position vector of the particle and its direction at any time t .

Sol. Coordinates of the particle are $x = at$ and $y = bt$

$\therefore$ Position vector of the particle at any time t is

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} = (at)\hat{\mathbf{i}} + (bt)\hat{\mathbf{j}}$$

Direction of $\mathbf{r}$, $\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{bt}{at}\right) = \tan^{-1}\left(\frac{b}{a}\right)$

Displacement vector

Consider a particle moving in XY -plane with a uniform velocity $\mathbf{v}$ and point O as an origin for measuring time and position of the particle. Let the particle be at positions A and B at timings t_1 and t_2 , respectively. The position vectors are $\mathbf{OA} = \mathbf{r}_1$ and $\mathbf{OB} = \mathbf{r}_2$.

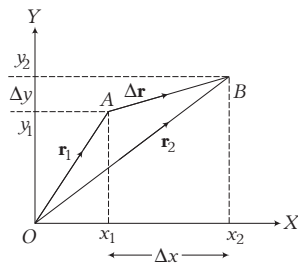


Fig. 4.2 Representation of displacement vector

Then, the displacement of the particle in time interval $(t_2 - t_1)$ is $\mathbf{AB}$. From triangle law of vector addition, we get

$$\mathbf{OA} + \mathbf{AB} = \mathbf{OB}$$

$$\Rightarrow \mathbf{AB} = \mathbf{OB} - \mathbf{OA}$$

$$\mathbf{AB} = \mathbf{r}_2 - \mathbf{r}_1 \quad \dots(i)$$

If the coordinates of the particle at points A and B are (x_1, y_1) and (x_2, y_2) , then

$$\mathbf{r}_1 = x_1\hat{\mathbf{i}} + y_1\hat{\mathbf{j}}$$

and

$$\mathbf{r}_2 = x_2\hat{\mathbf{i}} + y_2\hat{\mathbf{j}}$$

Substituting the values of $\mathbf{r}_1$ and $\mathbf{r}_2$ in Eq. (i), we get

$$\mathbf{AB} = (x_2\hat{\mathbf{i}} + y_2\hat{\mathbf{j}}) - (x_1\hat{\mathbf{i}} + y_1\hat{\mathbf{j}})$$

Displacement, $\mathbf{AB} = (x_2 - x_1)\hat{\mathbf{i}} + (y_2 - y_1)\hat{\mathbf{j}}$

or Displacement, $\Delta\mathbf{r} = \Delta x\hat{\mathbf{i}} + \Delta y\hat{\mathbf{j}}$

$$\Rightarrow \text{Magnitude of displacement, } |\Delta\mathbf{r}| = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

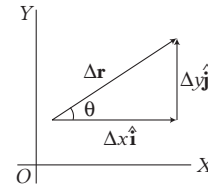


Fig. 4.3 Component of displacement

Direction of the displacement vector $\Delta\mathbf{r}$ is given by

$$\tan\theta = \frac{\Delta y}{\Delta x} \Rightarrow \theta = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right)$$

where, θ = angle made by $\Delta\mathbf{r}$ with X -axis.

Similarly, in three dimensions, the displacement vector can be represented as

$$\Delta\mathbf{r} = (x_2 - x_1)\hat{\mathbf{i}} + (y_2 - y_1)\hat{\mathbf{j}} + (z_2 - z_1)\hat{\mathbf{k}}$$

Note Magnitude of displacement (Δr) between two points is always less than or equal to distance (s) between corresponding points.
i.e. $\Delta r \leq s$

Example 4.2 An object moves from position $(3,4)$ to $(6,5)$ in the XY -plane. Find the magnitude and direction of displacement vector of the particle.

Sol. Position vectors of the particle are

$$\mathbf{r}_1 = x_1\hat{\mathbf{i}} + y_1\hat{\mathbf{j}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} \quad \text{and} \quad \mathbf{r}_2 = x_2\hat{\mathbf{i}} + y_2\hat{\mathbf{j}} = 6\hat{\mathbf{i}} + 5\hat{\mathbf{j}}$$

$$\therefore \text{Displacement vector, } \Delta\mathbf{r} = (x_2 - x_1)\hat{\mathbf{i}} + (y_2 - y_1)\hat{\mathbf{j}}$$

$$= (6 - 3)\hat{\mathbf{i}} + (5 - 4)\hat{\mathbf{j}} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}}$$

$\therefore$ Magnitude of displacement vector,

$$|\Delta\mathbf{r}| = \sqrt{(3)^2 + (1)^2} = \sqrt{10}$$

Direction of $\Delta\mathbf{r}$ with X -axis,

$$\theta = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right) = \tan^{-1}\left(\frac{1}{3}\right) \approx 18.43^\circ$$

Velocity vector

Velocity of an object in motion is defined as the ratio of displacement and the corresponding time interval taken by the object, i.e.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time interval}}$$

Velocity is a vector quantity as it has both the magnitude (speed) and direction.

It is of two types

- (i) Average velocity
- (ii) Instantaneous velocity

Average velocity

It is defined as the ratio of the displacement and the corresponding time interval.

Thus, average velocity = $\frac{\text{displacement}}{\text{time taken}}$

$$\text{Average velocity, } \mathbf{v}_{\text{av}} = \frac{\Delta \mathbf{r}}{\Delta t} = \frac{\mathbf{r}_2 - \mathbf{r}_1}{t_2 - t_1}$$

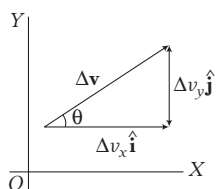


Fig. 4.4 Components of velocity

Velocity can be expressed in the component form as

$$\mathbf{v}_{\text{av}} = \frac{\Delta x}{\Delta t} \hat{\mathbf{i}} + \frac{\Delta y}{\Delta t} \hat{\mathbf{j}} = \Delta v_x \hat{\mathbf{i}} + \Delta v_y \hat{\mathbf{j}}$$

where, Δv_x and Δv_y are the components of average velocity along x -direction and y -direction, respectively.

The magnitude of $\mathbf{v}_{\text{av}}$ is given by

$$v_{\text{av}} = \sqrt{\Delta v_x^2 + \Delta v_y^2}$$

and the direction of $\mathbf{v}_{\text{av}}$ is given by angle θ

$$\tan \theta = \frac{\Delta v_y}{\Delta v_x} \quad (\text{From } X\text{-axis})$$

Example 4.3 A particle moves in XY -plane from position $(1\text{m}, 2\text{m})$ to $(3\text{m}, 4\text{m})$ in 2 s. Find the magnitude and direction of average velocity.

Sol. Given, position vectors of the particle are

$$\mathbf{r}_1 = x_1 \hat{\mathbf{i}} + y_1 \hat{\mathbf{j}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}}$$

$$\text{and } \mathbf{r}_2 = x_2 \hat{\mathbf{i}} + y_2 \hat{\mathbf{j}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$$

$$\text{Displacement, } \Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$$

$$\therefore \text{Average velocity, } \mathbf{v}_{\text{av}} = \frac{\Delta \mathbf{r}}{\Delta t}$$

$$\Rightarrow \mathbf{v}_{\text{av}} = \frac{2\hat{\mathbf{i}} + 2\hat{\mathbf{j}}}{2}$$

$$\Rightarrow \mathbf{v}_{\text{av}} = \Delta v_x \hat{\mathbf{i}} + \Delta v_y \hat{\mathbf{j}} = (\hat{\mathbf{i}} + \hat{\mathbf{j}}) \text{ ms}^{-1}$$

$$\Rightarrow |\mathbf{v}_{\text{av}}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2} \text{ ms}^{-1}$$

Direction of average velocity with X -axis,

$$\theta = \tan^{-1} \left(\frac{\Delta v_y}{\Delta v_x} \right) = \tan^{-1} \left(\frac{1}{1} \right) = 45^\circ$$

Instantaneous velocity

The velocity of the object at an instant of time (t) is known as instantaneous velocity. The average velocity will become instantaneous, if Δt approaches to zero.

$\therefore$ Instantaneous velocity,

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} = \frac{d\mathbf{r}}{dt}$$

Now, we can write

$$d\mathbf{r} = dx \hat{\mathbf{i}} + dy \hat{\mathbf{j}}$$

$$\therefore \mathbf{v} = \frac{dx \hat{\mathbf{i}} + dy \hat{\mathbf{j}}}{dt} = \frac{dx}{dt} \hat{\mathbf{i}} + \frac{dy}{dt} \hat{\mathbf{j}}$$

$$\Rightarrow \boxed{\mathbf{v} = v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}}}$$

Similarly, in three dimensions, we can write

$$\mathbf{v} = v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} + v_z \hat{\mathbf{k}}$$

where, $v_x = \frac{dx}{dt}$ is magnitude of instantaneous velocity in x -direction,

$v_y = \frac{dy}{dt}$ is magnitude of instantaneous velocity in y -direction

and $v_z = \frac{dz}{dt}$ is magnitude of instantaneous velocity in z -direction.

Magnitude of instantaneous velocity, $|\mathbf{v}| = \sqrt{v_x^2 + v_y^2}$

Direction of instantaneous velocity $\mathbf{v}$ with X -axis,

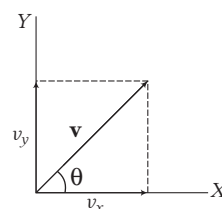


Fig. 4.5 Direction of instantaneous velocity

$$\tan \theta = \frac{v_y}{v_x} \Rightarrow \boxed{\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)}$$

Example 4.4 Position vector of a particle is given as

$$\mathbf{r} = 2t\hat{\mathbf{i}} + 3t^2\hat{\mathbf{j}}$$

where, t is in seconds and the coefficients have the proper units, for r to be in metres.

(i) Find instantaneous velocity $\mathbf{v}(t)$ of the particle.

(ii) Find magnitude and direction of $\mathbf{v}(t)$ at $t = 2\text{s}$.

Sol. Given, $\mathbf{r} = 2t\hat{\mathbf{i}} + 3t^2\hat{\mathbf{j}}$

$$(i) \text{ Instantaneous velocity, } \mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{d}{dt} (2t\hat{\mathbf{i}} + 3t^2\hat{\mathbf{j}})$$

$$\Rightarrow \mathbf{v} = v_x \hat{\mathbf{i}} + v_y \hat{\mathbf{j}} = 2\hat{\mathbf{i}} + 6t\hat{\mathbf{j}}$$

(ii) Magnitude of $\mathbf{v}(t)$,

$$|\mathbf{v}(t)| = \sqrt{v_x^2 + v_y^2} = \sqrt{(2)^2 + (6t)^2} = \sqrt{4 + 36t^2}$$

At $t = 2$ s, $|\mathbf{v}(t)| = \sqrt{4 + 36 \times 4} = \sqrt{148} \text{ ms}^{-1}$

Direction of $\mathbf{v}(t)$, $\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(\frac{6t}{2}\right) = \tan^{-1}(3t)$

At $t = 2$ s, $\theta = \tan^{-1}(3 \times 2) \approx \tan^{-1}(6) \text{ rad}$

Acceleration vector

It is defined as the rate of change of velocity. It can be expressed as

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}}$$

It is of two types as follows

- (i) Average acceleration
- (ii) Instantaneous acceleration

Average acceleration

It is defined as the change in velocity ($\Delta\mathbf{v}$) divided by the corresponding time interval (Δt). It can be expressed as

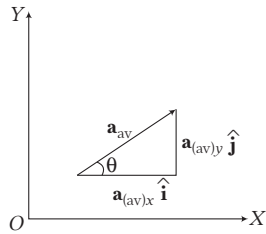


Fig. 4.6 Components of acceleration

$$\begin{aligned} \text{Average acceleration, } \mathbf{a}_{av} &= \frac{\Delta\mathbf{v}}{\Delta t} = \frac{\Delta v_x \hat{\mathbf{i}} + \Delta v_y \hat{\mathbf{j}}}{\Delta t} \\ &= \frac{\Delta v_x}{\Delta t} \hat{\mathbf{i}} + \frac{\Delta v_y}{\Delta t} \hat{\mathbf{j}} \end{aligned}$$

$$\text{Average acceleration} = \mathbf{a}_{(av)x} \hat{\mathbf{i}} + \mathbf{a}_{(av)y} \hat{\mathbf{j}}$$

which is expressed in component form

where, $\mathbf{a}_{(av)x} = \frac{\Delta v_x}{\Delta t}$ = average acceleration in x-direction

and $\mathbf{a}_{(av)y} = \frac{\Delta v_y}{\Delta t}$ = average acceleration in y-direction.

In three dimensions, we can write

$$\mathbf{a}_{av} = a_{(av)x} \hat{\mathbf{i}} + a_{(av)y} \hat{\mathbf{j}} + a_{(av)z} \hat{\mathbf{k}}$$

Magnitude of average acceleration is given by

$$|\mathbf{a}_{av}| = \sqrt{(a_{(av)x})^2 + (a_{(av)y})^2}$$

Angle θ made by average acceleration with X-axis is

$$\tan \theta = \frac{a_{(av)y}}{a_{(av)x}} \Rightarrow \theta = \tan^{-1}\left(\frac{a_{(av)y}}{a_{(av)x}}\right)$$

Example 4.5 Velocity of a particle changes from $(3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) \text{ m/s}$ to $(6\hat{\mathbf{i}} + 5\hat{\mathbf{j}}) \text{ m/s}$ in 2 s. Find magnitude and direction of average acceleration.

Sol. Given, velocity vectors of the particle,

$$\mathbf{v}_1 = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$$

and $\mathbf{v}_2 = 6\hat{\mathbf{i}} + 5\hat{\mathbf{j}}$

Change in velocity,

$$\Delta\mathbf{v}_x = (v_2)_x - (v_1)_x \hat{\mathbf{i}} = (6 - 3)\hat{\mathbf{i}} = 3\hat{\mathbf{i}}$$

$$\Delta\mathbf{v}_y = (v_2)_y - (v_1)_y \hat{\mathbf{j}} = (5 - 4)\hat{\mathbf{j}} = \hat{\mathbf{j}}$$

$$\therefore \Delta\mathbf{v} = \Delta\mathbf{v}_x + \Delta\mathbf{v}_y = 3\hat{\mathbf{i}} + \hat{\mathbf{j}}$$

Average acceleration,

$$\begin{aligned} \mathbf{a}_{av} &= \frac{\Delta\mathbf{v}}{\Delta t} = \frac{3\hat{\mathbf{i}} + \hat{\mathbf{j}}}{2} \quad (\because t = 2 \text{ s}) \\ &= 1.5\hat{\mathbf{i}} + 0.5\hat{\mathbf{j}} \end{aligned}$$

Direction of average acceleration,

$$\begin{aligned} \tan \theta &= \frac{a_{(av)y}}{a_{(av)x}} = \frac{0.5}{1.5} = \frac{1}{3} \\ \Rightarrow \theta &= \tan^{-1}\left(\frac{1}{3}\right) \\ &\approx 18.43^\circ \text{ with X-axis} \end{aligned}$$

Instantaneous acceleration

It is defined as the limiting value of the average acceleration as the time interval approaches to zero.

It can be expressed as

$$\mathbf{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt}$$

Instantaneous acceleration,

$$\mathbf{a} = a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}}$$

where,

$$a_x = \text{magnitude of instantaneous acceleration in x-direction} = \frac{dv_x}{dt}$$

$$a_y = \text{magnitude of instantaneous acceleration in y-direction} = \frac{dv_y}{dt}$$

The magnitude of instantaneous acceleration is given by

$$a = \sqrt{a_x^2 + a_y^2}$$

If acceleration $\mathbf{a}$ makes an angle θ with X -axis, then

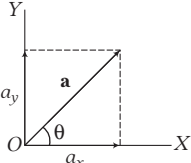
$$\tan \theta = \frac{a_y}{a_x} \Rightarrow \theta = \tan^{-1} \left(\frac{a_y}{a_x} \right)$$


Fig. 4.7 Direction of instantaneous acceleration

In three dimensions, we can write

$$\mathbf{a} = a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}} + a_z \hat{\mathbf{k}}$$

Example 4.6 The position of a particle is given by

$$\mathbf{r} = 3t\hat{\mathbf{i}} + 2t^2\hat{\mathbf{j}} + 8\hat{\mathbf{k}}$$

where, t is in seconds and the coefficients have the proper units for r to be in metres.

(i) Find $v(t)$ and $a(t)$ of the particle.

(ii) Find the magnitude and direction of $v(t)$ and $a(t)$ at $t = 1$ s.

Sol. Position of particle, $\mathbf{r} = 3t\hat{\mathbf{i}} + 2t^2\hat{\mathbf{j}} + 8\hat{\mathbf{k}}$

(i) As, $\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$ and $\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}$

$$\therefore \mathbf{v}(t) = \frac{d}{dt} (3t\hat{\mathbf{i}} + 2t^2\hat{\mathbf{j}} + 8\hat{\mathbf{k}}) = 3\hat{\mathbf{i}} + 4t\hat{\mathbf{j}}$$

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = 4\hat{\mathbf{j}}$$

(ii) $v(t) = \sqrt{v_x^2 + v_y^2}$ and $\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)$

Velocity, $\mathbf{v}(t) = 3\hat{\mathbf{i}} + 4t\hat{\mathbf{j}}$

At $t = 1$ s,

$$v = \sqrt{(3)^2 + (4)^2} = 5 \text{ ms}^{-1}$$

$$\Rightarrow \text{Direction of } v(t) = \theta = \tan^{-1} \left(\frac{4}{3} \right) = 53^\circ \text{ with } X\text{-axis}$$

Direction of $a(t)$, at $t = 1$ s,

$$\theta' = \tan^{-1} \left(\frac{a_y}{a_x} \right) = \tan^{-1} \left(\frac{4}{0} \right) = \tan^{-1}(\infty) = 90^\circ \text{ with } X\text{-axis}$$

Motion in plane with uniform acceleration

An object is said to be moving with uniform acceleration, if its velocity vector undergoes the same change in the same interval of time (however small).

Let an object is moving in XY -plane and its acceleration $\mathbf{a}$ is constant. At time $t = 0$, the velocity of an object be $\mathbf{v}_0$ (say) and $\mathbf{v}$ be the velocity at time t .

According to definition of average acceleration, we get

$$\mathbf{a} = \frac{\mathbf{v} - \mathbf{v}_0}{t - 0} = \frac{\mathbf{v} - \mathbf{v}_0}{t} \Rightarrow \mathbf{v} = \mathbf{v}_0 + \mathbf{a}t$$

In terms of rectangular components, we can express it as

$$\mathbf{v}_x = \mathbf{v}_{0x} + \mathbf{a}_x t \quad \text{and} \quad \mathbf{v}_y = \mathbf{v}_{0y} + \mathbf{a}_y t$$

It can be concluded that, each rectangular component of velocity of an object moving with uniform acceleration in a plane depends upon time as if it were the velocity vector of one dimensional uniformly accelerated motion.

Path of particle under constant acceleration

Now, we can also find the position vector ($\mathbf{r}$). Let $\mathbf{r}_0$ and $\mathbf{r}$ be the position vectors of the particle at time $t = 0$ and $t = t$ and their velocities at these instants be $\mathbf{v}_0$ and $\mathbf{v}$, respectively. Then, the average velocity is given by

$$\mathbf{v}_{av} = \frac{\mathbf{v}_0 + \mathbf{v}}{2}$$

Displacement is the product of average velocity and time interval.

It is expressed as

$$\mathbf{r} - \mathbf{r}_0 = \left(\frac{\mathbf{v} + \mathbf{v}_0}{2} \right) t = \left[\frac{(\mathbf{v}_0 + \mathbf{a}t) + \mathbf{v}_0}{2} \right] t$$

$$\Rightarrow \mathbf{r} - \mathbf{r}_0 = \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$$

$$\Rightarrow \mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$$

In terms of rectangular components, we get

$$x\hat{\mathbf{i}} + y\hat{\mathbf{j}} = x_0\hat{\mathbf{i}} + y_0\hat{\mathbf{j}} + (v_{0x}\hat{\mathbf{i}} + v_{0y}\hat{\mathbf{j}})t + \frac{1}{2}(a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}})t^2$$

Now, equating the coefficients of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$,

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \text{ along } X\text{-axis}$$

and

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \text{ along } Y\text{-axis}$$

Note Motion in a plane (two dimensional motion) can be treated as two separate simultaneous one dimensional motions with constant acceleration along two perpendicular directions.

Example 4.7 (i) What does $\left| \frac{d\mathbf{v}}{dt} \right|$ and $\frac{d|\mathbf{v}|}{dt}$ represent?

(ii) Can these be equal?

(iii) Can $\frac{d|\mathbf{v}|}{dt} = 0$ while $\left| \frac{d\mathbf{v}}{dt} \right| \neq 0$?

(iv) Can $\frac{d|\mathbf{v}|}{dt} \neq 0$ while $\left| \frac{d\mathbf{v}}{dt} \right| = 0$?

Sol. (i) $\left| \frac{d\mathbf{v}}{dt} \right|$ is the magnitude of total acceleration. While $\frac{d|\mathbf{v}|}{dt}$ represents the time rate of change of speed (called

the tangential acceleration, a component of total acceleration) as $|\mathbf{v}| = v$.

(ii) These two are equal in case of one dimensional motion (without change in direction).

(iii) In case of uniform circular motion, speed remains constant while velocity changes.

Hence, $\frac{d|\mathbf{v}|}{dt} = 0$ while $\left| \frac{d\mathbf{v}}{dt} \right| \neq 0$.

(iv) $\frac{d|\mathbf{v}|}{dt} \neq 0$ implies that speed of particle is not constant.

Velocity cannot remain constant, if speed is changing.

Hence, $\left| \frac{d\mathbf{v}}{dt} \right|$ cannot be zero in this case. So, it is not

possible to have $\left| \frac{d\mathbf{v}}{dt} \right| = 0$ while $\frac{d|\mathbf{v}|}{dt} \neq 0$.

Example 4.8 A particle starts from origin at $t = 0$ with a velocity of $15\hat{\mathbf{i}} \text{ ms}^{-1}$ and moves in XY -plane under the action of a force which produces a constant acceleration of $(15\hat{\mathbf{i}} + 20\hat{\mathbf{j}}) \text{ ms}^{-2}$. Find the y -coordinate of the particle at the instant when its x -coordinate is 180 m.

Sol. The position of the particle is given by

$$\begin{aligned}\mathbf{r}(t) &= \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2 = 15\hat{\mathbf{i}} t + \frac{1}{2} (15\hat{\mathbf{i}} + 20\hat{\mathbf{j}}) t^2 \\ &= (15t + 7.5t^2)\hat{\mathbf{i}} + 10t^2\hat{\mathbf{j}}\end{aligned}$$

$$\therefore x(t) = 15t + 7.5t^2 \quad \text{and} \quad y(t) = 10t^2$$

If $x(t) = 180 \text{ m}$, $t = ?$

$$180 = 15t + 7.5t^2 \Rightarrow t = 4 \text{ s}$$

$\therefore y$ -coordinate, $y(t) = 10 \times 16 = 160 \text{ m}$

Example 4.9 An object has a velocity $\mathbf{v} = (2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) \text{ ms}^{-1}$ at time $t = 0 \text{ s}$. It undergoes a constant acceleration $\mathbf{a} = (\hat{\mathbf{i}} - 3\hat{\mathbf{j}}) \text{ ms}^{-2}$ for 4s. Then,

(i) find the coordinates of the object, if it is at origin at $t = 0$.

(ii) find the magnitude of its velocity at the end of 4s.

Sol. (i) Here, initial position of the object,

$$\mathbf{r}_0 = x_0\hat{\mathbf{i}} + y_0\hat{\mathbf{j}} = 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}}$$

$$\text{Initial velocity, } \mathbf{v}_0 = v_{0x}\hat{\mathbf{i}} + v_{0y}\hat{\mathbf{j}} = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$$

$$\text{Acceleration, } \mathbf{a} = a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}} = \hat{\mathbf{i}} - 3\hat{\mathbf{j}}$$

and $t = 4 \text{ s}$

Let the final coordinates of the object be (x, y) . Then, according to the equation for the path of particle under constant acceleration,

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 = 0 + 2 \times 4 + \frac{1}{2}(1) \times 4^2$$

$$\Rightarrow x = 16 \text{ m} \quad \text{and} \quad y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$= 0 + 4 \times 4 + \frac{1}{2}(-3) \times 4^2 \Rightarrow y = -8 \text{ m}$$

Therefore, the object lies at $(16\hat{\mathbf{i}} - 8\hat{\mathbf{j}})$ at $t = 4 \text{ s}$.

(ii) Using equation

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{a} t$$

$$\Rightarrow \mathbf{v} = (2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) + (\hat{\mathbf{i}} - 3\hat{\mathbf{j}}) \times 4$$

$$= (2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) + (4\hat{\mathbf{i}} - 12\hat{\mathbf{j}}) = (2 + 4)\hat{\mathbf{i}} + (4 - 12)\hat{\mathbf{j}}$$

$$\Rightarrow \mathbf{v} = 6\hat{\mathbf{i}} - 8\hat{\mathbf{j}}$$

$$\therefore \text{Magnitude of velocity, } |\mathbf{v}| = \sqrt{6^2 + 8^2} = 10 \text{ ms}^{-1}$$

$$\text{Its direction with } X\text{-axis, } \theta = \tan^{-1}\left(\frac{-8}{6}\right) \approx -53^\circ$$

CHECK POINT 4.1

1. The x and y -components of a position vector $\mathbf{P}$ have numerical values 5 and 6, respectively. Direction and magnitude of vector $\mathbf{P}$ are

(a) $\tan^{-1}\left(\frac{6}{5}\right)$ and $\sqrt{61}$

(b) $\tan^{-1}\left(\frac{5}{6}\right)$ and $\sqrt{61}$

(c) 60° and 8

(d) 30° and 9

2. An object moves from positions (6, 8) to (12, 10) in the XY -plane. Magnitude and direction of displacement is

(a) $\sqrt{40}$ and 18.43°

(b) $\sqrt{40}$ and 71.56°

(c) 10 and 53°

(d) $\sqrt{244}$ and 53°

3. A particle moves in XY -plane from positions (2m, 4m) to (6m, 8m) in 2 s. Magnitude and direction of average velocity is

(a) $\sqrt{2} \text{ ms}^{-1}$ and 45°

(b) $2\sqrt{2} \text{ ms}^{-1}$ and 45°

(c) $4\sqrt{2} \text{ ms}^{-1}$ and 30°

(d) $3\sqrt{2} \text{ ms}^{-1}$ and 60°

4. The displacement of an object along the three axes are given by, $x = 2t^2$, $y = t^2 - 4t$ and $z = 3t - 5$. The initial velocity of the particle is

(a) 10 units

(b) 12 units

(c) 5 units

(d) 2 units

5. A particle moves along the positive branch of the curve

$$y = \frac{x^2}{2}, \text{ where } x = \frac{t^2}{2}, \text{ } x \text{ and } y \text{ are measured in metres and } t$$

in second. At $t = 2 \text{ s}$, the velocity of the particle is

(a) $(2\hat{\mathbf{i}} - 4\hat{\mathbf{j}}) \text{ ms}^{-1}$

(b) $(4\hat{\mathbf{i}} + 2\hat{\mathbf{j}}) \text{ ms}^{-1}$

(c) $(2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) \text{ ms}^{-1}$

(d) $(4\hat{\mathbf{i}} - 2\hat{\mathbf{j}}) \text{ ms}^{-1}$

6. The position vector of a particle is

$$\mathbf{r} = a \sin \omega t \hat{\mathbf{i}} + a \cos \omega t \hat{\mathbf{j}}$$

The velocity of the particle is

(a) parallel to position vector

(b) perpendicular to position vector

(c) directed towards the origin

(d) directed away from the origin

7. The position vector of an object at any time t is given by $3t^2\hat{i} + 6t\hat{j} + \hat{k}$. Its velocity (in m/s) along Y -axis has the magnitude
 (a) $6t$ (b) 6
 (c) 0 (d) 9
8. The coordinates of a moving particle at any time t are given by, $x = 2t^3$ and $y = 3t^3$. Acceleration of the particle is given by
 (a) $468t$ (b) $t\sqrt{468}$
 (c) $234t^2$ (d) $t\sqrt{234}$
9. The position of a particle moving in the XY -plane at any time t is given by $x = (3t^2 - 6t)$ m, $y = (t^2 - 2t)$ m. Select the correct statement about the moving particle from the following.
 (a) The acceleration of the particle is zero at $t = 0$ s
 (b) The velocity of the particle is zero at $t = 0$ s
 (c) The velocity of the particle is zero at $t = 1$ s
 (d) The velocity and acceleration of the particle are zero
10. A particle's velocity changes from $(2\hat{i} + 3\hat{j}) \text{ ms}^{-1}$ to $(3\hat{i} - 2\hat{j}) \text{ ms}^{-1}$ in 2 s. Its average acceleration (in ms^{-2}) is
 (a) $-(\hat{i} + 5\hat{j})$ (b) $(\hat{i} + 5\hat{j})/2$ (c) zero (d) $(\hat{i} - 5\hat{j})/2$
11. A particle has an initial velocity of $4\hat{i} + 3\hat{j}$ and an acceleration of $0.4\hat{i} + 0.3\hat{j}$. Its speed after 10 s is
 (a) 10 units (b) 7 units (c) $7\sqrt{2}$ units (d) 8.5 units
12. A body lying initially at point (3, 7) starts moving with a constant acceleration of $4\hat{i}$. Its position after 3 s is given by the coordinates
 (a) (7, 3) (b) (7, 18) (c) (21, 7) (d) (3, 7)
13. The initial position of an object at rest is given by $3\hat{i} - 8\hat{j}$. It moves with constant acceleration and reaches to the position $2\hat{i} + 4\hat{j}$ after 4 s. What is its acceleration?
 (a) $-\frac{1}{8}\hat{i} + \frac{3}{2}\hat{j}$ (b) $2\hat{i} - \frac{1}{8}\hat{j}$
 (c) $-\frac{1}{2}\hat{i} + 8\hat{j}$ (d) $8\hat{i} - \frac{3}{2}\hat{j}$

PROJECTILE

When an object or body released into the space with some initial velocity, moves freely under the effect of gravity is known as projectile.

Projectile motion

If a constant force (and hence, constant acceleration) acts on a particle at an angle θ ($\neq 0^\circ$ or 180°) with the direction of its initial velocity ($\neq$ zero), the path followed by the particle is parabolic and the motion of the particle is called projectile motion. It is a two dimensional motion, i.e. motion of the particle is constrained in a plane.

In other words, if a particle moves in horizontal as well as vertical motion simultaneously, the motion of the particle is known as projectile motion.

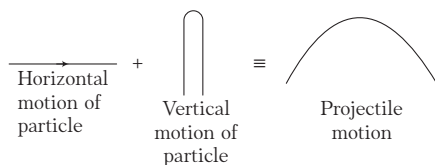


Fig. 4.8 Motion of particle

Note

- If the angle between acceleration and velocity is θ and where $0^\circ < \theta < 180^\circ$, then particle is executing projectile motion.
- In projectile motion, change in velocity of particle in magnitude and direction both act simultaneously.

When a particle is thrown obliquely near the earth's surface, it moves in a parabolic path, provided that the particle remains close to the surface of earth and the air resistance is negligible. This is an example of projectile motion.

Projectile projected obliquely on the surface of the earth

Let OX be a horizontal line on the ground, OY be a vertical line perpendicular to the ground and O be the origin for XY -axes on a plane.

Suppose an object is projected from point O with velocity (u), making an angle (θ) with the horizontal direction OX , such that $x_0 = 0$ and $y_0 = 0$ when $t = 0$.

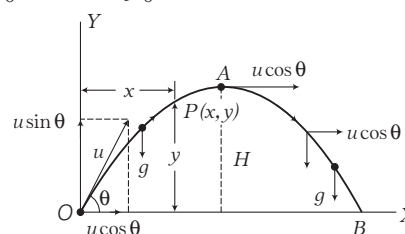


Fig. 4.9 Oblique projectile motion

While resolving velocity (u) into two components, we get

- $u \cos \theta$ along OX and
- $u \sin \theta$ along OY

As there is no force acting in horizontal direction, so the horizontal component of velocity ($u \cos \theta$) remains constant throughout the entire motion, so there is no acceleration in the horizontal direction (if air resistance is assumed to be zero).

However, the vertical components of velocity ($u \sin \theta$) decreases continuously with height from O to A , due to downward force of gravity and becomes zero at highest point A . At this point, the object attains maximum height, now it has only horizontal component of velocity. From point A , the object starts to fall down and reaches at point B on the ground.

Let us make ourselves familiar with certain terms used in projectile motion

1. Equation of path of projectile

Suppose at any time t , the object reaches at point $P(x, y)$. So, x is the horizontal distance travelled by object in time t and y is the vertical distance travelled by object in time t .

Motion along horizontal direction

The velocity of the object in horizontal direction, i.e. along OX is constant, so the acceleration a_x in horizontal direction is zero.

∴ Position of the object at time t along horizontal direction is given by, $x = x_0 + u_x t + \frac{1}{2} a_x t^2$

But $x_0 = 0$, $u_x = u \cos \theta$, $a_x = 0$ and $t = t$

∴ $x = u \cos \theta t$

$$\Rightarrow \boxed{\text{Time, } t = \frac{x}{u \cos \theta}} \quad \dots(i)$$

Motion along vertical direction

The vertical component of velocity of the object is decreasing from O to P due to gravity, so acceleration a_y is $-g$.

∴ Position of the object at any time t along the vertical direction, i.e. along OY is given by, $y = y_0 + u_y t + \frac{1}{2} a_y t^2$

But $y_0 = 0$, $u_y = u \sin \theta$, $a_y = -g$ and $t = t$

So, $y = u \sin \theta t + \frac{1}{2} (-g) t^2 = u \sin \theta t - \frac{1}{2} g t^2 \quad \dots(ii)$

Substituting the value of t from Eq. (i) in Eq. (ii), we get

$$\begin{aligned} y &= u \sin \theta \left(\frac{x}{u \cos \theta} \right) - \frac{1}{2} g \left(\frac{x}{u \cos \theta} \right)^2 \\ &= x \tan \theta - \frac{g}{2} \left(\frac{x}{u \cos \theta} \right)^2 \end{aligned}$$

Vertical displacement,
$$y = x \tan \theta - \left(\frac{1}{2} \frac{g}{u^2 \cos^2 \theta} \right) x^2$$

This equation is in the form of $y = ax - bx^2$, which represents a parabola and it is known as **equation of trajectory of a projectile**.

Example 4.10 A body is projected with a velocity of 20 ms^{-1} in a direction making an angle of 60° with the horizontal. Determine its (i) position after 0.5 s and (ii) the velocity after 0.5 s .

Sol. Given, $u = 20 \text{ ms}^{-1}$, $\theta = 60^\circ$, $t = 0.5 \text{ s}$

(i) Since, horizontal distance,
 $x = (u \cos \theta) t = (20 \cos 60^\circ) \times 0.5 = 5 \text{ m}$

Similarly, vertical distance,

$$\begin{aligned} y &= (u \sin \theta) t - \frac{1}{2} g t^2 \\ &= (20 \sin 60^\circ) \times 0.5 - \frac{1}{2} \times 9.8 \times (0.5)^2 \Rightarrow y = 7.43 \text{ m} \end{aligned}$$

(ii) Velocity along horizontal direction,

$$v_x = u \cos \theta = 20 \cos 60^\circ = 10 \text{ ms}^{-1}$$

Velocity along vertical direction,

$$v_y = u \sin \theta - g t = 20 \sin 60^\circ - 9.8 \times 0.5 = 12.42 \text{ ms}^{-1}$$

Example 4.11 A stone is thrown with a speed of 10 ms^{-1} at an angle of projection 60° . Find its height above the point of projection when it is at a horizontal distance of 3 m from the thrower? (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Considering the equation of trajectory,

$$y = (\tan \theta_0) x - \frac{g}{2(v_0^2 \cos^2 \theta_0)} x^2$$

Here, $\theta_0 = 60^\circ$, $v_0 = 10 \text{ ms}^{-1}$, $x = 3 \text{ m}$

$$\begin{aligned} \therefore y &= (\tan 60^\circ) \times 3 - \frac{10}{2(100 \cos^2 60^\circ)} (3)^2 \\ &= 3\sqrt{3} - \frac{9}{5} = \frac{15\sqrt{3} - 9}{5} \text{ m} = 3.396 \text{ m} \end{aligned}$$

2. Time of flight of projectile

It is defined as the total time for which projectile is in flight, i.e. time during the motion of projectile from O to B . It is denoted by T .

Time of flight consists of two parts such as

(i) Time taken by an object to go from point O to A . It is also known as **time of ascent** (t_a).

(ii) Time taken by an object to go from point A to B . It is also known as **time of descent** (t_d).

Total time can be expressed as

$$T = t_a + t_d = 2t \Rightarrow t = T/2 \quad (\because t_a = t_d = t)$$

The vertical component of velocity of the projectile becomes zero at the highest point H .

Let us consider vertical upward motion of the object from O to A , we get

$$u_y = u \sin \theta, a_y = -g, t = T/2 \text{ and } v_y = 0$$

$$\text{Since, } v_y = u_y + a_y t \Rightarrow 0 = u \sin \theta - g \frac{T}{2}$$

$$\therefore \boxed{\text{Time of flight, } T = \frac{2u \sin \theta}{g}}$$

Example 4.12 A cricket ball is thrown at a speed of 28 ms^{-1} in a direction 30° above the horizontal. Calculate the time taken by the ball to return to the same level.

Sol. Given, speed, $u = 28 \text{ ms}^{-1}$ and $\theta = 30^\circ$

∴ The time taken by the ball to return the same level is

$$T = \frac{2u \sin \theta}{g} = \frac{2 \times 28 \times \sin 30^\circ}{9.8} = \frac{28}{9.8} = 2.85 \approx 2.9 \text{ s}$$

3. Maximum height of a projectile

It is defined as the maximum vertical height attained by the projectile above the point of projection during its flight. It is denoted by H .

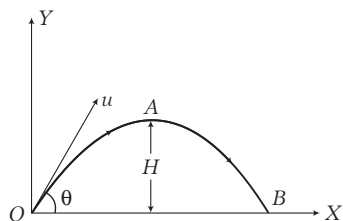


Fig. 4.10 Maximum height of projectile

Let us consider the vertical upward motion of the projectile from O to A .

We have,

$$u_y = u \sin \theta, a_y = -g, y_0 = 0, y = H,$$

$$t = \frac{T}{2} = \frac{u \sin \theta}{g}$$

Using this relation, $y = y_0 + u_y t + \frac{1}{2} a_y t^2$

$$\begin{aligned} \text{We have, } H &= 0 + u \sin \theta \left(\frac{u \sin \theta}{g} \right) + \frac{1}{2} (-g) \left(\frac{u \sin \theta}{g} \right)^2 \\ &= \frac{u^2}{g} \sin^2 \theta - \frac{1}{2} \frac{u^2 \sin^2 \theta}{g} \end{aligned}$$

Maximum height,
$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Example 4.13 Assume that a ball is kicked at an angle of 60° with the horizontal, so if the horizontal component of its velocity is 19.6 ms^{-1} , determine its maximum height.

Sol. Given, $\theta = 60^\circ$

Horizontal component of velocity $= u \cos 60^\circ = 19.6 \text{ ms}^{-1}$

$$\therefore u = \frac{19.6}{\cos 60^\circ} = \frac{19.6}{0.5} = 39.2 \text{ ms}^{-1}$$

Therefore, maximum height,

$$H = \frac{u^2 \sin^2 60^\circ}{2g} = \frac{(39.2)^2}{2 \times 9.8} \times \left(\frac{\sqrt{3}}{2} \right)^2 = 58.8 \text{ m}$$

4. Horizontal range of a projectile

The horizontal range of a projectile is defined as the horizontal distance covered by the projectile during its time of flight. It is denoted by R .

If the object having uniform velocity $u \cos \theta$ (i.e. horizontal component) and the time of flight is T , then the horizontal range covered by the projectile.

$$\text{i.e. } OB = R = u \cos \theta \times T = u \cos \theta \times 2u \frac{\sin \theta}{g}$$

$$\text{Horizontal range, } R = \frac{u^2 \sin 2\theta}{g} \quad (\because \sin 2\theta = 2 \sin \theta \cos \theta)$$

The horizontal range will be maximum, if angle of projection is 45° .

$$\therefore \text{Maximum horizontal range, } R_{\max} = \frac{u^2}{g}$$

For same value of initial velocity, horizontal range of projectile is same for complementary angles.

$$\text{So, } R_{30^\circ} = R_{60^\circ} \text{ or } R_{20^\circ} = R_{70^\circ}$$

Some important points related to projectile motion

(i) Velocity of the projectile at any instant t ,

$$\mathbf{v} = v_x \hat{i} + v_y \hat{j} = u \cos \theta \hat{i} + (u \sin \theta - gt) \hat{j}$$

$$|\mathbf{v}| = \sqrt{u^2 + g^2 t^2 - 2ugt \sin \theta}$$

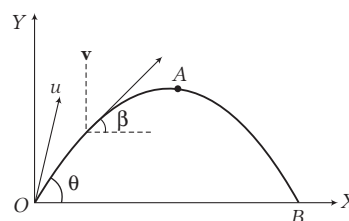


Fig. 4.11

- This velocity makes an angle β with the horizontal given by

$$\tan \beta = \frac{v_y}{v_x} = \frac{u \sin \theta - gt}{u \cos \theta} \quad \dots(i)$$

- When the projectile reaches at point B , substitute $t = T$ in Eq. (i).

(ii) When a projectile is thrown upward, its kinetic energy decreases, potential energy increases but the total energy always remains constant.

(iii) Total energy of projectile = Kinetic energy + Potential energy

$$= \frac{1}{2} mv^2 \cos^2 \theta + \frac{1}{2} mv^2 \sin^2 \theta = \frac{1}{2} mv^2$$

(iv) In projectile motion, speed (and hence, kinetic energy) is minimum at highest point of its trajectory and given as

Speed $= (\cos \theta)$ times the speed of projection
and kinetic energy $= (\cos^2 \theta)$ times the initial kinetic energy.

Here, θ = angle of projection.

- (v) In projectile motion, it is sometimes better to write the equations of H , R and T in terms of u_x and u_y as

$$T = \frac{2u_y}{g}, \quad H = \frac{u_y^2}{2g}$$

and
$$R = \frac{2u_x u_y}{g}$$

- (vi) In projectile motion $H = R$, when $u_y = 4u_x$ or $\tan \theta = 4$.

- (vii) Equation of trajectory can also be written as

$$y = x \left(1 - \frac{x}{R} \right) \tan \theta$$

where, R is horizontal range.

- (viii) All the above expressions for T , H and R are derived by neglecting air resistance. If air resistance is considered, then values may differ slightly.
- (ix) Due to air resistance when a net speed of projectile decreases, then R decreases, T increases and H decreases. Reverse is the case when net speed increases.

Example 4.14 An object is projected with a velocity of 30 ms^{-1} at an angle of 60° with the horizontal. Determine the horizontal range covered by the object.

Sol. Given, initial velocity, $u = 30 \text{ ms}^{-1}$

Angle of projection, $\theta = 60^\circ$

Therefore, the horizontal range (or distance) covered by the object will be given as

$$\begin{aligned} R &= \frac{u^2 \sin 2\theta}{g} = \frac{(30)^2 \sin 2(60^\circ)}{g} \\ &= \frac{(30)^2 2 \sin 60^\circ \cos 60^\circ}{9.8} \\ &= 79.53 \text{ m} \\ \Rightarrow R &= 79.53 \text{ m} \end{aligned}$$

Example 4.15 A projectile has a range of 40 m and reaches a maximum height of 10 m. Find the angle at which the projectile is fired.

Sol. Range of a projectile, $R = \frac{u_0^2 \sin 2\theta_0}{g} = 40 \text{ m}$... (i)

$$H = \frac{u_0^2 \sin^2 \theta_0}{2g} = 10 \text{ m} \quad \dots \text{(ii)}$$

On dividing Eq. (i) by Eq. (ii), we get

$$\begin{aligned} \frac{2(\sin 2\theta_0)}{\sin^2 \theta_0} &= 4 \\ \frac{4 \sin \theta_0 \cos \theta_0}{\sin^2 \theta_0} &= 4 \\ \Rightarrow \tan \theta_0 &= 1 \\ \Rightarrow \theta_0 &= 45^\circ \end{aligned}$$

Example 4.16 Find the angle of projection of a projectile for which the horizontal range and maximum height are equal.

Sol. Given, $R = H$

$$\therefore \frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2 \theta}{2g}$$

or $2 \sin \theta \cos \theta = \frac{\sin^2 \theta}{2}$

or $\frac{\sin \theta}{\cos \theta} = 4$ or $\tan \theta = 4$

$\therefore \theta = \tan^{-1}(4)$

Example 4.17 There are two angles of projection for which the horizontal range is the same. Show that the sum of the maximum heights for these two angles is independent of the angle of projection.

Sol. Let the angles of projection be α and $90^\circ - \alpha$ for which the horizontal range R is same.

Now, $H_1 = \frac{u^2 \sin^2 \theta}{2g}$

and $H_2 = \frac{u^2 \sin^2(90^\circ - \theta)}{2g} = \frac{u^2 \cos^2 \theta}{2g}$

Therefore, $H_1 + H_2 = \frac{u^2}{2g} (\sin^2 \theta + \cos^2 \theta) = \frac{u^2}{2g}$

Clearly, the sum of the heights for the two angles of projection is independent of the values of projection angles.

Example 4.18 Prove that the maximum horizontal range is four times the maximum height attained by the projectile; when fired at an inclination so as to have maximum horizontal range.

Sol. For $\theta = 45^\circ$, the horizontal range is maximum and is given by

$$R_{\max} = \frac{u^2}{g}$$

Maximum height attained,

$$H_{\max} = \frac{u^2 \sin^2 45^\circ}{2g} = \frac{u^2}{4g} = \frac{R_{\max}}{4}$$

or $R_{\max} = 4 H_{\max}$

Projectile fired at an angle with the vertical

Let a particle be projected vertically with an angle θ with vertical and its speed of projection is u .

Clearly, the angle made by the velocity of projectile at point of projection with horizontal is $(90^\circ - \theta)$.

In this case

$$(i) \text{ Time of flight} = \frac{2u \sin(90^\circ - \theta)}{g} = \frac{2u}{g} \cos \theta$$

$$(ii) \text{ Maximum height} = \frac{u^2 \sin^2(90^\circ - \theta)}{2g} = \frac{u^2 \cos^2 \theta}{2g}$$

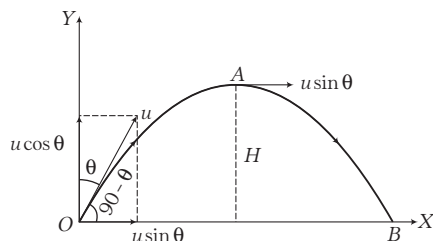


Fig. 4.12 Projectile fired at an angle with the vertical

(iii) Horizontal range

$$= \frac{u^2}{g} \sin 2(90^\circ - \theta) = \frac{u^2}{g} \sin (180^\circ - 2\theta) = \frac{u^2}{g} \sin 2\theta$$

(iv) Equation of path of projectile,

$$y = x \tan (90^\circ - \theta) - \frac{1}{2} \frac{gx^2}{u^2 \cos^2 (90^\circ - \theta)}$$

$$= x \cot \theta - \frac{gx^2}{2u^2 \sin^2 \theta}$$

(v) Velocity at any time t ,

$$v = \sqrt{u^2 + g^2 t^2 - 2ugt \sin (90^\circ - \theta)}$$

$$= \sqrt{u^2 + g^2 t^2 - 2ugt \cos \theta}$$

This velocity makes an angle β with the horizontal direction, then

$$\tan \beta = \frac{u \sin (90^\circ - \theta) - gt}{u \cos (90^\circ - \theta)} = \frac{u \cos \theta - gt}{u \sin \theta}$$

Example 4.19 A football is kicked at an angle of 30° with the vertical, so if the horizontal component of its velocity is 20 ms^{-1} , determine its maximum height.

Sol. Given, $\theta = 30^\circ$

Horizontal component of velocity $= u \sin 30^\circ = 20 \text{ ms}^{-1}$

$$\Rightarrow u = \frac{20}{\sin 30^\circ} = \frac{20}{1/2} = 40 \text{ ms}^{-1}$$

Therefore, maximum height,

$$H = \frac{u^2 \cos^2 30^\circ}{2g} = \frac{(40)^2}{2 \times 9.8} \times \left(\frac{\sqrt{3}}{2} \right)^2 = 61.22 \text{ m}$$

CHECK POINT 4.2

- At the top of the trajectory of a projectile, the directions of its velocity and acceleration are
 - parallel to each other
 - anti-parallel to each other
 - inclined to each other at an angle of 45°
 - perpendicular to each other
- The value of acceleration at the top of the trajectory of a particle when thrown obliquely is
 - maximum
 - minimum
 - zero
 - g
- In the motion of a projectile falling freely under gravity, its (neglect air friction)
 - total mechanical energy is conserved
 - momentum is conserved
 - mechanical energy and momentum both are conserved
 - None is conserved
- During the flight of a projectile thrown obliquely, which amongst the quantities remains constant?
 - Angular momentum
 - Linear momentum
 - Vertical component of velocity
 - Horizontal component of velocity
- A stone is projected with speed of 50 ms^{-1} at an angle of 60° with the horizontal. The speed of the stone at highest point of trajectory is
 - 75 ms^{-1}
 - 25 ms^{-1}
 - 50 ms^{-1}
 - Cannot find
- A football player throws a ball with a velocity of 50 ms^{-1} at an angle 30° from the horizontal. The ball remains in the air for (Take, $g = 10 \text{ ms}^{-2}$)
 - 2.5 s
 - 1.25 s
 - 5 s
 - 0.625 s
- A particle is projected with a velocity of 20 ms^{-1} at an angle of 60° to the horizontal. The particle hits the horizontal plane again during its journey. What will be the time of impact?
 - 3.53 s
 - 2.4 s
 - 1.7 s
 - 1 s
- If two balls are projected at angles 45° and 60° and the maximum heights reached are same, what is the ratio of their initial velocities?
 - $\sqrt{2} : \sqrt{3}$
 - $\sqrt{3} : \sqrt{2}$
 - 3 : 2
 - 2 : 3
- If the initial velocity of a projection is doubled, keeping the angle of projection same, the maximum height reached by it will
 - remain the same
 - be doubled
 - become four times
 - be halved
- For a projectile, the ratio of maximum height reached to the square of flight time is (Take, $g = 10 \text{ ms}^{-2}$)
 - 5 : 4
 - 5 : 2
 - 5 : 1
 - 10 : 1

11. A particle is projected from ground with speed u and at an angle θ with horizontal. If at maximum height from ground, the speed of particle is $1/2$ times of its initial velocity of projection, then find its maximum height attained.
- (a) $\frac{u^2}{g}$ (b) $\frac{2u^2}{g}$
 (c) $\frac{u^2}{2g}$ (d) $\frac{3u^2}{8g}$
12. A projectile thrown with velocity v_0 at an angle α to the horizontal, has a range R . It will strike a vertical wall at a distance $R/2$ from the point of projection with a speed of
- (a) v_0 (b) $v_0 \sin \alpha$
 (c) $v_0 \cos \alpha$ (d) $\sqrt{\frac{gR}{2}}$
13. A particle is projected at an angle of 45° with a velocity of 9.8 ms^{-1} . The horizontal range will be (Take, $g = 9.8 \text{ ms}^{-2}$)
- (a) 9.8 m (b) 4.9 m
 (c) $\frac{9.8}{\sqrt{2}}$ (d) $9.8\sqrt{2}$
14. Two projectiles A and B are projected with same speed at angles 30° and 60° to the horizontal, then which one is wrong?
- (a) $R_A = R_B$ (b) $H_B = 3H_A$
 (c) $T_B = \sqrt{3}T_A$ (d) None of these
15. A projectile fired with initial velocity u at some angle θ has a range R . If the initial velocity be doubled at the same angle of projection, then the range will be
- (a) $2R$ (b) $R/2$
 (c) R (d) $4R$
16. An object is thrown along a direction inclined at an angle of 45° with the horizontal direction. The horizontal range of the particle is equal to
- (a) vertical height
 (b) twice the vertical height
 (c) thrice the vertical height
 (d) four times the vertical height
17. An object is projected at an angle of 45° with the horizontal. The horizontal range and the maximum height reached will be in the ratio
- (a) 1 : 2 (b) 2 : 1
 (c) 1 : 4 (d) 4 : 1
18. The horizontal range of a projectile is $4\sqrt{3}$ times of its maximum height. The angle of projection will be
- (a) 60° (b) 37°
 (c) 16.1° (d) 45°
19. A ball is projected with a velocity $20\sqrt{3} \text{ ms}^{-1}$ at an angle 60° to the horizontal. The time interval after which the velocity vector will make an angle 30° to the horizontal is (Take, $g = 10 \text{ ms}^{-2}$)
- (a) 5 s (b) 2 s (c) 1 s (d) 3 s
20. A projectile is thrown with a velocity of 10 ms^{-1} at an angle of 60° with horizontal. The interval between the moments when speed is $\sqrt{5g} \text{ m/s}$ is (Take, $g = 10 \text{ ms}^{-2}$)
- (a) 1 s (b) 3 s
 (c) 2 s (d) 4 s

PROJECTILE FROM A POINT ABOVE THE GROUND

Projectile projected from a tower

When an object is at some height above the ground, then its projectile motion depends on the angle of projection.

Projectile projected horizontally from a tower

Assume that an object is thrown horizontally with some velocity u from point O , on a tower of height h above the ground level and after time t , it reaches ground at point E .

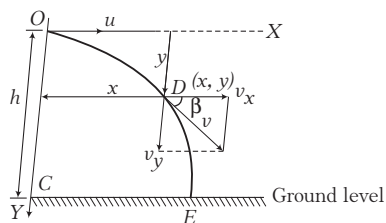


Fig. 4.13 Horizontal projectile

Horizontal components	Vertical components
$u_x = u$	$u_y = 0$
$a_x = 0 (\because F_x = 0)$	$a_y = -g$

Different terms related to this type of projectile motion are

(i) Equation of trajectory, $y = \frac{1}{2} \left(\frac{g}{u^2} \right) x^2$

(ii) Time of flight, $T = \sqrt{\frac{2h}{g}}$

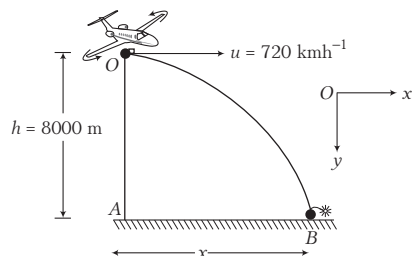
(iii) Horizontal range, $R = u \sqrt{\frac{2h}{g}}$

(iv) Velocity of projectile at any instant,

$$v = \sqrt{u^2 + g^2 t^2}$$

Example 4.20 A bomb is released from an aeroplane flying at a speed of 720 kmh^{-1} in the horizontal direction 8000 m above the ground. At what horizontal distance from the initial position of aeroplane, it strikes the ground?

Sol.



According to the figure, during motion of the bomb from O to B ,

$$u = 720 \times \frac{5}{18} = 200 \text{ ms}^{-1} \Rightarrow y = h = \frac{1}{2} g t^2$$

$$\Rightarrow 8000 = \frac{1}{2} \times 10 t^2 \Rightarrow t = 40 \text{ s}$$

$$\therefore x = ut = 200 \times 40 = 8000 \text{ m}$$

Example 4.21 A body is thrown horizontally from the top of a tower and strikes the ground after three seconds at an angle of 45° with the horizontal. Then, find

(i) the height of the tower.

(ii) the speed of projection of the body.

Sol. (i) Let H be the height of the tower.

$$\text{The time of flight, } T_f = \sqrt{\frac{2H}{g}} = 3 \text{ s}$$

$$\Rightarrow H = \frac{g \times (3)^2}{2} = \frac{9.8 \times 9}{2} = 44.1 \text{ m}$$

(ii) Let the speed of projection be v_0 .

Then, for horizontal projection,

$$v_x = v_0 \Rightarrow v_y = -gt$$

$$\text{At } t = T_f = 3 \text{ s, } v_y = -9.8 \times 3 = -29.4 \text{ ms}^{-1}$$

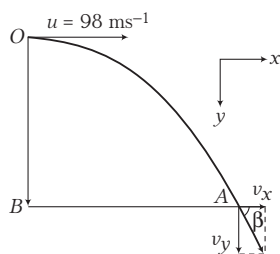
The angle which the final velocity makes with the horizontal $= \theta = 45^\circ$ (Given)

$$\Rightarrow \tan 45^\circ = \frac{-v_y}{v_x} \Rightarrow -v_y = v_x$$

$$\text{So, } v_x = 29.4 \text{ ms}^{-1}$$

Example 4.22 A projectile is fired horizontally with a velocity of 98 ms^{-1} from the top of a hill 490 m high. Find

(i) the time taken by the projectile to reach the ground,



(ii) the distance of the point, where the particle hits the ground from foot of the hill and

(iii) the velocity with which the projectile hits the ground.

(Take, $g = 9.8 \text{ ms}^{-2}$)

Sol. In this problem, we cannot apply the formulae of R , H and T directly. Here, it will be more convenient to choose x and y directions as shown in figure.

Here, $u_x = 98 \text{ ms}^{-1}$, $a_x = 0$, $u_y = 0$ and $a_y = g$.

(i) At A , $s_y = 490 \text{ m}$. So, applying

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$\Rightarrow 490 = 0 + \frac{1}{2} (9.8) t^2 \therefore t = 10 \text{ s}$$

(ii) $BA = s_x = u_x t + \frac{1}{2} a_x t^2$ or $BA = (98)(10) + 0$ ($\because a_x = 0$)
or $BA = 980 \text{ m}$

(iii) Horizontal velocity, $v_x = u_x = 98 \text{ ms}^{-1}$

Vertical velocity, $v_y = u_y + a_y t = 0 + (9.8)(10) = 98 \text{ ms}^{-1}$

$$\therefore \text{Resultant velocity, } v = \sqrt{v_x^2 + v_y^2} = \sqrt{(98)^2 + (98)^2}$$

$$= 98\sqrt{2} \text{ ms}^{-1}$$

$$\text{and } \tan \beta = \frac{v_y}{v_x} = \frac{98}{98} = 1 \therefore \beta = 45^\circ$$

Thus, the projectile hits the ground with a velocity $98\sqrt{2} \text{ ms}^{-1}$ at an angle of $\beta = 45^\circ$ with horizontal as shown in the given figure.

Projectile projected upward from a tower

Consider a projectile projected upward at an angle (θ) from point O which is situated on a tower at height h above the ground. Now, from the diagram, we have

$$u_x = u \cos \theta, a_x = 0$$

$$u_y = u \sin \theta, a_y = -g$$

(i) Equation of horizontal motion, $x = u \cos \theta t$... (i)

(ii) Equation of vertical motion,

$$-h = u \sin \theta t - \frac{1}{2} g t^2 \quad \dots \text{(ii)}$$

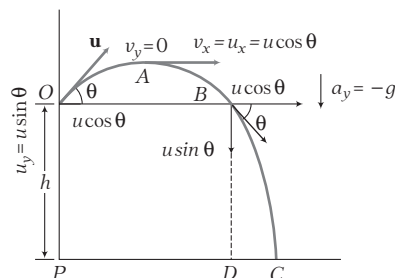


Fig. 4.14 Projectile projected upward from a tower

$$\text{(iii) Time of flight, } T = \frac{u \sin \theta}{g} \pm \sqrt{\frac{u^2 \sin^2 \theta}{g^2} + \frac{2h}{g}}$$

(iv) Horizontal distance covered (in time of flight T),

$$PC = (u \cos \theta)T$$

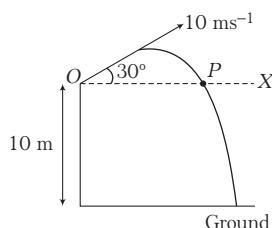
(v) Horizontal distance covered from the top of tower,

$$OB = \frac{u^2 \sin 2\theta}{g}$$

In such case for range PC to become maximum, θ should be 45° .

Example 4.23 A boy playing on the roof of a 10 m high building throws a ball with a speed of 10 ms^{-1} at an angle of 30° with the horizontal. How far from the throwing point will the ball be at the height of 10 m from the ground?

Sol. The ball will be at point P when it is at a height of 10 m from the ground. So, we have to find distance OP , which can be calculated by direct considering it, as a projectile on a level OX at a height h from the horizontal level.



$$OP = R = \frac{u^2 \sin 2\theta}{g}$$

$$\Rightarrow R = \frac{10^2 \times \sin(2 \times 30^\circ)}{10} = 8.66 \text{ m}$$

Projectile projected downward from a tower

Consider a projectile projected downward at an angle θ from point O which is on a tower of height h above the ground.

Now, from the diagram, we have

$$u_x = u \cos \theta, a_x = 0$$

$$\Rightarrow u_y = -u \sin \theta, a_y = -g$$

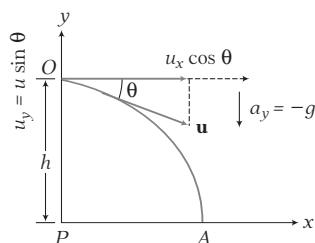


Fig. 4.15 Projectile projected downward from a tower

(i) Equation of motion, $-h = (-u \sin \theta)t + \frac{1}{2}(-g)t^2$

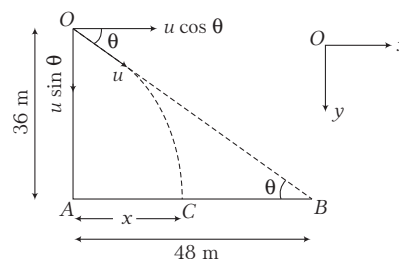
or $gt^2 + (2u \sin \theta)t - 2h = 0$

(ii) Time of flight, $T = \frac{-2u \sin \theta}{2g} \pm \frac{\sqrt{4u^2 \sin^2 \theta + 8gh}}{2g}$

(iii) Horizontal distance covered from the base of tower, $PA = (u \cos \theta)T$

Example 4.24 A boy standing on the top of a tower 36 m high has to throw a packet to his friend standing on the ground 48 m horizontally away. If he throws a packet directly aiming the friend with a speed of 10 ms^{-1} , how short will the packet fall?

Sol. The packet will strike at point C instead of reaching to point B because the packet will move in a parabolic path instead of straight line.



From geometry, $\tan \theta = \frac{36}{48} = \frac{3}{4}$

$$\Rightarrow \sin \theta = \frac{3}{5}, \cos \theta = \frac{4}{5}$$

For O to C , $s_y = u \sin \theta t + \frac{1}{2}gt^2$

$$\Rightarrow 36 = 10 \times \frac{3}{5}t + \frac{1}{2} \times 10t^2 = 6t + 5t^2$$

$$\Rightarrow 5t^2 + 6t - 36 = 0$$

$$\therefore t = 2.15 \text{ s,}$$

$$s_x = u \cos \theta t = 10 \times \frac{4}{5} \times 2.15$$

$$= 17.2 \text{ m} = AC$$

$$\therefore BC = AB - AC = 48 - 17.2 = 30.8 \text{ m}$$

The packet will fall at distance 30.8 m in front of his friend.

Shooting a freely falling target

Aiming a target situated at some height is also an example of projectile motion. For example, consider a stone of a slingshot (gulel) which aimed at a mango A as shown in figure.

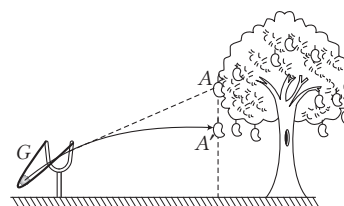
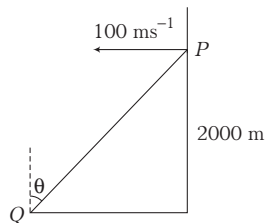


Fig. 4.16 Shooting a freely falling target

Suddenly the mango starts falling and at the same time stone is fired from the gulel. The stone hits the mango at points A' . Reason behind this is stone and mango falls through same height under same value of gravitational acceleration (g) in same time, hence stone hits the mango.

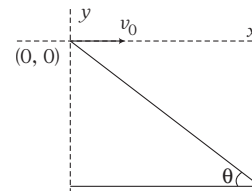
CHECK POINT 4.3

- A bomb is dropped from an aeroplane moving horizontally at constant speed. If air resistance is neglected, then the bomb
 - falls on the earth exactly below the aeroplane
 - falls on the earth behind the aeroplane
 - falls on the earth ahead of the aeroplane
 - flies with the aeroplane
- A body is projected horizontally with a velocity of 4 ms^{-1} from the top of a high tower. The velocity of the body after 0.7 s is nearly (Take, $g = 10 \text{ ms}^{-2}$)
 - 10 ms^{-1}
 - 8 ms^{-1}
 - 19.2 ms^{-1}
 - 11 ms^{-1}
- A particle is projected horizontally with speed 20 ms^{-1} from the top of a tower. After what time, velocity of particle will be at 45° angle from initial direction of projection?
 - 1 s
 - 2 s
 - 3 s
 - 4 s
- An aeroplane is travelling horizontally at a height of 2000 m from the ground. The aeroplane when at a point P , drops a bomb to hit a stationary target Q on the ground. In order that the bomb hits the target, what is the angle θ , the line PQ makes with the vertical? (Take, $g = 10 \text{ ms}^{-2}$)



- 45°
- 30°
- 60°
- 90°

- An aeroplane is flying at a constant height of 1960 m with speed 600 kmh^{-1} above the ground towards point directly over a person struggling in flood water. At what angle of sight with the vertical should the pilot release a survival kit, if it is to reach the person in water? (Take, $g = 9.8 \text{ ms}^{-2}$)
 - 45°
 - 30°
 - 60°
 - 90°
- A ball is projected horizontally from the top of a tower with a velocity v_0 . It will be moving at an angle of 60° with the horizontal after time,
 - $\frac{v_0}{\sqrt{3}g}$
 - $\frac{\sqrt{3} v_0}{g}$
 - $\frac{v_0}{g}$
 - $\frac{v_0}{2g}$
- A man standing on a hill top projects a stone horizontally with speed v_0 as shown in figure. Taking the coordinate system as given in the figure. The coordinates of the point, where the stone will hit the hill surface are



- $\left(\frac{2v_0^2 \tan \theta}{g}, -\frac{2v_0^2 \tan^2 \theta}{g} \right)$
- $\left(\frac{2v_0^2}{g}, -\frac{2v_0^2 \tan^2 \theta}{g} \right)$
- $\left(\frac{2v_0^2 \tan \theta}{g}, -\frac{2v_0^2}{g} \right)$
- $\left(\frac{2v_0^2 \tan^2 \theta}{g}, -\frac{2v_0^2 \tan \theta}{g} \right)$

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1** In a two dimensional motion, instantaneous speed v_0 is a positive constant. Then which of the following are necessarily true? [NCERT Exemplar]

(a) The average velocity is not zero at any time.
 (b) Average acceleration must always vanish.
 (c) Displacements in equal time intervals are equal.
 (d) Equal path lengths are traversed in equal intervals.

- 2** In a two dimensional motion, instantaneous speed v_0 is a positive constant. Then, which of the following are necessarily true? [NCERT Exemplar]

(a) The acceleration of the particle is zero.
 (b) The acceleration of the particle is bounded.
 (c) The acceleration of the particle is necessarily in the plane of motion.
 (d) The particle must be undergoing a uniform circular motion.

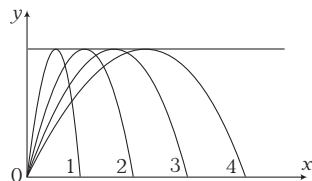
- 3** A particle velocity changes from $(2\hat{i} - 3\hat{j}) \text{ ms}^{-1}$ to $(3\hat{i} - 2\hat{j}) \text{ ms}^{-1}$ in 2 s. If its mass is 1 kg, the acceleration (ms^{-2}) is

(a) $-(\hat{i} + \hat{j})$ (b) $(\hat{i} + \hat{j})/2$
 (c) zero (d) $(\hat{i} - \hat{j})/2$

- 4** A projectile is thrown at an angle θ with the horizontal and its range is R_1 . It is then thrown at an angle θ with vertical and the range is R_2 , then

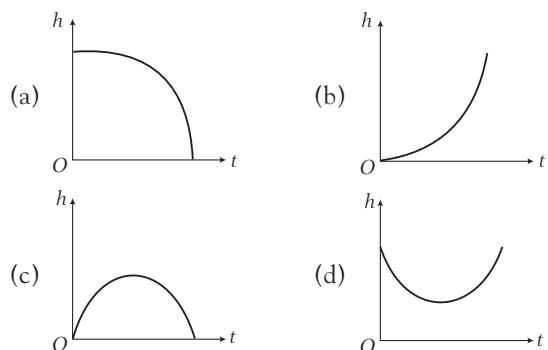
(a) $R_1 = 4R_2$ (b) $R_1 = 2R_2$
 (c) $R_1 = R_2$ (d) None of these

- 5** Figure shows four paths for a kicked football. Ignoring the effects of air on the flight, rank the paths according to initial horizontal velocity component highest first.

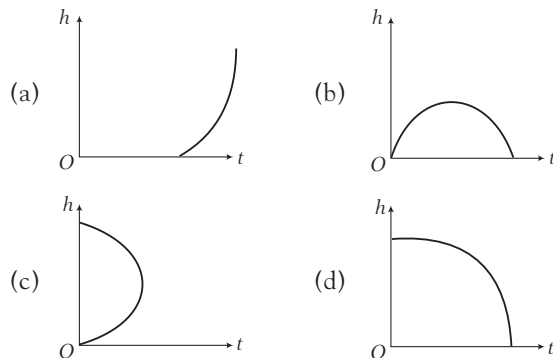


(a) 1, 2, 3, 4 (b) 2, 3, 4, 1
 (c) 3, 4, 1, 2 (d) 4, 3, 2, 1

- 6** Which of the following is the graph between the height (h) of a projectile and time (t), when it is projected from the ground?



- 7** Which of the following is the altitude-time graph for a projectile thrown horizontally from the top of the tower?



- 8** The horizontal range of a projectile fired at an angle of 15° is 50 m. If it is fired with the same speed at an angle of 45° , its range will be [NCERT Exemplar]

(a) 60 m (b) 71 m (c) 100 m (d) 141 m

- 9** A body is thrown horizontally from the top of a tower of height 5 m. It touches the ground at a distance of 10 m from the foot of the tower. The initial velocity of the body is (Take, $g = 10 \text{ ms}^{-2}$)

(a) 2.5 ms^{-1} (b) 5 ms^{-1} (c) 10 ms^{-1} (d) 20 ms^{-1}

- 10** Velocity and acceleration of a particle at some instant of time are $\mathbf{v} = (3\hat{i} + 4\hat{j}) \text{ ms}^{-1}$ and $\mathbf{a} = -(6\hat{i} + 8\hat{j}) \text{ ms}^{-2}$, respectively. At the same instant particle is at origin, maximum x-coordinate of particle will be

(a) 1.5 m (b) 0.75 m (c) 2.25 m (d) 4 m

- 11** Two paper screens A and B are separated by a distance of 100 m. A bullet pierces A and then B . The hole in B is 10 cm below the hole in A . If the bullet is travelling horizontally at the time of hitting A , then the velocity of the bullet at A is
- (a) 100 ms^{-1} (b) 200 ms^{-1}
(c) 600 ms^{-1} (d) 700 ms^{-1}

- 12** Two stones having different masses m_1 and m_2 are projected at an angle α and $(90^\circ - \alpha)$ with same speed from same point. The ratio of their maximum heights is
- (a) 1:1 (b) $1:\tan\alpha$ (c) $\tan\alpha:1$ (d) $\tan^2\alpha:1$

- 13** Two projectiles A and B are thrown from the same point with velocities v and $v/2$, respectively. If B is thrown at an angle 45° with horizontal, what is the inclination of A when their ranges are the same?

- (a) $\sin^{-1}\left(\frac{1}{4}\right)$ (b) $\frac{1}{2}\sin^{-1}\left(\frac{1}{4}\right)$
(c) $2\sin^{-1}\left(\frac{1}{4}\right)$ (d) $\frac{1}{2}\sin^{-1}\left(\frac{1}{8}\right)$

- 14** A particle moves in the XY -plane according to the law $x = kt$, $y = kt(1 - \alpha t)$, where k and α are positive constants and t is time. The trajectory of the particle is

- (a) $y = kx$ (b) $y = x - \frac{\alpha x^2}{k}$
(c) $y = -\frac{\alpha x^2}{k}$ (d) $y = \alpha x$

- 15** The maximum range of a gun on horizontal terrain is 1 km. If $g = 10 \text{ ms}^{-2}$, what must be the muzzle velocity of the shell?

- (a) 400 ms^{-1} (b) 200 ms^{-1}
(c) 100 ms^{-1} (d) 50 ms^{-1}

- 16** The equation of trajectory of a projectile is

$$y = \sqrt{3}x - \frac{g}{2}x^2, \text{ the angle of its projection is}$$

- (a) 90° (b) zero (c) 60° (d) 30°

- 17** A projectile is thrown upward with a velocity v_0 at an angle α to the horizontal. The change in velocity of the projectile when it strikes the same horizontal plane is

- (a) $v_0 \sin\alpha$ vertically downward
(b) $2v_0 \sin\alpha$ vertically downward
(c) $2v_0 \sin\alpha$ vertically upward
(d) zero

- 18** The maximum height attained by a projectile is increased by 10% by increasing its speed of projection, without changing the angle of projection.

The percentage increase in the horizontal range will be

- (a) 20% (b) 15% (c) 10% (d) 5%

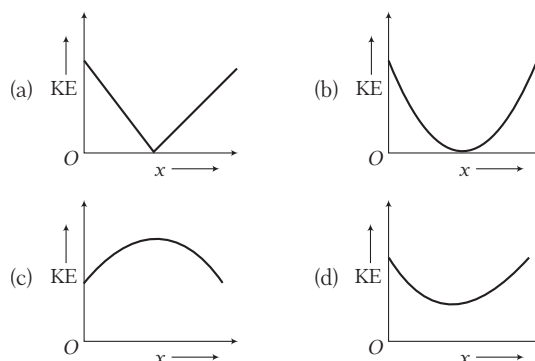
- 19** A body is projected at an angle of 30° with the horizontal with momentum p . At its highest point, the magnitude of the momentum is

- (a) $\frac{\sqrt{3}}{2}p$ (b) $\frac{2}{\sqrt{3}}p$ (c) p (d) $\frac{p}{2}$

- 20** A projectile is fired from ground level at an angle θ above the horizontal. The elevation angle ϕ of the highest point as seen from the launch point is related to θ by the relation

- (a) $\tan\phi = \frac{1}{4}\tan\theta$ (b) $\tan\phi = \tan\theta$
(c) $\tan\phi = \frac{1}{2}\tan\theta$ (d) $\tan\phi = 2\tan\theta$

- 21** A ball is thrown up with a certain velocity at an angle θ to the horizontal. The kinetic energy KE of the ball varies with horizontal displacement x as



- 22** A body projected with velocity u at projection angle θ has horizontal range R . For the same velocity and projection angle, its range on the moon surface will be ($g_{\text{moon}} = g_{\text{earth}}/6$)

- (a) $36R$ (b) $\frac{R}{36}$ (c) $\frac{R}{16}$ (d) $6R$

- 23** Three balls of same masses are projected with equal speeds at angle 15° , 45° , 75° , and their ranges are respectively R_1 , R_2 and R_3 , then

- (a) $R_1 > R_2 > R_3$ (b) $R_1 < R_2 < R_3$
(c) $R_1 = R_2 = R_3$ (d) $R_1 = R_3 < R_2$

- 24** A projectile is thrown with an initial velocity of $(a\hat{i} + b\hat{j}) \text{ ms}^{-1}$. If the range of the projectile is twice the maximum height reached by it, then

- (a) $a = 2b$ (b) $b = a$ (c) $b = 2a$ (d) $b = 4a$

- 25** The ratio of the speed of a projectile at the point of projection to the speed at the top of its trajectory is x . The angle of projection with the horizontal is

- (a) $\sin^{-1}x$ (b) $\cos^{-1}x$ (c) $\sin^{-1}(1/x)$ (d) $\cos^{-1}(1/x)$

- 26** A man can throw a stone such that it acquires maximum horizontal range 80 m. The maximum height to which it will rise for the same projectile (in metre) is

(a) 10 (b) 20 (c) 40 (d) 50

- 27** The velocity at the maximum height of a projectile is half of its initial velocity of projection (u). Its range on horizontal plane is

(a) $\frac{3u^2}{g}$ (b) $\frac{3}{2} \cdot \frac{u^2}{g}$ (c) $\frac{u^2}{3g}$ (d) $\frac{\sqrt{3}}{2} \cdot \frac{u^2}{g}$

- 28** A projectile is thrown from a point in a horizontal plane such that the horizontal and vertical velocities are 9.8 ms^{-1} and 19.6 ms^{-1} . It will strike the plane after covering distance of

(a) 39.2 m (b) 19.6 m (c) 9.8 m (d) 4.9 m

- 29** The greatest height to which a man can throw a stone is h . The greatest distance to which he can throw it will be

(a) $\frac{h}{2}$ (b) h (c) $2h$ (d) $3h$

- 30** Four bodies P , Q , R and S are projected with equal velocities having angles of projection 15° , 30° , 45° and 60° with the horizontal respectively. The body having shortest range is

(a) P (b) Q (c) R (d) S

- 31** A stone is projected in air. Its time of flight is 3 s and range is 150 m. Maximum height reached by the stone is (Take, $g = 10 \text{ ms}^{-2}$)

(a) 37.5 m (b) 22.5 m (c) 90 m (d) 11.25 m

- 32** A boy throws a ball with a velocity u at an angle θ with the horizontal. At the same instant, he starts running with uniform velocity to catch the ball before it hits the ground. To achieve this, he should run with a velocity of

(a) $u \cos \theta$ (b) $u \sin \theta$ (c) $u \tan \theta$ (d) $u \sec \theta$

- 33** Galileo writes that for angle of projection of a projectile at angle $(45^\circ - \theta)$ and $(45^\circ + \theta)$, the horizontal ranges described by the projectile are in the ratio of (if $\theta \leq 45^\circ$)

(a) 2:1 (b) 1:2 (c) 1:1 (d) 2:3

- 34** If time of flight of a projectile is 10 s. Range is 500 m. The maximum height attained by it will be

(a) 125 m (b) 50 m (c) 100 m (d) 150 m

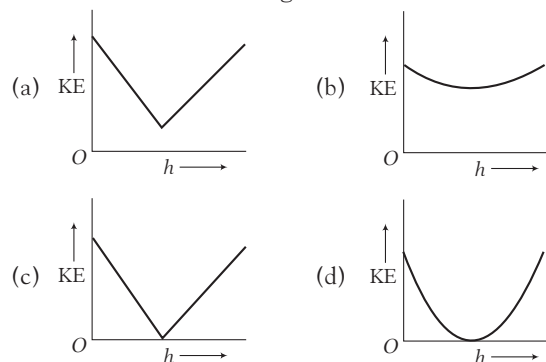
- 35** A body of mass m is thrown upwards at an angle θ with the horizontal with velocity v . While rising up the velocity of the mass after t seconds will be

(a) $\sqrt{(v \cos \theta)^2 + (v \sin \theta)^2}$ (b) $\sqrt{(v \cos \theta - v \sin \theta)^2 - gt}$
(c) $\sqrt{v^2 + g^2 t^2 - (2v \sin \theta) gt}$ (d) $\sqrt{v^2 + g^2 t^2 - (2v \cos \theta) gt}$

- 36** A stone is thrown at an angle θ with the horizontal, reaches a maximum height H . Then, the time of flight of stone will be

(a) $\sqrt{\frac{2H}{g}}$ (b) $2\sqrt{\frac{2H}{g}}$ (c) $\frac{2\sqrt{2H \sin \theta}}{g}$ (d) $\frac{\sqrt{2H \sin \theta}}{g}$

- 37** A ball is thrown up with a certain velocity at an angle θ to the horizontal. The kinetic energy KE of the ball varies with height h as



- 38** For a given velocity, a projectile has the same range R for two angles of projection. If t_1 and t_2 are the time of flight in the two cases, then $t_1 t_2$ is equal to

(a) $\frac{2R}{g}$ (b) $\frac{R}{g}$ (c) $\frac{4R}{g}$ (d) $\frac{R}{2g}$

- 39** A cricket ball is hit for a six by the bat at an angle of 45° to the horizontal with kinetic energy K . At the highest point, the kinetic energy of the ball is

(a) zero (b) K (c) $K/2$ (d) $K/\sqrt{2}$

- 40** The equation of motion of a projectile is

$$y = 12x - \frac{3}{4}x^2$$

What is the range of the projectile?

(a) 12 m (b) 16 m (c) 20 m (d) 24 m

- 41** A ball of mass m is projected from the ground with an initial velocity u making an angle of θ with the vertical. What is the change in velocity between the point of projection and the highest point?

(a) $u \cos \theta$ downward (b) $u \cos \theta$ upward
(c) $u \sin \theta$ upward (d) $u \sin \theta$ downward

- 42** The equation of projectile is $Y = \sqrt{3}X - \frac{1}{2}gX^2$. The velocity of projection is

(a) 1 ms^{-1} (b) 2 ms^{-1} (c) 3 ms^{-1} (d) 1.2 ms^{-1}

- 43** The range of a projectile when launched at an angle θ is same as when launched at an angle 2θ . What is the value of θ ?

(a) 15° (b) 30°
(c) 45° (d) 60°

- 44** A particle is thrown with a speed u at an angle θ with the horizontal. When the particle makes an angle ϕ with the horizontal, its speed changes to v , where

- (a) $v = u \cos \theta$
 (b) $v = u \cos \theta \cos \phi$
 (c) $v = u \cos \theta \sec \phi$
 (d) $v = u \sec \theta \cos \phi$

- 45** An aeroplane moving horizontally with a speed of 720 kmh^{-1} drops a food pocket, while flying at a height of 396.9 m . The time taken by a food pocket to reach the ground and its horizontal range is (Take $g = 9.8 \text{ ms}^{-2}$)

- (a) 3 s and 2000 m (b) 5 s and 500 m
 (c) 8 s and 1500 m (d) 9 s and 1800 m

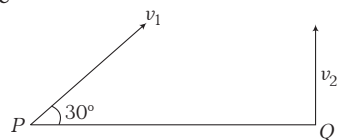
- 46** A bullet is to be fired with a speed of 2000 ms^{-1} to hit a target 200 m away on a level ground. If $g = 10 \text{ ms}^{-2}$, the gun should be aimed

- (a) directly at the target
 (b) 5 cm below the target
 (c) 5 cm above the target
 (d) 2 cm above the target

- 47** A projectile has the maximum range 500 m . If the projectile is thrown up a smooth inclined plane of 30° with the same (magnitude) velocity, the distance covered by it along the inclined plane till it stops will be

- (a) 250 m (b) 500 m
 (c) 750 m (d) 1000 m

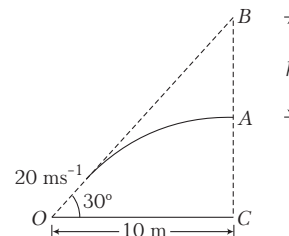
- 48** A projectile A is thrown at an angle 30° to the horizontal from point P . At the same time, another projectile B is thrown with velocity v_2 upwards from the point Q vertically below the highest point A would reach. For B to collide with A , the ratio $\frac{v_2}{v_1}$ should be



- (a) $\frac{\sqrt{3}}{2}$ (b) 2
 (c) $\frac{1}{2}$ (d) $\frac{2}{\sqrt{3}}$

- 49** A ball is thrown from a point O aiming a target at angle 30° with the horizontal, so that the ball hits the target at B but the ball hits at point A , a vertical distance h below B .

If the initial velocity of the ball is 20 ms^{-1} and the horizontal distance between O and C is 10 m . Find the value of h .



- (a) $\frac{g}{6}$ (b) $\frac{g}{10}$ (c) $\frac{g}{3}$ (d) $\frac{g}{12}$

- 50** A ball is thrown from a point with a speed v_0 at an angle of projection θ . From the same point and at the same instant, a person starts running with a constant speed $v_0/2$ to catch the ball. Will the person be able to catch the ball? If yes, what should be the angle of projection?

- (a) Yes, 60° (b) Yes, 30°
 (c) No (d) Yes, 45°

- 51** The initial velocity of a particle of mass 2 kg is $(4\hat{i} + 4\hat{j}) \text{ ms}^{-1}$. A constant force of $-20\hat{j} \text{ N}$ is applied on the particle. Initially, the particle was at $(0, 0)$. Find the x -coordinate of the point, where its y -coordinate is again zero.

- (a) 3.2 m (b) 6 m (c) 4.8 m (d) 1.2 m

- 52** An object of mass m is projected with a momentum p at such an angle that its maximum height is $1/4$ th of its horizontal range. Its minimum kinetic energy in its path will be

- (a) $\frac{p^2}{8m}$ (b) $\frac{p^2}{4m}$ (c) $\frac{3p^2}{4m}$ (d) $\frac{p^2}{m}$

- 53** The equation of motion of a projectile are given by $x = 36t \text{ m}$ and $2y = 96t - 9.8t^2 \text{ m}$. The angle of projection is

- (a) $\sin^{-1}\left(\frac{4}{5}\right)$ (b) $\sin^{-1}\left(\frac{3}{5}\right)$
 (c) $\sin^{-1}\left(\frac{4}{3}\right)$ (d) $\sin^{-1}\left(\frac{3}{4}\right)$

- 54** Two stones are projected so as to reach the same distance from the point of projection on a horizontal surface. The maximum height reached by one exceeds the other by an amount equal to half the sum of the height attained by them. Then, angle of projection of the stone which attains smaller height is

- (a) 45° (b) 60° (c) 30° (d) $\tan^{-1}(3/4)$

- 55** An arrow is shot into air. Its range is 200 m and its time of flight is 5 s. If $g = 10 \text{ ms}^{-2}$, then horizontal component of velocity and the maximum height will be respectively

(a) 20 ms^{-1} , 62.50 m (b) 40 ms^{-1} , 31.25 m
(c) 80 ms^{-1} , 62.5 m (d) None of these

- 56** A body is projected from the ground with a velocity $\mathbf{v} = (3\hat{i} + 10\hat{j}) \text{ ms}^{-1}$. The maximum height attained and the range of the body respectively are (Take, $g = 10 \text{ ms}^{-2}$)

(a) 5 m and 6 m (b) 3 m and 10 m
(c) 6 m and 5 m (d) 3 m and 5 m

- 57** A cricket fielder can throw the cricket ball with a speed v_0 . If he throws the ball while running with speed u at an angle θ to the horizontal, what is the effective angle to the horizontal at which the ball is projected in air as seen by a spectator?

[NCERT Exemplar]

(a) $\tan^{-1}\left(\frac{v_0 \cos \theta}{u + v_0 \sin \theta}\right)$
(b) $\tan^{-1}\left(\frac{v_0 \sin \theta}{u + v_0 \cos \theta}\right)$
(c) $\tan^{-1}\left(\frac{u}{v_0 \cos \theta + v_0 \sin \theta}\right)$
(d) $\tan^{-1}\left(\frac{v_0 \sin \theta + v_0 \cos \theta}{u}\right)$

- 58** A particle is projected with a velocity of 30 ms^{-1} , at an angle of $\theta_0 = \tan^{-1}\left(\frac{3}{4}\right)$. After 1 s, the particle is moving at an angle θ to the horizontal, where $\tan \theta$ will be equal to (Take, $g = 10 \text{ ms}^{-2}$)

(a) 1 (b) 2 (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

- 59** A bomber plane moves horizontally with a speed of 500 ms^{-1} and a bomb released from it, strikes the ground in 10 s. Angle at which it strikes the ground will be (Take, $g = 10 \text{ ms}^{-2}$)

(a) $\tan^{-1}\left(\frac{1}{5}\right)$ (b) $\tan^{-1}\left(\frac{1}{2}\right)$
(c) $\tan^{-1}(1)$ (d) $\tan^{-1}(5)$

- 60** Two balls are thrown simultaneously from ground with same velocity of 10 ms^{-1} but different angles of projection with horizontal. Both balls fall at same distance $5\sqrt{3} \text{ m}$ from point of projection. What is the time interval between balls striking the ground?
(a) $(\sqrt{3} - 1) \text{ s}$ (b) $(\sqrt{3} + 1) \text{ s}$ (c) $\sqrt{3} \text{ s}$ (d) 1 s

- 61** A particle moves along a parabolic path $y = -9x^2$ in such a way that the x-component of velocity remains constant and has a value $1/3 \text{ ms}^{-1}$. The acceleration of the particle is

(a) $\frac{1}{3} \text{ ms}^{-2}$ (b) 3 ms^{-2} (c) $\frac{2}{3} \text{ ms}^{-2}$ (d) 2 ms^{-2}

- 62** A projectile can have same range from two angles of projection with same initial speed. If h_1 and h_2 be the maximum heights, then

(a) $R = \sqrt{h_1 h_2}$ (b) $R = \sqrt{2 h_1 h_2}$
(c) $R = 2 \sqrt{h_1 h_2}$ (d) $R = 4 \sqrt{h_1 h_2}$

- 63** A ball is projected from ground with a speed of 20 ms^{-1} at an angle of 45° with horizontal. There is a wall of 25 m height at a distance of 10 m from the projection point. The ball will hit the wall at a height of
(a) 5 m (b) 7.5 m (c) 10 m (d) 12.5 m

- 64** A boy can throw a stone up to a maximum height of 10 m. The maximum horizontal distance that the boy can throw the same stone up to will be
(a) $20\sqrt{2} \text{ m}$ (b) 10 m (c) $10\sqrt{2} \text{ m}$ (d) 20 m

- 65** At the height 80 m, an aeroplane is moving with 150 ms^{-1} . A bomb is dropped from it so as to hit a target. At what distance from the target should the bomb be dropped? (Take, $g = 10 \text{ ms}^{-2}$)

(a) 605.3 m (b) 600 m (c) 80 m (d) 230 m

- 66** A ball is projected upwards from the top of a tower with velocity 25 ms^{-1} making an angle of 30° with the horizontal. If the height of the tower is 70 m, after what time from the instant of throwing, will the ball reach the ground? (Take, $g = 10 \text{ ms}^{-2}$)

(a) 2 s (b) 5 s (c) 7 s (d) 9 s

- 67** From the top of a tower of height 40 m, a ball is projected upwards with a speed of 20 ms^{-1} at an angle of elevation of 30° . The ratio of the total time taken by the ball to hit the ground to its time of flight (time taken to come back to the same elevation) is (Take, $g = 10 \text{ ms}^{-2}$)

(a) 2:1 (b) 3:1 (c) 3:2 (d) 1.5:1

- 68** The coordinates of a moving particle at any time t are given by $x = ct$ and $y = bt^2$. The speed of the particle is given by

(a) $2t \sqrt{b^2 - c^2}$ (b) $\sqrt{4b^2 t^2 + c^2}$
(c) $2t(b + c)$ (d) $2t(b - c)$

- 69** A ball rolls off the edge of a horizontal table top 4 m high. If it strikes the floor at a point 5 m horizontally away from the edge of the table, what was its speed at the instant it left the table?

(a) 2.5 ms^{-1} (b) 3.5 ms^{-1} (c) 5.55 ms^{-1} (d) 6.5 ms^{-1}

- 70** A ball is thrown from the ground to a wall 3 m high at a distance of 6 m and falls 18 m away from the wall, the angle of projection of ball is

(a) $\tan^{-1}\left(\frac{3}{2}\right)$ (b) $\tan^{-1}\left(\frac{2}{3}\right)$ (c) $\tan^{-1}\left(\frac{1}{2}\right)$ (d) $\tan^{-1}\left(\frac{3}{4}\right)$

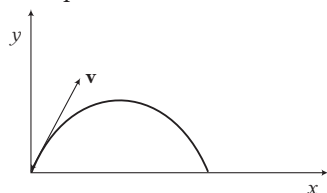
- 71** The horizontal range and maximum height attained by a projectile are R and H , respectively. If a constant horizontal acceleration $a = g/4$ is imparted to the projectile due to wind, then its horizontal range and maximum height will be

(a) $(R + H), \frac{H}{2}$ (b) $\left(R + \frac{H}{2}\right), 2H$
(c) $(R + 2H), H$ (d) $(R + H), H$

- 72** An object is projected with a velocity of 20 ms^{-1} making an angle of 45° with horizontal. The equation for trajectory is $h = Ax - Bx^2$, where h is height, x is horizontal distance, A and B are constants. The ratio $A : B$ is (Take, $g = 10 \text{ ms}^{-2}$)

(a) 1 : 5 (b) 5 : 1 (c) 1 : 40 (d) 40 : 1

- 73** If the instantaneous velocity of a particle projected as shown in figure is given by $\mathbf{v} = a\hat{i} + (b - ct)\hat{j}$, where a, b and c are positive constants, the range on the horizontal plane will be



(a) $2ab/c$ (b) ab/c (c) ac/b (d) $a/2bc$

- 74** Two particles are simultaneously projected in opposite directions horizontally from a given point in space, where gravity g is uniform. If u_1 and u_2 be their initial speeds, then the time t after which their velocities are mutually perpendicular is given by

(a) $\frac{\sqrt{u_1 u_2}}{g}$ (b) $\frac{\sqrt{u_1^2 + u_2^2}}{g}$
(c) $\frac{\sqrt{u_1 (u_1 + u_2)}}{g}$ (d) $\frac{\sqrt{u_2 (u_1 + u_2)}}{g}$

- 75** A projectile is fired at an angle of 30° to the horizontal such that the vertical component of its initial velocity is 80 ms^{-1} . Its time of flight is T . Its velocity at $t = T/4$ has a magnitude of nearly
- (a) 200 ms^{-1} (b) 300 ms^{-1}
(c) 100 ms^{-1} (d) None of these

- 76** A projectile is thrown with some initial velocity at an angle α to the horizontal. Its velocity when it is at the highest point is $(2/5)^{1/2}$ times the velocity when

it is at height half of the maximum height. Find the angle of projection α with the horizontal.

(a) 30° (b) 45°
(c) 60° (d) 37°

- 77** A very broad elevator is going up vertically with a constant acceleration of 2 ms^{-2} . At the instant when its velocity is 4 ms^{-1} , a ball is projected from the floor of the lift with a speed of 4 ms^{-1} relative to the floor at an elevation of 30° . The time taken by the ball to return the floor is (Take, $g = 10 \text{ ms}^{-2}$)

(a) $\frac{1}{2} \text{ s}$ (b) $\frac{1}{3} \text{ s}$ (c) $\frac{1}{4} \text{ s}$ (d) 1 s

- 78** A grasshopper can jump a maximum distance of 1.6 m. How far can it go in 10 s?

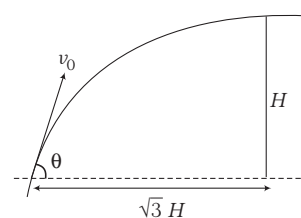
(a) $5\sqrt{2} \text{ m}$ (b) $10\sqrt{2} \text{ m}$ (c) $20\sqrt{2} \text{ m}$ (d) $40\sqrt{2} \text{ m}$

- 79** A body of mass 1 kg is projected with velocity 50 ms^{-1} at an angle of 30° with the horizontal. At the highest point of its path, a force 10 N starts acting on body for 5 s vertically upward besides gravitational force. What is the horizontal range of the body? (Take, $g = 10 \text{ ms}^{-2}$)

(a) $125\sqrt{3} \text{ m}$ (b) $200\sqrt{3} \text{ m}$
(c) 500 m (d) $250\sqrt{3} \text{ m}$

- 80** A projectile is thrown at an angle θ such that it is just able to cross a vertical wall at its highest point as shown in the figure.

The angle θ at which the projectile is thrown is given by

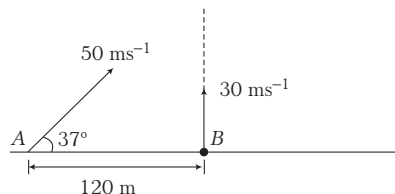


(a) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\tan^{-1}\sqrt{3}$
(c) $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (d) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

- 81** A jet aeroplane is flying at a constant height of 2 km with a speed 360 kmh^{-1} above the ground towards a target and releases a bomb. After how much time, it will hit the target and what will be the horizontal distance of the aeroplane from the target, so that the bomb should hit the target? (Take, $g = 10 \text{ ms}^{-2}$)

(a) 10 s, 1 km (b) 20 s, 2 km
(c) 30 s, 3 km (d) 40 s, 4 km

- 82** Balls A and B are thrown from two points lying on the same horizontal plane separated by a distance 120 m. Which of the following is true?



- (a) The two balls can never meet
 (b) The balls can meet, if the ball B is thrown 1 s later
 (c) The two balls meet at a height of 45 m
 (d) None of the above
- 83** Two particles are projected from the same point with same speed u at angles of projection α and β from horizontal. The maximum heights attained by them are h_1 and h_2 respectively, R is the range for both. If t_1 and t_2 are their times of flight respectively, then which amongst the option(s) is/are incorrect?

- (a) $\alpha + \beta = \frac{\pi}{2}$
 (b) $R = 4\sqrt{h_1 h_2}$
 (c) $\tan \alpha = \frac{t_1}{t_2} = \sqrt{h_1 h_2}$
 (d) $\tan \alpha = \frac{h_1}{h_2}$

- 84** A particle is projected from the ground at an angle θ with the horizontal with an initial speed u . Time after which velocity vector of the projectile is perpendicular to the initial velocity.

- (a) $u/g \sin \theta$ (b) $u/g \cos \theta$
 (c) $2u/g \sin \theta$ (d) $2u \tan \theta$

- 85** A particle is projected from horizontal making an angle of 53° with initial velocity of 100 ms^{-1} . The time taken by the particle to make angle 45° from horizontal is

- (a) 14 s (b) 2 s
 (c) Both (a) and (b) (d) None of these

- 86** A large number of bullets are fired in all directions with same speed v . What is the maximum area on the ground on which these bullets will spread?

- (a) $\pi \frac{v^2}{g}$ (b) $\pi \frac{v^4}{g^2}$
 (c) $\pi^2 \frac{v^4}{g^2}$ (d) $\pi^2 \frac{v^2}{g^2}$

- 87** For a ground-to-ground projectile, an object is at point A at $t = \frac{T}{3}$, at point B at $t = \frac{5T}{6}$ and reaches the ground at $t = T$. The difference in heights between points A and B is

- (a) $\frac{gT^2}{6}$ (b) $\frac{gT^2}{12}$ (c) $\frac{gT^2}{18}$ (d) $\frac{gT^2}{24}$

- 88** A cart is moving horizontally along a straight line with a constant speed of 30 ms^{-1} . A projectile is to be fired from the moving cart in such a way that it will return to the cart (at the same point on cart) after the cart has moved 80 m. At what velocity (relative to the cart) must projectile be fired? (Take, $g = 10 \text{ ms}^{-2}$)

- (a) 10 ms^{-1} (b) $\frac{20}{3} \text{ ms}^{-1}$
 (c) $\frac{40}{3} \text{ ms}^{-1}$ (d) $\frac{80}{3} \text{ ms}^{-1}$

- 89** Two second after projection, a projectile is travelling in a direction inclined at 30° with the horizontal. After 1 more second, it is travelling horizontally. Then, (Take, $g = 10 \text{ ms}^{-2}$)

- (a) the velocity of projection is $20\sqrt{3} \text{ ms}^{-1}$
 (b) the angle of projection is 30° with horizontal
 (c) Both (a) and (b) are correct
 (d) Both (a) and (b) are incorrect

- 90** A ball rolls off top of a stair way with a horizontal velocity $u \text{ ms}^{-1}$. If the steps are h metres high and b metres wide, the ball will just hit the edge of n th step, if n equals to

- (a) $\frac{hu^2}{gb^2}$ (b) $\frac{u^2g}{gb^2}$
 (c) $\frac{2hu^2}{gb^2}$ (d) $\frac{2u^2g}{hb^2}$

- 91** A hill is 500 m high. Supplies are to be sent across the hill using a canon that can hurl packets at a speed of 125 ms^{-1} over the hill. The canon is located at a distance of 800 m from the foot of hill and can be moved on the ground at a speed of 2 ms^{-1} , so that its distance from the hill can be adjusted. What is the shortest time in which a packet can reach on the ground across the hill? (Take, $g = 10 \text{ ms}^{-2}$)

[NCERT Exemplar]

- (a) 10 s (b) 25 s
 (c) 35 s (d) 45 s

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-3) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is correct but Reason is incorrect.
- (d) If Assertion is incorrect but Reason is correct.

1 Assertion In case of projectile motion, the magnitude of rate of change of velocity is variable.

Reason In projectile motion, magnitude of velocity first decreases and then increases during the motion.

2 Assertion At highest point of a projectile, dot product of velocity and acceleration is zero.

Reason At highest point, velocity and acceleration are mutually perpendicular.

3 Assertion If a particle is projected vertically upwards with velocity u , the maximum height attained by the particle is h_1 . The same particle is projected at angle 30° from horizontal with the same speed u . Now the maximum height is h_2 . Thus, $h_1 = 4h_2$.

Reason In first case, $v = 0$ at highest point and in second case, $v \neq 0$ at highest point.

Statement based questions

- 1** A particle (A) is dropped from a height and another particle (B) is thrown in horizontal direction with speed of 5 m/s from the same height. The correct statement is
 - (a) both particles will reach at ground simultaneously
 - (b) both particles will reach at ground with same speed
 - (c) particle (A) will reach at ground first with respect to particle (B)
 - (d) particle (B) will reach at ground first with respect to particle (A)
- 2** A ball is rolled off the edge of a horizontal table at a speed of 4 ms^{-1} . It hits the ground after 0.4 s. Which statement given below is true?
 - (a) It hits the ground at a horizontal distance 1.6 m from the edge of the table.
 - (b) The speed with which it hits the ground is 4.0 ms^{-1} .
 - (c) Height of the table 1m.
 - (d) It hits the ground at an angle of 60° to the horizontal.
- 3** I. In projectile motion, the angle between instantaneous velocity vector and acceleration

vector can be anything between 0 to π (excluding the limiting case.)

- II. In projectile motion, acceleration vector is always pointing vertically downwards. (Neglect air friction).

Which of the following statement(s) is/are correct?

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

- 4** I. Particle-1 is dropped from a tower and particle-2 is projected horizontal from the same tower, then both the particles reach the ground simultaneously.

- II. Both the particles strike the ground with different speeds.

Which of the following statement(s) is/are correct?

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

Match the columns

- 1** A particle is projected from ground with velocity u at angle θ from horizontal. Match the following two columns and mark the correct option from the codes given below.

Column I	Column II
(A) Average velocity between initial and final points	(p) $u \sin \theta$
(B) Change in velocity between initial and final points	(q) $u \cos \theta$
(C) Change in velocity between initial and peak points	(r) zero
(D) Average velocity between initial and highest points	(s) None

Codes

- | | | | | | | | |
|-------|---|---|---|-------|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) p | s | r | q | (b) p | r | q | s |
| (c) q | s | p | s | (d) r | p | q | s |

- 2** A particle is projected horizontally from a tower with velocity 10 ms^{-1} . Taking, $g = 10 \text{ ms}^{-2}$. Match the following two columns at time $t = 1 \text{ s}$ and mark the correct option from the codes given below.

Column I	Column II
(A) Horizontal component of velocity	(p) 5 SI units
(B) Vertical component of velocity	(q) 10 SI unit
(C) Horizontal displacement	(r) 15 SI unit
(D) Vertical displacement	(s) 20 SI unit

Codes

- | | | | | | | | |
|-------|---|---|---|-------|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) p | q | s | r | (b) q | q | s | p |
| (c) q | p | r | s | (d) q | q | q | p |

(c) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1** When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object is shot with the same velocity, it can travel x_2 distance. Then, $x_1 : x_2$ will be

[NEET 2019]

- (a) $\sqrt{2}:1$ (b) $1:\sqrt{3}$ (c) $1:2\sqrt{3}$ (d) $1:\sqrt{2}$

- 2** Two bullets are fired horizontally and simultaneously towards each other from roof tops of two buildings 100 m apart and of same height of 200 m with the same velocity of 25 ms^{-1} . When and where will the two bullets collide?

(Take, $g = 10 \text{ ms}^{-2}$)

[NEET (Odisha) 2019]

- (a) After 2s at a height of 180 m
(b) After 2s at a height of 20 m
(c) After 4s at a height of 120 m
(d) They will not collide

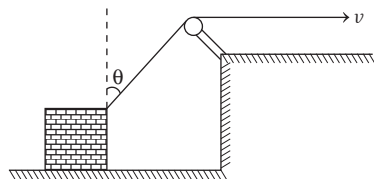
- 3 Assertion** The maximum height of projectile is always 25% of the maximum range.

Reason For maximum range, projectile should be projected at 90° .

[AIIMS 2018]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct and Reason is not the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Assertion is incorrect but Reason is correct.
- 4** What is the range of a projectile thrown with velocity 98 ms^{-1} with angle 30° from horizontal ?
- (a) $490\sqrt{3} \text{ m}$ (b) $245\sqrt{3} \text{ m}$ [JIPMER 2018]
(c) $980\sqrt{3} \text{ m}$ (d) 100 m

- 5** A block is dragged on a smooth plane with the help of a rope which moves with a velocity v as shown in the figure. The horizontal velocity of the block is



[AIIMS 2017]

- (a) $\frac{v}{\sin \theta}$ (b) $v \sin \theta$
(c) $\frac{v}{\cos \theta}$ (d) $v \cos \theta$

- 6 Assertion** When $\theta = 45^\circ$ or 135° , the value of R remains the same, only the sign changes.

$$\text{Reason } R = \frac{u^2 \sin 2\theta}{g}$$

[AIIMS 2017]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct and Reason is not the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Assertion is incorrect but Reason is correct.
- 7** The x and y -coordinates of a particle moving in a plane are given by $x(t) = a \cos(pt)$ and $y(t) = b \sin(pt)$, where $a, b (< a)$ and p are positive constants of appropriate dimensions and t is time. Then, which of the following is not true?

[JIPMER 2017]

- (a) The path of the particle is an ellipse.
(b) Velocity and acceleration of the particle are perpendicular to each other at $t = \frac{\pi}{2p}$.
(c) Acceleration of the particle is always directed towards a fixed point.
(d) Distance travelled by the particle in time interval between $t = 0$ and $t = \frac{\pi}{2p}$ is a .

- 8** A particle moves, so that its position vector is given by $\mathbf{r} = \cos \omega t \hat{\mathbf{x}} + \sin \omega t \hat{\mathbf{y}}$, where ω is a constant.

Which of the following is true? [NEET 2016]

- (a) Velocity and acceleration both are parallel to $\mathbf{r}$.
(b) Velocity is perpendicular to $\mathbf{r}$ and acceleration is directed towards to origin.
(c) Velocity is perpendicular to $\mathbf{r}$ and acceleration is directed away from the origin.
(d) Velocity and acceleration both are perpendicular to $\mathbf{r}$.
- 9** A particle is projected with an angle of projection θ to the horizontal line passing through the points (P, Q) and (Q, P) referred to horizontal and vertical axes (can be treated as X -axis and Y -axis, respectively).

The angle of projection can be given by [AIIMS 2015]

- (a) $\tan^{-1} \left[\frac{P^2 + PQ + Q^2}{PQ} \right]$ (b) $\tan^{-1} \left[\frac{P^2 + Q^2 - PQ}{PQ} \right]$
(c) $\tan^{-1} \left[\frac{P^2 + Q^2}{2PQ} \right]$ (d) $\sin^{-1} \left[\frac{P^2 + Q^2 + PQ}{2PQ} \right]$

- 10** An object is thrown towards the tower which is at a horizontal distance of 50 m with an initial velocity of 10 ms^{-1} and making an angle 30° with the horizontal. The object hits the tower at certain height. The height from the bottom of the tower, where the object hits the tower is (Take, $g = 10 \text{ ms}^{-2}$) [EAMCET 2015]

(a) $\frac{50}{\sqrt{3}} \left[1 - \frac{10}{\sqrt{3}} \right] \text{ m}$ (b) $\frac{50}{3} \left[1 - \frac{10}{\sqrt{3}} \right] \text{ m}$
 (c) $\frac{100}{\sqrt{3}} \left[1 - \frac{10}{\sqrt{3}} \right] \text{ m}$ (d) $\frac{100}{3} \left[1 - \frac{10}{\sqrt{3}} \right] \text{ m}$

- 11** The range of a projectile is R when the angle of projection is 40° . For the same velocity of projection and range, the other possible angle of projection is [Kerala CEE 2015]
 (a) 45° (b) 50° (c) 60° (d) 40°
 (e) 90°

- 12** A particle is projected with a velocity v , so that its horizontal range twice the greatest height attained. The horizontal range is [KCET 2015]

(a) $\frac{4v^2}{5g}$ (b) $\frac{v^2}{g}$ (c) $\frac{v^2}{2g}$ (d) $\frac{2v^2}{3g}$

- 13** If the angle of projection of a projector with same initial velocity exceed or fall short of 45° by equal amount α , then the ratio of horizontal ranges is [Kerala CEE 2014]
 (a) 1 : 2 (b) 1 : 3 (c) 1 : 4 (d) 1 : 1
 (e) $1 : \sqrt{2}$

- 14** A particle is moving such that its position coordinates (x, y) are $(2m, 3m)$ at time $t = 0$, $(6m, 7m)$ at time $t = 2 \text{ s}$ and $(13m, 14m)$ at time $t = 5 \text{ s}$. Average velocity vector (v_{av}) from $t = 0$ to 5 s is [CBSE AIPMT 2014]

(a) $\frac{1}{5} (13\hat{i} + 14\hat{j})$ (b) $\frac{7}{3} (\hat{i} + \hat{j})$
 (c) $2 (\hat{i} - \hat{j})$ (d) $\frac{11}{5} (\hat{i} + \hat{j})$

- 15** A cricket ball thrown across a field is at heights h_1 and h_2 from the point of projection at times t_1 and t_2 respectively after the throw. The ball is caught by a fielder at the same height as that of projection. The time of flight of the ball in this journey is [WB JEE 2014]

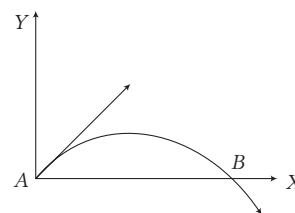
(a) $\left(\frac{h_1 t_2^2 - h_2 t_1^2}{h_1 t_1 - h_2 t_2} \right)$ (b) $\left(\frac{h_1 t_1^2 - h_2 t_2^2}{h_1 t_2 - h_2 t_1} \right)$
 (c) $\left(\frac{h_1 t_2^2 - h_2 t_1^2}{h_1 t_2 - h_2 t_1} \right)$ (d) None of these

- 16** For an object thrown at 45° to the horizontal, the maximum height H and horizontal range R are related as [UK PMT 2014]
 (a) $R = 16H$ (b) $R = 8H$ (c) $R = 4H$ (d) $R = 2H$

- 17** A body is projected horizontally from the top of a tower with a velocity of 10 ms^{-1} . If it hits the ground at an angle of 45° , then the vertical component of velocity when it hits ground (in ms^{-1}) is [EAMCET 2014]
 (a) $10\sqrt{2}$ (b) $5\sqrt{2}$ (c) 5 (d) 10

- 18** A body is projected with an angle θ . The maximum height reached is h . If the time of flight is 4 s and $g = 10 \text{ ms}^{-2}$, then value of h is [EAMCET 2014]
 (a) 40 m (b) 20 m (c) 5 m (d) 10 m

- 19** The velocity of a projectile at the initial point A is $(2\hat{i} + 3\hat{j}) \text{ ms}^{-1}$. Its velocity (in ms^{-1}) at point B is [NEET 2013]



(a) $-2\hat{i} - 3\hat{j}$ (b) $-2\hat{i} + 3\hat{j}$ (c) $2\hat{i} - 3\hat{j}$ (d) $2\hat{i} + 3\hat{j}$

- 20** A projectile is thrown with initial velocity u_0 and angle 30° with the horizontal. If it remains in the air for 1s, what was its initial velocity? [J & K CET 2013]
 (a) 19.6 ms^{-1} (b) 9.8 ms^{-1} (c) 4.9 ms^{-1} (d) 1 ms^{-1}

- 21** A projectile is projected at 10 ms^{-1} by making at an angle 60° to the horizontal. After some time, its velocity makes an angle of 30° to the horizontal. Its speed at this instant is [KCET 2013]
 (a) $\frac{10}{\sqrt{3}}$ (b) $10\sqrt{3}$ (c) $\frac{5}{\sqrt{3}}$ (d) $5\sqrt{3}$

- 22** Two particles are projected upwards with the same initial velocity v_0 in two different angles of projection such that their horizontal ranges are the same. The ratio of the heights of their highest point will be [UP CPMT 2013]
 (a) $\tan^2 \theta_1$ (b) $v_0^2 \sin^2 \theta_1$ (c) $v_0 \sin \theta_1$ (d) $v_0 / \cos \theta_1$

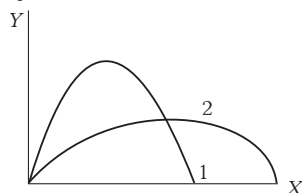
- 23** The velocity vector of the motion described by the position vector of a particle $r = 2t\hat{i} + t^2\hat{j}$ is given by [J & K CET 2013]

(a) $v = 2\hat{i} + 2t\hat{j}$ (b) $v = 2t\hat{i} + 2t\hat{j}$
 (c) $v = t\hat{i} + t^2\hat{j}$ (d) $v = 2\hat{i} + t^2\hat{j}$

- 24** The horizontal range and the maximum height of a projectile are equal. The angle of projection of the projectile is [CBSE AIPMT 2012]

(a) $\theta = \tan^{-1} \left(\frac{1}{4} \right)$ (b) $\theta = \tan^{-1}(4)$
 (c) $\theta = \tan^{-1}(2)$ (d) $\theta = 45^\circ$

- 25** Trajectories of two projectiles are shown in figure, let T_1 and T_2 be the periods and u_1 and u_2 are their speeds of projections, then [UP CPMT 2012]



- (a) $T_2 > T_1$ (b) $T_1 = T_2$
(c) $u_1 > u_2$ (d) $u_1 < u_2$

- 26** If for the same range, the two heights attained are 20 m and 80 m, then the range will be [BHU 2012]

- (a) 20 m (b) 40 m
(c) 120 m (d) 160 m

- 27** A ball thrown by one player reaches the other in 2 s. The maximum height attained by the ball above the point of projection will be (Take, $g = 10 \text{ ms}^{-2}$) [BHU 2012]

- (a) 2.5 m (b) 5 m
(c) 7.5 m (d) 10 m

ANSWERS

CHECK POINT 4.1

1. (a)	2. (a)	3. (b)	4. (c)	5. (c)	6. (b)	7. (b)	8. (b)	9. (c)	10. (d)
11. (a)	12. (c)	13. (a)							

CHECK POINT 4.2

1. (d)	2. (d)	3. (a)	4. (d)	5. (b)	6. (c)	7. (a)	8. (b)	9. (c)	10. (a)
11. (d)	12. (c)	13. (a)	14. (d)	15. (d)	16. (d)	17. (d)	18. (c)	19. (b)	20. (a)

CHECK POINT 4.3

1. (a)	2. (b)	3. (b)	4. (a)	5. (c)	6. (b)	7. (a)			
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(A) Taking it together

1. (d)	2. (c)	3. (b)	4. (c)	5. (d)	6. (c)	7. (d)	8. (c)	9. (c)	10. (b)
11. (d)	12. (d)	13. (d)	14. (b)	15. (c)	16. (c)	17. (b)	18. (c)	19. (a)	20. (c)
21. (d)	22. (d)	23. (d)	24. (c)	25. (d)	26. (b)	27. (d)	28. (a)	29. (c)	30. (a)
31. (d)	32. (a)	33. (c)	34. (a)	35. (c)	36. (b)	37. (a)	38. (a)	39. (c)	40. (b)
41. (a)	42. (b)	43. (b)	44. (c)	45. (d)	46. (c)	47. (b)	48. (c)	49. (a)	50. (a)
51. (a)	52. (b)	53. (a)	54. (c)	55. (b)	56. (a)	57. (b)	58. (d)	59. (a)	60. (a)
61. (d)	62. (d)	63. (b)	64. (d)	65. (a)	66. (c)	67. (a)	68. (b)	69. (c)	70. (b)
71. (d)	72. (d)	73. (a)	74. (a)	75. (d)	76. (c)	77. (b)	78. (c)	79. (d)	80. (c)
81. (b)	82. (c)	83. (d)	84. (a)	85. (c)	86. (b)	87. (d)	88. (c)	89. (a)	90. (c)
91. (d)									

(B) Medical entrance special format questions

• Assertion and reason

1. (d) 2. (a) 3. (b)

• Statement based questions

1. (a) 2. (a) 3. (c) 4. (c)

• Match the columns

1. (c) 2. (d)

(C) Medical entrances' gallery

1. (b)	2. (a)	3. (c)	4. (a)	5. (a)	6. (a)	7. (d)	8. (b)	9. (a)	10. (a)
11. (b)	12. (a)	13. (d)	14. (d)	15. (c)	16. (c)	17. (d)	18. (b)	19. (c)	20. (b)
21. (a)	22. (a)	23. (a)	24. (b)	25. (d)	26. (d)	27. (b)			

Hints & Explanations

• CHECK POINT 4.1

- 1 (a) Vector $\mathbf{P}$ is shown in the figure, then according to the given information,

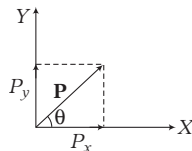
$$P_x = 5, P_y = 6$$

$$\therefore |\mathbf{P}| = \sqrt{P_x^2 + P_y^2} = \sqrt{25 + 36}$$

$$\Rightarrow |\mathbf{P}| = \sqrt{61}$$

$$\text{and } \tan \theta = \frac{P_y}{P_x} = \frac{6}{5} \Rightarrow \theta = \tan^{-1}\left(\frac{6}{5}\right)$$

Thus, $\mathbf{P}$ have magnitude of $\sqrt{61}$ and lies in XY -plane at an angle $\tan^{-1}\left(\frac{6}{5}\right)$ to the X -axis.



- 2 (a) Positions of the object are

$$\mathbf{r}_1 = 6\hat{\mathbf{i}} + 8\hat{\mathbf{j}} \text{ and } \mathbf{r}_2 = 12\hat{\mathbf{i}} + 10\hat{\mathbf{j}}$$

$$\therefore \Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 = 6\hat{\mathbf{i}} + 2\hat{\mathbf{j}} \Rightarrow |\Delta \mathbf{r}| = \sqrt{40}$$

$$\theta = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right) = \tan^{-1}\left(\frac{1}{3}\right) = 18.43^\circ$$

- 3 (b) Displacement, $\Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 = 4\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$

$$\therefore \mathbf{v}_{av} = \frac{\Delta \mathbf{r}}{\Delta t} = \frac{4\hat{\mathbf{i}} + 4\hat{\mathbf{j}}}{2} = 2(\hat{\mathbf{i}} + \hat{\mathbf{j}}) \text{ ms}^{-1}$$

$\Rightarrow$ Magnitude of velocity,

$$|\mathbf{v}_{av}| = 2\sqrt{1^2 + 1^2} = 2\sqrt{2} \text{ ms}^{-1}$$

$$\text{Direction, } \theta = \tan^{-1}\left(\frac{\Delta v_y}{\Delta v_x}\right) = \tan^{-1}\left(\frac{2}{2}\right) = \tan^{-1} 1 = 45^\circ$$

- 4 (c) $v_x = 4t, v_y = 2t - 4, v_z = 3$

$$\text{At } t = 0, v_x = 0, v_y = -4 \text{ and } v_z = 3$$

$$\therefore v = \sqrt{v_y^2 + v_z^2} = 5 \text{ units}$$

- 5 (c) As, $v_x = \frac{dx}{dt} = \frac{2t}{2} = t$

$$\text{At } t = 2 \text{ s, } v_x = 2 \text{ ms}^{-1}$$

$$\text{Further, } y = \frac{x^2}{2} = \frac{(t^2/2)^2}{2} = \frac{t^4}{8} \Rightarrow v_y = \frac{dy}{dt} = \frac{t^3}{2}$$

$$\text{At } t = 2 \text{ s, } v_y = 4 \text{ ms}^{-1}$$

So, velocity of particle is $(2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) \text{ ms}^{-1}$.

- 6 (b) Given, $\mathbf{r} = (a \sin \omega t) \hat{\mathbf{i}} + (a \cos \omega t) \hat{\mathbf{j}}$

$$\therefore \mathbf{v} = \frac{d\mathbf{r}}{dt} = (-a\omega \sin \omega t) \hat{\mathbf{j}} + (a\omega \cos \omega t) \hat{\mathbf{i}}$$

$$\mathbf{r} \cdot \mathbf{v} = 0 \Rightarrow \mathbf{r} \perp \mathbf{v}$$

- 7 (b) Position vector, $\mathbf{r} = 3t^2\hat{\mathbf{i}} + 6t\hat{\mathbf{j}} + \hat{\mathbf{k}}$

$$\text{Velocity vector, } \mathbf{v} = \frac{d\mathbf{r}}{dt} = 6t\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + 0 = 6t\hat{\mathbf{i}} + 6\hat{\mathbf{j}}$$

$\therefore$ Magnitude of the velocity of the object along Y -axis = 6 m/s

- 8 (b) Given, $x = 2t^3$

$$\therefore v_x = \frac{dx}{dt} = 6t^2 \Rightarrow a_x = \frac{dv_x}{dt} = 12t$$

$$\text{Also, } y = 3t^3 \Rightarrow v_y = \frac{dy}{dt} = 9t^2 \Rightarrow a_y = \frac{dv_y}{dt} = 18t$$

$$\therefore \text{Acceleration, } a = \sqrt{a_x^2 + a_y^2} = t\sqrt{468}$$

- 9 (c) $v_x = \frac{dx}{dt} = 6t - 6, a_x = \frac{dv_x}{dt} = 6 \text{ ms}^{-2}$

$$v_y = 2t - 2, a_y = \frac{dv_y}{dt} = 2 \text{ ms}^{-2}$$

At $t = 1 \text{ s}$, v_x and v_y both are zero. Hence, net velocity is zero.

- 10 (d) $\mathbf{a}_{av} = \frac{\Delta \mathbf{v}}{\Delta t} = \frac{3\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}}}{2} = \frac{\hat{\mathbf{i}} - 5\hat{\mathbf{j}}}{2}$

- 11 (a) $\mathbf{v} = \mathbf{u} + \mathbf{a}t = (4\hat{\mathbf{i}} + 3\hat{\mathbf{j}}) + (0.4\hat{\mathbf{i}} + 0.3\hat{\mathbf{j}})(10) = (8\hat{\mathbf{i}} + 6\hat{\mathbf{j}})$

$$\therefore v = \sqrt{(8)^2 + (6)^2} = 10 \text{ units}$$

- 12 (c) Position vector of the body, $\mathbf{r} = 3\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$

$\therefore$ Acceleration of the body,

$$\mathbf{a} = 4\hat{\mathbf{i}} \text{ and } t = 3 \text{ s}$$

$$\text{Using } \mathbf{s} = \mathbf{u}_0t + \frac{1}{2}\mathbf{a}t^2 = 0 \times 3 + \frac{4\hat{\mathbf{i}}}{2} \times 3 \times 3 \Rightarrow \mathbf{s} = 18\hat{\mathbf{i}}$$

$\therefore$ New position vector of the body,

$$\mathbf{r}_1 = 3\hat{\mathbf{i}} + 7\hat{\mathbf{j}} + 18\hat{\mathbf{i}} = 21\hat{\mathbf{i}} + 7\hat{\mathbf{j}}$$

Coordinates becomes (21, 7)

- 13 (a) Initial position vector, $\mathbf{r}_1 = 3\hat{\mathbf{i}} - 8\hat{\mathbf{j}}$

$$\text{Final position vector, } \mathbf{r}_2 = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$$

Change in position,

$$\Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 = 2\hat{\mathbf{i}} + 4\hat{\mathbf{j}} - 3\hat{\mathbf{i}} + 8\hat{\mathbf{j}} = -\hat{\mathbf{i}} + 12\hat{\mathbf{j}}$$

$$\text{Using } \mathbf{s} = \mathbf{u}_0t + \frac{1}{2}\mathbf{a}t^2$$

$$\Rightarrow -\hat{\mathbf{i}} + 12\hat{\mathbf{j}} = 0 + \frac{1}{2}\mathbf{a}(4)^2 \Rightarrow \mathbf{a} = \frac{-\hat{\mathbf{i}} + 12\hat{\mathbf{j}}}{8} = \frac{-\hat{\mathbf{i}}}{8} + \frac{3}{2}\hat{\mathbf{j}}$$

• CHECK POINT 4.2

- 1 (d) Velocity is horizontal and acceleration is vertically downward. Therefore, at the top of the trajectory of a projectile, the direction of its velocity and acceleration are perpendicular to each other.

- 2 (d) Acceleration throughout the projectile motion remains constant and equal to g .

- 3 (a) An external force of gravity is present throughout the motion. So, momentum will not be conserved. Its total mechanical energy is conserved.

4 (d) $a_x = 0 \therefore u_x = \text{constant}$

- 5 (b) At highest point, vertical component of velocity is zero.
Only horizontal component of velocity is present.

$$v_x = u \cos \theta = 50 \cos(60^\circ) = 25 \text{ ms}^{-1}$$

6 (c) Time of flight, $T = \frac{2u \sin \theta}{g} = \frac{2 \times 50 \times \sin 30^\circ}{10} = 5 \text{ s}$

- 7 (a) The particle hits the horizontal plane again at time,

$$T = \frac{2u \sin \theta}{g} = \frac{40 \times \sin 60^\circ}{9.8} = 3.53 \text{ s}$$

8 (b) $\frac{u_1^2 \sin^2 45^\circ}{2g} = \frac{u_2^2 \sin^2 60^\circ}{2g}$

$$\therefore \frac{u_1}{u_2} = \frac{\sin 60^\circ}{\sin 45^\circ} = \frac{\sqrt{3}/2}{(1/\sqrt{2})} = \sqrt{3} : \sqrt{2}$$

9 (c) $H = \frac{u^2 \sin^2 \theta}{2g} \Rightarrow H \propto u^2$

If initial velocity be doubled, then maximum height reached by the projectile will become four times.

10 (a) Maximum height, $H = \frac{u^2 \sin^2 \theta}{2g}$

and time of flight, $T = \frac{2u \sin \theta}{g}$

So, $\frac{H}{T^2} = \frac{u^2 \sin^2 \theta / 2g}{4u^2 \sin^2 \theta / g^2} = \frac{g}{8} = \frac{5}{4}$

- 11 (d) Given at maximum height,

$$u \cos \theta = \frac{1}{2} u \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$\therefore H = \frac{u^2 \sin^2 \theta}{2g} = \frac{u^2 \sin^2 60^\circ}{2g} = \frac{3u^2}{8g}$$

- 12 (c) Projectile will strike at highest point of its path with velocity $v_0 \cos \alpha$.

13 (a) Horizontal range, $R = \frac{u^2 \sin 2\theta}{g} = \frac{(9.8)^2 \sin 90^\circ}{9.8} = 9.8 \text{ m}$

14 (d) $T \propto \sin \theta$, $\frac{T_A}{T_B} = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{1}{\sqrt{3}}$ or $T_B = \sqrt{3} T_A$

$$H \propto \sin^2 \theta, \frac{H_A}{H_B} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \frac{1}{3} \text{ or } H_B = 3 H_A$$

As, $R_\theta = R_{90^\circ - \theta}$

$$\therefore R_A = R_B$$

15 (d) $R = \frac{u^2 \sin 2\theta}{g}$

$$\therefore R \propto u^2$$

If initial velocity be doubled at some angle of projection, then range will become four times.

16 (d) $\frac{R}{H} = \frac{\frac{u^2 \sin 2\theta}{g}}{\frac{u^2 \sin^2 \theta}{2g}} = \frac{2 \sin \theta \cos \theta}{\sin^2 \theta} \times \frac{2g}{\sin^2 \theta} = 4 \cot \theta$

$$\Rightarrow R = 4H \cot \theta; \text{ If } \theta = 45^\circ, \text{ then } R = 4H \cot(45^\circ) = 4H$$

17 (d) $R = 4H \cot \theta$

If $\theta = 45^\circ$, then $\frac{R}{H} = \frac{4}{1}$

18 (c) Given, range, $R = 4\sqrt{3}H$

$$\Rightarrow \frac{u^2 \sin \theta \cos \theta}{g} = 4\sqrt{3} \left(\frac{u^2 \sin^2 \theta}{2g} \right) \Rightarrow \tan \theta = \frac{1}{2\sqrt{3}}$$

$$\therefore \theta = 16.1^\circ$$

19 (b) $\tan 30^\circ = \frac{v_y}{v_x} = \frac{u_y - gt}{u_x} = \frac{(20\sqrt{3} \sin 60^\circ) - 10t}{(20\sqrt{3} \cos 60^\circ)}$

or $10 = 30 - 10t$

$$\therefore t = 2 \text{ s}$$

20. (a) $v^2 = v_y^2 + v_x^2$ or $5g = (u_y - gt)^2 + u_x^2$

Since, $v_y = u \sin 30^\circ = 10 \sin 30^\circ = 5\sqrt{3} \text{ ms}^{-1}$

$$v_x = u \cos 30^\circ = 10 \cos 30^\circ = 5 \text{ ms}^{-1}$$

or $50 = (5\sqrt{3} - 10t)^2 + (5)^2$

$$\therefore (5\sqrt{3} - 10t) = \pm 5$$

$$\Rightarrow t_1 = \frac{5\sqrt{3} + 5}{10} \text{ and } t_2 = \frac{5\sqrt{3} - 5}{10}$$

$$\therefore t_1 - t_2 = 1 \text{ s}$$

• CHECK POINT 4.3

- 1 (a) Horizontal component of velocity of the bomb is same as velocity of the aeroplane. Therefore, the bomb falls exactly below the aeroplane.

2 (b) $v_y = gt = 7 \text{ ms}^{-1}$, $v_x = 4 \text{ ms}^{-1}$

$$\therefore v = \sqrt{v_x^2 + v_y^2} \approx 8 \text{ ms}^{-1}$$

- 3 (b) Let horizontal direction is x and y is downward direction.

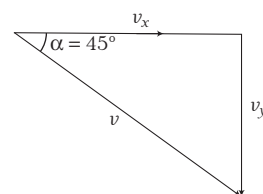
After time t, $v_x = u_x = 20 \text{ ms}^{-1}$

Velocity in y-direction,

$$v_y = u_y + a_y t = 0 + gt = gt$$

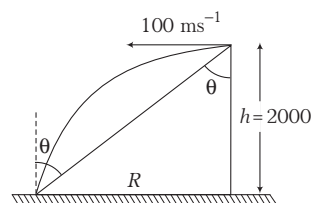
$$\tan \alpha = \frac{v_y}{v_x} = \frac{gt}{20}$$

$$\therefore 1 = \frac{10 \times t}{20} \Rightarrow t = 2 \text{ s}$$



- 4 (a) Let t be the time taken by bomb to hit the target.

$$h = 2000 = \frac{1}{2} gt^2 \Rightarrow t = 20 \text{ s}$$

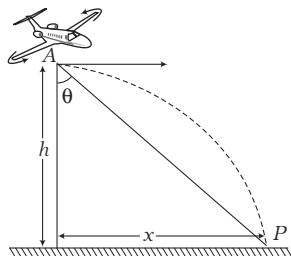


$$R = ut = (100)(20) = 2000 \text{ m}$$

$$\therefore \tan \theta = \frac{R}{h} = \frac{2000}{2000} = 1 \Rightarrow \theta = 45^\circ$$

- 5 (c) Plane is flying at a speed $= 600 \times \frac{5}{18} = \frac{500}{3} \text{ ms}^{-1}$

horizontally (at a height 1960 m)



Time taken by the kit to reach the ground,

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 1960}{9.8}} = 20 \text{ s}$$

In this time, the kit will move horizontally by

$$x = ut = \frac{500}{3} \times 20 = \frac{10000}{3} \text{ m}$$

So, the angle of sight,

$$\tan \theta = \frac{x}{h} = \frac{10000}{3 \times 1960} = \frac{10}{5.88}$$

$$= 1.7 \approx \sqrt{3} \text{ or } \theta = 60^\circ$$

- 6 (b) $\tan 60^\circ = \frac{v_V}{v_H} = \frac{gt}{v_0}$

$$\therefore t = \frac{\sqrt{3} v_0}{g}$$

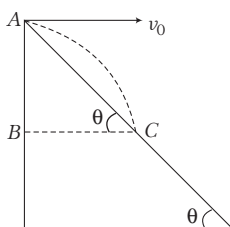
- 7 (a) $\frac{AB}{BC} = \frac{\frac{1}{2}gt^2}{v_0 t} = \tan \theta$

$$\therefore t = \frac{2v_0 \tan \theta}{g}$$

$$\text{Now, x-coordinate} = v_0 t = \frac{2v_0^2 \tan \theta}{g}$$

and y-coordinate

$$= -\frac{1}{2}gt^2 = -\frac{2v_0^2 \tan^2 \theta}{g}$$



(A) Taking it together

- 1 (d) We know that,

$$\text{Instantaneous speed, } v = \frac{ds}{dt}$$

Since, v is a positive constant, hence in this situation, equal path lengths are traversed in equal intervals of time by the moving object.

- 2 (c) As given motion is two dimensional motion and it is given that instantaneous speed v_0 is positive constant. Acceleration is rate of change of velocity (instantaneous speed), hence it will also be in the plane of motion.

$$3 (b) \mathbf{a} = \frac{\mathbf{v}_f - \mathbf{v}_i}{t} = \frac{(3\hat{i} - 2\hat{i}) - (2\hat{j} - 3\hat{j})}{2} = \frac{\hat{i} + \hat{j}}{2}$$

- 4 (c) We know that, $R_\theta = R_{90^\circ - \theta}$

$$\therefore R_1 = R_2$$

$$5 (d) R = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x u_y}{g}$$

$\therefore \text{Range} \propto \text{Horizontal initial velocity component } (u_x)$

In path 4, range is maximum, so football has maximum horizontal velocity component in this path and in path 1, range is minimum. So, horizontal velocity component is minimum.

- 8 (c) Given, $\theta = 15^\circ$ and $R = 50 \text{ m}$

$$\text{Range, } R = \frac{u^2 \sin 2\theta}{g}$$

Putting all the given values in the formula, we get

$$R = 50 \text{ m} = \frac{u^2 \sin (2 \times 15^\circ)}{g}$$

$$\Rightarrow 50 \times g = u^2 \sin 30^\circ = u^2 \times \frac{1}{2} \Rightarrow 50 \times g \times 2 = u^2$$

$$\Rightarrow u^2 = 50 \times 9.8 \times 2 = 980$$

$$\Rightarrow u = \sqrt{980}$$

$$= 31.304 \text{ ms}^{-1}$$

$$\text{For } \theta = 45^\circ, R = \frac{u^2 \sin 2 \times 45^\circ}{g} = \frac{u^2}{g} \quad (\because \sin 90^\circ = 1)$$

$$\Rightarrow R = \frac{(14\sqrt{5})^2}{g} = \frac{14 \times 14 \times 5}{9.8} = 100 \text{ m}$$

$$9 (c) s = u \times \sqrt{\frac{2h}{g}} \Rightarrow 10 = u \sqrt{2 \times \frac{5}{10}}$$

$$\Rightarrow u = 10 \text{ ms}^{-1}$$

- 10 (b) Given, $u_x = 3 \text{ ms}^{-1}$, $a_x = -6 \text{ ms}^{-1}$

$$\therefore X_{\max} = \frac{u_x^2}{2|a_x|} = \frac{9}{12} = 0.75 \text{ m}$$

- 11 (d) $h = \frac{1}{2}gt^2$ (In vertical direction)

$$\Rightarrow t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 0.1}{10}} = 0.141 \text{ s}$$

$$\text{Now, in horizontal direction, } v_x = \frac{s_x}{t} = \frac{100}{0.141} \approx 700 \text{ ms}^{-1}$$

- 12 (d) Maximum height, $H = \frac{u^2 \sin^2 \alpha}{2g} \Rightarrow H \propto \sin^2 \alpha$

$$\therefore \frac{H_1}{H_2} = \frac{\sin^2 \alpha}{\sin^2 (90^\circ - \alpha)} = \tan^2 \alpha$$

- 13 (d) Given, $R_A = R_B$

$$\Rightarrow \frac{v^2 \sin 2\theta}{g} = \frac{(v/2)^2 \sin 90^\circ}{g}$$

$$\Rightarrow \sin 2\theta = \frac{1}{4} \text{ or } \theta = \frac{1}{2} \sin^{-1} \left(\frac{1}{8} \right)$$

14 (b) $\because x = kt \Rightarrow t = \frac{x}{k}$

Now, $y = k \left(\frac{x}{k} \right) \left(1 - \alpha \cdot \frac{x}{k} \right)$

or $y = x - \frac{\alpha x^2}{k}$

15 (c) $R_{\max} = \frac{u^2}{g}$ at $\theta = 45^\circ$

$\therefore u = \sqrt{g R_{\max}} = 100 \text{ ms}^{-1}$

16 (c) Compare the given equation with

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$\Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$

17 (b) $\Delta v = a \Delta t$ (As, $a = \text{constant}$)

$$= (-g \hat{j}) \left(\frac{2v_0 \sin \alpha}{g} \right) = (-2v_0 \sin \alpha) \hat{j}$$

i.e. Change in velocity is $2v_0 \sin \alpha$, vertically downwards.

18 (c) H and R both are proportional to u^2 . Hence, percentage increases in horizontal range will also be 10%.

19 (a) At highest point, velocity will remain $v \cos 30^\circ$ or $\frac{\sqrt{3}v}{2}$.

Therefore, momentum will also remain $\frac{\sqrt{3}p}{2}$.

20 (c) $\tan \phi = \frac{H}{R/2} = \frac{2H}{R} = \frac{(u^2 \sin^2 \theta / g)}{u^2 \sin 2\theta / g}$

$\Rightarrow \tan \phi = \frac{\sin^2 \theta}{2 \sin \theta \cos \theta} = \frac{\tan \theta}{2}$

21. (d) The kinetic energy of projectile first decreases from ground to highest point and after that increases upto ground. At highest point, kinetic energy will be minimum but not zero.

22 (d) $R \propto \frac{1}{g}$

$\therefore R_{\text{moon}} = 6 R_{\text{earth}} \quad (\because g_{\text{moon}} = \frac{1}{6} g_{\text{earth}})$

23 (d) At 45° , range is maximum. At 15° and 75° , ranges are equal.

As, $R_\theta = R_{90^\circ - \theta}$

$\therefore R_{15^\circ} = R_{75^\circ} < R_{45^\circ}$

$\Rightarrow R_1 = R_3 < R_2$

24 (c) $R = 2H$ or $\frac{2u_x u_y}{g} = \frac{2u_y^2}{2g}$

or $2u_x = u_y$ or $2a = b$

25 (d) Speed of projectile at the point of projection = u

Speed of projectile at the top of its trajectory = $u \cos \theta$

$\therefore \frac{u}{u \cos \theta} = x$ or $\cos \theta = \frac{1}{x} \Rightarrow \theta = \cos^{-1} \left(\frac{1}{x} \right)$

26 (b) At $\theta = 45^\circ$, $R_{\max} = \frac{u^2}{g} = 80$

$\therefore u^2 = 800 \text{ ms}^{-2}$

Now, $H = \frac{u^2 \sin^2 45^\circ}{2g} = \frac{(800)(1/2)}{20} = 20 \text{ m}$

27 (d) Given, $u \cos \theta = \frac{u}{2} \Rightarrow \theta = 60^\circ$

Now, $R = \frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin 120^\circ}{g} = \frac{\sqrt{3} u^2}{2g}$

28 (a) $R = \frac{2u_x u_y}{g} = \frac{2 \times 9.8 \times 19.6}{9.8} = 39.2 \text{ m}$

29 (c) Maximum height, $H = \frac{u^2}{2g} = h$ (Given)

Now, $R_{\max} = \frac{u^2 \sin 90^\circ}{g} = \frac{u^2}{g} = 2h$

30 (a) When the angle of projection is very far from 45° , then range will be minimum. Therefore, the body P with angle of projection 15° will have a shortest range.

31 (d) Time of flight, $T = \frac{2u_y}{g} = 3$

$\therefore u_y = 15 \text{ ms}^{-1}$

Now, $H = \frac{u_y^2}{2g} = \frac{(15)^2}{20} = 11.25 \text{ m}$

32 (a) Velocity of boy should be equal to the horizontal component of velocity of ball, i.e. $u \cos \theta$.

33 (c) For angle $(45^\circ - \theta)$, $R = \frac{u^2 \sin(90^\circ - 2\theta)}{g} = \frac{u^2 \cos 2\theta}{g}$

For angle $(45^\circ + \theta)$, $R = \frac{u^2 \sin(90^\circ + 2\theta)}{g} = \frac{u^2 \cos 2\theta}{g}$

Thus, ratio becomes 1:1.

34 (a) Time of flight, $T = \frac{2u \sin \theta}{g} = 10 \text{ s}$

$\Rightarrow u \sin \theta = 50 \text{ ms}^{-1}$

$\therefore H = \frac{u^2 \sin^2 \theta}{2g} = \frac{(u \sin \theta)^2}{2g} = \frac{50 \times 50}{2 \times 10} = 125 \text{ m}$

35 (c) Instantaneous velocity of rising mass after t second will be

$$v_t = \sqrt{v_x^2 + v_y^2}$$

where, $v_x = v \cos \theta$ = horizontal component of velocity

$v_y = v \sin \theta - gt$ = vertical component of velocity.

$\Rightarrow v_t = \sqrt{(v \cos \theta)^2 + (v \sin \theta - gt)^2}$

$\Rightarrow v_t = \sqrt{v^2 + g^2 t^2 - (2v \sin \theta) gt}$

36 (b) $H = \frac{u^2 \sin^2 \theta}{2g}$ and $T = \frac{2u \sin \theta}{g} \Rightarrow T^2 = \frac{4u^2 \sin^2 \theta}{g^2}$

$\therefore \frac{T^2}{H} = \frac{8}{g} \Rightarrow T = \sqrt{\frac{8H}{g}} = 2 \sqrt{\frac{2H}{g}}$

- 37** (a) $K = K_0 - mgh$; Here K = kinetic energy at height h ,
 K_0 = initial kinetic energy. Variation of K with h is linear. At highest point, kinetic energy is not zero.

38 (a) $R = \frac{u^2 \sin 2\theta}{g}$ at angles θ and $90^\circ - \theta$
 Now, $t_1 = \frac{2u \sin \theta}{g}$ and $t_2 = \frac{2u \sin(90^\circ - \theta)}{g} = \frac{2u \cos \theta}{g}$
 $\therefore t_{t_2} = \frac{2}{g} \left(\frac{u^2 \sin 2\theta}{g} \right) = \frac{2R}{g}$

- 39** (c) At highest point, speed will remain $\frac{1}{\sqrt{2}}$ times ($= u \cos 45^\circ$).
 Therefore, kinetic energy will become $1/2$ times.

40 (b) Substitute $y = 0$, $0 = 12x - \frac{3}{4}x^2 \Rightarrow x = 16 \text{ m}$

41 (a) $\Delta v = a\Delta t = a \cdot \left(\frac{T}{2} \right) = (-g\hat{j}) \left(\frac{u \cos \theta}{g} \right) = (-u \cos \theta)\hat{j}$

Therefore, change in velocity is $u \cos \theta$ in downward direction.

42 (b) Comparing with $y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$, we have
 $\tan \theta = \sqrt{3}$ or $\theta = 60^\circ$ and $u^2 \cos^2 \theta = 1$

or $u = \sec \theta = \sec 60^\circ = 2 \text{ ms}^{-1}$

- 43** (b) Range of projectile launched at an angle θ is same as the range of projectile launched at angle 2θ .

$$\begin{aligned} \Rightarrow \frac{u^2 \sin 2(2\theta)}{g} &= \frac{u^2 \sin 2\theta}{g} \\ \Rightarrow \sin 2(2\theta) &= \sin 2\theta \\ \Rightarrow 2 \sin 2\theta \cos 2\theta &= 2 \sin \theta \cos \theta \\ 4 \sin \theta \cos \theta \cos 2\theta &= 2 \sin \theta \cos \theta \\ \Rightarrow 4 \cos 2\theta &= 2 \Rightarrow \cos 2\theta = \frac{1}{2} \\ \Rightarrow \cos 2\theta &= \cos 60^\circ \\ \Rightarrow 2\theta &= 60^\circ \\ \therefore \theta &= 30^\circ \end{aligned}$$

- 44** (c) Horizontal component of velocity remains unchanged.

Hence, $v \cos \phi = u \cos \theta$

$\therefore v = u \cos \theta \sec \phi$

45 (d) $T = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 396.9}{9.8}} = 9 \text{ s}$

and $u = 720 \text{ km/h} = 200 \text{ m/s}$

$\therefore R = u \times T = 200 \times 9 = 1800 \text{ m}$

- 46** (c) Let t be time taken by the bullet to hit the target.

$\therefore 200 \text{ m} = 2000 \text{ ms}^{-1}t$

$\Rightarrow t = \frac{200 \text{ m}}{2000 \text{ ms}^{-1}} = \frac{1}{10} \text{ s}$

For vertical motion, here $u = 0$

$\therefore h = \frac{1}{2}gt^2$

$$h = \frac{1}{2} \times 10 \times \left(\frac{1}{10} \right)^2 = \frac{1}{20} \text{ m} = 5 \text{ cm}$$

$\therefore$ Gun should be aimed 5 cm above the target.

47 (b) $R_{\max} = \frac{u^2}{g} = 500$

Now, $s = \frac{u^2}{2a} = \frac{u^2}{2g \sin 30^\circ} = \frac{u^2}{g} = 500 \text{ m}$

- 48** (c) Vertical component of velocity of projectile A should be equal to vertical velocity of projectile B.

or $v_1 \sin 30^\circ = v_2$ or $\frac{v_1}{2} = v_2$

$\therefore \frac{v_2}{v_1} = \frac{1}{2}$

- 49** (a) In moving horizontal distance 10 m, the ball will fall by distance $\frac{1}{2}gt^2$.

As, $R = u \cos \theta t$

$\Rightarrow 10 = 20 \times \cos 30^\circ t = 20 \times \frac{\sqrt{3}}{2} t \Rightarrow t = \frac{1}{\sqrt{3}} \text{ s}$

$$AB = h = \frac{1}{2}gt^2 = \frac{1}{2} \times g \times \left(\frac{1}{\sqrt{3}} \right)^2 = \frac{g}{6} \text{ m}$$

- 50** (a) Person will catch the ball, if his velocity will be equal to horizontal component of velocity of the ball.

$\Rightarrow \frac{v_0}{2} = v_0 \cos \theta \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$

- 51** (a) We can see that it is like a projectile motion, with $u_x = 4 \text{ ms}^{-1}$, $u_y = 4 \text{ ms}^{-1}$ and $a_y = 10 \text{ ms}^{-2}$.

x-coordinate = Range = $\frac{2u_x u_y}{g} = \frac{2 \times 4 \times 4}{10} = 3.2 \text{ m}$

- 52** (b) It is given that, $H = \frac{R}{4}$

$\therefore \frac{u^2 \sin^2 \theta}{2g} = \frac{2u^2 \sin \theta \cos \theta}{4g}$ or $\tan \theta = 1 \Rightarrow \theta = 45^\circ$

At highest point, momentum will remain $\frac{p}{\sqrt{2}}$.

$\therefore K = \frac{(p/\sqrt{2})^2}{2m} = \frac{p^2}{4m}$

- 53** (a) It is given that, $x = 36t$

$\therefore v_x = \frac{dx}{dt} = 36 \text{ ms}^{-1}$ and $y = 48t - 4.9t^2$

$\therefore v_y = 48 - 9.8t$

At $t = 0$ $v_x = 36 \text{ ms}^{-1}$ and $v_y = 48 \text{ ms}^{-1}$

So, angle of projection, $\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right)$
 $= \tan^{-1} \left(\frac{48}{36} \right) = \tan^{-1} \left(\frac{4}{3} \right)$

or $\theta = \sin^{-1} \left(\frac{4}{5} \right)$

- 54 (c) It is given that, $H_1 - H_2 = \frac{H_1 + H_2}{2}$ or $H_1 = 3H_2$

$$\therefore \frac{u^2 \sin^2 \theta}{2g} = 3 \left(\frac{u^2 \sin^2 (90^\circ - \theta)}{2g} \right)$$

$$\Rightarrow \tan^2 \theta = 3$$

$$\therefore \tan \theta = \sqrt{3} \text{ or } \theta = 60^\circ$$

Therefore, the other angle is $90^\circ - \theta$ or 30° .

- 55 (b) Time of flight, $T = \frac{2u_y}{g}$

$$\Rightarrow u_y = \frac{gT}{2} = 25 \text{ ms}^{-1}$$

$$\text{Now, } H = \frac{u_y^2}{2g} = \frac{(25)^2}{20} = 31.25 \text{ m}$$

$$\text{Further, } R = u_x T$$

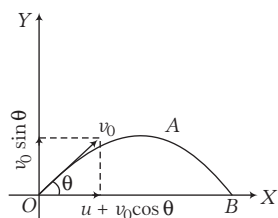
$$\Rightarrow u_x = \frac{R}{T} = 40 \text{ ms}^{-1}$$

- 56 (a) Given, $\mathbf{v} = (3\hat{i} + 10\hat{j}) \text{ ms}^{-1}$, $v_x = 3 \text{ ms}^{-1}$, $v_y = 10 \text{ ms}^{-1}$

$$H = \frac{v_y^2}{2g} = \frac{100}{2 \times 10} = 5 \text{ m}$$

$$R = v_x \times T = v_x \times \frac{2v_y}{g} = 6 \text{ m}$$

- 57 (b) Consider the adjacent diagram.



Initial velocity in x-direction, $u_x = u + v_0 \cos \theta$

Initial velocity in y-direction, $u_y = v_0 \sin \theta$

where, angle of projection is θ .

Now, we can write

$$\tan \theta = \frac{u_y}{u_x} = \frac{v_0 \sin \theta}{u + v_0 \cos \theta}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{v_0 \sin \theta}{u + v_0 \cos \theta} \right)$$

- 58 (d) $u_x = u \cos \theta_0 = 30 \times \frac{4}{5} = 24 \text{ ms}^{-1}$

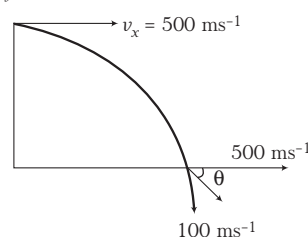
$$\text{and } u_y = u \sin \theta_0 = 30 \times \frac{3}{5} = 18 \text{ ms}^{-1}$$

After 1 s, u_x will remain same and u_y will decrease by 10 ms^{-1} or it will become 8 ms^{-1} .

$$\therefore \tan \theta = \frac{v_y}{v_x} = \frac{8}{24} = \frac{1}{3}$$

- 59 (a) Horizontal component of velocity, $v_x = 500 \text{ ms}^{-1}$ and vertical component of velocity while striking the ground,

$$v_y = 0 + 10 \times 10 = 100 \text{ ms}^{-1}$$



$\therefore$ Angle with which it strikes the ground,

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right) = \tan^{-1} \left(\frac{100}{500} \right) = \tan^{-1} \left(\frac{1}{5} \right)$$

- 60 (a) Given, $5\sqrt{3} = \frac{(10)^2 \sin 2\theta}{g}$ or $\sin 2\theta = \frac{\sqrt{3}}{2}$

$$\therefore 2\theta = 60^\circ \text{ or } \theta = 30^\circ$$

Two different angles of projection are therefore θ and $90^\circ - \theta$ or 30° and 60° .

$$\therefore T_1 = \frac{2u \sin 30^\circ}{g} = 1 \text{ s}$$

$$\therefore T_2 = \frac{2u \sin 60^\circ}{g} = \sqrt{3} \text{ s}$$

$$\text{Hence, } \Delta t = T_2 - T_1 = (\sqrt{3} - 1) \text{ s}$$

- 61 (d) Comparing with the trajectory of projectile in which particle is projected from certain height horizontally ($\theta = 0^\circ$).

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$$\text{Putting } \theta = 0^\circ \text{ and } g = a = 18u^2 = 2 \text{ ms}^{-2}$$

$$\left(\because y = -9x^2, u = \frac{1}{3} \text{ ms}^{-1} \right)$$

- 62 (d) $R = \frac{u^2 \sin \theta}{g}$ at angles θ and $90^\circ - \theta$

$$\text{Now, } h_1 = \frac{u^2 \sin^2 \theta}{2g} \text{ and } h_2 = \frac{u^2 \sin^2 (90^\circ - \theta)}{2g} = \frac{u^2 \cos^2 \theta}{2g}$$

$$h_1 h_2 = \left(\frac{u^2 \sin 2\theta}{g} \right)^2 \cdot \frac{1}{16} = \frac{R^2}{16}$$

$$\therefore R = 4 \sqrt{h_1 h_2}$$

- 63 (b) Since, $u \sin \theta \times t = 10 \text{ m}$

$$\Rightarrow 20 \sin 45^\circ t = 10$$

$$\Rightarrow t = \frac{10}{20 \sin 45^\circ} = \frac{1}{\sqrt{2}} \text{ s}$$

$$\text{Now, } y = (20 \sin 45^\circ) t - \frac{1}{2} g t^2$$

$$= 20 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times 10 \times \frac{1}{2} = 7.5 \text{ m}$$

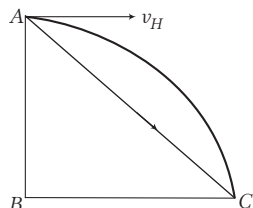
- 64 (d) $H_{\max} = \frac{u^2}{2g} = 10$

$$(\because \theta = 90^\circ)$$

$$\Rightarrow u^2 = 200 \Rightarrow R_{\max} = \frac{u^2}{g} = 20 \text{ m}$$

- 65** (a) The horizontal distance covered by bomb,

$$BC = v_H \times \sqrt{\frac{2h}{g}} = 150 \sqrt{\frac{2 \times 80}{10}} = 600 \text{ m}$$



$\therefore$ The distance of target from dropping point of bomb,

$$AC = \sqrt{(AB)^2 + (BC)^2} = \sqrt{(80)^2 + (600)^2} = 605.3 \text{ m}$$

- 66** (c) $s_y = u_y t + \frac{1}{2} a_y t^2$ or $-70 = 25t - 5t^2$

$$\Rightarrow 5t^2 - 25t - 70 = 0$$

$$\Rightarrow t^2 - 5t - 14 = 0$$

$$\Rightarrow t = 7, t = -2$$

$$\therefore t = 7 \text{ s}$$

- 67** (a) From $s_y = u_y t + \frac{1}{2} a_y t^2$, we have

$$-40 = (20 \sin 30^\circ) t - 5t^2 \text{ or } t^2 - 2t - 8 = 0$$

Solving this, we get $t = 4 \text{ s}$

$$T = \frac{2u \sin \theta}{g} = \frac{2 \times 20 \times \sin 30^\circ}{10} = 2 \text{ s}$$

$$\Rightarrow \frac{t}{T} = 2:1$$

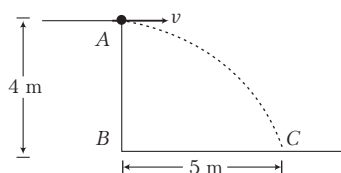
- 68** (b) Given, $x = ct \Rightarrow v_x = \frac{dx}{dt} = c$

$$\text{Also, } y = bt^2 \Rightarrow v_y = \frac{dy}{dt} = 2bt$$

$$\therefore \text{Speed} = |\mathbf{v}| = \sqrt{v_x^2 + v_y^2} = \sqrt{c^2 + (2bt)^2}$$

$$= \sqrt{c^2 + 4b^2 t^2}$$

- 69** (c) Using, $h = \frac{1}{2} g t^2$, we get



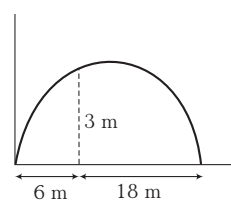
$$h_{AB} = \frac{1}{2} g t_{AC}^2$$

$$\text{or } t_{AC} = \sqrt{\frac{2h_{AB}}{g}} = \sqrt{\frac{2 \times 4}{9.8}} = 0.9 \text{ s}$$

Further, $BC = v t_{AC}$

$$\text{or } v = \frac{BC}{t_{AC}} = \frac{5}{0.9} = 5.55 \text{ ms}^{-1}$$

- 70** (b) Here, $R = \frac{u^2 \sin 2\theta}{g} = 24$... (i)



$$\therefore y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

$$3 = 6 \tan \theta - \frac{36g}{2u^2 \cos^2 \theta} \quad \dots (ii)$$

From Eq. (i), we get

$$\frac{g}{u^2} = \frac{\sin 2\theta}{24} = \frac{\sin \theta \cos \theta}{12}$$

Substituting in Eq. (ii), we get

$$3 = 6 \tan \theta - \frac{3}{2} \tan \theta = \frac{9}{2} \tan \theta \Rightarrow \theta = \tan^{-1} \left(\frac{2}{3} \right)$$

- 71** (d) $H' = H$ (as vertical component of acceleration has not changed)

$$R' = u_x T + \frac{1}{2} a_x T^2 = R + \frac{1}{2} \times \frac{g}{4} \times \frac{4u^2 \sin^2 \theta}{g^2}$$

$$= R + \frac{u^2 \sin^2 \theta}{2g} = (R + H)$$

- 72** (d) Standard equation of projectile motion,

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

Comparing with given equation,

$$A = \tan \theta \text{ and } B = \frac{g}{2u^2 \cos^2 \theta}$$

$$\text{So, } \frac{A}{B} = \frac{\tan \theta \times 2u^2 \cos^2 \theta}{g} = 40:1$$

$$(\text{As } \theta = 45^\circ, u = 20 \text{ ms}^{-1}, g = 10 \text{ ms}^{-2})$$

- 73** (a) $u_x = a, u_y = b, g = c$

$$\therefore \text{Horizontal range, } R = \frac{2u_x u_y}{g} = \frac{2ab}{c}$$

- 74** (a) Given, $\mathbf{v}_1 \perp \mathbf{v}_2$

$$\therefore \mathbf{v}_1 \cdot \mathbf{v}_2 = 0$$

$$\text{or } (u_1 \hat{i} - gt \hat{j}) \cdot (-u_2 \hat{i} - gt \hat{j}) = 0$$

$$\therefore g^2 t^2 = u_1 u_2 \text{ or } t = \frac{\sqrt{u_1 u_2}}{g}$$

- 75** (d) $\tan 30^\circ = \frac{u_y}{u_x}$

$$\therefore \frac{1}{\sqrt{3}} = \frac{80}{u_x} \text{ or } u_x = 80\sqrt{3} \text{ ms}^{-1}$$

$$T = \frac{2u_y}{g} = \frac{2 \times 80}{10} = 16 \text{ s}$$

At $t = \frac{T}{4} = 4 \text{ s} \Rightarrow v_x = u_x = 80\sqrt{3} \text{ ms}^{-1}$ and $v_y = u_y + a_y t$

or $v_y = 80 + (-10)(4) = 40 \text{ ms}^{-1}$

$\therefore$ Velocity $= \sqrt{(80\sqrt{3})^2 + (40)^2} = 144.2 \text{ ms}^{-1}$

76 (c) $(u \cos \alpha) = \sqrt{\frac{2}{5}} \sqrt{(u \cos \alpha)^2 + \{(u \sin \alpha)^2 - 2gh\}}$... (i)

Here, $h = \frac{H}{2} = \frac{u^2 \sin^2 \alpha}{4g}$... (ii)

Solving Eqs. (i) and (ii), we get

$\alpha = 60^\circ$

77 (b) The motion relative to elevator,

$a_r = a_b - a_e = (-10) - (+2) = -12 \text{ ms}^{-2}$

Now, $T = \frac{2u_y}{|a_r|} = \frac{2 \times u \sin \theta}{|a_r|} = \frac{2 \times 4 \times \sin 30^\circ}{12} = \frac{1}{3} \text{ s}$

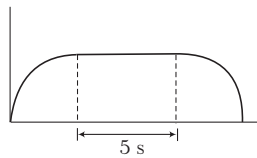
78 (c) $R_{\max} = \frac{u^2}{g} = 1.6 \text{ m}$ (At $\theta = 45^\circ$)

or $u = \sqrt{16} = 4 \text{ ms}^{-1}$

Now, $T = \frac{2u_y}{g} = \frac{2(4/\sqrt{2})}{10} = \frac{4}{5\sqrt{2}} \text{ s}$

$\therefore$ Total distance travelled in $10 \text{ s} = 1.6 \times \frac{10}{4/5\sqrt{2}} = 20\sqrt{2} \text{ m}$

79 (d) For 5 s , weight of the body is balanced by the given force. Hence, it will move in a straight line as shown in the figure.



$$R = \frac{u^2 \sin 2\theta}{g} + (u \cos \theta)(5)$$

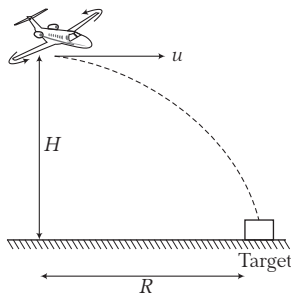
$$= \frac{(50)^2 \cdot \sin 60^\circ}{10} + (50 \times \cos 30^\circ)(5) = 250\sqrt{3} \text{ m}$$

80 (c) From the given diagram, $\frac{R/2}{H} = \frac{\sqrt{3}H}{H} = \sqrt{3}$

or $\frac{(v_0^2 \sin \theta \cos \theta)/g}{(v_0^2 \sin^2 \theta)/2g} = \sqrt{3} \Rightarrow 2 \cot \theta = \sqrt{3}$

or $\tan \theta = \frac{2}{\sqrt{3}}$ or $\theta = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

81 (b) Given, $H = 2000 \text{ m}$, $u = 360 \text{ kmh}^{-1}$



$u = 360 \times \frac{5}{18} = 100 \text{ ms}^{-1}$

The time taken by the bomb to hit the target is

$T = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2 \times 2000}{10}} = 20 \text{ s}$

The horizontal distance of the aeroplane from the target,

$R = u \sqrt{\frac{2H}{g}} = 100 \times 20 = 2000 \text{ m} = 2 \text{ km}$

82 (c) Two balls will meet, if $(50 \cos 37^\circ) t_A = 120$ or $t_A = 3 \text{ s}$

Vertical component of A is $50 \sin 37^\circ$ or 30 ms^{-1} , so they will meet, if thrown simultaneously.

$h_A = h_B = 30 \times 3 - \frac{1}{2} \times 10 \times (3)^2 = 45 \text{ m}$

83 (d) (a) Range becomes equal at complimentary angle. Hence,

$\beta = 90^\circ - \alpha$ or $\alpha + \beta = 90^\circ = \pi/2$

(b) $h_1 = \frac{u^2 \sin^2 \alpha}{2g}$

$\Rightarrow h_2 = \frac{u^2 \cos^2 \alpha}{2g}$ (As, $\beta = 90^\circ - \alpha$)

$\therefore 4\sqrt{h_1 h_2} = \frac{2u^2 \sin \alpha \cos \alpha}{g} = \frac{u^2 \sin 2\alpha}{g} = R$

(c) $\frac{t_1}{t_2} = \frac{(2u \sin \alpha / g)}{(2u \cos \alpha / g)} = \tan \alpha$

(d) $\sqrt{\frac{h_1}{h_2}} = \tan \alpha$

84 (a) Since, $\mathbf{v} \perp \mathbf{u}$ or $\mathbf{v} \cdot \mathbf{u} = 0$

or $(\mathbf{u} + g\mathbf{t}) \cdot \mathbf{u} = 0$

or $\mathbf{u} \cdot \mathbf{u} + (g \cdot \mathbf{u}) t = 0$

or $u^2 + gut \cos (90^\circ + \theta) = 0$

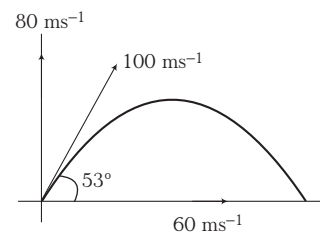
(Angle between $\mathbf{u}$ and g is $90^\circ + \theta$)

or $u - g t \sin \theta = 0$

$\therefore t = \frac{u}{g \sin \theta}$

85 (c) Component $(100 \cos 53^\circ) 60 \text{ ms}^{-1}$ will remain unchanged.

Velocity will make 45° with horizontal when vertical component also becomes $\pm 60 \text{ ms}^{-1}$.



Using $v = u + at$ (In vertical direction)

$+60 = 80 + (-10) t_1$

$\therefore t_1 = 2 \text{ s} \Rightarrow -60 = 80 + (-10) t_2$

$\therefore t_2 = 14 \text{ s}$

- 86 (b) Area in which bullet will spread = πr^2

For maximum area, $r = R_{\max} = \frac{v^2}{g}$ (When $\theta = 45^\circ$)

Maximum area = $\pi R_{\max}^2 = \pi \left(\frac{v^2}{g} \right)^2 = \frac{\pi v^4}{g^2}$

- 87 (d) As, $T = \frac{2u_y}{g} \Rightarrow u_y = \frac{gT}{2}$

$$\Rightarrow h_A = u_y t_A - \frac{1}{2} g t_A^2 = u_y \left(\frac{2u_y}{3g} \right) - \frac{1}{2} g \left(\frac{2u_y}{3g} \right)^2$$

$$= \frac{4}{9} \frac{u_y^2}{g} = \left(\frac{4}{9g} \right) \left(\frac{gT}{2} \right)^2 = \frac{gT^2}{9} \quad \left(\because u_y = \frac{gT}{2} \right)$$

$$\Rightarrow h_B = u_y \left(\frac{5}{6} \times \frac{2u_y}{g} \right) - \frac{1}{2} \times g \times \left(\frac{5}{6} \times \frac{2u_y}{g} \right)^2$$

$$= \frac{5}{18} \frac{u_y^2}{g} = \frac{5}{18g} \left(\frac{gT}{2} \right)^2 = \frac{5}{72} gT^2$$

$$\therefore h_A - h_B = \frac{gT^2}{24}$$

- 88 (c) The time taken by cart to cover 80 m,

$$\frac{s}{v} = \frac{80}{30} = \frac{8}{3} \text{ s}$$

The projectile must be fired (relative to cart) in vertically upward direction.

i.e. $a = -g = -10 \text{ ms}^{-2}$, $v' = 0$ and $t = \frac{8/3}{2} = \frac{4}{3} \text{ s}$

$$\therefore v' = u + at \text{ or } 0 = u - 10 \times \frac{4}{3} \text{ or } u = \frac{40}{3} \text{ ms}^{-1}$$

- 89 (a) $T/2 = 2 + 1 = 3 \text{ s}$ or $T = 6 \text{ s}$

$$\Rightarrow \frac{2u_y}{g} = 6$$

$$\therefore u_y = 30 \text{ ms}^{-1}$$

Further, $\tan 30^\circ = \frac{v_y}{v_x} = \frac{u_y - gt}{u_x} = \frac{30 - 20}{u_x}$

or $u_x = 10\sqrt{3} \text{ ms}^{-1}$

or $u = \sqrt{u_x^2 + u_y^2} = 20\sqrt{3} \text{ ms}^{-1}$

$$\tan \theta = \frac{u_y}{u_x} = \frac{30}{10\sqrt{3}} = \sqrt{3} \text{ or } \theta = 60^\circ$$

- 90 (c) If the ball hits the n th step, the horizontal and vertical distances traversed are nb and nh , respectively. Let t be the time taken by the ball for these horizontal and vertical displacement. Then, velocity along horizontal direction remains constant = u , initial vertical velocity is zero.

$$\therefore nb = ut \quad \dots(i)$$

$$nh = 0 + (1/2) gt^2 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get by eliminating t ,

$$nh = (1/2) g (nb/u)^2 \Rightarrow n = \frac{2hu^2}{gb^2}$$

- 91 (d) Given, speed of packets = 125 ms^{-1}

Height of the hill = 500 m.

To cross the hill, the vertical component of the velocity should be sufficient to cross such height.

$$u_y \geq \sqrt{2gh}$$

$$\geq \sqrt{2 \times 10 \times 500} \geq 100 \text{ ms}^{-1}$$

But $u^2 = u_x^2 + u_y^2$

$\therefore$ Horizontal component of initial velocity,

$$u_x = \sqrt{u^2 - u_y^2}$$

$$= \sqrt{(125)^2 - (100)^2}$$

$$= 75 \text{ ms}^{-1}$$

Time taken to reach the top of the hill,

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 500}{10}} = 10 \text{ s}$$

Time taken to reach the ground from the top of the hill,

$$t' = t = 10 \text{ s}$$

Horizontal distance travelled in 10 s,

$$x = u_x \times t = 75 \times 10$$

$$= 750 \text{ m}$$

$\therefore$ Distance through which canon has to be moved

$$= 800 - 750 = 50 \text{ m}$$

Speed with which canon can move = 2 ms^{-1}

$$\therefore \text{Time taken by canon, } t' = \frac{50}{2}$$

$$\Rightarrow = 25 \text{ s}$$

$\therefore$ Total time taken by a packet to reach on the ground

$$= t' + t + t'$$

$$= 25 + 10 + 10 = 45 \text{ s}$$

(B) Medical entrance special format questions

• Assertion and reason

- 1 (d) $\left| \frac{dv}{dt} \right| = |a| = 9.8 \text{ ms}^{-2} = \text{constant}$ and it is true that in case of projectile motion, the magnitude of velocity first decreases and then increases during the motion.

- 2 (a) At highest point, velocity is horizontal and acceleration is vertical, i.e. both are perpendicular to each other and hence, their dot product is zero.

3 (b) $h_1 = \frac{u^2}{2g}$, $h_2 = \frac{u^2 \sin^2 30^\circ}{2g} = \frac{u^2}{8g}$

$$\Rightarrow h_1 = 4h_2$$

Also, at highest point, $v = 0$ in first case

and $v \neq 0$ in second case.

• Statement based questions

- 1 (a) For both cases, $t = \sqrt{\frac{2h}{g}}$ = constant.

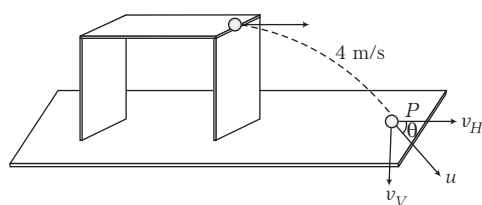
Because, initial vertical downward component of velocity will be zero for both the particles and both move under the effect of g .

- 2 (a) Vertical component of velocity of ball at point P .

$$u_V = 0 + gt \\ = 10 \times 0.4 = 4 \text{ ms}^{-1}$$

Horizontal component of velocity = initial velocity

$$\Rightarrow v_H = 4 \text{ m/s}$$



So, the speed with which it hits the ground,

$$v = \sqrt{v_H^2 + v_V^2} = 4\sqrt{2} \text{ m/s}$$

$$\text{and } \tan \theta = \frac{v_V}{v_H} = \frac{4}{4} = 1 \Rightarrow \theta = 45^\circ$$

It means the ball hits the ground at an angle of 45° to the horizontal.

$$\text{Height of the table } h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (0.4)^2 = 0.8 \text{ m}$$

Horizontal distance travelled by the ball from the edge of table $h = ut = 4 \times 0.4 = 1.6 \text{ m}$

- 3 (c) The angle between instantaneous velocity vector and acceleration vector before attaining the maximum height is acute (0 to $\pi/2$) and after is obtuse ($\pi/2$ to π). At the highest point, it is perpendicular ($\pi/2$).

- 4 (c) $t_1 = t_2 = \sqrt{\frac{2h}{g}}$ (where, h = height of tower)
 $v_1 = \sqrt{2gh}$

While $v_2 = \sqrt{v_1^2 + v_0^2}$ (where, v_0 = initial horizontal velocity)

Therefore, both statements are correct.

• Match the columns

- 2 (d) In horizontal projectile motion,
 Horizontal component of velocity,

$$u_x = u = 10 \text{ ms}^{-1}$$

Vertical component of velocity,

$$u_y = gt = 10 \times 1 = 10 \text{ ms}^{-1}$$

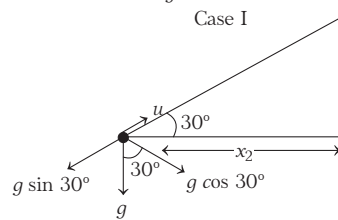
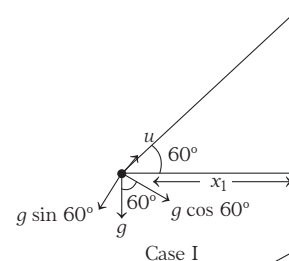
Horizontal displacement

$$= u \times t = 10 \times (1) = 10 \text{ m}$$

$$\text{Vertical displacement} = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (1)^2 = 5 \text{ m}$$

(C) Medical entrances' gallery

- 1 (b) The motion of the object shot in two cases can be depicted as below



Case II

Using third equation of motion, $v^2 = u^2 - 2gh$... (i)

As the object stops finally, so $v = 0$

For inclined motion, $g = g \sin \theta$ and $h = x$

Substituting these values in Eq. (i), we get

$$\Rightarrow u^2 = 2g \sin \theta x$$

$$\Rightarrow x = \frac{u^2}{2g \sin \theta}$$

$$\text{For case I } x_1 = \frac{u^2}{2g \sin 60^\circ}$$

$$\text{For case II } x_2 = \frac{u^2}{2g \sin 30^\circ}$$

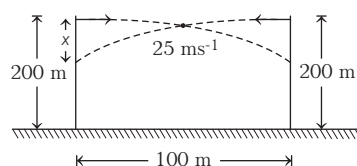
$$\therefore \frac{x_1}{x_2} = \frac{u^2}{2g \sin 60^\circ} \times \frac{2g \sin 30^\circ}{u^2} \\ = \frac{1}{2} \times \frac{2}{\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ or } 1 : \sqrt{3}$$

- 2 (a) Given, distance between the two buildings, $d = 100 \text{ m}$

Height of each tower, $h = 200 \text{ m}$

Speed of each bullet, $v = 25 \text{ ms}^{-1}$

The situation can be shown as below.



where, x be the vertical distance travelled from the top of the building and t be the time at which they collide.

As two bullets are fired toward each other, so their relative velocity will be

$$v_{\text{rel}} = 25 - (-25) = 50 \text{ ms}^{-1}$$

Then, time, $t = \frac{d}{v_{\text{rel}}} = \frac{100}{50} = 2 \text{ s}$

The distance or height at which they collide is calculated from equation of motion,

$$x = ut + \frac{1}{2}at^2$$

The bullet is initially at rest, i.e. $u = 0$ and as it is moving under the effect of gravity $a = -g$, so

$$x = -\frac{1}{2}gt^2$$

$$x = -\frac{1}{2} \times 10 (2)^2 = -20 \text{ m}$$

The negative sign shows that the bullets will collide 20 m below the top of tower, i.e. at a height of $(200 - 20) = 180 \text{ m}$ from the ground after 2 s.

- 3** (c) To obtain maximum range, angle of projection must be 45° , i.e. $\theta = 45^\circ$.

$$R_{\text{max}} = \frac{u^2 \sin(2 \times 45^\circ)}{g} = \frac{u^2}{g}$$

$$\therefore H_{\text{max}} = \frac{u^2 \sin^2 45^\circ}{2g} = \frac{u^2}{4g} = \frac{R_{\text{max}}}{4}$$

So, H_{max} is 25 % of R_{max} .

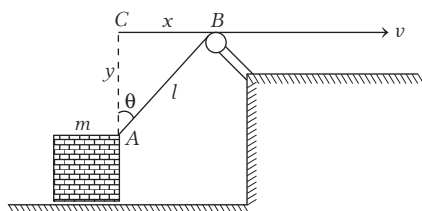
- 4** (a) Given, $u = 98 \text{ ms}^{-1}$ and $\theta = 30^\circ$

$\therefore$ Range of a projectile,

$$R = \frac{u^2 \sin(2\theta)}{g} = \frac{98 \times 98 \times \sin 60^\circ}{9.8}$$

$$R = 490\sqrt{3} \text{ m}$$

- 5** (a) Let at any instant of time, the length AB be l , here angle θ and length l vary with time, then using Pythagoras theorem in $\triangle ABC$,



$$x^2 + y^2 = l^2$$

On differentiating both sides w.r.t t , we get

$$\text{or } 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2l \frac{dl}{dt}$$

As, there is no vertical motion of the block, so

$$\frac{dy}{dt} = 0, \frac{dx}{dt} = v_x \text{ and } \frac{dl}{dt} = v$$

$$\therefore 2xv_x = 2lv$$

$$\text{or } v_x = \frac{l}{x}v \text{ or } v_x = \left(\frac{x}{l}\right)v = \frac{v}{\sin \theta}$$

- 6** (a) Horizontal range, $R = \frac{u^2 \sin 2\theta}{g}$

When $\theta = 45^\circ$, maximum horizontal range,

$$R_{\text{max}} = \frac{u^2}{g} \sin 90^\circ = \frac{u^2}{g}$$

When $\theta = 135^\circ$, maximum horizontal range,

$$R = \frac{u^2}{g} \sin 270^\circ = \frac{-u^2}{g}$$

Negative sign implies opposite direction.

- 7** (d) Given, $x = a \cos(pt)$, $y = b \sin(pt)$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ i.e. equation of ellipse}$$

$$\text{Now, } \mathbf{r} = x\hat{i} + y\hat{j} = a \cos(pt)\hat{i} + b \sin(pt)\hat{j}$$

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = -pa \sin(pt)\hat{i} + pb \cos(pt)\hat{j}$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = -p^2a \cos(pt)\hat{i} - p^2b \sin(pt)\hat{j}$$

$$\text{At } t = \frac{\pi}{2p},$$

$$\mathbf{v} = -p\hat{i} \text{ and } \mathbf{a} = -p^2\hat{j}$$

Thus, velocity is perpendicular to acceleration.

Also, $\mathbf{a} = -p^2\mathbf{b}$, i.e. directed towards a fixed point as p and b are positive constants.

$$\text{As, } v = \frac{dr}{dt} \Rightarrow \Delta r = \int_0^{\frac{\pi}{2p}} v dt$$

$$\Rightarrow \Delta r = [a \cos(pt)\hat{i} + b \sin(pt)\hat{j}]_0^{\frac{\pi}{2p}} = -a\hat{i} + b\hat{j}$$

$$\text{So, } |\mathbf{r}| = \sqrt{a^2 + b^2}$$

- 8** (b) Position vector of the particle is given by

$$\mathbf{r} = \cos \omega t \hat{x} + \sin \omega t \hat{y}$$

where, ω is a constant.

Velocity of the particle is

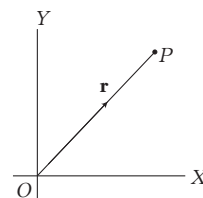
$$\begin{aligned} \mathbf{v} &= \frac{d\mathbf{r}}{dt} = \frac{d}{dt} (\cos \omega t \hat{x} + \sin \omega t \hat{y}) \\ &= (-\sin \omega t) \omega \hat{x} + (\cos \omega t) \omega \hat{y} \\ &= -\omega (\sin \omega t \hat{x} - \cos \omega t \hat{y}) \end{aligned}$$

Acceleration of the particle,

$$\begin{aligned} \mathbf{a} &= \frac{d\mathbf{v}}{dt} = \frac{d}{dt} (-\omega \sin \omega t \hat{x} + \omega \cos \omega t \hat{y}) \\ &= -\omega^2 \cos \omega t \hat{x} - \omega^2 \sin \omega t \hat{y} \end{aligned}$$

$$\Rightarrow \mathbf{a} = -\omega^2 \mathbf{r} = \omega^2(-\mathbf{r})$$

Assuming the particle is at P , then its position vector is directed as shown in the diagram.



Therefore, acceleration is directed towards $-\mathbf{r}$, i.e. towards O (origin).

$$\begin{aligned}\mathbf{v} \cdot \mathbf{r} &= -\omega(\sin \omega t \hat{\mathbf{x}} - \cos \omega t \hat{\mathbf{y}}) \cdot (\cos \omega t \hat{\mathbf{x}} + \sin \omega t \hat{\mathbf{y}}) \\ &= -\omega[\sin \omega t \cdot \cos \omega t + 0 + 0 - \sin \omega t \cdot \cos \omega t] \\ &= -\omega(0) = 0\end{aligned}$$

$$\Rightarrow \mathbf{v} \perp \mathbf{r}$$

Thus, velocity is perpendicular to $\mathbf{r}$.

- 9 (a) The equation of trajectory,

$$y = x \tan \alpha \left[1 - \frac{x}{R} \right] \quad (\text{Gives})$$

$$Q = P \tan \theta \left[1 - \frac{P}{R} \right] \quad \dots(i)$$

$$\text{and } P = Q \tan \theta \left[1 - \frac{Q}{R} \right] \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{Q^2}{P^2} = \frac{[1 - P/R]}{[1 - Q/R]}$$

$$\frac{1}{R}[P^3 - Q^3] = P^2 - Q^2$$

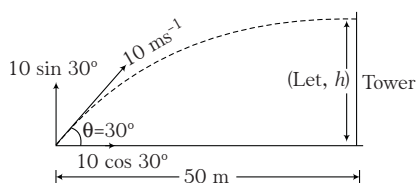
$$\Rightarrow R = \frac{P^3 - Q^3}{P^2 - Q^2} = \frac{P^2 + PQ + Q^2}{P + Q}$$

$$\begin{aligned}\text{Now, } \frac{Q}{P} &= \tan \theta \left[1 - \frac{P(P+Q)}{P^2 + PQ + Q^2} \right] \\ &= \tan \theta \left[\frac{P^2 + PQ + Q^2 - P^2 - PQ}{P^2 + PQ + Q^2} \right]\end{aligned}$$

$$\Rightarrow \tan \theta = \frac{P^2 + Q^2 + PQ}{PQ}$$

$$\Rightarrow \theta = \tan^{-1} \left[\frac{P^2 + PQ + Q^2}{PQ} \right]$$

- 10 (a) Consider the diagram



For horizontal motion, $50 = 10 \cos 30^\circ \times t$

$$\Rightarrow t = \frac{5}{\cos 30^\circ} = \frac{5}{\sqrt{3}/2} = \frac{10}{\sqrt{3}}$$

For vertical motion,

$$\begin{aligned}h &= 10 \sin 30^\circ \times \frac{10}{\sqrt{3}} - \frac{1}{2} \times 10 \times \left(\frac{10}{\sqrt{3}} \right)^2 \\ &= 10 \times \frac{1}{2} \times \frac{10}{\sqrt{3}} - \frac{100}{\sqrt{3}} \times \frac{5}{\sqrt{3}} = \frac{100}{\sqrt{3}} \left(\frac{1}{2} - \frac{5}{\sqrt{3}} \right) \\ &= \frac{50}{\sqrt{3}} \left[1 - \frac{10}{\sqrt{3}} \right] \text{ m}\end{aligned}$$

- 11 (b) Range of projectile is same for θ and $90^\circ - \theta$.

Given, $\theta = 40^\circ$

Another angle $= 90^\circ - \theta = 90^\circ - 40^\circ = 50^\circ$

- 12 (a) Given, a particle having horizontal range is twice the greatest height attained by it, i.e. $R = 2H$

$$\frac{v^2 \sin 2\theta}{g} = \frac{2v^2 \sin^2 \theta}{2g}$$

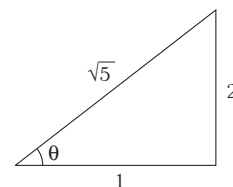
$$\Rightarrow 2 \sin \theta \cos \theta = \sin^2 \theta \Rightarrow \tan \theta = 2$$

$$\Rightarrow \sin \theta = \frac{2}{\sqrt{5}} \Rightarrow \cos \theta = \frac{1}{\sqrt{5}}$$

$$\therefore R = \frac{v^2 \sin 2\theta}{g} = \frac{v^2 \times 2 \sin \theta \cos \theta}{g}$$

$$\Rightarrow R = v^2 \times \frac{2}{g} \times \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}}$$

$$\text{i.e. } R = \frac{4v^2}{5g}$$



- 13 (d) Given, $\theta_1 = \theta_2 = 45^\circ$

$$\theta_1 + \theta_2 = (45^\circ + \alpha) + (45^\circ - \alpha) = 90^\circ$$

$$u_1 = u_2 = u \Rightarrow R_1 = R_2 = ?$$

$$R_1 = \frac{u_1^2 \sin(\theta_1 + \theta_2)}{g}$$

$$\Rightarrow R_1 = \frac{u^2 \sin 90^\circ}{g} = \frac{u^2}{g} \quad \dots(i)$$

$$\text{Similarly, } R_2 = \frac{u^2 \sin(\theta_1 + \theta_2)}{g}$$

$$\Rightarrow R_2 = \frac{u^2 \sin 90^\circ}{g} = \frac{u^2}{g} \quad \dots(ii)$$

The ratio of horizontal ranges

$$\frac{R_1}{R_2} = \frac{u^2/g}{u^2/g} = 1:1 \Rightarrow R_1 : R_2 = 1:1$$

- 14 (d) Average velocity, $\mathbf{v}_{av} = \frac{\text{Net displacement}}{\text{Time taken}}$

$$\begin{aligned}&= \frac{(13-2)\hat{\mathbf{i}} + (14-3)\hat{\mathbf{j}}}{5} \\ &= \frac{11\hat{\mathbf{i}} + 11\hat{\mathbf{j}}}{5} = \frac{11}{5}(\hat{\mathbf{i}} + \hat{\mathbf{j}})\end{aligned}$$

- 15 (c) For vertical motion, $h_1 = u \sin \theta t_1 - \frac{1}{2} g t_1^2$ (For h_1)

$$\Rightarrow t_1 = \frac{h_1 + \frac{1}{2} g t_1^2}{u \sin \theta} \quad \dots(i)$$

$$\Rightarrow h_2 = u \sin \theta t_2 - \frac{1}{2} g t_2^2 \quad (\text{For } h_2)$$

$$\Rightarrow t_2 = \frac{h_2 + \frac{1}{2} g t_2^2}{u \sin \theta} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{t_1}{t_2} = \frac{\left(h_1 + \frac{1}{2} g t_1^2 \right) / u \sin \theta}{\left(h_2 + \frac{1}{2} g t_2^2 \right) / u \sin \theta}$$

$$\Rightarrow h_{t_2} - h_{t_1} = \frac{g}{2} (t_2^2 - t_1^2)$$

The time of flight of the ball,

$$\begin{aligned} T &= \frac{2u \sin \theta}{g} = \frac{2}{g} (u \sin \theta) \\ &= \frac{2}{g} \left(\frac{h_1 + 1/2 g t_1^2}{t_1} \right) \quad [\text{From Eq. (i)}] \\ &= \frac{2}{g} \left(\frac{h_1}{t_1} + \frac{t_1}{2} \right) \\ &= \frac{h_1}{t_1} \times \frac{2}{g} + t_1 = \frac{h_1}{t_1} \times \left(\frac{t_2^2 - t_1^2}{h_{t_2} - h_{t_1}} \right) + t_1 \\ &= \frac{h_{t_2}^2 - h_{t_1}^2 + h_{t_1}^2 t_2 - h_{t_2}^3}{t_1 (h_{t_2} - h_{t_1})} = \left(\frac{h_{t_2}^2 - h_{t_1}^2}{h_{t_2} - h_{t_1}} \right) \end{aligned}$$

- 16 (c)** As we know that, maximum height of a projectile is given by

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$

where, u = initial velocity of projectile

g = acceleration due to gravity

and θ = angle of projection.

As per question,

$$H_{\max} = \frac{v^2 \sin^2 45^\circ}{2g} \quad \dots(i) \text{ (As, } u = v, \theta = 45^\circ)$$

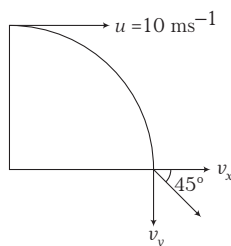
Now, range of a projectile is given by

$$\begin{aligned} R &= \frac{u^2 \sin 2\theta}{g} \\ \Rightarrow R &= \frac{v^2 \sin (2 \times 45^\circ)}{g} \\ \Rightarrow R &= \frac{v^2 \sin 90^\circ}{g} \quad \dots(ii) \end{aligned}$$

On dividing Eq. (i) by Eq. (ii), we get

$$\begin{aligned} \frac{H_{\max}}{R} &= \frac{v^2 \sin^2 45^\circ \times g}{2g \times v^2 \sin 90^\circ} = \frac{1}{4 \times 1} \\ \Rightarrow R &= 4H_{\max} = 4H \end{aligned}$$

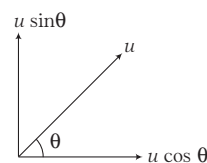
- 17 (d)** The horizontal component of velocity,
 $v_x = u = 10 \text{ ms}^{-1}$



$$\text{Now, } \tan \theta = \tan 45^\circ = \frac{v_y}{v_x} = \frac{v_y}{10}$$

$$\Rightarrow v_y = 10 \text{ ms}^{-1}$$

18 (b) As, $t = \frac{2u \sin \theta}{g} \Rightarrow 4 = \frac{2u \sin \theta}{g} \quad \dots(i)$



$$\Rightarrow u \sin \theta = 2g$$

and $h = \frac{u^2 \sin^2 \theta}{2g} = \frac{4g^2}{2g} \quad [\text{From Eq.(i)}]$

$$\Rightarrow h = 20 \text{ m}$$

- 19 (c)** From the question's figure, the x -component remain unchanged, while the y -component is reverse. Then, the velocity at point B is $(2\hat{i} - 3\hat{j}) \text{ ms}^{-1}$.

20 (b) Time of flight, $T = \frac{2u \sin \theta}{g}$

Given, $u = u_0$, $T = 1 \text{ s}$, $\theta = 30^\circ$, $g = 9.8 \text{ ms}^{-2}$

$$u_0 = g = 9.8 \text{ ms}^{-1}$$

- 21 (a)** As the velocity makes an angle of 30° with horizontal, so the horizontal component of velocity at the instant will be $v \cos 30^\circ$.

$$\Rightarrow v \cos 30^\circ = 5$$

$$\Rightarrow v = \frac{5}{\cos 30^\circ} = \frac{5}{\sqrt{3}/2} = \frac{10}{\sqrt{3}} \text{ ms}^{-1}$$

- 22 (a)** As the horizontal ranges are the same.

$$\therefore \frac{v_0^2 \sin 2\theta_1}{g} = \frac{v_0^2 \sin 2\theta_2}{g}$$

So, $\sin 2\theta_1 = \sin 2\theta_2$

or $2\theta_1 = \pi - 2\theta_2$

$$\Rightarrow \theta_1 + \theta_2 = \pi / 2$$

$$\therefore (h_1)_{\max} = \frac{v_0^2 \sin^2 \theta_1}{2g}$$

and $(h_2)_{\max} = \frac{v_0^2 \sin^2 \theta_2}{2g}$

$$\therefore \frac{(h_1)_{\max}}{(h_2)_{\max}} = \frac{\sin^2 \theta_1}{\sin^2 \theta_2} = \frac{\sin^2 \theta_1}{\cos^2 \theta_1} = \tan^2 \theta_1$$

$$\left[\because \sin^2 \theta_2 = \sin^2 \left(\frac{\pi}{2} - \theta_1 \right) = \cos^2 \theta_1 \right]$$

- 23 (a)** Given, $\mathbf{r} = 2t \hat{i} + t^2 \hat{j}$

Velocity vector, $\mathbf{v} = \frac{d\mathbf{r}}{dt}$ and using $\frac{d}{dx} x^n = nx^{n-1}$

We have, $\frac{d\mathbf{r}}{dt} = 2\hat{i} + 2t\hat{j}$

- 24 (b)** Given, $R = H$

$$\text{Range, } R = \frac{u^2 (2 \sin \theta \cos \theta)}{g}$$

$$\text{Height, } H = \frac{u^2 \sin^2 \theta}{2g}$$

$$\text{Hence, } \frac{u^2(2 \sin \theta \cos \theta)}{g} = \frac{u^2 \sin^2 \theta}{2g}$$

$$2 \cos \theta = \frac{\sin \theta}{2}$$

$$\Rightarrow \tan \theta = 4 \Rightarrow \theta = \tan^{-1}(4)$$

- 25** (d) Maximum height and time of flight depends upon the vertical component of initial velocity.

$$H_1 > H_2$$

$$\Rightarrow u_{y1} > u_{y2}$$

$$\text{Range } R_2 > R_1$$

$$\text{So, } u_2 > u_1$$

- 26** (d) According to given condition,

$$\frac{h_1}{h_2} = \frac{\sin^2 \theta}{\sin^2(90^\circ - \theta)} = \frac{20}{80}$$

$$\Rightarrow \tan^2 \theta = \frac{1}{4} \Rightarrow \tan \theta = \frac{1}{2}$$

$$\sin \theta = \frac{1}{\sqrt{5}} \text{ and } \cos \theta = \frac{2}{\sqrt{5}}$$

$$\text{So, } h = \frac{u^2 \sin^2 \theta}{2g}$$

$$\Rightarrow 20 = \frac{u^2}{10g} \text{ or } \frac{u^2}{g} = 200$$

$$\therefore \text{ The range, } R = \frac{u^2 \times 2 \sin \theta \cos \theta}{g} = \frac{200 \times 2 \times 2}{\sqrt{5} \times \sqrt{5}} \\ = \frac{200 \times 4}{5} = 160 \text{ m}$$

- 27** (b) Since, the ball reaches from one player to another in 2 s, so the time period of the flight, $T = 2$ s

$$\Rightarrow \frac{2u \sin \theta}{g} = 2 \text{ s}$$

Here, u is the initial velocity and θ is the angle of projection.

$$\Rightarrow u \sin \theta = g \quad \dots(i)$$

Now, we know that the maximum height of the projection,

$$H = \frac{u^2 \sin^2 \theta}{2g} \text{ or } H = \frac{(u \sin \theta)^2}{2g}$$

On putting the value of $u \sin \theta$ from Eq. (i), we get

$$H = \frac{g^2}{2g} = \frac{g}{2} \text{ or } H = \frac{g}{2} = \frac{10}{2} \text{ m or } H = 5 \text{ m}$$

CHAPTER 05

Laws of Motion

We normally observe around us, a number of objects or bodies at rest or in motion and find that, the objects at rest do not move by themselves or the objects in motion do not come to rest by themselves but they require some external force to do so. e.g. To move a book kept on a table, we need to push or pull it, or to stop a vehicle in motion, breaks are required.

The factor which is necessary for causing motion or change in motion is termed as force. This cause of motion (force) and effects of motion are governed by Newton's laws of motion.

In this chapter, we will discuss the motion of a body by taking into consideration the cause of motion, *i.e.* the external force which produces the motion or change the motion.

FORCE

Force is an effort in the form of push or pull causing or tending to cause motion, change in motion or deformations in a body.

There are basically two types of forces which are commonly observed

- (i) **Distant forces** (Field forces) The forces acting between two or more objects, which do not require the physical contact between the objects are called distant forces or field forces. Gravitational force between two bodies, electrostatic force between two charges, weak forces and nuclear forces are examples of distant forces.

Weight ($w = mg$) of a body also comes in this category.

- (ii) **Contact forces** The forces acting between two or more objects, which require the physical contact between the objects are called contact forces. Friction force, normal reaction, tension, spring force, etc., are some examples of contact forces.

Inertia

The term inertia means resistance or opposition to the change of state. It is defined as the inherent property of a body by virtue of which it remains in its state of rest or of uniform motion in a straight line. This term was first used by Galileo.

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There are three types of inertia

- (i) **Inertia of rest** It is defined as the tendency of a body to remain in its position of rest. *i.e.* A body at rest remains at rest and cannot start moving on its own.
- (ii) **Inertia of motion** It is defined as the tendency of a body to remain in its state of uniform motion along a straight line. *i.e.* A body in uniform motion can neither gets accelerated nor get retarded on its own, also it cannot stop on its own.
- (iii) **Inertia of direction** It is defined as inability of a body to change by itself its direction of motion.

Relation between mass and inertia

Mass of a body is the measurement of its inertia. A body with greater mass shows greater inertia, *i.e.* it is more difficult to change its state of rest or uniform motion as compared to that of a body having smaller mass. Quantitatively, the inertia of an object is measured by its mass. Thus, the SI unit for mass as well as inertia is kilogram (kg), whereas the units in the CGS system and in the British Imperial system for mass or inertia are gram (g) and slug (sl), respectively.

Momentum

Momentum of a body is the quantity of motion possessed by a moving body. It is measured as the product of mass and velocity of a body. It is represented by p .

$$i.e. \quad \text{Momentum } (p) = \text{Mass } (m) \times \text{Velocity } (v)$$

SI unit of momentum is $\text{kg}\cdot\text{ms}^{-1}$ and CGS unit of momentum is $\text{g}\cdot\text{cms}^{-1}$.

The dimensional formula of momentum is $[\text{MLT}^{-1}]$. It is a vector quantity.

NEWTON'S LAWS OF MOTION

Newton's laws of motion are the three physical laws that, together laid the foundation for classical mechanics. These three laws of motion were first proposed by Sir Isaac Newton.

Newton's 1st law of motion

This law states that, "every body continues in its state of rest or of uniform motion in a straight line unless it is compelled by some external force to change its state." Thus, it can be concluded that if the net external force on a body is zero, its acceleration is zero. Acceleration can be non-zero only if there is a net external force on the body. Newton's first law defines force qualitatively.

Newton's 2nd law of motion

This law states that, "the rate of change of momentum of a body is directly proportional to the external force applied on the body and the change takes place in the direction of the applied force."

We may also state Newton's second law of motion as

"If the unbalanced external force (net force) acts on a body, the body accelerates. The direction of acceleration is the same as the direction of the net force."

Calculating force with the help of Newton's 2nd law

Let $\mathbf{F}$ be external force applied on the body in the direction of motion of the body for time interval Δt , then the velocity of a body of mass m changes from $\mathbf{v}$ to $\mathbf{v} + \Delta \mathbf{v}$, *i.e.* change in momentum, $\Delta \mathbf{p} = m\Delta \mathbf{v}$.

According to Newton's second law,

$$\mathbf{F} \propto \frac{\Delta \mathbf{p}}{\Delta t} \quad \text{or} \quad \mathbf{F} = k \frac{\Delta \mathbf{p}}{\Delta t}$$

where, k is a constant of proportionality.

If limit $\Delta t \rightarrow 0$, then the term $\frac{\Delta \mathbf{p}}{\Delta t}$ becomes the derivative $\frac{d\mathbf{p}}{dt}$. Thus, $\mathbf{F} = k \frac{d\mathbf{p}}{dt}$

For a body of fixed mass m , we have

$$\mathbf{F} = k \frac{d(m\mathbf{v})}{dt} = km \frac{d\mathbf{v}}{dt} \Rightarrow \mathbf{F} = kma$$

Now, **a unit force may be defined as the force which produces unit acceleration in a body of unit mass.**

$$\text{So,} \quad F = 1, m = 1, a = 1 \Rightarrow k = 1$$

$$\text{So,} \quad \boxed{\text{Force, } \mathbf{F} = m\mathbf{a}}$$

In scalar form, this equation can be written as, $F = ma$.

If v is fixed and m is variable, then $F = kd \frac{(mv)}{dt} = \frac{kvd m}{dt}$

$$\therefore \quad k = 1, \text{ then } F = vdm/dt$$

The force is a vector quantity.

Note The slope of momentum-time graph is equal to force on the particle, e.g.

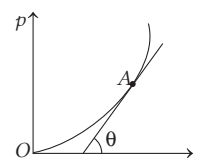


Fig. 5.1 Momentum-time graph

$$\text{At point A, } F = \frac{dp}{dt} = \text{slope} = \tan \theta$$

This graph depicts the motion of a body on which increasing force is acting.

Units of force

- In SI system, absolute unit of force is newton.
One newton is defined as that much force which produces an acceleration of 1 ms^{-2} in a body of mass 1 kg .

$$1 \text{ N} = 1 \text{ kg} \times 1 \text{ ms}^{-2}$$

$$1 \text{ N} = 1 \text{ kg ms}^{-2}$$

- In CGS system, absolute unit of force is dyne.
One dyne is that much of force which produces an acceleration of 1 cms^{-2} in a body of mass 1 g .

$$1 \text{ dyne} = 1 \text{ g} \times 1 \text{ cms}^{-2}$$

$$1 \text{ dyne} = 1 \text{ g cms}^{-2}$$

- In MKS system, gravitational unit of force is kilogram weight (kg-wt). One kg-wt is that much of force which produces an acceleration of 9.80 ms^{-2} in a body of mass 1 kg . It is also known as kilogram-force (kgf).

$$1 \text{ kg-wt} = 1 \text{ kgf} = 9.8 \text{ N}$$

- In CGS system, gravitational unit of force is gram weight (g-wt) or gram force (gf). It is defined as that force which produces an acceleration of 980 cms^{-2} in a body of mass 1 g .

$$1 \text{ g-wt} = 1 \text{ gf} = 980 \text{ g cm s}^{-2} \text{ or } 1 \text{ gf} = 980 \text{ dyne}$$

Relation between newton and dyne

$$1 \text{ N} = 1 \text{ kg} \times 1 \text{ ms}^{-2} = 1000 \text{ g} \times 100 \text{ cms}^{-2}$$

$$= 10^5 \text{ g cms}^{-2} \quad (1 \text{ dyne} = 1 \text{ g cms}^{-2})$$

$$1 \text{ N} = 10^5 \text{ dyne}$$

Example 5.1 If an electron is subjected to a force of 10^{-25} N in an X-ray machine, then find out the time taken by the electron to cover a distance of 0.2 m .

(Take, mass of an electron = 10^{-30} kg)

Sol. The acceleration of the electron, $a = \frac{F}{m} = \frac{10^{-25}}{10^{-30}} = 10^5 \text{ ms}^{-2}$

The time taken by the electron (t) to cover the distance (s) of 0.2 m can be given by

$$s = ut + \frac{1}{2} at^2$$

$$\Rightarrow 0.2 = 0 + \frac{1}{2} \times 10^5 \times t^2$$

$$\Rightarrow t^2 = 0.4 \times 10^{-5} = 4 \times 10^{-6} \Rightarrow t = 2 \times 10^{-3} \text{ s}$$

Example 5.2 A bullet of mass 0.06 kg moving with a speed of 90 ms^{-1} enters a heavy wooden block and is stopped after a distance of 60 cm . What is the average resistive force exerted by the block on the bullet?

Sol. The retardation a of the bullet is given by

$$a = -\frac{u^2}{2s} = \frac{-90 \times 90}{2 \times 0.6} \quad (\text{From, } v^2 = u^2 + 2as, v = 0)$$

$$(\because \text{Given, } s = 60 \text{ cm} = 0.6 \text{ m, } u = 90 \text{ ms}^{-1})$$

$$\Rightarrow a = -6750 \text{ ms}^{-2}$$

From second law of motion,

$$\text{Retarding force, } F = ma = 0.06 \times 6750$$

$$\Rightarrow F = 405 \text{ N}$$

Example 5.3 A stone of mass 1 kg is thrown with a velocity of 20 ms^{-1} across the frozen surface of a lake and it comes to rest after travelling a distance of 50 m . What is the magnitude of the force opposing the motion of the stone?

Sol. Given, $u = 20 \text{ ms}^{-1}$, $v = 0$, $s = 50 \text{ m}$ and $m = 1 \text{ kg}$

To calculate force, we have the formula $F = ma$, but we have to first calculate acceleration a .

Using the third equation of motion, i.e.

$$v^2 = u^2 + 2as$$

$$(0)^2 = (20)^2 + 2 \times a \times 50$$

$$\Rightarrow 100a = -400$$

$$\Rightarrow a = -4 \text{ ms}^{-2}$$

Acceleration $a = -4 \text{ ms}^{-2}$ (– ve sign shows that speed of the stone decreases, i.e. retardation)

$$\text{Now, } F = ma = (1 \text{ kg}) \times (-4 \text{ ms}^{-2})$$

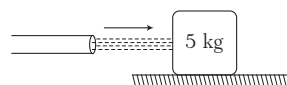
$$= -4 \text{ kg} \cdot \text{ms}^{-2} = -4 \text{ N}$$

Thus, force of opposition between the stone and the ice is -4 N .

The negative value of force shows that the opposing force acts in a direction opposite to the direction of motion.

Example 5.4 A block of 5 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 3 kgs^{-1} at a speed of 4 ms^{-1} . Calculate the acceleration of the block.

Sol. Force exerting on block, $F = v \frac{dm}{dt} = 4 \times 3 = 12 \text{ N}$



$$\text{So, acceleration of the block, } a = \frac{F}{m} = \frac{12}{5} = 2.4 \text{ ms}^{-2}$$

Resolution of force into different components

If the force applied, imparts acceleration a to a body, then its components in X , Y and Z -axis are $a = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$.

Components of force will be F_x , F_y and F_z

As,

$$F = ma$$

$$\Rightarrow F_x \hat{i} + F_y \hat{j} + F_z \hat{k} = m(a_x \hat{i} + a_y \hat{j} + a_z \hat{k})$$

$$\text{Thus, } F_x = ma_x = m \frac{dv_x}{dt} = m \frac{d^2x}{dt^2} = \frac{dp_x}{dt}$$

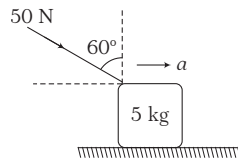
$$F_y = ma_y = m \frac{dv_y}{dt} = m \frac{d^2y}{dt^2} = \frac{dp_y}{dt}$$

$$F_z = ma_z = m \frac{dv_z}{dt} = m \frac{d^2z}{dt^2} = \frac{dp_z}{dt}$$

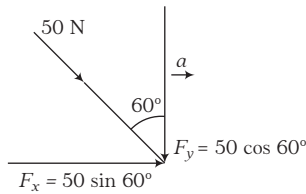
The component form of Newton's second law tells that if the applied force makes some angle with the velocity of the body, it changes the component of velocity along the direction of force.

The component of velocity normal to the force remains unchanged.

Example 5.5 A force of 50 N acts in the direction as shown in figure. The block of mass 5 kg, resting on a smooth horizontal surface. Find out the acceleration of the block.



Sol. Free body diagram



where, F_x = horizontal component of the force and F_y = vertical component of the force.

Horizontal component of the force = $50 \sin 60^\circ = \frac{50\sqrt{3}}{2}$ N

Acceleration of the block, a

$$= \frac{\text{Component of force in the direction of acceleration}}{\text{Mass}}$$

$$= \frac{50\sqrt{3}}{2} \times \frac{1}{5} = 5\sqrt{3} \text{ ms}^{-2}$$

Example 5.6 A force $\mathbf{F} = (6\hat{i} - 8\hat{j} + 10\hat{k})$ N produces acceleration of $\sqrt{2} \text{ ms}^{-2}$ in a body. Calculate the mass of the body.

Sol. $\therefore$ Acceleration, $a = \frac{|\mathbf{F}|}{m}$

$$\Rightarrow \text{Mass, } m = \frac{|\mathbf{F}|}{a} = \frac{\sqrt{6^2 + 8^2 + 10^2}}{\sqrt{2}} = 10 \text{ kg}$$

Example 5.7 A force $\mathbf{F} = (2t\hat{i} + 3t^2\hat{j})$ N acts on an object moving in XY-plane. Find magnitude of change in momentum of the object in time interval $t = 0$ to $t = 2$ s.

Sol. Given, $\mathbf{F} = (2t\hat{i} + 3t^2\hat{j})$ N $\Rightarrow \frac{d\mathbf{p}}{dt} = 2\hat{i} + 3t^2\hat{j}$

$$\Rightarrow d\mathbf{p} = 2t d\hat{i} + 3t^2 d\hat{j}$$

$$\Rightarrow \int d\mathbf{p} = 2 \int_0^2 t d\hat{i} + 3 \int_0^2 t^2 d\hat{j}$$

$$\Rightarrow \Delta\mathbf{p} = [t^2]_0^2 \hat{i} + [t^3]_0^2 \hat{j}$$

$$\Rightarrow \Delta\mathbf{p} = 4\hat{i} + 8\hat{j}$$

$$\Rightarrow |\Delta\mathbf{p}| = \sqrt{16 + 64} = \sqrt{80} \approx 9 \text{ kg ms}^{-1}$$

Resultant force

When two or more forces act on a body simultaneously, then the single force which produces the same effect as produced by all the forces acting together is known as the resultant force.

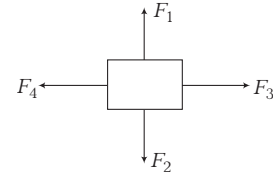


Fig. 5.2

Resultant force, $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4$

Balanced force

When a number of forces acting simultaneously on a body do not bring about any change in its state of rest or of uniform motion along a straight line, then the forces acting on the body are said to be balanced forces.

In other words, when different forces acting on a body give a zero resultant, then the forces are said to be balanced. Balanced forces do not produce any acceleration.

e.g. When two opposite forces having the same magnitude F act on a block placed on a smooth horizontal table, they fail to move the block.

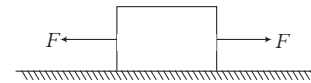


Fig. 5.3

This is because the net force is equal to zero. Similarly, two opposite forces having the same magnitude cannot change the speed of a moving body.

Unbalanced force

When a number of forces acting simultaneously on a body bring about a change in its state of rest or of uniform motion along a straight line, then these forces acting on the body are said to be unbalanced forces.

In this case, different forces acting on a body do not give zero resultant. If an unbalanced force is applied on the object, there will be a change either in its speed/velocity or in the direction of its motion.

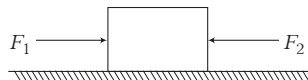
Thus, to accelerate an object, an unbalanced force is required. When net force on the body is not equal to zero, then the body at rest starts moving in the direction of resultant force.



Fig. 5.4

If $F_2 > F_1$, then $F_2 - F_1 > 0$ and as a result, the car accelerates in the direction of F_2 .

Example 5.8 Let us consider two forces F_1 and F_2 acting on a body of mass 2 kg as shown in the figure. $F_1 = 10\text{ N}$, $F_2 = 2\text{ N}$, what will be the acceleration?

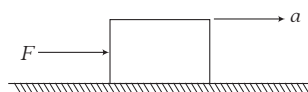


Sol. Unbalanced external force, $F = F_1 - F_2 = 10 - 2 = 8\text{ N}$

So,

$$F = ma$$

$$\Rightarrow \text{Acceleration, } a = \frac{F}{m} = \frac{8}{2} = 4\text{ ms}^{-2}$$



The body moves in direction of F_1 .

Impulse

When a large force acts on a body for very small time, then product of the average of total force for that small time period and the time period itself is called impulse.

$$\text{Impulse} = \text{Average force} \times \text{Time}$$

$$I = F\Delta t \quad \text{or} \quad \mathbf{I} = \int_{t_1}^{t_2} \mathbf{F} dt$$

It is a vector quantity and its direction is same as that of force. Dimensional formula of impulse is same as that of momentum, i.e. $[\text{MLT}^{-1}]$.

SI unit of impulse is N-s or kg ms^{-1} and CGS unit of it is g-cms^{-1} .

Relation between momentum and impulse

Suppose $\mathbf{F}$ is the value of force during impact at any time and $\mathbf{p}$ is the momentum of the body at that time, then according to Newton's IInd law of motion,

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} \quad \text{or} \quad \mathbf{F} \cdot dt = d\mathbf{p} \quad \dots(i)$$

Suppose that the impact lasts for a small time t and during this time, the momentum of the body changes from p_1 to p_2 . Then, integrating the above equation, we get

$$\int_0^t \mathbf{F} dt = \int_{p_1}^{p_2} d\mathbf{p} = \mathbf{p} \Big|_{p_1}^{p_2} \Rightarrow \underbrace{\int_0^t \mathbf{F} dt}_{\text{Impulse}} = p_2 - p_1$$

From this equation, we found that **impulse is equal to change in momentum of the body.**

Also, if $\mathbf{F}_{\text{av}}$ is the average force (constant) during the impact, then

$$\text{Impulse, } \mathbf{I} = \int_0^t \mathbf{F}_{\text{av}} dt = \mathbf{F}_{\text{av}} \int_0^t dt = \mathbf{p}_2 - \mathbf{p}_1$$

$$\text{or} \quad \mathbf{I} = \mathbf{F}_{\text{av}} \cdot t = \mathbf{p}_2 - \mathbf{p}_1 = \Delta \mathbf{p}$$

Thus, impulse is also equal to total change in momentum. This is known as **impulse-momentum theorem.**

Calculation of impulse: graphical method

- (i) **When applied force is constant**, then the graph between this force and the time of application of this force is a straight line parallel to time axis.

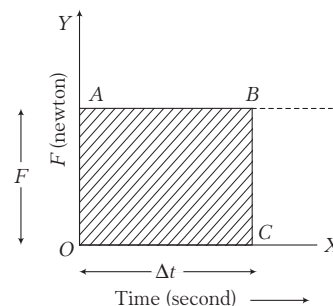


Fig. 5.5 Impulse of a constant force

Here, impulse is given by the area covered by the graph.

$$\text{i.e.} \quad I = F \times \Delta t$$

where, $F = OA$, $\Delta t = OC$

$$\therefore I = OA \times OC = \text{Area of rectangle } OABC.$$

- (ii) **When applied force is variable** for the time of application (Δt), then graph between force and time will be a curve as given in the figure below

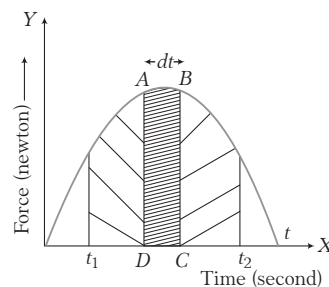


Fig. 5.6 Impulse of a variable force

Here, impulse = force $\times$ time

$$= F \times dt$$

$$= \text{Area of shaded region}$$

Total impulse for the force applied during period from t_1 to t_2

$$= \int_{t_1}^{t_2} F \cdot dt$$

$$= \text{Area under } F-t \text{ curve from } t_1 \text{ to } t_2$$

$$\text{Total impulse for the force applied} \\ = \text{Area covered between the curve and time-axis}$$

Example 5.9 A baseball player hits back the ball straight in the direction of the bowler without changing its initial speed of 12 ms^{-1} . If the mass of the ball is 0.15 kg , then find the impulse imparted to the ball. (Consider the ball in linear motion)

Sol. Given, $m = 0.15 \text{ kg}$, $v = 12 \text{ ms}^{-1}$ and $u = -12 \text{ ms}^{-1}$

Change in momentum,

$$p_2 - p_1 = m(v - u) = 0.15 [12 - (-12)] = 0.15 \times 24$$

$$p_2 - p_1 = 3.60 \text{ kg} \cdot \text{ms}^{-1}$$

Impulse, $I = p_2 - p_1 \Rightarrow I = 3.6 \text{ N} \cdot \text{s}$

Example 5.10 A hammer of mass 1 kg moving with a speed of 6 ms^{-1} strikes a wall and comes to rest in 0.1 s . Calculate

- impulse of the force,
- average retarding force that stops the hammer
- and average retardation of the hammer.

Sol. (i) Impulse $= F \times t = m(v - u) = 1(0 - 6) = -6 \text{ N} \cdot \text{s}$

(ii) Average retarding force that stops the hammer,

$$F = \frac{\text{Impulse}}{\text{Time}} = \frac{6}{0.1} = 60 \text{ N}$$

(iii) Average retardation, $a = \frac{F}{m} = \frac{60}{1} = 60 \text{ ms}^{-2}$

Example 5.11 A cricket ball of mass 150 g is moving with a velocity of 12 ms^{-1} and is hit by a bat, so that the ball is turned back with a velocity of 20 ms^{-1} . If the duration of contact between the ball and bat is 0.01 s , find the impulse and the average force exerted on the ball by the bat.

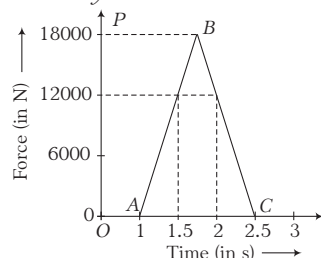
Sol. According to given question, change in momentum of the ball,

$$\Delta p = p_f - p_i = m(v - u) = 150 \times 10^{-3} [20 - (-12)] = 4.8 \text{ N} \cdot \text{s}$$

So, by impulse-momentum theorem, impulse, $I = \Delta p = 4.8 \text{ N} \cdot \text{s}$ and by time averaged definition of force in case of impulse

$$\Rightarrow F_{\text{av}} = \frac{I}{\Delta t} = \frac{\Delta p}{\Delta t} = \frac{4.80}{0.01} = 480 \text{ N}$$

Example 5.12 Figure shows an estimated force-time graph for a baseball struck by a bat.



From this curve, determine

- impulse delivered to the ball
- and average force exerted on the ball.

Sol. (i) Impulse = Area under F - t curve

$$= \text{Area of } \triangle ABC = \frac{1}{2} \times OP \times AC = \frac{1}{2} \times 18000 \times (2.5 - 1)$$

$$= 1.35 \times 10^4 \text{ kg} \cdot \text{ms}^{-1} \text{ or N} \cdot \text{s}$$

(ii) Average force $= \frac{\text{Impulse}}{\text{Time}} = \frac{1.35 \times 10^4}{(2.5 - 1)} = 9000 \text{ N}$

Newton's IIIrd law of motion

This law states that, “to every action, there is always an equal and opposite reaction or the mutual actions of two bodies upon each other are always directed to contrary parts.”

Analysis of forces

From Newton's IIIrd law, it can be analysed that we cannot produce a single isolated force in nature. Thus, forces occur in equal and opposite pairs. Whenever object A exerts a force on object B , object B must also exert a force on object A . The two forces are equal in magnitude and opposite in direction.

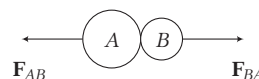


Fig. 5.7 Forces acting on bodies A and B

As shown in figure, if F_{BA} is the force exerted by body A on B and F_{AB} is the force exerted by B on A , then according to Newton's third law,

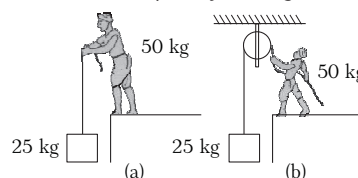
$$F_{AB} = -F_{BA}$$

Force on A by $B = -$ Force on B by A

Important features of Newton's IIIrd law of motion

- Newton's third law of motion is applicable irrespective of the nature of the forces** The forces of action and reaction may be mechanical, gravitational, electric or of any other nature.
- Action and reaction always act on two different bodies** If they act on the same body, the resultant force would be zero and there could never be accelerated motion.
- The force of action and reaction cannot cancel each other** This is because action and reaction, though equal and opposite forces always act on different bodies and so cannot cancel each other.
- No action can occur in the absence of a reaction** In a tug of war, one team can pull the rope only if the other team is pulling the other end of the rope; no force can be exerted, if the other end is free. One team exerts the force of action and the other team provides the force of reaction.

Example 5.13 A block of mass 25 kg is raised by a 50 kg man in two different ways as shown in figure. What is the action on the floor by the man in the two cases? If the floor yields to a normal force of 700 N , which mode should the man adopt to lift the block without the floor yielding? (Take, $g = 9.8 \text{ ms}^{-2}$)



Sol. In mode (a), the man applies a force equal to 25 kg weight in upward direction. According to Newton's third law of motion, there will be a downward force of reaction on the floor.

$$\begin{aligned}\therefore \text{Total action on the floor by the man} \\ &= 50 \text{ kg-wt} + 25 \text{ kg-wt} \\ &= 75 \text{ kg-wt} \\ &= 75 \times 9.8 \text{ N} = 735 \text{ N}\end{aligned}$$

In mode (b), the man applies a downward force on rope equal to 25 kg-wt. According to Newton's third law, the reaction will be in the upward direction by the rope on the man, so he becomes light by 25 kg-wt.

$$\begin{aligned}\therefore \text{Total action on the floor by the man} \\ &= 50 \text{ kg-wt} - 25 \text{ kg-wt} \\ &= 25 \text{ kg-wt} \\ &= 25 \times 9.8 \text{ N} = 245 \text{ N}\end{aligned}$$

As the floor yields to a downward force of 700 N, so the man should adopt mode (b).

LAW OF CONSERVATION OF LINEAR MOMENTUM

The total momentum of an isolated system (a system having no external force acting on it) of constant mass remains constant or conserved and does not change with time. If the momentum of two particles system of masses m_1 and m_2 are p_1 and p_2 respectively, then the net momentum of whole system is given by

$$p = p_1 + p_2 = \text{constant}$$

This principle is a consequence of Newton's second and third law of motion.

Conservation of linear momentum for a system of two or more particles

Force applied on particle 1 by particle 2 is F_{12} and force applied on particle 2 by particle 1 is F_{21} and their respective momentum are p_1 and p_2 .

From Newton's IInd law,

$$F_{12} = \frac{dp_1}{dt} \quad \text{and} \quad F_{21} = \frac{dp_2}{dt}$$

From Newton's IIIrd law,

$$F_{12} = -F_{21} \Rightarrow \frac{dp_1}{dt} = -\frac{dp_2}{dt}$$

$$\text{or} \quad \frac{dp_1}{dt} + \frac{dp_2}{dt} = 0 \Rightarrow \frac{d}{dt}(p_1 + p_2) = 0$$

Thus, $p_1 + p_2 = \text{constant}$

For n number of particles,

$$p_1 + p_2 + p_3 + \dots + p_n = \text{constant}$$

Conservation of linear momentum for the collision of two bodies

- (i) **Head-on collision** (collision in a straight line) Two bodies of masses m_1 and m_2 collide on frictionless surface moving in the same direction with respective velocities u_1 and u_2 . After collision, both the bodies separate with a variation in their velocities, i.e. v_1 and v_2 , respectively.

Initial momentum (before collision),

$$p_{1(\text{initial})} = m_1 u_1, \quad p_{2(\text{initial})} = m_2 u_2$$

Final momentum (after collision),

$$p_{1(\text{final})} = m_1 v_1, \quad p_{2(\text{final})} = m_2 v_2$$

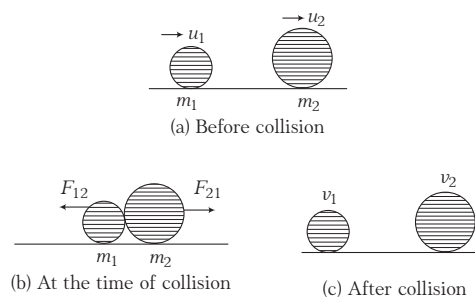


Fig. 5.8

During collision, particle 1 exerts a force F_{21} on particle 2 and simultaneously particle 2 exerts a force F_{12} on particle 1.

$$\begin{aligned}F_{12} &= \text{rate of change of momentum of particle 1} \\ &= \frac{m_1 v_1 - m_1 u_1}{t} = \frac{m_1 (v_1 - u_1)}{t}\end{aligned}$$

Similarly, F_{21} = rate of change of momentum of particle 2

$$= \frac{m_2 v_2 - m_2 u_2}{t} = \frac{m_2 (v_2 - u_2)}{t}$$

According to Newton's IIIrd law of motion,

$$\begin{aligned}F_{12} &= -F_{21} \\ \Rightarrow \frac{m_1 (v_1 - u_1)}{t} &= -\frac{m_2 (v_2 - u_2)}{t}\end{aligned}$$

$$\text{or} \quad m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

i.e. **Total momentum before collision remains same as total momentum after collision.**

- (ii) **Oblique collision** (collision of ball with wall) A ball of mass m strikes a wall with velocity u at an angle θ from the normal of wall and rebounds with the same speed in time t .

Here, initial momentum of the particle,

$$p_i = mu \cos \theta \hat{i} - mu \sin \theta \hat{j}$$

Final momentum of the particle,

$$\mathbf{p}_f = -mu \cos \theta \hat{\mathbf{i}} - mu \sin \theta \hat{\mathbf{j}}$$

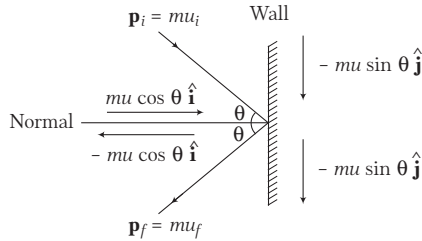


Fig. 5.9 Collision of ball with wall

Now, change in momentum, $\Delta \mathbf{p} = \mathbf{p}_f - \mathbf{p}_i$

$$\begin{aligned} &= -mu \cos \theta \hat{\mathbf{i}} - mu \sin \theta \hat{\mathbf{j}} - mu \cos \theta \hat{\mathbf{i}} + mu \sin \theta \hat{\mathbf{j}} \\ &= -2mu \cos \theta \hat{\mathbf{i}} \end{aligned}$$

It means that momentum changes only along the normal to the wall but not along the wall. This is so because force is acting on the ball only normal to the wall (this force is reaction force of wall) and no force acts parallel to the wall.

$$|\Delta \mathbf{p}| = 2mu \cos \theta$$

$$\begin{aligned} \text{Force, } \mathbf{F} &= \frac{\Delta \mathbf{p}}{\Delta t} = \frac{-2mu \cos \theta}{t} \hat{\mathbf{i}} \\ |\mathbf{F}| &= \frac{2mu \cos \theta}{t} \end{aligned}$$

There are two possible cases

Case I If $\theta = 0^\circ$, means ball is thrown perpendicular to the wall, then $\Delta \mathbf{p} = -2mu \cos 0^\circ \hat{\mathbf{i}}$

$$\Rightarrow \Delta \mathbf{p} = -2mu \hat{\mathbf{i}}$$

$$\therefore \mathbf{F} = \frac{\Delta \mathbf{p}}{\Delta t} = \frac{-2mu}{t} \hat{\mathbf{i}}$$

Case II If θ is the angle measured from the wall, then

$$\Delta \mathbf{p} = -2mu \sin \theta \hat{\mathbf{i}} \Rightarrow |\Delta \mathbf{p}| = 2mu \sin \theta$$

$$\mathbf{F} = \frac{-2mu \sin \theta}{t} \hat{\mathbf{i}} \Rightarrow |\mathbf{F}| = \frac{2mu \sin \theta}{t}$$

Newton's IIIrd law can be derived from principle of conservation of linear momentum

If two particles of masses m_1 and m_2 are moving under action of their mutually interacting forces with each other, such that no external force acts on the system. Then, momentum of system remains constant.

$$\text{i.e. } \Delta \mathbf{p}_1 + \Delta \mathbf{p}_2 = 0 \Rightarrow \Delta \mathbf{p}_1 = -\Delta \mathbf{p}_2 \Rightarrow \frac{\Delta \mathbf{p}_1}{\Delta t} = -\frac{\Delta \mathbf{p}_2}{\Delta t}$$

$$\Rightarrow \mathbf{F}_{12} = -\mathbf{F}_{21}$$

Force on 1st due to 2nd = - Force on 2nd due to 1st

Example 5.14 A bullet of mass 10 g is fired from a gun of mass 1 kg. If the recoil velocity is 5 ms^{-1} . Find the velocity of the muzzle.

Sol. From the law of conservation of momentum,

$$m_G v_G = m_B v_B$$

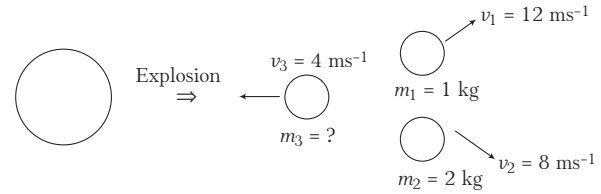
where, m_G, v_G = mass and velocity of gun

m_B, v_B = mass and velocity of bullet

$$\Rightarrow v_B = \frac{m_G v_G}{m_B} = \frac{1 \times 5}{10 \times 10^{-3}} = 500 \text{ ms}^{-1}$$

Example 5.15 On a mine site a rock is exploded. On explosion, rock breaks into three parts. Two parts go off at right angles to each other. Of these two, 1 kg first part is moving with a velocity of 12 ms^{-1} and 2 kg second part is moving with a velocity of 8 ms^{-1} . If the third part flies off with a velocity 4 ms^{-1} , what will be its mass?

Sol.



From the law of conservation of momentum,

$$p_3 = \sqrt{p_1^2 + p_2^2}$$

$$\Rightarrow m_3 \times 4 = \sqrt{(1 \times 12)^2 + (2 \times 8)^2} = 20$$

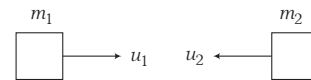
$$\Rightarrow m_3 = 5 \text{ kg}$$

Example 5.16 Two objects each of mass 5 kg are moving in the same straight line but in the opposite directions towards each other with same speed of 3 m/s. They stick to each other after collision. What will be the velocity of the combined object after collision?

Sol. Given, $m_1 = m_2 = 5 \text{ kg}$,

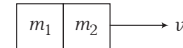
$$u_1 = 3 \text{ m/s}, u_2 = -3 \text{ m/s}$$

Before collision,



Let the velocity of the combined object be v .

Then, after collision,



Total momentum of the system before collision is

$$m_1 u_1 + m_2 u_2 = 5 \times 3 + 5 \times (-3) = 0$$

Total momentum of the system after collision is

$$m_1 v + m_2 v = (m_1 + m_2) v = (5 + 5) v = 10 v$$

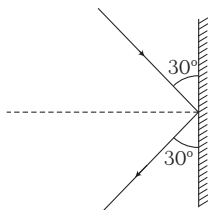
According to the law of conservation of momentum,

Momentum before collision = Momentum after collision

$$\therefore 0 = 10 v \Rightarrow v = 0$$

Hence, the velocity of the combined object after collision is zero.

Example 5.17 A ball of mass m strikes a rigid wall with speed v and gets reflected without any loss of speed, as shown in the figure.



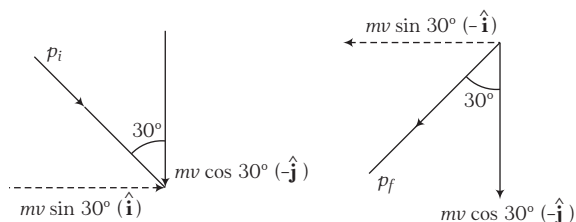
- What is the magnitude of the impulse imparted to the ball by the wall?
- What is the direction of the force on the wall due to the ball?

Sol. According to the question,

$$(i) \mathbf{p}_i = mv \sin 30^\circ \hat{\mathbf{i}} - mv \cos 30^\circ \hat{\mathbf{j}}, \quad \mathbf{p}_f = -mv \sin 30^\circ \hat{\mathbf{i}} - mv \cos 30^\circ \hat{\mathbf{j}}$$

$$\therefore \text{Impulse} = \Delta \mathbf{p} = \mathbf{p}_f - \mathbf{p}_i = -2mv \sin 30^\circ \hat{\mathbf{i}} = -mv \hat{\mathbf{i}}$$

$$\text{Magnitude of impulse} = |\Delta \mathbf{p}| = mv$$

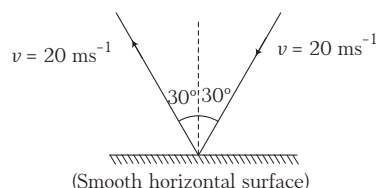


- Negative sign of the impulse shows that it is along negative x -direction. Since, impulse and force are in the same direction, the force on the ball is along the negative direction of X -axis. Hence, the force on the wall will be along positive X -axis.

CHECK POINT 5.1

- Inertia of an object is directly dependent on
 - impulse
 - momentum
 - mass
 - density
- A body of mass 6 kg is acted upon by a force, so that its velocity changes from 3 ms^{-1} to 5 ms^{-1} , then change in momentum is
 - 48 N-s
 - 24 N-s
 - 30 N-s
 - 12 N-s
- The momentum p (in $\text{kg}\cdot\text{ms}^{-1}$) of a particle is varying with time t (in second) as $p = 2 + 3t^2$. The force acting on the particle at $t = 3 \text{ s}$ will be
 - 18 N
 - 54 N
 - 9 N
 - 15 N
- A body is acted upon by balanced forces,
 - if it is in rest only
 - if it is moving with constant speed
 - if even number of forces are acting on it
 - if it is not accelerating
- A force of 72 dyne is inclined to the horizontal at an angle of 60° , find the acceleration of a mass of 9 g which moves in the effect of this force in a horizontal direction.
 - 5 cms^{-1}
 - 4 cms^{-1}
 - 2 ms^{-1}
 - 3 ms^{-1}
- A man of mass 60 kg is standing on a horizontal conveyor belt. When the belt is given an acceleration of 1 ms^{-2} , the man remains stationary with respect to the moving belt. If $g = 10 \text{ ms}^{-2}$, the net force acting on the man is
 - zero
 - 120 N
 - 60 N
 - 600 N
- A constant force acting on a body of mass 3 kg change its speed from 2 ms^{-1} to 3.5 ms^{-1} in 25 s, in the direction of the motion of the body. What is the magnitude and direction of the force?
 - 0.18 N in the direction of motion
 - 0.32 N in the direction of motion
 - 0.64 N in the direction of motion
 - 0.16 N in the direction of motion
- A body of mass 5 kg is acted upon by two perpendicular force 8 N and 6 N, find the magnitude and direction of the acceleration.
 - 3 ms^{-2} , $\theta = \cos^{-1}(0.8)$ from 8 N
 - 2 ms^{-2} , $\theta = \cos^{-1}(0.6)$ from 6 N
 - 3 ms^{-2} , $\theta = \cos^{-1}(0.9)$ from 6 N
 - 5 ms^{-2} , $\theta = \cos^{-1}(0.81)$ from 8 N
- A ball of mass m is moving towards a player with velocity v . If player stopped it, then impulse applied by the player is
 - $-mv$
 - $+mv$
 - $-2mv$
 - $+2mv$
- A constant retarding force of 50 N is applied to a body of mass 20 kg moving initially with a speed of 15 ms^{-1} . How much time does the body take to stop?
 - 6 s
 - 8 s
 - 9 s
 - 10 s
- A ball of mass 1 kg is dropped from height 9.8 m, strikes with ground and rebounds at height of 4.9 m, if the time of contact between ball and ground is 0.1 s, then find impulse and average force acting on ball.
 - 23.52 N-s, 235.2 N
 - 235.2 N-s, 23.53 N
 - 42.5 N-s, 525 N
 - 52.5 N-s, 525 N
- A body of mass 5 kg is moving with velocity of $\mathbf{v} = (2\hat{\mathbf{i}} + 6\hat{\mathbf{j}}) \text{ ms}^{-1}$ at $t = 0 \text{ s}$. After time $t = 2 \text{ s}$, velocity of body is $(10\hat{\mathbf{i}} + 6\hat{\mathbf{j}})$, then change in momentum of body is
 - $40\hat{\mathbf{i}} \text{ kg}\cdot\text{ms}^{-1}$
 - $20\hat{\mathbf{i}} \text{ kg}\cdot\text{ms}^{-1}$
 - $30\hat{\mathbf{i}} \text{ kg}\cdot\text{ms}^{-1}$
 - $(50\hat{\mathbf{i}} + 30\hat{\mathbf{j}}) \text{ kg}\cdot\text{ms}^{-1}$

13. If impulse I varies with time t as $F(\text{kg} \cdot \text{ms}^{-1}) = 20t^2 - 40t$. The change in momentum is minimum at
 (a) $t = 2$ s (b) $t = 1$ s
 (c) $t = \frac{1}{2}$ s (d) $t = \frac{3}{2}$ s
14. A ball of mass 0.5 kg moving with a velocity of 2 ms^{-1} strikes a wall normally and bounces back with the same velocity. If the time of contact between the ball and the wall is one millisecond, the average force exerted by the wall on the ball is
 (a) 2000 N (b) 1000 N
 (c) 5000 N (d) 125 N
15. An initially stationary device lying on a frictionless floor explodes into two pieces and slides across the floor. One piece is moving in positive x -direction, then other piece is moving in
 (a) positive y -direction (b) negative y -direction
 (c) negative x -direction (d) at angle from x -direction
16. A bullet of mass 0.1 kg is fired with a speed of 100 ms^{-1} . The mass of gun being 50 kg, then the velocity of recoil becomes
 (a) 0.05 ms^{-1} (b) 0.5 ms^{-1}
 (c) 0.1 ms^{-1} (d) 0.2 ms^{-1}
17. A ball is moving with speed 20 ms^{-1} collides with a smooth surface as shown in figure. The magnitude of change in velocity of the ball will be



- (a) $10\sqrt{3} \text{ ms}^{-1}$ (b) $20\sqrt{3} \text{ ms}^{-1}$
 (c) $40\sqrt{3} \text{ ms}^{-1}$ (d) 40 ms^{-1}

FORCES IN EQUILIBRIUM

Forces which have zero linear resultant will not cause any change in the motion of the object to which they are applied. Such forces (and the object) are said to be in equilibrium. For understanding the equilibrium of an object under two or more concurrent or coplanar forces, let us first discuss the resolution of force.

Resolution of a force

When a force is replaced by an equivalent set of components, it is said to be resolved. One of the most useful ways to resolve a force is to choose only two components (although a force may be resolved in three or more components also) which are at right angles also. The magnitude of these components can be very easily found using trigonometry.

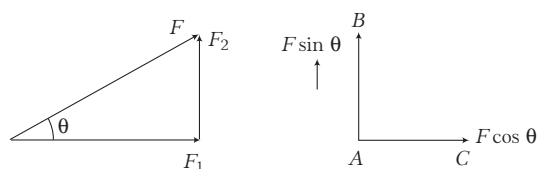


Fig. 5.10 Component of force

In the above figure,

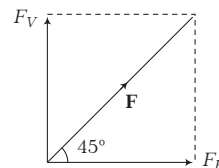
$$F_1 = F \cos \theta = \text{component of } \mathbf{F} \text{ along } AC$$

$$F_2 = F \sin \theta = \text{component of } \mathbf{F} \text{ perpendicular to } AC \text{ or along } AB$$

The component of a force in a direction perpendicular to itself is zero.

Example 5.18 Resolve horizontally and vertically a force $F = 8 \text{ N}$ which makes an angle of 45° with the horizontal.

Sol. Horizontal component of $\mathbf{F}$ is



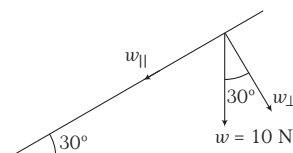
$$F_H = F \cos 45^\circ \\ = (8) \left(\frac{1}{\sqrt{2}} \right) = 4\sqrt{2} \text{ N}$$

and vertical component of $\mathbf{F}$ is

$$F_V = F \sin 45^\circ \\ = (8) \left(\frac{1}{\sqrt{2}} \right) = 4\sqrt{2} \text{ N}$$

Example 5.19 Resolve a weight of 10 N in two directions which are parallel and perpendicular to a slope inclined at 30° to the horizontal.

Sol. Component perpendicular to the plane,



$$w_\perp = w \cos 30^\circ = (10) \frac{\sqrt{3}}{2} = 5\sqrt{3} \text{ N}$$

and component parallel to the plane,

$$w_\parallel = w \sin 30^\circ = (10) \left(\frac{1}{2} \right) = 5 \text{ N}$$

Newton's first law for forces in equilibrium

When a particle is at rest or is moving with constant velocity in an inertial frame of reference, the net force acting on it, *i.e.* the vector sum of all the forces acting on it, must be zero, *i.e.* $\sum \mathbf{F} = 0$ (Particle in equilibrium).

We most often use this equation in component form,

$$\sum F_x = 0, \sum F_y = 0 \text{ and } \sum F_z = 0$$

Equilibrium under concurrent forces (*i.e.* those forces which act on same particle at same time) may be seen as

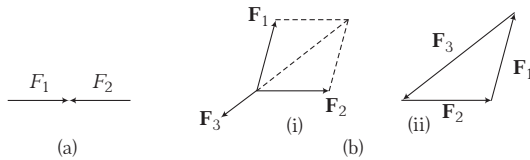


Fig. 5.11

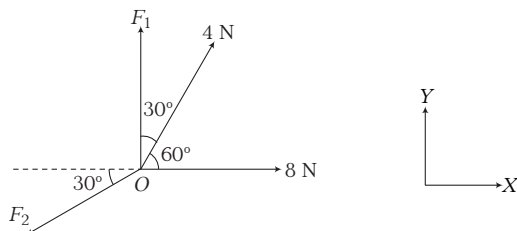
In Fig. (a)

$$\mathbf{F}_1 + \mathbf{F}_2 = 0$$

In Fig. (b)-(i) and (b)-(ii),

$$\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = 0$$

Example 5.20 An object is in equilibrium under four concurrent forces in the directions shown in figure. Find the magnitude of $\mathbf{F}_1$ and $\mathbf{F}_2$.



Sol. The object is in equilibrium.

$$\sum F_x = 0$$

$$\therefore 8 + 4 \cos 60^\circ - F_2 \cos 30^\circ = 0$$

$$\text{or } 8 + 2 - F_2 \frac{\sqrt{3}}{2} = 0$$

$$\text{or } F_2 = \frac{20}{\sqrt{3}} \text{ N}$$

$$\text{Similarly, } \sum F_y = 0$$

$$\therefore F_1 + 4 \sin 60^\circ - F_2 \sin 30^\circ = 0$$

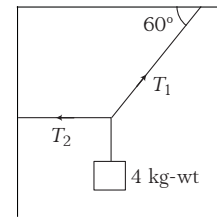
$$\text{or } F_1 + \frac{4\sqrt{3}}{2} - \frac{F_2}{2} = 0$$

$$\text{or } F_1 = \frac{F_2}{2} - 2\sqrt{3}$$

$$= \frac{10}{\sqrt{3}} - 2\sqrt{3}$$

$$\text{or } F_1 = \frac{4}{\sqrt{3}} \text{ N}$$

Example 5.21 Determine the tensions T_1 and T_2 in the strings as shown in figure.



Sol. Resolving the tension T_1 along horizontal and vertical directions. As the body is in equilibrium,

$$T_1 \sin 60^\circ = 4 \times 9.8 \text{ N} \quad \dots(i)$$

$$\text{and } T_1 \cos 60^\circ = T_2 \quad \dots(ii)$$

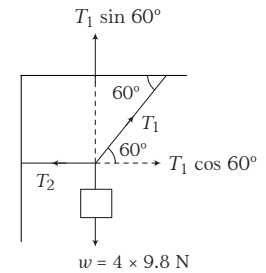
From Eq. (i), we get

$$T_1 = \frac{4 \times 9.8}{\sin 60^\circ} = \frac{4 \times 9.8 \times 2}{\sqrt{3}} = 45.26 \text{ N}$$

Putting this value in Eq. (ii), we get

$$T_2 = T_1 \cos 60^\circ = 45.26 \times 0.5 = 22.63 \text{ N}$$

Note Tension is a type of force produced in strings.



Lami's theorem

It states that, "if three forces acting on a particle are in equilibrium, then each force is proportional to the sine of the angle between the other two forces."

If an object O is in equilibrium under three coplanar forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ as shown in figure.

Then,

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

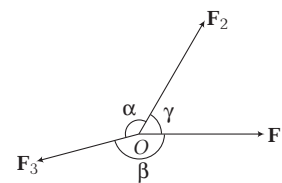


Fig. 5.12

If more than three forces are given in the problem, then solve the problem by using component approach. If three forces are in equilibrium, then the resultant of two forces is equal and opposite to the third.

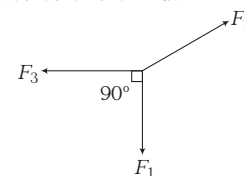


Fig. 5.13

$$F_2 = \sqrt{F_1^2 + F_3^2}$$

Working with Newton's laws

Normally any problem relating to Newton's laws is solved in following four steps

- First of all we decide the system on which the laws of motion are to be applied. The system may be a single particle, a block or a combination of two or more blocks, two blocks connected by a string, etc. The only restriction is that all parts of the system should have the same acceleration.
- Once the system is decided, we make the list of all the forces acting on the system. Any force applied by the system on other bodies is not included in the list of the forces.
- Then, we make a free body diagram of the system and indicate the magnitude and directions of all the forces listed in step (ii) in this diagram.
- In the last step, we choose any two mutually perpendicular axes say X and Y in the plane of the forces in case of coplanar forces. Choose the X -axis along the direction in which the system is known to have or is likely to have the acceleration. A direction perpendicular to it may be chosen as the Y -axis. If the system is in equilibrium, any mutually perpendicular directions may be chosen. Write the components of all the forces along the X -axis and equate their sum to the product of the mass of the system and its acceleration, i.e.

$$\Sigma F_x = ma \quad \dots(i)$$

This gives us one equation. Now, we write the components of the forces along the Y -axis and equate the sum to zero. This gives us another equation, i.e.

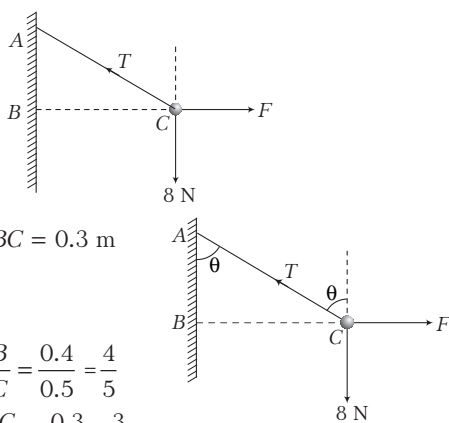
$$\Sigma F_y = 0 \quad \dots(ii)$$

- (a) If the system is in equilibrium, we will write the two equations as

$$\Sigma F_x = 0 \text{ and } \Sigma F_y = 0$$

- (b) If the forces are collinear, the second equation, i.e. $\Sigma F_y = 0$ is not needed.

Example 5.22 One end of a string 0.5 m long is fixed to a point A and the other end is fastened to a small object of weight 8 N. The object is pulled aside by a horizontal force F , until it is 0.3 m from the vertical through A . Find the magnitudes of the tension T in the string and the force F .



Sol. $AC = 0.5$ m, $BC = 0.3$ m

$$\therefore AB = 0.4 \text{ m}$$

$$\text{and if } \angle BAC = \theta$$

$$\text{Then, } \cos \theta = \frac{AB}{AC} = \frac{0.4}{0.5} = \frac{4}{5}$$

$$\text{and } \sin \theta = \frac{BC}{AC} = \frac{0.3}{0.5} = \frac{3}{5}$$

Here, the object is in equilibrium under three concurrent forces. So, we can apply Lami's theorem,

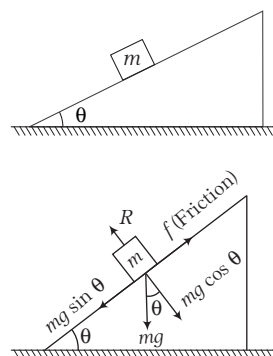
$$\frac{F}{\sin (180^\circ - \theta)} = \frac{8}{\sin (90^\circ + \theta)} = \frac{T}{\sin 90^\circ}$$

$$\text{or } \frac{F}{\sin \theta} = \frac{8}{\cos \theta} = T$$

$$\therefore T = \frac{8}{\cos \theta} = \frac{8}{4/5} = 10 \text{ N}$$

$$\text{and } F = \frac{8 \sin \theta}{\cos \theta} = \frac{(8)(3/5)}{(4/5)} = 6 \text{ N}$$

Example 5.23 A block of mass m is at rest on a rough wedge as shown in figure. What is the force exerted by the wedge on the block?



Sol.

Since, the block is permanently at rest, it is in equilibrium. Net force on it should be zero. In this case, only two forces are acting on the block.

- (i) Weight = mg (downwards)

- (ii) Contact force (resultant of normal reaction and friction force) applied by the wedge on the block.

For the block to be in equilibrium, these two forces should be equal and opposite. Therefore, force exerted by the wedge on the block is mg (upwards).

Alternate method From Newton's third law of motion, force exerted by the block on the wedge is also mg but downwards. The result can also be obtained in a different manner.

The normal force on the block is R or $N = mg \cos \theta$ and the friction force on the block is $f = mg \sin \theta$ (not $\mu mg \cos \theta$) because it is not the case of limiting friction.

These two forces are mutually perpendicular.

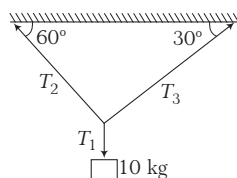
$$\therefore \text{Net contact force would be } \sqrt{N^2 + f^2}$$

$$\text{or } \sqrt{(mg \cos \theta)^2 + (mg \sin \theta)^2} \text{ which is equal to } mg.$$

CHECK POINT 5.2

- Four forces act on a point object. The object will be in equilibrium, if
 - they are opposite to each other in pairs
 - sum of x , y and z -components of forces is zero separately
 - they can be represented by a closed figure of 4 sides by direction and magnitude.
 - All of the above

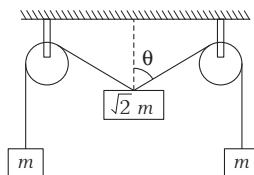
- A block of mass 10 kg is suspended by three strings as shown in the figure. The tension T_2 is



- (a) 100 N (b) $\frac{100}{\sqrt{3}}$ N (c) $\sqrt{3} \times 100$ N (d) $50\sqrt{3}$ N

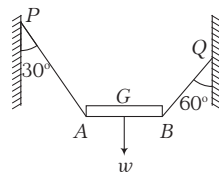
- An object is resting at the bottom of two strings which are inclined at an angle of 120° with each other. Each string can withstand a tension of 20 N. The maximum weight of the object that can be sustained without breaking the strings is
 - 10 N
 - 20 N
 - $20\sqrt{2}$ N
 - 40 N

- The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be



- (a) 0° (b) 30° (c) 45° (d) 60°

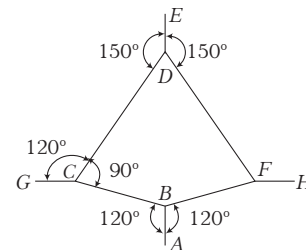
- A non-uniform rod AB of weight w is supported horizontally in a vertical plane by two light strings PA and QB as shown in the figure. G is the centre of gravity of the rod. If PA and QB make angles 30° and 60° respectively with the vertical, the ratio $\frac{AG}{BG}$ is



- (a) $\frac{1}{2}$ (b) $\sqrt{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{\sqrt{3}}$

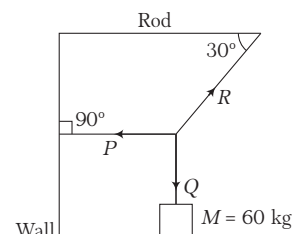
- A weight w is suspended from the mid-point of a rope, whose ends are at the same level. In order to make the rope perfectly horizontal, the force applied to each of its ends must be
 - less than w
 - equal to w
 - equal to $2w$
 - infinitely large

- The below figure is the part of a horizontally stretched net. Section AB is stretched with a force of 10 N. The tensions in the sections BC and BF are



- (a) 10 N, 11 N (b) 10 N, 6 N
(c) 10 N, 10 N (d) Cannot be calculated due to insufficient data

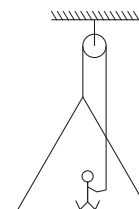
- A body of mass 60 kg suspended by means of three strings, P , Q and R as shown in the figure is in equilibrium. The tension in the string P is



- (a) 130.9 kgf (b) 60 kgf
(c) 50 kgf (d) 103.9 kgf

- A man of mass 50 g stands on a frame of mass 30 g. He pulls on a light rope which passes over a pulley. The other end of the rope is attached to the frame. For the system to be in equilibrium, what force man must exert on the rope?

- (a) 40 g (b) 80 g
(c) 30 g (d) 50 g



- Two particles of equal mass are connected to a rope AB of negligible mass, such that one is at end A and the other dividing the length of the rope in the ratio 1 : 2 from A . The rope is rotated about end B in a horizontal plane. Ratio of the tensions in the smaller part to the other is (ignore effect of gravity)

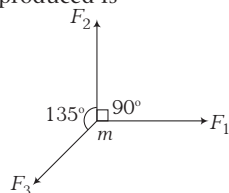
- (a) 4 : 3 (b) 1 : 4 (c) 1 : 2 (d) 1 : 3

- A body is under the action of two mutually perpendicular forces of 3 N and 4 N. The resultant force acting on the body is
 - 7 N
 - 1 N
 - 5 N
 - zero

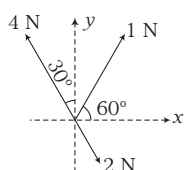
- Two equal forces are acting at a point with an angle of 60° between them. If the resultant force is equal to $40\sqrt{3}$ N, the magnitude of each force is

- (a) 40 N (b) 20 N
(c) 80 N (d) 30 N

13. When a force F acts on a body of mass m , the acceleration produced in the body is a . If three equal forces $F_1 = F_2 = F_3 = F$ act on the same body as shown in figure. The acceleration produced is

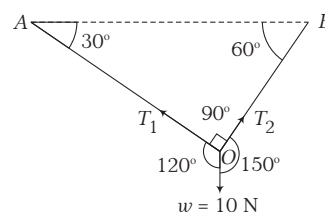


- (a) $(\sqrt{2} - 1)a$ (b) $(\sqrt{2} + 1)a$ (c) $\sqrt{2}a$ (d) a
14. Three forces acting on a body are shown in the figure. To have the resultant force only along the y -direction, the magnitude of the minimum additional force needed is



- (a) 0.5 N (b) 1.5 N
(c) $\frac{\sqrt{3}}{4}$ N (d) $\sqrt{3}$ N

15. A ball of mass 1 kg hangs in equilibrium from two strings OA and OB as shown in figure. What are the tensions in strings OA and OB? (Take, $g = 10 \text{ ms}^{-2}$)



- (a) 5 N, 5 N
(b) $5\sqrt{3}$ N, $5\sqrt{3}$ N
(c) 5 N, $5\sqrt{3}$ N
(d) $5\sqrt{3}$ N, 5 N

COMMON FORCES IN MECHANICS

Some of the common forces that we come across in mechanics are described below

1. Weight (w)

The weight of an object is equal to the gravitational force with which the earth pulls it downwards.

Weight of an object, $w = mg$

where, m = mass of the object and g = acceleration due to gravity.

2. Normal reaction (R or M)

It is a contact force between two surfaces in contact, which is always perpendicular to the surfaces in contact. The following diagrams show normal reaction between two surfaces

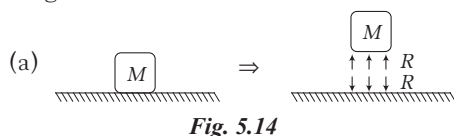


Fig. 5.14

Block pushes ground downward with force R and ground pushes the block back with force R , where R = normal reaction force.

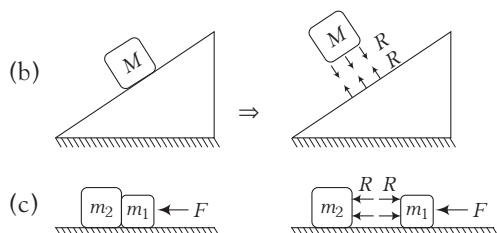


Fig. 5.15 Normal reaction between two surfaces

Here, m_1 pushes m_2 towards left by force R and m_2 pushes m_1 towards right by force R .

3. Tension (T)

When a body is connected through a string or rope, a force may act on the body by the string or rope due to the tendency of extension. This force is called tension.

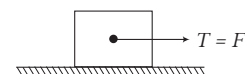


Fig. 5.16 Tension force

Regarding tension and string, the following points are important to remember

- Force of tension acts on a body in the direction away from the point of contact or tied ends of the string.
- If a string is inextensible, the magnitude of acceleration of any number of masses connected through string is always same.

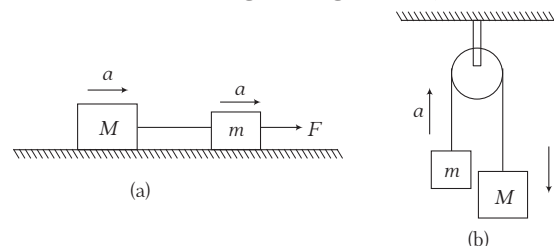


Fig. 5.17 Masses connected through string having same acceleration

- If a string is massless, the tension in it is same everywhere. However, if a string has a mass, tension at different points will be different.

- (iv) If there is friction between string and pulley, tension is different on two sides of the pulley, but if there is no friction between pulley and string, tension will be same on both sides of the pulley.

These points can be understood in diagram as follows

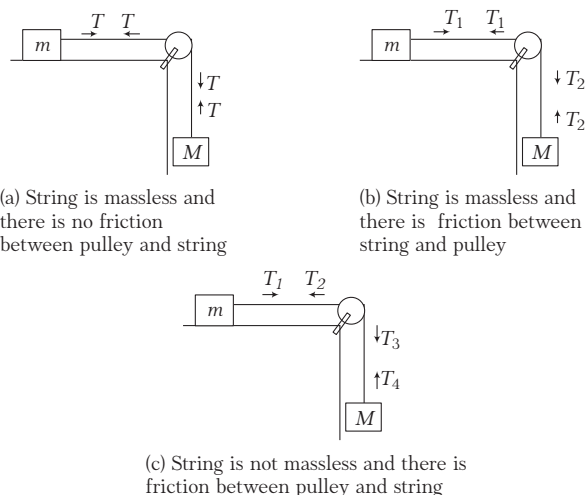


Fig. 5.18

- (v) If a force is directly applied on the string, the tension will be equal to the applied force irrespective of the motion of the pulling agent.

4. Spring force

The resistive force developed in a spring, when its length is changed is called spring force.

∴ Spring force, $F = -kx$

where, x = change in length of the spring

and k = spring constant.

Free body diagram

A free body diagram (FBD) consists of a diagrammatic representation of a single body or a sub-system of bodies isolated from its surroundings showing all the forces acting on it.

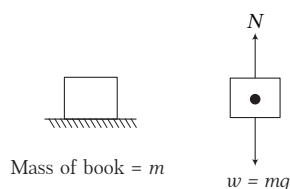
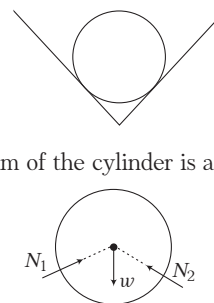


Fig. 5.19 Free body diagram

Consider, for example, a book lying on a horizontal surface.

A free body diagram of the book alone would consist of its weight ($w = mg$), acting through the centre of gravity and the reaction (N) exerted on the book by the surface.

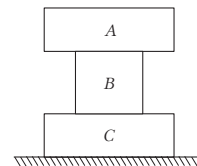
Example 5.24 A cylinder of weight w is resting on a V-groove as shown in figure. Draw its free body diagram.



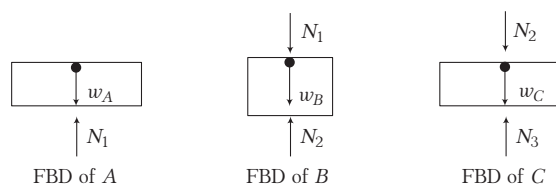
Sol. The free body diagram of the cylinder is as shown in figure.

Here, w = weight of cylinder and N_1 and N_2 are the normal reactions between the cylinder and the two inclined walls.

Example 5.25 Three blocks A, B and C are placed one over the other as shown in figure. Draw free body diagrams of all the three blocks.



Sol. Free body diagrams of A, B and C are shown below.

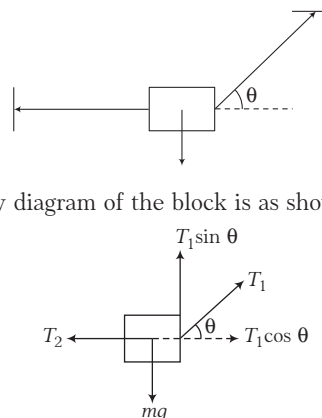


Here, N_1 = normal reaction between A and B,

N_2 = normal reaction between B and C

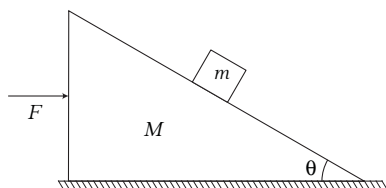
and N_3 = normal reaction between C and ground.

Example 5.26 A block of mass m is attached with two strings as shown in figure. Draw the free body diagram of the block.



Sol. The free body diagram of the block is as shown in figure.

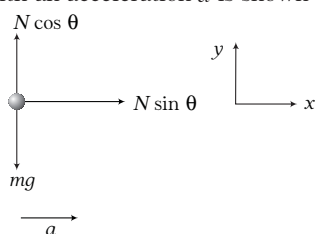
Example 5.27 All surfaces are smooth in following figure. Find F such that block remains stationary with respect to wedge.



Sol. Acceleration of (block + wedge), $a = \frac{F}{(M + m)}$

Let us solve the problem by both the methods

- (i) **From inertial frame of reference** (ground) FBD of block (only real forces) with respect to ground, which is moving with an acceleration a is shown below.



$$\therefore \Sigma F_y = 0$$

$$\Rightarrow N \cos \theta = mg \quad \dots(i)$$

$$\text{and } \Sigma F_x = ma$$

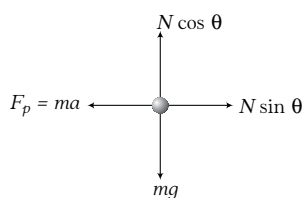
$$\Rightarrow N \sin \theta = ma \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$a = g \tan \theta$$

$$\therefore F = (M + m)a = (M + m)g \tan \theta$$

- (ii) **From non-inertial frame of reference** (wedge) FBD of block w.r.t. wedge (real forces + pseudo force) is shown below.



As, w.r.t. wedge, block is stationary.

$$\therefore \Sigma F_y = 0 \Rightarrow N \cos \theta = mg \quad \dots(iii)$$

$$\Sigma F_x = 0 \Rightarrow N \sin \theta = ma \quad \dots(iv)$$

In this way, the block's net acceleration becomes zero. Because all the forces acting on it balance each other for an observer on the wedge and not for an observer on the ground.

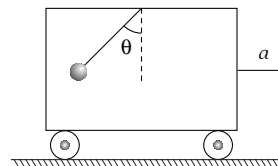
From Eqs. (iii) and (iv), we will get the same result

$$\text{i.e. } F = (M + m)g \tan \theta$$

Note In non-inertial frame, a force acting in a direction opposite to the direction of acceleration of frame, is called pseudo force.

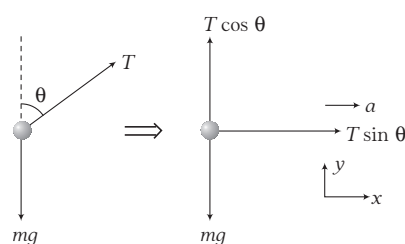
$$\mathbf{F}_p = -m\mathbf{a}$$

Example 5.28 A bob of mass m is suspended from the ceiling of a train moving with an acceleration a as shown in figure. Find the angle θ in equilibrium position.



Sol. This problem can also be solved by both the methods

- (i) **Inertial frame of reference** (ground) FBD of bob w.r.t. ground (only real forces), which is also moving with an acceleration a is shown below.



$$\therefore \Sigma F_x = ma$$

$$\Rightarrow T \sin \theta = ma \quad \dots(i)$$

$$\text{and } \Sigma F_y = 0$$

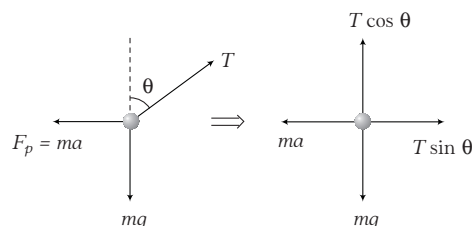
$$\Rightarrow T \cos \theta = mg \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\tan \theta = \frac{a}{g}$$

$$\text{or } \theta = \tan^{-1} \left(\frac{a}{g} \right)$$

- (ii) **Non-inertial frame of reference** (train) FBD of bob w.r.t. train (real forces + pseudo force) is shown below.



As, with respect to train, bob is in equilibrium.

$$\therefore \Sigma F_x = 0$$

$$\Rightarrow T \sin \theta = ma \quad \dots(iii)$$

$$\therefore \Sigma F_y = 0$$

$$\Rightarrow T \cos \theta = mg \quad \dots(iv)$$

From Eqs. (iii) and (iv), we get the same result, i.e.

$$\theta = \tan^{-1} \left(\frac{a}{g} \right)$$

APPARENT WEIGHT OF A MAN IN A LIFT

Let us consider a man of mass m is standing on a weighing machine placed in an elevator/lift. The actual weight mg of the man acts on the weighing machine and offers a reaction R given by the reading of the weighing machine.

This reaction R exerted by the surface of contact on the man is apparent weight of the person.

Now, we consider how R is related to mg in the different conditions.

- (i) When the lift moves upwards with acceleration a as shown in figure, the net upward force on the man is

$$R - mg = ma \Rightarrow R = ma + mg$$

$$\text{Apparent weight, } R = m(g + a)$$

So, **when a lift accelerates upwards, the apparent weight of the man inside it increases.**

- (ii) When the lift moves downwards with acceleration a as shown in figure, the net downward force on the man is

$$mg - R = ma$$

$$\text{Apparent weight, } R = m(g - a)$$

So, **when a lift accelerates downwards, the apparent weight of the man inside it decreases.**

- (iii) When the lift is at rest or moving with uniform velocity v downward or upward as shown in figure.

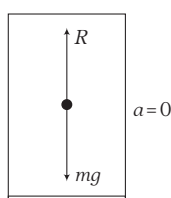


Fig. 5.22

Then, acceleration $a = 0$. So, net force on the man is

$$R - mg = m \times 0$$

$$\Rightarrow R = mg$$

or Apparent weight = Actual weight

So, **when the lift is at rest, the apparent weight of the man is his actual weight.**

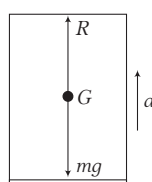


Fig. 5.20

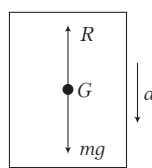


Fig. 5.21

- (iv) When the lift falls freely under gravity, if the supporting cable of the lift breaks. Then, $a = g$.

The net downward force on the man is

$$R = m(g - g) \Rightarrow R = 0$$

Thus, **the apparent weight of the man becomes zero.** This is because both the lift and man are moving downwards with the same acceleration g and so there is no force of action and reaction existing between the man and lift. Hence, a person develops a feeling of weightlessness when the lift falls freely under gravity.

Example 5.29 A spring balance is attached to the ceiling of an elevator. A boy hangs his bag on the spring and the spring reads 49 N, when the elevator is stationary. If the elevator moves downward with an acceleration of 5 ms^{-2} , what will be the reading of the spring balance?

Sol. When the elevator is stationary, then

$$w = mg$$

$$\Rightarrow 49 = m \times 9.8$$

$$\Rightarrow m = \frac{49}{9.8} = 5 \text{ kg}$$

When the elevator is moving downward with an acceleration,

$$R = m(9.8 - a) = 5(9.8 - 5) = 24 \text{ N}$$

Example 5.30 If in a stationary elevator, a man is standing with a bucket full of water. The bucket has a hole at its bottom. The rate of flow of water through this hole is R_0 . If the elevator starts to move up and then down with same acceleration, then the rate of flow of water are R_u and R_d . Find the relation between R_0 , R_u and R_d .

Sol. Rate of flow will be more when elevator will move in upward direction with some acceleration because the net downward pull will be more and vice-versa.

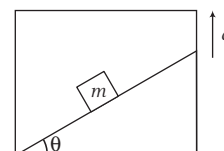
$$F_{\text{upward}} = R_u = m(g + a)$$

$$F_{\text{downward}} = R_d = m(g - a)$$

$$\Rightarrow F_{\text{at rest}} = R_0 = mg$$

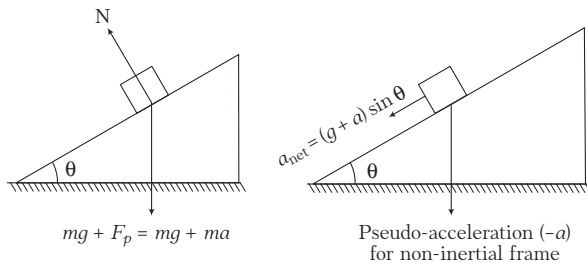
Thus, relation between R_u , R_0 and R_d is $R_u > R_0 > R_d$.

Example 5.31 In the adjoining figure, a wedge is fixed to an elevator moving upwards with an acceleration a . A block of mass m is placed over the wedge. Find the acceleration of the block with respect to wedge. Neglect friction.



Sol. Since, acceleration of block w.r.t. wedge (an accelerating or non-inertial frame of reference) is to be found out.

FBD of block w.r.t. wedge is shown in figure.



The acceleration would have been $g \sin \theta$ (down the plane), if the lift were stationary or when only weight (i.e. mg) acts downwards.

Here, downward force is $m(g + a)$

$\therefore$ Acceleration of the block (of course w.r.t. wedge) will be $(g + a) \sin \theta$ down the plane.

APPLICATIONS OF NEWTON'S LAWS OF MOTION

Consider two bodies of masses m_1 and m_2 placed in direct contact with each other on a smooth platform.

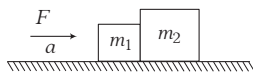


Fig. 5.23 Motion of two connected bodies

Suppose a horizontal force is applied and both the bodies move with acceleration a , then

$$F = (m_1 + m_2) a$$

Now, we can calculate normal reaction between the bodies, using free body diagram (FBD).

FBD of Ist body

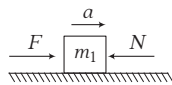


Fig. 5.24

$$\therefore F - N = m_1 a \quad (\text{From Newton's IInd law})$$

FBD of IInd body

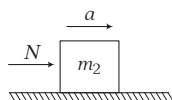


Fig. 5.25

$$\therefore N = m_2 a$$

In the same way, we can calculate the acceleration and normal reaction for three bodies in contact.

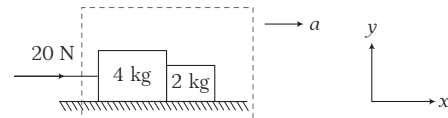
Example 5.32 Two blocks of masses 4 kg and 2 kg are placed side-by-side on a smooth horizontal surface as shown in the figure. A horizontal force of 20 N is applied on 4 kg block.

Find

(i) the acceleration of each block

(ii) and the normal reaction between two blocks.

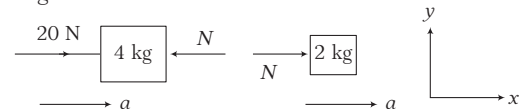
Sol. (i) Since, both the blocks will move with same acceleration (say a) in horizontal direction. Let us take both the blocks as a system. Net external force on the system is 20 N in horizontal direction.



Using $\Sigma F_x = ma_x$

$$20 = (4 + 2)a = 6a \quad \text{or} \quad a = \frac{10}{3} \text{ ms}^{-2}$$

(ii) The free body diagram of both the blocks are shown in the figure.



Using $\Sigma F_x = ma_x$

$$\text{For 4 kg block, } 20 - N = 4a = 4 \times \frac{10}{3}$$

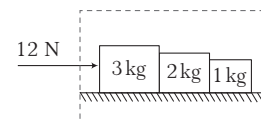
$$\Rightarrow N = 20 - \frac{40}{3} = \frac{20}{3} \text{ N}$$

$$\text{For 2 kg block, } N = 2a = 2 \times \frac{10}{3} = \frac{20}{3} \text{ N}$$

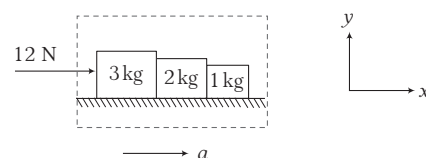
Here, N is the normal reaction between the two blocks.

Note In free body diagram of the blocks, we have not shown the forces acting on the blocks in vertical direction, because normal reaction between the blocks and acceleration of the system can be obtained without using $\Sigma F_y = 0$.

Example 5.33 Three blocks of masses 3 kg, 2 kg and 1 kg are placed side-by-side on a smooth surface as shown in figure. A horizontal force of 12 N is applied on 3 kg block. Find the net force on 2 kg block.



Sol. All the blocks will move with same acceleration (say a) in horizontal direction. Let us take all the blocks as a system, net external force on the system is 12 N in horizontal direction.



Using $\Sigma F_x = ma_x$, we get

$$12 = (3 + 2 + 1)a = 6a \quad \text{or} \quad a = \frac{12}{6} = 2 \text{ ms}^{-2}$$

Now, let F be the net force on 2 kg block in x-direction, then using $\Sigma F_x = ma_x$ for 2 kg block, we get

$$F = (2)(2) = 4 \text{ N}$$

Note Here, net force F on 2 kg block is the resultant of N_1 and

N_2 ($N_1 > N_2$), where N_1 = normal reaction between 3 kg and 2 kg block and N_2 = normal reaction between 2 kg and 1 kg block.

Thus, $F = N_1 - N_2$.

Motion of bodies connected through strings or springs

Horizontal motion

When two bodies are connected through an inextensible weightless string and a force is applied to impart an acceleration a in both the bodies, as given in the figure below, then equation of motion from the FBD can be given by

$$\text{For A,} \quad T = m_1 a \quad \dots(i)$$

$$\text{For B,} \quad F - T = m_2 a \quad \dots(ii)$$

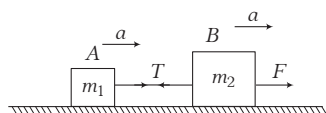


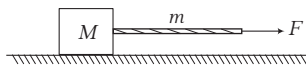
Fig. 5.26 Motion of two bodies connected through string

$$\text{From Eqs. (i) and (ii), } F = m_1 a + m_2 a \quad \dots(iii)$$

$$\text{From Eqs. (i) and (iii), } T = \frac{m_1 F}{(m_1 + m_2)}$$

In the same way, we can calculate the acceleration and tension in three bodies.

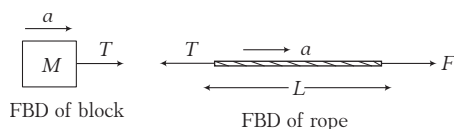
Example 5.34 A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m as shown in figure. A horizontal force F is applied to one end of the rope. Find (i) the acceleration of the rope and block, (ii) the force that the rope exerts on the block (iii) and tension in the rope at its mid-point.



Sol. (i) Acceleration, $a = \frac{F}{(m + M)}$

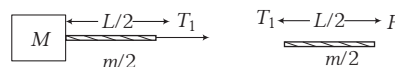
(ii) Force exerted by rope on the block is

$$T = Ma = \frac{M \cdot F}{(m + M)}$$

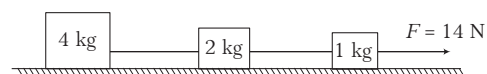


$$(iii) T_1 = \left(\frac{m}{2} + M\right)a = \left(\frac{m + 2M}{2}\right)\left(\frac{F}{m + M}\right)$$

$$\text{Tension in rope at mid-point, } T_1 = \frac{(m + 2M)F}{2(m + M)}$$



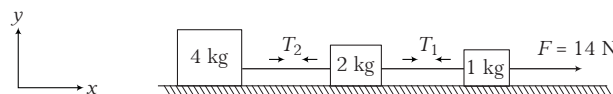
Example 5.35 In the arrangement shown in the figure, the strings are light and inextensible. The surface over which blocks are placed is smooth. Find



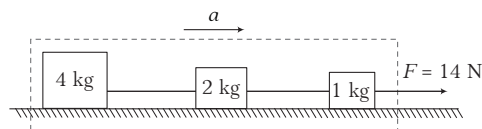
(i) the acceleration of each block

(ii) the tension in each string.

Sol. (i) Let a be the acceleration of each block and T_1 and T_2 be the tensions, in the two strings as shown in the figure.



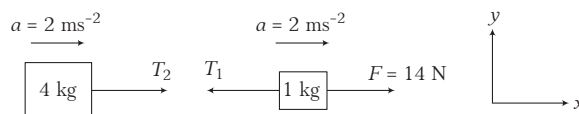
Taking the three blocks and the two strings as the system.



$$\text{Using } \Sigma F_x = ma_x$$

$$\text{or } 14 = (4 + 2 + 1)a \quad \text{or} \quad a = \frac{14}{7} = 2 \text{ ms}^{-2}$$

(ii) Free body diagram (showing the forces in x-direction only) of 4 kg block and 1 kg block are shown in figure.



$$\text{Using } \Sigma F_x = ma_x$$

$$\text{For 1 kg block, } F - T_1 = (1)(a) \quad \text{or} \quad 14 - T_1 = (1)(2) = 2$$

$$\therefore T_1 = 14 - 2 = 12 \text{ N}$$

$$\text{For 4 kg block, } T_2 = (4)(a)$$

$$\therefore T_2 = (4)(2) = 8 \text{ N}$$

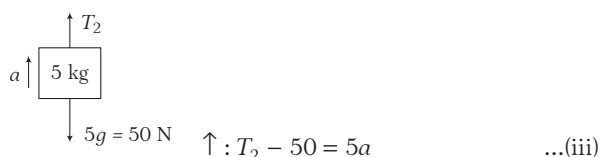
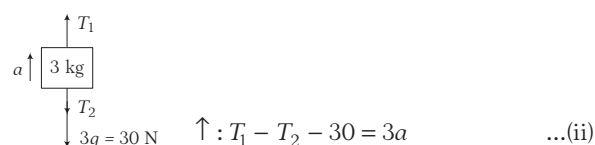
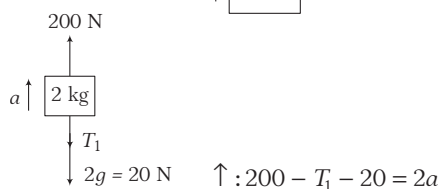
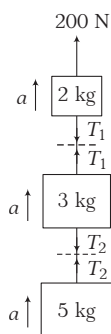
Vertical motion

In the same manner, as discussed above, we can calculate acceleration, tension in the string and common force in two or more blocks in vertical direction also.

Example 5.36 The blocks of masses 2 kg, 3 kg and 5 kg are connected by light, inextensible strings as shown in figure.

The system of blocks is raised vertically upwards by applying a force $F_0 = 200 \text{ N}$. Find the common acceleration and tension in the strings.

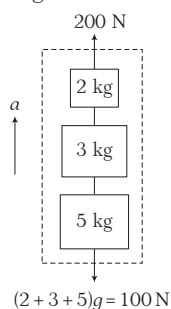


Sol.

Adding Eqs. (i), (ii) and (iii), we get

$$100 = 10a \Rightarrow a = 10 \text{ ms}^{-2}$$

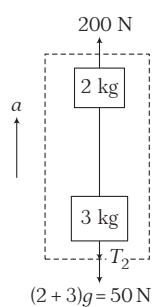
$$T_1 = 160 \text{ N} \text{ and } T_2 = 100 \text{ N}$$

Alternate Method Taking three blocks together as a system

$$\uparrow : 200 - 100 = (2 + 3 + 5)a$$

$$\Rightarrow a = 10 \text{ ms}^{-2}$$

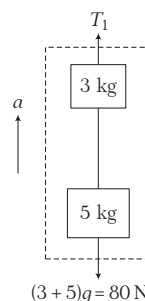
Two blocks at a time



$$200 - 50 - T_2 = (2 + 3)a$$

$$150 - T_2 = 5 \times 10$$

$$\Rightarrow T_2 = 100 \text{ N}$$



$$T_1 - 80 = (3 + 5)a$$

$$T_1 - 80 = 8 \times 10$$

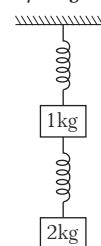
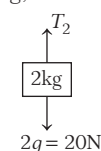
$$T_1 = 160 \text{ N}$$

For spring-block system

In the same manner, as discussed above, we can calculate the tension in a spring connected to a block of mass m , which produces an extension x in it, i.e. $T = kx$.

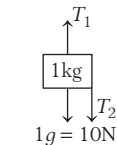
where, k = spring constant.

Example 5.37 Two masses are connected to two identical springs of spring constant 100 N/m as shown in figure. Find the extension in both the springs.

**Sol.** The FBD for mass 2 kg,

$$\Rightarrow T_2 - 20 = 0 \text{ or } T_2 = 20 \text{ N} \quad \dots (i)$$

The FBD for mass 1 kg,



$$\Rightarrow T_1 = T_2 + 10 \Rightarrow T_1 = 30 \text{ N} \quad [\text{Using Eq. (i)}]$$

As, these tensions in the springs are the spring forces,

$$\therefore T_1 = kx_1 \Rightarrow 30 = 100 \times x_1 \text{ or } x_1 = 0.3 \text{ m}$$

$$\text{and } T_2 = kx_2 \Rightarrow 20 = 100 \times x_2 \text{ or } x_2 = 0.2 \text{ m}$$

Bodies attached through pulley (using strings or springs)

Case I Consider two bodies of masses m_1 and m_2 ($m_2 > m_1$) connected to a massless and inextensible string which passes through a smooth pulley. If they are allowed to move freely, they move with common acceleration a . The tension in the string due to mass m_1 and m_2 is T .

From FBD,

For A, $T - m_1g = m_1a$... (i)

For B, $m_2g - T = m_2a$... (ii)

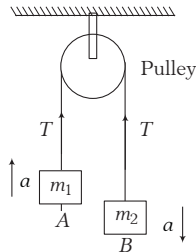


Fig. 5.27 Motion of connected bodies

On solving Eqs. (i) and (ii), we get

$$\Rightarrow a = \frac{(m_2 - m_1)}{(m_2 + m_1)} g$$

$$\Rightarrow T = \left(\frac{2m_1m_2}{m_1 + m_2} \right) g$$

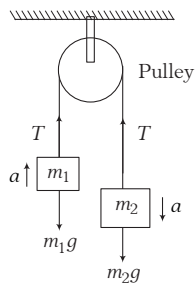
Reaction at suspension of pulley $R = 2T$.

Example 5.38 Two blocks of masses 2.6 kg and 4.1 kg are tied together by a light string looped over a frictionless pulley.

(a) What will the acceleration of each mass be?

(b) Find the value of tension in the string.

Sol. Here, $m_1 = 2.6$ kg, $m_2 = 4.1$ kg



FBD for m_1 , $T - m_1g = m_1a$ (upward) ... (i)

FBD for m_2 , $m_2g - T = m_2a$ (downward) ... (ii)

(a) Adding Eqs. (i) and (ii), we get

$$\Rightarrow 4.1g - 2.6g = (2.6 + 4.1)a$$

$$\Rightarrow a = \frac{1.5g}{6.7} = \frac{1.5 \times 10}{6.7}$$

$$= 2.2 \text{ ms}^{-2}$$

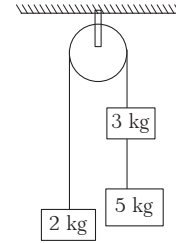
(b) Putting the value of a in Eq. (i), we get

$$\Rightarrow T - 2.6g = 2.6 \times 2.2$$

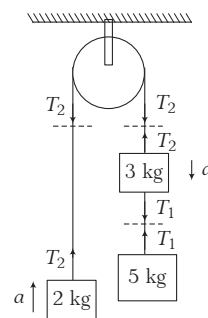
$$\Rightarrow T = 2.6 \times 2.2 + 2.6 \times 10$$

$$= 31.72 \text{ N}$$

Example 5.39 The pulley is light and smooth; the strings are inextensible and light. The system is released from rest. Find the acceleration of each block, tensions in the strings and reaction on pulley.



Sol.



FBD for 5 kg block:

Upward: T_1

Downward: $5g = 50 \text{ N}$

$$\downarrow : 50 - T_1 = 5a \quad \dots (i)$$

FBD for 3 kg block:

Upward: T_2

Downward: T_1

Downward: $3g = 30 \text{ N}$

$$\downarrow : 30 + T_1 - T_2 = 3a \quad \dots (ii)$$

FBD for 2 kg block:

Upward: T_2

Downward: $2g = 20 \text{ N}$

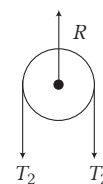
$$\uparrow : T_2 - 20 = 2a \quad \dots (iii)$$

Solving Eqs. (i), (ii) and (iii), we get

$$a = 6 \text{ ms}^{-2}, T_1 = 20 \text{ N}, T_2 = 32 \text{ N}$$

Forces on pulley, $R - 2T_2 = 0$ (Pulley is massless)

$$R = 2T_2 = 64 \text{ N}$$



where, R is the reaction on the pulley.

Case II When two bodies are attached through a pulley as given in the figure below.

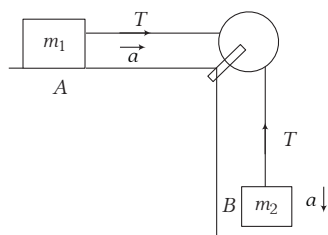


Fig. 5.28

Here, $m_2 > m_1$, then from FBD,

For A, $T = m_1 a$... (i)

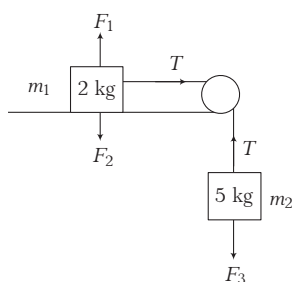
For B, $m_2 a = m_2 g - T$... (ii)

Solving Eqs. (i) and (ii), we get

$$a = \frac{m_2}{(m_1 + m_2)} g$$

$$\Rightarrow T = \frac{m_1 m_2}{(m_1 + m_2)} g$$

Example 5.40 A 2 kg mass placed on a level table is attached to a 5 kg mass by a string passing over the edge of a table as illustrated in the diagram.



(a) Calculate the magnitude of acceleration of the system.

(b) Calculate the tension in the string.

Sol. From FBD in m_1 ,

$$T - m_1 a = 0$$

$$\Rightarrow T - 2a = 0$$

FBD in m_2 ,

$$m_2 g - T = m_2 a$$

$$5g - T = 5a$$

Putting $T = 2a$ from Eq. (i), we get

$$5g - 2a = 5a$$

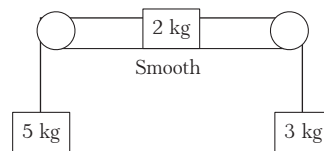
$$7a = 5g$$

(a) $a = \frac{5}{7} \times 10 = \frac{50}{7} \text{ ms}^{-2}$

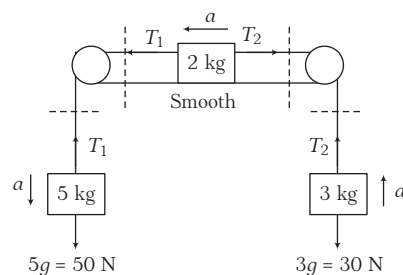
(b) Again, tension in the string,

$$T = 2a = 2 \times \frac{50}{7} = \frac{100}{7} \text{ N}$$

Example 5.41 The strings are inextensible and light; the pulleys are smooth and light. Find the acceleration of each block and tensions in the strings.



Sol.



5 kg : $\downarrow : 50 - T_1 = 5a$... (i)

2 kg : $\leftarrow : T_1 - T_2 = 2a$... (ii)

3 kg : $\uparrow : T_2 - 30 = 3a$... (iii)

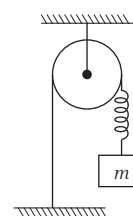
Adding Eqs. (i), (ii) and (iii), we get

$$20 = 10a \Rightarrow a = 2 \text{ ms}^{-2}$$

$$T_1 = 40 \text{ N and } T_2 = 36 \text{ N}$$

Example 5.42 Consider the situation shown in figure.

Initially the spring is unstretched when the block of mass is released from rest. Assume the pulley frictionless and light, the spring and string massless. Find the maximum extension of the spring.



Sol. The FBD of given mass m

$$\Rightarrow mg - kx = ma \text{ or } a = g - \frac{k}{m} x$$

$$v dv = \left(g - \frac{k}{m} x \right) dx$$

Integrating both sides, we get

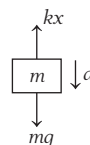
$$\int_0^v v dv = \int_0^x \left(g - \frac{k}{m} x \right) dx \Rightarrow \frac{v^2}{2} = gx - \frac{kx^2}{2m}$$

or

$$v = \left(2gx - \frac{kx^2}{m} \right)^{1/2}$$

When the block will stop, $v = 0$ (at maximum extension),

$$\left(2gx - \frac{kx^2}{m} \right)^{1/2} = 0 \Rightarrow 2gx = \frac{kx^2}{m} \Rightarrow x = x_m = \frac{2mg}{k}$$



Case III When pulley is attached to the edge of an inclined plane To understand this case, first of all we need to understand the motion of an object placed on a frictionless inclined plane.

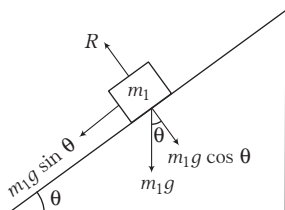


Fig. 5.29

Consider an object of mass m_1 is placed on a smooth inclined plane with angle of inclination θ . From the FBD of object, different forces acting on it, are

- Normal reaction R , acting perpendicular to the plane.
- Component of weight $m_1g \cos \theta$, acting perpendicular to plane.
- Component of weight $m_1g \sin \theta$ downward along the inclined plane.

Here, R and $m_1g \cos \theta$ will cancel each other (as here is no motion in vertically upward and vertical downward direction), so these forces can be ignored.

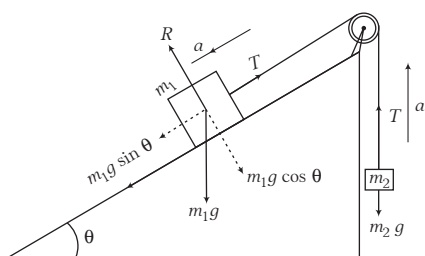


Fig. 5.30

Now, a pulley is connected at the edge of inclined plane and a block of mass m_2 ($m_1 > m_2$) is connected through a string passing over the pulley to mass m_1 as shown in Fig. 5.30.

Now, equation of motion for m_2 ,

$$T - m_2g = m_2a \quad \dots(i)$$

Equation of motion for m_1 ,

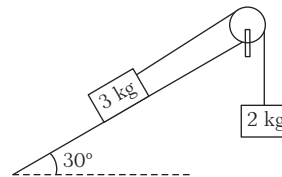
$$m_1g \sin \theta - T = m_1a \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

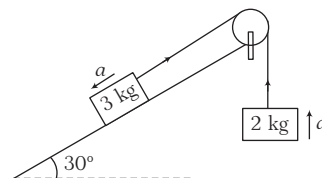
$$a = \frac{(m_1g \sin \theta - m_2g)}{(m_1 + m_2)}$$

$$T = \frac{m_1m_2}{m_1 + m_2} (1 + \sin \theta) g$$

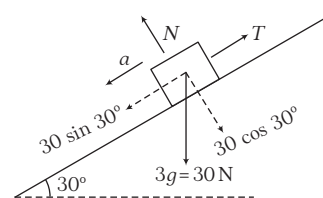
Example 5.43 Consider the situation shown in the figure. The surface is smooth and the string and the pulley are light. Find the acceleration of each block and tension in the string.



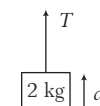
Sol.



Let 3 kg block be moving downward.



$$\text{Along the plane: } 30 \sin 30^\circ - T = 3a \quad \dots(i)$$



$$2g = 20 \text{ N}$$

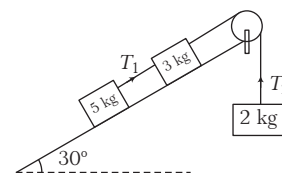
$$T - 20 = 2a \quad \dots(ii)$$

Adding Eqs. (i) and (ii), we get

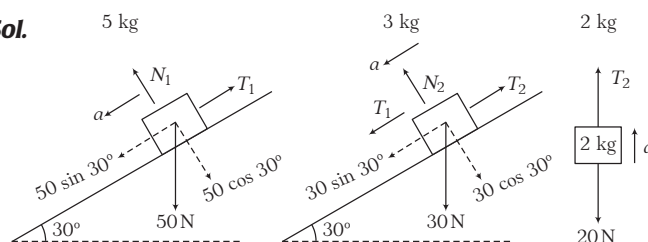
$$a = -1 \text{ ms}^{-2} \Rightarrow T = 18 \text{ N}$$

Since, the acceleration is negative, i.e. the block of mass 3 kg is moving upward.

Example 5.44 In the arrangement shown, inclined plane is smooth, strings and pulleys are massless. Find $\frac{T_1}{T_2}$.



Sol.



$$5 \text{ kg : along the plane : } 50 \sin 30^\circ - T_1 = 5a \quad \dots(i)$$

$$3 \text{ kg : along the plane : } 30 \sin 30^\circ + T_1 - T_2 = 3a \quad \dots(ii)$$

$$2 \text{ kg : } \uparrow : T_2 - 20 = 2a \quad \dots(iii)$$

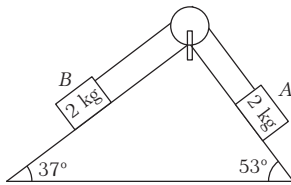
Adding Eqs. (i), (ii) and (iii), we get

$$20 = 10a \Rightarrow a = 2 \text{ ms}^{-2}$$

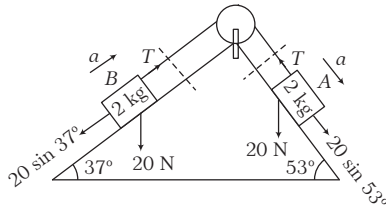
$$T_1 = 15 \text{ N} \quad \text{and} \quad T_2 = 24 \text{ N}$$

$$\therefore \frac{T_1}{T_2} = \frac{15}{24} = \frac{5}{8}$$

Example 5.45 In the arrangement shown, all the surfaces are smooth, strings and pulleys are light. Find the tension in the string.



Sol.



$$\text{Block A : along the plane, } 20 \sin 53^\circ - T = 2a \quad \dots(i)$$

$$\text{Block B : along the plane, } T - 20 \sin 37^\circ = 2a \quad \dots(ii)$$

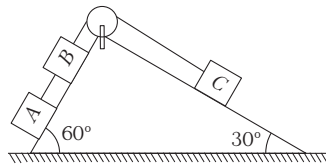
Adding Eqs. (i) and (ii), we get

$$20 (\sin 53^\circ - \sin 37^\circ) = 4a$$

$$20 \left(\frac{4}{5} - \frac{3}{5} \right) = 4 = 4a \Rightarrow a = 1 \text{ ms}^{-2}$$

$$T = 20 \sin 37^\circ + 2a = 20 \times \frac{3}{5} + 2 \times 1 = 14 \text{ N}$$

Example 5.46 In the adjacent figure, masses of A, B and C are 1 kg, 3 kg and 2 kg, respectively. Find



(i) the acceleration of the system

(ii) and tension in the string.

Neglect friction. ($g = 10 \text{ ms}^{-2}$)

Sol. (i) In this case, net pulling force

$$= m_A g \sin 60^\circ + m_B g \sin 60^\circ - m_C g \sin 30^\circ$$

$$= (1)(10) \left(\frac{\sqrt{3}}{2} \right) + (3)(10) \left(\frac{\sqrt{3}}{2} \right) - (2)(10) \left(\frac{1}{2} \right)$$

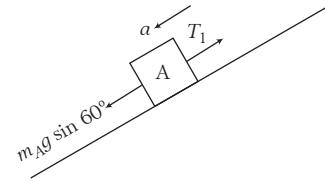
$$= 24.64 \text{ N}$$

$$\text{Total mass being pulled} = 1 + 3 + 2 = 6 \text{ kg}$$

$$\therefore \text{Acceleration of the system, } a = \frac{24.64}{6} = 4.10 \text{ ms}^{-2}$$

(ii) For the tension in the string between A and B

FBD of A



$$m_A g \sin 60^\circ - T_1 = (m_A)(a)$$

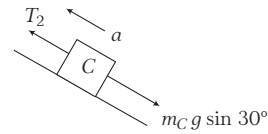
$$\therefore T_1 = m_A g \sin 60^\circ - m_A a$$

$$= m_A (g \sin 60^\circ - a)$$

$$\therefore T_1 = (1) \left(10 \times \frac{\sqrt{3}}{2} - 4.10 \right) = 4.56 \text{ N}$$

For the tension in the string between B and C

FBD of C



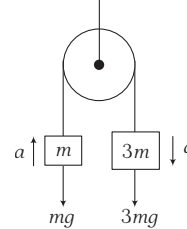
$$T_2 - m_C g \sin 30^\circ = (m_C)(a)$$

$$\therefore T_2 = m_C (a + g \sin 30^\circ)$$

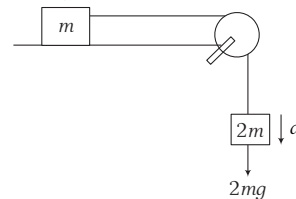
$$\therefore T_2 = 2 \left[4.10 + 10 \left(\frac{1}{2} \right) \right] = 18.2 \text{ N}$$

Example 5.47 Calculate the net acceleration produced in the arrangements shown below.

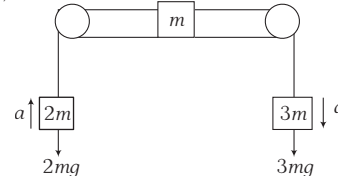
(i)



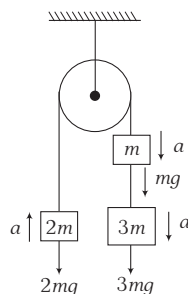
(ii)



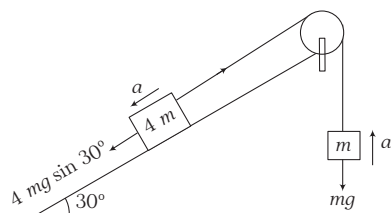
(iii)



(iv)



(v)



Sol. (i) Net pulling force = $3mg - mg = 2mg$
Total mass = $3m + m = 4m$
$$a = \frac{2mg}{4m} = \frac{g}{2}$$

(ii) Pulling force = $2mg$

Total mass = $3m$

$$a = \frac{2mg}{3m} = \frac{2g}{3}$$

(iii) Net pulling force = $3mg - 2mg = mg$

Total mass = $3m + m + 2m = 6m$

$$a = \frac{mg}{6m} = \frac{g}{6}$$

(iv) Net pulling force = $3mg + mg - 2mg$

= $2mg$

Total mass = $3m + m + 2m = 6m$

$$a = \frac{2mg}{6m} = \frac{g}{3}$$

(v) Net pulling force = $4mg \sin 30^\circ - mg$

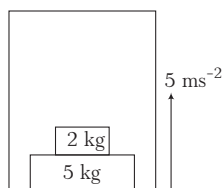
= mg

Total mass = $4m + m = 5m$

$$\Rightarrow a = \frac{mg}{5m} = \frac{g}{5}$$

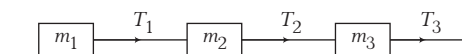
CHECK POINT 5.3

1. Find the force exerted by 5 kg block on floor of lift, as shown in figure. (Take, $g = 10 \text{ ms}^{-2}$)



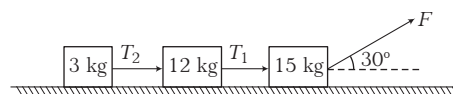
- (a) 100 N (b) 115 N
(c) 105 N (d) 135 N

2. A 50 kg boy stands on a platform having spring scale in a lift that is going down with a constant speed of 3 ms^{-1} . If the lift is brought to rest by a constant deceleration in a distance of 9 m, what does the scale read during this period? (Take, $g = 9.8 \text{ ms}^{-2}$)
(a) 500 N (b) 465 N
(c) 515 N (d) Zero
3. An elevator and its load have a total mass of 800 kg. If the elevator, originally moving downward at 10 ms^{-1} , is brought to rest with constant deceleration in a distance of 25 m, the tension in the supporting cable will be (Take, $g = 10 \text{ ms}^{-2}$)
(a) 8000 N (b) 6400 N
(c) 11200 N (d) 9600 N
4. Three blocks of masses m_1 , m_2 and m_3 are connected by massless strings as shown on a frictionless table. They are pulled with a force $T_3 = 40 \text{ N}$. If $m_1 = 10 \text{ kg}$, $m_2 = 6 \text{ kg}$ and $m_3 = 4 \text{ kg}$, the tension T_2 will be



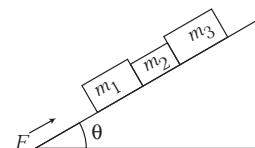
- (a) 20 N (b) 40 N (c) 10 N (d) 32 N

5. The surface is frictionless, the ratio between T_1 and T_2 is



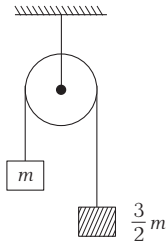
- (a) $\sqrt{3}:1$ (b) $1:\sqrt{3}$ (c) $1:5$ (d) $5:1$

6. Two blocks of masses 2 kg and 1 kg are in contact with each other on a frictionless table. When a horizontal force of 3 N is applied to the block of mass 2 kg, the value of the force of contact between the two blocks is
(a) 4 N (b) 3 N (c) 5 N (d) 1 N
7. Three blocks are placed at rest on a smooth inclined plane with force acting on m_1 parallel to the inclined plane. Find the contact force between m_2 and m_3 .

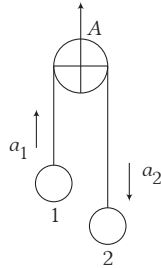


- (a) $\frac{(m_1 + m_2 + m_3) F}{m_3}$ (b) $\frac{m_3 F}{m_1 + m_2 + m_3}$
(c) $F - (m_1 + m_2) g$ (d) None of these

8. In the arrangement shown, the mass m will ascend with an acceleration (pulley and rope are massless)

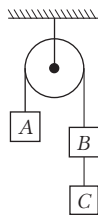


- (a) zero (b) $\frac{g}{5}$
(c) g (d) $2g$
9. Two masses are connected by a string which passes over a pulley accelerating upwards at a rate A as shown in figure. If a_1 and a_2 be the accelerations of bodies 1 and 2 respectively, then



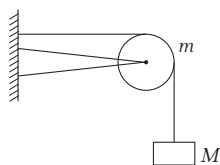
- (a) $A = a_1 - a_2$ (b) $A = a_1 + a_2$
(c) $A = \frac{a_1 - a_2}{2}$ (d) $A = \frac{a_1 + a_2}{2}$

10. Three equal weights A , B and C of mass 2 kg each are hanging on a string passing over a fixed pulley which is frictionless as shown in figure. The tension in the string connecting weight B and C is



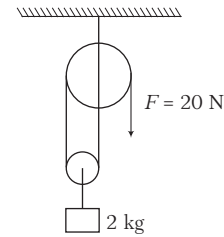
- (a) zero (b) 13 N (c) 3.3 N (d) 19.6 N

11. A light string going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by



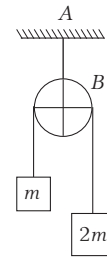
- (a) $\sqrt{2} Mg$ (b) $\sqrt{2} mg$
(c) $g \sqrt{(M + m)^2 + m^2}$ (d) $g \sqrt{(M + m)^2 + M^2}$

12. The acceleration of the 2 kg block, if the free end of string is pulled with a force of 20 N as shown, is



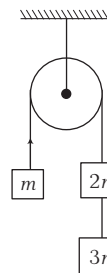
- (a) zero (b) 10 ms^{-2} upward
(c) 5 ms^{-2} upward (d) 5 ms^{-2} downward

13. In the arrangement shown in the figure, the pulley has a mass $3m$. Neglecting friction on the contact surface, the force exerted by the supporting rope AB on the ceiling is



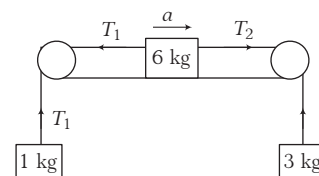
- (a) $6mg$ (b) $3mg$
(c) $4mg$ (d) None of these

14. In the figure given below, with what acceleration does the block of mass m will move? (Pulley and strings are massless and frictionless)



- (a) $\frac{g}{3}$ (b) $\frac{2g}{5}$ (c) $\frac{2g}{3}$ (d) $\frac{g}{2}$

15. Three masses of 1 kg , 6 kg and 3 kg are connected to each other with threads and are placed on a table as shown in figure. What is the acceleration with which the system is moving? (Take, $g = 10\text{ ms}^{-2}$)



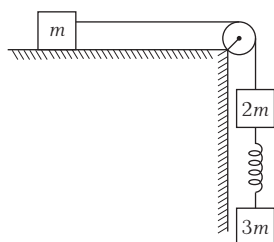
- (a) Zero (b) 2 ms^{-2}
(c) 4 ms^{-2} (d) 3 ms^{-2}

16. Two bodies of masses m_1 and m_2 are connected by a light string which passes over a frictionless massless pulley. If the pulley is moving upward with uniform acceleration $\frac{g}{2}$,

then tension in the string will be

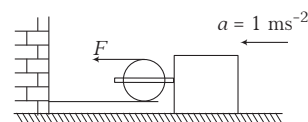
- (a) $\frac{3m_1 m_2}{m_1 + m_2} g$
 (b) $\frac{m_1 + m_2}{4m_1 m_2} g$
 (c) $\frac{2m_1 m_2}{m_1 + m_2} g$
 (d) $\frac{m_1 m_2}{m_1 + m_2} g$

17. In the system shown in figure, assume that all the surfaces are smooth, string and spring are massless. When masses connected are released, the acceleration of the system is



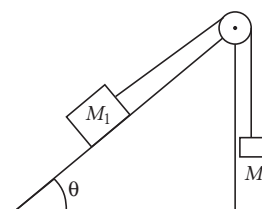
- (a) $\frac{17}{5} \text{ ms}^{-2}$
 (b) $\frac{50}{6} \text{ ms}^{-2}$
 (c) $\frac{60}{5} \text{ ms}^{-2}$
 (d) $\frac{60}{7} \text{ ms}^{-2}$

18. A block of mass 200 kg is set into motion on a frictionless horizontal surface with the help of frictionless pulley and a rope system as shown in figure. What horizontal force F should be applied to produce in the block an acceleration of 1 ms^{-2} ?



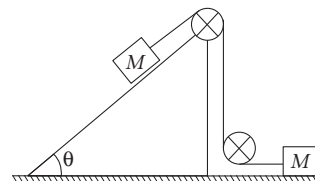
- (a) 50 N
 (b) 100 N
 (c) 200 N
 (d) 10 N

19. Two masses M_1 and M_2 are attached to the ends of a string which passes over a pulley attached to the top of an inclined plane. The angle of inclination of the plane is 30° and $M_1 = 10 \text{ kg}$, $M_2 = 5 \text{ kg}$. What is the acceleration of mass M_2 ?



- (a) 10 ms^{-2}
 (b) 5 ms^{-2}
 (c) Zero
 (d) None of these

20. Two blocks, each having a mass M , rest on frictionless surfaces as shown in the figure. If the pulleys are light and frictionless and M on the incline is allowed to move down, then the tension in the string will be



- (a) $\frac{2}{3} Mg \sin \theta$
 (b) $\frac{3}{2} Mg \sin \theta$
 (c) $\frac{Mg \sin \theta}{2}$
 (d) $2Mg \sin \theta$

FORCE OF FRICTION

Whenever a body moves or tends to move over the surface of another body, a force comes into play which acts parallel to the surface of contact and opposes the relative motion. This opposing force is called **friction**.

Consider a wooden block placed on a horizontal surface and is given a gentle push. The block slides through a small distance and comes to rest. According to Newton's second law, a retarding force must be acting on the block. This retarding force or opposing force is called friction or friction force.

As shown in the figure, the force of friction always acts tangential to the surface in contact and in a direction opposite to the direction of (relative) motion of the body.

If a body is at rest and no pulling force is acting on it, then force of friction on it is zero.

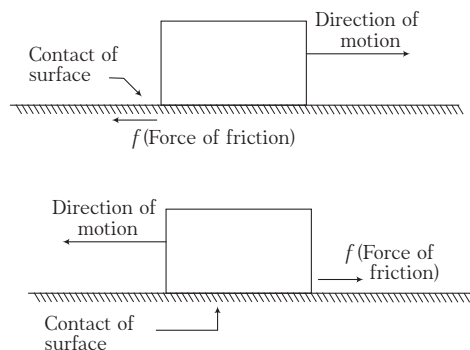


Fig. 5.31 Friction in different directions

Friction between the two bodies depends upon

- (i) **Nature of medium of contact between two bodies**, the roughness of medium increases the friction.
- (ii) **Normal reaction**, as normal reaction increases, the interlocking between two surfaces in contact increases because they press harder against each other, so friction increases.

Types of friction

Friction is mainly of two types

1. Static friction

It is the friction force which comes into existence when one object tends to move over the surface of other object but the actual force has not yet started.

If we keep on increasing the applied force to create movement, then a stage comes when the object is on the verge of moving over the other object.

Friction at this time is called **limiting friction**.

Limiting friction is maximum value of static friction.

Laws of static friction

- (i) The force of static friction is proportional to the normal force exerted by other surface normally on the former.
- (ii) The static frictional force is given by $f_s \leq \mu_s N$, where μ_s is the coefficient of static friction.
- (iii) The less than or equal to sign in the above equation represents the adjusting nature of the force of static friction.
- (iv) The equality sign in the equation holds only when the static frictional force has its maximum value.
- (v) It is independent of the area of contact between the two surfaces.

2. Kinetic friction

The friction force that comes into existence when one object is actually moving over the surface of other object, is called kinetic or dynamic friction.

Laws of kinetic friction

- (i) The force of kinetic friction is proportional to the normal force, which presses the two surfaces together. Mathematically, $f_k \leq \mu_k N$, where μ_k is the coefficient of kinetic friction.
- (ii) It is independent of the surface area of contact.

- (iii) It is independent of the speed of sliding, provided that the resulting heat does not alter the condition of the surface.

Note If more than two blocks are placed one over the other on a horizontal ground, then normal reaction between two blocks will be equal to the weight of the blocks over the common surface.

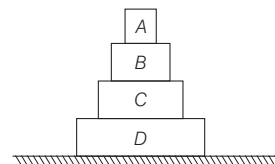


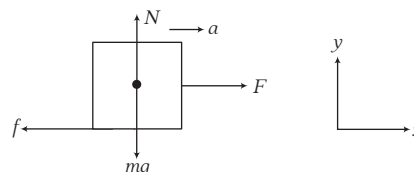
Fig. 5.32 Normal reaction between blocks

e.g. N_1 = normal reaction between A and B = $m_A g$
 N_2 = normal reaction between B and C
 $= (m_A + m_B) g$ and so on.

Example 5.48 Suppose a block of mass 1 kg is placed over a rough surface and a horizontal force F is applied on the block as shown in figure. What are the values of force of friction f , if the force F is gradually increased.

Given that, $\mu_s = 0.5$, $\mu_k = 0.4$ and $g = 10 \text{ ms}^{-2}$.

Sol. Free body diagram of block is



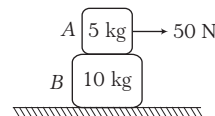
$$\Sigma F_y = 0$$

$$\therefore N - mg = 0 \text{ or } N = mg = (1)(10) = 10 \text{ N}$$

$$f_L = \mu_s N = (0.5)(10) = 5 \text{ N}$$

$$\text{and } f_k = \mu_k N = (0.4)(10) = 4 \text{ N}$$

Example 5.49 Blocks A and B of masses 5 kg and 10 kg are placed as shown in figure. If block A is pulled with 50 N, find out acceleration of the A and B. If coefficient of friction between A and B is 0.5 and between B and ground is 0.4.

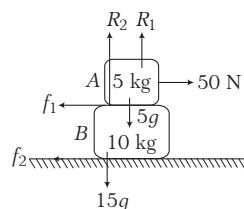


Sol. Limiting friction between A and B,

$$f_1 = \mu_1 R_1 = 0.5 \times 5g = 24.5 \text{ N}$$

Limiting friction between B and ground,

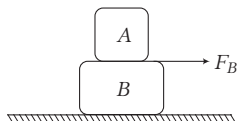
$$f_2 = \mu_2 R_2 = 0.4 \times 15g = 58.8 \text{ N}$$



$$\begin{aligned}\text{For block A, } 50 - f_1 &= 5 \times a \\ \Rightarrow 50 - 24.5 &= 5 \times a \\ \Rightarrow a &= 5.1 \text{ ms}^{-2}\end{aligned}$$

For block B, accelerating force $f_1 = 24.5 \text{ N}$ is less than limiting friction $f_2 = 58.8 \text{ N}$, so block B will remain at rest.

Example 5.50 12 N of force required to be applied on A of mass 4 kg to slip on B of mass 5 kg. Find the maximum horizontal force F to be applied on B, so that A and B move together.



Sol. Let μ be friction coefficient between A and B.

As 12 N force on A is required for slipping, so

$$\begin{aligned}\mu m_A g &= 12 \\ \Rightarrow \mu &= \frac{12}{4 \times 10} = 0.3\end{aligned}$$

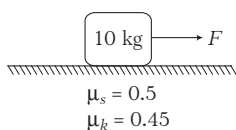
Maximum force (F_B) applied on B, so that A and B move together

$$F_B = (m_A + m_B) a$$

where, $a = \mu g$

$$\begin{aligned}\Rightarrow F_B &= (m_A + m_B) \mu g \\ &= (4 + 5)(0.3)(10) = 27 \text{ N}\end{aligned}$$

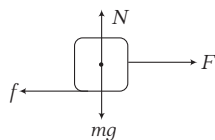
Example 5.51 A body of mass 10 kg is kept on a horizontal floor of coefficient of static friction $\mu_s = 0.5$ and coefficient of kinetic friction $\mu_k = 0.45$ as shown in figure.



Find the acceleration, force of friction and contact force on the body by the plane when the driving force is ($g = 10 \text{ ms}^{-2}$)

(i) 40 N (ii) 60 N

Sol. (i) FBD of the body



Normal reaction, $N = mg = 100 \text{ N}$

Limiting friction on the body,

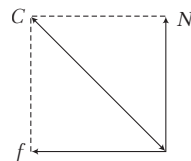
$$f_L = \mu_s N = 0.5 \times 100 \text{ N} = 50 \text{ N}$$

$F = 40 \text{ N}$ is less than the limiting friction, so the body is static, then $a = 0$.

Force of friction acting on the body is static friction,
 $f = \text{driving force} = 40 \text{ N}$

Contact force is the resultant of force of friction and normal reaction, so

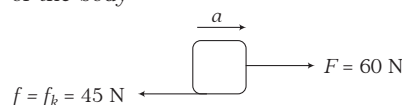
$$C = \sqrt{f^2 + N^2} = \sqrt{(40)^2 + (100)^2} = 107.7 \text{ N}$$



(ii) $F = 60 \text{ N}$ is greater than the limiting friction on the body, so body will start moving. Force of friction acting on the body

$$f_k = \text{kinetic friction} = \mu_k N = 0.45 \times 100 \text{ N} = 45 \text{ N}$$

FBD of the body

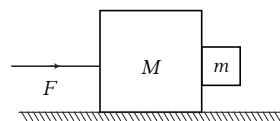


$\therefore$ Acceleration of the body is

$$a = \frac{F - f_k}{m} = \frac{60 - 45}{10} = 1.5 \text{ ms}^{-2}$$

$$\begin{aligned}\text{Contact force, } C &= \sqrt{f^2 + N^2} = \sqrt{f_k^2 + N^2} \\ &= \sqrt{(45)^2 + (100)^2} \approx 109.7 \text{ N}\end{aligned}$$

Example 5.52 In the adjoining figure, the coefficient of friction between wedge (of mass M) and block (of mass m) is μ . Find the minimum horizontal force F required to keep the block stationary with respect to wedge.

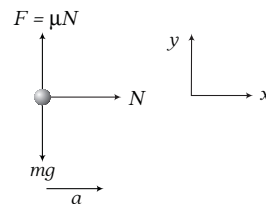


Sol. Such problems can be solved with or without using the concept of pseudo force. Let us solve the problem by both the methods.

a = Acceleration of (wedge + block) in horizontal direction

$$= \frac{F}{M + m}$$

(i) **Inertial frame of reference (ground)** FBD of block with respect to ground (only real forces have to be applied); with respect to ground block is moving with an acceleration a .



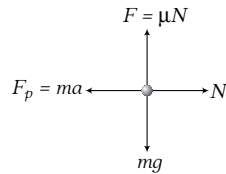
Therefore, $\Sigma F_y = 0$
and $\Sigma F_x = ma$

$$\Rightarrow mg = \mu N \text{ and } N = ma$$

$$\therefore a = \frac{g}{\mu}$$

$$\therefore F = (M + m)a = (M + m) \frac{g}{\mu}$$

(ii) **Non-inertial frame of reference (wedge)** FBD of m with respect to wedge (real + one pseudo force); with respect to wedge, block is stationary.



$$\therefore \Sigma F_x = 0 = \Sigma F_y$$

$$\therefore mg = \mu N \text{ and } N = ma$$

$$\therefore a = \frac{g}{\mu} \text{ and } F = (M + m)a = (M + m) \frac{g}{\mu}$$

From the above discussion, we can see that from both the methods results are same.

Example 5.53 If the coefficient of friction between an insect and bowl is μ and the radius of the bowl is r , find the maximum height to which the insect can crawl up in the bowl.

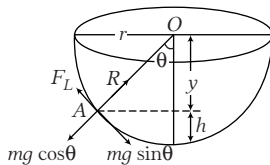
Sol. The insect will crawl up the bowl till the component of its weight along the bowl is balanced by limiting frictional force. So, resolving weight perpendicular to the bowl and along the bowl.

$$R = mg \cos \theta \quad \dots(i)$$

$$F_L = mg \sin \theta \quad \dots(ii)$$

Dividing Eq. (ii) by Eq. (i), we get

$$\tan \theta = \frac{F_L}{R} \text{ or } \tan \theta = \mu \quad (\because F_L = \mu R)$$



$$\text{From } \Delta OAB, \frac{\sqrt{(r^2 - y^2)}}{y} = \mu$$

$$\text{or } y = \frac{r}{\sqrt{1 + \mu^2}}$$

$$\text{So, } h = r - y = r \left[1 - \frac{1}{\sqrt{1 + \mu^2}} \right]$$

Angle of friction (λ)

The angle which the resultant of the force of limiting friction and normal reaction makes with the direction of normal reaction is known as angle of friction.

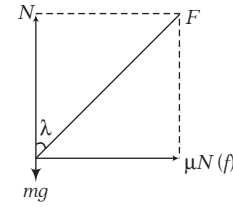


Fig. 5.33 Angle of friction

The resultant of these two forces is F and it makes an angle λ with the normal, where

$$\tan \lambda = \frac{\mu_s N}{N} = \mu_s = \frac{f}{N}$$

or

$$\lambda = \tan^{-1}(\mu_s) \quad \dots(i)$$

or

$$\mu_s = \tan \lambda$$

This angle λ is called the angle of friction.

Thus, the coefficient of static friction is equal to the tangent of the angle of friction.

Angle of repose (α)

The minimum angle of inclination of plane with the horizontal at which the body placed on the plane just begins to slide down, is known as angle of repose.

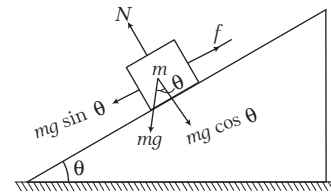


Fig. 5.34 Angle of repose

Suppose a block of mass m is placed on an inclined plane whose inclination θ can be increased or decreased. Let μ be the coefficient of friction between the block and the plane. At any angle of inclination,

Normal reaction, $N = mg \cos \theta$

Limiting friction, $f_L = \mu_s N = \mu_s mg \cos \theta$

and the driving force (or pulling force),

$$F = mg \sin \theta \quad (\text{Down the plane})$$

From these three equations; we see that when θ is increased from 0° to 90° , normal reaction N and hence, the limiting friction f_L is decreased while the driving force F is increased. There is a critical angle called angle of repose (α) at which these two forces are equal.

Now, if θ is further increased, then the driving force F becomes more than the limiting friction f_L and the block starts sliding.

Thus,

$$f_L = F$$

At

$$\theta = \alpha$$

$$\begin{aligned} \text{or } \mu mg \cos \alpha &= mg \sin \alpha \\ \text{or } \tan \alpha &= \mu \\ \text{or } \alpha &= \tan^{-1}(\mu) \end{aligned} \quad \dots(\text{ii})$$

From Eqs. (i) and (ii), we see that angle of friction (λ) is numerically equal to the angle of repose.

$$\text{or } \lambda = \alpha$$

From the above discussion, we can conclude that

- (i) if $\theta < \alpha$, $F < f_L$, the block is stationary.
- (ii) if $\theta = \alpha$, $F = f_L$, the block is on the verge of sliding,
- (iii) if $\theta > \alpha$, $F > f_L$, the block slides down with acceleration.

EQUATION OF MOTION ON A ROUGH INCLINED PLANE

- (i) When body is moving downward due to a acceleration (a) and the force of friction (f) between inclined plane and the body.

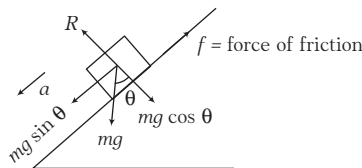


Fig. 5.35 Downward motion of a body (FBD)

$$\text{From FBD, } R = mg \cos \theta$$

As the body is moving downward with acceleration a , then net force in downward direction = ma

$$mg \sin \theta - f = ma$$

$$\text{We know that, } f = \mu R = \mu mg \cos \theta$$

$$\text{Thus, } a = (\sin \theta - \mu \cos \theta) g$$

- (ii) When body is moving upward due to a force F applied on it as shown in the figure.

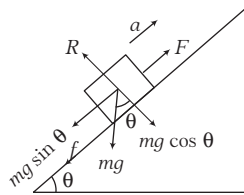


Fig. 5.36 Upward motion of a body (FBD)

$$\text{From FBD, } R = mg \cos \theta \quad \dots(\text{i})$$

$$F = mg \sin \theta + f \quad \dots(\text{ii})$$

$$f = \mu R = \mu mg \cos \theta \quad \dots(\text{iii})$$

From Eqs. (ii) and (iii), we get

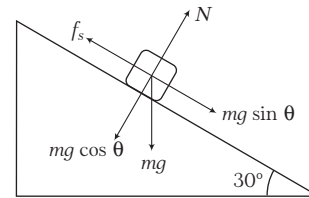
$$\Rightarrow F = mg (\sin \theta + \mu \cos \theta)$$

$$\Rightarrow a = g (\sin \theta + \mu \cos \theta)$$

Note To calculate the work done over a rough inclined surface, multiply force with distance travelled by the body, i.e.
 $W = F \times s = ma \times s = mgs [\sin \theta + \mu \cos \theta]$ (For body moving upwards) and $W = mgs (\sin \theta - \mu \cos \theta)$ (For body moving downwards).

Example 5.54 A block of mass 5 kg rests on an inclined plane at an angle of 30° with the horizontal. If the block just begins to slide, then what is the coefficient of static friction between the block and the surface?

Sol. In equilibrium, the resultant of these forces must be zero.



$$\therefore mg \sin \theta = f_s \quad \dots(\text{i})$$

$$\text{and } mg \cos \theta = N \quad \dots(\text{ii})$$

$$\therefore \tan \theta = \frac{f_s}{N} \quad [\text{Dividing Eq. (i) by Eq. (ii)}]$$

$$\text{As, } f_s = \mu_s N$$

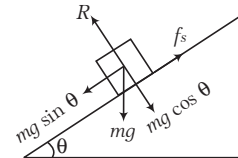
$$\Rightarrow \tan \theta = \mu_s$$

$$\therefore \mu_s = \tan 30^\circ$$

$$\Rightarrow \mu_s = 0.577$$

Example 5.55 A block of wood of 1 kg resting on an inclined plane of angle 30° , just starts moving down. If the coefficient of friction is 0.2, then find its velocity (in ms^{-1}) after 5 s. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Acceleration of block down the plane is given by



For the motion of body along the incline,

$$a = \frac{mg \sin \theta - \mu mg \cos \theta}{m} \quad (\text{As, } R = mg \cos \theta \text{ and } f_s = \mu R)$$

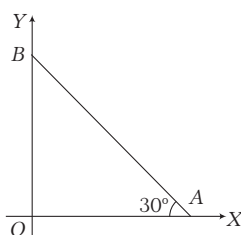
$$= 10 \times \sin 30^\circ - 0.2 \times 10 \cos 30^\circ$$

$$= 10 \times \frac{1}{2} - 2 \times \frac{\sqrt{3}}{2} = 3.268 \text{ ms}^{-2}$$

From first equation of motion, $v = u - at$

$$\text{Velocity after 5 s, } v = 0 + 3.268 \times 5 = 16.34 \text{ ms}^{-1}$$

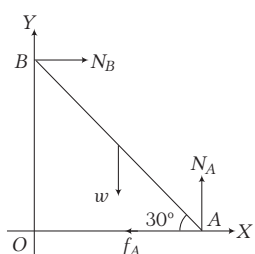
Example 5.56 A rod AB rests with the end A on rough horizontal ground and the end B against a smooth vertical wall. The rod is uniform and of weight w . If the rod is in equilibrium in the position shown in figure.



Find

- (a) frictional force at A ,
 (b) normal reaction at A
 (c) and normal reaction at B .

Sol. Let length of the rod be $2l$. Using the three conditions of equilibrium, anti-clockwise moment is taken as positive.



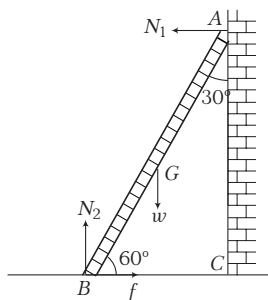
$$\begin{aligned} \therefore \quad \Sigma F_x &= 0 \\ \therefore \quad N_B - f_A &= 0 \\ \text{or} \quad N_B &= f_A \quad \dots(i) \\ \therefore \quad \Sigma F_y &= 0 \\ \therefore \quad N_A - w &= 0 \\ \text{or} \quad N_A &= w \quad \dots(ii) \\ \therefore \quad \Sigma \tau_0 &= 0 \\ \therefore \quad N_A(2l \cos 30^\circ) - N_B(2l \sin 30^\circ) - w(l \cos 30^\circ) &= 0 \\ \text{or} \quad \sqrt{3}N_A - N_B - \frac{\sqrt{3}}{2}w &= 0 \quad \dots(iii) \end{aligned}$$

Solving these three equations, we get

$$(a) f_A = \frac{\sqrt{3}}{2}w \quad (b) N_A = w \quad (c) N_B = \frac{\sqrt{3}}{2}w$$

Example 5.57 A 4 m long ladder weighing 25 kg rests with its upper end against a smooth wall and lower end on rough ground. What should be the minimum coefficient of friction between the ground and the ladder for it to be inclined at 60° with the horizontal without slipping? (Take, $g = 10 \text{ ms}^{-2}$)

Sol. In figure, AB is a ladder of weight w which acts at its centre of gravity G .



$$\angle ABC = 60^\circ$$

$$\angle BAC = 30^\circ$$

$\therefore$

Let N_1 be the reaction of the wall and N_2 be the reaction of the ground.

Force of friction f between the ladder and the ground acts along BC .

For horizontal equilibrium, $f = N_1$...(i)

For vertical equilibrium, $N_2 = w$...(ii)

Taking moments about B , we get for equilibrium,

$$N_1(4 \cos 30^\circ) - w(2 \cos 60^\circ) = 0 \quad \dots(iii)$$

Here,

$$w = 250 \text{ N}$$

Solving these three equations, we get

$$f = 72.17 \text{ N}$$

and

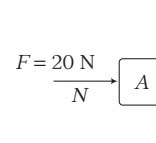
$$N_2 = 250 \text{ N}$$

$$\therefore \quad \mu = \frac{f}{N_2} = \frac{72.17}{250} = 0.288$$

Example 5.58 A block of mass 1 kg is pushed against a rough vertical wall with a force of 20 N (coefficient of static friction being $1/4$). Another horizontal force of 10 N is applied on the block in a direction parallel to the wall. Will the block move? If yes, in which direction? If no, find the frictional force exerted by the wall on the block. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Normal reaction on the block from the wall will be

$$N = F = 20 \text{ N}$$



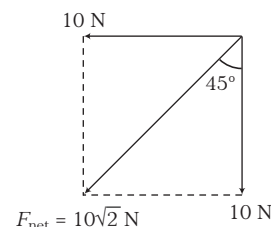
Therefore, limiting friction,

$$f_L = \mu N = \left(\frac{1}{4}\right)(20) = 5 \text{ N}$$

Weight of the block is

$$w = mg = (1)(10) = 10 \text{ N}$$

A horizontal force of 10 N is applied to the block. The resultant of these two forces will be $10\sqrt{2} \text{ N}$ in the direction shown in figure. Since, this resultant is greater than the limiting friction, so the block will move in the direction of $\mathbf{F}_{\text{net}}$ with acceleration,

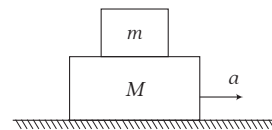


$$a = \frac{F_{\text{net}} - f_L}{m} = \frac{10\sqrt{2} - 5}{1} = 9.14 \text{ ms}^{-2}$$

CHECK POINT 5.4

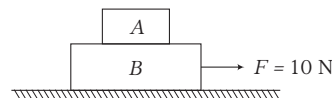
- The limiting value of static friction between two surfaces in contact is
 - proportional to normal force between the surfaces in contact
 - independent of area of contact
 - depends on the microscopic area of constant magnitude
 - All of the above
- A mass placed on an inclined plane is just in equilibrium. If μ is coefficient of friction of the surface, then maximum inclination of the plane with the horizontal is
 - $\tan^{-1} \mu$
 - $\tan^{-1} (\mu/2)$
 - $\sin^{-1} \mu$
 - $\cos^{-1} \mu$
- A 30 kg block rests on a rough horizontal surface. A force of 200 N is applied on the block. The block acquires a speed of 4 ms^{-1} , starting from rest in 2 s. What is the value of coefficient of friction?
 - $\frac{10}{3}$
 - $\frac{\sqrt{3}}{10}$
 - 0.47
 - 0.184
- A car having a mass of 1000 kg is moving at a speed of 30 ms^{-1} . Brakes are applied to bring the car to rest. If the frictional force between the tyres and the road surface is 5000 N, the car will come to rest in
 - 5 s
 - 10 s
 - 12 s
 - 6 s
- A 100 N force acts horizontally on a block of mass 10 kg placed on a horizontal rough table of coefficient of friction $\mu = 0.5$. If g at the place is 10 ms^{-2} , the acceleration of the block is
 - zero
 - 10 ms^{-2}
 - 5 ms^{-2}
 - 5.2 ms^{-2}
- A block of mass 2 kg is placed on the floor. The coefficient of static friction is 0.4. If a force of 2.8 N is applied on the block parallel to the floor, the force of friction between the block and floor is (Take, $g = 10 \text{ ms}^{-2}$)
 - 2.8 N
 - 8 N
 - 2 N
 - zero
- A body is projected along a rough horizontal surface with a velocity 6 ms^{-1} . If the body comes to rest after travelling 9 m, then coefficient of sliding friction is (Take, $g = 10 \text{ ms}^{-2}$)
 - 0.5
 - 0.4
 - 0.6
 - 0.2
- The coefficient of friction between the tyres and road is 0.4. The minimum distance covered before attaining a speed of 8 ms^{-1} starting from rest is nearly (Take, $g = 10 \text{ ms}^{-2}$)
 - 8 m
 - 4 m
 - 10 m
 - 16 m

- A block is gently placed on a conveyor belt moving horizontally with constant speed. After $t = 4 \text{ s}$, the velocity of the block becomes equal to the velocity of the belt. If the coefficient of friction between the block and the belt is $\mu = 0.2$, then the velocity of the conveyor belt is
 - 8 ms^{-1}
 - 4 ms^{-1}
 - 6 ms^{-1}
 - 18 ms^{-1}
- The breaking strength of the cable used to pull a body is 40 N. A body of mass 8 kg is resting on a table of coefficient of friction $\mu = 0.2$. The maximum acceleration which can be produced by the cable connected to the body is (Take, $g = 10 \text{ ms}^{-2}$)
 - 6 ms^{-2}
 - 3 ms^{-2}
 - 8 ms^{-2}
 - 8 ms^{-2}
- A block of mass m is placed on the top of another block of mass M as shown in the figure. The coefficient of friction between them is μ .



The maximum acceleration with which the block M may move, so that m also moves along with it, is

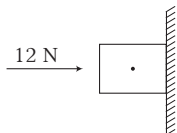
- μg
 - $\mu \frac{M}{m} g$
 - $\mu \frac{m}{M} g$
 - $\frac{g}{\mu}$
- In the shown arrangement, mass of $A = 1 \text{ kg}$ and mass of $B = 2 \text{ kg}$. Coefficient of friction between A and $B = 0.2$.



There is no friction between B and ground. The frictional force exerted by A on B equals

- 2 N
 - 3 N
 - 4 N
 - 5 N
- A block of mass 4 kg is placed on a rough horizontal plane. A time dependent force $F = kt^2$ acts on the block, where $k = 2 \text{ N s}^{-2}$ and coefficient of friction $\mu = 0.8$. Force of friction between block and the plane at $t = 2 \text{ s}$ is
 - 8 N
 - 4 N
 - 2 N
 - 32 N

- 14.** A block of weight 5 N is pushed against a vertical wall by a force 12 N. The coefficient of friction between the wall and block is 0.6. The magnitude of the force exerted by the wall on the block is



- (a) 12 N
(c) 7.2 N
- (b) 5 N
(d) 13 N
- 15.** A body of mass 10 kg is placed on rough surface, pulled by a force F making an angle of 30° above the horizontal. If the angle of friction is also 30° , then the minimum magnitude of force F required to move the body is equal to (Take, $g = 10\text{ms}^{-2}$)
- (a) 100 N
(c) $100\sqrt{2}$ N
- (b) $50\sqrt{2}$ N
(d) 50 N
- 16.** A block of mass 2 kg rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient

of static friction between the block and the plane is 0.7. The frictional force on the block is

- (a) 9.8 N
(c) $9.8 \times \sqrt{3}$ N
- (b) $0.7 \times 9.8 \times \sqrt{3}$ N
(d) 0.7×9.8 N
- 17.** A minimum force F is applied to a block of mass 102 kg to prevent it from sliding on a plane with an inclination angle 30° with the horizontal. If the coefficients of static and kinetic friction between the block and the plane are 0.4 and 0.3 respectively, then the force F is
- (a) 157 N
(c) 315 N
- (b) 224 N
(d) zero
- 18.** A box of mass 8 kg is placed on a rough inclined plane of inclination θ . Its downward motion can be prevented by applying an upward pull F and it can be made to slide upwards by applying a force $2F$. The coefficient of friction between the box and the inclined plane is
- (a) $\frac{1}{3} \tan \theta$
(c) $\frac{1}{2} \tan \theta$
- (b) $3 \tan \theta$
(d) $2 \tan \theta$

Chapter Exercises

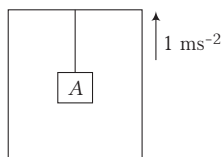
(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1** A metre scale is moving with uniform velocity. This implies
- the force acting on the scale is zero, but a torque about the centre of mass can act on the scale
 - the force acting on the scale is zero and the torque acting about centre of mass of the scale is also zero
 - the total force acting on it need not be zero but the torque on it is zero
 - Neither the force nor the torque need to be zero

- 2** Conservation of momentum in a collision between particles can be understood from
- conservation of energy
 - Newton's first law only
 - Newton's second law only
 - Both Newton's second and third law

- 3** If the elevator in the shown figure is moving upwards with constant acceleration 1 ms^{-2} , the tension in the string connected to block A of mass 6 kg would be



(Take, $g = 10 \text{ ms}^{-2}$)

- 60 N
 - 66 N
 - 54 N
 - 42 N
- 4** A block is placed on the top of a smooth inclined plane of inclination θ kept on the floor of a lift. When the lift is descending with a retardation a , the block is released. The acceleration of the block relative to the incline is
- $g \sin \theta$
 - $a \sin \theta$
 - $(g - a) \sin \theta$
 - $(g + a) \sin \theta$
- 5** A cricket ball of mass 150 g has an initial velocity $\mathbf{u} = (3\hat{i} + 4\hat{j}) \text{ ms}^{-1}$ and a final velocity $\mathbf{v} = -(3\hat{i} + 4\hat{j}) \text{ ms}^{-1}$, after being hit. The change in momentum (final momentum – initial momentum) is (in $\text{kg}\cdot\text{ms}^{-1}$)
- zero
 - $-(0.45\hat{i} + 0.6\hat{j})$
 - $-(0.9\hat{i} + 1.2\hat{j})$
 - $-5(\hat{i} + \hat{j})$
- 6** In the previous question (5), the magnitude of the momentum transferred during the hit is
- zero
 - $0.75 \text{ kg}\cdot\text{ms}^{-1}$
 - $1.5 \text{ kg}\cdot\text{ms}^{-1}$
 - $14 \text{ kg}\cdot\text{ms}^{-1}$

- 7** A block of mass m is placed on a smooth plane inclined at an angle θ with the horizontal. The force exerted by the plane on the block has a magnitude
- mg
 - $mg \sec \theta$
 - $mg \cos \theta$
 - $mg \sin \theta$

- 8.** A block has been placed on an inclined plane. The slope angle θ of the plane is such that the block slides down the plane at a constant speed. The coefficient of kinetic friction is equal to
- $\sin \theta$
 - $\cos \theta$
 - g
 - $\tan \theta$

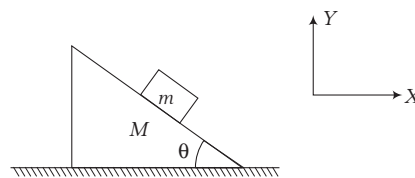
- 9** A body is projected up a 45° rough incline. If the coefficient of friction is 0.5, then the retardation of the block is

- $\frac{g}{2\sqrt{2}}$
- $\frac{g}{\sqrt{2}}$
- $\frac{3g}{2\sqrt{2}}$
- $\frac{g}{2}$

- 10** A rope of length L and mass M is hanging from a rigid support. The tension in the rope at a distance x from the rigid support is

- Mg
- $\left(\frac{L-x}{L}\right)Mg$
- $\left(\frac{L}{L-x}\right)Mg$
- $\frac{x}{L}Mg$

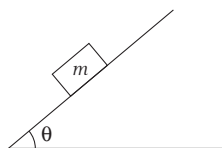
- 11** Consider the shown arrangement. Assume all surfaces to be smooth. If N represents magnitudes of normal reaction between block and wedge, then acceleration of M along horizontal is equal to



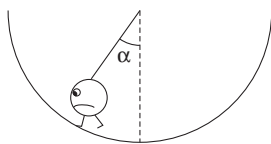
- $\frac{N \sin \theta}{M}$ along +ve X-axis
- $\frac{N \cos \theta}{M}$ along -ve X-axis
- $\frac{N \sin \theta}{M}$ along -ve X-axis
- $\frac{N \sin \theta}{m+M}$ along -ve X-axis

- 12** In the above problem, normal reaction between ground and wedge will have magnitude equal to
- $N \cos \theta + Mg$
 - $N \cos \theta + Mg + mg$
 - $N \cos \theta - Mg$
 - $N \sin \theta + Mg + mg$

- 13** A block of mass m is at rest on an inclined plane which is making angle θ with the horizontal. The coefficient of friction between the block and plane is μ . Then, frictional force acting between the surfaces is

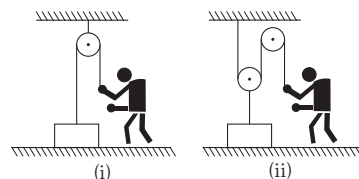


- (a) μmg (b) $\mu mg \sin \theta$
 (c) $\mu (mg \sin \theta - mg \cos \theta)$ (d) $mg \sin \theta$
- 14** If a body loses half of its velocity on penetrating 3 cm in a wooden block, then how much will it penetrate more before coming to rest?
- (a) 1 cm (b) 2 cm
 (c) 3 cm (d) 4 cm
- 15** An insect crawls up a hemispherical surface very slowly (see the figure). The coefficient of friction between the insect and the surface is $1/3$. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α is given by

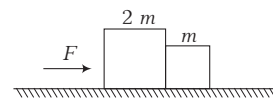


- (a) $\cot \alpha = 3$ (b) $\tan \alpha = 3$
 (c) $\sec \alpha = 3$ (d) $\operatorname{cosec} \alpha = 3$
- 16** A hockey player is moving northward and suddenly turns westward with the same speed to avoid an opponent. The force that acts on the player is
- (a) frictional force along westward
 (b) muscle force along southward
 (c) frictional force along south-west
 (d) muscle force along south-west
- 17** A car of mass m starts from rest and acquires a velocity along east $\mathbf{v} = v\hat{i}$ ($v > 0$) in two seconds. Assuming the car moves with uniform acceleration, the force exerted on the car is
- (a) $\frac{mv}{2}$ eastward and is exerted by the car engine
 (b) $\frac{mv}{2}$ eastward and is due to the friction on the tyres exerted by the road
 (c) more than $\frac{mv}{2}$ eastward exerted due to the engine and overcomes the friction of the road
 (d) $\frac{mv}{2}$ exerted by the engine

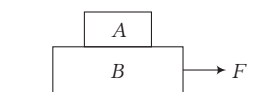
- 18** In the figure shown, a person wants to raise a block lying on the ground to a height h . In both the cases, if time required is the same, then in which case, he has to exert more force. (Assume pulleys and strings are light)



- (a) (i) (b) (ii)
 (c) Same in both (d) Cannot be determined
- 19** A body of mass 2 kg travels according to the relation $x(t) = pt + qt^2 + rt^3$, where $q = 4 \text{ ms}^{-2}$, $p = 3 \text{ ms}^{-1}$ and $r = 5 \text{ ms}^{-3}$. The force acting on the body at $t = 2 \text{ s}$ is
- (a) 136 N (b) 134 N (c) 158 N (d) 68 N
- 20** A body with mass 5 kg is acted upon by a force $\mathbf{F} = (-3\hat{i} + 4\hat{j}) \text{ N}$. If its initial velocity at $t = 0$ is $\mathbf{u} = (6\hat{i} - 12\hat{j}) \text{ ms}^{-1}$, the time at which it will just have a velocity along the Y-axis is
- (a) zero (b) 10 s (c) 2 s (d) 15 s
- 21** A 5000 kg rocket is set for vertical firing. The exhaust speed is 800 ms^{-1} . To give an initial upward acceleration of 20 ms^{-2} , the amount of gas ejected per second to supply the needed thrust will be (Take, $g = 10 \text{ ms}^{-2}$)
- (a) 127.5 kgs^{-1} (b) 187.5 kgs^{-1}
 (c) 185.5 kgs^{-1} (d) 137.5 kgs^{-1}
- 22** Two blocks are in contact on a frictionless table. One has mass m and the other $2m$. A force F is applied on $2m$ as shown in the figure. Now, the same force F is applied from the right on m . In the two cases, the ratio of force of contact between the two blocks will be

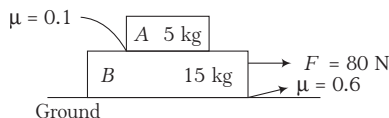


- (a) same (b) 1 : 2 (c) 2 : 1 (d) 1 : 3
- 23** A 4 kg block A is placed on the top of 8 kg block B which rests on a smooth table.

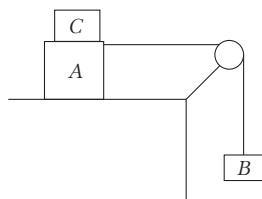


- A just slips on B when a force of 12 N is applied on A. Then, the maximum horizontal force F applied on B to make both A and B move together, is
- (a) 12 N (b) 24 N (c) 36 N (d) 48 N

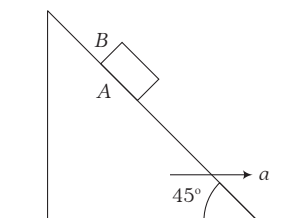
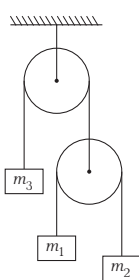
- 24** Find the value of friction forces between the blocks A and B ; and between B and ground. (Take, $g = 10 \text{ ms}^{-2}$)



- (a) 90 N, 5 N (b) 5 N, 90 N (c) 5 N, 75 N (d) 0 N, 80 N
- 25** A block of mass 5 kg is kept on a horizontal floor having coefficient of friction 0.09. Two mutually perpendicular horizontal forces of 3 N and 4 N act on this block. The acceleration of the block is (Take, $g = 10 \text{ ms}^{-2}$)
- (a) zero (b) 0.1 ms^{-2} (c) 0.2 ms^{-2} (d) 0.3 ms^{-2}
- 26** Two masses A and B of 10 kg and 5 kg respectively are connected with a string passing over a frictionless pulley fixed at the corner of a table as shown in figure. The coefficient of friction of A with the table is 0.2. The minimum mass of C that may be placed on A to prevent it from moving is equal to

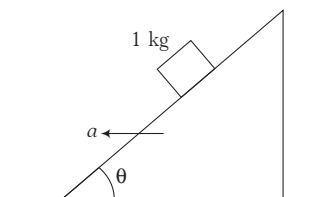


- (a) 15 kg (b) 10 kg
(c) 5 kg (d) 20 kg
- 27** In the figure, pulleys are smooth and strings are massless, $m_1 = 1 \text{ kg}$ and $m_2 = \frac{1}{3} \text{ kg}$. To keep m_3 at rest, mass m_3 should be
- (a) 1 kg (b) $\frac{2}{3} \text{ kg}$
(c) $\frac{1}{4} \text{ kg}$ (d) 2 kg
- 28** If the coefficient of friction between A and B is μ , the maximum acceleration of the wedge A for which B will remain at rest with respect to the wedge is



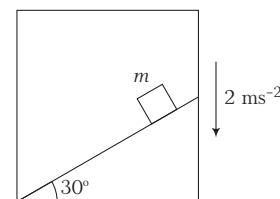
- (a) μg (b) $g \left(\frac{1+\mu}{1-\mu} \right)$ (c) $g \left(\frac{1-\mu}{1+\mu} \right)$ (d) $\frac{g}{\mu}$

- 29** A block of mass 1 kg is at rest relative to a smooth wedge moving leftwards with constant acceleration $a = 5 \text{ ms}^{-2}$.



Let N be the normal reaction between the block and the wedge. Then, (Take, $g = 10 \text{ ms}^{-2}$)

- (a) $N = 5\sqrt{5} \text{ N}$ (b) $N = 15 \text{ N}$
(c) $\tan \theta = \frac{1}{3}$ (d) $\tan \theta = 2$
- 30** A block of mass m is kept on an inclined plane of a lift moving down with acceleration of 2 ms^{-2} . What should be the coefficient of friction to let the block move down with constant velocity relative to lift?

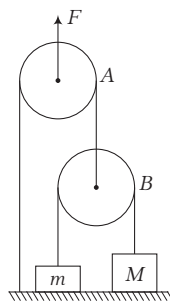


- (a) $\mu = \frac{1}{\sqrt{3}}$ (b) $\mu = 0.4$ (c) $\mu = 0.8$ (d) $\mu = \frac{\sqrt{3}}{2}$
- 31** Two blocks of mass 5 kg and 3 kg are attached to the ends of a string passing over a smooth pulley fixed to the ceiling of an elevator. The elevator is accelerated upwards. If the acceleration of the blocks is $\frac{9}{32}g$, the acceleration of the elevator is
- (a) $\frac{g}{3}$ (b) $\frac{g}{4}$ (c) $\frac{g}{8}$ (d) $\frac{g}{6}$
- 32** A balloon of weight w is falling vertically downward with a constant acceleration a ($< g$). The magnitude of the air resistance is
- (a) w (b) $w \left(1 + \frac{a}{g} \right)$
(c) $w \left(1 - \frac{a}{g} \right)$ (d) $w \frac{a}{g}$
- 33** A smooth inclined plane of length L , having an inclination θ with horizontal is inside a lift which is moving down with retardation a . The time taken by a block to slide down the inclined plane from rest will be

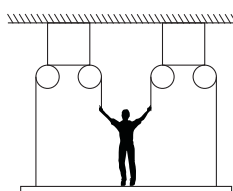
- (a) $\sqrt{\frac{2L}{\sqrt{a \sin \theta}}}$ (b) $\sqrt{\frac{2L}{g \sin \theta}}$
 (c) $\sqrt{\frac{2L}{(g-a) \sin \theta}}$ (d) $\sqrt{\frac{2L}{(g+a) \sin \theta}}$

- 34** A body of mass M at rest explodes into three pieces, two of which of mass $M/4$ each are thrown off in perpendicular directions with velocities of 3 ms^{-1} and 4 ms^{-1} , respectively. The third piece will be thrown off with a velocity of
 (a) 1.5 ms^{-1} (b) 2 ms^{-1} (c) 2.5 ms^{-1} (d) 3 ms^{-1}
- 35** A wooden box of mass 8 kg slides down an inclined plane of inclination 30° to the horizontal with a constant acceleration of 0.4 ms^{-2} . What is the force of friction between the box and inclined plane? (Take, $g = 10 \text{ ms}^{-2}$)
 (a) 36.8 N (b) 76.8 N
 (c) 65.6 N (d) None of these
- 36** A mass of 3 kg descending vertically downwards supports a mass of 2 kg by means of a light string passing over a pulley. At the end of 5 s , the string breaks. How much high from now the 2 kg mass will go? (Take, $g = 9.8 \text{ ms}^{-2}$)
 (a) 4.9 m (b) 9.8 m (c) 19.6 m (d) 2.45 m
- 37** The rear side of a truck is open and a box of mass 20 kg is placed on the truck 4 m away from the open end ($\mu = 0.15$ and $g = 10 \text{ ms}^{-2}$). The truck starts from rest with an acceleration of 2 ms^{-2} on a straight road. The box will fall off the truck when it is at a distance from the starting point equal to
 (a) 4 m (b) 8 m
 (c) 16 m (d) 32 m

- 38** Two blocks of masses $m = 5 \text{ kg}$ and $M = 10 \text{ kg}$ are connected by a string passing over a pulley B as shown. Another string connects the centre of pulley B to the floor and passes over another pulley A as shown. An upward force F is applied at the centre of pulley A . Both the pulleys are massless. The accelerations of blocks m and M , if F is 300 N are (Take, $g = 10 \text{ ms}^{-2}$)

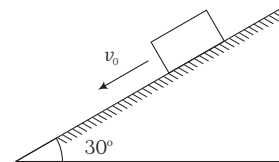


- (a) 5 ms^{-2} , zero (b) zero, 5 ms^{-2}
 (c) zero, zero (d) 5 ms^{-2} , 5 ms^{-2}
- 39** A man of mass m stands on a platform of equal mass m and pulls himself by two ropes passing over pulleys as shown in figure. If he pulls each rope with a force equal to half his



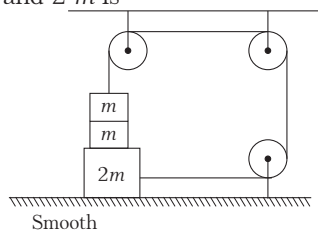
weight, his upward acceleration would be

- (a) $\frac{g}{2}$ (b) $\frac{g}{4}$
 (c) g (d) zero
- 40** A 40 N block supported by two ropes. One rope is horizontal and the other makes an angle of 30° with the ceiling. The tension in the rope attached to the ceiling is approximately
 (a) 80 N (b) 40 N
 (c) $40\sqrt{3} \text{ N}$ (d) $\frac{40}{\sqrt{3}} \text{ N}$
- 41** Starting from rest, a body slides down a 45° inclined plane in twice the time, it takes to slide down the same distance in the absence of friction. The coefficient of friction between the body and the inclined plane is
 (a) 0.2 (b) 0.25 (c) 0.75 (d) 0.5
- 42** A block of mass 0.1 kg is held against a wall applying a horizontal force of 5 N on the block. If the coefficient of friction between the block and the wall is 0.5 , the magnitude of the frictional force acting the block is
 (a) 2.5 N (b) 0.98 N (c) 4.9 N (d) 0.49 N
- 43** A block of mass m is given an initial downward velocity v_0 and left on an inclined plane (coefficient of friction $= 0.6$). The block will



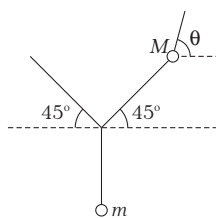
- (a) continue of move down the plane with constant velocity v_0
 (b) accelerate downward
 (c) decelerate and come to rest
 (d) first accelerate downward then decelerate
- 44** Pushing force making an angle θ to the horizontal is applied on a block of weight w placed on a horizontal table. If the angle of friction is ϕ , the magnitude of force required to move the body is equal to
 (a) $\frac{w \cos \phi}{\cos(\theta - \phi)}$ (b) $\frac{w \sin \phi}{\cos(\theta + \phi)}$
 (c) $\frac{w \tan \phi}{\sin(\theta - \phi)}$ (d) $\frac{w \sin \phi}{\tan(\theta - \phi)}$
- 45** In the arrangement shown in figure, there is a friction force between the blocks of masses m and $2m$ kept on a smooth horizontal surface. The mass of the suspended block is m . The block of mass m is stationary with respect to block of mass $2m$.

The minimum value of coefficient of friction between m and $2m$ is



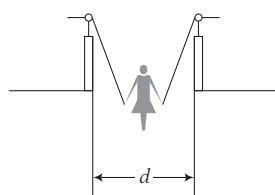
- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$

- 46** Two masses m and M are attached with strings as shown. For the system to be in equilibrium, we have



- (a) $\tan \theta = 1 + \frac{2M}{m}$ (b) $\tan \theta = 1 + \frac{2m}{M}$
 (c) $\tan \theta = 1 + \frac{M}{2m}$ (d) $\tan \theta = 1 + \frac{m}{2M}$

- 47** A man of mass m has fallen into a ditch of width d . Two of his friends are slowly pulling him out using a light rope and two fixed pulleys as shown in figure. Both the friends exert force of equal magnitudes F .



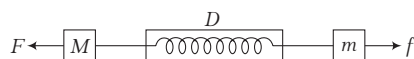
When the man is at a depth h , the value of F is

- (a) $\frac{mg}{4h} \sqrt{d^2 + 4h^2}$ (b) hmg
 (c) dmg (d) $\frac{mg}{2h} \sqrt{h^2 + d^2}$

- 48** Two weights w_1 and w_2 are suspended from the ends of a light string passing over a smooth fixed pulley. If the pulley is pulled up with an acceleration g . The tension in the string will be

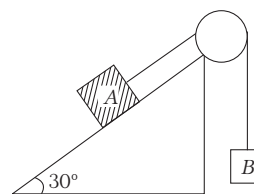
- (a) $\frac{4w_1 w_2}{w_1 + w_2}$ (b) $\frac{2w_1 w_2}{w_1 + w_2}$ (c) $\frac{w_1 - w_2}{w_1 + w_2}$ (d) $\frac{w_1 w_2}{2(w_1 + w_2)}$

- 49** A dynamometer D is attached to two bodies of masses $M = 6 \text{ kg}$ and $m = 4 \text{ kg}$. Forces $F = 20 \text{ N}$ and $f = 10 \text{ N}$ are applied to the masses as shown. The dynamometer reads



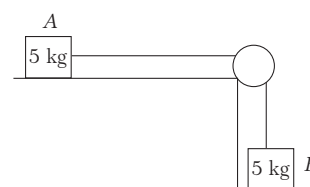
- (a) 10 N (b) 20 N (c) 6 N (d) 14 N

- 50** Two blocks are connected over a massless pulley as shown in figure. The mass of block A is 10 kg and the coefficient of kinetic friction is 0.2. Block A slides down the incline at constant speed. The mass of block B (in kg) is



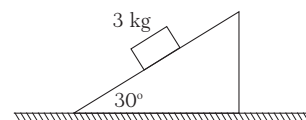
- (a) 5.4 (b) 3.3 (c) 4.2 (d) 6.8

- 51** A block of mass 5 kg resting on a horizontal surface is connected by a cord, passing over a light frictionless pulley to a hanging block of mass 5 kg. The coefficient of kinetic friction between the block and the surface is 0.5. Tension in the cord is (Take, $g = 9.8 \text{ ms}^{-2}$)



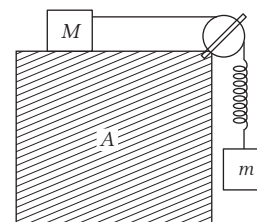
- (a) 49 N (b) zero (c) 36.75 N (d) 12.75 N

- 52** A block of mass 3 kg is at rest on a rough inclined plane as shown in the figure. The magnitude of net force exerted by the surface on the block will be



- (a) $15\sqrt{3} \text{ N}$ (b) 15 N (c) 10 N (d) 30 N

- 53** Two blocks of masses M and m are connected to each other by a massless string and spring of force constant k as shown in the figure. The spring passes over a frictionless pulley connected rigidly to the edge of a stationary block A. The coefficient of friction between block M and the plane horizontal surface of A is μ . The block M slides over the horizontal top surface of A and block m slides vertically downwards with the same speed. The mass M is equal to

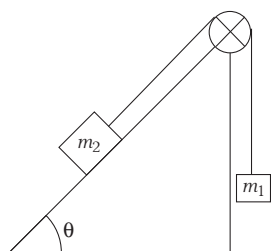


- (a) $\frac{m}{\mu}$ (b) $\frac{2m}{\mu}$ (c) $m\mu$ (d) μmg

- 54** A block of mass m , lying on a rough horizontal plane, is acted upon by a horizontal force P and another force Q , inclined at an angle θ to the vertical upwards. The block will remain in equilibrium, if minimum coefficient of friction between it and the surface is

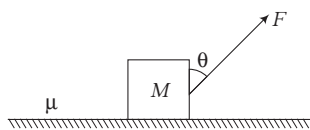
(a) $\frac{P + Q \sin \theta}{(mg - Q \cos \theta)}$ (b) $\frac{P \cos \theta + Q}{(mg - Q \sin \theta)}$
 (c) $\frac{P - Q \cos \theta}{(mg + Q \sin \theta)}$ (d) $\frac{P \sin \theta - Q}{(mg - Q \cos \theta)}$

- 55** In the figure shown, if coefficient of friction is μ , then m_2 will start moving upwards, if



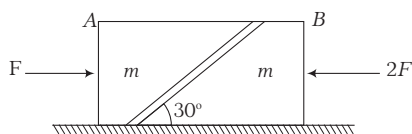
(a) $\frac{m_1}{m_2} > \sin \theta - \mu \cos \theta$ (b) $\frac{m_1}{m_2} > \sin \theta + \mu \cos \theta$
 (c) $\frac{m_1}{m_2} > \mu \sin \theta - \cos \theta$ (d) $\frac{m_1}{m_2} > \mu \sin \theta + \cos \theta$

- 56** A block of mass M rests on a rough horizontal surface as shown. Coefficient of friction between the block and the surface is μ . A force $F = Mg$ acting at angle θ with the vertical side of the block pulls it. In which of the following cases, the block can be pulled along the surface?



(a) $\tan \theta \geq \mu$ (b) $\tan \left(\frac{\theta}{2} \right) \geq \mu$
 (c) $\cos \theta \geq \mu$ (d) $\cot \left(\frac{\theta}{2} \right) \geq \mu$

- 57** Two blocks A and B each of mass m are placed on a smooth horizontal surface. Two horizontal force F and $2F$ are applied on the blocks A and B respectively as shown in figure. The block A does not slide on block B .



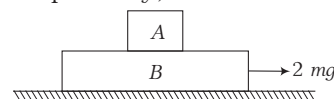
Then, the normal reaction acting between the two blocks is

(a) F (b) $\frac{F}{2}$ (c) $\frac{F}{\sqrt{3}}$ (d) $3F$

- 58** A uniform rope of length l lies on a table. If the coefficient of friction is μ , then the maximum length l_1 of the hanging part of the rope which can overhang from the edge of the table without sliding down is

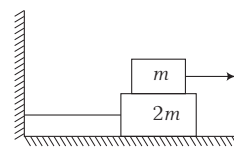
(a) l/μ (b) $l/(\mu + 1)$
 (c) $\mu l/(\mu + 1)$ (d) $\mu l/(\mu - 1)$

- 59** Block A of mass m rests on the plank B of mass $3m$ which is free to slide on a frictionless horizontal surface. The coefficient of friction between the block and plank is 0.2 . If a horizontal force of magnitude $2mg$ is applied to the plank B , the acceleration of A relative to the plank and relative to the ground respectively, are



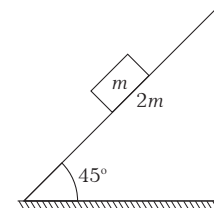
(a) $0, \frac{g}{2}$ (b) $0, \frac{2g}{3}$ (c) $\frac{3g}{5}, \frac{g}{5}$ (d) $\frac{2g}{5}, \frac{g}{5}$

- 60** Two blocks of masses m and $2m$ are placed one over the other as shown in figure. The coefficient of friction between m and $2m$ is μ and between $2m$ and ground is $\frac{\mu}{3}$. If a horizontal force F is applied on upper block and T is tension developed in string, then choose the incorrect alternative.



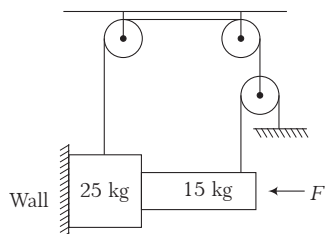
(a) If $F = \frac{\mu}{3} mg$, $T = 0$ (b) If $F = \mu mg$, $T = 0$
 (c) If $F = 2\mu mg$, $T = \frac{\mu mg}{3}$ (d) If $F = 3\mu mg$, $T = 0$

- 61** A block of mass m is placed on a wedge of mass $2m$ which rests on a rough horizontal surface. There is no friction between the block and the wedge. The minimum coefficient of friction between the wedge and the ground, so that the wedge does not move, is

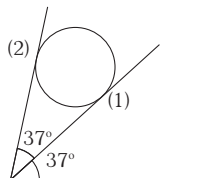


(a) 0.1 (b) 0.2
 (c) 0.3 (d) 0.4

- 62** If the coefficient of friction between all surfaces is 0.4, then find the minimum force F to have equilibrium of the system. (Take, $g = 10 \text{ ms}^{-2}$)

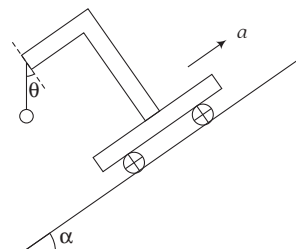


- (a) 62.5 N (b) 150 N (c) 135 N (d) 50 N
- 63** A sphere of mass m is held between two smooth inclined walls. For $\sin 37^\circ = \frac{3}{5}$, the normal reaction of the wall (2) is equal to



- (a) $\frac{16mg}{25}$ (b) $\frac{25mg}{39}$
(c) $\frac{39mg}{21}$ (d) None of these

- 64** A pendulum of mass m hangs from a support fixed to a trolley. The direction of the string (i.e. angle θ) when the trolley rolls up a plane of inclination α with acceleration a is



- (a) 0 (b) $\tan^{-1} \alpha$
(c) $\tan^{-1} \left(\frac{a + g \sin \alpha}{g \cos \alpha} \right)$ (d) $\tan^{-1} \frac{a}{g}$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-6) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) If Assertion is true but Reason is false.
(d) If Assertion is false but Reason is true.

1 Assertion When a person walks on a rough surface, the net force exerted by surface on the person is in the direction of his motion.

Reason It is the force exerted by the road on the person that causes the motion.

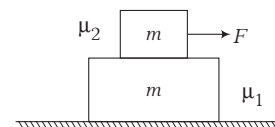
2 Assertion A body of mass 10 kg is placed on a rough inclined surface ($\mu = 0.7$). The surface is inclined to horizontal at angle 30° . Acceleration of the body down the plane will be zero.

Reason Force of friction is zero.

3 Assertion Static friction acting on a body is always greater than the kinetic friction acting on this body.

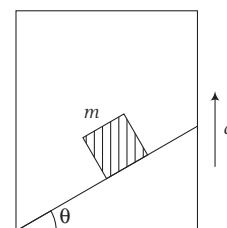
Reason Coefficient of static friction is more than the coefficient of kinetic friction.

4 Assertion In the system of two blocks of equal masses as shown, the coefficient of friction between the blocks (μ_2) is less than coefficient of friction (μ_1) between lower block and ground. For all values of force F applied on upper block, lower block remains at rest.



Reason Frictional force on lower block due to upper block is not sufficient to overcome the frictional force on lower block due to ground.

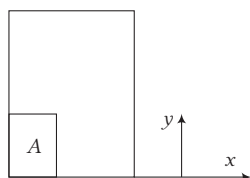
5 Assertion In the figure shown, block of mass m is stationary with respect to lift. Force of friction acting on the block is greater than $mg \sin \theta$.



Reason If lift moves with constant velocity, then force of friction is equal to $mg \sin \theta$.

6 Assertion Block A is resting on one corner of a box as shown in figure. Acceleration of box is $(2\hat{i} + 2\hat{j}) \text{ ms}^{-2}$. Let N_1 is the normal reaction on block from vertical wall and N_2 from ground to box. Then, $\frac{N_1}{N_2} = \frac{1}{6}$. (Neglect friction)

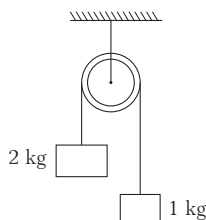
Reason $N_1 = 0$, if lift is stationary.



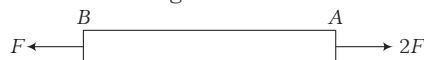
Statement based questions

- Which of the following statement (s) is/are correct?
 - The weighing machine measures the weight of a body.
 - During free fall of a person one feels weightlessness because his weight becomes zero.
 - During free fall, the person falls with an acceleration of g .
 - All of the above
- Which of the following statement (s) is/are incorrect?
 - Static friction is self-adjusting while kinetic friction is constant.
 - Friction always opposes the motion of two bodies.
 - Without friction, one can move on a smooth surface.
 - Some mechanical energy is lost in the form of heat due to air friction.
- Which of the following statement (s) is/are correct?
 - A string has a mass m . If it is accelerated, tension is non-uniform and if it is not accelerated, tension is uniform.
 - Tension force is an electromagnetic force.
 - Only I
 - Only II
 - Both I and II
 - None of these

- 4 In the diagram shown in figure, string is massless and pulley is smooth, then which of the following statement (s) is/are correct?



- Net force on 1 kg block is $\frac{10}{3}$ N.
 - Net force on both the blocks will be same.
- Only I
 - Only II
 - Both I and II
 - None of these
- 5 Two forces are acting on a rope lying on a smooth table as shown in figure.

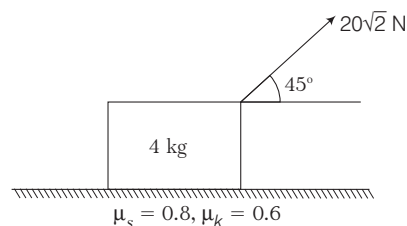


Which of the following statement (s) is/are correct?

- In moving from A to B, tension on string decreases from $2F$ to F .
 - Situation will become indeterminate, if we take it a massless string.
- Only I
 - Only II
 - Both I and II
 - None of these

Match the columns

- 1 In the diagram shown in figure, match the following columns (Take, $g = 10 \text{ ms}^{-2}$)

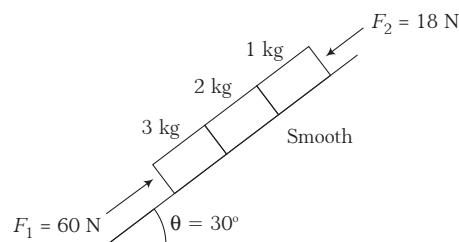


Column I	Column II
(A) Normal reaction	(p) 12 SI unit
(B) Force of friction	(q) 20 SI unit
(C) Acceleration of block	(r) zero
	(s) 2 SI unit

Codes

A	B	C	A	B	C
(a) p	q	r	(b) q	p	r
(c) q	p	s	(d) p	s	r

- 2 In the diagram shown in figure, match the following columns. (Take, $g = 10 \text{ ms}^{-2}$)

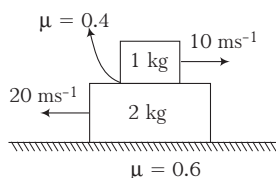


Column I	Column II
(A) Acceleration of 2 kg block	(p) 8 SI unit
(B) Net force on 3 kg block	(q) 25 SI unit
(C) Normal reaction between 2 kg and 1 kg	(r) 2 SI unit
(D) Normal reaction between 3 kg and 2 kg	(s) 45 N
	(t) None

Codes

A	B	C	D	A	B	C	D
(a) q	r	t	t	(b) p	s	r	t
(c) s	s	p	t	(d) r	t	q	t

- 3 In the diagram shown in figure. Match the following columns.

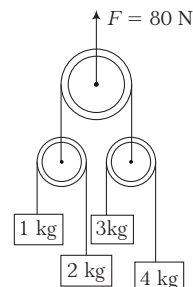


Column I	Column II
(A) Absolute acceleration of 1 kg block	(p) 11 ms^{-2}
(B) Absolute acceleration of 2 kg block	(q) 6 ms^{-2}
(C) Relative acceleration between the two	(r) 17 ms^{-2}
	(s) None

Codes

A	B	C	A	B	C
(a) p	q	r	(b) s	p	s
(c) r	q	p	(d) p	p	r

- 4 In the diagram shown in figure, all pulleys are smooth and massless and strings are light. Match the following columns.



Column I	Column II
(A) 1 kg block	(p) will remain stationary
(B) 2 kg block	(q) will move down
(C) 3 kg block	(r) will move up
(D) 4 kg block	(s) 5 ms^{-2}
	(t) 10 ms^{-2}

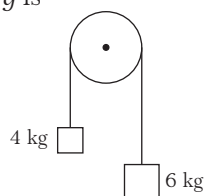
Codes

A	B	C	D	A	B	C	D
(a) r,t	p	q	q,s	(b) p	r,s	q,t	s
(c) p,s	r	q,s	t	(d) p	t	r	s

(C) Medical entrances' gallery

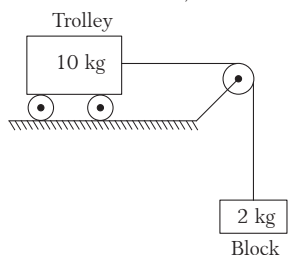
Collection of questions asked in NEET & various medical entrance exams

- 1 Two bodies of masses 4 kg and 6 kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity g is [NEET 2020]



- (a) $g/2$ (b) $g/5$ (c) $g/10$ (d) g

- 2 Calculate the acceleration of the block and trolley system shown in the figure. The coefficient of kinetic friction between the trolley and the surface is 0.05 ($g = 10 \text{ m/s}^2$, mass of the string is negligible and no other friction exists). [NEET 2020]



- (a) 1.25 m/s^2 (b) 1.50 m/s^2
(c) 1.66 m/s^2 (d) 1.00 m/s^2

- 3 A truck is stationary and has a bob suspended by a light string, in a frame attached to the truck. The truck, suddenly moves to the right with an acceleration of a . The pendulum will tilt

[NEET (Odisha) 2019]

- (a) to the left and the angle of inclination of the pendulum with the vertical is $\sin^{-1}\left(\frac{g}{a}\right)$
(b) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}\left(\frac{a}{g}\right)$
(c) to the left and angle of inclination of the pendulum with the vertical is $\sin^{-1}\left(\frac{a}{g}\right)$
(d) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}\left(\frac{g}{a}\right)$

- 4 A body of mass m is kept on a rough horizontal surface (coefficient of friction $= \mu$). Horizontal force is applied on the body, but it does not move.

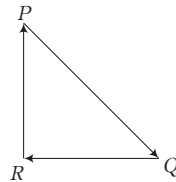
The resultant of normal reaction and the frictional force acting on the object is given $\mathbf{F}$, where $\mathbf{F}$ is

[NEET (Odisha) 2019]

- (a) $|\mathbf{F}| = mg + \mu mg$ (b) $|\mathbf{F}| = \mu mg$
 (c) $|\mathbf{F}| \leq mg \sqrt{1 + \mu^2}$ (d) $|\mathbf{F}| = mg$

- 5 A particle moving with velocity $\mathbf{v}$ is acted by three forces shown by the vector triangle PQR .

The velocity of the particle will [NEET 2019]



- (a) decrease
 (b) remain constant
 (c) change according to the smallest force QR
 (d) increase

- 6 **Assertion** A glass ball is dropped on concrete floor can easily get broken compared, if it is dropped on wooden floor.

Reason On concrete floor, glass ball will take less time to come to rest. [AIIMS 2019]

- (a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
 (c) Assertion is true, but Reason is false.
 (d) Both Assertion and Reason are false.

- 7 A gun applies a force F on a bullet which is given by $F = (100 - 0.5 \times 10^5 t)$ N. The bullet emerges out with speed 400 m/s. Then, find out the impulse exerted till force on bullet becomes zero. [AIIMS 2019]

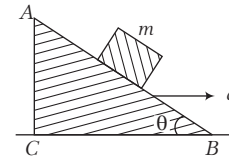
- (a) 0.2 N-s (b) 0.3 N-s
 (c) 0.1 N-s (d) 0.4 N-s

- 8 **Assertion** Even though net external force on a body is zero, momentum need not to be conserved.

Reason The internal interaction between particles of a body cancels out momentum of each other. [AIIMS 2019]

- (a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
 (c) Assertion is true, but Reason is false.
 (d) Both Assertion and Reason are false.

- 9 A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration a towards the right. The relation between a and θ for the block to remain stationary on the wedge is [NEET 2018]

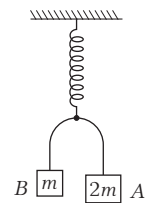


- (a) $a = g \cos \theta$ (b) $a = \frac{g}{\sin \theta}$
 (c) $a = \frac{g}{\operatorname{cosec} \theta}$ (d) $a = g \tan \theta$

- 10 Which one of the following statements is incorrect?

- (a) Frictional force opposes the relative motion. [NEET 2018]
 (b) Limiting value of static friction is directly proportional to normal reaction.
 (c) Rolling friction is smaller than sliding friction.
 (d) Coefficient of sliding friction has dimensions of length.

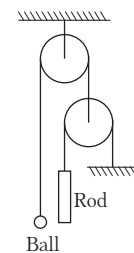
- 11 In the figure, block A and B of masses $2m$ and m are connected with a string and system is hanged vertically with the help of a spring. Spring has negligible mass. Find out magnitude of acceleration of masses $2m$ and m just after the instant when the string of mass m is cut [AIIMS 2018]



- (a) g, g (b) $g, \frac{g}{2}$ (c) $\frac{g}{2}, g$ (d) $\frac{g}{2}, \frac{g}{2}$

- 12 In the figure, mass of a ball is $\frac{9}{5}$ times mass of the

rod. Length of rod is 1 m. The level of ball is same as rod level. Find out time taken by the ball to reach at upper end of rod. [AIIMS 2018]



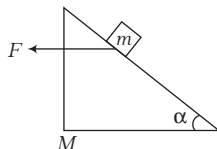
- (a) 1.4 s (b) 2.45 s
 (c) 3.25 s (d) 5 s

- 13 **Assertion** Angle of repose is equal to angle of limiting friction.

Reason When a body is just at the point of motion, the force of friction of this stage is called as limiting friction. [AIIMS 2018]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct, but Reason is incorrect.
 (d) Assertion is incorrect, but Reason is correct.

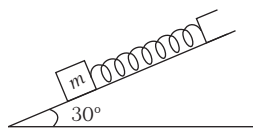
- 14** A wooden wedge of mass M and inclination angle α rest on a smooth floor. A block of mass m is kept on wedge. A force F is applied on the wedge as shown in the figure, such that block remains stationary with respect to wedge. The magnitude of force F is [AIIMS 2018]



- (a) $(M + m)g \tan \alpha$ (b) $g \tan \alpha$
(c) $mg \cos \alpha$ (d) $(M + m)g \operatorname{cosec} \alpha$
- 15** A piece of ice slides down a rough inclined plane at 45° inclination in twice the time that it takes to slide down an identical but frictionless inclined plane. What is the coefficient of friction between ice and incline? [AIIMS 2018]

- (a) $\frac{3}{7 \cot \theta}$ (b) $\frac{4}{7 \cot \theta}$ (c) $\frac{3}{4 \cot \theta}$ (d) $\frac{7}{9 \cot \theta}$

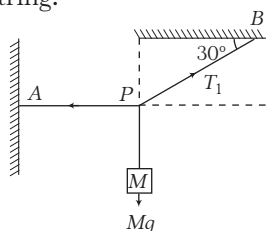
- 16** A body of mass 5 kg is suspended by a spring balance on an inclined plane as shown in figure.



So, force applied on spring balance is [AIIMS 2018]
(a) 50 N (b) 25 N (c) 500 N (d) 10 N

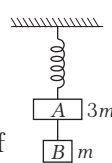
- 17** A force of 10N acts on a body of mass 0.5 kg for 0.25s starting from rest. What is its impulse? [JIPMER 2018]
(a) 0.25 N-s (b) 2.5 N-s (c) 0.5 N-s (d) 0.75 N-s

- 18** A mass M is hung with a light inextensible string as shown in the figure. Find the tension of the horizontal string. [JIPMER 2018]



- (a) $\sqrt{2} Mg$ (b) $\sqrt{3} Mg$ (c) $2 Mg$ (d) $3 Mg$

- 19** Two blocks A and B of masses $3m$ and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively [NEET 2017]

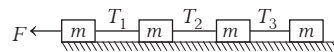


- (a) $g, \frac{g}{3}$ (b) $\frac{g}{3}, g$ (c) g, g (d) $\frac{g}{3}, \frac{g}{3}$

- 20** Four blocks of same mass connected by strings are pulled by a force F on a smooth horizontal surface as shown in figure.

The tension T_1, T_2 and T_3 will be

[AIIMS 2017]

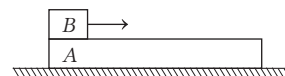


- (a) $T_1 = \frac{1}{4}F, T_2 = \frac{3}{2}F, T_3 = \frac{1}{4}F$
(b) $T_1 = \frac{1}{4}F, T_2 = \frac{1}{2}F, T_3 = \frac{1}{2}F$
(c) $T_1 = \frac{3}{4}F, T_2 = \frac{1}{2}F, T_3 = \frac{1}{4}F$
(d) $T_1 = \frac{3}{4}F, T_2 = \frac{1}{2}F, T_3 = \frac{1}{2}F$

- 21** A body of mass 5×10^{-3} kg is launched upon a rough inclined plane making an angle of 30° with the horizontal. Obtain the coefficient of friction between the body and the plane, if the time of ascent is half of the time of descent. [AIIMS 2017]

- (a) 0.346 (b) 0.921 (c) 1.926 (d) 2.912

- 22** A long block A of mass M is at rest on a smooth horizontal surface. A small block B of mass $M/2$ is placed on A at one end and projected along A with some velocity v . The coefficient of friction between the block is μ . Then, the accelerations of blocks A and B before reaching a common velocity will be respectively [JIPMER 2017]

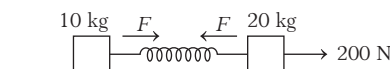


- (a) $\frac{\mu g}{2}$ (towards right), $\frac{\mu g}{2}$ (towards left)
(b) μg (towards right), μg (towards left)
(c) $\frac{\mu g}{2}$ (towards right), μg (towards left)
(d) μg (towards right), $\frac{\mu g}{2}$ (towards left)

- 23** A box of mass 8 kg is placed on a rough inclined plane of inclination 30° . Its downward motion can be prevented by applying a horizontal force F , then value of F for which friction between the block and the incline surface is minimum, is [JIPMER 2017]

- (a) $\frac{80}{\sqrt{3}}$ N (b) $40\sqrt{3}$ N (c) $\frac{40}{\sqrt{3}}$ N (d) $80\sqrt{3}$ N

- 24** Two masses 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. A force of 200 N acts on the 20 kg mass. At the instant shown in figure, the 10 kg mass has acceleration of 12 m/s^2 . The value of acceleration of 20 kg mass is [JIPMER 2017]



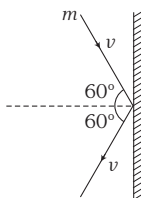
- (a) 4 m/s^2 (b) 10 m/s^2
(c) 20 m/s^2 (d) 30 m/s^2

- 25** The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half rough. A block starting from rest at the top of the plane will again come to rest at the bottom, if the coefficient of friction between the block and the lower half of the plane is given by [JIPMER 2017, NEET 2013]

(a) $\mu = 2 \tan \theta$ (b) $\mu = \tan \theta$
(c) $\mu = 2 / (\tan \theta)$ (d) $\mu = 1 / \tan \theta$

- 26** A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure. The value of impulse imparted by the wall on the ball will be [NEET 2016]

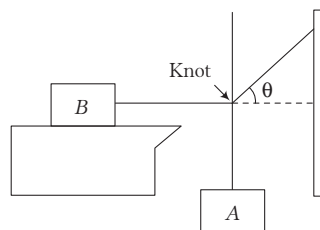
(a) mv (b) $2mv$
(c) $mv/2$ (d) $mv/3$



- 27** A body of mass 40 kg resting on rough horizontal surface is subjected to a force P which is just enough to start the motion of the body. If $\mu_s = 0.5$, $\mu_k = 0.4$, $g = 10 \text{ ms}^{-2}$ and the force P is continuously applied on the body, then the acceleration of the body is [AIIMS 2015]
- (a) zero (b) 1 ms^{-2} (c) 2 ms^{-2} (d) 24 ms^{-2}

- 28** A balloon of mass 10 kg is raising up with an acceleration of 20 ms^{-2} . If a mass of 4 kg is removed from the balloon, its acceleration becomes (Take, $g = 10 \text{ ms}^{-2}$) [EAMCET 2015]
- (a) 40 ms^{-2} (b) 20 ms^{-2}
(c) 30 ms^{-2} (d) 12 ms^{-2}

- 29** Block B lying on a table weighs w . The coefficient of static friction between the block and the table is μ . Assume that the cord between B and the knot is horizontal. The maximum weight of the block A for which the system will be stationary is [WB JEE 2015]



(a) $\frac{w \tan \theta}{\mu}$ (b) $\mu w \tan \theta$
(c) $\mu w \sqrt{1 + \tan^2 \theta}$ (d) $\mu w \sin \theta$

- 30** Three blocks A , B and C of masses 4 kg, 2 kg and 1 kg respectively are in contact on a frictionless surface as shown. If a force of 14 N is applied on the 4 kg block, then the contact force between A and B is [CBSE AIPMT 2015]

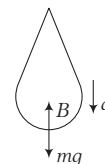


(a) 2 N (b) 6 N (c) 8 N (d) 18 N

- 31** A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pulley at the edge of table and from its other end, another block B of mass m_2 is suspended. The coefficient of kinetic friction between the block and the table is μ_k . When the block A is sliding on the table, the tension in the string is [CBSE AIPMT 2015]

(a) $\frac{(m_2 + \mu_k m_1)g}{(m_1 + m_2)}$ (b) $\frac{(m_2 - \mu_k m_1)g}{(m_1 + m_2)}$
(c) $\frac{m_1 m_2 (1 + \mu_k)g}{(m_1 + m_2)}$ (d) $\frac{m_1 m_2 (1 - \mu_k)g}{(m_1 + m_2)}$

- 32** A balloon with mass m is descending down with an acceleration a (where, $a < g$). How much mass should be removed from it, so that it starts moving up with an acceleration a ? [CBSE AIPMT 2014]

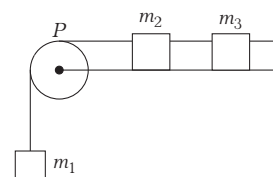


(a) $\frac{2ma}{g+a}$ (b) $\frac{2ma}{g-a}$ (c) $\frac{ma}{g+a}$ (d) $\frac{ma}{g-a}$

- 33** A bullet moving with a velocity of $30\sqrt{2} \text{ ms}^{-1}$ is fired into a fixed target. It penetrated into the target to the extent of s metre. If the same bullet is fired into a target of thickness $\frac{s}{2}$ metres and of the same material with the same velocity, then the bullet comes out of the target with velocity, is [EAMCET 2014]

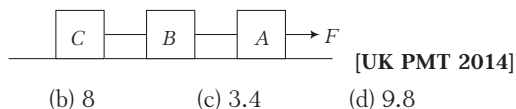
(a) 20 ms^{-1} (b) 30 ms^{-1}
(c) $20\sqrt{2} \text{ ms}^{-1}$ (d) $10\sqrt{2} \text{ ms}^{-1}$

- 34** A system consists of three masses m_1 , m_2 and m_3 connected by a string passing over a pulley P . The mass m_1 hangs freely and m_2 and m_3 are on a rough horizontal table (the coefficient of friction $= \mu$). The pulley is frictionless and of negligible mass. The downward acceleration of mass m_1 is (Assume, $m_1 = m_2 = m_3 = m$) [CBSE AIPMT 2014]



(a) $\frac{g(1 - g\mu)}{9}$ (b) $\frac{2g\mu}{3}$
(c) $\frac{g(1 - 2\mu)}{3}$ (d) $\frac{g(1 - 2\mu)}{2}$

- 35** Three identical blocks of masses $m = 2 \text{ kg}$ are drawn by a force 10.2 N on a frictionless surface. What is the tension (in newton) in the string between the blocks B and C ?

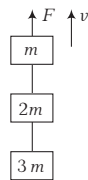


- (a) 9.2 (b) 8 (c) 3.4 (d) 9.8
- 36** A wooden block of mass 8 kg slides down an inclined plane of inclination 30° to the horizontal with constant acceleration 0.4 ms^{-2} . The force of friction between the block and inclined plane is (Take, $g = 10 \text{ ms}^{-2}$) [MHT CET 2014]
- (a) 12.2 N (b) 24.4 N
(c) 36.8 N (d) 48.8 N

- 37** To determine the coefficient of friction between a rough surface and a block, the surface is kept inclined at 45° and the block is released from rest. The block takes a time t in moving a distance d . The rough surface is then replaced by a smooth surface and the same experiment is repeated. The block now takes a time $t/2$ in moving down the same distance d . The coefficient of friction is [WB JEE 2014]
- (a) $3/4$ (b) $5/4$
(c) $1/2$ (d) $1/\sqrt{2}$

- 38** A body of mass m is placed on a rough surface with coefficient of friction μ inclined at θ . If the mass is in equilibrium, then [KCET 2014]
- (a) $\theta = \tan^{-1} \mu$ (b) $\theta = \tan^{-1} \left(\frac{1}{\mu} \right)$
(c) $\theta = \tan^{-1} \frac{m}{\mu}$ (d) $\theta = \tan^{-1} \frac{\mu}{m}$

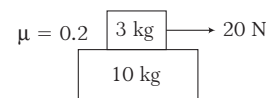
- 39** Three blocks with masses m , $2m$ and $3m$ are connected by strings as shown in the figure. After an upward force F is applied on block m , the masses move upward at constant speed v . What is the net force on the block of mass $2m$? (g is the acceleration due to gravity) [NEET 2013]
- (a) Zero (b) $2mg$ (c) $3mg$ (d) $6mg$



- 40** A balloon starting from rest ascends vertically with uniform acceleration to a height of 100 m in 10 s . The force on the bottom of the balloon by a mass of 50 kg is (Take, $g = 10 \text{ ms}^{-2}$) [EAMCET 2013]
- (a) 100 N (b) 300 N (c) 600 N (d) 400 N

- 41** A 3 kg block is placed over a 10 kg block and both are placed on a smooth horizontal surface. The coefficient of friction between the blocks is 0.2 . If a horizontal force of 20 N is applied to 3 kg block,

accelerations of the two blocks (in ms^{-2}) are (Take, $g = 10 \text{ ms}^{-2}$) [EAMCET 2013]

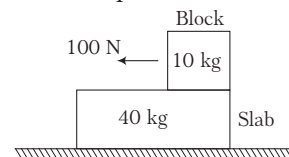


- (a) $\frac{13}{4}, 0.6$ (b) $\frac{14}{4}, 3$
(c) $\frac{13}{4}, 3$ (d) $\frac{14}{3}, 0.6$
- 42** A 60 kg person is weighed by a balance as 54 kg in a lift which is accelerated downwards. The acceleration of the lift is [Kerala CEE 2013]
- (a) 1.26 ms^{-2} (b) 1.76 ms^{-2}
(c) 1.98 ms^{-2} (d) 0.98 ms^{-2}
(e) None of these
- 43** A lift starting from rest with a constant upward acceleration moves 1.5 m in 0.4 s . If a person standing in the lift holds a packet of 2 kg by a string, then the tension in the string due to motion is [UP CPMT 2013]

- (a) 5.89 N (b) 57.1 N
(c) 67.1 N (d) None of these
- 44** A body of mass 0.25 kg is projected with muzzle velocity 100 ms^{-1} from a tank of mass 100 kg . What is the recoil velocity of the tank? [AIIMS 2012]
- (a) 5 ms^{-1} (b) 25 ms^{-1} (c) 0.5 ms^{-1} (d) 0.25 ms^{-1}
- 45** A rocket with a lift-off mass $3.5 \times 10^4 \text{ kg}$ is blast upward with an initial acceleration of 10 ms^{-2} . Then, the initial thrust of the blast is [AIIMS 2012]
- (a) $1.75 \times 10^5 \text{ N}$ (b) $3.5 \times 10^5 \text{ N}$
(c) $7 \times 10^5 \text{ N}$ (d) $14 \times 10^5 \text{ N}$

- 46** A marble block of mass 2 kg lying on ice when given a velocity of 6 ms^{-1} is stopped by friction in 10 s . Then, the coefficient of friction is [AIIMS 2012]
- (a) 0.01 (b) 0.02 (c) 0.03 (d) 0.06

- 47** A 40 kg slab rests on a frictionless floor. A 10 kg block rests on the top of the slab as shown in figure.



The static coefficient of friction between the block and the slab is 0.60 while the kinetic coefficient is 0.40 . The 10 kg block is acted upon by a horizontal force of 100 N . If $g = 9.8 \text{ ms}^{-2}$, the resulting acceleration of the slab will be [BHU 2012]

(a) 61 ms^{-2} (b) 152 ms^{-2}
(c) 147 ms^{-2} (d) 0.98 ms^{-2}

- 48** An inclined plane of height h and length l have the angle of inclination θ . The time taken by a body to come from the top to the bottom of this inclined plane will be [BCECE 2012]

(a) $\sin\theta \sqrt{\frac{2h}{g}}$ (b) $\frac{1}{\sin\theta} \sqrt{\frac{2h}{g}}$
 (c) $\sqrt{\frac{2h}{g}}$ (d) $\sqrt{\frac{2l}{g}}$

- 49** If a coin is dropped in a lift it takes t_1 time to reach the floor and takes t_2 time when lift is moving up with constant acceleration, then which one of the following relation is correct? [BCECE (Mains) 2012]

(a) $t_1 = t_2$ (b) $t_1 > t_2$
 (c) $t_2 > t_1$ (d) $t_1 \gg t_2$

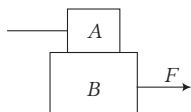
- 50** A 60 kg mass is pushed with a enough force to start it moving and the same force is continued to act afterwards. If the coefficient of static friction and sliding friction are 0.5 and 0.4 respectively, then the acceleration of the body will be [BCECE (Mains) 2012]

(a) 1ms^{-2} (b) 3.9ms^{-2}
 (c) 4.9ms^{-2} (d) 6ms^{-2}

- 51** A body is hanging from a rigid support by an inextensible string of length l . It is struck inelastically by an identical body of mass m with horizontal velocity $v = \sqrt{2gl}$, the tension in the string increases just after striking by [AFMC 2012]

(a) mg (b) $3mg$
 (c) $2mg$ (d) None of these

- 52** A block A of mass 100 kg rests on another block B of mass 200 kg and is tied to a wall as shown in the figure. The coefficient of friction between A and B is 0.2 and that between B and the ground is 0.3. The minimum force F required to move the block B is (Take, $g = 10\text{ ms}^{-2}$) [AFMC 2012]



(a) 900 N (b) 200 N
 (c) 1100 N (d) 700 N

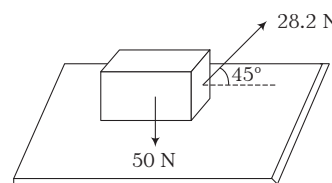
- 53** A machine gun fires a bullet of mass 40 g with a velocity 1200 ms^{-1} . The man holding it can exert a maximum force of 144 N on the gun. How many bullets can be fired per second at the most? [AFMC 2012]

(a) Only one
 (b) Three
 (c) Can fire any number of bullets
 (d) 144×48

- 54** A monkey of mass 25 kg is holding a vertical rope. The rope does not break, if a body of mass 30 kg is suspended from it, but the rope breaks, if the mass of the body suspended with the rope exceeds 30 kg. What will be the maximum acceleration with which the monkey can climb up along the rope? (Take, $g = 10\text{ ms}^{-2}$) [JCECE 2012]

(a) 2 ms^{-2} (b) 25 ms^{-2} (c) 3 ms^{-2} (d) 4 ms^{-2}

- 55** A body of weight 50 N placed on a horizontal surface is just moved by a force of 282 N. The frictional force and normal reaction are [JCECE 2012]



(a) 2 N, 3N (b) 5 N, 6 N
 (c) 10 N, 15 N (d) 20 N, 30 N

- 56** Diwali rocket is ejecting 50 g of gases/s at a velocity of 400 ms^{-1} . The accelerating force on the rocket will be [RPMT 2011]

(a) 22 dyne (b) 20 N (c) 20 dyne (d) 100 N

- 57** A rocket of mass 1000 kg is to be projected vertically upwards. The gases are exhausted vertically downwards with velocity 100 ms^{-1} with respect to the rocket. What is the minimum rate of burning of fuel, so as to just lift the rocket upwards against the gravitational attraction? (Take, $g = 10\text{ ms}^{-2}$) [AMU 2011]

(a) 50 kg s^{-1} (b) 100 kg s^{-1}
 (c) 200 kg s^{-1} (d) 400 kg s^{-1}

- 58** In a non-inertial frame, the second law of motion is written as [DUMET 2011]

(a) $\mathbf{F} = m\mathbf{a}$ (b) $\mathbf{F} = m\mathbf{a} + \mathbf{F}_p$ (c) $\mathbf{F} = m\mathbf{a} - \mathbf{F}_p$ (d) $\mathbf{F} = 2m\mathbf{a}$
 where, $\mathbf{F}_p$ is pseudo force while $\mathbf{a}$ is the acceleration of the body relative to the non-inertial frame.

- 59** A man of mass 60 kg is riding in a lift. The weight of the man when the lift is accelerating upwards and downwards at 2 ms^{-2} are respectively (Take, $g = 10\text{ ms}^{-2}$) [AMU 2011]

(a) 720 N and 480 N (b) 480 N and 720 N
 (c) 600 N and 600 N (d) None of these

- 60** An object is moving on a plane surface with uniform velocity 10 ms^{-1} in presence of a force 10 N. The frictional force between the object and the surface is [DUMET 2011]

(a) 1 N (b) -10 N (c) 10 N (d) 100 N

ANSWERS

• CHECK POINT 5.1

1. (c)	2. (d)	3. (a)	4. (a)	5. (b)	6. (c)	7. (a)	8. (b)	9. (a)	10. (a)
11. (a)	12. (a)	13. (b)	14. (a)	15. (c)	16. (d)	17. (b)			

• CHECK POINT 5.2

1. (d)	2. (d)	3. (b)	4. (c)	5. (c)	6. (d)	7. (c)	8. (d)	9. (a)	10. (d)
11. (c)	12. (a)	13. (a)	14. (a)	15. (c)					

• CHECK POINT 5.3

1. (c)	2. (c)	3. (d)	4. (d)	5. (d)	6. (d)	7. (b)	8. (b)	9. (c)	10. (b)
11. (d)	12. (b)	13. (d)	14. (c)	15. (b)	16. (a)	17. (b)	18. (b)	19. (c)	20. (c)

• CHECK POINT 5.4

1. (d)	2. (a)	3. (c)	4. (d)	5. (c)	6. (a)	7. (d)	8. (a)	9. (a)	10. (b)
11. (a)	12. (a)	13. (a)	14. (d)	15. (d)	16. (a)	17. (a)	18. (a)		

(A) Taking it together

1. (b)	2. (d)	3. (b)	4. (c)	5. (c)	6. (c)	7. (c)	8. (d)	9. (c)	10. (b)
11. (c)	12. (a)	13. (d)	14. (a)	15. (a)	16. (d)	17. (b)	18. (a)	19. (a)	20. (b)
21. (b)	22. (b)	23. (b)	24. (d)	25. (b)	26. (a)	27. (a)	28. (b)	29. (a)	30. (a)
31. (c)	32. (c)	33. (d)	34. (c)	35. (a)	36. (a)	37. (a)	38. (a)	39. (d)	40. (a)
41. (c)	42. (b)	43. (c)	44. (b)	45. (c)	46. (a)	47. (a)	48. (a)	49. (d)	50. (b)
51. (c)	52. (d)	53. (a)	54. (a)	55. (b)	56. (d)	57. (d)	58. (c)	59. (d)	60. (c)
61. (b)	62. (a)	63. (d)	64. (c)						

(B) Medical entrance special format questions

• Assertion and reason

1. (a)	2. (c)	3. (a)	4. (a)	5. (b)	6. (b)
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• Statement based questions

1. (d)	2. (b)	3. (b)	4. (a)	5. (c)
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• Match the columns

1. (c)	2. (d)	3. (b)	4. (a)
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(C) Medical entrances' gallery

1. (b)	2. (a)	3. (b)	4. (c)	5. (b)	6. (a)	7. (c)	8. (d)	9. (d)	10. (d)
11. (c)	12. (a)	13. (a)	14. (a)	15. (c)	16. (b)	17. (b)	18. (b)	19. (b)	20. (c)
21. (a)	22. (c)	23. (a)	24. (a)	25. (a)	26. (a)	27. (b)	28. (a)	29. (b)	30. (b)
31. (c)	32. (a)	33. (b)	34. (c)	35. (c)	36. (c)	37. (a)	38. (a)	39. (a)	40. (c)
41. (d)	42. (d)	43. (b)	44. (d)	45. (c)	46. (d)	47. (d)	48. (b)	49. (b)	50. (a)
51. (c)	52. (c)	53. (b)	54. (a)	55. (d)	56. (b)	57. (b)	58. (c)	59. (a)	60. (b)

Hints & Explanations

• CHECK POINT 5.1

2 (d) Given, mass, $m = 6 \text{ kg}$

Velocity, $\Delta v = v_2 - v_1 = 5 - 3 = 2 \text{ ms}^{-1}$

$\therefore$ Momentum, $\Delta p = m\Delta v = 6 \times 2 = 12 \text{ N-s}$

3 (a) Given, $p = 2 + 3t^2$

Differentiate w.r.t. t , we get

$$\frac{dp}{dt} = 0 + 3 \times 2t = 6t$$

$$\text{If } t = 3 \text{ s, then } \frac{dp}{dt} = 6 \times 3 = 18 \text{ N}$$

$\therefore$ Force acting on the particle = 18 N

5 (b) Given, $F = 72 \text{ dyne} \Rightarrow F = 72 \times 10^{-5} \text{ N}$,

$$\theta = 60^\circ, m = 9\text{g} = 9 \times 10^{-3} \text{ kg}$$

$$F' = ma$$

In this case, $F' = F \cos 60^\circ$

$$\begin{aligned} \Rightarrow a &= \frac{F'}{m} = \frac{F \cos 60^\circ}{9 \times 10^{-3}} \\ &= \frac{72 \times 10^{-5} \times \cos 60^\circ}{9 \times 10^{-3}} = 8 \times 10^{-2} \times \frac{1}{2} \\ &= 4 \times 10^{-2} \text{ ms}^{-1} = 4 \text{ cms}^{-1} \end{aligned}$$

6 (c) Given, $m = 60 \text{ kg}$ and $a = 1 \text{ ms}^{-2}$

$\therefore$ Net force, $F_{\text{net}} = \text{Mass} \times \text{Acceleration} = 60 \times 1 = 60 \text{ N}$

7 (a) Given, $m = 3 \text{ kg}$, $\Delta v = 3.5 - 2 = 1.5 \text{ ms}^{-1}$ and $\Delta t = 25 \text{ s}$

$$\text{Force, } F = ma = m \frac{\Delta v}{\Delta t} = 3 \times \frac{1.5}{25} = 0.18 \text{ N, in the direction of motion}$$

8 (b) Given, $m = 5 \text{ kg}$, $F_1 = 8 \text{ N}$ and $F_2 = 6 \text{ N}$

$$\text{Resultant force, } F = \sqrt{8^2 + 6^2} = 10 \text{ N}$$

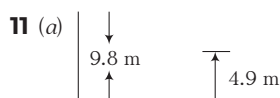
$$\therefore a = \frac{F}{m} = \frac{10}{5} = 2 \text{ ms}^{-2}$$

$$\text{and } \theta = \cos^{-1}\left(\frac{6}{10}\right) = \cos^{-1}(0.6), \text{ from } 6\text{N}$$

10 (a) Given, $F = 50 \text{ N}$, $m = 20 \text{ kg}$ and $v = 15 \text{ ms}^{-1}$

$$\text{Impulse} = F\Delta t = mv$$

$$\therefore \text{Time, } \Delta t = \frac{mv}{F} = \frac{20 \times 15}{50} = 6 \text{ s}$$



$$v^2 = u^2 + 2gh$$

$$v^2 = 0 + 2 \times 9.8 \times 9.8$$

$$v^2 = \sqrt{2 \times 9.8 \times 9.8}$$

$$\Rightarrow v = 9.8\sqrt{2} \text{ m/s}$$

$$\text{Again, } v^2 = u^2 - 2gh$$

$$0 = u^2 - 2 \times 9.8 \times 4.9$$

$$\Rightarrow u^2 = \sqrt{(9.8)^2} \Rightarrow u = 9.8 \text{ m/s}$$

$$\text{Impulse} = m[v - (-u)]$$

$$= 1[9.8\sqrt{2} + 9.8] = 9.8(\sqrt{2} + 1)$$

$$= 9.8 \times 2.4 = 23.52 \text{ N-s}$$

$$\text{Average force} = \frac{\text{Impulse}}{\text{Time}} = \frac{23.52}{0.1} = 235.2 \text{ N}$$

12 (a) Here, mass, $m = 5 \text{ kg}$

$$\text{Change in velocity, } \Delta \mathbf{v} = \mathbf{v}_f - \mathbf{v}_i = [(10 - 2)\hat{\mathbf{i}} + (6 - 6)\hat{\mathbf{j}}] = 8\hat{\mathbf{i}}$$

$$\text{Change in momentum} = m\Delta \mathbf{v} = 5[8\hat{\mathbf{i}}] = 40\hat{\mathbf{i}} \text{ kg-ms}^{-1}$$

13 (b) Impulse is defined as rate of change of momentum. For change in momentum to be minimum,

$$\frac{d}{dt}(20t^2 - 40t) = 0$$

$$40t - 40 = 0 \Rightarrow t = 1 \text{ s}$$

$$14 (a) F_{\text{av}} = ma_{\text{av}} = m \frac{\Delta v}{\Delta t} = \frac{0.5[2 - (-2)]}{10^{-3}} = 2000 \text{ N}$$

15 (c) From law of conservation of momentum, $p_i = p_f$

and initial momentum, $p_i = mu = m(0) = 0$

$\therefore p_f$ should also be zero.

Hence, other piece will move in negative x-direction.

16 (d) From the law of conservation of momentum,

total initial momentum = total final momentum

$$\Rightarrow m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\therefore 0.1 \times 0 + 50 \times 0 = 0.1 \times 100 + 50(-v_2)$$

$$\Rightarrow 0 = 10 - 50v_2$$

$$\therefore v_2 = \frac{10}{50} = 0.2 \text{ ms}^{-1}$$

17 (b) From the given figure,

$$\text{Initial velocity, } \mathbf{v}_i = (-20 \sin 30^\circ \hat{\mathbf{i}} - 20 \cos 30^\circ \hat{\mathbf{j}}) \text{ m/s}$$

$$\text{Final velocity, } \mathbf{v}_f = (-20 \sin 30^\circ \hat{\mathbf{i}} + 20 \cos 30^\circ \hat{\mathbf{j}}) \text{ m/s}$$

$$\text{Change in velocity, } \Delta \mathbf{v} = \mathbf{v}_f - \mathbf{v}_i$$

$$\Delta \mathbf{v} = (-20 \sin 30^\circ \hat{\mathbf{i}} + 20 \cos 30^\circ \hat{\mathbf{j}})$$

$$-(-20 \sin 30^\circ \hat{\mathbf{i}} - 20 \cos 30^\circ \hat{\mathbf{j}})$$

$$\Delta \mathbf{v} = 2 \times 20 \cos 30^\circ \hat{\mathbf{j}}$$

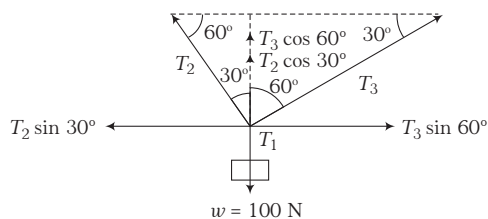
$$\Delta \mathbf{v} = 2 \times 20 \times \frac{\sqrt{3}}{2} \hat{\mathbf{j}} \Rightarrow \Delta \mathbf{v} = 20\sqrt{3} \hat{\mathbf{j}}$$

$$\text{Magnitude, } |\Delta \mathbf{v}| = 20\sqrt{3} \text{ m/s}$$

• CHECK POINT 5.2

2 (d) $T_3 \sin 60^\circ = T_2 \sin 30^\circ$

$\therefore T_3 = \frac{T_2}{\sqrt{3}}$



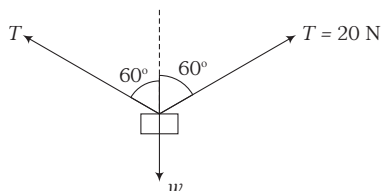
Now, $T_2 \cos 30^\circ + T_3 \cos 60^\circ = T_1 = 100$

or $\frac{\sqrt{3}T_2}{2} + \frac{T_2}{\sqrt{3}} \times \frac{1}{2} = 100$

or $\frac{8T_2}{4\sqrt{3}} = 100$

or $T_2 = 50\sqrt{3} \text{ N}$

3 (b) $2T \cos 60^\circ = w$



or $w = T$

or $w_{\max} = T_{\max} = 20 \text{ N}$

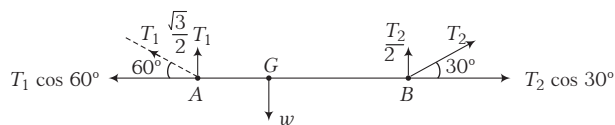
4 (c) Equilibrium of m : $T = mg$... (i)

Equilibrium of $\sqrt{2}m$: $2T \cos \theta = \sqrt{2}mg$... (ii)

Solving these two equations, we get $\theta = 45^\circ$

5 (c) $T_1 \cos 60^\circ = T_2 \cos 30^\circ$

$\Rightarrow \frac{T_1}{2} = \frac{\sqrt{3}T_2}{2}$



or $T_1 = \sqrt{3}T_2$... (i)

$\therefore \frac{\sqrt{3}T_1}{2} \times AG = \frac{T_2}{2} \times BG$

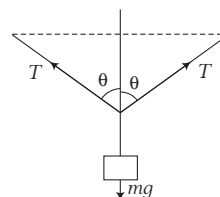
$\Rightarrow \frac{AG}{BG} = \frac{T_2}{\sqrt{3}T_1}$

But $\frac{T_2}{T_1} = \frac{1}{\sqrt{3}}$ [from Eq. (i)]

Hence, $\frac{AG}{BG} = \frac{1}{3}$

6 (d) For equilibrium of body,
 $mg = 2T \cos \theta$

$\Rightarrow T = \frac{mg}{2 \cos \theta}$



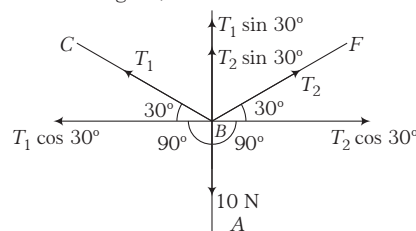
For the string to be horizontal,

$\theta = 90^\circ$

$T = \frac{mg}{2 \cos 90^\circ}$

$\Rightarrow T = \infty$

7 (c) As shown in figure,



$T_1 \cos 30^\circ = T_2 \cos 30^\circ$

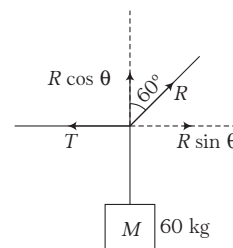
$\therefore T_1 = T_2 = T$ (Let)

Again, $T_1 \sin 30^\circ + T_2 \sin 30^\circ = 10 \Rightarrow 2T \sin 30^\circ = 10$

$\Rightarrow 2T \cdot \frac{1}{2} = 10 \Rightarrow T = 10 \text{ N}$

Thus, the tension in section BC and BF are 10 N and 10 N, respectively.

8 (d) The given figure can be drawn as



Taking component of forces,

$R \cos \theta = Mg \Rightarrow R \cos 60^\circ = Mg$... (i)

and $R \sin 60^\circ = T$... (ii)

By Eqs. (i) and (ii), we get

$\Rightarrow \tan 60^\circ = \frac{T}{Mg} \Rightarrow T = Mg \tan 60^\circ$

or $T = 60 \times g \times \sqrt{3} = 103.9 \text{ kgf}$

9 (a) Force exerted by man on rope transfers to it in the form of tension.

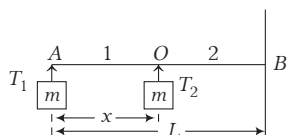
Net upward force on the system is $2T$ or $2F$.

Net downward force is $(50 + 30)g = 80g$.

For equilibrium of system, $2F = 80g$ or $F = 40g$

- 10 (d) Let x be distance from A to O and L be the total length of string. Then, ratio of tensions is

$$\frac{T_1}{T_2} = \frac{x}{L-x} = \frac{1}{3}$$



- 11 (c) The resultant force will be

$$F_R = |\mathbf{F}_R| = \sqrt{3^2 + 4^2 + 2 \times 3 \times 4 \cos 90^\circ} = 5 \text{ N}$$

- 12 (a) Let equal forces $F_1 = F_2 = F$ newton

Angle between the forces, $\theta = 60^\circ$

Resultant force, $R = 40\sqrt{3} \text{ N}$

$$\text{Now, } R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

$$\therefore 40\sqrt{3} = \sqrt{F^2 + F^2 + 2FF \cos 60^\circ}$$

$$\Rightarrow 40\sqrt{3} = \sqrt{2F^2 + 2F^2 \times \frac{1}{2}}$$

$$\Rightarrow F = 40 \text{ N}$$

- 13 (a) As, resultant of F_1 and F_2 ,

$$F_R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos 90^\circ}$$

$$\Rightarrow F = F\sqrt{2}$$

$\therefore$ Resultant of three forces F_1 , F_2 and F_3 will be $(\sqrt{2} - 1)F$.

As, $F = ma$

Therefore, acceleration of body is also $(\sqrt{2} - 1)a$.

- 14 (a) Minimum additional force needed

$$F = -(F_{\text{resultant}})_x$$

$$F_{\text{resultant}} = [(4 - 2)(\cos 30^\circ \hat{j} - \sin 30^\circ \hat{i}) + (\cos 60^\circ \hat{i} + \sin 60^\circ \hat{j})]$$

$$= \left[2 \left(\frac{\sqrt{3}}{2} \hat{j} - \frac{1}{2} \hat{i} \right) + \left(\frac{1}{2} \hat{i} + \frac{\sqrt{3}}{2} \hat{j} \right) \right]$$

$$= \left[\left(\sqrt{3} + \frac{\sqrt{3}}{2} \right) \hat{j} + \left(-\frac{1}{2} + \frac{1}{2} \right) \hat{i} \right]$$

$$= \left[-\frac{1}{2} \hat{i} + \frac{3\sqrt{3}}{2} \hat{j} \right] = -\frac{1}{2} \hat{i} + \frac{3\sqrt{3}}{2} \hat{j}$$

$$\Rightarrow F_x = \left(-\frac{1}{2} \right)$$

Hence, $F = |F_x| = 0.5 \text{ N}$

- 15 (c) Apply Lami's theorem at O ,

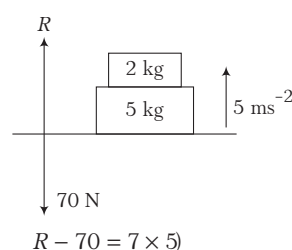
$$\frac{T_1}{\sin 150^\circ} = \frac{T_2}{\sin 120^\circ} = \frac{10}{\sin 90^\circ} = \frac{10}{1} = 10$$

$$\therefore T_1 = 10 \sin 150^\circ = 10 \times \frac{1}{2} = 5 \text{ N}$$

$$\begin{aligned} T_2 &= 10 \sin 120^\circ \\ &= 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3} \text{ N} \end{aligned}$$

• CHECK POINT 5.3

- 1 (c)



$$R - 70 = 7 \times 5$$

$$\therefore R = 105 \text{ N}$$

- 2 (c) Given, $v = 0$, $u = 3 \text{ ms}^{-1}$

So, using $v^2 = u^2 - 2as$

$$0 = (3)^2 - 2(a)(9)$$

$$\Rightarrow a = \frac{1}{2} = 0.5 \text{ ms}^{-2} \quad (\text{upwards})$$

Now, $N = m(g + a)$ (for deceleration)

$$= 50(9.8 + 0.5) = 515 \text{ N}$$

- 3 (d) Given, $u = 10 \text{ ms}^{-1}$, $v = 0$

So, using $v^2 = u^2 - 2as$

$$0 = (10)^2 - 2(a)(25)$$

$$\therefore a = 2 \text{ ms}^{-2} \quad (\text{upwards})$$

Now, $T - 800g = 800a$

$$\therefore T = 800(10 + 2) = 9600 \text{ N}$$

- 4 (d) Acceleration of the system,

$$a = \frac{T_3}{m_1 + m_2 + m_3} = \frac{40}{10 + 6 + 4} = 2 \text{ ms}^{-2}$$

Equation of motion of m_3 is

$$T_3 - T_2 = m_3 a$$

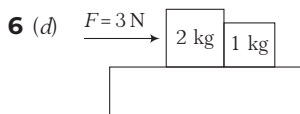
$$\therefore T_2 = T_3 - m_3 a = 40 - 4 \times 2 = 32 \text{ N}$$

$$\begin{aligned} 5 (d) \quad T_2 &= \frac{m_1 F \cos 30^\circ}{(m_1 + m_2 + m_3)} = \frac{3F \cos 30^\circ}{(3 + 12 + 15)} \\ &= \frac{3F \cos 30^\circ}{30} = \frac{F \cos 30^\circ}{10} \end{aligned}$$

$$\begin{aligned} \text{and } T_1 &= \frac{(m_1 + m_2) F \cos 30^\circ}{(m_1 + m_2 + m_3)} \\ &= \frac{(3 + 12) F \cos 30^\circ}{3 + 12 + 15} \\ &= \frac{15F \cos 30^\circ}{30} = \frac{F \cos 30^\circ}{2} \end{aligned}$$

The ratio between T_1 and T_2 is

$$T_1 : T_2 = \frac{F \cos 30^\circ / 2}{F \cos 30^\circ / 10} = 5 : 1$$



As the blocks are rigid under the action of a force F , so both will move with same acceleration.

$$a = \frac{F}{m + M} = \frac{3}{2 + 1} = \frac{3}{3} = 1 \text{ ms}^{-2}$$

The force is applied to 2 kg, then its action on 1 kg will be

$$F_N = ma = 1 \times 1 = 1 \text{ N}$$

7 (b) Acceleration of system, $a = \frac{\text{Net pushing force}}{\text{Total mass}}$

or
$$a = \frac{F - (m_1 + m_2 + m_3)g \sin \theta}{(m_1 + m_2 + m_3)}$$

Equation of motion for m_3 ,

$$N - m_3 g \sin \theta = m_3 a$$

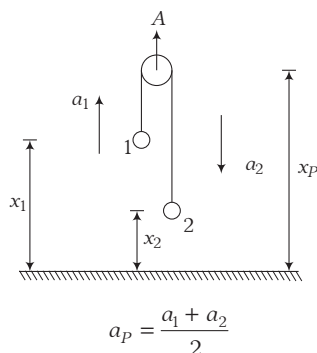
or
$$N = m_3 g \sin \theta + m_3 \left[\frac{F - (m_1 + m_2 + m_3)g \sin \theta}{(m_1 + m_2 + m_3)} \right]$$

$$= \frac{m_3 F}{m_1 + m_2 + m_3}$$

8 (b) Acceleration, $a = \left[\frac{m_2 - m_1}{m_1 + m_2} \right] g = \left[\frac{\frac{3}{2}m - m}{m + \frac{3}{2}m} \right] g = \frac{g}{5}$

9 (c) $(x_P - x_1) + (x_P - x_2) = \text{length of string} = \text{constant}$

Differentiating twice w.r.t. time, we get



Here $a_P = A$, a_1 is positive and a_2 is negative.

Hence, $A = \frac{a_1 - a_2}{2}$

10 (b) Acceleration of the system,

$$a = \frac{\text{Net pulling force}}{\text{Total mass}} = \frac{4g - 2g}{6} = \frac{g}{3}$$

Equation of motion of block C is

$$m_C g - T_{BC} = m_C a$$

$\therefore T_{BC} = m_C (g - a)$

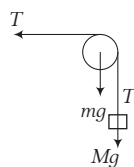
$$= 2 \left(9.8 - \frac{9.8}{3} \right) = 13 \text{ N}$$

11 (d) $T = Mg$

Force on the pulley (other than from clamp),

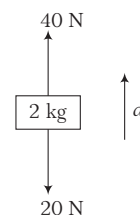
$$F_{\text{net}} = \sqrt{(T + mg)^2 + T^2}$$

$$= g \sqrt{(M + m)^2 + M^2}$$



Since, pulley is in equilibrium, clamp will exert the same amount of force in opposite direction or pulley will also exert this much force on clamp.

12 (b) Upward force on 2 kg block in upward direction will be 40 N ($= 2F$) in the form of tension.



$\therefore a = \frac{40 - 20}{2} = 10 \text{ ms}^{-2}$ (upward)

13 (d) Acceleration of the system, $a = \frac{\text{Net pulling force}}{\text{Total mass}}$

$$= \frac{2mg - mg}{3m} = \frac{g}{3}$$

Now, from equation of motion of m ,

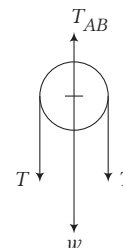
$$T - mg = ma = \frac{mg}{3}$$

$\therefore T = \frac{4mg}{3}$

For equilibrium of pulley,

$$T_{AB} = 2T + \text{Weight of pulley}$$

$$= \frac{8mg}{3} + 3mg = \frac{17mg}{3}$$



14 (c) Acceleration, $a = \left[\frac{m_2 - m_1}{m_2 + m_1} \right] g$

$$= \left[\frac{5m - m}{5m + m} \right] g$$

$$= \frac{4m}{6m} g = \frac{2}{3} g$$

15 (b) Acceleration of the system,

$$a = \frac{(m_2 - m_1)g}{(m_1 + m_2 + M)}$$

$$= \frac{(3 - 1)g}{(1 + 3 + 6)}$$

$$= \frac{2g}{10} = \frac{g}{5} = \frac{10}{5} = 2 \text{ ms}^{-2}$$

16 (a) Pulley is moving upward with acceleration a_0 ,

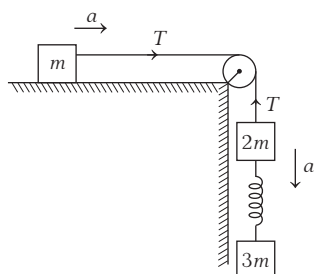
then tension, $T = 2 \left[\frac{m_1 m_2}{m_1 + m_2} \right] \left[g + a_0 \right]$

Here, $a_0 = \frac{g}{2}$

$\therefore T = 2 \left[\frac{m_1 m_2}{m_1 + m_2} \right] \left[g + \frac{g}{2} \right]$

$$= 3g \left[\frac{m_1 m_2}{m_1 + m_2} \right]$$

17 (b)



$$\text{For mass } m, \quad T = ma \quad \dots(i)$$

$$\text{For mass } 2m, \quad 2mg + kx - T = 2ma \quad \dots(ii)$$

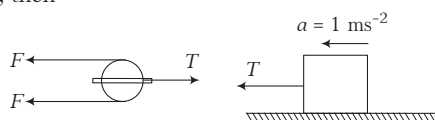
$$\text{For mass } 3m, \quad 3mg - kx = 3ma \quad \dots(iii)$$

Adding Eqs. (i), (ii) and (iii), we get

$$5mg = 6ma$$

$$\Rightarrow a = \frac{5g}{6} = \frac{50}{6} \text{ m/s}^2$$

18 (b) As shown in figure, when force F is applied at the end of the string, the tension in the lower part of the string is also F . If T is the tension in string connecting the pulley and the block, then



$$T = 2F$$

$$\text{But } T = ma = (200)(1) = 200 \text{ N}$$

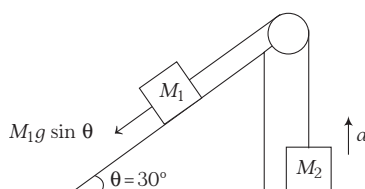
$$\therefore 2F = 200 \text{ N}$$

$$\text{or } F = 100 \text{ N}$$

19 (c) Since, $M_1 g \sin 30^\circ - M_2 g = (M_1 + M_2) a$

$$\Rightarrow 10 \times g \times \frac{1}{2} - 5g = 15a$$

$$\Rightarrow a = 0$$



20 (c) Acceleration of the system, $a = \frac{\text{Net pulling force}}{\text{Total mass}} = \frac{Mg \sin \theta}{2M}$

$$a = \frac{1}{2} g \sin \theta$$

Now, the block on ground is moving due to tension.

$$\text{Hence, } T = Ma = \frac{Mg \sin \theta}{2}$$

• CHECK POINT 5.4

2 (a) Maximum inclination of the plane with horizontal = angle of repose = $\tan^{-1}(\mu)$.

$$3 (c) a = \frac{v}{t} = \frac{4}{2} = 2 \text{ ms}^{-2} \Rightarrow F - f = ma$$

$$\text{or } 200 - \mu \times 30 \times 10 = 30 \times 2$$

$$\therefore \mu = 0.47$$

$$4 (d) \text{ Retardation, } a = \frac{F}{m} = \frac{5000}{1000} = 5 \text{ ms}^{-2}$$

$$\Rightarrow v = u + at \Rightarrow 0 = 30 - 5t$$

$$\therefore t = 6 \text{ s}$$

$$5 (c) a = \frac{F - \mu mg}{m} = \frac{100 - 0.5 \times 10 \times 10}{10} = 5 \text{ ms}^{-2}$$

$$6 (a) f_{\max} = \mu mg = 0.4 \times 2 \times 10 = 8 \text{ N}$$

Since, the applied force is less than $f_{\max}$, so force of friction will be equal to the applied force or 2.8 N.

$$7 (d) \text{ Retardation, } a = \frac{\mu mg}{m} = \mu g = 10 \mu$$

$$\text{Now, } v^2 = u^2 - 2as \text{ or } 0 = (6^2) - 2(10\mu)(9)$$

$$\therefore \mu = 0.2$$

$$8 (a) \text{ Retardation, } a = \frac{\mu mg}{m} = \mu g = 4 \text{ ms}^{-2}$$

$$\text{Using } v^2 = u^2 + 2as$$

$$(8)^2 = (0)^2 + 2(4)(s)$$

$$\therefore s = 8 \text{ m}$$

9 (a) Due to friction ($a = \mu g$), velocity of block will become equal to velocity of belt. Relative motion between two will stop.

$$\therefore v = at = \mu gt = 0.2 \times 10 \times 4 = 8 \text{ ms}^{-1}$$

$$10 (b) a_{\max} = \frac{T_{\max} - \mu mg}{m} = \frac{40 - 0.2 \times 8 \times 10}{8} = 3 \text{ ms}^{-2}$$

11 (a) m will move by friction,

$$f_{\max} = \mu mg \Rightarrow (a_m)_{\max} = \frac{\mu mg}{m} = \mu g$$

12 (a) Block A moves due to friction. Maximum acceleration of A can be $\frac{f_{\max}}{m}$ or $\frac{\mu mg}{m}$ or $\mu g = 0.2 \times 10 = 2 \text{ ms}^{-2}$. If both the blocks move together, then combined acceleration of A and B can be $\frac{10}{3} = 3.33 \text{ ms}^{-2}$. Since, this is more than the maximum acceleration of A. Slipping between them will take place and force of friction between A and B is $\mu m_A g = 2 \text{ N}$.

$$13 (a) f_{\max} = \mu mg = 0.8 \times 4 \times 10 = 32 \text{ N}$$

$$\text{At } t = 2 \text{ s, } F = kt^2 = (2)(2)^2 = 8 \text{ N}$$

Since, applied force $F < f_{\max}$, force of friction will be 8 N.

14 (d) N = applied force = 12 N

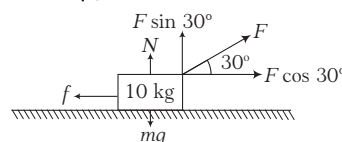
$$\therefore f_{\max} = \mu N = 7.2 \text{ N}$$

Since weight, $w < f_{\max}$

Force of friction, $f = 5 \text{ N}$

$$\therefore \text{Net contact force} = \sqrt{N^2 + f^2} = \sqrt{(12)^2 + (5)^2} = 13 \text{ N}$$

$$15 (d) \mu = \tan 30^\circ = \frac{1}{\sqrt{3}} \quad (\because \text{angle of friction} = 30^\circ)$$



$$N = mg - F \sin 30^\circ = \left(100 - \frac{F}{2}\right)$$

$$F \cos 30^\circ = \mu N$$

$$\Rightarrow \frac{\sqrt{3}F}{2} = \frac{1}{\sqrt{3}} \left(100 - \frac{F}{2}\right)$$

$$\Rightarrow \frac{3F}{2} = 100 - \frac{F}{2} \Rightarrow F = 50 \text{ N}$$

- 16 (a)** Angle of repose, $\theta_r = \tan^{-1}(\mu_s) = \tan^{-1}(0.7)$ or $\tan \theta_r = 0.7$

Angle of plane is $\theta = 30^\circ$, $\tan \theta = \tan 30^\circ = 0.577$

Since $\tan \theta < \tan \theta_r$, $\theta < \theta_r$

Block will not slide or $f = mg \sin \theta \neq \mu mg \cos \theta$

or $f = (2)(9.8) \sin 30^\circ = 9.8 \text{ N}$

- 17 (a)** $mg \sin \theta = (102)(10) \sin 30^\circ = 510 \text{ N}$

$$\mu_s mg \cos \theta = (0.4)(102)(10) \cos 30^\circ = 353 \text{ N}$$

$$F = 510 - 353 = 157 \text{ N}$$

- 18 (a)** $F = mg \sin \theta - \mu mg \cos \theta$... (i)

$$2F = mg \sin \theta + \mu mg \cos \theta$$
 ... (ii)

$$\Rightarrow 2mg \sin \theta - 2\mu mg \cos \theta = mg \sin \theta + \mu mg \cos \theta$$

$$\therefore \mu = \frac{1}{3} \tan \theta$$

(A) Taking it together

- 1 (b)** To solve this question, we have to apply Newton's second law of motion, in terms of force and change in momentum.

We know that, $F = \frac{dp}{dt}$

Given that, metre scale is moving with uniform velocity.

Hence, $dp = 0$. So, force, $F = 0$.

As all parts of the scale is moving with uniform velocity and total force is zero, hence torque on centre of mass will also be zero.

- 2 (d)** We know that for a system, $F_{\text{ext}} = \frac{dp}{dt}$
(from Newton's second law)

If $F_{\text{ext}} = 0$, $dp = 0 \Rightarrow p = \text{constant}$

Hence, momentum of a system will remain conserved, if external force on the system is zero.

In case of collision between particles equal and opposite forces will act on individual particles by Newton's third law.

Hence, total force on the system will be zero.

Note We should not confuse with system and individual particles. As total force on the system of both particles is zero, but force acts on individual particles.

- 3 (b)** The tension T in the string is $T = 6(g + a) = 6(10 + 1) = 66 \text{ N}$

- 4 (c)** If lift is accelerating, reading will be more, if it is decelerating, reading will be less and if it is moving with constant velocity, reading will be same.

Hence, the acceleration of the block relative to the incline is $(g - a) \sin \theta$.

- 5 (c)** Given, $\mathbf{u} = (3\hat{i} + 4\hat{j}) \text{ ms}^{-1}$ and $\mathbf{v} = -(3\hat{i} + 4\hat{j}) \text{ ms}^{-1}$

Mass of the ball = $150 \text{ g} = 0.15 \text{ kg}$

$\therefore \Delta \mathbf{p} = \text{Change in momentum}$

= Final momentum - Initial momentum = $m\mathbf{v} - m\mathbf{u}$

$$= m(\mathbf{v} - \mathbf{u}) = (0.15)[-(3\hat{i} + 4\hat{j}) - (3\hat{i} + 4\hat{j})]$$

$$= (0.15)[-6\hat{i} - 8\hat{j}] = -[0.15 \times 6\hat{i} + 0.15 \times 8\hat{j}]$$

$$= -(0.9\hat{i} + 1.20\hat{j})$$

Hence, $\Delta \mathbf{p} = -(0.9\hat{i} + 1.2\hat{j})$

- 6 (c)** By previous solution, $\Delta \mathbf{p} = -(0.9\hat{i} + 1.2\hat{j})$

$$\therefore \text{Magnitude} = |\Delta \mathbf{p}| = \sqrt{(0.9)^2 + (1.2)^2}$$

$$= \sqrt{0.81 + 1.44} = 1.5 \text{ kg} \cdot \text{ms}^{-1}$$

- 7 (c)** Force of friction is zero. Only contact force is the normal reaction which is $mg \cos \theta$.

- 8 (d)** Angle of plane is just equal to the angle of repose.

i.e. $\mu = \tan \theta$

Hence, the coefficient of kinetic friction, $\mu_k = \tan \theta$

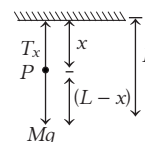
- 9 (c)** Acceleration, $a = (g \sin \theta + \mu g \cos \theta)$

$$= (g \sin 45^\circ + 0.5 \times g \times \cos 45^\circ)$$

$$= \frac{g}{\sqrt{2}} + 0.5 \times g \times \frac{1}{\sqrt{2}} = \frac{3g}{2\sqrt{2}}$$

- 10 (b)** Mass per unit length of rope = $\frac{M}{L}$

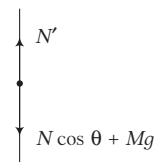
$$T_x = \text{mass of rope of length } (L - x) \times g = \frac{M}{L} (L - x)g$$



- 11 (c)** Wedge moves due to horizontal component of normal reaction.

Thus, $a = \frac{N_H}{M} = \frac{N \sin \theta}{M}$ (along -ve X-axis)

- 12 (a)** Net force on M in vertical direction should be zero.

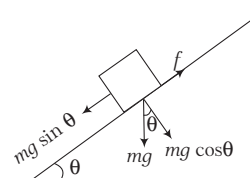


In vertically downward direction, two forces $N \cos \theta$ and Mg are acting. Therefore, N' the normal reaction from ground should be equal to $N \cos \theta + Mg$.

- 13 (d)** The block of mass m is at rest on an inclined plane.

Hence, frictional force acting between the surfaces is

$$f = mg \sin \theta$$



- 14 (a) Assuming the resistance force or retardation to be constant.

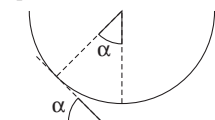
$$\left(\frac{v}{2}\right)^2 = v^2 - 2as_1 \quad \dots (i)$$

$$0 = \left(\frac{v}{2}\right)^2 - 2as_2 \quad \dots (ii)$$

Solving these two equations, we get

$$s_2 = \frac{s_1}{3} = \frac{3}{3} = 1 \text{ cm} \quad (\text{Given, } s_1 = 3 \text{ cm})$$

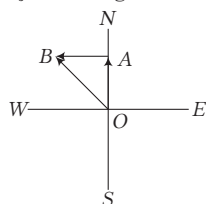
- 15 (a) α = angle of repose



$$\Rightarrow \tan \alpha = \mu = \frac{1}{3}$$

$$\therefore \cot \alpha = 3$$

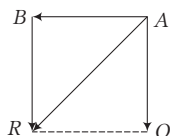
- 16 (d) Consider the adjacent diagram



Let $\mathbf{OA} = \mathbf{p}_1$ = Initial momentum of player northward

$\mathbf{AB} = \mathbf{p}_2$ = Final momentum of player towards west.

Clearly, $\mathbf{OB} = \mathbf{OA} + \mathbf{AB}$



Muscle force ($\mathbf{AR}$) = Change in momentum = $\mathbf{p}_2 - \mathbf{p}_1$

$$= \mathbf{AB} - \mathbf{OA} = \mathbf{AB} + (-\mathbf{OA})$$

Clearly, resultant $\mathbf{AR}$ will be along south-west.

- 17 (b) Given, mass of the car = m

As car starts from rest, $u = 0$

Velocity acquired along east, $\mathbf{v} = v\hat{\mathbf{i}}$

Duration, $t = 2 \text{ s}$

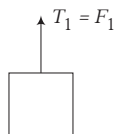
From first equation of motion,

$$v = u + at \Rightarrow v\hat{\mathbf{i}} = 0 + \mathbf{a} \times 2 \Rightarrow \mathbf{a} = \frac{v}{2} \hat{\mathbf{i}}$$

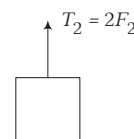
$$\text{Force, } \mathbf{F} = m\mathbf{a} = \frac{mv}{2} \hat{\mathbf{i}}$$

Hence, force acting on the car is $\frac{mv}{2}$ towards east due to friction on the tyres exerted by the road.

- 18 (a) In first case,



In second case,



$\therefore$ In first case, person will have to apply more force.

- 19 (a) Given, mass, $m = 2 \text{ kg}$

$$x(t) = pt + qt^2 + rt^3$$

$$\text{Velocity, } v = \frac{dx}{dt} = p + 2qt + 3rt^2$$

$$\text{Acceleration, } a = \frac{dv}{dt} = 0 + 2q + 6rt$$

$$\text{At } t = 2 \text{ s, } a = 2q + 6 \times r \times 2 = 2q + 12r \\ = 2 \times 4 + 12 \times 5 = 8 + 60 = 68 \text{ ms}^{-2}$$

$$\text{Force, } F = ma = 2 \times 68 = 136 \text{ N}$$

- 20 (b) Given, mass, $m = 5 \text{ kg}$

Acting force, $\mathbf{F} = (-3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) \text{ N}$

Initial velocity at $t = 0$, $\mathbf{u} = (6\hat{\mathbf{i}} - 12\hat{\mathbf{j}}) \text{ ms}^{-1}$

$$\text{Retardation, } \mathbf{a} = \frac{\mathbf{F}}{m} = \left(-\frac{3\hat{\mathbf{i}}}{5} + \frac{4\hat{\mathbf{j}}}{5}\right) \text{ ms}^{-2}$$

As final velocity is along Y-axis only, its x-component must be zero.

From $v = u + at$ for x-component only,

$$0 = 6 - \frac{3}{5}t \Rightarrow t = \frac{5 \times 6}{3} = 10 \text{ s}$$

- 21 (b) $F = u \frac{dm}{dt} = m(g + a)$

$$\Rightarrow \frac{dm}{dt} = \frac{m(g + a)}{u} = \frac{5000 \times (10 + 20)}{800} = 187.5 \text{ kgs}^{-1}$$

- 22 (b) Acceleration in both the cases will be same.

$$N_1 = ma, \text{ but } N_2 = (2m)a$$

$$\therefore \frac{N_1}{N_2} = \frac{1}{2}$$

- 23 (b) When force is applied on A, acceleration produced will be

$$\frac{F_A}{m_A + m_B} = \frac{\mu m_A g}{m_B} \quad \dots (i)$$

When force is applied on B, acceleration produced will be

$$\frac{F_B}{m_A + m_B} = \frac{\mu m_A g}{m_A} \quad \dots (ii)$$

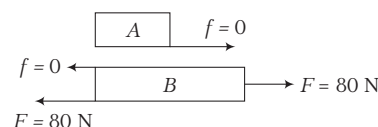
Dividing these two equations, we get

$$\frac{F_B}{F_A} = \frac{m_B}{m_A}$$

$$\therefore F_B = \frac{m_B}{m_A} \cdot F_A = \frac{8}{4} \times 12 = 24 \text{ N}$$

- 24 (d) Maximum force of friction between ground and B is

$$f_{\max} = (M + m)\mu g = (15 + 5)(0.6)(10) = 120 \text{ N}$$



As $f_{\max} > F$
 $\therefore$ Frictional force between B and ground will be 80 N and that between A and B is zero.

25 (b) Net external force, $F = \sqrt{(4)^2 + (3)^2} = 5 \text{ N}$

Maximum friction, $f_{\max} = \mu mg = (0.09)(5)(10) = 4.5 \text{ N}$

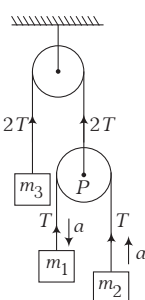
Since $F > f_{\max}$, block will move with an acceleration,

$$a = \frac{F - f_{\max}}{m} = \frac{5 - 4.5}{5} = 0.1 \text{ ms}^{-2}$$

26 (a) $\mu(m_A + m_C)g = m_B g$

$$\therefore m_C = \frac{m_B}{\mu} - m_A = \frac{5}{0.2} - 10 = 15 \text{ kg}$$

27 (a) m_3 is at rest. Therefore,



$$2T = m_3 g \quad \dots(i)$$

Further, if m_3 is at rest, then pulley P is also at rest.

Writing equations of motion, $m_1 g - T = m_1 a \quad \dots(ii)$

$$T - m_2 g = m_2 a \quad \dots(iii)$$

Solving Eqs. (i), (ii) and (iii), we get

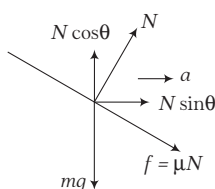
$$m_3 = 1 \text{ kg}$$

28 (b) $N \sin \theta + \mu N \cos \theta = ma$

$$N \cos \theta - \mu N \sin \theta = mg$$

Putting $\theta = 45^\circ$ and solving these two equations, we get

$$a = g \left(\frac{1 + \mu}{1 - \mu} \right)$$



29 (a) $N \sin \theta = ma = 1 \times 5 = 5$

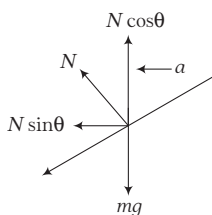
$$N \cos \theta = mg = 1 \times 10 = 10 \text{ N}$$

From these two equations, we get

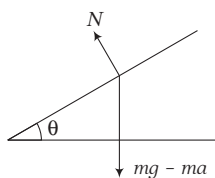
$$N = 5\sqrt{5} \text{ N}$$

(Adding and squaring them)

and $\tan \theta = \frac{1}{2}$

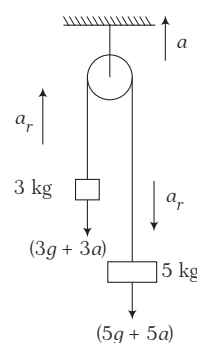


30 (a) $m(g - a) \sin \theta = \mu m(g - a) \cos \theta$



$$\Rightarrow \mu = \tan \theta = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

31 (c) Let acceleration of lift is a upwards. Then, with respect to lift,



$$a_r = \frac{\text{Net pulling force}}{\text{Total mass}} = \frac{(5g + 5a) - (3g + 3a)}{8}$$

$$\Rightarrow \frac{9}{32} g = \frac{(g + a)}{4} \quad (\text{Given})$$

$$\therefore a = \frac{g}{8}$$

32 (c) Suppose air resistance is F (upwards), then from equation of motion of balloon, we have

$$F' = \text{Mass} \times \text{Acceleration} = \frac{w}{g} \cdot a$$

$$\therefore F = w - F' = w \left(1 - \frac{a}{g} \right)$$

33 (d) Moving down with retardation a means, lift is accelerated upwards. With respect to lift, pseudo force on the block will be ma in downward direction, where m is the mass of block. So, downward force mg on the block will be replaced by $m(g + a)$. Therefore, acceleration of block relative to plane will be

$$a_r = (g + a) \sin \theta \quad (\text{Down the plane}) \quad \dots (i)$$

From $L = \frac{1}{2} a_r t^2 \quad \left(\because s = \frac{1}{2} a t^2 \right)$

$$\Rightarrow t = \sqrt{\frac{2L}{a_r}} = \sqrt{\frac{2L}{(g + a) \sin \theta}} \quad [\text{From Eq. (i)}]$$

34 (c) Momentum of one piece = $\frac{M}{4} \times 3$

Momentum of second piece = $\frac{M}{4} \times 4$

$$\therefore \text{Resultant momentum} = \sqrt{\frac{9M^2}{16} + M^2} = \frac{5M}{4}$$

The third piece should also have the same momentum. Let its velocity be v , then $\frac{5M}{4} = \frac{M}{2} \times v \Rightarrow v = \frac{5}{2} = 2.5 \text{ ms}^{-1}$

35 (a) $mg \sin \theta - f = ma$ (where, f = force of friction)

$$\text{or } f = m(g \sin 30^\circ - a) = 8 \left(10 \times \frac{1}{2} - 0.4 \right) = 36.8 \text{ N}$$

36 (a) Acceleration of system before breaking the string,

$$a = \frac{\text{Net pulling force}}{\text{Total mass}} = \frac{3g - 2g}{5} = \frac{g}{5}$$

After 5 s velocity of system, $v = at = \frac{g}{5} \times 5 = g \text{ ms}^{-1}$

Now, $h = \frac{v^2}{2g} = \frac{g^2}{2g} = \frac{g}{2} = 4.9 \text{ m}$

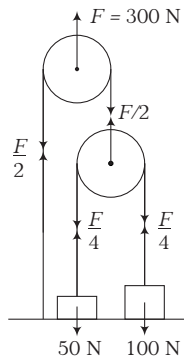
- 37** (a) Maximum acceleration of the box can be μg or 1.5 ms^{-2} , while acceleration of truck is 2 ms^{-2} . Therefore, relative acceleration of the box will be $a_r = 0.5 \text{ ms}^{-2}$ (backward). It will fall off the truck in a time,

$$t = \sqrt{\frac{2l}{a_r}} = \sqrt{\frac{2 \times 4}{0.5}} = 4 \text{ s} \quad \left(\because s = \frac{1}{2} at^2 \right)$$

Displacement of truck upto this instant,

$$s_r = \frac{1}{2} a_r t^2 = \frac{1}{2} \times 0.5 \times (4)^2 = 4 \text{ m}$$

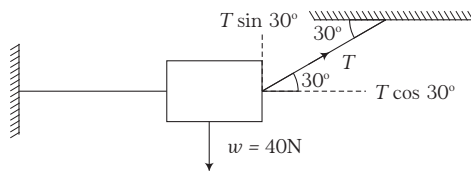
- 38** (a) $\frac{F}{4} = 75 \text{ N}$ or $\frac{F}{4} < 100 \text{ N}$



Therefore, $a_M = 0$, $a_m = \frac{75 - 50}{5} = 5 \text{ m/s}^2$

- 39** (d) Total upward force = $2 \left(\frac{mg}{2} \right) = mg$.
(Weight of man is balance by total tension acting upwards)
Total downward force is also mg .
 $\therefore F_{\text{net}} = 0 = a_{\text{net}}$

- 40** (a) $T \sin 30^\circ = w = 40 \text{ N}$



$\therefore \frac{T}{2} = 40$ or $T = 80 \text{ N}$

- 41** (c) For smooth surface, $a_1 = g \sin 45^\circ$, $t_1 = t$

For rough surface, $a_2 = g \sin 45^\circ - \mu g \cos 45^\circ$, $t_2 = 2t$

$\therefore s = \frac{1}{2} a_1 t_1^2 = \frac{1}{2} a_2 t_2^2 \quad \left(\because s = \frac{1}{2} at^2 \right)$

$\Rightarrow (g \sin 45^\circ) t^2 = (g \sin 45^\circ - \mu g \cos 45^\circ) (2t)^2$

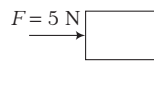
$\Rightarrow 1 = (1 - \mu) 4 \Rightarrow \mu = \frac{3}{4} = 0.75$

- 42** (b) Here, $N = F = 5 \text{ N}$, $\mu = 0.5$, $m = 0.1 \text{ kg}$

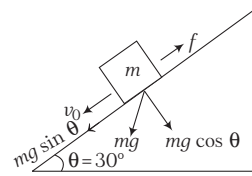
$\therefore f_{\text{max}} = \mu N = 2.5 \text{ N}$

Weight, $w = mg = 0.1 \times 9.8 = 0.98 \text{ N}$

Since $w < f_{\text{max}}$, force of friction will be 0.98 N .



- 43** (c) $f_{\text{max}} = \mu mg \cos \theta = 0.6 \times mg \times \frac{\sqrt{3}}{2} = 0.52mg$



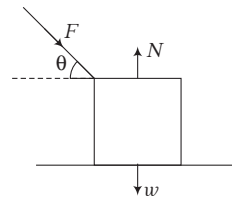
$$mg \sin \theta = \frac{mg}{2} = 0.5mg$$

Since, $f_{\text{max}} > mg \sin \theta$

Block will decelerate and come to rest.

- 44** (b) $N = w + F \sin \theta$

$\therefore f_{\text{max}} = \mu N = (\tan \phi)(w + F \sin \theta)$



To move the body, $F \cos \theta = f_{\text{max}} = (\tan \phi)(w + F \sin \theta)$

Solving this equation, we get

$$F = \frac{w \sin \phi}{\cos(\theta + \phi)}$$

- 45** (c) Maximum acceleration due to friction of mass m over mass $2m$ can be μg . Now, for the whole system,

$$a = \frac{\text{Net pulling force}}{\text{Total mass}}$$

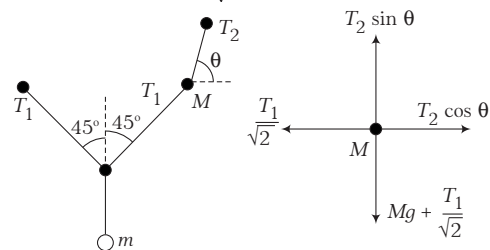
$\therefore \mu g = \frac{mg}{4m}$

or $\mu = \frac{1}{4}$

- 46** (a) $2T_1 \cos 45^\circ = mg$

$\therefore T_1 = \frac{mg}{\sqrt{2}}$

... (i)

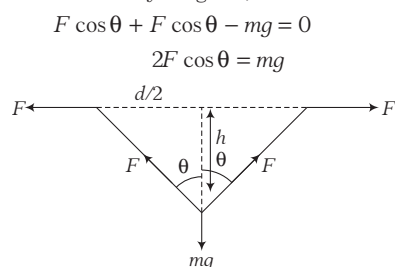


$$T_2 \cos \theta = \frac{T_1}{\sqrt{2}} = \frac{mg}{2}$$

$$T_2 \sin \theta = Mg + \frac{mg}{2}$$

$$\tan \theta = \frac{Mg + \frac{mg}{2}}{\frac{mg}{2}} = 1 + \frac{2M}{m}$$

- 47 (a) From the free body diagram,



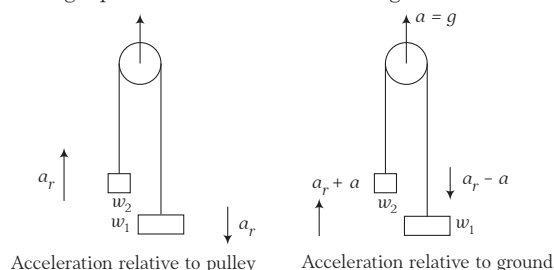
So, as the man moves up, θ increases, $\cos \theta$ decreases.

$$\therefore \cos \theta = \frac{h}{\sqrt{\left(\frac{d}{2}\right)^2 + h^2}}$$

Now, when the man is at depth h , then the force is

$$F = \frac{mg}{2 \cos \theta} = \frac{mg \sqrt{h^2 + \frac{d^2}{4}}}{2h} = \frac{mg}{4h} \sqrt{d^2 + 4h^2}$$

- 48 (a) Writing equation of motion for two weights



$$w_1 - T = \frac{w_1}{g} (a_r - a) \quad \dots(i)$$

$$T - w_2 = \frac{w_2}{g} (a_r + a) \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$T = \frac{4w_1 w_2}{w_1 + w_2} \text{ with } a = g$$

- 49 (d) Acceleration of system,

$$a = \frac{F - f}{M + m} = \frac{20 - 10}{6 + 4} = 1 \text{ ms}^{-2} \text{ (towards left)}$$

Let F_0 be the reading of dynamometer, then equation of motion of mass m would be

$$F_0 - f = ma$$

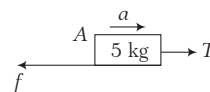
$$\Rightarrow F_0 = f + ma = 10 + (4)(1) = 14 \text{ N}$$

- 50 (b) Net pulling force on the system should be zero, as velocity is constant. Hence,

$$m_A g \sin 30^\circ = \mu m_A g \cos 30^\circ + m_B g$$

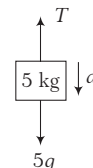
$$\therefore m_B = \left(\frac{m_A}{2} \right) - \left(\frac{\mu m_A \sqrt{3}}{2} \right) = 10 \left[\frac{1}{2} - 0.2 \times \frac{\sqrt{3}}{2} \right] = 3.3 \text{ kg}$$

- 51 (c)



$$\Rightarrow T - f = 5a, T - \mu mg = 5a$$

$$T - 0.5 \times 5 \times 9.8 = 5a \quad \dots(i)$$



$$\Rightarrow 5g - T = 5a \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$5 \times 9.8 - 0.5 \times 5 \times 9.8 = 10a$$

$$\Rightarrow 0.5 \times 5 \times 9.8 = 10a$$

$$\Rightarrow a = 2.45 \text{ m/s}^2$$

From Eq (ii), we get

$$5 \times 9.8 - 5 \times 2.45 = T$$

$$\Rightarrow T = 36.75 \text{ N}$$

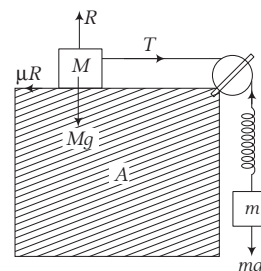
- 52 (d) Since, the block is at rest under two forces

(i) weight of block.

(ii) contact force from the plane (resultant of force of friction and normal reaction).

Contact force should be equal to weight (or 30 N) in upward direction. Because under the action of two forces, a body remains in equilibrium when both the forces are equal in magnitude, but opposite in direction, i.e. 30 N.

- 53 (a) When the system of two blocks of masses M and m is either at rest or moving with the same speed, no net force acts on them.



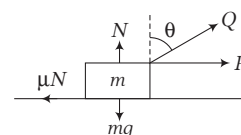
In this equilibrium state,

$$T = \mu R = \mu Mg \text{ and } T = mg$$

$$\therefore \mu Mg = mg \Rightarrow M = \frac{m}{\mu}$$

- 54 (a)

$$N = mg - Q \cos \theta$$



$$P + Q \sin \theta = \mu N = \mu (mg - Q \cos \theta)$$

$$\therefore \mu = \frac{(P + Q \sin \theta)}{(mg - Q \cos \theta)}$$

55 (b) $m_1 g > m_2 g \sin \theta + \mu m_2 g \cos \theta$

$$\Rightarrow \frac{m_1}{m_2} > \sin \theta + \mu \cos \theta$$

56 (d) $N = Mg - F \cos \theta = Mg - Mg \cos \theta = 2Mg \sin^2 \frac{\theta}{2}$

Further, block can be pulled, if $F \sin \theta \geq \mu N$

$$\Rightarrow 2Mg \sin \frac{\theta}{2} \cdot \cos \frac{\theta}{2} \geq 2\mu Mg \sin^2 \frac{\theta}{2}$$

$$\Rightarrow \cot\left(\frac{\theta}{2}\right) \geq \mu$$

57 (d) $a = \frac{2F - F}{2m} = \frac{F}{2m}$ (towards left)

Writing equation of right hand side block,

$$2F - N \sin 30^\circ = m a = \frac{F}{2}$$

$$\frac{N}{2} = 2F - \frac{F}{2} = \frac{3F}{2} \Rightarrow N = 3F$$

58 (c) In critical case, weight of hanging part = force of friction of the part of rope lying on table.

$$\therefore \frac{m}{l} \cdot l_1 g = \mu \frac{m}{l} (l - l_1) g$$

Solving above equation, we get

$$l_1 = \left(\frac{\mu}{1 + \mu} \right) l$$

59 (d) Block A moves due to friction. Maximum value of friction can be $\mu m_A g$. Therefore, maximum acceleration of A can be $\frac{\mu m_A g}{m_A}$ or $\mu g = 0.2g = \frac{g}{5}$. When force $2mg$ is applied on lower block, common acceleration (if both move together) will be

$$a = \frac{\text{Net force}}{\text{Total mass}} = \frac{2mg}{4m} = \frac{g}{2}$$

Since, $a = g/2$ is greater than maximum acceleration of A which can be given to it by friction. Therefore, slipping will take place.

$$a_A = 0.2g = \frac{g}{5}$$

$$a_B = \frac{2mg - 0.2mg}{3m} = 0.6g = \frac{3g}{5}$$

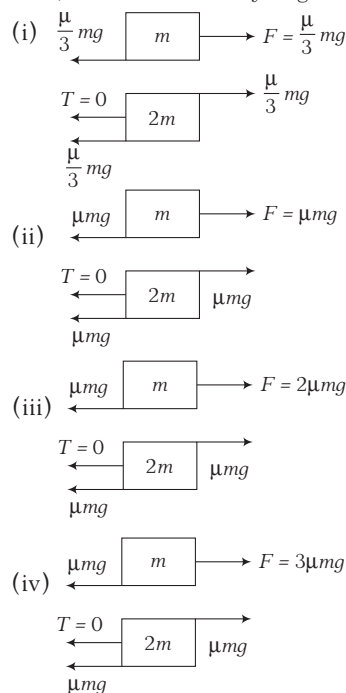
$$\begin{aligned} a_{AB} &= a_A - a_B = \left(\frac{g}{5} - \frac{3g}{5} \right) \\ &= \frac{-2g}{5} \text{ (in backward direction)} \end{aligned}$$

60 (c) f_1 = maximum value of friction between m and $2m = \mu mg$

f_2 = maximum value of friction between $2m$ and ground

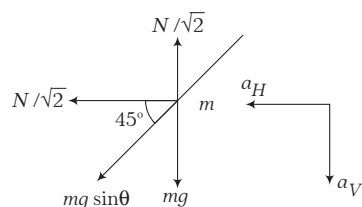
$$= \frac{\mu}{3} (3m)g = \mu mg$$

Now, let us draw free body diagram of m and $2m$ in all four cases.



As tension is zero in all four cases. So, option (c) is incorrect.

61 (b) For block, $a = g \sin \theta = \frac{g}{\sqrt{2}}$

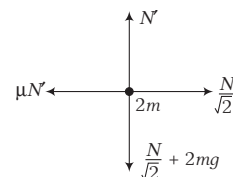


$$\begin{aligned} a_H &= a_V = a \cos 45^\circ \\ &= \frac{g}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{g}{2} \end{aligned}$$

$$\frac{N}{\sqrt{2}} = \frac{mg}{2} \quad \dots (i)$$

$$mg - \frac{N}{\sqrt{2}} = \frac{mg}{2} \quad \dots (ii)$$

For wedge,



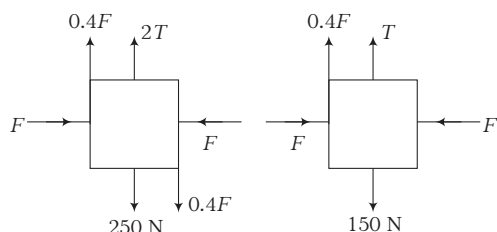
$$N' = \frac{N}{\sqrt{2}} + 2mg \quad \dots (iii)$$

$$\mu N' = \frac{N}{\sqrt{2}} \quad \dots (iv)$$

Solving above four equations, we get

$$\mu = 0.2$$

62 (a) $2T = 250$



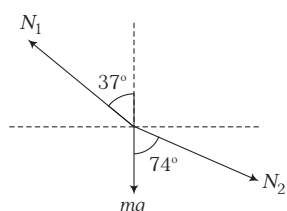
$\Rightarrow T = 125 \text{ N}$

Also, $T + 0.4F = 150$

$\Rightarrow F = 62.5 \text{ N}$

63 (d) $N_1 \sin 37^\circ = N_2 \sin 74^\circ$

$\Rightarrow N_1 = 2N_2 \cos 37^\circ$

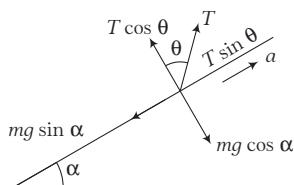


Now, $N_1 \cos 37^\circ - N_2 \cos 74^\circ = mg$

$\Rightarrow 2N_2 \cos^2 37^\circ - N_2 (2 \cos^2 37^\circ - 1) = mg$

$\Rightarrow 2N_2 \left(\frac{16}{25} \right) - N_2 \left(\frac{32}{25} - 1 \right) = mg \Rightarrow N_2 = mg$

64 (c) $T \sin \theta - mg \sin \alpha = ma$



and $T \cos \theta = mg \cos \alpha$

From these two equations, we get

$\tan \theta = \frac{a + g \sin \alpha}{g \cos \alpha} \Rightarrow \theta = \tan^{-1} \left(\frac{a + g \sin \alpha}{g \cos \alpha} \right)$

(B) Medical entrance special format questions

• Assertion and reason

2 (c) $mg \sin \theta = 10 \times 10 \times \frac{1}{2} = 50 \text{ N}$

$f_{\text{friction}} = \mu mg \cos \theta = 0.7 \times 10 \times 10 \times \frac{\sqrt{3}}{2} = 60.62 \text{ N}$

Since, $\mu mg \cos \theta$ is more, block will remain stationary.

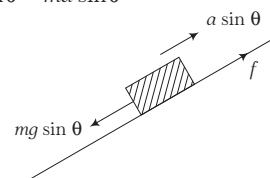
4 (a) $(f_2)_{\text{max}} = \mu_2 mg$

$(f_1)_{\text{max}} = \mu_1 (2m)g$

Since, $(f_1)_{\text{max}} > (f_2)_{\text{max}}$

Lower block will not move at all.

5 (b) $f - mg \sin \theta = ma \sin \theta$



$\therefore f = m(g + a) \sin \theta > mg \sin \theta$

6 (b) $N_1 = ma_x = 2m$

$N_2 = m(g + a_y) = 12m$

$\therefore \frac{N_1}{N_2} = \frac{1}{6}$

• Statement based questions

2 (b) The statement given in option (b) is incorrect, which can be corrected as,
Friction opposes the relative motion of the bodies in contact not the motion.

3 (b) Only Statement II is correct while Statement I is incorrect, which can be corrected as,
Tension is non-uniform even, if string is not accelerated.

4 (a) $F_{\text{net}} = ma$

For 1 kg mass $\uparrow T$
 $\downarrow g$
 $\Rightarrow T - g = a$

For 2 kg mass $\uparrow T$
 $\downarrow g$
 $\Rightarrow 2g - T = 2a$

Solving, we get $a = \frac{10}{3} \text{ m/s}^2$

Force on 1 kg, $F_1 = \frac{10}{3} \text{ N}$ and on 2 kg, $F_2 = \frac{20}{3} \text{ N}$

Hence, Statement I is correct while Statement II is incorrect

5 (c) As forces at two ends of string are different, so tension on it decreases from $2F$ to F in moving from A to B.

In case of massless string, $a = \frac{2F - F}{0} = \infty$

Hence, both statements are correct.

• Match the columns

1 (c) $N = mg - 20\sqrt{2} \sin 45^\circ = 20 \text{ N}$, $f = \mu_k N = 12 \text{ N}$

Since, $20\sqrt{2} \cos 45^\circ > \mu_k N$ block will move and its acceleration will be

$a = \frac{20\sqrt{2} \cos 45^\circ - \mu_k N}{m} = \frac{20 - 12}{4} = 2 \text{ ms}^{-2}$

Hence, A $\rightarrow$ q, B $\rightarrow$ p, C $\rightarrow$ s.

2. (d) Acceleration of system,

$a = \frac{60 - 18 - (m_1 + m_2 + m_3)g \sin 30^\circ}{(m_1 + m_2 + m_3)} = 2 \text{ ms}^{-2}$

Net force on 3 kg block $= m_3 a = 6 \text{ N}$

From free body diagram of 1 kg block, we have

$$N_{12} - m_1 g \sin 30^\circ - 18 = m_1 a$$

$$\Rightarrow N_{12} = 25 \text{ N}$$

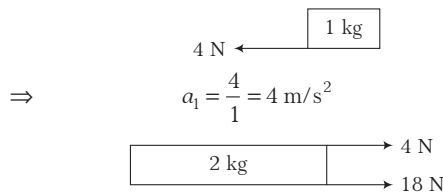
From free body diagram of 3 kg block, we have

$$60 - m_3 g \sin 30^\circ - N_{32} = m_3 a$$

$$\therefore N_{32} = 39 \text{ N}$$

Hence, A $\rightarrow$ r, B $\rightarrow$ t, C $\rightarrow$ q, D $\rightarrow$ t.

3 (b) Force diagram of both the blocks are as shown.



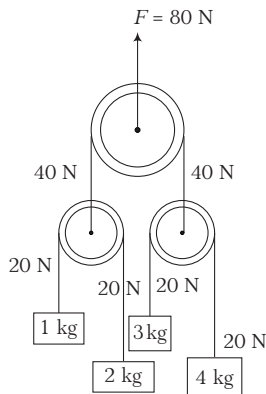
$$\Rightarrow a_1 = \frac{4}{1} = 4 \text{ m/s}^2$$

$$\Rightarrow a_2 = \frac{22}{2} = 11 \text{ m/s}^2$$

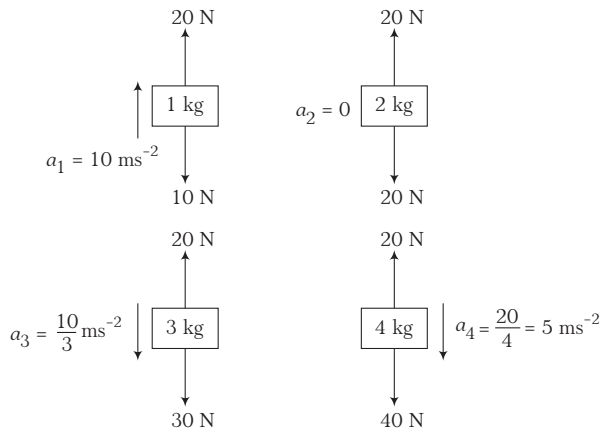
$$\Rightarrow a_{21} = a_2 - a_1 = 7 \text{ m/s}^2$$

Hence, A $\rightarrow$ s, B $\rightarrow$ p, C $\rightarrow$ s.

4 (a) Since, the pulleys are smooth, net force on each pulley should be zero. With this concept, tensions on all strings are shown below.



Now, we can draw free body diagrams of all the four blocks.

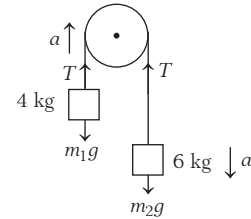


Hence, A $\rightarrow$ (r, t), B $\rightarrow$ p, C $\rightarrow$ q, D $\rightarrow$ (q, s).

(C) Medical entrances' gallery

1 (b) Given, $m_1 = 4 \text{ kg}$, $m_2 = 6 \text{ kg}$ and $a = ?$

The free body diagram of given system is shown below



The balancing equations for given system are

$$T - m_1 g = m_1 a \quad \dots(i)$$

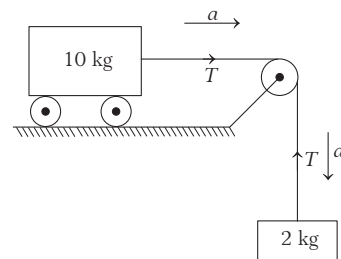
$$\text{and} \quad m_2 g - T = m_2 a \quad \dots(ii)$$

Adding Eqs. (i) and (ii), we get

$$m_2 g - m_1 g = m_1 a + m_2 a$$

$$\Rightarrow a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) \cdot g = \left(\frac{6 - 4}{4 + 6} \right) \times g = \frac{g}{5}$$

2 (a) The given situation is shown in the following diagram.



If a be the acceleration of the system,

then equation of motion of 10 kg trolley,

$$T - \mu R = 10a$$

$$\Rightarrow T - 0.05 \times 10g = 10a \quad (\text{Given, } \mu = 0.05, R = 10g)$$

$$\Rightarrow T - 5 = 10a \quad \dots (i)$$

Equation of motion of 2kg block,

$$2g - T = 2a$$

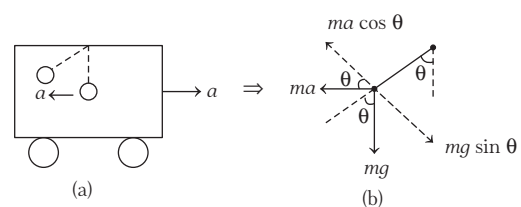
$$\Rightarrow 20 - T = 2a \quad \dots (ii)$$

Adding Eqs. (i) and (ii), we get

$$15 = 12a \Rightarrow a = \frac{15}{12} = \frac{5}{4} = 1.25 \text{ ms}^{-2}$$

3 (b) As the truck move to the right, so the bob will move to the left due to inertia of rest with acceleration a .

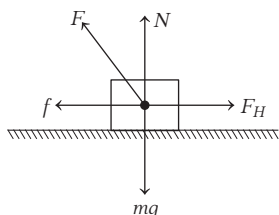
Thus, the given situation can be drawn as



Now, from equilibrium of forces in above diagram (b), we get

$$\begin{aligned} ma \cos \theta &= mg \sin \theta \\ \Rightarrow \frac{\sin \theta}{\cos \theta} &= \frac{ma}{mg} \\ \Rightarrow \tan \theta &= \frac{a}{g} \Rightarrow \theta = \tan^{-1} \left(\frac{a}{g} \right) \end{aligned}$$

- 4 (c) The situation can be drawn as



The frictional force, $f = \mu N = \mu mg$ ($\because N = mg$)

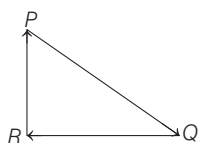
From free body diagram, the resultant force is

$$\begin{aligned} |\mathbf{F}| &= \sqrt{N^2 + f^2} \\ &= \sqrt{(mg)^2 + (\mu mg)^2} = mg\sqrt{1 + \mu^2} \end{aligned}$$

This is the minimum force required to move the object. But as the body is not moving.

$$\therefore |\mathbf{F}| \leq mg\sqrt{1 + \mu^2}$$

- 5 (b) As the three forces are represented by three sides of a triangle taken in order, then they will be in equilibrium.



$$\Rightarrow \mathbf{F}_{\text{net}} = \mathbf{F}_{PQ} + \mathbf{F}_{QR} + \mathbf{F}_{RP} = 0$$

$$\mathbf{F}_{\text{net}} = m \times \mathbf{a} = m \frac{d\mathbf{v}}{dt} = 0$$

$$\Rightarrow \frac{d\mathbf{v}}{dt} = 0 \text{ or } \mathbf{v} = \text{constant}$$

So, the velocity of particle remain constant.

- 6 (a) Force exerted by concrete floor is more as compared to wooden floor due to greater change in momentum.
Since on concrete floor, glass ball will take less time to come to rest, so a glass ball dropped on concrete floor can easily get broken compared, if it is dropped on wooden floor.
Hence, both Assertion and Reason are true and Reason is the correct explanation of Assertion.

- 7 (c) Force applied by gun, $F = (100 - 0.5 \times 10^5 t) \text{ N}$

Speed of bullet, $v = 400 \text{ m/s}$

When $F = 0$, then $100 - 0.5 \times 10^5 t = 0$

$$\Rightarrow t = 2 \times 10^{-3} \text{ s}$$

$$\text{Impulse, } I = \int F dt = \int_0^{2 \times 10^{-3}} (100 - 0.5 \times 10^5 t) dt$$

$$\begin{aligned} &= \left[100t - 0.5 \times 10^5 \cdot \frac{t^2}{2} \right]_0^{2 \times 10^{-3}} \\ &= 100 \times 2 \times 10^{-3} - \frac{0.5 \times 10^5}{2} \times (4 \times 10^{-6} - 0) \\ &= 0.2 - 0.1 = 0.1 \text{ N-s} \end{aligned}$$

- 8 (d) By Newton's second law of motion, net external force applied on a system is equal to rate of change of momentum.

$$\text{i.e. } \mathbf{F}_{\text{ext}} = \frac{d\mathbf{p}}{dt}$$

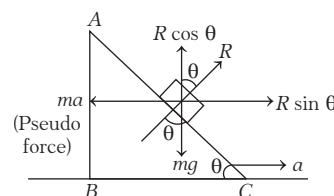
If $\mathbf{F}_{\text{ext}} = 0$, then $\frac{d\mathbf{p}}{dt} = 0 \Rightarrow \mathbf{p} = \text{constant}$

i.e. When $\mathbf{F}_{\text{ext}} = 0$, then momentum of the system remains conserved.

The internal interaction between particles of a body do not contribute to change in total momentum of body.

Hence, both Assertion and Reason are false.

- 9 (d) According to the question, the free body diagram of the given condition will be



Since, the wedge is accelerating towards right with a , thus a pseudo force acts in the left direction in order to keep the block stationary. As, the system is in equilibrium.

$$\therefore \Sigma F_x = 0 \text{ or } \Sigma F_y = 0$$

$$\Rightarrow R \sin \theta = ma \quad \dots(i)$$

$$\text{Similarly, } R \cos \theta = mg \quad \dots(ii)$$

Dividing Eq. (i) by Eq (ii), we get

$$\begin{aligned} \frac{R \sin \theta}{R \cos \theta} &= \frac{ma}{mg} \\ \Rightarrow \tan \theta &= \frac{a}{g} \text{ or } a = g \tan \theta \end{aligned}$$

- $\therefore$ The relation between a and g for the block to remain stationary on the wedge is $a = g \tan \theta$.

- 10 (d) The opposing force that comes into play when one body is actually sliding over the surface of the other body is called sliding friction.

The coefficient of sliding is given as $\mu_s = N/F_{\text{sliding}}$

where, N is the normal reaction and F_{sliding} is the sliding force.

As, the dimensions of N and F_{sliding} are same. Thus, μ_s is a dimensionless quantity. When body is rolling, then it reduces the area of contact of surfaces, hence rolling friction is smaller than sliding friction.

Hence, statement (d) is incorrect.

- 11 (c) When the system is in equilibrium, then the spring force is $3mg$. When the string is cut, then net force on block A

$$= 3mg - 2mg = mg$$

Hence, acceleration of block A at that instant,

$$a = \frac{\text{Force on block A}}{\text{Mass of block A}} = \frac{mg}{2m} = \frac{g}{2}$$

When string is cut, then block B falls freely with an acceleration equal to g .

- 12** (a) Let a_1 and a_2 be accelerations of a ball (upward) and rod (downward), respectively.

Clearly, from the diagram,

$$2a_1 = a_2 \quad \dots(i)$$

Now, for the ball,

$$2T - \frac{9}{5}mg = \frac{9}{5}ma_1 \quad \dots(ii)$$

and for the rod, $mg - T = ma_2$

...(iii)

On solving Eqs. (i), (ii) and (iii), we get

$$a_1 = \frac{g}{29} \text{ m/s}^2 \uparrow (\text{upward})$$

$$a_2 = \frac{2g}{29} \text{ m/s}^2 \downarrow (\text{downward})$$

So, acceleration of ball w.r.t. rod $= a_1 + a_2 = (3g/29) \text{ m/s}^2$

Now, displacement of ball w.r.t. rod when it reaches the upper end of rod is 1m.

Using second equation of motion, $s = ut + \frac{1}{2}at^2$

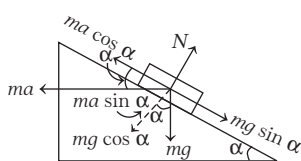
$$1 = 0 + \frac{1}{2} \times \frac{3 \times 10}{29} t^2$$

$$\Rightarrow t = \sqrt{58/30} = 1.4 \text{ s (approx)}$$

- 13** (a) Angle of repose is equal to angle of limiting friction as maximum value of static friction is called the limiting friction.

Hence, both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

- 14** (a) Since, $F = (M + m)a$... (i)



Normal, $N = (mg \cos \alpha + ma \sin \alpha)$

So, we apply pseudo force on the block by observing it from the wedge.

Now, from free body diagram of block, we get

$$ma \cos \alpha = mg \sin \alpha$$

$$\Rightarrow a = g \frac{\sin \alpha}{\cos \alpha} \Rightarrow a = g \tan \alpha \quad \dots(ii)$$

Now, from Eqs. (i) and (ii), we get

$$F = (M + m)g \tan \alpha$$

- 15** (c) Given, $\theta = 45^\circ$, $s_1 = s_2$, $u = 0$, $t_1 = 2t_2$

On the rough incline, $a_1 = g(\sin \theta - \mu \cos \theta)$

On the frictionless incline, $a_2 = g \sin \theta$

$$\text{As, } s = ut + \frac{1}{2}at^2$$

$$\text{So, } s_1 = 0 + \frac{1}{2}g(\sin \theta - \mu \cos \theta)t_1^2 \text{ and } s_2 = 0 + \frac{1}{2}g \sin \theta t_2^2$$

$$\text{As, } s_1 = s_2, \frac{1}{2}g(\sin \theta - \mu \cos \theta)t_1^2 = \frac{1}{2}g \sin \theta t_2^2$$

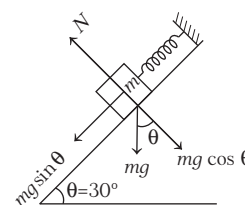
$$\Rightarrow \frac{\sin \theta - \mu \cos \theta}{\sin \theta} = \frac{t_2^2}{t_1^2}$$

$$\Rightarrow 1 - \mu \cot \theta = \frac{t_2^2}{(2t_2)^2} \quad (\because t_1 = 2t_2)$$

$$\Rightarrow 1 - \mu \cot \theta = 1/4 \Rightarrow \mu \cot \theta = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\therefore \mu = \frac{3}{4 \cot \theta}$$

- 16** (b) Acceleration of the body down the inclined plane $= g \sin \theta$.



$\therefore$ Force applied on spring balance

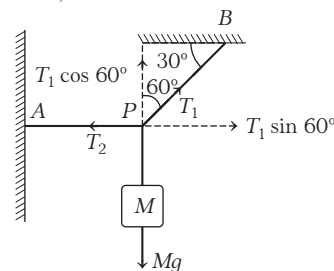
$$= mg \sin \theta = 5 \times 10 \times \sin 30^\circ = 5 \times 10 \times \frac{1}{2} = 25 \text{ N}$$

- 17** (b) Given, $F = 10 \text{ N}$, $v_i = 0$, $m = 0.5 \text{ kg}$ and $\Delta t = 0.25 \text{ s}$

As impulse, $I = p_f - p_i$

$$\text{Also, } I = F \cdot \Delta t = 10 \times 0.25 = 2.5 \text{ N-s}$$

- 18** (b) As there is a load at P, so tension in AP and PB will be different. Let these be T_2 and T_1 , respectively. For vertical equilibrium at P,



$$T_1 \cos 60^\circ = Mg, \text{ i.e. } T_1 = 2Mg \quad \dots(i)$$

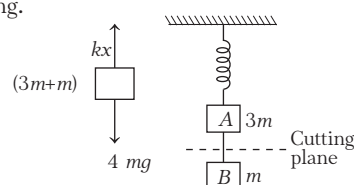
and for horizontal equilibrium at P,

$$T_2 = T_1 \sin 60^\circ = T_1 (\sqrt{3}/2)$$

Substituting the value of T_1 from Eq. (i), we get

$$T_2 = (2Mg) \times (\sqrt{3}/2) = \sqrt{3} Mg$$

- 19** (b) Initially system is in equilibrium with a total weight of $4mg$ over spring.



$$\therefore kx = 4mg$$

When string is cut at the location as shown above.

Force on mass m , $F_B = mg$

$\therefore$ Acceleration of mass m , $a_B = g$

For mass $3m$

If a_A = acceleration of block of mass $3m$, then $F_{\text{net}} = 4mg - 3mg$

$$\Rightarrow 3m \cdot a_A = mg \text{ or } a_A = \frac{g}{3}$$

So, accelerations of blocks A and B are $a_A = \frac{g}{3}$ and $a_B = g$

20 (c) Given situation can be represented as

$$F \leftarrow m_1 \rightarrow T_1 \leftarrow m_2 \rightarrow T_2 \leftarrow m_3 \rightarrow T_3 \leftarrow m_4$$

$$\Rightarrow T_1 = \frac{(m_2 + m_3 + m_4)}{m_1 + m_2 + m_3 + m_4} F$$

Given, $m_1 = m_2 = m_3 = m_4 = m$

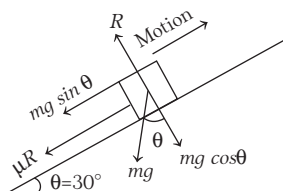
$$\therefore T_1 = \frac{3}{4} F$$

$$\text{Similarly, } T_2 = \frac{(m_3 + m_4)F}{m_1 + m_2 + m_3 + m_4}$$

$$\therefore T_2 = \frac{1}{2} F$$

$$\text{Also, } T_3 = \frac{m_4 F}{m_1 + m_2 + m_3 + m_4} \Rightarrow T_3 = \frac{1}{4} F$$

21 (a) According to question, the situation can be shown as



Case 1 For body projected up the plane, $v = 0$

Using third equation of motion, $v^2 = u^2 - 2a_1 s$

$$\Rightarrow 0 = u^2 - 2a_1 s$$

$$\therefore u = \sqrt{2a_1 s}$$

$$\text{Also, } v = u - a_1 t_1$$

$$\Rightarrow 0 = u - a_1 t_1$$

$$\Rightarrow t_1 = u / a_1$$

$$\therefore t_1 = \frac{\sqrt{2a_1 s}}{a_1} = \sqrt{\frac{2s}{a_1}} \quad \dots (i)$$

As, $ma_1 = \mu R + mg \sin \theta$

$$\therefore \text{Retardation, } a_1 = \frac{\mu R + mg \sin 30^\circ}{m}$$

$$\text{Also, } R = mg \cos 30^\circ$$

$$\begin{aligned} \Rightarrow a_1 &= \frac{\mu mg \cos 30^\circ + mg \sin 30^\circ}{m} \\ &= \mu g \frac{\sqrt{3}}{2} + \frac{g}{2} \quad \dots (ii) \end{aligned}$$

Case 2 For body coming down the plane,

$$s = ut + \frac{1}{2} a_2 t_2^2$$

$$\text{As, } u = 0$$

$$\Rightarrow t_2 = \sqrt{2s / a_2} \quad \dots (iii)$$

Downward acceleration,

$$\begin{aligned} a_2 &= \frac{mg \sin 30^\circ - \mu R}{m} \\ &= \frac{mg \sin 30^\circ - \mu mg \cos 30^\circ}{m} \\ &= \frac{g}{2} - \frac{\mu g \sqrt{3}}{2} \quad \dots (iv) \end{aligned}$$

$$\text{Given, } t_1 = t_2 / 2$$

Substituting values of t_1 and t_2 from Eqs. (i) and (iii), we get

$$\sqrt{\frac{2s}{a_1}} = \frac{1}{2} \sqrt{\frac{2s}{a_2}}$$

Now, squaring both sides, we get

$$a_1 = 4a_2 \quad \dots (v)$$

Substituting values of a_1 and a_2 from Eqs. (ii) and (iv) in Eq. (v), we get

$$\mu g \frac{\sqrt{3}}{2} + \frac{g}{2} = 4 \left(\frac{g}{2} - \frac{\mu g \sqrt{3}}{2} \right)$$

Solving for μ , we get

$$\mu = \frac{\sqrt{3}}{5} = \frac{1.732}{5}$$

$$\Rightarrow \mu = 0.346$$

22 (c) The force causing the motion of A is frictional force between A and B ,

$$\mu M_B g = M_A a_A$$

So, acceleration of A ,

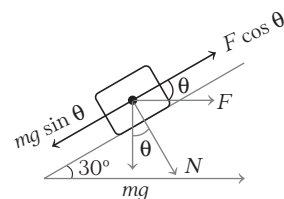
$$\begin{aligned} a_A &= \mu \left(\frac{M_B}{M_A} \right) g \\ &= \frac{\mu g}{2} \text{ (towards right)} \end{aligned}$$

Block B experiences friction force toward left,

$$M_B a_B = \mu M_B g$$

$$\Rightarrow a_B = \mu g \text{ (towards left)}$$

23 (a) For friction to be minimum,



$$F \cos \theta = mg \sin \theta$$

$$\Rightarrow F = mg \tan \theta = 8 \times 10 \times \tan 30^\circ = \frac{80}{\sqrt{3}} \text{ N}$$

- 24** (a) Given, $m_1 = 10 \text{ kg}$, $m_2 = 20 \text{ kg}$, $F_2 = 200 \text{ N}$, $a_1 = 12 \text{ m/s}^2$,

$$a_2 = ?$$

$$\text{For mass } m_1, F_1 = m_1 a_1 = 10 \times 12 = 120 \text{ N}$$

For mass m_2 ,

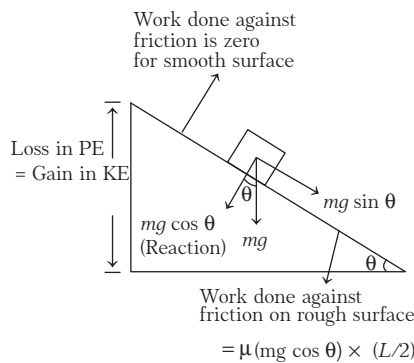
$$F_2 - F_1 = m_2 a_2$$

$$200 - 120 = 20 \times a_2$$

$$\Rightarrow a_2 = \frac{80}{20} = 4 \text{ ms}^{-2}$$

- 25** (a) Suppose length of plane is L . When block descends the plane, rise in kinetic energy = fall in potential energy

$$= mg L \sin \theta$$



Work done by friction

$$= \mu \times (\text{Reaction}) \times \text{Distance}$$

$$= 0 + \mu (mg \cos \theta) \times (L/2)$$

$$= \mu (mg \cos \theta) \times (L/2)$$

Now, work done = change in KE

$$mgL \sin \theta = \mu (mg \cos \theta) \times (L/2) \Rightarrow \tan \theta = \mu/2$$

$$\therefore \mu = 2 \tan \theta$$

- 26** (a) Impulse is imparted due to change in perpendicular components of momentum of ball,

$$J = \Delta p = mv_f - mv_i$$

$$= mv \cos 60^\circ - (-mv \cos 60^\circ)$$

$$= 2mv \cos 60^\circ = 2mv \times \frac{1}{2} = mv$$

- 27** (b) Force, $P = f_m = \mu_s mg$ (when body is at rest)

When the body starts moving with acceleration a , then

$$P - F_k = ma$$

$$\mu_s mg - \mu_k mg = ma$$

$$\Rightarrow a = (\mu_s - \mu_k)g = (0.5 - 0.4)10$$

$$= 0.1 \times 10 \text{ ms}^{-2} = 1 \text{ ms}^{-2}$$

- 28** (a) For the motion of balloon,

Let upward force is F , then $F - mg = ma$

$$\Rightarrow F - 10 \times 10 = 10 \times 20$$

$$\Rightarrow F - 100 = 200 \Rightarrow F = 300 \text{ N}$$

When 4 kg mass is removed, then weight of balloon

$$= 6 \times 10 = 60 \text{ N}$$

This is smaller than the upward force.

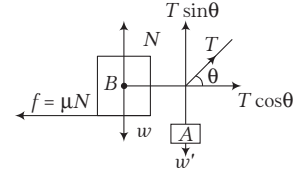
$$\text{So, } F - 60 = 6 \times a$$

where, a = new upward acceleration

$$\Rightarrow 300 - 60 = 6a \Rightarrow 240 = 6a$$

$$\therefore a = \frac{240}{6} = 40 \text{ ms}^{-2}$$

- 29** (b) Let weight of A is w' . From the free body diagram, for equilibrium of the system,



$$T \cos \theta = \mu N = \mu w \quad \dots(i)$$

$$T \sin \theta = w' \quad \dots(ii)$$

where, T = tension in the thread lying between knot and the support.

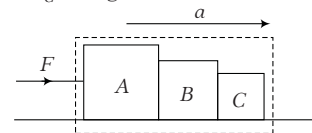
On dividing Eq. (ii) by Eq. (i), we get

$$\frac{T \sin \theta}{T \cos \theta} = \frac{w'}{\mu w} \Rightarrow \tan \theta = \frac{w'}{\mu w}$$

$$\Rightarrow w' = \mu w \tan \theta$$

- 30** (b) Given, $m_A = 4 \text{ kg}$, $m_B = 2 \text{ kg}$

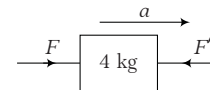
and $m_C = 1 \text{ kg}$



So, total mass, $M = 4 + 2 + 1 = 7 \text{ kg}$

$$\text{Now, } F = Ma \Rightarrow 14 = 7a \Rightarrow a = 2 \text{ ms}^{-2}$$

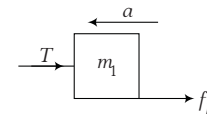
FBD of block A,



$$F - F' = 4a \Rightarrow F' = 14 - 4 \times 2 \Rightarrow F' = 6 \text{ N}$$

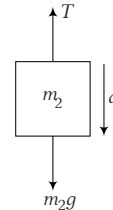
Hence, the contact force between A and B is 6N.

- 31** (c) FBD of block A,



$$T - m_1 a = f_k \quad \dots(i)$$

FBD of block B,



$$m_2 g - T = m_2 a \quad \dots(ii)$$

Adding Eqs. (i) and (ii), we get

$$m_2 g - m_1 a = m_2 a + f_k$$

$$\Rightarrow m_2 g - m_1 a = m_2 a + \mu_k m_1 g$$

$$\Rightarrow a = \frac{(m_2 - \mu_k m_1)g}{m_1 + m_2}$$

From Eq. (ii), we get

$$T = m_2(g - a) = m_2 \left[1 - \frac{(m_2 - \mu_k m_1)}{m_1 + m_2} \right] g$$

$$T = \frac{m_1 m_2 (1 + \mu_k)}{(m_1 + m_2)} g$$

- 32 (a)** When the balloon is descending down with acceleration a ,
 $mg - B = m \times a$... (i)

where, B = Buoyant force.

Here, we should assume that while removing some mass, the volume of balloon and hence, buoyant force will not change.

Let the new mass of the balloon is m' .

So, mass removed = $m - m'$

When the balloon is moving up with acceleration a , then

$$B - m'g = m' \times a$$
 ... (ii)

Adding Eqs. (i) and (ii), we get

$$\Rightarrow mg - m'g = ma + m'a$$

$$\Rightarrow (mg - ma) = m'(g + a)$$

$$\Rightarrow m(g - a) = m'(g + a)$$

$$\therefore m' = \frac{m(g - a)}{(g + a)}$$

So, mass removed,

$$\Delta m = m - m' = m \left[1 - \frac{(g - a)}{(g + a)} \right] = m \left[\frac{(g + a) - (g - a)}{(g + a)} \right]$$

$$= m \left[\frac{g + a - g + a}{g + a} \right] = \frac{2ma}{g + a}$$

- 33 (b)** Given, $u = 30\sqrt{2} \text{ ms}^{-1}$

Distance = s metre

Let a be the acceleration of the bullet.

According to the first condition, ($v = 0$)

From third equation of motion, $v^2 = u^2 - 2as$

$$0 = 900 \times 2 - 2as \Rightarrow a = \frac{900}{s} \text{ ms}^{-2}$$

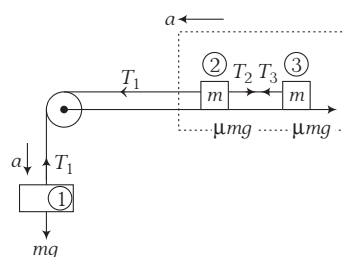
For second condition,

$$v^2 = u^2 - 2as = (30\sqrt{2})^2 - 2 \times \frac{900}{s} \times \frac{s}{2}$$

$$\Rightarrow v^2 = 1800 - 900$$

$$\Rightarrow v = \sqrt{900} = 30 \text{ ms}^{-1}$$

- 34 (c)** First of all consider the forces on the blocks



For the 1st block, $mg - T_1 = m \times a$ ($\because m_1 = m_2 = m_3$) ... (i)

$\Rightarrow$ Let us consider 2nd and 3rd blocks as a system.

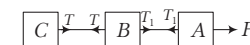
$$\text{So, } T_1 - 2\mu mg = 2m \times a$$
 ... (ii)

Adding Eqs. (i) and (ii), we get

$$mg(1 - 2\mu) = 3m \times a \Rightarrow a = \frac{g}{3}(1 - 2\mu)$$

- 35 (c)** Given, $m_1 = m_2 = m_3 = 2 \text{ kg}$, $F = 10.2 \text{ N}$

The FBD of given system is as shown below



$$\text{For body A, } F - T_1 = ma$$
 ... (i)

$$\text{For body B, } T_1 - T_2 = ma$$
 ... (ii)

$$\text{For body C, } T_2 = ma$$
 ... (iii)

Adding Eqs. (i), (ii) and (iii), we get

$$F = 3ma \Rightarrow a = \frac{10.2}{3 \times 2} = 1.7 \text{ ms}^{-2}$$

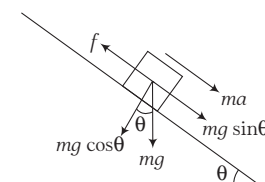
Alternately Acceleration can be found as net acceleration of a system, i.e.

$$a = \frac{\text{Total net force}}{\text{Total mass}} = \frac{10.2}{(2 + 2 + 2)} = \frac{10.2}{6} = 1.7 \text{ ms}^{-2}$$

So, net tension in string between the blocks B and C is

$$T = m \times a = 2 \times 1.7 = 3.4 \text{ N}$$

- 36 (c)** From FBD shown below, $mg \sin \theta - f = ma$

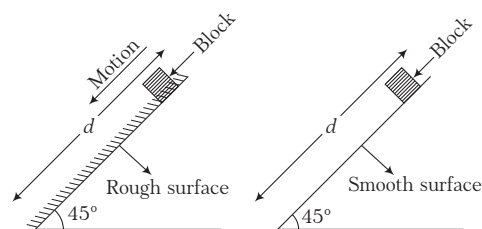


$$mg \sin \theta - ma = f$$

$$\Rightarrow 8 \times 10 \sin(30^\circ) - 8 \times 0.4 = f$$

$$\Rightarrow 40 - 3.2 = f \Rightarrow f = 36.8 \text{ N}$$

- 37 (a)** If the same wedge is made rough, then time taken by it to come down becomes n times more.



$$\text{For rough surface, } t = \sqrt{\frac{2s}{g \sin \theta - \mu g \cos \theta}}$$
 ... (i)

$$\text{For smooth surface, } \frac{t}{n} = \sqrt{\frac{2s}{g \sin \theta}}$$
 ... (ii)

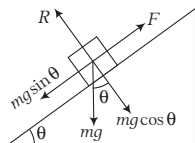
Squaring Eqs. (i) and (ii) and dividing them, we get

$$\mu = \left(1 - \frac{1}{n^2} \right) \tan \theta$$

$$\therefore \mu = \left[1 - \frac{1}{n^2} \right] \tan \theta = \left[1 - \frac{1}{2^2} \right] \tan 45^\circ$$

$$\Rightarrow \mu = \frac{4 - 1}{4} = \frac{3}{4}$$

- 38** (a) Consider a body of mass m placed on a rough inclined surface and it is just on the point of sliding down, with coefficient of friction μ inclined at angle θ , as shown in figure.



At the equilibrium point O ,

In case of limiting condition, $F = mg \sin \theta$... (i)

Normal force, $R = mg \cos \theta$... (ii)

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{F}{R} = \frac{mg \sin \theta}{mg \cos \theta}$$

$$\mu = \tan \theta \quad (\because \text{Force of friction, } F = \mu R)$$

$$\Rightarrow \theta = \tan^{-1}(\mu)$$

This is maximum value of θ for mass m to be at rest.

For smaller θ , body will be at rest, i.e. in equilibrium.

So, angle of repose, i.e. $\theta = \tan^{-1} \mu$.

- 39** (a) Since, all the blocks are moving with constant velocity, then the net force on all the blocks will be zero.

- 40** (c) Given, $h = 100$ m and $t = 10$ s

From second equation of motion, $h = \frac{1}{2}at^2$

$$\Rightarrow a = \frac{2h}{t^2} = \frac{2 \times 100}{(10)^2} = 2 \text{ ms}^{-2}$$

$$\text{Now, } F = m(g + a) = 50(10 + 2) = 600 \text{ N}$$

- 41** (d) We know that, $F - f = ma$

$$\text{Here, for small body, } a_1 = \frac{20 - 0.2 \times 3 \times 10}{3}$$

$$\Rightarrow a_1 = \frac{14}{3} \text{ ms}^{-2}$$

Similarly, $0.2 \times 3 \times 10 = 10 \times a_2$

$$\Rightarrow a_2 = \frac{6}{10} = 0.6 \text{ ms}^{-2}$$

- 42** (d) Here, $M_{\text{eff}} g = M(g - a)$

$$\text{or } \frac{g}{g - a} = \frac{M}{M_{\text{eff}}} \Rightarrow \frac{g}{g - a} = \frac{60}{54}$$

$$\Rightarrow \frac{g}{g - a} = \frac{10}{9} \Rightarrow 9g = 10g - 10a$$

$$\Rightarrow -g = -10a \Rightarrow g = 10a$$

$$\Rightarrow a = \frac{g}{10} \Rightarrow a = \frac{9.8}{10} = 0.98 \text{ ms}^{-2}$$

- 43** (b) Given, $u = 0$, $s = 1.5$ m, $t = 0.4$ s

From second equation of motion, $s = ut + \frac{1}{2}at^2$

$$\Rightarrow 1.5 = 0 + \frac{1}{2}a(0.4)^2$$

$$\Rightarrow a = \frac{1.5 \times 2}{(0.4)^2} = 18.75 \text{ ms}^{-2}$$

As the string is moving upwards with this acceleration.

$$\therefore T = m(g + a) = 2(9.8 + 18.75) = 57.1 \text{ N}$$

- 44** (d) Using law of conservation of momentum, we get

$$100 \times v = 0.25 \times 100 \Rightarrow v = 0.25 \text{ ms}^{-1}$$

- 45** (c) Initial thrust must be

$$m(g + a) = 3.5 \times 10^4 (10 + 10) = 7 \times 10^5 \text{ N}$$

- 46** (d) From first equation of motion, $v = u - at$

$$0 = u - \mu gt$$

$$\Rightarrow \mu = \frac{u}{gt} = \frac{6}{10 \times 10} = 0.06$$

- 47** (d) Force of limiting friction for block $= \mu_s mg$

$$= 0.60 \times 10 \times 9.8 = 58.8 \text{ N}$$

If the applied force is greater than 58.8 N, then the block will move over the slab.

Kinetic friction acting on the block towards right

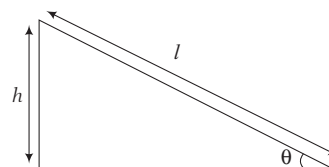
$$= \mu_k mg = 0.40 \times 10 \times 9.8 = 39.2 \text{ N}$$

This is also equal to the force of friction acting on slab towards left. This is the only force acting on slab.

$$\text{So, acceleration of the slab, } a = \frac{F}{m} = \frac{39.2}{40} = 0.98 \text{ ms}^{-2}$$

- 48** (b) Force down the plane $= mg \sin \theta$

$$\therefore \text{Acceleration down the plane} = g \sin \theta$$



Using the relation, $h = ut + \frac{1}{2}gt^2$

$$l = 0 + \frac{1}{2}(g \sin \theta)t^2$$

$$\Rightarrow t^2 = \frac{2l}{g \sin \theta} = \frac{2h}{g \sin^2 \theta} \quad \left(\because l = \frac{h}{\sin \theta} \right)$$

$$\text{or } t = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$$

- 49** (b) Time t_1 for stationary lift $= \sqrt{\frac{2h}{g}}$

When lift is moving up with constant acceleration, then

$$t_2 = \sqrt{\frac{2h}{g+a}} \quad [\because \text{Relative acceleration} = (g + a)]$$

$$\therefore t_1 > t_2$$

- 50** (a) Kinetic friction force $= \mu_k R = \mu_k mg = 0.4 \times 60 \times 10 = 240 \text{ N}$
and the limiting friction force

$$= \mu_s R = \mu_s mg = 0.5 \times 60 \times 10 = 300 \text{ N}$$

So, the force applied on the body is 300 N and if the body is moving afterwards with the same force, then

Net accelerating force = Applied force – Kinetic force

$$ma = 300 - 240$$

$$\Rightarrow a = \frac{60}{m} = \frac{60}{60} = 1 \text{ ms}^{-2}$$

51 (c) Tension in the string,

$$T = \frac{2m(v')^2}{l} + 2mg = \frac{m(2gl)}{2l} + 2mg$$

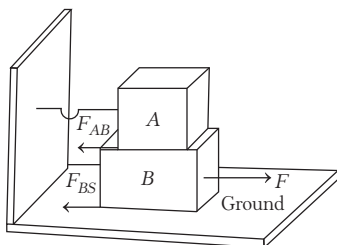
$$\left(\because \frac{2m(v')^2}{l} = \text{Centrifugal force and } v' = \frac{v}{2} = \sqrt{\frac{gl}{2}} \right)$$

$$T = 3mg$$

$\therefore$ Increase in tension = $3mg - mg = 2mg$

52 (c) Friction force between block A and block B; and between block B and surface will oppose the force F.

$$\therefore F = F_{AB} + F_{BS}$$



$$= \mu_{AB} m_A g + \mu_{BS} (m_A + m_B) g$$

$$= 0.2 \times 100 \times 10 + 0.3(100 + 200) 10$$

$$= 200 + 900 = 1100 \text{ N}$$

This is the required minimum force to move the block B.

53 (b) Given, $m_b = 40 \text{ g}$, $v_b = 1200 \text{ ms}^{-1}$,

$$F = 144 \text{ N}$$

$$\therefore F = nma = \frac{m_b v_b}{t} \times n$$

$$\Rightarrow 144 = \frac{40 \times 1200}{1000} \times n \Rightarrow n = 3$$

54 (a) The maximum weight which can be suspended with the rope without breaking it = 30 kg-wt.

$$\text{or } F_{\max} = 30 \times 10 = 300 \text{ N}$$

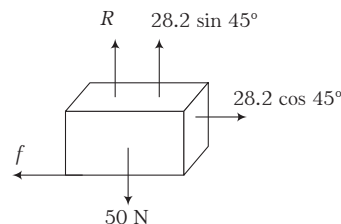
$$\therefore 300 = mg + ma$$

$$\Rightarrow ma = 300 - mg = 300 - 25 \times 10 = 50 \text{ N}$$

$$\Rightarrow a = \frac{50}{m} = \frac{50}{25} = 2 \text{ ms}^{-2}$$

55 (d) Frictional force, $f = 28.2 \cos 45^\circ$

$$= 28.2 \times \frac{1}{\sqrt{2}} = 20 \text{ N}$$



$$\text{Normal reaction, } R = 50 - 28.2 \sin 45^\circ$$

$$= 50 - 20 = 30 \text{ N}$$

56 (b) The accelerating force on the rocket = upward thrust

$$= \frac{\Delta m}{\Delta t} \cdot v$$

$$\text{Given, } \frac{\Delta m}{\Delta t} = 50 \times 10^{-3} \text{ kgs}^{-1}$$

$$\text{and } v = 400 \text{ ms}^{-1}$$

$$\text{So, accelerating force} = 50 \times 10^{-3} \times 400 = 20 \text{ N}$$

57 (b) The velocity of exhaust gases with respect to the rocket

$$= 100 \text{ ms}^{-1}$$

The minimum force on the rocket to lift it,

$$F_{\min} = mg = 1000 \times 10 = 10000 \text{ N}$$

Hence, minimum rate of burning of fuel is given by

$$\frac{dm}{dt} = \frac{F_{\min}}{v} = \frac{10000}{100} = 100 \text{ kgs}^{-1}$$

58 (c) In a non-inertial frame, the second law of motion is written as $\mathbf{F} = m\mathbf{a} - \mathbf{F}_p$

where, $\mathbf{F}_p$ is the pseudo force while a is the acceleration of the body relative to non-inertial frame.

59 (a) When the lift is accelerating upwards with a constant acceleration a , then the apparent weight,

$$w = m(g + a) = 60(10 + 2)$$

$$= 60 \times 12 = 720 \text{ N}$$

When the lift is accelerating downwards with acceleration a , then apparent weight,

$$w' = m(g - a) = 60(10 - 2)$$

$$= 60 \times 8 = 480 \text{ N}$$

60 (b) When an object moves in a plane surface with uniform velocity in the presence of a force, then the frictional force between the object and the surface has opposite value of the present force. So, the frictional force between the object and the surface is -10 N .

CHAPTER 06

Work, Energy and Power

The terms *work* and *energy* are quite familiar to us and we use these terms in many ways. In physics, work is said to be done on a body, when a force acts on it and the point of application of force actually moves.

Energy is the capacity to do work and the term *power* is usually associated with the time in which the work is done.

In this chapter, we are going to develop better understanding of these three physical quantities in detail.

WORK

Work is said to be done when a force acts on a body in such a way that the body is displaced through some distance in the direction of force.

We define the work done by a constant force on a body as the product of the force F and the displacement s through which the body is displaced in the direction of force.

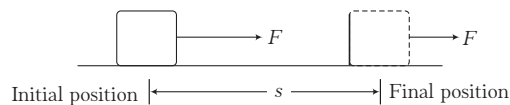


Fig. 6.1 Work done, when force and displacement are in the same direction

Then, the work done W is given by $W = F \cdot s$

On the other hand, in a situation, when the constant force F acting on the body makes angle θ with the horizontal and body is displaced through a distance s .

Then, F can be resolved into two components

- $F \cos\theta$ in the direction of displacement of body.
- $F \sin\theta$ in the perpendicular direction of the displacement of body.

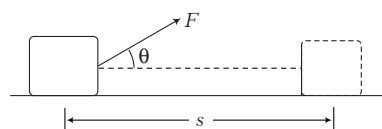


Fig. 6.2 Work done, when force and displacement are inclined to each other

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- 3 Non-conservative forces
- 4 Energy
 - Kinetic energy
 - Work-energy theorem
 - Potential energy
- 5 Equilibrium
- 6 Law of conservation of energy
- 7 Conversion of mass and energy
- 8 Power
 - Instantaneous power

Thus, in this case, work done by a constant force $\mathbf{F}$ is given by

$$W = (\text{component of force along the displacement}) \times (\text{displacement})$$

or $W = (F \cos \theta) (s)$

or $W = \mathbf{F} \cdot \mathbf{s}$ (From the definition of dot product)

So, work done is a scalar or dot product of $\mathbf{F}$ and $\mathbf{s}$.

Nature of work in different situations :

- (i) If θ is acute, then W is positive and force tries to increase the speed of the body.

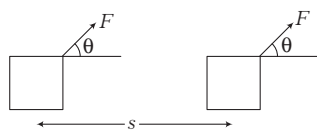


Fig. 6.3

- (ii) If $\theta = 90^\circ$, then W is zero and there is no change in speed of the body due the force.



Fig. 6.4

- (iii) If θ is obtuse, then W is negative and force tries to decrease the speed of the body.

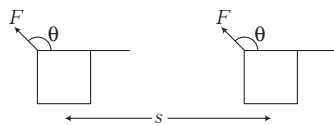


Fig. 6.5

Work is a scalar quantity, its dimensional formula is $[ML^2T^{-2}]$. SI unit of work is **joule** and CGS unit is **erg**.

$$1 \text{ joule} = 10^7 \text{ erg}$$

Gravitational unit of work done is **kg-m**.

$$1 \text{ kg-m} = 9.8 \text{ J}$$

Some other convenient units of work are eV (electron volt), MeV (mega electron volt) and kWh (kilowatt hour)

- $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}, 1 \text{ J} = 6.25 \times 10^{18} \text{ eV}$
- $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}, 1 \text{ J} = 6.25 \times 10^{12} \text{ MeV}$
- $1 \text{ kilowatt hour (kWh)} = 3.6 \times 10^6 \text{ J}$

Example 6.1 A lawn roller has been pushed by a gardener through a distance of 30 m. What will be the work done by him, if he applies a force of 30 kg-wt in the direction inclined at 60° to the ground?

Sol. Given, displacement, $s = 30 \text{ m}$

Force, $F = 30 \text{ kg-wt} = 300 \text{ N}$

$$\theta = 60^\circ$$

The work done by the gardener,

$$W = \mathbf{F} \cdot \mathbf{s} = Fs \cos \theta = 300 \times 30 \times \cos 60^\circ \\ = 300 \times 30 \times \frac{1}{2} \Rightarrow W = 4500 \text{ J}$$

Example 6.2 A body moves a distance of 10 m along a straight line under an action of 5 N force. If work done is 25 J, then find the angle between the force and direction of motion of the body.

Sol. Work is measured by the product of the applied force and the displacement of the body in the direction of the force.

$$\therefore \text{Work} = \text{Applied force} \times \text{Displacement}$$

$$W = (F \cos \theta) \times s = Fs \cos \theta$$

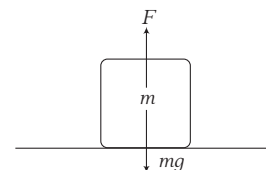
Given, $W = 25 \text{ J}, F = 5 \text{ N}, s = 10 \text{ m}$

$$\therefore \cos \theta = \frac{W}{F \cdot s} = \frac{25}{5 \times 10} = \frac{1}{2}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$$

Hence, angle between the force and direction of motion of the body is 60° .

Example 6.3 A block of mass $m = 2 \text{ kg}$ is pulled by a force $F = 40 \text{ N}$ upwards through a height $h = 2 \text{ m}$. Find the work done on the block by the applied force F and its weight mg . (Take, $g = 10 \text{ ms}^{-2}$)



Sol. Weight of the block, $w = mg = (2)(10) = 20 \text{ N}$

Work done by the applied force,

$$W_F = Fs \cos \theta = Fh \cos 0^\circ$$

(Here, $s = h = 2 \text{ m}$ and the angle between force and displacement is 0°)

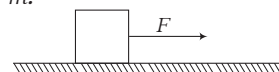
or $W_F = (40)(2)(1) = 80 \text{ J}$

Similarly, work done by its weight,

$$W_{mg} = (mg)(h) \cos 180^\circ$$

or $W_{mg} = (20)(2)(-1) = -40 \text{ J}$

Example 6.4 A 10 g block placed on a rough horizontal floor is being pulled by a constant force 50 N. Coefficient of kinetic friction between the block and the floor is 0.4. Find work done by each individual force acting on the block over displacement of 5 m.



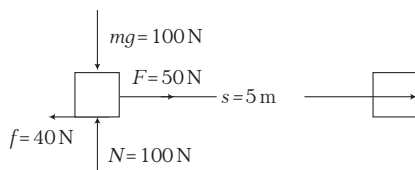
Sol. Given, mass, $m = 10 \text{ g}$

Constant force, $F = 50 \text{ N}$

Coefficient of kinetic friction, $\mu_k = 0.4$

Displacement, $s = 5 \text{ m}$

Forces acting on the block are; its weight ($mg = 100\text{ N}$), normal reaction ($N = 100\text{ N}$) from the ground, force due to kinetic friction ($f = \mu_k mg = 40\text{ N}$) and the applied force ($F = 50\text{ N}$) which are shown in the given figure.



Work done by the gravity, i.e. weight of the block,

$$W_g = 0\text{ J} \quad (\because mg \perp s)$$

Work done by the normal reaction,

$$W_N = 0\text{ J} \quad (\because N \perp s)$$

Work done by the applied force,

$$W_F = 50 \times 5 \times \cos 0^\circ = 250\text{ J} \quad (\because F \parallel s)$$

Work done by the force of kinetic friction,

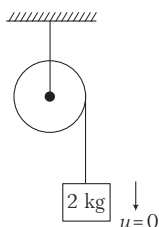
$$W_f = 40 \times 5 \times \cos 180^\circ = -200\text{ J} \quad (\because F \text{ and } s \text{ are anti-parallel})$$

Example 6.5 A block of mass 2 kg is being brought down by a string. If the block acquires a speed 1 ms^{-1} in dropping down 25 cm , find the work done by the string in the process.

Sol. Given, distance moved by block, $s = 25\text{ cm} = \frac{1}{4}\text{ m}$

Initial velocity of block, $u = 0$

Final velocity of block, $v = 1\text{ ms}^{-1}$

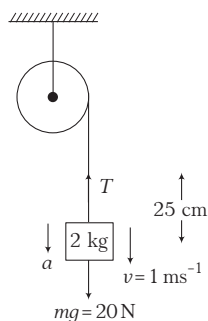


From the third equation of motion for block,

$$v^2 = u^2 + 2as \Rightarrow (1)^2 = 0 + 2a \times \frac{1}{4} \Rightarrow a = 2\text{ ms}^{-2}$$

$\therefore$ Acceleration of the block, $a = 2\text{ ms}^{-2}$

The free body diagram of the block is as shown below



For the motion of 2 kg block in downward direction, applying Newton's law, we get

$$mg - T = ma$$

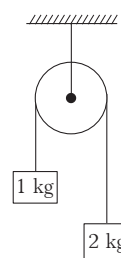
$$20 - T = 2a = 2 \times 2$$

$$\Rightarrow T = 16\text{ N}$$

$$\therefore \text{Work done by the string, } W = -Ts = -16 \times \frac{1}{4} = -4\text{ J}$$

Negative sign indicates the opposite direction of tension and displacement of the block.

Example 6.6 Two unequal masses of 1 kg and 2 kg are attached at the two ends of a light inextensible string passing over a smooth pulley as shown in figure. If the system is released from rest, find the work done by string on both the blocks in 1 s . (Take, $g = 10\text{ ms}^{-2}$)

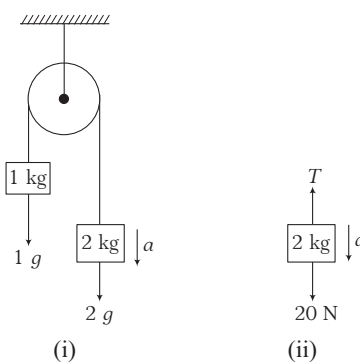


Sol. Net pulling force acting on the system,

$$F_{\text{net}} = 2g - 1g = 20 - 10 = 10\text{ N}$$

Total mass being pulled,

$$m = (1 + 2) = 3\text{ kg}$$



Therefore, acceleration of the system will be

$$a = \frac{F_{\text{net}}}{m} = \frac{10}{3}\text{ ms}^{-2}$$

Displacement of both the blocks in 1 s is

$$s = \frac{1}{2}at^2 = \frac{1}{2} \left(\frac{10}{3} \right) (1)^2 = \frac{5}{3}\text{ m}$$

Free body diagram of 2 kg block is shown in Fig. (ii).

Using $\Sigma F = ma$, we get

$$20 - T = 2a = 2 \left(\frac{10}{3} \right)$$

$$\text{or } T = 20 - \frac{20}{3} = \frac{40}{3}\text{ N}$$

∴ Work done by string (tension) on 1 kg block in 1 s is

$$W_1 = (T)(s) \cos 0^\circ = \left(\frac{40}{3}\right)\left(\frac{5}{3}\right)(1) = \frac{200}{9} \text{ J}$$

Similarly, work done by string on 2 kg block in 1 s will be

$$W_2 = (T)(s)(\cos 180^\circ) = \left(\frac{40}{3}\right)\left(\frac{5}{3}\right)(-1) = -\frac{200}{9} \text{ J}$$

Work done on a particle moving in three dimensional space

Let a particle moves in space under the action of a constant force given by

$$\mathbf{F} = F_x \hat{\mathbf{i}} + F_y \hat{\mathbf{j}} + F_z \hat{\mathbf{k}}$$

Let initial position of the particle be

$$\mathbf{s}_1 = x_1 \hat{\mathbf{i}} + y_1 \hat{\mathbf{j}} + z_1 \hat{\mathbf{k}}$$

and final position of the particle be

$$\mathbf{s}_2 = x_2 \hat{\mathbf{i}} + y_2 \hat{\mathbf{j}} + z_2 \hat{\mathbf{k}}$$

Then, work done by the force $\mathbf{F}$ given by

$$\begin{aligned} W &= \mathbf{F} \cdot \mathbf{s} = \mathbf{F} \cdot (\mathbf{s}_2 - \mathbf{s}_1) \\ &= [F_x \hat{\mathbf{i}} + F_y \hat{\mathbf{j}} + F_z \hat{\mathbf{k}}] \cdot [(x_2 - x_1) \hat{\mathbf{i}} \\ &\quad + (y_2 - y_1) \hat{\mathbf{j}} + (z_2 - z_1) \hat{\mathbf{k}}] \end{aligned}$$

$$\Rightarrow W = F_x(x_2 - x_1) + F_y(y_2 - y_1) + F_z(z_2 - z_1)$$

Example 6.7 A constant force $\mathbf{F} = (\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}) \text{ N}$ acts on a particle and displaces it from $(-1\text{m}, 2\text{m}, 1\text{m})$ to $(2\text{m}, -3\text{m}, 1\text{m})$. Find the work done by the force.

Sol. Given, $\mathbf{F} = (\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}) \text{ N}$

Initial position of the particle is given by

$$\mathbf{s}_1 = (-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}) \text{ m}$$

Final position, $\mathbf{s}_2 = (2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}) \text{ m}$

Displacement of the particle,

$$\begin{aligned} \mathbf{s} &= \mathbf{s}_2 - \mathbf{s}_1 = (2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}) - (-\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}) \\ &= 3\hat{\mathbf{i}} - 5\hat{\mathbf{j}} + 0\hat{\mathbf{k}} = (3\hat{\mathbf{i}} - 5\hat{\mathbf{j}}) \text{ m} \end{aligned}$$

Work done by the force $\mathbf{F}$,

$$W = \mathbf{F} \cdot \mathbf{s} = (\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}) \cdot (3\hat{\mathbf{i}} - 5\hat{\mathbf{j}}) = (3 - 15) = -12 \text{ J}$$

Example 6.8 A particle is shifted from point $(0, 0, 1\text{m})$ to $(1\text{m}, 1\text{m}, 2\text{m})$, under simultaneous action of several forces. Two of the forces are $\mathbf{F}_1 = (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}) \text{ N}$ and $\mathbf{F}_2 = (\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) \text{ N}$. Find work done by resultant of these two forces.

Sol. Work done by a constant force is equal to dot product of the force and displacement vectors.

$$\text{i.e. } W = \mathbf{F} \cdot \mathbf{s} \quad \dots(i)$$

Initial position of the particle, $\mathbf{s}_1 = \hat{\mathbf{k}}$ metre

Final position, $\mathbf{s}_2 = (\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}}) \text{ m}$

Displacement of the particle, $\mathbf{s} = \mathbf{s}_2 - \mathbf{s}_1 = \hat{\mathbf{i}} + \hat{\mathbf{j}} + 2\hat{\mathbf{k}} - \hat{\mathbf{k}}$

$$\Rightarrow \mathbf{s} = (\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \text{ m}$$

$$\begin{aligned} \text{and force, } \mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}} + \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}} \\ &= (3\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \text{ N} \end{aligned}$$

Substituting given values in Eq. (i), we get

$$W = (3\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) \cdot (\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}) = 3 + 1 + 1 = 5 \text{ J}$$

Work done by a variable force

The force is said to be variable force, if it changes its direction or magnitude or both. The work done by a variable force can be calculated as

$$W = \int_{r_1}^{r_2} \mathbf{F} \cdot d\mathbf{r}$$

where, integration is performed along the path of particle and $d\mathbf{r}$ is the position vector of the particle.

If the particle moves from $\mathbf{r}_1(x_1, y_1, z_1)$ to $\mathbf{r}_2(x_2, y_2, z_2)$, i.e. when the magnitude and direction of the force vary in three dimensions, then the work done by force $\mathbf{F}$, is given by

$$W = \int dW = \int_{x_1}^{x_2} F_x dx + \int_{y_1}^{y_2} F_y dy + \int_{z_1}^{z_2} F_z dz$$

where, F_x, F_y and F_z are the rectangular components of force in x, y and z -directions, respectively.

• If the motion is one dimensional, $W = \int_{x_i}^{x_f} F_x dx$

Here, F_x is the component of force along motion.

Example 6.9 A position dependent force $F = (7 - 2x + 3x^2) \text{ N}$ acts on a small body of mass 2 kg and displaces it from $x = 0$ to $x = 5 \text{ m}$. Calculate the work done (in joule).

Sol. Work done, $W = \int_{x_1}^{x_2} F dx = \int_0^5 (7 - 2x + 3x^2) dx$

Here, the body changes its position from $x = 0$ to $x = 5 \text{ m}$

$$\Rightarrow W = \left[7x - \frac{2x^2}{2} + \frac{3x^3}{3} \right]_0^5 = [7(5) - (5)^2 + (5)^3 - 0] = 135 \text{ J}$$

Example 6.10 A force $F = (2 + x)$ acts on a particle in x -direction, where F is in newton and x in metre. Find the work done by this force during a displacement from $x = 1 \text{ m}$ to $x = 2 \text{ m}$.

Sol. As the force is variable, we shall find the work done in a small displacement from x to $x + dx$ and then integrate it to find the total work. The work done in this small displacement is

$$dW = F dx = (2 + x) dx$$

$$\text{Thus, } W = \int_1^2 dW = \int_1^2 (2 + x) dx$$

$$= \left[2x + \frac{x^2}{2} \right]_1^2 = \left[\left(4 + \frac{4}{2} \right) - \left(2 + \frac{1}{2} \right) \right] = 3.5 \text{ J}$$

Example 6.11 A force $F = -\frac{k}{x^2}$ ($x \neq 0$) acts on a particle in x -direction. Find the work done by this force in displacing the particle from $x = +a$ to $x = +2a$. Here, k is a positive constant.

Sol. Given, $F = -\frac{k}{x^2}$, where x is the position of particle.

$$\begin{aligned}\text{Work done by this force, } W &= \int_{+a}^{+2a} F \, dx = \int_{+a}^{+2a} \left(-\frac{k}{x^2} \right) dx \\ &= \left[\frac{k}{x} \right]_{+a}^{+2a} = \frac{k}{2a} - \frac{k}{a} = -\frac{k}{2a}\end{aligned}$$

Note It is important to note that work comes out to be negative which is quite obvious as the force acting on the particle is in negative x -direction ($F = -\frac{k}{x^2}$) while displacement is along positive x -direction (from $x = a$ to $x = 2a$).

Calculation of work done by force-displacement graph

The area under force-displacement curve gives work done.

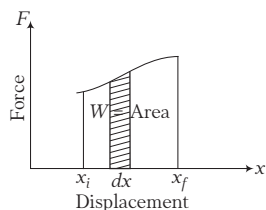


Fig. 6.6 Force-displacement graph

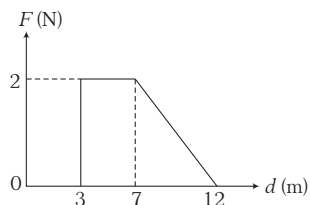
$$W = \int_{x_i}^{x_f} dW = \int_{x_i}^{x_f} F \cdot dx$$

where, x_i and x_f are the initial and final position, respectively.

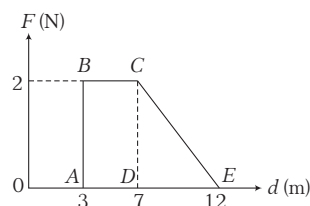
$$\therefore W = \int_{x_i}^{x_f} (\text{Area under curve})$$

$$\Rightarrow W = \text{Area under curve between } x_i \text{ and } x_f.$$

Example 6.12 Force F acting on a particle moving in a straight line varies with distance d as shown in the figure. Find the work done on the particle during its displacement of 12 m.

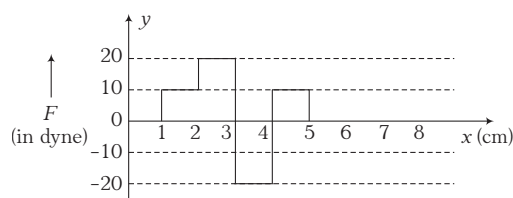


Sol. Work done = Area under force-displacement graph
 = Area of rectangle ABCD + Area of ΔDCE
 = Length $\times$ Breadth + $\frac{1}{2} \times$ Base $\times$ Height
 = $(AB \times AD) + \frac{1}{2} \times DE \times CD$



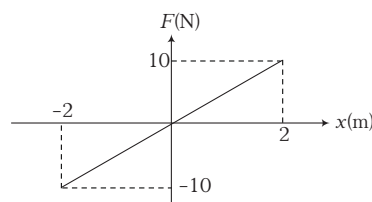
$$\begin{aligned}&= 2 \times (7 - 3) + \frac{1}{2} \times (12 - 7) \times 2 \\ &= 8 + \frac{1}{2} \times 10 = 8 + 5 = 13 \text{ J}\end{aligned}$$

Example 6.13 For the force-displacement graph shown below, calculate the work done by the force in displacing the body from $x = 1 \text{ cm}$ to $x = 5 \text{ cm}$.



Sol. Work done = Area under the curve and displacement axis
 = $10 \times 1 + 20 \times 1 - 20 \times 1 + 10 \times 1 = 20 \text{ erg}$

Example 6.14 A force F acting on a particle varies with the position x as shown in figure. Find the work done by this force in displacing the particle from



(i) $x = -2 \text{ m}$ to $x = 0$

(ii) $x = 0$ to $x = 2 \text{ m}$.

Sol. (i) From $x = -2 \text{ m}$ to $x = 0$, displacement of the particle is along positive x -direction while force acting on the particle is along negative x -direction. Therefore, work done is negative and is given by the area under F - x graph.

$$\therefore W = -\frac{1}{2} (2) (10) = -10 \text{ J}$$

(ii) From $x = 0$ to $x = 2 \text{ m}$, displacement of particle and force acting on the particle both are along positive x -direction. Therefore, work done is positive and is given by the area under F - x graph.

$$\therefore W = \frac{1}{2} (2) (10) = 10 \text{ J}$$

Spring block system

Consider an elastic spring of negligibly small mass having spring constant k with its one end attached to a rigid support and its other end is attached to a block of mass m that can slide over a smooth horizontal surface.

Suppose a force F is applied on the spring to stretch it from natural length to produce an elongation x in it.

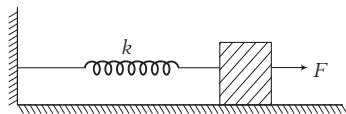


Fig. 6.7 Spring block system

The work done in stretching the spring by external applied force, $W = \int_0^x F dx = \int_0^x kx dx$

$$W = \frac{1}{2} kx^2$$

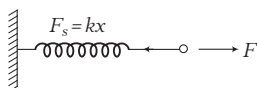
The work done by stretching or compressing force is positive. But the work done by the spring is negative because the force exerted by the spring is always opposite to elongation or contraction.

$$\therefore \text{Work done by the spring, } W = -\frac{1}{2} kx^2$$

When length of spring changes from $x = x_i$ to $x = x_f$

$$W = -\int_{x_i}^{x_f} F dx = -\int_{x_i}^{x_f} kx dx = \frac{1}{2} k(x_i^2 - x_f^2)$$

Example 6.15 The work done in extending a spring by x_0 is W_0 . Find the work done in further extension x_0 .



Sol. Force, $F = F_s = kx$

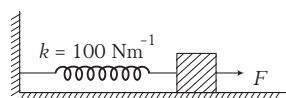
$$W_0 = \int F dx = \int_0^{x_0} kx dx = \frac{kx_0^2}{2}$$

Let W be the work done in extending a spring by $2x_0$.

$$W = \int F dx = \int_{x_0}^{2x_0} kx dx = \frac{k}{2} [(2x_0)^2 - x_0^2] = \frac{3}{2} kx_0^2$$

$$\Rightarrow W = 3W_0$$

Example 6.16 Consider a block connected to a light spring of spring constant 100 Nm^{-1} . Now, the block is displaced by applying a constant force F which gives zero resultant force when spring is stretched through 10 cm .



Evaluate

- work done by the spring force when the block attains equilibrium.
- net work done on the block when it attains maximum speed.

Sol. (i) Work done by the spring force, $W_s = -\frac{1}{2} kx^2$

Here, $k = 100 \text{ Nm}^{-1}$ and $x = 10 \text{ cm} = 0.1 \text{ m}$

$$\therefore W_s = -\frac{1}{2} \times 100 (0.1)^2 = -0.5 \text{ J}$$

Negative sign indicates that the work done by the spring force is negative.

(ii) When the block attains equilibrium, its speed is maximum.

$\therefore$ Work done on the block by external force F ,

$$W = F \cdot x = (10) \left(\frac{10}{100} \right) = 1 \text{ J}$$

$$[\because F = kx = 100 \times 0.1 = 10 \text{ N}]$$

Thus, net work done on the block,

$$W_N = W_s + W = -0.5 + 1 = 0.5 \text{ J}$$

Dependence of work done on frame of reference

Work depends on frame of reference. With change of frame of reference, inertial force does not change while displacement may change.

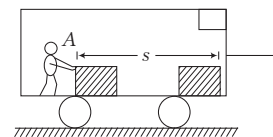


Fig. 6.8 Dependence of work done on frame of reference

So, the work done by a force will be different in different frames. *e.g.* If a person A is pushing a box inside a moving bus, then work done as seen by him from the frame of reference of bus is $\mathbf{F} \cdot \mathbf{s}$ while as seen by a person on the ground it is $\mathbf{F} \cdot (\mathbf{s} + \mathbf{s}_0)$. Here, $\mathbf{s}_0$ is the displacement of bus relative to ground.

Example 6.17 A train is moving with a speed of 90 kmh^{-1} . A passenger X inside the train displaces his 40 kg luggage slowly on the floor through 1 m in 10 s . Coefficient of friction of the floor of the train is 0.2 . Find the work done by this passenger X and the luggage as seen by

- a fellow passenger Y
- a person on the ground [Take, $g = 10 \text{ ms}^{-2}$].

Sol. Given, speed of the train,

$$v = 90 \text{ kmh}^{-1} = \frac{90 \times 1000}{60 \times 60} = 25 \text{ ms}^{-1}$$

(i) Displacement of the luggage with respect to the train, $s = 1 \text{ m}$.

As luggage is displaced slowly, the force applied on it must be same as frictional force on it by the floor,

$$f = \mu mg = (0.2) \times 40 \times 10 = 80 \text{ N}$$

Work done by the passenger X as seen by fellow passenger Y ,

$$W = fs = (80) (1) = 80 \text{ J}$$

(ii) The luggage is displaced for 10 s . Therefore, distance moved by train with respect to ground during this interval is

$$s_0 = 25 \times 10 = 250 \text{ m}$$

Therefore, work done by passenger X on the luggage as seen by a person on the ground,

$$W_G = f(s + s_0) = (80)(1 + 250) \\ = (251)(80) = 20.08 \text{ kJ}$$

CONSERVATIVE FORCES

A force is said to be conservative, if the work done by or against the force on a body is independent of path followed by the body and depends only on initial and final positions.

e.g. Gravitational force, spring force, coulomb force, etc.

Work done by conservative forces

One of the following two equivalent conditions on work done must be satisfied by conservative forces

- (i) Work done by or against a conservative force in moving a body from one position to another depends only on the initial and final positions of the body. It does not depend upon the nature of the path followed by the body in going from initial position to the final position.

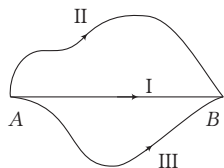


Fig. 6.9 Work done by conservative forces

Let W_1 , W_2 and W_3 denote the net work done in moving a body from A to B along three different paths I, II and III respectively, as shown in figure. If the force is conservative, then

$$W_I = W_{II} = W_{III} = \int \mathbf{F} \cdot d\mathbf{L} \\ \Rightarrow \int_A^B \mathbf{F} \cdot d\mathbf{L} = \int_A^B \mathbf{F} \cdot d\mathbf{L} = \int_A^B \mathbf{F} \cdot d\mathbf{L} \\ \text{Path I} \quad \text{Path II} \quad \text{Path III}$$

- (ii) Work done by or against a conservative force in moving a particle along a closed path (round trip) is zero.

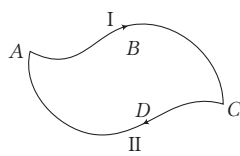


Fig. 6.10

Assume that a particle is moving along a closed path ABCDA as shown in figure.

If the force acting on the particle is conservative, then

$$W_{ABCD} = \int_{ABCD} \mathbf{F} \cdot d\mathbf{L} = \int_{ABC} \mathbf{F} \cdot d\mathbf{L} + \int_{CDA} \mathbf{F} \cdot d\mathbf{L} = 0 \\ \Rightarrow W_{ABC} + W_{CDA} = 0 \\ \Rightarrow W_{ABC} = -W_{CDA} \\ \Rightarrow W_I = -W_{II}$$

Example 6.18 An object is displaced from point A(2m, 3m, 4m) to a point B(1m, 2m, 3m) under a constant force $\mathbf{F} = (2\hat{i} + 3\hat{j} + 4\hat{k}) \text{ N}$. Find the work done by this force in this process.

Sol. Work done by force $\mathbf{F}$,

$$W = \int_{r_i}^{r_f} \mathbf{F} \cdot d\mathbf{s} = \int_{(2m, 3m, 4m)}^{(1m, 2m, 3m)} (2\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ = [2x + 3y + 4z]_{(2m, 3m, 4m)}^{(1m, 2m, 3m)} = -9 \text{ J}$$

Alternate Solution

Since, $\mathbf{F} = \text{constant}$, we can also use

$$W = \mathbf{F} \cdot \mathbf{s}$$

$$\text{Here, } \mathbf{s} = \mathbf{s}_f - \mathbf{s}_i = (\hat{i} + 2\hat{j} + 3\hat{k}) - (2\hat{i} + 3\hat{j} + 4\hat{k}) \\ = (-\hat{i} - \hat{j} - \hat{k}) \text{ m}$$

$$\therefore W = (2\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (-\hat{i} - \hat{j} - \hat{k}) \\ = -2 - 3 - 4 = -9 \text{ J}$$

Example 6.19 An object is displaced from position vector $\mathbf{r}_1 = (2\hat{i} + 3\hat{j}) \text{ m}$ to $\mathbf{r}_2 = (4\hat{i} + 6\hat{j}) \text{ m}$ under a force $\mathbf{F} = (3x^2\hat{i} + 2y\hat{j}) \text{ N}$. Find the work done by this force.

Sol. Work done, $W = \int_{r_1}^{r_2} \mathbf{F} \cdot d\mathbf{r} = \int_{r_1}^{r_2} (3x^2\hat{i} + 2y\hat{j}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ = \int_{r_1}^{r_2} (3x^2 dx + 2y dy) = [x^3 + y^2]_{(2,3)}^{(4,6)} = 83 \text{ J}$

Note In the above two examples, we saw that while calculating the work done we did not mention the path through which the object was displaced. Only initial and final coordinates were required. It shows that in both the examples, the work done is path independent or work done will be equal whichever path we follow.

Therefore, above two forces in which work done is path independent are examples of conservative forces.

NON-CONSERVATIVE FORCES

A force is said to be non-conservative, if work done by or against the force in moving a body depends upon the path between initial and final positions.

e.g. Frictional force, viscous force, air resistance, etc.

Work done by non-conservative forces

Let W_1 , W_2 and W_3 denote the net work done in moving a body from A to B along three different paths 1, 2 and 3 respectively, as shown in Fig. (a).

If the force is non-conservative, then $W_1 \neq W_2 \neq W_3$

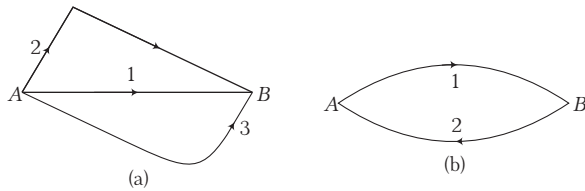


Fig. 6.11

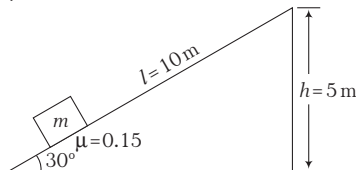
Work done by a non-conservative force in a round trip as shown in Fig. (b) is not zero.

i.e. $W_1 + W_2 \neq 0$

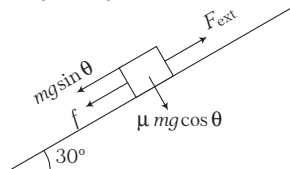
Example 6.20 A body of mass 0.3 kg is taken up an inclined plane of length 10 m and height 5 m, and then allowed to slide down the bottom again. The coefficient of friction between the body and the plane is 0.15. What is the

- work done by the applied force over the upward journey?
- work done by the gravitational force over the round trip?
- work done by the frictional force over the round trip?

Which of the above forces (except applied force) is/are conservative forces?



Sol. Upward journey Let us calculate work done by different forces over upward journey.



Work done by gravitational force,
 $W_1 = (mg \sin \theta)s \cos 180^\circ$

$$W_1 = 0.3 \times 10 \sin 30^\circ \times 10(-1) \quad [\because s = l = 10 \text{ m}]$$

$$W_1 = -15 \text{ J}$$

Work done by force of friction,

$$W_2 = (\mu mg \cos \theta)s \cos 180^\circ$$

$$W_2 = 0.15 \times 0.3 \times 10 \cos 30^\circ \times 10(-1)$$

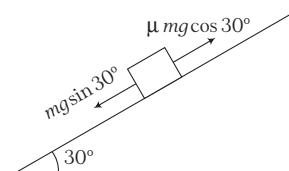
$$W_2 = -3.897 \text{ J}$$

Work done by external force, $W_3 = F_{\text{ext}} \times s \times \cos 0^\circ$

$$W_3 = (mg \sin \theta + \mu mg \cos \theta) \times 10 \times 1$$

$$W_3 = 18.897 \text{ J}$$

Downward journey



$$mg \sin 30^\circ > \mu mg \cos 30^\circ$$

Work done by the gravitational force,

$$W_4 = mg \sin 30^\circ \times s \cos 0^\circ$$

$$W_4 = 0.3 \times 10 \times \frac{1}{2} \times 10 \times 1 = +15 \text{ J}$$

Work done by the frictional force,

$$W_5 = \mu mg \cos 30^\circ \times s \cos 180^\circ$$

$$= 0.15 \times 0.3 \times 10 \times \frac{\sqrt{3}}{2} \times 10 \times (-1) = -3.897 \text{ J}$$

(i) Work done by applied force over upward journey,

$$W_3 = 18.897 \text{ J}$$

(ii) Work done by gravitational force over the round trip,

$$W_1 + W_4 = -15 + 15 = 0 \text{ J}$$

(iii) Work done by frictional force over the round trip,

$$W_2 + W_5 = -3.897 + (-3.897) = -7.794 \text{ J}$$

Work done by gravitational force over a closed path is zero but due to frictional force, it is non-zero.

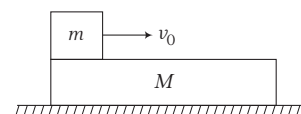
Therefore, gravitational force is conservative and frictional force is non-conservative.

CHECK POINT 6.1

- The net work done by kinetic friction
 - is always negative
 - is always zero
 - may be negative or positive
 - is always positive
- A man pushes a wall but fails to move it. He does
 - negative work
 - positive but not maximum work
 - maximum positive work
 - no work at all
- A gardener pushes a lawn roller through a distance 20 m. If he applies a force of 20 kg-wt in a direction inclined at 60° to the ground, the work done by him is
 - 1960 J
 - 196 J
 - 1.96 J
 - 196 kJ

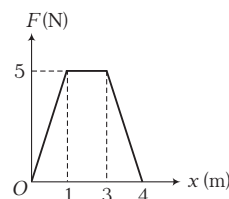
- How much work must be done by a force on 50 kg body in order to accelerate, it in the direction of force from rest to 20 ms^{-1} in 10 s?
 - 10^{-3} J
 - 10^4 J
 - $2 \times 10^3 \text{ J}$
 - $4 \times 10^4 \text{ J}$

- The coefficient of friction between the block and plank is μ and its value is such that block becomes stationary with respect to plank before it reaches the other end. Then, which of the following is not correct?

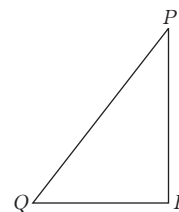


- (a) The work done by friction on the block is negative.
 (b) The work done by friction on the plank is positive.
 (c) The net work done by friction is negative.
 (d) Net work done by the friction is zero.
6. A horizontal force F pulls a 20 kg box at a constant speed along a rough horizontal floor. The coefficient of friction between the box and the floor is 0.25. The work done by force F on the block in displacing it by 2 m is
 (a) 49 J (b) 98 J
 (c) 147 J (d) 196 J
7. A force $(3\hat{i} + 4\hat{j})$ N acts on a body and displaces it by $(3\hat{i} + 4\hat{j})$ m. The work done by the force is
 (a) 10 J (b) 12 J
 (c) 16 J (d) 25 J
8. A particle moves from point $P(1\text{ m}, 2\text{ m}, 3\text{ m})$ to $Q(2\text{ m}, 1\text{ m}, 4\text{ m})$ under the action of a constant force $\mathbf{F} = (2\hat{i} + \hat{j} + \hat{k})$ N. Work done by the force is
 (a) 2 J (b) 4 J
 (c) 16 J (d) 8 J
9. Work done by a force $\mathbf{F} = (\hat{i} + 2\hat{j} + 3\hat{k})$ N acting on a particle in displacing it from the point $\mathbf{r}_1 = (\hat{i} + \hat{j} + \hat{k})$ m to the point $\mathbf{r}_2 = (\hat{i} - \hat{j} + 2\hat{k})$ m is
 (a) -3 J (b) -1 J
 (c) zero (d) 2 J
10. A body constrained to move in the y -direction, is subjected to a force $\mathbf{F} = (-2\hat{i} + 15\hat{j} + 6\hat{k})$ N. What is the work done by this force in moving the body through a distance of 10 m along the Y -axis?
 (a) 20 J (b) 150 J
 (c) 160 J (d) 190 J
11. A particle moves along the X -axis from $x = 0$ to $x = 5$ m under the influence of a force given by $F = 10 - 2x + 3x^2$. Work done in the process is
 (a) 70 units (b) 270 units
 (c) 35 units (d) 150 units
12. A force $F = Ay^2 + By + C$ acts on a body in the y -direction. The work done by this force during a displacement from $y = -a$ to $y = a$ is
 (a) $\frac{2Aa^3}{3}$
 (b) $\frac{2Aa^3}{3} + 2Ca$
 (c) $\frac{2Aa^3}{3} + \frac{Ba^2}{2} + Ca$
 (d) None of the above
13. Force acting on a particle is $(2\hat{i} + 3\hat{j})$ N. Work done by this force is zero, when a particle is moved on the line $3y + kx = 5$. Here, value of k is
 (a) 2 (b) 4
 (c) 6 (d) 8

14. The force F is acting on a particle moving in a straight line as shown in figure. What is the work done by the force on the particle in the 4 m of the trajectory?



- (a) 5 J (b) 10 J (c) 15 J (d) 2.5 J
15. A position dependent force F is acting on a particle and its force-position curve is shown in the figure. Work done on the particle, when its displacement is from 0 to 5 m is
 (a) 35 J (b) 25 J (c) 15 J (d) 5 J
16. A spring of force constant 800 Nm^{-1} has an extension of 5 cm. The work done in extending it from 5 cm to 15 cm is
 (a) 16 J (b) 8 J (c) 32 J (d) 24 J
17. A spring 40 mm long is stretched by the application of a force. If 10 N force is required to stretch the spring through 1 mm, then the work done in stretching the spring through 40 mm is
 (a) 84 J (b) 68 J (c) 23 J (d) 8 J
18. A porter with a suitcase of mass 20 kg on his head moves up a staircase upto a height of 4m. The amounts of work done by the upward lifting force relative to him and relative to a person on the ground respectively, are
 (a) 0 ; 800J (b) 400J ; 400J
 (c) 0 ; 400J (d) 800J ; 0
19. For the path PQR in a conservative force field, the amounts of work done in carrying a body from P to Q and from Q to R are 5J and 2J, respectively. The work done in carrying the body from P to R will be



- (a) 7J (b) 3J
 (c) $\sqrt{21}$ J (d) zero
20. Which of the following is a non-conservative force?
 (a) Spring force (b) Frictional force
 (c) Gravitational force (d) All of these

ENERGY

The energy of a body is defined as its capacity or ability for doing work. Like work, energy is a scalar quantity having magnitude only and no direction. The dimensions of energy are the same as the dimensions of work, *i.e.* $[ML^2 T^{-2}]$. It is measured in the same unit as work, *i.e.* joule in SI and erg in CGS system.

Energy can exist in various forms such as mechanical energy (potential energy and kinetic energy), sound energy, heat energy, light energy, etc.

Kinetic energy

The energy possessed by a body by virtue of its motion is called kinetic energy. Kinetic energy of a body can be calculated by the amount of work done in stopping the moving body or from the amount of work done in giving it same velocity from state of rest.

If an object of mass m has velocity $\mathbf{v}$, then its kinetic energy is given by $KE = \frac{1}{2}mv^2 = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v}$

The kinetic energy of a system having n particles is equal to the sum of the kinetic energies of all its constituent particles.

$$i.e. K = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \frac{1}{2}m_3v_3^2 + \dots + \frac{1}{2}m_nv_n^2$$

$$K = \sum_{i=1}^n \frac{1}{2}m_i v_i^2$$

Regarding the kinetic energy, the following two points are important to note

- Since, both m and v^2 are always positive. Hence, kinetic energy is always positive and does not depend on the direction of motion of the body.
- Kinetic energy depends on the frame of reference.
e.g. The kinetic energy of a person of mass m sitting in a train moving with speed v is zero in the frame of train but $(1/2)mv^2$ in the frame of earth.

Example 6.21 When a man increases his speed by 2 ms^{-1} , he finds that his kinetic energy is doubled. Find the original speed of the man.

Sol. Man possesses kinetic energy because of his velocity (v). If m is mass of man, then $K = \frac{1}{2}mv^2$

$$\text{Given, } v_1 = v, m_1 = m_2 = m$$

$$\text{When } v_2 = (v + 2) \text{ ms}^{-1}, \text{ then}$$

$$K_2 = 2K_1$$

$$\therefore \frac{K_1}{K_2} = \frac{v_1^2}{v_2^2} \Rightarrow \frac{K_1}{2K_1} = \frac{v^2}{(v+2)^2}$$

$$\Rightarrow v^2 - 4v - 4 = 0$$

$$\text{This gives, } v_1 = \frac{4 + \sqrt{16 + 16}}{2} = \frac{4 + \sqrt{32}}{2}$$

$$\Rightarrow v_1 = 2(\sqrt{2} + 1) \text{ ms}^{-1}$$

Example 6.22 In a ballistics demonstration, a police officer fires a bullet of mass 50 g with speed 200 ms^{-1} on soft plywood of thickness 2 cm . The bullet emerges with only 10% of its initial kinetic energy. What is the emergent speed of the bullet?

Sol. Initial kinetic energy, $K_i = \frac{1}{2} \times \frac{50}{1000} \times 200 \times 200 \text{ J} = 1000 \text{ J}$

$$\text{Final kinetic energy, } K_f = \frac{10}{100} \times 1000 \text{ J} = 100 \text{ J}$$

If v_f is emergent speed of the bullet, then

$$\frac{1}{2} \times \frac{50}{1000} \times v_f^2 = 100$$

$$\Rightarrow v_f^2 = 4000$$

$$\Rightarrow v_f = 63.2 \text{ ms}^{-1}$$

Example 6.23 A body of mass 0.8 kg has initial velocity $(3\hat{i} - 4\hat{j}) \text{ ms}^{-1}$ and final velocity $(-6\hat{j} + 2\hat{k}) \text{ ms}^{-1}$. Find change in kinetic energy of the body.

Sol. Change in kinetic energy,

$$\Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$\text{where, } v_f = \sqrt{6^2 + 2^2} = \sqrt{40} \text{ ms}^{-1}$$

$$\text{and } v_i = \sqrt{3^2 + 4^2} = \sqrt{25} \text{ ms}^{-1}$$

$$\begin{aligned} \therefore \Delta KE &= \frac{1}{2} \times 0.8 [\sqrt{40}]^2 - (\sqrt{25})^2 \\ &= 0.4[40 - 25] = 0.4(15) = 6 \text{ J} \end{aligned}$$

Relation between kinetic energy and linear momentum

The linear momentum of a body is given by $p = mv$, where m is the mass and v is the velocity of a body.

Then, kinetic energy of the body,

$$KE = \frac{1}{2}mv^2 = \frac{1}{2m}(m^2v^2)$$

$$KE = \frac{p^2}{2m} \text{ or } p^2 = 2m \text{ KE}$$

$$\Rightarrow \text{Linear momentum, } p = \sqrt{2m \text{ KE}}$$

Example 6.24 Two bodies A and B having masses in the ratio of 3 : 1 possess the same kinetic energy. Obtain the ratio of linear momentum of B to that of A.

Sol. Kinetic energy of the body is given by

$$E_K = \frac{1}{2}mv^2 \quad \dots(i)$$

and linear momentum, $p = mv$

$\dots(ii)$

From Eqs. (i) and (ii), we get

$$E_K = \frac{m^2v^2}{2m} = \frac{p^2}{2m}$$

Now,

$$E_{K_1} = E_{K_2}$$

$$\Rightarrow \frac{p_1^2}{2m_1} = \frac{p_2^2}{2m_2} \text{ or } \frac{p_1}{p_2} = \sqrt{\frac{m_1}{m_2}}$$

$$\text{or } \frac{p_1}{p_2} = \sqrt{\frac{3}{1}} \text{ or } \frac{p_1}{p_2} = \frac{1}{\sqrt{3}}$$

Example 6.25 Kinetic energy of a particle is increased by 300%. Find the percentage increase in momentum.

Sol. Kinetic energy, $E = \frac{1}{2}mv^2$

and momentum, $p = mv$

When E is increased by 300%,

$$E' = E + 3E = 4E$$

$$= 4\left(\frac{1}{2}mv^2\right) = 2mv^2$$

If v' is velocity of body, then

$$\frac{1}{2}m(v')^2 = 2mv^2$$

$$\Rightarrow v' = 2v$$

$$\text{So, } p' = mv' = 2mv$$

Hence, percentage change in momentum

$$= \frac{2mv - mv}{mv} \times 100 = 100\%$$

Work-energy theorem

This theorem states that work done by all the forces (conservative or non-conservative, external or internal) acting on a particle or an object is equal to the change in kinetic energy of it.

$$\therefore W_{\text{net}} = \Delta KE = K_f - K_i$$

$$W_{\text{net}} = \frac{1}{2}m(v_f^2 - v_i^2)$$

$$\Rightarrow W_{\text{conservative}} + W_{\text{non-conservative}} + W_{\text{ext. force}} = \Delta KE$$

If W_{net} is positive, then kinetic energy will increase and vice-versa.

Let $\mathbf{F}_1, \mathbf{F}_2, \dots$ be the individual forces acting on a particle. The resultant force is $\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \dots$ and the work done by the resultant force is

$$W = \int \mathbf{F} \cdot d\mathbf{r} = \int (\mathbf{F}_1 + \mathbf{F}_2 + \dots) \cdot d\mathbf{r}$$

$$= \int \mathbf{F}_1 \cdot d\mathbf{r} + \int \mathbf{F}_2 \cdot d\mathbf{r} + \dots$$

where, $\int \mathbf{F}_1 \cdot d\mathbf{r}$ is the work done on the particle by $\mathbf{F}_1$ and so on. Thus, work-energy theorem can also be stated as **work done by the resultant force is equal to the sum of the work done by the individual forces.**

Example 6.26 The position (x) of a particle of mass 1 kg moving along X-axis at time t is given by $\left(x = \frac{1}{2}t^2\right)$ metre.

Find the work done by force acting on it in time interval $t = 0$ to $t = 3$ s.

Sol. Given, $x = \frac{1}{2}t^2$

$$\Rightarrow v = \frac{dx}{dt} = \frac{1}{2}(2t) = t$$

$$\therefore \text{At } t = 0, v_i = 0$$

$$\Rightarrow \text{At } t = 3 \text{ s, } v_f = 3 \text{ ms}^{-1}$$

According to work-energy theorem,

$$W = \Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$= \left(\frac{1}{2} \times 1 \times 3^2\right) - \left(\frac{1}{2} \times 1 \times 0^2\right) = 4.5 \text{ J}$$

Example 6.27 A bullet weighing 10 g is fired with a velocity 800 ms^{-1} . After passing through a mud wall 1 m thick, its velocity decreases to 100 ms^{-1} . Find the average resistance offered by the mud wall.

Sol. According to work-energy theorem, work done by the average resistance offered by the wall = change in kinetic energy of the bullet.

$$\therefore W = F \cdot s = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$\Rightarrow F = \frac{m(v^2 - u^2)}{2s} = \frac{0.01(100^2 - 800^2)}{2 \times 1} = -3150 \text{ N}$$

$$\Rightarrow \text{Resistance offered} = 3150 \text{ N}$$

Example 6.28 An object of mass 5 kg falls from rest through a vertical distance of 20 m and attains a velocity of 10 ms^{-1} . How much work is done by the resistance of the air on the object? (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Applying work-energy theorem, work done by all the forces = change in kinetic energy

$$\text{or } W_{mg} + W_{\text{air}} = \frac{1}{2}mv^2$$

$$\therefore W_{\text{air}} = \frac{1}{2}mv^2 - W_{mg} = \frac{1}{2}mv^2 - mgh$$

$$= \frac{1}{2} \times 5 \times (10)^2 - (5) \times (10) \times (20)$$

$$= -750 \text{ J}$$

Example 6.29 A particle of mass m moves with velocity $v = a\sqrt{x}$, where a is a constant. Find the total work done by all the forces during a displacement from $x = 0$ to $x = d$.

Sol. Work done by all forces,

$$W = \Delta KE = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$\text{Here, } v_1 = a\sqrt{0} = 0, v_2 = a\sqrt{d}$$

$$\text{So, } W = \frac{1}{2}ma^2d - 0 = \frac{1}{2}ma^2d$$

Example 6.30 A vehicle of mass m is moving in x -direction with a constant speed. It is subjected to a retarding force $F = -0.1x \text{ Jm}^{-1}$ during its travel from $x = 20 \text{ m}$ to $x = 30 \text{ m}$. Evaluate the change in its kinetic energy.

Sol. According to work-energy theorem,

work done = change in kinetic energy of the vehicle

$$\therefore W = K_f - K_i \text{ or } F \cdot dx = K_f - K_i$$

$$\text{or } \int_{x=20}^{x=30} (-0.1)x \, dx = K_f - K_i$$

$$\text{or } -0.1 \left[\frac{x^2}{2} \right]_{x=20}^{x=30} = K_f - K_i$$

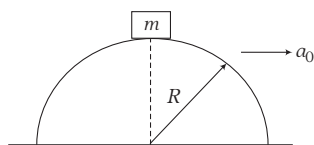
$$\text{or } -0.1 \left[\frac{(30)^2}{2} - \frac{(20)^2}{2} \right] = K_f - K_i$$

$$\text{or } K_f - K_i = -0.1(450 - 200)$$

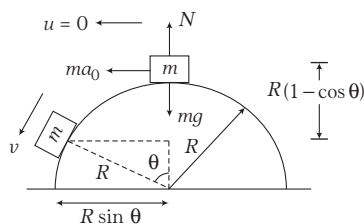
$$\therefore K_f - K_i = -25 \text{ J}$$

Example 6.31 A block is placed at the top of a smooth hemisphere of radius R . Now, the hemisphere is given a horizontal acceleration a_0 . Find the velocity of the block relative to the hemisphere as a function of θ as it slides down.

Sol. In the reference frame of hemisphere, apply a pseudo force on m .



From the work-energy theorem,



$$W_{\text{total}} = \Delta K$$

$$\Rightarrow W_{gr} + W_{\text{pseudo}} = \Delta K$$

$$mgR(1 - \cos \theta) + ma_0R \sin \theta = \frac{1}{2}mv^2 - 0$$

$$v = \sqrt{2R \{ g(1 - \cos \theta) + a_0 \sin \theta \}}$$

The work done by normal reaction N is zero because it is always perpendicular to displacement.

Potential energy

The energy possessed by a body or system by virtue of its position or configuration is known as the potential energy. *e.g.* A block attached to a compressed or elongated spring possesses some energy called **elastic potential energy**.

This block has a capacity to do work.

Similarly, a stone when released from a certain height also has energy in the form of **gravitational potential energy**. Two charged particles kept at certain distance has **electric potential energy**.

The units and dimensions of potential energy are same as that of kinetic energy.

There are mainly two types of potential energies as discussed below

1. Gravitational potential energy

Gravitational potential energy of a body is the energy associated with it due to the virtue of its position above the surface of earth. If an object of mass m is placed at height h above earth's surface, then

$$\boxed{\text{Gravitational potential energy, } U = mgh}$$

Example 6.32 A stone of mass 0.4 kg is thrown vertically up with a speed of 9.8 ms^{-1} . Find the potential energy after half second.

Sol. Here, mass, $m = 0.4 \text{ kg}$, speed, $u = 9.8 \text{ ms}^{-1}$,

$$\text{acceleration, } a = -g = -9.8 \text{ m/s}^2 \text{ and time, } t = \frac{1}{2} \text{ s}$$

From second equation of motion,

$$h = s = ut + \frac{1}{2}at^2$$

$$= \left(9.8 \times \frac{1}{2} \right) + \left(\frac{1}{2} \times (-9.8) \times \frac{1}{2} \times \frac{1}{2} \right) = 3.67 \text{ m}$$

$$\therefore \text{Potential energy} = mgh = (0.4)(9.8)(3.67) = 14.386 \text{ J}$$

2. Elastic potential energy of a spring

Potential energy of a spring is the energy associated with the state of compression or expansion of an elastic spring.

When the spring is stretched or compressed by an amount x from its unstretched position, then elastic potential energy of spring,

$$\boxed{U = \frac{1}{2}kx^2} \quad (\text{where, } k = \text{spring constant})$$

Note that elastic potential energy is always positive.

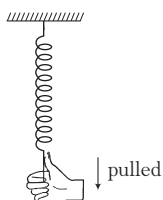
Natural length of spring indicates reference point, where potential energy of spring is taken zero.

Example 6.33 Two springs of spring constants 1500 Nm^{-1} and 3000 Nm^{-1} respectively are stretched with the same force, slowly. Compute the ratio of their potential energies.

Sol. The work done in pulling the string is stored as potential energy in the spring which is given by

$$U = \frac{1}{2} kx^2 \quad \dots(i)$$

where, k is spring constant and x is distance through which it is pulled.



In case of spring, applied force is given by

$$F = kx$$

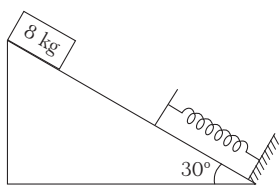
where, k is spring constant.

Putting $x = \frac{F}{k}$ in Eq. (i), we get

$$U = \frac{1}{2} k \left(\frac{F}{k} \right)^2 = \frac{F^2}{2k} \Rightarrow U \propto \frac{1}{k} \quad \dots(ii)$$

$$\therefore \frac{U_1}{U_2} = \frac{k_2}{k_1} = \frac{3000}{1500} = \frac{2}{1} \text{ or } U_1 : U_2 = 2 : 1$$

Example 6.34 A block of mass 8 kg is released from the top of an inclined smooth surface as shown in figure. If spring constant of spring is 200 Nm^{-1} and block comes to rest after compressing spring by 1 m , then find the distance travelled by block before it comes to rest.



Sol. Let d be the distance travelled by block along the plane and h be the height upto which it come in downward direction,
 $h = d \sin 30^\circ$

Now, when block will comes to rest, the decrease in its potential energy will be stored in spring in the form of its potential energy.

$$\therefore mgh = \frac{1}{2} ky^2 \quad (y \text{ is compression in spring})$$

$$\Rightarrow mgd \sin 30^\circ = \frac{1}{2} ky^2$$

$$\Rightarrow 8 \times 10 \times \frac{d}{2} = \frac{1}{2} \times 200 \times 1^2$$

$$\Rightarrow 40d = 100$$

$$\Rightarrow d = 2.5 \text{ m}$$

Change in potential energy

Potential energy is defined for a conservative force field only. For non-conservative forces, it has no meaning. The change in potential energy (dU) of a system corresponding to a conservative internal force is given by

$$dU = -\mathbf{F} \cdot d\mathbf{r} = -dW \quad \left(\because \mathbf{F} = -\frac{dU}{dr} \right)$$

$$\text{or} \quad \int_{U_i}^{U_f} dU = - \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r} \text{ or } U_f - U_i = - \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r}$$

We generally choose the reference point at infinity and assume potential energy to be zero there, i.e. if we take $r_i = \infty$ (infinite) and $U_i = 0$, then we can write

$$U = - \int_{\infty}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r} = -W$$

or **Potential energy of a body or system is the negative of work done by the conservative forces in bringing it from infinity to the present position.**

Regarding the potential energy it is worth noting that

(i) Potential energy should be considered to be a property of the entire system rather than assigning it to any specific particle.

(ii) Potential energy depends on frame of reference.

For three dimensional motion of the particle

$$\mathbf{F} = - \left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k} \right)$$

where, $\frac{\partial U}{\partial x}$ = partial derivative of U w.r.t. x ,

$\frac{\partial U}{\partial y}$ = partial derivative of U w.r.t. y

and $\frac{\partial U}{\partial z}$ = partial derivative of U w.r.t. z

Example 6.35 Calculate the work done in lifting a 300 N weight to a height of 10 m with an acceleration 0.5 ms^{-2} . (Take, $g = 10 \text{ ms}^{-2}$)

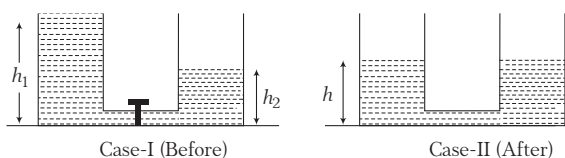
Sol. Given, weight, $w = 300 \text{ N}$, height, $h = 10 \text{ m}$ and acceleration, $a = 0.5 \text{ ms}^{-2}$

$$\therefore \text{Mass, } m = \frac{w}{g} = \frac{300}{10} = 30 \text{ kg, } a_{\text{net}} = g + a$$

$$\therefore \text{Work done, } W = -U = -m(a + g)h \\ = -30(0.5 + 10)10 = -3150 \text{ J}$$

Example 6.36 Two cylindrical vessels of equal cross-sectional area A contain water up to heights h_1 and h_2 . The vessels are inter-connected, so that the levels in them are equal. Calculate the work done by the force of gravity during the process. The density of water is ρ .

Sol. According to the question, we draw the following diagram.



Since, the mass of water remains same,

$$\rho A h_1 + \rho A h_2 = \rho A h + \rho A h \Rightarrow h = \frac{h_1 + h_2}{2}$$

Potential energy of first case,

$$U_1 = (\rho A h_1) g \frac{h_1}{2} + (\rho A h_2) g \frac{h_2}{2} = \frac{\rho A g}{2} (h_1^2 + h_2^2)$$

Potential energy of second case,

$$U_2 = (\rho A h) g \frac{h}{2} \times 2 = \frac{\rho A g (h_1 + h_2)^2}{4}$$

Since, the gravitational force is a conservative force, then

$$W_{gr} = U_1 - U_2 = \frac{\rho A g}{4} (h_1 - h_2)^2$$

Example 6.37 A particle is moving on frictionless XY-plane. It is acted upon a conservative force described by the potential energy function:

$$U(x, y, z) = \frac{1}{2} k(x^2 + y^2 + z^2)$$

where, $k = \text{constant}$

Derive an expression for the force acting on the particle.

Sol. Force acting on the particle in x-direction is given by

$$F_x = -\frac{\partial U}{\partial x} = -\frac{1}{2} k(2x + 0 + 0) = -kx$$

Similarly, $F_y = -ky$ and $F_z = -kz$

$\therefore$ Force acting on the particle is

$$\mathbf{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k} = -k(x\hat{i} + y\hat{j} + z\hat{k}) = -k\mathbf{r}$$

where, $\mathbf{r}$ = position vector of the particle = $x\hat{i} + y\hat{j} + z\hat{k}$

Work done in pulling a chain against gravity

Consider a chain of length L and mass M is held on a frictionless table with L/x of its length hanging over the edge. We have to find the work done in pulling the hanging portion of the chain on the table.

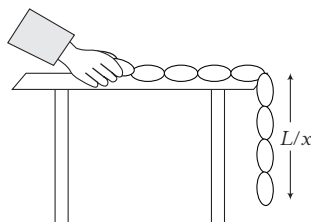


Fig. 6.12

For this, we can assume centre of mass of hanging portion of the chain at the middle.

$$\text{Mass of hanging part of chain} = \frac{M}{L} \times \frac{L}{x} = \frac{M}{x}$$

Now, assume hanging portion of the chain as a point mass centred at its centre of mass.

Therefore, work done to raise the centre of mass of the chain on the table,

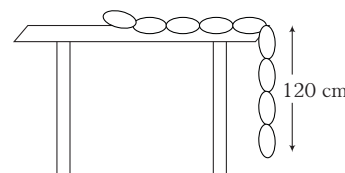
$$W = \frac{M}{x} \times g \times \frac{L}{2x} \text{ or } W = \frac{MgL}{2x^2}$$

Example 6.38 A uniform chain of length 4m is kept on a table such that a length of 120 cm hangs freely from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain on the table?

Sol. Fraction of length of the chain hanging from the table,

$$\frac{L}{x} = 120 \text{ cm} \Rightarrow x = \frac{10}{3}$$

Work done in pulling the chain on the table,



$$W = \frac{mgL}{2x^2} = \frac{4 \times 10 \times 4}{2 \times (10/3)^2} = 7.2 \text{ J}$$

Alternative Solution

Centre of mass of hanging part of the chain,

$$h = \frac{120}{2} = 60 \text{ cm} = 0.6 \text{ m}$$

Mass per unit length of chain, $\mu = \frac{M}{L} = \frac{4}{4} = 1 \text{ kg/m}$

Mass of hanging part of chain of length, $l = 120 \text{ cm} = 1.2 \text{ m}$

$$M' = \mu l = 1 \times 1.2 = 1.2 \text{ kg}$$

$\therefore$ Work done in pulling the entire chain on the table,

$$W = \text{Gravitational potential energy} \\ = M'gh = 1.2 \times 10 \times 0.6 = 7.2 \text{ J}$$

EQUILIBRIUM

If a large number of forces act on a system or on an object simultaneously in such a way that the resultant force on it is zero, then it is said to be in translational equilibrium. If the forces acting on the object are conservative and it is in equilibrium, then

$$F_{\text{net}} = 0 \Rightarrow -\frac{dU}{dr} = 0 \text{ or } \frac{dU}{dr} = 0$$

So, when force is conservative and object is in equilibrium, slope of U - r graph is zero or its potential energy is either minimum or maximum or constant.

On this basis, equilibrium of object or a system can be divided into three types

(i) **Stable equilibrium** An object is said to be in stable equilibrium, if on slight displacement from equilibrium position, it has tendency to come back. Here, potential energy in equilibrium position is minimum as compared to its neighbouring points or $\frac{d^2U}{dx^2}$ is positive.

(ii) **Unstable equilibrium** An object is said to be in unstable equilibrium, if on slight displacement from equilibrium position, it moves in the direction of displacement.

In unstable equilibrium, potential energy is maximum or $\frac{d^2U}{dx^2}$ is negative.

(iii) **Neutral equilibrium** An object is said to be in neutral equilibrium, if on displacement from its equilibrium position, it has neither the tendency to move in direction of displacement nor to come back to equilibrium position. In neutral equilibrium, potential energy of the object is constant or $\frac{d^2U}{dx^2} = 0$.

Potential energy curve for equilibrium

E_{mech} = constant is represented by horizontal dotted line in the following graph.

At x , kinetic energy, $K = E_{\text{mech}} - U(x)$

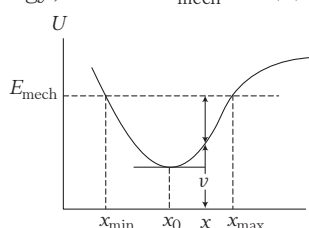


Fig. 6.13

The points $x = x_{\text{max}}$ and $x = x_{\text{min}}$ are called turning points. At these points, velocity of the particle decreases to zero and reverses.

From x_{min} to x_0 , slope of $U(x)$ is negative, $F = -\frac{dU}{dx}$ is positive and acts towards x_0 .

At x_0 , $F = 0$. So, x_0 is known as **stable equilibrium point**.

Beyond x_0 , slope is positive, indicating a negative force, towards x_0 .

From the PE and position graph,

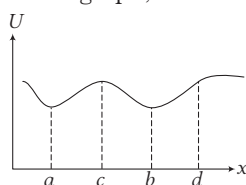


Fig. 6.14

At $x = a, b$, U is minimum $\Rightarrow$ stable equilibrium

$x = c$, U is maximum $\Rightarrow$ unstable equilibrium

$x = d$, U is constant $\Rightarrow$ neutral equilibrium

Example 6.39 The potential energy of a conservative system is given by $U = ax^2 - bx$ (where, a and b are positive constants). Find the equilibrium position and discuss whether the equilibrium is stable, unstable or neutral.

Sol. In a conservative field, $F = -\frac{dU}{dx}$

$$\therefore F = -\frac{d}{dx}(ax^2 - bx) = b - 2ax$$

$$\text{For equilibrium, } F = 0 \Rightarrow b - 2ax = 0$$

$$\therefore x = \frac{b}{2a}$$

From the given equation, we can see that $\frac{d^2U}{dx^2} = 2a$ (positive), i.e. U is minimum.

Therefore, $x = \frac{b}{2a}$ is the stable equilibrium position.

Example 6.40 The potential energy for a conservative force system is given by $U = ax^2 - bx$, where a and b are constants. Find out (i) an expression of force (ii) potential energy at equilibrium.

Sol. (i) For conservative force, $F = -\frac{dU}{dx} = -(2ax - b) = -2ax + b$

$$(ii) \text{ At equilibrium, } F = 0 \Rightarrow -2ax + b = 0 \Rightarrow x = \frac{b}{2a}$$

$$\therefore U = a\left(\frac{b}{2a}\right)^2 - b\left(\frac{b}{2a}\right) = \frac{b^2}{4a} - \frac{b^2}{2a} = -\frac{b^2}{4a}$$

LAW OF CONSERVATION OF ENERGY

Energy can neither be created nor be destroyed, it can only be transformed from one form to another form.

Conservation of mechanical energy The total mechanical energy (sum of kinetic energy and potential energy) of a system is conserved, if the forces acting on it are conservative.

Example 6.41 A body of mass 5 kg is thrown vertically up with a kinetic energy of 490 J. What will be height at which the kinetic energy of the body becomes half of the original value? (Take, acceleration due to gravity = 9.8 ms^{-2})

Sol. Given, $m = 5 \text{ kg}$ and $K_i = 490 \text{ J}$, $g = 9.8 \text{ ms}^{-2}$

$$\text{From the law of conservation of energy, } K_i + U_i = K_f + U_f$$

$$\Rightarrow K_i + 0 = \frac{K_i}{2} + mgh \quad \left(\because K_f = \frac{K_i}{2} \right)$$

$$490 = 245 + 5 \times 9.8 \times h$$

$$\Rightarrow h = \frac{490 - 245}{5 \times 9.8} = \frac{245}{49} = 5 \text{ m}$$

Example 6.42 A bullet of mass m moving with velocity v strikes a suspended wooden block of mass M and remains embedded in it. If the block rises to a height h , find the initial velocity of the bullet.

Sol. Initial kinetic energy of the block when the bullet strikes

$$= \frac{1}{2} (m + M) v^2$$

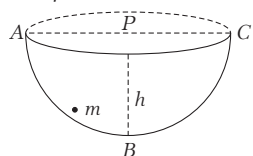
Due to this kinetic energy, the block will rise to a height h . Its potential energy $= (m + M) gh$

So, from the law of conservation of energy,

$$\frac{1}{2} (M + m) v^2 = (M + m) gh \Rightarrow \frac{v^2}{2} = gh$$

Initial velocity of the bullet, $v = \sqrt{2gh}$

Example 6.43 A particle of mass m makes SHM in a smooth hemispherical bowl ABC and it moves from A to C and back to A via ABC, so that $PB = h$. Find the speed of the ball when it just crosses the point B.



Sol. From law of conservation of energy, PE at A = KE at B

$$\text{or } mgh = \frac{1}{2} mv^2 \text{ or } v = \sqrt{2gh}$$

Example 6.44 A child is swinging on a swing. Minimum and maximum heights of swing from the earth's surface are 0.75 m and 2 m, respectively. What will be the maximum velocity of this swing?

Sol. From energy conservation,

gain in kinetic energy = loss in potential energy

$$\Rightarrow (1/2) mv_{\max}^2 = mg(H_2 - H_1)$$

Here, H_1 = minimum height of swing from earth's surface = 0.75 m and H_2 = maximum height of swing from earth's surface = 2 m

$$\therefore (1/2) mv_{\max}^2 = mg(2 - 0.75)$$

$$\text{or } v_{\max} = \sqrt{2 \times 10 \times 1.25} = \sqrt{25} = 5 \text{ ms}^{-1}$$

Example 6.45 A machine which is 75% efficient, uses 12 J of energy in lifting 1 kg mass through a certain height. The mass is then allowed to fall through the same height. Find the velocity at the end of its fall.

Sol. Potential energy of the mass at a height above the earth's surface $= \frac{75}{100} \times 12 = 9 \text{ J}$... (i)

Now, kinetic energy of the mass at the end of fall $= \frac{1}{2} mv^2$... (ii)

Applying law of conservation of energy,

$$\frac{1}{2} mv^2 = 9 \Rightarrow v = \sqrt{\frac{2 \times 9}{m}} = \sqrt{\frac{18}{1}} = \sqrt{18} \text{ ms}^{-1}$$

Example 6.46 Auto manufactures study the collision of cars with mounted springs of different spring constants. Consider a car of mass 1500 kg moving with a speed of 36 kmh^{-1} on a smooth road and colliding with a horizontally mounted spring of spring constant $7.5 \times 10^3 \text{ Nm}^{-1}$. Find the maximum compression of the spring.

Sol. At maximum compression, KE of car gets converted completely into PE of the spring.

$$\text{KE of car, } K = \frac{1}{2} mv^2 = \frac{1}{2} \times 1500 \times 10 \times 10$$

$$= 7.5 \times 10^4 \text{ J} \left(\because v = 36 \times \frac{5}{18} = 10 \text{ ms}^{-1} \right)$$

$$U = \frac{1}{2} kx_m^2 = K = 7.5 \times 10^4 \text{ J}$$

$$\Rightarrow x_m = \sqrt{\frac{2 \times 7.5 \times 10^4}{7.5 \times 10^3}} = 4.47 \text{ m}$$

Example 6.47 A spherical ball of mass 20 kg is stationary at the top of a hill of height 100 m. It rolls down a smooth surface to the ground, then climbs up another hill of height 30 m and finally rolls down to a horizontal base at a height of 20 m above the ground. Find the velocity attained by the ball, when moving at horizontal base.

Sol. According to conservation of energy,

$$mgH = \frac{1}{2} mv^2 + mgh_2 \Rightarrow mg(H - h_2) = \frac{1}{2} mv^2$$

where, H = height of the first hill,

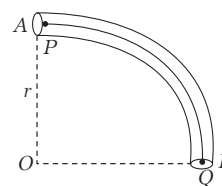
h_1 = height of the second hill,

h_2 = height of the horizontal base

and v = velocity attained by the ball.

$$\Rightarrow v = \sqrt{2g(100 - 20)} = \sqrt{2 \times 10 \times 80} = 40 \text{ ms}^{-1}$$

Example 6.48 A smooth narrow tube in the form of an arc AB of a circle of centre O and radius r is fixed, so that A is vertically above O and OB is horizontal. Particles P of mass m and Q of mass $2m$ with a light inextensible string of length $(\pi r/2)$ connecting them are placed inside the tube with P at A and Q at B and released from rest. Assuming the string remains taut during motion, find the speed of particles when P reaches B.



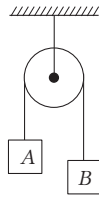
Sol. All surfaces are smooth. Therefore, mechanical energy of the system will remain conserved.

$\therefore$ Decrease in PE of both the blocks

= Increase in KE of both the blocks

$$\therefore (mgr) + (2mg) \left(\frac{\pi r}{2} \right) = \frac{1}{2} (m + 2m) v^2 \text{ or } v = \sqrt{\frac{2}{3} (1 + \pi) gr}$$

Example 6.49 In the arrangement shown in figure, string is light and inextensible and friction is absent everywhere.



Find the speed of both the blocks after the block A has ascended a height of 1 m. Given that, $m_A = 1$ kg and $m_B = 2$ kg (Take, $g = 10 \text{ ms}^{-2}$).

Sol. Friction is absent. Therefore, mechanical energy of the system will remain conserved. From constraint relations, we see that speed of both the blocks will be same. Suppose it is v . Here, gravitational potential energy of 2 kg block is decreasing while gravitational potential energy of 1 kg block is increasing.

Similarly, kinetic energy of both the blocks is also increasing.

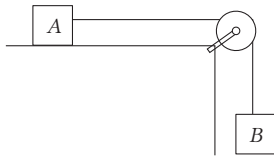
$$\therefore m_B gh = m_A gh + \frac{1}{2} m_A v^2 + \frac{1}{2} m_B v^2$$

$$\text{or } (2)(10)(1) = (1)(10)(1) + \frac{1}{2}(1)v^2 + \frac{1}{2}(2)v^2$$

$$\text{or } 20 = 10 + 0.5v^2 + v^2 \quad \text{or } 1.5v^2 = 10$$

$$\therefore v^2 = 6.67 \text{ m}^2\text{s}^{-2} \quad \text{or } v = 2.58 \text{ ms}^{-1}$$

Example 6.50 In the arrangement shown in figure, $m_A = 1$ kg, $m_B = 4$ kg. String is light and inextensible while pulley is smooth. Coefficient of friction between block A and the table is $\mu = 0.2$. Find the speed of both the blocks when block B has descended a height $h = 1$ m. (Take, $g = 10 \text{ ms}^{-2}$)



Sol. From constraint relation, we see that

$$v_A = v_B = v \text{ (say)}$$

Force of friction between block A and table will be

$$f = \mu m_A g = (0.2)(1)(10) = 2 \text{ N}$$

$$\therefore W_{nc} = \Delta U + \Delta K$$

$$\therefore -fs = -m_B gh + \frac{1}{2}(m_A + m_B)v^2$$

$$\text{or } (-2)(1) = -(4)(10)(1) + \frac{1}{2}(1+4)v^2$$

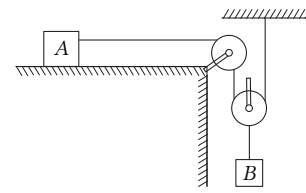
$$-2 = -40 + 2.5v^2$$

$$\text{or } 2.5v^2 = 38$$

$$\text{or } v^2 = 15.2 \text{ m}^2\text{s}^{-2} \quad \text{or } v = 3.9 \text{ ms}^{-1}$$

Example 6.51 In the arrangement shown in figure, $m_A = 4$ kg and $m_B = 1$ kg. The system is released from rest and block B is found to have a speed 0.3 m/s after it has descended through a distance of 1 m. Find the coefficient of friction

between the block and the table. Neglect friction elsewhere. (Take, $g = 10 \text{ ms}^{-2}$)



Sol. From constraint relations, we can see that $v_A = 2v_B$

$$\text{Therefore, } v_A = 2(0.3) \text{ (Given, } v_B = 0.3 \text{ ms}^{-1}) \\ = 0.6 \text{ ms}^{-1}$$

Applying $W_{nc} = \Delta U + \Delta K$, we get

$$-\mu m_A g s_A = -m_B g s_B + \frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2$$

$$\text{Here, } s_A = 2s_B \quad (\text{Given, } s_B = 1 \text{ m}) \\ = 2 \text{ m}$$

$$\therefore -\mu(4.0)(10)(2) = -(1)(10)(1) + \frac{1}{2}(4)(0.6)^2 + \frac{1}{2}(1)(0.3)^2$$

$$\text{or } -80\mu = -10 + 0.72 + 0.045$$

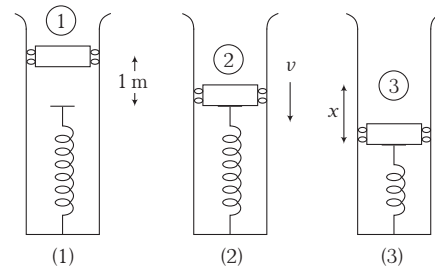
$$\text{or } 80\mu = 9.235 \quad \text{or } \mu = 0.115$$

Example 6.52 A 20 kg body is released from rest, so as to slide in between vertical rails and compresses a vertical spring ($k = 1920 \text{ Nm}^{-1}$) placed at a distance $h = 1$ m from the starting position of the body. The rails offer a frictional force of 40 N opposing the motion of body.

Find

- the velocity v of the body just before striking with the spring,
- the maximum compression of the spring and

Sol. According to the question, we draw the following diagram.



- Between diagrams (1) and (2),

Loss in PE = Gain in KE + Work done against friction

$$mgh = \frac{1}{2}mv^2 + fh \quad (\because h = s)$$

$$20 \times 10 \times 1 = \frac{1}{2} \times 20v^2 + 40 \times 1 \Rightarrow v = 4 \text{ m/s}$$

- Between diagrams (2) and (3),

Loss in PE + Loss in KE = Gain in spring energy + Work done against friction

$$mgx + \frac{1}{2}mv^2 = \frac{1}{2}kx^2 + fx$$

$$\begin{aligned}
 20 \times 10 \times x + \frac{1}{2} \times 20 \times 4^2 &= \frac{1}{2} \times 1920x^2 + 40x \\
 960x^2 - 160x - 160 &= 0 \\
 \Rightarrow 6x^2 - x - 1 &= 0 \\
 \Rightarrow x &= \frac{1 \pm \sqrt{1 + 4 \times 6 \times 1}}{12} = \frac{1 \pm 5}{12} = \frac{1}{2} \text{ m}
 \end{aligned}$$

CONVERSION OF MASS AND ENERGY

In 1905, Einstein discovered that mass can be converted into energy and *vice-versa*. He showed that mass and energy are equivalent and related by the relation

$$E = mc^2$$

where, m is mass that disappears, E is energy that appears and c is velocity of light in vacuum. Conversely, when energy E disappears, a mass $m (= E/c^2)$ appears.

Thus, according to modern physics, mass and energy are not conserved separately, but are conserved as a single entity called **mass energy**.

CHECK POINT 6.2

- If the force acting on a body is inversely proportional to its speed, the kinetic energy of the body is
 - constant
 - directly proportional to time
 - inversely proportional to time
 - directly proportional to square of time
- If the speed of a vehicle is increased by 1 ms^{-1} , its kinetic energy is doubled, then original speed of the vehicle is
 - $(\sqrt{2} + 1) \text{ ms}^{-1}$
 - $2(\sqrt{2} - 1) \text{ ms}^{-1}$
 - $2(\sqrt{2} + 1) \text{ ms}^{-1}$
 - $\sqrt{2}(\sqrt{2} + 1) \text{ ms}^{-1}$
- A running man has half the kinetic energy that of a boy whose mass is half the mass of the man. The man speeds up by 1 ms^{-1} and then has the same kinetic energy as that of boy. The original speeds of man and boy (in ms^{-1}) respectively, are
 - $(\sqrt{2} + 1), (\sqrt{2} - 1)$
 - $(\sqrt{2} + 1), 2(\sqrt{2} + 1)$
 - $\sqrt{2}, \sqrt{2}$
 - $(\sqrt{2} + 1), 2(\sqrt{2} - 1)$
- Two bodies of masses m_1 and m_2 have same momentum. The ratio of their kinetic energy is
 - $\sqrt{\frac{m_2}{m_1}}$
 - $\sqrt{\frac{m_1}{m_2}}$
 - $\frac{m_1}{m_2}$
 - $\frac{m_2}{m_1}$

So, the law of conservation of mass and law of conservation of energy have been unified by this relation into a single law of conservation of mass energy.

Nuclear energy

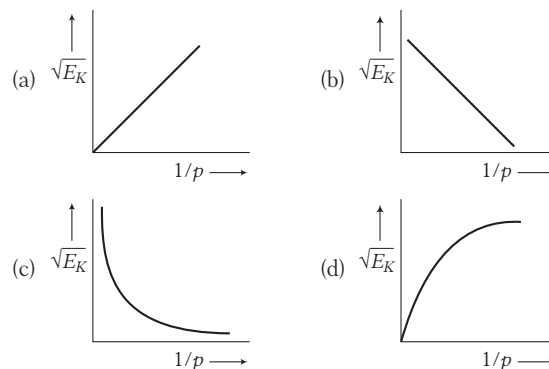
When U^{235} nucleus breaks up into lighter nuclei on being bombarded by a slow neutron, a tremendous amount of energy is released. Thus, the energy so released is called nuclear energy and this phenomenon is known as **nuclear fission**. Nuclear reactors and nuclear bombs are the sources of nuclear energy.

Example 6.53 Calculate the energy in MeV equivalent to the rest mass of an electron. Given that the rest mass of an electron, $m_0 = 9.1 \times 10^{-31} \text{ kg}$, $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$ and speed of light, $c = 3 \times 10^8 \text{ ms}^{-1}$.

Sol. According to the conversion of mass and energy,

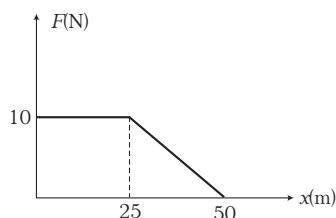
$$\begin{aligned}
 E &= m_0 c^2 = 9.1 \times 10^{-31} \times (3 \times 10^8)^2 \\
 &= 81.9 \times 10^{-15} \text{ J} \\
 &= \frac{81.9 \times 10^{-15}}{1.6 \times 10^{-13}} \\
 &= 0.512 \text{ MeV}
 \end{aligned}$$

- If the linear momentum is increased by 50%, then kinetic energy will be increased by
 - 50 %
 - 100 %
 - 125 %
 - 25 %
- The graph between $\sqrt{E_K}$ and $1/p$ is (E_K = kinetic energy and p = momentum)

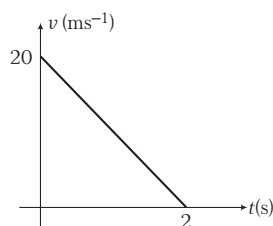


- The kinetic energy acquired by a body of mass m in travelling a certain distance starting from rest under a constant force is
 - directly proportional to m
 - directly proportional to $\sqrt{m}$
 - inversely proportional to $\sqrt{m}$
 - independent of m

8. Under the action of a force, a 2 kg body moves such that its position x as a function of time t is given by $x = \frac{t^3}{3}$, where x is in metre and t in second. The work done by the force in the first two seconds is
 (a) 1600 J (b) 160 J
 (c) 16 J (d) 1.6 J
9. An object of mass 5 kg is acted upon by a force that varies with position of the object as shown in the figure. If the object starts out from rest at a point $x = 0$, what is its speed at $x = 50\text{m}$?



- (a) 12.2 ms⁻¹ (b) 18.2 ms⁻¹
 (c) 16.4 ms⁻¹ (d) 20.4 ms⁻¹
10. A block of mass 20 kg is moving in x -direction with a constant speed of 10 ms⁻¹. It is subjected to a retarding force $F = (-0.1x)\text{N}$ during its travel from $x = 20\text{ m}$ to $x = 30\text{ m}$. Its final kinetic energy will be
 (a) 975 J (b) 450 J
 (c) 275 J (d) 250 J
11. Velocity-time graph of a particle of mass 2 kg moving in a straight line as shown in figure. Work done by all the forces on the particle is



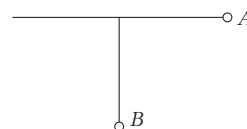
- (a) 400 J (b) -400 J (c) -200 J (d) 200 J
12. A particle of mass 0.01 kg travels with velocity given by $(4\hat{i} + 16\hat{k})\text{ms}^{-1}$. After sometime, its velocity becomes $(8\hat{i} + 20\hat{k})\text{ms}^{-1}$. The work done on particle during this interval of time is
 (a) 0.32 J (b) 6.9 J
 (c) 9.6 J (d) 0.96 J
13. A mass of 1 kg is acted upon by a single force $\mathbf{F} = (4\hat{i} + 4\hat{j})\text{N}$. Under this force, it is displaced from $(0, 0)$ to $(1\text{m}, 1\text{m})$. If initially, the speed of the particle was 2 ms⁻¹, its final speed should be
 (a) 6 ms⁻¹ (b) 4.5 ms⁻¹
 (c) 8 ms⁻¹ (d) 4 ms⁻¹

14. A body of mass 5 kg is raised vertically to a height of 10 m by a force 170 N. The velocity of the body at this height will be
 (a) 9.8 ms⁻¹ (b) 15 ms⁻¹
 (c) 22 ms⁻¹ (d) 37 ms⁻¹
15. A body of mass 0.1 kg moving with a velocity of 10 ms⁻¹ hits a spring (fixed at the other end) of force constant 1000 N m⁻¹ and comes to rest after compressing the spring. The compression in the spring is
 (a) 0.01 m (b) 0.1 m
 (c) 0.2 m (d) 0.5 m
16. A mass of 2 kg falls from a height of 40 cm on a spring of a force constant 1960 Nm⁻¹. The spring is compressed by (Take, $g = 9.8\text{ms}^{-2}$)
 (a) 10 cm (b) 1 cm
 (c) 20 cm (d) 5 cm
17. In which of the following cases, the potential energy is defined?
 (a) Both conservative and non-conservative forces
 (b) Conservative force only
 (c) Non-conservative force only
 (d) Neither conservative non-conservative forces
18. The potential energy of a system increases, if work is done
 (a) by the system against a conservative force
 (b) by the system against a non-conservative force
 (c) upon the system by a conservative force
 (d) upon the system by a non-conservative force
19. The potential energy for a conservative force system is given by

$$U = \frac{7}{2}x^2 - 3x$$

The potential energy at equilibrium is

- (a) + $\frac{9}{14}$ units (b) - $\frac{9}{14}$ units
 (c) + $\frac{13}{2}$ units (d) - $\frac{13}{2}$ units
20. A pendulum of length 2 m is left at A. When it reaches B, it loses 10% of its total energy due to air resistance. The velocity at B is



- (a) 6 ms⁻¹ (b) 1 ms⁻¹
 (c) 2 ms⁻¹ (d) 8 ms⁻¹
21. A body of mass m thrown vertically upwards attains a maximum height h . At what height will its kinetic energy be 75% of its initial value?
 (a) $\frac{h}{6}$ (b) $\frac{h}{5}$
 (c) $\frac{h}{4}$ (d) $\frac{h}{3}$

POWER

Power is defined as the rate at which work is done or energy is transferred. If a force does work W in time t , then its average power is given by

Average power (P_{av}) = Rate of doing work

$$\text{Average power } (P_{av}) = \frac{\text{Work done}}{\text{Time taken}} = \frac{W}{t}$$

Power is a scalar quantity because it is the ratio of two scalar quantities, work (W) and time (t). The dimensional formula of power is $[ML^2T^{-3}]$.

The SI unit of power is watt (W). The power of an agent is one watt, if it does work at the rate of 1 joule per second.

$$1 \text{ watt} = \frac{1 \text{ joule}}{1 \text{ second}} = 1 \text{ Js}^{-1}$$

Another popular units of power are kilowatt and horsepower.

1 kilowatt = 1000 watt or **1 kW** = **10³ W**

1 horsepower = 746 watt or **1 hp** = **746 W**

Horsepower is used to describe the output of automobiles, motorbikes, engines, etc.

Note

(i) Kilowatt hour (kWh) or Board Of Trade (BOT) is the commercial unit of electrical energy.

(ii) Relation between kWh and joule

$$1 \text{ kWh} = 1 \text{ kW} \times 1 \text{ h} = 1000 \text{ W} \times 1 \text{ h} \\ = 1000 \text{ Js}^{-1} \times 3600 \text{ s} \text{ or } 1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

(iii) Efficiency of an engine, $\eta = \frac{\text{Output power}}{\text{Input power}}$

Instantaneous power

The instantaneous power of an agent is defined as the limiting value of the average power of the agent in a small time interval, i.e. when the time interval approaches to zero. When work done by a force $\mathbf{F}$ for a small displacement $d\mathbf{r}$ is $dW = \mathbf{F} \cdot d\mathbf{r}$, then instantaneous power can be given as

$$P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

Now,

$$dW = \mathbf{F} \cdot d\mathbf{r}$$

$\therefore$

$$P = \mathbf{F} \cdot \frac{d\mathbf{r}}{dt}$$

Again $d\mathbf{r}/dt = \mathbf{v}$, instantaneous velocity of the agent. Therefore,

$$P = \mathbf{F} \cdot \mathbf{v} = Fv \cos \theta$$

where, θ is the angle between $\mathbf{F}$ and $\mathbf{v}$.

Power is zero, if force is perpendicular to velocity. e.g.

Power of a centripetal force is zero in a circular motion.

Example 6.54 A train has a constant speed of 40 ms^{-1} on a level road against resistive force of magnitude $3 \times 10^4 \text{ N}$. Find the power of the engine.

Sol. At constant speed, there is no acceleration, so the forces acting on the train are in equilibrium.

Therefore, $F = R = 3 \times 10^4 \text{ N}$, $v = 40 \text{ ms}^{-1}$

Now, power, $P = Fv$

$$P = 3 \times 10^4 \times 40 \\ = 1.2 \times 10^6 \text{ W}$$

Example 6.55 A machine gun fires 240 bullets per minute. If the mass of each bullet is 10 g and the velocity of the bullets is 600 ms^{-1} , then find power (in kW) of the gun.

Sol. Work done by the gun

= Total kinetic energy of the bullets

$$= \left(n \frac{1}{2} mv^2 \right) = 240 \times \frac{1}{2} \times 10 \times 10^{-3} \times (600)^2$$

$$= 120 \times 10 \times 10^{-3} \times 600 \times 600$$

$$\therefore \text{Power of gun} = \frac{\text{Work done}}{\text{Time taken}}$$

$$= \frac{120 \times 10 \times 10^{-3} \times 600 \times 600}{60}$$

$$= 7.2 \text{ kW}$$

Example 6.56 In unloading grain from the hold of a ship, an elevator lifts the grain through a distance of 12 m. Grain is discharged at the top of the elevator at a rate of 2 kg each second and the discharge speed of each grain particle is 3 ms^{-1} . Find the minimum horsepower of the motor that can elevate grain in this way. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. The work done by the motor each second, i.e.

$$\text{Power} = mgh + \frac{1}{2} mv^2, \text{ as } t = 1 \text{ s.}$$

Given, $m = 2 \text{ kg}$, $v = 3 \text{ ms}^{-1}$ and $h = 12 \text{ m}$

$$\therefore \text{Power} = 2 \times (10) \times (12) + \frac{1}{2} (2) (3)^2 = 249 \text{ W} = \frac{249}{746} \\ = 0.33 \text{ hp}$$

The motor must have an output of at least 0.33 hp.

Example 6.57 A pump can take out 7200 kg of water per hour from a well 100 m deep. Calculate the power of the pump, assuming that its efficiency is 50%. (Take, $g = 10 \text{ ms}^{-2}$)

$$\text{Sol. Output power} = \frac{mgh}{t} = \frac{7200 \times 10 \times 100}{3600} = 2000 \text{ W}$$

$$\text{Efficiency, } \eta = \frac{\text{Output power}}{\text{Input power}}$$

$$\text{Input power} = \frac{\text{Output power}}{\eta} \\ = \frac{2000 \times 100}{50} = 4 \text{ kW}$$

Example 6.58 A block of mass m is pulled by a constant power P placed on a rough horizontal plane. The coefficient of friction between the block and surface is μ . Find the maximum velocity of the block.

Sol. Power, $P = F \cdot v = \text{constant}$

$$F = \frac{P}{v} \text{ or } F \propto \frac{1}{v}$$

As v increases, F decreases.

When $F = \mu mg$, net force on block becomes zero, i.e. it has maximum or terminal velocity.

$$\therefore P = (\mu mg) v_{\max} \text{ or } v_{\max} = \frac{P}{\mu mg}$$

Example 6.59 The force required to row a boat at constant velocity is proportional to the speed. If a speed of 4 kmh^{-1} requires 7.5 kW , how much power does a speed of 12 kmh^{-1} require?

Sol. Let the force be $F = \alpha v$, where v is speed and α is a constant of proportionality. The power required is

$$P = Fv = \alpha v^2$$

Let P_1 be the power required for speed v_1 and P_2 be the power required for speed v_2 .

$$P_1 = 7.5 \text{ kW} \text{ and } v_2 = 3v_1$$

$$\Rightarrow P_2 = \left(\frac{v_2}{v_1}\right)^2 P_1$$

$$\Rightarrow P_2 = (3)^2 \times 7.5 \text{ kW} = 67.5 \text{ kW}$$

Example 6.60 An engine pumps 400 kg of water through height of 10 m in 40 s . Find the power of the engine, if its efficiency is 80% . (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Work done by engine against gravity,

$$W = mgh = 400 \times 10 \times 10 = 40 \text{ kJ}$$

Power used by engine (output power)

$$= \frac{W}{\Delta t} = \frac{40 \times 10^3}{40} \text{ W} = 1 \text{ kW}$$

If power of the engine is P (input power), then its efficiency,

$$\eta = \frac{\text{Output power}}{\text{Input power}}$$

$$\therefore P = \frac{1 \text{ kW}}{80\%}$$

$$\begin{aligned} \Rightarrow P &= \frac{1000 \times 100}{80} \text{ W} \\ &= \frac{100}{80} \text{ kW} \\ &= 1.25 \text{ kW} \end{aligned}$$

Example 6.61 An automobile of mass m accelerates, starting from rest. The engine supplies constant power P , show that

the velocity is given as a function of time by $v = \left(\frac{2Pt}{m}\right)^{1/2}$.

Sol. Power, $P = \text{constant}$

Work done upto time t , $W = Pt$

From work-energy theorem, $W = \Delta \text{KE}$ or $Pt = \frac{1}{2} mv^2$

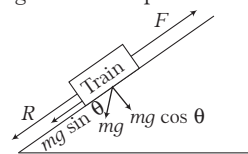
$$\therefore v = \left(\frac{2Pt}{m}\right)^{1/2}$$

Example 6.62 A train of mass $2 \times 10^5 \text{ kg}$ has a constant speed of 20 ms^{-1} up a hill inclined at $\theta = \sin^{-1}\left(\frac{1}{50}\right)$ to the horizontal when the engine is working at $8 \times 10^5 \text{ W}$. Find the resistance to motion of the train. (Take, $g = 9.8 \text{ ms}^{-2}$)

Sol. Power, $P = Fv$

$$\Rightarrow F = \frac{P}{v} = \frac{8 \times 10^5}{20} = 4 \times 10^4 \text{ N}$$

At constant speed, the forces acting on the train are in equilibrium. Resolving the forces parallel to the hill,



$$F = R + mg \sin \theta$$

$$\Rightarrow F = R + (2 \times 10^5) g \times \frac{1}{50}$$

$$4 \times 10^4 = R + 39200 \text{ or } R = 800 \text{ N}$$

Therefore, the resistance is 800 N .

Example 6.63 A small body of mass m moving with velocity v_0 on rough horizontal surface, finally stops due to friction. Find the mean power developed by the friction force during the motion of the body, if the frictional coefficient, $\mu = 0.27$, $m = 1 \text{ kg}$ and $v_0 = 1.5 \text{ ms}^{-1}$.

Sol. The retardation due to friction,

$$a = \frac{\text{Force of friction}}{\text{Mass}} = \frac{\mu mg}{m} = \mu g$$

Now, $v_0 = at$

$$\text{Therefore, } t = \frac{v_0}{a} = \frac{v_0}{\mu g} \quad \dots (i)$$

From work-energy theorem,

work done by force of friction = change in kinetic energy

$$\text{or } W = \frac{1}{2} mv_0^2 \quad \dots (ii)$$

$$\text{Mean power} = \frac{W}{t}$$

From Eqs. (i) and (ii), we get

$$P_{\text{mean}} = \frac{1}{2} \mu mg v_0$$

Substituting the values in above equation, we get

$$P_{\text{mean}} = \frac{1}{2} \times 0.27 \times 1.0 \times 9.8 \times 1.5 \approx 2 \text{ W}$$

CHECK POINT 6.3

1. A particle of mass M starting from rest undergoes uniform acceleration. If the speed acquired in time t is v , the power delivered to the particle is
 - (a) $\frac{Mv^2}{t}$
 - (b) $\frac{1}{2} \frac{Mv^2}{t^2}$
 - (c) $\frac{Mv^2}{t^2}$
 - (d) $\frac{1}{2} \frac{Mv^2}{t}$
 2. An engine develops 10 kW of power. How much time will it take to lift a mass of 200 kg to a height of 40 m? (Take, $g = 10 \text{ ms}^{-2}$)
 - (a) 4 s
 - (b) 5 s
 - (c) 8 s
 - (d) 10 s
 3. An engineer claims to make an engine delivering 10 kW power with fuel consumption of 1 gs^{-1} . The calorific value of fuel is 2 kcal g^{-1} . This claim is
 - (a) valid
 - (b) invalid
 - (c) dependent on engine design
 - (d) dependent on load
 4. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional forces are 10% of energy. How much power is generated by the turbine? (Take, $g = 10 \text{ ms}^{-2}$)
 - (a) 12.3 kW
 - (b) 7 kW
 - (c) 8.1 kW
 - (d) 10.2 kW
 5. If the heart pushes 1 cc of blood in one second under pressure 20000 Nm^{-2} , the power of heart is
 - (a) 0.02 W
 - (b) 400 W
 - (c) $5 \times 10^{-10} \text{ W}$
 - (d) 0.2 W
 6. A body of mass 10 kg moves with a constant speed v of 2 ms^{-1} along a circular path of radius 8 m. The power produced by the body will be
 - (a) 10 Js^{-1}
 - (b) 98 Js^{-1}
 - (c) 49 Js^{-1}
 - (d) zero
 7. A force of $(2\hat{i} + 3\hat{j} + 4\hat{k}) \text{ N}$ acting on a body for 4 s, produces a displacement of $(3\hat{i} + 4\hat{j} + 5\hat{k}) \text{ m}$. The power used is
 - (a) 9.5 W
 - (b) 7.5 W
 - (c) 6.5 W
 - (d) 4.5 W
 8. A body of mass 2 kg is projected at 20 ms^{-1} at an angle 60° above the horizontal. Power due to the gravitational force at its highest point is
 - (a) 200 W
 - (b) $100\sqrt{3} \text{ W}$
 - (c) 50 W
 - (d) zero
-

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1** A body is falling freely under the action of gravity alone in vacuum. Which of the following quantities remain constant during the fall? [NCERT Exemplar]
 (a) Kinetic energy (b) Potential energy
 (c) Total mechanical energy (d) Total linear momentum

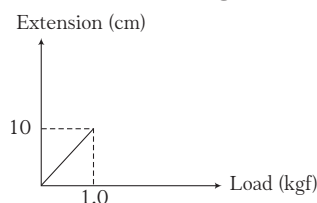
- 2** An electron and a proton are moving under the influence of mutual forces. In calculating the change in the kinetic energy of the system during motion, one ignores the magnetic force of one on another. This is because [NCERT Exemplar]
 (a) the two magnetic forces are equal and opposite, so they produce no net effect
 (b) the magnetic forces do not work on each particle
 (c) the magnetic forces do equal and opposite (but non-zero) work on each particle
 (d) the magnetic forces are necessarily negligible

- 3** An engine exerts a force $\mathbf{F} = (20\hat{i} - 3\hat{j} + 5\hat{k})$ N and moves with velocity $\mathbf{v} = (6\hat{i} + 20\hat{j} - 3\hat{k})$ ms⁻¹. The power of the engine (in watt) is
 (a) 45 (b) 75 (c) 20 (d) 10

- 4** A rod of mass m and length l is lying on a horizontal table. Work done in making it stand on one end will be
 (a) mgl (b) $\frac{mgl}{2}$ (c) $\frac{mgl}{4}$ (d) $2mgl$

- 5** A bicyclist comes to a skidding stop in 10 m. During this process, the force on the bicycle due to the road is 200 N and is directly opposed to the motion. The work done by the cycle on the road is [NCERT Exemplar]
 (a) + 2000 J (b) - 200 J (c) zero (d) - 20000 J

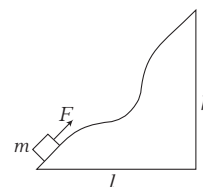
- 6** The pointer reading *versus* load graph for a spring balance is as shown in the figure.



The spring constant is

- (a) 15 kgf/cm (b) 5 kgf/cm
 (c) 0.1 kgf/cm (d) 10 kgf/cm

- 7** A body of mass m was slowly pulled up the hill by a force F which at each point was directed along the tangent of the trajectory. All surfaces are smooth. Find the work performed by this force.



- (a) mgl (b) $-mgl$
 (c) mgh (d) zero

- 8** Two masses of 1 g and 4 g are moving with equal kinetic energies. The ratio of the magnitudes of their momenta is

- (a) 4 : 1 (b) $\sqrt{2} : 1$ (c) 1 : 2 (d) 1 : 16

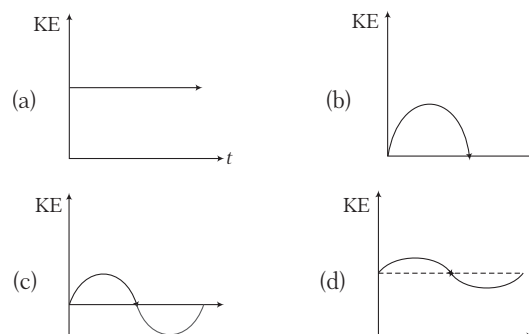
- 9** If v , p and E denote velocity, linear momentum and kinetic energy of the particle respectively, then

- (a) $p = \frac{dE}{dv}$ (b) $p = \frac{dE}{dt}$
 (c) $p = \frac{dv}{dt}$ (d) $p = \frac{dE}{dv} \times \frac{dE}{dt}$

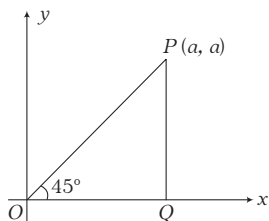
- 10** The energy required to accelerate a car from rest to 10 ms⁻¹ is W . The energy required to accelerate the car from 10 ms⁻¹ to 20 ms⁻¹ is

- (a) W (b) $2W$
 (c) $3W$ (d) $4W$

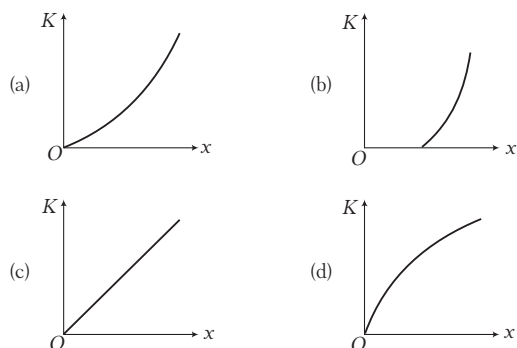
- 11** Which of the following diagrams most closely shows the variation in kinetic energy of the earth as it moves once around the sun in its elliptical orbit? [NCERT Exemplar]



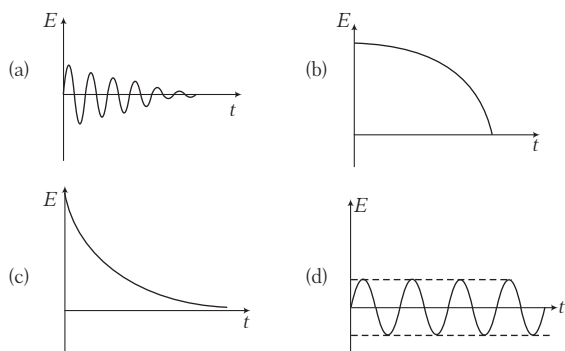
- 12** A particle is moved from $(0, 0)$ to (a, a) under a force $\mathbf{F} = (3\hat{i} + 4\hat{j})$ from two paths. Path 1 is OP and path 2 is OQP . Let W_1 and W_2 be the work done by this force in these two paths. Then,



- (a) $W_1 = W_2$ (b) $W_1 = 2W_2$ (c) $W_2 = 2W_1$ (d) $W_2 = 4W_1$
- 13** A body moves from rest with a constant acceleration. Which one of the following graphs represents the variation of its kinetic energy K with the distance travelled x ?



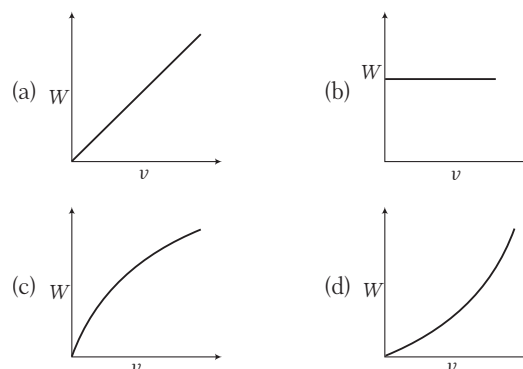
- 14** Which of the following diagrams represents variation of total mechanical energy of a pendulum oscillating in air as function of time? [NCERT Exemplar]



- 15** An object of mass m , initially at rest under the action of a constant force F attains a velocity v in time t . Then, the average power supplied to mass is
- (a) $\frac{mv^2}{2t}$ (b) $\frac{Fv}{2}$
- (c) Both (a) and (b) (d) None of these

- 16** A particle at rest on a frictionless table is acted upon by a horizontal force which is constant in magnitude and direction. A graph is plotted for the work done on the particle W against the speed of the particle v .

If there are no frictional forces acting on the particle, the graph will look like



- 17** A car moving with a speed of 40 kmh^{-1} can be stopped by applying brakes in 2 m. If the car is moving with a speed of 80 kmh^{-1} , the minimum stopping distance under similar brakes will be
- (a) 8 m (b) 2 m (c) 4 m (d) 6 m

- 18** A long spring is stretched by 2 cm. Its potential energy is U . If the spring is stretched by 10 cm, its potential energy would be

- (a) $\frac{U}{25}$ (b) $\frac{U}{5}$ (c) $5U$ (d) $25U$

- 19** A body is falling under gravity. When it loses a gravitational potential energy by U , its speed is v . The mass of the body will be

- (a) $\frac{2U}{v}$ (b) $\frac{U}{2v}$ (c) $\frac{2U}{v^2}$ (d) $\frac{U}{2v^2}$

- 20** How much mass is converted into energy per day in Tarapur nuclear power plant operated at 10^7 kW ?

- (a) 9.6 g (b) 9.63 kg (c) 8.6 g (d) 7 g

- 21** A proton is kept at rest. A positively charged particle is released from rest at a distance d in its field.

Consider two experiments: one in which the charged particle is also a proton and in another a positron. In the same time t , the work done on the two moving charged particles is [NCERT Exemplar]

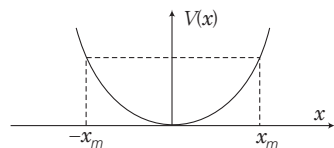
- (a) same as the same force law is involved in the two experiments
- (b) less for the case of a positron, as the positron moves away more rapidly and the force on it weakens
- (c) more for the case of a positron, as the positron moves away a larger distance
- (d) same as the work done by charged particle on the stationary proton

- 22** The potential energy function for a particle

executing linear SHM is given by $V(x) = \frac{1}{2} kx^2$

where, k is the force constant of the oscillator (see figure). For $k = 0.5 \text{ Nm}^{-1}$, the graph of $V(x)$ versus x is shown in the figure.

A particle of total energy E turns back when it reaches $x = \pm x_m$. If V and K indicate the PE and KE respectively of the particle at $x = +x_m$, then which of the following is correct? [NCERT Exemplar]



- (a) $V = 0, K = E$ (b) $V = E, K = 0$
(c) $V < E, K = 0$ (d) $V = 0, K < E$

23 The potential energy between the atoms in a

$$\text{molecule is given by } U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$$

where, a and b are positive constants and x is the distance between the atoms. The atom is in equilibrium when

- (a) $x = 0$ (b) $x = \left(\frac{a}{2b}\right)^{1/6}$
(c) $x = \left(\frac{2a}{b}\right)^{1/6}$ (d) $x = \left(\frac{11a}{5b}\right)^{1/6}$

24 A ball is thrown vertically upwards with a velocity of 10 ms^{-1} . It returns to the ground with a velocity of 9 ms^{-1} . If $g = 9.8 \text{ ms}^{-2}$, then the maximum height attained by the ball is nearly (assume air resistance to be uniform)

- (a) 5.1 m (b) 4.1 m (c) 4.61 m (d) 5 m

25 A body is moving down an inclined plane of slope 37° . The coefficient of friction between the body and plane varies as $\mu = 0.3x$, where x is the distance travelled down the plane by the body. The body will have maximum speed (Take, $g = 10 \text{ ms}^{-2}$, $\sin 37^\circ = \frac{3}{5}$)

- (a) at $x = 1.16 \text{ m}$ (b) at $x = 2 \text{ m}$
(c) at bottom of plane (d) at $x = 2.5 \text{ m}$

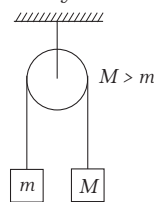
26 A mass-spring system oscillates such that the mass moves on a rough surface having coefficient of friction μ . It is compressed by a distance a from its normal length and on being released, it moves to a distance b from its equilibrium position. The decrease in amplitude for one half-cycle ($-a$ to b) is

- (a) $\frac{\mu mg}{k}$ (b) $\frac{2\mu mg}{k}$ (c) $\frac{\mu g}{k}$ (d) $\frac{k}{\mu mg}$

27 A block of mass 5 kg slides down a rough inclined surface. The angle of inclination is 45° . The coefficient of sliding friction is 0.20 . When the block slides 10 cm , the work done on the block by force of friction is

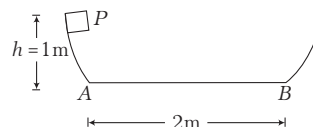
- (a) $-\frac{1}{\sqrt{2}} \text{ J}$ (b) 1 J (c) $-\sqrt{2} \text{ J}$ (d) -1 J

28 The system shown in the figure is released from rest. At the instant when mass M has fallen through a distance h , the velocity of m will be



- (a) $\sqrt{2gh}$ (b) $\frac{\sqrt{2ghM}}{m}$
(c) $\sqrt{\frac{2gh(M-m)}{m+M}}$ (d) $\sqrt{\frac{2gh(M+m)}{m-M}}$

29 The curved portions are smooth and horizontal surface is rough. The block is released from P . At what distance from A , it will stop (if $\mu = 0.2$)?



- (a) 1 m (b) 2 m (c) 3 m (d) 4 m

30 A particle moves on a rough horizontal ground with some initial velocity, say v_0 . If $\left(\frac{3}{4}\right)$ th of its kinetic energy is lost due to friction in time t_0 , then coefficient of friction between the particle and the ground is

- (a) $\frac{v_0}{2gt_0}$ (b) $\frac{v_0}{4gt_0}$ (c) $\frac{3v_0}{4gt_0}$ (d) $\frac{v_0}{gt_0}$

31 If a body of mass 200 g falls from a height 200 m and its total potential energy is converted into kinetic energy at the point of contact of the body with the surface, then decrease in potential energy of the body at the contact is (Take, $g = 10 \text{ ms}^{-2}$)

- (a) 900 J (b) 600 J
(c) 400 J (d) 200 J

32 A stone of mass 2 kg is projected upwards with kinetic energy of 98 J . The height at which the kinetic energy of the body becomes half of its original value will be (Take, $g = 9.8 \text{ ms}^{-2}$)

- (a) 5 m (b) 2.5 m (c) 1.5 m (d) 0.5 m

33 A toy gun uses a spring of very large value of force constant k . When charged before being triggered in the upward direction, the spring is compressed by a small distance x . If mass of shot is m , on being triggered, it will go upto a height of

- (a) $\frac{kx^2}{mg}$ (b) $\frac{x^2}{kmg}$ (c) $\frac{kx^2}{2mg}$ (d) $\frac{(kx)^2}{mg}$

- 34** Water falling from a 50 m high fall is to be used for generating electrical energy. If 1.8×10^5 kg of water falls per hour and half the gravitational potential energy can be converted into electrical energy, how many 100 W bulb can be lit?

(a) 50 (b) 125 (c) 150 (d) 200

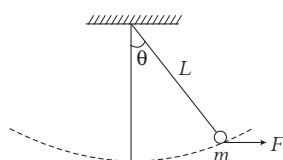
- 35** Given that the displacement of the body (in metre) is a function of time as follows

$$x = 2t^4 + 5$$

The mass of the body is 2 kg. What is the increase in its kinetic energy one second after the start of motion?

(a) 8 J (b) 16 J (c) 32 J (d) 64 J

- 36** An object of mass m is tied to a string of length L and a variable horizontal force is applied on it which starts at zero and gradually increases until the string makes an angle θ with the vertical.



Work done by the force F is

(a) $mgL(1 - \sin\theta)$ (b) mgL
(c) $mgL(1 - \cos\theta)$ (d) $mgL(1 + \cos\theta)$

- 37** A body is attached to the lower end of a vertical spiral spring and it is gradually lowered to its equilibrium position. This stretches the spring by a length d . If the same body attached to the same spring is allowed to fall suddenly, what would be the maximum stretching in this case?

(a) d (b) $2d$ (c) $3d$ (d) $(1/2)d$

- 38** Kinetic energy of a particle moving in a straight line varies with time t as $K = 4t^2$. The force acting on the particle

(a) is constant (b) is increasing
(c) is decreasing (d) first increases and then decreases

- 39** Three particles A , B and C are projected from the top of a tower with the same speed. A is thrown straight upwards, B straight down and C horizontally. They hit the ground with speeds v_A , v_B and v_C , then which of the following is correct?

(a) $v_A = v_B > v_C$ (b) $v_A = v_B = v_C$
(c) $v_A > v_B = v_C$ (d) $v_B > v_C > v_A$

- 40** A particle is moving in a conservative force field from point A to point B . U_A and U_B are the potential energies of the particle at points A and B ; and W_C is the work done by conservative forces in the process of taking the particle from A and B . Which of the following is true?

(a) $W_C = U_B - U_A$ (b) $W_C = U_A - U_B$
(c) $U_A > U_B$ (d) $U_B > U_A$

- 41** Power supplied to a particle of mass 2 kg varies with time as $P = \frac{3t^2}{2}$ W. Here, t is in second. If particle is

rest at $t = 0$, then velocity of particle at time $t = 2$ s will be

(a) 1 ms^{-1} (b) 4 ms^{-1} (c) 2 ms^{-1} (d) $2\sqrt{2} \text{ ms}^{-1}$

- 42** An open knife edge of mass m is dropped from a height h on a wooden floor. If the blade penetrates upto the depth d into the wood, the average resistance offered by the wood to the knife edge is

(a) $mg\left(1 + \frac{h}{d}\right)$ (b) $mg\left(1 + \frac{h}{d}\right)^2$
(c) $mg\left(1 - \frac{h}{d}\right)$ (d) $mg\left(1 + \frac{d}{h}\right)$

- 43** If v be the instantaneous velocity of the body dropped from the top of a tower, when it is located at height h , then which of the following remains constant?

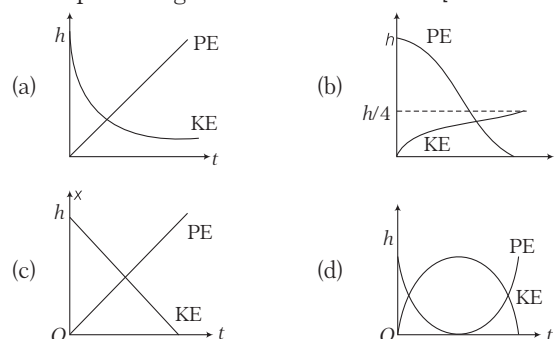
(a) $gh + v^2$ (b) $gh + \frac{v^2}{2}$ (c) $gh - \frac{v^2}{2}$ (d) $gh - v^2$

- 44** A block of mass 1 kg slides down a rough inclined plane of inclination 60° starting from its top. If coefficient of kinetic friction is 0.5 and length of the plane $d = 2$ m, then work done against friction is

(a) 2.45 J (b) 4.9 J (c) 9.8 J (d) 19.6 J

- 45** A raindrop falling from a height h above ground, attains a near terminal velocity when it has fallen through a height $(3/4)h$. Which of the diagrams shown in figure correctly shows the change in kinetic and potential energy of the drop during its fall up to the ground?

[NCERT Exemplar]

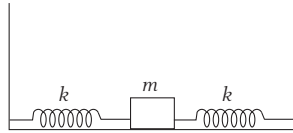


- 46** In a shotput event, an athlete throws the shotput of mass 10 kg with an initial speed of 1 ms^{-1} at 45° from a height 1.5 m above ground. Assuming air resistance to be negligible and acceleration due to gravity to be 10 ms^{-2} , the kinetic energy of the shotput when it just reaches the ground will be

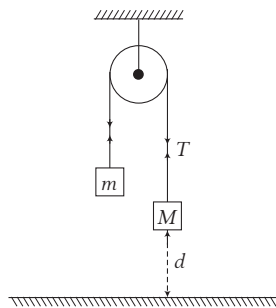
[NCERT Exemplar]

(a) 2.5 J (b) 5 J
(c) 52.5 J (d) 155 J

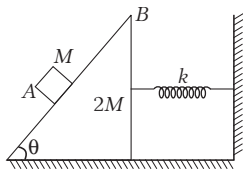
- 47** A block of mass m is attached to two unstretched springs of spring constants k , each as shown. The block is displaced towards right through a distance x and is released. The speed of the block as it passes through the mean position will be



- (a) $x\sqrt{\frac{m}{2k}}$ (b) $x\sqrt{\frac{2k}{m}}$ (c) $x\sqrt{\frac{m}{k}}$ (d) $x\frac{2k}{m}$
- 48** A particle of mass 1 g executes an oscillatory motion on the concave surface of a spherical dish of radius $2m$ placed on a horizontal plane. If the motion of the particle begins from a point on the dish at a height of 1 cm from the horizontal plane, the total distance covered by the particle before it comes to rest, is (Curved surface is smooth and $\mu = 0.01$ for horizontal surface)
- (a) 2 cm (b) 10 cm (c) 1 cm (d) 20 cm
- 49** The net work done by the tension in the figure when the bigger block of mass M touches the ground is

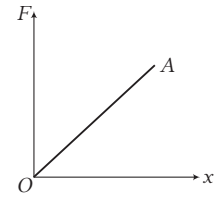


- (a) $+Mgd$ (b) $-(M+m)gd$ (c) $-mgd$ (d) zero
- 50** A block A of mass M rests on a wedge B of mass $2M$ and inclination θ . There is sufficient friction between A and B, so that A does not slip on B. If there is no friction between B and ground, the compression in spring is

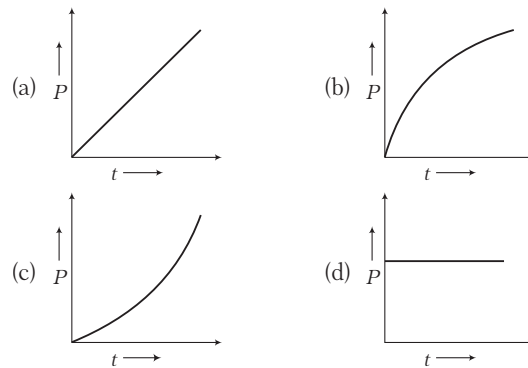


- (a) $\frac{Mg \cos \theta}{k}$ (b) $\frac{Mg \cos \theta \sin \theta}{k}$
 (c) $\frac{Mg \sin \theta}{k}$ (d) zero

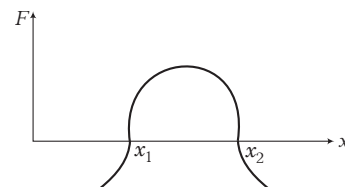
- 51** The force required to stretch a spring varies with the distance as shown in the figure. If the experiment is performed with the above spring of half the length, the line OA will



- (a) shift towards F -axis
 (b) shift towards X -axis
 (c) remain as it is
 (d) become double in length
- 52** A motor drives a body along a straight line with a constant force. The power P developed by the motor must vary with time t as

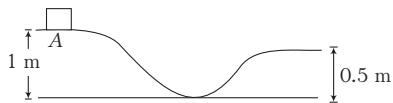


- 53** A force F acting on a body depends on its displacement s as $F \propto s^{-1/3}$. The power delivered by F will depend on displacement as
- (a) $s^{2/3}$ (b) $s^{-5/3}$ (c) $s^{1/2}$ (d) s^0
- 54** The force acting on a body moving along X -axis varies with the position of the particle as shown in the figure. The body is in stable equilibrium at



- (a) $x = x_1$ (b) $x = x_2$
 (c) Both x_1 and x_2 (d) Neither x_1 nor x_2
- 55** A body of mass 0.5 kg travels in a straight line with velocity $v = ax^{3/2}$, where $a = 5 \text{ m}^{-1/2} \text{ s}^{-1}$. The work done by the net force during its displacement from $x = 0$ to $x = 2 \text{ m}$ is [NCERT Exemplar]
- (a) 1.5 J (b) 50 J
 (c) 10 J (d) 100 J
- 56** The figure shows a particle sliding on a frictionless track, which terminates in a straight horizontal section.

If the particle starts slipping from the point A, how far away from the track will the particle hit the ground?

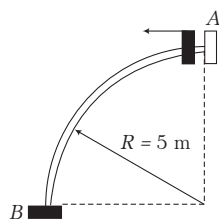


- (a) 1 m (b) 2 m (c) 3 m (d) 4 m

- 57** A uniform chain has a mass M and length L . It is placed on a frictionless table with length l_0 hanging over the edge. The chain begins to slide down. Then, the speed v with which the end slides down from the edge is given by

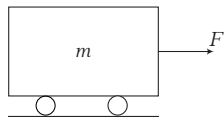
- (a) $v = \sqrt{\frac{g}{L}(L + l_0)}$ (b) $v = \sqrt{\frac{g}{L}(L - l_0)}$
 (c) $v = \sqrt{\frac{g}{L}(L^2 - l_0^2)}$ (d) $v = \sqrt{2g(L - l_0)}$

- 58** A bead of mass $1/2$ kg starts from rest from A to B move in a vertical plane along a smooth fixed quarter ring of radius 5 m, under the action of a constant horizontal force $F = 5$ N as shown in figure. The speed of bead as it reaches the point B is (Take, $g = 10 \text{ ms}^{-2}$)



- (a) 14.14 ms^{-1} (b) 7.07 ms^{-1} (c) 5 ms^{-1} (d) 25 ms^{-1}

- 59** A car of mass m is accelerating on a level smooth road under the action of a single force F . The power delivered to the car is constant and equal to P . If the velocity of the car at an instant is v , then after travelling how much distance will it become double?

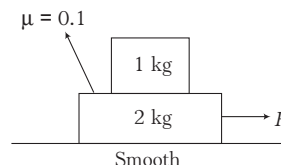


- (a) $\frac{7mv^3}{3P}$ (b) $\frac{4mv^3}{3P}$ (c) $\frac{mv^3}{P}$ (d) $\frac{18mv^3}{7P}$

- 60** A particle is released from a height H . At certain height, its kinetic energy is two times its potential energy. Height and speed of particle at that instant are

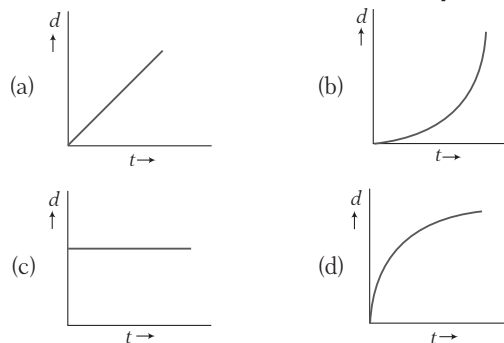
- (a) $\frac{H}{3}, \sqrt{\frac{2gH}{3}}$ (b) $\frac{H}{3}, 2\sqrt{\frac{gH}{3}}$
 (c) $\frac{2H}{3}, \sqrt{\frac{2gH}{3}}$ (d) $\frac{H}{3}, \sqrt{2gH}$

- 61** A force of $F = 0.5$ N is applied on lower block as shown in figure. The work done by lower block on upper block for a displacement of 3 m of the upper block with respect to ground is (Take, $g = 10 \text{ ms}^{-2}$)



- (a) -0.5 J (b) 0.5 J
 (c) 2 J (d) -2 J

- 62** A body is moving unidirectionally under the influence of a source of constant power supplying energy. Which of the following diagrams shown in figure correctly represents the displacement-time curve for its motion? [NCERT Exemplar]



- 63** A small block of mass m is kept on a rough inclined surface of inclination θ fixed in an elevator. The elevator goes up with a uniform velocity v and the block does not slide on the wedge.

The work done by the force of friction on the block in a time t will be

- (a) zero (b) $mgvt \cos^2 \theta$
 (c) $mgvt \sin^2 \theta$ (d) $\frac{1}{2} mgvt \sin 2\theta$

- 64** A pendulum of mass 1 kg and length $l = 1$ m is released from rest at angle $\theta = 60^\circ$. The power delivered by all the forces acting on the bob at angle $\theta = 30^\circ$ will be (Take, $g = 10 \text{ ms}^{-2}$)

- (a) 13.5 W (b) 20.4 W
 (c) 24.6 W (d) zero

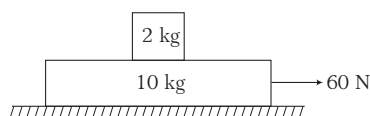
- 65** A uniform flexible chain of mass m and length l hangs in equilibrium over a smooth horizontal pin of negligible diameter. One end of the chain is given a small vertical displacement, so that the chain slips over the pin. The speed of chain when it leaves pin is

- (a) $\sqrt{\frac{gl}{2}}$ (b) $\sqrt{gl}$ (c) $\sqrt{2gl}$ (d) $\sqrt{3gl}$

- 66** The potential energy of a particle of mass 1 kg is $U = 10 + (x - 2)^2$. Here, U is in joule and x in metre on the positive X -axis. Particle travels upto $x = +6$ m. Choose the correct option.

- (a) On negative X -axis, particle travels upto $x = -2$ m
 (b) The maximum kinetic energy of the particle is 16 J
 (c) Both (a) and (b) are correct
 (d) Both (a) and (b) are incorrect

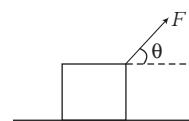
- 67** A plank of mass 10 kg and a block of mass 2 kg are placed on a horizontal plane as shown in the figure. There is no friction between plane and plank. The coefficient of friction between block and plank is 0.5. A force of 60 N is applied on plank horizontally. In first 2 s, the work done by friction on the block is



- (a) -100 J (b) 100 J
 (c) zero (d) 200 J

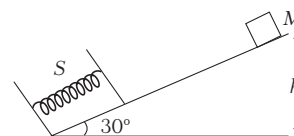
- 68** A block of mass m is pulled along a horizontal surface by applying a force at an angle θ with the horizontal. If the block travels with a uniform velocity and has a displacement d and the coefficient

of friction is μ , then the work done by the applied force is



- (a) $\frac{\mu mgd}{\cos \theta + \mu \sin \theta}$ (b) $\frac{\mu mgd \cos \theta}{\cos \theta + \mu \sin \theta}$
 (c) $\frac{\mu mgd \sin \theta}{\cos \theta + \mu \sin \theta}$ (d) $\frac{\mu mgd \cos \theta}{\cos \theta - \mu \sin \theta}$

- 69** An ideal massless spring S can be compressed 1 m by a force of 100 N in equilibrium. The same spring is placed at the bottom of a frictionless inclined plane inclined at 30° to the horizontal. A 10 kg block M is released from rest at the top of the incline and is brought to rest momentarily after compressing the spring by 2 m. If $g = 10 \text{ ms}^{-2}$, what is the speed of mass just before it touches the spring?



- (a) $\sqrt{20} \text{ ms}^{-1}$ (b) $\sqrt{30} \text{ ms}^{-1}$ (c) $\sqrt{10} \text{ ms}^{-1}$ (d) $\sqrt{40} \text{ ms}^{-1}$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-6) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) If Assertion is correct but Reason is incorrect.
 (d) If Assertion is incorrect but Reason is correct.

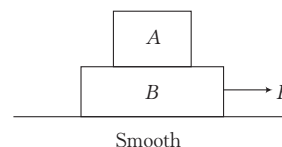
- 1 Assertion** At stable equilibrium position of a body, kinetic energy cannot be zero. Because it is maximum.

Reason During oscillations of a body, potential energy is minimum at stable equilibrium position.

- 2 Assertion** If work done by conservative force is negative, then potential energy associated with that force should increase.

Reason This is according to the relation, $\Delta U = -W$. Here, ΔU is change in potential energy and W is work done by conservative force.

- 3 Assertion** If the surface between the blocks A and B is rough, then work done by friction on block B is always negative.



Reason Total work done by friction in both the blocks is always zero.

- 4 Assertion** Force applied on a block moving in one dimension is producing a constant power, then the motion should be uniformly accelerated.

Reason This constant power multiplied with time is equal to the change in kinetic energy.

- 5 Assertion** Total work done by spring may be positive, negative or zero.

Reason Direction of spring force is always towards mean position.

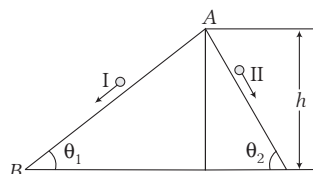
- 6 Assertion** At any instant, the magnitude of rate of change of potential energy of the projectile of mass

1 kg is numerically equal to magnitude of $a \cdot v$ (where, a is acceleration due to gravity and v is velocity at that instant)

Reason The graph representing power delivered by the gravitational force acting on the projectile with time will be straight line with negative slope.

Statement based questions

- 1 Mark out the correct statement(s).
 - (a) Total work done by internal forces on a system is always zero.
 - (b) Total work done by internal forces on a system may sometimes be zero.
 - (c) Total work done by friction can never be zero.
 - (d) Total work done by friction is always zero.
- 2 Two inclined frictionless tracks, one gradual and the other steep meet at A from where two stones are allowed to slide down from rest, one on each track as shown in figure. [NCERT Exemplar]
 Which of the following statement(s) is/are correct?



- (a) Both the stones reach the bottom at the same time but not with the same speed.
 - (b) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.
 - (c) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.
 - (d) Both the stones reach the bottom at different times and with different speeds.
- 3 In position A, kinetic energy of a particle is 60 J and potential energy is -20 J. In position B, kinetic energy is 100 J and potential energy is 40 J. Choose the correct statement, if the particle moves from A to B.
 - (a) Work done by conservative forces is 50 J.
 - (b) Work done by external forces is 40 J.
 - (c) Net work done by all the forces is 40 J.
 - (d) Net work done by all the forces is 100 J.
- 4 Which of the following statement(s) is/are correct?
 - I. Spring force is a conservative force.
 - II. Potential energy is defined only for conservative forces.
 - (a) Only I
 - (b) Only II
 - (c) Both I and II
 - (d) Neither I nor II

- 5 Which of the following statement(s) is/are correct?

- I. If momentum of a body increases by 50%, its kinetic energy will increase by 125%.
 - II. Kinetic energy is proportional to square of velocity.
- (a) Only I
 - (b) Only II
 - (c) Both I and II
 - (d) Neither I nor II

Match the columns

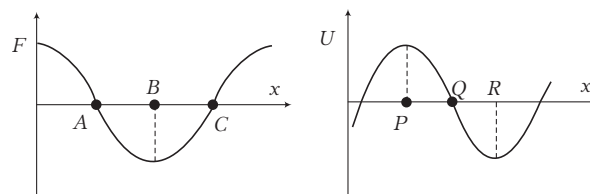
- 1 A body is moved along a straight line by a machine delivering a power proportional to time ($P \propto t$). Then, match the following columns and mark the correct option from the codes given below.

Column I		Column II	
(A)	Velocity is proportional to	(p)	t
(B)	Displacement is proportional to	(q)	t^2
(C)	Work done is proportional to	(r)	t^3

Codes

- | | | | | | |
|-------|---|---|-------|---|---|
| A | B | C | A | B | C |
| (a) p | q | r | (b) p | p | r |
| (c) p | q | q | (d) r | p | q |

- 2 F - x and corresponding U - x graphs are as shown in figures. Three points A, B and C in F - x graph may be corresponding to P, Q and R in the U - x graph. Match the following columns and mark the correct option from the codes given below.



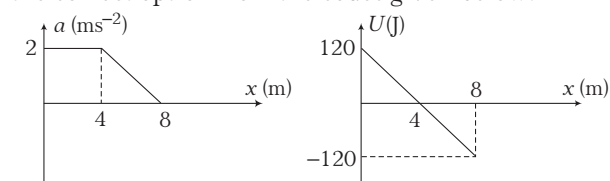
Column I		Column II	
(A)	A	(p)	P
(B)	B	(q)	Q
(C)	C	(r)	R
		(s)	None

Codes

- | | | |
|-------|---|---|
| A | B | C |
| (a) r | s | p |
| (b) p | s | r |
| (c) p | r | q |
| (d) q | r | s |

- 3 Acceleration *versus* x and potential energy *versus* x graph of a particle moving along X -axis are as shown below. Mass of the particle is 1 kg and velocity at $x = 0$ is 4 ms^{-1} .

Match the following columns for $x = 8$ m and mark the correct option from the codes given below.



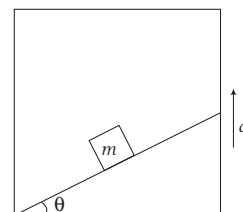
Column I	Column II
(A) Final kinetic energy	(p) 120 J
(B) Work done by conservative forces	(q) 240 J
(C) Total work done	(r) -120 J
(D) Work done by external forces	(s) 128 J

Codes

A	B	C	D	A	B	C	D
(a) p	s	r	q	(b) s	q	p	r
(c) s	p	r	q	(d) p	q	r	s

4. A block of mass m is stationary with respect to a rough wedge is shown in figure. Starting from rest in time t ($m = 1$ kg, $\theta = 30^\circ$, $a = 2$ ms $^{-2}$, $t = 4$ s).

Match the following columns for work done on the block and mark the correct option from the codes given below.



Column I	Column II
(A) By gravity	(p) 144 J
(B) By normal reaction	(q) 32 J
(C) By friction	(r) -160 J
(D) By all the forces	(s) 48 J

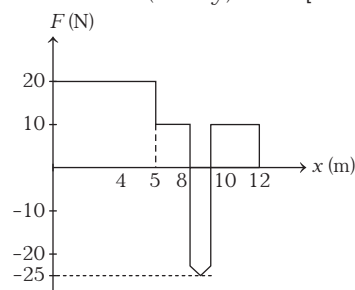
Codes

B	C	D
(a) r	q	s
(b) q	r	s
(c) p	q	r
(d) r	p	s

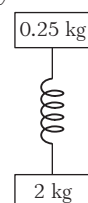
(C) Medical entrances' gallery

Collection of questions asked in NEET & Various Medical Entrance Exams

- 1 A force $F = 20 + 10y$ acts on a particle in y -direction, where F is in newton and y is in metre. Work done by this force to move the particle from $y = 0$ to $y = 1$ m is [NEET 2019]
 (a) 5 J (b) 25 J (c) 20 J (d) 30 J
- 2 When a block of mass M is suspended by a long wire of length L , the length of the wire becomes $(L + l)$. The elastic potential energy stored in the extended wire is [NEET 2019]
 (a) MgL (b) $\frac{1}{2}Mgl$ (c) $\frac{1}{2}MgL$ (d) Mgl
- 3 An object of mass 500 g, initially at rest acted upon by a variable force whose x component varies with x in the manner shown in figure. The velocities of the object at points $x = 8$ m and $x = 12$ m, would be the respective values of (nearly) [NEET (Odisha) 2019]



- (a) 18 ms $^{-1}$ and 24.4 ms $^{-1}$ (b) 23 ms $^{-1}$ and 24.4 ms $^{-1}$
 (c) 23 ms $^{-1}$ and 20.6 ms $^{-1}$ (d) 18 ms $^{-1}$ and 20.6 ms $^{-1}$
- 4 Initially spring in its natural length. Now, a block of mass 0.25 kg is released, then find out the value of maximum force by system on the floor. [AIIMS 2019]



- (a) 15 N (b) 20 N (c) 25 N (d) 30 N
- 5 The rate of decrease of kinetic energy is 9.6 Js $^{-1}$. Find the magnitude of force acting on particle when its speed is 3 ms $^{-1}$. [JIPMER 2019]
 (a) 3.2 N (b) 4.8 N (c) 2.4 N (d) 5.6 N
- 6 If a machine perform 4000 J output work and 1000 J is inside loss due to friction, then find the efficiency. [JIPMER 2018]
 (a) 80% (b) 30% (c) 25% (d) 60%
- 7 Kinetic energy of a particle is increased by 4 times. What will be the relation between initial and final momentum? [JIPMER 2018]
 (a) $p_2 = 2p_1$ (b) $p_2 = p_1/2$ (c) $p_2 = p_1$ (d) $p_2 = 4p_1$

- 8 1000 N force is required to lift a hook and 10000 N force is required to lift a load slowly. Find power required to lift hook with load with speed $v = 0.5 \text{ ms}^{-1}$. [JIPMER 2018]

(a) 5 kW (b) 1.5 kW (c) 5.5 kW (d) 4.5 kW

- 9 A spring of force constant k is cut into lengths of ratio 1 : 2 : 3. They are connected in series and the new force constant is k' . If they are connected in parallel and force constant is k'' , then $k' : k''$ is [NEET 2017]

(a) 1 : 6 (b) 1 : 9 (c) 1 : 11 (d) 1 : 14

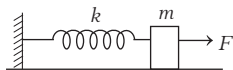
- 10 A force $\mathbf{F} = -k(y\hat{i} + x\hat{j})$, where k is a positive constant, acts on a particle moving in the XY -plane. Starting from the origin, the particle is taken along the positive X -axis to the point $(a, 0)$ and then parallel to the Y -axis to the point (a, a) . The total work done by the force on the particle is [AIIMS 2017]

(a) $-2ka^2$ (b) $2ka^2$ (c) $-ka^2$ (d) ka^2

- 11 **Assertion** A spring of force constant k is cut into two pieces having lengths in the ratio 1 : 2. The force constant of series combination of the two parts is $3k/2$.

Reason The spring connected in series are represented by $k = k_1 + k_2$. [AIIMS 2017]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Both Assertion and Reason are incorrect.
- 12 The figure shows a mass m on a frictionless surface. It is connected to rigid wall by the mean of a massless spring of its constant k . Initially, the spring is at its natural position. If a force of constant magnitude starts acting on the block towards right, then the speed of the block when the deformation in spring is x , will be [AIIMS 2017]



- (a) $\sqrt{\frac{2F \cdot x - kx^2}{m}}$ (b) $\sqrt{\frac{F \cdot x - kx^2}{m}}$
(c) $\sqrt{\frac{x(F - k)}{m}}$ (d) $\sqrt{\frac{F \cdot x - kx^2}{2m}}$

- 13 A person of weight 70 kg wants to lose 7 kg by going up and down 12m high stairs. Assume he burns twice as much fat while going up than going down. If 1 kg of fat is burnt on expending 9000 k-cal. How many times must he go up and down to reduce his 7 kg weight? (Take, $g = 10 \text{ ms}^{-2}$) [AIIMS 2017]

- (a) 18×10^3 times (b) 24×10^3 times
(c) 30×10^3 times (d) 21×10^3 times

- 14 A skier starts from rest at point A and slides down the hill without turning or breaking. The friction coefficient is μ . When he stops at point B, his horizontal displacement is s . What is the height difference between points A and B?

(The velocity of the skier is small, so that the additional pressure on the snow due to the curvature can be neglected. Neglect also the friction of air and the dependence of μ on the velocity of the skier.) [JIPMER 2017]

- (a) $h = \mu s$ (b) $h = \frac{\mu}{s}$
(c) $h = 2\mu s$ (d) $h = \mu s^2$

- 15 A body of mass 1 kg begins to move under the action of a time dependent force $\mathbf{F} = (2t\hat{i} + 3t^2\hat{j}) \text{ N}$, where $\hat{i}$ and $\hat{j}$ are unit vectors along X and Y -axis. What power will be developed by the force at the time (t) ?

- (a) $(2t^2 + 4t^4) \text{ W}$ (b) $(2t^3 + 3t^4) \text{ W}$ [NEET 2016]
(c) $(2t^3 + 3t^5) \text{ W}$ (d) $(2t + 3t^3) \text{ W}$

- 16 A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration, if the kinetic energy of the particle becomes equal to $8 \times 10^{-4} \text{ J}$ by the end of the second revolution after the beginning of the motion? [NEET 2016]

- (a) 0.15 ms^{-2} (b) 0.18 ms^{-2}
(c) 0.2 ms^{-2} (d) 0.1 ms^{-2}

- 17 A block of mass 10 kg moving in x -direction with a constant speed of 10 ms^{-1} , is subjected to a retarding force $F = -0.1x \text{ Jm}^{-1}$ during its travel from $x = 20 \text{ m}$ to 30 m . Its final KE will be [CBSE AIPMT 2015]

- (a) 475 J (b) 450 J
(c) 275 J (d) 250 J

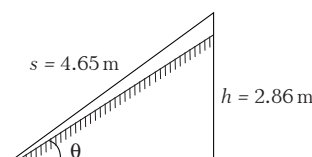
- 18 A particle of mass m is driven by a machine that delivers a constant power k watts. If the particle starts from rest, the force on the particle at time t is [CBSE AIPMT 2015]

- (a) $\sqrt{\frac{mk}{2}} t^{-1/2}$ (b) $\sqrt{mk} t^{-1/2}$
(c) $\sqrt{2mk} t^{-1/2}$ (d) $\frac{1}{2} \sqrt{mkt}^{-1/2}$

- 19 A force $F = (10 + 0.5x)$ acts on a particle in the x -direction. What would be the work done by this force during a displacement from $x = 0$ to $x = 2 \text{ m}$ (F is in newton and x in metre) [AIIMS 2015, UK PMT 2015]

- (a) 31.5 J (b) 63 J
(c) 21 J (d) 42 J

- 20** A block of mass $m = 11.7 \text{ kg}$ is to be pushed a distance of $s = 4.65 \text{ m}$ along an incline and raised to a distance of $h = 2.86 \text{ m}$. Assuming frictionless surface, calculate the work done in applying a force parallel to the incline to push the block up at a constant speed. (Take, $g = 9.8 \text{ ms}^{-2}$) [UK PMT 2015]



- (a) 328 J (b) 656 J
(c) 164 J (d) 530 J
- 21** An elevator weighing 500 kg is to be lifted up at a constant velocity of 0.20 ms^{-1} . What would be the minimum horse power of the motor to be used? [CGPMT 2015, UK PMT 2015]
- (a) 10.30 hp (b) 5.15 hp
(c) 2.62 hp (d) 1.31 hp
- 22** Two bodies of different masses are moving with same kinetic energy. Then, the ratio of their momenta is equal to the ratio of their [Kerala CEE 2015]
- (a) masses (b) square of masses
(c) square root of masses (d) cube root of masses
- 23** Two bodies of masses 1 kg and 2 kg moving with same velocities are stopped by the same force. Then, the ratio of their stopping distances is [Kerala CEE 2015]
- (a) 1 : 2 (b) 2 : 1
(c) $\sqrt{2} : 1$ (d) $1 : \sqrt{2}$
- 24** A bob of mass m accelerates uniformly from rest to v_1 in time t_1 . As a function of t , the instantaneous power delivered to the body is [Manipal 2015]
- (a) $\frac{mv_1 t}{t_1}$ (b) $\frac{mv_1^2 t}{t_1}$
(c) $\frac{mv_1^2 t^2}{t_1}$ (d) $\frac{mv_1^2 t}{t_1^2}$
- 25** A uniform chain of length l is placed on a smooth horizontal table, such that half of its length hangs over one edge. It is released from rest, the velocity with which it leaves the table is [EAMCET 2015]
- (a) $\sqrt{\frac{3gl}{4}}$ (b) $\sqrt{\frac{3gl}{2}}$
(c) $\sqrt{\frac{2gl}{3}}$ (d) $\sqrt{\frac{gl}{3}}$
- 26** The kinetic energy of a body of mass 4 kg and momentum 6 N-s will be [KCET 2015]
- (a) 4.5 J (b) 2.5 J
(c) 5.5 J (d) 3.5 J

- 27** A body of mass $m = 3.90 \text{ kg}$ slides on a horizontal frictionless table with a speed of $v = 1.20 \text{ ms}^{-1}$. It is brought to rest in compressing a spring in its path. How much does spring is compressed, if its force constant k is 135 Nm^{-1} ? [UK PMT 2015]

(a) 0.204 m (b) 0.408 m
(c) 0.804 m (d) 4.04 m

- 28** A string of length L and force constant k is stretched to obtain extension l . It is further stretched to obtain extension l_1 . The work done in second stretching is [MHT CET 2014]

(a) $\frac{1}{2} k l_1 (2l + l_1)$ (b) $\frac{1}{2} k l_1^2$
(c) $\frac{1}{2} k (l^2 + l_1^2)$ (d) $\frac{1}{2} k (l_1^2 - l^2)$

- 29** A body is initially at rest. It undergoes one dimensional motion with constant acceleration. The power delivered to it at time t is proportional to [Uttarakhand PMT 2014]

(a) $t^{1/2}$ (b) t
(c) $t^{3/2}$ (d) t^2

- 30** If two persons A and B take 2 s and 4 s respectively to lift an object to the same height h , then the ratio of their powers is [Kerala CEE 2014]

(a) 1 : 2 (b) 1 : 1
(c) 2 : 1 (d) 1 : 3
(e) 3 : 1

- 31** If a machine gun fires n bullets per second each with kinetic energy K , then the power of the machine gun is [Kerala CEE 2014]

(a) nK^2 (b) $\frac{K}{n}$
(c) $n^2 K$ (d) nK
(e) $\frac{n}{K}$

- 32** A uniform force of $(3\hat{i} + \hat{j}) \text{ N}$ acts on a particle of mass 2 kg . Hence, the particle is displaced from position $(2\hat{i} + \hat{k}) \text{ m}$ to position $(4\hat{i} + 3\hat{j} - \hat{k}) \text{ m}$. The work done by the force on the particle is [NEET 2013]

(a) 9 J (b) 6 J
(c) 13 J (d) 15 J

- 33** The power (P) of an engine lifting a mass of 100 kg upto a height of 10 m in 1 min is [J&K CET 2013]

(a) 163.3 W (b) 9800 W
(c) 10000 W (d) 5000 W

- 34** A body of mass 300 kg is moved through 10 m along a smooth inclined plane of angle 30° . The work done in moving (in joules) is (Take, $g = 9.8 \text{ ms}^{-2}$) [EAMCET 2013]

(a) 4900 (b) 9800
(c) 14700 (d) 2450

- 35** Force constants of two wires A and B of the same material are k and $2k$, respectively. If the two wires are stretched equally, then the ratio of work done in stretching $\left(\frac{W_A}{W_B}\right)$ is [EAMCET 2013]

- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{3}{2}$ (d) $\frac{1}{4}$

- 36** A truck accelerates from speed v to $2v$. Work done during this is [KCET 2013]
(a) three times as the work done in accelerating it from rest to v
(b) same as the work done in accelerating it from rest to v
(c) four times as the work done in accelerating it from rest to v
(d) less than the work done in accelerating it from rest to v

- 37** A block of 200 g mass is dropped from a height of 2 m on to a spring and compresses the spring to a distance of 50 cm. The force constant of the spring is [Kerala CET 2013]
(a) 20 Nm^{-1} (b) 40 Nm^{-1} (c) 30 Nm^{-1} (d) 60 Nm^{-1}
(e) 10 Nm^{-1}

- 38** The potential energy of a particle in a force field is $U = \frac{A}{r^2} - \frac{B}{r}$, where A and B are positive constants

and r is the distance of particle from the centre of the field. For stable equilibrium, the distance of the particle is [CBSE AIPMT 2012]

- (a) $B/2A$ (b) $2A/B$
(c) A/B (d) B/A

- 39** The particle of mass 50 kg is at rest. The work done to accelerate it by 20 ms^{-1} in 10 s is [AIIMS 2012]
(a) 10^3 J (b) 10^4 J
(c) $2 \times 10^3 \text{ J}$ (d) $4 \times 10^4 \text{ J}$

- 40** The slope of kinetic energy and displacement curve for a particle in motion will be [BCECE Mains 2012]
(a) equal to the acceleration of the particle
(b) directly proportional to the acceleration of the particle
(c) inversely proportional to the acceleration of the particle
(d) None of the above

- 41** Two masses m and $2m$ are attached to two ends of an ideal spring as shown in figure. When the spring is in the compressed state, the energy of the spring is 60 J, if the spring is released, then at its natural length, [BHU 2012]



- (a) energy of smaller body will be 20 J
(b) energy of smaller body will be 40 J
(c) energy of smaller body will be 10 J
(d) energy of both the bodies will be same

ANSWERS

- **CHECK POINT 6.1**

1. (c)	2. (d)	3. (a)	4. (b)	5. (d)	6. (b)	7. (d)	8. (a)	9. (b)	10. (b)
11. (d)	12. (b)	13. (a)	14. (c)	15. (d)	16. (b)	17. (d)	18. (a)	19. (a)	20. (b)

- **CHECK POINT 6.2**

1. (b)	2. (a)	3. (b)	4. (d)	5. (c)	6. (c)	7. (d)	8. (c)	9. (a)	10. (a)
11. (b)	12. (d)	13. (b)	14. (c)	15. (b)	16. (a)	17. (b)	18. (a)	19. (b)	20. (c)
21. (c)									

- **CHECK POINT 6.3**

1. (d) 2. (c) 3. (b) 4. (c) 5. (a) 6. (d) 7. (a) 8. (d)

(A) Taking it together

1. (c)	2. (b)	3. (a)	4. (b)	5. (c)	6. (c)	7. (c)	8. (c)	9. (a)	10. (c)
11. (d)	12. (a)	13. (c)	14. (c)	15. (c)	16. (d)	17. (a)	18. (d)	19. (c)	20. (a)
21. (c)	22. (b)	23. (c)	24. (c)	25. (d)	26. (b)	27. (a)	28. (c)	29. (a)	30. (a)
31. (c)	32. (b)	33. (c)	34. (b)	35. (d)	36. (c)	37. (b)	38. (a)	39. (b)	40. (b)
41. (c)	42. (a)	43. (b)	44. (b)	45. (b)	46. (d)	47. (b)	48. (b)	49. (d)	50. (d)
51. (a)	52. (a)	53. (d)	54. (b)	55. (b)	56. (a)	57. (c)	58. (a)	59. (a)	60. (b)
61. (b)	62. (b)	63. (c)	64. (a)	65. (a)	66. (c)	67. (b)	68. (b)	69. (a)	

(B) Medical entrance special format questions

- **Assertion and reason**

1. (d) 2. (a) 3. (c) 4. (d) 5. (b) 6. (c)

- **Statement based questions**

1. (b) 2. (c) 3. (c) 4. (c) 5. (c)

- **Match the columns**

1. (c) 2. (a) 3. (b) 4. (d)

(C) Medical entrances' gallery

1. (b)	2. (b)	3. (c)	4. (c)	5. (a)	6. (a)	7. (a)	8. (c)	9. (c)	10. (c)
11. (d)	12. (a)	13. (d)	14. (a)	15. (c)	16. (d)	17. (a)	18. (a)	19. (c)	20. (a)
21. (d)	22. (c)	23. (a)	24. (d)	25. (a)	26. (a)	27. (a)	28. (a)	29. (b)	30. (c)
31. (d)	32. (a)	33. (a)	34. (c)	35. (b)	36. (a)	37. (b)	38. (b)	39. (b)	40. (b)
41. (b)									

Hints & Explanations

• CHECK POINT 6.1

1 (c) Work done by kinetic friction may be positive or negative.

2 (d) As, work done, $W = \mathbf{F} \cdot \mathbf{s} = Fs \cos \theta$

When a man pushes a wall but fails to move it, then displacement of wall, $s = 0$

$\therefore$ Work done, $W = F \times 0 \times \cos \theta = 0$

Therefore, man does no work at all.

3 (a) Given, $F = 20 \text{ kg-wt} = 20 \times 9.8 \text{ N}$, $s = 20 \text{ m}$ and $\theta = 60^\circ$

$\therefore$ Work done $= F s \cos \theta = 20 \times 9.8 \times 20 \times \cos 60^\circ = 1960 \text{ J}$

4 (b) From first equation of motion,

$$v = u + at \Rightarrow 20 = 0 + a \times 10$$

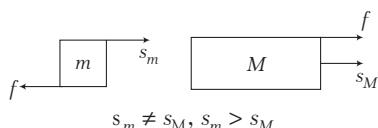
$$\Rightarrow a = 2 \text{ m/s}^2$$

$$\text{Now, distance, } s = ut + \frac{1}{2}at^2 \Rightarrow s = 0 + \frac{1}{2} \times 2 \times 10 \times 10$$

$$\text{or } s = 100 \text{ m}$$

$$\therefore \text{ Work done, } W = F \cdot s = mas = 50 \times 2 \times 100 = 10^4 \text{ J}$$

5 (d) The free body diagram is as shown below.



Therefore, net work done by the friction cannot be zero.

6 (b) Frictional force acting on the box,

$$F = \mu mg = 0.25 \times 20 \times 9.8 = 49 \text{ N}$$

This force must be same as applied force F , so that the box moves with constant speed.

$$\text{Work done by the applied force, } W = Fs = 49 \times 2 = 98 \text{ J}$$

7 (d) Given, force, $\mathbf{F} = (3\hat{i} + 4\hat{j}) \text{ N}$

and displacement, $\mathbf{s} = (3\hat{i} + 4\hat{j}) \text{ m}$

$$\therefore \text{ Work done, } W = \mathbf{F} \cdot \mathbf{s} = (3\hat{i} + 4\hat{j}) \cdot (3\hat{i} + 4\hat{j})$$

$$W = 9 + 16 = 25 \text{ J}$$

8 (a) Particle moves from $P(1\text{m}, 2\text{m}, 3\text{m})$ to $Q(2\text{m}, 1\text{m}, 4\text{m})$

$$\therefore \text{ Distance, } \mathbf{d} = PQ = (2\hat{i} + \hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= (\hat{i} - \hat{j} + \hat{k}) \text{ m}$$

$$\text{Force, } \mathbf{F} = (2\hat{i} + \hat{j} + \hat{k}) \text{ N}$$

Work done by the force,

$$W = \mathbf{F} \cdot \mathbf{d} = (2\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k}) = 2 - 1 + 1 = 2 \text{ J}$$

9 (b) $W = \mathbf{F} \cdot \mathbf{r} = \mathbf{F} \cdot (\mathbf{r}_2 - \mathbf{r}_1)$

$$= (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot [(\hat{i} - \hat{j} + 2\hat{k}) - (\hat{i} + \hat{j} + \hat{k})]$$

$$= (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (-2\hat{j} + \hat{k}) = -4 + 3 = -1 \text{ J}$$

10 (b) Work done, $W = (\text{y-component of force})$

$\times (\text{displacement along Y-axis})$

$$= 15 \times 10 = 150 \text{ J}$$

$$\begin{aligned} 11 (d) W &= \int_{x_1}^{x_2} F dx = \int_0^5 F dx = \int_0^5 (10 - 2x + 3x^2) dx \\ &= [10x - x^2 + x^3]_0^5 = 150 \text{ J} \end{aligned}$$

$$12 (b) W = \int_{-a}^{+a} F dy = \left[\frac{Ay^3}{3} + \frac{By^2}{2} + Cy \right]_{-a}^{+a} = \frac{2Aa^3}{3} + 2Ca$$

13 (a) Given, force, $\mathbf{F} = (2\hat{i} + 3\hat{j}) \text{ N}$

Displacement, $d\mathbf{s} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

$$\text{Work done, } W = \int \mathbf{F} \cdot d\mathbf{s} = \int (2dx + 3dy)$$

$$\text{Also, } 3y + kx = 5 \Rightarrow \frac{3dy}{dx} + k = 0$$

$$\Rightarrow 3dy = -kdx \Rightarrow W = \int (2dx - kdx) = 0$$

$$\Rightarrow 2x = kx \Rightarrow k = 2$$

Alternative Solution

$$\therefore m_1 m_2 = -1 \Rightarrow \frac{3}{2} \left(-\frac{k}{3} \right) = -1 \Rightarrow k = 2$$

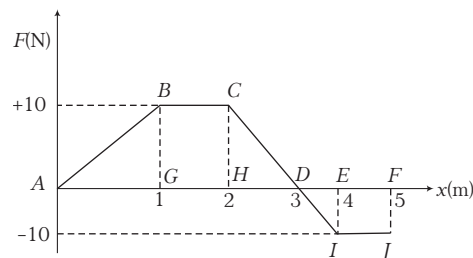
14 (c) $W = \text{Area under } F\text{-}x \text{ graph}$

$$\therefore W = \text{Area of trapezium} = \frac{1}{2} \times (4 + 2) \times 5 = 15 \text{ J}$$

15 (d) Work done = Area under $F\text{-}x$ graph

= Area of $\triangle ABG$ + Area of rectangle of $BGHC$

+ Area of $\triangle CDH$ + Area of $\triangle DEI$ + Area of rectangle $EFJI$



$$\begin{aligned} &= \left(\frac{1}{2} \times 10 \times 1 \right) + (10 \times 1) + \left(\frac{1}{2} \times 10 \times 1 \right) + \left(\frac{1}{2} \times (-10) \times 1 \right) \\ &\quad + (-10 \times 1) \end{aligned}$$

$$\Rightarrow W = 5 \text{ J}$$

$$16 (b) W_1 = \frac{1}{2} k(5)^2 \times 10^{-4} \text{ and } W_2 = \frac{1}{2} k(15)^2 \times 10^{-4}$$

where, W_1 and W_2 are the work done in extending spring to 5 cm and 15 cm, respectively.

$$W_{1 \rightarrow 2} = W_2 - W_1$$

$$= \frac{1}{2} k [(15)^2 - (5)^2] \times 10^{-4} = \frac{1}{2} \times 800(200) \times 10^{-4} = 8 \text{ J}$$

17 (d) Spring constant, $k = \frac{10 \text{ N}}{10^{-3} \text{ m}} = 10 \times 10^3 \text{ Nm}^{-1}$

Now, work done in stretching the spring through 40 mm ($= 40 \times 10^{-3} \text{ m}$), $W = \frac{1}{2} \times 10 \times 10^3 \times (40 \times 10^{-3})^2 = 8 \text{ J}$

18 (a) Work done by the force relative to porter will be zero because displacement relative to porter is zero.

Now, work done by the force relative to a person on the ground, $W_G = mgh = 20 \times 10 \times 4 = 800 \text{ J}$

19 (a) In case of a conservative force field, the work done is independent of the path followed.

$\therefore$ Required work done $= 5 \text{ J} + 2 \text{ J} = 7 \text{ J}$

20 (b) Amongst the given forces, frictional force is a non-conservative force, whereas spring and gravitational forces are conservative forces.

• CHECK POINT 6.2

1 (b) According to question, $F \propto \frac{1}{v}$

$\Rightarrow F = \frac{\lambda}{v}$ where, $\lambda = \text{constant}$

$$ma = m \frac{dv}{dt} = \frac{\lambda}{v} \Rightarrow m \int_0^v dv = \lambda \int_0^t dt$$

$$m \left[\frac{v^2}{2} \right]_0^v = \lambda \left[t \right]_0^t \Rightarrow \frac{1}{2} mv^2 = K = \lambda t$$

$$K \propto t$$

2 (a) It is given that, $K_f = 2K_i$

$$\text{or } \frac{1}{2} m(v+1)^2 = 2 \left(\frac{1}{2} mv^2 \right) \text{ or } v+1 = \sqrt{2} v$$

$$\text{or } v = \frac{1}{\sqrt{2}-1} = (\sqrt{2}+1) \text{ ms}^{-1}$$

3 (b) According to the question, $\frac{1}{2} mv_1^2 = \frac{1}{2} \left[\frac{1}{2} \times \frac{m}{2} \times v_2^2 \right]$

where, v_1 = speed of man and v_2 = speed of boy.

$$\text{Now, } \frac{1}{2} m(v_1+1)^2 = \frac{1}{2} \times \frac{m}{2} \times v_2^2$$

Solving these two equations, we get

$$v_1 = (\sqrt{2}+1) \text{ ms}^{-1} \text{ and } v_2 = 2(\sqrt{2}+1) \text{ ms}^{-1}$$

4 (d) Kinetic energy, $K = \frac{p^2}{2m} \Rightarrow K_1 = \frac{p^2}{2m_1}$ and $K_2 = \frac{p^2}{2m_2}$

$$\therefore \frac{K_1}{K_2} = \frac{m_2}{m_1}$$

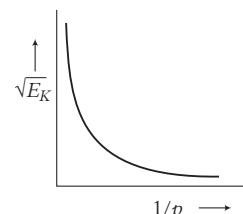
5 (c) $K = \frac{p^2}{2m}$, $K' = \frac{(1.5p)^2}{2m} = 2.25 \frac{p^2}{2m} = 2.25 K$

$$\therefore \% \text{ increase in kinetic energy} = \frac{2.25K - K}{K} \times 100 = 125\%$$

6 (c) $p = \sqrt{2mE_K}$

It is clear that $p \propto \sqrt{E_K}$

So, the graph between p and $\sqrt{E_K}$ will be straight line but graph between $1/p$ and $\sqrt{E_K}$ will be a hyperbola as shown below



7 (d) From work-energy theorem, $K_2 - K_1 = W = Fs$

$$K_2 - 0 = Fs$$

K is independent of m .

($\because$ Force is constant)

8 (c) Speed of the body, $v = \frac{dx}{dt} = \frac{d}{dt} \left(\frac{t^3}{3} \right) = t^2$

At $t = 0$, $v = 0$

At $t = 2 \text{ s}$, $v = 4 \text{ ms}^{-1}$

From work-energy theorem, $W = \text{Change in kinetic energy}$

$$= K_f - K_i = \frac{1}{2} m (v_f^2 - v_i^2) = \frac{1}{2} \times 2 \times (16 - 0) = 16 \text{ J}$$

9. (a) Change in kinetic energy = Work done

= Area under $F-x$ graph

$$\therefore \frac{1}{2} \times 5 \times v^2 = 10 \times 25 + \frac{1}{2} \times 25 \times 10 = 375$$

$$\therefore v = 12.2 \text{ ms}^{-1}$$

10 (a) Applying work-energy theorem, $K_f - K_i = W$

$$\begin{aligned} \Rightarrow K_f &= K_i + W = \frac{1}{2} \times 20 \times (10)^2 + \int_{20}^{30} (-0.1x) dx \\ &= 1000 - \frac{0.1}{2} (x^2)_{20}^{30} = 1000 - 25 = 975 \text{ J} \end{aligned}$$

11 (b) Work done by all forces = Change in kinetic energy

$$= \frac{1}{2} m (v_f^2 - v_i^2) = \frac{1}{2} \times 2(0 - 400) = -400 \text{ J}$$

12 (d) Work done on the particle, $W = \Delta KE = \frac{1}{2} m (v_f^2 - v_i^2)$

$$= \frac{1}{2} \times 0.01 [(64 + 400) - (16 + 256)] = 0.96 \text{ J}$$

13 (b) Given, force, $\mathbf{F} = (4\hat{i} + 4\hat{j}) \text{ N}$

$$\mathbf{r}_1 = 0\hat{i} + 0\hat{j}$$

$$\mathbf{r}_2 = \hat{i} + \hat{j}$$

$$\Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 = \hat{i} + \hat{j}$$

Initial speed, $v_1 = 2 \text{ ms}^{-1}$

From work-energy theorem, we have

$$\Delta W = \Delta K$$

$$\Rightarrow \mathbf{F} \cdot \Delta \mathbf{r} = \frac{1}{2} m (v_2^2 - v_1^2)$$

$$\begin{aligned} \Rightarrow (4\hat{i} + 4\hat{j}) \cdot (\hat{i} + \hat{j}) &= \frac{1}{2} \times 1 (\psi_2^2 - 4) \\ \Rightarrow 4 + 4 &= \frac{1}{2} (\psi_2^2 - 4) \\ \Rightarrow v_2^2 &= 20 \Rightarrow v_2 = 4.5 \text{ ms}^{-1} \end{aligned}$$

14 (c) Forces acting on the body are

$$\text{weight, } w = 5 \times 10 = 50 \text{ N}$$

$$\text{External force, } F = 170 \text{ N}$$

$$\text{Net upward force acting on the body, } F_{\text{net}} = 170 - 50 = 120 \text{ N}$$

$$\text{Now, applying work-energy theorem, } (F_{\text{net}}) h = \frac{1}{2} mv^2$$

$$\Rightarrow 120 \times 10 = \frac{1}{2} \times 5 \times v^2$$

$$\Rightarrow v^2 = \frac{2 \times 12 \times 100}{5} \Rightarrow v = 22 \text{ ms}^{-1}$$

15 (b) Decrease in kinetic energy = Increase in elastic potential energy

$$\therefore \frac{1}{2} mv^2 = \frac{1}{2} kx^2$$

$$\text{or } x = \sqrt{\frac{m}{k}} \cdot v = \sqrt{\frac{0.1}{1000}} \times 10 = 0.1 \text{ m}$$

16 (a) Decrease in gravitational potential energy

$$= \text{Increase in elastic potential energy}$$

$$\text{or } mgh(h+x) = \frac{1}{2} kx^2 \text{ or } 2 \times 9.8(0.4+x) = \frac{1}{2} \times 1960 \times x^2$$

Solving this equation, we get

$$x = 0.1 \text{ m or } 10 \text{ cm}$$

17 (b) Potential energy is only associated with conservative force.

$$\text{Force, } F = -\left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k}\right)$$

where, F = conservative force.

$$\left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k}\right) = \text{Partial derivative of potential energy}$$

w.r.t. x , y and z , respectively.

18 (a) The potential energy of a system increases, if work is done by the system against a conservative force.

$$-\Delta U = W_{\text{conservative force}}$$

19 (b) For conservative force, $F = -\frac{dU}{dx} = -(7x-3) = 3-7x$

$$\text{At equilibrium, } F = 0$$

$$\Rightarrow 3-7x=0 \Rightarrow x=\frac{3}{7}$$

$$\therefore U = \frac{7}{2} \left(\frac{3}{7}\right)^2 - 3\left(\frac{3}{7}\right) = \frac{9}{14} - \frac{9}{7} = -\frac{9}{14} \text{ units}$$

20 (c) Gain in KE = Loss in PE

$$\Rightarrow \frac{1}{2} mv^2 = (0.1)(mgh)$$

$$\therefore v = \sqrt{0.2gh} = \sqrt{0.2 \times 10 \times 2} = 2 \text{ ms}^{-1}$$

21 (c) From conservation of mechanical energy, we have

$$\Rightarrow \frac{1}{2} mv_i^2 = mgh$$

$$\therefore \frac{3}{4} mgh + mgh' = mgh \Rightarrow h' = \frac{h}{4}$$

• CHECK POINT 6.3

1 (d) From work-energy theorem, work done, $W = \frac{1}{2} Mv^2$

$$\therefore \text{Power delivered, } P = \frac{W}{t} = \frac{1}{2} \left(\frac{Mv^2}{t}\right)$$

2 (c) Power, $P = \frac{\text{Work done by the engine}}{\text{Time taken}} = \frac{W}{t}$

$$\Rightarrow t = \frac{W}{P} = \frac{mgh}{P} = \frac{200 \times 10 \times 40}{10 \times 10^3} = 8 \text{ s}$$

3 (b) Given, $P_{\text{output}} = 10 \text{ kW}$

$$\text{Now, } P_{\text{input}} = 2 \times 10^3 \text{ cal g}^{-1} \times 1 \text{ g s}^{-1}$$

$$= 2 \times 10^3 \text{ cal s}^{-1} = 2 \times 10^3 \times 4.2 \text{ Js}^{-1}$$

$$= 8.4 \text{ kW}$$

As $P_{\text{output}} > P_{\text{input}}$, hence the claim given in question is never possible.

4 (c) Power given to turbine, $P_{\text{in}} = \frac{mgh}{t}$

$$P_{\text{in}} = \left(\frac{m}{t}\right) \times g \times h$$

$$\Rightarrow P_{\text{in}} = 15 \times 10 \times 60 \Rightarrow P_{\text{in}} = 9000 \text{ W}$$

$$\Rightarrow P_{\text{in}} = 9 \text{ kW}$$

As efficiency of turbine is 90%, therefore power generated,

$$P_{\text{out}} = 90 \% \text{ of } 9 \text{ kW} = 9 \times \frac{90}{100}$$

$$\Rightarrow P_{\text{out}} = 8.1 \text{ kW}$$

5 (a) Power = $\frac{\text{Work done}}{\text{Time}} = \frac{\text{Pressure} \times \text{Change in volume}}{\text{Time}}$

$$\begin{aligned} &= \frac{20000 \times 1 \times 10^{-6}}{1} \\ &= 2 \times 10^{-2} = 0.02 \text{ W} \end{aligned}$$

6 (d) Power is defined as the rate of change of energy in a system or the time rate of doing work.

$$\Rightarrow P = \frac{dE}{dt} = \frac{dW}{dt}$$

$$\text{Also, work, } W = \text{force} \times \text{displacement} = F \times d$$

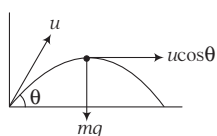
Since, the displacement is zero.

$$\therefore P = \frac{d}{dt} (F \times d) = \frac{d}{dt} \times 0 = 0$$

7 (a) $P = \mathbf{F} \cdot \mathbf{v} = \frac{\mathbf{F} \cdot \mathbf{s}}{t} = \frac{(2\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (3\hat{i} + 4\hat{j} + 5\hat{k})}{4}$

$$= \frac{38}{4} = 9.5 \text{ W}$$

8 (d) $P = \mathbf{F} \cdot \mathbf{v} = mg(u \cos \theta) \cos 90^\circ = 0$



(A) Taking it together

1 (c) As the body is falling freely under gravity, the potential energy decreases and kinetic energy increases but total mechanical energy (PE + KE) of the body and earth system will remain constant, as external force on the system is zero.

2 (b) When electron and proton are moving under the influence of their mutual forces, the magnetic forces will be perpendicular to their motion. Hence, no work is done by these forces.

3 (a) Power of the engine,

$$P = \mathbf{F} \cdot \mathbf{v} = (20\hat{i} - 3\hat{j} + 5\hat{k}) \cdot (6\hat{i} + 20\hat{j} - 3\hat{k})$$

$$= 120 - 60 - 15 = 45 \text{ W}$$

4 (b) Centre of mass of the rod moves a height, $h = \frac{l}{2}$.

$$\text{Work done, } W = mgh = mgl / 2$$

5 (c) Here, work is done by the frictional force on the cycle and is equal to $-200 \times 10 = -2000 \text{ J}$. As the road is not moving, hence work done by the cycle on the road = zero.

6 (c) $F = kx$ or $k = \text{slope of } F\text{-}x \text{ graph [when } F \text{ (load) is along } Y\text{-axis and extension (} x \text{) is along } X\text{-axis.]}$

Here, F is along X -axis.

$$\text{So, } k = \frac{1}{10} = 0.1 \text{ kgf/cm}$$

7 (c) $\therefore W_F + W_g = \Delta K = 0 \Rightarrow W_F = -W_g = mgh$

8 (c) $P = \sqrt{2Km}$ or $P \propto \sqrt{m}$

$$\therefore \frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

9 (a) Kinetic energy, $E = \frac{1}{2}mv^2$

$$\therefore \frac{dE}{dv} = mv \text{ or } p = \frac{dE}{dv} \quad (\because mv = p)$$

10 (c) $W = \frac{1}{2} \times m \times (10)^2 = 50m \quad \dots(i)$

$$\text{and } W' = \frac{1}{2} m (v_2^2 - v_1^2) = \frac{1}{2} m \times (400 - 100) = 150m = 3W$$

[from Eq. (i)]

11 (d) When the earth is closest to the sun, speed of the earth is maximum, hence kinetic energy is maximum. When the earth is farthest from the sun, speed is minimum, hence kinetic energy is minimum but it will never be zero and negative.

This variation is correctly represented by option (d).

12 (a) Given force is a constant force and work done by a constant force is always path independent.

$$\therefore W_1 = W_2$$

13 (c) From work-energy theorem,

work done = change in kinetic energy

$$\Rightarrow K \propto x \quad (\text{as } F = \text{constant because } a = \text{constant})$$

Therefore, K - x graph is a straight line passing through origin.

14 (c) When a pendulum oscillates in air, it will lose energy continuously in overcoming resistance due to air. Therefore, total mechanical energy of the pendulum decreases continuously with time.

The variation is correctly represented by curve (c).

$$15 (c) \therefore \text{Average power} = \frac{W}{t} = \frac{KE}{t} = \frac{mv^2}{2t}$$

$$\Rightarrow P = \frac{F\left(\frac{1}{2}at^2\right)}{t} = \frac{1}{2}Fv$$

16 (d) From work-energy theorem,

$$W = \text{Change in kinetic energy or } W = \frac{1}{2}mv^2$$

$\therefore W$ - v graph is a parabola.

17 (a) Speed is doubled. Therefore, kinetic energy will become four times. Hence, minimum stopping distance will also become four times, i.e. $4 \times 2 \text{ m} = 8 \text{ m}$.

18 (d) Potential energy of the spring, $U = \frac{1}{2}kx^2$ or $U \propto x^2$

Stretch is increased by 5 times. Therefore, stored potential energy will be increased by 25 times.

19 (c) Applying conservation of mechanical energy, we have

$$\frac{1}{2}mv^2 = U \text{ or } m = \frac{2U}{v^2}$$

20 (a) Here, $P = 10^7 \text{ kW} = 10^{10} \text{ W} = 10^{10} \text{ Js}^{-1}$

Time, $t = 1 \text{ day} = 24 \times 60 \times 60 \text{ s}$

Energy produced per day,

$$E = Pt = 10^{10} \times 24 \times 60 \times 60 = 864 \times 10^{12} \text{ J}$$

As, $E = mc^2$

$$\Rightarrow \text{Mass, } m = \frac{E}{c^2} = \frac{864 \times 10^{12} \text{ J}}{(3 \times 10^8)^2 \text{ ms}^{-1}}$$

$$= 9.6 \times 10^{-3} \text{ kg} = 9.6 \text{ g}$$

21 (c) Force between two protons is same as that of between proton and a positron.

As, positron is much lighter than proton, it moves away through much larger distance as compared to proton.

We know that, work done = force $\times$ distance.

As forces are same in case of proton and positron but distance moved by positron is larger, hence work done will be more.

22 (b) Total energy, $E = \text{PE} + \text{KE} \quad \dots(i)$

When particle is at $x = x_m$, i.e. at extreme position, it returns back. Hence, at $x = x_m$; $v = 0$; $\text{KE} = 0$

From Eq. (i), we get

$$E = \text{PE} + 0 = \text{PE} \Rightarrow E = V(x_m) = \frac{1}{2}kx_m^2$$

23 (c) At equilibrium, $F = -\frac{dU}{dx} = 0 \Rightarrow (-12)ax^{-13} + (6bx^{-7}) = 0$

$$\therefore x = \left(\frac{2a}{b}\right)^{1/6}$$

24 (c) For upward motion, $\frac{1}{2}m(10)^2 = mgh + R_a h$... (i)

where, R_a = air resistance.

For downward motion, $mgh = \frac{1}{2}m(9)^2 + R_a h$... (ii)

Solving Eqs. (i) and (ii), we get

$$2mgh = \frac{1}{2}m(100 + 81) \Rightarrow h = 4.61 \text{ m}$$

25 (d) Body will have maximum speed, where

$$mg \sin \theta = \mu mg \cos \theta$$

or $\sin 37^\circ = (0.3x) \cdot \cos 37^\circ$ or $x = 2.5 \text{ m}$

26 (b) From $-a$ to b , decrease in elastic potential energy

$$= \text{work done against friction.}$$

$$\therefore \frac{1}{2}ka^2 - \frac{1}{2}kb^2 = \mu mg(a+b) \text{ or } (a-b) = \frac{2\mu mg}{k}$$

$$\therefore \text{Decrease in amplitude} = \frac{2\mu mg}{k}$$

27 (a) Work done by frictional force, $W_f = fs \cos 180^\circ$

$$= (\mu mg \cos \theta)(-1)(s)$$

$$= -0.20 \times 5 \times 10 \times \frac{1}{\sqrt{2}} \times 0.1 \quad (\because \theta = 45^\circ)$$

$$= -\frac{1}{\sqrt{2}} \text{ J}$$

28 (c) From conservation of mechanical energy,

$$(M-m)gh = \frac{1}{2}(M+m)v^2$$

$$\therefore v = \sqrt{\frac{2gh(M-m)}{m+M}}$$

29 (a) The distance travelled by the block on the rough surface can be calculated from energy conservation,

$$K_i + U_i = K_f + U_f + W_f$$

where, W_f = work done by friction.

$$\Rightarrow 0 + mgh = 0 + 0 + \mu mgs \Rightarrow s = \frac{h}{\mu} = \frac{1}{0.2} = 5 \text{ m}$$

$$A \xrightarrow{\hspace{2cm}} B$$

$$A \xleftarrow{\hspace{2cm}} B$$

$$A \xrightarrow{1 \text{ m}} C$$

$$\xleftarrow{\hspace{2cm}} 2 \text{ m} \xrightarrow{\hspace{2cm}}$$

The block stops at distance 1 m from A.

30 (a) $\frac{3}{4}$ th of kinetic energy is lost. Hence, left kinetic energy is $\frac{1}{4}$ th.

$$\therefore v^2 = \frac{v_0^2}{4} \Rightarrow v = \frac{v_0}{2} = v_0 - at_0 = v_0 - \mu gt_0$$

or $\mu = \frac{v_0}{2gt_0}$

31 (c) Mass of the body, $m = 200 \text{ g} = 200 \times 10^{-3} \text{ kg}$

Height, $h = 200 \text{ m}$

$$\text{Potential energy, } U = mgh = 200 \times 10^{-3} \times 10 \times 200 = 400 \text{ J}$$

Therefore, decrease in potential energy at the surface = 400 J.

32 (b) At this height, half energy will be potential energy.

$$\therefore mgh = \frac{98}{2} \text{ or } 2 \times 9.8 \times h = 49 \text{ or } h = 2.5 \text{ m}$$

33 (c) Increase in gravitational potential energy

$$= \text{Decrease in elastic potential energy}$$

$$\therefore mgh = \frac{1}{2}kx^2 \Rightarrow h = \frac{kx^2}{2mg}$$

34 (b) Loss in potential energy = mgh

$$\text{Power generated} = \frac{mgh}{2 \times t}$$

$$\text{Number of bulbs, } n = \frac{mgh}{2 \times t \times 100} = \frac{1.8 \times 10^5 \times 10 \times 50}{2 \times 3600 \times 100} = 125$$

35 (d) Speed, $v = \frac{dx}{dt} = 8t^3$, $v_{0s} = 0$, $v_{1s} = 8 \text{ ms}^{-1}$

$$\therefore \Delta KE = \frac{1}{2}m(v_{1s}^2 - v_{0s}^2) = \frac{1}{2} \times 2 \times (64 - 0) \text{ J} = 64 \text{ J}$$

36 (c) W = Change in potential energy = $mgh = mgL(1 - \cos \theta)$

37 (b) In equilibrium, $kd = mg$ or $d = \frac{mg}{k}$... (i)

If allowed to fall suddenly, the body does not stop in its equilibrium position. In that case,

decrease in gravitational PE = increase in elastic PE

$$\text{or } mgd' = \frac{1}{2}kd'^2 \Rightarrow d' = \frac{2mg}{k} = 2d \quad [\text{From Eq. (i)}]$$

38 (a) $K = 4t^2$ or $v^2 \propto t^2$

$$\therefore v \propto t$$

v varies linearly with time when acceleration or force is constant.

39 (b) Change in potential energy for all three particles is same.

Hence, change in kinetic energy will also be same.

$$\text{or } v_A = v_B = v_C$$

40 (b) Work done by conservative force, $W_C = -\Delta U$

$$\Rightarrow W_C = -(U_B - U_A)$$

$$\Rightarrow W_C = U_A - U_B$$

Initial mechanical energy is same for all.

41 (c) Work done, $W = \int P dt$ or $\frac{1}{2}mv^2 = \int P dt = \int_0^2 \left(\frac{3t^2}{2}\right) dt = 4 \text{ J}$

$$\Rightarrow v = \sqrt{\frac{8}{2}} = 2 \text{ ms}^{-1}$$

- 42 (a) Work done by resistive force = Change in PE

$$\Rightarrow F \cdot d = mg(h + d) \Rightarrow F = mg \left(1 + \frac{h}{d} \right)$$

- 43 (b) $PE + KE = mgh + \frac{1}{2}mv^2 = \text{constant}$
 $= mgH$ (Here, H = initial height)
 or $gh + \frac{v^2}{2} = \text{constant}$

- 44 (b) Work done against friction = $(\mu mg \cos \theta) d$
 $= 0.5 \times 1 \times 9.8 \times \cos 60^\circ \times 2 = 4.9 \text{ J}$

- 45 (b) When drop falls, velocity first increases, hence kinetic energy also increases. After sometime, speed (velocity) is constant which is known as terminal velocity, hence kinetic energy also becomes constant. Potential energy decreases continuously as the drop is falling continuously.

The variation in potential energy and kinetic energy is best represented by (b).

- 46 (d) Given, $h = 1.5 \text{ m}$, $v = 1 \text{ ms}^{-1}$, $m = 10 \text{ kg}$, $g = 10 \text{ ms}^{-2}$

From conservation of mechanical energy,

$$(PE)_i + (KE)_i = (PE)_f + (KE)_f$$

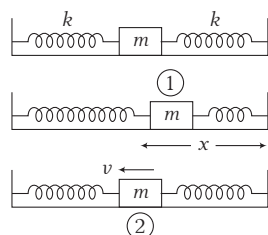
$$\Rightarrow mgh + \frac{1}{2}mv^2 = 0 + (KE)_f$$

$$\Rightarrow (KE)_f = mgh + \frac{1}{2}mv^2$$

$$\Rightarrow (KE)_f = 10 \times 10 \times 1.5 + \frac{1}{2} \times 10 \times (1)^2$$

$$= 150 + 5 = 155 \text{ J}$$

- 47 (b)

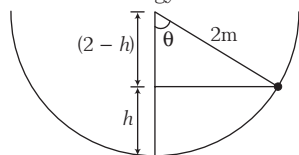


Total PE of spring = Total KE of block

$$\frac{1}{2}kx^2 + \frac{1}{2}kx^2 = \frac{1}{2}mv^2 \Rightarrow kx^2 = \frac{1}{2}mv^2$$

$$\Rightarrow \text{Speed of the block, } v = x \sqrt{\frac{2k}{m}}$$

- 48 (b) Loss in mechanical energy = Work done against friction.



$$mgh = \mu mg \cos \theta \times s \Rightarrow s = \frac{h}{\mu g \cos \theta}$$

$$= \frac{1 \times 10^{-2}}{0.01 \times 10 \times \left(\frac{2-h}{2} \right)} \left[\because \cos \theta = \frac{(2-h)}{2} \right]$$

$$= 0.1005 \text{ m}$$

$$s \approx 10 \text{ cm}$$

- 49 (d) Work done by tension on M is negative (force and displacement are in opposite directions). But work done by tension on m is positive. Net work done will be zero.

- 50 (d) On M , horizontal components of N and f are balanced (as Mg is vertical). Hence, on $2M$ also, they will be balanced.
 $\therefore$ Horizontal force kx on $2M$ should be zero.

- 51 (a) Spring constant, $k = \frac{F}{x}$ = slope of F - x graph.

$$k \propto \frac{1}{l} \quad (\because x \propto l)$$

Length is reduced to half. Therefore, k will become two times. Slope will increase. Hence, the line OA will shift towards F -axis.

- 52 (a) $F = \text{constant}$

$$\therefore a = \text{constant or } v = at$$

Now, $P = F \cdot v = F \cdot at$ or $P \propto t$ (As F and a both are constants)

Hence, P - t graph is a straight line passing through origin.

- 53 (d) $F \propto s^{-1/3}$ or $a \propto s^{-1/3}$

$$\text{or } v dv \propto s^{-1/3} ds \quad \left(\because a = v \cdot \frac{dv}{ds} \right)$$

On integrating, we have

$$v^2 \propto s^{2/3} \quad \text{or} \quad v \propto s^{1/3}$$

Now, power, $P \propto F \cdot v \propto s^{-1/3} \cdot s^{1/3}$ or $P \propto s^0$

- 54 (b) When displaced from x_2 in negative direction, force is positive. So, this force is restoring in nature and will bring the body back. Hence, at x_2 , body is in stable equilibrium position.

- 55 (b) Given, $v = ax^{3/2}$,

$$m = 0.5 \text{ kg}, a = 5 \text{ m}^{-1/2} \text{ s}^{-1}, \text{ work done, } W = ?$$

$$\text{Acceleration, } a_0 = \frac{dv}{dt} = v \frac{dv}{dx} = ax^{3/2} \frac{d}{dx} (ax^{3/2})$$

$$= ax^{3/2} \times a \times \frac{3}{2} \times x^{1/2} = \frac{3}{2} a^2 x^2$$

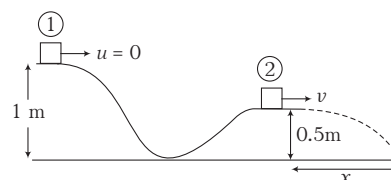
$$\text{Now, Force} = ma_0 = m \frac{3}{2} a^2 x^2$$

$$\text{Work done, } W = \int_{x=0}^{x=2} F dx = \int_0^2 \frac{3}{2} ma^2 x^2 dx$$

$$= \frac{3}{2} ma^2 \times \left[\frac{x^3}{3} \right]_0^2$$

$$= \frac{1}{2} ma^2 \times 8 = \frac{1}{2} \times (0.5) \times (25) \times 8 = 50 \text{ J}$$

- 56 (a) The given figure is



From law of conservation of energy, $mg(1) = \frac{1}{2}mv^2 + mg(0.5)$

$$\Rightarrow v^2 = 2g(1 - 0.5) = g$$

$$\text{Also, } t = \frac{x}{v}$$

$$\therefore y = \frac{1}{2}gt^2 \Rightarrow 0.5 = \frac{gx^2}{2v^2} = \frac{10x^2}{2(10)}$$

$$\Rightarrow x^2 = 1 \Rightarrow x = 1\text{m}$$

- 57** (c) Decrease in gravitational potential energy = increase in kinetic energy



$$U_1 = \left(\frac{Ml_0}{L}\right)g\frac{l_0}{2}$$

$$U_2 = -mg\left(\frac{L}{2}\right)$$

$$\frac{MgL}{2} - \frac{M}{L} \cdot l_0 g \frac{l_0}{2} = \frac{1}{2}Mv^2 \quad \text{or} \quad v = \sqrt{\frac{g}{L}(L^2 - l_0^2)}$$

- 58** (a) From work-energy theorem, $W_F + W_{mg} = \frac{1}{2}mv^2$

$$\Rightarrow F \cdot R + mgR = \frac{1}{2}mv^2$$

$$\Rightarrow 5 \times 5 + \frac{1}{2} \times 10 \times 5 = \frac{1}{2} \times \frac{1}{2} \times v^2$$

$$\Rightarrow v = \sqrt{200} = 14.14 \text{ ms}^{-1}$$

- 59** (a) Power, $P = F \cdot v = m\left(v \cdot \frac{dv}{ds}\right) \cdot v$

$$\therefore \int_v^{2v} v^2 \cdot dv = \frac{P}{m} \cdot \int_0^s ds$$

$$\left[\frac{v^3}{3}\right]_v^{2v} = \frac{Ps}{m} \quad \text{or} \quad s = \frac{7mv^3}{3P}$$

- 60** (b) Applying conservation of mechanical energy, we get $(mgH - mgh) = 2mgh$

$$\therefore h = \frac{H}{3} \Rightarrow v = \sqrt{2g(H - h)} = \sqrt{2g\left(\frac{2H}{3}\right)} = 2\sqrt{\frac{gH}{3}}$$

- 61** (b) Maximum acceleration of 1 kg block, $a_{\max} = \mu g = 1 \text{ ms}^{-2}$

Common acceleration without relative motion between two blocks,

$$a = \frac{0.5}{3} \text{ ms}^{-2}$$

Since, $a < a_{\max}$

There will be no relative motion and blocks will move with acceleration $\frac{0.5}{3} \text{ ms}^{-2}$.

Force of friction by lower block on upper block,

$$f = ma = (1)\left(\frac{0.5}{3}\right) = \frac{1}{6} \text{ N (towards right)}$$

$$\therefore \text{Work done, } W = f \times s = 0.5 \text{ J}$$

- 62** (b) Given, power = constant

$$\text{As, power, } P = \frac{dW}{dt} = \frac{\mathbf{F} \cdot d\mathbf{s}}{dt} = \frac{F ds \cos 0^\circ}{dt} = \frac{F ds}{dt}$$

($\because$ Body is moving unidirectionally)

$$P = \frac{Fds}{dt} = \text{constant} \quad (\because P = \text{constant})$$

Now, writing dimensions, $[F][v] = \text{constant}$

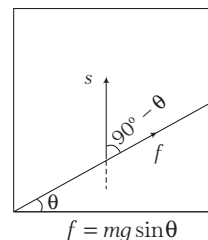
$$\Rightarrow [MLT^{-2}][LT^{-1}] = \text{constant}$$

$$\Rightarrow L^2 T^{-3} = \text{constant} \quad (\because \text{mass is constant})$$

$$\Rightarrow L \propto T^{3/2}$$

$$\Rightarrow \text{Displacement } (d) \propto t^{3/2}$$

- 63** (c) As block does not slide. Hence, force of friction,

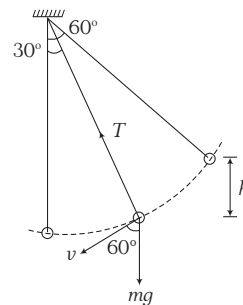


In time t , displacement, $s = vt$

$$\therefore W_f = f \cdot s \cdot \cos(90^\circ - \theta)$$

$$= (mg \sin \theta)(vt)(\sin \theta) = mgvt \sin^2 \theta$$

- 64** (a) Power due to tension = 0



$$\text{Power due to } mg = (mg)(v) \cos 60^\circ = \frac{mgv}{2}$$

$$\text{Here, } v = \sqrt{2gh} \quad \text{and} \quad h = l(\cos 30^\circ - \cos 60^\circ) = l\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)$$

$$\therefore v = \sqrt{2 \times 10 \times \left(\frac{\sqrt{3} - 1}{2}\right)} = \sqrt{10(\sqrt{3} - 1)} = 2.7 \text{ ms}^{-1}$$

$$\therefore \text{Power, } P = \frac{mgv}{2} = \frac{1 \times 10}{2} \times 2.7 = 13.5 \text{ W}$$

- 65** (a) Decrease in gravitational potential energy

= Increase in kinetic energy.

Initially, centre of mass of chain was at distance $\frac{l}{4}$ below the pin and in final position, it is at distance $\frac{l}{2}$ below the pin.

Hence, centre of mass has descended $\frac{l}{4}$.



$$\therefore mg \frac{l}{4} = \frac{1}{2}mv^2 \quad \text{or} \quad v = \sqrt{\frac{gl}{2}}$$

- 66 (c) At $x = 6$ m, $U = 26$ J (extreme position)

On the other side, $U = 26 = 10 + (x - 2)^2$ or $x - 2 = \pm 4$

$$\therefore x = 6 \text{ m and } x = -2 \text{ m}$$

$$U_{\min} = 10 \text{ J, at } x = 2 \text{ m}$$

$$KE_{\max} = E - U_{\min} = 16 \text{ J, at } x = 2$$

- 67 (b) Maximum acceleration of 2 kg block due to friction can be μg or 5 ms^{-2} .

Combined acceleration, if both move together with same acceleration, would be $a = \frac{60}{12} = 5 \text{ ms}^{-2}$

Since, both accelerations are equal, upper block will move with acceleration 5 ms^{-2} due to friction.

$$\text{In first two seconds, } s = \frac{1}{2}at^2 = \frac{1}{2} \times 5 \times 4 = 10 \text{ m}$$

and force of friction, $f = ma = 10 \text{ N}$

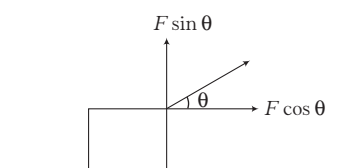
$$\therefore W_f = fs \cos 0^\circ = 100 \text{ J}$$

- 68 (b) Normal force on the block, $N = mg - F \sin \theta$

Block moves with uniform velocity. Hence, net force = 0

$$\text{or } F \cos \theta = \mu N = \mu(mg - F \sin \theta)$$

$$\therefore F = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$$



$$W = Fs \cos \theta = \frac{\mu mgd \cos \theta}{\cos \theta + \mu \sin \theta}$$

- 69 (a) $\therefore F = kx$

$$\Rightarrow k = \frac{F}{x} = \frac{100}{1} \text{ Nm}^{-1} = 100 \text{ Nm}^{-1}$$

Now, from energy conservation between natural length of spring and its maximum compression state.

$$\text{i.e. } \frac{1}{2}mv^2 + mgh = \frac{1}{2}kx_{\max}^2$$

$$\Rightarrow v = \sqrt{\frac{kx_{\max}^2}{m} - 2gh} = \sqrt{\frac{(100)(2)^2}{10} - (2)(10)(1)} = \sqrt{20} \text{ ms}^{-1}$$

(B) Medical entrance special format questions

• Assertion and reason

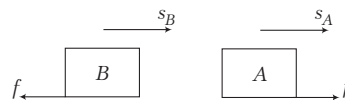
- 1 (d) If oscillations are not taking place, then kinetic energy may be zero at stable equilibrium position.

- 2 (a) For conservative forces, $\Delta U = -\Delta W$

where, ΔU = change in potential energy

and ΔW = work done by conservative force.

- 3 (c) Work done by friction on block A is positive and on B is negative. If there is no slipping between the two blocks, then $s_A = s_B$. Therefore, net work done will be zero, otherwise not.

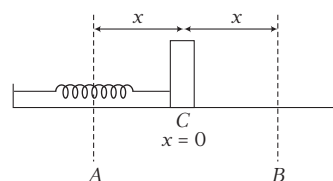


- 4 (d) $W = Pt = \frac{1}{2}mv^2$
 $v \propto t^{1/2}$

On differentiating, we get

$$a \propto t^{-1/2}$$

- 5 (b) $W_{AC} = +ve$



$$W_{CB} = -ve \text{ and } W_{ACB} = 0$$

Also,

$$F \propto (-x)$$

6. (c) $\frac{dU}{dt} = \frac{d}{dt}(mgh) = mg \frac{d}{dt} \left(u_y t - \frac{1}{2}gt^2 \right)$

$$= mg(u_y - gt) = mgv_y$$

$$\therefore \left| \frac{dU}{dt} \right| = mg|v_y|$$

$$\text{For } m = 1 \text{ kg, } \left| \frac{dU}{dt} \right| = g|v_y| = g(u_y - gt) \quad \dots(i)$$

$$a \cdot v = (-g\hat{j}) \cdot [u_x\hat{i} + (u_y - gt)\hat{j}] = -g(u_y - gt) = g|v_y| \quad \dots(ii)$$

From Eqs. (i) and (ii), we can see that two magnitudes are equal.

$$P = \mathbf{F} \cdot \mathbf{v} = (-mg\hat{j}) \cdot [u_x\hat{i} + (u_y - gt)\hat{j}] = -mgu_y + mg^2t$$

So, P versus t graph has the positive slope.

• Statement based questions

- 1 (b) Work done by friction may be positive, negative and zero. Work done by conservative, internal forces may be zero in a round trip but it is non-zero for non-conservative internal forces.

- 2 (c) As the given tracks are frictionless, hence mechanical energy will be conserved, as both the tracks having common height h .

From conservation of mechanical energy,

$$\frac{1}{2}mv^2 = mgh \quad (\text{For both tracks I and II})$$

$$\Rightarrow v = \sqrt{2gh}$$

Hence, speed is same for both stones.

For stone I, a_1 = acceleration along inclined plane = $g \sin \theta_1$

Similarly, for stone II, $a_2 = g \sin \theta_2$ as $\theta_2 > \theta_1$, hence $a_2 > a_1$.

And length for track II is also less, hence stone II reaches earlier than stone I.

$$\begin{aligned} 3 \text{ (c) Work done by conservative forces} &= U_i - U_f \\ &= -20 - 40 = -60 \text{ J} \end{aligned}$$

$$\text{Work done by external forces} = E_f - E_i = 140 - 40 = 100 \text{ J}$$

$$\begin{aligned} \text{and net work done by all the forces} &= K_f - K_i \\ &= 100 - 60 = 40 \text{ J} \end{aligned}$$

4 (c) Spring force is conservative in nature. Potential energy is only associated with conservative forces.

5 (c) As momentum of a body increases by 50% of its initial momentum, $p_2 = p_1 + 50\%$ of $p_1 = \frac{3}{2} p_1$

$$\therefore v_2 = \frac{3}{2} v_1$$

$$\text{As } K \propto v^2, \text{ so } K_2 = \frac{9}{4} K_1$$

$$\text{Increase in KE} = \frac{K_2 - K_1}{K_1} \times 100 = \frac{\frac{9}{4} K_1 - K_1}{K_1} \times 100 = 125\%$$

• Match the columns

1 (c) Power, $P \propto t$

$$\text{Work done, } W = \int P dt = \int \alpha t dt \quad \text{or } W \propto t^2$$

Since, work done is equal to change in kinetic energy.

$$\text{Hence, } v^2 \propto t^2 \text{ or } v \propto t$$

$$\text{Further, } v = \frac{ds}{dt}$$

$$\therefore \frac{ds}{dt} \propto t \text{ or } ds \propto t dt \text{ or } s \propto t^2 \quad (\text{By integration})$$

Hence, $A \rightarrow p, B \rightarrow q, C \rightarrow q$.

2 (a) A is the point of stable equilibrium, so potential energy is minimum, i.e. at R. Similarly, point C is the unstable equilibrium position, where potential energy should be maximum, i.e. at P.

Hence, $A \rightarrow r, B \rightarrow s, C \rightarrow p$.

3 (b) $v_f - v_i = \text{Area of } a-x \text{ graph} = 12 \text{ ms}^{-1}$

$$\therefore v_f = 12 + 4 = 16 \text{ ms}^{-1}$$

$$\Delta KE = \frac{1}{2} m(v_f^2 - v_i^2) = 120 \text{ J}$$

Total work done by all the forces $= \Delta KE = 120 \text{ J}$

$$\text{Final KE, } K_f = \frac{1}{2} m v_f^2 = 128 \text{ J}$$

$$\text{Work done by conservative forces} = U_i - U_f = 240 \text{ J}$$

Work done by external forces

$$= \text{Total work done} - \text{Work done by conservative forces}$$

$$= -120 \text{ J}$$

Hence, $A \rightarrow s, B \rightarrow q, C \rightarrow p, D \rightarrow r$.

4 (d) In $t = 4 \text{ s}, v = at = 8 \text{ ms}^{-1}$ and $s = \frac{1}{2} at^2 = 16 \text{ m}$

$$KE = \frac{1}{2} m v^2 = 32 \text{ J}$$

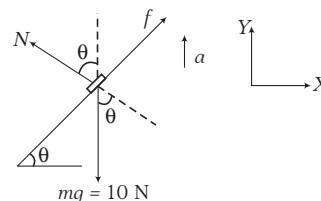
From work-energy theorem,

$$\text{Work done by all the forces} = \Delta KE = 32 \text{ J}$$

$$\text{Work done by gravity, } W_g = -mgh = -(1)(10)(16) = -160 \text{ J}$$

Writing equation of motion, we have

$$\Sigma F_y = ma$$



$$N \cos 30^\circ + f \sin 30^\circ - 10 = ma = 2$$

$$\text{or } \sqrt{3} N + f = 24 \quad \dots(i)$$

$$\Sigma F_x = 0$$

$$\therefore N \sin 30^\circ = f \cos 30^\circ \text{ or } N = \sqrt{3} f \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we have

$$f = 6 \text{ N} \quad \text{and} \quad N = 6\sqrt{3} \text{ N}$$

Now, work done by normal reaction, $W_N = (N \cos \theta)(s)$

$$= (6\sqrt{3}) \left(\frac{\sqrt{3}}{2} \right) (16) = 144 \text{ J}$$

$$\text{Work done by friction, } W_f = (f \sin \theta)(s) = (6) \left(\frac{1}{2} \right) (16) = 48 \text{ J}$$

Work done by all the forces,

$$W = W_g + W_N + W_f = -160 + 144 + 48 = 32 \text{ J}$$

Hence, $A \rightarrow r, B \rightarrow p, C \rightarrow s, D \rightarrow q$.

(C) Medical entrances' gallery

1 (b) Work done by a force F , which is variable in nature in

$$\text{moving a particle from } y_1 \text{ to } y_2 \text{ is given by } W = \int_{y_1}^{y_2} F \cdot dy \quad \dots(i)$$

Given, force, $F = 20 + 10y$, $y_1 = 0$ and $y_2 = 1 \text{ m}$

Substituting the given values in Eq. (i), we get

$$\begin{aligned} W &= \int_0^1 (20 + 10y) dy = \left[20y + \frac{10y^2}{2} \right]_0^1 \\ &= 20(1-0) + 5(1-0)^2 = 25 \text{ J} \end{aligned}$$

$\therefore$ Work done by the force will be 25 J.

2 (b) In stretching a wire, the work done against internal restoring force is stored as elastic potential energy in the wire and is given by

$$U = W = \frac{1}{2} \times \text{Force (F)} \times$$

Elongation (l)

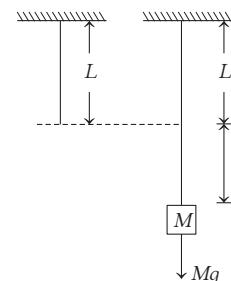
$$= \frac{1}{2} Fl = \frac{1}{2} \times Mg \times l = \frac{1}{2} Mgl$$

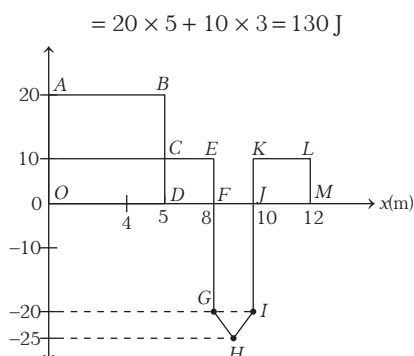
3 (c) The area under the

force-displacement curve gives the amount of work done.

From work-energy theorem, $W = \Delta KE$

$$\therefore \text{At } x = 8 \text{ m, } W = \text{Area } ABDO + \text{Area } CEFD \quad \dots(i)$$





Given, $m = 500 \text{ g} = 500 \times 10^{-3} \text{ kg}$

Using Eq. (i), we get

$$\Rightarrow 130 = \frac{1}{2}mv^2 = \frac{1}{2} \times 500 \times 10^{-3} \times v^2$$

$$\Rightarrow v = 2\sqrt{130} = 22.8 \text{ ms}^{-1} \approx 23 \text{ ms}^{-1}$$

At $x = 12 \text{ m}$,

$$W = \text{Area } ABDO + \text{Area } CEFD$$

$$+ \text{Area } FGHIJ + \text{Area } KLMJ$$

$$W = 20 \times 5 + 10 \times 3 + \left(-20 \times 2 + \frac{1}{2} \times (-5) \times 2 \right) + 10 \times 2$$

$$(\because \text{Area } FGHIJ = \text{Area } FGIJ + \text{Area } GHI)$$

$$= 100 + 30 - 40 - 5 + 20 = 105 \text{ J}$$

Using Eq. (i), we get

$$\therefore 105 = \frac{1}{2} \times 500 \times 10^{-3} \times v^2$$

$$\Rightarrow v = 2\sqrt{105} \approx 20.6 \text{ ms}^{-1}$$

- 4 (c) If x be the compression in the spring, when a block of 0.25 kg is released.

From law of conservation of energy,

Potential energy of spring = Potential energy of block

$$\frac{1}{2}kx^2 = mgx$$

where, k is force constant of spring.

$$kx = 2mg = 2 \times 0.25g$$

$$kx = 0.5g$$

$\therefore$ Force applied by the spring on the block, $F = kx = 0.5g$

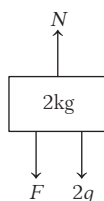
From free body diagram,

$\therefore$ Maximum force by system on the floor,

$$N = F + 2g = 0.5g + 2g = 2.5g$$

($\because g = 10 \text{ ms}^{-2}$)

$$= 25 \text{ N}$$



- 5 (a) Given, rate of decrease in kinetic energy,

$$\frac{dK}{dt} = 9.6 \text{ Js}^{-1}$$

When speed, $v = 3 \text{ ms}^{-1}$, then force $F = ?$

$$\text{Kinetic energy, } K = \frac{1}{2}mv^2$$

Differentiating above equation w.r.t. t , we get

$$\frac{dK}{dt} = \frac{1}{2}m2v \frac{dv}{dt} = vm \frac{dv}{dt} = v \frac{dp}{dt} \quad \left(\because m \frac{dv}{dt} = \frac{dp}{dt} \right)$$

$$\frac{dK}{dt} = vF \quad \left(\because \text{Force, } F = \frac{dp}{dt} \right)$$

$$\therefore F = \frac{1}{v} \cdot \frac{dK}{dt} = \frac{1}{3} \times 9.6 = 3.2 \text{ N}$$

- 6 (a) $\therefore$ Efficiency $= \frac{W_o}{W_i} \times 100$

Given, $W_o = 4000 \text{ J}$ and $W_i = 4000 + 1000 = 5000 \text{ J}$

$$\therefore \text{Efficiency } (\eta) = \frac{4000}{5000} \times 100 = 80 \%$$

- 7 (a) Relation between kinetic energy and momentum is

$$p_1 = \sqrt{2mK_1}$$

$\therefore$ Kinetic energy is increased by 4 times, then $K_2 = 4K_1$

$$\text{Hence, } p_2 = \sqrt{2mK_2} = \sqrt{2m(4K_1)}$$

$$p_2 = 2\sqrt{2mK_1} \text{ or } p_2 = 2p_1$$

- 8 (c) Net force required to lift a hook and load,

$$F_{\text{net}} = 1000 + 10000 = 11000 \text{ N}$$

$$\text{Power required to lift the hook, } P = \frac{W}{t}$$

As,

$$W = F_{\text{net}}d$$

$$\therefore P = \frac{F_{\text{net}}d}{t} = F_{\text{net}} \left(\frac{d}{t} \right) = F_{\text{net}}v \quad \left(\because v = \frac{d}{t} \right)$$

or

$$P = 11000 \times 0.5 = 5500 \text{ W} = 5.5 \text{ kW}$$

- 9 (c) When the spring is cut into pieces, they will have the new force constant. The spring is divided into 1 : 2 : 3 ratio.

When the pieces are connected in series, the resultant force constant will be given by

$$\frac{1}{k'} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

$$\frac{1}{k'} = \frac{1}{x} + \frac{1}{2x} + \frac{1}{3x} \Rightarrow k' = \frac{6x}{11}$$

In parallel, the net force constant,

$$k'' = x + 2x + 3x = 6x$$

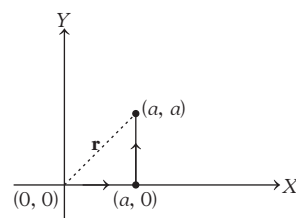
$$\text{The required ratio } \frac{k'}{k''} = \frac{6x/11}{6x} = 1:11$$

- 10 (c) Given, $\mathbf{F} = -k(\hat{\mathbf{y}} + \hat{\mathbf{x}})$

As, work done, $W = \int \mathbf{F} \cdot d\mathbf{r}$

$$\text{So, } W = -k \int (\hat{\mathbf{y}} + \hat{\mathbf{x}}) \cdot (dx\hat{\mathbf{i}} + dy\hat{\mathbf{j}})$$

$$\Rightarrow W = -k \int (ydx + xdy)$$



$$\Rightarrow W = -k \int_{(0,0)}^{(a,a)} d(xy) \quad [\because ydx + xdy = \int d(xy)]$$

$$\text{Hence, } W = -k(xy)_{(0,0)}^{(a,a)} = -ka^2$$

$$11 \text{ (d) As we know, } k = \frac{F}{l} \Rightarrow k \propto \frac{1}{l}$$

$$\Rightarrow \frac{k_2}{k_1} = \frac{l_1}{l_2} = \frac{1}{2}$$

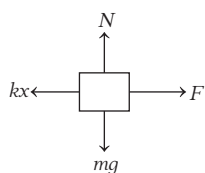
$$k_1 = 2k, k_2 = k$$

$$\text{In series, } \frac{1}{k'} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{1}{2k} + \frac{1}{k} = \frac{3}{2k}$$

$$\therefore k' = \frac{2k}{3}$$

So, both Assertion and Reason are incorrect.

12 (a) Free body diagram of block is as shown below.



Now, from law of conservation of energy, $W = \Delta K$

$$\Rightarrow W_F + W_{sp} = \frac{1}{2}mv^2$$

$$\Rightarrow F \cdot x - \frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

$$\therefore v = \sqrt{\frac{2F \cdot x - kx^2}{m}}$$

13 (d) Given, $m = 70 \text{ kg}$, $g = 10 \text{ ms}^{-2}$ and $h = 12 \text{ m}$

In going up and down once,

$$\begin{aligned} \text{Number of k-cal burnt} &= \left(mgh + \frac{mgh}{2} \right) \\ &= \frac{3}{2}mgh = \frac{3}{2} \times \frac{70 \times 10 \times 12}{4.2 \times 1000} \\ &= 3 \text{ k-cal} \end{aligned}$$

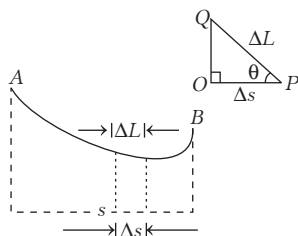
Total number of k-cal to be burnt to loose 7 kg of weight

$$= 7 \times 9000 = 63000 \text{ k-cal}$$

$\therefore$ Number of times the person has to go up and down the stairs

$$= \frac{63000}{3} = 21000 = 21 \times 10^3 \text{ times}$$

14 (a) According to question, the given situation is shown in the figure below.



For a sufficiently safe-horizontal displacement, Δs can be considered straight. If the corresponding length of path element is ΔL , the frictional force is given by $\Delta F = \mu mg(\Delta L \cos \theta)$

$$\text{From } \triangle OPQ, \cos \theta = \frac{\Delta s}{\Delta L}$$

$$\text{So, } \Delta F = \mu mg \Delta L \left(\frac{\Delta s}{\Delta L} \right) = \mu mg \Delta s$$

On adding up, we find that along the whole path, the total work done by the friction force is μmgs . By law of conservation of energy, this must equal to the decrease in potential energy of skier.

$$\therefore \mu mgs = mgh$$

$$\text{Hence, } h = \mu s$$

15 (c) According to question, a body of mass 1 kg begins to move under the action of time dependent force,

$$\mathbf{F} = (2t \hat{i} + 3t^2 \hat{j}) \text{ N}$$

$$m \frac{dv}{dt} = 2t \hat{i} + 3t^2 \hat{j}$$

$$\int dv = \int (2t \hat{i} + 3t^2 \hat{j}) dt \quad (\because m = 1 \text{ kg})$$

$$\mathbf{v} = t^2 \hat{i} + t^3 \hat{j}$$

$\therefore$ Power developed by the force at the time t will be given by

$$\begin{aligned} P &= \mathbf{F} \cdot \mathbf{v} = (2t \hat{i} + 3t^2 \hat{j}) \cdot (t^2 \hat{i} + t^3 \hat{j}) \\ &= (2t \cdot t^2 + 3t^2 \cdot t^3) \\ P &= (2t^3 + 3t^5) \text{ W} \end{aligned}$$

16 (d) From work-energy theorem,

$$\begin{aligned} W_{\text{tangential}} &= \Delta KE = K_f - K_i = K_f - 0 \\ (ma_T) \times S &= 8 \times 10^{-4} \end{aligned}$$

Here,

$$S = 2 \times 2\pi r$$

$$a_T = \frac{8 \times 10^{-4}}{10^{-2}(2\pi r \times 2)} = \frac{2 \times 10^{-2}}{\pi \times 6.4 \times 10^{-2}}$$

$$\approx 0.1 \text{ ms}^{-2}$$

17 (a) From work-energy theorem,

$$\text{Work done} = \text{Change in kinetic energy} \Rightarrow W = K_f - K_i$$

$$\Rightarrow K_f = W + K_i = \int_{x_1}^{x_2} F dx + \frac{1}{2}mv^2$$

$$= \int_{20}^{30} -0.1x dx + \frac{1}{2} \times 10 \times 10^2$$

$$= -0.1 \left[\frac{x^2}{2} \right]_{20}^{30} + 500 = -0.05 [30^2 - 20^2] + 500$$

$$= -0.05 [900 - 400] + 500$$

$$\Rightarrow K_f = -25 + 500 = 475 \text{ J}$$

18 (a) As, the machine delivers a constant power.

$$\text{So, } F \cdot v = \text{constant} = k \text{ (watts)}$$

$$\Rightarrow m \frac{dv}{dt} \cdot v = k \Rightarrow \int v dv = \frac{k}{m} \int dt$$

$$\Rightarrow \frac{v^2}{2} = \frac{k}{m} t \Rightarrow v = \sqrt{\frac{2k}{m}} t$$

Now, force on the particle is given by

$$F = m \frac{dv}{dt} = m \frac{d}{dt} \left(\frac{2kt}{m} \right)^{\frac{1}{2}}$$

$$= \sqrt{2km} \left(\frac{1}{2} t^{-\frac{1}{2}} \right) = \sqrt{\frac{mk}{2}} \cdot t^{-1/2}$$

- 19 (c) Given, force, $F = 10 + 0.5x = 10 + \frac{1}{2}x$

Let during small displacement, the work done by the force is $dW = Fdx$.

So, work done during displacement from $x = 0$ to $x = 2$ is

$$W = \int_0^x dW = \int_0^2 Fdx = \int_0^2 \left(10 + \frac{1}{2}x \right) dx$$

$$= 10 [x]_0^2 + \left[\frac{x^2}{4} \right]_0^2 = 21 \text{ J}$$

- 20 (a) Given, $m = 11.7 \text{ kg}$, $s = 4.65 \text{ m}$, $h = 2.86 \text{ m}$

From the figure, $\sin \theta = \frac{h}{s} = \frac{2.86}{4.65} = 0.615$

Also, component of mg along the incline is $mg \sin \theta$.

$\therefore$ Work done by this force $= mg \sin \theta \cdot s$

$$= 11.7 \times 9.8 \times 0.615 \times 4.65$$

$$= 11.7 \times 9.8 \times 2.86 = 327.9 \approx 328 \text{ J}$$

- 21 (d) Given, mass of elevator $= 500 \text{ kg}$

Velocity $= 0.20 \text{ ms}^{-1}$

Weight of elevator $= 500 \times 9.8 = F$

Now, power, $P = Fv = 500 \times 9.8 \times 0.20 = 980 \text{ W}$

Therefore, hp-rating of motor $= \frac{1}{746} \times 980 = 1.31 \text{ hp}$

- 22 (c) Let the kinetic energy is K and masses of bodies are m_1 and m_2 .

For first body, $p_1 = \sqrt{2m_1 K}$

$$\therefore p_1^2 = 2m_1 K \quad \dots(i)$$

$$\text{Similarly, for second body, } p_2^2 = 2m_2 K \quad \dots(ii)$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{p_1^2}{p_2^2} = \frac{2m_1 K}{2m_2 K}, \quad \frac{p_1}{p_2} = \sqrt{\frac{m_1}{m_2}}$$

$$\Rightarrow \frac{p_1}{p_2} = \text{square root of masses.}$$

- 23 (a) Given, $m_1 = 1 \text{ kg}$ and $m_2 = 2 \text{ kg}$

Let kinetic energies of bodies be K_1 and K_2 .

$$\text{For first body, } K_1 = Fs_1 \quad \dots(i)$$

$$\text{For second body, } K_2 = Fs_2 \quad \dots(ii)$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{s_1}{s_2} = \frac{K_1}{K_2} \Rightarrow \frac{s_1}{s_2} = \frac{m_1 v_1^2}{m_2 v_2^2}$$

$$\text{Given, } v_1 = v_2, \text{ so } \frac{s_1}{s_2} = \frac{m_1}{m_2}, \quad \frac{s_1}{s_2} = \frac{1}{2}$$

- 24 (d) From $v = u + at$, $v_1 = 0 + at_1$

$$F = ma = \frac{mv_1}{t_1} \quad \left(\because a = \frac{v_1}{t_1} \right)$$

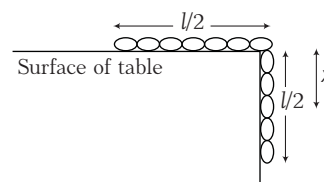
Velocity acquired in t second $= at \Rightarrow v = \frac{v_1}{t_1} t$

$$\text{Power} = Fv = \left(\frac{mv_1}{t_1} \right) \left(\frac{v_1}{t_1} t \right) = \left(\frac{mv_1^2}{t_1^2} \right) t$$

- 25 (a) Consider the diagram. Let the mass of chain be m .

$\therefore$ Mass per unit length $= \frac{m}{l}$

Consider an elementary length of the chain at a depth x below the table. The potential energy of their parts is given by



$$dU = - \frac{m}{l} g x dx$$

Now, potential energy of hanging part,

$$U_1 = \int dU = - \int_0^{l/2} \frac{m}{l} g x dx = - \frac{m}{l} g \left[\frac{x^2}{2} \right]_0^{l/2} = - \frac{mgl}{8}$$

When chain is completely leaves the table, then potential energy,

$$U_2 = - \int_0^l \frac{m}{l} g x dx = \frac{mg}{2l} [x^2]_0^l = - \frac{mgl}{2}$$

Now, loss in potential energy $= \frac{-mgl}{8} + \frac{mgl}{2}$

From law of conservation of energy,

Loss in potential energy = Gain in kinetic energy

$$\Rightarrow \frac{1}{2} mv^2 = \frac{3mgl}{8} \Rightarrow v = \sqrt{\frac{3gl}{4}}$$

- 26 (a) Given, $m = 4 \text{ kg}$, $p = 6 \text{ N-s}$

We know that, kinetic energy of a body,

$$K = \frac{p^2}{2m}, \quad K = \frac{(6)^2}{2 \times 4} \Rightarrow K = \frac{36}{2 \times 4} = 4.5 \text{ J}$$

- 27 (a) Given, $m = 3.90 \text{ kg}$, $v = 1.20 \text{ ms}^{-1}$

and k for spring $= 135 \text{ Nm}^{-1}$

From law of conservation of energy,

$$\frac{1}{2} mv^2 = \frac{1}{2} kx^2 \Rightarrow x = \sqrt{\frac{mv^2}{k}}$$

$$= \sqrt{\frac{3.90 \times 1.20 \times 1.20}{135}}$$

$$= 0.204 \text{ m}$$

- 28 (a) Work done in stretching a string to obtain an extension l ,

$$W_1 = \frac{1}{2} kl^2$$

Similarly, work done is stretching a string to obtain extension $(l + l_1)$, $W_2' = \frac{1}{2} k(l_1 + l)^2$

Now, work done in second stretching,

$$W_2 = W_2' - W_1 = \frac{1}{2} k(l_1 + l)^2 - \frac{1}{2} kl^2 = \frac{1}{2} kl_1(2l + l_1)$$

29 (b) For an instantaneous displacement dx , we can write

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \left(\frac{d\mathbf{x}}{dt} \right) = \mathbf{F} \cdot \mathbf{v}$$

$$P = (F)(at) = (ma)(at) \quad (\because F = ma) \\ = (ma^2)t \Rightarrow P \propto t \quad (\because ma^2 = \text{constant})$$

30 (c) Given, $t_1 = 2\text{ s}$, $t_2 = 4\text{ s}$ and $h_1 = h_2 = h$

$$\text{As, } P_A = \frac{mgh_1}{t_1} \quad \dots(i)$$

$$\text{and } P_B = \frac{mgh_2}{t_2} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$P_A : P_B = \frac{mgh_1/t_1}{mgh_2/t_2} = \left(\frac{h_1}{h_2} \right) \left(\frac{t_2}{t_1} \right) = \frac{t_2}{t_1} = \frac{4}{2} = \frac{2}{1} \quad (\because h_1 = h_2)$$

$$\Rightarrow P_A : P_B = 2 : 1$$

31 (d) Power of the machine gun

$$= \frac{\text{Total work done}}{\text{Time}} = \frac{n \cdot \frac{1}{2} mv^2}{t} \\ = n \cdot \frac{1}{2} \frac{mv^2}{t} \quad \left(\because K = \frac{1}{2} mv^2, t = 1\text{ s} \right)$$

$\therefore$ The power of the machine gun $= nK$

32 (a) Given, force, $\mathbf{F} = (3\hat{i} + \hat{j})\text{ N}$

$$\mathbf{r}_1 = (2\hat{i} + \hat{k})\text{ m and } \mathbf{r}_2 = (4\hat{i} + 3\hat{j} - \hat{k})\text{ m}$$

$$\therefore \mathbf{s} = \mathbf{r}_2 - \mathbf{r}_1 = (4\hat{i} + 3\hat{j} - \hat{k}) - (2\hat{i} + \hat{k})$$

$$= (2\hat{i} + 3\hat{j} - 2\hat{k})\text{ m}$$

$$\therefore W = \mathbf{F} \cdot \mathbf{s} = (3\hat{i} + \hat{j}) \cdot (2\hat{i} + 3\hat{j} - 2\hat{k})$$

$$= 3 \times 2 + 1 \times 3 + 0 = 6 + 3 = 9\text{ J}$$

33 (a) Power $= \frac{\text{Work done}}{\text{Time}}$

$$\text{Work done} = mgh = 100 \times 9.8 \times 10$$

$$\text{Time} = 1\text{ min} = 60\text{ s}$$

$$\therefore P = \frac{100 \times 9.8 \times 10}{60} = 163.3\text{ W}$$

34 (c) Work done, $W = mg \sin \theta \times s$

$$= 300 \times 9.8 \times \sin 30^\circ \times 10$$

$$= 300 \times 9.8 \times \frac{1}{2} \times 10 = 14700\text{ J}$$

35 (b) Work done in a stretched wire, $W = \frac{1}{2} kx^2$

$$\text{Given, } k_A = k \text{ and } k_B = 2k$$

$$\Rightarrow W_A = \frac{1}{2} kx^2$$

$$\text{and } W_B = \frac{1}{2} (2k)x^2 = kx^2$$

Hence, the ratio of work done in stretching the wires,

$$\frac{W_A}{W_B} = \frac{(1/2)kx^2}{kx^2} \Rightarrow \frac{W_A}{W_B} = \frac{1}{2}$$

36 (a) Work done to accelerate from 0 to v , $W_1 = \frac{1}{2} mv^2$

$$\text{Work done to accelerate from } v \text{ to } 2v, W_2 = \frac{1}{2} m (2v)^2$$

$$\Rightarrow \Delta W = W_2 - W_1 = \frac{1}{2} m (2v)^2 - \frac{1}{2} mv^2 \\ = 4W_1 - W_1 = 3W_1$$

37 (b) Given, $m = 200\text{ g} = 0.2\text{ kg}$, $g = 10\text{ ms}^{-2}$ and

$$H_{\text{eff}} = 2 + 0.5 = 2.5\text{ m}$$

$$\text{Here, } \frac{1}{2} kx^2 = mg H_{\text{eff}} \Rightarrow k = \frac{2mg H_{\text{eff}}}{x^2}$$

$$\Rightarrow k = \frac{2 \times 0.2 \times 10 \times 2.5}{(0.5)^2} \Rightarrow k = \frac{10}{2.5} = 40\text{ Nm}^{-1}$$

38 (b) Given, the potential energy of a particle in a force field,

$$U = \frac{A}{r^2} - \frac{B}{r}$$

$$\text{For stable equilibrium, } F = - \frac{dU}{dr} = 0$$

$$0 = - \frac{2A}{r^3} + \frac{B}{r^2} \quad \text{or} \quad \frac{2A}{r} = B$$

$$\text{The distance of particle from the centre of the field, } r = \frac{2A}{B}$$

39 (b) $W = \text{KE} = \frac{1}{2} mv^2 = \frac{1}{2} \times 50 \times (20)^2 = 10^4\text{ J}$

40 (b) Kinetic energy of a particle in motion, $E = \frac{1}{2} mv^2$

Differentiating above equation w.r.t. x , we get

$$\frac{dE}{dx} = \frac{1}{2} m \times 2v \frac{dv}{dx} = mv \times \frac{dv}{dt} \times \frac{dt}{dx} = mv \times \frac{a}{v} = ma$$

So, the slope of kinetic energy-displacement curve is directly proportional to the acceleration of the particle.

41 (b) According to law of conservation of momentum,

$$mv_1 + 2mv_2 = 0 \quad \text{or} \quad v_2 = - \frac{v_1}{2} \quad \dots(i)$$

According to law of conservation of energy,

$$\frac{1}{2} mv_1^2 + \frac{1}{2} 2mv_2^2 = 60$$

$$\Rightarrow \frac{1}{2} mv_1^2 + m \left(\frac{-v_1}{2} \right)^2 = 60$$

$$\Rightarrow \frac{1}{2} mv_1^2 + \frac{1}{4} mv_1^2 = 60$$

$$\Rightarrow \frac{3mv_1^2}{4} = 60$$

$$\Rightarrow \frac{1}{2} mv_1^2 = 40\text{ J}$$

CHAPTER 07

Circular Motion

Everyday in different activities of our daily life, we use circular motion. For example, to grind wheat from a flour mill, its wheel is given circular motion; to wash clothes, parts of a washing machine perform circular motion. Similarly, some natural phenomena like motion of the earth around the sun, motion of moon around the earth, etc., all are examples of circular motion. In this chapter, we will study circular motion in detail.

CIRCULAR MOTION AND ITS TYPES

In circular motion, an object moves along the perimeter or circumference of a circle.

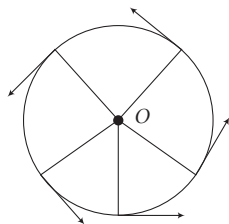


Fig. 7.1 Circular motion

Circular motion can be categorised into two types as follows

Uniform circular motion If a particle moves along a circular path with a constant speed (*i.e.* it covers equal distances along the circumference of the circle in equal intervals of time), then its motion is said to be uniform circular motion.

e.g. (i) Motion of a point on the rim of a wheel rotating uniformly.

(ii) Motion of the tip of the second hand of a clock.

Non-uniform circular motion If the speed of the particle in circular motion changes with respect to time, then its motion is said to be non-uniform circular motion.

e.g. Motion of a stone tied to a string moving in vertical circle.

KINEMATICS OF CIRCULAR MOTION

For a particle in circular motion, following variables are needed to describe its motion

Radius vector (r)

When a particle moves on a circular path, its distance from the centre is fixed and it is equal to radius of the circle. If the centre of the circle is taken as origin, then

Inside

- 1 Circular motion and its types
- 2 Kinematics of circular motion
 - Centripetal acceleration
 - Acceleration of a particle in non-uniform circular motion
- 3 Dynamics of circular motion
 - Centripetal force
 - Centrifugal force
 - Examples for obtaining centripetal force in daily life
 - Motion of a particle tied to a string in vertical circle

the vector joining centre to the particle is called radius vector. It is directed from centre to the particle and its magnitude is same as radius.

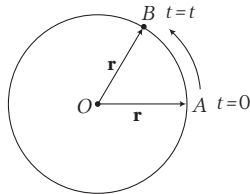


Fig. 7.2 Radius vector in circular motion

Angular position

Suppose a particle P is moving on a circular path of radius r and centre O , as shown in figure.

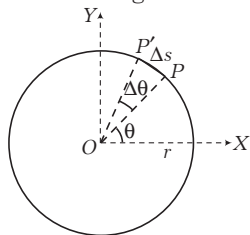


Fig. 7.3 Moving particle on a circle

The position of the particle P at a given instant may be described by the angle θ between OP and OX . This angle θ is called the angular position of the particle. As the particle moves on the circle, its angular position θ changes. Here, P and P' are given as $P(r, \theta)$ and $P'(r, \theta + \Delta\theta)$, respectively.

Angular displacement ($\Delta\theta$)

The angle traced out by the radius vector at the centre of the circular path in the given time is called angular displacement. It is denoted by $\Delta\theta$ and expressed in radians. In the Fig. 7.3, OP is the initial and OP' is the final position vectors of the particle.

Angular displacement

$$= \frac{\text{Linear displacement between two positions}}{\text{Radius vector}}$$

$$\Delta\theta = \frac{\Delta s}{r}$$

Small angular displacement is taken as a vector quantity and its SI unit is radian.

Direction of angular displacement Direction of $\Delta\theta$ is given by right handed (*i.e.* the direction, where screw advances) screw rule as shown in figure.

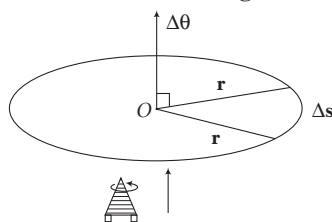


Fig. 7.4 Direction of advancement of screw

It is directed along a line passing through centre (O) and perpendicular to the plane of circular motion, containing r and Δs .

Note If a particle makes N revolutions, its angular displacement is

$$\theta = 2\pi N \text{ rad}$$

Angular velocity (ω)

Rate of change of angular displacement of a particle performing circular motion is called angular velocity.

$$\text{Angular velocity, } \omega = \frac{\Delta\theta}{\Delta t}$$

$$\text{Instantaneous angular velocity, } \omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

It is also a vector quantity. Its unit is rads^{-1} , rpm, rps, etc., and its dimensional formula is $[M^0 L^0 T^{-1}]$.

Direction of angular velocity Direction of angular velocity is given in the same way as that of angular displacement.

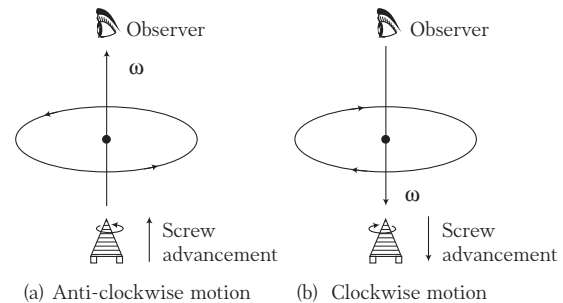


Fig. 7.5 Directions of angular velocities

Let the particle rotates in anti-clockwise direction as seen by observer, angular velocity is along a line passing through the centre and perpendicular to the plane of the circle and towards the observer. This is along a direction, where screw advances as shown in figure.

Similarly, when the particle rotates in clockwise direction as seen by the observer, angular velocity is away from the observer.

Important points regarding the angular velocity are given below

- (i) If the particle performing circular motion completes one rotation around the circular path in T seconds, then

$$\text{Angular velocity, } \omega = \frac{2\pi}{T} \text{ rad s}^{-1}$$

- (ii) If the particle performing circular motion makes n rotations per second around the circular path, then
Angular velocity, $\omega = 2\pi n \text{ rad s}^{-1}$

- (iii) If two particles are moving on the same circle in the same direction with different constant angular speeds ω_1 and ω_2 , then the angular speed of particle 2 with respect to 1 for an observer at centre will be

$$\omega = \omega_2 - \omega_1$$

(iv) Angular velocity depends on axis of rotation.

$$\omega_O = \frac{\alpha}{t}$$

$$\omega_P = \frac{\theta}{t} = \frac{\alpha/2}{t} = \frac{\omega_O}{2}$$

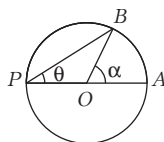


Fig. 7.6

Example 7.1 Calculate the average angular velocity of the hour hand of a clock.

Sol. The hour hand completes one round in 12 h. One round makes an angular displacement 2π .

$$\therefore \text{Average angular velocity, } \omega_{av} = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{ rad}}{12 \text{ h}}$$

$$= \frac{2\pi}{12 \times 3600} \text{ rad s}^{-1}$$

$$= \frac{\pi}{21600} \text{ rad s}^{-1}$$

Example 7.2 An object revolves uniformly in a circle of diameter 0.80 m and completes 100 rev min^{-1} . Find its time period and angular velocity.

Sol. Here, diameter = 0.80 m

$$\therefore \text{Radius, } r = \frac{\text{Diameter}}{2} = \frac{0.80}{2} = 0.4 \text{ m}$$

$$\text{Frequency, } n = 100 \text{ rev/min} = \frac{100}{60} \text{ rev/s}$$

$$\text{Therefore, time period, } T = \frac{1}{n} = \frac{60}{100} \text{ s} = 0.6 \text{ s}$$

$$\therefore \text{Angular velocity, } \omega = \frac{2\pi}{T} = \frac{2 \times 3.14}{0.6} \text{ rad/s}$$

$$= 10.467 \text{ rad/s}$$

Example 7.3 A threaded rod with 12 turns per cm and diameter 1.18 cm is mounted horizontally. A bar with a threaded hole to match the rod is screwed onto the rod. The bar spins at the rate of 216 rpm. Determine the velocity of the bar with which it will move the length of 1.50 cm along the rod. Also, find the time taken by it.

Sol. Given, frequency, $n = 216 \text{ rpm} = \frac{216}{60} \text{ rps}$

$$\text{Length of one turn} = \frac{1}{12} \text{ cm}$$

$\therefore$ Number of rotations required to move a distance of 1.5 cm,

$$N = \frac{\text{Distance}}{\text{Length of one turn}} = \frac{1.5}{1/12} = 18$$

Therefore, angular displacement, $\theta = 2\pi N = 2\pi \times 18 = 36\pi \text{ rad}$

$$\therefore \text{Angular velocity of the bar, } \omega = 2\pi n = 2\pi \times \frac{216}{60}$$

$$= 7.2\pi \text{ rad s}^{-1}$$

$$\therefore \text{Required time (t)} = \frac{\text{Angular displacement } (\theta)}{\text{Angular velocity } (\omega)}$$

$$= \frac{36\pi}{7.2\pi} = 5 \text{ s}$$

Relation between linear velocity and angular velocity

A particle performing circular motion also has linear velocity (as it covers linear displacement along circular path) along with angular velocity. If linear velocity of particle performing circular motion is v and angular velocity is ω , then both of these velocities are related as

$$v = r\omega$$

where, r is radius of circular path. In vector form,

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

Linear velocity is always along the tangent to the circular path.

Example 7.4 A particle moves in a circle of radius 4 m with a linear velocity of 20 ms^{-1} . Find the angular velocity.

Sol. Given, linear velocity, $v = 20 \text{ ms}^{-1}$

Radius, $r = 4 \text{ m}$

As, linear velocity, $v = r\omega$

$$\Rightarrow \text{Angular velocity, } \omega = \frac{v}{r} = \frac{20}{4} = 5 \text{ rad s}^{-1}$$

Example 7.5 If the length of the second's hand in a stop clock is 3 cm, find the angular velocity and linear velocity of the tip.

Sol. Given, radius, $r = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$

Time period of stop clock, $T = 60 \text{ s}$

$$\text{Angular velocity, } \omega = \frac{2\pi}{T} = \frac{2\pi}{60} = 0.1047 \text{ rad s}^{-1}$$

$$\text{and linear velocity, } v = \omega r = 0.1047 \times 3 \times 10^{-2}$$

$$= 0.00314 \text{ ms}^{-1}$$

Angular acceleration (α)

The rate of change of angular velocity of a particle performing circular motion is called angular acceleration.

$$\text{Angular acceleration, } \alpha = \frac{\Delta\omega}{\Delta t}$$

Instantaneous angular acceleration,

$$\alpha_{ins} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

The SI unit of angular acceleration is rad s^{-2} and its dimensional formula is $[M^0 L^0 T^{-2}]$. If $\alpha = 0$, circular motion is said to be uniform.

It has same characteristics as that of angular velocity.

Note Angular acceleration is an axial vector, when axis of rotation is fixed, angular acceleration and angular velocity vectors both lie along that axis.

Example 7.6 A point on the rim of a disc starts circular motion from rest and after time t , it gains an angular acceleration which is given by $\alpha = 3t - t^2$. Calculate the angular velocity after 2 s.

Sol. Angular acceleration, $\alpha = \frac{d\omega}{dt} = 3t - t^2$

$$\Rightarrow \int_0^{\omega} d\omega = \int_0^t (3t - t^2) dt \Rightarrow \omega = \frac{3t^2}{2} - \frac{t^3}{3}$$

$$\text{At } t = 2 \text{ s, } \omega = \frac{10}{3} \text{ rad s}^{-1}$$

Relation between angular variables

(Kinematic equations of circular motion)

If angular acceleration is constant, then we have kinematic equations of circular motion as follows

$$(i) \theta = \omega_0 t + \frac{1}{2} \alpha t^2 \quad (ii) \omega = \omega_0 + \alpha t$$

$$(iii) \omega^2 = \omega_0^2 + 2\alpha\theta \quad (iv) \theta_t = \omega_0 t + \frac{1}{2} \alpha (2t - 1)$$

$$(v) \theta = \left(\frac{\omega + \omega_0}{2} \right) t$$

Here, ω_0 and ω are the angular velocities at time $t = 0$ and t ; θ and θ_t are the angular displacements in time t and t th second, respectively.

Example 7.7 The wheel of a motor rotates with a constant acceleration of 4 rad s^{-2} . If the wheel starts from rest, how many revolutions will it make in the first 20 s?

Sol. The angular displacement in the first 20 s is given by

$$\begin{aligned} \theta &= \omega_0 t + \frac{1}{2} \alpha t^2 = \frac{1}{2} (4 \text{ rad s}^{-2})(20 \text{ s})^2 \\ &= 800 \text{ rad} \quad (\because \text{Angular velocity, } \omega_0 = 0) \end{aligned}$$

As, the wheel turns by 2π radian in each revolution, the

$$\begin{aligned} \text{number of revolutions in 20s is } N &= \frac{\theta}{2\pi} = \frac{800}{2\pi} \\ &= 127.38 \approx 127 \end{aligned}$$

Example 7.8 The wheel of a car accelerated uniformly from rest, rotates through 1.5 rad during the first second. Find the angle rotated during the next second.

Sol. As the angular acceleration is constant, we have

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2 = \frac{1}{2} \alpha t^2 \quad (\because \text{Angular velocity, } \omega_0 = 0)$$

$$\therefore 1.5 \text{ rad} = \frac{1}{2} \alpha (1)^2$$

Angular acceleration, $\alpha = 3 \text{ rad s}^{-2}$

The angular displacement in first two second is given by

$$\theta_1 = \frac{1}{2} \alpha (2)^2 = 6 \text{ rad}$$

Thus, the angle rotated during the 2nd second

$$= \theta_1 - \theta = 6 \text{ rad} - 1.5 \text{ rad} = 4.5 \text{ rad}$$

Centripetal acceleration

When a particle is in uniform circular motion, its speed remains constant but velocity changes continuously

because of change in direction of motion. Therefore, motion of the particle is accelerated. This acceleration is called centripetal acceleration and it is directed along radial direction towards centre of the circle. It is given by

$$a_r = a_c = \frac{v^2}{r}$$

where, v = magnitude of linear velocity of the particle and r = radius of circular path.

It is also called **radial acceleration**.

If we put, $v = r\omega$

where, ω is angular velocity, then centripetal acceleration,

$$a_r = a_c = r\omega^2$$

Example 7.9 Determine the magnitude of centripetal acceleration of a particle on the tip of a fan blade, 0.30 m in diameter, rotating at $1200 \text{ rev min}^{-1}$.

Sol. Given, diameter = 0.30 m

$$\therefore \text{Radius, } r = \frac{0.30}{2} = 0.15 \text{ m}$$

$$\text{and frequency, } n = 1200 \text{ rev min}^{-1} = \frac{1200}{60} = 20 \text{ rps}$$

$$\therefore \text{Angular velocity, } \omega = 2\pi n = 2\pi \times 20 = 40\pi \text{ rad s}^{-1}$$

Therefore, centripetal acceleration, $a_c = r\omega^2$

$$\Rightarrow a_c = 0.15 \times (40\pi)^2 \Rightarrow a_c = 2368.7 \text{ ms}^{-2}$$

Example 7.10 Find the acceleration of a particle placed on the surface of the earth at the equator, due to the earth's rotation. The radius of earth is 6400 km and time period of revolution of the earth about its axis is 24 h.

Sol. Given, radius of earth, $R_e = 6400 \text{ km} = 6400 \times 10^3 \text{ m}$

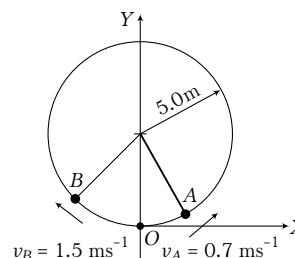
$$\text{Time period, } T = 24 \text{ h} = 24 \times 60 \times 60 \text{ s}$$

$$\text{Angular speed of the earth, } \omega = \frac{2\pi}{T} = \frac{2\pi}{24 \times 60 \times 60} \text{ rad s}^{-1}$$

Acceleration of the particle,

$$a_c = \omega^2 R_e = \left(\frac{2\pi}{24 \times 60 \times 60} \right)^2 \times (6400 \times 10^3) = 0.034 \text{ ms}^{-2}$$

Example 7.11 Two particles A and B start at the origin O and travel in opposite directions along the circular path at constant speeds $v_A = 0.7 \text{ ms}^{-1}$ and $v_B = 1.5 \text{ ms}^{-1}$, respectively. Determine the time when they collide and the magnitude of the acceleration of B just before this happens.



Sol. From the condition given in the question, it is clear that total distance (= velocity $\times$ time) will be equal to the circumference of circular path,

i.e. $v_B t + v_A t = \text{circumference of circular path}$

$$1.5t + 0.7t = 2\pi R = 10\pi$$

$$\therefore \text{Time, } t = \frac{10\pi}{2.2} = 14.3 \text{ s}$$

$$\text{Hence, acceleration, } a = \frac{v_B^2}{R} = \frac{(1.5)^2}{5} = 0.45 \text{ ms}^{-2}$$

Acceleration of a particle in non-uniform circular motion

If a particle is in non-uniform circular motion, i.e. its speed is not constant, then the particle has both radial and tangential components of acceleration.

Radial component (a_r)

This component of acceleration is towards the centre. This is responsible for change in direction of velocity. This is equal to $\frac{v^2}{r}$ or $r\omega^2$.

Thus,

$$a_r = \frac{v^2}{r} = r\omega^2$$

Note The radial acceleration (a_r) is also sometimes called **normal acceleration** (a_n).

Tangential component (a_t)

This is the component of acceleration in the direction of velocity, which is responsible for change in speed of particle. It is also equal to rate of change of speed.

Hence, $a_t = \text{component of } \mathbf{a} \text{ along } \mathbf{v}$

$$= \frac{dv}{dt} = \frac{d|\mathbf{v}|}{dt}$$

$$\Rightarrow \mathbf{a}_t = \frac{d\omega}{dt} \times \mathbf{r} \quad (\because \mathbf{v} = \omega \times \mathbf{r})$$

$$\Rightarrow \mathbf{a}_t = \alpha \times \mathbf{r}$$

$$\Rightarrow a_t = r\alpha$$

This component is tangential.

Example 7.12 A particle moves in a circle of radius 0.5 m at a speed that uniformly increases. Find the angular acceleration of particle, if its speed changes from 2 ms^{-1} to 4 ms^{-1} in 4 s.

Sol. The tangential acceleration of the particle is

$$a_t = \frac{dv}{dt} = \frac{4 - 2}{4} = 0.5 \text{ ms}^{-2}$$

$$\text{The angular acceleration, } \alpha = \frac{a_t}{r} = \frac{0.5}{0.5} = 1 \text{ rad s}^{-2}$$

Net acceleration

Consider the particle moving on circular path in anti-clockwise direction with increasing speed as shown in figure.

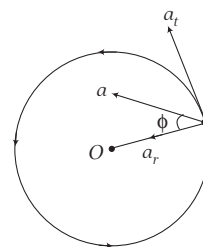


Fig. 7.7 Non-uniform circular motion

We know that, linear velocity, $\mathbf{v} = \omega \times \mathbf{r}$

Differentiating on both sides w.r.t. time t , we get

$$\text{Net acceleration, } \frac{d\mathbf{v}}{dt} = \frac{d\omega}{dt} \times \mathbf{r} + \omega \times \frac{d\mathbf{r}}{dt}$$

$$\Rightarrow \text{Net acceleration, } \mathbf{a} = \alpha \times \mathbf{r} + \omega \times \mathbf{v} \Rightarrow \boxed{\mathbf{a} = \mathbf{a}_t + \mathbf{a}_r}$$

$$(\because \mathbf{a}_t = \alpha \times \mathbf{r} \text{ and } \mathbf{a}_r = \omega \times \mathbf{v})$$

$$\text{Magnitude of net acceleration, } a = \sqrt{a_r^2 + a_t^2}$$

This resultant acceleration makes an angle ϕ with the

$$\text{radius, where } \tan \phi = \frac{a_t}{a_r}$$

Note

- (i) In accelerated circular motion, dv/dt is positive and hence, tangential acceleration of the particle is parallel to velocity $\mathbf{v}$.
- (ii) In decelerated circular motion, dv/dt is negative and hence, tangential acceleration is anti-parallel to velocity $\mathbf{v}$.

Example 7.13 A car is travelling along a circular curve that has a radius of 50 m. If its speed is 16 ms^{-1} and is increasing uniformly at 8 ms^{-2} , determine the magnitude of its acceleration at this instant.

Sol. Given, tangential acceleration $a_t = 8 \text{ ms}^{-2}$

Radius, $R = 50 \text{ m}$, speed, $v = 16 \text{ ms}^{-1}$

$$\therefore \text{Radial acceleration, } a_r = \frac{v^2}{R} = \frac{(16)^2}{50} = \frac{256}{50} \text{ ms}^{-2}$$

Magnitude of net acceleration of the car,

$$a = \sqrt{a_t^2 + a_r^2} = \sqrt{(8)^2 + \left(\frac{256}{50}\right)^2} = 9.5 \text{ ms}^{-2}$$

Example 7.14 The speed of a particle moving in a circle of radius $r = 2 \text{ m}$ varies with time t as $v = t^2$, where t is in second and v in ms^{-1} . Find the radial, tangential and net acceleration at $t = 2 \text{ s}$.

Sol. Given in the question, $v = t^2$

Linear speed of particle at $t = 2 \text{ s}$ is $v = (2)^2 = 4 \text{ ms}^{-1}$

$$\therefore \text{Radial acceleration, } a_r = \frac{v^2}{r} = \frac{(4)^2}{2} = 8 \text{ ms}^{-2}$$

The tangential acceleration is $a_t = \frac{dv}{dt} = \frac{d}{dt}(t^2) = 2t$

∴ Tangential acceleration at $t = 2$ s is $a_t = (2)(2) = 4 \text{ ms}^{-2}$

∴ Net acceleration of particle at $t = 2$ s is

$$a = \sqrt{(a_r)^2 + (a_t)^2} = \sqrt{(8)^2 + (4)^2} = \sqrt{64 + 16}$$

or $a = \sqrt{80} \text{ ms}^{-2}$

Example 7.15 A particle moves in a circle of radius 2 cm at a speed given by $v = 4t$, where v is in cms^{-1} and t in second.

(i) Find the tangential acceleration at $t = 1$ s.

(ii) Find total acceleration at $t = 1$ s.

Sol. Given, $v = 4t$

$$\text{Radial acceleration, } a_r = \frac{v^2}{R} = \frac{(4t)^2}{R} \text{ or } a_r = \frac{16t^2}{2} = 8t^2$$

$$\text{At } t = 1 \text{ s, } a_r = 8 \text{ cms}^{-2}$$

(i) Tangential acceleration, $a_t = \frac{dv}{dt}$ or $a_t = \frac{d}{dt}(4t) = 4 \text{ cms}^{-2}$

i.e. a_t is constant or tangential acceleration at $t = 1$ s is 4 cms^{-2} .

(ii) Total acceleration, $a = \sqrt{a_t^2 + a_r^2}$

$$\text{or } a = \sqrt{(4)^2 + (8)^2} = \sqrt{80} = 4\sqrt{5} \text{ cms}^{-2}$$

Example 7.16 A cyclist is riding with a speed of 18 kmh^{-1} .

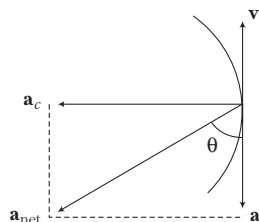
As he approaches a circular turn on the road of radius $25\sqrt{2} \text{ m}$, he applies brakes and reduces his speed at the

constant rate of 0.5 ms^{-1} . Determine the magnitude and direction of the net acceleration of the cyclist on the circular turn.

Sol. Linear speed of the cyclist, $v = 18 \text{ km h}^{-1} = 18 \times \frac{5}{18} = 5 \text{ ms}^{-1}$

and centripetal acceleration of the cyclist,

$$a_c = \frac{v^2}{R} = \frac{25}{25\sqrt{2}} = \frac{1}{\sqrt{2}} \text{ ms}^{-2}$$



Tangential acceleration of the cyclist, $a_t = \frac{dv}{dt} = \frac{1}{2} \text{ ms}^{-2}$

∴ Net acceleration of the cyclist,

$$a_{\text{net}} = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{\sqrt{3}}{2} = 0.86 \text{ ms}^{-2},$$

$$\tan \theta = \frac{a_c}{a_t} = \frac{1/\sqrt{2}}{1/2} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Angle made by resultant acceleration with tangential acceleration,

$$\theta = \tan^{-1}(\sqrt{2})$$

CHECK POINT 7.1

- The angular speed of a flywheel making 120 rev min^{-1} is
(a) $2\pi \text{ rad s}^{-1}$ (b) $4\pi^2 \text{ rad s}^{-1}$
(c) $\pi \text{ rad s}^{-1}$ (d) $4\pi \text{ rad s}^{-1}$
- The ratio of angular speeds of minute hand and hour hand of a watch is
(a) 1 : 12 (b) 6 : 1 (c) 12 : 1 (d) 1 : 6
- The wheel of a toy car rotates about a fixed axis. It slows down from 400 rps to 200 rps in 2 s. Then, its angular retardation (in rad s^{-2}) is (rps = revolutions per second)
(a) 200π (b) 100π
(c) 400π (d) None of these
- A wheel is rotating at 900 rpm about its axis. When the power is cut off, it comes to rest in 1 min. The angular retardation (in rad s^{-2}) is
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{8}$
- The motor of an engine is rotating about its axis with an angular velocity of 100 rev min^{-1} . It comes to rest in 15 s after being switched off, assuming constant angular deceleration. What is the number of revolutions made by it before coming to rest?
(a) 12.5 (b) 40 (c) 32.6 (d) 15.6
- A body is moving in a circular path with acceleration a . If its speed gets doubled, find the ratio of centripetal acceleration after and before the speed is changed.
(a) 1 : 4 (b) 1 : 2 (c) 2 : 1 (d) 4 : 1
- The circular orbit of two satellites have radii r_1 and r_2 respectively ($r_1 < r_2$). If angular velocities of satellites are same, then their centripetal accelerations are related as
(a) $a_1 > a_2$ (b) $a_1 = a_2$
(c) $a_1 < a_2$ (d) Data insufficient
- A particle is moving on a circular track of radius 30 cm with a constant speed of 6 ms^{-1} . Its acceleration is
(a) zero (b) 120 ms^{-2} (c) 1.2 ms^{-2} (d) 36 ms^{-2}
- A particle starts moving along a circle of radius $(20/\pi) \text{ m}$ with constant tangential acceleration. If velocity of the particle is 50 m/s at the end of the second revolution after motion has began, the tangential acceleration (in ms^{-2}) is
(a) 1.6 (b) 4 (c) 15.6 (d) 31.25
- Let a_r and a_t represent radial and tangential accelerations. The motion of a particle may be circular, if
(a) $a_r = 0, a_t = 0$ (b) $a_r = 0, a_t \neq 0$
(c) $a_r \neq 0, a_t = 0$ (d) None of these

11. A point starts from rest and moves along a circular path with a constant tangential acceleration. After one rotation, the ratio of its radial acceleration to its tangential acceleration will be equal to
 (a) 1 (b) 2π (c) $\frac{1}{2}\pi$ (d) 4π
12. A particle is moving on a circular path of 10 m radius. At any instant of time, its speed is 5 ms^{-1} and the speed is increasing at a rate of 2 ms^{-2} . At this instant, the magnitude of the net acceleration will be
 (a) 3.2 ms^{-2} (b) 2 ms^{-2} (c) 2.5 ms^{-2} (d) 4.3 ms^{-2}
13. A point on the rim of a flywheel has a peripheral speed of 10 ms^{-1} at an instant when it is decreasing at the rate of 60 ms^{-2} . If the magnitude of the total acceleration of the point at this instant is 100 ms^{-2} , the radius of the flywheel is
 (a) 1.25 m (b) 12.5 m (c) 25 m (d) 2.5 m
14. A particle moves in a circular path of radius R with an angular velocity $\omega = a - bt$, where a and b are positive constants and t is time. The magnitude of the acceleration of the particle after time $\frac{2a}{b}$ is
 (a) $\frac{a}{R}$ (b) $a^2 R$ (c) $R(a^2 + b)$ (d) $R\sqrt{a^4 + b^2}$
15. The distance of a particle moving on a circle of radius 12 m measured from a fixed point on the circle is given by $s = 2t^3$ (in metre). The ratio of its tangential to centripetal acceleration at $t = 2 \text{ s}$ is
 (a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 3 : 1
16. A body is moving on a circle of radius 80 m with a speed 20 m/s which is decreasing at the rate 5 ms^{-2} at an instant. The angle made by its acceleration with its velocity is
 (a) 45° (b) 90° (c) 135° (d) 0°

DYNAMICS OF CIRCULAR MOTION

In this section, we will start with the forces in circular motion and further we will discuss their practical utilisation in the applications of circular motion.

Centripetal force

When a body moves along a circular path with uniform speed, its direction changes continuously, *i.e.* velocity keeps on changing on account of a change in direction. According to Newton's first law of motion, a change in the direction of motion of the body can take place only if some external force acts on the body.

Thus, a particle performing circular motion is acted upon by a force directed along the radius towards the centre of the circle, this force is called the centripetal force.

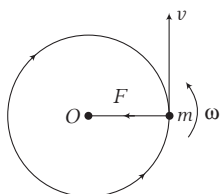


Fig. 7.8 Centripetal force on the particle

If m is mass of the particle performing circular motion as shown in Fig. 7.8, then magnitude of centripetal force is given by

Centripetal force = Mass $\times$ Centripetal acceleration

$$\therefore F = m \left(\frac{v^2}{r} \right) \quad (\text{In magnitude})$$

Putting $v = r\omega$, we get

$$F = mr\omega^2 \quad \text{or} \quad F = mv\omega$$

Putting

$$\omega = \frac{2\pi}{T}$$

or

$$\omega = 2\pi n$$

$$F = 2\pi m v n$$

$\Rightarrow$

$$F = \frac{2\pi m v}{T}$$

In vector form, centripetal force is given by

$$\mathbf{F} = -\frac{mv^2}{r} \hat{\mathbf{r}} = -m\omega^2 \mathbf{r}$$

Centripetal Force in Different Situations

Situation	The centripetal force
A particle tied to a string and whirled in a horizontal circle	Tension in the string
Vehicle taking a turn on a level road	Frictional force exerted by the road on the tyres
A vehicle on a speed breaker	Weight of the body or a component of weight
Revolution of earth around the sun	Gravitational force exerted by the sun
Electron revolving around the nucleus in an atom	Coulomb attraction exerted by the protons on electrons
A charged particle describing a circular path in a magnetic field	Magnetic force exerted by the magnetic field

Few important points related to circular motion

- (i) In non-uniform circular motion, the particle simultaneously possesses two forces

(a) Centripetal force, $F_c = ma_c = \frac{mv^2}{r} = mr\omega^2$

(b) Tangential force, $F_t = ma_t$

$$\therefore \text{Net force, } F_{\text{net}} = ma = m\sqrt{a_c^2 + a_t^2}$$

(ii) If a moving particle comes to stand still, then the particle will move along the radius towards the centre and if radial acceleration a_r is zero, the body will fly off along the tangent. So, a tangential velocity and a radial acceleration (hence, force) is a must for uniform circular motion.

(iii) The work done by the centripetal force is always zero as it is perpendicular to velocity and displacement.

Further, by work-energy theorem,

$$\begin{aligned} \text{Work done} &= \text{Change in kinetic energy} \\ \therefore \Delta K &= 0 \quad (\because \Delta W = 0) \\ \text{i.e. } K \text{ (kinetic energy) remains constant.} \end{aligned}$$

$$\text{Power} = \frac{\text{Work}}{\text{Time}} = \frac{0}{t} = 0$$

$$\text{or Power} = \mathbf{F}_c \cdot \mathbf{v} = F_c v \cos 90^\circ = 0$$

(iv) Total work done, $W = \mathbf{F}_t \cdot \mathbf{s} = F_t s \cos 0^\circ = F_t s$
(where, s is distance travelled by the particle)

Example 7.17 A ball of mass 0.25 kg attached to the ends of a string of length 1.96 m is rotating in a horizontal circle. The string will break, if tension is more than 25 N. What is the maximum velocity with which the ball can be rotated?

Sol. The centripetal force, $F = \frac{mv^2}{r}$

In the given situation, centripetal force (F) is provided by tension (T) in the string.

The string will break, if $F = T \geq 25 \text{ N}$

Hence, string will not break, if $F = T \leq 25$

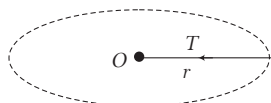
$$\Rightarrow \frac{mv^2}{r} \leq 25 \Rightarrow v^2 \leq \frac{25 \times r}{m} \Rightarrow v^2 \leq \frac{25 \times 1.96}{0.25} \Rightarrow v \leq 14$$

$$\therefore \text{Maximum velocity of the ball, } v_{\text{max}} = \sqrt{196} = 14 \text{ ms}^{-1}$$

Example 7.18 A ball of mass 0.12 kg is being whirled in a horizontal circle at the end of a string 0.5 m long. It is capable of making 231 revolutions in one minute. Find the breaking tension of the string.

Sol. Given, $m = 0.12 \text{ kg}$, $r = 0.5 \text{ m}$,

$$\omega = 231 \text{ rpm} = \frac{2\pi \times 231}{60} \text{ rad s}^{-1} = 24.2 \text{ rad s}^{-1}$$



When body performs circular motion, it is acted upon by a centripetal force, magnitude of which is given by

$$F = m\omega^2 = 0.12 \times 0.5 \times (24.2)^2 = 35.1 \text{ N}$$

This is the breaking tension of the string.

Example 7.19 A spaceman in training is rotated in a seat at the end of a horizontal arm of length 5 m. If he can withstand accelerations upto 9 g, then what is the maximum number of revolutions per second permissible? (Take, $g = 10 \text{ ms}^{-2}$)

Sol. In circular motion, necessary centripetal force to the spaceman is provided by effective weight of spaceman.

$$\begin{aligned} \therefore m \times 9g &= m\omega^2 = mr \times 4\pi^2 n^2 \\ \Rightarrow n &= \sqrt{\frac{9g}{4\pi^2 r}} \Rightarrow n = \sqrt{\frac{9 \times 10}{4 \times (3.14)^2 \times 5}} \\ &= 0.675 \text{ rev s}^{-1} \text{ or hertz (Hz)} \end{aligned}$$

Example 7.20 A string breaks under a load of 4.8 kg. A mass of 0.5 kg is attached to one end of the string 2 m long and is rotated in a horizontal circle. Determine the number of revolution per minute that the mass can make without breaking the string.

Sol. Given, $m = 0.5 \text{ kg}$, $r = 2 \text{ m}$, $g = 9.8 \text{ ms}^{-2}$

The maximum tension that the string can withstand,

$$F = 4.8 \text{ kg-wt} = (4.8 \times 9.8) \text{ N}$$

Let the maximum number of revolutions per second be n .

$$\begin{aligned} \therefore F &= m\omega^2 = mr(2\pi n)^2 \\ \Rightarrow n^2 &= \frac{F}{4\pi^2 mr} = \frac{9.8 \times 4.8}{4 \times (3.14)^2 \times 0.5 \times 2} = 1.193 \\ \Rightarrow n &= \sqrt{1.193} = 1.092 \text{ rps} \\ \Rightarrow n &= 1.092 \times 60 = 65.52 \text{ rpm} \end{aligned}$$

Centrifugal force

Centrifugal force can be defined as the radially directed outward force acting on a body in circular motion.

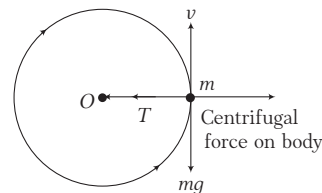


Fig. 7.9 Outward force on a body

Centrifugal force = Mass $\times$ Centrifugal acceleration

$$\text{or } F = \frac{mv^2}{r} = m\omega^2$$

$$\text{or } F = m\omega^2$$

This can be written in vector form as

$$\mathbf{F} = \frac{mv^2}{r} \hat{\mathbf{r}}$$

where, $\hat{r}$ is the unit vector acting along $\mathbf{r}$.

- (i) In an inertial frame, the centrifugal force does not act on the object.
- (ii) In non-inertial frame, pseudo force arises in the form of centrifugal force.

Example 7.21 A gramophone disc rotates at 60 rpm. If coin of mass 18 g is placed at a distance of 8 cm from the centre. Determine the centrifugal force on the coin.

Sol. Since, angular velocity, $\omega = 60 \times \frac{2\pi}{60} = 2\pi \text{ rad s}^{-1}$

$$\therefore \text{Centrifugal force, } F = m\omega^2 r = 18 \times (2\pi)^2 \times 8$$

$$= 18 \times 4 \times (3.14)^2 \times 8$$

$$\Rightarrow F = 5679.13 \text{ dyne} \approx 0.06 \text{ N}$$

Examples for obtaining centripetal force in daily life

Some examples for obtaining centripetal force in daily life are given below

1. Circular turning of roads

It is most popular example of circular motion. When vehicles go through turns, they travel along a nearly circular arc. There must be some force which will produce the required centripetal acceleration. If the vehicles travel on a horizontal circular path, this resultant force is also horizontal.

The necessary centripetal force is being provided to the vehicles by following three ways

- (i) By friction
- (ii) By banking of roads
- (iii) By friction and banking of roads

In practical, the necessary centripetal force is provided by friction and banking of roads. Now, let us write equations of motion in each of the three cases separately and see what are the constraints in each case.

(i) By friction : motion of a car on level road

Suppose, a car of mass m is moving with a speed v in a horizontal circular arc of radius r . In this case, the necessary centripetal force to the car will be provided by force of friction f acting towards centre.

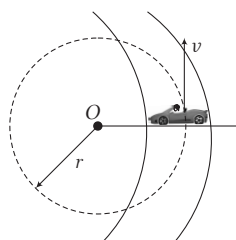


Fig. 7.10 Motion of a car on a level road

Required centripetal force, $f = \frac{mv^2}{r}$

Further, limiting value of f is μN

$$\text{or } f_L = \mu N = \mu mg \quad (\because N = mg)$$

Here, N = normal reaction force on the car by the road

and μ = coefficient of friction between road and tyre of the car.

Therefore, for a safe turn without sliding,

$$\frac{mv^2}{r} \leq f_L \quad \text{or} \quad \frac{mv^2}{r} \leq \mu mg \quad \text{or} \quad \mu \geq \frac{v^2}{rg}$$

or

$$v \leq \sqrt{\mu rg}$$

Here, two situations may arise. If μ and r are known to us, the speed of the vehicle should not exceed $\sqrt{\mu rg}$ and if v and r are known to us, the coefficient of friction should be greater than v^2/rg .

Maximum velocity for no skidding or slipping,

$$v_{\max} = \sqrt{\mu rg}$$

Note You might have seen that if the speed of the car is too high, car starts skidding outwards. With this radius of the circle increases or the necessary centripetal force is reduced ($\because \text{centripetal force} \propto \frac{1}{r}$).

Example 7.22 Determine the maximum speed at which a car can turn round a curve of 30 m radius on a level road, if the coefficient of friction between the tyres and the road is 0.4. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Given, $\mu = 0.4$, $r = 30 \text{ m}$, $g = 10 \text{ ms}^{-2}$

$$\text{Maximum speed, } v_{\max} = \sqrt{\mu rg}$$

$$\Rightarrow v_{\max} = \sqrt{0.4 \times 10 \times 30} = 10.95 \approx 11 \text{ ms}^{-1}$$

Example 7.23 A cyclist speeding at 4.5 km h^{-1} on a level road takes a sharp circular turn of radius 3 m without reducing the speed. The coefficient of static friction between the road and the tyres is 0.1. Will the cyclist slip while taking the turn (i) with a speed of 4.5 km h^{-1} and (ii) with a speed of 9 km h^{-1} ?

Sol. Frictional force provides the necessary centripetal force. He will slip, if the turn is too sharp (i.e. too small a radius) or if his speed is too large.

Maximum speed for no slipping is

$$v_{\max} = \sqrt{\mu_s rg} = \sqrt{0.1 \times 3 \times 9.8} = 1.72 \text{ ms}^{-1}$$

$$(i) \text{ If } v = 4.5 \text{ km h}^{-1} = 4.5 \times \frac{5}{18} = \frac{5}{4} \text{ ms}^{-1}$$

$$= 1.25 \text{ ms}^{-1} < 1.72 \text{ ms}^{-1}, \text{ hence he will not slip.}$$

$$(ii) \text{ If } v = 9 \text{ km h}^{-1} = 9 \times \frac{5}{18} = \frac{5}{2} \text{ ms}^{-1} = 2.5 \text{ ms}^{-1} > 1.72 \text{ ms}^{-1},$$

hence he will slip.

- (ii) **By banking of roads : motion of a car on banked road** Friction is not always reliable at circular turns, if high speed and sharp turns are involved. To avoid dependence on friction, the roads are banked by an angle (θ) at the turn, so that the outer part of the road is somewhat raised compared to the inner part. Applying Newton's second law along the radius and the first law in the vertical direction for the motion of a car on the banked road having mass m .

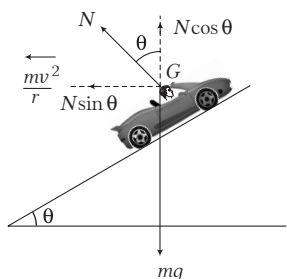


Fig. 7.11 Motion of a car on banked road

We get, $N \sin \theta = \frac{mv^2}{r}$ and $N \cos \theta = mg$

Here, r = radius of circular turn, θ = banking angle.
From these two equations, we get

$$\tan \theta = \frac{v^2}{rg} \quad \text{or} \quad v = \sqrt{rg \tan \theta}$$

Example 7.24 A turn of radius 600 m is banked for a vehicle of mass 200 kg going with a speed of 180 kmh⁻¹. Determine the banking angle of its path.

Sol. The turn is banked for speed,

$$v = 180 \text{ kmh}^{-1} = 180 \times \frac{5}{18} \text{ ms}^{-1} = 50 \text{ ms}^{-1}$$

and radius, $r = 600 \text{ m}$

$$\therefore \tan \theta = \frac{v^2}{rg} = \frac{50 \times 50}{600 \times 10}$$

$$\Rightarrow \tan \theta = \frac{25}{60} = 0.4167$$

$$\Rightarrow \text{Banking angle, } \theta = \tan^{-1}(0.4167) \Rightarrow \theta = 22.62^\circ$$

Example 7.25 A train has to describe a curve of radius 2000 m.

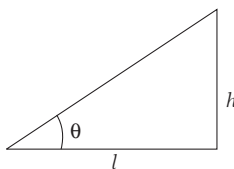
By how much should the outer rail be raised with respect to inner rail for a speed of 72 km h⁻¹. The distance between the rails is 1 m. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Given, $v = 72 \text{ km h}^{-1}$

$$= 72 \times \frac{5}{18} = 20 \text{ ms}^{-1},$$

$$l = 1 \text{ m}, r = 2000 \text{ m}, g = 10 \text{ ms}^{-2}$$

$$\text{We have, } \tan \theta = \frac{v^2}{rg}$$



$$\text{Also, } \tan \theta = \frac{h}{l} \Rightarrow \frac{v^2}{rg} = \frac{h}{l}$$

$$\Rightarrow h = \frac{v^2 l}{rg}$$

$$\therefore h = \frac{(20)^2 \times 1}{2000 \times 10} = \frac{1}{50} \text{ m} \\ = \frac{100}{50} = 2 \text{ cm}$$

- (iii) **By friction and banking of roads** If on a banked circular turning, there is a frictional force between car and road, then the vector sum of normal reaction force (N) and frictional force (f) provides the necessary centripetal force.

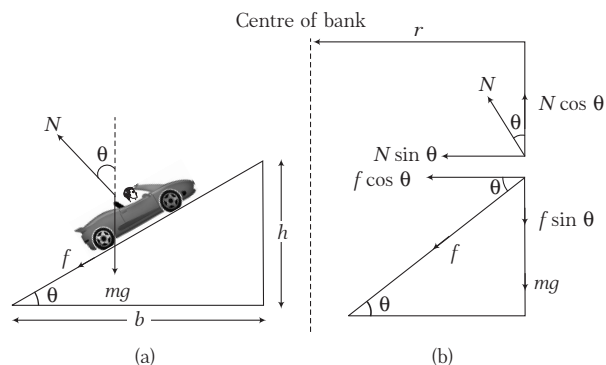


Fig. 7.12 Banked road with friction

$$\therefore N \sin \theta + f \cos \theta = \frac{mv^2}{r} \quad \dots(i)$$

$$\text{and } N \cos \theta = mg + f \sin \theta \quad \dots(ii)$$

($\because$ Vertical force is balanced)

Taking limiting condition of friction, we can write

$$f = \mu_s N \quad \dots(iii)$$

Here, μ_s = coefficient of static friction.

To obtain the value of N , solving above three equations properly, we get

$$N = \frac{mg}{\cos \theta - \mu_s \sin \theta}$$

After putting the value of N in Eq. (i), we get

$$v_{\max} = \left[\frac{rg(\sin \theta + \mu_s \cos \theta)}{\cos \theta - \mu_s \sin \theta} \right]^{1/2}$$

$$v_{\max} = \left[\frac{rg(\mu_s + \tan \theta)}{1 - \mu_s \tan \theta} \right]^{1/2}$$

If the vehicle is moving upward on inclined road, then we can find maximum speed for no skidding from the above formula.

If the vehicle is moving downward on inclined road, then **minimum velocity for no skidding is**

$$v_{\min} = \left[rg \frac{(\tan \theta - \mu_s)}{1 + \mu_s \tan \theta} \right]^{1/2}$$

Note

(i) For no slipping or skidding, we have $v_{\min} < v \leq v_{\max}$. This speed is greater than the maximum possible speed of a car on level road ($v = \sqrt{\mu g r}$).

(ii) If $\mu_s = 0$, $v_o = (g r \tan \theta)^{1/2}$

This speed is known as **optimum speed**.

Example 7.26 A circular race track of radius 300 m is banked at an angle of 15° . If the coefficient of friction between the wheels of a race car and the road is 0.2, then what will be the maximum permissible speed to avoid slipping? (Take, $\tan 15^\circ = 0.26$)

Sol. We know that, maximum permissible speed on a banked road to avoid slipping is

$$v_{\max} = \left[\frac{rg (\mu_s + \tan \theta)}{(1 - \mu_s \tan \theta)} \right]^{1/2}$$

Now, putting the values given in the question,

$$r = 300 \text{ m}, \theta = 15^\circ$$

$$g = 9.8 (\approx 10) \text{ ms}^{-2} \text{ and } \mu_s = 0.2$$

$$\begin{aligned} \text{We obtain, } v_{\max} &= \left[\frac{300 \times 9.8 (0.2 + \tan 15^\circ)}{1 - 0.2 \tan 15^\circ} \right]^{1/2} \\ &= \left[\frac{300 \times 9.8 (0.2 + 0.26)}{1 - 0.2 \times 0.26} \right]^{1/2} \end{aligned}$$

After solving this, we get

$$v_{\max} = 37.8 \text{ ms}^{-1}$$

2. Bending of a cyclist

To take a safe turn on a circular turning, cyclist bend himself inward. When the cyclist is inclined to the centre of the rounding off its path, the resultant of N , f and mg is directed horizontally to the centre of the circular path of the cycle. This resultant force imparts a centripetal acceleration to the cyclist.

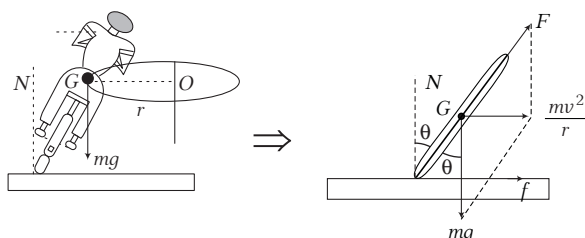


Fig. 7.13 Bending of a cyclist from vertical

From the above figure, we get

Resultant of normal reaction force (N) and frictional force (f), $F = \sqrt{N^2 + f^2}$

This resultant of N and f , i.e. F should pass through G , the centre of gravity of cyclist (for complete equilibrium, rotational as well as translational). Hence,

$$\tan \theta = \frac{f}{N}$$

$$\text{where, } f = \frac{mv^2}{r} \text{ and } N = mg \Rightarrow \tan \theta = \frac{v^2}{rg}$$

Note The angle through which cyclist should bend will be greater, if

- (i) the radius of the curve is small.
- (ii) the velocity of the cyclist is large.

Example 7.27 A cyclist speeding at 6 ms^{-1} in a circle of radius 18 m makes an angle θ with the vertical. Determine the value of θ . Also, determine the minimum possible value of coefficient of friction between the tyres and the ground.

Sol. Given, $v = 6 \text{ ms}^{-1}$, $r = 18 \text{ m}$, $g = 9.8 \text{ ms}^{-2}$

$$\text{Since, } \tan \theta = \frac{v^2}{rg} \Rightarrow \tan \theta = \frac{6 \times 6}{18 \times 9.8} = 0.2041$$

$$\Rightarrow \theta = \tan^{-1}(0.2041) \Rightarrow \theta = 11^\circ 53'$$

Also, minimum possible value of coefficient of friction,

$$\mu = \tan \theta = \frac{v^2}{rg} = 0.2041 \Rightarrow \mu = 0.2041$$

3. Conical pendulum

If a small particle of mass m tied to a string is whirled in a horizontal circle as shown in figure. The arrangement is called the conical pendulum.

In case of conical pendulum, the vertical component of tension balances the weight while its horizontal component provides the necessary centripetal force.

$$\text{Thus, } T \sin \theta = \frac{mv^2}{r} \quad \dots(i)$$

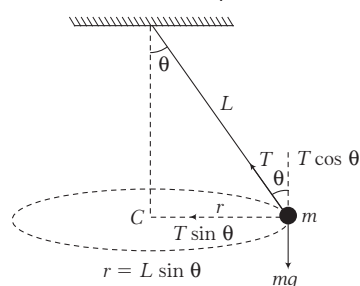


Fig. 7.14 Conical pendulum

$$\text{and } T \cos \theta = mg \quad \dots(ii)$$

From these two equations, we can find $v = \sqrt{rg \tan \theta}$

$$\therefore \text{Angular speed, } \omega = \frac{v}{r} = \sqrt{\frac{g \tan \theta}{r}}$$

So, the time period of pendulum is

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{r}{g \tan \theta}} = 2\pi \sqrt{\frac{L \cos \theta}{g}} \quad [\text{since, } r = L \sin \theta]$$

or

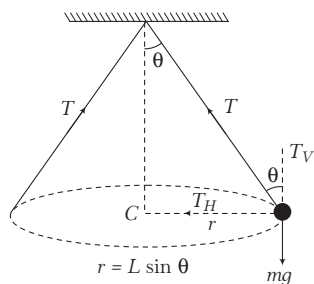
$$T = 2\pi \sqrt{\frac{L \cos \theta}{g}}$$

Example 7.28 A particle of mass 200 g tied to one end of string is revolved in a horizontal circle of radius 50 cm about a vertical axis passing through the point of suspension, with angular speed 60 rev per minute (rpm). Find (i) linear speed, (ii) the acceleration and (iii) horizontal component of tension in the string. What will happen, if string is broken? (Take, $\pi^2 = 10$)

Sol. Angular speed, $\omega = \frac{2\pi n}{60} = \frac{2\pi \times 60}{60} = 2\pi \text{ rad s}^{-1} = 6.28 \text{ rad s}^{-1}$

Horizontal component of the tension, $T_H = T \sin \theta$

Vertical component of tension, $T_V = T \cos \theta = mg$



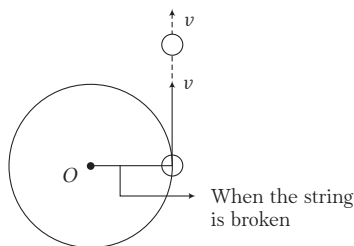
(i) Linear speed, $v = r\omega = 0.5 \times 2\pi = \pi \text{ ms}^{-1} = 3.14 \text{ ms}^{-1}$

(ii) Acceleration, $a_c = \frac{v^2}{r} = \frac{(\pi)^2}{0.5} \approx \frac{10}{0.5} = 20 \text{ ms}^{-2}$

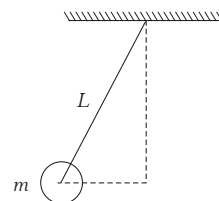
(iii) Horizontal component of tension,

$$T_H = \frac{mv^2}{r} = 0.2 \times 20 = 4 \text{ N}$$

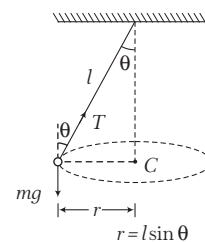
When the string is broken, tension T_H (i.e. centripetal force) vanishes and body moves along the tangent in a straight line with speed 3.14 ms^{-1} .



Example 7.29 A ball of mass (m) 0.5 kg is attached to the end of a string having length (L) 0.5 m. The ball is rotated on a horizontal circular path about vertical axis. The maximum tension that the string can bear is 324 N. Find the maximum possible value of angular velocity of ball (in rads^{-1}).



Sol. Consider the forces acting on the ball as shown in the figure.



The component $T \cos \theta$ will cancel mg .

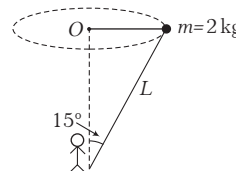
The component $T \sin \theta$ will provide necessary centripetal force to the ball towards centre C.

$$\therefore T \sin \theta = m\omega^2 r = m(l \sin \theta) \omega^2 \quad \text{or} \quad T = m\omega^2 l$$

Angular velocity, $\omega = \sqrt{\frac{T}{ml}}$

or $\omega_{\max} = \sqrt{\frac{T_{\max}}{ml}} = \sqrt{\frac{324}{0.5 \times 0.5}} = 36 \text{ rads}^{-1}$

Example 7.30 A boy whirls a stone of mass 2 kg in a horizontal circle of radius 1.5 m, which is attached to a string having length 1 m. What is the time period of the given system of stone and string? (Take, $\cos 15^\circ = 0.96$)



Sol. Given, $m = 2 \text{ kg}$, $r = 1.5 \text{ m}$,

$$L = 1 \text{ m}, \theta = 15^\circ, g = 10 \text{ ms}^{-2}$$

Time period of the given system of stone and string is given as

$$T = 2\pi \sqrt{\frac{L \cos \theta}{g}} = 2\pi \sqrt{\frac{1 \times \cos 15^\circ}{10}} = 2\pi \sqrt{\frac{0.96}{10}} = 2 \times 3.14 \times 0.31 = 1.95 \approx 2 \text{ s}$$

Thus, the time period is 2 s (approx).

4. 'Death well' or rotor

In case of death well, a person drives a bicycle on a vertical surface of a large wooden well, while in case of a rotor, at its certain angular speed, a person hangs resting against the wall without any support from the bottom. In death well, walls are at rest and person revolves while in case of rotor, person is at rest and the walls rotate.

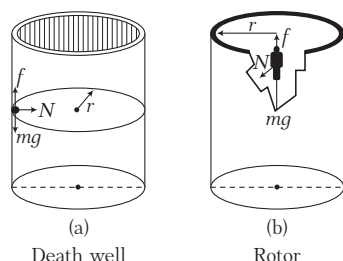


Fig. 7.15

In both cases, friction balances the weight of person while reaction provides the centripetal force for circular motion,

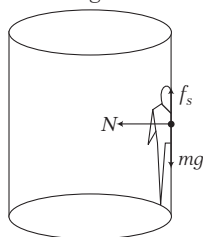
$$\text{i.e. } f = mg \quad \text{and} \quad N = \frac{mv^2}{r} = m\omega^2 r \quad (\because v = r\omega)$$

$$f \leq \mu N \Rightarrow mg \leq \mu mv^2/r \Rightarrow v^2 \geq \frac{gr}{\mu}$$

$$\therefore \text{Safe speed, } v = \sqrt{\frac{gr}{\mu}}$$

Example 7.31 In a rotor, a hollow vertical cylinder rotates about its axis and a person rest against the inner wall. At a particular speed of the rotor, the floor below the person is removed and the person hangs resting against the wall without any floor. If the radius of the rotor is 2 m and the coefficient of static friction between the wall and the person is 0.2. Find the minimum speed at which the floor may be removed.

Sol. The situation is shown in figure below.



When the floor is removed, the forces on the person are

- (i) weight mg downward.
- (ii) normal force N due to the wall towards the centre.
- (iii) frictional force f_s parallel to the wall, upwards.

The person is moving in a circle with a uniform speed, so its acceleration is v^2/r towards the centre.

For the minimum speed when the floor may be removed, the friction is limiting one and so equals $\mu_s N$.

This gives,

$$\mu_s N = mg$$

$$\text{or } \frac{\mu_s mv^2}{r} = mg \quad \left[\because N = \frac{mv^2}{r} \right]$$

$$\text{or } v = \sqrt{\frac{rg}{\mu_s}} = \sqrt{\frac{2 \times 10}{0.2}} = 10 \text{ ms}^{-1}$$

Motion of a particle tied to a string in vertical circle

Suppose a particle of mass m is attached to an inextensible light string of length R . This particle is moving in a vertical circle of radius R about a fixed point O .

It is imparted a velocity u in horizontal direction at lowest point A . Let v be its velocity at point B of the circle as shown in Fig. 7.16.

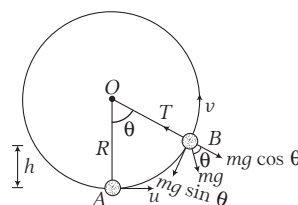


Fig. 7.16 Motion of a particle in vertical circle

Here, we have

$$h = R(1 - \cos \theta)$$

Now, from conservation of mechanical energy, we have

$$\frac{1}{2} m (u^2 - v^2) = mgh \quad \dots(i)$$

The necessary centripetal force is provided by the resultant of tension T and $mg \cos \theta$.

$$\therefore T - mg \cos \theta = \frac{mv^2}{R} \quad \dots(ii)$$

As speed of the particle decreases with height, tension in the string is maximum at the bottom. The particle will complete the circle, if the string does not slack even at the highest point.

Now, following conclusions can be made using above Eqs. (i) and (ii)

- (i) **Minimum velocity at highest point**, so that particle complete the circle $v_{\min} = \sqrt{gR}$, at this velocity, tension in the string is zero.
- (ii) **Minimum velocity at lowest point**, so that particle complete the circle $v_{\min} = \sqrt{5gR}$, at this velocity, tension in the string is $6mg$.
- (iii) **When string is horizontal**, then minimum velocity is $\sqrt{3Rg}$ and tension in this condition is $3mg$.
- (iv) If velocity at lowest point is less than $\sqrt{5gR}$, then tension in the string becomes zero before reaching

the highest point, now the particle will leave the circle and will move on parabolic path.

In this condition, if $\sqrt{2gR} < v < \sqrt{5gR}$, then tension in the string becomes zero but velocity is not zero, the particle will leave circle at $90^\circ < \theta < 180^\circ$ or $h > R$.

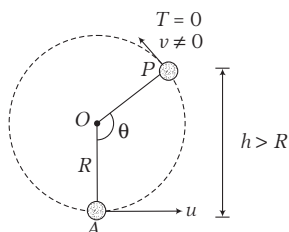


Fig. 7.17

- (v) If velocity at lowest point is $0 < v \leq \sqrt{2gR}$, the particle will oscillate. In this condition, velocity becomes zero but tension is not zero. The particle will oscillate in lower half of circle, i.e. $0^\circ < \theta < 90^\circ$.

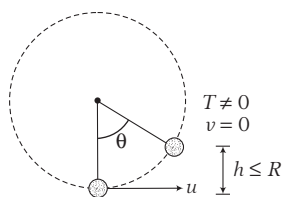


Fig. 7.18

Note The above points have been derived for a particle moving in a vertical circle attached to a string. The same conditions apply, if a particle moves inside a smooth spherical shell of radius R . The only difference is that the tension is replaced by the normal reaction N .

Example 7.32 One end of a string of length 1 m is tied to a body of mass 0.5 kg. It is whirled in a vertical circle with angular velocity 4 rad s^{-1} . Find the tension in the string when the body is at the lower most point of its motion. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. At lower most point, the tension in the string,

$$T = m\omega^2 r + mg = 0.5 \times (4)^2 \times 1 + 0.5 \times 10 = 13 \text{ N}$$

Example 7.33 A ball of mass 0.6 kg attached to a light inextensible string rotates in a vertical circle of radius 0.75 m such that it has speed of 5 ms^{-1} when the string is horizontal. Tension in string when it is horizontal on other side is (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Tension in the string when it makes angle θ with the vertical,

$$T = \frac{mv^2}{r} + mg \cos \theta$$

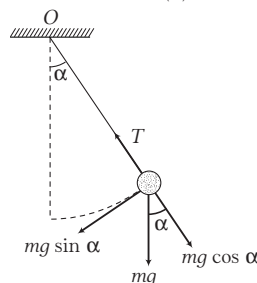
When the string is horizontal, $\theta = 90^\circ$

$$\begin{aligned} \therefore T &= \frac{mv^2}{r} + mg \times 0 = \frac{mv^2}{r} \\ &= \frac{0.6 \times (5)^2}{0.75} = 20 \text{ N} \end{aligned}$$

Example 7.34 A simple pendulum is constructed by attaching a bob of mass m to a string of length L fixed at its upper end. The bob oscillates in a vertical circle. It is found that the speed of the bob is v when the string makes an angle α with the vertical. Find the tension in the string and the magnitude of net force on the bob at that instant.

Sol. (i) The force acting on the bob are

- (a) the tension T (b) the weight mg



As the bob moves in a circle of radius L with centre at O . A centripetal force of magnitude $\frac{mv^2}{L}$ is required

towards O . This force will be provided by the resultant of T and $mg \cos \alpha$. Thus,

$$T - mg \cos \alpha = \frac{mv^2}{L}$$

or

$$T = m \left(g \cos \alpha + \frac{v^2}{L} \right)$$

$$\begin{aligned} \text{(ii) } |\mathbf{F}_{\text{net}}| &= \sqrt{(mg \sin \alpha)^2 + \left(\frac{mv^2}{L} \right)^2} \\ &= m \sqrt{g^2 \sin^2 \alpha + \frac{v^4}{L^2}} \end{aligned}$$

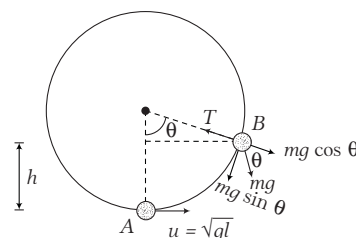
Example 7.35 A heavy particle hanging from a fixed point by a light inextensible string of length l is projected horizontally with speed $\sqrt{gl}$. Find the speed of the particle and the inclination of the string to the vertical at the instant of the motion when the tension in the string is equal to the weight of the particle.

Sol. Let $T = mg$ at angle θ as shown in figure.

$$h = l(1 - \cos \theta) \quad \dots(i)$$

Applying conservation of mechanical energy between points A and B , we get

$$\frac{1}{2} m(u^2 - v^2) = mgh$$



Here, $u^2 = gl$... (ii)

and v = speed of particle in position B

$$\therefore v^2 = u^2 - 2gh \quad \dots (iii)$$

Further, $T - mg \cos \theta = \frac{mv^2}{l}$

or $mg - mg \cos \theta = \frac{mv^2}{l} \quad (\because T = mg)$

or $v^2 = gl(1 - \cos \theta) \quad \dots (iv)$

Substituting the values of v^2 , u^2 and h from Eqs. (iv), (ii) and (i) in Eq. (iii), we get

$$gl(1 - \cos \theta) = gl - 2gl(1 - \cos \theta)$$

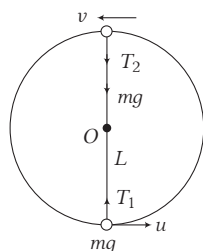
or $\cos \theta = \frac{2}{3}$ or $\theta = \cos^{-1}\left(\frac{2}{3}\right)$

Substituting $\cos \theta = \frac{2}{3}$ in Eq. (iv), we get

$$v = \sqrt{\frac{gl}{3}}$$

Example 7.36 A particle of mass m is attached to a string of length L and given velocity $\sqrt{10gL}$ in the horizontal direction at the lowest point. Find tension in the string when the particle is at (i) (a) lowest position (b) highest position; (ii) when the string makes an angle 60° with (a) lower vertical and (b) upper vertical.

Sol. (i)



(a) At the lowest position,

$$T_1 - mg = \frac{mv^2}{L} = \frac{m}{L} \times 10gL \Rightarrow T_1 = 11mg$$

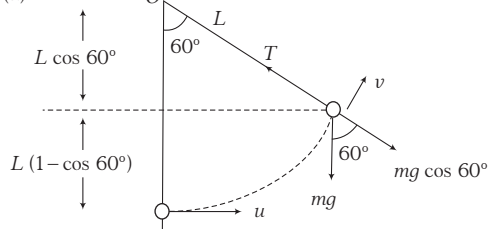
(b) At the highest position,

$$v^2 = u^2 - 2gh = u^2 - 2g \times 2L \\ = 10gL - 4gL = 6gL$$

$$\text{Also, } T_2 + mg = \frac{mv^2}{L} = \frac{m}{L} \times 6gL = 6mg$$

$$\Rightarrow T_2 = 5mg$$

(ii) (a)



In this situation, we can write

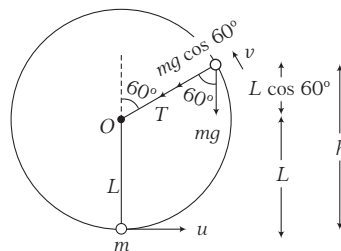
$$v^2 = u^2 - 2gh = u^2 - 2gL(1 - \cos 60^\circ) \\ = 10gL - 2gL\left(1 - \frac{1}{2}\right) = 9gL$$

$$\text{Also, } T - mg \cos 60^\circ = \frac{mv^2}{L} = \frac{m}{L} \times 9gL = 9mg$$

$$\Rightarrow T - \frac{mg}{2} = 9mg$$

$$\Rightarrow T = \frac{19}{2}mg$$

(b)



In this situation, we have

$$v^2 = u^2 - 2gh = 10gL - 2gL(1 + \cos 60^\circ) = 7gL$$

$$\text{Also, } T + mg \cos 60^\circ = \frac{mv^2}{L}$$

$$\Rightarrow T + \frac{mg}{2} = \frac{m}{L} \times 7gL = 7mg$$

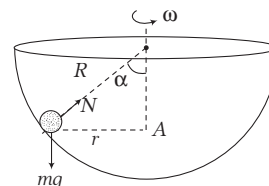
$$\Rightarrow T = \frac{13mg}{2}$$

Example 7.37 A hemispherical bowl of radius R is rotating about its axis of symmetry which is kept vertical. A small ball kept in the bowl rotates with the bowl without slipping on its surface. If the surface of the bowl is smooth and the angle made by the radius through the ball with the vertical is α . Find the angular speed at which the bowl is rotating.

Sol. Let ω be the angular speed of rotation of the bowl. Two forces are acting on the ball

(i) normal reaction N

(ii) weight mg



The ball is rotating in a circle of radius $r (= R \sin \alpha)$ with centre at A at an angular speed ω . Thus,

$$N \sin \alpha = m r \omega^2 = m R \omega^2 \sin \alpha$$

$$\Rightarrow N = m R \omega^2 \quad \dots (i)$$

$$\text{and } N \cos \alpha = mg \quad \dots (ii)$$

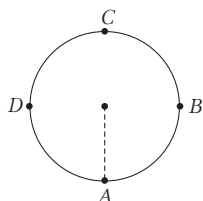
On dividing Eq. (i) by Eq. (ii), we get

$$\frac{1}{\cos \alpha} = \frac{\omega^2 R}{g} \Rightarrow \omega = \sqrt{\frac{g}{R \cos \alpha}}$$

CHECK POINT 7.2

- A particle of mass 2 kg is moving along a circular path of radius 1 m. If its angular speed is $2\pi \text{ rad s}^{-1}$, the centripetal force on it is
 (a) $4\pi \text{ N}$ (b) $8\pi \text{ N}$
 (c) $4\pi^4 \text{ N}$ (d) $8\pi^2 \text{ N}$
- Two particles of equal masses are revolving in circular paths of radii r_1 and r_2 respectively with the same speed. The ratio of their centripetal forces is
 (a) $\frac{r_2}{r_1}$ (b) $\sqrt{\frac{r_2}{r_1}}$
 (c) $\left(\frac{r_1}{r_2}\right)^2$ (d) $\left(\frac{r_2}{r_1}\right)^2$
- A particle of mass m is executing uniform circular motion on a path of radius r . If p is the magnitude of its linear momentum. The radial force acting on the particle is
 (a) pmr (b) $\frac{rm}{p}$
 (c) $\frac{mp^2}{r}$ (d) $\frac{p^2}{rm}$
- A stone of mass of 16 kg is attached to a string 144 m long and is whirled in a horizontal circle on a smooth surface. The maximum tension in the string that it can withstand is 16 N. The maximum velocity of revolution that can be given to the stone without breaking it, will be
 (a) 20 ms^{-1} (b) 16 ms^{-1}
 (c) 14 ms^{-1} (d) 12 ms^{-1}
- If mass, speed and radius of the circle, of a particle moving uniformly in a circular path are all increased by 50%, the necessary force required to maintain the body moving in the circular path will have to be increased by
 (a) 225% (b) 125% (c) 150% (d) 100%
- A string of length 0.1 m cannot bear a tension more than 100 N. It is tied to a body of mass 100 g and rotated in a horizontal circle. The maximum angular velocity can be
 (a) 100 rad s^{-1} (b) 1000 rad s^{-1}
 (c) 10000 s^{-1} (d) 0.1 rad s^{-1}
- A mass of 2 kg is whirled in a horizontal circle by means of a string at an initial speed of 5 rev min^{-1} . Keeping the radius constant the tension in the string is doubled. The new speed is nearly
 (a) $\frac{5}{\sqrt{2}} \text{ rpm}$ (b) 10 rpm
 (c) $10\sqrt{2} \text{ rpm}$ (d) $5\sqrt{2} \text{ rpm}$
- A mass of 100 g is tied to one end of a string 2 m long. The body is revolving in a horizontal circle making a maximum of 200 rev min^{-1} . The other end of the string is fixed at the centre of the circle of revolution. The maximum tension that the string can bear is (approximately)
 (a) 8.76 N (b) 8.94 N
 (c) 89.42 N (d) 87.64 N
- Radius of the curved road on national highway is R . Width of the road is b . The outer edge of the road is raised by h with respect to inner edge, so that a car with velocity v can pass safely over it. The value of h is
 (a) $\frac{v^2 b}{Rg}$ (b) $\frac{v}{Rg b}$
 (c) $\frac{v^2 R}{bg}$ (d) $\frac{v^2 b}{R}$
- A motor cyclist moving with a velocity of 72 km h^{-1} on a flat road takes a turn on the road at a point, where the radius of curvature of the road is 20 m. The acceleration due to gravity is 10 ms^{-2} . In order to avoid skidding, he must not bend with respect to the vertical plane by an angle greater than
 (a) $\theta = \tan^{-1}(6)$ (b) $\theta = \tan^{-1}(2)$
 (c) $\theta = \tan^{-1}(2592)$ (d) $\theta = \tan^{-1}(4)$
- A car of mass 1000 kg negotiates a banked curve of radius 90 m on a frictionless road. If the banking angle is 45° , the speed of the car is
 (a) 20 ms^{-1} (b) 30 ms^{-1}
 (c) 5 ms^{-1} (d) 10 ms^{-1}
- Keeping the angle of banking unchanged, if the radius of curvature is made four times, the percentage increase in the maximum speed with which a vehicle can travel on a circular road is
 (a) 25% (b) 50%
 (c) 75% (d) 100%
- A person wants to drive on the vertical surface of a large cylindrical wooden 'well' commonly known as 'death well' in a circus. The radius of the well is R and the coefficient of friction between the tyres of the motorcycle and the wall of the well is μ_s . The minimum speed, the motorcycle must have in order to prevent slipping, should be
 (a) $\sqrt{\frac{Rg}{\mu_s}}$ (b) $\sqrt{\frac{\mu_s}{Rg}}$
 (c) $\sqrt{\frac{\mu_s g}{R}}$ (d) $\sqrt{\frac{R}{\mu_s g}}$
- A motorcyclist wants to drive on the vertical surface of wooden 'well' of radius 5 m, with a minimum speed of $5\sqrt{5} \text{ ms}^{-1}$. The minimum value of coefficient of friction between the tyres and the wall of the well must be (Take, $g = 10 \text{ ms}^{-2}$)
 (a) 0.10 (b) 0.20
 (c) 0.30 (d) 0.40
- A block of mass m at the end of a string is whirled round in a vertical circle of radius R . The critical speed of the block at top of its swing below which the string would slacken before the block reaches the bottom is
 (a) $\sqrt{5Rg}$ (b) $\sqrt{3Rg}$
 (c) $\sqrt{2Rg}$ (d) $\sqrt{Rg}$

16. A stone is attached to one end of a string and rotated in a vertical circle. If string breaks at the position of maximum tension, it will break at



- (a) A (b) B
(c) C (d) D
17. A particle of mass m attached at the end of a string is being circulated on a vertical circle of radius r . If the speed of particle at the highest point be v , such that the string would not slacken before the string reached the bottom, then
- (a) $mg = \frac{mv^2}{r}$ (b) $mg > \frac{mv^2}{r}$
(c) $mg \leq \frac{mv^2}{r}$ (d) $mg \geq \frac{mv^2}{r}$
18. A particle is moving in a vertical circle. The tensions in the string when passing through two positions at angles 30° and 60° from vertical (lowest position) are T_1 and T_2 respectively, then
- (a) $T_1 = T_2$ (b) $T_2 > T_1$
(c) $T_1 > T_2$ (d) Data insufficient

19. A stone of mass 1 kg is tied to the end of a string 1 m long. It is whirled in a vertical circle. The velocity of the stone at the bottom of the circle is just sufficient to take it to the top of circle without slackening of the string. What is the tension in the string at the top of the circle? (Take, $g = 10 \text{ ms}^{-2}$)

(a) Zero (b) 1 N
(c) $\sqrt{10}$ N (d) 10 N

20. A small sphere of mass m is suspended by a thread of length l . It is raised up to the height of suspension with thread fully stretched and released. Then, the maximum tension in thread will be

(a) mg (b) $2mg$
(c) $3mg$ (d) $6mg$

21. A child is swinging a swing. Minimum and the maximum heights of swing from earth's surface are 0.75 m and 2 m, respectively. The maximum velocity of this swing is (Take, $g = 10 \text{ ms}^{-2}$)

(a) 5 ms^{-1} (b) 10 ms^{-1}
(c) 15 ms^{-1} (d) 20 ms^{-1}

22. A national roadway bridge over a canal is in the form of an arc of a circle of radius 49 m. What is the minimum speed with which a car can move without leaving the ground at the highest point? (Take, $g = 9.8 \text{ ms}^{-2}$)

(a) 19.6 ms^{-1}
(b) 40 ms^{-1}
(c) 22 ms^{-1}
(d) None of the above

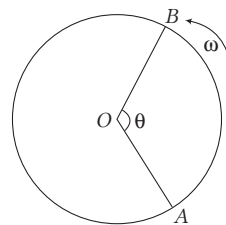
Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1 A particle moving along a circular path due to a centripetal force having constant magnitude is an example of motion with
 - (a) constant speed and velocity
 - (b) variable speed and variable velocity
 - (c) variable speed and constant velocity
 - (d) constant speed and variable velocity
- 2 A particle moving on a circular path makes 600 rpm. In how much time, it will complete one revolution?
 - (a) 0.2 s
 - (b) 0.1 s
 - (c) 0.4 s
 - (d) 0.3 s
- 3 A wheel rotates at 50 rpm about its axis. The angular retardation that can stop the wheel in one minute is
 - (a) $\frac{\pi}{36} \text{ rad s}^{-2}$
 - (b) $\frac{\pi}{18} \text{ rad s}^{-2}$
 - (c) $\frac{\pi}{72} \text{ rad s}^{-2}$
 - (d) $\frac{\pi}{9} \text{ rad s}^{-2}$
- 4 The speed of a particle moving in a circle is increasing. The dot product of its acceleration and velocity is
 - (a) negative
 - (b) zero
 - (c) positive
 - (d) may be positive or negative
- 5 A particle moves with constant angular velocity in a circle. During the motion, its
 - (a) energy is conserved
 - (b) momentum is conserved
 - (c) energy and momentum both are conserved
 - (d) None of the above
- 6 An object is moving in a circle of radius 100 m with a constant speed of 31.4 ms^{-1} . What is its average speed for one complete revolution?
 - (a) Zero
 - (b) 31.4 ms^{-1}
 - (c) 3.14 ms^{-1}
 - (d) $\sqrt{2} \times 31.4 \text{ ms}^{-1}$
- 7 A bucket full of water is rotated in a vertical circle of radius R . If the water does not split out, the speed of the bucket at topmost point will be
 - (a) $\sqrt{Rg}$
 - (b) $\sqrt{5gR}$
 - (c) $\sqrt{2Rg}$
 - (d) $\sqrt{\left(\frac{R}{g}\right)}$
- 8 A car when passes through a convex bridge with velocity v exerts a force on it at the topmost point is equal to
 - (a) $Mg + \frac{Mv^2}{r}$
 - (b) $\frac{Mv^2}{r}$
 - (c) $Mg - \frac{Mv^2}{r}$
 - (d) None of these
- 9 A stone tied to one end of rope and rotated in a circular motion. If the string suddenly breaks, then the stone travels
 - (a) in perpendicular direction
 - (b) in direction of centrifugal force
 - (c) towards centripetal force
 - (d) in tangential direction
- 10 A car is moving on a circular path and takes a turn. If R_1 and R_2 be the reactions on the inner and outer wheels respectively, then
 - (a) $R_1 = R_2$
 - (b) $R_1 < R_2$
 - (c) $R_1 > R_2$
 - (d) $R_1 \geq R_2$
- 11 An unbanked curve has a radius of 60 m. The maximum speed at which a car can make a turn, if the coefficient of static friction is 0.75, is
 - (a) 2.1 ms^{-1}
 - (b) 14 ms^{-1}
 - (c) 21 ms^{-1}
 - (d) 7 ms^{-1}
- 12 A motorcyclist riding at 36 kmh^{-1} has to turn a corner. Find the least radius of the curve, he should follow for safe travelling, if the coefficient of friction between the tyres and the road is 0.2.
 - (a) 10 m
 - (b) 25 m
 - (c) 50 m
 - (d) 100 m
- 13 A circular curve of a highway is designed for traffic moving at 72 kmh^{-1} . If the radius of the curved path is 100 m, the correct angle of banking of the road should be
 - (a) $\tan^{-1}\left(\frac{2}{5}\right)$
 - (b) $\tan^{-1}\left(\frac{3}{5}\right)$
 - (c) $\tan^{-1}\left(\frac{1}{5}\right)$
 - (d) $\tan^{-1}\left(\frac{1}{4}\right)$
- 14 A particle of mass m is circulating on a circle of radius r having angular momentum L about centre. Then, the centripetal force will be
 - (a) $\frac{L^2}{mr}$
 - (b) $\frac{L^2}{mr^2}$
 - (c) $\frac{L^2}{mr^3}$
 - (d) $\frac{L}{mr^2}$

- 15** A particle is moving along a circular path of radius 5 m with a uniform speed 5 ms^{-1} . What will be the average acceleration when the particle completes half revolution?
 (a) Zero (b) 10 ms^{-2}
 (c) $10\pi \text{ ms}^{-2}$ (d) $\frac{10}{\pi} \text{ ms}^{-2}$
- 16** A fan makes 2400 rpm. If after it is switched off, it comes to rest in 10 s, then find the number of times it will rotate before it comes to rest after it is switched off.
 (a) 400 (b) 100 (c) 200 (d) 50
- 17** The distance between the rails of the track is 1.67 m. How much the outer rail be elevated for curve of 0.5 km radius, so that a train moving with speed 54 kmh^{-1} can take safe turn on track.
 (a) 80 mm (b) 75 mm (c) 60 mm (d) 75 cm
- 18** A body of mass 1 kg is moving in a vertical circular path of radius 1 m. The difference between the kinetic energies at its highest and lowest positions is
 (a) 20 J (b) 10 J
 (c) $4\sqrt{5} \text{ J}$ (d) $10(\sqrt{5} - 1) \text{ J}$
- 19** If the banking angle of curved road is given by $\tan^{-1}\left(\frac{3}{5}\right)$ and the radius of curvature of the road is 6 m, then the safe driving speed should not exceed (Take, $g = 10 \text{ ms}^{-2}$)
 (a) 86.4 km h^{-1} (b) 43.2 km h^{-1}
 (c) 21.6 km h^{-1} (d) 30.4 km h^{-1}
- 20** A motorcyclist moving with a velocity of 144 kmh^{-1} on a flat road takes a turn on the road at a point, where the radius of curvature of the road is 40 m. The acceleration due to gravity is 10 ms^{-2} . In order to avoid sliding, he must bend with respect to the vertical plane by an angle
 (a) $\theta = \tan^{-1}(4)$ (b) $\theta = 45^\circ$
 (c) $\theta = \tan^{-1}(2)$ (d) $\theta = \tan^{-1}(6)$
- 21** A train has to negotiate a curve of radius 800 m. By how much height should the outer rail be raised with respect to inner rail for a speed of 96 kmh^{-1} ? The distance between the rails is 1 m.
 (a) 4.4 cm (b) 9 cm (c) 8.9 cm (d) 3.3 cm
- 22** A car wheel is rotated to uniform angular acceleration about its axis. Initially, its angular velocity is zero. It rotates through an angle θ_1 in the first 2 s. In the next 2 s, it rotates through an additional angle θ_2 , the ratio of $\frac{\theta_2}{\theta_1}$ is
 (a) 1 (b) 2 (c) 3 (d) 4
- 23** A pendulum bob on a 2m string is displaced 60° from the vertical and then released. What is the speed of the bob as it passes through the lowest point in its path?
 (a) $\sqrt{2} \text{ ms}^{-1}$ (b) $\sqrt{9.8} \text{ ms}^{-1}$
 (c) 4.43 ms^{-1} (d) $\frac{1}{\sqrt{2}} \text{ ms}^{-1}$
- 24** A pendulum bob having length of string 0.2 m describes an arc of circle in a vertical plane. If the tension in the cord is $\sqrt{3}$ times the weight of the bob when the cord makes an angle 30° with the vertical, the acceleration of the bob in that position is
 (a) g (b) $\frac{g}{2}$
 (c) $\frac{\sqrt{3}g}{2}$ (d) $\frac{g}{4}$
- 25** A jeep runs around a curve of radius 0.3 km at a constant speed of 60 ms^{-1} . The jeep covers a curve of 60° arc, then
 (a) resultant change in velocity of jeep is 60 ms^{-1}
 (b) instantaneous acceleration of jeep is 12 ms^{-2}
 (c) average acceleration of jeep is approximately 11.5 ms^{-2}
 (d) All are correct
- 26** A coin placed on a rotating turn-table slips, when it is placed at a distance of 9 cm from the centre. If the angular velocity of the turn-table is trippled, it will just slip, if its distance from the centre is
 (a) 27 cm (b) 9 cm (c) 3 cm (d) 1 cm
- 27** A circular disc of radius R is rotating about its axis O with a uniform angular velocity $\omega \text{ rad s}^{-1}$ as shown in the figure. The magnitude of the relative velocity of point A relative to point B on the disc is



- (a) zero (b) $R\omega \sin\left(\frac{\theta}{2}\right)$
 (c) $2R\omega \sin\left(\frac{\theta}{2}\right)$ (d) $\sqrt{3} R\omega \sin\left(\frac{\theta}{2}\right)$

- 28** When the angular velocity of a uniformly rotating body has increased thrice, the resultant of forces applied to it increases by 60 N. Find the accelerations of the body in the two cases. The mass of the body $m = 3 \text{ kg}$.
 (a) 2.5 ms^{-2} , 7.5 ms^{-2} (b) 7.5 ms^{-2} , 22.5 ms^{-2}
 (c) 5 ms^{-2} , 45 ms^{-2} (d) 2.5 ms^{-2} , 22.5 ms^{-2}

- 29** An automobile enters a turn of radius R . If the road is banked at an angle of 45° and the coefficient of friction is 1, the minimum speed with which the automobile can negotiate the turn without skidding is

(a) $\sqrt{\frac{rg}{2}}$ (b) $\frac{\sqrt{rg}}{2}$ (c) $\sqrt{rg}$ (d) zero

- 30** A mass is attached to the end of a string of length l which is tied to a fixed point O . The mass is released from the initial horizontal position of the string. Below the point O at what minimum distance, a peg P should be fixed, so that the mass turns about P and can describe a complete circle in the vertical plane?

(a) $\left(\frac{3}{5}\right)l$ (b) $\left(\frac{2}{5}\right)l$ (c) $\frac{l}{3}$ (d) $\frac{2l}{3}$

- 31** A stone is rotated in a vertical circle. Speed at bottommost point is $\sqrt{8gR}$, where R is the radius of circle. The ratio of tension at the top and the bottom is

(a) 1 : 2 (b) 1 : 3 (c) 2 : 3 (d) 1 : 4

- 32** A body is moving in a vertical circle of radius r such that the string is just taut at its highest point. The speed of the particle when the string is horizontal, is

(a) $\sqrt{gr}$ (b) $\sqrt{2gR}$ (c) $\sqrt{3gr}$ (d) $\sqrt{4gR}$

- 33** A small ball is pushed from a height h along a smooth hemispherical bowl of radius R . With what speed should the ball be pushed, so that it just reaches the top of the opposite end of the bowl?

(a) $\sqrt{2gh}$ (b) $\sqrt{2g(R+h)}$
(c) $\sqrt{2g(R-h)}$ (d) None of these

- 34** A 50 kg girl is swinging on a swing from rest. Then, the power delivered when moving with a velocity of 2 ms^{-1} upwards in a direction making an angle 60° with the vertical is

(a) 980 W (b) 490 W (c) $490\sqrt{3}$ W (d) 245 W

- 35** A simple pendulum of length l has a maximum angular displacement θ . The maximum kinetic energy of the bob of mass m will be

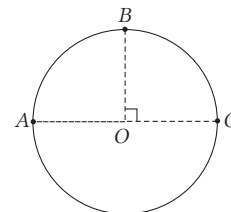
(a) $mgl(1 - \cos \theta)$ (b) $mgl \cos \theta$
(c) $mgl \sin \theta$ (d) None of these

- 36** Toy cart tied to the end of an unstretched string of length a , when revolved moves in a horizontal circle of radius $2a$ with a time period T . Now, the toy cart is speeded up until it moves in a horizontal circle of radius $3a$ with a period T' . If Hooke's law ($F = kx$) holds, then

(a) $T' = \sqrt{\frac{3}{2}} T$ (b) $T' = \left(\frac{\sqrt{3}}{2}\right) T$
(c) $T' = \left(\frac{3}{2}\right) T$ (d) $T' = T$

- 37** Three particles A , B and C move in a circle of radius $r = \frac{1}{\pi} \text{ m}$, in anti-clockwise direction with speed

1 ms^{-1} , 2.5 ms^{-1} and 2 ms^{-1} , respectively. The initial positions of A , B and C are as shown in figure.



The ratio of distance travelled by B and C by the instant A , B and C meet for the first time is

(a) 3 : 2 (b) 5 : 4
(c) 3 : 5 (d) 3 : 7

- 38** A stone tied to one end of spring 80 cm long is whirled in a horizontal circle with a constant speed. If stone makes 14 revolutions in 25 s, the magnitude of acceleration of stone is

(a) 850 cm s^{-2} (b) 992 cm s^{-2}
(c) 720 cm s^{-2} (d) 650 cm s^{-2}

- 39** A student whirls a stone in a horizontal circle of radius 3 m and at height 8 m above level ground. The string breaks, at lowest point and the stone flies off horizontally and strikes the ground after travelling a horizontal distance of 20 m. What is the magnitude of the centripetal acceleration of the stone while breaking off?

(a) 150 ms^{-2} (b) 140 ms^{-2}
(c) 81.4 ms^{-2} (d) 163 ms^{-2}

- 40** A stone is tied to a string of length l and is whirled in a vertical circle with the other end of the string as the centre. At a certain instant of time, the stone is at its lowest position and has a speed u . The magnitude of the change in velocity as it reaches a position, where the string is horizontal (g being acceleration due to gravity) is

(a) $\sqrt{2(u^2 - gl)}$ (b) $\sqrt{u^2 - gl}$
(c) $u - \sqrt{u^2 - 2gl}$ (d) $\sqrt{2gl}$

- 41** A ball suspended by a thread swings in a vertical plane, so that its acceleration at the extreme position and lowest position are equal. The angle θ of thread deflection in the extreme position will be

(a) $\tan^{-1}(2)$ (b) $\tan^{-1}(\sqrt{2})$
(c) $\tan^{-1}\left(\frac{1}{2}\right)$ (d) $2 \tan^{-1}\left(\frac{1}{2}\right)$

- 42** A body of mass m hangs at one end of a string of length l , the other end of which is fixed. It is given a horizontal velocity, so that the string would just reach, where it makes an angle of 60° with the vertical. The tension in the string at bottommost point position is

(a) $2mg$ (b) mg
(c) $3mg$ (d) $\sqrt{3}mg$

- 43** A simple pendulum oscillates in a vertical plane. When it passes through the bottommost point, the tension in the string is 3 times the weight of the pendulum bob. What is the maximum displacement of the pendulum of the string with respect to the vertical?

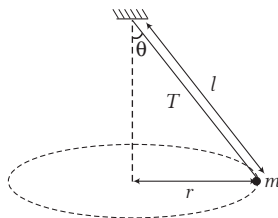
(a) 30° (b) 45° (c) 60° (d) 90°

- 44** A stone of mass 1 kg tied to a light inextensible string of length $L = \frac{10}{3}$ m, whirling in a circular path

in a vertical plane. The ratio of maximum tension to the minimum tension in the string is 4. If g is taken to be 10 ms^{-2} , the speed of the stone at the highest point of the circle is

(a) 10 ms^{-1} (b) $5\sqrt{2} \text{ ms}^{-1}$ (c) $10\sqrt{3} \text{ ms}^{-1}$ (d) 20 ms^{-1}

- 45** A string of length l fixed at one end carries a mass m at the other end. The string makes $\frac{2}{\pi} \text{ rev s}^{-1}$ around the axis through the fixed end as shown in the figure, the tension in the string is



(a) $16ml$ (b) $4ml$ (c) $8ml$ (d) $2ml$

- 46** A particle starts travelling on a circle with constant tangential acceleration. The angle between velocity vector and acceleration vector, at the moment when particle complete half the circular track, is

(a) $\tan^{-1}(2\pi)$ (b) $\tan^{-1}(\pi)$ (c) $\tan^{-1}(3\pi)$ (d) zero

- 47** A wet open umbrella is held vertical and it whirled about the handle at a uniform rate of 21 rev in 44 s. If the rim of the umbrella is a circle of diameter 1 m and the height of the rim above the floor is 4.9 m, then the locus of the drop on floor is a circle of radius

(a) $\sqrt{2.5}$ m (b) 1 m
(c) 3 m (d) 1.5 m

- 48** A heavy particle is tied to the end A of a string of length 1.6 m. Its other end O is fixed. It revolves as a conical pendulum with the string making 60° with the horizontal. Then,

(a) its period of revolution is $\frac{4\pi}{7}$ s

(b) the tension in the string is $1/\sqrt{3}$ times the weight of the particle

(c) the speed of the particle is $2.8\sqrt{3} \text{ ms}^{-1}$

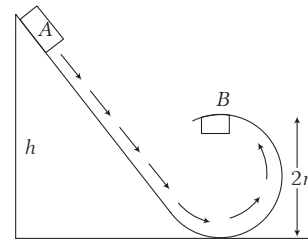
(d) the centripetal acceleration of the particle is $\frac{9.8}{\sqrt{3}} \text{ ms}^{-2}$

- 49** A pendulum bob has a speed of 3 ms^{-1} at its lowest position. The pendulum is 0.5 m long. The speed of the bob, when string makes an angle of 60° to the vertical is (Take, $g = 10 \text{ ms}^{-2}$)

(a) 2 ms^{-1} (b) $\frac{1}{2} \text{ ms}^{-1}$

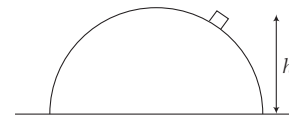
(c) 1 ms^{-1} (d) 2.5 ms^{-1}

- 50** A block is released from rest at the top of an inclined plane which later curves into a circular track of radius r as shown in figure. The minimum height h from where it should be released, so that it is able to complete the circle, is



(a) r (b) $2.5r$
(c) $1.5r$ (d) $0.5r$

- 51** A small body of mass m slides without friction from the top of a hemisphere of radius r . At what height will the body be detached from the centre of the hemisphere?

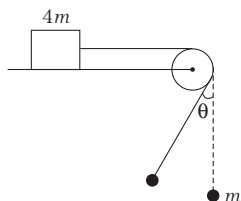


(a) $h = \frac{r}{2}$ (b) $h = \frac{r}{3}$ (c) $h = \frac{2r}{3}$ (d) $h = \frac{r}{4}$

- 52** The maximum tension that an inextensible ring of radius 1m and mass density 0.1 kg m^{-1} can bear is 40 N. The maximum angular velocity with which it can be rotated in a circular path is

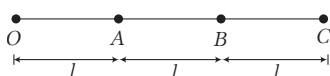
(a) 20 rad s^{-1} (b) 18 rad s^{-1}
(c) 16 rad s^{-1} (d) 15 rad s^{-1}

- 53** Two bodies of masses m and $4m$ are attached to a string as shown in the figure. The body of mass m hanging from a string of length l is executing periodic motion with amplitude $\theta = 60^\circ$ while other body is at rest on the surface.

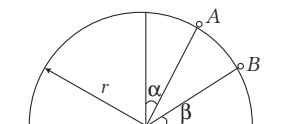


The minimum coefficient of friction between the mass $4m$ and the horizontal surface must be

- (a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{2}{3}$
- 54** A bullet of mass m moving with a horizontal velocity u strikes a stationary wooden block of mass M suspended by a string of length $L = 50$ cm. The bullet emerges out of the block with speed $\frac{u}{4}$. If $M = 6m$, the minimum value of u , so that the block can complete the vertical circle, is (Take, $g = 10 \text{ ms}^{-2}$)
- (a) 10 ms^{-1} (b) 20 ms^{-1} (c) 30 ms^{-1} (d) 40 ms^{-1}
- 55** Three identical particles are joined together by a thread as shown in figure. All the three particles are moving in a circle in horizontal plane. If the velocity of the outermost particle is v_0 , then the ratio of tensions in the three sections of the string is ($T_{BC} : T_{AB} : T_{OA}$)



- (a) $3 : 5 : 7$ (b) $3 : 4 : 5$ (c) $7 : 11 : 6$ (d) $3 : 5 : 6$
- 56** A particle moves from rest at A on the surface of a smooth circular cylinder of radius r as shown in the figure. At B, it leaves the cylinder. The equation relating α and β is



- (a) $3 \sin \alpha = 2 \cos \beta$ (b) $2 \sin \alpha = 3 \cos \beta$
(c) $3 \sin \beta = 2 \cos \alpha$ (d) $2 \sin \beta = 3 \cos \alpha$

- 57** A ball is placed on a smooth inclined plane of inclination $\theta = 30^\circ$ to the horizontal, which is rotating at frequency 0.5 Hz about a vertical axis passing through its lower end. At what distance from the lower end does the ball remain at rest?

- (a) 0.87 m (b) 0.33 m
(c) 0.5 m (d) 0.67 m

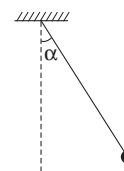
- 58** A particle suspended by a light inextensible thread of length l is projected horizontally from its lowest position with velocity $\sqrt{7gl/2}$. The string will slack after swinging through an angle equal to

- (a) 30° (b) 90°
(c) 120° (d) 150°

- 59** The kinetic energy K of a particle moving along a circle of radius R depends on the distance covered s as $K = as^2$. The force acting on the particle is

- (a) $\frac{2as^2}{R}$ (b) $2as \left(1 + \frac{s^2}{R^2}\right)^{1/2}$
(c) $as \left(1 + \frac{s^2}{R^2}\right)^{1/2}$ (d) None of these

- 60** A simple pendulum is vibrating with an angular amplitude of 90° as shown in the figure. For what value of α , is the acceleration directed?



- (i) Vertically upwards
(ii) Horizontally
(iii) Vertically downwards

- (a) $0^\circ, \cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 90^\circ$ (b) $90^\circ, \cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 0^\circ$
(c) $0^\circ, \cos^{-1}\sqrt{3}, 90^\circ$ (d) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right), 90^\circ, 0^\circ$

- 61.** A small block of mass m is released from the top of a smooth hemisphere of radius R with the horizontal speed u . What is the angle with vertical, where it loses contact with the hemisphere?

- (a) $\sin^{-1}\left(\frac{u^2}{3gR} + \frac{2}{3}\right)$ (b) $\cos^{-1}\left(\frac{u^2}{3gR} + \frac{2}{3}\right)$
(c) $\cos^{-1}\left(\frac{u^2}{6gR} + \frac{4}{3}\right)$ (d) $\sin^{-1}\left(\frac{u^2}{6gR} + \frac{2}{3}\right)$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1–5) *These questions consist of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses*

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is correct but Reason is incorrect.
- (d) If Assertion is incorrect but Reason is correct.

1 Assertion A particle is rotating in a circle of radius 1m. At some given instant, its speed is 2 ms^{-1} . Then, acceleration of particle at the given instant is 4 ms^{-2} .

Reason Centripetal acceleration at this instant is 4 ms^{-2} towards centre of circle.

2 Assertion When a car takes a circular turn on a horizontal road, then normal reaction on inner wheels is always greater than the normal reaction on outer wheels.

Reason This is for rotational equilibrium of car.

3 Assertion A ball tied by thread is undergoing circular motion (of radius R) in a vertical plane. (Thread always remains in vertical plane). The difference of maximum and minimum tension in thread is independent of speed (u) of ball at the lowest position ($u > \sqrt{5gR}$).

Reason For a ball of mass m tied by thread undergoing vertical circular motion (of radius R), difference in maximum and minimum magnitude of centripetal acceleration of the ball is independent of speed (u) of ball at the lowest position ($u > \sqrt{5gR}$).

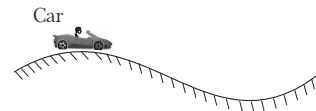
4 Assertion One end of a massless rod of length l is hinged, so that it is free to rotate in a vertical plane about a horizontal axis. If a particle is attached to the other end of the rod, then the minimum speed at lower most position of the particle is $\sqrt{5gl}$ to complete the circular motion.

Reason Work done by centripetal force on the particle is always zero.

5 Assertion A car moves along a road with uniform speed. The path of car lies in vertical plane and is shown in figure. The radius of curvature (R) of the path is same everywhere. If the car does not loose contact with road at the highest point, it can travel

the shown path without loosing contact with road anywhere else.

Reason For car to loose contact with road, the normal reaction between car and road should be zero.



Statement based questions

1 A particle moves in a uniform circular motion. Choose the incorrect statement.

- (a) The particle moves with constant speed.
- (b) The acceleration is always normal to the velocity.
- (c) The particle moves with uniform acceleration.
- (d) The particle moves with variable velocity.

2 Which of the following statement is incorrect?

- (a) Uniform circular motion is uniformly accelerated motion.
- (b) Acceleration in uniform circular motion is always towards centre.
- (c) In circular motion, dot product of $\mathbf{v}$ and $\boldsymbol{\omega}$ is always zero.
- (d) In any curvilinear path, average speed and average velocity are never equal.

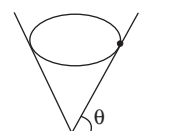
3 Which of the following statement(s) is/are correct?

- (a) Centripetal force mv^2 / R acts on a particle rotating in a circle.
- (b) If a particle is rotating in a circle, then centrifugal force is acting on the particle in radially outward direction.
- (c) Centrifugal force is equal and opposite to the centripetal force.
- (d) All of the above

4 Which of the following statement(s) is/are correct?

- I. When water in a bucket is whirled fast overhead, the water does not fall out at the top of the circular path.
 - II. The centripetal force in this position on water is more than the weight of water.
- (a) Only I
 - (b) Only II
 - (c) Both I and II
 - (d) None of these

5 A small block of mass m is rotating in a circle inside a smooth cone as shown in figure. Which of the following statement(s) is/are correct regarding this?



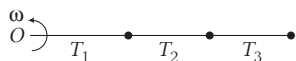
- I. In this case, the normal reaction, $N \neq mg \cos \theta$.

II. In this case, acceleration of the block is not along the surface of cone. It is horizontal.

- (a) Only I (b) Only II
(c) Both I and II (d) None of these

Match the columns

- 1 Three balls each of mass 1 kg are attached with three strings each of length 1 m as shown in figure. They are rotated in a horizontal circle with angular velocity $\omega = 4 \text{ rad s}^{-1}$ about point O. Match the following columns and mark the correct option from the codes given below.

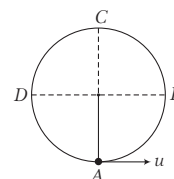


	Column I		Column II
(A)	T_1	(p)	Maximum
(B)	T_2	(q)	80 N
(C)	T_3	(r)	48 N
		(s)	90 N

Codes

- | | | | | | |
|-------|---|---|-------|---|---|
| A | B | C | A | B | C |
| (a) p | q | r | (b) s | q | p |
| (c) p | s | q | (d) s | r | p |

- 2 A particle is suspended from a string of length R . It is given a velocity $u = 3\sqrt{gR}$ at the bottom. Match the following columns and mark the correct option from the codes given below.



	Column I		Column II
(A)	Velocity at B	(p)	$7mg$
(B)	Velocity at C	(q)	$\sqrt{5gR}$
(C)	Tension in string at B	(r)	$\sqrt{7gR}$
(D)	Tension in string at C	(s)	$5mg$
		(t)	None

Codes

- | | | | | | | | |
|-------|---|---|---|-------|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) p | q | r | s | (b) r | q | p | t |
| (c) p | s | q | t | (d) t | q | p | s |

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1 A point mass m is moved in a vertical circle of radius r with the help of a string. The velocity of the mass is $\sqrt{7gr}$ at the lowest point. The tension in the string at the lowest point is [NEET 2020]

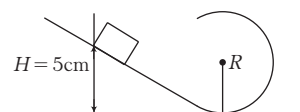
- (a) $6mg$ (b) $7mg$ (c) $8mg$ (d) $1mg$

- 2 A block of mass 10 kg is in contact against the inner wall of a hollow cylindrical drum of radius 1 m. The coefficient of friction between the block and the inner wall of the cylinder is 0.1. The minimum angular velocity needed for the cylinder to keep the block stationary when the cylinder is vertical and rotating about its axis, will be (Take, $g = 10 \text{ m/s}^2$) [NEET 2019]

- (a) $\frac{10}{2\pi} \text{ rad/s}$ (b) 10 rad/s (c) $10\pi \text{ rad/s}$ (d) $\sqrt{10} \text{ rad/s}$

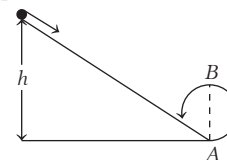
- 3 A mass m is attached to a thin wire and whirled in a vertical circle. The wire is most likely to break when [NEET 2019]
(a) the wire is horizontal
(b) the mass is at the lowest point
(c) inclined at an angle of 60° from vertical
(d) the mass is at the highest point

- 4 Find the maximum radius of circle, so that the block can complete the circular motion. [JIPMER 2019]



- (a) 5 cm (b) 3 cm
(c) 2 cm (d) 4 cm

- 5 A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter $AB = D$. The height h is equal to [NEET 2018]



- (a) $\frac{7}{5}D$ (b) D
(c) $\frac{3}{2}D$ (d) $\frac{5}{4}D$

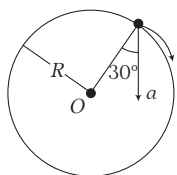
- 6** One end of the string of length l is connected to a particle of mass m and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed v , the net force on the particle (directed towards centre) will be (T represents the tension in the string) [NEET 2017]

(a) T (b) $T + \frac{mv^2}{l}$
 (c) $T - \frac{mv^2}{l}$ (d) zero

- 7 Assertion** For looping a vertical loop of radius r , the minimum velocity at lowest point should be $\sqrt{5gr}$.

Reason In this event, the velocity at the highest point will be zero. [AIIMS 2017]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Both Assertion and Reason are incorrect.
- 8** In the given figure, $a = 15 \text{ m/s}^2$ represents the total acceleration of a particle moving in the clockwise direction in a circle of radius $R = 2.5 \text{ m}$ at a given instant of time. The speed of the particle is [NEET 2016]



- (a) 4.5 m/s (b) 5.0 m/s
 (c) 5.7 m/s (d) 6.2 m/s

- 9** A car is negotiating a curved road of radius R . The road is banked at angle θ . The coefficient of friction between the tyres of the car and the road is μ_s . The maximum safe velocity on this road is [NEET 2016]

(a) $\sqrt{gR \left(\frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta} \right)}$ (b) $\sqrt{\frac{g}{R} \left(\frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta} \right)}$
 (c) $\sqrt{\frac{g}{R^2} \left(\frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta} \right)}$ (d) $\sqrt{gR^2 \left(\frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta} \right)}$

- 10** A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2 rad s^{-2} . Its net acceleration (in ms^{-2}) at the end of 2 s is approximately [NEET 2016]

- (a) 7 (b) 6
 (c) 3 (d) 8

- 11** What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R , so that it can complete the loop? [NEET 2016]

(a) $\sqrt{2gR}$ (b) $\sqrt{3gR}$
 (c) $\sqrt{5gR}$ (d) $\sqrt{gR}$

- 12** A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration, if the kinetic energy of the particle becomes equal to $8 \times 10^{-4} \text{ J}$ by the end of the second revolution after the beginning of the motion? [NEET 2016]

(a) 0.15 ms^{-2} (b) 0.18 ms^{-2}
 (c) 0.2 ms^{-2} (d) 0.1 ms^{-2}

- 13** A particle is moving in a curved path. Which of the following quantities may remain constant during its motion? [CG PMT 2015]
 (a) Acceleration (b) Velocity
 (c) Magnitude of acceleration (d) None of these

- 14** The ratio of angular speed of a second hand to the hour-hand of a watch is [KCET 2015]

(a) 3600 : 1 (b) 720 : 1
 (c) 72 : 1 (d) 60 : 1

- 15** If the length of second's hand of a clock is 10 cm, the speed of its tip (in cm s^{-1}) is nearly [Kerala CEE 2014]

(a) 2 (b) 0.5 (c) 1.5 (d) 1
 (e) 3

- 16** A particle is moving uniformly in a circular path of radius r . When it moves through an angular displacement θ , then the magnitude of the corresponding linear displacement will be [WB JEE 2014]

(a) $2r \cos \left(\frac{\theta}{2} \right)$ (b) $2r \cot \left(\frac{\theta}{2} \right)$
 (c) $2r \tan \left(\frac{\theta}{2} \right)$ (d) $2r \sin \left(\frac{\theta}{2} \right)$

- 17** A rotating wheel changes angular speed from 1800 rpm to 3000 rpm in 20 s. What is the angular acceleration assuming to be uniform? [KCET 2014]

(a) $60\pi \text{ rad s}^{-2}$ (b) $90\pi \text{ rad s}^{-2}$
 (c) $2\pi \text{ rad s}^{-2}$ (d) $40\pi \text{ rad s}^{-2}$

- 18** A stone tied to a rope is rotated in a vertical circle with uniform speed. If the difference between the maximum and minimum tensions in the rope is 20 N, mass of the stone (in kg) is (Take, $g = 10 \text{ ms}^{-2}$) [EAMCET 2013]

(a) 0.75 (b) 1.0
 (c) 1.5 (d) 0.5

- 19** A car of mass 1000 kg negotiates a banked curve of radius 90 m on a frictionless road. If the banking angle is 45° , the speed of the car is [CBSE AIPMT 2012]

(a) 20 ms^{-1}
 (b) 30 ms^{-1}
 (c) 5 ms^{-1}
 (d) 10 ms^{-1}

- 20** A car is moving in a circular horizontal track of radius 10.0 m with a constant speed of 10.0 ms^{-1} . A plumb bob is suspended from the roof of the car by a light rigid rod of length 10.0 m. The angle made by the rod with the track is (Take, $g = 10 \text{ ms}^{-2}$) [AFMC 2011]

(a) zero (b) 30° (c) 45° (d) 60°

ANSWERS

• CHECK POINT 7.1

1. (d)	2. (c)	3. (a)	4. (a)	5. (a)	6. (d)	7. (c)	8. (b)	9. (d)	10. (c)
11. (d)	12. (a)	13. (a)	14. (d)	15. (b)	16. (c)				

• CHECK POINT 7.2

1. (d)	2. (a)	3. (d)	4. (d)	5. (b)	6. (a)	7. (d)	8. (d)	9. (a)	10. (b)
11. (b)	12. (d)	13. (a)	14. (d)	15. (d)	16. (a)	17. (c)	18. (c)	19. (a)	20. (c)
21. (a)	22. (c)								

(A) Taking it together

1. (d)	2. (b)	3. (a)	4. (c)	5. (a)	6. (b)	7. (a)	8. (c)	9. (d)	10. (b)
11. (c)	12. (c)	13. (a)	14. (c)	15. (d)	16. (c)	17. (b)	18. (a)	19. (c)	20. (a)
21. (c)	22. (c)	23. (c)	24. (a)	25. (d)	26. (d)	27. (c)	28. (d)	29. (d)	30. (a)
31. (b)	32. (c)	33. (c)	34. (c)	35. (a)	36. (b)	37. (b)	38. (b)	39. (c)	40. (a)
41. (d)	42. (a)	43. (d)	44. (a)	45. (a)	46. (a)	47. (a)	48. (d)	49. (a)	50. (b)
51. (c)	52. (a)	53. (c)	54. (d)	55. (d)	56. (c)	57. (d)	58. (c)	59. (b)	60. (a)
61. (b)									

(B) Medical entrance special format questions

• Assertion and reason

1. (b) 2. (d) 3. (a) 4. (b) 5. (d)

• Statement based questions

1. (c) 2. (a) 3. (d) 4. (c) 5. (c)

• Match the columns

1. (a) 2. (b)

(C) Medical entrances' gallery

1. (c)	2. (b)	3. (b)	4. (c)	5. (d)	6. (a)	7. (c)	8. (c)	9. (a)	10. (d)
11. (c)	12. (d)	13. (c)	14. (b)	15. (d)	16. (d)	17. (c)	18. (b)	19. (b)	20. (c)

Hints & Explanations

• CHECK POINT 7.1

1 (d) $\omega = 120 \text{ rev/min} = 120 \times \frac{2\pi}{60} \text{ rad s}^{-1} = 4\pi \text{ rad s}^{-1}$

2 (c) $\omega_{\min} = \frac{2\pi}{60} \text{ rad/min}$ and $\omega_{\text{hr}} = \frac{2\pi}{12 \times 60} \text{ rad/min}$

$$\therefore \frac{\omega_{\min}}{\omega_{\text{hr}}} = \frac{2\pi / 60}{2\pi / 12 \times 60} = \frac{12}{1}$$

3 (a) $\alpha = \frac{\Delta\omega}{\Delta t}$ but $\omega = 2\pi f$

$$\therefore \text{Angular retardation, } \alpha = \frac{2\pi\Delta f}{\Delta t} = \frac{2\pi \times 200}{2} = (200\pi) \text{ rad s}^{-2}$$

4 (a) As, $\omega = \omega_0 + \alpha t$

Here, $\omega_0 = 900 \text{ rpm} = \frac{(2\pi \times 900)}{60} \text{ rad s}^{-1}$, $\omega = 0$

and $t = 60 \text{ s}$

$$\therefore 0 = \frac{2\pi \times 900}{60} + \alpha \times 60$$

$$\Rightarrow \alpha = -\frac{2\pi \times 900}{60 \times 60} = -\frac{\pi}{2} \text{ rad s}^{-2}$$

$$\therefore |\alpha| = \frac{\pi}{2} \text{ rad s}^{-2}$$

5 (a) From equation of motion (angular),

$$0 = \omega_0 - \alpha t \Rightarrow \alpha = \frac{\omega_0}{t} = \frac{(100 \times 2\pi)/60}{15} = 0.69 \text{ rad s}^{-2}$$

$$\cong 0.7 \text{ rads}^{-2}$$

Now, angle rotated before coming to rest is given by

$$\theta = \frac{\omega_0^2}{2\alpha} = \frac{\left(\frac{100 \times 2\pi}{60}\right)^2}{2 \times 0.7} = 78.25 \text{ rad}$$

$$\therefore \text{Number of revolutions} = \frac{\theta}{2\pi} = 12.5$$

6 (d) $a = \frac{v^2}{R}$ or $a \propto v^2 \Rightarrow \frac{a_{\text{after}}}{a_{\text{before}}} = \frac{(2v)^2}{v^2} = \frac{4}{1}$

7 (c) $a = r\omega^2$ or $a \propto r \Rightarrow$ so if $r_1 < r_2$, then $a_1 < a_2$

8 (b) $a_t = 0$, $a = a_n = \frac{v^2}{R} = \frac{(6)^2}{0.3} = 120 \text{ ms}^{-2}$

9 (d) As, $v = \omega r$

$$\Rightarrow 50 = \omega \left(\frac{20}{\pi}\right) \Rightarrow \omega = \frac{50\pi}{20} = \frac{5\pi}{2}$$

$$\Rightarrow \frac{2\pi}{T} = \frac{5\pi}{2}$$

$$\Rightarrow T = \frac{4}{5} \text{ s}$$

Given, $t = 2T = \frac{8}{5} \text{ s}$

$$\therefore \omega = \alpha t$$

$$\Rightarrow \frac{5\pi}{2} = \alpha \times \frac{8}{5}$$

$$\Rightarrow \alpha = \frac{25\pi}{16} \text{ rads}^{-2}$$

$$\therefore a_t = r(\alpha) = \left(\frac{20}{\pi}\right)\left(\frac{25\pi}{16}\right) = 31.25 \text{ ms}^{-2}$$

10 (c) In circular motion, a_r can never be zero. So, option (c) is correct.

11 (d) $\frac{a_r}{a_t} = \frac{v^2/R}{a}$ (Let $a_t = a$)

Here, $v^2 = 2al = 2a(2\pi R) = 4\pi aR$. Therefore, the ratio is $\frac{4\pi}{1}$.

12 (a) $a = \sqrt{a_t^2 + a_n^2}$

a_t = rate of change of speed = 2 ms^{-2}

$$a_n = \frac{v^2}{R} = \frac{(5)^2}{10} = 2.5 \text{ ms}^{-2}$$

$$\therefore a = \sqrt{a_t^2 + a_n^2} = \sqrt{(2)^2 + (2.5)^2} = 3.2 \text{ ms}^{-2}$$

13 (a) $a_n = \sqrt{a^2 - a_t^2} = \sqrt{(100)^2 - (60)^2} = 80 \text{ ms}^{-2}$

Now, $a_n = \frac{v^2}{R}$ or $R = \frac{v^2}{a_n} = \frac{(10)^2}{80} = 1.25 \text{ m}$

14 (d) Given, $\omega = a - bt$

$$\alpha = \frac{d\omega}{dt} = -b$$

$$a_t = R\alpha = -Rb$$

...(i)

At $t = \frac{2a}{b}$, $\omega = -a$

$$a_n = R\omega^2 = Ra^2$$

...(ii)

Now, $a = \sqrt{a_t^2 + a_n^2} = R\sqrt{a^4 + b^2}$

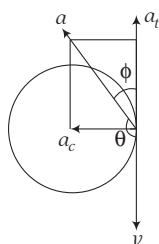
15 (b) The ratio of tangential to centripetal acceleration is

$$\left(\frac{a_t}{a_c}\right)_{t=t} = \frac{\frac{dv}{dt}}{\frac{v^2}{r}} = \frac{\frac{d}{dt}\left(\frac{ds}{dt}\right)}{\left(\frac{ds}{dt}\right)^2} r$$

$$= \frac{\frac{d}{dt}(6t^2)}{(6t^2)^2} r = \frac{(12t) \times 12}{36t^4} = \frac{4}{t^3}$$

$$\therefore \left(\frac{a_t}{a_c}\right)_{t=2s} = \frac{4}{(2)^3} = \frac{1}{2}$$

$$16 \text{ (c) } \tan \phi = \frac{a_c}{a_t} = \frac{\frac{v^2}{r}}{\left| \frac{dv}{dt} \right|} = \frac{\frac{20 \times 20}{80}}{5} = \frac{20 \times 20}{400} = 1$$



$$\Rightarrow \phi = 45^\circ$$

Angle between a (net) and v , $\theta = \pi - \phi = 135^\circ$

• CHECK POINT 7.2

$$1 \text{ (d) Centripetal force} = mR\omega^2 = (2) (1) (2\pi)^2 = 8\pi^2 \text{ N}$$

$$2 \text{ (a) } F = \frac{mv^2}{r}. \text{ If } m \text{ and } v \text{ are constants, then } F \propto \frac{1}{r}$$

$$\therefore \frac{F_1}{F_2} = \left(\frac{r_2}{r_1} \right)$$

$$3 \text{ (d) Radial force} = \frac{mv^2}{r} = \frac{m}{r} \left(\frac{p}{m} \right)^2 = \frac{p^2}{mr}$$

$$4 \text{ (d) Maximum tension} = \frac{mv^2}{r} = 16 \text{ N} \Rightarrow \frac{16 \times v^2}{144} = 16$$

$$\therefore v = 12 \text{ ms}^{-1}$$

$$5 \text{ (b) } F = \frac{mv^2}{r}, F' = \frac{(15m)(15v)^2}{(15r)} = 2.25 \frac{mv^2}{r} = 2.25 F$$

$$\text{Percentage increase} = \frac{2.25F - F}{F} \times 100 = 125\%$$

Therefore, F has to be increased by 125%.

$$6 \text{ (a) } T = 100 \text{ N} = m\omega_m^2 r$$

$$\Rightarrow 100 = 100 \times 10^{-3} \times \omega_m^2 \times 0.1$$

$$\Rightarrow \omega_m = 100 \text{ rads}^{-1}$$

$$7 \text{ (d) Tension in the string, } T = m\omega^2 r = 4\pi^2 n^2 mr$$

$$\therefore T \propto n^2 \Rightarrow \frac{n_2}{n_1} = \sqrt{\frac{T_2}{T_1}}$$

$$\therefore n_2 = 5\sqrt{\frac{2T}{T}} = 5\sqrt{2} \text{ rpm}$$

$$8 \text{ (d) Maximum tension} = m\omega^2 r = m \times 4\pi^2 \times n^2 \times r$$

By substituting the values, we get $T_{\max} = 87.64 \text{ N}$

$$9 \text{ (a) For safe turning,}$$

$$\tan \theta = \frac{v^2}{gr} \Rightarrow \frac{h}{b} = \frac{v^2}{gR} \Rightarrow h = \frac{v^2 b}{Rg}$$

$$10 \text{ (b) To avoid skidding, } \tan \theta = \frac{v^2}{gr} = \frac{20 \times 20}{10 \times 20} = 2$$

$$\Rightarrow \theta = \tan^{-1}(2)$$

$$11 \text{ (b) } \tan \theta = \frac{v^2}{rg} \Rightarrow \tan 45^\circ = \frac{v^2}{(90)(10)} \Rightarrow v = 30 \text{ m/s}$$

$$12 \text{ (d) } \tan \theta = \frac{v_1^2}{Rg} = \frac{v_2^2}{4Rg} \Rightarrow v_2 = 2v_1$$

$$\left(\frac{v_2}{v_1} - 1 \right) = 1$$

$$\% \text{ increase} = 100\%$$

$$13 \text{ (a) } N = \frac{mv^2}{R}, mg = \mu_s N = \frac{\mu_s mv^2}{R}$$

$$\therefore v = \sqrt{\frac{Rg}{\mu_s}}$$

$$14 \text{ (d) } N = \frac{mv^2}{R}, mg = \mu N = \frac{\mu mv^2}{R}$$

$\therefore$ Coefficient of friction between the tyres and wall of the well

$$\text{will be } \mu = \frac{Rg}{v^2} = \frac{5 \times 10}{(5\sqrt{5})^2} = 0.40$$

$$15 \text{ (d) } mg = \frac{mv_{\min}^2}{R} \text{ or } v_{\min} = \sqrt{Rg} \text{ (at topmost point)}$$

16 (a) Maximum tension will be at bottommost point. Therefore, the string will break at A.

$$17 \text{ (c) At the highest point, } T + mg = \frac{mv^2}{r} \Rightarrow mg \leq \frac{mv^2}{r}$$

$$18 \text{ (c) Tension, } T = \frac{mv^2}{r} + mg \cos \theta$$

$$\text{For } \theta = 30^\circ, T_1 = \frac{mv_1^2}{r} + mg \cos 30^\circ$$

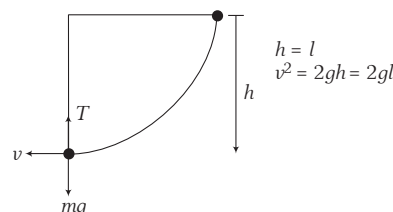
$$\text{For } \theta = 60^\circ, T_2 = \frac{mv_2^2}{r} + mg \cos 60^\circ$$

$$\therefore \cos 30^\circ > \cos 60^\circ$$

$$\therefore T_1 > T_2 \text{ as } v_1 > v_2$$

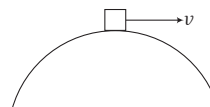
19 (a) In critical case, tension at topmost point is zero.

$$20 \text{ (c) } T - mg = \frac{mv^2}{l} \text{ or } T = 3mg$$



$$21 \text{ (a) } v_{\max} = \sqrt{2gh} = \sqrt{2g(h_{\max} - h_{\min})} = \sqrt{2 \times 10 \times 1.25} = 5 \text{ ms}^{-1}$$

$$22 \text{ (c) } v_{\min} = \sqrt{gR} = \sqrt{9.8 \times 49} = 21.9 \approx 22 \text{ ms}^{-1}$$



(A) Taking it together

- 2 (b)** Here, rpm mean rotation per minute, *i.e.* it is frequency of rotation,
 $f = 600 \text{ rpm} \Rightarrow f = \frac{600}{60} \text{ rps} = 10 \text{ Hz}$

$$\Rightarrow \text{Time period, } T = \frac{1}{f} = \frac{1}{10} = 0.1 \text{ s}$$

- 3 (a)** The initial angular velocity,

$$\omega_0 = 50 \text{ rpm} = \frac{5\pi}{3} \text{ rad s}^{-1}$$

$$\text{Using } \omega = \omega_0 + \alpha t$$

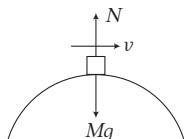
$$\therefore \alpha = \frac{\omega - \omega_0}{t} = \frac{0 - \frac{5\pi}{3}}{60} \text{ rad s}^{-2} = -\frac{\pi}{36} \text{ rad s}^{-2}$$

- 4 (c)** Speed of particle is increasing due to tangential component of acceleration. Hence, dot product of a and v is positive.

- 6 (b)** As the speed is constant throughout the circular motion, therefore its average speed is equal to instantaneous speed.

- 7 (a)** Minimum velocity at topmost point is $\sqrt{Rg}$.

$$\mathbf{8 (c)} \quad Mg - N = \frac{Mv^2}{r} \quad \text{or} \quad N = Mg - \frac{Mv^2}{r}$$



- 9 (d)** If the string suddenly breaks, the centripetal force will be zero as the tension of string become zero but the tangential force will be present due to tangential velocity, so the stone travels in tangential direction.

- 12 (c)** For safe circular turn, $v = \sqrt{\mu rg}$

$$\Rightarrow r = \frac{v^2}{\mu g} \Rightarrow r = \frac{(10)^2}{0.2 \times 10} = 50 \text{ m}$$

- 13 (a)** Given, $v = 72 \text{ kmh}^{-1} = 72 \times \frac{5}{18} \text{ ms}^{-1} = 20 \text{ ms}^{-1}$

$$R = 100 \text{ m}$$

$$\therefore \theta = \tan^{-1} \left(\frac{v^2}{Rg} \right) = \tan^{-1} \left(\frac{400}{100 \times 10} \right) = \tan^{-1} \left(\frac{2}{5} \right)$$

- 14 (c)** $L = mvr \Rightarrow v = \frac{L}{mr}$

$$\therefore F = \frac{mv^2}{r} = \frac{L^2}{mr^3}$$

- 15 (d)** $a_{av} = \frac{\Delta v}{\Delta t} = \frac{2 \times 5}{\pi} = \frac{10}{\pi} \text{ ms}^{-2}$

- 16 (c)** Given, $f = 2400 \text{ rpm} = \frac{2400}{60} = 40 \text{ Hz}$

$$\therefore \omega_0 = 2\pi f = 80\pi \text{ rad s}^{-2}$$

$$\text{Now, } \theta = \left[\frac{\omega_0 + \omega}{2} \right] t$$

$$\Rightarrow \theta = \left[\frac{80\pi + 0}{2} \right] \times 10 = 400\pi \text{ rad}$$

$\therefore$ Fan will turn angle 400π after it is switched off.

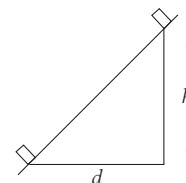
$$\Rightarrow \text{The number of rotation} = \frac{\theta}{2\pi} = \frac{400\pi}{2\pi} = 200$$

- 17 (b)** his elevation of outer rail over inner rail.

$$h = \frac{dv^2}{gr}$$

$$\Rightarrow h = \frac{1.67 \times (15)^2}{10 \times 0.5 \times 1000}$$

$$= 0.075 \text{ m} = 75 \text{ mm}$$



- 18 (a)** Difference between the kinetic energies at its highest and lowest positions is

$$= 2mgh = 2 \times 1 \times 10 \times 1 = 20 \text{ J}$$

- 19 (c)** $\tan \theta = \frac{v^2}{gr} \Rightarrow \frac{3}{5} = \frac{v^2}{10 \times 6}$

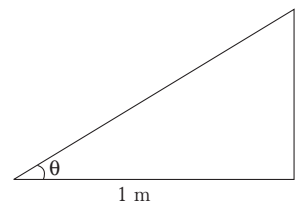
$$\Rightarrow v = 6 \text{ ms}^{-1} = 21.6 \text{ kmh}^{-1}$$

- 20 (a)** Given, $v = 144 \text{ kmh}^{-1} = 144 \times \frac{5}{18} \text{ ms}^{-1} = 40 \text{ ms}^{-1}$

$$\therefore \tan \theta = \frac{v^2}{Rg} = \frac{(40)^2}{40 \times 10} = \frac{1600}{400} = 4$$

$$\Rightarrow \theta = \tan^{-1}(4)$$

- 21 (c)** Given, $v = 96 \text{ kmh}^{-1} = 96 \times \frac{5}{18} \text{ ms}^{-1} = 26.67 \text{ ms}^{-1}$



$$\therefore \tan \theta = \frac{v^2}{Rg} = \frac{(26.67)^2}{800 \times 10} = 0.089$$

$$\Rightarrow 0.089 = \frac{h}{1} \therefore h = 0.089 \text{ m} = 8.9 \text{ cm}$$

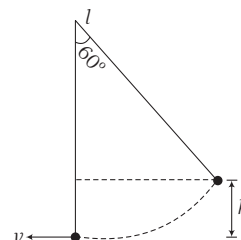
- 22 (c)** $\alpha = \text{constant}$

$$\therefore \theta = \frac{1}{2} \alpha t^2$$

$$\therefore \theta \propto t^2$$

$$\text{or } \frac{\theta_2 + \theta_1}{\theta_1} = \left(\frac{2+2}{2} \right)^2 = 4 \quad \text{or} \quad \frac{\theta_2}{\theta_1} = 3$$

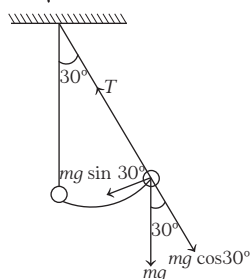
- 23 (c)** From figure, $h = l(1 - \cos 60^\circ) = \frac{l}{2} = 1 \text{ m}$



Applying conservation of mechanical energy, we get

$$v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 1} = 4.43 \text{ ms}^{-1}$$

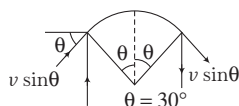
$$24 \text{ (a)} \quad a = \sqrt{a_n^2 + a_t^2} = \sqrt{\left(\frac{T - mg \cos 30^\circ}{m}\right)^2 + (g \sin 30^\circ)^2}$$



$$= g \sqrt{\left(\sqrt{3} - \frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{4}\right)} = g$$

$$25 \text{ (d)} \quad \Delta t = \frac{\text{Distance travelled}}{\text{Speed}} = \frac{(2\pi R / 6)}{v} = \frac{3.14 \times 300}{60 \times 3} = 5.23 \text{ s}$$

$$(i) \quad |\Delta \mathbf{v}| = |\mathbf{v}_f| - |\mathbf{v}_i| = \sqrt{v^2 + v^2 - 2v \cdot v \cos 60^\circ} = 2v \sin 30^\circ = 60 \text{ ms}^{-1}$$



$$(ii) \quad a_i = \frac{v^2}{R} = 12 \text{ ms}^{-2}$$

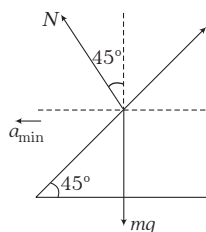
$$(iii) \quad |\mathbf{a}_{av}| = \frac{|\Delta \mathbf{v}|}{\Delta t} = \frac{60}{5.23} = 11.5 \text{ ms}^{-2}$$

26 (d) Necessary centripetal force to the coin is provided by friction. Thus, $mr\omega_{\max}^2 = \mu mg$ or $r = \frac{\mu g}{\omega_{\max}^2}$

$\omega_{\max}$ is made three times. Therefore, distance from centre r will become $\frac{1}{9}$ times, i.e. 1 cm.

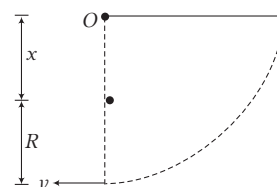
$$27 \text{ (c)} \quad |\mathbf{v}_{AB}| = |\mathbf{v}_A - \mathbf{v}_B| = \sqrt{(R\omega)^2 + (R\omega)^2 - 2(R\omega)(R\omega) \cos \theta} = 2R\omega \sin\left(\frac{\theta}{2}\right)$$

29 (d)



$$v_{\min} = \sqrt{\frac{rg(\tan \theta - \mu)}{1 + \mu \tan \theta}} \Rightarrow v_{\min} = 0, \text{ as } \mu = 1$$

30 (a) From conservation of mechanical energy, $v^2 = 2gl = 5gR$



$$\therefore R = 0.4l$$

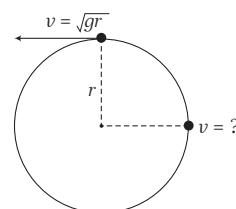
$$\text{or } x = l - R = 0.6l \text{ or } \frac{3}{5}l$$

$$31 \text{ (b)} \quad T = \frac{mu^2}{R} + mg \cos \theta; v^2 = u^2 - 2gh$$

$$T_{\text{top}} = 3mg \quad \text{and} \quad T_{\text{bottom}} = 9mg$$

$$\text{Therefore, the ratio } \frac{T_{\text{top}}}{T_{\text{bottom}}} = \frac{3mg}{9mg} = \frac{1}{3}$$

$$32 \text{ (c)} \quad v = \sqrt{gr}$$



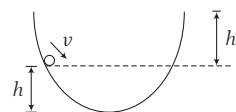
$$(\text{KE})_i + (\text{PE})_i = (\text{KE})_f + (\text{PE})_f$$

$$\Rightarrow \frac{1}{2}mv^2 + 0 = \frac{1}{2}m(\sqrt{gr})^2 + mgr$$

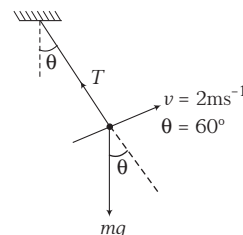
$$\Rightarrow v^2 = gr + 2gr \Rightarrow v = \sqrt{3gr}$$

$$33 \text{ (c)} \quad h' = R - h$$

$$\therefore v_{\min} = \sqrt{2gh'} \Rightarrow v_{\min} = \sqrt{2g(R - h)}$$



34 (c) Two forces are acting on girl, tension and weight. Power of tension will be zero and that of weight is,



$$P = mgv \cos(90^\circ + \theta)$$

$$= -mgv \sin 60^\circ = -50 \times 9.8 \times 2 \times \frac{\sqrt{3}}{2}$$

$$= -490\sqrt{3} \text{ W}$$

$$\therefore \text{Power delivered is } 490\sqrt{3} \text{ W.}$$

35 (a) Height, $h = l(1 - \cos \theta)$

$$v_m^2 = 2gh = 2gl(1 - \cos \theta)$$

$\therefore$ Maximum kinetic energy, $K_m = \frac{1}{2}mv_m^2 = mgl(1 - \cos \theta)$

36 (b) In the given problem, centripetal force will be equal to

$$\begin{aligned} F &= kx \\ \Rightarrow kx &= \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{kxr}{m}} \\ \Rightarrow T &= \frac{2\pi r}{v} = 2\pi \sqrt{\frac{mr}{kx}} \\ \Rightarrow T &\propto \sqrt{\frac{r}{x}} \\ \therefore T' &= T \sqrt{\frac{r'x}{rx'}} \\ &= T \sqrt{\frac{3a \times a}{2a \times 2a}} = \left(\frac{\sqrt{3}}{2}\right) T \end{aligned}$$

37 (b) $\therefore \frac{d_B}{d_C} = \frac{v_B t}{v_C t} = \frac{v_B}{v_C} \Rightarrow \frac{d_B}{d_C} = \frac{2.5}{2} = \frac{5}{4}$

38 (b) Given, radius of the horizontal circle, $r = 80 \text{ cm} = 0.80 \text{ m}$,
 $n = 14$ and $t = 25 \text{ s}$

Angular speed of revolution of the stone,

$$\begin{aligned} \omega &= \frac{2\pi n}{t} = 2\pi \left(\frac{n}{t}\right) \\ \omega &= 2 \times \frac{22}{7} \times \left(\frac{14}{25}\right) = \frac{88}{25} \text{ rads}^{-1} \end{aligned}$$

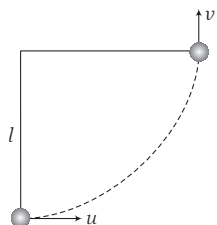
$\therefore$ Magnitude of centripetal acceleration $= r\omega^2$
 $= 80 \times \left(\frac{88}{25}\right)^2 = 9912 \approx 992 \text{ cms}^{-2}$

39 (c) $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 8}{9.8}} = 1.28 \text{ s}$

$\therefore v = \frac{20}{t} = 15.63 \text{ ms}^{-1}$

$\therefore a = \frac{v^2}{R} = 81.4 \text{ ms}^{-2}$

40 (a) From figure, we have $h = l$



$$\begin{aligned} \therefore v &= \sqrt{u^2 - 2gh} = \sqrt{u^2 - 2gl} \\ \Rightarrow |\Delta \mathbf{v}| &= |\mathbf{v}_f - \mathbf{v}_i| = \sqrt{v^2 + u^2 - 2vu \cos 90^\circ} \\ &= \sqrt{v^2 + u^2} = \sqrt{(u^2 - 2gl) + u^2} = \sqrt{2(u^2 - gl)} \end{aligned}$$

41 (d) $g \sin \theta = \frac{v^2}{R} = \frac{2gh}{R} = \frac{2gR(1 - \cos \theta)}{R}$

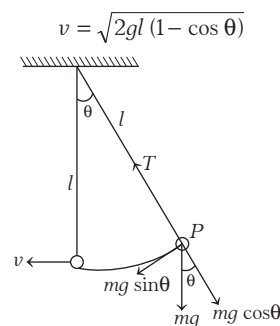
or $\sin \theta = 2(1 - \cos \theta)$

$$2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 2 \left(2 \sin^2 \frac{\theta}{2} \right)$$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{1}{2} \Rightarrow \frac{\theta}{2} = \tan^{-1} \left(\frac{1}{2} \right) \text{ or } \theta = 2 \tan^{-1} \left(\frac{1}{2} \right)$$

Note In extreme position of pendulum, only tangential component of acceleration ($a_t = g \sin \theta$) is present. In lowest position, only normal acceleration ($a_n = v^2/R$) is present.

42 (a) When body is released from the position P (inclined at angle θ from vertical), then velocity at mean position,



$\therefore$ Tension at the lowest point $= mg + \frac{mv^2}{l}$

$$= mg + \frac{m}{l} [2gl(1 - \cos 60^\circ)] = mg + mg = 2mg$$

43 (d) At mean position, $mg + \frac{mv^2}{r} = 3mg \Rightarrow v = \sqrt{2gl}$... (i)

and if the body displace by angle θ with the vertical, then

$$v = \sqrt{2gl(1 - \cos \theta)} \quad \dots (ii)$$

On comparing Eqs. (i) and (ii), we get

$$\cos \theta = 0 \Rightarrow \theta = 90^\circ$$

44 (a) Minimum tension is at topmost point (speed $= v$) and maximum tension at bottommost point (speed $= u$).

$$\therefore \frac{T_{\max}}{T_{\min}} = \frac{mg + \frac{mu^2}{L}}{-mg + \frac{mv^2}{L}} = 4$$

$$\Rightarrow u^2 = 4v^2 - 5gL \quad \dots (i)$$

$$\text{and } u^2 = v^2 + 2g(2L) \quad \dots (ii)$$

On solving Eqs. (i) and (ii), we get

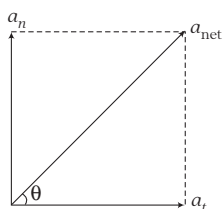
$$v = 10 \text{ ms}^{-1}$$

45 (a) Balancing horizontal forces, $T \sin \theta = m\omega^2 r$

or $T \sin \theta = m(l \sin \theta) \omega^2$

$$\begin{aligned} \therefore T &= m\omega^2 l = m(2\pi f)^2 l \\ &= ml \left(2\pi \times \frac{2}{\pi} \right)^2 = 16ml \end{aligned}$$

46 (a) $v = \sqrt{2a_t s} = \sqrt{2a_t (\pi R)}$

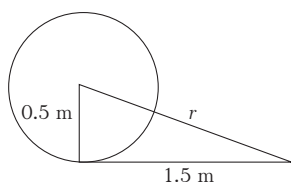


$\therefore a_n = \frac{v^2}{R} = 2\pi a_t$ or $\frac{a_n}{a_t} = 2\pi \Rightarrow \tan \theta = \frac{a_n}{a_t} = 2\pi$

$\therefore \theta = \tan^{-1}(2\pi)$

47 (a) $v = \frac{d}{t_1} = \frac{21 \times 2\pi \times 0.5}{44} = 1.5 \text{ ms}^{-1}$

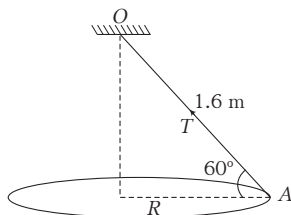
$t_2 = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 4.9}{9.8}} = 1 \text{ s}$



Horizontal distance travelled by drop $= vt_2 = 1.5 \text{ m}$

$\therefore r = \sqrt{(1.5)^2 + (0.5)^2} = \sqrt{2.5} \text{ m}$

48 (d) $R = 1.6 \cos 60^\circ = 0.8 \text{ m}$



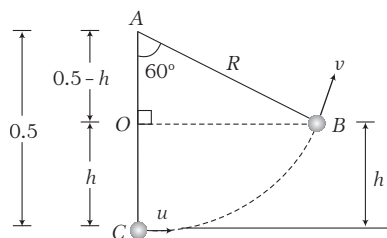
$T \sin 60^\circ = mg$ or $T = \frac{2mg}{\sqrt{3}}$ and $T \cos 60^\circ = \frac{mv^2}{R}$

$\therefore \frac{v^2}{R} = \frac{T}{2m} = \frac{g}{\sqrt{3}} = \frac{9.8}{\sqrt{3}} \text{ ms}^{-2}$

$\Rightarrow v = \sqrt{\frac{9.8}{\sqrt{3}}} \times 0.8 \text{ ms}^{-1}$

Time period $= 2\pi \frac{R}{v}$ and $a_c = \frac{v^2}{R} = \frac{9.8}{\sqrt{3}} \text{ ms}^{-2}$

49 (a) $h = R(1 - \cos 60^\circ) = \frac{R}{2}$



$\therefore$ In right angled $\triangle AOB$

$\cos 60^\circ = \frac{OA}{AB} \Rightarrow \frac{1}{2} = \frac{0.5 - h}{R} \Rightarrow R = 1 - 2h$

$\Rightarrow 0.5 = 1 - 2h$ ($\because R = 0.5 \text{ m}$)

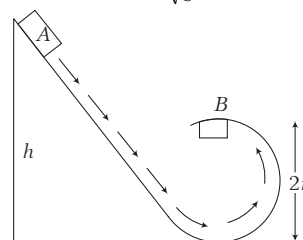
$\Rightarrow h = \frac{1}{4}$

$\therefore v = \sqrt{u^2 - 2gh} = \sqrt{9 - 2 \times 10 \times \frac{1}{4}} = \sqrt{9 - 5} = 2 \text{ ms}^{-1}$

$\therefore v = 2 \text{ ms}^{-1}$

50 (b) Complete the circle, the body should not loose contact with the track anywhere, so

$v \geq \sqrt{gr}$



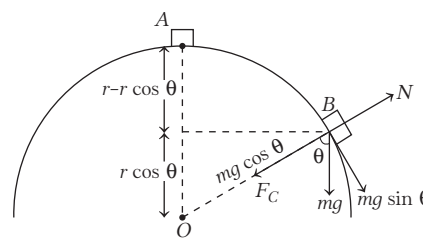
Applying law of conservation of mechanical energy between points A and B, i.e. magnitude of change in kinetic energy equals the magnitude of change in potential energy.

$\Rightarrow \frac{1}{2}mv^2 - 0 = mg(h - 2r)$

$\frac{1}{2}m(\sqrt{gr})^2 = mg(h - 2r) \Rightarrow h = \frac{5}{2}r$

Hence, h must be atleast equal to $2.5 r$.

51 (c) When released from top with zero velocity block leaves contact at point B.



According to diagram,

$mg \cos \theta - N = \frac{mv^2}{r}$

When the body is detached, then $N = 0$.

$\therefore mg \cos \theta = \frac{mv^2}{r}$

$\Rightarrow \cos \theta = \frac{v^2}{rg} \quad \dots(i)$

Applying the conservation of mechanical energy at position A and B,

$K_A + U_A = K_B + U_B$

$0 + mgr(1 - \cos \theta) = \frac{1}{2}mv^2 + 0$

$$2(1 - \cos \theta) = \frac{v^2}{rg} \quad \dots(ii)$$

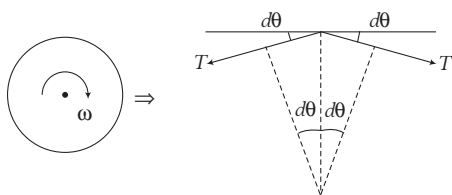
From Eqs. (i) and (ii), we get

$$2 - 2 \cos \theta = \cos \theta$$

$$\Rightarrow \cos \theta = \frac{2}{3}$$

$$\therefore h = r \cos \theta = r \times \frac{2}{3} = \frac{2r}{3}$$

- 52 (a)** To find tension in the ring, let us take an arc which subtends angle $2(d\theta)$ at centre. Tangential components of T cancel out each other, while inward components provide the necessary centripetal force. Thus,



$$2T \sin(d\theta) = (dm) R \omega^2 = (\lambda 2R \cdot d\theta) (R\omega^2)$$

Here, λ = linear mass density.

For small angles, $\sin d\theta \approx d\theta$

$$\therefore 2Td\theta = 2\lambda R^2 \omega^2 d\theta \quad \text{or} \quad T = \lambda R^2 \omega^2$$

$$\Rightarrow \omega_{\max} = \sqrt{\frac{T_{\max}}{\lambda}} \cdot \frac{1}{R} = \sqrt{\frac{40}{0.1}} \times \frac{1}{1} = 20 \text{ rad s}^{-1}$$

- 53 (c)** $h = l(1 - \cos 60^\circ) = \frac{l}{2}$, $v^2 = 2gh = gl$

$$\text{Now, } T_{\max} - mg = \frac{mv^2}{l} \quad (\text{at bottommost point})$$

$$\therefore T_{\max} = 2mg = \mu_s (4mg)$$

$$\therefore \mu_s = 0.5 \text{ or } \frac{1}{2}$$

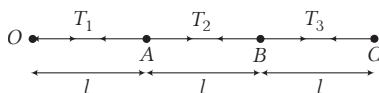
- 54 (d)** From momentum conservation,

$$mu = m \frac{u}{4} + (6m) \sqrt{5 \times 10 \times 0.5}$$

Solving, we get

$$u = 40 \text{ ms}^{-1}$$

- 55 (d)** Let ω is the angular speed of revolution.



$$T_3 = m \omega^2 (3l)$$

$$T_2 - T_3 = m \omega^2 2l$$

$$\Rightarrow T_2 = m \omega^2 (5l)$$

$$T_1 - T_2 = m \omega^2 l \Rightarrow T_1 = m \omega^2 (6l)$$

$$T_3 : T_2 : T_1 = 3 : 5 : 6$$

- 56 (c)** $h_{AB} = (r \cos \alpha - r \sin \beta)$

Velocity of particle at B,

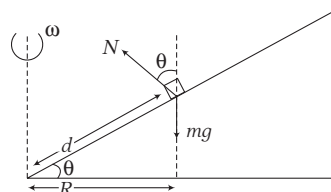
$$v = \sqrt{2gh_{AB}} = \sqrt{2g(r \cos \alpha - r \sin \beta)}$$

Particle will leave contact at B, if component of weight is just equal to centripetal force (towards centre).

$$\text{or } mg \sin \beta = \frac{mv^2}{r} \quad \text{or } \sin \beta = 2 \cos \alpha - 2 \sin \beta$$

$$\therefore 3 \sin \beta = 2 \cos \alpha$$

- 57 (d)** $\omega = 2\pi f = 2\pi(0.5) = \pi \text{ rad s}^{-1}$



From the figure,

$$N \sin \theta = mR \omega^2 \quad \text{and} \quad N \cos \theta = mg \quad \text{or} \quad \tan \theta = \frac{R \omega^2}{g}$$

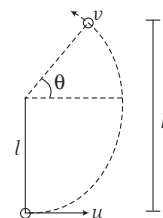
$$\therefore R = \frac{g \tan \theta}{\omega^2}$$

$$\Rightarrow R = \frac{10 \times \tan 30^\circ}{(\pi^2)} \approx \frac{1}{\sqrt{3}} \text{ m}$$

$$\text{Now, } d = \frac{R}{\cos \theta} = \frac{1/\sqrt{3}}{\cos 30^\circ} = \frac{1/\sqrt{3}}{\sqrt{3}/2} = \frac{2}{3} = 0.67 \text{ m}$$

- 58 (c)** $h = l + l \sin \theta = l(1 + \sin \theta)$

$$v^2 = u^2 - 2gh = u^2 - 2gl(1 + \sin \theta)$$



String will slack, where component of weight towards centre is just equal to centripetal force.

$$\text{or } mg \sin \theta = \frac{mv^2}{l} = \frac{m}{l} [u^2 - 2gl(1 + \sin \theta)]$$

$$\text{Substituting } u^2 = \frac{7gl}{2}, \text{ we get}$$

$$\sin \theta = \frac{1}{2} \quad \text{or} \quad \theta = 30^\circ$$

$\therefore$ The desired angle is $90^\circ + 30^\circ$ or 120° .

- 59 (b)** Given, $\frac{1}{2} mv^2 = as^2$

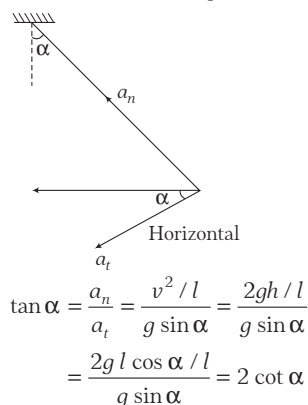
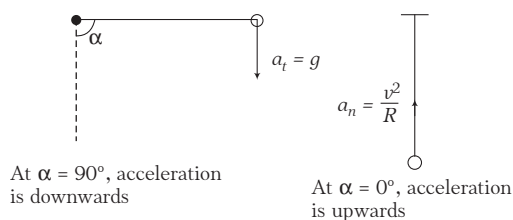
$$\text{or } F_n = \frac{mv^2}{R} = \frac{2as^2}{R} \quad \dots(i)$$

$$\begin{aligned} \text{Further, } v &= \sqrt{\frac{2a}{m}} \cdot s \quad \text{or} \quad a_t = \frac{dv}{dt} = \sqrt{\frac{2a}{m}} \cdot \frac{ds}{dt} = \sqrt{\frac{2a}{m}} \cdot v \\ &= \sqrt{\frac{2a}{m}} \cdot \sqrt{\frac{2a}{m}} \cdot s = \frac{2as}{m} \end{aligned}$$

$$\therefore F_t = ma_t = 2as \quad \dots(ii)$$

$$\therefore F_{\text{net}} = \sqrt{F_n^2 + F_t^2} = 2as \sqrt{1 + \frac{s^2}{R^2}}$$

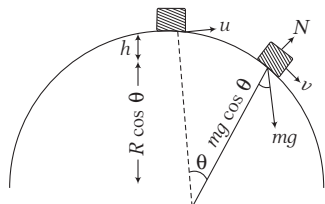
60 (a) When α is horizontal



$$\Rightarrow \tan \alpha = \sqrt{2} \text{ or } \cos \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

61 (b) $h = R - R \cos \theta$



$$v^2 = u^2 + 2gh = u^2 + 2gR(1 - \cos \theta) \quad \dots(i)$$

$$mg \cos \theta - N = \frac{mv^2}{R} \quad \dots(ii)$$

When the particle loses contact,

$$N = 0 \quad \dots(iii)$$

From Eqs. (i), (ii) and (iii), we get

$$\Rightarrow v^2 = gR \cos \theta = u^2 + 2gR(1 - \cos \theta)$$

$$\Rightarrow 3gR \cos \theta = u^2 + 2gR$$

$$\Rightarrow \cos \theta = \frac{u^2}{3gR} + \frac{2}{3} \Rightarrow \theta = \cos^{-1} \left(\frac{u^2}{3gR} + \frac{2}{3} \right)$$

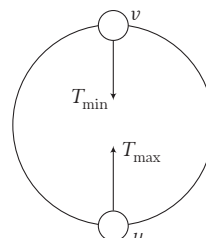
(B) Medical entrance special format questions

• Assertion and reason

1 (b) $a_c = \frac{v^2}{R} = \frac{(2)^2}{1} = 4 \text{ ms}^{-2}$

But no information is given for tangential acceleration a_t .

3 (a) Let the minimum and maximum tensions be $T_{\min}$ and $T_{\max}$ and the minimum and maximum speed be u and v .



$$\therefore T_{\max} = \frac{mu^2}{R} + mg$$

$$T_{\min} = \frac{mv^2}{R} - mg$$

$$\therefore \Delta T = m \left(\frac{u^2}{R} - \frac{v^2}{R} \right) + 2mg$$

From conservation of energy, $\frac{u^2}{R} - \frac{v^2}{R} = 4g$

$\Rightarrow \Delta T$ is independent of u and $\Delta T = 6mg$.

$\therefore$ Reason is the correct explanation of Assertion.

4 (b) In case of massless rod, minimum speed at lower most position of the particle is $\sqrt{5gl}$ to complete the circular motion.

Work done by centripetal force on the particle is always zero.

5 (d) The normal reaction is not least at topmost point, hence Assertion is false.

• Statement based questions

2 (a) Direction of acceleration continuously changes. Also, $\mathbf{v}$ is always perpendicular to $\boldsymbol{\omega}$.

5 (c) In vertical direction, $N \cos \theta = mg$

$$\text{In horizontal direction, } N \sin \theta = \frac{mv^2}{R}$$

• Match the columns

1 (a) $T_3 = (1) (3) (4)^2 = 48 \text{ N}$

($\because F = mR\omega^2$)

$$T_2 - T_3 = (1) (2) (4)^2 = 32 \text{ N}$$

$$\therefore T_2 = 80 \text{ N}$$

$$\Rightarrow T_1 - T_2 = (1) (1) (4)^2 = 16 \text{ N}$$

$$\therefore T_1 = 96 \text{ N}$$

Hence, $A \rightarrow p$, $B \rightarrow q$, $C \rightarrow r$.

2 (b) $v_B^2 = u_A^2 - 2gh_{AB} = (9gR) - (2gR) = 7gR$

$$\therefore v_B = \sqrt{7gR}$$

$$\text{Further, } T_B = \frac{mv_B^2}{R} = 7mg$$

$$\text{Again, } v_C^2 = v_A^2 - 2gh_{AC} = (9gR) - 2g(2R) = 5gR$$

$$\therefore v_C = \sqrt{5gR}$$

Further, $T_C + mg = \frac{mv_C^2}{R}$

$\therefore T_C = 4mg$

Hence, $A \rightarrow r$, $B \rightarrow q$, $C \rightarrow p$, $D \rightarrow t$.

(C) Medical entrances' gallery

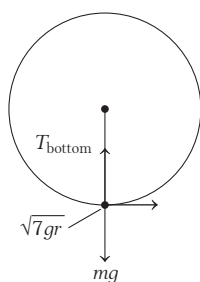
- 1 (c) Velocity of point mass in vertical circle at lowest point,

$v_l = \sqrt{7gr}$

$\therefore v_l = \sqrt{7gr} > \sqrt{5gr}$

Hence, point mass will have completed the vertical circular path.

We know that,



$$T_{\text{bottom}} - mg = \frac{mv^2}{r} = \frac{m}{r} (\sqrt{7gr})^2$$

$$\Rightarrow T_{\text{bottom}} - mg = 7mg$$

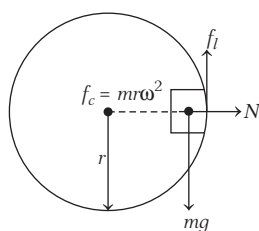
$$\Rightarrow T_{\text{bottom}} = 8mg$$

- 2 (b) Given, mass of cylinder, $m = 10 \text{ kg}$

radius of cylinder, $r = 1 \text{ m}$

and coefficient of friction, $\mu = 0.1$

The given situation can be as shown in the figure given below.



From the above figure, it can be concluded that the block will be stationary when the limiting friction (f_l) is equal to or greater than the downward force or weight of block,

i.e. $f_l \geq mg$... (i)

Also, the magnitude of limiting friction between two bodies is directly proportional to the normal reaction (N) between them,

i.e. $f_l \propto N$ or $f_l = \mu N$... (ii)

From Eqs. (i) and (ii), we get

$$\mu N \geq mg$$

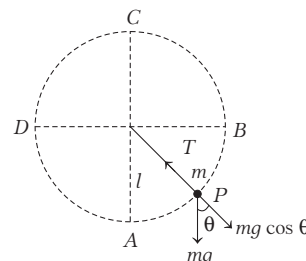
or $\mu(mr\omega^2) \geq mg$ ($\because N = mr\omega^2$)

$$\Rightarrow \omega \geq \sqrt{\frac{g}{r\mu}}$$

Thus, the minimum angular velocity,

$$\omega_{\min} = \sqrt{\frac{g}{r\mu}} = \sqrt{\frac{10}{1 \times 0.1}} = 10 \text{ rad/s}$$

- 3 (b) Let the mass m which is attached to a thin wire and is whirled in a vertical circle as shown in the figure below.



The tension in the string at any point P be T .

According to Newton's law of motion in equilibrium, net force towards the centre = centripetal force

$$\Rightarrow T - mg \cos \theta = \frac{mv^2}{l}$$

where, l = length of wire and v = linear velocity of the mass whirling in a circle.

$$\Rightarrow T = mg \cos \theta + \frac{mv^2}{l}$$

At point A, $\theta = 0^\circ \Rightarrow T_A = mg + \frac{mv^2}{l}$

At point B, $\theta = 90^\circ \Rightarrow T_B = \frac{mv^2}{l}$

At point C, $\theta = 180^\circ \Rightarrow T_C = -mg + \frac{mv^2}{l}$

At point D, $\theta = 90^\circ \Rightarrow T_D = T_B = \frac{mv^2}{l}$

So, from the above analysis, it can be concluded that the tension is maximum at point A, i.e. the lowest point of circle.

So, chance of breaking is maximum.

- 4 (c) From energy conservation,

$$mgH = \frac{1}{2}mv^2$$

$$\Rightarrow v^2 = 2gH \text{ or } v = \sqrt{2gH} \quad \dots (i)$$

To complete circular loop, minimum speed at bottom point should be $v_{\min} = \sqrt{5gR}$... (ii)

From Eqs. (i) and (ii), we get

$$\sqrt{2gH} \geq \sqrt{5gR} \Rightarrow R \leq \frac{2H}{5}$$

$$\Rightarrow R_{\max} = \frac{2H}{5} = \frac{2(5)}{5} = 2 \text{ cm}$$

- 5 (d) If a body is moving on a frictionless surface, then its total mechanical energy remains conserved.

According to the conservation of mechanical energy,

$$(TE)_{\text{initial}} = (TE)_{\text{final}}$$

$$\Rightarrow (KE)_i + (PE)_i = (KE)_f + (PE)_f$$

$$0 + mgh = \frac{1}{2}mv_A^2 + 0$$

$$\Rightarrow gh = \frac{v_A^2}{2}$$

or $h = \frac{v_A^2}{2g} \quad \dots(i)$

In order to complete the vertical circle, the velocity of the body at point A should be

$$v_A = v_{\min} = \sqrt{5gR}$$

where, R is the radius of the body.

Here, $R = \frac{AB}{2} = \frac{D}{2}$

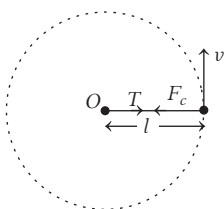
$$\Rightarrow v_{\min} = v_A = \sqrt{\frac{5}{2}gD}$$

Substituting the value of v_A in Eq. (i), we get

$$h = \frac{\left(\sqrt{\frac{5}{2}gD}\right)^2}{2g}$$

$$= \frac{5gD}{2 \times 2g} = \frac{5}{4}D$$

- 6 (a) Consider the string of length l connected to a particle as shown in the figure.



Speed of the particle is v . As, the particle is in uniform circular motion, net force on the particle must be equal to centripetal force which is provided by the tension (T).

$\therefore$ Net force = Centripetal force = Tension in string

$$\Rightarrow \frac{mv^2}{l} = T$$

- 7 (c) At the lowest point of a vertical circle, the minimum velocity, $v_{\min} = \sqrt{5gr}$

Velocity at highest point, $v = \sqrt{gr}$

So, Assertion is correct but Reason is incorrect.

- 8 (c) Centripetal acceleration of a particle moving on a circular path is given by

$$a_c = \frac{v^2}{R}$$

In the given figure, $a_c = a \cos 30^\circ = 15 \cos 30^\circ \text{ ms}^{-2}$

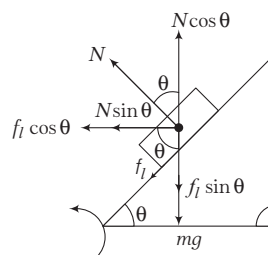
$$\Rightarrow \frac{v^2}{R} = 15 \cos 30^\circ$$

$$\Rightarrow v^2 = R \times 15 \times \frac{\sqrt{3}}{2}$$

$$= 2.5 \times 15 \times \frac{\sqrt{3}}{2}$$

$$\therefore v = 5.7 \text{ ms}^{-1}$$

- 9 (a) According to question, a car is negotiating a curved road of radius R . The road is banked at angle θ and the coefficient of friction between the tyres of car and the road is μ_s . So, the given situation can be drawn as shown in figure below.



Considering the case of vertical equilibrium,

$$N \cos \theta = mg + f_l \sin \theta$$

$$\Rightarrow mg = N \cos \theta - f_l \sin \theta \quad \dots(i)$$

Considering the case of horizontal equilibrium,

$$N \sin \theta + f_l \cos \theta = \frac{mv^2}{R} \quad \dots(ii)$$

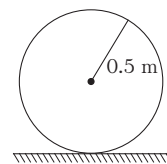
On dividing Eq. (ii) by Eq. (i), we get

$$\frac{v^2}{Rg} = \frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} \quad (\because f_l \propto \mu_s)$$

$$\Rightarrow v = \sqrt{Rg \left(\frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} \right)}$$

$$\Rightarrow v = \sqrt{Rg \left(\frac{\tan \theta + \mu_s}{1 - \mu_s \tan \theta} \right)}$$

- 10 (d) According to given question, a uniform circular disc of radius 50 cm at rest is free to turn about an axis perpendicular to its plane and passes through its centre. This situation can be shown by the figure.



Angular acceleration, $\alpha = 2 \text{ rad s}^{-2}$ (Given)

Angular speed, $\omega = \alpha t = 4 \text{ rad s}^{-1}$

$\therefore$ Centripetal acceleration, $a_c = \omega^2 r = (4)^2 \times 0.5 = 16 \times 0.5$

$$a_c = 8 \text{ ms}^{-2}$$

Hence, linear acceleration at the end of 2s,

$$a_t = \alpha r = 2 \times 0.5$$

$$\Rightarrow a_t = 1 \text{ ms}^{-2}$$

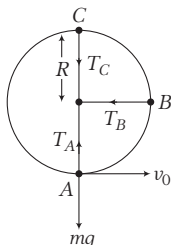
Therefore, the net acceleration at the end of 2 s is given by

$$a = \sqrt{a_c^2 + a_t^2}$$

$$a = \sqrt{(8)^2 + (1)^2} = \sqrt{65}$$

$$\Rightarrow a \approx 8 \text{ ms}^{-2}$$

11 (c) According to question, we have



Let the tension at point A be T_A . So, from Newton's second law,

$$T_A - mg = \frac{mv_C^2}{R}$$

$$\text{Energy at point A} = \frac{1}{2} mv_0^2 \quad \dots(i)$$

$$\text{Energy at point C} = \frac{1}{2} mv_C^2 + mg \times 2R \quad \dots(ii)$$

Applying Newton's second law at point C,

$$T_C + mg = \frac{mv_C^2}{R}$$

To complete the loop $T_C \geq 0$

$$\text{So, } mg = \frac{mv_C^2}{R}$$

$$\Rightarrow v_C = \sqrt{gR} \quad \dots(iii)$$

From Eqs. (i) and (ii) by conservation of energy,

$$\frac{1}{2} mv_0^2 = \frac{1}{2} mv_C^2 + 2mgR$$

$$\Rightarrow \frac{1}{2} mv_0^2 = \frac{1}{2} mgR + 2mgR \quad [\text{Using Eq. (iii)}]$$

$$\Rightarrow v_0^2 = gR + 4gR$$

$$\Rightarrow v_0 = \sqrt{5gR}$$

12 (d) Given, mass of particle, $m = 0.01 \text{ kg}$

Radius of circle along which particle is moving, $r = 6.4 \text{ cm}$

$\therefore$ Kinetic energy of particle, $\text{KE} = 8 \times 10^{-4} \text{ J}$

$$\Rightarrow \frac{1}{2} mv^2 = 8 \times 10^{-4} \text{ J}$$

$$\Rightarrow v^2 = \frac{16 \times 10^{-4}}{0.01} = 16 \times 10^{-2} \quad \dots(i)$$

As it is given that kinetic energy of particle is equal to $8 \times 10^{-4} \text{ J}$ by the end of second revolution after the beginning of motion of particle. It means, its initial velocity (u) is 0 ms^{-1} at this moment.

By Newton's third equation of motion,

$$v^2 = u^2 + 2a_t s$$

$$\Rightarrow v^2 = 2a_t s \text{ or } v^2 = 2a_t (4\pi r)$$

($\because$ particle covers 2 revolutions)

$$\Rightarrow a_t = \frac{v^2}{8\pi r} = \frac{16 \times 10^{-2}}{8 \times 3.14 \times 6.4 \times 10^{-2}}$$

[Using Eq. (i)]

$$\therefore a_t \approx 0.1 \text{ ms}^{-2}$$

13 (c) A particle moving in a curved path experiences an acceleration. Even if it moving around the perimeter of a circle with a constant speed, there is still a change in velocity and subsequently an acceleration. This acceleration is directed towards the centre of the circle. The quantity which may remain constant, here is magnitude of acceleration.

14 (b) We know that, angular speed of a watch,

$$\omega = \frac{\theta}{t} \Rightarrow \omega \propto \frac{1}{t}$$

$$\text{Hence, } \frac{\omega_{\text{second}}}{\omega_{\text{hour}}} = \frac{t_{\text{hour}}}{t_{\text{second}}} = \frac{12 \times 3600}{60} = 720$$

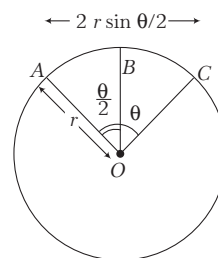
$$\text{i.e. } \frac{\omega_{\text{second}}}{\omega_{\text{hour}}} = \frac{720}{1}$$

15 (d) We know that, $\omega = \frac{\pi}{30} \text{ rad s}^{-1}$

$$\therefore v = r \omega = 10 \times \frac{\pi}{30} \quad (\because r = 10 \text{ cm})$$

$$= \frac{\pi}{3} = \frac{3.14}{3} = 1.046 \approx 1 \text{ cm s}^{-1}$$

16 (d) The figure is shown as below



$$\text{In } \triangle AOB, \sin \frac{\theta}{2} = \frac{AB}{AO}$$

$$\Rightarrow AB = AO \sin \frac{\theta}{2} \Rightarrow AB = r \sin \frac{\theta}{2} \quad (\because AO = r)$$

$$\therefore AC = AB + BC$$

$$AC = r \sin \frac{\theta}{2} + r \sin \frac{\theta}{2} \quad (\because AB = BC)$$

$$AC = 2r \sin \frac{\theta}{2}$$

So, the magnitude of the corresponding linear displacement will be $2r \sin \frac{\theta}{2}$.

17 (c) We know that, $\omega = 2\pi n$

Given, $n_1 = 1800 \text{ rpm}$, $n_2 = 3000 \text{ rpm}$, $\Delta t = 20 \text{ s}$

$$\omega_1 = 2\pi n_1 = 2\pi \times \frac{1800}{60} = 2\pi \times 30 = 60\pi \text{ rad/s}$$

Similarly, $\omega_2 = 2\pi n_2 = 2\pi \times \frac{3000}{60} = 2\pi \times 50 = 100\pi \text{ rad/s}$

If the angular velocity of a rotating wheel about an axis is changed by change in angular velocity in a time interval Δt , then the angular acceleration of rotating wheel about that axis is given by

$$\begin{aligned}\alpha &= \frac{\text{Change in angular velocity}}{\text{Time interval}} \\ &= \frac{\omega_2 - \omega_1}{\Delta t} = \frac{100\pi - 60\pi}{20} \\ &= \frac{40\pi}{20} = 2\pi \text{ rads}^{-2}\end{aligned}$$

- 18 (b)** We know that, the difference between the maximum and minimum tensions in the rope,

$$T_{\max} - T_{\min} = 2mg$$

$$\text{Here, } T_{\max} - T_{\min} = 20$$

$$\text{Now, mass of the stone, } m = \frac{20}{2g} = \frac{20}{2 \times 10} = 1 \text{ kg}$$

- 19 (b)** The angle of banking,

$$\tan \theta = \frac{v^2}{rg}$$

Given, $\theta = 45^\circ$, $r = 90 \text{ m}$ and $g = 10 \text{ ms}^{-2}$

$$\tan 45^\circ = \frac{v^2}{90 \times 10}$$

$$\Rightarrow v = \sqrt{90 \times 10 \times \tan 45^\circ}$$

Speed of car, $v = 30 \text{ ms}^{-1}$

- 20 (c)** If angle of banking is θ , then

$$\tan \theta = \frac{mv^2/r}{mg} \Rightarrow \tan \theta = \frac{v^2}{rg}$$

Given, $v = 10 \text{ ms}^{-1}$, $r = 10 \text{ m}$ and $g = 10 \text{ ms}^{-2}$

$$\text{So, } \tan \theta = \frac{(10)^2}{10 \times 10} = 1$$

$$\therefore \theta = 45^\circ$$

CHAPTER 08

COM, Conservation of Momentum and Collision

CENTRE OF MASS (COM)

For a system of particles or a body, the centre of mass is defined as a point at which the total mass of the system or of the body is supposed to be concentrated. On application of external forces, centre of mass of the system of particles moves in the same way as a point having mass equal to that of the whole system moves.

Position of centre of mass for a system of two particles

Consider two particles of masses m_1 and m_2 located at position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$.

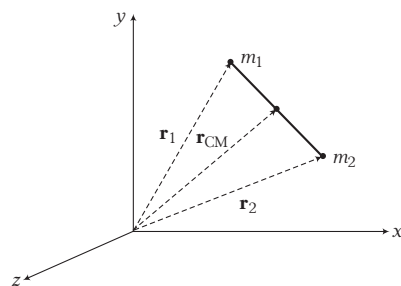


Fig. 8.1 System of two particles

Then, position of centre of mass $\mathbf{r}_{\text{CM}}$ is given as

$$\mathbf{r}_{\text{CM}} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}$$

$$\mathbf{r}_{\text{CM}} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m} = \frac{\sum m_i \mathbf{r}_i}{m}$$

where, $m = m_1 + m_2 =$ total mass of system.

Inside

- 1 Centre of mass (COM)
- 2 Motion of centre of mass
 - Linear momentum of a system of particles
- 3 Collision
 - Types of collisions
 - Newton's law of restitution

The x and y -coordinates of centre of mass can be written as

$$x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

and

$$y_{\text{CM}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

Hence, the centre of mass of two particles system lies between the two particles on the line joining them and the distance of the centre of mass from masses is in inverse ratio of masses of the particles.

$$\text{i.e. } r \propto \frac{1}{m} \Rightarrow \boxed{\frac{r_1}{r_2} = \frac{m_2}{m_1}}$$

Consider two particles of masses m_1 and m_2 at distance r from each other. Their centre of mass (CM) must lie in between them on the line joining them.

Let distances of these particles from the centre of mass be r_1 and r_2 .

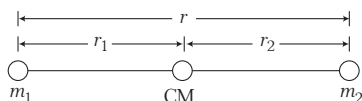


Fig. 8.2

$$\therefore r \propto \frac{1}{m} \Rightarrow m_1 r_1 = m_2 r_2$$

$$\boxed{r_1 = \frac{m_2 r_2}{m_1}} \text{ and } \boxed{r_2 = \frac{m_1 r_1}{m_2}}$$

If the two particles have the same mass, i.e. $m_1 = m_2 = m$, then

$$r_{\text{CM}} = \frac{mr_1 + mr_2}{2m} = \frac{r_1 + r_2}{2}$$

Thus, **for a system of two particles of equal mass, the centre of mass lies exactly midway between them.**

If $m_1 \neq m_2$, centre of mass is nearer to the particle of larger mass.

Example 8.1 Two bodies of masses 1 kg and 2 kg are located at (1, 2) and (−1, 3), respectively. Calculate the coordinates of centre of mass.

Sol. Let the coordinates of centre of mass be (x, y).

Given, mass, $m_1 = 1$ kg, $m_2 = 2$ kg

Coordinates, $x_1 = 1$, $x_2 = -1$, $y_1 = 2$ and $y_2 = 3$

$$\therefore x = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$\Rightarrow x = \frac{1 \times 1 + (2)(-1)}{1 + 2}$$

$$= \frac{1 - 2}{3} = -\frac{1}{3}$$

$$\text{Similarly, } y = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \frac{(1)(2) + (2)(3)}{1 + 2} = \frac{2 + 6}{3} = \frac{8}{3}$$

Therefore, the coordinates of centre of mass will be $\left(-\frac{1}{3}, \frac{8}{3}\right)$.

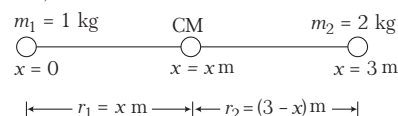
Example 8.2 Two particles of masses 1 kg and 2 kg are located at $x = 0$ and $x = 3$ m. Find the position of their centre of mass.

Sol. Since, both the particles lie on X-axis, so the CM will also lie on X-axis. Let the CM be located at x from 1 kg mass, then

r_1 = distance of CM from the particle of mass 1 kg = x

and r_2 = distance of CM from the particle of mass 2 kg

$$= (3 - x)$$



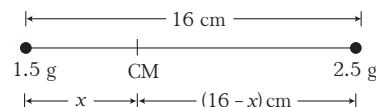
Using $\frac{r_1}{r_2} = \frac{m_2}{m_1}$, we get

$$\frac{x}{3 - x} = \frac{2}{1} \text{ or } x = 2 \text{ m}$$

Thus, the CM of the two particles is located at $x = 2$ m.

Example 8.3 Two point objects of masses 1.5 g and 2.5 g respectively are 16 cm apart, the centre of mass is at a distance x from the object of mass 1.5 g. Find the value of x .

Sol. As, centre of mass of two particles system lies between the two particles on the line joining them.



$$\therefore \text{From } \frac{r_1}{r_2} = \frac{m_2}{m_1} \Rightarrow \frac{x}{16 - x} = \frac{2.5}{1.5} \Rightarrow x = 10 \text{ cm}$$

Position of centre of mass for a system of large number of particles

If we have a system consisting of n particles of masses $m_1, m_2, \dots, m_n$ with $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n$ as their position vectors at a given instant of time. The position vector $\mathbf{r}_{\text{CM}}$ of the centre of mass of the system at that instant is given by

$$\mathbf{r}_{\text{CM}} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2 + \dots + m_n \mathbf{r}_n}{m_1 + m_2 + \dots + m_n} = \frac{\sum_{i=1}^n m_i \mathbf{r}_i}{\sum_{i=1}^n m_i}$$

$$\boxed{\mathbf{r}_{\text{CM}} = \frac{\sum_{i=1}^n m_i \mathbf{r}_i}{M}}$$

or

Here, $M = m_1 + m_2 + \dots + m_n$ and $\sum m_i \mathbf{r}_i$ is called the **first moment of the mass**.

Further, $\mathbf{r}_i = x_i \hat{i} + y_i \hat{j} + z_i \hat{k}$

and $\mathbf{r}_{CM} = x_{CM} \hat{i} + y_{CM} \hat{j} + z_{CM} \hat{k}$

So, the cartesian coordinates of the CM will be

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2 + \dots + m_n x_n}{m_1 + m_2 + \dots + m_n} = \frac{\sum_{i=1}^n m_i x_i}{\sum m_i}$$

or
$$x_{CM} = \frac{\sum_{i=1}^n m_i x_i}{M}$$

Similarly, $y_{CM} = \frac{\sum_{i=1}^n m_i y_i}{M}$ and $z_{CM} = \frac{\sum_{i=1}^n m_i z_i}{M}$

Example 8.4 The position vectors of three particles of masses

$m_1 = 1 \text{ kg}$, $m_2 = 2 \text{ kg}$ and $m_3 = 3 \text{ kg}$ are $\mathbf{r}_1 = (\hat{i} + 4\hat{j} + \hat{k}) \text{ m}$,

$\mathbf{r}_2 = (\hat{i} + \hat{j} + \hat{k}) \text{ m}$ and $\mathbf{r}_3 = (2\hat{i} - \hat{j} - 2\hat{k}) \text{ m}$, respectively. Find the position vector of their centre of mass.

Sol. The position vector of CM of the three particles will be given by

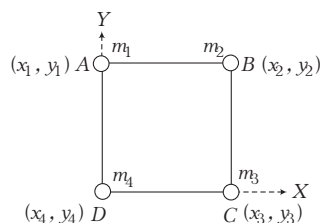
$$\mathbf{r}_{CM} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2 + m_3 \mathbf{r}_3}{m_1 + m_2 + m_3}$$

Substituting the given values in above equation, we get

$$\begin{aligned} \mathbf{r}_{CM} &= \frac{1(\hat{i} + 4\hat{j} + \hat{k}) + 2(\hat{i} + \hat{j} + \hat{k}) + 3(2\hat{i} - \hat{j} - 2\hat{k})}{1 + 2 + 3} \\ &= \frac{9\hat{i} + 3\hat{j} - 3\hat{k}}{6} \Rightarrow \mathbf{r}_{CM} = \frac{1}{2}(3\hat{i} + \hat{j} - \hat{k}) \text{ m} \end{aligned}$$

Example 8.5 Four particles of masses 1 kg, 2 kg, 3 kg and 4 kg are placed at the four vertices A, B, C and D of a square of side 1 m. Find the position of centre of mass of the particles.

Sol. Assuming D as the origin, DC as X-axis and DA as Y-axis, we have



$m_1 = 1 \text{ kg}$; $(x_1, y_1) = (0, 1 \text{ m})$

$m_2 = 2 \text{ kg}$; $(x_2, y_2) = (1 \text{ m}, 1 \text{ m})$

$m_3 = 3 \text{ kg}$; $(x_3, y_3) = (1 \text{ m}, 0)$

$m_4 = 4 \text{ kg}$; $(x_4, y_4) = (0, 0)$

and

Coordinates of their CM,

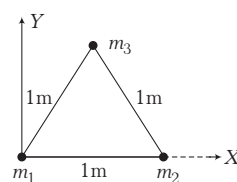
$$\begin{aligned} x_{CM} &= \frac{m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4}{m_1 + m_2 + m_3 + m_4} \\ &= \frac{1(0) + 2(1) + 3(1) + 4(0)}{1 + 2 + 3 + 4} = \frac{5}{10} = \frac{1}{2} \text{ m} = 0.5 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Similarly, } y_{CM} &= \frac{m_1 y_1 + m_2 y_2 + m_3 y_3 + m_4 y_4}{m_1 + m_2 + m_3 + m_4} \\ &= \frac{1(1) + 2(1) + 3(0) + 4(0)}{1 + 2 + 3 + 4} = \frac{3}{10} \text{ m} = 0.3 \text{ m} \end{aligned}$$

$\therefore$ Centre of mass is at $(x_{CM}, y_{CM}) = (0.5, 0.3)$

Example 8.6 Three point masses $m_1 = 2 \text{ kg}$, $m_2 = 4 \text{ kg}$ and $m_3 = 6 \text{ kg}$ are kept at the three corners of an equilateral triangle of side 1 m. Find the location of their centre of mass.

Sol. Assume m_1 to be at the origin and X-axis along the line joining m_1 and m_2 as shown in figure.



From the figure, it is clear that the coordinates of m_1 are $(x_1, y_1) = (0, 0)$ that of m_2 are $(x_2, y_2) = (1, 0)$ and that of m_3 are

$$(x_3, y_3) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

Coordinates of centre of mass are

$$x_{CM} = \frac{2 \times 0 + 4 \times 1 + 6 \times 1/2}{2 + 4 + 6} = \frac{7}{12} \text{ m}$$

$$\text{and } y_{CM} = \frac{2 \times 0 + 4 \times 0 + 6 \times \sqrt{3}/2}{2 + 4 + 6} = \frac{3\sqrt{3}}{12} = \frac{\sqrt{3}}{4} \text{ m}$$

$\therefore$ Centre of mass is at $(x_{CM}, y_{CM}) = \left(\frac{7}{12}, \frac{\sqrt{3}}{4}\right)$.

Position of centre of mass of continuous bodies

For a real body, which has a continuous distribution of matter, point masses are differential mass elements dm and their centre of mass is defined as

$$x_{CM} = \frac{\int x dm}{\int dm} = \frac{\int x dm}{M}$$

$$y_{CM} = \frac{\int y dm}{\int dm} = \frac{\int y dm}{M}$$

and

$$z_{CM} = \frac{\int z dm}{\int dm} = \frac{\int z dm}{M}$$

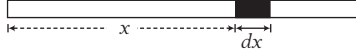
where, M is total mass of that real body.

If we choose the origin of coordinate axes at centre of mass, then

$$\int x dm = \int y dm = \int z dm = 0$$

Example 8.7 The linear density of a thin rod of length 1 m varies as $\lambda = (1 + 2x)$, where x is the distance from its one end. Find the distance of its centre of mass from this end.

Sol. Let the X -axis be along the length of the rod and origin at one of its end as shown in figure.



As, rod is along X -axis, for all points on it Y and Z will be zero, so $y_{CM} = 0$ and $z_{CM} = 0$, i.e. centre of mass will be on the rod. Now, consider an element of rod of length dx at a distance x from the origin.

Mass of this element, $dm = \lambda dx = (1 + 2x)dx$

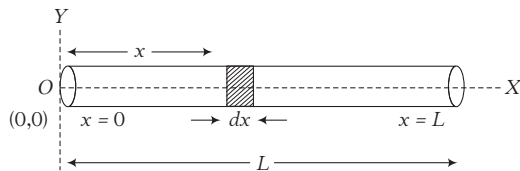
$$x_{CM} = \frac{\int x dm}{\int dm} = \frac{\int_0^1 x(1 + 2x)dx}{\int_0^1 (1 + 2x)dx}$$

$$= \frac{\left[\frac{x^2}{2} + \frac{2x^3}{3} \right]_0^1}{[x + x^2]_0^1}$$

$$= \frac{\left[\frac{(1)^2}{2} + \frac{2(1)^3}{3} - 0 - 0 \right]}{1 + (1)^2 - 0 - 0} = \frac{\frac{1}{2} + \frac{2}{3}}{2} = \frac{7}{12} \text{ m}$$

Example 8.8 A straight rod of length L has one of its ends at the origin and the other at $x = L$. If the mass per unit length of the rod is given by Ax , where A is constant, where is its centre of mass?

Sol. Let the mass of an element of length dx of the rod located at a distance x away from left end be $Ax dx$.



The x -coordinate of the centre of mass is given by

$$x_{CM} = \frac{1}{M} \int x dm = \frac{1}{M} \int_0^L x (Ax dx)$$

$$= \frac{\int_0^L x \cdot Ax dx}{\int_0^L Ax dx} = \frac{[Ax^3/3]_0^L}{[Ax^2/2]_0^L} = \frac{2}{3} L$$

The y -coordinate is $y_{CM} = \frac{1}{M} \int y dm = 0$

and similarly, $z_{CM} = 0$

Hence, the centre of mass is at $\left(\frac{2}{3}L, 0, 0\right)$ or at $\frac{2}{3}L$ from one end.

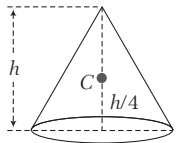
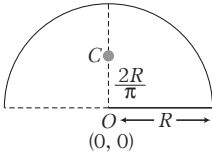
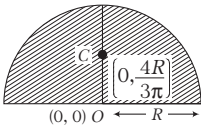
Position of centre of mass of symmetrical bodies

Given below are three points which are very important regarding the centre of mass of symmetrical bodies

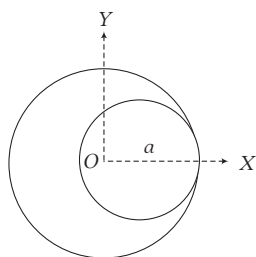
- For the bodies symmetrical about both the axes (X or Y) or all the three axes (X, Y, Z), the centre of mass will lie at point of intersection of symmetrical axes. There is no need to determine centre of mass.
- The bodies which are symmetrical about one axis, centre of mass will lie on that axis. Determine only that coordinate about which there is symmetry.
- If an arrangement or body is not symmetrical about any axis, then determine all the required coordinates.

Centre of mass of some symmetric bodies are given in a table below

Body	Figure	Position of CM
Uniform sphere (hollow or solid)		Centre of the sphere
Uniform circular ring		Centre of the ring
Uniform circular disc		Centre of the disc
Uniform rod		Centre of the rod
A plane square lamina		Point of intersection of diagonals
Triangular lamina		Point of intersection of the medians
Rectangular cubical block		Point of intersection of the diagonals
Cylinder (hollow or solid)		Middle point of the axis of the cylinder

Body	Figure	Position of CM
Cone or pyramid		On the axis of the cone at a distance $3h/4$ from the vertex, where h is the height of the cone.
Uniform semicircular wire		Coordinates of CM = $\left(0, \frac{2R}{\pi}\right)$ or $(0, 0.64R)$
Uniform semicircular plate or disc		Coordinates of CM = $\left(0, \frac{4R}{3\pi}\right)$ or $(0, 0.42R)$

Example 8.9 Find the position of centre of mass of the uniform lamina as shown in figure, if small disc of radius $a/2$ is cut from disc of radius a .



Sol. Here, A_1 = area of complete circle = πa^2

A_2 = area of small circle

$$= \pi \left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{4}$$

(x_1, y_1) = coordinates of centre of mass of large circle
= $(0, 0)$

(x_2, y_2) = coordinates of centre of mass of small circle = $\left(\frac{a}{2}, 0\right)$

Using $x_{CM} = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2}$, we get

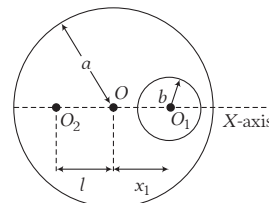
$$x_{CM} = \frac{\pi a^2 \times 0 - \frac{\pi a^2}{4} \left(\frac{a}{2}\right)}{\pi a^2 - \frac{\pi a^2}{4}}$$

$$= -\left(\frac{1}{8}\right)a = -\frac{a}{8}$$

and $y_{CM} = 0$ as y_1 and y_2 both are zero. Therefore, coordinates of CM of the lamina as shown in figure are $\left(-\frac{a}{8}, 0\right)$.

Example 8.10 A small disc of radius 2 cm is cut from a disc of radius 6 cm. If the distance between their centres is 3.2 cm, what is the shift in the centre of mass of the disc?

Sol. Let radius of complete disc be a and that of small disc be b . Also, let centre of mass now shifts to O_2 at a distance l from original centre.



The position of new centre of mass is given by

$$x_{CM} = \frac{A_2 x_2 - A_1 x_1}{A_2 - A_1} = \frac{-\sigma \pi b^2 x_1}{\sigma \pi a^2 - \sigma \pi b^2} \quad (\because x_2 = 0)$$

where, σ = mass per unit area.

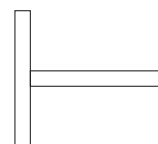
Here, $a = 6$ cm, $b = 2$ cm, $x_1 = 3.2$ cm

$$\text{Hence, } x_{CM} = \frac{-\sigma \times \pi (2)^2 \times 3.2}{\sigma \times \pi \times (6)^2 - \sigma \times \pi \times (2)^2}$$

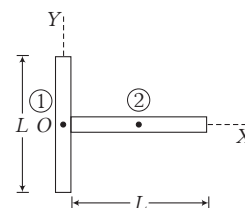
$$= -\frac{12.8\pi}{32\pi} = -0.4 \text{ cm}$$

Negative sign indicates the left side shift from the centre.

Example 8.11 Two identical rods each of mass m and length L are connected as shown in the figure. Locate the centre of mass.



Sol. This system is symmetrical about the X-axis. Hence, we need to find x_{CM} . Here, we will take coordinates of centre of mass of rods.



For rod 1, mass, $m_1 = m$

x -coordinate, $x_1 = 0$

For rod 2, mass, $m_2 = m$

x -coordinate, $x_2 = L/2$

$$\therefore x_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

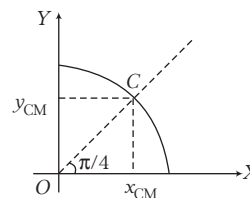
$$x_{\text{CM}} = \frac{m \times 0 + m \times \frac{L}{2}}{m + m} = \frac{m \frac{L}{2}}{2m} = \frac{L}{4}$$

Similarly, $y_{\text{CM}} = 0$

$$\therefore \text{Centre of mass} = \left(\frac{L}{4}, 0 \right)$$

Example 8.12 Find the coordinates of centre of mass of a quarter ring of radius r placed in the first quadrant of a cartesian coordinate system with centre at origin.

Sol. Distance OC of the centre of mass from the centre,

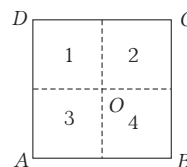
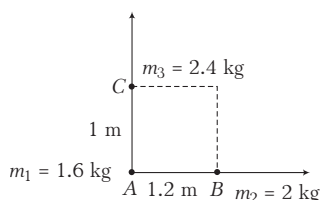


$$OC = \frac{r \sin \pi / 4}{\pi / 4} = \frac{2\sqrt{2}r}{\pi}$$

$\therefore$ Coordinates of centre of mass $(x_{\text{CM}}, y_{\text{CM}})$ are $\left(\frac{2r}{\pi}, \frac{2r}{\pi} \right)$.

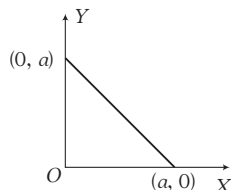
CHECK POINT 8.1

- Two bodies of masses 1 kg and 2 kg are lying in xy -plane at $(-1, 2)$ and $(2, 4)$, respectively. What are the coordinates of the centre of mass?
 - $\left(1, \frac{10}{3} \right)$
 - $(1, 0)$
 - $(0, 1)$
 - None of these
- The centre of mass of a system of two particles divides the distance between them
 - in inverse ratio of square of masses of particles
 - in direct ratio of square of masses of particles
 - in inverse ratio of masses of particles
 - in direct ratio of masses of particles
- In carbon monoxide molecules, the carbon and the oxygen atoms are separated by distance 1.2×10^{-10} m. The distance of the centre of mass from the carbon atom is
 - 0.48×10^{-10} m
 - 0.51×10^{-10} m
 - 0.56×10^{-10} m
 - 0.69×10^{-10} m
- The centre of mass of a system of particles does not depend on
 - masses of the particles
 - internal forces on the particles
 - position of the particles
 - relative distance between the particles
- All the particles of a body are situated at a distance R from the origin. The distance of centre of mass of the body from the origin is
 - $= R$
 - $\leq R$
 - $> R$
 - $\geq R$
- Three point masses m_1, m_2 and m_3 are placed at the corners of a thin massless rectangular sheet ($1.2 \text{ m} \times 1 \text{ m}$) as shown. Centre of mass will be located at the point
 - in region 1
 - in region 2
 - in region 3
 - at O
- Centre of mass of three particles of masses 1 kg, 2 kg and 3 kg lies at the point $(1, 2, 3)$ and centre of mass of another system of particles of total mass 3 kg lies at the point $(-1, 3, -2)$. Where should we put a particle of mass 5 kg, so that the centre of mass of the entire system lies at the centre of mass of first system?
 - $(0, 0, 0)$
 - $(1, 3, 2)$
 - $(-1, 2, 3)$
 - None of these
- Three identical spheres, each of mass 1 kg are placed touching each other with their centres on a straight line. Their centres are marked P, Q and R , respectively. The distance of centre of mass of the system from P is
 - $\frac{PQ + PR + QR}{3}$
 - $\frac{PQ + PR}{3}$
 - $\frac{PQ + QR}{3}$
 - None of the above
- Four rods AB, BC, CD and DA have masses $m, 2m, 3m$ and $4m$, respectively. The centre of mass of all the four rods lies
 - in region 1
 - in region 2
 - in region 3
 - at O



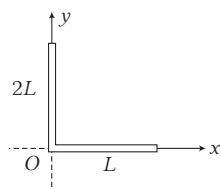
- The linear density of a rod of length L varies as $\lambda = A + Bx$, where x is the distance from one of its ends. The position of centre of mass will be
 - $\frac{L}{2} \left(\frac{3A + BL}{2A + 2BL} \right)$
 - $\frac{L}{3} \left(\frac{3A + 2BL}{2A + BL} \right)$
 - $\frac{L}{5} \left(\frac{2A + 3BL}{A + 3BL} \right)$
 - $\frac{L}{3} \left(\frac{2A + 3BL}{3A + 2BL} \right)$

11. Three rods of the same mass are placed as shown in the figure. What will be the coordinate of centre of mass of the system?



- (a) $\left(\frac{a}{2}, \frac{a}{2}\right)$ (b) $\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ (c) $\frac{2a}{3}, \frac{2a}{3}$ (d) $\left(\frac{a}{3}, \frac{a}{3}\right)$

12. Figure shows a composite system of two uniform rods of lengths as indicated. Then, the coordinates of the centre of mass of the system of rods are

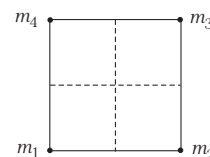


- (a) $\left(\frac{L}{2}, \frac{2L}{3}\right)$ (b) $\left(\frac{L}{4}, \frac{2L}{3}\right)$
(c) $\left(\frac{L}{6}, \frac{2L}{3}\right)$ (d) $\left(\frac{L}{6}, \frac{L}{3}\right)$

13. A uniform metal rod of length 1 m is bent at 90° , so as to form two arms of equal length. The centre of mass of this bent rod is

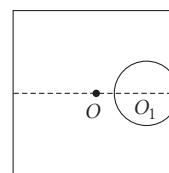
- (a) on the bisector of the angle, $\left(\frac{1}{\sqrt{2}}\right)$ m from vertex
(b) on the bisector of angle, $\left(\frac{1}{2\sqrt{2}}\right)$ m from vertex
(c) on the bisector of the angle, $\left(\frac{1}{2}\right)$ m from vertex
(d) on the bisector of the angle, $\left(\frac{1}{4\sqrt{2}}\right)$ m from vertex

14. Four particles of masses $m_1 = 2m$, $m_2 = 4m$, $m_3 = m$ and m_4 are placed at four corners of a square. What should be the value of m_4 , so that the centre of mass of all the four particles are exactly at the centre of the square?



- (a) $2m$ (b) $8m$
(c) $6m$ (d) None of these

15. A square plate of side 20 cm has uniform thickness and density. A circular part of diameter 8 cm is cut out symmetrically as shown in figure. The position of centre of mass of the remaining portion is



- (a) at O_1
(b) at O
(c) 0.54 cm from O on the left hand side
(d) None of the above

16. A uniform metal disc of radius R is taken and out of it a disc of diameter R is cut-off from the end. The centre of mass of the remaining part will be

- (a) $\frac{R}{4}$ from the centre
(b) $\frac{R}{3}$ from the centre
(c) $\frac{R}{5}$ from the centre
(d) $\frac{R}{6}$ from the centre

MOTION OF CENTRE OF MASS

Let us consider the motion of a system of n particles of individual masses $m_1, m_2, \dots, m_n$ and total mass M . It is assumed that no mass enters or leaves the system during its motion, so that M remains constant. Then,

velocity of centre of mass,

$$\mathbf{v}_{\text{CM}} = \frac{m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 + \dots + m_n \mathbf{v}_n}{M} \quad \dots(i)$$

or

$$\mathbf{v}_{\text{CM}} = \frac{\sum_{i=1}^n m_i \mathbf{v}_i}{M}$$

Differentiating Eq. (i) w.r.t. t , we get

Acceleration of centre of mass,

$$\mathbf{a}_{\text{CM}} = \frac{m_1 \mathbf{a}_1 + m_2 \mathbf{a}_2 + \dots + m_n \mathbf{a}_n}{M} \quad \dots(ii)$$

or

$$\mathbf{a}_{\text{CM}} = \frac{\sum_{i=1}^n m_i \mathbf{a}_i}{M}$$

Further, in accordance with Newton's second law of motion, $\mathbf{F} = m\mathbf{a}$. Hence, Eq. (ii) can be written as

$$\mathbf{F}_{\text{CM}} = \mathbf{F}_1 + \mathbf{F}_2 + \dots + \mathbf{F}_n$$

Force on centre of mass,

$$\mathbf{F}_{\text{CM}} = \sum_{i=1}^n \mathbf{F}_i \quad \dots(\text{iii})$$

From expression (iii), it is clear that the **centre of mass of a system of particles moves as though it is a particle of mass equal to that of the whole system with all the external forces acting directly on it.**

There are some important points related to motion of centre of mass

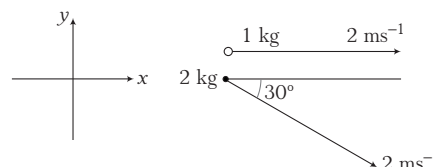
- If a system consists of more than one particle (or bodies) and net external force on the system in a particular direction is zero with centre of mass at rest. Then, the centre of mass will not move along that particular direction even though some particles (or bodies) of the system may move along that direction.
- Motion of centre of mass of a system or object is not affected by any of internal forces as they always make action-reaction pairs, so their net contribution to acceleration of centre of mass is zero.
- If two particles of masses m_1 and m_2 are placed on a smooth surface separated by distance r and they move towards each other due to the mutual attractive force, then
 - In the absence of any external force, the acceleration of CM is zero, irrespective of the individual acceleration of particles.
 - In the absence of any external force, the velocity of CM is also constant.
 - If initially the centre of mass is at rest, i.e. $\mathbf{v}_{\text{CM}} = 0$ and the external force is absent, i.e. $\mathbf{F}_{\text{ext}} = 0$, the location of CM is fixed.
 - Under the action of external forces, the CM moves just as all the mass were concentrated at that point, i.e. $\Sigma \mathbf{F}_{\text{ext}} = M\mathbf{a}_{\text{ext}}$

Example 8.13 Two blocks of masses 5 kg and 2 kg are placed on a frictionless surface and connected by a spring. An external kick gives a velocity of 14 ms^{-1} to the heavier block in the direction of lighter one. Calculate the velocity gained by the centre of mass.

Sol. Given, $m_1 = 5 \text{ kg}$, $m_2 = 2 \text{ kg}$, $v_1 = 14 \text{ ms}^{-1}$ and $v_2 = 0$

$$\begin{aligned} \therefore v_{\text{CM}} &= \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \\ &= \frac{5 \times 14 + 2 \times 0}{5 + 2} \\ &= \frac{70}{7} = 10 \text{ ms}^{-1} \text{ in the direction of lighter one.} \end{aligned}$$

Example 8.14 Find the velocity of centre of mass of the system shown in the figure.



Sol. Here, $m_1 = 1 \text{ kg}$, $\mathbf{v}_1 = 2\hat{\mathbf{i}}$,
 $m_2 = 2 \text{ kg}$, $\mathbf{v}_2 = 2 \cos 30^\circ \hat{\mathbf{i}} - 2 \sin 30^\circ \hat{\mathbf{j}}$

$$\begin{aligned} \mathbf{v}_{\text{CM}} &= \frac{m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2}{m_1 + m_2} \\ &= \frac{1 \times 2\hat{\mathbf{i}} + 2(2 \cos 30^\circ \hat{\mathbf{i}} - 2 \sin 30^\circ \hat{\mathbf{j}})}{1 + 2} \\ \mathbf{v}_{\text{CM}} &= \frac{2\hat{\mathbf{i}} + 2\sqrt{3}\hat{\mathbf{i}} - 2\hat{\mathbf{j}}}{3} = \left(\frac{2 + 2\sqrt{3}}{3} \right) \hat{\mathbf{i}} - \frac{2}{3} \hat{\mathbf{j}} \end{aligned}$$

Example 8.15 Two particles of masses 2 kg and 4 kg are approaching each other with acceleration 1 ms^{-2} and 2 ms^{-2} respectively, on a smooth horizontal surface. Find the acceleration of centre of mass of the system.

Sol. The acceleration of centre of mass of the system,

$$\mathbf{a}_{\text{CM}} = \frac{m_1 \mathbf{a}_1 + m_2 \mathbf{a}_2}{m_1 + m_2} = \frac{2 \times 1 - 4 \times 2}{(2 + 4)} = -1 \text{ ms}^{-2}$$

(Negative sign indicates that direction of 4 kg is opposite to that of 2 kg)

Since, $|m_2 a_2| > |m_1 a_1|$, so the direction of acceleration of centre of mass will be directed towards m_1 .

Example 8.16 Two particles of masses m_1 and m_2 are projected from the top of a tower. The particle m_1 is projected vertically downward with speed u and m_2 is projected horizontally with same speed. Find acceleration of CM of system of particles by neglecting the effect of air resistance.

Sol. As effect of air is neglected, therefore the only force acting on the particles is the gravitational force in downward direction.

Let the point of projection is taken as origin and downward direction as negative Y-axis, then

acceleration of 1st point mass, $\mathbf{a}_1 = -g\hat{\mathbf{j}}$

acceleration of 2nd point mass, $\mathbf{a}_2 = -g\hat{\mathbf{j}}$

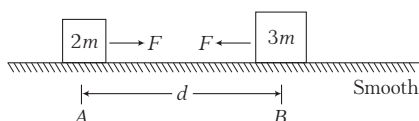
$$\therefore \mathbf{a}_{\text{CM}} = \frac{m_1 \mathbf{a}_1 + m_2 \mathbf{a}_2}{m_1 + m_2} = \frac{m_1(-g\hat{\mathbf{j}}) + m_2(-g\hat{\mathbf{j}})}{m_1 + m_2} = -g\hat{\mathbf{j}}$$

i.e. Acceleration of CM is equal to acceleration due to gravity and is in downward direction.

Note If large number of particles are projected under the effect of gravity only in different directions, then acceleration of CM is equal to the acceleration due to gravity irrespective of directions of projection of particles.

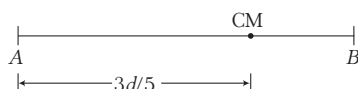
Example 8.17 Two particles of masses $2m$ and $3m$ separated by distance d are placed on a smooth surface. They move towards each other due to mutual attractive force. Find (i) acceleration of CM, (ii) velocity of CM when separation between particles becomes $d/3$ and (iii) at what distance from the initial position of mass $2m$ will the particles collide?

Sol. The given situation is shown below.



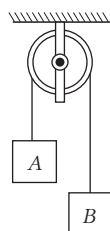
- (i) In the absence of any external force, the acceleration of CM is zero,
i.e. $\mathbf{F}_{\text{ext}} = 0 \Rightarrow \mathbf{a}_{\text{CM}} = 0$
- (ii) Initially, the particles are at rest, i.e. $v_1 = v_2 = 0$, therefore $v_{\text{CM}} = 0$. Since $\mathbf{F}_{\text{ext}} = 0$ the velocity of CM is constant and hence, v_{CM} is always zero whatever be the separation between the particles.
- (iii) The position of CM will be

$$x_{\text{CM}} = \frac{2m \times 0 + 3m \times d}{2m + 3m} = \frac{3d}{5}$$

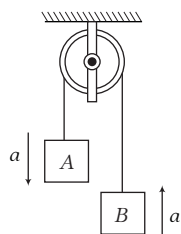


Since, the CM is at rest and its position is fixed, hence particles will meet at CM, i.e. at distance $3d/5$ from A.

Example 8.18 In the arrangement shown in figure, $m_A = 2 \text{ kg}$ and $m_B = 1 \text{ kg}$. String is light and inextensible. Find the acceleration of centre of mass of both the blocks. Neglect friction everywhere.



Sol. Net pulling force on the system $= (m_A - m_B)g$
 $= (2 - 1)g = g$



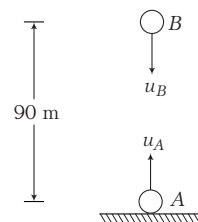
Total mass being pulled $= m_A + m_B = 3 \text{ kg}$

$$\therefore a = \frac{\text{Net pulling force}}{\text{Total mass}} = \frac{g}{3}$$

Now,

$$\begin{aligned} \mathbf{a}_{\text{CM}} &= \frac{m_A \mathbf{a}_A + m_B \mathbf{a}_B}{m_A + m_B} \\ &= \frac{2(a) - 1(a)}{1 + 2} = \frac{a}{3} = \frac{g}{9} \text{ (downwards)} \end{aligned}$$

Example 8.19 Two particles A and B of masses 1 kg and 2 kg respectively are projected in the directions as shown in figure with speeds $u_A = 200 \text{ ms}^{-1}$ and $u_B = 50 \text{ ms}^{-1}$. Initially, they were 90 m apart. Find the maximum height attained by the centre of mass of the particles. Assume acceleration due to gravity to be constant. (Take, $g = 10 \text{ ms}^{-2}$)



Sol. Using $m_A r_A = m_B r_B$ or $1(r_A) = 2(r_B)$

$$\text{or } r_A = 2r_B \quad \dots(i)$$

$$\text{and } r_A + r_B = 90 \text{ m} \quad \dots(ii)$$

Solving these two equations, we get

$$r_A = 60 \text{ m} \quad \text{and} \quad r_B = 30 \text{ m}$$

i.e. CM is at height 60 m from the ground at time $t = 0$.

$$\text{Further, } \mathbf{a}_{\text{CM}} = \frac{m_A \mathbf{a}_A + m_B \mathbf{a}_B}{m_A + m_B} = g = 10 \text{ ms}^{-2} \text{ (downwards)}$$

$$\text{As, } \mathbf{a}_A = \mathbf{a}_B = g \quad \text{(downwards)}$$

$$\begin{aligned} \mathbf{u}_{\text{CM}} &= \frac{m_A \mathbf{u}_A + m_B \mathbf{u}_B}{m_A + m_B} \\ &= \frac{1(200) - 2(50)}{1 + 2} = \frac{100}{3} \text{ ms}^{-1} \quad \text{(upwards)} \end{aligned}$$

Let h be the height attained by CM beyond 60 m .

$$\text{Using } \mathbf{v}_{\text{CM}}^2 = \mathbf{u}_{\text{CM}}^2 + 2\mathbf{a}_{\text{CM}}h$$

$$\text{or } 0 = \left(\frac{100}{3}\right)^2 - (2)(10)h$$

$$\text{or } h = \frac{(100)^2}{180} = 55.56 \text{ m}$$

Therefore, maximum height attained by the centre of mass,

$$\begin{aligned} H &= 60 + 55.56 \\ &= 115.56 \text{ m} \end{aligned}$$

Linear momentum of a system of particles

For a system of n -particles, total linear momentum is vector sum of linear momenta of individual particles, where linear momentum of an individual particle is product of its mass and velocity ($\mathbf{p} = m\mathbf{v}$).

So, linear momentum of system is given by

$$\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \dots + \mathbf{p}_n$$

$$\text{or } \mathbf{p} = m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 + m_3 \mathbf{v}_3 + \dots + m_n \mathbf{v}_n \quad \dots(i)$$

($\because p = mv$)

From the concept of centre of mass, we know that,

$$m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 + m_3 \mathbf{v}_3 + \dots + m_n \mathbf{v}_n = M \mathbf{v}_{\text{CM}} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\text{Total linear momentum, } \mathbf{p} = M \mathbf{v}_{\text{CM}} \quad \dots(iii)$$

Thus, **the total momentum of a system of particles is equal to the product of the total mass and velocity of its centre of mass.**

Note Relation between momentum and kinetic energy,

$$p = \sqrt{2mK}$$

Conservation of linear momentum for system of particles

According to law of conservation of linear momentum, total linear momentum of a system of particles remains constant or conserved in the absence of any external force.

Total linear momentum of system of particles,

$$p = Mv_{\text{CM}}$$

Differentiating both sides w.r.t. t , we get

$$\begin{aligned} \frac{dp}{dt} &= \frac{d}{dt} (Mv_{\text{CM}}) \\ &= M \frac{dv_{\text{CM}}}{dt} \end{aligned}$$

Here, $\frac{dv_{\text{CM}}}{dt} = a_{\text{CM}}$, acceleration of centre of mass.

$$\therefore \frac{dp}{dt} = Ma_{\text{CM}}$$

Here, from Newton's second law of motion, Ma_{CM} will be equal to the external force.

$$\therefore \frac{dp}{dt} = F_{\text{ext}}$$

$$\therefore \text{If } F_{\text{ext}} = 0, \text{ then}$$

$$\frac{dp}{dt} = 0 \text{ or } p = \text{constant}$$

$$\therefore Mv_{\text{CM}} = \text{constant}$$

$$v_{\text{CM}} = \text{constant}$$

So, we can conclude that, **if net external force on the system is zero, the linear momentum of the system is constant, hence centre of mass will move with constant velocity.**

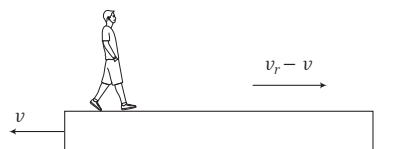
Here, rate of change of momentum is zero, i.e. momentum of system remains constant.

$$\text{So, } \mathbf{p}_{\text{initial}} = \mathbf{p}_{\text{final}}$$

Above expression represents the law of conservation of linear momentum for system of particles.

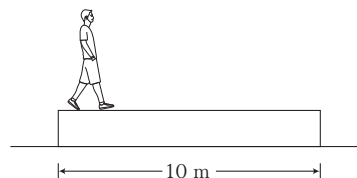
Example 8.20 A man of mass m_1 is standing on a platform of mass m_2 kept on a smooth horizontal surface. The man starts moving on the platform with a velocity v_r relative to the platform. Find the recoil velocity of platform.

Sol. Absolute velocity of man $= v_r - v$, where v = recoil velocity of platform. Taking the platform and the man as a system, net external force on the system in horizontal direction is zero. Initially, both the man and the platform were at rest, hence the linear momentum of the system remains constant.



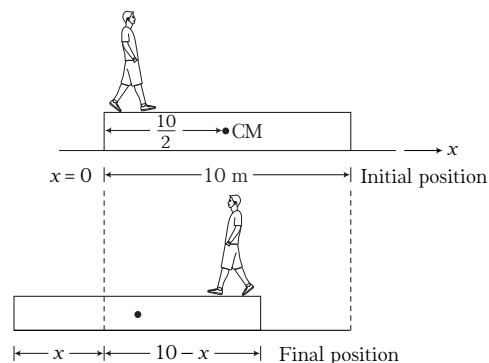
$$\text{Hence, } 0 = m_1(v_r - v) - m_2v \Rightarrow v = \frac{m_1 v_r}{m_1 + m_2}$$

Example 8.21 A wooden plank of mass 20 kg is resting on a smooth horizontal floor as shown in figure. A man of mass 60 kg starts moving from one end of the plank to the other end. The length of the plank is 10 m. Find the displacement of the plank over the floor when the man reaches the other end of the plank.



Sol. Here, the system is man + plank. Net force on this system in horizontal direction is zero and initially the centre of mass of the system is at rest. Therefore, the centre of mass does not move in horizontal direction.

Let x be the displacement of the plank. Assuming the origin, i.e. $x = 0$ at the position as shown in figure.



As, the centre of mass will not move in horizontal direction (X-axis). Therefore, for centre of mass to remain stationary,

$$x_i = x_f$$

$$\frac{60(0) + 20\left(\frac{10}{2}\right)}{60 + 20} = \frac{60(10 - x) + 20\left(\frac{10}{2} - x\right)}{60 + 20}$$

or

$$\frac{5}{4} = \frac{6(10 - x) + 2\left(\frac{10}{2} - x\right)}{8}$$

$$= \frac{60 - 6x + 10 - 2x}{8}$$

or

$$5 = 30 - 3x + 5 - x$$

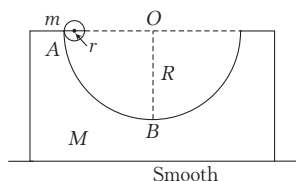
$$\Rightarrow 4x = 30$$

$$\Rightarrow x = \frac{30}{4} \text{ m}$$

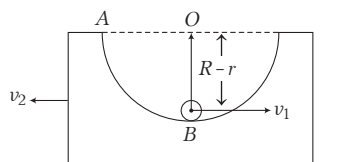
$$\Rightarrow x = 7.5 \text{ m}$$

Note The centre of mass of the plank lies at its centre.

Example 8.22 A block of mass M with a semi-circular track of radius R rests on a smooth floor. A sphere of mass m and radius r is released from rest at point A. Find the velocity of sphere and track when the sphere reaches at B.



Sol. According to the question,



When the sphere reaches at point B

Let v_1 = velocity of m

v_2 = velocity of M

Using conservation of momentum in the horizontal direction,

$$mv_1 = Mv_2 \quad \dots(i)$$

Applying the conservation of energy between A and B,

$$mg(R - r) = \frac{1}{2}(mv_1^2 + Mv_2^2) \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$v_1 = \sqrt{\frac{2Mg(R - r)}{M + m}} \text{ and } v_2 = \frac{m}{M} \sqrt{\frac{2Mg(R - r)}{M + m}}$$

Example 8.23 A disc of mass 100 g is kept floating horizontally in air by firing bullets, each of mass 5 g with the same velocity at the same rate of 10 bullets per second. The bullets rebound with the same speed in opposite direction. Find the velocity of each bullet at the time of impact.

Sol. From the law of conservation of momentum,

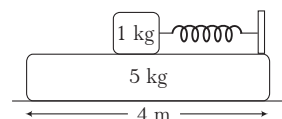
rate of change in momentum of bullets = weight of disc

$$2m'vn = mg$$

$$\therefore v = \frac{mg}{2m'n} = \frac{100 \times 980}{2 \times 5 \times 10}$$

$$= 980 \text{ cm/s}$$

Example 8.24 A plank of mass 5 kg is placed on a frictionless horizontal plane as shown in figure. Further, a block of mass 1 kg is placed over the plank. A massless spring of natural length 2 m is fixed to the plank by its one end. The other end of spring is compressed by the block by half of spring's natural length. The system is now released from the rest. What is the velocity of the plank when block leaves the plank? (The stiffness constant of spring is 100 Nm^{-1})



Sol. Let the velocities of the block and the plank, when the block leaves the spring, be u and v , respectively.

$$\text{By conservation of energy, } \frac{1}{2}kx^2 = \frac{1}{2}mu^2 + \frac{1}{2}Mv^2$$

(where, M = mass of the plank, m = mass of the block)

$$\Rightarrow 100 = u^2 + 5v^2 \quad \dots(i)$$

By conservation of momentum,

$$mu + Mv = 0 \Rightarrow u = -5v \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$30v^2 = 100$$

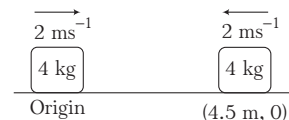
$$\Rightarrow v = \sqrt{\frac{10}{3}} \text{ ms}^{-1}$$

From this moment until block falls, both plank and block keep their velocity constant.

Thus, when block falls, velocity of plank = $\sqrt{10/3} \text{ ms}^{-1}$.

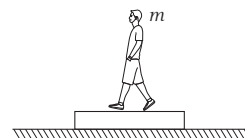
CHECK POINT 8.2

- A body falling vertically downwards under gravity breaks in two parts of unequal masses. The centre of mass of the two parts taken together shifts horizontally towards
 - heavier piece
 - lighter piece
 - does not shift horizontally
 - depends on the vertical velocity at the time of breaking
- Two balls are thrown simultaneously in air. The acceleration of the centre of mass of the two balls in air
 - depends on the direction of the motion of the balls
 - depends on the masses of the two balls
 - depends on the speeds of two balls
 - is equal to g
- Consider a system of two identical particles. One of the particles is at rest and the other has an acceleration $\mathbf{a}$. The centre of mass has an acceleration
 - zero
 - $\frac{1}{2}\mathbf{a}$
 - $\mathbf{a}$
 - $2\mathbf{a}$
- Two balls of equal mass are projected from a tower simultaneously with equal speeds, one at angle θ above the horizontal and the other at the same angle θ below the horizontal. The path of the centre of mass of the two balls is
 - a vertical straight line
 - a horizontal straight line
 - a straight line at angle α ($< \theta$) with horizontal
 - a parabola
- A ball kept in a closed box moves in the box making collisions with the walls. The box is kept on a smooth surface. The velocity of the centre of mass
 - of the box remains constant
 - of the box plus the ball system remains constant
 - of the ball remains constant
 - of the ball relative to the box remains constant
- Two blocks of masses 10 kg and 4 kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse gives a velocity of 14 ms^{-1} to the heavier block in the direction of the lighter block. The velocity of the centre of mass is
 - 30 ms^{-1}
 - 20 ms^{-1}
 - 10 ms^{-1}
 - 5 ms^{-1}
- A metre stick is placed vertically at the origin on a frictionless surface. A gentle push in $+x$ -direction is given to the top most point of the rod. When it has fallen completely, the x -coordinate of centre of rod is at
 - origin
 - -0.5 m
 - -1 m
 - $+0.5 \text{ m}$
- Two bodies having masses m_1 and m_2 and velocities v_1 and v_2 collide and form a composite system. If $m_1 v_1 + m_2 v_2 = 0$ ($m_1 \neq m_2$), the velocity of composite system will be
 - $v_1 - v_2$
 - $v_1 + v_2$
 - $\frac{v_1 + v_2}{2}$
 - zero
- Blocks A and B are resting on a smooth horizontal surface given equal speeds of 2 ms^{-1} in opposite sense as shown in the figure.

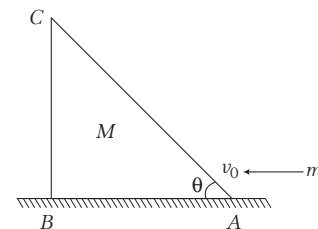


At $t = 0$, the position of blocks are shown, then the coordinates of centre of mass $t = 3 \text{ s}$ will be

- (1, 0)
 - (3, 0)
 - (5, 0)
 - (2.25, 0)
- Two particles of equal mass have coordinates $(4\text{m}, 4\text{m}, 6\text{m})$ and $(6\text{m}, 2\text{m}, 8\text{m})$. Of these, one particle has a velocity $\mathbf{v}_1 = (2\hat{i}) \text{ ms}^{-1}$ and another particle has velocity $\mathbf{v}_2 = (2\hat{j}) \text{ ms}^{-1}$ at time $t = 0$. The coordinates of their centre of mass at time $t = 1 \text{ s}$ will be
 - $(4\text{m}, 4\text{m}, 7\text{m})$
 - $(5\text{m}, 4\text{m}, 7\text{m})$
 - $(2\text{m}, 4\text{m}, 6\text{m})$
 - $(4\text{m}, 5\text{m}, 4\text{m})$
 - An isolated particle of mass m is moving in horizontal xy -plane, along the X -axis, at a certain height above the ground. It suddenly explodes into two fragments of masses $\frac{m}{4}$ and $\frac{3m}{4}$. An instant later, the smaller fragment is at $y = +15 \text{ cm}$. The larger fragment at this instant is at
 - $y = -5 \text{ cm}$
 - $y = +20 \text{ cm}$
 - $y = +5 \text{ cm}$
 - $y = -20 \text{ cm}$
 - Two particles A and B initially at rest, move towards each other under a mutual force of attraction. At the instant, when the speed of A is v and the speed of B is $2v$, the speed of centre of mass of the system is
 - zero
 - v
 - $1.5v$
 - $3v$
 - A man of mass m is standing on a plank of equal mass m resting on a smooth horizontal surface. The man starts moving on the plank with speed u relative to the plank. The speed of the man relative to the ground is



- $2u$
 - $\frac{u}{2}$
 - zero
 - $\frac{u}{4}$
- A shell of mass m is moving horizontally with velocity v_0 and collides with the wedge of mass M just above point A as shown in the figure. As a consequence, wedge starts to move towards left and the shell returns with a velocity in xy -plane. The principle of conservation of momentum can be applied for



- system $(m + M)$ along any direction
- system $(m + M)$ vertically
- system $(m + M)$ horizontally
- None of the above

15. A stationary bomb explodes into two parts of masses 3 kg and 1 kg. The total kinetic energy of the two parts after explosion is 2400 J. The kinetic energy of the smaller part is
(a) 600 J (b) 1800 J (c) 1200 J (d) 2160 J
16. An object of mass $3m$ splits into three equal fragments. Two fragments have velocities $v\hat{j}$ and $v\hat{i}$. The velocity of the third fragment is
(a) $v(\hat{j} - \hat{i})$ (b) $v(\hat{i} - \hat{j})$ (c) $-v(\hat{i} + \hat{j})$ (d) $\frac{v(\hat{i} + \hat{j})}{\sqrt{2}}$

COLLISION

A collision is an isolated event in which two or more colliding bodies exert strong forces on each other for a relatively short time. For a collision to take place, the actual physical contact is not necessary.

Total linear momentum is conserved in all collisions, i.e. the initial momentum of the system is equal to final momentum of the system.

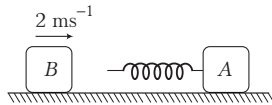
$\therefore$ Total momentum before collision = Total momentum after collision

$$m_1v_1 + m_2v_2 = m_1v'_1 + m_2v'_2$$

In the absence of any dissipative forces, the mechanical energy of the system will also remain conserved, i.e.

$$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}m_1v_1'^2 + \frac{1}{2}m_2v_2'^2$$

Example 8.25 Two blocks A and B of equal mass $m = 1 \text{ kg}$ are lying on a smooth horizontal surface as shown in figure. A spring of force constant $k = 200 \text{ Nm}^{-1}$ is fixed at one end of block A. Block B collides with block A with velocity $v_0 = 2 \text{ ms}^{-1}$. Find the maximum compression of the spring.



Sol. At maximum compression (x_m), velocity of both the blocks is same, say v . Applying conservation of linear momentum, we have

$$(m_A + m_B)v = m_Bv_0$$

$$\text{or } (1 + 1)v = (1)v_0$$

$$\text{or } v = \frac{v_0}{2} = \frac{2}{2} = 1 \text{ ms}^{-1}$$

Using conservation of mechanical energy, we have

$$\frac{1}{2}m_Bv_0^2 = \frac{1}{2}(m_A + m_B)v^2 + \frac{1}{2}kx_m^2$$

Substituting the given values in above equation, we get

$$\frac{1}{2} \times (1) \times (2)^2 = \frac{1}{2} \times (1 + 1) \times (1)^2 + \frac{1}{2} \times (200) \times x_m^2$$

$$\text{or } x_m = 0.1 \text{ m} = 10 \text{ cm}$$

17. A shell is fired from cannon with velocity v at an angle θ with the horizontal direction. At the highest point in its path, it explodes into two pieces of equal mass. One of the pieces retraces its path to the cannon and the speed (in ms^{-1}) of the other piece immediately after the explosion is
(a) $3v \cos \theta$ (b) $2v \cos \theta$
(c) $\frac{3}{2}v \cos \theta$ (d) $\sqrt{\frac{3}{2}}v \cos \theta$

Types of collisions

Collision between two bodies may be classified in two ways

Elastic and inelastic collision A collision is said to be **elastic**, if along with linear momentum, kinetic energy also remains conserved before and after collision.

A collision is said to be **inelastic**, if only linear momentum remains conserved but not the kinetic energy.

The collision is said to be perfectly inelastic, if approaching particles permanently stick to each other and move with common velocity.

Head on and oblique collision If velocity vectors of the colliding bodies are directed along the line of impact, the impact is called as **direct** or **head on collision**.

And if velocity vectors of both or any of the bodies are not along the line of impact, the impact is called **oblique collision**.

Head on elastic collision

Let the two balls of masses m_1 and m_2 collide with each other elastically with velocities v_1 and v_2 in the directions shown in figure below. Their velocities become v'_1 and v'_2 after the collision along the same line.

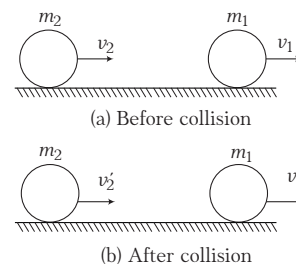


Fig. 8.3 Head on elastic collision

Applying law of conservation of linear momentum, we get

$$m_1v_1 + m_2v_2 = m_1v'_1 + m_2v'_2 \quad \dots(i)$$

In an elastic collision, kinetic energy before and after collision is also conserved. Hence,

$$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}m_1v_1'^2 + \frac{1}{2}m_2v_2'^2 \quad \dots(ii)$$

Solving Eqs. (i) and (ii) for v'_1 and v'_2 , we get

$$v'_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) v_1 + \left(\frac{2m_2}{m_1 + m_2} \right) v_2 \quad \dots(iii)$$

and
$$v'_2 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) v_2 + \left(\frac{2m_1}{m_1 + m_2} \right) v_1 \quad \dots(iv)$$

Special cases of head on elastic collision

(i) If $m_1 = m_2$, then from Eqs. (iii) and (iv), we can see that

$$v'_1 = v_2 \text{ and } v'_2 = v_1$$

i.e. When two particles of equal mass collide elastically in head on condition, they exchange their velocities.

(ii) If $m_1 \gg m_2$ and $v_1 = 0$, then $\frac{m_2}{m_1} \approx 0$

With these two substitutions $\left(v_1 = 0 \text{ and } \frac{m_2}{m_1} = 0 \right)$,

we get the following two results,

$$v'_1 \approx 0 \text{ and } v'_2 \approx -v_2$$

i.e. If a lighter particle collides with a heavier particle at rest, then heavier particle remains at rest but lighter particle bounces back with same velocity.

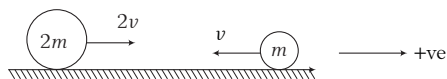
(iii) If $m_2 \gg m_1$ and $v_1 = 0$

With the substitution $m_1/m_2 \approx 0$ and $v_1 = 0$, we get the results, $v'_1 \approx 2v_2$ and $v'_2 \approx v_2$

i.e. If a heavier particle collides with a lighter particle at rest, then lighter particle moves with twice the velocity of heavier particle while velocity of heavier particle remains same.

Example 8.26 Two particles of masses m and $2m$ moving in opposite directions collide elastically with velocities v and $2v$. Find their velocities after collision.

Sol. Here, $v_1 = -v$, $v_2 = 2v$, $m_1 = m$ and $m_2 = 2m$.



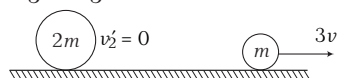
Substituting these values in Eqs. (iii) and (iv), we get

$$v'_1 = \left(\frac{m - 2m}{m + 2m} \right) (-v) + \left(\frac{4m}{m + 2m} \right) (2v)$$

or
$$v'_1 = \frac{v}{3} + \frac{8v}{3} = 3v$$

and
$$v'_2 = \left(\frac{2m - m}{2m + m} \right) (2v) + \left(\frac{2m}{m + 2m} \right) (-v)$$

or
$$v'_2 = \frac{2}{3}v - \frac{2}{3}v = 0$$



i.e. The second particle (of mass $2m$) comes to rest while the first (of mass m) moves with velocity $3v$ in the direction shown in figure given above.

Head on inelastic collision

In an inelastic collision, due to permanent deformation the kinetic energy of the particles no longer remains conserved. However, in the absence of external forces, law of conservation of linear momentum still holds good.

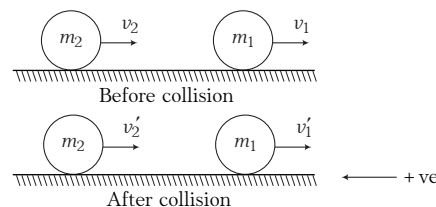


Fig. 8.4 Head on inelastic collision

Suppose the velocities of two particles of masses m_1 and m_2 before collision be v_1 and v_2 in the directions as shown in figure. Let v'_1 and v'_2 be their velocities after collision. The law of conservation of linear momentum gives

$$m_1v_1 + m_2v_2 = m_1v'_1 + m_2v'_2 \quad \dots(i)$$

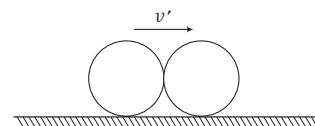


Fig. 8.5 After inelastic collision

Collision is said to be perfectly inelastic, if both the particles stick together after collision and move with same velocity, say v' as shown in figure. In this case, Eq. (i) can be written as

$$m_1v_1 + m_2v_2 = (m_1 + m_2)v'$$

or

$$v' = \frac{m_1v_1 + m_2v_2}{m_1 + m_2} \quad \dots(ii)$$

Loss in kinetic energy,

$$\begin{aligned} \Delta K &= \left(\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \right) - \frac{1}{2} (m_1 + m_2) v'^2 \\ &= \left(\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \right) - \frac{1}{2} (m_1 + m_2) \left(\frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} \right)^2 \end{aligned}$$

Solving above equation, we get

$$\Delta K = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (v_1 - v_2)^2$$

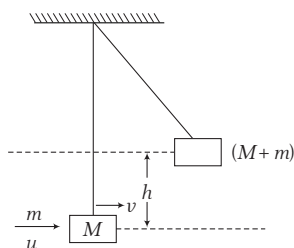
which is positive. Therefore, **some kinetic energy is always lost in an inelastic collision.**

Example 8.27 A simple pendulum of length 1 m has a wooden bob of mass 1 kg. It is struck by a bullet of mass 10^{-2} kg moving with a speed of $2 \times 10^2 \text{ ms}^{-1}$. The bullet gets embedded into the bob. Obtain the height to which the bob rises before swinging back.

Sol. Applying principle of conservation of linear momentum,

$$mu = (M + m)v \Rightarrow 10^{-2} \times (2 \times 10^2) = (1 + 0.01)v$$

$$\Rightarrow v = \frac{2}{1.01} \text{ ms}^{-1}$$



Kinetic energy of the block with bullet in it is converted into potential energy as it rises through a height h .

$$\therefore \frac{1}{2}(M + m)v^2 = (M + m)gh$$

$$\Rightarrow h = \frac{v^2}{2g} = \left(\frac{2}{1.01}\right)^2 \times \frac{1}{2 \times 9.8} = 0.2 \text{ m}$$

Example 8.28 A body falling on the ground from a height of 10 m, rebounds to a height 2.5 m, calculate

- the percentage loss in kinetic energy
- ratio of the velocities of the body just before and just after the collision.

Sol. Let v_1 and v_2 be the velocities of the body just before and just after the collision.

$$\text{KE}_1 = \frac{1}{2}mv_1^2 = mgh_1 \quad \dots(i)$$

$$\text{and} \quad \text{KE}_2 = \frac{1}{2}mv_2^2 = mgh_2 \quad \dots(ii)$$

$$(i) \text{ Percentage loss in kinetic energy} = \frac{mg(h_1 - h_2)}{mgh_1} \times 100$$

$$= \frac{10 - 2.5}{10} \times 100 = 75\%$$

$$(ii) \quad \frac{v_1^2}{v_2^2} = \frac{h_1}{h_2} = \frac{10}{2.5} = 4 \Rightarrow \frac{v_1}{v_2} = 2$$

Example 8.29 A pendulum bob of mass 10^{-2} kg is raised to a height 5×10^{-2} m and then released. At the bottom of its swing, it picks up a mass 10^{-3} kg. To what height will the combined mass rise? (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Velocity of pendulum bob in mean position,

$$v_1 = \sqrt{2gh} = \sqrt{2 \times 10 \times 5 \times 10^{-2}} = 1 \text{ ms}^{-1}$$

When the bob picks up a mass 10^{-3} kg at the bottom, then by conservation of linear momentum, the velocity of coalesced mass is given by

$$m_1v_1 + m_2v_2 = (m_1 + m_2)v$$

$$10^{-2} \times 1 + 10^{-3} \times 0 = (10^{-2} + 10^{-3})v$$

$$\text{or} \quad v = \frac{10^{-2}}{1.1 \times 10^{-2}} = \frac{10}{11} \text{ ms}^{-1}$$

$$\text{Now,} \quad h = \frac{v^2}{2g} = \frac{(10/11)^2}{2 \times 10} = 4.1 \times 10^{-2} \text{ m}$$

Example 8.30 An object of mass 40 kg having velocity 4 ms^{-1} collides with another object of mass 60 kg having velocity 2 ms^{-1} . What is the loss of energy during this process, if it is a perfectly inelastic collision?

Sol. Given, $m_1 = 40 \text{ kg}$, $m_2 = 60 \text{ kg}$, $v_1 = 4 \text{ ms}^{-1}$, $v_2 = 2 \text{ ms}^{-1}$

$$\therefore \Delta K = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (v_1 - v_2)^2 = \frac{1}{2} \left(\frac{40 \times 60}{40 + 60} \right) (4 - 2)^2 = 48 \text{ J}$$

Example 8.31 A particle of mass m moving with speed u collides perfectly inelastically with another particle of mass $2m$ at rest. Find loss of kinetic energy of system in the collision.

Sol. Let velocity of system $(m + 2m)$ of particles after collision be v .

$$\text{From law of conservation of momentum, } mu = 3mv \Rightarrow v = \frac{u}{3}$$

Now, loss of kinetic energy = $K_i - K_f$

$$\Rightarrow \Delta K = \frac{1}{2}mu^2 - \frac{1}{2}(3m)\left(\frac{u}{3}\right)^2$$

$$\Rightarrow \Delta K = \frac{1}{2}mu^2 - \frac{1}{6}mu^2$$

$$\Rightarrow \Delta K = \frac{1}{3}mu^2$$

Example 8.32 A railway carriage of mass 8000 kg moving with on speed of 54 km h^{-1} collides with an another stationary carriage of same mass. Determine the loss in kinetic energy in this process.

Sol. Given, $m_1 = 8000 \text{ kg}$, $v_1 = 54 \text{ km h}^{-1}$

$$= 54 \times \frac{5}{18} \text{ ms}^{-1} = 15 \text{ ms}^{-1}$$

$$m_2 = 8000 \text{ kg}, v_2 = 0$$

From law of conservation of momentum,

$$m_1v_1 + m_2v_2 = (m_1 + m_2)v$$

$$v = \frac{m_1v_1}{m_1 + m_2} \quad (\because v_2 = 0)$$

$$= \frac{8000 \times 15}{8000 + 8000} = 7.5 \text{ ms}^{-1}$$

Loss of kinetic energy = KE before collision - KE after collision

$$= \frac{1}{2}m_1v_1^2 - \frac{1}{2}(m_1 + m_2)v^2$$

$$= \frac{1}{2} \times 8000 \times (15)^2 - \frac{1}{2} (8000 + 8000)(7.5)^2$$

$$= 400 \times 225 - 8000 \times 56.25$$

$$= 450 \text{ kJ}$$

Newton's law of restitution

When two objects are in direct (head on) impact, the speed with which they separate after impact is usually less than or equal to their speed of approach before impact.

According to Newton's law of restitution, the ratio of relative velocity of separation after collision to relative velocity of approach before collision remains constant.

$$e = \frac{\text{Relative velocity of separation (after collision)}}{\text{Relative velocity of approach (before collision)}}$$

The ratio e is called the coefficient of restitution and is constant for two particular objects.

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

where, u_1 and u_2 are velocities of two bodies before collision and v_1 and v_2 are their velocities after collision.

For elastic collision, $e = 1$ and for inelastic collision, $0 < e < 1$ while for perfectly inelastic collision, $e = 0$.

Solving the following equations,

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ and $v_2 - v_1 = e(u_1 - u_2)$, we get

$$v_1 = \frac{m_1 - em_2}{m_1 + m_2} u_1 + \frac{(1+e)m_2}{m_1 + m_2} u_2$$

and
$$v_2 = \frac{m_2 - em_1}{m_1 + m_2} u_2 + \frac{(1+e)m_1}{m_1 + m_2} u_1$$

Putting $e = 1$, we will get formulae of v_1 and v_2 for an elastic collision.

Putting $e = 0$, we will get formulae of v_1 and v_2 for perfectly inelastic collision.

The loss in kinetic energy during an inelastic collision is

$$\Delta E = \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (1 - e^2) (u_1 - u_2)^2$$

However, if the target is massive (i.e. $m_2 \gg m_1$) and $u_2 = 0$, then the lighter body loses all its kinetic energy.

Note

In the situation shown in figure, if e is the coefficient of restitution between the ball and the ground, then after n th collision with the floor, the speed of ball will remain $e^n v_0$ and it will go upto a height $e^{2n} h$, i.e.

$$v_n = e^n v_0 = e^n \sqrt{2gh}$$

and $h_n = e^{2n} h$

Total distance travelled by the ball before it stops bouncing is

$H = h \left[\frac{1+e^2}{1-e^2} \right]$ and total time taken by the ball to stop bouncing will be

$$T = \left(\frac{1+e}{1-e} \right) \sqrt{\frac{2h}{g}}$$

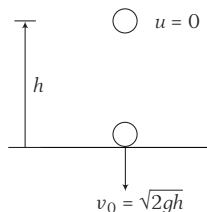
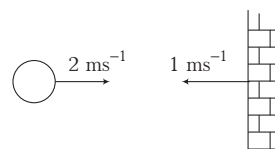
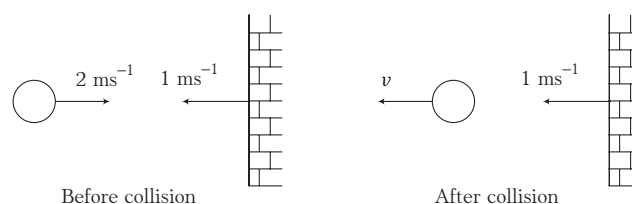


Fig. 8.6

Example 8.33 A ball is moving with velocity 2 ms^{-1} towards a heavy wall moving towards the ball with speed 1 ms^{-1} as shown in figure. Assuming collision to be elastic, find the velocity of ball immediately after the collision.



Sol. The speed of wall will not change after the collision. So, let v be the velocity of the ball after collision in the direction as shown in figure. Since, collision is elastic ($e = 1$),

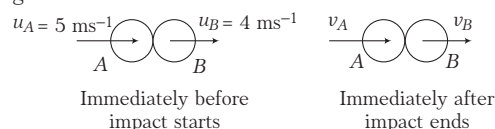


separation speed = approach speed

or $v - 1 = 2 + 1$ or $v = 4 \text{ ms}^{-1}$

Example 8.34 A ball of mass 2 kg moving with speed 5 ms^{-1} collides directly with another ball of mass 3 kg moving in the same direction with speed 4 ms^{-1} . The coefficient of restitution is $2/3$. Find their velocities after collision.

Sol. Denoting the first ball by A and the second ball by B , velocities immediately before and after the impact are shown in the figure.



Applying principle of conservation of momentum, we have

$$\begin{aligned} m_B v_B + m_A v_A &= m_A u_A + m_B u_B \\ \Rightarrow 3v_B + 2v_A &= 2 \times 5 + 3 \times 4 \\ 3v_B + 2v_A &= 22 \end{aligned} \quad \dots(i)$$

Applying equation of coefficient of restitution, we have

$$\begin{aligned} v_B - v_A &= e(u_A - u_B) \Rightarrow v_B - v_A = \frac{2}{3}(5 - 4) \\ 3v_B - 3v_A &= 2 \end{aligned} \quad \dots(ii)$$

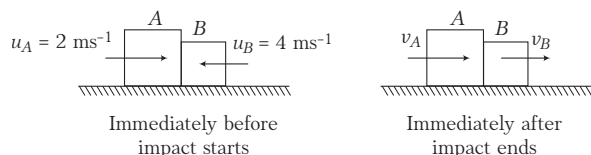
Solving Eqs. (i) and (ii), we get

$$v_A = 4 \text{ ms}^{-1} \text{ and } v_B = 4.67 \text{ ms}^{-1}$$

Example 8.35 A block of mass 5 kg moves from left to right with a velocity of 2 ms^{-1} and collides with another block of mass 3 kg moving along the same line in the opposite direction with velocity 4 ms^{-1} .

- If the collision is perfectly elastic, determine velocities of both the blocks after their collision.
- If coefficient of restitution is 0.6 , determine velocities of both the blocks after their collision.

Sol. Denoting the first block by A and the second block by B , velocities immediately before and after the impact are shown in the figure.



Applying principle of conservation of momentum, we have

$$\begin{aligned} m_B v_B + m_A v_A &= m_A u_A + m_B u_B \\ \Rightarrow 3 v_B + 5 v_A &= 5 \times 2 + 3 \times (-4) \\ 3 v_B + 5 v_A &= -2 \quad \dots(i) \end{aligned}$$

Applying equation of coefficient of restitution, we have

$$\begin{aligned} v_A - v_B &= e(u_A - u_B) \\ \Rightarrow v_B - v_A &= e\{2 - (-4)\} \\ v_B - v_A &= 6e \quad \dots(ii) \end{aligned}$$

(i) For perfectly elastic impact, $e = 1$. Using this value in Eq. (ii), we get

$$v_B - v_A = 6 \quad \dots(iii)$$

Now, solving Eqs. (i) and (iii), we obtain

$$v_A = -2.5 \text{ ms}^{-1} \quad \text{and} \quad v_B = 3.5 \text{ ms}^{-1}$$

(ii) For $e = 0.6$, Eq. (ii) is modified as

$$v_B - v_A = 3.6 \quad \dots(iv)$$

Now, solving Eqs. (i) and (iv), we obtain

$$v_A = -1.6 \text{ ms}^{-1} \quad \text{and} \quad v_B = 2 \text{ ms}^{-1}$$

Block A reverses back with speed 1.6 ms^{-1} and B also moves in opposite direction to its original direction with speed 2 ms^{-1} .

Example 8.36 A particle of mass 2 kg moving with a velocity $5 \hat{i} \text{ ms}^{-1}$ collides head on with another particle of mass 3 kg moving with a velocity $-2 \hat{i} \text{ ms}^{-1}$. After the collision, the first particle has speed of 1.6 ms^{-1} in negative x -direction. Find

- velocity of the centre of mass after the collision,
- velocity of the second particle after the collision,
- coefficient of restitution.

Sol. (i) $\mathbf{v}_c = \frac{m_1 \mathbf{u}_1 + m_2 \mathbf{u}_2}{m_1 + m_2} = 0.8 \hat{i} \text{ ms}^{-1}$

(Velocity of CM before and after collision will be same)

(ii) $\mathbf{v}_1 = -1.6 \hat{i} \text{ ms}^{-1}$

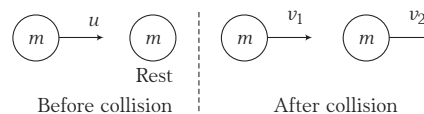
Using law of conservation of momentum,

$$m_1 \mathbf{u}_1 + m_2 \mathbf{u}_2 = m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 \Rightarrow \mathbf{v}_2 = 2.4 \hat{i} \text{ ms}^{-1}$$

(iii) $e = \frac{\mathbf{v}_2 - \mathbf{v}_1}{\mathbf{u}_1 - \mathbf{u}_2} = \frac{4}{7}$

Example 8.37 An object of mass m moving with speed u collides one dimensionally with another identical object at rest. Find their velocities after collision, if coefficient of restitution of collision is e .

Sol. Let v_1 and v_2 be the final velocities of 1st and 2nd object, respectively.



From law of conservation of momentum,

$$mu + m \times 0 = mv_1 + mv_2$$

$$\Rightarrow v_1 + v_2 = u \quad \dots(i)$$

$$\text{Now, } e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$\Rightarrow \frac{v_2 - v_1}{u} = e$$

$$\Rightarrow v_2 - v_1 = eu \quad \dots(ii)$$

Adding Eqs. (i) and (ii), we get

$$\begin{aligned} 2v_2 &= u + eu \\ \Rightarrow v_2 &= \left[\frac{1+e}{2} \right] u \end{aligned}$$

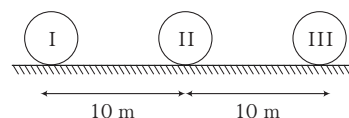
Subtracting Eq. (ii) from Eq. (i), we get

$$\begin{aligned} 2v_1 &= (1-e)u \\ \Rightarrow v_1 &= \left[\frac{1-e}{2} \right] u \end{aligned}$$

Example 8.38 Three identical balls, ball I, ball II and ball III are placed on a smooth floor in a straight line at the separation of 10 m between balls as shown in figure. Initially balls are stationary.

Ball I is given velocity of 10 ms^{-1} towards ball II, collision between balls I and II is inelastic with coefficient of restitution 0.5 but collision between balls II and III is perfectly elastic.

What is the time interval between two consecutive collisions between balls I and II?



Sol. Let velocity of 1st ball and 2nd ball after collision be v_1 and v_2 .

$$v_2 - v_1 = 0.5 \times 10 = 5 \quad \dots(i)$$

$$mv_2 + mv_1 = m \times 10 \quad \dots(ii)$$

$$\Rightarrow v_2 + v_1 = 10$$

Solving Eqs. (i) and (ii), we get

$$v_1 = 2.5 \text{ ms}^{-1} \quad \text{and} \quad v_2 = 7.5 \text{ ms}^{-1}$$

Ball II after moving 10 m collides with ball III elastically and stops. But ball I moves towards ball II. Time taken between two consecutive collisions,

$$t = \frac{10}{7.5} + \frac{10 - 10 \times \frac{2.5}{7.5}}{2.5} = 4 \text{ s}$$

CHECK POINT 8.3

- In an elastic collision,
 - both momentum and KE are conserved
 - only momentum is conserved
 - only KE is conserved
 - Neither KE nor momentum is conserved
- A ball hits the floor and rebounds after an inelastic collision. In this case,
 - the momentum of the ball just after the collision is the same as that just before the collision
 - the mechanical energy of the ball remains the same in the collision
 - the total momentum of the ball and the earth is conserved
 - the total energy of the ball and the earth is conserved
- If a body of mass m collides head on, elastically with velocity u with another identical body at rest. After collision, velocity of the second body will be
 - zero
 - u
 - $2u$
 - data insufficient
- A body of mass M_1 collides elastically with another body of mass M_2 at rest. There is maximum transfer of energy when
 - $M_1 > M_2$
 - $M_1 < M_2$
 - $M_1 = M_2$
 - same for all values of M_1 and M_2
- Two particles of masses m_A and m_B and velocities v_A and v_B respectively collides. After collision, they interchange their velocities, then ratio of $\frac{m_A}{m_B}$ is
 - $\frac{v_A}{v_B}$
 - $\frac{v_B}{v_A}$
 - $\frac{v_A + v_B}{v_B - v_A}$
 - 1
- Two perfectly elastic particles A and B of equal mass are travelling along the line joining them with velocities 15 ms^{-1} and 10 ms^{-1} . After collision, the respective velocities of A and B will be
 - 10 ms^{-1} , 10 ms^{-1}
 - 15 ms^{-1} , 15 ms^{-1}
 - 10 ms^{-1} , 15 ms^{-1}
 - 15 ms^{-1} , 10 ms^{-1}
- The collision of two balls of equal mass takes place at the origin of coordinates. Before collision, the components of velocities are $(v_x = 50 \text{ cms}^{-1}, v_y = 0)$ and $(v_x = -40 \text{ cms}^{-1}$ and $v_y = 30 \text{ cms}^{-1})$. The first ball comes to rest after collision. The velocity components v_x and v_y respectively of the second ball are
 - 10 and 30 cms^{-1}
 - 30 and 10 cms^{-1}
 - 5 and 15 cms^{-1}
 - 15 and 5 cms^{-1}
- A mass of 0.5 kg moving with a speed of 1.5 ms^{-1} on a horizontal smooth surface, collides with a nearly weightless spring of force constant $k = 50 \text{ Nm}^{-1}$. The maximum compression of the spring would be



- 0.12 m
 - 1.5 m
 - 0.5 m
 - 0.15 m
- A smooth sphere of mass M moving with velocity u directly collides elastically with another sphere of mass m at rest. After collision, their final velocities are v' and v , respectively. The value of v is
 - $\frac{2uM}{m}$
 - $\frac{2um}{M}$
 - $\frac{2u}{1 + \frac{m}{M}}$
 - $\frac{2u}{1 + \frac{M}{m}}$
 - A body of mass m_1 moving with velocity 3 ms^{-1} collides with another body at rest of mass m_2 . After collision, the velocities of the two bodies are 2 ms^{-1} and 5 ms^{-1} respectively along the direction of motion of m_2 . The ratio $\frac{m_1}{m_2}$ is
 - $\frac{5}{12}$
 - 5
 - $\frac{1}{5}$
 - $\frac{12}{5}$
 - A body of mass m moving with velocity v collides head on with another body of mass $2m$ which is initially at rest. The ratio of KE of colliding body before and after collision will be
 - 1 : 1
 - 2 : 1
 - 4 : 1
 - 9 : 1
 - The two diagrams show the situations before and after a collision between two spheres A and B of equal radii moving along the same straight line on a smooth horizontal surface. The coefficient of restitution e is

Before collision

8 ms^{-1} 2 ms^{-1}

After collision

2 ms^{-1} 5 ms^{-1}

 - $\frac{1}{3}$
 - $\frac{1}{2}$
 - $\frac{2}{3}$
 - $\frac{3}{4}$
 - Two balls of equal masses have a head on collision with speed 6 ms^{-1} each. If the coefficient of restitution is $1/3$, the relative speed of separation of balls after impact will be
 - 18 ms^{-1}
 - 4 ms^{-1}
 - 6 ms^{-1}
 - data insufficient
 - A block of mass m moving at a velocity v collides with another block of mass $2m$ at rest. The lighter block comes to rest after collision. Find the coefficient of restitution.
 - $\frac{1}{2}$
 - 1
 - $\frac{1}{3}$
 - $\frac{1}{4}$
 - A sphere of mass m moving with a constant velocity u hits another stationary sphere of same mass. If e is the coefficient of restitution, then ratio of velocities of the two spheres $\frac{v_1}{v_2}$ after collision will be

2 $\xrightarrow{u}$ 1

 - $\frac{1-e}{1+e}$
 - $\frac{1+e}{1-e}$
 - $\frac{e+1}{e-1}$
 - $\frac{e-1}{e+1}$

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- If the net external forces acting on the system of particles is zero, then which of the following may vary?
 - Momentum of the system
 - Velocity of centre of mass
 - Position of centre of mass
 - None of the above
- For which of the following does the centre of mass lie outside the body? **[NCERT Exemplar]**
 - A pencil
 - A shotput
 - A dice
 - A bangle
- Conservation of momentum in a collision between particles can be understood from **[NCERT Exemplar]**
 - Conservation of energy
 - Newton's first law
 - Newton's second law
 - Both Newton's second and third laws
- A body of mass a moving with velocity b strikes a body of mass c and gets embedded into it. The velocity of the system after collision is
 - $\frac{a+c}{ab}$
 - $\frac{ab}{a+c}$
 - $\frac{a}{b+c}$
 - $\frac{a}{a+b}$
- A cannon ball is fired with a velocity 200 ms^{-1} at an angle of 60° with the horizontal. At the highest point of its flight, it explodes into 3 equal fragments, one going vertically upwards with a velocity 100 ms^{-1} , the second one falling vertically downwards with a velocity 100 ms^{-1} . The third fragment will be moving with a velocity
 - 100 ms^{-1} in the horizontal direction
 - 300 ms^{-1} in the horizontal direction
 - 300 ms^{-1} in a direction making an angle of 60° with the horizontal
 - 200 ms^{-1} in a direction making an angle of 60° with the horizontal
- Two balls of equal mass have a head on collision with speed 4 ms^{-1} each travelling in opposite directions. If the coefficient of restitution is $1/2$, the speed of each ball after impact will be
 - 1 ms^{-1}
 - 2 ms^{-1}
 - 3 ms^{-1}
 - data insufficient
- A machine gun fires a steady stream of bullets at the rate of n per minute into a stationary target in which the bullets get embedded. If each bullet has a mass m and arrives at the target with a velocity v , the average force on the target is
 - $60 mnv$
 - $\frac{60v}{mn}$
 - $\frac{mnv}{60}$
 - $\frac{mv}{60n}$
- A machine gun fires a bullet of mass 40 g with a velocity 1200 ms^{-1} . The man holding it, can exert a maximum force of 144 N on the gun. How many bullets can be fired per second at the most?
 - One
 - Four
 - Two
 - Three
- A particle of mass m moving with speed v hits elastically another stationary particle of mass $2m$ inside a smooth horizontal circular tube of radius r . The time after which the second collision will take place is
 - $\frac{2\pi r}{v}$
 - $\frac{4\pi r}{v}$
 - $\frac{3\pi r}{2v}$
 - $\frac{\pi r}{v}$
- A bullet of mass 20 g moving with 600 ms^{-1} collides with a block of mass 4 kg hanging with the string of length 0.4 m . What is velocity of bullet when it comes out of block, if block rises to height 0.2 m after collision? (Take, $g = 10 \text{ ms}^{-2}$)
 - 200 ms^{-1}
 - 150 ms^{-1}
 - 400 ms^{-1}
 - 300 ms^{-1}
- A mass of 10 g moving horizontally with a velocity of 100 cms^{-1} strikes a pendulum bob of mass 10 g . Length of string is 50 cm . The two masses stick together. The maximum height reached by the system now is (Take, $g = 10 \text{ ms}^{-2}$)
 - 7.5 cm
 - 5 cm
 - 2.5 cm
 - 1.25 cm
- In a gravity free space, a man of mass M standing at a height h above the floor, throws a ball of mass m straight down with a speed u . When the ball reaches the floor, the distance of the man above the floor will be
 - $h \left(1 + \frac{m}{M}\right)$
 - $\left(1 + \frac{M}{m}\right) h$
 - h
 - $\frac{m}{M} h$

- 13** A cracker is thrown into air with a velocity of 10 ms^{-1} at an angle of 45° with the vertical. When it is at a height of $(1/2) \text{ m}$ from the ground, it explodes into a number of pieces which follow different parabolic paths. What is the velocity of centre of mass, when it is at a height of 1 m from the ground?

(Take, $g = 10 \text{ ms}^{-2}$)

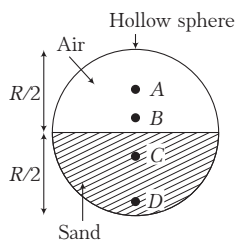
- (a) $4\sqrt{5} \text{ ms}^{-1}$ (b) $2\sqrt{5} \text{ ms}^{-1}$
(c) $5\sqrt{4} \text{ ms}^{-1}$ (d) 5 ms^{-1}
- 14** Two blocks of masses 10 kg and 30 kg are placed along a vertical line. The first block is raised through a height of 7 cm . By what distance should the second mass be moved to raise the centre of mass by 1 cm ?

- (a) 2 cm upward (b) 1 cm upward
(c) 2 cm downward (d) 1 cm downward

- 15** A cricket ball of mass 150 g moving with a speed of 126 kmh^{-1} hits at the middle of the bat, held firmly at its position by the batsman. The ball moves straight back to the bowler after hitting the bat. Assuming that collision between ball and bat is completely elastic and the two remain in contact for 0.001 s , the force that the batsman had to apply to hold the bat firmly at its place would be

- (a) 10.5 N (b) 21 N [NCERT Exemplar]
(c) $1.05 \times 10^4 \text{ N}$ (d) $2.1 \times 10^4 \text{ N}$

- 16** Which of the following points is the likely position of the centre of mass of the system as shown in figure? [NCERT Exemplar]



- (a) A (b) B
(c) C (d) D

- 17** A metal ball falls from a height of 32 m on a steel plate. If the coefficient of restitution is 0.5 , to what height will the ball rise after second bounce?

- (a) 2 m (b) 4 m
(c) 8 m (d) 16 m

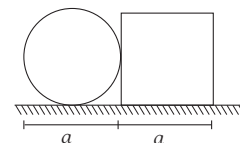
- 18** 10000 small balls, each weighing 1 g , strikes 1 cm^2 of area per second with a velocity 100 ms^{-1} in a normal direction and rebound with the same velocity. The value of pressure on the surface will be

- (a) $2 \times 10^3 \text{ Nm}^{-2}$ (b) $2 \times 10^5 \text{ Nm}^{-2}$
(c) 10^7 Nm^{-2} (d) $2 \times 10^7 \text{ Nm}^{-2}$

- 19** A particle of mass 1 kg is thrown vertically upward with speed 100 ms^{-1} . After 5 s , it explodes into two parts. One part of mass 400 g emerges with speed 25 ms^{-1} in downward direction, what is the velocity of other part just after explosion? (Take, $g = 10 \text{ ms}^{-2}$)

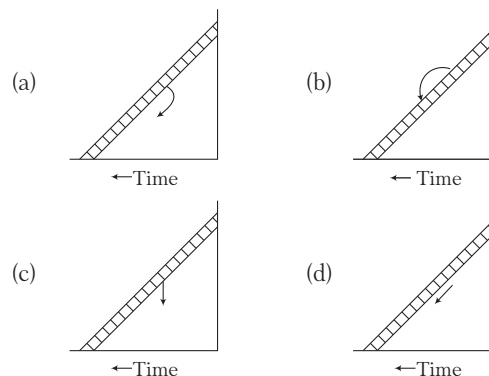
- (a) 100 ms^{-1} upward (b) 600 ms^{-1} upward
(c) 100 ms^{-1} downward (d) 300 ms^{-1} upward

- 20** A circular plate of diameter a is kept in contact with a square plate of edge a as shown in figure. The density of the material and the thickness are same everywhere. The centre of mass of the composite system will be

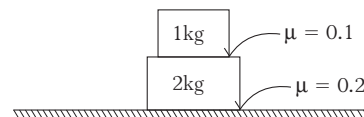


- (a) inside the circular plate
(b) inside the square plate
(c) at the point of contact
(d) outside the system

- 21** A ladder is leaned against a smooth wall and is allowed to slip on a frictionless floor. Which figure represents trace of its centre of mass?



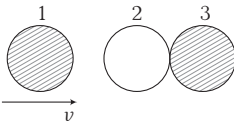
- 22** Both the blocks as shown in the given arrangement are given together a horizontal velocity towards right. If a_{CM} be the subsequent acceleration of the centre of mass of the system of blocks, then a_{CM} will be



- (a) zero (b) $\frac{5}{3} \text{ ms}^{-2}$
(c) $\frac{7}{3} \text{ ms}^{-2}$ (d) 2 ms^{-2}

- 23** In a free space, a rifle of mass M shoots a bullet of mass m at a stationary block of mass M at a distance D away from it. When the bullet has moved through a distance d towards the block, the centre of mass of the bullet-block system is at a distance of

- (a) $\frac{(D-d)m}{M+m}$ from the bullet
 (b) $\frac{md+MD}{M+m}$ from the block
 (c) $\frac{2md+MD}{M+m}$ from the block
 (d) $\frac{(D-d)M}{M+m}$ from the bullet
- 24** A man of mass M stands at one end of a plank of length L which lies at rest on a frictionless surface. The man walks to the other end of the plank. If the mass of the plank is $\frac{M}{3}$, the distance that the man moves relative to the ground is
 (a) $\frac{3L}{4}$ (b) $\frac{L}{4}$ (c) $\frac{4L}{5}$ (d) $\frac{L}{3}$
- 25** A 2 kg block of wood rests on a long table top. A 5 g bullet moving horizontally with a speed of 150 ms^{-1} is shot into the block and sticks to it. The block then slides 2.7 m along the table top and comes to a stop. The force of friction between the block and the table is
 (a) 0.052 N (b) 3.63 N (c) 2.50 N (d) 1.04 N
- 26** A particle falls from a height h upon a fixed horizontal plane and rebounds. If e is the coefficient of restitution, the total distance travelled before rebounding has stopped is
 (a) $h \left(\frac{1+e^2}{1-e^2} \right)$ (b) $h \left(\frac{1-e^2}{1+e^2} \right)$
 (c) $\frac{h}{2} \left(\frac{1-e^2}{1+e^2} \right)$ (d) $\frac{h}{2} \left(\frac{1+e^2}{1-e^2} \right)$
- 27** Two identical balls bearing in contact with each other and resting on a frictionless table are hit head on by another ball bearing the same mass moving initially with a speed v as shown in figure. If the collision is elastic, which of the following (figure) is a possible result after collision?



- (a) (b)
 (c) (d)

[NCERT Exemplar]

- 28** A bullet of mass m is fired into a block of wood of mass M which hangs on the end of pendulum and gets embedded into it. When the bullet strikes the wooden block, the pendulum starts to swing with maximum rise R . Then, the velocity of the bullet is given by

- (a) $\frac{M}{m+M} \sqrt{2gR}$ (b) $\frac{M+m}{m} \sqrt{2gR}$
 (c) $\frac{M}{m} \sqrt{2gR}$ (d) None of these

- 29** A ball falling freely from a height of 4.9 m, hits a horizontal surface. If $e = 3/4$, then the ball will hit the surface second time after

- (a) 1 s (b) 1.5 s (c) 2 s (d) 3 s

- 30** In a one dimensional collision between two identical particles A and B , B is stationary and A has momentum p before impact. During impact, B gives an impulse J to A . Then, coefficient of restitution between the two is

- (a) $\frac{2J}{p} - 1$ (b) $\frac{2J}{p} + 1$ (c) $\frac{J}{p} + 1$ (d) $\frac{J}{p} - 1$

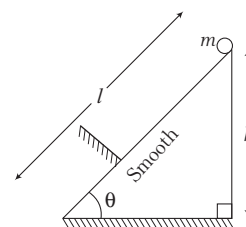
- 31** A particle of mass m kg moving with a velocity $(3\hat{i} + 2\hat{j}) \text{ ms}^{-1}$ collides with a stationary body of mass M kg and finally moves with a velocity $(-2\hat{i} + \hat{j}) \text{ ms}^{-1}$. If $\frac{m}{M} = \frac{1}{13}$, then

- (a) the impulse is $\pm m(5\hat{i} + \hat{j}) \text{ kg}\cdot\text{ms}^{-1}$
 (b) the velocity of the M is $\frac{1}{13}(5\hat{i} + \hat{j}) \text{ ms}^{-1}$
 (c) Both (a) and (b) are wrong
 (d) Both (a) and (b) are correct

- 32** A ball falls freely from a height of 45 m. When the ball is at a height of 25 m, it explodes into two equal pieces. One of them moves horizontally with a speed of 10 ms^{-1} . The distance between the two pieces on the ground is

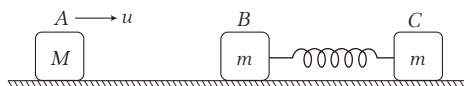
- (a) 20 m (b) 30 m (c) 40 m (d) 60 m

- 33** A ball of mass m is released from the top of an inclined plane of inclination θ as shown in figure. It strikes a rigid surface at a distance $\frac{3h}{4}$ from top elastically. Impulse imparted to ball by the rigid surface is



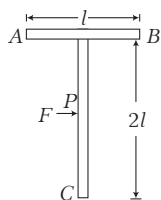
- (a) $m\sqrt{\frac{3}{2}gh}$ (b) $m\sqrt{3gh}$
 (c) $2m\sqrt{3gh}$ (d) $m\sqrt{6gh}$

- 34** A block A of mass M moving with speed u collides elastically with block B of mass m which is connected to block C of mass m with a spring.



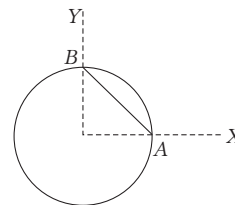
When the compression in spring is maximum, the velocity of block C with respect to block A is (Neglect the friction everywhere)

- (a) zero (b) $\left(\frac{M}{M+m}\right)u$ (c) $\left(\frac{m}{M+m}\right)u$ (d) $\left(\frac{m}{M}\right)u$
- 35** A particle of mass m moving with velocity u makes an elastic one dimensional collision with a stationary particle of mass m . They are in contact for a brief time T . Their force of interaction increases from zero to F_0 linearly in time $\frac{T}{2}$ and decreases linearly to zero in further time $\frac{T}{2}$. The magnitude of F_0 is
- (a) $\frac{mu}{T}$ (b) $\frac{2mu}{T}$ (c) $\frac{mu}{2T}$ (d) None of these
- 36** A T-shaped object with dimensions as shown in the figure, is lying on a smooth floor. A force F is applied at the point P parallel to AB, such that the object has only the translational motion without rotation. Find the location of P with respect to C.

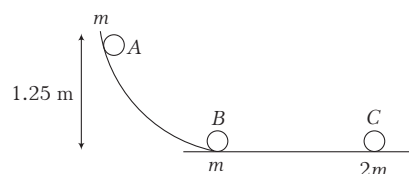


- (a) $\frac{4}{3}l$ (b) l
 (c) $\frac{2}{3}l$ (d) $\frac{3}{2}l$
- 37** A ball is projected vertically down with an initial velocity from a height of 20 m onto a horizontal floor. During the impact, it loses 50% of its energy and rebounds to the same height. The initial velocity of its projection is (Take, $g = 10 \text{ ms}^{-2}$)
- (a) 20 ms^{-1} (b) 15 ms^{-1} (c) 10 ms^{-1} (d) 5 ms^{-1}
- 38** An object comprises a uniform ring of radius R and its uniform chord AB (not necessarily made of the same material) as shown in figure.

Which of the following cannot be the coordinates of centre of mass of the object?



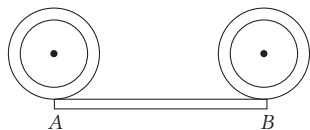
- (a) $\left(\frac{R}{3}, \frac{R}{3}\right)$ (b) $\left(\frac{R}{\sqrt{2}}, \frac{R}{\sqrt{2}}\right)$
 (c) $\left(\frac{R}{4}, \frac{R}{4}\right)$ (d) None of these
- 39** A circular ring of mass 6 kg and radius a is placed such that its centre lies at the origin. Two particles of masses 2 kg each are placed at the intersecting points of the circle with positive X-axis and positive Y-axis. Then, the angle made by the position vector of centre of mass of entire system with X-axis is
- (a) 45° (b) 60° (c) $\tan^{-1}\left(\frac{4}{5}\right)$ (d) 30°
- 40** A particle A of mass m initially at rest slides down a height of 1.25 m on a frictionless ramp, collides with and sticks to an identical particle B of mass m at rest as shown in the figure.



Then, particles A and B together collide elastically with particle C of mass $2m$ at rest. The speed of particle C after the collision with combined body (A + B) would be (Take, $g = 10 \text{ ms}^{-2}$)

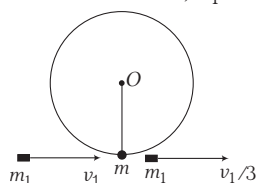
- (a) 2 ms^{-1} (b) 1.25 ms^{-1} (c) 2.5 ms^{-1} (d) 5 ms^{-1}
- 41** A man of mass m moves with a constant speed on a plank of mass M and length l kept initially at rest on a frictionless horizontal surface from one end to the other in time t . The speed of the plank relative to ground while man is moving, is
- (a) $\frac{l}{t}\left(\frac{M}{m}\right)$ (b) $\frac{l}{t}\left(\frac{m}{m+M}\right)$
 (c) $\frac{l}{t}\left(\frac{M}{M+m}\right)$ (d) None of these
- 42** You are supplied with three identical rods of same length and mass. If the length of each rod is 2π . Two of them are converted into rings and then placed over the third rod as shown in figure. If point A is considered as origin of the coordinate

system, the coordinates of the centre of mass will be (you may assume AB as X -axis of the coordinate system)

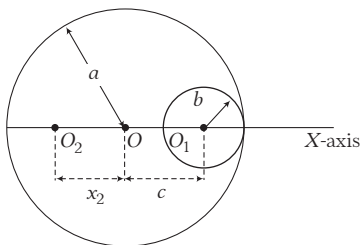


- (a) $\left(\frac{\pi}{2}, \frac{1}{3}\right)$ (b) $\left(\frac{\pi}{2}, \frac{2}{3}\right)$ (c) $\left(\pi, \frac{1}{3}\right)$ (d) $\left(\pi, \frac{2}{3}\right)$

- 43** A pendulum consists of a wooden bob of mass m and length l . A bullet of mass m_1 is fired towards the pendulum with a speed v_1 and it emerges from the bob with speed $\frac{v_1}{3}$. The bob just completes motion along a vertical circle. Then, v_1 is



- (a) $\frac{m}{m_1} \sqrt{5gl}$ (b) $\frac{3m}{2m_1} \sqrt{5gl}$
 (c) $\frac{2}{3} \left(\frac{m}{m_1}\right) \sqrt{5gl}$ (d) $\left(\frac{m_1}{m}\right) \sqrt{gl}$
- 44** From a circular disc of radius R , a square is cut out with a radius as its diagonal. The centre of mass of remaining portion is at a distance (from the centre)
- (a) $\frac{R}{(4\pi - 2)}$ (b) $\frac{R}{2\pi}$
 (c) $\frac{R}{(\pi - 2)}$ (d) $\frac{R}{(2\pi - 2)}$
- 45** A uniform circular disc of radius a is taken. A circular portion of radius b has been removed from it as shown in the figure. If the centre of hole is at a distance c from the centre of the disc, the distance x_2 of the centre of mass of the remaining part from the initial centre of mass O is given by



- (a) $\frac{\pi b^2}{(a^2 - c^2)}$ (b) $\frac{cb^2}{(a^2 - b^2)}$
 (c) $\frac{\pi c^2}{(a^2 - b^2)}$ (d) $\frac{ca^2}{(c^2 - b^2)}$

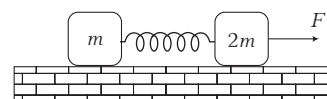
- 46** A girl throws a ball with initial velocity v at an inclination of 45° . The ball strikes the smooth vertical wall at a horizontal distance d from the girl and after rebounding returns to her hand. What is the coefficient of restitution between wall and the ball?

- (a) $v^2 - gd$ (b) $\frac{gd}{v^2 - gd}$
 (c) $\frac{gd}{v^2}$ (d) $\frac{v^2}{gd}$

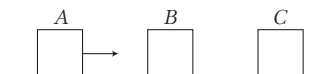
- 47** A disc of mass 10 g is kept floating horizontally by throwing 10 marbles per second against it from below. If the mass of each marble is 5 g. What will be velocity with which the marbles are striking the disc? Assume that, the marble strikes the disc normally and rebound downwards with the same speed.

- (a) 2.98 ms^{-1} (b) 0.98 ms^{-1}
 (c) 0.49 ms^{-1} (d) 1.96 ms^{-1}

- 48** Two blocks of masses m and $2m$ are kept on a smooth horizontal surface. They are connected by an ideal spring of force constant k . Initially, the spring is unstretched. A constant force is applied to the heavier block in the direction as shown in figure. Suppose at time t , displacement of smaller block is x , then displacement of the heavier block at this moment would be



- (a) $\frac{x}{2}$ (b) $\frac{Ft^2}{6m} + \frac{x}{3}$ (c) $\frac{x}{3}$ (d) $\frac{Ft^2}{4m} - \frac{x}{2}$
- 49** Three identical blocks A , B and C are placed on horizontal frictionless surface. The blocks B and C are at rest but A is approaching towards B with a speed 10 ms^{-1} .



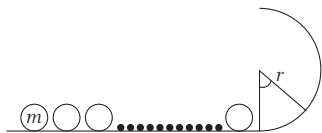
The coefficient of restitution for all collisions is 0.5. The speed of the block C just after collision is approximately

- (a) 5.6 ms^{-1} (b) 6.4 ms^{-1}
 (c) 3.2 ms^{-1} (d) 4.6 ms^{-1}

- 50** A train of mass M is moving on a circular track of radius R with constant speed v . The length of the train is half of the perimeter of the track. The linear momentum of the train will be

- (a) πMv (b) $\frac{2Mv}{\pi}$ (c) $\frac{\pi Mv}{2}$ (d) Mv

- 51** n elastic balls are placed at rest on a smooth horizontal plane which is circular at the ends with radius r as shown in the figure. The masses of the balls are $m, \frac{m}{2}, \frac{m}{2^2}, \dots, \frac{m}{2^{n-1}}$, respectively. What is the minimum velocity which should be imparted to the first ball of mass m such that this n th ball will complete the vertical circle?



- (a) $\left(\frac{3}{4}\right)^{n-1} \sqrt{5gr}$ (b) $\left(\frac{4}{3}\right)^{n-1} \sqrt{5gr}$
 (c) $\left(\frac{3}{2}\right)^{n-1} \sqrt{5gr}$ (d) $\left(\frac{2}{3}\right)^{n-1} \sqrt{5gr}$

- 52** A small ball rolls off the top landing of the staircase. It strikes the mid-point of the first step and then the mid-point of the second step. The steps are smooth, and identical in height and width. The coefficient of restitution between the ball and the first step is

- (a) 1 (b) $\frac{3}{4}$
 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

- 53** Two identical blocks A and B , each of mass m resting on smooth floor are connected by a light spring of natural length L and spring constant k with the spring at its natural length. A third identical block C (mass m) moving with a speed v along the line joining A and B collides with A , the maximum compression in the spring is

- (a) $v \sqrt{\frac{m}{2k}}$ (b) $m \sqrt{\frac{v}{2k}}$ (c) $\sqrt{\frac{mv}{k}}$ (d) $\frac{mv}{2k}$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-6) These questions consist of two statements each printed as Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) If Assertion is correct but Reason is incorrect.
 (d) If Assertion is incorrect but Reason is correct.

- 1 Assertion** The relative velocity of the two particles in head on elastic collision is unchanged both in magnitude and direction.

Reason The relative velocity is unchanged in magnitude but gets reversed in direction.

- 2. Assertion** If net force on a system is zero, then momentum of every individual body remains constant.

Reason If momentum of a system is constant, then kinetic energy of the system may change.

- 3. Assertion** Two bodies moving in opposite directions with same magnitude of linear momentum collide with each other. Then, after collision both the bodies will come to rest.

Reason Linear momentum of the system of bodies is zero.

- 4. Assertion** Two blocks A and B are connected at the two ends of an ideal spring as shown in figure. Initially spring was relaxed. Now, block B is pressed.

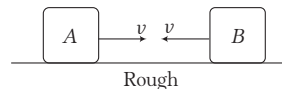
Linear momentum of the system will not remain constant till the spring reaches its initial natural length.



Reason An external force will act from the wall on block A .

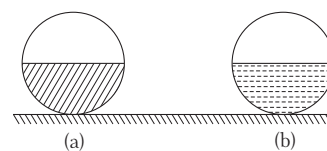
- 5. Assertion** Two blocks of masses m_A and m_B ($m_B > m_A$) are thrown towards each other with same speed over a rough ground. The coefficient of friction of both the blocks with ground is same.

Initial velocity of CM is towards left.



Reason Initial acceleration of centre of mass is towards right.

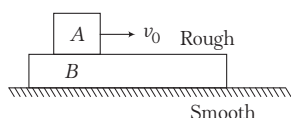
- 6. Assertion** Two identical spheres are half filled with two liquids of densities ρ_1 and ρ_2 ($> \rho_1$). The centre of mass of both the spheres lie at same level.



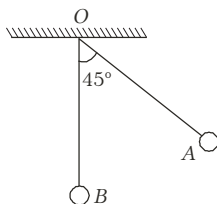
Reason The centre of mass will lie at centre of the sphere.

Statements based questions

- 1 Two trains A and B are running in the same direction on parallel rails such that A is faster than B . Packets of equal weight are transferred between them. Which of the following statement is correct?
- A will be accelerated, but B will be retarded.
 - B will be accelerated, but A will be retarded.
 - There will be no change in A , but B will be accelerated.
 - There will be no change in B , but A will be accelerated.
- 2 In a two block system, an initial velocity v_0 with respect to ground is given to block A . Which of the following statement(s) is/are correct?



- The momentum of block A is not conserved.
 - The momentum of system of blocks A and B is conserved.
 - The increase in momentum of B is equal to the decrease in momentum of block A .
 - All of the above
- 3 The bob A of a simple pendulum is released when the string makes an angle of 45° with the vertical. It hits another bob B of the same material and same mass kept at rest on a table. If the collision is elastic, which of the following statement is correct?
- Both A and B rise to the same height.
 - Both A and B come to rest at B .
 - Both A and B move with the same velocity of A .
 - A comes to rest and B moves with the velocity of A .



- 4 I. Linear momentum of the system remains constant.
II. Centre of mass of the system remains at rest.
Which of the following statement(s) is/are correct?
- I implies II and II implies I
 - I does not imply II and II does not imply I
 - I implies II but II does not imply I
 - II implies I but I does not imply II
- 5 I. Linear momentum of a system of particles is zero.
II. Kinetic energy of a system of particles is zero.
Which of the following statement(s) is/are correct?
- I implies II and II implies I
 - I does not imply II and II does not imply I
 - I implies II but II does not imply I
 - II implies I but I does not imply II
- 6 I. In elastic collision, initial kinetic energy is equal to the final kinetic energy.

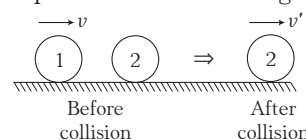
II. In elastic collision, kinetic energy during the collision time Δt is constant.

Which of the following statement(s) is/are correct?

- Only I
- Only II
- Both I and II
- Neither I nor II

Match the columns

- 1 In the diagram shown in figure, mass of both the balls is same. Match the following columns and mark the correct option from the codes given below.



Column I	Column II
(A) For $v' = v$	(p) $e = 0$
(B) For $v' = v/2$	(q) $e = 1$
(C) For $v' = (3/4)v$	(r) $e = 1/2$
	(s) Data is insufficient

Codes

- | | | | | | |
|-----|---|---|-----|---|---|
| A | B | C | A | B | C |
| (a) | p | q | (b) | q | p |
| (c) | s | r | (d) | s | p |

- 2 A particle of mass 1 kg has velocity $\mathbf{v}_1 = (2t)\hat{i}$ and another particle of mass 2 kg has velocity $\mathbf{v}_2 = (t^2)\hat{j}$. Match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Net force on centre of mass at 2 s	(p) $\frac{20}{9}$ unit
(B) Velocity of centre of mass at 2 s	(q) $\sqrt{68}$ unit
(C) Displacement of centre of mass in 2 s	(r) $\sqrt{80}/3$ unit
	(s) None

Codes

- | | | | | | |
|-----|---|---|-----|---|---|
| A | B | C | A | B | C |
| (a) | q | r | (b) | q | p |
| (c) | p | r | (d) | s | q |

- 3 A particle of mass m , kinetic energy K and momentum p collides head on elastically with another particle of mass $2m$ at rest. Match the following columns (after collision) and mark the correct option from the codes given below.

Column I	Column II
(A) Momentum of first particle	(p) $4p/3$
(B) Momentum of second particle	(q) $K/9$
(C) Kinetic energy of first particle	(r) $-p/3$
(D) Kinetic energy of second particle	(s) $8K/9$

- | | | | | | | | |
|-----|---|---|---|-----|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) | q | r | s | (b) | p | s | r |
| (c) | r | p | q | (d) | s | q | r |

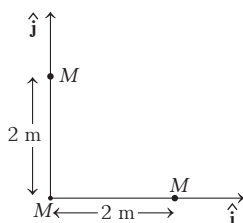
(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1** Two particles of masses 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of [NEET 2020]

(a) 50 cm (b) 67 cm
(c) 80 cm (d) 33 cm

- 2** Three identical spheres, each of mass M , are placed at the corners of a right angle triangle with the mutually perpendicular sides equal to 2 m (see figure). Taking the point of intersection of the two mutually perpendicular sides as the origin, find the position vector of centre of mass. [NEET 2020]



(a) $2(\hat{i} + \hat{j})$ (b) $(\hat{i} + \hat{j})$ (c) $\frac{2}{3}(\hat{i} + \hat{j})$ (d) $\frac{4}{3}(\hat{i} + \hat{j})$

- 3** Body A of mass $4m$ moving with speed u collides with another body B of mass $2m$ at rest. The collision is head on and elastic in nature. After the collision, the fraction of energy lost by the colliding body A is [NEET 2019]

(a) $\frac{8}{9}$ (b) $\frac{4}{9}$ (c) $\frac{5}{9}$ (d) $\frac{1}{9}$

- 4** An object flying in air with velocity $(20\hat{i} + 25\hat{j} - 12\hat{k})$ suddenly breaks in two pieces whose masses are in the ratio 1 : 5. The smaller mass flies off with a velocity $(100\hat{i} + 35\hat{j} + 8\hat{k})$. The velocity of the larger piece will be [NEET (Odisha) 2019]

(a) $4\hat{i} + 23\hat{j} - 16\hat{k}$ (b) $-100\hat{i} - 35\hat{j} - 8\hat{k}$
(c) $20\hat{i} + 15\hat{j} - 80\hat{k}$ (d) $-20\hat{i} - 15\hat{j} - 80\hat{k}$

- 5** A particle of mass $5m$ at rest suddenly breaks on its own into three fragments. Two fragments of mass m each move along mutually perpendicular directions each with speed v . The energy released during the process is [NEET (Odisha) 2019]

(a) $\frac{3}{5}mv^2$ (b) $\frac{5}{3}mv^2$ (c) $\frac{3}{2}mv^2$ (d) $\frac{4}{3}mv^2$

- 6** A body of mass 5×10^3 kg moving with speed 2 ms^{-1} collides with a body of mass 15×10^3 kg

inelastically and sticks to it. Then, loss in kinetic energy of the system will be [AIIMS 2019]

(a) 7.5 kJ (b) 15 kJ (c) 10 kJ (d) 5 kJ

- 7 Assertion** There is no loss in energy in elastic collision.

Reason Linear momentum is conserved in elastic collision. [AIIMS 2019]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.
(c) Assertion is correct, but Reason is incorrect.
(d) Both Assertion and Reason are incorrect.

- 8** One object of mass 20 kg is moving with speed 10 ms^{-1} in west direction and another object of mass 10 kg is moving with 15 ms^{-1} in north direction. Both collide and stick together. Choose the correct alternative. [JIPMER 2019]

- (a) Their kinetic energy is conserved as it is inelastic collision.
(b) Their kinetic energy is conserved as it is elastic collision.
(c) Their momentum is conserved as it is inelastic collision.
(d) Their momentum is conserved as it is elastic collision.

- 9** Two objects of mass m each moving with speed $u \text{ ms}^{-1}$ collide at 90° , then final momentum is (assume collision is inelastic) [JIPMER 2019]

(a) mu (b) $2mu$ (c) $\sqrt{2}mu$ (d) $2\sqrt{2}mu$

- 10** A moving block having mass m , collides with another stationary block having mass $4m$. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v , then the value of coefficient of restitution (e) will be [NEET 2018]

(a) 0.8 (b) 0.25 (c) 0.5 (d) 0.4

- 11** Three bodies having masses 5 kg, 4 kg and 2 kg are moving at the speeds of 5 ms^{-1} , 4 ms^{-1} and 2 ms^{-1} , respectively along X -axis. The magnitude of velocity of centre of mass is [AIIMS 2018]

(a) 1.0 ms^{-1} (b) 4 ms^{-1}
(c) 0.9 ms^{-1} (d) 1.3 ms^{-1}

- 12** Body of mass M is much heavier than the other body of mass m . The heavier body with speed v collides with the lighter body which was at rest initially elastically. The speed of lighter body after collision is [AIIMS 2018]

(a) $2v$ (b) $3v$ (c) v (d) $\frac{v}{2}$

- 13 Assertion** Two particles are moving in the same direction do not lose all their energy in completely inelastic collision.

Reason Principle of conservation of momentum holds true for all kinds of collisions. [AIIMS 2018]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Assertion is incorrect but Reason is correct.
- 14** A ball of 0.5 kg collided with wall at 30° and bounced back elastically. The speed of ball was 12ms^{-1} . The contact remained for 1s. What is the force applied by wall on ball? [JIPMER 2018]
- (a) $12\sqrt{3}\text{ N}$ (b) $\sqrt{3}\text{ N}$ (c) $6\sqrt{3}\text{ N}$ (d) $3\sqrt{3}\text{ N}$

- 15** A body of mass 4 kg moving with velocity 12 ms^{-1} collides with another body of mass 6 kg at rest. If two bodies stick together after collision, then the loss of kinetic energy of system is [AIIMS 2017]
- (a) zero (b) 288 J (c) 172.8 J (d) 144 J

- 16** Two masses of 6 and 2 unit, are at positions $(6\hat{i} - 7\hat{j})$ and $(2\hat{i} + 5\hat{j} - 8\hat{k})$, respectively. The coordinates of the centre of mass are [JIPMER 2017]
- (a) (2, -5, 3) (b) (5, -5, -3) (c) (5, -4, -2) (d) (5, -4, -4)

- 17** A block C of mass m is moving with velocity v_0 and collides elastically with block A of mass m and connected to another block B of mass $2m$ through spring of spring constant k . What is the value of k , if x_0 is compression of spring, when velocity of A and B is same? [JIPMER 2017]



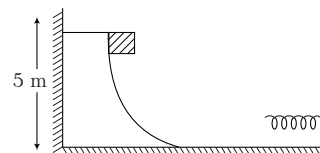
- (a) $\frac{mv_0^2}{x_0^2}$ (b) $\frac{mv_0^2}{2x_0^2}$ (c) $\frac{3}{2} \frac{mv_0^2}{x_0^2}$ (d) $\frac{2}{3} \frac{mv_0^2}{x_0^2}$
- 18.** Two particles of masses m_1 and m_2 move with initial velocities u_1 and u_2 . On collision, one of the particles get excited to higher level, after absorbing energy ϵ . If final velocities of particles be v_1 and v_2 , then we must have [CBSE AIPMT 2015]

- (a) $m_1^2 u_1 + m_2^2 u_2 - \epsilon = m_1^2 v_1 + m_2^2 v_2$
 (b) $\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \epsilon$
 (c) $\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 - \epsilon = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$
 (d) $\frac{1}{2} m_1^2 u_1^2 + \frac{1}{2} m_2^2 u_2^2 + \epsilon = \frac{1}{2} m_1^2 v_1^2 + \frac{1}{2} m_2^2 v_2^2$

- 19** Consider the situation as shown in figure. A spring of spring constant 400 Nm^{-1} is attached at one end

to a wedge fixed rigidly with the horizontal part. A 40 g mass is released from rest while situated at a height 5 m of the curved track. The minimum deformation in the spring is nearly equal to (Take, $g = 10\text{ ms}^{-2}$)

[AIIMS 2015]

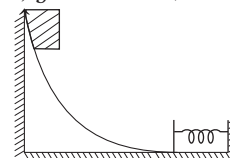


- (a) 9.8 m (b) 9.8 cm (c) 0.98 m (d) 0.009 km

- 20** A block having mass m collides with an another stationary block having mass $2m$. The lighter block comes to rest after collision. If the velocity of first block is v , then the value of coefficient of restitution will must be [AIIMS 2015]

- (a) 0.5 (b) 0.4 (c) 0.6 (d) 0.8

- 21** A smooth curved surface of height 10 m is ended horizontally. A spring of force constant 200 Nm^{-1} is fixed at the horizontal end as shown in figure. When an object of mass 10 g is released from the top, it travels along the curved path and collides with the spring. Then, the maximum compression in the spring is (Take, $g = 10\text{ ms}^{-2}$) [EAMCET 2015]



- (a) 10 m (b) 0.1 m (c) 1 m (d) 0.01 m

- 22** A frog sits on the end of a long board of length $L = 10\text{ cm}$. The board rests on a frictionless horizontal table. The frog wants to jump to the opposite end of the board. What is minimum take off speed v in ms^{-1} relative to the ground that the frog follows to do the trick? [Assume that, the board and frog have equal masses.] [UP CPMT 2015]

- (a) $2\sqrt{5}\text{ ms}^{-1}$ (b) 5 ms^{-1} (c) $5\sqrt{2}\text{ ms}^{-1}$ (d) $10\sqrt{2}\text{ ms}^{-1}$

- 23** A particle of mass m collides with another stationary particle of mass M . If the particle m stops just after collision, then the coefficient of restitution for collision is equal to [Manipal 2015]

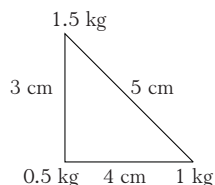
- (a) 1 (b) $\frac{m}{M}$ (c) $\frac{M-m}{M+m}$ (d) $\frac{m}{M+m}$

- 24** A body from height h is dropped, if the coefficient of restitution is e , then calculate the height achieved after one bounce. [Manipal 2015]

- (a) $h_1 = e^4 h$ (b) $h = e h_1$
 (c) $h_1 = e^2 h$ (d) $h = h_1 / e$

- 25** Three particles of masses 0.5 kg, 1 kg, 1.5 kg are placed at the three corners of a right angled triangle of sides 3 cm, 4 cm, 5 cm as shown in adjoining figure. What would be coordinates (x, y) of the centre of mass of system?

[UK PMT 2015, UP CPMT 2015]



- (a) (1.3, 1.5) (b) (2.3, 1.5)
(c) (1.3, 2.5) (d) (2.3, 2.5)
- 26** A large number of particles are placed around the origin, each at a distance R from the origin. The distance of the centre of mass of the system from the origin is [WB JEE 2015]
(a) equal to R (b) less than or equal to R
(c) greater than R (d) greater than or equal to R
- 27** A body of mass $4m$ is lying in xy -plane at rest. It suddenly explodes into three pieces. Two pieces each of mass m move perpendicular to each other with equal speeds v . The total kinetic energy generated due to explosion is [CBSE AIPMT 2014]
(a) mv^2 (b) $(3/2)mv^2$
(c) $2mv^2$ (d) $4mv^2$
- 28** The linear momentum of a particle varies with time t as $p = a + bt + ct^2$. Then, which of the following is correct? [EAMCET 2014]
(a) Velocity of particle is inversely proportional to time
(b) Displacement of the particle is independent of time
(c) Force varies with time in a quadratic manner
(d) Force is dependent linearly on time
- 29** The position of centre of mass of a system of particles does not depend upon the [Kerala CEE 2014]
(a) mass of particles
(b) symmetry of the body
(c) position of the particles
(d) nature of particles
(e) relative distance between the particles
- 30** An explosion breaks a rock into three parts in a horizontal plane. Two of them go off at right angles to each other. The first part of mass 1 kg moves with a speed of 12 ms^{-1} and the second part of mass 2 kg moves with speed of 8 ms^{-1} . If the third part flies off with speed of 4 ms^{-1} , then its mass is [NEET 2013]
(a) 3 kg (b) 5 kg
(c) 7 kg (d) 17 kg

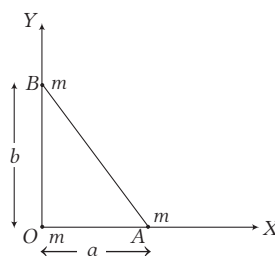
- 31** A gun fires a small bullet with kinetic energy K . Then, kinetic energy of the gun while recoiling is [KCET 2013]

(a) K (b) more than K
(c) less than K (d) $\sqrt{K}$

- 32** The linear momentum is conserved in [J&K CET 2013]

(a) elastic collisions (b) inelastic collisions
(c) Both (a) and (b) (d) Neither (a) nor (b)

- 33** Three particles, each of mass m are placed at the vertices of a right angled triangle as shown in figure. The position vector of the centre of mass of the system is (O is the origin and $\hat{i}$, $\hat{j}$, $\hat{k}$ are unit vectors) [EAMCET 2013]



- (a) $\frac{1}{3}(a\hat{i} - b\hat{j})$ (b) $\frac{2}{3}(a\hat{i} - b\hat{j})$
(c) $\frac{2}{3}(a\hat{i} + b\hat{j})$ (d) $\frac{1}{3}(a\hat{i} + b\hat{j})$

- 34** A ball of mass m moving with a horizontal velocity v strikes the bob of a pendulum at rest. Mass of the bob of the pendulum is also m . During this collision, the ball sticks with the bob of the pendulum. The height to which the combined mass rises will be (g = acceleration due to gravity) [EAMCET 2013]

(a) $\frac{v^2}{4g}$ (b) $\frac{v^2}{8g}$ (c) $\frac{v^2}{g}$ (d) $\frac{v^2}{2g}$

- 35** In an inelastic collision, [Kerala CEE 2013]

(a) momentum is not conserved
(b) momentum is conserved but kinetic energy is not conserved
(c) both momentum and kinetic energy are conserved
(d) neither momentum nor kinetic energy is conserved
(e) kinetic energy is conserved but not momentum

- 36** Two spheres A and B of masses m_1 and m_2 respectively collide. A is at rest initially and B is moving with velocity v along X -axis. After collision, B has a velocity $v/2$ in a direction perpendicular to the original direction. The mass A moves after collision in the direction [CBSE AIPMT 2012]

(a) same as that of B
(b) opposite to that of B
(c) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ to the X -axis
(d) $\theta = \tan^{-1}\left(\frac{-1}{2}\right)$ to the X -axis

- 37** Two persons of masses 55 kg and 65 kg respectively are at the opposite ends of a boat. The length of the boat is 3 m and weighs 100 kg. The 55 kg man walks up to the 65 kg man and sits with him. If the boat is in still water, the centre of mass of the system shifts by [CBSE AIPMT 2012]

(a) 3.0 m (b) 2.3 m
(c) zero (d) 0.75 m

- 38** A body of mass 0.25 kg is projected with muzzle velocity 100 ms^{-1} from a tank of mass 100 kg. What is the recoil velocity of the tank? [AIIMS 2012]

(a) 5 ms^{-1} (b) 25 ms^{-1}
(c) 0.5 ms^{-1} (d) 0.25 ms^{-1}

- 39** When a body of mass m_1 moving with uniform velocity 40 ms^{-1} collides with another body of mass m_2 at rest, then the two together begin to move with uniform velocity of 30 ms^{-1} . The ratio of the masses (i.e. m_1/m_2) of the two bodies will be [BCECE (Mains) 2012]

(a) 1 : 3 (b) 3 : 1
(c) 1 : 1.33 (d) 1 : 0.75

- 40** A ball moving with velocity 9 ms^{-1} collides with another similar stationary ball. If after the collision, both the balls move in directions making an angle of 30° with the initial direction, then their speeds after collision will be [BHU 2012]

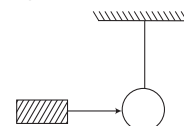
(a) 5.2 ms^{-1} (b) 0.52 ms^{-1}
(c) 52 ms^{-1} (d) 26 ms^{-1}

- 41** A body of mass $m_1 = 4 \text{ kg}$ moves at $5 \hat{i} \text{ ms}^{-1}$ and another body of mass $m_2 = 2 \text{ kg}$ moves at $10 \hat{i} \text{ ms}^{-1}$. The kinetic energy of centre of mass is [Manipal 2012]

(a) $\frac{200}{3} \text{ J}$ (b) $\frac{500}{3} \text{ J}$
(c) $\frac{400}{3} \text{ J}$ (d) $\frac{800}{3} \text{ J}$

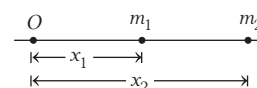
- 42** A mass of 10 g moving horizontally with a velocity of 100 cm s^{-1} strikes a pendulum bob of same mass. The two masses after collision stick together.

What will be the maximum height reached by the system now? (Take, $g = 10 \text{ ms}^{-2}$) [JCECE 2012]



(a) Zero (b) 1.25 cm (c) 2.5 cm (d) 5 cm

- 43** In the diagram shown below, m_1 and m_2 are the masses of two particles and x_1 and x_2 are their respective distances from the origin O . The centre of mass of the system is [J&K CET 2011]



(a) $\frac{m_1 x_2 + m_2 x_2}{m_1 + m_2}$ (b) $\frac{m_1 + m_2}{2}$
(c) $\frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$ (d) $\frac{m_1 m_2 + x_1 x_2}{m_1 + m_2}$

- 44** A bullet of mass m moving with velocity v strikes a suspended wooden block of mass M . If the block rises to a height h , then initial velocity of the block will be [Haryana PMT 2011]

(a) $\sqrt{2gh}$ (b) $\frac{M+m}{m} \sqrt{gh}$
(c) $\frac{m}{M+m} 2gh$ (d) $\frac{M+m}{M} \sqrt{2gh}$

- 45** A particle of mass m_1 moves with velocity v_1 and collides with another particle at rest of equal mass. The velocity of the second particle after the elastic collision is [DUMET 2011]

(a) $2v_1$ (b) v_1 (c) $-v_1$ (d) 0

ANSWERS

CHECK POINT 8.1

1. (a)	2. (c)	3. (d)	4. (b)	5. (b)	6. (c)	7. (d)	8. (b)	9. (a)	10. (b)
11. (d)	12. (c)	13. (d)	14. (d)	15. (d)	16. (d)				

CHECK POINT 8.2

1. (c)	2. (d)	3. (b)	4. (d)	5. (b)	6. (c)	7. (a)	8. (d)	9. (d)	10. (b)
11. (a)	12. (a)	13. (b)	14. (c)	15. (b)	16. (c)	17. (a)			

CHECK POINT 8.3

1. (a)	2. (c)	3. (b)	4. (c)	5. (d)	6. (c)	7. (a)	8. (d)	9. (c)	10. (b)
11. (d)	12. (b)	13. (b)	14. (a)	15. (b)					

(A) Taking it together

1. (c)	2. (d)	3. (d)	4. (b)	5. (b)	6. (b)	7. (c)	8. (d)	9. (a)	10. (a)
11. (d)	12. (a)	13. (a)	14. (d)	15. (c)	16. (c)	17. (a)	18. (d)	19. (a)	20. (b)
21. (a)	22. (d)	23. (d)	24. (b)	25. (a)	26. (a)	27. (b)	28. (b)	29. (b)	30. (a)
31. (d)	32. (a)	33. (d)	34. (c)	35. (b)	36. (a)	37. (a)	38. (b)	39. (a)	40. (c)
41. (b)	42. (d)	43. (b)	44. (a)	45. (b)	46. (b)	47. (b)	48. (d)	49. (a)	50. (b)
51. (a)	52. (b)	53. (a)							

(B) Medical entrance special format questions

• Assertion and reason

1. (d) 2. (d) 3. (d) 4. (a) 5. (b) 6. (c)

• Statement based questions

1. (b) 2. (d) 3. (d) 4. (d) 5. (d) 6. (a)

• Match the columns

1. (b) 2. (a) 3. (c)

(C) Medical entrances' gallery

1. (b)	2. (c)	3. (a)	4. (a)	5. (d)	6. (a)	7. (b)	8. (c)	9. (c)	10. (b)
11. (b)	12. (a)	13. (a)	14. (c)	15. (c)	16. (c)	17. (d)	18. (c)	19. (b)	20. (a)
21. (b)	22. (c)	23. (b)	24. (c)	25. (a)	26. (b)	27. (b)	28. (d)	29. (d)	30. (b)
31. (c)	32. (c)	33. (d)	34. (b)	35. (b)	36. (c)	37. (c)	38. (d)	39. (b)	40. (a)
41. (c)	42. (b)	43. (c)	44. (a)	45. (b)					

Hints & Explanations

CHECK POINT 8.1

- 1 (a) Let the coordinates of the centre of mass be (x, y) .

$$x = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{1 \times (-1) + 2 \times 2}{3} = \frac{-1 + 4}{3} = 1$$

$$\text{and } y = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \frac{1 \times 2 + 2 \times 4}{3} = \frac{2 + 8}{3} = \frac{10}{3}$$

Therefore, the coordinates of centre of mass be $\left(1, \frac{10}{3}\right)$.

- 2 (c) $r \propto \frac{1}{m}$

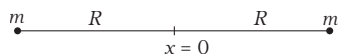
- 3 (d) Distance distributes in inverse ratio of masses.

$$\text{Hence, } \frac{r_C}{d - r_C} = \frac{m_O}{m_C} \Rightarrow \frac{r_C}{d - r_C} = \frac{16}{12}$$

$$\Rightarrow r_C = \frac{4}{7} \times d = \frac{4}{7} \times 1.2 \times 10^{-10} \\ = 0.69 \times 10^{-10} \text{ m}$$

- 4 (b) CM does not depend on the internal forces acting on the particles.

- 5 (b) For a single particle, distance of centre of mass from origin is R . For more than one particle, distance $\leq R$.



For example, for two particles of equal masses, kept as shown in figure, distance = 0.

- 6 (c) Given, $m_1 = 1.6 \text{ kg}$; $(x_1, y_1) = (0, 0)$

$$m_2 = 2 \text{ kg}; (x_2, y_2) = (1.2, 0)$$

$$m_3 = 2.4 \text{ kg}; (x_3, y_3) = (0, 1)$$

$\therefore$ Coordinates of centre of mass will be

$$x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3} \\ = \frac{(1.6)(0) + (2)(1.2) + (2.4)(0)}{1.6 + 2 + 2.4}$$

$$\Rightarrow x_{\text{CM}} = 0.4 \text{ m}$$

$$\text{and } y_{\text{CM}} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 + m_2 + m_3} \\ = \frac{(1.6)(0) + (2)(0) + (2.4)(1)}{1.6 + 2 + 2.4}$$

$$\Rightarrow y_{\text{CM}} = 0.4 \text{ m}$$

$\therefore$ Coordinates of centre of mass = $(0.4, 0.4) \text{ m}$

- 7 (d) Centre of mass of 1st system already lies at $(1, 2, 3)$.

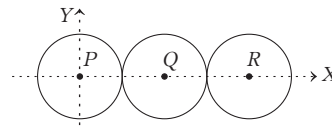
Therefore, centre of mass of 3 kg and 5 kg should lie at $(1, 2, 3)$.

$$\therefore \frac{3(-\hat{i} + 3\hat{j} - 2\hat{k}) + 5\mathbf{r}_5}{(3 + 5)} = (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\text{On solving, we get } \mathbf{r}_5 = \frac{11}{5}\hat{i} + \frac{7}{5}\hat{j} + 6\hat{k}$$

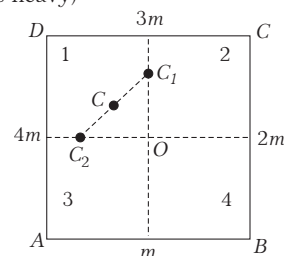
i.e. 5 kg mass should be kept at $(11/5, 7/5, 6)$

- 8 (b) X_{CM} (from P) = $\frac{m \times 0 + m \times PQ + m \times PR}{m + m + m}$ ($\because m = 1 \text{ kg}$)



$$\text{or } X_{\text{CM}} = \frac{PQ + PR}{3}$$

9. (a) $C_1 \rightarrow$ Position of centre of mass of rods AB and CD (nearer to CD, as it is heavy)

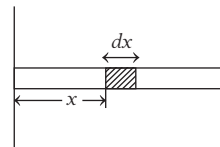


$C_2 \rightarrow$ Position of centre of mass of rods BC and DA.

$C \rightarrow$ Overall centre of mass of all four rods and it lies in region 1.

- 10 (b) Let the rod be along X-axis with origin at one of its ends.

As, the rod is along X-axis, so $y_{\text{CM}} = z_{\text{CM}} = 0$, i.e. centre of mass will be on the rod.



Now, consider an element of rod of length dx at a distance x from the origin.

Mass of element, $dm = \lambda dx = (A + Bx)dx$

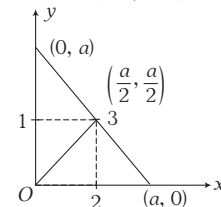
$$\therefore x_{\text{CM}} = \frac{\int_0^L x dm}{\int_0^L dm} = \frac{\int_0^L x(A + Bx) dx}{\int_0^L (A + Bx) dx} = \frac{\frac{AL^2}{2} + \frac{BL^3}{3}}{AL + \frac{BL^2}{2}} \\ = \frac{L(3A + 2BL)}{3(2A + BL)}$$

- 11 (d) Here, the coordinate of CM of inclined rod is $(a/2, a/2)$.

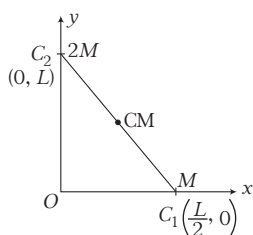
$$\therefore x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$= \frac{m \times 0 + (m) \left(\frac{a}{2}\right) + m \left(\frac{a}{2}\right)}{m + m + m} = \frac{a}{3}$$

$$\text{Similarly, } y_{\text{CM}} = \frac{a}{3}$$



- 12 (c) As rods are uniform, therefore centre of mass of both rods will be at their geometrical centres. The coordinates of CM of first rod, C_1 are $\left(\frac{L}{2}, 0\right)$ and second rod, C_2 are $(0, L)$.

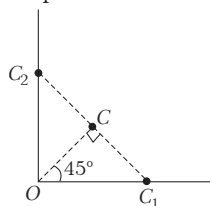


$$\therefore x_{CM} = \frac{M\left(\frac{L}{2}\right) + 2M(0)}{M + 2M} = \frac{L}{6}$$

$$y_{CM} = \frac{M(0) + 2M(L)}{M + 2M} = \frac{2L}{3}$$

Hence, coordinates of CM are $\left(\frac{L}{6}, \frac{2L}{3}\right)$.

- 13 (d) As, here, $OC_1 = \frac{1}{4} m$



$$\therefore OC = OC_1 \cos 45^\circ = \frac{1}{4\sqrt{2}} m$$

- 14 (d) Let centre of square is at origin.

$$x_{CM} = 0$$

$$\text{or } \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4} = 0$$

$$\text{or } (2m)(-a) + 4m(a) + m(a) + m_4(-a) = 0$$

$$\text{or } m_4 = 3m$$

Similarly, $y_{CM} = 0$

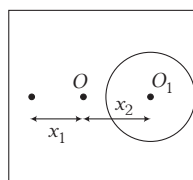
$$\text{or } (2m)(-a) + 4m(-a) + m(a) + m_4(a) = 0 \text{ or } m_4 = 5m$$

Since, value of m_4 is different to be satisfied by both $x_{CM} = 0$ and $y_{CM} = 0$.

Hence, it is not possible.

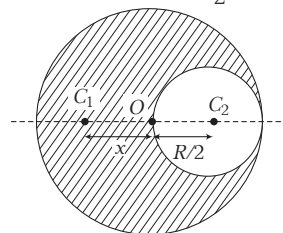
15. (d) $A_1x_1 = A_2x_2 \Rightarrow x_1 = \frac{A_2}{A_1} \cdot x_2 = \frac{\left(\frac{\pi}{4}\right)(8)^2}{(20)^2} \times 6$

$$= 0.75 \text{ cm from } O$$



- 16 (d) Centre of mass of complete disc should lie at point O. C_1 is the position of centre of mass of remaining portion and C_2 is the position of centre of mass of the removed disc.

$$\therefore x (\text{Area of remaining portion}) = \frac{R}{2} (\text{Area of removed disc})$$



$$\therefore x \left[\pi R^2 - \frac{\pi R^2}{4} \right] = \frac{R}{2} \left[\frac{\pi R^2}{4} \right]$$

$$\therefore x = \frac{R}{6}$$

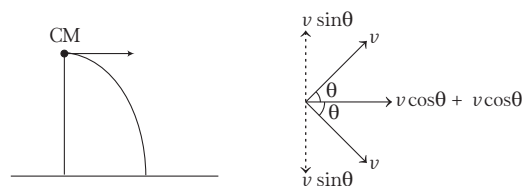
• CHECK POINT 8.2

- 1 (c) Centre of mass does not change its path during explosion. Therefore, it will keep on falling vertically and will not shift horizontally as, $F_x = 0$.

- 2 (d) Both the balls in air have acceleration g in downward direction. Hence, the acceleration of their centre of mass will also be g in downward direction.

- 3 (b) Acceleration of centre of mass, $a_{CM} = \frac{(m)(0) + (m)(a)}{m + m} = \frac{1}{2} a$

- 4 (d) Vertical component of velocity of CM is zero. Horizontal component of velocity of CM is non-zero. Acceleration of CM is g downwards. Hence, path of CM is a parabola as shown in figure.



- 5 (b) Net external force is zero. Hence, velocity of CM of the box and ball system will remain constant.

- 6 (c) Velocity of centre of mass, $v_{CM} = \frac{10(14) + 4(0)}{10 + 4} = 10 \text{ ms}^{-1}$

- 7 (a) Since, there is no external force on the rod in horizontal direction, the centre of mass of the rod will not move in horizontal direction.

In vertical direction, we have gravitational force as an external force, so CM of rod will come 0.5 m down, but will not move in x-direction. So, CM will be the origin.

Thus, option (a) is correct.

- 8 (d) $v_{CM} = \frac{m_1v_1 + m_2v_2}{m_1 + m_2} = 0$

As $m_1v_1 + m_2v_2 = 0$ is given. Hence, velocity of composite system will be zero.

- 9 (d) At $t = 0$, centre of mass is at mid-point or at $(2.25m, 0)$.
Velocity of centre of mass is zero. Hence, centre of mass will remain at this position all the time.

- 10 (b) After 1 s, coordinates of first particle will become $(4m, 4m, 6m)$ and coordinates of second particle will become $(6m, 4m, 8m)$.

$$\therefore X_{CM} = \frac{4 + 6}{2} = 5m$$

$$Y_{CM} = \frac{4 + 4}{2} = 4m$$

$$\text{and } Z_{CM} = \frac{6 + 8}{2} = 7m$$

- 11 (a) Centre of mass will not move along Y -axis.

$$\text{or } Y_{CM} = 0 \text{ (always)}$$

$$\therefore Y_{CM} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$0 = \frac{(m/4)15 + (3m/4)(y_2)}{(m/4 + 3m/4)}$$

$$\Rightarrow -\frac{15m}{4} = \frac{3m}{4}(y_2)$$

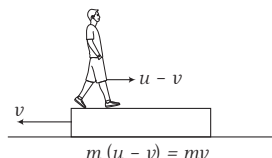
$$\Rightarrow y_2 = -5 \text{ cm}$$

- 12 (a) External force on system is zero, i.e. $p_i = p_f = 0$

$\therefore$ Centre of mass should be at rest at all instants.

- 13 (b) $m(u - v) = mv$ (using figure)

$$\therefore v = \frac{u}{2}$$



$$\therefore \text{Speed of man relative to ground} = u - v = \frac{u}{2}$$

14. (c) In horizontal direction, net force on the system is zero. Therefore, principle of conservation of momentum can be applied for system $(m + M)$ horizontally.

15. (b) $K_1 + K_2 = 2400$... (i)

$$\therefore \sqrt{2K_1 m_1} = \sqrt{2K_2 m_2}$$

$$\text{or } \frac{K_1}{K_2} = \frac{m_2}{m_1} = \frac{3}{1} \quad \dots \text{(ii)}$$

From Eqs. (i) and (ii), we get

$$\frac{K_1}{2400 - K_1} = \frac{3}{1}$$

$$\Rightarrow 7200 - 3K_1 = K_1$$

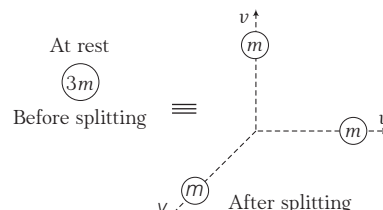
$$\Rightarrow 4K_1 = 7200$$

$$\Rightarrow K_1 = 1800$$

K_1 = Kinetic energy of smaller part = 1800 J

16. (c) Initial momentum of $3m$ mass = 0 ... (i)

Due to explosion, this mass splits into three fragments of equal masses.



Final momentum of system

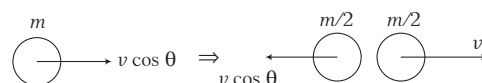
$$= mv + mv \hat{i} + mv \hat{j} \quad \dots \text{(ii)}$$

From law of conservation of linear momentum,

$$mv + mv \hat{i} + mv \hat{j} = 0$$

$$\Rightarrow \mathbf{v} = -v(\hat{i} + \hat{j})$$

- 17 (a) From conservation of linear momentum,



$$m(v \cos \theta) = \frac{m}{2}v' - \frac{m}{2}v \cos \theta$$

$$\therefore v' = 3v \cos \theta$$

• CHECK POINT 8.3

- 2 (c) Net force on ball and earth system is zero. Hence, total momentum of the ball and the earth is conserved.

- 3 (b) In elastic collision of two identical masses, velocities are interchanged after collision. Therefore, after collision, velocity of second body will be u .

- 5 (d) From law of conservation of momentum,

$$m_A v_A + m_B v_B = m_A v'_A + m_B v'_B$$

$$m_A(v_A - v_B) = m_B(v'_A - v'_B)$$

$$\Rightarrow \frac{m_A}{m_B} = 1$$

- 6 (c) In perfectly elastic collision between two bodies of equal masses, velocities are exchanged. So, after collision, particle A will move with 10 ms^{-1} and particle B with 15 ms^{-1} .

- 7 (a) Along x -direction, $m \times 50 - m \times 40 = m \times 0 + mv_x$

$$\Rightarrow v_x = 10 \text{ cm/s}$$

$$\text{Along } y\text{-direction, } m \times 0 + m \times 30 = m \times 0 + mv_y$$

$$\Rightarrow v_y = 30 \text{ cm/s}$$

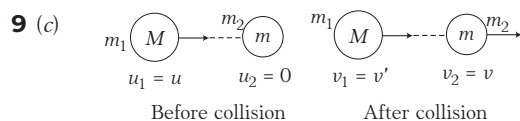
- 8 (d) From law of conservation of energy,

$$\frac{1}{2}mv^2 = \frac{1}{2} \times kx^2$$

$$\Rightarrow (0.5)(1.5)^2 = 50x^2$$

$$\Rightarrow x^2 = \frac{1.125}{50}$$

$$\Rightarrow x = 0.15 \text{ m}$$

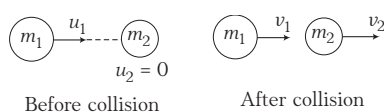


$$\therefore v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2 + \frac{2m_1 u_1}{m_1 + m_2}$$

$$\therefore v = \frac{2Mu}{M + m} \quad (\because u_2 = 0)$$

$$\Rightarrow v = \frac{2u}{1 + \frac{m}{M}}$$

10 (b) If target is at rest, then final velocity of bodies are



$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 \quad \dots(i)$$

$$\text{and} \quad v_2 = \frac{2m_1 u_1}{m_1 + m_2} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{v_1}{v_2} = \frac{m_1 - m_2}{2m_1} = \frac{2}{5} \Rightarrow \frac{m_1}{m_2} = 5$$

11. (d) $\therefore v'_2 = \left(\frac{m_2 - m_1}{m_2 + m_1} \right) v_2 + \left(\frac{2m_1}{m_1 + m_2} \right) v_1$

and $v_1 = 0$

$$\therefore \frac{v_2}{v'_2} = \left(\frac{m_2 + m_1}{m_2 - m_1} \right) = \left(\frac{m + 2m}{m - 2m} \right) = -3$$

$$\therefore \frac{K_2}{K'_2} = \left(\frac{v_2}{v'_2} \right)^2 = 9:1$$

12 (b) $e = \frac{\text{Relative velocity of separation}}{\text{Relative velocity of approach}} = \frac{5 - 2}{8 - 2} = \frac{1}{2}$

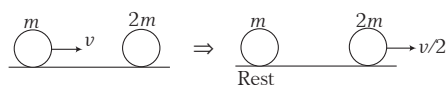
13 (b) Relative speed of approach is 12 ms^{-1} , i.e. $u_1 - u_2 = 12 \text{ ms}^{-1}$

$$e = \frac{1}{3} = \frac{v_2 - v_1}{u_1 - u_2}$$

$\Rightarrow v_2 - v_1 = \frac{12}{3} = 4 \text{ ms}^{-1}$

Therefore, relative speed of separation will be 4 ms^{-1} .

14 (a) From conservation of linear momentum, we can see that velocity of $2m$ will become $\frac{v}{2}$ after collision (as mass is double).



$$\text{Now, } e = \frac{\text{relative velocity of separation}}{\text{relative velocity of approach}} = \frac{v/2}{v} = \frac{1}{2}$$

15 (b) From conservation of linear momentum,

$$\begin{array}{c} (2) \xrightarrow{u} (1) \Rightarrow (2) \xrightarrow{v_2} (1) \xrightarrow{v_1} \end{array}$$

$$mu = mv_1 + mv_2 \quad \text{or} \quad u = v_1 + v_2 \quad \dots(i)$$

$$\text{From definition of } e, \quad v_1 - v_2 = eu \quad \dots(ii)$$

Solving these two equations, we get

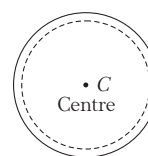
$$v_1 = \left(\frac{1+e}{2} \right) u \quad \text{and} \quad v_2 = \left(\frac{1-e}{2} \right) u$$

$$\therefore \frac{v_1}{v_2} = \left(\frac{1+e}{1-e} \right)$$

(A) Taking it together

1 (c) The centre of mass under the given condition may be at rest or may be moving with constant velocity, i.e. position of CM may be at rest or moving with constant velocity.

2 (d) A bangle is in the form of a ring as shown in the adjacent diagram. The centre of mass lies at the centre which is outside the body (boundary).



3. (d) We know that, for a system, $F_{\text{ext}} = \frac{dp}{dt}$

(i.e. from Newton's second law)

If $F_{\text{ext}} = 0$, $dp = 0 \Rightarrow p = \text{constant}$

Hence, momentum of a system will remain conserved, if external force on the system is zero.

In case of collision between particles, equal and opposite forces will act on individual particles as per Newton's third law.

Note We should not confuse with system and individual particles. As total force on the system of two particles is zero, but force acts on individual particles.

4 (b) As, $m_1 v_1 + m_2 v_2 = (m_1 + m_2) v$

$$\therefore a \cdot b + c \cdot 0 = (a + c) v \Rightarrow v = \frac{ab}{a + c}$$

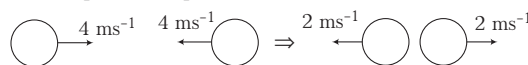
5 (b) At highest point, $\mathbf{p}_i = \mathbf{p}_f$ and perpendicular velocity = 0

$$\text{So, } (3m)(100\hat{i}) = m(100\hat{j}) - m(100\hat{j}) + m(\mathbf{v})$$

$$\therefore \mathbf{v} = 300\hat{i} \text{ (of third part)}$$

6 (b) Relative speed of approach is 8 ms^{-1} , $e = \frac{1}{2}$. Therefore,

relative speed of separation will be 4 ms^{-1} .



7 (c) $F = \text{rate of change of linear momentum}$

In 1 s, $\frac{n}{60}$ bullets are embedded. Momentum of each bullet is mv .

$$\therefore F = \left(\frac{n}{60} \right) mv$$

8 (d) $F = \frac{\Delta p}{\Delta t} = n(mv)$

Here, $n = \text{number of bullets fired per second}$.

$$\therefore n = \frac{F}{mv} = \frac{144}{0.04 \times 1200} = 3$$

- 9 (a) In elastic collision,

relative speed of separation = relative speed of approach = v

$$\therefore \text{Time of next collision} = \frac{2\pi r}{v}$$

- 10 (a) Velocity of block just after collision = $\sqrt{2gh} = \sqrt{2 \times 10 \times 0.2}$
 $= 2 \text{ ms}^{-1}$

Now, applying conservation of linear momentum just before and just after collision,

$$0.02 \times 600 = 4 \times 2 + 0.02 \times v$$

$$\therefore v = 200 \text{ ms}^{-1}$$

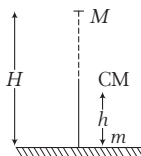
- 11 (d) From conservation of linear momentum, velocity of combined mass just after collision will be 50 cms^{-1} , as mass has doubled.

$$\text{Now, } H = \frac{u^2}{2g} = \frac{(0.5)^2}{20} \text{ m} = 1.25 \text{ cm}$$

- 12 (a) Centre of mass will remain at height h .

$$\therefore h_{\text{CM}} = \frac{m \times 0 + MH}{m + M} = h$$

$$\therefore H = h \left(1 + \frac{m}{M} \right)$$



- 13 (a) During explosion of a cracker, path of centre of mass does not change. At height 1 m,

$$v = \sqrt{u^2 - 2gh} \quad (\text{from conservation of mechanical energy})$$

$$= 4\sqrt{5} \text{ ms}^{-1}$$

- 14 (d) $y_{\text{CM}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$ or $+1 = \frac{(10)(7) + (30)y_2}{10 + 30}$

$$\therefore y_2 = -1 \text{ cm}$$

- 15 (c) Here, $u = -v = 126 \text{ kmh}^{-1} = 126 \times \frac{5}{18} = 35 \text{ m/s}$

Change in momentum of the ball,

$$\Delta p = m(v - u) = \frac{150}{1000} (-35 - 35)$$

$$= \frac{3}{20} (-70) = -\frac{21}{2} \text{ kg} \cdot \text{ms}^{-1}$$

$$\text{Now, force, } F = \frac{\Delta p}{\Delta t} = \frac{-21/2}{0.001} \text{ N} = -1.05 \times 10^4 \text{ N}$$

Here, negative sign shows that direction of force will be opposite to the direction of movement of the ball before hitting.

- 16 (c) The position of centre of the system shown in the given figure is likely to be at C.

This is because lower part of the sphere containing sand is heavier than upper part of the sphere containing air.

- 17 (a) $h_n = he^{2n} = 32 \left(\frac{1}{2} \right)^4 = \frac{32}{16} = 2 \text{ m}$ (here, $n = 2$, $e = 1/2$)

- 18 (d) In 1 cm^2 area, 10^4 balls are striking per second. Therefore, in 1 m^2 area, 10^8 balls will strike per second.

Change in momentum of each ball per second will be $2mu$.

$$\therefore \text{Pressure} = \frac{F}{A} = N \frac{(\Delta p / \Delta t)}{A} = n \times 2m \times u \quad [\because n = \frac{N}{A \Delta t} = 10^8]$$

$$= 10^8 \times 2 \times 10^{-3} \times 100$$

$$= 2 \times 10^7 \text{ Nm}^{-2}$$

- 19 (a) Velocity of particle after 5 s,

$$v = u - gt = 100 - 10 \times 5$$

$$= 100 - 50 = 50 \text{ ms}^{-1} \quad (\text{upwards})$$

Conservation of linear momentum gives

$$Mv = m_1 v_1 + m_2 v_2 \quad \dots(i)$$

Taking upward direction positive,

$$v_1 = -25 \text{ ms}^{-1}, v = 50 \text{ ms}^{-1}$$

$$M = 1 \text{ kg}, m_1 = 400 \text{ g} = 0.4 \text{ kg}$$

$$m_2 = M - m_1 = 1 - 0.4 = 0.6 \text{ kg}$$

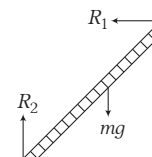
From Eq. (i), we get

$$1 \times 50 = 0.4 \times (-25) + 0.6 v_2 \quad \text{or } v_2 = 100 \text{ ms}^{-1} (\text{upwards})$$

- 20 (b) Area of circle, $A_1 = \frac{\pi}{4} a^2$ and area of square $A_2 = a^2$. Since,

$A_2 > A_1$, so centre of mass will lie inside the square plate.

- 21 (a) Due to net force in downward direction and towards left, centre of mass will move downward and will shift left horizontally.



Hence, the correct trace of centre of mass is shown in option (a).

- 22 (d) $a_{\text{CM}} = \frac{\text{External force}}{\text{Total mass}} = \frac{\text{Force of friction from ground}}{\text{Total mass}}$
 $= \frac{0.2 \times (2+1)(10)}{1+2} = 2 \text{ ms}^{-2}$

- 23 (d) Distance between bullet and block at this instant is $(D - d)$.

Distance of CM from bullet

$$= \frac{M(D - d) + m \times 0}{M + m} = \frac{M}{M + m} (D - d)$$

$$\text{Similarly, distance of CM from block} = \frac{m}{M + m} (D - d)$$

Hence, option (d) is correct.

- 24 (b) Let plank moves x distance in opposite direction. Then, displacement of man relative to ground will be $(L - x)$.

Applying law of conservation of momentum,

$$m_R \frac{x_R}{t} = m_L \frac{x_L}{t} \quad \text{or } M(L - x) = \frac{M}{3} x$$

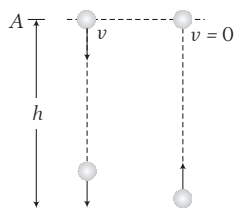
$$\text{Solving this equation, we get } x = \frac{3L}{4}$$

$$\therefore \text{Displacement of man relative to ground} = L - \frac{3L}{4} = \frac{L}{4}$$

Taking D as origin, $y_{CM} = \frac{m \times 0 + 2ml}{m + 2m} = \frac{2l}{3}$

From C , $CP = 2l - \frac{2l}{3} = \frac{4l}{3}$

- 37 (a)** Let the ball be projected vertically downward with velocity v from height h .



Total energy at point $A = \frac{1}{2}mv^2 + mgh$

During collision, loss of energy is 50% and the ball rises up to same height. This means it possesses only potential energy at same level.

$$50\% \left(\frac{1}{2}mv^2 + mgh \right) = mgh$$

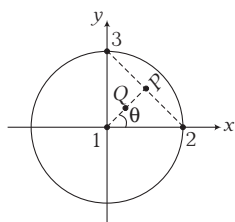
$$\frac{1}{2} \left(\frac{1}{2}mv^2 + mgh \right) = mgh \Rightarrow v = \sqrt{2gh} = \sqrt{2 \times 10 \times 20}$$

$$\therefore v = 20 \text{ ms}^{-1}$$

- 38 (b)** The centre of mass of the object must lie on the line segment joining $(0, 0)$ and $(R/\sqrt{2}, R/\sqrt{2})$. Here, $(0, 0)$ is the centre of mass of the ring and $(R/\sqrt{2}, R/\sqrt{2})$ is the centre of mass of the chord.

Hence, here option (b) cannot be the coordinate of CM of the system.

- 39 (a)** P is the position of centre of mass of particles at 2 and 3. Q is position of centre of mass of all three particles.



$$\begin{aligned} \tan \theta &= \frac{y_{CM}}{x_{CM}} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{m_1 x_1 + m_2 x_2 + m_3 x_3} \\ &= \frac{6 \times 0 + 2 \times 0 + 2 \times a}{6 \times 0 + 2 \times a + 2 \times 0} = 1 \text{ or } \theta = 45^\circ \end{aligned}$$

- 40 (c)** Velocity of A just before collision
 $= \sqrt{2gh} = \sqrt{2 \times 10 \times 1.25} = 5 \text{ ms}^{-1}$

Velocity of $(A+B)$ just after collision $= \frac{5}{2} = 2.5 \text{ ms}^{-1}$

In elastic collision between two bodies of equal masses, velocities are interchanged.

Hence, velocity of C will become 2.5 ms^{-1} .

- 41 (b)** We know that, $v_r = \frac{l}{t}$

Now, $m(v_r - v) = Mv$ (v = speed of plank)

$$\therefore v = \frac{mv_r}{M+m} = \frac{l}{t} \left(\frac{m}{M+m} \right)$$

- 42 (d)** Here, $2\pi R = 2\pi \Rightarrow R = 1$

$$\therefore y_{CM} = \frac{m \times 0 + m \times 1 + m \times 1}{m + m + m} = \frac{2}{3}$$

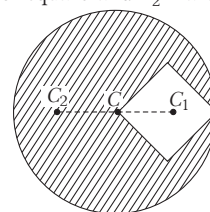
$$\text{and } x_{CM} = \frac{m(\pi) + m(0) + m(2\pi)}{m + m + m} = \pi$$

- 43 (b)** From conservation of linear momentum,

$$m_1 v_1 = m \sqrt{5gl} + m_1 \frac{v_1}{3} \text{ or } v_1 = \frac{3}{2} \frac{m}{m_1} \sqrt{5gl}$$

- 44 (a)** Here, $A_1(CC_1) = A_2(CC_2)$

where, A = area of square and A_2 = area of remaining portion.



Side of square will be $\frac{R}{\sqrt{2}}$.

$$\therefore CC_2 = \frac{A_1}{A_2} (CC_1) = \frac{(R/\sqrt{2})^2}{\pi R^2 - (R/\sqrt{2})^2} \left(\frac{R}{2} \right) = \left(\frac{R}{4\pi - 2} \right)$$

- 45 (b)** Centre of mass of remaining portion was at point O_2 .

Hence, x_2 (area of remaining portion) = c (area of removed disc)

$$\therefore x_2(\pi a^2 - \pi b^2) = c(\pi b^2) \Rightarrow x_2 = \frac{cb^2}{a^2 - b^2}$$

- 46 (b)** $T = \frac{d}{v/\sqrt{2}} + \frac{d}{ev/\sqrt{2}} = \left(1 + \frac{1}{e} \right) \frac{\sqrt{2}d}{v}$

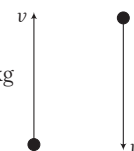
$$\text{or } \frac{2v/\sqrt{2}}{g} = \left(1 + \frac{1}{e} \right) \frac{\sqrt{2}d}{v} \text{ or } e = \frac{gd}{v^2 - gd}$$

- 47 (b)** $F = \frac{\Delta p}{\Delta t} = \frac{\Delta(mn v)}{\Delta t}$

Here, m = mass of one marble $= 5 \text{ g} = 5 \times 10^{-3} \text{ kg}$

$\frac{n}{\Delta t}$ = number of molecules striking per second
 $= 10$

$$\therefore Mg = m \left(\frac{n}{\Delta t} \right) |\Delta v|$$



$$(\because |\Delta v| = 2v)$$

Here, M = mass of disc

$$(10 \times 10^{-3}) (9.8) = (5 \times 10^{-3}) (10) 2v$$

$$\text{or } v = 0.98 \text{ ms}^{-1}$$

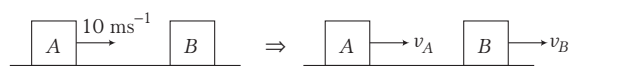
48 (d) $X_{CM} = \frac{mx + 2mx_{2m}}{m + 2m}$

$$\therefore \frac{1}{2} a_{CM} t^2 = \frac{x + 2x_{2m}}{3}$$

$$\therefore \frac{3}{2} \left(\frac{F}{3m} \right) t^2 = x + 2x_{2m} \quad [\because F = Ma_{CM}]$$

$$\therefore x_{2m} = \frac{Ft^2}{4m} - \frac{x}{2}$$

49 (a) For collision between A and B,



$$m \times 10 = mv_A + mv_B \Rightarrow v_B + v_A = 10 \quad \dots(i)$$

$$e = \frac{1}{2} = \frac{v_B - v_A}{10} \text{ or } v_B - v_A = 5 \quad \dots(ii)$$

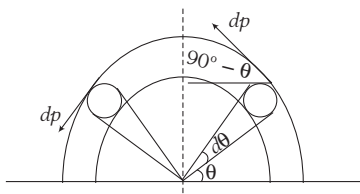
Solving Eqs. (i) and (ii), we get

$$v_B = 7.5 \text{ ms}^{-1}$$

Hence, A has given 75% of its speed to B and B will also transfer its 75% speed to C.

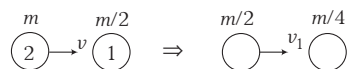
$$\therefore v_C = \frac{75}{100} \times 7.5 = 5.625 \text{ ms}^{-1} \approx 5.6 \text{ ms}^{-1}$$

50 (b) $p_{\text{net}} = \int dp \sin \theta = \int_0^\pi \left(\frac{M}{\pi} \cdot d\theta \right) v \cdot \sin \theta = \frac{2Mv}{\pi}$



51 (a) In head on elastic collision,

$$v_1 = \left(\frac{m/2 - m}{m/2 + m} \right) (0) + \left[\frac{2(m)}{m/2 + m} \right] (v) = \frac{4}{3} v$$

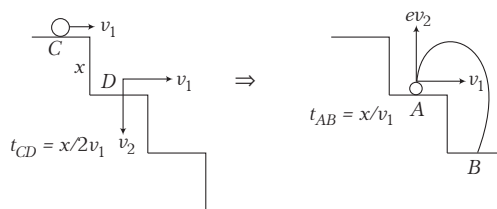
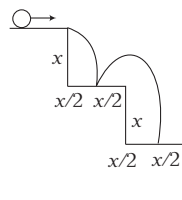


$$\frac{m}{2} \rightarrow v \quad \frac{m}{4} \rightarrow v_1$$

Finally, $v_n = \left(\frac{4}{3} \right)^{n-1} \cdot v = \sqrt{5gr}$

$$\therefore v = \left(\frac{3}{4} \right)^{n-1} \sqrt{5gr}$$

52 (b) The given situation is shown below



From C to D, $v_2 = \sqrt{2gx} = gt \Rightarrow t = \frac{v_2}{g}$

From A to B, time will become two times.

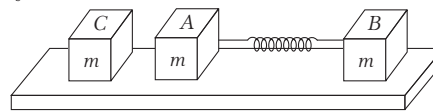
Applying $s = ut + \frac{1}{2} at^2$ in vertical direction, we have

$$-x = (ev_2)(2t) - \frac{1}{2} \times g \times (2t)^2$$

$$-x = \frac{2ev_2^2}{g} - \frac{2v_2^2}{g} \Rightarrow -x = 2e(2x) - 2(2x)$$

$$\therefore e = \frac{3}{4}$$

53 (a) After striking with A, the block C comes to rest and block A moves with velocity v . When compression in spring is maximum, both A and B will be moving with common velocity v .



From law of conservation of linear momentum,

$$mv = (m + m)V \Rightarrow V = \frac{v}{2} \quad \dots(i)$$

From law of conservation of energy,

KE of block C = KE of system + PE of system

$$\frac{1}{2} mv^2 = \frac{1}{2} (2m) V^2 + \frac{1}{2} kx^2$$

$$\Rightarrow \frac{1}{2} mv^2 = \frac{1}{2} (2m) \left(\frac{v}{2} \right)^2 + \frac{1}{2} kx^2 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow kx^2 = \frac{1}{2} mv^2 \Rightarrow x = v \sqrt{\frac{m}{2k}}$$

(B) Medical entrance special format questions

• Assertion and reason

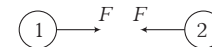
1 (d) $e = \frac{|\text{RVOS, i.e. relative velocity of separation}|}{|\text{RVOA, i.e. relative velocity of approach}|}$

In elastic collision, $e = 1$

$$\therefore |\text{RVOS}| = |\text{RVOA}|$$

2 (d) If two bodies are released from rest in space, net force on the system is zero. Momentum of system is constant but momentum of individual body is not constant. Further, kinetic energy of system is also increasing.

3 (d) Only in case of perfectly inelastic collision, they will come to rest.



- 4 (a) Linear momentum will not remain constant till spring will remain compressed. Therefore, a force will act on block A from the wall.

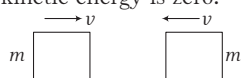
5 (b) $v_{CM} = \frac{m_A(+v) + m_B(-v)}{m_A + m_B} = -ve$

Both A and B will have same acceleration (μg) on A towards left and on B towards right. Since, B have more mass, so acceleration of CM will be towards right.

- 6 (c) In this case, centre of mass of half filled sphere will depend only on radius and not on density of liquid inside. Since, both spheres are of same radius, so both will have CM at the same level.

• Statement based questions

- 1 (b) Packet from train A falls with greater momentum on train B. Therefore, train B is slightly accelerated while A will be retarded.
- 2 (d) Force of friction on A is backward and force of friction on B is forward. Net external force on the system is zero. Hence, momentum of system will remain conserved. As the momentum of system is conserved, so increase in momentum of B is equal to decrease in momentum of A.
- 3 (d) Due to the same mass of A and B as well as due to elastic collision, velocities of spheres get interchanged after the collision. So, A comes to rest and B moves with the velocity of A.
- 4 (d) If centre of mass is at rest, it definitely means momentum of the system is constant. But if momentum of the system is constant, it does not mean centre of mass is at rest.
5. (d) If kinetic energy of the system is zero, it definitely means momentum is zero. But if momentum of the system is zero, it does not mean kinetic energy is zero.



- 6 (a) Statement I is correct and Statement II is incorrect and it can be corrected as,

During collision time, some kinetic energy is stored as potential energy in the form of deformation.

• Match the columns

- 1 (b) When $e = 1$, collision is elastic and equal masses exchange their velocities.

For $e = 0$, collision is perfectly inelastic. Hence, velocity of each will remain half.

In the last case, when $v_2 = v' = \frac{3}{4}v$.

Then, $v_1 = \frac{v}{4}$ (from conservation of momentum)

$$\therefore e = \frac{\frac{3v}{4} - \frac{v}{4}}{v} = \frac{1}{2}$$

Hence, A $\rightarrow$ q, B $\rightarrow$ p, C $\rightarrow$ r.

- 2 (a) $\mathbf{F}_{CM} = \mathbf{F}_1 + \mathbf{F}_2 = m_1\mathbf{a}_1 + m_2\mathbf{a}_2 = (2\hat{i} + 8\hat{j})$

$$\therefore |\mathbf{F}_{CM}| = \sqrt{4 + 64} \Rightarrow |\mathbf{F}_{CM}| = \sqrt{68} \text{ unit}$$

$$\mathbf{v}_{CM} = \frac{m_1\mathbf{v}_1 + m_2\mathbf{v}_2}{m_1 + m_2} = \frac{(1)(4\hat{i}) + (2)(4\hat{j})}{3} = \frac{4\hat{i} + 8\hat{j}}{3}$$

$$\therefore |\mathbf{v}_{CM}| = \frac{1}{3} \sqrt{16 + 64} = \frac{\sqrt{80}}{3} \text{ unit}$$

$$\mathbf{s}_1 = \int_0^2 \mathbf{v}_1 dt = (4\hat{i})$$

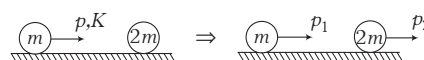
$$\mathbf{s}_2 = \int_0^2 \mathbf{v}_2 dt = \left(\frac{8}{3}\hat{j}\right)$$

$$\text{Now, } \mathbf{s}_{CM} = \frac{m_1\mathbf{s}_1 + m_2\mathbf{s}_2}{m_1 + m_2} = \frac{(1)(4\hat{i}) + 2\left(\frac{8}{3}\hat{j}\right)}{3} = \left(\frac{4}{3}\hat{i} + \frac{16}{9}\hat{j}\right)$$

$$\therefore |\mathbf{s}_{CM}| = \sqrt{\frac{16}{9} + \frac{256}{81}} = \frac{20}{9} \text{ unit}$$

Hence, A $\rightarrow$ q, B $\rightarrow$ r, C $\rightarrow$ p.

- 3 (c) $p_1 + p_2 = p$... (i)



Further, $K_1 + K_2 = K$

$$\text{or } \frac{p_1^2}{2m} + \frac{p_2^2}{4m} = \frac{p^2}{2m} \text{ or } 2p_1^2 + p_2^2 = 2p^2 \quad \dots (ii)$$

Solving Eqs. (i) and (ii), we get

$$p_2 = \frac{4}{3}p \quad \text{and} \quad p_1 = -\frac{p}{3}, K_1 = \frac{K}{9} \quad \text{and} \quad K_2 = \frac{8K}{9}$$

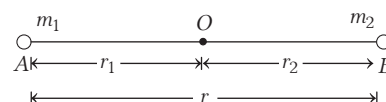
Hence, A $\rightarrow$ r, B $\rightarrow$ p, C $\rightarrow$ q, D $\rightarrow$ s.

(C) Medical entrances' gallery

- 1 (b) Given, $m_1 = 5 \text{ kg}$, $m_2 = 10 \text{ kg}$

and $r = 1 \text{ m} = 100 \text{ cm}$

Let the centre of mass lies at origin O.

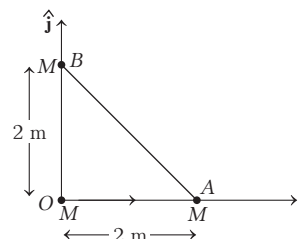


$$\therefore \frac{m_1 r_1 - m_2 r_2}{m_1 + m_2} = 0 \Rightarrow 5r_1 - 10r_2 = 0 \Rightarrow r_2 = \frac{r_1}{2}$$

$$\text{Also, } r_1 + r_2 = 100 \Rightarrow r_1 + \frac{r_1}{2} = 100$$

$$\Rightarrow 3r_1 = 200 \Rightarrow r_1 = \frac{200}{3} \approx 67 \text{ cm}$$

- 2 (c) The given situation as shown in the figure.



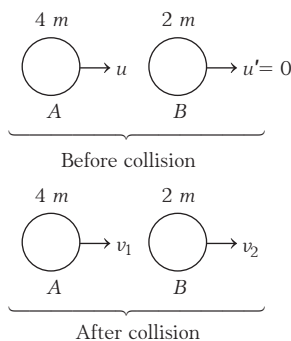
$$\mathbf{OA} = 2\hat{i}$$

$$\mathbf{OB} = 2\hat{j}$$

Position vector of centre of mass,

$$\begin{aligned} \mathbf{R}_{\text{CM}} &= \frac{M_1 \mathbf{r}_1 + M_2 \mathbf{r}_2 + M_3 \mathbf{r}_3}{M_1 + M_2 + M_3} = \frac{M(\mathbf{OA}) + M(\mathbf{OB})}{M + M + M} \\ &= \frac{M \times 2\hat{\mathbf{i}} + M \times 2\hat{\mathbf{j}}}{3M} = \frac{2}{3}(\hat{\mathbf{i}} + \hat{\mathbf{j}}) \end{aligned}$$

3 (a) The given situation of collision can be drawn as



Applying law of conservation of linear momentum,

Initial momentum of system = Final momentum of system

$$\Rightarrow (4m)u + (2m)u' = (4m)v_1 + (2m)v_2$$

$$4mu + (2m) \times 0 = 4mv_1 + 2mv_2$$

or

$$2u = 2v_1 + v_2 \quad \dots (i)$$

The kinetic energy of A before collision,

$$(\text{KE})_A = \frac{1}{2}(4m)u^2 = 2mu^2$$

Kinetic energy of B before collision,

$$(\text{KE})_B = 0$$

The kinetic energy of A after collision,

$$(\text{KE}')_A = \frac{1}{2}(4m)v_1^2 = 2mv_1^2$$

Kinetic energy of B after collision,

$$(\text{KE}')_B = \frac{1}{2}(2m)v_2^2 = mv_2^2$$

As, initial kinetic energy of the system = final kinetic energy of the system

$$\Rightarrow (\text{KE})_A + (\text{KE})_B = (\text{KE}')_A + (\text{KE}')_B$$

$$2mu^2 + 0 = 2mv_1^2 + mv_2^2$$

$$2mu^2 = 2mv_1^2 + mv_2^2$$

or

$$2u^2 = 2v_1^2 + v_2^2 \quad \dots (ii)$$

Solving Eqs. (i) and (ii), we get

$$v_1 = \frac{1}{3}u \quad \text{and} \quad v_2 = \frac{4}{3}u$$

or the final velocity of A can be directly calculated by using the formula,

$$\begin{aligned} v_1 &= \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \frac{2m_2 u_2}{m_1 + m_2} \\ &= \left(\frac{4m - 2m}{4m + 2m} \right) u + \frac{2(2m) \times 0}{(4m + 2m)} \quad (\because u_2 = u' = 0) \\ &= \frac{2m}{6m} u = \frac{1}{3}u \end{aligned}$$

\(\therefore\) Net decrease in kinetic energy of A,

$$\begin{aligned} \Delta \text{KE} &= (\text{KE})_A - (\text{KE}')_A = 2mu^2 - 2mv_1^2 \\ &= 2m(u^2 - v_1^2) \end{aligned}$$

Substituting the value of v_1 , we get

$$\Delta \text{KE} = 2m \left(u^2 - \frac{u^2}{9} \right) = \frac{16mu^2}{9}$$

\(\therefore\) The fractional decrease in kinetic energy,

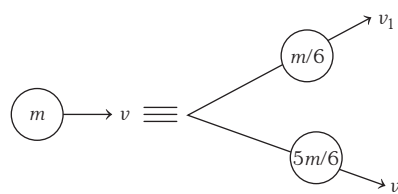
$$\frac{\Delta \text{KE}}{(\text{KE})_A} = \frac{16mu^2}{9} \times \frac{1}{2mu^2} = \frac{8}{9}$$

4 (a) Let m be the mass of an object flying with velocity v in air.

When it gets split into two pieces of masses in ratio 1 : 5, the

mass of smaller piece is $m/6$ and of bigger piece is $\frac{5m}{6}$.

This situation can be interpreted diagrammatically as below.



As, the object breaks in two pieces, so the momentum of the system will remain conserved, i.e. the total momentum (before breaking) = total momentum (after breaking)

$$m\mathbf{v} = \frac{m}{6}\mathbf{v}_1 + \frac{5m}{6}\mathbf{v}_2 \Rightarrow \mathbf{v} = \frac{\mathbf{v}_1}{6} + \frac{5\mathbf{v}_2}{6} \quad \dots (i)$$

Given, $\mathbf{v} = 20\hat{\mathbf{i}} + 25\hat{\mathbf{j}} - 12\hat{\mathbf{k}}$

and $\mathbf{v}_1 = 100\hat{\mathbf{i}} + 35\hat{\mathbf{j}} + 8\hat{\mathbf{k}}$

Putting these values in Eq. (i), we get

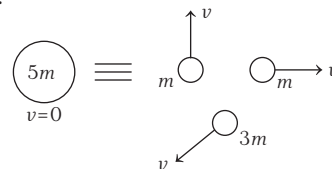
$$(20\hat{\mathbf{i}} + 25\hat{\mathbf{j}} - 12\hat{\mathbf{k}}) = \frac{(100\hat{\mathbf{i}} + 35\hat{\mathbf{j}} + 8\hat{\mathbf{k}})}{6} + \frac{5\mathbf{v}_2}{6}$$

$$\Rightarrow (120\hat{\mathbf{i}} + 150\hat{\mathbf{j}} - 72\hat{\mathbf{k}}) = (100\hat{\mathbf{i}} + 35\hat{\mathbf{j}} + 8\hat{\mathbf{k}}) + 5\mathbf{v}_2$$

$$\Rightarrow \mathbf{v}_2 = \frac{1}{5}(20\hat{\mathbf{i}} + 115\hat{\mathbf{j}} - 80\hat{\mathbf{k}})$$

$$= 4\hat{\mathbf{i}} + 23\hat{\mathbf{j}} - 16\hat{\mathbf{k}}$$

5 (d) The particle of mass $5m$ breaks into three fragments of masses m , m and $3m$, respectively. Two fragments of mass m each move in perpendicular directions with velocity v and the left fragment will move in a direction with velocity $\mathbf{v}'$ such that the total momentum of the system must remain conserved.



By law of conservation of momentum,

$$5m \times 0 = mv\hat{\mathbf{i}} + mv\hat{\mathbf{j}} + 3m\mathbf{v}'$$

$$\Rightarrow \mathbf{v}' = -\frac{v}{3}\hat{\mathbf{i}} - \frac{v}{3}\hat{\mathbf{j}}$$

$$\therefore |\mathbf{v}'| = \sqrt{\left(-\frac{v}{3}\right)^2 + \left(-\frac{v}{3}\right)^2} = \frac{v\sqrt{2}}{3}$$

$\therefore$ Energy released,

$$\begin{aligned} E &= \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2} \times 3m \left(\frac{v\sqrt{2}}{3}\right)^2 \\ &= mv^2 + \frac{mv^2}{3} = \frac{4}{3}mv^2 \end{aligned}$$

- 6** (a) Given, mass of body, $m_1 = 5 \times 10^3$ kg

Velocity, $v_1 = 2 \text{ ms}^{-1}$

and mass of another body, $m_2 = 15 \times 10^3$ kg

For perfectly inelastic collision, $e = 0$.

$\therefore$ Loss in kinetic energy of system,

$$\begin{aligned} \Delta E_K &= \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} \times v_1^2 \\ &= \frac{1}{2} \times \frac{5 \times 10^3 \times 15 \times 10^3}{5 \times 10^3 + 15 \times 10^3} \times (2)^2 \\ &= 7.5 \times 10^3 \text{ J} \\ &= 7.5 \text{ kJ} \end{aligned}$$

- 7** (b) In elastic collision, total energy, kinetic energy and momentum remain conserved, therefore no loss in energy occurs in elastic collision.

Hence, both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

- 8** (c) When object of mass 20 kg moving with speed 10 ms^{-1} in west direction collides with object of mass 10 kg and both of them stick together, hence it is perfectly inelastic collision. In inelastic collision, only momentum is conserved.

- 9** (c) Speed of objects $= u \text{ ms}^{-1}$

Since, both objects collide at 90° .

Hence, by the law of conservation of momentum,

Total momentum before collision

$$\begin{aligned} &= \text{Total momentum after collision} \\ |mu\hat{i} + mu\hat{j}| &= p_f \\ \sqrt{m^2u^2 + m^2u^2} &= p_f \\ \Rightarrow p_f &= \sqrt{2} mu \end{aligned}$$

- 10** (b) Since, the collision mentioned is an elastic head on collision. Thus, according to the law of conservation of linear momentum, we have

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

where, m_1 and m_2 are the masses of the two blocks respectively, u_1 and u_2 are their initial velocities and v_1 and v_2 are their final velocities, respectively.

Given, $m_1 = m$, $m_2 = 4m$

$$u_1 = v, u_2 = 0 \text{ and } v_1 = 0$$

$$\therefore mv + 4m \times 0 = 0 + 4mv_2$$

$$\Rightarrow mv = 4mv_2 \text{ or } v_2 = \frac{v}{4} \quad \dots(i)$$

Now, the coefficient of restitution,

$$e = \frac{\text{relative velocity of separation}}{\text{relative velocity of approach}}$$

$$= -\frac{v_2 - v_1}{u_2 - u_1} = -\frac{\frac{v}{4} - 0}{0 - v} \quad [\text{from Eq. (i)}]$$

$$= 1/4$$

$$\therefore e = 0.25$$

- 11** (b) Velocity of centre of mass,

$$v_{CM} = \frac{mv_1 + m_2v_2 + m_3v_3}{m_1 + m_2 + m_3}$$

Given, $m_1 = 5 \text{ kg}$, $v_1 = 5 \text{ ms}^{-1}$,

$m_2 = 4 \text{ kg}$, $v_2 = 4 \text{ ms}^{-1}$, $m_3 = 2 \text{ kg}$ and $v_3 = 2 \text{ ms}^{-1}$

Substituting all these values in above equation, we get

$$v_{CM} = \frac{5 \times 5 + 4 \times 4 + 2 \times 2}{5 + 4 + 2} = \frac{25 + 16 + 4}{11}$$

$$\therefore v_{CM} = \frac{45}{11} = 4.09 \approx 4 \text{ ms}^{-1}$$

- 12.** (a) From law of conservation of momentum,

$$Mv + m \times 0 = Mv_1 + mv_2$$

$$\Rightarrow M(v - v_1) = mv_2 \quad \dots(i)$$

Again, from the conservation of kinetic energy (as collision is of elastic nature),

$$\frac{1}{2}Mv^2 + \frac{1}{2}m \times 0 = \frac{1}{2}Mv_1^2 + \frac{1}{2}mv_2^2$$

$$\Rightarrow M(v^2 - v_1^2) = mv_2^2 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$\frac{M(v - v_1)}{M(v + v_1)(v - v_1)} = \frac{mv_2}{mv_2^2}$$

$$v_2 = v + v_1 \quad \dots(iii)$$

Now, solving Eqs. (i) and (iii), we get

$$v_1 = \frac{(M - m)v}{(M + m)}$$

$$\text{and } v_2 = \frac{2Mv}{(M + m)}$$

As, $M \gg m$

So, $v_1 = v$ and $v_2 = 2v$

Hence, velocity of lighter body (m) is $2v$.

- 13.** (a) If two particles are initially moving in the same direction, then their resultant momentum will not be zero. Therefore, their resultant momentum cannot be zero after a completely inelastic collision.

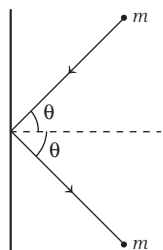
As, kinetic energy is directly proportional to the square of the momentum, hence kinetic energy cannot be zero. This implies, not all the energy in inelastic collision is lost.

Hence, both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

- 14** (c) Given, $m = 0.5 \text{ kg}$, $v = 12 \text{ ms}^{-1}$, $\Delta t = 1 \text{ s}$ and $\theta = 30^\circ$

Force applied by wall on ball,

$$F = \frac{\Delta p}{\Delta t} \text{ or } F = \frac{(p_f)_H - (p_i)_H}{\Delta t}$$



$\therefore$ In this elastic collision, final and initial velocities will be same but direction changes

$$(p_f)_H = mv \cos \theta$$

$$\text{and } (p_i)_H = -mv \cos \theta$$

$$\therefore F = \frac{mv \cos \theta + mv \cos \theta}{\Delta t} = \frac{2mv \cos \theta}{\Delta t}$$

$$F = \frac{2 \times 0.5 \times 12 \times \cos 30^\circ}{1} = 6\sqrt{3} \text{ N}$$

$$15 \text{ (c) Loss of kinetic energy} = \frac{1}{2} m_1 u_1^2 - \frac{m_1^2 u_1^2}{2(m_1 + m_2)} = \frac{m_1 m_2 u_1^2}{2(m_1 + m_2)}$$

$$\text{Given, } m_1 = 4 \text{ kg, } u_1 = 12 \text{ ms}^{-1},$$

$$m_2 = 6 \text{ kg and } u_2 = 0$$

$$\therefore \Delta \text{KE} = \frac{1}{2} \times \frac{4 \times 6}{(4 + 6)} (12)^2 = \frac{1}{2} \times \frac{24}{10} \times (12)^2$$

$$= \frac{12}{10} \times 144 = 172.8 \text{ J}$$

16 (c) Given, masses, $m_1 = 6$ unit and $m_2 = 2$ unit

$$\text{Positions} = 6\hat{i} - 7\hat{j} \text{ and } 2\hat{i} + 5\hat{j} - 8\hat{k}$$

Coordinates of centre of mass are calculated below

$$x_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{6 \times 6 + 2 \times 2}{6 + 2} = \frac{36 + 4}{8} = 5\hat{i}$$

$$y_{\text{CM}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \frac{6 \times (-7) + 2 \times (5)}{6 + 2} = \frac{-42 + 10}{8} = -4\hat{j}$$

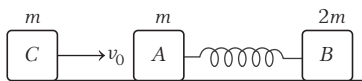
$$\text{and } z_{\text{CM}} = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2} = \frac{6 \times (0) + 2 \times (-8)}{6 + 2} = \frac{-16}{8} = -2\hat{k}$$

$$\therefore \text{Centre of mass lies at } 5\hat{i} - 4\hat{j} - 2\hat{k}.$$

$\Rightarrow$ Coordinates of centre of mass are (5, -4, -2).

17 (d) Using the law of conservation of linear momentum, we have

$$mv_0 = mv + 2mv \Rightarrow v = v_0 / 3$$



Using law of conservation of energy, we have

$$\frac{1}{2} mv_0^2 = \frac{1}{2} kx_0^2 + \frac{1}{2} (3m)v^2$$

where, x_0 is compression in the spring.

$$\therefore mv_0^2 = kx_0^2 + (3m)\frac{v_0^2}{9}$$

$$\Rightarrow kx_0^2 = mv_0^2 - \frac{mv_0^2}{3} \Rightarrow kx_0^2 = \frac{2mv_0^2}{3}$$

$$\therefore k = \frac{2mv_0^2}{3x_0^2}$$

$$18 \text{ (c) Total initial energy} = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

Since, after collision one particle absorbs energy ϵ .

$$\therefore \text{Total final energy} = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 + \epsilon$$

From conservation of energy,

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 + \epsilon$$

$$\Rightarrow \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 - \epsilon = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

19 (b) Applying the law of conservation of momentum to the

$$\text{system, } mgh = \frac{1}{2} kx^2$$

$$\text{Given, } m = 0.04 \text{ kg, } h = 5 \text{ m, } k = 400 \text{ Nm}^{-1}$$

and x = deformation (compression) in the spring

$$\Rightarrow x = \sqrt{\frac{2mgh}{k}} = \sqrt{\frac{2 \times 0.04 \times 10 \times 5}{400}} = \frac{1}{10} \text{ m} = 10 \text{ cm} \approx 9.8 \text{ cm}$$

20 (a) Let the velocity of block of mass $2m$ after the collision be v' , then from law of conservation of momentum,

$$mv = 2mv' \Rightarrow v' = \frac{v}{2}$$

Now, the coefficient of restitution,

$$e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = \frac{v'}{v} = \frac{v/2}{v} = \frac{1}{2} = 0.5$$

21 (b) Let the maximum compression in the spring be x . From

$$\text{law of conservation of energy, } mgh = \frac{1}{2} kx^2$$

$$10 \times 10^{-3} \times 10 \times 10 = \frac{1}{2} \times 200 \times x^2$$

$$\Rightarrow x^2 = 10 \times 10^{-3} = 10^{-2}$$

$$\Rightarrow x = \sqrt{\frac{1}{100}} = \frac{1}{10} \text{ m} = 0.1 \text{ m}$$

Thus, compression in the spring is 0.1m.

22 (c) Let the speed of the board be u and frog jumps with angle of inclination to the board θ , then from law of conservation of momentum in horizontal direction,

$$mv \cos \theta - mu = 0, u = v \cos \theta \quad \dots(i)$$

Let distance moved by board be x .

$$\text{So, } L - x = ut \quad \dots(ii)$$

$$\text{and } x = v \cos \theta t \quad \dots(iii)$$

Solving above equations, we get

$$x = \frac{L}{2}$$

Also, $x = \frac{v^2 \sin 2\theta}{g}$

$$\Rightarrow \frac{L}{2} = \frac{v^2 \sin 2\theta}{g}$$

$$\Rightarrow v = \sqrt{\frac{gL}{2 \sin 2\theta}}$$

Hence, v should be minimum for $\sin 2\theta = 1$ (i.e. maximum)

$$\Rightarrow v_{\min} = \sqrt{\frac{gL}{2}} = \sqrt{10 \times \frac{10}{2}} = \sqrt{50}$$

$$= 5\sqrt{2} \text{ ms}^{-1}$$

- 23 (b)** As, net horizontal force acting on the system is zero, hence momentum must remain conserved.

Hence, $mu + 0 = 0 + Mv_2$

$$\Rightarrow v_2 = \frac{mu}{M}$$

Coefficient of restitution, $e = \frac{|v_2 - v_1|}{|u_2 - u_1|} = \frac{|v_2 - 0|}{|0 - u|}$

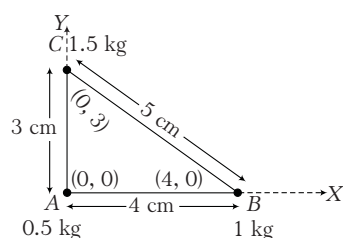
$$= \frac{v_2}{u} = \frac{\frac{mu}{M}}{u} = \frac{m}{M}$$

- 24 (c)** When a body falls from height h , it strikes the ground with a velocity $u = \sqrt{2gh}$. Let it rebounds with a velocity v and rise to a height h_1 .

$$v = \sqrt{2gh_1} \Rightarrow e = \frac{v}{u} = \sqrt{\frac{h_1}{h}}$$

Clearly, $h_1 = e^2 h$

- 25 (a)** The figure given in question is



Let X and Y -axes be along AB and AC , respectively. The coordinates of centre of mass would be

$$X_{\text{CM}} = \frac{1.5 \times 0 + 1 \times 4 + 1.5 \times 0}{1.5 + 0.5 + 1} = \frac{4}{3} = 1.3$$

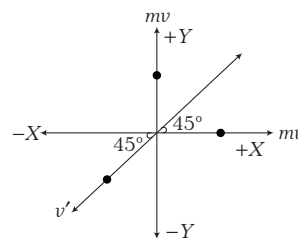
and $Y_{\text{CM}} = \frac{0.5 \times 0 + 4 \times 0 + 1.5 \times 3}{1.5 + 1 + 0.5} = \frac{1.5 \times 3}{3} = 1.5$

Hence, $(X_{\text{CM}}, Y_{\text{CM}}) = (1.3, 1.5)$

- 26 (b)** As large number of particles are situated at a distance R from the origin. If particles are uniformly distributed and make a circular boundary around the origin, then centre of mass will be at the origin.

While, if the particles are not uniformly distributed, then centre of mass will lie between particle and origin. This implies that the distance between centre of mass and origin is always less than or equal to R .

- 27 (b)**



$$\sqrt{2}(mv) = \text{Resultant momentum of two small masses}$$

$$\sqrt{2}(mv) = (2m) \times v' \Rightarrow v' = \frac{v}{\sqrt{2}}$$

So, total kinetic energy generated by the explosion

$$= \frac{1}{2} mv^2 + \frac{1}{2} mv^2 + \frac{1}{2} (2m) v'^2$$

$$= mv^2 + m \times \left(\frac{v}{\sqrt{2}}\right)^2 = mv^2 + \frac{mv^2}{2} = \frac{3}{2} mv^2$$

- 28 (d)** Given, $p = a + bt + ct^2$

Differentiating with respect to t , we get

$$\frac{dp}{dt} = 0 + b + 2ct$$

From Newton's second law of motion, $F \propto \frac{dp}{dt}$

$\Rightarrow F \propto t$ or force is dependent linearly on time.

- 29 (d)** The position of centre of mass of a system of particles does not depend upon the nature of particles.

$$\mathbf{r}_{\text{CM}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}$$

- 30 (b)** We have, $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 = 0$

$$(\because p = mv)$$

$$\therefore 1 \times 12 \hat{i} + 2 \times 8 \hat{j} + \mathbf{p}_3 = 0$$

$$\Rightarrow 12 \hat{i} + 16 \hat{j} + \mathbf{p}_3 = 0 \Rightarrow \mathbf{p}_3 = -(12 \hat{i} + 16 \hat{j})$$

$$\therefore p_3 = \sqrt{(12)^2 + (16)^2} = \sqrt{144 + 256}$$

$$= 20 \text{ kg} \cdot \text{ms}^{-1}$$

Now, $p_3 = m_3 v_3 \Rightarrow m_3 = \frac{p_3}{v_3} = \frac{20}{4} = 5 \text{ kg}$

- 31 (c)** After firing, the momentum of gun and bullet is same.

Therefore, by the relation, $K = \frac{p^2}{2m}$

We have, $K \propto \frac{1}{m}$ (As p is same)

As, the mass of gun is greater, hence its kinetic energy will be less.

- 32 (c)** In all types of collisions, total linear momentum of colliding particles remains conserved.

- 33 (d)** Given, mass = m

Position of centre of mass, $r_{\text{CM}} = \frac{ma \hat{i} + mb \hat{j} + m(0) \hat{k}}{3m}$

$$\Rightarrow r_{\text{CM}} = \frac{1}{3} (a \hat{i} + b \hat{j})$$

- 34 (b)** From the law of conservation of momentum,

$$mv + m \times 0 = (m + m)v'$$

$$\Rightarrow v' = \frac{mv}{(m + m)} \Rightarrow v' = \frac{v}{2}$$

$$\text{and we know that, } h = \frac{u^2}{2g} \quad (\because v^2 - u^2 = 2gh)$$

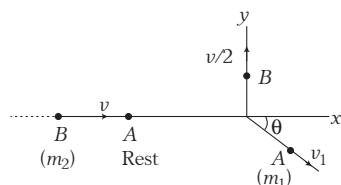
$$\text{Here, } u = v'$$

$$\text{So, } h = \frac{(v')^2}{2g} \quad \left(\because v' = \frac{v}{2} \right)$$

$$\Rightarrow h = \frac{v^2}{8g}$$

- 35 (b)** In an inelastic collision, kinetic energy before collision is not equal to kinetic energy after collision. But the linear momentum is conserved in all types of collisions.

- 36 (c)** Initial momentum, $\mathbf{p}_i = m_2 v \hat{\mathbf{i}} + m_1 \times 0$



$$\text{Final momentum, } \mathbf{p}_f = m_2(v/2)\hat{\mathbf{j}} + m_1 \times \mathbf{v}_1$$

From law of conservation of momentum,

$$\mathbf{p}_i = \mathbf{p}_f$$

$$m_2 v \hat{\mathbf{i}} = m_2 \frac{v}{2} \hat{\mathbf{j}} + m_1 \times \mathbf{v}_1$$

$$\mathbf{v}_1 = \frac{m_2 v}{m_1} \hat{\mathbf{i}} - \frac{m_2 v}{m_1} \frac{\hat{\mathbf{j}}}{2}$$

$$\text{From this equation, we can find, } \tan \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{1}{2} \right) \text{ to the X-axis.}$$

- 37 (c)** Here, net external force on the entire system is zero, hence position of centre of mass remains unchanged.

- 38 (d)** Using law of conservation of momentum, we get

$$100 \times v = 0.25 \times 100$$

$$\Rightarrow v = 0.25 \text{ ms}^{-1}$$

- 39 (b)** Initial momentum of the system

$$= m_1 \times 40 + m_2 \times 0 = 40 m_1$$

Final momentum of the system

$$= (m_1 + m_2) \times 30$$

By the law of conservation of momentum,

$$\text{Initial momentum} = \text{Final momentum}$$

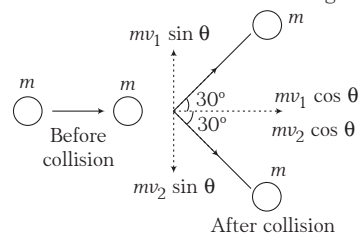
$$40 m_1 = (m_1 + m_2) 30$$

$$40 m_1 - 30 m_1 = 30 m_2$$

$$\Rightarrow 10 m_1 = 30 m_2$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{3}{1}$$

- 40 (a)** The given condition as shown in the figure below.



$$\text{After collision, } mv_1 \sin \theta = mv_2 \sin \theta \Rightarrow v_1 = v_2 = v$$

$$\therefore 2mv \cos \theta = m \times 9$$

$$\Rightarrow 2v \cos 30^\circ = 9$$

$$\Rightarrow v = \frac{9}{\sqrt{3}} = 3\sqrt{3} = 5.2 \text{ ms}^{-1}$$

- 41 (c)** Velocity of centre of mass,

$$v_{\text{CM}} = \frac{m_1 \frac{dr_1}{dt} + m_2 \frac{dr_2}{dt}}{m_1 + m_2} = \frac{4 \times 5 \hat{\mathbf{i}} + 2 \times 10 \hat{\mathbf{i}}}{4 + 2} = \frac{40 \hat{\mathbf{i}}}{6} = \frac{20}{3} \hat{\mathbf{i}}$$

The kinetic energy of centre of mass,

$$K = \frac{1}{2} m v_{\text{CM}}^2 = \frac{1}{2} \times (4 + 2) \times \frac{20 \times 20}{3 \times 3}$$

$$= \frac{1}{2} \times 6 \times \frac{20 \times 20}{3 \times 3} = \frac{400}{3} \text{ J}$$

- 42 (b)** From law of conservation of momentum,

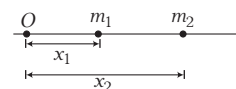
$$mu = (m + m)v \Rightarrow v = \frac{u}{2} = \frac{1}{2} \text{ m/s}$$

Now, maximum height reached by the system,

$$h = \frac{v^2}{2g} = \frac{1}{8g} = \frac{1}{80} \times 100 = \frac{5}{4} = 1.25 \text{ cm}$$

- 43 (c)** The centre of mass of the system,

$$r_{\text{CM}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$



- 44 (a)** Final kinetic energy of the block along with the bullet when the bullet strikes $= \frac{1}{2} (m + M)v^2$

Due to this kinetic energy, the block will rise to a height h .

Its potential energy $= (m + M)gh$

So, from the law of conservation of energy,

$$\frac{1}{2} (M + m)v^2 = (M + m)gh \Rightarrow v = \sqrt{2gh}$$

- 45 (b)** Given, mass, $m_1 = m_2$

Velocity, $u_1 = v_1$ and $u_2 = 0$

$$\text{For elastic collision, } v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2 + \frac{2m_1 u_1}{m_1 + m_2}$$

After putting the given values, we get

$$v_2 = \frac{2m_1 v_1}{2m_1} \Rightarrow v_2 = v_1$$

CHAPTER 09

Rotation

In this chapter, we will discuss rotational motion, *i.e.* the motion of an object around its own axis and the variables which describe rotational motion. Motion of wheels, gears, motors, planets, the hands of a clock, the rotor of jet engines, blades of helicopters, etc. are examples of rotational motion.

Rigid body

Ideally, a body is said to be a rigid body when it has a definite shape and size. For a rigid body, it is considered that its constituent particles do not change their mutual distances even when an external force is applied on it.

e.g. A wheel can be considered as rigid body by ignoring a little change in its shape.

Motion of rigid body

A rigid body can possess pure **translational** motion, pure **rotational** motion or a combination of both of these motions.

In previous chapters, we have discussed translational motion, *i.e.* the motion of an object along a line (both straight and curved). So, here we will study about the rotational motion of a body.

ROTATIONAL MOTION (FIXED AXIS OF ROTATION)

In any kind of motion of a body, there is either change in position or change in orientation or change in both. If a body changes its orientation during its motion, it is said to have rotational motion.

In pure rotational motion, every particle of the rigid body moves in circles of different radii about a fixed line, which is known as **axis of rotation**.

e.g. In case of motion of a ceiling fan, the vertical rod supporting the fan (here, axis of rotation) remains stationary and all the particles on the fan move in circular paths. The centres of circular paths are on the axis of rotation.

Also, it is not necessary that axis of rotation passes through the body.

Physical quantities required to describe rotational motion and equations of rotational motion

All the physical quantities required to describe rotational motion are same as in circular motion.

Similarly, equations of rotational motion are same as that of circular motion.

Inside

1 Rotational motion (Fixed axis of rotation)

- Moment of inertia
- Radius of gyration
- Theorems on moment of inertia

2 Torque or moment of force

3 Angular momentum

- Angular momentum of a particle about some point
- Angular momentum of a rigid body rotating about a fixed axis
- Law of conservation of angular momentum

4 Rotational kinetic energy of the particles

- Work-energy theorem in rotational motion

5 Rolling motion

- Classification of rolling motion
- Kinetic energy of rolling motion
- Rolling down on a rough inclined plane

Moment of inertia

Moment of inertia plays the same role in rotational motion as mass plays in translational motion. It is the property of a body due to which it opposes any change in its state of rest or uniform rotation.

The moment of inertia is calculated about some axis (usually the rotational axis) and it depends on the mass as well as its distribution about that axis.

Moment of inertia of a single particle

The moment of inertia of a single particle about an axis is given by

$$I = mr^2 \quad \dots(i)$$

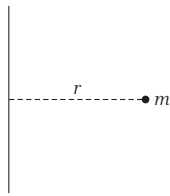


Fig. 9.1

Here, m is the mass of the particle and r is distance from the axis under consideration.

The SI unit of moment of inertia is kg-m^2 . Its dimensions are $[\text{ML}^2]$.

Moment of inertia of a system of particles

The moment of inertia of a system of particles about an axis is given by

$$I = \sum_i m_i r_i^2 \quad \dots(ii)$$

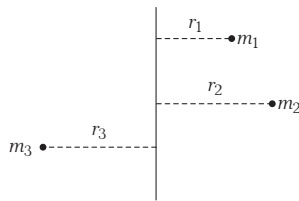


Fig. 9.2

where, r_i is the perpendicular distance from the axis of the i th particle of mass m_i .

Moment of inertia of rigid bodies

For a continuous mass distribution found in a rigid body, we replace the summation of Eq. (ii) by an integral. If the system is divided into infinitesimal elements of mass dm and if r is the distance from a mass element to the axis of rotation, then moment of inertia is

$$I = \int r^2 dm$$

where, the integral is taken over the system.

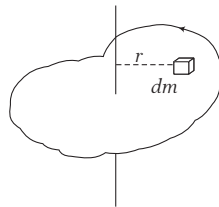


Fig. 9.3

Example 9.1 A ring of diameter 0.4 m and of mass 10 kg is rotating about its geometrical axis at the rate of 35 rotations/second. Find the moment of inertia.

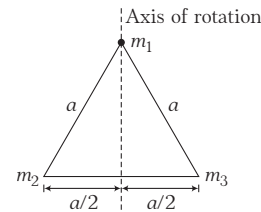
Sol. Moment of inertia of the ring about its axis,

$$I = MR^2$$

$$I = 10 \times (0.2)^2 = 10 \times 0.04 = 0.4 \text{ kg-m}^2$$

Example 9.2 Three point masses m_1 , m_2 and m_3 are located at the vertices of an equilateral triangle of side a . Determine the moment of inertia of the system about an axis along the altitude of the triangle passing through m_1 .

Sol. As shown in the figure given below, the axis of rotation passes through mass m_1 . The distances of m_1 , m_2 and m_3 from the axis of rotation are 0, $a/2$, $a/2$, respectively.



Therefore, moment of inertia of the system about the axis passing through m_1 is

$$\begin{aligned} I &= m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 \\ &= m_1 (0)^2 + m_2 \left(\frac{a}{2}\right)^2 + m_3 \left(\frac{a}{2}\right)^2 \end{aligned}$$

$$\Rightarrow I = \frac{a^2}{4} (m_2 + m_3)$$

Example 9.3 Calculate the moment of inertia w.r.t. rotational axis XX' in following figures.

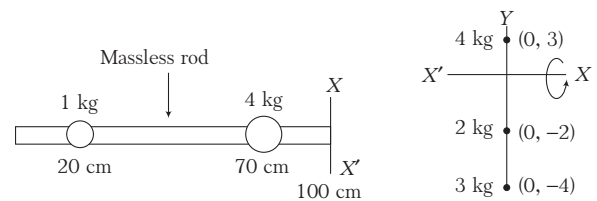


Fig. (i)

Fig. (ii)

Sol. (i) Moment of inertia about axis XX' ,

$$I_{XX'} = m_1 r_1^2 + m_2 r_2^2 = 4 \times (1 - 0.7)^2 + 1 \times (1 - 0.2)^2$$

$$I_{XX'} = 4 \times (0.3)^2 + 1 \times (0.8)^2 = 1 \text{ kg-m}^2$$

(ii) Moment of inertia about axis XX' ,

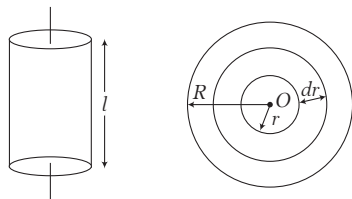
$$I_{XX'} = 4 \times (3)^2 + 2 \times (2)^2 + 3 \times (4)^2 = 92 \text{ kg-m}^2$$

Example 9.4 Find the moment of inertia of a uniform cylinder about an axis through its centre of mass and perpendicular to its base. Mass of the cylinder is M and radius is R .

Sol. The cylinder can be divided into annular shells of width dr and length l as shown in figure.

The moment of inertia of one of these shells,

$$I = \int r^2 dm = r^2 (\rho dV)$$



Here, ρ = density, dV = volume of shell = $2\pi r l dr$

$$\therefore I = 2\pi \rho l r^3 dr$$

The cylinder's moment of inertia is found by integrating this expression between 0 and R

$$\text{So, } I = 2\pi \rho l \int_0^R r^3 dr = \frac{\pi \rho l}{2} R^4 \quad \dots(i)$$

The density ρ of the cylinder is the mass divided by the volume,

$$\text{i.e. } \rho = \frac{M}{\pi R^2 l} \quad \dots(ii)$$

From Eqs. (i) and (ii), we have

$$I = \frac{1}{2} MR^2$$

Radius of gyration

The radius of gyration of a body about a given axis is the perpendicular distance of a point from the axis at which the whole mass of the body could be concentrated without any change in the moment of inertia of the body about that axis.

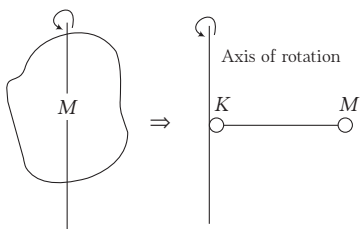


Fig. 9.4

If a body has mass M and radius of gyration K , then moment of inertia,

$$I = MK^2 \Rightarrow K = \sqrt{I/M}$$

Radius of gyration is also defined as the root-mean-square-value of distances of all the particles about the axis of rotation.

$$\text{i.e. } K = \sqrt{\frac{r_1^2 + r_2^2 + r_3^2 + \dots + r_n^2}{n}}$$

It depends on the shape and size of the body, position and configuration of the axis of rotation, distribution of mass of the body with respect to the axis of rotation.

Radius of gyration does not depend on the mass of body.

Example 9.5 A wheel of mass 8 kg has moment of inertia equals to 0.5 kg-m^2 . Determine its radius of gyration.

Sol. Given, mass, $M = 8 \text{ kg}$

Moment of inertia, $I = 0.5 \text{ kg-m}^2$

$$\therefore I = MK^2$$

$$\Rightarrow K^2 = \frac{I}{M}$$

$$\Rightarrow K = \sqrt{\frac{I}{M}}$$

$$\Rightarrow \text{Radius of gyration, } K = \sqrt{\frac{0.5}{8}}$$

$$\Rightarrow K = 0.25 \text{ m}$$

$$\text{or } K = 25 \text{ cm}$$

Example 9.6 The moment of inertia of a solid cylinder about its axis of rotation is $\frac{MR^2}{2}$. What is the value of the radius of gyration of the cylinder about this axis?

Sol. Moment of inertia of solid cylinder,

$$I = \frac{MR^2}{2} \quad \dots(i)$$

Also, moment of inertia,

$$I = MK^2 \quad \dots(ii)$$

Comparing Eqs. (i) and (ii), we get

$$K^2 = \frac{R^2}{2}$$

$$\Rightarrow K = \frac{R}{\sqrt{2}}$$

Theorems on moment of inertia

There are two important theorems on moment of inertia, which enable calculation of the moment of inertia of a body about an axis, if its moment of inertia about some other axis is known. Let us now discuss both of them.

Theorem of parallel axes

It states that moment of inertia of a rigid body about any axis is equal to the sum of its moment of inertia about a parallel axis through its centre of mass, the product of the mass of the body and the square of the perpendicular distance between the two axes.

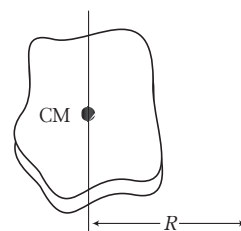


Fig. 9.5

Two such axes are shown in figure for a body of mass M . If R is the distance between the axes and I_{CM} and I are the respective moments of inertia about these axes, then

$$I = I_{CM} + MR^2$$

Note From the above theorem, we can conclude that among many parallel axes, moment of inertia is least about an axis which passes through centre of mass, e.g. I_2 is least among I_1, I_2 and I_3 . Similarly, I_5 is least among I_4, I_5 and I_6 .

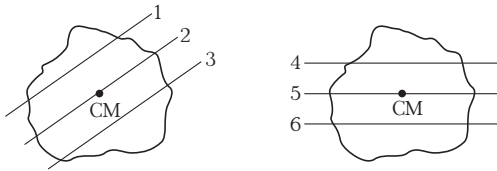


Fig. 9.6

Theorem of perpendicular axes

It states that the moment of inertia of a plane lamina about an axis perpendicular to its plane is equal to the sum of the moments of inertia of the lamina about any two mutually perpendicular axes in its plane and intersecting each other at the point, where the perpendicular axis passes through it.

Let X and Y -axes be chosen in the plane of the body and Z -axis perpendicular to this plane, three axes being mutually perpendicular, then according to the theorem,

$$I_Z = I_X + I_Y$$

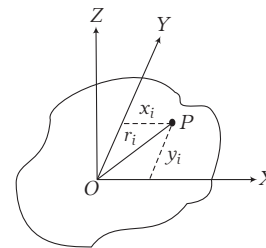


Fig. 9.7

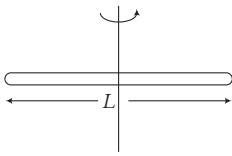
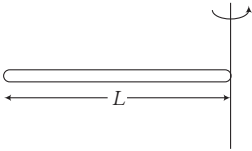
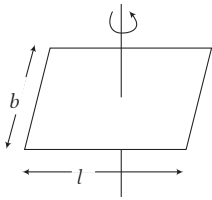
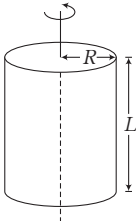
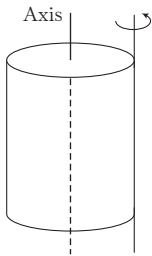
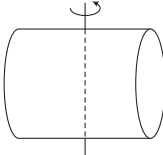
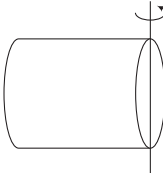
where, I_X, I_Y and I_Z are the moments of inertia about the X, Y and Z -axes respectively.

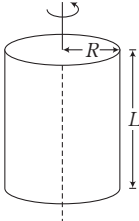
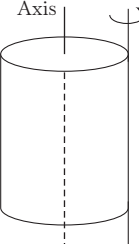
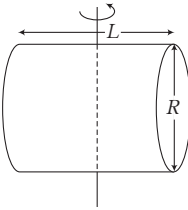
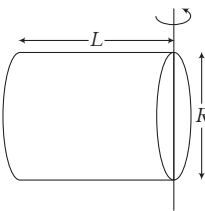
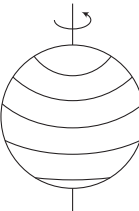
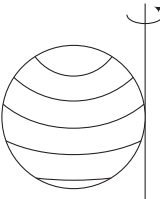
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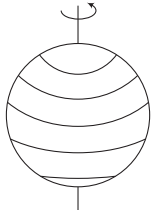
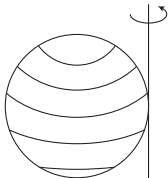
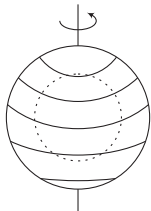
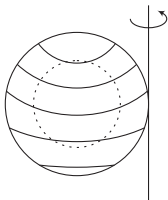
- (i) Theorem of parallel axes is applicable for any type of rigid body whether it is a two dimensional or three dimensional, while the theorem of perpendicular axes is applicable for laminar type or two dimensional bodies only.
- (ii) In theorem of perpendicular axes, the point of intersection of the three axes (X, Y and Z) may be any point on the plane of body (it may even lie outside the body). This point may or may not be the centre of mass of the body.

Moment of Inertia of Simple Geometrical Objects

S.No.	Body	Axis of Rotation	Figure	Moment of Inertia	K	K^2/R^2
(i)	Thin circular ring, radius R	(a) About an axis passing through CG and perpendicular to its plane		MR^2	R	1
		(b) About its diameter		$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$	$\frac{1}{2}$
(ii)	Circular disc, radius R	(a) Perpendicular to disc at centre		$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$	$\frac{1}{2}$
		(b) About its diameter		$\frac{1}{4}MR^2$	$\frac{R}{2}$	$\frac{1}{4}$

S.No.	Body	Axis of Rotation	Figure	Moment of Inertia	K	K^2/R^2
(iii)	Long uniform thin rod	(a) About an axis passing through its centre of mass and perpendicular to the rod		$\frac{ML^2}{12}$	$\frac{L}{\sqrt{12}}$	
		(b) About an axis passing through its edge and perpendicular to the rod		$\frac{ML^2}{3}$	$\frac{L}{\sqrt{3}}$	
(iv)	Rectangular lamina	Passing through the centre of mass and perpendicular to the plane		$\frac{M}{12}[l^2 + b^2]$	$\sqrt{\frac{l^2 + b^2}{12}}$	
(v)	Hollow cylinder, radius R	About its own axis		MR^2	R	1
(vi)	Cylindrical shell	(a) Tangential line passing through its surface parallel to its axis		$2MR^2$	$\sqrt{2}R$	2
		(b) About an axis passing through its CG and perpendicular to its own axis		$M\left[\frac{L^2}{12} + \frac{R^2}{2}\right]$	$\sqrt{\frac{L^2}{12} + \frac{R^2}{2}}$	
		(c) About the diameter of one of faces of the cylinder		$M\left[\frac{L^2}{3} + \frac{R^2}{2}\right]$	$\sqrt{\frac{L^2}{3} + \frac{R^2}{2}}$	

S.No.	Body	Axis of Rotation	Figure	Moment of Inertia	K	K^2/R^2
(vii)	Solid cylinder	(a) About its own axis		$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$	$\frac{1}{2}$
		(b) Tangential line passing through its surface and parallel to its axis		$\frac{3}{2}MR^2$	$\sqrt{\frac{3}{2}}R$	$\frac{3}{2}$
		(c) About an axis passing through its CG and perpendicular to its own axis		$M\left[\frac{L^2}{12} + \frac{R^2}{4}\right]$	$\sqrt{\frac{L^2}{12} + \frac{R^2}{4}}$	
		(d) About the diameter of one of the faces of the cylinder		$M\left[\frac{L^2}{3} + \frac{R^2}{4}\right]$	$\sqrt{\frac{L^2}{3} + \frac{R^2}{4}}$	
(viii)	Spherical shell	(a) About its diametric axis		$\frac{2}{3}MR^2$	$\sqrt{\frac{2}{3}}R$	$\frac{2}{3}$
		(b) About a tangential axis		$\frac{5}{3}MR^2$	$\sqrt{\frac{5}{3}}R$	$\frac{5}{3}$

S.No.	Body	Axis of Rotation	Figure	Moment of Inertia	K	K^2/R^2
(ix)	Solid sphere, radius R	(a) About its diametric axis		$\frac{2}{5}MR^2$	$\sqrt{\frac{2}{5}}R$	$\frac{2}{5}$
		(b) About a tangential axis		$\frac{7}{5}MR^2$	$\sqrt{\frac{7}{5}}R$	$\frac{7}{5}$
(x)	Hollow sphere of inner radius R_1 and outer radius R_2	(a) About its diametric axis		$\frac{2}{5}M \left[\frac{R_2^5 - R_1^5}{R_2^3 - R_1^3} \right]$		
		(b) Tangential		$\frac{2}{5}M \left[\frac{R_2^5 - R_1^5}{R_2^3 - R_1^3} \right] + MR_2^2$		

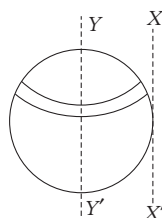
Example 9.7 Find the moment of inertia of a sphere about a tangent to the sphere, while the mass of the sphere is M and the radius of the sphere is R .

Sol. The moment of inertia of a sphere about its own axis (YY') is $\frac{2MR^2}{5}$.

Applying the theorem of parallel axes, the moment of inertia of the sphere about tangent (XX'),

$$I_t = I + MR^2$$

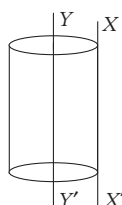
$$I_t = \frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$$



Example 9.8 The mass of the cylinder is 25 kg and radius of cylinder is 5 m. Find the moment of inertia of a solid cylinder about a tangent to the cylinder.

Sol. The moment of inertia of a cylinder about its own axis (YY')

$$= \frac{1}{2}MR^2$$



Applying the theorem of parallel axes, the moment of inertia of the cylinder about tangent (XX'),

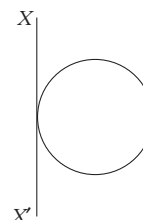
$$I_t = I + MR^2 = \frac{1}{2}MR^2 + MR^2$$

$$\Rightarrow I_t = \frac{3}{2}MR^2$$

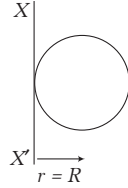
Here, $M = 25$ kg and $R = 5$ m

$$\begin{aligned} \therefore I_t &= \frac{3}{2}(25)(5)^2 \\ &= 937.5 \text{ kg-m}^2 \end{aligned}$$

Example 9.9 Find the moment of inertia of a solid sphere of mass 5 kg and radius 2 m about an axis XX' as shown in figure.



Sol. From theorem of parallel axes, the moment of inertia of solid sphere is



$$I_{XX'} = I_{CM} + MR^2$$

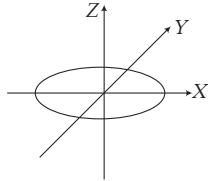
$$= \frac{2}{5} MR^2 + MR^2 = \frac{7}{5} MR^2 = \frac{7}{5} (5)(2)^2 = 28 \text{ kg-m}^2$$

Example 9.10 What is the moment of inertia of a

- (i) uniform circular ring of mass 2 kg about its 4 m diameter?
- (ii) a thin disc of mass M and radius R about an axis coinciding with a diameter?

Sol. (i) The moment of inertia of a uniform ring about an axis passing through its centre and perpendicular to its plane is given by $I = MR^2$.

According to the theorem of perpendicular axis, $I_Z = I_X + I_Y$



Now, X and Y -axes are along the diameter of the disc. By symmetry, $I_X = I_Y \Rightarrow I_Z = 2I_X \Rightarrow I_Z = MR^2$

The moment of inertia of ring about of its diameter,

$$I_X = \frac{MR^2}{2} = \frac{(2)(2)^2}{2} = 4 \text{ kg-m}^2$$

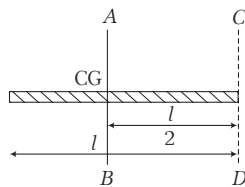
(ii) Similarly, moment of inertia about the diameter of disc,

$$I = \frac{1}{2} \left(\frac{1}{2} MR^2 \right) = \frac{1}{4} MR^2$$

Example 9.11 Calculate the moment of inertia of a rod of mass 2 kg and length 5 m about an axis perpendicular to it and passing through one of its ends.

Sol. For the rod of mass M and length l , the moment of inertia of the rod about an axis AB passing through its centre of mass,

$$I_{AB} = \frac{Ml^2}{12}$$



According to the parallel axes theorem,

$$I_{CD} = I_{AB} + M \left(\frac{l}{2} \right)^2 = \frac{Ml^2}{12} + \frac{Ml^2}{4}$$

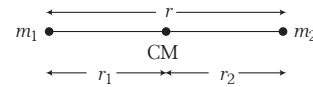
$$= \frac{Ml^2 + 3Ml^2}{12} = \frac{4Ml^2}{12} = \frac{Ml^2}{3}$$

$$I_{CD} = \frac{Ml^2}{3} = \frac{2(5)^2}{3} = \frac{50}{3} \text{ kg-m}^2$$

Example 9.12 Two masses m_1 and m_2 are placed at a distance r from each other. Find out the moment of inertia of system about an axis passing through their centre of mass.

Sol. We have, $m_1 r_1 = m_2 r_2$ and $r_1 + r_2 = r$

$$\Rightarrow r_1 = \frac{m_2 r}{m_1 + m_2}, r_2 = \frac{m_1 r}{m_1 + m_2}$$



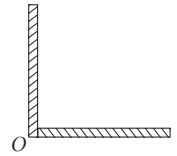
Moment of inertia of the system about an axis passing through their centre of mass,

$$I = m_1 r_1^2 + m_2 r_2^2 = m_1 \left(\frac{m_2 r}{m_1 + m_2} \right)^2 + m_2 \left(\frac{m_1 r}{m_1 + m_2} \right)^2$$

$$= \left(\frac{m_1 m_2}{m_1 + m_2} \right) r^2$$

Note Here, $I = \mu r^2$, where μ is called **reduced mass** which is equal to $\frac{m_1 m_2}{m_1 + m_2}$.

Example 9.13 Two identical rods each of mass M and length L are kept according to figure. Find the moment of inertia of rods about an axis passing through O and perpendicular to the plane of rods.

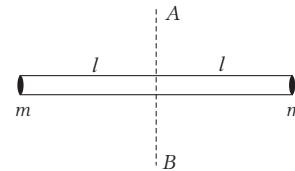


Sol. Moment of inertia of each rod about an axis passing through an end = $\frac{ML^2}{3}$

According to perpendicular axes theorem,

$$\therefore I_{\text{given system}} = \frac{ML^2}{3} + \frac{ML^2}{3} = \frac{2ML^2}{3}$$

Example 9.14 Consider a uniform rod of mass m and length $2l$ with two particles of mass m each at its ends. Let AB be a line perpendicular to the length of the rod and passing through its centre. Find the moment of inertia of the system about AB .

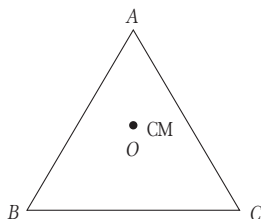


Sol. Moment of inertia of the system about AB ,

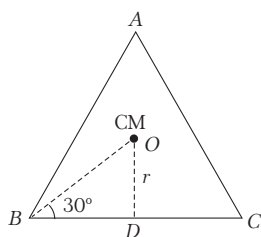
$$I_{AB} = I_{\text{rod}} + I_{\text{both particles}}$$

$$= \frac{m(2l)^2}{12} + 2(ml^2) = \frac{7}{3} ml^2$$

Example 9.15 Three rods each of mass m and length l are joined together to form an equilateral triangle as shown in figure. Find the moment of inertia of the system about an axis passing through its centre of mass and perpendicular to the plane of the triangle.



Sol. Moment of inertia of rod BC about an axis perpendicular to plane of triangle ABC and passing through the mid-point of rod BC (i.e. D) is



$$I_1 = \frac{ml^2}{12}, \text{ in } \triangle BDO, AD = \sqrt{l^2 - \frac{l^2}{4}} = \frac{\sqrt{3}}{2} l$$

$$r = OD = \frac{AD}{3} = \frac{\sqrt{3} l}{3 \times 2} = \frac{l}{2\sqrt{3}}$$

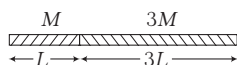
From theorem of parallel axes, moment of inertia of this rod about the axis passing through CM and perpendicular to plane ABC is

$$I_2 = I_1 + mr^2 = \frac{ml^2}{12} + m \left(\frac{l}{2\sqrt{3}} \right)^2 = \frac{ml^2}{6}$$

$\therefore$ Moment of inertia of all the three rods,

$$I = 3I_2 = 3 \left(\frac{ml^2}{6} \right) = \frac{ml^2}{2}$$

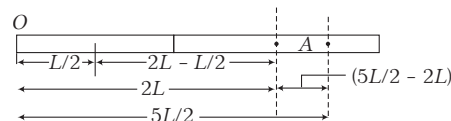
Example 9.16 Two thin uniform rods A (M, L) and B ($3M, 3L$) are joined as shown in figure. Find the moment of inertia about an axis passing through the centre of mass of the system of rods and perpendicular to the length.



Sol. Taking origin at O , the centre of mass will be located at

$$x_{CM} = \frac{M \times \frac{L}{2} + 3M \times \left(L + \frac{3L}{2} \right)}{M + 3M} = 2L$$

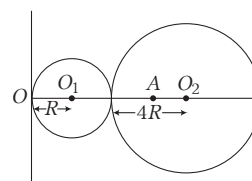
A is CM of system of rods.



So, using theorem of parallel axes,

$$\begin{aligned} I_A = I_1 + I_2 &= \left[\frac{ML^2}{12} + M \left(2L - \frac{L}{2} \right)^2 \right] \\ &\quad + \left[\frac{3M(3L)^2}{12} + 3M \left(\frac{5L}{2} - 2L \right)^2 \right] \\ &= ML^2 \left[\left\{ \frac{1}{12} + \frac{9}{4} \right\} + \left\{ \frac{27}{12} + \frac{3}{4} \right\} \right] \\ &= \frac{16ML^2}{3} \end{aligned}$$

Example 9.17 Two thin uniform rings made of same material and of radii R and $4R$ are joined as shown in figure. The mass of smaller ring is m . Find the moment of inertia about an axis passing through the centre of mass of system of rings and perpendicular to the plane.



Sol. Here, $m_1 = m$ and $m_2 = 4m$ ($\because m \propto \text{radius}$)

Location of the centre of mass from O ,

$$x_{CM} = \frac{m \times R + 4m(2R + 4R)}{m + 4m} = 5R$$

A is CM of system of rings, so using theorem of parallel axes,

$$\begin{aligned} I_A = I_1 + I_2 &= \{mR^2 + m(5R - R)^2\} \\ &\quad + \{4m(4R)^2 + 4m(6R - 5R)^2\} \\ &= mR^2 [1 + 16] + (64 + 4) \\ I_A &= 85mR^2 \end{aligned}$$

CHECK POINT 9.1

- A wheel rotates with a constant angular velocity of 300 rpm. The angle through which the wheel rotates in 1 s is
(a) π rad (b) 5π rad
(c) 10π rad (d) 20π rad
- A motor is rotating at a constant angular velocity of 500 rpm. The angular displacement per second is
(a) $\frac{3}{50\pi}$ rad (b) $\frac{3\pi}{50}$ rad
(c) $\frac{25\pi}{3}$ rad (d) $\frac{50\pi}{3}$ rad
- A body rotating with uniform angular acceleration covers 100π rad in the first 5 s after the start. Its angular speed at the end of 5 s (in rad/s) is
(a) 40π (b) 30π
(c) 20π (d) 10π
- A rotating wheel changes angular speed from 1800 rpm to 3000 rpm in 20 s. What is the angular acceleration, assuming it to be uniform?
(a) $60\pi \text{ rad s}^{-2}$ (b) $90\pi \text{ rad s}^{-2}$
(c) $2\pi \text{ rad s}^{-2}$ (d) $40\pi \text{ rad s}^{-2}$
- The angular velocity of a wheel increases from 100 to 300 rad/s in 10 s. The number of revolutions made during that time is
(a) 600 (b) 1500
(c) 1000 (d) 318
- A wheel has angular acceleration of 3 rad s^{-2} and an initial angular speed of 2 rad s^{-1} . In a time of 2 s, it has rotated through an angle (in rad) of
(a) 6 (b) 10 (c) 12 (d) 4
- A body rotates about a fixed axis with an angular acceleration of 3 rad s^{-2} . The angle rotated by it during the time when its angular velocity increases from 10 rad s^{-1} to 20 rad s^{-1} (in rad) is
(a) 50 (b) 100
(c) 150 (d) 200
- A wheel which is initially at rest is subjected to a constant angular acceleration about its axis. It rotates through an angle of 15° in time t second. The increase in angle through which it rotates in the next $2t$ second is
(a) 90° (b) 120°
(c) 30° (d) 45°
- Analogue of mass in rotational motion is
(a) moment of inertia (b) angular momentum
(c) radius of gyration (d) None of these
- Moment of inertia of a body depends upon
(a) axis of rotation (b) torque
(c) angular momentum (d) angular velocity
- A particle of mass 1 kg is kept at (1m, 1m, 1m). The moment of inertia of this particle about Z-axis would be
(a) 1 kg-m^2 (b) 2 kg-m^2
(c) 3 kg-m^2 (d) None of these
- Three thin rods each of length L and mass M are placed along X , Y and Z -axes such that one end of each rod is at origin. The moment of inertia of this system about Z-axis is
(a) $\frac{2}{3} ML^2$ (b) $\frac{4ML^2}{3}$
(c) $\frac{5ML^2}{3}$ (d) $\frac{ML^2}{3}$
- The radius of gyration of a solid sphere of radius R about its tangential axis is
(a) $\sqrt{\frac{7}{5}} R$ (b) $\sqrt{\frac{2}{5}} R$
(c) $\sqrt{\frac{5}{7}} R$ (d) R
- Moment of inertia of a rod of mass m and length l about its one end is I . If one-fourth of its length is cut away, then moment of inertia of the remaining rod about its one end will be
(a) $\frac{3}{4} I$ (b) $\frac{9}{16} I$
(c) $\frac{27}{64} I$ (d) $\frac{I}{16}$
- One circular ring and one circular disc both having the same mass and radius. The ratio of their moments of inertia about the axes passing through their centres and perpendicular to planes will be
(a) 1 : 1 (b) 2 : 1
(c) 1 : 2 (d) 4 : 1
- The moment of inertia of a solid cylinder of mass M , length $2R$ and radius R about an axis passing through the centre of mass and perpendicular to the axis of the cylinder is I_1 . For the same cylinder, its value about an axis passing through one end of the cylinder and perpendicular to the axis of cylinder is I_2 . The relation between I_1 and I_2 is
(a) $I_2 - I_1 = MR^2$ (b) $I_2 = I_1$
(c) $\frac{I_2}{I_1} = \frac{19}{12}$ (d) $I_1 - I_2 = MR^2$
- I_1 is the moment of inertia of a thin rod about an axis perpendicular to its length and passing through its centre of mass. If I_2 is the moment of inertia of the ring formed by bending the rod about an axis perpendicular to the plane of ring and passing through its centre, then
(a) $\frac{I_1}{I_2} = \frac{3}{\pi^2}$ (b) $\frac{I_1}{I_2} = \frac{2}{\pi^2}$
(c) $\frac{I_1}{I_2} = \frac{\pi^2}{2}$ (d) $\frac{I_1}{I_2} = \frac{\pi^2}{3}$

TORQUE OR MOMENT OF FORCE

Torque is a quantity which measures the capability of a force to rotate a body. Torque due to a force is also known as the moment of a force. Torque is the rotational analogue of force of linear motion.

It is defined as the product of the force and the perpendicular distance between the line of action of the force and the axis of rotation. This perpendicular distance is known as the **force arm**.

Torque, $\tau = \text{Force arm} \times \text{Force}$

$$\tau = \mathbf{r} \times \mathbf{F}$$

$$\tau = rF \sin \theta$$

where, θ is the angle between $\mathbf{r}$ and $\mathbf{F}$.

The SI unit of torque is N-m and its dimensions are $[\text{ML}^2\text{T}^{-2}]$.

Torque is an axial vector, i.e. its direction is always perpendicular to the plane containing vectors $\mathbf{r}$ and $\mathbf{F}$ in accordance with right-hand screw rule.

There is a mathematical relation between torque (τ), angular acceleration (α) and moment of inertia (I) and that can be given as $\tau = I\alpha$

The above expression is similar to the expression of Newton's second law for translational motion with constant mass.

Note

- If $\theta = 0^\circ$ or 180° , the torque of force is zero.
- $\tau = \mathbf{r} \times \mathbf{F}$ is a vector product, so on reversing the direction of $\mathbf{F}$, the direction of torque is reversed. If directions of both $\mathbf{r}$ and $\mathbf{F}$ are reversed, the direction of τ remains the same.
- If the total torque acting on the rigid body vanishes, i.e. the vector sum of the torques on the rigid body is zero.

$$\tau_1 + \tau_2 + \tau_3 \dots \tau_n = \sum_{i=1}^n \tau_i = 0$$

Then, the body is said to be in **rotational equilibrium**.

Example 9.18 Find the torque of a force $\mathbf{F} = (\hat{i} + 2\hat{j} - 3\hat{k})\text{N}$ about a point O . The position vector of point of application of force about O is $\mathbf{r} = (2\hat{i} + 3\hat{j} - \hat{k})\text{m}$.

Sol. Torque, $\tau = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 1 & 2 & -3 \end{vmatrix}$

$$\tau = \hat{i}(-9+2) + \hat{j}(-1+6) + \hat{k}(4-3)$$

or $\tau = (-7\hat{i} + 5\hat{j} + \hat{k})\text{N-m}$

Example 9.19 A force $10(-\hat{k})\text{N}$ acts on the origin of the coordinate system. Find the torque about the point $(1\text{ m}, -1\text{ m}, 0\text{ m})$.

Sol. Torque, $\tau = \mathbf{r} \times \mathbf{F} = [(1-0)\hat{i} + (-1-0)\hat{j} + (0-0)\hat{k}] \times (-10)\hat{k}$
 $= 10(\hat{i} - \hat{j}) \times (-\hat{k}) = 10(\hat{i} + \hat{j})\text{N-m}$

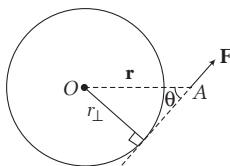
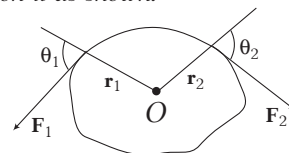


Fig. 9.8

Example 9.20 The body in figure shown is pivoted at O and two forces act on it as shown.



- Find an expression of the net torque on the body about the pivot.
- If $r_1 = 1.30\text{ m}$, $r_2 = 2.15\text{ m}$, $F_1 = 4.20\text{ N}$, $F_2 = 4.90\text{ N}$, $\theta_1 = 75^\circ$ and $\theta_2 = 60^\circ$, what is the net torque about the pivot?

Sol. (i) Take a torque that tends to cause a counter clockwise rotation from rest to be positive and a torque that tends to cause a clockwise rotation from rest to be negative. Thus, a positive torque of magnitude $r_1 F_1 \sin \theta_1$ is associated with $\mathbf{F}_1$ and a negative torque of magnitude $r_2 F_2 \sin \theta_2$ is associated with $\mathbf{F}_2$. Both of these are about O . The net torque about O is $\tau = r_1 F_1 \sin \theta_1 - r_2 F_2 \sin \theta_2$.

(ii) Substituting the given values in above equation, we get

$$\tau = (1.30\text{ m})(4.20\text{ N}) \sin 75^\circ - (2.15\text{ m})(4.90\text{ N}) \sin 60^\circ$$

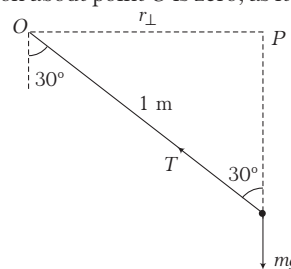
$$= -3.85\text{ N-m}$$

Example 9.21 A small ball of mass 1 kg is attached to one end of a 1 m long massless string and the other end of the string is hung from a point. When the resulting pendulum is 30° from the vertical, what is the magnitude of torque about the point of suspension? [Take, $g = 10\text{ ms}^{-2}$]

Sol. Two forces are acting on the ball

- tension (T)
- weight (mg)

Torque of tension about point O is zero, as it passes through O .



$$\tau_{mg} = F \times r_{\perp}$$

Here, $r_{\perp} = OP = 1 \sin 30^\circ = 0.5\text{ m}$

$\therefore \tau_{mg} = (mg)(0.5) = (1)(10)(0.5) = 5\text{ N-m}$

Example 9.22 A solid flywheel of mass 20 kg and radius 120 mm revolves at 600 rev min^{-1} . With what force must a brake lining be pressed against it for the flywheel to stop in 3 s , if the coefficient of friction is 0.1 ?

Sol. Given, revolutions per minute, $n_0 = 600\text{ rev min}^{-1}$

$$= \frac{600}{60}\text{ rev s}^{-1}$$

Revolutions per second, $n_0 = 10\text{ rev s}^{-1}$

So, initial angular velocity, $\omega_0 = 2\pi n_0 = (2\pi)(10) = 20\pi \text{ rad s}^{-1}$

Let α be the constant angular retardation, then applying

$$\omega = \omega_0 - \alpha t$$

$$\text{or } 0 = (20\pi) - 3(\alpha) \text{ or } \alpha = \frac{20}{3} \pi \text{ rad s}^{-2}$$

$$\text{Further, } \alpha = \frac{\tau}{I}$$

$$\text{Here, torque } \tau = \mu NR \quad (R = \text{radius})$$

$$\text{or } \tau = \mu FR \quad (F = \text{applied force})$$

$$\text{as } N = F \text{ and } I = \frac{1}{2} mR^2$$

From the above equations, we get

$$\frac{20}{3} \pi = \frac{\mu FR}{\frac{1}{2} mR^2} = \frac{2\mu F}{mR}$$

$$\text{or Force, } F = \frac{10 \pi mR}{3\mu}$$

Substituting the values, we have

$$F = \frac{10 \times 22 \times 20 \times 0.12}{3 \times 7 \times 0.1} = 251.43 \text{ N}$$

Work done by torque

Consider a rigid body acted upon by a force F at a perpendicular distance r from the axis of rotation. Let under the action of this force, the body rotates through an angle $d\theta$.

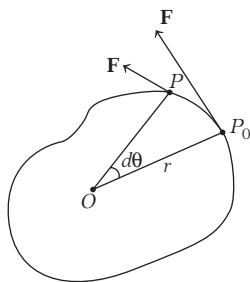


Fig. 9.9

$$\text{Work done} = \text{Torque} \times \text{Displacement} = \tau d\theta$$

The power associated with the work done by a torque acting on a rotating body is given by

$$\text{Power} = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau \omega$$

where, $\tau \omega$ is also known as **instantaneous rotational power**.

Example 9.23 To maintain a rotor at a uniform angular speed of 100 rad s^{-1} , an engine needs to transmit a torque of 200 N-m . What is the power required by engine? Assume efficiency of engine is 100%.

Sol. Work done by torque in turning rotor by an angle $d\theta$ is $\tau d\theta$, so power delivered by engine,

$$P = \frac{\text{Work done}}{\text{Time taken}} = \tau \frac{d\theta}{dt} \quad (dt = \text{time taken for turning by angle } d\theta)$$

$$\text{or } P = \tau \omega$$

$$\text{So, power required} = 200 \times 100 = 20000 \text{ W} \\ = 20 \text{ kW} \quad [\because 1 \text{ kW} = 1000 \text{ W}]$$

Example 9.24 The power output of an automobile engine is advertised to be 200 HP at 6000 rpm . What is the corresponding torque?

$$\text{Sol. } P = 200 \text{ HP} = 200 \times 746 \quad [\because 1 \text{ HP} = 746 \text{ W}] \\ = 1.49 \times 10^5 \text{ W} \\ \text{and } \omega = 6000 \text{ rev/min} \\ = 6000 \times \frac{2\pi}{60} = 628 \text{ rad/s} \\ \tau = \frac{P}{\omega} = \frac{1.49 \times 10^5}{628} = 237.3 \text{ N-m}$$

Example 9.25 A flywheel of moment of inertia 10 kg-m^2 is rotating at 50 rad/s . It must be brought to rest in 10 s .

(i) How much work must be done to stop it?

(ii) What is the required average power?

Sol. (i) Given, $\omega_0 = 50 \text{ rad/s}$ and $t = 10 \text{ s}$

From the equation, $\omega = \omega_0 - \alpha t$

$$0 = 50 - \alpha \times 10$$

$$\alpha = 5 \text{ rad/s}^2$$

$$\text{Again, } \theta = \omega_0 t - \frac{1}{2} \alpha t^2 = 50 \times 10 - \frac{1}{2} \times 5 \times (10)^2 \\ = 500 - 250 = 250 \text{ rad}$$

$$\text{Retarding torque, } \tau = I\alpha = 10 \times 5 = 50 \text{ N-m}$$

Work done by retarding torque,

$$W = \tau \theta = 50 \times 250 = 12500 \text{ J}$$

(ii) Required average power,

$$P = \frac{W}{t} = \frac{12500}{10} = 1250 \text{ W}$$

ANGULAR MOMENTUM

A mass moving in a straight line has linear momentum ($\mathbf{p}$). When a mass rotates about some point/axis, there is momentum associated with rotational motion called the **angular momentum ($\mathbf{L}$)**.

Just as net external force is required to change the linear momentum of an object, a net external torque is required to change the angular momentum of an object.

Angular momentum of a particle about some point

Angular momentum of a particle (of mass m) about a point is defined as the moment of its linear momentum ($\mathbf{p}$) about the point.

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times (m\mathbf{v}) = m(\mathbf{r} \times \mathbf{v})$$

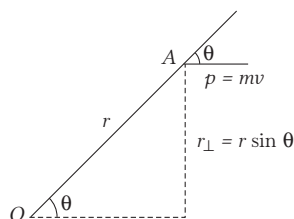


Fig. 9.10

Here, $\mathbf{r}$ is the radius vector of particle A about O at that instant of time. The magnitude of $\mathbf{L}$ is

$$L = mvr \sin \theta = mvr_{\perp}$$

Here, $r_{\perp} = r \sin \theta$ is the perpendicular distance of line of action of velocity $\mathbf{v}$ from point O. The direction of $\mathbf{L}$ is same as that of $\mathbf{r} \times \mathbf{v}$. The SI unit of angular momentum is $\text{kg-m}^2\text{s}^{-1}$ and its dimensions are $[\text{ML}^2\text{T}^{-1}]$.

Important concept related to angular momentum

Angular momentum is given by $\mathbf{L} = \mathbf{r} \times \mathbf{p}$

We can obtain the three rectangular components of angular momentum, i.e. L_x, L_y and L_z as we did in the case of torque.

$$\mathbf{L} = L_x \hat{i} + L_y \hat{j} + L_z \hat{k}$$

$$\mathbf{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$\mathbf{p} = p_x \hat{i} + p_y \hat{j} + p_z \hat{k}$$

Using determinant method, we get

$$\mathbf{L} = (L_x \hat{i} + L_y \hat{j} + L_z \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix}$$

$$= (yp_z - zp_y) \hat{i} + (zp_x - xp_z) \hat{j} + (xp_y - yp_x) \hat{k}$$

Comparing the components of both sides, we get

$$L_x = (yp_z - zp_y) \quad \dots(i)$$

$$L_y = (zp_x - xp_z) \quad \dots(ii)$$

$$L_z = (xp_y - yp_x) \quad \dots(iii)$$

If the particle is moving in yz-plane, it has the angular momentum component only in x-direction,

i.e. $L_y = L_z = 0$ and $L_x \neq 0$.

Similarly, if the particle is moving in xz-plane, $L_y \neq 0, L_x = L_z = 0$ and if the particle is moving in xy-plane only, $L_z \neq 0, L_x = L_y = 0$.

Example 9.26 A car of mass 300 kg is travelling on a circular track of radius 100 m with a constant speed of 60 m/s. Calculate the angular momentum.

Sol. Momentum, $p = mv = 300 \times 60 = 18000 \text{ kg-m/s}$

(directed along the tangent to the circle)

Angular momentum, $\mathbf{L} = \mathbf{r} \times \mathbf{p}$

$$L = rp \sin \theta \quad (\text{Here, } \theta = 90^\circ)$$

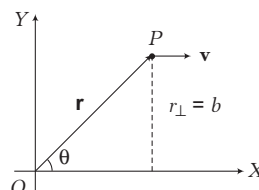
$$\therefore L = rp = 100 \times 18000 = 18 \times 10^5 \text{ kg-m}^2\text{s}^{-1}$$

Example 9.27 A particle of mass m is moving along the line $y = b, z = 0$ with constant speed v . State whether the angular momentum of particle about origin is increasing, decreasing or constant.

Sol. Angular momentum of particle,

$$|\mathbf{L}| = mvr \sin \theta$$

$$= mvr_{\perp} = mvb \quad (\text{Given } y = r_{\perp} = b)$$



$\therefore |\mathbf{L}| = \text{constant}$, as m, v and b all are constants.

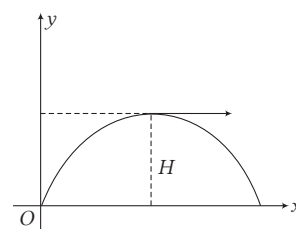
Direction of $\mathbf{r} \times \mathbf{v}$ also remains the same. Therefore, angular momentum of particle about origin remains constant with time.

Note In this problem, $|\mathbf{r}|$ is increasing, θ is decreasing but $r \sin \theta$, i.e. b remains constant. Hence, the angular momentum remains constant.

Example 9.28 A particle of mass m is projected with velocity v at an angle θ with the horizontal. Find its angular momentum about the point of projection when it is at the highest point of its trajectory.

Sol. At the highest point, it has only horizontal velocity

$$v_x = v \cos \theta \quad \dots(i)$$



Length of the perpendicular to the horizontal velocity from O is the maximum height, where

$$H_{\max} = \frac{v^2 \sin^2 \theta}{2g}$$

Angular momentum $= mvr = mv_x H_{\max}$

$$= mv_x \left(\frac{v^2 \sin^2 \theta}{2g} \right)$$

$$= \frac{mv^2 \sin^2 \theta}{2g} v \cos \theta \quad [\text{from Eq. (i)}]$$

$$= \frac{mv^3 \sin^2 \theta \cos \theta}{2g}$$

Example 9.29 Find the components along the X, Y, Z-axes of the angular momentum J of a particle, whose position vector is $\mathbf{r}$ with components x, y and z and linear momentum is $\mathbf{p}$ with components p_x, p_y and p_z . Show that if the particle moves only in the xy -plane, then the angular momentum has only a z component.

Sol. The position vector $\mathbf{r}$ and the linear momentum $\mathbf{p}$ of the particle are written in terms of x, y, z components as

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

and

$$\mathbf{p} = p_x\hat{\mathbf{i}} + p_y\hat{\mathbf{j}} + p_z\hat{\mathbf{k}}$$

By definition, the angular momentum of the particle is

$$\begin{aligned} \mathbf{J} &= \mathbf{r} \times \mathbf{p} = (x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}) \times (p_x\hat{\mathbf{i}} + p_y\hat{\mathbf{j}} + p_z\hat{\mathbf{k}}) \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix} \\ &= \hat{\mathbf{i}}(yp_z - zp_y) + \hat{\mathbf{j}}(zp_x - xp_z) + \hat{\mathbf{k}}(xp_y - yp_x) \quad \dots(i) \end{aligned}$$

The angular momentum may be written in terms of x, y, z components as

$$\mathbf{J} = J_x\hat{\mathbf{i}} + J_y\hat{\mathbf{j}} + J_z\hat{\mathbf{k}} \quad \dots(ii)$$

Comparing Eqs. (i) and (ii), we have

$$J_x = yp_z - zp_y,$$

$$J_y = zp_x - xp_z,$$

and

$$J_z = xp_y - yp_x.$$

If the particle moves only in the xy -plane, then $z = 0$ and $p_z = 0$. Now, Eq. (i) becomes

$$\mathbf{J} = (xp_y - yp_x)\hat{\mathbf{k}}$$

i.e. $\mathbf{J}$ has only z -component.

Angular momentum of a rigid body rotating about a fixed axis

The angular momentum of rigid body, i.e. a system of particles is the sum of angular momentum of all the particles within the system.

Suppose a particle P of mass m is going in a circle of radius r and at some instant the speed of the particle is v . For finding the angular momentum of the particle about the axis of rotation, the origin may be chosen anywhere on the axis. We choose it at the centre of the circle. In this case, $\mathbf{r}$ and $\mathbf{p}$ are perpendicular to each other and $\mathbf{r} \times \mathbf{p}$ is along the axis. Thus, component of $\mathbf{r} \times \mathbf{p}$ along the axis is mvr itself. The angular momentum of the whole rigid body about AB is the sum of components of all particles, i.e.

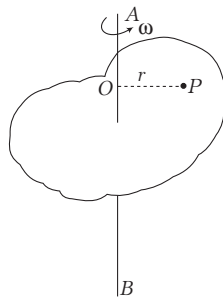


Fig. 9.11

$$L = \sum_i m_i r_i v_i$$

Here,

$$v_i = r_i \omega$$

$\therefore$

$$L = \sum_i m_i r_i^2 \omega \quad \text{or} \quad L = \omega \sum_i m_i r_i^2$$

or

$$L = I\omega \quad (\text{as } \sum_i m_i r_i^2 = I)$$

Here, I is the moment of inertia of the rigid body about AB .

- The relation between torque and angular momentum can be given as follows

As angular momentum, $\mathbf{L} = \mathbf{r} \times \mathbf{p}$

$$\begin{aligned} \therefore \frac{d\mathbf{L}}{dt} &= \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt} \\ &= \mathbf{v} \times m\mathbf{v} + \mathbf{r} \times \mathbf{F} = 0 + \mathbf{r} \times \mathbf{F} = \boldsymbol{\tau}_{\text{ext}} \\ \Rightarrow \frac{d\mathbf{L}}{dt} &= \boldsymbol{\tau}_{\text{ext}} \end{aligned}$$

Thus, **rate of change of angular momentum is equal to the torque due to an external force.**

- Angular impulse = $\int \boldsymbol{\tau} dt = \int d\mathbf{L} = \mathbf{L}_f - \mathbf{L}_i$

Hence, **angular impulse of torque is equal to total change in angular momentum of the body in given time.**

Example 9.30 A 40 kg flywheel in the form of a uniform circular disc of 1m radius is making 120 rpm. Determine the angular momentum.

Sol. Here, $M = 40 \text{ kg}$, $R = 1 \text{ m}$, $v = 120 \text{ rpm} = \frac{120}{60} \text{ rps} = 2 \text{ rps}$

Now, moment of inertia, $I = \frac{1}{2}MR^2 = \frac{1}{2} \times 40 \times (1)^2 = 20 \text{ kg-m}^2$

and $\omega = 2\pi v = 2\pi \times 2 = 4\pi \text{ rad s}^{-1}$

So, angular momentum, $L = I\omega = 20 \times 4\pi = 20 \times 4 \times 3.14$

$$L = 251.2 \text{ kg-m}^2\text{s}^{-1}$$

Example 9.31 A torque of 10 N-m is applied on a wheel having angular momentum of $2 \text{ kg-m}^2\text{s}^{-1}$, calculate the angular momentum of the wheel after 4 s.

Sol. Torque, $\tau = \frac{dL}{dt}$

$$\Rightarrow \tau = \frac{L_2 - L_1}{t}$$

Angular momentum of the wheel,

$$\begin{aligned} L_2 &= \tau t + L_1 = 10 \times 4 + 2 \\ &= 40 + 2 = 42 \text{ kg-m}^2\text{s}^{-1} \end{aligned}$$

Example 9.32 The diameter of a solid disc is 0.5 m and its mass is 16 kg. What torque will increase its angular velocity from 0 to 120 rotations/min in 8 s?

Sol. Moment of inertia of solid disc,

$$I = \frac{1}{2}MR^2 = \frac{1}{2} \times 16 \times \left(\frac{0.5}{2}\right)^2 = \frac{1}{2} \text{ kg-m}^2$$

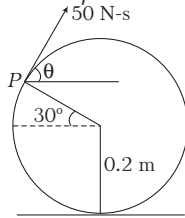
Angular velocity, $\omega = \frac{120 \times 2\pi}{60} = 4\pi$ rad/s and change in

angular momentum, $\Delta L = I\omega - 0 = I\omega$

Angular impulse, $\tau \times t = \Delta L$

$$\Rightarrow \tau = \frac{\Delta L}{t} = \frac{I\omega}{t} = \frac{1}{8} \times \frac{1}{2} \times 4\pi = \frac{\pi}{4} \text{ N-m}$$

Example 9.33 A solid ball of radius 0.2 m and mass 1 kg is given an instantaneous impulse of 50 N-s at point P as shown in figure. Find the number of rotations made by the ball about its diameter before hitting the ground. Initially, the ball is kept on smooth surface.



Sol. Impulse gives translational velocity,

$$u = \frac{\text{Impulse}}{\text{Mass}} \text{ (along impulse)}$$

$$= \frac{50}{1} = 50 \text{ m/s}$$

T = time of flight of projectile

$$= \frac{2u \sin \theta}{g} = \frac{2 \times 50 \times \sin 60^\circ}{10} = 5\sqrt{3} \text{ s}$$

Impulse also gives angular velocity,

$$\omega = \frac{\text{Impulse} \times R}{I} \text{ or } \omega = \frac{\text{Impulse} \times R}{\frac{2}{5} mR^2}$$

$$= \frac{5 \times \text{Impulse}}{2 \times mR} = \frac{5 \times 50}{2 \times 1 \times 0.2} = 625 \text{ rad/s}$$

$$\text{Number of rotations, } n = \frac{\omega T}{2\pi} = \frac{3125\sqrt{3}}{2\pi}$$

Law of conservation of angular momentum

It states that when no external torque acts on an object or isolated system of objects, then no change of angular momentum occurs. Hence, the angular momentum before an event involving only internal torques or no torques is equal to the angular momentum after the event.

We know that, $\tau_{\text{ext}} = \frac{dL}{dt}$

If $\tau_{\text{ext}} = 0$, then $\frac{dL}{dt} = 0$ or L must be a constant.

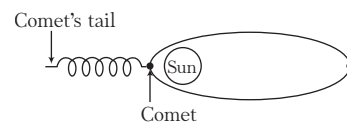
Therefore, **in the absence of any external torque, the total angular momentum of a system must remain conserved.**

As $L = I\omega$, the law of conservation of momentum leads us to the following conclusion.

For an isolated system, $I\omega = \text{constant}$ or $I_1\omega_1 = I_2\omega_2$

This principle is often used by gymnast, swimmers, circus acrobats and ballet dancers cat while catching birds, etc. Constant angular momentum is extremely useful when dealing with the orbits of planets and satellites and also when analysing the Bohr's model of the atom.

Example 9.34 The maximum and minimum distances of a comet from sun are $1.4 \times 10^{12} \text{ m}$ and $7 \times 10^{10} \text{ m}$, respectively. If velocity nearest to sun is $6 \times 10^4 \text{ ms}^{-1}$, what is the velocity of comet when it is farthest from the sun?



Sol. Consider comet as a point mass and its path as circular.

Applying conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$

$$mr_1^2 \times \frac{v_1}{r_1} = mr_2^2 \times \frac{v_2}{r_2} \Rightarrow v_1 r_1 = v_2 r_2$$

$$\Rightarrow v_2 = \frac{v_1 r_1}{r_2} = \frac{6 \times 10^4 \times 1.4 \times 10^{12}}{7 \times 10^{10}} = 3000 \text{ ms}^{-1}$$

Example 9.35 A wheel of moment of inertia I and radius R is rotating about its axis at an angular speed ω_0 . It picks up a stationary particle of mass m at its edge. Find the new angular speed of the wheel.

Sol. Net external torque on the system is zero. Therefore, angular momentum will remain conserved, thus

$$I_1\omega_1 = I_2\omega_2 \text{ or } \omega_2 = \frac{I_1\omega_1}{I_2}$$

Here,

$$I_1 = I, \omega_1 = \omega_0, I_2 = I + mR^2$$

$\therefore$

$$\omega_2 = \frac{I\omega_0}{I + mR^2}$$

Example 9.36 If the radius of the earth contracts to half of its present value without change in its mass, what will be the new duration of the day?

Sol. Present angular momentum of earth, $L_1 = I\omega = \frac{2}{5} MR^2\omega$

New angular momentum due to change in radius,

$$L_2 = \frac{2}{5} M \left(\frac{R}{2} \right)^2 \omega'$$

If external torque is zero, then angular momentum must be conserved, i.e. $L_1 = L_2$

$$\frac{2}{5} MR^2\omega = \frac{1}{4} \times \frac{2}{5} MR^2\omega'$$

$$\text{i.e. } \omega' = 4\omega$$

$$\therefore \text{New duration of the day, } T' = \frac{1}{4} T \quad \left(\because \omega = \frac{2\pi}{T} \right)$$

$$= \frac{1}{4} \times 24 = 6 \text{ h}$$

ROTATIONAL KINETIC ENERGY OF THE PARTICLES

The energy due to rotational motion of a body is known as rotational kinetic energy.

Let us consider a body rotating about a fixed axis. For a rotating rigid body, kinetic energy is the sum of kinetic energies of individual particles.

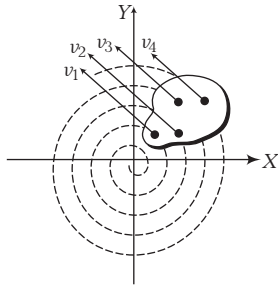


Fig. 9.12 Rotation of a rigid body

So, kinetic energy of a rotating body,

$$K = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \dots + \frac{1}{2}m_nv_n^2$$

or

$$K = \sum_{i=1}^n \frac{1}{2}m_iv_i^2$$

Substituting $v = r\omega$, $K = \sum_{i=1}^n \frac{1}{2}m_ir_i^2\omega^2$

where, m_i is the mass of i th particle and v_i is its linear velocity.

As, the angular velocity ω is same for all the particles of rigid body.

$$\text{So, } K = \frac{1}{2}\omega^2 \left(\sum_{i=1}^n m_ir_i^2 \right) \quad \dots(i)$$

The quantity $\sum (m_ir_i^2)$ depends on distribution of mass around axis of rotation. This quantity is called moment of inertia (I).

$$\text{Kinetic energy, } K = \frac{1}{2}I\omega^2 \quad \dots(ii)$$

When $\omega = 1$, rotational kinetic energy $= \frac{1}{2}I$

or Moment of inertia, $I = 2 \times \text{Rotational kinetic energy}$

Hence, **the moment of inertia of a rigid body about an axis of rotation is numerically equal to twice the rotational kinetic energy of the body when it is rotating with unit angular velocity about that axis.**

Rotational kinetic energy depends upon axis of rotation.

Example 9.37 A wheel is rotating at a rate of 1000 rpm and its kinetic energy is 10^6 J. Determine the moment of inertia of the wheel about its axis of rotation.

Sol. Given, $v = 1000 \text{ rpm} = \frac{1000}{60} \text{ rps} = \frac{50}{3} \text{ rps}$

$$\therefore \omega = 2\pi v = \frac{100\pi}{3} \text{ rads}^{-1}$$

$$\therefore \text{Rotational kinetic energy} = \frac{1}{2}I\omega^2$$

$$\therefore 10^6 = \frac{1}{2} \times I \times \left(\frac{100\pi}{3} \right)^2$$

$$\Rightarrow I = \frac{2 \times 10^6 \times 9}{(100)^2 \pi^2} = \frac{200 \times 9}{9.87} = 182.4 \text{ kg-m}^2$$

Example 9.38 A flywheel of mass 0.2 kg and radius 10 cm is rotating with $(5/\pi)$ rev/s about an axis perpendicular to its plane passing through its centre. Calculate angular momentum and kinetic energy of flywheel.

Sol. Angular velocity, $\omega = 2\pi v = 2\pi \times \frac{5}{\pi} = 10 \text{ rad/s}$

and moment of inertia, $I = mr^2 = (0.2)(0.1)^2 = 2 \times 10^{-3} \text{ kg-m}^2$

$$\begin{aligned} \text{Angular momentum} &= I\omega = 2 \times 10^{-3} \times 10 = 2 \times 10^{-2} \text{ J-s} \\ &= 2 \times 10^{-2} \text{ kg-m}^2\text{s}^{-1} \end{aligned}$$

$$\text{Kinetic energy} = \frac{1}{2}I\omega^2 = \frac{1}{2} \times 2 \times 10^{-3} \times (10)^2 = 0.1 \text{ J}$$

Work-energy theorem in rotational motion

According to work-energy theorem, the change in the rotational kinetic energy of a rigid body is equal to the work done by external torques acting on the body.

Work done by torque = Change in rotational kinetic energy

$$W = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$$

Example 9.39 An energy of 484 J is spent in increasing the speed of a flywheel from 60 rpm to 360 rpm. Calculate moment of inertia of flywheel.

Sol. Given, energy spent, $W = 484 \text{ J}$

$$\text{Initial speed, } \omega_1 = 60 \text{ rpm} = \frac{60}{60} \times 2\pi = 2\pi \text{ rad s}^{-1}$$

$$\text{Final speed, } \omega_2 = 360 \text{ rpm} = \frac{360}{60} \times 2\pi = 12\pi \text{ rad s}^{-1}$$

Moment of inertia, $I = ?$

$$\begin{aligned}
 \text{Energy spent, } W &= E_2 - E_1 = \frac{1}{2} I \omega_2^2 - \frac{1}{2} I \omega_1^2 \\
 &= \frac{1}{2} I [(12\pi)^2 - (2\pi)^2] = 70 I \pi^2 \\
 \Rightarrow I &= \frac{484}{70 \times \pi^2} = 0.7 \text{ kg-m}^2
 \end{aligned}$$

Example 9.40 If the rotational kinetic energy of a body is increased by 300%, then determine percentage increase in its angular momentum.

Sol. Percentage increase in angular momentum,

$$= \frac{L_2 - L_1}{L_1} \times 100$$

$$\text{As, } L \propto \sqrt{E} \text{ and } E_1 = E$$

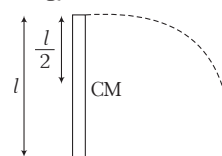
$$\Rightarrow E_2 = E + \frac{300}{100} E$$

$$\Rightarrow E_2 = 4E$$

$$\begin{aligned}
 \text{Increase in momentum} &= \frac{\sqrt{E_2} - \sqrt{E_1}}{\sqrt{E_1}} \times 100 \\
 &= \frac{\sqrt{4E} - \sqrt{E}}{\sqrt{E}} \times 100 = 100\%
 \end{aligned}$$

Example 9.41 A thin metre scale is kept vertical by placing its one end on floor. Keeping the end in contact stationary, it is allowed to fall. Calculate the velocity of its upper end when it hits the floor.

Sol. Loss in potential energy = Gain in rotational kinetic energy



$$\Rightarrow \frac{mgl}{2} = \frac{1}{2} I \omega^2 = \frac{1}{2} \frac{ml^2}{3} \times \frac{v^2}{l^2} \Rightarrow v = \sqrt{3gl}$$

CHECK POINT 9.2

- The torque of a force $\mathbf{F} = -6\hat{\mathbf{i}}$ acting at a point $\mathbf{r} = 4\hat{\mathbf{j}}$ about origin will be
 (a) $-24\hat{\mathbf{k}}$ (b) $24\hat{\mathbf{k}}$
 (c) $24\hat{\mathbf{j}}$ (d) $24\hat{\mathbf{i}}$
- Moment of a force of magnitude 20 N acting along positive x-direction at point (3m, 0, 0) about the point (0, 2m, 0) has magnitude of (in N-m)
 (a) 20 (b) 60
 (c) 40 (d) 30
- The torque of force $\mathbf{F} = -3\hat{\mathbf{i}} + \hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ acting on a point $\mathbf{r} = 7\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ about origin will be
 (a) $14\hat{\mathbf{i}} - 38\hat{\mathbf{j}} + 16\hat{\mathbf{k}}$ (b) $4\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 6\hat{\mathbf{k}}$
 (c) $-14\hat{\mathbf{i}} + 38\hat{\mathbf{j}} - 16\hat{\mathbf{k}}$ (d) $-21\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$
- A door 1.6 m wide requires a force of 1 N to be applied at the free end to open or close it. The force that is required at a point 0.4 m distant from the hinges for opening or closing the door is
 (a) 1.2 N (b) 3.6 N
 (c) 2.4 N (d) 4 N
- A flywheel of moment of inertia 2 kg-m^2 is rotated at a speed of 30 rad/s. A tangential force at the rim stops the wheel in 15 s. Average torque of the force is
 (a) 4 N-m (b) 2 N-m
 (c) 8 N-m (d) 1 N-m
- A mass of 10 kg connected at the end of a rod of negligible mass is rotating in a circle of radius 30 cm with an angular velocity of 10 rad/s. If this mass is brought to rest in 10 s by a brake, what is the magnitude of the torque applied?
 (a) 0.9 N-m (b) 1.2 N-m
 (c) 2.3 N-m (d) 0.5 N-m
- A disc is rotating with angular velocity ω . A force $\mathbf{F}$ acts at a point whose position vector with respect to the axis of rotation is $\mathbf{r}$. The power associated with torque due to the force is given by
 (a) $(\mathbf{r} \times \mathbf{F}) \cdot \omega$ (b) $(\mathbf{r} \times \mathbf{F}) \times \omega$ (c) $\mathbf{r} \times (\mathbf{F} \cdot \omega)$ (d) $\mathbf{r} \cdot (\mathbf{F} \times \omega)$
- Angular momentum is
 (a) moment of momentum
 (b) product of mass and angular velocity
 (c) product of moment of inertia and velocity
 (d) moment in angular motion
- The unit mass has $\mathbf{r} = 8\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$ and $\mathbf{v} = 8\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$. Its angular momentum is
 (a) 64 units in $-\hat{\mathbf{k}}$ direction
 (b) 64 units in $+\hat{\mathbf{k}}$ direction
 (c) 64 units in $+\hat{\mathbf{j}}$ direction
 (d) 64 units in $+\hat{\mathbf{i}}$ direction
- If the earth is a point mass of $6 \times 10^{24} \text{ kg}$ revolving around the sun at a distance of $1.5 \times 10^8 \text{ km}$ and in time $T = 3.14 \times 10^7 \text{ s}$, then the angular momentum of the earth around the sun is
 (a) $1.2 \times 10^{18} \text{ kg-m}^2\text{s}^{-1}$ (b) $1.8 \times 10^{29} \text{ kg-m}^2\text{s}^{-1}$
 (c) $1.5 \times 10^{37} \text{ kg-m}^2\text{s}^{-1}$ (d) $2.7 \times 10^{40} \text{ kg-m}^2\text{s}^{-1}$
- A particle with the position vector $\mathbf{r}$ has linear momentum $\mathbf{p}$. Which of the following statement is true in respect of its angular momentum $\mathbf{L}$ about the origin?
 (a) $\mathbf{L}$ acts along $\mathbf{p}$.
 (b) $\mathbf{L}$ acts along $\mathbf{r}$.
 (c) $\mathbf{L}$ is maximum when $\mathbf{p}$ and $\mathbf{r}$ are parallel.
 (d) $\mathbf{L}$ is maximum when $\mathbf{p}$ is perpendicular to $\mathbf{r}$.

- 12.** By keeping moment of inertia of a body constant, if we double the time period, then angular momentum of body
 (a) remains constant (b) becomes half
 (c) doubles (d) quadruples
- 13.** If torque is zero, then
 (a) angular momentum is conserved
 (b) linear momentum is conserved
 (c) energy is conserved
 (d) angular momentum is not conserved
- 14.** The angular momentum of a rotating body changes from A_0 $\text{kg m}^2\text{min}^{-1}$ to $4A_0$ $\text{kg m}^2\text{min}^{-1}$ in 4 min. The torque acting on the body is
 (a) $\frac{3}{4}A_0$ (b) $4A_0$ (c) $3A_0$ (d) $\frac{3}{2}A_0$
- 15.** A disc of mass 2 kg and radius 0.2 m is rotating with angular velocity 30 rad s^{-1} . What is angular velocity, if a mass of 0.25 kg is put on periphery of the disc?
 (a) 24 rad s^{-1} (b) 36 rad s^{-1}
 (c) 15 rad s^{-1} (d) 26 rad s^{-1}
- 16.** If the radius of earth contracts $1/n$ of its present day value, the length of the day will be approximately
 (a) $\frac{24}{n} \text{ h}$ (b) $\frac{24}{n^2} \text{ h}$
 (c) $24n \text{ h}$ (d) $24n^2 \text{ h}$
- 17.** A thin circular ring of mass M and radius R is rotating about its axis with a constant angular velocity ω . Two objects each of mass m are attached gently to the ring. The wheel now rotates with an angular velocity is
 (a) $\frac{\omega M}{(m + M)}$ (b) $\frac{\omega (M - 2m)}{(M + 2m)}$
 (c) $\frac{\omega M}{(M + 2m)}$ (d) $\frac{\omega (M + 2m)}{M}$
- 18.** A circular disc of mass 2 kg and radius 1 m is rotating about an axis perpendicular to its plane and passing through its centre of mass with a rotational kinetic energy of 8 J. The angular momentum (in J-s) is
 (a) 8 (b) 4
 (c) 2 (d) 1

ROLLING MOTION

When a body performs translatory motion as well as rotational motion, then this type of motion is known as rolling motion. *e.g.* Motion of football rolling on a surface. In rolling motion, the velocity of centre of mass represents linear motion while angular velocity represents rotational motion.

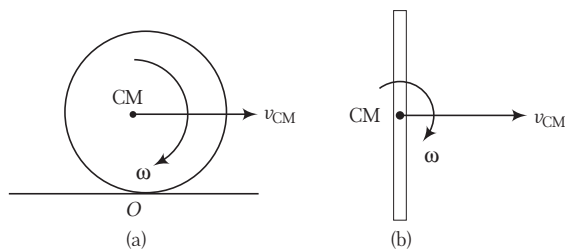


Fig. 9.13

The kinetic energy of a rolling body is equal to the sum of kinetic energies of translational and rotational motion.

$\therefore$ Total kinetic energy of a rolling body = Rotational kinetic energy + Translational kinetic energy

$$= \frac{1}{2} I \omega^2 + \frac{1}{2} m v_{\text{CM}}^2$$

Example 9.42 A disc of mass 5 kg and radius 50 cm rolls on the ground at the rate of 10 ms^{-1} . Determine the kinetic energy of the disc.

Sol. Given, mass of the disc, $M = 5 \text{ kg}$

Radius of the disc, $R = 50 \text{ cm} = \frac{1}{2} \text{ m}$

Linear velocity of the disc, $v = 10 \text{ ms}^{-1}$

$$\therefore v = R\omega$$

$$\Rightarrow 10 = \frac{1}{2} \omega$$

$$\Rightarrow \omega = 10 \times 2 = 20 \text{ rad s}^{-1}$$

Also, moment of inertia of disc about an axis through its centre,

$$I = \frac{1}{2} MR^2$$

$$\Rightarrow I = \frac{1}{2} \times 5 \times \left(\frac{1}{2}\right)^2 = \frac{5}{8} \text{ kg-m}^2$$

$$\text{Translational kinetic energy} = \frac{1}{2} Mv^2 = \frac{1}{2} \times 5 \times (10)^2 = 250 \text{ J}$$

$$\begin{aligned} \text{Rotational kinetic energy} &= \frac{1}{2} I \omega^2 \\ &= \frac{1}{2} \times \frac{5}{8} \times (20)^2 = 125 \text{ J} \end{aligned}$$

$$\therefore \text{Total kinetic energy} = (250 + 125) \text{ J} = 375 \text{ J}$$

Classification of rolling motion

Depending on the fact that relative velocity of point of contact of the body undergoing rolling motion, with the platform is zero or non-zero. Rolling motion is classified into two categories, which are as follows

- Pure rolling or rolling without slipping/sliding or perfect rolling motion.
- Impure rolling or rolling with slipping/sliding or imperfect rolling motion.

Pure rolling

If the velocity of point of contact with respect to the surface is zero, then it is known as pure rolling.

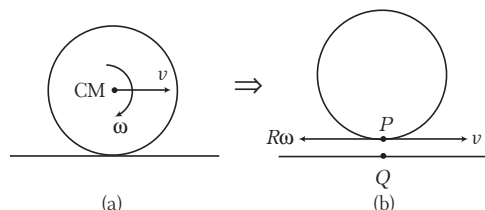


Fig. 9.14

For pure rotational motion, velocity of centre of mass is zero given by the relation, $v = R\omega$

In the above figure, when we observe the motion of contact point P on the disc, we find that it has two simultaneous motions

- Linear forward velocity v because whole body is moving forward.
- Linear backward velocity ωR due to rotation of the disc as a whole in clockwise direction. In case of pure rolling, these two velocities become equal.

This is the condition of pure rolling on a stationary ground. Sometimes, it is simply known as rolling.

Suppose the base over which the disc is rolling is also moving with some velocity (say v_0), then in that case condition of pure rolling is different.

e.g. In the above figure, $v_P = v_Q$ or $v - R\omega = v_0$

Thus, in this case $v - R\omega \neq 0$, but $v - R\omega = v_0$. **In uniform pure rolling, v and ω are constants**, i.e. they are neither increasing nor decreasing.

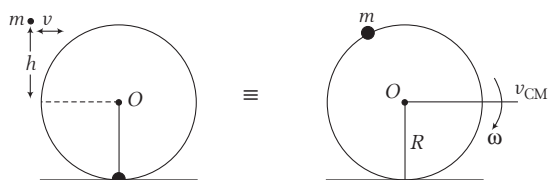
Impure rolling

If the velocity of point of contact with respect to the surface is not zero, then it is known as impure rolling. In impure rolling, the body slides on the surface in contact and it depends on the nature of surface.

In above case shown in Fig. 9.13, if $v > \omega R$, disc will start slipping on the ground.

Example 9.43 A hollow sphere of mass M lies on a rough horizontal plane when a particle of mass m travelling with speed v collides and sticks with it. If line of motion of the particle is at height h above the centre of sphere, find h if the body rolls without slipping after collision.

Sol. Mass m moving with velocity v creating rotation in sphere.



Applying conservation of linear momentum, we get

$$mv = (M + m)v_{CM}$$

$$\Rightarrow v_{CM} = \frac{mv}{(M + m)}$$

Now, applying conservation of angular momentum about O , we get

$$mvh = I_0\omega = \left(\frac{2}{3}MR^2 + mR^2\right)\omega$$

$$\omega = \frac{mvh}{\left(\frac{2}{3}M + m\right)R^2}$$

For pure rolling,

$$v_{CM} = \omega R$$

$$\Rightarrow \frac{mv}{M + m} = \frac{mvh}{\left(\frac{2}{3}M + m\right)R}$$

$$\Rightarrow h = \frac{\left(\frac{2}{3}M + m\right)R}{(M + m)}$$

$$= \frac{2R}{3} \quad (\text{Since, } m \ll M)$$

Kinetic energy of rolling motion

The kinetic energy of a rolling body is the sum of kinetic energy of translational motion and kinetic energy of rotational motion.

$\therefore$ Kinetic energy of rolling motion,

$$E = \frac{1}{2}I\omega^2 + \frac{1}{2}Mv_{CM}^2$$

Substituting the value of $I = MK^2$, where K is radius of gyration and $v_{CM} = R\omega$, we get

$$E = \frac{1}{2}MK^2\left(\frac{v_{CM}}{R}\right)^2 + \frac{1}{2}Mv_{CM}^2$$

$$\Rightarrow E = \frac{1}{2}Mv_{CM}^2\left(1 + \frac{K^2}{R^2}\right)$$

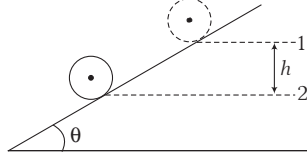
The ratio of translational energy, rotational energy and total energy can be calculated as

$$E_{\text{translational}} : E_{\text{rotational}} : E_{\text{total}}$$

$$1 : \frac{K^2}{R^2} : 1 + \frac{K^2}{R^2}$$

Example 9.44 A solid cylinder of mass m and radius r starts rolling down an inclined plane of inclination θ . Friction is enough to prevent slipping. Find the speed of its centre of mass when its centre of mass has fallen a height h .

Sol. Considering the two shown positions of the cylinder. As it does not slip, hence total mechanical energy will be conserved.



Energy at position 1, $E_1 = mgh$

Energy at position 2, $E_2 = \frac{1}{2}mv_{CM}^2 + \frac{1}{2}I_{CM}\omega^2$

$$\therefore \frac{v_{CM}}{r} = \omega \quad \text{and} \quad I_{CM} = \frac{mr^2}{2}$$

$$\Rightarrow E_2 = \frac{3}{4}mv_{CM}^2$$

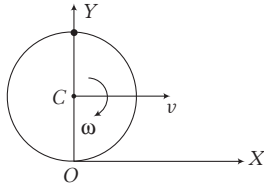
From law of conservation of energy,

$$E_1 = E_2$$

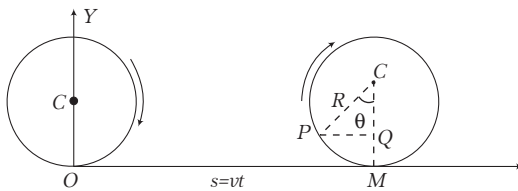
$$\Rightarrow mgh = \frac{3}{4}mv_{CM}^2$$

$$\Rightarrow v_{CM} = \sqrt{\frac{4}{3}gh}$$

Example 9.45 A disc of radius R starts at time $t = 0$ moving along the positive X -axis with linear speed v and angular speed ω . Find the x and y -coordinates of the bottom most point at any time t .



Sol. At time t , the bottom most point will rotate at an angle $\theta = \omega t$ with respect to the centre of the disc C . The centre C will travel a distance, $s = vt$.



In the figure, $PQ = R \sin \theta = R \sin \omega t$

and $CQ = R \cos \theta = R \cos \omega t$

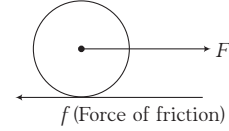
Coordinates of point P at time t are

$$x = OM - PQ = vt - R \sin \omega t$$

and $y = CM - CQ = R - R \cos \omega t$

$$\therefore (x, y) \equiv (vt - R \sin \omega t, R - R \cos \omega t)$$

Example 9.46 A horizontal force F acts on the hollow sphere at its centre as shown in figure. Coefficient of friction between ground and sphere is μ . What is the maximum value of F for which there is no slipping?



Sol. As,

$$F - f = Ma \quad \dots(i)$$

$\therefore$

$$\tau = fR = I\alpha$$

$\Rightarrow$

$$f \cdot R = \frac{2}{5}MR^2 \frac{a}{R} \quad (\because a = R\alpha)$$

$\Rightarrow$

$$f = \frac{2}{5}Ma$$

$\Rightarrow$

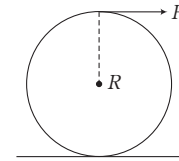
$$Ma = \frac{5}{2}f$$

$\Rightarrow$

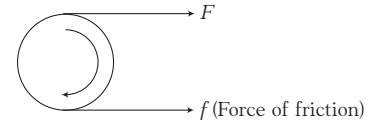
$$f = \frac{2}{7}F \quad [\text{using Eq. (i)}]$$

$$\therefore \text{For no slipping, } \frac{2}{7}F \leq \mu mg \Rightarrow F \leq \frac{7}{2}\mu mg \quad (\because f = \mu mg)$$

Example 9.47 A tangential force F acts at the top of a thin spherical shell of mass m and radius R . Find the acceleration of the shell, if it rolls without slipping.



Sol. Let f be the force of friction between the shell and the horizontal surface.



For translational motion, $F + f = ma$

$\dots(i)$

For rotational motion,

$$FR - fR = I\alpha = I \frac{a}{R} \quad (\because a = R\alpha \text{ for pure rolling})$$

$\Rightarrow$

$$F - f = I \frac{a}{R^2}$$

$\dots(ii)$

Adding Eqs. (i) and (ii), we get

$$2F = \left(m + \frac{I}{R^2}\right)a = \left(m + \frac{2}{3}m\right)a \quad \left(\because I_{\text{shell}} = \frac{2}{3}mR^2\right)$$

$$= \frac{5}{3}ma \text{ or } F = \frac{5}{6}ma \Rightarrow a = \frac{6F}{5m}$$

Rolling down on a rough inclined plane

A body of mass m and radius R rolling down a plane inclined at an angle θ with the horizontal. If the body rolls without slipping, then the centre of mass of the body moves in a straight line.

Different forces acting on the body are

- Frictional force (f) acting upwards at the point of contact with inclined plane and parallel to the inclined plane.
- Normal reaction of inclined plane (N).
- Weight of the body (mg) acting vertically downwards through the centre of mass of body.

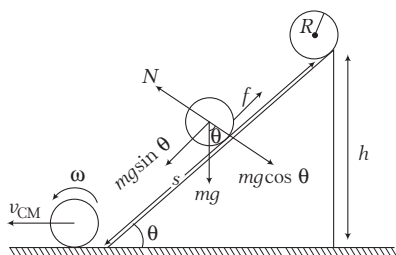


Fig. 9.15

For rolling without slipping, the friction force is static and no work is done by frictional force.

So, from law of conservation of mechanical energy,

$$mgh = \frac{1}{2}mv_{CM}^2 + \frac{1}{2}I\omega^2$$

Putting $I = mK^2$, where K = radius of gyration and $v_{CM} = R\omega$, we get

$$v_{\text{rolling}} = \sqrt{\frac{2gh}{1 + K^2/R^2}}$$

From Fig. 9.15, $h = s \sin \theta$

$$\begin{aligned} \therefore v_{\text{rolling}} &= \sqrt{\frac{2gs \sin \theta}{1 + K^2/R^2}} \\ &= \sqrt{\frac{2gs \sin \theta}{1 + n^2}} \quad (\text{where, } K = nR) \end{aligned}$$

If body slides, then $\frac{K^2}{R^2} = 0$

$$\Rightarrow v_{\text{sliding}} = \sqrt{2gh} = \sqrt{2gs \sin \theta}$$

From the above formula, we see that **velocity of rolling motion is independent of its mass (M) and radius (R).** It means all uniform solid spheres or any other similar

object will reach the bottom with same speed because the ratio K^2/R^2 will be same for similar bodies.

$$\text{Acceleration, } a_{\text{rolling}} = \frac{g \sin \theta}{1 + (K^2/R^2)} = \frac{g \sin \theta}{1 + n^2}$$

It is also independent of mass and radius.

If body slides, then $\frac{K^2}{R^2} = 0$ (For frictionless surface)

$$\therefore a_{\text{sliding}} = g \sin \theta$$

Time taken by rolling body to reach the bottom,

$$t = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g} \left(1 + \frac{K^2}{R^2} \right)}$$

A body with smaller value of K^2/R^2 will take less time to reach the bottom.

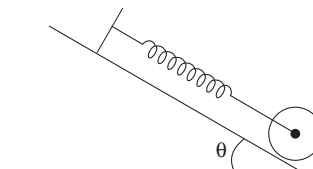
Change in kinetic energy due to rolling ($v_2 > v_1$)

$$= \frac{1}{2}m \left(1 + \frac{K^2}{R^2} \right) (v_2^2 - v_1^2)$$

Note

- For pure rolling motion, there should be friction on surface because it provides torque to the body. On smooth surface, rolling is not possible.
- Velocity of falling and sliding bodies from inclined plane are equal and is more than that of rolling body.
- For rolling along inclined plane, acceleration is maximum in case of falling and minimum in case of rolling. Falling body reaches the bottom first and rolling body in the last.
- When ring (R), disc (D), hollow sphere (H) and solid sphere (S) rolls on same inclined plane, then $v_S > v_D > v_H > v_R \Rightarrow a_S > a_D > a_H > a_R \Rightarrow t_S < t_D < t_H < t_R$

Example 9.48 A sphere of mass m attached to a spring on inclined plane as shown in figure is held in unstretched position of spring. Suddenly sphere is left free, what is the maximum extension of spring, if friction allows only rolling of sphere about horizontal diameter?



Sol. At the point of maximum extension, sphere is at rest. No work is done by frictional force in rolling.

$\Rightarrow$ Loss in gravitational potential energy = Gain in potential energy of spring

$$\Rightarrow \frac{1}{2}kx^2 = mgx \sin \theta$$

$$\Rightarrow x = \frac{2mg \sin \theta}{k}$$

Example 9.49 Calculate the kinetic energy of rolling ring of mass 0.2 kg about an axis passing through its centre of mass and perpendicular to it, if its centre of mass is moving with a velocity of 3 m/s.

Sol. Kinetic energy, $KE = \frac{1}{2}mv_{CM}^2 \left(1 + \frac{K^2}{R^2}\right)$

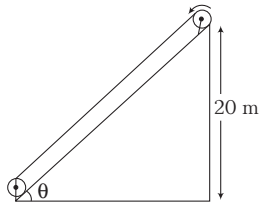
Here, for ring $\frac{K^2}{R^2} = 1$,

$m = 0.2 \text{ kg}$ and $v_{CM} = 3 \text{ m/s}$

$\therefore KE = \frac{1}{2} \times 0.2 \times 9(1 + 1) = 0.2 \times 9 = 1.8 \text{ kg-m}^2\text{s}^{-2}$

Example 9.50 A solid sphere is rolling down an inclined plane without slipping of height 20 m. Calculate the maximum velocity with which it will reach the bottom of the plane. (Take, $g = 10 \text{ m/s}^2$)

Sol. When the sphere rolls down, its potential energy changes to kinetic energy of rotation. Therefore, kinetic energy = potential energy



$$\frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2}\right) = mgh \Rightarrow v = \sqrt{\frac{2gh}{1 + K^2/R^2}}$$

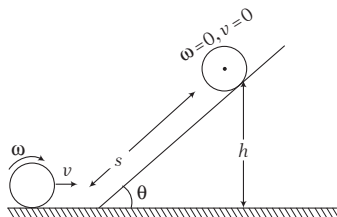
Moment of inertia of solid sphere $= \frac{2}{5}MR^2$

$$\therefore K^2 = \frac{2}{5}R^2$$

$$\begin{aligned} \text{Hence, } v &= \sqrt{\frac{2gh}{1 + \frac{2}{5}}} = \sqrt{\frac{2gh \times 5}{7}} = \sqrt{\frac{2 \times 10 \times 20 \times 5}{7}} \\ &= \sqrt{285.714} = 16.9 \text{ m/s} \end{aligned}$$

Example 9.51 A solid sphere is rolling without slipping on a level surface at a constant speed of 2 ms^{-1} . How far can it roll up on a ramp inclined at 30° before it stops?

Sol. The moment of inertia of the sphere about its axis of rotation is $I = \frac{2}{5}mr^2$. Suppose it rises to height h before stopping. Using the conservation of energy, we get



$$\begin{aligned} mgh &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right)\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{5}mv^2 = \frac{7}{10}mv^2 \quad \text{or} \quad h = \frac{7v^2}{10g} \end{aligned}$$

$\therefore$ The distance rolled up on the ramp,

$$s = \frac{h}{\sin \theta} = \frac{7v^2}{10g \sin 30^\circ} = \frac{7(2 \text{ ms}^{-1})^2}{10(9.8 \text{ ms}^{-2}) \times 1/2} = \frac{4}{7} \text{ m} \approx 57 \text{ cm}$$

Example 9.52 When a body is under pure rolling, the fraction of its total kinetic energy which is the purely rotational is $2/5$. Identify the body.

Sol. In these type of questions, calculate the expression for moment of inertia I and that helps in identifying the body.

$$\text{Rotational kinetic energy} = \frac{1}{2}I\omega^2$$

$$\text{Translational kinetic energy} = \frac{1}{2}Mv^2$$

$$\therefore \text{Total kinetic energy} = \frac{1}{2}I\omega^2 + \frac{1}{2}Mv^2$$

$$\text{Given, } \frac{2}{5} = \frac{\text{Rotational kinetic energy}}{\text{Total kinetic energy}}$$

$$\text{or } \frac{2}{5} = \frac{(1/2)I\omega^2}{(1/2)I\omega^2 + (1/2)Mv^2}$$

$$\text{For pure rolling, put } v = \omega R \Rightarrow \frac{2}{5} = \frac{I\omega^2}{I\omega^2 + M\omega^2 R^2}$$

$$\therefore I = \frac{2MR^2}{3}$$

So, the rolling body is hollow sphere.

Example 9.53 A uniform sphere of mass m and radius R rolls without slipping down an inclined plane set at an angle θ to the horizontal.

Find

- the friction coefficient when slipping is absent and
- the kinetic energy of the sphere t seconds after the beginning of motion.

Sol. (i) If a body rolls without slipping,

$$\mu_{\min} = \frac{\tan \theta}{1 + \frac{R^2}{K^2}} = \frac{\tan \theta}{1 + \frac{5}{2}} = \frac{2}{7} \tan \theta \quad \left(\text{For sphere, } \frac{K^2}{R^2} = \frac{2}{5} \right)$$

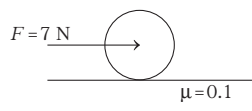
$$(ii) \text{ For pure rolling, } a = \frac{g \sin \theta}{1 + \frac{K^2}{R^2}} = \frac{g \sin \theta}{1 + \frac{5}{2}} = \frac{5}{7} g \sin \theta$$

$$\text{Linear velocity after time } t, v = u + at = 0 + \frac{5g \sin \theta}{7} t$$

In pure rolling, kinetic energy is given by

$$\begin{aligned} K &= \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2}\right) = \frac{1}{2}mv^2 \left(1 + \frac{2}{5}\right) \\ &= \frac{7}{10}mv^2 = \frac{7}{10}m \left(\frac{5g \sin \theta t}{7}\right)^2 = \frac{5}{14}mg^2 \sin^2 \theta t^2 \end{aligned}$$

CHECK POINT 9.3

- At any instant, a rolling body may be considered to be in pure rotation about an axis through the point of contact. This axis is translating forward with speed
 (a) equal to speed of centre of mass
 (b) zero
 (c) twice of speed of centre of mass
 (d) None of the above
- A wheel of radius R rolls on the ground with a uniform velocity v . The velocity of top most point relative to the bottom most point is
 (a) v
 (b) $2v$
 (c) $v/2$
 (d) zero
- The centre of a wheel rolling on a plane surface moves with a speed v_0 . A particle on the rim of the wheel at the same level as the centre will be moving at speed
 (a) zero
 (b) v_0
 (c) $\sqrt{2}v_0$
 (d) $2v_0$
- A sphere is rolling down a plane of inclination θ to the horizontal. The acceleration of its centre down the plane is
 (a) $g \sin \theta$
 (b) less than $g \sin \theta$
 (c) greater than $g \sin \theta$
 (d) zero
- Two uniform solid spheres having unequal masses and unequal radii are released from rest from the same height on a rough incline. If the spheres roll without slipping,
 (a) the heavier sphere reaches the bottom first
 (b) the bigger sphere reaches the bottom first
 (c) Both the spheres reach the bottom together
 (d) the information given is not sufficient to tell which sphere will reach the bottom first
- A solid sphere of mass m rolls without slipping on an inclined plane of inclination θ . The linear acceleration of the sphere is
 (a) $\frac{7}{5} g \sin \theta$
 (b) $\frac{2}{7} g \sin \theta$
 (c) $\frac{3}{7} g \sin \theta$
 (d) $\frac{5}{7} g \sin \theta$
- An inclined plane makes an angle of 30° with the horizontal. A solid sphere rolling down this inclined plane from rest without slipping, has a linear acceleration equal to
 (a) $\frac{g}{3}$
 (b) $\frac{2g}{3}$
 (c) $\frac{5g}{7}$
 (d) $\frac{5g}{14}$
- A thin uniform circular ring is rolling down an inclined plane of inclination 30° without slipping. Its linear acceleration along the inclined plane is
 (a) g
 (b) $\frac{g}{2}$
 (c) $\frac{g}{3}$
 (d) $\frac{g}{4}$
- The ratio of the time taken by a solid sphere and that taken by a disc of the same mass and radius to roll down a rough inclined plane from rest, from the same height is
 (a) $15 : 14$
 (b) $\sqrt{15} : \sqrt{14}$
 (c) $14 : 15$
 (d) $\sqrt{14} : \sqrt{15}$
- A solid sphere, a hollow sphere and a disc, all having same mass and radius, are placed at the top of an inclined plane and released. The coefficients of friction between the objects and the incline are same and not sufficient to allow pure rolling. Least time will be taken in reaching the bottom by
 (a) the solid sphere
 (b) the hollow sphere
 (c) the disc
 (d) All will take same time
- The speed of a uniform spherical shell after rolling down an inclined plane of vertical height h from rest is
 (a) $\sqrt{\frac{10gh}{7}}$
 (b) $\sqrt{\frac{6gh}{5}}$
 (c) $\sqrt{\frac{4gh}{5}}$
 (d) $\sqrt{2gh}$
- A solid sphere, a hollow sphere and a disc, all having same mass and radius, are placed at the top of a smooth inclined and released. Least time will be taken in reaching the bottom by
 (a) the solid sphere
 (b) the hollow sphere
 (c) the disc
 (d) all will take same time
- A solid cylinder rolls down an inclined plane of height 3 m and reaches the bottom of plane with angular velocity $2\sqrt{2} \text{ rad s}^{-1}$. The radius of cylinder must be (Take, $g = 10 \text{ ms}^{-2}$)
 (a) 5 cm
 (b) 0.5 cm
 (c) $\sqrt{10}$ cm
 (d) $\sqrt{5}$ m
- A uniform solid spherical ball is rolling down a smooth inclined plane from a height h . The velocity attained by the ball when it reaches the bottom of the inclined plane is v . If the ball is now thrown vertically upwards with the same velocity v , the maximum height to which the ball will rise is
 (a) $\frac{5h}{8}$
 (b) $\frac{3h}{5}$
 (c) $\frac{5h}{7}$
 (d) $\frac{7h}{9}$
- A solid sphere of mass 10 kg is placed on a rough surface having coefficient of friction, $\mu = 0.1$. A constant force $F = 7 \text{ N}$ is applied along a line passing through the centre of the sphere as shown in figure. The frictional force acting on the sphere will be

 (a) 1 N
 (b) 2 N
 (c) 3 N
 (d) 7 N

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1 The angular momentum of a system of particles is conserved
 - (a) when no external force acts upon the system
 - (b) when no external torque acts upon the system
 - (c) when no external impulse acts upon the system
 - (d) when axis of rotation remains same
- 2 If a person standing on a rotating disc stretches out his hands, the angular speed will
 - (a) increase
 - (b) decrease
 - (c) remain same
 - (d) None of these
- 3 A diver in a swimming pool bends his head before diving, because it
 - (a) decreases his moment of inertia
 - (b) decreases his angular velocity
 - (c) increases his moment of inertia
 - (d) decreases his linear velocity
- 4 A body is under the action of two equal and oppositely directed forces and the body is rotating with constant non-zero angular acceleration. Which of the following cannot be the separation between the lines of action of the forces?
 - (a) 1 m
 - (b) 0.4 m
 - (c) 0.25 m
 - (d) zero
- 5 Five particles of masses 2 kg each are attached to the rim of a circular disc of radius 0.1 m and negligible mass. Moment of inertia of the system about the axis passing through the centre of the disc and perpendicular to its plane is
 - (a) 1 kg-m^2
 - (b) 0.1 kg-m^2
 - (c) 2 kg-m^2
 - (d) 0.2 kg-m^2
- 6 The rotational kinetic energy of a body is E and its moment of inertia is I . The angular momentum is
 - (a) EI
 - (b) $2\sqrt{EI}$
 - (c) $\sqrt{2EI}$
 - (d) $\frac{E}{I}$
- 7 A wheel is rotating at 900 rpm about its axis. When power is cut-off, it comes to rest in 1 min. The angular retardation (in rad s^{-2}) is
 - (a) $\frac{\pi}{2}$
 - (b) $\frac{\pi}{4}$
 - (c) $\frac{\pi}{6}$
 - (d) $\frac{\pi}{8}$
- 8 A wheel is subjected to uniform angular acceleration about its axis. Initially, its angular velocity is zero. In the first 2 s, it rotates through an angle θ_1 , in the next 2 s, it rotates through an angle θ_2 . The ratio of $\frac{\theta_2}{\theta_1}$ is
 - (a) 1 : 1
 - (b) 2 : 1
 - (c) 3 : 1
 - (d) 5 : 1
- 9 A wheel is at rest. Its angular velocity increases uniformly and becomes 80 rad s^{-1} after 5 s. The total angular displacement is
 - (a) 800 rad
 - (b) 400 rad
 - (c) 200 rad
 - (d) 100 rad
- 10 A ring of diameter 0.4 m and mass 10 kg is rotating about its axis at the rate of 1200 rpm. The angular momentum of the ring is
 - (a) $60.28 \text{ kg-m}^2\text{s}^{-1}$
 - (b) $55.26 \text{ kg-m}^2\text{s}^{-1}$
 - (c) $40.28 \text{ kg-m}^2\text{s}^{-1}$
 - (d) $50.28 \text{ kg-m}^2\text{s}^{-1}$
- 11 A body is in pure rotation. The linear speed v of a particle, the distance r of the particle from the axis and the angular velocity ω of the body are related as $\omega = \frac{v}{r}$. Thus,
 - (a) $\omega \propto \frac{1}{r}$
 - (b) $\omega \propto r$
 - (c) $\omega = 0$
 - (d) ω is independent of r
- 12 A particle performs uniform circular motion with an angular momentum L . If the frequency of the particle motion is doubled, the angular momentum becomes
 - (a) $2L$
 - (b) $4L$
 - (c) $\frac{L}{2}$
 - (d) $\frac{L}{4}$
- 13 A particle of mass 2 kg located at the position $(\hat{i} + \hat{j}) \text{ m}$ has a velocity $2(\hat{i} - \hat{j} + \hat{k}) \text{ ms}^{-1}$. Its angular momentum about Z-axis (in $\text{kg-m}^2\text{s}^{-1}$) is
 - (a) + 4
 - (b) + 8
 - (c) - 4
 - (d) - 8
- 14 A constant torque of 1000 N-m turns a wheel of moment of inertia 200 kg-m^2 about an axis through its centre. Its angular velocity after 3 s is
 - (a) 1 rad s^{-1}
 - (b) 5 rad s^{-1}
 - (c) 10 rad s^{-1}
 - (d) 15 rad s^{-1}
- 15 A flywheel having a radius of gyration of 2 m and mass 10 kg rotates at an angular speed of 5 rad s^{-1} about an axis perpendicular to it through its centre. The kinetic energy of rotation is
 - (a) 500 J
 - (b) 2000 J
 - (c) 1000 J
 - (d) 250 J
- 16 A rod is placed along the line $y = 2x$ with its centre at origin. The moment of inertia of the rod is maximum about
 - (a) X-axis
 - (b) Y-axis
 - (c) Z-axis
 - (d) Data insufficient

- 17** Let I_A and I_B be moments of inertia of a body about two axes A and B , respectively. The axis A passes through the centre of mass of the body but B does not. Choose the correct option.

(a) $I_A < I_B$
 (b) $I_A < I_B$, whether the axes are parallel or not parallel
 (c) If the axes are parallel, $I_A < I_B$
 (d) If the axes are not parallel, $I_A \geq I_B$

- 18** A rigid body rotates with an angular momentum L . If its rotational kinetic energy is made 4 times, its angular momentum will become

(a) $4L$ (b) $16L$ (c) $\sqrt{2}L$ (d) $2L$

- 19** Work done by friction in case of pure rolling

(a) is always zero
 (b) is always positive
 (c) is always negative
 (d) may be positive, negative or zero

- 20** A disc of mass m and radius R is rolling on horizontal ground with linear velocity v . What is the angular momentum of the disc about an axis passing through bottom most point and perpendicular to the plane of motion?

(a) $\frac{3}{2}mvR$ (b) mvR (c) $\frac{1}{2}mvR$ (d) $\frac{4}{3}mvR$

- 21** A sphere cannot roll without applying external force on

(a) a smooth inclined surface
 (b) a smooth horizontal surface
 (c) a rough inclined surface
 (d) a rough horizontal surface

- 22** A solid sphere of mass 2 kg rolls up a 30° incline with an initial speed of 10 ms^{-1} . The maximum height reached by the sphere is (Take, $g = 10 \text{ ms}^{-2}$)

(a) 3.5 m (b) 7 m (c) 10.5 m (d) 14 m

- 23** A solid sphere and a solid cylinder of same mass are rolled down on two inclined planes of heights h_1 and h_2 , respectively. If at the bottom of the plane, the two objects have same linear velocities, then the ratio of $h_1 : h_2$ is

(a) 2 : 3 (b) 7 : 5 (c) 14 : 15 (d) 15 : 14

- 24** If $\mathbf{F}$ be a force acting on a particle having the position vector $\mathbf{r}$ and $\boldsymbol{\tau}$ be the torque of this force about the origin, then

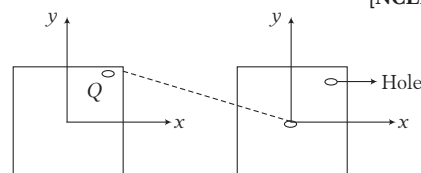
(a) $\mathbf{r} \cdot \boldsymbol{\tau} = 0$ and $\mathbf{F} \cdot \boldsymbol{\tau} = 0$ (b) $\mathbf{r} \times \boldsymbol{\tau} = 0$ and $\mathbf{F} \times \boldsymbol{\tau} \neq 0$
 (c) $\mathbf{r} \cdot \boldsymbol{\tau} \neq 0$ and $\mathbf{F} \cdot \boldsymbol{\tau} \neq 0$ (d) $\mathbf{r} \times \boldsymbol{\tau} \neq 0$ and $\mathbf{F} \times \boldsymbol{\tau} = 0$

- 25** A uniform disc of radius a and mass m is rotating freely with angular speed ω in a horizontal plane about a smooth fixed vertical axis through its centre. A particle of mass m is suddenly attached to the rim of the disc and rotates with it. The new angular speed is

(a) $\frac{\omega}{6}$ (b) $\frac{\omega}{3}$ (c) $\frac{\omega}{2}$ (d) $\frac{\omega}{5}$

- 26** A uniform square plate has a small piece Q of an irregular shape removed and glued to the centre of the plate leaving a hole behind as shown in figure. The moment of inertia about the Z -axis is

[NCERT Exemplar]



(a) increased
 (b) decreased
 (c) the same
 (d) changed in unpredicted manner

- 27** Moment of a force of magnitude 10 N acting along positive y -direction at point $(2\text{m}, 0, 0)$ about the point $(0, 1\text{m}, 0)$ (in N-m) is

(a) 10 (b) 20 (c) $10\sqrt{2}$ (d) 30

- 28** A rigid body rotates about a fixed axis with variable angular velocity equal to $\alpha - \beta t$, at the time t , where α, β are constants. The angle through which it rotates before it stops is

(a) $\frac{\alpha^2}{2\beta}$ (b) $\frac{\alpha^2 - \beta^2}{2\alpha}$ (c) $\frac{\alpha^2 - \beta^2}{2\beta}$ (d) $\frac{(\alpha - \beta)\alpha}{2}$

- 29** The radius of gyration of a uniform rod of length L about an axis passing through its centre of mass is

(a) $\frac{L}{2\sqrt{3}}$ (b) $\frac{L^2}{12}$ (c) $\frac{L}{\sqrt{3}}$ (d) $\frac{L}{\sqrt{2}}$

- 30** The motor of an engine is rotating about its axis with an angular velocity of 100 rpm. It comes to rest in 15 s, after being switched OFF. Assuming constant angular deceleration, what are the numbers of revolutions made by it before coming to rest?

(a) 12.5 (b) 40 (c) 32.6 (d) 15.6

- 31** A particle of mass 5 g is moving with a uniform speed of $3\sqrt{2} \text{ cm s}^{-1}$ in the XY -plane along the line $y = 2\sqrt{5} \text{ cm}$. The magnitude of its angular momentum about the origin (in $\text{g-cm}^2\text{s}^{-1}$) is

(a) zero (b) 30
 (c) $30\sqrt{2}$ (d) $30\sqrt{10}$

- 32** A flywheel is in the form of a uniform circular disc of radius 1 m and mass 2 kg. The work which must be done on it to increase its frequency of rotation from 5 rps to 10 rps is approximately

(a) $1.5 \times 10^2 \text{ J}$ (b) $3.0 \times 10^2 \text{ J}$
 (c) $1.5 \times 10^3 \text{ J}$ (d) $3.0 \times 10^3 \text{ J}$

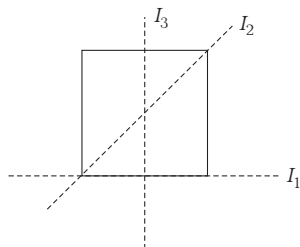
- 33** A thin uniform circular disc of mass M and radius R is rotating in a horizontal plane about an axis passing through its centre and perpendicular to its plane with an angular velocity ω . Another disc of same dimensions but of mass $\frac{1}{4}M$ is placed gently on the first disc co-axially. The angular velocity of the system is

(a) $\frac{2}{3}\omega$ (b) $\frac{4}{5}\omega$ (c) $\frac{3}{4}\omega$ (d) $\frac{1}{3}\omega$

- 34** The ratio of the radii of gyration of a hollow sphere and a solid sphere of the same radii about a tangential axis is

(a) $\sqrt{\frac{7}{3}}$ (b) $\frac{5}{\sqrt{21}}$ (c) $\sqrt{\frac{21}{5}}$ (d) $\frac{25}{9}$

- 35** A square lamina is as shown in figure. The moment of inertia of the frame about the three axes as shown in figure are I_1 , I_2 and I_3 , respectively. Select the correct alternative.



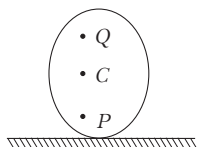
(a) $I_2 = I_3 > I_1$ (b) $I_1 > I_2 > I_3$
(c) $I_2 = I_3 < I_1$ (d) $I_1 < I_2 < I_3$

- 36** A merry-go-round made of a ring-like platform of radius R and mass M , is revolving with angular speed ω . A person of mass M is standing on it. At one instant, the person jumps off the round, radially away from the centre of the round (as seen from the round). The speed of the round afterwards is

[NCERT Exemplar]

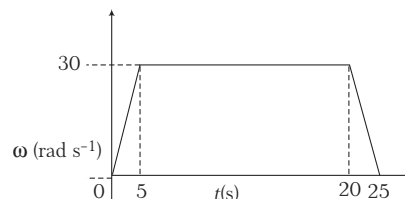
(a) 2ω (b) ω (c) $\frac{\omega}{2}$ (d) 0

- 37** A disc is rolling without slipping on a horizontal surface with C as its centre and Q and P the two points equidistant from C . Let v_P , v_Q and v_C be the magnitudes of velocities of points P , Q and C respectively, then



(a) $v_Q > v_C > v_P$ (b) $v_Q < v_C < v_P$
(c) $v_Q = v_P, v_C = \frac{1}{2}v_P$ (d) $v_Q < v_C > v_P$

- 38** The figure shows the angular velocity *versus* time graph of a flywheel. The angle (in radians) through which the flywheel turns during 25 s is



(a) 120 (b) 480 (c) 600 (d) 750

- 39** If a disc of mass m and radius r is reshaped into a ring of radius $2r$, the mass remaining the same, the radius of gyration about centroidal axis perpendicular to plane goes up by a factor of

(a) $\sqrt{2}$ (b) 2 (c) $2\sqrt{2}$ (d) 4

- 40** A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is K . If radius of the ball be R , then the fraction of total energy associated with its rotational energy will be

(a) $\frac{K^2}{K^2 + R^2}$ (b) $\frac{R^2}{K^2 + R^2}$ (c) $\frac{K^2 + R^2}{R^2}$ (d) $\frac{K^2}{R^2}$

- 41** Two discs have same mass and thickness. Their materials are made of densities d_1 and d_2 . The ratio of their moments of inertia about an axis passing through the centre and perpendicular to the plane is

(a) $d_1 : d_2$ (b) $d_2 : d_1$ (c) $\left(\frac{d_1}{d_2}\right)^2$ (d) $\left(\frac{d_2}{d_1}\right)^2$

- 42** If earth suddenly shrinks to $\frac{1}{8}$ th of its original volume while mass remaining the same, then duration of day (in hour) will be

(a) 6 (b) 8
(c) 12 (d) 3

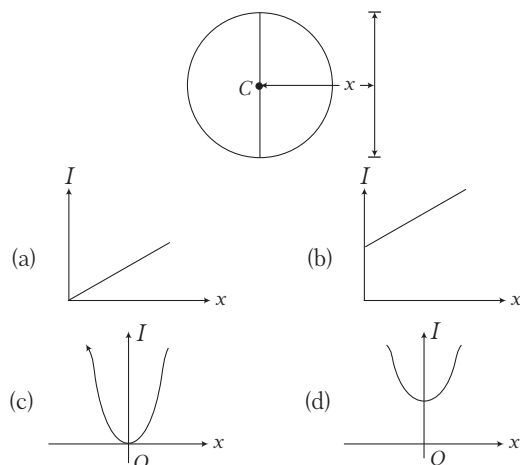
- 43** The speed of a homogeneous solid sphere after rolling down an inclined plane of vertical height h , from rest without sliding is

(a) $\sqrt{gh}$ (b) $\sqrt{\left(\frac{g}{5}\right)gh}$
(c) $\sqrt{\left(\frac{4}{3}\right)gh}$ (d) $\sqrt{\left(\frac{10}{7}\right)gh}$

- 44** The moment of inertia of a cube of mass m and side a about one of its edges is equal to

(a) $\frac{2}{3}ma^2$ (b) $\frac{4}{3}ma^2$
(c) $3ma^2$ (d) $\frac{8}{3}ma^2$

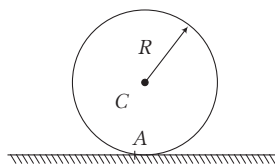
- 45** Figure represents the moment of inertia of the solid sphere about an axis parallel to the diameter of the solid sphere and at a distance x from it. Which one of the following represents the variations of I with x ?



- 46** Two uniform, thin identical rods each of mass M and length l are joined together to form a cross. What will be the moment of inertia of the cross about an axis passing through the point at which the two rods are joined and perpendicular to the plane of the cross?

(a) $\frac{Ml^2}{12}$ (b) $\frac{Ml^2}{6}$ (c) $\frac{Ml^2}{4}$ (d) $\frac{Ml^2}{3}$

- 47** A disc of radius R rolls on a rough horizontal surface. The distance covered by the point A in one revolution is



(a) $2\pi R$ (b) $2R$ (c) $8R$ (d) πR

- 48** When a body is projected at an angle with the horizontal in a uniform gravitational field of the earth, the angular momentum of the body about the point of projection, as it proceeds along its path
- (a) remains constant
(b) increases
(c) decreases
(d) initially decreases and after its highest point increases

- 49** A particle of mass $m = 5$ units is moving with a uniform speed $v = 3\sqrt{2}$ units in the XY -plane along the line $y = x + 4$. The magnitude of the angular momentum about origin is
- (a) zero (b) 60 units (c) 7.5 units (d) $40\sqrt{2}$ units

- 50** A sphere rolls without slipping on a rough horizontal surface with centre of mass speed v_0 . If mass of the

sphere is M and its radius is R , then what is the angular momentum of the sphere about the point of contact?

(a) $\frac{5}{2} Mv_0 R$ (b) $\frac{7}{5} Mv_0 R$ (c) $\frac{3}{5} Mv_0 R$ (d) $\frac{1}{2} Mv_0 R$

- 51** The ratio of the radii of gyration of a circular disc and a circular ring of the same radii about a tangential axis perpendicular to plane of disc or ring is

(a) 1 : 2 (b) $\sqrt{5} : \sqrt{6}$ (c) 2 : 3 (d) $\sqrt{3} : 2$

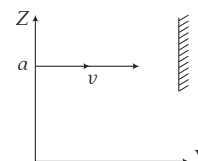
- 52** The ratio of the radii of gyration of a circular disc and a circular ring of the same radius about a tangential axis in the plane is

(a) $\sqrt{3} : \sqrt{4}$ (b) $\sqrt{5} : \sqrt{6}$
(c) $\sqrt{6} : \sqrt{5}$ (d) $\sqrt{4} : \sqrt{3}$

- 53** A particle of mass m is projected with a velocity v making an angle of 45° with the horizontal. The magnitude of angular momentum of projectile about the point of projection when the particle is at its maximum height h is

(a) zero (b) $\frac{mvh}{\sqrt{2}}$ (c) mvh (d) $\sqrt{2} mvh$

- 54** A particle of mass m is moving in YZ -plane with a uniform velocity v with its trajectory running parallel to +ve Y -axis and intersecting Z -axis at $z = a$ as shown in figure. The change in its angular momentum about the origin as it bounces elastically from a wall at $y = \text{constant}$ is



[NCERT Exemplar]

(a) $mva \hat{e}_x$ (b) $2mva \hat{e}_x$ (c) $ymv \hat{e}_x$ (d) $2ymv \hat{e}_x$

- 55** A ring is kept on a rough inclined surface. But the coefficient of friction is less than the minimum value required for pure rolling. At any instant of time, let K_T and K_R be the translational and rotational kinetic energies of the ring respectively, then

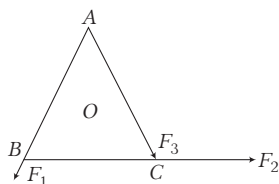
(a) $K_R = K_T$ (b) $K_R > K_T$ (c) $K_T > K_R$ (d) $K_R = 0$

- 56** A ring and a disc of different masses are rotating with the same kinetic energy. If we apply a retarding torque τ on the ring, it stops after making n revolutions. After how many revolutions will the disc stop, if the retarding torque on it is also τ ?

(a) $\frac{n}{2}$ (b) n
(c) $2n$ (d) Data insufficient

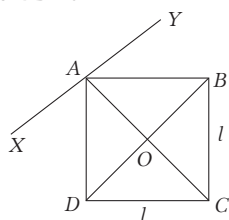
- 57** O is the centre of an equilateral triangle ABC . F_1 , F_2 and F_3 are three forces acting along the sides AB , BC and AC respectively as shown in figure.

What should be the magnitude of F_3 , so that the total torque about O is zero?



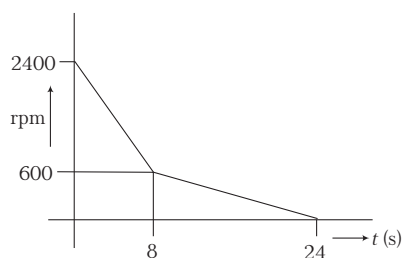
- (a) $\frac{F_1 + F_2}{2}$ (b) $F_1 - F_2$ (c) $F_1 + F_2$ (d) $2(F_1 + F_2)$

- 58** Four point masses each of mass m are placed at the corners of a square $ABCD$ of side l . The moment of inertia of the system about an axis passing through A and parallel to BD is



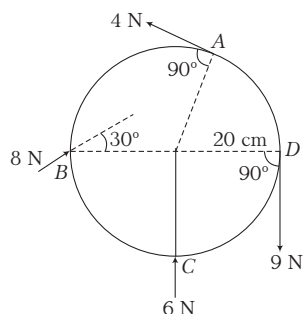
- (a) $\sqrt{3} ml^2$ (b) $3 ml^2$ (c) ml^2 (d) $2 ml^2$

- 59** A table fan rotating at a speed of 2400 rpm, is switched OFF and the resulting variation of the rpm with time as shown in the figure. The total number of revolutions of the fan before it comes to rest is



- (a) 420 (b) 190 (c) 280 (d) 380

- 60** Forces are applied on a wheel of radius 20 cm as shown in the figure. The torque produced by the forces 4 N at A , 8 N at B , 6 N at C and 9 N at D at angles indicated is

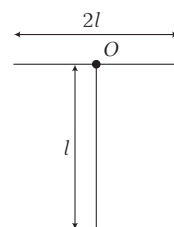


- (a) 5.4 N-m anti-clockwise (b) 1.80 N-m clockwise
(c) 2.0 N-m clockwise (d) 3.6 N-m clockwise

- 61** A uniform rod of mass 2 kg and length 1 m lies on a smooth horizontal plane. A particle of mass 1 kg moving at a speed of 2 ms^{-1} perpendicular to the length of the rod strikes it at a distance $\frac{1}{4}$ m from the centre and stops. What is the angular velocity of the rod about its centre just after the collision?

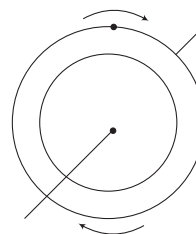
- (a) 3 rad s^{-1} (b) 4 rad s^{-1} (c) 1 rad s^{-1} (d) 2 rad s^{-1}

- 62** For the uniform T shaped structure with mass $3M$, moment of inertia about an axis normal to the plane and passing through O would be



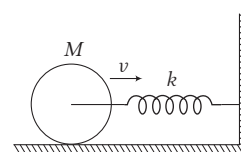
- (a) $\frac{2}{3} Ml^2$ (b) Ml^2
(c) $\frac{Ml^2}{3}$ (d) None of these

- 63** A disc is free to rotate about a smooth horizontal axis passing through its centre of mass. A particle is fixed at the top of the disc. A slight push is given to the disc and it starts rotating. During the process,



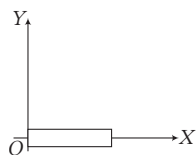
- (a) only mechanical energy is conserved
(b) only angular momentum (about the axis of rotation) is conserved
(c) Both mechanical energy and angular momentum are conserved
(d) Neither the mechanical energy nor the angular momentum are conserved

- 64** A solid sphere rolls without slipping and presses a spring of spring constant k as shown in figure. Then, the compression in the spring will be

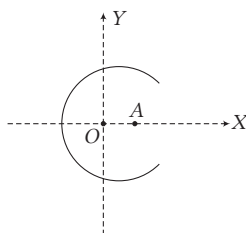


- (a) $v \sqrt{\frac{2M}{3k}}$ (b) $v \sqrt{\frac{2M}{5k}}$ (c) $v \sqrt{\frac{5k}{7M}}$ (d) $v \sqrt{\frac{7M}{5k}}$

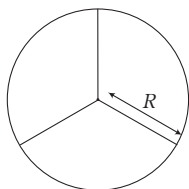
- 65** The figure shows a uniform rod lying along the X -axis. The locus of all the points lying on the XY -plane, about which the moment of inertia of the rod is same as that about O is



- (a) an ellipse
(b) a circle
(c) a parabola
(d) a straight line
- 66** A portion of a ring of radius R has been removed as shown in figure. Mass of the remaining portion is m . Centre of the ring is at origin O . Let I_A and I_O be the moments of inertia passing through points A and O are perpendicular to the plane of the ring. Then,



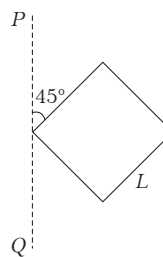
- (a) $I_O = mR^2$ (b) $I_O = I_A$
(c) $I_O > I_A$ (d) $I_A > I_O$
- 67** A wheel comprises a ring of radius R and mass M and three spokes each of mass m . The moment of inertia of the wheel about its axis is



- (a) $\left(M + \frac{m}{4}\right)R^2$ (b) $(M + m)R^2$
(c) $(M + 3m)R^2$ (d) $\left(\frac{M + m}{2}\right)R^2$
- 68** A thin bar of mass m and length l is free to rotate about a fixed horizontal axis through a point at its end. The bar is brought to a horizontal position ($\theta = 90^\circ$) and then released. The angular velocity when it reaches the lowest point is
- (a) directly proportional to its length and inversely proportional to its mass
(b) independent of mass and inversely proportional to the square root of its length
(c) dependent only upon the acceleration due to gravity and the mass of the bar
(d) directly proportional to its length and inversely proportional to the acceleration due to gravity

- 69** A square is made by joining four rods each of mass M and length L . Its moment of inertia about an axis PQ , in its plane and passing through one of its corner is

- (a) $6ML^2$ (b) $\frac{4}{3}ML^2$
(c) $\frac{8}{3}ML^2$ (d) $\frac{10}{3}ML^2$



- 70** A rod of length L whose lower end is fixed along the horizontal plane starts to topple from the vertical position. The velocity of the upper end of the rod when it hits the ground is

- (a) $\sqrt{3gL}$ (b) $\sqrt{2gL}$ (c) $\sqrt{gL}$ (d) $\sqrt{5gL}$

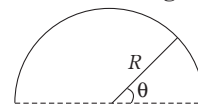
- 71** A cord is wound around the circumference of wheel of radius r . The axis of the wheel is horizontal and moment of inertia is I . A weight mg is attached to the end of the cord and falls from rest. After falling through a distance h , the angular velocity of the wheel will be

- (a) $\sqrt{\frac{2gh}{I + mr^2}}$ (b) $\sqrt{\frac{2mgh}{I + mr^2}}$
(c) $\sqrt{\frac{2mgh}{I + 2mr^2}}$ (d) $\sqrt{2gh}$

- 72** A particle P is moving in a circle of radius a with uniform speed u . C is the centre of the circle and AB is its diameter. The angular velocities of P about A and C are in the ratio

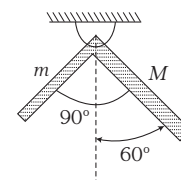
- (a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 4 : 1

- 73** The moment of inertia of a semicircular ring of mass M and radius R about an axis which is passing through its centre and at an angle θ with the line joining its ends as shown in figure, is



- (a) $\frac{MR^2}{4}$, if $\theta = 0^\circ$ (b) $\frac{MR^2}{2}$, if $\theta = 0^\circ$
(c) $\frac{MR^2}{2}$, if $\theta = \text{any angle}$ (d) $\frac{MR^2}{2}$, if $\theta = 90^\circ$

- 74** Two uniform rods of equal length but different masses are rigidly joined to form an L -shaped body, which is then pivoted as shown in figure. If in equilibrium, the body is in the shown configuration, ratio M/m will be



- (a) 2 (b) 3 (c) $\sqrt{2}$ (d) $\sqrt{3}$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) These questions consists of two statements each printed as Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses.

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is correct but Reason is incorrect.
- (d) If Assertion is incorrect but Reason is correct.

1 Assertion A body is moving along a circle with a constant speed. Its angular momentum about the centre of the circle remains constant.

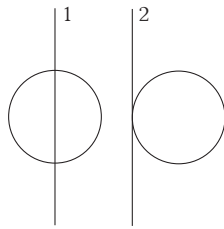
Reason In this situation, a constant non-zero torque acts on the body.

2 Assertion If a particle moves with a constant velocity, then angular momentum of this particle about any point remains constant.

Reason Angular momentum has the units of Planck's constant.

3 Assertion Two identical solid spheres are rotated from rest to same angular velocity ω about two different axes as shown in figure. More work has to be done to rotate the sphere in case 2.

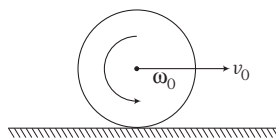
Reason Moment of inertia in case 2 is more.



4 Assertion A solid sphere cannot roll without slipping on smooth horizontal surface.

Reason If the sphere is left free on smooth inclined surface, it cannot roll without slipping.

5 Assertion A uniform disc of radius R is performing impure rolling motion on a rough horizontal plane as shown in figure. After some time, the disc comes to rest. It is possible only when $v_0 = \frac{\omega_0 R}{2}$.



Reason For a body performing pure rolling motion, the angular momentum is conserved about any point in space.

Statement based questions

1 When a disc rotates with uniform angular velocity, choose the incorrect statement. [NCERT Exemplar]

- (a) The sense of rotation remains same.
- (b) The orientation of the axis of rotation remains same.
- (c) The speed of rotation is non-zero and remains same.
- (d) The angular acceleration is non-zero and remains same.

2 A particle is moving in a circular orbit with constant speed. Choose the incorrect statement.

- (a) Its linear momentum is conserved.
- (b) Its angular momentum is conserved.
- (c) It is moving with variable velocity.
- (d) It is moving with variable acceleration.

3 A sphere can roll on a surface inclined at an angle θ , if the friction coefficient $\mu > (2/7) g \sin \theta$. Now, suppose the friction coefficient is $(1/7) g \sin \theta$ and the sphere is released from rest on the incline, then choose the correct statement.

- (a) The sphere will stay at rest.
- (b) The sphere will make pure translational motion.
- (c) The sphere will translate and rotate about the centre.
- (d) The angular momentum of the sphere about its centre will remain constant.

4 Consider three solid spheres, sphere (i) has radius r and mass m , sphere (ii) has radius r and mass $3m$, sphere (iii) has radius $3r$ and mass m . All can be placed at the same point on the same inclined plane, where they will roll without slipping to the bottom. If allowed to roll down the incline, then at the bottom of the incline, choose the correct statement.

- (a) Sphere (i) will have the largest speed.
- (b) Sphere (ii) will have the largest speed.
- (c) Sphere (iii) will have the largest kinetic energy.
- (d) All the spheres will have equal speeds.

5 A solid homogeneous sphere is moving on a rough horizontal surface, partly rolling and partly sliding. Then, choose the correct statement.

- (a) Total kinetic energy is conserved.
- (b) Angular momentum of the sphere about the point of contact with the plane is conserved.
- (c) Only the rotational kinetic energy about centre of mass is conserved.
- (d) Angular momentum about centre of mass is conserved.

- 6 I. Angular momentum of a particle moving in a straight line is always constant with respect to fixed point.
 II. Moment of inertia of a body remains same irrespective of position of axis of rotation.
 Which of the statement(s) is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 7 I. Speed of any point on rigid body executing rolling motion can be calculated by the expression $v = r\omega$, where r is the distance of point from instantaneous centre of rotation.
 II. Rolling motion of rigid body can be considered as a pure rotation about instantaneous centre of rotation.
 Which of the above statement(s) is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

Match the columns

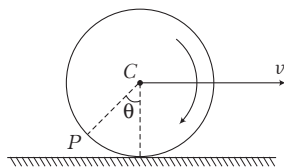
- 1 If radius of earth is reduced to half without changing its mass, then match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Angular velocity of earth	(p) will become two times
(B) Time period of rotation of earth	(q) will become four times
(C) Rotational kinetic energy of earth	(r) will remain constant
	(s) None

Codes

A	B	C	A	B	C
(a) r	s	p	(b) q	r	p
(c) q	s	q	(d) q	p	s

- 2 A disc rolls on ground without slipping. Velocity of centre of mass is v . There is a point P on the circumference of disc at angle θ . Suppose v_P is the speed of this point. Then, match the following columns and mark the correct option from the codes given below.

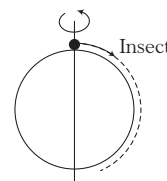


Column I	Column II
(A) If $\theta = 60^\circ$	(p) $v_P = \sqrt{2} v$
(B) If $\theta = 90^\circ$	(q) $v_P = v$
(C) If $\theta = 120^\circ$	(r) $v_P = 2v$
(D) If $\theta = 180^\circ$	(s) $v_P = \sqrt{3} v$

Codes

A	B	C	D	A	B	C	D
(a) r	s	p	q	(b) q	p	s	r
(c) s	r	q	p	(d) p	r	q	s

- 3 A solid sphere is rotating about an axis as shown in figure. An insect follows the dotted path on the circumference of sphere as shown in the figure.



Then, match the following columns and mark the correct option from the codes given below.

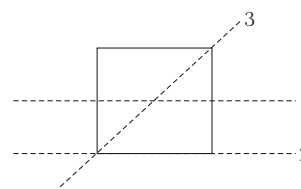
Column I	Column II
(A) Moment of inertia	(p) will remain constant
(B) Angular velocity	(q) will first increase, then decrease
(C) Angular momentum	(r) will first decrease, then increase
(D) Rotational kinetic energy	(s) will continuously decrease
	(t) will continuously increase

Codes

A	B	C	D	A	B	C	D
(a) p	r	q	t	(b) s	q	r	p
(c) s	t	r	p	(d) q	r	p	r

- 4 Four rods of equal length l and each of mass m form a square as shown in figure.

Moments of inertia about three axes 1, 2 and 3 are say I_1 , I_2 and I_3 . Then, match the following columns and mark the correct option from the codes given below.



Column I	Column II
(A) I_1	(p) $\frac{4}{3} ml^2$
(B) I_2	(q) $\frac{2}{3} ml^2$
(C) I_3	(r) $\frac{1}{2} ml^2$
	(s) None

Codes

A	B	C	A	B	C
(a) q	s	p	(b) p	s	q
(c) r	p	q	(d) q	s	q

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1** Find the torque about the origin when a force of $3\hat{j}$ N acts on the particle whose position vector is $2\hat{k}$ m. [NEET 2020]

(a) $6\hat{j}$ N-m (b) $-6\hat{i}$ N-m (c) $6\hat{k}$ N-m (d) $6\hat{i}$ N-m

- 2** The angular speed of the wheel of a vehicle is increased from 360 rpm to 1200 rpm in 14 s. Its angular acceleration is [NEET 2020]

(a) 2π rad/s² (b) 28π rad/s²
(c) 120π rad/s² (d) 1 rad/s²

- 3** A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is [NEET 2019]

(a) 2×10^{-3} N-m (b) 12×10^{-4} N-m
(c) 2×10^6 N-m (d) 2×10^{-6} N-m

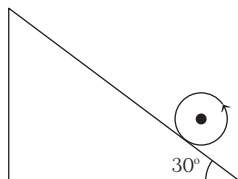
- 4** A disc of radius 2 m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm s^{-1} . How much work is needed to stop it? [NEET 2019]

(a) 30 kJ (b) 2 J (c) 1 J (d) 3 J

- 5** A solid cylinder of mass 2 kg and radius 50 cm rolls up an inclined plane of angle inclination 30° . The centre of mass of cylinder has speed of 4 ms^{-1} . The distance travelled by the cylinder on the inclined surface will be

(Take, $g = 10 \text{ ms}^{-2}$) [NEET (Odisha) 2019]
(a) 2.2 m (b) 1.6 m (c) 1.2 m (d) 2.4 m

- 6** A sphere pure rolls on a rough inclined plane with initial velocity 2.8 ms^{-1} . Find the maximum distance on the inclined plane. [AIIMS 2019]



(a) 2.74 m (b) 5.48 m (c) 1.38 m (d) 3.2 m

- 7** Two discs have mass ratio 1 : 2 and diameter ratio 2 : 1, then find the ratio of their moments of inertia. [JIPMER 2019]

(a) 2 : 1 (b) 1 : 1 (c) 1 : 2 (d) 2 : 3

- 8** If an object starts from rest and covers angle of 60 rad in 10 s in circular motion, then magnitude of its angular acceleration will be [JIPMER 2019]

(a) 1.2 rad s^{-2} (b) 1.5 rad s^{-2}
(c) 2 rad s^{-2} (d) 2.5 rad s^{-2}

- 9** The moment of the force $\mathbf{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at $(2, 0, -3)$, about the point $(2, -2, -2)$ is given by [NEET 2018]

(a) $-7\hat{i} - 8\hat{j} - 4\hat{k}$ (b) $-4\hat{i} - \hat{j} - 8\hat{k}$
(c) $-8\hat{i} - 4\hat{j} - 7\hat{k}$ (d) $-7\hat{i} - 4\hat{j} - 8\hat{k}$

- 10** A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere? [NEET 2018]

(a) Rotational kinetic energy
(b) Moment of inertia
(c) Angular velocity
(d) Angular momentum

- 11** A solid sphere is in rolling motion. In rolling motion, a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is [NEET 2018]

(a) 10 : 7 (b) 5 : 7
(c) 7 : 10 (d) 2 : 5

- 12** A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a point near the rim of disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, the angular speed of the disc [AIIMS 2018]

(a) continuously decreases
(b) continuously increases
(c) first increases and then decreases
(d) remains unchanged

- 13** Find the ratio of radius of gyration of a disc and ring of same radii at their tangential axis in plane. [JIPMER 2018]

(a) $\sqrt{\frac{5}{6}}$ (b) $\sqrt{\frac{5}{3}}$ (c) 1 (d) $\frac{2}{3}$

- 14** Which of the following statements are correct? [NEET 2017]

I. Centre of mass of a body usually coincides with the centre of gravity of the body.
II. Centre of mass of a body is the point at which the total gravitational torque on the body is zero.
III. A couple on a body produces both translational and rotational motion in a body.
IV. Mechanical advantage greater than one means that small effort can be used to lift a large load.
(a) II and IV (b) I and II
(c) II and III (d) III and IV

- 15** Two discs of same moment of inertia rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2 . They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is [NEET 2017]

(a) $\frac{1}{2} I(\omega_1 + \omega_2)^2$ (b) $\frac{1}{4} I(\omega_1 - \omega_2)^2$
 (c) $I(\omega_1 - \omega_2)^2$ (d) $\frac{I}{8} (\omega_1 - \omega_2)^2$

- 16** A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder, if the rope is pulled with a force of 30 N? [NEET 2017]

(a) 25 ms^{-2} (b) 0.25 rad s^{-2} (c) 25 rad s^{-2} (d) 5 ms^{-2}

- 17 Assertion** The total kinetic energy of a rolling solid sphere is the sum of translational and rotational kinetic energies.

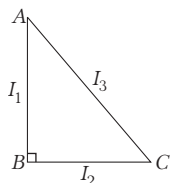
Reason For all solid bodies, total kinetic energy is always twice of translational kinetic energy.

[AIIMS 2017]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Both Assertion and Reason are incorrect.
- 18** A thin hollow sphere of mass m is completely filled with a liquid of mass m . When the sphere rolls with a velocity v , kinetic energy of the system is (neglect friction) [AIIMS 2017]

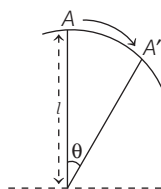
(a) $(1/2)mv^2$ (b) mv^2 (c) $(4/3)mv^2$ (d) $(4/5)mv^2$

- 19** ABC is right angled triangular plane of uniform thickness. The sides are such that $AB > BC$ as shown in figure. I_1, I_2, I_3 are moments of inertia about AB, BC and AC, respectively. Then, which of the following relations is correct? [JIPMER 2017]



(a) $I_1 = I_2 = I_3$ (b) $I_2 > I_1 > I_3$
 (c) $I_3 < I_2 < I_1$ (d) $I_3 > I_1 > I_2$

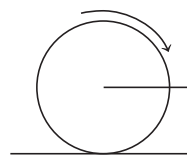
- 20** A uniform rod of length l is free to rotate in a vertical plane about a fixed horizontal axis through B. The rod begins rotating from rest from its unstable equilibrium position.



When it has turned through an angle θ , its angular velocity ω is given by [JIPMER 2017]

(a) $\sqrt{\left(\frac{6g}{l}\right)} \sin \frac{\theta}{2}$ (b) $\sqrt{\left(\frac{6g}{l}\right)} \cos \frac{\theta}{2}$
 (c) $\sqrt{\left(\frac{6g}{l}\right)} \sin \theta$ (d) $\sqrt{\left(\frac{6g}{l}\right)} \cos \theta$

- 21** A bicycle wheel rolls without slipping on a horizontal floor. Which one of the following is true about the motion of points on the rim of the wheel, relative to the axis at the wheel's centre? [JIPMER 2017]

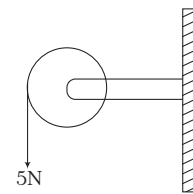


- (a) Points near the top move faster than points near the bottom
 (b) Points near the bottom move faster than points near the top
 (c) All points on the rim move with the same speed
 (d) All points have the velocity vectors that are pointing in the radial direction towards the centre of the wheel

- 22** A cylinder rolls up an inclined plane, reaches some height and then rolls down (without slipping throughout these motions). The directions of the frictional force acting on the cylinder are [JIPMER 2017]

- (a) up the incline while ascending and down the incline while descending
 (b) up the incline while ascending as well as descending
 (c) down the incline while ascending and up the incline while descending
 (d) down the incline while ascending as well as descending

- 23** A wheel of radius 10 cm can rotate freely about its centre as shown in figure. A string wrapped over its rim is pulled by a force of 5N. It is observed that the torque produces angular acceleration of 2 rad s^{-2} in the wheel. What is the moment of inertia of the wheel? [NEET 2016]



(a) 0.25 kg-m^2 (b) 0.45 kg-m^2
 (c) 0.15 kg-m^2 (d) 0.16 kg-m^2

- 24** From a disc of radius R and mass M , a circular hole of diameter R , whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre? [NEET 2016]

(a) $13 MR^2/32$ (b) $11 MR^2/32$
 (c) $9 MR^2/32$ (d) $15 MR^2/32$

- 25** A disc and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first? [NEET 2016]

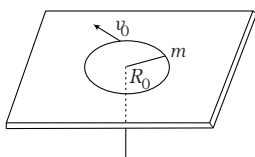
(a) Sphere
(b) Both reach at the same time
(c) Depends on their masses
(d) Disc

- 26** A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2 rad s^{-2} . Its net acceleration (in ms^{-2}) at the end of 2 s is approximately [NEET 2016]

(a) 7 (b) 6
(c) 3 (d) 8

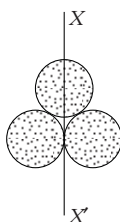
- 27** A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0 . The mass is attached to a string which passes through a smooth hole in the plane as shown in the figure.

The tension in the string is increased gradually and finally m moves in a circle of radius $R_0/2$. The final value of the kinetic energy is [CBSE AIPMT 2015]



(a) mv_0^2 (b) $\frac{1}{4}mv_0^2$ (c) $2mv_0^2$ (d) $\frac{1}{2}mv_0^2$

- 28** Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX' , which is touching the two shells and passing through diameter of third shell. Moment of inertia of the system consisting of these three spherical shells about XX' axis is [CBSE AIPMT 2015]

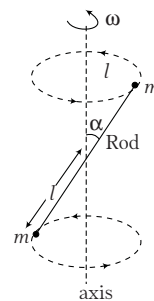


(a) $\frac{11}{5}mr^2$ (b) $3mr^2$
(c) $\frac{16}{5}mr^2$ (d) $4mr^2$

- 29** A uniform metallic rod rotates about its perpendicular bisector with constant angular speed. If it is heated uniformly to raise its temperature slightly, then [AIIMS 2015]

(a) its speed of rotation increases
(b) its speed of rotation decreases
(c) its speed of rotation remains same
(d) its speed increases because its moment of inertia increases

- 30** A massless rod S having length $2l$ has equal point masses attached to its two ends as shown in figure. The rod is rotating about an axis passing through its centre and making an angle α with the axis. The magnitude of change of momentum of rod, i.e. $\left| \frac{dL}{dt} \right|$



equals [AIIMS 2015]

(a) $2m l^3 \omega^2 \sin \theta \cdot \cos \theta$ (b) $ml^2 \omega^2 \sin 2\theta$
(c) $ml^2 \sin 2\theta$ (d) $m^{1/2} l^{1/2} \omega \sin \theta \cdot \cos \theta$

- 31** A uniform sphere of mass 500 g rolls without slipping on a plane surface, so that its centre moves at a speed of 0.02 ms^{-1} . The total kinetic energy of rolling sphere would be (in joule) [AIIMS 2015]

(a) 1.4×10^{-4} (b) 0.75×10^{-3}
(c) 5.75×10^{-3} (d) 4.9×10^{-5}

- 32** A particle travels in a circle of radius 20 cm at a uniformly increasing speed. If the speed changes from 8 ms^{-1} to 9 ms^{-1} in 2s, what would be the angular acceleration (in rad s^{-2})? [UK PMT 2015]

(a) 1.5 rad s^{-2} (b) 2.5 rad s^{-2}
(c) 3.5 rad s^{-2} (d) 4.5 rad s^{-2}

- 33** A solid sphere of radius r is rolling on a horizontal surface. The ratio between the rotational kinetic energy and total energy

[EAMCET 2015, AIIMS 2012, JCECE 2010]

(a) $\frac{5}{7}$ (b) $\frac{2}{7}$ (c) $\frac{1}{2}$ (d) $\frac{1}{7}$

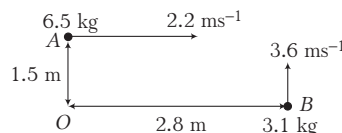
- 34** If two circular discs A and B are of same mass but of radii r and $2r$ respectively, then the moment of inertia of A is [Kerala CEE 2015]

(a) one-fourth that of B (b) twice that of B
(c) four times that of B (d) half that of B
(e) same as that of B

- 35** Choose the wrong statement. [Kerala CEE 2015]

(a) The centre of mass of a uniform circular ring is at its geometric centre.
(b) Moment of inertia is a tensor quantity.
(c) Radius of gyration is a vector quantity.
(d) Force in translational motion is analogous to torque in rotational motion.
(e) For same mass and radius, the moment of inertia of a ring is twice that of a uniform disc.

- 36** Two particles A and B are moving as shown in the figure.



Their total angular momentum about the point O is
[WB JEE 2015]

- (a) $9.8 \text{ kg-m}^2\text{s}^{-1}$ (b) zero
(c) $52.7 \text{ kg-m}^2\text{s}^{-1}$ (d) $37.9 \text{ kg-m}^2\text{s}^{-1}$

37 A rod of mass 5 kg is connected to the string at point B . The span of rod is along horizontal. The other end of the rod is hinged at point A . If the string is massless, then the reaction of hinge at the instant when string is cut, is (Take, $g = 10 \text{ ms}^{-2}$) [UP CPMT 2015]

- (a) 10.1 N (b) 12.5 N
(c) 5 N (d) 15 N

38 A particle moves with a constant velocity parallel to the X -axis. Its angular momentum with respect to the origin [UP CPMT 2015]

- (a) is zero (b) remains constant
(c) goes on increasing (d) goes on decreasing

39 A ring of radius 0.5 m and mass 10 kg is rotating about its diameter with angular velocity of 20 rad s^{-1} . Its kinetic energy is [Manipal 2015]

- (a) 10 J (b) 100 J (c) 500 J (d) 250 J

40 Moment of inertia of ring about its diameter is I . The moment of inertia of the same ring about that axis perpendicular to its plane and passing through centre is [KCET 2014]

- (a) $\frac{I}{2}$ (b) $2I$ (c) $\frac{I}{4}$ (d) $4I$

41 Two bodies have their moments of inertia I and $2I$ respectively, about their axis of rotation. If their kinetic energies of rotation are equal, their angular velocities will be in the ratio [UK PMT 2014]

- (a) $2:1$ (b) $1:2$ (c) $\sqrt{2}:1$ (d) $1:\sqrt{2}$

42 A body having a moment of inertia about its axis of rotation equal to 3 kg-m^2 is rotating with angular velocity of 3 rad s^{-1} . Kinetic energy of this rotating body is same as that of a body of mass 27 kg moving with a velocity v . The value of v is [KCET 2014]

- (a) 1 ms^{-1} (b) 0.5 ms^{-1}
(c) 2 ms^{-1} (d) 1.5 ms^{-1}

43 A uniform solid spherical ball is rolling down a smooth inclined plane from a height h . The velocity attained by the ball when it reaches the bottom of the inclined plane is v . If the ball is now thrown vertically upwards with the same velocity v , the maximum height to which the ball will rise is [WB JEE 2014]

- (a) $\frac{5h}{8}$ (b) $\frac{3h}{5}$ (c) $\frac{5h}{7}$ (d) $\frac{7h}{9}$

44 A rod PQ of mass M and length L is hinged at end P . The rod is kept horizontal by a massless string tied

to point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is



- (a) $\frac{3g}{2L}$ (b) $\frac{g}{L}$ (c) $\frac{2g}{L}$ (d) $\frac{2g}{3L}$

[NEET 2013]

45 A small object of uniform density rolls up a curved surface with an initial velocity v' . It reaches upto a maximum height of $3v'^2/4g$ with respect to the initial position. The object is [NEET 2013]

- (a) ring (b) solid sphere
(c) hollow sphere (d) disc

46 The conservation of angular momentum demands that [J&K CET 2013]

- (a) the external force on the system must be zero
(b) the external torque on the system must be zero
(c) Both the external force as well as the external torque must be zero
(d) Neither of them must be zero

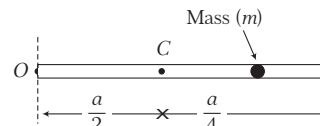
47 The moment of inertia (I) and the angular momentum (L) are related by the expression [J&K CET 2013]

- (a) $I = L\omega$ (b) $L = I\omega$ (c) $L = I^2\omega$ (d) $\omega = LI$

48 The moment of inertia (I) of a sphere of radius R and mass M is given by [J&K CET 2013]

- (a) $I = MR^2$ (b) $I = (1/2)MR^2$
(c) $I = (4/3)MR^2$ (d) $I = (2/5)MR^2$

49 A particle of mass m is attached to a thin uniform rod of length a at a distance of $\frac{a}{4}$ from the mid-point C as shown in the figure. The mass of the rod is $4m$. The moment of inertia of the combined system about an axis passing through O and perpendicular to the rod is [EAMCET 2013]

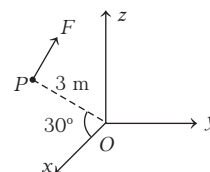


- (a) $\frac{91}{48}ma^2$ (b) $\frac{27}{48}ma^2$ (c) $\frac{51}{48}ma^2$ (d) $\frac{64}{48}ma^2$

50 A particle moving in a circular path has an angular momentum of L . If the frequency of rotation is halved, then its angular momentum becomes [Kerala CEE 2013]

- (a) $\frac{L}{2}$ (b) L (c) $\frac{L}{3}$ (d) $\frac{L}{4}$
(e) $2L$

- 51** The torque of a force $\mathbf{F} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ acting at a point whose position vector $\mathbf{r} = 3\hat{i} - 3\hat{j} + 5\hat{k}$ about the origin is [Kerala CEE 2013]
 (a) $-3\hat{i} + 5\hat{k}$ (b) $-5\hat{j} + 3\hat{k}$
 (c) $-5\hat{j} - 3\hat{k}$ (d) $3\hat{i} - 5\hat{j} + 3\hat{k}$
 (e) $-\hat{j} + 2\hat{k}$
- 52** Moment of inertia of a disc of radius R about a diametric axis is 25 kg m^{-2} . The moment of inertia of the disc about a parallel axes at a distance $R/2$ from the centre is [MPPMT 2013]
 (a) 31.25 kg-m^2 (b) 37.5 kg-m^2
 (c) 50 kg-m^2 (d) 62.5 kg-m^2
- 53** A thin hollow sphere of mass m is completely filled with a liquid of mass m . When the sphere rolls with a velocity v , kinetic energy of the system is (neglect friction) [AIIMS 2012]
 (a) $\frac{1}{2}mv^2$ (b) mv^2 (c) $\frac{4}{3}mv^2$ (d) $\frac{4}{5}mv^2$
- 54** The moment of inertia of a circular loop of radius R , at a distance of $R/2$ around a rotating axis parallel to horizontal diameter of loop is [AIIMS 2012]
 (a) MR^2 (b) $\frac{1}{2}MR^2$ (c) $2MR^2$ (d) $\frac{3}{4}MR^2$
- 55** A wheel having moment of inertia 2 kg-m^2 about its vertical axis, rotates at the rate of 60 rpm about this axis. The torque which can stop the wheel's rotation in one minute would be [UP CPMT 2012]
 (a) $\frac{\pi}{15} \text{ N-m}$ (b) $\frac{2\pi}{15} \text{ N-m}$ (c) $\frac{\pi}{18} \text{ N-m}$ (d) $\frac{\pi}{12} \text{ N-m}$
- 56** What is the moment of inertia of solid sphere of density ρ and radius R about its diameter? [UP CPMT 2012]
 (a) $\frac{105}{176}R^5\rho$ (b) $\frac{105}{176}R^2\rho$ (c) $\frac{176}{105}R^5\rho$ (d) $\frac{186}{105}R^2\rho$
- 57** The radius of gyration of a body depends upon [UP CPMT 2012]
 (a) shape and size of body
 (b) nature of mass distribution of body
 (c) choice of axis of rotation
 (d) All of the above
- 58** If two discs have same mass and thickness but their densities are ρ_1 and ρ_2 , then the ratio of their moments of inertia about central axis will be [BCECE (Mains) 2012]
 (a) $1 : \rho_1 \rho_2$ (b) $\rho_1 \rho_2 : 1$ (c) $\rho_1 : \rho_2$ (d) $\rho_2 : \rho_1$
- 59** If a disc starting from rest acquires an angular velocity of 240 rev min^{-1} in 10 s, then its angular acceleration will be [BCECE (Mains) 2012]
 (a) 1.52 rad s^{-1} (b) 3.11 rad s^{-1}
 (c) 2.51 rad s^{-1} (d) 1.13 rad s^{-1}
- 60** A rod of length L is composed of a uniform length $1/2 L$ of wood whose mass is m_w and a uniform length $1/2 L$ of brass whose mass is m_b . The moment of inertia I of the rod about an axis perpendicular to the rod and through its centre is equal to [AMU 2012]
 (a) $(m_w + m_b)L^2 / 12$ (b) $(m_w + m_b)L^2 / 6$
 (c) $(m_w + m_b)L^2 / 3$ (d) $(m_w + m_b)L^2 / 2$
- 61** The instantaneous angular position of a point on a rotating wheel is given by the equation $\theta(t) = 2t^3 - 6t^2$. The torque on the wheel becomes zero at [CBSE AIPMT 2011]
 (a) $t = 0.5 \text{ s}$ (b) $t = 0.25 \text{ s}$
 (c) $t = 2 \text{ s}$ (d) $t = 1 \text{ s}$
- 62** The moment of inertia of a thin uniform rod of mass M and length L about an axis passing through its mid-point and perpendicular to its length is I_0 . Its moment of inertia about an axis passing through one of its ends and perpendicular to its length is [CBSE AIPMT 2011]
 (a) $I_0 + ML^2/4$ (b) $I_0 + 2ML^2$
 (c) $I_0 + ML^2$ (d) $I_0 + ML^2/2$
- 63** A solid sphere of mass M rolls without slipping on an inclined plane of inclination θ . What should be the minimum coefficient of friction, so that the sphere rolls down without slipping? [NEET 2011]
 (a) $\frac{2}{5} \tan \theta$ (b) $\frac{2}{7} \tan \theta$
 (c) $\frac{5}{7} \tan \theta$ (d) $\tan \theta$
- 64** A force $F = 2.0 \text{ N}$ acts on a particle P in the XZ -plane. The force F is parallel to X -axis. The particle P (as shown in the figure) is at a distance 3 m and the line joining P with the origin makes angle 30° with the X -axis. The magnitude of torque on P w.r.t. origin O (in N-m) is [AMU 2011]



- (a) 2 (b) 3
 (c) 4 (d) 5

- 65** One solid sphere A and another hollow sphere B are of same mass and same outer radius. Their moments of inertia about their diameters are respectively, I_A and I_B such that [J&K CET 2011]
 (a) $I_A = I_B$ (b) $I_A > I_B$
 (c) $I_A < I_B$ (d) None of these

ANSWERS

CHECK POINT 9.1

1. (c)	2. (d)	3. (c)	4. (c)	5. (d)	6. (b)	7. (a)	8. (b)	9. (a)	10. (a)
11. (b)	12. (a)	13. (a)	14. (c)	15. (b)	16. (a)	17. (d)			

CHECK POINT 9.2

1. (b)	2. (c)	3. (a)	4. (d)	5. (a)	6. (a)	7. (a)	8. (a)	9. (b)	10. (d)
11. (d)	12. (b)	13. (a)	14. (a)	15. (a)	16. (b)	17. (c)	18. (b)		

CHECK POINT 9.3

1. (a)	2. (b)	3. (c)	4. (b)	5. (c)	6. (d)	7. (d)	8. (d)	9. (d)	10. (d)
11. (b)	12. (a)	13. (d)	14. (c)	15. (b)					

(A) Taking it together

1. (b)	2. (b)	3. (a)	4. (d)	5. (b)	6. (c)	7. (a)	8. (c)	9. (c)	10. (d)
11. (d)	12. (a)	13. (d)	14. (d)	15. (a)	16. (c)	17. (c)	18. (d)	19. (a)	20. (a)
21. (b)	22. (b)	23. (c)	24. (a)	25. (b)	26. (b)	27. (b)	28. (a)	29. (a)	30. (a)
31. (d)	32. (c)	33. (b)	34. (b)	35. (c)	36. (a)	37. (a)	38. (c)	39. (c)	40. (a)
41. (b)	42. (a)	43. (d)	44. (a)	45. (d)	46. (b)	47. (c)	48. (b)	49. (b)	50. (b)
51. (d)	52. (b)	53. (b)	54. (b)	55. (c)	56. (b)	57. (c)	58. (b)	59. (c)	60. (b)
61. (a)	62. (b)	63. (a)	64. (d)	65. (b)	66. (d)	67. (b)	68. (b)	69. (c)	70. (a)
71. (b)	72. (b)	73. (c)	74. (d)						

(B) Medical entrance special format questions

• Assertion and reason

1. (c) 2. (b) 3. (a) 4. (d) 5. (c)

• Statement based questions

1. (d) 2. (a) 3. (c) 4. (d) 5. (b) 6. (a) 7. (a)

• Match the columns

1. (c) 2. (b) 3. (d) 4. (d)

(C) Medical entrance's gallery

1. (b)	2. (a)	3. (d)	4. (d)	5. (d)	6. (c)	7. (a)	8. (a)	9. (d)	10. (d)
11. (b)	12. (c)	13. (a)	14. (b)	15. (b)	16. (c)	17. (c)	18. (c)	19. (b)	20. (a)
21. (a)	22. (b)	23. (a)	24. (a)	25. (a)	26. (d)	27. (c)	28. (d)	29. (b)	30. (b)
31. (a)	32. (b)	33. (b)	34. (a)	35. (c)	36. (a)	37. (b)	38. (b)	39. (d)	40. (b)
41. (c)	42. (a)	43. (c)	44. (a)	45. (d)	46. (b)	47. (b)	48. (d)	49. (a)	50. (a)
51. (c)	52. (b)	53. (c)	54. (d)	55. (a)	56. (c)	57. (d)	58. (d)	59. (c)	60. (a)
61. (d)	62. (a)	63. (b)	64. (b)	65. (c)					

Hints & Explanations

• CHECK POINT 9.1

- 1 (c) Given, frequency of wheel, $v = 300 \text{ rpm} = \frac{300}{60} \text{ rps} = 5 \text{ rps}$

Angle described by wheel in one rotation $= 2\pi \text{ rad}$

Therefore, angle described by wheel in $1 \text{ s} = 2\pi \times 5 = 10\pi \text{ rad}$

- 2 (d) Angular displacement per second,

$$\theta = \omega t = \frac{500 \times 2\pi}{60} = \frac{50\pi}{3} \text{ rad}$$

- 3 (c) Given, $\theta = 100\pi \text{ rad}$, $t = 5 \text{ s}$, $\omega = ?$

$$\text{Angular speed, } \omega = \frac{\theta}{t} = \frac{100\pi}{5} = 20\pi$$

- 4 (c) We know that, $\omega = 2\pi n \Rightarrow \omega_1 = 2\pi n_1$

where, $n_1 = 1800 \text{ rpm}$, $n_2 = 3000 \text{ rpm}$

$$\omega_1 = 2\pi \times \frac{1800}{60} = 2\pi \times 30 = 60\pi \text{ rad s}^{-1}$$

$$\begin{aligned} \text{Similarly, } \omega_2 &= 2\pi n_2 = 2\pi \times \frac{3000}{60} \\ &= 2\pi \times 50 = 100\pi \text{ rad s}^{-1} \end{aligned}$$

If the angular velocity of a rotating wheel about an axis changes due to change in angular velocity in a time interval Δt , then the angular acceleration of rotating wheel about that axis is

$$\alpha = \frac{\text{Change in angular velocity}}{\text{Time interval}}$$

$$\alpha = \frac{\omega_2 - \omega_1}{\Delta t} \Rightarrow \alpha = \frac{100\pi - 60\pi}{20}$$

$$\alpha = \frac{40\pi}{20} = 2\pi \text{ rad s}^{-2}$$

- 5 (d) Angular displacement during given time, $\theta = (\omega_2 - \omega_1)t$

Here, $\omega_1 = 100 \text{ rad/s}$,

$\omega_2 = 300 \text{ rad/s}$ and $t = 10 \text{ s}$

$$\Rightarrow \theta = (300 - 100) \times 10 = 2000$$

Therefore, number of revolutions made during this time

$$= \frac{2000}{2\pi} = 318.31 \approx 318$$

- 6 (b) The kinematical equation for rotational motion is

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

Here, $\alpha = 3 \text{ rad s}^{-2}$, $\omega_0 = 2 \text{ rad s}^{-1}$ and $t = 2 \text{ s}$

$$\text{Hence, } \theta = 2 \times 2 + \frac{1}{2} \times 3 \times (2)^2 \text{ or } \theta = 4 + 6 = 10 \text{ rad}$$

- 7 (a) Given, $\alpha = 3 \text{ rad s}^{-2}$, $\omega_1 = 10 \text{ rad s}^{-1}$, $\omega_2 = 20 \text{ rad s}^{-1}$

$$\text{Angle, } \theta = \frac{\omega_2^2 - \omega_1^2}{2\alpha} = \frac{(20)^2 - (10)^2}{2 \times 3} = 50 \text{ rad}$$

- 8 (b) If angular acceleration is constant, we have

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

Given, $\theta = 15^\circ$, $\omega_0 = 0$

For the 1st condition (time $= t$ second),

$$15^\circ = \frac{1}{2} \alpha t^2 \quad \dots(i)$$

For the 2nd condition (time $= 3t$ second),

$$\theta_1 = \frac{1}{2} \alpha (3t)^2 = \frac{1}{2} (\alpha) 9t^2 \quad \dots(ii)$$

So, the increase in angle through which it rotates in the next 2 s,

$$\begin{aligned} \Delta\theta &= \theta_1 - \frac{1}{2} \alpha t^2 = 9 \times \frac{1}{2} \alpha t^2 - \frac{1}{2} \alpha t^2 = 8 \left(\frac{1}{2} \alpha t^2 \right) \\ &= 8 \times 15^\circ = 120^\circ \quad [\text{from Eq. (i)}] \end{aligned}$$

- 9 (a) The role of moment of inertia in the study of rotational motion is analogous to that of mass in the study of translational motion.

- 10 (a) Moment of inertia of a body depends on position and orientation of the axis of rotation with respect to the body.

- 11 (b) Perpendicular distance from Z-axis

$$= \sqrt{(1)^2 + (1)^2} = \sqrt{2} \text{ m}$$

$$\therefore I = Mr^2 = (1)(\sqrt{2})^2 = 2 \text{ kg-m}^2$$

- 12 (a) Moment of inertia of the rod lying along Z-axis will be zero. Moment of inertia of the rods along X and Y-axes will be $\frac{ML^2}{3}$ each about Z-axis. Hence, total moment of inertia about Z-axis is $\frac{2}{3} ML^2$.

- 13 (a) Radius of gyration of solid sphere,

$$K = \sqrt{\frac{I}{M}} = \sqrt{\frac{\frac{7}{5} MR^2}{M}} = \sqrt{\frac{7}{5}} R$$

- 14 (c) Moment of inertia, $I = \frac{ml^2}{3} \quad \dots(i)$

New moment of inertia,

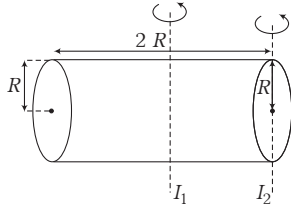
$$\begin{aligned} I' &= \frac{(3m/4)(3l/4)^2}{3} \\ &= \frac{27}{64} \left(\frac{ml^2}{3} \right) = \frac{27}{64} I \quad [\text{from Eq. (i)}] \end{aligned}$$

- 15 (b) Moments of inertia of circular ring and circular disc about the axes passing through their centres and perpendicular to their planes are MR^2 and $\frac{1}{2} MR^2$, respectively. So, the desired

$$\text{ratio is } \frac{MR^2}{(1/2) MR^2} = 2:1$$

- 16 (a) From theorem of parallel axes, $I_2 = I_1 + MR^2$

$$I_2 - I_1 = MR^2$$



- 17 (d) According to the question,

$$l = 2\pi R \Rightarrow R = \frac{l}{2\pi}$$

$$\therefore I_1 = \frac{Ml^2}{12}$$

$$\text{and } I_2 = MR^2 = \frac{Ml^2}{4\pi^2}$$

$$\therefore \frac{I_1}{I_2} = \frac{\pi^2}{3}$$

• CHECK POINT 9.2

- 1 (b) Torque, $\tau = \mathbf{r} \times \mathbf{F}$

$$\Rightarrow \tau = 4\hat{j} \times (-6\hat{i}) = -(-24\hat{k}) = 24\hat{k}$$

- 2 (c) Torque, $\tau = \mathbf{r} \times \mathbf{F} = [(0-3)\hat{i} + (2-0)\hat{j} + (0-0)\hat{k}] \times [20\hat{i}]$
 $= [-3\hat{i} + 2\hat{j}] \times [20\hat{i}] = -40\hat{k}$

- 3 (a) Torque of the force, $\tau = \mathbf{r} \times \mathbf{F}$

$$\tau = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 1 \\ -3 & 1 & 5 \end{vmatrix} = (14\hat{i} - 38\hat{j} + 16\hat{k})$$

- 4 (d) Torque required to open/close the door will remain constant.

$$\therefore r_1 \times F_1 = r_2 \times F_2$$

$$1.6 \times 1 = 0.4 \times F_2 \Rightarrow F_2 = 4 \text{ N}$$

- 5 (a) Given, $I = 2 \text{ kg-m}^2$, $\omega = 30 \text{ rad s}^{-1}$ and $t = 15 \text{ s}$

$$\text{Average torque of the force, } \tau = I\alpha = I \frac{\omega}{t} = 2 \times \frac{30}{15} = 4 \text{ N-m}$$

- 6 (a) $\omega_1 = 10 \text{ rad s}^{-1}$, $\omega_2 = 0$, $t = 10 \text{ s}$

$$\therefore \alpha = \frac{\omega_2 - \omega_1}{t} = \frac{0 - 10}{10} = -1 \text{ rad s}^{-2}$$

Negative sign indicates retardation.

$$\text{Now, } I = MR^2 = 10 \times (0.3)^2 = 0.9 \text{ kg-m}^2$$

$$\therefore \text{Magnitude of torque, } \tau = I\alpha = 0.9 \times (1) = 0.9 \text{ N-m}$$

- 8 (a) As $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, so angular momentum is the moment of momentum.

- 9 (b) For a body of mass m rotating with velocity v in a circle of radius r , angular momentum is given by $\mathbf{L} = m(\mathbf{r} \times \mathbf{v})$
 For unit mass, $m = 1$

$$\therefore |\mathbf{L}| = [(8\hat{i} - 4\hat{j}) \times (8\hat{i} + 4\hat{j})]$$

$$\Rightarrow \mathbf{L} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & -4 & 0 \\ 8 & 4 & 0 \end{vmatrix}$$

$$\Rightarrow \mathbf{L} = \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(32+32) \Rightarrow \mathbf{L} = 64\hat{k} \text{ units}$$

- 10 (d) Angular momentum, $L = mvr = m\omega r^2 = m \times \frac{2\pi}{T} \times r^2$
 $= \frac{6 \times 10^{24} \times 2 \times 3.14 \times (1.5 \times 10^{11})^2}{3.14 \times 10^7} = 2.7 \times 10^{40} \text{ kg-m}^2 \text{ s}^{-1}$

- 11 (d) As, $\mathbf{L} = \mathbf{r} \times \mathbf{p} = rp \sin\theta$

Angular momentum $\mathbf{L}$ is maximum when $\mathbf{p}$ is perpendicular to $\mathbf{r}$.

- 12 (b) The angular momentum of the body is given by

$$L = I\omega \text{ or } L = I \times \frac{2\pi}{T} \text{ or } L \propto \frac{1}{T} \Rightarrow \frac{L_1}{L_2} = \frac{T_2}{T_1}$$

$$\Rightarrow \frac{L}{L_2} = \frac{2T}{T} \quad (\because T_2 = 2T \text{ and } L_1 = L)$$

$$\Rightarrow L_2 = \frac{L}{2}$$

Thus, on doubling the time period, angular momentum of body becomes half.

- 13 (a) Torque, $\tau = \frac{dL}{dt}$

where, L = angular momentum.

If $\tau = 0$, then $\frac{dL}{dt} = 0$.

i.e. L = constant or conserved.

- 14 (a) Torque = Rate of change of angular momentum = $\frac{dL}{dt}$

$$\text{Given, } dL = 4A_0 - A_0 = 3A_0, dt = 4 \text{ min}$$

$$\Rightarrow \text{Torque acting on the body, } \tau = \frac{3}{4}A_0$$

- 15 (a) If no external torque acts on a system of particles, then angular momentum of the system remains constant, i.e. $\tau = 0$.

$$\Rightarrow \frac{dL}{dt} = 0 \Rightarrow L = I\omega = \text{constant}$$

$$\Rightarrow I_1\omega_1 = I_2\omega_2 \Rightarrow \frac{1}{2}Mr^2\omega_1 = \frac{1}{2}(M+2m)r^2\omega_2 \quad \dots(i)$$

Here, $M = 2 \text{ kg}$, $m = 0.25 \text{ kg}$, $r = 0.2 \text{ m}$, $\omega_1 = 30 \text{ rad s}^{-1}$

Substituting the given values in Eq. (i), we get

$$\frac{1}{2} \times 2 \times (0.2)^2 \times 30 = \frac{1}{2} \times (2 + 2 \times 0.25) (0.2)^2 \times \omega_2$$

$$\text{or } 60 = 2.5\omega_2 \text{ or } \omega_2 = 24 \text{ rad s}^{-1}$$

- 16 (b) According to law of conservation of angular momentum,

$$I\omega = \text{constant}$$

$$\text{or } \frac{R^2}{T} = \text{constant} \quad \left(\because I \propto R^2 \text{ and } \omega \propto \frac{1}{T} \text{ and } I \propto \frac{1}{\omega} \right)$$

$$\begin{aligned} \therefore T &\propto R^2 \\ \text{As, } R' &= \frac{1}{n} R \\ \Rightarrow \frac{T}{T'} &= \left(\frac{R}{R'}\right)^2 \\ \Rightarrow \frac{T}{T'} &= \left(\frac{R \times n}{R}\right)^2 \\ \therefore T' &= \frac{T}{n^2} = \frac{24}{n^2} \text{ h} \end{aligned}$$

- 17 (c)** From conservation of angular momentum,

$$\begin{aligned} \omega_1 I_1 &= \omega_2 I_2 \\ \Rightarrow \omega_2 &= \frac{\omega_1 I_1}{I_2} \\ \Rightarrow \omega_2 &= \omega \left(\frac{MR^2}{MR^2 + 2mR^2} \right) = \omega \left(\frac{M}{M + 2m} \right) \end{aligned}$$

- 18 (b)** Rotational kinetic energy $= \frac{1}{2} I \omega^2 = 8 \text{ J}$

$$\Rightarrow \frac{1}{2} \times \frac{1}{2} m r^2 \omega^2 = 8 \quad \text{or} \quad \frac{1}{4} \times 2 \times (1)^2 \omega^2 = 8$$

$$\text{or} \quad \omega^2 = 16 \quad \text{or} \quad \omega = 4 \text{ rad s}^{-1}$$

$$\text{Angular momentum, } L = I \omega = \frac{1}{2} m r^2 \omega$$

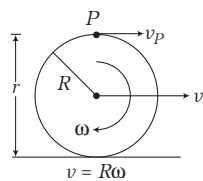
$$= \frac{1}{2} \times 2 \times (1)^2 \times 4 = 4 \text{ J-s}$$

• CHECK POINT 9.3

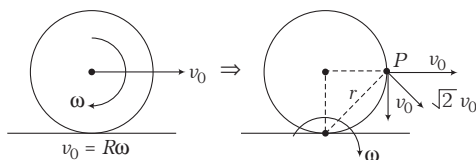
- 1 (a)** In this case, instantaneous axis of rotation is always below the centre of mass. This is possible only when point of contact on this axis moves with a speed equal to that of centre of mass.

- 2 (b)** Required velocity of the particle at top most point,

$$v_P = r \omega = (2R) \omega = 2v$$



- 3 (c)** Speed of particle, $v_P = r \omega = (\sqrt{2} R) \omega = \sqrt{2} v_0$



- 4 (b)** The acceleration of its centre down the plane,

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} \quad \text{or} \quad a < g \sin \theta$$

- 5 (c)** In case of pure rolling,

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}; \text{ for solid sphere, } \frac{I}{mR^2} = \frac{2}{5}$$

$$\therefore a = \frac{5}{7} g \sin \theta$$

Since, a is independent of m and R , both the spheres reach the bottom together.

- 6 (d)** The linear acceleration of the sphere,

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} = \frac{g \sin \theta}{1 + \frac{2}{5}} = \frac{5}{7} g \sin \theta$$

- 7 (d)** The linear acceleration,

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}} \quad \text{and} \quad \frac{I}{mR^2} = \frac{2}{5} \quad (\text{For solid sphere})$$

$$\therefore a = \frac{g \sin 30^\circ}{1 + \frac{2}{5}} = \frac{5g}{14}$$

- 8 (d)** In case of pure rolling,

$$\text{Acceleration, } a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

$$\text{For ring, } \frac{I}{mR^2} = 1$$

$$\therefore a = \frac{g}{4} \quad (\because \theta = 30^\circ)$$

- 9 (d)** In case of pure rolling, $a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$

$$\frac{I}{mR^2} = \frac{2}{5} \quad (\text{for solid sphere}) \quad \text{and} \quad \frac{I}{mR^2} = \frac{1}{2} \quad (\text{for disc}).$$

$$\therefore \text{For solid sphere, } a_1 = \frac{5}{7} g \sin \theta$$

$$\text{Similarly, for disc, } a_2 = \frac{2}{3} g \sin \theta$$

$$\text{From } s = \frac{1}{2} at^2, t = \sqrt{\frac{2s}{a}}$$

$$\Rightarrow \frac{t_1}{t_2} = \sqrt{\frac{a_2}{a_1}} = \sqrt{\frac{2/3}{5/7}} = \sqrt{\frac{14}{15}}$$

- 10 (d)** For all bodies, $a = g \sin \theta - \mu g \cos \theta$

As, μ is same for all bodies.

Hence, all will take same time.

- 11 (b)** In pure rolling, mechanical energy remains conserved.

Therefore, at bottom most point, total kinetic energy will be mgh .

$$K_{\text{total}} = K_{\text{rot}} + K_{\text{trans}}$$

$$= \frac{1}{2} I \omega^2 + \frac{1}{2} m v^2 = \frac{1}{2} \left(\frac{2}{3} m R^2 \right) \left(\frac{v^2}{R^2} \right) + \frac{1}{2} m v^2$$

$$K_{\text{total}} = \frac{5}{6}mv^2$$

$$\text{Now, } \frac{5}{6}mv^2 = mgh$$

∴ The speed of a uniform spherical shell,

$$v = \sqrt{\frac{6gh}{5}}$$

- 12 (a)** Time taken by rolling body to reach the bottom,

$$t = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g} \left(1 + \frac{K^2}{R^2} \right)}$$

A body with smaller value of $\frac{K^2}{R^2}$ will take less time to reach the bottom. Hence, solid sphere will take least time to reach the bottom.

- 13 (d)** The relation between linear velocity (v) and angular velocity (ω) is

$$v = r\omega \Rightarrow \omega = \frac{v}{r}$$

$$\begin{aligned} \text{Total kinetic energy} &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{2} \times \left(\frac{1}{2}mr^2 \right) \left(\frac{v^2}{r^2} \right) \\ &= \frac{1}{2}mv^2 + \frac{1}{4}mv^2 = \frac{3}{4}mv^2 \\ \frac{3}{4}mv^2 &= mg(3) \Rightarrow v = 2\sqrt{10} \text{ ms}^{-1} \end{aligned}$$

$$\text{Now, } r = \frac{v}{\omega} = \frac{2\sqrt{10}}{2\sqrt{2}} = \sqrt{5} \text{ m}$$

- 14 (c)** Total kinetic energy of a body which is rolling without slipping is given by $K_{\text{total}} = K_{\text{rot}} + K_{\text{trans}}$

$$\text{For solid spherical ball, } I = \frac{2}{5}mR^2 \text{ and } v = R\omega$$

(Along the diameter)

where, R is radius of spherical ball.

$$\begin{aligned} \text{So, } K_{\text{total}} &= \frac{1}{2} \left(\frac{2}{5}mR^2 \right) \omega^2 + \frac{1}{2}mR^2\omega^2 \\ &= \frac{7}{10}mR^2\omega^2 \Rightarrow K_{\text{total}} = \frac{7}{10}mv^2 \end{aligned}$$

Potential energy = Kinetic energy

$$mgh = \frac{7}{10}mv^2 \Rightarrow v^2 = \frac{10}{7}gh \quad \dots(i)$$

For vertical projection,

$$\begin{aligned} v'^2 &= v^2 - 2gh' \Rightarrow 0 = v^2 - 2gh' \Rightarrow v^2 = 2gh' \\ \Rightarrow \frac{10}{7}gh &= 2gh' \Rightarrow h' = \frac{5}{7}h \end{aligned}$$

- 15 (b)** Let f be the frictional force towards left for pure rolling.

$$\text{Now, linear acceleration, } a = \frac{F-f}{m} = \frac{7-f}{10} \quad \dots(ii)$$

$$\text{Angular acceleration, } \alpha = \frac{\tau}{I} = \frac{fR}{\frac{2}{5}mR^2} = \frac{5f}{2mR}$$

$$\begin{aligned} \text{Also, } a &= R\alpha \Rightarrow a = R \left(\frac{5f}{2mR} \right) = \frac{5f}{2m} \\ a &= \frac{5f}{20} \quad \dots(ii) \end{aligned}$$

Equating Eqs. (i) and (ii), we get

$$\frac{7-f}{10} = \frac{5f}{20} \Rightarrow f = 2N$$

(A) Taking it together

- 1 (b)** Torque, $\tau = \frac{dL}{dt}$. When $\tau = 0$, then $\frac{dL}{dt} = 0$

$\Rightarrow L = \text{constant, i.e. } L \text{ is conserved.}$

- 2 (b)** We have, $I\omega = L = \text{constant} \Rightarrow \omega \propto 1/I$

On stretching hands, I will increase, therefore ω will decrease.

- 3 (a)** On bending the head before diving in swimming pool, the moment of inertia of the diver decreases.

- 4 (d)** It cannot be zero, otherwise net torque or net angular acceleration will become zero.

- 5 (b)** Moment of inertia of the system,

$$I = 5mR^2 = 5 \times 2 \times (0.1)^2 = 0.1 \text{ kg-m}^2$$

- 6 (c)** Rotational kinetic energy, $E = \frac{1}{2}I\omega^2 \Rightarrow \omega = \sqrt{\frac{2E}{I}}$

$$\text{Now, angular momentum, } L = I\omega = \sqrt{2EI}$$

- 7 (a)** Angular velocity, $\omega_0 = \frac{900 \times 2\pi}{60} \text{ rad s}^{-1} = 30\pi \text{ rad s}^{-1}$

$$\text{Now, } 0 = \omega_0 - \alpha t$$

$$\therefore \text{Angular retardation, } \alpha = \frac{\omega_0}{t} = \frac{30\pi}{60} \text{ rad s}^{-2} = \frac{\pi}{2} \text{ rad s}^{-2}$$

- 8 (c)** We have, $\theta_1 = \frac{1}{2}(\alpha)(2)^2 = 2\alpha$

$$\text{and } \theta_2 = \frac{1}{2}\alpha(4)^2 - \frac{1}{2}\alpha(2)^2 = 6\alpha$$

$$\therefore \frac{\theta_2}{\theta_1} = \frac{3}{1}$$

- 9 (c)** Angular velocity, $\omega = \alpha t$

$$\therefore \alpha = \frac{\omega}{t} = \frac{80}{5} = 16 \text{ rad s}^{-2}$$

$$\text{Now, } \theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(16)(5)^2 = 200 \text{ rad}$$

- 10 (d)** Here, $r = 0.2 \text{ m}$, $M = 10 \text{ kg}$, $v = 1200 \text{ rpm} = 20 \text{ rps}$

$$\therefore \text{Angular momentum, } L = I\omega = (Mr^2)(2\pi v)$$

$$= 10 \times (0.2)^2 \times 2 \times \frac{22}{7} \times 20 = 50.28 \text{ kg-m}^2\text{s}^{-1}$$

- 11 (d)** If the distance of point from axis (r) will increase, then corresponding v of the point will also increase accordingly. Hence, ω is constant or independent of r .

- 12 (a) Angular momentum, $L = I\omega = I(2\pi f)$

Frequency f is doubled. Hence, angular momentum will become $2L$.

- 13 (d) About origin, angular momentum will be

$$L = m(\mathbf{r} \times \mathbf{v}) = 2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 2 & -2 & 2 \end{vmatrix} = 4\hat{i} - 4\hat{j} - 8\hat{k}$$

Therefore, component of angular momentum about Z -axis would be $-8 \text{ kg-m}^2\text{s}^{-1}$.

- 14 (d) Angular acceleration, $\alpha = \frac{\tau}{I} = \frac{1000}{200} = 5 \text{ rad s}^{-2}$

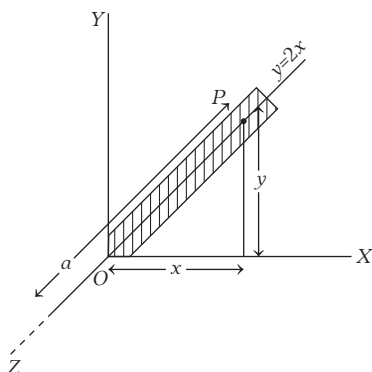
Now, angular velocity, $\omega = \alpha t = (5)(3) = 15 \text{ rad s}^{-1}$

- 15 (a) $\therefore I = mK^2 = 10(2)^2 = 40 \text{ kg-m}^2$

Kinetic energy of rotation,

$$K_R = \frac{1}{2} I \omega^2 = \frac{1}{2} (40) (5)^2 = 500 \text{ J}$$

- 16 (c) The given situation of rod is shown below.



Suppose a mass particle situated at point $P(x, y)$.

Hence, distance of mass particle from X -axis $= x$, distance of mass particle from Y -axis $= y$ and distance of mass particle from Z -axis, $a = \sqrt{x^2 + y^2}$

Clearly, $a > x$

$$a > y$$

Since, moment of inertia, $I \propto (\text{distance})^2$.

Hence, moment of inertia of the rod is maximum about Z -axis.

- 17 (c) I_{CM} is less than I about any other axis not passing through centre of mass only when two axes are parallel.

- 18 (d) Angular momentum, $L = \sqrt{2KI}$ (just like $p = \sqrt{2Km}$)

$$\Rightarrow L \propto \sqrt{K}$$

If K is made 4 times, L will become 2 times.

- 19 (a) Work done by friction in pure rolling is always zero. It is because there is no displacement by frictional force. Friction works only in case of rolling with sliding.

- 20 (a) Angular momentum, $L = mvR + I_C\omega = mvR + \frac{1}{2}mvR$
 $= \frac{3}{2}mvR$

- 21 (b) On inclined surface, gravitational force provides torque. On rough surface, friction provides torque for rolling but on horizontal smooth surface, external force is needed to create torque.

- 22 (b) In case of pure rolling of sphere, $\frac{K_R}{K_T} = \frac{2}{5}$

$\therefore$ Total kinetic energy $= 7/5$ times the translational kinetic energy. At highest point, whole of the kinetic energy will be converted into potential energy.

$$\therefore mgh = \frac{7}{5} \left(\frac{1}{2} mv^2 \right) \text{ or } h = \frac{7v^2}{10g} = \frac{7(10)^2}{10 \times 10} = 7 \text{ m}$$

- 23 (c) According to the question,

$$(K_T)_{\text{sphere}} = (K_T)_{\text{cylinder}} \quad (m \text{ and } v \text{ are same})$$

$$\therefore \frac{5}{7} (mgh_1) = \frac{2}{3} (mgh_2)$$

$$\Rightarrow \frac{h_1}{h_2} = \frac{14}{15}$$

- 24 (a) Torque, $\tau = \mathbf{r} \times \mathbf{F}$, i.e. τ is perpendicular to both $\mathbf{r}$ and $\mathbf{F}$.

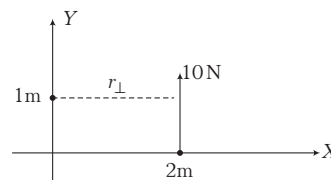
$$\therefore \mathbf{r} \cdot \tau = 0 \text{ and } \mathbf{F} \cdot \tau = 0$$

- 25 (b) From the conservation of angular momentum, $I_1\omega_1 = I_2\omega_2$

$$\Rightarrow \omega_2 = \frac{I_1}{I_2} \omega_1 = \frac{(1/2 ma^2)}{(1/2 ma^2) + ma^2} \omega = \frac{\omega}{3}$$

- 26 (b) In the given diagrams, when the small piece Q removed and glued to the centre of the plate, the mass comes closer to the Z -axis, hence moment of inertia decreases.

- 27 (b) Moment of force $= \text{Force} \times r_{\perp} = 10 \times 2 = 20 \text{ N-m}$



- 28 (a) $\omega = \alpha - \beta t$

Comparing with $\omega = \omega_0 - \alpha t$, we get

Initial angular velocity $= \alpha$

Angular retardation $= \beta$

$\therefore$ Angle rotated before it stops is

$$0 = \alpha^2 - 2\beta\theta \Rightarrow \theta = \frac{\alpha^2}{2\beta}$$

- 29 (a) For uniform rod, the moment of inertia about an axis passing through its centre is $mL^2/12$.

$$\therefore \text{Radius of gyration, } K = \sqrt{\frac{I}{m}} = \sqrt{\frac{(mL^2/12)}{m}} = \frac{L}{2\sqrt{3}}$$

- 30 (a) From the equation, $0 = \omega_0 - \alpha t$

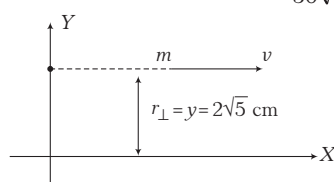
$$\therefore \alpha = \frac{\omega_0}{t} = \frac{(100 \times 2\pi) / 60}{15} = 0.7 \text{ rad s}^{-2}$$

Now, angle rotated before coming to rest, $\theta = \frac{\omega_0^2}{2\alpha}$

$$\text{or } \theta = \frac{(100 \times 2\pi / 60)^2}{2 \times 0.7} = 78.33 \text{ rad}$$

Now, number of rotations, $n = \frac{\theta}{2\pi} = 12.47 \approx 12.5$

- 31 (d) The angular momentum, $L = mvr_{\perp} = (5)(3\sqrt{2})(2\sqrt{5})$
 $= 30\sqrt{10} \text{ g-cm}^2\text{s}^{-1}$



- 32 (c) $W = \text{Change in rotational kinetic energy} = \frac{1}{2} I (\omega_f^2 - \omega_i^2)$

$$= \frac{1}{2} \left(\frac{1}{2} mR^2 \right) [(2\pi f_f)^2 - (2\pi f_i)^2]$$

$$= \left(\frac{1}{4} mR^2 \right) [4\pi^2 (f_f^2 - f_i^2)]$$

$$= \frac{1}{4} \times 2 \times (1)^2 \times 4 \times \pi^2 (100 - 25)$$

$$= 1.48 \times 10^3 \text{ J} \approx 1.5 \times 10^3 \text{ J}$$

- 33 (b) $\therefore I_1 = \frac{MR^2}{2}$ and $I_2 = \frac{MR^2}{2} + \frac{1}{2} \left(\frac{M}{4} \right) R^2 = \frac{5}{8} MR^2$

By conservation of angular momentum,

$$I_1 \omega_1 = I_2 \omega_2 \Rightarrow \omega_2 = \frac{I_1}{I_2} \omega_1$$

$$= \frac{(1/2) MR^2}{(5/8) MR^2} \omega \Rightarrow \omega_2 = \frac{4}{5} \omega$$

- 34 (b) For hollow sphere, $K_H = \sqrt{\frac{I}{m}} = \sqrt{\frac{\frac{2}{3} mR^2 + mR^2}{m}} = \sqrt{\frac{5}{3}} R$

$$\text{For solid sphere, } K_S = \sqrt{\frac{I}{m}} = \sqrt{\frac{\frac{2}{5} mR^2 + mR^2}{m}} = \sqrt{\frac{7}{5}} R$$

$$\therefore \frac{K_H}{K_S} = \sqrt{\frac{25}{21}} = \frac{5}{\sqrt{21}}$$

- 35 (c) For square lamina, $I_2 = I_3$. These values will be less than I_1 because distribution of masses are nearer to the axes corresponding to I_2 and I_3 .

- 36 (a) As no external torque acts on the system, angular momentum should be conserved.

Hence, $I\omega = \text{constant}$

...(i)

where, I is moment of inertia of the system and ω is angular velocity of the system.

From Eq. (i), $I_1 \omega_1 = I_2 \omega_2$

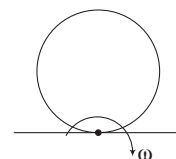
(where, ω_1 and ω_2 are angular velocities before and after jumping)

$$\Rightarrow I\omega = \frac{I}{2} \times \omega_2$$

(As mass reduced to half, hence moment of inertia also reduced to half)

$$\Rightarrow \omega_2 = 2\omega$$

- 37 (a) Velocity, $v = r\omega$ or $v \propto r$

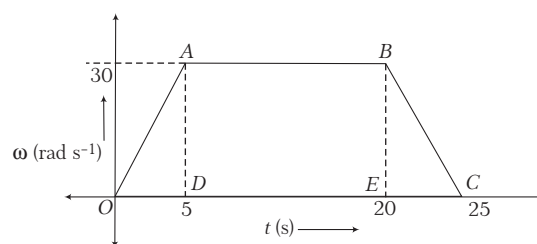


$$r_Q > r_C > r_P \Rightarrow v_Q > v_C > v_P$$

- 38 (c) Angle rotated, $\theta = \text{Area under } \omega\text{-}t \text{ graph.}$

Area under $\omega\text{-}t$ graph = Area of trapezium $OABCO$.

$\therefore \text{Area of } (\triangle OABCO) = \frac{1}{2} (\text{Sum of parallel sides}) \times \text{Distance between them}$



$$= \frac{1}{2} (AB + OC) \times AD = \frac{1}{2} (15 + 25) \times 30 = 600 \text{ rad}$$

- 39 (c) Radius of gyration of disc, $K_D = \sqrt{\frac{I_D}{m}} = \sqrt{\frac{\frac{1}{2} mr^2}{m}} = \frac{r}{\sqrt{2}}$

$$\text{Radius of gyration of ring, } K_R = \sqrt{\frac{I_R}{m}} = \sqrt{\frac{m(2r)^2}{m}} = 2r$$

$$\therefore \frac{K_R}{K_D} = 2\sqrt{2}$$

- 40 (a) Moment of inertia, $I = mK^2$

$\therefore$ Ball is rolling without slipping, $v = R\omega$

$$\text{Translational kinetic energy, } K_T = \frac{1}{2} mv^2$$

$$\text{Rotational kinetic energy, } K_R = \frac{1}{2} I\omega^2 = \frac{mK^2 v^2}{2R^2}$$

$$\therefore \text{Total kinetic energy} = \frac{1}{2} mv^2 + \frac{mK^2 v^2}{2R^2}$$

$$\text{Required fraction} = \frac{\frac{mK^2 v^2}{2R^2}}{\frac{1}{2} mv^2 + \frac{mK^2 v^2}{2R^2}} = \frac{K^2}{R^2 + K^2}$$

- 41 (b)** We have, $(\pi R^2 t d) = m$ ($\because$ Mass = Density $\times$ Volume)

$$\therefore R^2 = \frac{m}{\pi t d}$$

$$\text{Now, } I = \frac{1}{2} m R^2 = \frac{m^2}{2 \pi t d}$$

$$\text{or } I \propto \frac{1}{d}$$

$$\therefore \text{The ratio of moments of inertia, } \frac{I_1}{I_2} = \frac{d_2}{d_1}$$

- 42 (a)** Original volume of the earth,

$$V = \frac{4}{3} \pi R^3$$

Volume of earth after shrinking

$$V' = \frac{V}{4}$$

$$\Rightarrow \frac{4}{3} \pi R'^3 = \frac{1}{4} \times \frac{4}{3} \pi R^3$$

$$R' = \frac{R}{2}$$

By conservation of angular momentum,

$$I' \omega' = I \omega$$

$$\Rightarrow \frac{2}{5} M R'^2 \times \frac{2\pi}{T'} = \frac{2}{5} M R^2 \times \frac{2\pi}{T}$$

$$\Rightarrow \frac{T}{T'} = \left(\frac{R}{R'} \right)^2$$

$$\Rightarrow \frac{T}{T'} = \left(\frac{R}{R/2} \right)^2$$

$$\Rightarrow \frac{T}{T'} = 4$$

$$\Rightarrow T' = \frac{T}{4} = \frac{24}{4} \quad [\because T = 24 \text{ h}]$$

$$= 6 \text{ h}$$

- 43 (d)** At bottom most point, total kinetic will be mgh .

$$\begin{aligned} \text{Total kinetic energy} &= \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 \\ &= \frac{1}{2} m v^2 + \frac{1}{2} \left(\frac{2}{5} m R^2 \right) \left(\frac{v}{R} \right)^2 = \frac{7}{10} m v^2 \end{aligned}$$

$$\text{Now, } \frac{7}{10} m v^2 = mgh \Rightarrow v = \sqrt{\frac{10}{7} gh}$$

- 44 (a)** From theorem of perpendicular axes, we have

$$I = I_C + m \left(\frac{a}{\sqrt{2}} \right)^2 = \left(\frac{m a^2}{12} + \frac{m a^2}{12} \right) + \frac{m a^2}{2} = \frac{2}{3} m a^2$$

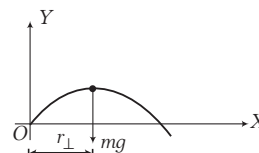
- 45 (d)** Moment of inertia, $I = I_{CM} + m x^2$

i.e. I - x graph is a parabola not passing through origin.

- 46 (b)** Moment of inertia, $I = 4 \left[\frac{M}{2} \frac{(l/2)^2}{3} \right] = \frac{M l^2}{6}$

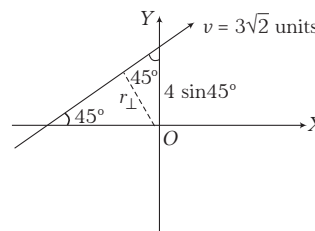
- 47 (c)** Path of A is cycloid and distance travelled in one rotation is $8R$.

- 48 (b)** Torque, $\tau_0 = mg \times r_{\perp}$



As, $r_{\perp}$ is continuously increasing or torque is continuously increasing on the particle. Hence, angular momentum is continuously increasing.

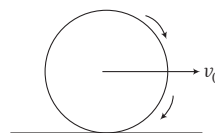
- 49 (b)** From the figure, $r_{\perp} = 4 \sin 45^\circ = 2\sqrt{2}$ units



The magnitude of the angular momentum,

$$L = m v r_{\perp} = 5(3\sqrt{2})(2\sqrt{2}) = 60 \text{ units}$$

- 50 (b)** About bottom most point,



The angular momentum of sphere,

$$L = M R v_0 + \frac{2}{5} M R^2 \left(\frac{v_0}{R} \right) = \frac{7}{5} M R v_0$$

- 51 (d)** Radius of gyration, $K = \sqrt{\frac{I}{m}}$

$$\therefore K_{\text{disc}} = \sqrt{\frac{\frac{1}{2} m R^2 + m R^2}{m}} = \sqrt{\frac{3}{2}} R$$

$$\text{and } K_{\text{ring}} = \sqrt{\frac{m R^2 + m R^2}{m}} = \sqrt{2} R$$

$$\therefore \frac{K_{\text{disc}}}{K_{\text{ring}}} = \frac{\sqrt{3/2}}{\sqrt{2}} = \frac{\sqrt{3}}{2}$$

- 52 (b)** Radius of gyration of a circular disc about a tangential axis,

$$K_1 = \sqrt{\frac{I_1}{m_1}} = \sqrt{\frac{(5/4) m_1 R^2}{m_1}} = \sqrt{\frac{5}{4}} R$$

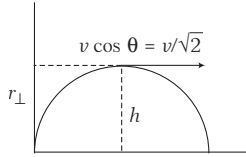
Radius of gyration of a circular ring about a tangential axis,

$$K_2 = \sqrt{\frac{I_2}{m_2}} = \sqrt{\frac{(3/2) m_2 R^2}{m_2}} = \sqrt{\frac{3}{2}} R$$

$$\therefore \frac{K_1}{K_2} = \frac{\sqrt{5}}{\sqrt{6}}$$

- 53 (b) The magnitude of angular momentum, $L = mvr_{\perp}$

$$\Rightarrow L = \frac{mvh}{\sqrt{2}}$$



- 54 (b) The initial velocity is $\mathbf{v}_i = v\hat{\mathbf{e}}_y$ and after reflection from the wall, the final velocity is $\mathbf{v}_f = -v\hat{\mathbf{e}}_y$. The trajectory is described as position vector $\mathbf{r} = y\hat{\mathbf{e}}_y + a\hat{\mathbf{e}}_z$.
Hence, the change in angular momentum

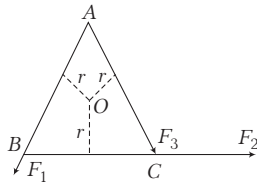
$$= \mathbf{r} \times m(\mathbf{v}_f - \mathbf{v}_i)$$

$$= (y\hat{\mathbf{e}}_y + a\hat{\mathbf{e}}_z) \times (-mv\hat{\mathbf{e}}_y - mv\hat{\mathbf{e}}_y)$$

$$= (y\hat{\mathbf{e}}_y + a\hat{\mathbf{e}}_z) \times (-2mv\hat{\mathbf{e}}_y) \quad [\because \hat{\mathbf{e}}_y \times \hat{\mathbf{e}}_y = 0 \text{ and } \hat{\mathbf{e}}_z \times \hat{\mathbf{e}}_y = -\hat{\mathbf{e}}_x]$$

$$= 2mva\hat{\mathbf{e}}_x$$
- 55 (c) Friction in this case is backward. If it is insufficient, it is the case of forward slipping or $K_T > K_R$.
- 56 (b) Work done by retarding torque = Change in rotational kinetic energy
 Since, rotational kinetic energy of both ring and disc are equal, both will rotate with same number of revolutions.

- 57 (c) For total torque about O to be zero,



$$F_1 r + F_2 r - F_3 r = 0$$

$$\Rightarrow F_3 = F_1 + F_2$$

- 58 (b) From the figure, $AC = BD = \sqrt{l^2 + l^2} = \sqrt{2}l$

Moment of inertia of four point masses about BD,

$$I_{BD} = m\left(\frac{l\sqrt{2}}{2}\right)^2 + m \times 0 + m\left(\frac{l\sqrt{2}}{2}\right)^2 + m \times 0$$

$$I_{BD} = ml^2$$

Applying the theorem of parallel axes,

$$I_{XY} = I_{BD} + 4m(AO)^2 = ml^2 + 4m\left(\frac{l}{\sqrt{2}}\right)^2 = 3ml^2$$

- 59 (c) Area under $n-t$ graph will give total number of revolution before coming to stop.

Area = Total number of revolutions

$$= \frac{1}{2} \times 8 \times \left(\frac{1800}{60}\right) + 8 \times \frac{600}{60} + \frac{1}{2} \times 16 \times \left(\frac{600}{60}\right)$$

$$= 120 + 80 + 80 = 280$$

- 60 (b) Force 8 N can also be resolved in tangential and radial directions.

$$\tau_{\text{net}} = (9 + 8 \sin 30^\circ - 4)(0.2) \text{ N-m clockwise}$$

$$= 1.80 \text{ N-m clockwise}$$

- 61 (a) From law of conservation of angular momentum, (about centre of mass of the rod) $L_i = L_f$

$$mvr = \frac{ML^2}{12} \cdot \omega$$

$$\Rightarrow \omega = \frac{12mvr}{ML^2}$$

$$= \frac{(12)(1)(2)(1/4)}{(2)(1)^2} = 3 \text{ rad s}^{-1}$$

- 62 (b) Moment of inertia,

$$I = 3 \left[\frac{Ml^2}{3} \right] = Ml^2$$

- 63 (a) Angular momentum will not remain conserved due to the torque produced by weight of particle acting about axis of rotation. Hence, only mechanical energy is conserved.

- 64 (d) In case of pure rolling, ratio of rotational to translational kinetic energy is 2/5. Therefore, total kinetic energy is 7/5 times the translational kinetic energy. At maximum compression, whole of energy is elastic potential. Hence,

$$\frac{7}{5} \left(\frac{1}{2} Mv^2 \right) = \frac{1}{2} kx_{\text{max}}^2$$

$\therefore$ The compression of the spring,

$$x_{\text{max}} = v \sqrt{\frac{7M}{5k}}$$

- 65 (b) From theorem of parallel axes, moment of inertia about an axis passing through a distance $l/2$ from centre in all directions will have moment of inertia $\frac{ml^2}{3}$. All such points at same distance from centre will lie in a circle.

- 66 (d) Whole mass has equal distance from the centre O. Hence, $I_O = mR^2$. Further, centre of mass of the remaining portion will be to the left of point O. More the distance of axis from centre of mass, more is the moment of inertia. Hence, $I_A > I_O$.

- 67 (b) The moment of inertia,

$$I = I_{\text{ring}} + 3I_{\text{spoke}}$$

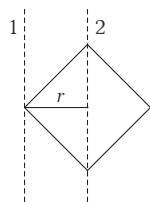
$$= MR^2 + 3 \left(\frac{mR^2}{3} \right) = (M + m)R^2$$

- 68 (b) Decrease in gravitational potential energy = Increase in rotational kinetic energy about an axis passing through end point.

$$\therefore mg \frac{l}{2} = \frac{1}{2} \left(\frac{ml^2}{3} \right) \omega^2$$

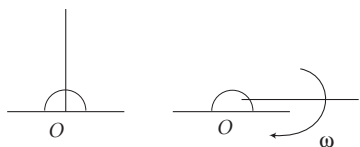
$$\Rightarrow \text{Angular velocity, } \omega = \sqrt{\frac{3g}{l}} \text{ or } \omega \propto \sqrt{\frac{g}{l}}$$

- 69 (c) By theorem of parallel axes, $I_1 = I_2 + (4M)r^2$



$$= 4 \left[\frac{ML^2}{3} \sin^2 45^\circ \right] + 4M \left[\frac{\sqrt{2}L}{2} \right]^2 = \frac{8}{3} ML^2$$

- 70 (a) Decrease in gravitational potential energy = Increase in rotational kinetic energy about point O.



$$\text{or } mg \frac{L}{2} = \frac{1}{2} \left(\frac{mL^2}{3} \right) \omega^2$$

$$\text{or } \omega = \sqrt{\frac{3g}{L}}$$

$$\text{Now, } v = r\omega \quad (\text{In pure rotation})$$

$$\therefore v = L \sqrt{\frac{3g}{L}} \quad (\because r = L)$$

$$= \sqrt{3gL}$$

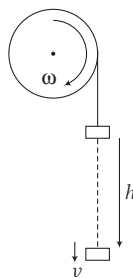
- 71 (b) Velocity, $v = r\omega$

Decrease in gravitational potential energy = Increase in kinetic energy

$$\therefore mgh = \frac{1}{2} I\omega^2 + \frac{1}{2} mv^2$$

$$= \frac{1}{2} I\omega^2 + \frac{1}{2} m(r\omega)^2$$

$$\therefore \text{Angular velocity, } \omega = \sqrt{\frac{2mgh}{I + mr^2}}$$



- 72 (b) From the property of a circle, if an arc subtends an angle θ at any point A on circumference, then it will subtend an angle 2θ at centre C. So, in the same time interval, a particle rotating in a circle turns double the angle with respect to centre point C compared to point A.

$$\therefore \omega_C = 2\omega_A$$

$$\text{or } \frac{\omega_A}{\omega_C} = \frac{1}{2}$$

- 73 (c) If we complete the ring, its mass will become $2M$.

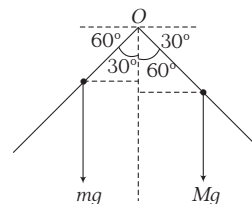
$$\therefore I_{\text{whole ring}} = \frac{1}{2} (2M) (R^2) = MR^2$$

($I_{\text{whole ring}}$ = moment of inertia about any diameter)

$$\therefore I_{\text{half ring}} = \frac{1}{2} MR^2 \quad (\text{About any radius})$$

This value is independent of angle θ .

- 74 (d) Net torque about O should be zero.



$$\Rightarrow Mg \frac{l}{2} \sin 30^\circ = mg \frac{l}{2} \sin 60^\circ$$

$$\frac{M}{m} = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$$

(B) Medical entrance special format questions

• Assertion and reason

- 1 (c) L about centre = $mvr = \text{constant}$

As, $\tau = \frac{dL}{dt}$ and derivative of any constant is zero.

$$\therefore \tau_{\text{centre}} = 0$$

2. (b) $L = mvr \sin \theta$ or $mvr_{\perp}$

In case of constant velocity, m , v and $r_{\perp}$ all are constants. Therefore, angular momentum is constant.

Further, $L = n \frac{h}{2\pi}$ (in Bohr's theory).

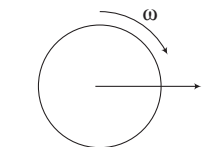
Therefore, L and h have same units.

- 3 (a) Moment of inertia of sphere in case 1 is $\frac{2}{5} MR^2$.

Moment of inertia of sphere in case 2 is $\frac{7}{5} MR^2$.

So, more work has to be done to rotate the sphere in case 2.

- 4 (d) Sphere can roll without slipping on surface, if $v = r\omega$ on an inclined plane. It is friction which creates rotation of sphere. So, smooth surface cannot create rotation.



- 5 (c) Angular momentum about bottom

$$\text{most point, } L = mv_0 R - \frac{1}{2} mR^2 \omega_0$$

If $L = 0$ or $v_0 = \frac{\omega_0 R}{2}$, disc will come to rest after some time.

In case of pure rotation, since frictional force is acting, therefore angular momentum is not conserved.

• Statement based questions

- 1 (d) As, angular acceleration,

$$\alpha = \frac{d\omega}{dt}$$

where, ω is angular velocity of the disc.

Given, $\omega = \text{constant}$

$$\Rightarrow \alpha = \frac{d\omega}{dt} = \frac{0}{dt} = 0$$

Hence, angular acceleration is zero.

2 (a) Direction of linear velocity always keeps on changing in circular orbit. Hence, linear momentum is varying.

3 (c) Here, $\mu < \mu_{\min}$ but $\mu \neq 0$. Therefore, sphere will roll with forward slipping or translation.

4 (d) Acceleration, $a = \frac{g \sin \theta}{1 + \frac{I}{mr^2}}$

For a sphere, $\frac{I}{mr^2} = \frac{2}{5}$

Hence, $a = \frac{5}{7} g \sin \theta = \text{constant}$.

Hence, speeds of all spheres will be same at the bottom.

Sphere (ii) has the largest mass. Hence, it will have the maximum kinetic energy.

5 (b) Since, the sphere is sliding, maximum force of friction will act on passing through the bottom most point. Torque of this force about point of contact is zero. Hence, angular momentum will remain conserved about the point of contact but not about the centre of mass because torque of friction will not be zero about centre of mass.

6 (a) Statement I is correct while Statement II is incorrect.

The corrected statement is;

Moment of inertia changes with axis of rotation.

7 (a) Statement I is correct while Statement II is incorrect.

The corrected statement is;

Every rolling motion cannot be considered as pure rotation.

For pure rotation, the condition is

linear velocity, $v = \omega r$

• Match the columns

1 (c) $L_1 = I\omega = \frac{2}{5}MR^2\omega$

$$L_2 = \frac{2}{5}M\left(\frac{R}{2}\right)^2\omega'$$

As, $L_1 = L_2 \Rightarrow \omega' = 4\omega$

$$\therefore \frac{2\pi}{T'} = 4\left(\frac{2\pi}{T}\right) \Rightarrow T' = \frac{T}{4}$$

Further, $K = \frac{L^2}{2I}$

Since, angular momentum is constant and I has become (1/4)th. Therefore, kinetic energy will become 4 times.

Hence, A $\rightarrow$ q, B $\rightarrow$ s, C $\rightarrow$ q.

2 (b) In general, $v_P = 2v \sin\left(\frac{\theta}{2}\right)$

(A) If $\theta = 60^\circ$, then $v_P = 2v \sin\left(\frac{60^\circ}{2}\right) = 2v \times \frac{1}{2} = v$

(B) If $\theta = 90^\circ$, then $v_P = 2v \sin\left(\frac{90^\circ}{2}\right) = 2v \times \frac{1}{\sqrt{2}} = \sqrt{2}v$

(C) If $\theta = 120^\circ$, then $v_P = 2v \sin\left(\frac{120^\circ}{2}\right) = 2v \times \frac{\sqrt{3}}{2} = \sqrt{3}v$

(D) If $\theta = 180^\circ$, then $v_P = 2v \sin\left(\frac{180^\circ}{2}\right) = 2v$

Hence, A $\rightarrow$ q, B $\rightarrow$ p, C $\rightarrow$ s, D $\rightarrow$ r.

3 (d) Torque, $\tau = 0$

$$\therefore L = \text{constant}, K = \frac{L^2}{2I} \text{ and } I\omega = \text{constant. The insect first}$$

moves away from the axis, then towards it. Hence, I will first increase and then decrease.

Hence, A $\rightarrow$ q, B $\rightarrow$ r, C $\rightarrow$ p, D $\rightarrow$ r.

4 (d) (A) $I_1 = 2\left(\frac{ml^2}{12}\right) + 2(m)\left(\frac{l}{2}\right)^2 = \frac{2}{3}ml^2$

(B) $I_2 = 0 + 2\left(\frac{ml^2}{3}\right) + ml^2 = \left(\frac{5}{3}\right)ml^2$

(C) $I_3 = 4\left[\frac{ml^2}{3} \sin^2 45^\circ\right] = \frac{2}{3}ml^2 = I_1$

Hence, A $\rightarrow$ q, B $\rightarrow$ s, C $\rightarrow$ q.

(C) Medical entrances' gallery

1 (b) Given, position vector, $\mathbf{r} = 2\hat{\mathbf{k}}$ m

Force, $\mathbf{F} = 3\hat{\mathbf{j}}$ N

As, torque, $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = 2\hat{\mathbf{k}} \times 3\hat{\mathbf{j}} = 6(-\hat{\mathbf{i}}) = -6\hat{\mathbf{i}}$ N-m

2 (a) Initial angular speed of wheel,

$$\omega_0 = 2\pi f_0 = 2\pi \times \frac{360}{60} \text{ rad/s} = 12\pi \text{ rad/s}$$

Final angular speed of wheel,

$$\omega = 2\pi f = 2\pi \times \frac{1200}{60} \text{ rad/s}$$

$$= 40\pi \text{ rad/s}$$

$$t = 14 \text{ s}$$

From the equation of rotational motion, $\omega = \omega_0 + \alpha t$

$$\Rightarrow \alpha = \frac{\omega - \omega_0}{t} = \frac{40\pi - 12\pi}{14} = \frac{28\pi}{14} = 2\pi \text{ rad/s}^2$$

3 (d) Given, mass of cylinder, $m = 2 \text{ kg}$

Radius of cylinder, $r = 4 \text{ cm} = 4 \times 10^{-2} \text{ m}$

Rotational velocity, $\omega = 3 \text{ rpm} = 3 \times \frac{2\pi}{60} = \frac{\pi}{10} \text{ rad s}^{-1}$ and

$\theta = 2\pi \times \text{revolution} = 2\pi \times 2\pi = 4\pi^2 \text{ rad}$

The work done in rotating an object by an angle θ from rest is given by $W = \tau \theta$

As the cylinder is brought to rest, so the work done will be negative.

According to work-energy theorem,

Work done = Change in rotational kinetic energy

$$-\tau\theta = \frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2 = \frac{1}{2}I(\omega_f^2 - \omega_i^2)$$

$$\Rightarrow \tau = \frac{I(\omega_i^2)}{2\theta} \quad (\because \omega_f = 0)$$

$$= \frac{1}{2} \left(\frac{1}{2}mr^2 \right) \frac{\omega_i^2}{\theta} \quad \left[\because I = \frac{1}{2}mr^2 \text{ (for cylinder)} \right]$$

$$= \frac{1}{4}mr^2 \frac{\omega^2}{\theta} \quad (\because \omega_i = \omega)$$

$$= \frac{1}{4} \times 2 \times (4 \times 10^{-2})^2 \times \left(\frac{\pi}{10} \right)^2 \times \frac{1}{4\pi^2}$$

$$= \frac{1}{4} \times 2 \times 16 \times 10^{-4} \times \frac{\pi^2}{100} \times \frac{1}{4\pi^2}$$

$$= \frac{2}{100} \times 10^{-4} = 2 \times 10^{-6} \text{ N-m}$$

4 (d) Given, radius, $R = 2 \text{ m}$, mass, $m = 100 \text{ kg}$

and $v_{\text{CM}} = v = 20 \text{ cm s}^{-1} = 20 \times 10^{-2} \text{ m s}^{-1}$

Then, according to work-energy theorem, the work done in stopping the disc is equal to the change in its kinetic energy, i.e.

$$W = (\text{KE})_f - (\text{KE})_i \quad \dots(i)$$

As, the disc stops at the end, so final velocity is zero.

Thus, $(\text{KE})_f = 0$

Since, the disc is rolling, so its initial kinetic energy would have both rotational and translational kinetic energy components.

$$(\text{KE})_i = (\text{KE})_r + (\text{KE})_t = \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

$$= \frac{1}{2} \left(\frac{1}{2}mR^2 \right) \omega^2 + \frac{1}{2}mv^2 \quad (\because \text{For disc, } I = \frac{1}{2}mR^2)$$

$$= \frac{1}{4}mR^2\omega^2 + \frac{1}{2}mv^2$$

$$= \frac{1}{4}mR^2 \frac{v^2}{R^2} + \frac{1}{2}mv^2 \quad (\because v = r\omega)$$

$$= \frac{3}{4}mv^2$$

$$\therefore W = |\text{KE}_i| = \frac{3}{4}mv^2 \quad [\text{From Eq. (i)}]$$

Substituting the given values in above equation, we get

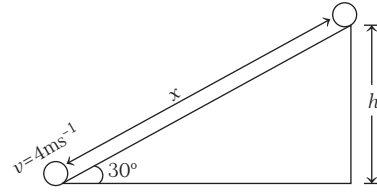
$$W = \frac{3}{4} \times 100 \times (20 \times 10^{-2})^2$$

$$= \frac{3}{4} \times 400 \times 100 \times 10^{-4} = 3 \text{ J}$$

5 (d) When a body rolls, i.e. have rotational motion, the total kinetic energy of the system will be

$$\text{KE} = \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2} \right)$$

where, m = mass of body, v = velocity and K = radius of gyration.



Given, $m = 2 \text{ kg}$, $\theta = 30^\circ$ and $v = 4 \text{ ms}^{-1}$

Let h be the height of the inclined plane, then from law of conservation of energy,

KE = PE

$$\frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2} \right) = mgh$$

Substituting the given values in the above equation, we get

$$\frac{1}{2} \times 2 \times 16 \left(1 + \frac{1}{2} \right) = 2 \times 10 \times h \quad \left(\text{For cylinder, } \frac{K^2}{R^2} = \frac{1}{2} \right)$$

$$\Rightarrow 8 \times \frac{3}{2} = 10h \Rightarrow h = 1.2 \text{ m}$$

From the above diagram, $\sin\theta = \frac{h}{x}$

$$\Rightarrow x = \frac{h}{\sin\theta} = \frac{1.2}{\sin 30^\circ}$$

$$= 1.2 \times 2 = 2.4 \text{ m}$$

6 (c) Given, initial velocity of sphere, $u = 2.8 \text{ ms}^{-1}$

Acceleration on the inclined plane,

$$a = \frac{g \sin\theta}{1 + \frac{K^2}{R^2}} \quad \dots(i)$$

where, K and R are the radius of gyration and radius of sphere, respectively.

Moment of inertia of sphere, $I = mK^2 = \frac{2}{5}mR^2$

For sphere, $\frac{K^2}{R^2} = \frac{2}{5}$

Putting this value in Eq. (i), we get

$$a = \frac{g \sin 30^\circ}{1 + \frac{2}{5}} \quad [\because \theta = 30^\circ \text{ (given)}]$$

$$= \frac{5g}{14} = \frac{25}{7} \text{ ms}^{-2} \quad [\because g = 10 \text{ ms}^{-2}]$$

If s be the maximum distance travelled by the sphere, then at maximum distance, $v = 0$.

$\therefore$ From third equation of motion, $v^2 = u^2 - 2as$

$$0 = u^2 - 2as$$

$$\Rightarrow s = \frac{u^2}{2a} = \frac{(2.8)^2 \times 7}{2 \times 25} = 1.09 \text{ m} \approx 1.38 \text{ m}$$

- 7 (a) Given, mass ratio of two discs,

$$m_1 : m_2 = 1 : 2 \quad \text{i.e.} \quad \frac{m_1}{m_2} = \frac{1}{2}$$

and diameter ratio, $\frac{d_1}{d_2} = \frac{2}{1} \Rightarrow \frac{r_1}{r_2} = \frac{2}{1}$

$\therefore$ Ratio of their moments of inertia,

$$\frac{I_1}{I_2} = \frac{\frac{m_1 r_1^2}{2}}{\frac{m_2 r_2^2}{2}} = \frac{m_1}{m_2} \cdot \left(\frac{r_1}{r_2}\right)^2 = \frac{1}{2} \left(\frac{2}{1}\right)^2 = \frac{2}{1}$$

$$\therefore I_1 : I_2 = 2 : 1$$

- 8 (a) Given, initial angular velocity of object, $\omega_0 = 0$

Angular displacement, $\theta = 60$ rad

and $\Delta t = 10$ s

From equation of rotational motion, $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

$$60 = 0 \times t + \frac{1}{2} \times \alpha \times 10^2$$

$$\Rightarrow \alpha = \frac{60}{50} \Rightarrow \alpha = 1.2 \text{ rad s}^{-2}$$

- 9 (d) Moment of force is defined as the cross product of the force and the force arm.

Given, $\mathbf{F} = 4\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 6\hat{\mathbf{k}}$, $\mathbf{r}_1 = 2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$

and $\mathbf{r}_2 = 2\hat{\mathbf{i}} + 0\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

Moment of force = $\mathbf{r} \times \mathbf{F} = (\mathbf{r}_2 - \mathbf{r}_1) \times \mathbf{F}$

$$= [(2\hat{\mathbf{i}} + 0\hat{\mathbf{j}} - 3\hat{\mathbf{k}}) - (2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} - 2\hat{\mathbf{k}})] \times (4\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 6\hat{\mathbf{k}})$$

$$= (0\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 1\hat{\mathbf{k}}) \times (4\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 6\hat{\mathbf{k}}) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 2 & -1 \\ 4 & 5 & -6 \end{vmatrix}$$

$$= \hat{\mathbf{i}}[(-6 \times 2) - (-1 \times 5)] - \hat{\mathbf{j}}[(-6 \times 0) - (-1 \times 4)] + \hat{\mathbf{k}}[(0 \times 5) - 2 \times 4]$$

$$= -7\hat{\mathbf{i}} - 4\hat{\mathbf{j}} - 8\hat{\mathbf{k}}$$

- 10 (d) As we know that, external torque, $\tau_{\text{ext}} = \frac{dL}{dt}$

where, L is the angular momentum.

Since, in the given condition, $\tau_{\text{ext}} = 0 \Rightarrow \frac{dL}{dt} = 0$

$$\Rightarrow L = \text{constant}$$

Hence, when the radius of the sphere is increased keeping its mass same, only the angular momentum remains constant. But other quantities like moment of inertia, rotational kinetic energy and angular velocity changes.

- 11 (b) Translational kinetic energy of a rolling body,

$$K_t = \frac{1}{2} m v_{\text{CM}}^2 \quad \dots(i)$$

Total kinetic energy of a rolling body

$$= \text{Rotational KE} + \text{Translational KE} = K_r + K_t$$

$$= \frac{1}{2} I \omega^2 + \frac{1}{2} m v_{\text{CM}}^2 \quad \dots(ii)$$

For a solid sphere, moment of inertia about its diametric axis,

$$I = \frac{2}{5} m R^2$$

Substituting the value of I in Eq. (ii), we get

$$K_r + K_t = \frac{1}{2} \left(\frac{2}{5} m R^2 \right) \omega^2 + \frac{1}{2} m v_{\text{CM}}^2$$

$$= \frac{1}{2} \left(\frac{2}{5} m R^2 \right) \left(\frac{v_{\text{CM}}}{R} \right)^2 + \frac{1}{2} m v_{\text{CM}}^2 \quad [\because v_{\text{CM}} = R\omega]$$

$$= \frac{1}{5} m v_{\text{CM}}^2 + \frac{1}{2} m v_{\text{CM}}^2$$

$$= \left(\frac{1}{5} + \frac{1}{2} \right) m v_{\text{CM}}^2 = \frac{7}{10} m v_{\text{CM}}^2 \quad \dots(iii)$$

$$\therefore \text{Ratio, } \frac{K_t}{K_t + K_r} = \frac{\frac{1}{2} m v_{\text{CM}}^2}{\frac{7}{10} m v_{\text{CM}}^2} = \frac{1}{2} \times \frac{10}{7} = \frac{5}{7}$$

$$\therefore K_t : (K_t + K_r) = 5 : 7$$

- 12 (c) Moment of inertia of the insect-disc system,

$$MI = \frac{1}{2} M R^2 + m x^2$$

where, m = mass of insect

and x = distance of insect from centre.

Clearly, as the insect moves along the diameter of the disc. Moment of inertia first decreases and then increases.

By conservation of angular momentum, angular speed first increases and then decreases.

- 13 (a) For disc, moment of inertia about the diameter, $I_d = m r^2 / 4$

and moment of inertia about the tangential axis,

$$I_{\text{disc}} = I_d + m r^2$$

$$I_{\text{disc}} = \frac{m r^2}{4} + m r^2 = \frac{5 m r^2}{4}$$

Let the radius of gyration of disc be K_{disc} .

$$I_{\text{disc}} = m K_{\text{disc}}^2$$

$$\Rightarrow \frac{5 m r^2}{4} = m K_{\text{disc}}^2 \Rightarrow K_{\text{disc}}^2 = \frac{5}{4} r^2$$

For ring, moment of inertia about the diameter,

$$I_a = \frac{m r^2}{2}$$

and moment of inertia about the tangential axis,

$$I_{\text{ring}} = I_a + m r^2 = \frac{m r^2}{2} + m r^2 = \frac{3}{2} m r^2$$

Let the radius of gyration of ring be K_{ring} .

$$I_{\text{ring}} = m K_{\text{ring}}^2$$

$$\Rightarrow \frac{3}{2} m r^2 = m K_{\text{ring}}^2 \Rightarrow K_{\text{ring}}^2 = \frac{3}{2} r^2$$

$$\therefore \frac{K_{\text{disc}}}{K_{\text{ring}}} = \frac{\sqrt{\frac{5}{4}} r}{\sqrt{\frac{3}{2}} r} = \sqrt{\frac{5}{6}}$$

- 14** (b) Usually, the centre of gravity coincides with the centre of mass for bodies, when the gravitational field is uniform for bodies of small height.

Since, the gravitational force can be taken to be centred or confined at the centre of gravity of body, so the torque of gravitational force about the centre of gravity must be zero.

A couple always produces the rotation without translation. Mechanical advantage is the ratio of the force produced by a machine to the force applied on it. It is used in assessing the performance of a machine. So, option (b) is correct.

- 15** (b) When no external torque acts on system, then angular momentum of system remains constant.

Angular momentum before contact

$$= I\omega_1 + I_2\omega_2 = I\omega_1 + I\omega_2 = I(\omega_1 + \omega_2)$$

Angular momentum after the discs brought into contact

$$= I_{\text{net}}\omega = (I_1 + I_2)\omega = 2I\omega$$

From law of conservation of angular momentum,

$$I(\omega_1 + \omega_2) = 2I\omega \Rightarrow \omega = \frac{\omega_1 + \omega_2}{2}$$

Now, to calculate loss of energy, we subtract initial and final energies of system.

$$\text{Loss of energy} = \frac{1}{2}I\omega_1^2 + \frac{1}{2}I\omega_2^2 - \frac{1}{2}(2I)\omega^2 = \frac{1}{4}I(\omega_1 - \omega_2)^2$$

- 16** (c) Torque (τ) acting on a body and angular acceleration (α) produced in it are related as

$$\tau = I\alpha$$

Consider a hollow cylinder around which a rope is wound.

Torque acting on the cylinder due to the force F is $\tau = Fr$.

Now, we have $\tau = I\alpha$

where, I = moment of inertia of the cylinder about the axis through the centre.

$$\Rightarrow I = mr^2$$

Given, $F = 30$ N, $m = 3$ kg and $r = 40$ cm = 40×10^{-2} m

Angular acceleration,

$$\begin{aligned}\alpha &= \frac{\tau}{I} = \frac{Fr}{mr^2} \\ &= \frac{F}{mr} = \frac{30}{3 \times 40 \times 10^{-2}} \\ &= \frac{100}{4} = 25 \text{ rad s}^{-2}\end{aligned}$$

- 17** (c) The kinetic energy of a rolling solid sphere

$$\begin{aligned}&= K_{\text{trans}} + K_{\text{rot}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{2} \times \frac{2}{5}mR^2\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{5}mv^2 \quad \left(\because \omega = \frac{v}{R} \right) \\ &= 7/10 mv^2\end{aligned}$$

Hence, total kinetic energy is sum of translation and rotational kinetic energies.

For different solid bodies, total kinetic energy is not always twice of translation kinetic energy. Hence, Assertion is correct but Reason is incorrect.

- 18** (c) Total kinetic energy (KE) of the system

$$= \text{KE (translational)} + \text{KE (rotational)}$$

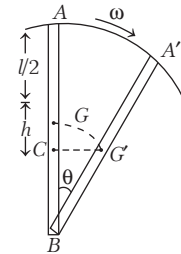
$$= \frac{1}{2}(2m)v^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}(2m)v^2 + \frac{1}{2}\left(\frac{2}{3}mR^2\right)\frac{v^2}{R^2}$$

$$= \frac{1}{2}(2m)v^2 + \frac{1}{3}mv^2 = \frac{4}{3}mv^2$$

- 19** (b) The moment of inertia of a body about an axis depends not only on the mass of the body but also on the distribution of mass about the axis. For a given body, mass is same, so it will depend only on the distribution of mass about the axis. The mass is farthest from axis BC , so I_2 is maximum, mass is nearest to axis AC , so I_3 is minimum.

Hence, the correct sequence will be $I_2 > I_1 > I_3$.

- 20** (a) When the rod rotates through an angle θ , the centre of gravity falls through a distance h .



$$\text{In } \triangle BCG', \cos \theta = \frac{(l/2) - h}{l/2} \text{ or } h = \frac{l}{2}(1 - \cos \theta)$$

$$\text{Decrease in potential energy} = mgh = \frac{1}{2}mg l (1 - \cos \theta) \quad \dots(i)$$

The decrease in potential energy is equal to the kinetic energy of rotation.

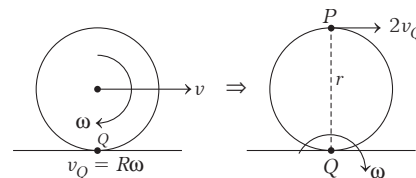
$$\begin{aligned}\therefore (\text{KE})_{\text{rotational}} &= \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2 \quad \dots(ii) \\ &\left(\because I = \frac{ml^2}{3} \right)\end{aligned}$$

From Eqs. (i) and (ii), we get

$$\frac{1}{2}\left(\frac{ml^2}{3}\right)\omega^2 = \frac{1}{2}mg l (1 - \cos \theta) \Rightarrow \omega = \sqrt{\frac{6g}{l}} \sin \frac{\theta}{2}$$

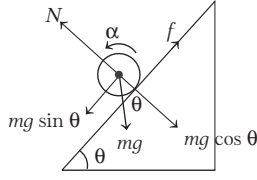
- 21** (a) Velocity of the particle at Q , $v_Q = r\omega = R\omega$

Velocity of the particle at P , $v_P = r\omega = (2R)\omega = 2v_Q$



Hence, points near the top move faster than points near the bottom.

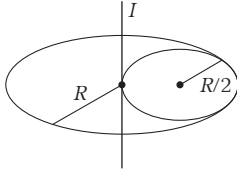
- 22 (b)** The direction of the frictional force will be up the inclined plane in both the motions. The friction acts opposite to the relative motion of the point of contact on ground. In both cases, the force acting on the point of cylinder in contact with the plane is $mg \sin \theta$. As there is pure rolling, the velocity and acceleration of the point of contact must be zero, so friction acts upwards along the inclined plane in both the motions (ascending or descending).



- 23 (a)** Torque, $\tau = I\alpha$ or moment of inertia of wheel, $I = \frac{\tau}{\alpha}$

$$\text{Moment of inertia of wheel, } I = \frac{5 \times 10 \times 10^{-2}}{2} \text{ kg-m}^2 \\ = 0.25 \text{ kg-m}^2$$

- 24 (a)** Considering the information given in the question, let us draw the figure.



If the above figure is considered, then moment of inertia of disc will be given as

$$I = I_{\text{remaining}} + I_{(R/2)} \\ \Rightarrow I_{\text{remaining}} = I - I_{(R/2)} \\ \text{Putting the values, we get} \\ = \frac{MR^2}{2} - \left[\frac{M \left(\frac{R}{2} \right)^2}{2} + \frac{M \left(\frac{R}{2} \right)^2}{4} \right] \\ = \frac{MR^2}{2} - \left[\frac{MR^2}{32} + \frac{MR^2}{16} \right] = \frac{MR^2}{2} - \left[\frac{MR^2 + 2MR^2}{32} \right] \\ = \frac{MR^2}{2} - \frac{3MR^2}{32} = \frac{16MR^2 - 3MR^2}{32} \\ I_{\text{remaining}} = \frac{13MR^2}{32}$$

- 25 (a)** Acceleration of an object rolling down an inclined plane is given by

$$a = \frac{g \sin \theta}{1 + I/mr^2}$$

where, θ = angle of inclination of the inclined plane,

m = mass of the object

and I = moment of inertia about the axis through centre of mass.

Type of sphere is not mentioned in the question. Therefore, we will assume the given sphere as solid sphere.

$$\text{For disc, } \frac{I}{mr^2} = \frac{1/2 mr^2}{mr^2} = \frac{1}{2}$$

$$\text{For solid sphere, } \frac{I}{mr^2} = \frac{2/5 mr^2}{mr^2} = \frac{2}{5}$$

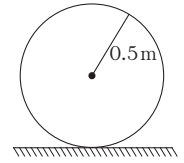
$$\therefore a_{\text{disc}} = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta = 0.66 g \sin \theta$$

$$a_{\text{solid sphere}} = \frac{g \sin \theta}{1 + \frac{2}{5}} = \frac{5}{7} g \sin \theta = 0.71 g \sin \theta$$

Clearly, $a_{\text{solid sphere}} > a_{\text{disc}}$

Hence, sphere gets to the bottom of the plane first.

- 26 (d)** According to given question, a uniform circular disc of radius 50 cm at rest is free to turn about an axis having perpendicular to its plane and passes through its centre. This situation can be shown by the given figure.



$$\therefore \text{Angular acceleration, } \alpha = 2 \text{ rad s}^{-2}$$

(Given)

$$\text{Angular speed, } \omega = \alpha t = 4 \text{ rad s}^{-1}$$

$$\therefore \text{Centripetal acceleration, } a_c = \omega^2 r = (4)^2 \times 0.5 = 16 \times 0.5$$

$$a_c = 8 \text{ ms}^{-2}$$

$$\therefore \text{Linear acceleration at the end of 2 s,}$$

$$a_t = \alpha r = 2 \times 0.5 \Rightarrow a_t = 1 \text{ ms}^{-2}$$

Therefore, the net acceleration at the end of 2 s is given by

$$a = \sqrt{a_c^2 + a_t^2} \\ a = \sqrt{(8)^2 + (1)^2} = \sqrt{65} \Rightarrow a \approx 8 \text{ ms}^{-2}$$

- 27 (c)** According to law of conservation of angular momentum,

$$L_i = L_f \Rightarrow mv_0 R_0 = mv' \left(\frac{R_0}{2} \right) \Rightarrow v' = 2v_0$$

So, final kinetic energy of the particle,

$$K_f = \frac{1}{2} mv'^2 = \frac{1}{2} m (2v_0)^2 = 4 \times \frac{1}{2} mv_0^2 = 2mv_0^2$$

- 28 (d)** The total moment of inertia of the system,

$$I = I_1 + I_2 + I_3 \quad \dots(i)$$

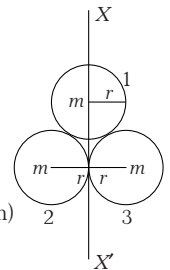
Here,

$$I_1 = \frac{2}{3} mr^2$$

$$I_2 = I_3 = \frac{2}{3} mr^2 + mr^2$$

(From parallel axes theorem)

$$= \frac{5}{3} mr^2$$



From Eq. (i), we get

$$I = \frac{2}{3} mr^2 + 2 \times \frac{5}{3} mr^2 = mr^2 \left(\frac{2}{3} + \frac{10}{3} \right) \Rightarrow I = 4mr^2$$

- 29 (b)** When a metallic rod is heated, it expands. Its moment of inertia (I) about a perpendicular bisector increases. According to law of conservation of angular momentum, its angular speed (ω) decreases, since $\omega \propto \frac{1}{I}$

- 30 (b)** The radius of the circle followed by the masses,

$$r = l \sin \alpha$$

As, angular momentum, $\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\mathbf{v}$

$$\Rightarrow |\mathbf{L}| = l \sin \theta (m \omega l \sin \theta)$$

On differentiating both sides, we get

$$\frac{d|\mathbf{L}|}{dt} = m\omega l^2 2 \sin \theta \cdot \cos \theta \frac{d\theta}{dt}$$

$$\Rightarrow \left| \frac{d\mathbf{L}}{dt} \right| = 2ml^2 \omega^2 \sin \theta \cdot \cos \theta = ml^2 \omega^2 \sin 2\theta$$

- 31 (a)** Let the velocity of centre of sphere be v .

The angular speed of the sphere about its centre is given by

$$\omega = \frac{v}{r}$$

where, r = radius of sphere.

The total kinetic energy = Translational kinetic energy
+ Rotational kinetic energy

$$= \frac{1}{2} I \omega^2 + \frac{1}{2} M v^2 = \frac{1}{2} \cdot \frac{2}{5} M r^2 \omega^2 + \frac{1}{2} M v^2$$

$$= \frac{1}{5} M v^2 + \frac{1}{2} M v^2 = \frac{7}{10} M v^2 = \frac{7}{10} \times 0.5 \times (0.02)^2 = 1.4 \times 10^{-4} \text{ J}$$

- 32 (b)** Given, radius of the circle = 20 cm = 20×10^{-2} m

When speeds are 8 ms^{-1} and 9 ms^{-1} , then angular speeds are

$$\text{respectively, } \omega_1 = \frac{8}{20 \times 10^{-2}} = 40 \text{ rad s}^{-1}$$

$$\text{and } \omega_2 = \frac{9}{20 \times 10^{-2}} = 45 \text{ rad s}^{-1}$$

$$\therefore \text{Angular acceleration} = \frac{\omega_2 - \omega_1}{t} = \frac{45 - 40}{2} = 2.5 \text{ rad s}^{-2}$$

- 33 (b)** The rotational kinetic energy, $K_r = \frac{1}{2} I \omega^2$

where, I = moment of inertia of solid sphere.

The total kinetic energy is the sum of rotational kinetic energy and translational kinetic energy.

$$\begin{aligned} K_T &= \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 \\ &= \frac{1}{2} m \omega^2 r^2 + \frac{1}{2} I \omega^2 \quad (\text{For pure rolling, } v = \omega r) \end{aligned}$$

$$\text{Now, } \frac{K_r}{K_T} = \frac{\frac{1}{2} I \omega^2}{\frac{1}{2} \omega^2 (m r^2 + I)} = \frac{I}{m r^2 + I}$$

$$\text{As, } I = \frac{2}{5} m r^2 \quad (\text{For solid sphere})$$

$$\Rightarrow \frac{K_r}{K_T} = \frac{\frac{2}{5} m r^2}{m r^2 + \frac{2}{5} m r^2} = \frac{\frac{2}{5} m r^2}{\frac{7}{5} m r^2} = \frac{2}{5} \times \frac{5}{7} = \frac{2}{7}$$

- 34 (a)** Given, radius of circular disc A , $r_1 = r$

Radius of circular disc B , $r_2 = 2r$

For disc A , $I_A = m r^2$... (i)

For disc B , $I_B = m(2r)^2 = 4m r^2$... (ii)

On dividing Eq. (i) by Eq. (ii), we get

$$\therefore \frac{I_A}{I_B} = \frac{m r^2}{4m r^2} \Rightarrow \frac{I_A}{I_B} = \frac{1}{4} \Rightarrow I_A = \frac{1}{4} I_B$$

So, moment of inertia of A is one-fourth that of B .

- 35 (c)** Radius of gyration is a scalar quantity.

Hence, statement of option (c) is incorrect.

- 36 (a)** Total angular momentum about O is given as

$$\begin{aligned} L &= L_1 + L_2 = m_1 v_1 r_1 + m_2 v_2 r_2 \\ &= -6.5 \times 2.2 \times 1.5 + 3.1 \times 3.6 \times 2.8 \\ &= -21.45 + 31.248 = 9.8 \text{ kg-m}^2 \text{ s}^{-1} \end{aligned}$$

- 37 (b)** When string is cut, the weight of the rod constitutes a

torque about the hinge point A . So, $\tau_A = mg \frac{l}{2}$... (i)

Also, from Newton's law, $\tau_A = I \alpha$... (ii)

where, α = angular acceleration of the rod

and I = moment of inertia of rod.

From Eqs. (i) and (ii), we get

$$I \alpha = mg \frac{l}{2} \Rightarrow \alpha = mg \frac{l}{2} / I$$

$$\text{As, } I = m \frac{l^2}{3}$$

$$\text{So, } \alpha = \frac{mg(l/2)}{\frac{ml^2}{3}} = \frac{3g}{2l}$$

Now, acceleration of centre of mass of rod, $a_{\text{CM}} = \alpha \cdot r$

Here, r = distance between hinge point A and centre of the rod.

$$\Rightarrow a_{\text{CM}} = \frac{3g}{2l} \cdot \frac{l}{2} = \frac{3g}{4}$$

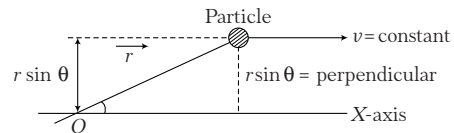
$$\Rightarrow mg - R_A = m a_{\text{CM}}$$

where, R_A = reaction at A .

$$\Rightarrow mg - R_A = m \times \frac{3g}{4}$$

$$\Rightarrow R_A = \frac{mg}{4} = \frac{5 \times 10}{4} = 12.5 \text{ N}$$

- 38 (b)** Consider the diagram.



As, the particle goes with constant linear velocity, the perpendicular distance of velocity vector from the axis is constant. So, angular momentum $mvr \sin \theta$ is constant.

- 39 (d)** Moment of inertia of a ring about its diameter, $I = \frac{1}{2} m r^2$

and kinetic energy is given by

$$K = \frac{1}{2} I \omega^2 \Rightarrow K = \frac{1}{4} m r^2 \omega^2 = \frac{1}{4} \times 10 \times (0.5)^2 \times (20)^2 = 250 \text{ J}$$

40 (b) For a ring, $I_z = MR^2$

From perpendicular axes theorem, $I_x + I_y = I_z$... (i)

Given, $I_x = I_y = I$

From Eq. (i), we get

$$2I = I_z$$

41 (c) Consider two bodies A and B having moment of inertia I and $2I$.

Kinetic energy of rotation for the body A, $K_A = (1/2)I\omega_A^2$

where, ω_A is angular speed of the body A.

Kinetic energy of rotation for the body B, $K_B = \frac{1}{2}(2I)\omega_B^2$

According to the question,

$$K_A = K_B$$

$$\Rightarrow \frac{1}{2}I\omega_A^2 = \frac{1}{2}(2I)\omega_B^2$$

$$\Rightarrow \omega_A^2 = 2\omega_B^2 \Rightarrow \left(\frac{\omega_A}{\omega_B}\right)^2 = 2$$

$$\Rightarrow \frac{\omega_A}{\omega_B} = \sqrt{2}$$

$$\Rightarrow \omega_A : \omega_B = \sqrt{2} : 1$$

42 (a) We know that, the kinetic energy, $K = \frac{1}{2}mv^2 = \frac{1}{2}I\omega^2$

where, $m = 27 \text{ kg}$ (mass of the body),

$\omega = 3 \text{ rads}^{-1}$ (angular velocity)

and $I = 3 \text{ kg-m}^2$ (moment of inertia).

$$v = ? \text{ or } mv^2 = I\omega^2$$

$$\Rightarrow v^2 = \frac{I\omega^2}{m} \Rightarrow v^2 = \frac{3 \times 3^2}{27} \Rightarrow v^2 = \frac{27}{27} = 1$$

$$\Rightarrow \text{Velocity of body, } v = \sqrt{1} = 1 \text{ ms}^{-1}$$

43 (c) Total kinetic energy of a body which is rolling without slipping is given by

$$K_{\text{total}} = K_{\text{rot}} + K_{\text{trans}}$$

For solid spherical ball, $I = \frac{2}{5}mR^2$ and $v = R\omega$
(Along the diameter)

where, R is radius of spherical ball.

$$\text{So, } K_{\text{total}} = \frac{1}{2}\left(\frac{2}{5}mR^2\right)\omega^2 + \frac{1}{2}mR^2\omega^2 = \frac{7}{10}mR^2\omega^2 = \frac{7}{10}mv^2$$

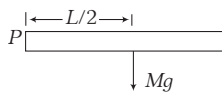
Now, potential energy = kinetic energy

$$\Rightarrow mgh = \frac{7}{10}mv^2 \Rightarrow v^2 = \frac{10}{7}gh \quad \dots(i)$$

For vertical projection,

$$v^2 = u^2 + 2gh' \Rightarrow \frac{10}{7}gh = 0 + 2gh' \Rightarrow h' = \frac{5}{7}h$$

44 (a) When the string is cut, the rod can be shown as



Torque on the rod = Moment of weight of the rod about P,

$$\tau = Mg \frac{L}{2} \quad \dots(i)$$

$$\therefore \text{Moment of inertia of rod about P, } I = \frac{ML^2}{3} \quad \dots(ii)$$

As, $\tau = I\alpha$

From Eqs. (i) and (ii), we get

$$Mg \frac{L}{2} = \frac{ML^2}{3} \alpha$$

$$\therefore \text{The initial angular acceleration of the rod, } \alpha = \frac{3g}{2L}$$

45 (d) Velocity, $v = \sqrt{\frac{2gh}{1 + K^2/R^2}}$

$$\text{Given, } h = \frac{3v^2}{4g}$$

$$\Rightarrow v^2 = \frac{2gh}{1 + \frac{K^2}{R^2}} = \frac{2g \cdot 3v^2}{4g \left(1 + \frac{K^2}{R^2}\right)}$$

$$1 = \frac{3}{2 \left(1 + \frac{K^2}{R^2}\right)} \text{ or } 1 + \frac{K^2}{R^2} = \frac{3}{2}$$

$$\text{or } \frac{K^2}{R^2} = \frac{3}{2} - 1 = \frac{1}{2}$$

$$\Rightarrow K^2 = \frac{1}{2}R^2 \quad (\text{Equation of disc})$$

Hence, the object is disc.

46 (b) From law of conservation of angular momentum, if no external torque acts on a system, then total vector sum of angular momentum of different particles of the system remains conserved.

$$\text{i.e. if } \tau_{\text{ext}} = 0, \text{ then } \frac{d\mathbf{L}}{dt} = 0$$

$$\Rightarrow L = \text{constant.}$$

47 (b) Angular momentum, $L = I\omega$

48 (d) Moment of inertia about its diameter, $I = \frac{2}{5}MR^2$

49 (a) In given condition, the moment of inertia of the combine system about an axis passing through O is

$$I = \frac{4}{3}ma^2 + m\left(\frac{3a}{4}\right)^2$$

$$\therefore I = \frac{91}{48}ma^2$$

50 (a) We know that, $L = I\omega = 2\pi mr^2f$

Now, $\omega' = \omega / 2$

$$\text{Hence, } L' = I\omega' = mr^2 \frac{\omega}{2} = \pi mr^2 f$$

$$\therefore \frac{L}{L'} = \frac{2\pi mr^2 f}{\pi mr^2 f} \Rightarrow L' = \frac{L}{2}$$

- 51 (c) We know that, torque, $\tau = \mathbf{r} \times \mathbf{F}$

Given, $\mathbf{F} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$

$\Rightarrow \mathbf{r} = 3\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$

Now, $\tau = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & -3 & 5 \\ 2 & -3 & 5 \end{vmatrix}$

$\tau = (-15 + 15)\hat{\mathbf{i}} - (15 - 10)\hat{\mathbf{j}} - (9 + 6)\hat{\mathbf{k}}$

$\tau = 0 - 5\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

$\Rightarrow \tau = -5\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

- 52 (b) Moment of inertia of a disc about a parallel axes,

$$I = I_v + md^2 = \frac{ma^2}{2} + md^2 \quad (\because a = R \text{ and } d = R)$$

$\Rightarrow$ Moment of inertia,

$$I = \frac{3}{2}ma^2 = \frac{3}{2}(25)$$

$\therefore I = \frac{75}{2} = 37.5 \text{ kg-m}^2$

- 53 (c) Total energy = kinetic energy + Rotational kinetic energy

$$= \frac{1}{2}(2m)v^2 + \frac{1}{3}mv^2 = \frac{4}{3}mv^2$$

- 54 (d) According to theorem of parallel axes,

$$I = I_{\text{CM}} + M\left(\frac{R}{2}\right)^2$$

$\Rightarrow I = \frac{1}{2}MR^2 + \frac{MR^2}{4}$

Moment of inertia, $I = \frac{3}{4}MR^2$

- 55 (a) We have, $\omega_2 = \omega_1 + \alpha t$

$\Rightarrow \omega_1 = \omega_2 - \alpha t$

Here, $\omega_1 = 0$

and $\omega_2 = 60 \text{ rpm}$

$$= \frac{60 \times 2\pi}{60} \text{ rad s}^{-1}$$

$\Rightarrow 0 = 2\pi \times \frac{60}{60} - \alpha \cdot 60$

or $\alpha = \frac{2\pi}{60}$

$\therefore$ Torque, $\tau = I\alpha = 2 \times \frac{2\pi}{60} = \frac{\pi}{15} \text{ N-m}$

- 56 (c) $I = \frac{2}{5}MR^2 = \frac{2}{5}\left(\frac{4}{3}\pi R^3\rho\right)R^2 = \frac{8}{15}\pi R^5\rho$

$$I = \frac{176}{105}R^5\rho \quad \left(\because \pi = \frac{22}{7}\right)$$

- 57 (d) The radius of gyration does not depend on the total mass of a body but depends upon the shape and size of body, distribution of mass within the body and choice of rotation of axis.

- 58 (d) Moment of inertia of disc, $I = \frac{1}{2}MR^2 = \frac{1}{2}M\left(\frac{M}{\pi\rho t}\right)$

$$\left(\because \rho = \frac{M}{V} = \frac{M}{\pi R^2 t}, \therefore R^2 = \frac{M}{\pi\rho t}\right)$$

$$I = \frac{1}{2} \frac{M^2}{\pi\rho t}$$

$\therefore I \propto \frac{1}{\rho}$ (As, M and t are constants)

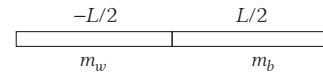
$\therefore \frac{I_1}{I_2} = \frac{\rho_2}{\rho_1}$

- 59 (c) Here, $\omega_2 = 2\pi n_2 = 2 \times \frac{240}{60}$ and $\omega_1 = 2\pi n_1 = 0$

The angular acceleration, $\alpha = \frac{\omega_2 - \omega_1}{t}$

$\therefore \alpha = \frac{2\pi\left(\frac{240}{60} - 0\right)}{10} = 2.51 \text{ rad s}^{-1}$

- 60 (a) The given situation as shown in the figure.



Moment of inertia for the wooden part,

$$M_1 = m_w \times \frac{(L/2)^2}{3} = m_w \frac{L^2}{12}$$

Moment of inertia for the brass part,

$$M_2 = m_b \times \frac{(L/2)^2}{3} = m_b \times \frac{L^2}{12}$$

Total moment of inertia,

$$M = M_1 + M_2 = m_w \frac{L^2}{12} + m_b \times \frac{L^2}{12} = (m_w + m_b) \frac{L^2}{12}$$

- 61 (d) According to question,

torque, $\tau = I\alpha = 0$

$\Rightarrow \alpha = 0$ or $\alpha = \frac{d^2\theta}{dt^2} = 0$

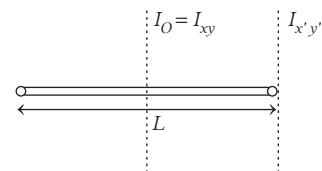
Given, $\theta(t) = 2t^3 - 6t^2$

So, $\frac{d\theta}{dt} = 6t^2 - 12t$

$\Rightarrow \frac{d^2\theta}{dt^2} = 12t - 12 \quad \left(\because \alpha = \frac{d^2\theta}{dt^2} = 0\right)$

$$12t - 12 = 0 \Rightarrow t = 1 \text{ s}$$

- 62 (a) According to parallel axes theorem, we have



$$I_{x'y'} = I_{xy} + Md^2$$

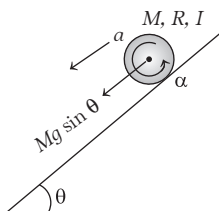
$$= I_0 + M \left(\frac{L}{2} \right)^2$$

$$\Rightarrow I_{x'y'} = I_0 + \frac{ML^2}{4}$$

- 63** (b) This is the case of rolling on rough inclined plane. Force of friction in this case will be backward.

$$\text{Hence, } a = \frac{Mg \sin \theta - f}{M} \quad \dots(i)$$

$$\text{and } \alpha = \frac{fR}{I} \quad \dots(ii)$$



For pure rolling to take place, $a = R\alpha$... (iii)

Solving Eqs. (i), (ii) and (iii), we get

$$f = \frac{Mg \sin \theta}{1 + \frac{MR^2}{I}} \quad \dots(iv)$$

Further, the force of friction calculated in Eq. (iv) for pure rolling to take place should be less than or equal to the maximum friction,

$$\text{i.e. } \frac{Mg \sin \theta}{1 + \frac{MR^2}{I}} \leq \mu Mg \cos \theta \quad \left(\because I = \frac{2}{5} MR^2 \right)$$

$$\Rightarrow \mu_{\min} \leq \frac{\tan \theta}{1 + \frac{MR^2}{I}} = \frac{\tan \theta}{1 + \frac{5}{2}} = \frac{2}{7} \tan \theta$$

- 64** (b) As, torque, $\tau = F \cdot r \sin \theta$

$$= 2 \times 3 \times \sin 30^\circ \quad (\text{Given})$$

$$\Rightarrow \tau = 6 \times \frac{1}{2} = 3 \text{ N-m}$$

- 65** (c) Let same mass and same outer radii of solid sphere and hollow sphere be M and R , respectively. The moment of inertia of solid sphere A about its diameter,

$$I_A = \frac{2}{5} MR^2 \quad \dots(i)$$

The moment of inertia of hollow sphere (spherical shell) B

$$\text{about its diameter, } I_B = \frac{2}{3} MR^2 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$I_A < I_B$$

Gravitation

Each body in this universe attracts other bodies towards itself with a force known as gravitational force, thus gravitation is the phenomena of mutual attraction between any two bodies in the universe. It is the force of gravitation due to which earth attracts everything towards itself.

In this chapter, we will study the basic laws which govern gravitational interactions.

Planets

The celestial objects which moves in fixed semi-circular orbits around the sun are known as planets. Including the earth mainly we have 8 planets in our solar system. These planets move around the sun in their fixed semi-circular orbits with increasing distance from the sun, name of these planets are

- | | | |
|--------------|----------------|-------------|
| (i) Mercury | (ii) Venus | (iii) Earth |
| (iv) Mars | (v) Jupiter | (vi) Saturn |
| (vii) Uranus | (viii) Neptune | |

Satellites

Celestial objects which are moving around the planets are known as satellites.

There are two types of satellites namely; natural and artificial satellites.

For example, the moon is a natural satellite while INSAT-1B is an artificial satellite of the earth.

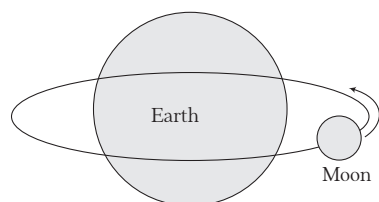


Fig. 10.1 Earth and moon

KEPLER'S LAWS OF PLANETARY MOTION

Kepler's laws of planetary motion consist of three scientific laws describing motion of planets around the sun, which are as given below

Kepler's first law of orbits

According to this law, "all planets move in elliptical orbits around the sun, situated at one of the foci of the ellipse."

Inside

1 Kepler's laws of planetary motion

2 Newton's law of gravitation

- Vector form of Newton's law of gravitation
- Principle of superposition of gravitational forces

3 Gravity

- Acceleration due to gravity on earth
- Variation in the value of acceleration due to gravity

4 Gravitational field

- Gravitational field intensity for different bodies

5 Gravitational potential

- Gravitational potential for different bodies
- Relation between gravitational field and potential

6 Gravitational potential energy

7 Binding energy

- Escape velocity
- Escape energy

8 Motion of satellites

9 Artificial satellite

This law identifies that the distance between the sun (S) and planet (P) is constantly changing as the earth revolve around its orbit as shown in figure.

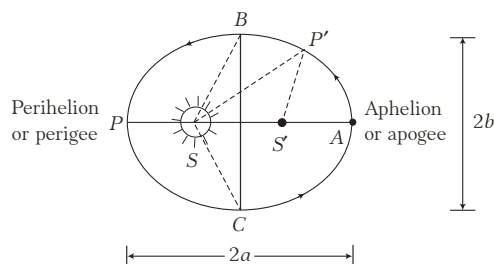


Fig. 10.2 Law of orbit

Above figure shows an ellipse traced out by a planet around the sun S . The closest point P is called **perihelion** (perigee) and the farthest point A is called **aphelion** (apogee).

It is the characteristics of an ellipse that the sum of the distances of any point (P') on it from its two foci (S and S') is constant, i.e. $SP' + S'P' = \text{constant}$.

Note Shorter the orbit of the planet around the sun, shorter is the time to cover one complete revolution.

Kepler's second law of areas

According to this law, "the line joining the planet sweeps out equal areas in equal interval of time." It means that areal velocity is constant.

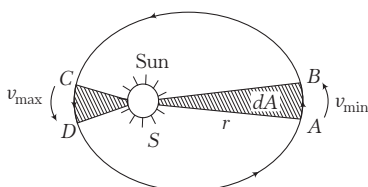


Fig. 10.3 Law of area

In the given figure, if the line joining the sun and the planet swept out area dA in a small time interval dt , then

$$\begin{aligned} \text{Areal velocity} &= \frac{dA}{dt} = \frac{d}{dt} \left[\frac{1}{2} r(vdt) \right] = \frac{1}{2} rv \\ &= \frac{mvr}{2m} = \frac{L}{2m} \end{aligned}$$

where, L = angular momentum of the earth-planet system
= constant (for gravitational force between the sun and the planet)

$\therefore$

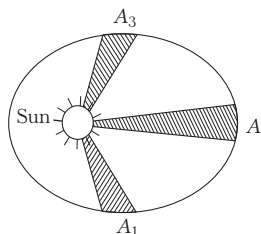
$$\text{Areal velocity} = \frac{dA}{dt} = \text{constant}$$

Also, if the planet moves from A to B in some time interval t and from C to D in the same time interval t , then

$$\text{area (ASB)} = \text{area (CSD)}$$

In simple terms, this law states that the planet will move slowly ($v_{\min}$) only when it is farthest from the sun and move rapidly ($v_{\max}$) when it is nearest to the sun. This law is similar to law of conservation of angular momentum.

Example 10.1 A planet moving around sun sweeps area A_1 in 2 days, A_2 in 3 days and A_3 in 6 days, then find the relation between A_1 , A_2 and A_3 .



Sol. According to Kepler's second law, when a planet revolves around the sun, its areal velocity is constant.

$$\therefore \frac{dA}{dt} = \text{constant}$$

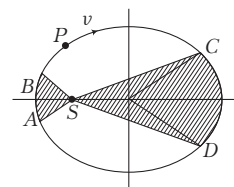
$$\text{Hence, } \frac{A_1}{t_1} = \frac{A_2}{t_2} = \frac{A_3}{t_3}$$

Given, $t_1 = 2$ days, $t_2 = 3$ days and $t_3 = 6$ days

$$\therefore \frac{A_1}{2} = \frac{A_2}{3} = \frac{A_3}{6} \Rightarrow 3A_1 = 2A_2 = A_3$$

This is the relation between A_1 , A_2 and A_3 .

Example 10.2 The figure shows an elliptical orbit of a planet P about the sun S . The shaded area CSD is twice the shaded area ASB . If t_1 is the time taken by the planet to move from C to D and t_2 is the time to move from A to B , determine the ratio t_1/t_2 .



Sol. According to Kepler's second law, the areal velocity of the planet remains constant.

$$\therefore \frac{A_1}{t_1} = \frac{A_2}{t_2}$$

$$\text{Hence, } \frac{t_1}{t_2} = \frac{A_1}{A_2} = \frac{\text{Area CSD}}{\text{Area ASB}} = 2$$

Kepler's third law of period

According to this law, "the square of the time period (T) of revolution of a planet around the sun is proportional to the cube of the semi-major axis (a) of its elliptical orbit."

$$\text{i.e. } T^2 \propto a^3 \quad \dots(i)$$

As shown in figure, we have

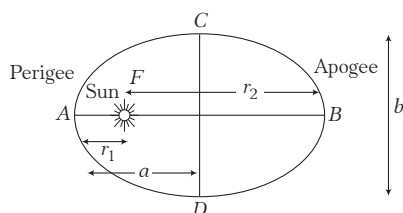


Fig. 10.4 Law of period

$$AB = AF + FB \Rightarrow 2a = r_1 + r_2$$

$$\therefore a = \frac{r_1 + r_2}{2}$$

So, substituting the value of a in Eq. (i), we get

$$T^2 \propto \left(\frac{r_1 + r_2}{2} \right)^3$$

where, r_1 = shortest distance of planet from sun (perigee) and r_2 = longest distance of planet from sun (apogee).

It can be shown that, if e is the eccentricity of an elliptical orbit of the planet, then $\frac{r_2}{r_1} = \frac{1+e}{1-e}$.

Note Kepler's laws are applicable not only to the solar system but also to artificial satellites as well as to the moon revolving around the planets.

Example 10.3 The planet neptune travels around the sun with a period of 165 yr. What is the radius of the orbit approximately, if the orbit is considered as circular?

Sol. As, $T_1 = T_{\text{earth}} = 1 \text{ yr}$ and $T_2 = T_{\text{neptune}} = 165 \text{ yr}$

Let R_1 and R_2 be the radii of the circular orbits of the earth and neptune, respectively.

According to Kepler's third law,

$$\frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3}$$

$$\Rightarrow R_2^3 = \frac{R_1^3 T_2^2}{T_1^2}$$

$$\Rightarrow R_2^3 = \frac{R_1^3 \times (165)^2}{1^2}$$

$$\Rightarrow R_2^3 = 165^2 R_1^3 \Rightarrow R_2 \approx 30 R_1$$

Example 10.4 A satellite of time period 24 h is orbiting the earth at a height $6R$ above the surface of earth, where R is radius of earth. What will be the time period of another satellite at a height $2.5 R$ from the surface of earth?

Sol. By Kepler's third law, $T^2 \propto R^3$

$$\therefore \frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3} \text{ or } \left(\frac{T_2}{T_1} \right)^2 = \left(\frac{R_2}{R_1} \right)^3$$

$$\Rightarrow T_2^2 = T_1^2 \left(\frac{3.5R}{7R} \right)^3$$

$$(\because R_2 = R + 2.5R \text{ and } R_1 = R + 6R)$$

$$\text{or } T_2^2 = \frac{T_1^2}{8} \Rightarrow T_2 = \frac{T_1}{2\sqrt{2}}$$

$$\Rightarrow T_2 = \frac{24}{2\sqrt{2}} \quad (\because T_1 = 24 \text{ h})$$

$$\therefore T_2 = 6\sqrt{2} \text{ h}$$

NEWTON'S LAW OF GRAVITATION

According to this law, "every particle in the universe attracts every other particle with a force whose magnitude is directly proportional to the product of their masses and inversely proportional to the square of distances between their centres." This tendency of particles (or bodies) to attract each other is called gravitation.

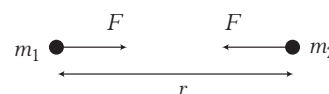


Fig. 10.5 Force between two point masses

Thus, the magnitude of gravitational force F between the two particles of masses m_1 and m_2 placed at distance r can be given as

$$F \propto \frac{m_1 m_2}{r^2} \text{ or } F = \frac{G m_1 m_2}{r^2} \quad \dots(i)$$

where, G is the universal gravitational constant.

The magnitude of G in MKS or SI is $6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ and in CGS, it is $6.67 \times 10^{-8} \text{ dyne-cm}^2 \text{ g}^{-2}$.

The dimensional formula of G is $[M^{-1} L^3 T^{-2}]$.

Important points about force of gravitation

- The gravitational constant G is called a universal constant because it has same value for any two bodies anywhere in the space.
- Gravitational force is always attractive in nature and is independent of the nature of medium between two masses and presence or absence of other bodies.
- Gravitational forces are central forces as they act along the line joining the centre of gravity of two bodies.
- Gravitational forces are conservative in nature, so work done by gravitational forces does not depend upon the path.

- (v) For the gravitational force on a body outside a spherically symmetric body (solid spheres or spherical shells), the entire mass of the body is assumed to be concentrated at its centre.
- (vi) The force of attraction due to a hollow spherical shell of uniform density on a point mass situated inside it is zero.
- (vii) In calculating the gravitational force between two non-spherical bodies, we cannot assume their masses to be concentrated at their centres of mass but if the separation between them is very large in comparison to their sizes, then the bodies may be considered as particles. That means if we have to find gravitational force between a sphere and a circular ring at infinite separation, then the mass of sphere can be assumed at its centre.
- Generally, the gravitational force between non-spherical bodies are found by integrating the individual forces on individual particles within the bodies.
- (viii) Force developed between any two masses is called gravitational force and the force between earth and any other body is called force of gravity.

Example 10.5 Two heavy particles of masses 40 kg and 60 kg attracts each other with a force of 4×10^{-5} N. If G is $6 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$, calculate the distance between them.

Sol. Given, $m_1 = 40 \text{ kg}$, $m_2 = 60 \text{ kg}$, $F = 4 \times 10^{-5} \text{ N}$,

$$G = 6 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}, g = 10 \text{ ms}^{-2}$$

According to Newton's law of gravitation, $F = \frac{Gm_1m_2}{r^2}$

$$\therefore \text{Distance, } r = \sqrt{\frac{Gm_1m_2}{F}} = \sqrt{\frac{6 \times 10^{-11} \times 40 \times 60}{4 \times 10^{-5}}} \\ = 0.06 \text{ m} = 6 \text{ cm}$$

Example 10.6 If the distance between the two spherical bodies is increased to four times, then by how many times, the mass of one of the bodies is to be changed to maintain the same gravitational force?

Sol. Let $F_1 = \frac{Gm_1m_2}{r^2}$ and $F_2 = \frac{GM_1M_2}{R^2}$

Now, $R = 4r$

Also, let m_1 remains unchanged, so $m_1 = M_1$

$$\therefore \frac{Gm_1m_2}{r^2} = \frac{GM_1M_2}{R^2}$$

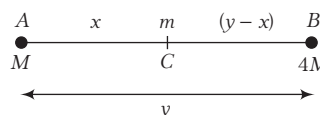
$$\Rightarrow \frac{m_2}{r^2} = \frac{M_2}{(4r)^2} \Rightarrow m_2 = \frac{M_2}{16}$$

$$\therefore M_2 = 16 m_2$$

Hence, the mass of one body should be increased to 16 times to maintain the same gravitational force.

Example 10.7 Two particles A and B having masses M and $4M$ respectively are kept at a distance 2.73 m apart. Another small particle of mass m is to be placed, so that the net gravitational force on it is zero. What will be its distance from body A?

Sol. Two particles A and B are placed as shown below and a third particle is placed at C.



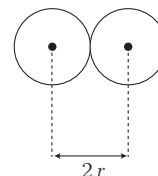
Now, according to the given conditions,

$$F_{CA} = F_{CB} \\ \therefore \frac{GMm}{x^2} = \frac{G4Mm}{(y-x)^2} \\ \Rightarrow \frac{1}{x^2} = \frac{4}{(y-x)^2} \\ \Rightarrow y-x = 2x \\ \Rightarrow y = 3x \\ \therefore x = \frac{y}{3} = \frac{2.73}{3} \Rightarrow x = 0.91 \text{ m}$$

Hence, a particle of mass m should be placed at 0.91 m apart from body A.

Example 10.8 Spheres of the same material and of same radius r are touching each other. Show that gravitational force between them is directly proportional to r^4 .

Sol. Two spheres are touching as shown below.



According to the question,

$$m_1 = m_2 = (\text{volume})(\text{density}) = \left(\frac{4}{3}\pi r^3\right)\rho \\ \therefore \text{Gravitational force, } F = \frac{Gm_1m_2}{r^2} = \frac{G\left(\frac{4}{3}\pi r^3\right)\left(\frac{4}{3}\pi r^3\right)\rho^2}{r^2} \\ \Rightarrow F \propto r^4$$

Example 10.9 Force between two objects of equal masses is F . If 25% mass of one object is transferred to the other object, then find the new force.

Sol. Suppose mass of one object, $m = 100 \text{ kg}$

$\therefore$ Masses of both objects are equal.

$$\therefore m_1 = m_2 = 100 \text{ kg}$$

According to Newton's law of gravitation,

$$F = G \frac{m_1m_2}{R^2}$$

$$\therefore F = \frac{G \times 100 \times 100}{R^2} = \frac{G \times 10000}{R^2}$$

$$\Rightarrow \frac{G}{R^2} = \frac{F}{10000} \quad \dots(i)$$

If 25% mass of one object is transferred to the other object, then

$$m_1 = 100 - 25 = 75 \text{ kg}$$

$$m_2 = 100 + 25 = 125 \text{ kg}$$

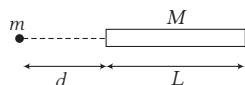
$$\therefore \text{New force, } F' = \frac{G \times 75 \times 125}{R^2} = \frac{G \times 9375}{R^2}$$

$$\Rightarrow \frac{G}{R^2} = \frac{F'}{9375} \quad \dots(ii)$$

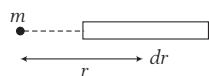
From Eqs. (i) and (ii), we get

$$F' = \frac{15}{16} F$$

Example 10.10 A particle of mass m is placed at a distance d from one end of a uniform rod with length L and mass M as shown in the figure. Find the magnitude of the gravitational force on the particle due to the rod.



Sol. Let us consider an elementary mass dm of length dr at a distance r from the particle of mass m . Here, $dm = (M/L)dr$. The gravitational force dF on m due to this elementary mass dm is



$$dF = \frac{Gm}{r^2} dm = \frac{GmM}{Lr^2} dr$$

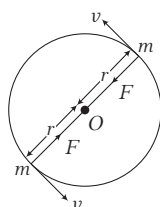
$$\therefore \text{Force on the particle, } F = \frac{GmM}{L} \int_d^{L+d} \frac{dr}{r^2}$$

$$= \frac{GmM}{L} \left[-\frac{1}{r} \right]_d^{L+d}$$

$$= -\frac{GmM}{L} \left[\frac{1}{L+d} - \frac{1}{d} \right] = \frac{GmM}{d(L+d)}$$

Example 10.11 Two particles of equal mass m are moving round a circle of radius r due to their mutual gravitational interaction. Find the time period of each particle.

Sol. Two particles will always remain on diametrically opposite points, so that the gravitational force is centripetal. Here, mutual gravitational force is



$$F = \frac{Gmm}{(2r)^2} = \frac{Gm^2}{4r^2} \quad \dots(i)$$

If the speed of each particle is v , then the centripetal force,

$$F = \frac{mv^2}{r} \quad \dots(ii)$$

Equating Eqs. (i) and (ii), we get

$$v = \sqrt{\frac{Gm}{4r}}$$

$$\therefore \text{Time period, } T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{\frac{Gm}{4r}}} = \frac{4\pi}{\sqrt{Gm}} r^{3/2}$$

Vector form of Newton's law of gravitation

The vector form of Newton's law of gravitation signifies that the gravitational forces acting between the two particles form action and reaction pair.

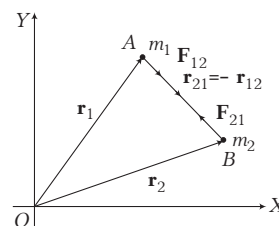


Fig. 10.6 Vector form of gravitational force ($\mathbf{F}_{12}$) on m_1 due to m_2

In figure, it can be seen that the two particles of masses m_1 and m_2 are placed at a distance r , therefore according to Newton's law of gravitation, force on m_1 due to m_2 ,

$$\mathbf{F}_{12} = -\frac{Gm_1m_2}{|\mathbf{r}_{12}|^2} \cdot \hat{\mathbf{r}}_{12} \quad \dots(i)$$

$$\mathbf{F}_{12} = -Gm_1m_2 \frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|^3} \quad \left(\because \hat{\mathbf{r}}_{12} = \frac{\mathbf{r}_1 - \mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|} \right)$$

where, $\hat{\mathbf{r}}_{12}$ is a unit vector pointing from m_2 to m_1 .

The negative sign in Eq.(i) indicates that the direction of force $\mathbf{F}_{12}$ is opposite to that of $\hat{\mathbf{r}}_{12}$.

Similarly, force on m_2 due to m_1 ,

$$\mathbf{F}_{21} = -\frac{Gm_1m_2}{|\mathbf{r}_{21}|^2} \cdot \hat{\mathbf{r}}_{21} \quad \dots(ii)$$

where, $\hat{\mathbf{r}}_{21}$ is a unit vector pointing from m_1 to m_2 .

From Eqs. (i) and (ii), we get

$$\mathbf{F}_{12} = -\mathbf{F}_{21}$$

$$\Rightarrow |\mathbf{F}_{12}| = |\mathbf{F}_{21}| \quad \dots(iii)$$

As $\mathbf{F}_{12}$ and $\mathbf{F}_{21}$ are directed towards the centres of the two particles, so gravitational force is conservative in nature.

Principle of superposition of gravitational forces

Suppose $F_1, F_2, \dots, F_n$ be the individual forces due to the masses $m_1, m_2, m_3, \dots, m_n$ which are given by the universal law of gravitation, then from the principle of superposition, each of these forces acts independently and uninfluenced by the other bodies as shown in figure.

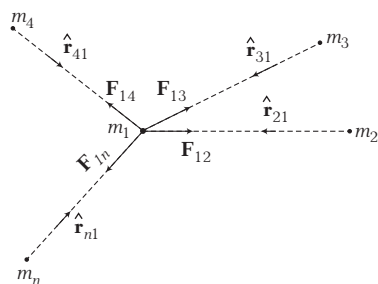


Fig. 10.7 Gravitational forces on m_1 due to $m_2, m_3, \dots, m_n$ using principle of superposition of gravitational forces

So, the resultant force F on mass m_1 can be expressed in vector addition of various forces

$$F = F_{12} + F_{13} + F_{14} + \dots + F_{1n}$$

It states that, the resultant gravitational force F acting on a particle due to number of point masses is equal to the vector sum of forces exerted by the individual masses on the given particle.

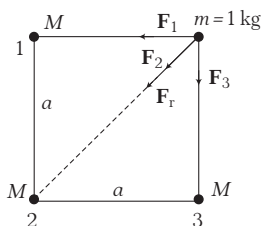
$$\text{Clearly, } F = -G \frac{m_1 m_2}{r_{12}^2} \hat{r}_{21} - G \frac{m_1 m_3}{r_{13}^2} \hat{r}_{31} - \dots - G \frac{m_1 m_n}{r_{1n}^2} \hat{r}_{n1}$$

$$\text{Resultant force, } F = -Gm_1 \left[\frac{m_2}{r_{12}^2} \hat{r}_{21} + \frac{m_3}{r_{13}^2} \hat{r}_{31} + \dots + \frac{m_n}{r_{1n}^2} \hat{r}_{n1} \right]$$

Example 10.12 Three masses each equal to M are placed at the three corners of a square of side a . Calculate the force of attraction on unit mass at the fourth corner.

Sol. Force on m due to masses at 1 and 3 are F_1 and F_3 ,

$$F_1 = F_3 = \frac{GM}{a^2} \quad (\because m = 1)$$



Resultant of F_1 and F_3 is $F_r = \sqrt{2} \frac{GM}{a^2}$ and its direction is along the diagonal, i.e. towards 2.

Force on m due to mass M at corner 2, $F_2 = \frac{GM}{(\sqrt{2}a)^2} = \frac{GM}{2a^2}$ and F_r and F_2 act in the same direction.

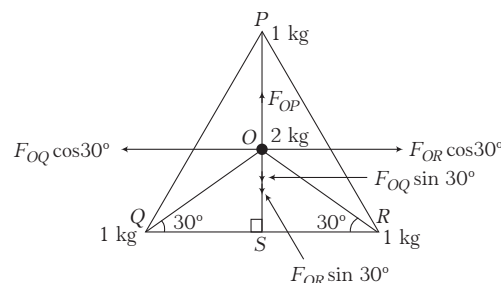
Resultant of these two gives net force,

$$F_r = \frac{\sqrt{2}GM}{a^2} + \frac{GM}{2a^2} = \frac{GM}{a^2} \left[\sqrt{2} + \frac{1}{2} \right]$$

It is directed along the diagonal as shown in figure.

Example 10.13 Three equal masses of 1 kg each are placed at the vertices of an equilateral ΔPQR and a mass of 2 kg is placed at the centroid O of the triangle which is at a distance of $\sqrt{2}$ m from each of the vertices of the triangle. Find the force (in newton) acting on the mass of 2 kg.

Sol. Given, $OP = OQ = OR = \sqrt{2}$ m



The gravitational force on mass 2 kg due to mass 1 kg at P ,

$$F_{OP} = G \frac{2 \times 1}{(\sqrt{2})^2} = G \text{ along } OP$$

$$\text{Similarly, } F_{OQ} = G \frac{2 \times 1}{(\sqrt{2})^2} = G \text{ along } OQ$$

$$\text{and } F_{OR} = G \frac{2 \times 1}{(\sqrt{2})^2} = G \text{ along } OR$$

$F_{OQ} \cos 30^\circ$ and $F_{OR} \cos 30^\circ$ are equal and acting in opposite directions, hence cancel out each other.

Force on mass 2 kg along OS

$$\begin{aligned} &= F_{OQ} \sin 30^\circ + F_{OR} \sin 30^\circ \\ &= 2F_{OQ} \sin 30^\circ \quad (\because F_{OQ} = F_{OR}) \\ &= F_{OQ} \end{aligned}$$

The resultant force on the mass 2 kg at O ,

$$\begin{aligned} F &= F_{OP} - F_{OQ} \\ &= G - G = 0 \text{ (zero)} \end{aligned}$$

CHECK POINT 10.1

- Kepler's second law is based on
 - Newton's first law
 - Newton's second law
 - special theory of relativity
 - conservation of angular momentum
- For a satellite in elliptical orbit, which of the following quantities does not remain constant?
 - Angular momentum
 - Momentum
 - Areal velocity
 - All of these
- A planet moves around the sun. It is closest to sun at a distance d_1 and have velocity v_1 . At farthest distance d_2 , its speed will be
 - $\frac{d_1^2 v_1}{d_2^2}$
 - $\frac{d_2 v_1}{d_1}$
 - $\frac{d_1 v_1}{d_2}$
 - $\frac{d_2^2 v_1}{d_1^2}$
- Kepler's law states that, square of the time period of any planet moving around the sun in an elliptical orbit of semi-major axis (R) is directly proportional to
 - R
 - $\frac{1}{R}$
 - R^3
 - $\frac{1}{R^3}$
- The distance of two planets from the sun are 10^{13} m and 10^{12} m, respectively. The ratio of the periods of the planet is
 - 100
 - $\frac{1}{\sqrt{10}}$
 - $\sqrt{10}$
 - $10\sqrt{10}$
- The period of revolution of planet A round the sun is 8 times that of B . The distance of A from the sun is how many times greater than that of B from the sun?
 - 5
 - 4
 - 3
 - 2
- The ratio of mean distances of three planets from the sun are $0.5 : 1 : 1.5$, then the square of time periods are in the ratio of
 - $1 : 4 : 9$
 - $1 : 9 : 4$
 - $1 : 8 : 27$
 - $2 : 1 : 3$
- A body is orbiting around earth at a mean radius which is two times greater than the parking orbit of a satellite, the period of body is
 - 4 days
 - 16 days
 - $2\sqrt{2}$ days
 - 64 days
- Which of the following statements about the gravitational constant is true?
 - It is a force.
 - It has no unit.
 - It has same value in all systems of unit.
 - It does not depend on the nature of the medium in which the bodies are kept.
- Two point masses each equal to 1 kg attract one another with a force of 10^{-9} kg-wt. The distance between the two point masses is approximately ($G = 6.67 \times 10^{-11}$ MKS units)
 - 8 cm
 - 0.8 cm
 - 80 cm
 - 0.08 cm
- Gravitational force between a point mass m and M separated by a distance r is F . Now, if a point mass $2m$ is placed next to m in contact with it, the force on M due to m and the total force on M are
 - $2F, F$
 - $F, 2F$
 - $F, 3F$
 - F, F
- The distance of the centres of moon and earth is D . The mass of earth is 81 times the mass of the moon. At what distance from the centre of the earth, the gravitational force on a particle will be zero?
 - $\frac{D}{2}$
 - $\frac{2D}{3}$
 - $\frac{4D}{3}$
 - $\frac{9D}{10}$
- A spherical planet far out in space has mass $2M$ and radius a . A particle of mass m is falling freely near its surface. What will be the acceleration of that particle?
 - $\frac{GM}{a^2}$
 - $\frac{3GM}{a^2}$
 - $\frac{2GM}{a^2}$
 - $\frac{4GM}{a^2}$
- Two balls each of radius R having equal mass and density are placed in contact, then the force of gravitation between them is proportional to
 - $F \propto \frac{1}{R^2}$
 - $F \propto R$
 - $F \propto R^4$
 - $F \propto \frac{1}{R}$
- If the distance between the sun and the earth is increased by three times, then attraction between two will
 - remain constant
 - decrease by 63%
 - increase by 63%
 - decrease by 89%
- Three equal masses of 2 kg each are placed at the vertices of an equilateral triangle and a mass of 4 kg is placed at the centroid of the triangle which is at a distance of $\sqrt{2}$ m from each of the vertices of the triangle. The force (in newton) acting on the mass of 4 kg is
 - 2
 - $\sqrt{2}$
 - 1
 - zero

GRAVITY

In Newton's law of gravitation, the force of attraction between the two bodies is known as gravitation. If one of these bodies is earth, then gravitation is called gravity. Hence, gravity is another force by which earth attracts another body towards its centre.

Acceleration due to gravity (g) on earth

The force of gravity acting on a body having unit mass placed on or near the surface of the earth is known as acceleration due to gravity.

Or

When a body is dropped from a certain height above the ground, it begins to fall towards the centre of the earth under the effect of gravity. The acceleration produced in the body due to gravity, is known as acceleration due to gravity.

It is a vector quantity and denoted by g . Its average value on the earth's surface at mean sea level is 9.8 ms^{-2} .

Relation between acceleration due to gravity (g) and gravitational constant (G)

Suppose that the mass of the earth is M , its radius is R , then the force of attraction acting on a body of mass m close to the surface of the earth is

$$F = \frac{GMm}{R^2}$$

According to Newton's second law, the acceleration due to gravity,

$$g = \frac{F}{m} = \frac{GM}{R^2} \quad \dots(i)$$

This expression is free from m . It means acceleration due to gravity does not depend on the mass of the object, thus it is same for all. If in Eq. (i), we put the value of G as $6.67 \times 10^{-11} \text{ N-m}^2\text{kg}^{-2}$, M as $6 \times 10^{24} \text{ kg}$ and R as $6.4 \times 10^6 \text{ m}$, then on solving it, we will get the value of g as 9.8 ms^{-2} .

If two bodies of different masses are allowed to fall freely, then they will have the same acceleration, i.e. if they are allowed to fall from the same height, then they will reach the earth simultaneously. The dimensions of acceleration due to gravity are $[\text{LT}^{-2}]$.

Note

- (i) The value of g is independent of mass, shape, size, etc., of the body and depends upon the mass and radius of the earth.
- (ii) The value of g on the surface of the moon is equal to $1/6$ times the value of g on the surface of the earth.

Example 10.14 The acceleration due to gravity at the moon's surface is 1.67 ms^{-2} . If the radius of the moon is $1.74 \times 10^6 \text{ m}$, calculate the mass of the moon.

Sol. The acceleration due to gravity, $g = \frac{GM}{R^2}$ or $M = \frac{gR^2}{G}$

This relation is true not only for the earth but also for any heavenly body which is assumed to be spherical.

Now, $g = 1.67 \text{ ms}^{-2}$, $R = 1.74 \times 10^6 \text{ m}$

and $G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$

$$\begin{aligned} \therefore \text{Mass of the moon, } M &= \frac{1.67 \times (1.74 \times 10^6)^2}{6.67 \times 10^{-11}} \text{ kg} \\ &= 7.58 \times 10^{22} \text{ kg} \end{aligned}$$

Example 10.15 Assume that, if the earth were made of lead of relative density 11.3, then what would be the value of acceleration due to gravity on the earth's surface?

Sol. Since, density of earth can be given as

$$\rho = \text{Relative density} \times \text{Density of water} = 11.3 \times 10^3 \text{ kgm}^{-3}$$

$\therefore$ Acceleration due to gravity on the earth's surface,

$$\begin{aligned} g &= \frac{GM}{R^2} = \frac{G}{R^2} \cdot \frac{4}{3} \pi R^3 \cdot \rho = \frac{4}{3} \pi GR \cdot \rho \\ &= \frac{4}{3} \times \frac{22}{7} \times 6.67 \times 10^{-11} \times 6.4 \times 10^6 \times 11.3 \times 10^3 \\ &= 20.21 \text{ ms}^{-2} \end{aligned}$$

Example 10.16 What will be the relation between the acceleration due to gravity on the surface of the earth and on a planet respectively, whose mass and radius are four times that of the earth?

Sol. Let the acceleration due to gravity of the planet be g_p .

$$\text{As, } g_p = \frac{GM_p}{R_p^2}$$

According to the question,

$$\text{Mass of planet, } M_p = 4M_e$$

$$\text{and radius of planet, } R_p = 4R_e$$

$$\therefore g_p = \frac{G4M_e}{16R_e^2}$$

$$\therefore g_p = \frac{1}{4} g_e \quad \left(\because g_e = \frac{GM_e}{R_e^2} \right)$$

Variation in the value of acceleration due to gravity (g)

The value of acceleration due to gravity varies due to following factors

- (i) Height above the surface of earth
- (ii) Depth below the surface of earth
- (iii) Shape of the earth
- (iv) Axial rotation of the earth

1. Variation in g due to height above the surface of earth

Consider a body of mass m lying on the surface of the earth of mass M and radius R . Acceleration due to gravity at the surface of the earth, $g = \frac{GM}{R^2}$

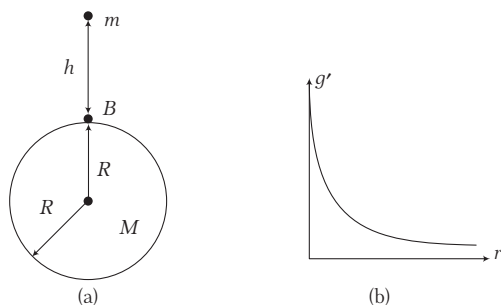


Fig. 10.8 Variation of g with altitude

The force of gravity on an object of mass m at a height h above the surface of the earth is

$$F = \frac{GMm}{(R+h)^2}$$

$\therefore$ Acceleration due to gravity at this height will be

$$g' = \frac{F}{m} = \frac{GM}{(R+h)^2}$$

This can also be written as

$$g' = \frac{GM}{R^2 \left(1 + \frac{h}{R}\right)^2} \Rightarrow \boxed{g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}} \quad \left(\because \frac{GM}{R^2} = g\right)$$

Thus,

$$g' < g$$

i.e. The value of acceleration due to gravity g goes on decreasing as we go above the surface of the earth.

Also, if $(R+h) = r$

$$g' = \frac{GM}{(R+h)^2} = \frac{gR^2}{r^2}; \quad g' \propto \frac{1}{r^2}$$

Further, by using Binomial theorem, we get

$$g' = g \left(1 + \frac{h}{R}\right)^{-2} \Rightarrow \boxed{g' \approx g \left(1 - \frac{2h}{R}\right)} \quad (\because h \ll R)$$

Note With height h , the decrease in the value of g is $g - g' = \frac{2gh}{R}$

$$\therefore \text{Fractional decreases in the value of } g \text{ is } \frac{g - g'}{g} = \frac{2h}{R}$$

$$\therefore \text{Percentage decreases in the value of } g = \left(\frac{g - g'}{g}\right) \times 100$$

$$= \frac{2h}{R} \times 100\%$$

Example 10.17 At what altitude, the acceleration due to gravity reduces to one-fourth of its value as that on the surface of the earth? (Take, radius of earth as 6.4×10^6 m and g on the surface of the earth as 9.8 ms^{-2})

Sol. Acceleration due to gravity at height h , $g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$

If acceleration due to gravity reduced to one-fourth of its value, then

$$\Rightarrow \frac{g}{4} = \frac{g}{\left(1 + \frac{h}{R}\right)^2} \Rightarrow \left(1 + \frac{h}{R}\right)^2 = 4 \Rightarrow 1 + \frac{h}{R} = 2$$

$$\Rightarrow \frac{h}{R} = 1 \Rightarrow h = R = 6.4 \times 10^6 \text{ m}$$

$\therefore$ Acceleration due to gravity is reduced to one-fourth of its value on the earth's surface at an altitude of 6.4×10^6 m.

Example 10.18 At what height, the acceleration due to gravity decreases by 51% of its value on the surface of the earth?

Sol. As, $g_h = g \left(\frac{R}{R+h}\right)^2$ and $g_h = g - 51\% \text{ of } g = g - \frac{51}{100}g$

$$\Rightarrow g_h = \frac{49}{100}g$$

$$\frac{49}{100}g = g \left(\frac{R}{R+h}\right)^2$$

$$\frac{7}{10} = \frac{R}{R+h}$$

$$7R + 7h = 10R$$

$$\therefore 7h = 3R \Rightarrow h = \frac{3R}{7}$$

2. Variation in g due to depth below the surface of earth

Let an object of mass m is situated at a depth d below the earth's surface. Its distance from the centre of the earth is $(R-d)$. This mass is situated at the outer surface of the inner solid sphere. The gravitational force of attraction on a mass inside a spherical shell is always zero. Therefore, the object experiences gravitational attraction only due to inner solid sphere.

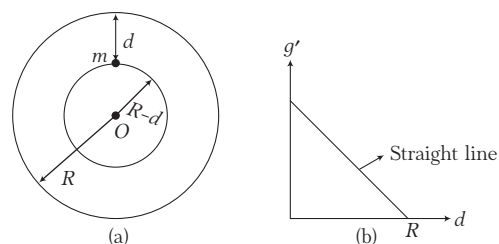


Fig. 10.9 Variation of g with depth

The mass of this sphere is $M' = \left\{ \frac{M}{(4/3)\pi R^3} \right\} \frac{4}{3} \pi (R-d)^3$

Here, M being the mass of whole sphere of radius R .

Therefore, $M' = \frac{(R-d)^3}{R^3} M$

The force of gravity on an object of mass M at a depth d below the surface of earth,

$$F = \frac{GM'm}{(R-d)^2} = \frac{GMm(R-d)}{R^3} \text{ and } g' = \frac{F}{m} = \frac{GM}{R^3} (R-d) \quad \dots(i)$$

Since, $g = \frac{GM}{R^2}$

Substituting the values in Eq. (i), we get

$$g' = g \left(1 - \frac{d}{R} \right)$$

Since, g is a constant at a given place of the earth and R is also a constant.

$$\therefore (g - g') \propto d$$

Hence, **the acceleration due to gravity decreases as we move down from the surface of the earth.**

Note Decrease in the value of g with depth d is $g - g' = \frac{g}{R}d$

$$\therefore \text{Fractional decrease in the value of } g \text{ with depth} = \frac{g - g'}{g} = \frac{d}{R}$$

$$\therefore \text{Percentage decrease in the value of } g = \frac{g - g'}{g} \times 100 = \frac{d}{R} \times 100$$

We can see from this equation that $g' = 0$ at $d = R$, i.e. acceleration due to gravity is zero (minimum value) at the centre of the earth.

If r is distance from the centre of earth.

For $r \leq R$, $g' = g \left(1 - \frac{d}{R} \right) = \frac{gr}{R} \quad (\because R - d = r)$

or

For $r > R$, $g' = \frac{g}{\left(1 + \frac{h}{R} \right)^2} = \frac{gR^2}{r^2} \text{ or } g' \propto \frac{1}{r^2}$

The graphical representation of change in the value of g' with height and depth is as follows

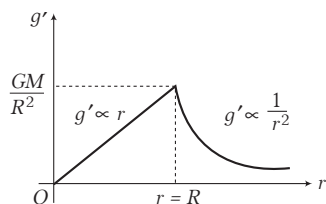


Fig. 10.10 Graphical representation for variation of g'

Note The acceleration due to gravity is maximum (having standard value of 9.8 ms^{-2}) at the earth's surface. It decreases either when we go at higher altitudes or we move below the surface.

Example 10.19 At what depth from the surface of the earth, the acceleration due to gravity will be half the value of g on the surface of the earth?

Sol. As, $g' = g \left(1 - \frac{d}{R_e} \right)$

According to the question,

$$g' = \frac{g}{2}$$

$$\Rightarrow \frac{g}{2} = g \left(1 - \frac{d}{R_e} \right)$$

$$\therefore \frac{1}{2} = 1 - \frac{d}{R_e}$$

$$\Rightarrow \frac{d}{R_e} = 1 - \frac{1}{2}$$

$$\Rightarrow d = 0.5 R_e$$

Example 10.20 At what depth from earth's surface, acceleration due to gravity is decreased by 1%?

Sol. Fractional decrease in the value of g with depth,

$$\frac{\Delta g_d}{g} = \frac{d}{R_e}$$

$$\Rightarrow \frac{1}{100} = \frac{d}{6400}$$

$$\Rightarrow d = 64 \text{ km}$$

Example 10.21 Assuming earth to be a sphere of uniform mass density, how much would a body weigh half-way down the centre of the earth, if it weighed 100 N on the surface?

Sol. Given, $w = mg = 100 \text{ N}$

As, $g' = g \left(1 - \frac{d}{R} \right)$

It is also given that, $\frac{d}{R} = \frac{1}{2}$

$$\therefore g' = g \left(1 - \frac{1}{2} \right) = \frac{g}{2}$$

Weight of body half-way down the centre of the earth,

$$w' = mg' = \frac{mg}{2} = \frac{100}{2} = 50 \text{ N}$$

Example 10.22 Determine the decrease in the weight of a body when it is taken 32 km below the earth's surface. (Take, radius of the earth as 6400 km)

Sol. Given, $R = 6400 \text{ km}$, $d = 32 \text{ km}$

$$\therefore g' = g \left(1 - \frac{d}{R} \right) = g \left(1 - \frac{32}{6400} \right) = \left(\frac{199}{200} \right) g$$

$$\therefore g - g' = g - \frac{199}{200} g = \frac{g}{200}$$

The percentage decrease in the weight of the body

$$\begin{aligned}
 &= \frac{mg - mg'}{mg} \times 100 \\
 &= \frac{g - g'}{g} \times 100 \\
 &= \frac{g}{200} \times 100 \\
 &= \frac{1}{200} \times 100 = 0.5\%
 \end{aligned}$$

3. Variation in g due to the shape of the earth

The earth is not a perfect sphere. It is somewhat flat at the two poles. The equatorial radius (R_{eq}) is approximately 21 km more than the polar radius (R_p).

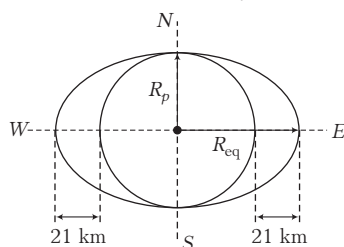


Fig. 10.11 Variation in g due to shape of the earth

Now, acceleration due to gravity at poles,

$$g_p = \frac{GM}{R_p^2}$$

Acceleration due to gravity at equator,

$$g_{eq} = \frac{GM}{R_{eq}^2}$$

$$\because R_p < R_{eq}$$

$$\therefore g_p > g_{eq}$$

Difference in g at poles and equator due to shape,

$$(\Delta g) = g_p - g_{eq} = 0.02 \text{ ms}^{-2}$$

The value of g is minimum at the equator and maximum at the poles. So, this is the reason why the weight of the body increases when it is taken from equator to the pole.

Note If weight = constant, i.e. $mg = \text{constant}$, then $m \propto \frac{1}{g}$.

It means that, it will be profitable to buy sugar at the equator in comparison to poles because more mass (amount) will be obtained there.

Example 10.23 Imagine a new planet having the same density as that of earth but it is 3 times bigger than the earth in size. If the acceleration due to gravity on the surface of earth is g and that on the surface of the new planet is g' , then find the relation between g and g' .

Sol. The acceleration due to gravity on the new planet can be written using the relation,

$$g = \frac{GM}{R^2} \quad \dots(i)$$

But $M = (4/3)\pi R^3 \rho$, where ρ be the density.

Thus, Eq. (i) becomes

$$g = \frac{G \times (4/3)\pi R^3 \rho}{R^2} = G \times \frac{4}{3} \pi R \rho$$

$$\Rightarrow g \propto R \quad (\because \rho = \text{constant})$$

$$\therefore \frac{g'}{g} = \frac{R'}{R} \Rightarrow \frac{g'}{g} = \frac{3R}{R}$$

$$\Rightarrow \frac{g'}{g} = 3 \Rightarrow g' = 3g$$

4. Variation in g due to axial rotation of the earth

Suppose the earth is rotating on its axis with angular velocity ω . Consider a particle P at rest on the surface of the earth, at latitude ϕ , then the pseudo force acting on the particle is $m\omega^2 r$ in outward direction. The acceleration due to gravity g is acting towards the centre O of the earth. Thus, the effective acceleration due to gravity g' is the resultant of g and $\omega^2 r$.

After derivation, we can find the following relation

$$g' = g - R\omega^2 \cos^2 \phi$$

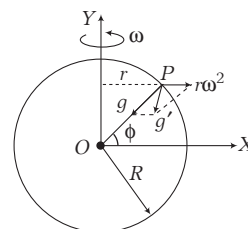


Fig. 10.12 Axial rotation of the earth

Following conclusions can be drawn from the above discussion

- The effective value of g is not truly vertical.
- The effect of centrifugal force due to rotation of the earth is to reduce the effective value of g .
- At equators $\phi = 0^\circ$.

Therefore, $g' = g - R\omega^2$ and at poles $\phi = 90^\circ$.

Therefore, $g' = g$

Thus, at equator g' is minimum while at poles g' is maximum.

Note

- If the angular velocity of earth increases, the value of g will decrease at all places except at poles.
- Since, earth rotates west to east, we also project our rockets from west to east, so that effective acceleration due to gravity ($g' = g - R\omega^2$) on the rocket is less.

Example 10.24 Find the imaginary angular velocity of the earth for which the effective acceleration due to gravity at the equator shall be zero. (Take, $g = 10 \text{ m/s}^2$ for the acceleration due to gravity, if the earth were at rest and radius of earth equal to 6400 km and $\phi = 60^\circ$)

Sol. Acceleration due to gravity, $g' = g - \omega^2 R \cos^2 \phi$

$$0 = g - \omega^2 R \cos^2 60^\circ \quad (\because g' = 0)$$

$$0 = g - \frac{\omega^2 R}{4}$$

$$\omega = 2\sqrt{\frac{g}{R}} = 2\sqrt{\frac{10}{6400 \times 1000}}$$

$$\Rightarrow \omega = \frac{1}{400} = 2.5 \times 10^{-3} \text{ rad/s}$$

Example 10.25 Calculate the angular speed of rotation of the earth, so that the apparent g at the equator becomes half of its value at the surface. Also, calculate the length of the day in this situation.

Sol. The apparent acceleration due to gravity,

$$g = g_0 - \omega^2 R = g_0/2$$

$$\Rightarrow \omega = \sqrt{\frac{g_0}{2R}} = \sqrt{\frac{9.8}{6.4 \times 10^6 \times 2}} = 8.75 \times 10^{-4} \text{ rad/s}^{-1}$$

The length of the day = Time period of rotation of the earth

$$= \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{2R}{g_0}} = 2\pi \sqrt{\frac{2 \times 6.4 \times 10^6}{9.8}} \\ = 7180 \text{ s} \approx 2 \text{ h}$$

Example 10.26 Find the value of angular velocity of axial rotation of the earth, such that weight of a person at equator becomes (3/4)th of its weight at pole. Radius of the earth at equator is 6400 km.

Sol. Let acceleration due to gravity at pole is g and at equator, it is g' . If mass of the person is m and angular velocity of rotation of the earth is ω , then

$$g' = g - \omega^2 R \Rightarrow mg' = mg - m\omega^2 R$$

$$\Rightarrow \frac{3}{4} mg = mg - m\omega^2 R$$

$$\Rightarrow \omega^2 R = \frac{1}{4} g$$

$$\Rightarrow \omega = \sqrt{\frac{g}{4R}} = \sqrt{\frac{9.8}{4 \times 6400 \times 10^3}}$$

$$\therefore \omega = \frac{1}{2} \times \frac{1}{80} \times \frac{1}{10^2} \times \sqrt{98} = 6.18 \times 10^{-4} \text{ rad/s}^{-1}$$

Example 10.27 Suppose the earth increases its speed of rotation. At what new time period, will the weight of a body on the equator becomes zero? (Take, $g = 10 \text{ ms}^{-2}$ and radius of earth $R = 6400 \text{ km}$)

Sol. The weight will become zero, when

$$g' = 0 \text{ or } g - R\omega^2 = 0$$

(On the equator, $g' = g - R\omega^2$)

$$\text{or } \omega = \sqrt{\frac{g}{R}}$$

$$\therefore \frac{2\pi}{T} = \sqrt{\frac{g}{R}} \text{ or } T = 2\pi \sqrt{\frac{R}{g}}$$

Substituting the given values in above expression, we get

$$T = \frac{2\pi \sqrt{6400 \times 10^3}}{3600} \text{ h or } T \approx 1.4 \text{ h}$$

Thus, the new time period should be 1.4 h instead of 24 h for the weight of a body to be zero at the equator.

CHECK POINT 10.2

- The mass of a planet is twice the mass of earth and diameter of the planet is thrice the diameter of the earth, then the acceleration due to gravity on the planet's surface is
(a) $g/2$ (b) $2g$
(c) $2g/9$ (d) $3g/\sqrt{2}$
- If the earth suddenly shrinks (without changing mass) to half of its present radius, then acceleration due to gravity will be
(a) $g/2$ (b) $4g$
(c) $g/4$ (d) $2g$
- If G is universal gravitational constant and g is acceleration due to gravity, then the unit of the quantity $\frac{G}{g}$ is
(a) kg-m^2 (b) kgm^{-1}
(c) kgm^{-2} (d) $\text{m}^2 \text{kg}^{-1}$
- The acceleration due to gravity g and mean density of earth ρ are related by which of the following relations?
(G = gravitational constant and R = radius of earth)
(a) $\rho = \frac{4\pi g R^2}{3G}$ (b) $\rho = \frac{4\pi g R^3}{3G}$ (c) $\rho = \frac{3g}{4\pi G R}$ (d) $\rho = \frac{3g}{4\pi G R^3}$
- The diameters of two planets are in the ratio 4:1 and their mean densities in the ratio of 1:2. The acceleration due to gravity on the planets will be in ratio
(a) 1 : 2 (b) 2 : 3
(c) 2 : 1 (d) 4 : 1
- If density of earth increased 4 times and its radius reduced to half of what it is, our weight will
(a) be four times its present value
(b) be doubled
(c) remain same
(d) be halved

7. A body has a weight 72 N. When it is taken to a height $h = R =$ radius of earth, it would weight
 (a) 72 N (b) 36 N (c) 18 N (d) zero
8. The height above the surface of the earth, where acceleration due to gravity is $(1/64)$ th of its value at surface of the earth is approximately
 (a) 45×10^6 m (b) 54×10^6 m
 (c) 102×10^6 m (d) 72×10^6 m
9. The depth d , at which the value of acceleration due to gravity becomes $1/n$ times the value at the surface is (Here, $R =$ radius of the earth)
 (a) $\frac{R}{n}$ (b) $R\left(\frac{n-1}{n}\right)$ (c) $\frac{R}{n^2}$ (d) $R\left(\frac{n}{n+1}\right)$
10. If the change in the value of g at a height h above the surface of earth is the same as at a depth d below it (both h and d are much smaller than the radius of the earth), then
 (a) $d = h$ (b) $d = 2h$
 (c) $d = h/2$ (d) $d = h^2$
11. At what depth below the surface of the earth, acceleration due to gravity will be half of its value at 1600 km above the surface of the earth?
 (a) 4.3×10^6 m (b) 2.4×10^6 m
 (c) 3.2×10^6 m (d) 1.6×10^6 m
12. The weight of a body at the centre of the earth is
 (a) zero
 (b) infinite
 (c) same as on the surface of earth
 (d) None of the above
13. Weight of a body is maximum at
 (a) poles (b) equator
 (c) centre of earth (d) at latitude 45°
14. When a body is taken from the equator to the poles, its weight
 (a) remains constant
 (b) increases
 (c) decreases
 (d) increase at N -pole and decrease at S -pole
15. The angular speed of earth in rad s^{-1} , so that the object on equator may appear weightless, is (radius of earth = 6400 km)
 (a) 1.23×10^{-3} (b) 6.20×10^{-3} (c) 1.56 (d) 1.23×10^{-5}
16. Earth is supposed to be sphere of radius R . If g_{30° is value of acceleration due to gravity at latitude of 30° and g at the equator, then value of $g - g_{30^\circ}$ is
 (a) $\frac{1}{4}\omega^2 R$ (b) $\frac{3}{4}\omega^2 R$ (c) $\omega^2 R$ (d) $\frac{1}{2}\omega^2 R$

GRAVITATIONAL FIELD

An invisible field in space around a body due to its mass, in which any other body experiences a force of attraction is called the gravitational field of the first body.

In case of our planet earth, since it is surrounded by a gravitational field, so any body brought in this field experiences a force of attraction towards the centre of earth.

Gravitational field intensity

The intensity of gravitational field at any point in the gravitational field due to a given mass is defined as the force experienced by a unit mass placed at that point provided the presence of that unit mass does not alter the original gravitational field.

Usually, it is denoted by E or I .

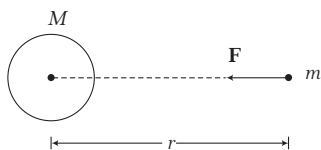


Fig. 10.13 Gravitational field intensity

Thus, if $\mathbf{F}$ be the gravitational force experienced by the test mass m , then the gravitational field intensity at any point will be

$$\mathbf{E} = \frac{\mathbf{F}}{m}$$

Its SI unit is ms^{-2} or Nkg^{-1} and dimensions are $[\text{M}^0 \text{L} \text{T}^{-2}]$.

It is a vector quantity and is always directed towards the centre of gravity of body whose gravitational field intensity is to be considered.

Note

- Inertial mass** of a body can be measured as the ratio of the magnitude of the external force applied, to the magnitude of acceleration produced in its motion.
- Gravitational mass** of a body is defined as the gravitational pull experienced by the body in a gravitational field of unit intensity.

Gravitational field intensity for different bodies

1. Field due to a point mass

Suppose a point mass M is placed at point O and we want to find the intensity of gravitational field E at point P , at a distance r from O .

So, the magnitude of force F acting on the particle of mass m placed at P is

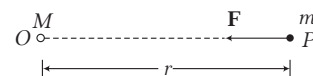


Fig. 10.14 Gravitational field due to a point mass

$$F = \frac{GMm}{r^2} \Rightarrow E = \frac{F}{m} = \frac{GM}{r^2}$$

or

$$E = \frac{GM}{r^2}$$

In vector form, $\mathbf{E} = \frac{GM}{r^2}(-\hat{\mathbf{r}}) = -\frac{GM}{r^3} \mathbf{r}$

The direction of force $\mathbf{F}$ and $\mathbf{E}$ is from P to O as shown in the figure above.

Here, the negative sign shows the attractive force on body.

Note

- (i) If $r = \infty$, $E = 0$. It means the intensity of gravitational field is zero only at infinite distance from the body.
- (ii) If a unit mass (m) is placed on the surface of earth, then the gravitational force acting on the test mass m will be equal to the weight w of the test mass.

$$\text{Now, } E = \frac{w}{m} = \frac{mg}{m} = g \quad \text{or} \quad E = g$$

- (iii) We can also apply the principle of superposition to gravitational field intensity in the same way as that of gravitational force.

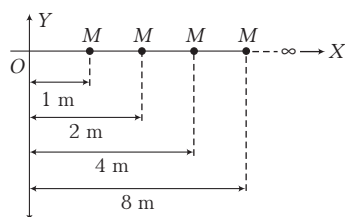
$$\text{i.e. } \mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \dots + \mathbf{E}_n$$

Example 10.28 Two point masses of mass 10 kg and 1000 kg are at a distance 1 m apart. At which points on the line joining them, will the gravitational field intensity be zero?

Sol. Let the resultant gravitational intensity be zero at a distance x from the mass of 10 kg on the line joining the centre of two bodies. At this point, the gravitational intensities due to the two bodies must be equal and opposite.

$$\begin{aligned} \therefore \quad \frac{G \times 10}{x^2} &= \frac{G \times 1000}{(1-x)^2} \\ \Rightarrow \quad 100x^2 &= (1-x)^2 \\ \Rightarrow \quad 10x &= 1-x \\ \Rightarrow \quad 11x &= 1 \\ \Rightarrow \quad x &= \left(\frac{1}{11}\right) \text{ m} \end{aligned}$$

Example 10.29 Figure shows a system of point masses placed on X -axis. Find the net gravitational field intensity at the origin.



(Take, sum of an infinite GP as $S = \frac{a}{1-r}$,

where a = first term and r = least common ratio.)

Sol. Net gravitational field intensity at the origin,

$$\mathbf{E}_{\text{net}} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \mathbf{E}_4 + \dots \infty \text{ terms}$$

$$= \frac{GM}{(1)^2} \hat{\mathbf{i}} + \frac{GM}{(2)^2} \hat{\mathbf{i}} + \frac{GM}{(4)^2} \hat{\mathbf{i}} + \dots \infty \text{ terms}$$

$$= GM \hat{\mathbf{i}} \left(1 + \frac{1}{4} + \frac{1}{16} + \dots \infty \right) \quad \left(\text{Here, } a = 1 \text{ and } r = \frac{1}{4} \right)$$

$$\text{So, } \mathbf{E}_{\text{net}} = GM \left(\frac{a}{1-r} \right) \hat{\mathbf{i}}$$

$$= GM \hat{\mathbf{i}} \left(\frac{1}{1-1/4} \right) = GM \hat{\mathbf{i}} \left(\frac{4}{3} \right)$$

$$\Rightarrow \mathbf{E}_{\text{net}} = \frac{4}{3} GM \hat{\mathbf{i}}$$

2. Field due to a uniform solid sphere

Field at an external point A uniform solid sphere may be treated as a single particle of same mass placed at its centre (O) for calculating the gravitational field at an external point (P).

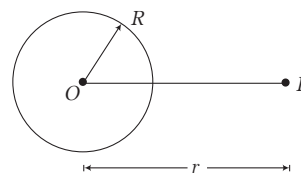


Fig. 10.15 Gravitational field at an external point P

$$\text{Thus, } E(r) = \frac{GM}{r^2}$$

$$\text{For } r \geq R \quad \text{or} \quad E(r) \propto \frac{1}{r^2}$$

Here, r is the distance of the point from the centre of the sphere and R is the radius of sphere.

Field at an internal point The gravitational field due to a uniform sphere at an internal point (P) is proportional to the distance (r) of the point from the centre of the sphere.

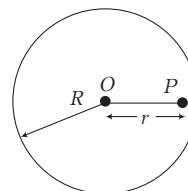


Fig. 10.16 Gravitational field at an internal point P

The gravitational field at the centre is zero and at surface is $\frac{GM}{R^2}$, where R is the radius of the sphere.

$$\text{Thus, } E(r) = \frac{GM}{R^3} \cdot r$$

$$\text{For } r \leq R \quad \text{or} \quad E(r) \propto r$$

Thus, E versus r graph is as shown in figure below

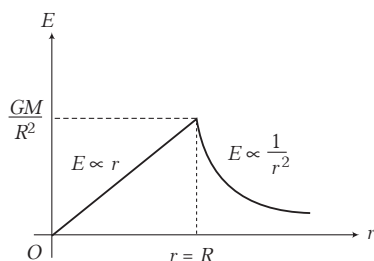
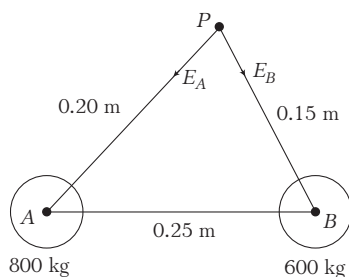


Fig. 10.17 E versus r graph for solid sphere

Example 10.30 Two solid spheres of radius 10 cm and masses 800 kg and 600 kg, are at a distance 0.25 m apart. Calculate the magnitude of the gravitational field intensity at a point distance 0.20 m from the 800 kg sphere and 0.15 m from the 600 kg sphere and does not lie on the line joining their centres. (Take, $G = 6.6 \times 10^{-11} \text{ N-m}^2\text{kg}^{-2}$)

Sol. Let E_A be the gravitational field intensity at P due to 800 kg sphere at A .



$$\text{Then, } E_A = G \frac{800}{(0.2)^2} = 2 \times 10^4 G, \text{ along } PA$$

Let E_B be the gravitational field intensity at P due to 600 kg sphere at B .

$$\text{Then, } E_B = G \frac{600}{(0.15)^2} = \frac{80000 G}{3}, \text{ along } PB$$

The angle between E_A and E_B is 90° . ($\because AB^2 = AP^2 + BP^2$)

If E be the magnitude of the resultant intensity, then

$$\begin{aligned} E &= \sqrt{E_A^2 + E_B^2} = \sqrt{(2 \times 10^4 G)^2 + \left(\frac{80000 G}{3}\right)^2} \\ &= G \sqrt{4 \times 10^8 + \frac{64}{9} \times 10^8} \\ &= 6.67 \times 10^{-11} \times 2 \times 10^4 \sqrt{1 + \frac{16}{9}} \\ &= 6.67 \times 2 \times \frac{5}{3} \times 10^{-7} \text{ N kg}^{-1} \end{aligned}$$

$$\Rightarrow E = 2.22 \times 10^{-6} \text{ N kg}^{-1}$$

Example 10.31 Gravitational field at the surface of a solid sphere is $1.5 \times 10^{-4} \text{ N kg}^{-1}$. Find the gravitational field at a point situated inside the sphere at a distance equal to half of the radius of the solid sphere.

Sol. Let mass of the solid sphere is M and radius is R .

It is given that,

$$E_{\text{surface}} = \frac{GM}{R^2} = 1.5 \times 10^{-4} \text{ N kg}^{-1}$$

$$\therefore E(r = R/2) = \frac{GM}{R^3} (R/2)$$

$$= \frac{GM}{2R^2} = \frac{1.5 \times 10^{-4}}{2} = 7.5 \times 10^{-5} \text{ N kg}^{-1}$$

3. Field due to a uniform spherical shell

At an external point For an external point (P), the shell may be treated as a single particle of same mass placed at its centre (O).

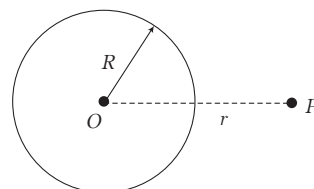


Fig. 10.18 Gravitational field at an external point P

Thus, at an external point the gravitational field is given by

$$E(r) = \frac{GM}{r^2} \quad (\text{for } r \geq R)$$

At $r = R$ (the surface of shell),

$$E = \frac{GM}{R^2}$$

and otherwise, $E \propto \frac{1}{r^2}$

At an internal point The field inside a uniform spherical shell is zero.

Thus, E versus r graph is as shown in figure below

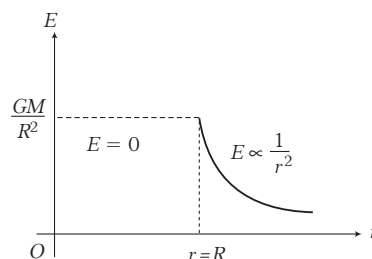
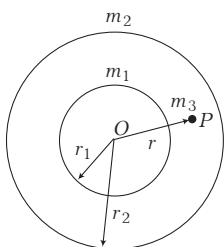


Fig. 10.19 E versus r graph for spherical shell

Example 10.32 Two concentric spherical shells have masses m_1 and m_2 and radii r_1 and r_2 ($r_2 > r_1$). What is the force exerted by this system on a particle of mass m_3 , if it is placed at a distance r ($r_1 < r < r_2$) from the centre?

Sol. The outer shell will have no contribution in the gravitational field at point P .

$$\therefore E_P = \frac{Gm_1}{r^2}$$



Thus, force on mass m_3 placed at P is,

$$F = m_3 E_P \text{ or } F = \frac{Gm_1 m_3}{r^2}$$

The field E_P and the force F both are towards centre O .

4. Field due to a uniform circular ring at a point on the axis

Gravitational field intensity at a point P on the axis of a circular ring of radius R and mass M is given by

$$E(r) = \frac{GMr}{(R^2 + r^2)^{3/2}}$$

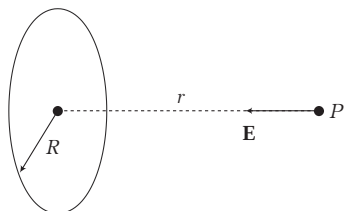


Fig. 10.20 Gravitational field at a point P on the axis of ring

This is directed towards the centre of the ring. It is zero at the centre of the ring and maximum at $r = R/\sqrt{2}$

(can be obtained by putting $dE/dr = 0$).

The maximum value is
$$E_{\max} = \frac{2GM}{3\sqrt{3}R^2}$$

Thus, E versus r graph is as shown in figure below

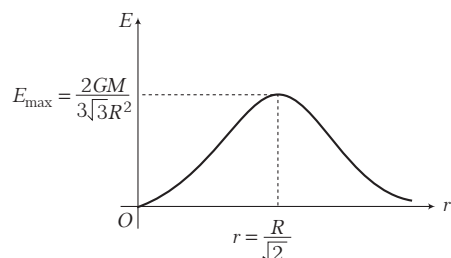


Fig. 10.21 E versus r graph for a circular ring

Example 10.33 Mass of 2 kg is distributed uniformly over a ring of radius 2m. Find the gravitational field at a point lying on the axis of the ring at a distance of $2\sqrt{3}$ m from the centre.

Sol. Given, mass of the ring, $M = 2$ kg

Radius of the ring, $R = 2$ m

Distance of the point, $r = 2\sqrt{3}$ m

$\therefore$ Required gravitational field,

$$\begin{aligned} E(r = 2\sqrt{3}) &= \frac{GMr}{(R^2 + r^2)^{3/2}} = \frac{G \times 2 \times 2\sqrt{3}}{(\sqrt{4 + 12})^3} \\ &= \frac{4\sqrt{3}G}{64} = \frac{\sqrt{3}G}{16} = \frac{6.67 \times 10^{-11} \times \sqrt{3}}{16} \\ &= 0.72 \times 10^{-11} = 7.2 \times 10^{-12} \text{ Nkg}^{-1} \end{aligned}$$

GRAVITATIONAL POTENTIAL

The work done in bringing a unit mass slowly from reference point (infinity) to a given point in the gravitational field, is called the gravitational potential at that point.

This work is done by the external agent in bringing the mass. The gravitational potential is denoted by V .

Let W joule of work is to be done by external agent in bringing a test mass m from infinity to some point, then gravitational potential at the point will be

$$V = \frac{W}{m}$$

The SI unit of gravitational potential is J kg^{-1} or $\text{m}^2 \text{s}^{-2}$ and it is a scalar quantity because it is defined as work done per unit mass. Its dimensions are $[\text{M}^0 \text{L}^2 \text{T}^{-2}]$.

Note Negative work means force is opposite to displacement.

Gravitational potential for different bodies

1. Potential due to a point mass

Suppose a point mass M is situated at a point O . We want to find the gravitational potential due to this mass at a point P situated at a distance r from O .

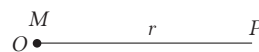


Fig. 10.22 Gravitational potential due to a point mass

For this let us find work done in taking the unit mass from P to infinity. This will be

$$W = \int_r^\infty F \cdot dr = \int_r^\infty \frac{GM}{r^2} dr = \frac{GM}{r}$$

Hence, the work done in bringing unit mass from infinity to P will be $-\frac{GM}{r}$. Thus, the gravitational potential at P will be

$$V = -\frac{GM}{r}$$

Gravitational potential at a point is always negative.

Note

(i) When $r = \infty$ from above formula, then $V = 0$, hence gravitational potential is maximum (zero) at infinity.

(ii) At surface of the earth $r = R$, then $V = -\frac{GM}{R}$ at the surface of the earth.

Example 10.34 A point of mass 15 kg is placed at the origin of coordinate axis. Find the gravitational potential at a point located at $x = 5$ m on X -axis.

Sol. Given, $M = 15$ kg

Gravitational potential at $x = 5$ m,

$$V = \frac{-GM}{x} = \frac{-6.67 \times 10^{-11} \times 15}{5} \\ = -20.01 \times 10^{-11} \text{ J kg}^{-1}$$

2. Gravitational potential due to a system of more than one point mass

Consider a system of point masses $M_1, M_2, M_3, \dots, M_n$. A point P is situated at a distances $r_1, r_2, r_3, \dots, r_n$ respectively from these points.

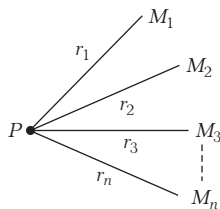


Fig. 10.23 Potential due to system of point masses

Gravitational potential at this point P due to these point masses,

$$V_P = -\left[\frac{GM_1}{r_1} + \frac{GM_2}{r_2} + \frac{GM_3}{r_3} + \dots + \frac{GM_n}{r_n} \right]$$

Example 10.35 Two heavy point masses of mass 10^3 kg and 10^5 kg are separated by a distance of 200 m. What will be the potential at the mid-point of the line joining them?

Sol. The total potential at A ,

$$V = V_P + V_Q = -\frac{GM_P}{r} - \frac{GM_Q}{r}$$

$$\therefore V = \frac{-6.67 \times 10^{-11}}{100} [10^3 + 10^5]$$

$$\text{Hence, } V = -6.73 \times 10^{-8} \text{ J kg}^{-1}$$

Example 10.36 Infinite number of bodies, each of mass 2 kg are situated on X -axis at distance 1 m, 2 m, 4 m and 8 m respectively from the origin. What is the resulting gravitational potential due to this system at the origin?

Sol. The resulting gravitational potential,

$$V = -2G \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \dots \right) \\ = -2G \left(1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} \dots \right)$$

$$\Rightarrow V = \frac{-2G}{\left(1 - \frac{1}{2}\right)} = \frac{-2G}{\frac{1}{2}} = -4G$$

3. Potential due to a uniform solid sphere

Potential at an external point The gravitational potential due to a uniform solid sphere at an external point P is same as that due to a single particle of same mass placed at its centre.

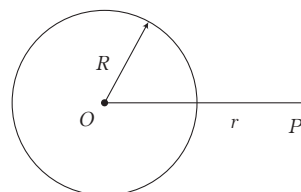


Fig. 10.24 Potential at an external point P

$$\text{Thus, } V(r) = -\frac{GM}{r}, \quad r \geq R$$

$$\text{At the surface, } r = R \quad \text{and} \quad V = -\frac{GM}{R}$$

Potential at an internal point At some internal point, potential at a distance r from the centre is given by

$$V(r) = -\frac{GM}{R^3} (1.5R^2 - 0.5r^2), \quad r \leq R$$

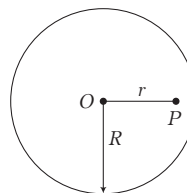


Fig. 10.25 Potential at an internal point P

$$\text{At } r = R, \quad V = -\frac{GM}{R}$$

$$\text{while at } r = 0, \quad V = -\frac{1.5 GM}{R}$$

i.e. At the centre of the sphere, the potential is 1.5 times the potential at surface.

The V versus r graph is as shown in figure below.

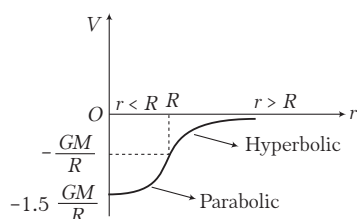


Fig. 10.26 V versus r graph for solid sphere

Example 10.37 The radius of the earth is 6.37×10^6 m and its mean density is 5.5×10^3 kg m $^{-3}$ and $G = 6.67 \times 10^{-11}$ N-m 2 kg $^{-2}$. Find the gravitational potential on the surface of the earth.

Sol. Given, radius of the earth, $R = 6.37 \times 10^6$ m

Density, $\rho = 5.5 \times 10^3$ kg m $^{-3}$, $G = 6.67 \times 10^{-11}$ N-m 2 kg $^{-2}$

$\therefore$ Mass of the earth, $M = \text{Volume} \times \text{Density} = \frac{4}{3} \pi R^3 \rho$

$\therefore$ Gravitational potential on the earth's surface,

$$\begin{aligned} V &= \frac{-GM}{R} = \frac{-G}{R} \times \frac{4}{3} \pi R^3 \rho = \frac{-4}{3} \pi G R^2 \rho \\ &= \frac{-4}{3} \times 3.14 \times 6.67 \times 10^{-11} \times (6.37 \times 10^6)^2 \times 5.5 \times 10^3 \\ \Rightarrow V &= -6.23 \times 10^7 \text{ J kg}^{-1} \end{aligned}$$

Example 10.38 At a point above the surface of the earth, the gravitational potential is -5.12×10^7 J/kg and the acceleration due to gravity is 6.4 m/s 2 . Assuming the mean radius of the earth to be 6400 km, calculate the height of the point above the earth's surface.

Sol. If r is the distance of the given point from the centre of the earth, then gravitational potential at the point,

$$V = -\frac{GM}{r} = -5.12 \times 10^7 \text{ J/kg}$$

Acceleration due to gravity at this point, $g = \frac{GM}{r^2} = 6.4$ m/s 2

Clearly,
$$\frac{|V|}{g} = \frac{GM/r}{GM/r^2} = r$$

$$\begin{aligned} \text{Thus, } r &= \frac{5.12 \times 10^7 \text{ J/kg}}{6.4 \text{ m/s}^2} \\ &= 8 \times 10^6 \text{ m} = 8000 \text{ km} \end{aligned}$$

Obviously, height of the point from the earth's surface
 $= (r - R) = 8000 \text{ km} - 6400 \text{ km} = 1600 \text{ km}$

4. Potential due to a thin spherical shell

Potential at an external point To calculate the potential at an external point P , a uniform spherical shell may be

treated as a point mass of same magnitude at its centre. Thus, potential at a distance r is given by

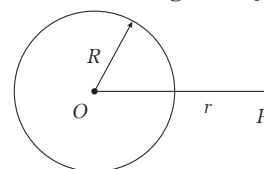


Fig. 10.27 Potential at an external point P

$$V(r) = \frac{-GM}{r}, \text{ where } r \geq R$$

$$\text{At } r = R, V = -\frac{GM}{R}$$

Potential at an internal point The potential due to a uniform spherical shell is constant throughout at any point inside the shell and this is equal to $-\frac{GM}{R}$.

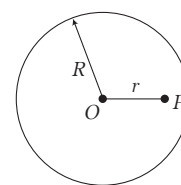


Fig. 10.28 Potential at an internal point P

Thus, V versus r graph for a spherical shell is as shown in figure below

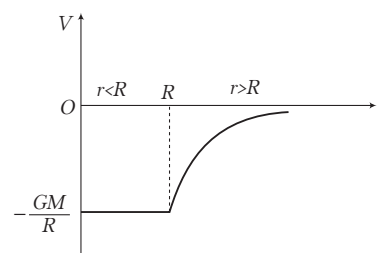


Fig. 10.29 V versus r graph for spherical shell

Example 10.39 A particle of mass 1 kg is kept on the surface of a uniform thin spherical shell of mass 20 kg and radius 1 m. Find the work to be done against the gravitational force between them to take the particle away from the thin spherical shell.

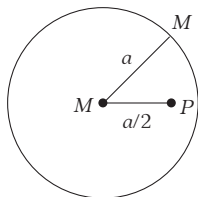
Sol. Potential at the surface of thin spherical shell,

$$\begin{aligned} V &= -\frac{GM}{R} \\ &= -\frac{(6.67 \times 10^{-11})(20)}{1} \text{ J kg}^{-1} \\ &= -1.334 \times 10^{-9} \text{ J kg}^{-1} \end{aligned}$$

i.e. 1.334×10^{-9} J work is obtained to bring a mass of 1 kg from infinity to the surface of spherical shell. Hence, the same amount of work will have to be done to take the particle away from the surface of spherical shell. Thus, $W = 1.334 \times 10^{-9}$ J.

Example 10.40 A particle of mass M is placed at the centre of a spherical shell of same mass and radius a . What will be the magnitude of the gravitational potential at a point situated at $a/2$ distance from the centre?

Sol. The given condition is as shown in figure below.



Therefore, the magnitude of gravitational potential at a point situated at $a/2$ distance from the centre can be given as

$$V_P = V_{\text{sphere}} + V_{\text{particle}} = \frac{-GM}{a} - \frac{GM}{a/2} = \frac{-3GM}{a}$$

5. Potential due to a uniform ring at a point on its axis

The gravitational potential at a distance r from the centre on the axis of a ring of mass M and radius R is given by

$$V(r) = -\frac{GM}{\sqrt{R^2 + r^2}}, \quad 0 \leq r \leq \infty$$

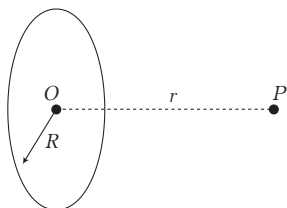


Fig. 10.30 Potential due to a ring

At $r = 0$, $V = -\frac{GM}{R}$, i.e. at the centre of the ring,

gravitational potential is $-\frac{GM}{R}$.

Thus, V versus r graph is as shown in figure below.

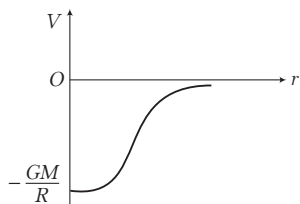


Fig. 10.31 V versus r graph for a ring

Example 10.41 Mass of 1 kg is distributed uniformly over a ring of radius 1 m. Find the gravitational potential at a point lying on the axis of the ring at a distance of 1 m from the centre.

Sol. Mass of the ring, $M = 1$ kg

Radius of the ring, $R = 1$ m

Distance of the point, $r = 1$ m

$\therefore$ Required gravitational potential,

$$\begin{aligned} V(r=1) &= \frac{-GM}{\sqrt{R^2 + r^2}} = \frac{-G \times 1}{\sqrt{(1)^2 + (1)^2}} \\ &= \frac{-G}{\sqrt{2}} = \frac{-6.67 \times 10^{-11}}{\sqrt{2}} \text{ Jkg}^{-1} \\ &= -4.716 \times 10^{-11} \text{ Jkg}^{-1} \end{aligned}$$

Relation between gravitational field and potential

If change in gravitational potential at a point is dV , gravitational field intensity is E , then during displacement $d\mathbf{r}$ in the field,

$$dV = -\mathbf{E} \cdot d\mathbf{r}$$

where, $\mathbf{E} = E_x \hat{\mathbf{i}} + E_y \hat{\mathbf{j}} + E_z \hat{\mathbf{k}}$

$$d\mathbf{r} = dx \hat{\mathbf{i}} + dy \hat{\mathbf{j}} + dz \hat{\mathbf{k}}$$

$$\therefore dV = -E_x dx - E_y dy - E_z dz$$

Also we can write, $E_x = -\frac{\partial V}{\partial x}$, $E_y = -\frac{\partial V}{\partial y}$ and $E_z = -\frac{\partial V}{\partial z}$

where, $\frac{\partial}{\partial x}$ represents partial differentiation with respect to

x considering y and z as constants. Similarly, $\frac{\partial}{\partial y}$ and $\frac{\partial}{\partial z}$

are partial differentiations with respect to y and z respectively considering other variables as constants.

Note If field is given, the potential can be obtained as

$$V(r_2) - V(r_1) = -\int_{r_1}^{r_2} \mathbf{E} \cdot d\mathbf{r}$$

Example 10.42 If gravitational potential is $V = xy^2$, find the gravitational field at $(2, 1)$.

Sol. Given, $V = xy^2$

Gravitational field intensity in x -direction,

$$E_x = -\frac{\partial V}{\partial x} = -\frac{\partial}{\partial x}(xy^2) = -y^2$$

Gravitational field intensity in y -direction,

$$E_y = -\frac{\partial V}{\partial y} = -\frac{\partial}{\partial y}(xy^2) = -2yx$$

$$\therefore \mathbf{E} = E_x \hat{\mathbf{i}} + E_y \hat{\mathbf{j}} = (-y^2) \hat{\mathbf{i}} + (-2xy) \hat{\mathbf{j}}$$

$$\therefore \mathbf{E}(x=2, y=1) = -(1)^2 \hat{\mathbf{i}} + (-2 \times 2 \times 1) \hat{\mathbf{j}} = -\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$$

$$\begin{aligned} \Rightarrow |\mathbf{E}| &= \sqrt{(-1)^2 + (-4)^2} \\ &= \sqrt{17} \text{ unit} \end{aligned}$$

Example 10.43 If gravitational field is given by $\mathbf{E} = -x\hat{i} - 2y^2\hat{j}$. When gravitational potential is zero at (0,0), find potential at (2, 1).

Sol. Given, gravitational field,

$$\mathbf{E} = -x\hat{i} - 2y^2\hat{j}, E_x = -x$$

$$\text{and } E_y = -2y^2$$

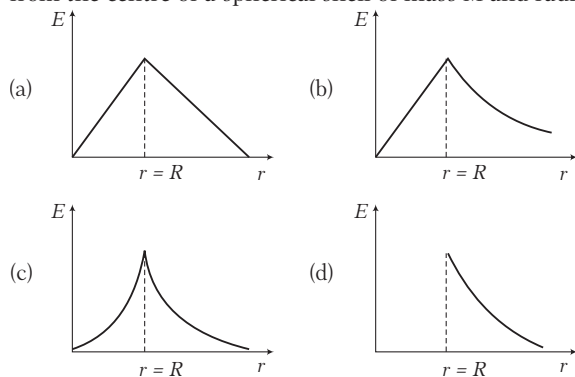
As we know that,

$$V(\mathbf{r}_2) - V(\mathbf{r}_1) = - \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{E} \cdot d\mathbf{r}$$

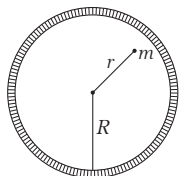
$$\begin{aligned} \Rightarrow V(2, 1) - V(0, 0) &= - \left[\int_0^2 E_x dx + \int_0^1 E_y dy \right] \\ &= - \left[\int_0^2 -x dx + \int_0^1 -2y^2 dy \right] \\ &= - \left[\left(\frac{x^2}{2} \right)_0^2 + \left(\frac{-2y^3}{3} \right)_0^1 \right] = - \left[\frac{4}{2} + \frac{-2}{3} \right] = \frac{8}{3} \text{ unit} \\ \Rightarrow V(2, 1) &= \frac{8}{3} + V(0, 0) = \frac{8}{3} + 0 = \frac{8}{3} \text{ unit} \end{aligned}$$

CHECK POINT 10.3

- Three particles each of mass m are kept at vertices of an equilateral triangle of side L . The gravitational field at centre due to these particles is
(a) zero (b) $\frac{3GM}{L^2}$ (c) $\frac{9GM}{L^2}$ (d) $\frac{12GM}{\sqrt{3}L^2}$
- At what height, the gravitational field reduces by 75% the gravitational field at the surface of earth? (where, R is the radius of earth)
(a) R (b) $2R$ (c) $3R$ (d) $4R$
- Gravitational field due to a solid sphere
(a) remains constant throughout the sphere
(b) increases inside the sphere and decreases outside the sphere
(c) increases throughout with distance from the centre
(d) decreases throughout with distance from the centre
- Which one of the following graphs represents correctly the variation of the gravitational field (E) with the distance (r) from the centre of a spherical shell of mass M and radius R ?

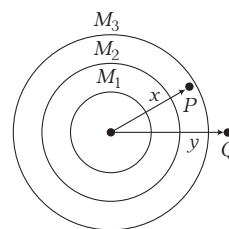


- A mass m is placed inside a hollow sphere of mass M as shown in figure. The gravitational force on mass m is



- (a) $\frac{GMm}{R^2}$ (b) $\frac{GMm}{r^2}$ (c) $\frac{GMm}{(R-r)^2}$ (d) zero

- A uniform solid sphere of mass m and radius r is surrounded symmetrically by a uniform thin spherical shell of radius $2r$ and mass m .
(a) The gravitational field at a distance of $1.5r$ from the centre is $\frac{2Gm}{9r^2}$.
(b) The gravitational field at a distance of $2.5r$ from the centre is $\frac{8Gm}{25r^2}$.
(c) The gravitational field at a distance of $1.5r$ from the centre is zero.
(d) The gravitational field between the sphere and spherical shell is uniform.
- Consider three concentric shells of masses M_1 , M_2 and M_3 having radii a , b and c respectively are situated as shown in figure.

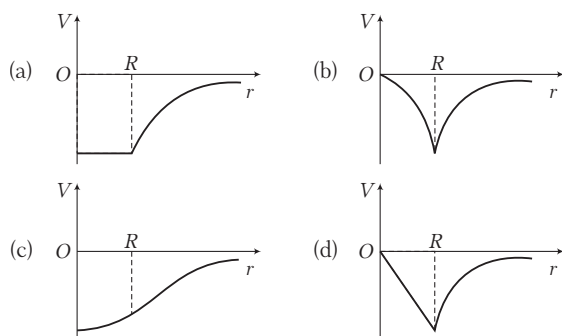


Gravitational field at a point located at Q and P is

- (a) $\frac{G(M_1 + M_2)}{y^2}, \frac{G(M_1 + M_2)}{y^2}$
(b) $\frac{G(M_1 + M_2 + M_3)}{y^2}, \frac{G(M_1 + M_2)}{x^2}$
(c) $\frac{G(M_1 + M_2 + M_3)}{a^2}, \frac{GM_1}{a^2}$
(d) $\frac{G(M_1 + M_2 + M_3)}{c^2}, \frac{GM_2}{b^2}$

- For a uniform ring of mass M and radius R at its centre,
(a) field and potential both are zero
(b) field is zero but potential is $\frac{GM}{R}$
(c) field is zero but potential is $-GM/R$
(d) magnitude of field is $\frac{GM}{R^2}$ and potential $-\frac{GM}{R}$

9. The diagram showing the variation of gravitational potential of earth with distance from the centre of earth is



10. If V is the gravitational potential on the surface of the earth, then what is its value at the centre of the earth?

(a) $2V$ (b) $3V$ (c) $\frac{3}{2}V$ (d) $\frac{2}{3}V$

11. A thin rod of length L is bent to form a semi-circle. The mass of rod is M . What will be the gravitational potential at the centre of the circle?

(a) $-\frac{GM}{L}$ (b) $-\frac{GM}{2\pi L}$
(c) $-\frac{\pi GM}{2L}$ (d) $-\frac{\pi GM}{L}$

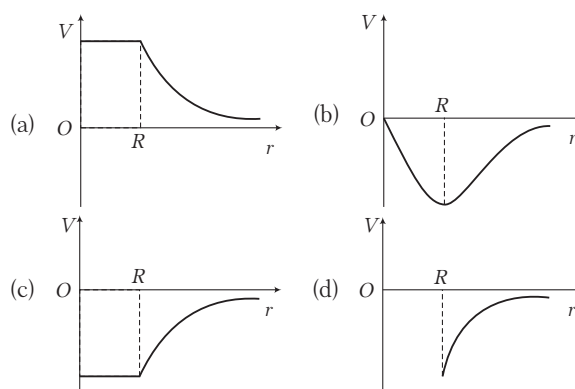
12. Two particles of masses m and M are placed a distance d apart. The gravitational potential at the position, where the gravitational field due to them is zero is V . Then,

(a) $V = -\frac{G}{d}(m + M)$ (b) $V = -\frac{Gm}{d}$
(c) $V = -\frac{GM}{d}$ (d) $V = -\frac{G}{d}(\sqrt{m} + \sqrt{M})^2$

13. A particle is kept on the surface of a uniform sphere of mass 100 kg and radius 10 cm. Find the work to be done per unit mass against the gravitational force between them, to take the particle far away from the sphere (you may take $h = 6.67 \times 10^{-11} \text{ N-m}^2\text{kg}^{-2}$).

(a) $13.34 \times 10^{-10} \text{ J}$ (b) $3.33 \times 10^{-10} \text{ J}$
(c) $6.67 \times 10^{-9} \text{ J}$ (d) $6.67 \times 10^{-8} \text{ J}$

14. By which curve will be variation of gravitational potential of a hollow sphere of radius R with distance be depicted?



GRAVITATIONAL POTENTIAL ENERGY

Gravitational potential energy of a body at a point is defined as the amount of work done in bringing the given body from infinity to that point against the gravitational force. The change in potential energy (dU) of a system corresponding to a conservative internal force ($\mathbf{F}$) is given by $dU = -\mathbf{F} \cdot d\mathbf{r}$

$$\text{or} \quad \int_i^f dU = - \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r}$$

$$\text{or} \quad U_f - U_i = - \int_{\mathbf{r}_i}^{\mathbf{r}_f} \mathbf{F} \cdot d\mathbf{r}$$

We generally choose the reference point at infinity and assume potential energy to be zero there, i.e. if we take $r_i = \infty$ (infinite) and $U_i = 0$, then expression can be written as

$$U_f = U = - \int_{\infty}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r} = -W$$

Hence, potential energy of a body or system is negative of the work done by the conservative forces in bringing it from infinity to the particular position.

Gravitational potential energy of a two particles system

The gravitational potential energy of two particles of masses m_1 and m_2 separated by a distance r is given by

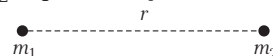


Fig. 10.32 System of two point masses separated by a distance r

$$U = - \frac{Gm_1m_2}{r}$$

This is actually the negative of work done in bringing these masses from infinity to a distance r apart by the gravitational forces between them.

$$\text{Gravitational potential energy} = \left(- \frac{Gm_1}{r} \right) \times m_2$$

$\therefore$ Gravitational potential energy =

Gravitational potential $\times$ Mass of the body

Gravitational potential energy is a scalar quantity. Its SI unit is joule and dimensional formula is $[\text{ML}^2\text{T}^{-2}]$.

Example 10.44 Two point masses 1 kg and 4 kg are separated by a distance of 10 cm. Find gravitational potential energy of the two point masses.

Sol. Given, $m_1 = 1$ kg, $m_2 = 4$ kg, $r = 10$ cm $= 10 \times 10^{-2}$ m

$\therefore$ Gravitational potential energy,

$$U = \frac{-Gm_1m_2}{r} = \frac{-6.67 \times 10^{-11} \times 1 \times 4}{10 \times 10^{-2}} \\ = -26.68 \times 10^{-10} \text{ J}$$

Gravitational potential energy for a system of more than two particles

The gravitational potential energy for a system of particles (say m_1, m_2, m_3 and m_4) is given by

$$U = -G \left[\frac{m_4m_3}{r_{43}} + \frac{m_4m_2}{r_{42}} + \frac{m_4m_1}{r_{41}} + \frac{m_3m_2}{r_{32}} + \frac{m_3m_1}{r_{31}} + \frac{m_2m_1}{r_{21}} \right]$$

Thus, for a n particle system there are $\frac{n(n-1)}{2}$ pairs and

the potential energy is calculated for each pair and added to get the total potential energy of the system.

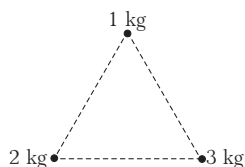
Example 10.45 Three masses of 1 kg, 2 kg and 3 kg are placed at the vertices of an equilateral triangle of side 1 m. Find the gravitational potential energy of this system. (Take, $G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$)

Sol. Gravitational potential energy of the three particles system

$$U = -G \left(\frac{m_3m_2}{r_{32}} + \frac{m_3m_1}{r_{31}} + \frac{m_2m_1}{r_{21}} \right)$$

Here, $r_{32} = r_{31} = r_{21} = 1$ m,

$m_1 = 1$ kg, $m_2 = 2$ kg and $m_3 = 3$ kg



Substituting in above, we get

$$U = -(6.67 \times 10^{-11}) \left(\frac{3 \times 2}{1} + \frac{3 \times 1}{1} + \frac{2 \times 1}{1} \right)$$

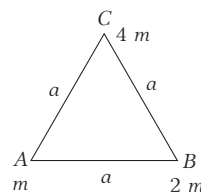
or $U = -7.337 \times 10^{-10} \text{ J}$

Example 10.46 Three particles of masses $m, 2m$ and $4m$ are placed at the corners of an equilateral triangle of side a . Calculate

(i) the potential energy of the system.

(ii) the work done on the system, if the side of the triangle is changed from a to $2a$. Assume the potential energy to be zero when the separation is infinity.

Sol. (i) The potential energy of the system = Sum of the potential energies of all the three possible distinct pairs.



$$\text{i.e. } U = U_{AB} + U_{BC} + U_{CA} \\ = \frac{-G(m)(2m)}{a} + \frac{-G(2m)(4m)}{a} + \frac{-G(4m)(m)}{a} \\ = \frac{-14Gm^2}{a}$$

(ii) When the side a is changed to $2a$, the potential energy

$$U' = \frac{-14Gm^2}{2a}$$

The work done on the system = The work done against the gravitational force

= The work done by the external force

$$= U' - U = \frac{14Gm^2}{2a}$$

Gravitational potential energy of an object in different conditions

Object placed at earth's surface Let an object of mass m is placed at a point in the gravitational field of the earth of mass M . Then, gravitational potential energy of an object placed at earth's surface,

$$U = -\frac{GMm}{R}$$

Note

(i) If the body of mass m is moved from a point at distance r_a to a point at distance r_b ($r_a > r_b$), then change in potential energy,

$$\Delta U = GMm \left[\frac{1}{r_a} - \frac{1}{r_b} \right]$$

(ii) Gravitational potential energy at the centre of earth relative to infinity is

$$U_{\text{centre}} = -\frac{3}{2} \left(\frac{GMm}{R} \right)$$

Object placed at a point, above the earth's surface Let an object of mass m is placed at height h above the surface of earth, then change in gravitational potential energy when the object is taken from the centre of the earth to the height h is

$$\Delta U = \frac{mgh}{1 + h/R}$$

ΔU is positive, it means the potential energy of a body increases as its height above earth surface increases.

For $h < R$, $\Delta U \approx mgh$

Thus, what we observe the mgh is actually the difference in potential energy (not the absolute potential energy), that too for $h \ll R$.

Note Gravitational potential energy of a body at height x from the earth's surface is

$$U_x = \frac{-mgR}{1+x/R}$$

Example 10.47 The mass of the earth is 6×10^{24} kg. The constant of gravitation $G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$. The potential energy of the earth and moon system is -7.79×10^{28} J. Determine the mean distance between earth and moon. (Mass of moon is 7.4×10^{22} kg)

Sol. The mass of moon, $m = 7.4 \times 10^{22}$ kg

Potential energy of the earth and moon system,

$$U = \frac{-GMm}{r}$$

$$\Rightarrow r = \frac{-GMm}{U} = \frac{-6.67 \times 10^{-11} \times 6 \times 10^{24} \times 7.4 \times 10^{22}}{-7.79 \times 10^{28}}$$

$$\Rightarrow r = 3.8 \times 10^8 \text{ m}$$

Example 10.48 A body of mass m is raised to a height $10R$ from the surface of the earth, where R is the radius of the earth. Find the increase in potential energy. (G = universal constant of gravitation, M = mass of the earth and g = acceleration due to gravity)

Sol. Potential energy at the earth's surface = $\frac{-GMm}{R}$

Potential energy at a height h above the earth's surface

$$= \frac{-GMm}{(R+h)}$$

$$\therefore \text{Change in potential energy} = \frac{-GMm}{(R+h)} - \left(-\frac{GMm}{R} \right)$$

$$= \frac{GMmh}{R(R+h)}$$

Substituting, $h = 10R$

$$\Delta U = \frac{GMm \times 10R}{R(R+10R)}$$

$$= \frac{10GMm}{11R}$$

Example 10.49 Find the change in the gravitational potential energy when a body of mass m is raised to a height nR above the surface of the earth. (Here, R is the radius of the earth)

Sol. Gravitational potential energy of mass m at earth's surface,

$$U_e = -\frac{GMm}{R}$$

Gravitational potential energy of same mass at a height nR from the earth's surface,

$$U_h = -\frac{GMm}{(R+nR)} = -\frac{GMm}{R(n+1)}$$

Thus, magnitude of the change in gravitational potential energy,

$$\Delta U = U_h - U_e$$

$$= \frac{GMm}{R} \left(1 - \frac{1}{(n+1)} \right)$$

$$= \left(\frac{n}{n+1} \right) \frac{GMm}{R}$$

$$= \left(\frac{n}{n+1} \right) mgR \quad (\because GM = gR^2)$$

Example 10.50 An object is dropped from height $h = 2R$ on the surface of earth. Find the speed with which it will collide with ground by neglecting effect of air. (where, R is radius of earth and M is mass of earth.)

Sol. The initial potential energy of object, $U_i = -\frac{GMm}{3R}$

$$\text{Final potential energy, } U_f = -\frac{GMm}{R}$$

By law of conservation of energy, $\Delta KE = -\Delta PE$

$$\Rightarrow \frac{1}{2}mv^2 = -(U_f - U_i) = U_i - U_f$$

$$\Rightarrow \frac{1}{2}mv^2 = -\frac{GMm}{3R} + \frac{GMm}{R}$$

$$\Rightarrow \frac{1}{2}v^2 = \frac{2GM}{3R}$$

$$\Rightarrow v = \sqrt{\frac{4GM}{3R}} = 2\sqrt{\frac{GM}{3R}}$$

CHECK POINT 10.4

- Consider the two identical particles shown in the given figure. They are released from rest and may move towards each other under the influence of mutual gravitational force. The gravitational potential energy of the two particle system
 - zero
 - constant ($\neq 0$)
 - decreases as the separation decreases
 - increases as the separation decreases
- The magnitude of gravitational potential energy of a body at a distance r from the centre of earth is u . Its weight at a distance $2r$ from the centre of earth is
 - $\frac{u}{r}$
 - $\frac{u}{4r}$
 - $\frac{u}{2r}$
 - $\frac{4r}{u}$
- A body of mass m is kept at a small height h above the ground. If the radius of the earth is R and its mass is M , the potential energy of the body and earth system (with $h = \infty$ being the reference position) is
 - $\frac{GMm}{R} + mgh$
 - $-\frac{GMm}{R} + mgh$
 - $\frac{GMm}{R} - mgh$
 - $-\frac{GMm}{R} - mgh$
- The gravitational potential energy of a body of mass m at a distance r from the centre of the earth is U . What is the weight of the body at this distance?
 - U
 - Ur
 - $\frac{U}{r}$
 - $\frac{U}{2r}$
- Find the potential energy of four-particles, each of mass 1 kg placed at the four vertices of a square of side length 1 m.
 - + 4.0 G
 - 7.5 G
 - 5.4 G
 - + 6.3 G
- Energy required in moving a body of mass m from a distance $2R$ to $3R$ from centre of earth of mass M is
 - $\frac{GMm}{12R^2}$
 - $\frac{GMm}{3R^2}$
 - $\frac{GMm}{8R}$
 - $\frac{GMm}{6R}$
- When a body is lifted from surface of earth at a height equal to radius of earth, then the change in its potential energy is
 - mgR
 - $2mgR$
 - $\frac{1}{2}mgR$
 - $4mgR$
- If an object of mass m is taken from the surface of earth (radius R) to a height $5R$, then the work done is
 - $2mgR$
 - mgR
 - $\frac{5}{6}mgR$
 - $\frac{3}{2}mgR$
- If a body of mass m has to be taken from the surface of the earth to a height $h = 4R$, then the amount of energy required is (R = radius of the earth)
 - mgR
 - $\frac{mgR}{5}$
 - $\frac{4mgR}{5}$
 - $\frac{mgR}{12}$

BINDING ENERGY

The minimum energy required to take a particle to an infinite distance from the earth is called the binding energy of the earth particle system.

Binding energy of a particle on the surface of the earth

Suppose the mass m is placed on the surface of earth. The radius of the earth is R and its mass is M . Then, the kinetic energy of the particle, $K = 0$ and potential energy of the particle is $U = -\frac{GMm}{R}$.

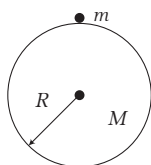


Fig. 10.33 A particle on the surface of the earth

Therefore, the total mechanical energy of the particle,

$$E = K + U = 0 - \frac{GMm}{R}$$

or

$$E = -\frac{GMm}{R}$$

$$\therefore \text{Binding energy, } BE = |E| = \frac{GMm}{R}$$

Due to binding energy, the particle is attached to the earth. If this much energy is supplied to the particle in any form (normally kinetic) the particle no longer remains bound to the earth. It goes out of the gravitational field of earth.

Example 10.51 Find the binding energy of a particle of mass m on the surface of earth in term of g . (Here, R = radius of earth and g = acceleration due to gravity)

Sol. As, binding energy $= -E = \frac{GMm}{R}$

$$\text{But } g = \frac{GM}{R^2} \Rightarrow GM = gR^2$$

$$\therefore \text{Binding energy} = |E| = \frac{gmR^2}{R} = mgR$$

Escape velocity

The minimum velocity with which a body must be projected vertically upwards in order that it may just escape the gravitational field of the earth (*i.e.* $v = 0$) is called escape velocity.

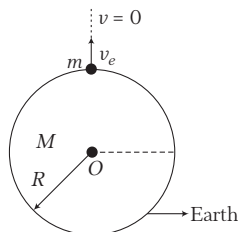


Fig. 10.34 Escape velocity of a body

A body projected from the surface of the earth with speed v_e , such that $v_f = v = 0$.

As we have discussed that the binding energy of a particle on the surface of earth kept at rest is GMm/R . If this much energy in the form of kinetic energy is supplied to the particle, it leaves the gravitational field of the earth. So, if v_e is the escape velocity of the particle, then from conservation of mechanical energy, we have

$$\frac{1}{2}mv_e^2 = \frac{GMm}{R}$$

$$\text{or} \quad v_e = \sqrt{\frac{2GM}{R}} \quad \dots(i)$$

$$\text{As,} \quad g = \frac{GM}{R^2}$$

$$\Rightarrow \quad v_e = \sqrt{2gR}$$

Substituting the value of $g (= 9.8 \text{ ms}^{-2})$ and $R (= 6.4 \times 10^6 \text{ m})$, we get

$$v_e \approx 11.2 \text{ kms}^{-1}$$

Thus, the minimum velocity needed to take a particle infinitely away from the earth is called the escape velocity. On the surface of earth, its value is 11.2 kms^{-1} .

$$\text{From Eq. (i),} \quad v_e = \sqrt{\frac{2G}{R} \times \frac{4}{3}\pi R^3 \times \rho}$$

where, ρ is the mean density of the earth.

$$\text{Escape velocity, } v_e = R \sqrt{\frac{8}{3}\pi G\rho}$$

Important points related to escape velocity

- (i) The escape velocity does not depend on angle of projection from the earth's surface. But as the earth rotates about its axis, so it becomes easier to attain

escape velocity, if the body is projected in the direction in which the earth is moving.

- (ii) As the escape velocity depends on the mass and radius of the planet from the surface of which the body is projected, so value of escape velocity is different for different planets.
- (iii) A planet will have atmosphere, if the root mean square velocity of its atmospheric molecules is less than the escape velocity for the given planet. That is why moon has no atmosphere ($v_e = 2.3 \text{ kms}^{-1}$) while jupiter has a thick atmosphere ($v_e = 60 \text{ kms}^{-1}$). Even the lightest hydrogen cannot escape from its surface.
- (iv) If a body falls freely from infinite height, then it will reach the surface of the earth with a speed approximately equal to the average escape speed $v_e = 11.2 \text{ kms}^{-1}$.
- (v) In calculating the escape speed, the air resistance and the gravitational effect due to other celestial bodies are neglected.
- (vi) For earth to become black hole, the escape speed must become equal to or more than speed of light, *i.e.* $\sqrt{\frac{2GM}{R}} \geq c$. In that case, radius of the earth will become that of the size of a grape.

Note If the escape velocity of a body is equal to the velocity of light, then from such bodies nothing can escape, not even light. Such bodies are called **black hole**.

Example 10.52 Calculate the escape velocity from the surface of a planet of mass $14.8 \times 10^{22} \text{ kg}$. It is given that radius of the planet is $3.48 \times 10^6 \text{ m}$.

Sol. Given, mass of planet,

$$M_p = 14.8 \times 10^{22} \text{ kg}$$

and radius of the planet,

$$R_p = 3.48 \times 10^6 \text{ m}$$

Escape velocity from the surface of planet,

$$v_e = \sqrt{\frac{2GM_p}{R_p}}$$

Substituting the values, we get

$$\begin{aligned} v_e &= \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 14.8 \times 10^{22}}{3.48 \times 10^6}} \\ &= 2.4 \times 10^3 \text{ ms}^{-1} \text{ or } 2.4 \text{ kms}^{-1} \end{aligned}$$

Example 10.53 The ratio of the masses and radii of two planets are 4 : 6 and 8 : 18. What is the ratio of the escape velocity at their surfaces?

Sol. Given, $\frac{M_1}{M_2} = \frac{4}{6} = \frac{2}{3}$ and $\frac{R_1}{R_2} = \frac{8}{18} = \frac{4}{9}$

As we know, escape velocity, $v = \sqrt{\frac{2GM}{R}}$

$$\begin{aligned} \therefore \frac{v_1}{v_2} &= \sqrt{\frac{M_1}{M_2} \times \frac{R_2}{R_1}} \\ &= \sqrt{\frac{2}{3} \times \frac{9}{4}} = \sqrt{\frac{3}{2}} \end{aligned}$$

Hence, $v_1 : v_2 = \sqrt{3} : \sqrt{2}$

Example 10.54 Jupiter has a mass 318 times that of earth, and its radius is 11.2 times the earth's radius. Estimate the escape velocity of a body from jupiter's surface, given that the escape velocity from the earth's surface is 11.2 kms^{-1} .

Sol. Escape velocity from the earth's surface is

$$v_e = \sqrt{\frac{2GM}{R}} = 11.2 \text{ kms}^{-1}$$

Escape velocity from jupiter's surface will be

$$v_e = \sqrt{\frac{2GM'}{R'}}$$

Given, $M' = 318 M$ and $R' = 11.2R$

$$\therefore v'_e = \sqrt{\frac{2G(318M)}{11.2R}} = \sqrt{\frac{2GM}{R}} \times \frac{318}{11.2}$$

$$= v'_e \times \sqrt{\frac{318}{11.2}} = 11.2 \times \sqrt{\frac{318}{11.2}}$$

$$\Rightarrow v'_e = 59.7 \text{ kms}^{-1}$$

Example 10.55 At what temperature, hydrogen molecules will escape from the earth's surface? (Take, radius of earth $R_e = 6.4 \times 10^6 \text{ m}$, mass of hydrogen molecule,

$m = 0.34 \times 10^{-26} \text{ kg}$, Boltzmann constant

$k = 1.38 \times 10^{-23} \text{ JK}^{-1}$, acceleration due to gravity

$= 9.8 \text{ ms}^{-2}$ and rms speed of gas as $v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$)

Sol. The root-mean-square-velocity of gas is $v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$... (i)

Escape velocity of gas molecules is $v_e = \sqrt{2gR_e}$... (ii)

As the root-mean-square velocity of gas molecules must be equal to the escape velocity.

$$\therefore \text{From Eqs. (i) and (ii), } \sqrt{\frac{3kT}{m}} = \sqrt{2gR_e}$$

$$\Rightarrow T = \frac{2gR_e m}{3k}$$

$$\Rightarrow T = \frac{2 \times 9.8 \times 6.4 \times 10^6 \times 0.34 \times 10^{-26}}{3 (1.38 \times 10^{-23})} \approx 10^4 \text{ K}$$

Therefore, 10^4 K is the temperature at which hydrogen molecules will escape from earth's surface.

Escape energy

Minimum energy given to a particle in form of kinetic energy, so that it can escape from earth's gravitational field, is called escape energy.

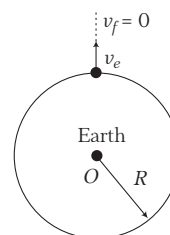


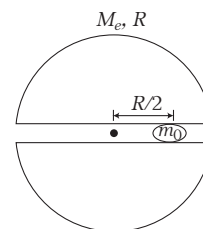
Fig. 10.35 Particle projected from earth surface

$$\text{Escape energy} = \frac{GMm}{R}$$

If escape velocity is v_e , then

$$\text{Escape energy} = \frac{1}{2} m v_e^2$$

Example 10.56 A very small groove is made in the earth, and a particle of mass m_0 is placed at $R/2$ distance from its centre as shown in figure. Find the escape energy of the particle from that place.



Sol. Suppose we projected the particle with speed v_e , so that it just reaches at $r \rightarrow \infty$.

Applying energy conservation, $K_i + U_i = K_f + U_f$

$$\Rightarrow \frac{1}{2} m_0 v_e^2 + m_0 \left[-\frac{GM_e}{2R^3} \left\{ 3R^2 - \left(\frac{R}{2} \right)^2 \right\} \right] = 0$$

$$\Rightarrow \text{Escape energy} = \frac{1}{2} m_0 v_e^2 = \frac{11Gm_0 M_e}{8R}$$

Example 10.57 A body is projected upwards with a velocity of $4 \times 11.2 \text{ km s}^{-1}$ from the surface of earth. What will be the velocity of the body when it escapes from the gravitational pull of earth?

Sol. Initial kinetic energy of the body $= \frac{1}{2} m v^2$

Here, $v = 4 \times 11.2 \text{ kms}^{-1}$

So, kinetic energy, $\text{KE} = \frac{1}{2} \times m (4 \times 11.2)^2 = 16 \times \frac{1}{2} m v_e^2$

As, $\frac{1}{2}mv_e^2$ energy is used up in coming out from the gravitational pull of the earth, so final kinetic energy should be $15 \times \frac{1}{2}mv_e^2$.

$$\text{Hence, } \frac{1}{2}mv'^2 = 15 \times \frac{1}{2}mv_e^2$$

$$\therefore v'^2 = 15v_e^2 \Rightarrow v' = \sqrt{15}v_e = \sqrt{15} \times 11.2 \text{ kms}^{-1}$$

Example 10.58 The earth is assumed to be a sphere of radius R . A platform is arranged at a height R from the surface of the earth. The escape velocity of a body from this platform is $f v_e$, where v_e is its escape velocity from the surface of the earth. Find the value of f .

Sol. For a platform at a height h ,
escape energy = binding energy of platform

$$\Rightarrow \frac{1}{2}mv_e'^2 = \frac{GMm}{R+h} \Rightarrow v_e' = \sqrt{\frac{2GM}{R+h}} = \sqrt{\frac{2GM}{2R}} \quad (\because h = R)$$

$$\text{But at surface of earth, } v_e = \sqrt{\frac{2GM}{R}}$$

$$\text{As given, } v_e' = f v_e$$

$$\text{Hence, } \sqrt{\frac{2GM}{2R}} = f \sqrt{\frac{2GM}{R}} \text{ or } \frac{1}{2R} = \frac{f^2}{R} \Rightarrow f = \frac{1}{\sqrt{2}}$$

Maximum height attained by a particle

Suppose a particle of mass m is projected vertically upwards with a speed v and we want to find the maximum height h attained by the particle.

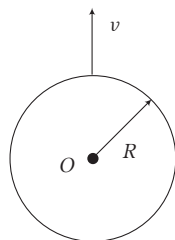


Fig. 10.36 A particle projected vertically upwards from the surface of the earth

Then, we apply conservation of mechanical energy, i.e.
decrease in kinetic energy = increase in gravitational potential energy of particle

$$\therefore \frac{1}{2}mv^2 = \Delta U \text{ or } \frac{1}{2}mv^2 = \frac{mgh}{1+h/R}$$

Solving this, we get

$$h = \frac{v^2}{2g - v^2/R}$$

From this, we can obtain that

$$(i) \text{ If } v = v_e \text{ or } v^2 = v_e^2 = 2gR, \text{ then } h = \infty.$$

(ii) If $v < v_e$, body will attain a certain maximum height and then may either move in an orbit around the planet or may fall down back to the planet.

(iii) If $v > v_e$, body will move in interplanetary or interstellar space with velocity $\sqrt{v^2 - v_e^2}$.

Example 10.59 A particle is projected from the surface of earth with an initial speed of 4 km/s. Find the maximum height attained by the particle. (Take, radius of earth = 6400 km and $g = 9.8 \text{ m/s}^2$)

Sol. The maximum height attained by the particle is, $h = \frac{v^2}{2g - \frac{v^2}{R}}$

Substituting the values, we get

$$h = \frac{(4 \times 10^3)^2}{2 \times 9.8 - \frac{(4.0 \times 10^3)^2}{6.4 \times 10^6}} = 9.35 \times 10^5 \text{ m or } h \approx 935 \text{ km}$$

Example 10.60 A particle is projected vertically upwards from the surface of the earth (radius R_e) with a speed equal to one-fourth of escape velocity. What is the maximum height attained by it from the surface of the earth?

Sol. From conservation of mechanical energy,

$$\frac{1}{2}mv^2 = \frac{GMm}{R_e} - \frac{GMm}{R} \quad \dots(i)$$

where, R = maximum height (distance) from centre of the earth.

$$\text{Also, } v = \frac{1}{4}v_e = \frac{1}{4}\sqrt{\frac{2GM}{R_e}}$$

Substituting values in Eq (i), we get

$$\begin{aligned} \Rightarrow \frac{1}{2}m \times \frac{1}{16} \times \frac{2GM}{R_e} &= \frac{GMm}{R_e} - \frac{GMm}{R} \\ \Rightarrow R &= \frac{16}{15}R_e \Rightarrow h = R - R_e = \frac{R_e}{15} \end{aligned}$$

Example 10.61 With what velocity must a body be thrown from earth's surface, so that it may reach a maximum height of $4R_e$ above the earth's surface? (Take, radius of the earth, $R_e = 6400 \text{ km}$, $g = 9.8 \text{ ms}^{-2}$)

Sol. From conservation of mechanical energy,

$$\begin{aligned} \frac{1}{2}m_0v^2 - \frac{GMm_0}{R_e} &= 0 - \frac{GMm_0}{(R_e + 4R_e)} \\ \Rightarrow \frac{1}{2}m_0v^2 &= -\frac{GMm_0}{5R_e} + \frac{GMm_0}{R_e} \\ \Rightarrow \frac{1}{2}m_0v^2 &= \frac{4}{5} \frac{GMm_0}{R_e} \\ \Rightarrow v^2 &= \frac{8}{5} \frac{GM}{R_e} = \frac{8}{5} \frac{gR_e^2}{R_e} \\ \Rightarrow v^2 &= \frac{8}{5} \times 9.8 \times 6400 \times 10^3 = 10^8 \\ \therefore v &= 10 \text{ kms}^{-1} \end{aligned}$$

CHECK POINT 10.5

- In a gravitational field, if a body is bound with earth, then total mechanical energy is
 - positive
 - zero
 - negative
 - may be positive, negative or zero
- The binding energy of an object of mass m placed on the surface of the earth is (R = radius of earth, g = acceleration due to gravity)
 - $\frac{mgR}{2}$
 - $\frac{mgR}{4}$
 - $\frac{mgR}{8}$
 - $\frac{mgR}{8}$
- When escape velocity is given to a particle on surface of earth, its total energy is (Take, M = mass of earth, m = mass of particle and R = radius of earth)
 - zero
 - greater than zero
 - less than zero
 - $-GMm/2R$
- The escape velocity for a body of mass 1 kg from the earth surface is 11.2 kms^{-1} . The escape velocity for a body of mass 100 kg would be
 - $11.2 \times 10^2 \text{ kms}^{-1}$
 - 112 kms^{-1}
 - 11.2 kms^{-1}
 - $11.2 \times 10^{-2} \text{ kms}^{-1}$
- The escape velocity of a body projected vertically upward from the earth's surface is 11.2 kms^{-1} . If the body is projected in a direction making 30° angle to the vertical, its escape velocity in this case will be
 - 11.2 kms^{-1}
 - $\frac{11.2}{2} \text{ kms}^{-1}$
 - $11.2 \times \frac{\sqrt{3}}{2} \text{ kms}^{-1}$
 - $\frac{11.2}{3} \text{ kms}^{-1}$
- Escape velocity on earth is 11.2 kms^{-1} , what would be the escape velocity on a planet whose mass is 1000 times and radius is 10 times that of earth?
 - 112 kms^{-1}
 - 11.2 kms^{-1}
 - 1.12 kms^{-1}
 - 3.7 kms^{-1}
- The ratio of the radius of the earth to that of the moon is 10. The ratio of the acceleration due to gravity on the earth to that on the moon is 6. The ratio of the escape velocity from the earth's surface to that from the moon is
 - 4
 - 6
 - 12
 - None of these
- There are two planets and the ratio of radius of the two planets is K but ratio of acceleration due to gravity of both planets is g . What will be the ratio of their escape velocities?
 - $(Kg)^{1/2}$
 - $(Kg)^{-1/2}$
 - $(Kg)^2$
 - $(Kg)^{-2}$
- The escape velocity from earth is v_e . A body is projected with velocity $2v_e$. With what constant velocity will it move in the inter planetary space?
 - v_e
 - $\sqrt{2}v_e$
 - $\sqrt{3}v_e$
 - $\sqrt{5}v_e$
- A particle is projected vertically upwards from the surface of earth (radius R) with a kinetic energy equal to half of the minimum value needed for it to escape. The height to which it rises above the surface of earth is
 - R
 - $2R$
 - $3R$
 - $4R$
- The kinetic energy required to project a body of mass m from the earth surface (radius R) to infinity is
 - $\frac{mgR}{2}$
 - $2mgR$
 - mgR
 - $\frac{mgR}{2}$
- A body is projected upwards with a velocity of $3 \times 11.2 \text{ km s}^{-1}$ from the surface of earth. What will be the velocity of the body when it escapes from the gravitational pull of earth?
 - 11.2 kms^{-1}
 - $\sqrt{2} \times 11.2 \text{ kms}^{-1}$
 - $\sqrt{3} \times 11.2 \text{ kms}^{-1}$
 - $\sqrt{8} \times 11.2 \text{ kms}^{-1}$
- With what velocity should a particle be projected, so that its maximum height attained becomes equal to radius of earth?
 - $\left(\frac{GM}{R}\right)^{1/2}$
 - $\left(\frac{8GM}{R}\right)^{1/2}$
 - $\left(\frac{2GM}{R}\right)^{1/2}$
 - $\left(\frac{4GM}{R}\right)^{1/2}$
- A body is projected vertically upwards from the surface of a planet of radius R with a velocity equal to 1/3rd the escape velocity for the planet. The maximum height attained by the body is
 - $R/2$
 - $R/3$
 - $R/5$
 - $R/9$

MOTION OF SATELLITES

Just as the planets revolve around the sun, in the same way few celestial bodies revolve around these planets. These bodies are called **satellites**. *e.g.* Moon is a natural satellite of earth. The paths of these satellites are elliptical with the centre of earth at a focus.

The difference in major and minor axes is not large, so we can assume circular orbits.

Orbital velocity of satellites

The velocity with which a satellite moves in its closed orbit around the earth is known as orbital velocity.

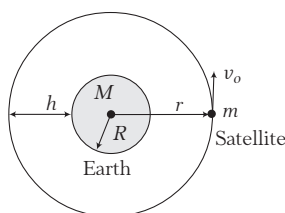


Fig. 10.37 Motion of a satellite

The necessary centripetal force to the satellite is provided by the gravitational force exerted by the earth on the

satellite. Thus,

$$\frac{mv_o^2}{r} = \frac{GMm}{r^2}$$

$$\therefore v_o = \sqrt{\frac{GM}{r}}$$

or
$$v_o \propto \frac{1}{\sqrt{r}}$$

Here $r = R + h$, where R is radius of earth and r is the orbital radius.

$$\therefore v_o = \sqrt{\frac{GM}{R + h}}$$

Further, we know that, $GM = gR^2$

$$\therefore v_o = \sqrt{\frac{gR^2}{R + h}}$$

$$\Rightarrow v_o = R \sqrt{\frac{g}{(R + h)}}$$

Hence, the orbital speed (v_o) of the satellite decreases as the orbital radius (r) of the satellite increases. Further, the orbital speed of a satellite close to the earth's surface ($r = R + h \approx R$) is,

$$v_o = \sqrt{\frac{GM}{R}} = \sqrt{gR}$$

Substituting the values of g and R , we get

$$v_o = 7.9 \text{ kms}^{-1} \approx 8 \text{ kms}^{-1}$$

Note

- Orbital velocity of a satellite is independent of mass of the satellite.
- It decreases with increase in the radius of the orbit and with increase in the height of the satellite.
- It depends on the mass and radius of the planet about which the satellite revolves.
- The angular momentum of a satellite of mass m moving with velocity v_o in an orbit of radius r ($= R + h$) is given by $L = \sqrt{GMm^2r}$.

Example 10.62 A satellite circled around the earth at a distance of 3400 km. Determine its orbital velocity, if the radius of the earth is 6400 km and $g = 9.8 \text{ ms}^{-2}$.

Sol. Given, radius of earth, $R = 6400 \text{ km} = 64 \times 10^5 \text{ m}$

Distance of satellite from earth, $h = 3400 \text{ km} = 34 \times 10^5 \text{ m}$

$\therefore$ Orbital velocity of an artificial satellite can be given as

$$v_o = R \sqrt{\frac{g}{R + h}} = 64 \times 10^5 \sqrt{\frac{9.8}{(64 + 34) \times 10^5}}$$

$$\Rightarrow v_o = 6400 \text{ ms}^{-1}$$

Relation between orbital velocity and escape velocity

The escape velocity of a body from the earth's surface,

$$v_e = \sqrt{2gR}$$

The orbital velocity of a satellite revolving close to earth's surface,

$$v_o = \sqrt{gR}$$

$$\therefore \frac{v_e}{v_o} = \frac{\sqrt{2gR}}{\sqrt{gR}} = \sqrt{2}$$

$$\Rightarrow v_e = \sqrt{2} v_o \quad \dots(i)$$

So, Eq. (i) represents that the escape velocity of a body from the earth's surface is $\sqrt{2}$ times its orbital velocity in a circular orbit just above the earth's surface.

Example 10.63 A satellite is launched into a circular orbit close to the earth's surface. What additional velocity has now to be imparted to the satellite in the orbit to overcome the gravitational pull?

Sol. Since, orbital velocity near earth's surface, $v_o = \sqrt{gR_e}$

Escape velocity, $v_e = \sqrt{2gR_e} = 1.414 \sqrt{gR_e}$

Therefore, additional velocity required

$$= v_e - v_o = (1.414 - 1) \sqrt{gR_e}$$

$$= 0.414 \sqrt{9.8 \times 6400 \times 10^3}$$

$$= 3.278 \times 10^3 \text{ ms}^{-1} = 3.278 \text{ kms}^{-1}$$

Time period of revolution of satellite

The time taken by a satellite to complete one revolution around the earth, is known as time period of revolution of satellite.

The period of revolution (T) is given by

$$T = \frac{2\pi r}{v_o} \text{ or } T = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

or
$$T = 2\pi \sqrt{\frac{r^3}{GM}}$$

or
$$T = 2\pi \sqrt{\frac{r^3}{gR^2}}$$

or
$$T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}} \quad (\because GM = gR^2 \text{ and } r = R+h)$$

Also,
$$T = \frac{2\pi r}{\sqrt{\frac{GM}{r}}} = \frac{2\pi (R+h)}{v_o}$$

From this expression of T , we can make the following conclusions

(i) $T \propto r^{3/2}$ or $T^2 \propto r^3$

which is also the Kepler's third law.

(ii) Time period of a satellite very close to earth's surface ($r \approx R$) is,

$$T = 2\pi \sqrt{\frac{R}{g}}$$

Substituting the values, we get

$$T \approx 84.6 \text{ min} \approx 1.4 \text{ h}$$

(iii) Suppose the height of a satellite is such that the time period of the satellite is 24 h and it moves in the same sense as the earth. The satellite will always be overhead a particular place on the equator. As seen from the earth, this satellite will appear to be stationary. Such a satellite is called a **geo-stationary satellite**.

Putting $T = 24 \text{ h}$ in the expression of T , the radius of geo-stationary satellite comes out to be $r = 4.2 \times 10^4 \text{ km}$. The height above the surface of earth is about $3.6 \times 10^4 \text{ km}$.

Example 10.64 A satellite revolves around the earth at a height of 1000 km. The radius of the earth is $6.38 \times 10^3 \text{ km}$. Mass of the earth is $6 \times 10^{24} \text{ kg}$ and $G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$. Determine its orbital velocity and period of revolution.

Sol. Given, height of satellite from the earth, $h = 1000 \text{ km} = 10^6 \text{ m}$,

$$R = 6.38 \times 10^3 \text{ km} = 6.38 \times 10^6 \text{ m}$$

$$\Rightarrow h + R = 7.38 \times 10^6 \text{ m}, M = 6 \times 10^{24} \text{ kg}$$

$$\begin{aligned} \therefore \text{Orbital velocity, } v_o &= \sqrt{\frac{GM}{R+h}} \\ &= \sqrt{\frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{7.38 \times 10^6}} \\ &= 7364 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned} \therefore \text{Period of revolution, } T &= \frac{2\pi (R+h)}{v_o} \\ &= \frac{2\pi \times 7.38 \times 10^6}{7364} \\ \Rightarrow T &\approx 6297 \text{ s} \approx 1.749 \text{ h} \end{aligned}$$

Height of satellite

As it is known that the time period of satellite,

$$T = 2\pi \sqrt{\frac{r^3}{GM}} = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}} \quad \dots(i)$$

By squaring both sides of Eq. (i), we get

$$\begin{aligned} T^2 &= 4\pi^2 \frac{(R+h)^3}{gR^2} \\ \Rightarrow \frac{gR^2 T^2}{4\pi^2} &= (R+h)^3 \end{aligned}$$

$$\Rightarrow h = \left(\frac{T^2 g R^2}{4\pi^2} \right)^{1/3} - R$$

By knowing the value of time period, the height of the satellite from the earth's surface can be calculated.

Example 10.65 A satellite forms a circle around the earth in 90 min. Determine the height of the satellite above the earth's surface.

Sol. Given, $T = 90 \text{ min} = 5400 \text{ s}$,

$$R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$$

$$\therefore \text{Time period, } T = 2\pi \sqrt{\frac{(R+h)^3}{gR^2}}$$

$$\Rightarrow R+h = \left[\frac{gR^2 T^2}{4\pi^2} \right]^{1/3}$$

$$\Rightarrow R+h = 6668.13 \text{ km}$$

$$\Rightarrow h = 6668.13 - R = 6668.13 - 6400$$

$$\Rightarrow h = 268.13 \text{ km} \approx 268 \text{ km}$$

Total energy of satellite

Assume that a satellite of mass m moving around the earth with velocity v_o in an orbit of radius r due to gravitational pull of the earth.

The potential energy of the satellite,

$$U = -\frac{GMm}{r}$$

The kinetic energy of the satellite,

$$K = \frac{1}{2} m v_o^2 = \frac{1}{2} m \left(\frac{GM}{r} \right)$$

or

$$K = \frac{1}{2} \frac{GMm}{r}$$

The total energy, $E = K + U = -\frac{GMm}{2r}$

$\Rightarrow$

$$E = -\frac{GMm}{2r}$$

This energy is constant and negative, i.e. the system is closed. The farther the satellite from the earth, the greater is its total energy.

Variation of energy with distance can be represented by graphs as shown in figure.

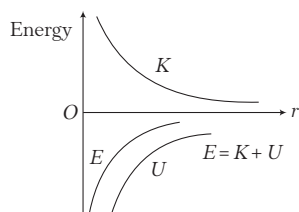


Fig.10.38 Variation of energy of a satellite

Important points to be remembered

- The total mechanical energy of a satellite is negative, if the reference level of zero potential energy is taken when separation is infinity.
- For a satellite $K = -E$, $E = \frac{U}{2}$
and $U = -2K$
- If we consider elliptical path of a satellite, then maximum and minimum energies are as shown in figure.

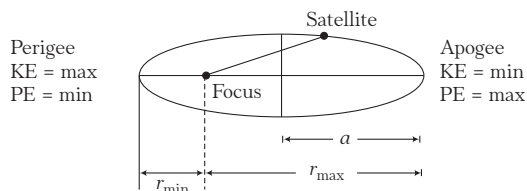


Fig. 10.39 Motion of satellite

Binding energy of a satellite

The energy required by a satellite to leave its orbit around the earth and escape to infinity, is known as binding energy of satellite.

From above, we know that, the total energy of the satellite is $-GMm/2r$. So, if the satellite has to leave the earth's orbit and escape to infinity, then an extra energy equal to $+GMm/2r$ is to be supplied to it, so that its total energy becomes zero.

$$\therefore \text{Binding energy of a satellite} = \frac{GMm}{2r}$$

Note The total energy of the satellite is negative only when it is bound to the earth. It means that its orbit is an ellipse or circle. It is positive when satellite is not bound to any star or its equivalent. In this case, the path covered by satellite will be hyperbola.

Example 10.66 If a spaceship orbits the earth at a height of 500 km from its surface, then determine its (i) kinetic energy, (ii) potential energy, (iii) total energy and (iv) binding energy. (Take, mass of the satellite = 300 kg, mass of the earth = 6×10^{24} kg, radius of the earth = 6.4×10^6 m, $G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$). Will your answer alter, if the earth were to shrink suddenly to half its size?

Sol. Given, height, $h = 500 \text{ km} = 500 \times 10^3 \text{ m}$

Mass of spaceship, $m = 300 \text{ kg}$

Mass of the earth, $M = 6 \times 10^{24} \text{ kg}$

Radius of the earth, $R = 6.4 \times 10^6 \text{ m}$

and $G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$

- Kinetic energy $= \frac{1}{2} mv^2$
 $= \frac{1}{2} m \cdot \frac{GM}{R+h}$ $\left(\because v = \sqrt{\frac{GM}{R+h}} \right)$
 $= \frac{1}{2} \times \frac{300 \times 6.67 \times 10^{-11} \times 6 \times 10^{24}}{6.4 \times 10^6 + 500 \times 10^3}$
 $= 8.7 \times 10^9 \text{ J}$
- Potential energy $= -\frac{GMm}{(R+h)}$
 $= -\frac{6.67 \times 10^{-11} \times 6 \times 10^{24} \times 300}{6.4 \times 10^6 + 500 \times 10^3}$
 $= -17.4 \times 10^9 \text{ J}$
- Total energy $= \text{KE} + \text{PE}$
 $= 8.7 \times 10^9 - 17.4 \times 10^9$
 $= -8.7 \times 10^9 \text{ J}$
- Binding energy $= -(\text{Total energy}) = 8.7 \times 10^9 \text{ J}$

No, there is no change in answer, because $r = R + h$ will remain constant.

ARTIFICIAL SATELLITE

Those man-made objects which revolve around the earth in a fixed orbit are known as artificial satellites.

These satellites are launched from the earth. Artificial satellites are used for telecommunication, weather forecast and other applications. These satellites are of two types : Geo-stationary satellite and polar satellite.

Launching of an artificial satellite around the earth

Let a satellite be projected from point A with velocity v in the direction AB . For different values of v , the paths are different.

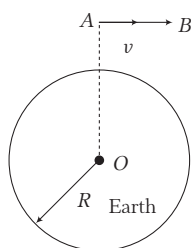


Fig. 10.40 A satellite projected from A

Here, are the possible cases

- (i) If $v = 0$, path is a straight line from A to O .
- (ii) If $0 < v < v_o$, path is an ellipse with centre O of the earth as a focus.
- (iii) If $v = v_o$, path is a circle with O as the centre.
- (iv) If $v_o < v < v_e$, path is again an ellipse with O as a focus.

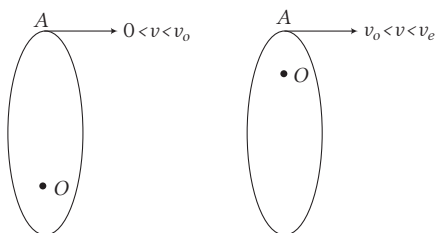


Fig. 10.41 Elliptical orbit with focus at lower side and upper side

- (v) If $v = v_e$, satellite escapes from the gravitational pull of the earth and path is a parabola.
- (vi) If $v > v_e$, satellite again escapes from the gravitational pull of earth but now the path is a hyperbola.

Here, v_o = orbital speed $\left(\sqrt{\frac{GM}{r}} \right)$ at A

and v_e = escape velocity at A.

The figure given below shows the path of an artificial satellite launched from A, with different possible velocities discussed in the above cases.

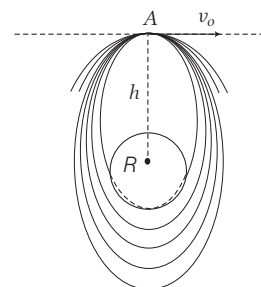


Fig. 10.42 Path of an artificial satellite

Note

- (i) From case (i) to (iv), total energy of the satellite is negative. Hence, these are the closed orbits. For case (v), total energy is zero and for case (vi), total energy is positive. In these two cases, orbits are open.
- (ii) If v is not very large the elliptical orbit will intersect the earth and the satellite will fall back to earth.

Weightlessness in a satellite

A satellite does not produce its own gravity but moves around the earth in a circular orbit under the action of gravity. The acceleration of satellite is GM/r^2 towards the centre of the earth as shown in figure.

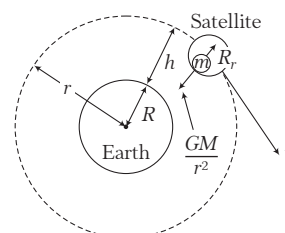


Fig. 10.43 Weightlessness in a satellite

If a body of mass m placed on a surface inside a satellite moving around the earth.

Then, forces on the body are as follows

The gravitational pull of the earth = $\frac{GMm}{r^2}$

The reaction by the surface = R_r

By Newton's law, $\frac{GMm}{r^2} - R_r = ma$

$$\frac{GMm}{r^2} - R_r = m \left(\frac{GM}{r^2} \right)$$

$$\Rightarrow R_r = 0$$

Thus, the surface does not exert any force on the body and hence, its apparent weight is zero.

Note Condition of weightlessness can be experienced only when the mass of satellite is negligible, so that it does not produce its own gravity. e.g. Moon is a satellite of the earth but due to its own weight, it applies gravitational force of attraction on the body placed on its surface and hence, weight of the body will not be equal to zero at the surface of the moon.

Example 10.67 What will be the mass of an astronaut in a space satellite if at earth's surface the astronaut weighs 850 N? (Take, $g = 10 \text{ ms}^{-2}$)

Sol. In a satellite, the astronaut feels weightlessness and therefore, weight becomes zero but mass remains unchanged.

Therefore, mass of astronaut can be calculated by using weight at the surface of the earth.

$$\begin{aligned} \therefore \quad & mg = 850 \text{ N} \\ \Rightarrow \quad & m = \frac{850}{10} \\ \Rightarrow \quad & m = 85 \text{ kg} \end{aligned}$$

CHECK POINT 10.6

- The centripetal force on a satellite orbiting round the earth and the gravitational force of earth acting on the satellite both equals to F . The net force on the satellite is
(a) zero (b) F
(c) $F\sqrt{2}$ (d) $2F$
- The orbital velocity of a body close to the earth's surface is
(a) 8 kms^{-1} (b) 11.2 kms^{-1}
(c) $3 \times 10^8 \text{ ms}^{-1}$ (d) $2.2 \times 10^3 \text{ kms}^{-1}$
- Two satellites A and B moves round a planet P in circular orbits having radii $4R$ and R , respectively. If the velocity of the satellite A is $3v$, the velocity of the satellite B will be
(a) $12v$ (b) $6v$ (c) $\frac{4}{3}v$ (d) $\frac{3}{2}v$
- The orbital velocity of an artificial satellite in a circular orbit just above the earth's surface is v . For a satellite orbiting at an altitude of half the earth's radius, the orbital velocity is
(a) $\frac{3}{2}v$ (b) $\sqrt{\frac{3}{2}}v$ (c) $\sqrt{\frac{2}{3}}v$ (d) $\frac{2}{3}v$
- The period of a satellite in a circular orbit around a planet is independent of
(a) the mass of the planet
(b) the radius of the planet
(c) the mass of the satellite
(d) All the three parameters (a), (b) and (c)
- Which of the following quantities does not depend upon the orbital radius of a satellite? (Take, R = radius of orbit)
(a) $\frac{T}{R}$ (b) $\frac{T^2}{R}$
(c) $\frac{T^2}{R^2}$ (d) $\frac{T^2}{R^3}$
- A satellite is revolving in circular orbit of radius r around the earth of mass M . Time of revolution of satellite is
(a) $T \propto \frac{r^5}{GM}$ (b) $T \propto \sqrt{\frac{r^3}{GM}}$
(c) $T \propto \sqrt{\frac{r}{GM^2}}$ (d) $T \propto \sqrt{\frac{r^3}{GM}}$
- If mean radius of earth is R , its angular velocity is ω and the acceleration due to gravity at the surface of the earth is g , then the cube of the radius of the orbit of geo-stationary satellite will be
(a) $\frac{R^2 g}{\omega^2}$ (b) $\frac{R^2 \omega^2}{g}$ (c) $\frac{R^2 g}{\omega}$ (d) $\frac{Rg}{\omega^2}$
- For a satellite orbiting very close to earth's surface, total energy is
(a) zero (b) $\frac{GMm}{R}$ (c) $-\frac{GMm}{R}$ (d) $-\frac{GMm}{2R}$
- An artificial satellite moving in a circular orbit around the earth has total mechanical energy E_0 . Its kinetic energy is
(a) $-2E_0$ (b) $1.5E_0$ (c) $2E_0$ (d) $-E_0$
- In case of an orbiting satellite, if the radius of orbit is decreased,
(a) its kinetic energy decreases
(b) its potential energy increases
(c) Both (a) and (b) are correct
(d) Both (a) and (b) are incorrect
- Two satellites A and B , ratio of masses $3 : 1$ are in circular orbits of radii r and $4r$. Then, ratio of total mechanical energy of A and B is
(a) $1:3$ (b) $3:1$ (c) $3:4$ (d) $12:1$
- Two satellites P and Q , ratio of masses $3 : 1$ are in circular orbits of radii r and $8r$. Then, ratio of total mechanical energy of A to B is
(a) $1:3$ (b) $3:1$ (c) $3:4$ (d) $24:1$
- What is the energy required to launch a m kg satellite from earth's surface to a circular orbit at an altitude of $7R$? (R = radius of the earth)
(a) $\frac{12}{13}mgR$ (b) mgR (c) $\frac{15}{16}mgR$ (d) $\frac{1}{9}mgR$
- Two identical satellites are orbiting at distances R and $7R$ from the surface of the earth, R being the radius of the earth. The ratio of their
(a) kinetic energies is 4 (b) potential energies is 4
(c) total energies is 4 (d) All of these

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1 Reason of weightlessness in a satellite is
 - (a) zero gravity
 - (b) no atmosphere
 - (c) zero reaction force by satellite surface
 - (d) None of the above
- 2 The acceleration due to gravity near the surface of a planet of radius R and density d is proportional to
 - (a) $\frac{d}{R^2}$
 - (b) dR^2
 - (c) dR
 - (d) $\frac{d}{R}$
- 3 A satellite of the earth is revolving in a circular orbit with a uniform speed v . If gravitational force suddenly disappears, the satellite will
 - (a) continue to move with speed v along the original orbit
 - (b) move with the velocity v tangentially to the original orbit
 - (c) fall downward with increasing velocity
 - (d) ultimately come to rest somewhere on the original orbit
- 4 A body is projected from earth's surface to become its satellite, its time period of revolution will not depend upon
 - (a) mass of earth
 - (b) its own mass
 - (c) gravitational constant
 - (d) radius of orbit
- 5 A planet is revolving round the sun in an elliptical orbit, If $\mathbf{v}$ is the velocity of the planet when its position vector from the sun is $\mathbf{r}$, then areal velocity of the planet is
 - (a) $|\mathbf{v} \times \mathbf{r}|$
 - (b) $2|\mathbf{r} \times \mathbf{v}|$
 - (c) $\left| \frac{1}{2}(\mathbf{r} \times \mathbf{v}) \right|$
 - (d) None of these
- 6 The required kinetic energy of an object of mass m , so that it may escape, will be
 - (a) $\frac{1}{4}mgR$
 - (b) $\frac{1}{2}mgR$
 - (c) mgR
 - (d) $2mgR$
- 7 A planet of mass m is in an elliptical orbit about the sun with an orbital period T . If A be the area of orbit, then its angular momentum would be
 - (a) $\frac{2mA}{T}$
 - (b) mAT
 - (c) $\frac{mA}{2T}$
 - (d) $2mAT$
- 8 In our solar system, the inter-planetary region has chunks of matter (much smaller in size compared to planets) called asteroids. They [NCERT Exemplar]
 - (a) will not move around the sun, since they have very small masses compared to the sun
 - (b) will move in an irregular way because of their small masses and will drift away into outer space
 - (c) will move around the sun in closed orbits but not obey Kepler's laws
 - (d) will move in orbits like planets and obey Kepler's laws
- 9 Two satellites of same mass are launched in the same orbit of radius r around the earth so as to rotate opposite to each other. If they collide inelastically and stick together as wreckage, the total energy of the system just after collision is
 - (a) $-\frac{2GMm}{r}$
 - (b) $-\frac{GMm}{r}$
 - (c) $\frac{GMm}{2r}$
 - (d) zero
- 10 Energy required in moving a body of mass m from a distance $2R$ to $4R$ from centre of earth of mass M is
 - (a) $\frac{GMm}{12R^2}$
 - (b) $\frac{GMm}{3R^2}$
 - (c) $\frac{GMm}{8R}$
 - (d) $\frac{GMm}{4R}$
- 11 Earth orbiting satellite will escape, if
 - (a) its speed is increased by 41%
 - (b) its kinetic energy is doubled
 - (c) Both (a) and (b) are correct
 - (d) Both (a) and (b) are incorrect
- 12 What is the fractional decrease in the value of free fall acceleration g for a particle when it is lifted from the surface to an elevation h ? ($h \ll R$)
 - (a) $\frac{-h}{R}$
 - (b) $\frac{-2h}{R}$
 - (c) $\frac{-3h}{R}$
 - (d) $\frac{-4h}{R}$
- 13 The earth is an approximate sphere. If the interior contained matter which is not of the same density everywhere, then on the surface of the earth, the acceleration due to gravity [NCERT Exemplar]
 - (a) will be directed towards the centre but not the same everywhere
 - (b) will have the same value everywhere but not directed towards the centre
 - (c) will be same everywhere in magnitude directed towards the centre
 - (d) cannot be zero at any point

- 14** Satellites orbiting the earth have a finite life and sometimes debris of satellites fall to the earth. This is because [NCERT Exemplar]

(a) the solar cells and batteries in satellites run out
 (b) the laws of gravitation predict a trajectory spiralling inwards
 (c) of viscous forces causing the speed of satellite and hence, height to gradually decrease
 (d) of collisions with other satellites

- 15** Both the earth and the moon are subject to the gravitational force of the sun. As observed from the sun, the orbit of the moon [NCERT Exemplar]

(a) will be elliptical
 (b) will not be strictly elliptical because the total gravitational force on it is not central
 (c) is not elliptical but will necessarily be a closed curve
 (d) deviates considerably from being elliptical due to influence of planets other than the earth

- 16** As observed from the earth, the sun appears to move in an approximate circular orbit. For the motion of another planet like mercury as observed from the earth, this would [NCERT Exemplar]

(a) be similarly true
 (b) not be true because the force between the earth and mercury is not inverse square law
 (c) not be true because the major gravitational force on mercury is due to the sun
 (d) not be true because mercury is influenced by forces other than gravitational forces

- 17** Different points in the earth are at slightly different distances from the sun and hence, experience different forces due to gravitation.

For a rigid body, we know that, if various forces act at various points in it, the resultant motion is as if a net force acts on the CM (centre of mass) causing translation and a net torque at the CM causing rotation around an axis through the CM. For the earth-sun system (approximating the earth as a uniform density sphere) [NCERT Exemplar]

(a) the torque is zero
 (b) the torque causes the earth to spin
 (c) the rigid body result is not applicable, since the earth is not even approximately a rigid body
 (d) the torque causes the earth to move around the sun

- 18** Suppose the gravitational attraction varies inversely as the distance from the earth. The orbital velocity of a satellite in such a case varies as n th power of distance, where n is equal to

(a) -1 (b) zero (c) $+1$ (d) $+2$

- 19** A body attains a height equal to the radius of the earth. The velocity of the body with which it was projected is

$$\begin{array}{ll} \text{(a)} \sqrt{\frac{GM}{R}} & \text{(b)} \sqrt{\frac{2GM}{R}} \\ \text{(c)} \sqrt{\frac{1}{4} \frac{GM}{R}} & \text{(d)} \sqrt{\frac{GM}{2R}} \end{array}$$

- 20** Compute the additional velocity required by a satellite orbiting around earth with radius $2R$ to become free from earth's gravitational field. Mass of earth is M .

$$\begin{array}{ll} \text{(a)} \sqrt{\frac{2GM}{R}}(\sqrt{2}-1) & \text{(b)} \sqrt{\frac{GM}{2R}}(\sqrt{2}-1) \\ \text{(c)} \sqrt{\frac{GM}{R}}(\sqrt{3}-1) & \text{(d)} \sqrt{\frac{GM}{R}}(\sqrt{2}+1) \end{array}$$

- 21** The rotation of the earth about its axis speeds up such that a man on the equator becomes weightless. In such a situation, what would be the duration of one day?

$$\begin{array}{llll} \text{(a)} 2\pi\sqrt{\frac{R}{g}} & \text{(b)} \frac{1}{2\pi}\sqrt{\frac{R}{g}} & \text{(c)} 2\pi\sqrt{Rg} & \text{(d)} \frac{1}{2\pi}\sqrt{Rg} \end{array}$$

- 22** The rotation of the earth having radius R about its axis speeds up to a value such that a man at latitude angle 60° feels weightlessness. The duration of the day in such a case is

$$\begin{array}{llll} \text{(a)} \pi\sqrt{\frac{R}{g}} & \text{(b)} \frac{\pi}{2}\sqrt{\frac{R}{g}} & \text{(c)} \frac{\pi}{3}\sqrt{\frac{R}{g}} & \text{(d)} \pi\sqrt{\frac{g}{R}} \end{array}$$

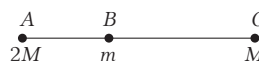
- 23** An earth's satellite of mass m revolves in a circular orbit at a height h from the surface of the earth. R is the radius of the earth and g is acceleration due to gravity at the surface of the earth. The velocity of the satellite in the orbit is given by

$$\begin{array}{llll} \text{(a)} \frac{gR^2}{R+h} & \text{(b)} gR & \text{(c)} \frac{gR}{R+h} & \text{(d)} \sqrt{\frac{gR^2}{R+h}} \end{array}$$

- 24** A planet of mass m moves around the sun of mass M in an elliptical orbit. The maximum and minimum distance of the planet from the sun are r_1 and r_2 , respectively. The time period of the planet is proportional to

$$\begin{array}{llll} \text{(a)} r_1^{3/2} & \text{(b)} (r_1 + r_2)^{3/2} & \text{(c)} (r_1 - r_2)^{3/2} & \text{(d)} r_1^{3/2} \end{array}$$

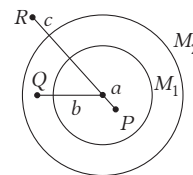
- 25** Particles of masses $2M$, m and M are respectively at points A , B and C with $AB = 1/2(BC)$. m is much-much smaller than M and at time $t = 0$, they are all at rest as given in figure.



At subsequent times before any collision takes place

- (a) m will remain at rest [NCERT Exemplar]
 (b) m will move towards M
 (c) m will move towards $2M$
 (d) m will have oscillatory motion

- 26** Two particles of equal mass m go round a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is
 (a) $v = \sqrt{\frac{Gm}{R}}$ (b) $v = \sqrt{\frac{Gm}{2R}}$ (c) $v = \frac{1}{2} \sqrt{\frac{Gm}{R}}$ (d) $v = \sqrt{\frac{4Gm}{R}}$
- 27** Suppose a smooth tunnel is dug along a straight line joining two points on the surface of the earth and a particle is dropped from rest at its one end. Assume that mass of earth is uniformly distributed over its volume. Then,
 (a) the particle will emerge from the other end with velocity $\sqrt{\frac{GM_e}{2R_e}}$, where M_e and R_e are earth's mass and radius, respectively
 (b) the overlapping will come to rest at centre of the tunnel
 (c) potential energy of the particle will be equal to zero at centre of tunnel, if it is along a diameter
 (d) acceleration of the particle will be proportional to its distance from mid-point of the tunnel.
- 28** A satellite is moving in a circular orbit round the earth with a diameter of orbit $2R$. At a certain point a rocket fixed to the satellite is fired such that it increases the velocity of the satellite tangentially. The resulting orbit of the satellite would be
 (a) same as before
 (b) circular orbit with diameter greater than $2R$
 (c) elliptical orbit with minimum distance from the earth equal to R
 (d) elliptical orbit with maximum distance from the earth equal to R
- 29** Assuming the radius of the earth to be 6.4×10^6 m. What is the time period T and speed of satellite for equatorial orbit at 1.4×10^3 km above the surface of the earth?
 (a) 6842 s and 7163 ms^{-1} (b) 34155 s and 3204 ms^{-1}
 (c) 6831 s and 2144 ms^{-1} (d) 2431 s and 3514 ms^{-1}
- 30** A body which is initially at rest at a height R above the surface of the earth of radius R , falls freely towards the earth, then its velocity on reaching the surface of the earth is
 (a) $\sqrt{2gR}$ (b) $\sqrt{gR}$ (c) $\sqrt{\frac{3}{2}gR}$ (d) $\sqrt{4gR}$
- 31** A rocket is launched vertical from the surface of the earth of radius R with an initial speed v . If atmospheric resistance is neglected, then maximum height attained by the rocket is
 (a) $h = \frac{R}{\left(\frac{2gR}{v^2} - 1\right)}$ (b) $h = \frac{R}{\left(\frac{2gR}{v^2} + 1\right)}$
 (c) $h = \frac{R^2}{\left(\frac{2gR}{v^2} - 1\right)}$ (d) $h = \frac{R^2}{\left(\frac{2gR}{v^2} + 1\right)}$
- 32** If the mass of moon is $\frac{M}{81}$, where M is the mass of earth, find the distance of the point, where gravitational field due to earth and moon cancel each other, from the centre of moon. Given that distance between centres of earth and moon is $60R$, where R is the radius of earth.
 (a) $4R$ (b) $8R$ (c) $12R$ (d) $6R$
- 33** The orbital angular momentum of a satellite revolving at a distance r from the centre is L . If the distance is increased to $16r$, then the new angular momentum will be
 (a) $16L$ (b) $64L$ (c) $\frac{L}{4}$ (d) $4L$
- 34** Two spheres of masses m and $2m$ are separated by distance d . A particle of mass $\frac{m}{5}$ is projected straight from $2m$ towards m with a velocity v_0 . Which of the following statements is correct?
 (a) Velocity of the particle decreases constantly.
 (b) Velocity of the particle increase constantly.
 (c) Acceleration of the particle may become momentarily zero.
 (d) The particle never retraces its path.
- 35** A ring of mass m_1 and radius R is fixed in space at some location. An external agent brings a point mass m_2 from infinity to centre of the ring. Work done by the external agent will be
 (a) $-\frac{Gm_1m_2}{R}$ (b) $\frac{Gm_1m_2}{R}$
 (c) $\frac{G\sqrt{m_1^2 + m_2^2}}{R}$ (d) $\frac{Gm_1m_2}{R(m_1^2 + m_2^2)}$
- 36** If an artificial satellite is moving in a circular orbit around the earth with a speed equal to half the magnitude of the escape velocity from the earth, the height of the satellite above the surface of the earth is
 (a) $2R$ (b) $\frac{R}{2}$ (c) R (d) $\frac{R}{4}$
- 37** Two concentric shells of masses M_1 and M_2 are concentric as shown in the figure. Which of the following is correct regarding the gravitational force on m due to M_1 and M_2 at points P, Q and R .



- (a) $F_P = 0$ (b) $F_Q = \frac{GM_1m}{b^2}$
 (c) $F_R = \frac{G(M_1 + M_2)m}{c^2}$ (d) All are correct

- 38** Energy of a satellite in circular orbit is E_0 . The energy required to move the satellite in a circular orbit of 3 times the radius of the initial orbit is

(a) $\frac{2}{3}E_0$ (b) $2E_0$ (c) $\frac{E_0}{3}$ (d) $\frac{3}{2}E_0$

- 39** What is the energy required to launch a m kg satellite from earth's surface in a circular orbit at an altitude of $2R$? (R = radius of the earth)

(a) $\frac{2}{3}mgR$ (b) mgR (c) $\frac{5}{6}mgR$ (d) $\frac{1}{3}mgR$

- 40** A person brings a mass of 1 kg from infinity to a point A. Initially, the mass was at rest but it moves at a speed of 2 ms^{-1} as it reached at A. The work done by the person on the mass is -3 J . The potential at A is

(a) -3 J kg^{-1} (b) -12 J kg^{-1}
(c) -5 J kg^{-1} (d) None of these

- 41** If gravitational attraction between two point masses

be given by $F = G \frac{m_1 m_2}{r^3}$. Then, the period of a

satellite in a circular orbit will be proportional to

(a) r (b) r^2
(c) $\frac{1}{r^2}$ (d) independent of r

- 42** Suppose the gravitational force varies inversely as the n th power of distance. Then, the time period of a planet in circular orbit of radius r around the sun will be proportional to

(a) $r^{\frac{1}{2}(n+1)}$ (b) $r^{\frac{1}{2}(n-1)}$ (c) r^n (d) $r^{\frac{1}{2}(n-2)}$

- 43** A body is projected vertically upwards from the surface of earth with a velocity equal to half the escape velocity. If R be the radius of earth, maximum height attained by the body from the surface of earth is

(a) $\frac{R}{6}$ (b) $\frac{R}{3}$ (c) $\frac{2R}{3}$ (d) R

- 44** What impulse need to be given to a body of mass m , released from the surface of earth along a straight tunnel passing through centre of earth, at the centre of earth, to bring it to rest? (Mass of earth M , radius of earth R)

(a) $m \sqrt{\frac{GM}{R}}$ (b) $\sqrt{\frac{GMm}{R}}$ (c) $m \sqrt{\frac{GM}{2R}}$ (d) Zero

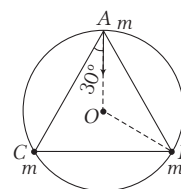
- 45** Four equal masses (each of mass M) are placed at the corners of a square of side a . The escape velocity of a body from the centre O of the square is

(a) $4\sqrt{\frac{2GM}{a}}$ (b) $\sqrt{\frac{8\sqrt{2}GM}{a}}$
(c) $\frac{4GM}{a}$ (d) $\sqrt{\frac{4\sqrt{2}GM}{a}}$

- 46** Let E be the energy required to raise a satellite to height h above earth's surface and E' be the energy required to put the same satellite into orbit at that height. Then, E/E' is equal to

(a) $\frac{2h}{(R+2h)}$ (b) $\frac{2h}{(2R+3h)}$ (c) $\frac{R}{R+h}$ (d) $\frac{2R}{2h+R}$

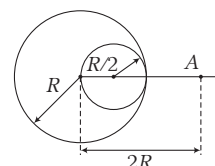
- 47** Three particles each of mass m are located at the vertices of an equilateral triangle of side a . At what speed must they move, if they all revolve under the influence of their gravitational force of attraction in a circular orbit circumscribing the triangle while still preserving the equilateral triangle?



(a) $\sqrt{\frac{Gm}{a}}$ (b) $\sqrt{\frac{2Gm}{a}}$ (c) $\sqrt{\frac{3Gm}{a}}$ (d) $\sqrt{\frac{4Gm}{a}}$

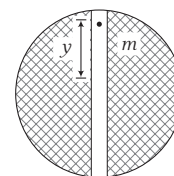
- 48** A solid sphere of uniform density and radius R applies a gravitational force of attraction equal to F_1 on a particle placed at a distance $2R$ from the centre of the sphere. A spherical cavity of radius $R/2$ is now made in the sphere as shown in the figure. The sphere with cavity now applies a gravitational force

F_2 on the same particle. The ratio $\frac{F_2}{F_1}$ is



(a) $\frac{5}{9}$ (b) $\frac{7}{8}$ (c) $\frac{3}{4}$ (d) $\frac{7}{9}$

- 49** Suppose a vertical tunnel is along the diameter of earth (here earth is assumed to be a sphere of uniform mass density ρ). If a body of mass m is thrown in this tunnel, its acceleration at a distance y from the centre is given by



(a) $\frac{4\pi}{3} G \rho y m$ (b) $\frac{3}{4} \pi \rho y$
(c) $\frac{4}{3} \pi \rho y$ (d) $\frac{4}{3} \pi G \rho y$

- 50** A solid sphere of mass M and radius R has a spherical cavity of radius $R/2$ such that the centre of cavity is at a distance $R/2$ from the centre of the sphere. A point mass m is placed inside the cavity at a distance $R/4$ from the centre of sphere. The gravitational force on mass m is
- (a) $\frac{11GMm}{R^2}$ (b) $\frac{14GMm}{R^2}$ (c) $\frac{GMm}{2R^2}$ (d) $\frac{GMm}{R^2}$
- 51** Two particles of masses m and M are initially at rest at infinite distance. Find their relative velocity of approach due to gravitational attraction when d is their separation at any instant.
- (a) $\sqrt{\frac{2G(M+m)}{d}}$ (b) $\sqrt{\frac{G(M+m)}{d}}$
 (c) $\sqrt{\frac{G(M+m)}{2d}}$ (d) $\sqrt{\frac{G(M+m)}{4d}}$
- 52** A satellite is revolving round the earth with orbital speed v_o . If it stops suddenly, the speed with which it will strike the surface of earth would be (v_e = escape velocity of a particle on earth's surface)
- (a) $\frac{v_e^2}{v_o}$ (b) $2v_o$ (c) $\sqrt{v_e^2 - v_o^2}$ (d) $\sqrt{v_e^2 - 2v_o^2}$
- 53** Three point masses each of mass m rotate in a circle of radius r with constant angular velocity ω due to their mutual gravitational attraction. If at any instant, the masses are on the vertex of an equilateral triangle of side a , then the value of ω is
- (a) $\sqrt{\frac{Gm}{a^3}}$ (b) $\sqrt{\frac{3Gm}{a^3}}$
 (c) $\sqrt{\frac{Gm}{3a^3}}$ (d) None of these
- 54** Pertaining to two planets, the ratio of escape velocities from respective surfaces is 1:2, the ratio of the time period of the same simple pendulum at their respective surfaces is 2:1 (in same order). Then the ratio of their average densities is
- (a) 1:1 (b) 1:2 (c) 1:4 (d) 8:1
- 55** The ratio of energy required to raise a satellite to a height h above the earth's surface to that required to put it into the orbit is
- (a) $h:2R$ (b) $2h:R$
 (c) $R:h$ (d) $h:R$
- 56** A body of supercondense material with mass twice the mass of earth but size very small compared to the size of earth starts from rest from $h \ll R$ above the earth's surface. It reaches earth in time
- (a) $t = \sqrt{\frac{h}{g}}$ (b) $t = \sqrt{\frac{2h}{g}}$ (c) $t = \sqrt{\frac{2h}{3g}}$ (d) $t = \sqrt{\frac{4h}{3g}}$
- 57** Four particles each of mass M , move along a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is
- (a) $\frac{GM}{R}$ (b) $\sqrt{2\sqrt{2}} \frac{GM}{R}$
 (c) $\sqrt{\frac{GM}{R} (2\sqrt{2} + 1)}$ (d) $\sqrt{\frac{GM}{R} \left(\frac{2\sqrt{2} + 1}{4} \right)}$
- 58** An object is released from a height twice the radius of the earth on the surface of earth. Find the speed with which it will collide with ground by neglecting effect of air. (Take, R as radius of earth and M as mass of earth)
- (a) $2\sqrt{\frac{GM}{3R}}$ (b) $3\sqrt{\frac{GM}{2R}}$
 (c) $2\sqrt{\frac{GM}{R}}$ (d) $3\sqrt{\frac{GM}{R}}$
- 59** A planet of mass m revolves in an elliptical orbit around the sun, so that its maximum and minimum distance from the sun are equal to r_a and r_p , respectively. The angular momentum of this planet relative to the sun is
- (a) $m \sqrt{\frac{2GMr_p r_a}{(r_p + r_a)}}$ (b) $m \sqrt{\frac{4GMr_p r_a}{(r_p + r_a)}}$
 (c) $m \sqrt{\frac{GMr_p r_a}{(r_p + r_a)}}$ (d) $m \sqrt{\frac{GMr_p r_a}{2(r_p + r_a)}}$
- 60** The magnitudes of the gravitational force at distances r_1 and r_2 from the centre of a uniform sphere of radius R and mass M are F_1 and F_2 , respectively. Then,
- (a) $\frac{F_1}{F_2} = \frac{r_1}{r_2}$; if $r_1 < R$ and $r_2 < R$
 (b) $\frac{F_1}{F_2} = \frac{r_1^2}{r_2^2}$; if $r_1 > R$ and $r_2 > R$
 (c) $\frac{F_1}{F_2} = \frac{r_2}{r_1}$; if $r_1 < R$ and $r_2 < R$
 (d) $\frac{F_1}{F_2} = \frac{r_2^2}{r_1^2}$; if $r_1 > R$ and $r_2 < R$
- 61** Two identical thin rings each of radius R are coaxially placed at a distance R . If the rings have a uniform mass distribution and each has masses $2m$ and $4m$ respectively, then the work done in moving a mass m from centre of one ring to that of the other is
- (a) zero (b) $\frac{\sqrt{2}Gm^2}{R} (1 - \sqrt{2})$
 (c) $\frac{Gm^2}{\sqrt{2}R} (\sqrt{2} - 1)$ (d) $\frac{Gm^2}{\sqrt{2}R}$

- 62** A mass m is at a distance a from centre of a uniform rod of length l and M . The gravitational force on the mass due to the rod is

(a) $\frac{4GMm}{(a+l)^2}$ (b) $\frac{4GmM}{4a^2 - l^2}$
 (c) $\frac{GMm}{a^2}$ (d) $\frac{GmM}{2(l+a)^2}$

- 63** A particle would take a time t to move down a straight tube from the surface of earth (supposed to be a homogeneous sphere) to its centre. If gravity were to remain constant, then the time would be t' . The ratio of t/t' will be

(a) $\frac{\pi}{2\sqrt{2}}$ (b) $\frac{\pi}{\sqrt{2}}$ (c) $\sqrt{2}\pi$ (d) $\frac{\pi}{\sqrt{2}}$

(B) Medical entrance special format questions

Assertion and reason

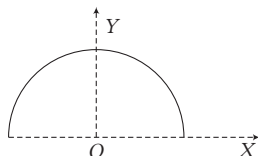
Directions (Q. Nos. 1-5) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) If Assertion is correct but Reason is correct.
 (d) If Assertion is incorrect but Reason is correct.

- 1 Assertion** Gravitational force between two masses in air is F . If they are immersed in water, force will remain F .

Reason Gravitational force does not depend on the medium between the masses.

- 2 Assertion** The centre of semicircular ring of mass m and radius R is the origin O . The potential at origin is $-\frac{Gm}{R}$.



Reason The gravitational field strength is towards Y -axis.

- 3 Assertion** Areal velocity of a planet around sun will become two times, if mass of planet is halved.

Reason Areal velocity $= \frac{L}{2m}$, where L is angular momentum of planet about centre of sun.

- 4 Assertion** If gravitational potential at some point is zero, then gravitational field strength at that point will also be zero.

Reason Except at infinity gravitational potential due to a system of point masses at some finite distance cannot be zero.

- 5 Assertion** If radius of earth suddenly shrinks to half its present value without changing its mass, then the period of an earth's satellite will not change.

Reason Time period of a satellite does not depend upon the mass of earth.

Statement based questions

- 1** Which of the following statement is correct?
 (Take, earth as a sphere of uniform density)
 (a) Acceleration due to gravity increases with altitude.
 (b) Acceleration due to gravity decreases with latitude.
 (c) Acceleration due to gravity depends on the mass of the object.
 (d) Acceleration due to gravity decreases with increasing depth.
- 2** When the radius and mass of earth are increased by 0.5%. Which of the following statement is correct at the surface of the earth?
 (a) g will decrease.
 (b) g will increase.
 (c) Escape velocity varies by 15%.
 (d) Potential energy varies by 19.99%.
- 3** Which one of the following statement is correct?
 (a) A geo-stationary satellite goes around the earth in east-west direction.
 (b) A geo-stationary satellite goes around the earth in west-east direction.
 (c) The first half of the rotation of geo-stationary satellite goes around the earth in north-south direction and then in south-north direction.
 (d) The first half of the rotation of geo-stationary satellite goes around the earth in east-west direction, then in west direction.
- 4** Which one of the following statement is in correct?
 (a) Inertial mass is a measure of difficulty of accelerating a body by an external force, whereas the gravitational mass is relevant in determining the gravitational force on it by an external mass.
 (b) According to an experimental result, gravitational mass and inertial mass are equal.
 (c) Acceleration due to gravity on the earth is the same for all bodies is due to equality of gravitational mass and inertial mass.
 (d) Gravitational mass of a particle like proton can depend on the presence of neighbouring heavy objects but the inertial mass cannot.

- 5 According to Newton's law of gravitation, the magnitude of the force F on point masses m_1 and m_2 (say) is given by $|F| = \frac{Gm_1m_2}{r^2}$.

- I. This equation is not directly applicable for the gravitational force between extended object (like the earth) and a point mass.
 II. This equation can be applied to find the gravitational force between an extended object and a point mass provided r is the distance of the point mass from the geometric centre of the extended objects.

Which of the following statement(s) about this equation is/are correct?

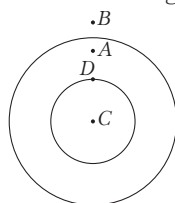
- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I and II
- 6 I. The geo-stationary satellite always has circular orbit.
 II. The time period of satellite depends on mass of the satellite, mass of the earth and height of satellite from the surface of earth.

Which of the following statement(s) is/are incorrect?

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

Match the columns

- 1 Two concentric spherical shells as shown in figure. Match the Column I with Column II and mark the correct option from the codes given below.



Column I	Column II
(A) Potential at A	(p) greater than B
(B) Gravitational field at A	(q) less than B
(C) As one moves from C to D	(r) potential remains constant
(D) As one moves from D to A	(s) gravitational field decreases
	(t) None

Codes

A	B	C	D	A	B	C	D
(a) p	r	q	s	(b) p	t	r	s
(c) q	t	r	s	(d) q	s	t	p

- 2 On the surface of earth acceleration due to gravity is g and gravitational potential is V . Match the Column I with Column II and mark the correct option from the codes given below.

Column I	Column II
(A) At height $h = R$, value of g	(p) decreases by a factor $\frac{1}{4}$
(B) At depth $h = \frac{R}{2}$, value of g	(q) decrease by a factor $\frac{1}{2}$
(C) At height $h = R$, value of V	(r) decrease by a factor $\frac{11}{8}$
(D) At depth $h = \frac{R}{2}$, value of V	(s) increases by a factor 2
	(t) None

Codes

A	B	C	D
(a) p	t	s	r
(b) q	t	q	r
(c) q	r	p	s
(d) p	q	q	r

- 3 In elliptical orbit of a planet, as the planet moves from apogee position to perigee position, match the Column I with Column II and mark the correct option from the codes given below.

Column I	Column II
(A) Speed of planet	(p) remains same
(B) Distance of planet from centre of sun	(q) decreases
(C) Potential energy	(r) increases
(D) Angular momentum about centre of sun	(s) cannot say

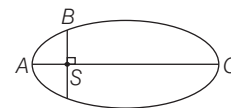
Codes

A	B	C	D
(a) p	r	q	s
(b) p	s	q	r
(c) r	q	q	p
(d) r	p	q	s

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1 A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half of radius of the earth? [NEET 2020]
(a) 32 N (b) 30 N (c) 24 N (d) 48 N
- 2 What is the depth at which the value of acceleration due to gravity becomes $1/n$ times the value that the surface of earth? (Radius of earth = R) [NEET 2020]
(a) R/n^2 (b) $R(n-1)/n$
(c) $Rn/(n-1)$ (d) R/n
- 3 A body weighs 200 N on the surface of the earth. How much will it weigh half way down to the centre of the earth? [NEET 2019]
(a) 200 N (b) 250 N
(c) 100 N (d) 150 N
- 4 The work done to raise a mass m from the surface of the earth to a height h , which is equal to the radius of the earth, is [NEET 2019]
(a) $2mgR$ (b) $\frac{1}{2}mgR$
(c) $\frac{3}{2}mgR$ (d) mgR
- 5 The time period of a geo-stationary satellite is 24 h, at a height $6R_E$ (R_E is the radius of earth) from surface of earth. The time period of another satellite whose height is $2.5 R_E$ from surface will be [NEET (Odisha) 2019]
(a) $6\sqrt{2}$ h (b) $12\sqrt{2}$ h (c) $\frac{24}{2.5}$ h (d) $\frac{12}{2.5}$ h
- 6 Find gravitational field at a distance of 2000 km from the centre of earth. (Take, $R_{\text{earth}} = 6400$ km, $r = 2000$ km, $M_{\text{earth}} = 6 \times 10^{24}$ kg) [AIIMS 2019]
(a) 1.53 ms^{-2} (b) 7.12 ms^{-2}
(c) 3.06 ms^{-2} (d) 1.8 ms^{-2}
- 7 If escape velocity on earth surface is 11.1 kmh^{-1} , then find the escape velocity on moon surface. If mass of moon is $\frac{1}{81}$ times of mass of earth and radius of moon is $\frac{1}{4}$ times radius of earth. [JIPMER 2019]
(a) 2.46 kmh^{-1} (b) 3.46 kmh^{-1}
(c) 4.4 kmh^{-1} (d) None of these
- 8 The kinetic energies of a planet in an elliptical orbit about the sun at positions A, B and C are K_A , K_B and K_C , respectively. AC is the major axis and SB is perpendicular to AC at the position of the sun S as shown in the figure. Then, [NEET 2018]



- (a) $K_B < K_A < K_C$ (b) $K_A > K_B > K_C$
(c) $K_A < K_B < K_C$ (d) $K_B > K_A > K_C$
- 9 Two satellites A and B revolve round the same planet in coplanar circular orbits lying in the same plane. Their periods of revolutions are 1 h and 8 h, respectively. The radius of the orbit of A is 10^4 km. The speed of B relative to A, when they are close in km/h is [AIIMS 2018]
(a) $3\pi \times 10^4$ (b) zero (c) $2\pi \times 10^4$ (d) $\pi \times 10^4$
- 10 A planet is revolving around the sun in a circular orbit with a radius r . The time period is T . If the force between the planet and star is proportional to $r^{-3/2}$, then the square of time period is proportional to [AIIMS 2018]
(a) $r^{3/2}$ (b) r^2 (c) r (d) $r^{5/2}$
- 11 If R is the radius of earth, ω is its angular velocity and g_p is the value of g at the poles. The effective value of g at a latitude $\lambda = 60^\circ$ is [JIPMER 2018]
(a) $g_p - \frac{1}{4}R\omega^2$ (b) $g_p + \frac{1}{4}R\omega^2$
(c) $g_p - \frac{1}{2}R\omega^2$ (d) $g_p + \frac{1}{2}R\omega^2$
- 12 At what distance (in metre) from the centre of the moon, the intensity of gravitational field will be zero? (Take, mass of earth and moon as 5.98×10^{24} kg and 7.35×10^{22} kg respectively and the distance between moon and earth is 3.855×10^8 m.) [JIPMER 2018]
(a) zero (b) 3.85×10^7 (c) 8×10^8 (d) 3.46×10^8
- 13 Two astronauts are floating in gravitation free space after having lost contact with their spaceship. The two will [NEET 2017]
(a) keep floating at the same distance between them
(b) move towards each other
(c) move away from each other
(d) will become stationary
- 14 The acceleration due to gravity at a height 1 km above the earth is the same as at a depth d below the surface of earth. Then, [NEET 2017]
(a) $d = \frac{1}{2}$ km (b) $d = 1$ km
(c) $d = \frac{3}{2}$ km (d) $d = 2$ km

- 15** A spaceship is launched into a circular orbit close to earth's surface. What additional velocity has now to be imparted to the spaceship in the orbit to overcome the gravitational pull?

(Take, radius of earth = 6400 km and $g = 9.8 \text{ ms}^{-2}$)

[AIIMS 2017]

- (a) 3.28 kms^{-1} (b) 12 kms^{-1}
(c) 10 kms^{-1} (d) 40 kms^{-1}

- 16** What is the maximum height attained by a body projected with a velocity equal to one-third of the escape velocity from the surface of the earth?

(Radius of the earth = R)

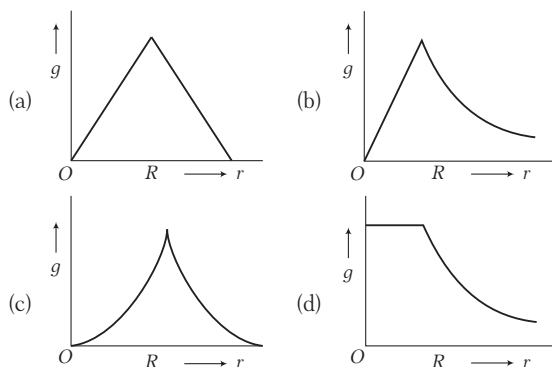
[AIIMS 2017]

- (a) $R/2$ (b) $R/3$ (c) $R/5$ (d) $R/8$

- 17** The planets with radii R_1 and R_2 have densities ρ_1 and ρ_2 , respectively. Their atmospheric pressures are p_1 and p_2 , respectively. Therefore, the ratio of masses of their atmospheres, neglecting variation of g within the limits of atmosphere is [JIPMER 2017]

- (a) $\rho_1 R_2 p_1 / \rho_2 R_1 p_2$ (b) $\rho_1 R_2 p_2 / \rho_2 R_1 p_1$
(c) $\rho_1 R_1 p_1 / \rho_2 R_2 p_2$ (d) $\rho_1 R_1 p_2 / \rho_2 R_2 p_1$

- 18** Starting from the centre of the earth having radius R , the variation of g (acceleration due to gravity) is shown by [NEET 2016]



- 19** A satellite of mass m is orbiting the earth (of radius R) at a height h from its surface. The total energy of the satellite in terms of g_0 , the value of acceleration due to gravity at the earth's surface is [NEET 2016]

- (a) $\frac{mg_0 R^2}{2(R+h)}$ (b) $-\frac{mg_0 R^2}{2(R+h)}$ (c) $\frac{2mg_0 R^2}{R+h}$ (d) $-\frac{2mg_0 R^2}{R+h}$

- 20** At what height from the surface of earth, the gravitation potential and the value of g are $-5.4 \times 10^7 \text{ J kg}^{-2}$ and 6 ms^{-2} , respectively? (Take, the radius of earth is 6400 km) [NEET 2016]

- (a) 1600 km (b) 1400 km (c) 2000 km (d) 2600 km

- 21** The ratio of escape velocity at earth (v_e) to the escape velocity at a planet (v_p) whose radius and mean density are twice as that of earth is [NEET 2016]

- (a) $1:2\sqrt{2}$ (b) $1:4$ (c) $1:\sqrt{2}$ (d) $1:2$

- 22** Kepler's third law states that square of period of revolution (T) of a planet around the sun is proportional to third power of average distance r between the sun and planet, i.e. $T^2 = Kr^3$, here K is constant. If the masses of the sun and planet are M and m respectively, then as per Newton's law of gravitational force of attraction between them is $F = \frac{GMm}{r^2}$, here G is gravitational constant. The relation between G and K is described as [CBSE AIPMT 2015]

- (a) $GK = 4\pi^2$ (b) $GK = 4\pi^2$ (c) $K = G$ (d) $K = I/G$

- 23** Two spherical bodies of masses M and $5M$ and radii R and $2R$ are released in free space with initial separation between their centres equal to $12R$. If they attract each other due to gravitational force only, then the distance covered by the smaller body before collision is [CBSE AIPMT 2015]

- (a) $2.5R$ (b) $4.5R$ (c) $7.5R$ (d) $1.5R$

- 24** The reading of a spring balance corresponds to 100 N while situated at the north pole and a body is kept on it. The weight record on the same scale, if it is shifted to the equator, is (Take, $g = 10 \text{ ms}^{-2}$ and radius of the earth, $R = 6.4 \times 10^6 \text{ m}$) [AIIMS 2015]

- (a) 99.66 N (b) 110 N (c) 97.66 N (d) 106 N

- 25** The gravitational field due to a uniform solid sphere of mass M and radius a at the centre of the sphere is [UK PMT 2015]

- (a) $\frac{GM}{a^2}$ (b) $\frac{GM}{a^3}$ (c) $\frac{GM}{2a}$ (d) zero

- 26** What would be the value of acceleration due to gravity at a point 5 km below the earth's surface? (Take, $R_E = 6400 \text{ km}$, $g_E = 9.8 \text{ ms}^{-2}$) [UK PMT 2015]

- (a) 9.6 ms^{-2} (b) 9.79 ms^{-2} (c) 9.89 ms^{-2} (d) 10 ms^{-2}

- 27** Two particles of equal masses go round a circle of radius R under the action of their mutual gravitational attraction. What would be the speed of each particle? [UK PMT 2015]

- (a) $\sqrt{\frac{GM}{2R}}$ (b) $\sqrt{\frac{GM}{R}}$ (c) $\sqrt{\frac{GM}{4R}}$ (d) $\sqrt{\frac{2GM}{R}}$

- 28** What would be the escape velocity from the moon, if the mass of the moon is $7.4 \times 10^{22} \text{ kg}$ and its radius is 1740 km? [UK PMT 2015]

- (a) 2.4 ms^{-1} (b) 2.4 kms^{-1} (c) 240 kms^{-1} (d) 0.24 kms^{-1}

- 29** Two spheres of masses 16 kg and 4 kg are separated by a distance 30 m on a table. Then, the distance from sphere of mass 16 kg at which the net gravitational force becomes zero is [EAMCET 2015]

- (a) 10 m (b) 20 m (c) 15 m (d) 5 m

- 30** Orbital velocity of earth satellite does not depend on [Kerala CEE 2015]

(a) mass of the earth
(b) mass of the satellite
(c) radius of the earth
(d) acceleration due to gravity

- 31** Gravitational potential energy of a body of mass m at a height of h above the surface of earth (M = mass of earth R = radius of earth) is [Kerala CEE 2015]

(a) $\frac{GMm}{h}$ (b) $\frac{GMm}{(R+h)}$
(c) $-\frac{GM}{(R+h)}$ (d) $-\frac{GMm}{(R+h)}$

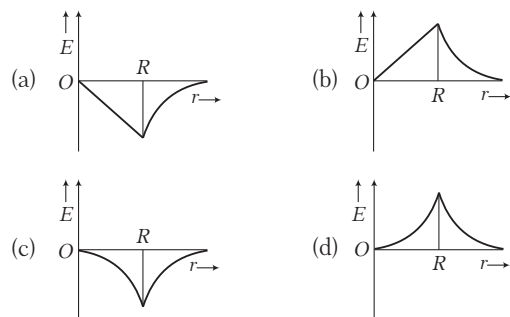
- 32** According to Kepler's law of planetary motion, if T represents time period and r is orbital radius, then for two planets these are related as [Manipal 2015]

(a) $\left(\frac{T_1}{T_2}\right)^{3/2} = \frac{r_1}{r_2}$ (b) $\frac{T_1}{T_2} = \left[\frac{r_1}{r_2}\right]$
(c) $\left[\frac{T_1}{T_2}\right]^2 = \left[\frac{r_1}{r_2}\right]^3$ (d) $\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$

- 33** If the mass of a body is M on the surface of the earth, then mass of the same body on the surface of the moon is [KCET 2015]

(a) $6M$ (b) $\frac{M}{6}$
(c) zero (d) M

- 34** Dependence of intensity of gravitational field (E) of earth with distance (r) from centre of earth is correctly represented by [CBSE AIPMT 2014]



- 35** The force of gravitation is [UK PMT 2014]

(a) repulsive (b) electrostatic
(c) conservative (d) non-conservative

- 36** Keeping the mass of the earth as constant, if its radius is reduced to $1/4$ th of its initial value, then the period of revolution of the earth about its own axis and passing through the centre (in hours) is (assume the earth to be a solid sphere and its initial period of rotation as 24 h) [EAMCET 2014]

(a) 12 (b) 3
(c) 6 (d) 1.5

- 37** A body of mass m is raised to a height $10R$ from the surface of the earth, where R is the radius of the earth. The increase in potential energy is (G = universal constant of gravitation, M = mass of the earth and g = acceleration due to gravity) [MHT CET 2014]

(a) $\frac{GMm}{11R}$ (b) $\frac{GMm}{10R}$ (c) $\frac{mgR}{11G}$ (d) $\frac{10GMm}{11R}$

- 38** An artificial satellite moves in a circular orbit around the earth. Total energy of the satellite is given by E . The potential energy of the satellite is [WB JEE 2014]

(a) $-2E$ (b) $2E$ (c) $\frac{2E}{3}$ (d) $-\frac{2E}{3}$

- 39** What is a period of revolution of the earth satellite? Ignore the height of satellite above the surface of the earth. Given,

(i) the value of gravitational acceleration, $g = 10 \text{ ms}^{-2}$.

(ii) radius of the earth, $R_e = 6400 \text{ km}$

(Take, $\pi = 3.14$)

[KCET 2014]

(a) 85 min (b) 156 min (c) 83.73 min (d) 90 min

- 40** The time period of the earth's satellite revolving at a height of 35800 km is [Kerala CEE 2014]

(a) 24 h (b) 100 min (c) 12 h (d) 48 h

- 41** At a height H from the surface of earth, the total energy of a satellite is equal to the potential energy of a body of equal mass at a height $3R$ from the surface of the earth (R = radius of the earth). The value of H is [EAMCET 2014]

(a) R (b) $\frac{4R}{3}$ (c) $3R$ (d) $\frac{R}{3}$

- 42** A body of mass m taken from the earth's surface to the height equal to twice the radius (R) of the earth. The change in potential energy of body will be [NEET 2013]

(a) $mg2R$ (b) $\frac{2}{3}mgR$ (c) $3mgR$ (d) $\frac{1}{3}mgR$

- 43** Infinite number of bodies, each of mass 2 kg are situated on X-axis at distance 1m, 2 m, 4 m, 8 m respectively from the origin. The resulting gravitational potential due to this system at the origin will be [NEET 2013]

(a) $-G$ (b) $-(8/3)G$ (c) $-(4/3)G$ (d) $-4G$

- 44** The value of acceleration due to gravity at the surface of earth [J&K CET 2013]

(a) is maximum at the poles
(b) is maximum at the equator
(c) remains constant everywhere on the surface of the earth
(d) is maximum at the international time line

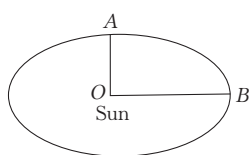
- 45** The escape velocity of a particle from the surface of the earth is given by [J&K CET 2013]

(a) $(gR)^{1/2}$ (b) $(2gR)^{1/2}$
(c) $(3gR)^{1/2}$ (d) $(gR/2)^{1/2}$

- 46** If earth were to rotate on its own axis such that the weight of a person at the equator becomes half the weight at the poles, then its time period of rotation is (g = acceleration due to gravity near the poles and R is the radius of earth) (Ignore equatorial bulge) [EAMET 2013]

(a) $2\pi\sqrt{\frac{2R}{g}}$ (b) $2\pi\sqrt{\frac{R}{2g}}$ (c) $2\pi\sqrt{\frac{R}{3g}}$ (d) $2\pi\sqrt{\frac{R}{g}}$

- 47** Earth is moving around the sun in elliptical orbit as shown. The ratio of OB and OA is R . Then, the ratio of earth's velocities at A and B is [KCET 2013]



(a) R^{-1} (b) $\sqrt{R}$ (c) R (d) $R^{2/3}$

- 48** If two planets of radii R_1 and R_2 have densities d_1 and d_2 , then the ratio of their respective acceleration due to gravity is [Kerala CEE 2013]

(a) $R_1 d_1 : R_2 d_2$ (b) $R_1^2 d_1 : R_2^2 d_2$
(c) $R_1^3 d_1 : R_2^3 d_2$ (d) $R_1 d_1^2 : R_2 d_2^2$

- 49** The weight of an object is 90 kg at the surface of the earth. If it is taken to a height equal to half of the radius of the earth, then its weight will become [MPPET 2013]

(a) 135 kg (b) 45 kg
(c) 60 kg (d) 40 kg

- 50** The escape velocity on earth is 11.2 kms^{-1} . If the body is projected out with twice this velocity, then the speed of the body far away from the earth, ignoring the presence of any other object in universe, will be [MP PET 2013]

(a) 11.2 kms^{-1} (b) 22.4 kms^{-1}
(c) 19.4 kms^{-1} (d) 15.2 kms^{-1}

- 51** A satellite of mass m is circulating around the earth with constant angular velocity. If the radius is R_0 and mass of earth is M , then the angular momentum about the centre of the earth is [UP CPMT 2013]

(a) $m\sqrt{GM/R_0}$ (b) $M\sqrt{GmR_0}$
(c) $m\sqrt{GMR_0}$ (d) $M\sqrt{GM/R_0}$

- 52** A spherical planet has a mass M_p and diameter D_p . A particle of mass m falling freely near the surface of this planet will experience an acceleration due to gravity, equal to [CBSE AIPMT 2012]

(a) $4GM_p/D_p^2$ (b) $GM_p m/D_p^2$
(c) $GM_p m/D_p^2$ (d) $4GM_p m/D_p^2$

- 53** A geo-stationary satellite is orbiting the earth at a height of $5R$ above that surface of the earth, R being the radius of the earth. The time period of another satellite (in hours) at a height of $2R$ from the surface of the earth is [CBSE AIPMT 2012]

(a) 5 (b) 10 (c) $6\sqrt{2}$ (d) $6/\sqrt{2}$

- 54** Two identical thin rings A and B each of radius R are co-axially placed at a distance R . If mass of rings are m_1, m_2 respectively, then the work done in moving a mass m from centre of one ring to that of the other is [UP CPMT 2012]

(a) $\frac{Gm}{m_2 R}(\sqrt{2} + 1)m$ (b) $\frac{Gm(m_1 - m_2)}{\sqrt{2}R}(\sqrt{2} - 1)$
(c) $\frac{Gm\sqrt{2}}{R}(m_1 + m_2)$ (d) zero

- 55** The distance between the sun and the earth be r , then the angular momentum of the earth around the sun is proportional to [UP CPMT 2012]

(a) $\sqrt{r}$ (b) $r^{3/2}$
(c) r (d) None of these

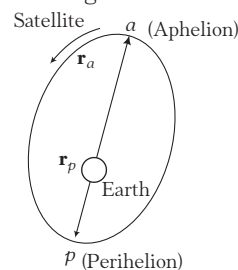
- 56** When a satellite is moving around the earth with velocity v , then to make the satellite escape, the minimum percentage increase in its velocity should be [BHU 2012]

(a) 100% (b) 82.4%
(c) 41.4% (d) None of these

- 57** A launching vehicle carrying an artificial satellite of mass m is set for launch on the surface of the earth of mass M and radius R . If the satellite is intended to move in a circular orbit of radius $7R$, the minimum energy required to be spent by the launching vehicle on the satellite is (Gravitational constant = G) [Manipal 2012]

(a) $\frac{GMm}{R}$ (b) $-\frac{13GMm}{14R}$ (c) $\frac{GMm}{7R}$ (d) $\frac{GMm}{14R}$

- 58** Consider a satellite orbiting the earth as shown in the figure below. Let $\mathbf{L}_a$ and $\mathbf{L}_p$ represent the angular momentum of the satellite about the earth when at aphelion and perihelion, respectively. Consider the following relations. [AMU 2012]



(i) $\mathbf{L}_a = \mathbf{L}_p$
(ii) $\mathbf{L}_a = -\mathbf{L}_p$
(iii) $\mathbf{r}_a \times \mathbf{L}_a = \mathbf{r}_p \times \mathbf{L}_p$

Which of the above relation(s) is/are true?

- (a) (i) only (b) (ii) only
(c) (iii) only (d) (i) and (iii)

59 A body is projected vertically upward from the surface of the earth with a velocity equal to half the escape velocity. If R is radius of the earth, then maximum height attained by the body is [AMU 2012]

- (a) $R/6$ (b) $R/3$
(c) $2R/3$ (d) R

60 The imaginary angular velocity of the earth for which the effective acceleration due to gravity at the equator shall be zero is equal to [AMU 2012]

(Take, $g = 10 \text{ ms}^{-2}$ for the acceleration due to gravity, if the earth were at rest, radius of earth equal to 6400 km and $\lambda = 60$)

- (a) $1.25 \times 10^{-3} \text{ rad s}^{-1}$ (b) $2.50 \times 10^{-3} \text{ rad s}^{-1}$
(c) $3.75 \times 10^{-3} \text{ rad s}^{-1}$ (d) $5 \times 10^{-3} \text{ rad s}^{-1}$

61 The mass of the moon is $1/81$ of the earth but the gravitational pull is $1/6$ of the earth. It is due to the fact that [JCECE 2012]

- (a) the radius of earth is $\frac{9}{\sqrt{6}}$ of the moon
(b) the radius of moon is $\frac{81}{6}$ of the earth
(c) moon is the satellite of the earth
(d) None of the above

62 The height vertically above the earth's surface at which the acceleration due to gravity becomes 1% of its value at the surface is [WB JEE 2011]

- (a) $8R$ (b) $9R$ (c) $10R$ (d) $20R$

63 Two satellites of mass m and $9m$ are orbiting a planet in orbits of radius R . Their periods of revolution will be in the ratio of [KCET 2011]

- (a) 9:1 (b) 3:1
(c) 1:1 (d) 1:3

ANSWERS

CHECK POINT 10.1

1. (d)	2. (b)	3. (c)	4. (c)	5. (d)	6. (b)	7. (c)	8. (c)	9. (d)	10. (a)
11. (c)	12. (d)	13. (c)	14. (c)	15. (d)	16. (d)				

CHECK POINT 10.2

1. (c)	2. (b)	3. (d)	4. (c)	5. (c)	6. (b)	7. (c)	8. (a)	9. (b)	10. (b)
11. (a)	12. (a)	13. (a)	14. (b)	15. (a)	16. (b)				

CHECK POINT 10.3

1. (a)	2. (a)	3. (b)	4. (d)	5. (d)	6. (b)	7. (b)	8. (c)	9. (c)	10. (c)
11. (d)	12. (d)	13. (d)	14. (c)						

CHECK POINT 10.4

1. (c)	2. (b)	3. (b)	4. (c)	5. (c)	6. (d)	7. (c)	8. (c)	9. (c)	
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CHECK POINT 10.5

1. (c)	2. (b)	3. (a)	4. (c)	5. (a)	6. (a)	7. (d)	8. (a)	9. (c)	10. (a)
11. (c)	12. (d)	13. (a)	14. (d)						

CHECK POINT 10.6

1. (b)	2. (a)	3. (b)	4. (c)	5. (c)	6. (d)	7. (b)	8. (a)	9. (d)	10. (d)
11. (d)	12. (d)	13. (d)	14. (c)	15. (d)					

(A) Taking it together

1. (c)	2. (c)	3. (b)	4. (b)	5. (c)	6. (c)	7. (a)	8. (d)	9. (a)	10. (d)
11. (c)	12. (b)	13. (d)	14. (c)	15. (b)	16. (c)	17. (a)	18. (b)	19. (a)	20. (a)
21. (a)	22. (a)	23. (d)	24. (b)	25. (c)	26. (c)	27. (d)	28. (c)	29. (a)	30. (b)
31. (a)	32. (d)	33. (d)	34. (c)	35. (a)	36. (c)	37. (d)	38. (a)	39. (c)	40. (c)
41. (b)	42. (a)	43. (b)	44. (a)	45. (b)	46. (a)	47. (a)	48. (d)	49. (d)	50. (c)
51. (a)	52. (d)	53. (b)	54. (c)	55. (b)	56. (c)	57. (d)	58. (a)	59. (a)	60. (a)
61. (b)	62. (b)	63. (c)							

(B) Medical entrance special format questions**• Assertion and reason**

1. (a) 2. (b) 3. (d) 4. (d) 5. (c)

• Statement based questions

1. (d) 2. (a) 3. (b) 4. (d) 5. (a) 6. (a)

• Match the columns

1. (c) 2. (d) 3. (c)

(C) Medical entrances' gallery

1. (a)	2. (b)	3. (c)	4. (b)	5. (a)	6. (c)	7. (a)	8. (b)	9. (d)	10. (d)
11. (a)	12. (b)	13. (b)	14. (d)	15. (a)	16. (d)	17. (d)	18. (b)	19. (b)	20. (d)
21. (a)	22. (b)	23. (c)	24. (a)	25. (d)	26. (b)	27. (c)	28. (b)	29. (b)	30. (b)
31. (d)	32. (c)	33. (d)	34. (b)	35. (c)	36. (a)	37. (d)	38. (a)	39. (c)	40. (a)
41. (a)	42. (b)	43. (d)	44. (a)	45. (b)	46. (a)	47. (c)	48. (a)	49. (d)	50. (c)
51. (c)	52. (a)	53. (c)	54. (b)	55. (a)	56. (c)	57. (b)	58. (a)	59. (b)	60. (b)
61. (a)	62. (b)	63. (c)							

Hints & Explanations

• CHECK POINT 10.1

- 1 (d) Kepler's second law is based on conservation of angular momentum, $\frac{dA}{dt} = \frac{L}{2m} = \text{constant}$.

- 2 (b) In elliptical orbit, velocity both in magnitude and direction keep on changing. Therefore, momentum does not remain constant.

- 3 (c) Since, Kepler's second law is same as the law of conservation of angular momentum.

$d_1 v_1 = d_2 v_2$ (from conservation of angular momentum at these two points)

$$\therefore v_2 = \frac{d_1 v_1}{d_2}$$

- 4 (c) Kepler's third law states that square of the time period of any planet about the sun is directly proportional to R^3 .

- 5 (d) $T \propto r^{3/2}$

$$\therefore \frac{T_1}{T_2} = \left(\frac{10^{13}}{10^{12}} \right)^{3/2} = 10\sqrt{10}$$

- 6 (b) According to Kepler's third law, $T^2 \propto r^3$

$$\therefore T_A = 8T_B \Rightarrow \frac{T_A}{T_B} = \frac{8}{1} \quad (\text{Given})$$

$$\Rightarrow \left(\frac{T_A}{T_B} \right)^2 = \left(\frac{r_A}{r_B} \right)^3 \Rightarrow \left(\frac{8}{1} \right)^2 = \left(\frac{r_A}{r_B} \right)^3 \Rightarrow \frac{r_A}{r_B} = 4$$

$$\Rightarrow r_A = 4r_B$$

- 7 (c) According to Kepler's third law, $T^2 \propto r^3$

$$\begin{aligned} T_1^2 : T_2^2 : T_3^2 &= r_1^3 : r_2^3 : r_3^3 \\ &= \left(\frac{1}{2} \right)^3 : (1)^3 : \left(\frac{3}{2} \right)^3 = 1 : 8 : 27 \end{aligned}$$

- 8 (c) By Kepler's third law of planetary motion,

$$T^2 \propto a^3$$

Given, $T_1 = 1 \text{ day}$ (geostationary)

$$a_1 = a, a_2 = 2a$$

$$\therefore \frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \Rightarrow T_2^2 = \frac{a_2^3}{a_1^3} T_1^2 = \frac{(2a)^3}{a^3} \times 1 = 8$$

$$\Rightarrow T_2 = 2\sqrt{2} \text{ days}$$

- 9 (d) Gravitational constant does not depend on the nature of the medium in which bodies are kept.

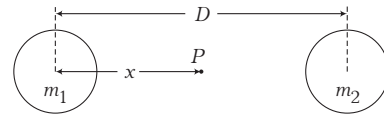
$$10 (a) F = \frac{Gm_1 m_2}{r^2}$$

$$\begin{aligned} \therefore r &= \sqrt{\frac{Gm_1 m_2}{F}} = \sqrt{\frac{6.67 \times 10^{-11} \times 1 \times 1}{9.8 \times 10^{-9}}} \\ &= 0.082 \text{ m} = 8.2 \text{ cm} \approx 8 \text{ cm} \end{aligned}$$

$$11 (c) \therefore F = \frac{Gm_1 m_2}{r^2} \text{ or } F \propto m_1 m_2$$

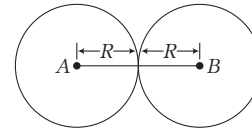
On M due to m force is F and due to mass $2m$ force is $2F$. Therefore, net force is $3F$.

- 12 (d) Force will be zero at the point P located at



$$x = \frac{\sqrt{m_1}}{\sqrt{m_1} + \sqrt{m_2}} D = \frac{\sqrt{81M}}{\sqrt{81M} + \sqrt{M}} D = \frac{9}{10} D$$

- 14 (c) Let masses of two balls be $m_1 = m_2 = m$ (given) and the density is ρ . Distance between their centres = $AB = 2R$.



Thus, the magnitude of the gravitational force F that two balls separated by a distance $2R$ exert on each other is

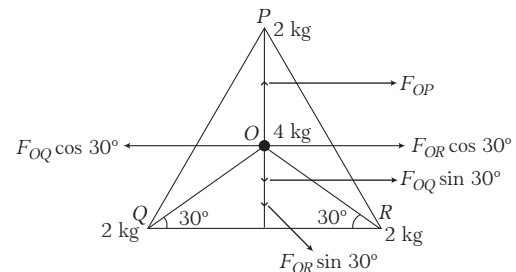
$$F = G \frac{(m)(m)}{(2R)^2} = G \frac{m^2}{4R^2} = G \frac{\left(\frac{4}{3} \pi R^3 \rho \right)^2}{4R^2}$$

$$\therefore F \propto R^4$$

- 15 (d) As we know, $F = \frac{Gm_1 m_2}{r^2}$ and $F' = \frac{Gm_1 m_2}{(3r)^2} = \frac{F}{9}$

$$\text{Percent decrease in } F = \left(\frac{F - F'}{F} \right) \times 100 = \frac{8}{9} \times 100 = 89\%$$

- 16 (d) Consider the equilateral triangle as PQR with centroid at O . Given, $OP = OQ = OR = \sqrt{2} \text{ m}$



The gravitational force on mass 2 kg due to mass 1 kg at P ,

$$F_{OP} = G \frac{4 \times 2}{(\sqrt{2})^2} = 4G \text{ along } OP$$

$$\text{Similarly, } F_{OQ} = G \frac{4 \times 2}{(\sqrt{2})^2} = 4G \text{ along } OQ$$

$$\text{and } F_{OR} = G \frac{4 \times 1}{(\sqrt{2})^2} \\ = 4G \text{ along } OR$$

$F_{OQ} \cos 30^\circ$ and $F_{OR} \cos 30^\circ$ are equal and acting in opposite directions, hence cancel out each other. Then, the resultant force on the mass 4 kg at O,
 $F = F_{OP} - (F_{OQ} \sin 30^\circ + F_{OR} \sin 30^\circ) = 0$ (zero).

• CHECK POINT 10.2

1 (c) $g = \frac{GM}{R^2}$ or $g \propto \frac{M}{R^2}$

Given, mass is 2 times and diameter is 3 times. Hence,
 $g' = \frac{2g}{9}$.

2 (b) $g = \frac{GM}{R^2}$. If radius shrinks to half of its present value, then g will become four times.

3 (d) $g = \frac{GM}{R^2}$ or $\frac{G}{g} = \frac{R^2}{m}$
 $\therefore \frac{G}{g}$ will have the unit $\frac{\text{m}^2}{\text{kg}}$.

4. (c) $g = \frac{GM}{R^2} = \frac{G \left(\frac{4}{3} \pi R^3 \right) \rho}{R^2}$
 $\therefore \rho = \frac{g}{G \cdot 4\pi \frac{R}{3}} = \frac{3g}{4\pi GR}$

5 (c) $g = \frac{4}{3} G \pi R \rho \Rightarrow \frac{g_1}{g_2} = \frac{\rho_1 R_1}{\rho_2 R_2} = \frac{1}{2} \times \frac{4}{1} = \frac{2}{1}$

6 (b) $g \propto \rho R$. If density of earth is increased four times and its radius become half, then the value of acceleration due to gravity becomes double due to which our weight get doubled.

7 (c) Given, $g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2} = \frac{g}{4}$ (At $h = R$)

$\therefore w' = \frac{72}{4} = 18\text{N}$

8 (a) Since, it is known that $g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$

Given, $g' = \frac{g}{64}$

where, g' is the value of acceleration due to gravity at height h .

$\therefore 1 + \frac{h}{R} = 8$ or $\frac{h}{R} = 7$

$\therefore h = 7R = 44.8 \times 10^6 \text{ m} \approx 45 \times 10^6 \text{ m}$

9 (b) At depth, $g' = g \left(1 - \frac{h}{R}\right)$ or $g \left(1 - \frac{d}{R}\right)$

$\Rightarrow \frac{g'}{g} = \frac{1}{n} = \left(1 - \frac{d}{R}\right)$

or $d = R \left(\frac{n-1}{n}\right)$

10 (b) According to the question,

$$g \left(1 - \frac{2h}{R}\right) = \left(1 - \frac{d}{R}\right) \quad (\text{At } h \ll R)$$

or $1 - \frac{2h}{R} = 1 - \frac{d}{R}$

$\therefore d = 2h$

11 (a) Given, $g \left(1 - \frac{d}{R}\right) = \frac{1}{2} \frac{g}{\left(1 + \frac{h}{R}\right)^2}$

or $1 - \frac{d}{R} = \frac{1}{2 \left(1 + \frac{1600}{6400}\right)^2} = \frac{16}{50} \Rightarrow \frac{d}{R} = \frac{34}{50}$

$\therefore d = (6.4 \times 10^6) \left(\frac{34}{50}\right) = 4.3 \times 10^6 \text{ m}$

12 (a) At centre of earth, value of g is zero. Hence, weight is zero.

13 (a) At poles, value of g is maximum. Since, there is no effect of rotation of earth.

14 (b) When a body is taken from equator to poles, its weight increases, because acceleration due to gravity increases.

15 (a) At equator, $g' = g - R\omega^2 \Rightarrow 0 = g - R\omega^2$

$\therefore \omega = \sqrt{\frac{g}{R}} = \sqrt{\frac{9.8}{6400 \times 10^3}} = 1.23 \times 10^{-3} \text{ rad s}^{-1}$

16 (b) Acceleration due to gravity at latitude ϕ is given by

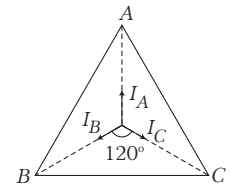
$$g' = g - R\omega^2 \cos^2 \phi$$

At $\phi = 30^\circ$, $g_{30^\circ} = g - R\omega^2 \cos^2 30^\circ = g - \frac{3}{4} R\omega^2$

$\therefore g - g_{30^\circ} = \frac{3}{4} \omega^2 R$

• CHECK POINT 10.3

1 (a) Due to three particles, net intensity at the centre,
 $I = I_A + I_B + I_C = 0$. Because these three intensities are equal in magnitude and the angle between each other is 120° .



2 (a) Gravitational field reduces by 75%, means 25% is left.

$$\frac{GM}{r^2} = \frac{1}{4} \left(\frac{GM}{R^2} \right)$$

or $r = 2R$

$\therefore h = r - R = 2R - R = R$

3 (b) Gravitational field due to a solid sphere,

$$E(r) = \frac{GM}{R^3} r \quad (\therefore r \leq R)$$

$\Rightarrow E(r) \propto r$, i.e. $E(r)$ increases inside the sphere,

$$E(r) = \frac{GM}{r^2} \quad (\therefore r > R)$$

$\Rightarrow E(r) \propto \frac{1}{r^2}$, i.e. $E(r)$ decreases outside the sphere.

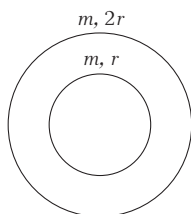
- 4 (d) Intensity will be zero inside the spherical shell.

$$E = 0 \text{ upto } r = R \text{ and } E \propto \frac{1}{r^2} \text{ when } r > R$$

This is best shown in option (d).

- 5 (d) Inside a shell, field strength is zero. Therefore, force on a particle is zero.

- 6 (b) At distance $2.5r$ or $\frac{5}{2}r$



$$E = \frac{G(m+m)}{\left(\frac{5}{2}r\right)^2} = \frac{8Gm}{25r^2}$$

- 7 (b) Gravitational field at an external point due to spherical shell of mass M is $\left(\frac{GM}{r^2}\right)$ while at an internal point is zero.

- (i) Point Q is external to shell M_1 , M_2 and M_3

So, field at Q will be

$$E_Q = \frac{GM_1}{y^2} + \frac{GM_2}{y^2} + \frac{GM_3}{y^2} = \frac{G}{y^2} (M_1 + M_2 + M_3)$$

- (ii) Field at P will be

$$E_P = \frac{GM_1}{x^2} + \frac{GM_2}{x^2} + 0 = \frac{G}{x^2} (M_1 + M_2)$$

- 8 (c) For a uniform ring of mass M and radius R at its centre, the field is zero but potential is $-\frac{GM}{R}$.

- 9 (c) $V_{in} = \frac{-Gm}{2R} \left[3 - \left(\frac{r}{R}\right)^2 \right]$, $V_{surface} = \frac{-GM}{R}$, $V_{out} = \frac{-GM}{r}$

- 10 (c) Gravitational potential at centre, $V_c = -\frac{3}{2} \frac{GM}{R}$

$$\text{Gravitational potential on surface, } V = -\frac{GM}{R}$$

$$\therefore V_c = \frac{3}{2}V$$

- 11 (d) Since, length of rod is equal to the circumference of semi-circle,

$$\pi R = L \Rightarrow R = \frac{L}{\pi}$$

Therefore, the gravitational potential at the centre of circle will be

$$V = -\frac{GM}{R} = -\frac{\pi GM}{L}$$

- 12 (d) $E_{net} = 0$

$$\therefore \frac{Gm}{x^2} = \frac{GM}{(d-x)^2}, \text{ where } x \text{ is distance from } m.$$

$$\text{Hence, } \frac{x}{d-x} = \frac{\sqrt{m}}{\sqrt{M}} \Rightarrow x = \frac{\sqrt{m}}{\sqrt{m} + \sqrt{M}} \cdot d$$

$$\text{and } d-x = \frac{\sqrt{M}}{\sqrt{m} + \sqrt{M}} \cdot d$$

Since, gravitational potential between two bodies of masses M and m respectively, is given by

$$\begin{aligned} V &= -\frac{Gm}{x} - \frac{GM}{d-x} \\ &= -\frac{Gm(\sqrt{m} + \sqrt{M})}{\sqrt{m} \cdot d} - \frac{GM(\sqrt{m} + \sqrt{M})}{\sqrt{M} \cdot d} \\ &= -\frac{G}{d} (\sqrt{m} + \sqrt{M})^2 \end{aligned}$$

- 13 (d) Gravitational potential, $V_i = -\frac{GM}{r}$

$$V_i = -\frac{6.67 \times 10^{-11} \times 100}{0.1}$$

$$V_i = -\frac{6.67 \times 10^{-9}}{0.1} = -6.67 \times 10^{-8} \text{ J}$$

$$\therefore V_f = 0$$

$\therefore$ Work done per unit mass,

$$W = \Delta V = (V_f - V_i) = 6.67 \times 10^{-8} \text{ J}$$

- 14 (c) For hollow sphere,

$$V_{in} = \frac{-GM}{R}, V_{surface} = \frac{-GM}{R}, V_{out} = \frac{-GM}{r}$$

i.e. Gravitational potential remains constant inside the sphere and it is equal to potential at the surface. Its value increases when the point moves away from the surface of sphere.

• CHECK POINT 10.4

- 1 (c) When particles are released from rest their separation decreases. Therefore, gravitational potential energy of the system decreases.

- 2 (b) Potential energy, $u = \frac{GMm}{r}$

$$\text{At distance } 2r, g = \frac{GM}{(2r)^2} = \frac{GM}{4r^2} = \frac{u}{4mr}$$

$$\text{Now, } w = mg = \frac{u}{4r}$$

- 3 (b) On surface of earth, $U = -\frac{GMm}{R}$

At height $h (<< R)$, increase in potential energy is mgh .

$\therefore$ The potential energy of body and the earth is given by

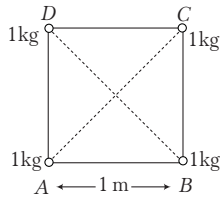
$$U_h = -\frac{GMm}{R} + mgh$$

- 4 (c) Magnitude of gravitational potential energy, $U = \frac{GMm}{r}$

$$\text{or } U = \frac{GMm}{r^2} \times r \text{ or } U = g \times mr \quad \left(\because g = \frac{GM}{r^2} \right)$$

$$\therefore U = (mg)r \text{ or } mg = \frac{U}{r}$$

- 5 (c) Here, $AB = BC = CD = DA = 1\text{ m}$



$$BD = AC = \sqrt{1^2 + 1^2} = \sqrt{2}\text{ m}$$

Total potential energy,

$$\begin{aligned} U &= \left[\frac{-G \times 1 \times 1}{AB} \right] + \left[\frac{-G \times 1 \times 1}{BC} \right] + \left[\frac{-G \times 1 \times 1}{CD} \right] \\ &\quad + \left[\frac{-G \times 1 \times 1}{DA} \right] + \left[\frac{-G \times 1 \times 1}{BD} \right] + \left[\frac{-G \times 1 \times 1}{AC} \right] \\ &= 4 \times \left[\frac{-G \times 1 \times 1}{1} \right] + 2 \left[\frac{-G \times 1 \times 1}{\sqrt{2}} \right] \\ &= -5.4G \end{aligned}$$

- 6 (d) Change in potential energy in displacing a body from r_1 to r_2 is given by

$$\begin{aligned} \Delta U &= GMm \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = GMm \left(\frac{1}{2R} - \frac{1}{3R} \right) \quad (\because r_1 = 2R, r_2 = 3R) \\ &= \frac{GMm}{6R} \end{aligned}$$

- 7 (c) Since, potential energy, $\Delta U = \frac{mgh}{\left(1 + \frac{h}{R}\right)}$, $h = R$ (given)

$$\therefore \Delta U = \frac{mgR}{2}$$

- 8 (c) Work done, $W = \Delta U = \frac{mgh}{1 + \left(\frac{h}{R}\right)}$

Substituting $h = 5R$ in above equation, we get

$$\Delta U = \frac{mg \times 5R}{1 + 5} = \frac{5mgR}{6}$$

- 9 (c) Initial gravitational potential energy, $U_i = -\frac{GM_e m}{R}$... (i)

$$\text{Final gravitational potential energy, } U_f = -\frac{1}{5} \cdot \frac{GM_e m}{R} \quad \dots \text{(ii)}$$

$$\begin{aligned} \therefore \text{Energy required, } E &= U_f - U_i = -\frac{GM_e m}{5R} + \frac{GM_e m}{R} \\ &= \frac{5GM_e m - GM_e m}{5R} = \frac{4GM_e m}{5R} \end{aligned}$$

But, acceleration due to gravity (g) in terms of gravitational constant (G) is

$$g = \frac{GM_e}{R^2} \quad \dots \text{(iii)}$$

$$\begin{aligned} \therefore \frac{GM_e}{R^2} \times R^2 \times \frac{4m}{5R} &= g \times R^2 \times \frac{4m}{5R} \quad [\text{From Eq. (iii)}] \\ &= \frac{4gmR}{5} \quad \text{or} \quad \frac{4mgR}{5} \end{aligned}$$

• CHECK POINT 10.5

- 1 (c) Total mechanical energy of any closed system is always negative.

- 2 (b) The energy required to remove an object from the surface of the earth to infinity is called binding energy of the object. It is equal to negative of gravitational potential energy of the object.

$$\text{Thus, binding energy} = -E = \frac{GMm}{R}$$

$$\text{But, } g = \frac{GM}{R^2} \Rightarrow GM = gR^2 \Rightarrow \text{BE} = \frac{gmR^2}{R} = mgR$$

- 3 (a) At $v < v_e$, total energy is negative

$v = v_e$, total energy is zero

and $v > v_e$ total energy is positive

So, with reference to the above given cases, it can be said that when the escape velocity is given to a particle, its total energy is zero.

- 4 (c) Escape velocity is independent of mass of projectile which is projected. Therefore, escape velocity of body of mass 100 kg will also be 11.2 kms^{-1} .

- 5 (a) Escape velocity is independent of angle from which particle is projected. Therefore, when the body is projected in a direction making 30° angle to the vertical, then its escape velocity will remain same, i.e. 11.2 kms^{-1} .

- 6 (a) $v_e = \sqrt{2gR} = \sqrt{2 \cdot \frac{GM}{R^2} \cdot R}$ or $v_e \propto \sqrt{\frac{M}{R}}$

Mass is 1000 times and radius is 10 times that of earth,

$$v'_e = \sqrt{\frac{(1000M)G}{10R}} \Rightarrow v'_e = 10 \sqrt{\frac{GM}{R}}$$

$$\Rightarrow v'_e = 10v_e \Rightarrow v'_e = 10 \times 11.2 \Rightarrow v'_e = 112 \text{ kms}^{-1}$$

- 7 (d) Escape velocity, $v = \sqrt{2gR}$

$$\therefore \frac{v_e}{v_m} = \sqrt{\frac{g_e \times \frac{R_e}{R_m}}{\frac{g_m}{1} \times \frac{10}{1}}} = \sqrt{\frac{6 \times \frac{10}{1}}{1 \times 1}} \quad \left(\because \frac{R_e}{R_m} = \frac{10}{1}, \frac{g_e}{g_m} = \frac{6}{1} \right)$$

$$= \sqrt{60}$$

- 8 (a) Escape velocity, $v_e = \sqrt{2gR}$

where, g is acceleration due to gravity and R is radius.

$$\text{Given, } \frac{R_1}{R_2} = K, \frac{g_1}{g_2} = g$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{\sqrt{g_1 R_1}}{\sqrt{g_2 R_2}} = \sqrt{Kg} \Rightarrow \frac{v_1}{v_2} = (Kg)^{1/2}$$

- 9 (c) $U_i + K_i = U_f + K_f$

$$\Rightarrow -\frac{GMm}{R} + \frac{1}{2} m (2v_e)^2 = 0 + \frac{1}{2} mv^2$$

$$\text{or } -\frac{GM}{R} + 2v_e^2 = \frac{1}{2} v^2 \Rightarrow -\frac{2GM}{R} + \frac{8GM}{R} = v^2$$

$$\begin{aligned} \therefore v &= \sqrt{\frac{6GM}{R}} = \sqrt{3 \left(\frac{2GM}{R} \right)} \\ &= \sqrt{3(2gR)} = \sqrt{3} v_e \end{aligned}$$

- 10 (a) Decrease in kinetic energy = Increase in potential energy

$$\therefore \frac{1}{2} m \left(\frac{v_e}{\sqrt{2}} \right)^2 = \frac{mgh}{1 + \frac{h}{R}} \quad \text{or} \quad \frac{v_e^2}{4} = \frac{gh}{1 + \frac{h}{R}}$$

$$\text{or} \quad \frac{2gR}{4} = \frac{gh}{1 + \frac{h}{R}} \Rightarrow \frac{R}{2} = \frac{h}{1 + \frac{h}{R}} \quad (\because v_e = \sqrt{2gR})$$

Solving this equation, we get $h = R$.

Note Kinetic energy is half the value required to escape. Therefore, speed is $1/\sqrt{2}$ times the value required to escape.

- 11 (c) The kinetic energy required to project a body of mass m from the earth surface to infinity is given by

$$\text{KE} = \frac{1}{2} mv_e^2 = \frac{1}{2} m 2Rg$$

$$(\because \text{Escape velocity, } v_e = \sqrt{2Rg} \Rightarrow v_e^2 = 2Rg)$$

$$\Rightarrow \text{KE} = mgR$$

- 12 (d) Initial kinetic energy of the body = $\frac{1}{2} mv^2$

$$\text{Here, } v = 3 \times 11.2 \text{ kms}^{-1}$$

$$\text{So, kinetic energy, KE} = \frac{1}{2} \times m (3 \times 11.2)^2 = 9 \times \frac{1}{2} mv_e^2$$

As, $\frac{1}{2} mv_e^2$ energy is used up in coming out from the gravitational pull of the earth, so final kinetic energy should be $8 \times \frac{1}{2} mv_e^2$.

$$\text{Hence, } \frac{1}{2} mv'^2 = 8 \times \frac{1}{2} mv_e^2$$

$$\therefore v'^2 = 8v_e^2$$

$$\Rightarrow v' = \sqrt{8}v_e = \sqrt{8} \times 11.2 \text{ kms}^{-1}$$

- 13 (a) Decrease in kinetic energy = Increase in potential energy

$$\therefore \frac{1}{2} mv^2 = \frac{mgh}{1 + \frac{h}{R}}$$

$$\therefore v^2 = \frac{2gR}{2} \quad (\text{Given, } h = R)$$

$$= gR = \frac{GM}{R} \quad \left(\because g = \frac{GM}{R^2} \right)$$

$$\text{or} \quad v = \sqrt{\frac{GM}{R}}$$

- 14 (d) Let h be the maximum height attained by the body, then from third equation of motion, $v^2 = u^2 + 2gh$

$$\text{When } u = 0, \text{ then}$$

$$v = \sqrt{2gh}$$

$$\text{Given, } v = \frac{v_e}{3}, \text{ where } v_e = \sqrt{2gR}$$

$$\Rightarrow \sqrt{2gh} = \frac{1}{3} \sqrt{2gR}$$

On squaring both sides of equation, we get

$$h = \frac{R}{9}$$

CHECK POINT 10.6

- 1 (b) The gravitational force provides the centripetal force. Therefore, the net force on the satellite will be F .

- 2 (a) Since, orbital velocity of body close to earth can be given as
- $$v_o = \sqrt{rg} = \sqrt{6400 \times 10^3 \times 10} = \sqrt{64 \times 10^6} = 8 \times 10^3 \text{ m s}^{-1}$$
- $$v_o = 8 \text{ kms}^{-1}$$

- 3 (b) Velocity v of a satellite varies inversely as the square root of radius R of the orbit.

$$\text{i.e.} \quad v \propto \frac{1}{\sqrt{R}}$$

$$\therefore \frac{v_A}{v_B} = \sqrt{\frac{R_B}{R_A}} = \sqrt{\frac{R}{4R}}$$

$$\Rightarrow \frac{3v}{v_B} = \frac{1}{2} \Rightarrow v_B = 6v$$

- 4 (c) Since, orbital velocity, $v = \sqrt{\frac{GM}{r}}$ or $v \propto \frac{1}{\sqrt{r}}$

$$\therefore \frac{v_2}{v_1} = \sqrt{\frac{r_1}{r_2}} \Rightarrow v_2 = \sqrt{\frac{r_1}{r_2}} \cdot v_1 = \sqrt{\frac{R}{R + \frac{R}{2}}} \cdot v = \sqrt{\frac{2}{3}} v$$

- 5 (c) The time period of satellite in a circular orbit around a planet is independent of mass of the satellite.

- 6 (d) From Kepler's third law, $T^2 \propto R^3$

$$\therefore \frac{T^2}{R^3} = \text{constant}$$

- 7 (b) The time period of revolution of satellite is given by

$$T = 2\pi \sqrt{\frac{r^3}{GM}} \Rightarrow T \propto \sqrt{\frac{r^3}{GM}}$$

- 8 (a) Time period of satellite, $T = \frac{2\pi r^{3/2}}{\sqrt{GM}}$ or $\sqrt{GM} = \frac{2\pi}{T} \cdot r^{3/2}$

$$\therefore \omega r^{3/2} = \sqrt{GM} \quad \left(\because \frac{2\pi}{T} = \omega \right)$$

$$\Rightarrow r^3 = \frac{GM}{\omega^2} = \frac{gR^2}{\omega^2}$$

9. (d) Energy of a satellite is given by $E = -\frac{GMm}{2r}$

For a satellite very close to earth, $r = R$

$$\therefore E = -\frac{GMm}{2R}$$

- 10 (d) Kinetic energy is negative of the mechanical energy, i.e. $-E_0$.

- 11 (d) As, kinetic energy, $K = \frac{GMm}{2r}$

$$\text{Potential energy, } U = -\frac{GMm}{r}$$

$$\text{Total energy, } E = -\frac{GMm}{2r}$$

If r is decreased, K will increase but U and E will decrease.

12 (d) Energy of satellite is given by $E = -\frac{GMm}{2r}$ or $E \propto \frac{m}{r}$

$$\Rightarrow \frac{E_A}{E_B} = \frac{m_A}{m_B} \times \frac{r_B}{r_A} = \frac{3}{1} \times \frac{4r}{r} \quad \left(\because \frac{m_A}{m_B} = \frac{3}{1}, \frac{r_B}{r_A} = \frac{4r}{r} \right)$$

$$\Rightarrow \frac{E_A}{E_B} = \frac{12}{1}$$

13 (d) Total mechanical energy of satellite,

$$E = \frac{-GMm}{2r} \Rightarrow \frac{E_P}{E_Q} = \frac{m_P}{m_Q} \times \frac{r_Q}{r_P}$$

$$\Rightarrow \frac{E_P}{E_Q} = \frac{3}{1} \times \frac{8r}{r} = \frac{24}{1}$$

14 (c) E = Energy of satellite – Energy of satellite on surface of earth

$$= -\frac{GMm}{2(8R)} - \left(-\frac{GMm}{R} \right)$$

$$= \frac{15}{16} \frac{GMm}{R} = \frac{15}{16} mgR \quad \left(\because \frac{GM}{R} = gR \right)$$

15 (d) Let r_1 and r_2 be the respective distance of two satellites from the earth's surface.

$$r_1 = R + R = 2R \text{ and } r_2 = 7R + R = 8R$$

($\because R$ is the radius of the earth)

Now, $K = \frac{GMm}{2r}$

$$\Rightarrow \frac{K_1}{K_2} = \frac{r_2}{r_1} = 4$$

$$U = -\frac{GMm}{r}, \frac{U_1}{U_2} = \frac{r_2}{r_1} = 4$$

and $E = -\frac{GMm}{2r}$

or $\frac{E_1}{E_2} = \frac{r_2}{r_1} = 4$

(A) Taking it together

1 (c) Reason of weightlessness in a satellite is a zero reaction force by satellite surface.

2 (c) $g = \frac{4}{3} \pi \rho GR \Rightarrow g \propto dR \quad (\because \rho = d \text{ given in the problem})$

3 (b) The satellite is revolving around earth because the centripetal force is balanced by earth's gravitational pull. If the gravitational pull disappears, the satellite is free of centripetal force.

So, it will travel with its instantaneous velocity, i.e. in the direction tangential to the circular path.

4 (b) $T = \frac{2\pi(r)^{3/2}}{\sqrt{GM}}$, so T is independent of m , the mass of satellite.

5 (c) According to Kepler's second law, $\frac{dA}{dt} = \frac{vr \sin \theta}{2} = \frac{1}{2} |\mathbf{r} \times \mathbf{v}|$

6 (c) Here, $KE = \frac{1}{2} mv_e^2 = \frac{1}{2} m(2gR) = mgR$

7 (a) Considering Kepler's second law, we know that,

$$\frac{A}{T} = \frac{L}{2m} \Rightarrow L = \frac{2mA}{T}$$

8 (d) Asteroids are also being acted upon by central gravitational forces, hence they are moving in circular orbits like planets and obey Kepler's laws.

9 (a) If they collide inelastically and stick together as wreckage, then

KE = 0, only PE is present.

$$\therefore \text{Total energy, } E = U = 2 \left(-\frac{GMm}{r} \right) = -\frac{2GMm}{r}$$

10 (d) Change in potential energy in displacing a body from r_1 to r_2 is given by

$$\Delta U = GMm \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = GMm \left(\frac{1}{2R} - \frac{1}{4R} \right) = \frac{GMm}{4R}$$

11 (c) Escape velocity, $v_e = \sqrt{2}v_o = 1.414v_o$

Further speed is to increase $\sqrt{2}$ times or orbital speed is to be increased by 41% and kinetic energy is to increase two times.

12 (b) We know that, $g = \frac{GM}{R^2} \Rightarrow \frac{dg}{dR} = \frac{-2GM}{R^3}$

$$\Rightarrow \frac{dg}{dR} = \frac{-2GM}{R^2} \cdot \frac{1}{R}$$

$$\Rightarrow \frac{dg}{g} = -2 \left(\frac{h}{R} \right) \quad (\because dR \approx h)$$

13 (d) If we assume the earth as a sphere of uniform density, then it can be treated as point mass placed at its centre. In this case, acceleration due to gravity $g = 0$, at the centre.

It is not so, if the earth is considered as a sphere of non-uniform density, in that case, value of g will be different at different points and cannot be zero at any point.

14 (c) As the total energy of the earth satellite bounded system is negative $\left(-\frac{GM}{2a} \right)$, where a is radius of the satellite and M is

mass of the earth. Due to the viscous force acting on satellite, energy decreases continuously and radius of the orbit or height decreases gradually.

15 (b) As observed from the sun, two types of forces are acting on the moon : one is due to gravitational attraction between the sun and the moon and the other is due to gravitational attraction between the earth and the moon. Hence, total force on the moon is not central.

16 (c) As observed from the earth, the sun appears to move in an approximate circular orbit. The gravitational force of attraction between the earth and the sun always follows inverse square law. Due to relative motion between the earth and mercury, the orbit of mercury as observed from the earth will not be approximately circular, since the major gravitational force on mercury is due to the sun.

17 (a) As the earth is revolving around the sun in a circular motion due to gravitational attraction. The force of attraction will be of radial nature, i.e. angle between position vector $\mathbf{r}$ and force $\mathbf{F}$ is zero.

So, torque, $|\boldsymbol{\tau}| = |\mathbf{r} \times \mathbf{F}| = rF \sin 0^\circ = 0$

18 (b) $F = \frac{k}{r}$ (where, $k = \text{constant}$)

$$\therefore \frac{mv^2}{r} = \frac{k}{r} \text{ or } v \propto r^0 \Rightarrow n = 0$$

19 (a) Applying conservation of mechanical energy, we get

$$\frac{1}{2}mv^2 = \frac{mgh}{1 + h/R}$$

Substituting $h = R$, we get

$$v = \sqrt{gR} = \sqrt{\frac{GM}{R}}$$

20 (a) Required additional velocity,

$$\begin{aligned} \Delta v &= v_e - v_o = \sqrt{4gR} - \sqrt{2gR} \\ &= (2 - \sqrt{2})\sqrt{gR} = (\sqrt{2} - 1)\sqrt{\frac{2GM}{R}} \end{aligned}$$

21 (a) At equator, $g' = g - R\omega^2 = 0$

$$\therefore \omega = \sqrt{\frac{g}{R}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{g}{R}} \Rightarrow T = 2\pi\sqrt{\frac{R}{g}}$$

22 (a) $0 = g - R\omega^2 \cos^2 60^\circ$

$$\begin{aligned} \text{or } \omega^2 &= \frac{4g}{R} \text{ or } \omega = 2\sqrt{\frac{g}{R}} \\ \Rightarrow \frac{2\pi}{T} &= 2\sqrt{\frac{g}{R}} \Rightarrow T = \pi\sqrt{\frac{R}{g}} \end{aligned}$$

23 (d) Orbital velocity, $v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{R+h}}$

24 (b) Semi-major axis, $a = \frac{r_1 + r_2}{2}$

$$\begin{aligned} \text{Now, } T^2 &\propto a^3 \\ \Rightarrow T &\propto a^{3/2} \propto \left(\frac{r_1 + r_2}{2}\right)^{3/2} \Rightarrow T \propto (r_1 + r_2)^{3/2} \end{aligned}$$

25 (c) Force on B due to A = $F_{BA} = \frac{G(2Mm)}{(AB)^2}$ towards BA

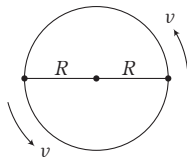
$$\text{Force on B due to C} = F_{BC} = \frac{GMm}{(BC)^2} \text{ towards BC}$$

As, $(BC) = 2AB$

$$\Rightarrow F_{BC} = \frac{GMm}{(2AB)^2} = \frac{GMm}{4(AB)^2} < F_{BA}$$

Hence, m will move towards $2M$.

26 (c) As gravitational force provides the centripetal force to the satellite to revolve around the earth,



$$\frac{G \cdot m \cdot m}{4R^2} = \frac{mv^2}{(R)} \Rightarrow v = \frac{1}{2}\sqrt{\frac{Gm}{R}}$$

28 (c) The initial position will become the perigee (nearest) position.

So, the resulting orbit of the satellite will become elliptical orbit with minimum distance from the earth equal to R .

29 (a) $r = R_{\text{earth}} + h = (6.4 \times 10^6 + 1.4 \times 10^6) \text{ m}$

$$\Rightarrow r = 7.8 \times 10^6 \text{ m}$$

$$\therefore T = \left[\frac{4\pi^2 r^3}{GM_{\text{earth}}} \right]^{1/2} = 6842 \text{ s}$$

$$\text{Speed of satellite, } v = \sqrt{\frac{GM_{\text{earth}}}{r}} = 7162.93 \approx 7163 \text{ ms}^{-1}$$

30 (b) Increase in kinetic energy = Decrease in potential energy

$$\therefore \frac{1}{2}mv^2 = \frac{mgR}{1 + R/R} = \frac{mgR}{2} \quad \left(\because \Delta U = \frac{mgh}{1 + h/R} \right)$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{mgR}{2} \Rightarrow v = \sqrt{gR}$$

31 (a) $\frac{1}{2}mv^2 = \frac{mgh}{1 + \frac{h}{R}} \dots(i)$

On solving Eq. (i) for h , we get

$$\therefore h = \frac{v^2}{2g - \frac{v^2}{R}} = \frac{R}{\left(\frac{2gR}{v^2}\right) - 1}$$

32 (d) Equating gravitational field intensities, $\frac{GM}{r^2} = \frac{GM}{(60R - r)^2}$

Solving above equation, we get

$$60R - r = 9r \Rightarrow r = 6R$$

33 (d) $L = mvr = m\sqrt{\frac{GM}{r}}r = m\sqrt{GM}r$

$$\therefore L \propto \sqrt{r}$$

$$\Rightarrow \frac{L_1}{L_2} = \sqrt{\frac{r_1}{r_2}} \quad \left(\because \begin{matrix} L_1 = L, \\ r_1 = r, \\ r_2 = 16r \end{matrix} \right)$$

$$\Rightarrow \frac{L}{L_2} = \sqrt{\frac{r}{16r}} \Rightarrow \frac{L}{L_2} = \frac{1}{4} \Rightarrow L_2 = 4L$$

34 (c) Acceleration between them is zero, where force between $2m$ and $\frac{m}{5}$ is equal to the force between m and $\frac{m}{5}$ or the net

force on $\frac{m}{5}$ is zero. Suppose this point is P . Now, if velocity v_0 is less than the value necessary to cross point P , it will retrace its path otherwise not.

35 (a) Work done, $W = \Delta U = U_f - U_i = U_f$ ($\because U_i = 0$)

$$\Rightarrow W = -(Gm_1m_2/R)$$

36 (c) We know that, $v_o = \frac{1}{2}v_e \dots(i)$

$$\therefore v_o = \sqrt{\frac{GM_e}{R_e + h}} \text{ and } v_e = \sqrt{\frac{2GM_e}{R_e}}$$

Putting these values in Eq. (i), we get

$$\begin{aligned}\sqrt{\frac{GM_e}{R_e + h}} &= \frac{1}{2} \sqrt{\frac{2GM_e}{R_e}} \\ \Rightarrow \frac{GM_e}{R_e + h} &= \frac{1}{2} \times \frac{1}{2} \times \frac{2GM_e}{R_e} \\ \Rightarrow R_e + h &= 2R_e \\ \Rightarrow h &= R_e \text{ or } R\end{aligned}$$

37 (d) At P, $F_P = 0$

$$\text{At Q, } F_Q = \frac{GM_P m}{b^2}$$

$$\text{At R, } F_R = \frac{G(M_1 + M_2)m}{c^2}$$

38 (a) $E_i = E_o = -\frac{GMm}{2r}$

$$\text{or } E_f = -\frac{GMm}{2(3r)} = -\frac{GMm}{6r}$$

$$\therefore W = E_f - E_i = -\frac{GMm}{3r} = \frac{2}{3}E_o$$

39 (c) $E = \text{Energy of satellite} - \text{Energy of satellite on surface of earth}$

$$\begin{aligned}&= -\frac{GMm}{2(3R)} - \left[-\frac{GMm}{R} \right] \quad \left(\because h = 2R \right. \\ &\quad \left. \therefore \text{total distance} = 3R \right) \\ &= \frac{5}{6} \frac{GMm}{R} = \frac{5}{6} mgR \quad \left(\because \frac{GM}{R} = gR \right)\end{aligned}$$

40 (c) $W = E_A - E_\infty = (U_A + K_A) - (U_\infty + K_\infty)$

$$\therefore -3 = \left[(1)V_A + \frac{1}{2} \times (1)(2)^2 \right] - (0 + 0) \quad (\because W = -3\text{J})$$

$$\therefore V_A = -5 \text{ J kg}^{-1}$$

41 (b) As, $\frac{mv^2}{r} = \frac{Gm_1 m_2}{r^3}$ or $v \propto r^{\left(\frac{1-3}{2}\right)}$

$$\text{Now, } T = \frac{2\pi r}{v} \text{ or } T \propto \frac{r}{v}$$

$$\therefore T \propto \frac{r}{r^{\left(\frac{1-3}{2}\right)}} \Rightarrow T \propto r^{\frac{1+3}{2}} = r^2$$

42 (a) As, gravitational force provides the necessary centripetal force

$$\frac{mv^2}{r} \propto r^{-n}$$

$$\Rightarrow v \propto r^{(1-n)/2}$$

$$\text{Also, } T = \frac{2\pi r}{v} \text{ or } T \propto rv^{-1}$$

$$\text{or } T \propto r \cdot r^{(n-1)/2}$$

$$\text{or } T \propto r^{(n+1)/2}$$

43 (b) Applying conservation of mechanical energy,

$$\frac{1}{2} m \left(\frac{v_e}{2} \right)^2 = \frac{mgh}{1 + \frac{h}{R}} \text{ or } \frac{v_e^2}{8} = \frac{gh}{1 + \frac{h}{R}}$$

$$\text{or } \frac{2gR}{8} = \frac{gh}{1 + h/R} \quad \dots(i) \quad (\because v_e = \sqrt{2gR})$$

$$\text{On solving Eq. (i), we get } h = \frac{R}{3}$$

44 (a) Conservation of mechanical energy gives,

$$\frac{1}{2} mv^2 = m \left(-\frac{GM}{R} + \frac{3GM}{2R} \right) = \frac{GMm}{2R}$$

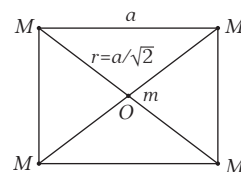
($\because$ Increase in KE = Decrease in PE)

$$\Rightarrow v = \sqrt{\frac{GM}{R}}$$

$$\text{Momentum, } mv = m \sqrt{\frac{GM}{R}}$$

$$\therefore \text{Impulse required} = m \sqrt{\frac{GM}{R}}$$

45 (b) Potential at centre = $-\frac{4GM}{r} = -\frac{4GM}{a/\sqrt{2}} = -\frac{4\sqrt{2}GM}{a}$



$$\text{Potential energy} = \frac{-4\sqrt{2}GMm}{a}$$

$$\text{Binding energy} = \frac{4\sqrt{2}GMm}{a}$$

$$\therefore \frac{1}{2} mv_e^2 = \frac{4\sqrt{2}GMm}{a}$$

$$\text{or } v_e = \sqrt{\frac{8\sqrt{2}GM}{a}}$$

46 (a) $E = \Delta U = \frac{mgh}{1 + h/R} = \frac{mghR}{(R + h)}$

$E' = \text{Energy of satellite} - \text{Energy of satellite on surface of earth}$

$$= -\frac{GMm}{2(R + h)} - \left(-\frac{GMm}{R} \right)$$

$$= mgR^2 \left[\frac{1}{R} - \frac{1}{2(R + h)} \right] = \frac{mg(R + 2h)R}{2(R + h)} \quad (\because GM = gR^2)$$

$$\text{Now, } \frac{E}{E'} = \frac{2h}{(R + 2h)}$$

47 (a) From the figure,

$$\mathbf{F}_A = \mathbf{F}_{AB} + \mathbf{F}_{AC} = 2 \left[\frac{Gm^2}{a^2} \right] \cos 30^\circ = \left[\frac{Gm^2}{a^2} \cdot \sqrt{3} \right]$$

$$\text{Also, } r = \frac{a}{\sqrt{3}}$$

$$\text{Now, } \frac{mv^2}{r} = |\mathbf{F}_A| \text{ or } \frac{mv^2 \sqrt{3}}{a} = \frac{Gm^2}{a^2} \sqrt{3}$$

$$\therefore v = \sqrt{\frac{Gm}{a}}$$

$$48 \text{ (d)} F_1 = \frac{GMm}{(2R)^2} = \frac{GMm}{4R^2}$$

$$F_2 = F_1 - F_{\text{cavity}} = \frac{GMm}{4R^2} - \frac{G\left(\frac{M}{8}\right)(m)}{\left(\frac{3}{2}R\right)^2}$$

$$\Rightarrow F_2 = \frac{7GMm}{36R^2}$$

$$\therefore \frac{F_2}{F_1} = \frac{7}{9}$$

$$49 \text{ (d)} a = g' = g \left(1 - \frac{h}{R}\right) = g \left(\frac{R-h}{R}\right) = \frac{g}{R} \cdot y \quad (\because R-h=y)$$

$$= \frac{\left(\frac{GM}{R^2}\right)}{R} \cdot y = \frac{G}{R^3} \left(\frac{4}{3} \pi R^3 \rho\right) \cdot y = \frac{4}{3} \pi \rho G y$$

50 (c) Field strength is uniform inside the cavity. Let us find its value at centre.

$$E_T = E_O + E_C \quad (T = \text{Total}, O = \text{Remaining}, C = \text{Cavity})$$

$$\therefore E_O = E_T - E_C = \frac{GM}{R^3} \cdot \frac{R}{2} - 0 = \frac{GM}{2R^2}$$

$$\therefore F = mF_R = \frac{GMm}{2R^2}$$

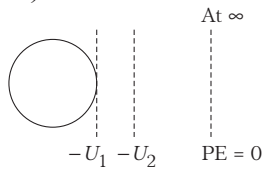
$$51 \text{ (a)} \frac{1}{2} \mu v_r^2 = U_i - U_f$$

$$\text{Here, } \mu = \text{reduced mass} = \frac{mM}{M+m}$$

$$\therefore \frac{1}{2} \left(\frac{mMv_r^2}{m+M} \right) = 0 - \left(-\frac{GMm}{d} \right)$$

$$\text{Hence, relative velocity, } v_r = \sqrt{\frac{2G(M+m)}{d}}$$

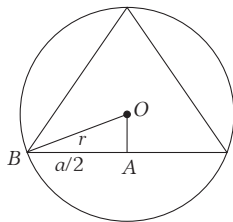
$$52 \text{ (d)} U_2 = 2 \left(\frac{1}{2} mv_o^2 \right)$$



$$\text{Now, } \frac{1}{2} mv^2 = -U_2 + U_1 = U_1 - U_2 = \frac{mv_o^2}{2} - mv_o^2$$

$$\therefore v = \sqrt{v_o^2 - 2v_o^2}$$

$$53 \text{ (b)} \text{ In right angled } \triangle AOB, \cos 30^\circ = \frac{AB}{OB}$$



$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{a}{2r} \Rightarrow r = \frac{a}{\sqrt{3}}$$

$$\sqrt{3}F = mr\omega^2$$

$$\sqrt{3} \left(\frac{Gmm}{a^2} \right) = m \frac{a}{\sqrt{3}} \cdot \omega^2$$

$$\therefore \omega = \sqrt{\frac{3Gm}{a^3}}$$

54 (c) Time period of simple pendulum,

$$T = 2\pi \sqrt{\frac{l}{g}} \quad \text{or} \quad T \propto \frac{1}{\sqrt{g}}$$

$$\therefore \frac{T_1}{T_2} = \sqrt{\frac{g_2}{g_1}} = \frac{2}{1} \Rightarrow \frac{g_2}{g_1} = 4 \quad \dots(i)$$

$$\therefore v_e = \sqrt{2gR} \quad \text{or} \quad v_e \propto \sqrt{gR}$$

$$\Rightarrow \frac{v_{e1}}{v_{e2}} = \sqrt{\frac{g_1 \cdot R_1}{g_2 \cdot R_2}} = 1/2$$

$$\therefore \sqrt{\frac{1}{4} \times \frac{R_1}{R_2}} = \frac{1}{2} \quad \left(\text{from Eq. (i), } \frac{g_2}{g_1} = 4 \right)$$

$$\text{Hence, } \frac{R_1}{R_2} = 1$$

$$\text{Further, } g = \frac{Gm}{R^2} = \frac{G\left(\frac{4}{3} \pi R^3 \rho\right)}{R^2} \quad \text{or} \quad g \propto R\rho$$

$$\therefore \frac{g_1}{g_2} = \frac{R_1}{R_2} \cdot \frac{\rho_1}{\rho_2} \quad \text{or} \quad \left(\frac{1}{4} \right) = (1) \frac{\rho_1}{\rho_2} \quad \text{or} \quad \frac{\rho_1}{\rho_2} = \frac{1}{4}$$

$$55 \text{ (b)} E_1 = \Delta U = \left(\frac{mgh}{1+h/R} \right)$$

$E_2 = \text{Energy of satellite} - \text{Energy of satellite on surface of earth}$

$$= -\frac{GMm}{2(R+h)} + \frac{GMm}{R}$$

$$= mgR \left[1 - \frac{1}{2(1+h/R)} \right] \quad (\because GM = gR^2)$$

$$\Rightarrow E_2 = \frac{mgR \left(\frac{2h}{R} + 1 \right)}{2 \left(1 + \frac{h}{R} \right)}$$

$$\therefore \frac{E_1}{E_2} = \frac{mgh}{1 + \frac{h}{R}} \times \frac{2 \left(1 + \frac{h}{R} \right)}{mgR} = \frac{2h}{R} \quad \left(\because h < R \right)$$

$$\therefore 1 + \frac{2h}{R} \approx 1$$

56 (c) Acceleration of supercondense material = g

But acceleration of earth = $2g$ (towards the mass)

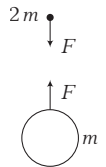
$\therefore$ Relative acceleration = $3g$

Since, we know that, $h = ut + \frac{1}{2} a_r t^2$

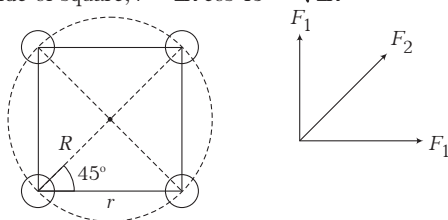
But the body starts from rest.

So, $u = 0$

$$\text{Now, } h = \frac{1}{2} a_r t^2 \quad \text{or} \quad h = \frac{1}{2} (3g) t^2 \quad \text{or} \quad t = \sqrt{\frac{2h}{3g}}$$



- 57 (d) Side of square, $r = 2R \cos 45^\circ = \sqrt{2}R$



$$\begin{aligned} \text{Now, } 2F_1 \cos 45^\circ + F_2 &= \frac{Mv^2}{R} \\ \Rightarrow v &= \sqrt{\frac{R}{M} (\sqrt{2} F_1 + F_2)} \\ \Rightarrow v &= \sqrt{\frac{R}{M} \left[\sqrt{2} \frac{GMM}{(\sqrt{2}R)^2} + \frac{GMM}{(2R)^2} \right]} \\ &= \sqrt{\frac{GM}{R} \left(\frac{2\sqrt{2} + 1}{4} \right)} \end{aligned}$$

- 58 (a) The initial potential energy (U_i) of object, $U_i = -\frac{GMm}{3R}$

$$\text{Final potential energy, } U_f = -\frac{GMm}{R}$$

By law of conservation of energy, $\Delta KE = -\Delta PE$

$$\Rightarrow \frac{1}{2}mv^2 = -(U_f - U_i) = U_i - U_f$$

$$\Rightarrow \frac{1}{2}mv^2 = -\frac{GMm}{3R} + \frac{GMm}{R}$$

$$\Rightarrow \frac{1}{2}v^2 = \frac{2GM}{3R}$$

$$\Rightarrow v = \sqrt{\frac{4GM}{3R}} = 2\sqrt{\frac{GM}{3R}}$$

- 59 (a) Using conservation of angular momentum,

$$mv_p r_p = mv_a r_a$$

As velocities are perpendicular to the radius vectors at apogee and perigee.

$$\Rightarrow v_p r_p = v_a r_a$$

Using conservation of energy,

$$-\frac{GMm}{r_p} + \frac{1}{2}mv_p^2 = -\frac{GMm}{r_a} + \frac{1}{2}mv_a^2$$

By solving the above equations,

$$v_p = \sqrt{\frac{2GM r_a}{r_p(r_p + r_a)}}$$

Hence,

$$L = mv_p r_p = m \sqrt{\frac{2GM r_p r_a}{(r_p + r_a)}}$$

- 60 (a) As, $g = \frac{4}{3}\pi\rho Gr$

$$\therefore g \propto r, \text{ if } r < R$$

$$g = \frac{GM}{r^2}$$

$$\therefore g \propto \frac{1}{r^2}, \text{ if } r > R$$

$$\text{If } r_1 < R \text{ and } r_2 < R, \text{ then } \frac{F_1}{F_2} = \frac{g_1}{g_2} = \frac{r_1}{r_2}$$

$$\text{If } r_1 > R \text{ and } r_2 > R, \text{ then } \frac{F_1}{F_2} = \frac{g_1}{g_2} = \left(\frac{r_2}{r_1}\right)^2$$

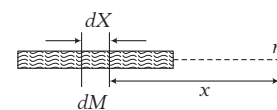
- 61 (b) Work done, $W = U_2 - U_1 = mv_2 - mv_1$

$$\begin{aligned} &= m \left[\left(-\frac{Gm_2}{R} - \frac{Gm_1}{\sqrt{2}R} \right) - \left(-\frac{Gm_1}{R} - \frac{Gm_2}{\sqrt{2}R} \right) \right] \\ &= \frac{Gm(m_1 - m_2)}{\sqrt{2}R} (\sqrt{2} - 1) = \frac{\sqrt{2}Gm^2}{R} (1 - \sqrt{2}) \end{aligned}$$

- 62 (b) $dF = \frac{GmdM}{x^2}$

$$\therefore F = \int_{x=a}^{x=a+l} dF$$

$$= \int_{a-l/2}^{a+l} \frac{G \cdot m \left(\frac{M}{l} \cdot dx \right)}{x^2} = \frac{4GmM}{4a^2 - l^2}$$



- 63 (c) The force on particle for varying g is given by

$$F = \frac{GMmx}{R^3} = m\omega^2 x$$

$$\Rightarrow \omega^2 = \frac{GM}{R^3}$$

$$\left(\frac{2\pi}{t} \right)^2 = \frac{GM}{R^3} \quad \left(\because \omega = \frac{2\pi}{t} \right)$$

$$\Rightarrow t = 2\pi \sqrt{\frac{R^3}{GM}}$$

When g is constant,

$$s = \frac{1}{2}gt'^2 \quad (\text{For initial velocity, } u = 0)$$

$$\Rightarrow R = \frac{1}{2} \left(\frac{GM}{R^2} \right) (t')^2 \quad \left(\because g = \frac{GM}{R^2} \right)$$

$$(t')^2 = \frac{2R^3}{GM} \Rightarrow t' = \sqrt{2} \sqrt{\frac{R^3}{GM}}$$

$$\therefore \frac{t}{t'} = \frac{2\pi \sqrt{R^3/GM}}{\sqrt{2} \sqrt{R^3/GM}} = \frac{2\pi}{\sqrt{2}} = \sqrt{2} \pi$$

(B) Medical entrance special format questions

• Assertion and reason

3 (d) $\frac{dA}{dt} = \frac{L}{2m} = \text{constant} = \frac{mvr \sin \theta}{2m} = \frac{vr \sin \theta}{2}$

i.e. $\frac{dA}{dt}$ is independent of m .

- 4 (d) Gravitational potential due to a point mass at some finite distance is always negative and non-zero.

- 5 (c) $T = \frac{2\pi}{\sqrt{GM}} r^{3/2}$. Time period does not depend upon the radius of earth (R) but it depends upon its mass (M).

Statement based questions

- 1 (d) Acceleration due to gravity at altitude h ,

$$g_h = \frac{g}{(1+h/R)^2} \approx g \left(1 - \frac{2h}{R}\right) \quad \dots (i)$$

At depth d ,

$$g_d = g \left(1 - \frac{d}{R}\right) \quad \dots (ii)$$

In both Eqs. (i) and (ii), if h and d increases, g decreases.

At latitude ϕ , $g_\phi = g - \omega^2 R \cos^2 \phi$

As, ϕ increases, g_ϕ also increases.

Thus, option (d) is correct but rest are incorrect.

- 2 (a) As we know that, $g = \frac{GM}{R^2}$, $v_e = \sqrt{\frac{2GM}{R}}$

and

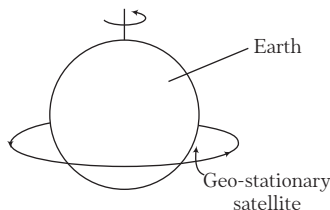
$$U = \frac{-GMm}{R}$$

$$\therefore g \propto \frac{M}{R^2}, v_e \propto \sqrt{\frac{M}{R}} \text{ and } U \propto \frac{M}{R}$$

If both mass and radius are increased by 0.5%, then escape velocity and potential energy remains unchanged, whereas g decreases by 0.5%.

Thus, option (a) is correct but rest are incorrect.

- 3 (b) A geo-stationary satellite have same rotation as that of earth, i.e. in west-east direction



- 4 (d) Gravitational mass of proton is equivalent to its inertial mass and is independent of presence of neighbouring heavy objects.

- 5 (a) For the gravitational force between an extended object (like the earth) and a point mass, the Newton's universal law of gravitation is not directly applicable.

Thus, Statement I is correct but Statement II is incorrect.

- 6 (a) Time period of satellite = $\frac{2\pi(R_E + h)^{3/2}}{\sqrt{GM_E}}$

From the above equation, it is clear that the time period of satellite depends on the mass of earth (M_E), radius of orbit ($r = R_E + h$) and height of the satellite from the surface of earth (h).

The geo-stationary satellite can have circular or elliptical orbit.

Thus, only Statement I is incorrect.

Match the columns

- 1 (c) Inside a shell, gravitational potential, $V = -\frac{GM}{R} = \text{constant}$

and gravitational field intensity, $E = 0$

Outside the shell, $V = -\frac{GM}{r}$ and $E = \frac{GM}{r^2}$

As r increase, V increase and E decreases.

Therefore, potential at A will be less than B . Gravitational field inside the shell will be zero. As one moves from C to D , the gravitational potential remains constant and when one moves from D to A , gravitational field decreases.

Hence, $A \rightarrow q$, $B \rightarrow t$, $C \rightarrow r$, $D \rightarrow s$.

- 2 (d) As $g = \frac{GM}{R^2}$, $V = -\frac{GM}{R}$

$$\text{At height } h = R, g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2} = \frac{g}{4} \Rightarrow V' = -\frac{GM}{2R}$$

i.e. g decrease by a factor $1/4$ and V' also decreases by a factor of $1/2$.

At depth $h = \frac{R}{2}$,

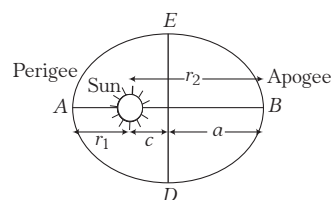
$$g' = g \left(1 - \frac{h}{R}\right) = g \left(1 - \frac{1}{2}\right) = \frac{g}{2}$$

$$V' = -\frac{GM}{R^3} (15R^2 - 0.5r^2) = -\frac{GM}{R^3} \left(15R^2 - \frac{1}{8}R^2\right) = -\frac{11}{8} \frac{GM}{R}$$

i.e. g decrease by a factor $\frac{1}{2}$ and V' also decreases by a factor $\frac{11}{8}$.

Hence, $A \rightarrow p$, $B \rightarrow q$, $C \rightarrow q$, $D \rightarrow r$.

- 3 (c) In the figure given below, it can be seen that at perigee position planet is nearest to the sun. So, as the planet moves in the elliptical orbit from apogee position to perigee position, then speed of planet increases, distance of planet from centre of sun decreases, potential energy decreases. Angular momentum about centre of sun remains the same.



Hence, $A \rightarrow r$, $B \rightarrow q$, $C \rightarrow q$, $D \rightarrow p$.

(C) Medical entrance gallery

- 1 (a) Given, $w = mg = 72 \text{ N}$ (on the surface of earth)

At height equal to half of radius of the earth, (i.e. $h = \frac{R}{2}$)

Acceleration due to gravity,

$$g' = g \left(\frac{R}{R+h}\right)^2 = g \left(\frac{R}{R+R/2}\right)^2 = g \left(\frac{4R^2}{9R^2}\right)$$

$$\Rightarrow g' = \frac{4}{9} g$$

Gravitational force on body at height h ,

$$F = mg' = \frac{4}{9} mg = \frac{4}{9} \times 72 = 32 \text{ N}$$

or $w' = 32 \text{ N}$

2 (b) Radius of earth = R

Let at depth d , acceleration due to gravity be g_d which becomes $\frac{g}{n}$.

$$\text{i.e. } g_d = \frac{g}{n} \Rightarrow g \left(1 - \frac{d}{R}\right) = \frac{g}{n}$$

$$\Rightarrow 1 - \frac{d}{R} = \frac{1}{n} \Rightarrow 1 - \frac{1}{n} = \frac{d}{R}$$

$$\Rightarrow \frac{n-1}{n} = \frac{d}{R} \Rightarrow d = \left(\frac{n-1}{n}\right)R$$

3 (c) Given, weight of the body, $w = 200 \text{ N}$

As we know $w = mg$, where m is the mass of the body and g ($\approx 10 \text{ ms}^{-2}$) is acceleration due to gravity of the body at the surface of the earth.

Since, mass m remains constant irrespective of the position of the body on the earth. However, g is not constant and its value at a depth d below the earth's surface is given as

$$g' = g \left(1 - \frac{d}{R}\right) \quad \dots (i)$$

where, R is the radius of the earth.

Multiplying m on both sides of Eq. (i), we get

$$mg' = mg \left(1 - \frac{d}{R}\right)$$

Now, the weight of the body at half-way down (i.e. $d = \frac{R}{2}$) to the centre of the earth is

$$\begin{aligned} mg' &= 200 \times \left(1 - \frac{R/2}{R}\right) \\ &= 200 \left(1 - \frac{1}{2}\right) = 200 \times \frac{1}{2} = 100 \text{ N} \end{aligned}$$

$\therefore$ The body will weigh 100 N half-way down to the centre of the earth.

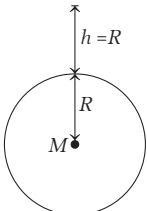
4 (b) As we know, the potential energy of body of mass m on the surface of earth,

$$U_1 = -\frac{GMm}{R} \quad \dots (i)$$

where, G = gravitational constant,

M = mass of earth and R = radius of earth.

When the mass is raised to a height h from the surface of the earth, then the potential energy of the body becomes,



$$U_2 = -\frac{GMm}{(R+h)}$$

Here, $h = R$

(Given)

$$\Rightarrow U_2 = -\frac{GMm}{2R} \quad \dots (ii)$$

Thus, the change in potential energy, $\Delta U = U_2 - U_1$

Substituting the values from Eqs. (i) and (ii), we get

$$\begin{aligned} \Delta U &= -\frac{GMm}{2R} + \frac{GMm}{R} \\ &= \frac{GMm}{2R} = \frac{gR^2m}{2R} \quad \left(\because g = \frac{GM}{R^2}\right) \\ &= \frac{mgR}{2} \end{aligned}$$

Thus, the work done in raising the mass to a height R is equals to $\frac{mgR}{2}$.

5 (a) From Kepler's third law, the time period of revolution of satellite around earth is

$$T^2 \propto r^3 \quad \text{or} \quad T \propto r^{3/2} \quad \dots (i)$$

where, r is the radius of satellite's orbit.

Here, $r_1 = 6R_E + R_E$, $T_1 = 24 \text{ h}$

$$r_2 = 2.5R_E + R_E, \quad T_2 = ?$$

So, from Eq. (i), we get

$$\begin{aligned} \frac{T_1}{T_2} &= \left(\frac{r_1}{r_2}\right)^{3/2} \\ \frac{24}{T_2} &= \left(\frac{6R_E + R_E}{2.5R_E + R_E}\right)^{3/2} = \left(\frac{7}{3.5}\right)^{3/2} \\ \Rightarrow T_2 &= \frac{24}{(2)^{3/2}} = \frac{24}{2\sqrt{2}} = \frac{12}{\sqrt{2}} = 6\sqrt{2} \text{ h} \end{aligned}$$

6 (c) Given, radius of earth, $R_e = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$

Distance from centre of earth, $r = 2000 \text{ km}$

Distance from the surface of earth,

$$\begin{aligned} d &= R_e - r = 6400 - 2000 \\ &= 4400 \text{ km} = 4.4 \times 10^6 \text{ m} \end{aligned}$$

If g' be the gravitational field below the surface of earth at a depth d , then

$$\begin{aligned} g' &= g \left(1 - \frac{d}{R_e}\right) = 9.8 \left(1 - \frac{4.4 \times 10^6}{6.4 \times 10^6}\right) \quad (\because g = 9.8 \text{ ms}^{-2}) \\ &= 9.8 \times 0.3125 = 3.06 \text{ ms}^{-2} \end{aligned}$$

7 (a) Given, escape velocity on the surface of earth,

$$v_e = 11.1 \text{ km/h}$$

$$\text{i.e. } v_e = \sqrt{2gR_e}$$

where, R_e is radius of earth.

$$\text{As, } v_e = \sqrt{\frac{2GM_e}{R_e}} \quad \left(\because g = \frac{GM_e}{R_e^2}\right)$$

where, M_e is mass of earth.

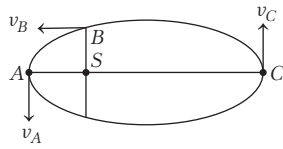
$$\text{Mass of moon, } M_m = \frac{M_e}{81}$$

Radius of moon, $R_m = \frac{R_e}{4}$

∴ Escape velocity on the surface of moon,

$$\begin{aligned} v_m &= \sqrt{\frac{2GM_m}{R_m}} = \sqrt{\frac{2G \frac{M_e}{81}}{\frac{R_e}{4}}} \\ &= \frac{2\sqrt{2}}{9} \sqrt{\frac{GM_e}{R_e}} = \frac{2}{9} \sqrt{\frac{2GM_e}{R_e}} = \frac{2}{9} v_e \\ &= \frac{2}{9} \times 11.1 = 2.46 \text{ km h}^{-1} \end{aligned}$$

8 (b) According to the question,



The figure above shows an ellipse traced by a planet around the sun S. The closest point A is known as perihelion and the farthest point C is known as aphelion.

Since, as per the Kepler's second law of area, the planet will move slowly ($v_{\min}$) only, when it is farthest from the sun and more rapidly ($v_{\max}$), when it is nearest to the sun.

Thus, $v_A = v_{\max}$, $v_C = v_{\min}$

Therefore, we can write, $v_A > v_B > v_C$... (i)

Kinetic energy of the planet at any point is given as

$$K = \frac{1}{2}mv^2$$

Thus, at point A, $K_A = \frac{1}{2}mv_A^2$

At point B, $K_B = \frac{1}{2}mv_B^2$

At point C, $K_C = \frac{1}{2}mv_C^2$

From Eq. (i), we can write

$$K_A > K_B > K_C$$

9 (d) Given, $T_A = 1\text{h}$, $T_B = 8\text{h}$

and $r_A = 10^4 \text{ km}$

From Kepler's law, $\frac{T_A^2}{T_B^2} = \frac{r_A^3}{r_B^3} \Rightarrow \frac{1^2}{8^2} = \frac{(10^4)^3}{r_B^3}$

$$\Rightarrow r_B^3 = 64 \times (10^4)^3$$

$$\therefore r_B = 4 \times 10^4 \text{ km}$$

$$\text{Speed of satellite A, } v_A = \frac{2\pi r_A}{T_A} = \frac{2\pi \times 10^4}{1}$$

$$= 2\pi \times 10^4 \text{ km/h}$$

$$\text{Speed of satellite B, } v_B = \frac{2\pi r_B}{T_B} = \frac{2\pi \times 4 \times 10^4}{8}$$

$$= \pi \times 10^4 \text{ km/h}$$

The speed of B relative to A when they are close,

$$v_{BA} = v_A - v_B = 2\pi \times 10^4 - \pi \times 10^4$$

$$= \pi \times 10^4 \text{ km/h}$$

10 (d) As force, $F = \frac{GMm}{r^{3/2}} = m\omega^2 r$

$$\Rightarrow \frac{GMm}{r^{3/2}} = \frac{4\pi^2 mr}{T^2} \quad \left(\because \omega = \frac{2\pi}{T} \right)$$

$$\Rightarrow T^2 = \left(\frac{4\pi^2}{GM} \right) \cdot r^{5/2} \Rightarrow T^2 \propto r^{5/2}$$

11 (a) The acceleration due to gravity at a latitude λ is

$$g' = g - R\omega^2 \cos^2 \lambda$$

At the poles, $\lambda = 90^\circ$

$$\therefore g' = g = g_p \quad (\because \cos 90^\circ = 0)$$

For $\lambda = 60^\circ$, $g' = g_p - R\omega^2 \cos^2 \lambda$

$$\text{or } g' = g_p - R\omega^2 \cos^2 60^\circ$$

$$\text{Hence, } g' = g_p - R\omega^2 \left(\frac{1}{2} \right)^2 = g_p - \frac{1}{4} R\omega^2$$

12 (b) Given, $M_e = 5.98 \times 10^{24} \text{ kg}$, $M_m = 7.35 \times 10^{22} \text{ kg}$ and

$$r = 3.855 \times 10^8 \text{ m}$$

Let x be the distance of the point from the centre of earth, where gravitational intensity is zero. Therefore,

$$\frac{GM_e}{x^2} = \frac{GM_m}{(3.855 \times 10^8 - x)^2}$$

$$\text{or } \frac{x}{3.855 \times 10^8 - x} = \sqrt{\frac{M_e}{M_m}} = \sqrt{\frac{5.98 \times 10^{24}}{7.35 \times 10^{22}}} \cong 9$$

$$\text{or } \frac{x}{9} + x = 3.855 \times 10^8$$

$$\text{or } x = 9 \times 3.855 \times 10^8 / 10 = 3.469 \times 10^8 \text{ m}$$

$$\text{Distance from moon} = 3.855 \times 10^8 - 3.469 \times 10^8$$

$$= 3.86 \times 10^7 \text{ m}$$

$$\cong 3.85 \times 10^7 \text{ m}$$

13 (b) In gravitation free space, there is no external gravity. Due to masses of the astronauts, there will be small gravitational attractive force between them. Thus, these astronauts will move towards each other.

14 (d) Acceleration due to gravity at height h above earth's surface,

$$g_h = g \left(1 - \frac{2h}{R} \right) \quad (\because h \ll R)$$

Acceleration due to gravity at depth d below earth's surface,

$$g_d = g \left(1 - \frac{d}{R} \right)$$

Given, $h = 1 \text{ km}$ and $g_d = g_h$

$$\text{or } g \left(1 - \frac{d}{R} \right) = g \left(1 - \frac{2h}{R} \right)$$

$$\Rightarrow d = 2h \quad \text{or } d = 2 \text{ km}$$

- 15 (a)** The orbital velocity of spaceship in circular orbit, $v_o = \sqrt{\frac{GM}{r}}$

If spaceship is very close to earth surface, then $r =$ radius of earth (R).

$$\therefore v_o = \sqrt{\frac{GM}{R}} = \sqrt{\frac{R^2 g}{R}} = \sqrt{Rg} \quad \left(\text{Since, } g = \frac{GM}{R^2} \right)$$

Here, $R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$ and $g = 9.8 \text{ ms}^{-2}$

$$\therefore v_o = \sqrt{6.4 \times 10^6 \times 9.8} \\ = 7.92 \times 10^3 \text{ m/s} = 7.92 \text{ km s}^{-1}$$

The escape velocity of spaceship,

$$v_e = \sqrt{2Rg} = \sqrt{2} v_o = \sqrt{2} \times 7.92 = 11.20 \text{ km s}^{-1}$$

$\therefore$ Additional velocity required

$$= 11.20 - 7.92 = 3.28 \text{ km s}^{-1}$$

Therefore, velocity of 3.28 kms^{-1} must be added to orbital velocity of spaceship to overcome the gravitational pull.

- 16 (d)** From conservation of energy,

$$\frac{1}{2}mv_1^2 - \frac{GM_em}{R} = \frac{1}{2}mv_2^2 - \frac{GM_em}{R+H}$$

Here, $v_2 = 0$ (At maximum height)

$$\therefore \frac{1}{2}mv_1^2 - \frac{GM_em}{R} = -\frac{GM_em}{R+H}$$

Solving, we get

$$H = \frac{R}{\frac{2gR}{v_1^2} - 1} = \frac{R}{\frac{v_e^2}{v_1^2} - 1} \quad (\because v_e = \sqrt{2gR})$$

where, v_e is escape velocity.

$$\text{Here, } v_1 = \frac{1}{3}v_e \Rightarrow \frac{v_e}{v_1} = 3$$

$$\therefore H = \frac{R}{9-1} = \frac{R}{8}$$

- 17 (d)** Acceleration due to gravity,

$$g = \frac{GM}{R^2} = \frac{4}{3}\pi\rho GR \quad \left(\because \rho = \frac{M}{V} \text{ and } V = \frac{4}{3}\pi R^3 \right)$$

$$\therefore \frac{g_1}{g_2} = \frac{\rho_1 R_1}{\rho_2 R_2} \quad \dots(i)$$

$$\text{Atmospheric pressure can be given by } p = \frac{w}{S} \quad \dots(ii)$$

where, $w =$ weight of atmosphere

and $S =$ surface area of the planet.

From $w = mg$, we get

$$\begin{aligned} \frac{m_1}{m_2} &= \frac{w_1 g_2}{w_2 g_1} \\ \therefore \frac{m_1}{m_2} &= \frac{p_1 S_1 g_2}{p_2 S_2 g_1} \quad [\text{From Eq. (ii)}] \\ &= \frac{p_1 \cdot (4\pi R_1^2) \cdot \rho_2 R_2}{p_2 (4\pi R_2^2) \cdot \rho_1 R_1} \quad [\text{From Eq. (i)}] \\ &= \frac{p_1 R_1 \rho_2}{p_2 R_2 \rho_1} \end{aligned}$$

- 18 (b)** Acceleration due to gravity at a depth d below the surface of the earth is given by

$$g_{\text{depth}} = g_{\text{surface}} \left(1 - \frac{d}{R} \right)$$

Also, for a point at height h above surface of the earth,

$$g_{\text{height}} = g_{\text{surface}} \left(\frac{R^2}{(R+h)^2} \right)$$

Therefore, we can say that value of g increases from centre of earth and it is maximum at its surface and then decreases with height.

Hence, graph given in option (b) is correct.

- 19 (b)** Total energy of a satellite at height h is

$$\begin{aligned} &= \text{KE} + \text{PE} = \frac{GMm}{2(R+h)} - \frac{GMm}{(R+h)} \\ &= -\frac{GMm}{2(R+h)} = -\frac{GMmR^2}{2R^2(R+h)} = -\frac{mg_0 R^2}{2(R+h)} \quad \left(\because g_0 = \frac{GM}{R^2} \right) \end{aligned}$$

- 20 (d)** Gravitational potential at some height h from the surface of the earth is given by

$$V = -\frac{GM}{R+h} \quad \dots(i)$$

and acceleration due to gravity at some height h from the surface of the earth can be given as

$$g = \frac{GM}{(R+h)^2} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{|V|}{g} = \frac{GM}{(R+h)} \times \frac{(R+h)^2}{GM} \Rightarrow \frac{|V|}{g} = R+h \quad \dots(iii)$$

$$\therefore V = -5.4 \times 10^7 \text{ J kg}^{-2}, \quad g = 6 \text{ ms}^{-2}$$

$$\text{and } R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$$

Substituting these values in Eq. (iii), we get

$$\frac{5.4 \times 10^7}{6} = 6.4 \times 10^6 + h$$

$$\Rightarrow h = (9 - 6.4) \times 10^6 = 2.6 \times 10^6 \text{ m} \Rightarrow h = 2600 \text{ km}$$

- 21 (a)** Since, the escape velocity of the earth can be given as

$$v_e = \sqrt{2gR} = \sqrt{2 \left(\frac{GM}{R^2} \right) R} = R \sqrt{\frac{8}{3} \pi G \rho} \quad [\because \text{mass of earth, } m = \rho \left(\frac{4}{3} \pi R^3 \right)]$$

$$\Rightarrow v_e = R \sqrt{\frac{8}{3} \pi G \rho} \quad \dots(i)$$

As it is given that, the radius and mean density of planet are twice as that of the earth. So, escape velocity at planet will be

$$v_p = 2R \sqrt{\frac{8}{3} \pi G 2\rho} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{v_e}{v_p} = \frac{R \sqrt{\frac{8}{3} \pi G \rho}}{2R \sqrt{\frac{8}{3} \pi G 2\rho}} \Rightarrow \frac{v_e}{v_p} = \frac{1}{2\sqrt{2}}$$

- 22 (b)** The gravitational force of attraction between the planet and sun provides the centripetal force,

$$\text{i.e.} \quad \frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

The time period of planet will be

$$T = \frac{2\pi r}{v}$$

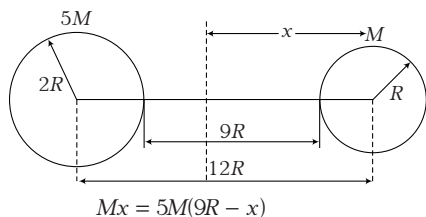
$$\Rightarrow T^2 = \frac{4\pi^2 r^2}{GM} = \frac{4\pi^2 r^3}{GM} \quad \dots(i)$$

Also, from the Kepler's third law, $T^2 = Kr^3$...(ii)

From Eqs. (i) and (ii), we get

$$\frac{4\pi^2 r^3}{GM} = Kr^3 \Rightarrow GMK = 4\pi^2$$

- 23 (c)** Let the smaller body cover a distance x before collision, then



$$Mx = 5M(9R - x)$$

or $x = 45R - 5x$ or $x = \frac{45R}{6} = 7.5R$

- 24 (a)** At the pole, the weight is same as the actual weight.

$$\text{Thus,} \quad 100 \text{ N} = m(10 \text{ ms}^{-2}) \Rightarrow m = 10 \text{ kg}$$

At the equator, the apparent weight is given by

$$mg' = mg - m\omega^2 R$$

Also, the angular speed of an equatorial point on the earth's surface is

$$\omega = (2\pi / 24 \times 60 \times 60)$$

$$\Rightarrow \omega = 7.27 \times 10^{-5} \text{ rad/s}$$

$$\text{Hence, now,} \quad mg' = 100 - 10(7.27 \times 10^{-5})^2 \times 6.4 \times 10^6$$

$$\approx 99.66 \text{ N}$$

- 25 (d)** The gravitational field at the centre of a solid sphere of mass M is zero.

- 26 (b)** The value of g at a depth d from the earth's surface,

$$g = g_E \left(1 - \frac{d}{R_E}\right)$$

where, g_E is acceleration due to gravity at the earth's surface and R_E is the radius of earth.

$$\Rightarrow g = 9.8 \left(1 - \frac{5}{6400}\right) = 9.8 \left(\frac{6395}{6400}\right) = 9.79 \text{ ms}^{-2}$$

- 27 (c)** The gravitational force between the particles will give the required centripetal force.

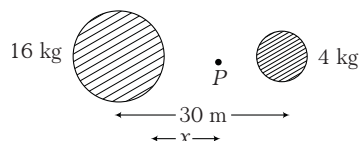
$$\text{So,} \quad \frac{GM^2}{(2R)^2} = \frac{Mv^2}{R} \Rightarrow \frac{GM^2}{4R^2} = \frac{Mv^2}{R} \Rightarrow v = \sqrt{\frac{GM}{4R}}$$

- 28 (b)** The escape velocity is given by

$$v_e = \sqrt{\frac{2GM}{R}} \quad (\text{By putting value})$$

$$v_e = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 7.4 \times 10^{22}}{1740 \times 10^3}} \approx 2.4 \text{ kms}^{-1}$$

- 29 (b)** Consider the diagram as shown below.



Let P be a point at a distance x from 16 kg sphere, where the net gravitation field (and hence, force) due to two spheres is zero, then

$$\frac{16G}{x^2} - \frac{4G}{(30-x)^2} = 0$$

$$\Rightarrow \frac{4}{x^2} = \frac{1}{(30-x)^2}$$

$$\Rightarrow 3600 + 3x^2 - 240x = 0$$

$$\Rightarrow x^2 + 1200 - 80x = 0$$

$$\Rightarrow x = \frac{80 \pm \sqrt{(-80)^2 - 4 \times 1200 \times 1}}{2}$$

$$\Rightarrow x = \frac{80 \pm \sqrt{1600}}{2} = \frac{80 \pm 40}{2} = 60 \text{ m or } 20 \text{ m}$$

As, $x = 60 \text{ m}$ is not possible, so $x = 20 \text{ m}$

- 30 (b)** Orbital velocity of earth, $v_o = \sqrt{gR}$, so the orbital velocity of earth satellite does not depend on mass of the satellite.

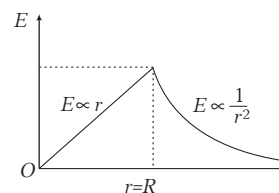
- 31 (d)** Gravitational potential energy of a body of mass m at a height h above the surface of earth is given by $-\frac{GMm}{(R+h)}$.

- 32 (c)** By Kepler's third law of planetary motion, $T^2 \propto r^3$

$$\left[\frac{T_1}{T_2}\right]^2 = \left[\frac{r_1}{r_2}\right]^3$$

- 33 (d)** Mass of a body is its inherent property. It is independent of location of the body. Thus, mass of the body remains same (i.e. M), on the surface of the earth or on the surface of the moon.

- 34 (b)** The dependence of intensity of gravitational field (E) is as shown in the figure.



- 35 (c)** As the work done by the gravitational force in closed path is zero, so it is conservative in nature.

36 (a) As we know that,

Time period of revolution of earth at initial condition,

$$T = 2\pi \sqrt{\frac{R_e}{g}} \quad \dots(i)$$

So, if radius is reduced, $R' = \frac{R_e}{4}$

$$\text{Time period becomes } T' = 2\pi \sqrt{\frac{R_e}{4g}} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\begin{aligned} \frac{T}{T'} &= \frac{2\pi \sqrt{R_e/g}}{2\pi \sqrt{R_e/4g}} = 2 \\ \Rightarrow T' &= \frac{T}{2} = \frac{24}{2} = 12 \text{ h} \end{aligned}$$

37 (d) Potential energy at the earth surface $= -\frac{GMm}{R}$

Potential energy at a height h above the earth's surface

$$= -\frac{GMm}{(R+h)} \quad \dots(i)$$

$$\begin{aligned} \therefore \text{Change in potential energy, } \Delta PE &= -\frac{GMm}{(R+h)} - \left(-\frac{GMm}{R}\right) \\ &= \frac{GMmh}{R(R+h)} \quad \dots(ii) \end{aligned}$$

Substituting the value of $h = 10R$ in Eq. (i), we get

$$\Delta PE = \frac{GMm \times 10R}{R(R+10R)} = \frac{10 GMm}{11R}$$

38 (a) We know that, the potential energy of the satellite,

$$U = -\frac{GM_e m}{R_e}$$

The kinetic energy of the satellite, $K = \frac{1}{2} \frac{GM_e m}{R_e}$

The total energy, $E = U + K = -\frac{GM_e m}{R_e} + \frac{1}{2} \frac{GM_e m}{R_e} = -\frac{1}{2} \frac{GM_e m}{R_e}$

$$\Rightarrow 2E = -\frac{GM_e m}{R_e} \Rightarrow -2E = U$$

So, potential energy $= -2$ (total energy)

$$\Rightarrow PE = -2E$$

39 (c) Given, $R_e = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$, $\pi = 3.14$, $g = 10 \text{ ms}^{-2}$

We know that, the period of revolution of the earth satellite,

$$T = 2\pi \sqrt{\frac{(R_e + h)^3}{gR_e^2}} \quad [\text{if } h \ll R_e, \text{ then } (R_e + h) = R_e]$$

$$\begin{aligned} \text{So, } T &= 2\pi \sqrt{\frac{R_e^3}{gR_e^2}} = 2\pi \sqrt{\frac{R_e}{g}} = 2 \times 3.14 \sqrt{\frac{6.4 \times 10^6}{10}} \\ &= 5.024 \times 10^3 = 5024 \text{ s} \\ &= 83.73 \text{ min} \end{aligned}$$

40 (a) Given, $H = 35800 \text{ km}$ which is height of geo-stationary orbit.

Therefore, $T =$ time period of earth's satellite at this height $= 24 \text{ h}$

41 (a) We know that, the total energy of the satellite,

$$E = -\frac{1}{2} \frac{GM_e m}{R}$$

At a height H from the surface of the earth, the total energy

$$\text{of satellite, } E = -\frac{1}{2} \frac{GM_e m}{(R+H)} \quad \dots(i)$$

and the potential energy of the satellite,

$$U = -\frac{GM_e m}{(R+3R)} = -\frac{GM_e m}{4R} \quad \dots(ii)$$

$$\text{From given condition, } E = U \Rightarrow -\frac{1}{2} \frac{GM_e m}{(R+H)} = -\frac{GM_e m}{4R}$$

$$\Rightarrow \frac{1}{2R+2H} = \frac{1}{4R} \Rightarrow 2H = 4R - 2R \Rightarrow H = R$$

42 (b) Change in potential energy of body,

$$\begin{aligned} \Delta U &= -\frac{GMm}{(R+2R)} - \left(-\frac{GMm}{R}\right) \\ &= -\frac{GMm}{3R} + \frac{GMm}{R} = \frac{2GMm}{3R} = \frac{2}{3} mgR \quad \left(\because g = \frac{GM}{R^2}\right) \end{aligned}$$

43 (d) The resulting gravitational potential,

$$V = -2G \left[\frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right]$$

$$\Rightarrow V = -2G \left[1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} \dots \right]$$

Using Binomial expansion, $V = -2G \left(1 + \frac{1}{2} \right)^{-1}$

$$\Rightarrow V = -\frac{2G}{\left(1 - \frac{1}{2} \right)} = -\frac{2G}{\left(\frac{1}{2} \right)} = -4G$$

44 (a) Value of acceleration due to gravity is $g = \frac{GM}{R^2}$

At poles, R is minimum and at equator R is maximum, hence g is maximum at the poles.

45 (b) Escape velocity, $v_e = \sqrt{2gR}$

46 (a) Here, $g_0 = g' - R\omega^2$

Given, $g_0 = \frac{g}{2}$ and $g' = g$

$$\text{So, } \frac{g}{2} = g - R\omega^2 \Rightarrow \omega = \sqrt{\frac{g}{2R}}$$

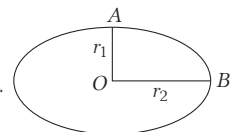
$$\text{We have, } T = 2\pi \left(\frac{r}{v} \right) \Rightarrow T = \frac{2\pi}{\omega} \quad \left(\because \omega = \frac{v}{r} \right)$$

$$\text{Hence, } T = 2\pi \sqrt{\frac{2R}{g}}$$

47 (c) In central force motion, the angular momentum of body remains conserved.

i.e. Angular momentum, $\mathbf{L} = \mathbf{r} \times \mathbf{p} = m\mathbf{v}r$ (At minor and major axis, $\mathbf{v}$ and $\mathbf{r}$ are perpendicular)

So, if v_1 and v_2 be the speed of earth at positions A and B , respectively.



$$\begin{aligned} \text{Then, } mv_1r_1 &= mv_2r_2 \\ \Rightarrow \frac{v_1}{v_2} &= \frac{r_2}{r_1} = \frac{OB}{OA} \Rightarrow \frac{v_1}{v_2} = R \end{aligned}$$

- 48 (a)** We know that gravity, acceleration due to gravity,
 $g = \frac{4}{3} \pi G R d$

Here, given densities be d_1 and d_2 and radii be R_1 and R_2 .

$$\text{So, } \frac{g_1}{g_2} = \frac{\frac{4}{3} \pi G R_1 d_1}{\frac{4}{3} \pi G R_2 d_2} \Rightarrow \frac{g_1}{g_2} = \frac{R_1 d_1}{R_2 d_2}$$

- 49 (d)** Weight of the object at height h above the surface of the earth,
 $w' = \frac{w}{\left(1 + \frac{h}{R}\right)^2} \Rightarrow w' (h = R/2) = \frac{w}{(1 + 1/2)^2}$

$$\Rightarrow w' = w \times \frac{4}{9} = 90 \times \frac{4}{9} = 40 \text{ kg}$$

- 50 (c)** Using energy conservation law,

$$\begin{aligned} (\text{PE})_i + (\text{KE})_i &= (\text{PE})_f + (\text{KE})_f \\ -\frac{GMm}{R} + \frac{1}{2} m (2v_e)^2 &= 0 + \frac{1}{2} m (v')^2 \\ \Rightarrow -m \left(\frac{1}{2} v_e^2 \right) + \frac{1}{2} m (2v_e)^2 &= \frac{1}{2} m (v')^2 \quad \left(\because \frac{GM}{R} = \frac{v_e^2}{2} \right) \\ \Rightarrow v' &= \sqrt{3} v_e = \sqrt{3} \times 11.2 = 19.4 \text{ km/s} \end{aligned}$$

- 51 (c)** Angular momentum = Linear momentum $\times$ Perpendicular distance from the axis of rotation
 = Mass of satellite $\times$ Orbital velocity $\times$ Radius of earth

$$= m \times \sqrt{\frac{GM}{R_0}} \times R_0 = m \sqrt{GM R_0}$$

- 52 (a)** Force due to particle, $F = \frac{GM_e m}{R_p^2}$

$$\text{Here, } F = \frac{GM_p m}{(D_p/2)^2} \Rightarrow F = \frac{4GM_p m}{D_p^2}$$

$$\text{Also, } F = mg$$

$$\therefore \text{Acceleration due to gravity, } g = \frac{F}{m} = \frac{4GM_p}{D_p^2}$$

- 53 (c)** From Kepler's third law, $T^2 \propto r^3$

$$\text{Hence, } T_1^2 \propto r_1^3 \text{ and } T_2^2 \propto r_2^3$$

$$\text{So, } \frac{T_2^2}{T_1^2} = \frac{r_2^3}{r_1^3} = \frac{(3R)^3}{(6R)^3} \text{ or } \frac{T_2^2}{T_1^2} = \frac{1}{8}$$

$$\Rightarrow T_2^2 = \frac{1}{8} T_1^2$$

$$\Rightarrow T_2 = \frac{24}{2\sqrt{2}} = 6\sqrt{2} \text{ h}$$

- 54 (b)** V_B = Potential at B due to A + Potential at A due to B

$$V_B = \frac{-Gm_2}{R} - \left(\frac{-Gm_1}{\sqrt{2}R} \right)$$

$$\Rightarrow V_A = \frac{-Gm_1}{R} - \left(\frac{-Gm_2}{\sqrt{2}R} \right)$$

$$W_{A \rightarrow B} = m (V_B - V_A) = \frac{Gm (m_1 - m_2)}{\sqrt{2}R} (\sqrt{2} - 1)$$

- 55 (a)** According to Kepler's third law,

$$T^2 \propto r^3 \Rightarrow T \propto r^{3/2}$$

$$\text{We have, } \omega = \frac{2\pi}{T} \Rightarrow \omega \propto r^{-3/2}$$

$$\text{Now, } L = mr^2 \omega \Rightarrow L \propto r^2 \text{ and } L \propto \omega$$

$$\Rightarrow L \propto r^2 \cdot r^{-3/2}$$

$$\therefore L \propto \sqrt{r}$$

- 56 (c)** The orbital velocity of a satellite moving around the earth,

$$v_o = \sqrt{gr}$$

The escape velocity is given by $v_e = \sqrt{2gr}$

$$\begin{aligned} \therefore \text{Increase in velocity} &= \sqrt{2gr} - \sqrt{gr} \\ &= (\sqrt{2} - 1) \sqrt{gr} = 0.414 \sqrt{gr} \end{aligned}$$

So, percentage increase in velocity

$$= \frac{0.414}{\sqrt{gr}} \sqrt{gr} \times 100 = 41.4\%$$

- 57 (b)** The energy of artificial satellite at the surface of the earth,

$$E_1 = -\frac{GMm}{R}$$

When the satellite is intended to move in a circular orbit of radius $7R$, then energy of artificial satellite,

$$E_2 = -\frac{1}{2} \frac{GMm}{7R}$$

The minimum energy required,

$$\begin{aligned} E &= E_1 - E_2 = -\frac{GMm}{R} + \frac{1}{2} \left(\frac{GMm}{7R} \right) \\ &= \frac{-14GMm + GMm}{14R} \\ &= -\frac{13GMm}{14R} \end{aligned}$$

- 58 (a)** Using the law of conservation of angular momentum,

Angular momentum of planet at p

= Angular momentum of planet at a

i.e.

$$\mathbf{L}_p = \mathbf{L}_a$$

But $\mathbf{r}_a > \mathbf{r}_p$. So, $\mathbf{r}_a \times \mathbf{L}_a > \mathbf{r}_p \times \mathbf{L}_p$

- 59 (b)** Maximum height attained by body,

$$h = R \left(\frac{v^2}{v_e^2 - v^2} \right)$$

$$\Rightarrow h = R \left[\frac{v_e^2/4}{v_e^2 - v_e^2/4} \right] \quad \left(\because v = \frac{v_e}{2} \right)$$

$$= R \left[\frac{v_e^2/4}{3v_e^2/4} \right] \Rightarrow h = \frac{R}{3}$$

- 60 (b) Acceleration due to gravity,

$$g' = g - \omega^2 R \cos^2 \lambda$$

$$\Rightarrow 0 = g - \omega^2 R \cos^2 60^\circ$$

$$0 = g - \frac{\omega^2 R}{4}$$

$$\Rightarrow \omega = 2\sqrt{\frac{g}{R}} = 2\sqrt{\frac{10}{6400 \times 1000}}$$

$$\text{Angular velocity, } \omega = \frac{1}{400} = 2.50 \times 10^{-3} \text{ rad s}^{-1}$$

- 61 (a) Gravitational pull depends upon the acceleration due to gravity on that planet.

$$M_m = \frac{1}{81} M_e, \quad g_m = \frac{1}{6} g_e$$

$$\text{By the relation, } g = \frac{GM}{R^2}$$

$$\therefore \frac{R_e}{R_m} = \left(\frac{M_e}{M_m} \times \frac{g_m}{g_e} \right)^{1/2} = \left(81 \times \frac{1}{6} \right)^{1/2}$$

$$\therefore R_e = \frac{9}{\sqrt{6}} R_m$$

- 62 (b) Acceleration due to gravity at a height h from earth's surface,

$$g' = \frac{GM}{(R+h)^2}$$

$$\text{Given, } g' = 1\% \text{ of } g = \frac{g}{100}$$

$$\Rightarrow \frac{g}{100} = \frac{GM}{(R+h)^2}$$

$$\frac{(R+h)^2}{100} = \frac{GM}{g} \quad \text{or} \quad \frac{(R+h)^2}{100} = R^2 \quad \left(\because g = \frac{GM}{R^2} \right)$$

$$\therefore R+h = 10R$$

$$h = 9R$$

where, R is the radius of the earth.

- 63 (c) As we know,

$$\text{From Kepler's law, time period of satellite is } T^2 = \frac{4\pi^2 R^3}{GM}$$

$$T_{\text{sat}} = 2\pi \sqrt{\frac{(R+h)^2}{GM_e}}$$

As two satellites of masses m and $9m$ are orbiting a planet of radius R , their time periods will remain same, i.e. in the ratio 1 : 1, because time period of satellite does not depend on its mass.

Simple Harmonic Motion

In this chapter, we are going to study periodic and a basic type of oscillatory motion called simple harmonic motion and its various aspects with the help of some fundamental concepts like period, frequency, amplitude and phase, etc.

PERIODIC AND OSCILLATORY MOTION

On the basis of path traced and various other properties, motion can be classified into several categories.

Two main categories of motion are given below

Periodic motion

A motion which repeats itself after a regular intervals of time is termed as periodic motion. *e.g.* Circular motion, motion of a bouncing ball, motion of planets around the sun, rotation of earth about its polar axis, motion of a pendulum, etc.

For a particle executing periodic motion, its displacement can be represented by a mathematical function of time. One of the simplest periodic function is given by

$$f(t) = A \cos \omega t$$

If the argument of this function ωt is increased by an integral multiple of 2π rad, the value of the function remains the same.

Then, the function $x(t)$ is periodic and its period is given as $T = \frac{2\pi}{\omega}$.

Thus, any function $x(t)$ is periodic with period T , if $x(t) = x(t + T)$.

Further, a linear combination of sine and cosine functions like

$$f(t) = A \sin \omega t + B \cos \omega t \quad \dots(i)$$

is also periodic function with the same time period T .

Take, $A = D \cos \phi$ and $B = D \sin \phi$

Eq. (i) can be written as $f(t) = D \sin(\omega t + \phi)$

where, D and ϕ are constants which are given as

$$D = \sqrt{A^2 + B^2} \text{ and } \tan^{-1} \left(\frac{B}{A} \right), \text{ respectively.}$$

Thus, any periodic function can be expressed as a superposition of sine and cosine functions of different time periods with suitable coefficients.

Inside

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- Spring block system
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5 SHM in some special cases

- Free oscillations
- Damping and damped oscillations
- Maintained oscillations
- Forced oscillations
- Resonance

Oscillatory (or vibratory) motion

A motion in which a particle moves to and fro (back and forth or up and down) about a fixed point after regular interval of time, is termed as oscillatory (or vibratory) motion. The fixed point about which the particle oscillates (or vibrates) is called **mean position** or **equilibrium position**.

Note Every oscillatory motion is periodic, but every periodic motion need not to be oscillatory. e.g. Motion of the planet around the sun is periodic but not oscillatory.

SIMPLE HARMONIC MOTION

Of all the oscillatory motions, the most important and simplest form of motion is **simple harmonic motion** (SHM). In this type of oscillatory motion, displacement, velocity, acceleration and force vary (w.r.t. time) in a way that can be described by either the sine or the cosine function collectively called **sinusoids**.

Note Any oscillatory motion that cannot be described, in terms of single harmonic function, (i.e. sine or cosine function) is called **anharmonic oscillation**.

The particle executing simple harmonic motion oscillates such that its acceleration is always directed towards the mean position and magnitude of acceleration is proportional to the displacement of the particle from mean position.

Mathematically, a simple harmonic motion can be expressed as

$$y = A \sin \omega t = A \sin 2\pi t/T \quad \dots(i)$$

$$\text{or} \quad y = A \cos \omega t = A \cos 2\pi t/T \quad \dots(ii)$$

Here, y = displacement of body from mean position at any instant t ,

A = amplitude or maximum displacement of the body,

ω = angular frequency

and T = time period of SHM.

Types of SHM

SHM are of two types as follows

- (i) **Linear SHM** When a particle moves to and fro about a fixed point (called equilibrium position) along a straight line, then its motion is called linear simple harmonic motion. e.g. Motion of a mass connected to spring.

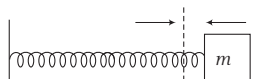


Fig. 11.1 Motion of a spring block system

- (ii) **Angular SHM** When a system oscillates angularly with respect to a fixed axis, then its motion is called angular simple harmonic motion.

e.g. Motion of a bob of simple pendulum.

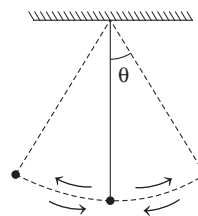


Fig. 11.2 Motion of a pendulum

Some basic terms related to SHM

- (i) **Restoring force** (F) The force acting on the particle (executing SHM) which tends to bring it towards its mean position, is called restoring force. Restoring force is always directed towards mean position or acts in the direction opposite to that of displacement. Restoring force, $F = -kx$

where, k is a constant and x is displacement about mean position.

- (ii) **Time period** (T) The time taken by the particle to complete one oscillation is called time period. Its unit is second.

- (iii) **Frequency** (n) The number of oscillations made by the particle (executing SHM) in one second is called frequency.

$$\text{Frequency, } n = \frac{1}{\text{Time period}(T)}$$

Its unit is cycles per second or hertz (Hz).

- (iv) **Angular frequency** (ω) The rate of change of phase angle of particle (executing SHM) with respect to time is called angular frequency. Its unit is radian/second.

$$\begin{aligned} \text{Angular frequency, } \omega &= \frac{2\pi}{\text{Time period}(T)} \\ &= 2\pi \times \text{Frequency}(n) \end{aligned}$$

- (v) **Amplitude** (A) The maximum displacement of particle from its mean position is defined as amplitude.

- (vi) **Phase** (ϕ) Time varying quantity, which can fully describe displacement and velocity of particle is called phase. Its unit is radian.

In the given graph, two SHM in one second having same amplitude A and same angular frequency ω are shown.

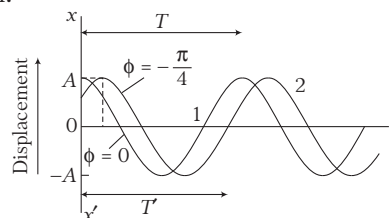


Fig. 11.3 The curves 1 and 2 are for $\phi = 0$ and $-\pi/4$, respectively. Their phase angle ϕ are different.

Example 11.1 A nurse in a hospital, noted for a patient that heart was beating 75 times a minute. Find its frequency and time period.

Sol. Number of oscillations (beats) in 60 s = 75

$$\text{Number of oscillations (beats) in 1 s} = \frac{75}{60} = 1.25$$

$$\therefore \text{Beat frequency of heart, } n = 1.25 \text{ s}^{-1} = 1.25 \text{ Hz}$$

$$\therefore \text{Time period, } T = \frac{1}{n} = \frac{1}{1.25} = 0.8 \text{ s}$$

Necessary conditions to execute SHM

- (i) Motion of particle should be oscillatory.
- (ii) Total mechanical energy of particle should be conserved (Kinetic energy + Potential energy = constant).
- (iii) Extreme position should be well defined.
- (iv) **In linear SHM** The restoring force (or acceleration) acting on the particle should always be proportional to the displacement of the particle and directed towards the equilibrium position.

$$\therefore F \propto -y \text{ or } a \propto -y$$
 Negative sign shows that direction of force and acceleration is towards equilibrium position and y is displacement of particle from equilibrium position.
- (v) **In angular SHM** The restoring torque (or angular acceleration) acting on the particle should always be proportional to the angular displacement of the particle and directed towards the equilibrium position.

$$\therefore \tau \propto -\theta \text{ or } \alpha \propto -\theta$$

Kinematics of SHM

Consider a particle P moving along a circle of radius A with uniform angular velocity ω in anti-clockwise sense. Then, motion of N (foot of the perpendicular drawn from the point P) about O represents simple harmonic motion. In the given figure, the particle P is called **reference particle** and the circle along which the particle P revolves is called **circle of reference**.

Thus, SHM can be defined as, “the straight line motion of the foot of perpendicular drawn from the particle on the diameter of the circle.”

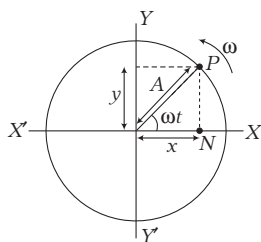


Fig. 11.4 A particle moving along a circle

1. Displacement

The displacement of a particle executing SHM at any instant (t) is given by

$$x = A \sin \omega t \text{ or } x = A \cos \omega t$$

First relation is valid when the time is measured from mean position and second relation is valid when time is measured from extreme position.

As shown in figure, consider a particle P moving in anti-clockwise direction with uniform angular velocity ω along a circle of radius A and centre O . Let at time $t = 0$, it makes an angle ϕ with X -axis. In time t , it covers an angle ωt . Then, at position P' , it makes an angle $(\omega t + \phi)$ with positive X -axis. The position of P' on the axis as the particle moves on the circle is given by

$$x = A \sin(\omega t + \phi) \text{ or } x = A \cos(\omega t + \phi)$$

The term $(\omega t + \phi)$ is called phase and ϕ is called initial phase or ‘epoch’.

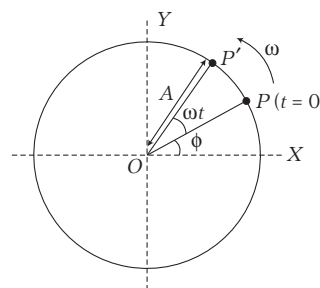


Fig. 11.5 Circle of reference

Example 11.2 An object performs SHM of amplitude 5 cm and time period 4 s. If timing is started when the object is at the centre of the oscillation, i.e. $x = 0$, then calculate

- (i) frequency of oscillation,
- (ii) displacement at 0.5 s.

Sol. (i) Frequency, $n = \frac{1}{T} = \frac{1}{4} = 0.25 \text{ Hz}$

(ii) The displacement equation of object,

$$x = A \sin \omega t$$

$$= A \sin(2\pi nt)$$

So, at $t = 0.5 \text{ s}$,

$$x = 5 \sin(2\pi \times 0.25 \times 0.5)$$

$$= 5 \sin \frac{\pi}{4} = \frac{5}{\sqrt{2}} \text{ cm}$$

Example 11.3 A particle executes SHM from extreme position and covers a distance equal to half of its amplitude in 1 s. Determine the time period of motion.

Sol. For particle starting SHM from extreme position,

$$x = A \cos \omega t$$

At $t = 1 \text{ s}$,

$$x = \frac{A}{2}$$

$$\Rightarrow \frac{A}{2} = A \cos(\omega \times 1)$$

$$\Rightarrow \cos \omega = 1/2$$

$$\Rightarrow \cos \omega = \cos \frac{\pi}{3} \Rightarrow \omega = \frac{\pi}{3}$$

$$\therefore \text{Time period, } T = \frac{2\pi}{\omega} = \frac{2\pi \times 3}{\pi} = 6 \text{ s}$$

Example 11.4 A harmonic oscillation is represented by $x = 0.34 \cos(3000t + 0.74)$, where x and t are in mm and second, respectively. Deduce (i) amplitude (ii) angular frequency (iii) initial phase.

Sol. Given, $x = 0.34 \cos(3000t + 0.74)$

Comparing with general equation $x = A \cos(\omega t + \phi)$, we get

(i) Amplitude, $A = 0.34 \text{ mm} = 0.034 \text{ cm}$

(ii) Angular frequency, $\omega = 3000 \text{ rad s}^{-1}$

(iii) Initial phase, $\phi = 0.74 \text{ rad}$

Example 11.5 Find the equation of simple harmonic motion of a particle whose amplitude is 0.04 and whose frequency is 50 Hz. The initial phase is $\pi/3$. Assume that motion of particle is started from mean position.

Sol. From equation of SHM, $x = A \sin(\omega t + \phi)$

Here, $A = 0.04 \text{ m}$, $\nu = 50 \text{ Hz}$, $\phi = \frac{\pi}{3}$

$$\therefore x = 0.04 \sin\left(2\pi \times 50t + \frac{\pi}{3}\right) \quad (\because \omega = 2\pi\nu)$$

$$\text{or } x = 0.04 \sin\left(100\pi t + \frac{\pi}{3}\right)$$

2. Velocity

The velocity of a particle executing SHM at any instant, is defined as the time rate of change of its displacement at that instant, i.e.

$$v(t) = dx(t)/dt$$

or $\boxed{\text{Velocity, } v = \omega \sqrt{A^2 - x^2}}$

Special Cases

(i) When the particle is at the mean position, i.e. $x = 0$, then its velocity is maximum.

$$\therefore v_{\max} = \omega A$$

(ii) When the particle is at the extreme position, i.e. $x = \pm A$, then its velocity is zero.

$$\therefore v = \omega \sqrt{A^2 - A^2} = 0$$

Therefore, the **velocity of a particle executing SHM is zero at the either of its extreme positions.**

3. Acceleration

The acceleration of the particle executing SHM at any instant is defined as the time rate of change of its velocity at that instant, i.e.

$$a(t) = dv(t)/dt$$

or $\boxed{\text{Acceleration, } a = -\omega^2 x}$

Special Cases

(i) When the particle is at the mean position, i.e. $x = 0$, then acceleration is zero.

$$\therefore a_{\text{mean position}} = 0$$

(ii) When the particle is at the extreme position, i.e. $x = A$, then acceleration is maximum.

$$\therefore a_{\text{(extreme position)}} = -\omega^2 A$$

Graphical representation of displacement, velocity and acceleration in SHM

If a particle is executing SHM and is passing through its positive extreme position ($x = +A$) at $t = 0$, then its displacement equation can be written as

$$x = A \cos \omega t$$

$$\text{Velocity, } v = \frac{dx}{dt} = -\omega A \sin \omega t$$

$$= \omega A \cos\left(\omega t + \frac{\pi}{2}\right)$$

$$\text{Acceleration, } a = \frac{dv}{dt} = -\omega^2 A \cos \omega t$$

$$= \omega^2 A \cos(\omega t + \pi)$$

The following table gives the different values of displacement, velocity and acceleration for various values of t for one complete cycle

Time, t	0	$\frac{T}{4}$	$\frac{T}{2}$	$\frac{3T}{4}$	T
Phase angle, $\omega t = \frac{2\pi}{T}t$	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
Displacement, $x(t)$	$+A$ (max)	0 (min)	$-A$ (max)	0 (min)	$+A$ (max)
Velocity, $v(t)$	0 (min)	$-\omega A$ (max)	0 (min)	$+\omega A$ (max)	0 (min)
Acceleration, $a(t)$	$-\omega^2 A$ (max)	0 (min)	$+\omega^2 A$ (max)	0 (min)	$-\omega^2 A$ (max)

Thus, on the basis of the above mentioned values, following graphs are plotted between $x(t)$ versus t , $v(t)$ versus t and $a(t)$ versus t for different values of t ,

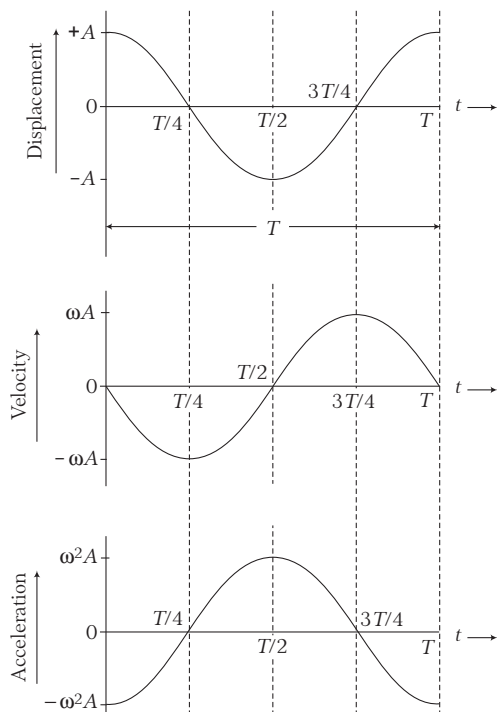


Fig. 11.6 Relation between displacement, velocity and acceleration in SHM

Important points regarding kinematics of SHM

- In linear SHM, the length of SHM path is $2A$.
- The total work done and displacement in one complete oscillation is zero but total traversed length is $4A$.
- The velocity and acceleration varies with the same frequency as displacement.
- Velocity is ahead of displacement by phase angle of $(\pi/2)$ rad, acceleration is ahead of velocity by phase angle of $(\pi/2)$ rad and acceleration is ahead of displacement by phase angle of π rad.

Example 11.6 The particle is executing SHM on a line 4 cm long. If its velocity at mean position is 12 cm/s, then determine its frequency.

Sol. Given, amplitude, $A = 2$ cm

At mean position, velocity is maximum, $v_{\max} = A\omega = 12$

$$\Rightarrow 2 \times 2\pi n = 12 \Rightarrow n = (3/\pi) \text{ Hz}$$

Example 11.7 If a SHM is represented by the equation

$$x = 10 \sin \left(\pi t + \frac{\pi}{6} \right) \text{ in SI units. Determine its amplitude,}$$

time period and maximum velocity $v_{\max}$.

Sol. Comparing the above equation with $x = A \sin (\omega t + \phi)$, we get

$$\text{Amplitude, } A = 10 \text{ m, } \omega = \pi \text{ s}^{-1} \text{ and } \phi = \frac{\pi}{6}$$

$$\therefore \text{Time period, } T = \frac{2\pi}{\omega} \Rightarrow T = 2 \text{ s}$$

$$\therefore \text{Maximum velocity, } v_{\max} = \omega A = 10 \pi \text{ ms}^{-1}$$

Example 11.8 Amplitude of a harmonic oscillator is A .

When particle's velocity of this oscillator is half of maximum velocity, then determine position of particle.

Sol. As, $v = \omega \sqrt{A^2 - x^2}$ but it is given that $v = \frac{v_{\max}}{2} = \frac{A\omega}{2}$

$$\Rightarrow \frac{A\omega}{2} = \omega \sqrt{A^2 - x^2} \Rightarrow A^2 = 4(A^2 - x^2)$$

$$\Rightarrow x^2 = \frac{4A^2 - A^2}{4} \Rightarrow x = \pm \frac{\sqrt{3}}{2} A$$

Example 11.9 A particle executes SHM with a time period of

2 s and amplitude 10 cm. Find the magnitude of its (i) displacement (ii) velocity (iii) acceleration after $(1/6)$ s, starting from mean position.

Sol. (i) Given, $T = 2$ s, $A = 10$ cm

$$\begin{aligned} \text{Displacement, } x &= A \sin \frac{2\pi}{T} t \\ &= 10 \sin \frac{2\pi}{2} \times \frac{1}{6} = 10 \sin \frac{\pi}{6} = 5 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{(ii) Velocity, } v &= \frac{dx}{dt} = A \frac{2\pi}{T} \cos \frac{2\pi}{T} t \\ &= 10 \times \frac{2\pi}{2} \cos \frac{2\pi}{2} \times \frac{1}{6} \\ &= 10 \times 3.14 \times \frac{\sqrt{3}}{2} = 27.19 \text{ cm/s} \end{aligned}$$

$$\begin{aligned} \text{(iii) Acceleration, } a &= \frac{dv}{dt} = \frac{4\pi^2}{T^2} A \sin \frac{2\pi}{T} t \\ &= \frac{4 \times (3.14)^2}{4} \times 10 \times \sin \frac{2\pi}{2} \times \frac{1}{6} \\ &= 49.3 \text{ cm/s}^2 \end{aligned}$$

Example 11.10 A body oscillates with SHM according to the equation, $x = (5 \text{ m}) \cos [(2\pi \text{ rad s}^{-1}) t + \pi/4]$.

At $t = 1.5$ s, calculate the (i) displacement, (ii) speed and (iii) acceleration of the body.

Sol. The angular frequency ω of the body $= 2\pi \text{ rad s}^{-1}$ and its time period, $T = 1$ s

At $t = 1.5$ s,

$$\begin{aligned} \text{(i) Displacement} &= (5 \text{ m}) \cos [(2\pi \text{ rad s}^{-1}) \times 1.5 \text{ s} + \pi/4] \\ &= (5 \text{ m}) \cos [3\pi + \pi/4 \text{ rad}] \\ &= -5 \times 0.707 \text{ m} = -3.535 \text{ m} \end{aligned}$$

(ii) Using $v = -\omega A \sin (\omega t + \phi)$,

Speed of the body

$$= - (5 \text{ m}) (2\pi \text{ s}^{-1}) \sin \left[(2\pi \text{ rad s}^{-1}) \times 1.5 \text{ s} + \frac{\pi}{4} \right]$$

$$= -(5 \text{ m}) (2\pi \text{ s}^{-1}) \sin \left[\left(3\pi + \frac{\pi}{4} \right) \right]$$

$$= 10\pi \times 0.707 \text{ ms}^{-1} = 22.2 \text{ ms}^{-1}$$

(iii) Acceleration of the body

$$= -(2\pi \text{ rads}^{-1})^2 \times \text{Displacement}$$

$$= -(2\pi \text{ rads}^{-1})^2 \times (-3.535 \text{ m}) = 139.41 \text{ ms}^{-2}$$

Example 11.11 A particle executes simple harmonic motion about the point $x = 0$. At time $t = 0$, it has displacement $x = 2 \text{ cm}$ and zero velocity. If the frequency of motion is 0.25 s^{-1} , find (i) the time period, (ii) angular frequency, (iii) the amplitude, (iv) maximum speed, (v) the displacement at $t = 3 \text{ s}$, (vi) the velocity at $t = 3 \text{ s}$ and (vii) maximum acceleration.

Sol. (i) Time period, $T = \frac{1}{f} = \frac{1}{0.25 \text{ s}^{-1}} = 4 \text{ s}$

(ii) Angular frequency, $\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ rad s}^{-1} = 1.57 \text{ rad s}^{-1}$

(iii) Amplitude is the maximum displacement from mean position. Hence, $A = 2 - 0 = 2 \text{ cm}$.

(iv) Maximum speed,

$$v_{\max} = A\omega = 2 \cdot \frac{\pi}{2} = \pi \text{ cms}^{-1} = 3.14 \text{ cms}^{-1}$$

(v) The displacement is given by

$$x = A \sin (\omega t + \phi)$$

Initially at $t = 0$, $x = 2 \text{ cm}$, then

$$2 = 2 \sin \phi$$

$$\text{or} \quad \sin \phi = 1 = \sin 90^\circ$$

$$\text{or} \quad \phi = 90^\circ$$

Now at $t = 3 \text{ s}$,

$$x = 2 \sin \left(\frac{\pi}{2} \times 3 + \frac{\pi}{2} \right) = 0$$

(vi) At $t = 3 \text{ s}$, $x = 0$

As, velocity at $x = 0$ is $v_{\max}$.

$$\therefore v_{\max} = \omega A = \left(\frac{\pi}{2} \right) (2) = 3.14 \text{ cms}^{-1}$$

(vii) Maximum acceleration,

$$a = -\omega^2 A = -(1.57)^2 \times 2 = -4.93 \text{ cms}^{-2}$$

Example 11.12 Two particles move parallel to X-axis about the origin with the same amplitude and frequency. At a certain instant, they are found at distance $A/3$ from the origin on opposite sides but their velocities are found to be in the same direction. What is the phase difference between the two?

Sol. Let equations of two SHM be

$$x_1 = A \sin \omega t \quad \dots (i)$$

$$x_2 = A \sin (\omega t + \phi) \quad \dots (ii)$$

$$\text{Given,} \quad \frac{A}{3} = A \sin \omega t$$

$$\text{and} \quad -\frac{A}{3} = A \sin (\omega t + \phi)$$

$$\text{which gives,} \quad \sin \omega t = \frac{1}{3} \quad \dots (iii)$$

$$\sin (\omega t + \phi) = -\frac{1}{3} \quad \dots (iv)$$

From Eq. (iv), we get

$$\sin \omega t \cos \phi + \cos \omega t \sin \phi = -\frac{1}{3}$$

$$\Rightarrow \quad \frac{1}{3} \cos \phi + \sqrt{1 - \frac{1}{9}} \sin \phi = -\frac{1}{3}$$

Solving this equation, we get

$$\cos \phi = -1, \frac{7}{9}$$

$$\Rightarrow \quad \phi = \pi \quad \text{or} \quad \cos^{-1} \left(\frac{7}{9} \right)$$

Differentiating Eqs. (i) and (ii), we obtain

$$v_1 = A\omega \cos \omega t$$

$$\text{and} \quad v_2 = A\omega \cos (\omega t + \phi)$$

If we put $\phi = \pi$, we find v_1 and v_2 are of opposite signs.

Hence, $\phi = \pi$ is not acceptable.

$$\therefore \quad \phi = \cos^{-1} \left(\frac{7}{9} \right)$$

Force law for SHM

The acceleration of a particle undergoing SHM is

$$a(t) = -\omega^2 x(t)$$

From Newton's second law of motion, force acting on the particle of mass m in SHM is $F(t) = ma(t)$, we get

$$F(t) = -m\omega^2 x(t) = -kx(t)$$

$$\text{where, } k = m\omega^2 \text{ or } \omega = \sqrt{\frac{k}{m}}$$

Thus, in SHM, the force is directly proportional and opposite to the displacement and is always directed towards the mean position.

This force is called **restoring force**.

$$\text{Further, since } \omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$$

$$\therefore \text{ Time period, } T = 2\pi \sqrt{\frac{m}{k}}$$

$$\text{and frequency, } n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

In different types of SHM, the quantities m and k will go on taking different forms and names. In general, m is called **inertial factor** and k is called **spring factor**.

In linear SHM, the general formula for time period is

$$T = 2\pi \sqrt{\frac{\text{Inertial factor}}{\text{Spring factor}}}$$

In linear SHM, the spring factor stands for force per unit displacement and inertial factor for mass of the body executing SHM.

Thus, in linear SHM,

$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{\text{Force}}{\text{Displacement}}}$$

or

$$T = 2\pi \sqrt{\frac{\text{Displacement}}{\text{Acceleration}}}$$

Example 11.13 Frequency of oscillation of a body is 6 Hz when force F_1 is applied and 8 Hz when F_2 is applied. If both forces F_1 and F_2 are applied together, then find out the frequency of oscillation.

Sol. According to question, $F_1 = -k_1x$ and $F_2 = -k_2x$

$$\text{So, } n_1 = \frac{1}{2\pi} \sqrt{\frac{k_1}{m}} = 6 \text{ Hz,}$$

$$n_2 = \frac{1}{2\pi} \sqrt{\frac{k_2}{m}} = 8 \text{ Hz}$$

$$\text{Now, } F = F_1 + F_2 = -(k_1 + k_2)x$$

$$\text{Therefore, } n = \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}}$$

$$\Rightarrow n = \frac{1}{2\pi} \sqrt{\frac{4\pi^2 n_1^2 m + 4\pi^2 n_2^2 m}{m}} \quad (\text{As, } k = 4\pi^2 n^2 m)$$

$$= \sqrt{n_1^2 + n_2^2} = \sqrt{6^2 + 8^2} = 10 \text{ Hz}$$

CHECK POINT 11.1

- The motion which is not simple harmonic is
 - vertical oscillations of a spring
 - motion of simple pendulum
 - motion of a planet around the sun
 - oscillation of liquid column in a U-tube
- Select the wrong statement regarding simple harmonic motion.
 - The body is uniformly accelerated.
 - The velocity of the body changes smoothly at all instants.
 - The amplitude of oscillation is symmetric about the equilibrium position.
 - The frequency of oscillation is independent of amplitude.
- The amplitude and the time period in a SHM is 0.5 cm and 0.4 s, respectively. If the initial phase is $\pi/2$ rad, then the equation of SHM will be
 - $y = 0.5 \sin 5\pi t$
 - $y = 0.5 \sin 4\pi t$
 - $y = 0.5 \sin 2.5\pi t$
 - $y = 0.5 \cos 5\pi t$
- A particle is moving with constant angular velocity along the circumference of a circle. Which of the following statement is true?
 - The particle executes SHM
 - The projection of the particle on any one of the diameters executes SHM
 - The projection of the particle on any of the diameters executes SHM
 - None of the above
- The displacement of a particle in SHM is indicated by equation $y = 10 \sin(20t + \pi/3)$, where y is in metre. The value of time period of vibration will be (in second)
 - $10/\pi$
 - $\pi/10$
 - $2\pi/10$
 - $10/2\pi$
- The equation of a simple harmonic motion is $x = 0.34 \cos(3000t + 0.74)$, where x and t are in mm and second, respectively. The frequency of the motion (in Hz) is
 - 3000
 - $3000/2\pi$
 - $0.74/2\pi$
 - $3000/\pi$
- A body is executing simple harmonic motion with an angular frequency of 2 rad s^{-1} . The velocity of the body at 20 mm displacement when the amplitude of the motion is 60 mm, is
 - 131 mms^{-1}
 - 118 mms^{-1}
 - 113 mms^{-1}
 - 90 mms^{-1}
- The maximum velocity of a simple harmonic motion represented by $y = 3 \sin\left(100t + \frac{\pi}{6}\right)$ is given by
 - 300 units
 - $\frac{3\pi}{6}$ units
 - 100 units
 - $\frac{\pi}{6}$ units
- A body is executing SHM with an amplitude of 0.1 m. Its velocity while passing through the mean position is 3 ms^{-1} . Its frequency (in Hz) is
 - 15
 - $\frac{15}{\pi}$
 - 30π
 - 25π
- Velocity at mean position of a particle executing SHM is v . Velocity of the particle at a distance equal to half of the amplitude will be
 - $\frac{v}{2}$
 - $\frac{v}{\sqrt{2}}$
 - $\frac{\sqrt{3}}{2} v$
 - $\frac{\sqrt{3}}{4} v$

11. A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position, the velocity of the particle is 10 cms^{-1} . The distance of the particle from the mean position when its speed becomes 5 cms^{-1} is

(a) $\sqrt{3} \text{ cm}$ (b) $\sqrt{5} \text{ cm}$
(c) $2\sqrt{3} \text{ cm}$ (d) $2\sqrt{5} \text{ cm}$

12. When a particle executes SHM, there is always a constant ratio between its displacement and

(a) velocity (b) acceleration
(c) mass (d) time period

13. What is the maximum acceleration of the particle executing SHM whose equation of motion is given below?

$$y = 2 \sin \left[\frac{\pi t}{2} + \phi \right] \text{ where } y \text{ is in cm and } t \text{ is in second.}$$

(a) $\frac{\pi}{2} \text{ cms}^{-2}$ (b) $\frac{\pi^2}{2} \text{ cms}^{-2}$
(c) $\frac{\pi}{4} \text{ cms}^{-2}$ (d) $\frac{\pi^2}{4} \text{ cms}^{-2}$

14. The maximum acceleration of a body moving in SHM is a_0 and maximum velocity is v_0 . The amplitude is given by

(a) $\frac{v_0^2}{a_0}$ (b) $a_0 v_0$
(c) $\frac{a_0^2}{v_0}$ (d) $\frac{1}{a_0 v_0}$

15. A particle executing simple harmonic motion has an amplitude of 6 cm. Its acceleration at a distance of 2 cm from the mean position is 8 cms^{-2} . The maximum speed of the particle is

(a) 8 cms^{-1} (b) 12 cms^{-1}
(c) 16 cms^{-1} (d) 24 cms^{-1}

16. The average acceleration of a particle performing SHM over one complete oscillation is

(a) $\frac{\omega^2 A}{2}$ (b) $\frac{\omega^2 A}{\sqrt{2}}$
(c) zero (d) $A\omega^2$

17. In SHM, the acceleration is ahead of velocity by a phase angle

(a) 0° (b) $\frac{\pi}{2}$
(c) π (d) 2π

3. ENERGY IN SHM

The particle executing SHM has both types of energy: potential energy and kinetic energy. When a body is displaced from its equilibrium position by doing work upon it, the body acquires potential energy. However, when the body is released, it begins to move back with a velocity, thus acquiring kinetic energy.

Kinetic energy

The relation for kinetic energy of a particle executing SHM can be expressed in two forms, *i.e.* in terms of displacement and in terms of time.

- (i) **In terms of displacement** If mass of the particle executing SHM is m and its displacement at any

instant is x , then
$$\text{KE} = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

where, A = amplitude of SHM.

As, $\omega = \sqrt{\frac{k}{m}} \Rightarrow m\omega^2 = k \therefore \text{KE} = \frac{1}{2} k (A^2 - x^2)$

Special cases

- Kinetic energy is maximum at mean position, *i.e.* at $x = 0$

$$\text{KE}_{\max} = \frac{1}{2} m \omega^2 A^2$$

- Kinetic energy is minimum (zero) at extreme position, *i.e.* at $x = \pm A$, $\text{KE}_{\min} = \text{zero}$

- (ii) **In terms of time** Kinetic energy of a particle executing SHM at any time t is given by

$$K = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

i.e. kinetic energy of a particle executing SHM is a periodic function of time.

Kinetic energy versus time graph is given below

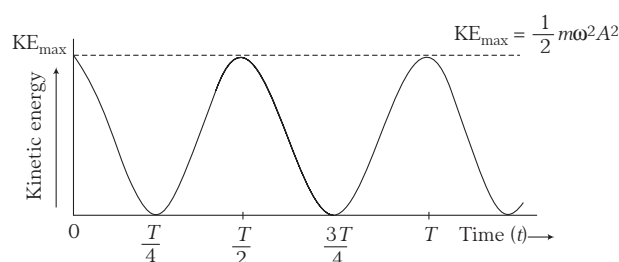


Fig. 11.7 Kinetic energy versus time graph for one oscillation

From the graph 11.7, it is clear that at $t = 0, T/2, T$, etc., kinetic energy has its maximum value. For $x = \pm A$ or $t = \frac{T}{4}, \frac{3T}{4}$, the kinetic energy has its minimum value.

Clearly, twice in each cycle, kinetic energy acquires its peak values. It is a periodic function of time, where the time period equals to $T/2$.

Potential energy

The relation for potential energy of a particle executing SHM in terms of displacement and in terms of time are as follows

- (i) **In terms of displacement** If the displacement of particle executing SHM at any instant is x , then

$$\text{Potential energy, } U = \frac{1}{2} m \omega^2 x^2 = \frac{1}{2} k x^2$$

Special cases

- Potential energy is maximum at extreme position, *i.e.* at $x = \pm A$

$$U_{\max} = \frac{1}{2} m \omega^2 A^2$$

- Potential energy is minimum (zero) at mean position, *i.e.* at $x = 0$, $U_{\min} = 0$

- (ii) **In terms of time** Potential energy of a particle executing SHM at any time t is given by

$$U = \frac{1}{2} m \omega^2 A^2 \sin^2 \omega t$$

Potential energy *versus* time graph is given below

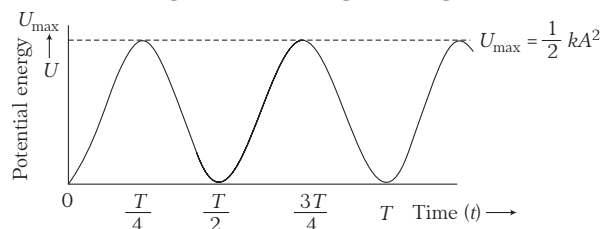


Fig. 11.8 Potential energy versus time graph for one oscillation

From the graph, it is clear that at mean position (*i.e.* $x = 0$ or $t = 0, T/2, T \dots$), potential energy is minimum and at extreme position (*i.e.* $x = \pm A$ or $t = T/4, 3T/4$), its value is maximum. Clearly, twice in each cycle, potential energy acquires its peak values.

Hence, the potential energy is the periodic function of time with a period of $T/2$.

Total energy

Total energy can be obtained by adding potential and kinetic energies. Therefore,

$$E = K + U = \frac{1}{2} m \omega^2 (A^2 - x^2) + \frac{1}{2} m \omega^2 x^2$$

$$E = \frac{1}{2} m \omega^2 A^2 \quad \text{or} \quad E = \frac{1}{2} k A^2$$

which is a **constant** quantity, *i.e.* it remains same at all instant and at all displacement.

Fig. 11.9 shows the variation of kinetic energy (K), potential energy (U) and total energy (E) as a function of time [shown in Fig. 11.9(a)] and displacement [shown in Fig. 11.9(b)] for a particle executing SHM.

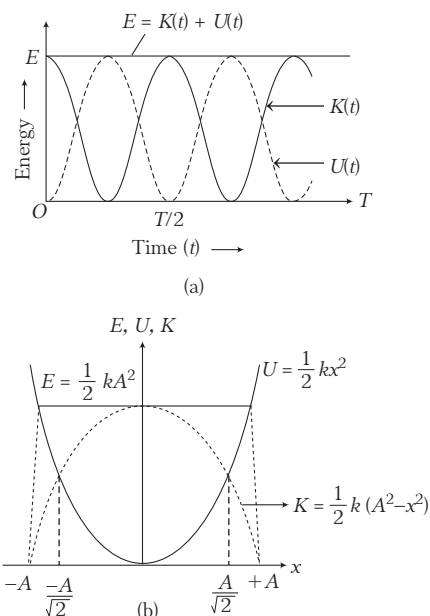


Fig. 11.9 K , U and E as functions of time and displacement x for a particle executing SHM

$\therefore$ From the above graph, it can be concluded that

- The total energy remains constant at all t or x . So, it is shown by straight line parallel to time.
- Also, $U = \frac{1}{2} k x^2$ and $K = \frac{1}{2} k (A^2 - x^2)$, so the graphs of U and K *versus* x are parabolic.
- Both kinetic energy and potential energy reach maximum value twice, during each period of SHM.
- In the course of motion, kinetic energy increases at the expense of potential energy or *vice-versa*. Therefore, we may say that during an oscillation, there is a continuous exchange of kinetic and potential energies.

Important points regarding energy in SHM

- The frequency of oscillation of potential energy and kinetic energy is twice as that of displacement or velocity or acceleration.
- The average value of kinetic energy and potential energy is

$$K_{\text{av}} = U_{\text{av}} = \frac{1}{2} E = \frac{1}{4} k A^2$$

Example 11.14 The potential energy of a particle executing SHM is 2.5 J, when its displacement is half of the amplitude, then determine total energy of particle.

Sol. As, potential energy = $\frac{1}{2}kx^2$

$$\Rightarrow \frac{1}{2}k\frac{A^2}{4} = 2.5 \text{ J} \quad \left(\because x = \frac{A}{2} \right)$$

where, x is the displacement of the particle and A is the amplitude.

$$\Rightarrow \text{Total energy} = \frac{1}{2}kA^2 = 2.5 \times 4 = 10 \text{ J}$$

Example 11.15 A harmonic oscillator of force constant $4 \times 10^6 \text{ Nm}^{-1}$ and amplitude 0.01 m has total energy 240 J. What is maximum kinetic energy and minimum potential energy?

Sol. Given, $k = 4 \times 10^6 \text{ N/m}$, $A = 0.01 \text{ m}$, total energy = 240 J
Maximum kinetic energy

$$= \frac{1}{2}m\omega^2 A^2 = \frac{1}{2}kA^2 = \frac{1}{2} \times 4 \times 10^6 \times (0.01)^2 = 200 \text{ J}$$

Minimum potential energy

$$= \text{Total energy} - \text{Maximum kinetic energy} = 40 \text{ J}$$

Example 11.16 A linear harmonic oscillator has a total mechanical energy of 200 J. Potential energy of it at mean position is 50 J. Find

- the minimum potential energy,
- the maximum kinetic energy,
- the potential energy at extreme positions.

Sol. (i) At mean position, potential energy is minimum.

$$\text{Hence, } U_{\min} = 50 \text{ J}$$

(ii) At mean position, kinetic energy is maximum.

$$\therefore K_{\max} = E - U_{\min} = 200 - 50 = 150 \text{ J}$$

(iii) At extreme positions, kinetic energy is zero and potential energy is maximum.

$$\therefore U_{\max} = E = 200 \text{ J}$$

(From law of conservation of energy)

Example 11.17 A particle executes SHM, at what value of displacement are the kinetic and potential energies equal?

Sol. We know that, kinetic energy = $\frac{1}{2}m\omega^2(A^2 - x^2)$ and

$$\text{potential energy } U = \frac{1}{2}m\omega^2 x^2$$

$$\text{Since, } K = U$$

$$\Rightarrow \frac{1}{2}m\omega^2(A^2 - x^2) = \frac{1}{2}m\omega^2 x^2 \quad \text{or } 2x^2 = A^2$$

$$\text{or } x = \frac{A}{\sqrt{2}} = 0.707A$$

Example 11.18 A particle starts oscillating simple harmonically from its equilibrium position with time period T . Determine ratio of kinetic energy and potential energy of the particle at time $t = \frac{T}{12}$.

Sol. At $t = \frac{T}{12}$, $x = A \sin \frac{2\pi}{T} \times \frac{T}{12} = A \sin \frac{\pi}{6} = \frac{A}{2}$

$$\text{So, kinetic energy, } KE = \frac{1}{2}k(A^2 - x^2) = \frac{3}{4} \times \frac{1}{2}kA^2$$

$$\text{and potential energy, } PE = \frac{1}{2}kx^2 = \frac{1}{4} \times \frac{1}{2}kA^2$$

$$\therefore \frac{KE}{PE} = \frac{3}{1}$$

Example 11.19 A particle of mass 0.2 kg is executing SHM of amplitude 0.2 m. When it passes through the mean position, its kinetic energy is $64 \times 10^{-3} \text{ J}$. Obtain the equation of motion of this particle, if the initial phase of oscillation is $\pi/4$.

Sol. Here, $A = 0.2 \text{ m}$, $\phi = \frac{\pi}{4}$, $KE = 64 \times 10^{-3} \text{ J}$

$$\text{As we know that, } KE = \frac{1}{2}m\omega^2 A^2$$

where, m = mass of particle = 0.2 kg

$$\Rightarrow 64 \times 10^{-3} = \frac{1}{2} \times 0.2 \times \omega^2 \times (0.2)^2$$

$$\Rightarrow \omega^2 = \frac{128 \times 10^{-3}}{0.2 \times 0.2 \times 0.2} \quad \text{or } \omega = 4 \text{ s}^{-1}$$

$\therefore$ Equation of motion can be written as

$$x = A \sin(\omega t + \phi) = 0.2 \sin\left(4t + \frac{\pi}{4}\right)$$

Example 11.20 A particle executes SHM with amplitude A and time period T . When the displacement from the equilibrium position is half the amplitude, what fractions of the total energy are kinetic and potential?

Sol. Total energy, $E = \frac{1}{2}kA^2$

When $x = \pm \frac{A}{2}$, then

$$\text{Potential energy, } U = \frac{1}{2}kx^2 = \frac{1}{2}k \cdot \left(\pm \frac{A}{2}\right)^2$$

$$= \frac{1}{4} \cdot \frac{1}{2}kA^2 = \frac{1}{4}E = 25\% \text{ of } E$$

$$\therefore \text{Kinetic energy, } E - U = E - \frac{1}{4}E = \frac{3}{4}E = 75\% \text{ of } E$$

Example 11.21 The potential energy of a particle oscillating on X -axis is given as, $U = 20 + (x - 2)^2$. Here, U is in joule and x is in metre. Total mechanical energy of the particle is 36 J.

(i) State whether the motion of the particle is simple harmonic or not.

(ii) Find the mean position.

(iii) Find the maximum kinetic energy of the particle.

Sol. (i) $F = -\frac{dU}{dx} = -2(x - 2)$

By assuming $x - 2 = X$, we get

$$F = -2X$$

Since, $F \propto -X$

The motion of the particle is simple harmonic.

- (ii) The mean position of the particle is $X = 0$ or $x - 2 = 0$, which gives $x = 2$ m

- (iii) At $x = 2$ m (mean position),

$$U_{\min} = 20 + (2 - 2)^2 = 20 \text{ J}$$

$\therefore$ Maximum kinetic energy of the particle,

$$K_{\max} = E - U_{\min} = 36 - 20 = 16 \text{ J}$$

Note $U_{\min}$ is 20 J at mean position or at $x = 2$ m.

CHECK POINT 11.2

- A body of mass 1 kg is executing simple harmonic motion. Its displacement y (cm) at t seconds is given by $y = 6 \sin \left(100t + \frac{\pi}{4} \right)$. Its maximum kinetic energy is
(a) 6 J (b) 18 J (c) 24 J (d) 36 J
- The total vibrational energy of a particle in SHM is E . Its kinetic energy at half the amplitude from mean position will be
(a) $E/2$ (b) $E/3$
(c) $E/4$ (d) $3E/4$
- A particle undergoing SHM has the equation $x = A \sin (\omega t + \phi)$, where x represents the displacement of the particle. The kinetic energy oscillates with time period
(a) $\frac{2\pi}{\omega}$ (b) $\frac{\pi}{\omega}$
(c) $\frac{4\pi}{\omega}$ (d) None of these
- A particle executes SHM on a line 8 cm long. Its kinetic energy and potential energy will be equal when its distance from the mean position is
(a) 4 cm (b) 2 cm (c) $2\sqrt{2}$ cm (d) $\sqrt{2}$ cm
- A particle starts SHM from the mean position. Its amplitude is A and total energy E . At one instant, its kinetic energy is $\frac{3E}{4}$. Its displacement at that instant is
(a) $\frac{A}{\sqrt{2}}$ (b) $\frac{A}{2}$ (c) $\sqrt{3} \frac{A}{2}$ (d) zero
- A body executes simple harmonic motion. The potential energy (PE), the kinetic energy (KE) and total energy (TE) are measured as a function of displacement x . Which of the following statement is true?
(a) PE is maximum when $x = 0$
(b) KE is maximum when $x = 0$
(c) TE is zero when $x = 0$
(d) KE is maximum when x is maximum
- In SHM for how many times, potential energy is equal to kinetic energy during one complete period?
(a) 1 (b) 2 (c) 4 (d) 8
- In a simple oscillating pendulum, the work done by the pendulum in one oscillation will be
(a) equal to the total energy of the pendulum
(b) equal to the kinetic energy of the pendulum
(c) equal to the potential energy of the pendulum
(d) zero
- The total energy of a vibrating particle in SHM is E . If its amplitude and time period are doubled, its total energy will be
(a) $16E$ (b) $8E$ (c) $4E$ (d) E
- Energy of particle executing SHM depends upon
(a) amplitude only (b) amplitude and frequency
(c) velocity only (d) frequency only
- A particle of mass 0.10 kg executes SHM with an amplitude 0.05 m and frequency 20 vib/s. Its energy of oscillation is
(a) 2 J (b) 4 J (c) 1 J (d) zero
- Amplitude of a particle in SHM is 6 cm. If instantaneous potential energy is half the total energy, then distance of particle from its mean position is
(a) 3 cm (b) 4.2 cm (c) 5.8 cm (d) 6 cm

SOME SYSTEMS EXECUTING SHM

A pure simple harmonic motion (SHM) is not possible unless some conditions are not applied on it. Let us consider some of the examples of pure simple harmonic motion under certain conditions.

1. Simple pendulum

It consists of a point mass suspended by a massless and inextensible string tied from a rigid support. Consider a simple pendulum as shown in figure.

A small bob of mass m is tied to an inextensible massless string of length l . When the bob is pulled sideways and released, it oscillates in a circular arc with the centre at the point of suspension O .

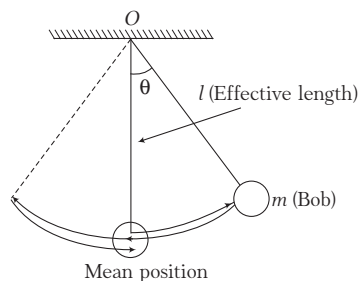


Fig. 11.10 SHM of simple pendulum in one oscillation

Let θ be the angle which the string makes with the vertical, when it is displaced from its mean position.

At mean position $\theta = 0^\circ$ and v is maximum. At extreme position, v is minimum. For the smaller value of θ , the motion of bob is simple harmonic.

Time period of simple pendulum,
$$T = 2\pi\sqrt{\frac{l}{g}}$$

Note that the time period is independent of the mass of the pendulum. This formula is valid only when radius of point mass of pendulum is negligible as compared to length of pendulum.

Variation in time period of simple pendulum in different cases

(i) Effect of acceleration

- Above or below the surface of earth, g decreases. Hence, time period of simple pendulum increases.
- At the centre of earth or in a satellite or in a freely falling lift, the value of g is zero, so time period of simple pendulum becomes infinity. That means, it will not oscillate and remain as it is, where it is left.

- If a simple pendulum is suspended at the ceiling of a lift which is accelerating upwards with acceleration a , then time period of simple pendulum, $T' = 2\pi\sqrt{\frac{l}{g+a}}$,

hence time period will decrease.

- If a simple pendulum is suspended at the ceiling of a lift which is accelerating downwards with acceleration a , then time period of simple pendulum,

$$T' = 2\pi\sqrt{\frac{l}{g-a}} \text{ i.e. Time period will increase.}$$

- If a simple pendulum in a car which is accelerating horizontally with acceleration ' a ', then time period of pendulum will be $T' = 2\pi\sqrt{\frac{l}{g_{\text{eff}}}}$

where $g_{\text{eff}} = \sqrt{g^2 + a^2}$, i.e. time period will decrease.

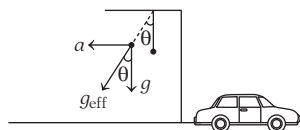


Fig. 11.11

In this case, equilibrium position will not be vertical but will make an angle $\theta = \tan^{-1}(a/g)$ to the vertical in a direction opposite to the acceleration.

(ii) Effect of buoyancy

If the bob of simple pendulum of density ρ is made to oscillate in a liquid of density σ , then its time period will increase as compared to that of air.

$$T' = 2\pi\sqrt{\frac{l}{\left(1 - \frac{\sigma}{\rho}\right)g}}$$

(iii) Effect of change in length

- As $T \propto \sqrt{l}$, so if a person stand on a swing while swinging, l decreases and thus T decreases.
- If the length of simple pendulum (l) is comparable to radius of earth (R), then time period,

$$T = 2\pi\sqrt{\frac{1}{g\left(\frac{1}{l} + \frac{1}{R}\right)}}$$

If length becomes infinite, then time period, $T = 2\pi\sqrt{R/g}$ which is approximately equal to 84.6 min.

If the length is comparable to R , (i.e. $l \approx R$), then

$$\text{time period, } T = 2\pi\sqrt{\frac{R}{2g}} \approx 1 \text{ h}$$

(iv) Effect of temperature

Suppose the thread of simple pendulum is replaced by a metal wire.

Now,
$$T = 2\pi\sqrt{\frac{l}{g}} \text{ and } T' = 2\pi\sqrt{\frac{l(1+\alpha t)}{g}}$$

Here, t represents the rise in temperature and α represents the coefficient of linear expansion.

Now,
$$\frac{T'}{T} = (1 + \alpha t)^{1/2}$$

Using Binomial theorem, $(1+x)^n = 1 + nx + \dots$

(Neglecting higher terms)

$$\frac{T'}{T} = 1 + \frac{1}{2}\alpha t \text{ or } \frac{T' - T}{T} = \frac{1}{2}\alpha t$$

Oscillations of a simple pendulum in an electric field

If the bob of simple pendulum has positive charge q and is made to oscillate in a uniform electric field $\mathbf{E}$, then

- (i) If the field is acting in upward direction, then

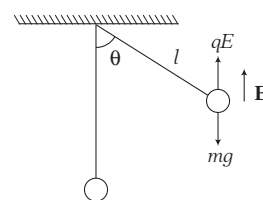


Fig. 11.12 A pendulum in upward electric field

$$\text{Time period of pendulum, } T = 2\pi \sqrt{\frac{l}{g - \frac{qE}{m}}}$$

Hence, the time period will increase.

(ii) If the field is in downward direction, then

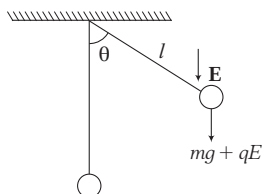


Fig. 11.13 A pendulum in downward electric field

$$\text{Time period of pendulum, } T = 2\pi \sqrt{\frac{l}{g + \frac{qE}{m}}}$$

Hence, the time period will decrease.

(iii) If the direction of electric field is horizontal, then time period of pendulum,

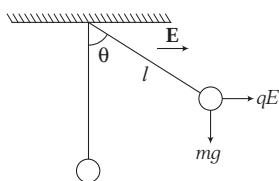


Fig. 11.14 A pendulum in horizontal electric field

$$T = 2\pi \sqrt{\frac{l}{\left[g^2 + \left(\frac{qE}{m}\right)^2\right]^{1/2}}}$$

Hence, the time period will decrease.

Second pendulum

If the time period of a simple pendulum is 2 second, it is called second pendulum.

$$\text{Time period of second pendulum, } T = 2 \text{ s} = 2\pi \sqrt{\frac{l}{g}}$$

As, on the earth's surface, $g = 9.8 \text{ ms}^{-2} \approx \pi^2 \text{ ms}^{-2}$

$\therefore$ The length of second pendulum on earth's surface is 1 m.

Example 11.22 Two pendulums whose lengths differ by 22 cm oscillate at the same place, so that one of them makes 30 oscillations and the other 36 oscillations during same time. Find the lengths of the pendulum.

Sol. According to the question,

$$30 \times 2\pi \sqrt{\frac{l_1}{g}} = 36 \times 2\pi \sqrt{\frac{l_2}{g}}$$

$$\text{or } \frac{l_1}{l_2} = \frac{36 \times 36}{30 \times 30} = \frac{36}{25}$$

$$\text{Also, } l_1 - l_2 = 22 \text{ cm, } \frac{l_1}{l_2} - 1 = \frac{22}{l_2}$$

$$\text{or } \frac{22}{l_2} = \frac{36}{25} - 1 \quad \text{or } \frac{l_2}{22} = \frac{25}{11}$$

$$\text{or } l_2 = 50 \text{ cm}$$

$$\therefore l_1 = 22 + l_2 = (22 + 50) \text{ cm} = 72 \text{ cm}$$

Example 11.23 The bob of a simple pendulum executes SHM in water with a period t . The period of oscillation of the bob in air is t_0 . What is the relation between t and t_0 ?

Given, density of the bob is $\frac{4000}{3} \text{ kg m}^{-3}$. Neglect frictional force of water.

Sol. Here, $\rho = \frac{4000}{3} \text{ kgm}^{-3}$, $\sigma = 1000 \text{ kgm}^{-3}$

$$\therefore g' = \left(1 - \frac{\sigma}{\rho}\right)g = \left(1 - \frac{1000 \times 3}{4000}\right)g = \frac{g}{4}$$

$$\therefore t_0 = 2\pi \sqrt{\frac{l}{g'}} \quad \dots(i)$$

$$\text{and } t = 2\pi \sqrt{\frac{l \times 4}{g}} = 2 \times 2\pi \sqrt{\frac{l}{g}}$$

$$\text{or } t = 2t_0 \quad [\text{from Eq. (i)}]$$

Example 11.24 Determine the period of small oscillations of a simple pendulum, whose bob is suspended by a thread $l = 20 \text{ cm}$ in length, such that it is located in a liquid whose density is $\eta = 3$ times less than that of the bob. The resistance of the liquid is to be neglected.

Sol. Let V = volume of bob,

ρ = density of material of the bob

and g' = effective value of acceleration due to gravity.

Now, effective weight = weight of bob – upthrust of liquid

$$V\rho g' = V\rho g - V \frac{\rho}{\eta} g$$

$$\Rightarrow g' = \left(1 - \frac{1}{\eta}\right)g \quad \text{or } g' = \left(\frac{\eta - 1}{\eta}\right)g$$

$$\text{Again, } T = 2\pi \sqrt{\frac{l}{g'}} \quad \text{or } T = 2\pi \sqrt{\frac{l}{\left(\frac{\eta - 1}{\eta}\right)g}}$$

$$\text{or } T = 2\pi \sqrt{\frac{\eta l}{(\eta - 1)g}}$$

Substituting the given values, we get

$$T = 2 \times \frac{22}{7} \sqrt{\frac{3 \times 20 \times 10^{-2}}{(3 - 1)9.8}} \text{ s} = \frac{44}{7} \times 0.175 \text{ s} = 1.1 \text{ s}$$

Example 11.25 The length of a simple pendulum is 16 cm. It is suspended by the roof of a lift which is moving upwards with an acceleration of 6.2 ms^{-2} . Find the time period of pendulum.

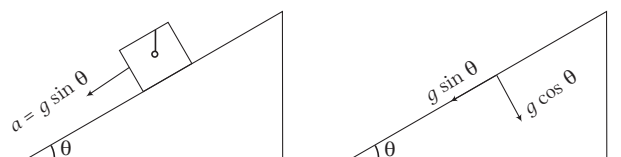
Sol. Given, length of the pendulum, $l = 16 \text{ cm} = 0.16 \text{ m}$

Acceleration of the lift, $a = 6.2 \text{ ms}^{-2}$

$$\begin{aligned} \therefore \text{Time period, } T &= 2\pi \sqrt{\frac{l}{(g+a)}} = 2 \times 3.14 \sqrt{\frac{0.16}{(9.8+6.2)}} \\ &= 6.28 \times \sqrt{\frac{0.16}{16}} = 6.28 \times \sqrt{\frac{1}{100}} = \frac{6.28}{10} \\ &= 0.628 \text{ s} \end{aligned}$$

Example 11.26 A simple pendulum of length l is suspended from the ceiling of a cart which is sliding without friction on an inclined plane of inclination θ . What will be the time period of the pendulum?

Sol. Here, point of suspension has an acceleration, $a = g \sin \theta$ (down the plane). Further, g can be resolved into two components $g \sin \theta$ (along the plane) and $g \cos \theta$ (perpendicular to plane).



$$\therefore g_{\text{eff}} = g \cos \theta \quad (\text{perpendicular to plane})$$

$$\Rightarrow T = 2\pi \sqrt{\frac{l}{|g_{\text{eff}}|}} = 2\pi \sqrt{\frac{l}{g \cos \theta}}$$

Note If $\theta = 0^\circ$, $T = 2\pi \sqrt{\frac{l}{g}}$ which is quite obvious.

Example 11.27 A bob of simple pendulum is suspended by a metallic wire. If α is the coefficient of linear expansion and $d\theta$ is the change in temperature, then prove that percentage change in time period is $50\alpha d\theta$.

Sol. With change in temperature $d\theta$, the effective length of wire becomes $l' = l(1 + \alpha d\theta)$

$$T' = 2\pi \sqrt{\frac{l'}{g}} \quad \text{and} \quad T = 2\pi \sqrt{\frac{l}{g}}$$

$$\text{Hence, } \frac{T'}{T} = \sqrt{\frac{l'}{l}} = (1 + \alpha d\theta)^{1/2}$$

$$= 1 + \frac{1}{2} \alpha d\theta$$

Percentage increase in time period

$$= \left(\frac{T' - T}{T} \right) \times 100 = \left(\frac{T'}{T} - 1 \right) \times 100$$

$$= \left(1 + \frac{\alpha d\theta}{2} - 1 \right) \times 100$$

$$= 50\alpha d\theta$$

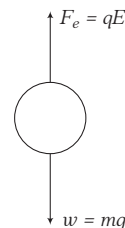
Example 11.28 A simple pendulum consists of a small sphere of mass m suspended by a thread of length l . The sphere carries a positive charge q . The pendulum is placed in a uniform electric field of strength E directed vertically upwards. With what period will pendulum oscillate, if the electrostatic force acting on the sphere is less than the gravitational force?

Sol. The two forces acting on the bob are shown in figure.

$$g_{\text{eff}} \text{ in this case will be } \frac{w - F_e}{m}$$

$$\text{or } g_{\text{eff}} = \frac{mg - qE}{m} = g - \frac{qE}{m}$$

$$\therefore T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}} = 2\pi \sqrt{\frac{l}{g - \frac{qE}{m}}}$$



2. Spring block system

Let us consider a massless spring placed on a frictionless horizontal surface. Its one end is attached to a rigid support and other end to a body of mass m . When it is pulled on one side through a small distance x and released, it moves to and fro about a mean position. Let $x = 0$ be the position of the centre of the body when the spring is at mean position.

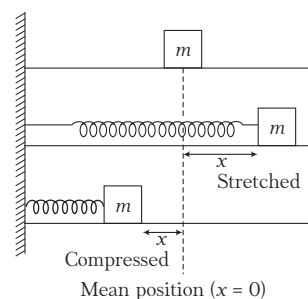


Fig. 11.15

By Hooke's law, a restoring force F is developed in the spring on stretching and compressing it through a small displacement x ,

$$F = -kx$$

$$\Rightarrow a = -\frac{k}{m}x$$

The time period of such oscillations is given by

$$T = 2\pi \sqrt{\frac{m}{k}}$$

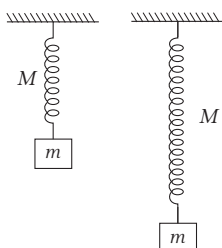
where, k is force constant of spring and m is mass of the object connected to the spring.

If the stretch in a vertically loaded spring is y , then for

equilibrium of mass m , $T = 2\pi \sqrt{y/g}$

Special cases

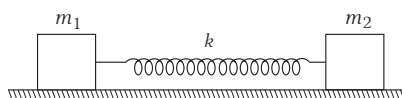
- (i) If the spring has a mass M and a mass m is suspended from it, then time period is given by

**Fig. 11.16**

$$T = 2\pi\sqrt{\frac{M_{\text{eff}}}{k}}, \text{ where } M_{\text{eff}} = m + \frac{M}{3}$$

- (ii) If two masses m_1 and m_2 are connected by a spring of spring constant k and made to oscillate on horizontal surface. Then, time period is given by

$$T = 2\pi\sqrt{\frac{\mu}{k}}$$

**Fig. 11.17** Two blocks connected by a spring

Here, μ = reduced mass = $\frac{m_1 m_2}{m_1 + m_2}$

Example 11.29 A 2.5 kg collar is attached to a spring of spring constant 250 Nm^{-1} . It slides without friction over a horizontal surface. It is displaced from its equilibrium position by 20 cm and released. Calculate the period of oscillation and the maximum speed.

Sol. The period of oscillation,

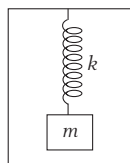
$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2.5}{250}}$$

$$= \frac{2 \times 3.14}{10} = 0.63 \text{ s}$$

Maximum speed, $v_m = A\omega = 0.2\sqrt{\frac{k}{m}} = 0.2 \times \sqrt{\frac{250}{2.5}}$

$$= 0.2 \times 10 = 2 \text{ ms}^{-1}$$

Example 11.30 A spring mass system is hanging from the ceiling of an elevator in equilibrium. The elevator suddenly starts accelerating upwards with acceleration a , find



- (i) the frequency and
(ii) the amplitude of the resulting SHM.

Sol. (i) Frequency = $2\pi\sqrt{m/k}$

(Frequency is independent of g in spring mass system)

(ii) Initial extension in spring in equilibrium = $\frac{mg}{k}$

Extension in spring in equilibrium when lift starts accelerating upwards = $\frac{m(g+a)}{k}$

$$\therefore \text{Amplitude} = \frac{m(g+a)}{k} - \frac{mg}{k} = \frac{ma}{k}$$

Example 11.31 A body of mass m attached to a spring which is oscillating with time period 4 s. If the mass of the body is increased by 4 kg, its time period increases by 2 s. Determine the value of initial mass m .

Sol. In 1st case, $T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow 4 = 2\pi\sqrt{\frac{m}{k}}$... (i)

In 2nd case, $6 = 2\pi\sqrt{\frac{m+4}{k}}$... (ii)

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{4}{6} = \sqrt{\frac{m}{m+4}} \Rightarrow \frac{16}{36} = \frac{m}{m+4} \Rightarrow m = 3.2 \text{ kg}$$

Example 11.32 A block with a mass of 3 kg is suspended from an ideal spring having negligible mass and stretches the spring upto 0.2 m.

- (i) What is the force constant of the spring?
(ii) What is the period of oscillation of the block, if it is pulled down and released?

Sol. (i) In equilibrium, $kl = mg$

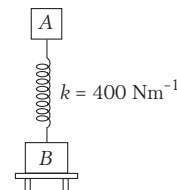
$$\therefore k = \frac{mg}{l}$$

Substituting the given values, we get

$$k = \frac{(3)(9.8)}{0.2} = 147 \text{ Nm}^{-1}$$

(ii) $T = 2\pi\sqrt{\frac{l}{g}} = 2\pi\sqrt{\frac{0.2}{9.8}} = 0.897 \text{ s}$

Example 11.33 Two rigid bodies A and B of masses 1 kg and 2 kg respectively are rigidly connected to a spring of force constant 400 Nm^{-1} . The body B rests on a horizontal table. From the rest position, the body A is compressed by 2 cm and then released. Deduce (i) the frequency of oscillation, (ii) total oscillation energy, (iii) the amplitude of the harmonic vibration of the reaction of the table on body B.



Sol. Since, it is the body A which is oscillating, $m = 1 \text{ kg}$

(i) Frequency of oscillation, $\nu = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

or $\nu = \frac{1}{2\pi}\sqrt{\frac{400}{1}} = \frac{10}{\pi} = 3.185 \text{ Hz}$

(ii) Total oscillation energy, $E = \frac{1}{2} mA^2 \omega^2$

or $E = \frac{1}{2} mA^2 \left(\frac{k}{m} \right) = \frac{1}{2} kA^2$
 $= \frac{1}{2} \times 400 \times (0.02)^2 \quad (\because A = 2 \text{ cm} = 0.02 \text{ m})$
 $= 0.08 \text{ J}$

(iii) Total force acting on the table

$$= (1 + 2) \text{ kg-wt} = 3 \text{ kg-wt} = 3 \times 9.8 \text{ N} = 29.4 \text{ N}$$

This is the mean upward reaction. Now, due to oscillations, the maximum tension developed in the spring is given by

$$F = kA = 400 \times 0.02 \text{ N} = 8 \text{ N}$$

Therefore, the net reaction of the table will vary from $(29.4 + 8) \text{ N}$ to $(29.4) \text{ N}$.

Example 11.34 A block with mass M attached to a horizontal spring with force constant k is moving with simple harmonic motion having amplitude A_1 . At the instant when the block passes through its equilibrium position, a lump of putty with mass m is dropped vertically on the block from a very small height and sticks to it.

(i) Find the new amplitude and period.

(ii) Repeat part (i) for the case in which the putty is dropped on the block when it is at one end of its path.

Sol. (i) Before the lump of putty is dropped the total mechanical energy of the block and spring is $E_1 = \frac{1}{2} kA_1^2$

Since, the block is at the equilibrium position, $U = 0$ and the energy is purely kinetic. Let v_1 be the speed of the block at the equilibrium position, we get

$$E_1 = \frac{1}{2} Mv_1^2 = \frac{1}{2} kA_1^2 \Rightarrow v_1 = \sqrt{\frac{k}{M}} A_1$$

During the process, momentum of the system in horizontal direction is conserved. Let v_2 be the speed of the combined mass, then $(M + m)v_2 = Mv_1$

$$\therefore v_2 = \frac{M}{M + m} v_1 = \frac{M}{M + m} \left(\sqrt{\frac{k}{M}} A_1 \right)$$

Now, let A_2 be the amplitude afterwards. Then,

$$E_2 = \frac{1}{2} kA_2^2 = \frac{1}{2} (M + m)v_2^2$$

Substituting the value of v_2 , we get

$$A_2 = A_1 \sqrt{\frac{M}{M + m}}$$

Note $E_2 < E_1$, as some energy is lost into heating up the block and putty. Further, $T_2 = 2\pi \sqrt{\frac{M + m}{k}}$.

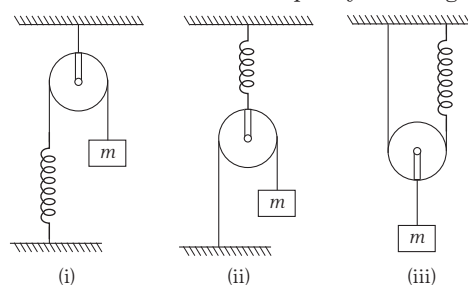
(ii) When the putty drops on the block, the block is instantaneously at rest. All the mechanical energy is stored in the spring as potential energy. Again, the momentum in horizontal direction is conserved during the process. But now it is zero just before and after putty is dropped. So, in this case, adding the extra mass of the putty has no effect on the mechanical energy, i.e.

$$E_2 = E_1 = \frac{1}{2} kA_1^2$$

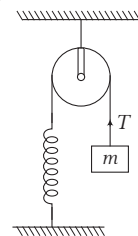
and the amplitude is still A_1 .

$$\text{Thus, } A_2 = A_1 \text{ and } T_2 = 2\pi \sqrt{\frac{M + m}{k}}.$$

Example 11.35 In the following arrangements, block is slightly displaced vertically downwards from its equilibrium position and released. Find time period of vertical oscillations. Assume the pulley to be light.



Sol. (i) Let at some instant extension of spring is x , tension in the string $T = kx$



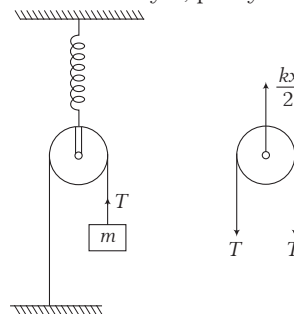
Restoring force, $F = T = -kx$

(Hooke's law)

where, k = spring constant

$$\text{Time period, } T_0 = 2\pi \sqrt{\frac{m}{k}}$$

(ii) Let block move down by x , pulley will move by $x/2$.



$$2T = \frac{kx}{2} \Rightarrow T = \frac{kx}{4}$$

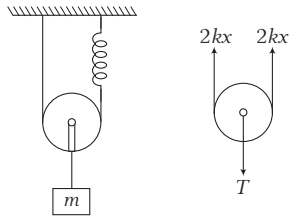
Restoring force, $F = -\frac{k}{4}x$

Comparing it with equation $F = -kx$, we get

$$k' = k/4 \therefore T_0 = 2\pi \sqrt{\frac{m}{k'}}$$

$$\Rightarrow T_0 = 2\pi \sqrt{\frac{m}{k/4}} = 4\pi \sqrt{\frac{m}{k}}$$

- (iii) If block moves by x , pulley also moves x and hence, spring is stretched by $2x$.



Tension produced in spring, $T = 4kx$

Restoring force, $F = -4kx$

Comparing it with equation $F = -k'x$, we get

$$k' = +4k$$

$$\therefore T_0 = 2\pi\sqrt{\frac{m}{k'}} \Rightarrow T_0 = 2\pi\sqrt{\frac{m}{4k}} = \pi\sqrt{\frac{m}{k}}$$

Oscillations of spring combination

Series combination Two springs of spring constants k_1 and k_2 are joined in series as shown in figure.

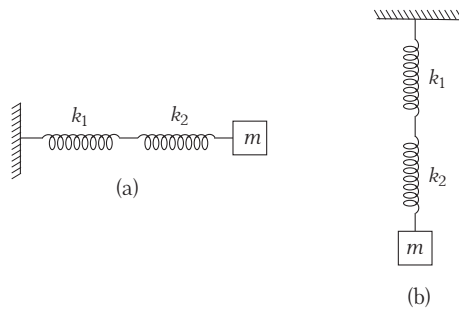


Fig. 11.18 Series combination of springs

Then, resultant spring constant of the combination is given by

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{k_1 + k_2}{k_1 k_2} \Rightarrow k_s = \frac{k_1 k_2}{k_1 + k_2}$$

and hence,

$$T = 2\pi\sqrt{\frac{m}{k_s}} = 2\pi\sqrt{\frac{(k_1 + k_2)m}{k_1 k_2}}$$

Parallel combination Two springs of spring constants k_1 and k_2 are joined in parallel as shown in figure.

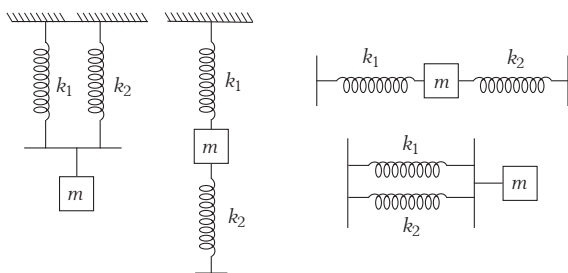


Fig. 11.19 Parallel combination of springs

Then, spring constant, $k_p = k_1 + k_2$

$$\text{Time period, } T = 2\pi\sqrt{\frac{m}{k_p}} = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$$

Important points regarding spring block system

- If the length of the spring is made n times, then the effective force constant becomes $1/n$ times and the time period becomes $\sqrt{n}$ times.
- If a spring of spring constant k is divided into n equal parts, the spring constant of each part becomes nk and time period becomes $\frac{1}{\sqrt{n}}$ times.
- The force constant of a stiffer spring is more than that of a soft spring.

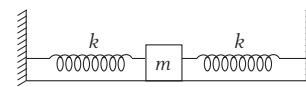
Example 11.36 One body is suspended from a spring of length l , spring constant k and has time period T . Now, if spring is divided in two equal parts which are joined in parallel and the same body is suspended from this arrangement, then determine new time period.

Sol. When a spring of spring constant k is divided into 2 parts, the spring constant of each part becomes $2k$.

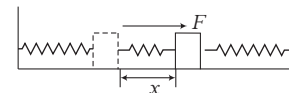
$\therefore$ Spring constant in parallel combination $k' = 2k + 2k = 4k$

$$\begin{aligned} \therefore T &= 2\pi\sqrt{\frac{m}{k'}} = 2\pi\sqrt{\frac{m}{4k}} \\ &= 2\pi\sqrt{\frac{m}{k}} \times \frac{1}{\sqrt{4}} = \frac{T}{\sqrt{4}} = \frac{T}{2} \end{aligned}$$

Example 11.37 Two identical springs of spring constant k are attached to a block of mass m and to fixed supports as shown in figure. Show that when the mass is displaced from its equilibrium position on either side, it executes a simple harmonic motion. Find the period of oscillations.



Sol. Let the mass be displaced by a small distance x to the right side of the equilibrium position as shown in figure.



Under this situation, the spring on the left side gets elongated by a length equal to x and that on the right side gets compressed by the same length. The forces acting on the mass are then

$F_1 = -kx$ (force exerted by the spring on the left side, trying to pull the mass towards the mean position)

$F_2 = -kx$ (force exerted by the spring on the right side, trying to push the mass towards the mean position)

The net force F acting on the mass is then given by, $F = -2kx$. Hence, the force acting on the mass is proportional to the displacement and is directed towards the mean position, therefore the motion executed by the mass is simple

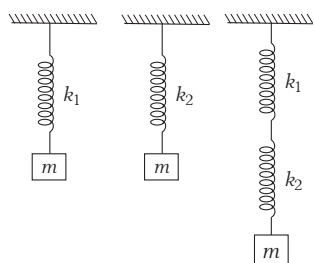
harmonic. The time period of oscillations is $T = 2\pi\sqrt{\frac{m}{2k}}$.

Example 11.38 Periodic time of oscillation T_1 is obtained when a mass is suspended from a spring and if another spring is used with same mass, then periodic time of oscillation is T_2 . Now, if this mass is suspended from series combination of above springs, then calculate the time period.

Sol. As, $T_1 = 2\pi\sqrt{\frac{m}{k_1}}$

$$\Rightarrow T_1^2 = 4\pi^2 \frac{m}{k_1} \Rightarrow k_1 = \frac{4\pi^2 m}{T_1^2}$$

$$\text{and } T_2 = 2\pi\sqrt{\frac{m}{k_2}} \Rightarrow T_2^2 = 4\pi^2 \frac{m}{k_2} \Rightarrow k_2 = \frac{4\pi^2 m}{T_2^2}$$



Now,

where,

$$\begin{aligned} T &= 2\pi\sqrt{\frac{m}{k'}} \\ \frac{1}{k'} &= \frac{1}{k_1} + \frac{1}{k_2} \\ \Rightarrow k' &= \frac{k_1 k_2}{k_1 + k_2} = \frac{\left(\frac{4\pi^2 m}{T_1^2}\right)\left(\frac{4\pi^2 m}{T_2^2}\right)}{\frac{4\pi^2 m}{T_1^2} + \frac{4\pi^2 m}{T_2^2}} \\ \Rightarrow k' &= \frac{4\pi^2 m \left(\frac{4\pi^2 m}{T_1^2 T_2^2}\right)}{4\pi^2 m \left(\frac{1}{T_1^2} + \frac{1}{T_2^2}\right)} = \frac{4\pi^2 m}{T_1^2 + T_2^2} \\ \therefore T &= 2\pi\sqrt{\frac{m}{k'}} = 2\pi\sqrt{\frac{m}{\frac{4\pi^2 m}{T_1^2 + T_2^2}}} = \sqrt{T_1^2 + T_2^2} \end{aligned}$$

3. The physical pendulum

Any rigid body suspended from a rigid support capable of oscillating about a horizontal axis passing through it constitutes a physical pendulum (or compound pendulum). The figure given below shows a rigid body pivoted at O , so

that it can oscillate without friction about an axis passing through O .

In equilibrium, the centre of gravity (G) is directly below O . In the position shown in figure, the body is displaced from equilibrium by an angle θ . The distance from O to the centre of gravity is l . Let, the moment of inertia of the body about the axis of rotation through O is I and the total mass is m . In the displaced position, the weight mg causes a restoring torque, $\tau = -(mg)(l \sin \theta)$

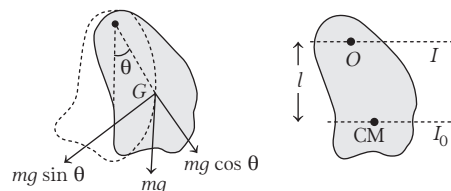


Fig. 11.20 Physical pendulum

The negative sign shows that the restoring torque is clockwise when the displacement is counterclockwise and vice-versa.

For small oscillations,

$$\sin \theta \approx \theta \text{ and } \Sigma \tau = I\alpha \Rightarrow -(mgl) \theta = I\alpha$$

As α is proportional to $-\theta$, the motion is simple harmonic, the time period of which is

$$T = 2\pi \sqrt{\left| \frac{\theta}{\alpha} \right|} \text{ or } T = 2\pi \sqrt{\frac{I}{mgl}}$$

Note Time period of some common physical pendulum

- (i) Bar, $T = 2\pi\sqrt{\frac{2l}{3g}}$ (ii) Ring, $T = 2\pi\sqrt{\frac{2R}{g}}$
(iii) Disc, $T = 2\pi\sqrt{\frac{3R}{2g}}$

Example 11.39 A uniform circular disc of radius R oscillates in a vertical plane about a horizontal axis. Find the distance of the axis of rotation from the centre for which the period is minimum. What is the value of this period?

Sol. The time period of a compound pendulum is minimum when its length is equal to the radius of gyration about its centre of gravity, i.e. $l = K$

Since, the moment of inertia of a disc about an axis perpendicular to its plane and passing through its centre is

$$\text{equal to, } I = MK^2 = \frac{1}{2}MR^2 \Rightarrow K = \frac{R}{\sqrt{2}}$$

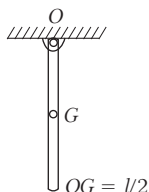
Thus, the disc will oscillate with the minimum time period when the distance of the axis of rotation from the centre is $\frac{R}{\sqrt{2}}$.

And the value of this minimum time period will be

$$T_{\min} = 2\pi\sqrt{\frac{2R/\sqrt{2}}{g}} = 2\pi\sqrt{\frac{\sqrt{2}R}{g}} \text{ or } T_{\min} \approx 2\pi\sqrt{\frac{1.414R}{g}}$$

Example 11.40 Find the period of small oscillations of a uniform rod with length l , pivoted at one end.

Sol. Time period of a uniform rod which is same as physical pendulum,



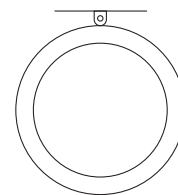
$$T = 2\pi \sqrt{\frac{I_0}{mg(OG)}}$$

Here, $I_0 = \frac{1}{3}ml^2$ and $OG = \frac{l}{2}$

$$\therefore T = 2\pi \sqrt{\frac{\left(\frac{1}{3}ml^2\right)}{(m)(g)\left(\frac{l}{2}\right)}}$$

or $T = 2\pi \sqrt{\frac{2l}{3g}}$

Example 11.41 A ring of radius r is suspended from a point on its circumference. Determine its angular frequency of small oscillations.



Sol. It is a physical pendulum, the time period of which is,

$$T = 2\pi \sqrt{\frac{I}{mgl}}$$

Here, I = moment of inertia of the ring about point of suspension
 $= mr^2 + mr^2 = 2mr^2$

and l = distance of point of suspension from centre of gravity
 $= r$

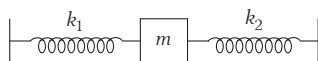
$$\therefore T = 2\pi \sqrt{\frac{2mr^2}{mgr}} = 2\pi \sqrt{\frac{2r}{g}}$$

Hence, angular frequency, $\omega = \frac{2\pi}{T}$ or $\omega = \sqrt{\frac{g}{2r}}$

CHECK POINT 11.3

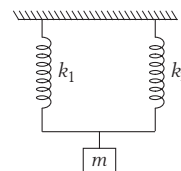
- If the metal bob of a simple pendulum is replaced by a wooden bob, then its time period will
 - increase
 - decrease
 - remains the same
 - may increase or decrease
- The length of a simple pendulum is $(39.2/\pi^2)\text{m}$. If $g = 9.8 \text{ m/s}^2$, the value of time period is
 - 4 s
 - 8 s
 - 2 s
 - 3 s
- The length of a simple pendulum is increased four times of its initial value, its time period with respect to its previous value will
 - become twice
 - not be different
 - be halved
 - be $\sqrt{2}$ times
- The length of a simple pendulum executing simple harmonic motion is increased by 21%. The percentage increase in the time period of the pendulum of increased length is
 - 11%
 - 21%
 - 42%
 - 10.5%
- A cabin is falling freely under gravity, what is the time period of a pendulum attached to its ceiling?
 - zero
 - ∞
 - 1 s
 - 2 s
- A simple pendulum is suspended from the roof of a trolley which moves in a horizontal direction with an acceleration a , then the time period T is given by $T = 2\pi \sqrt{l/g'}$, where g' is equal to
 - g
 - $g - a$
 - $g + a$
 - $\sqrt{g^2 + a^2}$
- A man measures the period of a simple pendulum inside a stationary lift and finds it to be T second. If the lift accelerates upwards with an acceleration $g/4$, then the period of the pendulum will be
 - T
 - $\frac{T}{4}$
 - $\frac{2T}{\sqrt{5}}$
 - $2T\sqrt{5}$
- A simple pendulum is made of a body which is a hollow sphere containing mercury suspended by means of a wire. If a little mercury is drained off, the period of pendulum will
 - remains unchanged
 - increase
 - decrease
 - become erratic
- The time period of a simple pendulum of infinite length is (R = radius of earth)
 - infinite
 - $2\pi \sqrt{\frac{R}{g}}$
 - $2\pi \sqrt{\frac{g}{R}}$
 - $\frac{1}{2g} \sqrt{\frac{R}{g}}$

10. An object of mass m is suspended from a spring and it executes SHM with frequency n . If the mass is increased 4 times, the new frequency will be
 (a) $2n$ (b) $n/2$ (c) n (d) $n/4$
11. A mass m is suspended from a spring. Its frequency of oscillation is f . The spring is cut into two halves and the same mass is suspended from one of the two pieces of the spring. The frequency of oscillation of the mass will be
 (a) $\sqrt{2} f$ (b) $\frac{f}{2}$ (c) f (d) $2f$
12. Three masses 0.1 kg, 0.3 kg and 0.4 kg are suspended at end of a spring. When the 0.4 kg mass is removed, the system oscillates with a period 2 s. When the 0.3 kg mass is also removed, the system will oscillate with a period
 (a) 1 s (b) 2 s (c) 3 s (d) 4 s
13. Time period of a spring mass system is T . If this spring is cut into two parts whose lengths are in the ratio 1 : 3 and the same mass is attached to the longer part, the new time period will be
 (a) $\sqrt{\frac{3}{2}} T$ (b) $\frac{T}{\sqrt{3}}$ (c) $\frac{\sqrt{3} T}{2}$ (d) $\sqrt{3} T$
14. If both spring constants k_1 and k_2 are increased to $4k_1$ and $4k_2$ respectively, what will be the new frequency, if f was the original frequency?



- (a) f (b) $2f$ (c) $\frac{1}{2} f$ (d) $4f$

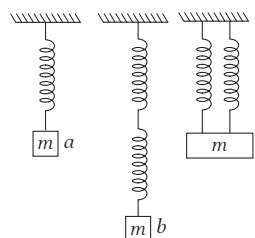
15. The spring constants of two springs of same length are k_1 and k_2 as shown in figure. If an object of mass m is suspended and set in vibration, the time period will be



- (a) $2\pi \sqrt{\frac{mk_1}{k_2}}$ (b) $2\pi \sqrt{\frac{m}{k_1 k_2}}$
 (c) $2\pi \sqrt{\frac{m}{k_1 - k_2}}$ (d) $2\pi \sqrt{m/(k_1 + k_2)}$

16. Five identical springs are used in the three configurations as shown in figure.

The time periods of vertical oscillations in configurations (a), (b) and (c) are in the ratio



- (a) $1 : \sqrt{2} : \frac{1}{\sqrt{2}}$ (b) $2 : \sqrt{2} : \frac{1}{\sqrt{2}}$ (c) $\frac{1}{\sqrt{2}} : 2 : 1$ (d) $2 : \frac{1}{\sqrt{2}} : 1$

SHM IN SOME SPECIAL CASES

Here, we will discuss motion of some objects being reduced to simple harmonic motion under specific condition

- (i) **Vertical motion of a straight wire** If a mass m is suspended from a massless wire of length l , cross-section area A and Young's modulus Y and is pulled along the length of the wire, then restoring force will be developed by the elasticity of the wire.

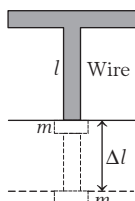


Fig. 11.21 Mass suspended from a wire

Here, restoring force is linear, so motion is simple harmonic in nature, where time period is given by

$$T = 2\pi \sqrt{\frac{ml}{YA}}$$

- (ii) **Motion of a liquid in U-shape tube** Let an incompressible and non-viscous liquid is filled in a U-tube of uniform cross-section. If the liquid is pressed in one limb, then it starts to perform SHM along limb of the tube, where time period is given by

$$T = 2\pi \sqrt{\frac{h}{g}}$$

Here, h is height of liquid in one limb.

- (iii) **Motion of ball in bowl** If a small ball of mass m is placed at a small distance from O inside the smooth concave surface of radius R of a bowl and released, then it will oscillate about O .

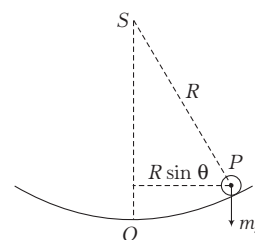


Fig. 11.22 A ball is moving in a bowl

Time period of such oscillations is

$$T = 2\pi\sqrt{\frac{R}{g}}$$

- (iv) **Motion of a ball in a tunnel through the earth** If a ball moves through a tunnel along a diameter of earth, then due to gravitational force between ball and earth a restoring force is set up, due to which the ball performs SHM, whose time period is given by

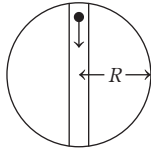


Fig. 11.23 A ball in a tunnel through the earth

$$T = 2\pi\sqrt{R/g}$$

where, R = radius of earth.

Example 11.42 A liquid of mass m is set into oscillations in a U-tube of cross-section A . Its time period recorded is T , where $T = 2\pi\sqrt{l/2g}$, here l is the length of liquid column. If the liquid of same mass is set into oscillations in U-tube of cross-section $A/16$, then determine time period of oscillation.

Sol. As mass is constant.

$$\therefore \text{Volume} \times \text{density} = \text{constant} \quad \left(\because \text{Density} = \frac{\text{Mass}}{\text{Volume}} \right)$$

$$\Rightarrow V_1 d = V_2 d$$

where, V_1 and V_2 are the volume of liquid in U-tube in two cases-respectively.

$$(Al)d = \left| \frac{A}{16} l' \right| d \Rightarrow l' = 16l$$

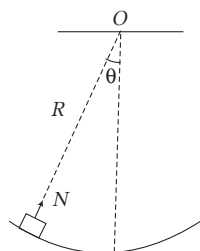
$$\therefore T = 2\pi\sqrt{\frac{l}{g}}$$

$$\therefore \frac{T'}{T} = \sqrt{\frac{l'}{l}} = \sqrt{\frac{16l}{l}} = 4 \Rightarrow T' = 4T$$

Example 11.43 A small block oscillates back and forth on a smooth concave surface of radius R . Find the time period of small oscillation.



Sol. For the oscillation of small block on a smooth concave surface, following figure can be drawn, where R is the radius of the concave surface.



$\therefore$ Restoring torque, $\tau = -mgR \sin \theta$

As $I = mR^2 \Rightarrow \tau = -mgR\theta$ (Since, $\sin \theta \approx \theta$)

$$\therefore I\alpha = -mgR\theta \Rightarrow \alpha = -\frac{mgR}{I}\theta = -\omega^2\theta$$

Time period of such oscillation is

$$T = 2\pi\sqrt{\frac{I}{mgR}} = 2\pi\sqrt{\frac{mR^2}{mgR}} \Rightarrow T = 2\pi\sqrt{\frac{R}{g}}$$

Example 11.44 If the earth were a homogeneous sphere and a straight hole was bored in it through its centre, so when a body is dropped in the hole, it will execute SHM. Determine the time period of its oscillation. Radius of the earth is 6.4×10^5 m and $g = 9.8 \text{ ms}^{-2}$

Sol. Given, $R = 6.4 \times 10^5$ m, $g = 9.8 \text{ ms}^{-2}$

$\therefore$ Time period of oscillation of body will be

$$T = 2\pi\sqrt{\frac{R}{g}} = 2\pi\sqrt{\frac{6.4 \times 10^5}{9.8}} = 1604.86 \text{ s}$$

Free oscillations

When a body oscillates with its own natural frequency, it is said to be executing **free oscillations**. The frequency and the time period of free oscillations depends on the dimensions of the body and the force constant, i.e. the inertia factor and the spring factor.

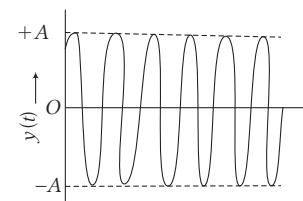


Fig. 11.24 Free oscillations

The frequency of free oscillations (or natural oscillations) is given by

$$\nu_0 = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

e.g. The vibrations of string of a sitar when pulled aside and released, etc.

Damping and damped oscillations

Damping is an influence within or upon an oscillatory system, which has the effect of reducing, restricting and preventing its oscillations. In physical systems, damping is produced by processes that dissipates the energy stored in the oscillation.

The oscillations in which the amplitude decreases gradually with the passage of time are called damped oscillations. Most of the oscillations occurring in viscous media like water, air, etc., are damped oscillations.

So, in this type of oscillations part of the energy is dissipated in overcoming resistive forces.

Consequently, the amplitude of oscillations goes on decreasing exponentially with time and eventually, these oscillations die out. *e.g.* Oscillations of simple pendulum in air, a swinging ball left to itself will eventually stop, etc.

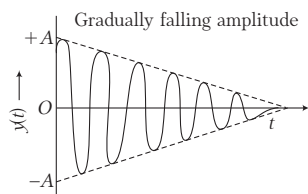


Fig. 11.25 Damped oscillations

Amplitude of damped oscillations $A = A_0 e^{-\gamma t}$

Energy of damped oscillations $E = E_0 e^{-2\gamma t}$

where, γ = damping coefficient $= \frac{b}{2m}$

Here, m is the mass of body undergoing damped oscillations.

Damping force $F_d = -bv$

where, b = damped constant and v = velocity of oscillation.

Equation of damped oscillations

$$m \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + ky = 0$$

Displacement of damped oscillations

$$y = y_m e^{-bt/2m} \sin(\omega' t + \phi)$$

where, ω' = angular frequency of damped oscillations

$$= \sqrt{\omega_0^2 - (b/2m)^2}$$

Maintained oscillations

The oscillations of a system are said to be maintained or sustained oscillations, if the system vibrates with its own natural frequency and with constant amplitude even in the presence of damping. If energy is supplied to the oscillator at the same rate at which it is dissipated, the amplitude of oscillations remains unchanged.

e.g. Balance wheel of watch, electronic oscillator, electrically maintained tuning fork.

Forced oscillations

When a body is maintained in a state of oscillation by a strong periodic force of frequency other than the natural frequency of the body, the oscillations are called forced

oscillations. The external agent which exerts the periodic force is called the driver and the oscillating system under consideration is called the driven body.

e.g.

- (i) When the stem of a vibrating tuning fork is pressed against a table, a loud sound is heard. This is because the particles of table are forced to vibrate with the frequency of the tuning fork.
- (ii) When the free end of the string of a simple pendulum is held in hand and the pendulum is made to oscillate by giving jerks by the hand, the pendulum executes forced oscillations.

Resonance

When a body oscillates with its own natural frequency, with the help of an external periodic force whose frequency is equal to the natural frequency of the body, then the amplitude of the body increases and this phenomenon is known as resonance.

Condition for resonance $\nu_0 = \nu_d$

where, ν_0 = natural frequency

and ν_d = frequency of external periodic force.

Example 11.45 The amplitude of a damped oscillator becomes half in one minute. The amplitude after 3 min will be $1/x$ times of the original. Determine the value of x .

Sol. Amplitude of damped oscillations is $A = A_0 e^{-\gamma t}$

As, $A = A_0/2$ at $t = 1$ min

$$\text{So, } \frac{A_0}{2} = A_0 e^{-\gamma}$$

$$\text{or } e^{\gamma} = 2$$

After 3 min, the amplitude will be A_0/x .

$$\text{So, } \frac{A_0}{x} = A_0 e^{-\gamma 3}$$

$$\text{or } x = e^{3\gamma} = (e^{\gamma})^3 = 2^3 = 8$$

Vector method of combining two or more simple harmonic motions in same direction

A simple harmonic motion is produced when a force (called restoring force) proportional to the displacement acts on a particle. If a particle is acted upon by two such forces, the resultant motion of the particle is a combination of two simple harmonic motions. Suppose the two individual motions are represented by

$$x_1 = A_1 \sin \omega t$$

$$\text{and } x_2 = A_2 \sin(\omega t + \phi)$$

Both the simple harmonic motions have same angular frequency ω .

The resultant displacement of the particle is given by

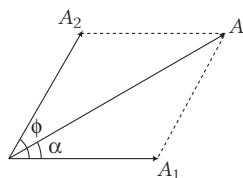


Fig. 11.26

$$\begin{aligned} x &= x_1 + x_2 \\ &= A_1 \sin \omega t + A_2 \sin (\omega t + \phi) \\ &= A \sin (\omega t + \alpha) \end{aligned}$$

Here, $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$

and

$$\tan \alpha = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}$$

Thus, we can see that this is similar to the vector addition. Hence, the same method of vector addition can be applied to the combination of more than two simple harmonic motions.

Note For SHM's moving in perpendicular directions, if

- (i) $\phi = \pi$, the resultant motion is along a straight line.
- (ii) $\phi = \pi/4$, the resultant motion is in oblique ellipse.
- (iii) $\phi = \pi/2$, the resultant motion is along a circle.

Example 11.46 Find the displacement equation of the simple harmonic motion obtained by combining the motions.

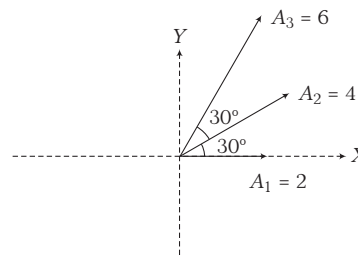
$$x_1 = 2 \sin \omega t, \quad x_2 = 4 \sin \left(\omega t + \frac{\pi}{6} \right)$$

and $x_3 = 6 \sin \left(\omega t + \frac{\pi}{3} \right)$

Sol. Let, the resultant equation is,

$$x = A \sin (\omega t + \phi)$$

$$\Sigma A_x = 2 + 4 \cos 30^\circ + 6 \cos 60^\circ = 8.5$$



and $\Sigma A_y = 4 \sin 30^\circ + 6 \sin 60^\circ = 7.2$

$$\begin{aligned} \therefore A &= \sqrt{(\Sigma A_x)^2 + (\Sigma A_y)^2} \\ &= \sqrt{(8.5)^2 + (7.2)^2} \\ &= 11.11 \end{aligned}$$

and $\tan \phi = \frac{\Sigma A_y}{\Sigma A_x}$

$$= \frac{7.2}{8.5} = 0.85$$

or $\phi = \tan^{-1} (0.85) = 40.4^\circ$

Thus, the displacement equation of the combined motion is,

$$x = A \sin (\omega t + \phi)$$

$$x = 11.11 \sin (\omega t + \phi)$$

where $\phi = 40.4^\circ$

CHECK POINT 11.4

1. A point mass m is suspended at the end of a massless wire of length L and cross-sectional area A . If Y is the Young's modulus of the wire. Then, the frequency of the simple harmonic oscillation along the vertical direction is

- (a) $\frac{1}{2\pi} \sqrt{\frac{LA}{mY}}$
- (b) $\frac{1}{2\pi} \sqrt{\frac{LAm}{Y}}$
- (c) $\frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$
- (d) $\frac{1}{2\pi} \sqrt{\frac{mY}{AL}}$

2. A rectangular block of mass m and area of cross-section A floats in a liquid of density ρ . If it is given small vertical displacement from equilibrium, it undergoes oscillation with a time period T , then

- (a) $T \propto m$
- (b) $T \propto \rho$
- (c) $T \propto \frac{1}{A}$
- (d) $T \propto \sqrt{\frac{1}{\rho}}$

3. Imagine a narrow tunnel between the two diametrically opposite points of the earth. A particle of mass m is released in this tunnel. The time period of oscillation is

- (a) $\pi \sqrt{\frac{R}{g}}$
- (b) $\frac{\pi}{2} \sqrt{\frac{R}{g}}$
- (c) $2\pi \sqrt{\frac{R}{g}}$
- (d) $\frac{2}{\pi} \sqrt{\frac{R}{g}}$

4. In damped oscillations, damping force is directly proportional to speed of oscillator. If amplitude becomes half of its maximum value in 1s, then after 2 s, amplitude will be (A_0 = initial amplitude)

- (a) $\frac{1}{4} A_0$
- (b) $\frac{1}{2} A_0$
- (c) A_0
- (d) $\frac{\sqrt{3}}{2} A_0$

5. In damped oscillations, the amplitude of oscillations is reduced to one-third of its initial value a_0 at the end of 100 oscillations. When the oscillator completes 200 oscillations, its amplitude must be

- (a) $a_0/2$
- (b) $a_0/4$
- (c) $a_0/6$
- (d) $a_0/9$

6. A weakly damped harmonic oscillator of frequency n_1 is driven by an external periodic force of frequency n_2 . When the steady state is reached, the frequency of the oscillator will be
- (a) n_1 (b) n_2
(c) $\frac{n_1 + n_2}{2}$ (d) $(n_1 + n_2)$
7. In the case of sustained forced oscillations, the amplitude of oscillations
- (a) decreases linearly
(b) decreases sinusoidally
(c) decreases exponentially
(d) always remains constant
8. In forced oscillations, a particle oscillates simple harmonically with a frequency equal to
- (a) frequency of driving force
(b) natural frequency of body
(c) difference of frequency of driving force and natural frequency
(d) mean of frequency of driving force and natural frequency
9. In case of a forced vibration, the resonance wave becomes very sharp when the
- (a) applied periodic force is small
(b) quality factor is small
(c) damping force is small
(d) restoring force is small
10. Two sources of sound are in resonance when
- (a) they look alike
(b) they are situated at a particular distance from each other
(c) they produce the sound of same frequency
(d) they are excited by the same exciting device
11. During the phenomenon of resonance,
- (a) the amplitude of oscillation becomes large
(b) the frequency of oscillation becomes large
(c) the time period of oscillation becomes large
(d) All of the above
-

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1 Which of the following quantities is always negative in SHM?

(a) $\mathbf{F} \cdot \mathbf{a}$ (b) $\mathbf{v} \cdot \mathbf{s}$ (c) $\mathbf{a} \cdot \mathbf{s}$ (d) $\mathbf{F} \cdot \mathbf{v}$

Here, $\mathbf{s}$ is displacement from mean position.

- 2 In a simple pendulum, the period of oscillation T is related to length of the pendulum l as

(a) $\frac{l}{T} = \text{constant}$ (b) $\frac{l^2}{T} = \text{constant}$

(c) $\frac{l}{T^2} = \text{constant}$ (d) $\frac{l^2}{T^2} = \text{constant}$

- 3 A particle executing simple harmonic motion along Y-axis has its motion described by the equation $y = A \sin(\omega t) + B$. The amplitude of the simple harmonic motion is

(a) A (b) B (c) $A + B$ (d) $\sqrt{A^2 + B^2}$

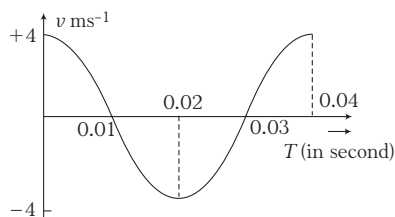
- 4 Two particles are executing SHMs. The equations of their motions are

$$y_1 = 10 \sin\left(\omega t + \frac{\pi}{4}\right) \text{ and } y_2 = 5 \sin\left(\omega t + \frac{\sqrt{3}\pi}{4}\right)$$

What is the ratio of their amplitudes?

(a) 1 : 1 (b) 2 : 1
(c) 1 : 2 (d) None of these

- 5 The velocity-time graph of a harmonic oscillator is shown in the figure below. The frequency of oscillation is



(a) 25 Hz (b) 50 Hz
(c) 12.25 Hz (d) 33.3 Hz

- 6 The ratio of frequencies of two pendulums are 2 : 3, then their lengths are in ratio

(a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{\frac{3}{2}}$
(c) $\frac{4}{9}$ (d) $\frac{9}{4}$

- 7 A body oscillates in SHM according to the equation

(in SI unit), $x = 5 \cos\left(2\pi t + \frac{\pi}{4}\right)$. Its instantaneous displacement at $t = 1$ s is

(a) $\frac{\sqrt{2}}{5}$ m (b) $\frac{1}{\sqrt{3}}$ m (c) $\frac{5}{\sqrt{2}}$ m (d) $\frac{1}{2}$ m

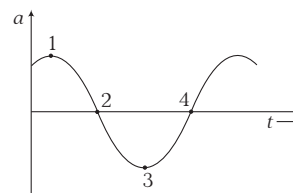
- 8 A body is vibrating in simple harmonic motion. If its acceleration is 12 cms^{-2} at a displacement 3 cm from the mean position, then time period is

(a) 6.28 s (b) 3.14 s (c) 1.57 s (d) 2.57 s

- 9 A horizontally placed spring mass system has time period T . The same system is now placed on a car moving with acceleration a in horizontal direction. Then,

(a) time period will increase
(b) time period will decrease
(c) time period will remain constant
(d) no conclusion can be drawn

- 10 The acceleration a of a particle undergoing SHM is shown in the figure. Which of the labelled points corresponds to the particle being at $-x_{\text{max}}$?



(a) 4 (b) 3
(c) 2 (d) 1

- 11 Two particles are executing simple harmonic motion.

At an instant of time t , their displacements are

$$y_1 = A \cos(\omega t) \text{ and } y_2 = A \sin(\omega t)$$

Then, the phase difference between y_1 and y_2 is

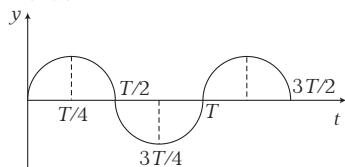
(a) 120° (b) 90° (c) 180° (d) zero

- 12 A particle executing SHM of amplitude 4 cm and $T = 4$ s. The time taken by it to move from positive extreme position to half the amplitude is

(a) 1 s (b) $\frac{1}{3}$ s
(c) $\frac{2}{3}$ s (d) $\sqrt{\frac{3}{2}}$ s

- 13** The maximum acceleration of a particle in SHM is made two times keeping the maximum speed to be constant. It is possible when
- amplitude of oscillation is doubled while frequency remains constant
 - amplitude is doubled while frequency is halved
 - frequency is doubled while amplitude is halved
 - frequency of oscillation is doubled while amplitude remains constant
- 14** Under the action of a force $F = -kx^3$, the motion of a particle is (k = a positive constant)
- simple harmonic motion
 - uniformly accelerated motion
 - not periodic
 - periodic but not simple harmonic
- 15** A light spring of force constant 8 Nm^{-1} is cut into two equal halves and the two are connected in parallel; the equivalent force constant of the system is
- 16 Nm^{-1}
 - 32 Nm^{-1}
 - 8 Nm^{-1}
 - 24 Nm^{-1}
- 16** In order that the resultant path on superimposing two mutually perpendicular SHM be a circle, the conditions are that
- the amplitudes on both SHM should be equal and they should have a phase difference of $\frac{\pi}{2}$
 - the amplitudes should be in the ratio 1 : 2 and the phase difference should be zero
 - the amplitudes should be in the ratio 1 : 2 and the phase difference should be $\frac{\pi}{2}$
 - the amplitudes should be equal and the phase difference should be zero
- 17** The period of a particle executing SHM is 8 s. At $t = 0$, it is at the mean position. The ratio of the distances covered by the particle in the 1st second to the 2nd second is
- $\frac{1}{\sqrt{2} + 1}$
 - $\sqrt{2}$
 - $\frac{1}{\sqrt{2}}$
 - $\sqrt{2} + 1$
- 18** Force constant of a weightless spring is 16 Nm^{-1} . A body of mass 1 kg suspended from it is pulled down through 5 cm from its mean position and then released. The maximum kinetic energy of the system (spring + body) will be
- $2 \times 10^{-2} \text{ J}$
 - $4 \times 10^{-2} \text{ J}$
 - $8 \times 10^{-2} \text{ J}$
 - $16 \times 10^{-2} \text{ J}$

- 19** The graph shows the variation of displacement of a particle executing SHM with time. We infer from this graph that

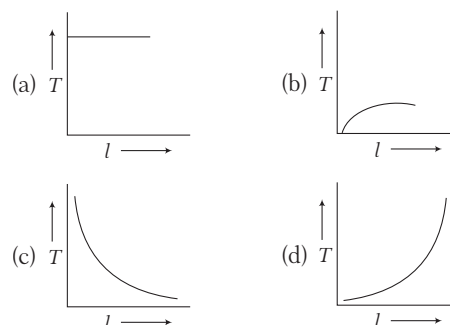


- the force is zero at time $\frac{3T}{4}$
- the velocity is maximum at time $\frac{T}{2}$
- the acceleration is maximum at time T
- the potential energy is equal to total energy at time $\frac{T}{2}$

- 20** The time period of a simple pendulum inside a stationary lift is $\sqrt{5} \text{ s}$. What will be the time period when the lift moves upward with an acceleration $\frac{g}{4}$?
- $\sqrt{5} \text{ s}$
 - $2\sqrt{5} \text{ s}$
 - $(2 + \sqrt{5}) \text{ s}$
 - 2 s

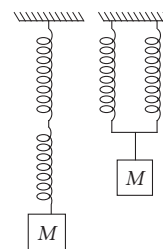
- 21** A linear harmonic oscillator of force constant $2 \times 10^6 \text{ Nm}^{-1}$ and amplitude 0.01 m has a total mechanical energy of 160 J. Its
- maximum potential energy is 160 J
 - maximum potential energy is 100 J
 - minimum potential energy is zero
 - minimum potential energy is 100 J

- 22** In case of a simple pendulum, time period *versus* length is depicted by



- 23** A second pendulum is moved to moon, where acceleration due to gravity is $1/6$ times that of the earth, the length of the second pendulum on moon would be
- 6 times
 - 12 times
 - $\frac{1}{6}$ times
 - $\frac{1}{12}$ times

- 24** Two identical springs of spring constant k each are connected in series and parallel as shown in figure. A mass M is suspended from them. The ratio of their frequencies of vertical oscillation will be



- 1 : 2
- 2 : 1
- 4 : 1
- 1 : 4

- 25** The relation between acceleration and displacement of four particles are given below [NCERT Exemplar]

(a) $a_x = +2x$ (b) $a_x = +2x^2$
 (c) $a_x = -2x^2$ (d) $a_x = -2x$

Which one of the particle is exempting simple harmonic motion?

- 26** A particle executing SHM has a maximum speed of 30 cm s^{-1} and a maximum acceleration of 60 cm s^{-2} . The period of oscillation is [NCERT Exemplar]

(a) $\pi \text{ s}$ (b) $\frac{\pi}{2} \text{ s}$
 (c) $2\pi \text{ s}$ (d) $\frac{\pi}{t} \text{ s}$

- 27** A particle is attached to a vertical spring and is pulled down a distance 0.04 m below its equilibrium position and is released from rest. The initial upward acceleration of the particle is 0.30 ms^{-2} . The period of the oscillation is

(a) 4.08 s (b) 1.92 s
 (c) 3.90 s (d) 2.29 s

- 28** A disc of radius R and mass M is pivoted at the rim and is set for small oscillations about an axis perpendicular to plane of disc. If a simple pendulum has to have the same time period as that of the disc, the length of the pendulum should be

(a) $\frac{5}{4} R$ (b) $\frac{2}{3} R$
 (c) $\frac{3}{4} R$ (d) $\frac{3}{2} R$

- 29** If the length of second's pendulum is decreased by 2%, how many seconds, it will lose per day?

(a) 3927s (b) 3727s
 (c) 3427s (d) 864s

- 30** A clock which keeps correct time at 20°C , is subjected to 40°C . If coefficient of linear expansion of the pendulum is $12 \times 10^{-6} / ^\circ\text{C}$. How much will it gain or lose in time?

(a) 10.3 s/day (b) 20.6 s/day
 (c) 5 s/day (d) 20 min/day

- 31** The equation of a damped simple harmonic motion is $m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = 0$. Then, the angular frequency of oscillation is

(a) $\omega = \left(\frac{k}{m} - \frac{b^2}{4m^2} \right)^{1/2}$ (b) $\omega = \left(\frac{k}{m} - \frac{b}{4m} \right)^{1/2}$
 (c) $\omega = \left(\frac{k}{m} - \frac{b^2}{4m} \right)^{1/2}$ (d) $\omega = \left(\frac{k}{m} - \frac{b^2}{4m^2} \right)^{-1/2}$

- 32** The motion of a particle is given

$x = a \sin \omega t + b \cos \omega t$. The motion of the particle is

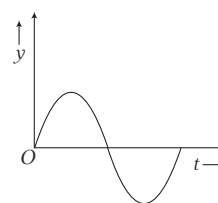
[NCERT Exemplar]

- (a) not simple harmonic
 (b) simple harmonic with amplitude $a + b$
 (c) simple harmonic with amplitude $\frac{(a+b)}{2}$
 (d) simple harmonic with amplitude $\sqrt{a^2 + b^2}$

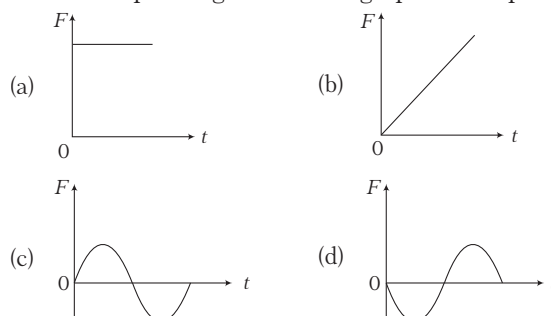
- 33** The displacement equation of a particle is $x = 3 \sin 2t + 4 \cos 2t$. The amplitude and maximum velocity will be respectively

(a) 5, 10 (b) 3, 2 (c) 4, 2 (d) 3, 4

- 34** The displacement-time graph of a particle executing SHM as shown in the figure.



The corresponding force-time graph of the particle is



- 35** A tunnel has been dug through the centre of the earth and a ball is released in it. It will reach the other end of the tunnel after

(a) 84.6 min
 (b) 42.3 min
 (c) 1 day
 (d) will not reach the other end

- 36** Two bodies M and N of equal masses are suspended from two separate massless springs of spring constants k_1 and k_2 , respectively. If the two bodies oscillate vertically such that their maximum velocities are equal, the ratio of the amplitude of vibration of M to that on N is

(a) $\frac{k_1}{k_2}$ (b) $\sqrt{\frac{k_1}{k_2}}$
 (c) $\frac{k_2}{k_1}$ (d) $\sqrt{\frac{k_2}{k_1}}$

- 37** A particle is executing simple harmonic motion with a period of T seconds and amplitude in metre. The shortest time it takes to reach a point $\frac{a}{\sqrt{2}}$ m from its mean position in seconds is

(a) T (b) $\frac{T}{4}$ (c) $\frac{T}{8}$ (d) $\frac{T}{16}$

- 38** A mass M is suspended from a massless spring. An additional mass m stretches the spring further by a distance x . The combined mass will oscillate with a period

(a) $2\pi \sqrt{\frac{(M+m)x}{mg}}$ (b) $2\pi \sqrt{\frac{mg}{(M+m)x}}$
 (c) $2\pi \sqrt{\frac{(M+m)}{mgx}}$ (d) $\frac{\pi}{2} \sqrt{\frac{mg}{(M+m)x}}$

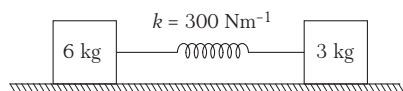
- 39** A particle executes linear simple harmonic motion with an amplitude of 2 cm. When the particle is at 1 cm from the mean position the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is

(a) $\frac{1}{2\pi\sqrt{3}}$ (b) $2\pi\sqrt{3}$
 (c) $\frac{2\pi}{\sqrt{3}}$ (d) $\frac{\sqrt{3}}{2\pi}$

- 40** A spring has a natural length of 50 cm and a force constant of $2 \times 10^3 \text{ Nm}^{-1}$. A body of mass 10 kg is suspended from it and the spring is stretched. If the body is pulled down to a length of 58 cm and released, it executes simple harmonic motion. What is the net force on the body when it is at its lowermost position of its oscillation? (Take, $g = 10 \text{ ms}^{-2}$)

(a) 20 N (b) 40 N (c) 60 N (d) 80 N

- 41** Two point masses of 3 kg and 6 kg are attached to opposite ends of horizontal spring whose spring constant is 300 Nm^{-1} as shown in the figure. The natural vibration frequency of the system is approximately



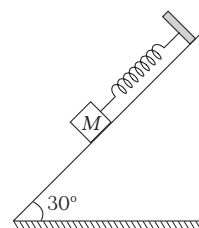
(a) 4 Hz (b) 3 Hz (c) 2 Hz (d) 1 Hz

- 42** In a spring-mass system, the length of the spring is L , and it has a mass M attached to it and oscillates with an angular frequency ω . The spring is then cut into two parts, one (i) with relaxed length αL and the other (ii) with relaxed length $(1 - \alpha)L$. The force constants of the two springs A and B are

(a) $\frac{k}{1-\alpha}$ and $\frac{k}{\alpha}$ (b) $\frac{k}{\alpha}$ and $\frac{k}{1-\alpha}$
 (c) αk and $(1-\alpha)k$ (d) k and k

- 43** A block of mass 0.2 kg which slides without friction on a 30° incline, is connected to the top of the incline by a massless spring of force constant 80 Nm^{-1} as shown in figure.

If the block is pulled slightly from its mean position, what is the period of oscillations?



(a) $\pi \text{ s}$ (b) $\frac{\pi}{10} \text{ s}$ (c) $\frac{2\pi}{5} \text{ s}$ (d) $\frac{\pi}{2} \text{ s}$

- 44** Two simple harmonic motions are represented by the following equations

$$y_1 = 40 \sin \omega t$$

and $y_2 = 10 (\sin \omega t + C \cos \omega t)$

If their displacement amplitudes are equal, then the value of C (in appropriate units) is

(a) $\sqrt{13}$ (b) $\sqrt{15}$
 (c) $\sqrt{17}$ (d) 4

- 45** The potential energy of a particle of mass 2 kg in SHM is $(9x^2)$ joule. Here, x is the displacement from mean position. If total mechanical energy of the particle is 36 J, then maximum speed of the particle is

(a) 4 ms^{-1} (b) 2 ms^{-1}
 (c) 6 ms^{-1} (d) 10 ms^{-1}

- 46** A particle of mass 0.1 kg is executing SHM of amplitude 0.1 m. When the particle passes through the mean position, its kinetic energy is $8 \times 10^{-3} \text{ J}$. Find the equation of motion of the particle, if the initial phase of oscillation is 45° .

(a) $y = 0.1 \cos \left(3t + \frac{\pi}{4} \right)$ (b) $y = 0.1 \sin \left(6t + \frac{\pi}{4} \right)$
 (c) $y = 0.1 \sin \left(4t + \frac{\pi}{4} \right)$ (d) $y = 0.1 \cos \left(4t + \frac{\pi}{4} \right)$

- 47** Molten-wax of mass m drops on a block of mass M , which is oscillating on a frictionless table. Select the correct option.

(a) amplitude does not change
 (b) amplitude increases
 (c) time period decreases
 (d) time period increases

- 48** The displacement of a particle is represented by the equation $y = 3 \cos\left(\frac{\pi}{4} - 2\omega t\right)$. The motion of the

particle is [NCERT Exemplar]

- (a) simple harmonic with period $2\pi / \omega$
 (b) simple harmonic with period π / ω
 (c) periodic but not simple harmonic
 (d) non-periodic

- 49** The displacement of a particle is represented by the equation $y = \sin^3 \omega t$. The motion is

[NCERT Exemplar]

- (a) non-periodic
 (b) periodic but not simple harmonic
 (c) simple harmonic with period $2\pi / \omega$
 (d) simple harmonic with period π / ω

- 50** Motion of an oscillating liquid column in a U-tube is

[NCERT Exemplar]

- (a) periodic but not simple harmonic
 (b) non-periodic
 (c) simple harmonic and time period is independent of the density of the liquid
 (d) simple harmonic and time period is directly proportional to the density of the liquid

- 51** A particle is acted simultaneously by mutually perpendicular simple harmonic motion $x = a \cos \omega t$ and $y = a \sin \omega t$. The trajectory of motion of the particle will be

[NCERT Exemplar]

- (a) an ellipse (b) a parabola
 (c) a circle (d) a straight line

- 52** The equation of motion of a particle is

$x = a \cos (\alpha t)^2$. The motion is [NCERT Exemplar]

- (a) periodic but not oscillatory
 (b) periodic and oscillatory
 (c) oscillatory but not periodic
 (d) Neither periodic nor oscillatory

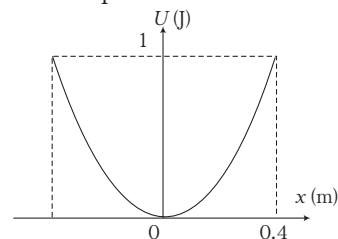
- 53** A block of mass m , attached to a spring of spring constant k , oscillates on a smooth horizontal table. The other end of the spring is fixed to a wall. The block has a speed v when the spring is at its natural length. Before coming to an instantaneous rest, if the block moves a distance x from the mean position, then

- (a) $x = \sqrt{\frac{m}{k}}$ (b) $x = \frac{1}{v} \sqrt{\frac{m}{k}}$ (c) $x = v \sqrt{\frac{m}{k}}$ (d) $x = \sqrt{\frac{mv}{k}}$

- 54** In SHM, potential energy of a particle at mean position is E_1 and kinetic energy is E_2 , then

- (a) $E_1 = E_2$
 (b) total potential energy at $x = \frac{\sqrt{3}A}{2}$ is $E_1 + \frac{3E_2}{4}$
 (c) total kinetic energy at $x = \frac{\sqrt{3}A}{2}$ is $\frac{3E_2}{4}$
 (d) total kinetic energy at $x = \frac{A}{\sqrt{2}}$ is $\frac{E_2}{4}$

- 55** A particle of mass 2 kg moves in simple harmonic motion and its potential energy U varies with position x as shown in the figure. The period of oscillation of the particle is



- (a) $\frac{2\pi}{5}$ s (b) $\frac{2\sqrt{2}\pi}{5}$ s (c) $\frac{\sqrt{2}\pi}{5}$ s (d) $\frac{4\pi}{5}$ s

- 56** Maximum kinetic energy of a particle of mass 1 kg in SHM is 8 J. Time period of SHM is 4 s. Maximum potential energy during the motion is 10 J. Then,

- (a) amplitude of oscillations is approximately 4.53 m
 (b) minimum potential energy of the particle is 4 J
 (c) maximum acceleration of the particle is approximately 6.3 ms^{-2}
 (d) minimum kinetic energy of the particle is 2 J

- 57** A simple pendulum has a time period T in vacuum. Its time period when it is completely immersed in a liquid of density one-eighth of the density of material of the bob is

- (a) $\sqrt{\frac{7}{9}}T$ (b) $\sqrt{\frac{5}{8}}T$ (c) $\sqrt{\frac{3}{8}}T$ (d) $\sqrt{\frac{8}{7}}T$

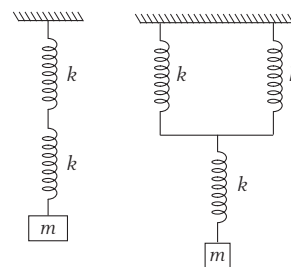
- 58** A simple pendulum has time period T . The bob is given negative charge and surface below it is given positive charge. The new time period will be

- (a) less than T (b) greater than T
 (c) equal to T (d) infinite

- 59** Two simple pendulums whose lengths are 100 cm and 121 cm are suspended side by side. Their bobs are pulled together and then released. After how many minimum oscillations of the longer pendulum, will the two be in phase again?

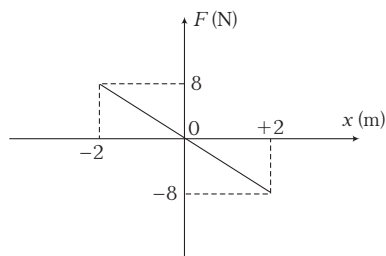
- (a) 11 (b) 10 (c) 21 (d) 20

- 60** Some springs are combined in series and parallel arrangement as shown in the figure and a mass m is suspended from them. The ratio of their frequencies will be



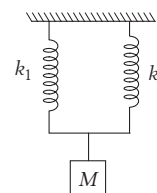
- (a) 1 : 1 (b) 2 : 1 (c) $\sqrt{3}$: 2 (d) 4 : 1

- 61 A body of mass 0.01 kg executes simple harmonic motion (SHM) about $x = 0$ under the influence of a force as shown in figure. The period of the SHM is



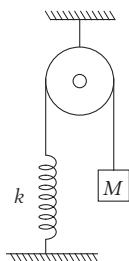
- (a) 1.05 s (b) 0.52 s
(c) 0.25 s (d) 0.31 s
- 62 The vertical motion of a ship at sea is described by the equation $\frac{d^2x}{dt^2} = -4x$, where x is the vertical height of the ship (in metre) above its mean position. If it oscillates through a height of 1 m, its maximum vertical
- (a) speed will be 1 ms^{-1}
(b) speed will be 2 ms^{-1}
(c) acceleration is 2 ms^{-2}
(d) acceleration is 1 ms^{-2}
- 63 A simple pendulum is suspended from the ceiling of a car and its period of oscillation is T when the car is at rest. The car starts moving on a horizontal road with a constant acceleration g (equal to the acceleration due to gravity, in magnitude) in the forward direction. To keep the time period same, the length of the pendulum will have to be
- (a) increased by $\sqrt{2} l$
(b) increased by $(\sqrt{2} - 1) l$
(c) decreased by $\sqrt{2} l$
(d) decreased by $(\sqrt{2} - 1) l$
- 64 A small ball is dropped from a certain height on the surface of a non-viscous liquid of density less than the density of ball. The motion of the ball is
- (a) SHM
(b) periodic but not SHM
(c) not periodic
(d) SHM for half a period and non-periodic for rest half of the period
- 65 Two particles undergoing SHM of the same amplitude and frequency along the same straight line. They pass one another when going in opposite directions each time their displacement is half of their amplitude. What is the phase difference between them?
- (a) 60° (b) 30° (c) 90° (d) 120°

- 66 A mass is suspended separately by two springs of spring constants k_1 and k_2 in successive order. The time periods of oscillations in the two cases are T_1 and T_2 , respectively. If the same mass be suspended by connecting the two springs in parallel (as shown in figure), then the time period of oscillations is T . The correct relation is

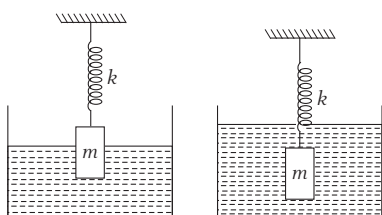


- (a) $T^2 = T_1^2 + T_2^2$ (b) $T^{-2} = T_1^{-2} + T_2^{-2}$
(c) $T^{-1} = T_1^{-1} + T_2^{-1}$ (d) $T = T_1 + T_2$
- 67 A horizontal platform with an object placed on it is executing SHM in the vertical direction. The amplitude of oscillation is $4 \times 10^{-3} \text{ m}$. What must be least period of these oscillations, so that the object is not detached from the platform? (Take, $g = 10 \text{ ms}^{-2}$)
- (a) $\frac{\pi}{25} \text{ s}$ (b) $\frac{\pi}{5} \text{ s}$
(c) $\frac{\pi}{10} \text{ s}$ (d) $\frac{\pi}{50} \text{ s}$
- 68 Two masses 8 kg and 4 kg are suspended together by a massless spring of spring constant 1000 Nm^{-1} . When the masses are in equilibrium 8 kg is removed without disturbing the system. The amplitude of oscillation is
- (a) 0.5 m (b) 0.08 m
(c) 0.4 m (d) 0.04 m
- 69 A particle executes SHM with amplitude of 20 cm and time period of 12 s. What is the minimum time required for it to move between two points 10 cm on either side of the mean position?
- (a) 1 s (b) 2 s
(c) 3 s (d) 4 s
- 70 A uniform spring whose unstretched length is l has a force constant k . The spring is cut into two pieces of unstretched lengths l_1 and l_2 , where $l_1 = nl_2$ and n is being an integer. Now, a mass m is made to oscillate with first spring. The time period of its oscillation would be
- (a) $T = 2\pi \sqrt{\frac{mn}{k(n+1)}}$ (b) $T = 2\pi \sqrt{\frac{m}{k(n+1)}}$
(c) $T = 2\pi \sqrt{\frac{m}{nk}}$ (d) $T = 2\pi \sqrt{\frac{m(n+1)}{nk}}$

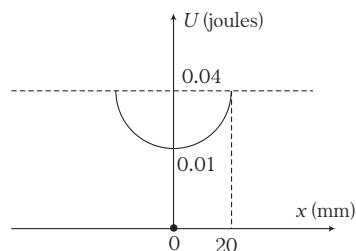
- 71** A mass $M = 5 \text{ kg}$ is attached to a spring as shown in the figure and held in position, so that the spring remains unstretched. The spring constant is 200 Nm^{-1} . The mass M is then released and begins to undergo small oscillations. The amplitude of oscillation is



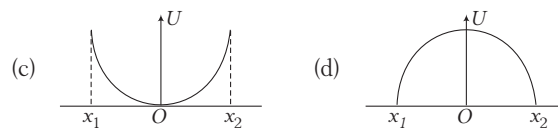
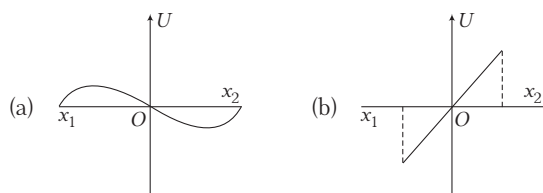
- (a) 0.5 m (b) 0.25 m
(c) 0.2 m (d) 0.1 m
- 72** Period of small oscillations in the two cases shown in figure is T_1 and T_2 , respectively. Assume that, fluid does not have any viscosity, then



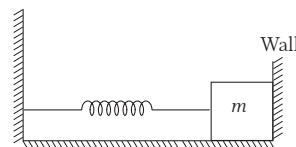
- (a) $T_1 = T_2$ (b) $T_1 < T_2$
(c) $T_1 > T_2$ (d) Cannot say anything
- 73** The variation of potential energy of harmonic oscillator as shown in figure. The spring constant is



- (a) $1 \times 10^2 \text{ Nm}^{-1}$ (b) $1.5 \times 10^2 \text{ Nm}^{-1}$
(c) $2 \times 10^2 \text{ Nm}^{-1}$ (d) $3 \times 10^2 \text{ Nm}^{-1}$
- 74** A particle of mass m oscillates with simple harmonic motion between points x_1 and x_2 , the equilibrium position being at O . Its potential energy is plotted. It will be as given below in the graph.



- 75** A solid cube floats in water half immersed and has small vertical oscillations of time period $\frac{\pi}{5} \text{ s}$. Its mass (in kg) is (Take, $g = 10 \text{ ms}^{-2}$)
- (a) 4 (b) 2 (c) 1 (d) 0.5
- 76** In the figure, the block of mass m attached to the spring of stiffness k is in contact with the completely elastic wall, and the compression in the spring is e . The spring is compressed further by e by displacing the block towards left and is then released. If the collision between the block and the wall is completely elastic, then the time period of oscillations of the block will be



- (a) $\frac{2\pi}{3} \sqrt{\frac{m}{k}}$ (b) $2\pi \sqrt{\frac{m}{k}}$ (c) $\frac{\pi}{3} \sqrt{\frac{m}{k}}$ (d) $\frac{\pi}{6} \sqrt{\frac{m}{k}}$
- 77** A cubical block of mass M vibrates horizontally with amplitude of 4 cm and a frequency of 2 Hz. A small block of mass m is placed on the bigger block. In order that the smaller block does not slide on the bigger block, the minimum value of the coefficient of static friction between the two blocks is
- (a) 0.36 (b) 0.40 (c) 0.64 (d) 0.72

- 78** Two pendulums of time periods 3 s and 7 s respectively, start oscillating simultaneously from two opposite extreme positions. After how much time they will be in same phase?

- (a) $\frac{21}{8} \text{ s}$ (b) $\frac{21}{4} \text{ s}$ (c) $\frac{21}{2} \text{ s}$ (d) $\frac{21}{10} \text{ s}$

- 79** A particle under the action of a SHM has a period of 3 s and under the effect of another it has a period 4 s. What will be its period under the combined action of both the SHM's in the same direction?
- (a) 7 s (b) 5 s (c) 2.4 s (d) 0.4 s

- 80** A particle performs SHM in a straight line. In the first second starting from rest, it travels a distance a and in the next second, it travels a distance b on the same side of mean position. The amplitude of the SHM is

- (a) $a - b$ (b) $\frac{2a - b}{3}$
(c) $\frac{2a^2}{3a - b}$ (d) None of these

- 81** A particle is in linear SHM of amplitude A and time period T . If v refers to its average speed during any interval of $T/3$, then the maximum possible value of v is

(a) $\frac{3\sqrt{3}}{T}A$ (b) $\frac{\sqrt{3}}{T}A$ (c) $\frac{2\sqrt{3}}{T}A$ (d) $\frac{3A}{T}$

- 82** A block whose mass is 2 kg is fastened to a spring whose spring constant is 100 Nm^{-1} . It is pulled to a distance of 0.1 m from over a frictionless surface and is released at $t = 0$. Calculate the kinetic energy of the block when it is 0.05 m away from its mean position.

(a) 3 J (b) 0.37 J (c) 4.16 J (d) Zero

- 83** A uniform rod of length l is suspended from any point P . It is then made to undergo small oscillations. If P is the

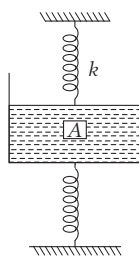
(a) centre of mass, T is zero

(b) end point, T is $2\pi\sqrt{\frac{l}{3g}}$

(c) centre of mass, $T = 2\pi\sqrt{\frac{l}{g}}$

(d) end point, T is infinite

- 84** The system shown in figure is in equilibrium. The mass of the container with liquid is M , density of liquid in the container is ρ and the volume of the block is V . If the container is now displaced downwards through a distance x_0 and released such that the block remains well inside the liquid, then during subsequent motion,



(a) time period of SHM of the container will be $2\pi\sqrt{\frac{M}{k}}$

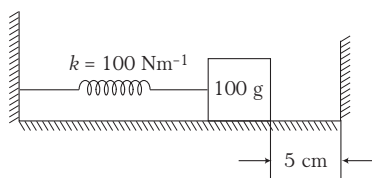
(b) time period of SHM of the container

will be $2\pi\sqrt{\frac{M + \rho V}{k}}$

(c) amplitude of SHM of the container is x_0

(d) amplitude of SHM of the container is $2x_0$

- 85** A block of mass 100 g attached to a spring of spring constant 100 Nm^{-1} is lying on a frictionless floor as shown in the figure. The block is moved to compress the spring by 10 cm and then released. If the collisions with the wall in front are elastic, then the time period of the motion is



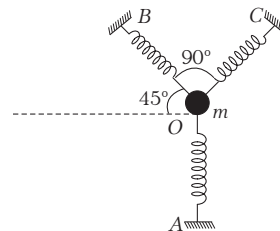
(a) 0.2 s (b) 0.1 s (c) 0.15 s (d) 0.132 s

- 86** One end of a long metallic wire of length L is tied to the ceiling. The other end is tied to massless spring of spring constant k . A mass m hangs freely from the free end of the spring. The area of cross-section and Young's modulus of the wire are A and Y , respectively. If the mass is slightly pulled down and released, it will oscillate with a time period T equal to

(a) $2\pi\sqrt{\frac{m}{k}}$ (b) $2\pi\sqrt{\frac{m(YA + kL)}{YAk}}$

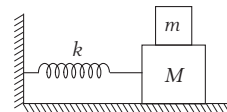
(c) $2\pi\sqrt{\frac{m(YA + kL)}{Ak}}$ (d) $2\pi\sqrt{\frac{m(Y + kL)}{YAk}}$

- 87** A particle of mass m is attached to three identical springs A, B and C each of force constant k as shown in figure. If the particle of mass m is pushed slightly against the spring A and released, then the time period of oscillation is



(a) $2\pi\sqrt{\frac{2m}{k}}$ (b) $2\pi\sqrt{\frac{m}{2k}}$ (c) $2\pi\sqrt{\frac{m}{k}}$ (d) $2\pi\sqrt{\frac{m}{3k}}$

- 88** A mass M is attached to a horizontal spring of force constant k fixed one side to a rigid support as shown in figure. The mass oscillates on a frictionless surface with time period T and amplitude A . When the mass M is in equilibrium position, another mass m is gently placed on it. What will be the new amplitude of oscillations?

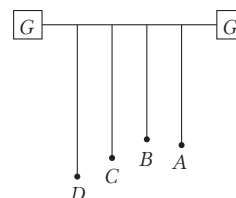


(a) $\sqrt{\frac{(M+m)}{M}}A$ (b) $\sqrt{\frac{(M-m)}{M}}A$

(c) $\sqrt{\frac{M}{(M+m)}}A$ (d) $\sqrt{\frac{M}{(M-m)}}A$

- 89** Four pendulums A, B, C and D are suspended from the same elastic support as shown in figure.

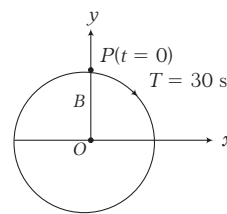
[NCERT Exemplar]



A and C are of the same length, while B is smaller than A and D is larger than A . If A is given a transverse displacement,

- (a) D will vibrate with maximum amplitude
- (b) C will vibrate with maximum amplitude
- (c) B will vibrate with maximum amplitude
- (d) All the four will oscillate with equal amplitude

- 90** Figure shows the circular motion of a particle. The radius of the circle, the period, sense of revolution and the initial position are indicated on the figure. The simple harmonic motion of the x -projection of the radius vector of the rotating particle P is
- [NCERT Exemplar]



- (a) $x(t) = B \sin\left(\frac{2\pi t}{30}\right)$
- (b) $x(t) = B \cos\left(\frac{\pi t}{15}\right)$
- (c) $x(t) = B \sin\left(\frac{\pi t}{15} + \frac{\pi}{2}\right)$
- (d) $x(t) = B \cos\left(\frac{\pi t}{15} + \frac{\pi}{2}\right)$

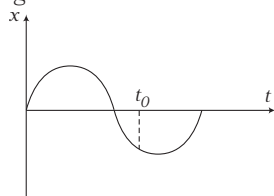
(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) These questions consist of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is correct but Reason is incorrect.
- (d) If Assertion is incorrect but Reason is correct.

- 1 Assertion** In the x - t graph of a particle in SHM, acceleration of particle at time t_0 is positive but velocity is negative.



Reason $a \propto -x$ and velocity is slope of x - t graph.

- 2 Assertion** If a pendulum is suspended in a lift and lift accelerates upwards, then its time period will decrease.

Reason Effective value of g will be

$$g_{\text{eff}} = g + a$$

- 3 Assertion** If amplitude of SHM is increased, time period of SHM will increase.

Reason If amplitude is increased, body will have to travel more distance in one complete oscillation.

- 4 Assertion** In SHM, v - x graph is an ellipse, where v is velocity and x is displacement from mean position.

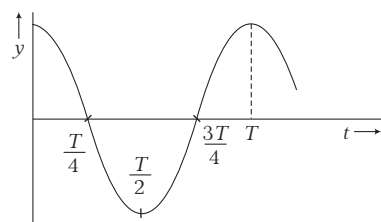
Reason Equation between v and x is $\frac{v^2}{\omega^2} + \frac{x^2}{1} = A^2$.

- 5 Assertion** Time period of a spring block system is T . If length of spring is decreased, time period will decrease.

Reason If length is decreased, then the block will have to travel less distance and it will take less time.

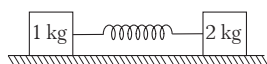
Statement based questions

- 1** The displacement-time graph of a particle executing SHM as shown in figure. Which of the following statement(s) is/are false?



- (a) The acceleration is maximum at $t = T$.
 - (b) The force is zero at $t = \frac{3T}{4}$.
 - (c) The potential energy equals the total oscillation energy at $t = \frac{T}{2}$.
 - (d) None of the above
- 2** A mass of 0.2 kg is attached to the lower end of a massless spring of force constant 200 Nm^{-1} , the upper end of which is fixed to a rigid support. Which of the following statement(s) is/are true?
- (a) In equilibrium, the spring will be stretched by 1 cm.
 - (b) If the mass is raised till the spring becomes unstretched and then released, it will go down by 2 cm before moving upwards.
 - (c) The frequency of oscillation will be nearly 5 Hz.
 - (d) All of the above

- 3** The speed (v) of a particle moving along a straight line, when it is at a distance (x) from a fixed point on the line, is given by $v^2 = 144 - 9x^2$. Which of the following statement(s) is/are correct?
- (a) The magnitude of acceleration at a distance 3 units from the fixed point is 27 units.
 (b) The motion is simple harmonic with $T = \frac{2\pi}{3}$ units.
 (c) The maximum displacement from the fixed point is 4 units.
 (d) All of the above
- 4** The potential energy of a particle of mass 0.1 kg, moving along the X -axis, is given by $U = 5x(x - 4)$ J, where x is in metre. Which of the following statement(s) is/are incorrect?
- (a) The speed of the particle is maximum at $x = 2$ m.
 (b) The particle executes simple harmonic motion.
 (c) The period of oscillation of the particle is $\frac{\pi}{7}$ s.
 (d) All are correct
- 5** In the two block spring system, force constant of spring is $k = 6 \text{ Nm}^{-1}$. Spring is stretched by 12 cm towards left.



Now, choose the correct statement(s).

- (a) Angular frequency of oscillation is 3 rad s^{-1} .
 (b) Maximum kinetic energy of 1 kg is 4.8 mJ.
 (c) Maximum kinetic energy of 2 kg is 4.8 mJ.
 (d) Both (a) and (c)
- 6** A particle of mass m is dropped from a great height h above the hole in the earth dug along its diameter. Then, which of the following statement is correct?
- (a) The motion of the particle is simple harmonic.
 (b) The motion of the particle is periodic.
 (c) The speed of the particle at the centre of earth equals $\sqrt{\frac{2GM}{(R+h)}}$, where R and M are the radius and mass of the earth, respectively.
 (d) The speed of the particle at the centre of earth equals $\sqrt{\frac{GM(R+3h)}{R(R+h)}}$, where R and M are the radius and mass of the earth, respectively.
- 7** I. In SHM to find time taken in moving from one point to another point, we cannot apply the relation, $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

II. In SHM, speed is constant.

Which of the following statement(s) is/are correct?

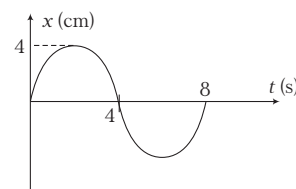
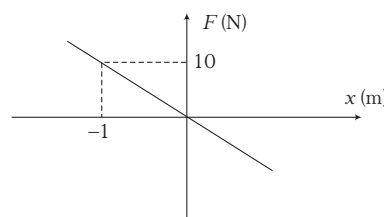
- (a) Both I and II (b) Only I
 (c) Only II (d) Neither I nor II
- 8** I. Average kinetic energy in one oscillation during SHM of a body is $\frac{1}{4} m\omega^2 A^2$.
 II. Maximum kinetic energy is $\frac{1}{2} m\omega^2 A^2$.
- Which of the following statement(s) is/are correct?
- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 9** I. In a spring block system, if length of spring and mass both are halved, time period of oscillation will remain unchanged.

II. Angular frequency of SHM is $\omega = \sqrt{\frac{k}{m}}$.

where, k is spring constant and m is mass of block.

Which of the following statement(s) is/are correct?

- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 10.** F - x and x - t graph of a particle in SHM as shown in figure.



Now, study the following statements.

- I. Mass of the particle is $\left(\frac{160}{\pi^2}\right)$ SI unit.
 II. Maximum kinetic energy of the particle is 6×10^{-3} SI unit.
- Which of the following statement(s) is/are correct?
- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

Match the columns

- 1 In SHM, match the Column I with Column II and mark the correct option from the codes given below.

Column I	Column II
(A) Displacement and velocity	(p) Phase difference of zero
(B) Displacement and acceleration	(q) Phase difference of $\frac{\pi}{2}$
(C) Velocity and acceleration	(r) Phase difference of π

Codes

A	B	C	A	B	C
(a) r	p	q	(b) q	r	q
(c) p	q	r	(d) p	q	p

- 2 In SHM, match the Column I with Column II and mark the correct option from the codes given below.

Column I	Column II
(A) Acceleration-displacement graph	(p) Parabola
(B) Velocity-acceleration graph	(q) Straight line
(C) Velocity-time graph	(r) Circle
(D) Acceleration-time graph	(s) None

Codes

A	B	C	D	A	B	C	D
(a) p	q	r	s	(b) s	q	p	r
(c) q	s	s	s	(d) r	p	s	p

- 3 In case of second's pendulum, match the Column I with Column II and mark the correct option from the codes given below. (Consider shape of earth also).

Column I	Column II
(A) At pole	(p) $T > 2s$
(B) On a satellite	(q) $T < 2s$
(C) At mountain	(r) $T = 2s$
(D) At centre of earth	(s) $T = 0$
	(t) $T = \infty$

Codes

A	B	C	D
(a) q	t	p	t
(b) r	p	s	t
(c) p	q	r	s
(d) t	s	q	p

(C) Medical entrances' gallery

Collection of questions asked in NEET & Various Medical Entrance Exams

- 1 Identify the function which represents a periodic motion. [NEET 2020]
 (a) $e^{\omega t}$ (b) $\log_e(\omega t)$
 (c) $\sin \omega t + \cos \omega t$ (d) $e^{-\omega t}$
- 2 The phase difference between displacement and acceleration of a particle in a simple harmonic motion is [NEET 2020]
 (a) $\frac{3\pi}{2}$ rad (b) $\frac{\pi}{2}$ rad (c) zero (d) π rad
- 3 The distance covered by a particle undergoing SHM in one time period is (amplitude = A) [NEET (Odisha) 2019]
 (a) zero (b) A (c) $2A$ (d) $4A$
- 4 The displacement of a particle executing simple harmonic motion is given by
 $y = A_0 + A \sin \omega t + B \cos \omega t$ [NEET 2019]
 Then, the amplitude of its oscillation is given by
 (a) $\sqrt{A^2 + B^2}$ (b) $\sqrt{A_0^2 + (A + B)^2}$
 (c) $A + B$ (d) $A_0 + \sqrt{A^2 + B^2}$
- 5 Average velocity of a particle executing SHM in one complete vibration is [NEET 2019]
 (a) $A\omega$ (b) $\frac{A\omega^2}{2}$ (c) zero (d) $\frac{A\omega}{2}$
- 6 A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is [NEET 2018]
 (a) 2 s (b) π s (c) 2π s (d) 1 s
- 7 A block of rectangular size of mass m and area of cross-section A , floats in a liquid of density ρ . If we give a small vertical displacement from equilibrium, it undergoes SHM with time period T , then [AIIMS 2018]
 (a) $T^2 \propto \frac{1}{\rho}$ (b) $T^2 \propto \rho$
 (c) $T^2 \propto m^{-1}$ (d) $T^2 \propto \frac{1}{A^2}$
- 8 A particle executing SHM having amplitude 5cm, mass 0.5 kg and angular frequency 5 rad/s is at 1cm from mean position. Find potential energy and kinetic energy. [JIPMER 2018]
 (a) $KE = 6.25 \times 10^{-4} \text{ J}$, $PE = 150 \times 10^{-3} \text{ J}$
 (b) $KE = 150 \times 10^{-4} \text{ J}$, $PE = 6.25 \times 10^{-4} \text{ J}$
 (c) $KE = 6.25 \times 10^{-4} \text{ J}$, $PE = 6.25 \times 10^{-4} \text{ J}$
 (d) $KE = 150 \times 10^{-3} \text{ J}$, $PE = 150 \times 10^{-4} \text{ J}$

- 9 Given equation of SHM is $Y = 5 \cos \frac{\pi}{2} (100t - 2x)$, what is time period? [JIPMER 2018]

(a) 0.04 s (b) 1 s
(c) 0.06 s (d) 0.02 s

- 10 A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then, its time period in seconds is [NEET 2017]

(a) $\frac{\sqrt{5}}{\pi}$ (b) $\frac{\sqrt{5}}{2\pi}$ (c) $\frac{4\pi}{\sqrt{5}}$ (d) $\frac{2\pi}{\sqrt{3}}$

- 11 The displacement of a particle along the X-axis is given by $x = a \sin^2 \omega t$. The motion of the particle corresponds to [JIPMER 2017]

(a) simple harmonic motion of frequency ω / π
(b) simple harmonic motion of frequency $3\omega/2\pi$
(c) non-simple harmonic motion
(d) simple harmonic motion of frequency $\omega / 2\pi$

- 12 A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3 s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m (in kg) is [NEET 2016]

(a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{16}{9}$ (d) $\frac{9}{16}$

- 13 Two similar springs P and Q have spring constants k_P and k_Q , such that $k_P > k_Q$. They are stretched, first by the same amount (case a), then by the same force (case b). The work done by the springs W_P and W_Q are related as, in case (a) and case (b), respectively [CBSE AIPMT 2015]

(a) $W_P = W_Q$; $W_P > W_Q$ (b) $W_P = W_Q$; $W_P = W_Q$
(c) $W_P > W_Q$; $W_Q > W_P$ (d) $W_P < W_Q$; $W_Q < W_P$

- 14 A particle is executing SHM along a straight line. Its velocities at distances x_1 and x_2 from the mean position are v_1 and v_2 , respectively. Its time period is [CBSE AIPMT 2015; EAMCET 2013]

(a) $2\pi \sqrt{\frac{x_1^2 + x_2^2}{v_1^2 + v_2^2}}$ (b) $2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$
(c) $2\pi \sqrt{\frac{v_1^2 + v_2^2}{x_1^2 + x_2^2}}$ (d) $2\pi \sqrt{\frac{v_1^2 - v_2^2}{x_1^2 - x_2^2}}$

- 15 The velocity vector v and displacement vector x of a particle executing SHM are related as

$$\frac{v dv}{dx} = -\omega^2 x \text{ with the initial condition } v = v_0 \text{ at } x = 0.$$

The velocity v when displacement is x , is [AIIMS 2015]

(a) $v = \sqrt{v_0^2 + \omega^2 x^2}$ (b) $v = \sqrt{v_0^2 - \omega^2 x^2}$
(c) $v = \sqrt[3]{v_0^3 + \omega^3 x^3}$ (d) $v = v_0 - (\omega^3 x^3 e^{x^3})^{1/3}$

- 16 A particle of mass 200 g executes simple harmonic motion. The restoring force is provided by a spring of spring constant 80 Nm^{-1} . What would be the time period? [UK PMT 2015]

(a) 0.93 s (b) 0.62 s
(c) 0.31 s (d) None of these

- 17 A body of mass 40 g executing simple harmonic motion of amplitude 2 cm. If the time period is 0.20 s, what would be the total mechanical energy of the system? [UK PMT 2015]

(a) $14.8 \times 10^{-3} \text{ J}$ (b) $7.9 \times 10^{-3} \text{ J}$
(c) $23.7 \times 10^{-3} \text{ J}$ (d) $3.9 \times 10^{-3} \text{ J}$

- 18 A copper sphere attached to the bottom of a vertical spring is oscillating with time period 10 s. If the copper sphere is immersed in a fluid (assume the viscosity of the fluid is negligible) of specific gravity $\frac{1}{4}$ of that of the copper, then time period of the oscillation is [EAMCET 2015]

(a) 5 s (b) 10 s (c) 2.5 s (d) 20 s

- 19 Out of the following functions representing motion of a particle, which functions/function represent SHM?

A. $y = \sin \omega t - \cos \omega t$ B. $y = \sin^3 \omega t$

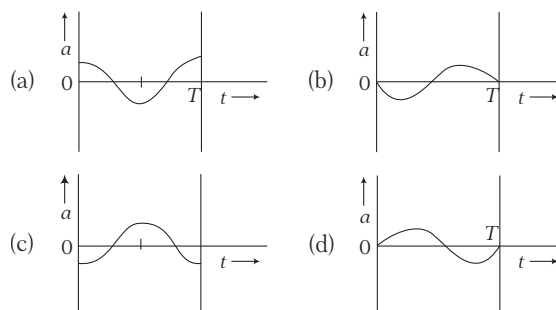
C. $y = 5 \cos \left(\frac{3\pi}{4} - 3\omega t \right)$ D. $y = 1 + \omega t + \omega^2 t^2 + \omega^3 t^3$ [CG PMT 2015]

(a) A and B (b) A and C
(c) A only (d) A, B and C not D

- 20 The oscillation of a body on a smooth horizontal surface is represented by the equation $x = A \cos \omega t$, where, x = displacement at time t and ω = frequency of oscillation.

Which one of the following graphs shows correctly the variation of a with t ? [CBSE AIPMT 2014]

Here, a = acceleration at time t and T = time period.



- 21** The ratio of kinetic energy to the potential energy of a particle executing SHM at a distance equal to half its amplitude, the distance being measured from its equilibrium position is [KCET 2015; Kerala CEE 2014]
 (a) 2 : 1 (b) 3 : 1 (c) 8 : 1 (d) 4 : 1
 (e) 1 : 1
- 22** A body oscillates with SHM according to the equation (in SI unit), $x = 5 \cos \left(2\pi t + \frac{\pi}{4} \right)$. Its instantaneous displacement at $t = 1$ s is [Kerala CEE 2014]
 (a) $\frac{\sqrt{2}}{5}$ m (b) $\frac{1}{\sqrt{3}}$ m (c) $\frac{1}{\sqrt{2}}$ m (d) $\frac{5}{\sqrt{2}}$ m
 (e) $\frac{1}{2}$ m
- 23** The amplitude of a particle executing SHM is 4 cm. At the mean position, the speed of the particle is 16 cms^{-1} . The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3} \text{ cms}^{-1}$ will be [AFMC 2014]
 (a) $2\sqrt{3}$ cm (b) $\sqrt{3}$ cm (c) 1 cm (d) 2 cm
- 24** What is the phase difference between two simple harmonic motions represented by $x_1 = A \sin \left(\omega t + \frac{\pi}{6} \right)$ and $x_2 = A \cos \omega t$? [WB JEE 2014]
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$
- 25** When a particle executing SHM oscillates with a frequency ν , then the kinetic energy of the particle [WB JEE 2014]
 (a) changes periodically with a frequency of ν
 (b) changes periodically with a frequency of 2ν
 (c) changes periodically with a frequency of $\nu/2$
 (d) remains constant
- 26** If a body is executing simple harmonic motion and its current displacement is $\sqrt{3}/2$ times the amplitude from its mean position, then the ratio between potential energy and kinetic energy is [EAMCET 2014]
 (a) 3 : 2 (b) 2 : 3 (c) $\sqrt{3} : 1$ (d) 3 : 1
- 27** A 10 kg metal block is attached to a spring of spring constant 1000 Nm^{-1} . A block is displaced from equilibrium position by 10 cm and released. The maximum acceleration of the block is [KCET 2014]
 (a) 10 ms^{-2} (b) 100 ms^{-2} (c) 200 ms^{-2} (d) 0.1 ms^{-2}
- 28** A block resting on the horizontal surface executes SHM in horizontal plane with amplitude A . The frequency of oscillation for which the block just starts to slip is (μ = coefficient of friction, g = gravitational acceleration) [MHT CET 2014]
 (a) $\frac{1}{2\pi} \sqrt{\frac{\mu g}{A}}$ (b) $\frac{1}{4\pi} \sqrt{\frac{\mu g}{A}}$
 (c) $2\pi \sqrt{\frac{A}{\mu g}}$ (d) $4\pi \sqrt{\frac{A}{\mu g}}$
- 29** The displacement of a particle in a periodic motion is given by $y = 4 \cos^2(t/2) \sin(1000t)$. This displacement may be considered as the result of superposition of n independent harmonic oscillations. Here, n is [WB JEE 2014]
 (a) 1 (b) 2 (c) 3 (d) 4
- 30** The time period of a simple pendulum of length 9.8 m is [JK CET 2013]
 (a) 0.159 s (b) 3.14 s (c) 6.5 s (d) 6.28 s
- 31** A particle executing simple harmonic motion covers a distance equal to half its amplitude in one second. Then, the time period of the simple harmonic motion is [Kerala CEE 2013]
 (a) 4 s (b) 6 s (c) 8 s (d) 12 s
 (e) 20 s
- 32** The displacement, velocity and acceleration in a simple harmonic motion are related as the [JK CET 2013]
 (a) displacement, velocity and acceleration all act in the same direction
 (b) displacement and velocity act in the same direction but acceleration in the opposite direction
 (c) velocity and acceleration are parallel and both are perpendicular to the displacement
 (d) displacement and acceleration are anti-parallel and both perpendicular to the velocity
- 33** If T_1 and T_2 are the time periods of oscillation of a simple pendulum on the surface of earth (of radius R) and at a depth d , the d is equal to [Kerala CEE 2013]
 (a) $\left(1 - \frac{T_1^2}{T_2^2} \right) R$ (b) $\left(1 - \frac{T_2^2}{T_1^2} \right) R$
 (c) $\left(1 - \frac{T_1}{T_2} \right) R$ (d) $\left(1 - \frac{T_2}{T_1} \right) R$
 (e) $\frac{T_1^2}{T_2^2}$
- 34** In case of a forced vibration, the resonance wave becomes very sharp when the [UP CPMT 2013]
 (a) restoring force is small
 (b) damping force is small
 (c) quality factor is small
 (d) applied periodic force is small
- 35** When a boy is playing on a swing in the sitting position, the time period of oscillations of the swing is T . If the boy stands up, the time period of oscillation of the spring will be [UP CPMT 2013]
 (a) more than T (b) less than T
 (c) equal to T (d) Cannot be predicted

- 36** To make the frequency of an oscillator double, one has to [UP CPMT 2013]
 (a) half the mass
 (b) quadruple the mass
 (c) double the mass
 (d) reduce the mass to one-fourth
- 37** A simple harmonic oscillator consists of a particle of mass m and an ideal spring with spring constant k . The particle oscillates with a time period T . The spring is cut into two equal parts. If one part oscillates with the same particle, the time period will be [AIIMS 2012]
 (a) $2T$ (b) $\sqrt{2}T$
 (c) $\frac{T}{\sqrt{2}}$ (d) $\frac{T}{2}$
- 38** A SHM is given by $y = 5[\sin(3\pi t) + \sqrt{3}\cos(3\pi t)]$. What is the amplitude of the motion of y in metre? [UP CPMT 2012]
 (a) 10 (b) 20 (c) 1 (d) 5
- 39** If a simple pendulum is taken to a place, where g decreases by 2%, then time period [UP CPMT 2012]
 (a) increase by 5%
 (b) increase by 1%
 (c) increase by 2%
 (d) decrease by 5%
- 40** If a simple pendulum executing SHM falls freely along with the support, then [BCECE 2012]
 (a) it does not oscillate at all
 (b) its periodic time increases
 (c) its periodic time decreases
 (d) None of the above
- 41** If the equation $y_1 = A\sin\omega t$ and $y_2 = \frac{A}{2}\sin\omega t + \frac{A}{2}\cos\omega t$ represents SHM, then the ratio of the amplitudes of the two motions is [BCECE 2012]
 (a) 0.5 (b) $\sqrt{2}$ (c) 1 (d) 2
- 42 Assertion** If a bob of a simple pendulum is kept in a horizontal electric field, its period of oscillation will remain same.
Reason If bob is charged and kept in horizontal electric field, then the time period will be decreased. [BCECE 2012]
 (a) If both Assertion and Reason are correct but Reason is the correct explanation of Assertion.
 (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) If Assertion is correct and Reason is incorrect.
 (d) If Assertion is incorrect and Reason is correct.
- 43** The displacements of two particles of same mass executing SHM are represented by the equations $x_1 = 4\sin\left(10t + \frac{\pi}{6}\right)$ and $x_2 = 5\cos(\omega t)$.
 The value of ω for which the energies of both the particles remain same is [Manipal 2012]
 (a) 16 unit (b) 6 unit (c) 4 unit (d) 8 unit
- 44** The amplitude and the periodic time of a SHM are 5 cm and 6 s, respectively. At a distance of 2.5 cm away from the mean position, the phase will be [JCECE 2012]
 (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{5\pi}{12}$

ANSWERS

• CHECK POINT 11.1

1. (c)	2. (a)	3. (d)	4. (c)	5. (b)	6. (b)	7. (c)	8. (a)	9. (b)	10. (c)
11. (c)	12. (b)	13. (b)	14. (a)	15. (b)	16. (c)	17. (b)			

• CHECK POINT 11.2

1. (b)	2. (d)	3. (b)	4. (c)	5. (b)	6. (b)	7. (c)	8. (d)	9. (d)	10. (b)
11. (a)	12. (b)								

• CHECK POINT 11.3

1. (c)	2. (a)	3. (a)	4. (d)	5. (b)	6. (d)	7. (c)	8. (b)	9. (b)	10. (b)
11. (a)	12. (a)	13. (b)	14. (b)	15. (d)	16. (a)				

• CHECK POINT 11.4

1. (c)	2. (d)	3. (c)	4. (a)	5. (d)	6. (b)	7. (d)	8. (a)	9. (c)	10. (c)
11. (a)									

(A) Taking it together

1. (c)	2. (c)	3. (a)	4. (b)	5. (a)	6. (d)	7. (c)	8. (b)	9. (c)	10. (d)
11. (b)	12. (c)	13. (c)	14. (d)	15. (b)	16. (a)	17. (c)	18. (a)	19. (d)	20. (d)
21. (a)	22. (b)	23. (c)	24. (a)	25. (d)	26. (a)	27. (d)	28. (d)	29. (d)	30. (a)
31. (a)	32. (d)	33. (a)	34. (d)	35. (b)	36. (d)	37. (c)	38. (a)	39. (c)	40. (c)
41. (c)	42. (b)	43. (b)	44. (b)	45. (c)	46. (c)	47. (d)	48. (b)	49. (b)	50. (c)
51. (c)	52. (c)	53. (c)	54. (b)	55. (d)	56. (c)	57. (d)	58. (a)	59. (b)	60. (c)
61. (d)	62. (b)	63. (a)	64. (b)	65. (d)	66. (b)	67. (a)	68. (b)	69. (b)	70. (a)
71. (b)	72. (b)	73. (b)	74. (c)	75. (a)	76. (a)	77. (c)	78. (a)	79. (c)	80. (c)
81. (a)	82. (b)	83. (b)	84. (b)	85. (d)	86. (b)	87. (b)	88. (c)	89. (b)	90. (a)

(B) Medical entrance special format questions

• Assertion and reason

1. (a)	2. (a)	3. (d)	4. (a)	5. (c)
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• Statement based questions

1. (d)	2. (d)	3. (d)	4. (c)	5. (d)	6. (d)	7. (b)	8. (c)	9. (b)	10. (a)
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• Match the columns

1. (b)	2. (c)	3. (a)
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(C) Medical entrances' gallery

1. (c)	2. (d)	3. (d)	4. (a)	5. (c)	6. (b)	7. (a)	8. (b)	9. (a)	10. (c)
11. (c)	12. (d)	13. (c)	14. (b)	15. (b)	16. (c)	17. (b)	18. (b)	19. (b)	20. (c)
21. (b)	22. (d)	23. (d)	24. (b)	25. (b)	26. (d)	27. (a)	28. (a)	29. (c)	30. (d)
31. (d)	32. (b)	33. (a)	34. (b)	35. (b)	36. (d)	37. (c)	38. (a)	39. (b)	40. (a)
41. (b)	42. (b)	43. (d)	44. (c)						

Hints & Explanations

• CHECK POINT 11.1

1 (c) The motion of planets around the sun is periodic but not simple harmonic motion.

2 (a) The body is non-uniformly accelerated in case of simple harmonic motion.

$$3 (d) y = A \sin(\omega t + \phi) = A \sin\left(\frac{2\pi}{T}t + \phi\right)$$

$$\Rightarrow y = 0.5 \sin\left(\frac{2\pi}{0.4}t + \frac{\pi}{2}\right)$$

$$y = 0.5 \sin\left(5\pi t + \frac{\pi}{2}\right) = 0.5 \cos 5\pi t$$

4 (c) When the particle moves along the circumference of a circle with constant angular velocity, the projection of the particle on any of the diameters executes SHM.

5 (b) Given equation, $y = 10 \sin(20t + \pi/3)$... (i)
Comparing Eq. (i) with general equation of motion of simple harmonic $x = A \sin(\omega t + \phi)$, we get

$$A = 10 \text{ m} \quad \omega = 20 \text{ rad s}^{-1}$$

$$\text{As,} \quad \omega = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi}{\omega} = \frac{\pi}{10} \text{ second}$$

6 (b) Given equation, $x = 0.34 \cos(3000t + 0.74)$... (i)

Comparing Eq. (i) with general equation of motion of simple harmonic $x = A \cos(\omega t + \phi)$, we get

$$A = 0.34 \text{ mm}, \omega = 3000 \text{ rad s}^{-1}$$

$$\therefore \text{Frequency, } \nu = \frac{\omega}{2\pi} = \frac{3000}{2\pi} \text{ Hz}$$

7 (c) $\therefore$ Velocity, $v = \omega \sqrt{A^2 - x^2} = 2 \sqrt{(60)^2 - (20)^2} \approx 113 \text{ mms}^{-1}$

8 (a) $\therefore$ Maximum velocity, $v_{\max} = A\omega = 3 \times 100 = 300 \text{ units}$

9 (b) Here, $\omega A = 3 \text{ ms}^{-1}$ or $\omega = \frac{3}{A} = \frac{3}{0.1} = 30 \text{ rad s}^{-1}$

$$\therefore \text{Frequency, } f = \frac{\omega}{2\pi} = \frac{30}{2\pi} = \left(\frac{15}{\pi}\right) \text{ Hz}$$

10 (c) Velocity at mean position $v = A\omega$, velocity at a distance of half amplitude,

$$v' = \omega \sqrt{A^2 - y^2} = \omega \sqrt{A^2 - \frac{A^2}{4}} \Rightarrow v' = \frac{\sqrt{3}}{2} A\omega = \frac{\sqrt{3}}{2} v$$

11 (c) $\therefore v_{\max} = A\omega \Rightarrow \omega = \frac{v_{\max}}{A} = \frac{10}{4}$

$$\text{Now,} \quad v = \omega \sqrt{A^2 - y^2}$$

$$\Rightarrow v^2 = \omega^2 (A^2 - y^2)$$

$$\Rightarrow y^2 = A^2 - \frac{v^2}{\omega^2}$$

$$\Rightarrow y = \sqrt{A^2 - \frac{v^2}{\omega^2}} = \sqrt{4^2 - \frac{5^2}{(10/4)^2}} = 2\sqrt{3} \text{ cm}$$

12 (b) $\therefore a = -\omega^2 x$

$$\therefore \frac{x}{a} = -\frac{1}{\omega^2} = \text{constant}$$

13 (b) Given, $\omega = \frac{\pi}{2} \text{ rad s}^{-1}$ and $A = 2 \text{ cm}$

$$\therefore \text{Maximum acceleration} = \omega^2 A = \frac{\pi^2}{2} \text{ cms}^{-2}$$

14 (a) $\therefore \omega^2 A = a_0$... (i)

$$\text{and} \quad \omega A = v_0$$

... (ii)

On solving Eqs. (i) and (ii), we get

$$A = \frac{v_0^2}{a_0}$$

15 (b) $a = \omega^2 y \Rightarrow \omega = \sqrt{\frac{a}{y}} = \sqrt{\frac{8}{2}} = 2 \text{ rad s}^{-1}$

$$\text{Now,} \quad v_{\max} = A\omega = 6 \times 2 = 12 \text{ cms}^{-1}$$

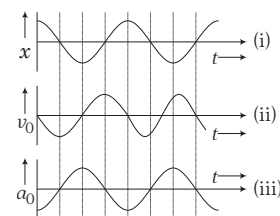
16 (c) The average acceleration of a particle performing SHM over one complete oscillation is zero. Because acceleration is a function of sine and average value of sine for a complete oscillation is zero.

17 (b) As, $x = A \cos \omega t$

Velocity, $v = A\omega \cos\left(\omega t + \frac{\pi}{2}\right)$ and the acceleration,

$$a = A\omega^2 \cos(\omega t + \pi)$$

Here, A = amplitude.



$\therefore$ In SHM, the acceleration is ahead of velocity by a phase angle $\frac{\pi}{2}$.

• CHECK POINT 11.2

1 (b) Given, $A = 6 \text{ cm}$, $\omega = 100 \text{ rad s}^{-1}$

Maximum kinetic energy, $K_{\max} = \frac{1}{2} m \omega^2 A^2$

$$= \frac{1}{2} \times 1 \times (100)^2 \times (6 \times 10^{-2})^2 = 18 \text{ J}$$

2 (d) $\text{KE} = \frac{1}{2} m \omega^2 (A^2 - x^2) \Rightarrow \text{KE} = \frac{1}{2} m \omega^2 \left(A^2 - \frac{A^2}{4}\right) \left(\because x = \frac{A}{2}\right)$

$$= \frac{3}{4} \times \frac{1}{2} m \omega^2 A^2 = \frac{3}{4} E$$

- 3** (b) Frequency of oscillation of kinetic energy is doubled, i.e. 2ω .

$$\therefore T = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$$

- 4** (c) Amplitude, $A = 4$ cm

For, KE = PE

$$\Rightarrow \frac{1}{2} m \omega^2 (A^2 - x^2) = \frac{1}{2} m \omega^2 x^2$$

$$\Rightarrow x^2 = \frac{A^2}{2} \Rightarrow x = \frac{A}{\sqrt{2}}$$

$$\therefore x = \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ cm}$$

5 (b) $\frac{K}{E} = \frac{\frac{1}{2} m \omega^2 (A^2 - y^2)}{\frac{1}{2} m \omega^2 A^2} = \frac{A^2 - y^2}{A^2} = 1 - \frac{y^2}{A^2}$

So, $\left(\frac{3E}{4}\right) = 1 - \frac{y^2}{A^2}$

$$\Rightarrow \frac{y^2}{A^2} = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow y = \frac{A}{2}$$

- 6** (b) Potential energy is minimum and kinetic energy is maximum, at mean position, i.e. at $x = 0$.

- 7** (c) $U = K$ or $U = \frac{E}{2}$

$$\therefore \frac{1}{2} kx^2 = \frac{1}{2} \left(\frac{1}{2} kA^2 \right)$$

i.e. at $x = \pm \frac{A}{\sqrt{2}}$, $U = K$

and this situation will occur four times in one complete period.

- 8** (d) The work done by the pendulum in one oscillation will be zero because displacement is zero.

- 9** (d) As, $E = \frac{1}{2} m \omega^2 A^2$

or $E = \frac{1}{2} m \left(\frac{2\pi}{T} \right)^2 A^2 \quad \left(\because \omega = \frac{2\pi}{T} \right)$

As, $T' = 2T$ and $A' = 2A$

$$\therefore E' = \frac{1}{2} m \times \frac{1}{4} \left(\frac{2\pi}{T} \right)^2 \times 4 (A)^2 = E$$

- 10** (b) Energy in SHM, $E = \frac{1}{2} m \omega^2 A^2$

i.e. E depends on ω and A both.

11 (a) $E = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} m (2\pi f)^2 A^2 = 2\pi^2 f^2 m A^2$
 $= 2\pi^2 (20)^2 (0.10)(0.05)^2 \approx 2 \text{ J}$

- 12** (b) If at any instant, displacement is y , then it is given that

$$U = \frac{1}{2} \times E$$

$$\Rightarrow \frac{1}{2} m \omega^2 y^2 = \frac{1}{2} \times \left(\frac{1}{2} m \omega^2 A^2 \right)$$

$$\Rightarrow y = \frac{A}{\sqrt{2}} = \frac{6}{\sqrt{2}} = 4.2 \text{ cm}$$

• CHECK POINT 11.3

- 1** (c) Time period will remain the same because time period T of simple pendulum is independent of mass of the bob.

2 (a) $T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow T = 2\pi \sqrt{\frac{39.2}{\pi^2 \times 9.8}} = 4 \text{ s}$

- 3** (a) According to question, $l' = 4l$

$$\therefore T' = 2\pi \sqrt{\frac{4l}{g}} = 2T$$

4 (d) As, $T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l}$

Given, $\frac{\Delta l}{l} = 21\%$

$$\Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \times 21\% = 10.5\%$$

- 5** (b) $T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}}$. Here, $g_{\text{eff}} = g - g$, which is zero in a freely falling lift. Hence, $T = \infty$

- 6** (d) The pendulum is suspended from the roof of a trolley which is moving in horizontal direction with acceleration a . Hence, the force acting will be

$$\mathbf{F} = m\mathbf{a}\hat{\mathbf{i}} - m\mathbf{g}\hat{\mathbf{j}}$$

Let θ angle between the horizontal and vertical, then

$$\tan \theta = \frac{ma}{mg} = \frac{a}{g}$$

Then, the value of $g' = \sqrt{g^2 + a^2}$.

- 7** (c) In stationary lift, $T = 2\pi \sqrt{\frac{l}{g}}$

In upward motion of lift,

$$T' = 2\pi \sqrt{\frac{l}{(g+a)}} \quad (a = \text{acceleration of lift})$$

$$\therefore \frac{T'}{T} = \sqrt{\frac{g}{g+a}} = \sqrt{\frac{g}{\left(g + \frac{g}{4}\right)}} \quad \left(\text{given, } a = \frac{g}{4} \right)$$

$$= \sqrt{\frac{4}{5}} \Rightarrow T' = \frac{2T}{\sqrt{5}}$$

- 8** (b) When a little mercury is drained off, the position of centre of mass of ball falls (with respect to fixed end), so that effective length of pendulum increases, hence T increases.

- 9** (b) $T = 2\pi \sqrt{\frac{R}{g}} = 84.6 \text{ min}$ (If l is infinite)

$$10 \text{ (b)} \quad n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \Rightarrow n \propto \frac{1}{\sqrt{m}}$$

$$\therefore n' = n/2$$

$$(\because \sqrt{m'} = \sqrt{4m})$$

$$11 \text{ (a)} \quad k \propto \frac{1}{l} \text{ and } f \propto \sqrt{k}$$

Spring constant of each part will be $2k$.

$$\therefore f' = \sqrt{2} f$$

$$12 \text{ (a)} \quad T = 2\pi \sqrt{\frac{m}{k}} \text{ or } T \propto \sqrt{m}$$

$$\therefore \frac{T}{T'} = \sqrt{\frac{m_1}{m_2}}$$

$$\Rightarrow \frac{T}{T'} = \sqrt{\frac{(0.8 - 0.4)}{(0.8 - 0.4 - 0.3)}} = \frac{2}{T'} \quad (\because \text{Given, } T = 2 \text{ s})$$

$$\Rightarrow \sqrt{\frac{0.4}{0.1}} = \frac{2}{T'} \Rightarrow \frac{2}{T'} = 2 \Rightarrow T' = 1 \text{ s}$$

13 (b) The spring mass system oscillates in SHM, its time period is given by

$$T = 2\pi \sqrt{\frac{m}{k}}$$

When spring is cut into ratio 1 : 3, the new time constant is $k' = 3k$

$$\therefore \frac{T}{T'} = \sqrt{\frac{3k}{k}} \Rightarrow T' = \frac{T}{\sqrt{3}}$$

$$14 \text{ (b)} \quad f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \text{ or } f \propto \sqrt{k}$$

$$k = k_1 + k_2 \text{ and } k' = 4(k_1 + k_2) = 4k$$

$$\text{So, } f \propto \sqrt{k'} \Rightarrow f' \propto \sqrt{4k}$$

$$f' \propto 2\sqrt{k} \Rightarrow f' \propto 2f$$

15 (d) The springs are connected in parallel. So, the effective force constant will be $k_{\text{eff}} = k_1 + k_2$

$$\therefore T = 2\pi \sqrt{\frac{m}{k_{\text{eff}}}} = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$$

$$16 \text{ (a)} \quad T_a = 2\pi \sqrt{\frac{m}{k}}$$

$$T_b = 2\pi \sqrt{\frac{m}{(k/2)}} \quad (\because \text{Springs are connected in series})$$

$$T_c = 2\pi \sqrt{\frac{m}{2k}} \quad (\because \text{Springs are connected in parallel})$$

• CHECK POINT 11.4

$$1 \text{ (c)} \text{ Here, } k = \frac{YA}{L}$$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$$

$$2 \text{ (d)} \quad T = 2\pi \sqrt{\frac{m}{\rho Ag}} \Rightarrow T \propto \frac{1}{\sqrt{\rho}}$$

$$3 \text{ (c)} \text{ The time period in this case is given by } T = 2\pi \sqrt{\frac{R}{g}}$$

where, R is the radius of earth.

4 (a) Amplitude of damped oscillation is $A = A_0 e^{-\gamma t}$

$$\text{Given, } A = \frac{A_0}{2}, \text{ at } t = 1 \text{ s}$$

$$\text{So, } \frac{A_0}{2} = A_0 e^{-\gamma t} \text{ or } e^{\gamma} = 2$$

$$\text{After 2 s, } A = A_0 e^{-\gamma^2} \Rightarrow A = \frac{A_0}{4}$$

5 (d) In damped oscillation, amplitude goes on decaying exponentially as $a = a_0 e^{-bt}$, where b = damping coefficient.

$$\text{Initially, } \frac{a_0}{3} = a_0 e^{-b \times 100T}, T = \text{time of one oscillation}$$

$$\text{or } \frac{1}{3} = e^{-100bT} \quad \dots(i)$$

$$\text{Finally, } a = a_0 e^{-b \times 200T}$$

$$\text{or } a = a_0 [e^{-100bT}]^2 \text{ or } a = a_0 \times \left[\frac{1}{3}\right]^2 \quad [\text{From Eq. (i)}]$$

$$\text{or } a = a_0/9$$

6 (b) With weak damping frequency (n_1) of the oscillator, the system reaches frequency of driving force (n_2) at steady state.

7 (d) Amplitude always remains constant in case of sustained forced oscillations.

8 (a) In forced oscillations, particle oscillates with a frequency equal to the frequency of driving force.

9 (c) In resonant vibrations of a body, the frequency of external force applied on the body is equal to its natural frequency. If on increasing or decreasing the frequency of external force from the natural frequency by a large factor, the amplitude of vibrations reduces very much. In this case, sharp resonance will take place.

But if it reduces by a small factor, then flat resonance will take place. The sharp and flat resonance will depend on damping present in the body executing resonant vibrations. Lesser the damping, greater will be sharpness.

10 (c) Two sources of sound are in resonance when they produce the sound of same frequency.

11 (a) During the phenomenon of resonance, the amplitude of oscillation becomes large. Because applied frequency is equal to natural frequency.

(A) Taking it together

1 (c) **a** and **s** are always in opposite directions.

$$2 \text{ (c)} \quad T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow \frac{l}{T^2} = \frac{g}{4\pi^2} = \text{constant}$$

3 (a) The amplitude is the maximum displacement of the particle from the mean position, i.e. A .

4 (b) There is no effect of phase angle on amplitude. Hence,

$$\frac{A_1}{A_2} = \frac{10}{5} = \frac{2}{1}$$

$$5 \text{ (a)} \text{ Frequency of oscillation, } f = \frac{1}{T} = \frac{1}{0.04} = 25 \text{ Hz}$$

6 (d) Frequency, $n \propto \frac{1}{\sqrt{l}}$

$$\Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{l_2}{l_1}} \Rightarrow \frac{l_2}{l_1} = \frac{n_1^2}{n_2^2} \Rightarrow \frac{l_2}{l_1} = \frac{(2)^2}{(3)^2}$$

$$\Rightarrow \frac{l_2}{l_1} = \frac{4}{9} \Rightarrow \frac{l_1}{l_2} = \frac{9}{4}$$

7 (c) Given, $x = 5 \cos \left(2\pi t + \frac{\pi}{4} \right)$ (where, x is displacement)

$$\Rightarrow x = 5 \cos \left(2\pi + \frac{\pi}{4} \right) \quad (\text{At } t = 1\text{ s})$$

$$\Rightarrow x = 5 \cos \frac{\pi}{4} \quad [\because \cos(360^\circ + \theta) = \cos \theta]$$

$$\Rightarrow x = 5 \times \frac{1}{\sqrt{2}} = \frac{5}{\sqrt{2}} \text{ m}$$

8 (b) $|a| = \omega^2 x$

$$12 = \omega^2 (3)$$

$$\therefore \omega = 2 \text{ rad s}^{-1} = \frac{2\pi}{T}$$

$$\text{or } T = (\pi) \text{ s or } 3.14 \text{ s}$$

9 (c) In case of spring block system, time period does not change because restoring force is still kx .

10 (d) Using acceleration, $a = -\omega^2 x$

At $-x_{\max}$, a will be maximum and positive.

11 (b) $y_1 = A \cos \omega t = A \sin \left(\omega t + \frac{\pi}{2} \right)$

$$y_2 = A \sin \omega t$$

$$\therefore \Delta\phi = \omega t + \frac{\pi}{2} - \omega t = \frac{\pi}{2} = 90^\circ$$

12 (c) Equation of motion, $y = A \cos \omega t$

$$\Rightarrow \frac{A}{2} = A \cos \omega t \Rightarrow \omega t = \frac{\pi}{3}$$

$$\Rightarrow \frac{2\pi t}{T} = \frac{\pi}{3} \Rightarrow t = \frac{\frac{\pi}{3} \times T}{2\pi} = \frac{4}{3 \times 2} = \frac{2}{3} \text{ s}$$

13 (c) $\omega A = \text{constant}$ and $\omega^2 A$ is made two times.

This is possible when frequency (ω) is doubled and amplitude (A) is halved.

14 (d) At $x = 0$, $F = 0$. For $x > 0$, force is in negative direction and for $x < 0$, force is in positive x -direction. Therefore, motion of the particle is periodic about mean position $x = 0$.

15 (b) $k \propto \frac{1}{l}$, hence k of individual halves will be 16 Nm^{-1} . When they are connected in parallel, effective value of k will become 32 Nm^{-1} .

16 (a) Suppose $x = A \sin \omega t$ and $y = A \cos \omega t = A \sin (\omega t + \pi/2)$

Then, by squaring and adding these two equations, we get

$$x^2 + y^2 = A^2 \quad (\text{circle})$$

Thus, option (a) is correct.

17 (c) $\omega = \frac{2\pi}{T} = \frac{\pi}{4} \text{ s}$. Therefore, $y = A \sin \frac{\pi}{4} t$. Now, put $t = 1 \text{ s}$

$$\text{and then } t = 2 \text{ s, we get } \frac{y_1}{y_2} = \frac{1}{\sqrt{2}}.$$

18 (a) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{16}{1}} = 4 \text{ rad s}^{-1}$

$$\text{Now, } K_{\max} = \frac{1}{2} m \omega^2 A^2$$

$$= \frac{1}{2} \times 1 \times (4)^2 (5 \times 10^{-2})^2 = 2 \times 10^{-2} \text{ J}$$

19 (d) At time $\frac{T}{2}$, $v = 0$

$\therefore$ Total energy = Potential energy

20 (d) $T = 2\pi \sqrt{\frac{l}{g}}$ and $T' = 2\pi \sqrt{\frac{l}{\left(g + \frac{g}{4}\right)}}$ or $T' = \frac{2}{\sqrt{5}} T = 2 \text{ s}$

21 (a) $\frac{1}{2} k A^2 = \frac{1}{2} \times 2 \times 10^6 \times (0.01)^2 = 100 \text{ J}$

At mean positions,

$$U = 60 \text{ J and } K = 100 \text{ J}$$

At extreme positions,

$$U = 160 \text{ J and } K = 0$$

22 (b) $T = 2\pi \sqrt{\frac{l}{g}}$ or $T^2 \propto l$

i.e. T - l graph is a parabola.

23 (c) For second's pendulum at the surface of earth,

$$2 = 2\pi \sqrt{\frac{l_e}{g_e}} \quad \dots(i)$$

For second's pendulum at the surface of moon,

$$= 2\pi \sqrt{\frac{l_m}{g_m}} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{l_e}{g_e} = \frac{l_m}{g_m} \Rightarrow l_m = \left(\frac{g_m}{g_e} \right) l_e$$

$$\text{As, } g_m = \frac{1}{6} g_e \Rightarrow l_m = \frac{l_e}{6} = \frac{1}{6} \text{ times } (\because l_e = 1 \text{ m})$$

24 (a) $k_s = \frac{k \times k}{k + k} = \frac{k}{2}$

$$k_p = k + k = 2k$$

$$\therefore \frac{n_s}{n_p} = \frac{\frac{1}{2\pi} \sqrt{\frac{k_s}{M}}}{\frac{1}{2\pi} \sqrt{\frac{k_p}{M}}} = \sqrt{\frac{k_s}{k_p}} = \sqrt{\frac{k/2}{2k}} = \frac{1}{2}$$

25 (d) For motion to be SHM, acceleration of the particle must be proportional to negative of displacement.

$$\text{i.e. } a \propto -y \text{ (or } x)$$

$$\Rightarrow a = -[y \text{ (or } x)]$$

i.e. x (or y) has to be linear.

- 26** (a) Let equation of a SHM is represented by $y = A \sin \omega t$

$$v = \frac{dy}{dt} = A \omega \cos \omega t$$

$$\Rightarrow (v)_{\max} = A \omega = 30 \quad \dots(i)$$

$$\text{Acceleration } (a) = \frac{dv}{dt} = -A \omega^2 \sin \omega t$$

$$a_{\max} = \omega^2 A = 60 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\omega (\omega A) = 60 \Rightarrow \omega (30) = 60$$

$$\Rightarrow \omega = 2 \text{ rad s}^{-1}$$

$$\Rightarrow \frac{2\pi}{T} = 2 \text{ rad s}^{-1} \Rightarrow T = \pi \text{ s}$$

- 27** (d) Here, $A = 0.04 \text{ m}$, $\omega^2 A = 0.30 \text{ ms}^{-2}$

$$\therefore \omega = 2.74 \text{ rads}^{-1}$$

$$\text{Now, } T = \frac{2\pi}{\omega} = 2.29 \text{ s}$$

- 28** (d) $T = 2\pi \sqrt{\frac{I}{MgL}} = 2\pi \sqrt{\frac{l}{g}}$

$$\text{or } l = \frac{I}{ML} = \frac{(3/2 MR^2)}{MR} = \frac{3}{2} R$$

- 29** (d) $T \propto \sqrt{l} \Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l} = \frac{0.02}{2} = 0.01$

$$\Rightarrow \Delta T = 0.01 T$$

$$\text{Loss of time per day} = 0.01 \times 24 \times 60 \times 60 = 864 \text{ s}$$

- 30** (a) Time period, $T \propto \sqrt{l}$

Also, according to thermal expansion, $l' = l(1 + \alpha \Delta \theta)$

$$\frac{\Delta l}{l} = \alpha \Delta \theta$$

$$\text{Hence, } \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l} = \frac{1}{2} \alpha \Delta \theta$$

$$= \frac{1}{2} \times 12 \times 10^{-6} \times (40 - 20) = 12 \times 10^{-5}$$

$$\Rightarrow \Delta T = 12 \times 10^{-5} \times 86400 \text{ s/day}$$

$$\therefore \Delta T \approx 10.3 \text{ s/day}$$

- 31** (a) Displacement of damped oscillations will be

$$x = x_m e^{-bt/2m} \sin(\omega_d t + \phi)$$

$$\text{where, } \omega = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$$

- 32** (d) According to the question, the displacement,

$$y = a \sin \omega t + b \cos \omega t$$

$$\text{Let } a = A \sin \phi \text{ and } b = A \cos \phi$$

$$\text{Now, } a^2 + b^2 = A^2 \sin^2 \phi + A^2 \cos^2 \phi$$

$$= A^2 \Rightarrow A = \sqrt{a^2 + b^2}$$

$$\therefore y = A \sin \phi \cdot \sin \omega t + A \cos \phi \cdot \cos \omega t$$

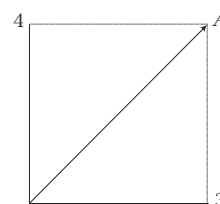
$$= A \sin(\omega t + \phi) \Rightarrow \frac{dy}{dt} = A \omega \cos(\omega t + \phi)$$

$$\Rightarrow \frac{d^2 y}{dt^2} = -A \omega^2 \sin(\omega t + \phi) = -A y \omega^2 = (-A \omega^2) y$$

$$\Rightarrow \frac{d^2 y}{dt^2} \propto (-y)$$

Hence, it is an equation of SHM with amplitude, $A = \sqrt{a^2 + b^2}$.

- 33** (a) $x = 3 \sin 2t + 4 \cos 2t$. From given equation,



$$A_1 = 3, A_2 = 4 \text{ and } \phi = \frac{\pi}{2}$$

$$\therefore A = \sqrt{A_1^2 + A_2^2} = \sqrt{3^2 + 4^2} = 5$$

$$\Rightarrow v_{\max} = A \omega = 5 \times 2 = 10$$

- 34** (d) $\because y$ is sinusoidal function.

$$\text{Acceleration} = -\omega^2 y, \text{ so } F = -m \omega^2 y.$$

So, F will also be sinusoidal function with phase difference π .

- 35** (b) Ball will execute SHM inside the tunnel with time period,

$$T = 2\pi \sqrt{R/g} = 84.63 \text{ min}$$

Hence, time taken by the ball to reach from one end to the

$$\text{other end of the tunnel, } t = \frac{84.63}{2} = 42.3 \text{ min}$$

- 36** (d) Given, $\omega_1 A_1 = \omega_2 A_2$

$$\Rightarrow \frac{A_1}{A_2} = \frac{\omega_2}{\omega_1} = \sqrt{\frac{k_2}{k_1}} \quad \left(\because \omega = \sqrt{\frac{k}{m}} \right)$$

- 37** (c) $y = a \sin \frac{2\pi}{T} t$

$$\Rightarrow \frac{a}{\sqrt{2}} = a \sin \frac{2\pi}{T} \cdot t$$

$$\Rightarrow \sin \frac{2\pi}{T} t = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4}$$

$$\Rightarrow \frac{2\pi}{T} t = \frac{\pi}{4}$$

$$\therefore t = \frac{T}{8}$$

- 38** (a) $mg = kx$

$$\Rightarrow k = \frac{mg}{x}$$

$$\therefore T = 2\pi \sqrt{\frac{M+m}{k}} = 2\pi \sqrt{\frac{(M+m)x}{mg}}$$

- 39** (c) Velocity, $v = \omega \sqrt{A^2 - x^2}$

$$\text{and magnitude of acceleration} = \omega^2 x$$

$$\text{Now given, } \omega^2 x = \omega \sqrt{A^2 - x^2}$$

$$\Rightarrow \omega^2 \cdot 1 = \omega \sqrt{2^2 - 1^2}$$

$$\Rightarrow \omega = \sqrt{3}$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{3}}$$

- 40** (c) At its lowermost point, spring is stretched by 8 cm or 8×10^{-2} m.

$$\begin{aligned} \therefore F_{\text{net}} &= kx - mg \\ &= (2 \times 10^3 \times 8 \times 10^{-2}) - (10 \times 10) = 60 \text{ N} \end{aligned}$$

- 41** (c) Reduced mass of two blocks,

$$\mu = \frac{m_1 m_2}{m_1 + m_2} = 2 \text{ kg}$$

$$\text{Now, } f = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} = \frac{1}{2\pi} \sqrt{\frac{300}{2}} \approx 2 \text{ Hz}$$

- 42** (b) Force constant, $k \propto \frac{1}{\text{Length of spring}}$

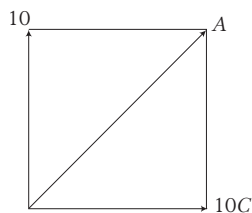
$$\therefore \frac{k_A}{k} = \frac{L}{\alpha L}$$

$$\therefore k_A = \frac{k}{\alpha}$$

$$\text{Similarly, } k_B = \frac{k}{1 - \alpha}$$

- 43** (b) $T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.2}{80}} = \frac{\pi}{10} \text{ s}$

- 44** (b) For y_2 ,



$$A = \sqrt{(10)^2 + (10C)^2} \text{ or } 10\sqrt{1 + C^2} = 40$$

$$\therefore C = \sqrt{15}$$

- 45** (c) At $x = 0$, $U = 0$. Therefore, total mechanical energy is equal to the maximum kinetic energy.

$$\therefore \frac{1}{2} m v_{\text{max}}^2 = 36$$

$$\text{or } v_{\text{max}} = \sqrt{\frac{72}{m}} = \sqrt{\frac{72}{2}} = 6 \text{ ms}^{-1}$$

- 46** (c) $\frac{1}{2} m \omega^2 A^2 = 8 \times 10^{-3} \text{ J}$

$$\therefore \frac{1}{2} \times 0.1 \times \omega^2 \times (0.1)^2 = 8 \times 10^{-3} \text{ or } \omega = 4 \text{ rad s}^{-1}$$

Therefore, the equation of motion of the particle, if the initial phase of the oscillation is 45° , will be

$$y = 0.1 \sin\left(4t + \frac{\pi}{4}\right)$$

- 47** (d) $T = 2\pi \sqrt{\frac{m}{k}}$

$$\text{or } T \propto \sqrt{m}$$

Mass is increasing, therefore time period will increase.

- 48** (b) Given, $y = 3 \cos\left(\frac{\pi}{4} - 2\omega t\right)$

Velocity of the particle,

$$\begin{aligned} v &= \frac{dy}{dt} = \frac{d}{dt} \left[3 \cos\left(\frac{\pi}{4} - 2\omega t\right) \right] \\ &= 3(-2\omega) \left[-\sin\left(\frac{\pi}{4} - 2\omega t\right) \right] \\ &= 6\omega \sin\left(\frac{\pi}{4} - 2\omega t\right) \end{aligned}$$

$$\begin{aligned} \text{Acceleration, } a &= \frac{dv}{dt} = \frac{d}{dt} \left[6\omega \sin\left(\frac{\pi}{4} - 2\omega t\right) \right] \\ &= 6\omega \times (-2\omega) \cos\left(\frac{\pi}{4} - 2\omega t\right) \\ &= -12\omega^2 \cos\left(\frac{\pi}{4} - 2\omega t\right) \\ &= -4\omega^2 \left[3 \cos\left(\frac{\pi}{4} - 2\omega t\right) \right] \end{aligned}$$

$$\Rightarrow a = -4\omega^2 y$$

As acceleration, $a \propto -y$

Hence, due to negative sign motion is SHM.

Clearly, from the equation

$$\omega' = 2\omega \quad [\because \text{Standard equation } y = a \cos(\omega t + \phi)]$$

$$\Rightarrow \frac{2\pi}{T'} = 2\omega \Rightarrow T' = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$$

So, motion is SHM with period $\frac{\pi}{\omega}$.

- 49** (b) Given equation of displacement of the particle,

$$y = \sin^3 \omega t = (3 \sin \omega t - 4 \sin 3\omega t)/4$$

$$(\because \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta)$$

$$\Rightarrow \frac{dy}{dt} = \left[\frac{d}{dt} (3 \sin \omega t) - \frac{d}{dt} (4 \sin 3\omega t) \right] / 4$$

$$\Rightarrow 4 \frac{dy}{dt} = 3\omega \cos \omega t - 4 \times (3\omega \cos 3\omega t)$$

$$\Rightarrow 4 \times \frac{d^2 y}{dt^2} = -3\omega^2 \sin \omega t + 12\omega^2 \sin 3\omega t$$

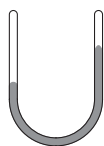
$$\Rightarrow \frac{d^2 y}{dt^2} = -\frac{3\omega^2 \sin \omega t + 12\omega^2 \sin 3\omega t}{4}$$

$$\Rightarrow \frac{d^2 y}{dt^2} \text{ is not proportional to } y.$$

Hence, motion is not SHM.

As, the expression is involving sine functions, hence it will be periodic.

- 50 (c)** Consider the diagram in which a liquid column oscillates. In this case, restoring force acts on the liquid due to gravity. Acceleration of the liquid column can be calculated in terms of restoring force.



Restoring force, $f = \text{Weight of liquid column of height } 2y$

$$\Rightarrow f = -(A \times 2y \times \rho) \times g = -2A\rho g y \quad (\because m = \rho v)$$

$$\Rightarrow f \propto -y \Rightarrow \text{Motion is SHM with force constant,}$$

$$k = 2A\rho g$$

$$\Rightarrow \text{Time period, } T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{A \times 2h \times \rho}{2A\rho g}} = 2\pi\sqrt{\frac{h}{g}}$$

$$T = 2\pi\sqrt{\frac{l}{g}}, \text{ where } l = h$$

which is independent of the density of the liquid.

51 (c) Given, $x = a \cos \omega t$... (i)

$y = a \sin \omega t$... (ii)

Squaring and adding Eqs. (i) and (ii), we get

$$x^2 + y^2 = a^2 \quad (\because \cos^2 \omega t + \sin^2 \omega t = 1)$$

This is the equation of a circle.

Clearly, the locus is a circle of constant radius a .

- 52 (c)** As the given equation is

$$x = a \cos (\alpha t)^2$$

is a cosine function. Hence, it is an oscillatory motion.

Now, putting $(t + T)$ in place of t ,

$$x(t + T) = a \cos [\alpha(t + T)]^2 \quad [\because x(t) = a \cos (\alpha t)^2]$$

$$= a \cos [\alpha t^2 + \alpha T^2 + 2\alpha t T] \neq x(t)$$

where, T is supposed as period of the function $\omega(t)$.

Hence, it is not periodic.

53 (c) $\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

$$\therefore x = v \sqrt{\frac{m}{k}}$$

54 (b) $U = U_{\text{mean}} + \frac{1}{2}kx^2$

Given, $U_{\text{mean}} = E_1$ and $\frac{1}{2}kA^2 = E_2 = \text{maximum kinetic energy}$

at mean position

$$\therefore \text{At } x = \frac{\sqrt{3}A}{2}, U = E_1 + \frac{1}{2}k\left(\frac{\sqrt{3}A}{2}\right)^2 = E_1 + \frac{3}{4}E_2$$

55 (d) $\therefore \frac{1}{2}kA^2 = 1 \text{ J}$

$$\therefore k = \frac{2}{A^2} = \frac{2}{(0.4)^2} = \frac{25}{2} \text{ Nm}^{-1}$$

$$\therefore T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2}{25/2}} = \frac{4\pi}{5} \text{ s}$$

- 56 (c)** Maximum kinetic energy = Energy of oscillation in SHM

$$\therefore 8 = \frac{1}{2}kA^2$$

$$\Rightarrow kA^2 = 16 \quad \dots (i)$$

Further, $2\pi\sqrt{\frac{m}{k}} = 4 \quad (\because T = 4 \text{ s})$

$$\therefore \frac{1}{k} = \frac{4}{\pi^2} \text{ or } k = \frac{\pi^2}{4} \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$k \approx 2.5 \text{ Nm}^{-1}$$

and

$$A \approx 2.53 \text{ m}$$

Maximum acceleration of the particle will be

$$a_{\text{max}} = \omega^2 A = \frac{k}{m} A$$

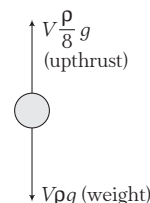
$$= \left(\frac{2.5}{1}\right) (2.53) \approx 6.3 \text{ ms}^{-2}$$

57 (d) In vacuum, $T = 2\pi\sqrt{\frac{l}{g}}$

Let V be the volume and ρ the density of the mass of the bob.

Net downward force acting on the bob inside the liquid

$$= \text{Weight} - \text{Upthrust} = V\rho g - V \cdot \frac{\rho}{8} g = \frac{7}{8} V\rho g$$

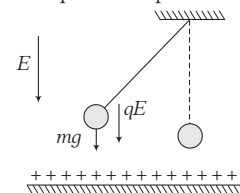


i.e. Effective value of g is $\frac{7}{8}g$.

So, time period of the bob inside the liquid,

$$T_1 = 2\pi\sqrt{\frac{l}{(7/8)g}} = 2\pi\sqrt{\frac{l}{g}} \times \sqrt{\frac{8}{7}} = \sqrt{\frac{8}{7}} T$$

- 58 (a)** In this case, time period of pendulum becomes



$$T' = 2\pi\sqrt{\frac{l}{\left(g + \frac{qE}{m}\right)}} \Rightarrow T' < T \quad \left(\because T = 2\pi\sqrt{\frac{l}{g}}\right)$$

- 59 (b)** Let T_1 and T_2 be the time period of the two pendulums,

$$T_1 = 2\pi\sqrt{\frac{100}{g}} \text{ and } T_2 = 2\pi\sqrt{\frac{121}{g}} \quad (T_1 < T_2 \text{ because } l_1 < l_2)$$

Let at $t = 0$, they start swinging together. Since, their time periods are different, the swinging will not be same always.

Only when number of completed oscillation differs by an integer, the two pendulum will again begin to swing together.

Let longer length pendulum complete n oscillation and shorter length pendulum complete $(n + 1)$ oscillation, for the two be in same phase, then $(n + 1)T_1 = nT_2$.

$$(n + 1) \times 2\pi \sqrt{\frac{100}{g}} = n \times 2\pi \sqrt{\frac{121}{g}}$$

$$\Rightarrow n = 10$$

60 (c) For series arrangement, $f_s = \frac{1}{2\pi} \sqrt{\frac{k_s}{m}} = \frac{1}{2\pi} \sqrt{\frac{k}{2m}}$

$$\left(\because k_s = \frac{k_1 k_2}{k_1 + k_2} = \frac{k}{2} \right)$$

For parallel and series arrangement,

$$f_p = \frac{1}{2\pi} \sqrt{\frac{2k}{3m}} \quad \left(\because k_p = \frac{2k k}{2k + k} = \frac{2k}{3} \right)$$

$$\therefore \frac{f_s}{f_p} = \sqrt{\frac{k}{2m}} \times \sqrt{\frac{3m}{2k}} = \frac{\sqrt{3}}{2} \Rightarrow \sqrt{3}:2$$

61 (d) From graph, slope $k = \frac{F}{x} = \frac{8}{2} = 4$

$$\therefore T = 2\pi \sqrt{\frac{m}{k}} \Rightarrow T = 2\pi \sqrt{\frac{0.01}{4}} = 0.31 \text{ s}$$

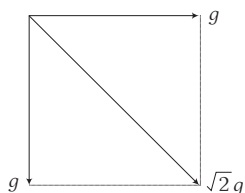
62 (b) Comparing with $\frac{d^2x}{dt^2} = -\omega^2 x$, we have

$$\omega = 2 \text{ rads}^{-1},$$

$$\therefore v_{\max} = \omega A = 2 \text{ ms}^{-1}$$

$$\text{and } a_{\max} = \omega^2 A = 4 \text{ ms}^{-2}$$

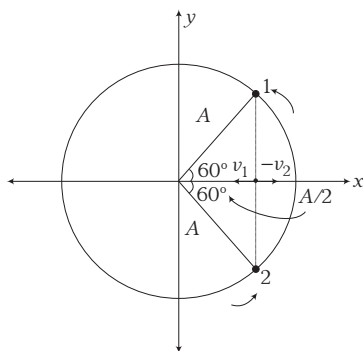
63 (a)



Hence, to keep time period same, $l' = \sqrt{2} l$

64 (b) After attaining terminal velocity, ball will move with constant velocity. Therefore, the motion of the ball is periodic but not SHM.

65 (d) According to question, the situation can be represented as



Hence, here $\phi = 60^\circ + 60^\circ = 120^\circ$

66 (b) $T = 2\pi \sqrt{\frac{m}{k}}$ or $T \propto \frac{1}{\sqrt{k}}$ or $k = \frac{\alpha}{T^2}$

Now,

$$k_e = k_1 + k_2$$

$$\therefore \frac{\alpha}{T^2} = \frac{\alpha}{T_1^2} + \frac{\alpha}{T_2^2}$$

$$\text{or } T^{-2} = T_1^{-2} + T_2^{-2}$$

67 (a) As, $F = N$ or $m\omega^2 A = mg \Rightarrow \omega^2 A = g$

$$\therefore \omega = \sqrt{\frac{g}{A}} \text{ or } T = 2\pi \sqrt{\frac{A}{g}}$$

$$= 2\pi \sqrt{\frac{4 \times 10^{-3}}{10}} = 2\pi \times 2 \times 10^{-2}$$

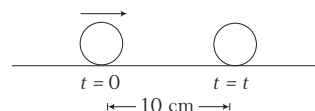
$$\Rightarrow T = \frac{4\pi}{100} = \frac{\pi}{25} \text{ s}$$

68 (b) $x_{12} = \frac{12 \times 10}{1000} = 0.12 \text{ m}$

$$x_4 = \frac{4 \times 10}{1000} = 0.04 \text{ m}$$

$$\therefore A = x_{12} - x_4 = 0.08 \text{ m}$$

69 (b) Let $x = A \sin \omega t$



or

$$10 = 20 \sin \omega t$$

$$\therefore \omega t = \left(\frac{\pi}{6} \right) \Rightarrow t = \frac{\pi}{6\omega}$$

The desired time will be $2t$.

$$\text{or } \frac{\pi}{3\omega} = \frac{\pi}{3(2\pi/T)} = \frac{T}{6} = \frac{12}{6} = 2 \text{ s}$$

70 (a) $k \propto \frac{1}{l}$

$$\therefore \frac{l_1}{l_2} = n$$

$$\Rightarrow l_1 = \left(\frac{n}{n+1} \right) l \text{ or } k' = \left(\frac{n+1}{n} \right) k$$

$$\therefore T = 2\pi \sqrt{\frac{mn}{k(n+1)}}$$

71 (b) Amplitude of oscillation, $A = \frac{Mg}{k} = \frac{5 \times 10}{200} = 0.25 \text{ m}$

72 (b) $T_1 = 2\pi \sqrt{\frac{m}{k + \rho Ag}}$ but $T_2 = 2\pi \sqrt{\frac{m}{k}}$

Hence, $T_1 < T_2$.

73 (b) $\frac{1}{2} kA^2 = (0.04 - 0.01) \text{ J} = 0.03 \text{ J}$

$$\therefore k = \frac{0.06}{A^2} = \frac{0.06}{(0.02)^2} = 150 \text{ Nm}^{-1} = 1.5 \times 10^2 \text{ Nm}^{-1}$$

74 (c) Potential energy graph is parabolic with its minimum value at mean position.

- 75** (a) Since, the cube is half immersed. The density of cube should be half the density of water, i.e. 500 kgm^{-3}

$$T = 2\pi \sqrt{\frac{m}{\rho_w Ag}} \quad (\because k = \rho_w Ag)$$

$$\frac{\pi}{5} = 2\pi \sqrt{\frac{a^3 \times \rho}{\rho_w \times a^2 \times g}} \quad (\because \text{Mass} = \text{Density} \times \text{Volume})$$

$$\therefore \frac{a\rho}{g\rho_w} = \frac{1}{100}$$

$$\text{or } \frac{a}{20} = \frac{1}{100} \text{ or } a = (0.2) \text{ m}$$

$$\text{Now, } m = a^3 \rho = 4 \text{ kg}$$

- 76** (a) From e to $2e$ or $\frac{A}{2}$ to A , $t = \frac{T}{6}$

$$\therefore \text{Time period of oscillation} = 2t = \frac{T}{3} = \frac{2\pi}{3} \sqrt{\frac{m}{k}}$$

- 77** (c) $\mu g >$ maximum acceleration in SHM, i.e. $\omega^2 A$

$$\therefore \mu > \frac{\omega^2 A}{g}$$

$$\Rightarrow \mu > \frac{(2\pi f)^2 \times A}{g}$$

$$\Rightarrow \mu > \frac{(2 \times 3.14 \times 2)^2 \times 4 \times 10^{-2}}{10}$$

$$\Rightarrow \mu > 0.631 \approx 0.64$$

- 78** (a) $y_1 = A \sin\left(\omega_1 t + \frac{\pi}{2}\right) \Rightarrow y_2 = A \sin\left(\omega_2 t - \frac{\pi}{2}\right)$

$$\text{Now, } \omega_1 t + \frac{\pi}{2} = \omega_2 t - \frac{\pi}{2}$$

$$\text{or } t = \frac{\pi}{\omega_2 - \omega_1} = \frac{\pi}{(2\pi / T_2) - (2\pi / T_1)}$$

$$= \frac{T_1 T_2}{2(T_2 - T_1)} = \frac{3 \times 7}{2(7 - 3)} = \frac{21}{8} \text{ s}$$

- 79** (c) $T = 2\pi \sqrt{\frac{m}{k}}$ or $T \propto \frac{1}{\sqrt{k}}$

$$T \text{ has increased } \frac{4}{3} \text{ times. Hence, } k' = \frac{9}{16} k$$

When both are combined,

$$k_{\text{net}} = k + k' = \frac{25}{16} k \text{ or } \frac{25}{16} \text{ times}$$

Hence, new time period will become $\frac{4}{5}$ times of 3 s or 2.4 s.

- 80** (c) Particle starts from rest. Hence, $x = A \cos \omega t$

$$a = A - x = A - A \cos (\omega \times 1)$$

$$\text{or } \cos \omega = \frac{A - a}{A} = \left(1 - \frac{a}{A}\right)$$

$$\Rightarrow a + b = A - A \cos (\omega \times 2)$$

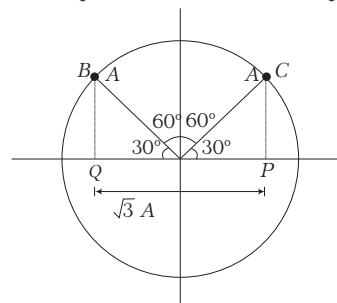
$$= A - A(2 \cos^2 \omega - 1)$$

$$= 2A - 2A \left(1 - \frac{a}{A}\right)^2$$

On solving, we get

$$A = \frac{2a^2}{3a - b}$$

- 81** (a) $\frac{T}{3}$ means 120° in reference circle. Maximum possible value of v is near mean position between P and Q .



$$\therefore v = \frac{\text{Distance}}{\text{Time}} = \frac{\sqrt{3}A}{(T/3)} = \frac{3\sqrt{3}A}{T}$$

- 82** (b) The block executes SHM, so its angular frequency,

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{100 \text{ Nm}^{-1}}{2 \text{ kg}}} = 7.07 \text{ rad s}^{-1}$$

Its displacement at any time t is

$$x(t) = a \cos \omega t = 0.1 \cos(7.07t)$$

When the particle is 0.05m away from the mean position.

$$0.05 = 0.1 \cos(7.07t)$$

$$\text{or } \cos(7.07t) = 0.5$$

$$\text{or } \sin(7.07t) = \frac{\sqrt{3}}{2} = 0.866$$

Velocity of the block at $x = 0.05 \text{ m}$ is

$$v = A\omega \sin \omega t$$

$$= 0.1 \times 7.07 \times 0.866 = 0.61 \text{ ms}^{-1}$$

$$\text{Hence, } KE = \frac{1}{2} mv^2 = \frac{1}{2} \times 2 (0.61)^2 = 0.37 \text{ J}$$

- 83** (b) $T = 2\pi \sqrt{\frac{I}{mgl}}$. Here, l is distance of point P from centre of mass.

When P is the centre, $l = 0$. Therefore, $T = \infty$, i.e. rod will not oscillate.

$$\text{When } P \text{ is end point, } I = \frac{ml^2}{3}$$

$$\therefore T = 2\pi \sqrt{\frac{ml^2}{3mgl}} = 2\pi \sqrt{\frac{l}{3g}}$$

- 84** (b) Here, effective mass of container,

$$M_e = \text{Mass of container} + \frac{\text{Upthrust}}{g}$$

$$= M + \frac{\rho Vg}{g} = M + \rho V$$

$$\text{Hence, } T = 2\pi \sqrt{\frac{M_e}{k}} = 2\pi \sqrt{\frac{M + \rho V}{k}}$$

$$85 \text{ (d)} T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.1}{100}} = 0.198 \text{ s}$$

From $x = A \sin \omega t$ or $5 = 10 \sin \omega t$

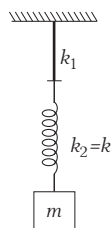
$$\Rightarrow \omega t = \frac{\pi}{6} \text{ or } \frac{2\pi}{T} t = \frac{\pi}{6} \text{ or } 2t = \frac{T}{6}$$

The desired time in the question will be $\frac{T}{2} + 2t$

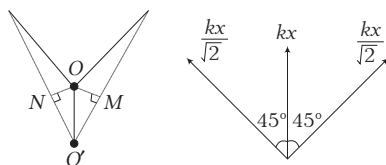
$$\text{or } \frac{T}{2} + \frac{T}{6} = \frac{2T}{3} = \left(\frac{2}{3}\right)(0.198) = 0.132 \text{ s}$$

$$86 \text{ (b)} \text{ Here, } k_{\text{eq}} = \frac{k_1 k_2}{k_1 + k_2} = \frac{\frac{YA}{L}}{\frac{YA}{L} + k}$$

$$= \frac{YA}{YA + Lk} = 2\pi \sqrt{\frac{m(YA + Lk)}{YA k}}$$



87 (b)



$OO' = x$ (say)

Then $O'M = O'N \approx \frac{x}{\sqrt{2}}$

i.e. Elongation in spring B and C is $x/\sqrt{2}$, while compression in spring A is x .

Net restoring force,

$$F = -\left[kx + \frac{2kx}{\sqrt{2}} \cos 45^\circ\right] = -2kx$$

$$\therefore a = \frac{F}{m} = -\frac{2k}{m} x$$

$$T = 2\pi \sqrt{\frac{x}{a}} = 2\pi \sqrt{\frac{m}{2k}}$$

88 (c) According to law of conservation of momentum,

$$p_i = p_f$$

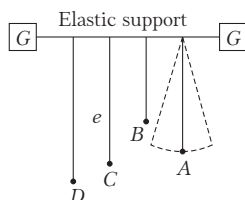
$$Mv_{\text{max}} = (m + M)v'_{\text{max}}$$

$$\Rightarrow v'_{\text{max}} = \frac{Mv_{\text{max}}}{(m + M)}$$

$$\therefore v'_{\text{max}} = A'\omega'$$

$$\Rightarrow \frac{M \cdot A}{(m + M)} \sqrt{\frac{k}{M}} = A' \sqrt{\frac{k}{(m + M)}} \Rightarrow A' = A \sqrt{\frac{M}{(m + M)}}$$

89 (b) According to the question, A is given a transverse displacement.



Through the elastic support, the disturbance is transferred to all the pendulums.

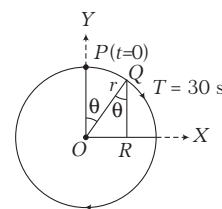
A and C are having same length, hence they will be in resonance, because their time period of oscillation,

$$T = 2\pi \sqrt{\frac{l}{g}} \text{ and hence, frequency will be same.}$$

So, amplitude of C will be maximum.

Note In this problem, we have assumed that the support is perfectly elastic and there is no damping. Hence, oscillation is considered as undamped.

90 (a) Let angular velocity of the particle executing circular motion is ω and when it is at Q, it makes an angle θ as shown in the diagram.



Clearly, $\theta = \omega t$

Now, we can write OR

$$= OQ \cos (90^\circ - \theta)$$

$$= OQ \sin \theta = OQ \sin \omega t$$

$$= r \sin \omega t$$

$$(\because OQ = r)$$

$$\Rightarrow x = r \sin \omega t = B \sin \omega t$$

$$(\because r = B)$$

$$= B \sin \frac{2\pi}{T} t = B \sin \left(\frac{2\pi}{30} t\right)$$

Clearly, this equation represents SHM.

(B) Medical entrance special format questions

• Assertion and reason

1 (a) At given time, x is negative. Therefore, acceleration is positive ($a \propto -x$). Further slope is negative. Therefore, velocity is negative.

$$2 \text{ (a)} T = 2\pi \sqrt{\frac{l}{g_e}}$$

Here, $g_{\text{eff}} = g + a$, if the lift accelerates upwards.

3 (d) Time period does not depend on amplitude of oscillation.

$$4 \text{ (a)} v = \omega \sqrt{A^2 - x^2}$$

$$\therefore \frac{v^2}{\omega^2} + \frac{x^2}{A^2} = A^2, \text{ i.e. ellipse.}$$

$$5 \text{ (c)} \text{ Force constant of a spring, } k \propto \frac{1}{l} \text{ and } T \propto \frac{1}{\sqrt{k}}$$

$$\therefore T \propto \sqrt{l}$$

• Statement based questions

1 (d) Acceleration is maximum when displacement is maximum. Force is zero when displacement is zero and potential energy is maximum when displacement is maximum.

2 (d) At equilibrium, $x = \frac{mg}{k} = \frac{0.2 \times 10}{200} = 0.01 \text{ m} = 1 \text{ cm}$

$$\therefore f = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{200}{0.2}} \approx 5 \text{ Hz}$$

3 (d) $v = \omega \sqrt{A^2 - x^2}$ (In SHM)

$$\text{or } v^2 = \omega^2 A^2 - \omega^2 x^2$$

Comparing the given equation with the above equation, we get

$$\omega^2 = 9$$

$$\therefore \omega = 3 = \frac{2\pi}{T}$$

$$\therefore T = \frac{2\pi}{3} \text{ units}$$

Also, $\omega^2 A^2 = 144$

$$\therefore A = 4 \text{ units, } |a| = \omega^2 x = (9)(3) = 27 \text{ units}$$

Displacement $\leq$ Distance

4 (c) $U = 5x^2 - 20x \Rightarrow \frac{dU}{dx} = 10x - 20$

$$\therefore F = -\frac{dU}{dx} = (-10x + 20)$$

Let us suppose $x = (X + 2)$. Then $F = -10X$, $F = 0$ at $X = 0$ or $x = 2$, i.e. $x = 2\text{m}$ is the mean position about which particle is in SHM. Here, $k = 10$

$$\therefore T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.1}{10}} = \frac{\pi}{5} \text{ s. At mean position, kinetic}$$

energy is maximum and force acting on particle in SHM is variable.

5 (d) $\omega = \sqrt{\frac{k}{\mu}}$

Here, $\mu = \text{reduced mass} = \frac{m_1 m_2}{m_1 + m_2} = \frac{2}{3} \text{ kg}$

$$\therefore \omega = \sqrt{\frac{6}{2/3}} = 3 \text{ rad s}^{-1}$$

Amplitude 12 cm distributes in inverse ratio of mass.

$$\therefore A_1 = 8 \text{ cm and } A_2 = 4 \text{ cm}$$

Now, maximum kinetic energy,

$$K_1 = \frac{1}{2} m_1 \omega^2 A_1^2 = \frac{1}{2} \times 1 \times 3 \times (8 \times 10^{-2})^2 = 9.6 \text{ mJ}$$

$$K_2 = \frac{1}{2} \times 2 \times 3 \times (4 \times 10^{-2})^2 = 4.8 \text{ mJ}$$

6 (d) The motion is simple harmonic only inside earth. Further

$$\frac{1}{2} mv^2 = -\frac{GMm}{(R+h)} + \frac{3}{2} \frac{GMm}{R} \Rightarrow v = \sqrt{\frac{GM(R+3h)}{R(R+h)}}$$

7 (b) In simple harmonic motion, speed is not constant. Thus, in order to calculate the time taken by the particle executing SHM, we cannot use the following relation,

$$\text{i.e. Time} = \frac{\text{Distance}}{\text{Speed}}$$

8 (c) If $v = \omega A \sin \omega t$

Then, $v_{\max} = \omega A$

$$\therefore K_{\max} = \frac{1}{2} mv_{\max}^2 = \frac{1}{2} m\omega^2 A^2$$

$$\text{Further, } \langle K \rangle = \left\langle \frac{1}{2} m\omega^2 A^2 \sin^2 \omega t \right\rangle \\ = \frac{1}{2} m\omega^2 A^2 \langle \sin^2 \omega t \rangle$$

$$\text{But } \langle \sin^2 \omega t \rangle = \frac{1}{2} \text{ in one oscillation}$$

$$\therefore \langle K \rangle = \frac{1}{4} m\omega^2 A^2$$

9 (b) $k \propto \frac{1}{\text{Length of spring}}$

If length is halved, k will become two times.

$$\text{From } T = 2\pi \sqrt{\frac{m}{k}}$$

Time period T will remain half for the given conditions.

10 (a) $T = 8 \text{ s}$

$$\therefore \omega = \frac{2\pi}{T} = \frac{\pi}{4} \text{ rad s}^{-1}$$

$$k = -\text{slope of } F\text{-}x \text{ graph} = 10 \text{ Nm}^{-1}$$

$$\text{From } T = 2\pi \sqrt{\frac{m}{k}}, \text{ we have}$$

$$8 = 2\pi \sqrt{\frac{m}{10}}$$

$$\therefore m = \frac{160}{\pi^2} \text{ kg}$$

$$K_{\max} = \frac{1}{2} m\omega^2 A^2 \\ = \frac{1}{2} \times \frac{160}{\pi^2} \times \frac{\pi^2}{16} \times (4 \times 10^{-2})^2 \\ = 8.0 \times 10^{-3} \text{ J}$$

• Match the columns

1 (b) Suppose $x = A \sin \omega t$

$$\text{Then, } v = \frac{dx}{dt} = \omega A \cos \omega t = \omega A \sin (\omega t + \pi/2)$$

$$\text{and } a = \frac{dv}{dt} = -\omega^2 A \sin \omega t = \omega^2 A \sin (\omega t + \pi)$$

Hence, $A \rightarrow q$, $B \rightarrow r$, $C \rightarrow p$.

2 (c) $a = -\omega^2 x$, i.e. a - x graph is straight line passing through origin.

If $v = v_0 \sin \omega t$, then

$$a = \frac{dv}{dt} = v_0 \omega \cos \omega t$$

$$= v_0 \omega \sqrt{1 - \sin^2 \omega t}$$

$$= v_0 \omega \sqrt{1 - \frac{v^2}{v_0^2}} \Rightarrow a = \omega \sqrt{v_0^2 - v^2}$$

So, a - v graph is neither a straight line nor a parabola. Further, acceleration and velocity-time graphs are sine or cosine functions.

Hence, $A \rightarrow q$, $B \rightarrow s$, $C \rightarrow s$, $D \rightarrow s$.

- 3** (a) On a satellite and at centre of earth, $g' = 0$

$$\therefore T = \infty$$

At pole, value of g is more than the normal value.

Thus, $T < 2s$

At mountain or at some height, value of g is less.

Thus, $T > 2s$

Hence, $A \rightarrow q$, $B \rightarrow t$, $C \rightarrow p$, $D \rightarrow t$.

(C) Medical entrances' gallery

- 1** (c) $\sin \omega t$ and $\cos \omega t$ both are periodic function of period $\frac{2\pi}{\omega}$.

We know that, sum of two periodic functions is also a periodic function, hence $\sin \omega t + \cos \omega t$ represents periodic motion.

- 2** (d) In SHM, equation of displacement of a particle is

$$y = a \sin \omega t$$

and equation of acceleration of a particle is

$$A = -a\omega^2 \sin \omega t = a\omega^2 \sin(\omega t + \pi)$$

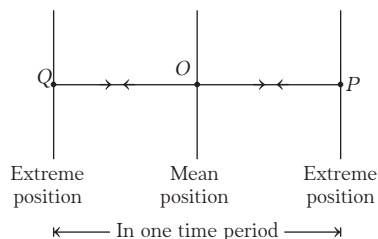
$\therefore$ Phase difference between displacement and acceleration of a particle is

$$= (\omega t + \pi) - \omega t = \pi \text{ rad}$$

- 3** (d) In a simple harmonic motion (SHM), the particle oscillates about its mean position on a straight line.

As shown in the figure below, the particle moves from its mean position (O) to an extreme position (P) and then return to its mean position covering same distance of A .

Then, by the conservative force, it is moved in opposite direction to a point Q by distance A and then back to mean position covering a distance of A . This comprises of one time period as shown below.



Hence, in one time period, it covers a distance of

$$x = OP + PO + OQ + QO \\ = A + A + A + A = 4A$$

- 4** (a) The displacement of given particle is

$$y = A_0 + A \sin \omega t + B \cos \omega t \quad \dots (i)$$

The general equation of SHM can be given as

$$x = a \sin \omega t + b \cos \omega t \quad \dots (ii)$$

So, from Eqs. (i) and (ii), we can say that A_0 be the value of mean position, at which $y = 0$.

$$\therefore \text{Amplitude, } R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

As two functions sine and cosine have phase difference of 90° .

$$\therefore R = \sqrt{A^2 + B^2} \quad (\because \cos 90^\circ = 0)$$

- 5** (c) The average velocity of a particle executing simple harmonic motion (SHM) is

$$v_{av} = \frac{\text{Total displacement}}{\text{Time interval}} = \frac{x_f - x_i}{T}$$

where, x_f and x_i are the initial and final position of the particle executing SHM.

As in SHM, the particle executes motion about its mean position. So, after one complete oscillation by the particle, it will reach its initial position, i.e.

Displacement, $x_f - x_i = 0$

$$\therefore v_{av} = \frac{0}{T}$$

Hence, the average velocity is zero.

- 6** (b) The acceleration of particle/body executing SHM at any instant (at position x) is given as

$$a = -\omega^2 x$$

where, ω is the angular frequency of the body.

$$\Rightarrow |a| = \omega^2 x \quad \dots (i)$$

Here, $x = 5 \text{ m}$, $|a| = 20 \text{ ms}^{-2}$

Substituting the given values in Eq. (i), we get

$$20 = \omega^2 \times 5$$

$$\Rightarrow \omega^2 = \frac{20}{5} = 4 \quad \text{or} \quad \omega = 2 \text{ rad s}^{-1}$$

As we know that, time period, $T = 2\pi / \omega \quad \dots (ii)$

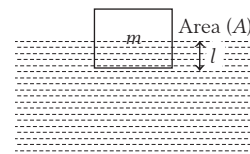
$\therefore$ Substituting the value of ω in Eq. (ii), we get

$$T = \frac{2\pi}{2} = \pi \text{ s}$$

- 7** (a) For a rectangular block, floating in a liquid as shown in the figure below, then

At equilibrium, $mg = V\rho g = A l \rho g$

$$\Rightarrow m = A \rho l$$



where, l = length of part immersed in liquid.

When it is given in downward displacement y , restoring force (upward direction) on block is

$$F = -[A(l + y)\rho g - mg] \\ = -[A(l + y)\rho g - A l \rho g] = -A \rho g y$$

i.e. $F \propto -y$ or $a \propto -y$, so it executes SHM.

$$\therefore \text{Time period} = 2\pi \sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}}$$

$$T = 2\pi \sqrt{\frac{m}{A\rho g}} \Rightarrow T^2 \propto \frac{m}{A\rho}$$

or $T^2 \propto \frac{1}{\rho}$

- 8** (b) Given, $m = 0.5 \text{ kg}$, $\omega = 5 \text{ rad/s}$,
 $x = 1 \text{ cm} = 10^{-2} \text{ m}$,
 $A = 5 \text{ cm} = 5 \times 10^{-2} \text{ m}$

$$\therefore \text{Potential energy, PE} = \frac{1}{2} m \omega^2 x^2$$

$$\text{or PE} = \frac{1}{2} \times 0.5 \times (5)^2 \times (10^{-2})^2$$

$$= \frac{25}{4} \times 10^{-4} \text{ J} = 6.25 \times 10^{-4} \text{ J}$$

$$\therefore \text{Kinetic energy, KE} = \frac{1}{2} m \omega^2 (A^2 - x^2)$$

$$\text{or KE} = \frac{1}{2} \times 0.5 \times (5)^2 (25 - 1) \times 10^{-4}$$

$$= 150 \times 10^{-4} \text{ J}$$

- 9** (a) Given,

$$Y = 5 \sin \frac{\pi}{2} (100t - 2\pi) = 5 \sin (50\pi t - \pi x) \quad \dots(i)$$

General equation of SHM is

$$Y = A \sin (\omega t - kx) \quad \dots(ii)$$

After comparing with Eqs. (i) and (ii),

$$\omega = 50\pi$$

$$\therefore \text{Time period, } T = \frac{2\pi}{\omega} = \frac{2\pi}{50\pi} = 0.04 \text{ s}$$

- 10** (c) For a particle executing SHM, magnitude of velocity of particle when it is at displacement x from mean position

$$= \omega \sqrt{A^2 - x^2}$$

Also, magnitude of acceleration of particle in SHM $= \omega^2 x$

Given, when $x = 2 \text{ cm}$

$$\text{Magnitude of velocity} = \text{Magnitude of acceleration}$$

$$\Rightarrow \omega \sqrt{A^2 - x^2} = \omega^2 x$$

$$\Rightarrow \omega = \frac{\sqrt{A^2 - x^2}}{x} = \frac{\sqrt{9 - 4}}{2}$$

$$\Rightarrow \text{Angular velocity, } \omega = \frac{\sqrt{5}}{2}$$

$\therefore$ Time period of motion,

$$T = \frac{2\pi}{\omega} = \frac{4\pi}{\sqrt{5}} \text{ s}$$

- 11** (c) For a SHM, acceleration, $a = \frac{d^2x}{dt^2} = -\omega^2 x \quad \dots(i)$

According to the question,

$$x = a \sin^2 \omega t = a \left(\frac{1 - \cos 2\omega t}{2} \right)$$

Differentiating the above expression w.r.t. t , we get

$$\frac{dx}{dt} = a\omega \sin 2\omega t$$

Again, differentiating x w.r.t. t , we get

$$\frac{d^2x}{dt^2} = 2a\omega^2 \cos 2\omega t$$

$$\text{or } \frac{d^2x}{dt^2} = 2a\omega^2 (1 - 2\sin^2 \omega t)$$

$$\Rightarrow \frac{d^2x}{dt^2} = 2a\omega^2 - 4a\omega^2 \sin^2 \omega t = 2a\omega^2 - 4\omega^2 x \quad \dots(ii)$$

Therefore, after comparing Eqs. (i) and (ii), we can say that the motion of the particle is non-simple harmonic motion.

- 12** (d) Time period,

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$\text{Case I} \quad T_1 = 2\pi \sqrt{\frac{m}{k}} \quad \dots(i)$$

Case II When the mass m is increased by 1 kg, then net mass $= m + 1$

$$\Rightarrow T_2 = 2\pi \sqrt{\frac{m+1}{k}} \quad \dots(ii)$$

On dividing Eq. (ii) by Eq. (i), we get

$$\frac{T_2}{T_1} = \sqrt{\frac{m+1}{m}} \Rightarrow \frac{5}{3} = \sqrt{\frac{m+1}{m}}$$

$$\Rightarrow \frac{25}{9} = \frac{m+1}{m} \Rightarrow \frac{25}{9} = 1 + \frac{1}{m}$$

$$\Rightarrow \frac{1}{m} = \frac{16}{9}$$

$$\Rightarrow m = \frac{9}{16} \text{ kg}$$

- 13** (c) Given, $k_P > k_Q$

In case (a), the elongation is same,

$$\text{i.e. } x_1 = x_2 = x$$

$$\text{So, } W_P = \frac{1}{2} k_P x^2 \text{ and } W_Q = \frac{1}{2} k_Q x^2$$

$$\therefore \frac{W_P}{W_Q} = \frac{k_P}{k_Q} > 1 \quad (\because k_P > k_Q)$$

$$\Rightarrow W_P > W_Q$$

In case (b), the spring force is same,

$$\text{i.e. } F_1 = F_2 = F$$

$$\text{So, } x_1 = \frac{F}{k_P}, x_2 = \frac{F}{k_Q}$$

$$\therefore W_P = \frac{1}{2} k_P x_1^2 = \frac{1}{2} k_P \frac{F^2}{k_P^2} = \frac{1}{2} \frac{F^2}{k_P}$$

$$\text{and } W_Q = \frac{1}{2} k_Q x_2^2 = \frac{1}{2} k_Q \frac{F^2}{k_Q^2} = \frac{1}{2} \frac{F^2}{k_Q}$$

$$\therefore \frac{W_P}{W_Q} = \frac{k_Q}{k_P} < 1 \Rightarrow W_P < W_Q$$

- 14** (b) Let A be the amplitude of oscillation, then

$$v_1^2 = \omega^2(A^2 - x_1^2) \quad \dots (i)$$

$$v_2^2 = \omega^2(A^2 - x_2^2) \quad \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$v_1^2 - v_2^2 = \omega^2(x_2^2 - x_1^2)$$

$$\Rightarrow \omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}} \Rightarrow T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

- 15** (b) Given, $v \frac{dv}{dx} = -\omega^2 x$

On integrating within the limit, $\int_{v_0}^v v dv = \int_0^x -\omega^2 x dx$

$$\Rightarrow \left[\frac{v^2}{2} \right]_{v_0}^v = -\omega^2 \left[\frac{x^2}{2} \right]_0^x$$

$$\Rightarrow v^2 - v_0^2 = -\omega^2 x^2$$

$$\Rightarrow v = \sqrt{v_0^2 - \omega^2 x^2}$$

- 16** (c) Time period, $T = 2\pi \sqrt{\frac{m}{k}}$

$$\Rightarrow T = 2 \times 3.14 \sqrt{\frac{200 \times 10^{-3}}{80}}$$

$$\text{or } T = 2 \times 3.14 \sqrt{\frac{1}{400}} \quad \text{or } T = \frac{2 \times 3.14}{20}$$

$$\text{or } T = 0.314 \text{ s}$$

- 17** (b) Total mechanical energy of the system,

$$E = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} \times 40 \times 10^{-3} \left(\frac{2\pi}{T} \right)^2 \times (2)^2$$

$$= \frac{1}{2} \times 40 \times 10^{-3} \left(\frac{2 \times 3.14}{20} \right)^2 \times (2)^2 = 7.9 \times 10^{-3} \text{ J}$$

- 18** (b) Given, time period, $T = 2\pi \sqrt{\frac{m}{k}} = 10 \text{ s}$

When the spring system is immersed in liquid with specific gravity $\frac{1}{4}$ th of the copper, but as the liquid is non-viscous in nature, so the time period would not be changed because there will be no change in m and k .

- 19** (b) For a simple harmonic motion, $a = \frac{d^2 y}{dt^2} \propto (-y)$

So, only equations, $y = \sin \omega t - \cos \omega t$

$$\text{and } y = 5 \cos \left(\frac{3\pi}{4} - 3\omega t \right)$$

are satisfying this condition. Thus, they represent SHM.

- 20** (c) $a = -\omega^2 x = -\omega^2 A \cos \omega t$

So, at $t = 0$,

$$a = -\omega^2 A$$

$$\text{At } t = \frac{T}{2},$$

$$a = -\omega^2 A \cos \left(\frac{2\pi}{T} \right) \left(\frac{T}{2} \right) = -\omega^2 A \cos \pi = \omega^2 A$$

At $t = T$,

$$a = -\omega^2 A \cos \left(\frac{2\pi}{T} \right) (T)$$

$$= -\omega^2 A \cos 2\pi = -\omega^2 A$$

$\therefore$ Correct graph is depicted in option (c).

- 21** (b) Let the amplitude of SHM be A .

Now, potential energy of SHM, $U = \frac{1}{2} kx^2$

$$\text{Here, } x = \frac{A}{2} \Rightarrow U = \frac{1}{2} k \frac{A^2}{4} \quad \dots (i)$$

$$\text{Kinetic energy, } K = \frac{1}{2} kA^2 - \frac{1}{2} k \frac{A^2}{4} \Rightarrow K = \frac{3}{8} kA^2 \quad \dots (ii)$$

On dividing Eq. (ii) by Eq. (i), we get

$$\frac{K}{U} = \frac{3}{8} \times \frac{8}{1} \Rightarrow \frac{K}{U} = \frac{3}{1}$$

- 22** (d) Given, displacement, $x = 5 \cos \left(2\pi t + \frac{\pi}{4} \right)$

$$\Rightarrow x = 5 \cos \left(2\pi + \frac{\pi}{4} \right) \quad (\text{At } t = 1 \text{ s})$$

$$\Rightarrow x = 5 \cos \frac{\pi}{4} \quad [\because \cos(2\pi + \theta) = \cos \theta]$$

$$\Rightarrow x = 5 \times \frac{1}{\sqrt{2}}$$

$$\therefore x = \frac{5}{\sqrt{2}} \text{ m}$$

- 23** (d) At mean position, velocity is maximum,

$$v_{\max} = \omega A \Rightarrow \omega = \frac{v_{\max}}{A} = \frac{16}{4} = 4 \text{ rad/s}$$

$$\therefore v = \omega \sqrt{A^2 - y^2} \Rightarrow 8\sqrt{3} = 4\sqrt{4^2 - y^2}$$

$$\Rightarrow 192 = 16(16 - y^2) \Rightarrow 12 = 16 - y^2 \Rightarrow y = 2 \text{ cm}$$

- 24** (b) Given, $x_1 = A \sin \left(\omega t + \frac{\pi}{6} \right)$ and $x_2 = A \cos \omega t$

$$\Rightarrow x_2 = A \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$\text{Phase difference, } \Delta\phi = \phi_2 - \phi_1 \Rightarrow \Delta\phi = \frac{\pi}{2} - \frac{\pi}{6}$$

$$\Rightarrow \Delta\phi = \frac{3\pi - \pi}{6} = \frac{\pi}{3}$$

- 25** (b) We know that, kinetic energy, $K = \frac{1}{2} mv^2$

$$\text{where, } v = \frac{dy}{dt} = \omega A \cos \omega t$$

$$\text{So, } K = \frac{1}{2} m \omega^2 A^2 \cos^2 \omega t$$

Hence, kinetic energy varies periodically with double the frequency of SHM. So, when a particle executing SHM oscillates with a frequency ν , then the kinetic energy of particle changes periodically with a frequency of 2ν .

- 26** (d) The current displacement is $\frac{\sqrt{3}}{2}$ times the amplitude,

$$\text{i.e. } y = \frac{\sqrt{3}}{2} A$$

We know that, potential energy of a body in SHM is given by

$$U = \frac{1}{2} m \omega^2 y^2$$

where, m and ω are constants.

$$U = \frac{1}{2} m \omega^2 \left(\frac{\sqrt{3}}{2} A \right)^2 = \frac{1}{2} m \omega^2 \times \frac{3A^2}{4} \quad \dots(i)$$

Similarly, kinetic energy of a body in SHM is given by

$$\begin{aligned} K &= \frac{1}{2} m \omega^2 (A^2 - y^2) \\ &= \frac{1}{2} m \omega^2 \left(A^2 - \frac{3A^2}{4} \right) = \frac{1}{2} m \omega^2 \frac{A^2}{4} \quad \dots(ii) \end{aligned}$$

So, the ratio of PE and KE,

$$\text{PE} : \text{KE} = \frac{1}{2} m \omega^2 \times \frac{3A^2}{4} : \frac{1}{2} m \omega^2 \frac{A^2}{4} = 3 : 1$$

- 27** (a) We know that in spring SHM, the restoring force is proportional to displacement,

$$\text{i.e. } F = -m\omega^2 y \quad \dots(i)$$

$$F = -ky \quad \dots(ii)$$

where, k = force constant of the spring.

Given, $m = 10 \text{ kg}$, $A = 10 \text{ cm} = 0.1 \text{ m}$, $k = 1000 \text{ Nm}^{-1}$

Comparing both equations, we get

$$\omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{1000}{10}} \Rightarrow \omega = 10 \text{ rads}^{-1}$$

and acceleration in SHM,

$$\begin{aligned} a_{\max} &= -\omega^2 \cdot y = -10^2 \times (0.1) = -10 \text{ ms}^{-2} \\ \Rightarrow |a_{\max}| &= 10 \text{ ms}^{-2} \end{aligned}$$

- 28** (a) When restoring force will become equal to the frictional force block will start to slip.

$\therefore$ Restoring force = Friction force

$$\Rightarrow kA = \mu mg \quad (\because F_{\text{restoring}} = -kx) \quad \dots(i)$$

$$\text{Now, frequency, } f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

From Eq. (i), we get

$$f = \frac{1}{2\pi} \sqrt{\frac{\mu g}{A}}$$

- 29** (c) $y = 4 \cos^2\left(\frac{t}{2}\right) \sin 1000t$
- $$\begin{aligned} &= 4 \left(\frac{1 + \cos t}{2} \right) \sin 1000t \\ &= 2 \sin 1000t + 2 \cos t \sin 1000t \\ &= 2 \sin 1000t + \frac{1}{2} (\sin 2t + \sin 2000t) \end{aligned}$$

$\therefore$ It is a superposition of three waves, i.e. $n = 3$.

- 30** (d) Time period, $T = 2\pi \sqrt{\frac{l}{g}}$

Given, $g = 9.8 \text{ ms}^{-2}$, $l = 9.8 \text{ m}$

$$\Rightarrow T = 2\pi \sqrt{\frac{9.8}{9.8}} = 2\pi = 2 \times 3.14 = 6.28 \text{ s}$$

- 31** (d) We know that, equation of SHM, $y = a \sin \omega t$

Here in given condition, $\frac{a}{2} = a \sin \omega \times 1$

$$\frac{1}{2} = \sin \omega \Rightarrow \omega = \frac{\pi}{6}$$

$$\Rightarrow \omega = \frac{2\pi}{T} = \frac{\pi}{6}$$

Hence, time period of SHM, $T = 6 \times 2 \Rightarrow T = 12 \text{ s}$

- 32** (b) For SHM $a = -\omega^2 y$, where a is acceleration, ω is angular velocity and y is displacement. Hence, it is clear that acceleration is directly proportional to displacement and is always opposite to displacement.

While, velocity and displacement act in same direction.

- 33** (a) We know that, the time period of a simple pendulum,

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\Rightarrow T \propto \frac{1}{\sqrt{g}}$$

Here, in given condition,

$$T_1 = \frac{K}{\sqrt{g}} \quad \dots(i)$$

$$\text{and } T_2 = \frac{K}{\sqrt{g \left(1 - \frac{d}{R}\right)}} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{T_1}{T_2} = \frac{K/\sqrt{g}}{K/\sqrt{g \left(1 - \frac{d}{R}\right)}}$$

$$\text{or } \frac{T_1}{T_2} = \sqrt{\left(1 - \frac{d}{R}\right)} \quad \text{or } \frac{T_1^2}{T_2^2} = 1 - \frac{d}{R}$$

$$\Rightarrow d = \left(1 - \frac{T_1^2}{T_2^2}\right) R$$

- 35** (b) As the boy stands up, the centre of gravity of the pendulum is raised up, thereby decreasing the effective length of the pendulum of the swing. Hence, the time period T decreases.

- 36** (d) Frequency of oscillator, $n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$

$$\text{We get, } \frac{n_2}{n_1} = \sqrt{\frac{m_1}{m_2}}$$

$$\therefore 2 = \sqrt{\frac{m_1}{m_2}} \quad \text{or } m_2 = \frac{1}{4} m_1$$

37 (c) Mass of the particle = m

Spring constant = k

The time period of oscillator, $T = 2\pi\sqrt{\frac{m}{k}}$

As $k \propto \frac{1}{l}$ (where, l is the length of spring)

$$\therefore k' = 2k$$

$$\therefore T' = 2\pi\sqrt{\frac{m}{2k}} = \frac{1}{\sqrt{2}} T$$

38 (a) Given equation,

$$y = 5 (\sin 3\pi t + \sqrt{3} \cos 3\pi t)$$

$$\text{or } y = 5 \times 2 \left[\frac{1}{2} \sin 3\pi t + \frac{\sqrt{3}}{2} \cos 3\pi t \right]$$

$$\text{or } y = 10 \left[\sin 3\pi t \cos \frac{\pi}{3} + \cos 3\pi t \sin \frac{\pi}{3} \right]$$

$$\text{or } y = 10 \sin \left(3\pi t + \frac{\pi}{3} \right)$$

39 (b) $T = 2\pi\sqrt{\frac{l}{g}}$, $\log T = \log 2\pi + \frac{1}{2} \log l - \frac{1}{2} \log g$

$$\text{Differentiating, } \frac{dT}{T} = 0 + \frac{dl}{2l} - \frac{1}{2} \frac{dg}{g}$$

$$\therefore l = \text{constant}$$

$$\therefore dl = 0$$

$$100 \times \frac{dT}{T} = \frac{-1}{2} \frac{dg}{g} \times 100$$

$$= \frac{-1}{2} \times \frac{(-2) \times 100}{100} = +1\% \text{ increase}$$

40 (a) Time period of a simple pendulum,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

For freely falling system, effective $g = 0$

So, $T = \infty$

$\therefore$ Pendulum does not oscillate at all.

41 (b) Given, $y_1 = A \sin \omega t$ and $y_2 = \frac{A}{2} \sin \omega t + \frac{A}{2} \cos \omega t$

$$\text{or } y_2 = \frac{A}{2} (\sin \omega t + \cos \omega t)$$

$$= \frac{A}{2} \sqrt{2} [\sin(\omega t + 45^\circ)]$$

$$= \frac{A}{\sqrt{2}} \sin(\omega t + 45^\circ)$$

$\therefore$ The ratio of amplitudes of two motions,

$$\frac{A_1}{A_2} = \frac{A}{A/\sqrt{2}} = \sqrt{2}$$

42 (b) When the bob is placed in an electric field, the time period of simple pendulum will remain same as the bob is not charged. On contrary, if simple pendulum having charged bob is placed in a horizontal electric field, then the period will be decreased because there will be an increase in restoring force. Hence, Assertion and Reason both are correct but Reason is not the correct explanation of Assertion.

43 (d) The equation of displacement,

$$x_1 = 4 \sin \left(10t + \frac{\pi}{6} \right)$$

The energy of this equation,

$$E_1 = \frac{1}{2} m \omega_1^2 A_1^2 = \frac{1}{2} m \times 10 \times 10 \times 4 \times 4$$

The second equation of displacement,

$$x_2 = 5 \cos(\omega t)$$

The energy of this equation,

$$E_2 = \frac{1}{2} m \omega_2^2 A_2^2 = \frac{1}{2} m \omega_2^2 \times 5 \times 5$$

According to question,

$$E_2 = E_1$$

$$\frac{1}{2} m \omega_2^2 \times 5 \times 5 = \frac{1}{2} m \times 10 \times 10 \times 4 \times 4$$

$$\Rightarrow \omega_2^2 = \frac{10 \times 10 \times 4 \times 4}{5 \times 5}$$

$$\Rightarrow \omega_2^2 = 2 \times 2 \times 4 \times 4$$

$$\Rightarrow \omega_2 = \sqrt{2 \times 2 \times 4 \times 4} = 2 \times 4 = 8 \text{ unit}$$

44 (c) The equation of motion is given as

$$y = 5 \sin \frac{2\pi t}{6} \quad \left(\because \omega = \frac{2\pi}{T} \right)$$

Here, $y = 2.5 \text{ cm}$

$$\therefore 2.5 = 5 \sin \frac{2\pi t}{6}$$

$$\Rightarrow \frac{\pi}{6} = \frac{2\pi t}{6} \Rightarrow t = \frac{1}{2} \text{ s}$$

$$\therefore \text{Phase, } \phi = \frac{2\pi t}{6} = \frac{2\pi}{6} \times \frac{1}{2} = \frac{\pi}{6}$$

CHAPTER 12

Elasticity

A solid has a definite shape and size. In order to change the shape or size of a body, a force is required, such a force is called **deforming force**. Whenever a load is attached to a thin hanging wire, it elongates and the load moves downwards. When load is removed, the wire attains its original size and shape. The property of a body by virtue of which it tends to regain its original size and shape, when applied force is removed, is known as **elasticity** and the deformation caused is known as **elastic deformation**. In this chapter, we will study about the elastic behaviour and mechanical properties of solids.

Some terms related to elasticity are discussed below

- (i) **Perfectly elastic body** A body is said to be perfectly elastic, if it returns back completely to its original size on removing the external force(s).
- (ii) **Plastic body** If a body remains in the deformed shape and does not even partially regain its original shape, after removal of external force, is called plastic body or perfectly inelastic body.
- (iii) **Elastic limit** It is the upper limit of deforming force up to which if deforming force is removed, the body regains its original form completely and beyond which if deforming force is increased, the body loses its property of elasticity and gets permanently deformed.

Note

- (i) Elastic limit is the property of body, whereas elasticity is the property of material of a body.
- (ii) All rigid bodies are elastic to some extent which means we can change their dimensions slightly by pulling or pushing them.
- (iii) No body is perfectly elastic or perfectly plastic. All the bodies found in nature lie between these two limits. When the elastic behaviour of body decreases, its plastic behaviour increases or vice-versa.

STRESS AND STRAIN

When an external force is applied to a body, then at each cross-section of the body, an internal restoring force is developed which tends to restore the body back to its original position. **The internal restoring force acting per unit area of cross-section of the deformed body is called stress.**

Thus,

$$\text{Stress} = \frac{\text{Restoring force}}{\text{Area}}$$

If there is no permanent change in configuration of the body, the restoring force is equal and opposite to the external deforming force applied.

Inside

1 Stress and strain

2 Hooke's law

3 Poisson's ratio

- Stress-strain curve
- Work done or potential energy stored in a stretched wire
- Thermal stresses and strains

So, stress can be given as

$$\text{Stress} = \frac{\text{External deforming force}}{\text{Area}}$$

In SI system, the unit of stress is Nm^{-2} or pascal (Pa) and in CGS system, it is dyne cm^{-2} .

The dimensions of stress are $[\text{ML}^{-1}\text{T}^{-2}]$.

Note The minimum value of stress required to break a wire is called **breaking stress**. It depends on nature of material, temperature and impurities.

Due to external force, shape of the body changes and it is said to have strain.

Strain is defined as the ratio of change in shape of an object to the original shape.

$$\text{Strain} = \frac{\text{Change in configuration of the object}}{\text{Original configuration of the object}}$$

Strain is a pure number, it has no unit. There are three types of stress resulting in three types of strain

1. Longitudinal stress

When the stress is normal to the surface of an object, then it is known as longitudinal stress.

It is of two types

- (i) **Tensile stress** If the stress produced in an object is due to increase in its length, then it is called tensile stress.

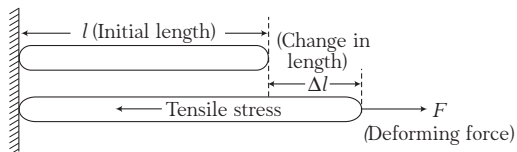


Fig. 12.1 An object under tensile stress

- (ii) **Compressive stress** If the stress produced in an object is due to decrease in its length, then it is called compressive stress.

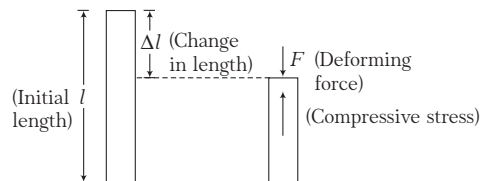


Fig. 12.2 An object under compressive stress

Longitudinal strain

It is defined as the change in length per unit length of the object on application of deforming force.

$$\text{Longitudinal strain} = \frac{\text{Change in length } (\Delta l)}{\text{Original length } (l)}$$

2. Volumetric stress

If equal normal forces are applied on an object all over its surface, then its volume changes. The restoring force acting per unit area inside the object opposing change in volume is called volumetric stress.

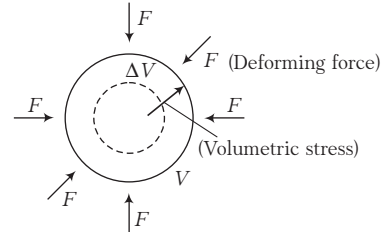


Fig. 12.3 An object under effect of normal deforming forces

Volumetric strain

It is defined as the change in volume per unit volume of the object on application of deforming force.

$$\text{Volumetric strain} = \frac{\text{Change in volume } (\Delta V)}{\text{Original volume } (V)}$$

3. Shearing stress

When a force is applied on an object along the tangential direction of the surface of the object, then stress produced in the object is called shearing stress. Such force tends to change the shape of object. For this, the opposite surface is kept fixed.

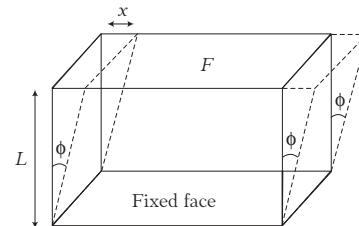


Fig. 12.4 An object under tangential deforming forces

Tangential or shearing strain

It is defined as the ratio of displacement (x) of the upper surface to the distance (L) between two faces on the application of deforming force.

$$\text{Shearing strain} = \frac{\text{Displacement } (x)}{\text{Distance } (L)}$$

Example 12.1 A rod has a radius of 100 mm and a length of 10 cm. A 100 N force compresses along its length. Calculate the longitudinal stress developed in the rod.

Sol. Given, radius of the rod, $r = 100 \text{ mm} = 100 \times 10^{-3} \text{ m}$

Length of the rod, $l = 10 \text{ cm} = 10 \times 10^{-2} \text{ m}$

Force, $F_n = 100 \text{ N}$

Longitudinal stress developed in the rod

$$= \frac{F_n}{A} = \frac{100}{\pi (100 \times 10^{-3})^2} \approx 3185 \text{ Nm}^{-2}$$

Example 12.2 Find the greatest length of steel wire that can remain hanged vertically without breaking. Breaking stress of steel = $8 \times 10^8 \text{ Nm}^{-2}$. Density of steel = $8 \times 10^3 \text{ kgm}^{-3}$. (Take, $g = 10 \text{ ms}^{-2}$)

Sol. Let l be the length of the wire that can hang vertically without breaking. The stretching force on it is equal to its own weight.

If A is the area of cross-section and ρ is the density, then

$$\text{maximum stress or breaking stress } (S_m) = \frac{\text{weight}}{A} \quad \left(\because \text{Stress} = \frac{\text{Force}}{\text{Area}} \right)$$

$$\text{Weight} = mg = V\rho g \quad (\because \text{Mass} = \text{Volume} \times \text{Density})$$

$$= A\rho g$$

$$(\because \text{Volume} = \text{Area of cross-section} \times \text{Length})$$

$$\text{or} \quad S_m = \frac{(A\rho)g}{A}$$

$$\therefore \text{Length of the steel wire, } l = \frac{S_m}{\rho g} = \frac{8 \times 10^8}{(8 \times 10^3)(10)} = 10^4 \text{ m}$$

Example 12.3 Consider a rod of steel having radius of 8 mm and the length of 2 m. If a force of 150 kN stretches it along its length, then calculate the stress, percentage strain in the rod, if the elongation in length is 7.46 mm.

Sol. Given, radius of rod, $r = 8 \text{ mm} = 8 \times 10^{-3} \text{ m}$, length, $L = 2 \text{ m}$

Applied force, $F = 150 \text{ kN} = 15 \times 10^4 \text{ N}$

Cross-section area of wire,

$$A = \pi r^2 = \pi \times (8 \times 10^{-3})^2 = 201 \times 10^{-6} \text{ m}^2$$

$$\Delta L = 7.46 \text{ mm} = 7.46 \times 10^{-3} \text{ m, percentage strain} = ?$$

$$\begin{aligned} \text{Stress in rod} &= \frac{F}{A} = \frac{15 \times 10^4}{201 \times 10^{-6}} \\ &= 7.46 \times 10^8 \text{ Nm}^{-2} \end{aligned}$$

$$\text{Longitudinal strain} = \frac{\Delta L}{L} = \frac{7.46 \times 10^{-3}}{2} = 3.73 \times 10^{-3}$$

$$\text{Percentage strain} = 3.73 \times 10^{-3} \times 100 = 0.37\%$$

Example 12.4 If the angle of shear is 30° for a cubical body and the change in length is 250 cm, then what must be the volume of this cubical body?

Sol. Given, angle of shear, $\phi = 30^\circ$

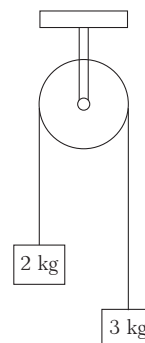
and change in length, $\Delta L = 250 \text{ cm} = 2.5 \text{ m}$

$$\therefore \text{Shear strain, } \tan \phi = \frac{\Delta L}{L} \Rightarrow \tan 30^\circ = \frac{2.5}{L}$$

$$\Rightarrow L = \frac{2.5}{\tan 30^\circ} = \frac{2.5}{0.577} = 4.332$$

$$\text{Volume of cubical body, } V = L^3 = 81.295 \text{ m}^3$$

Example 12.5 Two blocks of masses 2 kg and 3 kg are connected by a metal wire going over a smoother pulley as shown in figure. The breaking stress of the metal is $2 \times 10^9 \text{ Nm}^{-2}$. What would be the minimum radius of the wire used, if it is not to break? (Take, $g = 10 \text{ ms}^{-2}$)

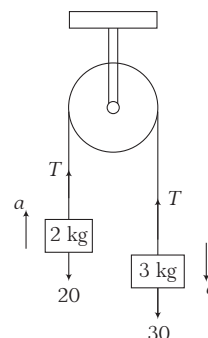


Sol. Equation of motion for 3 kg mass,

$$30 - T = 3a \quad \dots(i)$$

Equation of motion for 2 kg mass,

$$T - 20 = 2a \quad \dots(ii)$$



On dividing Eq. (i) by Eq. (ii), we get

$$\frac{30 - T}{T - 20} = 3/2$$

$$\Rightarrow 2(30 - T) = 3(T - 20)$$

$$\Rightarrow 60 - 2T = 3T - 60$$

$$5T = 120 \Rightarrow T = 24 \text{ N}$$

$$\text{Breaking stress, } S = \frac{T}{A} \quad \left(\because \text{Stress} = \frac{\text{Force}}{\text{Area}} \right)$$

$$\Rightarrow 2 \times 10^9 = \frac{24}{A}$$

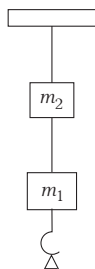
$$\text{Area, } A = 12 \times 10^{-9} \text{ m}^2 \Rightarrow \pi r^2 = 12 \times 10^{-9} \quad (\because A = \pi r^2)$$

$\therefore$ Minimum radius of the wire,

$$r = \left(\frac{12 \times 10^{-9}}{3.14} \right)^{1/2} = 6.18 \times 10^{-5} \text{ m}$$

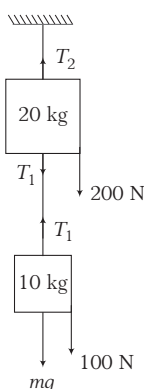
Example 12.6 The two wires shown in figure are made of the same material which has a breaking stress of $8 \times 10^8 \text{ Nm}^{-2}$. The area of cross-section of the upper wire is 0.006 cm^2 and that of the lower wire is 0.003 cm^2 . The mass $m_1 = 10 \text{ kg}$, $m_2 = 20 \text{ kg}$ and the hanger is light.

Find the maximum load that can be put on the hanger without breaking a wire. Which wire will break first, if the load is increased?



Sol. Maximum tension that the upper wire tolerate
 $= \text{Breaking stress} \times \text{Area}$

$$T_2 = 8 \times 10^8 \times 0.006 \times 10^{-4} = 480 \text{ N}$$



Maximum tension that the lower wire has

$$T_1 = 8 \times 10^8 \times 0.003 \times 10^{-4} = 240 \text{ N}$$

We have, $T_2 = T_1 + 200$ and $T_1 = 100 + mg$

$$\therefore T_2 = 300 + mg$$

$$\therefore T_2 \leq 480$$

$$\therefore 300 + mg \leq 480 \quad \text{or} \quad mg \leq 180 \quad \text{or} \quad m \leq 18 \text{ kg}$$

$$\text{Again,} \quad T_1 \leq 240 \quad \text{or} \quad 100 + mg \leq 240$$

$$\text{or} \quad m \leq 14 \text{ kg}$$

Maximum mass, $m_{\max} = 14 \text{ kg}$. When load is increased, T_1 increases, hence lower wire will break first.

Example 12.7 Consider a solid cube which is subjected to a pressure of $6 \times 10^5 \text{ Nm}^{-2}$. Due to this pressure, the each side of the cube is shortened by 2%. Find out the volumetric strain of the cube.

Sol. Let L be the initial length of the each side of the cube.

$$\text{Volume, } V = L \times L \times L = L^3 = \text{Initial volume (} V_i \text{ say)}$$

If each side of the cube is shortened by 2%, then final length of the cube $= L - 2\% \text{ of } L$

$$= \left(L - \frac{2L}{100} \right) = L \left(1 - \frac{2}{100} \right)$$

$$\therefore \text{Final volume, } V_f = L^3 \left(1 - \frac{2}{100} \right)^3 = V \left(1 - \frac{2}{100} \right)^3$$

$$\text{Change in volume, } \Delta V = V_f - V_i$$

$$= V \left(1 - \frac{2}{100} \right)^3 - V = V \left[\left(1 - \frac{2}{100} \right)^3 - 1 \right]$$

$$\Rightarrow \frac{\Delta V}{V} = \left(1 - \frac{2}{100} \right)^3 - 1 \approx \left[1 - \frac{2 \times 3}{100} \right] - 1$$

$$[\because (1-x)^n \approx 1 - nx \text{ for } x \ll 1]$$

$$\therefore \text{Volumetric strain, } \frac{\Delta V}{V} = 1 - 0.06 - 1 = 0.06$$

(Take positive sign)

CHECK POINT 12.1

- Breaking stress depends on
 - length of wire
 - area of cross-section of wire
 - Both (a) and (b)
 - independent of length and area of cross-section
- The lower surface of a cube is fixed. On its upper surface, force is applied at an angle of 30° from its surface. The change will be in its
 - shape
 - size
 - volume
 - Both shape and size
- When a spiral spring is stretched by force, strain produced is
 - longitudinal strain
 - volumetric strain
 - shear strain
 - Both (a) and (c)
- A metallic cube whose each side is 10 cm is subjected to a shearing force of 100 kgf. Calculate the shearing stress produced.
 - $9.8 \times 10^4 \text{ Nm}^{-2}$
 - 10 Nm^{-2}
 - 9.8 Nm^{-2}
 - $9.8 \times 10^2 \text{ Nm}^{-2}$
- A wire of length 2.5 m has a percentage strain of 0.012% under a tensile force. The extension produced in the wire will be
 - 0.03 mm
 - 0.3 mm
 - 0.3 m
 - 0.03 m
- A cube is subjected to a uniform volume compression. If the side of the cube decreases by 2%, the bulk strain is
 - 0.02
 - 0.03
 - 0.04
 - 0.06
- A metallic cube of side 10 cm is subjected to a shearing force of 300 kgf. The top face is displaced through 0.25 cm with respect to the bottom? Calculate the shearing strain produced.
 - 0.25
 - 2.5
 - 0.025
 - 0.08

HOOKE'S LAW

According to Hooke's law,

"For small deformation, the stress in a body is proportional to the corresponding strain."

i.e. Stress $\propto$ Strain or Stress = (E) (Strain)

Here, $E = \frac{\text{stress}}{\text{strain}}$ is a constant called the **modulus of elasticity**.

The SI unit of modulus of elasticity is Nm^{-2} and its dimensions are $[\text{ML}^{-1}\text{T}^{-2}]$.

The modulus of elasticity depends on the type of material and temperature. It does not depend upon the values of stress and strain.

Depending upon the nature of force applied on the body, the modulus of elasticity is classified in the following three types

1. Young's modulus of elasticity (Y)

When a wire is acted upon by two equal and opposite forces in the direction of its length, the length of the body

is changed. The change in length per unit length $\left(\frac{\Delta l}{l}\right)$ is

called the **longitudinal strain** and the restoring force (which is equal to the applied force in equilibrium) per unit area of cross-section of the wire is called the **longitudinal stress**.

For a small change in the length of the wire, the ratio of the longitudinal stress to the corresponding strain is called the Young's modulus of elasticity (Y) of the wire.

$$\text{Thus, } Y = \frac{\frac{F}{A}}{\frac{\Delta l}{l}} \text{ or } Y = \frac{Fl}{A\Delta l}$$

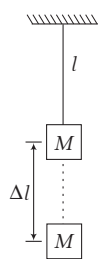


Fig. 12.5

Let there be a wire of length l and radius r . Its one end is clamped to a rigid support and a mass M is attached at the other end. Then,

$$F = Mg \text{ and } A = \pi r^2$$

Substituting in above equation, we have

$$Y = \frac{Mgl}{(\pi r^2)\Delta l}$$

Young's modulus is only defined for solids, and not for liquids and gases. Larger the value of Y for a material, more elastic it would be. For this reason, steel is more elastic than rubber.

The material which has smaller value of Y is more ductile, i.e. it offers less resistance in framing it into wire. Thus, for making wire, we choose a material having less value of Y .

Note For a perfectly rigid body, value of Y is infinite.

Force constant of wire

Force required to produce unit elongation in a wire is called force constant of material of wire. It is denoted by k .

$$k = \frac{F}{\Delta l} = \frac{YA}{l} \quad \left(\because \frac{F}{\Delta l} = \frac{YA}{l} \right)$$

If a spring is stretched or compressed by an amount Δl , the restoring force produced in it, is

$$F_s = k \Delta l \quad \dots(i)$$

Here, k = force constant of spring.

Similarly, if a wire is stretched by an amount Δl , the restoring force produced in it, is

$$F = \left(\frac{YA}{l} \right) \Delta l \quad \dots(ii)$$

Comparing Eqs. (i) and (ii), we can see that force constant of a wire,

$$k = \frac{YA}{l} \quad \dots(iii)$$

i.e. **A wire is just like a spring of force constant $\frac{YA}{l}$.**

So, all formulae which we use in case of a spring can be applied to a wire also.

Example 12.8 If a wire of length 4 m and cross-sectional area of 2 m^2 is stretched by a force of 3 kN, then determine the change in length due to this force. (Take, Young's modulus of material of wire is $110 \times 10^9 \text{ Nm}^{-2}$)

Sol. Given, area of cross-section, $A = 2 \text{ m}^2$

Applied force, $F = 3 \text{ kN} = 3 \times 10^3 \text{ N}$, length, $L = 4 \text{ m}$

Young's modulus, $Y = 110 \times 10^9 \text{ Nm}^{-2}$

$$\Rightarrow Y = \frac{FL}{A\Delta L}$$

$\therefore$ Change in length,

$$\Delta L = \frac{FL}{AY} = \frac{3 \times 10^3 \times 4}{2 \times 110 \times 10^9} = 0.0545 \times 10^{-6} \text{ m}$$

$$\Rightarrow \Delta L = 54.5 \times 10^{-6} \text{ mm}$$

Example 12.9 A steel wire of length 4 m and diameter 5 mm is stretched by 5 kg-wt. Find the increase in its length, if the Young's modulus of steel is 2.4×10^{12} dyne cm^{-2} .

Sol. Hence, length of the wire, $l = 4 \text{ m} = 400 \text{ cm}$

Diameter of wire, $2r = 5 \text{ mm}$

or radius of wire, $r = 2.5 \text{ mm} = 0.25 \text{ cm}$

Force, $F = 5 \text{ kg-wt} = 5000 \text{ g-wt}$

$$= 5000 \times 980 \text{ dyne}$$

Young's modulus, $Y = 2.4 \times 10^{12} \text{ dyne cm}^{-2}$

$$\text{As we have, } Y = \frac{F}{\pi r^2} \times \frac{l}{\Delta l}$$

$$\begin{aligned} \text{Increase in length, } \Delta l &= \frac{Fl}{\pi r^2 Y} \\ &= \frac{(5000 \times 980) \times 400}{(22/7) \times (0.25)^2 \times 2.4 \times 10^{12}} = 0.0041 \text{ cm} \end{aligned}$$

Example 12.10 Determine the elongation of the steel bar 1 m long and 1.5 cm^2 cross-sectional area when subjected to a pull of $1.5 \times 10^4 \text{ N}$. (Take, $Y = 2 \times 10^{11} \text{ Nm}^{-2}$)

Sol. Young's modulus, $Y = \frac{F/A}{\Delta l/l}$

$$\therefore \text{Elongation of the steel wire, } \Delta l = \frac{Fl}{AY}$$

$$\begin{aligned} \Delta l &= \frac{(1.5 \times 10^4)(1)}{(1.5 \times 10^{-4})(2 \times 10^{11})} \\ &= 0.5 \times 10^{-3} \text{ m} \end{aligned}$$

$$\text{or } \Delta l = 0.5 \text{ mm}$$

Example 12.11 A cable is replaced by another one of same length and material but twice the diameter. How will this effect the elongation under a given load? How does this effect the maximum load, it can support without exceeding the elastic limit?

Sol. Young's modulus, $Y = \frac{Mgl}{\pi r^2 \cdot \Delta l} = \frac{Mgl}{\pi \left(\frac{D}{2}\right)^2 \Delta l} = \frac{4Mgl}{\pi D^2 \cdot \Delta l}$

where, D is the diameter of the wire.

$$\therefore \text{Elongation, } \Delta l = \frac{4Mgl}{\pi D^2 \cdot Y}, \text{ i.e. } \Delta l \propto \frac{1}{D^2}$$

Clearly, if the diameter is doubled, the elongation will become one-fourth.

$$\text{Also, maximum stress, } S_m = \frac{M_m g}{(\pi D^2/4)}$$

$$\text{or } M_m \propto D^2$$

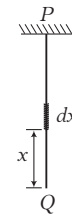
Clearly, if the diameter is doubled, the wire can support four times the original load.

Example 12.12 A bar of mass m and length l is hanging from point A as shown in figure. Find the increase in its length due to its own weight. The Young's modulus of

elasticity of the wire is Y and area of cross-section of the wire is A .



Sol. Consider a small section dx of the bar at a distance x from Q. The weight of the bar for a length x is,



$$w = \left(\frac{mg}{l}\right)x$$

$$\text{Elongation in section } dx \text{ will be, } dl = \left(\frac{w}{AY}\right)dx = \left(\frac{mg}{lAY}\right)x dx$$

Total elongation in the bar can be obtained by integrating this expression for $x = 0$ to $x = l$.

$$\therefore \Delta l = \int_{x=0}^{x=l} dl = \left(\frac{mg}{lAY}\right) \int_0^l x dx$$

$$\text{or Increase in length, } \Delta l = \frac{mgl}{2AY}$$

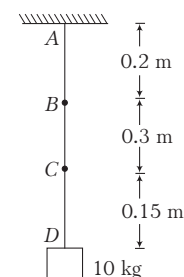
Example 12.13 A rod AD consisting of three segments AB, BC and CD joined together is hanging vertically from a fixed support at A. The lengths of the segments are respectively 0.2 m, 0.3 m and 0.15 m. The cross-section of the rod is uniformly 10^{-4} m^2 . A weight of 10 kg is hung from D. Calculate the displacements of points B, C and D, if $Y_{AB} = 3.5 \times 10^{10} \text{ Nm}^{-2}$, $Y_{BC} = 5 \times 10^{10} \text{ Nm}^{-2}$, $Y_{CD} = 2 \times 10^{10} \text{ Nm}^{-2}$. (Neglect the weight of the rod)

Sol. Given, area, $A = 10^{-4} \text{ m}^2$,

$$Y_{AB} = 3.5 \times 10^{10} \text{ Nm}^{-2}$$

$$Y_{BC} = 5 \times 10^{10} \text{ Nm}^{-2}$$

$$Y_{CD} = 2 \times 10^{10} \text{ Nm}^{-2}$$



Increase in length,

$$\Delta L = \frac{FL}{AY} = \frac{MgL}{AY} = \frac{10 \times 10L}{10^{-4}Y} = 10^6 \frac{L}{Y}$$

Now, increase in length of AB segment,

$$(\Delta L)_{AB} = 10^6 \times \frac{L_{AB}}{Y_{AB}} = 10^6 \times \frac{0.2}{3.5 \times 10^{10}} = 5.7 \times 10^{-6} \text{ m}$$

Increase in length of BC segment, $(\Delta L)_{BC} = 10^6 \times \frac{L_{BC}}{Y_{BC}}$

$$= 10^6 \times \frac{0.3}{5 \times 10^{10}} = 6 \times 10^{-6} \text{ m}$$

Increase in length of CD segment,

$$(\Delta L)_{CD} = 10^6 \frac{L_{CD}}{Y_{CD}} = 10^6 \times \frac{0.15}{2 \times 10^{10}} = 7.5 \times 10^{-6} \text{ m}$$

So, displacement of B = $\Delta L_{AB} = 5.7 \times 10^{-6} \text{ m}$

Displacement of C = $\Delta L_{AB} + \Delta L_{BC} = 11.7 \times 10^{-6} \text{ m}$

Displacement of D = $\Delta L_{AB} + \Delta L_{BC} + \Delta L_{CD} = 19.2 \times 10^{-6} \text{ m}$

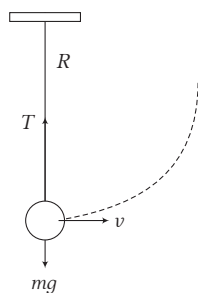
Example 12.14 A block of weight 10 N is fastened to one end of wire of cross-sectional area 4 mm^2 and is rotated in a vertical circle of radius 30 cm. The speed of the block at the bottom of the circle is 3 ms^{-1} . Find the elongation of the wire when the block is at the bottom of the circle. (Take, Young's modulus of the material of the wire = $2 \times 10^{11} \text{ Nm}^{-2}$ and $g = 10 \text{ ms}^{-2}$)

Sol. Given, weight, $w = 10 \text{ N}$

Area, $A = 4 \text{ mm}^2 = 4 \times 10^{-6} \text{ m}^2$

Radius, $R = 30 \text{ cm} = 0.3 \text{ m}$

Speed of block, $v = 3 \text{ ms}^{-1}$



Tension of the wire, $T - mg = \frac{mv^2}{R}$

$$\text{or } T = mg + \frac{mv^2}{R} = 100 + \frac{10 \times 3^2}{0.3} = 400 \text{ N}$$

The elongation of the wire when the block is at the bottom of the circle,

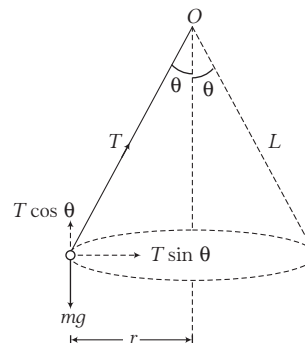
$$\Delta L = \frac{FR}{AY}$$

Here, force (F) is equal to the tension in the wire.

$$\text{So, } \Delta L = \frac{TR}{AY} = \frac{400 \times 0.3}{4 \times 10^{-6} \times 2 \times 10^{11}} = 1.5 \times 10^{-4} \text{ m}$$

Example 12.15 A 2 kg mass is suspended by a rubber cord 2 m long and of cross-section 0.5 cm^2 . It is made to describe a horizontal circle of radius 50 cm, 4 times in a second. Find the extension of the cord. (Take, Young's modulus, $Y = 7 \times 10^8 \text{ Nm}^{-2}$)

Sol. Given, $m = 2 \text{ kg}$, $\omega = 2\pi f = 2\pi \times 4 = 8\pi \text{ rad s}^{-1}$, $r = 0.5 \text{ m}$



From the figure, $T \cos \theta = mg$

and $T \sin \theta = \frac{mv^2}{r} = m\omega^2 r$

Tension in cord, $T = \sqrt{(T \sin \theta)^2 + (T \cos \theta)^2}$

$$= \sqrt{(m\omega^2 r)^2 + (mg)^2}$$

$$T = \sqrt{[2 \times (8\pi)^2 \times 0.5]^2 + [2 \times 10]^2} \approx 631 \text{ N}$$

The extension of the cord,

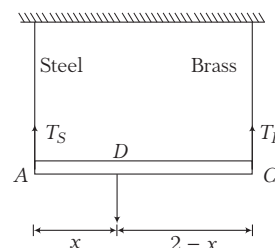
$$\Delta L = \frac{TL}{AY} = \frac{631 \times 2}{0.5 \times 10^{-4} \times 7 \times 10^8} = 360 \times 10^{-4} \text{ m} = 3.60 \text{ cm}$$

Example 12.16 A light rod of length 2 m is suspended from the ceiling horizontally by means of two vertical wires of equal length tied to its ends. One of the wires is made of steel and is of cross-section 10^{-3} m^2 and the other is of brass of cross-section $2 \times 10^{-3} \text{ m}^2$. Find out the position along the rod at which a weight may be hang to produce

- equal stresses in both wires,
- equal strains on both wires.

(Young's modulus for steel is $2 \times 10^{11} \text{ Nm}^{-2}$ and for brass is 10^{11} Nm^{-2})

Sol. (i) Given, stress in steel = stress in brass



$$\therefore \frac{T_S}{A_S} = \frac{T_B}{A_B}$$

$$\Rightarrow \frac{T_S}{T_B} = \frac{A_S}{A_B} = \frac{10^{-3}}{2 \times 10^{-3}} = \frac{1}{2} \quad \dots(i)$$

As the system is in equilibrium, taking moments about D , we have

$$\begin{aligned} T_S \cdot x &= T_B (2 - x) \\ \therefore \frac{T_S}{T_B} &= \frac{2 - x}{x} \quad \dots(ii) \end{aligned}$$

From Eqs. (i) and (ii), we get

$$x = 1.33 \text{ m}$$

$$(ii) \therefore \text{Strain} = \frac{\text{Stress}}{Y}$$

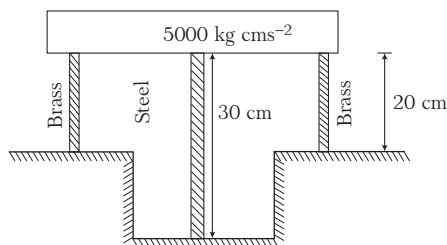
Given, strain in steel = strain in brass

$$\begin{aligned} \therefore \frac{T_S/A_S}{Y_S} &= \frac{T_B/A_B}{Y_B} \\ \therefore \frac{T_S}{T_B} &= \frac{A_S Y_S}{A_B Y_B} = \frac{(1 \times 10^{-3})(2 \times 10^{11})}{(2 \times 10^{-3})(10^{11})} = 1 \quad \dots(iii) \end{aligned}$$

From Eqs. (ii) and (iii), we have

$$x = 1 \text{ m}$$

Example 12.17 A steel rod of cross-sectional area 16 cm^2 and two brass rods each of cross-sectional area 10 cm^2 together support a load of 5000 kg cms^{-2} as shown in figure. Find the stress in the rods. (Take, Young's modulus for steel $= 2 \times 10^6 \text{ kg cm}^{-2}$ and for brass $= 1 \times 10^6 \text{ kg cm}^{-2}$)



Sol. Given, area of steel rod, $A_S = 16 \text{ cm}^2$

Area of two brass rods, $A_B = 2 \times 10 = 20 \text{ cm}^2$

Load, $F = 5000 \text{ kg cms}^{-2}$

Young's modulus for the steel, $Y_S = 2 \times 10^6 \text{ kg cm}^{-2}$

Young's modulus for the brass, $Y_B = 1 \times 10^6 \text{ kg cm}^{-2}$

Length of the steel rod, $l_S = 30 \text{ cm}$

Length of the brass rod, $l_B = 20 \text{ cm}$

Let S_S = stress in steel and S_B = stress in brass

Decrease in length of the steel rod

= Decrease in length of the brass rod

$$\text{or} \quad \frac{S_S}{Y_S} \times l_S = \frac{S_B}{Y_B} \times l_B$$

$$\text{or} \quad S_S = \frac{Y_S}{Y_B} \times \frac{l_B}{l_S} \times S_B = \frac{2 \times 10^6}{1 \times 10^6} \times \frac{20}{30} \times S_B$$

$$\therefore S_S = \frac{4}{3} S_B \quad \dots(i)$$

Now, using the relation,

$$F = S_S A_S + S_B A_B$$

$$\text{or} \quad 5000 = S_S \times 16 + S_B \times 20 \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$S_B = 120.9 \text{ kg cm}^{-2}$$

$\therefore$ From Eq. (i), we get

$$S_S = \frac{4}{3} \times 120.9 = 161.2 \text{ kg cm}^{-2}$$

2. Bulk modulus of elasticity (B)

When a uniform pressure (normal force) is applied all over the surface of a body, the volume of the body changes.

The change in per unit volume of the body is called the **volume strain** and the normal force acting per unit area of the surface (pressure) is called the normal stress or **volume stress**. For small strains, the ratio of the volume stress to the volume strain is called the bulk modulus of the material of the body. It is denoted by B .

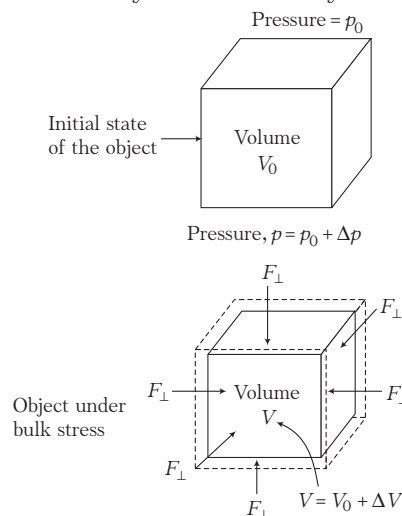


Fig. 12.6

Consider a solid cube body of volume V and surface area A . In order to compress the body, let a force F be applied normally on the entire surface of the body and its volume decreases by ΔV , then volumetric strain $= -\frac{\Delta V}{V}$.

Here, negative sign shows the volume is decreasing when normal force is applied.

$$\text{Stress} = F/A$$

$$\therefore B = \frac{F/A}{-\Delta V/V}$$

Then,

$$B = \frac{-p}{\Delta V/V}$$

Here, negative sign implies that when the pressure increases, volume decreases and *vice-versa*.

The SI unit of bulk modulus is Nm^{-2} and its dimensions are $[\text{ML}^{-1}\text{T}^{-2}]$. The relation between density ρ , pressure Δp and bulk modulus (B) is given by

$$\rho' = \rho \left(1 + \frac{\Delta p}{B} \right)$$

In case of gas The isothermal bulk modulus of the gas (B_T) is equal to the pressure of the gas at that instant.

$$i.e. \quad B_T = p$$

The adiabatic bulk modulus of gas (B_s) is equal to γ times the pressure of the gas at that instant of time, i.e. $B_s = \gamma p$.

Note For a perfectly rigid body, $B = \infty$

Example 12.18 How much should the pressure on a litre of water be changed to compress it by 0.10%? (Take, bulk modulus of water = $2.2 \times 10^9 \text{ Pa}$)

Sol. Bulk modulus, $|B| = \frac{p}{\frac{\Delta V}{V}}$

$$\text{Here, } \frac{\Delta V}{V} = 0.10\% = \frac{0.10}{100} = 1 \times 10^{-3}$$

$$\begin{aligned} \text{Change in pressure, } p &= |B| \left(\frac{\Delta V}{V} \right) = (2.2 \times 10^9 \text{ Pa}) \times 1 \times 10^{-3} \\ &= 2.2 \times 10^6 \text{ Pa} \end{aligned}$$

Compressibility

The reciprocal of the bulk modulus of the material of a body is called the compressibility of that material. Thus,

$$\text{Compressibility} = \frac{1}{|B|} = \frac{\Delta V}{pV} = k$$

The SI unit of compressibility is N^{-1}m^2 and its dimensions are $[\text{M}^{-1}\text{LT}^2]$.

The bulk modulus is defined for all solids, liquids and gases. The value of B for solids are much larger than for liquids and that for liquids are much larger than for gases. Thus, solids are least compressible while gases are most compressible.

Example 12.19 What is the density of lead under a pressure of $2 \times 10^8 \text{ Nm}^{-2}$, if the bulk modulus of lead is $8 \times 10^9 \text{ Nm}^{-2}$ and initially the density of lead is 11.4 g cm^{-3} ?

Sol. The changed density, $\rho' = \rho \left(1 + \frac{\Delta p}{B} \right)$

Substituting the values, we have

$$\rho' = 11.4 \left(1 + \frac{2 \times 10^8}{8 \times 10^9} \right)$$

$$\text{or } \rho' = 11.69 \text{ g cm}^{-3}$$

Example 12.20 The bulk modulus of water is $2.3 \times 10^9 \text{ Nm}^{-2}$.

(i) Find its compressibility.

(ii) How much pressure in atmospheres is needed to compress a sample of water by 0.1%?

Sol. Here, $|B| = 2.3 \times 10^9 \text{ Nm}^{-2}$

$$= \frac{2.3 \times 10^9}{1.01 \times 10^5} = 2.27 \times 10^4 \text{ atm}$$

$$\begin{aligned} \text{(i) Compressibility} &= \frac{1}{|B|} = \frac{1}{2.27 \times 10^4} \\ &= 4.4 \times 10^{-5} \text{ atm}^{-1} \end{aligned}$$

$$\text{(ii) Here, } \frac{\Delta V}{V} = -0.1\% = -0.001$$

$\therefore$ Required increase in pressure,

$$\begin{aligned} \Delta p &= B \times \left(-\frac{\Delta V}{V} \right) \\ &= 2.27 \times 10^4 \times 0.001 \\ &= 22.7 \text{ atm} \end{aligned}$$

Example 12.21 What will be the decrease in volume of 100 cm^3 of water under pressure of 100 atm, if the compressibility of water is 4×10^{-5} per unit atmospheric pressure?

Sol. Bulk modulus, $|B| = \frac{1}{\text{Compressibility}} = \frac{1}{k}$

$$\begin{aligned} &= \frac{1}{4 \times 10^{-5}} = 0.25 \times 10^5 \text{ atm} \\ &= 0.25 \times 10^5 \times 1.013 \times 10^5 \text{ Nm}^{-2} \\ &= 2.533 \times 10^9 \text{ Nm}^{-2} \end{aligned}$$

$$\text{Volume, } V = 100 \text{ cm}^3 = 10^{-4} \text{ m}^3$$

$$\begin{aligned} \text{Pressure, } p &= 100 \text{ atm} = 100 \times 1.013 \times 10^5 \text{ Nm}^{-2} \\ &= 1.013 \times 10^7 \text{ Nm}^{-2} \end{aligned}$$

$$\text{Now, apply } \frac{1}{|B|} = k = \frac{\Delta V}{pV}$$

$$\therefore \Delta V = \frac{pV}{|B|} = \frac{1.013 \times 10^7 \times 10^{-4}}{2.533 \times 10^9}$$

Decrease in volume,

$$\begin{aligned} \Delta V &\approx 0.4 \times 10^{-6} \text{ m}^3 \\ &= 0.4 \text{ cm}^3 \end{aligned}$$

Example 12.22 Find the decrease in the volume of a sample of water from the following data. Initial volume = 2000 cm^3 , initial pressure = 10^5 Nm^{-2} , final pressure = 10^6 Nm^{-2} and compressibility of water = $25 \times 10^{-11} \text{ N}^{-1} \text{ m}^2$.

Sol. Bulk modulus,

$$|B| = \frac{1}{\text{Compressibility}} = \frac{1}{25 \times 10^{-11}} = 4 \times 10^9$$

$$\text{and } |B| = \frac{-\Delta p}{\Delta V/V}$$

∴ Decrease in the volume,

$$\begin{aligned}\Delta V &= \frac{\Delta p V}{|B|} = \frac{(10^6 - 10^5)(2000 \times 10^{-6})}{4 \times 10^9} \\ &= 4.5 \times 10^{-7} \text{ m}^3 = 4.5 \times 10^{-7} \times 10^6 \text{ cm}^3 \\ &= 0.45 \text{ cm}^3\end{aligned}$$

3. Modulus of rigidity (η)

When a body is acted upon by an external force tangential to a surface of the body, where the opposite surface is kept fixed, it suffers a change in its shape but the volume remains unchanged. Then, the body is said to be sheared.

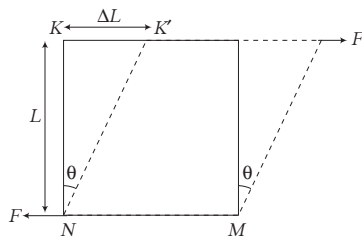


Fig. 12.7

The ratio of the displacement of a layer in the direction of the tangential force to the distance of the layer from the fixed surface is called the **shearing strain** and the tangential force acting per unit area of the surface is called the **shearing stress**.

For small strain, the ratio of the shearing stress to the shearing strain is called the **modulus of rigidity** of the material of the body. It is denoted by η .

Thus,
$$\eta = \frac{F/A}{KK'/KN}$$

Here,
$$\frac{KK'}{KN} = \tan \theta \approx \theta$$

∴
$$\eta = \frac{F/A}{\theta}$$

or
$$\eta = \frac{F}{A\theta} \quad \dots(i)$$

Since, $\theta \approx \tan \theta = \frac{KK'}{KN} = \frac{\Delta L}{L}$

Therefore, Eq. (i) will become

$$\eta = \frac{F}{A} \cdot \frac{L}{\Delta L}$$

The SI unit of modulus of rigidity is Nm^{-2} and its dimensions are $[\text{ML}^{-1}\text{T}^{-2}]$.

Shear modulus (or modulus of rigidity) is generally less than Young's modulus (from table). For most material, $\eta \approx Y/3$. Shear modulus is defined for solids only.

A solid will have all the three moduli of elasticity Y , B and η . But in case of liquid or gas only B can be defined as liquid or gas cannot be framed into a wire or no shear force can be applied on them.

Example 12.23 A 4 cm cube has its upper face displaced by 0.1 mm by a tangential force of 8 kN. Calculate the shear modulus of the cube.

Sol. Here, each side of the cube, $L = 4 \text{ cm}$

Area of the face over which the force is applied,

$$A = L^2 = 16 \text{ cm}^2$$

Displacement, $\Delta L = 0.1 \text{ mm} = 0.01 \text{ cm}$

Force applied, $F = 8 \text{ kN} = 8000 \times 10^5 \text{ dyne}$

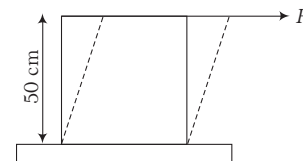
$$= 8 \times 10^8 \text{ dyne}$$

As,
$$\eta = \frac{FL}{A\Delta L}$$

Shear modulus of the cube,

$$\eta = \frac{8 \times 10^8 \times 4}{16 \times 0.01} = 2 \times 10^{10} \text{ dyne/cm}^2$$

Example 12.24 A square lead slab of side 50 cm and thickness 10.0 cm is subjected to a shearing force (on its narrow face) of magnitude $9.0 \times 10^4 \text{ N}$. The lower edge is riveted to the floor as shown in figure. How much is the upper edge is displaced, if the shear modulus of lead is $5.6 \times 10^9 \text{ Pa}$?



Sol. Here, $L = 50 \text{ cm} = 50 \times 10^{-2} \text{ m}$,

$$\eta = 5.6 \times 10^9 \text{ Pa}, F = 9.0 \times 10^4 \text{ N}$$

Area of the face on which force is applied,

$$A = 50 \times 10 \text{ cm}^2 = 500 \text{ cm}^2 = 0.05 \text{ m}^2$$

If ΔL is the displacement of the upper edge of the slab due to tangential force F applied, then

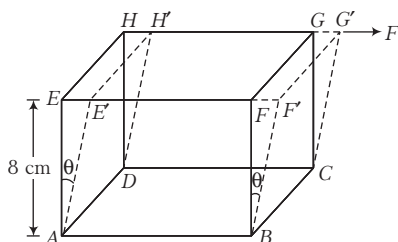
$$\eta = \frac{FL}{A\Delta L}$$

$$\Rightarrow \Delta L = \frac{FL}{\eta A} = \frac{9 \times 10^4 \times 50 \times 10^{-2}}{5.6 \times 10^9 \times 0.05}$$

$$\Rightarrow \Delta L = 1.6 \times 10^{-4} \text{ m}$$

Example 12.25 Consider an Indian rubber cube having modulus of rigidity of $2 \times 10^7 \text{ dyne cm}^{-2}$ and of side 8 cm. If one side of the rubber is fixed, while a tangential force equal to the weight of 300 kg is applied to the opposite face, then find out the shearing strain produced and distance through which the strained side moves.

Sol. Given, modulus of rigidity, $\eta = 2 \times 10^7$ dyne cm^{-2}



Side of the cube, $l = 8$ cm

Area, $A = l^2 = 64 \text{ cm}^2$

Force or load, $F = 300 \text{ kgf} = 300 \times 1000 \times 981$ dyne

Modulus of rigidity, $\eta = \frac{F}{A\theta} \Rightarrow \theta = \frac{F}{A\eta}$

Shearing strain, $\theta = \frac{300 \times 1000 \times 981}{64 \times 2 \times 10^7} = 0.23 \text{ rad}$

As, $\eta = \frac{F}{A} \frac{l}{\Delta l} \Rightarrow \frac{\Delta l}{l} = \theta$

Distance through which the strained side moves,

$$\Delta l = l\theta = 8 \times 0.23$$

$$\Rightarrow \Delta l = 1.84 \text{ cm}$$

Factors affecting elasticity

- (i) **Effect of temperature** Almost for all materials, the modulus of elasticity decreases with the rise in temperature, but the elasticity of invar remains unchanged with the change in temperature.

CHECK POINT 12.2

- Amongst the following, which option correctly represents the increasing order of coefficient of elasticity for Copper, Steel, Glass and Rubber.
(a) Steel, Rubber, Copper, Glass
(b) Rubber, Copper, Glass, Steel
(c) Rubber, Glass, Steel, Copper
(d) Rubber, Glass, Copper, Steel
- The Young's modulus of the material of a wire is equal to the
(a) stress required to increase its length four times
(b) stress required to produce unit strain
(c) strain produced in it
(d) half the strain produced in it
- A rubber cord 10 m is suspended vertically. How much does it stretch under its own weight? (Density of rubber is 1500 kg m^{-3} , $Y = 5 \times 10^8 \text{ Nm}^{-2}$ and $g = 10 \text{ ms}^{-2}$)
(a) $15 \times 10^{-4} \text{ m}$
(b) $7.5 \times 10^{-4} \text{ m}$
(c) $12 \times 10^{-4} \text{ m}$
(d) $25 \times 10^{-4} \text{ m}$
- Two similar wires under the same load yield elongation of 0.1 mm and 0.05 mm, respectively. If the area of

- (ii) **Effect of impurities** The addition of impurities affects the elastic properties depending on whether impurities are themselves more or less elastic.

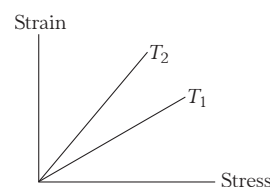
When carbon is added to iron and potassium to gold, their elasticities are strengthened.

- (iii) **Effect of annealing** By annealing (*i.e.* heating and then cooling gradually) large crystal grains are formed and hence, the elasticity of the material decreases.

- (iv) **Effect of hammering and rolling** By hammering and rolling, crystal grains break up into smaller units and hence, the elasticity of the material increases.

Example 12.26 The stress-strain graph for a metallic wire is shown at two different temperatures T_1 and T_2 , then which temperature is high T_1 or T_2 ?

Sol. If the slope of stress-strain curve with strain at X-axis gives the value of Young's modulus.



In the above graph, strain is taken along Y-axis. Therefore, the slope of graph at temperature T_1 is less than the slope of graph at temperature T_2 .

Now, as we know with increase in temperature, the value of modulus of elasticity decreases. Therefore, temperature T_1 is less than temperature T_2 .

cross-section of the first wire is 4 mm^2 , then the area of cross-section of the second wire is

(a) 6 mm^2 (b) 8 mm^2 (c) 10 mm^2 (d) 12 mm^2

- A 5 m long aluminium wire ($Y = 7 \times 10^{10} \text{ Nm}^{-2}$) of diameter 3 mm supports a 40 kg mass. In order to have the same elongation in a copper wire ($Y = 1.2 \times 10^{10} \text{ Nm}^{-2}$) of the same length under the same weight, the diameter should now be (in mm)
(a) 1.75 (b) 1.5 (c) 2.29 (d) 5.0
- Which one of the following statement(s) is/are incorrect?
(a) Young's modulus for a perfectly rigid body is zero.
(b) Bulk modulus is relevant for solids, liquids and gases.
(c) Rubber is less elastic than steel.
(d) The Young's modulus and shear modulus are relevant for solids.
- Choose the incorrect statement.
(a) The bulk modulus for solids is much larger than for liquids.
(b) Gases are least compressible.
(c) The incompressibility of the solids is due to the tight coupling.
(d) The reciprocal of the bulk modulus is called compressibility.

POISSON'S RATIO

When a wire is loaded, its length increases but its diameter decreases. The strain produced in the direction of applied force is called **longitudinal strain** and strain produced in the perpendicular direction is called **lateral strain**. Within the elastic limit, the ratio of lateral strain to the longitudinal strain is called **Poisson's ratio**. It is represented by (σ).

Let the length of the loaded wire increases from L to $L + \Delta L$ and its diameter decreases from D to $D - \Delta D$.

$$\text{Longitudinal strain} = \frac{\Delta L}{L}$$

$$\text{Lateral strain} = \frac{-\Delta D}{D}$$

$$\text{Poisson's ratio is } \sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{-\Delta D / D}{\Delta L / L}$$

$$\text{Poisson's ratio, } \sigma = \frac{-L}{D} \cdot \frac{\Delta D}{\Delta L}$$

The negative sign indicates that longitudinal and lateral strain are in opposite sense. It has no units and dimensions.

$$\begin{cases} -1 < \sigma < 0.5, & \text{for theoretical purpose} \\ 0 < \sigma < 0.5, & \text{for practical purpose} \end{cases}$$

Relation between Y , B , η and σ

There are four relations between Y , B , η and σ , which can be given as follows

$$\begin{aligned} \text{(i)} \quad B &= \frac{Y}{3(1-2\sigma)} & \text{(ii)} \quad \eta &= \frac{Y}{(1+\sigma)2} \\ \text{(iii)} \quad \sigma &= \frac{3B-2\eta}{2\eta+6B} & \text{(iv)} \quad \frac{9}{Y} &= \frac{1}{B} + \frac{3}{\eta} \end{aligned}$$

Example 12.27 Determine the Poisson's ratio of the material of a wire whose volume remains constant under an external normal stress.

Sol. Volume of a wire, $V = \frac{\pi D^2}{4} l$

As volume remains constant, the differentiation of above equation gives

$$\begin{aligned} 0 &= \frac{\pi l}{4} \cdot 2D \cdot dD + \frac{\pi D^2}{4} \cdot dl \\ -2ldD &= Ddl \\ \Rightarrow \quad \frac{dD}{D} &= -\frac{1}{2} \frac{dl}{l} \end{aligned}$$

$\therefore$ By definition, Poisson's ratio,

$$\sigma = \frac{-dD / D}{dl / l} = -\frac{\left(-\frac{1}{2} \frac{dl}{l}\right)}{\frac{dl}{l}} = \frac{1}{2} = 0.5$$

Example 12.28 A material is having Poisson's ratio 0.2. A load is applied on it due to which it suffers the longitudinal strain 3×10^{-3} , then find out the percentage change in its volume.

Sol. Given, Poisson's ratio, $\sigma = 0.2$

$$\text{Longitudinal strain, } \frac{\Delta L}{L} = 3 \times 10^{-3}$$

$$\therefore \text{Poisson's ratio, } \sigma = \frac{-\Delta D / D}{\Delta L / L}$$

$$\Rightarrow \sigma = \frac{-\Delta R / R}{\Delta L / L}$$

$$\Rightarrow \frac{\Delta R}{R} = -\sigma \times \frac{\Delta L}{L} = -0.6 \times 10^{-3}$$

$$\therefore \text{Volume, } V = \pi R^2 L$$

Therefore, percentage change in volume,

$$\begin{aligned} \left(\frac{\Delta V}{V} \times 100\right) &= \left(\frac{2\Delta R}{R} + \frac{\Delta L}{L}\right) \times 100 \\ &= [2 \times (-0.6 \times 10^{-3}) + 3 \times 10^{-3}] \times 100 = 0.18\% \end{aligned}$$

Stress-strain curve

When a wire is stretched by a load as shown in Fig. 12.9 (a), it is seen that for small value of load, the extension produced in the wire is proportional to the load.

$$\therefore \text{Stress} \propto \text{Strain}$$

When a graph between stress and strain is plotted, then

(i) The stress is found to be proportional to strain (% elongation) up to point A. Thus, Hooke's law is fully obeyed in this region, the point A is known as point of **proportional limit**.

(ii) When stress is increased beyond A, then for small stress, there is a large strain in the wire upto point B.

(iii) When the load is gradually removed between points O to B, the wire return to its original length.

The wire regains its original dimension only when load applied is less than or equal to certain limit. This limit is called **elastic limit**. The point B on stress-strain curve is known as **elastic limit** or **yield point**. The material of the wire in the region OB shows the elastic behaviour, hence known as **elastic region**.

(iv) If the stress or load increases beyond point B, the strain further increases. This increase in strain represented by BC part of the curve. Now, if the load

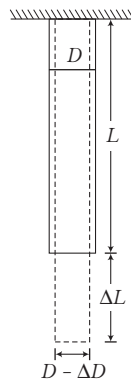


Fig. 12.8

is removed, the wire does not regain its original length. But the increase in the length of the wire is permanent.

In other words, there is permanent strain equal to OO' in the wire even when the stress is zero. This permanent strain in the wire is known as permanent set.

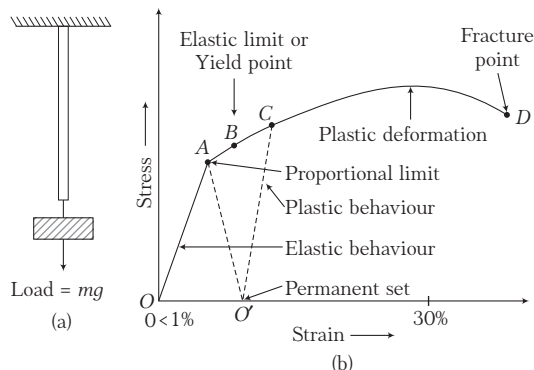


Fig. 12.9 A typical stress-strain curve

- (v) Now, as the stress beyond C is increased, there is huge strain in the wire. This huge increase in the strain for small stress is represented by CD part of the curve. The wire breaks at point D which is also known as **fracture point**.

The material of the wire from point C to point D shows the **plastic behaviour** or **plastic deformation**. The stress needed to cause the actual fracture of the material is known as **breaking stress** or **ultimate tensile strength**.

Note

- Elastic limit and limit of proportionality are very close to each other, so that Hooke's law is nearly applicable upto elastic limit.
- In the yield region, strains is 15 to 20 times than those strains that takes place upto the proportional limit that occur during yielding.

Breaking Stress of Some Materials

Material	Breaking stress (in Nm^{-2})
Aluminium	2.2×10^8
Iron	3.0×10^8
Brass	4.7×10^8
Phosphor bronze	5.6×10^8
Steel	5 to 20×10^8
Glass	10×10^8

Different nature of the stress-strain curve

The stress-strain curve differ considerably for different materials. The steeper curve indicates a stiffer material, line parallel to the strain axis indicates a ductile material and in the absence of yield point or plastic behaviour refer to a brittle material.

Ductile and brittle materials

On the basis of elastic and plastic properties, materials can be classified in two ways; ductile materials and brittle materials.

The materials which have large plastic range of extension are called **ductile materials**. As shown in the stress-strain curve, the fracture point is widely separated from the elastic limit. Such materials undergo an irreversible increase in length before snapping. So, they can be drawn into thin wires, e.g. copper, silver, iron, aluminium, etc.

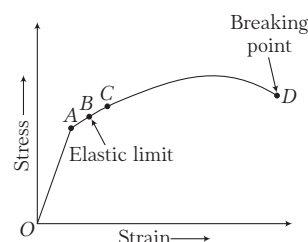


Fig. 12.10 Stress-strain curve for a ductile material

The materials which have very small range of plastic extension are called **brittle materials**. Such materials break as soon as the stress is increased beyond the elastic limit.

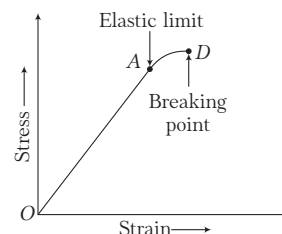


Fig. 12.11 Stress-strain curve for a brittle material

Their breaking point lies just close to their elastic limit as shown in figure, e.g. cast iron, glass, ceramics, etc.

Malleability

When a solid is compressed, a stage is reached beyond which it cannot regain its original shape after the deforming force is removed.

This is the elastic limit point (A') for compression. The solid then behaves like a plastic body.

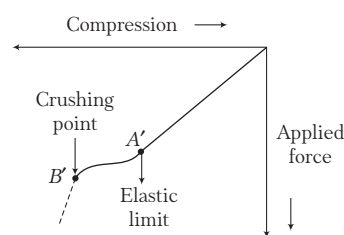


Fig. 12.12 Load compression curve for a metal

The yield point (B') obtained under compression is called **crushing point**.

After this stage, metals are said to be **malleable**, *i.e.* they can be hammered or rolled into thin sheets, *e.g.* gold, silver, lead, etc.

Elastomers

The materials which can be elastically stretched to large values of strain are called elastomers. *e.g.* Rubber can be stretched to several times its original length but still it can regain its original length when the applied force is removed. There is no well defined plastic region, rubber just breaks when pulled beyond a certain limit.

Elastic region in such cases is very large, but the material does not obey Hooke's law.

In our body, the elastic tissue of aorta (the large blood vessel carrying blood from the heart) is an elastomer.

For which the stress-strain curve is shown in figure.

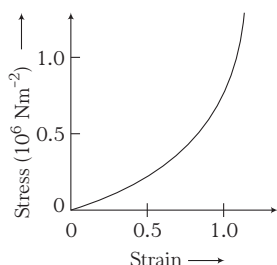


Fig. 12.13 Stress-strain curve

Elastic hysteresis

As a natural consequence of the elasticity after effect, the strain in the body tends to lag behind the stress applied to the body, so that during a rapidly changing stress, the strain is greater for the same value of stress. This lag of strain behind the stress is called elastic hysteresis.

Due to elastic hysteresis, the original curve (OAB) is not retraced when the deforming force is removed, although the body finally acquires natural length. The figure clearly indicates that the work done by the material in returning to its original shape is less than the work done by the deforming force. Hence, some amount of energy is absorbed by the material in the cycle which appears as heat.

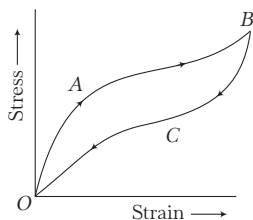
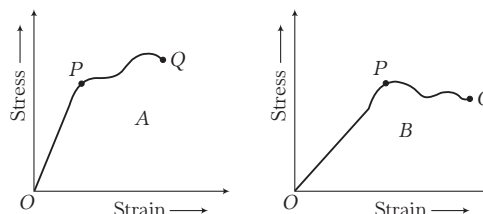


Fig. 12.14 Vulcanised rubber

The magnitude of the energy absorbed is proportional to the area of the loop. The material having low elastic hysteresis have low elastic relaxation time.

Example 12.29 The stress-strain graphs for two materials A and B are shown in the figure. The graphs are drawn to the same scale.



- Which material has a greater Young's modulus?
- Which material is more ductile?
- Which material is more brittle?
- Which of the two is the stronger material?

- Sol.**
- Material A has a greater Young's modulus because the slope of the linear portion of the stress-strain curve is greater for the material A.
 - Material A is more ductile because it has a greater plastic region (from elastic limit P to breaking point Q) than material B.
 - Material B is more brittle because it has a smaller plastic region.
 - Material A is stronger because it can withstand a greater stress before it breaks.

Work done or potential energy stored in a stretched wire

When a wire is stretched, work is done against the interatomic forces. This work is stored in the wire in the form of elastic potential energy.

Thus, stored elastic energy is

$$U = \frac{1}{2} \text{Stress} \times \text{Strain} \times \text{Volume}$$

or
$$U = \frac{1}{2} Y (\text{Strain})^2 \times \text{Volume}$$

$$U = \frac{1}{2} \text{Load} \times \text{Elongation}$$

Example 12.30 Calculate the work done in stretching a steel wire of Young's modulus of $2 \times 10^{11} \text{ Nm}^{-2}$, length of 200 cm and area of cross-section 0.06 cm^2 slowly by applying a load of 40 kg without the elastic limit being reached.

Sol. Here, force, $F = mg = 40 \times 9.8 \text{ N}$

Length of wire, $l = 200 \text{ cm} = 2 \text{ m}$

Area of cross-section, $A = 0.06 \text{ cm}^2 = 0.06 \times 10^{-4} \text{ m}^2$

Young's modulus, $Y = 2 \times 10^{11} \text{ Nm}^{-2}$

$$\text{Work done} = \frac{1}{2} \times \text{Stretching force} \times \text{Extension} = \frac{1}{2} F \Delta l$$

$$= \frac{1}{2} F \times \frac{F l}{A Y}$$

$$\Rightarrow W = \frac{F^2 l}{2 A Y} = \frac{(40 \times 9.8)^2 \times 2}{2 \times 0.06 \times 10^{-4} \times 2 \times 10^{11}} = 0.128 \text{ J}$$

Example 12.31 A steel wire 4 m in length is stretched through 2 mm. The cross-sectional area of the wire is 2.0 mm^2 . If Young's modulus of the steel is $2 \times 10^{11} \text{ Nm}^{-2}$. Find

(i) the energy density of wire,

(ii) the elastic potential energy stored in the wire.

Sol. Here, length of wire, $l = 4.0 \text{ m}$, $\Delta l = 2 \times 10^{-3} \text{ m}$

$$\text{Area, } A = 2.0 \times 10^{-6} \text{ m}^2$$

$$\text{Young's modulus, } Y = 2.0 \times 10^{11} \text{ Nm}^{-2}$$

(i) The energy density of stretched wire,

$$u = \frac{1}{2} \times \text{Stress} \times \text{Strain} = \frac{1}{2} \times Y \times (\text{Strain})^2$$

$$= \frac{1}{2} \times 2.0 \times 10^{11} \times \left[\frac{(2 \times 10^{-3})^2}{(4)^2} \right]$$

$$= 0.25 \times 10^5 \text{ Jm}^{-3}$$

$$= 2.5 \times 10^4 \text{ Jm}^{-3}$$

(ii) Elastic potential energy = Energy density $\times$ Volume

$$= 2.5 \times 10^4 \times (2.0 \times 10^{-6}) \times 4.0 \text{ J}$$

$$= 20 \times 10^{-2} = 0.20 \text{ J}$$

Example 12.32 A 45 kg boy whose leg bones are 5 cm^2 in area and 50 cm long falls through a height of 2 m without breaking his leg bones. If the bones can stand a stress of $0.9 \times 10^8 \text{ Nm}^{-2}$, then what will be the Young's modulus for the material of the bone?

Sol. Here, mass, $m = 45 \text{ kg}$, height of leg bones, $h = 2 \text{ m}$

$$\text{Length, } L = 0.50 \text{ m, area of bones, } A = 5 \times 10^{-4} \text{ m}^2$$

Loss in gravitational energy

= Gain in elastic energy in both leg bones

$$\text{So, } mgh = 2 \times \left(\frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume} \right)$$

$$\text{Here, volume} = AL = 5 \times 10^{-4} \times 0.50 = 2.5 \times 10^{-4} \text{ m}^3$$

$$\therefore 45 \times 10 \times 2 = 2 \times \left[\frac{1}{2} \times 0.9 \times 10^8 \times \text{strain} \times 2.5 \times 10^{-4} \right]$$

$$\text{or } \text{Strain} = \frac{45 \times 10 \times 2}{0.9 \times 2.5 \times 10^4} = 0.04$$

$\therefore$ Young's modulus of bones material,

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{0.9 \times 10^8}{0.04}$$

$$= 2.25 \times 10^9 \text{ Nm}^{-2}$$

Example 12.33 A rubber cord has a cross-sectional area 1 mm^2 and total unstretched length 10.0 cm . It is stretched to 12.5 cm and then released to project a missile of mass 5.0 g . (Take, Young's modulus Y for rubber as $5.0 \times 10^8 \text{ Nm}^{-2}$). Calculate the velocity of projection.

Sol. Equivalent force constant of rubber cord,

$$k = \frac{YA}{l} = \frac{(5.0 \times 10^8)(1.0 \times 10^{-6})}{(0.1)} = 5.0 \times 10^3 \text{ Nm}^{-1}$$

Now, from conservation of mechanical energy, elastic potential energy of cord = kinetic energy of missile

$$\therefore \frac{1}{2} k (\Delta l)^2 = \frac{1}{2} mv^2$$

$$\therefore v = \left(\sqrt{\frac{k}{m}} \right) \Delta l = \left(\sqrt{\frac{5.0 \times 10^3}{5.0 \times 10^{-3}}} \right) (12.5 - 10.0) \times 10^{-2}$$

$$= 25 \text{ ms}^{-1}$$

Note Following assumptions have been made in this example

(i) k has been assumed constant, even though it depends on the length (l).

(ii) The whole of the elastic potential energy is converting into kinetic energy of missile.

Example 12.34

(i) A wire 4 m long and 0.3 mm in diameter is stretched by a force of 100 N . If extension in the wire is 0.3 mm , calculate the potential energy stored in the wire.

(ii) Find the work done in stretching a wire of cross-section 1 mm^2 and length 2 m through 0.1 mm . Young's modulus for the material of wire is $2.0 \times 10^{11} \text{ Nm}^{-2}$.

Sol.

(i) Energy stored, $U = \frac{1}{2} (\text{Stress})(\text{Strain})(\text{Volume})$

$$\text{or } U = \frac{1}{2} \left(\frac{F}{A} \right) \left(\frac{\Delta l}{l} \right) (Al) = \frac{1}{2} F \cdot \Delta l$$

$$= \frac{1}{2} (100)(0.3 \times 10^{-3}) = 0.015 \text{ J}$$

(ii) Work done = Potential energy stored

$$= \frac{1}{2} k (\Delta l)^2 = \frac{1}{2} \left(\frac{YA}{l} \right) (\Delta l)^2 \quad \left(\because k = \frac{YA}{l} \right)$$

Substituting the values, we have

$$W = \frac{1}{2} \frac{(2.0 \times 10^{11})(10^{-6})}{(2)} (0.1 \times 10^{-3})^2$$

$$= 5.0 \times 10^{-4} \text{ J}$$

Thermal stresses and strains

Whenever there is some increase or decrease in the temperature of the body, it causes the body to expand or contract. If the body is allowed to expand or contract freely, with the rise or fall in the temperature, no stresses are induced in the body. But, if the deformation of the body is prevented, some stresses are induced in the body.

Such stresses are called thermal stresses or **temperature stresses**.

The corresponding strains are called **thermal strains** or **temperature strains**.

Consider a rod AB fixed at two supports as shown in figure.

Let l = length of the rod,

A = area of cross-section of the rod,

Y = Young's modulus of elasticity of the rod

and α = thermal coefficient of linear expansion of the rod.

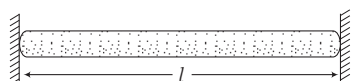


Fig. 12.15 Rod fixed between two supports

Let the temperature of the rod be increased by an amount t . The length of the rod would have increased by an amount Δl , if it were not fixed at two supports. Here,

$$\Delta l = l\alpha t$$

But since, the rod is fixed at the supports, so a compressive strain will be produced in the rod due to which a force is applied on the supports, given

$$F = YA\alpha\Delta t$$

Example 12.35 A steel wire of length 20 cm and uniform cross-section 1 mm^2 is tied rigidly at both the ends. If temperature of the wire is altered from 40°C to 20°C , calculate the change in tension. (Take, coefficient of linear expansion of steel is $1.1 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and Young's modulus for steel is $2.0 \times 10^{11} \text{ Nm}^{-2}$)

Sol. The change in length l of the wire when wire is cooled by temperature Δt is given by

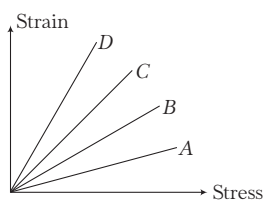
$$\Delta l = l\alpha\Delta t \text{ or } \Delta l/l = \alpha\Delta t \quad \dots(i)$$

where, α is the coefficient of linear expansion. Change in the tension of the wire when it is cooled is given by

$$\begin{aligned} F &= YA\Delta l/l = YA\alpha\Delta t \quad [\text{From Eq. (i)}] \\ &= (2.0 \times 10^{11}) \times (10^{-6}) \times (1.1 \times 10^{-5}) \times (40^\circ - 20^\circ) \\ &= 44 \text{ N} \end{aligned}$$

CHECK POINT 12.3

- There is no change in the volume of a wire due to change in its length on stretching. The Poisson's ratio of the material of the wire is
(a) +0.50 (b) -0.50 (c) 0.25 (d) -0.25
- For a given material, the Young's modulus is 2.4 times that of rigidity modulus. Its Poisson's ratio is
(a) 2.4 (b) 1.2 (c) 0.4 (d) 0.2
- Stress-strain curve for four metals are shown in figure. The maximum Young's modulus of elasticity for metal, is



- (a) A (b) B (c) C (d) D
- A metallic rod of length l and cross-sectional area A is made of a material of Young's modulus Y . If the rod is elongated by an amount y , then the work done is proportional to
(a) y (b) $1/y$ (c) y^2 (d) $1/y^2$
- The work done per unit volume to stretch the length by 1% of a wire with cross-sectional area of 1 mm^2 will be (Take, $Y = 9 \times 10^{11} \text{ Nm}^{-2}$)
(a) $9 \times 10^{11} \text{ J}$ (b) $4.5 \times 10^7 \text{ J}$
(c) $9 \times 10^7 \text{ J}$ (d) $4.5 \times 10^{11} \text{ J}$
- When a load of 10 kg is hung from the wire, then extension of 2 m is produced. Then, work done by restoring force is
(a) 200 J (b) 100 J (c) 50 J (d) 25 J

- If a spring extends by x on loading, then the energy stored by the spring is (if T is tension in the spring and k is spring constant)
(a) $\frac{T^2}{2x}$ (b) $\frac{T^2}{2k}$ (c) $\frac{2x}{T^2}$ (d) $\frac{2T^2}{k}$
- A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower end. The weight stretches the wire by 1 mm. Then, the energy stored in the wire is
(a) 0.1 J (b) 0.2 J (c) 10 J (d) 20 J
- Wires A and B are made from the same material. A has twice the diameter and three times the length of B. If the elastic limits are not reached, when each is stretched by the same tension, the ratio of energy stored in A to that in B is
(a) 2 : 3 (b) 3 : 4 (c) 3 : 2 (d) 6 : 1
- A wire of length 50 cm and cross-sectional area of 1 sq. mm is extended by 1 mm. The required work will be (Take, $Y = 2 \times 10^{10} \text{ Nm}^{-2}$)
(a) $6 \times 10^{-2} \text{ J}$ (b) $4 \times 10^{-2} \text{ J}$
(c) $2 \times 10^{-2} \text{ J}$ (d) $1 \times 10^{-2} \text{ J}$
- When a force is applied on a wire of uniform cross-sectional area $3 \times 10^{-6} \text{ m}^2$ and length 4 m, the increase in length is 1 mm. Energy stored in it will be (Take, $Y = 2 \times 10^{11} \text{ Nm}^{-2}$)
(a) 6250 J (b) 0.177 J
(c) 0.075 J (d) 0.150 J
- When a 4 kg mass is hung vertically on a light spring that obeys Hooke's law, the spring stretches by 2 cms. The work required to be done by an external agent in stretching this spring by 5 cm will be (Take, $g = 9.8 \text{ m s}^{-2}$)
(a) 4.9 J (b) 2.4 J
(c) 0.495 J (d) 0.24 J

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1** The maximum load, a wire can withstand without breaking, when its length is reduced to half of its original length, will [NCERT Exemplar]

(a) be double (b) be half
(c) be four times (d) remain same

- 2** For steel $Y = 2 \times 10^{11} \text{ Nm}^{-2}$. The force required to double the length of a steel wire of area 1 cm^2 is

(a) $2 \times 10^7 \text{ N}$ (b) $2 \times 10^6 \text{ N}$
(c) $2 \times 10^8 \text{ N}$ (d) $2 \times 10^5 \text{ N}$

- 3** Longitudinal stress of 1 kg mm^{-2} is applied on a wire. The percentage increase in length is (Take, $Y = 10^{11} \text{ Nm}^{-2}$)

(a) 0.002 (b) 0.001
(c) 0.003 (d) 0.01

- 4.** The Young's modulus of a wire is Y , if the energy per unit volume is E , then the strain will be

(a) $\sqrt{\frac{2E}{Y}}$ (b) $E\sqrt{2Y}$
(c) EY (d) $\frac{E}{Y}$

- 5** When a force is applied at one end of an elastic wire, it produces a strain E in the wire. If Y is the Young's modulus of the material of the wire, the amount of energy stored per unit volume of the wire is given by

(a) $Y \times E$ (b) $\frac{1}{2} (Y \times E)$
(c) $Y \times E^2$ (d) $\frac{1}{2} (Y \times E^2)$

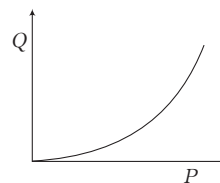
- 6** Young's modulus of the material of a wire of length L and radius r is $Y \text{ Nm}^{-2}$. If the length is reduced to $\frac{L}{2}$ and radius to $\frac{r}{2}$, the Young's modulus will be

(a) Y (b) $2Y$ (c) $\frac{Y}{4}$ (d) $\frac{Y}{2}$

- 7.** The Young's modulus of a wire is numerically equal to the stress which will

(a) not change the length of the wire
(b) double the length of the wire
(c) increase the length by 50%
(d) change the radius of the wire to half

- 8** The graph shows the behaviour of a length of wire in the region for which the substance obeys Hooke's law. P and Q represent



(a) P = applied force, Q = extension
(b) P = extension, Q = applied force
(c) P = extension, Q = stored elastic energy
(d) P = stored elastic energy, Q = extension

- 9** A metal block is experiencing an atmospheric pressure of 10^5 Nm^{-2} . When the same block is placed in a vacuum chamber, the fractional change in its volume, is (the bulk modulus of metal is $1.25 \times 10^{11} \text{ Nm}^{-2}$)

(a) 4×10^{-7} (b) 2×10^{-7}
(c) 8×10^{-7} (d) 1×10^{-7}

- 10** The increase in length on stretching a wire is 0.05%. If its Poisson's ratio is 0.4, then its diameter

(a) reduce by 0.02% (b) reduce by 0.1%
(c) increase by 0.02% (d) decrease by 0.4%

- 11** An elastic material with Young's modulus Y is subjected to a tensile stress S , elastic energy stored per unit volume of the material is

(a) $\frac{YS}{2}$ (b) $\frac{S^2}{Y}$ (c) $\frac{S^2}{2Y}$ (d) $\frac{S}{2Y}$

- 12** When a metal wire elongates by hanging a load Mg on it, the gravitational potential energy of mass M decreases by Mgl . This energy appears

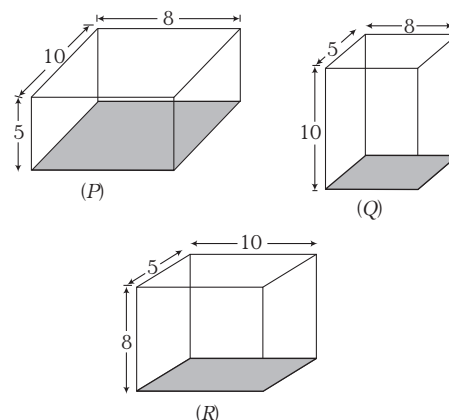
(a) as elastic potential energy completely
(b) as thermal energy completely
(c) half as elastic potential energy and half as thermal energy
(d) as kinetic energy of the load completely

- 13** A 5 m long wire is fixed to the ceiling. A weight of 10 kg is hang at the lower end and is 1m above the floor. The wire was elongated by 1mm. The energy stored in the wire due to stretching is

(a) 0.01 J (b) 0.05 J
(c) 0.02 J (d) 0.04 J

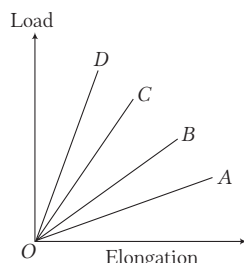
- 14** If the Young's modulus of the material is 3 times its modulus of rigidity, then its volume elasticity will be
 (a) zero (b) infinity
 (c) $2 \times 10^{10} \text{ N/m}^2$ (d) $3 \times 10^{10} \text{ N/m}^2$
- 15** A cable that can support a load w is cut into two equal parts. The maximum load that can be supported by either part is
 (a) $\frac{w}{4}$ (b) $\frac{w}{2}$ (c) w (d) $2w$
- 16** The longitudinal extension of any elastic material is very small. In order to have an appreciable change, the material must be in the form of
 (a) long thick wire (b) short thick wire
 (c) long thin wire (d) short thin wire
- 17** Two wires of same diameter of the same material having the length l and $2l$. If the force F is applied on each, the ratio of the work done in the two wires will be
 (a) 1 : 2 (b) 1 : 4 (c) 2 : 1 (d) 1 : 1
- 18** The temperature of a wire of length 1 m and area of cross-section 1 cm^2 is increased from 0°C to 100°C . If the rod is not allowed to increase in length, the force required will be ($\alpha = 10^{-5} / ^\circ\text{C}$ and $Y = 10^{11} \text{ Nm}^{-2}$)
 (a) 10^3 N (b) 10^4 N (c) 10^5 N (d) 10^9 N
- 19** The temperature of a wire is doubled. The Young's modulus of elasticity [NCERT Exemplar]
 (a) will also double (b) will become four times
 (c) will remain same (d) will decrease
- 20** When a pressure of 100 atm is applied on a spherical ball, then its volume reduces to 0.01%. The bulk modulus of the material of the rubber (in dyne cm^{-2}) is
 (a) 10×10^{12} (b) 100×10^{12} (c) 1×10^{12} (d) 20×10^{12}
- 21** Two wires of copper having the length in the ratio 4 : 1 and their radii ratio as 1 : 4 are stretched by the same force. The ratio of longitudinal strain in the two will be
 (a) 1 : 16 (b) 16 : 1 (c) 1 : 64 (d) 64 : 1
- 22** The interatomic distance for a metal is $3 \times 10^{-10} \text{ m}$. If the interatomic force constant is $3.6 \times 10^{-9} \text{ N/\AA}$, then the Young's modulus (in Nm^{-2}) will be
 (a) 1.2×10^{11} (b) 4.2×10^{11} (c) 10.8×10^8 (d) 2.4×10^{10}
- 23** Young's modulus of rubber is 10^4 Nm^{-2} and area of cross-section is 2 cm^2 . If force of $2 \times 10^5 \text{ dyne}$ is applied along its length, then its initial length l becomes
 (a) $3l$ (b) $4l$
 (c) $2l$ (d) None of these

- 24** A rectangular block of size $10 \text{ cm} \times 8 \text{ cm} \times 5 \text{ cm}$ is kept in three different positions P , Q and R in turn as shown in the figure. In each case, the shaded area is rigidly fixed and a definite force F is applied tangentially to the opposite face to deform the block. The displacement of the upper face will be

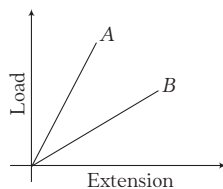


- (a) same in all the three cases
 (b) maximum in P position
 (c) maximum in Q position
 (d) maximum in R position
- 25** A force of 20 N is applied at one end of a wire of length 2 m and having area of cross-section 10^{-2} cm^2 . The other end of the wire is rigidly fixed. If the coefficient of linear expansion of the wire $\alpha = 8 \times 10^{-6} / ^\circ\text{C}$ and Young's modulus $Y = 2.2 \times 10^{11} \text{ Nm}^{-2}$ and its temperature is increased by 5°C , then the increase in the tension of the wire will be
 (a) 4.2 N (b) 4.4 N
 (c) 2.4 N (d) 8.8 N
- 26** A load suspended by a massless spring produces an extension of $x \text{ cm}$ in equilibrium. When it is cut into two unequal parts, the same load produces an extension of 7.5 cm when suspended by the larger part of length 60 cm. When it is suspended by the smaller part, the extension is 5.0 cm. Then,
 (a) $x = 12.5 \text{ cm}$
 (b) $x = 3.0 \text{ cm}$
 (c) the length of the original spring is 90 cm
 (d) the length of the original spring is 80 cm
- 27** A steel wire of cross-sectional area $3 \times 10^{-6} \text{ m}^2$ can withstand a maximum strain of 10^{-3} . Young's modulus of steel is $2 \times 10^{11} \text{ Nm}^{-2}$. The maximum mass this wire can hold is
 (a) 40 kg (b) 60 kg
 (c) 80 kg (d) 100 kg

- 28** The load *versus* elongation graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line

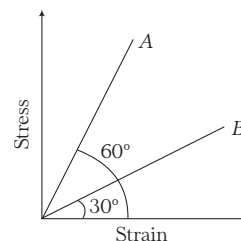


- (a) OD (b) OC (c) OB (d) OA
- 29** Two wires of same material and length but diameter in the ratio 1 : 2 are stretched by the same force. The potential energy per unit volume for the two wires when stretched will be in the ratio
- (a) 16 : 1 (b) 4 : 1 (c) 2 : 1 (d) 1 : 1
- 30** A stress of 10^6 Nm^{-2} is required for breaking a material. If the density of the material is $3 \times 10^3 \text{ kgm}^{-3}$, then what should be the length of the wire made of this material, so that it breaks under its own weight?
- (a) 10 m (b) 33.3 m
(c) 5 m (d) 66.6 m
- 31** A cable is replaced by another one of the same length and material but of twice the diameter. The maximum load that the new wire can support without exceeding the elastic limit, as compared to the load that the original wire could support, is
- (a) half (b) double
(c) four times (d) one-fourth
- 32** A uniform steel bar of cross-sectional area A and length L is suspended, so that it hangs vertically. The stress at the middle point of the bar is (ρ is the density of steel)
- (a) $\frac{L}{2A} \rho g$ (b) $\frac{L\rho g}{2}$ (c) $\frac{LA}{\rho g}$ (d) $L\rho g$
- 33** A thick rope of rubber of length 8 m and density $1.5 \times 10^3 \text{ kgm}^{-3}$ has Young's modulus $5 \times 10^6 \text{ Nm}^{-2}$. When hung from ceiling of a room, the increase in length due to its own weight is
- (a) $96 \times 10^{-3} \text{ m}$ (b) $19.2 \times 10^{-5} \text{ m}$
(c) 9.4 cm (d) 9.6 mm
- 34** In the given figure, if the dimensions of the wires are the same and materials are different, Young's modulus is more for

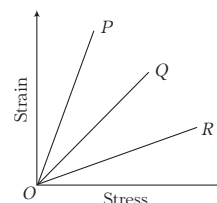


- (a) A (b) B
(c) Both (d) None of these

- 35** The stress *versus* strain graphs for wires of two materials A and B are as shown in the figure. If Y_A and Y_B are the Young's moduli of the materials, then

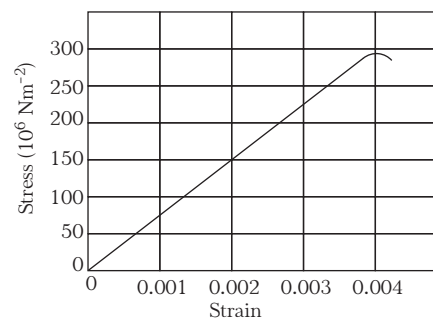


- (a) $Y_B = 2Y_A$ (b) $Y_A = Y_B$ (c) $Y_B = 3Y_A$ (d) $Y_A = 3Y_B$
- 36** An elevator cable is to have a maximum stress of $7 \times 10^7 \text{ Nm}^{-2}$ to allow for appropriate safety factors. Its maximum upward acceleration is 1.5 ms^{-2} . If the cable has to support the total weight of 2000 kg of a loaded elevator, the area of cross-section of the cable should be
- (a) 3.23 cm^2 (b) 2.38 cm^2 (c) 0.32 cm^2 (d) 8.23 cm^2
- 37** The strain-stress curves of three wires of different materials are shown in the figure. P, Q and R are elastic limits of the wires.



The figure shows that

- (a) elasticity of wire P is maximum
(b) elasticity of wire Q is maximum
(c) tensile strength of R is maximum
(d) None of the above
- 38** Figure shows the strain-stress curve for a given material. What are (i) Young's modulus and (ii) approximate yield strength for this material?

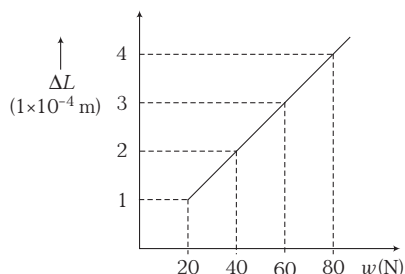


- (a) $5 \times 10^5 \text{ Nm}^{-2}$, $3 \times 10^6 \text{ Nm}^{-2}$
(b) $2 \times 10^{-6} \text{ Nm}^{-2}$, $4 \times 10^4 \text{ Nm}^{-2}$
(c) $7.5 \times 10^{10} \text{ Nm}^{-2}$, $3 \times 10^8 \text{ Nm}^{-2}$
(d) $7.5 \times 10^{10} \text{ Nm}^{-2}$, $3 \times 10^6 \text{ Nm}^{-2}$

- 39** Consider two cylindrical rods of identical dimensions, one of rubber and the other of steel. Both the rods are fixed rigidly at one end to the roof. A mass M is attached to each of the free ends at the centre of the rods. Then, [NCERT Exemplar]

- (a) both the rods will elongate but there shall be no perceptible change in shape
 (b) the steel rod will elongate and change shape but the rubber rod will only elongate
 (c) the steel rod will elongate without any perceptible change in shape, but the rubber rod will elongate and the shape of the bottom edge will change to an ellipse
 (d) the steel rod will elongate without any perceptible change in shape, but the rubber rod will elongate with the shape of the bottom edge tapered to a tip at the centre

- 40** The graph shown below gives the extension (ΔL) of a wire of length 1 m suspended from the top of a roof at one end with a load w connected to the other end. If the cross-sectional area of the wire is 10^{-6} m^2 , calculate the Young's modulus of the material of the wire.



- (a) $2 \times 10^{11} \text{ Nm}^{-2}$ (b) $2 \times 10^{-11} \text{ Nm}^{-2}$
 (c) $3 \times 10^{-12} \text{ Nm}^{-2}$ (d) $2 \times 10^{-13} \text{ Nm}^{-2}$
- 41** A brass rod of length 2 m and cross-sectional area 2 cm^2 is attached end-to-end to a steel rod of length L and cross-sectional area 1 cm^2 . The compound rod is subjected to equal and opposite pulls of magnitude $5 \times 10^4 \text{ N}$ at its ends. If the elongations of the two rods are equal, the length of the steel rod (L) is ($Y_{\text{brass}} = 1 \times 10^{11} \text{ Nm}^{-2}$ and $Y_{\text{steel}} = 2 \times 10^{11} \text{ Nm}^{-2}$)
- (a) 1.5 m (b) 1.8 m
 (c) 1 m (d) 2 m
- 42** One end of a uniform wire of length L and weight w is attached rigidly to a point in roof and a weight w_1 is suspended from its lower end. If S is the area of cross-section of the wire, the stress in the wire at a height of $\frac{3L}{4}$ from its lower end is

- (a) $\frac{w_1}{S}$ (b) $\frac{w_1 + w}{S}$ (c) $\left(\frac{w_1 + \frac{3w}{4}}{S} \right)$ (d) $\frac{w_1 + w}{S}$

- 43** A rigid bar of mass M is supported symmetrically by three wires each of length L . Those at each end are of copper and the middle one is of iron. The ratio of their diameters, if each is to have the same tension, is equal to

- (a) $\frac{\sqrt{Y_{\text{copper}}}}{Y_{\text{iron}}}$ (b) $\sqrt{\frac{Y_{\text{iron}}}{Y_{\text{copper}}}}$ (c) $\frac{Y_{\text{iron}}^2}{Y_{\text{copper}}^2}$ (d) $\frac{Y_{\text{iron}}}{Y_{\text{copper}}}$

- 44** One end of a long metallic wire of length L is tied to the ceiling, the other end is tied to a massless spring of spring constant k . A mass m hangs freely from the free end of the spring. The area of cross-section and Young's modulus of the wire are A and Y , respectively. If the mass is slightly pulled down and released, it will oscillate with a time period T equal to

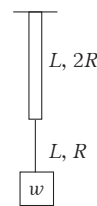
- (a) $2\pi\sqrt{\frac{m}{k}}$ (b) $2\pi\sqrt{\frac{m(YA + kL)}{YAk}}$
 (c) $2\pi\sqrt{\frac{mYA}{kL}}$ (d) $2\pi\sqrt{\frac{mL}{YA}}$

- 45** The length of an elastic string is a metre when the tension is 4 N and b metre when the tension is 5 N. The length, (in metre), when the tension is 9 N, is
 (a) $(a + b)$ (b) $(4b - 5a)$ (c) $(5b - 4a)$ (d) $(9b - 9a)$

- 46** A uniform elastic plank moves due to a constant force F placed over a smooth surface. The area of end face is S and Young's modulus of the material is E . What is average strain produced in the direction of the force?

- (a) $\frac{F}{SE}$ (b) $\frac{F}{2SE}$ (c) $\frac{F}{4SE}$ (d) Zero

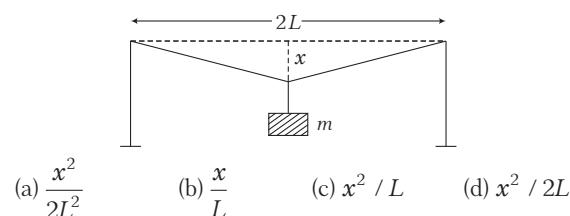
- 47** Two wires of the same material (Young's modulus = Y) and same length L but radii R and $2R$ respectively, are joined end-to-end and a weight w is suspended from the combination as shown in the figure.



The elastic potential energy in the system is

- (a) $\frac{3w^2L}{4\pi R^2Y}$ (b) $\frac{3w^2L}{8\pi R^2Y}$ (c) $\frac{5w^2L}{8\pi R^2Y}$ (d) $\frac{w^2L}{\pi R^2Y}$

- 48** A mild steel wire of length $2L$ and cross-sectional area A is stretched, well within elastic limit, horizontally between two pillars (figure). A mass m is suspended from the mid-point of the wire. Strain in the wire is [NCERT Exemplar]



- (a) $\frac{x^2}{2L^2}$ (b) $\frac{x}{L}$ (c) x^2 / L (d) $x^2 / 2L$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) *These questions consist of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.*

- (a) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (c) If Assertion is true but Reason is false.
- (d) If Assertion is false but Reason is true.

1 Assertion Upto the elastic limit, strain $\propto$ stress.

Reason Upto elastic limit, material returns to its original shape and size, when external force is removed.

2 Assertion If length of a wire is halved, its Young's modulus of elasticity will become two times.

Reason The ratio of longitudinal stress and longitudinal strain is called Young's modulus of elasticity.

3 Assertion If a wire is stretched, only half of the work done in stretching the wire remains stored as elastic potential energy.

Reason Potential energy stored in the wire is $(1/2) \times \text{stress} \times \text{strain}$.

4 Assertion Bulk modulus of an incompressible fluid is infinite.

Reason Density of incompressible fluid remains constant.

5 Assertion If radius of cylindrical wire is doubled, then this wire can bear four times stress.

Reason By doubling the radius, area of cross-section will become four times.

Statement based questions

- 1** The length of an iron wire is L and area of cross-section is A . The increase in length is l on applying the force F on its two ends. Which of the following statement(s) is/are correct?
- (a) Increase in length is inversely proportional to its length L .
 - (b) Increase in length is directly proportional to area of cross-section A .
 - (c) Increase in length is inversely proportional to A .
 - (d) Increase in length is directly proportional to Young's modulus.
- 2** When a steel wire fixed at one end is pulled by a constant force F at its other end, its length increases by l .

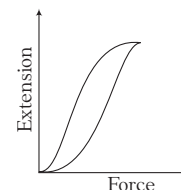
Which of the following statement(s) is/are incorrect?

- (a) Work done by the external force is Fl .
- (b) Some heat is produced in the wire in the process.
- (c) The elastic potential energy of the wire is $\frac{Fl}{2}$.
- (d) None of the above

3 A copper wire ($Y = 10^{11} \text{ Nm}^{-2}$) of length 8 m and a steel wire ($Y = 2 \times 10^{11} \text{ Nm}^{-2}$) of length 4 m, each of 0.5 cm^2 cross-section are fastened end-to-end and stretched with a tension of 500 N. Choose the correct statement.

- (a) Elongation in copper wire is 0.8 mm.
- (b) Elongation in steel is $\frac{1}{4}$ th the elongation in copper wire.
- (c) Total elongation is 1.0 mm.
- (d) All of the above

4 The diagram shows a force-extension graph for a rubber band. Consider the following statements.



- I. It will be easier to compress this rubber than expand it.
- II. Rubber does not return to its original length after it is stretched.
- III. The rubber band will get heated, if it is stretched and released.

Which of the following statement(s) is/are correct regarding the graph?

- (a) Only III
- (b) II and III
- (c) I and III
- (d) Only I

5 Which of the following statement(s) is/are correct?

- I. Incompressible liquids have finite value of bulk modulus of elasticity.
- II. Compressibility is inverse of bulk modulus of elasticity.

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) None of these

6 Which of the following statement(s) is/are correct?

- I. The materials having low value of Young's modulus of elasticity are more ductile.
- II. If Young's modulus is less, then they can be easily stretched as wires.

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) None of the above

Match the columns

- 1 Match the following columns and choose the correct option from the codes given below.

Column I	Column II
(A) Longitudinal stress	(p) volume changes
(B) Shear stress	(q) shape changes
(C) Volumetric stress	(r) volume does not changes
(D) Tensile stress	(s) shapes does not change

Codes

A	B	C	D
(a) p,s	q,r	p,s	p,s
(b) q,r	p,s	p,s	p,s
(c) p,s	p,s	q,r	p,s
(d) p,s	p,s	p,s	q,r

- 2 Match the following columns and choose the correct option from the codes given below.

Column I	Column II
(A) Stress $\times$ Strain	(p) J
(B) $\frac{YA}{l}$	(q) Nm^{-1}
(C) Yl^3	(r) Jm^{-3}
(D) $\frac{Fl}{AY}$	(s) m

Codes

A	B	C	D
(a) r	q	p	s
(b) q	r	p	s
(c) p	s	r	q
(d) q	r	s	p

- 3 With regard to dependence of quantities given in Columns I and II, match the following columns and choose the correct option from the codes given below.

Column I	Column II
(A) Young's modulus of a substance	(p) depends on temperature
(B) Bulk modulus of a substance	(q) depends on length
(C) Modulus of rigidity of a substance	(r) depends on area of cross-section
(D) Volume of a substance	(s) depends on the nature of material

Codes

A	B	C	D
(a) p,s	q,s	p,s	p,q
(b) p,s	p,s	p,s	p,q
(c) p,s	p,s	p,s	p,q,r
(d) p,q,r	p,s	p,s	p,r

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1 A wire of length L , area of cross-section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is [NEET 2020]
 (a) $\frac{Mg(L_1 - L)}{AL}$ (b) $\frac{MgL}{AL_1}$ (c) $\frac{MgL}{A(L_1 - L)}$ (d) $\frac{MgL_1}{AL}$
- 2 The stress-strain curves are drawn for two different materials X and Y . It is observed that, the ultimate strength point and the fracture point are close to each other for material X but are far apart for material Y . We can say that, materials X and Y are likely to be (respectively) [NEET (Odisha) 2019]
 (a) ductile and brittle (b) brittle and ductile
 (c) brittle and plastic (d) plastic and ductile
- 3 Two wires of same material having radius in ratio 2 : 1 and lengths in ratio 1 : 2. If same force is applied on them, then ratio of their change in length will be [JIPMER 2019]
 (a) 1 : 1 (b) 1 : 2 (c) 1 : 4 (d) 1 : 8
- 4 Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area $3A$. If the length of the first wire is increased by Δl on applying a force F , how much force is needed to stretch the second wire by the same amount? [NEET 2018]
 (a) $4F$ (b) $6F$
 (c) $9F$ (d) F
- 5 If compressibility of a material is 4×10^{-5} per atm, pressure is 100 atm and volume is 100 cm^3 , then find the value of ΔV . [JIPMER 2018]
 (a) 0.4 cm^3 (b) 0.8 cm^3
 (c) 0.6 cm^3 (d) 0.2 cm^3
- 6 The bulk modulus of a spherical object is B . If it is subjected to uniform pressure p , the fractional decrease in radius is [NEET 2017]
 (a) $\frac{p}{B}$ (b) $\frac{B}{3p}$ (c) $\frac{3p}{B}$ (d) $\frac{p}{3B}$

- 7** The length of a metal wire is l_1 when the tension in it is T_1 and is l_2 when the tension is T_2 . The natural length of the wire is [AIIMS 2015, WBJEE 2015, Manipal 2015]

(a) $\frac{l_1 + l_2}{2}$ (b) $\sqrt{l_1 l_2}$
 (c) $\frac{l_1 T_2 - l_2 T_1}{T_2 - T_1}$ (d) $\frac{l_1 T_2 + l_2 T_1}{T_1 + T_2}$

- 8** A load of 4 kg is suspended from a ceiling through a steel wire of length 2 m and radius 2 mm. It is found that, the length of the wire increases by 0.031 mm as equilibrium is achieved. What would be the Young's modulus of steel? (Take, $g = 3.1 \pi \text{ ms}^{-2}$)

[UK PMT 2015]

(a) $2.0 \times 10^{11} \text{ Nm}^{-2}$ (b) $2.82 \times 10^{11} \text{ Nm}^{-2}$
 (c) $0.20 \times 10^{11} \text{ Nm}^{-2}$ (d) $0.028 \times 10^{11} \text{ Nm}^{-2}$

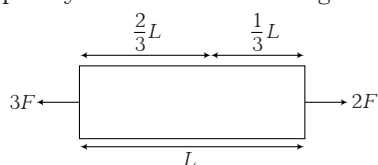
- 9** Two wires of same length and same material but of radii r and $2r$ are stretched by forces F and f respectively to produce equal elongation. The ratio F to f is [Kerala CEE 2015]

(a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 1 : 4
 (e) 4 : 1

- 10** The elastic potential energy of a stretched wire is given by [CG PMT 2015]

(a) $U = \frac{AL}{2Y} l^2$ (b) $U = \frac{AY}{2L} l^2$
 (c) $U = \frac{1}{2} \left(\frac{AL}{Y} \right) l$ (d) $U = \frac{1}{2} \cdot \frac{YL}{2A} \cdot l$

- 11** A uniform cylindrical rod of length L and cross-sectional area A having Young's modulus Y is acted upon by forces as shown in figure. [CG PMT 2015]



The elongation produced in the rod is

(a) $\frac{3FL}{8AY}$ (b) $\frac{3FL}{5AY}$ (c) $\frac{8FL}{3AY}$ (d) $\frac{5FL}{3AY}$

- 12** For most of the material, Young's modulus (Y) and rigidity modulus (G) are related as [UK PMT 2014]

(a) $G = 3Y$ (b) $G = \frac{Y}{3}$ (c) $G = \frac{3}{2}Y$ (d) $G = \frac{Y}{8}$

- 13** The following four wires are made of the same material. Which of these will have the largest extension when the same tension is applied? [NEET 2013]

- (a) Length = 50 cm, diameter = 0.5 mm
 (b) Length = 100 cm, diameter = 1 mm
 (c) Length = 200 cm, diameter = 2 mm
 (d) Length = 300 cm, diameter = 3 mm

- 14** The Young's modulus of a rope of 10 m length and having diameter of 2 cm is $20 \times 10^{11} \text{ dyne cm}^{-2}$. If the elongation produced in the rope is 1 cm, the force applied on the rope is [J&K CET 2013]

(a) $6.28 \times 10^5 \text{ N}$ (b) $6.28 \times 10^4 \text{ N}$
 (c) $6.28 \times 10^4 \text{ dyne}$ (d) $6.28 \times 10^5 \text{ dyne}$

- 15** A steel wire of length l and cross-section area A is stretched by 1 cm under a given load. When the same load is applied to another steel wire of double its length and half of its cross-section area, the amount of stretching is [Kerala CEE 2013]

(a) 0.5 cm (b) 2 cm (c) 4 cm (d) 1.5 cm
 (e) 2.5 cm

- 16** Theoretically, the value of Poisson's ratio σ lies between [MP PMT 2013]

(a) $0 < \sigma < 1$ (b) $-1 < \sigma < 0.5$
 (c) $0.2 < \sigma < 0.4$ (d) $-1 < \sigma < 1$

- 17** A steel ring of radius r and cross-sectional area A is fitted on to a wooden disc of radius R ($R > r$). If Young's modulus of the steel is Y , then the force with which the steel ring is expanded is [MP PMT 2013]

(a) $\frac{AYr}{(R-r)}$ (b) $\frac{Y(R-r)}{r}$ (c) $\frac{AY(R-r)}{R}$ (d) $\frac{AY(R-r)}{r}$

- 18** A rubber cord L metre long and having $A \text{ m}^2$ area of cross-section is suspended vertically. If the wire extends 1 m under its own weight, then change in length (l) is (Take, density of rubber = $D \text{ kgm}^{-3}$ and Young's modulus of rubber = $E \text{ Nm}^{-2}$). [UP CPMT 2013]

(a) $\frac{L^2 Dg}{2E}$ (b) $\frac{L^2 Dg}{E}$ (c) $\frac{L^2 Dg}{4E}$ (d) L

- 19** The density of a metal at normal pressure is ρ . Its density when it is subjected to an excess pressure p is ρ' . If B is the bulk modulus of the metal, then the ratio ρ'/ρ is [UP CPMT 2013]

(a) $1 + p/B$ (b) $1 + B/p$ (c) $\frac{1}{(1 - B/p)}$ (d) $\frac{1}{(1 - p/B)}$

- 20** In designing, a beam for its use to support a load. The depression at centre is proportional to (where, Y is Young's modulus). [AIIMS 2012]

(a) Y^2 (b) Y (c) $\frac{1}{Y}$ (d) $\frac{1}{Y^2}$

- 21** If to break a wire of 1 m length, minimum 40 kg-wt is required, then to break a wire of same material 6 m in length and double in radius, the breaking weight required will be [BCECE (Mains) 2012, Haryana PMT 2010]

(a) 40 kg-wt (b) 80 kg-wt
 (c) 160 kg-wt (d) 320 kg-wt

- 22** If Poisson's ratio σ for a material is $-\frac{1}{2}$, then the material is [BCECE (Mains) 2012]

(a) elastic fatigue (b) incompressible
(c) compressible (d) None of these

- 23** Two block of masses of 1 kg and 2 kg are connected by a metal wire going over a smooth pulley. The breaking stress of metal is $\frac{40}{3\pi} \times 10^6 \text{ Nm}^{-2}$. What should be the minimum radius of wire used, if it should not break? (Take, $g = 10 \text{ ms}^{-2}$). [Manipal 2012]
- (a) 0.5 mm (b) 1 mm (c) 1.5 mm (d) 2 mm

- 24** There is some change in length when a 33000 N tensile force is applied on a steel rod of area of

cross-section 10^{-3} m^{-2} . The change of temperature required to produce the same elongation, if the steel rod is heated is (Take, modulus of elasticity is $3 \times 10^{11} \text{ Nm}^{-2}$ and coefficient of linear expansion of steel is $11 \times 10^{-5}/^\circ\text{C}$.) [AFMC 2012]

(a) 20°C (b) 15°C (c) 10°C (d) 0°C

- 25** A stress of $6 \times 10^6 \text{ Nm}^{-2}$ required for breaking a material. The density ρ of the material is $3 \times 10^3 \text{ kg m}^{-3}$. If the wire is to break under its own weight, the length of the wire made of that material should be (Take, $g = 10 \text{ ms}^{-2}$) [Manipal 2011]
- (a) 20 m (b) 200 m
(c) 100 m (d) 2000 m

ANSWERS

• CHECK POINT 12.1

1. (d)	2. (d)	3. (d)	4. (a)	5. (b)	6. (d)	7. (c)
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• CHECK POINT 12.2

1. (d)	2. (b)	3. (a)	4. (b)	5. (c)	6. (a)	7. (b)
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• CHECK POINT 12.3

1. (a)	2. (d)	3. (a)	4. (c)	5. (b)	6. (b)	7. (b)	8. (a)	9. (b)	10. (c)
11. (c)	12. (b)								

(A) Taking it together

1. (d)	2. (a)	3. (b)	4. (a)	5. (d)	6. (a)	7. (b)	8. (c)	9. (c)	10. (a)
11. (c)	12. (c)	13. (b)	14. (b)	15. (c)	16. (c)	17. (a)	18. (b)	19. (d)	20. (c)
21. (b)	22. (a)	23. (c)	24. (c)	25. (d)	26. (a)	27. (b)	28. (a)	29. (a)	30. (b)
31. (c)	32. (b)	33. (c)	34. (a)	35. (d)	36. (a)	37. (d)	38. (c)	39. (d)	40. (a)
41. (d)	42. (c)	43. (b)	44. (b)	45. (c)	46. (b)	47. (c)	48. (a)		

(B) Medical entrance special format questions

• Assertion and reason

1. (d) 2. (d) 3. (c) 4. (b) 5. (d)

• Statement based questions

1. (c) 2. (d) 3. (d) 4. (a) 5. (b) 6. (c)

• Match the columns

1. (a) 2. (a) 3. (c)

(C) Medical entrances' gallery

1. (c)	2. (b)	3. (d)	4. (c)	5. (a)	6. (d)	7. (c)	8. (a)	9. (d)	10. (b)
11. (c)	12. (b)	13. (a)	14. (b)	15. (c)	16. (b)	17. (d)	18. (a)	19. (d)	20. (c)
21. (c)	22. (b)	23. (c)	24. (c)	25. (b)					

Hints & Explanations

• CHECK POINT 12.1

2 (d) This is the case of shear strain in which both shape and size changes.

3 (d) On stretching a spiral spring, both types of strains are produced, *i.e.* longitudinal and shear strain.

4 (a) Here, $L = 10 \text{ cm} = 10 \times 10^{-2} \text{ m}$

and $F = 100 \text{ kgf} = 100 \times 9.8 \text{ N}$

$$\therefore \text{Shearing stress} = \frac{F}{L^2} = \frac{100 \times 9.8}{(10 \times 10^{-2})^2} = 9.8 \times 10^4 \text{ Nm}^{-2}$$

5 (b) Here, original length, $L = 2.5 \text{ m}$

$$\text{Strain} = \frac{\Delta L}{L} = 0.012\% = \frac{0.012}{100}$$

$$\therefore \Delta L = \text{Strain} \times L$$

$$\text{or } \Delta L = \text{Extension} = \frac{0.012}{100} \times L$$

$$= \frac{0.012 \times 2.5}{100} = 3 \times 10^{-4} \text{ m} = 0.3 \text{ mm}$$

6 (d) We have, $V = l^3$

$$\frac{\Delta V}{V} = \frac{3\Delta l}{l} = 3 \left(\frac{2}{100} \right) = 0.06$$

7 (c) Here, $L = 10 \text{ cm} = 10 \times 10^{-2} \text{ m}$

$$\Delta x = 0.25 \text{ cm} = 0.25 \times 10^{-2} \text{ m}$$

$$\therefore \text{Shearing strain} = \frac{\text{Displacement } (\Delta x)}{\text{Distance } (L)} = \frac{0.25 \times 10^{-2}}{10 \times 10^{-2}} = 0.025$$

• CHECK POINT 12.2

2 (b) Young's modulus of material,

$$Y = \frac{\text{Linear stress}}{\text{Longitudinal strain}}$$

If longitudinal strain is equal to unity, then $Y = \text{linear stress produced}$.

3 (a) Elongation in rubber cord,

$$l = \frac{L^2 dg}{2Y} = \frac{(10)^2 \times 1500 \times 10}{2 \times 5 \times 10^8} = 15 \times 10^{-4} \text{ m}$$

4 (b) Elongation, $\Delta l = \frac{FL}{AY}$

$$\therefore \Delta l \propto \frac{1}{A} \quad (\text{If } F, L \text{ and } Y \text{ are constants})$$

$$\text{or } \frac{A_2}{A_1} = \frac{\Delta l_1}{\Delta l_2}$$

$$\Rightarrow A_2 = A_1 \left(\frac{0.1}{0.05} \right) = 2A_1 = 2 \times 4 = 8 \text{ mm}^2$$

$$\mathbf{5 (c)} \quad \Delta l = \frac{FL}{\pi r^2 Y}$$

$$\Rightarrow r^2 \propto \frac{1}{Y} \quad (\text{If } F, L \text{ and } \Delta l \text{ are constants})$$

$$\frac{r_2}{r_1} = \left(\frac{Y_1}{Y_2} \right)^{1/2} = \left(\frac{7 \times 10^{10}}{1.2 \times 10^{10}} \right)^{1/2}$$

$$\text{Radius, } r_2 = 1.5 \times \left(\frac{7}{12} \right)^{1/2} = 1.145 \text{ mm}$$

$$\therefore d = 2.29 \text{ mm}$$

6 (a) For a perfectly rigid body, Young's modulus is infinite.

7 (b) Solids are least compressible, whereas gases are most compressible.

• CHECK POINT 12.3

1 (a) We know that, $\frac{dV}{V} = (1 - 2\sigma) \frac{dL}{L}$

$$\text{If } \frac{dV}{V} = 0 \Rightarrow 1 - 2\sigma = 0$$

$$\text{or } \sigma = \frac{1}{2}$$

$$\therefore \sigma = + 0.50$$

2 (d) We have,

$$Y = 2\eta (1 + \sigma)$$

$$2.4\eta = 2\eta (1 + \sigma)$$

$$1.2 = (1 + \sigma)$$

Poisson's ratio, $\sigma = 0.2$

3 (a) As, $Y = \frac{\text{Stress}}{\text{Strain}}$

$$\therefore \text{Slope of strain and stress graph} = \frac{1}{Y}$$

Thus, A has maximum Young's modulus of elasticity as its slope is minimum.

4 (c) Volume (V) = Cross-sectional area (A) $\times$ Length (l) or $V = Al$

$$\text{Strain} = \frac{\text{Elongation}}{\text{Original length}} = \frac{y}{l}$$

$$\text{Young's modulus, } Y = \frac{\text{Stress}}{\text{Strain}}$$

$$\text{Work done, } W = \frac{1}{2} \times \text{Stress} \times \text{Strain} \times \text{Volume}$$

$$W = \frac{1}{2} \times Y \times (\text{Strain})^2 \times Al$$

$$= \frac{1}{2} \times Y \times \left(\frac{y}{l} \right)^2 \times Al = \frac{1}{2} \left(\frac{YA}{l} \right) y^2$$

$$\Rightarrow W \propto y^2$$

- 5 (b) The work done per unit volume,

$$u = \frac{1}{2} \times Y \times (\text{Strain})^2 = \frac{1}{2} \times 9 \times 10^{11} \times \left(\frac{1}{100}\right)^2$$

$$= 4.5 \times 10^7 \text{ J}$$

- 6 (b) $W = \frac{1}{2} (\text{Maximum stretching force}) \times (\text{Elongation produced})$
- $$= \frac{1}{2} F \Delta l$$

Here, $F = mg = 10 \times 10, \Delta l = 2 \text{ m}$

$$\therefore W = \frac{1}{2} (10 \times 10) \times 2 = 100 \text{ J}$$

- 7 (b) Energy stored by the spring,

$$U = \frac{F^2}{2k} = \frac{T^2}{2k}$$

- 8 (a) The energy stored in the wire,

$$U = \frac{1}{2} \times F \times l = \frac{1}{2} \times 200 \times 10^{-3} = 0.1 \text{ J}$$

- 9 (b) $U = \frac{1}{2} F \Delta l = \frac{F^2 l}{2AY}$

$$U \propto \frac{l}{r^2} \quad (\because F \text{ and } Y \text{ are constants})$$

$$\therefore \frac{U_A}{U_B} = \left(\frac{l_A}{l_B}\right) \times \left(\frac{r_B}{r_A}\right)^2 = (3) \times \left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

- 10 (c) Required work, $W = \frac{YA(\Delta L)^2}{2L}$
- $$= \frac{2 \times 10^{10} \times 10^{-6} \times (10^{-3})^2}{2 \times 50 \times 10^{-2}}$$
- $$= 2 \times 10^{-2} \text{ J}$$

- 11 (c) Energy stored, $U = \frac{1}{2} \times \frac{YA(\Delta L)^2}{L}$
- $$= \frac{1}{2} \times \frac{2 \times 10^{11} \times 3 \times 10^{-6} \times (1 \times 10^{-3})^2}{4}$$
- $$= 0.075 \text{ J}$$

- 12 (b) We have, $k = \frac{F}{x} = \frac{mg}{x}$
- $$= \frac{4 \times 9.8}{2 \times 10^{-2}} = 19.6 \times 10^2$$

$$\text{Work done} = \frac{1}{2} kx^2 = \frac{1}{2} \times 19.6 \times 10^2 \times (0.05)^2 = 2.4 \text{ J}$$

(A) Taking it together

- 1 (d) We know that,

$$\text{Breaking stress} = \frac{\text{Breaking force}}{\text{Area of cross-section}} \quad \dots(i)$$

When length of the wire changes, area of cross-section remains same.

Hence, breaking force will be same when length changes.

- 2 (a) $Y = \frac{F}{A}$ or $F = Y \cdot A = 2 \times 10^{11} \times 10^{-4} = 2 \times 10^7 \text{ N}$

- 3 (b) Longitudinal strain $= \frac{\text{Stress}}{Y} = \frac{10^6}{10^{11}} = 10^{-5}$

$$\text{Percentage increase in length} = 10^{-5} \times 100 = 0.001\%$$

- 4 (a) Energy per unit volume, $E = \frac{1}{2} \times Y \times (\text{Strain})^2$

$$\therefore \text{Strain} = \sqrt{\frac{2E}{Y}}$$

- 5 (d) Energy per unit volume $= \frac{1}{2} \times \text{stress} \times \text{strain}$

$$= \frac{1}{2} \times (Y \times \text{strain}) \times \text{strain} = \frac{1}{2} \times Y \times E^2$$

- 6 (a) Young's modulus is the property of material. So, it is independent of the length.

- 7 (b) $Y = \frac{\text{Stress}}{\text{Strain}}, Y = \text{Stress for unit strain or } \Delta l = l$

Hence, stress will double the length of the wire.

- 8 (c) Graph between applied force and extension will be straight line because in elastic range.

Applied force $\propto$ Extension

But the graph given is parabolic in nature, so it will be between extension and stored elastic energy.

$$\text{As, } U = (1/2) kx^2 \text{ or } U \propto x^2$$

- 9 (c) As, $|B| = \frac{\Delta p}{\Delta V/V}$

$$\therefore \frac{\Delta V}{V} = \frac{\Delta p}{|B|} = \frac{10^5}{1.25 \times 10^{11}} = 8 \times 10^{-7}$$

- 10 (a) Poisson's ratio $= \frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$$\text{Lateral strain} = 0.4 \times \frac{0.05}{100}$$

So, it will be reduced by 0.02 %.

- 11 (c) Strain $= \frac{\text{Stress}}{Y} = \frac{S}{Y}$

$$\text{Now, energy stored per unit volume} = \frac{1}{2} \times \text{stress} \times \text{strain}$$

$$= \frac{1}{2} \times S \times \frac{S}{Y} = \frac{S^2}{2Y}$$

- 13 (b) The energy stored in the wire,

$$U = \frac{1}{2} \times F \times l = \frac{1}{2} mg l = \frac{1}{2} \times 10 \times 10 \times 1 \times 10^{-3} = 0.05 \text{ J}$$

- 14 (b) $Y = 2\eta(1 + \sigma)$

$$\Rightarrow 3\eta = 2\eta(1 + \sigma) \Rightarrow \sigma = \frac{3}{2} - 1 = \frac{1}{2}$$

Now, substituting the value of σ in the following expression

$$Y = 3K(1 - 2\sigma)$$

$$\Rightarrow K = \frac{Y}{3(1 - 2\sigma)} = \infty$$

- 15 (c) $(\text{Stress})_1 = (\text{Stress})_2$

Maximum load does not depend on the length.

$\therefore$ The maximum load that can be supported by either part is w .

- 16 (c) We have, $\Delta l = \frac{Fl}{AY}$ or $\Delta l \propto \frac{l}{A}$... (i)

As per the information given in the question, the longitudinal extension of any material of wire is small.

It means that if Δl is small, then A will be large, which indicates that in order to have an appreciable change, the material must be in the form of a long thin wire.

- 17 (a) Work done, $W = \frac{1}{2} Fl$

$\therefore$ Work done, $W \propto l$ (If F is constant)

$$\Rightarrow \frac{W_1}{W_2} = \frac{l_1}{l_2} = \frac{l}{2l} = \frac{1}{2}$$

- 18 (b) Force developed, $F = YA\alpha (\Delta\theta)$

$$= 10^{11} \times 10^{-4} \times 10^{-5} \times 100 = 10^4 \text{ N}$$

- 19 (d) We know that, with increase in temperature, length of a wire changes as

$$L_t = L_o (1 + \alpha \Delta T)$$

where, ΔT is change in the temperature, L_o is original length, α is coefficient of linear expansion and L_t is length at temperature T .

Now, we can write, $\Delta L = L_t - L_o = L_o \alpha \Delta T$

$$\text{Young's modulus } (Y) = \frac{\text{Stress}}{\text{Strain}} = \frac{FL_o}{A \times \Delta L} = \frac{FL_o}{AL_o \alpha \Delta T} \propto \frac{1}{\Delta T}$$

$$\text{As, } Y \propto \frac{1}{\Delta T}$$

When temperature (ΔT) increases, hence Y decreases.

- 20 (c) $\therefore$ Bulk modulus of the material of the rubber,

$$|B| = \frac{\Delta p}{\Delta V/V} = \frac{100}{0.01/100} = 10^6 \text{ atm} = 10^{11} \text{ Nm}^{-2} \\ = 10^{12} \text{ dyne cm}^{-2}$$

- 21 (b) Strain $\propto$ Stress $\propto \frac{F}{A}$

$$\text{Ratio of strain} = \frac{A_2}{A_1} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{4}{1}\right)^2 = \frac{16}{1}$$

- 22 (a) Young's modulus, $Y = \frac{3.6 \times 10^{-9} \text{ N/\AA}}{3 \times 10^{-10} \text{ m}}$
 $= \frac{3.6 \times 10^{-9}}{3 \times 10^{-10} \times 10^{-10}} = 1.2 \times 10^{11} \text{ Nm}^{-2}$

- 23 (c) Elongation of the wire,

$$\Delta l = \frac{Fl}{AY} = \frac{(2)l}{(2 \times 10^{-4})(10^4)} = l \quad (\because 1 \text{ dyne} = 10^{-5} \text{ N})$$

$\therefore$ New length, $l' = l + \Delta l = l + l = 2l$

- 24 (c) $\eta = \frac{F/A}{x/L} \Rightarrow x = \frac{L}{\eta} \times \frac{F}{A}$

If η and F are constants, then $x \propto \frac{L}{A}$.

For maximum displacement, area at which force applied should be minimum and vertical side of the block should be maximum. This condition is satisfied when the block is kept in position (Q).

- 25 (d) Increase in tension of wire

$$= YA \alpha \Delta\theta$$

$$= 2.2 \times 10^{11} \times 10^{-2} \times 10^{-4} \times 8 \times 10^{-6} \times 5 = 8.8 \text{ N}$$

- 26 (a) Elongation of the wire, $\Delta l = \frac{Fl}{AY}$

(Can also be applied for a spring)

$$\therefore \Delta l \propto l$$

$$\frac{7.5}{5.0} = \frac{60}{l_2} \Rightarrow l_2 = 40 \text{ cm}$$

$\therefore$ Length of original spring is $(60 + 40) \text{ cm} = 100 \text{ cm}$

$$\text{Now, } \frac{x}{7.5} = \frac{100}{60}$$

$$\therefore x = 12.5 \text{ cm}$$

- 27 (b) Maximum stress $= Y \times (\text{Maximum strain})$

$$\frac{Mg}{A} = 2 \times 10^{11} \times 10^{-3} = 2 \times 10^8 \text{ Nm}^{-2}$$

$\Rightarrow$ Maximum mass of this wire,

$$M = \frac{2 \times 10^8 \times 3 \times 10^{-6}}{10} = 60 \text{ kg}$$

- 28 (a) Elongation of the wire, $\Delta l = \frac{FL}{AY}$

$$\therefore \Delta l \propto \frac{1}{r^2} \quad (\text{If } Y, L \text{ and } F \text{ are constants})$$

i.e. For the same load, thickest wire will show minimum elongation. So, line OD represents the thickest wire.

- 29 (a) $\Delta l = \frac{Fl}{AY} \Rightarrow \frac{\Delta l}{l} = \left(\frac{F}{AY}\right)$

Potential energy per unit volume $= \frac{1}{2} \times \text{Stress} \times \text{Strain}$

$$u = \frac{1}{2} \times \frac{F}{A} \times \frac{F}{AY} \text{ or } u \propto \frac{1}{A^2} \text{ or } u \propto \frac{1}{d^4}$$

$$\therefore \frac{u_1}{u_2} = \left(\frac{d_2}{d_1}\right)^4 = \frac{16}{1}$$

- 30 (b) Maximum breaking stress $= \frac{F}{A} = \frac{mg}{A} = \frac{(Al\rho)g}{A} = S_{\max}$

$$\therefore l_{\max} = \frac{S_{\max}}{\rho g} = \frac{10^6}{(3 \times 10^3)(10)} = 33.33 \text{ m}$$

- 31 (c) We have, $S = \frac{w}{\pi r^2} = \frac{w'}{\pi (2r)^2}$

$\therefore$ Load, $w' = 4w$

- 32 (b) At middle, T = weight of half the length of steel bar

$$= \left(\frac{L}{2}\right) A \rho g$$

$$\therefore \text{Stress} = \frac{T}{A} = \frac{L \rho g}{2}$$

$$\begin{aligned}
 33. (c) \text{ Due to own weight, } \Delta l &= \frac{mgl}{2AY} = \frac{(lAp)gl}{2AY} \\
 &= \frac{l^2 pg}{2Y} = \frac{(8)^2 (1.5 \times 10^3) (9.8)}{2 \times 5 \times 10^6} \\
 &= 9.4 \times 10^{-2} \text{ m} = 9.4 \text{ cm}
 \end{aligned}$$

$$34. (a) F = \left(\frac{YA}{l} \right) \cdot \Delta l, \text{ i.e. } F \cdot \Delta l \text{ graph is a straight line with slope } \frac{YA}{l}, \text{ i.e. slope is proportional to } Y.$$

$$(\text{Slope})_A > (\text{Slope})_B$$

$$\therefore Y_A > Y_B$$

$$\begin{aligned}
 35. (d) \frac{Y_A}{Y_B} &= \frac{\tan \theta_A}{\tan \theta_B} = \frac{\tan 60^\circ}{\tan 30^\circ} \\
 &= \frac{\sqrt{3}}{1/\sqrt{3}} = 3 \Rightarrow Y_A = 3Y_B
 \end{aligned}$$

$$36. (a) \text{ Tension, } T_{\max} = m(g + a) = (2000)(9.8 + 1.5) = 22600 \text{ N}$$

$$\therefore \text{Maximum stress} = \frac{T_{\max}}{\text{Area}}$$

$$\begin{aligned}
 \therefore \text{Area} &= \frac{T_{\max}}{\text{Maximum stress}} = \frac{22600}{7 \times 10^7} \\
 &= 3.23 \times 10^{-4} \text{ m}^2 = 3.23 \text{ cm}^2
 \end{aligned}$$

37. (d) As stress is shown on X-axis and strain on Y-axis.

$$\text{So, we can say that } Y = \cot \theta = \frac{1}{\tan \theta} = \frac{1}{\text{slope}}$$

So, elasticity of wire *P* is minimum and of wire *R* is maximum.

38. (c) The slope of straight line portion of strain-stress curve for a given material represents its Young's modulus.

(i) Young's modulus of the given material (*Y*)

= Slope of strain-stress curve

$$Y = \frac{150 \times 10^6}{0.002} = 75 \times 10^9 = 7.5 \times 10^{10} \text{ Nm}^{-2}$$

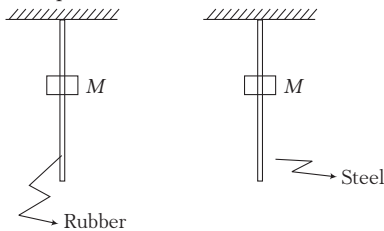
(ii) Yield strength of the given material

= Maximum stress that material can sustain

$$= 300 \times 10^6 = 3 \times 10^8 \text{ Nm}^{-2}$$

39. (d) Consider the diagram.

A mass *M* is attached at the centre of the two cylindrical rods as shown. As the mass is attached to both the rods, both rods will be elongated, but due to different elastic properties of material rubber changes shape also and bottom edge becomes tapered to a tip at the centre.



$$40. (a) \text{ We have, } \frac{\Delta L}{w} = \frac{(4-1) \times 10^{-4}}{80-20} = \frac{10^{-4}}{20}$$

$$\text{Elongation, } \Delta L = \frac{wL}{AY}$$

$$\Rightarrow Y = \frac{L}{A} \frac{w}{\Delta L}$$

$$\text{Young's modulus, } Y = \frac{1}{10^{-6}} \times \frac{20}{10^{-4}} = 2 \times 10^{11} \text{ Nm}^{-2}$$

41. (d) $\therefore$ Since, the elongations of the two rods are equal

$$(\Delta L)_b = (\Delta L)_s$$

$$F \leftarrow \boxed{\hspace{2cm}} \rightarrow F$$

$$\Rightarrow \left(\frac{FL}{AY} \right)_b = \left(\frac{FL}{AY} \right)_s \quad (\because F_b = F_s)$$

$$\text{or } \left(\frac{L}{AY} \right)_b = \left(\frac{L}{AY} \right)_s$$

$\therefore$ The length of the steel rod,

$$L_s = \left(\frac{A_s Y_s}{A_b Y_b} \right) L_b = \left(\frac{1.0 \times 2.0 \times 10^{11}}{2.0 \times 1.0 \times 10^{11}} \right) (2) = 2 \text{ m}$$

42. (c) At length $\frac{3L}{4}$ from lower end, tension in the wire,

$$T = \text{suspended load} + \frac{3}{4} \times \text{weight of wire} = w_1 + \frac{3w}{4}$$

$$\therefore \text{Stress} = \frac{T}{S} = \frac{w_1 + \frac{3w}{4}}{S}$$

43. (b) We know that, Young's modulus,

$$\begin{aligned}
 Y &= \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta L/L} = \frac{F}{A} \times \frac{L}{\Delta L} \\
 &= \frac{F}{\pi (D/2)^2} \times \frac{L}{\Delta L} = \frac{4FL}{\pi D^2 \Delta L}
 \end{aligned}$$

$$\Rightarrow D^2 = \frac{4FL}{\pi \Delta L Y}$$

$$\Rightarrow D = \sqrt{\frac{4FL}{\pi \Delta L Y}}$$

As *F* and $\frac{L}{\Delta L}$ are constants.

$$\text{Hence, } D \propto \sqrt{\frac{1}{Y}}$$

Now, we can find ratio as,

$$\frac{D_{\text{copper}}}{D_{\text{iron}}} = \sqrt{\frac{Y_{\text{iron}}}{Y_{\text{copper}}}}$$

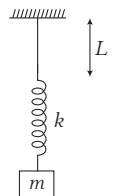
44. (b) Wire is equivalent to a spring of spring

$$\text{constant, } k' = \frac{AY}{L}$$

k' and *k* are in series,

$$k_{\text{eq}} = \frac{k'k}{k' + k} = \frac{(AY/L)k}{(AY/L) + k} = \frac{AYk}{AY + kL}$$

$$T = 2\pi \sqrt{\frac{m}{k_{\text{eq}}}} = 2\pi \sqrt{\frac{m(AY + kL)}{AYk}}$$



- 45 (c) Let l be the natural length and $k \left(= \frac{YA}{l} \right)$ be the force

constant of wire. Then,

$$a = l + \frac{4}{k} \quad \dots(i) \left(\because F = k \Delta l \text{ or } \Delta l = \frac{F}{k} \right)$$

$$\text{and} \quad b = l + \frac{5}{k} \quad \dots(ii)$$

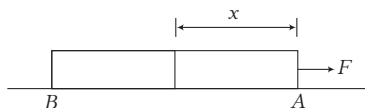
On solving Eqs. (i) and (ii), we get

$$\text{or} \quad \frac{1}{k} = (b - a) \quad \text{and} \quad l = (5a - 4b)$$

Now, when $T = 9 \text{ N}$

$$l' = l + \frac{9}{k} = (5a - 4b) + 9(b - a) = (5b - 4a)$$

- 46 (b) Tension at distance x from end A, $T_x = F - \frac{F}{L} x$



$$\text{Total change in length, } \Delta L = \int_0^L \frac{T_x dx}{ES} = \frac{FL}{2ES}$$

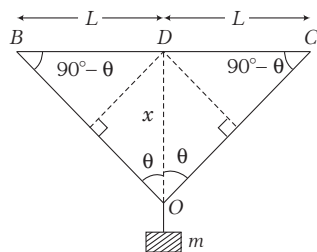
$$\therefore \text{Average strain} = \frac{\Delta L}{L} = \frac{F}{2ES}$$

- 47 (c) We have, $\Delta l_1 = \frac{wL}{(4\pi R^2)Y}$ and $\Delta l_2 = \frac{wL}{\pi R^2 Y}$

The elastic potential energy in the system,

$$\begin{aligned} U &= \frac{1}{2} k_1 (\Delta l_1)^2 + \frac{1}{2} k_2 (\Delta l_2)^2 \\ &= \frac{1}{2} \times \frac{Y(4\pi R^2)}{L} \times \left[\frac{wL}{4\pi R^2 Y} \right]^2 + \frac{1}{2} \times \frac{Y(\pi R^2)}{L} \times \left[\frac{wL}{\pi R^2 Y} \right]^2 \\ &= \frac{5w^2 L}{8\pi R^2 Y} \quad \left(\because k = \frac{YA}{L} \right) \end{aligned}$$

- 48 (a) Consider the diagram below



$$\begin{aligned} \text{Hence, change in length, } \Delta L &= BO + OC - (BD + DC) \\ &= 2BO - 2BD \quad (\because BO = OC, BD = DC) \\ &= 2[BO - BD] = 2[(x^2 + L^2)^{1/2} - L] \\ &= 2L \left[\left(1 + \frac{x^2}{L^2} \right)^{1/2} - 1 \right] \approx 2L \left[1 + \frac{1}{2} \frac{x^2}{L^2} - 1 \right] = \frac{x^2}{L} \quad (\because x \ll L) \\ \therefore \text{Strain} &= \frac{\Delta L}{2L} = \frac{x^2/L}{2L} = \frac{x^2}{2L^2} \end{aligned}$$

(B) Medical entrance special format questions

• Assertion and reason

- 1 (d) Upto proportional limit, strain $\propto$ stress.

A wire regains its original dimension only when load applied is less than its elastic limit.

- 2 (d) Young's modulus is the property of material. It does not depend on length of the wire.

- 3 (c) Potential energy per unit volume is $\frac{1}{2} \times \text{stress} \times \text{strain}$.

- 4 (b) Incompressible fluid is that which cannot be compressed ($\Delta V = 0$) by applying pressure on it. Thus, bulk modulus for these types of fluid is infinite.

- 5 (d) Stress = $\frac{\text{Restoring force}}{\text{Area}}$

$$S = \frac{\text{Force}}{\pi r^2} = \frac{F}{\pi r^2} \quad \dots(i)$$

When the radius of cylindrical wire is doubled,

$$S' = \frac{F}{\pi (2r)^2}$$

$$\Rightarrow S' = \frac{F}{4\pi r^2} = \frac{S}{4} \quad [\text{From Eq. (i)}]$$

$$\text{Now, } A = \pi r^2 \quad \dots(ii)$$

By doubling the radius, area of cross-section,

$$A' = \pi (2r)^2 = 4\pi r^2$$

$$\Rightarrow A' = 4A \quad [\text{From Eq. (ii)}]$$

Therefore, area of cross-section will become four times.

• Statement based questions

- 1 (c) Elongation, $\Delta l = \frac{FL}{YA}$

$$\Rightarrow \Delta l \propto \frac{1}{A}$$

- 3 (d) Elongation in the copper wire,

$$\begin{aligned} (\Delta l)_C &= \left(\frac{Fl}{AY} \right)_C = \frac{500 \times 8}{0.5 \times 10^{-4} \times 10^{11}} \\ &= 0.8 \times 10^{-3} \text{ m} = 0.8 \text{ mm} \end{aligned}$$

Elongation in the steel wire,

$$\begin{aligned} (\Delta l)_S &= \left(\frac{Fl}{AY} \right)_S = \frac{500 \times 4}{0.5 \times 10^{-4} \times 2 \times 10^{11}} \\ &= 0.2 \times 10^{-3} \text{ m} = 0.2 \text{ mm} \end{aligned}$$

$$(\Delta l)_S = \frac{1}{4} (\Delta l)_C$$

and total elongation, $\Delta l = 0.8 + 0.2 = 1.0 \text{ mm}$

- 4 (a) Energy loss $\propto$ Area of hysteresis loop

Also, energy loss will appear as heat.

For rubber, it is difficult to compress than expand.

5 (b) Bulk modulus, $B = \frac{-\Delta p}{\Delta V/V}$

For incompressible fluids, $\Delta V = 0$

$\therefore B = \infty$

Apart from this, it is true to say that compressibility is inverse of bulk modulus of elasticity.

• Match the columns

- 1 (a) Volume changes in case of volumetric stress, longitudinal stress and tensile stress, whereas shape gets changed in shear stress.

Hence, A $\rightarrow$ p,s; B $\rightarrow$ q,r; C $\rightarrow$ p,s; D $\rightarrow$ p,s.

2 (a) (A) Stress $\times$ Strain = $\text{Nm}^{-2} \cdot \frac{\text{m}^3}{\text{m}^3} = \text{Jm}^{-3}$

(B) $\frac{YA}{l} = \frac{\text{Nm}^{-2}\text{m}^2}{\text{m}} = \text{Nm}^{-1}$

(C) $Yl^3 = \text{Nm}^{-2}\text{m}^3 = \text{Nm} = \text{J}$

(D) $\frac{Fl}{AY} = \frac{\text{Nm}}{\text{m}^2\text{Nm}^{-2}} = \text{m}$

Hence, A $\rightarrow$ r, B $\rightarrow$ q, C $\rightarrow$ p, D $\rightarrow$ s.

- 3 (c) Modulus of elasticity (Y , B , η) depends on the temperature and nature of material. Volume of a substance depends on temperature, length and area of cross-section.

Hence, A $\rightarrow$ p,s; B $\rightarrow$ p,s; C $\rightarrow$ p,s; D $\rightarrow$ p,q,r.

(C) Medical entrances' gallery

- 1 (c) Here, change in length, $\Delta L = (L_1 - L)$

Area = A

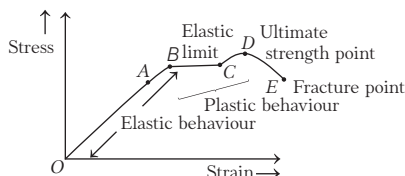
Force, $F = Mg$

Young's modulus,

$$Y = \frac{\text{Normal stress}}{\text{Longitudinal strain}}$$

$$\Rightarrow Y = \frac{(F/A)}{\left(\frac{\Delta L}{L}\right)} = \frac{\frac{Mg}{A}}{\left(\frac{L_1 - L}{L}\right)} = \frac{MgL}{A(L_1 - L)}$$

- 2 (b) The stress-strain curve for a material is shown in figure.



This curve specifies the behaviour of material.

For the material X, as the distance between strength point and fracture point is small, so it is brittle and will break easily on the application of some extra stress after point D.

For material Y, as the distance between strength point and fracture point is large, so it is a ductile material and can withstand for some extra stress beyond point D.

- 3 (d) Given, ratio of radius of two wires, $\frac{r_1}{r_2} = \frac{2}{1}$

and ratio in their lengths, $\frac{l_1}{l_2} = \frac{1}{2}$

We know that, Young's modulus,

$$Y = \frac{Fl}{A\Delta l}$$

$$\Rightarrow \Delta l = \frac{Fl}{AY}$$

$$\Delta l \propto \frac{l}{A}$$

When same force is applied on them, then ratio of change in their lengths,

$$\frac{\Delta l_1}{\Delta l_2} = \frac{l_1}{l_2} \cdot \frac{A_2}{A_1} = \frac{l_1 A_2}{l_2 A_1}$$

$$\Rightarrow \frac{\Delta l_1}{\Delta l_2} = \frac{l_1}{l_2} \cdot \frac{r_2^2}{r_1^2} = \frac{1}{2} \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{8}$$

- 4 (c) According to the question,

For wire 1 Area of cross-section = A_1

Force applied = F_1

and increase in length = Δl

From the relation of Young's modulus of elasticity,

$$Y = \frac{Fl}{A\Delta l}$$

Substituting the values for wire 1 in the above relation, we get

$$\Rightarrow Y_1 = \frac{F_1 l_1}{A_1 \Delta l} \quad \dots(i)$$

For wire 2 Area of cross-section = A_2

Force applied = F_2

and increase in length = Δl

$$\text{Similarly, } Y_2 = \frac{F_2 l_2}{A_2 \Delta l} \quad \dots(ii)$$

$$\because \text{Volume, } V = Al \text{ or } l = \frac{V}{A}$$

Substituting the value of l in Eqs. (i) and (ii), we get

$$Y_1 = \frac{F_1 V}{A_1^2 \Delta l} \text{ and } Y_2 = \frac{F_2 V}{A_2^2 \Delta l}$$

As it is given that the wires are made up of same material,

i.e.

$$Y_1 = Y_2$$

$$\Rightarrow \frac{F_1 V}{A_1^2 \Delta l} = \frac{F_2 V}{A_2^2 \Delta l}$$

$$\Rightarrow \frac{F_1}{F_2} = \frac{A_1^2}{A_2^2} = \frac{A^2}{9A^2} = \frac{1}{9} \quad (\because A_1 = A \text{ and } A_2 = 3A)$$

or

$$F_2 = 9F_1 = 9F \quad (\text{Given, } F_1 = F)$$

- 5 (a) Given, $k = 4 \times 10^{-5}$ per atm,

$$p = 100 \text{ atm}$$

$$\text{and } V = 100 \text{ cm}^3$$

$$= 100 \times 10^{-6} \text{ m}^3$$

$$\therefore \text{Compressibility} = \frac{1}{|\text{Bulk modulus of elasticity}|}$$

$$k = \frac{1}{\Delta p \frac{\Delta V}{V}} = \frac{\Delta V}{\Delta p V}$$

$$\therefore \Delta V = k (\Delta p V) = 4 \times 10^{-5} (100 \times 100 \times 10^{-6}) \\ = 0.4 \times 10^{-6} \text{ m}^3 = 0.4 \text{ cm}^3$$

- 6 (d) The object is spherical and the bulk modulus is represented by B . It is the ratio of normal stress to the volumetric strain.

$$\text{Hence, } B = \frac{F/A}{\Delta V/V} \quad \left(\because p = \frac{F}{A} \right)$$

$$\Rightarrow \frac{\Delta V}{V} = \frac{p}{B} \Rightarrow \left| \frac{\Delta V}{V} \right| = \frac{p}{B}$$

Here, p is applied pressure on the object and $\frac{\Delta V}{V}$ is volume strain.

$$\text{Fractional decrease in volume, } \frac{\Delta V}{V} = 3 \frac{\Delta R}{R} \quad (\because V = \frac{4}{3} \pi R^3)$$

Volume of the sphere decreases due to the decrease in its radius, hence

$$\frac{\Delta V}{V} = \frac{3\Delta R}{R} = \frac{p}{B} \Rightarrow \frac{\Delta R}{R} = \frac{p}{3B}$$

- 7 (c) As, Young's modulus of elasticity, $Y = \frac{\text{Stress}}{\text{Strain}}$

$$\Rightarrow Y = \frac{T_1 \times L}{A \times (l_1 - L)} \quad \dots (i)$$

where, L is the original length of the wire and also

$$Y = \frac{T_2 \times L}{A \times (l_2 - L)} \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$\frac{T_1 L}{A (l_1 - L)} = \frac{T_2 L}{A (l_2 - L)} \Rightarrow L = \frac{T_2 l_1 - T_1 l_2}{T_2 - T_1}$$

- 8 (a) Given, mass of load = 4 kg

Length of wire, $l = 2 \text{ m}$

Radius of wire (cross-section) = 2 mm = $2 \times 10^{-3} \text{ m}$

and change in length, $\Delta l = 0.031 \text{ mm} = 0.031 \times 10^{-3} \text{ m}$

Young's modulus,

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{\frac{mg}{A}}{\left(\frac{\Delta l}{l} \right)} = \frac{\frac{4 \times 3.1 \pi}{\pi (2 \times 10^{-3})^2}}{(\Delta l / l)}$$

$$\Rightarrow Y = \frac{4 \times 3.1 \pi}{\pi \times 4 \times 10^{-6}} \times \frac{2}{0.031 \times 10^{-3}} \\ = \frac{3.1 \times 2}{0.031} \times 10^9 = 2 \times 10^{11} \text{ Nm}^{-2}$$

- 9 (d) Given, $r_1 = r, r_2 = 2r$

$$\text{Now, we know that, } Y = \frac{F}{A} \cdot \frac{l}{\Delta l} = \frac{F}{\pi r^2} \cdot \frac{l}{\Delta l}$$

$$\text{So, } \frac{F}{\pi r_1^2} = \frac{f}{\pi r_2^2} \quad (\because l, \Delta l \text{ and } Y \text{ are same})$$

$$\Rightarrow \frac{F}{f} = \frac{r_1^2}{r_2^2} \Rightarrow \frac{F}{f} = \frac{r^2}{4r^2} \Rightarrow \frac{F}{f} = \frac{1}{4}$$

- 10 (b) Elastic potential energy of a stretched wire is given by

$$U = \frac{1}{2} \times \text{Stress} \times \text{Strain} \times \text{Volume}$$

$$= \frac{1}{2} \times Y \times \text{Strain} \times \text{Strain} \times \text{Volume}$$

$$= \frac{1}{2} \times Y \times \left(\frac{l}{L} \right)^2 \times AL$$

$$\therefore U = \frac{AYl^2}{2L}$$

- 11 (c) Elongation produced, if the $\frac{2}{3}$ part of length L of rod is stretched by a force of $3F$, is given by

$$\Delta L_1 = \frac{FL_1}{AY} = \frac{3F \times \frac{2}{3}L}{AY} = 2 \frac{FL}{AY}$$

Elongation produced, if the remaining $(1/3)$ part of length L of rod is stretched by a force of $2F$, is given by

$$\Delta L_2 = \frac{FL_2}{AY} = \frac{2F \times \frac{1}{3}L}{AY} = \frac{2}{3} \frac{FL}{AY}$$

Total elongation produced in the rod,

$$\Delta L = \Delta L_1 + \Delta L_2 = 2 \frac{FL}{AY} + \frac{2}{3} \frac{FL}{AY}$$

$$\therefore \Delta L = \frac{8}{3} \frac{FL}{AY}$$

- 12 (b) We know that,

Relation between Y and G

$$Y = 2G(1 + \sigma) \Rightarrow G = \frac{Y}{2(1 + \sigma)} \Rightarrow G \cong \frac{Y}{3} \quad (\because \sigma \approx 0.5)$$

$$13 (a) \quad \Delta L = \frac{FL}{AY} \text{ or } \Delta L \propto \frac{L}{d^2} \quad \left(\because A = \frac{\pi d^2}{4} \right)$$

Therefore, ΔL will be maximum for that wire for which $\frac{L}{d^2}$ is maximum. Here by checking options, we get option (a) has maximum extension.

- 14 (b) Young's modulus of a rope, $Y = \frac{FL}{A\Delta l}$

Given, $L = 10 \text{ m}, A = \pi r^2 = \pi (1)^2 = \pi$

$$Y = 20 \times 10^{11} \text{ dyne cm}^{-2}, \Delta l = 1 \text{ cm}$$

$$\Rightarrow F = \frac{Y \cdot A \cdot \Delta l}{L} \Rightarrow F = \frac{20 \times 10^{11} \times \pi \times 1}{10 \times 10^2}$$

$$F = 6.28 \times 10^9 \text{ dyne}$$

$$\text{or } F = 6.28 \times 10^4 \text{ N}$$

- 15 (c) We know that, $\Delta l = \frac{FL}{AY}$

$$\text{Here, } \Delta l \propto \frac{L}{A}$$

According to question, $\frac{\Delta l_1}{\Delta l_2} = \frac{L_1}{L_2} \times \frac{A_2}{A_1}$

$$\Rightarrow \frac{\Delta l_1}{\Delta l_2} = \frac{1}{2} \times \frac{1}{2} \Rightarrow \frac{\Delta l_1}{\Delta l_2} = \frac{1}{4}$$

$$\Rightarrow \Delta l_2 = 4 \Delta l_1 \Rightarrow \Delta l_2 = 4 \times 1 = 4 \text{ cm}$$

17 (d) Change in length of ring = $2\pi R - 2\pi r$

$$\text{Longitudinal strain} = \frac{2\pi R - 2\pi r}{2\pi r} = \frac{R - r}{r}$$

$$\text{Young's modulus, } Y = \frac{F/A}{\frac{R - r}{r}}$$

$$\Rightarrow Y = \frac{Fr}{A(R - r)} \Rightarrow F = \frac{YA(R - r)}{r}$$

18 (a) Volume of the rubber cord = AL

$$\text{Weight of the rubber cord} = ALDg$$

$$\therefore \text{Stress} = \frac{ALDg}{A} = LDg$$

The weight of the rubber cord acts at the centre of gravity which is $\frac{L}{2}$ from the top.

$$\therefore \text{Strain} = \frac{l}{L/2} = \frac{2l}{L}$$

$$\text{So, Young's modulus, } E = \frac{\text{Stress}}{\text{Strain}} = \frac{LDg}{2l/L} = \frac{L^2 Dg}{2l} \text{ or } l = \frac{L^2 Dg}{2E}$$

19 (d) Magnitude of bulk modulus,

$$B = \frac{p}{dV/V} \text{ or } dV = \frac{pV}{B}$$

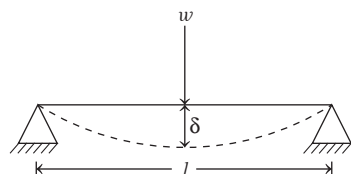
$$\therefore \rho' = \frac{M}{V - dV} = \frac{M}{V - (pV/B)}$$

$$= \frac{M}{V(1 - p/B)} = \frac{\rho}{(1 - p/B)} \quad \left(\because \rho = \frac{M}{V} \right)$$

$$\therefore \frac{\rho'}{\rho} = \frac{1}{(1 - p/B)}$$

20 (c) We have, $\delta = \frac{wl}{4Ybd^3}$

where, b and d are cross-sectional dimensions.



or $\delta \propto \frac{1}{Y}$

21 (c) Breaking force = Breaking stress $\times$ Area of cross-section of wire

$$\therefore \text{Breaking force} \propto r^2$$

If radius becomes doubled, then breaking force becomes 4 times, i.e. $40 \times 4 = 160 \text{ kg-wt}$

22 (b) We know that, compressibility, $C = \frac{1}{|B|}$

$$\text{Also, } \frac{dV}{V} = (1 + 2\sigma) \frac{dL}{L}$$

$$\text{Given, } \sigma = -\frac{1}{2}, \text{ then}$$

$$\frac{dV}{V} = 0, \text{ i.e. volume is constant.}$$

or $B = \infty$, i.e. the material is incompressible.

23 (c) Breaking stress = $\frac{F}{A} = \frac{40}{3\pi} \times 10^6 \text{ Nm}^{-2}$

According to question,

$$\frac{40}{3\pi} \times 10^6 = \frac{3 \times 10}{\pi r^2}$$

$$\Rightarrow r^2 = \frac{9}{4 \times 10^6}$$

$$\Rightarrow r = 1.5 \times 10^{-3} \text{ m} \\ = 1.5 \text{ mm}$$

24 (c) Modulus of elasticity = $\frac{\text{Force}}{\text{Area}} \times \frac{l}{\Delta l}$

$$3 \times 10^{11} = \frac{33000}{10^{-3}} \times \frac{l}{\Delta l}$$

$$\frac{\Delta l}{l} = \frac{33000}{10^{-3}} \times \frac{1}{3 \times 10^{11}} \\ = 11 \times 10^{-5}$$

$$\text{Change in length, } \frac{\Delta l}{l} = \alpha \Delta T$$

$$11 \times 10^{-5} = 1.1 \times 10^{-5} \times \Delta T$$

$$\Rightarrow \Delta T = 10^\circ\text{C}$$

25 (b) The minimum stress after which the wire breaks, i.e.

$$\text{Stress} = \frac{\text{Force}}{\text{Area}} = \frac{mg}{A} \\ = \frac{V\rho g}{A} = \frac{LA\rho g}{A} = L\rho g$$

$$\text{Given, stress} = 6 \times 10^6 \text{ Nm}^{-2},$$

$$\rho = 3 \times 10^3 \text{ kgm}^{-3} \text{ and } g = 10 \text{ ms}^{-2}$$

$$\Rightarrow L = \frac{\text{Stress}}{\rho g} = \frac{6 \times 10^6}{3 \times 10^3 \times 10} \\ = 2 \times 10^2 = 200 \text{ m}$$

CHAPTER 13

Fluid Mechanics

The term ‘fluid’ refers to a substance which cannot withstand shear, so it has no definite shape of its own and it has the ability to flow. Thus, the term fluid applies to both liquids and gases. In this chapter, we will study some important properties of fluids like pressure, viscosity, surface tension, etc.

PRESSURE

Normal force acting per unit area of a surface is called pressure. It is generally represented by p .

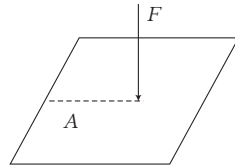


Fig. 13.1

If surface area is A and normal force acting on the surface is F (shown in figure), then pressure on the surface is

$$p = \frac{\text{Normal force } (F)}{\text{Area } (A)}$$

If area is very small, then $p = \lim_{\Delta A \rightarrow 0} \frac{\Delta F}{\Delta A} = \frac{dF}{dA}$

It is clear from the above formula that pressure acting on a surface depends upon the normal force acting on the surface and its area. Thus, force of same magnitude exerts different pressure on different area that is for the same force, if area is lesser, then pressure will be greater.

SI unit of pressure is Nm^{-2} or pascal (Pa). Pressure is a scalar quantity and its dimensional formula is $[\text{ML}^{-1}\text{T}^{-2}]$.

Pressure in a fluid

When a fluid (either liquid or gas) is at rest, it exerts a force perpendicular to any surface in contact with it, such as a container wall or a body immersed in the fluid.

While the fluid as a whole is at rest, the molecules or atoms that make up the fluid are in motion. These molecules or atoms collide continuously with the

walls of container. Pressure of a fluid is due to molecules colliding with the container walls.

The pressure applied by a fluid can be defined as the magnitude of the normal force (applied by fluid) acting per unit surface area.

If the pressure is same at all points of a finite plane surface with area A , then

$$p = \frac{F_{\perp}}{A}$$

where, $F_{\perp}$ is the normal force on the surface.

Unit of fluid pressure French scientist **Pascal** contributed many researches in the field of fluid pressure. Thus, in his honour, SI unit of pressure is known as pascal which is denoted by Pa. $1 \text{ Pa} = 1 \text{ Nm}^{-2}$

One unit used practically in meteorology is the **bar** which is equal to 10^5 Pa . $1 \text{ bar} = 10^5 \text{ Pa}$

Atmospheric pressure (p_{atm})

It is the pressure of the earth's atmosphere. This changes with weather and elevation. Normal atmospheric pressure at sea level (an average value) is $1.013 \times 10^5 \text{ Pa}$. Thus,

$$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$$

Other practical units of atmospheric pressure are bar and torr (mm of Hg). $1 \text{ atm} = 1.01 \text{ bar} = 760 \text{ torr}$

Absolute pressure and gauge pressure

The excess pressure above atmospheric pressure is usually called **gauge pressure** and the total pressure is called **absolute pressure**. Thus,

$$\text{Gauge pressure} = \text{Absolute pressure} - \text{Atmospheric pressure}$$

Absolute pressure is always greater than or equal to zero. While gauge pressure can be negative also.

Example 13.1 An open container has dimensions of $4.0 \text{ m} \times 5.0 \text{ m}$ and height of 3.0 m .

- Find the weight of the air in the container at 20°C .
- What is the weight of an equal volume of water? Also, find pressure at the base of container due to this weight of water.
- What is the total downward force on the base of the container due to air pressure of 1.0 atm ?
(Take, the densities of air and water as 1.2 kg/m^3 and 10^3 kg/m^3 , respectively.)

Sol. It is given that, density of air, $\rho_{\text{air}} = 1.2 \text{ kg/m}^3$

and density of water, $\rho_{\text{water}} = 10^3 \text{ kg/m}^3$

$$\begin{aligned} \text{(i) The volume of air in the container} \\ &= (4.0)(5.0)(3.0) = 60 \text{ m}^3 \end{aligned}$$

$$\begin{aligned} \therefore \text{The mass of the air is } m_{\text{air}} &= \rho_{\text{air}} \cdot V_{\text{air}} \\ &= (1.2)(60) = 72 \text{ kg} \end{aligned}$$

The weight of the air is $w_{\text{air}} = m_{\text{air}}g = (72)(9.8) = 705.6 \text{ N}$

$$\begin{aligned} \text{(ii) The weight of water, } w_{\text{water}} &= m_{\text{water}}g = \rho_{\text{water}} \cdot V_{\text{water}}g \\ &= (10^3)(60)(9.8) = 5.9 \times 10^5 \text{ N} \end{aligned}$$

Pressure at the base due to this weight,

$$p = \frac{F (= w)}{A} = \frac{5.9 \times 10^5}{4 \times 5} = 2.95 \times 10^4 \text{ Nm}^{-2}$$

$$\begin{aligned} \text{(iii) The downward force on the base is} \\ &= (\text{Air pressure})(\text{Surface area}) \\ &= (1.013 \times 10^5)(4.0 \times 5.0) \approx 2.0 \times 10^6 \text{ N} \end{aligned}$$

Density of a liquid

Density (ρ) of a liquid or any substance is defined as the mass per unit volume.

$$\therefore \text{Density, } \rho = \frac{\text{Mass}}{\text{Volume}} = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV}$$

SI unit of density is kgm^{-3} and its dimensions are $[\text{ML}^{-3}\text{T}^0]$ and CGS unit is gcm^{-3} with $1 \text{ gcm}^{-3} = 10^3 \text{ kg m}^{-3}$.

Relative density (RD)

In case of a liquid (or a substance), sometimes an another term **relative density** (RD) or **specific gravity** is defined. It is the ratio of density of the liquid (or a substance) to the density of water at 4°C .

$$\text{Hence, } \text{RD} = \frac{\text{Density of liquid or substance}}{\text{Density of water at } 4^\circ \text{C}}$$

Relative density is a pure ratio, so it has no units.

Density of water at 4°C in CGS is 1 gcm^{-3} . Therefore, numerically the relative density and density of a substance in CGS are equal. In SI unit, the density of water at 4°C is 1000 kgm^{-3} .

Example 13.2 Relative density of an oil is 0.8. Find the absolute density of oil in CGS and SI units.

Sol. Density of oil (in CGS) = (RD) $\text{gcm}^{-3} = 0.8 \text{ gcm}^{-3}$
and in SI unit = 800 kgm^{-3}

Density of a mixture of two or more liquids

Case I Suppose two liquids of densities ρ_1 and ρ_2 having masses m_1 and m_2 are mixed together. Then, the density of the mixture will be

$$\begin{aligned} \rho &= \frac{\text{Total mass}}{\text{Total volume}} \\ &= \frac{(m_1 + m_2)}{(V_1 + V_2)} = \frac{(m_1 + m_2)}{\left(\frac{m_1}{\rho_1} + \frac{m_2}{\rho_2}\right)} \end{aligned}$$

If $m_1 = m_2$, then $\rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2}$.

Case II If two liquids of densities ρ_1 and ρ_2 having volumes V_1 and V_2 are mixed, then the density of the mixture,

$$\rho = \frac{\text{Total mass}}{\text{Total volume}} = \frac{m_1 + m_2}{V_1 + V_2} = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2}$$

If $V_1 = V_2$, then $\rho = \frac{\rho_1 + \rho_2}{2}$

Example 13.3 Two liquids of densities ρ and 3ρ having volumes $3V$ and V are mixed together. Find density of the mixture.

Sol. Given, density of first liquid, $\rho_1 = \rho$

Density of second liquid, $\rho_2 = 3\rho$

Volume of first liquid, $V_1 = 3V$

Volume of second liquid, $V_2 = V$

$\therefore$ Density of the mixture,

$$\begin{aligned}\rho_m &= \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2} \\ &= \frac{\rho \times 3V + 3\rho \times V}{3V + V} = \frac{6\rho}{4} = \left(\frac{3}{2}\right)\rho\end{aligned}$$

Effect of temperature on density

When the temperature of a liquid is increased, the mass remains the same while the volume is increased and

hence, the density of the liquid decreases $\left(\text{as } \rho \propto \frac{1}{V}\right)$.

$$\begin{aligned}\text{Thus, } \frac{\text{Density after increase of temperature by } \Delta\theta}{\text{Initial density}} &= \frac{\rho'}{\rho} \\ &= \frac{V}{V'} = \frac{V}{V + dV} = \frac{V}{V + V\gamma\Delta\theta} \quad \text{or} \quad \frac{\rho'}{\rho} = \frac{1}{1 + \gamma\Delta\theta}\end{aligned}$$

Here, γ = thermal coefficient of volume expansion
and $\Delta\theta$ = rise in temperature.

Therefore,
$$\rho' = \frac{\rho}{1 + \gamma\Delta\theta}$$

Effect of pressure on density

As pressure is increased, volume decreases and hence, density will increase. Thus, $\rho \propto \frac{1}{V}$

$$\begin{aligned}\therefore \frac{\text{Density after increase of pressure}}{\text{Initial density}} &= \frac{\rho'}{\rho} \\ &= \frac{V}{V'} = \frac{V}{V + dV} = \frac{V}{V - \left(\frac{dp}{B}\right)V}\end{aligned}$$

or
$$\frac{\rho'}{\rho} = \frac{1}{1 - \frac{dp}{B}}$$

Here, dp = change in pressure

and B = bulk modulus of elasticity of the liquid.

Therefore,
$$\rho' = \frac{\rho}{1 - \frac{dp}{B}}$$

Example 13.4 The thermal coefficient of volume expansion of a liquid is $5 \times 10^{-4} \text{ K}^{-1}$. If its temperature is increased by 30°C , find the ratio of new density to the previous one.

Sol. Given, thermal coefficient of volume expansion,

$$\gamma = 5 \times 10^{-4} \text{ K}^{-1}$$

Rise in temperature, $\Delta\theta = 30^\circ \text{C}$

$$\therefore \rho' = \frac{\rho}{1 + \gamma\Delta\theta}$$

$$\begin{aligned}\Rightarrow \frac{\rho'}{\rho} &= \frac{1}{1 + \gamma\Delta\theta} \\ &= \frac{1}{1 + (5 \times 10^{-4})(30)} \approx 0.98\end{aligned}$$

Example 13.5 The bulk modulus of a liquid is $8 \times 10^9 \text{ Nm}^{-2}$ and its density is 11 g cm^{-3} . What will be the density of liquid under a pressure of $20,000 \text{ Ncm}^{-2}$?

Sol. Given, bulk modulus of liquid, $B = 8 \times 10^9 \text{ Nm}^{-2}$

Density of liquid, $\rho = 11 \text{ g cm}^{-3}$

and pressure, $p = 20000 \text{ Ncm}^{-2}$

$$\begin{aligned}\therefore \text{New density of liquid, } \rho' &= \frac{\rho}{\left(1 - \frac{dp}{B}\right)} \quad \text{or} \quad \frac{\rho}{\left(1 - \frac{p}{B}\right)} \\ &= \frac{11}{1 - \frac{20000}{8 \times 10^9 \times 10^{-4}}} \\ &= \frac{11 \times 40}{39} = \frac{440}{39} \text{ gcm}^{-3}\end{aligned}$$

Variation of fluid pressure with depth

Consider a fluid of density ρ is kept at rest, in a cylindrical vessel of height h as shown in figure.

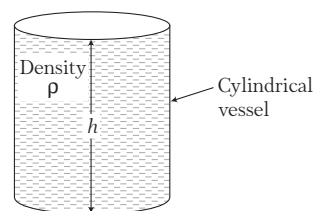


Fig. 13.2

Pressure applied by liquid at the bottom of the container is

$$p = h\rho g \quad \dots(i)$$

From the above formula, we can say that pressure applied by a liquid column is proportional to height of liquid column.

If the container is open, then atmospheric pressure will also work on the top surface, so total pressure on the bottom will be equal to

$$p_{\text{total}} = p_0 (\text{atmospheric pressure}) + \rho gh$$

From Eq. (i), we can also find the difference of pressure between any two points in the liquid column.

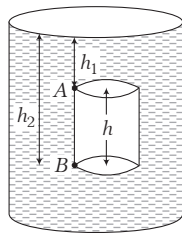


Fig. 13.3

Consider two points A and B in a liquid column situated at h_1 and h_2 respectively, below free surface of liquid.

Pressure due to liquid column at point A, $p_A = h_1\rho g$

Pressure due to liquid column at point B, $p_B = h_2\rho g$

$\therefore$ Pressure difference

$$p_B - p_A = h_2\rho g - h_1\rho g = (h_2 - h_1)\rho g$$

$$p_B - p_A = h\rho g$$

Hence, the pressure difference depends on the vertical height (i.e. distance between points A and B), density of the fluid and the acceleration due to gravity.

According to Eq. (i), pressure increases linearly with depth, if ρ and g are uniform.

A graph between p and h is shown below

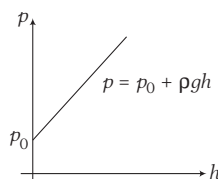


Fig. 13.4 Graph between pressure and height

Important points related to fluid pressure

Important points related to fluid pressure are given below

- (i) At a point in the liquid column, the pressure applied on it is same in all directions.
- (ii) In a liquid, pressure will be same at all points at the same level.

- (iii) The pressure exerted by a liquid depends only on the height of fluid column and is independent of the shape of the containing vessel.

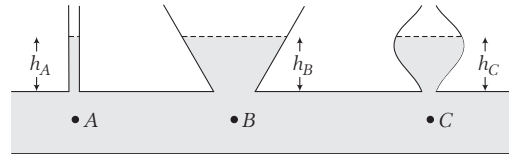


Fig. 13.5 The three vessels A, B and C containing different amounts of liquids

If $h_A = h_B = h_C$, then $p_A = p_B = p_C$

- (iv) Consider following shapes of vessels

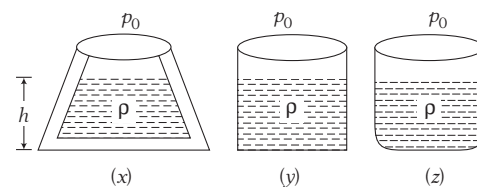


Fig. 13.6

Pressure at the base of each vessel,

$$p_x = p_y = p_z = p_0 + \rho gh \text{ but weight, } w_x \neq w_y \neq w_z$$

where, ρ = density of liquid in each vessel,

h = height of liquid in each vessel

and p_0 = atmospheric pressure.

- (v) In the figure, a block of mass m floats over a fluid surface

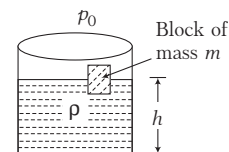


Fig. 13.7

If ρ = density of the liquid,

A = area of the block, then

Pressure at the base of the vessel is

$$p = p_0 + \rho gh + \frac{mg}{A}$$

Example 13.6 Find the pressure exerted below a column of water, open to the atmosphere, at depth

(i) 10 m

(ii) 30 m

(Take, density of water = $1 \times 10^3 \text{ kgm}^{-3}$, $g = 10 \text{ ms}^{-2}$)

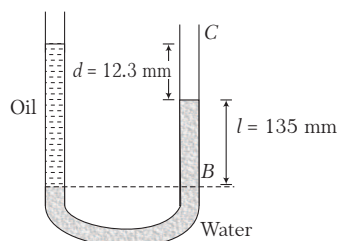
Sol. (i) Pressure at depth of 10 m,

$$\begin{aligned} p &= p_a + \rho gh \\ &= 1.013 \times 10^5 \text{ Pa} + (1 \times 10^3 \text{ kgm}^{-3})(10 \text{ ms}^{-2})(10 \text{ m}) \\ &= 1.013 \times 10^5 \text{ Pa} + 1 \times 10^5 \text{ Pa} \\ &= 2.013 \times 10^5 \text{ Pa} \\ &\approx 2 \text{ atm} \end{aligned}$$

(ii) Pressure at depth of 30 m,

$$\begin{aligned} p &= p_a + \rho gh \\ &= 1.013 \times 10^5 \text{ Pa} + (1 \times 10^3 \text{ kgm}^{-3})(10 \text{ ms}^{-2})(30 \text{ m}) \\ &= 1.013 \times 10^5 \text{ Pa} + 3 \times 10^5 \text{ Pa} \\ &= 4.013 \times 10^5 \text{ Pa} \approx 4 \text{ atm} \end{aligned}$$

Example 13.7 For the arrangement shown in the figure, what is the density of oil?



Sol. According to the figure, for the equilibrium of liquid,

Net downward force = Net upward force

$$\begin{aligned} p_0 + \rho_w gl &= p_0 + \rho_{\text{oil}} (l + d) g \\ \Rightarrow \rho_{\text{oil}} &= \frac{\rho_w l}{l + d} = \frac{1000 \times (135)}{(135 + 12.3)} = 916 \text{ kgm}^{-3} \end{aligned}$$

Example 13.8 A curved glass vessel full of water upto a height of 10 cm has a bottom of area 10 cm^2 , top of area 30 cm^2 and volume 1 L.

- Find the force exerted by the water on the bottom.
- Find the resultant force exerted by the sides of the glass vessel on the water.
- If the glass vessel is covered by a jar and the air inside the jar is completely pumped out, what will be the answers to parts (i) and (ii)?
- If a glass vessel of different shape is used provided the height, the bottom area and the volume are unchanged, will the answers of parts (i) and (ii) change?
(Take, $g = 10 \text{ ms}^{-2}$, density of water $= 10^3 \text{ kgm}^{-3}$ and atmospheric pressure $= 1.01 \times 10^5 \text{ Nm}^{-2}$)

Sol. (i) Force exerted by the water on the bottom,

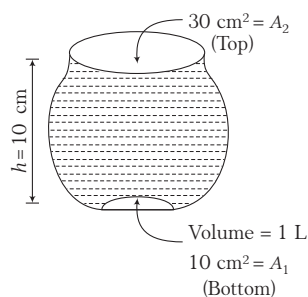
$$F_1 = (p_0 + \rho gh)A_1 \quad \dots(i)$$

Here, p_0 = atmospheric pressure $= 1.01 \times 10^5 \text{ Nm}^{-2}$

ρ = density of water $= 10^3 \text{ kgm}^{-3}$

$g = 10 \text{ ms}^{-2}$, $h = 10 \text{ cm} = 0.1 \text{ m}$

and A_1 = area of base $= 10 \text{ cm}^2 = 10^{-3} \text{ m}^2$



Substituting these values in Eq. (i), we get

$$F_1 = (1.01 \times 10^5 + 10^3 \times 10 \times 0.1) \times 10^{-3}$$

or $F_1 = 102 \text{ N}$ (downwards)

(ii) Force exerted by atmosphere on water, $F_2 = (p_0)A_2$

Here, A_2 = area of top $= 30 \text{ cm}^2 = 3 \times 10^{-3} \text{ m}^2$

$$\therefore F_2 = (1.01 \times 10^5)(3 \times 10^{-3}) = 303 \text{ N} \text{ (downwards)}$$

Force exerted by bottom on the water,

$$F_3 = -F_1 \text{ or } F_3 = 102 \text{ N (upwards)}$$

Weight of water, $w = (\text{Volume})(\text{Density})(g)$

$$= (10^{-3})(10^3)(10) = 10 \text{ N (downwards)}$$

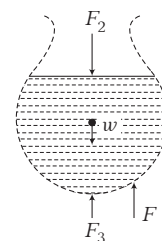
Let F be the force exerted by side walls on the water (upwards). Then, for equilibrium of water (FBD),

Net upward force = Net downward force

$$\text{or } F + F_3 = F_2 + w$$

$$\therefore F = F_2 + w - F_3 = 303 + 10 - 102$$

$$\text{or } F = 211 \text{ N (upwards)}$$



(FBD of water)

(iii) If the air inside the jar is completely pumped out,

$$\begin{aligned} F_1 &= (\rho gh)A_1 \quad (\because p_0 = 0) \\ &= (10^3)(10)(0.1)(10^{-3}) \\ &= 1 \text{ N (downwards)} \end{aligned}$$

In this case, $F_2 = 0$

and $F_3 = 1 \text{ N (upwards)}$

$$\begin{aligned} \therefore F &= F_2 + w - F_3 = 0 + 10 - 1 \\ &= 9 \text{ N (upwards)} \end{aligned}$$

(iv) No, the answers will remain the same. Because the answers depend upon p_0 , ρ , g , h , A_1 and A_2 .

Pascal's law

It states that, "pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and the walls of the containing vessel."

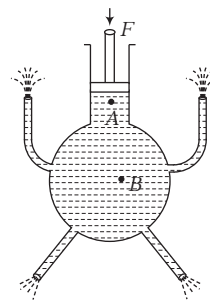


Fig. 13.8 Pressure applied to an enclosed fluid

Consider an example in which a flask fitted with a piston is filled with a liquid. Let an external force F is applied on the piston. If the cross-sectional area of the piston is A , the pressure just below the piston is increased by F/A . By Pascal's law, the pressure applied by piston will be transmitted equally at all points in the flask. If we make some holes in the flask, then the liquid from all the holes will emerge out with same intensity.

Hydraulic lift

A well known application of Pascal's law is the hydraulic lift used to support or lift heavy objects. It is schematically illustrated in figure.

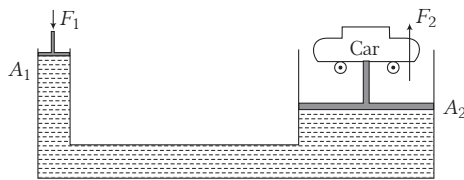


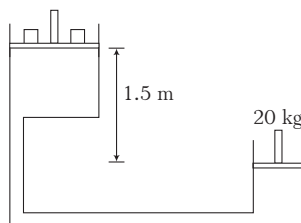
Fig. 13.9 Hydraulic lift

A piston with small cross-section area A_1 exerts a force F_1 on the surface of a liquid such as oil. The applied pressure $p = F_1/A_1$ is transmitted through the connecting pipe to a larger piston of area A_2 . It pushes larger piston with force F_2 . The applied pressure is the same in both cylinders, so

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \text{ or } F_2 = \frac{A_2}{A_1} \cdot F_1$$

Now, since $A_2 \gg A_1$, therefore $F_2 \gg F_1$. Thus, hydraulic lift is a force multiplying device with a multiplication factor equal to the ratio of the areas of the two pistons. Dentist's chairs, car lifts and jacks, hydraulic elevators and hydraulic brakes all use this principle.

Example 13.9 Figure shows a hydraulic press with the larger piston of diameter 35 cm at a height of 1.5 m relative to the smaller piston of diameter 10 cm. The mass on the smaller piston is 20 kg. What is the force exerted on the load by the larger piston? The density of oil in the press is 750 kgm^{-3} . (Take, $g = 9.8 \text{ ms}^{-2}$)



Sol. Pressure on the smaller piston = $\frac{20 \times 9.8}{\pi \times (5 \times 10^{-2})^2} \text{ Nm}^{-2}$

$$\text{Pressure on the larger piston} = \frac{F}{\pi \times (17.5 \times 10^{-2})^2} \text{ Nm}^{-2}$$

The difference between the two pressures = $h\rho g$

where, $h = 1.5 \text{ m}$

and

$$\rho = 750 \text{ kgm}^{-3}$$

$$\text{Thus, } 1.5 \times 750 \times 9.8 = \frac{20 \times 9.8}{\pi \times (5 \times 10^{-2})^2} - \frac{F}{\pi \times (17.5 \times 10^{-2})^2}$$

which gives, $F = 1.3 \times 10^3 \text{ N}$

Note Atmospheric pressure is common to both pistons and has been ignored.

Example 13.10 Two pistons of a hydraulic machine have diameters 20 cm and 2 cm. Find the force exerted on the larger piston when 50 kg-wt is placed on the smaller piston. When the smaller piston moves in through 5 cm, by what distance, the other piston moves out?

Sol. For smaller piston, area, $A_1 = \pi \times (1)^2$

For larger piston, area, $A_2 = \pi \times (10)^2$

$\therefore$ Force exerted on the larger piston,

$$\begin{aligned} F_2 &= \frac{A_2}{A_1} \times F_1 \\ &= \frac{\pi(10)^2}{\pi(1)^2} \times 50 \times 9.8 \\ &= 100 \times 50 \times 9.8 \\ &\approx 5 \times 10^4 \text{ N} \end{aligned}$$

This is the force exerted on the larger piston. The liquids are considered incompressible. Therefore, volume covered by movement of smaller piston inwards equal to the outward movement of larger piston.

$$\therefore L_1 A_1 = L_2 A_2$$

$$\begin{aligned} \Rightarrow L_2 &= \left(\frac{A_1}{A_2} \right) L_1 \\ &= \frac{\pi(1 \text{ cm})^2}{\pi(10 \text{ cm})^2} \times 5 \text{ cm} \\ &= \frac{1}{100} \times 5 \text{ cm} \\ &= 0.05 \text{ cm} \end{aligned}$$

So, the distance moved out by the larger piston is 0.05 cm.

Measurement of pressure

Pressure can be measured by using following two devices

- (i) **Barometer** It is a device used to measure atmospheric pressure. In principle, any liquid can be used to fill the barometer, but mercury is the substance of choice because its high density makes possible an instrument of reasonable size.

A barometer is an inverted evacuated tube, put over a mercury volume. Outside pressure pushes mercury into tube till the weight of liquid column equalises the force due to external pressure. In given diagram, in equilibrium,

$$p_1 = p_2$$

Here, $p_1 = \text{atmospheric pressure } (p_0)$
and $p_2 = 0 + \rho gh = \rho gh$

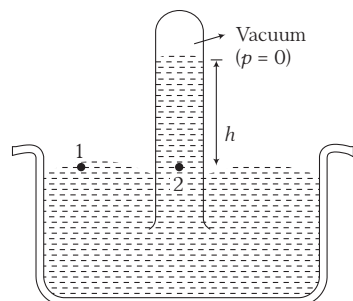


Fig. 13.10 Barometer

where, $\rho = \text{density of mercury}$

$$\therefore p_0 = \rho gh$$

Thus, the mercury barometer reads the atmospheric pressure (p_0) directly from the height of the mercury column.

e.g. If the height of mercury in a barometer is 760 mm, then atmospheric pressure will be

$$\begin{aligned} p_0 &= \rho gh \\ &= (13.6 \times 10^3)(9.8)(0.760) \\ &= 1.01 \times 10^5 \text{ Nm}^{-2} \end{aligned}$$

- (ii) **Manometer** It is a device used to measure the pressure of a gas inside a container.

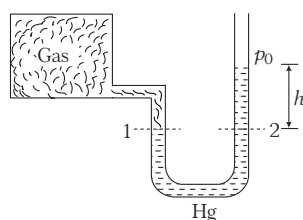


Fig. 13.11 Manometer

The U-shaped tube often contains mercury. Let 1 and 2 are points on same horizontal level, then

$$p_1 = p_2$$

Here, $p_1 = \text{pressure of the gas in the container } (p)$
and $p_2 = \text{atmospheric pressure } (p_0) + \rho gh$

$$\therefore p = p_0 + \rho gh$$

This can also be written as

$$p - p_0 = \text{gauge pressure} = \rho gh$$

Here, ρ is the density of the liquid used in U-tube.
Thus, by measuring h , we can find absolute (or gauge) pressure in the vessel.

Example 13.11 What will be the length of mercury column in a barometer tube, when the atmospheric pressure is 76 cm of mercury and the tube is inclined at an angle of 30° with the horizontal direction?

Sol. Here, $h = 76 \text{ cm}$, $\theta = 30^\circ$

If l is the length of mercury column in a barometer tube, then

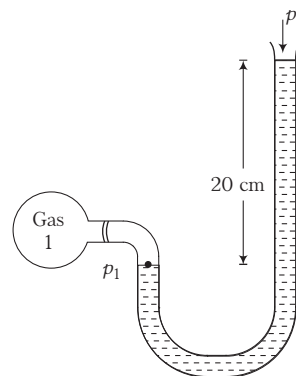
$$\begin{aligned} \frac{h}{l} &= \sin 30^\circ \\ \Rightarrow \frac{76 \text{ cm}}{l} &= \frac{1}{2} \\ \Rightarrow l &= 2 \times 76 = 152 \text{ cm} \end{aligned}$$

Example 13.12 A manometer tube contains a liquid of density $3 \times 10^3 \text{ kg m}^{-3}$. When connected to a vessel containing a gas, the liquid level in the other arm of the tube is higher by 20 cm. When connected to another sample of enclosed gas, the liquid level in the other arm of the manometer tube falls 8 cm below the liquid level in the first arm. Which of the two samples exerts more pressure and by what amount?

Sol. For sample 1

Difference in level of liquids, $h_1 = 20 \text{ cm} = 0.2 \text{ m}$

Pressure of the gas in the left arm, $p_1 = p_a + \rho gh_1$... (i)



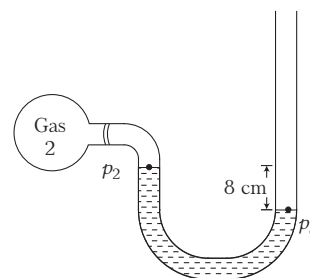
For sample 2

In this case, level of the liquid in the left arm is higher than that in the right arm by 8 cm.

$\therefore$ Atmospheric pressure p_a is greater than the pressure exerted by the sample, i.e.

$$p_a = p_2 + \rho gh_2 \Rightarrow p_2 = p_a - \rho gh_2 \quad \dots (ii)$$

Comparing Eqs. (i) and (ii), it is clear that $p_1 > p_2$.



Therefore, the gas in sample 1 exerts greater pressure than that in sample 2.

The difference in the two pressures is

$$\begin{aligned} p_1 - p_2 &= (p_a + \rho g h_1) - (p_a + \rho g h_2) \\ &= \rho g (h_1 + h_2) = \rho g (28 \text{ cm}) \\ &= (3 \times 10^3 \text{ kg m}^{-3}) \times (9.8 \text{ ms}^{-2}) (0.28 \text{ m}) \\ &= 8.23 \times 10^3 \text{ Pa} \approx 8 \text{ kPa} \end{aligned}$$

Archimedes' principle

If a heavy object is immersed in water, it seems to weigh less than when it is in air. This is because the water exerts a net upward normal force called **buoyant force**.

The Archimedes principle gives the magnitude of buoyant force on a body. It states that, "a body wholly or partially submerged in a fluid experiences an upward force which is equal to the weight of the displaced fluid."

Thus, the magnitude of buoyant force (F) which is also called upthrust is given by

$$F = V_i \rho_L g$$

Fig. 13.12 Net upward force on immersed object

where, V_i = immersed volume of solid,
 ρ_L = density of liquid
 and g = acceleration due to gravity.

The upthrust act vertically upwards through the centre of gravity of displaced fluid.

Apparent weight of a body inside a liquid

If a body is completely immersed in a liquid, its effective weight gets decreased. The decrease in its weight is equal to the upthrust on the body.

Hence, apparent weight of body = $w_{\text{app}} = w_{\text{actual}} - \text{upthrust}$

$$\text{or } w_{\text{app}} = V \rho_S g - V \rho_L g$$

Here, V = total volume of the body,

ρ_S = density of body and ρ_L = density of liquid.

$$\text{Thus, } w_{\text{app}} = V g (\rho_S - \rho_L)$$

$$\text{or } w_{\text{app}} = V \rho_S g \left(1 - \frac{\rho_L}{\rho_S} \right)$$

$$\therefore w_{\text{app}} = w_{\text{actual}} \left(1 - \frac{\rho_L}{\rho_S} \right)$$

Special cases Three possibilities may now arise

- (i) If $\rho_S < \rho_L$, then in this condition, the upthrust applied by the liquid will be greater than the weight of the body. That means, if the body is completely immersed in liquid, it will experience a net upward force. When released, the body comes up to the fluid surface till the upthrust becomes equal to the weight of the body, at this point, the body floats partially immersed in the fluid.

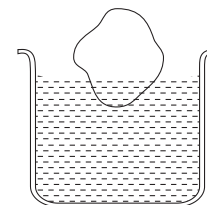


Fig. 13.13

- (ii) If $\rho_S = \rho_L$, then $w_{\text{app}} = 0$ in this condition, upthrust of the liquid balances the weight of the body. The body floats completely submerged just below the surface of the fluid.

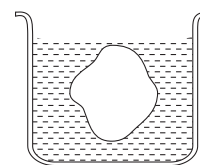


Fig. 13.14

- (iii) If $\rho_S > \rho_L$, then in this condition, the upthrust of the liquid is less than the weight of the liquid, i.e. it is not sufficient to balance the weight of the body, so the body sinks.

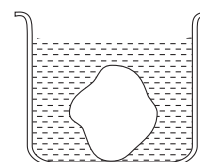


Fig. 13.15

Example 13.13 An ornament weighing 50 g in air weights only 46 g in water. Assuming that some copper is mixed with gold to prepare the ornament. Find the amount of copper in it. Specific gravity of gold is 20 and that of copper is 10.

Sol. Let m be the mass of the copper in ornament.

Then, mass of gold in it is $(50 - m)$.

$$\text{Volume of copper, } V_1 = \frac{m}{10} \quad \left(\because \text{Volume} = \frac{\text{Mass}}{\text{Density}} \right)$$

$$\text{and volume of gold, } V_2 = \frac{50 - m}{20}$$

When immersed in water ($\rho_w = 1 \text{ g cm}^{-3}$),

decrease in weight = upthrust

$$\therefore (50 - 46)g = (V_1 + V_2)\rho_w g$$

$$\text{or } 4 = \frac{m}{10} + \frac{50 - m}{20} \quad \text{or } 80 = 2m + 50 - m$$

$$\therefore m = 30 \text{ g}$$

Example 13.14 Density of ice is 900 kg m^{-3} . A piece of ice is floating in water (of density 1000 kg m^{-3}). Find the fraction of volume of the piece of ice outside the water.

Sol. Let V be the total volume and V_i the volume of ice piece immersed in water. For equilibrium of ice piece,

weight = upthrust

$$\therefore V \rho_i g = V_i \rho_w g \quad \dots(i)$$

Here, ρ_i = density of ice = 900 kg m^{-3}

and ρ_w = density of water = 1000 kg m^{-3}

Substituting in Eq. (i), we get

$$\frac{V_i}{V} = \frac{\rho_i}{\rho_w} = \frac{900}{1000} = 0.9$$

i.e. The fraction of volume outside the water, $f = 1 - 0.9 = 0.1$

Example 13.15 A piece of ice is floating in a glass vessel filled with water. How will the level of water in the vessel changes when the ice melts?

Sol. Let m be the mass of ice piece floating in water.

In equilibrium, weight of ice piece = upthrust

$$\text{or } mg = V_i \rho_w g \quad \text{or } V_i = \frac{m}{\rho_w} \quad \dots(i)$$

Here, V_i is the volume of ice piece immersed in water.

When the ice melts, let V be the volume of water formed by m mass of ice.

$$\text{Then, } V = \frac{m}{\rho_w} \quad \dots(ii)$$

From Eqs. (i) and (ii), we see that, $V_i = V$

Hence, the level will not change.

Example 13.16 A piece of ice having a stone frozen in it, floats in a glass vessel filled with water. How will the level of water in the vessel changes when the ice melts?

Sol. Let m_1 = mass of ice, m_2 = mass of stone,

ρ_s = density of stone and ρ_w = density of water.

In equilibrium, when the piece of ice floats in water,

weight of (ice + stone) = upthrust

$$(m_1 + m_2)g = V_i \rho_w g$$

$$\therefore V_i = \frac{m_1}{\rho_w} + \frac{m_2}{\rho_w} \quad \dots(i)$$

Here, V_i = volume of ice immersed.

When the ice melts, m_1 mass of ice converts into water and stone of mass m_2 is completely submerged.

Volume of water formed by m_1 mass of ice, $V_1 = \frac{m_1}{\rho_w}$

Volume of stone (which is also equal to the volume of water displaced),

$$V_2 = \frac{m_2}{\rho_s}$$

Since, $\rho_s > \rho_w$, so $V_1 + V_2 < V_i$ or the level of water will decrease.

Buoyant force in accelerating fluids

If a body is dipped inside a liquid of density ρ_L placed in an elevator moving upward with an acceleration a as shown in figure.

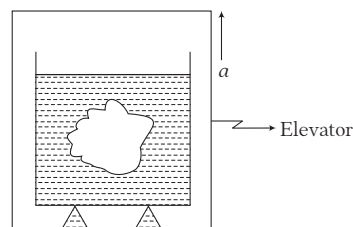


Fig. 13.16 A body dipped in a liquid is placed inside an elevator

Now, consider the force acting on the liquid replaced by the body. For upward motion of the replaced liquid, we can write,

$$F - w = ma$$

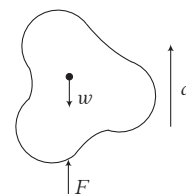


Fig. 13.17 Force acting on the displaced liquid

where, w = weight of the displaced liquid

F = buoyant force acting on the body

$$\therefore F = w + ma = m(g + a) = \rho_L V(g + a)$$

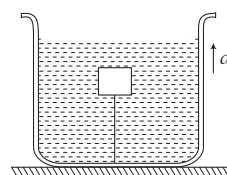
$$F = V \rho_L g_{\text{eff}}$$

Here,

$$g_{\text{eff}} = |g + a|$$

e.g. If the lift is **moving upwards** with an acceleration a , the value of g_{eff} is $g + a$ and if it is **moving downwards** with acceleration a , the g_{eff} is $g - a$. In a **freely falling** lift, g_{eff} is **zero** (as $a = g$) and hence, net buoyant force is zero. This is why, in a freely falling vessel filled with some liquid, the air bubbles do not rise up (which otherwise move up due to buoyant force).

Example 13.17 The tension in a string holding a solid block below the surface of a liquid (of density greater than that of solid) as shown in figure is T_0 when the system is at rest. What will be the tension T in the string, if the system has an upward acceleration a ?



Sol. Let m be the mass of block.

Initially for the equilibrium of block, $F = T_0 + mg$... (i)

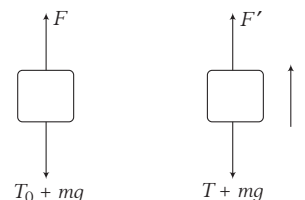
Here, F is the upthrust on the block.

When the lift is accelerated upwards, g_{eff} becomes $g + a$ instead of g .

Hence, $F' = F \left(\frac{g + a}{g} \right)$... (ii)

From Newton's second law,

$$F' - T - mg = ma \quad \dots \text{(iii)}$$

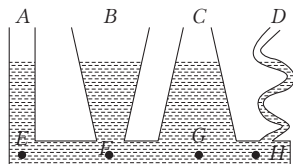


Solving Eqs. (i), (ii) and (iii), we get

$$T = T_0 \left(1 + \frac{a}{g} \right)$$

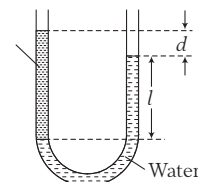
CHECK POINT 13.1

- Pins and nails are made to have pointed end because
 - it transmits very small pressure
 - it transmits a large pressure
 - it provide a large area
 - None of the above
- The two thigh bones each of cross-sectional area 15 cm^2 support the upper part of a person of mass 70 kg . The pressure sustained by these thigh bones is
 - $2.5 \times 10^5 \text{ N m}^{-2}$
 - $1.33 \times 10^5 \text{ N m}^{-2}$
 - $4.66 \times 10^5 \times 10^5 \text{ Nm}^{-2}$
 - $2.33 \times 10^5 \text{ N m}^{-2}$
- Which of the following is correct?
 - Gauge pressure = Absolute pressure + Atmospheric pressure
 - Absolute pressure = Gauge pressure - Atmospheric pressure
 - Gauge pressure = Absolute pressure - Atmospheric pressure
 - Absolute pressure = Atmospheric pressure - Gauge pressure
- At a depth of 500 m in an ocean, what is the absolute pressure? Given that the density of sea water is $1.03 \times 10^3 \text{ kg m}^{-3}$ and $g = 10 \text{ ms}^{-2}$.
 - 40 atm
 - 52 atm
 - 32 atm
 - 62 atm
- Four vessels A, B, C and D have different shapes and hold different amount of water. Which of the following is correct?



- $p_E > p_F > p_G > p_H$
 - $p_E < p_F < p_G < p_H$
 - $p_E = p_F = p_G = p_H$
 - $p_E = p_F > p_G = p_H$
- Two liquids of densities 2ρ and ρ having their volumes in the ratio $3 : 2$ are mixed together. Density of the mixture will be
 - $\frac{2\rho}{3}$
 - $\frac{\rho}{2}$
 - $\frac{8\rho}{5}$
 - $\frac{4\rho}{5}$

- The U-tube in figure contains two different liquids in static equilibrium, water in the right arm and oil of unknown density ρ_x in the left. If $l = 135 \text{ mm}$ and $d = 15 \text{ mm}$. Density of the oil is



- 1000 kg m^{-3}
 - 920 kg m^{-3}
 - 895 kg m^{-3}
 - 900 kg m^{-3}
- Increase in pressure at one point of the enclosed liquid in equilibrium at rest is transmitted equally to all other points. This is as per
 - impulse
 - Pascal's law
 - conservation of momentum
 - None of the above
 - In a vehicle lifter, the enclosed gas exerts a force F on a small piston having a diameter of 8 cm . This pressure is transmitted to a second piston of diameter 24 cm . If the mass of the vehicle to be lifted is 1400 kg , then value of F is
 - 1200 N
 - 1800 N
 - 1600 N
 - 700 N
 - What will be the length of mercury column in a barometer tube when the atmospheric pressure is $x \text{ cm}$ of mercury and the tube is inclined at an angle ϕ with the vertical direction?
 - $\frac{x}{\sin \phi}$
 - $\frac{x}{\cos \phi}$
 - $\frac{x}{\tan \phi}$
 - x
 - An iron casting containing a number of cavities weights 6000 N in air and 4000 N in water. What is the volume of the cavities in the casting? Density of iron is 7.87 g cm^{-3} . (Take, $g = 9.8 \text{ ms}^{-2}$ and density of water $= 10^3 \text{ kg m}^{-3}$)
 - 0.16 m^3
 - 0.2 m^3
 - 0.12 m^3
 - 0.14 m^3

FLOW OF FLUIDS

Till now we have studied about the fluids at rest. In this section, we will study fluids in motion. When a fluid is in flow, its motion can either be smooth or irregular depending on its velocity of flow.

Steady flow

Consider a liquid flowing in a tube as shown in the figure given below.

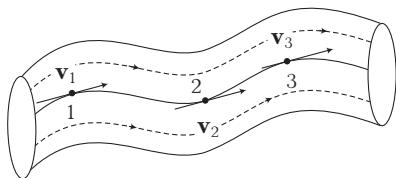


Fig. 13.18 Steady flow of liquid through a tube

If the velocity of fluid particles at any point does not vary with time, the flow is said to be steady. Steady flow is also called streamlined or laminar flow. The velocity at different points may be different. Hence, in the figure, we can say that velocity of fluid particles at different points 1, 2 and 3 remains same with time.

i.e. $\mathbf{v}_1 = \text{constant}$, $\mathbf{v}_2 = \text{constant}$, $\mathbf{v}_3 = \text{constant}$,
but $\mathbf{v}_1 \neq \mathbf{v}_2 \neq \mathbf{v}_3$

Streamlines The path followed by a fluid particle in steady flow is called streamline. Velocity of fluid particle at any point of the streamline is along tangent to the curve at that point.

Properties of streamline are given below

- In streamline flow, no two streamlines can cross each other. If they do so the particles of the liquid at the point of intersection will have two different directions for their flow, which will decrease the steady nature of the liquid flow.
- The greater is the crowding of streamlines at a place greater is the velocity of the liquid particles at that place and *vice-versa*.
- Streamline flow is possible only if the liquid velocity does not exceed a limiting value called **critical velocity**.
- The bundle of streamlines forming a tubular region is called a tube of flow.

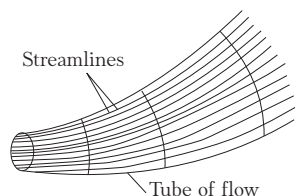


Fig. 13.19 Tube of flow formed by streamlines

Turbulent flow

In rivers and canals, where speed of water is quite high or the boundary surfaces cause abrupt changes in velocity of the flow, then the flow becomes irregular. Such flow of liquid is known as turbulent flow.

Thus, the flow of fluid in which velocity of all particles crossing a given point is not same and the motion of the fluid becomes irregular or disordered. This is called turbulent flow.

A few examples of turbulent flow are

- a jet of air striking a flat plate.
- the smoke rising from a burning stock of wood.
- when a fast flowing stream encounters rocks, foamy whirlpool like regions are formed.

Reynolds number

Reynolds number (R_e) is a dimensionless number, whose value gives us an idea whether the flow would be laminar (streamlined) or turbulent.

$$R_e = \frac{\text{Inertial force}}{\text{Viscous force}}$$

Thus, R_e represents the ratio of inertial force of moving fluid to viscous force offered by the fluid.

The expression for R_e is

$$R_e = \frac{\rho v D}{\eta}$$

Here, ρ = density of fluid,

v = speed of fluid,

D = diameter of tube

and η = viscosity of fluid (later on it is discussed in detail)

- For $R_e < 1000$, flow is streamline or laminar.
- For $R_e \rightarrow 1000$ to 2000 , flow is unsteady.
- For $R_e > 2000$, flow is turbulent.

Example 13.18 Water is flowing in a pipe of diameter 6 cm with an average velocity 7.5 cm s^{-1} and its density is 10^3 kg m^{-3} . What is the nature of flow? Given, coefficient of viscosity of water is $10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$.

Sol. Reynold's number for the given situation is given as

$$R_e = \frac{\rho v D}{\eta}$$

Here, density, $\rho = 10^3 \text{ kg m}^{-3}$

Coefficient of viscosity, $\eta = 10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$

Average velocity of water, $v = 7.5 \text{ cm s}^{-1}$
 $= 0.075 \text{ m s}^{-1}$

Diameter of pipe, $D = 6 \text{ cm} = 0.06 \text{ m}$

$$\begin{aligned} \text{Hence, } R_e &= \frac{10^3 \times 0.075 \times 0.06}{10^{-3}} \\ &= 10^6 \times 0.0045 = 4500 \end{aligned}$$

$$\therefore R_e > 2000$$

Therefore, the flow is turbulent.

Equation of continuity

When an incompressible and non-viscous liquid flows in a streamlined motion through a tube of non-uniform cross-section, then mass flow rate is same at every section of the tube.

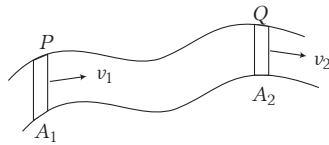


Fig. 13.20 A liquid is flowing through a tube of non-uniform cross-section

Thus,

$$A_1 v_1 = A_2 v_2$$

$$\text{or } Av = \text{constant} \quad \text{or } v \propto \frac{1}{A}$$

This is called equation of continuity which is basically derived from the law of conservation of mass.

We can generalise the continuity equation for the case in which the fluid is not incompressible. If ρ_1 and ρ_2 are the densities at sections 1 and 2, then

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

So, this is the continuity equation for an incompressible fluid.

Note The product Av is the volume flow rate $\frac{dV}{dt}$, the rate at which volume crosses a section of the tube. Hence,

$$\frac{dV}{dt} = \text{volume flow rate} = Av$$

The mass flow rate is the mass flow per unit time through a cross-section. This is equal to $\frac{dm}{dt} = \rho \cdot \frac{dV}{dt}$

Example 13.19 Water is flowing through a horizontal tube of non-uniform cross-section. At a place, the radius of the tube is 1.0 cm and the velocity of water is 2 ms^{-1} . What will be the velocity of water, where the radius of the pipe is 2.0 cm?

Sol. Using equation of continuity,

$$A_1 v_1 = A_2 v_2 \quad \text{or } v_2 = \left(\frac{A_1}{A_2} \right) v_1$$

$$\text{or } v_2 = \left(\frac{\pi r_1^2}{\pi r_2^2} \right) v_1 = \left(\frac{r_1}{r_2} \right)^2 v_1$$

$$\text{Given, } r_1 = 1 \text{ cm} = 1 \times 10^{-2} \text{ m, } r_2 = 2 \text{ cm} = 2 \times 10^{-2} \text{ m,}$$

$$v_1 = 2 \text{ ms}^{-1}$$

Substituting the above values, we get

$$v_2 = \left(\frac{1.0 \times 10^{-2}}{2.0 \times 10^{-2}} \right)^2 (2) \quad \text{or } v_2 = 0.5 \text{ ms}^{-1}$$

BERNOULLI'S THEOREM

Bernoulli's theorem is based on the law of conservation of energy and applied to ideal fluids.

It states that the sum of pressure energy per unit volume, kinetic energy per unit volume and potential energy per unit volume of an incompressible, non-viscous fluid in a streamlined irrotational flow remains constant at every cross-section throughout the liquid flow.

Mathematically, it can be expressed as

$$p + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

where, p represents the pressure energy per unit volume (or pressure), $\frac{1}{2} \rho v^2$ the kinetic energy per unit volume

and ρgh the potential energy per unit volume.

Proof Consider an ideal fluid having streamline flow through a pipe of varying area of cross-section as shown in figure.

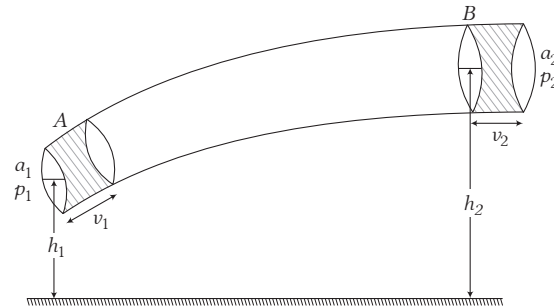


Fig. 13.21 Flow of an ideal fluid

If p_1 and p_2 are the pressures at two ends of the tube respectively, the work done by pressure difference in pushing the volume ΔV of fluid from the points A to B through the tube,

$$W = p \Delta V = (p_1 - p_2) \Delta V \quad \dots(i)$$

The change in potential energy of mass Δm (of volume ΔV),

$$\Delta U = \Delta mg(h_2 - h_1) \quad \dots(ii)$$

The change in kinetic energy,

$$\Delta K = \frac{1}{2} \Delta m (v_2^2 - v_1^2) \quad \dots(iii)$$

By conservation of energy, $W = \Delta K + \Delta U$

Putting the values from Eqs. (i), (ii) and (iii), we get

$$(p_1 - p_2) \Delta V = \frac{1}{2} \Delta m (v_2^2 - v_1^2) + \Delta mg (h_2 - h_1)$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \rho gh_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho gh_2 \quad \left(\because \rho = \frac{\Delta m}{\Delta V} \right)$$

$$p + \frac{1}{2} \rho v^2 + \rho gh = \text{constant} \quad \dots(\text{iv})$$

This is called Bernoulli's equation.

On dividing both sides of Eq. (iv) by ρg , we get

$$\frac{p}{\rho g} + h + \frac{v^2}{2g} = \frac{\text{constant}}{\rho g} = \text{new constant} \quad \dots(\text{v})$$

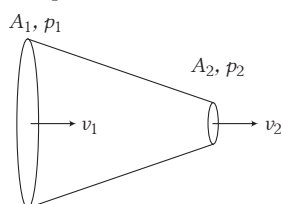
Here, $p/\rho g$ is called **pressure head**, h is called **gravitational head** and $v^2/2g$ is called **velocity head**.

Note *Bernoulli's equation for the fluid at rest* When a fluid is at rest, i.e. the velocity is zero everywhere, then the Bernoulli's equation becomes

$$p_1 + \rho gh_1 = p_2 + \rho gh_2 \Rightarrow p_1 - p_2 = \rho g (h_2 - h_1)$$

Example 13.20 Calculate the rate of flow of glycerine of density $1.25 \times 10^3 \text{ kg m}^{-3}$ through the conical section of a horizontal pipe, if the radii of its ends are 0.1 m and 0.04 m and the pressure drop across its length is 10 Nm^{-2} .

Sol. According to the question, we draw the following diagram.



From continuity equation,

$$A_1 v_1 = A_2 v_2 \quad \text{or} \quad \frac{v_1}{v_2} = \frac{A_2}{A_1} = \frac{\pi r_2^2}{\pi r_1^2}$$

$$\frac{v_1}{v_2} = \left(\frac{r_2}{r_1} \right)^2 = \left(\frac{0.04}{0.1} \right)^2 = \frac{4}{25} \quad \dots(\text{i})$$

From Bernoulli's equation, $p_1 + \frac{1}{2} \rho v_1^2 = p_2 + \frac{1}{2} \rho v_2^2$ (As $h_1 = h_2$)

or
$$v_2^2 - v_1^2 = \frac{2(p_1 - p_2)}{\rho}$$

or
$$v_2^2 - v_1^2 = \frac{2 \times 10}{1.25 \times 10^3} = 1.6 \times 10^{-2} \text{ m}^2 \text{ s}^{-2} \quad \dots(\text{ii})$$

Solving Eqs. (i) and (ii), we get

$$v_2 = \sqrt{\frac{1.6 \times 10^{-2} \times 625}{(625 - 16)}} \approx 0.128 \text{ m s}^{-1}$$

$\therefore$ Rate of volume flow through the tube,

$$Q = A_2 v_2 = (\pi r_2^2) v_2 = \pi (0.04)^2 (0.128)$$

$$= 6.43 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$$

Applications based on Bernoulli's theorem

Applications based on Bernoulli's theorem are given below

1. Venturimeter

Figure shows a venturimeter used to measure flow speed in a pipe. We apply Bernoulli's equation to the wide (point 1) and narrow (point 2) parts of a horizontal pipe

with $h_1 = h_2$, $p_1 + \frac{1}{2} \rho v_1^2 = p_2 + \frac{1}{2} \rho v_2^2$

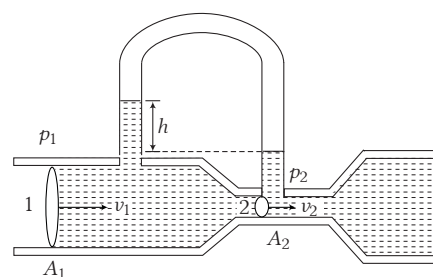


Fig. 13.22 Venturimeter

From the continuity equation, $v_2 = \frac{A_1 v_1}{A_2}$

Substituting and rearranging, we get

$$p_1 - p_2 = \frac{1}{2} \rho v_1^2 \left(\frac{A_1^2}{A_2^2} - 1 \right) \quad \dots(\text{i})$$

Because A_1 is greater than A_2 , v_2 is greater than v_1 and hence, the pressure p_2 is less than p_1 . A net force to the right accelerates the fluid as it enters the narrow part of the tube (called throat) and a net force to the left slows as it leaves. The pressure difference is also equal to ρgh , where h is the difference in liquid level in the two limbs of a manometer tube. Substituting in Eq. (i), we get

$$v_1 = \sqrt{\frac{2gh}{\left(\frac{A_1}{A_2} \right)^2 - 1}}$$

Note

(i) The discharge or volume flow rate can be obtained as

$$\frac{dV}{dt} = A_1 v_1 = A_1 \sqrt{\frac{2gh}{\left(\frac{A_1}{A_2} \right)^2 - 1}}$$

(ii) The venturi effect can be used to give a qualitative understanding of the lift of an airplane wing and the path of a pitcher's curve ball. An airplane wing is designed, so that air moves faster over the top of the wing than it does under the wing, thus making the air pressure less on top than underneath. This difference in pressure results in a net upward lifting force on the wing.

Example 13.21 The flow of blood in a large artery of an anaesthetised dog is diverted through a venturimeter. The wider part of the meter has a cross-sectional area equal to that of the artery, $A = 16 \text{ mm}^2$. The narrower part has an area $a = 8 \text{ mm}^2$. The pressure drop in the artery is 24 Pa . What is the speed of the blood in the artery?

Sol. Bernoulli's equation for the horizontal flow of blood is

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

By equation of continuity,

$$Av_1 = av_2 \quad \text{or } v_2 = Av_1/a$$

$$\begin{aligned} \therefore p_1 - p_2 &= \frac{1}{2}\rho \frac{A^2 v_1^2}{a^2} - \frac{1}{2}\rho v_1^2 \\ &= \frac{1}{2}\rho v_1^2 \left[\frac{A^2}{a^2} - 1 \right] \end{aligned}$$

Here, $p_1 - p_2 = 24 \text{ Pa}$

$$\rho (\text{blood}) = 1.06 \times 10^3 \text{ kg m}^{-3}, A/a = 16/8 = 2$$

$$\begin{aligned} \therefore v_1 &= \sqrt{\frac{2(p_1 - p_2)}{\rho \left(\frac{A^2}{a^2} - 1 \right)}} \\ &= \sqrt{\frac{2 \times 24}{1.06 \times 10^3 \times (2^2 - 1)}} \\ &= 0.123 \text{ ms}^{-1} \end{aligned}$$

2. Speed of efflux

The outflow of a fluid is called efflux and the speed of the liquid coming out is called speed of efflux.

Consider a closed vessel filled with a liquid upto height H and a small hole is made in the wall of the vessel at a depth h below the surface of liquid. Then, the speed of efflux is,

$$v = \sqrt{2gh}$$

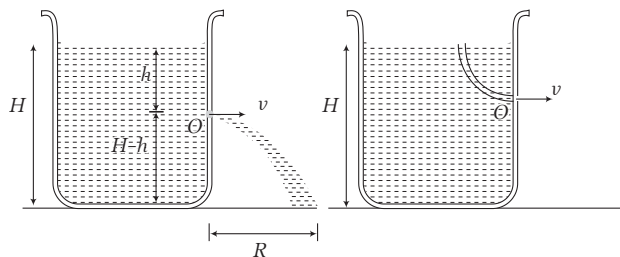


Fig. 13.23 Liquid coming out from the orifice of a tank

From the above formula, it is clear that

“The velocity of efflux of a liquid issuing out of an orifice is the same as it would attain, if allowed to fall freely through the vertical height between the liquid surface and orifice.”

The above statement is also known as **Torricelli's theorem**.

Range The range of liquid (R), i.e. the horizontal distance covered by liquid coming out of the hole is given by

$$R = 2\sqrt{h(H-h)}$$

$$R^2 = 4(Hh - h^2)$$

For R to be maximum, $\frac{dR^2}{dh} = 0$

$$\text{or } H - 2h = 0 \quad \text{or } h = \frac{H}{2}$$

i.e. R is maximum at $h = \frac{H}{2}$

$$\text{and } R_{\max} = 2\sqrt{\frac{H}{2} \left(H - \frac{H}{2} \right)}$$

$$\Rightarrow R_{\max} = H$$

i.e. The maximum horizontal distance covered by liquid coming out of a hole is equal to the height of the liquid column.

Time taken to empty a tank is given by the formula

$$t = \frac{A}{a} \sqrt{\frac{2H}{g}}$$

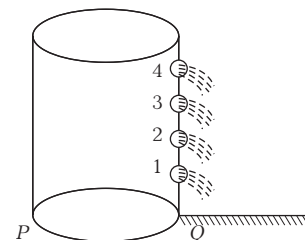
Example 13.22 If the water emerge from an orifice in a tank in which the gauge pressure is $4 \times 10^5 \text{ Nm}^{-2}$ before the flow starts, then what will be the velocity of the water emerging out? (Take, density of water is 1000 kg m^{-3})

Sol. Here, $p = 4 \times 10^5 \text{ Nm}^{-2}$ and $\rho = 1000 \text{ kg m}^{-3}$, $g = 10 \text{ ms}^{-2}$

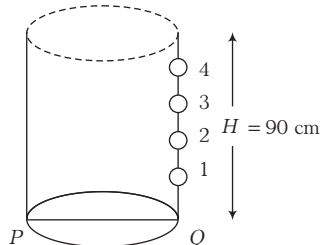
$$\text{Apply, } p = h\rho g \Rightarrow h = \frac{p}{\rho g} = \frac{4 \times 10^5}{1000 \times 10}$$

$$\begin{aligned} \text{Velocity of efflux, } v &= \sqrt{2gh} = \sqrt{\frac{2 \times 10 \times 4 \times 10^5}{1000 \times 10}} \\ &= \sqrt{800} = 28.28 \text{ ms}^{-1} \end{aligned}$$

Example 13.23 A cylindrical vessel of 90 cm height is kept filled with water upto the rim. It has four holes 1, 2, 3, 4 which are respectively at heights of 10 cm , 20 cm , 45 cm and 50 cm from the horizontal floor PQ . Through which of the holes water is falling at the maximum horizontal distance?



Sol. Different holes of the cylindrical vessel are shown in the figure. Water comes out from different holes with different speeds. Therefore, range will be different corresponding to different holes.



Horizontal range will be maximum when $h = \frac{H}{2} = \frac{90}{2} = 45$ cm

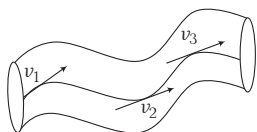
This height corresponds to hole 3.

Example 13.24 A tank is filled with a liquid upto a height H . A small hole is made at the bottom of this tank. Let t_1 be the time taken to empty first half of the tank and t_2 the time taken to empty rest half of the tank.

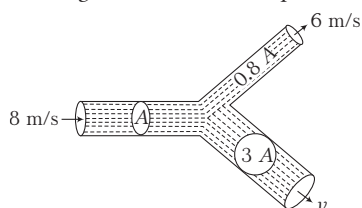
Then, find the value of $\frac{t_1}{t_2}$.

CHECK POINT 13.2

1. Consider streamline flow of a liquid flowing through a tube as shown in the figure, which of the following is correct regarding velocities of liquid at different points?



- (a) $v_1 = \text{constant}$, $v_2 = \text{constant}$, $v_3 = \text{constant}$
 (b) $v_1 \neq v_2 \neq v_3$
 (c) $v_1 = v_2 = v_3$
 (d) Both (a) and (b) are correct
2. If R_e is the Reynold's number, then which of the following is incorrect?
 (a) For $R_e < 1000$, flow is laminar
 (b) For $1000 < R_e < 2000$, flow is steady
 (c) For $R_e > 2000$, flow is turbulent
 (d) All are incorrect
3. An incompressible liquid is flowing through a horizontal pipe as shown in figure. The value of speed v is



- (a) 1.1 ms^{-1} (b) 2.1 ms^{-1} (c) 3.1 ms^{-1} (d) 5.1 ms^{-1}
4. Bernoulli's theorem is a consequence of
 (a) conservation of mass
 (b) conservation of energy
 (c) conservation of linear momentum

Sol. As we know that, the time in which level of liquid in a tank falls from H_1 to H_2 is

$$t = \frac{A}{a} \sqrt{\frac{2}{g}} (\sqrt{H_1} - \sqrt{H_2})$$

where, A = area of cross-section of the tank
 and a = area of hole.

It is given that, $H_1 = H$ and $H_2 = \frac{H}{2}$

$$\therefore t_1 = \frac{A}{a} \sqrt{\frac{2}{g}} \left(\sqrt{H} - \sqrt{\frac{H}{2}} \right) \quad \dots(i)$$

Similarly, time taken to empty the rest half of the tank,

$$t_2 = \frac{A}{a} \sqrt{\frac{2(H/2)}{g}} = \frac{A}{a} \sqrt{\frac{H}{g}} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{t_1}{t_2} = \sqrt{2} - 1 \text{ or } \frac{t_1}{t_2} = 0.414$$

Note From here we see that $t_1 < t_2$. This is because initially the pressure is high and the liquid comes out with greater speed.

(d) conservation of angular momentum

5. Water is in streamline flows along a horizontal pipe with non-uniform cross-section. At a point in the pipe, where the area of cross-section is 10 cm^2 , the velocity of water is 1 ms^{-1} and the pressure is 2000 Pa .

The pressure at another point, where the cross-sectional area is 5 cm^2 , is

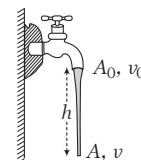
- (a) 4000 Pa (b) 2000 Pa (c) 1000 Pa (d) 500 Pa

6. If the velocity head of a stream of water is equal to 10 cm , then its speed of flow is approximately

- (a) 1.0 ms^{-1} (b) 1.4 ms^{-1} (c) 140 ms^{-1} (d) 10 ms^{-1}

7. Water falls from a tap with $A_0 = 4 \text{ m}^2$, $A = 1 \text{ m}^2$ and $h = 2 \text{ m}$, then volume rate at velocity v is

- (a) 2.5 ms^{-1} (b) 6.5 ms^{-1}
 (c) 4.5 ms^{-1} (d) 1.5 ms^{-1}



8. The lift of an aeroplane is based on
 (a) Torricelli's theorem (b) Bernoulli's principle
 (c) law of gravitation (d) continuity equation

9. The velocity of efflux of a liquid through an orifice in the bottom of a tank does not depend upon

- (a) density of liquid
 (b) height of the liquid column above orifice
 (c) acceleration due to gravity
 (d) None of the above

10. A tank is filled to a height H . The range of water coming out of a hole which is a depth $H/4$ from the surface of water level is

- (a) $\frac{2H}{\sqrt{3}}$ (b) $\frac{\sqrt{3}H}{2}$ (c) $\sqrt{3}H$ (d) $\frac{3H}{4}$

VISCOSITY

The characteristic of fluid by virtue of which relative motion between different layers is opposed is known as viscosity.

Viscosity is internal friction in a fluid which occurs due to diffusion of molecules of one layer into another. Viscous forces opposes the motion of one portion of a fluid relative to the other. The simplest example of viscous flow is motion of a fluid between two parallel plates.

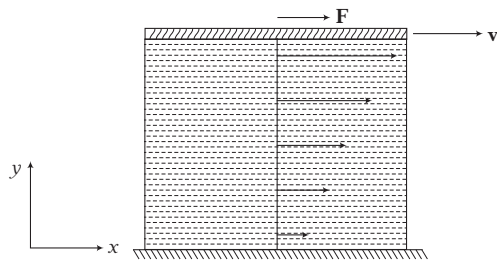


Fig. 13.24

The bottom plate is stationary and the top plate moves with constant velocity $\mathbf{v}$. The fluid in contact with each surface has same velocity as that of the surface. The flow speeds of intermediate layers of fluid increase uniformly from bottom to top, as shown by arrows. So, the fluid layers slide over one another.

According to Newton, the frictional force F (or viscous force) between two layers depends upon the following factors

- Force F is directly proportional to the area (A) of the layers in contact, i.e. $F \propto A$.
- Force F is directly proportional to the velocity gradient $\left(\frac{dv}{dy}\right)$ between the layers.

Combining these two, we have

$$F \propto A \frac{dv}{dy}$$

or

$$F = -\eta A \frac{dv}{dy}$$

Here, η is constant of proportionality and is called **coefficient of viscosity**. Its value depends on the nature of the fluid.

The negative sign in the above equation shows that the direction of viscous force F is opposite to the direction of relative velocity of the layer.

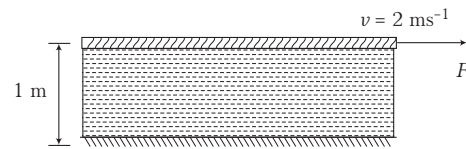
The SI unit of η is Nsm^{-2} . It is also called decapoise or pascal second.

Thus, $1 \text{ decapoise} = 1 \text{ Nsm}^{-2} = 1 \text{ Pa-s} = 10 \text{ poise}$
Dimensions of η are $[\text{ML}^{-1}\text{T}^{-1}]$.

Coefficient of viscosity of water at 10°C is $\eta = 1.3 \times 10^{-3} \text{ Nsm}^{-2}$. Experiments show that coefficient of viscosity of a liquid decreases as its temperature rises.

Example 13.25 A plate of area 2m^2 is made to move horizontally with a speed of 2 ms^{-1} by applying a horizontal tangential force over the free surface of a liquid. The depth of the liquid is 1 m and the liquid in contact with the bed is stationary. Coefficient of viscosity of liquid is 0.01 poise . Find the tangential force needed to move the plate.

Sol. Velocity gradient $= \frac{2 - 0}{1 - 0} = 2 \text{ s}^{-1}$



From Newton's law of viscous force,

$$\begin{aligned} |F| &= \eta A \frac{\Delta v}{\Delta y} = (0.01 \times 10^{-1}) (2) (2) \\ &= 4 \times 10^{-3} \text{ N} \end{aligned}$$

So, to keep the plate moving a force of $4 \times 10^{-3} \text{ N}$ must be applied.

Flow of liquid through a cylindrical pipe

Figure shows the flow speed profile for laminar flow of a viscous fluid in a long cylindrical pipe.

The speed is greatest along the axis and zero at the pipe walls.

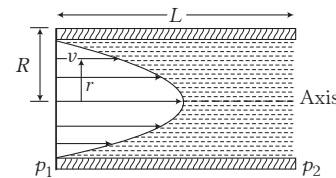


Fig. 13.25 A viscous fluid flowing in a long cylindrical pipe

The flow speed v at a distance r from the axis of a pipe of radius R is

$$v = \frac{p_1 - p_2}{4\eta L} (R^2 - r^2)$$

where, p_1 and p_2 are the pressure at the two ends of a pipe with length L . The flow is always in the direction of decreasing pressure.

At $r = R$ (along the walls), $v = 0$

The velocity of viscous fluid along the wall of the pipe is

zero and at $r = 0$ (along the axis), $v = \frac{(p_1 - p_2)R^2}{4\eta L} = v_{\text{max}}$

The velocity of viscous fluid flowing along a pipe is maximum along its axis.

Volume flow rate $\left(Q \text{ or } \frac{dV}{dt}\right)$: Poiseuille equation

To find the total volume flow rate through the pipe, we consider a ring with inner radius r , outer radius $r + dr$ and cross-sectional area $dA = 2\pi r dr$. The volume flow rate through this element is $v dA$. The total volume flow rate is found by integrating from $r = 0$ to $r = R$.

The result is
$$Q = \frac{dV}{dt} = \frac{\pi}{8} \left(\frac{R^4}{\eta} \right) \left(\frac{p_1 - p_2}{L} \right)$$

The relation was first derived by Poiseuille and is called Poiseuille's equation.

Some important points related to Poiseuille equation are given below

- (i) Poiseuille's equation can also be written as

$$Q = \frac{p_1 - p_2}{\left(\frac{8\eta L}{\pi R^4} \right)} = \frac{\Delta p}{X}, \text{ here } X = \frac{8\eta L}{\pi R^4}$$

This equation can be compared with the current equation through a resistance, i.e. $i = \frac{\Delta V}{R}$.

Here, ΔV = potential difference

and R = electrical resistance.

For current flow through a resistance, potential difference is a requirement, similarly for flow of liquid through a pipe, pressure difference is must.

- (ii) Problems of series and parallel combination of pipes can be solved in the similar manner as is done in case of an electrical circuit. *The only difference is*

- (a) Potential difference (ΔV) is replaced by the pressure difference (Δp).

- (b) The electrical resistance $R \left(= \rho \frac{L}{A} \right)$ is replaced by

$$\text{flow resistance } X \left(= \frac{8\eta L}{\pi R^4} \right).$$

- (c) The electrical current i is replaced by volume flow rate Q or dV/dt .

Combination of capillary (thin) tubes

(i) Series combination

- (a) The rate of flow (Q) will be same.
(b) Pressure difference across each tube is different.

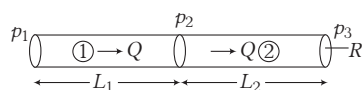


Fig. 13.26

By analogy, total flow resistance is

$$X_{eq} = X_1 + X_2 = \frac{8\eta}{\pi R^4} (L_1 + L_2)$$

Also, $p_1 - p_2 = QX_1$, $p_2 - p_3 = QX_2$

and $p_1 - p_3 = QX_{eq}$

(ii) Parallel combination

- (a) The pressure difference across each tube is same.

- (b) The rate of flow across each tube is different.

Flow resistance of tubes are

$$X_1 = \frac{8\eta L}{\pi R_1^4}$$

$$\text{and } X_2 = \frac{8\eta L}{\pi R_2^4}$$

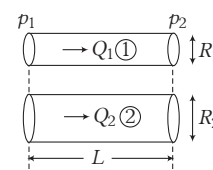


Fig. 13.27

By electrical analogy, total flow

resistance of parallel tubes is given by, $\frac{1}{X_{eq}} = \frac{1}{X_1} + \frac{1}{X_2}$,

$$p_1 - p_2 = Q_1 X_1 = Q_2 X_2$$

$$\text{Also, } p_1 - p_2 = (Q_1 + Q_2) X_{eq}$$

Following examples illustrates above discussion.

Example 13.26 Water is flowing through a horizontal tube 8 cm in diameter and 4 km in length at the rate of 20 litre/s. Assuming only viscous resistance. Find the pressure required to maintain the flow in terms of mercury column. (Take, coefficient of viscosity of water is 0.001 Pa-s)

Sol. Here, $2r = 8 \text{ cm} = 0.08 \text{ m}$ or $r = 0.04 \text{ m}$, $l = 4 \text{ km} = 4000 \text{ m}$,

$$V = 20 \text{ litre/s} = 20 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}, \eta = 0.001 \text{ Pa-s}, p = ?$$

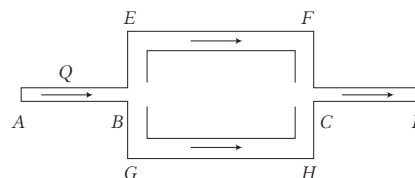
$$\text{As, } V = \frac{\pi p r^4}{8\eta l} \text{ or } p = \frac{8V\eta l}{\pi r^4}$$

$$= \frac{8 \times (20 \times 10^{-3}) \times 0.001 \times 4000}{\left(\frac{22}{7} \right) \times (0.04)^4} = 7.954 \times 10^4 \text{ Pa}$$

$\therefore$ Height of mercury column for pressure difference p will be

$$h = \frac{p}{\rho g} = \frac{7.954 \times 10^4}{(13.6 \times 10^3) \times 9.8} = 0.5968 \text{ m} = 59.68 \text{ cm}$$

Example 13.27 A liquid is flowing through horizontal pipes as shown in figure.



Length of different pipes has the following ratio

$$L_{AB} = L_{CD} = \frac{L_{EF}}{2} = \frac{L_{GH}}{2}$$

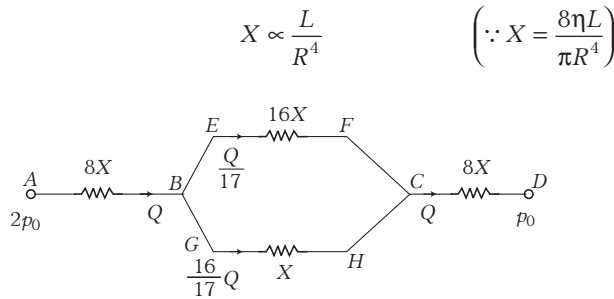
Similarly, radii of different pipes has the ratio,

$$R_{AB} = R_{EF} = R_{CD} = \frac{R_{GH}}{2}$$

Pressure at A is $2p_0$ and pressure at D is p_0 . The volume flow rate through the pipe AB is Q. Find

- volume flow rates through EF and GH
- pressure at E and F.

Sol. The equivalent electrical circuit can be drawn as under,



$$\therefore X_{AB} : X_{CD} : X_{EF} : X_{GH} = \frac{1}{2} : \frac{1}{2} : \frac{1}{2} : \frac{1}{2} = 1 : 1 : 1 : 1$$

$$= 8 : 8 : 16 : 16$$

- As the current is distributed in the inverse ratio of the resistance (in parallel), the Q will be distributed in the inverse ratio of X.

Thus, volume flow rate through EF will be $\frac{Q}{17}$ and that from GH will be $\frac{16}{17}Q$.

$$(ii) \quad X_{net} = 8X + \left[\frac{(16X)(X)}{(16X) + (X)} \right] + 8X = \frac{288}{17}X$$

$$\therefore Q = \frac{\Delta p}{X_{net}} \quad \left(\because i = \frac{\Delta V}{R} \right)$$

$$= \frac{(2p_0 - p_0)}{\frac{288}{17}X} = \frac{17p_0}{288X}$$

Now, let p_1 be the pressure at E, then

$$2p_0 - p_1 = 8QX = \frac{8 \times 17p_0}{288}$$

$$\therefore p_1 = \left(2 - \frac{17 \times 8}{288} \right) p_0$$

$$= 1.53 p_0$$

Similarly, if p_2 be the pressure at F, then

$$p_2 - p_0 = 8QX$$

$$\therefore p_2 = p_0 + \frac{8 \times 17}{288} p_0$$

$$= 1.47 p_0$$

or

STOKES' LAW

When an object moves through a fluid, it experiences a viscous force which acts in opposite direction of its velocity.

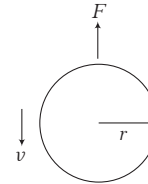


Fig. 13.28 An object is moving through a fluid

The formula for the viscous force on a sphere was first derived by the English physicist G. Stokes in 1843. According to him, a spherical object of radius r moving at velocity v experiences a viscous force given by

$$F = 6\pi\eta rv \quad (\text{where, } \eta = \text{coefficient of viscosity})$$

This law is called **Stokes' law**.

Terminal velocity (v_T)

Consider a small sphere falling from rest through a large column of viscous fluid.

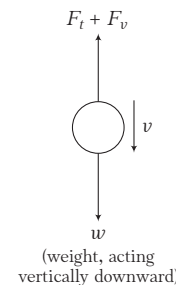


Fig. 13.29 Force acting on a small sphere falling through a liquid

The forces acting on the sphere are

- weight w of the sphere acting vertically downwards.
- upthrust buoyant force F_t acting vertically upwards.
- viscous force F_v acting vertically upwards, i.e. in a direction opposite to velocity of the sphere.

Initially, $F_v = 0$ ($\because v = 0$)

and $w > F_t$

and the sphere accelerates downwards. As the velocity of the sphere increases, F_v increases. Eventually a stage is reached when $w = F_t + F_v$... (i)

After this net force on the sphere is zero and it moves downwards with a constant velocity called **terminal velocity (v_T)**.

It can be defined as the maximum constant velocity acquired by a body while falling through a viscous medium.

Substituting proper values in Eq. (i), we get

$$\frac{4}{3}\pi r^3 \rho g = \frac{4}{3}\pi r^3 \sigma g + 6\pi \eta r v_T \quad \dots(ii)$$

Here, ρ = density of sphere, σ = density of fluid
and η = coefficient of viscosity of fluid.

From Eq. (ii), we get

$$v_T = \frac{2}{9} \frac{r^2(\rho - \sigma)g}{\eta}$$

Figure shows the variation of the velocity v of the sphere with time

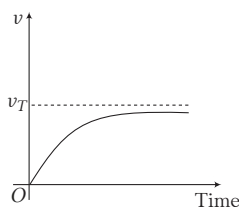


Fig. 13.30 Variation of velocity of a sphere falling through a liquid, with time

Note From the above expression, we can see that terminal velocity of a spherical body is directly proportional to the difference in the densities of the body and the fluid ($\rho - \sigma$).

If the density of fluid is greater than that of body (i.e. $\sigma > \rho$), the terminal velocity is negative. This means that the body instead of falling, moves upward. This is why air bubbles rise up in water.

Example 13.28 With what terminal velocity will an air bubble 0.8 mm in diameter rise in a liquid of viscosity 0.15 Nsm^{-2} and specific gravity 0.9 kgm^{-3} ? (Take, density of air is 1.293 kg m^{-3})

Sol. The terminal velocity of the bubble is given by

$$v_T = \frac{2}{9} \frac{r^2(\rho - \sigma)g}{\eta}$$

Here, $r = 0.4 \times 10^{-3} \text{ m}$, $\sigma = 0.9 \times 10^3 \text{ kgm}^{-3}$

$\rho = 1.293 \text{ kgm}^{-3}$, $\eta = 0.15 \text{ Nsm}^{-2}$

and $g = 9.8 \text{ ms}^{-2}$

Substituting the values, we get

$$v_T = \frac{2}{9} \times \frac{(0.4 \times 10^{-3})^2 (1.293 - 0.9 \times 10^3) \times 9.8}{0.15} \\ = -0.0021 \text{ ms}^{-1} \text{ or } v_T = -0.21 \text{ cms}^{-1}$$

Note Here, negative sign implies that the bubble will rise up.

Example 13.29 Two spherical raindrops of equal size are falling vertically through air with a terminal velocity of 1 ms^{-1} . What would be the terminal speed, if these two drops were to coalesce to form a large spherical drop?

Sol. As, $v_T \propto r^2 \quad \dots(i)$

Let r be the radius of small raindrops and R the radius of large drop.

Equating the volumes, we get

$$\frac{4}{3}\pi R^3 = 2 \left(\frac{4}{3}\pi r^3 \right)$$

$$\therefore R = (2)^{1/3} \cdot r \text{ or } \frac{R}{r} = (2)^{1/3}$$

$$\therefore \frac{v'_T}{v_T} = \left(\frac{R}{r} \right)^2 = (2)^{2/3}$$

$$\therefore v'_T = (2)^{2/3} (1.0) \text{ ms}^{-1} = 1.587 \text{ ms}^{-1}$$

CHECK POINT 13.3

1. The rate of flow of liquid in a tube of radius r , length l , whose ends are maintained at a pressure difference p is

$$V = \frac{\pi N p r^4}{\eta l}, \text{ where } \eta \text{ is coefficient of the viscosity and}$$

N is

- (a) 8 (b) $\frac{1}{8}$
(c) 16 (d) $\frac{1}{16}$

2. Two capillary tubes of the same length but different radii r_1 and r_2 are fitted in parallel to the bottom of a vessel. The pressure head is p . What should be the radius of a single tube that can replace the two tubes, so that the rate of flow is same as before?

- (a) $r_1 + r_2$ (b) $\frac{r_1 r_2}{r_1 + r_2}$
(c) $\frac{r_1 + r_2}{2}$ (d) None of these

3. The terminal velocity v of a small steel ball of radius r falling under gravity through a column of viscous liquid of coefficient of viscosity η depends on mass of the ball m , acceleration due to gravity g , coefficient of viscosity η and radius r . Which of the following relations is dimensionally correct?

- (a) $v \propto \frac{mgr}{\eta}$ (b) $v \propto mg\eta r$ (c) $v \propto \frac{mg}{\eta r}$ (d) $v \propto \frac{\eta mg}{r}$

4. The ratio of the terminal velocities of two drops of radii R and $R/2$ is

- (a) 2 (b) 1
(c) $1/2$ (d) 4

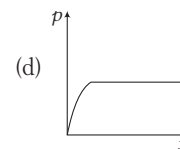
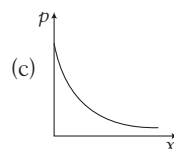
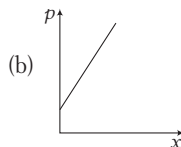
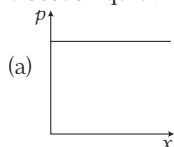
5. An air bubble rises from the bottom of a lake of large depth. The rising speed of air bubble will

- (a) go on increasing till it reaches surface
(b) go on decreasing till it reaches surface
(c) increases in the beginning, then will become constant
(d) be constant all throughout

6. Two equal drops of water are falling through air with a steady velocity v . If the drops coalesce, the new velocity will be

(a) $2v$ (b) $\sqrt{2}v$ (c) $2^{2/3}v$ (d) $\frac{v}{\sqrt{2}}$

7. From amongst the following curves, which one shows the variation of the velocity v with time t for a small sized spherical body falling vertically in a long column of a viscous liquid?



8. A spherical ball of radius 3.0×10^{-4} m and density 10^4 kg m^{-3} falls freely under gravity through a distance h before entering a tank of water. If after entering the water, the velocity of the ball does not change, what is the value of h (Take, viscosity of water is $9.8 \times 10^{-6} \text{ N s m}^{-2}$)?

(a) 1.65×10^3 m (b) 2.65×10^2 m
(c) 3.65×10^4 m (d) 1.45×10^2 m

SURFACE TENSION

It is the property of liquid at rest by virtue of which a liquid surface tends to occupy a minimum surface area and behaves like stretched membrane.

A small drop of liquid is spherical in shape, because for given volume, sphere has minimum surface area.

This stretched behaviour is due to net downward force acting on a molecule on the surface of liquid as shown in figure.

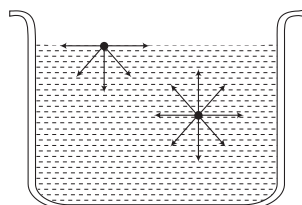


Fig. 13.31 Forces between molecules of a liquid

A molecule well inside the liquid experiences no net force.

Now, consider a line AB on the free surface of the liquid. The small elements of the surface on this line are in equilibrium because they are acted upon by equal and opposite forces, acting perpendicular to the line from either side as shown in figure.

The force acting on this line is proportional to the length of this line. If l is the length of imaginary line and F the total force on either side of the line, then $F \propto l \Rightarrow F = Sl$

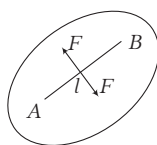


Fig. 13.32 Force on a line at the surface of a liquid

or

$$\text{Surface tension, } S = \frac{\text{Force}}{\text{Length}}$$

From this expression, **surface tension can be defined as the force acting per unit length of an imaginary line drawn on the liquid surface, the direction of force being perpendicular to this line and tangential to the liquid surface.**

It is denoted by S and it is a scalar quantity.

Units and dimension of surface tension

SI unit of surface tension = Nm^{-1} , CGS unit of surface tension = dyne cm^{-1} ,

Dimensions of surface tension

$$= \frac{\text{Force}}{\text{Length}} = \frac{[\text{MLT}^{-2}]}{[\text{L}]} = [\text{ML}^0\text{T}^{-2}]$$

Note The surface tension of a particular liquid usually decreases as temperature increases. To wash cloth thoroughly, water must be forced through the tiny spaces between the fibres. This requires increasing the surface area of the water, which is difficult to do because of surface tension. Hence, hot water and soapy water is better for washing.

Example 13.30 A liquid is kept in a beaker of radius 4 cm.

Consider a diameter of the beaker on the surface of the water.

Find the force by which the surface on one side of the diameter pulls the surface on the other side. (Take, surface tension of liquid = 0.075 Nm^{-1})

Sol. The length of the diameter, $l = 2r = 8 \text{ cm} = 0.08 \text{ m}$

The surface tension is $S = F / l$.

Thus, $F = Sl = (0.075 \text{ Nm}^{-1}) \times (0.08 \text{ m}) = 6 \times 10^{-3} \text{ N}$

Surface energy

The free surface of a liquid always has a tendency to contract and possess minimum surface area. To increase the surface area of the liquid work has to be done. This work done is stored in the surface film of the liquid as its potential energy.

This potential energy per unit area of the surface film is called the surface energy.

Hence, **the surface energy may be defined as the amount of work done in increasing the area of the surface film through unity.** Thus,

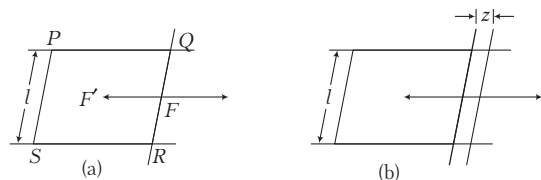
$$\text{Surface energy} = \frac{\text{Work done in increasing the surface area}}{\text{Increase in surface area}}$$

The SI unit of surface energy is J m^{-2} .

Relation between surface energy and surface tension

Consider a rectangular frame $PQRS$. Here, wire QR is movable. A soap film is formed on the frame. The film pulls the movable wire QR inward due to surface tension.

As, surface tension = $\frac{\text{force}}{\text{length}} = \frac{F'}{2l} \Rightarrow F' = S \times 2l$



Equilibrium position of the film Stretched position of the film

Fig. 13.33

If QR is moved through a distance z by an external force F very slowly, then some work has to be done against this force.

$\therefore$ External work done = Force $\times$ Distance
 $= S \times 2l \times z \quad (\because F' = F)$

Increase in surface area of film = $2l \times z$
 (As soap film has two sides)

$$\text{Surface energy} = \frac{\text{Work done}}{\text{Surface area}} = \frac{S2lz}{2lz} = S$$

So, **value of surface energy of liquid is numerically equal to the value of surface tension.**

Note

(i) Work done in forming a drop, $W = S \times 4\pi r^2$
 (As, $\Delta A = 4\pi r^2 - 0 = 4\pi r^2$)

(ii) Work done in forming a bubble, $W = S \times 4\pi r^2 \times 2$ (As, bubble in air has two surfaces)

Example 13.31 How much work will be done in increasing the diameter of a soap bubble from 2 cm to 5 cm. (Take, surface tension of soap solution is $3.0 \times 10^{-2} \text{ Nm}^{-2}$)

Sol. Soap bubble has two surfaces.

Hence, $W = T \Delta A$

Here, $\Delta A = 2[4\pi\{(2.5 \times 10^{-2})^2 - (1.0 \times 10^{-2})^2\}]$
 $= 1.32 \times 10^{-2} \text{ m}^2$

$\therefore$ Work done, $W = (3.0 \times 10^{-2})(1.32 \times 10^{-2}) \text{ J}$
 $= 3.96 \times 10^{-4} \text{ J}$

Example 13.32 Calculate the energy released when 1000 small water drops each of radius 10^{-7} m coalesce to form one large drop. (Take, surface tension of water is $7.0 \times 10^{-2} \text{ Nm}^{-1}$)

Sol. Let r be the radius of smaller drops and R of bigger one. Equating the initial and final volumes, we get

$$\frac{4}{3}\pi R^3 = (1000)\left(\frac{4}{3}\pi r^3\right)$$

or $R = 10r = (10)(10^{-7}) \text{ m}$ or $R = 10^{-6} \text{ m}$

Further, the water drops have only one free surface.

Therefore, $\Delta A = 4\pi R^2 - (1000)(4\pi r^2)$
 $= 4\pi [(10^{-6})^2 - (10^3)(10^{-7})^2]$
 $= -36\pi(10^{-12}) \text{ m}^2$

Here, negative sign implies that surface area is decreasing.

Hence, energy released in the process,

$$U = T|\Delta A| = (7 \times 10^{-2})(36\pi \times 10^{-12}) \text{ J}$$

$$= 7.9 \times 10^{-12} \text{ J}$$

Excess pressure

Due to surface tension, a drop or bubble tends to contract and so compresses the matter enclosed. This in turn increases the internal pressure which prevents further contraction and equilibrium is achieved. So, in equilibrium, the pressure inside a bubble or drop is greater than outside and the difference of pressure between two sides of the liquid surface is called excess pressure.

Following ways are used to calculate the excess pressure inside a bubble or drop

Excess pressure inside a soap bubble A soap bubble consists of two spherical surface films with a thin layer of liquid between them. Because of surface tension, the film tend to contract in an attempt to minimize their surface area. But as the bubble contracts, it compresses the inside air, eventually increasing the interior pressure to a level that prevents further contraction.

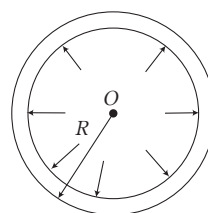


Fig. 13.34 A soap bubble of radius R

Excess pressure inside the soap bubble in terms of its radius R and the surface tension T of the liquid is given by

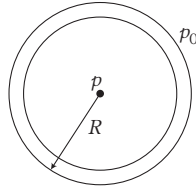


Fig. 13.35 Pressures inside and outside of a soap bubble

or

$$\Delta p = p - p_0 = \frac{4T}{R}$$

Example 13.33 0.04 cm liquid column balances the excess pressure inside a soap bubble of radius 6 mm. Evaluate density of the liquid. (Take, surface tension of soap solution = 0.03 Nm^{-1})

Sol. The excess pressure inside a soap bubble,

$$\begin{aligned} \Delta p &= 4T / R \\ &= \frac{4 \times 0.03 \text{ Nm}^{-1}}{6 \times 10^{-3} \text{ m}} = 20 \text{ Nm}^{-2} \end{aligned}$$

The pressure due to 0.04 cm of the liquid column,

$$\Delta p = h\rho g = (0.04 \times 10^{-2} \text{ m})\rho(10 \text{ ms}^{-2})$$

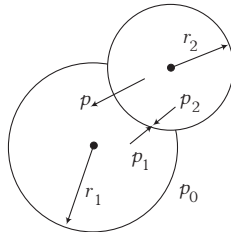
Thus, $20 \text{ Nm}^{-2} = (0.04 \times 10^{-2} \text{ m})\rho(10 \text{ ms}^{-2})$

$\therefore$ Density of the liquid, $\rho = 5 \times 10^3 \text{ kgm}^{-3}$

Example 13.34 Two separate soap bubbles (radii 0.004 m and 0.002 m) formed of the same liquid (surface tension 0.07 Nm^{-1}) come together to form a double bubble. Find the radius and the sense of curvature of the internal film surface common to both the bubbles.

Sol. Excess pressure inside first soap bubble, $p_1 = p_0 + \frac{4T}{r_1}$

Excess pressure inside second soap bubble, $p_2 = p_0 + \frac{4T}{r_2}$



$\therefore r_2 < r_1$, therefore $p_2 > p_1$

i.e. Pressure inside the smaller bubble will be more. The excess pressure,

$$p = p_2 - p_1 = 4T \left(\frac{r_1 - r_2}{r_1 r_2} \right) \quad \dots(i)$$

This excess pressure acts from concave to convex side, the interface will be concave towards smaller bubble and convex towards larger bubble. Let R be the radius of interface, then

$$p = \frac{4T}{R} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$R = \frac{r_1 r_2}{r_1 - r_2} = \frac{(0.004)(0.002)}{(0.004 - 0.002)} = 0.004 \text{ m}$$

Excess pressure inside a liquid drop A liquid drop has only one surface film, so excess pressure inside the liquid drop is given by

$$\Delta p = \frac{2T}{R}$$

Some Important points regarding excess pressure

- (i) If we have an air bubble inside a liquid, a single surface is formed. There is air on the concave side and liquid on the convex side. The pressure in the concave side (i.e. in the air) is greater than the pressure in the convex side (i.e. in the liquid) by an amount $2T/R$.

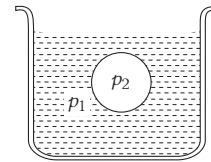


Fig. 13.36 Excess pressure in an air bubble inside a liquid

$$\therefore p_2 - p_1 = \frac{2T}{R}$$

The above expression has been written by assuming p_1 to be constant from all sides of the bubble. For small size bubbles this can be assumed.

- (ii) From the above discussion, we can make a general statement. The pressure on the concave side of a spherical liquid surface is greater than the convex side by $2T/R$ as shown below.

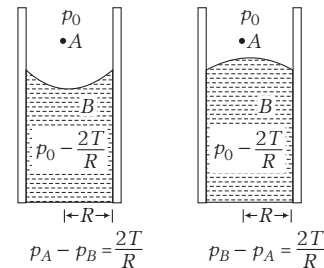


Fig. 13.37

- (iii) Excess pressure inside a soap bubble can also be understood in terms of excess pressure inside a curved surface as shown below.

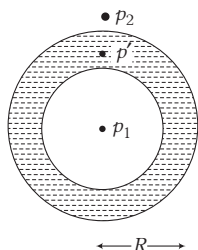


Fig. 13.38

$$p_1 - p' = \frac{2T}{R}$$

$$\Rightarrow p' - p_2 = \frac{2T}{R}$$

$$\text{Excess pressure, } \Delta p = p_1 - p_2 = \frac{4T}{R}$$

- (iv) If two bubbles of different sizes are connected with each other through a thin tube (as shown in figure), then the air will rush from smaller to larger bubble (pressure inside smaller bubble will be greater than pressure inside larger bubble), so that the smaller will shrink while the larger will expand till the smaller bubble reduces to a droplet.

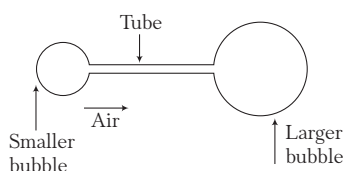


Fig. 13.39

Example 13.35 What should be the pressure inside a small air bubble of 0.1 mm radius situated just below the water surface? (Take, surface tension of water = $72 \times 10^{-2} \text{ Nm}^{-1}$ and atmospheric pressure = $1.013 \times 10^5 \text{ Nm}^{-2}$)

Sol. Surface tension of water,

$$T = 7.2 \times 10^{-2} \text{ Nm}^{-1}$$

$$\text{Radius of air bubble, } R = 0.1 \text{ mm} \\ = 10^{-4} \text{ m}$$

The excess pressure inside the air bubble is given by

$$p_2 - p_1 = \frac{2T}{R}$$

$$\therefore \text{Pressure inside the air bubble, } p_2 = p_1 + \frac{2T}{R}$$

Substituting the values, we get

$$p_2 = (1.013 \times 10^5) + \frac{(2 \times 7.2 \times 10^{-2})}{10^{-4}} \\ = 1.027 \times 10^5 \text{ Nm}^{-2}$$

Angle of contact

The surface of liquid near the plane of contact with another medium is in curved shape.

The angle of contact is defined as the angle that the tangent to the liquid surface at the point of contact makes with the solid surface. It is denoted by θ .

The value of contact angle determines whether a liquid will spread on the surface of a solid or it will form droplets on it.

When a liquid is put on a solid surface, then there are three interfacial tensions at all the three interfaces; liquid-air, solid-air and solid-liquid denoted by S_{la} , S_{sa} and S_{sl} .

At the line of contact, the surfaces between the three media must be in equilibrium.

The following cases arise

- (i) If the surface tension at the solid-liquid S_{sl} interface is greater than the surface tension at the liquid-air S_{la} interface, i.e. $S_{sl} > S_{la}$, then $\theta > 90^\circ$ (the angle of contact is **obtuse angle**).

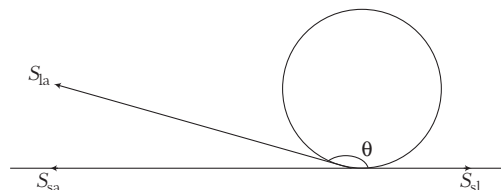


Fig. 13.40

The molecules of a liquid are attracted strongly to themselves and weakly to those of solid. It costs a lot of energy to create a liquid-solid surface. The liquid then does not wet the solid.

e.g. Water-leaf or glass-mercury interface.

- (ii) If the surface tension at the solid-liquid S_{sl} interface is less than the surface tension at the liquid-air S_{la} interface, i.e. $S_{sl} < S_{la}$, then $0 < 90^\circ$ (the angle of contact is **acute angle**).

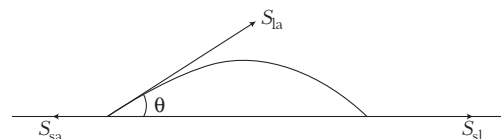


Fig. 13.41

The molecules of the liquid are strongly attracted to those of solid and weakly attracted to themselves. It costs less energy to create a liquid-solid surface and liquid wets the solid.

e.g. When soap or detergent is added to water, the angle of contact becomes small.

Shape of liquid surface

The curved surface of the liquid is called meniscus. The shape of the meniscus (convex or concave) is determined by the relative strengths of cohesive and adhesive forces. The force between the molecules of the same material is known as **cohesive force** and the force between the molecules of different kinds of material is called **adhesive force**.

When the adhesive force between solid and liquid molecules is more than the cohesive force between liquid-liquid molecules (as with water and glass), shape of the meniscus is concave and the angle of contact θ is less than 90° . In this case, the liquid wets or adheres to the solid surface.

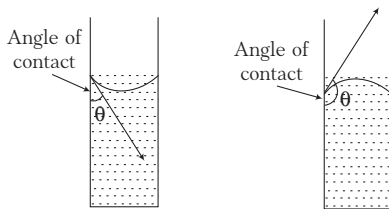


Fig. 13.42

When the adhesive force between solid and liquid molecules is less than the cohesive force between liquid-liquid molecules than shape of meniscus is convex.

Note The angle of contact between water and clean glass is zero and that between mercury and clean glass is 137° .

Capillarity

The term capilla means hair which is Latin word. A tube of very fine (hair-like) bore is called a **capillary tube**.

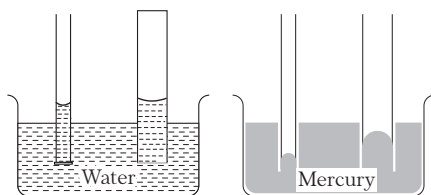


Fig. 13.43 Elevation and depression of liquid in a capillary tube

If a capillary tube of glass is dipped in liquid like water, the liquid rises in the tube, but when the capillary tube is dipped in a liquid like mercury, the level of liquid falls in the tube.

This phenomenon of rise or fall of a liquid in the capillary is called **capillarity**.

Some examples are

- Small capillaries in fibres of towels soaks water from our skin.
- In trees sap rises in stem due to capillary action.

Formula of capillary rise

When a capillary tube is dipped in a liquid, then the level of liquid in capillary tube rises or falls w.r.t. free surface of liquid outside the capillary.

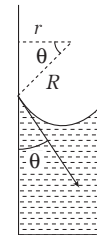


Fig. 13.44 Shape of meniscus

This phenomena of rises or fall of liquid is called capillary action.

The formula in capillary motion is

$$h = \frac{2T \cos \theta}{r\rho g}$$

where, h = height of liquid column rises or falls,

r = radius of capillary tube,

ρ = density of liquid,

g = acceleration due to gravity,

θ = angle of contact and T = surface tension.

The result has following notable features

- If the contact angle θ is greater than 90° , the term $\cos \theta$ becomes negative and hence, h is negative. The result, then gives the depression of the liquid in the tube.
- Suppose a capillary tube is held vertically in a liquid which has a concave meniscus, then capillary rise is given by

$$h = \frac{2T \cos \theta}{r\rho g} = \frac{2T}{R\rho g} \quad \left(\because R = \frac{r}{\cos \theta} \right)$$

or

$$hR = \frac{2T}{\rho g}$$

Note

- If the tube is of a length l less than h ($l < h$), the liquid does not overflow. The angle made by liquid surface with the tube adjusts in such a way that rise will stop when it is equal to length of tube.
- If the tube makes an angle ϕ units with vertical, rise will be

$$h' = \frac{h}{\cos \phi}$$

Example 13.36 A capillary tube whose inside radius is 0.5 mm is dipped in water having surface tension $7.0 \times 10^{-2} \text{ Nm}^{-1}$. To what height is the water raised above the normal water level? Angle of contact of water with glass is 0° . Density of water is 10^3 kg m^{-3} and $g = 9.8 \text{ ms}^{-2}$.

Sol. Height raised, $h = \frac{2T \cos \theta}{\rho g}$

Substituting the proper values, we get

$$h = \frac{(2)(7.0 \times 10^{-2}) \cos 0^\circ}{(0.5 \times 10^{-3})(10^3)(9.8)} = 2.86 \times 10^{-2} \text{ m}$$

$$= 2.86 \text{ cm}$$

Example 13.37 A glass tube of radius 0.4 mm is dipped vertically in water. Find upto what height, the water will rise in the capillary. If the tube is inclined at an angle of 60° with the vertical, how much length of the capillary is occupied by water? (Take, surface tension of water = $7.0 \times 10^{-2} \text{ Nm}^{-1}$, density of water = 10^3 kgm^{-3})

CHECK POINT 13.4

- Which of the following is not the unit of surface tension?
 - N/m
 - J/m²
 - kg/s²
 - W/m
- A square frame of side L is dipped in a liquid. On taking out, a membrane is formed. If the surface tension of the liquid is T , the force acting on one side of the frame will be
 - $2TL$
 - $4TL$
 - TL
 - $\frac{TL}{2}$
- A 10 cm long wire is placed horizontally on the surface of water and is gently pulled up with a force of $2 \times 10^{-2} \text{ N}$ to keep the wire in equilibrium. The surface tension (in Nm^{-1}) of water is
 - 0.1
 - 0.2
 - 0.001
 - 0.002
- If temperature increases, the surface tension of a liquid
 - increases
 - decreases
 - remains the same
 - first increases then decreases
- If work W is done in blowing a bubble of radius R from soap solution, then the work done in blowing a bubble of radius $2R$ from the same solution is
 - $W/2$
 - $2W$
 - $4W$
 - $2\frac{1}{3}W$
- If two soap bubbles of equal radii r coalesce, then the radius of curvature of interface between two bubbles will be
 - r
 - 0
 - infinity
 - $\frac{r}{2}$
- The surface tension of soap solution is $25 \times 10^{-3} \text{ Nm}^{-1}$. The excess pressure inside a soap bubble of diameter 1 cm is
 - 10 Pa
 - 20 Pa
 - 5 Pa
 - None of these
- Water rises in a capillary tube to a height of 2.0 cm. In another capillary tube whose radius is one-third of it, how much the water will rise?
 - 6.0 cm
 - 2.0 cm
 - 4.0 cm
 - 8.0 cm

Sol. For glass-water, angle of contact, $\theta = 0^\circ$.

Now, height of water in capillary,

$$h = \frac{2T \cos \theta}{\rho g} = \frac{(2)(7.0 \times 10^{-2}) \cos 0^\circ}{(0.4 \times 10^{-3})(10^3)(9.8)}$$

$$= 3.57 \times 10^{-2} \text{ m}$$

$$= 3.57 \text{ cm}$$

Length of capillary occupied after tilting it is,

$$l = \frac{h}{\cos 60^\circ} = \frac{3.57}{1/2}$$

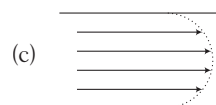
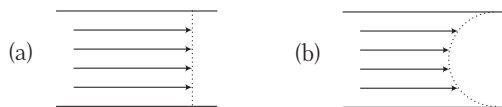
$$= 7.14 \text{ cm}$$

Chapter Exercises

(A) Taking it together

Assorted Questions of the Chapter for Advanced Level Practice

- 1 Hair of shaving brush cling together when it is removed from water due to
 - (a) force of attraction between hair
 - (b) surface tension
 - (c) viscosity of water
 - (d) characteristic property of hair
- 2 Coatings used on raincoat are waterproof because they
 - (a) increases angle of contact
 - (b) decreases angle of contact
 - (c) does not alters angle of contact
 - (d) forms a smooth surface
- 3 Small droplets of liquid are usually more spherical in shape than larger drops of the same liquid because
 - (a) force of surface tension is equal and opposite to the force of gravity
 - (b) force of surface tension predominates the force of gravity
 - (c) force of gravity predominates the force of surface tension
 - (d) force of gravity and force of surface tension act in the same direction and are equal
- 4 Water does not wet an oily glass because
 - (a) cohesive force of oil > adhesive force between oil and glass
 - (b) cohesive force of oil > cohesive force of water
 - (c) oil repels water
 - (d) cohesive force of water > adhesive force between water and oil molecules
- 5 If two identical mercury drops are combined to form a single drop, then its temperature will
 - (a) decrease
 - (b) increase
 - (c) remains the same
 - (d) None of these
- 6 A water proofing agent changes the angle of contact
 - (a) from acute to 90°
 - (b) from obtuse to 90°
 - (c) from an acute to obtuse value
 - (d) from an obtuse to acute value
- 7 A viscous fluid is flowing through a cylindrical tube. The velocity distribution of the fluid is best represented by the diagram



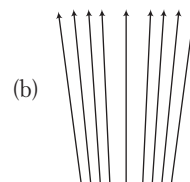
(c)

(d) None of these

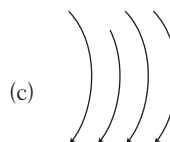
- 8 Which of the following diagrams does not represent a streamline flow?



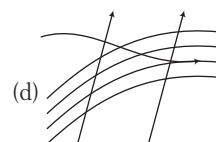
(a)



(b)

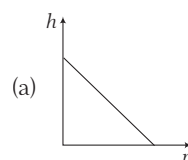


(c)

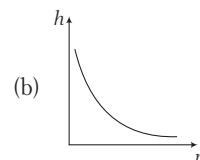


(d)

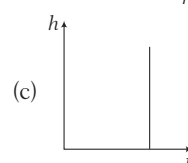
- 9 A body measures 5 N in air and 2 N when put in water. The buoyant force is
 - (a) 7 N
 - (b) 9 N
 - (c) 3 N
 - (d) None of these
- 10 The reading of spring balance when a block is suspended from it in air, is 60 N. This reading is changed to 40 N when the block is immersed in water. The specific gravity of the block is
 - (a) 3
 - (b) 2
 - (c) 6
 - (d) $3/2$
- 11 An iron block is on a boat which floats in a pond. The block is thrown into the water. The level of water in the pond will be
 - (a) equal to the earlier level
 - (b) less than the earlier level
 - (c) more than the earlier level
 - (d) depends on how large the block is
- 12 The correct curve between the height or depression h of liquid in a capillary tube and its radius r is



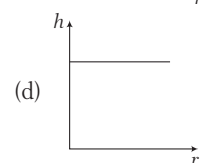
(a)



(b)



(c)



(d)

- 13** When two capillary tubes of different diameters are dipped vertically, the rise of the liquid is
 (a) same in both the tubes
 (b) more in the tube of larger diameter
 (c) less in the tube of smaller diameter
 (d) more in the tube of smaller diameter
- 14** The surface tension of liquid at its boiling point
 (a) becomes zero
 (b) becomes infinity
 (c) is equal to the value at room temperature
 (d) is half to the value at the room temperature
- 15** When the temperature is increased, the angle of contact of a liquid
 (a) increases
 (b) decreases
 (c) remains the same
 (d) first increases and then decreases
- 16** The surface tension of a soap solution is $2 \times 10^{-2} \text{ Nm}^{-1}$. To blow a bubble of radius 1 cm, the work done is
 (a) $4\pi \times 10^{-6} \text{ J}$
 (b) $8\pi \times 10^{-6} \text{ J}$
 (c) $12\pi \times 10^{-6} \text{ J}$
 (d) $16\pi \times 10^{-6} \text{ J}$
- 17** A liquid rises in a vertical tube. The relation between the weight of the liquid in the tube, surface tension of the liquid T and radius of the tube r is given by (if the angle of contact is zero)
 (a) $w = \pi r^2 T$
 (b) $w = 2\pi r T$
 (c) $w = 2r^2 \pi T$
 (d) $w = \frac{3}{4} \pi r^3 T$
- 18** If the angle of contact is less than 90° , the pressure just inside the surface of a meniscus
 (a) is less than atmospheric pressure
 (b) is greater than atmospheric pressure
 (c) is same as the atmospheric pressure
 (d) None of the above
- 19** If two glass plates are quite nearer to each other in water, then there will be force of
 (a) attraction
 (b) repulsion
 (c) attraction or repulsion
 (d) Neither attraction nor repulsion
- 20** The density of ice is $x \text{ g cm}^{-3}$ and that of water is $y \text{ g cm}^{-3}$. When m gram of ice melts, then the change in volume is
 (a) $m(y - x)$
 (b) $\frac{y - x}{m}$
 (c) $my(y - x)$
 (d) $\frac{m}{y} - \frac{m}{x}$
- 21** If a capillary tube is dipped into liquid and the levels of the liquid inside and outside are same, then the angle of contact is
 (a) 120°
 (b) 90°
 (c) 45°
 (d) 30°
- 22** A barometer kept in an elevator reads 76 cm when it is at rest. If the elevator goes up with increasing speed, the reading will be
 (a) zero
 (b) 76 cm
 (c) $> 76 \text{ cm}$
 (d) $< 76 \text{ cm}$
- 23** A hole is made at the bottom of the tank filled with water (density = 1000 kg m^{-3}). If the total pressure at the bottom of the tank is three atmospheres ($1 \text{ atmosphere} = 10^5 \text{ Nm}^{-2}$), then the velocity of efflux is nearest to
 (a) $\sqrt{400} \text{ ms}^{-1}$
 (b) $\sqrt{200} \text{ ms}^{-1}$
 (c) $\sqrt{600} \text{ ms}^{-1}$
 (d) $\sqrt{500} \text{ ms}^{-1}$
- 24** Assuming that the atmosphere has the same density anywhere as at sea level ($\rho = 1.3 \text{ kg m}^{-3}$) and g to be constant ($g = 10 \text{ ms}^{-2}$). What should be the approximate height of atmosphere?
 ($p_0 = 1.01 \times 10^5 \text{ Nm}^{-2}$)
 (a) 6 km
 (b) 8 km
 (c) 12 km
 (d) 18 km
- 25** Two capillary tubes P and Q are dipped in water. The height of water level in capillary P is $2/3$ to the height in Q capillary. The ratio of their diameters is
 (a) 2 : 3
 (b) 3 : 2
 (c) 3 : 4
 (d) 4 : 3
- 26** In a surface tension experiment with capillary tube, water rises upto 0.1 m. If the same experiment is repeated on an artificial satellite, which is revolving around the earth, water will rise in the capillary tube upto a height of
 (a) 0.1 m
 (b) 0.2 m
 (c) 0.98 m
 (d) full length of the capillary tube
- 27** Two parallel glass plates are dipped partly in the liquid of density d keeping them vertical. If the distance between the plates is x , surface tension for the liquid is T and angle of contact θ , then rise of liquid between the plates due to capillary will be
 (a) $\frac{T \cos \theta}{xd}$
 (b) $\frac{2T \cos \theta}{xdg}$
 (c) $\frac{2T}{xdg \cos \theta}$
 (d) $\frac{T \cos \theta}{xdg}$
- 28** A balloon has volume of 2000 m^3 . It is filled with hydrogen ($\rho = 0.009 \text{ g L}^{-1}$). If the density of air is 1.29 g L^{-1} , it can lift a total weight of
 (a) 600 kg
 (b) 2400 kg
 (c) 300 kg
 (d) 1800 kg
- 29** A raft of mass $M = 600 \text{ kg}$ floats in calm water with 7 cm submerged. When a man stands on the raft, it is 1.4 cm more submerged, then the man's mass is
 (a) 30 kg
 (b) 60 kg
 (c) 90 kg
 (d) 120 kg

- 30** A piston of cross-sectional area 100 cm^2 is used in a hydraulic pressure to exert a force of 10^7 dyne on the water. The cross-sectional area of the other piston which supports an object having a mass of 2000 kg is

(a) $9.8 \times 10^2 \text{ cm}^2$ (b) $9.8 \times 10^3 \text{ cm}^2$
(c) $1.96 \times 10^3 \text{ cm}^2$ (d) $1.96 \times 10^4 \text{ cm}^2$

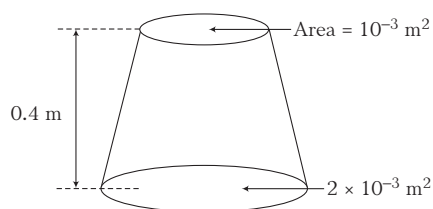
- 31** An ideal fluid flows through two pipes of circular cross-section with diameters 2.5 cm and 3.75 cm connected one after another. The ratio of the velocities in the two pipes is

(a) $9 : 4$ (b) $3 : 2$
(c) $\sqrt{3} : \sqrt{2}$ (d) $\sqrt{2} : \sqrt{3}$

- 32** The angle of contact at the interface of water-glass is 0° , ethyl alcohol-glass is 0° , mercury-glass is 140° and methyl iodide-glass is 30° . A glass capillary is put in a trough containing one of these four liquids. It is observed that the meniscus is convex. The liquid in the trough is

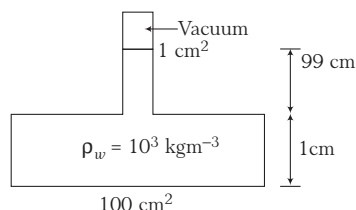
(a) water (b) ethylalcohol
(c) mercury (d) methyl iodide

- 33** A uniformly tapered vessel is filled with a liquid of density 900 kgm^{-3} . The force that acts on the base of the vessel due to the liquid (excluding atmospheric force) is ($g = 10 \text{ ms}^{-2}$)



(a) 3.6 N (b) 7.2 N
(c) 9.0 N (d) 12.6 N

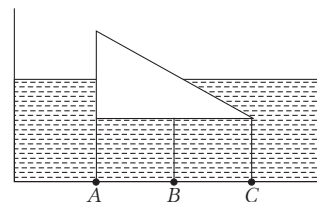
- 34** For the arrangement shown in the figure, the force at the bottom of the vessel is



(a) 200 N (b) 100 N (c) 20 N (d) 2 N

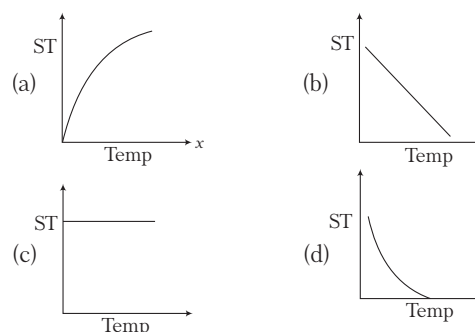
- 35** An object of uniform density is allowed to float in water kept in a beaker. The object has triangular cross-section as shown in the figure.

If the water pressure measured at the three points A , B and C below the object are p_A , p_B and p_C , respectively. Then,



(a) $p_A > p_B > p_C$ (b) $p_A > p_B < p_C$
(c) $p_A = p_B = p_C$ (d) $p_A = p_B < p_C$

- 36** Which graph represents the variation of surface tension with temperature over small temperature ranges for water?



- 37** A solid shell loses half of its weight in water. If relative density of shell is 5, then what fraction of its volume is hollow?

(a) $\frac{3}{5}$ (b) $\frac{2}{5}$ (c) $\frac{1}{5}$ (d) $\frac{4}{5}$

- 38** A tank full of water has a small hole at its bottom. Let t_1 be the time taken to empty the first half of the tank and t_2 be the time needed to empty the rest half of the tank, then

(a) $t_1 = t_2$ (b) $t_1 > t_2$
(c) $t_1 < t_2$ (d) $t_1 = 0.523t_2$

- 39** An open U-tube contains mercury. When 11.2 cm of water is poured into one of the arms of the tube, how high does the mercury rise in the other arm from its initial level?

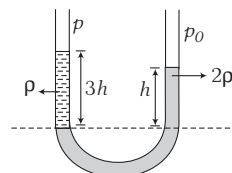
(a) 0.82 cm (b) 1.35 cm
(c) 0.41 cm (d) 2.32 cm

- 40** A small ball (mass m) falling under gravity in a viscous medium experiences a drag force proportional to the instantaneous speed u such that $F_{\text{drag}} = ku$. Then, the terminal speed of ball within viscous medium is

(a) $\frac{k}{mg}$ (b) $\frac{mg}{k}$ (c) $\sqrt{\frac{mg}{k}}$ (d) None of these

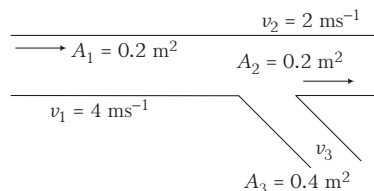
- 41** If a capillary tube of radius r is immersed in water, the mass of water rise in capillary is M . If the radius of capillary be doubled, the mass of water rise in the capillary will be
 (a) $M/2$ (b) M (c) $2M$ (d) $4M$
- 42** The pressure of water in a pipe when tap is closed is $5.5 \times 10^5 \text{ Nm}^{-2}$. When tap gets open, pressure reduces to $5 \times 10^5 \text{ Nm}^{-2}$. The velocity with which water comes out on opening the tap is
 (a) 10 ms^{-1} (b) 5 ms^{-1}
 (c) 20 ms^{-1} (d) 15 ms^{-1}
- 43** The level of water in a tank is 5 m high. A hole of area 1 cm^2 is made at the bottom of the tank. The rate of leakage of water from the hole is (Take, $g = 10 \text{ ms}^{-2}$)
 (a) $10^{-3} \text{ m}^3 \text{ s}^{-1}$ (b) $10^{-4} \text{ m}^3 \text{ s}^{-1}$
 (c) $10 \text{ m}^3 \text{ s}^{-1}$ (d) $10^{-2} \text{ m}^3 \text{ s}^{-1}$
- 44** Water is flowing through two horizontal pipes of different diameters which are connected together. The diameters of the two pipes are 3 cm and 6 cm, respectively. If the speed of water in the narrower tube is 4 ms^{-1} , then the speed of water in the wider tube is
 (a) 16 ms^{-1} (b) 1 ms^{-1} (c) 4 ms^{-1} (d) 2 ms^{-1}
- 45** A block of wood floats in water with $(4/5)$ th of its volume submerged. If the same block just floats in a liquid, the density of the liquid is (in kgm^{-3})
 (a) 1250 (b) 600 (c) 400 (d) 800
- 46** Water flows along a horizontal pipe of non-uniform cross-section. The pressure is 1 cm of Hg, where the velocity is 35 cms^{-1} . At a point, where the velocity is 65 cms^{-1} , the pressure will be
 (a) 0.89 cm of Hg (b) 0.62 cm of Hg
 (c) 0.5 cm of Hg (d) 1 cm of Hg
- 47** Three liquids of equal masses are taken in three identical cubical vessels A, B and C. Their densities are ρ_A , ρ_B and ρ_C respectively but $\rho_A < \rho_B < \rho_C$. The force exerted by the liquid on the base of the cubical vessel is
 (a) maximum in vessel C (b) minimum in vessel C
 (c) the same in all the vessels (d) maximum in vessel A
- 48** An object weighs m_1 in a liquid of density d_1 and that in liquid of density d_2 is m_2 . The density of the object is
 (a) $\frac{m_2 d_2 - m_1 d_1}{m_2 - m_1}$ (b) $\frac{m_1 d_1 - m_2 d_2}{m_2 - m_1}$
 (c) $\frac{m_2 d_1 - m_1 d_2}{m_1 - m_2}$ (d) $\frac{m_1 d_2 - m_2 d_1}{m_1 - m_2}$

- 49** In the figure shown,

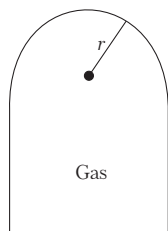


- (a) $p_0 > p$ (b) $p > p_0$ (c) $p = p_0$ (d) $p = 0$
- 50** A boat having a length 3 m and breadth 2 m is floating on a lake. The boat sinks by 1 cm when a man gets on it. The mass of the man is
 (a) 60 kg (b) 72 kg (c) 52 kg (d) 65 kg
- 51** A small block of wood of relative density 0.5 is submerged in water. When the block is released, it starts moving upwards, the acceleration of the block is (Take, $g = 10 \text{ ms}^{-2}$)
 (a) 5 ms^{-2} (b) 10 ms^{-2} (c) 7.5 ms^{-2} (d) 15 ms^{-2}
- 52** A raft of wood of mass 120 kg floats in water. The weight that can be put on the raft to make it just sink, should be (Take, $d_{\text{raft}} = 600 \text{ kgm}^{-3}$)
 (a) 80 kg (b) 50 kg (c) 60 kg (d) 30 kg
- 53** A stone of relative density K is released from rest on the surface of a lake. If viscous effects are ignored, the stone sinks in water with an acceleration of
 (a) $g(1 - K)$ (b) $g(1 + K)$
 (c) $g\left(1 - \frac{1}{K}\right)$ (d) $g\left(1 + \frac{1}{K}\right)$
- 54** A body floats in a liquid contained in a beaker. The whole system shown in figure is falling under gravity, the upthrust on the body due to the liquid is
 (a) zero
 (b) equal to weight of the body in air
 (c) equal to weight of liquid displaced
 (d) equal to the weight of the immersed part of the body
-
- 55** The relative density of ice is 0.9 and that of sea water is 1.125. What fraction of the whole volume of an iceberg appears above the surface of the sea?
 (a) $1/5$ (b) $2/5$
 (c) $3/5$ (d) $4/5$
- 56** A metallic sphere floats in immiscible mixture of water (density 10^3 kgm^{-3}) and a liquid (density $8 \times 10^3 \text{ kgm}^{-3}$) such that its $(2/3)$ part is in water and $(1/3)$ part in the liquid. The density of the metal is
 (a) $\frac{5000}{3} \text{ kgm}^{-3}$ (b) $\frac{10000}{3} \text{ kgm}^{-3}$
 (c) 5000 kgm^{-3} (d) 2000 kgm^{-3}

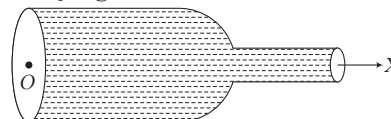
- 57 In the given figure, the velocity v_3 will be



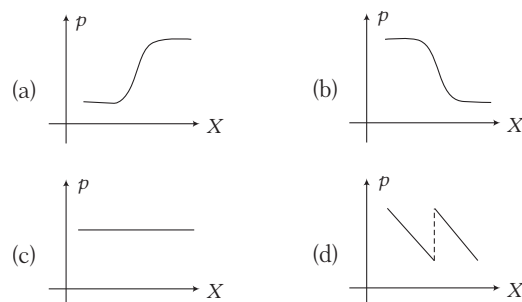
- (a) 2 ms^{-1} (b) 4 ms^{-1}
(c) 1 ms^{-1} (d) 3 ms^{-1}
- 58 A water tank standing on the floor has two small holes punched in the vertical wall one above the other. The holes are 2.4 cm and 7.6 cm above the floor. If the jet of water from the holes hit the floor at the same point, then the height of water in the tank is
- (a) 10 cm (b) 5 cm (c) 20 cm (d) 4.8 cm
- 59 A body of volume V and density ρ is initially submerged in a non-viscous liquid of density $\sigma (> \rho)$. If it rises by itself through a height h in the liquid, its kinetic energy will
- (a) increase by $hV(\sigma - \rho)g$ (b) increase by $hV(\rho + \sigma)g$
(c) increase by $\frac{hV\rho g}{\sigma}$ (d) decrease by $\frac{hV\rho g}{\sigma}$
- 60 The pressure of the gas in a cylindrical chamber is p_0 . The vertical force exerted by the gas on its hemispherical end is



- (a) $p_0 r^2$ (b) $4p_0 \pi r^2$
(c) $2p_0 \pi r^2$ (d) $p_0 \pi r^2$
- 61 Water rises in a capillary tube to a certain height such that the upward force due to surface tension is balanced by $75 \times 10^{-4} \text{ N}$ force due to the weight of the liquid. If the surface tension of water is $6 \times 10^{-2} \text{ N m}^{-1}$, the inner circumference of the capillary must be
- (a) $1.25 \times 10^{-2} \text{ m}$ (b) $0.50 \times 10^{-2} \text{ m}$
(c) $6.5 \times 10^{-2} \text{ m}$ (d) $12.5 \times 10^{-2} \text{ m}$
- 62 Water rises to a height of 30 mm in a capillary tube. If the radius of the capillary tube is made $3/4$ of its previous value. The height to which the water will rise in the tube is
- (a) 30 mm (b) 20 mm (c) 40 mm (d) 10 mm
- 63 A vessel whose bottom has round holes with diameter of 1.0 mm is filled with water. The maximum height to which the water can be filled without leakage is (Take, surface tension of water $= 75 \text{ dyne cm}^{-1}$, $g = 1000 \text{ cms}^{-2}$)
- (a) 10 cm (b) 7.5 cm (c) 5 cm (d) 3 cm
- 64 If a water drop is kept between two glass plates, then its shape is
- (a) (b)
(c) (d) None of these
- 65 Two small drops of mercury each of radius r form a single large drop. The ratio of surface energy before and after this change is
- (a) $2 : 2^{2/3}$ (b) $2^{2/3} : 1$ (c) $2 : 1$ (d) $1 : 2$
- 66 A non-viscous liquid is flowing through a frictionless duct of varying cross-section as shown in figure.



Which of the following graph represents the variation of pressure p along the axis of tube?



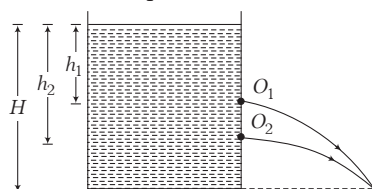
- (a) (b)
(c) (d)
- 67 There is hole of area a at the bottom of a cylinder of area A . Water is filled upto a height h and water flows out in t second. If water is filled to a height $4h$, it will flow out in time
- (a) $\frac{t}{4}$ (b) $2t$ (c) $4t$ (d) $\frac{t}{2}$
- 68 A container has a small hole at its bottom. Area of cross-section of the hole is A_1 and that of the container is A_2 . Liquid is poured in the container at a constant rate $Q \text{ m}^3 \text{ s}^{-1}$. The maximum level of liquid in the container will be
- (a) $\frac{Q^2}{2g A_1 A_2}$ (b) $\frac{Q^2}{2g A_1^2}$ (c) $\frac{Q}{2g A_1 A_2}$ (d) $\frac{Q^2}{2g A_2^2}$

- 69** Air stream flows horizontally past an aeroplane wing of surface area 4 m^2 . The speed of air over the top surface is 60 ms^{-1} and under the bottom surface is 40 ms^{-1} . The force of lift on the wing is (Take, density of air = 1 kgm^{-3})
 (a) 800 N (b) 1000 N (c) 4000 N (d) 3200 N

- 70** Water from a tap emerges vertically down with an initial speed of 1.0 ms^{-1} . The cross-sectional area of tap is 10^{-4} m^2 . Assume that, the pressure is constant throughout the stream of water and that the flow is steady. The cross-sectional area of the stream 0.15 m below the tap is

- (a) $5.0 \times 10^{-4} \text{ m}^2$ (b) $1.0 \times 10^{-5} \text{ m}^2$
 (c) $5.0 \times 10^{-5} \text{ m}^2$ (d) $2.0 \times 10^{-5} \text{ m}^2$

- 71** There are two holes O_1 and O_2 in a tank of height H . The water emerging from O_1 and O_2 strikes the ground at the same points as shown in figure. Then,



- (a) $H = h_1 + h_2$ (b) $H = h_2 - h_1$
 (c) $H = \sqrt{h_1 h_2}$ (d) None of these

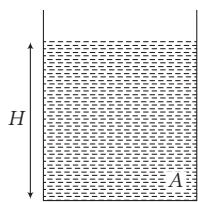
- 72** Two rain drops of same radii r falling with terminal velocity v merge and form a bigger drop of radius R . The terminal velocity of the bigger drop will be

- (a) $v \frac{R}{r}$ (b) $v \frac{R^2}{r^2}$ (c) v (d) $2v$

- 73** A cubical block of steel of each side equal to l is floating on mercury in a vessel. The densities of steel and mercury are ρ_s and ρ_m . The height of the block above the mercury level is given by

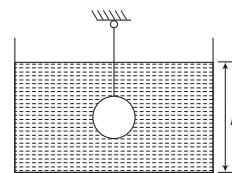
- (a) $l \left(1 + \frac{\rho_s}{\rho_m} \right)$ (b) $l \left(1 - \frac{\rho_s}{\rho_m} \right)$ (c) $l \left(1 + \frac{\rho_m}{\rho_s} \right)$ (d) $l \left(1 - \frac{\rho_m}{\rho_s} \right)$

- 74** A cylindrical tank contains water up to a height H . If the tank is accelerated upwards with acceleration a , the pressure at the point A is p_1 . If the tank is accelerated downwards with acceleration a , the pressure at A is p_2 . Then,



- (a) $p_1 < p_2$ (b) $p_1 = p_2$
 (c) $p_1 > p_2$ (d) Data insufficient

- 75** A metal sphere connected by a string is dipped in a liquid of density ρ as shown in figure. The pressure at the bottom of the vessel will be (p_0 = atmospheric pressure)



- (a) $p = p_0 + \rho gh$ (b) $p > p_0 + \rho gh$
 (c) $p < p_0 + \rho gh$ (d) p_0

- 76** An air bubble of radius 1 mm is formed inside water at a depth 10 m below free surface (where, air pressure is 10^5 Nm^{-2}). The pressure inside the bubble is (surface tension of water = $7 \times 10^{-2} \text{ Nm}^{-1}$)

- (a) $2.28 \times 10^5 \text{ Nm}^{-2}$ (b) $2.0028 \times 10^5 \text{ Nm}^{-2}$
 (c) $2.14 \times 10^5 \text{ Nm}^{-2}$ (d) $2.0014 \times 10^5 \text{ Nm}^{-2}$

- 77** A block of mass 4 kg and volume $5 \times 10^{-4} \text{ m}^3$ is suspended by a spring balance in a lift which is accelerating. The apparent weight shown by the spring balance is 3 kg. Now, the block is immersed in water in a container inside the lift. The apparent weight (in kg) shown by the spring balance is
 (a) 2.375 (b) 0.25 (c) 2.5 (d) 3.125

- 78** A cubical block of wood of specific gravity 0.5 and chunk of concrete of specific gravity 2.5 are fastened together. The ratio of the mass of wood to the mass of concrete, which makes the combination to float with its entire volume submerged under water is
 (a) 1/5 (b) 1/3 (c) 3/5 (d) 2/3

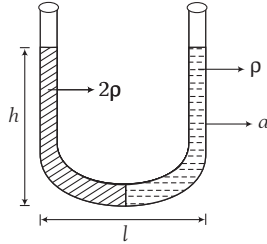
- 79** Two identical cylindrical vessels, each of base area A , have their bases at the same horizontal level. They contain a liquid of density ρ . In one vessel, the height of the liquid is h_1 and in the other h_2 ($> h_1$). When the two vessels are connected, the work done by gravity in equalising the levels is

- (a) $2\rho Ag (h_2 - h_1)^2$ (b) $\rho Ag (h_2 - h_1)^2$
 (c) $\frac{1}{2} \rho Ag (h_2 - h_1)^2$ (d) $\frac{1}{4} \rho Ag (h_2 - h_1)^2$

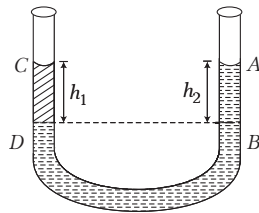
- 80** A small ball of density ρ is immersed in a liquid of density σ ($\sigma > \rho$) to a depth h and then released. The height above the surface of water up to which the ball will jump is

- (a) $\frac{\sigma h}{\rho}$ (b) $\left(\frac{\sigma}{\rho} - 1 \right) h$
 (c) $\left(1 - \frac{\sigma h}{\rho} \right) h$ (d) $\frac{\rho h}{\sigma}$

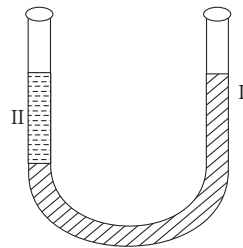
- 81** A U-tube of base length l filled with same volume of two liquids of densities ρ and 2ρ is moving with an acceleration a on the horizontal plane. If the height difference between the two surfaces (open to atmosphere) becomes zero, then height h is given by



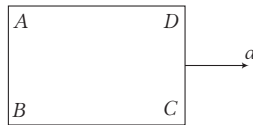
- (a) $\frac{al}{g}$ (b) $\frac{3al}{2g}$ (c) $\frac{2al}{3g}$ (d) $\frac{al}{2g}$
- 82** In a U-tube experiment, a column AB of water is balanced by a column CD of paraffin. The relative density of paraffin is



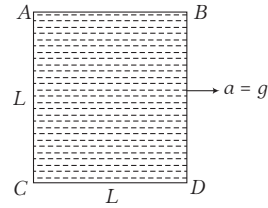
- (a) $\frac{h_2}{h_1}$ (b) $\frac{h_1}{h_2}$ (c) $\frac{h_2 - h_1}{h_1}$ (d) $\frac{h_2}{h_1 + h_2}$
- 83** A U-tube of uniform cross-section shown in figure is partially filled with liquid I. Another liquid II which does not mix with I is poured into one side. The liquid levels of the two sides are found to be same, while the level of liquid I has risen by 2 cm. If the specific gravity of liquid I is 1.1, then specific gravity of liquid II must be



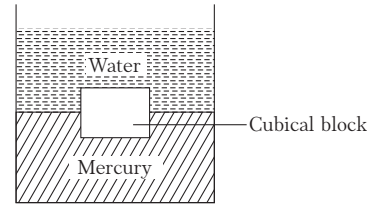
- (a) 1.2 (b) 1.1 (c) 1.3 (d) 1.0
- 84** A closed rectangular tank is completely filled with water and is accelerated horizontally with an acceleration towards right. Pressure is
- (i) maximum at and (ii) minimum at
- (a) (i) B (ii) D (b) (i) C (ii) D
(c) (i) B (ii) C (d) (i) B (ii) A



- 85** The liquid inside the container has density ρ . Choose the correct option.

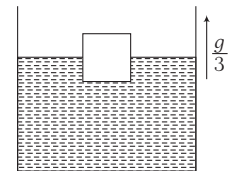


- (a) $p_A - p_C = 2\rho gL$ (b) $p_C - p_B = \sqrt{2}\rho gL$
(c) $p_C - p_D = \rho gL$ (d) $p_A - p_D = 0$
- 86** A tank contains water on top of mercury as shown in figure. A cubical block of side 10 cm is in equilibrium inside the tank. The depth of the block inside mercury is (Take, relative density of the material of block = 8.56, relative density of mercury = 13.6)

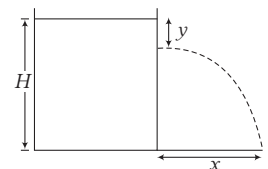


- (a) 6 cm (b) 5 cm (c) 7 cm (d) 8 cm
- 87** If T is the surface tension of a liquid, the energy needed to break a liquid drop of radius R into 64 drops is
- (a) $6\pi R^2 T$ (b) $\pi R^2 T$ (c) $12\pi R^2 T$ (d) $8\pi R^2 T$
- 88** Two solid spheres of same metal but of mass M and $8M$ fall simultaneously on a viscous liquid and their terminal velocities are v and nv , then value of n is
- (a) 16 (b) 8 (c) 4 (d) 2

- 89** A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of $g/3$, the fraction of volume immersed in the liquid will be
- (a) $1/2$ (b) $3/8$ (c) $2/3$ (d) $3/4$



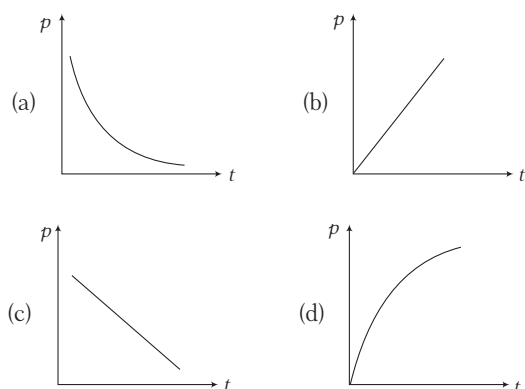
- 90** A cylindrical vessel is filled with a liquid upto a height H . A small hole is made in the vessel at a distance y below the liquid surface as shown in figure. The liquid emerging from the hole strike the ground at distance x , then
- (a) x is equal, if hole is at depth y or $H - y$
(b) x is maximum for $y = H/2$
(c) Both (a) and (b) are correct
(d) Both (a) and (b) are incorrect



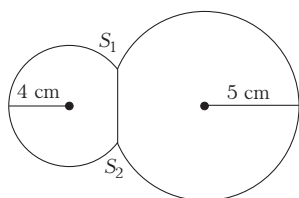
- 91 Two capillary of lengths L and $2L$ and of radius R and $2R$ are connected in series. The net rate of flow of fluid through them will be (given rate of the flow through single capillary, $X = \pi p R^4 / 8 \eta L$)

(a) $\frac{8}{9}X$ (b) $\frac{9}{8}X$ (c) $\frac{5}{7}X$ (d) $\frac{7}{5}X$

- 92 A soap bubble is blown with the help of a mechanical pump at the mouth of a tube. The pump produces a constant increase per minute in the radius of the bubble, irrespective of its internal pressure. The graph between the excess pressure inside the soap bubble and time t will be



- 93 Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be

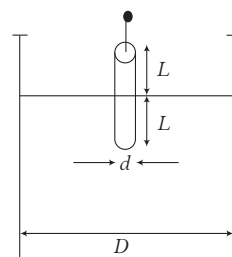


- (a) 4 cm (b) 20 cm (c) 5 cm (d) 4.5 cm
- 94 Two substances of relative densities ρ_1 and ρ_2 are mixed in equal volume and the relative density of mixture is 4. When they are mixed in equal masses, the relative density of the mixture is 3. The values of ρ_1 and ρ_2 are
- (a) 6 and 2 (b) 3 and 5
(c) 12 and 4 (d) None of these
- 95 A large block of ice 10 cm thick with a vertical hole drilled through it is floating in a lake. The minimum length of the rope required to scoop out a bucket full of water through the hole is (Take, density of ice = 0.9 g cm^{-3})
- (a) 0.5 cm (b) 1.0 cm (c) 1.2 cm (d) 1.8 cm

- 96 A body of density ρ is dropped from rest from a height h into a lake of density σ ($\sigma > \rho$). The maximum depth to which the body sinks inside the liquid is (neglect viscous effect of liquid)

(a) $\frac{h\rho}{\sigma - \rho}$ (b) $\frac{h\sigma}{\sigma - \rho}$ (c) $\frac{h\rho}{\sigma}$ (d) $\frac{h\sigma}{\rho}$

- 97 A candle of diameter d is floating on a liquid in a cylindrical container of diameter D ($D \gg d$) as shown in figure. If it is burning at the rate of 2 cm h^{-1} , then the top of the candle will



- (a) remain at the same height
(b) fall at the rate 1 cm h^{-1}
(c) fall at the rate of 2 cm h^{-1}
(d) go up at the rate of 1 cm h^{-1}
- 98 A spherical object of mass 1 kg and radius 1 m is falling vertically downward inside a viscous liquid in a gravity free space. At a certain instant, the velocity of the sphere is 2 m s^{-1} . If the coefficient of viscosity of the liquid is $\frac{1}{6\pi}$ SI units, then velocity of ball will become 0.5 m s^{-1} after a time is
- (a) $\ln(4) \text{ s}$ (b) $2 \ln(4) \text{ s}$
(c) $3 \ln(4) \text{ s}$ (d) $3 \ln(2) \text{ s}$
- 99 A wooden block of mass 8 kg is tied to a string attached to the bottom of the tank. In the equilibrium, the block is completely immersed in water. If relative density of wood is 0.8 and $g = 10 \text{ ms}^{-2}$, the tension T in the string is
- (a) 120 N (b) 100 N (c) 80 N (d) 20 N
- 100 A metal ball immersed in alcohol weighs w_1 at 0°C and w_2 at 59°C . The coefficient of cubical expansion of the metal is less than that of alcohol. Assuming that the density of the metal is large compared to that of alcohol, it can be shown that
- (a) $w_1 > w_2$ (b) $w_1 = w_2$
(c) $w_1 < w_2$ (d) $w_1 = (w_2/2)$
- 101 The surface energy of a liquid drop is S . It is sprayed into 1000 equal droplets, then its surface energy becomes
- (a) S (b) $10 S$ (c) $100 S$ (d) $1000 S$

- 102** An open tank containing non-viscous liquid to a height of 5 m is placed over the ground. A heavy spherical ball falls from height 40 m over the ground in the tank. Ignoring air resistance, find the height to which ball will go back. Collision between ball and bottom of tank is perfectly elastic
- (a) 45 m (b) 35 m
(c) 40 m (d) 20 m

- 103** A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth $4y$ from the top. When the tank is completely filled with water, the quantities of water flowing out per second from holes are the same. Then, R is equal to
- (a) $L/\sqrt{2\pi}$ (b) $2\pi L$
(c) L (d) $L/2\pi$

- 104** A piece of steel has a weight w in air, w_1 when completely immersed in water and w_2 when completely immersed in an unknown liquid. The relative density (specific gravity) of liquid is
- (a) $\frac{w - w_1}{w - w_2}$ (b) $\frac{w - w_2}{w - w_1}$ (c) $\frac{w_1 - w_2}{w - w_1}$ (d) $\frac{w_1 - w_2}{w - w_2}$

- 105** Two cylinders of same cross-section and length L but made of two materials of densities d_1 and d_2 are connected together to form a cylinder of length $2L$. The combination floats in a liquid of density d with a length $L/2$ above the surface of the liquid. If $d_1 < d_2$, then

- (a) $d_1 < \frac{3}{4}d$ (b) $\frac{d}{2} > d_1$ (c) $\frac{d}{4} > d_1$ (d) $d_1 > \frac{d}{4}$

- 106** A block of wood is floating on the surface of water in a beaker. The beaker is covered with a bell jar and the air is evacuated. What will happen to the block?

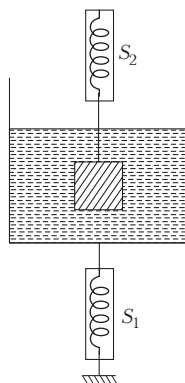
- (a) Sink a little (b) Rise a little
(c) Remain unchanged (d) Sink completely

- 107** A beaker containing water is kept on a spring scale. The mass of water and beaker is 5 kg. A block of mass 2 kg and specific gravity 10 is suspended by means of thread from a spring balance as shown in figure.

The readings of scales S_1 and S_2 are respectively

(Take, $g = 10 \text{ ms}^{-2}$)

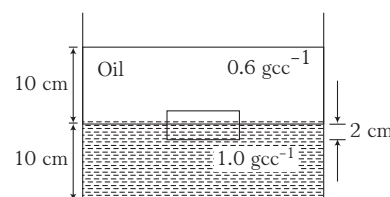
- (a) 52 N and 20 N
(b) 50 N and 18 N
(c) 52 N and 18 N
(d) 52 N and 22 N



- 108** A sphere of solid material of specific gravity 8 has a concentric spherical cavity and just sinks in water. Then, the ratio of the radius of the cavity to the outer radius of the sphere must be

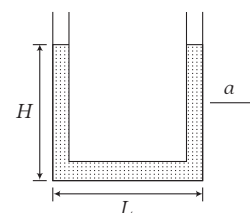
- (a) $\frac{\sqrt[3]{3}}{2}$ (b) $\frac{\sqrt[3]{5}}{2}$ (c) $\frac{\sqrt[3]{7}}{2}$ (d) $\frac{2}{\sqrt[3]{7}}$

- 109** A cubical block of side 10 cm floats at the interface of an oil and water. The pressure above that of atmosphere at the lower face of the block is



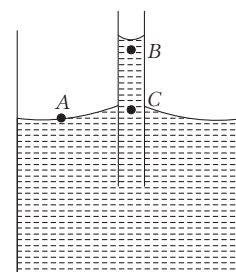
- (a) 200 Nm^{-2} (b) 680 N m^{-2}
(c) 400 Nm^{-2} (d) 800 Nm^{-2}

- 110** A liquid stands at the plane level in U-tube when at rest. If areas of cross-section of both the limbs are equal, what will be the difference in heights h of the liquid in the two limbs of U-tube, when the system is given an acceleration a in horizontal direction towards right as shown in figure?



- (a) $\frac{g L^2}{a H}$ (b) $\frac{L a}{g}$ (c) $\frac{L^2 a}{H g}$ (d) $\frac{L g}{a}$

- 111** A capillary tube is dipped in a liquid. Let pressure at points A, B and C be p_A , p_B , p_C respectively, then



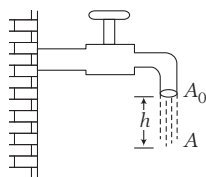
- (a) $p_A = p_B = p_C$ (b) $p_A = p_B < p_C$
(c) $p_A = p_C < p_B$ (d) $p_A = p_C > p_B$

- 112** The volume of an air bubble becomes three times as it rises from the bottom of a lake to its surface. Assuming temperature to be constant and

atmospheric pressure to be 75 cm of Hg and the density of water to be $1/10$ of the density of the mercury, the depth of the lake is

- (a) 5 m (b) 10 m
(c) 15 m (d) 20 m

- 113** Figure shows how the stream of water emerges from a faucet necks down as it falls. The area changes from A_0 to A through a fall of h . At what rate does the water flow from the tap?



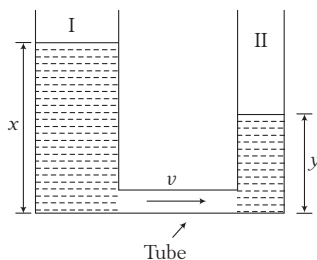
- (a) $A_0 \sqrt{\frac{2ghA^2}{A_0^2 - A^2}}$ (b) $2A_0 \sqrt{\frac{ghA^2}{A_0^2 - A^2}}$
(c) $A_0 \sqrt{\frac{gh}{2}}$ (d) $2A \sqrt{\frac{ghA_0^2}{A_0^2 - A^2}}$

- 114** A pump is designed as a horizontal cylinder with a piston area A and an outlet orifice arranged near the axis of the cylinder. Find the velocity of outflow of liquid from pump, if the piston moves with a constant velocity under the action of a constant force F and the density of liquid is ρ .



- (a) $\sqrt{\frac{F}{A\rho}}$ (b) $\sqrt{\frac{2F}{A\rho}}$ (c) $\sqrt{\frac{A\rho}{F}}$ (d) $\sqrt{\frac{A\rho}{2F}}$

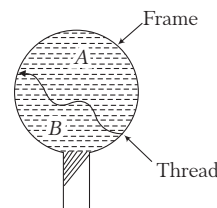
- 115** If cross-sectional area of limb I is A_1 and that of limb II is A_2 , then velocity of the liquid in the tube will be (cross-sectional area of tube is very small)



- (a) $\sqrt{2g(x-y)}$ (b) $\frac{A_1}{A_2} \sqrt{2g(x-y)}$
(c) $\frac{A_2}{A_1} \sqrt{2g(x-y)}$ (d) None of these

- 116** Two capillaries of same length and radii in the ratio 1:2 are connected in series. A liquid flows through them in streamlined condition. If the pressure across the two extreme ends of the combination is 1 m of water, then pressure difference across first capillary is
- (a) 9.4 m (b) 4.9 m (c) 0.49 m (d) 0.94 m

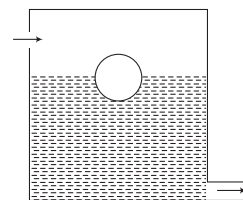
- 117** A thread is tied slightly loose to a wire frame as shown in figure and the frame is dipped into a soap solution and taken out. The frame is completely covered with the film.



When A is pricked

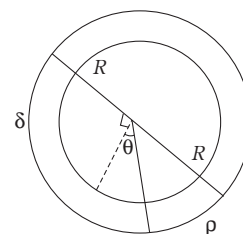
- (a) thread will become concave on seeing from side A
(b) thread will become concave on seeing from side B
(c) thread will become straight
(d) thread will remain as it is

- 118** A tank filled with water has two taps to exhaust and pour. A hollow spherical ball is half submerged in water. Through one tap, water is taken out and through another tap, a liquid of density double the density of water is poured in tank such that volume of liquid in tank remains constant. Sphere will



- (a) go down (b) go up
(c) maintain same height (d) sink to bottom

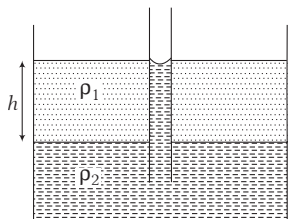
- 119** A uniform long tube is bent into a circle of radius R and it lies in vertical plane. Two liquids of same volume but densities ρ and δ fill half the tube. The angle θ is



- (a) $\tan^{-1} \left(\frac{\rho - \delta}{\rho + \delta} \right)$ (b) $\tan^{-1} \left(\frac{\rho}{\delta} \right)$
(c) $\tan^{-1} \left(\frac{\delta}{\rho} \right)$ (d) $\tan^{-1} \left(\frac{\rho + \delta}{\rho - \delta} \right)$

- 120** A container has two immiscible liquids of densities ρ_1 and ρ_2 ($> \rho_1$). A capillary tube of radius r is inserted in the liquid, so that its bottom reaches upto the denser liquid. The denser liquid rises in the capillary and attains a height h from the interface

of the liquids, which is equal to the column length of the lighter liquid. Assuming angle of contact to be zero, the surface tension of denser liquid is



- (a) $2\pi r \rho_2 gh$ (b) $\frac{\rho_2 r gh}{2}$
 (c) $\frac{r}{2} (\rho_2 - \rho_1) gh$ (d) $2\pi r (\rho_2 - \rho_1) gh$

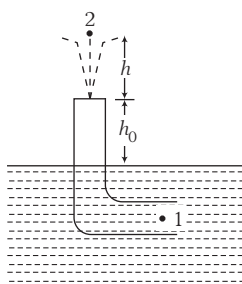
- 121** A ball of mass 1 kg falls from a height of 5m above the free surface of water. The relative density of the solid ball is $s = 2/3$. The ball travels a distance of 2m under water and becomes stationary. The work done by the resistive forces of water is

- (a) -50 J (b) -20 J (c) -40 J (d) -30 J

- 122** Under isothermal condition, two soap bubbles of radii r_1 and r_2 coalesce to form a single bubble of radius r . The external pressure is p_0 . The surface tension of the soap in terms of the given parameters is

- (a) $\frac{2p_0(r^3 - r_1^3 - r_2^3)}{4(r_1^2 + r_2^2 - r^2)}$ (b) $\frac{p_0(r^3 - r_1^3 - r_2^3)}{4(r_1^2 + r_2^2 - r^2)}$
 (c) $\frac{p_0(r^3 + r_1^2 + r_2^3)}{4(r_1^2 + r_2^2 + r^2)}$ (d) None of these

- 123** A bent tube is lowered into the stream as shown in the figure below. The velocity of the stream relative to the tube equal to v . The closed upper end of the tube located at the height h_0 . To what height, h will the water jet spurt?

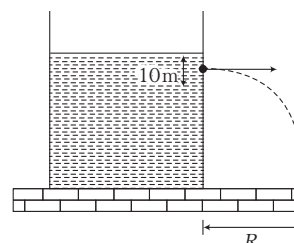


- (a) $\frac{v^2}{2g}$ (b) $\frac{v^2}{2g} + h_0$
 (c) $\frac{v^2}{2g} - h_0$ (d) $\frac{v^2}{4g} - h_0$

- 124** A solid ball of density half that of water falls freely under gravity from a height of 19.6 m and then enters water. How much time will it take to come again to the water surface? Neglect air resistance and viscosity effects in water. (Take, $g = 9.8 \text{ ms}^{-2}$)

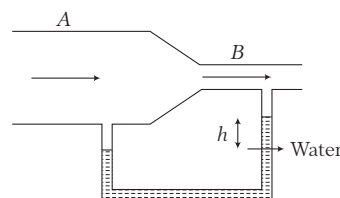
- (a) 4 s (b) 8 s
 (c) 6 s (d) 2 s

- 125** A large tank is filled with water (density $= 10^3 \text{ kgm}^{-3}$). A small hole is made at a depth 10 m below water surface. The range of water flowing out of the hole is R on ground. What extra pressure must be applied on the water surface, so that the range become $2R$? (Take, $1 \text{ atm} = 10^5 \text{ Pa}$ and $g = 10 \text{ ms}^{-2}$)



- (a) 1 atm (b) 2 atm
 (c) 4 atm (d) 3 atm

- 126** Air is blown through a pipe AB at a rate of 15 Lmin^{-1} . The cross-sectional area of the broad portion of the pipe AB is 2 cm^2 and that of the narrow portion is 0.5 cm^2 . The difference in water level h is (Take, density of air $= 1.32 \text{ kg m}^{-3}$)



- (a) 16 mm (b) 1.5 mm
 (c) 10 mm (d) 3.2 mm

(B) Medical entrance special format questions

Assertion and reason

Direction (Q. Nos. 1-4) *These questions consist of two statements each printed as Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses.*

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is correct but Reason is incorrect.
- (d) If Assertion is incorrect but Reason is correct.

1 Assertion If angle of contact is 90° , then liquid will neither rise nor fall in a capillary.

Reason When angle of contact is 90° , surface is neither convex nor concave inside the capillary. It is flat.

2 Assertion When an ideal fluid flows through a pipe of non-uniform cross-section, then pressure is more at that section, where area is more, if the pipe is horizontal.

Reason According to Bernoulli's theorem, speed at broader cross-section will be less.

3 Assertion A solid is floating in a liquid of density ρ_1 . When the solid melts its density becomes ρ_2 in liquid state.

If $\rho_1 > \rho_2$ level of liquids will increase after melting.

Reason In liquid state, volume always increases after a solid melts.

4 Assertion A solid sphere and a hollow sphere both of same material are immersed in a liquid, then change in weight in both the spheres will be same.

Reason Upthrust depends upon the volume of the solid immersed not the mass.

2 Which of the following statement is true in case when two water drops coalesce and make a bigger drop?

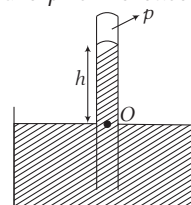
- (a) Energy is released.
- (b) Energy is absorbed.
- (c) The surface area of the bigger drop is greater than the sum of the surface areas of both the drops.
- (d) The surface area of the bigger drop is same that of the sum of the surface areas of both the drops.

3 Which of the following statement(s) is/are correct?

- I. An ice ball is floating in water. Some stone pieces are embedded inside the ice. When ice will melt, level of water will fall.
- II. In floating condition, stone pieces will displace more liquid compared to the condition when they sink.
- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) None of the above

Match the columns

1 A tube is inverted in a mercury vessel as shown in figure. If pressure p is increased, then



Column I		Column II	
(A)	Height h	(p)	will increase
(B)	Pressure at O	(q)	will decrease
(C)	Pressure at 1 cm above O	(r)	will remain same

Match the columns and marks the correct option from the codes given below.

Codes

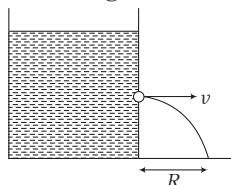
- | | | |
|-----|---|---|
| A | B | C |
| (a) | p | q |
| (b) | p | r |
| (c) | q | r |
| (d) | r | p |

Statement based questions

1 Which of the following statement is incorrect regarding surface tension?

- (a) Dancing of a camphor piece over the surface of water.
- (b) Small mercury drop itself becomes spherical.
- (c) A liquid surface comes at rest after stirring.
- (d) Mercury does not wet the glass vessel.

- 2 In the figure shown velocity of liquid which comes out is v , time of liquid to fall to ground is t and range on ground is R . If the vessel is taken to a mountain, match the following (consider all cases which might possible) columns and mark the correct option from the codes given below.

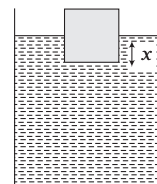


Column I		Column II	
(A)	v	(p)	will increase
(B)	t	(q)	will decrease
(C)	R	(r)	will remain same

Codes

A	B	C	A	B	C
(a) p	q	r	(b) q	p	r
(c) p	r	q	(d) q	r	p

- 3 A cube is floating in liquid as shown in figure. Match the following columns and mark the correct option from the codes given below.



Column I		Column II	
(A)	If density of liquid decreases, x will	(p)	increase
(B)	If size of cube is increased, x will	(q)	decrease
(C)	If the whole system is accelerated upwards, x will	(r)	remain same

Codes

A	B	C	A	B	C
(a) p	p	r	(b) q	r	p
(c) r	p	r	(d) p	r	p

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1 A capillary tube of radius r is immersed in water and water rises in it to a height h . The mass of the water in the capillary tube is 5 g. Another capillary tube of radius $2r$ is immersed in water. The mass of water that will rise in this tube is [NEET 2020]

- (a) 5.0 g (b) 10.0 g
(c) 20.0 g (d) 2.5 g

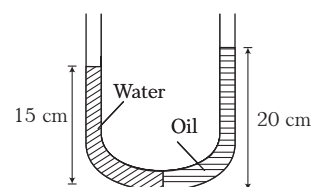
- 2 A liquid does not wet the solid surface, if the angle of contact is [NEET 2020]

- (a) equal to 45°
(b) equal to 60°
(c) greater than 90°
(d) zero

- 3 A barometer is constructed using a liquid (density = 760 kg/m^3). What would be the height of the liquid column, when a mercury barometer reads 76 cm? (Take, density of mercury = 13600 kg/m^3) [NEET 2020]

- (a) 1.36 m (b) 13.6 m
(c) 136 m (d) 0.76 m

- 4 In a U-tube as shown in the figure, water and oil are in the left side and right side of the tube, respectively. The heights from the bottom for water and oil columns are 15 cm and 20 cm, respectively. The density of the oil is (Take, $\rho_{\text{water}} = 1000 \text{ kg/m}^3$) [NEET (Odisha) 2019]

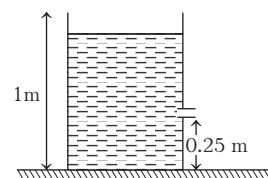


- (a) 1200 kgm^{-3} (b) 750 kgm^{-3}
(c) 1000 kgm^{-3} (d) 1333 kgm^{-3}

- 5 Two small spherical metal balls having equal masses are made from materials of densities ρ_1 and ρ_2 ($\rho_1 = 8\rho_2$) and have radii of 1 mm and 2 mm, respectively. They are made to fall vertically (from rest) in viscous medium whose coefficient of viscosity equals η and whose density is $0.1 \rho_2$. The ratio of their terminal velocities would be [NEET (Odisha) 2019]

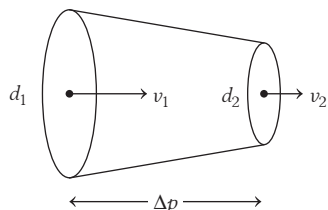
- (a) $\frac{79}{72}$ (b) $\frac{19}{36}$ (c) $\frac{39}{72}$ (d) $\frac{79}{36}$

- 6 If a small orifice is made at a height of 0.25 m from the ground, the horizontal range of water stream will be [AIIMS 2019]

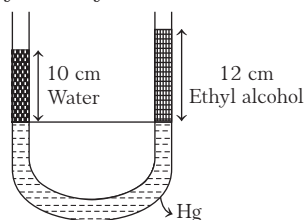


- (a) 46.5 cm (b) 56.6 cm (c) 76.6 cm (d) 86.6 cm

- 7 Determine the pressure difference in tube of non-uniform cross-sectional area as shown in figure.
 $\Delta p = ?$, $d_1 = 5 \text{ cm}$, $v_1 = 4 \text{ m/s}$, $d_2 = 2 \text{ cm}$, $v_2 = ?$ [AIIMS 2019]

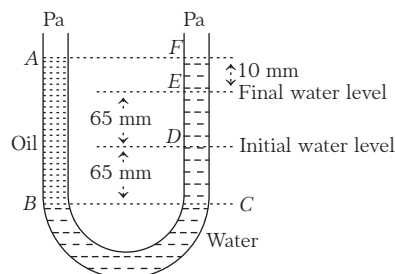


- (a) 304200 Pa (b) 304500 Pa
 (c) 302500 Pa (d) 303500 Pa
- 8 **Assertion** Sometimes insects can walk on water.
Reason The gravitational force on insect is balanced by force due to surface tension. [AIIMS 2019]
 (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Both Assertion and Reason are incorrect.
 (d) Assertion is incorrect but Reason is correct.
- 9 **Assertion** Water drops take spherical shape when falling freely.
Reason Water has minimum surface tension among all liquids. [AIIMS 2019]
 (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Both Assertion and Reason are incorrect.
 (d) Assertion is incorrect but Reason is correct.
- 10 In an isothermal process, 2 water drops of radius 1 mm are combined to form a bigger drop. Find the energy change in this process, if $T = 0.1 \text{ N/m}$. [AIIMS 2019]
 (a) 1 μJ (b) 0.5 μJ (c) 0.25 μJ (d) 0.75 μJ
- 11 A small sphere of radius r falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to [NEET 2018]
 (a) r^5 (b) r^2 (c) r^3 (d) r^4
- 12 Find density of ethyl alcohol. [JIPMER 2018]



- (a) 0.83 g cm^{-3} (b) 0.5 g cm^{-3} (c) 1.83 g cm^{-3} (d) 0.12 g cm^{-3}

- 13 A solid floats with $(1/4)$ th of its volume above the surface of water, the density of the solid is [JIPMER 2018]
 (a) 750 kg m^{-3} (b) 650 kg m^{-3}
 (c) 560 kg m^{-3} (d) 450 kg m^{-3}
- 14 A rain drop of radius 0.3 mm has a terminal velocity in air 1 ms^{-1} . The viscosity of air is 18×10^{-5} poise. Find the viscous force on the rain drops. [JIPMER 2018]
 (a) $2.05 \times 10^{-7} \text{ N}$ (b) $1.018 \times 10^{-7} \text{ N}$
 (c) $1.05 \times 10^{-7} \text{ N}$ (d) $2.058 \times 10^{-7} \text{ N}$
- 15 At what speed will the velocity of a stream of water be equal to 20 cm of mercury column? (Take, $g = 10 \text{ ms}^{-2}$) [JIPMER 2018]
 (a) 6.4 ms^{-1} (b) 7.3756 ms^{-1}
 (c) 6.4756 ms^{-1} (d) None of these
- 16 Water flows through a horizontal pipe of variable cross-section at the rate of 20 L min^{-1} . What will be the velocity of water at a point, where diameter is 4 cm? [JIPMER 2018]
 (a) 0.2639 ms^{-1} (b) 0.5639 ms^{-1}
 (c) 0.4639 ms^{-1} (d) 0.3639 ms^{-1}
- 17 The work done in blowing a soap bubble of surface tension 0.06 Nm^{-1} from 2 cm radius to 5 cm radius is [JIPMER 2018]
 (a) 0.004168 J (b) 0.003168 J
 (c) 0.003158 J (d) 0.004568 J
- 18 A U-tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 10 mm above the water level on the other side. Meanwhile the water rises by 65 mm from its original level (see diagram). The density of the oil is [NEET 2017]



- (a) 650 kg m^{-3} (b) 425 kg m^{-3}
 (c) 800 kg m^{-3} (d) 928 kg m^{-3}
- 19 An ice-berg of density 900 kg m^{-3} is floating in water of density 1000 kg m^{-3} . The percentage of volume of ice-berg outside the water is [JIPMER 2017]
 (a) 20% (b) 35% (c) 10% (d) 11%

- 20** A liquid is flowing into a tube of truncated cone shape. Identify the correct statement from the following. [JIPMER 2017]

(a) The speed is high at the wider end and low at the narrow end.
 (b) The speed is low at the wider end and high at the narrow end.
 (c) The speed is same at both ends in a streamline flow.
 (d) The liquid flows with uniform velocity in the tube.

- 21** Two soap bubbles coalesce. It is noticed that, while joined together, the radii of the two bubbles are a and b , where $a > b$. Then, the radius of curvature of interface between the two bubbles will be [JIPMER 2017]

(a) $a - b$ (b) $a + b$ (c) $\frac{ab}{(a - b)}$ (d) $\frac{ab}{(a + b)}$

- 22** A rectangular film of liquid is extended from $(4 \text{ cm} \times 2 \text{ cm})$ to $(5 \text{ cm} \times 4 \text{ cm})$. If the work done is $3 \times 10^{-4} \text{ J}$, the value of the surface tension of the liquid is [NEET 2016]

(a) 0.250 Nm^{-1} (b) 0.125 Nm^{-1}
 (c) 0.2 Nm^{-1} (d) 8.0 Nm^{-1}

- 23** Three liquids of densities ρ_1, ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$), having the same value of surface tension T , rise to the same height in three identical capillaries. The angles of contact θ_1, θ_2 and θ_3 obey [NEET 2016]

(a) $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geq 0$
 (b) $0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$
 (c) $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$
 (d) $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$

- 24** Two non-mixing liquids of densities ρ and $n\rho$ ($n > 1$) are put in a container. The height of each liquid is h . A solid cylinder of length L and density d is put in this container.

The cylinder floats with its axis vertical and length pL ($p < 1$) in the denser liquid. The density d is equal to [NEET 2016]

(a) $[2 + (n + 1)p]\rho$ (b) $[2 + (n - 1)p]\rho$
 (c) $[1 + (n - 1)p]\rho$ (d) $[1 + (n + 1)p]\rho$

- 25** A wind with speed 40 ms^{-1} blows parallel to the roof of a house. The area of the roof is 250 m^2 . Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind

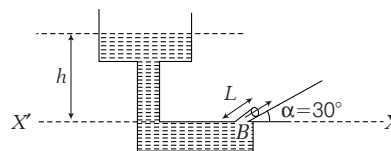
on the roof and the direction of the force will be (Take, $\rho_{\text{air}} = 1.2 \text{ kgm}^{-3}$) [CBSE AIPMT 2015]

(a) $4.8 \times 10^5 \text{ N}$, downwards
 (b) $4.8 \times 10^5 \text{ N}$, upwards
 (c) $2.4 \times 10^5 \text{ N}$, upwards
 (d) $2.4 \times 10^5 \text{ N}$, downwards

- 26** The approximate depth of an ocean is 2700 m . The compressibility of water is $45.4 \times 10^{-11} \text{ Pa}^{-1}$ and density of water is 10^3 kgm^{-3} . What fractional compression of water will be obtained at the bottom of the ocean? [CBSE AIPMT 2015]

(a) 0.8×10^{-2} (b) 1×10^{-2}
 (c) 1.2×10^{-2} (d) 1.4×10^{-2}

- 27** Determine the height above the dashed line XX' attained by the water stream coming out through the hole situated at point B in the diagram given below. Given that $h = 10 \text{ m}$, $L = 2 \text{ m}$ and $\alpha = 30^\circ$. [AIIMS 2015]



(a) 10 m (b) 7.1 m
 (c) 5 m (d) 3.2 m

- 28** A water drop of radius 1 cm is broken into 1000 equal droplets. What would be the gain in surface energy, if the surface tension of water is 0.075 Nm^{-1} ? [UK PMT 2015]

(a) $16.96 \times 10^{-4} \text{ J}$ (b) $8.48 \times 10^{-4} \text{ J}$
 (c) $4.24 \times 10^{-4} \text{ J}$ (d) $2.12 \times 10^{-4} \text{ J}$

- 29** The lower end of a capillary tube is dipped into water and it is seen that the water rises through 7.5 cm in capillary tube. What would be the radius of capillary tube, if surface tension of water is $7.5 \times 10^{-2} \text{ Nm}^{-1}$? The contact angle between water and glass is 0° and $g = 10 \text{ ms}^{-2}$. [UK PMT 2015]

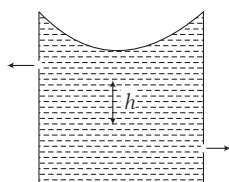
(a) 0.2 cm (b) 0.1 cm
 (c) 0.4 mm (d) 0.2 mm

- 30** A soap bubble of diameter a is produced using the soap solution of surface tension T . Find the energy required to double the radius of the bubble without change of temperature. [EAMCET 2015]

(a) $2 \pi a^2 T$ (b) $6 \pi a^2 T$
 (c) $3 \pi a^2 T$ (d) $12 \pi a^2 T$

- 31** A boat carrying a few number of big stones floats in a water tank. If the stones are unloaded into water, the water level [Kerala CEE 2015]
 (a) rises till half the number of stones are unloaded and then begins to fall
 (b) rises
 (c) fall till half the number of stones are unloaded and then begins to rise
 (d) falls
 (e) remains unchanged
- 32** Choose the correct statement. [Kerala CEE 2015]
 (a) Terminal velocities of rain drops are proportional to square of their radii.
 (b) Water proof agents decrease the angle of contact between water and fibres.
 (c) Detergents increase the surface tension of water.
 (d) Hydraulic machines work on the principle of Torricelli's law.
 (e) Venturimeter measures the flow speed of compressible fluids.
- 33** A 20 cm long capillary tube is dipped vertically in water and the liquid rises up to 10 cm. If the entire system is kept in a freely falling platform, the length of the water column in the tube will be [WB JEE 2015]
 (a) 5 cm (b) 10 cm
 (c) 15 cm (d) 20 cm
- 34** By sucking through a straw, a student can reduce the pressure in his lungs to 750 mm of Hg (density 13.6 g cm^{-3}). Using the straw, he can drink water from a glass upto a maximum depth of [UP CPMT 2015]
 (a) 10.2 cm (b) 75.3 cm
 (c) 13.6 cm (d) 1.96 cm
- 35** If 1000 drops are combined to form a larger drop, then the ratio of surface energy of smaller drop to the larger drop will be [UP CPMT 2015]
 (a) 1 : 100 (b) 1 : 1000
 (c) 1 : 10 (d) 0.1 : 1
- 36** A solid floats such that its $1/3$ part is above the water surface. Then, the density of solid is [UP CPMT 2015]
 (a) 744 kg m^{-3} (b) $\frac{1000}{3} \text{ kg m}^{-3}$
 (c) $\frac{2000}{3} \text{ kg m}^{-3}$ (d) 910 kg m^{-3}
- 37** The amount of work done in blowing a soap bubble such that its diameter increases from d to D is (S = surface tension of the solution) [Manipal 2015]
 (a) $2\pi (D^2 - d^2) S$
 (b) $\pi (D^2 - d^2) S$
 (c) $4\pi (D^2 - d^2) S$
 (d) $8\pi (D^2 - d^2) S$
- 38** A flow of liquid is streamline, if the Reynolds' number is [CBSE AIPMT 2014]
 (a) less than 1000
 (b) greater than 1000
 (c) between 2000 to 3000
 (d) between 4000 to 5000
- 39** A certain number of spherical drops of a liquid of radius r coalesce to form a single drop of radius R and volume V . If T is the surface tension of the liquid, then [CBSE AIPMT 2014]
 (a) energy $= 4VT\left(\frac{1}{r} - \frac{1}{R}\right)$ is released
 (b) energy $= 3VT\left(\frac{1}{r} + \frac{1}{R}\right)$ is absorbed
 (c) energy $= 3VT\left(\frac{1}{R} - \frac{1}{r}\right)$ is released
 (d) energy is neither released nor absorbed
- 40** A bubble is at the bottom of the lake of depth h . As the bubble comes to sea level, its radius increases three times. If atmospheric pressure is equal to l metre of water column, then h is equal to [UK PMT 2014]
 (a) $26 l$ (b) l
 (c) $25 l$ (d) $30 l$
- 41** A wooden block is floating on water kept in a beaker. 40% of the block is above the water surface. Now, the beaker is kept inside a lift that starts going upward with acceleration equal to $g/2$. The block will then [WB JEE 2014]
 (a) sink
 (b) float with 10% above the water surface
 (c) float with 40% above the water surface
 (d) float with 70% above the water surface
- 42** A drop of some liquid of volume 0.04 cm^3 is placed on the surface of a glass slide. Then, another glass slide is placed on it in such a way that the liquid forms a thin layer of area 20 cm^2 between the surfaces of the two slides. To separate the slides a force of 16×10^5 dyne has to be applied normal to the surfaces. The surface tension of the liquid is (in dyne cm^{-1}) [WB JEE 2014]
 (a) 60 (b) 70
 (c) 80 (d) 90
- 43** The wettability of a surface by a liquid depends primarily on [NEET 2013]
 (a) viscosity
 (b) surface tension of liquid and air
 (c) density
 (d) angle of contact between the surface and the liquid

- 44** There are two identical small holes on the opposite sides of a tank containing a liquid. The tank is open at the top. The difference in height between the two holes is h . As the liquid comes out of the two holes, the tank will experience a net horizontal force proportional to [J&KCET 2013]



- (a) $\sqrt{h}$ (b) h (c) $h^{3/2}$ (d) h^2
- 45** Water rises in a capillary tube up to a height of 10 cm, whereas mercury depresses in it by 3.42 cm. If the angle of contact and density of mercury are 135° and 13.6 gcc^{-1} respectively, then the ratio of the surface tension of water and mercury will be nearly [EAMCET 2013]
- (a) 13 : 2 (b) 5 : 16
(c) 16 : 5 (d) 2 : 13
- 46** Two capillary tubes of lengths in the ratio 2 : 1 and radii in the ratio 1 : 2 are connected in series. Assume the flow of the liquid through the tube is steady. Then, the ratio of pressure difference across the tubes is [EAMCET 2013]
- (a) 1 : 8 (b) 1 : 16
(c) 32 : 1 (d) 1 : 1
- 47** Two spherical soap bubbles of diameters 10 cm and 6 cm are formed, one at each end of a narrow horizontal glass tube. If the surface tension of the soap solution is 0.03 Nm^{-1} , then the pressure difference (in pascal) between the two ends of the tube is [Kerala CEE 2013]
- (a) 16 (b) 1.6
(c) 0.016 (d) 0.08
(e) 0.16
- 48** The excess pressure inside one soap bubble is three times that inside a second bubble. The ratio of the volume of first bubble to that of the second [MPPMT 2013]
- (a) 1 : 27 (b) 27 : 1 (c) 1 : 9 (d) 9 : 1
- 49** Water rises to a height of 20 mm in a capillary. If the radius of the capillary is made one-third of its previous value, then the new value of capillary rise will be [MPPMT 2013]
- (a) $\frac{20}{3}$ mm (b) 60 mm
(c) $\frac{20}{9}$ mm (d) 180 mm

- 50** The unit of viscosity in the CGS system is poise (P) and that in SI is poiseuille (Pl). Which of the following statement is correct? [UP CPMT 2013]

(a) $1 \text{ P} = 1 \text{ Pl}$ (b) $1 \text{ P} = 10 \text{ Pl}$
(c) $10 \text{ P} = 1 \text{ Pl}$ (d) None of these

- 51** Liquid rises to a height of 2 cm in a capillary tube and the angle of contact between the solid and the liquid is zero. If the tube is depressed more now, so that top of capillary is only 1 cm above the liquid, then the apparent angle of contact between the solid and the liquid is [UP CPMT 2013]

(a) 0° (b) 30°
(c) 60° (d) 90°

- 52 Assertion** The water rises higher in a capillary tube of small diameter than in the capillary tube of large diameter.

Reason Height through which liquid rise in capillary tube inversely proportional to the diameter of capillary tube. [AIIMS 2012]

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) If Assertion is correct but Reason is incorrect.
(d) If both Assertion and Reason are incorrect.
- 53** The ratio of radii of two bubbles is 2 : 1. What is the ratio of excess pressures inside them? [UP CPMT 2012]
- (a) 1 : 2 (b) 1 : 4
(c) 2 : 1 (d) 4 : 1

- 54** 10 min are taken to empty a rectangular vessel of height h through an orifice in its bottom. How much time will it take to be emptied the vessel when half filled? [BCECE (Mains) 2012]
- (a) 3 min (b) 5 min
(c) 7 min (d) 9 min

- 55** Water is moving with a speed of 5.18 ms^{-1} through a pipe with a cross-sectional area of 4.20 cm^2 . The water gradually descends 9.66 m as the cross-sectional area of pipe is increased to 7.60 cm^2 . The speed of the flow of water at lower level is [BCECE (Mains) 2012]
- (a) 2.86 ms^{-1} (b) 3.0 ms^{-1}
(c) 3.82 ms^{-1} (d) 5.7 ms^{-1}

- 56** A block of ice in which a piece of stone is embedded is floating on water contained in a beaker. When all the ice melts, the level of water in the beaker [BHU 2012]
- (a) rises
(b) falls
(c) remains unchanged
(d) None of the above

- 57** The excess pressure inside a spherical soap bubble of radius 1 cm is balanced by a column of oil (specific gravity = 0.8), 2 mm high, the surface tension of the bubble is [Manipal 2012]
 (a) 3.92 Nm^{-1} (b) 0.0392 Nm^{-1}
 (c) 0.392 Nm^{-1} (d) 0.00392 Nm^{-1}
- 58** Water from a tap emerges vertically downwards with initial velocity 4 ms^{-1} . The cross-sectional area of the tap is A . The flow is steady and pressure is constant throughout the stream of water. The distance h vertically below the tap, where the cross-sectional area of the stream becomes $(2/3) A$, is (Take, $g = 10 \text{ m s}^{-2}$) [Manipal 2012]
 (a) 2 m (b) 1 m
 (c) 0.5 m (d) 4 m
- 59** The terminal speed attained by an aluminium sphere of radius 1 mm falling through water at 20°C will be close to [AMU 2012]
 (a) 9.2 ms^{-1} (b) 6.9 ms^{-1}
 (c) 4.6 ms^{-1} (d) 2.3 ms^{-1}
 (Assume laminar flow, specific gravity of Al = 2.7 and $\eta_{\text{water}} = 8 \times 10^{-4} \text{ PI}$)
- 60** Water flows in a streamlined manner through a capillary tube of radius a , the pressure difference being p and the rate of flow Q . If the radius is reduced to $a/2$ and the pressure increased to $2p$. The rate of flow becomes [AMU 2012]
 (a) $4Q$ (b) Q
 (c) $\frac{Q}{4}$ (d) $\frac{Q}{8}$
- 61** If two soap bubbles of different radii are connected by a tube, then [Manipal 2012]
 (a) air flow from bigger bubble to the smaller bubble till sizes becomes equal
 (b) air flow from bigger bubble to the smaller bubble till sizes are interchanged
 (c) air flow from smaller bubble to bigger bubble
 (d) there is no flow of air
- 62** A body floats in water with 40% of its volume outside water. When the same body floats in an oil, 60% of its volume remains outside oil. The relative density of oil is [WB JEE 2011]
 (a) 0.9 (b) 1
 (c) 1.2 (d) 1.5
- 63** A vertical tank with depth H is full with water. A hole is made on one side of the walls at a depth h below the water surface. At what distance from the foot of the wall does the emerging stream of water strike the foot? [DUMET 2011]
 (a) $\sqrt{h(H-h)}$ (b) $2\sqrt{h(H-h)}$
 (c) $2(H-h)\sqrt{h/(H-h)}$ (d) $\sqrt{2h/(H-h)}$

ANSWERS

CHECK POINT 13.1

1. (b)	2. (d)	3. (c)	4. (b)	5. (c)	6. (c)	7. (d)	8. (b)	9. (c)	10. (b)
11. (c)									

CHECK POINT 13.2

<i>1. (d)</i>	<i>2. (b)</i>	<i>3. (a)</i>	<i>4. (b)</i>	<i>5. (d)</i>	<i>6. (b)</i>	<i>7. (b)</i>	<i>8. (b)</i>	<i>9. (a)</i>	<i>10. (b)</i>
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CHECK POINT 13.3

1. (b) 2. (d) 3. (c) 4. (d) 5. (c) 6. (c) 7. (d) 8. (a)

CHECK POINT 13.4

1. (d)	2. (a)	3. (a)	4. (b)	5. (c)	6. (c)	7. (b)	8. (a)
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(A) Taking it together

1. (b)	2. (a)	3. (b)	4. (d)	5. (b)	6. (c)	7. (c)	8. (d)	9. (c)	10. (a)
11. (b)	12. (b)	13. (d)	14. (a)	15. (b)	16. (d)	17. (b)	18. (a)	19. (a)	20. (d)
21. (b)	22. (d)	23. (a)	24. (b)	25. (b)	26. (d)	27. (b)	28. (b)	29. (d)	30. (d)
31. (a)	32. (c)	33. (b)	34. (b)	35. (c)	36. (b)	37. (a)	38. (c)	39. (c)	40. (d)
41. (c)	42. (a)	43. (a)	44. (b)	45. (d)	46. (a)	47. (c)	48. (d)	49. (a)	50. (a)
51. (b)	52. (a)	53. (c)	54. (a)	55. (a)	56. (b)	57. (c)	58. (a)	59. (a)	60. (d)
61. (d)	62. (c)	63. (d)	64. (c)	65. (a)	66. (b)	67. (b)	68. (b)	69. (c)	70. (c)
71. (a)	72. (b)	73. (b)	74. (c)	75. (a)	76. (d)	77. (b)	78. (c)	79. (d)	80. (b)
81. (b)	82. (a)	83. (b)	84. (a)	85. (c)	86. (a)	87. (c)	88. (c)	89. (a)	90. (c)
91. (a)	92. (a)	93. (b)	94. (a)	95. (b)	96. (a)	97. (b)	98. (a)	99. (d)	100. (c)
101. (b)	102. (c)	103. (a)	104. (b)	105. (a)	106. (c)	107. (c)	108. (c)	109. (d)	110. (b)
111. (d)	112. (c)	113. (a)	114. (b)	115. (a)	116. (d)	117. (a)	118. (b)	119. (a)	120. (c)
121. (c)	122. (b)	123. (c)	124. (a)	125. (d)	126. (b)				

(B) Medical entrance special format questions

- **Assertion and reason**

1. (b) 2. (b) 3. (c) 4. (a)

- **Statement based questions**

1. (c) 2. (a) 3. (c)

- **Match the columns**

1. (c) 2. (b) 3. (a)

(C) Medical entrances' gallery

1. (d)	2. (c)	3. (b)	4. (b)	5. (d)	6. (d)	7. (b)	8. (a)	9. (c)	10. (b)
11. (a)	12. (a)	13. (a)	14. (b)	15. (b)	16. (a)	17. (b)	18. (d)	19. (c)	20. (b)
21. (c)	22. (b)	23. (b)	24. (c)	25. (c)	26. (c)	27. (d)	28. (b)	29. (d)	30. (a)
31. (d)	32. (a)	33. (d)	34. (c)	35. (a)	36. (c)	37. (a)	38. (a)	39. (c)	40. (a)
41. (b)	42. (c)	43. (d)	44. (b)	45. (d)	46. (c)	47. (b)	48. (a)	49. (b)	50. (c)
51. (c)	52. (a)	53. (a)	54. (c)	55. (a)	56. (b)	57. (b)	58. (b)	59. (c)	60. (d)
61. (c)	62. (d)	63. (b)							

Hints & Explanations

• CHECK POINT 13.1

1 (b) Pointed ends have very small area, therefore transmits a large pressure.

$$2 \text{ (d) Pressure, } p = \frac{F}{A} = Mg/A = \frac{70 \times 10}{15 \times 10^{-4} \times 2} = \frac{7 \times 10^4}{30} \times 10^2 \\ = 2.33 \times 10^5 \text{ N m}^{-2}$$

3 (c) Gauge pressure = Absolute pressure – Atmospheric pressure

4 (b) Absolute pressure, $p = p_a + \rho gh$

Here, $p_a = 1.01 \times 10^5 \text{ Pa}$,

$$\rho = 1.03 \times 10^3 \text{ kg m}^{-3}, h = 500 \text{ m}, g = 10 \text{ ms}^{-2}$$

$$\therefore p = 1.01 \times 10^5 \text{ Pa} + 1.03 \times 10^3 \times 10 \times 500 \\ = 1.01 \times 10^5 \text{ Pa} + 51.5 \times 10^5 \text{ Pa} \\ = 52.5 \times 10^5 \text{ Pa} \approx 52 \text{ atm}$$

5 (c) The shape of the container does not affect the pressure. The four vessels A, B, C and D have different shapes and hold different amount of water. The free surface of water in all vessels are in same horizontal plane. The pressure at the four points E, F, G and H lying in the same horizontal plane will be same.

$$6 \text{ (c) } \therefore \rho = \frac{\rho_1 V_1 + \rho_2 V_2}{V_1 + V_2} = \frac{(2\rho)(1) + (\rho)(2/3)}{1 + 2/3} = \frac{8\rho}{5}$$

7 (d) If the pressure at the oil-water interface in the left arm is p , then the pressure in the right arm at the level of the interface will also be p .

$$\text{In the left arm, } p = p_0 + \rho_x g(l + d) \quad \dots(i)$$

$$\text{In the right arm, } p = p_0 + \rho_w gl \quad \dots(ii)$$

Equating Eqs. (i) and (ii), we get

$$\rho_x = \frac{l}{l + d} \rho_w = \frac{135 \text{ mm}}{135 \text{ mm} + 15 \text{ mm}} \times 1000 \text{ kg m}^{-3} = 900 \text{ kg m}^{-3}$$

8 (b) It is due to Pascal's law.

9 (c) Here, $r_1 = 4 \text{ cm}$ and $r_2 = 12 \text{ cm}$

$$F_1 = F \text{ and } F_2 = mg = 1400 \times 10 \text{ N}$$

According to Pascal's law,

$$F_1 = F_2 \times \frac{r_1^2}{r_2^2} = 14 \times 10^3 \times \left(\frac{4}{12}\right)^2 \\ = 14 \times 10^3 \times \frac{1}{9} \approx 1600 \text{ N}$$

10 (b) Given, $h = x \text{ cm}$, $\theta = \phi$

Let l = length of mercury column.

$$\therefore \frac{h}{l} = \sin(90^\circ - \phi)$$

$$\Rightarrow \frac{x}{l} = \cos \phi \Rightarrow l = x / \cos \phi$$

11 (c) Let v be the volume of cavities and V be the volume of solid iron.

$$\text{Then, } V = \frac{\text{mass}}{\text{density}} = \left(\frac{6000 / 9.8}{7.87 \times 10^3}\right) = 0.078 \text{ m}^3$$

Further, decrease in weight = upthrust

$$\therefore (6000 - 4000) = (V + v) \rho_w g$$

$$\text{or } 2000 = (0.078 + v) \times 10^3 \times 9.8$$

$$\text{or } 0.078 + v \approx 0.2$$

$$\therefore v = 0.12 \text{ m}^3$$

• CHECK POINT 13.2

1 (d) In the streamline flow,

$$v_1 = \text{constant}, v_2 = \text{constant}, v_3 = \text{constant}$$

$$\text{but, } v_1 \neq v_2 \neq v_3$$

2 (b) For $1000 < R_e < 2000$, flow is unsteady.

3 (a) By equation of continuity,

$$Av = \text{constant}$$

$$\therefore Ap_1 = Ap_2 + Ap_3$$

$$\Rightarrow A \times 8 = 0.8A \times 6 + 3A \times v$$

$$\Rightarrow 4A = 2.4A + 1.5Av \Rightarrow 4 = 2.4 + 1.5v$$

$$\Rightarrow v = \frac{1.6}{1.5} = 1.1 \text{ ms}^{-1}$$

4 (b) Bernoulli's theorem is based upon conservation of mechanical energy.

5 (d) According to equation of continuity,

$$A_1 v_1 = A_2 v_2$$

$$10 \times 1 = 5 \times v_2$$

$$\Rightarrow v_2 = 2 \text{ ms}^{-1}$$

Now, applying Bernoulli's equation, we get

$$\frac{p_1}{\rho} + \frac{v_1^2}{2} = \frac{p_2}{\rho} + \frac{v_2^2}{2}$$

$$\Rightarrow \frac{2000}{1000} + \frac{1^2}{2} = \frac{p_2}{1000} + \frac{2^2}{2} \Rightarrow p_2 = 500 \text{ Pa}$$

6 (b) Given, $\frac{v^2}{2g} = h = 10 \text{ cm} = 0.1 \text{ m}$

$$\therefore \text{Speed, } v = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.1} = \sqrt{2} \text{ ms}^{-1} = 1.4 \text{ ms}^{-1}$$

7 (b) As water falls from a tap, the velocity of water goes on increasing, hence from the equation of continuity, the cross-section of water-stream goes on decreasing. If A_0 is area of cross-section of the mouth of the tap and v_0 is the velocity of water at this point and A, v are corresponding quantities at depth h , then

$$A_0 v_0 = Av = \frac{dV}{dt} \text{ (volume rate of flow)}$$

$$\text{and } v^2 - v_0^2 = 2gh$$

$$\begin{aligned}\therefore \frac{dV}{dt} &= Av = A_0 A \sqrt{\frac{2gh}{A_0^2 - A^2}} = 4 \times 1 \sqrt{\frac{2 \times 10 \times 2}{16 - 1}} \\ &= 8 \sqrt{\frac{2}{3}} \text{ ms}^{-1} = 6.5 \text{ ms}^{-1}\end{aligned}$$

8 (b) The lifting of an aeroplane is based on Bernoulli's principle. When airplane moves through air, pressure difference is created due to higher speed of air in the region above the wings.

9 (a) Velocity of efflux $= \sqrt{2gh}$, which is independent of density of liquid.

10 (b) Range of water $= 2\sqrt{h(H-h)} = 2\sqrt{\frac{H}{4} \times \frac{3H}{4}} = \frac{\sqrt{3H}}{2}$

• CHECK POINT 13.3

1 (b) Rate of flow through a tube is given by $V = \frac{\pi p r^4}{8 \eta l}$

$$\therefore N = 1/8$$

2 (d) When the tubes are fitted in parallel, $V = V_1 + V_2$

$$\Rightarrow \frac{\pi p r^4}{8 \eta l} = \frac{\pi p r_1^4}{8 \eta l} + \frac{\pi p r_2^4}{8 \eta l}$$

$$\Rightarrow r^4 = r_1^4 + r_2^4$$

$$\therefore r = (r_1^4 + r_2^4)^{1/4}$$

3 (c) $F = 6\pi\eta r v$, i.e. v has the dimensions of $\frac{F}{\eta r}$ or $\frac{mg}{\eta r}$.

4 (d) Terminal velocity, $v_T \propto r^2$

$$\therefore \frac{(v_T)_1}{(v_T)_2} = \left(\frac{R}{R/2}\right)^2 = 4$$

5 (c) Velocity first increases and then becomes constant when terminal velocity is attained.

6 (c) Let r be the radius of smaller drop and R the radius of bigger drop. By equating the volume, we get

$$2\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi R^3 \text{ or } R = (2)^{1/3} \cdot r$$

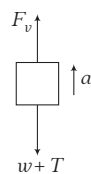
Now, terminal velocity $\propto (\text{radius})^2$

$$\therefore v' = (2)^{2/3} v$$

7 (d) When the small sized spherical body is falling through the viscous liquid first, its velocity increases and then becomes constant which is called terminal velocity as shown in (d).

8 (a) Before entering the water, the velocity of ball is $\sqrt{2gh}$. If after entering the water, this velocity does not change, then this value should be equal to the terminal velocity. Therefore,

$$\begin{aligned}\sqrt{2gh} &= \frac{2}{9} \frac{r^2(\rho - \sigma)g}{\eta} \\ \therefore h &= \frac{\left[\frac{2}{9} \frac{r^2(\rho - \sigma)g}{\eta}\right]^2}{2g} = \frac{2}{81} \times \frac{r^4(\rho - \sigma)^2 g}{\eta^2} \\ &= \frac{2}{81} \times \frac{(3 \times 10^{-4})^4 (10^4 - 10^3)^2 \times 9.8}{(9.8 \times 10^{-6})^2} = 1.65 \times 10^3 \text{ m}\end{aligned}$$



• CHECK POINT 13.4

1 (d) Surface tension $T = F/l$ or $\frac{\Delta W}{\Delta A}$, i.e. unit of surface tension is N/m or J/m².

$$\frac{\text{N}}{\text{m}} = \frac{\text{kg} \cdot \text{ms}^{-2}}{\text{m}} = \frac{\text{kg}}{\text{s}^2}$$

2 (a) Force on each side $= 2TL$ (due to two surfaces)

3 (a) Surface tension, $T = \frac{F}{2l} = \frac{2 \times 10^{-2}}{2 \times 10 \times 10^{-2}} = 0.1 \text{ Nm}^{-1}$

5 (c) $W = 8\pi R^2 T$

$$\therefore W \propto R^2$$

If radius is doubled, then work done will become four times.

6 (c) Radius of curvature, $r = \frac{r_1 r_2}{r_2 - r_1} = \infty$. ($\because r_1 = r_2$)

7 (b) Excess pressure, $\Delta p = \frac{4T}{r} = \frac{4 \times 2 \times 25 \times 10^{-3}}{1 \times 10^{-2}} = 20 \text{ Nm}^{-2} = 20 \text{ Pa}$ ($\because r = d/2$)

8 (a) $h = \frac{2T \cos \theta}{\rho g}$

$$\therefore hr = \frac{2T \cos \theta}{\rho g} = \text{constant} \Rightarrow h_1 r_1 = h_2 r_2 \text{ or } h_2 = \frac{h_1 r_1}{r_2}$$

Substituting the values, we get

$$h_2 = (2.0)(3) = 6.0 \text{ cm} \quad \left(\because \frac{r_2}{r_1} = \frac{1}{3}\right)$$

(A) Taking it together

1 (b) As brush is removed, water surface formed over bristle tends to take smallest area due to surface tension.

2 (a) Waterproof surface coatings are made up of materials which increases angle of contact and water does not wet the surface.

3 (b) For small drops, force of surface tension predominates gravitational force and so, they are more spherical.

4 (d) Cohesive force of water is greater than adhesive force and hence, water does not wets the surface.

5 (b) Surface energy of combined drop will be lowered, so excess surface energy will raise the temperature of the drop.

7 (c) Velocity of different layers of a flowing fluid decreases with distance from the axis of the tube. It is maximum at centre and minimum at walls. So, correct graph is (c).

8 (d) In a streamline flow at any given point, the velocity of each passing fluid particles remains constant. If we consider a cross-sectional area, then a point on the area cannot have different velocities at the same time, hence two streamlines of flow cannot cross each other.

9 (c) $F = W_{\text{air}} - W_{\text{water}} = 5\text{N} - 2\text{N} = 3\text{N}$

10 (a) Specific gravity or relative density $= \frac{\text{Weight in air}}{\text{Change of weight in water}} = \frac{60}{20} = 3$

- 11 (b)** When iron block was floating, it displaces V_1 volume of water such that weight of this V_1 volume of water is equal to weight of block. But since density of iron is more than density of water, this volume V_1 will be greater than volume V_2 of iron block. When iron block sinks in the pond, it displaces water of volume equal to its own volume V_2 .

Since $V_1 > V_2$, displaced volume in first case was more than displaced volume in second case. Hence, level will fall.

- 12 (b)** $h = \frac{2T \cos \theta}{rdg} \Rightarrow h \propto \frac{1}{r}$. So, graph will be hyperbolic.
- 13 (d)** $h \propto \frac{1}{r}$, so for less r , h will be more.
- 14 (a)** Surface tension decreases with temperature and becomes zero at boiling point.
- 15 (b)** Cohesive force decreases with increase in temperature, so angle of contact of a liquid decreases.
- 16 (d)** Work done, $W = 8\pi R^2 T$
 $= 8 \times \pi \times (10^{-2})^2 \times 2 \times 10^{-2} = 16\pi \times 10^{-6} \text{ J}$
- 17 (b)** Weight of liquid in tube is balanced by force of surface tension.
 $\therefore w = 2\pi r T$
- 18 (a)** If the angle of contact is less than 90° , the surface tension of liquid-surface interface is less than that of air-liquid interface. And the pressure difference is proportional to the surface tension, so pressure inside liquid is less than atmospheric pressure.
- 19 (a)** Due to force of attraction between water molecules, it is not easier to separate the two glass plates.
- 20 (d)** Volume of ice > Volume of water
 $\therefore \Delta V = V_w - V_i = \frac{m}{\rho_w} - \frac{m}{\rho_i}$

- 22 (d)** Initially, upthrust $p_0 A$ supports the weight of mercury column in barometer. While in second case, it provides acceleration too. So, the reading will decrease.

- 23 (a)** Neglecting gravitational head, difference in pressure energy = difference in kinetic energy

$$\therefore (3p_0 - p_0) = \frac{1}{2} \rho v^2$$

$$\text{or } v = 2\sqrt{\frac{p_0}{\rho}} = 2\sqrt{\frac{10^5}{10^3}} = 20 \text{ ms}^{-1} \text{ or } \sqrt{400} \text{ ms}^{-1}$$

- 24 (b)** 1 atmosphere = $1.01 \times 10^5 \text{ N m}^{-2} = \rho gh$

$$\therefore h = \frac{1.01 \times 10^5}{1.3 \times 10} = 7769 \text{ m} \approx 8 \text{ km}$$

- 25 (b)** $r \propto \frac{1}{h} \Rightarrow \frac{r_P}{r_Q} = \frac{h_Q}{h_P} \Rightarrow \frac{r_P}{r_Q} = \frac{h}{\frac{2}{3}h} = \frac{3}{2}$

- 26 (d)** In artificial satellite, gravity is absent, so capillary will be fully filled.

- 27 (b)** Let the width of each plate is b and due to surface tension liquid will rise upto height h , then upward force due to surface tension

$$= 2Tb \cos \theta \quad \dots(i)$$

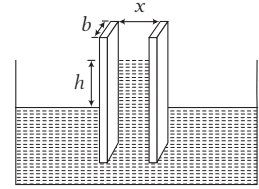
Weight of the liquid rises in between the plates

$$= Vdg = (b x h)dg \quad \dots(ii)$$

Equating Eqs. (i) and (ii), we get

$$2Tb \cos \theta = b x h d g$$

$$\therefore h = \frac{2T \cos \theta}{x d g}$$



- 28 (b)** $1 \text{ gL}^{-1} = 1 \text{ kgm}^{-3}$

Upthrust = Total downward force

$$V\rho_0 g = V\rho g + mg$$

$$\therefore 2000 \times 1.29 \times g = (2000 \times 0.09 \times g + mg)$$

$$\therefore m = 2400 \text{ kg}$$

- 29 (d)** Upthrust on 7 cm = 600 kg

$$\therefore \text{Upthrust on further 1.4 cm} = \frac{600}{7} \times 1.4 = 120 \text{ kg}$$

- 30 (d)** $\therefore \frac{F_1}{A_1} = \frac{F_2}{A_2}$ (Pascal's law)

$\Rightarrow$ Cross-sectional area of the other piston,

$$A_2 = \frac{F_2}{F_1} \cdot A_1 = \left(\frac{2000 \times 9.8 \times 10^5}{10^7} \right) (100) = 1.96 \times 10^4 \text{ cm}^2$$

- 31 (a)** As given

d_1 = Diameter of Ist pipe is 2.5 cm.

d_2 = Diameter of IInd pipe is 3.75 cm.

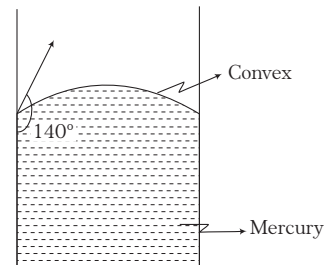
Applying equation of continuity for cross-sections A_1 and A_2 ,

$$A_1 v_1 = A_2 v_2$$

$$\Rightarrow \frac{v_1}{v_2} = \frac{A_2}{A_1} = \frac{\pi(r_2^2)}{\pi(r_1^2)} = \left(\frac{r_2}{r_1} \right)^2$$

$$= \left(\frac{3.75}{2.5} \right)^2 = \left(\frac{3.75}{2.5} \right)^2 = \frac{9}{4}$$

- 32 (c)** According to the question, the observed meniscus is of convex shape. Which is only possible when angle of contact is obtuse. Hence, the combination will be of mercury-glass (140°).



- 33 (b)** Force on the base,

$$F = \rho ghA = 900 \times 10 \times 0.4 \times 2 \times 10^{-3} = 7.2 \text{ N}$$

- 34 (b)** Force at the bottom,

$$F = p \times A = (\rho gh_{\text{total}})A$$

$$= (10^3 \times 10 \times 1.0) (100 \times 10^{-4}) = 100 \text{ N}$$

35 (c) In horizontal direction, pressure remains constant unless the fluid volume is accelerated.

36 (b) $T_c = T_0 (1 - \alpha t)$, i.e. surface tension decreases linearly with increase in temperature.

37 (a) Loss of weight = Upthrust

$$\frac{w}{2} = \text{Total volume} \times \rho_w \times g$$

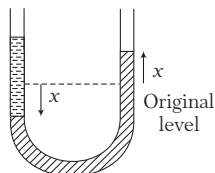
$$\text{or } \frac{(V - V_C) 5 \times \rho_w \times g}{2} = V \times \rho_w \times g \quad (V_C = \text{volume of cavity})$$

$$\text{or } V - V_C = \frac{2}{5}V \Rightarrow V_C = \frac{3}{5}V$$

38 (c) Initially, due to more pressure at the hole, the velocity of efflux will be more. So, the liquid drains out more quickly for first half.

39 (c) Equating pressure, we get

$$(11.2)(1) = (13.6)(2x)$$



$$\therefore x = 0.41 \text{ cm}$$

40 (d) Weight – Upthrust = Drag force.

But no information is given regarding the upthrust.

$$\mathbf{41} \text{ (c) } h = \frac{2T \cos \theta}{r\rho g} \text{ or } h \propto \frac{1}{r}$$

$$m = (\pi r^2 h) \rho \text{ or } m \propto r^2 h$$

$$\text{or } m \propto r$$

$$\therefore M_2 = 2M$$

42 (a) Given, $p_1 = 5.5 \times 10^5 \text{ Nm}^{-2}$ and $p_2 = 5 \times 10^5 \text{ Nm}^{-2}$

$$\therefore \Delta p = p_1 - p_2 = (5.5 - 5) \times 10^5$$

$$= 0.5 \times 10^5 \text{ Nm}^{-2}$$

Decrease in pressure energy = Increase in kinetic energy

$$\text{or } \Delta p = \frac{1}{2} \rho v^2$$

$$\therefore v = \sqrt{\frac{2(\Delta p)}{\rho}} = \sqrt{\frac{2 \times 0.5 \times 10^5}{10^3}} = 10 \text{ ms}^{-1}$$

43 (a) Rate of leakage of water from the hole

$$= Av = A\sqrt{2gh}$$

$$= 10^{-4} \sqrt{2 \times 10 \times 5} = 10^{-3} \text{ m}^3 \text{ s}^{-1}$$

44 (b) From continuity equation, $v \propto \frac{1}{A}$ or $v \propto \frac{1}{d^2}$
(d = diameter of pipe)

$$\therefore \frac{v_{\text{narrow}}}{v_{\text{wider}}} = \left(\frac{d_w}{d_N} \right)^2 = \left(\frac{6}{3} \right)^2 = 4$$

$$\Rightarrow v_{\text{wider}} = \frac{v_{\text{narrow}}}{4} = \frac{4}{4} = 1 \text{ ms}^{-1}$$

45 (d) Submerged part = Replaced water

$$\frac{4}{5} h \rho_w = h \times \rho_l$$

$$\therefore \rho_l = \frac{4}{5} \times \rho_w = \frac{4}{5} \times 1000 \Rightarrow \rho_l = 800 \text{ kgm}^{-3}$$

46 (a) Applying Bernoulli's equation, we get

$$p_1 + \frac{1}{2} \rho v_1^2 = p_2 + \frac{1}{2} \rho v_2^2$$

$$\text{or } (10^{-2} \times 13.6 \times 10^3 \times 10) + \frac{1}{2} \times 10^3 \times (0.35)^2$$

$$= p_2 + \frac{1}{2} \times 10^3 \times (0.65)^2$$

$$\text{or } p_2 = 1210 \text{ Nm}^{-2} = \frac{1210 \times 10^2}{13.6 \times 10^3 \times 10} \text{ cm of Hg}$$

$$= 0.89 \text{ cm of Hg}$$

47 (c) Force exerted by the liquid on the base of the vessel is

$$F = mg$$

$$\text{Here, } m_A = m_B = m_C$$

$$\therefore F_A = F_B = F_C$$

48 (d) Given that an object weights m_1 in a liquid of density d_1 and m_2 in a liquid of density d_2 , so when the density of the object is d , then we get

$$V(d - d_1) = m_1 \text{ and } V(d - d_2) = m_2$$

$$\text{Thus, } \frac{d - d_1}{d - d_2} = \frac{m_1}{m_2}$$

$$\text{So, we get } d = \frac{m_1 d_2 - m_2 d_1}{m_1 - m_2}$$

49 (a) Pressure at same level in two limbs will be same.

$$\therefore p + \rho g(3h) = p_0 + 2\rho g(h)$$

$$\therefore p = p_0 - \rho gh \text{ or } p < p_0$$

50 (a) Weight of man = Extra upthrust

$$\therefore mg = V\rho_w g = (3 \times 2 \times 10^{-2})(10)^3(10)$$

$$\text{or } m = 60 \text{ kg}$$

$$\mathbf{51} \text{ (b) } a = \frac{\text{Upthrust} - \text{Weight}}{\text{Mass}} = \frac{V\rho_w g - V\rho g}{V\rho}$$

$$= \left(\frac{\rho_w - \rho}{\rho} \right) g = \left(\frac{1 - 0.5}{0.5} \right) (10) = 10 \text{ ms}^{-2}$$

52 (a) For just sink, $(m_{\text{raft}} + m)g = V_{\text{raft}}\rho_w g$

$$\therefore m = V_{\text{raft}}\rho_w - m_{\text{raft}}$$

$$= \frac{120}{600} \times 1000 - 120 = 80 \text{ kg}$$

53 (c) The net force acting on the stone in a downward direction would be equal to $F = ma$, where m is the mass of the stone. The force acting downwards would be equal to $V\sigma g$ due to its weight and the force acting upwards would be equal to $V\rho g$ due to the buoyant force.

$$\text{Hence, we get } F = V\sigma g - V\rho g = V\sigma g \left(1 - \frac{\rho}{\sigma} \right)$$

$$ma = mg \left(1 - \frac{\rho}{\sigma} \right) = mg \left(1 - \frac{1}{K} \right)$$

Thus, $a = g \left(1 - \frac{1}{K} \right)$

- 54** (a) In a freely falling vessel upthrust ($V\rho g_{\text{eff}}$) is zero, because effective value of g is zero.

55 (a) Fraction immersed = $\frac{\rho_{\text{solid}}}{\rho_{\text{liquid}}} = \frac{0.9}{1.125} = 0.8$

Therefore, fraction outside the water = $0.2 = 1/5$

- 56** (b) Weight of sphere = Upthrust from water + Upthrust from liquid

$$\therefore V\rho g = \left(\frac{2}{3}V \times 10^3 \times g \right) + \left(\frac{V}{3} \times 8 \times 10^3 \times g \right)$$

or $\rho = \frac{10000}{3} \text{ kgm}^{-3}$

- 57** (c) $A_1v_1 = A_2v_2 + A_3v_3$

$$\therefore v_3 = \frac{A_1v_1 - A_2v_2}{A_3} = \frac{(0.2 \times 4) - (0.2 \times 2)}{0.4} = 1 \text{ ms}^{-1}$$

- 58** (a) $R_h = R_{H-h}$, since range from both the holes is same.

$$h = 2.4 \text{ cm and } H - h = 7.6 \text{ cm}$$

Adding these two, we get $H = 10 \text{ cm}$

- 59** (a) Work done by all the forces, $W = \text{Change in kinetic energy}$
Here, two forces acting are weight and upthrust.

$$\therefore W = F \times h = (\sigma - \rho)g(hV)$$

As, $\sigma > \rho$, so kinetic energy will increase.

- 60** (d) Force exerted by gas on hemispherical end = Pressure of gas $\times$ Projected area = $p_0 (\pi r^2)$

- 61** (d) $6 \times 10^{-2} \times \text{Circumference} = \text{Force}$

$$\therefore \text{Circumference} = \frac{75 \times 10^{-4}}{6 \times 10^{-2}} = 12.5 \times 10^{-2} \text{ m}$$

- 62** (c) $\therefore h_1r_1 = h_2r_2$

$$\Rightarrow h_2 = h_1 \left(\frac{r_1}{r_2} \right) = 30 \times 4/3 = 40 \text{ mm}$$

- 63** (d) $h = \frac{2T}{rdg} \Rightarrow h = \frac{2 \times 75}{0.05 \times 1 \times 10^3} = 3 \text{ cm}$

$$\left(\because r = \frac{1 \text{ mm}}{2} = 0.05 \text{ cm} \right)$$

- 64** (c) Angle of contact is acute. So, the pulling force on glass will attract the water towards them and the water drop takes concave shape.

- 65** (a) Suppose R be the radius of bigger drop. Then, by equating volumes, we get

$$2(4/3\pi r^3) = (4/3)\pi R^3 \text{ or } R = (2)^{1/3} r$$

Now, surface energy $\propto$ surface area

$$\therefore \frac{U_1}{U_2} = \frac{A_1}{A_2} = \frac{2 \cdot r^2}{R^2} = \frac{2}{2^{2/3}}$$

- 66** (b) As we know according to equation of continuity, when cross-section of duct decreases, the velocity of flow of liquid increases and in accordance with Bernoulli's theorem in a horizontal pipe, graph (b) correctly represents the variation of pressure.

- 67** (b) Time to empty a tank, $t = \frac{A}{a} \sqrt{\frac{2h}{g}}$

$$\text{or } t \propto \sqrt{h} \Rightarrow t_2/t_1 = \sqrt{\frac{h_2}{h_1}} = \sqrt{\frac{4h}{h}} = 2 \Rightarrow t_2 = 2t$$

- 68** (b) Level in the container will become maximum, when rate of inflow = rate of outflow

$$Q = A_1v = A_1\sqrt{2gh_{\text{max}}}$$

$$\therefore h_{\text{max}} = \frac{Q^2}{2gA_1^2}$$

- 69** (c) Pressure difference, $\Delta p = \Delta \text{KE} = \frac{1}{2} \times \rho(v_1^2 - v_2^2)$

$$\text{or } \Delta p = \frac{1}{2} \times 1.0 \times (3600 - 1600) = 1000 \text{ Nm}^{-2}$$

$$\Rightarrow F = (\Delta p) \times A = 4000 \text{ N}$$

- 70** (c) Decrease in potential energy = Increase in kinetic energy

$$\therefore \rho gh = \frac{1}{2} \rho (v_f^2 - v_i^2)$$

$$\text{or } 2(10)(0.15) = v_f^2 - (1.0)^2$$

$$\text{or } v_f = 2 \text{ ms}^{-1}$$

Now, from continuity equation, $A_1v_1 = A_2v_2$ or $A \propto \frac{1}{v}$

Velocity has become two times. Hence, area of cross-section will become half, i.e. $0.5 \times 10^{-4} \text{ m}^2$ or $5 \times 10^{-5} \text{ m}^2$.

- 71** (a) $\therefore$ Time of flight, $t = \sqrt{2g(H-h)}$

$$\therefore \text{Range, } R = 2g\sqrt{h(H-h)}$$

$$\text{If } R_1 = R_2 \Rightarrow h_1(H-h_1) = h_2(H-h_2)$$

$$\Rightarrow H(h_1 - h_2) = h_1^2 - h_2^2$$

$$\Rightarrow h_1 + h_2 = H$$

- 72** (b) Terminal velocity, $v = \frac{2}{9} \frac{r^2(\rho - \sigma)g}{\eta}$

$$\therefore v \propto r^2$$

$$\Rightarrow \frac{v}{v'} = \frac{r^2}{R^2}$$

$$\Rightarrow v' = \frac{vR^2}{r^2}$$

- 73** (b) $\frac{h_i}{l} = \frac{\rho_{\text{solid}}}{\rho_{\text{liquid}}} = \frac{\rho_s}{\rho_m}$

$$\therefore h_i = \frac{\rho_s}{\rho_m} \cdot l$$

$$\therefore \text{Height above the mercury level} = \left(1 - \frac{\rho_s}{\rho_m} \right) l$$

- 74 (c)** When moving upwards $\frac{dp}{dh} = \rho(g + a)$ and when moving

$$\text{downwards } \frac{dp}{dh} = \rho(g - a).$$

In first case, pressure decreases with h more rapidly.

- 75 (a)** Upthrust on sphere from the liquid makes equal and opposite pair of forces, so $p = p_0 + \rho gh$.

- 76 (d)** Pressure inside the bubble

$$\begin{aligned} &= \text{Pressure outside it} + \frac{2T}{r} = (p_0 + \rho gh) + \frac{2T}{r} \\ &= 10^5 + (10^3 \times 10 \times 10) + \frac{2 \times 7 \times 10^{-2}}{10^{-3}} \\ &= 2.0014 \times 10^5 \text{ Nm}^{-2} \end{aligned}$$

- 77 (b)** $\therefore F = mg - ma$

$$\Rightarrow 3g = 4(g - a) \Rightarrow g - a = \frac{3g}{4}$$

$$\text{Upthrust force} = v \times \rho (g - a) = (5 \times 10^{-4}) (10^3) \left(\frac{3g}{4} \right)$$

$$= 3.75 \text{ kg}$$

$$\text{Apparent weight} = (4 - 3.75) = 0.25 \text{ kg}$$

- 78 (c)** $w = \text{Upthrust}$

$$\text{or } (m_1 + m_2)g = [m_1/(0.5 \times \rho_w) + m_2/(2.5 \times \rho_w)] \rho_w g$$

$$\text{or } m_1 + m_2 = 2m_1 + 0.4m_2$$

$$\text{or } m_1 = 0.6m_2$$

$$\therefore \frac{m_1}{m_2} = 0.6 = \frac{3}{5}$$

- 79 (d)** Common height after they are connected can be determined by equating the volumes, hence

$$(A_1 + A_2)h = A_1h_1 + A_2h_2 \quad (\because A_1 = A_2 = A)$$

$$\therefore h = \frac{h_1 + h_2}{2}$$

$$\text{The decrease in height } h_1, \Delta h = h_1 - \left(\frac{h_1 + h_2}{2} \right)$$

$$= \frac{h_1 - h_2}{2}$$

$$\text{Mass of liquid in } \Delta h \text{ column, } m = \left(\frac{h_1 - h_2}{2} \right) \rho A$$

$$\begin{aligned} \therefore \text{Work done by gravity} &= mg\Delta h = \left(\frac{h_1 - h_2}{2} \right) \rho Ag \left(\frac{h_1 - h_2}{2} \right) \\ &= \frac{\rho Ag}{4} (h_2 - h_1)^2 \end{aligned}$$

- 80 (b)** $2a_1h_1 = 2a_2h_2$

Here, a_1 = acceleration inside the liquid

$$= \frac{\text{upthrust} - \text{weight}}{\text{mass}} = \frac{V\sigma g - V\rho g}{V\rho} = \left(\frac{\sigma}{\rho} - 1 \right) g$$

$$a_2 = \text{retardation in air} = g$$

$$\therefore h_2 = \frac{a_1}{a_2} \cdot h_1 = \left(\frac{\sigma}{\rho} - 1 \right) h$$

- 81 (b)** Writing pressure equation from one end to other end of tube,

$$p_0 + 2\rho gh - 2\rho \cdot \frac{l}{2} \cdot a - \rho \cdot \frac{l}{2} \cdot a - \rho gh = p_0$$

$$\therefore h = \frac{3al}{2g}$$

- 82 (a)** Pressure is same in a horizontal level, $p_D = p_B$

$$\therefore p_0 + \rho_1 gh_1 = p_0 + \rho_w gh_2$$

$$\text{or } \frac{\rho_1}{\rho_w} = \text{relative density of paraffin} = \frac{h_2}{h_1}$$

- 83 (b)** As, liquids in the two limbs are at same level, so pressure is equal and $\rho_1 = \rho_2 = 1.1$.

- 84 (a)** $\frac{dp}{dx} = \rho a$ (along horizontal towards right), i.e. in the

direction of acceleration along horizontal, the pressure will decrease or $p_{AB} > p_{CD}$

$$\frac{dp}{dy} = \rho g$$

i.e. Along vertically upwards direction pressure will decrease.

Hence, pressure at B is maximum and pressure at D is minimum.

- 85 (c)** $\frac{\Delta p}{\Delta x} = \rho a$ (in horizontal direction)

Pressure decreases in the direction of acceleration in horizontal.

$$\therefore p_D < p_C$$

$$\Rightarrow p_C - p_D = \rho gL$$

$$\text{Also, } \frac{\Delta p}{\Delta x} = \rho g \quad (\text{in vertical direction})$$

Pressure increases with depth in vertical direction.

$$\therefore p_C - p_A = \rho gL$$

- 86 (a)** Weight = Upthrust from mercury + Upthrust from water

$$\therefore 10 \times 8.56 = (x \times 13.6) + (10 - x) \times 1.0$$

Here, x = depth of block inside mercury.

Solving, we get

$$x = 6 \text{ cm}$$

- 87 (c)** Total volume of 64 drops = Volume of big drop

$$64 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 \Rightarrow R^3 = 64r^3$$

$$\Rightarrow R = 4r \Rightarrow r = \frac{R}{4}$$

$$\text{Energy of big drop, } E_1 = T \cdot 4\pi R^2$$

$$\begin{aligned} \text{Energy of 64 drops, } E_2 &= 64(T \times 4\pi r^2) = 64 \left[T \times 4\pi \left(\frac{R}{4} \right)^2 \right] \\ &= 16\pi TR^2 \end{aligned}$$

$\therefore$ Energy needed to break the drop into 64 droplets,

$$\Delta W = E_2 - E_1 = 12\pi R^2 T$$

- 88 (c)** The terminal velocity is given by v , where $v \propto r^2$ (r = radius of the sphere).

The mass of the sphere can be given by $m = \frac{4}{3}\pi r^3 \rho$.

Thus, $m \propto r^3$

So, according to the question, $\frac{m_1}{m_2} = \frac{1}{8} = \left(\frac{r_1}{r_2}\right)^3 \Rightarrow \frac{r_1}{r_2} = \frac{1}{2}$

and since, $\frac{v_1}{v_2} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4} \Rightarrow \frac{v}{nv} = \frac{1}{4} \Rightarrow n = 4$

- 89** (a) Let x be the fraction of volume immersed in first case and y the fraction in second case.

Then, in first case, $W = \text{Upthrust}$

$$\text{or } V\rho_b g = (xV)\rho_l g \quad \text{or } x = \frac{\rho_b}{\rho_l} \quad \dots(i)$$

In second case,

Upthrust – Weight = Mass $\times$ Acceleration

$$\text{or } (yV)\rho_l (g + g/3) - (V\rho_b)g = (V\rho_b)\left(\frac{g}{3}\right) \quad \text{or } y = \frac{\rho_b}{\rho_l} \quad \dots(ii)$$

From Eqs. (i) and (ii), we see that $x = y$ or fraction of volume immersed does not change by the acceleration of vessel.

So, fraction of volume immersed in liquid = $1/2$.

- 90** (c) At $y = \frac{H}{2}$, maximum range is obtained, which is equal to

H . Further $x_y = x_{H-y}$.

- 91** (a) Fluid resistance is given by $R = \frac{8\eta l}{\pi r^4}$.

When two capillary tubes of same size are joined in series, then equivalent fluid resistance is

$$R_e = R_1 + R_2 = \frac{8\eta L}{\pi R^4} + \frac{8\eta \times 2L}{\pi (2R)^4} = \left(\frac{8\eta L}{\pi R^4}\right) \times \frac{9}{8}$$

Equivalent resistance becomes $9/8$ times, so rate of flow will be $(8/9)X$.

- 92** (a) $\Delta p = \frac{4T}{r} \Rightarrow \Delta p \propto \frac{1}{r}$

As, radius of soap bubble increases constantly with time.

$$\therefore \Delta p \propto \frac{1}{t}$$

Hence, the graph will be as shown in option (a).

- 93** (b) $\therefore \frac{2T}{r_2} + \frac{2T}{R} = \frac{2T}{r_1}$

$$\Rightarrow \frac{1}{r_2} + \frac{1}{R} = \frac{1}{r_1} \Rightarrow \frac{1}{5} + \frac{1}{R} = \frac{1}{4} \Rightarrow \frac{1}{R} = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$$

$$\Rightarrow R = 20 \text{ cm}$$

- 94** (a) Density of the mixture,

$$4 = \frac{\rho_1 V + \rho_2 V}{2V} \quad \text{or } \rho_1 + \rho_2 = 8 \quad \dots(i)$$

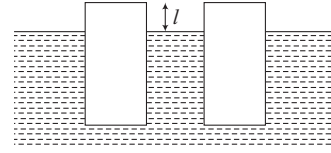
$$3 = \frac{m + m}{(m/\rho_1) + (m/\rho_2)}$$

$$\text{or } \frac{\rho_1 \rho_2}{\rho_1 + \rho_2} = \frac{3}{2} \quad \dots(ii)$$

Solving these two equations, we get

$$\rho_1 = 6 \text{ and } \rho_2 = 2$$

- 95** (b) In equilibrium, weight = upthrust, i.e. $V\rho_i g = V\rho_w g$



$$\text{or } \frac{V_i}{V} = \frac{\rho_i}{\rho_w} = \frac{0.9}{1.0} = 0.9$$

i.e. 90% or 9 cm of ice block is inside the water. Therefore, minimum length l required to scoop out a bucket of water is $10 - 9 = 1.0 \text{ cm}$.

- 96** (a) $v = \sqrt{2gh}$

$$\text{Retardation in liquid, } a = \frac{V\sigma g - V\rho g}{V\rho} = \left(\frac{\sigma - \rho}{\rho}\right)g$$

$$\therefore d_{\max} = \frac{v^2}{2a} = \frac{2gh}{2 \times \left(\frac{\sigma - \rho}{\rho}\right)g} = \frac{\rho h}{\sigma - \rho}$$

- 97** (b) Fraction of length will remain unchanged.

$$\frac{y_i}{y} = \text{constant} = \frac{1}{2}$$

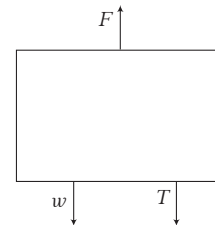
$\therefore$ If total melting rate is 2 cm h^{-1} , then top of candle will fall at the rate of 1 cm h^{-1} .

- 98** (a) $F = (6\pi\eta rv) = 6\pi \left(\frac{1}{6\pi}\right) (l) (v) = v$

$$\therefore \text{Retardation, } a = -\frac{F}{m} = -v \quad \text{or } \frac{dv}{dt} = -v$$

$$\Rightarrow \int_0^t dt = -\int_2^{0.5} \frac{dv}{v} \quad \text{or } t = \ln(4) \text{ s}$$

- 99** (d) Upthrust, $F = w + T$



$$\Rightarrow T = F - w = V\rho_w g - V\rho_b g = (\rho_w - \rho_b)Vg$$

$$= (1000 - 800) \left(\frac{8}{800}\right) (10) = 20 \text{ N}$$

- 100** (c) Upthrust = Volume of metal ball $\times$ Density of liquid $\times g$.

With increase in temperature, volume of ball will increase and density of liquid will decrease.

But coefficient of cubical expansion of liquid is more.

Hence, second effect is more dominating.

Therefore, upthrust at higher temperatures will be less or apparent weight will be more.

- 101** (b) Let r be the radius of smaller drop and R the radius of bigger drop. Then, by equating volume, we get

$$1000 \left(\frac{4}{3} \pi r^3 \right) = \frac{4}{3} \pi R^3 \quad \text{or} \quad r = \frac{R}{10}$$

Surface energy $\propto$ Surface area

$$S_i = 4\pi R^2 \quad \text{or} \quad S_f = 1000 (4\pi r^2) = 10 (4\pi R^2) = 10 S_i$$

As surface area has increased 10 times, surface energy will also become ten times.

- 102** (c) From conservation of mechanical energy, it will return to same height.

103 (a) $A_1 v_1 = A_2 v_2$ or $A_1 \sqrt{2gh_1} = A_2 \sqrt{2gh_2}$

or $A_1^2 h_1 = A_2^2 h_2 \Rightarrow (L^2)^2 (y) = (\pi R^2)^2 (4y)$ or $R = \frac{L}{\sqrt{2\pi}}$

104 (b) Relative density of steel = $\frac{\text{Weight in air}}{\text{Change in weight in water}}$

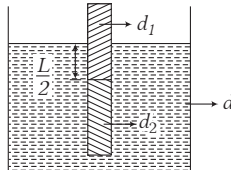
$$= \frac{w}{w - w_1}$$

Now, change in weight in the given liquid = upthrust in this liquid

or $w - w_2 = \left[\frac{w/g}{(w/w - w_1) \rho_w} \right] d_2 \times \rho_w \times g$

$\therefore d_2 = \frac{w - w_2}{w - w_1} = \text{relative density of given liquid}$

- 105** (a) Weight of combined cylinder = Upthrust

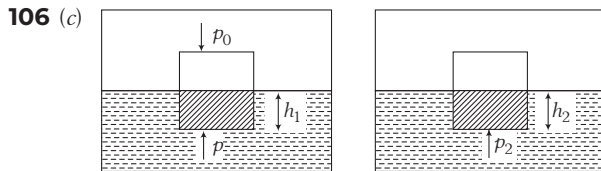


$$\therefore d_1 L A g + d_2 L A g = d \left(\frac{3L}{2} \right) A g$$

$$\Rightarrow d_1 + d_2 = \frac{3}{2} d$$

But since $d_1 < d_2$

$$\Rightarrow d_1 < \frac{3}{4} d$$



In the first case,

Initially $(p - p_0)A = \text{weight of block}$

or $[(p_0 + \rho g h_1) - p_0]A = w$ or $h_1 = \frac{w}{A \rho g}$... (i)

In the second case,

$$p_2 A = w \quad \text{or} \quad (\rho g h_2)A = w$$

or $h_2 = \frac{w}{\rho A g}$... (ii)

From Eqs. (i) and (ii), we see that $h_1 = h_2$.

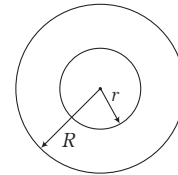
107 (c) Upthrust = $\frac{2}{10} \times 1 \times 10 = 2 \text{ N}$

Reading of S_1 will increase by 2N, i.e. it will become

50 + 2 = 52 N, while that of S_2 will decrease by 2N, i.e.

20 - 2 = 18 N.

- 108** (c) Weight = Upthrust



$$\left(\frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 \right) (8 \times \rho_w \times g) = \left(\frac{4}{3} \pi R^3 \right) \rho_w g$$

or $8R^3 - 8r^3 = R^3$

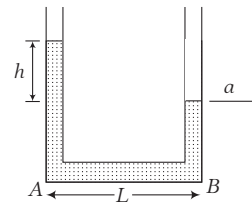
$\therefore 8r^3 = 7R^3$ or $\frac{r}{R} = \left(\frac{7}{8} \right)^{1/3} = \frac{\sqrt[3]{7}}{2}$

- 109** (d) Pressure, $p = p_0 + \rho_1 g h_1 + \rho_2 g h_2$

or $p - p_0 = \rho_1 g h_1 + \rho_2 g h_2$

$$= (600)(10)(0.1) + (1000)(10)(0.02) = 800 \text{ Nm}^{-2}$$

- 110** (b) Change in pressure along horizontal direction, $\frac{dp}{dX} = \rho a$



$$\therefore \Delta p_{AB} = \rho a L = \rho g h$$

$$\therefore h = \frac{La}{g}$$

- 111** (d) As, A and C are at the same level inside the liquid, thus the pressure must be equal at these points, i.e. $p_A = p_C$.

Also, pressure at C, $p_C = p_B + \rho g h$, where ρ is the density of the liquid.

$$\therefore p_B < p_C$$

$$\Rightarrow p_B < p_A = p_C$$

- 112** (c) If temperature remains constant, then

$$p_1 V_1 = p_2 V_2$$

$$(75 + h) V = (75) (3V)$$

$$\therefore h = 150 \text{ cm of Hg} = 1.5 \text{ m of Hg or } 15 \text{ m of water}$$

- 113** (a) As, $A_0 v_0 = A v$ and $v^2 = v_0^2 + 2gh$

$$\Rightarrow v_0 = \sqrt{\frac{2ghA^2}{A_0^2 - A^2}}$$

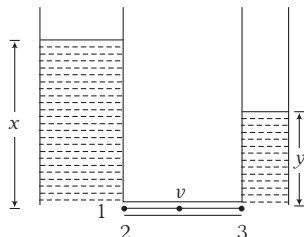
or $R = A_0 v_0 = A_0 \sqrt{\frac{2ghA^2}{A_0^2 - A^2}}$

- 114 (b) We know that, $\Delta p = \frac{F}{A}$.

Difference in pressure energy is equal to difference in kinetic

energy, i.e. $\frac{F}{A} = \frac{1}{2} \rho v^2$ or $v = \sqrt{\frac{2F}{\rho A}}$

- 115 (a) $p_2 = p_3$ and $p_1 > p_2$



$$p_1 = p_0 + \rho g x \Rightarrow p_2 = p_0 + \rho g y$$

$$\therefore \Delta p = p_1 - p_2 = \rho g (x - y)$$

Between 1 and 2,

Difference in pressure energy = Difference in kinetic energy

$$\text{or } \rho g (x - y) = \frac{1}{2} \rho v^2 \text{ or } v = \sqrt{2g(x - y)}$$

- 116 (d) Given, $l_1 = l_2 = 1\text{ m}$ and $\frac{r_1}{r_2} = \frac{1}{2}$

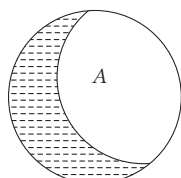
$$V = \frac{\pi p r_1^4}{8 \eta l} = \frac{\pi p r_2^4}{8 \eta l}$$

$$\Rightarrow \frac{p_1}{p_2} = \left(\frac{r_2}{r_1} \right)^4 = 16 \Rightarrow p_1 = 16 p_2$$

Since, both tubes are connected in series, hence pressure difference across combination,

$$p = p_1 + p_2 \Rightarrow p = \frac{17 p_1}{16} \Rightarrow p_1 = \frac{16}{17} p = 0.94 \text{ m}$$

- 117 (a) Due to surface tension, B will keep its area as minimum as possible.



So, the thread will become concave on seeing from side A .

- 118 (b) $w = \text{Upthrust} = V_i \rho_l g$ or $V_i = \frac{w}{\rho_l g}$

If density of liquid ρ_l is increased, immersed volume V_i will decrease or the ball will go up.

- 119 (a) From the given figure, we get

$$\delta g R (\cos \theta + \sin \theta) = \rho g R (\cos \theta - \sin \theta)$$

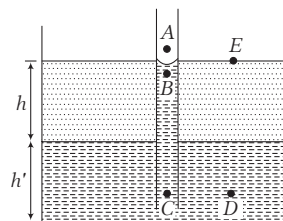
$$\Rightarrow \delta \cos \theta + \delta \sin \theta = \rho \cos \theta - \rho \sin \theta$$

$$\Rightarrow \sin \theta (\delta + \rho) = \cos \theta (\rho - \delta)$$

$$\Rightarrow \tan \theta = \frac{\rho - \delta}{\rho + \delta}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{\rho - \delta}{\rho + \delta} \right)$$

- 120 (c) Let us write pressure equation in path $ABCDE$.



$$p_0 - \frac{2T}{r} + \rho_2 g (h + h') - \rho_2 g h' - \rho_1 g h = p_0$$

$$\therefore T = \frac{r}{2} (\rho_2 - \rho_1) g h$$

- 121 (c) From work-energy theorem,

work done by all the forces = change in kinetic energy.

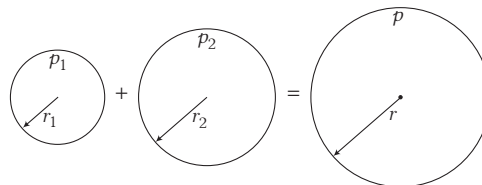
$\therefore$ (Work done by gravity for 7 m) + (Work done by upthrust for 2 m) + (Work done by resistive forces) = 0 as $\Delta \text{KE} = 0$

$$\therefore (1)(10)(7) \cos 0^\circ + \left(\frac{1}{2/3 \times \rho_w} \right) (\rho_w g) (2) \cos 180^\circ$$

+ Work done by resistive forces = 0

$\therefore$ Work done by resistive forces = -40 J

- 122 (b) As, mass of the air is conserved,



$$\therefore n_1 + n_2 = n \quad (\because pV = nRT)$$

$$\therefore \frac{p_1 V_1}{RT_1} + \frac{p_2 V_2}{RT_2} = \frac{pV}{RT}$$

As temperature is constant,

$$T_1 = T_2 = T$$

$$\therefore p_1 V_1 + p_2 V_2 = pV$$

$$\therefore \left(p_0 + \frac{4S}{r_1} \right) \left(\frac{4}{3} \pi r_1^3 \right) + \left(p_0 + \frac{4S}{r_2} \right) \left(\frac{4}{3} \pi r_2^3 \right) = \left(p_0 + \frac{4S}{r} \right) \left(\frac{4}{3} \pi r^3 \right)$$

Solving this, we get

$$S = \frac{p_0 (r^3 - r_1^3 - r_2^3)}{4(r_1^2 + r_2^2 - r^2)}$$

Note To avoid confusion with the temperature, surface tension here is represented by S .

- 123 (c) Let tube's entrance be at depth y below the surface. Take point 1 at entry, 2 at the maximum height of the fountain. Applying Bernoulli's theorem,

$$p_1 + \rho g h_1 + \frac{1}{2} \rho v_1^2 = p_2 + \rho g h_2 + \frac{1}{2} \rho v_2^2 \quad \dots(i)$$

Taking, $h_1 = y$, $h_2 = (y + h_0 + h)$, $v_1 = v$, $v_2 = 0$, $p_2 = p_1 = p_0$

Hence, pressure term on both sides are equal to atmospheric pressure.

Substituting these values in Eq. (i), we get

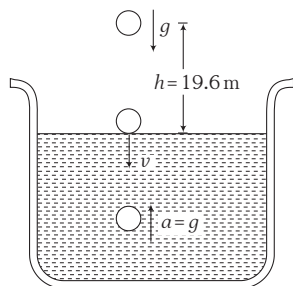
$$p_0 + \rho g y + \frac{1}{2} \rho v^2 = p_0 + \rho g (y + h_0 + h)$$

$$\Rightarrow \frac{1}{2} \rho v^2 = \rho g (h_0 + h)$$

$$\Rightarrow h = \left(\frac{v^2}{2g} - h_0 \right)$$

124 (a) $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 19.6} = 19.6 \text{ ms}^{-1}$

Let ρ be the density of ball and 2ρ the density of water. Net retardation inside the water,



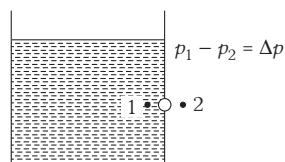
$$a = \frac{\text{upthrust} - \text{weight}}{\text{mass}} = \frac{V(2\rho)g - V(\rho)g}{V(\rho)}$$

(where, V = volume of ball)

$$= g = 9.8 \text{ ms}^{-2}$$

Hence, the ball will go upto the same depth 19.6 m below the water surface. Further, time taken by the ball to come back to water surface is $t = 2\left(\frac{v}{a}\right) = 2\left(\frac{19.6}{9.8}\right) = 4 \text{ s}$

125 (d) $\Delta p = \frac{1}{2} \rho v^2$ or $v = \sqrt{\frac{2(\Delta p)}{\rho}}$



$$R = vt \text{ or } R \propto \sqrt{\Delta p}$$

To make R two times Δp should be 4 times. Initially,

$$\Delta p = \rho_w g h = 10^3 \times 10 \times 10$$

$$= 10^5 \text{ Nm}^{-2} = 1 \text{ atm}$$

In second case, $\rho_w g h + \text{extra pressure} = 4(\Delta p) = 4 \text{ atm}$
or $1 \text{ atm} + \text{extra pressure} = 4 \text{ atm}$
 $\therefore$ Extra pressure = 3 atm

126 (b) $Q = A_1 v_1 = A_2 v_2 = 15 \text{ Lmin}^{-1} = \frac{15 \times 10^{-3}}{60} = 2.5 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$

$$v_1 = \frac{2.5 \times 10^{-4}}{2 \times 10^{-4}} = 1.25 \text{ ms}^{-1} \text{ and } v_2 = \frac{2.5 \times 10^{-4}}{0.5 \times 10^{-4}} = 5 \text{ ms}^{-1}$$

Now, difference in pressure = difference in kinetic energy

or $h \rho_w g = \frac{1}{2} \rho_{\text{air}} (v_2^2 - v_1^2)$

$$\therefore h = \frac{\rho_{\text{air}}}{2 \rho_w g} (v_2^2 - v_1^2)$$

$$= \frac{1.32}{2 \times 10^3 \times 10} [(5)^2 - (1.25)^2]$$

$$= 1.5 \times 10^{-3} \text{ m} \approx 1.5 \text{ mm}$$

(B) Medical entrance special format questions

• Assertion and reason

1 (b) $h = \frac{2T \cos \theta}{r \rho g} \Rightarrow h \propto \cos \theta \Rightarrow h = 0$, when $\theta = 90^\circ$

Also, when angle of contact is 90° , then surface is flat.

4 (a) Upthrust = Weight of fluid displaced
= Volume of body $\times \rho_{\text{fluid}} \times g$

Hence, both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

• Statement based Questions

- 1** (c) The liquid comes to rest due to viscosity.
- 2** (a) Energy is released in process of coalescing of water drops.
- 3** (c) Water level fall as stone pieces displaces more liquid when embedded in ice compared to condition when they sinks.

• Match the columns

- 1** (c) Pressure at O is the atmospheric pressure which will remain same.

Also, as pressure is increased the height h decreases but the pressure at 1 cm above O will remain same.

Hence, $A \rightarrow q$, $B \rightarrow r$, $C \rightarrow r$.

2 (b) $v = \sqrt{2gh}$, $t = \sqrt{\frac{2h}{g}}$ and $R = 2 \sqrt{h(H-h)}$

At mountain, value of g will be less. Hence, v will decrease, t will increase and R will remain same.

Hence, $A \rightarrow q$, $B \rightarrow p$, $C \rightarrow r$.

- 3** (a) In equilibrium, upthrust = weight

If density of liquid is decreasing, more volume should be immersed in the liquid, so x will increase.

Similarly, if size of cube is increased, volume of cube increases and again it immerses more, so x will increase.

The x remains same, when whole system is accelerated upwards.

Hence, $A \rightarrow p$, $B \rightarrow p$, $C \rightarrow r$.

(C) Medical entrances' gallery

- 1** (d) Relation for height of water in capillary tube is

$$h = \frac{2S \cos \theta}{\rho g r} \Rightarrow h \propto \frac{1}{r}$$

$$\Rightarrow \frac{h_1}{h_2} = \frac{r_2}{r_1} = \frac{2r}{r} = 2$$

$$\begin{aligned} \text{As, } m &= A \cdot h \cdot \rho \\ \therefore \frac{m_2}{m_1} &= \frac{Ah_2\rho}{Ah_1\rho} = \frac{h_2}{h_1} = \frac{1}{2} \\ \Rightarrow m_2 &= \frac{m_1}{2} = \frac{5}{2} = 2.5 \text{ g} \end{aligned}$$

- 2 (c)** A liquid does not wet the solid surface, if the angle of contact is obtuse, i.e. $\theta > 90^\circ$.

- 3 (b)** Given, density of liquid, $\rho_l = 760 \text{ kgm}^{-3}$

Density of mercury, $\rho_m = 13600 \text{ kgm}^{-3}$

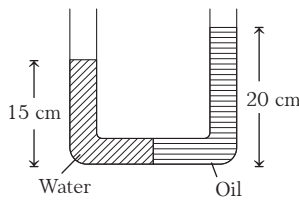
Height of liquid column in mercury barometer,

$$h_m = 76 \text{ cm} = 0.76 \text{ m}$$

If height of liquid in liquid column be h_l , then

$$\begin{aligned} \Rightarrow p_{\text{liquid}} &= p_{\text{mercury}} \\ \Rightarrow h_l \rho_l g &= h_m \rho_m g \\ \Rightarrow h_l &= \frac{h_m \rho_m}{\rho_l} = \frac{0.76 \times 13600}{760} = 13.6 \text{ m} \end{aligned}$$

- 4 (b)** In the given situation as shown in the figure below.



From Pascal's law,

Pressure due to water column of height 15 cm
= Pressure due to oil column of height 20 cm

$$\begin{aligned} \Rightarrow h_w \rho_w g &= h_o \rho_o g \\ 15 \rho_w &= 20 \rho_o \Rightarrow \rho_o = \frac{15}{20} \rho_w \\ \Rightarrow \rho_o &= \frac{15}{20} \times 1000 = 750 \text{ kgm}^{-3} \end{aligned}$$

- 5 (d)** The terminal velocity achieved by ball in a viscous fluid is

$$v_t = \frac{2(\rho - \sigma)r^2g}{9\eta}$$

where, ρ = density of metal ball,

σ = density of viscous medium,

r = radius of ball

and η = coefficient of viscosity of medium.

Terminal velocity of first ball,

$$\begin{aligned} v_{t_1} &= \frac{2(\rho_1 - \sigma)r_1^2g}{9\eta} \\ &= \frac{2(8\rho_2 - \sigma)r_1^2g}{9\eta} \quad \dots(i) \text{ (Given, } \rho_1 = 8\rho_2) \end{aligned}$$

Similarly, for second ball,

$$v_{t_2} = \frac{2(\rho_2 - \sigma)r_2^2g}{9\eta} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{v_{t_1}}{v_{t_2}} = \frac{2(8\rho_2 - \sigma)r_1^2g}{2(\rho_2 - \sigma)r_2^2g} \times \frac{9\eta}{9\eta}$$

$$= \left(\frac{8\rho_2 - 0.1\rho_2}{\rho_2 - 0.1\rho_2} \right) \left(\frac{r_1}{r_2} \right)^2 \quad \dots(iii)$$

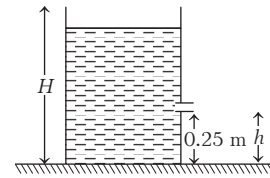
(Given, $\sigma = 0.1\rho_2$)

Here, $r_1 = 1 \text{ mm}$ and $r_2 = 2 \text{ mm}$

Substituting these values in Eq. (iii), we get

$$\Rightarrow \frac{v_{t_1}}{v_{t_2}} = \left(\frac{7.9\rho_2}{0.9\rho_2} \right) \left(\frac{1}{2} \right)^2 = \frac{79}{36}$$

- 6 (d)** Given, height of small orifice from ground, $h = 0.25 \text{ m}$



Total height of water tank, $H = 1 \text{ m}$

$\therefore$ Range of water stream,

$$\begin{aligned} R &= 2\sqrt{h(H-h)} \\ &= 2\sqrt{(1-0.25)0.25} = 2\sqrt{0.75 \times 0.25} \\ &= 0.866 \text{ m} = 86.6 \text{ cm} \end{aligned}$$

- 7 (b)** Given, diameter of tube at first end, $d_1 = 5 \text{ cm}$

$$\begin{aligned} \therefore \text{Radius, } r_1 &= \frac{d_1}{2} = 2.5 \text{ cm} \\ &= 2.5 \times 10^{-2} \text{ m} \end{aligned}$$

Diameter of tube at second end, $d_2 = 2 \text{ cm}$

$$\therefore \text{Radius, } r_2 = \frac{d_2}{2} = 1 \text{ cm} = 1 \times 10^{-2} \text{ m}$$

Velocity of fluid at first end, $v_1 = 4 \text{ m/s}$

By the principle of continuity,

$$A_1 v_1 = A_2 v_2$$

$$\pi r_1^2 v_1 = \pi r_2^2 v_2$$

$$(2.5 \times 10^{-2})^2 \times 4 = (10^{-2})^2 \cdot v_2$$

$$\Rightarrow v_2 = 25 \text{ m/s}$$

From Bernoulli's theorem,

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

$$p_1 - p_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

$$= \frac{1}{2} \times 10^3 (25^2 - 4^2)$$

$$(\because \text{Density of water, } \rho = 10^3 \text{ kg/m}^3)$$

$$= 304500 \text{ Pa}$$

- 8 (a)** Sometimes insects can walk on the surface of water due to surface tension, when legs of insects are not being wet. In this situation, the gravitational force on insect is balanced by force due to surface tension.

Hence, both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

- 9 (c)** Surface tension is responsible for the shape of tiny water drops. When falling freely in the absence of other forces like gravity, shape of water drops is spherical because droplets of

water tend to be pulled into a spherical shape by the cohesive forces of the surface layer.

In the presence of gravity, the shape of heavy water drops is not spherical when falling freely. It is oval shaped. Surface tension of methyl alcohol is lower than surface tension of water.

Hence, water does not have minimum surface tension among all liquids.

Therefore, both Assertion and Reason are incorrect.

- 10** (b) Given, surface tension of water, $T = 0.1 \text{ N/m}$

Radius of small drops, $r = 1 \text{ mm} = 10^{-3} \text{ m}$

If R be the radius of bigger drops, then volume remains conserved.

$$\begin{aligned} \text{i.e.} \quad V_1 &= V_2 \\ \frac{4}{3} \pi R^3 &= 2 \times \frac{4}{3} \pi r^3 \\ R^3 &= 2r^3 = 2 \times (10^{-3})^3 \end{aligned}$$

$$\therefore R = 2^{1/3} \times 10^{-3} \text{ m}$$

$\therefore$ Change in energy during isothermal combination,

$$\begin{aligned} \Delta E &= T \cdot \Delta A = T(A_2 - A_1) \\ &= 0.1(2 \times 4\pi r^2 - 4\pi R^2) \\ &= 0.1 \times 4\pi(2r^2 - R^2) \\ &= 0.4\pi(2 \times 10^{-6} - 2^{2/3} \times 10^{-6}) \\ &= 0.4\pi \times 10^{-6} \times 0.41 \\ &= 0.52 \times 10^{-6} \text{ J} \\ &= 0.52 \mu\text{J} \approx 0.5 \mu\text{J} \end{aligned}$$

- 11** (a) The rate of heat generation is equal to the rate of work done by the viscous force which in turn is equal to its power.

Rate of heat produced,

$$\frac{dQ}{dt} = F \times v_T$$

where, F is the viscous force and v_T the terminal velocity.

$$\text{As,} \quad F = 6\pi\eta r v_T$$

$$\Rightarrow \quad \frac{dQ}{dt} = 6\pi\eta r v_T \times v_T = 6\pi\eta r v_T^2 \quad \dots(\text{i})$$

From the relation of terminal velocity,

$$\begin{aligned} v_T &= \frac{2}{9} \cdot \frac{r^2(\rho - \sigma)}{\eta} g, \text{ we get} \\ v_T &\propto r^2 \quad \dots(\text{ii}) \end{aligned}$$

From Eq. (ii), we can rewrite Eq. (i) as

$$\frac{dQ}{dt} \propto r \cdot (r^2)^2 \quad \text{or} \quad \frac{dQ}{dt} \propto r^5$$

- 12** (a) Pressure at left hand of a U-tube,

$$p_1 = p_0 + \rho_1 g h_1 = p_0 + \rho_1 \times 10 \quad (10 \times 10^{-2})$$

$$\text{or} \quad p_1 = p_0 + \rho_1$$

Pressure at right hand of a U-tube,

$$p_2 = p_0 + \rho_2 g h_2 = p_0 + \rho_2 \times 10 \quad (12 \times 10^{-2})$$

$$\text{or} \quad p_2 = p_0 + 1.2 \rho_2$$

The mercury column in both arms of U-tube are at same level, therefore pressure in both arms will be same.

$$\text{or} \quad p_1 = p_2 \Rightarrow p_0 + \rho_1 = p_0 + 1.2 \rho_2$$

$\therefore$ Density of ethyl alcohol,

$$\rho_2 = \frac{\rho_1}{1.2} = \frac{1000}{1.2} = 833.3 \text{ kg m}^{-3} = 0.83 \text{ g cm}^{-3}$$

- 13** (a) Let V and ρ be volume and density of solid respectively and ρ' be the density of water,

$$\text{i.e.} \quad \rho' = 10^3 \text{ kg m}^{-3}$$

Weight of body $= V\rho g$

Volume of solid body outside water $= V/4$

$\therefore$ Volume of solid body inside water $= V - V/4 = 3V/4$

$$\text{Weight of water displaced by solid} = \frac{3V}{4} \times 10^3 \times g$$

As solid body is floating, then

Weight of body = Weight of water displaced by it

$$V\rho g = \frac{3V}{4} \times 10^3 g$$

$$\Rightarrow \quad \rho = \frac{3}{4} \times 1000 = 750 \text{ kg m}^{-3}$$

- 14** (b) Here, $r = 0.3 \text{ mm} = 0.3 \times 10^{-3} \text{ m}$, $v = 1 \text{ ms}^{-1}$,
 $\eta = 18 \times 10^{-5} \text{ poise} = 18 \times 10^{-6} \text{ decapoise}$

Viscous force, $F = 6\pi\eta r v$

$$\begin{aligned} &= 6 \times \frac{22}{7} \times (18 \times 10^{-6}) \times (0.3 \times 10^{-3}) \times 1 \\ &= 1.018 \times 10^{-7} \text{ N} \end{aligned}$$

- 15** (b) Here, velocity head $= 20 \text{ cm of Hg} = 20 \times 13.6 \text{ cm of water}$

As, velocity head $= v^2/2g$

$$\therefore \quad 20 \times 13.6 = \frac{v^2}{2 \times 1000}$$

$$\begin{aligned} \Rightarrow \quad v &= \sqrt{20 \times 13.6 \times 2 \times 1000} \\ &= 737.56 \text{ cms}^{-1} = 7.3756 \text{ ms}^{-1} \end{aligned}$$

- 16** (a) Volume of the water flowing per second, $V = 20 \text{ L min}^{-1}$

$$= \frac{20 \times 1000}{60 \times (100)^3} \text{ m}^3 \text{ s}^{-1} = \frac{1}{3} \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$$

$$\text{Radius of the pipe, } r = \frac{4}{2} = 2 \text{ cm} = 0.02 \text{ m}$$

$$\text{Area of cross-section, } A = \pi r^2 = \frac{22}{7} \times (0.02)^2 \text{ m}^2$$

Let v be the velocity of the flow of water at the given point. Clearly, $V = Av$

$$\Rightarrow \quad \frac{1}{3} \times 10^{-3} = \frac{22}{7} \times (0.02)^2 \times v$$

$$\begin{aligned} \Rightarrow \quad v &= \frac{7 \times 10^{-3}}{3 \times 22 \times (0.02)^2} \\ &\approx 0.2639 \text{ ms}^{-1} \end{aligned}$$

- 17** (b) Here, $S = 0.06 \text{ Nm}^{-1}$,

$$r_1 = 2 \text{ cm} = 0.02 \text{ m}, r_2 = 5 \text{ cm} = 0.05 \text{ m}$$

Since, bubble has two surfaces.

Initial surface area of the bubble

$$= 2 \times 4\pi r_1^2 = 2 \times 4\pi (0.02)^2 \\ = 32\pi \times 10^{-4} \text{ m}^2$$

Final surface area of the bubble

$$= 2 \times 4\pi r_2^2 = 2 \times 4\pi (0.05)^2 = 200\pi \times 10^{-4} \text{ m}^2$$

Increase in surface area

$$= 200\pi \times 10^{-4} - 32\pi \times 10^{-4} \\ = 168\pi \times 10^{-4} \text{ m}^2$$

$\therefore$ Work done = Surface tension $\times$ Increase in surface area

$$= 0.06 \times 168\pi \times 10^{-4} \\ = 0.003168 \text{ J}$$

- 18** (d) Pressure of two points lie in the same horizontal level should be same and $p = h\Delta g$.

Both ends of the U-tube are open, so the pressure on both the free surfaces must be equal,

i.e. $p_1 = p_2$

$$h_{\text{oil}} \cdot S_{\text{oil}} \cdot g = h_{\text{water}} \cdot S_{\text{water}} \cdot g$$

where, S_{oil} = specific density of oil.

$$\Rightarrow S_{\text{oil}} = \frac{h_{\text{water}} \cdot S_{\text{water}} \cdot g}{h_{\text{oil}} \cdot g}$$

$$\text{From figure, } S_{\text{oil}} = \frac{(65 + 65) \times 1000}{(65 + 65 + 10)} \\ = 928 \text{ kgm}^{-3}$$

- 19** (c) Let the volume of ice-berg is V and its density is ρ . If this ice-berg floats in water with volume V_{in} inside it, then

$$V_{\text{in}} \sigma g = V \rho g$$

$$\Rightarrow V_{\text{in}} = \left(\frac{\rho}{\sigma} \right) V$$

$$\Rightarrow V_{\text{out}} = V - V_{\text{in}} = \left(\frac{\sigma - \rho}{\sigma} \right) V \\ = \left(\frac{1000 - 900}{1000} \right) V = \frac{V}{10}$$

$$\Rightarrow \frac{V_{\text{out}}}{V} = 0.1 = 10\%$$

- 20** (b) For an incompressible liquid, equation of continuity,

$$Av = \text{constant} \quad \text{or} \quad A \propto \frac{1}{v}$$

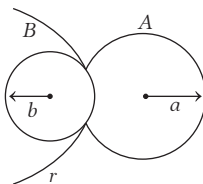
where, A = area of cross-section of tube
and v = speed of liquid.

So, the speed of liquid flow is not uniform, but changes with area of cross-section of tube.

Therefore, at the wider end speed will be low and at the narrow end speed will be high.

- 21** (c) According to given figure,

Let the radius of curvature of the common internal film surface of the double bubble formed by two bubbles be r . Excess of pressure as compared to atmosphere inside A is $(4s/a)$. In double bubble, the pressure difference



between A and B on either side of the common surface

$$= \frac{4S}{b} - \frac{4S}{a}$$

This will be equal to $(4S/r)$.

$$\therefore \frac{4S}{r} = \frac{4S}{b} - \frac{4S}{a} \quad \text{or} \quad \frac{1}{r} = \left(\frac{1}{b} - \frac{1}{a} \right) = \frac{a-b}{ba}$$

$$\text{or} \quad r = \frac{ab}{a-b}$$

- 22** (b) Increase in surface area,

$$\Delta A = (5 \times 4 - 4 \times 2) \times 2 \quad (\because \text{Film has two surfaces}) \\ = (20 - 8) \times 2 \text{ cm}^2 \\ = 24 \text{ cm}^2 = 24 \times 10^{-4} \text{ m}^2$$

So, work done, $W = T \Delta A$

$$3 \times 10^{-4} = T \times 24 \times 10^{-4}$$

$$\Rightarrow T = \frac{1}{8} = 0.125 \text{ N m}^{-1}$$

- 23** (b) Ascent formula for capillary tube,

$$h = \frac{2T \cos \theta}{\rho g r}$$

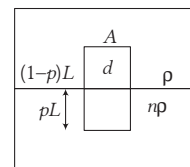
$$\therefore \frac{\cos \theta_1}{\rho_1} = \frac{\cos \theta_2}{\rho_2} = \frac{\cos \theta_3}{\rho_3}$$

Thus, $\cos \theta \propto \rho$

$$\therefore \cos \theta_1 > \cos \theta_2 > \cos \theta_3 \quad (\because \rho_1 > \rho_2 > \rho_3)$$

$$0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$$

- 24** (c) According to question, the situation can be drawn as following.



Applying Archimedes' principle,

weight of cylinder = (upthrust)₁ + (upthrust)₂

$$\text{i.e. } ALdg = (1-p)LA\rho g + (pLA)n\rho g$$

$$\Rightarrow d = (1-p)\rho + p n \rho = \rho - p\rho + n p \rho \\ = \rho + (n-1)p\rho = \rho [1 + (n-1)p]$$

- 25** (c) From Bernoulli's theorem, $p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$

where, p_1, p_2 are pressure inside and outside the roof and v_1, v_2 are velocities of wind inside and outside the roof.

Neglect the width of the roof. Pressure difference is

$$p_1 - p_2 = \frac{1}{2}\rho (v_2^2 - v_1^2) = \frac{1}{2} \times 1.2 (40^2 - 0) = 960 \text{ Nm}^{-2}$$

Force acting on the roof is given by

$$F = (p_1 - p_2)A = 960 \times 250 = 24 \times 10^4 \text{ N} = 2.4 \times 10^5 \text{ N}$$

As the pressure inside the roof is more than outside to it. So, the force will act in the upward direction.

$$\text{i.e. } F = 2.4 \times 10^5 \text{ N (upward).}$$

- 26 (c)** Given, $d = 2700 \text{ m}$, $\rho = 10^3 \text{ kg m}^{-3}$

$$\text{Compressibility} = 45.4 \times 10^{-11} \text{ Pa}^{-1}$$

The pressure at the bottom of ocean is given by

$$p = \rho g d = 10^3 \times 10 \times 2700 = 27 \times 10^6 \text{ Pa}$$

So, fractional compression = compressibility $\times$ pressure

$$= 45.4 \times 10^{-11} \times 27 \times 10^6 = 1.2 \times 10^{-2}$$

- 27 (d)** Let the velocity at point B is v_B .

From conservation of total mechanical energy,

$$mg(h - L \sin \alpha) = \frac{1}{2} m v_B^2$$

$$\Rightarrow v_B^2 = 2g(h - L \sin \alpha) = 2g\left(10 - 2 \times \frac{1}{2}\right)$$

$$\Rightarrow v_B^2 = 18g$$

Now, let maximum height attained by water stream be H .

$$\therefore H = L \sin \alpha + \frac{v_B^2 \sin^2 \alpha}{2g}$$

$$\Rightarrow H = 2 \times \frac{1}{2} + \frac{18g \left(\frac{1}{2}\right)^2}{2g} = 1 + \frac{18}{8} = 3.2 \text{ m}$$

- 28 (b)** Gain in surface energy,

$$\begin{aligned} E &= 4\pi R^2 (n^{1/3} - 1)T \\ &= 4 \times 3.14 \times (10^{-2})^2 [(1000)^{1/3} - 1] \times 0.075 \\ &= 8.48 \times 10^{-4} \text{ J} \end{aligned}$$

- 29 (d)** Given, height raised by liquid,

$$h = 7.5 \text{ cm} = 7.5 \times 10^{-2} \text{ m}$$

Surface tension, $S = 7.5 \times 10^{-2} \text{ Nm}^{-1}$

and contact angle, $\theta = 0^\circ$

$$\text{As, } r = \frac{2S \cos \theta}{h\rho g}$$

where, ρ = density of water = 1000 kg m^{-3}

$$\begin{aligned} \Rightarrow r &= \frac{2 \times 7.5 \times 10^{-2} \times \cos 0^\circ}{7.5 \times 10^{-2} \times 1000 \times 10} \\ &= \frac{2}{10^4} = 2 \times 10^{-4} \text{ m} = \frac{2}{10} \times 10^{-3} \text{ m} = 0.2 \text{ mm} \end{aligned}$$

- 30 (a)** The area of soap bubble = $4\pi \left(\frac{a}{2}\right)^2 = 4\pi \frac{a^2}{4} \Rightarrow A = \pi a^2$

$$\text{As, surface energy} = \frac{\text{work done}}{\text{area}} = \frac{TA}{A} = T$$

$\therefore$ Surface energy is equal to surface tension.

Total energy associated with surface, $U = TA$

As soap bubbles are of two layers, so

$$U = T \times 2A = T \times 2\pi a^2 = 2\pi a^2 T$$

- 31 (d)** When the stones are unloaded into water, the water level falls because the volume of the water displaced by stones in water will be less than the volume of water displaced when stones are in the boat.

- 32 (a)** Statement given in option (a) is correct, while rest are incorrect whose corrected form is as

Water proof agents increase the angle of contact between water and fibres.

Detergents decrease the surface tension of water.

Hydraulic machines work on the Pascal's law.

Venturimeter measures the flow speed of fluid in a pipe, which can be compressible or incompressible.

- 33 (d)** In freely falling platform, a body experience weightlessness. So, the liquid will rise up to length of the capillary, *i.e.* height raised by the liquid will be 20 cm.

- 34 (c)** Pressure difference between atmosphere and lungs = Pressure difference at a depth h of fluid

$$\begin{aligned} \Rightarrow 760 - 750 \text{ mm} \times \rho_{\text{Hg}} \times g &= h \rho_w g \\ h &= 10 \text{ mm} \times \frac{13.6}{1} = 13.6 \text{ cm} \end{aligned}$$

- 35 (a)** Let R = radius of bigger drop

and r = radius of smaller drop.

$$\text{Then, } \frac{4}{3} \pi R^3 = 1000 \times \frac{4}{3} \pi r^3 \Rightarrow R = 10r$$

The surface energy of smaller drop, $E_1 = S \times 4\pi r^2$ and the surface energy of bigger drop,

$$E_2 = S \times 4\pi (10r)^2 \Rightarrow \frac{E_1}{E_2} = \frac{1}{100}$$

- 36 (c)** Volume of part above the water surface = $\frac{V}{3}$

$$\text{Volume of solid inside the water surface} = V - \frac{V}{3} = \frac{2V}{3}$$

$$\text{Weight of displaced water, } w = \frac{2V}{3} \times 10^3 \times g$$

Now, weight of body = weight of displaced water

$$\Rightarrow V\rho g = \frac{2V}{3} \times 10^3 g$$

$$\Rightarrow \rho = \frac{2}{3} \times 1000 = \frac{2000}{3} \text{ kg m}^{-3}$$

- 37 (a)** Change in surface area

$$= 2 \times 4\pi \left[\left(\frac{D}{2}\right)^2 - \left(\frac{d}{2}\right)^2 \right] = 2\pi (D^2 - d^2)$$

$$\begin{aligned} \therefore \text{Work done} &= \text{Surface tension} \times \text{Change in area} \\ &= 2\pi S (D^2 - d^2) \end{aligned}$$

- 38 (a)** $R_e \leq 1000$: Streamlined flow

$1000 < R_e < 2000$; Unstable flow

$R_e \geq 2000$; Turbulent flow

- 39 (c)** Here, if the surface area changes, it will change the surface energy as well. If the surface area decreases, it means that energy is released and *vice-versa*.

$$\text{Change in surface energy} = \Delta A \times T \quad \dots(i)$$

Let we have n number of drops initially.

$$\text{So, } \Delta A = 4\pi R^2 - n(4\pi r^2) \quad \dots(ii)$$

Volume is constant.

$$\text{So, } n \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3 = V \quad \dots(iii)$$

From Eqs. (ii) and (iii), we get

$$\Delta A = \frac{3}{R} \frac{4\pi}{3} \times R^3 - \frac{3}{r} \left(n \frac{4\pi}{3} r^3 \right) = \frac{3}{R} \times V - \frac{3}{r} V$$

$$\Rightarrow \Delta A = 3V \left(\frac{1}{R} - \frac{1}{r} \right)$$

As $R > r$, so ΔA is negative. It means that surface area is decreased, so energy must be released.

$$\text{Energy released} = \Delta A \times T = 3VT \left(\frac{1}{R} - \frac{1}{r} \right)$$

40 (a) Assuming temperature is constant.

As, from the Boyle's law $pV = \text{constant}$.

So, from the two cases at depth h and at the surface,

$$p_1 V_1 = p_2 V_2 \quad \dots(i)$$

where, p_1 = pressure at depth h , V_1 = volume at depth h
 p_2 = pressure at surface, V_2 = volume at surface.

$$\Rightarrow p_1 = (h + l) \rho g, V_1 = \frac{4}{3} \pi r^3$$

$$p_2 = l \rho g, V_2 = \frac{4}{3} \pi (3r)^3$$

Putting these values in Eq. (i), we get

$$(h + l) \rho g \left(\frac{4}{3} \pi r^3 \right) = l \rho g \times \left(\frac{4}{3} \pi (3r)^3 \right)$$

$$\text{So, } (h + l) = 27l$$

$$\Rightarrow h = 26l$$

41 (b) According to Archimedes' principle, weight of the wooden block = weight of the liquid displaced

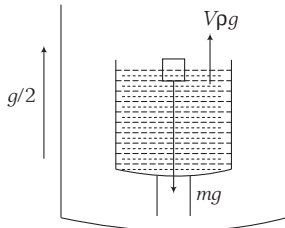
$$\Rightarrow mg = \left(\frac{60}{100} \right) V \times \rho_l \times g = \frac{3}{5} V \times \rho_l \times g \quad \dots(ii)$$

[Here, V = total volume of block, ρ_l = density of liquid]

When beaker is kept in a lift,

Net weight of the block = $mg + mg/2$

$$= \frac{3mg}{2} \quad \left(\because R - mg = \frac{mg}{2} \right)$$



Again, for float condition,

Net weight of the block = Buoyant force

$$\Rightarrow \frac{3mg}{2} = V_l \times \rho_l \times g \quad \dots(iii)$$

Here, V_l = volume of block inside liquid.

From Eqs. (i) and (ii), we get

$$\left(\frac{3}{2} \right) \times \frac{3}{5} (V \times \rho_l \times g) = V_l \times \rho_l \times g$$

$$\text{or } V_l = \frac{9}{10} V$$

So, volume of the block above liquid

$$= \frac{1}{10} V = 10\% \text{ of total volume.}$$

42 (c) Let thickness of layer be x .

So, volume, $V = \text{area} \times x$

$$\Rightarrow V = A \times x$$

$$x = V / A \Rightarrow 2r = \frac{V}{A} \quad (\because x = 2r)$$

$$\Rightarrow r = \frac{V}{2A} \quad \dots(i)$$

$$\text{and } \Delta p = \frac{T}{r}$$

We know that, $F = \Delta p \times A = \frac{T}{r} \times A$

$$\Rightarrow F = \left(\frac{T}{\frac{V}{2A}} \right) \times A \quad [\text{From Eq. (i)}]$$

$$\Rightarrow T = \frac{F \times V}{2A^2}$$

where, $F = 16 \times 10^5$ dyne, $V = 0.04$ cm³, $A = 20$ cm²

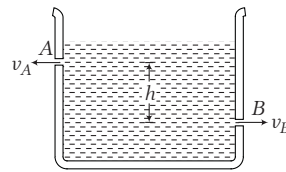
$$\therefore T = \frac{16 \times 10^5 \times 0.04}{2 \times 20^2}$$

$$= \frac{8 \times 10^5 \times 4}{20^2 \times 100} = \frac{8 \times 10^5 \times 4}{400 \times 100}$$

$$= 8 \times 10^5 \times 10^{-4} = 80 \text{ dyne cm}^{-1}$$

43 (d) The value of angle of contact determines whether a liquid will spread on the surface.

44 (b)



$$\text{Net horizontal force (reaction), } F = F_B - F_A = \frac{dp_B}{dt} - \frac{dp_A}{dt}$$

$$= av_B \rho \times v_B - av_A \rho \times v_A$$

$$\therefore F = a \rho (v_B^2 - v_A^2) \quad \dots(i)$$

According to Bernoulli's theorem,

$$p_A + \frac{1}{2} \rho v_A^2 + \rho gh = p_B + \frac{1}{2} \rho v_B^2 + (p_A - p_B)$$

$$\Rightarrow \frac{1}{2} \rho (v_B^2 - v_A^2) = \rho gh \Rightarrow v_B^2 - v_A^2 = 2gh$$

From Eq. (i), we get

$$F = 2a \rho gh \Rightarrow F \propto h$$

45 (d) Given, $h_1 = 10$ cm, $h_2 = 3.42$ cm,

$$\theta_2 = 135^\circ \text{ and } d_2 = 13.6 \text{ gcc}^{-1}$$

We know that, angle of contact of water, $\theta_1 = 0^\circ$.

$$\begin{aligned} \text{Here, } \frac{T_1}{T_2} &= \frac{h_1 d_1 \cos \theta_2}{h_2 d_2 \cos \theta_1} \\ \Rightarrow \frac{T_1}{T_2} &= \frac{10 \times 1}{3.42 \times 13.6} \frac{\cos 135^\circ}{\cos 0^\circ} \\ \Rightarrow \frac{T_1}{T_2} &\approx \frac{2}{13} \end{aligned}$$

46 (c) We know that, pressure, $p \propto \frac{r^4}{l}$

Given, $l_1 : l_2 = 2 : 1$ and $r_1 : r_2 = 1 : 2$

Let $l_1 = 2l$, $l_2 = l$ and $r_1 = r$, $r_2 = 2r$

$$\begin{aligned} \text{Now, } \frac{p_1}{p_2} &= \frac{r_1^4}{l_1} \times \frac{l_2}{r_2^4} \Rightarrow \frac{p_1}{p_2} = \frac{r^4}{2l} \times \frac{l}{(2r)^4} \\ \Rightarrow \frac{p_1}{p_2} &= \frac{1}{2 \times (2)^4} \Rightarrow \frac{p_1}{p_2} = \frac{1}{32} \end{aligned}$$

$\therefore$ Pressure across the tubes has ratio of 1: 32 or 32:1.

47 (b) We know that, the pressure at the end of tube, $p = \frac{4T}{r}$

Here, the pressure difference at the both end of tube,

$$\Delta p = \Delta p_1 - \Delta p_2 = \frac{4T}{r_1} - \frac{4T}{r_2} = 4T \left[\frac{r_2 - r_1}{r_1 r_2} \right]$$

Given, $r_1 = 5 \text{ cm} = 0.05 \text{ m}$ and $r_2 = 3 \text{ cm} = 0.03 \text{ m}$

$$\begin{aligned} \Delta p &= 4 \times 0.03 \left[\frac{0.05 - 0.03}{0.05 \times 0.03} \right] \\ &= \frac{4 \times 0.03 \times 0.02}{0.05 \times 0.03} = \frac{0.0024}{0.0015} = 1.6 \text{ Pa} \end{aligned}$$

48 (a) We know that, for soap bubble, $p = \frac{4T}{r}$

Given, $p_1 = 3p_2$

$$\text{Here, } \frac{4T}{r_1} = 3 \left(\frac{4T}{r_2} \right) \Rightarrow r_2 = 3r_1 \quad \dots(i)$$

$$\begin{aligned} V_1 &= \frac{4}{3} \pi r_1^3 \text{ and } V_2 = \frac{4}{3} \pi r_2^3 \\ &= \frac{4}{3} \pi (3r_1)^3 = \frac{4}{3} \pi (27r_1^3) \end{aligned}$$

$$\Rightarrow \frac{V_1}{V_2} = \frac{(4/3) \pi r_1^3}{(4/3) \pi 27r_1^3} = \frac{1}{27}$$

49 (b) We know that, capillary rises,

$$h = \frac{2S \cos \theta}{\rho g r} \Rightarrow h \propto \frac{1}{r}$$

$$\Rightarrow h' = h \frac{r}{r'} = 20 \times 3 = 60 \text{ mm}$$

50 (c) We have, $1 \text{ P} = 1 \text{ g cm}^{-1} \text{ s}^{-1}$

$$\therefore 1 \text{ Pl} = 10 \text{ g cm}^{-1} \text{ s}^{-1} = 10 \text{ P}$$

51 (c) If a liquid can rise to a height h , but the tube has insufficient height h' , then the angle of contact increases from

$$\theta \text{ to } \theta' \text{ given by } \frac{h}{\cos \theta} = \frac{h'}{\cos \theta'}$$

Here, $h = 2 \text{ cm}$, $h' = 1 \text{ cm}$, $\theta = 0^\circ$

$$\begin{aligned} \therefore \frac{2}{\cos 0^\circ} &= \frac{1}{\cos \theta'} \\ \Rightarrow \frac{2}{1} &= \frac{1}{\cos \theta'} \\ \text{or } \cos \theta' &= \frac{1}{2} \\ \Rightarrow \theta' &= 60^\circ \end{aligned}$$

52 (a) The height of capillary rise is inversely proportional to radius (or diameter) of capillary tube, i.e. $h \propto \frac{1}{r}$.

So, for smaller r , the value of h is higher.

Hence, both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

53 (a) The excess pressure inside the bubble,

$$p = \frac{4T}{r}, \text{ then } p_1 = \frac{4T}{r_1} \quad \dots(i)$$

$$\text{and } p_2 = \frac{4T}{r_2} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{p_1}{p_2} = \frac{4T/r_1}{4T/r_2} = \frac{r_2}{r_1} = \frac{1}{2}$$

54 (c) Time taken to empty a vessel of height h ,

$$t = \sqrt{2h/g}$$

$\therefore$ Time taken in emptying a vessel of height $h/2$,

$$t' = \sqrt{\frac{2h/2}{g}} = \sqrt{\frac{h}{g}}$$

$$\therefore \frac{t'}{t} = \frac{1}{\sqrt{2}}$$

$$\text{or } t' = \frac{t}{\sqrt{2}} = \frac{10}{\sqrt{2}} \approx 7 \text{ min}$$

55 (a) From the principle of continuity, $A_1 v_1 = A_2 v_2$

$$\Rightarrow v_2 = \frac{A_1 v_1}{A_2} = \frac{4.20 \times 5.18}{7.60} = 2.86 \text{ ms}^{-1}$$

56 (b) Volume of liquid displaced is more than the volume of water formed. So, the level of water in the beaker falls.

57 (b) The excess pressure of soap bubble,

$$p = \frac{4T}{R}$$

$$\text{or } h \rho g = \frac{4T}{R} \quad (\because p = h \rho g)$$

$$\begin{aligned} \Rightarrow T &= \frac{R h \rho g}{4} = \frac{1 \times 10^{-2} \times 2 \times 10^{-3} \times 0.8 \times 10^3 \times 9.8}{4} \\ &= 0.0392 \text{ Nm}^{-1} \end{aligned}$$

58 (b) The equation of continuity,

$$A_1 v_1 = A_2 v_2$$

$$\Rightarrow A \times 4 = \frac{2}{3} A \times v_2$$

$$\Rightarrow v_2 = 6 \text{ ms}^{-1}$$

From Bernoulli's theorem,

$$p + \rho gh_1 + \frac{1}{2} \rho v_1^2 = p + \rho gh_2 + \frac{1}{2} \rho v_2^2$$

$$\text{or } g(h_1 - h_2) = \frac{1}{2} (v_2^2 - v_1^2)$$

$$\text{or } g \times h = \frac{1}{2} [(6)^2 - (4)^2] \quad (\because h_1 - h_2 = h)$$

$$\text{or } 10 \times h = \frac{1}{2} [36 - 16]$$

$$\text{or } h = \frac{20}{20} = 1 \text{ m}$$

59 (c) Specific gravity of Al = 2.7

Density of Al = $2.7 \times 10^3 \text{ kg m}^{-3}$

$$\text{Terminal velocity, } v = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

$$v = \frac{2 \times (1 \times 10^{-3})^2 (2.7 \times 10^3 - 1000) \times 9.8}{9 \times 8 \times 10^{-4}}$$

$$\Rightarrow v = 4.6 \text{ ms}^{-1}$$

60 (d) As, $V = \frac{\pi pr^4}{8\eta l}$

$$\therefore V \propto pr^4 \quad (\eta \text{ and } l \text{ are constants})$$

$$\therefore \frac{V_2}{V_1} = \left(\frac{p_2}{p_1} \right) \left(\frac{r_2}{r_1} \right)^4 = 2 \times \left(\frac{1}{2} \right)^4 = \frac{1}{8}$$

$$\Rightarrow V_2 = \frac{Q}{8}$$

61 (c) Air flow from smaller bubble to bigger bubble as the pressure in smaller bubble is higher.

62 (d) Using the law of buoyancy, we get

$V\sigma g = 0.6 V\sigma_1 g$, for the part of body inside water and

$V\sigma g = 0.4 V\sigma_2 g$, for the part of body inside oil, hence we get

$$0.6 V\sigma_1 g = 0.4 V\sigma_2 g$$

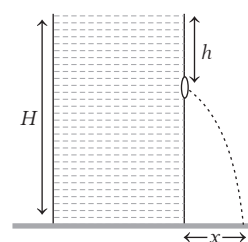
$$\Rightarrow 1 = \frac{0.6\sigma_1}{0.4\sigma_2}$$

So, we get

$$\frac{\sigma_2}{\sigma_1} = \frac{6}{4} = \frac{3}{2} = 1.5$$

63 (b) Time taken by water to reach the foot,

$$t = \sqrt{\frac{2(H-h)}{g}}$$



Velocity of water coming out of hole,

$$v = \sqrt{2gh}$$

$\therefore$ Emerging stream of water strike the foot at distance,

$$\begin{aligned} x &= v \times t \\ &= \sqrt{2gh} \times \sqrt{\frac{2(H-h)}{g}} \\ &= 2\sqrt{h(H-h)} \end{aligned}$$

Thermometry, Thermal Expansion and Kinetic Theory of Gases

The branch of science which deals with the measurement of temperature of a substance is known as **thermometry**.

Almost all substances (solids, liquids and gases) expand on heating and contract on cooling. Few exceptions are: water from 0°C to 4°C and silver iodide from 80°C to 140°C contract on heating and expand on cooling. The expansion of a substance on heating is called **thermal expansion of that substance**.

Further, in this chapter, we will study the **kinetic theory of gases**, which relates the macroscopic properties (pressure, temperature, etc.) of gases to the microscopic properties (velocity, speed, etc.) of gas molecules.

TEMPERATURE

The degree of hotness or coldness of a body is known as the temperature of the body. An object at higher temperature is said to be hotter than the one at lower temperature.

Scales of temperature

The measurement of temperature is done by some specified scales. In any scale of temperature, two fixed points are considered: one is ice point (Lower Fixed Point-LFP) and other is steam point (Upper Fixed Point-UFP). Generally, these points are **freezing point** and **boiling point** of water at 1 atm.

Some of the commonly used temperature scales are discussed below

- (i) **The Celsius scale (°C)** This scale was designed by Anders Celsius. It defines the ice point temperature as 0°C and the steam point temperature as 100°C. The space between 0°C and 100°C is divided into 100 equal intervals and each interval is known as 1°C.

$$0^{\circ}\text{C} \xleftarrow{100^{\circ}\text{C}} 100^{\circ}\text{C} \text{ (100 equal parts)}$$

Inside

1 Temperature

- Scales of temperature
- Thermometers

2 Thermal expansion

- Types of thermal expansion
- Variation of density with temperature

3 Kinetic theory of ideal gases

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- Ideal gas equation
- Molecular velocities
- Mean free path (λ)

4 Ideal gas laws

- Boyle's law
- Charles' law
- Gay Lussac's law or pressure law
- Dalton's law of partial pressure
- Graham's law of diffusion
- Behaviour of real gases
- van der Waals' equation

5 Degree of freedom

- Maxwell's law of equipartition of energy
- Critical temperature, pressure and volume

- (ii) **The Fahrenheit scale (°F)** This scale was designed by Fahrenheit. It defines the ice point temperature as 32°F and the steam point as 212°F. The space between 32°F and 212°F is divided into 180 equal intervals and each interval represents 1°F.

$$32^\circ \text{ F} \xrightarrow{180^\circ \text{ F}} 212^\circ \text{ F} \text{ (180 equal parts)}$$

- (iii) **The Kelvin scale (K)** This scale was designed by Kelvin. It defines the ice point temperature as 273.15 K and the steam point temperature as 373.15 K. The space between 273.15 K and 373.15 K is divided into 100 equal intervals and each interval represents 1 K. Kelvin is the SI unit of temperature.

$$273.15 \text{ K} \xrightarrow{100 \text{ K}} 373.15 \text{ K} \text{ (100 equal parts)}$$

Conversion of temperature from one scale to another

In order to convert the temperature from one scale to another, the following relation should be kept in mind, *i.e.*

$$\frac{\text{Temperature on one scale} - \text{LFP (ice point)}}{\text{UFP (steam point)} - \text{LFP (ice point)}} = \frac{\text{Temperature on other scale} - \text{LFP (ice point)}}{\text{UFP (steam point)} - \text{LFP (ice point)}}$$

In simple terms, the temperature measured by different scales is given as

$$\frac{C - 0}{100} = \frac{F - 32}{180} = \frac{K - 273}{100}$$

Temperature difference measured by different scales is given as

$$\Delta C = \Delta K = (5/9) \Delta F$$

Example 14.1 Express the temperature of 60°F in degree celsius and in kelvin.

Sol. Temperature in degree celsius, $\frac{C - 0}{100} = \frac{F - 32}{180}$

$$C = \frac{5}{9} (F - 32) = \frac{5}{9} (60^\circ - 32^\circ) = 15.55^\circ \text{ C}$$

$$\begin{aligned} \text{Temperature in kelvin, } K &= C + 273.15 \\ &= 15.55^\circ \text{ C} + 273.15 \\ &= 288.7 \text{ K} \end{aligned}$$

Example 14.2 The temperature of an iron piece is raised from 30°C to 90°C. What is the change in its temperature on the Fahrenheit scale and on the Kelvin scale?

Sol. Given, $\Delta C = 90^\circ - 30^\circ = 60^\circ \text{ C}$

Temperature difference on Fahrenheit scale,

$$\Delta F = \frac{9}{5} \Delta C = \frac{9}{5} (60^\circ \text{ C}) = 108^\circ \text{ F}$$

Temperature difference on Kelvin scale, $\Delta K = \Delta C = 60 \text{ K}$

Example 14.3 On a new scale of temperature (which is linear) and called the W scale, the freezing and boiling points of water are 39° W and 239° W, respectively. What will be the temperature on the new scale, corresponding to a temperature of 39°C on the Celsius scale?

Sol. In general, whenever we have to go from any known scale to any unknown scale, then we follow the equation

$$\begin{aligned} &\frac{(\text{Temperature on one scale}) - (\text{LFP for known scale})}{(\text{UFP} - \text{LFP})_{\text{known}}} \\ &= \frac{(\text{Temperature on other scale}) - (\text{LFP for unknown scale})}{(\text{UFP} - \text{LFP})_{\text{unknown}}} \\ \Rightarrow &\frac{39 - 0}{100 - 0} = \frac{t - 39}{239 - 39} \Rightarrow t = 117^\circ \text{ W} \end{aligned}$$

Example 14.4 An arbitrary scale has the ice point at -20° and the steam point at 180°. When the thermometer reads 5°, then find the reading of centigrade thermometer.

Sol. In order to convert the temperature from one scale to another,

$$\frac{C - 0}{100 - 0} = \frac{t - (-20)}{180 - (-20)}$$

Here, $t = 5^\circ$

$$\Rightarrow \frac{C}{100} = \frac{5 + 20}{200}$$

$\therefore$ Reading of centigrade thermometer,
 $C = 12.5^\circ \text{ C}$

Example 14.5 A faulty scale has its fixed points marked as 5° and 95°. The temperature of a body as measured by the faulty scale is 59°. Find the correct temperature of the body on Celsius scale.

Sol. Let θ_0 be the lower fixed point of the faulty scale and n be the number of divisions between its lower and the upper fixed points. If a temperature C on Celsius scale corresponds to temperature θ on the faulty scale, then

$$\frac{C - 0}{100} = \frac{\theta - \theta_0}{n} \quad \dots(i)$$

Here, $\theta_0 = 5^\circ$, $n = 95 - 5 = 90$ and $\theta = 59^\circ$

Therefore, the Eq. (i) becomes

$$\frac{C - 0}{100} = \frac{59 - 5}{90}$$

$$\text{or } C = \frac{54}{90} \times 100 = 60^\circ \text{ C}$$

Thermometers

The thermometers work on the thermometric property, *i.e.* the property which changes with temperature like any physical quantity such as length, volume, pressure, resistance, etc. which varies linearly with a certain range of temperature. Let X denotes the thermometric physical

quantity and X_0 , X_{100} and X_t be its values at 0°C , 100°C and $t^\circ\text{C}$ respectively. Then,

$$\text{Temperature, } t = \left(\frac{X_t - X_0}{X_{100} - X_0} \right) \times 100^\circ\text{C}$$

Different types of thermometers

Some different types of thermometers are given below

- (i) **Mercury thermometer** In this thermometer, the length of a mercury column from some fixed point is taken as thermometric property. If length of mercury column at 0° and 100° are l_0 and l_{100} respectively and at t° , the length of mercury is l_t . Then,

$$t = \left(\frac{l_t - l_0}{l_{100} - l_0} \right) \times 100^\circ\text{C}$$

- (ii) **Constant volume gas thermometer** It works on the principle of change in pressure with temperature when the volume is kept constant. If p_0 , p_{100} and p_t are the pressures of gas at temperatures 0°C , 100°C and unknown temperature ($t^\circ\text{C}$) respectively keeping the volume constant, then

$$t = \left(\frac{p_t - p_0}{p_{100} - p_0} \times 100 \right)^\circ\text{C}$$

Note For constant pressure gas thermometers, $t = \frac{V_t - V_0}{V_{100} - V_0} \times 100^\circ\text{C}$

- (iii) **Platinum resistance thermometer** It works on the principle of variation of resistance of metals with temperature.

If R_0 , R_{100} and R_t are the resistances of a platinum wire at temperature 0°C , 100°C and unknown temperature ($t^\circ\text{C}$), respectively.

$$\text{Then, } t = \left(\frac{R_t - R_0}{R_{100} - R_0} \times 100 \right)^\circ\text{C} = \left(\frac{R_t}{R_{tr}} \times 273.16 \right) \text{K}$$

where, R_{tr} = resistance of wire at 273.16 K .

Here, temperature coefficient of resistance (α) is given by

$$\alpha = \frac{R_{100} - R_0}{R_0 \times 100}$$

- (iv) **Pyrometers** Pyrometers are the devices used to measure the temperature by measuring the

intensity of radiations received from the body. They are based on the fact that the amount of radiation emitted from a body per unit area per second is directly proportional to the fourth power of temperature, *i.e.* Stefan's law. These can be used to measure temperature ranging from 800°C to 6000°C .

Example 14.6 The pressures of the gas filled in the bulb of a constant volume gas thermometer are 66 cm and 88 cm of mercury column at 0°C and 100°C , respectively. When its bulb is immersed in a liquid placed in a vessel, its pressure is 82.5 cm of mercury column. Calculate the temperature of the liquid.

Sol. Given, pressure at 0°C , $p_0 = 66 \text{ cm}$ of Hg (mercury) column

Pressure at 100°C , $p_{100} = 88 \text{ cm}$ of Hg column

Pressure at unknown temperature t , $p_t = 82.5 \text{ cm}$ of Hg column

According to formula, the unknown temperature is

$$\begin{aligned} t &= \left(\frac{p_t - p_0}{p_{100} - p_0} \right) \times 100^\circ\text{C} \\ &= \left(\frac{82.5 - 66}{88 - 66} \right) \times 100^\circ\text{C} \\ &= \frac{16.5}{22} \times 100^\circ\text{C} = 75^\circ\text{C} \end{aligned}$$

Example 14.7 The following observations were recorded on a platinum resistance thermometer. Resistance at melting point of ice = 3.70Ω , resistance at boiling point of water at normal pressure = 4.71Ω and resistance at $t^\circ\text{C}$ = 5.29Ω . Calculate

- (i) temperature coefficient of resistance of platinum.
(ii) value of temperature t .

Sol. Given, resistance at melting point of ice, $R_0 = 3.70 \Omega$

Resistance at boiling point of water at normal pressure,

$$R_{100} = 4.71 \Omega$$

Resistance at $t^\circ\text{C}$, $R_t = 5.29 \Omega$

- (i) According to the formula, temperature coefficient of resistance is given by

$$\alpha = \frac{R_{100} - R_0}{R_0 \times 100} = \frac{4.71 - 3.70}{3.70 \times 100} = \frac{1.01}{370} = 2.73 \times 10^{-3}/^\circ\text{C}$$

- (ii) According to the formula, for temperature t , we have

$$\begin{aligned} t &= 100^\circ\text{C} \times \frac{R_t - R_0}{R_{100} - R_0} \\ &= 100^\circ\text{C} \times \frac{5.29 - 3.70}{4.71 - 3.70} = 100^\circ\text{C} \times \frac{1.59}{1.01} = 157.4^\circ\text{C} \end{aligned}$$

CHECK POINT 14.1

- A device used to measure temperature is
(a) hygrometer (b) thermometer
(c) barometer (d) calorimeter
- A difference of temperature of 25°C is equivalent to a difference of
(a) 45°F (b) 72°F (c) 32°F (d) 25°F
- At what temperature, does the temperature in Celsius and Fahrenheit equalise?
(a) -40° (b) 40°
(c) 36.6° (d) 38°
- The freezing point on a thermometer is marked as -20° and the boiling point as 130° . A temperature of human body (34°C) on this thermometer will be read as
(a) 31° (b) 51°
(c) 20° (d) None of these
- The temperature of a body on Kelvin scale is found to be x Kelvin. When it is measured by Fahrenheit thermometer, it is found to be $x^{\circ}\text{Fahrenheit}$, then the value of x is
(a) 40 (b) 313
(c) 574.25 (d) 301.25
- A faulty thermometer has its fixed points marked 5° and 95° . If the temperature of a body as shown on the Celsius scale is 40°C , then its temperature shown on this faulty thermometer is
(a) 39° (b) 40°
(c) 41° (d) 44.4°
- The readings of a bath on Celsius and Fahrenheit thermometers are in the ratio 2 : 5. The temperature of the bath is
(a) -26.66°C (b) 40°C
(c) 45.71°C (d) 26.66°C
- The reading of air thermometer at 0°C and 100°C are 50 cm and 75 cm of mercury column, respectively. The temperature at which its reading is 80 cm of mercury column is
(a) 105°C (b) 110°C
(c) 115°C (d) 120°C

THERMAL EXPANSION

When matter is heated without any change in its state, it usually expands. This phenomenon of expansion of matter on heating is called thermal expansion.

On heating the matter, the energy of atoms increases and hence, the average distance between them also increases which results in thermal expansion. Thermal expansion is minimum in solids and maximum in gases because intermolecular forces are maximum in solids and minimum in gases.

Types of thermal expansion

Solids can expand in one dimension (linear expansion), two dimensions (superficial expansion) and three dimensions (volume expansion) while liquids and gases usually expand only in three dimensions.

Thermal expansion in solids

There are three types of thermal expansion

1. Linear expansion

Suppose the temperature of a thin uniform rod of length l is changed from T to $T + \Delta T$, then its length becomes $l + \Delta l$.



Fig. 14.1 Thermal expansion in rod

Then, increase in length, $\Delta l = l_0 \alpha \Delta T \Rightarrow \alpha = \frac{\Delta l}{l_0 \Delta T}$

Therefore, final length, $l = l_0 + \Delta l$

$$l = l_0 [1 + \alpha \Delta T]$$

Here, the constant α is called the **coefficient of linear expansion** of the material of the rod and its unit is K^{-1} or $(^{\circ}\text{C})^{-1}$. Its dimensions are $[\text{M}^0 \text{L}^0 \text{T}^0 \theta^{-1}]$.

Note Test tubes, beakers and cubicles are made up of pyrex-glass or silica because they have very low value of coefficient of linear expansion.

Example 14.8 A steel ruler exactly 20 cm long is graduated to give correct measurements at 20°C .

- Will it give readings that are too long or too short at lower temperatures?
- What will be the actual length of the ruler be when it is used in the desert at a temperature of 40°C ?
(Take, $\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ } ^{\circ}\text{C}^{-1}$)

Sol. (i) If the temperature decreases, the length of the ruler also decreases due to thermal contraction. Below 20°C , each centimetre division is actually somewhat shorter than 1.0 cm, so the steel ruler gives readings that are too long.

(ii) Given, change in temperature,

$$\Delta T = T_2 - T_1 = 40^{\circ} - 20^{\circ} = 20^{\circ}\text{C}$$

$$\text{Length, } l = 20 \text{ cm}$$

At 40°C , the increase in length of the ruler,

$$\Delta l = l \alpha \Delta T = (20) (1.2 \times 10^{-5}) (20)$$

$$= 0.48 \times 10^{-2} \text{ cm}$$

∴ The actual length of the ruler,

$$l' = l + \Delta l = 20 + 0.48 \times 10^{-2} \\ = 20.0048 \text{ cm}$$

Example 14.9 The length of a steel rod is 5 cm longer than that of a brass rod. If this difference in their lengths is to remain same at all temperatures, then find the length of brass rod. (Coefficients of linear expansion for steel and brass are $12 \times 10^{-6} / ^\circ\text{C}$ and $18 \times 10^{-6} / ^\circ\text{C}$, respectively.)

Sol. Given, $\Delta l_s - \Delta l_b = \Delta l$

Let $l_s = l \text{ cm} \Rightarrow l_b = (l - 5) \text{ cm}$
 (∵ Steel rod is 5 cm longer than that of a brass rod)
 $\alpha_s = 12 \times 10^{-6} / ^\circ\text{C}$ and $\alpha_b = 18 \times 10^{-6} / ^\circ\text{C}$

As, $\Delta l_s = l_s \alpha_s t$

$$\therefore \alpha_s = \frac{\Delta l_s}{l_s \times t}$$

$$\Rightarrow 12 \times 10^{-6} = \frac{\Delta l}{l \times t} \quad \dots(i)$$

Similarly, $\Delta l_b = l_b \alpha_b t$

$$\Rightarrow \alpha_b = \frac{\Delta l_b}{l_b \times t}$$

$$\Rightarrow 18 \times 10^{-6} = \frac{\Delta l}{(l - 5) \times t} \quad \dots(ii)$$

Dividing Eq. (i) by Eq. (ii), we get

$$\frac{12 \times 10^{-6}}{18 \times 10^{-6}} = \frac{\Delta l / l \times t}{\Delta l / (l - 5) \times t}$$

$$\Rightarrow \frac{2}{3} = \frac{l - 5}{l}$$

$$\Rightarrow 2l = 3l - 15 \Rightarrow l = 15 \text{ cm}$$

So, $l_s = 15 \text{ cm}$ and $l_b = l - 5 = 15 - 5 = 10 \text{ cm}$

2. Superficial (areal) expansion

Suppose the temperature of a metal sheet of an initial surface area A_0 is changed from T to $T + \Delta T$, then its area becomes $A + \Delta A$.

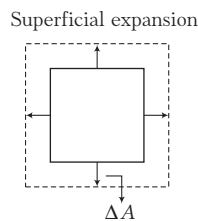


Fig. 14.2 Thermal expansion in thin sheet

Now, increase in area, $\Delta A = A_0 \beta \Delta T$

Therefore, final area, $A = A_0 + \Delta A$

$$A = A_0 [1 + \beta \Delta T]$$

where, β is the **coefficient of superficial (areal) expansion** and its unit is $(^\circ\text{C})^{-1}$ or K^{-1} .

Example 14.10 A metal ball having a diameter of 0.4 m is heated from 273 K to 360 K. If the coefficient of areal expansion of the material of the ball is 0.000034 K^{-1} , then determine the increase in surface area of the ball.

Sol. Given, diameter = 0.4 m and radius, $r = \frac{0.4}{2} = 0.2 \text{ m}$

$$\therefore \text{Area of ball, } A_0 = 4\pi r^2 = 4 \times \pi \times (0.2)^2 = 0.5024 \text{ m}^2$$

$$\text{Change in temperature, } \Delta T = T_2 - T_1 = 360 \text{ K} - 273 \text{ K} = 87 \text{ K}$$

$$\text{Coefficient of areal expansion, } \beta = 0.000034 \text{ K}^{-1}$$

$$\text{Increase in surface area, } \Delta A = \beta A_0 \Delta T$$

$$\Rightarrow \Delta A = 0.000034 \times 0.5024 \times 87 = 0.001486 \\ = 1.486 \times 10^{-3} \text{ m}^2$$

3. Volume expansion

Suppose the temperature of a metal block of volume V_0 is increased from T to $T + \Delta T$, then its volume becomes $V + \Delta V$.

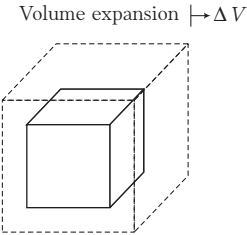


Fig. 14.3 Thermal expansion in solid cube

Now, increase in volume, $\Delta V = V_0 \gamma \Delta T$

Therefore, new volume, $V = V_0 + \Delta V = V_0 [1 + \gamma \Delta T]$

Here, γ is called the **coefficient of volume expansion** and its unit is K^{-1} or $(^\circ\text{C})^{-1}$.

Relation between α , β and γ

For solids, α , β and γ are related as

$$\alpha : \beta : \gamma = 1 : 2 : 3$$

i.e. $\gamma = 3\alpha, \beta = 2\alpha, 2\gamma = 3\beta$

or Percentage change in area = $2 \times$ Percentage change in length

Percentage change in volume = $3 \times$ Percentage change in length

Example 14.11 On heating a glass block of 10000 cm^3 from 25°C to 40°C , its volume increases by 4 cm^3 . Determine the coefficient of linear expansion of glass.

Sol. Given, volume, $V = 10000 \text{ cm}^3$

$$\text{Increase in temperature, } \Delta T = (40^\circ - 25^\circ) = 15^\circ\text{C}, \Delta V = 4 \text{ cm}^3$$

$$\therefore \text{Coefficient of cubical expansion, } \gamma = \frac{\Delta V}{V \cdot \Delta T} = \frac{4}{10000 \times 15} \\ = 26.67 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$$

∴ Coefficient of linear expansion,

$$\alpha = \frac{\gamma}{3} = \frac{26.67 \times 10^{-6}}{3} = 8.89 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$$

Example 14.12 The volume of mercury in the bulb of a thermometer is 10^{-6} m^3 . The area of cross-section of the capillary tube is $2 \times 10^{-7} \text{ m}^2$. If the temperature is raised by 100°C , then find the increase in the length of the mercury column. (Take, $\gamma_{\text{Hg}} = 18 \times 10^{-5} / ^\circ \text{C}$)

Sol. From volume expansion relation, $\Delta V = V \times \gamma \Delta T$... (i)

where, $\gamma = 18 \times 10^{-5} / ^\circ \text{C}$,

initial volume, $V = 10^{-6} \text{ m}^3$

and increase in temperature, $\Delta T = 100^\circ \text{C}$

Putting these values in Eq. (i), we get

$$\Delta V = 10^{-6} \times 18 \times 10^{-5} \times 10^2 = 18 \times 10^{-9} \text{ m}^3$$

$$\text{Now, } \Delta V = A \times \Delta l \Rightarrow 18 \times 10^{-9} = 2 \times 10^{-7} \times \Delta l$$

$$\text{or } 9 \times 10^{-2} = \Delta l \text{ or } \Delta l = 9 \text{ cm}$$

Some effects of thermal expansion in solids

Some effects of thermal expansion in solids are given below

- (i) If a **bimetallic strip** (strip made up of two strips of different materials placed one over another and rigidly joined) is heated, then it bends such that the strip of material having high value of α will be on convex side.

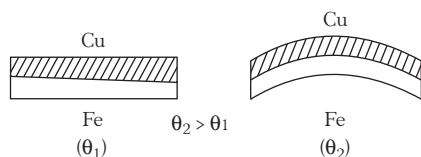


Fig. 14.4 A bimetallic strip is heated from θ_1 to θ_2

Since $\alpha_{\text{Cu}} > \alpha_{\text{Fe}}$, increase in length of copper strip will be greater, hence strip of copper will be on convex side.

- (ii) If the **temperature of a pendulum** increases, then its length increases, hence its time period changes.

Time period of pendulum,

$$T = 2\pi\sqrt{l/g} \Rightarrow \frac{\Delta T}{T} = \frac{\Delta l}{2l} = \frac{l\alpha\Delta\theta}{2l}$$

Here, $\Delta\theta$ is change in temperature and α is coefficient of linear expansion of material of pendulum.

$$\text{Change in time period, } \Delta T = \frac{1}{2} \alpha \Delta\theta T$$

Time lost by the clock in a day ($T = 86400 \text{ s}$),

$$\Delta t = \frac{1}{2} \times \alpha \Delta\theta \times T = \frac{1}{2} \times \alpha \Delta\theta \times 86400 = 43200 \alpha \Delta\theta \text{ s}$$

Change in time (shown by pendulum clock),

$$\Delta t = \frac{\Delta T}{T} t$$

Hence, if temperature increases, then time period increases and *vice-versa*. On increasing the temperature, pendulum clock runs slowly and *vice-versa*.

- (iii) If a **metal rod** fixed at both the ends with rigid support is heated, then due to thermal expansion, it exerts force on rigid supports.

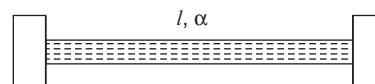


Fig. 14.5 A rod with a rigid support

$$\text{Now, } Y = \frac{F/A}{\Delta l/l} = \frac{F}{A} \times \frac{l}{\Delta l} = \frac{F}{A} \times \frac{l}{l\alpha\Delta T}$$

where, ΔT is rise in temperature, α is coefficient of linear expansion, Y is Young's modulus of material of rod, l is length of rod and A is area of cross-section of rod.

Force on rigid supports,

$$F = YA\alpha\Delta T$$

Example 14.13 A clock which keeps correct time at 20°C is subjected to 40°C . If coefficient of linear expansion of the pendulum is $12 \times 10^{-6} / ^\circ \text{C}$. Determine the gain or loss in time period.

Sol. Given, coefficient of linear expansion, $\alpha = 12 \times 10^{-6} / ^\circ \text{C}$

Change in temperature, $\Delta\theta = \theta_2 - \theta_1 = 40^\circ - 20^\circ = 20^\circ \text{C}$

Initial time period, $T = 24 \times 60 \times 60 = 24 \times 3600 = 86400 \text{ s}$

Now, change in time period,

$$\begin{aligned} \Delta T &= \frac{1}{2} T \alpha \Delta\theta = \frac{1}{2} \times 86400 \times 12 \times 10^{-6} \times 20 \\ &= 12 \times 10^{-5} \times 86400 = 10368 \times 10^{-3} \\ &= 10.4 \text{ s}^{-1} \end{aligned}$$

Example 14.14 A second's pendulum clock has a steel wire. The clock is calibrated at 20°C . How much time does the clock loss or gain in one week when the temperature is increased to 30°C ? (Take, $\alpha_{\text{steel}} = 1.2 \times 10^{-5} / ^\circ \text{C}$)

Sol. The time period of second's pendulum is 2 s. As the temperature increases, length and hence, time period increases. Clock becomes slow and it loses the time. The change in time period,

$$\begin{aligned} \Delta T &= \frac{1}{2} T \alpha \Delta\theta = \left(\frac{1}{2}\right)(2)(1.2 \times 10^{-5})(30 - 20) \\ &= 1.2 \times 10^{-4} \text{ s} \end{aligned}$$

$\therefore$ New time period,

$$T' = T + \Delta T = (2 + 1.2 \times 10^{-4}) = 2.00012 \text{ s}$$

$\therefore$ Time lost in one week,

$$\Delta t = \left(\frac{\Delta T}{T'}\right)t = \frac{(1.2 \times 10^{-4})}{(2.00012)}(7 \times 24 \times 3600) = 36.28 \text{ s}$$

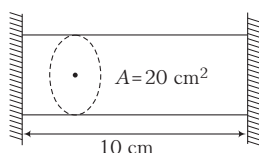
Example 14.15 An aluminium cylinder 10 cm long, with a cross-sectional area of 20 cm^2 , is to be used as a spacer between two steel walls. At 17.2°C , it just slips in between the walls. When it warms to 22.3°C , calculate the stress in the cylinder and the total force it exerts on each wall. Assuming that, the walls are perfectly rigid and a constant distance apart. (For aluminium, $\alpha = 2.4 \times 10^{-5} \text{ K}^{-1}$)

Sol. As we know, for aluminium, $Y = 7.0 \times 10^{10} \text{ Pa}$

$$\text{and } \alpha = 2.4 \times 10^{-5} \text{ K}^{-1}$$

$$\text{Change in temperature, } \Delta T = T_2 - T_1 = (22.3 - 17.2)^\circ\text{C} = 5.1^\circ\text{C} = 5.1 \text{ K}$$

$$\text{Area, } A = 20 \text{ cm}^2 = 20 \times 10^{-4} \text{ m}^2$$



$$\therefore \text{Stress} = \frac{F}{A} = Y\alpha\Delta T = (0.70 \times 10^{11} \text{ Pa})(2.4 \times 10^{-5} \text{ K}^{-1})(5.1 \text{ K}) = 1.68 \times 5.1 \times 10^6 = 8.6 \times 10^6 \text{ Pa}$$

The total force F is the product of cross-sectional area and the stress.

$$F = \text{Stress} \times \text{Area} = (8.6 \times 10^6 \text{ Pa})(20 \times 10^{-4} \text{ m}^2) = 1.7 \times 10^4 \text{ N}$$

Thermal expansion of liquids

Liquids do not have linear and superficial expansion but they only have volumetric expansion.

Since, liquids are always heated in a vessel, so initially on heating the system (liquid + vessel), the level of liquid in vessel falls (as vessel expands more since it absorbs heat and liquid expands less) but later on, it starts rising due to faster expansion of the liquid.

Thus, liquids have two coefficients of volume expansion

- (i) **Coefficient of real expansion** (γ_r), which is due to the actual increase in volume of liquid due to heating.

$$\gamma_r = \frac{\text{Real increase in volume}}{\text{Original volume} \times \text{Change in temperature}}$$

- (ii) **Coefficient of apparent expansion** (γ_a), which is due to apparent increase in the volume of liquid, if expansion of vessel containing the liquid is also taken into account.

$$\gamma_a = \frac{\text{Observed (or apparent) increase in volume}}{\text{Original volume} \times \text{Change in temperature}}$$

$$\text{Now, } \gamma_r = \gamma_a + \gamma_g$$

Here, γ_g = expansion of vessel.

Anomalous behaviour of water

Generally, with increasing temperature, the coefficient of volume expansion of liquids is about ten times greater than that of solids. Water is an exception to this rule. From 0°C to

4°C , water contracts and beyond 4°C , it expands. Thus, density of water reaches a maximum value of 1000 kgm^{-3} at 4°C .

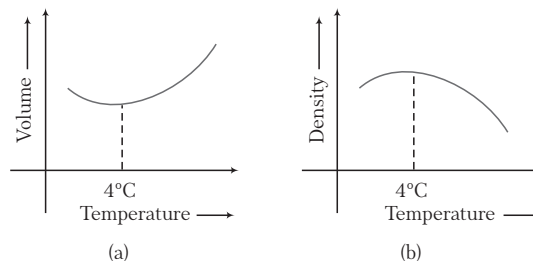


Fig. 14.6 (a) Thermal expansion of water
(b) Variation of density of water with temperature

Example 14.16 A non-conducting body floats in a liquid at 25°C with one-third of its volume immersed in the liquid. When liquid temperature is increased to 105°C , half of body's volume is immersed in the liquid. Then, find the coefficient of real expansion of the liquid (neglecting the expansion of container of the liquid).

Sol. Coefficient of real expansion, $\gamma_r = \frac{V_2 - V_1}{V_1(T_2 - T_1)} \dots(i)$

$$\text{Here, } V_2 = \frac{1}{2}, V_1 = \frac{1}{3}$$

$$\text{and } (T_2 - T_1) = (105 - 25) = 80^\circ\text{C}$$

Putting these values in Eq. (i), we get

$$\gamma_r = \frac{\left(\frac{1}{2} - \frac{1}{3}\right)}{\frac{1}{3}(80)} = \frac{1}{160} = 62.5 \times 10^{-4}^\circ\text{C}^{-1}$$

Example 14.17 The coefficient of apparent expansion of a liquid when determined using two different vessels A and B are γ_1 and γ_2 , respectively. If the coefficient of linear expansion of vessel A is α . Find the coefficient of linear expansion of the vessel B.

Sol. From thermal expansion in liquid, $\gamma_{\text{real}} = \gamma_{\text{app}} + \gamma_{\text{vessel}}$

$$\text{Given, } \gamma_{\text{vessel}} = 3\alpha$$

$$\text{For vessel A, } \gamma_{\text{real}} = \gamma_1 + 3\alpha_A$$

$$\text{For vessel B, } \gamma_{\text{real}} = \gamma_2 + 3\alpha_B$$

$$\text{Hence, } \gamma_1 + 3\alpha_A = \gamma_2 + 3\alpha_B$$

$$\Rightarrow \alpha_B = \frac{\gamma_1 - \gamma_2}{3} + \alpha_A$$

Example 14.18 A glass vessel of volume V_0 is completely filled with a liquid and its temperature is raised by ΔT . What volume of the liquid will overflow? Coefficient of linear expansion of glass = α_g and coefficient of volume expansion of the liquid = γ_l .

Sol. Volume of the liquid that will overflow

$$= \text{Increase in the volume of the liquid}$$

$$- \text{Increase in the volume of the vessel}$$

$$= [V_0(1 + \gamma_l \Delta T) - V_0] - [V_0(1 + \gamma_g \Delta T) - V_0]$$

$$= V_0 \Delta T (\gamma_l - \gamma_g) = V_0 \Delta T (\gamma_l - 3\alpha_g) \quad (\because \gamma_g \approx 3\alpha_g)$$

Example 14.19 Assume that one early morning when the temperature is 10°C , a driver of an automobile gets his gasoline tank which is made of steel, filled with 75 L of gasoline, which is also at 10°C . During the day, the temperature rises to 30°C , how much gasoline will overflow? (Take, α for steel $= 1.2 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ and γ for gasoline $= 9.5 \times 10^{-4} \text{ }^\circ\text{C}^{-1}$)

Sol. Change in volume of gasoline, $\Delta V_g = \gamma_g V \Delta T$
 $= 9.5 \times 10^{-4} \times V \times 20 = 190 \times 10^{-4} V$
 Change in volume of steel tanks, $\Delta V_s = \gamma_s V \Delta T$
 $= 3\alpha_s V \Delta T = 3 \times 1.2 \times 10^{-5} \times V \times 20 = 7.2 \times 10^{-4} V$
 $\therefore$ Volume of gasoline that will overflow
 $= \Delta V_g - \Delta V_s = (190 \times 10^{-4} - 7.2 \times 10^{-4}) V$
 $= 182.8 \times 10^{-4} \times 75 = 1.37 \text{ L}$

Thermal expansion of gases

On heating, gases expand more than solids or liquids and equal volumes of different gases expand equally, when heated by the same amount.

All gases have coefficient of volume expansion γ_V with volume variation given by

$$V = V_0 (1 + \gamma_V \Delta T)$$

and pressure variation given by $p = p_0 (1 + \gamma_p \Delta T)$

Variation of density with temperature

Most of the substances expand when they are heated, i.e. volume of a given mass of a substance increases on heating, so the density should decrease $\left(\text{as } \rho \propto \frac{1}{V}\right)$.

Let us see how the density (ρ) varies with increase in temperature.

$$\rho = \frac{m}{V} \text{ or } \rho \propto \frac{1}{V} \quad (\text{For a given mass})$$

$$\therefore \frac{\rho'}{\rho} = \frac{V}{V'} = \frac{V}{V + \Delta V} = \frac{V}{V + \gamma V \Delta T} = \frac{1}{1 + \gamma \Delta T}$$

$$\therefore \rho' = \frac{\rho}{1 + \gamma \Delta T}$$

This expression can also be written as

$$\rho' = \rho (1 + \gamma \Delta T)^{-1}$$

As γ is small, $(1 + \gamma \Delta T)^{-1} \approx 1 - \gamma \Delta T$

$$\therefore \rho' \approx \rho (1 - \gamma \Delta T)$$

Example 14.20 A vertical column whose density is 50 g cm^{-3} at 50°C balances another column of same liquid whose density is 60 g cm^{-3} at 100°C . Find the coefficient of absolute expansion of the liquid.

Sol. Given, density of vertical column, $\rho = 50 \text{ g cm}^{-3}$

Temperature, $T_1 = 50^\circ\text{C}$

Density of another column, $\rho' = 60 \text{ g cm}^{-3}$

Temperature, $T_2 = 100^\circ\text{C}$

Now, variation of density with temperature is given by

$$\begin{aligned} \rho' &= \rho(1 - \gamma \Delta T) \\ 60 &= 50[1 - \gamma(100 - 50)] \\ \Rightarrow \frac{60}{50} &= [1 - \gamma(50)] \Rightarrow 50\gamma = 1 - \frac{6}{5} \\ \Rightarrow \gamma &= -\frac{1}{5 \times 50} = -\frac{1}{250} \\ \Rightarrow \gamma &= -0.004 / ^\circ\text{C} \end{aligned}$$

Example 14.21 A sphere of diameter 7 cm and mass 266.5 g floats in a bath of liquid. As the temperature is raised, the sphere just sinks at a temperature of 35°C . If the density of the liquid at 0°C is 1.527 g/cm^3 , find the coefficient of cubical expansion of the liquid.

Sol. The sphere will sink in the liquid at 35°C when its density becomes equal to the density of liquid at 35°C

The density of sphere,

$$\begin{aligned} \rho_{35} &= \frac{\text{Mass}}{\text{Volume}} = \frac{266.5}{\frac{4}{3} \times \left(\frac{22}{7}\right) \times \left(\frac{7}{2}\right)^3} \\ &= 1.483 \text{ g/cm}^3 \end{aligned}$$

Now,

$$\begin{aligned} \rho_0 &= \rho_{35} [1 + \gamma \Delta T] \\ 1.527 &= 1.483 [1 + \gamma \times 35] \\ 1.029 &= 1 + \gamma \times 35 \\ \Rightarrow \gamma &= \frac{1.029 - 1}{35} \\ &= 0.00083 / ^\circ\text{C} \end{aligned}$$

Example 14.22 The coefficient of volume expansion of glycerine is $49 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$. What is the fractional change in its density for a 30°C rise in temperature?

Sol. Given, coefficient of volume expansion,

$$\gamma = 49 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$$

Temperature, $T = 30^\circ\text{C}$

Let ρ_0 is density at 0°C and ρ_t is density at $t^\circ\text{C}$.

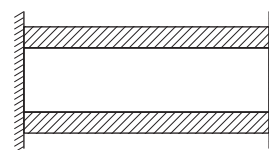
$\therefore$ Fractional change in density,

$$\begin{aligned} \frac{\rho_0 - \rho_t}{\rho_0} &= \gamma \Delta T \\ &= 49 \times 10^{-5} \times 30 = 0.0147 \end{aligned}$$

CHECK POINT 14.2

- A bar of iron is 10 cm at 20°C. At 19°C, it will be ($\alpha_{\text{Fe}} = 11 \times 10^{-6}/^\circ\text{C}$)
 (a) 11×10^{-6} cm longer (b) 11×10^{-6} cm shorter
 (c) 11×10^{-5} cm shorter (d) 11×10^{-5} cm longer
- ABCD is a thin iron sheet. A hole with radius r is made in it at the middle. If we heat up the sheet, the radius of the hole
 (a) will increase
 (b) will decrease
 (c) will remain constant
 (d) can increase or decrease depending upon size
- A steel tape gives correct measurement at 20°C. A piece of wood is being measured with the steel tape at 0°C. The reading is 25 cm on the tape, the real length of the given piece of wood must be
 (a) 25 cm (b) < 25 cm
 (c) > 25 cm (d) Cannot say
- Two rods of lengths l_1 and l_2 are made of materials whose coefficients of linear expansions are α_1 and α_2 , respectively. If the difference between two lengths is independent of temperature, then
 (a) $\frac{l_1}{l_2} = \frac{\alpha_1}{\alpha_2}$ (b) $\frac{l_1}{l_2} = \frac{\alpha_2}{\alpha_1}$ (c) $l_2^2 \alpha_1 = l_1^2 \alpha_2$ (d) $\frac{\alpha_1^2}{l_1} = \frac{\alpha_2^2}{l_2}$
- The radius of a ring is R and its coefficient of linear expansion is α . If the temperature of ring increases by θ , then its circumference will increase by
 (a) $\pi R \alpha \theta$ (b) $2\pi R \alpha \theta$
 (c) $\pi R \alpha \frac{\theta}{2}$ (d) $\pi R \alpha \frac{\theta}{4}$
- A metre rod of silver at 0°C is heated to 100°C. Its length is increased by 0.19 cm. Coefficient of volume expansion of the silver rod is
 (a) $5.7 \times 10^{-5}/^\circ\text{C}$ (b) $0.63 \times 10^{-5}/^\circ\text{C}$
 (c) $1.9 \times 10^{-5}/^\circ\text{C}$ (d) $16.1 \times 10^{-5}/^\circ\text{C}$
- A uniform metal rod is used as a bar pendulum. If the room temperature rises by 10°C and the coefficient of linear expansion of the metal of the rod is $2 \times 10^{-6}/^\circ\text{C}$, the period of the pendulum will have percentage increase of
 (a) -2×10^{-3} (b) -1×10^{-3}
 (c) 2×10^{-3} (d) 1×10^{-3}
- If two rods of lengths L and $2L$ having coefficients of linear expansion α and 2α respectively are connected end-to-end, the average coefficient of linear expansion of the composite rod is equal to
 (a) $\frac{3}{2}\alpha$ (b) $\frac{5}{2}\alpha$
 (c) $\frac{5}{3}\alpha$ (d) None of these
- Two rods of different materials having coefficients of thermal expansions α_1 , α_2 and Young's moduli Y_1 , Y_2 respectively are fixed between two rigid massive walls. The rods are heated such that they undergo the same increase

in temperature. There is no bending of the rods. If $\alpha_1 : \alpha_2 = 2 : 3$, the thermal stresses developed in the two rods are equal provided $Y_1 : Y_2$ is equal to



- (a) 2 : 3 (b) 1 : 1 (c) 3 : 2 (d) 4 : 9
- A steel rod of diameter 1 cm is clamped firmly at each end when its temperature at 25°C, so that it cannot contract on cooling. The tension in the rod at 0°C is approximately ($\alpha = 10^{-5}/^\circ\text{C}$, $Y = 2 \times 10^{11} \text{ Nm}^{-2}$)
 (a) 4000 N (b) 7000 N (c) 7400 N (d) 4700 N
 - A solid ball of metal has a spherical cavity inside it. If the ball is heated, the volume of the cavity will
 (a) increase (b) decrease
 (c) remain unchanged (d) Data insufficient
 - The coefficient of linear expansion of crystal in one direction is α_1 and that in other two directions perpendicular to it is α_2 . The coefficient of cubical expansion is
 (a) $\alpha_1 + \alpha_2$ (b) $2\alpha_1 + \alpha_2$
 (c) $\alpha_1 + 2\alpha_2$ (d) None of these
 - Ratio among linear expansion coefficient (α), area expansion coefficient (β) and volume expansion coefficient (γ) is
 (a) 1:2:3 (b) 3:2:1 (c) 4:3:2 (d) All of these
 - On heating a liquid having coefficient of volume expansion α in a container having coefficient of linear expansion $\alpha/2$, the level of the liquid in the container would
 (a) rise
 (b) fall
 (c) remains almost stationary
 (d) Cannot be predicted
 - A beaker is completely filled with water at 4°C. It will overflow, if
 (a) heated above 4°C
 (b) cooled below 4°C
 (c) Both heated and cooled above and below 4°C, respectively
 (d) None of the above
 - A glass flask of volume 200 cm³ is just filled with mercury at 20°C. The amount of mercury that will overflow when the temperature of the system is raised to 100°C is ($\gamma_{\text{glass}} = 1.2 \times 10^{-5}/^\circ\text{C}$ and $\gamma_{\text{mercury}} = 1.8 \times 10^{-4}/^\circ\text{C}$)
 (a) 2.15 cm³ (b) 2.69 cm³ (c) 2.52 cm³ (d) 2.25 cm³
 - Coefficient of volume expansion of mercury is $0.18 \times 10^{-3}/^\circ\text{C}$. If the density of mercury at 0°C is 13.6 g/cc, then its density at 200°C is
 (a) 13.11 g/cc (b) 52.11 g/cc
 (c) 16.11 g/cc (d) 26.11 g/cc

KINETIC THEORY OF IDEAL GASES

Kinetic theory of gases explains the behaviour of gases based on the idea that the gas consists of rapidly moving atoms or molecules. It correlates the macroscopic properties of gases such as pressure, temperature, etc., to the microscopic properties of gas molecules such as speed, momentum, kinetic energy of molecules, etc.

Postulates of kinetic theory

The kinetic theory of an ideal gas is based on the following postulates

- (i) A gas consists of large number of tiny particles, either atoms or molecules moving around randomly.
- (ii) The volume of the particles themselves is negligible as compared to the total volume of the gas. Most of the volume of a gas is empty space.
- (iii) The gas particles act independently of one another; there are no attractive or repulsive forces between particles.
- (iv) Collisions of the gas particles, either with other particles or with the walls of the container, are elastic, *i.e.* the total kinetic energy of the gas particles is constant at constant temperature. Particles have no energy in the form of potential energy because particles do not have mutual attraction or repulsion except gravitational attraction which is very feeble considering their masses.
- (v) The average kinetic energy of the gas particles is proportional to the temperature of the sample (in kelvin).
- (vi) The pressure exerted by a gas is the result of collisions of the particles with the walls of the container.
- (vii) At any particular time, different particles in a gas have different speeds and hence, different kinetic energy.
However, average kinetic energy of the molecules remains constant at a particular temperature.

Pressure of an ideal gas (Concept of pressure)

On the basis of kinetic theory of gases, we can do mathematical calculations to find the expression for pressure exerted by a gas.

Consider an ideal gas consisting of N molecules each of mass m in a container of volume V having edges of length l .

Due to the continuous movement, the gas molecules collide with the walls of the container, thus applying pressure on the walls.

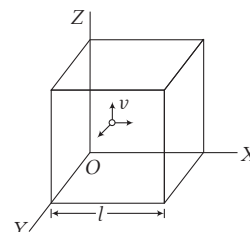


Fig. 14.7 A cubical box with sides of length l containing an ideal gas. The molecule shown moves with velocity v .

The pressure exerted by gas molecules on the wall of container is given by

$$p = \frac{1}{3} \frac{mN}{V} \bar{v}^2 = \frac{1}{3} \rho \bar{v}^2$$

Here, m is mass of one gas molecule, N is total number of gas molecules, V is volume of gas, $\bar{v}$ is root mean square velocity of gas molecules and ρ is density of gas.

The above expression can be written as

$$p = \frac{1}{3} m \left(\frac{N}{V} \right) \bar{v}^2 = \frac{2}{3} \left(\frac{1}{2} m \bar{v}^2 \right) \frac{N}{V}$$

Here, $\frac{1}{2} m \bar{v}^2$ is the kinetic energy of one gas molecule.

$$\therefore p = \frac{2}{3} \frac{N}{V} (\text{KE}) \quad \text{or} \quad p = \frac{2}{3} \left(\frac{N}{V} \frac{m \bar{v}^2}{2} \right)$$

or

$$p = \frac{2}{3} \left(\frac{1}{2} \rho \bar{v}^2 \right)$$

Here, $\frac{1}{2} \rho \bar{v}^2$ is **average kinetic energy of gas molecules per unit volume**.

$$\therefore p = \frac{2}{3} \times \text{average kinetic energy per unit volume}$$

$$\therefore p = \frac{2}{3} E$$

If E_{mol} be the energy of one mole of gas, then

$$p = \frac{2}{3} \frac{E_{\text{mol}}}{V}$$

where, V is volume of one mole gas which is equal to 22.4 L.

Now, $pV = \frac{2}{3}E_{\text{mol}} = RT$
 where, gas constant, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$.

$$\Rightarrow E_{\text{mol}} = \frac{3}{2}RT$$

$$\Rightarrow E_{\text{mol}} \propto T$$

Example 14.23 If the mass of each molecule of a gas is halved and speed is doubled. Find the ratio of initial and final pressure.

Sol. Given, $m_1 = m$, $v_1 = v$, $V_1 = V$, $m_2 = m/2$, $v_2 = 2v$, $V_2 = V_1 = V$

$\therefore$ Pressure exerted by first gas molecule,

$$p_1 = \frac{1}{3} \frac{m_1}{V_1} v_1^2$$

and pressure exerted by second gas molecule,

$$p_2 = \frac{1}{3} \frac{m_2}{V_2} v_2^2$$

$$\therefore \frac{p_1}{p_2} = \frac{m_1}{m_2} \times \frac{V_2}{V_1} \times \frac{v_1^2}{v_2^2}$$

Putting all the values in above equation, we get

$$\frac{p_1}{p_2} = \frac{m}{m/2} \times \frac{V}{V} \times \left(\frac{v}{2v}\right)^2 = 2 \times \frac{1}{4} = \frac{1}{2}$$

Ideal gas equation

Pressure of an ideal gas is given by

$$p = \frac{1}{3} \frac{mN}{V} \bar{v}^2 \quad \text{or} \quad pV = \frac{2N}{3} \left(\frac{1}{2} m \bar{v}^2 \right) \quad \dots(i)$$

For ideal gas, kinetic energy of gas molecules is proportional to temperature of the gas.

$$\therefore \frac{1}{2} m \bar{v}^2 \propto T \quad \text{or} \quad \bar{v}^2 \propto T \Rightarrow \bar{v}^2 = k_B T$$

where, k_B is a constant.

$$\text{From Eq. (i), } pV = \frac{2}{3} N \left(\frac{1}{2} m k_B T \right) = RT = N \frac{R}{N} T$$

We can write the above equation as

$$pV = Nk_B T \quad \text{where, } k_B = \frac{R}{N}$$

where, k_B is called **Boltzmann's constant**.

Example 14.24 Oxygen is filled in a closed metal jar of volume 10^{-3} m^3 at a pressure of $1.5 \times 10^5 \text{ Pa}$ and temperature 400 K. The jar has a small leak in it. The atmospheric pressure is 10^5 Pa and the atmospheric temperature is 300 K. Find the mass of the gas that leaks out by the time, the pressure and the temperature inside the jar equalise with the surrounding.

Sol. Given, volume, $V = 10^{-3} \text{ m}^3$

Pressure, $p_1 = 1.5 \times 10^5 \text{ Pa}$, $p_2 = 10^5 \text{ Pa}$

Temperature, $T_1 = 400 \text{ K}$, $T_2 = 300 \text{ K}$

From ideal gas equation, $p_1 V = n_1 R T_1$

$$\Rightarrow n_1 = \frac{p_1 V}{R T_1}$$

$$\text{Similarly, } p_2 V = n_2 R T_2 \Rightarrow n_2 = \frac{p_2 V}{R T_2}$$

$\therefore$ Number of moles leaked,

$$\begin{aligned} \Delta n &= n_1 - n_2 = \left(\frac{p_1}{T_1} - \frac{p_2}{T_2} \right) \frac{V}{R} \\ &= \left(\frac{1.5}{400} - \frac{1}{300} \right) \times \frac{10^5 \times 10^{-3}}{8.3} \\ &= 5.02 \times 10^{-3} \end{aligned}$$

$$\text{Mass of gas leaked} = \Delta n M_0 = 5.02 \times 10^{-3} \times 32 = 0.16 \text{ g}$$

Example 14.25 Find the coefficient of volume expansion for an ideal gas at constant pressure and at temperature T .

Sol. For an ideal gas, $pV = nRT$

As p is constant, we have

$$p \cdot dV = nRdT$$

$$\therefore \frac{dV}{dT} = \frac{nR}{p}$$

According to volume expansion, $dV = V\gamma dT$

$$\text{or} \quad \gamma = \frac{1}{V} \cdot \frac{dV}{dT} = \frac{nR}{pV} = \frac{nR}{nRT} = \frac{1}{T}$$

$$\therefore \gamma = \frac{1}{T}$$

Molecular velocities

In gaseous state, molecules tend to move in different directions with different speeds.

Based upon the calculations, there are three types of molecular velocities

1. Average velocity (v or v_{av})

It is the arithmetic mean of velocities of the different molecules of a gas.

$$\text{Mathematically, } \bar{v} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{n}$$

where, $v_1, v_2, v_3, \dots, v_n$ are the velocities of the individual molecules and $n = n_1 + n_2 + n_3 \dots$ is the total number of molecules.

This is related to temperature T of the gas as

$$\bar{v} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8p}{\pi \rho}} = \sqrt{\frac{8k_B T}{\pi M}}$$

where, M is molar mass in kg/mol and ρ is density.

Example 14.26 Find the average speed of nitrogen molecules at 27°C.

Sol. Mass of 1 mole of nitrogen gas, $M = 14 \text{ g} = 14 \times 10^{-3} \text{ kg}$

$$\text{Average speed, } \bar{v} = \sqrt{\frac{8RT}{\pi M}}$$

where, T = temperature of the gas (in kelvin)

$$= 273 + 27^\circ = 300 \text{ K}$$

R = gas constant = $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

$$\therefore \bar{v} = \sqrt{\frac{8 \times 8.314 \times 300}{3.14 \times 14 \times 10^{-3}}} = 673.72 \text{ ms}^{-1}$$

2. Root-mean-square velocity (u or v_{rms})

It is the square root of the mean of the squares of the velocities of the molecules of gas.

$$\text{Mathematically, } v_{\text{rms}} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2}{n}}$$

From kinetic gas equation,

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3pV}{M}} = \sqrt{\frac{3p}{M/V}} = \sqrt{\frac{3p}{\rho}}$$

where, p = pressure (RT/V) and ρ = density (M/V).

Example 14.27 In a container, two gases neon and argon are filled. Find the ratio of the root-mean-square speed of the molecules of the two gases. Given, $M_{\text{Ne}} = 20.2 \mu$ and $M_{\text{Ar}} = 39.9 \mu$. The temperature of the system is 30°C.

Sol. From kinetic gas equation,

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\therefore v_{\text{rms}}^2 \propto \frac{1}{M}$$

$$\therefore \frac{(v_{\text{rms}})_{\text{Ne}}}{(v_{\text{rms}})_{\text{Ar}}} = \sqrt{\frac{(M)_{\text{Ar}}}{(M)_{\text{Ne}}}} = \sqrt{\frac{39.9}{20.2}} \approx 1.41$$

Example 14.28 What is rms velocity of O_2 gas at 127°C? The molecular weight of oxygen is 32.

Sol. As, $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$... (i)

Given, $R = 8.3143 \text{ J mol}^{-1} \text{ K}^{-1}$,

$$T = 127^\circ \text{C} = (273 + 127) \text{ K} = 400 \text{ K}$$

$$M = 32 \text{ g mol}^{-1} = 32 \times 10^{-3} \text{ kg mol}^{-1}$$

Putting these values in Eq. (i), we get

$$v_{\text{rms}} = \sqrt{\frac{3 \times 8.3143 \times 400}{32 \times 10^{-3}}} = 5.5838 \times 10^2 \text{ ms}^{-1}$$

Example 14.29 Determine the rms velocity of oxygen molecules at STP. The molecular weight of oxygen is 32.

Sol. At STP, pressure, $p = 1 \text{ atm} = 1.013 \times 10^5 \text{ Nm}^{-2}$

Mass, $M = 32 \text{ g} = 32 \times 10^{-3} \text{ kg}$

Molar volume, $V = 22.4 \text{ L}$

$$= 22.4 \times 10^{-3} \text{ m}^3$$

Root-mean-square-velocity of oxygen molecules at STP,

$$\begin{aligned} v_{\text{rms}} &= \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3pV}{M}} \\ &= \sqrt{\frac{3 \times 1.013 \times 10^5 \times 22.4 \times 10^{-3}}{32 \times 10^{-3}}} \\ v_{\text{rms}} &= 461.23 \text{ ms}^{-1} \end{aligned}$$

Example 14.30 The molecules of a given mass of a gas have root-mean-square-speed of 100 ms^{-1} at 27°C and 1.00 atm pressure. What will be the root-mean-square-speed of the molecules of the gas at 127°C and 2.0 atm pressure?

Sol. Given, at 27°C, root-mean-square-speed,

$$(v_{\text{rms}})_1 = 100 \text{ ms}^{-1}$$

$$(v_{\text{rms}})_1 = \sqrt{\frac{3p_1}{\rho_1}} = \sqrt{\frac{3p_1V_1}{M}} \quad \dots(i)$$

According to ideal gas equation, $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$

$$\Rightarrow \frac{V_1}{V_2} = \frac{p_2T_1}{p_1T_2} = \frac{2 \times 300}{400} = \frac{3}{2}$$

$\therefore$ At 127°C, root mean square speed,

$$(v_{\text{rms}})_2 = \sqrt{\frac{3p_2}{\rho_2}} = \sqrt{\frac{3p_2V_2}{M}} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\begin{aligned} (v_{\text{rms}})_2^2 &= (v_{\text{rms}})_1^2 \times \frac{V_2}{V_1} \times \frac{p_2}{p_1} \\ &= (100)^2 \times \frac{2}{3} \times 2 \end{aligned}$$

$$\text{or } (v_{\text{rms}})_2 = \frac{200}{\sqrt{3}} \text{ ms}^{-1}$$

3. Most probable velocity (v_{mp} or u_{mp})

It is the velocity possessed by the largest number of molecules of gas (Maxwell speed distribution law).

From kinetic gas equation,

$$v_{\text{mp}} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2pV}{M}} = \sqrt{\frac{2p}{\rho}}$$

Note The ratio in different molecular velocities is

$$v_{\text{rms}} : v_{\text{av}} : v_{\text{mp}} = \sqrt{3} : \sqrt{\frac{8}{\pi}} : \sqrt{2} = \sqrt{3} : \sqrt{2.5} : \sqrt{2}$$

Example 14.31 Calculate the root mean square, average and most probable speeds of oxygen molecules at 27° C.

Sol. Given, molar mass of oxygen,

$$M = 32 \text{ g mol}^{-1} = 0.032 \text{ kg mol}^{-1}$$

$$\text{Temperature, } T = 27^\circ \text{ C} = (27 + 273) \text{ K} = 300 \text{ K}$$

$$\begin{aligned} \text{(i) Root mean square speed, } v_{\text{rms}} &= \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.314)(300)}{(0.032)}} \\ &= 483.6 \text{ ms}^{-1} \end{aligned}$$

$$\text{(ii) Average speed, } \bar{v} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8(8.314)(300)}{(3.14)(0.032)}} = 445.6 \text{ ms}^{-1}$$

(iii) Most probable speed,

$$v_{\text{mp}} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2(8.314)(300)}{(0.032)}} = 394.8 \text{ ms}^{-1}$$

Kinetic interpretation of temperature

According to kinetic theory of gases, $pV = \frac{2}{3} E$,

but according to equation of state for an ideal gas,

$$pV = RT$$

$$\Rightarrow \frac{2}{3} E = RT = Nk_B T$$

$$\therefore \text{Total internal energy, } E = \frac{3}{2} Nk_B T$$

where, k_B is the Boltzmann's constant. Its value is $1.38 \times 10^{-23} \text{ J mol}^{-1} \text{ K}^{-1}$.

Mean translational kinetic energy, $E = \frac{3}{2} k_B T$

i.e. The mean translational kinetic energy of a gas molecule depends only on its temperature and is independent of its nature or mass.

On this basis, we can say that **absolute zero temperature is the temperature at which translational kinetic energy of a gas molecule becomes zero, i.e. at which the molecular motion ceases altogether.**

Example 14.32 At what temperature, the kinetic energy of a gas molecule is half of the value at 27°C?

Sol. Given, $T_1 = 27^\circ \text{ C} = 27 + 273 = 300 \text{ K}$

Kinetic energy of a gas molecule,

$$E = \frac{3}{2} k_B T$$

where, k is Boltzmann's constant.

$$\therefore E \propto T$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{T_1}{T_2}$$

$$\Rightarrow \frac{E}{(E/2)} = \frac{300}{T_2}$$

$$\begin{aligned} \Rightarrow T_2 &= 150 \text{ K} \\ &= 150 - 273 = -123^\circ \text{ C} \end{aligned}$$

Example 14.33 Prove that the pressure of an ideal gas is numerically equal to two-third of the mean translational kinetic energy per unit volume of the gas.

Sol. Translational kinetic energy per unit volume,

$$E = \frac{1}{2} (\text{Mass per unit volume})(\bar{v}^2)$$

$$= \frac{1}{2} (\rho) \left(\frac{3p}{\rho} \right) = \frac{3}{2} p$$

$$\text{or } p = \frac{2}{3} E$$

Example 14.34 A balloon has 5.0 moles of helium at 7°C.

Calculate

(i) the number of atoms of helium in the balloon.

(ii) the total internal energy of the system.

(Avogadro's number, $N_A = 6.02 \times 10^{23}$

and Boltzmann's constant, $k_B = 1.38 \times 10^{-23} \text{ J mol}^{-1} \text{ K}^{-1}$).

Sol Given, $n = 5.0$, $T = 7^\circ \text{ C} = 7 + 273 = 280 \text{ K}$

$$\begin{aligned} \text{(i) Number of atoms} &= nN_A = 5.0 \times 6.02 \times 10^{23} \\ &\approx 30 \times 10^{23} \end{aligned}$$

$$\text{(ii) Average kinetic energy per molecule} = \frac{3}{2} k_B T$$

$$\begin{aligned} \therefore \text{Total internal energy} &= \frac{3}{2} k_B T \times N \\ &= \frac{3}{2} \times 1.38 \times 10^{-23} \times 280 \times 30 \times 10^{23} \\ &= \frac{3}{2} \times 30 \times 280 \times 1.38 = 1.74 \times 10^4 \text{ J} \end{aligned}$$

Example 14.35 A tank used for filling helium balloons has a volume of 3.0 m^3 and contains 2.0 moles of helium gas at 20.0° C . Assuming that the helium behaves like an ideal gas,

(i) what is the total translational kinetic energy of the molecules of the gas?

(ii) what is the average kinetic energy per molecule?

Sol. (i) Using $(\text{KE})_{\text{trans}} = \frac{3}{2} nRT$

Given, $n = 2.0 \text{ mol}$ and $T = 293 \text{ K}$

$$\therefore (\text{KE})_{\text{trans}} = \frac{3}{2} (2.0) (8.31) (293) = 7.3 \times 10^3 \text{ J}$$

(ii) The average kinetic energy per molecule,

$$\begin{aligned} (\text{KE})_{\text{av}} &= \frac{3}{2} k_B T = \frac{3}{2} (1.38 \times 10^{-23}) (293) \\ &= 6.07 \times 10^{-21} \text{ J} \end{aligned}$$

Mean free path (λ)

Every gas consists of a large number of molecules undergoing frequent collisions, therefore they cannot move straight or unhindered, these molecules are in a state of continuous and random motion. Also, they undergo perfectly elastic collision against each other.

The average distance travelled by a molecule between two successive collisions is known as the mean free path of the molecule.

$$\text{Mean free path, } \lambda = \frac{1}{\sqrt{2} n \pi d^2} \quad \dots(i)$$

where, d = diameter of each molecule
and n = number of molecules per unit volume.

Relation between mean free path (λ), temperature (T) and pressure (p)

For the derivation of Eq. (i), we have assumed the other molecules to be at rest. Taking into consideration the motion of all the gas molecules, the mean free path comes out to be

$$\lambda = \frac{1}{\sqrt{2} n \pi d^2}$$

If m is the mass of each gas molecule, then the density of the gas is

$$\rho = mn \quad \text{or} \quad n = \frac{\rho}{m}$$

$$\therefore \lambda = \frac{m}{\sqrt{2} \pi d^2 \rho}$$

For one mole of a gas, $pV = RT$

$$\text{or} \quad p = \frac{RT}{V} = \frac{N}{V} \times \frac{R}{N} \times T = nk_B T$$

$$\text{or} \quad n = \frac{p}{k_B T}$$

$$\Rightarrow \lambda = \frac{k_B T}{\sqrt{2} \pi d^2 p}$$

Example 14.36 Calculate the ratio of the mean free path of the molecules of two gases, if the ratio of the numbers density per cm^3 of the gases is 5 : 3 and the ratio of the diameters of the molecules of the gases is 4 : 5.

Sol. Given, ratio of the numbers density,

$$\frac{n_1}{n_2} = \frac{5}{3}$$

Ratio of the diameters,

$$\frac{d_1}{d_2} = \frac{4}{5}$$

Mean free path of the molecules of first gas,

$$\lambda_1 = \frac{1}{\sqrt{2} \pi n_1 d_1^2}$$

Similarly, mean free path of the second gas,

$$\lambda_2 = \frac{1}{\sqrt{2} \pi n_2 d_2^2}$$

$\therefore$ Ratio of λ_1 and λ_2 ,

$$\frac{\lambda_1}{\lambda_2} = \frac{n_2 d_2^2}{n_1 d_1^2} = \frac{3}{5} \times \frac{5}{4} \times \frac{5}{4} = \frac{15}{16}$$

$$\Rightarrow \lambda_1 : \lambda_2 = 15 : 16$$

Example 14.37 For a molecule of an ideal gas, $n = 3 \times 10^8 \text{ cm}^{-3}$ and mean free path is 10^{-2} cm . Calculate the diameter of the molecule.

Sol. Given, $n = 3 \times 10^8 \text{ cm}^{-3}$, $\lambda = 10^{-2} \text{ cm}$

Mean free path is given by

$$\lambda = \frac{1}{\sqrt{2} \pi n d^2}$$

$$\Rightarrow d^2 = \frac{1}{\sqrt{2} \pi n \lambda}$$

$$\Rightarrow d^2 = \frac{1}{\sqrt{2} \times 3.14 \times 3 \times 10^8 \times 10^{-2}}$$

$$= \frac{10^{-6}}{1.414 \times 3.14 \times 3}$$

$$\Rightarrow d = \sqrt{7.5 \times 10^{-8}} = \sqrt{7.5} \times 10^{-4} \text{ cm}$$

$$\Rightarrow d = 2.7 \times 10^{-4} \text{ cm}$$

Hence, the diameter of the molecule of the gas is $2.7 \times 10^{-4} \text{ cm}$.

CHECK POINT 14.3

- Which one of the following is not an assumption in the kinetic theory of gases?
 - The volume occupied by the molecules of the gas is negligible
 - The force of attraction between the molecules is negligible
 - The collisions between molecules are elastic
 - All molecules have same speed
- A closed vessel is maintained at a constant temperature. It is first evacuated and then vapour is injected into it continuously. The pressure of the vapour in the vessel
 - increases continuously
 - first increases and then becomes constant
 - first increases and then decreases
 - None of the above
- A vessel contains 1 mole of O_2 gas (molar mass 32) at a temperature T . The pressure of the gas is p . An identical vessel containing one mole of He gas (molar mass 4) at temperature $2T$ has a pressure of
 - $p/8$
 - p
 - $2p$
 - $8p$
- If Maxwell distribution is valid and v_p denotes the most probable speed, v_{av} is the average speed and v_{rms} is the root-mean-square speed, then
 - $v_{av} < v_p < v_{rms}$
 - $v_{av} < v_{rms} < v_p$
 - $v_p < v_{av} < v_{rms}$
 - $v_p < v_{rms} < v_{av}$
- What will be the temperature when the rms velocity is double of that at 300 K?
 - 300 K
 - 600 K
 - 900 K
 - 1200 K
- By what factor the rms velocity will change, if the temperature is raised from 27°C to 327°C ?
 - $\sqrt{2}$
 - 2
 - $2\sqrt{2}$
 - 1
- The velocities of three molecules are $3v$, $4v$ and $5v$, respectively. Their rms speed will be
 - $\sqrt{\frac{50}{3}} v$
 - $\sqrt{\frac{5}{2}} v$
 - $\frac{7}{2} v$
 - $\frac{5}{2} v$
- The temperature at which the root-mean-square speed of a gas will be half of its value at 0°C is (assume the pressure remains constant)
 - -86.4°C
 - -204.75°C
 - -104.75°C
 - -68.25°C
- Four molecules of a gas have speeds (in kms^{-1}) 1, 2, 3 and 4. The value of the root-mean-square speed of the gas molecules is
 - $\frac{1}{2}\sqrt{15} \text{ kms}^{-1}$
 - $\frac{1}{2}\sqrt{10} \text{ kms}^{-1}$
 - 25 kms^{-1}
 - $\sqrt{15/2} \text{ kms}^{-1}$
- The root-mean-square (rms) speed of oxygen molecules (O_2) at a certain absolute temperature is v . If the temperature is doubled and the oxygen gas dissociates into atomic oxygen, the rms speed would be
 - v
 - $\sqrt{2} v$
 - $2 v$
 - $2\sqrt{2} v$
- The molecules of a given mass of gas have rms speed 200 ms^{-1} at 27°C and 10^5 Nm^{-2} pressure. When the absolute temperature is doubled and the pressure is halved, the rms speed of the molecules of the same gas is
 - 200 ms^{-1}
 - 400 ms^{-1}
 - $200\sqrt{2} \text{ ms}^{-1}$
 - $400\sqrt{2} \text{ ms}^{-1}$
- The velocities of ten particles (in ms^{-1}) are 0, 2, 3, 4, 4, 4, 5, 5, 6, 9. The most probable speed is
 - 2 ms^{-1}
 - 4 ms^{-1}
 - 5 ms^{-1}
 - 9 ms^{-1}
- If at the same temperature and pressure, the densities of two diatomic gases are d_1 and d_2 , respectively. The ratio of mean kinetic energy per molecule of gases will be
 - 1 : 1
 - $d_1 : d_2$
 - $\sqrt{d_1} : \sqrt{d_2}$
 - $\sqrt{d_2} : \sqrt{d_1}$
- Two vessels A and B having equal volume contain equal masses of hydrogen in A and helium in B at 300 K. Then, mark the correct statement.
 - The pressure exerted by hydrogen is half of that exerted by helium.
 - The pressure exerted by hydrogen is equal to that exerted by helium.
 - Average kinetic energy of the molecules of hydrogen is half the average kinetic energy of the molecules of helium.
 - The pressure exerted by hydrogen is twice that exerted by helium.
- A sealed container with negligible coefficient of volumetric expansion contains helium (a monoatomic gas). When it is heated from 300 K to 600 K, the average kinetic energy of helium atoms is
 - halved
 - unchanged
 - doubled
 - increased by factor $\sqrt{2}$
- The gases carbon monoxide (CO) and nitrogen at the same temperature have translational kinetic energies E_1 and E_2 , respectively. Then,
 - $E_1 = E_2$
 - $E_1 > E_2$
 - $E_1 < E_2$
 - None of these
- Pressure of an ideal gas is increased by keeping temperature constant. What is the effect on kinetic energy of molecules?
 - Increases
 - Decreases
 - No change
 - Cannot be determined
- Some gas at 300 K is enclosed in a container. Now, the container is placed on a fast moving train. While the train is in motion, the temperature of the gas
 - rises above 300 K
 - falls below 300 K
 - remains unchanged
 - becomes unsteady
- The mean free path for air molecule with average speed 18.5 ms^{-1} at STP is ($d = 2 \times 10^{-10} \text{ m}$ and $n = 2.7 \times 10^{25} \text{ m}^{-3}$)
 - $3.5 \times 10^{-7} \text{ m}$
 - $4 \times 10^{-7} \text{ m}$
 - $2.08 \times 10^{-7} \text{ m}$
 - $5 \times 10^{-7} \text{ m}$

IDEAL GAS LAWS

Ideal gases irrespective of their nature obey the following laws

Boyle's law

According to this law, for a given mass of a gas, the volume of a gas at constant temperature (called isothermal process) is inversely proportional to its pressure, i.e.

$$V \propto \frac{1}{p} \quad (T = \text{constant})$$

or

$$pV = \text{constant} \quad \text{or} \quad p_i V_i = p_f V_f$$

Thus, p - V graph in an isothermal process is a rectangular hyperbola or pV versus p or V graph is a straight line parallel to p or V -axis.

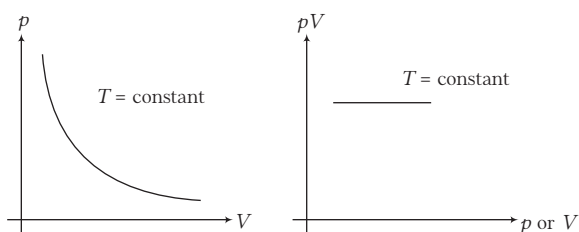


Fig. 14.8 p versus V graph and pV versus p or V graph

Example 14.38 A gas at 27°C in a cylinder has a volume of 4 L and pressure 100 Nm^{-2} . If the gas is first compressed at constant temperature, so that the pressure is 150 Nm^{-2} . Estimate the change in volume.

Sol. Given, $V_1 = 4 \text{ L}$, $V_2 = ?$, $p_1 = 100 \text{ Nm}^{-2}$,

$$p_2 = 150 \text{ Nm}^{-2}, \Delta V = ?$$

Using Boyle's law for constant temperature, we have

$$p_1 V_1 = p_2 V_2$$

$$\Rightarrow V_2 = \frac{p_1 V_1}{p_2} = \frac{100 \times 4}{150} = 2.667 \text{ L}$$

$$\begin{aligned} \therefore \text{Change in volume, } \Delta V &= V_1 - V_2 \\ &= 4 - 2.667 \\ &= 1.33 \text{ L} \end{aligned}$$

Example 14.39 The pressure of a given mass of a gas filled in a vessel of volume V at constant temperature is reduced to one-third of its initial value. Calculate the percentage change in its volume.

Sol. According to the question, $p_2 = \frac{1}{3} p_1$

Applying Boyle's law, $p_1 V_1 = p_2 V_2$

$$\therefore p_1 V_1 = \frac{1}{3} p_1 V_2$$

Then,

$$V_2 = 3V_1$$

Change in volume, $\Delta V = V_2 - V_1 = 2V_1$

$$\text{Percentage change} = \frac{\Delta V}{V} \times 100 = \frac{2V_1}{V_1} \times 100 = 200\%$$

Charles' law

According to this law, for a given mass of a gas, the volume of a gas at constant pressure (called **isobaric process**) is directly proportional to its absolute temperature, i.e.

$$V \propto T \quad (p = \text{constant})$$

or

$$\frac{V}{T} = \text{constant}$$

or

$$\frac{V_i}{T_i} = \frac{V_f}{T_f}$$

Thus, V - T graph in an isobaric process is a straight line passing through origin or V/T versus V or T graph is a straight line parallel to V or T axis.

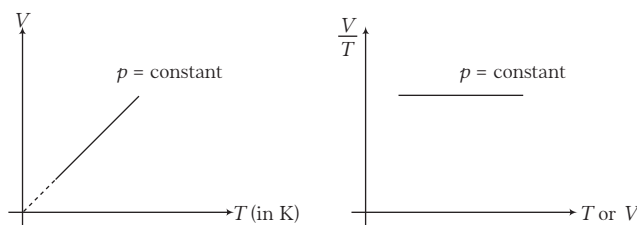


Fig. 14.9 V versus T graph and V/T versus T or V graph

If volume of ideal gas at 0°C is V_0 and at $t^\circ\text{C}$ is V_t , then

$$V_t = V_0 (1 + \alpha t)$$

where, α is volume coefficient of the gas. For all gases, its value is nearly $\frac{1}{273} / ^\circ\text{C}$.

$$\therefore V_t = V_0 \left(1 + \frac{t}{273} \right)$$

Example 14.40 Air is filled in a container at 333 K .

Calculate the temperature upto which it should be heated, so that one-third of air may escape out of the vessel.

Sol Given, $T_1 = 333 \text{ K}$, $T_2 = ?$, $V_1 = V$, $V_2 = V + \frac{V}{3}$

($\therefore$ According to question, one-third of air may escape out the vessel)

$$= \frac{4}{3} V$$

According to Charles' law, we get

$$\frac{V_2}{V_1} = \frac{T_2}{T_1}$$

$$\Rightarrow T_2 = T_1 \times \frac{V_2}{V_1} = 333 \times \frac{4}{3} = 444 \text{ K}$$

Gay Lussac's law or pressure law

According to this law, for a given mass of a gas, the pressure of a gas at constant volume (called **isochoric** process) is directly proportional to its absolute temperature.

$$\text{i.e.} \quad p \propto T \quad (V = \text{constant})$$

$$\text{or} \quad \frac{p}{T} = \text{constant}$$

$$\text{or} \quad \frac{p_i}{T_i} = \frac{p_f}{T_f}$$

Thus, p - T graph in an isochoric process is a straight line passing through origin or p/T versus p or T graph is a straight line parallel to p or T -axis.

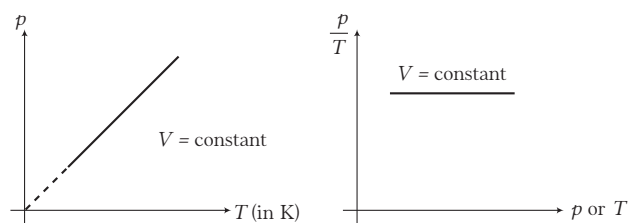


Fig. 14.10 p versus T graph and p/T versus p or T graph

If pressure of an ideal gas at 0°C is p_0 and at $t^\circ\text{C}$ is p_t , then

$$p_t = p_0 (1 + \beta t)$$

where, β is pressure coefficient of gas. For all gases, its volume is nearly $\frac{1}{273}^\circ\text{C}$.

$$\therefore \quad p_t = p_0 \left(1 + \frac{t}{273} \right)$$

Example 14.41 The pressure of a gas filled in a closed container increases by 1% when temperature is increased by 1°C . Find the initial temperature of the gas.

Sol. Given, $p_1 = p$, $T_1 = T$ K

$$\text{Increased pressure, } p_2 = p + \frac{1}{100}p = \frac{101}{100}p$$

$$\text{Increased temperature, } T_2 = (T + 1) \text{ K}$$

According to the Gay Lussac's law, we get

$$\frac{p_1}{p_2} = \frac{T_1}{T_2}$$

$$\Rightarrow \quad \frac{101}{100}p = \frac{T}{(T + 1)}$$

$$\Rightarrow \quad \frac{100}{101} = \frac{T}{T + 1}$$

$$\Rightarrow \quad 100T + 100 = 101T$$

$$\Rightarrow \quad T = 100 \text{ K}$$

Dalton's law of partial pressure

It states that, the total pressure of a mixture of non-interacting ideal gases is the sum of partial pressures exerted by individual gases in the mixture.

$$\text{i.e.} \quad p = p_1 + p_2 + p_3 + \dots$$

Consider a mixture of non-interacting ideal gases. Let gas 1 has μ_1 moles, gas 2 has μ_2 moles and so on.

Suppose the net volume of the mixture is V , temperature of mixture is T and pressure of mixture is p .

Equation of state for the mixture,

$$pV = (\mu_1 + \mu_2 + \dots) RT$$

$$p = \frac{\mu_1 RT}{V} + \frac{\mu_2 RT}{V} + \dots$$

$$\text{Dalton's law, } p = p_1 + p_2 + p_3 + \dots$$

Here, $p_1 = \frac{\mu_1 RT}{V}$ is the pressure of gas 1 exerted at the

same condition of volume and temperature, if no other gases are present.

Example 14.42 One mole each of hydrogen, oxygen and nitrogen are mixed in a close container of volume 10 L and temperature 27°C . Calculate the pressure exerted by the mixture. ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

Sol. Given, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

$$\text{Temperature, } T = 27^\circ\text{C} = 27 + 273 = 300 \text{ K}$$

$$\text{Volume, } V = 10 \text{ L} = 10 \times 10^{-3} \text{ m}^3$$

According to Dalton's law of partial pressure,

$$p = p_1 + p_2 + p_3$$

$$\Rightarrow \quad p = \frac{\mu_1 RT}{V} + \frac{\mu_2 RT}{V} + \frac{\mu_3 RT}{V} = \frac{3\mu RT}{V}$$

$$\text{Here, } \mu = 1 \text{ mol}$$

$$\therefore \quad p = \frac{3RT}{V} = \frac{3 \times 8.314 \times 300}{10 \times 10^{-3}} = 7.48 \times 10^5 \text{ Nm}^{-2}$$

Example 14.43 4 g hydrogen is mixed with 11.2 L of helium at STP in a container of volume 20 L. If the final temperature is 300 K, find the pressure.

Sol. $\therefore$ 4 g hydrogen = 2 moles hydrogen.

$$\therefore 11.2 \text{ L helium at STP} = \frac{1}{2} \text{ mole of helium}$$

According to Dalton's law of partial pressure,

$$\begin{aligned}
 p &= p_H + p_{He} = (n_H + n_{He}) \frac{RT}{V} \\
 &= \left(2 + \frac{1}{2}\right) \frac{8.31 \times (300 \text{ K})}{(20 \times 10^{-3}) \text{ m}^3} \\
 &= 3.12 \times 10^5 \text{ N/m}^2
 \end{aligned}$$

Graham's law of diffusion

It states that the rate of diffusion of a gas is inversely proportional to the square root of its density, i.e. $r \propto \frac{1}{\sqrt{\rho}}$

where, r = rate of diffusion and ρ = density of the gas. Let r_1 and r_2 be the rates of diffusion of gases 1 and 2 respectively, then according to Graham's law,

$$\frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}}$$

where, ρ_1 is density of gas 1 and ρ_2 is density of gas 2.

$$\frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}} = \sqrt{\frac{M_2}{M_1}}$$

If V is the volume of gas diffused in t seconds, then

$$r = \frac{V}{t} \Rightarrow \frac{r_1}{r_2} = \frac{V_1}{V_2} \times \frac{t_2}{t_1}$$

Behaviour of real gases

If a gas obeys the gas laws at all conditions, it is called an **ideal gas** and if it does not, it is called the **real gas**.

- (i) All the existing gases are real and the ideal gases are hypothetical.
- (ii) Real gases obey the gas laws under the moderate conditions of temperature and pressure. At low temperature and high pressure, the real gases deviate from the ideal gas behaviour significantly (i.e. start disobeying gas laws).
- (iii) Real gases behave ideally at high temperature and low pressure.

Reasons for deviation of real gas from ideal gas behaviour

Real gases deviate from ideal gas behaviour because some of the assumptions of kinetic theory of gases do not hold good for real gases.

- (i) For ideal gas, it has been considered that there is no force of attraction between gas molecules. But this assumption is not valid for all pressures and temperatures.

At low temperature and high pressure, gas molecules are close to each other, so intermolecular attraction cannot be ignored between them.

- (ii) For ideal gas, volume of gas molecules is negligible as compared to volume of gas.

But at high pressure and low temperature, total volume of gas decreases (total volume changes but volume of molecules remains same), so volume of gas molecules is not negligible as compared to volume of gas.

van der Waals' equation

Based on the deviation in behaviour of real gas from ideal gas, van der Waals' modified ideal gas equation by taking into consideration the intermolecular atomic forces and the volume occupied by the gas molecules and introduced a new equation known as van der Waals' equation which explains behaviour of real gas correctly.

van der Waals' equation is

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

Here, a and b are called van der Waals' constants. In this equation, p , V , T and n represent the measured values of pressure, volume, temperature (expressed on the absolute scale) and number of moles respectively, just as in the ideal gas equation.

Here, the term $\frac{an^2}{V^2}$ is called pressure correction for

molecular attraction and term nb is called volume correction for finite size of molecules.

Significance of van der Waals' constants

The quantities a and b are experimentally derived constants that differ for different gases. When a and b are both zero, the van der Waals' equation reduces to the ideal gas equation ($pV = nRT$).

- (i) Larger molecules have greater value of b . For instance, H_2 (a diatomic molecule) has greater value of b than that of He (a monoatomic smaller molecule).
- (ii) $\frac{n^2 a}{V^2}$ affects internal pressure of the gas and a is a measure of force of attraction between gas molecules.
- (iii) Greater the value of a , greater the liquefaction of the gas. That is why, H_2 and He having smaller value of a are difficult to liquify.
- (iv) b is also called co-volume or excluded volume, i.e. $b = 4VN_0$

where, V is the volume of a spherical molecule of radius r , $V = (4/3) \pi r^3$.

(v) a is expressed in $\text{atm L}^2 \text{mol}^{-2}$ and b is expressed in L mol^{-1} .

Example 14.44 If pressure of CO_2 (real gas) in a container is given by $p = \frac{RT}{2V-b} - \frac{a}{4b^2}$, then compute the mass of the gas in container.

Sol. van der Waals' gas equation for n mole of real gas is

$$\left(p + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

$$p + \frac{n^2 a}{V^2} = \frac{nRT}{V - nb}$$

$$\therefore p = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

$$\text{Given equation, } p = \left(\frac{RT}{2V-b} - \frac{a}{4b^2}\right)$$

On comparing the given equation with the standard equation, we get

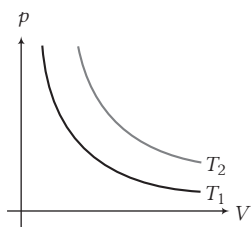
$$n = 1/2$$

$$\text{As, } n = \frac{m}{M}$$

$$\Rightarrow \text{Mass of gas, } m = nM = \frac{1}{2} \times 44 \quad (\because \text{Molar mass of } \text{CO}_2 = 44 \text{ g})$$

$$= 22 \text{ g}$$

Example 14.45 The p - V diagrams of same mass of a gas are drawn at two different temperatures T_1 and T_2 . Explain whether $T_1 > T_2$ or $T_2 > T_1$.



Sol. The ideal gas equation is

$$pV = nRT$$

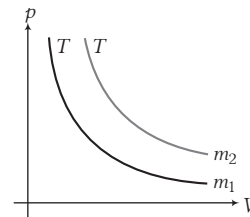
$$\text{or } T = \frac{pV}{nR}$$

$T \propto pV$, if number of moles of the gas are kept constant. Here, mass of the gas is constant, which implies that number of moles are constant, i.e. $T \propto pV$.

In the given diagram, product of p and V for T_2 is more than T_1 at all points (keeping either p or V same for both graphs).

Hence, $T_2 > T_1$.

Example 14.46 The p - V diagram of two different masses m_1 and m_2 are drawn (as shown) at constant temperature T . State whether $m_1 > m_2$ or $m_2 > m_1$?



Sol. The ideal gas equation is

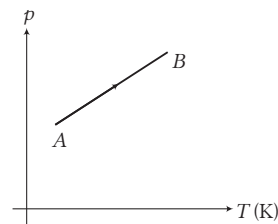
$$pV = nRT = \frac{m}{M} RT$$

$$\therefore m = (pV) \left(\frac{M}{RT}\right)$$

or $m \propto pV$, if $T = \text{constant}$

From the graph, we can see that $p_2 V_2 > p_1 V_1$ (for same p or V). Therefore, $m_2 > m_1$.

Example 14.47 The p - T graph for the given mass of an ideal gas is shown in figure. What inference can be drawn regarding the change in volume (whether it is constant, increasing or decreasing)?



Sol. From the given graph, we can write the p - T equation as

$$p = aT + b \quad (\because y = mx + c)$$

Here, a and b are positive constants. Further,

$$\frac{p}{T} = a + \frac{b}{T}$$

Now, $T_B > T_A$

$$\therefore \frac{b}{T_B} < \frac{b}{T_A}$$

$$\text{or } \left(\frac{p}{T}\right)_B < \left(\frac{p}{T}\right)_A$$

$$\text{or } \left(\frac{T}{p}\right)_B > \left(\frac{T}{p}\right)_A \text{ or } V_B > V_A$$

Thus, as we move from A to B , volume of the gas is increasing.

DEGREE OF FREEDOM

The number of independent ways in which a molecule or an atom can exhibit motion or have energy is called its degree of freedom. A rigid body of finite size (gas molecule) can have both translational and rotational motions. Like translational motion, rotational motion can be resolved into mutually perpendicular components.

Hence, a rigid body has 6 degrees of freedom; three for translational motion and three for rotational motion. Out of three possible rotational degrees, only two are applicable for linear molecules. For non-linear molecules, third rotational degree is allowed. Apart from 3 rotational and 3 translational, there are $3N - 6$ degree of freedom which are vibrational.

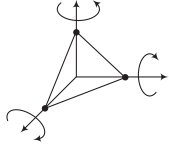
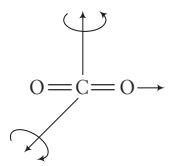
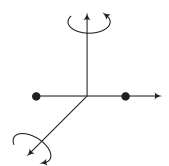
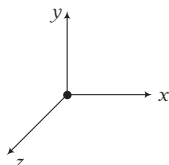
A very important relation regarding degree of freedom is

$$\gamma = 1 + \frac{2}{f}; \quad \gamma = C_p / C_v$$

where, f = degree of freedom

Degrees of freedom for different gases according to atomicity of gas at low temperature is given below

Atomicity of gas	Translational	Rotational	Total
Monoatomic, <i>e.g.</i> Ar, Ne, ideal gas, etc.	3	0	3
Diatomic, <i>e.g.</i> O ₂ , Cl ₂ , N ₂ , etc.	3	2	5
Triatomic (linear), <i>e.g.</i> CO ₂ , C ₂ H ₂ , etc.	3	2	5
Triatomic (Non-linear) or Polyatomic, <i>e.g.</i> H ₂ O, NH ₃ , CH ₄ , etc.	3	3	6



Note Degree of freedom for different gases can also be given by the formula, $f = 3NK$

where, N = number of particles in the system

and K = number of independent relations between the particles.

Maxwell's law of equipartition of energy

The total kinetic energy of a gas molecule is equally distributed among its all degrees of freedom and the energy associated with each degree of freedom at absolute temperature T is $\frac{1}{2}k_B T$, where k_B is Boltzmann's constant.

For a gas in thermal equilibrium at temperature T , the average value of translational energy of the molecule is

$$\langle E_t \rangle = \left\langle \frac{1}{2}mv_x^2 \right\rangle + \left\langle \frac{1}{2}mv_y^2 \right\rangle + \left\langle \frac{1}{2}mv_z^2 \right\rangle$$

Translational energy of the molecules, (for monoatomic gas)

$$\langle E_t \rangle = \frac{3}{2}k_B T$$

It is same as the case with an ideal gas. It distributes its internal energy equally in all degrees of freedom. In each degree of freedom, energy of one mole of an ideal gas is $\frac{1}{2}RT$, where T is the absolute temperature of the gas.

Thus, if f be the number of degrees of freedom, the internal energy of 1 mole of the gas will be $\frac{f}{2}RT$ or

internal energy of n moles of the gas will be $\frac{n}{2}fRT$. Thus,

$$\text{Internal energy, } U = \frac{n}{2}fRT$$

Equipartition of energy of gases is given below

Types of gases	f	Translational energy of one molecule	Internal energy of one mole
Monoatomic gas	3	$\frac{3}{2}k_B T$	$\frac{3}{2}RT$
Diatomic gas			
(a) excluding vibration	5	$\frac{5}{2}k_B T$	$\frac{5}{2}RT$
(b) including vibration	7	$\frac{7}{2}k_B T$	$\frac{7}{2}RT$
Polyatomic gas like CO ₂ , H ₂ O	6	$\frac{6}{2}k_B T = 3k_B T$	$\frac{6}{2}RT = 3RT$

Different energies of a system of degree of freedom f are as follows

(i) Total energy associated with each molecule = $\frac{f}{2}k_B T$

- (ii) Total energy associated with N -molecule $= \frac{f}{2} N k_B T$
- (iii) Total energy associated with 1 mole $= \frac{f}{2} R T$
- (iv) Total energy associated with μ moles $= \frac{f}{2} \mu R T$
- (v) Total energy associated with each gram $= \frac{f}{2} r T$
- (vi) Total energy associated with m gram $= \frac{f}{2} m r T$

Example 14.48 Calculate the ratio of kinetic energy of molecule of oxygen and neon gas at 27°C .

Sol. For oxygen gas, number of degrees of freedom is 5.

$$\therefore (\text{KE})_{\text{oxygen}} = \frac{5}{2} k_B T$$

Neon is monoatomic, therefore it has 3 degrees of freedom.

$$\therefore (\text{KE})_{\text{neon}} = \frac{3}{2} k_B T$$

Hence, the ratio of their kinetic energy

$$= \frac{(\text{KE})_{\text{oxygen}}}{(\text{KE})_{\text{neon}}} = \frac{(5/2) k_B T}{(3/2) k_B T} = \frac{5}{3}$$

Example 14.49 Calculate the average kinetic energy of oxygen molecule at 0°C . ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, $N_A = 6.023 \times 10^{23}$)

Sol. Given, $T = 0^\circ\text{C} = 273 \text{ K}$

Oxygen is diatomic molecule, therefore it has 5 degrees of freedom, i.e. 3 translational and 2 rotational.

$$\therefore \text{KE} = \frac{5}{2} \frac{RT}{N_A} = \frac{5}{2} \times \frac{8.314}{6.023 \times 10^{23}} \times 273 = 9.4 \times 10^{-21} \text{ J}$$

Example 14.50 Find the average kinetic energy per molecule at temperature T for an equimolar mixture of two ideal gases A and B , where A is monoatomic and B is diatomic.

Sol. Number of degrees of freedom per molecule for $A = 3$

Number of degrees of freedom per molecule for $B = 5$

Since, the mixture is equimolar, the average kinetic energy per molecule will be the simple average of the two values, i.e.

$$\left(\frac{3+5}{2} \right) k_B T = 4 k_B T, \text{ where } k_B \text{ is Boltzmann's constant.}$$

Example 14.51 Calculate the change in internal energy of 3.0 moles of helium gas when its temperature is increased by 2.0 K .

Sol. Helium is a monoatomic gas.

$$\text{Internal energy of } n\text{-moles of the gas, } U = \frac{3}{2} n R T$$

$$\therefore \Delta U = \frac{3}{2} n R (\Delta T)$$

$$\text{Substituting the values, } \Delta U = \left(\frac{3}{2} \right) (3) (8.31) (2.0) = 74.8 \text{ J}$$

Example 14.52 Given, Avogadro's number, $N = 6.02 \times 10^{23}$ and Boltzmann's constant, $k_B = 1.38 \times 10^{-23} \text{ J/K}$. Calculate (i) the average kinetic energy of translation of the molecules of an ideal gas at 0°C and at 100°C

(ii) and the corresponding energies per mole of the gas.

Sol. (i) According to the kinetic theory, the average kinetic energy of translation per molecule of an ideal monoatomic gas at temperature T (in kelvin) is $\left(\frac{3}{2} \right) k_B T$,

where k_B is Boltzmann's constant.

At 0°C ($T = 273 \text{ K}$),

$$\text{the kinetic energy of translation} = \frac{3}{2} k_B T$$

$$= \frac{3}{2} \times (1.38 \times 10^{-23}) \times 273$$

$$= 5.65 \times 10^{-21} \text{ J/molecule}$$

At 100°C ($T = 373 \text{ K}$), the energy is

$$= \frac{3}{2} \times (1.38 \times 10^{-23}) \times 373 = 7.72 \times 10^{-21} \text{ J/molecule}$$

(ii) 1 mole of gas contains $N (= 6.02 \times 10^{23})$ mol

Therefore, at 0°C , the kinetic energy of translation of 1 mole of the gas,

$$= (5.65 \times 10^{-21}) (6.02 \times 10^{23}) \approx 3401 \text{ J/mol}$$

At 100°C , the kinetic energy of translation of 1 mol of gas

$$= (7.72 \times 10^{-21}) (6.02 \times 10^{23}) \approx 4647 \text{ J/mol}$$

Example 14.53 One mole of an ideal monoatomic gas is taken at a temperature of 300 K . Its volume is doubled keeping its pressure constant. Find the change in internal energy.

Sol. Since, pressure is constant, $V \propto T \Rightarrow \frac{V_i}{T_i} = \frac{V_f}{T_f}$

$$\text{or } T_f = \frac{V_f}{V_i} T_i \Rightarrow T_f = 2 T_i = 600 \text{ K}$$

$$\therefore \Delta U = \frac{f}{2} n \cdot R \Delta T = \frac{3}{2} R (600 - 300) = 450 R$$

Critical temperature, pressure and volume

Gases cannot be liquified above a temperature called critical temperature (T_C), howsoever large the pressure may be.

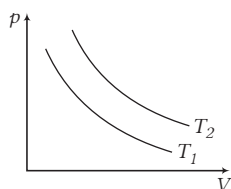
The pressure required to liquify the gas at critical temperature is called critical pressure (p_C) and the volume of the gas at critical temperature and pressure is called critical volume (V_C). Value of critical constants in terms of van der Waals' constants a and b are as follows

$$V_C = 3b, \quad p_C = \frac{a}{27b^2} \quad \text{and} \quad T_C = \frac{8a}{27Rb}$$

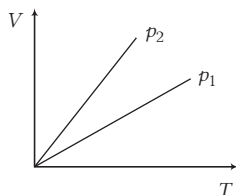
Further, $\frac{RT_C}{p_C V_C} = \frac{8}{3}$ is called **critical coefficient** and is same for all gases.

CHECK POINT 14.4

- If temperature of the gas remains constant, but pressure is decreased by 20%. The percentage change in volume
 - increases by 20%
 - decreases by 20%
 - increases by 25%
 - decreases by 25%
- The adjoining figure shows graph of pressure and volume of a gas at two temperatures T_1 and T_2 . Which of the following is correct?



- $T_1 > T_2$
 - $T_1 = T_2$
 - $T_1 < T_2$
 - Nothing can be said about temperatures
- A perfect gas at 27°C is heated at constant pressure, so as to double its volume. The increase in temperature of the gas will be
 - 600°C
 - 327°C
 - 54°C
 - 300°C
 - One litre of an ideal gas at 27°C is heated at a constant pressure till it reaches 297°C . Then, the final volume is the approximately
 - 1.2 L
 - 1.9 L
 - 19 L
 - 2.4 L
 - In the adjacent V - T diagram, what is the relation between p_1 and p_2 ?



- $p_2 = p_1$
 - $p_2 > p_1$
 - $p_2 < p_1$
 - Cannot be predicted
- 16 g of oxygen, 14 g of nitrogen and 11 g of carbon dioxide are mixed in an enclosure of volume 5 L and temperature 27°C . The pressure exerted by the mixture is
 - $4 \times 10^5 \text{ Nm}^{-2}$
 - $5 \times 10^5 \text{ Nm}^{-2}$
 - $6 \times 10^5 \text{ Nm}^{-2}$
 - $9 \times 10^5 \text{ Nm}^{-2}$
 - What is the degree of freedom in case of a monoatomic gas?
 - 1
 - 3
 - 5
 - None of these
 - The degrees of freedom of a molecule of a non-linear triatomic gas is (ignore vibrational motion)
 - 2
 - 4
 - 6
 - 8

- The average translational kinetic energy of O_2 (molar mass 32) at a particular temperature is 0.048 eV. The average translational kinetic energy of N_2 (molar mass 28) molecules (in eV) at the same temperature is
 - 0.0015
 - 0.003
 - 0.048
 - 0.768

- A gas has volume V and pressure p . The total translational kinetic energy of all the molecules of the gas is
 - $\frac{3}{2}pV$ only if the gas is monoatomic
 - $\frac{3}{2}pV$ only if the gas is diatomic
 - $\frac{3}{2}pV$ in all cases
 - None of the above

- A vessel contains a mixture of one mole of oxygen and two moles of nitrogen at 300 K. The ratio of the average rotational kinetic energy per O_2 molecule to that per N_2 molecule is
 - 1 : 1
 - 1 : 2
 - 2 : 1
 - 8 : 7

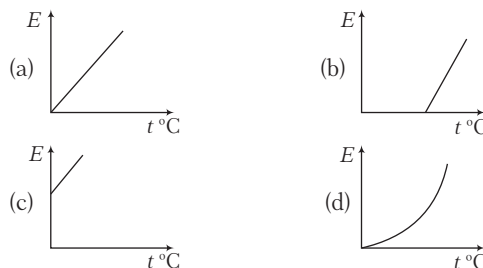
- Two monoatomic gases are at absolute temperatures 300 K and 350 K, respectively. Ratio of average kinetic energy of their molecules is
 - 7 : 6
 - 6 : 7
 - 36 : 49
 - 49 : 36

- At 27°C temperature, the kinetic energy of an ideal gas is E_1 . If the temperature is increased to 327°C , then the kinetic energy will be

- $\frac{E_1}{\sqrt{2}}$
- $\sqrt{2} E_1$
- $2E_1$
- $\frac{E_1}{2}$

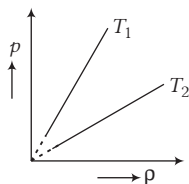
- The gas in a vessel is subjected to a pressure of 20 atm at a temperature 27°C . The pressure of the gas in a vessel after one-half of the gas is released from the vessel and the temperature of the remaining gas is raised by 50°C is
 - 8.5 atm
 - 10.8 atm
 - 11.7 atm
 - 17 atm

- The graph which represents the variation of mean kinetic energy of molecules with temperature $t^\circ\text{C}$ is

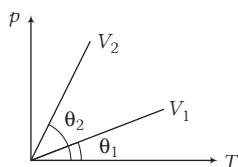


- A balloon is filled at 27°C and 1 atm pressure by 500 m^3 He. At -3°C and 0.5 atm pressures, the volume of He-gas contained in balloon will be
 - 700 m^3
 - 900 m^3
 - 1000 m^3
 - 500 m^3

17. The figure shows graphs of pressure *versus* density for an ideal gas at two temperatures T_1 and T_2 , then



- (a) $T_1 > T_2$ (b) $T_1 = T_2$
 (c) $T_1 < T_2$ (d) None of these
18. From the given p - T graph, what conclusion can be drawn?

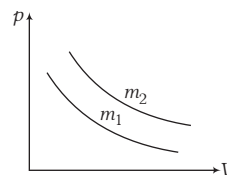


- (a) $V_2 = V_1$ (b) $V_2 < V_1$
 (c) $V_2 > V_1$ (d) None of these
19. A cylinder contains 20 kg of N_2 gas ($M_1 = 28 \text{ kg K}^{-1} \text{ mol}^{-1}$) at a pressure of 5 atm. The mass of hydrogen

($M_2 = 2 \text{ kg K}^{-1} \text{ mol}^{-1}$) at a pressure of 3 atm contained in the same cylinder at same temperature is

- (a) 1.08 kg (b) 0.86 kg
 (c) 0.68 kg (d) 1.68 kg

20. Two different isotherms representing the relationship between pressure p and volume V at a given temperature of the same ideal gas are shown for masses m_1 and m_2 , then



- (a) $m_1 > m_2$ (b) $m_1 = m_2$
 (c) $m_1 < m_2$ (d) Nothing can be predicted
21. A gas is found to obey the law $p^2V = \text{constant}$. The initial temperature and volume are T_0 and V_0 . If the gas expands to a volume $3V_0$, its final temperature becomes
- (a) $\frac{T_0}{3}$ (b) $\frac{T_0}{\sqrt{3}}$
 (c) $3T_0$ (d) None of these

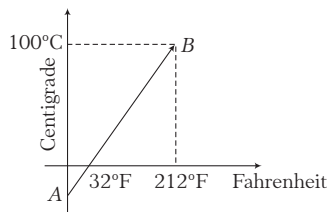
Chapter Exercises

(A) Taking it together

Associated questions of the chapter for advanced level practice

- 1 The absolute zero temperature in Fahrenheit scale is
 (a) -273°F (b) -32°F
 (c) -460°F (d) -132°F
- 2 On which of the following scales of temperature, the temperature is never negative?
 (a) Celsius (b) Fahrenheit
 (c) Reaumur (d) Kelvin
3. Boyle's law is applicable for an [NCERT Exemplar]
 (a) adiabatic process (b) isothermal process
 (c) isobaric process (d) isochoric process
4. Heat is associated with [NCERT Exemplar]
 (a) kinetic energy of random motion of molecules
 (b) kinetic energy of orderly motion of molecules
 (c) total kinetic energy of random and orderly motion of molecules
 (d) kinetic energy of random motion in some cases and kinetic energy of orderly motion in other
- 5 A centigrade and a Fahrenheit thermometers are dipped in boiling water. The water temperature is lowered until the Fahrenheit thermometer registers 140° . What is the fall in temperature as registered by the centigrade thermometer?
 (a) 30° (b) 40°
 (c) 60° (d) 80°
- 6 A constant volume gas thermometer shows pressure reading of 50 cm and 90 cm of mercury at 0°C and 100°C , respectively. When the pressure reading is 60 cm of mercury, the temperature is
 (a) 25°C (b) 40°C
 (c) 15°C (d) 12.5°C
- 7 The volume of gas at 20°C is 100 cm^3 at normal pressure. If it is heated to 100°C , its volume becomes 125 cm^3 at the same pressure, then volume coefficient of the gas at normal pressure is
 (a) $0.0015/^{\circ}\text{C}$ (b) $0.0045/^{\circ}\text{C}$
 (c) $0.0025/^{\circ}\text{C}$ (d) $0.0033/^{\circ}\text{C}$
- 8 Coefficient of real expansion of mercury is $0.18 \times 10^{-3}/^{\circ}\text{C}$. If the density of mercury at 0°C is 13.6 g/cc , its density at 200°C is
 (a) 13.11 g/cc (b) 26.22 g/cc
 (c) 52.11 g/cc (d) None of these
- 9 A gas at the temperature 250 K is contained in a closed vessel. If the gas is heated through 1 K, then the percentage increase in its pressure will be
 (a) 0.4% (b) 0.2% (c) 0.1% (d) 0.8%
- 10 An electron tube was sealed off during manufacture at a pressure of 1.2×10^{-7} mm of mercury at 27°C . Its volume is 100 cm^3 . The number of molecules that remain in the tube is
 (a) 2×10^{16} (b) 3×10^{15}
 (c) 3.94×10^{11} (d) 5×10^{11}
- 11 At what temperature is the root mean square velocity of the gaseous hydrogen molecules equal to that of oxygen molecules at 47°C ?
 (a) 20 K (b) 80 K (c) -73 K (d) 3 K
- 12 The root-mean-square velocity of the molecules in a sample of helium is $(5/7)$ th that of the molecules in a sample of hydrogen. If the temperature of hydrogen sample is 0°C , then the temperature of the helium sample is about
 (a) 5.57°C (b) 0°C
 (c) 273°C (d) 100°C
- 13 At room temperature, the rms speed of the molecules of certain diatomic gas is found to be 1930 ms^{-1} . The gas is
 (a) H_2 (b) F_2
 (c) O_2 (d) Cl_2
- 14 In a mercury thermometer, the ice point (lower fixed point) is marked as 10° and the steam point (upper fixed point) is marked as 130° . At 40°C temperature, what will this thermometer read?
 (a) 78° (b) 66°
 (c) 62° (d) 58°
- 15 A clock with a metal pendulum beating seconds keeps correct time at 0°C . If it loses 12.5 s a day at 25°C , the coefficient of linear expansion of metal pendulum is
 (a) $\frac{1}{86400}/^{\circ}\text{C}$ (b) $\frac{1}{43200}/^{\circ}\text{C}$
 (c) $\frac{1}{14400}/^{\circ}\text{C}$ (d) $\frac{1}{28800}/^{\circ}\text{C}$

- 16** The graph AB shown in figure is a plot of temperature of a body in degree celsius and degree fahrenheit, then



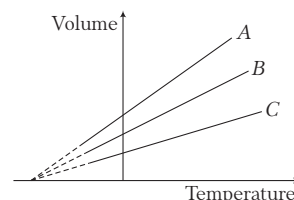
- (a) slope of line AB is $9/5$
 (b) slope of line AB is $5/9$
 (c) slope of line AB is $1/9$
 (d) slope of line AB is $3/9$
- 17** A chamber containing a gas was evacuated till the vacuum attained was 10^{-14} m of Hg. If the temperature of the chamber was 30°C , the number of molecules that remain in it per cubic metre is
 (a) 3.2×10^{11} (b) 3.2×10^{12}
 (c) 2.3×10^{12} (d) 2.3×10^{10}
- 18** A bimetallic strip is made of aluminium and steel ($\alpha_{\text{Al}} > \alpha_{\text{steel}}$). On heating, the strip will
 (a) remains straight
 (b) get twisted
 (c) will bend with aluminium on concave side
 (d) will bend with steel on concave side
- 19** A uniform metallic rod rotates about its perpendicular bisector with constant angular speed. If it is heated uniformly to raise its temperature slightly,
 (a) its speed of rotation increases
 (b) its speed of rotation decreases
 (c) its speed of rotation remains same
 (d) its speed increases because its moment of inertia increases
- 20** As the temperature is increased, the period of a pendulum
 (a) increases as its effective length increases even though its centre of mass still remains at the centre of the bob
 (b) decreases as its effective length increases even though its centre of mass still remains at the centre of the bob
 (c) increases as its effective length increases due to shifting to centre of mass below the centre of the bob
 (d) decreases as its effective length remains same but the centre of mass shifts above the centre of the bob
- 21** A cubic vessel (with face horizontal + vertical) contains an ideal gas at NTP. The vessel is being carried by a rocket which is moving at a speed of 500 ms^{-1} in vertical direction. The pressure of the gas inside the vessel as observed by us on the ground

- (a) remains the same because 500 ms^{-1} is very much smaller than v_{rms} of the gas
 (b) remains the same because motion of the vessel as a whole does not affect the relative motion of the gas molecules and the walls
 (c) will increase by a factor equal to $(v_{\text{rms}}^2 + (500)^2) / v_{\text{rms}}^2$ where v_{rms} was the original mean square velocity of the gas.
 (d) will be different on the top wall and bottom wall of the vessel

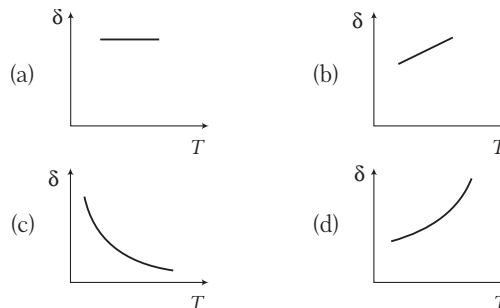
- 22** The steam point and the ice point of a mercury thermometer are marked as 80° and 10° . At what temperature on centigrade scale will the reading of this thermometer be 59° ?
 (a) 70°C (b) 60°C
 (c) 40°C (d) 50°C

- 23** Two uniform brass rods A and B of lengths l and $2l$ and radii $2r$ and r respectively are heated to the same temperature. The ratio of the increase in volume of A to that of B is
 (a) 1:1 (b) 1:2
 (c) 2:1 (d) 1:4

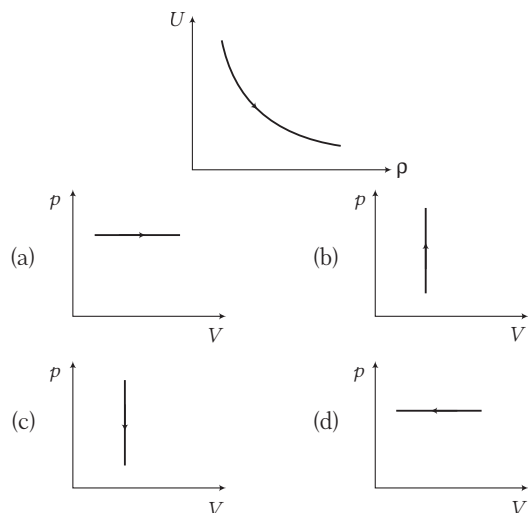
- 24** The expansion of an ideal gas of mass m at a constant pressure p is given by the straight line B . Then, the expansion of the same ideal gas of mass $2m$ at a pressure $2p$ is given by the straight line



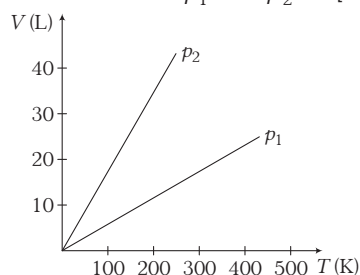
- (a) C (b) A
 (c) B (d) None of these
- 25** An ideal gas is initially at temperature T and volume V . Its volume is increased by ΔV due to an increase in temperature ΔT , pressure remaining constant. The quantity $\delta = \Delta V / (V \Delta T)$ varies with temperature as



- 26** Variation of internal energy with density of one mole of monoatomic gas is depicted in the adjoining figure. Corresponding variation of pressure with volume can be depicted as (Assuming the curve is rectangular hyperbola)



- 27** Volume *versus* temperature graphs for a given mass of an ideal gas are shown in figure. At two different values of constant pressure, what can be inferred about relation between p_1 and p_2 ? [NCERT Exemplar]



- (a) $p_1 > p_2$ (b) $p_1 = p_2$
(c) $p_1 < p_2$ (d) Data is insufficient
- 28** The radius of a metal sphere at room temperature T is R and the coefficient of linear expansion of the metal is α . The sphere is heated a little by a temperature ΔT , so that its new temperature is $T + \Delta T$. The increase in the volume of the sphere is approximately, [NCERT Exemplar]
- (a) $2\pi R\alpha\Delta T$ (b) $\pi R^2\alpha\Delta T$ (c) $4\pi R^3\alpha\Delta T/3$ (d) $4\pi R^3\alpha\Delta T$
- 29** An aluminium sphere is dipped into water. Which of the following statement is true? [NCERT Exemplar]
- (a) Buoyancy will be less in water at 0°C than that in water at 4°C .
(b) Buoyancy will be more in water at 0°C than that in water at 4°C .
(c) Buoyancy in water at 0°C will be same as that in water at 4°C .
(d) Buoyancy may be more or less in water at 4°C depending on the radius of the sphere.

- 30** A cylindrical steel plug is inserted into a circular hole of diameter 2.6 cm in a brass plate. When the plug and the plates are at a temperature of 20°C , the diameter of the plug is 0.01 cm smaller than that of the hole. The temperature at which the plug will just fit in it is

$$\left(\text{Given, } \alpha_{\text{steel}} = \frac{11 \times 10^{-6}}{^\circ\text{C}} \text{ and } \alpha_{\text{brass}} = \frac{19 \times 10^{-6}}{^\circ\text{C}} \right)$$

- (a) -48°C (b) -20°C (c) -10°C (d) -458°C

- 31** The apparent coefficient of expansion of a liquid when heated in a copper vessel is C and when heated in a silver vessel, it is S . If A is the linear coefficient of expansion of copper, then the linear coefficient of expansion of silver is

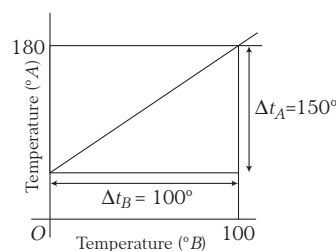
(a) $\frac{C + S - 3A}{3}$ (b) $\frac{C - S + 3A}{3}$
(c) $\frac{S + 3A - C}{3}$ (d) $\frac{C + S + 3A}{3}$

- 32** Two identical containers joined by a small pipe initially contain the same gas at pressure p_0 and absolute temperature T_0 . One container is now maintained at the same temperature while the other is heated to $2T_0$. The common pressure of the gases will be

(a) $\frac{3}{2}p_0$ (b) $\frac{4}{3}p_0$ (c) $\frac{5}{3}p_0$ (d) $2p_0$

- 33** The graph between two temperature scales A and B as shown in figure. Between upper fixed point and lower fixed point, there are 150 equal divisions on scale A and 100 on scale B . The relationship for conversion between the two scales is given by

[NCERT Exemplar]



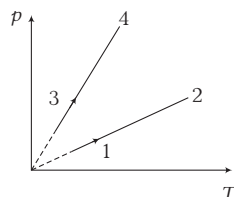
(a) $\frac{t_A - 180}{100} = \frac{t_B}{150}$ (b) $\frac{t_A - 30}{150} = \frac{t_B}{100}$
(c) $\frac{t_B - 180}{150} = \frac{t_A}{100}$ (d) $\frac{t_B - 40}{100} = \frac{t_A}{180}$

- 34** 1 mole of H_2 gas is contained in a box of volume $V = 100 \text{ m}^3$ at $T = 300 \text{ K}$. The gas is heated to a temperature of $T = 3000 \text{ K}$ and the gas gets converted to a gas of hydrogen atoms. The final pressure would be (considering all gases to be ideal)

[NCERT Exemplar]

- (a) same as the pressure initially
 (b) 2 times the pressure initially
 (c) 10 times the pressure initially
 (d) 20 times the pressure initially

35 Pressure *versus* temperature graph of an ideal gas of equal number of moles of different volumes are plotted as shown in figure. Choose the correct alternative.



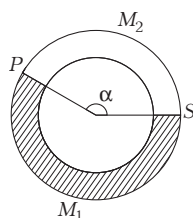
- (a) $V_1 = V_2$, $V_3 = V_4$ and $V_2 > V_3$
 (b) $V_1 = V_2$, $V_3 = V_4$ and $V_2 < V_3$
 (c) $V_1 = V_2 = V_3 = V_4$
 (d) $V_4 > V_3 > V_2 > V_1$

36. An inflated rubber balloon contains one mole of an ideal gas, has a pressure p , volume V and temperature T . If the temperature rises to $1.1 T$, and the volume is increased to $1.05 V$, the final pressure will be

[NCERT Exemplar]

- (a) $1.1 p$ (b) p
 (c) less than p (d) between p and $1.1 p$

37 A ring shaped tube contains two ideal gases with equal masses and relative molar masses $M_1 = 32$ and $M_2 = 28$. The gases are separated by one fixed partition P and another movable stopper S which can move freely without friction inside the ring.

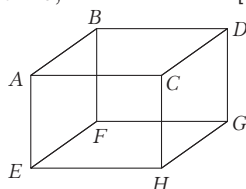


The angle α in equilibrium as shown in the figure (in degrees) is

- (a) 291 (b) 219 (c) 129 (d) 192

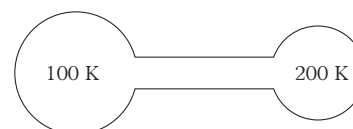
38 1 mole of an ideal gas is contained in a cubical volume V , $ABCDEFGH$ at 300 K (see figure). One face of the cube ($EFGH$) is made up of a material which totally absorbs any gas molecule incident on it. At any given time,

[NCERT Exemplar]



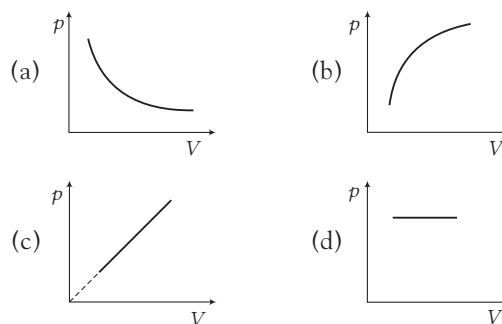
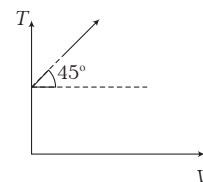
- (a) the pressure on $EFGH$ would be zero
 (b) the pressure on all the faces will be equal
 (c) the pressure of $EFGH$ would be double the pressure on $ABCD$
 (d) the pressure on $EFGH$ would be half that on $ABCD$

39 Figure shows two flasks connected to each other. The volume of the flask 1 is twice that of flask 2. The system is filled with an ideal gas at temperatures 100 K and 200 K, respectively. If the mass of the gas in flask 1 is m , then what is the mass of the gas in flask 2?



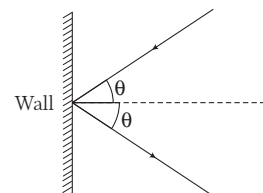
- (a) m (b) $m/2$ (c) $m/4$ (d) $m/8$

40 The given curve represents the variations of temperature as a function of volume for one mole of an ideal gas. Which of the following curves best represents the variation of pressure as a function of volume?



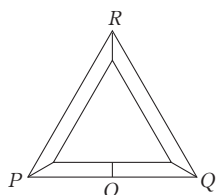
41 The mass of hydrogen molecule is 3.32×10^{-27} kg.

If 10^{23} hydrogen molecules strike per second at 2 cm^2 area of a rigid wall at an angle of 45° from the normal and rebound back with a speed of 1000 ms^{-1} , then the pressure exerted on the wall is



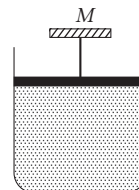
- (a) $2.34 \times 10^3 \text{ Pa}$ (b) $0.23 \times 10^6 \text{ Pa}$
 (c) $0.23 \times 10^3 \text{ Pa}$ (d) $23.4 \times 10^3 \text{ Pa}$

- 42** Three rods of equal length l are joined to form an equilateral triangle PQR . O is the mid-point of PQ . Distance OR remains same for small change in temperature. Coefficient of linear expansion for PR and RQ is same, i.e. α_2 but that for PQ is α_1 . Then,

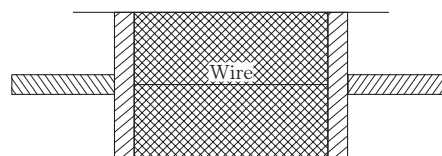


- (a) $\alpha_2 = 3\alpha_1$ (b) $\alpha_2 = 4\alpha_1$
 (c) $\alpha_1 = 3\alpha_2$ (d) $\alpha_1 = 4\alpha_2$
- 43** A piece of metal weighs 46 g in air. When it is immersed in a liquid of specific gravity 1.24 at 27°C , it weighs 30 g. When the temperature is raised to 42°C , the metal piece weighs 30.5 g. If the specific gravity of the liquid at 42°C is 1.20, the coefficient of linear expansion of the metal is
- (a) $1.6 \times 10^{-3}/^\circ\text{C}$ (b) $2.3 \times 10^{-5}/^\circ\text{C}$
 (c) $4.3 \times 10^{-3}/^\circ\text{C}$ (d) $3.4 \times 10^{-3}/^\circ\text{C}$
- 44** A vertical cylinder closed at both ends is fitted with a smooth piston dividing the volume into two parts each containing one mole of air. At the equilibrium temperature of 320 K, the upper and lower parts are in the ratio 4 : 1. The ratio will become 3 : 1 at a temperature of
- (a) 450 K (b) 228 K (c) 420 K (d) 570 K

- 45** A cylinder containing an ideal gas is in vertical position and has a piston of mass M that is able to move up or down without friction (see figure). If the temperature is increased, then [NCERT Exemplar]



- (a) both p and V of the gas will change
 (b) only p will increase according to Charles' law
 (c) V will change but not p
 (d) p will change but not V
- 46** A cylindrical tube of uniform cross-sectional area A is fitted with two air tight frictionless pistons.



The pistons are connected to each other by a metallic wire. Initially, the pressure of the gas is p_0 and temperature is T_0 , atmospheric pressure is also p_0 . Now, the temperature of the gas is increased to $2T_0$, the tension in the wire will be

- (a) $2p_0A$ (b) p_0A (c) $\frac{p_0A}{2}$ (d) $4p_0A$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-4) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) If Assertion is correct but Reason is incorrect.
 (d) If Assertion is incorrect but Reason is correct.

- 1 Assertion** In summers, a metallic scale will read more than the actual.

Reason In summers, length of metallic scale increases.

- 2 Assertion** Translational kinetic energy of any gas at temperature T would be $\frac{1}{2}mv_{\text{rms}}^2$.

Reason Translational kinetic energy of any type of gas at temperature T would be $\frac{3}{2}RT$ for one mole.

- 3. Assertion** Degree of freedom of a monoatomic gas is always three, whether we consider vibrational effects or not.

Reason At all temperatures (low or high), vibrational kinetic energy of an ideal gas is zero.

- 4. Assertion** Total internal energy of oxygen gas at a given temperature is E . Of this energy, $\frac{3}{5}E$ is translational kinetic energy and $\frac{2}{5}E$ is rotational kinetic energy.

Reason Potential energy of an ideal gas is zero.

Statement based questions

- 1 From the following statements, concerning ideal gas at any given temperature T , which of the following statement is incorrect?
 - (a) The coefficient of volume expansion at constant pressure is same for all ideal gases.
 - (b) The average translational kinetic energy per molecule of oxygen gas is $k_B T$ (k_B being Boltzmann constant).
 - (c) In a gaseous mixture, the average translational kinetic energy of the molecules of each component is same.
 - (d) The mean free path of molecules increases with the decrease in pressure.
- 2 A bimetallic strip is formed out of two identical strips, one of copper and other of brass. The coefficients of linear expansion of the two metals are α_c and α_b . On heating, the temperature of the strip goes up by ΔT and the strip bends to form an arc of radius of curvature R . Then, which of the following statement is correct about the radius of curvature?
 - (a) It is proportional to ΔT .
 - (b) It is inversely proportional to ΔT .
 - (c) It is proportional to $|\alpha_b - \alpha_c|$.
 - (d) It is inversely proportional to $|\alpha_b + \alpha_c|$.
3. I. The pressure of an ideal gas depends on the volume of the gas.
 II. The pressure of an ideal gas depends on the temperature of the gas.
 Which of the following statement(s) is/are correct?
 - (a) Only I
 - (b) Only II
 - (c) Both I and II
 - (d) Neither I nor II
- 4 I. Atomic theory relates molecular properties with bulk properties of a gas.
 II. According to kinetic theory, the interatomic forces are negligible in gases.
 III. According to kinetic theory, the gas consists of rapidly moving atoms.
 Which of the following statement(s) is/are correct?
 - (a) Both I and II
 - (b) Only III
 - (c) Both II and III
 - (d) I, II and III

Match the columns

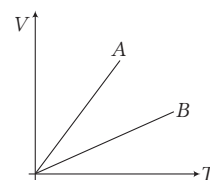
- 1 In the process $T^2 \propto 1/V$, pressure of the gas increases from p_0 to $4p_0$. Match the Column I with Column II and mark the correct option from the codes given below.

Column I	Column II
(A) Temperature of the gas	(p) Positive
(B) Volume of the gas	(q) Negative
(C) Energy of the gas	(r) Two times
	(s) None of these

Codes

A	B	C	A	B	C
(a) s	r	p	(b) r	s	p
(c) r	p	s	(d) s	p	r

- 2 In the V - T graph shown in the figure. Match the Column I with Column II and mark the correct option from the codes given below.



Column I	Column II
(A) Gas A is	(p) Monoatomic
(B) p_A/p_B is	(q) Diatomic
(C) n_A/n_B is	(r) > 1
(D) Gas B is	(s) < 1
	(t) Cannot say anything

Codes

A	B	C	D
(a) t	p	r	t
(b) t	s	t	t
(c) t	t	t	t
(d) s	t	p	t

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1 The average thermal energy for a monoatomic gas is (where, k_B is Boltzmann constant and T is absolute temperature.) [NEET 2020]
 - (a) $\frac{3}{2} k_B T$
 - (b) $\frac{5}{2} k_B T$
 - (c) $\frac{7}{2} k_B T$
 - (d) $\frac{1}{2} k_B T$
- 2 A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C . Its density is (Take, $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$) [NEET 2020]
 - (a) 0.2 kg/m^3
 - (b) 0.1 kg/m^3
 - (c) 0.02 kg/m^3
 - (d) 0.5 kg/m^3
- 3 The mean free path λ for a gas, with molecular diameter d and number of density n can be expressed as [NEET 2020]
 - (a) $\frac{1}{\sqrt{2} n \pi d^2}$
 - (b) $\frac{1}{\sqrt{2} n^2 \pi d^2}$
 - (c) $\frac{1}{\sqrt{2} n^2 \pi^2 d^2}$
 - (d) $\frac{1}{\sqrt{2} n \pi d}$

- 4 An ideal gas equation can be written as

$$p = \frac{\rho RT}{M_0}$$

where, ρ and M_0 are respectively, [NEET 2020]

- (a) mass density, mass of the gas
 (b) number density, molar mass
 (c) mass density, molar mass
 (d) number density, mass of the gas
- 5 The mean free path l for a gas molecule depends upon diameter d of the molecule as [NEET 2020]

(a) $l \propto \frac{1}{d^2}$ (b) $l \propto d$ (c) $l \propto d^2$ (d) $l \propto \frac{1}{d}$

- 6 A copper rod of length 88 cm and an aluminium rod of unknown length have their increase in length independent of increase in temperature. The length of aluminium rod is [NEET (National) 2019]

- (a) 113.9 cm (b) 88 cm
 (c) 68 cm (d) 6.8 cm

- 7 Increase in temperature of a gas filled in a container would lead to [NEET 2019]

- (a) increase in its kinetic energy
 (b) decrease in its pressure
 (c) decrease in intermolecular distance
 (d) increase in its mass

- 8 An ideal gas initially at pressure 1 bar is being compressed from 30 m^3 to 10 m^3 volume and its temperature decreases from 320 K to 280 K, then find the value of final pressure of the gas. [AIIMS 2019]

- (a) 2.625 bar (b) 3.4 bar
 (c) 1.325 bar (d) 4.5 bar

- 9 **Assertion** NH_3 is liquified more easily than CO_2 .

Reason Critical temperature of NH_3 is more than CO_2 . [AIIMS 2019]

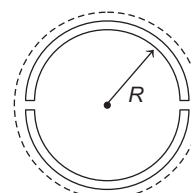
- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Both Assertion and Reason are incorrect.

- 10 **Assertion** Vibrational degree of freedom of a diatomic gas molecule appears at every high temperature.

Reason Diatomic gas has two vibrational degree of freedom in one direction. [AIIMS 2019]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Both Assertion and Reason are incorrect.

- 11 The wooden wheel of radius R is made of two semicircular parts (see figure). The two parts are held together by a ring made of a metal strip of cross-sectional area S and length L . L is slightly less than $2\pi R$. To fit the ring on the wheel, it is heated, so that its temperature rises by ΔT and it just steps over the wheel. As it cools down to surrounding temperature, it presses the semicircular parts together. If the coefficient of linear expansion of the metal is α and its Young's modulus is Y , the force that one part of the wheel applies on the other part is [JIPMER 2019]



- (a) $2\pi SY\alpha\Delta T$ (b) $SY\alpha\Delta T$
 (c) $\pi SY\alpha\Delta T$ (d) $2SY\alpha\Delta T$

- 12 At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the earth's atmosphere? [NEET 2018]

(Take, mass of oxygen molecule, $m = 2.76 \times 10^{-26} \text{ kg}$ and Boltzmann's constant, $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$)

- (a) $5.016 \times 10^4 \text{ K}$ (b) $8.326 \times 10^4 \text{ K}$
 (c) $2.508 \times 10^4 \text{ K}$ (d) $1.254 \times 10^4 \text{ K}$

- 13 The coefficient of volume expansion of glycerine is $49 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$. What is the fractional change in density for a 30°C rise in temperature? [JIPMER 2018, UP CPMT 2015]

- (a) 0.0155 (b) 0.0145
 (c) 0.0255 (d) 0.0355

- 14 A given sample of an ideal gas occupies a volume V at a pressure p and absolute temperature T . The mass of each molecule of the gas is m . Which of the following gives the density of the gas? [NEET 2017]

- (a) $\frac{p}{kT}$ (b) $\frac{pm}{kT}$ (c) $\frac{p}{kTV}$ (d) mkT

- 15 **Assertion** The molecules of a monoatomic gas have three degrees of freedom.

Reason The molecules of diatomic gas have five degrees of freedom. [NEET 2017]

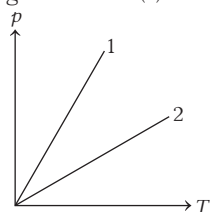
- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Both Assertion and Reason are incorrect.

- 16** A gas mixture consists of 2 mol of O_2 and 4 mol of Ar at temperature T . Neglecting all vibrational modes, the total internal energy of the system is [NEET 2017]
 (a) $4 RT$ (b) $15 RT$ (c) $9 RT$ (d) $11 RT$

- 17** The coefficient of cubical expansion of mercury is $0.00018/^\circ C$ and that of brass $0.00006/^\circ C$. If a barometer having a brass scale were to read 74.5 cm at $30^\circ C$, find the true barometric height at $0^\circ C$. The scale is supposed to be correct at $15^\circ C$. [AIIMS 2017]
 (a) 74.122 cm (b) 79.152 cm (c) 42.161 cm (d) 142.39 cm

- 18** Mercury boils at $367^\circ C$. However, mercury thermometers are made such that they can measure temperature upto $500^\circ C$. This is done by [JIPMER 2017]
 (a) maintaining vacuum above mercury column in the stem of the thermometer
 (b) filling nitrogen gas at high pressure above the mercury column
 (c) filling oxygen gas at high pressure above the mercury column
 (d) filling nitrogen gas at low pressure above the mercury column

- 19** A graph between pressure p (along Y-axis) and absolute temperature T (along X-axis) for equal moles of two gases has been drawn. Given that volume of second gas is more than volume of first gas. Which of the following statement(s) is/are correct? [JIPMER 2017]



- (a) Slope of gas 1 is less than gas 2.
 (b) Slope of gas 1 is more than gas 2.
 (c) Both have some slopes.
 (d) None of the above
- 20.** Coefficient of linear expansion of brass and steel rods are α_1 and α_2 . Lengths of brass and steel rods are l_1 and l_2 , respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which one of the following relations holds good? [NEET 2016]
 (a) $\alpha_1 l_2^2 = \alpha_2 l_1^2$ (b) $\alpha_1^2 l_2 = \alpha_2^2 l_1$ (c) $\alpha_1 l_1 = \alpha_2 l_2$ (d) $\alpha_1 l_2 = \alpha_2 l_1$
- 21** The molecules of a given mass of a gas have rms velocity of 200 ms^{-1} at $27^\circ C$ and $10 \times 10^5 \text{ Nm}^{-2}$ pressure. When the temperature and pressure of the gas are respectively, $127^\circ C$ and $0.05 \times 10^5 \text{ Nm}^{-2}$, the rms velocity of its molecules (in ms^{-1}) is [NEET 2016]
 (a) $\frac{400}{\sqrt{3}}$ (b) $\frac{100\sqrt{2}}{3}$ (c) $\frac{100}{3}$ (d) $100\sqrt{2}$

- 22** The rms speed of oxygen molecule in a gas at $27^\circ C$ would be given by [UK PMT 2015]
 (a) 483 ms^{-1} (b) 966 ms^{-1}
 (c) 4.83 ms^{-1} (d) 9.66 ms^{-1}

- 23** The pressure of an ideal gas is directly proportional to [UK PMT 2015]
 (a) total kinetic energy (b) translational kinetic energy
 (c) rotational kinetic energy (d) vibrational kinetic energy

- 24** Two metal rods of lengths L_1 and L_2 and coefficients of linear expansion α_1 and α_2 respectively are welded together to make a composite rod of length $(L_1 + L_2)$ at $0^\circ C$. Find the effective coefficient of linear expansion of the composite rod. [EAMCET 2015]
 (a) $\frac{L_1 \alpha_1^2 - L_2 \alpha_2^2}{L_1^2 + L_2^2}$ (b) $\frac{L_1^2 \alpha_1 - L_2^2 \alpha_2}{L_1^2 + L_2^2}$
 (c) $\frac{L_1 \alpha_1 + L_2 \alpha_2}{L_1 - L_2}$ (d) $\frac{L_1 \alpha_1 + L_2 \alpha_2}{L_1 + L_2}$

- 25** The ratio of rms speed of an ideal gas molecules at pressure p to that at pressure $2p$ is [Kerala CEE 2015]
 (a) 1 : 2 (b) 2 : 1 (c) $1 : \sqrt{2}$ (d) $\sqrt{2} : 1$
 (e) 1 : 1

- 26** The rms speed of oxygen is v at a particular temperature. If the temperature is doubled and oxygen molecules dissociate into oxygen atoms, the rms speed becomes [WB JEE 2015]
 (a) v (b) $\sqrt{2}v$ (c) $2v$ (d) $4v$

- 27** The deviation of a real gas from the ideal one is minimum at [UP CPMT 2015]
 (a) low pressure and high temperature
 (b) low temperature and low pressure
 (c) high pressure and low temperature
 (d) high pressure and high temperature

- 28** The density of hydrogen gas at STP is 0.09 kg m^{-3} . The mean kinetic energy of one gram molecule of hydrogen gas is approx [UP CPMT 2015]
 (a) 3403 J (b) 3500 J
 (c) 3704 J (d) 3207 J

- 29** At constant pressure, the ratio of increase in volume of an ideal gas per degree rise in kelvin temperature to its volume is [Manipal 2015]
 (a) $1/T$ (b) $1/T^2$
 (c) T (d) T^2

- 30** Water is heated from $0^\circ C$ to $10^\circ C$, then its volume [KCET 2015]
 (a) does not change
 (b) decreases
 (c) first decreases and then increases
 (d) increases

- 31** In anomalous expansion of water, at what temperature, the density of water is maximum? [KCET 2014]
 (a) 4°C (b) $< 4^{\circ}\text{C}$ (c) $> 4^{\circ}\text{C}$ (d) 10°C
- 32** A metal rod is fixed rigidly at two ends so as to prevent its thermal expansion. If L , α and Y respectively denote the length of the rod, coefficient of linear thermal expansion and Young's modulus of its material, then for an increase in temperature of the rod by ΔT , the longitudinal stress developed in the rod is [AMU 2014]
 (a) inversely proportional to α
 (b) inversely proportional to Y
 (c) directly proportional to $\Delta T / Y$
 (d) independent of L
- 33** Which one of the following is a wrong statement in kinetic theory of gases? [Kerala CEE 2014]
 (a) The gas molecules are in random motion.
 (b) The collision between molecules is inelastic.
 (c) The volume occupied by the molecules of a gas is negligible.
 (d) The force of attraction between the molecules is negligible.
 (e) Both (b) and (c)
- 34** The average pressure of an ideal gas is [J&K CET 2013]
 (a) $p = (1/3) mn V_{av}^2$ (b) $p = (1/2) mn V_{av}$
 (c) $p = (1/4) mn V_{av}^2$ (d) $p = (1/3) mn V_{av}$
- 35** According to equipartition law of energy, each particle in a system of particles have thermal energy E equal to [J&K CET 2013]
 (a) $k_B T$ (b) $(1/2) k_B T$
 (c) $3 k_B T$ (d) $(3/2) k_B T$
- 36** When a liquid is heated in a glass vessel, its coefficient of apparent expansion is $1.03 \times 10^{-3} / ^{\circ}\text{C}$. When the same liquid is heated in a copper vessel, its coefficient of apparent expansion is $1.006 \times 10^{-3} / ^{\circ}\text{C}$. If the coefficient of linear expansion of copper is $17 \times 10^{-6} / ^{\circ}\text{C}$, then the coefficient of linear expansion of glass is [EAMCET 2013]
 (a) $8.5 \times 10^{-4} / ^{\circ}\text{C}$ (b) $9 \times 10^{-6} / ^{\circ}\text{C}$
 (c) $27 \times 10^{-6} / ^{\circ}\text{C}$ (d) $10 \times 10^{-4} / ^{\circ}\text{C}$
- 37** A horizontal uniform tube, open at both ends, is containing a liquid of certain length at some temperature. When the temperature is changed, the length of the liquid in the tube is not changed. If α is the coefficient of linear expansion of the material of the tube and γ is the coefficient of volume expansion of the liquid, then [EAMCET 2013]
 (a) $\gamma = 2\alpha$ (b) $\gamma = 3\alpha$ (c) $\gamma = 4\alpha$ (d) $\gamma = \alpha$
- 38** The average kinetic energy of a gas molecule at absolute temperature T is [Kerala CEE 2013]
 (a) directly proportional to T^2
 (b) inversely proportional to T^2
 (c) directly proportional to T
 (d) inversely proportional to T
 (e) inversely proportional to T^3
- 39** At what temperature, the rms velocity of gas molecules would be double of its value at NTP, if pressure is remaining constant? [MP PMT 2013]
 (a) 819°C (b) 819 K (c) 546 K (d) 546°C
- 40** The average kinetic energy per mole of hydrogen at given temperature is [MP PMT 2013]
 (a) equal of that of oxygen (b) 16 times that of oxygen
 (c) $\frac{1}{16}$ times that of oxygen (d) $\frac{1}{8}$ times that of oxygen
- 41** In kinetic theory of gases, it is assumed that molecular collisions are [UP CPMT 2013]
 (a) for negligible duration
 (b) inelastic
 (c) one-dimensional (head on)
 (d) unable to exert mutual force
- 42** The number of molecules in a litre of a gas at temperature of 27°C and a pressure of 10^6 dyne cm^{-2} is [UP CPMT 2013]
 (a) 2.4×10^{20} (b) 2.4×10^{21} (c) 2.4×10^{22} (d) 2.4×10^{23}
- 43** A non-conducting body floats in a liquid at 20°C with $\frac{2}{3}$ of its volume immersed in the liquid. When liquid temperature is increased to 100°C , $\frac{3}{4}$ of body's volume is immersed in the liquid. Then, the coefficient of real expansion of the liquid is (neglecting the expansion of container of the liquid) [AIIMS 2012]
 (a) $15.6 \times 10^{-4} / ^{\circ}\text{C}$ (b) $156 \times 10^{-4} / ^{\circ}\text{C}$
 (c) $1.56 \times 10^{-4} / ^{\circ}\text{C}$ (d) $0.156 \times 10^{-4} / ^{\circ}\text{C}$
- 44** During an experiment an ideal gas is found to obey an additional law $Vp^2 = \text{constant}$. The gas is initially at temperature T and volume V . When it expands to volume $2V$, the resulting temperature is [UP CPMT 2012]
 (a) $\frac{T}{2}$ (b) $2T$ (c) $\sqrt{2} T$ (d) $\frac{T}{\sqrt{2}}$
- 45** Two temperature scales A and B are related by $\frac{A - 42}{110} = \frac{B - 72}{220}$. At which temperature, two scales have the same readings? [WB JEE 2011]
 (a) -42° (b) -72° (c) $+12^{\circ}$ (d) -40°

ANSWERS

CHECK POINT 14.1

1. (b)	2. (a)	3. (a)	4. (a)	5. (c)	6. (c)	7. (c)	8. (d)
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CHECK POINT 14.2

1. (c)	2. (a)	3. (b)	4. (b)	5. (b)	6. (a)	7. (d)	8. (c)	9. (c)	10. (a)
11. (a)	12. (c)	13. (a)	14. (b)	15. (c)	16. (b)	17. (a)			

CHECK POINT 14.3

1. (d)	2. (b)	3. (c)	4. (c)	5. (d)	6. (a)	7. (a)	8. (b)	9. (d)	10. (c)
11. (c)	12. (b)	13. (a)	14. (d)	15. (c)	16. (a)	17. (c)	18. (c)	19. (c)	

CHECK POINT 14.4

1. (c)	2. (c)	3. (d)	4. (b)	5. (c)	6. (c)	7. (b)	8. (c)	9. (c)	10. (c)
11. (a)	12. (b)	13. (c)	14. (c)	15. (c)	16. (b)	17. (a)	18. (c)	19. (b)	20. (c)
21. (d)									

(A) Taking it together

1. (c)	2. (d)	3. (b)	4. (a)	5. (b)	6. (a)	7. (d)	8. (a)	9. (a)	10. (c)
11. (a)	12. (a)	13. (a)	14. (d)	15. (a)	16. (b)	17. (a)	18. (d)	19. (b)	20. (a)
21. (b)	22. (a)	23. (c)	24. (c)	25. (c)	26. (d)	27. (a)	28. (d)	29. (a)	30. (d)
31. (b)	32. (b)	33. (b)	34. (d)	35. (a)	36. (d)	37. (d)	38. (d)	39. (c)	40. (a)
41. (a)	42. (d)	43. (b)	44. (a)	45. (c)	46. (b)				

(B) Medical entrance special format questions

• Assertion and reason

1. (d) 2. (b) 3. (c) 4. (b)

• Statement based questions

1. (b) 2. (b) 3. (c) 4. (c)

• Match the columns

1. (b) 2. (c)

(C) Medical entrances' gallery

1. (a)	2. (a)	3. (a)	4. (a)	5. (a)	6. (c)	7. (a)	8. (a)	9. (a)	10. (b)
11. (d)	12. (b)	13. (b)	14. (b)	15. (b)	16. (d)	17. (a)	18. (b)	19. (b)	20. (c)
21. (a)	22. (a)	23. (b)	24. (d)	25. (c)	26. (c)	27. (a)	28. (a)	29. (a)	30. (c)
31. (a)	32. (c)	33. (b)	34. (a)	35. (b)	36. (b)	37. (b)	38. (c)	39. (a)	40. (a)
41. (a)	42. (c)	43. (a)	44. (c)	45. (c)					

Hints & Explanations

• CHECK POINT 14.1

2 (a) $\Delta T_F = \frac{9}{5} \Delta T_C = \frac{9}{5} \times 25^\circ \text{C} = 45^\circ \text{F}$

3 (a) Let the required temperature is θ .

$$\therefore \frac{\theta - 0}{100} = \frac{\theta - 32}{180} \Rightarrow \theta = -40^\circ$$

4 (a) On this thermometer, 34°C will read as,

$$\frac{t - (-20)}{130 - (-20)} = \frac{34 - 0}{100 - 0}$$

$$\Rightarrow t = -20 + \frac{150}{100} \times 34 = 31^\circ$$

5 (c) $\frac{F - 32}{9} = \frac{K - 273}{5} \Rightarrow \frac{x - 32}{9} = \frac{x - 273}{5}$

$$\Rightarrow x = 574.25$$

6 (c) Temperature shown on faulty thermometer will be

$$t = 5 + \left(\frac{95 - 5}{100} \right) \times 40 = 41^\circ$$

7 (c) Given, $\frac{T_C}{T_F} = \frac{2}{5} \Rightarrow T_F = \frac{5}{2} T_C$

$$\therefore \frac{T_C}{100} = \frac{(5/2) T_C - 32}{180}$$

$$\Rightarrow 9T_C = \frac{25}{2} T_C - 160$$

$$\Rightarrow T_C = 45.71^\circ \text{C}$$

8 (d) $t = \left(\frac{80 - 50}{75 - 50} \right) \times 100^\circ \text{C}$

$$\therefore t = 120^\circ \text{C}$$

• CHECK POINT 14.2

1 (c) Change in length, $\Delta l = l_0 \alpha \Delta T = 10 \times 11 \times 10^{-6} (19 - 20)$
 $= -11 \times 10^{-5} \text{cm}$

The negative sign shows that, it is shorter in size.

2 (a) As, metal expands on heating in all directions, so the radius of the hole will also increase in size.

3 (b) With decreasing temperature, length of the steel tape decreases and hence the tape gives reading greater than the real reading. So, real length of wood must be less than 25 cm.

4 (b) $\Delta l_1 = \Delta l_2 \Rightarrow l_1 \alpha_1 \Delta \theta = l_2 \alpha_2 \Delta \theta \Rightarrow \frac{l_1}{l_2} = \frac{\alpha_2}{\alpha_1}$

5 (b) Original value of circumference, $l = 2\pi R$

$$\therefore \Delta l = l \alpha \theta = (2\pi R) \alpha \theta$$

6 (a) As, $\Delta L = L \alpha \Delta T \Rightarrow 0.19 = 100 \times \alpha \times 100$

$$\Rightarrow \alpha = 0.19 \times 10^{-4} = 1.9 \times 10^{-5} ^\circ \text{C}^{-1}$$

$$\Rightarrow \gamma = 3\alpha = 5.7 \times 10^{-5} / ^\circ \text{C}$$

7 (d) Fractional change in period,

$$\frac{\Delta T}{T} = \frac{1}{2} \alpha \Delta \theta = \frac{1}{2} \times 2 \times 10^{-6} \times 10 = 10^{-5}$$

$$\% \text{ change} = \frac{\Delta T}{T} \times 100 = 10^{-5} \times 100 = 10^{-3} \%$$

8 (c) $L \alpha \Delta \theta + 2L (2\alpha) (\Delta \theta) = (3L) (\alpha_{av}) \Delta \theta$

$$\frac{L}{\alpha} + \frac{2L}{2\alpha} \Rightarrow \frac{3L}{\alpha_{av}}$$

or $\alpha_{av} = \frac{5}{3} \alpha$

9 (c) Strain = $\frac{\Delta l}{l} = \frac{l \alpha \Delta \theta}{l} = \alpha \Delta \theta$

Stress = $Y \times \text{Strain} = Y \alpha \Delta \theta$
 $(\text{Stress})_1 = (\text{Stress})_2$

$$\therefore Y_1 \alpha_1 = Y_2 \alpha_2 \quad (\because \Delta \theta \rightarrow \text{same})$$

or $\frac{Y_1}{Y_2} = \frac{\alpha_2}{\alpha_1} = \frac{3}{2}$

10 (a) Strain = $\frac{\Delta l}{l} = \alpha \Delta \theta$

Stress = $Y \times \text{Strain} = Y \alpha \Delta \theta$

$\therefore$ Force or tension, $T = \text{Stress} \times \text{Area} = Y \alpha \Delta \theta$

$$= \frac{\pi Y \alpha d^2 \Delta \theta}{4} \quad \left(\because A = \frac{\pi d^2}{4} \right)$$

or $T = \frac{\pi \times 2 \times 10^{11} \times 10^{-5} \times 10^{-4} \times 25}{4}$
 $= 3925 \text{ N} \approx 4000 \text{ N}$

11 (a) Volumetric expansion will take place. So, if the ball is heated, then the volume of the cavity will increase.

12 (c) $V = V_0 (1 + \gamma \Delta \theta)$

$$L^3 = L_0 (1 + \alpha_1 \Delta \theta) L_0^2 (1 + \alpha_2 \Delta \theta)^2$$

$$= L_0^3 (1 + \alpha_1 \Delta \theta) (1 + \alpha_2 \Delta \theta)^2$$

Since, $L_0^3 = V_0$ and $L^3 = V$

Hence, $1 + \gamma \Delta \theta = (1 + \alpha_1 \Delta \theta) (1 + \alpha_2 \Delta \theta)^2$

$$\cong (1 + \alpha_1 \Delta \theta) (1 + 2\alpha_2 \Delta \theta)$$

(Using Binomial expansion)

$$\cong (1 + \alpha_1 \Delta \theta + 2\alpha_2 \Delta \theta)$$

$$\Rightarrow \gamma = \alpha_1 + 2\alpha_2$$

14 (b) Coefficient of volume expansion of container will become

$$3 \left(\frac{\alpha}{2} \right), \text{ which is greater than the coefficient of volume}$$

expansion of liquid. Hence, container will expand more and the level of the liquid in the container would fall.

- 15** (c) Water has maximum density at 4°C, so if the water is heated above 4°C or cooled below 4°C, density decreases, *i.e.* volume increases. In other words, it expands and overflows in both the cases.

- 16** (b) Volume of mercury that will overflow,

$$\Delta V = V\Delta T(\gamma_{\text{mercury}} - \gamma_{\text{glass}})$$

$$= 200 \times 80(18 \times 10^{-4} - 0.12 \times 10^{-4}) = 2.688 \approx 2.69 \text{ cm}^3$$

- 17** (a) $\rho_{200} = \rho_0(1 - \gamma_m \Delta T) = 13.6(1 - 0.18 \times 10^{-3} \times 200) = 13.11 \text{ g/cc}$

• CHECK POINT 14.3

- 1** (d) All molecules have same speed. This statement is true but this is not an assumption of kinetic theory of gases.

- 2** (b) Total pressure inside the container is given as

$$p = \frac{1}{3} \frac{mN}{V} v_{\text{rms}}^2 \Rightarrow p \propto m \quad \dots(i)$$

(When volume and temperature are constants)

From Eq. (i),

Pressure first increases and then becomes constant.

- 3** (c) $pV = nRT \Rightarrow p \propto nT$ ($\because V$ and R are constants)

$$\Rightarrow \frac{p_2}{p_1} = \frac{n_2}{n_1} \times \frac{T_2}{T_1}$$

$$\Rightarrow \frac{p_{\text{He}}}{p} = \frac{1}{1} \times \frac{2T}{T} \Rightarrow p_{\text{He}} = 2p$$

- 4** (c) $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$, $v_{\text{av}} = \sqrt{\frac{8RT}{\pi M}}$ and $v_p = \sqrt{\frac{2RT}{M}}$

$$\Rightarrow v_{\text{rms}} : v_{\text{av}} : v_p = \sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$$

$$\therefore v_{\text{rms}} > v_{\text{av}} > v_p$$

- 5** (d) As, $v_{\text{rms}} \propto \sqrt{T}$, so v_{rms} will become two times when temperature is made four times.

- 6** (a) $T_1 = 273 + 27 = 300 \text{ K} \Rightarrow T_2 = 273 + 327 = 600 \text{ K}$

$$T_2 = 2T_1 \text{ and } v_{\text{rms}} \propto \sqrt{T}$$

$$\Rightarrow \frac{v'_{\text{rms}}}{v_{\text{rms}}} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{T_1}{2T_1}}$$

$$\Rightarrow \frac{v'_{\text{rms}}}{v_{\text{rms}}} = \frac{1}{\sqrt{2}} \Rightarrow v'_{\text{rms}} = \frac{v_{\text{rms}}}{\sqrt{2}}$$

- 7** (a) $v_{\text{rms}} = \sqrt{\frac{(3v)^2 + (4v)^2 + (5v)^2}{3}} = \sqrt{\frac{50}{3}} v$

- 8** (b) $v_{\text{rms}} \propto \sqrt{T}$

root-mean-square velocity will become half, if temperature

becomes $(1/4)$ th or $\left(\frac{273}{4} - 273\right)^\circ \text{C}$ or -204.75°C .

- 9** (d) $v_{\text{rms}} = \sqrt{\frac{(1)^2 + (2)^2 + (3)^2 + (4)^2}{4}} = \sqrt{\frac{15}{2}} \text{ kms}^{-1}$

- 10** (c) $v_{\text{rms}} \propto \sqrt{\frac{T}{M}}$

T is doubled and M has become half. Therefore, v_{rms} will become two times.

- 11** (c) $v_{\text{rms}} \propto \sqrt{T}$. If temperature is doubled, rms speed will become $\sqrt{2}$ times.

$$\therefore v_{\text{rms}} = 200 \sqrt{2} \text{ ms}^{-1}$$

- 12** (b) From the given velocities, the most probable velocity is 4 ms^{-1} .

- 13** (a) At a given temperature (T), all the ideal gas molecules no matter what their masses, have the same average translational kinetic energy.

$$\text{i.e. } E = \frac{3}{2} kT$$

So, E does not depend upon density.

($\because \text{Mass} = \rho V$)

$$\frac{E_1}{E_2} = 1 : 1$$

- 14** (d) As, $pV = nRT$

$$\Rightarrow p_{\text{H}_2} = \frac{m}{M_{\text{H}_2}} \cdot \frac{RT}{V} \text{ and } p_{\text{He}} = \frac{m}{M_{\text{He}}} \cdot \frac{RT}{V}$$

$$\therefore \frac{p_{\text{H}_2}}{p_{\text{He}}} = \frac{M_{\text{He}}}{M_{\text{H}_2}} = \frac{4 \times 10^{-3}}{2 \times 10^{-3}} = 2 \Rightarrow p_{\text{H}_2} = 2p_{\text{He}}$$

- 15** (c) The kinetic energy of gas is given as

$$K = \frac{1}{2} m v_{\text{rms}}^2$$

$$\text{Also, } v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\text{Now, } v_{\text{rms}} \propto \sqrt{T} \Rightarrow \frac{K_2}{K_1} = \left[\frac{(v_{\text{rms}})_2}{(v_{\text{rms}})_1} \right]^2 = \left[\frac{\sqrt{T_2}}{\sqrt{T_1}} \right]^2$$

$$\Rightarrow K_2 = K_1 \left(\frac{T_2}{T_1} \right) \Rightarrow K_2 = K_1 \left(\frac{600}{300} \right)$$

$$\therefore K_2 = 2K_1$$

- 16** (a) The gases carbon monoxide (CO) and nitrogen (N_2) have equal translational kinetic energy as both are diatomic gases, *i.e.*

$$E_1 = E_2$$

- 17** (c) Kinetic energy of an ideal gas depends only on its temperature. Hence, it remains constant whether its pressure is increased or decreased.

- 18** (c) The kinetic energy of gas molecules is the internal property, which is independent of the motion of frame. The temperature depends on the kinetic energy, so it will remain unchanged, when the container is kept in a moving train.

- 19** (c) For air at STP, $n = 27 \times 10^{25} \text{ m}^{-3}$

$$\text{and } d = 2 \times 10^{-10} \text{ m}$$

$$\text{Mean free path, } \lambda = \frac{1}{\sqrt{2} n \pi d^2}$$

On putting values, we get

$$\lambda = \frac{1}{\sqrt{2} \times 27 \times 10^{25} \times 3.14 \times (2 \times 10^{-10})^2}$$

$$= 2.08 \times 10^{-7} \text{ m}$$

CHECK POINT 14.4

- 1 (c) According to Boyle's law, $p_1 V_1 = p_2 V_2$

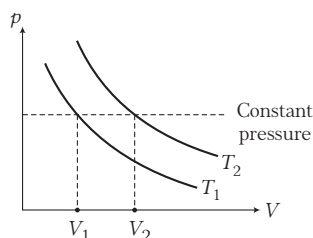
As, the pressure is decreased by 20%, therefore

$$p_2 = \frac{80}{100} p_1 \Rightarrow p_1 V_1 = \frac{80}{100} p_1 V_2 \Rightarrow V_1 = \frac{80}{100} V_2$$

$\therefore$ Percentage increase in volume

$$= \frac{V_2 - V_1}{V_1} \times 100 = \frac{100 - 80}{80} \times 100 = 25\%$$

- 2 (c) For a given pressure, volume will be more, if temperature is more (Charles' law).



From the graph, it is clear that $V_2 > V_1 \Rightarrow T_2 > T_1$

- 3 (d) For perfect gas, $V \propto T$

$$\therefore \frac{V_1}{V_2} = \frac{T_1}{T_2} \quad \dots(i)$$

According to the question,

$$V_1 = V, \text{ then } V_2 = 2V \text{ and } T_1 = 300 \text{ K}$$

$$\therefore \frac{1}{2} = \frac{300}{T_2} \Rightarrow T_2 = 600 \text{ K} \Rightarrow T_2 = 327^\circ \text{C}$$

$$\text{So, } \Delta T = 327 - 27 = 300^\circ \text{C}$$

- 4 (b) From Charles' law, $\frac{V_1}{V_2} = \frac{T_1}{T_2}$

$$\Rightarrow V_2 = \frac{V_1 T_2}{T_1}$$

$$\therefore V_2 = 1 \times \frac{(297 + 273)}{(27 + 273)} = \frac{570}{300} = 1.9 \text{ L}$$

- 5 (c) V - T graph is a straight line passing through origin. Hence, process is isobaric.

$$V = \left(\frac{nR}{p} \right) T \Rightarrow \text{Slope} = \frac{nR}{p}$$

Slope of 2 > Slope of 1

$$\therefore p_2 < p_1$$

- 6 (c) $p = p_1 + p_2 + p_3 = \left(\frac{nRT}{V} \right)_1 + \left(\frac{nRT}{V} \right)_2 + \left(\frac{nRT}{V} \right)_3$
- $$= (n_1 + n_2 + n_3) \frac{RT}{V}$$
- $$= \frac{(0.5 + 0.5 + 0.25)(8.31)(300)}{5 \times 10^{-3}}$$
- $$= 6.23 \times 10^5 \text{ Nm}^{-2} \approx 6 \times 10^5 \text{ Nm}^{-2}$$

- 7 (b) $f = 3N - K$

For monoatomic gas, $N = 1, K = 0$

$$\therefore f = 3 \times 1 - 0 = 3$$

where, N = number of particles in the system

and K = number of independent relations between the particles.

- 8 (c) $f = 3N - K$

For non-linear triatomic gas, $f = 3 \times 3 - 3 = 6$

- 9 (c) Average translational kinetic energy per molecule is $(3/2)kT$ which depends only on temperature and not on molecular mass.

- 10 (c) Translational degrees of freedom for any type of gas is three.

$$\therefore \text{Total translational kinetic energy} = 3 \left(\frac{1}{2} nRT \right) = \frac{3}{2} pV$$

- 11 (a) Both O_2 and N_2 are diatomic gases. Rotational degree of freedom in both cases is two. Therefore, average rotational kinetic energy per molecule for each of them is $2 \left(\frac{1}{2} k_B T \right)$ or the ratio is 1 : 1.

- 12 (b) In monoatomic gases, only translational kinetic energy is present, which is proportional to temperature and is given by $(3/2)k_B T$.

$$\therefore \text{Ratio of average kinetic energy} = \frac{300}{350} = \frac{6}{7}$$

- 13 (c) $E = \frac{3}{2} kT \Rightarrow E \propto T$

$$\therefore \frac{E_2}{E_1} = \frac{T_2}{T_1} = \frac{600}{300} = 2 \Rightarrow E_2 = 2E_1$$

- 14 (c) As, $pV = \mu RT = \frac{m}{M} RT \Rightarrow 20 \times V = \frac{m}{M} \times R \times 300 \quad \dots(i)$

When $m = m/2$ and $p = p'$, then

$$\text{As, } p' \times V = \left(\frac{m/2}{M} \right) R \times 350 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$p' = \frac{140}{12} \approx 11.7 \text{ atm}$$

- 15 (c) Mean kinetic energy of gas molecules,

$$E = \frac{f}{2} k_B T = \frac{f}{2} k_B (t + 273) = \left(\frac{f}{2} k_B \right) t + \frac{f}{2} \times 273 k_B$$

Comparing it with standard equation of straight line

$$y = mx + c, \text{ we get}$$

$$m = \frac{f}{2} k \text{ and } c = \frac{f}{2} 273 k. \text{ So, the graph between } E \text{ and } t \text{ will}$$

be straight line with positive intercept on E -axis and positive slope with t -axis.

- 16 (b) As, $\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$ or $V_2 = \frac{p_1 V_1 T_2}{p_2 T_1}$

$$\therefore \text{Volume of He, } V_2 = \frac{1 \times 500 \times (273 - 3)}{0.5 \times (273 + 27)}$$

$$= \frac{1 \times 500 \times 270}{0.5 \times 300} = 900 \text{ m}^3$$

- 17 (a) As, $pV = nRT = \frac{m}{M} RT$

where, M = molecular weight.

$$\Rightarrow \frac{p}{m/V} = \frac{RT}{M} \Rightarrow \frac{p}{\rho} = \frac{RT}{M} \Rightarrow \frac{p}{\rho} \propto T$$

($\because$ for a given gas, M is constant)

Temperature is directly proportional to the slope of p - ρ graph, so $T_1 > T_2$.

18 (c) As $\theta_2 > \theta_1 \Rightarrow \tan \theta_2 > \tan \theta_1$

$$\Rightarrow V_2 > V_1$$

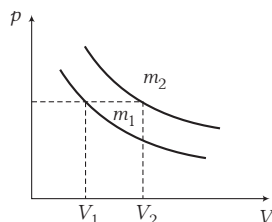
19 (b) $\frac{V}{T} = \frac{nR}{p}$

V and T for both cases are same.

Hence, $\frac{n_1}{p_1} = \frac{n_2}{p_2}$ or $\frac{m_1}{p_1 M_1} = \frac{m_2}{p_2 M_2}$

or $m_2 = \frac{p_2 M_2}{p_1 M_1} \cdot m_1 = \frac{(3)(2)}{(5)(28)} \cdot 20 = 0.86 \text{ kg}$

20 (c) $pV = \mu RT = \frac{m}{M} RT$



For 1st plot, $p = \frac{m_1}{M} \frac{RT}{V_1}$... (i)

For 2nd plot, $p = \frac{m_2}{M} \frac{RT}{V_2}$... (ii)

From Eqs. (i) and (ii), we get

$$\frac{m_1}{m_2} = \frac{V_1}{V_2} \Rightarrow m \propto V$$

As, $V_2 > V_1 \Rightarrow m_1 < m_2$

21 (d) $p^2 V = \text{constant}$

$$p = \frac{nRT}{V} \text{ or } p = \frac{CT}{V} \quad (C = \text{constant})$$

$$\therefore \frac{T^2}{V} = \text{constant} \text{ or } T \propto \sqrt{V}$$

V is increased to 3 times. Hence, T will increase $\sqrt{3}$ times.

(A) Taking it together

1 (c) $\frac{F - 32}{9} = \frac{K - 273}{5} \Rightarrow \frac{F - 32}{9} = \frac{0 - 273}{5}$

$$F = -459.4^\circ \text{F} \approx -460^\circ \text{F}$$

2 (d) Zero kelvin = -273°C (absolute temperature). As no matter can attain this temperature, hence temperature can never be negative on Kelvin scale.

3 (b) Boyle's law is applicable when temperature is constant.
i.e. $pV = nRT = \text{constant}$

$$\Rightarrow pV = \text{constant} \quad (\text{At constant temperature})$$

i.e. $p \propto \frac{1}{V}$ (where, p = pressure and V = volume)

So, this process can be called as isothermal process.

4 (a) We know that as temperature increases, vibration of molecules about their mean position increases. Hence, kinetic energy associated with random motion of molecules increases.

5 (b) $\frac{\Delta T_C}{100} = \frac{\Delta T_F}{180} = \frac{212 - 140}{180}$

i.e. $\Delta T_C = 100 \times \frac{72}{180} = 40^\circ \text{C}$

$\therefore$ Fall in temperature = 40°

6 (a) $t = \frac{(p_t - p_0)}{(p_{100} - p_0)} \times 100^\circ \text{C} = \frac{(60 - 50)}{(90 - 50)} \times 100^\circ \text{C} = 25^\circ \text{C}$

7 (d) $\frac{V_1}{V_2} = \frac{1 + \gamma T_1}{1 + \gamma T_2} \Rightarrow \frac{100}{125} = \frac{1 + \gamma \times 20}{1 + \gamma \times 100}$

$$\Rightarrow \gamma = 0.0033 / ^\circ \text{C}$$

8 (a) $\rho = \rho_0(1 - \gamma \Delta \theta) = 13.6[1 - 0.18 \times 10^{-3}(473 - 273)]$

$$= 13.6[1 - 0.036] = 13.11 \text{ g/cc}$$

9 (a) $p \propto T \Rightarrow \frac{p_1}{p_2} = \frac{T_1}{T_2} \Rightarrow \frac{p_2 - p_1}{p_1} = \frac{T_2 - T_1}{T_1}$

$$\Rightarrow \left(\frac{\Delta p}{p} \right) \% = \left(\frac{251 - 250}{250} \right) \times 100 = 0.4\%$$

10 (c) Gas equation for N molecules $pV = NkT$

$$\Rightarrow N = \frac{pV}{kT} = \frac{h\rho gV}{kT} = \frac{1.2 \times 10^{-10} \times 13.6 \times 10^3 \times 10 \times 10^{-4}}{1.38 \times 10^{-23} \times 300}$$

$$= 3.94 \times 10^{11}$$

11 (a) $v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \Rightarrow T \propto M$ ($\because v_{\text{rms}}$ and R are constants)

$$\Rightarrow \frac{T_{\text{H}_2}}{T_{\text{O}_2}} = \frac{M_{\text{H}_2}}{M_{\text{O}_2}} \Rightarrow \frac{T_{\text{H}_2}}{(273 + 47)} = \frac{2}{32} \Rightarrow T_{\text{H}_2} = 20 \text{ K}$$

12 (a) $v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \Rightarrow v_{\text{rms}} \propto \sqrt{\frac{T}{M}}$

$$\frac{v_{\text{He}}}{v_{\text{H}_2}} = \frac{5}{7} = \sqrt{\frac{T_{\text{He}}}{M_{\text{He}}} \times \frac{M_{\text{H}_2}}{T_{\text{H}_2}}}$$

$$\Rightarrow T_{\text{He}} = \frac{25}{49} \times \frac{4}{2} \times 273 = 278.57 \text{ K} \approx 5.57^\circ \text{C}$$

13 (a) $v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \Rightarrow M = \frac{3RT}{v_{\text{rms}}^2}$

At room temperature, $T = 27^\circ \text{C} = 300 \text{ K}$

$$\therefore M = \frac{3 \times 8.3 \times 300}{(1930)^2} \approx 2 \times 10^{-3} \text{ kg} = 2 \text{ g}$$

$\Rightarrow$ Gas is hydrogen.

14 (d) Using $\frac{T_C - 0}{100 - 0} = \frac{T - 10}{130 - 10}$

$$\Rightarrow \frac{40 - 0}{100 - 0} = \frac{T - 10}{130 - 10}$$

$$\Rightarrow \frac{40}{100} = \frac{T-10}{120} \text{ or } \frac{2}{5} = \frac{T-10}{120}$$

$$\Rightarrow 5T - 50 = 240$$

$$\Rightarrow 5T = 290 \Rightarrow T = \frac{290}{5} = 58^\circ$$

- 15 (a)** Number of seconds lost in a day, $\Delta t = \frac{1}{2} \alpha \Delta \theta \times 86400$

The coefficient of linear expansion of metal pendulum,

$$\alpha = \frac{2\Delta t}{\Delta \theta \times 86400} = \frac{2 \times 12.5}{25 \times 86400}$$

$$\Rightarrow \alpha = \frac{1}{86400} / ^\circ \text{C}$$

- 16 (b)** Relation between Celsius and Fahrenheit scale of temperature is $\frac{C}{5} = \frac{F-32}{9}$

$$\Rightarrow C = \frac{5}{9} F - \frac{160}{9}$$

Equating above equation with standard equation of the line

$$y = mx + c, \text{ we get}$$

$$\text{Slope of the line } AB \text{ is } m = \frac{5}{9}.$$

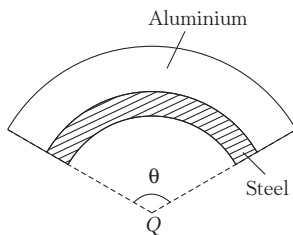
- 17 (a)** $pV = nRT$ or $n = \frac{pV}{RT}$

$$\text{Now, } N = n \times 6.02 \times 10^{23} = \frac{pV}{RT} \times 6.02 \times 10^{23}$$

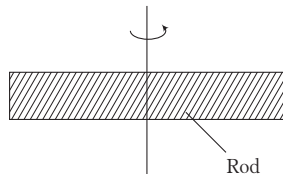
$$\text{or } N = \frac{(10^{-14} \times 13.6 \times 10^3 \times 10) (1)}{(8.31) (273 + 30)} \times 6.02 \times 10^{23}$$

$$= 3.2 \times 10^{11}$$

- 18 (d)** As $\alpha_{\text{Al}} > \alpha_{\text{steel}}$, aluminium will expand more. So, it should have larger radius of curvature. Hence, aluminium will be on convex side and steel on concave side.



- 19 (b)** As the rod is heated, it expands. No external torque is acting on the system, so angular momentum should be conserved.



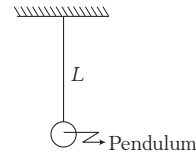
$$\text{Angular momentum, } L = I\omega = \text{constant} \Rightarrow I_1\omega_1 = I_2\omega_2$$

Due to expansion of the rod, $I_2 > I_1$

$$\Rightarrow \frac{\omega_2}{\omega_1} = \frac{I_1}{I_2} < 1 \Rightarrow \omega_2 < \omega_1$$

So, angular velocity (speed of rotation) decreases.

- 20 (a)** As the temperature is increased, length of the pendulum increases but its centre of mass remains at the centre of bob. The time period of pendulum,



$$T = 2\pi \sqrt{\frac{L}{g}} \Rightarrow T \propto \sqrt{L}$$

So, as L increases, time period (T) will also increase.

- 21 (b)** As the motion of the vessel as a whole does not affect the relative motion of the gas molecules with respect to the walls of the vessel, hence pressure of the gas inside the vessel, as observed by us, on the ground remains the same.

- 22 (a)** Let the relation between the thermometer reading and centigrade scale be $y = ax + b$.

Given, at $x = 100$, $y = 80$ and at $x = 0$, $y = 10$

$$\therefore 80 = 100a + b, 10 = b \Rightarrow a = 0.7$$

Now, we have to find x when $y = 59$

$$59 = 0.7x + 10 \Rightarrow x = 70$$

$\therefore$ At 70°C , the reading of given thermometer be 59° .

- 23 (c)** Let the original temperature be 0°C .

Volume of A , $V_1 = l \times \pi(2r)^2$; $V_1' = V_1(1 + \gamma \Delta T)$

$$\frac{(V_1' - V_1)}{V_1} = \gamma \Delta T \Rightarrow (V_1' - V_1) \propto V_1$$

$$\text{Similarly, } \frac{(V_2' - V_2)}{V_2} = \gamma \Delta T \Rightarrow (V_2' - V_2) \propto V_2$$

$$\therefore \frac{\Delta V_1}{\Delta V_2} = \frac{l(2r)^2}{2lr^2} = \frac{2}{1}$$

$$\mathbf{24 (c)} \quad V = \left(\frac{nR}{p} \right) T = \left(\frac{mR}{Mp} \right) T$$

$$\therefore \text{Slope of } V-T \text{ graph } \propto \frac{m}{p}$$

Since, ratio $\frac{m}{p}$ is same in both cases. So, the expansion will be represented by the same straight line B .

- 25 (c)** From ideal gas equation, $pV = RT$... (i)

$$\text{or } p\Delta V = R\Delta T \quad \dots \text{(ii)}$$

On dividing Eq. (ii) by Eq. (i), we get

$$\frac{\Delta V}{V} = \frac{\Delta T}{T} \Rightarrow \frac{\Delta V}{V \Delta T} = \frac{1}{T} = \delta \quad (\text{Given})$$

$$\therefore \delta = \frac{1}{T}$$

So, the graph between δ and T will be rectangular hyperbola.

- 26 (d)** $U \propto \frac{1}{\rho}$ or $T \propto \frac{1}{(1/V)} \Rightarrow T \propto V$

Hence, from ideal gas equation ($pV = nRT$), p is constant.

Further, ρ is increasing in U - ρ , therefore V should decrease.

- 27** (a) We know for an ideal gas,

$$pV = nRT \Rightarrow V = \left(\frac{nR}{p} \right) T$$

where, p = pressure, V = volume,

n = number of moles of gases

R = gas constant

and T = temperature.

Slope of the V - T graph, $m = \frac{dV}{dT} = \frac{nR}{p}$

$$\Rightarrow m \propto \frac{1}{p} \quad (\because nR = \text{constant})$$

$$\Rightarrow p \propto \frac{1}{m}$$

Hence, $\frac{p_1}{p_2} = \frac{m_2}{m_1} > 1$

where, m_1 is slope of the graph corresponding to p_1 and m_2 is slope corresponding to p_2 .

$$\Rightarrow p_2 < p_1 \text{ or } p_1 > p_2$$

- 28** (d) Let the radius of the sphere be R . As the temperature increases, radius of the sphere increases as shown in figure.

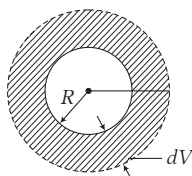
Original volume, $V = \frac{4}{3} \pi R^3$

Coefficient of linear expansion = α

$\therefore$ Coefficient of volume expansion = 3α

$$\therefore \frac{1}{V} \frac{dV}{dT} = 3\alpha \Rightarrow dV = 3V\alpha dT \approx 4\pi R^3 \alpha dT$$

= Increase in the volume



- 29** (a) Let volume of the sphere be V and ρ be its density, then we can write buoyant force as

$$F = V \rho g \quad (g = \text{acceleration due to gravity})$$

$$\Rightarrow F \propto \rho \quad (\because V \text{ and } g \text{ both are constants})$$

$$\Rightarrow \frac{F_{4^\circ\text{C}}}{F_{0^\circ\text{C}}} = \frac{\rho_{4^\circ\text{C}}}{\rho_{0^\circ\text{C}}} > 1 \quad (\because \rho_{4^\circ\text{C}} > \rho_{0^\circ\text{C}})$$

Hence, buoyancy will be less in water at 0°C than that in water at 4°C .

- 30** (d) Diameter of brass plate = 2.6 cm

and diameter of steel plate = $(2.6 - 0.01) \text{ cm} = 2.59 \text{ cm}$

Now,

$$(d + \Delta d)_b = (d + \Delta d)_s$$

$$\text{or } d_b (1 + \alpha \Delta \theta)_b = d_s (1 + \alpha \Delta \theta)_s$$

$$\therefore 2.6 (1 + 19 \times 10^{-6} \Delta \theta) = 2.59 (1 + 11 \times 10^{-6} \Delta \theta)$$

$$\Delta \theta \approx -478^\circ\text{C} \text{ or } \theta_f = -478^\circ + 20^\circ = -458^\circ\text{C}$$

- 31** (b) $\gamma_r = \gamma_a + \gamma_v$, where γ_r = coefficient of real expansion,

γ_a = coefficient of apparent expansion

and γ_v = coefficient of expansion of vessel.

For copper, $\gamma_r = C + 3\alpha_{\text{Cu}} = C + 3A$

For silver, $\gamma_r = S + 3\alpha_{\text{Ag}}$

$$\Rightarrow C + 3A = S + 3\alpha_{\text{Ag}}$$

$$\Rightarrow \alpha_{\text{Ag}} = \frac{C - S + 3A}{3}$$

- 32** (b) $(n_1 + n_2)_i = (n_1 + n_2)_f$

$$\frac{p_0 V}{RT_0} + \frac{p_0 V}{RT_0} = \frac{pV}{RT_0} + \frac{pV}{2RT_0}$$

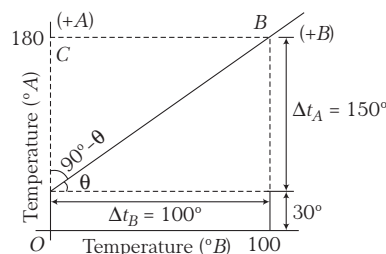
$$\therefore p = (4/3) p_0$$

- 33** (b) It is clear from the graph that lowest point for scale A is 30° and lowest point for scale B is 0° . Highest point for the scale A is 180° and for scale B is 100° . Hence, correct relation is

$$\frac{t_A - (\text{LFP})_A}{(\text{UFP})_A - (\text{LFP})_A} = \frac{t_B - (\text{LFP})_B}{(\text{UFP})_B - (\text{LFP})_B}$$

where, LFP = lower fixed point

and UFP = upper fixed point.



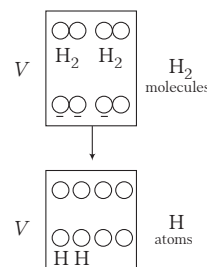
$$\Rightarrow \frac{t_A - 30}{180 - 30} = \frac{t_B - 0}{100 - 0} \Rightarrow \frac{t_A - 30}{150} = \frac{t_B}{100}$$

- 34** (d) Consider the diagram, when the molecules breaks into atoms, the number of moles would become twice.

Now, from ideal gas equation,

$$pV = nRT$$

Here, p = pressure of gas, n = number of moles, R = gas constant, T = temperature and V = volume of the container.



As, number of moles becomes twice of initial, so $n_2 = 2n_1$.

So,

$$p \propto nT$$

$$\Rightarrow \frac{p_2}{p_1} = \frac{n_2 T_2}{n_1 T_1} = \frac{(2n_1)(3000)}{n_1(300)} = 20$$

$$\Rightarrow p_2 = 20 p_1$$

Hence, final pressure of the gas would be 20 times the pressure initially.

- 35** (a) From ideal gas equation, $pV = \mu RT$

$$\Rightarrow \text{Slope of } p\text{-}T \text{ curve, } \frac{p}{T} = \frac{\mu R}{V} \Rightarrow \text{Slope} \propto \frac{1}{V}$$

It means line of smaller slope represents greater volume of gas. In the given problems 1 and 2 are on the same line, so they will represent same volume, i.e. $V_1 = V_2$.

Similarly, points 3 and 4 are on the same line, so they will represent same volume, i.e. $V_3 = V_4$.

Also, slope of line 1-2 is less than that of line 3-4.

Hence, $(V_1 = V_2) > (V_3 = V_4)$

36 (d) For an ideal gas, $pV = nRT$

Here, n = number of moles, p = pressure, V = volume,

R = gas constant, T = temperature $\Rightarrow n = \frac{pV}{RT}$

As number of moles of the gas remains fixed, hence we can write

$$\begin{aligned} \frac{p_1 V_1}{RT_1} &= \frac{p_2 V_2}{RT_2} \Rightarrow p_2 = (p_1 V_1) \left(\frac{T_2}{V_2 T_1} \right) \\ &= \frac{(p)(V)(1.1T)}{(1.05V)(T)} \left[\text{Given, } p_1 = p, V_2 = 1.05V \right] \\ &\quad \left[\text{and } T_2 = 1.1T \right] \\ &= p \times \left(\frac{1.1}{1.05} \right) = p(1.0476) \approx 1.05p \end{aligned}$$

Hence, final pressure p_2 lies between p and $1.1p$.

37 (d) Pressure on both sides should be same.

$$\begin{aligned} \Rightarrow \frac{n_1 RT}{V_1} &= \frac{n_2 RT}{V_2} \Rightarrow \frac{m}{M V_1} = \frac{m}{M V_2} \\ \Rightarrow \frac{m}{32(360^\circ - \alpha)V} &= \frac{m}{28\alpha V} \\ \Rightarrow 28\alpha &= 32(360^\circ - \alpha) \Rightarrow \alpha = \frac{32 \times 360}{60} \text{ or } \alpha = 192^\circ \end{aligned}$$

38 (d) In an ideal gas, when a molecule collides elastically with a wall, the momentum or pressure transferred to each molecule will be twice the magnitude of its normal momentum. As face $EFGH$ absorbs the molecule incident on it, so for the face $EFGH$, it transfers only half of that on other faces.

39 (c) $pV = \frac{m}{M} RT \Rightarrow V \propto mT \Rightarrow \frac{V_1}{V_2} = \frac{m_1}{m_2} \cdot \frac{T_1}{T_2}$

$$\Rightarrow \frac{2V}{V} = \frac{m}{m_2} \times \frac{100}{200} \Rightarrow m_2 = \frac{m}{4}$$

40 (a) Slope of line is 1. Therefore, T - V equation can be written as $T = V + T_0$

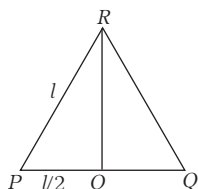
$$\therefore \frac{pV}{nR} = V + T_0 \Rightarrow p = (nR) + \frac{nRT_0}{V}$$

$\therefore p$ versus V graph is a sort of rectangular hyperbola.

41 (a) F = rate of change of momentum $= \frac{\Delta p}{\Delta t} = (nm)(2v \cos \theta)$

$$\begin{aligned} \text{Pressure, } p &= \frac{F}{A} = \frac{(nm)(2v \cos \theta)}{A} \\ &= \frac{10^{23} \times 3.32 \times 10^{-27} \times 2 \times 1000 \times \cos 45^\circ}{2 \times 10^{-4}} \\ &= 2.34 \times 10^3 \text{ Nm}^{-2} \text{ or Pa} \end{aligned}$$

42 (d) $(OR)^2 = (PR)^2 - (PO)^2 = l^2 - \left(\frac{l}{2}\right)^2$



$$\Rightarrow (OR)^2 = [l(1 + \alpha_2 t)]^2 - \left[\frac{l}{2}(1 + \alpha_1 t) \right]^2$$

$$l^2 - \frac{l^2}{4} = l^2(1 + \alpha_2^2 t^2 + 2\alpha_2 t) - \frac{l^2}{4}(1 + \alpha_1^2 t^2 + 2\alpha_1 t)$$

Neglecting $\alpha_2^2 t^2$ and $\alpha_1^2 t^2$, we get

$$0 = l^2(2\alpha_2 t) - \frac{l^2}{4}(2\alpha_1 t)$$

$$\Rightarrow 2\alpha_2 = \frac{2\alpha_1}{4}$$

$$\Rightarrow \alpha_1 = 4\alpha_2$$

43 (b) Change in weight = Upthrust (F)

where, $F = V_s \rho_l g$

$\Rightarrow F' = V'_s \rho'_l g$

$$\therefore \frac{F'}{F} = \frac{V'_s}{V_s} \cdot \frac{\rho'_l}{\rho_l}$$

$$\text{or } \frac{F'}{F} = (1 + \gamma_s \Delta \theta) \frac{\rho'_l}{\rho_l}$$

$$\left(\frac{46 - 30.5}{46 - 30} \right) = (1 + 3 \times \alpha_s \times 15) \left(\frac{1.20}{1.24} \right)$$

$$\text{or } \alpha_s = 2.3 \times 10^{-5} / ^\circ \text{C}$$

44 (a) $(p_2 - p_1)A = mg$

$$\text{or } \frac{mg}{A} = \frac{RT_i}{V_1} - \frac{RT_i}{4V_1} = \frac{3RT_i}{4V_1} \quad \dots(i)$$

Similarly, in second case,

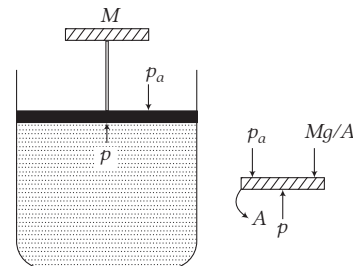
$$\frac{mg}{A} = \frac{RT_f}{V_2} - \frac{RT_f}{3V_2} = \frac{2RT_f}{3V_2} \quad \dots(ii)$$

Further, $5V_1 = 4V_2$

Equating Eqs. (i) and (ii), we get

$$\begin{aligned} \Rightarrow \frac{3T_i}{4V_1} &= \frac{2T_f}{3V_2} \\ T_f &= \frac{9}{8} \times \frac{V_2}{V_1} \times T_i \\ &= \frac{9}{8} \times \frac{5}{4} \times 320 \\ &= 450 \text{ K} \end{aligned}$$

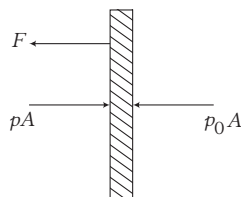
45 (c) Consider the diagram, where an ideal gas is contained in a cylinder, having a piston of mass M . Friction is absent.



The pressure inside the gas will be $p = p_a + Mg/A$ where, p_a = atmospheric pressure,

A = area of cross-section of the piston
 and Mg = weight of piston.
 Hence, p = constant
 As, $pV = nRT$
 $\Rightarrow$ When temperature increases, volume (V) increases at constant pressure.

46 (b) Volume of the gas V is constant.



$\therefore p \propto T$, i.e. pressure will be doubled when temperature is doubled.

$\therefore p = 2p_0$

Now, let F be the tension in the wire. Then, equilibrium of any one piston will give

$$\begin{aligned}
 F &= (p - p_0) A \\
 &= (2p_0 - p_0) A \\
 &= p_0 A
 \end{aligned}$$

(B) Medical entrance special format questions

• Assertion and reason

1 (d) In summers, a metallic scale will read less than the actual because length of metallic scale increases in summers.

2 (b) The translational kinetic energy is $\frac{1}{2}mv_{\text{rms}}^2$.

Further, translational degree of freedom of any type of gas is three. Therefore, translational kinetic energy of any type of gas (of one mole) is $\frac{3}{2}RT$.

3 (c) In ideal gases, there is no inter-particle interaction. So, no vibrational energy is there.

$\therefore$ Only the kinetic energy contribute to the internal energy.

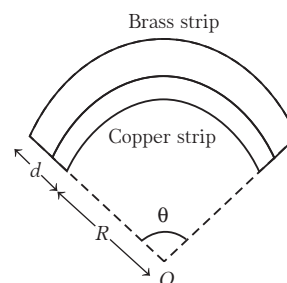
4 (b) Translational degree of freedom of a diatomic gas (O_2) is three and rotational degree of freedom is two. Therefore, according to law of equipartition of energy, $\frac{K_T}{K_R} = \frac{3}{2}$.

Further, intermolecular force between ideal gas molecules is zero. Hence, potential energy is zero.

• Statement based questions

1 (b) All statements are correct except the statement given in option (b) and it can be corrected as,
 The average translational kinetic energy per molecule is same for molecules of all gases like here O_2 and for each molecule, it is $\frac{3}{2}k_B T$.

2 (b) Let L_0 be the initial length of each strip before heating.



Length after heating will be

$$L_b = L_0(1 + \alpha_b \Delta T) = (R + d)\theta \quad \dots (i)$$

($\because$ Length of an arc = Radius $\times$ Angle)

$$L_c = L_0(1 + \alpha_c \Delta T) = R\theta \quad \dots (ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\Rightarrow \frac{R + d}{R} = \frac{1 + \alpha_b \Delta T}{1 + \alpha_c \Delta T} \Rightarrow 1 + \frac{d}{R} = 1 + (\alpha_b - \alpha_c) \Delta T$$

$$\Rightarrow R = \frac{d}{(\alpha_b - \alpha_c) \Delta T}$$

$$\Rightarrow R \propto \frac{1}{\Delta T}$$

$$\text{and } R \propto \frac{1}{(\alpha_b - \alpha_c)}$$

i.e. R is inversely proportional to ΔT and $(\alpha_b - \alpha_c)$.

3 (c) An ideal gas is that which satisfy the equation,

$$pV = \mu RT$$

where, p = pressure, V = volume,

μ = number of moles, R = gas constant

and T = temperature.

Here, μ and R are constants for a gas, so

$$p \propto \frac{T}{V}$$

Hence, pressure of an ideal gas depends on both the temperature and volume of the gas. Thus, both the statements are correct.

4 (c) Statements II and III are correct but statement I is incorrect and it can be corrected as,

The kinetic theory relates the molecular properties like viscosity, conduction and diffusion with bulk properties of a gas.

• Match the columns

$$\mathbf{1 (b)} \quad T^2 \propto \frac{1}{V} \text{ or } T^2 \propto \frac{1}{1/p} \text{ or } T \propto \sqrt{p}$$

p has become four times. Therefore, T will become two times or V will become $\frac{1}{4}$ times.

Further, T is increasing, hence ΔU will be positive.

Hence, $A \rightarrow r$, $B \rightarrow s$, $C \rightarrow p$.

- 2 (c) From the given V - T graph, we cannot tell the nature of gas.

$$V = \frac{nRT}{p} = \left(\frac{nR}{p} \right) T$$

If p is constant, V - T graph is a straight line. Slope of this line is $\frac{nR}{p}$.

Here, $(\text{Slope})_A > (\text{Slope})_B$

$$\Rightarrow \left(\frac{nR}{p} \right)_A > \left(\frac{nR}{p} \right)_B$$

So, either $\frac{n_A}{n_B}$ may be greater than 1 or $\frac{p_A}{p_B}$ may be less than 1.

But it is certain that $\left(\frac{n}{p} \right)_A > \left(\frac{n}{p} \right)_B$.

Hence, $A \rightarrow t$, $B \rightarrow t$, $C \rightarrow t$, $D \rightarrow t$.

(C) Medical entrances' gallery

- 1 (a) The average thermal energy of a system with degree of freedom f is equal to its average energy, which is given as

$$= \frac{f}{2} k_B \cdot T$$

For monoatomic gas, $f = 3$

$$\therefore \text{Average thermal energy} = \frac{3}{2} k_B \cdot T$$

- 2 (a) Given, pressure, $p = 249 \text{ kPa} = 249 \times 10^3 \text{ Pa}$

Temperature, $T = 27^\circ \text{C} = 273 + 27 = 300 \text{ K}$

Density, $\rho = ?$

From ideal gas equation, $pV = nRT$

$$\Rightarrow pV = \frac{m}{M} RT \quad \left(\because n = \frac{m}{M} \right)$$

$$\Rightarrow pVM = mRT$$

$$\Rightarrow pM = \frac{m}{V} RT = \rho RT \quad \left(\because \frac{m}{V} = \rho \right)$$

$$\Rightarrow \rho = \frac{pM}{RT} = \frac{249 \times 10^3 \times 2 \times 10^{-3}}{8.3 \times 300}$$

($\because$ For hydrogen gas, $M = 2 \text{ g} = 2 \times 10^{-3} \text{ kg}$)

$$\Rightarrow \rho = 0.2 \text{ kg/m}^3$$

- 4 (a) Ideal gas equation is given as

$$p = \frac{\rho RT}{M_0} \Rightarrow p \cdot \frac{M_0}{\rho} = RT$$

$$\Rightarrow pV = RT \quad \left(\because V = \frac{M_0}{\rho} \right)$$

Here, ρ and M_0 are mass density and mass of the gas, respectively.

- 5 (a) The mean free path l for a gas molecule is given as

$$l = \frac{1}{\sqrt{2} \pi n d^2} \Rightarrow l \propto \frac{1}{d^2}$$

where, d = diameter of molecule of gas.

- 6 (c) Due to change in temperature, the thermal strain produced in a rod of length L is given by

$$\frac{\Delta L}{L} = \alpha \Delta T \Rightarrow \Delta L = L \alpha \Delta T$$

where, L = original length of rod and α = coefficient of linear expansion of solid rod.

As, the change in length (Δl) of the two given rods of copper and aluminium are independent of temperature change, i.e. ΔT is same for both copper and aluminium.

$$L_{\text{Cu}} \alpha_{\text{Cu}} = L_{\text{Al}} \alpha_{\text{Al}} \quad \dots (i)$$

Here, $\alpha_{\text{Cu}} = 17 \times 10^{-5} \text{ K}^{-1}$

$$\alpha_{\text{Al}} = 22 \times 10^{-5} \text{ K}^{-1}$$

and $L_{\text{Cu}} = 88 \text{ cm}$

Substituting the given values in Eq. (i), we get

$$L_{\text{Al}} = \frac{\alpha_{\text{Cu}} L_{\text{Cu}}}{\alpha_{\text{Al}}} = \frac{17 \times 10^{-5} \times 88}{22 \times 10^{-5}} \approx 68 \text{ cm}$$

- 7 (a) The kinetic energy of gas molecules is given by

$$\text{KE} = \frac{3}{2} RT$$

$$\Rightarrow \text{KE} \propto T$$

Thus, increase in temperature would lead to increase the kinetic energy of molecules.

Option (b) is incorrect as increase in temperature will lead to increase in pressure as $p \propto T$. Other options (c) and (d) are also incorrect as molecular distance increases while mass remains the same for increase in the temperature.

- 8 (a) Given, initial pressure of ideal gas,

$$p_1 = 1 \text{ bar} = 1 \text{ Nm}^{-2}$$

Initial volume, $V_1 = 30 \text{ m}^3$

Final volume, $V_2 = 10 \text{ m}^3$

Initial temperature, $T_1 = 320 \text{ K}$

Final temperature, $T_2 = 280 \text{ K}$

Final pressure of gas, $p_2 = ?$

From ideal gas equation,

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

$$\Rightarrow \frac{1 \times 30}{320} = \frac{p_2 \times 10}{280}$$

$$\Rightarrow p_2 = 2.625 \text{ bar}$$

- 9 (a) A gas can be liquified by applying pressure only when it is cooled below critical temperature. Critical temperature of NH_3 is more than CO_2 , i.e. $T_{\text{NH}_3} = 405 \text{ K}$ and $T_{\text{CO}_2} = 304.1 \text{ K}$.

Therefore, NH_3 is liquified more easily than CO_2 .

- 10 (b) There are two vibrational degree of freedom of diatomic gas molecules at high temperature. Hence, vibrational degree of freedom of a diatomic gas molecule appears at very high temperature because vibration in gas molecules is directly proportional to the square root of its temperature.

- 11 (d) Elongation due to change in temperature,

$$\Delta l = L \alpha \Delta T \quad \dots (i)$$

where, α = coefficient of linear expansion,

which is compensated by elastic strain.

When temperature becomes normal, then

$$\Delta l = \frac{TL}{YS} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{TL}{YS} = L\alpha\Delta T$$

$$\Rightarrow T = YS\alpha\Delta T$$

At equilibrium, force exerted by one-half on other,

$$F = 2T = 2YS\alpha\Delta T$$

- 12** (b) The minimum velocity with which the body must be projected vertically upwards, so that it could escape from the earth's atmosphere is its escape velocity (v_e).

$$\text{As, } v_e = \sqrt{2gR}$$

Substituting the value of g (9.8 ms^{-2}) and radius of earth ($R = 6.4 \times 10^6 \text{ m}$), we get

$$v_e = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} \\ \cong 11.2 \text{ km s}^{-1} = 11.2 \times 10^3 \text{ m s}^{-1}$$

Let the temperature of molecule be T when it attains v_e .

According to the question, $v_{\text{rms}} = v_e$

where, v_{rms} is the rms speed of the oxygen molecule.

$$\Rightarrow \sqrt{\frac{3k_B T}{m_{\text{O}_2}}} = 11.2 \times 10^3$$

$$\text{or } T = \frac{(11.2 \times 10^3)^2 (m_{\text{O}_2})}{(3k_B)}$$

Substituting the given values, *i.e.*

$$k_B = 1.38 \times 10^{-23} \text{ JK}^{-1}$$

$$\text{and } m_{\text{O}_2} = m = 2.76 \times 10^{-26} \text{ kg,}$$

$$\text{we get } T = \frac{(11.2 \times 10^3)^2 (2.76 \times 10^{-26})}{(3 \times 1.38 \times 10^{-23})} \cong 8.326 \times 10^4 \text{ K}$$

- 13** (b) Given, $\gamma = 49 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$,

$$\Delta T = 30^\circ\text{C}$$

$$V' = V + \Delta V = V(1 + \gamma \Delta T)$$

$$\therefore V' = V(1 + 49 \times 10^{-5} \times 30) = 1.0147V$$

$$\therefore \rho = \frac{m}{V} \text{ and } \rho' = \frac{m}{V'} = \frac{m}{1.0147V} = 0.9855\rho$$

Hence, fractional change in density

$$= \frac{\rho - \rho'}{\rho} = \frac{\rho - 0.9855\rho}{\rho} = 0.0145$$

- 14** (b) We know that, $pV = RT \Rightarrow V = \frac{RT}{p} \quad \dots(i)$

$$\therefore \text{Density, } \rho = \frac{M}{V} = \frac{mN_A}{RT} \quad [\text{From Eq. (i)}]$$

$$= \frac{mpN_A}{kN_A T} \quad (\because R = kN_A)$$

$$\Rightarrow \rho = \frac{pm}{kT}$$

- 15** (b) A monoatomic gas molecules like He consists of single atom. It can have translational motion in any direction in space.

Thus, it has three translational degrees of freedom $f = 3$. It can also rotate but due to its small moment of inertia, rotational kinetic energy is neglected.

The molecules of a diatomic gas (like O_2 , CO_2 , H_2) cannot only move but also rotate about any one of the three coordinates.

Hence, it can have two rotational degree of freedom.

Thus, a diatomic molecule has 5 degree of freedom, *i.e.* 3 translational and 2 rotational.

- 16** (d) Total internal energy of system

= Internal energy of oxygen molecules + Internal energy of argon molecules

$$= \frac{f_1}{2} n_1 RT + \frac{f_2}{2} n_2 RT \\ = \frac{5}{2} \times 2RT + \frac{3}{2} \times 4RT = 11RT$$

- 17** (a) As, $\alpha_{\text{brass}} = \frac{\gamma_{\text{brass}}}{3} = \frac{0.00006}{3}$

$$= 0.00002 = 2 \times 10^{-5} / ^\circ\text{C}$$

The brass scale is true at 15°C , therefore at 30° , its graduations will increase in length and hence, observed reading will be less than actual reading at 30° .

$\therefore$ The change in reading,

$$\Delta l = l\alpha_{\text{brass}}(\Delta T) = 74.5 \times 2 \times 10^{-5}(30 - 15) \\ = 0.02235 \text{ cm}$$

$\therefore$ Actual reading at 30°C ,

$$l_{30} = l_{\text{observed}} + \Delta l = 74.5 + 0.02235 = 74.522 \text{ cm}$$

Assuming area of cross-section to be constant, we have

$$V_0 \rho_0 = V_{30} \rho_{30}$$

$$\text{or } ah_0 \rho_0 = ah_{30} \rho_{30}$$

Therefore, true height at 0°C ,

$$h_0 = h_{30} \frac{\rho_{30}}{\rho_0} = \frac{h_{30}}{(1 + \gamma_{\text{Hg}} \Delta T)} \\ = \frac{74.522}{1 + 0.00018 \times 30} = \frac{74.522}{1.0054} = 74.122 \text{ cm}$$

- 18** (b) If we fill nitrogen gas at high pressure above the mercury level, the boiling point of mercury will get increased which can extend to the range upto 500°C .

- 19** (b) According to ideal gas equation, $pV = nRT$

$$\Rightarrow \frac{p}{T} = \frac{nR}{V}$$

where, $\frac{p}{T}$ represents slope of the graph.

As, the number of moles are same for the two gases.

$$\Rightarrow \frac{p}{T} \propto \frac{1}{V}$$

$$\therefore V_2 > V_1$$

$$\therefore (\text{Slope})_2 < (\text{Slope})_1 \text{ or } (\text{Slope})_1 > (\text{Slope})_2$$

(Given)

20 (c) According to question,

Coefficient of linear expansion of brass = α_1

Coefficient of linear expansion of steel = α_2

Length of brass and steel rods are l_1 and l_2 , respectively.

As the increase in length ($l_2 - l_1$) is same for all temperatures.

So, $l_2 - l_1 = l_2 - l_1$

$$\Rightarrow l_2 (1 + \alpha_2 \Delta t) - l_1 (1 + \alpha_1 \Delta t) = l_2 - l_1 \Rightarrow l_2 \alpha_2 = l_1 \alpha_1$$

21 (a) It is given that,

$$(v_{\text{rms}})_1 = 200 \text{ ms}^{-1}, T_1 = 300 \text{ K}, p_1 = 10^5 \text{ Nm}^{-2}$$

$$T_2 = 400 \text{ K}, p_2 = 0.05 \times 10^5 \text{ Nm}^{-2}, (v_{\text{rms}})_2 = ?$$

As, rms velocity of gas molecules,

$$v_{\text{rms}} \propto \sqrt{T} \quad \left(\because v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \right)$$

For two different cases,

$$\Rightarrow \frac{(v_{\text{rms}})_1}{(v_{\text{rms}})_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{200}{(v_{\text{rms}})_2} = \sqrt{\frac{300}{400}} = \sqrt{\frac{3}{4}}$$

$$\Rightarrow (v_{\text{rms}})_2 = \frac{2}{\sqrt{3}} \times 200 = \frac{400}{\sqrt{3}} \text{ ms}^{-1}$$

22 (a) The rms speed is given by

$$v_{\text{rms}} = \sqrt{\frac{3RT}{\text{Molecular weight}}}$$

$$= \sqrt{\frac{3 \times 8.3 \times (273 + 27)}{2 \times 16 \times 10^{-3}}} = \sqrt{\frac{3 \times 8.3 \times 300}{32 \times 10^{-3}}}$$

$$\Rightarrow v_{\text{rms}} = 483 \text{ ms}^{-1}$$

23 (b) Translational kinetic energy,

$$K = \frac{1}{2} M v_{\text{rms}}^2 \quad \dots(i)$$

$$\text{and} \quad v_{\text{rms}} = \sqrt{\frac{3pV}{M}} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$K = \frac{1}{2} M \frac{3pV}{M} = \frac{3}{2} pV$$

$$\Rightarrow p = \frac{2K}{3V} \Rightarrow p \propto K \quad (\text{If, } V = \text{constant})$$

24 (d) Given, the lengths of rods are L_1 and L_2

and coefficients of linear expansions are α_1 and α_2 .

As, $\Delta L_1 = \alpha_1 L_1 \Delta t$ and $\Delta L_2 = \alpha_2 L_2 \Delta t$

and for combined rod,

$$\Delta L = \alpha (L_1 + L_2) \Delta t$$

Now, $\Delta L = \Delta L_1 + \Delta L_2$

$$\Rightarrow \alpha (L_1 + L_2) \Delta t = \alpha_1 L_1 \Delta t + \alpha_2 L_2 \Delta t$$

$$\Rightarrow \alpha (L_1 + L_2) = \alpha_1 L_1 + \alpha_2 L_2$$

$$\Rightarrow \alpha = \frac{\alpha_1 L_1 + \alpha_2 L_2}{L_1 + L_2}$$

25 (c) According to the question,

$$\text{In the first condition, } p = \frac{1}{3} \rho v_{\text{rms}}^2 \quad \dots(i)$$

In the second condition,

$$2p = \frac{1}{3} \rho v'_{\text{rms}} \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{v_{\text{rms}}^2}{v'^2_{\text{rms}}} = \frac{1}{2} \Rightarrow \frac{v_{\text{rms}}}{v'_{\text{rms}}} = \frac{1}{\sqrt{2}}$$

26 (c) The rms speed is given by

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\Rightarrow v_{\text{rms}} \propto \sqrt{\frac{T}{M}}$$

When temperature is doubled and molecules dissociate into atoms, then

$$\frac{(v_{\text{rms}})_1}{(v_{\text{rms}})_2} = \frac{\sqrt{\frac{T}{M}}}{\sqrt{\frac{2T}{M/2}}} = \frac{\sqrt{\frac{T}{M}}}{\sqrt{\frac{4T}{M}}} = \frac{\sqrt{T}}{\sqrt{4T}} = \frac{1}{2}$$

If $(v_{\text{rms}})_1$ is v , then $(v_{\text{rms}})_2$ will be $2v$.

28 (a) Given, $\rho = 0.09 \text{ kg m}^{-3}$

Pressure, p (at STP) = $1.01 \times 10^5 \text{ Pa}$

According to kinetic theory of gas,

$$p = (1/3) \rho v^2$$

$$\Rightarrow v = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3 \times 1.01 \times 10^5}{0.09}}$$

$$= 1834.85 \text{ ms}^{-1}$$

Now, volume occupied by one mole of hydrogen at STP

$$= 22.4 \text{ L} = 22.4 \times 10^{-3} \text{ m}^3$$

$\therefore$ Mass of hydrogen = Volume $\times$ Density

$$= 22.4 \times 10^{-3} \times 0.09$$

$$= 2.016 \times 10^{-3} \text{ kg}$$

$\therefore$ Average kinetic energy per mol = $\frac{1}{2} m v^2$

$$= \frac{1}{2} \times 2.016 \times 10^{-3} \times (1834.85)^2$$

$$\approx 3403 \text{ J}$$

29 (a) According to the ideal gas law,

$$\Rightarrow pV = RT \Rightarrow V = \left(\frac{R}{p} \right) T$$

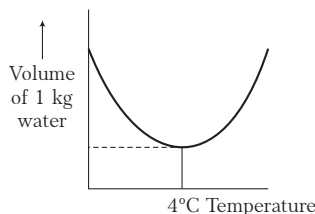
$$\Rightarrow V \propto T \Rightarrow \frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$\Rightarrow \frac{V_2}{V_1} - 1 = \frac{T_2}{T_1} - 1$$

$$\Rightarrow \frac{V_2 - V_1}{V_1} = \frac{T_2 - T_1}{T_1}$$

$$\Rightarrow \frac{V_2 - V_1}{V_1} = \frac{1}{T_1} = \frac{1}{T}$$

- 30** (c) Water is heated from 0°C to 10°C , then its volume first decreases and then increases.



- 31** (a) When cooled from room temperature, liquid water becomes dense as compared to other substances, but at approximately 4°C (39°F), pure water reaches its maximum density. If it is cooled further, it expands to become less dense.

- 32** (c) $\text{Strain} = \alpha \cdot \Delta T \Rightarrow \text{Stress} \times Y \propto \Delta T$

where, L = length of the rod,

α = coefficient of linear thermal expansion

and Y = Young's modulus of its material.

So, the longitudinal stress developed in the rod is directly proportional to $\frac{\Delta T}{Y}$.

- 33** (b) The kinetic theory of gases assume the collision between molecules as perfectly elastic.

- 35** (b) From law of equipartition of energy, for any system in thermal equilibrium, the total energy is equally distributed among its various degrees of freedom and each degree of freedom is associated with energy $(1/2) k_B T$.

- 36** (b) Here, $\gamma_r = \gamma_{ag} + 3\alpha_g = \gamma_{ac} + 3\alpha_c$

$$\Rightarrow 1.03 \times 10^{-3} + 3\alpha_g = 1.006 \times 10^{-3} + 3 \times (17 \times 10^{-6})$$

$$\Rightarrow \alpha_g = 9 \times 10^{-6} / ^\circ\text{C}$$

- 38** (c) We know that, kinetic energy, $\text{KE} = \frac{3}{2} k_B T$

where, $\frac{3}{2} k_B$ is constant, so $\text{KE} \propto T$.

Now, we can say that the average kinetic energy of a gas molecule at absolute temperature T is directly proportional to T .

- 39** (a) From gas equation, $v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$

$$\text{In given condition, } \frac{(v_{\text{rms}})_T}{(v_{\text{rms}})_0} = \sqrt{\frac{T}{T_0}}$$

$$\Rightarrow 4 = \sqrt{\frac{T}{T_0}}$$

$$\Rightarrow T = 4 T_0 = 4 \times 273 = 1092 \text{ K} = 819^\circ\text{C}$$

- 40** (a) For an ideal gas, the mean kinetic energy is proportional to the absolute temperature. It does not depend on the kind of gas.

- 41** (a) In kinetic theory of gases, it is assumed that time taken in a collision is negligible compared to the average time taken in free travel between any two collisions, to be precise.

- 42** (c) The number of molecules (n) in a volume (V) at pressure p and temperature T is given by

$$n = \frac{pV}{k_B T}$$

Here, $p = 10^6 \text{ dyne cm}^{-2}$, $V = 1000 \text{ cm}^3$,

$$T = 273 + 27 = 300 \text{ K},$$

$$k_B = 1.38 \times 10^{-16} \text{ erg per kelvin}$$

$$\therefore n = \frac{(10^6)(1000)}{1.38 \times 10^{-16} \times 300} \approx 2.4 \times 10^{22}$$

- 43** (a) Coefficient of real expansion,

$$\gamma_R = \frac{V_2 - V_1}{V_1(T_2 - T_1)}$$

$$\text{Here, } V_2 = \frac{3}{4}, V_1 = \frac{2}{3}$$

$$\text{and } (T_2 - T_1) = (100 - 20) = 80^\circ\text{C}$$

$$\therefore \gamma_R = \frac{\left(\frac{3}{4} - \frac{2}{3}\right)}{\frac{2}{3}(80)} = \frac{1}{640} = 15.6 \times 10^{-4} ^\circ\text{C}^{-1}$$

- 44** (c) $pV = RT = \text{constant}$

$$\Rightarrow p = \frac{RT}{V}$$

$$Vp^2 = \text{constant}$$

$$\Rightarrow V \left(\frac{RT}{V} \right)^2 = \text{constant}$$

$$\Rightarrow \frac{R^2 T^2}{V} = \text{constant}$$

$$\Rightarrow \frac{T^2}{V} = \text{constant} \Rightarrow \frac{T^2}{V} = \frac{T'^2}{2V}$$

$$\Rightarrow T'^2 = 2T^2 \Rightarrow T' = \sqrt{2} T$$

- 45** (c) The relation between the temperature scales A and B is given as

$$\frac{A - 42}{110} = \frac{B - 72}{220}, \text{ so for the two scales to have the same reading, } A = B$$

$$\Rightarrow \frac{A - 42}{110} = \frac{A - 72}{220}$$

$$\Rightarrow \frac{A - 42}{1} = \frac{A - 72}{2}$$

$$\Rightarrow 2A - 84 = A - 72$$

$$\Rightarrow 2A - A = 84 - 72$$

$$\Rightarrow A = 12^\circ$$

Laws of Thermodynamics

Thermodynamics is the branch of Physics that deals with the concepts of heat, work and interconversion of heat and other forms of energy. In this chapter, we will study the laws of thermodynamics, various thermodynamic processes, basic theory of heat engines, refrigerators and Carnot engine.

THERMODYNAMIC TERMS

Thermodynamics studies transformation of energy taking place with physical and chemical changes. In order to understand these transformations, we need to understand the terms given below

1. Thermodynamic system

It is an assembly of an extremely large number of particles (atoms or molecules) which is capable of exchange of energy with its surroundings. A thermodynamic system may contain matter in solid, liquid or gaseous form or a combination of two or more of these states.

Everything outside the system which has a direct effect on the system is called its **surroundings**. All space in universe outside the system is surroundings. *e.g.* Environment.

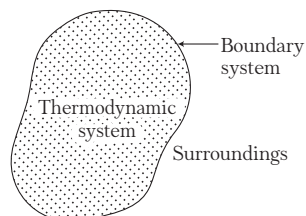


Fig. 15.1 A thermodynamic system having boundary and surroundings

A system is separated from its surrounding by a **boundary**. It is a real or imaginary two-dimensional closed surface that enclose the volume or region that a thermodynamic system occupies.

Inside

1 Thermodynamic terms

2 Zeroth law of thermodynamics

3 First law of thermodynamics

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- Applications of first law of thermodynamics

4 Heat capacity

- Thermodynamic parameters for a mixture of gases

5 Thermodynamic processes

- Isobaric process
- Isochoric process
- Isothermal process
- Adiabatic process
- Cyclic process

6 Heat engine

7 Refrigerator

8 Second law of thermodynamics

- Reversible and irreversible processes
- Carnot engine
- Entropy

A thermodynamic system can be classified by the nature of the transfer of heat that are allowed to occur across its boundary

- (i) **Open system** It exchanges both heat (or energy) and matter with its surroundings.

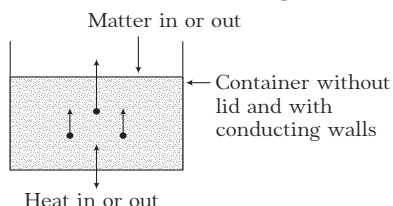


Fig. 15.2 Open system

- (ii) **Closed system** It exchanges only heat (or energy), but no matter with its surroundings.

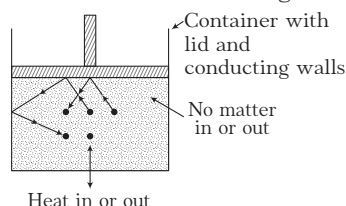


Fig. 15.3 Closed system

- (iii) **Isolated system** It exchanges neither heat (or energy) nor matter with its surroundings.

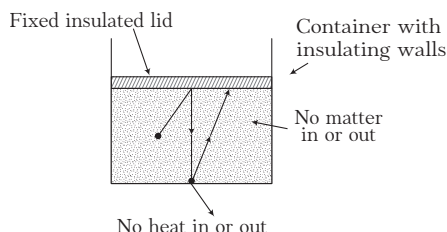


Fig. 15.4 Isolated system

2. Thermodynamic or state variables

The quantities like pressure (p), volume (V) and temperature (T) which help us to completely specify state of thermodynamic system are called thermodynamic variables. Thermodynamic variables may be extensive or intensive in nature.

Extensive variables are those which depend on the size of the system and **intensive variables** are those which do not depend on the size of the system. *e.g.* Internal energy (U), volume (V) and total mass (M) are extensive variables and pressure (p), temperature (T) and density (ρ) are intensive variables.

3. Equation of state

The equation which represents the relationship between the state variables of a system is called its equation of state. *e.g.* For an ideal gas, the equation of state is $pV = \mu RT$. For a fixed amount of gas given, there are only two independent variables, say p and V or T and V .

4. Thermodynamic process

When state of a system changes, then thermodynamic variables associated with the system like pressure p , volume V and temperature T also changes. This process is known as thermodynamic process.

5. Thermal equilibrium

Two systems are said to be in thermal equilibrium, if there is no net flow of heat between them when they are brought into thermal contact.

6. Adiabatic and diathermic wall

Adiabatic wall is an insulating wall between two thermodynamic systems which does not allow flow of energy (or heat) from one system to another system. Diathermic wall is a conducting wall between two thermodynamic systems which allows energy flow (or heat) from one system to another system.

ZEROth LAW OF THERMODYNAMICS

The Zeroth law of thermodynamics states that, if two systems A and B are separately in thermal equilibrium with a third system C , then A and B are in thermal equilibrium with each other.

Hence, from Zeroth law, if systems A and B are separately in thermal equilibrium with C , then

$$T_A = T_C \quad \text{and} \quad T_B = T_C \Rightarrow T_A = T_B$$

So, systems A and B will also be in thermal equilibrium with each other. Now, to understand Zeroth law of thermodynamics, let us consider two systems A and B separated by a fixed adiabatic wall.

The two systems A and B are in contact with a third system C through diathermic wall. The macroscopic variables of A and B will vary until both A and B come in thermal equilibrium with the third system C .

This shows that two systems A and B are separately in thermal equilibrium with a third system C .

This condition is shown in Fig. (a).

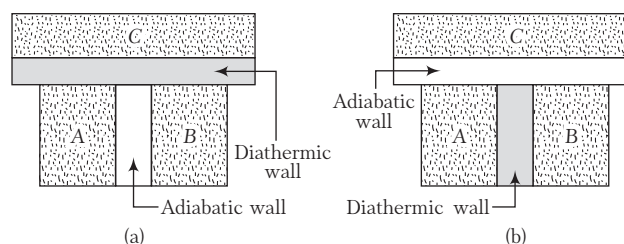


Fig. 15.5

Let the adiabatic wall between A and B is replaced by a diathermic wall and an adiabatic wall insulates C from A and B . Now, A and B are found in thermal equilibrium with each other. This condition is shown in Fig. (b). This experiment proves the Zeroth law of thermodynamics.

FIRST LAW OF THERMODYNAMICS

First law of thermodynamics is a statement of conservation of energy applied to any system in which energy transfer from or to the surroundings is taken into account.

It states that, if some quantity of heat is supplied to a system capable of doing external work, then the quantity of heat absorbed by the system is equal to the sum of the increase in the internal energy of the system and the external work done by the system.

$$\text{i.e.} \quad \boxed{\text{Heat supplied, } \Delta Q = \Delta U + \Delta W} \quad \dots(i)$$

where, ΔQ = heat supplied to the system by the surroundings. It is a path dependent function.

ΔW = Work done by the system on the surroundings. It is a path dependent function.

ΔU = Change in internal energy of the system. It depends only on the initial and final states.

Often the first law is used in its differential form, which is

$$\boxed{dU = dQ - dW}$$

This can also be written as

$$\boxed{dQ = dU + dW}$$

Following sign conventions are used in the measurement of heat, work and internal energy

Thermodynamic Sign Conventions

Process	Convention
Heat added to the system	$\Delta Q > 0$, i.e. positive
Heat removed from the system	$\Delta Q < 0$, i.e. negative
Work done by the system (volume increases)	$\Delta W > 0$, i.e. positive
Work done on the system (volume decreases)	$\Delta W < 0$, i.e. negative
When temperature of the system rises	$\Delta U > 0$, i.e. positive
When temperature of the system falls	$\Delta U < 0$, i.e. negative

- In the Eq. (i), all three quantities ΔQ , ΔU and ΔW must be expressed in same unit either in joule or in calorie.

Explanation of the terms used in first law

First law of thermodynamics basically revolves around the three terms ΔQ , ΔU and ΔW . If you substitute these three terms correctly with proper signs in the equation $\Delta Q = \Delta U + \Delta W$, then you are able to solve most of the problems of first law.

Let us understand each term one by one, for an ideal gas system

1. Work done by a thermodynamic system

By work, we mean work done by the system or on the system. Suppose a gas is confined in a cylinder with a movable piston, if p is pressure on the piston and A is area of piston, then force exerted by the gas on the piston of cylinder, $F = pA$.

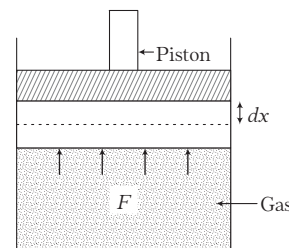


Fig. 15.6 Work done during expansion

When the piston is pushed outward by the gas through an infinitesimal distance dx , then the work done by the gas

$$dW = Fdx = pA dx = pdV$$

and for a finite volume change from V_i to V_f ,

$$\text{Work done, } W = \int_{V_i}^{V_f} dW$$

$$\boxed{\text{Work done, } W = \int_{V_i}^{V_f} p dV}$$

Here, p is constant.

From the equation of state, $pV = nRT$

$\therefore$

$$p\Delta V = nR\Delta T$$

$$\text{Work done, } W = p\Delta V = nR\Delta T$$

or

$$\boxed{\text{Work done, } W = nR \int_{T_i}^{T_f} \Delta T = nR(T_f - T_i)}$$

When variation of pressure with the volume is known to us, then the area bounded by p - V graph (indicator diagram) represents the work done.

$$\boxed{\text{Work} = \int_{V_i}^{V_f} p dV = \text{Area under } p\text{-}V \text{ graph}}$$

- Work done is a path dependent quantity.
So, work done is not a state function.
- If **volume is constant** in a thermodynamic process, then **work done is zero**.
- If volume increases, then work done is **positive** otherwise **negative**.
- For a cyclic process, work done is **positive**, if cycle is **clockwise** and **negative**, if cycle is **anti-clockwise**.

Example 15.1 Two moles of an ideal gas are in a rubber balloon at 30°C in equilibrium at 1 atm pressure. The balloon is fully expanded and can be assumed to require no energy in its expansion. The temperature of the balloon increases slowly to 35°C . What is the work done of expansion of balloon? (Take, $R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$).

Sol. If the balloon is fully expanded (i.e. does not develop significant tension as it expands), the pressure inside the balloon is always nearly equal to the pressure outside the balloon. And the pressure outside the balloon is 1 atm (constant). So, the pressure inside the balloon is constant at about 1 atm. According to the question, the limits of temperature are 30°C and 35°C .

$$\therefore T_1 = 30^\circ\text{C} = 30 + 273 = 303 \text{ K}$$

$$T_2 = 35^\circ\text{C} = 35 + 273 = 308 \text{ K}$$

We know that, when the limits of temperature are given and pressure is constant, then

$$\begin{aligned} \text{Work done, } W &= nR\Delta T = nR(T_2 - T_1) = 2 \times 8.31 [308 - 303] \\ &(\because \text{Number of moles of the gas, } n = 2) \\ &= 2 \times 8.31 \times (5) = 83.1 \text{ J} \end{aligned}$$

Example 15.2 (i) 2 mol of an ideal gas is heated at constant pressure, so that its temperature increases from 27°C to 127°C . Find work done by the gas.

(ii) For the same gas in a process, if its volume increases from 1000 cc to 2000 cc at constant pressure 100 kPa. Find work done by the gas.

Sol. Given, pressure, $p = 100 \text{ kPa} = 100 \times 10^3 \text{ Pa}$

$$\text{Temperature } T_1 = 27^\circ\text{C}, T_2 = 127^\circ\text{C}$$

$$\text{Volume, } V_1 = 1000 \text{ cc}, V_2 = 2000 \text{ cc}$$

According to the question, pressure is constant.

$$\therefore \Delta W = p\Delta V = nR\Delta T$$

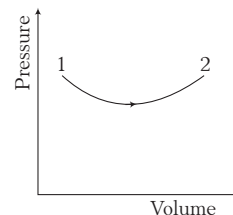
(i) Work done by the gas,

$$\begin{aligned} \Delta W &= nR\Delta T = nR(T_2 - T_1) \\ &= 2 \times 8.3 \times (127 - 27) \\ &(\because \text{Number of moles of the gas, } n = 2) \\ &= 1660 \text{ J} \end{aligned}$$

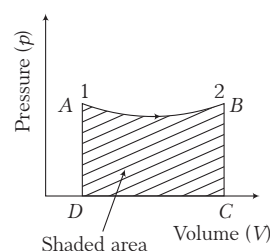
(ii) Work done by the gas,

$$\begin{aligned} \Delta W &= p\Delta V = p(V_2 - V_1) \\ &= 100 \times 10^3 (2000 - 1000) \times 10^{-6} \\ &= 100 \text{ J} \end{aligned}$$

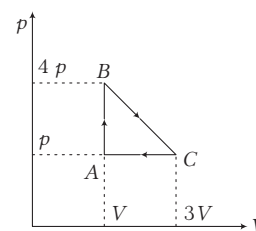
Example 15.3 Consider the process on a system shown in figure below. During the process, predict the nature of work done by the system.



Sol. We know that, the area under the p - V diagram is equal to the work done. So, if we move from 1 to 2, the shaded area (ABCD) under the graph is continuously increasing, which means the work done by the system continuously increases.



Example 15.4 An ideal gas is taken around the cycle ABCA as shown in the p - V diagram.



Evaluate the total work done by the gas during the cycle.

Sol. Work done = Area enclosed by triangle under the p - V diagram

$$= \frac{1}{2} AB \times AC$$

Here, side AB of the triangle is parallel to pressure (p) axis, therefore

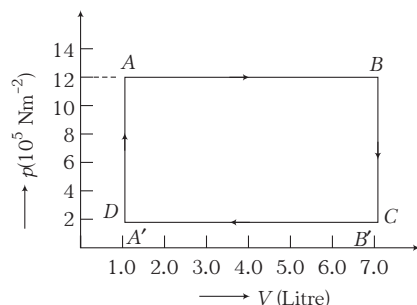
$$\begin{aligned} AB &= \text{Change in pressure, } \Delta p = p_B - p_A \\ &= 4p - p = 3p \end{aligned}$$

Similarly, side AC of the triangle is parallel to volume (V) axis, therefore

$$\begin{aligned} AC &= \text{Change in volume, } \Delta V = V_C - V_A \\ &= 3V - V = 2V \end{aligned}$$

$$\therefore \text{Work done, } W = \frac{1}{2} 3p \times 2V = 3pV$$

Example 15.5 The figure shows a p - V graph of the thermodynamic behaviour of an ideal gas. Find out from this graph (i) work done by the gas in the process $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$ and $D \rightarrow A$, (ii) work done by the gas in complete cycle $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$.



Sol. (i) The work done in a thermodynamic process is equal to the area enclosed between the p - V curve and the volume axis.

Work done by the gas in the process $A \rightarrow B$ is

$$\begin{aligned} W_1 &= \text{area of } ABB'A' = AB \times A'A \\ &= (7 - 1) \text{ litre} \times (12 \times 10^5) \text{ Nm}^{-2} \\ &= 6 \times 10^{-3} \text{ m}^3 \times (12 \times 10^5) \text{ Nm}^{-2} \\ &= 7200 \text{ N-m} \\ &= 7200 \text{ J} \end{aligned}$$

Work done in the process $B \rightarrow C$ is zero, since volume remains constant.

Work done by the gas in the process $C \rightarrow D$ is

$$\begin{aligned} W_2 &= \text{area of } DCB'A' \\ &= DC \times A'D \\ &= (-6 \times 10^{-3}) \times (2 \times 10^5) \\ &= -1200 \text{ J} \end{aligned}$$

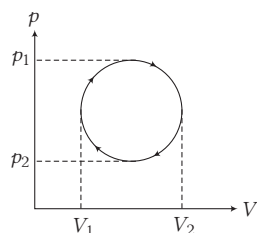
(Negative sign is taken because volume decreases)

Work done in the process $D \rightarrow A$ is also zero, because volume remains constant.

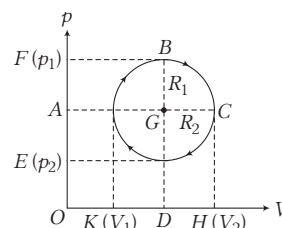
(ii) Work W_1 is positive, while W_2 is negative. Hence, the net work done in the whole cycle by the gas is

$$\begin{aligned} W &= W_1 + W_2 \\ &= 7200 - 1200 \\ &= 6000 \text{ J} \end{aligned}$$

Example 15.6 Find the work done during the perfectly circular cyclic process as shown in the figure.



Sol. As we know that, area of a circle of radius R is πR^2 .



From the diagram, it is clear that,

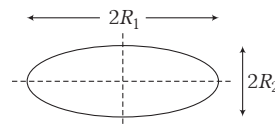
Vertical radius of the circle,

$$\begin{aligned} R_1 &= GB \\ &= AF = \frac{EF}{2} = \frac{OF - OE}{2} = \frac{p_1 - p_2}{2} \end{aligned}$$

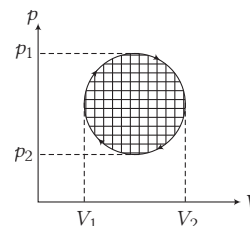
Horizontal radius of the circle,

$$\begin{aligned} R_2 &= GC \\ &= DH = \frac{KH}{2} = \frac{OH - OK}{2} = \frac{V_2 - V_1}{2} \end{aligned}$$

We know that, the area of an ellipse is $\pi R_1 R_2$, where R_1 and R_2 are semi-major and semi-minor axis respectively as shown in figure.



If $R_1 = R_2$, the ellipse becomes a circle as shown in figure.



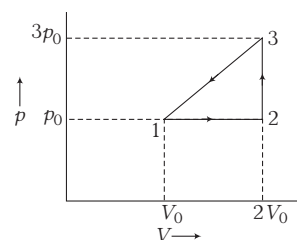
So, here the work done = area of the circle

= area of shaded portion

$$= \pi R_1 R_2 = \pi \left(\frac{p_1 - p_2}{2} \right) \left(\frac{V_2 - V_1}{2} \right)$$

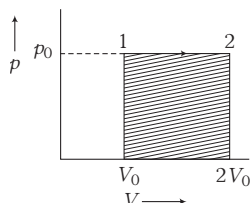
The positive sign is due to the fact that the cycle is clockwise.

Example 15.7 Calculate the work done by a gas, when it is taken from 1 to 2, 2 to 3 and 3 to 1 as shown in the figure. Also, calculate the work done in cyclic process $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$.



Sol. The work done in a process is the area below the process in p - V diagram.

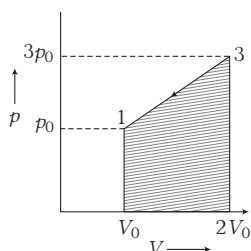
Process 1 $\rightarrow$ 2



$$\Delta W_{1 \rightarrow 2} = p_0(2V_0 - V_0) = p_0V_0$$

In the process 2 $\rightarrow$ 3, volume remains same, $\Delta W_{2 \rightarrow 3} = 0$

Process 3 $\rightarrow$ 1



$$\Delta W_{3 \rightarrow 1} = \text{area below the process } 3 \rightarrow 1$$

$$= -\frac{1}{2}(p_0 + 3p_0)(2V_0 - V_0) = -2p_0V_0$$

Work done is negative because volume is decreasing.

Work done in cyclic process,

$$\begin{aligned} \Delta W_{\text{cyclic}} &= \Delta W_{1 \rightarrow 2} + \Delta W_{2 \rightarrow 3} + \Delta W_{3 \rightarrow 1} \\ &= p_0V_0 + 0 - 2p_0V_0 = -p_0V_0 \end{aligned}$$

2. Internal energy of a thermodynamic system

Internal energy of a system is defined as the total energy possessed by the system due to molecular motion and molecular configuration.

It is represented by U . The energy due to molecular motion is called **internal kinetic energy** U_K .

The molecular motion includes translation, rotation and vibrational motion of the molecules.

The energy due to molecular configuration is called **internal potential energy** U_P .

Then, internal energy, $U = U_K + U_P$

- Internal energy is sum of the kinetic energies and potential energies of all the constituent particles of the system.
- Internal energy is a state variable which does not depend on the path taken to arrive at that state.

Internal energy of an ideal gas

In an ideal gas, there are no intermolecular forces, hence the gas does not possess intermolecular potential energy, thus $U_P = 0$.

So, internal energy of an ideal gas is just the sum of kinetic energies associated with various random motions (*i.e.* rotational, translational and vibrational) of its molecules.

Therefore, internal energy of an ideal gas depends only on its temperature and is given by $U = U_K = \frac{3}{2}nRT$

$$\therefore \text{Change in internal energy, } \Delta U = \frac{3}{2}nR\Delta T$$

Note For any gas, $\Delta U = nC_V\Delta T = n\left(\frac{f}{2}\right)R\Delta T = \frac{nR}{\gamma-1}\Delta T$

Here, $f = \text{degree of freedom, } \gamma = \frac{C_P}{C_V}$

Change in internal energy is independent of path followed by a thermodynamic system.

$$\therefore \text{Change in internal energy, } \Delta U = U_f - U_i$$

Example 15.8 1 kg of water at 373 K is converted into steam at same temperature. Volume of 1 cm^3 of water becomes 1671 cm^3 on boiling. What is the change in the internal energy of the system, if the latent heat of vaporisation of water is $5.4 \times 10^5 \text{ cal kg}^{-1}$?

Sol. Volume of 1 kg of water = $1000 \text{ cm}^3 = 10^{-3} \text{ m}^3$

Volume of 1 kg of steam = $10^3 \times 1671 \text{ cm}^3 = 1.671 \text{ m}^3$

Change in volume, $\Delta V = (1.671 - 10^{-3}) \text{ m}^3 = 1.670 \text{ m}^3$

Pressure, $p = 1 \text{ atm} = 1.01 \times 10^5 \text{ Nm}^{-2}$

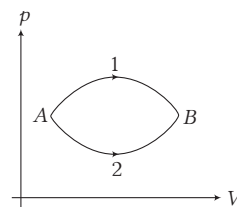
In expansion, work done, $\Delta W = p\Delta V = 1.01 \times 10^5 \times 1.670 \text{ J}$

$$= \frac{1.686 \times 10^5}{4.2} \text{ cal} = 4.015 \times 10^4 \text{ cal}$$

But $\Delta U = \Delta Q - \Delta W$ (From first law of thermodynamics)

$$\text{or } \Delta U = (5.4 \times 10^5 - 0.4015 \times 10^5) \text{ cal} = 4.9985 \times 10^5 \text{ cal}$$

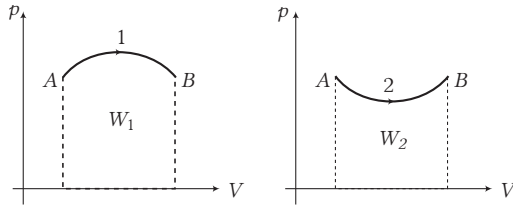
Example 15.9 A certain amount of an ideal gas passes from state A to state B first by means of process 1, then by means of process 2. In which of the process is the amount of heat absorbed by the gas greater?



Sol. From the first law of thermodynamics,

$$\Delta Q_1 = \Delta W_1 + \Delta U_1 \text{ and } \Delta Q_2 = \Delta W_2 + \Delta U_2$$

U is a state function. Hence, ΔU depends only on the initial and final positions. Therefore,



$$\Delta U_1 = \Delta U_2$$

But

$$\Delta W_1 > \Delta W_2$$

As the area under process 1 is greater than area under process 2.

Hence,

$$\Delta Q_1 > \Delta Q_2$$

Example 15.10 When a system goes from state A to state B, it is supplied with 400 J of heat and it does 100 J of work.

- For this transition, what is the system's change in internal energy?
- If the system moves from B to A, what is the change in internal energy?
- If in moving from A to B along a different path in which $W'_{AB} = 400$ J of work is done on the system, how much heat does it absorb?

Sol. (i) From the first law of thermodynamics,

$$\Delta U_{AB} = Q_{AB} - W_{AB} = (400 - 100) \text{ J} = 300 \text{ J}$$

- (ii) Consider a closed path that passes through the state A and state B. Internal energy is a state function, so ΔU is zero for a closed path.

$$\text{Thus, } \Delta U = \Delta U_{AB} + \Delta U_{BA} = 0$$

$$\text{or } \Delta U_{BA} = -\Delta U_{AB} = -300 \text{ J}$$

- (iii) The change in internal energy is the same for any path, so

$$\Delta U_{AB} = \Delta U'_{AB} = Q'_{AB} - W'_{AB}$$

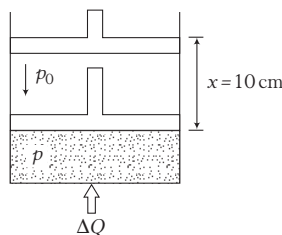
$$300 \text{ J} = Q'_{AB} - (-400 \text{ J})$$

and the heat exchanged is, $Q'_{AB} = -100 \text{ J}$

The negative sign indicates that the system loses heat in this transition.

Example 15.11 A gas is enclosed in a cylindrical vessel fitted with a frictionless piston. The gas is slowly heated for sometime. During the process, 10 J of heat is supplied and the piston is found to move out 10 cm. Find the increase in the internal energy of the gas. The area of cross-section of the cylinder = 4 cm² and the atmospheric pressure = 100 kPa.

Sol. In given process, piston moves against atmospheric process, so



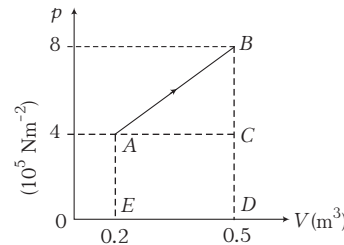
$$\begin{aligned} \text{Work done by the gas} &= \Delta W = p \Delta V = p_0 \Delta x \\ &= 100 \times 10^3 \times 4 \times 10^{-4} \times 0.1 = 4 \text{ J} \end{aligned}$$

From first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W \Rightarrow 10 = \Delta U + 4$$

$\therefore$ Increase in internal energy, $\Delta U = 6 \text{ J}$

Example 15.12 The pressure in monoatomic gas increases linearly from $4 \times 10^5 \text{ Nm}^{-2}$ to $8 \times 10^5 \text{ Nm}^{-2}$, when its volume increases from 0.2 m³ to 0.5 m³. Calculate



- work done by the gas.
- increase in the internal energy.
- amount of heat supplied.

Sol. Given, $p_1 = 4 \times 10^5 \text{ Nm}^{-2}$, $p_2 = 8 \times 10^5 \text{ Nm}^{-2}$, $V_1 = 0.2 \text{ m}^3$, $V_2 = 0.5 \text{ m}^3$

- (i) Work done by the gas, $\Delta W = \text{Area under } p\text{-}V \text{ graph}$
(Area ABCDEA)

$$\begin{aligned} &= \frac{1}{2} (AE + BD) \times AC \\ &= \frac{1}{2} (4 \times 10^5 + 8 \times 10^5) \times (0.5 - 0.2) \\ &= \frac{1}{2} \times 12 \times 10^5 \times 0.3 = 1.8 \times 10^5 \text{ J} \end{aligned}$$

- (ii) Increase in the internal energy,

$$\Delta U = C_V(T_2 - T_1) = \frac{C_V}{R} R(T_2 - T_1) = \frac{C_V}{R} (p_2 V_2 - p_1 V_1)$$

For monoatomic gas, $C_V = \frac{3}{2} R$

$$\begin{aligned} \therefore \Delta U &= \frac{3}{2} [(8 \times 10^5 \times 0.5) - (4 \times 10^5 \times 0.2)] \\ &= \frac{3}{2} [4 \times 10^5 - 0.8 \times 10^5] = 4.8 \times 10^5 \text{ J} \end{aligned}$$

- (iii) Amount of heat supplied,

$$\Delta Q = \Delta U + \Delta W = 4.8 \times 10^5 + 1.8 \times 10^5 = 6.6 \times 10^5 \text{ J}$$

Example 15.13 A thermodynamical process is shown in figure with

$$p_A = 3 \times 10^4 \text{ Nm}^{-2},$$

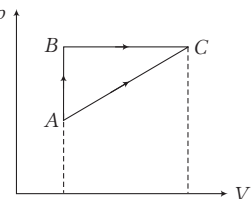
$$V_A = 2 \times 10^{-3} \text{ m}^3,$$

$$p_B = 8 \times 10^4 \text{ Nm}^{-2},$$

$$V_C = 5 \times 10^{-3} \text{ m}^3. \text{ In the processes}$$

AB and BC, 600 J and 200 J of

heat is added to the system, respectively. Find the change in internal energy of the system in process AC.



Sol. Since, internal energy is a state function, so the change in internal energy during the process AC will be same as that during the total processes AB plus BC. Calculate the total

work ΔW during AB plus BC and total heat ΔQ during these processes. Then, calculate $\Delta U = \Delta Q - \Delta W$.

During the process AB , the volume does not change, so work done is zero.

During BC , the pressure is constant, so total work done

$$\begin{aligned} &= p_B(V_C - V_B) = p_B(V_C - V_A) \quad (\because V_B = V_A) \\ &= 8 \times 10^4 \text{ Nm}^{-2} (5 \times 10^{-3} \text{ m}^3 - 2 \times 10^{-3} \text{ m}^3) \\ &= 240 \text{ J} \end{aligned}$$

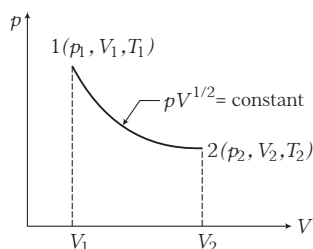
$\therefore \Delta W$ during process AB plus $BC = 240 \text{ J}$

Now, given during the same whole process,

$$\Delta Q = 600 + 200 = 800 \text{ J}$$

$\therefore \Delta U = \Delta Q - \Delta W = 800 - 240 = 560 \text{ J}$

Example 15.14 Consider a p - V diagram in which the path followed by one mole of perfect gas in a cylindrical container as shown in figure.



(i) Find the work done when the gas is taken from state 1 to state 2.

(ii) What is the ratio of temperatures T_1/T_2 , if $V_2 = 2V_1$?

(iii) Given that internal energy for one mole of gas at temperature T is $(3/2)RT$, find the change in internal energy and work done by the gas when it is taken from states 1 to 2, with $V_2 = 2V_1$.

Sol. (i) It is given that

$$pV^{1/2} = \text{constant}$$

Let this constant is A ,

$$\therefore pV^{1/2} = A$$

$$\Rightarrow p = \frac{A}{V^{1/2}} = \frac{A}{\sqrt{V}}$$

Also, we can write

$$p_1V_1^{1/2} = p_2V_2^{1/2} = A$$

$\therefore$ Work done by the gas,

$$\begin{aligned} \Delta W &= \int_{V_1}^{V_2} p dV = A \int_{V_1}^{V_2} \frac{dV}{\sqrt{V}} = A \left[\frac{\sqrt{V}}{1/2} \right]_{V_1}^{V_2} \\ &= 2A (\sqrt{V_2} - \sqrt{V_1}) \\ &= 2p_1V_1^{1/2} (V_2^{1/2} - V_1^{1/2}) \end{aligned}$$

(ii) Since, $T = pV/nR = \frac{A}{nR} \cdot \sqrt{V}$

$$\text{Thus, } \frac{T_2}{T_1} = \frac{\sqrt{V_2}}{\sqrt{V_1}} = \sqrt{2}$$

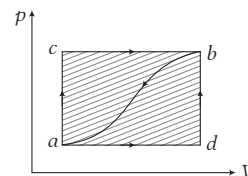
(iii) Then, the change in internal energy,

$$\Delta U = U_2 - U_1 = \frac{3}{2}R(T_2 - T_1) = \frac{3}{2}RT_1(\sqrt{2} - 1)$$

Work done by the gas,

$$\Delta W = 2A\sqrt{V_1}(\sqrt{2} - 1) = 2RT_1(\sqrt{2} - 1)$$

Example 15.15 When a system is taken from state a to state b (in figure) along the path $a \rightarrow c \rightarrow b$, 60 J of heat flow into the system and 30 J of work is done.



(i) How much heat flows into the system along the path $a \rightarrow d \rightarrow b$, if the work done is 10 J?

(ii) When the system is returned from b to a along the curved path, the work done on the system is 20 J. Does the system absorb or liberate heat and how much?

Sol. For the path $a \rightarrow c \rightarrow b$,

$$\Delta U = \Delta Q - \Delta W = 60 - 30 = 30 \text{ J or } U_b - U_a = 30 \text{ J}$$

(i) Along the path $a \rightarrow d \rightarrow b$,

$$\Delta Q = \Delta U + \Delta W = 30 + 10 = 40 \text{ J}$$

(ii) Along the curved path $b \rightarrow a$,

$$\Delta Q = (U_a - U_b) + \Delta W = (-30) + (-20) = -50 \text{ J, heat flows out of the system.}$$

Example 15.16 In the previous example, if $U_a = 0$ and $U_d = 22 \text{ J}$, then find the heat absorbed in the process $a \rightarrow d$ and $d \rightarrow b$.

Sol. For the path $a \rightarrow d \rightarrow b$

$$W_{adb} = W_{a \rightarrow d} + W_{d \rightarrow b} = W_{a \rightarrow d} + 0$$

$$\Rightarrow W_{a \rightarrow d} = W_{adb} = 10 \text{ J}$$

Now, applying first law of thermodynamics for the path $a \rightarrow d$,

$$\begin{aligned} \Delta Q_{a \rightarrow d} &= \Delta U_{a \rightarrow d} + \Delta W_{a \rightarrow d} \\ &= U_d - U_a + 10 = 22 - 0 + 10 = 32 \text{ J} \end{aligned}$$

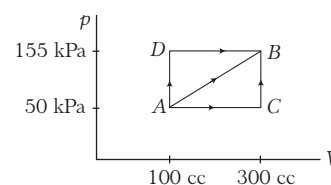
Similarly, for the path $d \rightarrow b$,

$$\Delta Q_{d \rightarrow b} = \Delta U_{d \rightarrow b} + \Delta W_{d \rightarrow b} = U_b - U_d + 0 = U_b - U_d$$

Given, $U_b - U_a = 30 \text{ J} \Rightarrow U_b - 0 = 30 \Rightarrow U_b = 30 \text{ J}$

$$\therefore Q_{d \rightarrow b} = 30 - 22 = 8 \text{ J}$$

Example 15.17 50 cal of heat should be supplied to take a system from the state A to the state B through the path ACB as shown in figure. Find the quantity of heat to be supplied to take it from A to B via ADB .



Sol. In process ACB ,

$$\begin{aligned}\Delta W_{ACB} &= \Delta W_{AC} + \Delta W_{CB} \\ &= 50 \times 10^3 \times (300 - 100) \times 10^{-6} + 0 \\ &= 10 \text{ J} = \frac{10}{4.2} \approx 2.4 \text{ cal}\end{aligned}$$

Given, $\Delta Q = 50 \text{ cal}$

From the first law of thermodynamics,

$$\begin{aligned}\Delta Q &= \Delta U + \Delta W = U_B - U_A + \Delta W \\ \Rightarrow 50 &= (U_B - U_A) + 2.4 \\ \Rightarrow U_B - U_A &= 47.6 \text{ cal}\end{aligned}$$

In process ADB ,

$$\begin{aligned}\Delta W_{ADB} &= \Delta W_{AD} + \Delta W_{DB} \\ &= 0 + 155 \times 10^3 \times (300 - 100) \times 10^{-6} \\ &= 31 \text{ J} = \frac{31}{4.2} \approx 7.4 \text{ cal}\end{aligned}$$

$\therefore$ Quantity of heat supplied,

$$\begin{aligned}\Delta Q &= \Delta U + \Delta W = U_B - U_A + 7.4 \\ &= 47.6 + 7.4 = 55 \text{ cal}\end{aligned}$$

Applications of first law of thermodynamics

Applications of first law of thermodynamics are given below

Melting process

When a substance is heated and it melts, the change in volume (dV) is very small and can, therefore be neglected. The temperature of the substance remains unchanged during melting process.

Heat absorbed during melting process, $\Delta Q = mL$

where, L is latent heat of fusion.

By the first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

$$\Rightarrow \boxed{mL = \Delta U} \quad (\because \Delta W = p\Delta V = p \times 0 = 0)$$

So, the internal energy increases by mL during the melting process.

Example 15.18 At normal pressure and 0°C temperature when 2 kg of ice melts, its volume decreases by 182 cm^3 . Latent heat of ice for melting is $3.4 \times 10^5 \text{ J kg}^{-1}$. Find out by what value, internal energy of the ice changes when it melts at normal pressure and 0°C temperature. (Normal pressure = $1.01 \times 10^5 \text{ Nm}^{-2}$)

Sol. Heat required to melt 2 kg of ice, $Q = mL$

$$= 2 \text{ kg} \times (3.4 \times 10^5) \text{ J kg}^{-1} = 6.8 \times 10^5 \text{ J}$$

Decrease in volume of ice = $\Delta V = 182 \text{ cm}^3$

Therefore, work done by external force on the ice,

$$\begin{aligned}W &= -p \times \Delta V = (1.01 \times 10^5 \text{ Nm}^{-2}) \times (-182 \times 10^{-6} \text{ m}^3) \\ &= -18.4 \text{ J}\end{aligned}$$

Because work is done on the ice, therefore it is negative.

According to first law of thermodynamics, change in internal energy of ice,

$$\begin{aligned}\Delta U &= Q - W = 6.8 \times 10^5 \text{ J} - (-18.4 \text{ J}) \\ &= (6.8 \times 10^5 + 18.4) \text{ J} \approx 6.8 \times 10^5 \text{ J}\end{aligned}$$

ΔU is positive. Therefore, internal energy of ice increases when it melts.

Boiling process

When a liquid is heated, it changes into vapour at constant temperature (called boiling point) and pressure. Let us consider the vaporisation of liquid of mass m . Let V_l and V_v be the volumes of liquid and vapours, respectively.

The work done in expanding at constant temperature and pressure p ,

$$\Delta W = p\Delta V = p(V_v - V_l)$$

Let the latent heat of vaporisation = L

$\therefore$ Heat absorbed during boiling process, $\Delta Q = mL$

Let U_l and U_v be the internal energies of the liquid and vapours respectively, then $\Delta U = U_v - U_l$

According to the first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

$\therefore$

$$\boxed{mL = (U_v - U_l) + p(V_v - V_l)}$$

Example 15.19 When water is boiled at 2 atm pressure, then latent heat of vaporisation is $2.2 \times 10^6 \text{ J kg}^{-1}$ and boiling point is 120°C . At this temperature, volume of 1 kg of water is 10^{-3} m^3 and that of 1 kg vapours is 0.824 m^3 . When 1 kg water is converted to the vapour at 120°C temperature, then find the work done and increase in internal energy. (1 atm pressure = $1.013 \times 10^5 \text{ Nm}^{-2}$)

Sol. Heat required to convert water into vapour at 120°C is

$$\Delta Q = m \times L = 1 \text{ kg} \times 2.2 \times 10^6 \text{ J kg}^{-1} = 2.2 \times 10^6 \text{ J}$$

Change in volume of water = $(0.824 - 0.001) \text{ m}^3 = 0.823 \text{ m}^3$

Therefore, work done against atmospheric pressure,

$$\begin{aligned}\Delta W &= p\Delta V \\ &= (2 \times 1.013 \times 10^5 \text{ Nm}^{-2}) \times (0.823 \text{ m}^3) \\ &= 166739.8 \text{ N-m} = 1.67 \times 10^5 \text{ J}\end{aligned}$$

From first law of thermodynamics, $\Delta Q = \Delta W + \Delta U$

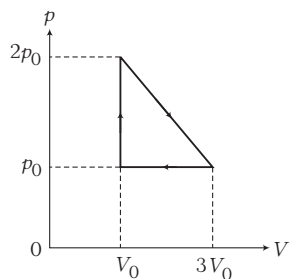
$$\begin{aligned}\text{or } \Delta U &= \Delta Q - \Delta W = 2.2 \times 10^6 \text{ J} - 166739.8 \text{ J} \\ &= 2033260.2 = 2.03 \times 10^6 \text{ J}\end{aligned}$$

$\therefore$ Required work = $1.67 \times 10^5 \text{ J}$

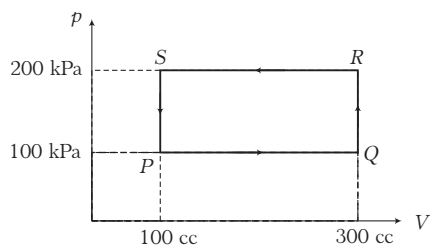
and change in internal energy = $2.03 \times 10^6 \text{ J}$

CHECK POINT 15.1

- First law of thermodynamics corresponds to
 - conservation of energy
 - conservation of momentum
 - law of conservation of angular momentum
 - conservation of mass
- In a process for an ideal gas, $\Delta W = 0$ and $\Delta Q < 0$. Then, for the gas,
 - the temperature will decrease
 - the volume will increase
 - the pressure will remain constant
 - the temperature will increase
- A thermodynamic system is taken from state (p_1, V_1) to (p_2, V_2) by two different process. The quantity which will remain same will be
 - ΔQ
 - ΔW
 - $\Delta Q + \Delta W$
 - $\Delta Q - \Delta W$
- An ideal monoatomic gas is taken around the cycle as shown in p - V diagram. The work done during the cycle is given by

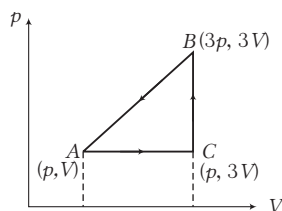


- $\frac{1}{2} p_0 V_0$
 - $p_0 V_0$
 - $2p_0 V_0$
 - None of these
- A thermodynamic system is taken through the cyclic process $PQRSP$. The net work done by the system is



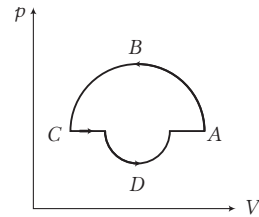
- 20 J
- 20 J
- 400 J
- 400 J

- An ideal gas is taken around $ABCA$ as shown in the p - V diagram. The work done during the cycle is

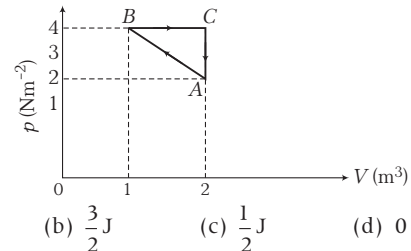


- $2pV$
- pV
- $\frac{pV}{2}$
- zero

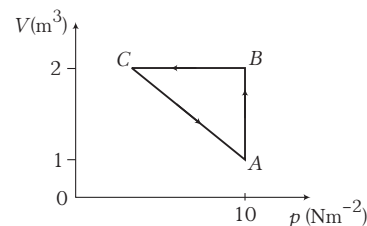
- Work done by the gas in the process as shown in figure, is



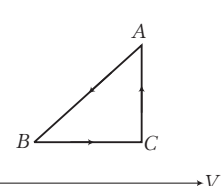
- positive
 - negative
 - zero
 - may be positive or negative
- In thermodynamic process, 200 J of heat is given to a gas and 100 J of work is also done on it. The change in internal energy of the gas is
 - 100 J
 - 300 J
 - 419 J
 - 24 J
 - Corresponding to the process $ABCA$ as shown in figure, what is the heat given to the gas?



- 1 J
 - $\frac{3}{2}$ J
 - $\frac{1}{2}$ J
 - 0
- An ideal gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$ as shown in the figure. If the net heat supplied to the gas in the cycle is 5 J, the work done on the gas in the process $C \rightarrow A$ is



- 5 J
 - 10 J
 - 15 J
 - 20 J
- The p - V diagram of a system undergoing thermodynamic transformation is shown in figure. The work done on the system in going from $A \rightarrow B \rightarrow C$ is 50 J and 20 cal heat is given to the system. The change in internal energy between A and C is
 - 34 J
 - 70 J
 - 84 J
 - 134 J



- A closed system undergoes a change of state by process $1 \rightarrow 2$ for which $Q_{12} = 10$ J and $W_{12} = -5$ J. The system is now returned to its initial state by a different path $2 \rightarrow 1$ for which $Q_{21} = -3$ J. The work done by the gas in the process $2 \rightarrow 1$ is
 - 8 J
 - zero
 - 2 J
 - +5 J

HEAT CAPACITY

If an amount of heat ΔQ is needed to change the temperature of a body by ΔT , then heat capacity of the material of the body is given by

$$\text{Heat capacity, } S = \frac{\Delta Q}{\Delta T}$$

Heat capacity is equal to the heat energy required to change the temperature of a body by unity. Its unit is JK^{-1} .

Note Heat capacity of a body is not constant over a large temperature range. Different amounts of heat may be needed for a unit rise in temperature at different temperature values.

Specific heat capacity

The amount of heat required to change the temperature of unit mass of a substance by unity is known as specific heat capacity of the substance.

$$\text{Specific heat capacity, } s = \frac{S}{m} = \frac{1}{m} \frac{\Delta Q}{\Delta T}$$

$$\text{Specific heat capacity, } s = \frac{1}{m} \frac{\Delta Q}{\Delta T}$$

It depends on the nature of substance and its temperature. Its unit is $\text{J/kg}\cdot\text{K}$.

Molar specific heat capacity

The amount of heat required to raise the temperature of one gram mole of a substance by unity is called molar specific heat capacity of the substance.

In other words, heat capacity per mole is known as molar specific heat capacity or molar heat capacity.

$$C = \frac{S}{n} = \frac{1}{n} \left(\frac{\Delta Q}{\Delta T} \right)$$

$$\text{Molar specific heat capacity, } C = \frac{1}{n} \frac{\Delta Q}{\Delta T}$$

where, n = number of moles. It depends on the nature of the substance, its temperature and the conditions under which heat is supplied. Its unit is $\text{J/mol}\cdot\text{K}$.

On the basis of condition under which the heat is supplied to a gas, molar specific heat capacity is of two types

(i) **Molar heat capacity at constant pressure**

$$C_p = \left(\frac{\Delta Q}{n\Delta T} \right)_{\text{constant pressure}}$$

(ii) **Molar heat capacity at constant volume**

$$C_v = \left(\frac{\Delta Q}{n\Delta T} \right)_{\text{constant volume}}$$

Note In case of solids and liquids, volume expansion is small (i.e. $\Delta W = 0$), therefore we define only one specific heat for solids and liquids.

Relation between C_p and C_v

For an ideal gas, the relation between C_p and C_v is

$$C_p - C_v = R$$

Here, C_p and C_v are molar specific heat capacities of an ideal gas at constant pressure and volume respectively and R is the universal gas constant. This relation is known as **Mayer's formula**.

The ratio of C_p and C_v is notified by γ .

$$\gamma = C_p / C_v$$

It is known as **heat capacity ratio**.

From the above two relation between C_p and C_v , we get

$$C_v = \frac{R}{\gamma - 1}$$

and

$$C_p = \frac{\gamma R}{\gamma - 1}$$

The heat capacity ratio in terms of degrees of freedom is

$$\gamma = 1 + \frac{2}{f}$$

where, f = degrees of freedom.

Values of f , C_v , C_p and γ for different types of gases are given below

Atomicity of gas	f	C_v	C_p	γ
Monoatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} = 1.67$
Diatomic, triatomic and triatomic linear (at normal temperature)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
Polyatomic triangular (Non-linear)	6	$\frac{6}{2}R = 3R$	$\frac{8}{2}R = 4R$	$\frac{4}{3} = 1.33$

For n moles of a gas, internal energy at temperature T is

$$U = \frac{nfRT}{2} \quad \text{or} \quad U = nC_vT$$

and change in internal energy, $\Delta U = nC_v\Delta T$.

Example 15.20 5 mol of oxygen is heated at constant volume from 10°C to 20°C . What will be change in the internal energy of the gas? (The gram molecular specific heat of oxygen at constant pressure is $C_p = 8 \text{ cal mol}^{-1}^\circ\text{C}^{-1}$ and $R = 8.36 \text{ J/mol}^\circ\text{C}$)

Sol. According to the Mayer's formula, $C_V = C_p - R$
 $= 8 - 2 = 6 \text{ cal/mol}^\circ\text{C}$
 $(\because R = 8.36 \text{ J/mol}^\circ\text{C} \approx 2 \text{ cal/mol}^\circ\text{C})$

$\therefore$ Heat absorbed by 5 mol of oxygen at constant volume,

$$Q = nC_V\Delta T = 5 \times 6 (20 - 10) = 30 \times 10 = 300 \text{ cal}$$

At constant volume, $\Delta V = 0$

$$\therefore \Delta W = 0$$

From first law of thermodynamics, $Q = \Delta U + \Delta W$

$$\Rightarrow 300 = \Delta U + 0 \Rightarrow \Delta U = 300 \text{ cal}$$

Example 15.21 At normal temperature and pressure (NTP), density of air is 1.293 kg m^{-3} and specific heat at constant volume is $169 \text{ cal kg}^{-1}\text{K}^{-1}$. Find the value of specific heat at constant pressure.

Sol. Density = $\frac{\text{Mass}}{\text{Volume}}$ or Volume = $\frac{\text{Mass}}{\text{Density}}$

$$\text{Volume of 1 kg of air, } V = \frac{1}{1.293} \text{ m}^3$$

Normal pressure, i.e. 76 cm mercury column pressure,

$$p = h\rho g = 76 \times 10^{-2} \times 13.6 \times 10^3 \times 9.8 \text{ Nm}^{-2}$$

Normal temperature, $T = 273 \text{ K}$

For 1 kg air, $pV = RT$

$$\text{or } R = \frac{pV}{T} = \frac{(76 \times 10^{-2} \times 13.6 \times 10^3 \times 9.8) \left(\frac{1}{1.293} \right)}{273}$$

$$= \frac{76 \times 136 \times 98 \times 10^{-1}}{273 \times 1.293} = 286.96 \text{ J kg}^{-1} \text{ K}^{-1}$$

From Mayer's formula,

$$C_p = C_V + \frac{R}{J} \quad (\text{where, } J = 4.2 \text{ J cal}^{-1})$$

$$= 169 + \frac{286.96}{4.2} = 169 + 68.32$$

$$= 237.32 \text{ cal kg}^{-1} \text{ K}^{-1}$$

Example 15.22 Find the work done in increasing temperature of 0.2 mol nitrogen at constant pressure from 37°C to 337°C . If $C_p = 7 \text{ cal / mol}^\circ\text{C}$. In this process, what is the work done by the gas?

Sol. Heat given to increase temperature of n mole of a gas by dT ,

$$\Delta Q = nC_p dT$$

$$= 0.2 \times 7(337 - 37)$$

$$= 0.2 \times 7 \times 300 = 420 \text{ cal}$$

Equivalent work done on nitrogen,

$$\Delta W = J\Delta Q = 4.2 \times 420 = 1764 \text{ J}$$

Change in internal energy,

$$\Delta U = nC_V\Delta T$$

$$\Rightarrow \Delta U = 0.2 \times 5 \times (337 - 37)$$

$$(\because C_V = C_p - R = 7 - 2 = 5 \text{ cal / mol}^\circ\text{C})$$

$$= 0.2 \times 5 \times 300 = 300 \text{ cal}$$

Work done by the gas,

$$\Delta W = \Delta Q - \Delta U = 420 - 300 = 120 \text{ cal}$$

$$= 120 \times 4.2 = 504 \text{ J}$$

Example 15.23 An ideal gas has a specific heat or molar heat capacity at constant pressure $C_p = 5R/2$. The gas is kept in a closed vessel of volume 0.0083 m^3 at a temperature of 300 K and a pressure of $1.6 \times 10^6 \text{ Nm}^{-2}$. An amount of $2.49 \times 10^4 \text{ J}$ heat energy is supplied to the gas. Calculate the final temperature and pressure of the gas.

Sol. Given, $C_p = 5R/2$, $\Delta V = 0$, $T_1 = 300 \text{ K}$,

$$V = 0.0083 \text{ m}^3, p_1 = 1.6 \times 10^6 \text{ Nm}^{-2} \text{ and } Q = 2.49 \times 10^4 \text{ J}$$

According to the Mayer's formula,

$$C_V = C_p - R = \frac{5R}{2} - R = \frac{3R}{2}$$

From first law of thermodynamics, $Q = \Delta U + p\Delta V$

$$\Rightarrow \Delta U = Q = 2.49 \times 10^4 \text{ J}$$

$$\text{From gas equation, } n = \frac{p_1 V}{RT_1} = \frac{1.6 \times 10^6 \times 0.0083}{8.3 \times 300} = \frac{16}{3}$$

$$\text{As, } \Delta U = nC_V\Delta T$$

$$\Rightarrow \Delta T = \frac{\Delta U}{nC_V} = \frac{2.49 \times 10^4 \times 6}{3 \times 8.3 \times 16} = 375 \text{ K}$$

Final temperature = $300 + 375 = 675 \text{ K}$

$$\text{As volume remains constant, } p \propto T \Rightarrow \frac{p_2}{p_1} = \frac{T_2}{T_1}$$

$$\Rightarrow p_2 = \frac{T_2}{T_1} \times p_1 = \frac{1.6 \times 10^6 \times 675}{300}$$

$$= 3.6 \times 10^6 \text{ Nm}^{-2}$$

Example 15.24 Find molar specific heat for the process $p = \frac{a}{T}$ for a monoatomic gas. $\left(a = \text{constant}, C_V = \frac{3}{2}R \right)$.

Sol. From first law of thermodynamics,

$$\Delta Q = \Delta W + \Delta U$$

$$\Rightarrow C = \frac{\Delta Q}{\Delta T} = \frac{\Delta W}{\Delta T} + \frac{\Delta U}{\Delta T}$$

$$\text{or } C = \frac{\Delta W}{\Delta T} + C_V \quad (\because \Delta U = C_V \Delta T)$$

$$= \frac{p\Delta V}{\Delta T} + C_V$$

From gas equation,

$$pV = RT$$

$$\Rightarrow V = \frac{RT}{p} = \frac{RT^2}{a} \quad \left(\because p = \frac{a}{T} \right)$$

$$\Rightarrow \frac{\Delta V}{\Delta T} = \frac{dV}{dT} = \frac{2RT}{a}$$

$$\text{So, } C = \frac{p\Delta V}{\Delta T} + C_V = p \left(\frac{2RT}{a} \right) + C_V$$

$$= \frac{a}{T} \left(\frac{2RT}{a} \right) + C_V$$

$$\Rightarrow C = 2R + C_V = 2R + \frac{3}{2}R = \frac{7}{2}R$$

Thermodynamic parameters for a mixture of gases

Thermodynamic parameters for a mixture of gases are given below

- (i) **Equivalent molar mass** When n_1 moles of a gas with molar mass M_1 are mixed with n_2 moles of a gas with molar mass M_2 , the equivalent molar mass of the mixture is given by $M = \frac{n_1 M_1 + n_2 M_2}{n_1 + n_2}$
- (ii) **Internal energy of the mixture** The total energy of the mixture is $U = U_1 + U_2$... (i)
- (iii) **C_V of the mixture**
 As, $U = U_1 + U_2 \Rightarrow \Delta U = \Delta U_1 + \Delta U_2$
 or $nC_V \Delta T = n_1 C_{V1} \Delta T + n_2 C_{V2} \Delta T$... (ii)
 or $(n_1 + n_2) C_V = n_1 C_{V1} + n_2 C_{V2}$ (As, $n = n_1 + n_2$)
 $\therefore C_V = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2}$
- (iv) **C_p of the mixture**
 As, $C_p = C_V + R$ or $C_p = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2} + R$

$$= \frac{n_1 (C_{V1} + R) + n_2 (C_{V2} + R)}{n_1 + n_2} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 + n_2}$$

$$\text{Thus, } C_p = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 + n_2}$$

(v) **γ of the mixture**

From Eq. (i),

$$(n_1 + n_2) C_V = n_1 C_{V1} + n_2 C_{V2}$$

$$\text{or } \frac{(n_1 + n_2) R}{\gamma - 1} = \frac{n_1 R}{\gamma_1 - 1} + \frac{n_2 R}{\gamma_2 - 1}$$

$$\text{or } \frac{n_1 + n_2}{\gamma - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

Thus, γ of the mixture is given by above equation.

Example 15.25 A gaseous mixture enclosed in a vessel

consists of 1g mole of a gas A $\left(\gamma = \frac{5}{3} \right)$ and some amount of

gas B $\left(\gamma = \frac{7}{5} \right)$ at a temperature T . The gases A and B do not

react with each other and are assumed to be ideal. Find the

number of gram moles of the gas B, if γ for the gaseous

mixture is $\left(\frac{19}{13} \right)$.

Sol. Using formula for γ of a mixture of two gases, we get

$$\Rightarrow \frac{n_A + n_B}{\gamma - 1} = \frac{n_A}{\gamma_A - 1} + \frac{n_B}{\gamma_B - 1}$$

$$\Rightarrow \frac{1 + n_B}{\left(\frac{19}{13} - 1 \right)} = \frac{1}{\left(\frac{5}{3} - 1 \right)} + \frac{n_B}{\left(\frac{7}{5} - 1 \right)}$$

$$\Rightarrow 13 + 13n_B = 9 + 15n_B$$

$$\Rightarrow n_B = 2 \text{ g-mol}$$

CHECK POINT 15.2

- If 10000 cal of heat is added to 4 mol of a monoatomic gas without changing its volume, then change in temperature of the gas is
 (a) 233 K (b) 833 K (c) 123 K (d) 433 K
- The specific heat at constant pressure is greater than that of the same gas at constant volume because
 (a) at constant volume, work is done in expanding the gas
 (b) at constant pressure, work is done in expanding the gas
 (c) the molecular attraction is more at constant pressure
 (d) the molecular vibration is more at constant pressure
- For hydrogen gas $C_p - C_V = a$ and for oxygen gas $C_p - C_V = b$, then the relation between a and b is (where, C_p and C_V are gram specific heats)
 (a) $a = 16b$ (b) $b = 16a$
 (c) $a = b$ (d) None of these
- The molar specific heats of an ideal gas at constant pressure and volume are denoted by C_p and C_V , respectively. If $\gamma = C_p/C_V$ and R is the universal gas constant, then C_V is equal to
 (a) $\frac{1 + \gamma}{1 - \gamma}$ (b) $\frac{R}{(\gamma - 1)}$ (c) $\frac{(\gamma - 1)}{R}$ (d) γR
- The molar heat capacity of a gas at constant volume is C_V . If n moles of the gas undergo ΔT change in temperature, its internal energy will change by $nC_V \Delta T$
 (a) only if the change of temperature occurs at constant volume
 (b) only if the change of temperature occurs at constant pressure
 (c) in any process which is not adiabatic
 (d) in any process

6. Two moles of a gas at temperature T and volume V are heated to twice its volume at constant pressure. If $C_p/C_V = \gamma$, then increase in internal energy of the gas is

(a) $\frac{RT}{\gamma-1}$ (b) $\frac{2RT}{\gamma-1}$ (c) $\frac{2RT}{3(\gamma-1)}$ (d) $\frac{2T}{\gamma-1}$

7. A gas mixture consists of 2 mol of oxygen and 4 mol of argon at temperature T . Neglecting all vibrational modes, the total internal energy of the system is

(a) $4RT$ (b) $15RT$ (c) $9RT$ (d) $11RT$

8. Five moles of helium are mixed with two moles of hydrogen to form a mixture. Take, molar mass of helium $M_1 = 4$ g and that of hydrogen $M_2 = 2$ g. The equivalent molar mass of the mixture is

(a) 6 g (b) $\frac{13}{7}$ g (c) $\frac{18}{7}$ g (d) $\frac{24}{7}$ g

9. The equivalent value of γ in the above question is

(a) 1.59 (b) 1.53
(c) 1.56 (d) None of these

10. A gaseous mixture enclosed in a vessel consists of 3 g mole of a gas X with $\gamma = (5/3)$ and some amount of gas Y with $\gamma = (7/5)$ at a temperature T . The gases X and Y do not react with each other and are assumed to be ideal. If γ for the gaseous mixture is $\left(\frac{29}{23}\right)$, then the number of gram moles of

the gas Y is

(a) 0.5 (b) 0.4 (c) 0.8 (d) 1

THERMODYNAMIC PROCESSES

When state of a thermodynamic system changes or the state variables change with time, then the process is known as **thermodynamic process**.

Few of thermodynamic processes are discussed below

1. Isobaric process

A thermodynamic process occurring at constant pressure, is known as isobaric process. *e.g.* Boiling of water in an open container.

Equation of state In this process, volume V and temperature T changes but pressure p remains constant.

$$\frac{V}{T} = \text{constant} \quad (\text{Charles' law})$$

Graphs Graphs related to isobaric process are given below

- (i) **p - V graph** In an isobaric process, pressure of the gas remains constant.

Therefore, p - V graph associated with this process is a straight line parallel to V -axis (having slope, $\frac{dp}{dV} = 0$) as shown below.

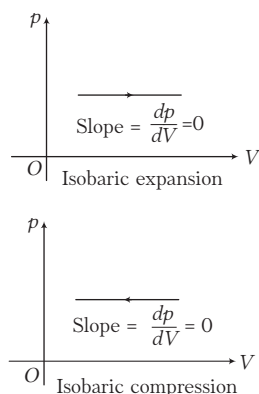


Fig. 15.7 p - V graph for isobaric expansion and compression

- (ii) **V - T graph** In isobaric process, $\frac{V}{T} = \text{constant}$

$$\therefore V \propto T$$

i.e. V - T graph is a straight line passing through origin as shown below.

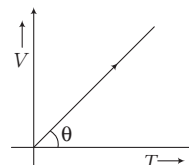


Fig. 15.8 V - T graph for isobaric process

Note Slope of the V - T graph, $\tan\theta = \frac{V}{T} = \frac{nR}{p} = \text{constant}$.

- (iii) **p - T graph** In isobaric process, pressure remains constant, so volume changes resulting in change of temperature as shown below.

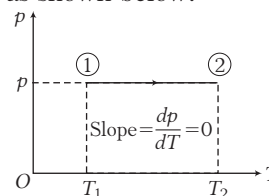


Fig. 15.9 p - T graph for isobaric process

Work done Suppose the pressure p of a gas remains constant and its volume changes from V_i to V_f , then the work done by the gas is

$$\text{Work done, } W = p(V_f - V_i)$$

$$\text{or Work done, } W = nR(T_f - T_i)$$

First law of thermodynamics for isobaric process

For isobaric process, $\Delta W = p\Delta V$ or $\Delta W = nR\Delta T$

$$\text{From } \Delta Q = \Delta U + \Delta W$$

$$\text{So, } \Delta Q = \Delta U + nR\Delta T$$

Example 15.26 Find the ratio of $\frac{\Delta Q}{\Delta U}$ and $\frac{\Delta Q}{\Delta W}$ in an isobaric process. The ratio of molar heat capacities $\frac{C_p}{C_v} = \gamma$.

Sol. In an isobaric process, $p = \text{constant}$. Therefore, $C = C_p$.

$$\text{Now, } \frac{\Delta Q}{\Delta U} = \frac{nC_p \Delta T}{nC_v \Delta T} = \frac{C_p}{C_v} = \gamma$$

$$\begin{aligned} \text{and } \frac{\Delta Q}{\Delta W} &= \frac{\Delta Q}{\Delta Q - \Delta U} \\ &= \frac{nC_p \Delta T}{nC_p \Delta T - nC_v \Delta T} \\ &= \frac{C_p}{C_p - C_v} = \frac{C_p / C_v}{C_p / C_v - 1} = \frac{\gamma}{\gamma - 1} \end{aligned}$$

Example 15.27 Suppose 1.0 g of water vaporises isobarically at atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$). Its volume in the liquid state is $V_i = V_{\text{liquid}} = 1.0 \text{ cm}^3$ and its volume in vapour state is $V_f = V_{\text{vapour}} = 1671 \text{ cm}^3$. Find the work done in the expansion and the change in internal energy of the system. Ignore any heat transfer of the steam and the surrounding air. (Take, latent heat of vaporisation, $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$)

Sol. Because expansion takes place at constant pressure, the work done is

$$\begin{aligned} W &= \int_{V_i}^{V_f} p_0 dV = p_0 \int_{V_i}^{V_f} dV = p_0 (V_f - V_i) \\ &= (1.01 \times 10^5) (1671 \times 10^{-6} - 1.0 \times 10^{-6}) \approx 169 \text{ J} \end{aligned}$$

Heat of vaporisation,

$$Q = mL_v = (1.0 \times 10^{-3}) (2.26 \times 10^6) = 2260 \text{ J}$$

Hence, from the first law of thermodynamics, the change in internal energy,

$$\Delta U = Q - W = 2260 - 169 = 2091 \text{ J}$$

Note The positive value of ΔU indicates that the internal energy of the system increases. We see that most $\left(\frac{2091 \text{ J}}{2260 \text{ J}} = 93\%\right)$ of the energy transferred to the liquid goes into increasing the internal energy of the system only $\frac{169 \text{ J}}{2260 \text{ J}} = 7\%$ leaves the system, by work done through the steam in pushing the surrounding atmosphere.

2. Isochoric process

A thermodynamic process occurring at constant volume, is known as isochoric process. *e.g.* Heat given to a system with fixed walls.

Equation of state In this process, p and T changes but V remains constant.

$$p \propto T \Rightarrow \boxed{\frac{p}{T} = \text{constant}} \quad (\text{Gay-Lussac's law})$$

Graph Graphs related to isochoric process are given below

(i) **p - V graph** In an isochoric process, volume of the gas remains constant. Therefore, p - V graph is a straight line parallel to p -axis with slope,

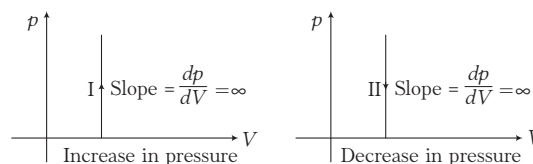
$$\frac{dp}{dV} = \tan \frac{\pi}{2} = \infty \text{ as shown below.}$$


Fig. 15.10 Graph for isochoric process

(ii) **p - T graph** For isochoric process, $\frac{p}{T} = \text{constant}$.

$$\therefore p \propto T$$

i.e. p - T graph is a straight line passing through origin.

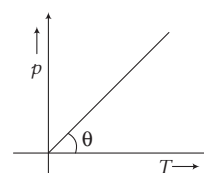


Fig. 15.11 p - T graph for isochoric process

Note Slope of the p - T graph, $\tan \theta = \frac{p}{T} = \frac{nR}{V} = \text{constant}$.

(iii) **V - T graph** In isochoric process, volume remains constant. So, pressure changes resulting in change of temperature as shown below.

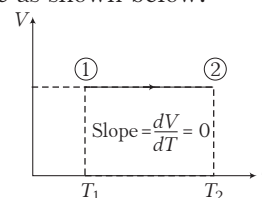


Fig. 15.12 V - T graph for isochoric process

Work done As volume remains constant,

$$\text{i.e. } V = \text{constant} \Rightarrow \Delta V = 0$$

$$\therefore \text{Work done, } \Delta W = p \Delta V = 0$$

First law of thermodynamics for isochoric process

$$\text{For isochoric process, } \Delta Q = nC_v \Delta T$$

From first law of thermodynamics, we get

$$\Delta Q = \Delta U + \Delta W$$

$$\text{As } dW = 0 \text{ (for isochoric process), } \Delta Q = \Delta U = nC_v \Delta T$$

$$\text{For one mole of a gas, } n = 1 \Rightarrow \Delta Q = C_v \Delta T = \Delta U$$

Example 15.28 What is the heat input needed to raise the temperature of 2 mol of helium gas from 0°C to 100°C

(i) at constant volume? (ii) at constant pressure?

(iii) What is the work done by the gas in part (ii)?

Give your answer in terms of R .

Sol. Helium is monoatomic gas.

Therefore, $C_V = \frac{3R}{2}$ and $C_p = \frac{5R}{2}$

(i) At constant volume, $Q = nC_V\Delta T = (2)\left(\frac{3R}{2}\right)(100) = 300R$

(ii) At constant pressure, $Q = nC_p\Delta T = (2)\left(\frac{5R}{2}\right)(100)$
 $= 500R$

(iii) At constant pressure, $W = Q - \Delta U = nC_p\Delta T - nC_V\Delta T$
 $= nR\Delta T = (2)(R)(100) = 200R$

3. Isothermal process

A thermodynamic process occurring at constant temperature is known as isothermal process. *e.g.* Freezing of water at 0°C to form ice at 0°C .

Equation of state For an ideal gas, the equation of state for isothermal process is given by

$$pV = \text{constant}$$

So, gas follows Boyle's law,

$$\Rightarrow p_i V_i = p_f V_f, \text{ for isothermal process}$$

Note Slow processes are treated as isothermal.

Graph Graphs related to isothermal process are given below

(i) **p - V graph** For isothermal process $pV = \text{constant}$.

Therefore, p - V graph is a rectangular hyperbola as shown below.

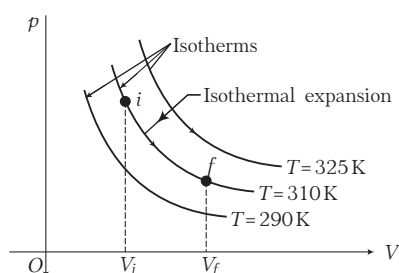


Fig. 15.13 An isothermal expansion process

Note An isotherm is a curve that connects points of same temperature.

(ii) **V - T graph** Temperature of the gas remains constant during isothermal process. Thus, V - T graph is parallel to V -axis as shown in figure.

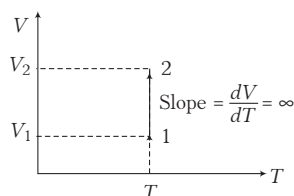


Fig. 15.14 V - T graph for isothermal process

(iii) **p - T graph** Similar to V - T graph, p - T graph is parallel to p -axis.

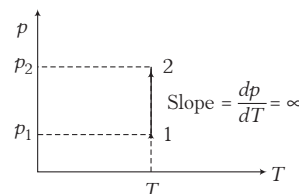


Fig. 15.15 p - T graph for isothermal process

Slope of p - V graph For isothermal process,
 $pV = \text{constant}$

On differentiating, we get

$$pdV + Vdp = 0 \Rightarrow \frac{dp}{dV} = -p/V$$

$$\text{Slope of graph} = -\tan\phi = \frac{dp}{dV} = -\frac{p}{V}$$

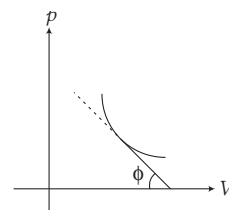


Fig. 15.16 Slope of isothermal curve

Specific and molar heat capacity During isothermal process, $\Delta T = 0$

$$\therefore \text{Specific heat capacity, } s = \frac{\Delta Q}{m\Delta T} = \frac{\Delta Q}{m(0)} = \infty$$

$$\therefore \text{Molar heat capacity, } C_{\text{iso}} = \frac{\Delta Q}{n\Delta T} = \infty$$

$$\begin{aligned} \text{Work done } W &= \int_{V_i}^{V_f} p dV = \int_{V_i}^{V_f} \left(\frac{nRT}{V} \right) dV \\ &\quad \left(\because p = \frac{nRT}{V} \right) \\ &= nRT \int_{V_i}^{V_f} \frac{dV}{V} \quad (\because T = \text{constant}) \end{aligned}$$

$$W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

or

$$W = nRT \ln \left(\frac{p_i}{p_f} \right) \quad (\because p_i V_i = p_f V_f)$$

Note

(i) If $V_f > V_i$, then $W > 0$, hence $Q > 0$, i.e. gas absorbs heat and work is done by the gas on the surrounding during expansion.

(ii) If $V_f < V_i$, then $W < 0$, hence $Q < 0$, i.e. during compression, work is done on the gas by the surrounding and heat is released.

First law of thermodynamics for isothermal

process As temperature remains constant in isothermal process,

$$\Delta U = 0$$

So, by first law of thermodynamics, $\Delta Q = \Delta W + \Delta U$

$$\Rightarrow \Delta Q = \Delta W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

$$= 2.3026 nRT \log_{10} \left(\frac{V_f}{V_i} \right)$$

where, n = number of moles,

R = gas constant and T = temperature.

Example 15.29 Five moles of an ideal gas at constant temperature 1200K is compressed from 200 L to 20L. Evaluate the work done in the process.

Sol. It is given that, number of moles, $n = 5$

Temperature of the gas, $T = 1200$ K

$\therefore$ Work done in an isothermal process,

$$W = 2.3026 nRT \log_{10} \left(\frac{V_2}{V_1} \right)$$

$$= 2.3026 \times 5 \times 8.3 \times 1200 \log_{10} \left(\frac{20}{200} \right)$$

$$\cong -11.5 \times 10^4 \text{ J}$$

Example 15.30 If 1 g mole of a gas is expanded isothermally at 127°C such that its volume increases two times, then find the work done and heat absorbed by the gas.

Sol. Given, temperature of the gas,

$$T = 127^\circ\text{C} = 127 + 273 = 400 \text{ K}$$

Initial volume, $V_1 = V$

Final volume, $V_2 = 2V$

$$\text{Work done, } W = 2.303 nRT \log_{10} \left(\frac{V_2}{V_1} \right)$$

$$= 2.303 \times 1 \times 8.3 \times 400 \log_{10} \left(\frac{2V}{V} \right)$$

$$= 2.303 \times 8.3 \times 400 \times 0.3010 \cong 2302 \text{ J}$$

For isothermal process, $dU = 0$

$$\Delta Q = dU + \Delta W = 0 + 2302 = 2302 \text{ J}$$

$$\Rightarrow \Delta Q = \frac{2302}{4.2} \cong 548 \text{ cal}$$

Example 15.31 2 g mol ideal gas is expanded isothermally at 27°C. Its volume increases three times of initial volume. Find the work done and heat absorbed by the gas. (Take, $R = 8.3 \text{ J mol}^{-1}\text{K}^{-1}$)

Sol. Work done by n moles of an ideal gas at isothermal temperature T , when its volume changes from V_i to V_f is

$$W = 2.3026 nRT \log_{10} \frac{V_f}{V_i}$$

According to the question,

$$n = 2 \text{ g-mol}$$

$$T = 273 + 27 = 300 \text{ K}$$

and $V_f / V_i = 3$

$$W = 2.3026 \times 2 \times 8.3 \times 300 \times \log_{10} 3$$

$$= 2.3026 \times 2 \times 8.3 \times 300 \times 0.4771$$

$$= 5.47 \times 10^3 \text{ J}$$

Now, $W = JQ$

$$\therefore \text{ Absorbed heat, } Q = \frac{W}{J} = \frac{5.47 \times 10^3 \text{ J}}{4.18 \text{ J cal}^{-1}}$$

$$= 1.31 \times 10^3 \text{ cal}$$

4. Adiabatic process

A thermodynamic process which takes place in such a way that there is no transfer of heat into or out of a system is known as adiabatic process.

e.g. Rapid compression like filling of a cycle tube by a hand pump, rapid expansion.

Equation of state For an ideal gas, equation of state for adiabatic process is

$$pV^\gamma = \text{constant}$$

Proof of the p - V relation In adiabatic process, $dQ = 0$

and

$$dW = -dU$$

$$\therefore pdV = -C_V dT \quad (\text{For } n = 1)$$

$$\Rightarrow dT = -\frac{pdV}{C_V} \quad \dots(i)$$

Also, for 1 mole of an ideal gas,

$$d(pV) = d(RT)$$

$$\text{or } pdV + Vdp = RdT$$

$$\text{or } dT = \frac{pdV + Vdp}{R} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$C_V V dp + (C_V + R) pdV = 0$$

$$\text{or } C_V V dp + C_p pdV = 0$$

Dividing this equation by pV , we are left with

$$C_V \frac{dp}{p} + C_p \frac{dV}{V} = 0$$

$$\text{or } \frac{dp}{p} + \gamma \frac{dV}{V} = 0$$

$$\text{or } \int \frac{dp}{p} + \gamma \int \frac{dV}{V} = 0$$

$$\text{or } \ln(p) + \gamma \ln(V) = \text{constant}$$

$$\text{or } \ln(p) + \ln(V^\gamma) = \text{constant}$$

or $\ln(pV^\gamma) = \text{constant}$

We can write this in the form

$$pV^\gamma = \text{constant}$$

This equation is the condition that must be obeyed by an ideal gas in an adiabatic process. *e.g.* If an ideal gas makes an adiabatic transition from a state with pressure and volume p_i and V_i to a state with p_f and V_f , then

$$p_i V_i^\gamma = p_f V_f^\gamma$$

The equation $pV^\gamma = \text{constant}$ can be written in terms of other pairs of thermodynamic variables by combining it with the ideal gas law ($pV = nRT$).

Putting $p = \frac{RT}{V}$, we get (From ideal gas equation)

$$TV^{\gamma-1} = \text{constant}$$

Again putting $V = \frac{RT}{p}$, we get (From ideal gas equation)

$$T^\gamma p^{1-\gamma} = \text{constant}$$

Graphs Graphs related to adiabatic process are given below

- (i) **p - V graph** For no heat exchange in adiabatic process, pressure decreases exponentially with increase in volume as shown below.

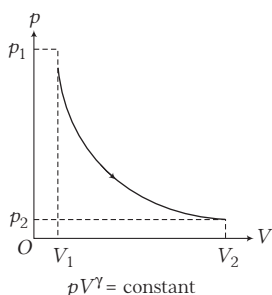


Fig. 15.17 p - V graph for adiabatic process

- (ii) **V - T graph** In adiabatic process with increase in temperature, volume decreases exponentially as shown below.

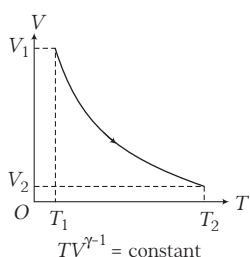


Fig. 15.18 V - T graph for adiabatic process

- (iii) **p - T graph** For adiabatic process with decrease in temperature, pressure decreases as shown below.

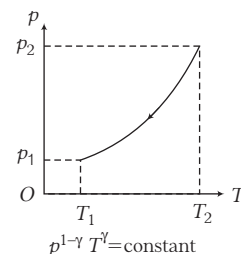


Fig. 15.19 p - T graph for adiabatic process

Slope of p - V graph In an adiabatic process ($pV^\gamma = \text{constant}$), the slope of p - V diagram at any point is

$$\begin{aligned} \frac{dp}{dV} &= -\tan \phi = \frac{d}{dV} \left(\frac{\text{constant}}{V^\gamma} \right) \\ &= -\gamma \left(\frac{p}{V} \right) \end{aligned}$$

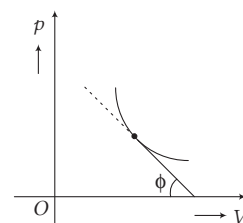


Fig. 15.20

Thus,

$$(\text{Slope})_{\text{adiabatic}} = -\gamma \left(\frac{p}{V} \right)$$

Also, we know that, in an isothermal process,

$(\text{Slope})_{\text{isothermal}} = -\frac{p}{V}$. Because $\gamma > 1$, the isothermal curve is not as steep as that for the adiabatic expansion.

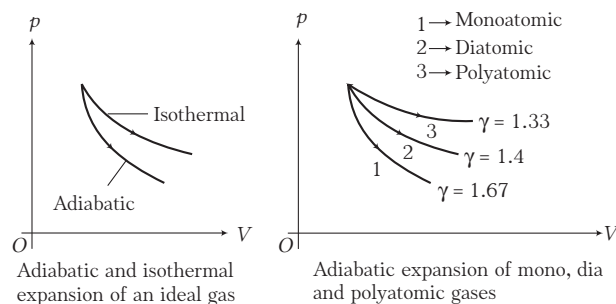


Fig. 15.21 Adiabatic and isothermal expansion

In similar way for adiabatic and isothermal compression of an ideal gas, p - V diagram as shown, below implies that, the curve which is more vertical is adiabatic.

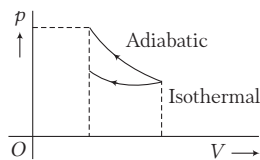


Fig. 15.22

Specific and molar heat capacity During adiabatic process, $\Delta Q = 0$

$$\therefore \text{Specific heat capacity, } s = \frac{\Delta Q}{m\Delta T} = 0$$

$$\therefore \text{Molar heat capacity, } C_{\text{adi}} = \frac{\Delta Q}{n\Delta T} = 0$$

Work done Let initial state of gas is (p_1, V_1, T_1) and after adiabatic change final state of gas is (p_2, V_2, T_2) , then

$$\text{work done } W = \frac{p_1 V_1 - p_2 V_2}{\gamma - 1}$$

On putting $pV = nRT$, we get

$$\Rightarrow W = \frac{nR}{(\gamma - 1)} (T_1 - T_2)$$

Note

- (i) If work is done by the gas in an adiabatic process ($W > 0$), then $T_2 < T_1$.
- (ii) If work is done on the gas ($W < 0$), we get $T_2 > T_1$, i.e. the temperature of the gas rises.

First law of thermodynamics for adiabatic process

Now, by first law of thermodynamics, $\Delta Q = \Delta U + \Delta W$

$$\text{As, } \Delta Q = 0 \Rightarrow \Delta U = -\Delta W$$

So, if work is done by the system, on the expense of internal energy and so the temperature of system falls, i.e. $W > 0$, $T_f < T_i$. Conversely, if work is done on the system, the internal energy and so the temperature of system increases, i.e. $W < 0$, $T_f > T_i$.

Example 15.32 Initial pressure of a gas is $5 \times 10^5 \text{ Nm}^{-2}$. Under an adiabatic process, its volume becomes $1/9$ of its initial value. What is the pressure of the gas in this situation? (Take, $\gamma = 3/2$)

Sol. For adiabatic process of the gas, $p_1 V_1^\gamma = p_2 V_2^\gamma \Rightarrow p_2 = p_1 \left(\frac{V_1}{V_2} \right)^\gamma$

$$\text{Given, } \gamma = 3/2, V_1 = V, V_2 = \frac{V}{9} \text{ and } p_1 = 5 \times 10^5 \text{ Nm}^{-2}$$

$$\begin{aligned} \therefore p_2 &= 5 \times 10^5 \times \left(\frac{V}{V/9} \right)^{3/2} \\ &= 5 \times 10^5 \times (9)^{3/2} = 5 \times 10^5 \times (3^2)^{3/2} \\ &= 5 \times 10^5 \times 3^3 = 135 \times 10^5 \text{ Nm}^{-2} \end{aligned}$$

Example 15.33 Two samples of a gas initially at same temperature and pressure are compressed from a volume V to $V/2$. One sample is compressed isothermally and the other adiabatically. In which sample is the pressure greater?

Sol. Let initial volume, $V_1 = V$ and pressure, $p_1 = p$

$$\text{Final volume, } V_2 = \frac{V}{2} \text{ and final pressure, } p_2 = ?$$

$$\text{For isothermal compression, } p_2 V_2 = p_1 V_1$$

$$\Rightarrow p_2 = \frac{p_1 V_1}{V_2} = \frac{pV}{V/2} = 2p$$

$$\text{For adiabatic compression, } p_2' = p_1 \left(\frac{V_1}{V_2} \right)^\gamma$$

$$\Rightarrow p_2' = p \left(\frac{V}{V/2} \right)^\gamma = 2^\gamma p \Rightarrow p_2' = 2^\gamma p$$

$$\therefore \gamma > 1 \therefore 2^\gamma > 2 \text{ and } p_2' > p_2$$

Pressure during adiabatic compression is greater than the pressure during isothermal compression.

Example 15.34 An ideal monoatomic gas at 300 K expands adiabatically to twice its volume. What is the final temperature?

Sol. For an ideal monoatomic gas, $\gamma = 5/3$

$$\text{In an adiabatic process, } TV^{\gamma-1} = \text{constant}$$

$$\therefore T_f V_f^{\gamma-1} = T_i V_i^{\gamma-1}$$

$$\text{or } T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1} = (300) \left(\frac{1}{2} \right)^{\frac{5}{3}-1} \approx 189 \text{ K}$$

Bulk modulus of gas

Bulk modulus of gas is given by

$$B = - \frac{dp}{(dV/V)}$$

or

$$B = -V \left(\frac{dp}{dV} \right)$$

where, $\frac{dp}{dV}$ is slope of p - V graph.

For isothermal process, $\frac{dp}{dV} = -\frac{p}{V}$

$$\therefore \text{Bulk modulus of gas, } B = -V \left(-\frac{p}{V} \right)$$

$$\boxed{B = p}$$

For adiabatic process, $\frac{dp}{dV} = -\frac{\gamma p}{V}$

$$\therefore \text{Bulk modulus of gas, } B = -V \left(-\frac{\gamma p}{V} \right) \text{ or } B = \gamma p$$

Therefore, isothermal bulk modulus is equal to pressure of the gas and adiabatic bulk modulus is γ times the pressure of gas.

p - V Diagram of Different Processes

p - V diagram of different processes is shown as follows. We can compare the work done in each process from area under p - V curve.

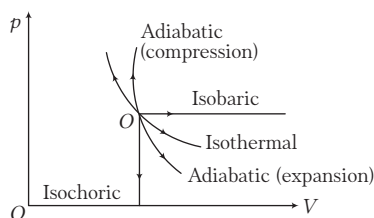


Fig. 15.23 Combined p - V diagram of different processes

We know that, in an isothermal process, $\frac{\Delta p}{\Delta V} = -\frac{p}{V}$.

For an adiabatic process, $\frac{\Delta p}{\Delta V} = -\frac{\gamma p}{V}$ it means that at a particular point, slope (value) of adiabatic curve is more than that for isotherm or we can say adiabatic curve is steeper than an isotherm for expansion and just reverse for compression. It is clear from the figure that for expansion that occurs within same limits.

$$W_{\text{isobaric}} > W_{\text{isothermal}} > W_{\text{adiabatic}} > W_{\text{isochoric}}$$

Note In general many thermodynamic processes follow a rule of type, $pV^n = \text{constant}$.

These processes are called polytropic process and 'n' is called polytropic index.

Process	n
$p = \text{constant}$	0
$V = \text{constant}$	∞
$T = \text{constant}$	1
$\Delta Q = 0$	γ

(i) Work done in a polytropic ($1 \rightarrow 2$) process,

$$\Delta W = \int p dV = \frac{p_1 V_1 - p_2 V_2}{n-1}$$

(ii) Heat transfer in a polytropic process,

$$Q = \frac{R(T_1 - T_2)}{n-1} - C_V(T_1 - T_2)$$

$$\text{where, } C_V = \frac{R}{\gamma - 1}$$

$$\Rightarrow Q = \left(\frac{\gamma - n}{\gamma - 1} \right) W = C_n(T_2 - T_1)$$

(iii) Specific heat capacity of gas in a polytropic process,

$$C_n = C_V + \frac{R}{1-n} = C_V \left(\frac{\gamma - n}{1-n} \right)$$

Summary of ΔQ , ΔU and ΔW for different processes

Name of the process	ΔQ	ΔU	ΔW
Isothermal	$\Delta Q = \Delta W$	0	$nRT \ln \left(\frac{V_f}{V_i} \right) = nRT \ln \left(\frac{p_i}{p_f} \right)$
Adiabatic	0	$nC_V \Delta T$	$\frac{p_i V_i - p_f V_f}{\gamma - 1} = -\Delta U$
Isobaric	$n C_p \Delta T$	$nC_V \Delta T$	$p(V_f - V_i)$
Isochoric	$\Delta Q = \Delta U$ $= nC_V \Delta T$	$nC_V \Delta T$	0

Example 15.35 14 g of nitrogen gas is compressed in an adiabatic process such that its temperature increases by 60°C . Find the work done on the gas.

Sol. Given, $m = 14 \text{ g}$, $T_2 - T_1 = 60^\circ\text{C}$, $\gamma = 1.5$

$$\text{Work done on the gas, } W = \frac{nR}{\gamma - 1} (T_1 - T_2)$$

$$\therefore n = \frac{m}{M} = \frac{14}{28} = \frac{1}{2}$$

$$\therefore W = \frac{1}{2} \times \frac{8.3}{1.5 - 1} \times (-60) = -498 \text{ J}$$

Negative sign shows that work is done on the gas.

Example 15.36 Temperature of a 5 mol of hydrogen gas decreases from 273 K to 260 K while expanded through adiabatic process. Find the work done by the gas. What is the change in internal energy of the gas?

Sol. Work done by the gas, $W = \frac{nR}{\gamma - 1} (T_i - T_f)$

According to the question,

$$n = 5 \text{ mol, } R = 8.31 \text{ J/mol-K,}$$

$$\gamma = 1.4, T_i = 273 \text{ K, } T_f = 260 \text{ K}$$

$$\text{Work done, } W = \frac{5 \times 8.31}{1.4 - 1} (273 - 260) \text{ K} \approx 1350 \text{ J}$$

According to first law of thermodynamics,

$$\Delta U = \Delta Q - W = 0 - (1350) = -1350 \text{ J}$$

Example 15.37 2 m^3 volume of a gas at a pressure of $4 \times 10^5 \text{ Nm}^{-2}$ is compressed adiabatically, so that its volume becomes 0.5 m^3 . Find the new pressure. Compare this with the pressure that would result, if the compression was isothermal. Calculate work done in each process. (Take, $\gamma = 1.4$)

Sol. Given, $V_1 = 2 \text{ m}^3$, $p_1 = 4 \times 10^5 \text{ Nm}^{-2}$, $V_2 = 0.5 \text{ m}^3$

In adiabatic process, $p_1 V_1^\gamma = p_2 V_2^\gamma$

$$\Rightarrow p_2 = 4 \times 10^5 \left[\frac{2}{0.5} \right]^{1.4} = 4 \times 10^5 (4)^{1.4} \\ \approx 2.8 \times 10^6 \text{ Nm}^{-2}$$

In isothermal process, $p_1V_1 = p_2V_2$

$$\Rightarrow p_2 = \frac{p_1V_1}{V_2} = \frac{4 \times 10^5 \times 2}{0.5} = 1.6 \times 10^6 \text{ Nm}^{-2}$$

Now, work done in adiabatic process,

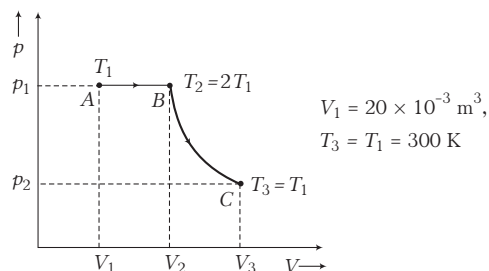
$$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1} = \frac{(4 \times 10^5 \times 2 - 1.6 \times 10^6 \times 0.5)}{1.4 - 1} = -1.5 \times 10^6 \text{ J}$$

Work done in isothermal process,

$$\begin{aligned} W &= 2.3026 RT \log \frac{V_2}{V_1} = 2.3026 p_1V_1 \log \frac{V_2}{V_1} \\ &= 2.3026 \times 4 \times 10^5 \times 2 \times \log \left(\frac{0.5}{2.0} \right) \\ &= 2.3026 \times 4 \times 10^5 \times 2 \log \left(\frac{1}{4} \right) \\ &= -1.1 \times 10^6 \text{ J} \end{aligned}$$

Example 15.38 Two moles of helium gas ($\gamma = 5/3$), assumed ideal, are initially at 27°C and occupy a volume of 20 L. The gas is first expanded at constant pressure till its volume is doubled. It then undergoes an adiabatic change until the temperature returns to its initial value. Determine the final pressure and volume of the gas. Also, calculate the work done under isobaric and adiabatic processes. (Take, $R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$)

Sol.



For process AB ($p = \text{constant}$),

$$T_2 = 2T_1 = 600 \text{ K} \quad (\because V_2 = 2V_1)$$

For process BC, $TV^{\gamma-1} = \text{constant}$

$$\text{Therefore, } (2T_1)(V_2)^{5/3-1} = (T_1)(V_3)^{5/3-1}$$

$$\Rightarrow V_3 = (V_2)(2)^{3/2} = 2\sqrt{2}V_2 = 80\sqrt{2} \times 10^{-3} \text{ m}^3 = 80\sqrt{2} \text{ L}$$

$$\Rightarrow p_3 = \frac{nRT_3}{V_3} = \frac{(2)(8.3)(300)}{80\sqrt{2} \times 10^{-3}} = 0.44 \times 10^5 \text{ Nm}^{-2}$$

$$\begin{aligned} \text{Work done under isobaric process} &= p\Delta V = p_1(V_2 - V_1) \\ &= \frac{nRT_1}{V_1}(V_2 - V_1) = \frac{2 \times 8.3 \times 300}{20 \times 10^{-3}}(40 \times 10^{-3} - 20 \times 10^{-3}) = 4980 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{Work done under adiabatic process} &= \frac{nR(T_2 - T_3)}{(\gamma - 1)} \\ &= \frac{2 \times 8.3 \times (300)}{(5/3 - 1)} = 7470 \text{ J} \end{aligned}$$

5. Cyclic process

A single process or a series of processes in which after certain interchanges of heat and work, the system is restored to its initial state are known as a **cyclic process**.

As both initial and final states are same in a cyclic process,

$$\Delta U = U_f - U_i = 0$$

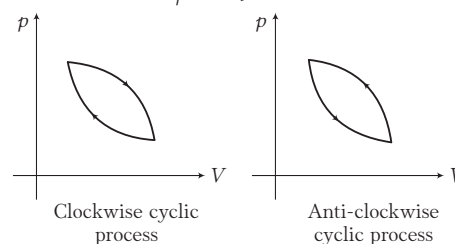


Fig. 15.24 Cyclic processes

For a cyclic process, p - V graph is a closed curve and area enclosed by the curve is equal to the work done.

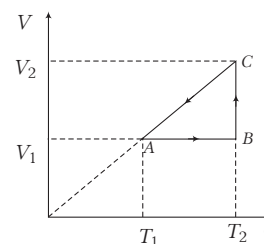
From first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

For cyclic process, $\Delta U = 0$ or $\Delta Q = \Delta W$

So, heat supplied to the system is converted into work done and *vice-versa*. Heat is supplied to the system for clockwise cycle and heat is taken from the system for anti-clockwise cycle.

Example 15.39 Figure shows a process ABCA performed on an ideal gas. Find the net heat given to the system during the process.



Sol. Since, the process is cyclic, hence the change in internal energy is zero.

The heat given to the system is then equal to the work done by it.

The work done in part AB is $W_1 = 0$ (the volume remains constant). The part BC represents an isochoric process, so the work done by the gas during this part is

$$W_2 = nRT_2 \ln \frac{V_2}{V_1}$$

During the part CA, $V \propto T$. So, V/T is constant and hence,

$$p = \frac{nRT}{V} \text{ is constant.}$$

The work done by the gas during the part CA is

$$\begin{aligned} W_3 &= p(V_1 - V_2) = nRT_1 - nRT_2 \\ &= -nR(T_2 - T_1) \end{aligned}$$

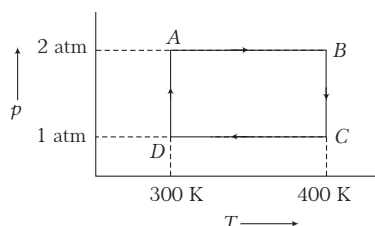
The net work done by the gas in the process ABCA is

$$W = W_1 + W_2 + W_3 = nR \left[T_2 \ln \frac{V_2}{V_1} - (T_2 - T_1) \right]$$

The same amount of heat is given to the gas.

Example 15.40 Two moles of helium gas undergo a cyclic process as shown in figure. Assuming the gas to be ideal, calculate the following quantities in the process:

- the net work done,
- the net change in internal energy,
- the net change in heat energy. (Take, $R = 8.32 \text{ J mol}^{-1}\text{K}^{-1}$)



Sol. Given, $n = 2$, $C_V = 3R/2$, $C_p = 5R/2$,

$$T_A = T_D = 300 \text{ K}, T_B = T_C = 400 \text{ K}$$

- (i) $A \rightarrow B$ (isobaric process):

$$\Delta W_{A \rightarrow B} = p \Delta V = nR \Delta T = 2R(400 - 300) = 200R$$

$B \rightarrow C$ (isothermal expansion):

$$\Delta W_{B \rightarrow C} = nRT_B \ln(p_B/p_C) = 2R \times 400 \ln(2/1) = 800R \ln 2$$

$C \rightarrow D$ (isobaric process):

$$\Delta W_{C \rightarrow D} = nR \Delta T = 2R(300 - 400) = -200R$$

$D \rightarrow A$ (isothermal compression):

$$\Delta W_{D \rightarrow A} = nRT_A \ln(p_D/p_A) = 2R \times 300 \ln(1/2) = -600R \ln 2$$

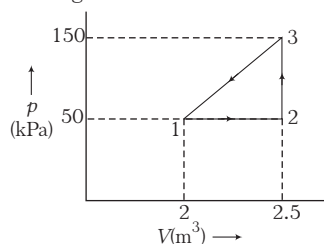
$$\begin{aligned} \Delta W_{\text{cyclic}} &= \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CD} + \Delta W_{DA} \\ &= 200R + 800R \ln 2 - 200R - 600R \ln 2 \\ &= 200R \ln 2 = 200 \times 8.3 \times 0.693 \approx 1150 \text{ J} \end{aligned}$$

- (ii) For cyclic process, $\Delta T = 0$, $\Delta U = 0$

- (iii) For cyclic process, $\Delta Q = \Delta W = 1150 \text{ J}$

Example 15.41 A gas is initially at a pressure of 50 kPa. Its pressure is kept constant and the volume is changed from 2 m^3 to 2.5 m^3 . Its volume is now kept constant and the pressure is increased from 50 kPa to 150 kPa. The gas is brought back to its initial state via a process in which, the pressure varies linearly with its volume. Whether the heat is supplied to or extracted from the gas in the complete cycle?

Sol. First draw p - V diagram as shown below.



According to the p - V diagram, work done

$$\begin{aligned} \Delta W_{\text{cyclic}} &= -\frac{1}{2} (150 - 50) \times 10^3 \times (2.5 - 2.0) \\ &= -25000 \text{ J} \end{aligned}$$

In the cyclic process,

$$\Delta Q = \Delta W = -25000 \text{ J}$$

Negative sign shows that heat is extracted from the gas.

Efficiency of a cycle

Efficiency (η) of a cycle can be defined as

$$\eta = \left(\frac{\text{Work done by the working substance during a cycle}}{\text{Heat supplied to the gas during the cycle}} \right) \times 100$$

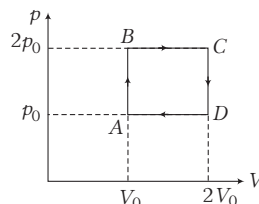
$$\eta = \frac{W_{\text{total}}}{|Q_{\text{absorbed}}|} \times 100 = \frac{|Q_{\text{absorbed}}| - |Q_{\text{released}}|}{|Q_{\text{absorbed}}|} \times 100$$

$$\text{Thus, } \eta = \frac{W_{\text{total}}}{|Q_{\text{absorbed}}|} \times 100 = \left\{ 1 - \frac{|Q_{\text{released}}|}{|Q_{\text{absorbed}}|} \right\} \times 100$$

There cannot be a cycle whose efficiency is 100%. Hence, η is always less than 100%.

Thus, $W_{\text{total}} \neq Q_{\text{absorbed}}$

Example 15.42 N moles of a monoatomic gas is carried round the rectangular cycle ABCDA as shown in the diagram. The temperature at A is T_0 . Find the thermodynamic efficiency of the cycle.



Sol. First of all note that point C is at highest temperature and A is at lowest temperature. So, during process A to B and B to C, heat is being added to the cycle. Work output can be calculated by calculating the area under the cycle of p - V diagram. Then, efficiency can be calculated as work output upon heat input. Note that the temperature at A is T_0 , that at B is $2T_0$ and that at C is $4T_0$. Also, note C_V for monoatomic gas is $3R/2$.

$$\begin{aligned} \text{For A-B, } \Delta Q &= NC_V \Delta T \quad (\because \text{constant volume process}) \\ &= N \frac{3R}{2} (2T_0 - T_0) = \frac{N}{2} 3RT_0 \end{aligned}$$

$$\text{For B-C, } \Delta Q = NC_p \Delta T \quad (\because \text{constant pressure process})$$

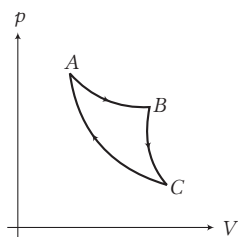
$$= N \frac{5R}{2} (4T_0 - 2T_0) = 5NRT_0$$

$$\text{Now, total heat input, } Q = \left(\frac{3}{2} + 5 \right) NRT_0 = \frac{13}{2} NRT_0$$

Work output = Area under the cycle

$$\begin{aligned}
 &= (2p_0 - p_0)(2V_0 - V_0) \\
 &= p_0 V_0 = NRT_0 \\
 \therefore \text{Efficiency} &= \frac{W}{Q} = \frac{NRT_0}{\frac{13}{2}NRT_0} = \frac{2}{13} \text{ or } 15.38\%
 \end{aligned}$$

Example 15.43 An ideal gas expands isothermally along AB and does 700 J of work.



- (i) How much heat does the gas exchange along AB?
(ii) The gas then expands adiabatically along BC and does 400 J of work. When the gas returns to A along CA, it exhausts 100 J of heat to its surroundings. How much work is done on the gas along this path?

Sol. (i) AB is an isothermal process.

$$\begin{aligned}
 \text{Hence, } \Delta U_{AB} &= 0 \\
 \text{and } Q_{AB} &= W_{AB} = 700 \text{ J}
 \end{aligned}$$

(ii) BC is an adiabatic process.

$$\text{Hence, } Q_{BC} = 0 \Rightarrow W_{BC} = 400 \text{ J} \quad (\text{given})$$

$$\therefore \Delta U_{BC} = -W_{BC} = -400 \text{ J}$$

ABCA is a cyclic process and internal energy is a state function.

Therefore, $(\Delta U)_{\text{whole cycle}} = 0$

$$= \Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA}$$

and from first law of thermodynamics,

$$Q_{AB} + Q_{BC} + Q_{CA} = W_{AB} + W_{BC} + W_{CA}$$

Substituting the values,

$$700 + 0 - 100 = 700 + 400 + \Delta W_{CA}$$

$$\therefore \Delta W_{CA} = -500 \text{ J}$$

Negative sign implies that work is done on the gas.

Different values in different processes

Process	Q (J)	W (J)	ΔU (J)
AB	700	700	0
BC	0	400	-400
CA	-100	-500	400
For complete cycle	600	600	0

Note Total work done is 600 J, which implies that area of the closed curve is also 600 J.

Example 15.44 In example 15.43, find the efficiency of the given cycle.

Sol. From table of the previous example, we can see that Q_{absorbed} during the cycle is 700 J, while the total work done in the cycle is 600 J.

$$\therefore \eta = \frac{W_{\text{total}}}{|Q_{\text{absorbed}}|} \times 100 = \left(\frac{600}{700} \right) \times 100 = 85.71\%$$

CHECK POINT 15.3

1. Choose the incorrect statement related to an isobaric process.

(a) $\frac{V}{T} = \text{constant}$

(b) $W = p \Delta V$

(c) Heat given to a system is used up in raising the temperature only

(d) None of the above

2. When 1 g of water at 0°C and $1 \times 10^5 \text{ Nm}^{-2}$ pressure is converted into ice of volume 1.091 cm^3 , the work done will be

(a) 0.0091 J

(b) 0.0182 J

(c) 0.091 J

(d) Data insufficient

3. An ideal gas of volume $1.5 \times 10^{-3} \text{ m}^3$ and at pressure $1.0 \times 10^5 \text{ Pa}$ is supplied with 70 J of energy. The volume increases to $1.7 \times 10^{-3} \text{ m}^3$, the pressure remaining constant. The internal energy of the gas is

(a) increased by 90 J

(b) increased by 70 J

(c) increased by 50 J

(d) decreased by 50 J

4. In an isochoric process, if $T_1 = 27^\circ\text{C}$ and $T_2 = 127^\circ\text{C}$, then $\frac{p_1}{p_2}$

will be equal to

(a) $\frac{9}{59}$

(b) $\frac{2}{3}$

(c) $\frac{3}{4}$

(d) None of these

5. In an isothermal expansion of an ideal gas, select the wrong statement.

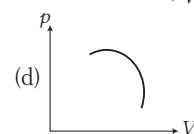
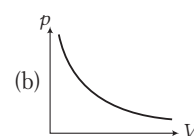
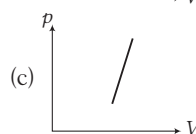
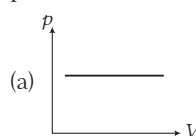
(a) There is no change in the temperature of the gas.

(b) There is no change in the internal energy of the gas.

(c) The work done by the gas is equal to the heat supplied to the gas.

(d) The work done by the gas is equal to the change in its internal energy.

6. Which of the p - V diagrams best represents an isothermal process?



7. A cycle tyre bursts suddenly. This represents an
 (a) isothermal process (b) isobaric process
 (c) isochoric process (d) adiabatic process
8. A gas for which $\gamma = 1.5$ is suddenly compressed to $(1/4)$ th of the initial volume. Then, the ratio of the final to the initial pressure is
 (a) 1 : 16 (b) 1 : 8 (c) 1 : 4 (d) 8 : 1
9. For a gas $\gamma = \frac{5}{3}$ and 640 cc of this gas is suddenly compressed to 80 cc. If the initial pressure is p , then the final pressure will be
 (a) $8p$ (b) $32p$ (c) $16p$ (d) $64p$
10. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The ratio $\frac{C_p}{C_v} = \gamma$ for the gas is
 (a) 2 (b) $\frac{3}{2}$ (c) $\frac{5}{3}$ (d) $\frac{4}{3}$
11. Two identical samples of a gas are allowed to expand
 (i) isothermally (ii) adiabatically. Work done is
 (a) more in the isothermal process
 (b) more in the adiabatic process
 (c) zero in both of them
 (d) equal in both processes
12. The volume of a gas is reduced adiabatically to $(1/4)$ th of its volume at 27°C . If $\gamma = 1.4$, the new temperature will be
 (a) $(300) 2^{0.4}$ K (b) $(300) 4^{1.4}$ K (c) $(300) 4^{0.4}$ K (d) $(300) 2^{1/4}$ K
13. An ideal gas at a pressure of 1 atm and temperature of 27°C is compressed adiabatically until its pressure becomes 8 times the initial pressure, then the final temperature is ($\gamma = 3/2$)
 (a) 627°C (b) 527°C (c) 427°C (d) 327°C
14. In an adiabatic expansion, a gas does 25 J of work while in an adiabatic compression 100 J of work is done on a gas. The change of internal energy in the two processes respectively are
 (a) 25 J and -100 J (b) -25 J and 100 J
 (c) -25 J and -100 J (d) 25 J and 100 J

HEAT ENGINE

A heat engine is a device which converts heat energy into mechanical energy.

The essential parts of a heat engine are as follows

Source It is a reservoir of heat at high temperature and infinite thermal capacity. Any amount of heat can be extracted from it at constant temperature.

Sink It is a reservoir of heat at low temperature and infinite thermal capacity. Any amount of heat can be given to the sink at constant temperature.

Working substance A material which absorbs heat energy from the source and converts it into mechanical energy (useful work) by rejecting some of the heat to sink is called working substance. *e.g.* Steam, petrol, etc.

Working

The working substance absorbs heat Q_1 (as shown in figure) from the source, does an amount of work W , returns the remaining amount of heat Q_2 to the sink and comes back to its original state and there occurs no change in its internal energy.

Source is always at higher temperature than sink.

By repeating the same cycle over and over again, work can be continuously obtained from an engine,

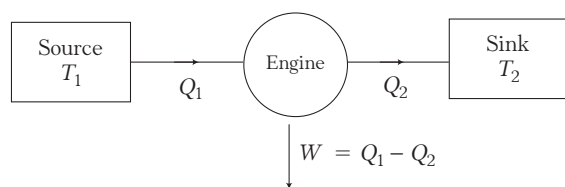


Fig. 15.25 Working of heat engine

Thermal efficiency

Thermal efficiency of a heat engine is defined as the ratio of net work done per cycle by the engine to the total amount of heat absorbed per cycle by the working substance from the source. It is denoted by η .

$$\text{Thus, } \eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\text{Thermal efficiency, } \eta = 1 - \frac{Q_2}{Q_1}$$

where,

Q_1 = heat absorbed by the working substance,

Q_2 = heat released by the working substance to the sink,

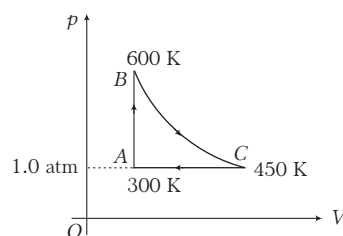
η = thermal efficiency of heat engine

and W = work done.

For an ideal heat engine, $Q_2 = 0$

So $\eta = 1$, i.e. 100% (practically not possible)

Example 15.45 The p - V diagram of 0.2 mol of an ideal diatomic gas used as working substance in heat engine is as shown in figure. Process BC is adiabatic. The value of γ for this gas is 1.4.



- (i) Find the pressure and volume at points A, B and C.
 (ii) Calculate ΔQ , ΔW and ΔU for each of the three processes.
 (iii) Find the thermal efficiency of the cycle.

(Take, $1 \text{ atm} = 1.01 \times 10^5 \text{ Nm}^{-2}$)

Sol. (i) Given, $p_A = p_C = 1 \text{ atm} = 1.01 \times 10^5 \text{ Nm}^{-2}$

Process AB is an isochoric process.

$$\therefore p \propto T \text{ or } \frac{p_B}{p_A} = \frac{T_B}{T_A}$$

$$\therefore p_B = \left(\frac{T_B}{T_A} \right) p_A = \left(\frac{600}{300} \right) (1 \text{ atm}) = 2 \text{ atm}$$

$$= 2.02 \times 10^5 \text{ Nm}^{-2}$$

From ideal gas equation, $V = \frac{nRT}{p}$

$$\therefore V_A = V_B = \frac{nRT_A}{p_A} = \frac{(0.2)(8.31)(300)}{(1.01 \times 10^5)}$$

$$\approx 5.0 \times 10^{-3} \text{ m}^3 = 5 \text{ L}$$

$$\text{and } V_C = \frac{nRT_C}{p_C} = \frac{(0.2)(8.31)(450)}{(1.01 \times 10^5)}$$

$$\approx 7.4 \times 10^{-3} \text{ m}^3 = 7.4 \text{ L}$$

State	p	V
A	1 atm	5 L
B	2 atm	5 L
C	1 atm	7.4 L

(ii) **Process AB** is an isochoric process.

Hence, $\Delta W_{AB} = 0$

$$\Delta Q_{AB} = \Delta U_{AB} = nC_V \Delta T = n \left(\frac{5}{2} R \right) (T_B - T_A)$$

$$= (0.2) \left(\frac{5}{2} \right) (8.31) (600 - 300) \approx 1246 \text{ J}$$

Process BC is an adiabatic process.

Hence, $\Delta Q_{BC} = 0$

$$\therefore \Delta W_{BC} = -\Delta U_{BC}$$

$$\Delta U_{BC} = nC_V \Delta T = nC_V (T_C - T_B)$$

$$= (0.2) \left(\frac{5}{2} R \right) (450 - 600)$$

$$= (0.2) \left(\frac{5}{2} \right) (8.31) (-150) \text{ J} \approx -623 \text{ J}$$

$$\therefore \Delta W_{BC} = -\Delta U_{BC} = 623 \text{ J}$$

Process CA is an isobaric process.

$$\text{Hence, } \Delta Q_{CA} = nC_p \Delta T = n \left(\frac{7}{2} R \right) (T_A - T_C)$$

$$= (0.2) \left(\frac{7}{2} \right) (8.31) (300 - 450) \approx -872 \text{ J}$$

$$\Delta U_{CA} = nC_V \Delta T = \frac{\Delta Q_{CA}}{\gamma} \quad \left(\because \gamma = \frac{C_p}{C_V} \right)$$

$$= -\frac{872}{1.4} \approx -623 \text{ J}$$

$$\therefore \Delta W_{CA} = \Delta Q_{CA} - \Delta U_{CA} = -872 \text{ J} + 623 \text{ J} = -249 \text{ J}$$

Process	ΔQ (in J)	ΔW (in J)	ΔU (in J)
AB	1246	0	1246
BC	0	623	-623
CA	-872	-249	-623
Total	374	374	0

(iii) Efficiency of the cycle,

$$\eta = \frac{W_{\text{total}}}{|Q_{\text{ve}}|} \times 100 = \frac{374}{1246} \times 100 = 30\%$$

REFRIGERATOR

It is a device which works in the reverse direction of heat engine. Refrigerator takes heat Q_2 (as shown in figure) from a cold body (sink), work (W) is done on it and the work done together with the heat absorbed (i.e. Q_1) is rejected to the source.

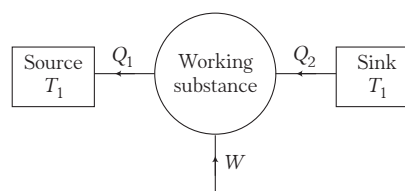


Fig. 15.26 Working of refrigerator

Coefficient of performance (COP)

The performance of a refrigerator is expressed by means of coefficient of performance β which is defined as the ratio of the heat extracted from the cold body to the work needed to transfer it to the hot body.

$$\text{i.e. } \beta = \frac{\text{Heat extracted}}{\text{Work done}}$$

$$= \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$$

$$\text{Coefficient of performance, } \beta = \frac{Q_2}{Q_1 - Q_2}$$

- A perfect refrigerator is one which transfers heat from cold to hot body without doing work.
 i.e. $W = 0$, so that $Q_1 = Q_2$ and hence, $\beta = \infty$

- For an ideal refrigerator or heat engine, we can take

$$\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

- Air conditioners are basically refrigerators whose refrigerated space is a room or a building.
- In domestic refrigerator, inside portion acts as a cold reservoir and the surrounding act as hot reservoir.

Relation between coefficient of performance and efficiency of heat engine

We know, $\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{Q_2/Q_1}{1 - Q_2/Q_1} \quad \dots(i)$

But $\eta = 1 - \frac{Q_2}{Q_1}$ or $\frac{Q_2}{Q_1} = 1 - \eta \quad \dots(ii)$

From Eqs. (i) and (ii), we get $\beta = \frac{1 - \eta}{\eta}$

Example 15.46 An ideal refrigerator runs between -23°C and 27°C . Find the heat rejected to atmosphere for every joule of work input.

Sol. Let the heat rejected to the atmosphere, $Q_1 = x$

and given $W = 1\text{ J}$. Now, $Q_2 = Q_1 - W = x - 1$

Given, hot temperature, $T_1 = 273 + 27 = 300\text{ K}$

and cold temperature, $T_2 = 273 - 23 = 250\text{ K}$

For an ideal refrigerator, we know that,

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$\therefore \frac{x}{x-1} = \frac{300}{250}$$

or $x = 6\text{ J}$

Example 15.47 Calculate the amount of work that must be done to freeze one gram of water at 0°C by means of an ideal refrigerator. Temperature of surroundings is 27°C . How much heat is passed to the surroundings in this process. (Take, latent heat of fusion, $L = 80\text{ cal g}^{-1}$.)

Sol. Heat released during freezing,

$$Q_2 = mL = 1 \times 80 = 80\text{ cal}$$

$$T_2 = 0^\circ\text{C} = 273\text{ K}$$

and $T_1 = 27^\circ\text{C} = 300\text{ K}$

Since, $\frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}$

$$\therefore \text{Work done, } W = \frac{Q_2(T_1 - T_2)}{T_2}$$

$$= \frac{80(300 - 273)}{273} = 7.91\text{ cal}$$

$\Rightarrow$ Heat passed to the surrounding,

$$Q_1 = Q_2 + W = (80 + 7.91) = 87.91\text{ cal}$$

SECOND LAW OF THERMODYNAMICS

The second law of thermodynamics gives a fundamental limitation to the efficiency of a heat engine and the coefficient of performance of a refrigerator. It says that efficiency of a heat engine can never be unity (or 100%). This implies that heat released to the cold reservoir can never be made zero.

For a refrigerator, the second law says that the coefficient of performance can never be infinite. This implies that external work can never be zero.

Second law of thermodynamics may be stated in various ways, but two statements are worth mentioning which are as follows

Kelvin-Planck statement

No process is possible which can completely convert heat absorbed from a reservoir into work.

Clausius statement

No process is possible which can transfer heat from a cold body to a hot body without doing any external work.

Reversible and irreversible processes

Reversible process

A process which could be reversed in such a way that the system and its surrounding returns exactly to their initial states with no other changes in the surroundings is known as reversible process.

e.g. If heat is absorbed in the direct process, then same amount of heat should be given out in the reverse process.

If work is done on the working substance in the direct process, then the same amount of work should be done by the working substance in the reverse process.

The conditions for reversibility are as follows

- There must be complete absence of dissipative forces such as friction, viscosity, electric resistance, etc.
- The direct and reverse processes must take place infinitely slowly.
- The temperature of the system must not differ appreciably from its surroundings.

Irreversible process

Any process which is not reversible exactly is an irreversible process. All natural processes such as conduction, radiation, radioactive decay, etc. are irreversible processes.

Carnot engine

The reversible engine which operates between two temperatures of source and sink is known as Carnot heat engine. The designed engine is a theoretical engine which is free from all the defects of a practical engine.

This engine cannot be realised in actual practice, however this can be taken as a standard against which the performance of an actual engine can be judged.

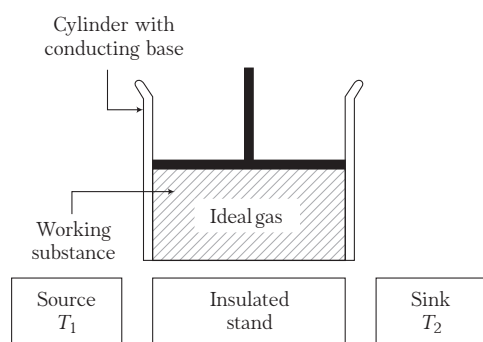


Fig. 15.27 A Carnot engine

The main parts of Carnot engine are as follows

Cylinder The cylinder has conducting base and insulating walls. It contains the gas as a working substance and an insulating and frictionless piston is attached with it as shown in the figure.

Source It is a hot reservoir at a temperature T_1 with conducting walls. It has infinite thermal capacity. Any amount of heat can be taken from it without changing the temperature.

Sink It is a cold reservoir at temperature T_2 . It has an infinite thermal capacity. So, any amount of heat can be rejected to it without changing the temperature of the sink.

Working substance We use an ideal gas as a working substance in the cylinder.

Insulating stand The base of the cylinder could be placed on the insulated stand, to isolate it completely from the surroundings.

Carnot cycle

As the engine works, the working substance of the engine undergoes a cycle known as Carnot cycle.

The Carnot cycle consists of the following four strokes

First stroke (Isothermal expansion) (Curve AB) The cylinder containing one mole of an ideal gas as working substance is allowed to expand slowly at the constant temperature T_1 by putting it on the source.

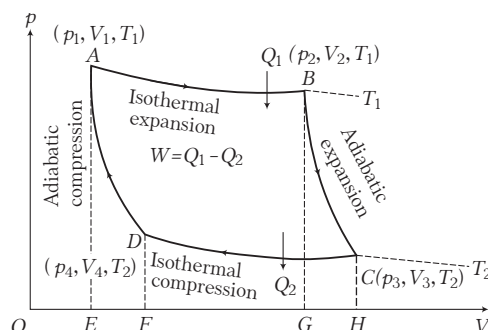


Fig. 15.28 Various processes in Carnot cycle

Work done = Heat absorbed by the gas

$$W_1 = Q_1 = \int_{V_1}^{V_2} p dV = RT_1 \log_e \left(\frac{V_2}{V_1} \right)$$

$$= \text{Area of ABGE}$$

Second stroke (Adiabatic expansion) (Curve BC) The cylinder is then placed on the non-conducting stand and the gas is allowed to expand adiabatically till the temperature falls from T_1 to T_2 .

$$W_2 = \int_{V_2}^{V_3} p dV = \frac{R}{(\gamma - 1)} (T_1 - T_2)$$

$$= \text{Area of BCHG}$$

Third stroke (Isothermal compression) (Curve CD) The cylinder is placed on the sink and the gas is compressed at constant temperature T_2 .

Work done = Heat released by the system

$$W_3 = Q_2 = - \int_{V_3}^{V_4} p dV = - RT_2 \log_e \frac{V_4}{V_3}$$

$$= RT_2 \log_e \frac{V_3}{V_4} = \text{Area of CDFH}$$

Fourth stroke (Adiabatic compression) (Curve DA) Finally, the cylinder is again placed on non-conducting stand and the compression is continued, so that gas returns to its initial stage.

$$W_4 = - \int_{V_4}^{V_1} p dV = - \frac{R}{\gamma - 1} (T_2 - T_1)$$

$$= \frac{R}{\gamma - 1} (T_1 - T_2) = \text{Area of ADFE}$$

Efficiency of the Carnot cycle

The efficiency of engine is defined as the ratio of net work done to the heat supplied, i.e.

$$\eta = \frac{\text{Net work done}}{\text{Heat input}} = \frac{W_{\text{net}}}{Q_1}$$

$$= \frac{W_1 + W_2 + W_3 + W_4}{Q_1}$$

$$= 1 - \frac{Q_2}{Q_1}$$

On further calculation, we get

$$\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

$$\therefore \text{Efficiency of Carnot engine, } \boxed{\eta = 1 - \frac{T_2}{T_1}}$$

- The efficiency of the Carnot engine depends only on the temperatures of source and sink.
- The efficiency of Carnot engine does not depend on the nature of the working substance.
- The efficiency of Carnot engine will be 100%, if the temperature of the sink is 0 K. As practically, we cannot attain a sink at 0 K, so it is not possible to have 100% efficiency.

Carnot theorem

According to Carnot theorem,

- A heat engine working between the two given temperatures T_1 of hot reservoir, i.e. source and T_2 of cold reservoir, i.e. sink cannot have efficiency more than that of the Carnot engine.
- The efficiency of the Carnot engine is independent of the nature of working substance.

Example 15.48 A Carnot engine takes in 3000 kcal of heat from a reservoir at 627°C and gives a part of it to a sink at 27°C . Find the work done by the engine.

Sol. Here, $T_1 = 273 + 627 = 900\text{ K}$ and $T_2 = 273 + 27 = 300\text{ K}$

$$\text{Now, } \eta = \frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow \frac{W}{3000\text{ kcal}} = 1 - \frac{300}{900}$$

$$\therefore \text{Work done by the engine, } W = 2000\text{ kcal} \\ = 2000 \times 4.2\text{ kJ} = 8.4 \times 10^6\text{ J}$$

Example 15.49 Carnot engine takes in a thousand kilo calories of heat from a reservoir at 827°C and exhausts it to a sink at 27°C . How much work does it perform? What is the efficiency of the engine?

Sol. Given, $Q_1 = 10^6\text{ cal}$, $T_1 = (827 + 273) = 1100\text{ K}$

and $T_2 = (27 + 273) = 300\text{ K}$

$$\therefore \frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

$$\Rightarrow Q_2 = \frac{T_2}{T_1} \cdot Q_1 = \left(\frac{300}{1100}\right)(10^6) = 2.73 \times 10^5\text{ cal}$$

$$\text{Efficiency of the engine, } \eta = \left(1 - \frac{T_2}{T_1}\right) \times 100$$

$$\text{or } \eta = \left(1 - \frac{300}{1100}\right) \times 100 = 72.72\%$$

Example 15.50 The efficiency of a Carnot engine at a particular source and sink temperature is $1/2$. When the sink temperature is reduced by 100°C , the engine's efficiency becomes $2/3$. Find the source temperature.

Sol. According to Carnot engine, $1 - \frac{T_2}{T_1} = \frac{1}{2}$... (i)

(T_2 = sink temperature, T_1 = source temperature)

$$\text{and } 1 - \frac{(T_2 - 100)}{T_1} = \frac{2}{3} \quad \dots (ii)$$

$$\text{or, } \frac{T_2}{T_1} = \frac{1}{2} \quad [\text{From Eq. (i)}]$$

$$\text{and } \frac{(T_2 - 100)}{T_1} = \frac{1}{3} \quad [\text{From Eq. (ii)}]$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{T_2}{T_2 - 100} = \frac{3}{2} \quad \text{or } T_2 = 300\text{ K}$$

$$\text{So, } T_1 = 600\text{ K}$$

Example 15.51 If a Carnot engine whose heat sink is at 27°C has an efficiency of 40%. By how many degrees should the temperature of the source be changed to increase the efficiency by 10% of the original efficiency?

Sol. Given, $T_2 = 27 + 273 = 300\text{ K}$, $\eta = 40\%$

$$\text{As, } \eta = 1 - \frac{T_2}{T_1}$$

$$\therefore \frac{T_2}{T_1} = 1 - \eta = 1 - \frac{40}{100} = \frac{60}{100} = \frac{3}{5}$$

$$\Rightarrow T_1 = \frac{5}{3} \times T_2 = \frac{5}{3} \times 300 = 500\text{ K}$$

$$\therefore \text{New efficiency, } \eta' = 40 + 4 = 44\%$$

($\because$ Increase in efficiency = 10% of 40 = 4%)

Let the new temperature of the source be $T_1'\text{ K}$, then

$$\eta' = 1 - \frac{T_2}{T_1'}$$

$$\Rightarrow \frac{44}{100} = 1 - \frac{300}{T_1'}$$

$$\Rightarrow \frac{300}{T_1'} = 1 - \frac{44}{100} = \frac{56}{100}$$

$$\Rightarrow T_1' = \frac{100 \times 300}{56} = 535.7\text{ K}$$

$$\therefore \text{Increase in the temperature of the source,} \\ = 535.7 - 500 \\ = 35.7\text{ K or } 35.7^\circ\text{ C}$$

Example 15.52 A scientist claims to have developed 60% efficient engine while working between 27°C and 327°C . Does he claim right?

Sol. Here, sink temperature, $T_2 = 273 + 27 = 300\text{ K}$

(Low temperature happens to be sink temperature)

and source temperature, $T_1 = 327 + 273 = 600\text{ K}$

So, within this temperature difference, the maximum possible efficiency is $\eta_{\text{max}} = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{600} = 0.5 = 50\%$

But the scientist is claiming more efficiency. So, as per Carnot theorem his claim is not correct.

Entropy

Like pressure, volume, temperature, internal energy, etc., we have another thermodynamic variable of a system, named entropy. It is related to the disorder of molecular motion of the system. Greater the randomness or disorder, greater is the entropy.

The change in entropy,

$$\text{i.e. } \Delta S = \frac{\text{Heat absorbed by the system}}{\text{Absolute temperature}} = \frac{\Delta Q}{T}$$

This relation is also called mathematical form of second law of thermodynamics.

SI unit of ΔS is J/K and its dimensional formula is $[\text{ML}^2\text{T}^{-2}\text{K}^{-1}]$.

Note We always define change in entropy rather than its absolute value.

Change in entropy for solids and liquids

- (i) When heat is supplied to a solid and its state changes such that temperature remains constant, then

$$\text{Change in entropy, } \Delta S = \frac{\Delta Q}{T}$$

$$\Rightarrow dS = \frac{dQ}{T} = \frac{\pm |mL|}{T}$$

Positive sign is used for heat absorption and negative sign is used for heat rejection.

- (ii) When temperature of a substance changes from T_1

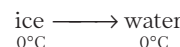
$$\begin{aligned} \text{to } T_2, \text{ then } dS &= \int \frac{dQ}{T} = ms \int_{T_1}^{T_2} \frac{dT}{T} = ms \ln \left(\frac{T_2}{T_1} \right) \\ &= 2.303 ms \log_{10} \left(\frac{T_2}{T_1} \right) \end{aligned}$$

where, m = mass of the substance

and s = specific heat of the substance.

Example 15.53 200 g of ice at 0°C is converted into water vapour at 100°C . Evaluate the change in entropy.

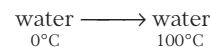
Sol. Consider conversion of ice into water



Change in entropy during this conversion,

$$\begin{aligned} \Delta S_1 &= \frac{\Delta Q}{T_1} = \frac{mL_{\text{ice}}}{T_1} = \frac{200 \times 80}{273} \\ &= 58.6 \text{ cal/K} \end{aligned}$$

Again, consider rise of temperature of water from 0°C to 100°C .



Change in entropy for this increase in temperature of the water,

$$\begin{aligned} \Delta S_2 &= ms_w \ln \frac{T_2}{T_1} = 200 \times 1 \times \ln \left(\frac{373}{273} \right) \\ &= 200 \ln (1.4) \text{ cal/K} \end{aligned}$$

Again, consider conversion of water to vapour



Change in entropy for this conservation is

$$\begin{aligned} \Delta S_3 &= \frac{\Delta Q}{T_2} = \frac{mL_{\text{steam}}}{T_2} = \frac{200 \times 540}{373} \\ &= 289.5 \text{ cal/K} \end{aligned}$$

Net change in entropy, $\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3$

$$\begin{aligned} &= 58.6 + 200 \ln(1.4) + 289.5 \\ &= 348.1 + 200 \times 2.303 \log_{10} (1.4) \\ &= 415.4 \text{ cal/K} \end{aligned}$$

CHECK POINT 15.4

1. A system undergoes a cyclic process in which it absorbs Q_1 heat and gives out Q_2 heat. The efficiency of the process is η and work done is W . Select the correct statement.
(a) $W = \frac{Q_1 - Q_2}{Q_2}$
(b) $\eta = \frac{W}{Q_1}$
(c) Both (a) and (b) are correct
(d) Both (a) and (b) are wrong
2. An ideal heat engine is operating between 227°C and 127°C . It absorbs 10^4 J amount of heat at the higher temperature. The amount of heat converted into work is
(a) 2000 J (b) 4000 J
(c) 8000 J (d) 5600 J
3. The inside and outside temperature of a refrigerator are 273 K and 303 K, respectively. Assuming that refrigerator cycle is reversible, for every joule of work done, the heat delivered to the surroundings will be
(a) 10 J (b) 20 J
(c) 30 J (d) 50 J
4. A Carnot engine has an efficiency of 50% at sink temperature 50°C . Calculate the temperature of source.
(a) 133°C (b) 143°C
(c) 100°C (d) 373°C
5. If a Carnot engine functions at source temperature = 127°C and sink temperature = 87°C , what is its efficiency?
(a) 10% (b) 25%
(c) 40% (d) 50%
6. For which combination of temperatures, the efficiency of Carnot's engine is highest?
(a) 80 K, 60 K
(b) 100 K, 80 K
(c) 60 K, 40 K
(d) 40 K, 20 K
7. A Carnot engine is made to work between 200°C and 0°C first and then between 0° and -200°C . The ratio of efficiencies $\left(\frac{\eta_2}{\eta_1}\right)$ of the engine in the two cases is
(a) 1 : 1.5 (b) 1 : 1
(c) 1 : 2 (d) 1.73 : 1
8. A Carnot engine with its cold body at 17°C has 50% efficiency. If the temperature of its hot body is now increased by 145°C , the efficiency becomes
(a) 55% (b) 60%
(c) 40% (d) 45%
9. The heat reservoir of an ideal Carnot engine is at 800 K and its sink is at 400 K. The amount of heat taken in it in one second to produce useful mechanical work at the rate of 750 J is
(a) 2250 J (b) 1125 J
(c) 1500 J (d) 750 J
10. Entropy of a system is related with
(a) disorder of molecular motion of the system
(b) volume of the system
(c) pressure of the system
(d) None of the above

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1** Two identical containers joined by a small pipe initially contain the same gas at pressure p_0 and absolute temperature T_0 . One container is now maintained at the same temperature while the other is heated to $2T_0$. The common pressure of the gases will be

(a) $\frac{3}{2} p_0$ (b) $\frac{4}{3} p_0$ (c) $\frac{5}{3} p_0$ (d) $2p_0$

- 2** An ideal monoatomic gas is compressed (no heat being added or removed in the process), so that its volume is halved. The ratio of the new pressure to the original pressure is

(a) $(2)^{3/5}$ (b) $(2)^{4/3}$ (c) $(2)^{3/4}$ (d) $(2)^{5/3}$

- 3** If a Carnot's engine functions at source temperature = 137°C and sink temperature = 97°C , what is its efficiency?

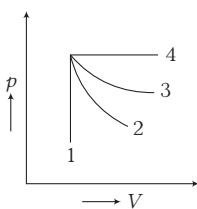
(a) 9.75% (b) 10.5% (c) 40% (d) 50%

- 4** A gas is compressed at a constant pressure of 50 Nm^{-2} from a volume of 10 m^3 to a volume of 4 m^3 . Energy of 100 J then added to the gas by heating. Its internal energy is

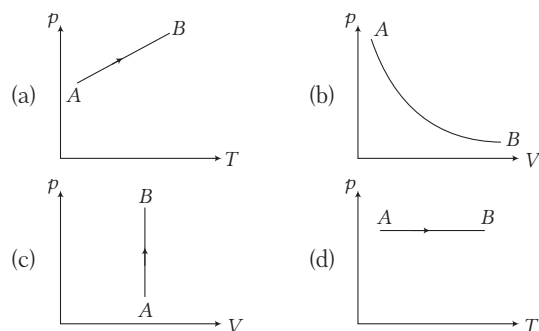
(a) increased by 400 J (b) increased by 200 J
(c) increased by 100 J (d) decreased by 200 J

- 5** An ideal gas undergoes four different processes from the same initial state (figure). Four processes are adiabatic, isothermal, isobaric and isochoric. Out of 1, 2, 3 and 4 which one is adiabatic?

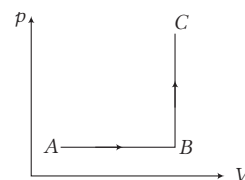
(a) 4 (b) 3 (c) 2 (d) 1



- 6** The process $\Delta U = 0$, for an ideal gas can be best represented in the form of a graph



- 7** The p - V diagram of a system undergoing thermodynamic transformation as shown in figure. The work done by the system is going from $A \rightarrow B \rightarrow C$ is 30 J and 40 J heat is given to the system. The change in internal energy between A and C is



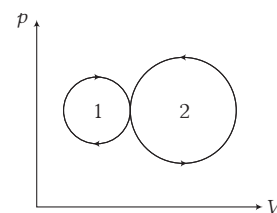
(a) 10 J (b) 70 J (c) 84 J (d) 134 J

- 8** A thermally insulated rigid container contain an ideal gas heated by a filament of resistance $100\ \Omega$ through a current of 1 A for 5 min , then change in internal energy is

(a) 0 kJ (b) 10 kJ (c) 20 kJ (d) 30 kJ

- 9** In the p - V diagram as shown in figure, the net amount of work done will be

(a) positive
(b) negative
(c) zero
(d) infinity



- 10** An ideal gas is allowed to expand freely against vacuum in a rigid insulated container. The gas undergoes

(a) an increase in its internal energy
(b) a decrease in its internal energy
(c) Neither increase nor decrease in temperature or internal energy
(d) an increase in temperature

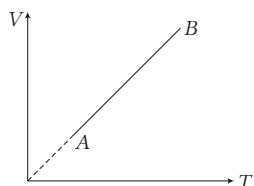
- 11** An ideal gas mixture filled inside a balloon expands according to the relation $pV^{2/3} = \text{constant}$. The temperature inside the balloon is

(a) increasing (b) decreasing
(c) constant (d) Cannot be defined

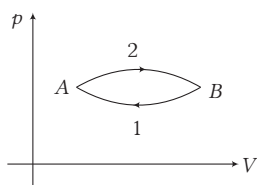
- 12** The internal energy of a gas is given by $U = 2pV$. It expands from V_0 to $2V_0$ against a constant pressure p_0 . The heat absorbed by the gas in the process is

(a) $2p_0V_0$ (b) $4p_0V_0$ (c) $3p_0V_0$ (d) p_0V_0

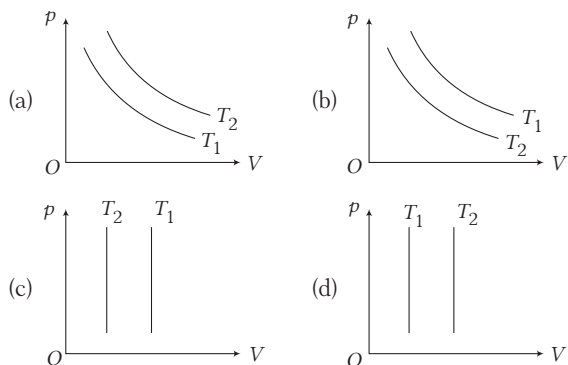
- 13** An ideal monoatomic gas undergoes the process AB as shown in the figure. If the heat supplied and the work done in the process are ΔQ and ΔW , respectively. The ratio $\Delta Q : \Delta W$ is



- (a) 2.50 (b) 1.67
(c) 1.25 (d) 0.40
- 14** The figure shows two paths for the change of state of a gas from A to B . The ratio of molar heat capacities in path 1 and path 2 is



- (a) < 1 (b) > 1
(c) 1 (d) data insufficient
- 15.** An ideal gas expands in such a manner that its pressure and volume can be related by equation $pV^2 = \text{constant}$. During this process, the gas is
- (a) heated
(b) cooled
(c) neither heated nor cooled
(d) first heated and then cooled
- 16** p - V diagram of a diatomic gas is a straight line passing through origin. The molar heat capacity of the gas in the process will be
- (a) $4R$ (b) $2.5R$
(c) $3R$ (d) $\frac{4R}{3}$
- 17** Identify the graph(s) which correctly represents an isotherm at two temperatures T_1 and T_2 ($T_2 > T_1$).



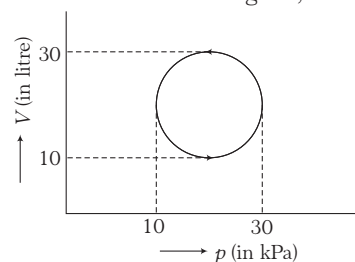
- 18** One mole of a perfect gas in a cylinder fitted with a piston has a pressure p , volume V and temperature T . If the temperature is increased by 1 K keeping pressure constant, the increase in volume is

(a) $\frac{2V}{273}$ (b) $\frac{V}{91}$ (c) $\frac{V}{273}$ (d) V

- 19** Ideal monoatomic gas is taken through a process $dQ = 2dU$. The molar heat capacity for the process is (where, dQ is heat supplied and dU is change in internal energy)

(a) $2.4R$ (b) $3R$ (c) R (d) $2R$

- 20** Heat energy absorbed by a system is going through a cyclic process as shown in figure, is



- (a) $10^7 \pi \text{ J}$ (b) $10^4 \pi \text{ J}$ (c) $10^2 \pi \text{ J}$ (d) $10^3 \pi \text{ J}$
- 21** Unit mass of a liquid with volume V_1 is completely changed into a gas of volume V_2 at a constant external pressure p at temperature T . If the latent heat of evaporation is L , then the increase in the internal energy of the system is
- (a) zero (b) $p(V_2 - V_1)$
(c) $L - p(V_2 - V_1)$ (d) L

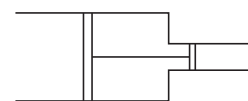
- 22** A monoatomic ideal gas, initially at temperature T_1 , is enclosed in a cylinder fitted with a frictionless piston. The gas is allowed to expand adiabatically to a temperature T_2 by releasing the piston suddenly. If L_1 and L_2 are lengths of the gas column before and after expansion respectively, then T_1/T_2 is given by

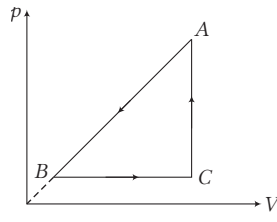
(a) $\left(\frac{L_1}{L_2}\right)^{2/3}$ (b) $\left(\frac{L_1}{L_2}\right)^{5/3}$ (c) $\left(\frac{L_2}{L_1}\right)^{5/3}$ (d) $\left(\frac{L_2}{L_1}\right)^{2/3}$

- 23** A refrigerator, whose coefficient of performance β is 4, extracts heat from the cooling compartment at the rate of 400 J per cycle. How much work per cycle is required to operate the refrigerator?

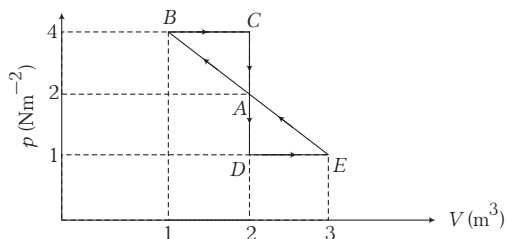
(a) 100 J (b) 200 J
(c) 50 J (d) 25 J

- 24** Two pistons can move freely inside a horizontal cylinder having two sections of unequal cross-sections. The pistons are joined by an inextensible, light string and some gas is enclosed between the pistons.

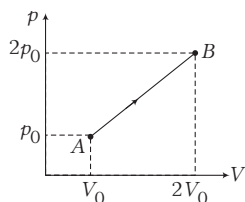


- On heating the system, the piston will
- move to the left
 - move to the right
 - remain stationary
 - (a) or (c) depending upon the initial pressure of the gas
- 25** Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 , if the process is isothermal W_2 , if isobaric and W_3 , if adiabatic, then
- $W_2 > W_1 > W_3$
 - $W_2 > W_3 > W_1$
 - $W_1 > W_2 > W_3$
 - $W_1 > W_3 > W_2$
- 26** The pressure ($1 \times 10^5 \text{ Nm}^{-2}$) of the air filled in a vessel is decreased adiabatically so much as to increase its volume three times.
- The air pressure is (γ for air = 1.4, $\log_{10} 3 = 0.4771$, $\log_{10} 2.148 = 0.33206$)
- $2.14 \times 10^4 \text{ Nm}^{-2}$
 - $2.19 \times 10^5 \text{ Nm}^{-2}$
 - $1.14 \times 10^5 \text{ Nm}^{-2}$
 - $3.14 \times 10^4 \text{ Nm}^{-2}$
- 27** The equation of a state of a gas is given by $p(V - b) = nRT$. 1 mol of a gas is isothermally expanded from volume V to $2V$, the work done during the process is
- $RT \ln \left(\frac{2V - b}{V - b} \right)$
 - $RT \ln \left(\frac{V - b}{V} \right)$
 - $RT \ln \left(\frac{V - b}{2V - b} \right)$
 - $RT \ln \left(\frac{V}{V - b} \right)$
- 28** A monoatomic gas is supplied the heat Q very slowly keeping the pressure constant. The work done by the gas will be
- $\frac{2}{3} Q$
 - $\frac{3}{5} Q$
 - $\frac{2}{5} Q$
 - $\frac{1}{5} Q$
- 29** 70 cal of heat are required to raise the temperature of 2 mol of an ideal gas at constant pressure from 30°C to 35°C . The amount of heat required in calories to raise the temperature of the same gas through the same range (30°C - 35°C) at constant volume is
- 30 cal
 - 50 cal
 - 40 cal
 - 90 cal
- 30** The temperature of a hypothetical gas increases to $\sqrt{2}$ times when compressed adiabatically to half the volume. Its equation can be written as
- $pV^{3/2} = \text{constant}$
 - $pV^{5/2} = \text{constant}$
 - $pV^{7/3} = \text{constant}$
 - $pV^{4/3} = \text{constant}$
- 31** A gas mixture consists of 4 mol of oxygen and 6 mol of argon at temperature T . Neglecting all vibrational modes, the total internal energy of the system is
- $4RT$
 - $14RT$
 - $8RT$
 - $19RT$
- 32** One mole of an ideal monoatomic gas is at 360 K and a pressure of 10^5 Pa . It is compressed at constant pressure until its volume is halved. Taking R as $8.3 \text{ J mol}^{-1} \text{ K}^{-1}$ and the initial volume of the gas as $3.0 \times 10^{-2} \text{ m}^3$, the work done on the gas is
- -1500 J
 - $+1500 \text{ J}$
 - -3000 J
 - $+3000 \text{ J}$
- 33** If an average person jogs, he produces $145 \times 10^3 \text{ cal/min}$. This is removed by the evaporation of sweat. The amount of sweat evaporated per minute (assuming 1 kg requires $580 \times 10^3 \text{ cal}$ for evaporation) is
- 0.25 kg
 - 2.25 kg
 - 0.05 kg
 - 0.20 kg
- 34** An ideal monoatomic gas undergoes a process in which its internal energy U and density ρ vary as $U\rho = \text{constant}$. The ratio of change in internal energy and the work done by the gas is
- $\frac{3}{2}$
 - $\frac{-2}{3}$
 - $\frac{1}{3}$
 - $\frac{3}{5}$
- 35** The molar heat capacity in a process of a diatomic gas, if it does a work of $Q/4$ when a heat of Q is supplied to it is
- $\frac{2}{5} R$
 - $\frac{5}{2} R$
 - $\frac{10}{3} R$
 - $\frac{6}{7} R$
- 36** The amount of heat required to raise the temperature of 1 mole of a monoatomic gas from 20°C to 30°C at constant volume is H . Then, the amount of heat required to raise the temperature of 2 mol of a diatomic gas from 20°C to 25°C at constant pressure is
- $2H$
 - $\frac{4}{3} H$
 - $\frac{5}{3} H$
 - $\frac{7}{3} H$
- 37** p - V diagram of a cyclic process $ABCA$ is as shown in figure. Choose the correct option.
- $\Delta Q_{A \rightarrow B} = \text{negative}$
 - $\Delta U_{B \rightarrow C} = \text{positive}$
 - $\Delta W_{CAB} = \text{negative}$
 - All of the above
- 
- 38** Two different ideal diatomic gases A and B are initially in the same state. A and B are then expanded to same final volume through adiabatic and isothermal process, respectively. If p_A , p_B and T_A , T_B represents the final pressure and temperatures of A and B respectively, then
- $p_A < p_B$ and $T_A < T_B$
 - $p_A > p_B$ and $T_A > T_B$
 - $p_A > p_B$ and $T_A < T_B$
 - $p_A < p_B$ and $T_A > T_B$

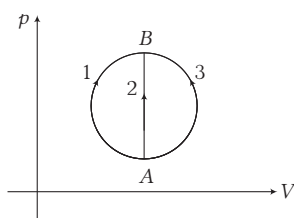
- 39** One mole of a monoatomic gas is carried along process $ABCDEA$ as shown in the diagram. Find the net work done by gas



- (a) $\frac{3}{2}$ J (b) 1 J (c) $\frac{1}{2}$ J (d) 0 J
- 40** The p - V diagram of 2 g of helium gas for a certain process $A \rightarrow B$ is shown in the figure. What is the heat given to the gas during the process $A \rightarrow B$?



- (a) $4p_0V_0$ (b) $6p_0V_0$ (c) $3p_0V_0$ (d) $2p_0V_0$
- 41** A gas undergoes A to B through three different processes 1, 2, and 3 as shown in the figure. The heat supplied to the gas is Q_1 , Q_2 and Q_3 respectively, then



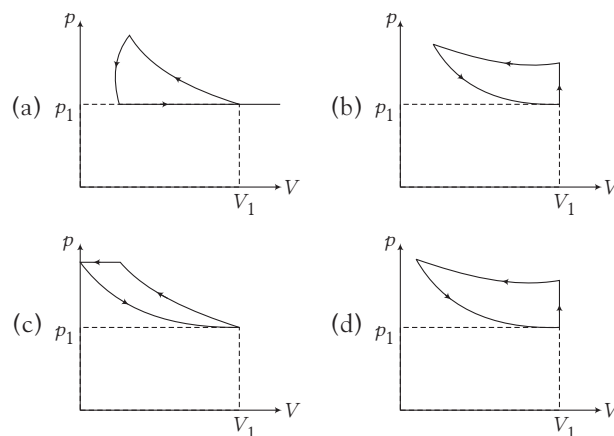
- (a) $Q_1 = Q_2 = Q_3$ (b) $Q_1 < Q_2 < Q_3$
(c) $Q_1 > Q_2 > Q_3$ (d) $Q_1 = Q_3 > Q_2$
- 42** When an air bubble rises from the bottom to the surface of a lake, its radius becomes double. Find the depth of the lake. (Given that the atmospheric pressure is equal to the pressure due to a column of water 10 m high. Assume constant temperature and disregard surface tension.)
- (a) 30 m (b) 40 m (c) 70 m (d) 80 m
- 43** A container of volume 1 m^3 is divided into two equal compartments by a partition. One of these compartments contains an ideal gas at 300 K. The other compartment is vacuum. The whole system is thermally isolated from its surroundings. The partition is removed in the gas container. Its temperature now would be

- (a) 300 K (b) greater than 300 K
(c) less than 300 K (d) Data insufficient

- 44** A Carnot engine whose low temperature reservoir is at 7°C has efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased?
- (a) 933 K (b) 432 K
(c) 373 K (d) 267 K

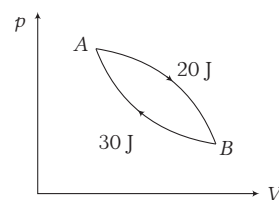
- 45** A certain mass of an ideal gas is at pressure p_1 and volume V_1 . It is compressed isothermally and then allowed to expand adiabatically until its pressure returns to p_1 . The gas is then allowed to expand its original volume.

Upon which of the following p - V graphs are these processes correctly shown?



- 46** Two different masses of a gas m and $2m$ are heated separately in vessels of equal volume. The T - p curve for mass $2m$ makes angle α with T -axis and that for mass m makes angle β with T -axis, then
- (a) $\tan \alpha = \tan \beta$ (b) $\tan \alpha = 2 \tan \beta$
(c) $\tan \beta = 2 \tan \alpha$ (d) None of these

- 47** In a cyclic process shown in the figure, an ideal gas is adiabatically taken from B to A , the work done on the gas during the process $B \rightarrow A$ is 30 J, when the gas is taken from $A \rightarrow B$, the heat absorbed by the gas is 20 J. The change in internal energy of the gas in the process $A \rightarrow B$ is



- (a) 20 J (b) -30 J
(c) 50 J (d) -10 J

- 48** n moles of an ideal gas undergo a process in which the temperature changes with volume as $T = KV^2$. The work done by the gas as the temperature changes from T_0 to $4T_0$ is

(a) $3nRT_0$ (b) $\left(\frac{5}{2}\right)nRT_0$
 (c) $\left(\frac{3}{2}\right)nRT_0$ (d) zero

- 49** One mole of a gas expands with temperature T such that its volume $V = kT^2$, where k is a constant. If the temperature of the gas changes by 60°C , then the work done by the gas is

(a) $120R$ (b) $R \ln 60$ (c) $kR \ln 60$ (d) $60kR$

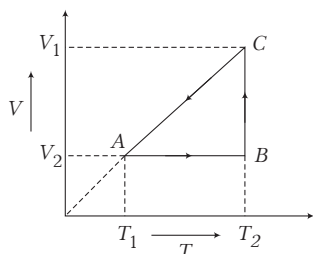
- 50** An ideal monoatomic gas undergoes a process in which the gas volume relates to temperature as $VT = \text{constant}$. Then, molar specific heat of gas in this process is

(a) $\frac{R}{2}$ (b) R
 (c) $\frac{3R}{2}$ (d) None of these

- 51** One mole of an ideal gas with heat capacity at constant pressure C_p undergoes the process $T = T_0 + \alpha V$, where T_0 and α are constants. If its volume increases from V_1 to V_2 , the amount of heat transferred to the gas is

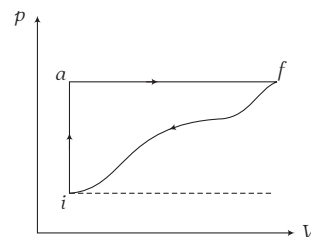
(a) $C_p RT_0 \ln \left| \frac{V_2}{V_1} \right|$
 (b) $\alpha C_p (V_2 - V_1) - RT_0 \ln \left| \frac{V_2}{V_1} \right|$
 (c) $\alpha C_p (V_2 - V_1) + RT_0 \ln \left| \frac{V_2}{V_1} \right|$
 (d) $RT_0 \ln \left| \frac{V_2}{V_1} \right| - \alpha C_p (V_2 - V_1)$

- 52** A cyclic process for 1 mol of an ideal gas is shown in the V - T diagram. The work done in AB , BC and CA respectively are



(a) $Q, RT_2 \ln \left| \frac{V_1}{V_2} \right|, R(T_2 - T_1)$ (b) $R(T_1 - T_2), Q, RT_1 \ln \left| \frac{V_1}{V_2} \right|$
 (c) $Q, RT_2 \ln \left| \frac{V_1}{V_2} \right|, R(T_1 - T_2)$ (d) $Q, RT_2 \ln \left| \frac{V_2}{V_1} \right|, R(T_2 - T_1)$

- 53** When a system is taken from state i to a state f along path iaf , $Q = 50 \text{ J}$ and $W = 20 \text{ J}$. If $W = -13 \text{ J}$ for the curved return path fi , Q in this path is

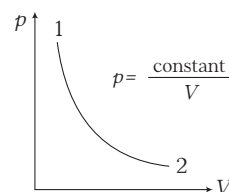


(a) 33 J (b) 23 J (c) -7 J (d) -43 J

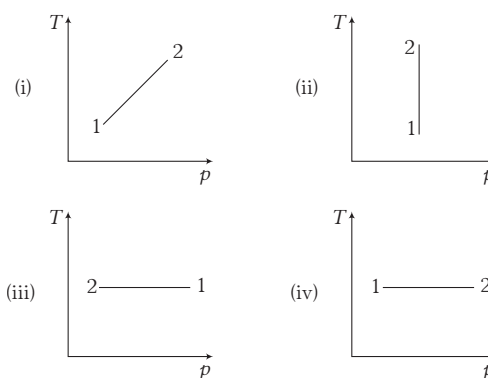
- 54** Two Carnot's engines A and B are operated in succession. The first one, A receives heat from a source at $T_1 = 800 \text{ K}$ and rejects to sinks at $T_2 \text{ K}$. The second engine B receives heat rejected by the first engine and rejects to another sink at $T_3 = 300 \text{ K}$. If the work outputs of two engines are equal, then the value of T_2 is

(a) 100 K (b) 300 K
 (c) 550 K (d) 700 K

- 55** Consider p - V diagram for an ideal gas shown in figure.



Out of the following diagrams, which figure represents the T - p diagram?



(a) (iv) (b) (ii) (c) (iii) (d) (i)

- 56** A sample of 100 g water is slowly heated from 27°C to 87°C . What is the change in the entropy of the water? (Take, specific heat capacity of water $= 4200 \text{ J/kg-K}$)

(a) $420 \ln(1.2) \text{ JK}^{-1}$ (b) $220 \ln(1.4) \text{ JK}^{-1}$
 (c) $120 \ln(1.5) \text{ JK}^{-1}$ (d) $320 \ln(1.2) \text{ JK}^{-1}$

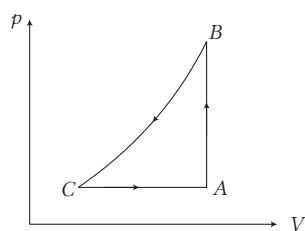
- 57** An ideal gas is taken through a cyclic thermodynamic process through four steps. The amounts of heat involved in these steps are $Q_1 = 5960 \text{ J}$, $Q_2 = -5585 \text{ J}$, $Q_3 = 2980 \text{ J}$ and $Q_4 = 3645 \text{ J}$, respectively. The corresponding works involved are $W_1 = 2200 \text{ J}$, $W_2 = -825 \text{ J}$, $W_3 = -1100 \text{ J}$ and $W_4 = 482.4 \text{ J}$, respectively.

What is the efficiency of the cycle?

- (a) 10.82 % (b) 11.42 %
(c) 9.32 % (d) 5.82 %
- 58** Consider two containers *A* and *B* containing identical gases at the same pressure, volume and temperature. The gas in container *A* is compressed to half of its original volume isothermally while the gas in container *B* is compressed to half of its original value adiabatically. The ratio of final pressure of gas in *B* to that of gas in *A* is

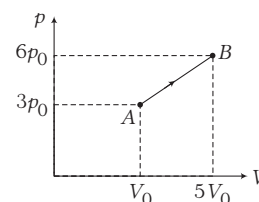
- (a) $2^{\gamma-1}$ (b) $\left(\frac{1}{2}\right)^{\gamma-1}$ (c) $\left(\frac{1}{1-\gamma}\right)^2$ (d) $\left(\frac{1}{\gamma-1}\right)^2$

- 59** A sample of an ideal gas is taken through a cycle as shown in figure. It absorbs 50 J of energy during the process *AB*, no heat during *BC*, rejects 70 J during *CA*. 40 J of work is done on the gas during *BC*. Internal energy of gas at *A* is 1500 J, the internal energy at *C* would be



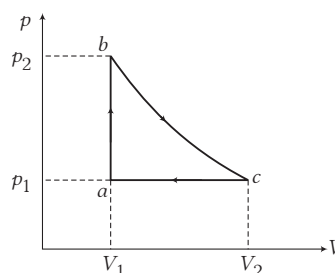
- (a) 1590 J (b) 1620 J (c) 1540 J (d) 1570 J

- 60** One mole of a monoatomic ideal gas undergoes the process $A \rightarrow B$ in the given p - V diagram. The molar heat capacity for this process is



- (a) $\frac{3R}{2}$ (b) $\frac{13R}{6}$ (c) $\frac{5R}{2}$ (d) $2R$

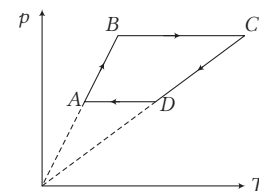
- 61** Carbon monoxide is carried around a closed cycle *abca* in which *bc* is an isothermal process as shown in the figure. The gas absorbs 7000 J of heat as its temperature increases from 300 K to 1000 K in going from *a* to *b*.



The quantity of heat rejected by the gas during the process *ca* is approximately

- (a) 4200 J (b) 5000 J
(c) 9000 J (d) -9800 J

- 62** Six moles of an ideal gas performs a cycle as shown in figure. If the temperatures are $T_A = 600 \text{ K}$, $T_B = 800 \text{ K}$, $T_C = 2200 \text{ K}$ and $T_D = 1200 \text{ K}$, the work done per cycle is approximately



- (a) 20 kJ (b) 30 kJ (c) 40 kJ (d) 60 kJ

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-4) These questions consists of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) If Assertion is correct but Reason is incorrect.
(d) If Assertion is incorrect but Reason is correct.

1 Assertion During melting of ice, work done by surrounding on (ice + water) system is positive.

Reason Volume of the given system decreases on melting of ice.

2 Assertion If initial and final volumes are equal, then work done by gas is zero.

Reason In isochoric process, initial and final volumes are equal and work done by gas is zero.

3 Assertion If volume of a gas is increasing but temperature of the gas is decreasing, then heat given to the gas may be positive, negative or zero.

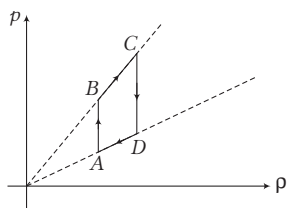
Reason Heat given to a gas is a state function. It is not path function.

4 Assertion There are two processes : Process-1 is $pV = \text{constant}$ and process-2 is $pV^2 = \text{constant}$. In both the processes, volume of gas is increased from V_1 to V_2 . Initial coordinates (p_1, V_1) of the gas were same. Then, more work is done by the gas in process-1.

Reason In second process, pressure drops more rapidly with increase in volume.

Statement based questions

1 Pressure *versus* density graph of an ideal gas is shown in figure. Which of the following statement is correct?

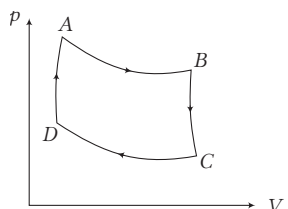


- (a) During the process, AB work done by the gas is positive.
- (b) During the process, AB work done by the gas is negative.
- (c) During the process, BC internal energy of the gas is increasing.
- (d) None of the above

2 At 27°C , two moles of an ideal monoatomic gas occupy a volume V . The gas expands adiabatically to a volume $2V$. Which of the following statement(s) is/are correct?

- (a) Final temperature of the gas is 189 K .
- (b) Change in its internal energy, -2767.235 .
- (c) The work done by the gas during the process is 2767.235 .
- (d) All are correct

3 In the cyclic process $ABCD$ as shown in the figure, consider the following statements.



- I. Area $ABCD$ = Work done on the gas
- II. Area $ABCD$ = Net heat absorbed

III. Change in the internal energy in cycle = 0

Which of these are correct?

- (a) Only I
- (b) Only II
- (c) II and III
- (d) I, II and III

4 Which of the following statement(s) is/are correct?

- I. Efficiency of any heat engine cannot be greater than the efficiency of Carnot engine.
- II. Heat flow never takes place from a body at lower temperature to a body at higher temperature.
- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) None of these

Match the columns

1 Match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Adiabatic bulk modulus	(p) $-(p/V)$
(B) Slope of p - V graph in isothermal process	(q) $\frac{2}{\gamma-1}$
(C) Degree of freedom	(r) γp
(D) Molar heat capacity at constant pressure divided by R	(s) $\frac{\gamma}{\gamma-1}$

Codes

- A B C D
- (a) s r p q
- (b) r p q s
- (c) r s q p
- (d) s p q r

2. For one mole of a monoatomic gas, match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Isothermal bulk modulus	(p) $-\frac{RT}{V^2}$
(B) Adiabatic bulk modulus	(q) $-\frac{5p}{3V}$
(C) Slope of p - V graph in isothermal process	(r) $\frac{T}{V}$
(D) Slope of p - V graph in adiabatic process	(s) $\frac{4T}{3V}$
	(t) None

Codes

- A B C D
- (a) p q r s
- (b) p s r q
- (c) t t p q
- (d) t s r p

(C) Medical entrances' gallery

Collection of questions asked in NEET & Various Medical Entrance Exams

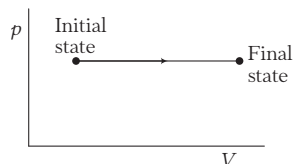
- 1** Two cylinders *A* and *B* of equal capacity are connected to each other *via* a stop cock. *A* contains an ideal gas at standard temperature and pressure. *B* is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is [NEET 2020]

(a) adiabatic (b) isochoric
(c) isobaric (d) isothermal

- 2** The efficiency of a Carnot engine depends upon [NEET 2020]

(a) the temperature of the sink only
(b) the temperatures of the source and sink
(c) the volume of the cylinder of the engine
(d) the temperature of the source only

- 3** The *p-V* diagram for an ideal gas in a piston cylinder assembly undergoing a thermodynamic process is shown in the figure. The process is [NEET 2020]



(a) adiabatic (b) isochoric (c) isobaric (d) isothermal

- 4** 1g of water of volume 1 cm^3 at 100°C is converted into steam at same temperature under normal atmospheric pressure ($\approx 1 \times 10^5 \text{ Pa}$). The volume of steam formed equals 1671 cm^3 . If the specific latent heat of vaporisation of water is 2256 J/g , the change in internal energy is [NEET (Odisha) 2019]

(a) 2423 J (b) 2089 J
(c) 167 J (d) 2256 J

- 5** In which of the following processes, heat is neither absorbed nor released by a system? [NEET 2019]

(a) Adiabatic (b) Isobaric
(c) Isochoric (d) Isothermal

- 6** The value of $\gamma \left(= \frac{C_p}{C_v} \right)$, for hydrogen, helium and

another ideal diatomic gas *X* (whose molecules are not rigid but have an additional vibrational mode) are respectively equal to [NEET (Odisha) 2019]

(a) $\frac{7}{5}, \frac{5}{3}, \frac{9}{7}$ (b) $\frac{5}{3}, \frac{7}{5}, \frac{9}{7}$ (c) $\frac{5}{3}, \frac{7}{5}, \frac{7}{5}$ (d) $\frac{7}{5}, \frac{5}{3}, \frac{7}{5}$

- 7 Assertion** In adiabatic process, work is independent of path.

Reason In adiabatic process, work done is equal to negative of change in internal energy. [AIIMS 2019]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) If Assertion is correct but Reason is incorrect.
(d) Both Assertion and Reason are incorrect.

- 8** The temperature of food material in refrigerator is 4°C and temperature of environment is 15°C . If Carnot cycle is used in its working gas, then find its Carnot efficiency. [AIIMS 2019]

(a) 0.038 (b) 0.028 (c) 0.053 (d) 0.072

- 9** If 7 g N_2 is mixed with 20 g Ar, the $\frac{C_p}{C_v}$ of mixture will be [AIIMS 2019]

(a) $\frac{17}{6}$ (b) $\frac{11}{7}$ (c) $\frac{17}{11}$ (d) $\frac{17}{13}$

- 10** In an isobaric process, the work done by a di-atomic gas is 10 J, the heat given to the gas will be [AIIMS 2019]

(a) 35 J (b) 30 J (c) 45 J (d) 60 J

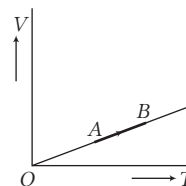
- 11** A sample of 0.1 g of water at 100°C and normal pressure ($1.013 \times 10^5 \text{ Nm}^{-2}$) requires 54 cal of heat energy to convert to steam at 100°C . If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample is [NEET 2018]

(a) 42.2 J (b) 208.7 J (c) 104.3 J (d) 84.5 J

- 12** The efficiency of an ideal heat engine working between the freezing point and boiling point of water is [NEET 2018]

(a) 6.25% (b) 20% (c) 26.8% (d) 12.5%

- 13** The volume (*V*) of a monoatomic gas varies with its temperature (*T*) as shown in the graph. The ratio of work done by the gas, to the heat absorbed by it, when it undergoes a change from state *A* to state *B*, is [NEET 2018]



(a) 1/3 (b) 2/3 (c) 2/5 (d) 2/7

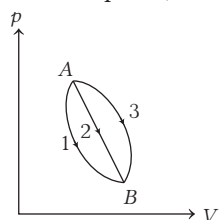
- 14 Assertion** In isothermal process, whole of the heat energy supplied to the body is converted into internal energy.

Reason According to the first law of thermodynamics,

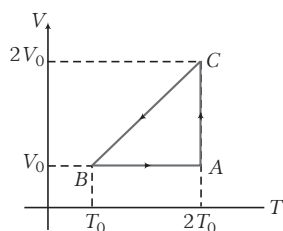
$$\Delta Q = \Delta U + \Delta W \quad [\text{AIIMS 2018}]$$

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Assertion is incorrect but Reason is correct.

- 15** An ideal gas of mass m in a state A goes to another state B via three different processes as shown in figure. If Q_1 , Q_2 and Q_3 denote the heat absorbed by the gas along the three paths, then [AIIMS 2018]

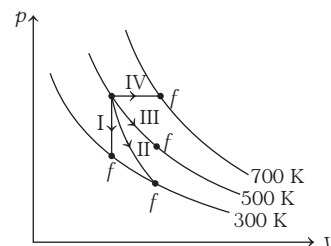


- (a) $Q_1 < Q_2 < Q_3$ (b) $Q_1 < Q_2 = Q_3$
 (c) $Q_1 = Q_2 > Q_3$ (d) $Q_1 > Q_2 > Q_3$
- 16** A gas consisting of a rigid diatomic molecules was initially under standard condition. Then, gas was compressed adiabatically to one-fifth of its initial volume. What will be the mean kinetic energy of a rotating molecule in the final state? [AIIMS 2018]
- (a) 1.44 J (b) 4.55 J
 (c) 787.98×10^{-23} J (d) 757.3×10^{-23} J
- 17** The efficiency of an ideal gas with adiabatic exponent γ for the shown cyclic process would be [JIPMER 2018]



- (a) $\frac{(2\ln 2 - 1)}{\gamma / (\gamma - 1)}$ (b) $\frac{(1 - \ln 2)}{\gamma / (\gamma - 1)}$ (c) $\frac{(2\ln 2 + 1)}{\gamma / (\gamma - 1)}$ (d) $\frac{(2\ln 2 - 1)}{\gamma / (\gamma + 1)}$
- 18** A Carnot engine absorbs 6×10^5 cal at 227°C . The work done per cycle by the engine, if its sink is maintained at 127°C is [JIPMER 2018]
- (a) 15×10^8 J (b) 15×10^4 J
 (c) 5×10^5 J (d) 2×10^4 J
- 19** If the efficiency of an engine is 50% and its work output is 500 J, then find the value of input work. [JIPMER 2018]
- (a) 1000 J (b) 500 J (c) 100 J (d) 250 J
- 20** The efficiency of a heat engine is $1/6$. Its efficiency doubled when the temperature of sink decreased by 62°C , then what is the temperature of sink? [JIPMER 2018]
- (a) 470 K (b) 372 K (c) 542 K (d) 1042 K

- 21** Thermodynamic processes are indicated in the following diagram



Match the Column I with Column II and mark the correct option from the codes given below.

Column I	Column II
A. Process I	p. Adiabatic
B. Process II	q. Isobaric
C. Process III	r. Isochoric
D. Process IV	s. Isothermal

[NEET 2017]

Codes

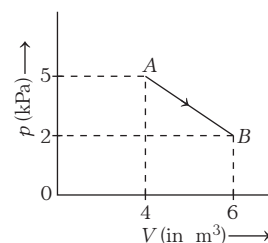
- A B C D A B C D
 (a) p r s q (b) r p s q
 (c) r s q p (d) s q p r

- 22** A Carnot engine having an efficiency of $\frac{1}{10}$ as heat

engine, is used as a refrigerator. If the work done on the system is 10 J, then the amount of energy absorbed from the reservoir at lower temperature is [NEET 2017]

- (a) 1 J (b) 90 J (c) 99 J (d) 100 J

- 23** One mole of an ideal diatomic gas undergoes transition from A to B along a path AB as shown below. [AIIMS 2017]



The change in internal energy of the gas during the transition is

- (a) 20 kJ (b) -12 kJ (c) -20 kJ (d) 20 J

- 24** **Assertion** Molar heat capacity cannot be defined for isothermal process.

Reason In isothermal process, p - V versus T graph is a dot. [AIIMS 2017]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Reason is correct but Assertion is incorrect.

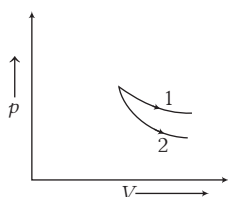
25 Assertion In adiabatic expansion process, the product of p and V always decreases.

Reason In adiabatic expansion process, work is done by the gas at the cost of internal energy of gas.

[AIIMS 2017]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) Assertion is correct but Reason is incorrect.
- (d) Reason is correct but Assertion is incorrect.

26 p - V plots for two gases during adiabatic processes as shown in figure. Plots 1 and 2 should correspond respectively to [JIPMER 2017]



- (a) He and O_2
- (b) O_2 and He
- (c) He and Ar
- (d) O_2 and N_2

27 The temperature of source and sink of a heat engine are 127°C and 27°C , respectively. An inventor claims its efficiency to be 26%, then [JIPMER 2017]

- (a) it is impossible
- (b) it is possible with high probability
- (c) it is possible with low probability
- (d) Data are insufficient

28 One mole of an ideal monoatomic gas undergoes a process described by the equation $pV^3 = \text{constant}$. The heat capacity of the gas during this process is [NEET 2016]

- (a) $\frac{3}{2}R$
- (b) $\frac{5}{2}R$
- (c) $2R$
- (d) R

29 The temperature inside a refrigerator is $t_2^\circ\text{C}$ and the room temperature is $t_1^\circ\text{C}$. The amount of heat delivered to the room for each joule of electrical energy consumed ideally will be [NEET 2016]

- (a) $\frac{t_1}{t_1 - t_2}$
- (b) $\frac{t_1 + 273}{t_1 - t_2}$
- (c) $\frac{t_2 + 273}{t_1 - t_2}$
- (d) $\frac{t_1 + t_2}{t_1 + 273}$

30 A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then [NEET 2016]

- (a) compressing the gas through adiabatic process will require more work to be done
- (b) compressing the gas isothermally or adiabatically will require the same amount of work

(c) which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas

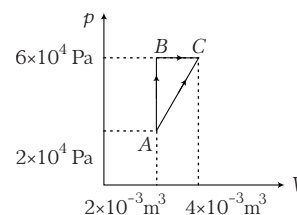
(d) compressing the gas isothermally will require more work to be done

31 A refrigerator works between 4°C and 30°C . It is required to remove 600 cal of heat every second in order to keep the temperature of the refrigerated space constant. The power required is (Take, 1 cal = 4.2 J) [NEET 2016]

- (a) 23.65 W
- (b) 236.5 W
- (c) 2365 W
- (d) 2.365 W

32 Figure below shows two paths that may be taken by a gas to go from a state A to a state C.

In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be [CBSE AIPMT 2015]



- (a) 380 J
- (b) 500 J
- (c) 460 J
- (d) 300 J

33 The ratio of the specific heats $\frac{C_p}{C_v} = \gamma$ in terms of degrees of freedom (n) is given by [CBSE AIPMT 2015]

- (a) $\left(1 + \frac{1}{n}\right)$
- (b) $\left(1 + \frac{n}{3}\right)$
- (c) $\left(1 + \frac{2}{n}\right)$
- (d) $\left(1 + \frac{n}{2}\right)$

34 The molar specific heat of a gas as given from the kinetic theory is $\frac{5}{2}R$. If it is not specified whether it is C_p or C_v , one could conclude that the molecules of the gas [AIIMS 2015]

- (a) are definitely monoatomic
- (b) are definitely rigid diatomic
- (c) are definitely non-rigid diatomic
- (d) can be monoatomic or rigid diatomic

35 If ΔU represents the increases in internal energy and W the work done by the thermodynamic system, then [Kerala CEE 2015]

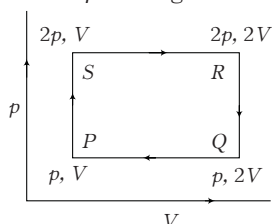
- (a) $\Delta U = -W$ is an adiabatic process
- (b) $\Delta U = W$ is an isothermal process
- (c) $\Delta U = W$ is an adiabatic process
- (d) $\Delta U = W$ is an isochoric process

- 36** If the energy input to a Carnot engine is thrice the work it performs, then the fraction of energy rejected to the sink is [Kerala CEE 2015]

(a) $1/3$ (b) $1/4$ (c) $2/5$ (d) $2/3$
(e) $1/6$

- 37** A gas with $\gamma = 1.4$ undergoes the adiabatic process of compression, if the pressure is increased by 0.5%, then the volume decreases by (about) [UP CPMT 2015]
- (a) 0.25% (b) 0.1% (c) 0.5% (d) 0.36%

- 38** An ideal monoatomic gas is taken round the cycle PQRS as shown in p - V diagram.



The work done during the cycle is [CG PMT 2015]

(a) pV (b) $2pV$ (c) p (d) zero

- 39** The efficiency of a Carnot engine which operates between the two temperatures $T_1 = 500$ K and $T_2 = 300$ K is [KCET 2015]
- (a) 75 % (b) 50% (c) 40% (d) 25%

- 40** An ideal diatomic gas is heated at constant pressure. What fraction of heat energy is utilised to increase its internal energy? [EAMCET 2015]

(a) $\frac{5}{7}$ (b) $\frac{2}{5}$ (c) $\frac{3}{5}$ (d) $\frac{3}{7}$

- 41** How much heat (in joules) is required to raise the temperature of 1g of water from 14.5°C to 15.5°C at 1 atmospheric pressure? [UK PMT 2015]

(a) 41.86 J (b) 4.186 J
(c) 418.6 J (d) None of these

- 42** The increase in internal energy of 1kg of water at 100°C when it is converted into steam at the same temperature and at 1 atmospheric pressure will be (Take, density of steam is 0.6 kg m^{-3} , latent heat of vaporisation of water = $2.25 \times 10^6 \text{ J kg}^{-1}$) [UK PMT 2015]

(a) $1.04 \times 10^6 \text{ J}$ (b) $2.08 \times 10^6 \text{ J}$
(c) $4.16 \times 10^6 \text{ J}$ (d) $8.32 \times 10^6 \text{ J}$

- 43** Specific heats of one mole of hydrogen at constant pressure and at constant volume are 450 JK^{-1} and 300 JK^{-1} , respectively. Then, what is the density of the gas at STP? (Take, $p_{\text{atm}} = 1.013 \times 10^5 \text{ Nm}^{-2}$) [EAMCET 2015]

(a) 2.47 kg m^{-3} (b) 3 kg m^{-3}
(c) 3.58 kg m^{-3} (d) 4 kg m^{-3}

- 44** A monoatomic gas at a pressure p , having a volume V expands isothermally to a volume $2V$ and then adiabatically to a volume $16V$. The final pressure of the gas is (Take, $\gamma = \frac{5}{3}$) [CBSE AIPMT 2014]

(a) $64 p$ (b) $32 p$
(c) $\frac{p}{64}$ (d) $16 p$

- 45** If for hydrogen $C_p - C_v = m$ and for nitrogen

$C_p - C_v = n$, where C_p and C_v refer to specific heat per unit mass respectively at constant pressure and constant volume, the relation between m and n is (Molecular weight of hydrogen = 2 and molecular weight of nitrogen = 28) [AIIMS, UP CPMT 2014]

(a) $n = 14 m$ (b) $n = 7m$
(c) $m = 7n$ (d) $m = 14n$

- 46 Assertion** For monoatomic gas atom, the number of degrees of freedom is 3.

Reason $\frac{C_p}{C_v} = \gamma = \frac{5}{3}$

[AIIMS 2014]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Assertion is incorrect but Reason is correct.

- 47** The change in internal energy of a thermodynamic system which has absorbed 2 kcal of heat and done 400 J of work is (Take, 1 cal = 4.2 J) [Kerala CEE 2014]

(a) 2 kJ (b) 8 kJ
(c) 3.5 kJ (d) 5.5 kJ
(e) 2.5 kJ

- 48** The p - T relation for an adiabatic expansion is

[UK PMT 2014]

(a) $p^\gamma T^{\gamma-1} = \text{constant}$ (b) $p^{\gamma-1} T^\gamma = \text{constant}$
(c) $p^\gamma T^{1-\gamma} = \text{constant}$ (d) $p^{1-\gamma} T^\gamma = \text{constant}$

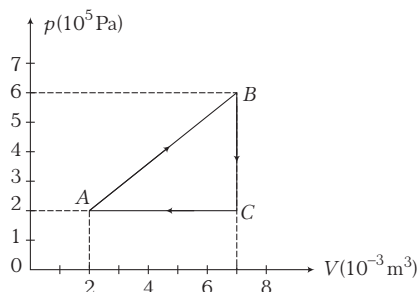
- 49** A Carnot refrigerator extracts heat from water at 0°C and rejects it to room at 24.4°C . The work required by the refrigerator for every 1 kg of water converted into ice (latent heat of ice = 336 kJ kg^{-1}) [EAMCET 2014]

(a) 30 kJ (b) 336 kJ
(c) 11.2 kJ (d) 24.4 kJ

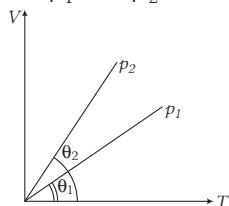
- 50** A Carnot engine working between 300 K and 600 K has work output of 800 J cycle^{-1} . The amount of heat energy supplied from the source of engine in each cycle is [WB JEE 2014]

(a) 800 J (b) 1600 J
(c) 3200 J (d) 6400 J

- 51 A gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$, as shown. What is the net work done by the gas? [NEET 2013]



- (a) 2000 J (b) 1000 J (c) Zero (d) -2000 J
- 52 During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of $\frac{C_p}{C_v}$ for the gas is [NEET 2013]
- (a) $\frac{4}{3}$ (b) 2 (c) $\frac{5}{3}$ (d) $\frac{3}{2}$
- 53 In the given V - T diagram, what is the relation between pressures p_1 and p_2 ? [NEET 2013]



- (a) $p_2 = p_1$ (b) $p_2 > p_1$
(c) $p_2 < p_1$ (d) Cannot be predicted
- 54 The amount of heat energy required to raise the temperature of 1 g of helium at NTP, from T_1 K to T_2 K is [NEET 2013]
- (a) $\frac{3}{8} N_A k_B (T_2 - T_1)$ (b) $\frac{3}{2} N_A k_B (T_2 - T_1)$
(c) $\frac{3}{4} N_A k_B (T_2 - T_1)$ (d) $\frac{3}{4} N_A k_B \left(\frac{T_2}{T_1} \right)$
- 55 The efficiency of a Carnot engine kept at the temperatures of 27°C and 127°C is [J & K CET 2013]
- (a) 20% (b) 25% (c) 30% (d) 40%
56. Match the Column I with Column II and mark the correct option from the codes given below. [EAMCET 2013]

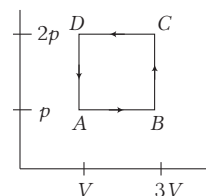
Column I	Column II
(A) Work done in isobaric process	(p) $nRT \log_e (V_2/V_1)$
(B) Work done in isothermal process	(q) $p(V_2 - V_1)$
(C) Work done in adiabatic process	(r) $\frac{nR(T_1 - T_2)}{\gamma - 1}$
	(s) zero

Codes

	A	B	C
(a)	q	p	r
(c)	q	p	s

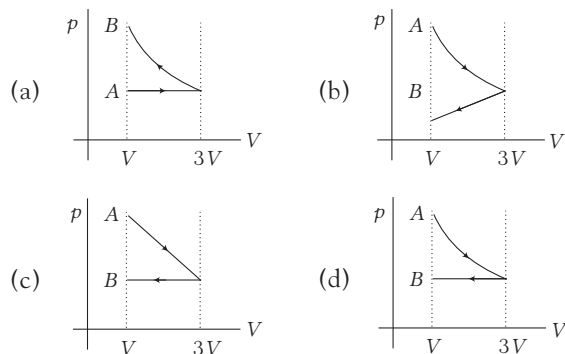
	A	B	C
(b)	p	s	q
(d)	s	p	q

- 57 Two cylinders A and B fitted with pistons contain equal amounts of an ideal diatomic gas at 300 K. Piston A is free to move and piston of B is fixed. Same amount of heat is given to the gases in the two cylinders. Temperature of the gas in cylinder A increases by 30 K, then increase in temperature of the gas in the cylinder B is ($\gamma = 1.4$ for diatomic gas) [EAMCET 2013]
- (a) 24 K (b) 36 K (c) 54 K (d) 42 K
- 58 For which combination of working temperatures of source and sink, the efficiency of Carnot's heat engine is maximum? [KCET 2013]
- (a) 600 K, 400 K (b) 400 K, 200 K
(c) 500 K, 300 K (d) 300 K, 100 K
- 59 During an adiabatic process of an ideal gas, if p is proportional to $\frac{1}{V^{1.5}}$, then the ratio of specific heat capacities at constant pressure to that at constant volume for the gas is [Kerala CEE 2013]
- (a) 1.5 (b) 0.25
(c) 0.75 (d) 0.4
(e) 1.3
- 60 One litre of a gas (with $\gamma = \frac{5}{3}$) at NTP is compressed adiabatically to one cubic centimetre, then the resulting pressure is [MP PMT 2013]
- (a) 10 atm (b) 10^3 atm
(c) 10^5 atm (d) 100 atm
- 61 The specific heat of an ideal gas for an isothermal process is [MP PMT 2013]
- (a) zero (b) $\frac{3}{2}R$
(c) $\frac{5}{2}R$ (d) infinite
- 62 A thermodynamic system is taken through the cycle $ABCD$ as shown in figure. Heat rejected by the gas during the cycle is [CBSE AIPMT 2012]

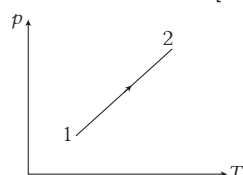


- (a) $2pV$ (b) $4pV$
(c) $\frac{1}{2}pV$ (d) pV

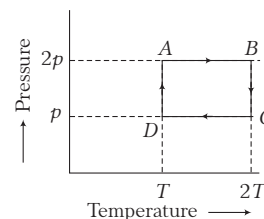
- 63** One mole of an ideal gas goes from an initial state A to final state B via two processes. It first undergoes isothermal expansion from volume V to $3V$ and then its volume is reduced from $3V$ to V at constant pressure. The correct p - V diagram representing the two processes is [CBSE AIPMT 2012]



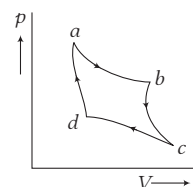
- 64** If an ideal gas changes from state 1 to state 2 as shown in figure, then the work done by the gas in the process will be [BHU (Screening) 2012]



- (a) positive (b) negative
(c) zero (d) infinite
- 65** The molar specific heats of an ideal gas at constant pressure and volume are denoted by C_p and C_v , respectively. If $\frac{C_p}{C_v} = \gamma$ and R is the gas constant for 1 g mole of a gas, then C_p is equal to [BHU 2012]
- (a) γR (b) $\frac{\gamma R}{\gamma - 1}$ (c) $\frac{\gamma R}{\gamma - 1}$ (d) R
- 66** If a gas expands with temperature according to the relation $V = KT^{2/3}$, then what will be work done when the temperature changes by 30°C ? [BCECE (Mains) 2012]
- (a) $40 R$ (b) $30 R$
(c) $20 R$ (d) $10 R$
- 67** When heat is supplied to a diatomic gas at constant pressure, then the ratio of $\Delta Q : \Delta U : \Delta W$ will be [BCECE (Mains) 2012]
- (a) $5 : 2 : 3$ (b) $7 : 2 : 5$
(c) $5 : 3 : 2$ (d) $7 : 5 : 2$
- 68** An ideal monoatomic gas is taken through the thermodynamic states $A \rightarrow B \rightarrow C \rightarrow D$ via the paths shown in the figure. If U_A, U_B, U_C and U_D represent the internal energy of the gas in states A, B, C and D respectively, then which of the following is not true? [Manipal 2012]



- (a) $U_A = U_D$ (b) $U_B < U_A$ (c) $U_B = U_C$ (d) $U_C > U_D$
- 69** During an adiabatic process, the cube of the pressure is found to be inversely proportional to the fourth power of the volume. Then, the ratio of specific heats is [Manipal 2012]
- (a) 1 (b) 1.33 (c) 1.67 (d) 1.4
- 70** A Carnot's engine has an efficiency of 50% at sink temperature 50°C . Calculate the temperature of source. [Manipal 2012]
- (a) 133°C (b) 143°C (c) 100°C (d) 373°C
- 71** The water of volume 4 m^3 at the height 20 m is pressed by $2 \times 10^5 \text{ N}$ pressure. The work done by motor is [Manipal 2012]
- (a) $8 \times 10^5 \text{ J}$ (b) $16 \times 10^5 \text{ J}$
(c) $12 \times 10^5 \text{ J}$ (d) $32 \times 10^5 \text{ J}$
- 72** In adiabatic process, the work done by system is 50 J, then [Manipal 2012]
- (a) the temperature of the system will increase by 50 J
(b) the temperature of the system will be constant
(c) the internal energy of the system will increase by 50 J
(d) the internal energy of the system will decrease by 50 J
- 73** 100 mol of an ideal gas is heated from 10° to 20°C keeping its (i) volume constant (ii) pressure constant. Let ΔU_a and ΔU_b denote the change in the internal energy of the gas due to process (i) and (ii), respectively. Then, which of the following shall be true? [AMU 2012]
- (a) $\Delta U_a = \Delta U_b$ (b) $\Delta U_a > \Delta U_b$
(c) $\Delta U_a < \Delta U_b$ (d) $\Delta U_a = \Delta U_b = 0$
- 74** Refer to the Carnot cycle of an ideal gas as shown in the figure. Let $W_{a \rightarrow b}, W_{b \rightarrow c}, W_{c \rightarrow d}$ and $W_{d \rightarrow a}$ represent the work done by the system during the processes $a \rightarrow b, b \rightarrow c, c \rightarrow d$ and $d \rightarrow a$, respectively. Consider the following relations [AMU 2012]



- (i) $W_{a \rightarrow b} + W_{b \rightarrow c} + W_{c \rightarrow d} + W_{d \rightarrow a} > 0$
- (ii) $W_{a \rightarrow b} + W_{b \rightarrow c} + W_{c \rightarrow d} + W_{d \rightarrow a} < 0$
- (iii) $W_{a \rightarrow b} + W_{c \rightarrow d} > 0$
- (iv) $W_{b \rightarrow c} + W_{d \rightarrow a} = 0$

Which of the above relation(s) is/are true?

- (a) (i) and (iii) (b) (iii) and (iv)
 - (c) (i) (iii) and (iv) (d) (ii), (iii) and (iv)
- 75** An ideal gas undergoes a reversible isothermal expansion from a state i to f . The change in entropy $\Delta S(i \rightarrow f)$ is [AMU 2012]
- (a) 0 (b) > 0
 - (c) < 0 (d) not defined for the process
- 76** One mole of an ideal gas is heated at a constant pressure of 1 atm from 0°C to 100°C . Work done by the gas is [UP CPMT 2012]
- (a) $8.31 \times 10^3 \text{ J}$ (b) $8.31 \times 10^{-3} \text{ J}$
 - (c) $8.31 \times 10^{-3} \text{ J}$ (d) $8.31 \times 10^2 \text{ J}$
- 77** The efficiency of a Carnot engine is 60%. If the temperature of source is 127°C . The sink must be maintained at [UP CPMT 2012]
- (a) 113 K (b) $+113^\circ\text{C}$
 - (c) -113°C (d) -113 K

- 78** A perfect gas goes from state A to state B by absorbing $8 \times 10^5 \text{ J}$ of heat and doing $6.5 \times 10^5 \text{ J}$ of external work. It is now transferred between the same two states in another process in which it absorbs 10^5 J of heat. In the second process,

[AIIMS 2011]

- (a) work done on the gas is 10^5 J
- (b) work done on the gas is $0.5 \times 10^5 \text{ J}$
- (c) work done by the gas is 10^5 J
- (d) work done by the gas is $0.5 \times 10^5 \text{ J}$

- 79** A cylinder of fixed capacity (of 44.8 L) contains 2 mol of helium gas at STP. What is the amount of heat needed to raise the temperature of the gas in the cylinder by 20°C ? (Take, $R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$)

[AMU 2011]

- (a) 996 J (b) 831 J (c) 499 J (d) 374 J

- 80** A Carnot engine has efficiency 25%. It operates between reservoirs of constant temperatures with temperature difference of 80 K. What is the temperature of the low temperature reservoir?

[DUMET 2011]

- (a) -25°C (b) 25°C (c) -33°C (d) 33°C

ANSWERS

CHECK POINT 15.1

1. (a)	2. (a)	3. (d)	4. (b)	5. (b)	6. (a)	7. (b)	8. (b)	9. (a)	10. (a)
11. (d)	12. (a)								

CHECK POINT 15.2

1. (b)	2. (b)	3. (a)	4. (b)	5. (d)	6. (b)	7. (d)	8. (d)	9. (c)	10. (a)
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CHECK POINT 15.3

1. (c)	2. (a)	3. (c)	4. (c)	5. (d)	6. (b)	7. (d)	8. (d)	9. (b)	10. (b)
11. (a)	12. (c)	13. (d)	14. (b)						

CHECK POINT 15.4

1. (b)	2. (a)	3. (a)	4. (d)	5. (a)	6. (d)	7. (d)	8. (b)	9. (c)	10. (a)
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(A) Taking it together

1. (b)	2. (d)	3. (a)	4. (a)	5. (c)	6. (b)	7. (a)	8. (d)	9. (b)	10. (c)
11. (a)	12. (c)	13. (a)	14. (a)	15. (b)	16. (c)	17. (a)	18. (c)	19. (b)	20. (c)
21. (c)	22. (d)	23. (a)	24. (a)	25. (a)	26. (a)	27. (a)	28. (c)	29. (b)	30. (a)
31. (d)	32. (a)	33. (a)	34. (a)	35. (c)	36. (d)	37. (d)	38. (a)	39. (c)	40. (b)
41. (c)	42. (c)	43. (a)	44. (c)	45. (a)	46. (b)	47. (b)	48. (c)	49. (a)	50. (a)
51. (c)	52. (c)	53. (d)	54. (c)	55. (c)	56. (a)	57. (a)	58. (a)	59. (a)	60. (b)
61. (d)	62. (c)								

(B) Medical entrance special format questions

• Assertion and reason

1. (a) 2. (d) 3. (c) 4. (b)

• Statement based questions

1. (d) 2. (a) 3. (c) 4. (a)

• Match the columns

1. (b) 2. (c)

(C) Medical entrances' gallery

1. (a)	2. (b)	3. (c)	4. (b)	5. (a)	6. (a)	7. (a)	8. (a)	9. (c)	10. (a)
11. (b)	12. (c)	13. (c)	14. (d)	15. (a)	16. (c)	17. (a)	18. (c)	19. (a)	20. (b)
21. (b)	22. (b)	23. (c)	24. (b)	25. (a)	26. (a)	27. (a)	28. (d)	29. (b)	30. (a)
31. (b)	32. (c)	33. (c)	34. (d)	35. (a)	36. (d)	37. (d)	38. (a)	39. (c)	40. (a)
41. (b)	42. (b)	43. (a)	44. (c)	45. (d)	46. (b)	47. (b)	48. (d)	49. (a)	50. (b)
51. (b)	52. (d)	53. (c)	54. (a)	55. (b)	56. (a)	57. (d)	58. (d)	59. (a)	60. (c)
61. (d)	62. (a)	63. (d)	64. (c)	65. (c)	66. (c)	67. (d)	68. (b)	69. (b)	70. (d)
71. (b)	72. (d)	73. (b)	74. (a)	75. (a)	76. (d)	77. (c)	78. (b)	79. (c)	80. (c)

Hints & Explanations

• CHECK POINT 15.1

2 (a) $\Delta U = \Delta Q - \Delta W \Rightarrow \Delta U = \Delta Q$ ($\because \Delta W = 0$)

$\therefore \Delta U < 0$, so the temperature will decrease.

3 (d) Change in internal energy does not depend upon path, so $\Delta U = \Delta Q - \Delta W$ remain constant.

4 (b) $W = \text{Area under } p\text{-}V \text{ diagram}$

$$W = \frac{1}{2} \times (3V_0 - V_0) \times (2p_0 - p_0) = \frac{1}{2} \times 2V_0 \times p_0 = p_0 V_0$$

5 (b) Work done by the system = Area enclosed by the cycle
 $= (300 - 100) 10^{-6} \times (200 - 100) \times 10^3 = 20 \text{ J}$.

As, cycle is anti-clockwise. Hence, work done is negative.

6 (a) Work done = Area enclosed by triangle ABC

$$= \frac{1}{2} AC \times BC = \frac{1}{2} \times (3V - V) \times (3p - p) = 2pV$$

7 (b) In cyclic process, work done is area enclosed by the cycle. It is positive when cycle is clockwise and negative when cycle is anti-clockwise (with p on Y -axis and V on X -axis).

8 (b) From first law of thermodynamics, $\Delta Q = \Delta U + \Delta W$

Here, $\Delta Q = 200 \text{ J}$ and $\Delta W = -100 \text{ J}$

$\therefore \Delta U = \Delta Q - \Delta W = 200 - (-100) = 300 \text{ J}$

9 (a) Cycle is clockwise. Hence, heat will be given to the gas, as work done will be positive.

$$Q = W = \text{Area of cycle} = \frac{1}{2} \times 2 \times 1 = 1 \text{ J}$$

10 (a) For cyclic process, total work done $= W_{AB} + W_{BC} + W_{CA}$

$$\Delta W_{AB} = p \Delta V = 10 (2 - 1) = 10 \text{ J}$$

and $\Delta W_{BC} = 0$ (As, $V = \text{constant}$)

From first law of thermodynamics, $\Delta Q = \Delta U + \Delta W$

$$\Delta U = 0 \text{ (Process } ABCA \text{ is cyclic)}$$

$$\Rightarrow \Delta Q = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA}$$

$$\Rightarrow 5 = 10 + 0 + \Delta W_{CA} \Rightarrow \Delta W_{CA} = -5 \text{ J}$$

11 (d) Heat given, $\Delta Q = 20 \text{ cal} = 20 \times 4.2 = 84 \text{ J}$

Work done, $\Delta W = -50 \text{ J}$ (As process is anti-clockwise)

By first law of thermodynamics,

$$\Delta U = \Delta Q - \Delta W = 84 - (-50) = 134 \text{ J}$$

12 (a) In a cyclic process, $\Delta U = 0$

$$\therefore Q_{\text{total}} = W_{\text{total}} \text{ or } 10 - 5 = -3 + W_{21} \text{ or } W_{21} = 8 \text{ J}$$

• CHECK POINT 15.2

1 (b) For monoatomic gas, $C_V = \frac{3}{2}R$

Thus,

$$\begin{aligned} C_V &= \frac{3}{2} \times 8.31 \text{ J mol}^{-1} \text{ K}^{-1} \\ &= \frac{3}{2} \times \frac{8.31}{4.18} \approx 3 \text{ cal (mol}^{-1} \text{ K}^{-1}) \end{aligned}$$

Let ΔT be the rise in temperature when n moles of the gas is given Q cal of heat at constant volume. Then,

$$\begin{aligned} Q &= nC_V \Delta T \\ \Rightarrow \Delta T &= \frac{Q}{nC_V} = \frac{10000 \text{ cal}}{4.0 \text{ mol} \times 3 \text{ cal (mol}^{-1} \text{ K}^{-1})} \approx 833 \text{ K} \end{aligned}$$

2 (b) When a gas is heated at constant volume, all the heat supplied is used to increase the temperature or internal energy of the gas. At constant pressure, an additional amount of heat equivalent to work done is utilised in expanding the gas. Hence, $C_p > C_V$.

3 (a) Given, $C_p - C_V = a$ (for hydrogen gas)

$$C_p - C_V = b \quad (\text{for oxygen gas})$$

$$\therefore C_p - C_V = a = \frac{R}{1} \text{ (H}_2\text{)} \quad \dots(i)$$

$$C_p - C_V = b = \frac{R}{16} \text{ (O}_2\text{)} \quad \dots(ii)$$

Comparing Eqs. (i) and (ii), we get

$$a = 16b$$

4 (b) As we know that, $\gamma = \frac{C_p}{C_V}$ and $C_p - C_V = R$

Then, value of molar specific heat of an ideal gas at constant volume, i.e. $C_V = \frac{R}{\gamma - 1}$

6 (b) Increase in internal energy,

$$\Delta U = \mu C_V \Delta T = \mu \left(\frac{R}{\gamma - 1} \right) \Delta T$$

$$\Rightarrow \Delta U = \frac{p \Delta V}{\gamma - 1} = \frac{p (2V - V)}{\gamma - 1} = \frac{pV}{\gamma - 1}$$

As, $pV = nRT = 2RT$ ($\because n = 2$)

$$\therefore \Delta U = \frac{2RT}{\gamma - 1}$$

7 (d) For 2 mol of oxygen and 4 mol of argon,

$$\Delta U = 2C_{V_O} T + 4C_{V_{Ar}} T = 2T (C_{V_O} + 2C_{V_{Ar}})$$

$$= 2T \left(\frac{R}{2/5} + 2 \times \frac{R}{2/3} \right) = 2TR \left(\frac{5+6}{2} \right) = 11RT$$

8 (d) Molar mass of mixture $= \frac{n_1 M_1 + n_2 M_2}{n_1 + n_2} = \frac{5(4) + 2(2)}{5 + 2} = \frac{24}{7} \text{ g}$

$$9 (c) C_{V(\text{mix})} = \frac{n_1 C_V + n_2 C_V}{n_1 + n_2} = \frac{5 \times \frac{3}{2} R + 2 \times \frac{5}{2} R}{5 + 2}$$

$$= \frac{\frac{15}{2} R + 5R}{7} = \frac{15R + 10R}{2 \times 7} = \frac{25R}{14}$$

$$\therefore \gamma_{\text{mix}} = 1 + \frac{R}{C_{V(\text{mix})}} = 1 + \frac{R}{25R} \times 14 = 1.56$$

10 (a) As, $C_V = \frac{R}{(\gamma - 1)}$

$\therefore (C_V)_1 = \frac{R}{\left(\frac{5}{3}\right) - 1} = \frac{3}{2}R$

and $(C_V)_2 = \frac{R}{\left(\frac{7}{5}\right) - 1} = \frac{5}{2}R$

$\Rightarrow (C_V)_{\text{mix}} = \frac{R}{\left(\frac{29}{23}\right) - 1} = \frac{23}{6}R$

$\therefore C_{V_{\text{mix}}} = \frac{n_1(C_V)_1 + n_2(C_V)_2}{n_1 + n_2}$

We have, $\frac{23}{6}R = \frac{3 \times \frac{3}{2}R + n_2 \frac{5}{2}R}{1 + n_2} = \frac{(9 + 5n_2)R}{2(1 + n_2)}$

or $23 + 23n_2 = 27 + 15n_2$

i.e. $n_2 = 0.5 \text{ g-mol}$

• CHECK POINT 15.3

1 (c) In isobaric process, heat is used in both, in raising the temperature and in doing work.

2 (a) Work done $= p(V_2 - V_1) = 1 \times 10^5 (1.091 - 1) \times 10^{-6}$
 $= 0.0091 \text{ J}$

3 (c) Work done by gas, $W = p(V_f - V_i)$
 $= 1.0 \times 10^5 (1.7 \times 10^{-3} - 1.5 \times 10^{-3}) = 20 \text{ J}$

$\therefore \Delta U = Q - W = 70 - 20 = 50 \text{ J}$

4 (c) At constant volume, $p \propto T$

$\Rightarrow \frac{p_1}{p_2} = \frac{T_1}{T_2} \Rightarrow \frac{p_1}{p_2} = \frac{300}{400} = \frac{3}{4}$

5 (d) During isothermal change, $T = \text{constant}$

$\Rightarrow \Delta U = 0$. Also, from first law of thermodynamics,
 $\Delta Q = \Delta W$.

6 (b) In isothermal process, $p \propto \frac{1}{V}$

Hence, graph between p and V is a rectangular hyperbola.

7 (d) The process is very fast, so the gas fails to gain or lose heat. Hence, this process is adiabatic.

8 (d) For adiabatic compression,

$$p_1 V_1^\gamma = p_2 V_2^\gamma \Rightarrow \frac{p_2}{p_1} = \left(\frac{V_1}{V_2}\right)^\gamma = \left(\frac{4}{1}\right)^{3/2} = \frac{8}{1}$$

9 (b) Fast processes are adiabatic. Hence, $p_1 V_1^\gamma = p_2 V_2^\gamma$

$\therefore p_2 = p_1 \left(\frac{V_1}{V_2}\right)^\gamma = p \left(\frac{640}{80}\right)^{5/3}$
 $= p(8)^{5/3} = 32p$

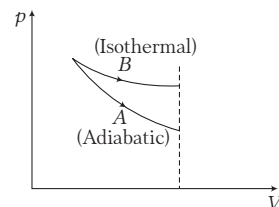
10 (b) Given, $p \propto T^3$

In adiabatic process, $p^{1-\gamma} T^\gamma = \text{constant}$

or $p \propto T^{\frac{\gamma}{\gamma-1}}$, i.e. $\frac{\gamma}{\gamma-1} = 3$

Solving this equation, we get $\gamma = \frac{3}{2}$

11 (a) Work done = Area under p - V diagram.



$(\text{Area})_{\text{iso}} > (\text{Area})_{\text{adi}} \Rightarrow W_{\text{iso}} > W_{\text{adi}}$

12 (c) In adiabatic process, $TV^{\gamma-1} = \text{constant}$

or $T_2 V_2^{\gamma-1} = T_1 V_1^{\gamma-1}$ or $T_2 = \left(\frac{V_1}{V_2}\right)^{\gamma-1} T_1$
 $= (4)^{1.4-1} (300)$
 $= (300)(4)^{0.4} \text{ K}$

13 (d) Using relation $\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}} = (8)^{\frac{3/2-1}{3/2}} = 2$

$\Rightarrow T_2 = 2T_1 \Rightarrow T_2 = 2(273 + 27) = 600 \text{ K} = 327^\circ \text{C}$

14 (b) In adiabatic process, $\Delta U = -\Delta W$

In the first process, $\Delta W_1 = +25 \text{ J}$

$\therefore \Delta U_1 = -25 \text{ J}$

In second process, $\Delta W_2 = -100 \text{ J}$

$\therefore \Delta U_2 = +100 \text{ J}$

• CHECK POINT 15.4

1 (b) $\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$

2 (a) $\frac{W}{Q_1} = \left(1 - \frac{T_2}{T_1}\right)$

or $W = Q_1 \left(1 - \frac{T_2}{T_1}\right) = 10000 \left(1 - \frac{400}{500}\right) = 2000 \text{ J}$

3 (a) Coefficient of performance, $\beta = \frac{Q_2}{W} \approx \frac{T_L}{T_H - T_L}$

$Q_2 = \frac{273 \times 1}{303 - 273} = \frac{273}{30} \approx 9 \text{ J}$

Heat delivered to the surroundings,

$Q_1 = Q_2 + W = 9 + 1 = 10 \text{ J}$

4 (d) Thermal efficiency of a Carnot engine is given by

$\eta = 1 - \frac{T_2}{T_1} \quad \dots(i)$

where, T_2 is temperature of sink and T_1 is the temperature of the source.

Here, $\eta = 50\%$, $T_2 = 50^\circ\text{C} = 273 + 50 = 323\text{ K}$

Putting these values in Eq. (i), we get

$$\frac{50}{100} = 1 - \frac{323}{T_1} \quad \text{or} \quad \frac{323}{T_1} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\therefore T_1 = 646\text{ K} = 373^\circ\text{C}$$

$$5 \text{ (a)} \quad \eta = \left(1 - \frac{T_2}{T_1}\right) \times 100 = \left(1 - \frac{(87 + 273)}{(127 + 273)}\right) \times 100 \approx 10\%$$

$$6 \text{ (d)} \quad \eta = 1 - \frac{T_2}{T_1}; \text{ for } \eta \text{ to be maximum ratio, } \frac{T_2}{T_1} \text{ should be minimum.}$$

$$7 \text{ (d)} \quad \left(\frac{\eta_2}{\eta_1}\right) = \frac{\left(1 - \frac{73}{273}\right)}{\left(1 - \frac{273}{473}\right)} = \frac{200}{273} \times \frac{473}{200} = \frac{1.73}{1}$$

$$8 \text{ (b)} \quad \text{Efficiency of the engine, } \eta = 1 - \frac{T_2}{T_1} = 0.5 \text{ or } T_1 = 2T_2$$

$$= 2(17 + 273) = 580\text{ K}$$

Temperature of hot body is increased by 145°C or 145 K .

$$\therefore T_1' = (580 + 145) = 725\text{ K}$$

$$\text{and } T_2 = (17 + 273) = 290\text{ K}$$

$$\therefore \eta = \left(1 - \frac{290}{725}\right) \times 100 = 60\%$$

$$9 \text{ (c)} \quad \text{Given, } \frac{W}{Q} = 1 - \frac{T_2}{T_1} = 1 - \frac{400}{800} = 0.5$$

$$\text{or Amount of heat taken, } Q = \frac{W}{0.5} = \frac{750}{0.5} = 1500\text{ J}$$

(A) Taking it together

$$1 \text{ (b)} \quad (n_1 + n_2)_i = (n_1 + n_2)_f$$

$$\text{or } \frac{p_0 V}{RT_0} + \frac{p_0 V}{RT_0} = \frac{pV}{RT_0} + \frac{pV}{2RT_0}$$

$$\text{or } \frac{2p_0 V}{RT_0} = \frac{3pV}{2RT_0}$$

$$\therefore p = \frac{4}{3} p_0$$

$$2 \text{ (d)} \quad \text{Process is adiabatic, as } \Delta Q = 0$$

$$p_1 V_1^\gamma = p_2 V_2^\gamma \Rightarrow \frac{p_2}{p_1} = \left(\frac{V_1}{V_2}\right)^\gamma = (2)^{5/3}$$

$$3 \text{ (a)} \quad \text{Efficiency,}$$

$$\eta = 1 - \frac{T_2}{T_1} \times 100 = 1 - \left(\frac{273 + 97}{273 + 137}\right) \times 100 = 9.75\%$$

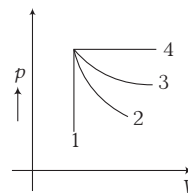
$$4 \text{ (a)} \quad \text{From first law of thermodynamics,}$$

$$\Delta Q = \Delta U + \Delta W = \Delta U + p \Delta V$$

$$\Rightarrow 100 = \Delta U + 50 \times (4 - 10)$$

$$\Rightarrow \Delta U = 400\text{ J}$$

5 (c) For the curve 4, pressure is constant, so this is an isobaric process.



For the curve 1, volume is constant, so it is isochoric process. Between curves 3 and 2, curve 2 is steeper, so it is adiabatic and 3 is isothermal.

$$6 \text{ (b)} \quad \Delta U = 0$$

$$\therefore T = \text{constant or } p \propto \frac{1}{V}$$

$$7 \text{ (a)} \quad \text{Heat given, } \Delta Q = 40\text{ J and work done, } \Delta W = 30\text{ J}$$

$$\Rightarrow \Delta U = \Delta Q - \Delta W = 40 - 30 = 10\text{ J}$$

$$8 \text{ (d)} \quad \text{Since, volume of the ideal gas is constant, so } W = 0.$$

Using first law of thermodynamics,

$$\Delta Q = \Delta U \Rightarrow \Delta U = i^2 \pi$$

$$= 1^2 \times 100 \times 5 \times 60 = 30 \times 10^3 = 30\text{ kJ}$$

9 (b) Cyclic process 1 is clockwise, whereas process 2 is anti-clockwise. Since, negative area (2) > positive area (1), hence net work done is negative.

$$10 \text{ (c)} \quad \text{Since, vessel is insulated. Therefore, } Q = 0$$

$$\text{Against vacuum, } W = 0$$

$$\therefore \Delta U \text{ or } \Delta T = 0$$

$$11 \text{ (a)} \quad pV^{2/3} = \text{constant}$$

$$\therefore \left(\frac{T}{V}\right) \cdot V^{2/3} = \text{constant} \quad \text{(Using Charles' law)}$$

$$\text{or } TV^{-1/3} = \text{constant or } T \propto V^{1/3}$$

So, with increase in V , temperature T will also increase.

$$12 \text{ (c)} \quad \text{Heat absorbed, } Q = W + \Delta U$$

$$= p_0(2V_0 - V_0) + (2p_0 \cdot 2V_0 - 2p_0 V_0)$$

$$= p_0 V_0 + 2p_0 V_0 = 3p_0 V_0$$

$$13 \text{ (a)} \quad V-T \text{ graph is a straight line passing through origin.}$$

$$\text{or } V \propto T$$

Hence, the given process is isobaric.

$$\frac{\Delta Q}{\Delta W} = \frac{nC_p \Delta T}{nR \Delta T} = \frac{C_p}{R} = \frac{5}{2} = 2.5$$

$$14 \text{ (a)} \quad \text{Using first law of thermodynamics, we get}$$

$$\Delta Q_1 < \Delta Q_2$$

$$\therefore \frac{\Delta Q_1}{\Delta T} < \frac{\Delta Q_2}{\Delta T} \quad \text{or } C_1 < C_2 \quad \text{or } \frac{C_1}{C_2} < 1$$

$$15 \text{ (b)} \quad pV^2 = \text{constant}$$

$$\therefore pV \propto \frac{1}{V} \text{ or } T \propto \frac{1}{V}$$

So, as volume increases, the temperature of gas decreases or it gets cooled.

- 16 (c)** p - V diagram of the gas is a straight line passing through origin. Hence, $p \propto V$ or $pV^{-1} = \text{constant}$

Molar heat capacity in the process $pV^x = \text{constant}$ is

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - x}$$

Here, $\gamma = 1.4$ (For diatomic gas) and $x = -1$

$$\Rightarrow C = \frac{R}{1.4 - 1} + \frac{R}{1 + 1} \Rightarrow C = 3R$$

- 17 (a)** In isothermal process, p - V diagram is a rectangular hyperbola. As $T_2 > T_1$, then $(pV)_2 > (pV)_1$.

- 18 (c)** For isobaric process, $\frac{V_2}{V_1} = \frac{T_2}{T_1} \Rightarrow V_2 = V \times \frac{274}{273}$

$$\text{Increase in volume} = \frac{274}{273}V - V = \frac{V}{273}$$

- 19 (b)** $\Delta Q = 2 \cdot \Delta U$ (Given)

$$\therefore nC \cdot \Delta T = 2(nC_V \Delta T) \text{ or } C = 2C_V = 2 \times \frac{3}{2}R = 3R$$

- 20 (c)** In cyclic process, $\Delta U = 0$

$$\begin{aligned} \therefore \text{Heat absorbed} &= \text{Work done} = \text{Area under } p\text{-}V \text{ graph} \\ &= (\pi)(R_x)(R_y) \\ &= \pi(10 \times 10^3)(10 \times 10^{-3}) = 10^2 \pi \text{ J} \end{aligned}$$

- 21 (c)** First law of thermodynamics given, $\Delta Q = \Delta U + p \Delta V$

$$\Rightarrow mL = \Delta U + p(V_2 - V_1)$$

$$\Rightarrow \Delta U = L - p(V_2 - V_1) \quad (\because m = 1)$$

- 22 (d)** In adiabatic process, $TV^{\gamma-1} = \text{constant}$

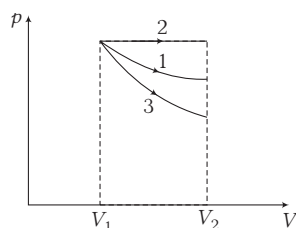
$$\therefore \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} = \left(\frac{AL_2}{AL_1}\right)^{5/3-1} = \left(\frac{L_2}{L_1}\right)^{2/3}$$

- 23 (a)** $\beta = \frac{Q_2}{W} \Rightarrow 4 = \frac{400}{W}$

$$\Rightarrow W = 100 \text{ J}$$

- 24 (a)** On heating, volume will increase. Hence, piston will move to the left.

- 25 (a)** $W = \text{Area under } p\text{-}V \text{ graph}$



$$A_2 > A_1 > A_3$$

$$\therefore W_2 > W_1 > W_3$$

- 26 (a)** Using $p_1 V_1^\gamma = p_2 V_2^\gamma$, we get

$$p_2 = p_1 \left(\frac{V_1}{V_2}\right)^\gamma = 1 \times 10^5 \left(\frac{1}{3}\right)^{1.4} = 2.14 \times 10^4 \text{ Nm}^{-2}$$

- 27 (a)** $p = \frac{nRT}{V-b}$

$$W = \int_V^{2V} p dV = nRT \int_V^{2V} \frac{dV}{V-b} \quad (\text{As, } T = \text{constant})$$

$$= nRT \ln \left(\frac{2V-b}{V-b} \right) = RT \ln \left(\frac{2V-b}{V-b} \right) \quad (\text{As, } n = 1)$$

- 28 (c)** $\Delta Q = \Delta U + \Delta W$

$$\Rightarrow \Delta W = (\Delta Q)_p - (\Delta U)_V = (\Delta Q)_p \left[1 - \frac{(\Delta U)_V}{(\Delta Q)_p} \right]$$

$$= (\Delta Q)_p \left[1 - \frac{C_V}{C_p} \right] = Q \left[1 - \frac{3}{5} \right] = \frac{2}{5} Q$$

$$(\because (\Delta Q)_p = Q \text{ and } \gamma = \frac{5}{3} \text{ for monoatomic gas})$$

- 29 (b)** $Q_1 = nC_p \Delta T \Rightarrow Q_2 = nC_V \Delta T$

$$\therefore Q_1 - Q_2 = nR \Delta T$$

$$\text{or } Q_2 = Q_1 - nR \Delta T = (70) - (2) \left(\frac{8.31}{4.18} \right) (5) \equiv 50 \text{ cal}$$

- 30 (a)** For adiabatic process, $TV^{\gamma-1} = \text{constant}$

$$\therefore \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1} \text{ or } \left(\frac{1}{2}\right)^{\gamma-1} = \sqrt{\frac{1}{2}}$$

$$\therefore \gamma - 1 = \frac{1}{2} \text{ or } \gamma = \frac{3}{2} \therefore pV^{3/2} = \text{constant}$$

- 31 (d)** Oxygen is diatomic gas, hence its energy for four moles

$$= 4 \times \frac{5}{2} RT = 10 RT$$

Argon is a monoatomic gas, hence its internal energy for 6 mol = $6 \times (3/2) RT = 9 RT$

Total internal energy = $(9 + 10) RT = 19 RT$

- 32 (a)** Work done by the gas should be positive, but work done on the gas should be negative.

$$W = p(V_f - V_i) \quad (\text{On the gas})$$

$$= 10^5 (1.5 \times 10^{-2} - 3.0 \times 10^{-2}) = -1500 \text{ J}$$

- 33 (a)** Amount of sweat evaporated/minute

$$\begin{aligned} &= \frac{\text{Sweat produced / minute}}{\text{Number of calories required for evaporation / kg}} \\ &= \frac{\text{Amount of heat produced per minute in jogging}}{\text{Latent heat (in cal / kg)}} \end{aligned}$$

$$= \frac{145 \times 10^3}{580 \times 10^3} = \frac{145}{580} = 0.25 \text{ kg}$$

- 34 (a)** $Up = \text{constant}$

$$\therefore \frac{T}{V} = \text{constant}$$

$$\therefore p = \text{constant}$$

$$\therefore \frac{\Delta U}{\Delta W} = \frac{\Delta U}{\Delta Q - \Delta U} = \frac{C_V}{C_p - C_V}$$

$$= \frac{1}{C_p / C_V - 1} = \frac{1}{(5/3) - 1} = \frac{3}{2}$$

$$35 \text{ (c) } \Delta U = C_V \Delta T = \left(\frac{5}{2}R\right) \Delta T \text{ or } \Delta T = \frac{2(\Delta U)}{5R}$$

From first law of thermodynamics,

$$\Delta U = \Delta Q - \Delta W = Q - \frac{Q}{4} = \frac{3Q}{4}$$

$$\text{Now, molar heat capacity, } C = \frac{\Delta Q}{\Delta T} = \frac{Q}{\frac{2(\Delta U)}{5R}} = \frac{5RQ}{2\left(\frac{3Q}{4}\right)} = \frac{10}{3}R$$

$$36 \text{ (d) At constant volume, } \Delta Q_V = C_V \Delta T_V \text{ (for monoatomic gas)}$$

At constant pressure, $\Delta Q_p = 2C_p \Delta T_p$ (for diatomic gas)

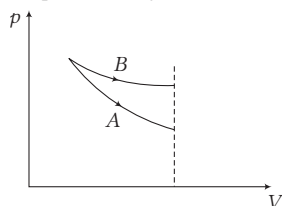
$$\begin{aligned} \therefore \frac{\Delta Q_p}{\Delta Q_V} &= 2 \frac{C_p}{C_V} \times \frac{\Delta T_p}{\Delta T_c} \\ &= 2 \times \frac{7}{3} \times \frac{5}{10} = \frac{7}{3} \\ \Rightarrow \Delta Q_p &= \frac{7}{3} (\Delta Q_V) = \frac{7}{3} (H) \end{aligned}$$

37 (d) During process A to B , pressure and volume both are decreasing. Therefore, temperature and hence, internal energy of the gas will decrease ($T \propto pV$)

or $\Delta U_{A \rightarrow B}$ = negative. Further $\Delta W_{A \rightarrow B}$ is also negative as the volume of the gas is decreasing. Thus, $\Delta Q_{A \rightarrow B}$ is negative.

In process B to C , pressure of the gas is constant, while volume is increasing. Hence, temperature should increase or $\Delta U_{B \rightarrow C}$ = positive. During process CAB , volume of the gas is decreasing. Hence, work done by the gas is negative.

38 (a) In adiabatic process, pressure will be less. Further temperature is the product of p and V .



Hence $p_A < p_B$ and $T_A < T_B$.

39 (c) Net work done = Area ABC - Area AED

$$= \frac{1}{2} \times 2 \times 1 - \frac{1}{2} \times 1 \times 1 = \frac{1}{2} \text{ J}$$

$$40 \text{ (b) Number of moles} = \frac{2}{4} = \frac{1}{2}$$

$$\begin{aligned} W_{A \rightarrow B} &= \text{Area under } p\text{-}V \text{ graph} \\ &= \frac{1}{2} \times 3p_0 \times V_0 = \frac{3}{2} p_0 V_0 \end{aligned}$$

$$\Delta U_{A \rightarrow B} = nC_V \Delta T = \frac{1}{2} \left(\frac{3}{2}R\right) \left(\frac{8p_0 V_0}{R} - \frac{2p_0 V_0}{R}\right) = \frac{9}{2} p_0 V_0$$

$$\therefore Q_{A \rightarrow B} = W_{A \rightarrow B} + \Delta U_{A \rightarrow B} = 6p_0 V_0$$

41 (c) ΔU = Same in all processes

$$W_1 = + \text{ve}, W_2 = 0 \text{ and } W_3 = - \text{ve}$$

$$\therefore Q_1 > Q_2 > Q_3$$

$$42 \text{ (c) } p_1 V_1 = p_2 V_2 \text{ or } p \propto \frac{1}{V}$$

Radius has become two times. Hence, volume will become eight times, i.e. pressure at surface is $\frac{1}{8}$ times the pressure at depth h . On surface, pressure is equal to 10 m of water. Hence, pressure at depth h is equal to 80 m of water. Of this 10 m is atmospheric pressure. Therefore, h is 70 m.

43 (a) This is the case of free expansion and in this case $\Delta W = 0$, $\Delta U = 0$ so temperature remains same, i.e. 300 K.

$$44 \text{ (c) Initially, } \eta = \frac{T_1 - T_2}{T_1} \Rightarrow 0.5 = \frac{T_1 - (273 + 7)}{T_1}$$

$$\Rightarrow \frac{1}{2} = \frac{T_1 - 280}{T_1} \Rightarrow T_1 = 560 \text{ K}$$

$$\text{Finally, } \eta'_1 = \frac{T'_1 - T_2}{T'_1}$$

$$\Rightarrow 0.7 = \frac{T'_1 - (273 + 7)}{T'_1}$$

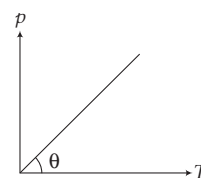
$$\Rightarrow T'_1 = 933 \text{ K}$$

$\therefore$ Increase in temperature = $933 - 560 = 373 \text{ K}$

$$46 \text{ (b) As, } p = \left(\frac{nR}{V}\right) T = \left(\frac{mR}{MV}\right) T$$

At constant volume, $p \propto T$

i.e. p - T graph is a straight line passing through origin. Slope of this line is proportional to m , mass of gas.



$$\therefore \tan \alpha = 2 \tan \beta$$

47 (b) In adiabatic process, $\Delta U = -W$

$$\therefore U_A - U_B = -(-30) = 30 \text{ J}$$

$$\text{or } U_B - U_A = -30 \text{ J}$$

48 (c) Given, $T = KV^2$

$$\therefore dT = (2KV) \cdot dV$$

$$\text{or } dV = \frac{dT}{2KV} \text{ and } p = \frac{nRT}{V}$$

$$\Rightarrow p dV = \frac{nRT}{2KV^2} \cdot dT = \frac{nRT}{2(T)} \cdot dT = \frac{nR}{2} \cdot dT$$

$$\Rightarrow W = \int p dV = \int_{T_0}^{4T_0} \frac{nR}{2} \cdot dT = \frac{3}{2} nRT_0$$

49 (a) Given, $V = kT^2$

$$\therefore dV = 2kT \cdot dT$$

$$\text{and } p = \frac{RT}{V} = \frac{RT}{kT^2} = \frac{R}{kT} \text{ or } p dV = 2R dT$$

$$W = \int p dV = 2R \int dT = 2R (60) = 120R$$

- 50 (a) $VT = \text{constant} \Rightarrow V(pV) = \text{constant}$ (As, $T \propto pV$)

$$\therefore pV^2 = \text{constant}$$

Molar heat capacity of a gas in the process

$pV^x = \text{constant}$ is

$$C = C_V + \frac{R}{1-x}$$

Here,

$$x = 2$$

$$\therefore C = \frac{3}{2}R + \frac{R}{1-2} = \frac{R}{2}$$

51 (c) $p = \frac{RT}{V} = \frac{R(T_0 + \alpha V)}{V}$

$$\therefore W = \int_{V_1}^{V_2} p \cdot dV = \int_{V_1}^{V_2} \frac{R(T_0 + \alpha V)}{V} \cdot dV$$

$$= RT_0 \ln \frac{V_2}{V_1} + R\alpha (V_2 - V_1)$$

$$\Delta U = nC_V \Delta T = (C_p - R)[(T_0 + \alpha V_2) - (T_0 + \alpha V_1)]$$

$$= \alpha (C_p - R)(V_2 - V_1)$$

Now, $Q = \Delta U + W$

$$= \alpha C_p (V_2 - V_1) + RT_0 \ln \left| \frac{V_2}{V_1} \right|$$

- 52 (c) Process AB is isochoric. Hence, $W_{AB} = 0$

Process BC is isothermal. Hence, $W_{BC} = nRT_2 \ln \left| \frac{V_f}{V_i} \right|$

or $W_{BC} = RT_2 \ln \left| \frac{V_1}{V_2} \right|$

Process CA is isobaric (As, $V \propto T$)

$$\therefore W_{CA} = p(V_f - V_i) = nR(T_f - T_i) = R(T_1 - T_2)$$

- 53 (d) ΔU remains same for both paths.

For path iaf, $\Delta U = \Delta Q - \Delta W = 50 - 20 = 30 \text{ J}$

For path fi, $\Delta U = -30 \text{ J}$ and $\Delta W = -13 \text{ J}$

$$\Rightarrow \Delta Q = -30 - 13 = -43 \text{ J}$$

54 (c) $\eta_A = \frac{T_1 - T_2}{T_1} = \frac{W_A}{Q_1}$

and $\eta_B = \frac{T_2 - T_3}{T_2} = \frac{W_B}{Q_2}$

As, $W_A = W_B$

$$\therefore \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \times \frac{T_2 - T_3}{T_1 - T_2} = \frac{T_1}{T_2} \quad \left(\because \frac{Q_1}{Q_2} = \frac{T_1}{T_2} \right)$$

$$\therefore T_2 = \frac{T_1 + T_3}{2} = \frac{800 + 300}{2} = 550 \text{ K}$$

- 55 (c) According to the question, given that $pV = \text{constant}$

Hence, we can say that the gas is going through an isothermal process.

It is clear from the graph that between processes 1 and 2 temperature is constant and the gas expands and pressure decreases, i.e. $p_2 < p_1$ which corresponds to diagram (iii).

- 56 (a) $\Delta Q = ms\Delta T$

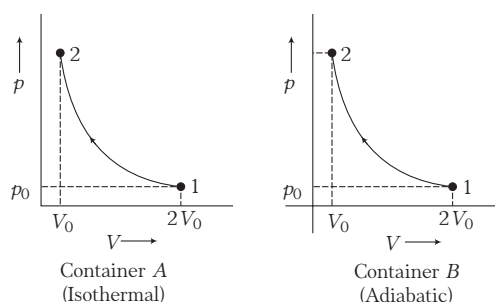
$$\Rightarrow \Delta S = \frac{\Delta Q}{T} = \frac{ms\Delta T}{T}$$

The total change in entropy as temperature rises from T_1 to T_2 ,

$$\begin{aligned} S_2 - S_1 &= \int_{T_1}^{T_2} ms \frac{dT}{T} = ms \ln \left(\frac{T_2}{T_1} \right) \\ &= 0.1 \times 4200 \ln \left(\frac{87 + 273}{27 + 273} \right) \\ &= 420 \ln(1.2) \text{ JK}^{-1} \end{aligned}$$

57 (a) $\eta = \frac{\Sigma \Delta W}{\Sigma \Delta Q} \times 100 = 10.82\%$

- 58 (a) Consider the p - V diagram shown for the container A (isothermal) and for container B (adiabatic).



Both the process involve compression of the gas.

For isothermal process (gas A) (during $1 \rightarrow 2$),

$$p_1 V_1 = p_2 V_2 \Rightarrow p_0 (2V_0) = p_2 (V_0) \Rightarrow p_2 = 2p_0$$

For adiabatic process (gas B) (during $1 \rightarrow 2$),

$$\begin{aligned} p_1 V_1^\gamma &= p_2 V_2^\gamma \\ \Rightarrow p_0 (2V_0)^\gamma &= p_2 (V_0)^\gamma \\ \Rightarrow p_2 &= \left(\frac{2V_0}{V_0} \right)^\gamma p_0 = (2)^\gamma p_0 \end{aligned}$$

Hence, $\frac{(p_2)_B}{(p_2)_A}$ = Ratio of final pressure

$$= \frac{(2)^\gamma p_0}{2p_0} = 2^{\gamma-1}$$

where, γ is ratio of specific heat capacities for the gas.

- 59 (a) $\Delta W_{AB} = 0$ (As, $V = \text{constant}$)

$$\therefore \Delta Q_{AB} = \Delta U_{AB} = 50 \text{ J} \quad (\text{Given})$$

$$U_A = 1500 \text{ J}$$

$$\therefore U_B = (1500 + 50) \text{ J} = 1550 \text{ J}$$

$$\Delta W_{BC} = -\Delta U_{BC} = -40 \text{ J} \quad (\text{Given})$$

$$\therefore \Delta U_{BC} = 40 \text{ J}$$

$$\therefore U_C = (1550 + 40) \text{ J} = 1590 \text{ J}$$

- 60 (b) $W_{A \rightarrow B}$ = Area under p - V diagram $= 18 p_0 V_0$

$$\Delta U_{A \rightarrow B} = nC_V \Delta T$$

$$= (1) \left(\frac{3}{2} R \right) \left(\frac{30 p_0 V_0}{R} - \frac{3 p_0 V_0}{R} \right) = \frac{81}{2} p_0 V_0$$

$$\therefore Q_{A \rightarrow B} = W_{A \rightarrow B} + \Delta U_{A \rightarrow B} = \frac{117}{2} p_0 V_0$$

$$\text{Molar heat capacity, } C = \frac{Q}{n \Delta T} = \frac{\frac{117}{2} p_0 V_0}{\left(\frac{30 p_0 V_0}{R} \right) - \left(\frac{3 p_0 V_0}{R} \right)} = \frac{13}{6} R$$

61 (d) For path ab , $(\Delta U)_{ab} = 7000 \text{ J}$

By using $\Delta U = n C_V \Delta T$

$$7000 = n \times \frac{5}{2} R \times 700 \Rightarrow n = 4/R$$

For path ca , $(\Delta Q)_{ca} = (\Delta U)_{ca} + (\Delta W)_{ca}$... (i)

$$\therefore (\Delta U)_{ab} + (\Delta U)_{bc} + (\Delta U)_{ca} = 0$$

$$\therefore 7000 + 0 + (\Delta U)_{ca} = 0$$

$$\Rightarrow (\Delta U)_{ca} = -7000 \text{ J} \quad \dots (ii)$$

$$\begin{aligned} \text{Also, } (\Delta W)_{ca} &= p_1(V_1 - V_2) = nR(T_1 - T_2) \\ &= (4/R) \times R \times (300 - 1000) \\ &= -2800 \text{ J} \quad \dots (iii) \end{aligned}$$

By solving Eqs. (i), (ii) and (iii), we get

$$\begin{aligned} (\Delta Q)_{ca} &= -7000 - 2800 \\ &= -9800 \text{ J} \end{aligned}$$

62 (c) Processes A to B and C to D are parts of straight line graphs passing through origin.

$p \propto T$. So, volume remains constant for the graphs AB and CD .

So, no work is done during processes for A to B and C to D ,

$$\text{i.e. } W_{AB} = W_{CD} = 0$$

$$\begin{aligned} \text{and } W_{BC} &= p_2(V_C - V_B) = nR(T_C - T_B) \\ &= 6R(2200 - 800) = 6R \times 1400 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{Also, } W_{DA} &= p_1(V_A - V_D) = nR(T_A - T_D) \\ &= 6R(600 - 1200) = -6R \times 600 \text{ J} \end{aligned}$$

Hence, work done in complete cycle,

$$\begin{aligned} W &= W_{AB} + W_{BC} + W_{CD} + W_{DA} \\ &= 0 + 6R \times 1400 + 0 - 6R \times 600 \\ &= 6 \times 8.3 \times 800 \approx 40 \text{ kJ} \end{aligned}$$

(B) Medical entrance special format questions

• Assertion and reason

1 (a) During melting, volume of given system decreases, so work done by the system on surrounding is negative. But the work done on the system by the surrounding is positive.

2 (d) If initial and final volumes are equal, then work done may or may not be zero. For work done by the gas to be zero, volume should remain constant throughout the process. For example in isochoric process.

3 (c) Volume is increasing. Therefore, ΔW is positive.

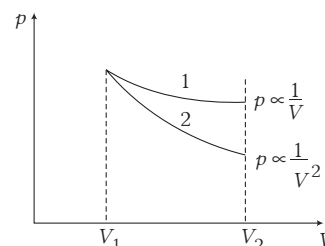
Therefore, ΔU is negative.

$$\text{Now, } \Delta Q = \Delta W + \Delta U$$

Therefore, Q may be positive, negative or zero.

Further, heat given to gas is path function.

4 (b) Area under p - V graph in process-1 is more.



• Statement based questions

1 (d) In process AB , density is constant. Hence, volume is constant. Therefore, work done will be zero.

In process BC , $p \propto \rho$ but $\rho \propto \frac{1}{V} \therefore p \propto \frac{1}{V}$

or the process is isothermal. Hence, internal energy is constant.

2 (a) Using $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$, we get

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 300 \left(\frac{1}{2} \right)^{5/3-1} \approx 189 \text{ K}$$

3 (c) Work is done by the gas (As, cyclic process is clockwise)

$$\Delta W = \text{Area } ABCD$$

So, from the first law of thermodynamics ΔQ (net heat absorbed) $= \Delta W = \text{Area } ABCD$

As change in internal energy in cycle $\Delta U = 0$.

4 (a) For given temperatures T_1 and T_2 , efficiency of Carnot engine is maximum.

Heat can flow from a body at lower temperature to a body at higher temperature by means of external agency.

• Match the columns

1 (b) For adiabatic process bulk modulus, $B = \gamma p$

(where, B is constant)

For isothermal process, temperature is constant.

$$\therefore \text{Slope} = -\frac{p}{V}$$

$$\therefore \gamma = 1 + \frac{2}{f} \Rightarrow f = \frac{2}{\gamma-1} \Rightarrow C_p = \frac{\gamma}{\gamma-1} R$$

Hence, $A \rightarrow r$, $B \rightarrow p$, $C \rightarrow q$, $D \rightarrow s$.

2 (c) Isothermal bulk modulus, $B_T = p = \frac{RT}{V}$ (As, $n = 1$)

$$\text{and adiabatic bulk modulus, } B_S = \gamma p = \frac{5}{3} \frac{RT}{V}$$

$$\text{Slope of } p\text{-}V \text{ graph in isothermal process} = -\frac{p}{V} = -\frac{RT}{V^2}$$

$$\text{and slope of } p\text{-}V \text{ graph in adiabatic process} = -\frac{\gamma p}{V}$$

$$= -\frac{5}{3} \frac{RT}{V^2} = -\frac{5}{3} \frac{p}{V}$$

Hence, $A \rightarrow t$, $B \rightarrow t$, $C \rightarrow p$, $D \rightarrow q$.

(C) Medical entrances' gallery

1 (a) Since, the entire system is thermally insulated. Thus, there will be no transfer of heat between the system and the surrounding. Hence, when the stop cock is suddenly opened, a sudden process of expansion will take place with no heat transfer. Thus, the given process will be adiabatic, as it is a process that occurs without transferring heat or mass between a thermodynamic system and its surroundings.

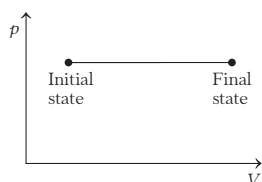
2 (b) Efficiency of Carnot engine is given as $\eta = 1 - \frac{T_2}{T_1}$

where, T_2 = temperature of sink

and T_1 = temperature of source.

Hence, η depends upon the temperatures of the source and sink both.

3 (c) pV -diagram for an ideal gas is given as



From the p - V diagram, it is clear that pressure of an ideal gas in the piston cylinder is constant during thermodynamic process. Hence, this process is isobaric.

4 (b) Given, mass of water, $m = 1\text{g}$

Volume of 1 g of water = $1\text{cm}^3 = 10^{-6}\text{ m}^3$

Volume of 1 g of steam = $1671\text{cm}^3 = 1671 \times 10^{-6}\text{ m}^3$

Pressure, $p = 1 \times 10^5\text{ Pa}$

Latent heat of vaporisation of water, $L = 2256\text{J/g}$

Change in volume, $\Delta V = (1671 - 1) \times 10^{-6}\text{ m}^3$
 $= 1670 \times 10^{-6}\text{ m}^3$

Heat supplied, $\Delta Q = mL = 1 \times 2256 = 2256\text{J}$

As the steam expands, so the work done in expansion is

$$\Delta W = p\Delta V = 1 \times 10^5 \times 1670 \times 10^{-6}$$

$$= 167\text{J}$$

According to first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

$$\Rightarrow \Delta U = \Delta Q - \Delta W = 2256 - 167$$

$$= 2089\text{J}$$

5 (a) In an adiabatic process, the system is completely insulated from the surroundings. Thus, heat is neither absorbed nor released by the system to the surroundings, so $\Delta Q = 0$.

Sudden processes are adiabatic like bursting of cycle tyre, etc.

If the pressure of gas is kept constant, then the process is called isobaric, i.e. $\Delta p = 0$.

If the temperature of the system remains constant, then it is called isothermal process, i.e. $\Delta T = 0$.

If the volume of gas is constant in a system, then it is called isochoric process, i.e. $\Delta V = 0$.

6 (a) The heat capacity ratio, $\gamma = \frac{C_p}{C_v}$... (i)

where, C_p = molar heat capacity at constant pressure and C_v = molar heat capacity at constant volume.

Also, $C_p = C_v + R$ (From Mayer's relation)

$$\therefore C_v = \frac{f}{2}R$$

where, f = degree of freedom.

$$\Rightarrow C_p = \left(\frac{f}{2} + 1\right)R$$

So, Eq. (i) becomes $\gamma = 1 + \frac{2}{f}$

For hydrogen gas, which is diatomic, the degree of freedom is 5 (3 translational, 2 rotational).

$$\therefore \gamma = 1 + \frac{2}{5} = \frac{7}{5}$$

For helium gas, which is monoatomic, the degree of freedom is 3 (3 translational only).

$$\therefore \gamma = 1 + \frac{2}{3} = \frac{5}{3}$$

The diatomic gas X also have vibrational motion, so degree of freedom is 7 (3 translational, 2 rotational and 2 vibrational).

$$\therefore \gamma = 1 + \frac{2}{7} = \frac{9}{7}$$

7 (a) In adiabatic process, heat transfer to the thermodynamic system is zero, i.e. $Q = 0$.

Hence, by first law of thermodynamics, $Q = 0 = \Delta U + W$

$$\therefore W = -\Delta U$$

So, work done is equal to negative of change in internal energy.

Since, the internal energy is a function of thermodynamic state of the system.

It does not depend on the path, therefore work done is independent of path in adiabatic process.

8 (a) Given, temperature of food material in refrigerator,

$$T_2 = 4^\circ\text{C} = 273 + 4 = 277\text{K}$$

Temperature of environment, $T_1 = 15^\circ\text{C} = 273 + 15 = 288\text{K}$

$$\therefore \text{Carnot efficiency, } \eta = 1 - \frac{T_2}{T_1} = 1 - \frac{277}{288} = 0.038$$

9 (c) For N_2 gas, degree of freedom, $f = 5$ (For di-atomic gas)

$$\therefore C_{V1} = \frac{f}{2} \cdot R = \frac{5}{2}R \text{ and } C_{p1} = C_{V1} + R = \frac{7}{2}R$$

Amount of N_2 gas, $m_1 = 7\text{g}$

$\therefore$ Number of moles of N_2 ,

$$n_1 = \frac{m_1}{28} = \frac{7}{28} = \frac{1}{4}$$

For Ar gas, degree of freedom, $f = 3$ (For monoatomic gas)

$$\therefore C_{V2} = \frac{3}{2}R$$

$$\therefore C_{p2} = C_{V2} + R = \frac{3}{2}R + R = \frac{5}{2}R$$

Amount of Ar gas, $m_2 = 20$ g

∴ Number of moles of Ar gas,

$$n_2 = \frac{m_2}{40} = \frac{20}{40} = \frac{1}{2}$$

$$\therefore (C_p)_{\text{mix}} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 + n_2} = \frac{\frac{1}{4} \times \frac{7}{2} R + \frac{1}{2} \times \frac{5}{2} R}{\frac{1}{4} + \frac{1}{2}} = \frac{17R}{6}$$

$$(C_v)_{\text{mix}} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2} = \frac{\frac{1}{4} \times \frac{5}{2} R + \frac{1}{2} \times \frac{3}{2} R}{\frac{1}{4} + \frac{1}{2}} = \frac{11R}{6}$$

$$\therefore \left(\frac{C_p}{C_v} \right)_{\text{mix}} = \gamma_{\text{mix}} = \frac{\frac{17R}{6}}{\frac{11R}{6}} = \frac{17}{11}$$

10 (a) By ideal gas equation, at constant pressure, $p \Delta V = nR \Delta T$... (i)

$$\therefore \text{Work done, } W = p \Delta V$$

$$W = nR \Delta T \quad \dots (ii)$$

At constant pressure, heat given to gas is given by $Q = nC_p \Delta T$... (iii)

where, C_p = specific heat at constant pressure.

From Eqs. (ii) and (iii), we get

$$\frac{W}{Q} = \frac{nR \Delta T}{nC_p \Delta T} = \frac{R}{C_p} = \frac{R}{\frac{(f+2)R}{2}}$$

$$\left[\therefore C_p = \frac{(f+2)R}{2} \right]$$

$$\frac{W}{Q} = \frac{2}{f+2}$$

$$\frac{10}{Q} = \frac{2}{5+2} \quad (\text{For diatomic gas, } f = 5)$$

$$\Rightarrow Q = 35 \text{ J}$$

11 (b) According to the question,

heat spent during the conversion of sample of water at 100°C to steam is

$$\Delta Q = 54 \text{ cal} = 54 \times 4.18 \text{ J} = 225.72 \text{ J}$$

Normal pressure, $p = 1.013 \times 10^5 \text{ Nm}^{-2}$

Net work done during the conversion would be given as

$$\Delta W = p \Delta V = p (V_{\text{steam}} - V_{\text{water}})$$

Here, $V_{\text{steam}} = 167.1 \text{ cc}$

$$= 167.1 \times 10^{-6} \text{ m}^3$$

$$V_{\text{water}} = 0.1 \text{ g} = 0.1 \text{ cc}$$

$$= 0.1 \times 10^{-6} \text{ m}^3$$

$$\therefore \Delta W = 1.013 \times 10^5 [(167.1 - 0.1) \times 10^{-6}]$$

$$= 1.013 \times 167 \times 10^{-1}$$

$$= 16.917 \text{ J}$$

Now, by the first law of thermodynamics,

$$\Delta Q = \Delta U + \Delta W$$

where, ΔU is the change in internal energy of the sample.

$$\Rightarrow \Delta U = \Delta Q - \Delta W \quad \dots (i)$$

Substituting the values in Eq. (i), we get

$$\Delta U = 225.72 - 16.917 \approx 208.7 \text{ J}$$

12 (c) Efficiency of an ideal heat engine is given as

$$\eta = 1 - \frac{T_2}{T_1}$$

where, T_1 is the temperature of the source and T_2 the temperature of the sink.

Here, $T_1 = 100 + 273 = 373 \text{ K}$

$$T_2 = 0 + 273 = 273 \text{ K}$$

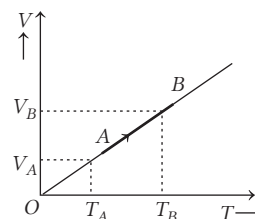
$$\Rightarrow \eta = 1 - \frac{273}{373}$$

$$= \frac{373 - 273}{373} = \frac{100}{373} = 0.268$$

$$\therefore \eta \% = 0.268 \times 100 = 26.8 \%$$

13 (c) According to the given graph,

volume (V) $\propto$ temperature (T) or $\frac{V}{T} = \text{constant}$



Thus, the process is isobaric.

$$\therefore \text{Work done, } \Delta W = p \Delta V$$

$$= nR \Delta T = nR(T_B - T_A) \quad \dots (i)$$

$$\text{Heat absorbed, } \Delta Q = nC_p \Delta T$$

$$= nC_p(T_B - T_A) \quad \dots (ii)$$

As, $C_p = \frac{\gamma R}{\gamma - 1}$ (where, $\gamma = 1 + \frac{2}{f}$)

For a monoatomic gas, $f = 3$

$$\Rightarrow C_p = \left(R + \frac{3}{2} R \right) = \frac{5}{2} R$$

Substituting the value of C_p in Eq. (ii), we get

$$\Delta Q = n \left(\frac{5}{2} R \right) (T_B - T_A)$$

$$\text{Hence, } \frac{\Delta W}{\Delta Q} = \frac{nR(T_B - T_A)}{n \left(\frac{5}{2} R \right) (T_B - T_A)} = \frac{2}{5}$$

14 (d) As in isothermal process, temperature remains constant, internal energy ($\Delta U = 0$), hence there is no change in internal energy of the system. Hence, the energy taken by the gas is utilised by doing work against external pressure.

Hence, whole heat energy given to the body in isothermal process gets converted into work done.

But $\Delta Q = \Delta U + \Delta W$ is the relation of first law of thermodynamics.

- 15 (a) Initial and final states are same in all the process.

Hence, $\Delta U = 0$ is same for each case.

$\therefore \Delta Q = \Delta W = \text{Area enclosed by curve with volume axis}$

$\therefore (\text{Area})_1 < (\text{Area})_2 < (\text{Area})_3$

$\therefore Q_1 < Q_2 < Q_3$

- 16 (c) For diatomic gas, $\gamma = 1.4 = \frac{7}{5}$

At standard condition, $T = 27^\circ \text{C} = 300 \text{ K}$ (Here, $T = T_1$)

For adiabatic process, $TV^{\gamma-1} = \text{constant}$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$(300 V_1^{\frac{7}{5}-1} = T_2 \left(\frac{V_1}{5}\right)^{\frac{7}{5}-1} \quad (\text{Given})$$

$$\Rightarrow T_2 = \frac{300 \times V_1^{2/5}}{V_1^{2/5} \times \left(\frac{1}{5}\right)^{2/5}} = 300 \times 5^{2/5} = 300 \times 1.903 = 571 \text{ K}$$

Mean kinetic energy of rotating molecules

$$= kT = 1.38 \times 10^{-23} \times 571$$

$$= 787.98 \times 10^{-23} \text{ J}$$

- 17 (a) As we know, $W = p\Delta V = nR\Delta T$

$$W_{BC} = -nR(2T_0 - T_0) = -nRT_0$$

and $W_{CA} = + 2nRT_0 \ln 2$

$$\text{Work done} = W_{CA} + W_{BC} = 2nRT_0 \ln 2 - nRT_0 = nRT_0 (2\ln 2 - 1)$$

Also, input heat,

$$\Delta Q_{BC} = nC_p \Delta T = \frac{nR\gamma T_0}{\gamma - 1}$$

$$\text{Efficiency} = \frac{\text{Work}}{\text{Input heat}} = \frac{(2\ln 2 - 1)}{\gamma/(\gamma - 1)}$$

- 18 (c) Here, $Q_1 = 6 \times 10^5 \text{ cal}$,

$$T_1 = 227^\circ \text{C} = 227 + 273 = 500 \text{ K}$$

$$T_2 = 127^\circ \text{C} = 127 + 273 = 400 \text{ K}$$

Work done/cycle, $W = ?$

$$\text{As, } \frac{Q_2}{Q_1} = \frac{T_2}{T_1} \text{ or } Q_2 = \frac{T_2}{T_1} \times Q_1 = \frac{400}{500} \times 6 \times 10^5 = 4.8 \times 10^5 \text{ cal}$$

$$\text{As, } W = Q_1 - Q_2 = 6 \times 10^5 - 4.8 \times 10^5 = 1.2 \times 10^5 \text{ cal} = 1.2 \times 10^5 \times 4.2 \text{ J} \quad (\because 1 \text{ cal} = 4.2 \text{ J})$$

$$\Rightarrow W = 5.04 \times 10^5 \text{ J} \approx 5 \times 10^5 \text{ J}$$

- 19 (a) $\therefore$ Efficiency of heat engine is $\eta = \frac{\text{Work done}}{\text{Heat input}} \times 100$

$$\text{or } \frac{\eta}{100} = \frac{\text{Work done}}{\text{Heat input}} = \frac{500}{\text{Heat input}}$$

$$\text{Heat input} = \frac{500 \times 100}{50} = 1000 \text{ J}$$

- 20 (b) $\therefore$ Efficiency of a heat engine,

$$\eta = 1 - \frac{T_2}{T_1} \text{ or } \frac{1}{6} = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow \frac{T_2}{T_1} = 1 - \frac{1}{6} = \frac{5}{6} \Rightarrow T_2 = \frac{5T_1}{6} \quad \dots(i)$$

According to question, its efficiency is doubled when temperature of sink decreased by 62°C .

$$\therefore 2\eta = 1 - \frac{T_2 - 62}{T_1}$$

$$\Rightarrow 2\left(\frac{1}{6}\right) = 1 - \frac{\left(\frac{5T_1}{6} - 62\right)}{T_1} \quad [\text{From Eq. (i)}]$$

$$\Rightarrow \frac{5T_1}{6} - 62 = T_1 \left(1 - \frac{2}{6}\right) = \frac{2T_1}{3}$$

$$\text{or } \frac{5T_1}{6} - \frac{2T_1}{3} = 62 \Rightarrow \frac{T_1}{6} = 62$$

$$\text{or } T_1 = 372 \text{ K}$$

- 21 (b) In isochoric process, the curve is parallel to Y-axis because volume is constant, i.e. for process I.

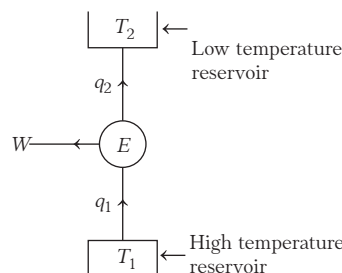
Isobaric curve is parallel to X-axis because pressure is constant, i.e. for process IV.

In processes II and III, the curve II is steeper than curve III.

So, process II is adiabatic and process III is isothermal.

Hence, $A \rightarrow r$, $B \rightarrow p$, $C \rightarrow s$, $D \rightarrow q$.

- 22 (b) Consider schematic diagram for a Carnot engine as shown below.



In case of engine, engine efficiency

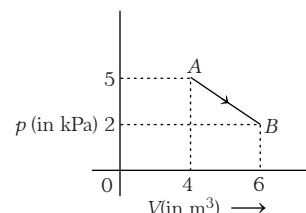
$$= \frac{\text{work}}{\text{heat absorbed}} = \frac{W}{q_1}$$

$$\therefore \frac{W}{q_1} = \frac{1}{10} \Rightarrow \frac{10}{q_1} = \frac{1}{10} \text{ or } q_1 = 100 \text{ J}$$

When this engine is reversed, it takes in work W and heat q_2 from cold reservoir and ejects 100 J of heat to hot reservoir.

$$\therefore W + q_2 = q_1 \Rightarrow 10 + q_2 = 100 \text{ or } q_2 = 90 \text{ J}$$

- 23 (c) For a diatomic gas,



$$C_V = \frac{5}{2}R$$

The change in internal energy of gas from A to B is

$$\begin{aligned}\Delta U &= nC_V dT = n \left(\frac{5R}{2} \right) (T_B - T_A) \\ &= nR \frac{5}{2} \left(\frac{p_B V_B}{nR} - \frac{p_A V_A}{nR} \right) \\ &= \frac{5}{2} (2 \times 10^3 \times 6 - 5 \times 10^3 \times 4) \\ &= \frac{5}{2} \times (-8 \times 10^3) = -\frac{4 \times 10^4}{2} \\ &= -20 \text{ kJ}\end{aligned}$$

24 (b) As, molar heat capacity, $C = \frac{Q}{m\Delta T} = \frac{Q}{0} = \infty$

In isothermal process $\Delta T = 0$, therefore C is not defined further. In isothermal process, pV and T both are constants.

25 (a) In adiabatic expansion, $\Delta Q = 0$

$$\Rightarrow W + \Delta U = 0, \text{ so } W = \text{positive and } \Delta U = \text{negative.}$$

Thus, work is done by the gas at the cost of internal energy of gas.

$$\therefore U, T \text{ or product of } pV \text{ will decrease because } U \propto T \propto pV.$$

26 (a) Slope of adiabatic process of p - V diagram $\left(\frac{dp}{dV} \right) = \gamma \frac{p}{V}$,

$$\text{i.e. slope} \propto \gamma$$

$$\text{From graph, (slope)}_2 > (\text{slope})_1$$

$$\Rightarrow \gamma_2 > \gamma_1$$

Also, γ for monoatomic gas (Ar, He) is greater than γ for diatomic gas (O_2 , N_2).

27 (a) Efficiency of heat engine, $\eta = 1 - \frac{T_2}{T_1}$ or $\eta = \frac{T_1 - T_2}{T_1}$

$$\text{Here, } T_1 = 273 + 127 = 400 \text{ K}$$

$$T_2 = 273 + 27 = 300 \text{ K}$$

$$\therefore \eta = \frac{400 - 300}{400} = \frac{100}{400} = 0.25 = 25\%$$

Hence, 26% efficiency is impossible for the given heat engine.

28 (d) For polytropic process of index n ,

$$\text{specific heat capacity} = C_V + \frac{R}{1-n}$$

$$\therefore \text{Process, } pV^3 = \text{constant} \Rightarrow n = 3$$

$$\therefore C = C_V + \frac{R}{1-n} = \frac{fR}{2} + \frac{R}{1-3}$$

$$\text{For monoatomic gas, } f = 3$$

$$\text{So, } C_V = \frac{fR}{2} = \frac{3R}{2}$$

$$\Rightarrow C = \frac{3R}{2} - \frac{R}{2} = R$$

29 (b) For a refrigerator, we know that,

$$\frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2} = \frac{T_1}{T_1 - T_2}$$

where, Q_1 = amount of heat delivered to the room,

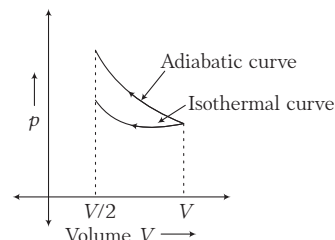
W = electrical energy consumed,

$$T_1 = \text{room temperature} = t_1 + 273$$

$$\text{and } T_2 = \text{temperature of sink} = t_2 + 273.$$

$$\therefore \frac{Q_1}{1} = \frac{t_1 + 273}{t_1 + 273 - (t_2 + 273)} \Rightarrow Q_1 = \frac{t_1 + 273}{t_1 - t_2}$$

- 30 (a)** The solution of this question can be understood by plotting a p - V graph for the compression of a gas isothermally and adiabatically simultaneously to half of its initial volume.



Since, the isothermal curve is less steeper than the adiabatic curve. So, area under the p - V curve for adiabatic process has more magnitude than isothermal curve. Hence, work done in adiabatic process will be more than that of isothermal process.

31 (b) Given, temperature of source, $T_1 = 30^\circ \text{C} = 30 + 273$

$$T_1 = 303 \text{ K}$$

$$\text{Temperature of sink, } T_2 = 4^\circ \text{C} = 4 + 273 \Rightarrow T_2 = 277 \text{ K}$$

As we know that,

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \Rightarrow \frac{Q_2 + W}{Q_2} = \frac{T_1}{T_2} \quad (\because W = Q_1 - Q_2)$$

where, Q_2 is the amount of heat drawn from the sink (at T_2), W is work done on working substance and Q_1 is amount of heat rejected to source (at room temperature T_1).

$$\Rightarrow WT_2 + T_2 Q_2 = T_1 Q_2 \Rightarrow WT_2 = T_1 Q_2 - T_2 Q_2$$

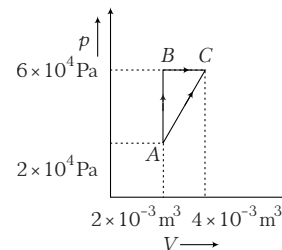
$$\Rightarrow WT_2 = Q_2 (T_1 - T_2)$$

$$\Rightarrow W = Q_2 \left(\frac{T_1}{T_2} - 1 \right) \Rightarrow W = 600 \times 4.2 \times \left(\frac{303}{277} - 1 \right)$$

$$\Rightarrow W = 600 \times 4.2 \times \left(\frac{26}{277} \right) \Rightarrow W = 236.5 \text{ J}$$

$$\therefore \text{Power} = \frac{\text{Work done}}{\text{Time}} = \frac{W}{t} = \frac{236.5}{1} = 236.5 \text{ W}$$

32 (c) Since, initial and final points are same



$$\text{So, } \Delta U_{A \rightarrow B \rightarrow C} = \Delta U_{A \rightarrow C} \quad \dots(i)$$

Also, $A \rightarrow B$ is isochoric process.

$$\text{So, } dW_{A \rightarrow B} = 0 \text{ and } dQ = dU + dW$$

$$\text{So, } dQ_{A \rightarrow B} = dU_{A \rightarrow B} = 400 \text{ J}$$

Next $B \rightarrow C$ is isobaric process.

$$\text{So, } dQ_{B \rightarrow C} = dU_{B \rightarrow C} + dW_{B \rightarrow C}$$

$$= dU_{B \rightarrow C} + p \Delta V_{B \rightarrow C}$$

$$\Rightarrow 100 = dU_{B \rightarrow C} + 6 \times 10^4 (2 \times 10^{-3})$$

$$\Rightarrow dU_{B \rightarrow C} = 100 - 120 = -20 \text{ J}$$

From Eq. (i), we get

$$\Delta U_{A \rightarrow B \rightarrow C} = \Delta U_{A \rightarrow C}$$

$$\Rightarrow \Delta U_{A \rightarrow B} + \Delta U_{B \rightarrow C} = dQ_{A \rightarrow C} - dW_{A \rightarrow C}$$

$$\Rightarrow 400 + (-20) = dQ_{A \rightarrow C} - (p \Delta V_A + \text{Area of } \Delta ABC)$$

$$\Rightarrow dQ_{A \rightarrow C} = 380 + \left(\frac{2 \times 10^4 \times 2 \times 10^{-3}}{2} + \frac{1}{2} \times 2 \times 10^{-3} \times 4 \times 10^4 \right)$$

$$= 380 + (40 + 40)$$

$$\Rightarrow dQ_{A \rightarrow C} = 460 \text{ J}$$

- 33** (c) The specific heat of gas at constant volume in terms of degree of freedom n is

$$C_V = \frac{n}{2} R$$

$$\text{Also, } C_p - C_V = R$$

$$\text{So, } C_p = \frac{n}{2} R + R = R \left(1 + \frac{n}{2} \right)$$

$$\text{Now, } \gamma = \frac{C_p}{C_V} = \frac{R \left(1 + \frac{n}{2} \right)}{\frac{n}{2} R} = \frac{2}{n} + 1$$

- 34** (d) A monoatomic gas has 3 degrees of freedom and diatomic gas has 5 degrees of freedom.

For monoatomic gas,

$$C_V = \frac{3}{2} R \quad \text{and} \quad C_p = C_V + R = \frac{3}{2} R + R = \frac{5}{2} R$$

For a diatomic gas,

$$C_V = \frac{5}{2} R \quad \text{and} \quad C_p = C_V + R = \frac{5}{2} R + R = \frac{7}{2} R$$

Hence, we can conclude that the molecules of the gas can be monoatomic or rigid diatomic.

- 35** (a) From the first law of thermodynamics, $Q = \Delta U + W$

In an adiabatic process, $Q = 0$

$$\therefore \Delta U = -W$$

- 36** (d) We know that, $\eta = \frac{W \text{ (output)}}{\text{Heat (input)}} = \frac{W}{3W} = \frac{1}{3}$

Now, efficiency of engine,

$$\eta = 1 - \frac{Q_2}{Q_1} \Rightarrow \frac{1}{3} = 1 - \frac{Q_2}{Q_1}$$

$$\Rightarrow \frac{Q_2}{Q_1} = 1 - \frac{1}{3} \Rightarrow \frac{Q_2}{Q_1} = \frac{2}{3}$$

- 37** (d) As for adiabatic process, $pV^\gamma = \text{constant}$

Differentiating above equation, we get

$$p\gamma V^{\gamma-1} dV + dpV^\gamma = 0$$

$$\Rightarrow -\frac{dV}{V} = \frac{dp}{\gamma p}$$

$$\Rightarrow -\frac{dV}{V} = \frac{0.5}{14} \times 100 \approx 0.36\%$$

- 38** (a) The work done during the given cyclic process $PSRQ$ is

$$\Delta W = \text{Area } PQRS$$

$$= (2V - V)(2p - p) = pV$$

- 39** (c) Given, $T_1 = 500 \text{ K}$, $T_2 = 300 \text{ K}$

We know that, efficiency of a Carnot engine,

$$\eta = \left(1 - \frac{T_2}{T_1} \right) \times 100 = \left(1 - \frac{300}{500} \right) \times 100$$

$$= 0.4 \times 100 = 40\%$$

- 40** (a) The $\gamma = \left(\frac{C_p}{C_V} \right)$ for an ideal diatomic gas is 1.40 or $\frac{7}{5}$.

As pressure is constant, so the process is isobaric.

$\therefore$ The fraction of heat energy utilised to increase the internal energy of gas is $\frac{\Delta U}{\Delta Q} = \frac{1}{\gamma} = \frac{5}{7}$.

- 41** (b) By definition of calorie, we know that heat required to raise temperature of 1g of water from 14.5°C to 15.5°C is called 1 calorie and

$$1 \text{ cal} = 4.186 \text{ J}$$

- 42** (b) Internal energy of steam,

$$\Delta U = \Delta Q - W = mL - pV$$

$$= 1 \times 2.25 \times 10^6 - 1.01 \times \frac{10^5 \times 1}{0.6}$$

$$= 2.25 \times 10^6 - \frac{1.01 \times 10^6}{6}$$

$$= 2.25 \times 10^6 - 0.168 \times 10^6 = 2.08 \times 10^6 \text{ J}$$

- 43** (a) Given, $C_p = 450 \text{ JK}^{-1}$ and $C_V = 300 \text{ JK}^{-1}$

$$\therefore C_p - C_V = R$$

$$\Rightarrow R = 450 - 300 = 150 \text{ JK}^{-1}$$

$$\text{At STP, } T = 273 \text{ K}$$

$$p = p_{\text{atm}} = 1.013 \times 10^5 \text{ Nm}^{-2}$$

From ideal equation of gas,

$$pV = nRT$$

$$\Rightarrow p \frac{M}{\rho} = \frac{M}{m} RT$$

$$\Rightarrow \rho = \frac{pm}{RT}$$

$$\text{or } \rho = \frac{1013 \times 10^5 \times 1}{150 \times 273}$$

$$\text{or } \rho = 2.47 \text{ kg/m}^3$$

- 44** (c) For isothermal expansion,

$$pV = p' \times 2V$$

$$(\because V' = 2V)$$

$$\Rightarrow p' = \frac{p}{2}$$

For adiabatic expansion, $pV^\gamma = \text{constant}$

$$p'V'^\gamma = p''V''^\gamma$$

$$\frac{p}{2}(2V)^{5/3} = p''(16V)^{5/3} \quad \left(\because \gamma = \frac{5}{3}\right)$$

$$p'' = \frac{p}{2} \left(\frac{2V}{16V} \right)^{5/3} = \frac{p}{2} \left(\frac{1}{8} \right)^{5/3}$$

$$\therefore \text{Final pressure of the gas} = \frac{p}{2} \left(\frac{1}{2} \right)^5 = p/64$$

45 (d) For hydrogen, $C_p - C_v = \frac{R}{M_{H_2}} = \frac{R}{2}$,

but given that $C_p - C_v = m \Rightarrow m = R/2$

For nitrogen, $C_p - C_v = n$

$$\therefore n = R/28$$

$$\frac{m}{n} = \frac{R/2}{R/28} = 14$$

Hence, $m = 14n$

46 (b) Using the relation, $\gamma = 1 + \frac{2}{f}$

(where, f is the number of degrees of freedom for gas atom)

For monoatomic gas, $f = 3$

Hence, $\gamma = \frac{C_p}{C_v} = 1 + \frac{2}{3} = \frac{5}{3}$

47 (b) According to first law of thermodynamics,

$$Q = \Delta U + W$$

$$Q = 2 \text{ kcal} = 2000 \times 4.2 = 8400 \text{ J}$$

$$W = 400 \text{ J}$$

$$\Rightarrow 8400 = \Delta U + 400$$

$$\Delta U = 8400 - 400$$

$$\Delta U = 8000 \text{ J}$$

Change in internal energy, $\Delta U = 8 \text{ kJ}$

48. (d) For an adiabatic process, pressure (p) and volume (V) of the system are related as

$$pV^\gamma = \text{constant} \quad \dots(i)$$

where, γ is the ratio of specific heat capacities at constant pressure and constant volume.

For an ideal gas, $pV = nRT$ (Ideal gas equation)

$$\Rightarrow V = \frac{nRT}{p} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$p \left(\frac{nRT}{p} \right)^\gamma = \text{constant}$$

$$\Rightarrow p^{1-\gamma} T^\gamma = \text{constant.}$$

[$\because$ Number of moles (n) and gas constant (R) are constants]

49. (a) Given, $m = 1 \text{ kg}$,

$$T_1 = 24.4 + 273 = 297.4 \text{ K}, T_2 = 0 + 273 = 273 \text{ K},$$

$$L = 336 \text{ kJ kg}^{-1}$$

Total heat produced per kg of mass,

i.e. $Q_2 = mL = 1 \times 336 \text{ kJ}$

We know that, $\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

$$\Rightarrow Q_1 = \left(\frac{T_1}{T_2} \right) Q_2 = \frac{297.4}{273} \times 336 = \frac{99926.4}{273} \text{ kJ} = 366 \text{ kJ}$$

Work required by refrigerator,

$$W = Q_1 - Q_2 = 366 - 336 = 30 \text{ kJ}$$

50 (b) The efficiency of Carnot engine is given by

$$\eta = 1 - \frac{T_2}{T_1}$$

Given, $T_1 = 600 \text{ K}, T_2 = 300 \text{ K}$

$$\therefore \eta = 1 - \frac{300}{600}$$

$$\Rightarrow \eta = 1 - \frac{1}{2} \Rightarrow \eta = \frac{1}{2} = 50\%$$

Also, $\eta = \frac{W}{Q} \Rightarrow \frac{800}{Q} = \frac{1}{2}$

Thus, amount of heat supplied from source to engine is

$$Q = 800 \times 2 = 1600 \text{ J}$$

51 (b) Net work done = Area enclosed in p - V curve, i.e. ΔABC

$$= \frac{1}{2} \times 5 \times 10^{-3} \times 4 \times 10^5 \text{ J}$$

$$= 10^3 \text{ J} = 1000 \text{ J}$$

52 (d) According to question, $p \propto T^3$

$$pV = nRT \Rightarrow pV \propto T$$

$$p \propto (pV)^3 \Rightarrow p^2 V^3 = \text{constant}$$

$$pV^{3/2} = \text{constant}$$

$$\therefore pV^\gamma = \text{constant} \quad (\text{For adiabatic process})$$

$$\therefore \gamma = \frac{C_p}{C_v} = \frac{3}{2}$$

53 (c) The slope of the graph $\propto \frac{1}{p}$. Hence, $p_2 < p_1$.

54 (a) We know that, $Q = \frac{f}{2} nR \Delta T$

For helium, $f = 3$ and $n = \frac{1}{4}$

$\therefore$ Amount of heat required,

$$Q = \frac{3}{2} \times \frac{1}{4} \times k_B N_A \Delta T$$

$$= \frac{3}{8} N_A k_B (T_2 - T_1)$$

55 (b) Efficiency, $\eta = 1 - \frac{T_2}{T_1}$

Given, $T_2 = 27^\circ \text{C} = 27 + 273 = 300 \text{ K}$

$$T_1 = 127^\circ \text{C} = 127 + 273 = 400 \text{ K}$$

$$\eta = 1 - \frac{3}{4} = 25\%$$

56 (a) We know that,

(i) The work done in isobaric process is $p(V_2 - V_1)$.

(ii) The work done in isothermal process is $nRT \log_e \left(\frac{V_2}{V_1} \right)$.

(iii) The work done in adiabatic process is $\frac{nR(T_1 - T_2)}{\gamma - 1}$.

Hence, $A \rightarrow q$, $B \rightarrow p$, $C \rightarrow r$.

57 (d) We know that, $dQ = nC_p dT_A = nC_V dT_B$

$$\therefore nC_p dT_A = nC_V dT_B$$

$$\Rightarrow \frac{C_p}{C_V} = \frac{dT_B}{dT_A}$$

$$\Rightarrow \gamma = \frac{dT_B}{dT_A} \quad \left(\because \gamma = \frac{C_p}{C_V} \right)$$

$$\Rightarrow dT_B = \gamma dT_A = 1.4 \times 30 = 42 \text{ K}$$

58 (d) The efficiency of Carnot's heat engine, $\eta = \frac{T_1 - T_2}{T_1}$

For given combinations $\eta_1 = \frac{1}{3}$, $\eta_2 = \frac{1}{2}$, $\eta_3 = \frac{2}{5}$, $\eta_4 = \frac{2}{3}$

So, it is maximum for $T_1 = 300 \text{ K}$ and $T_2 = 100 \text{ K}$.

59 (a) Given, $p \propto \frac{1}{V^{1.5}}$

So, $pV^{1.5} \propto \text{constant}$

As the value of $\gamma = 1.5$ for adiabatic process, so

$$\frac{C_p}{C_V} = \gamma = 1.5$$

60 (c) We know that, the gas equation in adiabatic process,

$$pV^\gamma = \text{constant}$$

$$\text{Here in given condition, } p \propto \frac{1}{V^\gamma} \Rightarrow \frac{p_1}{p_2} = \left(\frac{V_2}{V_1} \right)^\gamma$$

$$\frac{1}{p_2} = \left(\frac{1}{10^3} \right)^{\frac{5}{3}} = \frac{1}{10^5}$$

$$\Rightarrow p_2 = 10^5 \text{ atm}$$

61 (d) Specific heat, $C = \frac{dQ}{dT} = \frac{dQ}{0} = \infty$ ($\because T = \text{constant}$)

62 (a) For given cyclic process, $\Delta U = 0$

$$\therefore Q = W$$

Also, $W = \text{Area enclosed by the curve}$

$$= AB \times AD = (2p - p)(3V - V) = p \times 2V$$

$\therefore$ Heat rejected $= 2pV$

63 (d) According to question, first gas goes from volume V to $3V$ and after this volume is reduced from $3V$ to V at constant pressure.

In isothermal expansion, p - V curve is rectangular hyperbola.

Hence, correct graph is (d).

64 (c) From the figure, $p \propto T$

So, the volume of the gas remains constant. Hence, work done is zero.

65 (c) We know that, $C_p - C_V = R$

$$\text{Given, } \frac{C_p}{C_V} = \gamma \Rightarrow C_V = \frac{C_p}{\gamma}$$

$$\therefore C_p - \frac{C_p}{\gamma} = R \text{ or } C_p \left(1 - \frac{1}{\gamma} \right) = R$$

$$\text{or } C_p \left(\frac{\gamma - 1}{\gamma} \right) = R \Rightarrow C_p = \frac{\gamma R}{\gamma - 1}$$

66 (c) Work done, $W = \int p dV = \int \frac{RT}{V} dV$

$$\text{Since, } V = KT^{-2/3} \Rightarrow dV = \frac{2}{3} KT^{-1/3} dT$$

$$\text{Eliminating } K, \text{ we find } \frac{dV}{V} = \frac{2}{3} \frac{dT}{T}$$

$$\text{Hence, } W = \int_{T_1}^{T_2} \frac{2}{3} \frac{RT}{T} dT$$

$$= \frac{2}{3} R(T_2 - T_1) = \frac{2}{3} R(30) = 20 R$$

67 (d) We know that, $\Delta Q = \mu C_p \Delta T = \frac{7}{2} \mu R \Delta T$ ($\because C_p = \frac{7}{2} R$)

$$\text{and } \Delta U = \mu C_V \Delta T = \frac{5}{2} \mu R \Delta T \quad \left(\because C_V = \frac{5}{2} R \right)$$

$$\text{Also, } \Delta W = \Delta Q - \Delta U = \mu R \Delta T$$

$$\therefore \Delta Q : \Delta U : \Delta W = 7 : 5 : 2$$

68 (b) Internal energy of a gas, $U = \frac{3}{2} nRT$

where, R is constant.

Thus, for a given number of moles of the same gas

$$U \propto T$$

From the graphs, we conclude that

$$\Rightarrow \begin{aligned} U_A &= U_D \text{ and } U_B = U_C \\ U_B &> U_A \text{ and } U_C > U_D \end{aligned}$$

69 (b) Equation of an adiabatic process is $pV^\gamma = k$... (i)

$$\text{Given, } p^3 = \frac{k}{V^4}$$

$$\Rightarrow p^3 V^4 = k \text{ or } pV^{4/3} = k$$

Comparing Eqs. (i) and (ii), we get

$$\gamma = \frac{4}{3} = 1.33$$

70 (d) Efficiency of Carnot's engine, $\eta = 1 - \frac{T_2}{T_1}$... (i)

where, T_2 is temperature of sink and T_1 is the temperature of the source.

Here, $\eta = 50\%$, $T_2 = 50^\circ \text{C} = 50 + 273 = 323 \text{ K}$

Putting these values in Eq. (i), we get

$$\frac{50}{100} = 1 - \frac{323}{T_1}$$

$$\Rightarrow T = 646 \text{ K} = 373^\circ \text{C}$$

71 (b) Work done, $W = mgh + \Delta pV = V\rho gh + \Delta pV$

$$= 4 \times 10^3 \times 10 \times 20 + (2 \times 10^5) \times 4 = 16 \times 10^5 \text{ J}$$

72 (d) From first law of thermodynamics, $dQ = dU + dW$

From adiabatic process, $dQ = 0$

$$\therefore dU + dW = 0$$

$$\Rightarrow dU = -dW = -50 \text{ J}$$

Therefore, the internal energy of system will decrease.

73 (b) (i) Volume is constant,

$$\text{i.e. } \Delta V = 0$$

$$\Rightarrow \Delta U_a = Q - p\Delta V$$

$$\Rightarrow \Delta U_a = Q$$

(ii) Pressure is constant,

$$\text{i.e. } \Delta p = 0$$

$$\Rightarrow \Delta U_b = Q - p\Delta V$$

$$\therefore \Delta U_a > \Delta U_b$$

74 (a) W_{a-b} is isothermal

W_{b-c} is adiabatic

W_{c-d} is isothermal (Negative) and

W_{d-a} is adiabatic (Negative)

Net work done during the complete cycle,

$$W = W_{a-b} + W_{b-c} + (-W_{c-d}) + (-W_{d-a})$$

$$W = W_{a-b} - W_{c-d} \quad (\text{As, } W_{b-c} = W_{d-a})$$

$$\text{As, } W > 0$$

$$\Rightarrow W_{a-b} + W_{c-d} > 0$$

$$W_{a-b} + W_{b-c} + W_{c-d} + W_{d-a} > 0$$

$$\text{and } W_{b-c} + (-)W_{d-a} = 0$$

75 (a) Change in entropy, $dS = \frac{dQ}{T}$

Entropy remains constant for reversible process, i.e.

$$dS = 0$$

76 (d) $dU = dQ - dW$

$$\Rightarrow dW = dQ - dU$$

For ideal gas, $dU = 0$

$$\therefore dW = dQ = R(T_2 - T_1)$$

$$= 8.31 \times 100 = 8.31 \times 10^2 \text{ J}$$

$$\mathbf{77} \text{ (c) } \eta = 1 - \frac{T_2}{T_1}$$

where, T_2 = sink temperature and T_1 = source temperature.

$$0.6 = 1 - \frac{T_2}{400} \Rightarrow \frac{T_2}{400} = 0.4$$

$$\Rightarrow T_2 = 160 \text{ K}$$

$$T_2 = 160 \text{ K} = -113^\circ \text{ C}$$

78 (b) According to first law of thermodynamics,

$$dU = dQ - dW$$

$$dU = 8 \times 10^5 - 6.5 \times 10^5 = 1.5 \times 10^5 \text{ J}$$

In the second process, dU remains the same.

$$\therefore dW = dQ - dU = 1 \times 10^5 - 1.5 \times 10^5$$

$$dW = -0.5 \times 10^5 \text{ J}$$

$\therefore$ Work done on the gas is $0.5 \times 10^5 \text{ J}$.

79 (c) Since, the volume of cylinder is fixed and helium is a monoatomic gas.

The internal energy = $\mu C_V \Delta T$

$$\Rightarrow U = \frac{1}{2} \mu R \Delta T \quad \left(\because C_V = \frac{1}{2} f R \right)$$

$$U = \frac{1}{2} \times 3 \times 2 \times 8.31 \times 20 \quad (\because f = 3)$$

$$= 498.6 \approx 499 \text{ J}$$

80 (c) Efficiency of a Carnot engine, i.e.

$$\eta = 1 - \frac{T_2}{T_1} = \frac{T_1 - T_2}{T_1}$$

$$\Rightarrow \frac{25}{100} = \frac{T_1 - (T_1 - 80)}{T_1}$$

$$\Rightarrow \frac{1}{4} = \frac{80}{T_1} \Rightarrow T_1 = 320 \text{ K}$$

$$T_1 = 320 - 273 = 47^\circ \text{ C}$$

Temperature of the low temperature reservoir

$$= 47 - 80 = -33^\circ \text{ C}$$

CHAPTER 16

Calorimetry and Heat Transfer

In our day-to-day life, we have noticed that when we touch a utensil placed on fire, it appears hot. When steam is passed through water, the water becomes warm. On the other hand, when we touch a piece of ice, it appears cold. To understand the cause of these phenomena, let us study in detail about heat and temperature. Further, we will also study about the various processes by which the heat flows from one body to another.

HEAT

Practically all the bodies around us either release or absorb energy often in the form of heat. The form of energy which is exchanged among various bodies or system on account of temperature difference is defined as **heat**. Heat is a scalar quantity and its SI unit is joule (J). Other practical units of heat are erg, cal, BTU, etc.

$$1 \text{ J} = 10^7 \text{ erg}$$

$$1 \text{ cal} = 4.18 \text{ J}$$

$$1 \text{ BTU} = 252 \text{ cal} = 1055 \text{ J}$$

Heat capacity

The heat capacity of substances is defined as the amount of heat required to raise its temperature by unity, *i.e.* 1°C (or 1K). It is denoted by S . If temperature of the substance changes from T to $T + \Delta T$, when it exchanges an amount of heat ΔQ with its surroundings, then its

$$\text{Heat capacity, } S = \frac{\Delta Q}{\Delta T}$$

The SI unit of heat capacity is JK^{-1} and its dimensional formula is $[\text{ML}^2\text{T}^{-2}\text{K}^{-1}]$.

Specific heat capacity

The amount of heat required to raise the temperature of unit mass of a substance by unity is known as specific heat capacity of the substance. It is denoted by c .

Inside

1 Heat

- Heat capacity
- Calorimetry

2 Change of state (or phase)

- Latent heat
- Heating curve

3 Heat transfer

- Conduction of heat in conducting rod
- Combinations of rods with different thermal conductivities

4 Radiation

- Absorption, emission, transmission and reflection of heat radiations
- Black body radiation
- Kirchhoff's radiation law
- Stefan's law
- Cooling by radiation
- Newton's law of cooling
- Wien's displacement law

If the temperature of a substance of mass m changes from T to $T + \Delta T$, when it exchanges an amount of heat ΔQ with its surroundings, then its

$$\text{Specific heat capacity, } c = \frac{1}{m} \frac{\Delta Q}{\Delta T}$$

$\therefore$ Heat, $\Delta Q = mc\Delta T$
The SI unit of specific heat capacity is $\text{J kg}^{-1} \text{K}^{-1}$ and its dimensional formula is $[\text{M}^0 \text{L}^2 \text{T}^{-2} \text{K}^{-1}]$.

In differential form, we may write $c = \frac{1}{m} \frac{dQ}{dT}$

Therefore, if c varies with temperature over an interval, then

$$\text{Heat, } Q = m \int_{T_1}^{T_2} c dT$$

Important points related to specific heat capacity are given below

- The specific heat capacity of water is approximately $1 \text{ cal/g}^\circ\text{C}$.
- Specific heat capacity is constant for a particular material.
- If a substance is undergoing change of state which takes place at constant temperature, i.e. $\Delta T = 0$, then specific heat, $c = \frac{\Delta Q}{m\Delta T} = \frac{Q}{m \times 0} = \infty$.
- If the temperature of a body changes without transfer of heat with the surrounding (i.e. $Q = 0$), then specific heat, $c = \frac{\Delta Q}{m\Delta T} = \frac{0}{m \times \Delta T} = 0$.
- Specific heat of a substance can be negative.
- Specific heat of a substance depends on temperature, nature of material of body and state of substance, i.e. solid, liquid or gas.

Note

(i) Specific heat of water is very high in comparison to other liquids and solids. For this reason, water is used as a heater in hot water bags as well as a coolant in automobile radiators. Owing to its high specific heat, the water warms up much more slowly than the land during summer and consequently wind from the sea has a cooling effect. This is the reason for cool temperature (climate) of coastal areas.

(ii) Specific heat of water $= 1 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1} = 4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$.

Relation between heat capacity and specific heat capacity

As we know that,

$$\text{heat capacity, } S = \frac{\Delta Q}{\Delta T}$$

$$\text{and specific heat capacity, } c = \frac{1}{m} \frac{\Delta Q}{\Delta T}$$

$$\therefore S = m \times c$$

i.e. Heat capacity = Mass $\times$ Specific heat capacity

Example 16.1 When 400 J of heat are added to a 0.1 kg sample of metal, its temperature increases by 20°C . What is the specific heat of the metal?

Sol. Given, heat, $\Delta Q = 400 \text{ J}$, mass $m = 0.1 \text{ kg}$ and temperature, $\Delta T = 20^\circ\text{C}$

$$\therefore \text{Specific heat, } c = \frac{1}{m} \cdot \frac{\Delta Q}{\Delta T} = \left(\frac{1}{0.1} \right) \left(\frac{400}{20} \right) = 200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$$

Example 16.2 A 10 kW drilling machine is used to drill a bore in a small aluminium block of mass 8.0 kg . How much is the rise in temperature of the block in 2.5 min ? Assuming 50% of power is used up in heating the machine itself or lost to the surroundings. Specific heat of aluminium $= 0.91 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$.

Sol. Given, power, $P = 10 \text{ kW} = 10 \times 10^3 \text{ W}$

Time, $t = 2.5 \text{ min} = 2.5 \times 60 \text{ s}$

As power is defined as rate at which energy is consumed,

$$\text{i.e. } P = \frac{E}{t}$$

$\therefore$ Total energy used,

$$E = Pt = 10 \times 10^3 \times 2.5 \times 60 = 1.5 \times 10^6 \text{ J}$$

Energy absorbed by aluminium block,

$$Q = 50\% \text{ of the total energy} = 50\% \text{ of } 1.5 \times 10^6 = 0.75 \times 10^6 \text{ J}$$

$$\text{Also, } m = 8.0 \text{ kg} = 8.0 \times 10^3 \text{ g, } c = 0.91 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$$

$$\text{As, } Q = mc\Delta T$$

$$\therefore \Delta T = \frac{Q}{mc} = \frac{0.75 \times 10^6}{8.0 \times 10^3 \times 0.91} = 103.02^\circ\text{C}$$

Example 16.3 A geyser heats water flowing at the rate of 3.0 L per minute from 27°C to 77°C . If the geyser operates on a gas burner, what is the rate of combustion of fuel, if its heat of combustion is $4.0 \times 10^4 \text{ J g}^{-1}$?

Sol. Given, volume of water heated,

$$V = 3.0 \text{ L min}^{-1} = 3 \times 10^{-3} \text{ m}^3 \text{ min}^{-1}$$

$$\text{As, density } (\rho) = \frac{\text{Mass } (m)}{\text{Volume } (V)}$$

$\therefore$ Mass of water heated,

$$m = \rho V = 10^3 \times 3 \times 10^{-3} \text{ kg min}^{-1} = 3 \text{ kg min}^{-1}$$

Rise in temperature of water,

$$\Delta T = T_2 - T_1 = 77^\circ - 27^\circ = 50^\circ\text{C}$$

Specific heat of water, $c = 4.2 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$

$$= 4.2 \times 10^3 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$$

Heat required by water, $\Delta Q = mc\Delta T$

$$= (3 \times 4.2 \times 10^3 \times 50) \text{ J min}$$

$$= (63 \times 10^4) \text{ J min}^{-1} \quad \dots(i)$$

$$\text{Heat of combustion of fuel} = 4.0 \times 10^4 \text{ J g}^{-1} = 4.0 \times 10^7 \text{ J kg}^{-1}$$

Let $m \text{ kg min}^{-1}$ be the rate of combustion of fuel.

$$\therefore \text{Heat supplied by gas burner or combustion of fuel} \\ = m \times 4 \times 10^7 \text{ J min}^{-1} \quad \dots(ii)$$

$\therefore$ From Eqs. (i) and (ii), we get

$$63 \times 10^4 = m \times 4 \times 10^7$$

$$\text{or} \quad m = \frac{63 \times 10^4}{4 \times 10^7} \\ = 15.75 \times 10^{-3} \text{ kg min}^{-1}$$

$$\text{or rate of combustion of fuel} \approx 16 \text{ g min}^{-1}$$

Molar specific heat capacity

The amount of heat required to raise the temperature of one mole of a substance (usually a gas) by unity is known as the **molar heat capacity** of that substance. It is denoted by C . Its SI unit is $\text{J mol}^{-1}\text{K}^{-1}$.

$$\therefore \quad \text{Molar specific heat capacity, } C = \frac{S}{n} = \frac{\Delta Q}{n\Delta T}$$

where n = number of moles of substance (gas).

There are two types of molar specific heat capacities for a gas

- (i) **Molar specific heat capacity at constant pressure (C_p)** It is molar heat capacity of a gas at constant pressure. The amount of heat required to raise the temperature of 1 mole of a gas by unity at constant pressure is known as molar heat capacity at constant pressure.

$$\therefore \quad C_p = \left[\frac{\Delta Q}{\Delta T} \right]_{p=\text{constant}}$$

$$\Rightarrow \quad \Delta Q = C_p \Delta T$$

For n moles of a gas $\Delta Q = nC_p \Delta T$

- (ii) **Molar specific heat capacity at constant volume (C_v)** It is molar heat capacity of a gas at constant volume. The amount of heat required to raise the temperature of 1 mole of gas by unity at constant volume is known as molar specific heat capacity at constant volume.

$$\therefore \quad C_v = \left(\frac{\Delta Q}{\Delta T} \right)_{V=\text{constant}}$$

$$\Rightarrow \quad \Delta Q = C_v \Delta T$$

For n moles of a gas $\Delta Q = nC_v \Delta T$

Note When a solid substance is heated in a small range of temperature, its volume almost remains constant. Thus, specific heat capacity of a solid can be taken as specific heat at constant volume C_v .

Relation between specific heat capacity and molar specific heat capacity

$$\text{As, number of moles, } n = \frac{m}{M}$$

$$\Rightarrow \quad m = nM$$

where, m = mass of the substance and M = molecular mass.

$$\Rightarrow \quad C = \frac{1}{n} \left[\frac{\Delta Q}{\Delta T} \right] = M \left[\frac{\Delta Q}{m\Delta T} \right]$$

$$\text{But,} \quad \frac{\Delta Q}{m\Delta T} = c$$

$$\Rightarrow \quad C = Mc$$

where, c = specific heat capacity,

M = molecular mass of the substance

and C = molar specific heat capacity.

Hence, molar specific heat capacity
= molecular mass $\times$ specific heat capacity.

Note

(i) If a gas is compressed or expanded suddenly such that no exchange of heat takes place with the surrounding.

$$\text{i.e.} \quad \Delta Q = 0$$

$$\text{Then,} \quad C = \frac{\Delta Q}{n\Delta T} = 0$$

(ii) If a gas is allowed to expand or compressed very slowly, such that $\Delta T = 0$, then

$$C = \frac{\Delta Q}{n\Delta T} = \infty$$

Example 16.4 0.64 g of oxygen is kept in a rigid container and is heated. Find the amount of heat required to raise the temperature from 20°C to 30°C . The molar heat capacity of oxygen at constant volume is $20 \text{ J K}^{-1} \text{ mol}^{-1}$.

Sol. Mass of the oxygen, $m = 0.64 \text{ g}$

The molecular weight of oxygen, $M = 32 \text{ g mol}^{-1}$

$\therefore$ Number of moles of the gas,

$$n = \frac{\text{Mass (m)}}{\text{Molecular weight (M)}} = \frac{0.64}{32} = 0.02$$

Molar heat capacity at constant volume, $C_v = 20 \text{ J K}^{-1} \text{ mol}^{-1}$

$\therefore$ The amount of heat required,

$$\Delta Q = nC_v \Delta T = (0.02)(20)(30 - 20) = 4 \text{ J}$$

Example 16.5 A certain substance has a mass of 50 g/mol. When 300 J of heat is added to 25 g of sample of this material, its temperature rises from 25°C to 45°C . Calculate (i) thermal capacity, (ii) specific heat and (iii) molar heat capacity of the sample.

Sol. Given, mass, $m = 25 \text{ g}$,

Temperature, $\Delta T = 45^\circ - 25^\circ = 20^\circ \text{C}$,

Heat, $\Delta Q = 300 \text{ J}$

and molar mass, $M = 50 \text{ g/mol}$.

(i) Thermal capacity, $S = \frac{\Delta Q}{\Delta T} = \frac{300}{20} = 15 \text{ J/}^\circ\text{C}$

(ii) Specific heat, $c = \frac{\Delta Q}{\Delta T \times m} = 15 \times \frac{1}{25} = 0.6 \text{ J/g } ^\circ\text{C}$
 $= 0.6 \times 10^3 \text{ J/kg } ^\circ\text{C} = 600 \text{ J/kg } ^\circ\text{C}$

(iii) Number of moles, $n = \frac{\text{Mass (m)}}{\text{Molecular mass (M)}} = \frac{25}{50} = \frac{1}{2}$

$\therefore$ Heat, $\Delta Q = nC\Delta T$

$\therefore$ Molar heat capacity, $C = \frac{\Delta Q}{n\Delta T} = \frac{300}{(1/2)(20)} = 30 \text{ J/mol } ^\circ\text{C}$

Calorimetry

The branch of physics that deals with the measurement of heat is called calorimetry.

Principle of calorimetry

Whenever two bodies at different temperatures are placed in contact with each other, heat flows from the body at higher temperature to the body at lower temperature till the two bodies attain the same temperature.

Thus, the principle of calorimetry states that total heat lost by the hotter body equals to the total heat gained by the colder body, provided there is no exchange of heat with the surroundings.

i.e. $\text{Heat lost by hotter body} = \text{Heat gained by colder body}$

The principle of calorimetry is based on the law of conservation of energy.

Calculations based on principle of calorimetry

If there are two bodies of masses m_1 and m_2 having specific heat capacities as c_1 and c_2 , respectively. Let these bodies be at temperatures T_1 and T_2 respectively, such that $T_1 > T_2$.

When they are placed in contact with each other, they attain common temperature T , then according to principle of calorimetry,

Heat lost by first body = Heat gained by second body

$\Rightarrow m_1c_1(T_1 - T) = m_2c_2(T - T_2)$

$\therefore$ Common temperature of the mixture at equilibrium,

$$T = \frac{m_1c_1T_1 + m_2c_2T_2}{m_1c_1 + m_2c_2}$$

Example 16.6 10 g of water at 70°C is mixed with 5 g of water at 30°C . Find the temperature of the mixture in equilibrium.

Sol. Given, mass, $m_1 = 10 \text{ g}$, $m_2 = 5 \text{ g}$,

Temperature, $T_1 = 70^\circ\text{C}$, $T_2 = 30^\circ\text{C}$

Let $T^\circ\text{C}$ be the temperature of the mixture.

From principle of calorimetry, common temperature of the mixture at equilibrium is given by

$$\begin{aligned} T &= \frac{m_1c_wT_1 + m_2c_wT_2}{m_1c_w + m_2c_w} = \frac{c_w(m_1T_1 + m_2T_2)}{c_w(m_1 + m_2)} \\ &= \frac{m_1T_1 + m_2T_2}{m_1 + m_2} = \frac{10 \times 70 + 5 \times 30}{10 + 5} = \frac{700 + 150}{15} \\ &= \frac{850}{15} = 56.67^\circ\text{C} \end{aligned}$$

Example 16.7 Liquids A and B are at 30°C and 20°C . When mixed in equal masses, the temperature of the mixture is found to be 26°C . Find the ratio of their specific heat capacities.

Sol. Let the specific heats of A and B be c_A and c_B ,

respectively. Now, heat lost by A in cooling from 30°C to 26°C will be same as liquid B warming up from 20°C to 26°C . Since, the masses are same, then from the principle of calorimetry,

Heat lost by A = Heat gained by B
 $\therefore m \cdot c_A(30^\circ\text{C} - 26^\circ\text{C}) = m \cdot c_B(26^\circ\text{C} - 20^\circ\text{C})$

or $4c_A = 6c_B$ or $\frac{c_A}{c_B} = \frac{6}{4} = \frac{3}{2}$

Example 16.8 The temperatures of equal masses of three different liquids A, B and C are 12°C , 19°C and 28°C respectively. The temperature when A and B are mixed is 16°C and when B and C are mixed, it is 23°C . What will be the temperature when A and C are mixed?

Sol. Given, temperature, $T_A = 12^\circ\text{C}$, $T_B = 19^\circ\text{C}$ and $T_C = 28^\circ\text{C}$

Let specific heat of liquids A, B and C are c_A , c_B and c_C , respectively.

When A and B are mixed, then from principle of calorimetry,

$$mc_B(19 - 16) = mc_A(16 - 12)$$

$$3c_B = 4c_A \Rightarrow c_A = \frac{3}{4}c_B$$

When B and C are mixed, then from principle of calorimetry,

$$mc_C(28 - 23) = mc_B(23 - 19)$$

$$5c_C = 4c_B \Rightarrow c_C = \frac{4}{5}c_B$$

When A and C are mixed, let equilibrium temperature be T .

From principle of calorimetry, common temperature of the mixture at equilibrium,

$$\begin{aligned} T &= \frac{mc_C T_1 + mc_A T_2}{mc_C + mc_A} = \frac{\frac{4}{5}c_B(28) + \frac{3}{4}c_B \times 12}{\frac{4}{5}c_B + \frac{3}{4}c_B} \\ &= \frac{c_B \left(\frac{4}{5} \times 28 + \frac{3}{4} \times 12 \right)}{\left(\frac{4}{5} + \frac{3}{4} \right) c_B} = \frac{\frac{112}{5} + 9}{\frac{16 + 15}{20}} = \frac{(112 + 45) 20}{(31) \times 5} \\ &= \frac{157 \times 20}{155} = \frac{3140}{155} = 20.3^\circ\text{C} \end{aligned}$$

Water equivalent of a substance

Water equivalent of an object is the amount of water that absorbs or gives out the same amount of heat as is done by the substance when heated or cooled through same temperature difference as that of object.

To attain the same temperature as that of water, the calorimeter gains some heat.

Water equivalent of an object = Mass of the object
 $\times$ Specific heat of material of object

or $W = mc$

Water equivalent is usually expressed in gram.

Example 16.9 A piece of iron of mass 100 g is kept inside a furnace for a long time and then put in a calorimeter of water equivalent 10 g containing 240 g of water at 20°C .

The mixture attains an equilibrium temperature of 60°C .

Find the temperature of the furnace.

(Specific heat of iron = $470\text{ J/kg}^\circ\text{C}$)

Sol. Let temperature of furnace be t .

Given, $m_i = 100\text{ g}$, $m_w = 240\text{ g}$,

water equivalent $m'_w = 10\text{ g}$,

Equilibrium temperature = 60°C

Then from principle of calorimetry,

Heat given = Heat taken

$$m_i s_i (t - 60) = (m'_w + m_w) s_w (60 - 20)$$

$$100 \times 470 (t - 60) = (10 + 240) \times 4200 \times 40$$

$$t - 60 = \frac{250 \times 4200 \times 40}{100 \times 470} = 893.6$$

Therefore, $t = 953.6^\circ\text{C}$

CHECK POINT 16.1

- If specific heat of a substance is infinite, it means
 - heat is given out
 - heat is taken in
 - no change in temperature takes place whether heat is taken in or given out
 - All of the above
- How much heat energy is gained when 5 kg of water at 20°C is brought to its boiling point, (Specific heat of water = $4.2\text{ kJ kg}^{-1}^\circ\text{C}^{-1}$)
 - 1680 kJ
 - 1700 kJ
 - 1720 kJ
 - 1740 kJ
- A metal block is made from a mixture of 2.4 kg of aluminium, 1.6 kg of brass and 0.8 kg of copper. The amount of heat required to raise the temperature of this block from 20°C to 80°C is (specific heats of aluminium, brass and copper are 0.216, 0.0917 and $0.0931\text{ cal kg}^{-1}^\circ\text{C}^{-1}$, respectively)
 - 96.2 cal
 - 44.4 cal
 - 86.2 cal
 - 62.8 cal
- 50 g of copper is heated to increase its temperature by 10°C . If the same quantity of heat is given to 10 g of water, the rise in its temperature is (Specific heat of copper = $420\text{ J kg}^{-1}^\circ\text{C}^{-1}$ and water = $4200\text{ J kg}^{-1}^\circ\text{C}^{-1}$)
 - 5°C
 - 6°C
 - 7°C
 - 8°C
- A bubble of 8 mole of helium is submerged at a certain depth in water. The temperature of water increases by 30°C .

How much heat is added approximately to helium during expansion?

- 4000 J
 - 3000 J
 - 3500 J
 - 5000 J
- A beaker contains 200 g of water. The heat capacity of the beaker is equal to that of 20 g of water. The initial temperature of water in the beaker is 20°C . If 440 g of hot water at 92°C is poured in it, the final temperature (neglecting radiation loss) will be nearest to
 - 58°C
 - 68°C
 - 73°C
 - 78°C
 - 100 g of water at 0°C is mixed with 100 g of water at 100°C . What will be the final temperature of the mixture?
 - 50°C
 - 20°C
 - 30°C
 - 0°C
 - A liquid of mass m and specific heat c is heated to a temperature $2T$. Another liquid of mass $m/2$ and specific heat $2c$ is heated to a temperature T . If these two liquids are mixed, the resulting temperature of the mixture is
 - $\left(\frac{2}{3}\right)T$
 - $\left(\frac{8}{5}\right)T$
 - $\left(\frac{3}{5}\right)T$
 - $\left(\frac{3}{2}\right)T$
 - 0.1 m^3 of water at 80°C is mixed with 0.3 m^3 of water at 60°C . The final temperature of the mixture is
 - 70°C
 - 65°C
 - 60°C
 - 75°C

CHANGE OF STATE (OR PHASE)

Matter exists in three states: solid, liquid and gas. These states (or phases) can be changed into one another by heating or cooling. This process of conversion of one state of a substance into the another state, is known as change of state. The common changes of states of matter are as follows

(i) Solid to liquid (and *vice-versa*)

(ii) Liquid to gas (and *vice-versa*)

During the change of state of matter, the exchange of heat takes place between the substance and its surroundings.

Latent heat

The amount of heat required to change the state of a unit mass of a substance without change in its temperature and pressure is known as latent heat.

If a mass m of a substance undergoes a change from one state to the another, then the quantity of heat Q required is given by

$$Q = mL$$

i.e.

Latent heat, $L = Q/m$

It is denoted by L and its SI unit is J kg^{-1} .

Hence, during the phase change, the heat required by the substance depends on the mass m of the substance and heat of transformation Q .

There are two types of latent heat of materials

Latent heat of fusion or melting It is the quantity of heat required to change the state of a substance from solid to liquid state at its melting point. It is denoted by L_f .

Latent heat of vaporisation It is the quantity of heat required to change the state of unit mass of a substance from liquid to vapour state at its boiling point. It is denoted by L_v .

Note For water at 1 atm, latent heat of fusion is 80.0 cal g^{-1} . This simply means 80.0 cal of heat are required to melt 1.0 g of ice or 80.0 cal heat is liberated when 1.0 g of water freezes at 0°C . Similarly, latent heat of vaporisation for water at 1 atm is 540 cal/g .

Example 16.10 At 1 atm pressure, 1.000 g of water having a volume of 1.000 cm^3 becomes 1671 cm^3 of steam when boiled. The heat of vaporisation of water at 1 atm is 539 cal g^{-1} . What is the change in internal energy during the process?

Sol. Heat spent during vaporisation,

$$Q = mL = 1.000 \times 539 = 539 \text{ cal}$$

$$\text{Work done } W = p(V_v - V_l)$$

$$= 1.013 \times 10^5 \times (1671 - 1.000) \times 10^{-6}$$

$$= 169.2 \text{ J} = \frac{169.2}{4.18} \text{ cal} = 40.5 \text{ cal}$$

$$\therefore \text{Change in internal energy, } U = (539 - 40.5) \text{ cal} \\ = 498.5 \text{ cal}$$

Example 16.11 Find the quantity of heat required to convert 40 g of ice at -20°C into water at 20°C . (Given, $L_{ice} = 0.336 \times 10^6 \text{ J/kg}$, specific heat of ice = 2100 J/kg-K and specific heat of water = 4200 J/kg-K)

Sol. Given, mass, $m = 40 \text{ g} = 40 \times 10^{-3} \text{ kg} = 0.04 \text{ kg}$

$$\text{Temperature, } T_1 = -20^\circ\text{C} \text{ and } T_2 = 20^\circ\text{C}$$

$$\text{Latent heat of ice, } L_{ice} = 0.336 \times 10^6 \text{ J/kg}$$

$$\text{Specific heat of ice, } c_i = 2100 \text{ J/kg-K}$$

$$\text{Specific heat of water, } c_w = 4200 \text{ J/kg-K}$$

$$\text{Heat required to raise the temperature of ice from } -20^\circ\text{C} \text{ to } 0^\circ\text{C} = mc_i T_1 = 0.04 \times 2100 \times 20 = 1680 \text{ J}$$

$$\text{Heat required to convert the ice into water at } 0^\circ\text{C}$$

$$= mL_{ice} = 0.04 \times 0.336 \times 10^6 = 13440 \text{ J}$$

$$\text{Heat required to heat water from } 0^\circ\text{C} \text{ to } 20^\circ\text{C}$$

$$= mc_w T_2 = 0.04 \times 4200 \times 20 = 3360 \text{ J}$$

$$\text{Total heat required} = 1680 + 13440 + 3360 = 18480 \text{ J}$$

Example 16.12 In a container of negligible mass, 30 g of steam at 100°C is added to 200 g of water that has a temperature of 40°C . If no heat is lost to the surroundings, what is the final temperature of the system? Also, find masses of water and steam in equilibrium. (Take, $L_v = 539 \text{ cal g}^{-1}$ and $c_{water} = 1 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$)

Sol. Given, mass, $m = 200 \text{ g}$, $m_1 = 30 \text{ g}$

$$\text{Temperature, } T_1 = 100^\circ\text{C}, T_2 = 40^\circ\text{C}$$

$$\text{Latent heat, } L_v = 539 \text{ cal g}^{-1}$$

$$\text{Specific heat, } c_w = 1 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$$

Let Q be the heat required to convert 200 g of water at 40°C into 100°C , then $Q = mc\Delta T$

$$= (200)(1.0)(100 - 40)$$

$$= 12000 \text{ cal}$$

Now, suppose m_0 mass of steam converts into water to liberate this much amount of heat, then

$$m_0 = \frac{Q}{L} = \frac{12000}{539} = 22.26 \text{ g}$$

Since, it is less than 30 g , the temperature of the mixture is 100°C .

Mass of steam in the mixture

$$= m_1 - m_0 = 30 - 22.26 = 7.74 \text{ g}$$

and mass of water in the mixture = $m + m_0$

$$= 200 + 22.26$$

$$= 222.26 \text{ g}$$

Example 16.13 5 g of water at 30°C and 5 g of ice at -20°C are mixed together in a calorimeter. Find the final temperature of mixture. Water equivalent of calorimeter is negligible, specific heat of ice = $0.5 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$ and latent heat of ice = 80 cal g^{-1} .

Sol. In this case, heat is given by water and taken by ice.

Heat available with water to cool from 30°C to 0°C

$$= mc\Delta T = 5 \times 1 \times 30 = 150 \text{ cal}$$

Heat required by 5 g ice to increase its temperature up to 0°C

$$= mc\Delta T = 5 \times 0.5 \times 20 = 50 \text{ cal}$$

Out of 150 cal heat available, 50 cal is used for increasing temperature of ice from -20°C to 0°C . The remaining heat 100 cal is used for melting the ice.

If mass of ice melted is m g then $m \times 80 = 100 \Rightarrow m = 1.25 \text{ g}$

Thus, 1.25 g ice out of 5 g melts and mixture of ice and water is at 0°C .

Example 16.14 100 g ice at 0°C is mixed with 10 g steam at 100°C . Find the final temperature and composition of mixture.

Sol. Heat required to melt ice at 0°C ,

$$Q_1 = m_i L_f = 100 \times 80 \text{ cal g}^{-1} = 8000 \text{ cal}$$

Heat required to raise the temperature of water from 0°C to 100°C

$$Q_2 = m_i c_w \Delta T = 100 \times 1 \times 100 = 10000 \text{ cal}$$

Ice can absorb maximum heat from steam to reach boiling point

$$= Q_1 + Q_2 = 18000 \text{ cal}$$

Heat rejected by steam on complete condensation,

$$Q_3 = m_s L_v = 10 \text{ g} \times 540 \text{ cal g}^{-1} = 5400 \text{ cal}$$

Heat rejected by water at 100°C to cooled to 0°C

$$Q_4 = m_s c_w \Delta T = 10 \text{ g} \times 1 \text{ cal g}^{-1} \times 100^{\circ}\text{C} = 1000 \text{ cal}$$

Maximum heat of steam can be supply to ice at 0°C

$$= Q_3 + Q_4 = (5400 + 1000) \text{ cal} = 6400 \text{ cal}$$

To melt the ice, 8000 cal heat is required, but maximum heat supplied by steam, 6400 cal is insufficient to melt the ice. So, resulting temperature of mixture is 0°C .

Example 16.15 5 g ice at 0°C is mixed with 1 g steam at 100°C . Find the final temperature and composition of the mixture.

Sol. Given, mass of the steam, $m = 1 \text{ g}$

Temperature, $\Delta T = T_2 - T_1 = 100 - 0 = 100^{\circ}\text{C}$

Heat required to melt ice at 0°C

$$Q_1 = m_i L_f \quad (\text{where, } m_i = \text{mass of ice}) = 5 \times 80 = 400 \text{ cal}$$

Heat required to raise the temperature of water from 0 to 100°C

$$Q_2 = m_i c_w \Delta T = 5 \times 1 \times 100 = 500 \text{ cal}$$

Maximum heat ice can absorb for steam to reach boiling point is

$$Q_1 + Q_2 = 900 \text{ cal}$$

Heat rejected by steam on complete condensation

$$Q_3 = m_s L_v = 1 \text{ g} \times 540 = 540 \text{ cal}$$

This heat rejected can melt the ice completely but cannot raise the temperature of water from 0 to 100°C , as there is a deficiency of heat $= Q_1 + Q_2 - Q_3 = (900 - 540) \text{ cal} = 360 \text{ cal}$

$\therefore$ Resulting temperature

$$= 100^{\circ}\text{C} - \frac{\text{Heat deficient}}{\text{Thermal capacity of system, i.e. 6 g water}} = 100^{\circ}\text{C} - \frac{360 \text{ cal}}{6 \text{ g} \times 1 \text{ cal g}^{-1} \text{ } ^{\circ}\text{C}^{-1}} = 100^{\circ}\text{C} - \frac{360^{\circ}\text{C}}{6} = 100^{\circ}\text{C} - 60^{\circ}\text{C} = 40^{\circ}\text{C}$$

Example 16.16 A lead bullet penetrates into a solid object and melts. Assuming that 25% of its kinetic energy was used to heat it, calculate the initial speed of the bullet. The initial temperature of the bullet is 27°C and its melting point is 327°C . Latent heat of fusion of lead $= 2.5 \times 10^4 \text{ J kg}^{-1}$ and specific heat capacity of lead $= 125 \text{ J kg}^{-1} \text{ K}^{-1}$.

Sol. Let the mass of the bullet $= m$.

Heat required to take the bullet from 27°C to 327°C

$$Q_1 = mc(T_2 - T_1) = m \times 125 \times (327 - 27) = m \times (125)(300) = m \times (3.75 \times 10^4 \text{ J kg}^{-1})$$

Heat required to melt the bullet,

$$Q_2 = mL = m \times (2.5 \times 10^4 \text{ J kg}^{-1})$$

If the initial speed be v , the kinetic energy is $\frac{1}{2}mv^2$

$\therefore$ 25% or $\frac{1}{4}$ part of kinetic energy convert into heat.

Hence, the heat developed is $\frac{1}{4} \left(\frac{1}{2}mv^2 \right) = \frac{1}{8}mv^2$.

Thus, $\frac{1}{8}mv^2 = Q_1 + Q_2 = (m \times 3.75 + m \times 2.5) \times 10^4 \text{ J kg}^{-1}$

Therefore, $v = 707.11 \text{ ms}^{-1}$

Example 16.17 Suppose you want to cool 0.25 kg of cola (mostly water), at 25°C by adding ice initially at -20°C . How much ice should you add, so that the final temperature will be 0°C with all the ice melt? Neglect the heat capacity of the container. Specific heat of ice is $2000 \text{ J kg}^{-1} \text{ K}^{-1}$ and $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$. (Take, specific heat of cola $= 4160 \text{ J kg}^{-1} \text{ K}^{-1}$)

Sol. Heat lost by cola is

$$Q_{\text{cola}} = m_{\text{cola}} c_{\text{cola}} \Delta T_{\text{cola}} = (0.25 \text{ kg})(4160 \text{ J kg}^{-1} \text{ K}^{-1})(25^{\circ}\text{C} - 0^{\circ}\text{C}) = 26000 \text{ J} \quad (\because \Delta T^{\circ}\text{C} = \Delta \text{TK})$$

Let the mass of the required ice be m_{ice} , then the heat needed to warm it from -20°C to 0°C is

$$Q_1 = m_{\text{ice}} c_{\text{ice}} \Delta T_{\text{ice}} = m_{\text{ice}} (2000 \text{ J kg}^{-1} \text{ K}^{-1})(0^{\circ}\text{C} - (-20^{\circ}\text{C})) = m_{\text{ice}} (4.0 \times 10^4 \text{ J kg}^{-1})$$

The heat needed to melt this mass of ice is

$$Q_2 = m_{\text{ice}} L_f = m_{\text{ice}} (3.34 \times 10^5 \text{ J kg}^{-1})$$

$\therefore$ Total energy gained by ice is

$$Q_1 + Q_2 = m_{\text{ice}} (3.74 \times 10^5 \text{ J kg}^{-1})$$

Using principle of calorimetry, heat lost = heat gained, we get

$$26000 \text{ J} = m_{\text{ice}} (3.74 \times 10^5 \text{ J kg}^{-1}) \quad \text{or} \quad m_{\text{ice}} = 0.07 \text{ kg} = 70 \text{ g}$$

Example 16.18 An ice cube of mass 0.1 kg at 0°C is placed in an isolated container which is at 227°C . The specific heat c of the container varies with temperature T according to the empirical relation $c = A + BT$, where $A = 100 \text{ cal/kg-K}$ and $B = 2 \times 10^{-2} \text{ cal/kg-K}^2$. If the final temperature of the container is 27°C , determine the mass of the container. (Latent heat of fusion of water = $8 \times 10^4 \text{ cal/kg}$ and specific heat of water = 10^3 cal/kg-K)

Sol. Given,

Initial temperature of container, $T_1 = 227 + 273 = 500 \text{ K}$

Final temperature of container, $T_2 = 27 + 273 = 300 \text{ K}$

Specific heat of container, $c = A + BT = 100 + 2 \times 10^{-2} T$

Let m is mass of container.

$\therefore$ Heat lost by container

$$\begin{aligned} \Delta Q &= \int_{T_1}^{T_2} mcdT = m \int_{500}^{300} (100 + 2 \times 10^{-2} T) dT \\ &= m \left[100 T + 2 \times 10^{-2} \frac{T^2}{2} \right]_{500}^{300} \\ &= m [100(300 - 500) + 10^{-2} (300^2 - 500^2)] \\ &= m [-2 \times 10^4 + 10^{-2} (300 + 500)(300 - 500)] \\ &= m [-2 \times 10^4 - 1600] = -21600 m \end{aligned}$$

Heat gained by ice = $mL + mc_w \Delta T$

$$= 0.1 \times 8 \times 10^4 + 0.1 \times 10^3 (27 - 0) = 10700 \text{ cal}$$

Heat lost by container = Heat gained by ice

$$\begin{aligned} 21600 m &= 10700 \\ &= 0.495 \text{ kg} \end{aligned}$$

Heating curve

The graph between temperature and time is called heating curve.

Suppose the heat is supplied to the solid at constant rate P , then a graph is plotted between temperature and time as shown

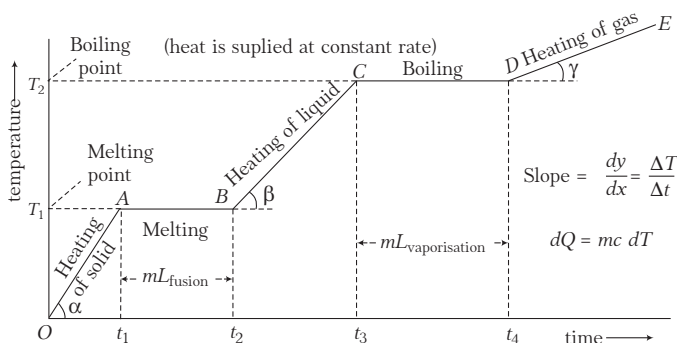


Fig. 16.1 Heating curve

We can study this graph by dividing it into different regions

Region OA (Temperature increases)

From the graph, it is clear that in this region temperature rises with time. Heat given, $Q = mc \Delta T$

If rate at which heat is supplied is P , then heat supplied in time Δt is $Q = P \Delta t$

$$\therefore P \Delta t = mc \Delta T \quad \text{or} \quad \frac{\Delta T}{\Delta t} = \frac{P}{mc} = \frac{P}{\text{Heat capacity}}$$

Here, $\frac{\Delta T}{\Delta t}$ is slope of temperature-time graph. So slope of

temperature-time graph is inversely proportional to heat capacity or specific heat capacity.

In this region, the solid absorbs heat till it reaches melting point temperature T_1 .

Region AB (Temperature constant)

In this region, temperature is constant so change of state is taking place here. This change is indicated by the horizontal line AB. At point A, solid starts melting and at point B all solids gets converted into liquid. If latent heat of fusion is L_f , then heat supplied in this state is

$$Q = mL_f = P \Delta t$$

Therefore, latent heat of fusion is directly proportional to the time taken in fusion of heat. It means latent heat of fusion is directly proportional to the length of line AB (line of zero slope).

Region BC (Temperature increases)

In this region, temperature rises again (now the substance is in liquid state), therefore

Slope of temperature-time graph (line BC) $\propto 1/\text{heat capacity or specific heat of liquid}$

Greater the heat capacity, lesser will be the slope and hence there will be small change in temperature. In this region, the liquid absorbs heat till it reaches vaporising point temperature T_2 .

Region CD (Temperature constant)

In this region, change in temperature is constant, so change of state from liquid to gas takes place. This change is indicated by horizontal line CD. So (like region AB) latent heat of vaporisation is proportional to the length of line CD.

At point C, liquid starts to convert into vapour and at point D whole of the liquid is converted into vapour.

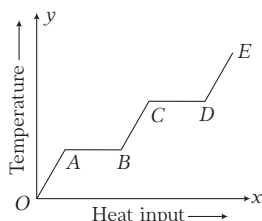
Region DE (Temperature increases)

Line DE represents absolute vapour state of the substance. Here, also the specific heat is inversely proportional to the slope of temperature versus time graph.

From the graph, it can be concluded that the total heat required to convert a solid into vapour form will be the sum of all the heats given from O to D.

Example 16.19 A solid material is supplied with heat at a constant rate. The temperature of the material is changing with the heat input as shown in figure. Study the graph carefully and answer the following questions.

- What do the horizontal regions AB and CD represent?
- If $CD = 2BA$, what do you infer?
- What does slope DE represent?
- The slope of $OA >$ the slope of BC . What does this indicate?



Sol. (i) In horizontal portions AB and CD, temperature remains same, hence these portions show change of state.

AB : conversion from solid to liquid state

CD : conversion from liquid to vapour state

- (ii) When change of state takes place, $\Delta Q = mL$

Length of line $\propto$ Latent heat

$$CD = 2AB \Rightarrow \Delta Q_{CD} = 2\Delta Q_{AB}$$

$$mL_V = 2mL_f \Rightarrow L_V = 2L_f$$

where L_f and L_V are the latent heat of fusion and vaporisation, respectively.

- (iii) Heat, $\Delta Q = mc \Delta T$

$$\frac{\Delta T}{\Delta Q} = \text{Slope of inclined line}$$

$$\Rightarrow = \frac{1}{mc} = \frac{1}{\text{Heat capacity}}$$

The slope of line DE represents reciprocal of heat capacity.

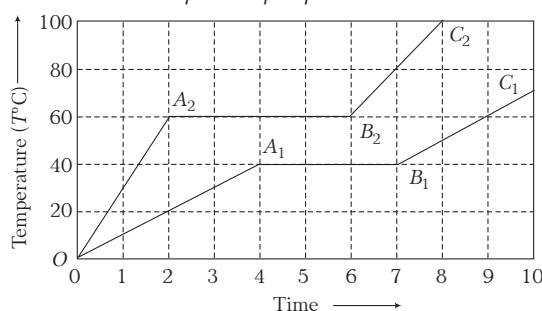
- (iv) Slope of $OA >$ slope of BC

$$\frac{1}{(mc)_s} > \frac{1}{(mc)_l} \Rightarrow c_l > c_s$$

The specific heat of liquid is greater than the specific heat of solid.

Example 16.20 Two bodies of equal mass m are heated at a uniform rate under identical conditions. Their change in temperature are shown graphically in figure.

- What are their melting points?
- What is the ratio of their latent heats?
- What is the ratio of their specific heats?



Sol. (i) A_1B_1 and A_2B_2 represent change of state for substances 1 and 2. During change of state, temperature remains same.

Melting point of substance 1 : 40°C

Melting point of substance 2 : 60°C

- (ii) Heat, $\Delta Q \propto \text{time}$

Since, heat is supplied at constant rate

$$\therefore \Delta Q = kt$$

where $k = \text{constant rate (J/s)}$.

$$\text{Also, } \Delta Q = mL$$

$$\Rightarrow mL = kt$$

$$\Rightarrow L \propto t$$

$$\frac{L_1}{L_2} = \frac{t_1}{t_2} = \frac{\text{time during change of state from } A_1 \text{ to } B_1}{\text{time during change of state from } A_2 \text{ to } B_2} = \frac{(7-4)}{(8-6)} = \frac{3}{2}$$

The latent heat of fusion of two substances are in the ratio 3 : 2.

- (iii) Heat supplied, $\Delta Q = mc \Delta T$

$$\Rightarrow kt = mc(T - T_0)$$

$$T = \frac{kt}{mc} + T_0, T \text{ versus } t \text{ is straight line}$$

So, $\frac{k}{mc}$ represents slope of inclined line.

$$\text{In solid state, } \frac{\left(\frac{k}{mc}\right)_1}{\left(\frac{k}{mc}\right)_2} = \frac{\text{slope of } OA_1}{\text{slope of } OA_2} = \frac{40/4}{60/2} = \frac{1}{3}$$

$$\therefore \text{Ratio of specific heat, } \frac{c_2}{c_1} = \frac{1}{3}$$

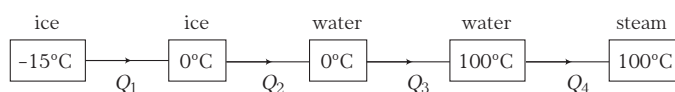
In liquid state,

$$\frac{\left(\frac{k}{mc}\right)_1}{\left(\frac{k}{mc}\right)_2} = \frac{\text{slope of } B_1C_1}{\text{slope of } B_2C_2} = \frac{(70-40)/(10-7)}{(100-60)/(8-6)} = \frac{1}{2}$$

$$\therefore \text{Ratio of specific heat, } \frac{c_2}{c_1} = \frac{1}{2}$$

Example 16.21 How much heat is required to convert 8.0 g of ice at -15°C to steam at 100°C ?

(Given, $c_{\text{ice}} = 0.53 \text{ cal g}^{-1} ^\circ\text{C}^{-1}$, $L_f = 80 \text{ cal g}^{-1}$ and $L_V = 539 \text{ cal g}^{-1}$ and $c_{\text{water}} = 1 \text{ cal g}^{-1} ^\circ\text{C}^{-1}$)



Sol. Heat, $Q_1 = mc_{\text{ice}} (T_f - T_i)$
 $= (8.0) (0.53) [0 - (-15)]$
 $= 63.6 \text{ cal}$
 $Q_2 = mL_f = (8) (80) = 640 \text{ cal}$
 $Q_3 = mc_{\text{water}} (T_f - T_i)$
 $= (8.0) (1.0) [100 - 0]$
 $= 800 \text{ cal}$
 $Q_4 = mL_V = (8.0) (539)$
 $= 4312 \text{ cal}$
 $\therefore$ Net heat required,
 $Q = Q_1 + Q_2 + Q_3 + Q_4$
 $= 5815.6 \text{ cal}$

Phase diagram and triple point

The temperature of a substance remains constant during its phase change (or change of state) and this temperature depends on the pressure.

The graph between temperature and pressure of substance is called **phase diagram** or **p - T diagram**. The phase diagram (drawn for water and CO_2) shows p - T plane which is divided into three regions, i.e. solid region, liquid region and vapour region which are separated by **sublimation curve**, **fusion curve** and **vaporisation curve**.

These curves are described below

- A curve which shows the variation of boiling point with pressure is called **vaporisation curve** (OC). The points on this curve represents the states in which liquid and vapour phases co-exist. The vaporisation line is called **steam line** in case of water.
- A curve which shows the variation of melting point with pressure is called the **fusion curve** (OA). The points on this curve represents the states in which solid and liquid phases co-exist. The fusion line is called **ice line** in case of water.
- A curve which shows the variation of pressure with temperature at which a solid changes directly into vapour state is called **sublimation curve** (OB). The points on this curve represents the state in which solid and vapour phases co-exist. The sublimation line is called the **hoar-frost line** in case of water.
- The temperature and pressure at which these three curves co-exist simultaneously is known as the **triple point** of the substance.

e.g. The triple point for water is represented by temperature 273.16 K and pressure $6.11 \times 10^{-3} \text{ Pa}$.

Two figures below show the phase diagram with triple point for (a) water and (b) CO_2

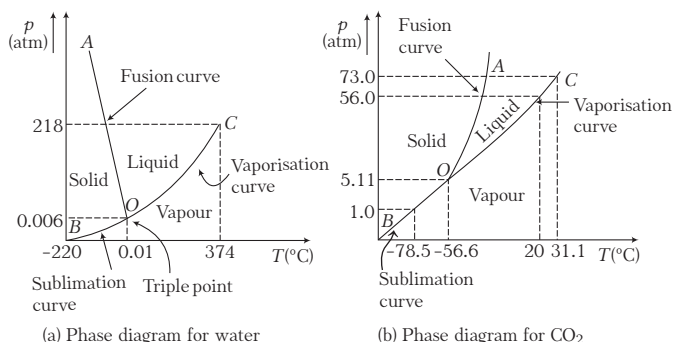
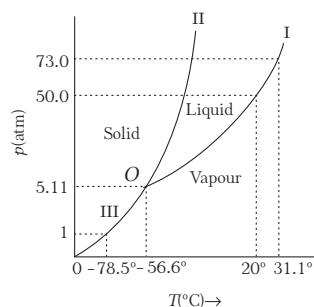


Fig. 16.2

- At pressure below that of triple point, no substance can exist as a liquid.

Example 16.22 Answer the following questions based on the p - T phase diagram of carbon dioxide as shown in the figure.



- At what temperature and pressure, can the solid, liquid and vapour phases of CO_2 co-exist in equilibrium?
- What is the effect of decrease of pressure on the fusion and boiling point of CO_2 ?
- What are the critical temperature and pressure for CO_2 ? What is their significance?
- Is CO_2 solid, liquid or gas at (a) -70°C under 1 atm (b) -60°C under 10 atm (c) 15°C under 56 atm?

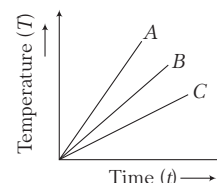
Sol. (i) The solid, liquid and vapour phases of CO_2 co-exist in equilibrium at its triple point O for which

$$p_{tr} = 5.11 \text{ atm and } T_{tr} = -56.6^\circ \text{C}$$

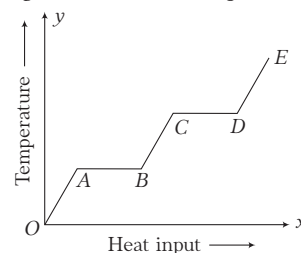
- The vaporisation curve I and fusion curve II show that both the boiling point and fusion point of CO_2 decrease with decrease in pressure.
- For CO_2 , $p_c = 73.0 \text{ atm}$ and $T_c = 31.1^\circ \text{C}$. Above its critical temperature, CO_2 gas cannot be liquified, however large pressure may be applied.
- (a) -70°C under 1 atm. This point lies in vapour region. Therefore, at -70°C under 1 atm, CO_2 is vapour.
 (b) -60°C under 10 atm. This point lies in solid region. Therefore, CO_2 is solid at -60°C under 10 atm.
 (c) 15°C under 56 atm. This point lies in liquid region. Therefore, CO_2 is liquid at 15°C under 56 atm.

CHECK POINT 16.2

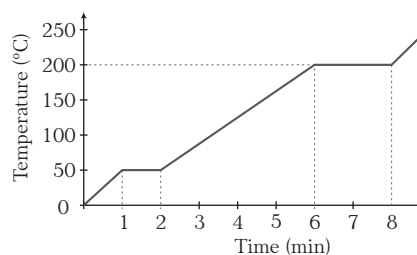
- A substance of mass m kg requires a power input of P watts to remain in the molten state at its melting point. When the power is turned off, the sample completely solidifies in time t second. What is the latent heat of fusion of the substance?
 (a) $\frac{Pm}{t}$ (b) $\frac{Pt}{m}$ (c) $\frac{m}{Pt}$ (d) $\frac{t}{Pm}$
- 50 g of ice at 0°C is mixed with 50 g of water at 60°C , final temperature of mixture will be
 (a) 0°C (b) 40°C
 (c) 10°C (d) 15°C
- 80 g of water at 30°C are poured on a large block of ice at 0°C . The mass of ice that melts is
 (a) 30 g (b) 80 g
 (c) 1600 g (d) 150 g
- An iron ball of mass 0.2 kg is heated to 100°C and put on a block of ice at 0°C , 25 g of ice melts. If the latent heat of fusion of ice is 80 cal g^{-1} , then the specific heat of iron in $\text{cal g}^{-1} ^\circ\text{C}^{-1}$ is
 (a) 1 (b) 0.1
 (c) 0.8 (d) 0.08
- 4 kg of ice at -15°C are added to 5 kg of water at 15°C . The temperature of the resulting mixture equals
 (a) -15°C (b) 0°C
 (c) 5°C (d) 15°C
- Steam at 100°C is passed into 1.1 kg of water contained in a calorimeter of water equivalent 0.02 kg at 15°C till the temperature of calorimeter and its temperature rises to 80°C . The mass of the steam condensed in kilogram is ($c_w = 10^3 \text{ cal kg}^{-1} ^\circ\text{C}^{-1}$ and $L_v = 540 \times 10^3 \text{ cal kg}^{-1}$)
 (a) 0.130 (b) 0.065
 (c) 0.260 (d) 0.135
- A lead bullet of 10 g travelling at 300 ms^{-1} strikes against a block of wood and comes to rest. Assuming 50% of heat is absorbed by the bullet, the increase in its temperature is (take, $c_{\text{lead}} = 150 \text{ J kg}^{-1} ^\circ\text{C}^{-1}$)
 (a) 100°C (b) 125°C
 (c) 150°C (d) 200°C
- The temperatures *versus* time graph is shown in figure. Which of the substances A, B and C has the lowest heat capacity, if heat is supplied to all of them at equal rates?



- (a) A (b) B
(c) C (d) All have equal specific heat
9. A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represent?



- (a) Latent heat of liquid
 (b) Latent heat of vapour
 (c) Heat capacity of vapour
 (d) Inverse of heat capacity of vapour
10. A student takes 50 g wax (specific heat = $0.6 \text{ kcal/kg}^\circ\text{C}$) and heats it till it boils. The graph between temperature and time is as follows. Heat supplied to the wax per minute and boiling point are respectively



- (a) 500 cal, 50°C (b) 1000 cal, 100°C
 (c) 1500 cal, 200°C (d) 1000 cal, 200°C

HEAT TRANSFER

As we have already studied, heat is a form of energy which transfers from a body at higher temperature to a body at lower temperature. The transfer of heat from one body to the other may take place by any of three possible processes namely

- (i) Conduction (ii) Convection
 (iii) Radiation

In the first two processes, a medium is necessary for the heat transfer. Radiation, however, does not have this restriction.

This is also the fastest mode of heat transfer, in which heat is transferred from one place to the other in the form of electromagnetic radiations.

- (i) **Conduction** The transfer of heat taking place due to molecular vibrations (*i.e.* molecular collisions) is known as heat conduction. In this process, there is no change in average position of the molecule and hence there is no mass movement of matter.

In solids, heat is transferred by the process of conduction.

- (ii) **Convection** It is the process in which heat is transferred from one point to another by the actual motion of matter from a region of high temperature to a region of lower temperature. This process of heat transfer takes place only in fluids.
- (iii) **Radiation** It is a mode of heat transfer from one place to another without heating the intervening medium. The heat is transferred by the means of thermal radiations, radiant energy or simply radiation. The heat from the sun reaches to the earth by radiation after travelling millions of kilometers of empty space (*i.e.* without any material medium).

Conduction of heat in conducting rod

Figure shows a rod whose ends are in thermal contact with a hot reservoir at temperature T_1 and a cold reservoir at temperature T_2 . The sides of the rod are covered with insulating medium, so the transport of heat is along the rod and not through the sides.

Heat is transferred from the hot end to the cold end due to molecular collisions. Molecules at hot end transfer their kinetic energy to neighbouring molecules by collision. In this way, heat transfers from one end to other. During heat transfer, the molecules do not change their equilibrium position.

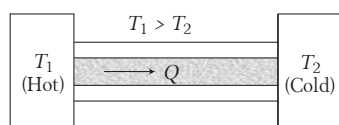


Fig. 16.3

Steady state

Consider a section ab of a rod as shown in figure. Suppose Q_1 heat enters into the section at a and Q_2 leaves at b , then $Q_2 < Q_1$. Part of the energy $Q_1 - Q_2$ is utilised in raising the temperature of section ab and the remaining is lost to atmosphere through ab .

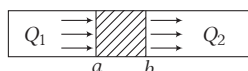


Fig. 16.4

If heat is continuously supplied from the left end of the rod, a stage comes when temperature of the section becomes constant. This is called the steady state condition. In that case, $Q_1 = Q_2$, if rod is insulated from the surroundings (or heat loss through ab is zero).

In steady state, temperature of different sections of the rod remains constant with time.

In the given figure,

$$T_1 = \text{constant}, T_2 = \text{constant etc.}$$

and

$$T_1 > T_2 > T_3 > T_4$$

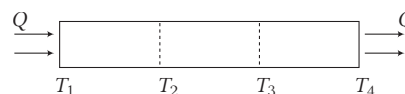


Fig. 16.5 Insulated rod in steady state

For a conductor in steady state, neither absorption nor emission takes place.

Rate of heat transfer and temperature gradient

When a quantity of heat dQ is transferred through the rod in a time dt , the rate of heat flow is $\frac{dQ}{dt}$.

This rate is also called the **heat current** denoted by H . Experiments show that the **heat current is proportional to the cross-section area A of the rod and to the temperature gradient dT/dx** , which is the rate of change of temperature with distance along the bar as shown in the figure. In general,

$$H = \frac{dQ}{dt} = -KA \frac{dT}{dx} \quad \dots(i)$$

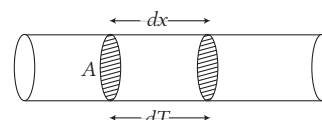


Fig. 16.6

The negative sign indicates that temperature decreases along the direction of flow of heat.

SI unit of temperature gradient is Km^{-1} or $^{\circ}\text{Cm}^{-1}$.

Thermal conductivity The constant K (in the above expression) is called the thermal conductivity, which is a measure of the ability of a material to conduct heat.

A substance with a large thermal conductivity K is a good heat conductor. Let us apply Eq. (i) to a rod of length L and constant cross sectional area A in which a steady state has been reached. In a steady state, the temperature at each point of rod is constant in time. Hence,

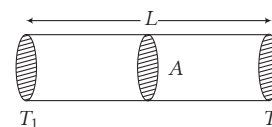


Fig. 16.7

$$\frac{dT}{dx} = \frac{T_1 - T_2}{L}$$

Therefore, the heat ΔQ transferred in time Δt is

$$\Delta Q = KA \left(\frac{T_1 - T_2}{L} \right) \Delta t \quad \dots(ii)$$

If $A = 1$, $T_1 - T_2 = 1$, $L = 1$, $\Delta t = 1$, then $\Delta Q = K$.

Hence, the coefficient of thermal conductivity of a material may be defined as the quantity of heat that flows per unit time through a unit cube of the material when its opposite faces are kept at a temperature difference of one degree.

The SI unit of K is $\text{Js}^{-1} \text{m}^{-1} \text{K}^{-1}$ or $\text{Wm}^{-1} \text{K}^{-1}$ and its dimensions are $[\text{MLT}^{-3} \text{K}^{-1}]$.

Note

- (i) A substance which is a good conductor of heat (e.g. silver) is also a good conductor of electricity. For such substances at a given temperature T , the ratio of thermal conductivity to electrical conductivity ($K/\sigma T$) is constant. This is known as **Wiedemann-Franz law**.
- (ii) The ratio of the coefficient of thermal conductivity (K) to the thermal capacity per unit volume (ρc) of a material is called **thermometric conductivity or diffusivity**.

Thermal resistance (R) In differential form, rate of heat flow can be written as

$$\frac{dQ}{dt} = H = \frac{\Delta T}{l/K A} = \frac{\Delta T}{R} \quad \dots(iii)$$

Here, ΔT = temperature difference

and $R = \frac{l}{KA}$ = thermal resistance of the rod.

Also, we can write $R = \frac{\Delta T}{H}$

Thus, thermal resistance is the the ratio of the temperature difference between the ends of conductors to the heat current.

SI unit of thermal resistance is KW^{-1} (Kelvin per watt) and dimensional formula of R is $[\text{M}^{-1} \text{L}^{-2} \text{T}^3 \text{K}]$.

Example 16.23 Thickness of ice on a lake is 5 cm and the temperature of air is -20°C . If the rate of cooling of water inside the lake be $20000 \text{ cal min}^{-1}$ through each square metre surface, find K for ice.

Sol. Thickness of ice on a lake, $L = 5 \text{ cm} = 5 \times 10^{-2} \text{ m}$

Change in temperature, $\Delta T = 0 - (-20) = 20^\circ \text{C}$

Rate of cooling of water = $\frac{\Delta Q}{\Delta t} = 20000 \text{ cal min}^{-1}$

$$= \frac{20000}{60} \text{ cal s}^{-1}$$

$$= \frac{20000}{60} \times 4.2 = 1400 \text{ Js}^{-1}$$

$$\text{As,} \quad \frac{\Delta Q}{\Delta t} = KA \left(\frac{\Delta T}{L} \right)$$

where, K is thermal conductivity

$$\therefore K = \frac{\Delta Q}{\Delta t} \cdot \frac{L}{\Delta T A}$$

On putting values, we get

$$K = \frac{1400 \times 5 \times 10^{-2}}{1 \times 20} = 3.5 \text{ Wm}^{-1} \text{C}^{-1} \quad (\text{where } A = 1 \text{ m}^2)$$

Example 16.24 One end of a copper rod of length 1m and area of cross-section $4.0 \times 10^{-4} \text{ m}^2$ is maintained at 100°C . At the other end of the rod, ice is kept at 0°C . Neglecting the loss of heat from the surroundings, find the mass of ice melted in 1 h. (Given, $K_{\text{Cu}} = 401 \text{ Wm}^{-1} \text{K}^{-1}$ and $L_f = 3.35 \times 10^5 \text{ Jkg}^{-1}$)

Sol. Thermal resistance of the rod,



$$R = \frac{l}{KA} = \frac{1.0}{(401)(4 \times 10^{-4})} = 6.23 \text{ KW}^{-1}$$

$$\therefore \text{Heat current, } H = \frac{\text{Temperature difference}}{\text{Thermal resistance}} = \frac{(100 - 0)}{6.23} = 16 \text{ W}$$

$$\text{Heat transferred in 1h, } Q = Ht \quad \left(\because H = \frac{Q}{t} \right)$$

$$= (16)(3600) = 57600 \text{ J}$$

Now, let m be the mass of ice that melts in 1 h, then

$$m = \frac{Q}{L_f} \quad (Q = mL_f)$$

$$= \frac{57600}{3.35 \times 10^5} = 0.172 \text{ kg or } 172 \text{ g}$$

Example 16.25 An iron boiler is 1 cm thick and has a heating area 2.5 m^2 . The two surfaces of the boiler are at 230°C and 100°C , respectively. If the latent heat of the steam is 540 kcal kg^{-1} and thermal conductivity of iron is $1.6 \times 10^{-2} \text{ kcal s}^{-1} \text{m}^{-1} \text{K}^{-1}$, then how much water will be evaporated into steam per minute?

Sol. Here, thickness of iron boiler, $\Delta x = 1 \text{ cm} = 0.01 \text{ m}$

Heating area, $A = 2.5 \text{ m}^2$

Temperature, $\Delta T = T_1 - T_2 = 230 - 100 = 130^\circ \text{C}$

Latent heat of steam, $L = 540 \text{ kcal kg}^{-1}$

Thermal conductivity of iron, $K = 1.6 \times 10^{-2} \text{ kcal s}^{-1} \text{m}^{-1} \text{K}^{-1}$

$$\Delta t = 1 \text{ min} = 60 \text{ s}$$

Let m kg of the mass be evaporated in 1 min, then

$$\text{Heat conducted, } \Delta Q = \frac{KA(T_1 - T_2)\Delta t}{\Delta x} = mL$$

$$\Rightarrow m = \frac{KA(T_1 - T_2)\Delta t}{\Delta x L}$$

$$= \frac{(1.6 \times 10^{-2}) \times (2.5) \times 130 \times 60}{0.01 \times 540} = 57.78 \text{ kg}$$

Example 16.26 A copper rod 2 m long has a circular cross-section of radius 1 cm. One end is kept at 100°C and the other at 0°C and the surface is insulated, so that negligible heat is lost through the surface. Find

- the thermal resistance of the bar,
- the thermal current H ,
- the temperature gradient dT/dx
- and the temperature at 25 cm from the hot end.

Thermal conductivity of copper is $401 \text{ Wm}^{-1}\text{K}^{-1}$.

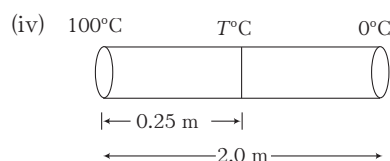
Sol. (i) Thermal resistance, $R = \frac{l}{KA} = \frac{l}{K(\pi r^2)}$

or
$$R = \frac{(2)}{(401)(\pi)(10^{-2})^2} = 15.9 \text{ KW}^{-1}$$

(ii) Thermal current, $H = \frac{\Delta T}{R} = \frac{100}{15.9}$ or $H = 6.3 \text{ W}$

(iii) Temperature gradient,

$$\frac{\Delta T}{dx} = \frac{0 - 100}{2} = -50 \text{ Km}^{-1} = -50^\circ\text{Cm}^{-1}$$



Let $T^\circ\text{C}$ be the temperature at 25 cm from the hot end, then

$$(T - 100) = (\text{temperature gradient}) \times (\text{distance})$$

or $T - 100 = (-50)(0.25)$ or $T = 87.5^\circ\text{C}$

Combinations of rods with different thermal conductivities

If two or more rods or thermal conductors are connected with one another, the equivalent thermal conductivity as a whole may be altered.

Now we will discuss different combinations of rods and heat flow associated with them

1. Series combination

Consider a series combination of rods with sides insulated, so that heat flow across length of each rod. In series combination when steady state is reached, heat current is same and temperature difference across each rod is different. Let us assume that n -slabs each of cross-sectional area A , lengths $L_1, L_2, L_3, \dots, L_n$ and thermal conductivities $K_1, K_2, K_3, \dots, K_n$ respectively are connected in series as shown in the figure below.

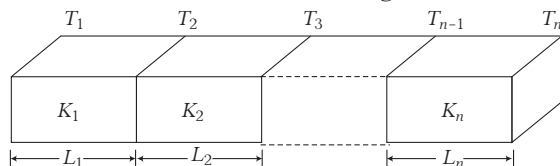


Fig. 16.8 Series combination

(a) **Heat current** As heat current is the same in all the conductors of series combination in steady state.

So, we have $\frac{Q}{t} = H_1 = H_2 = H_3 \dots = H_n = H$

$$\Rightarrow \frac{K_1 A (T_1 - T_2)}{L_1} = \frac{K_2 A (T_2 - T_3)}{L_2} = \frac{K_n A (T_{n-1} - T_n)}{L_n}$$

(b) **Equivalent thermal resistance** Equivalent thermal resistance of the combination is given by

$$R_{\text{eq}} = R_1 + R_2 + \dots + R_n$$

(c) **Equivalent thermal conductivity**

As, $R_{\text{eq}} = R_1 + R_2 + \dots + R_n$

$$\Rightarrow \frac{L_1 + L_2 + L_3 + \dots + L_n}{K_{\text{eq}} A} = \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A} + \dots + \frac{L_n}{K_n A}$$

So, equivalent thermal conductivity is given by,

$$K_{\text{eq}} = \frac{L_1 + L_2 + \dots + L_n}{\frac{L_1}{K_1} + \frac{L_2}{K_2} + \dots + \frac{L_n}{K_n}} = \frac{\sum_{i=1}^n L_i}{\sum_{i=1}^n \frac{L_i}{K_i}}$$

Case I If slabs are of equal length,

i.e. $L_1 = L_2 = L_3 = \dots = L_n = L$, then

$$K_{\text{eq}} = \frac{n}{\frac{1}{K_1} + \frac{1}{K_2} + \dots + \frac{1}{K_n}}$$

Case II If two slabs of equal lengths are combined in series, then

$$K_{\text{eq}} = \frac{2K_1 K_2}{K_1 + K_2}$$

(d) **Temperature of common interface of junction of two composite rods** Let temperature at two ends be T_1 and T_2 and temperature of the junction be T .

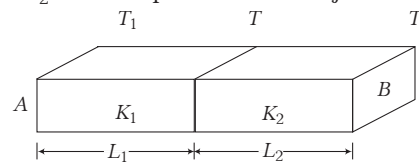


Fig. 16.9

As the two rods are connected in series, heat current must be same through both the rods

$$\therefore \frac{K_1 A (T_1 - T)}{L_1} = \frac{K_2 A (T - T_2)}{L_2}$$

Solving, we get

$$T = \frac{\frac{K_1}{L_1} T_1 + \frac{K_2}{L_2} T_2}{\frac{K_1}{L_1} + \frac{K_2}{L_2}}$$

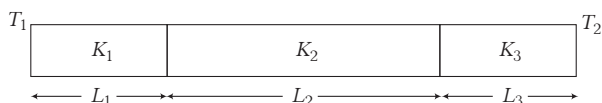
Case I If $L_1 = L_2 = L$, then

$$T = \frac{K_1 T_1 + K_2 T_2}{K_1 + K_2}$$

Case II If $K_1 = K_2 = K$ and $L_1 = L_2 = L$, then

$$T = \frac{T_1 + T_2}{2}$$

Example 16.27 Three rods of same cross-section but different length and conductivity are joined in series. If the temperature of the two extreme ends are T_1 and T_2 ($T_1 > T_2$), find the rate of heat transfer H .



Sol. Total thermal resistance,

$$R = R_1 + R_2 + R_3 = \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A} + \frac{L_3}{K_3 A}$$

Now, heat transfer, $H = \frac{T_1 - T_2}{R}$

$$\therefore H = \frac{(T_1 - T_2)}{\frac{L_1}{K_1 A} + \frac{L_2}{K_2 A} + \frac{L_3}{K_3 A}}$$

Example 16.28 A composite slab is prepared by pasting two plates of thicknesses L_1 and L_2 and thermal conductivities K_1 and K_2 . The slabs have equal cross-sectional area. Find the equivalent conductivity of the slab.

Sol. The situation can be shown as



Total thermal resistance, $R_{eq} = R_1 + R_2$

$$\therefore \frac{(L_1 + L_2)}{K_{eq} A} = \frac{L_1}{K_1 A} + \frac{L_2}{K_2 A}$$

$$K_{eq} = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}}$$

Example 16.29 The three bars of equal lengths and of equal areas of cross-section are connected in series. Their thermal conductivities are in the ratio of 2 : 4 : 3. If the open ends of the first and the last bars are at temperatures 200°C and 18°C respectively in the steady state, calculate the temperatures of both the junctions.

Sol. Let the temperature of the junctions B and C be θ_1 and θ_2 , respectively. In the steady state, the rate of flow of heat through all rods will be the same. Hence, the figure can be shown as



Therefore, heat transfer, $H = (2K) A \frac{(200 - \theta_1)}{L}$

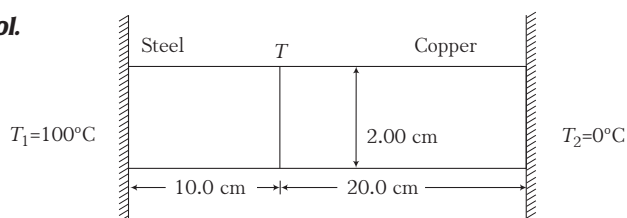
$$= (4K) A \frac{(\theta_1 - \theta_2)}{L} = (3K) A \frac{(\theta_2 - 18)}{L}$$

or $2(200 - \theta_1) = 4(\theta_1 - \theta_2) = 3(\theta_2 - 18)$

Solving, $\theta_1 = 116^\circ\text{C}$ and $\theta_2 = 74^\circ\text{C}$

Example 16.30 A steel bar 10.0 cm long is welded end-to-end to a copper bar 20.0 cm long. Both bars are insulated perfectly on their sides. Each bar has a square cross-section, 2.00 cm on a side. The free end of the steel bar is maintained at 100°C by placing it in contact with ice. Find the temperature at the junction of the two bars and the total rate of heat flow. Thermal conductivity of steel = $50.2 \text{ Wm}^{-1}\text{K}^{-1}$ and thermal conductivity of copper = $385 \text{ Wm}^{-1}\text{K}^{-1}$.

Sol.



In the steady state, the two heat currents are equal. We have,

$$H_{\text{steel}} = \frac{K_{\text{steel}} A (100^\circ\text{C} - T)}{L_{\text{steel}}}$$

and $H_{\text{copper}} = \frac{K_{\text{copper}} A (T - 0^\circ\text{C})}{L_{\text{copper}}}$

Substituting $L_{\text{steel}} = 0.100 \text{ m}$, $L_{\text{copper}} = 0.200 \text{ m}$

$$K_{\text{steel}} = 50.2 \text{ Wm}^{-1}\text{K}^{-1}$$

and $K_{\text{copper}} = 385 \text{ Wm}^{-1}\text{K}^{-1}$

Also, both rods have same area.

So, $\frac{(50.2)(100^\circ\text{C} - T)}{0.100} = \frac{(385)(T - 0^\circ\text{C})}{0.200}$

Solving for T , we obtain $T = 20.7^\circ\text{C}$.

We can find the total heat current by substituting this value for T ,

$$H_{\text{steel}} = \frac{(50.2)(0.0200)(100 - 20.7)}{0.100}$$

$$H_{\text{steel}} = 796.17 \text{ W}$$

$\therefore H_{\text{steel}} = H_{\text{copper}}$

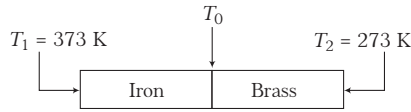
$\therefore H_{\text{copper}} = 796.17 \text{ W}$

Example 16.31 An iron bar having length $L_1 = 0.1 \text{ m}$, area of cross-section $A_1 = 0.02 \text{ m}^2$, thermal conductivity $K_1 = 79 \text{ Wm}^{-1}\text{K}^{-1}$ and brass bar having length $L_2 = 0.1 \text{ m}$, area of cross-section, $A_2 = 0.02 \text{ m}^2$ and thermal conductivity $K_2 = 109 \text{ Wm}^{-1}\text{K}^{-1}$ are soldered end-to-end as shown in figure.

The terminal ends of two rods are maintained at 373 K and 273 K , respectively. Find the expression and compute

(i) the temperature of the junction of two bars.

- (ii) equivalent thermal conductivity of composite bar,
 (iii) and the heat current through the composite bar



Sol. Given, length, $L_1 = L_2 = L = 0.1 \text{ m}$

Area, $A_1 = A_2 = A = 0.02 \text{ m}^2$

Thermal conductivity, $K_1 = 79 \text{ W m}^{-1} \text{ K}^{-1}$, $K_2 = 109 \text{ W m}^{-1} \text{ K}^{-1}$

Temperature, $T_1 = 373 \text{ K}$ and $T_2 = 273 \text{ K}$

At steady state, heat transferred from any section of thermal conductor is same, i.e. $H_1 = H_2 = H$

$$\Rightarrow \frac{K_1 A_1 (T_1 - T_0)}{L_1} = \frac{K_2 A_2 (T_0 - T_2)}{L_2} \quad \dots(i)$$

For $A_1 = A_2 = A$ and $L_1 = L_2 = L$, Eq. (i) becomes

$$K_1 (T_1 - T_0) = K_2 (T_0 - T_2) \Rightarrow T_0 = \frac{K_1 T_1 + K_2 T_2}{K_1 + K_2}$$

Therefore, heat current through other bar

$$H = \frac{K_1 A (T_1 - T_0)}{L} = \frac{K_2 A (T_0 - T_2)}{L} = \frac{A(T_1 - T_2)}{L \left(\frac{1}{K_1} + \frac{1}{K_2} \right)} \quad \dots(ii)$$

Now, heat current through composite bar of length, $L_1 + L_2 = 2L$ and equivalent thermal conductivity K' , can be given by

$$H' = \frac{K' A (T_1 - T_2)}{2L} = H \quad \dots(iii)$$

Thermal conductivity, $K' = \frac{2K_1 K_2}{K_1 + K_2}$ [From Eqs. (ii) and (iii)]

(i) So, the temperature of the junction of two bars is

$$T_0 = \frac{K_1 T_1 + K_2 T_2}{K_1 + K_2} = \frac{79 \text{ W m}^{-1} \text{ K}^{-1} \times 373 \text{ K} + 109 \text{ W m}^{-1} \text{ K}^{-1} \times 273 \text{ K}}{79 \text{ W m}^{-1} \text{ K}^{-1} + 109 \text{ W m}^{-1} \text{ K}^{-1}} = 315 \text{ K}$$

(ii) Equivalent thermal conductivity, $K' = \frac{2K_1 K_2}{K_1 + K_2}$

$$= \frac{2 \times 79 \text{ W m}^{-1} \text{ K}^{-1} \times 109 \text{ W m}^{-1} \text{ K}^{-1}}{79 \text{ W m}^{-1} \text{ K}^{-1} + 109 \text{ W m}^{-1} \text{ K}^{-1}} = 91.6 \text{ W m}^{-1} \text{ K}^{-1}$$

(iii) Heat current through the composite bar,

$$H' = H = \frac{K' A (T_1 - T_2)}{2L} = \frac{91.6 \text{ W m}^{-1} \text{ K}^{-1} \times 0.02 \text{ m}^2 \times (373 - 273) \text{ K}}{2 \times 0.1 \text{ m}} = 916 \text{ W}$$

2. Parallel combination

In a parallel combination of rods, one end of each of rod is kept at higher temperature and other end of each of rod is kept at lower temperature, as shown in Fig. 16.10. In this way rods, provide parallel path for flow of heat energy.

In parallel, heat currents are different for different rod but temperature difference across each rod is same.

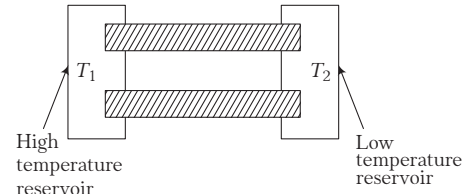


Fig. 16.10

(a) **Heat current** It will be different in each rod. Net heat current will be given by sum of heat currents through individual rod.

$$i.e. \quad H = H_1 + H_2 + \dots + H_n$$

(b) **Equivalent resistance** $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$

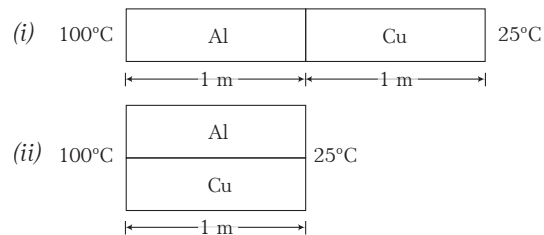
$$\text{For two rods, } R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

(c) **Temperature gradient** It is same across each rod and is given by $\frac{T_1 - T_2}{L}$.

(d) **Equivalent thermal conductivity**

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2 + \dots + K_n A_n}{A_1 + A_2 + \dots + A_n}$$

Example 16.32 An aluminium rod and copper rod of equal length 1 m and cross-sectional area 1 cm^2 are welded as shown in figure. Given, $K_{Al} = 200 \text{ W / m}^\circ \text{C}$, $K_{Cu} = 400 \text{ W / m}^\circ \text{C}$. Find heat current.



Sol. Resistance, $R_{Al} = \frac{l}{K_{Al} A} = \frac{1}{200 \times 10^{-4}} = 50^\circ \text{C/W}$

$$R_{Cu} = \frac{l}{K_{Cu} A} = \frac{1}{400 \times 10^{-4}} = 25^\circ \text{C/W}$$

(i) Equivalent resistance, $R_{eq} = R_{Al} + R_{Cu} = 50 + 25 = 75^\circ \text{C/W}$
 Heat current, $H = \frac{100 - 25}{R_{eq}} = \frac{75}{75} = 1 \text{ W}$

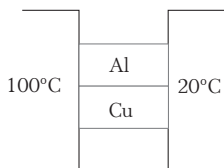
(ii) Equivalent resistance,

$$R_{eq} = \frac{R_{Al} \cdot R_{Cu}}{R_{Al} + R_{Cu}} = \frac{50 \times 25}{50 + 25} = \frac{50}{3}^\circ \text{C/W}$$

$$\text{Heat current, } H = \frac{100 - 25}{50/3} = 4.5 \text{ W}$$

Example 16.33 Two metal cubes with 3 cm edges of copper and aluminium are arranged as shown in figure. Find

(i) the total thermal current from one reservoir to the other and



(ii) the ratio of the thermal current carried by the copper cube to that carried by the aluminium cube. Thermal conductivity of copper is $401 \text{ Wm}^{-1}\text{K}$ and that of aluminium is $237 \text{ Wm}^{-1}\text{K}$.

Sol. (i) Thermal resistance of aluminium cube, $R_1 = \frac{l}{KA}$

$$\text{or } R_1 = \frac{(3.0 \times 10^{-2})}{(237)(3.0 \times 10^{-2})^2} = 0.14 \text{ KW}^{-1}$$

and thermal resistance of copper cube, $R_2 = \frac{l}{KA}$

$$\text{or } R_2 = \frac{(3.0 \times 10^{-2})}{(401)(3.0 \times 10^{-2})^2} = 0.08 \text{ KW}^{-1}$$

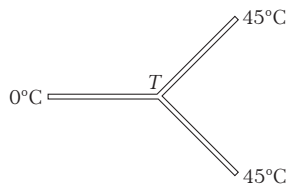
As these two resistances are in parallel, their equivalent resistance will be

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{(0.14)(0.08)}{(0.14) + (0.08)} = 0.05 \text{ KW}^{-1}$$

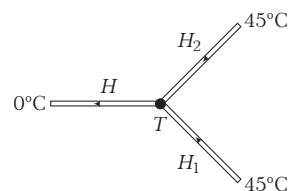
$$\therefore \text{Thermal current, } H = \frac{\text{Temperature difference}}{\text{Thermal resistance}} = \frac{(100 - 20)}{0.05} = 1.6 \times 10^3 \text{ W}$$

(ii) In parallel, thermal current distributes in the inverse ratio of resistance. Hence, $\frac{H_{\text{Cu}}}{H_{\text{Al}}} = \frac{R_{\text{Al}}}{R_{\text{Cu}}} = \frac{R_1}{R_2} = \frac{0.14}{0.08} = 1.75$

Example 16.34 Three identical rods have been joined at a junction to make it a Y shape structure. If two free ends are maintained at 45°C and the other end is at 0°C , then what is the junction temperature T ?



Sol. According to the question,



Rods are in steady state, so $H = H_1 + H_2$.

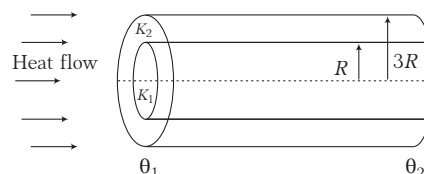
$$\Rightarrow \frac{KA(T - 0)}{L} = \frac{KA(45 - T)}{L} + KA \frac{(45 - T)}{L}$$

$$\Rightarrow T = 45 - T + 45 - T$$

$$\Rightarrow 3T = 90 \Rightarrow T = 30^\circ\text{C}$$

Example 16.35 A cylinder of radius R made of a material of thermal conductivity $K_1 = K$ is surrounded by cylindrical shell of inner radius R and outer radius $3R$ made of a material of thermal conductivity $K_2 = 2K$. The two ends of the combined system are maintained at two different temperatures.

There is no loss of heat across the cylindrical surface and system is in steady state. What is the effective thermal conductivity of the system?



Sol. In this situation, a rod of length L and area of cross-section πR^2 and another of same length L and area of cross-section $\pi[(3R)^2 - R^2] = 8\pi R^2$ will conduct heat simultaneously, so total heat flowing per second will be

$$\begin{aligned} \frac{dQ}{dt} &= \frac{dQ_1}{dt} + \frac{dQ_2}{dt} \\ &= \frac{K_1 \pi R^2 (T_1 - T_2)}{L} + \frac{K_2 8\pi R^2 (T_1 - T_2)}{L} \end{aligned} \quad \dots(i)$$

Now, if the equivalent conductivity is K .

$$\text{Then, } \frac{dQ}{dt} = K \frac{9\pi R^2 (T_1 - T_2)}{L} \quad (\text{As, } A = \pi(3R)^2) \quad \dots(ii)$$

So, from Eqs. (i) and (ii), we get

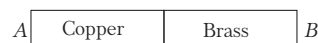
$$9K = K_1 + 8K_2$$

$$\text{i.e. } K = \frac{(K_1 + 8K_2)}{9} = \frac{K + 8 \times 2K}{9} = \frac{17K}{9}$$

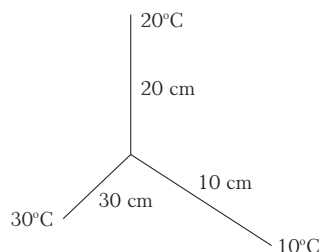
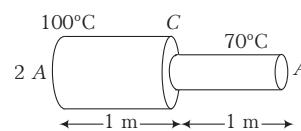
CHECK POINT 16.3

- The layers of atmosphere are heated through
 - convection
 - conduction
 - radiation
 - Both (a) and (c)
- Mud houses are cooler in summer and warmer in winter because
 - mud is superconductor of heat
 - mud is good conductor of heat
 - mud is bad conductor of heat
 - None of the above
- Snow is more heat insulating than ice, because
 - air is filled in pores of snow
 - ice is more bad conductor than snow
 - air is filled in pores of ice
 - density of ice is more
- On heating one end of a rod, the temperature of whole rod will be uniform when
 - $K = 1$
 - $K = 0$
 - $K = 100$
 - $K = \infty$
- If the temperature difference on the two sides of a wall increases from 100°C to 200°C , its thermal conductivity
 - remains unchanged
 - is doubled
 - is halved
 - becomes four times
- Wires A and B have identical lengths and have circular cross-sections. The radius of A is twice the radius of B , i.e. $r_A = 2r_B$. For a given temperature difference between the two ends, both wires conduct heat at the same rate. The relation between the thermal conductivities is given by
 - $K_A = 4K_B$
 - $K_A = 2K_B$
 - $K_A = \frac{K_B}{2}$
 - $K_A = \frac{K_B}{4}$
- The ends of two rods of different materials with their thermal conductivities, radii of cross-section and lengths all in the ratio $1 : 2$ are maintained at the same temperature difference. If the rate of flow of heat in the larger rod is 4 cal s^{-1} , then that in the shorter rod in cal s^{-1} will be
 - 1
 - 2
 - 8
 - 16
- The length of the two rods made up of the same metal and having the same area of cross-section are 0.6 m^2 and 0.8 m^2 , respectively. The temperature between the ends of first rod is 90°C and 60°C and that for the other rod is 150°C and 110°C . For which rod, the rate of conduction will be greater?
 - First
 - Second
 - Same for both
 - None of these
- Consider a compound slab consisting of two different materials having equal thickness and thermal conductivities K and $2K$ in series. The equivalent conductivity of the slab is
 - $\frac{2}{3}K$
 - $\sqrt{2}K$
 - $3K$
 - $\frac{4}{3}K$

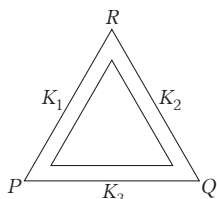
- Equal temperature differences exist between the ends of two metallic rods 1 and 2 of equal lengths. Their thermal conductivities are K_1 and K_2 and area of cross-section are A_1 and A_2 , respectively. The condition of equal rates of heat transfer is
 - $K_1 A_2 = K_2 A_1$
 - $K_1 A_1 = K_2 A_2$
 - $K_1 A_1^2 = K_2 A_2^2$
 - $K_1^2 A_2 = K_2^2 A_1$
- In a steady state, the temperatures at the end A and B of 20 cm long rod AB are 100°C and 0°C . The temperature of a point 9 cm from A is
 - 45°C
 - 55°C
 - 5°C
 - 65°C
- Two rods of copper and brass ($K_C > K_B$) of same length and area of cross-section are joined as shown in figure. End A is kept at 100°C and end B at 0°C . The temperature at the junction
 - will be more than 50°C
 - will be less than 50°C
 - will be 50°C
 - may be more or less than 50°C depending upon the size of rods



- A wall has two layers A and B , each made of different material. Both the layers have the same thickness. The thermal conductivity for A is twice that of B and under steady condition, the temperature difference across the wall is 36°C . The temperature difference across the layer A is
 - 6°C
 - 12°C
 - 24°C
 - 18°C
- A metal rod of length 2 m has cross-sectional areas $2A$ and A as shown in figure. The ends are maintained at temperatures 100°C and 70°C . The temperature at middle point C is
 - 80°C
 - 85°C
 - 90°C
 - 95°C
- Three rods made of the same material and having same cross-sectional area but different lengths 10 cm , 20 cm and 30 cm are joined as shown in figure. The temperature of the junction is
 - 10.8°C
 - 14.6°C
 - 16.4°C
 - 18.2°C



16. Three rods of same dimensions are arranged as shown in figure. They have thermal conductivities K_1 , K_2 and K_3 . The points P and Q are maintained at different temperatures. For the heat flow at the same rate along PRQ and PQ , which of the following option is correct?



- (a) $K_3 = \frac{1}{2}(K_1 + K_2)$
 (b) $K_3 = K_1 + K_2$
 (c) $K_3 = \frac{K_1 K_2}{K_1 + K_2}$
 (d) $K_3 = 2(K_1 + K_2)$

17. Two rods of same length and material transfer a given amount of heat in 12 s, when they are joined end-to-end. But when they are joined lengthwise, then they will transfer same heat in same conditions in

- (a) 24 s (b) 3 s
 (c) 1.5 s (d) 48 s

RADIATION

Radiation is the transfer of heat energy in form of electromagnetic waves. In this mode, transfer of heat takes place without heating the intervening medium.

All bodies even at lower temperature, emit energy in the form of the electromagnetic radiation. Heat energy is carried by infrared waves with wavelength greater than 750 nm.

Absorption, emission, transmission and reflection of heat radiations

When heat radiations are incident on a surface, then part of radiations is absorbed, part of it is reflected, part of it is emitted and part of radiation is transmitted as shown in figure.

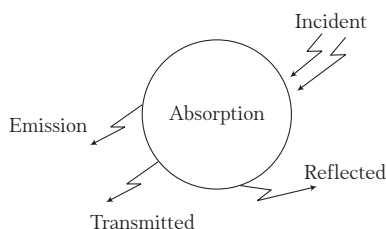


Fig. 16.11

Absorption

The process of absorbing the incident radiation by the surface of a body is called absorption. Due to absorption, temperature of surface of the body increases.

Absorptive power (a)

It is defined as the ratio of the radiant energy absorbed by a body in a given time to the total radiant energy incident on it in the same interval of time.

$$\text{Absorptive power, } a = \frac{\text{Energy absorbed } (Q_a)}{\text{Energy incident } (Q_i)}$$

It is a dimensionless quantity.

Spectral absorptive power (a_λ) The absorptive power a refers to radiations of all wavelengths (or the total energy) while the spectral absorptive power is the ratio of radiant energy absorbed by a surface to the radiant energy incident on it for a particular wavelength λ . It may have different values for different wavelengths for a given surface.

Let us take an example, suppose $a = 0.6$, $a_\lambda = 0.4$ for 1000 Å and $a_\lambda = 0.7$ for 2000 Å for a given surface. Then, it means that this surface will absorb only 60% of the total radiant energy incident on it. Similarly, it absorbs 40% of the energy incident on it corresponding to 1000 Å and 70% corresponding to 2000 Å.

The spectral absorptive power a_λ is related to absorptive power a through the relation

$$a = \int_0^\infty a_\lambda d\lambda$$

Emission

The process of emitting radiation by the surface of a body is called emission. Due to emission, temperature of surface of the body decreases.

Radiation energy emitted by a surface depends upon following factors

- (i) Wavelength of radiation
- (ii) Nature of the surface
- (iii) Surface area
- (iv) Temperature of the surface

Emissive power (e)

For a given surface, it is defined as the total radiant energy emitted per second per unit area of the surface.

$$\text{Emissive power, } e = \frac{1}{A} \frac{\Delta Q}{\Delta t}$$

SI unit of emissive power is $\text{Js}^{-1}\text{m}^{-2}$ or Wm^{-2} .

Object having lesser emissive power is called **bad emitter** and object having greater emissive power is called **good emitter**.

Spectral emissive power (e_λ) It is emissive power for a particular wavelength λ and is defined as the total amount of thermal energy emitted per unit time, per unit area of the body for a particular wavelength.

Thus,

$$e = \int_0^\infty e_\lambda d\lambda$$

Its SI unit is $\text{J/m}^2\text{s}$ or W/m^2 and dimensional formula is $[\text{MT}^{-3}]$.

Emissivity (ϵ)

The ratio of emissive power (e) of a body to the emissive power (E) of a black body (a body that absorbs all the radiations falling on it) at the same temperature is called emissivity. It is denoted by ϵ .

Thus, emissivity,
$$\epsilon = \frac{e}{E}$$

It is a dimensionless quantity.

- (i) For highly polished body, $\epsilon = 0$.
- (ii) For practical bodies, $0 < \epsilon < 1$.

Transmittance (Transmittivity)

The ratio of heat energy transmitted by a body in a given time to the total radiation energy incident on it in the same interval of time is called transmittance or transmissivity.

It is denoted by t .

$\therefore$
$$\text{Transmittance, } t = \frac{Q_{\text{transmitted}}}{Q_{\text{incident}}} = \frac{Q_t}{Q_i}$$

It is a dimensionless quantity.

Reflectivity

The ratio of heat energy reflected by a body in a given time to the total radiation energy incident on it in the same interval of time is called reflectivity.

It is denoted by r .

$\therefore$
$$\text{Reflectivity, } r = \frac{Q_{\text{reflected}}}{Q_{\text{incident}}} = \frac{Q_r}{Q_i}$$

It is a dimensionless quantity.

Relation between Absorptive Power (a), Transmittance (t) and Reflectivity (r)

$\therefore$
$$Q_a + Q_t + Q_r = Q$$

Dividing both sides by Q , we get

$$\frac{Q_a}{Q} + \frac{Q_t}{Q} + \frac{Q_r}{Q} = 1 \Rightarrow a + t + r = 1$$

For a perfect reflector, $a = t = 0$, $r = 1$.

For a perfect transmitter, $a = r = 0$, $t = 1$.

Black body radiation

A body that absorbs almost all the radiations falling on it is known as a black body. It emits the radiations at the fastest rate. The radiations emitted by a black body is known as black body radiation. The black body is also called as **ideal radiator**.

Perfectly black body

A body that completely absorbs the radiation of all the wavelengths incident on it is called perfectly black body.

Properties of perfectly black body Few properties are given below

- (i) It does not reflect any radiation.
 $\therefore$ Reflectivity, $r = 0$
- (ii) It does not transmit any radiation.
 $\therefore$ Transmittivity, $t = 0$
- (iii) A perfectly black body absorbs all radiations incident on it, the absorptive power of a perfectly black body is maximum and unity, *i.e.* $a = 1$.
- (iv) A perfectly black body does not reflect any wavelength, so it appears black, whatever be the colour of radiations incident on it.
- (v) When it is heated to a suitable high temperature, it emits radiation of all possible wavelengths.
e.g. Sun is at very high temperature ($\approx 6 \times 10^3 \text{ K}$), it emits all possible radiations, so it is an example of black body.
- (vi) For perfectly black body, $\epsilon = 1$.

Materials like black velvet or lamp black come close to being ideal black bodies, but the best practical realisation of an ideal black body is a small hole leading into a cavity, as this absorbs 98% of the radiation incident on them and called **ferry's black body**.

Kirchhoff's radiation law

According to this law, “the ratio of emissive power (e) to absorptive power (a) is same for all surfaces at the same temperature and is equal to emissive power (E) of a black body at that temperature.”

Then, $\left(\frac{e}{a}\right)_{\text{for any surface}} = \text{constant} = E$

Similarly, for a particular wavelength λ ,

$$\left(\frac{e_{\lambda}}{a_{\lambda}}\right)_{\text{for any body}} = E_{\lambda}$$

Here, E_{λ} = emissive power of black body.

Stefan's law

All objects radiate energy continuously in the form of electromagnetic waves. The rate at which an object radiates energy is proportional to the fourth power of its absolute temperature. This is known as the Stefan's law and is expressed in equation form as

$$P = \sigma A e T^4$$

Here, P is the power in watts (Js^{-1}) radiated by the object, A is the surface area in m^2 , T is the absolute temperature in kelvin (K), e is a fraction between 0 and 1 called the **emissivity of the object** and σ is a universal constant called **Stefan's constant**, which has the value

$$\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2} \text{K}^{-4}$$

This law is also known as **Stefan-Boltzmann law**.

Note For a black body (i.e. $e = 1$), net loss of energy per second per unit area, $E = \sigma T^4$.

Example 16.36 A black body at 227°C radiates heat at a rate of $7 \text{ cal cm}^{-2}\text{s}^{-1}$. At a temperature of 727°C , what will be the rate of heat radiated in the same units?

Sol. According to Stefan's law, $E = \sigma T^4$

$$\therefore 7 = \sigma (227 + 273)^4 = \sigma \times (500)^4$$

$$\text{and } x = \sigma (727 + 273)^4 = \sigma \times (1000)^4$$

$$\text{Hence, } \frac{x}{7} = \frac{(1000)^4}{(500)^4} = 16 \Rightarrow x = 16 \times 7 = 112 \text{ cal cm}^{-2}\text{s}^{-1}$$

Cooling by radiation

Consider a hot body at temperature T placed in an environment at a lower temperature T_0 . The body emits more radiation than it absorbs and cools down, while the surroundings absorb radiation from the body and warm up. The body is losing energy by emitting radiations at a rate,

$$P_1 = eA\sigma T^4$$

and is receiving energy by absorbing radiations at a rate,

$$P_2 = aA\sigma T_0^4$$

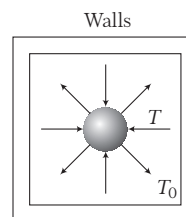


Fig. 16.12 Diagram showing cooling by radiation

Here, a is a pure number between 0 and 1 indicating the relative ability of the surface to absorb radiation from its surroundings. Note that, this a is different from the absorptive power a . In thermal equilibrium, both the body and the surrounding have the same temperature (say T_c) and

$$P_1 = P_2 \quad \text{or} \quad eA\sigma T_c^4 = aA\sigma T_c^4 \quad \text{or} \quad e = a$$

Thus, when $T > T_0$, the net rate of heat transfer from the body to the surroundings is $\frac{dQ}{dt} = eA\sigma (T^4 - T_0^4)$

$$\text{or} \quad mc \left(-\frac{dT}{dt} \right) = eA\sigma (T^4 - T_0^4) \quad (\because dQ = mcdT)$$

$$\text{Rate of cooling, } \left(-\frac{dT}{dt} \right) = \frac{eA\sigma}{mc} (T^4 - T_0^4)$$

$$\text{or} \quad \left(-\frac{dT}{dt} \right) \propto (T^4 - T_0^4)$$

So, we can say that rate of cooling of a body by radiation is proportional to the difference of fourth power of temperature of body and its environment.

Example 16.37 A cubical block of mass 1.0 kg and edge 5.0 cm is heated to 227°C . It is kept in an evacuated chamber maintained at 27°C . Assuming that the block emits radiation like a black body, find the rate at which the temperature of the block will decrease. Specific heat capacity of the material of the block is 400 J/kgK . (Take, $\sigma = 6 \times 10^{-8} \text{ Wm}^{-2} \text{K}^{-4}$.)

Sol. As, rate of cooling, $\frac{dT}{dt} = \frac{e\sigma A}{ms} (T^4 - T_0^4)$

Given, $e = 1$

$$\text{Area, } A = 6a^2 = 6 \times (5 \times 10^{-2})^2 = 150 \times 10^{-4} \text{ m}^2$$

Temperature, $T = 227 + 273 = 500 \text{ K}$

$$T_0 = 27 + 273 = 300 \text{ K}$$

Mass, $m = 1 \text{ kg}$ and $s = 400 \text{ J/kgK}$

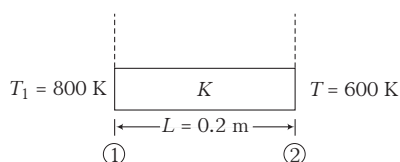
$$\therefore \frac{dT}{dt} = \frac{1 \times 6 \times 10^{-8} \times 150 \times 10^{-4} [(500)^4 - (300)^4]}{1 \times 400}$$

$$= 2.25 \times 10^{-12} \times (625 - 81) \times 10^8 = 0.12^\circ\text{C/s}$$

Example 16.38 One end of a rod of length 0.2 m is inserted in a furnace at 800 K. The sides of the rod are covered with an insulating material and the other end emits the radiation like a black body. The temperature of this end is 600 K in the steady state and the temperature of the surrounding air is 300 K. Assuming energy is lost only by radiation from the open end of the rod, find the thermal conductivity of the material of the rod.

(Stefan's constant, $\sigma = 6 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$)

Sol. According to the question,



Energy received per second at end (1),

$$Q_1 = \frac{KA(T_1 - T)}{L} \quad \dots(i)$$

Energy radiated per second at end (2),

$$Q_2 = \sigma A(T^4 - T_0^4) \quad \dots(ii)$$

In steady state, $Q_1 = Q_2$

$$\frac{KA(T_1 - T)}{L} = \sigma A(T^4 - T_0^4)$$

$$\begin{aligned} \therefore K &= \frac{\sigma L(T^4 - T_0^4)}{(T_1 - T)} \\ &= \frac{6 \times 10^{-8} \times 0.2 [(600)^4 - (300)^4]}{(800 - 600)} \\ &= \frac{1.2 \times 10^{-8} \times (6^4 - 3^4) \times 10^8}{200} \\ &= 6 \times 10^{-3} (1296 - 81) \\ &= 7.29 \text{ W/m}^\circ \text{C} \end{aligned}$$

Newton's law of cooling

It states that the rate of cooling of a body (R) is directly proportional to the temperature difference between the body and its surroundings, provided the temperature difference is small.

i.e. Rate of cooling (R) $\propto$ Temperature difference between the body and its surroundings

$$-\frac{dT}{dt} \propto (T - T_0)$$

Rate of cooling, (R) = $-\frac{dT}{dt} = k(T - T_0)$

where, k is a constant which depends upon surface area and nature of the surface. It can also be written as

$$\left(\frac{T_1 - T_2}{t} \right) = k \left[\left(\frac{T_1 + T_2}{2} \right) - T_0 \right]$$

Here, T_1 and T_2 are temperatures of the body in time interval t and T_0 is surrounding temperature.

Example 16.39 A body cools in 10 min from 60°C to 40°C.

What will be its temperature after next 10 min?

The temperature of the surroundings is 10°C.

Sol. According to Newton's law of cooling,

$$\left(\frac{T_1 - T_2}{t} \right) = k \left[\left(\frac{T_1 + T_2}{2} \right) - T_0 \right]$$

For the given conditions,

$$\frac{60 - 40}{10} = k \left[\left(\frac{60 + 40}{2} \right) - 10 \right] \quad \dots(i)$$

Let T be the temperature after next 10 min and now take $T_1 = 40^\circ \text{C}$, then

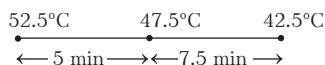
$$\frac{40 - T}{10} = k \left[\left(\frac{40 + T}{2} \right) - 10 \right] \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$T = 28^\circ \text{C}$$

Example 16.40 A body cools down from 52.5°C to 47.5°C in 5 min and to 42.5°C in 7.5 min. Find the temperature of the surroundings.

Sol. Here,



Using Newton's law of cooling,

$$\frac{T_1 - T_2}{t} = k \left[\left(\frac{T_1 + T_2}{2} \right) - T_0 \right]$$

$$\text{we get } \frac{52.5^\circ \text{C} - 47.5^\circ \text{C}}{5 \text{ min}} = k \left[\left(\frac{52.5^\circ \text{C} + 47.5^\circ \text{C}}{2} \right) - T_0 \right]$$

$$\text{or } \frac{5^\circ \text{C}}{5 \text{ min}} = k(50^\circ \text{C} - T_0) \quad \dots(i)$$

$$\text{and } \frac{47.5^\circ \text{C} - 42.5^\circ \text{C}}{7.5 \text{ min}} = k \left[\left(\frac{47.5^\circ \text{C} + 42.5^\circ \text{C}}{2} \right) - T_0 \right]$$

$$\text{or } \frac{5^\circ \text{C}}{7.5 \text{ min}} = k(45^\circ \text{C} - T_0) \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{50^\circ \text{C} - T_0}{45^\circ \text{C} - T_0} = \frac{7.5}{5} = \frac{3}{2} \quad \text{or } T_0 = 35^\circ \text{C}$$

Cooling curves

(i) Curve of $\log(T - T_0)$ versus time

$$\therefore \frac{dT}{dt} = -K(T - T_0)$$

$$\Rightarrow \int \frac{dT}{T - T_0} = - \int K dt$$

$$\Rightarrow \log_e (T - T_0) = -Kt + \log_e A$$

where, $\log_e A$ is a constant.

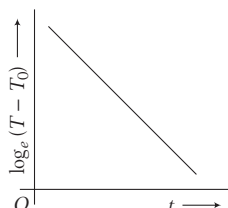


Fig. 16.13 Graph $\log_e(T - T_0)$ versus t

Thus **graph is a straight line with negative slope.**

It means rate of cooling of the body decreases with time.

(ii) **Curve of body temperature versus time**

We know that, temperature of the body at any time t is given by $T = T_0 + (T_i - T_0)e^{-\alpha t}$. From this expression, we can say that $T = T_i$ at $t = 0$ and $T = T_0$ at $t = \infty$, i.e. **temperature of the body varies exponentially with time from T_i to T_0 ($< T_i$)**. The temperature versus time graph is as shown in figure

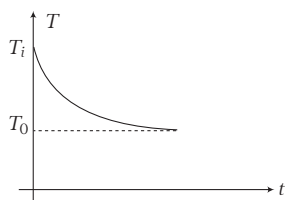


Fig. 16.14 Graph T versus t

(iii) **Curve of rate of cooling versus body temperature**

$$\text{Rate of cooling } (R) = - \frac{dT}{dt} = -K(T - T_0)$$

Graph of R versus T is a straight line with intercept $-KT_0$ as shown in figure.

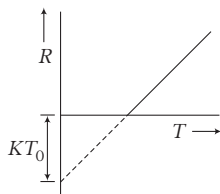


Fig. 16.15 Graph R versus T

(iv) **Curve of rate of cooling versus temperature difference between body temperature (T) and surrounding (T_0).**

$$\text{Rate of cooling } (R) = - \frac{dT}{dt} = K(T - T_0)$$

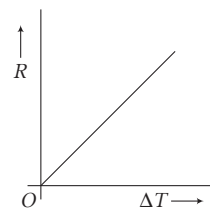


Fig. 16.16 Graph R versus ΔT

Clearly, $R \propto (T - T_0)$

i.e. Rate of cooling (R) $\propto$ temperature difference (ΔT)

Therefore, graph between R and ΔT is a straight line passing through origin as shown in figure.

Wien's displacement law

At ordinary temperatures (below 600°C), the thermal radiation emitted by a body is not visible, most of it is concentrated in wavelengths much longer than those of visible light as when temperature increases, wavelength shift to shorter wavelength. According to Wien's displacement law, **the wavelength corresponding to maximum emission of radiation decreases with increasing temperature.**

$$\lambda_{\max} \propto \frac{1}{T}$$

or

$$\lambda_{\max} T = b$$

Here, b is a constant called Wien's constant. The value of this constant in SI unit is $2.898 \times 10^{-3} \text{ m}\cdot\text{K}$. Thus, the graph between intensity I versus λ can be drawn as

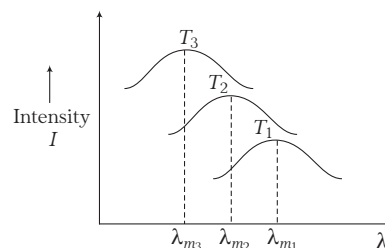


Fig. 16.17 Graph I versus λ

This law is used to find temperature of the sun and stars.

For example in the spectrum of the sun,

$$\lambda_m = 4753 \times 10^{-10} \text{ m}$$

$\therefore$ Temperature of the surface of sun,

$$T = \frac{b}{\lambda_m} = \frac{2.898 \times 10^{-3}}{4753 \times 10^{-10}} = 6097 \text{ K}$$

Example 16.41 The spectral energy distribution of the sun has maxima at $4753 \text{ \AA}$. Find the temperature of a star for which spectral distribution has maxima at $10350 \text{ \AA}$. (Take, temperature of sun is 6000 K)

Sol. According to Wien's displacement law,

$$\lambda_m \propto \frac{1}{T}$$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{T_1}{T_2} \Rightarrow T_2 = T_1 \frac{\lambda_1}{\lambda_2}$$

$$\Rightarrow T_2 = 6000 \times \frac{4753}{10350}$$

$$\Rightarrow T_2 = 2755.4 \text{ K}$$

Example 16.42 Two bodies A and B have thermal emissivities of 0.01 and 0.81, respectively. The outer surface areas of the two bodies are same. The two bodies emit total radiant power at the same rate. The wavelength λ_B corresponding to maximum spectral radiance from B is shifted from the wavelength corresponding to maximum spectral radiance in the radiation from A by $1.0 \mu\text{m}$. If the temperature of A is 5802 K , calculate

(i) the temperature of B (ii) and wavelength λ_B .

Sol. (i) Given, $P_A = P_B$

$$\therefore e_A \sigma A_A T_A^4 = e_B \sigma A_B T_B^4$$

$$\therefore T_B = \left(\frac{e_A}{e_B} \right)^{1/4} T_A \quad (\text{As, } A_A = A_B)$$

Substituting the values, we get

$$T_B = \left(\frac{0.01}{0.81} \right)^{1/4} (5802) = 1934 \text{ K}$$

(ii) According to Wien's displacement law,

$$\lambda_A T_A = \lambda_B T_B \Rightarrow \lambda_B = \left(\frac{5802}{1934} \right) \lambda_A$$

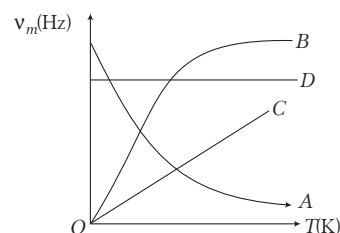
or

$$\lambda_B = 3\lambda_A$$

Also, $\lambda_B - \lambda_A = 1 \mu\text{m}$

$$\text{or } \lambda_B - \left(\frac{1}{3} \right) \lambda_B = 1 \mu\text{m} \text{ or } \lambda_B = 1.5 \mu\text{m}$$

Example 16.43 Which of the following is ν_m versus T graph for perfectly black body? Here, ν_m is the frequency of radiations with maximum intensity and T is the absolute temperature.



Sol. According to Wien's displacement law,

$$\lambda_m T = b = \text{Wien's constant}$$

If ν_m is the frequency corresponding to wavelength λ_m , then

$$\left(\frac{c}{\nu_m} \right) T = b \text{ or } \nu_m = \frac{c}{b} T$$

i.e.

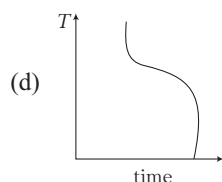
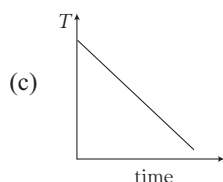
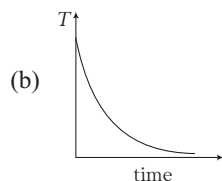
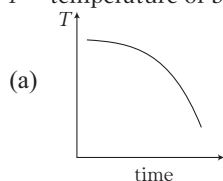
$$\nu_m \propto T$$

$\therefore \nu_m$ versus T graph is straight line inclined to time-axis shown by the curve C.

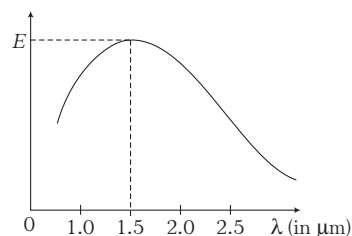
CHECK POINT 16.4

- A hot and a cold body are kept in vacuum separated from each other. Which of the following causes decrease in temperature of the hot body?
 - Radiation
 - Convection
 - Conduction
 - Temperature remains same
- Which of the following law states that "good absorbers of heat are good emitters"?
 - Stefan's law
 - Kirchhoff's law
 - Planck's law
 - Wien's law
- If e_λ and a_λ be the emissive and absorptive power of a body, then according to Kirchhoff's law, which is true (E_λ = emissive power of perfectly black body)
 - $e_\lambda = a_\lambda = E_\lambda$
 - $e_\lambda E_\lambda = a_\lambda$
 - $e_\lambda = a_\lambda E_\lambda$
 - $e_\lambda a_\lambda E_\lambda = \text{constant}$
- There is a black spot on a body. If the body is heated and carried in dark room, then it glows more. This can be explained on the basis of
 - Newton's law of cooling
 - Wien's law
 - Kirchhoff's law
 - Stefan's law
- In MKS system, Stefan's constant is denoted by σ . In CGS system, multiplying factor of σ will be
 - 1
 - 10^3
 - 10^5
 - 10^2
- If a black body radiates 10 calories per second at 227°C , then how much heat it will radiate at 727°C
 - 10 cal s^{-1}
 - 80 cal s^{-1}
 - 160 cal s^{-1}
 - None of these
- Two spherical black bodies of radii r_1 and r_2 and with surface temperatures T_1 and T_2 respectively radiate the same power. Then, $\frac{r_1}{r_2}$ must be equal to
 - $\left(\frac{T_1}{T_2} \right)^2$
 - $\left(\frac{T_2}{T_1} \right)^2$
 - $\left(\frac{T_1}{T_2} \right)^4$
 - $\left(\frac{T_2}{T_1} \right)^4$
- A sphere has a surface area of 1.0 m^2 and a temperature of 400 K and the power radiated from it is 150 W . Assuming the sphere is black body radiator, the power in kilowatt radiated when the area expands to 2.0 m^2 and the temperature changes to 800 K ,
 - 6.2
 - 9.6
 - 4.8
 - 16

9. Two spheres of the same material have radii 1 m and 4 m, temperatures 4000 K and 2000 K, respectively. Then, the ratio of energy radiated per second by the first sphere as compared to that by the second is
 (a) 4 : 1 (b) 2 : 1
 (c) 1 : 1 (d) 1 : 4
10. The area of the hole of heat furnace is 10^{-4} m^2 . It radiates $1.58 \times 10^5 \text{ cal}$ of heat per hour. If the emissivity of the furnace is 0.80, then its temperature is
 (a) 1500 K (b) 2000 K
 (c) 2500 K (d) 3000 K
11. If a body cools down from 80°C to 60°C in 10 min when the temperature of the surrounding is 30°C . Then, the temperature of the body after next 10 min will be
 (a) 50°C (b) 48°C
 (c) 30°C (d) None of these
12. A block of metal is heated to a temperature much higher than the room temperature and allowed to cool in a room free from air currents. Which of the following curves correctly represents the cooling? (T = temperature of block)



13. If wavelength of maximum intensity of radiation emitted by sun and moon are $0.5 \times 10^{-6} \text{ m}$ and 10^{-4} m respectively, then the ratio of their temperatures is
 (a) 2000 (b) 1000
 (c) 100 (d) 200
14. The maximum wavelength of radiations emitted at 900 K is $4 \mu\text{m}$. What will be the maximum wavelength of radiations emitted at 1200 K?
 (a) $3 \mu\text{m}$ (b) $0.3 \mu\text{m}$
 (c) $1 \mu\text{m}$ (d) None of these
15. The maximum energy in thermal radiation from a source occurs at wavelength $4000 \text{ \AA}$. The effective temperature of the source is approximately
 (a) 7000 K (b) 8000 K
 (c) 10^4 K (d) 10^6 K
16. The intensity of radiation emitted by the sun has its maximum value at a wavelength of 510 nm and that emitted by the North star has the maximum value at 350 nm. If these stars behave like black bodies, then the ratio of the surface temperature of the sun and North star is
 (a) 1.46 (b) 0.69
 (c) 1.21 (d) 0.83
17. In the figure, the distribution of energy density of the radiation emitted by a black body at a given temperature is as shown in figure. The possible temperature of the black body is



- (a) 1500 K (b) 2000 K (c) 2500 K (d) 3000 K

Chapter Exercises

(A) Taking it together

Assorted questions of the chapter for advanced level practice

- 1 The temperature of body is increased from 27°C to 127°C . The radiation emitted by it increases by a factor of

(a) $\frac{256}{81}$ (b) $\frac{15}{9}$ (c) $\frac{4}{5}$ (d) $\frac{12}{27}$

- 2 The calories of heat developed in 200 W heater in 7 min is estimated

(a) 15000 (b) 100 (c) 1000 (d) 20000

- 3 A sphere, a cube and a thin circular plate are heated to the same temperature. All are made of the same material and have the equal masses. If t_1 , t_2 and t_3 are the respective time taken by the sphere, cube and the circular plate in cooling down to a common temperature, then

(a) $t_1 > t_2 > t_3$ (b) $t_1 < t_2 < t_3$
(c) $t_2 > t_1 > t_3$ (d) $t_1 = t_2 = t_3$

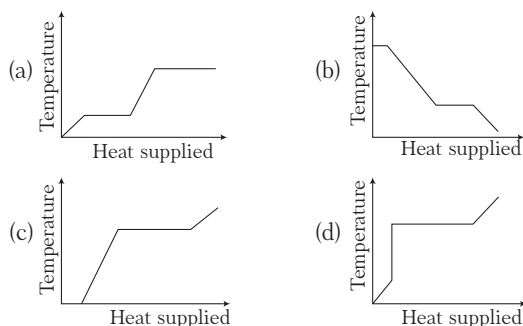
- 4 A spherical black body with radius 12 cm radiates 640 W power at 500 K. If the radius is halved and the temperature is doubled, the power radiated (in W) would be

(a) 5120 (b) 640
(c) 2560 (d) 1280

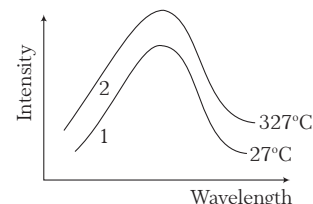
- 5 Rate of heat flow through a cylindrical rod is H_1 . Temperatures of ends of rod are T_1 and T_2 . If all the dimensions of rod become double and temperature difference remains same and rate of heat flow becomes H_2 . Then,

(a) $H_2 = 2H_1$ (b) $H_2 = H_1$ (c) $H_2 = \frac{H_1}{4}$ (d) $H_2 = 4H_1$

- 6 A block of ice at -10°C is slowly heated and converted to steam at 100°C . Which of the following curves represent the phenomenon qualitatively?



- 7 The spectrum of a black body at two temperatures 27°C and 327°C as shown in the figure. Let A_1 and A_2 be the areas under the two curves, respectively.



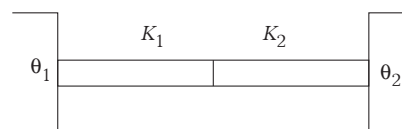
The value of A_2/A_1 is

(a) 1 : 16 (b) 4 : 1 (c) 2 : 1 (d) 16 : 1

- 8 If a body coated black at 600 K surrounded by atmosphere at 300 K has cooling rate r_0 , the same body at 900 K, surrounded by the same atmosphere, will have cooling rate equal to

(a) $\frac{16}{3} r_0$ (b) $\frac{8}{16} r_0$ (c) $16 r_0$ (d) $4 r_0$

- 9 Two identical rods are made of different materials whose thermal conductivities are K_1 and K_2 .



They are placed end-to-end between two heat reservoirs at temperatures θ_1 and θ_2 . The temperature of the junction of the rod is

(a) $\frac{\theta_1 + \theta_2}{2}$ (b) $\frac{K_1\theta_1 + K_2\theta_2}{K_1 + K_2}$
(c) $\frac{K_1\theta_2 + K_2\theta_1}{K_1 + K_2}$ (d) $\frac{K_1\theta_1 + K_2\theta_2}{|K_1 - K_2|}$

- 10 Certain amount of heat is given to 100 g of copper to increase its temperature by 21°C . If the same amount of heat is given to 50 g of water, then the rise in its temperature is

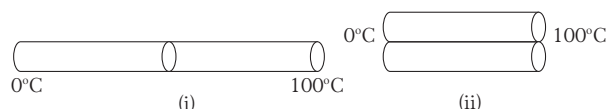
(Take, specific heat capacity of copper = $400 \text{ J kg}^{-1}\text{K}^{-1}$ and that for water = $4200 \text{ J kg}^{-1}\text{K}^{-1}$)

(a) 4°C (b) 5.25°C (c) 8°C (d) 6°C

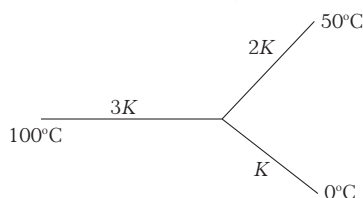
- 11 2 g of steam condenses when passed through 40 g of water initially at 25°C . The condensation of steam raises the temperature of water to 54.3°C . What is the latent heat of steam?

(a) 540 cal/g (b) 536 cal/g (c) 270 cal/g (d) 480 cal/g

- 12** Two identical straight rods of metal are welded end-to-end as shown in Fig. (i), 20 cal of heat flows through it in 4 min. If the rods are welded as shown in Fig. (ii), the same amount of heat will flow through the rods in



- (a) 1 min (b) 2 min
(c) 4 min (d) 16 min
- 13** Three rods of same dimensions have thermal conductivities $3K$, $2K$ and K . They are arranged as shown in figure, with their ends at 100°C , 50°C and 0°C . The temperature of their junction is

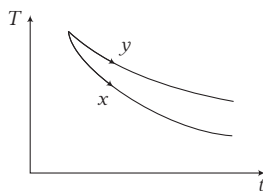


- (a) 75°C (b) $\frac{200}{3}^\circ\text{C}$ (c) 40°C (d) $\frac{100}{3}^\circ\text{C}$
- 14** A wall has two layers A and B each made of different materials. The layer A is 10 cm thick and B is 20 cm thick. The thermal conductivity of A is thrice that of B . Under thermal equilibrium, temperature difference across the wall is 35°C . The difference of temperature across the layer A is
- (a) 20°C (b) 10°C (c) 15°C (d) 5°C
- 15** A body cools from 50°C to 40°C in 5 min. The surrounding temperature is 20°C . In what further time (in minute) will it cool to 30°C ?
- (a) 5 (b) $\frac{15}{2}$ (c) $\frac{25}{3}$ (d) 10

- 16** A liquid cools from 50°C to 45°C in 5 min and from 45°C to 41.5°C in the next 5 min. The temperature of the surrounding is

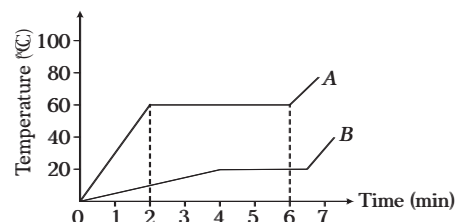
- (a) 27°C (b) 40.3°C
(c) 23.3°C (d) 33.3°C

- 17** The graph shown in the adjacent diagram, represents the variation of temperature (T) of two bodies, x and y having same surface area with time (t) due to the emission of radiation. Find the correct relation between the emissivity (e) and absorptivity (a) of two bodies.



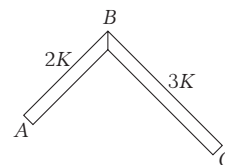
- (a) $e_x > e_y$ and $a_x < a_y$ (b) $e_x < e_y$ and $a_x > a_y$
(c) $e_x > e_y$ and $a_x > a_y$ (d) $e_x < e_y$ and $a_x < a_y$

- 18** Two substances A and B of equal mass m are heated at a uniform rate of 6 cal s^{-1} under similar conditions. A graph between temperature and time as shown in figure. Ratio of heat absorbed $\frac{H_A}{H_B}$ by them for complete fusion is



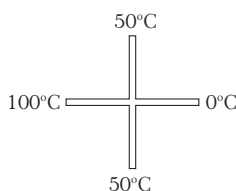
- (a) $\frac{9}{4}$ (b) $\frac{4}{9}$
(c) $\frac{24}{5}$ (d) $\frac{5}{8}$
- 19** Sun rays are allowed to fall on a lens of diameter 20 cm. They are then brought to focus on a calorimeter containing 20 g of ice. If the absorption by the lens is negligible, the time required to melt all the ice is (Take, solar constant = $1.9 \text{ cal min}^{-1} \text{ cm}^{-2}$ and $L = 80 \text{ cal g}^{-1}$)
- (a) 6.4 min (b) 3.2 min
(c) 7.2 min (d) 2.7 min

- 20** In the figure, ABC is a conducting rod whose lateral surfaces are insulated. The length of the section AB is one-half of that of BC , and the respective thermal conductivities of the two sections are given in the figure. If the ends A and C are maintained at 0°C and 70°C respectively, the temperature of junction B in the steady state is



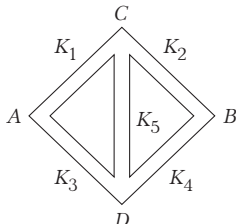
- (a) 30°C (b) 40°C
(c) 50°C (d) 60°C
- 21** 0.3 kg of hot coffee which is at 70°C , is poured into a cup of mass 0.12 kg. Find the final equilibrium temperature. (Take, room temperature is 20°C , $c_{\text{coffee}} = 4080 \text{ J/kg-K}$ and $c_{\text{cup}} = 1020 \text{ J/kg-K}$)
- (a) 45.5°C (b) 55.5°C
(c) 65.5°C (d) 40.5°C
- 22** Work done in converting one gram of ice at -10°C into steam at 100°C is
- (a) 3045 J (b) 6056 J
(c) 721 J (d) 616 J

- 23** Two similar rods are joined as shown in figure. Then, temperature of junction is (assume no heat loss through lateral surface of rod and temperatures at the ends are shown in steady state)



- (a) 50°C (b) 75°C (c) 66.6°C (d) 33.3°C

- 24** Five rods of same dimensions are arranged as shown in figure. They have thermal conductivities K_1, K_2, K_3, K_4 and K_5 . When points A and B are maintained at different temperatures, no heat flows through the central rod, if



- (a) $K_1 = K_4$ and $K_2 = K_3$ (b) $K_1 K_4 = K_2 K_3$
(c) $K_1 K_2 = K_3 K_4$ (d) None of these

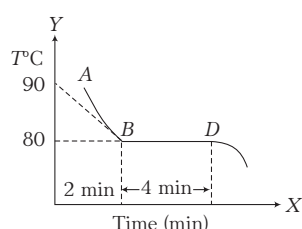
- 25** A calorimeter contains 10 g of water at 20°C. The temperature falls to 15°C in 10 min. When calorimeter contains 20 g of water at 20°C, it takes 15 min for the temperature to become 15°C. The water equivalent of the calorimeter is

- (a) 5 g (b) 10 g (c) 25 g (d) 50 g
- 26** 19 g of water at 30°C and 5 g of ice at -20°C are mixed together in a calorimeter. What is the final temperature of the mixture? (Take, specific heat of ice = 0.5 cal g⁻¹ °C⁻¹ and latent heat of fusion of ice = 80 cal g⁻¹)

- (a) 0°C (b) -5°C (c) 5°C (d) 10°C
- 27** Two rigid boxes containing different ideal gases are placed on a table. Box A contains one mole of nitrogen at temperature T_0 , while Box B contains one mole of helium at temperature $(7/3)T_0$. The boxes are then put into thermal contact with each other and heat flows between them until the gases reach a common final temperature (ignore the heat capacity of boxes). Then, the final temperature of the gases T_f , in terms of T_0 is

- (a) $T_f = \frac{7}{3}T_0$ (b) $T_f = \frac{3}{2}T_0$ (c) $T_f = \frac{5}{2}T_0$ (d) $T_f = \frac{3}{7}T_0$

- 28** The figure given below shows the cooling curve of pure wax material after heating. It cools from A to B and solidifies along BD . If L and c are respectively values of latent heat and the specific heat of the liquid wax, the ratio L/c is

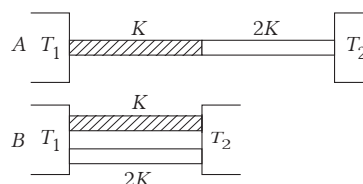


- (a) 40 (b) 80 (c) 100 (d) 20

- 29** Three conducting rods of same material and cross-section are shown in figure. Temperatures of A, D and C are maintained at 20°C, 90°C and 0°C. The ratio of lengths BD and BC , if there is no heat flow in AB is

- (a) 2/7 (b) 7/2 (c) 9/2 (d) 2/9

- 30** Two rods with the same dimensions have thermal conductivities in the ratio 1 : 2. They are arranged between heat reservoirs with the same temperature difference, in two different configurations, A and B . The rates of heat flow in A and B are I_A and I_B , respectively. The ratio $\frac{I_A}{I_B}$ is equal to



- (a) 1 : 2 (b) 1 : 3 (c) 2 : 5 (d) 2 : 9

- 31** Two identical conducting rods are first connected independently to two vessels, one containing water at 100°C and the other containing ice at 0°C. In the second case, the rods are joined end-to-end and connected to the same vessels. Let q_1 and q_2 gs⁻¹ be the rate of melting of ice in two cases, respectively.

The ratio of $\frac{q_1}{q_2}$ is

- (a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{4}{1}$ (d) $\frac{1}{4}$

- 32** Three rods of identical area of cross-section and made from the same metal form the sides of an isosceles triangle ABC , right angled at B . The points A and B are maintained at temperatures T and $\sqrt{2}T$, respectively. In the steady state, the temperature of the point C is T_C .

Assuming that only heat conduction takes place, T_C/T is equal to

- (a) $\frac{1}{(\sqrt{2}+1)}$ (b) $\frac{3}{(\sqrt{2}+1)}$ (c) $\frac{1}{2(2\sqrt{2}-1)}$ (d) $\frac{1}{\sqrt{3}(\sqrt{2}-1)}$

- 33** 2 kg of ice at -20°C is mixed with 5 kg of water at 20°C in an insulating vessel having a negligible heat capacity. Calculate the final mass of water remaining in the container. It is given that the specific heats of water and ice are 1 kcal kg⁻¹ °C⁻¹ and 0.5 kcal kg⁻¹ °C⁻¹ while the latent heat of fusion of ice is 80 kcal kg⁻¹.

- (a) 7 kg (b) 6 kg (c) 4 kg (d) 2 kg

- 34** 10 g of ice cubes at 0°C are released in a tumbler (water equivalent 55 g) at 40°C . Assuming that negligible heat is taken from the surroundings, the temperature of water in the tumbler becomes nearly (Take, $L = 80 \text{ cal/g}$)
 (a) 31°C (b) 22°C (c) 19°C (d) 15°C

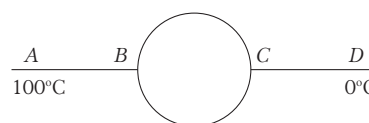
- 35** A cylindrical metallic rod in thermal contact with two reservoirs of heat at its two ends conducts an amount of heat Q in time t . The metallic rod is melted and the material is formed into a rod of half the radius of the original rod. What is the amount of heat conducted by the new rod when placed in thermal contact with the two reservoirs in time t ?
 (a) $\frac{Q}{4}$ (b) $\frac{Q}{16}$ (c) $2Q$ (d) $\frac{Q}{2}$

- 36** Equal masses of three liquids A , B and C have temperatures 10°C , 25°C and 40°C , respectively. If A and B are mixed, the mixture has a temperature of 15°C . If B and C are mixed, the mixture has a temperature of 30°C . If A and C are mixed, the mixture will have a temperature of
 (a) 16°C (b) 20°C (c) 25°C (d) 29°C

- 37** In an industrial process, 10 kg of water per hour is to be heated from 20°C to 80°C . To do this, steam at 150°C is passed from a boiler into a copper coil immersed in water. The steam condenses in the coil and is returned to the boiler as water at 90°C , how many kg of steam is required per hour?
 (Take, specific heat of steam = $1 \text{ cal per } g^\circ\text{C}$, latent heat of vaporisation = $540 \text{ cal } g^{-1}$)

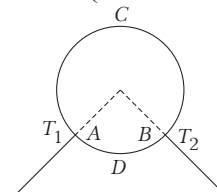
- (a) 1 g (b) 1 kg
 (c) 10 g (d) 10 kg

- 38** Two identical conducting rods AB and CD are connected to a circular conducting ring at two diametrically opposite points B and C . The radius of the ring is equal to the length of rods AB and CD . The area of cross-section, thermal conductivity of the rod and ring are equal. Points A and D are maintained at temperatures of 100°C and 0°C . Temperature at point C will be



- (a) 62°C (b) 37°C (c) 28°C (d) 45°C

- 39** A ring consisting of two parts ADB and ACB of same conductivity K carries an amount of heat H . The ADB part is now replaced with another metal keeping the temperatures T_1 and T_2 constant. The heat carried increases to $2H$. What should be the conductivity of the new ADB part? (Take, $\frac{ACB}{ADB} = 3$)



- (a) $\frac{7}{3}K$ (b) $2K$ (c) $\frac{5}{2}K$ (d) $3K$

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) These questions consist of two statements each printed as Assertion and Reason. While answering these questions you are required to choose any one of the following four responses.

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) If Assertion is correct but Reason is incorrect.
 (d) If Assertion is incorrect but Reason is correct.

- 1 Assertion** A body that is a good radiator is also a good absorber of radiation at a given wavelength.

Reason According to Kirchhoff's law, the absorptivity of a body is equal to its emissivity at a given wavelength.

- 2 Assertion** Temperature near the sea coast are moderate.

Reason Water has a high thermal conductivity.

- 3 Assertion** Blue star is at high temperature than red star.

Reason Wien's displacement law states that $T \propto \left(\frac{1}{\lambda_m}\right)$.

- 4 Assertion** Good conductors of electricity are also good conductors of heat.

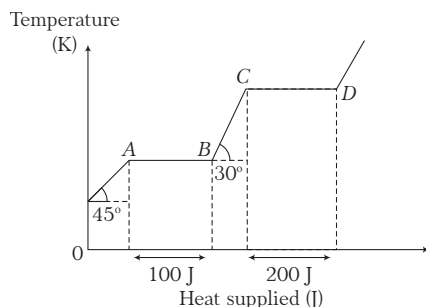
Reason In good conductors of electricity, there are large numbers of free electrons.

- 5 Assertion** Water vapours at 100°C will burn you more than water at 100°C .

Reason Heat required to convert total mass of any substance from one state to another state is called latent heat.

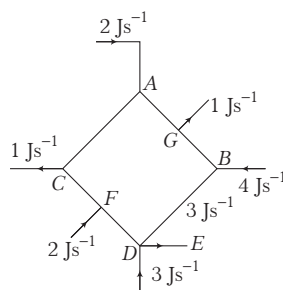
Statement based questions

- 1 The temperature change *versus* heat supplied curve is given for 1 kg of a solid block. Then, which of the following statement is correct?



- (a) Specific heat capacity of the solid is $2 \text{ J kg}^{-1} \text{ K}^{-1}$.
 (b) Specific heat capacity of liquid phase is $\sqrt{3} \text{ J kg}^{-1} \text{ K}^{-1}$.
 (c) Latent heat of vaporisation is 100 J kg^{-1} .
 (d) Latent heat of vaporisation is 200 J kg^{-1} .

- 2 The rate of flow of heat through 12 identical conductors made of same material as shown in the figure. Then, which of the following statement is correct?



- (a) The rate of flow of heat through rod DE is 8 Js^{-1} .
 (b) Junctions C and F are at the same temperature.
 (c) Junctions A and G are at the same temperature.
 (d) The rate of flow of heat through CF is 5 Js^{-1} .

- 3 Which of the following statement(s) is/are correct?

- I. For higher temperature, the peak emission wavelength of a black body shifts to lower wavelengths.
 II. Peak emission wavelength of a black body is proportional to the fourth power of temperature.

- (a) Only I (b) Only II
 (c) Both I and II (d) None of these

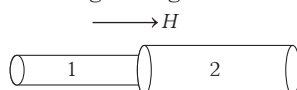
- 4 Which of the following statement(s) is/are correct?

- I. Gravity plays very important role in the process of natural convection.

- II. Convection mainly takes place in liquids and gases.

- (a) Only I (b) Only II
 (c) Both I and II (d) None of these

- 5 Two conduction rods of same material and same lengths are joined end-to-end as shown in figure. Heat current H is flowing through them as shown in figure.



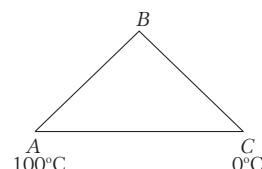
Which of the following statement(s) is/are correct relating to above information?

- I. Temperature difference across rod-1 is more than the temperature difference across rod-2.
 II. Thermal resistance of rod-1 is less compared to rod-2.

- (a) Only I (b) Only II
 (c) Both I and II (d) None of these

Match the columns

- 1 Three rods of equal length of same material are joined to form an equilateral triangle ABC as shown in figure. Area of cross-section of rod AB is S , of rod BC is $2S$ and that of AC is S . Then, match the following columns and mark the correct option from the codes given below.



Column I	Column II
(A) Temperature of junction B	(p) Greater than 50°C
(B) Heat current in AB	(q) Less than 50°C
(C) Heat current in BC	(r) Is equal to heat current in BC
	(s) Is $\frac{2}{3}$ times heat current in AC
	(t) None

Codes

- A B C A B C
 (a) p r s (b) q r s
 (c) p r t (d) None of these

- 2 Three liquids A, B and C having same specific heat and masses m , $2m$ and $3m$ have temperatures 20°C , 40°C and 60°C , respectively. Temperature of the mixture when

Column I	Column II
(A) A and B are mixed	(p) 33.3°C
(B) A and C are mixed	(q) 52°C
(C) B and C are mixed	(r) 50°C
(D) A, B and C all three are mixed	(s) 45°C
	(t) None

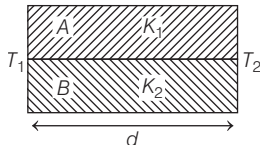
Match the following columns and mark the correct option from the codes given below.

Codes

- A B C D A B C D
 (a) p r t s (b) p r q t
 (c) s p r q (d) t p s r

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1 The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 ($r_1 = 1.5 r_2$) through 1 K are in the ratio
[NEET 2020]
(a) $\frac{9}{4}$ (b) $\frac{3}{2}$ (c) $\frac{5}{3}$ (d) $\frac{27}{8}$
- 2 Three stars A, B, C have surface temperatures T_A, T_B, T_C , respectively. Star A appears bluish, star B appears reddish and star C , appears yellowish. We can conclude that
[NEET 2020]
(a) $T_A > T_B > T_C$ (b) $T_B > T_C > T_A$
(c) $T_C > T_B > T_A$ (d) $T_A > T_C > T_B$
- 3 A deep rectangular pond of surface area A , containing water (density = ρ , specific heat capacity = s), is located in a region, where the outside air temperature is a steady value at -26°C . The thickness of the frozen ice layer in this pond at a certain instant is x .
Taking the thermal conductivity of ice as K and its specific latent heat of fusion as L , the rate of increase of the thickness of ice layer at this instant would be given by
[NEET (Odisha) 2019]
(a) $26K/\rho r(L - 4s)$ (b) $26K/(\rho x^2 - L)$
(c) $26K/(\rho xL)$ (d) $26K/\rho r(L + 4s)$
- 4 The unit of thermal conductivity is
[NEET (National) 2019]
(a) $\text{J m}^{-1} \text{K}^{-1}$ (b) W m K^{-1}
(c) $\text{W m}^{-1} \text{K}^{-1}$ (d) J m K^{-1}
- 5 An object kept in a large room having air temperature of 25°C takes 12 min to cool from 80°C to 70°C . The time taken to cool for the same object from 70°C to 60°C would be nearly
[NEET (Odisha) 2019]
(a) 10 min (b) 12 min (c) 20 min (d) 15 min
- 6 Calculate radiation power for sphere whose temperature is 227°C , radius 2 m and emissivity 0.8.
[AIIMS 2019]
(a) 142.5 kW (b) 1500 W
(c) 1255 W (d) 1575 W
- 7 If temperature of sun = 6000 K , radius of sun is $7.2 \times 10^5 \text{ km}$, radius of earth = 6000 km and distance between earth and sun = $15 \times 10^7 \text{ km}$. Find the intensity of light on earth (in watt).
[AIIMS 2019]
(a) 19.2×10^{16} (b) 12.2×10^{16}
(c) 18.3×10^{16} (d) 9.2×10^{16}
- 8 The power radiated by a black body is P and it radiates maximum energy at wavelength λ_0 . If the temperature of the black body is now changed, so that it radiates maximum energy at wavelength $\frac{3}{4} \lambda_0$, the power radiated by it becomes nP . The value of n is
[NEET 2018]
(a) $\frac{256}{81}$ (b) $\frac{4}{3}$ (c) $\frac{3}{4}$ (d) $\frac{81}{256}$
- 9 Two rods A and B of different materials are welded together as shown in figure. Their thermal conductivities are K_1 and K_2 . The thermal conductivity of the composite rod will be [NEET 2017]

(a) $\frac{K_1 + K_2}{2}$ (b) $\frac{3(K_1 + K_2)}{2}$
(c) $K_1 + K_2$ (d) $2(K_1 + K_2)$
- 10 A spherical black body with a radius of 12 cm radiates 450 W power at 500 K. If the radius were halved and the temperature doubled, the power radiated (in watt) would be
[NEET 2017]
(a) 225 (b) 450 (c) 1000 (d) 1800
- 11 A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is U_1 , at wavelength 500 nm is U_2 and that at 1000 nm is U_3 . Wien's constant, $b = 2.88 \times 10^6 \text{ nmK}$. Which of the following is correct?
[NEET 2016]
(a) $U_3 = 0$ (b) $U_1 > U_2$ (c) $U_2 > U_1$ (d) $U_1 = 0$
- 12 A piece of ice falls from a height h , so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is (Take, latent heat of ice is $3.4 \times 10^5 \text{ J/kg}$ and $g = 10 \text{ N/kg}$)
[NEET 2016]
(a) 544 km (b) 136 km (c) 68 km (d) 34 km
- 13 The two ends of a metal rod are maintained at temperatures 100°C and 110°C . The rate of heat flow in the rod is found to be 4 Js^{-1} . If the ends are maintained at temperatures 200°C and 210°C , the rate of heat flow will be
[CBSE AIPMT 2015]
(a) 44 Js^{-1} (b) 16.8 Js^{-1} (c) 8 Js^{-1} (d) 4 Js^{-1}

- 14** On observing light from three different stars P , Q and R , it was found that intensity of violet colour is maximum in the spectrum of P , the intensity of green colour is maximum in the spectrum of R and the intensity of red colour is maximum in the spectrum of Q . If T_P , T_Q and T_R are the respective absolute temperatures of P , Q and R , then it can be concluded from the above observations that

[CBSE AIPMT 2015]

- (a) $T_P > T_Q > T_R$ (b) $T_P > T_R > T_Q$
(c) $T_P < T_R < T_Q$ (d) $T_P < T_Q < T_R$

- 15** The black body spectrum of an object O_1 is such that its radiant intensity (*i.e.* intensity per unit wavelength interval) is maximum at a wavelength of 200 nm. Another object O_2 has the maximum radiant intensity at 600 nm. The ratio of power emitted per unit area by source O_1 to that of source O_2 is

[AIIMS 2015]

- (a) 1 : 81 (b) 1 : 9 (c) 9 : 1 (d) 81 : 1

- 16** Two plates of same area are placed in contact. Their thickness as well as thermal conductivities are in the ratio 2:3. The outer surface of one plate is maintained at 10 °C and that of the other at 0 °C. What is the temperature at the common surface?

[CG PMT 2015]

- (a) 0 °C (b) 2.5 °C (c) 5 °C (d) 6.5 °C

- 17** For maximum radiant energy from the moon, the corresponding wavelength is 14 micron. If Wien's constant is $b = 2892 \times 10^{-6}$ mK, then temperature of the moon is

[CG PMT 2015]

- (a) 207 K (b) 270 K (c) 207 °C (d) 270 °C

- 18** A black body with surface area 0.001 m^2 is heated upto a temperature 400 K and is suspended in a room temperature 300 K. The initial rate of loss of heat from the body to room is

[UP CPMT 2015]

- (a) 10 W (b) 1 W
(c) 0.1 W (d) 0.5 W

- 19** Identify the meaningless

[UP CPMT 2015]

- (a) temperature of the surface of a body
(b) the heat of a body
(c) the calorific value of fuel
(d) the heat transferred to a body

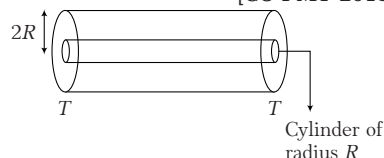
- 20** Water is boiled in flat bottom kettle placed on a stove. The area of the bottom is 3000 cm^2 and the thickness is 2 mm. If the amount of steam produced is 1 g min^{-1} . Then, the difference of temperature between inner and outer surfaces of the bottom is (K for the material of kettle is $0.5 \text{ cal}^\circ\text{C}^{-1}\text{s}^{-1}\text{cm}^{-1}$)

[UP CPMT 2015]

- (a) $0.12 \times 10^{-5} \text{ K}$ (b) $1.9 \times 10^{-3} \text{ K}$
(c) $1.3 \times 10^{-4} \text{ }^\circ\text{C}$ (d) $1.2 \times 10^{-3} \text{ K}$

- 21** A cylinder of radius R of thermal conductivity K_1 is surrounded by a cylindrical shell of inner radius R and outer radius $2R$ made of thermal conductivity K_2 . The two ends of the whole system are maintained at two different temperatures T_1 and T_2 ($T_1 < T_2$). The system is in steady state and there is no loss of heat across cylindrical surface. The electrical thermal conductivity of the system is

[CG PMT 2015, KCET 2013]



- (a) $K = K_1 + K_2$ (b) $K = \frac{K_1 K_2}{K_1 + K_2}$
(c) $K = \frac{K_1 + 3K_2}{4}$ (d) $K = \frac{3K_1 + K_2}{4}$

- 22** A piece of ice of mass 100 g and at temperature 0°C is put in 200 g of water at 25°C. How much ice will melt as the temperature of water reaches 0°C?

(Take, latent heat of fusion of ice = $3.4 \times 10^5 \text{ J kg}^{-1}$)

- (a) 128 g (b) 185.4 g [UK PMT 2015]
(c) 92.8 g (d) 61.8 g

- 23** A pan filled with hot food cools from 94°C to 86°C in 2 min. When the room temperature is at 20°C, the time taken to cool from 71°C to 69°C is

[EAMCET 2015]

- (a) 50 s (b) 40 s (c) 38 s (d) 42 s

- 24** Steam at 100°C is passed into 20 g of water at 10°C. When water acquires a temperature of 80°C, the mass of water present will be

(Take, specific heat of water = $1 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$ and latentheat of steam = 540 cal g^{-1}) [CBSE AIPMT 2014]

- (a) 24 g (b) 31.5 g (c) 42.5 g (d) 22.5 g

- 25** Certain quantity of water cools from 70°C to 60°C in the first 5 min and to 54°C in the next 5 min. The temperature of the surrounding is

[CBSE AIPMT 2014]

- (a) 45°C (b) 20°C (c) 42°C (d) 10°C

- 26** A 10 W electric heater is used to heat a container filled with 0.5 kg of water. It is found that the temperature of water and container rises by 3K in 15 min. The container is then emptied, dried and filled with 2 kg of oil. The same heater now raises the temperature of container oil system by 2K in 20 min. Assume there is no heat loss in the process and the specific heat of water is $4200 \text{ J kg}^{-1}\text{K}^{-1}$, the specific heat of oil in the same limit is equal to

- (a) 1.50×10^3 (b) 2.55×10^3 [WB JEE 2014]
(c) 3.00×10^3 (d) 2.10×10^3

- 27** A block of ice of mass 50 kg is sliding on a horizontal plane. It starts with speed 5 ms^{-1} and stops after moving through some distance. The mass of ice that has melted due to friction between the block and the surface is (Assuming that no energy is lost and latent heat of fusion of ice is 80 cal g^{-1} , $J = 4.2 \text{ J cal}^{-1}$) [WB JEE 2014]

(a) 2.86 g (b) 3.86 g
(c) 0.86 g (d) 1.86 g

- 28** Same quantity of ice is filled in each of the two metal containers P and Q having the same size, shape and wall thickness but made of different materials. The containers are kept in identical surroundings. The ice in P melts completely in time t_1 , whereas in Q takes a time t_2 . The ratio of thermal conductivities of the materials of P and Q is [WB JEE 2014]

(a) $t_2 : t_1$ (b) $t_1 : t_2$
(c) $t_1^2 : t_2^2$ (d) $t_2^2 : t_1^2$

- 29** Two identical rods are connected between two containers. One of them is at 100°C . If rods are connected in parallel, then the rate of melting of ice is $g_1 \text{ gs}^{-1}$. If they are connected in series, then the rate is $g_2 \text{ gs}^{-1}$. The ratio g_2/g_1 is [UK PMT 2014]
- (a) 2 (b) 4
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

- 30** A piece of iron is heated in a flame. If at first it becomes dull red, then becomes reddish, yellow and finally turns to white hot. The correct explanation for the above observation is possible by using [NEET 2013]

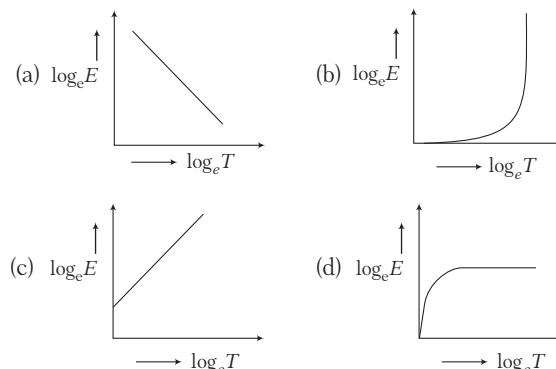
(a) Stefan's law (b) Wien's displacement law
(c) Kirchhoff's law (d) Newton's law of cooling

- 31** The temperature at the two ends A and B of a rod of length 25 cm and circular cross-section are 100°C and 0°C , respectively. In the steady state, the temperature at a point 10 cm from the end B is (Ignore loss of heat from curved surface of the rod) [EAMCET 2013]
- (a) 60°C (b) 80°C (c) 90°C (d) 40°C

- 32** Two stars A and B radiate maximum energy at the wavelength of 360 nm and 480 nm, respectively. Then, the ratio of the surface temperatures of A and B is [KCET 2013]
- (a) 3 : 4 (b) 81 : 256 (c) 4 : 3 (d) 256 : 81

- 33** In a room, where the temperature is 30°C , a body cools from 61°C to 59°C in 4 min. The time taken by the body to cool from 51°C to 49°C will be [MP PMT 2013]
- (a) 4 min (b) 5 min (c) 6 min (d) 8 min

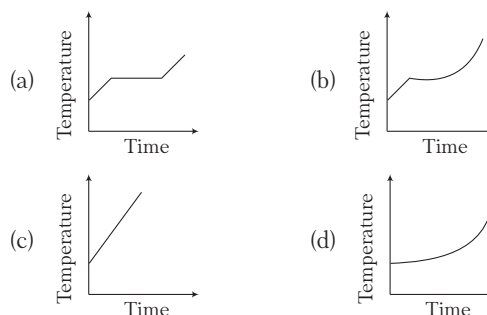
- 34** The emissive power of a body at temperature $T(\text{K})$ is E . Then, the graph between $\log_e E$ and $\log_e T$ will be [MP PMT 2013]



- 35** When 500 kg of water is heated from 20°C to 100°C , then the increase in mass of the water will be [UP CPMT 2013]
- (a) $3.2 \times 10^{-9} \text{ kg}$ (b) $1.87 \times 10^{-9} \text{ kg}$
(c) $0.96 \times 10^{-9} \text{ kg}$ (d) $2.8 \times 10^{-9} \text{ kg}$

- 36** If the radius of a star is R and it acts as a black body, what would be the temperature of the star, in which the rate of energy production is Q ? [CBSE AIPMT 2012]
- (a) $Q/4\pi R^2\sigma$ (b) $(Q/4\pi R^2\sigma)^{-1/2}$
(c) $(4\pi R^2Q/\sigma)^{1/4}$ (d) $(Q/4\pi R^2\sigma)^{1/4}$

- 37** Liquid oxygen at 50 K is heated to 300 K at constant pressure of 1 atm. The rate of heating is constant. Which one of the following graphs represents the variation of temperature with time? [CBSE AIPMT 2012]



- 38** A rod AB is 1 m long. The temperature of its one end A is maintained at 100°C and other end B at 10°C , the temperature at a distance of 60 cm from point B is [AIIMS 2012]
- (a) 64°C (b) 36°C (c) 46°C (d) 72°C

- 39** Two slabs A and B of different materials but of the same thickness are joined end-to-end to form a composite slab. The thermal conductivities of A and B are K_1 and K_2 , respectively. A steady temperature difference of 12°C is maintained across the

- composite slab. If $K_1 = \frac{K_2}{2}$, the temperature difference across slab A is [AIIMS 2012]
(a) 4°C (b) 6°C
(c) 8°C (d) 10°C
- 40** To cool the car engines in radiators, water is used because it is having [UP CPMT 2012]
(a) high value of specific heat (b) high density
(c) low surface tension (d) low density
- 41** A body cools from 60°C to 50°C in 10 min. If room temperature is 25°C . Temperature of body at the end of next 10 min, will be [UP CPMT 2012]
(a) 38.5°C (b) 40°C
(c) 45°C (d) 42.85°C
- 42** The temperature at which a black body of unit area losses its energy at the rate of 1 J s^{-1} will be [BHU 2012]
(a) 65°C (b) -65°C
(c) 65 K (d) 72 K
- 43** A piece of blue glass heated to a high temperature and a piece of red glass at room temperature are taken inside a dark room, then [AFMC 2012]
(a) the blue piece will look blue and red will look as usual
(b) red looks brighter red and blue looks ordinary blue
(c) blue shines brighter compared to the red piece
(d) Both the pieces will look equally red
- 44** The temperature gradient of a rod of 0.5 m long is 80°C m^{-1} . If the temperature of hotter end of the rod is 30°C , then the temperature of the cooler end is [JCECE 2012]
(a) 0°C (b) -10°C
(c) 10°C (d) 40°C
- 45** The thickness of a metallic plate is 0.4 cm. The temperature between its two surfaces is 20°C . The quantity of heat flowing per second is 50 cal from 5 cm^2 area. The coefficient of thermal conductivity (in CGS system) will be [BCECE (Mains) 2012]
(a) 0.2 (b) 0.3 (c) 0.4 (d) 0.5
- 46** Three identical rods A , B and C are placed end-to-end. A temperature difference is maintained between the free ends of A and C . The thermal conductivity of B is thrice that of C and half of that of A . The effective thermal conductivity of the system will be (K_A is the thermal conductivity of rod A) [KCET 2011]
(a) $\frac{1}{3} K_A$ (b) $3K_A$
(c) $2K_A$ (d) $\frac{2}{3} K_A$

ANSWERS

CHECK POINT 16.1

1. (c)	2. (a)	3. (b)	4. (a)	5. (d)	6. (b)	7. (a)	8. (d)	9. (b)
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CHECK POINT 16.2

1. (b)	2. (a)	3. (a)	4. (b)	5. (b)	6. (a)	7. (c)	8. (a)	9. (d)	10. (c)
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CHECK POINT 16.3

1. (d)	2. (c)	3. (a)	4. (d)	5. (a)	6. (d)	7. (a)	8. (c)	9. (d)	10. (b)
11. (b)	12. (a)	13. (b)	14. (c)	15. (c)	16. (c)	17. (b)			

CHECK POINT 16.4

1. (a)	2. (b)	3. (c)	4. (c)	5. (b)	6. (c)	7. (b)	8. (c)	9. (c)	10. (c)
11. (b)	12. (b)	13. (d)	14. (a)	15. (a)	16. (b)	17. (b)			

(A) Taking it together

1. (a)	2. (d)	3. (a)	4. (c)	5. (a)	6. (a)	7. (d)	8. (a)	9. (b)	10. (a)
11. (a)	12. (a)	13. (b)	14. (d)	15. (c)	16. (d)	17. (c)	18. (c)	19. (d)	20. (a)
21. (c)	22. (a)	23. (a)	24. (b)	25. (b)	26. (c)	27. (b)	28. (d)	29. (b)	30. (d)
31. (c)	32. (b)	33. (b)	34. (b)	35. (b)	36. (a)	37. (b)	38. (c)	39. (a)	

(B) Medical entrance special format questions

• Assertion and reason

1. (a)	2. (c)	3. (a)	4. (a)	5. (c)
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• Statement based questions

1. (d)	2. (b)	3. (a)	4. (c)	5. (a)
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• Match the columns

1. (d)	2. (b)
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(C) Medical entrances' gallery

1. (d)	2. (d)	3. (c)	4. (c)	5. (d)	6. (a)	7. (a)	8. (a)	9. (a)	10. (d)
11. (c)	12. (b)	13. (d)	14. (b)	15. (d)	16. (c)	17. (a)	18. (b)	19. (b)	20. (d)
21. (c)	22. (d)	23. (d)	24. (d)	25. (a)	26. (b)	27. (d)	28. (a)	29. (d)	30. (b)
31. (d)	32. (c)	33. (c)	34. (c)	35. (b)	36. (d)	37. (a)	38. (a)	39. (c)	40. (a)
41. (d)	42. (c)	43. (c)	44. (b)	45. (a)	46. (a)				

Hints & Explanations

• CHECK POINT 16.1

1 (c) $Q = mc\Delta T \Rightarrow c = \frac{Q}{m\Delta T}$

When $\Delta T = 0 \Rightarrow c = \infty$

2 (a) Heat energy gained,

$$Q = mc\Delta T = 5 \times (4.2 \times 10^3) \times (100 - 20) \\ = 1680 \times 10^3 \text{ J} = 1680 \text{ kJ}$$

3 (b) Heat required,

$$Q = m_1c_1\Delta T + m_2c_2\Delta T + m_3c_3\Delta T \\ = (2.4 \times 0.216 + 1.6 \times 0.0917 + 0.8 \times 0.0931) (60) \\ = 44.376 \text{ cal} \approx 44.44 \text{ cal}$$

4 (a) Same amount of heat is supplied to copper and water, so

$$m_Cc_C\Delta T_C = m_Wc_W\Delta T_W \\ \Rightarrow \Delta T_W = \frac{m_Cc_C(\Delta T)_C}{m_Wc_W} \\ = \frac{50 \times 10^{-3} \times 420 \times 10}{10 \times 10^{-3} \times 4200} = 5^\circ \text{C}$$

5 (d) Given, $n = 8 \text{ mol}$ and $\Delta T = 30^\circ \text{C}$

$$Q = nC_p\Delta T \Rightarrow Q = 8 \times \frac{5}{2} \times 8.31 \times 30 \approx 5000 \text{ J}$$

6 (b) Heat lost by hot water = Heat gained by cold water in beaker + Heat absorbed by beaker

$$\Rightarrow 440(92 - T) = 200 \times (T - 20) + 20 \times (T - 20) \\ \Rightarrow T = 68^\circ \text{C}$$

7 (a) Heat lost by water = Heat gained by ice.

$$(100 \times 1 \times T) = 100 \times 1 \times (100 - T)$$

or $T = 50^\circ \text{C}$

8 (d) Temperature of mixture,

$$\theta_{\text{mix}} = \frac{m_1c_1T_1 + m_2c_2T_2}{m_1c_1 + m_2c_2} = \frac{m \times c \times 2T + \frac{m}{2}(2c)T}{m \times c + \frac{m}{2}(2c)} = \frac{3}{2}T$$

9 (b) Density of water = 10^3 kg m^{-3} .

Let the final temperature of the mixture be t .

Assuming, no heat transfer to or from container.

$$\text{Heat lost by water} = 0.1 \times 10^3 \times S_{\text{water}} \times (80 - t)$$

$$\text{Heat gained by water} = 0.3 \times 10^3 \times S_{\text{water}} (t - 60)$$

According to principle of calorimetry,

Heat lost = Heat gain

$$0.1 \times 10^3 \times S_{\text{water}} \times (80 - t) = 0.3 \times 10^3 \times S_{\text{water}} \times (t - 60)$$

$$\Rightarrow 1 \times (80 - t) = 3 \times (t - 60)$$

$$\Rightarrow t = 65^\circ \text{C}$$

• CHECK POINT 16.2

1 (b) Heat lost in t second = mL

$$\text{or heat lost per second} = \frac{mL}{t}$$

This must be the heat supplied for keeping the substance in molten state per second

$$\therefore \frac{mL}{t} = P \quad \text{or} \quad L = \frac{Pt}{m}$$

2 (a) Heat liberated when 50 g of water at 60°C converts into water at 0°C .

$$Q = mc\Delta\theta = 50 \times 1 \times 60 = 3000 \text{ cal}$$

From this heat, mass of ice which can be melted is

$$m = \frac{Q}{L} = \frac{3000}{80} = 37.5 \text{ g}$$

i.e. whole ice is not melted and temperature of mixture will be 0°C .

3 (a) If m g ice melts, then Heat lost = Heat gain

$$80 \times 1 \times (30 - 0) = m \times 80 \Rightarrow m = 30 \text{ g}$$

4 (b) Heat taken = Heat given

$$\text{or} \quad mc\Delta\theta = mL$$

$$\therefore (200)(c)(100) = 25 \times 80$$

$$\text{or} \quad c = 0.1 \text{ cal g}^{-1}\text{C}^{-1}$$

5 (b) Heat released when 5 kg of water at 15°C converts into 0°C of water is, $Q = mc\Delta\theta = (5000)(1)(15) = 75000 \text{ cal}$. This

$$\text{heat can melt mass of ice } m = \frac{Q}{L} = \frac{75000}{80} = 937.5 \text{ g or}$$

0.9375 kg. Since, whole ice is not melting, temperature of mixture will be 0°C .

6 (a) Heat required,

$$Q_1 = (1.1 + 0.02) \times 10^3 \times (80 - 15) = 72800 \text{ cal}$$

Suppose m kg of steam is condensed and it also reaches to 80°C . Then heat rejected,

$$Q_2 = mL + mc\Delta\theta \\ = m(540 \times 10^3) + m \times 10^3 \times (100 - 80) \\ = m 560 \times 10^3$$

Equating Q_1 and Q_2 , we get

$$m (560 \times 10^3) = 72800$$

$$m = 0.13 \text{ kg}$$

7 (c) Heat absorbed, $mc\Delta T = \frac{1}{2} \left(\frac{1}{2} mv^2 \right)$

$$\therefore \Delta T = \frac{v^2}{4c} = \frac{(300)^2}{4 \times 150} = 150^\circ \text{C}$$

8 (a) From phase diagram, slope $\propto \frac{1}{(\text{heat capacity})}$

9 (d) As we know, $\Delta Q = mc\Delta T$

$$\therefore \text{Slope of inclined line } \frac{\Delta T}{\Delta Q} = \frac{1}{mc} = \frac{1}{\text{Heat capacity}}$$

10 (c) From $t = 0$ to $t = 1$ min,

$$\Delta Q = mc \Delta T = \frac{50}{1000} \times 0.6 \times 1000 \times (50 - 0) = 1500 \text{ cal}$$

From graph, boiling point (slope = 0) is 200°C .

CHECK POINT 16.3

1 (d) Radiation from sun and convection currents, are responsible for heating.

2 (c) Mud is bad conductor of heat. So it prevents the flow of heat between surrounding and inside.

3 (a) Snow contains air packets while ice does not enclose packets of air.

4 (d) $\frac{dQ}{dT} = -KA \frac{d\theta}{dx}$; when $K = \infty$, $\frac{d\theta}{dx} = 0$

i.e. θ is independent of x , i.e. constant or uniform.

5 (a) Thermal conductivity is independent of temperatures of the wall, it is a constant for the material, so it will remain unchanged.

6 (d) Rate of heat flow, $\frac{Q}{t} = \frac{KA\Delta\theta}{l} \Rightarrow K \propto \frac{1}{A}$

$$\Rightarrow \frac{K_A}{K_B} = \frac{A_B}{A_A} = \left(\frac{\pi r_B}{\pi r_A}\right)^2 = \frac{1}{4} \Rightarrow K_A = \frac{K_B}{4}$$

7 (a) Rate of heat flow, $\frac{dQ}{dt} = \frac{K(\pi r^2) \Delta\theta}{l}$

$$\Rightarrow \left(\frac{dQ}{dt}\right)_2 = \frac{K_2 \times r_2^2 \times l_1}{K_1 \times r_1^2 \times l_2} = \frac{1}{2} \times \frac{1}{4} \times \frac{2}{1}$$

$$\Rightarrow \left(\frac{dQ}{dt}\right)_2 = \frac{\left(\frac{dQ}{dt}\right)_1}{4} = \frac{4}{4} = 1$$

8 (c) Rate of heat transfer, $\frac{Q_1}{t} = \frac{KA(90 - 60)}{0.6} = 50 KA$

$$\text{and } \frac{Q_2}{t} = \frac{KA(150 - 110)}{0.8} = 50 KA$$

9 (d) $K_{eq} = \frac{2K_1K_2}{K_1 + K_2} = \frac{2(K)(2K)}{K + 2K} = \frac{4}{3} K$

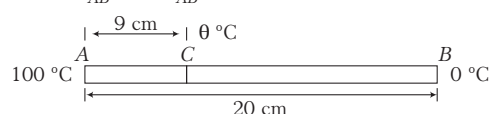
10 (b) $\therefore H_1 = H_2$

$$\therefore \left(\frac{\Delta T_1}{R_1}\right) = \left(\frac{\Delta T_2}{R_2}\right)$$

Given, $R_1 = R_2$, $\Delta T_1 = \Delta T_2$

$$\therefore \frac{l}{K_1 A_1} = \frac{l}{K_2 A_2} \text{ or } K_1 A_1 = K_2 A_2$$

11 (b) $H = \frac{100 - 0}{R_{AB}} = \frac{100}{R_{AB}}$



$$100 - \theta_C = H \cdot R_{AC} = \frac{100}{R_{AB}} \cdot R_{AC} = \frac{100 \times 9}{20} = 45$$

$$\therefore \theta_C = 100 - 45 = 55^\circ\text{C}$$

12. (a) $K_C > K_B$

$$\therefore R_C < R_B$$

Temperature difference across copper $(TD)_C = HR_C$... (i)

and temperature difference across brass $(TD)_B = HR_B$... (ii)

Heat current H will be same in both, as they are in series.

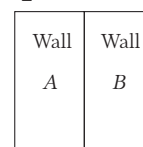
Since, $R_C < R_B$

$$\therefore (TD)_C < (TD)_B \quad [\text{From Eqs. (i) and (ii)}]$$

or temperature of junction is more than 50°C .

13 (b) $\therefore K_A = 2K_B$

$$\therefore R_A = \frac{R_B}{2} \quad \left(\because R = \frac{l}{KA}\right)$$



Suppose $R_A = R$, then $R_B = 2R$

$$\text{Heat current, } H = \frac{36}{R + 2R} = \frac{36}{3R} = \frac{12}{R}$$

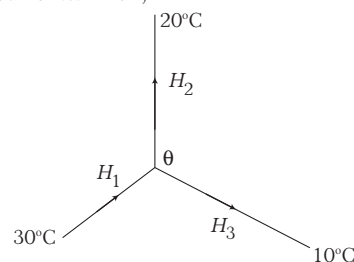
$$\therefore \text{Temperature difference across } A = H \times R_A = \frac{12}{R} \times R = 12^\circ\text{C}$$

14 (c) Let T be temperature of middle point C and in series, rate of heat flow is same.

$$\Rightarrow K(2A)(100 - T) = KA(T - 70)$$

$$\Rightarrow 200 - 2T = T - 70 \Rightarrow 3T = 270 \Rightarrow T = 90^\circ\text{C}$$

15 (c) Let θ be the temperature of junction and H_1 , H_2 and H_3 be the heat currents. Then,



$$H_1 = H_2 + H_3$$

$$\text{or } \frac{30 - \theta}{(30 / KA)} = \frac{\theta - 20}{(20 / KA)} + \frac{\theta - 10}{(10 / KA)}$$

$$\text{or } 2(30 - \theta) = 3(\theta - 20) + 3(\theta - 10)$$

$$\text{or } \theta = 16.36^\circ\text{C} \approx 16.4^\circ\text{C}$$

16 (c) For the heat flow at the same rate along PRQ and PQ ,

$$R_{PR} + R_{RQ} = R_{PQ}$$

$$\therefore \frac{l}{K_1 A} + \frac{l}{K_2 A} = \frac{l}{K_3 A} \text{ or } K_3 = \frac{K_1 K_2}{K_1 + K_2}$$

17 (b) Let the heat transferred be Q .

When rods are joined end-to-end (series). Heat transferred by each rod,

$$Q = \frac{K(A) \Delta\theta}{2l} \times 12 \quad \dots(i)$$

When rods are joined lengthwise (parallel),

$$Q = \frac{K2A\Delta\theta}{l} t \quad \dots(ii)$$

As the same amount of heat is transferred

$\therefore$ From Eqs. (i) and (ii), we get $t = 3s$

• CHECK POINT 16.4

- 1 (a) Radiation occurs without any medium, hence the cause involved is radiation.

- 3 (c) According to Kirchhoff's law, the ratio of emissive power to absorptive power is same for all bodies, which is equal to the emissive power of a perfectly black body, i.e.

$$\left(\frac{e}{a}\right)_{\text{body}} = E_{\text{black body}}$$

For a particular wavelength,

$$\left(\frac{e_{\lambda}}{a_{\lambda}}\right)_{\text{body}} = (E_{\lambda})_{\text{black body}} \Rightarrow e_{\lambda} = a_{\lambda} E_{\lambda}$$

- 4 (c) According to Kirchhoff's law, a good emitter is also a good absorber.

5 (b) In MKS system, unit of σ is $\frac{\text{J}}{\text{m}^2 \times \text{s} \times \text{K}^4}$

$$\Rightarrow \frac{\text{J}}{\text{m}^2 \times \text{s} \times \text{K}^4} = \frac{10^7 \text{ erg}}{10^4 \text{ cm}^2 \times \text{s} \times \text{K}^4} = 10^3 \frac{\text{erg}}{\text{cm}^2 \times \text{s} \times \text{K}^4}$$

- 6 (c) As, $E \propto T^4$,

$$T_1 = 500 \text{ K and } T_2 = 1000 \text{ K} \Rightarrow T_2 = 2T_1$$

$$\therefore E_2 = (2)^4 E_1 = 16E_1 = 160 \text{ cal s}^{-1}$$

- 7 (b) According to Stefan's law, if $P_1 = P_2$

$$4\pi r_1^2 T_1^4 = 4\pi r_2^2 T_2^4$$

$$\therefore \frac{r_1}{r_2} = \frac{T_2}{T_1}$$

- 8 (c) $\therefore P \propto AT^4$

Area and temperature both are doubled. Hence, power will become $(2)^5$ or 32 times.

$$\therefore P' = 32P = \frac{32 \times 150}{1000} \text{ kW} = 4.8 \text{ kW}$$

9 (c) $\frac{P_1}{P_2} = \frac{A_1 T_1^4}{A_2 T_2^4} = \frac{4\pi (1)^2 (4000)^4}{4\pi (4)^2 (2000)^4} = \frac{1}{1}$ or 1:1

- 10 (c) According to Stefan's law, $E = \sigma \epsilon AT^4$

$$\Rightarrow \frac{1.58 \times 10^5 \times 4.2}{60 \times 60} = 5.67 \times 10^{-8} \times 10^{-4} \times 0.8 \times T^4$$

Therefore, $T = 2500 \text{ K}$

11 (b) Applying $\frac{\theta_1 - \theta_2}{t} = \alpha \left[\frac{\theta_1 + \theta_2}{2} - \theta_0 \right]$, where α is constant

$\therefore$ We get

$$\frac{80 - 60}{10} = \alpha \left[\frac{80 + 60}{2} - 30 \right] \quad \dots(i)$$

$$\frac{60 - \theta}{10} = \alpha \left[\frac{60 + \theta}{2} - 30 \right] \quad \dots(ii)$$

Solving these two equations, we get

$$\theta = 48^\circ \text{C}$$

- 13 (d) According to Wien's law, $\lambda_m T = \text{constant}$

$$\therefore (\lambda_m T)_s = (\lambda_m T)_m$$

$$\text{or } \frac{T_s}{T_m} = \frac{(\lambda_m)_m}{(\lambda_m)_s} = \frac{10^{-4}}{0.5 \times 10^{-6}} = 200$$

14 (a) As, $\frac{\lambda_2}{\lambda_1} = \frac{T_1}{T_2} \Rightarrow \lambda_2 = \frac{T_1}{T_2} \times \lambda_1 = \frac{900}{1200} \times 4 = 3 \mu\text{m}$

- 15 (a) Applying Wien's law, we get

$$\lambda_m = \frac{b}{T}$$

$$\Rightarrow T = \frac{b}{\lambda_m} = \frac{2.89 \times 10^{-3}}{4000 \times 10^{-10}} = 7225 \text{ K} \approx 7000 \text{ K}$$

- 16 (b) From Wien's law,

$$\frac{T_S}{T_N} = \frac{(\lambda_N)_{\text{max}}}{(\lambda_S)_{\text{max}}} = \frac{350}{510} = 0.69$$

- 17 (b) $\lambda_m T = b$, where $b = 2.89 \times 10^{-3} \text{ mK}$

$$\Rightarrow T = \frac{b}{\lambda_m} = \frac{2.89 \times 10^{-3}}{1.5 \times 10^{-6}} \approx 2000 \text{ K}$$

(A) Taking it together

- 1 (a) $E \propto T^4$, $T_1 = 300 \text{ K}$, $T_2 = 400 \text{ K}$

$$\therefore \frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{400}{300}\right)^4 = \frac{256}{81}$$

2 (d) $Q = \frac{200 \times 7 \times 60}{4.18} \text{ cal}$

$$= 20095.7 \approx 20000 \text{ cal}$$

- 3 (a) A body having maximum surface area will cool at fastest rate, hence $t_1 > t_2 > t_3$.

4 (c) Ratio of powers, $\frac{P_1}{P_2} = \frac{A_1 T_1^4}{A_2 T_2^4}$

$$\Rightarrow P_2 = \frac{A_2}{A_1} \left(\frac{T_2}{T_1}\right)^4 P_1 = \left(\frac{1}{2}\right)^2 (2)^4 (640) = 2560 \text{ W}$$

5 (a) Rate of heat flow, $H = \frac{Q}{t} = \frac{KA(T_1 - T_2)}{l}$

$$\text{or } H \propto \frac{A}{l} \text{ or } H_2 = 2H_1$$

- 6 (a) Temperature first increases from -10°C to 0°C . Then, it remains constant at 0°C till whole ice is melted. Then, it increases from 0°C to 100°C . Again, it remains constant at 100°C till whole water converts into steam.

- 7 (d) Area $\propto$ Total energy radiated $\propto T^4$

$$T_1 = 300 \text{ K and } T_2 = 600 \text{ K}$$

Temperature is doubled. Therefore, area will become 16 times.

- 8 (a) Cooling rate $\propto T^4 - T_0^4$

$$r = \left[\frac{(900)^4 - (300)^4}{(600)^4 - (300)^4} \right] r_0 = \frac{16}{3} r_0$$

- 9 (b) Heat currents in the two rods will be same.

$$\therefore H_1 = H_2$$

$$\text{or } \frac{\theta_1 - \theta}{(l/K_1A)} = \frac{\theta - \theta_2}{(l/K_2A)} \quad (\text{where, } \theta = \text{temperature of junction})$$

$$\text{or } K_1(\theta_1 - \theta) = K_2(\theta - \theta_2)$$

$$\therefore \theta = \frac{K_1\theta_1 + K_2\theta_2}{K_1 + K_2}$$

- 10 (a) The amount of heat supplied is given by the relation,

$$Q = mc\Delta T$$

$$\text{Here, } m = 100 \text{ g} = 0.1 \text{ kg,}$$

$$c = 400 \text{ J/kg-K,}$$

$$\Delta T = 21^\circ\text{C or } 21\text{ K for copper.}$$

$$\text{Thus, } Q = 0.1 \times 400 \times 21 = 840 \text{ J}$$

$$\text{Hence, } 840 = 0.05 \times 4200 \times \Delta T$$

$$\Rightarrow \Delta T = 4 \text{ K or } 4^\circ\text{C for water}$$

- 11 (a) Let L be the latent heat, then using principle of calorimetry,

$$2L + 2(100 - 54.3) = 40 \times (54.3 - 25)$$

$$\Rightarrow L = 540.3 \text{ cal/g}$$

- 12 (a) Rate of heat flow, $\frac{Q}{t} = \frac{KA\Delta T}{l} = \frac{\Delta T}{(l/KA)} = \frac{\Delta T}{R}$

$$(\text{where, } R = \text{thermal resistance})$$

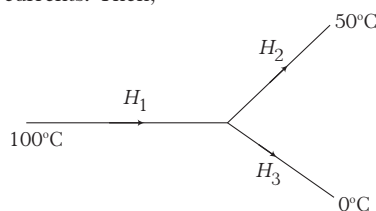
$$\Rightarrow t \propto R \quad (\because Q \text{ and } \Delta T \text{ are same})$$

$$\Rightarrow \frac{t_p}{t_s} = \frac{R_p}{R_s} \quad (\text{Series resistance, } R_s = R_1 + R_2)$$

$$\text{and parallel resistance, } R_p = \frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{R/2}{2R} = \frac{1}{4} \Rightarrow t_p = \frac{t_s}{4} = \frac{4}{4} = 1 \text{ min}$$

- 13 (b) Suppose T be the temperature of junction. H_1 , H_2 and H_3 are heat currents. Then,



$$H_1 = H_2 + H_3$$

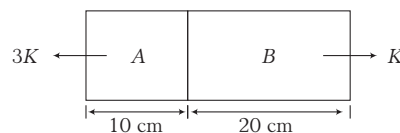
$$\text{or } \frac{100 - T}{(l/3KA)} = \frac{T - 50}{(l/2KA)} + \frac{T - 0}{(l/KA)}$$

$$3(100 - T) = 2(T - 50) + T$$

On solving, we get

$$T = \frac{200}{3}^\circ\text{C}$$

- 14 (d) Here, $R_A = \frac{10}{3KA}$ $\left(\because R = \frac{l}{KA} \right)$



$$\text{and } R_B = \frac{20}{KA} \therefore \frac{R_A}{R_B} = \frac{1}{6}$$

$$\text{So let } R_A = R, \text{ then } R_B = 6R$$

$$\text{Heat current, } H = \frac{\text{Temperature difference}}{\text{Total resistance}} = \frac{35}{7R} = \frac{5}{R}$$

Now, temperature difference across A

$$= H \times R_A = \frac{5}{R} \times R = 5^\circ\text{C}$$

- 15 (c) Applying, $\frac{T_1 - T_2}{t} = \alpha \left(\frac{T_1 + T_2}{2} - T_0 \right)$, we get

$$\frac{50 - 40}{5} = \alpha \left[\frac{50 + 40}{2} - 20 \right] \quad \dots(i)$$

$$\text{and } \frac{40 - 30}{t} = \alpha \left[\frac{40 + 30}{2} - 20 \right] \quad \dots(ii)$$

Solving these two equations, we get

$$t = \frac{25}{3} \text{ min}$$

- 16 (d) Applying $\frac{\theta_1 - \theta_2}{t} = \alpha \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$, we get

(where, α is constant)

$$\frac{50 - 45}{5} = \alpha \left(\frac{50 + 45}{2} - \theta_0 \right) \quad \dots(i)$$

$$\text{and } \frac{45 - 41.5}{5} = \alpha \left(\frac{45 + 41.5}{2} - \theta_0 \right) \quad \dots(ii)$$

Solving these two equations, we get

$$\theta_0 = \text{temperature of surrounding} = 33.3^\circ\text{C}$$

- 17 (c) Rate of cooling, $\left(-\frac{dT}{dt} \right) \propto \text{emissivity } (e)$

$$\text{From graph, } \left(-\frac{dT}{dt} \right)_x > \left(-\frac{dT}{dt} \right)_y$$

$$\Rightarrow e_x > e_y$$

Further emissivity (e) $\propto$ absorptive power (a)

$$\Rightarrow a_x > a_y \quad (\because \text{Good absorbers are good emitters})$$

- 18 (c) From given curve,

melting point for substance A = 60°C

and melting point for substance B = 20°C

Time taken by A for fusion = $(6 - 2) = 4 \text{ min}$

Time taken by B for fusion = $(6.5 - 4) = 2.5 \text{ min}$

$$\text{Then, } \frac{H_A}{H_B} = \frac{6 \times 4 \times 60}{6 \times 2.5 \times 20} = \frac{24}{5}$$

- 19 (d) Heat required to melt whole ice is

$$Q = mL = 20 \times 80 = 1600 \text{ cal}$$

Energy incident per minute is

$$P = 1.9 \times \frac{\pi}{4} \times d^2 = 1.9 \times \frac{\pi}{4} \times (20)^2 = 597 \text{ cal min}^{-1}$$

Therefore, time required to melt whole ice is $t = \frac{Q}{P} \approx 2.7 \text{ min}$

- 20 (a) Heat currents in both the rods are equal.

$$\text{or } H_{CB} = H_{BA}$$

$$\text{or } \frac{\theta_C - \theta_B}{(2l / 3KA)} = \frac{\theta_B - \theta_A}{(l / 2KA)}$$

$$\therefore \frac{3}{2} (70 - \theta_B) = 2(\theta_B - 0)$$

$$\text{or } \theta_B = 30^\circ\text{C}$$

- 21 (c) Let T be the final equilibrium temperature.

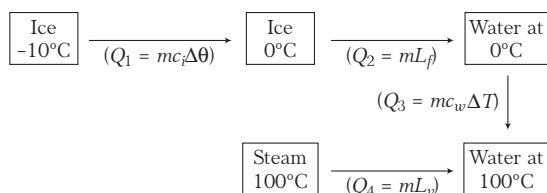
Heat lost by coffee = Heat gained by cup

$$0.3 \times c_{\text{coffee}} \times (70 - T) = 0.12 \times c_{\text{cup}} \times (T - 20)$$

$$\Rightarrow 0.3 \times 4080 \times (70 - T) = 0.12 \times 1020 \times (T - 20)$$

$$\Rightarrow 4 \times 70 - 4T = 0.4T - 8 \Rightarrow T = \frac{288}{4.4} = 65.5^\circ\text{C}$$

- 22 (a) Ice (-10°C) converts into steam as follows



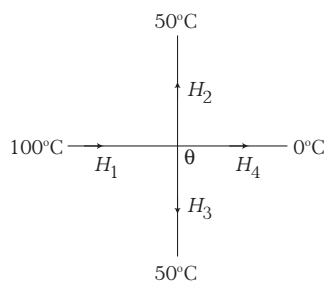
(Here, c_i = specific heat of ice, c_w = specific heat of water)

Total heat required, $Q = Q_1 + Q_2 + Q_3 + Q_4$

$$\Rightarrow Q = 1 \times 0.5(10) + 1 \times 80 + 1 \times 1 \times (100 - 0) + 1 \times 540 = 725 \text{ cal}$$

Hence, work done, $W = JQ = 4.2 \times 725 = 3045 \text{ J}$

- 23 (a) Let θ be the temperature of junction.



H_1, H_2, H_3 and H_4 be the heat currents. Then,

$$H_1 = H_2 + H_3 + H_4$$

$$\text{or } \frac{100 - \theta}{R} = \frac{\theta - 50}{R} + \frac{\theta - 50}{R} + \frac{\theta - 0}{R}$$

(where, R = thermal resistance of each part of two rods)

Solving the above equations, we get

$$\theta = 50^\circ\text{C}$$

- 24 (b) For no current flow between C and D ,

$$\left(\frac{Q}{t}\right)_{AC} = \left(\frac{Q}{t}\right)_{CB} \Rightarrow \frac{K_1 A (\theta_A - \theta_C)}{l} = \frac{K_2 A (\theta_C - \theta_B)}{l} \Rightarrow \frac{\theta_A - \theta_C}{\theta_C - \theta_B} = \frac{K_2}{K_1} \dots(i)$$

$$\text{Also, } \left(\frac{Q}{t}\right)_{AD} = \left(\frac{Q}{t}\right)_{DB} \Rightarrow \frac{K_3 A (\theta_A - \theta_D)}{l} = \frac{K_4 A (\theta_D - \theta_B)}{l} \Rightarrow \frac{\theta_A - \theta_D}{\theta_D - \theta_B} = \frac{K_4}{K_3} \dots(ii)$$

It is given that $\theta_C = \theta_D$, hence from Eqs. (i) and (ii), we get

$$\frac{K_2}{K_1} = \frac{K_4}{K_3} \Rightarrow K_1 K_4 = K_2 K_3$$

- 25 (b) Let water equivalent be x .

$$\therefore (10 + x) c \frac{\Delta T}{\Delta t} = -k \left(\frac{T_1 + T_2}{2} - T_0 \right)$$

where, c is the specific heat of water,

ΔT = change in temperature

and Δt = time at which temperature falls

$$\therefore (10 + x) c \frac{(20 - 15)}{10} = -k \left(\frac{20 + 15}{2} - T_0 \right) \dots(i)$$

$$\text{and } (20 + x) c \frac{(20 - 15)}{15} = -k \left(\frac{20 + 15}{2} - T_0 \right) \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{10 + x}{10} = \frac{20 + x}{15} \Rightarrow 150 + 15x = 200 + 10x \Rightarrow 5x = 50 \Rightarrow x = 10 \text{ g}$$

- 26 (c) Here, specific heat of ice, $c_{\text{ice}} = 0.5 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$

Specific heat of water, $c_{\text{water}} = 1 \text{ cal g}^{-1} \text{ } ^\circ\text{C}^{-1}$

Latent heat of fusion of ice, $L_{\text{ice}} = 80 \text{ cal g}^{-1}$

Here, ice will absorb heat while hot water will release it.

Let T be the final temperature of the mixture.

Assuming water equivalent of calorimeter to be neglected.

$$\text{Heat given by water, } Q_1 = m_{\text{water}} c_{\text{water}} \Delta T = 19 \times 1 \times (30 - T) = 570 - 19T \dots(i)$$

Heat absorbed by ice,

$$Q_2 = m_{\text{ice}} \times c_{\text{ice}} \times [0 - (-20)] + m_{\text{ice}} \times L_{f \text{ ice}} + m_{\text{ice}} \times c_{\text{water}} \times (T - 0) = 5 \times 0.5 \times 20 + 5 \times 80 + 5 \times 1 \times T = 450 + 5T \dots(ii)$$

According to principle of calorimetry, $Q_1 = Q_2$

$$570 - 19T = 450 + 5T \Rightarrow T = \frac{120}{24} = 5^\circ\text{C}$$

- 27 (b) When two gases are mixed together, then

Heat lost by the helium gas = Heat gained by the nitrogen gas

$$n_B \times (c_v)_{\text{He}} \times \left(\frac{7}{3} T_0 - T_f \right) = n_A \times (c_v)_{\text{N}_2} \times (T_f - T_0)$$

Box A	Box B
1 mole N_2	1 mole He
Temperature = T_0	Temperature = $\frac{7}{3}T_0$

$$\Rightarrow 1 \times \frac{3}{2}R \times \left(\frac{7}{3}T_0 - T_f\right) = 1 \times \frac{5}{2}R \times (T_f - T_0)$$

By solving, we get

$$T_f = \frac{3}{2}T_0$$

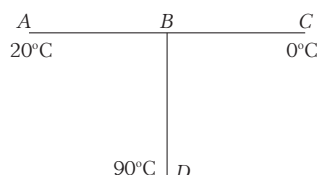
- 28** (d) Let the quantity of heat supplied per minute be Q , then quantity of heat supplied in 2 min = $mc(90 - 80)$

In 4 min, heat supplied = $2mc(90 - 80)$

$$\therefore mL = 2mc(90 - 80)$$

$$\Rightarrow \frac{L}{c} = 20$$

- 29** (b) There will be no heat flow through AB , if temperature of B is also 20°C and heat current in DB is equal to heat current in BC .



$$\therefore H_{DB} = H_{BC} \text{ or } \frac{70}{(l_{BD}/KA)} = \frac{20}{(l_{BC}/KA)} \text{ or } \frac{l_{BD}}{l_{BC}} = \frac{7}{2}$$

- 30** (d) Rate of heat flow, $I \propto \frac{\Delta T}{R}$

As, ΔT is same.

$$\therefore I \propto \frac{1}{\text{Thermal resistance}}$$

$$\Rightarrow \frac{I_A}{I_B} = \frac{R_B}{R_A} = \frac{(R)(2R)/(R+2R)}{(R+2R)} = \frac{2}{9}$$

- 31** (c) Initially the rods are in parallel, $\frac{Q}{t} = \frac{(\theta_1 - \theta_2)}{R}$

$$\Rightarrow \left(\frac{Q}{t}\right)_1 = \frac{mL}{t} = q_1 L = \frac{(100 - 0)}{R/2} \quad \dots(i)$$

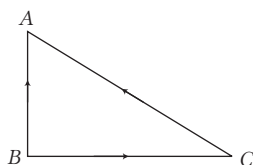
Finally when rods are in series,

$$\Rightarrow \left(\frac{Q}{t}\right)_2 = \frac{mL}{t} = q_2 L = \frac{(100 - 0)}{2R} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{q_1}{q_2} = \frac{4}{1}$$

- 32** (b) $\because T_B > T_A \Rightarrow$ Heat will flow B to A via two paths. from B to A and along BCA as shown below,



Rate of flow of heat in path BCA will be same,

$$\text{i.e.} \quad \left(\frac{Q}{t}\right)_{BC} = \left(\frac{Q}{t}\right)_{CA}$$

$$\Rightarrow \frac{K(\sqrt{2}T - T_C)A}{a} = \frac{K(T_C - T)A}{\sqrt{2}a} \Rightarrow \frac{T_C}{T} = \frac{3}{1 + \sqrt{2}}$$

- 33** (b) Initially ice will absorb the heat to raise its temperature at 0°C , then its melting takes places.

If m_i = initial mass of ice, m'_i = mass of ice that melts and m_w = initial mass of water.

By law of mixture, heat gained by ice = heat lost by water

$$\Rightarrow m_i \times c \times (20) + m'_i \times L = m_w c_w (20)$$

$$\Rightarrow 2 \times 0.5(20) + m'_i \times 80 = 5 \times 1 \times 20 \Rightarrow m'_i = 1 \text{ kg}$$

So, final mass of water = initial mass of water

$$+ \text{mass of ice that melts} \\ = 5 + 1 = 6 \text{ kg}$$

- 34** (b) Let T be the temperature when thermal equilibrium has reached.

Heat gained by ice to be converted to water at 0°C

$$= mL + m \times c \times (T - 0) = 10 \times 80 + 10 \times 1 \times T$$

Heat lost by tumbler and its contents = $55 \times (40 - T)$

Using principle of calorimetry, heat gained = heat lost

$$10 \times 80 + 10 \times 1 \times T = 55 \times (40 - T)$$

$$\Rightarrow 65T = 2200 - 800 = 1400$$

$$\Rightarrow T = \frac{1400}{65} \approx 21.5^\circ\text{C} \approx 22^\circ\text{C}$$

- 35** (b) In steady state, the amount of heat flowing from one face to the other face in time t is given by $Q = \frac{KA(\theta_1 - \theta_2)t}{l}$, where K is coefficient of thermal conductivity of material of rod.

$$\Rightarrow Q \propto \frac{A}{l} \propto \frac{r^2}{l} \quad \dots(i)$$

As the metallic rod is melted and the material is formed into a rod of half the radius,

$$V_1 = V_2 \Rightarrow \pi r_1^2 l_1 = \pi r_2^2 l_2 \Rightarrow l_1 = \frac{l_2}{4} \quad \dots(ii)$$

Now, from Eqs. (i) and (ii), we get

$$\frac{Q_1}{Q_2} = \frac{r_1^2}{r_2^2} \times \frac{l_2}{l_1} = \frac{r_1^2}{r_2^2} \times \frac{4l_1}{(r_1/2)^2}$$

$$\Rightarrow Q_1 = 16 Q_2 \Rightarrow Q_2 = \frac{Q}{16}$$

- 36** (a) When A and B are mixed,

$$mc_A(15 - 10) = mc_B(25 - 15) \text{ or } \frac{c_A}{c_B} = 2 \quad \dots(i)$$

When B and C are mixed,

$$mc_B(30 - 25) = mc_C(40 - 30) \text{ or } \frac{c_B}{c_C} = 2 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{c_A}{c_C} = 4$$

Now, let θ be the equilibrium temperature when A and C are mixed. Then, $mc_A(\theta - 10) = mc_C(40 - \theta)$

$$\text{or } 4(\theta - 10) = 40 - \theta \quad \left(\text{Using } \frac{c_A}{c_C} = 4 \right)$$

$$\text{or } \theta = 16^\circ\text{C}$$

37 (b) Suppose m gm steam is required per hour.

Heat is released by steam in following three steps:

(i) When steam at $150^\circ\text{C} \xrightarrow{Q_1}$ steam at 100°C

$$Q_1 = mc_{\text{steam}} \Delta\theta = m \times 1 \times (150 - 100) = 50m \text{ cal}$$

(ii) When steam at $100^\circ\text{C} \xrightarrow{Q_2}$ water at 100°C

$$Q_2 = mL_V = m \times 540 = 540m \text{ cal}$$

(iii) When water at $100^\circ\text{C} \xrightarrow{Q_3}$ water at 90°C

$$Q_3 = mc_w \Delta\theta = m \times 1 \times (100 - 90) = 10m \text{ cal}$$

Hence, total heat given by the steam,

$$Q = Q_1 + Q_2 + Q_3 = 600m \text{ cal} \quad \dots(i)$$

Heat taken by 10 kg water,

$$Q' = mc_w \Delta\theta = 10 \times 10^3 \times 1 \times (80 - 20) = 600 \times 10^3 \text{ cal}$$

$$\text{As, } Q = Q' \Rightarrow 600m = 600 \times 10^3 \Rightarrow m = 10^3 \text{ g} = 1 \text{ kg}$$

38 (c) $R_{AB} = R_{CD} = R$ (say)

Length of semi-circle = πR

$$\therefore \text{Resistance of semi-circle} = \pi R \quad (\because R \propto l)$$

$$R_{BC} = \frac{\pi R}{2} \text{ (in parallel)}$$

$$\text{Heat current, } H = \frac{100 - 0}{R + R + \frac{\pi}{2} \cdot R} = \frac{28}{R} \text{ units}$$

$$\text{Now, } \theta_C - 0^\circ = HR = \frac{28}{R} R$$

$$\therefore \theta_C = 28^\circ\text{C}$$

39 (a) Heat carried increases to 2 times. Therefore, new net thermal resistance will become half.

$$\therefore R' = \frac{R}{2} = \frac{R_1 R_2}{2(R_1 + R_2)} \quad \left(\because R = \frac{R_1 R_2}{(R_1 + R_2)} \right)$$

$$\text{or } \frac{(3l / KA) (l / K' A)}{(3l / KA) + (l / K' A)} = \frac{(3l / KA) (l / KA)}{2 \left(\frac{3l}{KA} + \frac{l}{KA} \right)}$$

$$\text{Solving this, we get } K' = \frac{7}{3} K$$

(B) Medical entrance special format questions

• Assertion and reason

1 (a) According to Kirchhoff's law, good radiators are also good absorbers, i.e. $\alpha \propto e$.

2 (c) Water has high specific heat due to which it does not losses heat easily.

3 (a) $T \propto \frac{1}{\lambda_m} \propto f \Rightarrow \text{As } f_{\text{blue}} > f_{\text{red}}$. So, blue stars are hotter.

4 (a) Both heat and electricity are affected by free electrons.

5 (c) Water in vapour state has more heat content.

• Statement based questions

1 (d) Given, mass, $m = 1 \text{ kg}$

Heat supplied, $dQ = mc \cdot dT$

$$\therefore c = \frac{dQ}{dT} = \text{Slope of } Q-T \text{ graph}$$

$$c_{\text{solid}} = \tan 45^\circ = 1 \text{ J kg}^{-1} \text{ K}^{-1}$$

$$c_{\text{liquid}} = \tan 30^\circ = \frac{1}{\sqrt{3}} \text{ J kg}^{-1} \text{ K}^{-1}$$

Latent heat of vaporisation

$$= \text{Heat required to melt complete } 1 \text{ kg} = 200 \text{ J kg}^{-1}$$

2 (b) No heat is flowing through C and F . Therefore, they are at same temperatures.

3 (a) As maximum wavelength, $\lambda_m \propto \frac{1}{T}$. So, with increase in temperature, the wavelength of radiation of black body shifts to lower wavelength.

4 (c) Due to gravity hot (but lighter) material rises up and cold (but heavier) material comes down.

Thus, gravity plays important role in the process of natural convection, which mainly takes place in liquids and gases.

$$\textbf{5} \text{ (a) } R = \frac{1}{KA} \text{ or } R \propto \frac{1}{A}$$

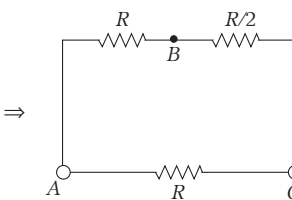
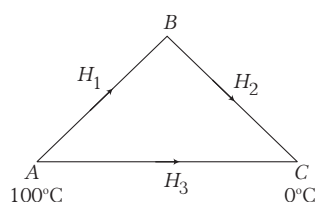
$$\therefore R_1 > R_2$$

Further, $TD = H \times R$ or $TD \propto R$

$$\therefore (TD)_1 > (TD)_2$$

• Match the columns

1 (d) Thermal resistance $\propto \frac{1}{\text{Area}}$



$$\text{Let } R_{AB} = R_{AC} = R$$

$$\text{Then, } R_{BC} = R/2$$

$$H_1 = H_2$$

$$\therefore \frac{100 - \theta_B}{R} = \frac{\theta_B - 0}{R/2}$$

$$\therefore \theta_B = 33.3^\circ\text{C}$$

$$R_{ABC} = \frac{3}{2} R$$

$$\text{and } R_{AC} = R$$

$$\therefore H_1 = H_2 = \frac{2}{3} H_3$$

- 2 (b) When A and B are mixed together,

$$(m)(c)(\theta - 20) = (2m)(c)(40 - \theta)$$

$$\text{or } \theta = \frac{100}{3} = 33.3^\circ\text{C}$$

When A and C are mixed together,

$$(m)(c)(\theta - 20) = (3m)(c)(60 - \theta) \text{ or } \theta = 50^\circ\text{C}$$

When B and C are mixed together,

$$(2m)(c)(\theta - 40) = (3m)(c)(60 - \theta) \Rightarrow \theta = 52^\circ\text{C}$$

When A, B and C all are mixed,

$$(m)(c)(\theta - 20) + (2m)(c)(40 - \theta) = (3m)(c)(60 - \theta)$$

$$\text{or } \theta = 60^\circ\text{C}$$

Hence, $A \rightarrow p$, $B \rightarrow r$, $C \rightarrow q$, $D \rightarrow t$.

(C) Medical entrances' gallery

- 1 (d) Since, heat required, $Q = mc\Delta T$

$$= \left(\frac{4}{3} \pi r^3 \cdot \rho \right) c \Delta T \quad (\because m = V_{\text{sphere}} \rho)$$

Since, π , ρ , c and T are constants.

$$\Rightarrow Q \propto r^3$$

$$\text{or } \frac{Q_1}{Q_2} = \frac{r_1^3}{r_2^3} = \left(\frac{1.5r_2}{r_2} \right)^3 = \frac{27}{8}$$

- 2 (d) According to Wien's displacement law,

$$\lambda \propto \frac{1}{T} \quad \dots (i)$$

We know that, $\lambda_{\text{blue}} < \lambda_{\text{yellow}} < \lambda_{\text{red}}$

$$\text{or } \lambda_A < \lambda_C < \lambda_B \quad \dots (ii)$$

Hence, using Eqs. (i) and (ii), we get

$$T_A > T_C > T_B$$

- 3 (c) Given, $\theta_0 = 0^\circ\text{C}$, $\theta_1 = -26^\circ\text{C}$

The rate of increase of thickness can be calculated from equation,

$$\begin{aligned} \frac{dQ}{dt} &= \frac{KA(\theta_0 - \theta_1)}{x} \\ \Rightarrow \frac{\rho A dx L}{dt} &= \frac{KA(\theta_0 - \theta_1)}{x} \quad (\because Q = mL) \\ \Rightarrow \frac{dx}{dt} &= \frac{KA(\theta_0 - \theta_1)}{\rho AxL} \\ &= \frac{K[0 - (-26)]}{\rho xL} = \frac{26K}{\rho xL} \end{aligned}$$

- 4 (c) The rate of heat flow through a conductor of length L and area of cross-section A is given by

$$\frac{dQ}{dt} = KA \frac{\Delta T}{L} \text{ J/s or watt}$$

where, K = coefficient of thermal conductivity

and ΔT = change in temperature.

$$\Rightarrow K = \frac{L}{A \Delta T} \frac{dQ}{dt}$$

$$\therefore \text{Unit of } K = \frac{\text{metre}}{(\text{metre})^2 \times \text{kelvin}} \times \text{watt} = \text{Wm}^{-1}\text{K}^{-1}$$

- 5 (d) When the object cool from 80°C to 70°C in 12 min, then from Newton's law of cooling,

$$\frac{80^\circ - 70^\circ}{12} = \frac{1}{K} \left(\frac{80^\circ + 70^\circ}{2} - 25^\circ \right) \quad (\because T_0 = 25^\circ\text{C})$$

$$\Rightarrow \frac{5}{6} = \frac{1}{K} 50 \quad \dots (i)$$

Similarly, when object cool from 70°C to 60°C , we get

$$\frac{70^\circ - 60^\circ}{t} = \frac{1}{K} \left(\frac{70^\circ + 60^\circ}{2} - 25^\circ \right)$$

$$\Rightarrow \frac{10}{t} = \frac{1}{K} 40 \quad \dots (ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{5}{6} \times \frac{t}{10} = \frac{50}{40}$$

$$\Rightarrow t = \frac{5}{4} \times 12 = 15 \text{ min}$$

- 6 (a) Temperature of sphere, $T = 227^\circ\text{C} = 273 + 227 = 500 \text{ K}$

Radius, $r = 2 \text{ m}$

Emissivity, $e = 0.8$

$\therefore$ Radiation power of sphere

= Radiated energy per second

$$= \sigma AeT^4 = 5.67 \times 10^{-8} \times 4\pi \times 2^2 \times 0.8 \times (500)^4$$

$$= 142502.6 \text{ W}$$

$$= 142.5 \text{ kW}$$

- 7 (a) Given, temperature of sun, $T_s = 6000 \text{ K}$

Radius of sun, $R_s = 7.2 \times 10^5 \text{ km} = 7.2 \times 10^8 \text{ m}$

Radius of earth, $R_e = 6000 \text{ km} = 6000 \times 10^3 \text{ m}$

Distance between earth and sun,

$$d = 15 \times 10^7 \text{ km} = 15 \times 10^{10} \text{ m}$$

Intensity of light on the earth,

$$\begin{aligned} I &= \frac{\text{Total energy emitted by sun}}{4\pi d^2} \times (\pi R_e^2) \\ &= \frac{\sigma(T_s^4 \cdot 4\pi R_s^2) \times \pi R_e^2}{4\pi d^2} = \frac{\sigma T_s^4 R_s^2 \times \pi R_e^2}{d^2} \\ &= \frac{5.67 \times 10^{-8} \times (6000)^4 \times (7.2 \times 10^8)^2}{(15 \times 10^{10})^2} \\ &= \frac{\times 3.14 \times (6000 \times 10^3)^2}{(15 \times 10^{10})^2} \\ &= 19.2 \times 10^{16} \text{ W} \end{aligned}$$

- 8 (a) According to Wien's law, $\lambda_{(\text{max})} \propto 1/T$

$$\text{i.e. } \lambda_{(\text{max})} T = \text{constant}$$

where, $\lambda_{(\text{max})}$ is the maximum wavelength of the radiation emitted at temperature T .

$$\therefore \lambda_{(\text{max})_1} T_1 = \lambda_{(\text{max})_2} T_2$$

$$\text{or } \frac{T_1}{T_2} = \frac{\lambda_{(\text{max})_2}}{\lambda_{(\text{max})_1}} \quad \dots (i)$$

$$\text{Here, } \lambda_{(\text{max})_1} = \lambda_0 \text{ and } \lambda_{(\text{max})_2} = \frac{3}{4} \lambda_0$$

Substituting the above values in Eq. (i), we get

$$\frac{T_1}{T_2} = \frac{\frac{3}{4}\lambda_0}{\lambda_0} = \frac{3}{4} \quad \text{or} \quad \frac{T_1}{T_2} = \frac{3}{4} \quad \dots(ii)$$

As we know that, from Stefan's law, the power radiated by a body at temperature T is given as

$$P = \sigma A e T^4$$

$$i.e. \quad P \propto T^4$$

($\because$ The quantity $\sigma A e$ is constant for a body)

$$\Rightarrow \quad \frac{P_1}{P_2} = \frac{T_1^4}{T_2^4} = \left(\frac{T_1}{T_2}\right)^4$$

From Eq. (i), we get

$$\frac{P_1}{P_2} = \left(\frac{3}{4}\right)^4 = \frac{81}{256}$$

Given, $P_1 = P$ and $P_2 = nP$

$$\Rightarrow \quad \frac{P_1}{P_2} = \frac{P}{nP} = \frac{81}{256} \quad \text{or} \quad n = \frac{256}{81}$$

- 9 (a)** In parallel arrangement of n rods, equivalent thermal conductivity is given by

$$K_{eq} = \frac{K_1 A_1 + K_2 A_2 + \dots + K_n A_n}{A_1 + A_2 + \dots + A_n}$$

If rods are of same area, then $K_{eq} = \frac{K_1 + K_2 + \dots + K_n}{n}$

Now, in the question, it is not given that rods are of same area. But we can judge that from given diagram.

$\therefore$ Equivalent thermal conductivity of the system of two rods,

$$K_{eq} = \frac{K_1 + K_2}{2}$$

- 10 (d)** Radiated power of a black body, $P = \sigma A T^4$... (i)

where, A = surface area of the body,

T = temperature of the body

and σ = Stefan's constant.

When radius of the sphere is halved, new area, $A' = A/4$

$\therefore$ Power radiated,

$$P' = \sigma \left(\frac{A}{4}\right) (2T)^4 = \frac{16}{4} \cdot (\sigma A T^4) = 4P \quad [\text{From Eq. (i)}]$$

$$= 4 \times 450 = 1800 \text{ W}$$

- 11 (c)** Given, temperature, $T_1 = 5760 \text{ K}$

Since, it is given that energy of radiation emitted by the body at wavelength 250 nm is U_1 , at wavelength 500 nm is U_2 and that at 1000 nm is U_3 .

$\therefore$ According to Wien's law, we get

$$\lambda_m T = b$$

where, b = Wien's constant = $2.88 \times 10^6 \text{ nmK}$

$$\Rightarrow \quad \lambda_m = \frac{b}{T} \Rightarrow \lambda_m = \frac{2.88 \times 10^6 \text{ nmK}}{5760 \text{ K}} \Rightarrow \lambda_m = 500 \text{ nm}$$

$\therefore \lambda_m$ = wavelength corresponding to maximum energy, so $U_2 > U_1$.

- 12 (b)** According to the conservation of energy, energy gained by the ice during its fall from height h is given by

$$E = mgh$$

As given, only one quarter of its energy is absorbed by the ice.

$$\text{So,} \quad \frac{mgh}{4} = mL_f$$

$$\Rightarrow \quad h = \frac{mL_f \times 4}{mg} = \frac{L_f \times 4}{g} = \frac{3.4 \times 10^5 \times 4}{10} = 13.6 \times 10^4$$

$$= 136000 \text{ m} = 136 \text{ km}$$

- 13 (d)** Here, $\Delta T_1 = 110 - 100 = 10^\circ \text{C}$

$$\frac{dQ_1}{dt} = 4 \text{ Js}^{-1}$$

As the rate of heat flow is directly proportional to the temperature difference and the temperature difference in both the cases is same, i.e. $\Delta T_2 = 210 - 200 = 10^\circ \text{C}$. So, the same rate of heat will flow in the second case.

$$\text{Hence,} \quad \frac{dQ_2}{dt} = 4 \text{ Js}^{-1}$$

- 14 (b)** We know from Wien's displacement law, $\lambda_m T = \text{constant}$

$$\text{So,} \quad T \propto \frac{1}{\lambda_m}$$

$$\text{As,} \quad \lambda_r > \lambda_g > \lambda_v$$

$$\text{So,} \quad T_r < T_g < T_v$$

$$\text{Hence,} \quad T_Q < T_R < T_P$$

$$i.e. \quad T_P > T_R > T_Q$$

- 15 (d)** From Wien's displacement law,

$$\lambda_m T = b \Rightarrow \lambda_m = \frac{b}{T} \quad \dots(i)$$

From Stefan's law, $E \propto T^4$

$$\Rightarrow \quad \frac{E_1}{E_2} = \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{\lambda_{m_2}}{\lambda_{m_1}}\right)^4 \quad [\text{By Eq. (i)}]$$

$$\Rightarrow \quad \frac{E_1}{E_2} = \left(\frac{600}{200}\right)^4 \Rightarrow E_1 : E_2 = 81 : 1$$

- 16 (c)** If T is the temperature of the interface of two plates of same area, then

$$\frac{dQ}{dt} = \frac{K_1 A (10 - T)}{\Delta x_1} = \frac{K_2 A (T - 0)}{\Delta x_2}$$

$$\text{or} \quad \frac{K_1 A (10 - T)}{\Delta x_1} = \frac{3}{2} \frac{K_1 A (T - 0)}{\Delta x_1} \quad \left(\because \frac{K_1}{K_2} = \frac{2}{3} \text{ and } \frac{\Delta x_1}{\Delta x_2} = \frac{2}{3} \right)$$

$$\text{or} \quad 10 - T = T - 0 \quad \text{or} \quad T + T = 10 \quad \text{or} \quad 2T = 10$$

$$\therefore \quad T = 5^\circ \text{C}$$

- 17 (a)** Given, $\lambda_m = 14 \text{ micron} = 14 \mu\text{m} = 14 \times 10^{-6} \text{ m}$

and Wien constant, $b = 2892 \times 10^{-6} \text{ mK}$

According to Wien's displacement law,

$$\lambda_m = \frac{b}{T} \quad \text{or} \quad T = \frac{b}{\lambda_m} = \frac{2892 \times 10^{-6} \text{ mK}}{14 \times 10^{-6} \text{ m}}$$

$$\therefore \quad T = 206.6 \text{ K} \approx 207 \text{ K}$$

- 18** (b) The rate of heat emission, $u = \sigma AT^4$.

At room temperature, the rate of absorption is $u_0 = \sigma AT_0^4$

The net rate of heat loss, $u - u_0 = \sigma A(T^4 - T_0^4)$

Given, $A = 0.001 \text{ m}^2$,

$T = 400 \text{ K}$ and $T_0 = 300 \text{ K}$

Thus, $u - u_0 = 5.67 \times 10^{-8} \times 0.001 \times [(400)^4 - (300)^4]$
 $= 0.99 \text{ W} \approx 1 \text{ W}$

- 19** (b) If we talk about heat of a body, it is meaningless.

- 20** (d) Rate of production of mass of steam $= \frac{dm}{dt} = \frac{1}{60} \text{ gs}^{-1}$

Rate of heat transformed by stove,

$$\frac{dQ}{dt} = L \frac{dm}{dt} = 540 \times \frac{1}{60} = 9 \text{ cal } ^\circ\text{C}^{-1}\text{s}^{-1}\text{cm}^{-1}$$

$$\text{Also, } \frac{dQ}{dt} = \frac{KA d\theta}{dx} \Rightarrow L \frac{dm}{dt} = \frac{KA d\theta}{dx}$$

$$\Rightarrow 9 = \frac{0.5 \times 3000 \times d\theta}{0.2}$$

$$\Rightarrow d\theta = 1.2 \times 10^{-3} ^\circ\text{C} = 1.2 \times 10^{-3} \text{ K}$$

- 21** (c) At steady state, $\frac{dQ}{dt} = \frac{dQ_1}{dt} + \frac{dQ_2}{dt}$

$$\text{or } \frac{K(4\pi R^2)(T_2 - T_1)}{L} = \frac{K_1(\pi R^2)(T_2 - T_1)}{L} + \frac{K_2(3\pi R^2)(T_2 - T_1)}{L}$$

$$\text{or } 4K = K_1 + 3K_2 \Rightarrow K = \frac{K_1 + 3K_2}{4}$$

- 22** (d) Given, mass of ice-piece, $m_1 = 100 \text{ g} = 0.1 \text{ kg}$

Mass of water, $m_2 = 200 \text{ g} = 0.2 \text{ kg}$

When temperature of mixture is 0°C , then heat lost by water,

$$Q = m_2 c (\Delta\theta) = 0.2 \times 4200(25 - 0)$$

($\because$ Specific heat of water = 4200 J)

$$\Rightarrow Q = 21000 \text{ J}$$

$$\text{Now, mass of melted ice, } m' = \frac{Q}{L} = \frac{21000}{3.4 \times 10^5} \approx 61.8 \text{ g}$$

- 23** (d) From the Newton's law of cooling,

$$\frac{\theta_1 - \theta_2}{t} \propto \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

where, θ_0 = temperature of surrounding

$$\therefore \frac{94 - 86}{t_1} \propto \left(\frac{94 + 86}{2} - 20 \right)$$

$$\frac{8}{t_1} \propto (70) \quad \dots(i)$$

$$\text{and } \frac{71 - 69}{t_2} \propto \left(\frac{71 + 69}{2} - 20 \right)$$

$$\frac{2}{t_2} \propto 50 \quad \dots(ii)$$

On dividing Eq (i) by Eq (ii), we get

$$\frac{t_2}{t_1} = \frac{70}{50 \times 4} \Rightarrow t_2 = \frac{70 \times 2}{50 \times 4}$$

$$\Rightarrow t_2 = 0.7 \text{ min} \Rightarrow t_2 = 42 \text{ s}$$

- 24** (d) Heat lost by system = Heat gained by water

Let m' be amount of heat converted into water

$$m' \times L = mc\Delta t$$

$$m' \times 540 = 20 \times 1 \times (80 - 10)$$

$$\Rightarrow m' = \frac{20 \times 70}{540} = 2.5 \text{ g}$$

Now, net mass of water = $20 + 2.5 = 22.5 \text{ g}$

- 25** (a) Let temperature of surrounding be Q_0 , then from Newton's law of cooling

$$\text{Case I } \frac{Q_2 - Q_1}{t} = K \left(\frac{Q_2 + Q_1}{2} - Q_0 \right)$$

$$\frac{70 - 60}{5} = K(65 - Q_0) \quad \dots(i)$$

$$\text{Case II } \frac{60 - 54}{5} = (57 - Q_0) K \quad \dots(ii)$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{10}{6} = \frac{65 - Q_0}{57 - Q_0} \Rightarrow Q_0 = 45^\circ\text{C}$$

- 26** (b) As we know, work done by an electric heater,

$$\text{i.e. } m_1 c_1 \Delta t + m_2 c_2 \Delta t = \text{work done}$$

$$\text{So, } m_1 c_1 \Delta t + m_2 c_2 \Delta t = P t_1$$

where, $m_1 = 0.5 \text{ kg}$, specific heat, $c_1 = 4200 \text{ Jkg}^{-1}\text{K}^{-1}$,

$$\Delta t = \Delta t_1 = \Delta t_2 = 3 \text{ K},$$

$$P_1 = 10 \text{ W},$$

$$t_1 = 15 \times 60 = 900 \text{ s}$$

c_2 = specific heat capacity of container.

$$\text{So, } 0.5 \times 4200 \times (3 - 0) + m_2 c_2 \times (3 - 0) = 10 \times 15 \times 60$$

$$\Rightarrow 2100 \times 3 + m_2 c_2 \times 3 = 9000$$

$$\Rightarrow m_2 c_2 = \frac{9000 - 6300}{3} = 900$$

Similarly, in the case of oil, $m_1 c_1 \Delta t + m_2 c_2 \Delta t = P_2 t_2$

where, c_o = specific heat capacity of oil, $P_2 = 10 \text{ W}$

$$2 \times c_o \times 2 + 900 \times 2 = 10 \times 20 \times 60$$

$$\Rightarrow 4c_o + 1800 = 12000$$

$$\Rightarrow c_o = 2550 = 2.55 \times 10^3 \text{ J kg}^{-1} \text{K}^{-1}$$

- 27** (d) Given, $m = 50 \text{ kg}$, $v = 5 \text{ ms}^{-1}$, $L = 80 \text{ calg}^{-1}$,

$$J = 4.2 \text{ J cal}^{-1} \text{ and } m = ?$$

Heat lost, i.e. Q_1 = work done = K (kinetic energy)

$$= \frac{1}{2} mv^2 = \frac{1}{2} \times 50 \times 5^2 = 625 \text{ J}$$

and heat gained, i.e. $Q_2 = m' \times L = 80 \times m' \times 4.2$

$$\therefore Q_1 = Q_2$$

$$625 = m' \times 8 \times 42 \Rightarrow m' = 1.86 \text{ g}$$

- 28** (a) We know that, relation between the temperature gradient (TG) and thermal conductivity (K),

$$\frac{dQ}{dt} = -KA \frac{d\theta}{dx} = -KA \times (\text{TG})$$

$$\text{i.e. } \text{TG} \propto \frac{1}{K} \quad \left(\because \frac{dQ}{dt} = \text{constant} \right)$$

or $K \propto \frac{1}{t}$

$\Rightarrow K_1 \propto \frac{1}{t_1}$

and $K_2 \propto \frac{1}{t_2}$

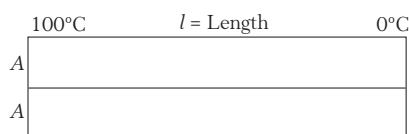
From Eqs. (i) and (ii), we get

$$\frac{K_1}{K_2} = \frac{1/t_1}{1/t_2}$$

$\Rightarrow K_1 : K_2 = t_2 : t_1$

29 (d) According to the question,

When two identical rods are connected in parallel,



So, rate of heat transfer is given by

$$\frac{dQ}{dt} = \frac{2KA(T_1 - T_2)}{l}$$

$$= \frac{2KA(100 - 0)}{l} = \frac{200KA}{l}$$

where, net area = $2A$

K = thermal conductivity constant,

T_1 = temperature at one end,

T_2 = temperature at other end

and l = length of rod.

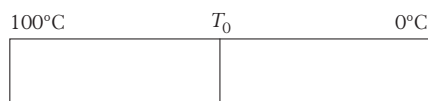
As, $\frac{dQ}{dt}$ is rate of heat transfer = $\frac{mL}{t}$

So, $\frac{dQ}{dt} = \left(\frac{m}{t}\right)L = \frac{200KA}{l}$

So, for parallel connection,

$$\frac{m}{t} = g_1 = \frac{200KA}{Ll}$$

Similarly, for series connection



So, $100 - T_0 = T_0 - 0 \Rightarrow 2T_0 = 100$

$\Rightarrow T_0 = 50$

So, $\frac{dQ}{dt} = \left(\frac{m}{t}\right)L = \frac{KA(50 - 0)}{l}$

and it can be written as

$$g_2 = \frac{KA 50}{Ll}$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{g_1}{g_2} = \frac{200}{50} = 4 \Rightarrow \frac{g_2}{g_1} = \frac{1}{4}$$

30 (b) The equation of Wien's displacement law, i.e. $\lambda_m T = \text{constant}$.

31 (d) From the equation, $\frac{\theta}{x} = \frac{\Delta t}{l}$

Given, $t_1 = 100^\circ\text{C}$, $t_2 = 0^\circ\text{C}$, $l = 25\text{ cm}$ and $x = 10\text{ cm}$

$$\frac{\theta}{10} = \frac{100 - 0}{25} \Rightarrow \frac{\theta}{10} = 4 \Rightarrow \theta = 40^\circ\text{C}$$

32 (c) According to Wien's displacement law,

$$\lambda T = \text{constant}$$

$$\therefore \frac{T_1}{T_2} = \frac{\lambda_2}{\lambda_1} = \frac{480}{360} = \frac{4}{3}$$

33 (c) According to Newton's law of cooling,

$$\frac{61 - 59}{4} = K \left(\frac{61 + 59}{2} - 30 \right)$$

$$\Rightarrow \frac{2}{4} = K \times 30 \quad \dots(i)$$

$$\text{Similarly, } \frac{51 - 49}{t} = K \left(\frac{51 + 49}{2} - 30 \right)$$

$$\Rightarrow \frac{2}{t} = K \times 20 \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$t = 6\text{ min}$$

34 (c) According to Stefan-Boltzmann law, the emissive power is directly proportional to the fourth power of temperature. So, the given graph will be a straight line.

35 (b) Given, $m = 500\text{ kg}$

Rise in temperature, $dT = (100^\circ - 20^\circ)\text{C} = 80^\circ\text{C}$

Specific heat of water, $c = 10^3\text{ cal kg}^{-1}^\circ\text{C}^{-1}$

Heat energy gained by water, $E = mcdT$

$$= 500 \times 10^3 \times 80\text{ cal} = 4 \times 10^7 \times 4.2\text{ J} = 16.8 \times 10^7\text{ J}$$

From $E = mc^2$

Increase in mass of water,

$$m = \frac{E}{c^2} = \frac{16.8 \times 10^7}{(3 \times 10^8)^2}$$

$$= 1.87 \times 10^{-9}\text{ kg}$$

36 (d) From Stefan' law, $E = \sigma T^4$

So, the rate of energy production,

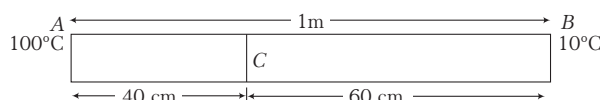
$$Q = E \times A \Rightarrow Q = \sigma T^4 \times 4\pi R^2$$

$$\text{Temperature of star, } T = \left(\frac{Q}{4\pi R^2 \sigma} \right)^{1/4}$$

37 (a) Graph (a) shows the variation of temperature with time.

At first, temperature will increase, then there will be state change from liquid to gas.

38 (a) Let the temperature at distance 60 cm from point B be θ .



When K = coefficient of thermal conductivity, A = area of rod,

then $\frac{KA(100 - \theta)}{40} = \frac{KA(\theta - 10)}{60}$

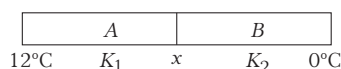
$$\text{or } \frac{100 - \theta}{2} = \frac{\theta - 10}{3}$$

$$\text{or } 300 - 3\theta = 2\theta - 20$$

$$\text{or } 5\theta = 320$$

$$\text{or } \theta = 64^\circ \text{C}$$

39 (c) The given situation can be shown as below.



Rate of flow of heat will be equal in both the slabs,

$$(12 - x)K_1 = K_2(x - 0)$$

$$12 - x = 2x$$

$$\Rightarrow x = 4^\circ \text{C}$$

The temperature difference across slab,

$$A = (12 - x) = (12 - 4) = 8^\circ \text{C}$$

40 (a) The heat absorbed by a substance is directly proportional to its specific heat. Therefore, a coolant having high specific heat will absorb a large amount of heat from the engine. So, water is used as a coolant to cool the car engines in radiators.

41 (d) Newton's cooling law is given as

$$\frac{\theta_1 - \theta_2}{t} = k \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

Given, $\theta_0 = 25^\circ \text{C}$, $\theta_1 = 60^\circ \text{C}$, $t = 10$ min and $\theta_2 = 50^\circ \text{C}$

$$\therefore \frac{60 - 50}{10} = k \left(\frac{60 + 50}{2} - 25 \right)$$

We get, $k = \frac{1}{30}$ units and $t_2 = 10$ min

Again, $\theta_2 = \theta$, $\theta_1 = 50^\circ \text{C}$, $\theta_0 = 25^\circ \text{C}$

$$\therefore \left(\frac{50 - \theta}{10} \right) = \frac{1}{30} \left(\frac{50 + \theta}{2} - 25 \right)$$

$$\Rightarrow \theta = 42.85^\circ \text{C}$$

42 (c) According to Stefan's law, $E = \sigma T^4$

$$\Rightarrow 1 = 5.6 \times 10^{-8} \times T^4$$

$$\text{or } T^4 = \frac{1}{5.6 \times 10^{-8}}$$

$$\Rightarrow T = \left(\frac{1}{5.6 \times 10^{-8}} \right)^{1/4} = 65 \text{ K}$$

43 (c) According to Stefan's law, the energy emitted by a body per second is directly proportional to the fourth power of the temperature of the body. Here, the temperature of blue glass is more than that of red glass, so it will look brighter.

44 (b) Temperature gradient = $\frac{\theta_1 - \theta_2}{l}$

Here, $\theta_1 = 30^\circ \text{C}$ and $l = 0.5$ m

$$\therefore 80 = \frac{30 - \theta_2}{0.5}$$

$$\Rightarrow \theta_2 = 30 - 80 \times 0.5$$

$$\Rightarrow \theta_2 = 30 - 40 = -10^\circ \text{C}$$

45 (a) We know that,

$$\frac{Q}{t} = \frac{KA(\Delta\theta)}{l}$$

$$50 = \frac{5 \times 20K}{0.4}$$

$$\text{or } K = \frac{50 \times 0.4}{5 \times 20} = 0.2$$

46 (a) Given, $K_B = K_A / 2$

and $K_B = 3K_C$

$$\therefore K_C = K_A / 6$$

A	B	C
---	---	---

Rods are in series form, so

$$\frac{L}{K} = \frac{l_1}{K_A} + \frac{l_2}{K_B} + \frac{l_3}{K_C}$$

$$\frac{3l}{K} = \frac{l}{K_A} + \frac{l}{K_A / 2} + \frac{l}{K_A / 6}$$

$$(\because l = l_1 = l_2 = l_3)$$

$$\text{or } \frac{3l}{K} = \frac{9l}{K_A} \text{ or } K = \frac{K_A}{3}$$

CHAPTER 17

Wave Motion

A wave is a disturbance which transfers energy from one part of a medium to another part without actual transfer or flow of matter as a whole. We are all familiar with water waves, sound waves and light waves. These waves occur when a system is disturbed from its equilibrium position and this disturbance travels from one place to another.

In other words, “a wave is any disturbance from a normal or equilibrium condition that propagates without the transport of matter. In general, a wave transports both energy and momentum.”

In this chapter, we will discuss wave motion and the factors affecting the speed of waves along with superposition of waves and Doppler’s effect in sound.

WAVE MOTION

A wave motion is a means of transferring energy and momentum from one point to another without actual transport of matter between two points. Transmission of energy over considerable distances is possible through wave motion.

According to the number of dimensions in which waves propagate energy, can be classified as one, two or three dimensional. *e.g.* Waves developed in a stretched string are one dimensional, ripples on water are two dimensional, while sound and light waves are three dimensional.

Necessary properties of medium for wave motion are as given below

- (i) The medium must possess **inertia**, so that its particles can store kinetic energy.
- (ii) In order to make the particles return to their original position after getting disturbed, the medium must possess **elasticity**.
- (iii) There should be **minimum frictional force** between the particles of medium.

Characteristics of wave motion are as given below

- (i) In a wave motion, the disturbance travels through the medium due to repeated periodic oscillations of the particles of the medium about their mean positions.
- (ii) The energy is transferred from one place to another without any actual transfer of the particles of the medium.

Inside

1 Wave motion

- Definitions related to the wave motion
- Types of waves

2 Sound waves

- Speed of a longitudinal wave (sound wave) in a medium
- Factors affecting the speed of sound in air
- Characteristics of sound
- Shock waves

3 Progressive wave

- Plane progressive harmonic wave
- Energy in wave

4 Principle of superposition of waves

- Interference of waves
- Beats

5 Reflection of waves

- Echo
- Standing/stationary waves

6 Vibrations of air columns in organ pipes

- Standing waves in a closed organ pipe
- Standing waves in an open organ pipe

7 Vibrations of stretched string

8 Doppler’s effect

- Doppler’s effect in sound
- Doppler’s effect in light

- (iii) There is a regular phase difference between one particle and the next particle because each particle receives disturbance a little later than its preceding particle.
- (iv) The velocity with which a wave travels is different from the velocity of the particles with which they vibrate about their mean positions.
- (v) In a given medium, the wave velocity remains constant, while the particle velocity changes continuously during its vibration about the mean position. The particle velocity is maximum at the mean position and zero at the extreme position.
- (vi) For the propagation of a mechanical wave, the medium must possess the properties of inertia, elasticity and minimum friction among its particles.

Definitions related to the wave motion

Some important definitions related to the wave motion are given below

- (i) **Amplitude** The maximum displacement suffered by the particles of the medium from their mean position is called amplitude. It is denoted by A .
- (ii) **Time period** The time in which a particle of medium completes one vibration to and fro about its mean position is known as time period of a wave. It is denoted by T .
- (iii) **Frequency** The number of waves produced per unit time in the given medium is known as frequency of a wave. It is denoted by ν .
- (iv) **Wavelength** The distance covered by a wave during the time in which a particle of medium completes one vibration about its mean position is known as wavelength of a wave. It is denoted by λ .
- (v) **Wave Number** The number of waves present in a unit distance along the direction of propagation is known as wave number. It is equal to the reciprocal of wavelength (λ). It is denoted by $\bar{\nu}$.

$$\therefore \bar{\nu} = \frac{1}{\lambda}$$

The SI unit of wave number is m^{-1} .

- (vi) **Phase** The position of a point in time on a waveform is known as the phase of a wave. Phase can also be expressed as relative displacement between two corresponding peaks of a waveform.
- (vii) **Path Difference** The difference in the path traversed by the two waves measured in the terms of wavelength of the associated waves is called path difference.

- (viii) **Speed of a Wave** Speed of a wave is the measure of how fast the wave travels. It is calculated as the ratio of how far a wave travels to the time taken by the wave to travel that distance.

- (ix) **Wave Pulse** It is a short wave produced in a medium when the disturbance is created for a short time. If we give an upward jerk to one end of a long rope whose opposite end is fixed, a single wave pulse is formed as shown in figure and it travels along the rope with a fixed speed.

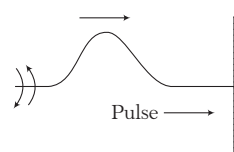


Fig. 17.1 Wave pulse

- (x) **Wave Train** A series of wave pulses is called wave train. If we continuously give periodic up and down jerks to one end of a rope, we get sinusoidal wave (called wave train) on the rope as shown below.

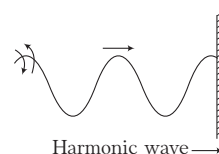
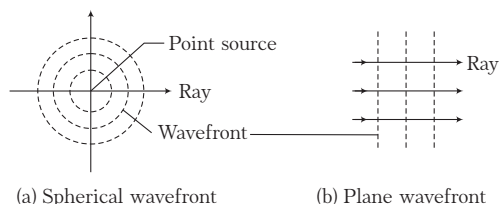


Fig. 17.2 Wave train

- (xi) **Wavefront** A wavefront is a line or surface on which the disturbance has the same phase at all points. Ripples in a pond are the examples of wavefronts. Two types of wavefronts are shown below.



(a) Spherical wavefront (b) Plane wavefront

Fig. 17.3 Spherical and plane wavefronts

Relation between frequency, speed and wavelength

When a particle of the medium completes one oscillation about its mean position in periodic time (T), then the wave travels a distance equal to the wavelength (λ). So,

$$\text{Wave speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\Rightarrow \boxed{v = \frac{\lambda}{T}} \Rightarrow \boxed{v = \nu \times \lambda} \left(\because \nu = \frac{1}{T} \right)$$

i.e. Wave speed = Frequency $\times$ Wavelength

Example 17.1 The speed of a wave in a medium is 960 ms^{-1} . If 3600 waves pass through a point in the medium in 1 minute, then determine its wavelength.

Sol. Given, speed of the wave, $v = 960 \text{ ms}^{-1}$

$$\text{Frequency of the wave, } \nu = 3600 \text{ min}^{-1} \\ = \frac{3600}{60} = 60 \text{ s}^{-1}$$

$$\therefore \text{Wavelength, } \lambda = \frac{v}{\nu} = \frac{960}{60} = 16 \text{ m} \Rightarrow \lambda = 16 \text{ m}$$

Types of waves

There are mainly three types of waves

- (i) **Mechanical waves** Mechanical waves are those waves which require a medium for their propagation. They cannot travel through vacuum. *e.g.* Sound waves, water waves, etc.
- (ii) **Electromagnetic waves** Electromagnetic waves are those waves which do not require medium for their propagation. These waves travel in the form of oscillating electric and magnetic fields. *e.g.* X-rays, radio waves, etc.
- (iii) **Matter waves** Moving microscopic particles such as electrons, protons, neutrons, atoms, molecules, etc., sometimes behave like waves. These are called matter waves.

In this chapter, we will discuss mechanical waves and their characteristic properties in detail.

Types of mechanical waves

The energy of the mechanical waves is the kinetic and potential energy of the matter. In the propagation of mechanical waves, elasticity and inertia of the medium play an important role. That is why mechanical waves are also called **elastic waves**.

Depending on the relationship between the direction of oscillation of individual particles and the direction of wave propagation, the mechanical waves are classified into two categories

1. Transverse wave

In transverse wave, the particles of the medium oscillate perpendicular to the direction in which the wave travels. Transverse wave travels through a medium in the form of crests and troughs.

A **crest** is a portion of the medium which is raised temporarily above the normal position of rest of the particles of the medium, when a transverse wave passes through it.

A **trough** is a portion of the medium which is depressed temporarily below the normal position of rest of the particles of the medium, when a transverse wave passes through it.

Travelling waves on a tight rope are transverse waves. If one end of the rope is rigidly fixed and the other end is given periodic up and down jerks, the disturbance propagates along the length of the rope, but the particles of the rope oscillate up and down as shown in figure.

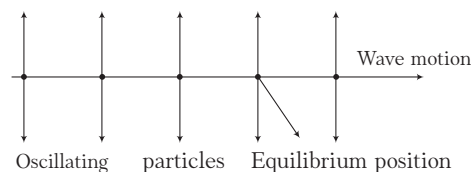


Fig. 17.4 Transverse wave

Vertical arrows in the above figure show the oscillation of particles and the horizontal line arrows show the direction of the motion.

Disturbance travels along the rope in the form of crests (upward peaks) and troughs (downward peaks). Distance between two consecutive crests or two consecutive troughs is equal to the wavelength.

This type of waves are possible in media which possess elasticity, *e.g.* solids and liquids, etc.

The stretched strings of sitar, violin, sonometer, etc. execute transverse vibrations. Further, all the electromagnetic waves are also transverse in nature.

2. Longitudinal wave

In longitudinal wave, the particles of the medium oscillate about their mean or equilibrium position along the direction of propagation of the wave motion itself.

Longitudinal wave travels in the form of compressions and rarefactions.

A **compression** is a region of the medium in which particles come to distances less than the normal distance between them.

A **rarefaction** is a region of the medium in which particles of the medium go apart to distances greater than the normal distance between them.

When a tuning fork is set into vibrations, its prong compresses the air medium just in front of it. As a wave form of compression and rarefaction progresses in the air along horizontal, the particles of the air medium also execute periodic motion along horizontal as shown in Fig. 17.5 (a).

The longitudinal wave can also be set in a clamped rod (by pulling it along its length) or in an organ pipe.

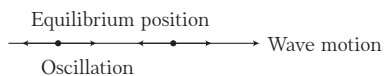


Fig. 17.5(a) Longitudinal wave

In the figure shown above, horizontal arrows represent the oscillation of particles along the direction of wave motion.

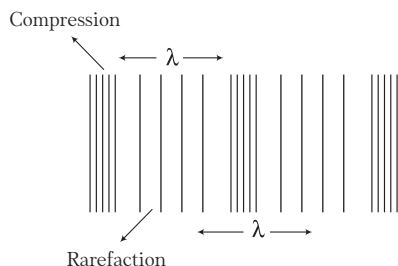


Fig. 17.5 (b) Longitudinal wave set up in a tuning fork

It is found that in case of a longitudinal wave, wavelength is equal to the distance between two consecutive compressions or rarefactions.

Longitudinal wave motion is possible in a medium possessing elasticity of volume, *i.e.* in solids, liquids and gases etc.

SOUND WAVES

Sound is a form of energy which produces a sensation of hearing in our ears. Sound waves are longitudinal in nature.

Sound waves can be classified in three groups according to their range of frequencies

- (i) **Infrasonic waves** Longitudinal waves having frequencies below 20 Hz are called infrasonic waves. They cannot be heard by human beings.
- (ii) **Audible waves** Longitudinal waves having frequencies lying between 20-20000 Hz are called audible waves.
- (iii) **Ultrasonic waves** Longitudinal waves having frequencies above 20000 Hz are called ultrasonic waves. They are produced and heard by bats. They have a large energy content.

Ultrasonic waves have a large range of applications. Some of their applications are as follows

- (i) The fine internal cracks in a metal can be detected by ultrasonic waves.
- (ii) Ultrasonic waves can be used for determining the depth of the sea, lakes, etc.
- (iii) Ultrasonic waves can be used to detect submarines, icebergs, etc.

(iv) These waves can be used to clean clothes, fine machinery parts, etc.

(v) Ultrasonic waves can be used to kill smaller animals like rats, fish, frogs, etc.

Speed of a longitudinal wave (sound wave) in a medium

In a longitudinal wave, the constituents of the medium oscillate forward and backward in the direction of wave propagation.

Speed of longitudinal waves in a medium is

$$v = \sqrt{\frac{B}{\rho}}$$

where, bulk modulus of the medium,

$$B = -\frac{\Delta p}{\Delta V/V}$$

where, Δp = change in pressure, ΔV = change in volume and V = initial volume.

- The speed of a longitudinal wave (sound wave) in a medium depends on its elastic properties and inertial properties.
- For a linear medium like a solid bar, the modulus of elasticity will be Young's modulus (Y).

Note The speed of sound is greater in solids and liquids than in gases even though they are denser than gases. This happens because they are much more difficult to compress than gases and hence, have much higher values of bulk modulus. This factor compensates for their higher densities than gases.

So, the speed of longitudinal waves in a solid bar is given by

$$v = \sqrt{\frac{Y}{\rho}}$$

where, Y is the Young's modulus of the material of the bar and ρ is density. Also, the speed of a longitudinal wave through a solid of bulk modulus B , modulus of rigidity η and density ρ is given by

$$v = \sqrt{\frac{B + (4/3)\eta}{\rho}}$$

Speed of longitudinal waves in gases,

$$v = \sqrt{\frac{p}{\rho}}$$

where, p is the pressure exerted by gas.

For gases, bulk modulus is equal to the pressure of the gas. This formula is known as **Newton's formula**. Newton assumed that when sound waves are propagated through a gas, the temperature variations in the layers of compression and rarefaction are negligible (isothermal process).

Laplace correction

The result obtained for the speed of sound in air at STP from above formula is 280 ms^{-1} , which is about 15% smaller as compared to the experimental value, i.e. 331 ms^{-1} .

The mistake in the formula was pointed out by Laplace and he told that the changes in pressure and volume of a gas, when sound waves are propagated through it, are not isothermal, but adiabatic.

He modified the formula as

$$\text{Speed of sound in gases, } v = \sqrt{\frac{\gamma p}{\rho}}$$

This modification in Newton's formula is referred to as the Laplace correction.

For air, $\gamma = 7/5$.

By this formula, speed of sound comes out to be 331.3 ms^{-1} , which agrees with the practical value.

Example 17.2 If the Young's modulus of elasticity for a steel rod is $2.9 \times 10^{11} \text{ Nm}^{-2}$ and density is $8 \times 10^3 \text{ kg m}^{-3}$. Determine the velocity of longitudinal waves in the steel rod.

Sol. Given, Young's modulus, $Y = 2.9 \times 10^{11} \text{ Nm}^{-2}$

Density, $\rho = 8 \times 10^3 \text{ kg m}^{-3}$

$\therefore$ Velocity of longitudinal waves in steel,

$$v = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2.9 \times 10^{11}}{8 \times 10^3}} \\ = 6.02 \times 10^3 \text{ ms}^{-1}$$

Example 17.3 The bulk modulus and modulus of rigidity for aluminium are $7.5 \times 10^{10} \text{ Nm}^{-2}$ and $2.1 \times 10^{10} \text{ Nm}^{-2}$, respectively. Determine the velocity of the waves in the medium. Density of aluminium is $2.7 \times 10^3 \text{ kg m}^{-3}$.

Sol. Given, bulk modulus, $B = 7.5 \times 10^{10} \text{ Nm}^{-2}$

Modulus of rigidity, $\eta = 2.1 \times 10^{10} \text{ Nm}^{-2}$

Density, $\rho = 2.7 \times 10^3 \text{ kg m}^{-3}$

Velocity of longitudinal waves in aluminium,

$$v = \sqrt{\frac{B + \frac{4}{3}\eta}{\rho}} = \sqrt{\frac{7.5 \times 10^{10} + \frac{4}{3} \times 2.1 \times 10^{10}}{2.7 \times 10^3}} \\ v = 6.18 \times 10^3 \text{ ms}^{-1}$$

Example 17.4 The volumetric strain of water at a pressure of 10^5 Nm^{-2} is 5×10^{-5} . Determine the speed of sound in water. Density of water is 10^3 kg m^{-3} .

Sol. Bulk modulus of water, $B = \frac{\text{Normal stress}}{\text{Volumetric strain}} = \frac{10^5}{5 \times 10^{-5}} \\ = 2 \times 10^9 \text{ Nm}^{-2}$

$\therefore$ Density of water, $\rho = 10^3 \text{ kg m}^{-3}$ (Given)

$\therefore$ Speed of sound in water, $v = \sqrt{\frac{B}{\rho}}$

$$\Rightarrow v = \sqrt{\frac{2 \times 10^9}{10^3}} = 1.41 \times 10^3 \text{ ms}^{-1}$$

Example 17.5 Determine the speed of sound waves in water and find the wavelength of a wave having a frequency of 242 Hz. (Take, $B_{\text{water}} = 2 \times 10^9 \text{ Pa}$)

Sol. Speed of sound wave in water,

$$v = \sqrt{\frac{B_{\text{water}}}{\rho}} = \sqrt{\frac{(2 \times 10^9)}{10^3}} = 1414 \text{ ms}^{-1}$$

$$\text{Wavelength, } \lambda = \frac{v}{f} = \frac{1414}{242} = 5.84 \text{ m}$$

Example 17.6 At normal temperature and pressure, the speed of sound in air is 332 m/s. What will be the speed of sound in hydrogen at normal temperature and pressure? (Air is 16 times heavier than hydrogen)

Sol. The speed sound in a gas is given by

$$v = \sqrt{\gamma p / \rho} \Rightarrow \frac{v_a}{v_H} = \sqrt{\frac{\rho_H}{\rho_a}}$$

$$\Rightarrow \frac{\rho_H}{\rho_a} = 1/16 \Rightarrow \frac{v_a}{v_H} = \sqrt{\frac{1}{16}} = \frac{1}{4}$$

$$\Rightarrow v_H = 4v_a = 4 \times 332 = 1328 \text{ m/s}$$

Example 17.7 The speed of sound in air at NTP is 332 ms^{-1} . Calculate the percentage error in speed of sound as calculated from Newton's formula. Given that the density of air at 1 atm pressure is 1.293 kgm^{-3} .

Sol. From Newton's formula, $v = \sqrt{\frac{p}{\rho}}$

$$\text{At NTP, } p = 1.01 \times 10^5 \text{ Pa} \Rightarrow v = \sqrt{\frac{1.01 \times 10^5}{1.293}} = 280 \text{ ms}^{-1}$$

$$\text{Difference in speed} = 332 - 280 = 52 \text{ ms}^{-1}$$

$$\% \text{ error} = \frac{52}{332} \times 100 = 15.7\%$$

Example 17.8 Find speed of sound in hydrogen gas at 27°C , if C_p/C_v for H_2 is 1.4. Gas constant, $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$.

Sol. The speed sound in a gas, $v = \sqrt{\gamma p / \rho}$

$$\therefore v = \sqrt{\frac{\gamma RT}{M}} \quad (\because pV = RT \text{ and } \rho = \frac{M}{V})$$

Here, $T = 27 + 273 = 300 \text{ K}$, $\gamma = \frac{C_p}{C_v} = 1.4$,

$$R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1},$$

$$M = 2 \times 10^{-3} \text{ kg mol}^{-1}$$

$$\therefore v = \sqrt{\frac{1.4 \times 8.31 \times 300}{2 \times 10^{-3}}} = 1321 \text{ ms}^{-1}$$

Note In the above formula, put $M = 2 \times 10^{-3} \text{ kg mol}^{-1}$. Do not put it 2 kg.

Example 17.9 At what temperature will the speed of sound in hydrogen be the same as in oxygen at 100°C ? Molar masses of oxygen and hydrogen are in the ratio 16 : 1.

Sol. As we know, $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v_{\text{H}_2} = v_{\text{O}_2}$

$$\therefore \sqrt{\frac{\gamma_{\text{H}_2} R T_{\text{H}_2}}{M_{\text{H}_2}}} = \sqrt{\frac{\gamma_{\text{O}_2} R T_{\text{O}_2}}{M_{\text{O}_2}}}$$

$$\gamma_{\text{H}_2} = \gamma_{\text{O}_2} \quad (\text{As both are diatomic})$$

$$\therefore T_{\text{H}_2} = \left(\frac{M_{\text{H}_2}}{M_{\text{O}_2}} \right) (T_{\text{O}_2}) = \left(\frac{1}{16} \right) (100 + 273)$$

$$= 23.31 \text{ K} \approx -249.7^\circ \text{C}$$

Factors affecting the speed of sound in air

The following factors affect the speed of sound in air (or gas)

1. Effect of temperature

As,
$$v = \sqrt{\frac{\gamma p}{\rho}}$$

We can write

$$v = \sqrt{\frac{\gamma RT}{M}} \quad (\because pV = RT)$$

where, M = molecular mass of gas, T is absolute temperature and R is gas constant.

$$\Rightarrow v \propto \sqrt{T}$$

or
$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

At STP, the temperature is 0°C or 273 K . If the speed of sound at 0°C is v_0 , its value at $T^\circ \text{C}$ will satisfy

$$\frac{v_t}{v_0} = \sqrt{\frac{273 + T}{273}} = \left(1 + \frac{T}{273} \right)^{1/2} = 1 + \frac{T}{546}$$

$$\therefore v_t = v_0 \left(1 + \frac{T}{546} \right) \quad (\text{For smaller value of } T)$$

If the speed of sound in air at 0°C (v_0) be taken as 332 ms^{-1} , then

$$v_t = 332 \left(1 + \frac{T}{546} \right)$$

or

$$v_t = 332 + 0.61T$$

Thus, the velocity of sound in air increases roughly by 0.61 ms^{-1} with per degree centigrade rise in temperature. It means sound travels faster in warm air.

Example 17.10 The density of air at NTP is 1.29 kg m^{-3} . Assume air to be diatomic with $\gamma = 1.4$. What will be the velocity of sound at 127°C ?

Sol. According to Laplace formula, speed of sound in air at NTP,

$$v = \sqrt{\frac{\gamma p}{\rho}} = \sqrt{\frac{1.4 \times 1.013 \times 10^5}{1.29}}$$

$$= 331.6 \text{ ms}^{-1}$$

As, the velocity of sound is proportional to the square root of absolute temperature.

$$\Rightarrow \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} \Rightarrow v_2 = v_1 \sqrt{\frac{T_2}{T_1}} = 331.6 \sqrt{\frac{273 + 127}{273 + 27}}$$

$$v_2 = 382.8 \text{ ms}^{-1}$$

Example 17.11 In given gaseous medium, at what temperature, the velocity of sound will be double that of velocity at 27°C .

Sol. Given, velocity of sound, $v_2 = 2v_1$

$$\text{Temperature, } T_1 = 27^\circ \text{C} = (27 + 273) = 300 \text{K}$$

$$\text{As, } v \propto \sqrt{T}$$

$$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{v_1}{2v_1} = \sqrt{\frac{300}{T_2}}$$

$$\Rightarrow T_2 = 1200 \text{K} \text{ or } T_2 = 927^\circ \text{C}$$

Example 17.12 At what temperature does the velocity of sound in air increases by 2% in comparison with velocity at 0°C ?

Sol. Let $t^\circ \text{C}$ be the required temperature, v_0 the speed of sound at 0°C and v_t the speed at $t^\circ \text{C}$.

$$\text{As, } v \propto \sqrt{T}$$

$$\Rightarrow \frac{v_t}{v_0} = \sqrt{\frac{t + 273}{273}}$$

$$\text{But } v_t = v_0 + \frac{2}{100} v_0 = v_0 (1 + 0.02) = (1.02)v_0$$

$$\therefore \frac{(1.02)v_0}{v_0} = \sqrt{\frac{t + 273}{273}}$$

$$t + 273 = (1.02)^2 \times 273 = 1.0404 \times 273$$

$$\Rightarrow t = 284.02 - 273 \Rightarrow t = 11.02^\circ \text{C}$$

2. Effect of pressure

The speed of sound in a gas, $v = \sqrt{\frac{\gamma p}{\rho}}$. From this formula, it appears that $v \propto \sqrt{p}$. But actually, it is not so.

Because $\frac{p}{\rho} = \frac{RT}{M} = \text{constant}$ at constant temperature.

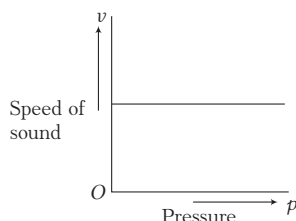


Fig. 17.6 Variation of pressure and speed of sound

Hence, **if the temperature of a gas remains constant, then there is no effect of pressure change on the speed of sound in the gas.**

Example 17.13 A sample of oxygen at NTP has volume V and a sample of hydrogen at NTP has volume $4V$. Both the gases are mixed and the mixture is maintained at NTP. If the speed of sound in hydrogen at NTP is 1270 ms^{-1} , then calculate the speed of sound in mixture.

Sol.
$$p_{\text{mix}} = \frac{p_{\text{O}_2}V_{\text{O}_2} + p_{\text{H}_2}V_{\text{H}_2}}{V_{\text{O}_2} + V_{\text{H}_2}} = \frac{16 \times V + 1 \times 4V}{V + 4V} = 4$$

As temperature remains same, therefore pressure will remain constant.

$$\therefore \frac{v_{\text{mix}}}{v_{\text{H}_2}} = \sqrt{\frac{p_{\text{H}_2}}{p_{\text{mix}}}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$$\Rightarrow v_{\text{mix}} = \frac{v_{\text{H}_2}}{2} = \frac{1270}{2} = 635 \text{ ms}^{-1}$$

3. Effect of humidity

The density of moist air (i.e. air mixed with water vapour) is less than the density of dry air. It is clear from the formula, $v = \sqrt{\gamma p / \rho}$ that the **speed of sound in moist air is slightly greater than that in dry air**. So, speed of sound increases with increase in humidity.

4. Effect of frequency

The speed of sound in air is independent of its frequency. Sound waves of different frequencies travel with the same speed in air but their wavelengths in air are different.

Example 17.14 A sound wave propagating in air has a frequency of 4000 Hz . Calculate the percentage change in wavelength when the wavefront initially in a region, where $T = 27^\circ \text{C}$, enters a region, where the temperature decreases to 10°C .

Sol.
$$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{273 + 10}{273 + 27}} = \sqrt{\frac{283}{300}} = 0.97$$

As, frequency remains unchanged, so

$$\frac{v_2}{v_1} = \frac{v\lambda_2}{v\lambda_1} = \frac{\lambda_2}{\lambda_1} = 0.97 \quad (\because v = v\lambda)$$

Percentage change in wavelength,

$$\frac{\lambda_1 - \lambda_2}{\lambda_1} \times 100 = \left(1 - \frac{\lambda_2}{\lambda_1}\right) \times 100 = (1 - 0.97) \times 100 = 3\%$$

Relation between speed of sound in a gas and root mean square speed of molecules

If sound travels in a gaseous medium, then velocity of

sound in gas will be
$$v_{\text{sound}} = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{\gamma p}{\rho}}$$

where, M is the molecular mass of the gas.

rms speed of gas molecule is given by,
$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \text{ or } \sqrt{\frac{3p}{\rho}}$$

So,
$$\frac{v_{\text{rms}}}{v_{\text{sound}}} = \sqrt{\frac{3}{\gamma}} \text{ or } v_{\text{sound}} = [\gamma / 3]^{1/2} v_{\text{rms}}$$

$$\therefore \sqrt{\frac{\gamma}{3}} < 1 \Rightarrow v_{\text{sound}} < v_{\text{rms}}$$

Hence, speed of sound in a gas is less than its rms speed of molecules.

Example 17.15 Compute the ratio of speed of sound in hydrogen gas to the rms speed of hydrogen molecules.

Sol. As hydrogen is a diatomic gas, $\gamma = \frac{7}{5}$

Speed of sound in hydrogen, $(v_{\text{sound}})_{\text{H}} = \sqrt{\frac{\gamma p}{\rho}}$

rms speed of hydrogen molecules, $(v_{\text{rms}})_{\text{H}} = \sqrt{\frac{3p}{\rho}}$

$$\therefore \frac{(v_{\text{sound}})_{\text{H}}}{(v_{\text{rms}})_{\text{H}}} = \sqrt{\frac{\gamma}{3}} = \sqrt{\frac{7/5}{3}} = \sqrt{\frac{7}{15}}$$

Characteristics of sound

Sound has following three characteristics

- (i) **Loudness** Loudness is related to the intensity $\left(= \frac{\text{energy}}{\text{time} \times \text{area}} \right)$ of sound and is measured in the unit 'bel' or decibel.

In decibels,
$$L_1 - L_2 = 10 \log_{10} \frac{i}{i_0}$$

where, i is the intensity of sound to be measured and i_0 is a constant reference intensity, i.e. 10^{-12} Wm^{-2} .

The reference intensity i_0 roughly represents the minimum intensity that is just audible at frequency around 1 kHz.

Some important points related to loudness are given below

- 'bel' is a large unit, so it is more commonly represented as 'decibel' (dB). As one metre is 10 decimeter, so 1 bel is 10 decibel. Hence, loudness in decibel will be represented as $10\log_{10}(i/i_0)$.
 - Loudness of a sound of given intensity may be different for different listeners. So, two sounds of equal intensity but different frequency may not appear to be equally loud even to the same listener because the sensitivity of the ear is different for different frequencies.
 - When intensity of a sound doubles, its loudness increases by almost 3dB and if the intensity quadruples, the loudness increases by 6dB and so on.
 - Some common sound levels in decibel are as follows.
Whispering is at around 10 decibel. Normal talk is at around 60 decibel. Car horn is at around 100 decibel and maximum tolerable sound is at around 120 decibel.
- (ii) **Pitch** Pitch is related to the frequency of sound. A shrill or sharp sound has high frequency, hence higher pitch and a grave or dull sound has low frequency, hence lower pitch. For example, the buzzing of a bee or humming of a mosquito has high pitch but low loudness while the roar of a lion has large loudness but lower pitch.
- (iii) **Quality** The property of sound due to which difference between two sounds can be made is quality of sound. A particular sound may have a number of frequencies and overtones attached to it, that defines its quality. We can recognise a person by listening to his sound as it has a definite quality.

Example 17.16 What is the intensity of sound of 70 decibel?
(Take, reference intensity, $i_0 = 10^{-12} \text{ Wm}^{-2}$)

Sol. Loudness (in decibel) $= 10\log_{10} \frac{i}{i_0}$
or $70 = 10\log_{10} \frac{i}{i_0}$ or $\frac{i}{i_0} = 10^7$
or $i = i_0 10^7 = 10^{-12} \times 10^7 = 10^{-5} \text{ Wm}^{-2}$

Example 17.17 A window whose area is 2 m^2 opens on a street, where the street noise results at the window with an intensity level of 60 dB. How much acoustic power enters the window through sound waves? Now, if a sound absorber is fitted at the window, how much energy from the street will it collect in a day?

Sol. By definition, sound level $= 10\log \frac{i}{i_0} = 60$ or $\frac{i}{i_0} = 10^6$

$$i = 10^{-12} \times 10^6 = 1\mu \text{ Wm}^{-2}$$

Power entering the room $= 1 \times 2 = 2\mu \text{ W}$

Energy collected in a day $= 2 \times 10^{-6} \times 86400 = 0.173 \text{ J}$

Example 17.18 A point source of sound emits a constant power with intensity inversely proportional to the square of the distance from the source. By how many decibels does the sound intensity level drop when you move from point P_1 to P_2 . Distance of P_2 from the source is two times the distance of source from P_1 .

Sol. Let the two points be labelled as 1 and 2. The difference in sound intensity level is given by

$$\begin{aligned}\beta_2 - \beta_1 &= (10 \text{ dB}) \left(\log \frac{i_2}{i_0} - \log \frac{i_1}{i_0} \right) \\ &= (10 \text{ dB}) [(\log i_2 - \log i_0) - (\log i_1 - \log i_0)] \\ &= (10 \text{ dB}) \log (i_2 / i_1)\end{aligned}$$

$$\text{Now, } i \propto \frac{1}{r^2} \quad (\text{Given})$$

$$\therefore \frac{i_2}{i_1} = \left(\frac{r_1}{r_2} \right)^2 = \frac{1}{4} \quad (\because r_2 = 2r_1)$$

$$\therefore \beta_2 - \beta_1 = (10 \text{ dB}) \log \left(\frac{1}{4} \right) = -6.0 \text{ dB}$$

Shock waves

If the speed of a body in air is greater than the speed of the sound, then it is called supersonic speed. Such a body leaves behind a conical region of disturbance (high energy pressure waves) which spreads continuously. Such a disturbance is called a shock waves.

As the source moves, it drags a cone with it. When an observer on ground is intercepted by the cone surface, a large sound is heard called **sonic boom**.

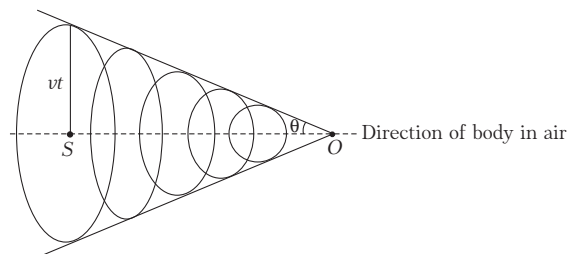


Fig. 17.7 Shock wave

This sound is so high that if it strikes a building it may damage it. Supersonic speed is measured in **Mach number**. One mach number is the ratio of speed of moving source to the speed of sound.

$$\text{Mach number} = \frac{\text{Speed of moving source}}{\text{Speed of sound}}$$

CHECK POINT 17.1

- The number of waves contained in unit length of the medium is called
(a) wave speed (b) wave number
(c) angular frequency (d) wavelength
- A line or surface on which the disturbance has the same phase at all points is called
(a) wavefront (b) wave train
(c) wave pulse (d) path difference
- The waves associated with the motion of microscopic particles such as electrons, protons, neutrons are called
(a) matter waves
(b) mechanical waves
(c) electromagnetic waves
(d) cosmic waves
- The relation between frequency n , wavelength λ and velocity v of propagation of wave is
(a) $n = v\lambda$ (b) $n = \frac{\lambda}{v}$ (c) $n = \frac{v}{\lambda}$ (d) $n = \frac{1}{v}$
- The speed of wave in a medium is 60 m/s. If 1200 waves are passing through a point in the medium in 1 min, then wavelength is
(a) 4.0 m (b) 6.0 m
(c) 3.0 m (d) 7.0 m
- Movement of string of a sitar or violin is an example of
(a) electromagnetic waves
(b) transverse waves
(c) matter waves
(d) longitudinal waves
- Vibrations of air column above the surface of water in the tube of a resonance apparatus are
(a) electromagnetic (b) transverse
(c) longitudinal (d) Both (b) and (c)
- Distance between a compression and adjoining rarefaction in a pressure wave is
(a) λ (b) $\frac{\lambda}{2}$ (c) $\frac{\lambda}{4}$ (d) $\frac{\lambda}{3}$
- Frequency range of the audible sound is
(a) 0 Hz-30 Hz (b) 20 Hz-20 kHz
(c) 20 kHz-20000 kHz (d) 20 kHz-20 MHz
- Ultrasonic waves can be used to
(a) detect submarines and icebergs
(b) clean clothes and fine machinery parts
(c) to kill smaller animals like rats, fish, frogs, etc.
(d) All of the above
- Sound waves transfer
(a) only energy not momentum
(b) energy
(c) momentum
(d) Both (b) and (c)
- Water waves produced by a motorboat sailing in water are
[NCERT Exemplar]
(a) Neither longitudinal nor transverse
(b) Both longitudinal and transverse
(c) only longitudinal
(d) only transverse
- The velocity of sound through a diatomic gaseous medium of molecular weight M at 0°C , is
(a) $\sqrt{\frac{R}{M}}$ (b) $\sqrt{\frac{3R}{M}}$ (c) $\sqrt{\frac{382R}{M}}$ (d) $\sqrt{\frac{273R}{M}}$
- A pulse of sound wave travels a distance l in helium gas in time T at a particular temperature. If at the same temperature, a pulse of sound wave is propagated in oxygen gas, it will cover the same distance l in time
(a) $4.36 T$ (b) $0.23 T$
(c) $3 T$ (d) $0.46 T$
- The ratio of the speed of sound in nitrogen gas to that in helium gas at 300 K is
(a) $\sqrt{\frac{2}{7}}$ (b) $\sqrt{\frac{1}{7}}$ (c) $\frac{\sqrt{3}}{5}$ (d) $\frac{\sqrt{6}}{5}$
- If the temperature of the gaseous medium drops by 1%, the velocity of sound in that medium
(a) increases by 5% (b) remains unchanged
(c) decreases by 0.5% (d) decreases by 2%
- When the temperature of an ideal gas is increased by 600 K, the velocity of sound in the gas become $\sqrt{3}$ times the initial velocity in it. The initial temperature of the gas is
(a) -73°C (b) 27°C (c) 127°C (d) 327°C
- A tuning fork of frequency 220 Hz produces sound waves of wavelength 1.5 m in air at STP. The increase in wavelength, when temperature of air is 27°C , is
(a) 0.07 m (b) 0.08 m
(c) 0.09 m (d) 0.10 m
- Change in temperature of the medium changes
(a) frequency of sound waves [NCERT Exemplar]
(b) amplitude of sound waves
(c) wavelength of sound waves
(d) loudness of sound waves
- Velocity of sound in air
I. increases with temperature
II. decreases with temperature
III. increases with pressure
IV. is independent of pressure
V. is independent of temperature
Choose the correct option.
(a) I and II are true (b) I and III are true
(c) II and III are true (d) I and IV are true
- What does not change when sound goes from one medium to another?
(a) Wavelength (b) Speed
(c) Frequency (d) None of these
- Speed of sound wave in air [NCERT Exemplar]
(a) is independent of temperature
(b) increases with pressure
(c) increases with increase in humidity
(d) decreases with increase in humidity
- In a medium, sound travels 2 km in 3 s and in air, it travels 3 km in 10 s. The ratio of the wavelengths of sound in the two media is
(a) 1 : 8 (b) 1 : 18
(c) 8 : 1 (d) 20 : 9

24. In a mixture of gases, the average number of degrees of freedom per molecules is 6. The rms speed of the molecule of the gas is c . The velocity of sound in the gas is
 (a) $\frac{c}{\sqrt{3}}$ (b) $\frac{c}{\sqrt{2}}$ (c) $\frac{2c}{3}$ (d) $\frac{3c}{4}$
25. The ratio of speed of source to the speed of sound is called
 (a) Reynold's number (b) Mach number
 (c) Complex number (d) None of these
26. Intensity level of a sound of intensity I is 30 dB. The ratio I/I_0 is (where, I_0 is the threshold of hearing)
 (a) 30 (b) 300
 (c) 1000 (d) 3000
27. A sound has an intensity of $2 \times 10^{-8} \text{ Wm}^{-2}$. Its intensity level in decibels is (Take, $\log_{10} 2 = 0.3$)
 (a) 23 (b) 4.3
 (c) 43 (d) None of these
28. How many times is 90 dB sound more intense than 40 dB sound?
 (a) 5 (b) 50
 (c) 500 (d) 10^5
29. The intensity level of a sound wave is 4 dB. If the intensity of the wave is doubled, then the intensity level of the sound, as expressed (in dB), would be
 (a) 8 (b) 16
 (c) 7 (d) 14
30. The pitch of a sound wave is related to its
 (a) frequency (b) amplitude
 (c) velocity (d) beats

PROGRESSIVE WAVE

A wave that travels from one point of the medium to another is called a progressive wave. It can be transverse or longitudinal.

Plane progressive harmonic wave

During the propagation of a wave through a medium, if the particles of the medium vibrate simple harmonically about their mean positions, then the wave is said to be plane progressive harmonic wave.

Displacement relation for a wave travelling in + x-direction is given by

$$y = A \sin \left\{ 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \pm \phi \right\}$$

while displacement relation for a wave travelling in - x-direction is given by

$$y = A \sin \left\{ 2\pi \left(\frac{t}{T} + \frac{x}{\lambda} \right) \pm \phi \right\}$$

In terms of speed of wave (v), above relation can be written as

$$\text{For + x-direction, } y = A \sin \left\{ \frac{2\pi}{\lambda} (vt - x) \pm \phi \right\}$$

$$\text{For -x-direction, } y = A \sin \left\{ \frac{2\pi}{\lambda} (vt + x) \pm \phi \right\}$$

In the above equation, v represents speed of wave and not that of the particle.

Other forms of displacement relation for progressive waves are as follows

$$\begin{aligned} y &= A \sin \{(\omega t \pm kx) \pm \phi\} \\ &= A \sin \{k(vt \mp x) \pm \phi\} \\ &= A \sin \left\{ \omega \left(t \mp \frac{x}{v} \right) \pm \phi \right\} \end{aligned}$$

Here, $\omega = \frac{2\pi}{T}$ is the angular frequency

$$\text{and } k = \frac{2\pi}{\lambda} = \frac{\omega}{v}$$

k is called **wave number** or **propagation constant**.

Note

(i) Equation of the form $y = f(at \pm bx)$ represents progressive or travelling waves.

Some common progressive wave equations are

- $y = A \log (at + bx)$ • $y = \sqrt{(ax + bt)}$
- $y = (ax - bt)^2$ • $y = A \sin (ax - bt)^2$
- $y = a \cos^2 (\omega t - kx)$
- $y = a \cos \Delta \omega t \sin (\omega t - kx)$

(ii) Equations like $y = A \sin (2x^2 - 3t^2)$

$$\text{or } y = A \sin (ax^2 + bt) \text{ or } y = \cos kx \sin \omega t$$

do not represent progressive waves.

(iii) Equation of the form $y = \frac{A}{r} \sin (\omega t - kr)$ represents **spherical**

progressive waves while equations of the form $y = \frac{A}{\sqrt{r}} \sin (\omega t - kr)$

represents **cylindrical progressive waves**.

(iv) A sound wave can be considered as pressure wave,

$p = p_0 \cos(\omega t - kx)$. The pressure wave is 90° out of phase w.r.t. displacement wave, so displacement will be maximum when pressure is minimum and vice-versa. The amplitude of pressure wave, $p_0 = BKA$, where B is bulk modulus, K is propagation constant $= 2\pi/\lambda$ and A is displacement amplitude.

Description of different terms used in displacement relation of plane progressive waves

- y is the **displacement** of medium particles from its mean position. It is measured in centimetre or metre.
- A represents the **amplitude** or **maximum displacement** of the medium particle from the mean position. It is measured in centimetre or metre.
- ω is known as **angular** or **circular frequency** of medium particle in oscillation under SHM. Its unit is radian per

second. $\omega = 2\pi\nu = \frac{2\pi}{T}$, where ν is **natural frequency**

given in hertz or per second or cycle per second, T is **time period** given in second.

- t is the time elapsed from when the wave begins its motion. It is measured in second.
- Negative sign between t and x indicates that the wave is travelling in positive x -direction.
- k is called **propagation constant** or **wave number** and given by $k = \frac{2\pi}{\lambda}$, where λ is the **wavelength**.
- x is the **position** of the particle with respect to the point of origin of wave at $t = 0$.
- The constant ϕ is called **phase constant** or **initial phase**.

Graphs of plane progressive harmonic wave

As the displacement relation for a wave travelling in $+x$ -direction is given by

$$y = A \sin \left\{ 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \pm \phi \right\}$$

or $y = A \sin(\omega t - kx) \pm \phi$... (i)

So at $t = 0$, the position of different particles at same instant of time in a string can be plotted by substituting $x = x$ and $t = 0$ in the Eq. (i), we get

$$y(x, t = 0) = A \sin kx = A \sin \left(\frac{2\pi}{\lambda} x \right)$$

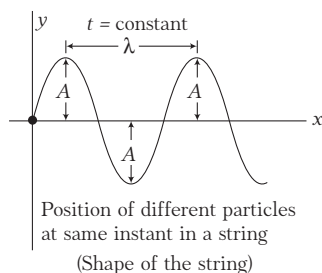


Fig. 17.8 Graph y versus x

For displacement of a particle at different instant of time, we can substitute $x = 0$ and $t = t$ in the Eq. (i), we get

$$y(x = 0, t) = A \sin \omega t = A \sin \frac{2\pi}{T} t$$

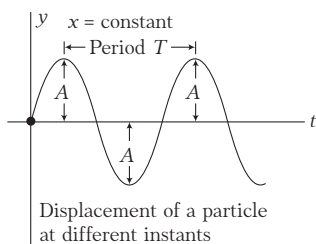


Fig. 17.9 Graph y versus t

Particle velocity and acceleration in plane progressive harmonic wave

As particle displacement, $y = A \sin(\omega t - kx)$

$$\therefore \text{Particle velocity, } v = \frac{dy}{dt} = A\omega \cos(\omega t - kx)$$

Again, differentiating the above equation, we get

Particle acceleration,

$$a = \frac{d^2 y}{dt^2} = -A\omega^2 \sin(\omega t - kx) = -\omega^2 y$$

Consider the motion of a string along X -axis. Its one end is connected to a source executing SHM of amplitude A , time period, $T = 2\pi/\omega$ and frequency of oscillation, $f = 1/T$. The wave produced by such a vibrating source is sine wave as shown in figure below.

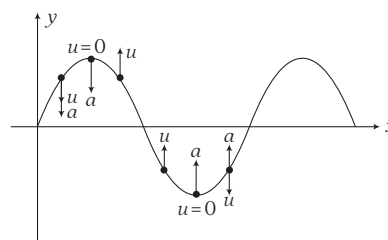


Fig. 17.10 Direction of particle velocity and acceleration

Now, slope of string, $\frac{dy}{dx} = -Ak \cos(\omega t - kx)$

$$\therefore \frac{dy/dt}{dy/dx} = -\frac{\omega}{k} \Rightarrow \frac{dy}{dt} = -v \frac{dy}{dx}$$

$$\Rightarrow \text{Particle velocity} = -\text{Wave velocity} \times \text{Slope}$$

$$\text{Similarly, } \frac{d^2 y}{dt^2} \times \frac{dx^2}{d^2 y} = \frac{\omega^2}{k^2} = v^2$$

$$\Rightarrow \boxed{\frac{d^2 y}{dt^2} = v^2 \frac{d^2 y}{dx^2}} \quad (\text{Wave equation})$$

The direction of particle velocity and acceleration at various position along the motion of the string is shown in the graph with the help of upward and downward arrows.

- If the slope of the curve is positive while displacement is positive, v_p is in downward direction (negative) and a_p will also be negative (downward).
- If slope of the curve is negative while displacement is positive. Hence, v_p will be positive (upwards) and a_p is negative (downwards).

Example 17.19 Equation of a transverse wave travelling in a rope is given by $y = 5 \sin(4.0t - 0.02x)$, where y and x are expressed in cm and time in seconds. Calculate

- the amplitude, frequency, velocity and wavelength of the wave.
- the maximum transverse velocity speed and acceleration of a particle in the rope.

Sol. (i) Comparing the given equation with the standard equation of wave motion, $y = A \sin \left(2\pi ft - \frac{2\pi}{\lambda} x \right)$

where, A , f and λ are amplitude, frequency and wavelength, respectively.

Thus, amplitude, $A = 5$ cm, $2\pi f = 4$

$$\Rightarrow \text{Frequency, } f = \frac{4}{2\pi} = 0.637 \text{ Hz}$$

$$\text{Again, } \frac{2\pi}{\lambda} = 0.02$$

$$\Rightarrow \text{Wavelength, } \lambda = \frac{2\pi}{0.02} = 100\pi \text{ cm}$$

$$\text{Velocity of the wave, } v = f\lambda = \frac{4}{2\pi} \frac{2\pi}{0.02} = 200 \text{ cm s}^{-1}$$

(ii) Transverse velocity of the particle,

$$u = \frac{dy}{dt} = 5 \times 4 \cos(4.0t - 0.02x) \\ = 20 \cos(4.0t - 0.02x)$$

Maximum velocity of the particle = 20 cm s^{-1}

Particle acceleration,

$$a = \frac{d^2y}{dt^2} = -20 \times 4 \sin(4.0t - 0.02x)$$

Maximum particle acceleration = 80 cm s^{-2} .

Example 17.20 The equation of a wave is

$$y(x, t) = 0.05 \sin \left(\frac{\pi}{2} (10x - 40t) - \frac{\pi}{4} \right) \text{ m}$$

Find

- the wavelength, frequency and wave velocity,
- the particle velocity and acceleration at $x = 0.5$ m and $t = 0.05$ s.

Sol. (i) The given equation may be rewritten as,

$$y(x, t) = 0.05 \sin \left(5\pi x - 20\pi t - \frac{\pi}{4} \right) \text{ m}$$

Comparing this with standard equation of plane progressive harmonic wave,

$$y(x, t) = A \sin(kx - \omega t + \phi), \text{ we get}$$

$$\text{Wave number, } k = \frac{2\pi}{\lambda} = 5\pi \text{ rad m}^{-1}$$

$$\therefore \lambda = 0.4 \text{ m}$$

Angular frequency,

$$\omega = 2\pi f = 20\pi \text{ rad s}^{-1} \Rightarrow f = 10 \text{ Hz}$$

Wave velocity,

$$v = f\lambda = \frac{\omega}{k} = 4 \text{ ms}^{-1} \text{ in } +x\text{-direction}$$

(ii) The particle velocity and acceleration at $x = 0.5$ m and $t = 0.05$ s are

$$\frac{dy}{dt} = -(20\pi)(0.05) \cos \left(\frac{5\pi}{2} - \pi - \frac{\pi}{4} \right) = 2.22 \text{ ms}^{-1}$$

$$\frac{d^2y}{dt^2} = -(20\pi)^2(0.05) \sin \left(\frac{5\pi}{2} - \pi - \frac{\pi}{4} \right) = 140 \text{ ms}^{-2}$$

Example 17.21 A simple harmonic wave of amplitude 8 units travels along positive X -axis. At any given instant of time, for a particle at a distance of 10 cm from the origin, the displacement is +6 units, and for a particle at a distance of 25 cm from the origin, the displacement is +4 units. Calculate the wavelength.

$$\text{Sol. As, } y = A \sin \frac{2\pi}{\lambda} (vt - x) \text{ or } \frac{y}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

$$\text{In the first case, } \frac{y_1}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{x_1}{\lambda} \right)$$

$$\text{Here, } y_1 = +6, A = 8 \text{ cm, } x_1 = 10 \text{ cm}$$

$$\therefore \frac{6}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right) \quad \dots(i)$$

Similarly, in the second case,

$$\frac{4}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right) \quad \dots(ii)$$

From Eq. (i), we get

$$2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right) = \sin^{-1} \left(\frac{6}{8} \right) = 0.85 \text{ rad}$$

$$\text{or } \frac{t}{T} - \frac{10}{\lambda} = 0.14 \quad \dots(iii)$$

Similarly, from Eq. (ii), we get

$$2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right) = \sin^{-1} \left(\frac{4}{8} \right) = \frac{\pi}{6} \text{ rad}$$

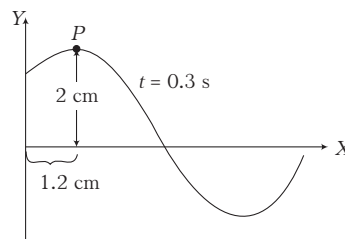
$$\text{or } \frac{t}{T} - \frac{25}{\lambda} = 0.08 \quad \dots(iv)$$

Subtracting Eq. (iv) from Eq. (iii), we get

$$\frac{15}{\lambda} = 0.06$$

$$\therefore \lambda = 250 \text{ cm}$$

Example 17.22 Figure shows a snapshot of a sinusoidal travelling wave taken at $t = 0.3$ s. The wavelength is 7.5 cm and the amplitude is 2 cm. If the crest P was at $x = 0$ at $t = 0$, write the equation of travelling wave.



Sol. Given, $A = 2$ cm, $\lambda = 7.5$ cm

$$\therefore k = \frac{2\pi}{\lambda} = 0.84 \text{ cm}^{-1}$$

The wave has travelled a distance of 1.2 cm in 0.3 s. Hence, speed of the wave,

$$v = \frac{1.2}{0.3} = 4 \text{ cm s}^{-1}$$

$\therefore$ Angular frequency, $\omega = (v)(k) = 3.36 \text{ rad s}^{-1}$

Since, the wave is travelling along positive x -direction and crest (maximum displacement) is at $x = 0$ at $t = 0$, we can write the wave equation as

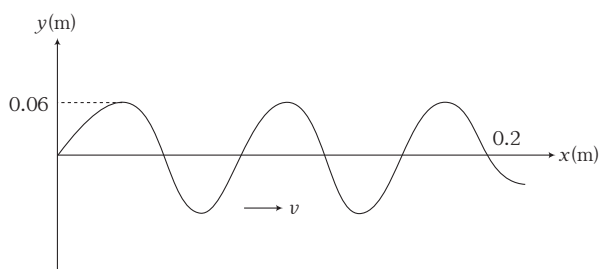
$$y(x, t) = A \cos(kx - \omega t)$$

or $y(x, t) = A \cos(\omega t - kx)$ [$\because \cos(-\theta) = \cos \theta$]

Therefore, the desired equation is

$$y(x, t) = 2 \cos(0.84x - 3.36 t) \text{ cm}$$

Example 17.23 For the wave shown in figure, write the equation of this wave, if its position is shown at $t = 0$. Speed of wave is $v = 300 \text{ ms}^{-1}$.



Sol. From figure, $\frac{5}{2} \lambda = 0.2 \text{ m}$

$$\therefore \lambda = 0.08 \text{ m}$$

$$\Rightarrow f = \frac{v}{\lambda} = \frac{300}{0.08} = 3750 \text{ Hz}$$

$$\text{Also, } k = \frac{2\pi}{\lambda} = 78.5 \text{ m}^{-1}$$

$$\text{and } \omega = 2\pi f = 23562 \text{ rad s}^{-1}$$

At $t = 0$, $x = 0$, $\frac{dy}{dx}$ = positive

and the given curve is a sine curve.

Hence, equation of wave travelling in positive x -direction should have the form,

$$y(x, t) = A \sin(kx - \omega t)$$

Substituting all the values in above equation, we get

$$y(x, t) = 0.06 \sin(78.5x - 23562 t) \text{ m}$$

Example 17.24 The displacement of a standing wave on a string is given by

$$y(x, t) = 0.4 \sin(0.5x) \cos(30t)$$

where, x and y are in centimetres.

(i) Find the frequency, amplitude and wave speed of the component waves.

(ii) What is the particle velocity at $x = 2.4 \text{ cm}$ and $t = 0.8 \text{ s}$?

Sol. (i) The given wave can be written as the sum of two component waves as

$$y(x, t) = 0.2 \sin(0.5x - 30t) + 0.2 \sin(0.5x + 30t)$$

The two component waves are

$$y_1(x, t) = 0.2 \sin(0.5x - 30t)$$

(Travelling in positive x -direction)

$$\text{and } y_2(x, t) = 0.2 \sin(0.5x + 30t)$$

(Travelling in negative x -direction)

Now, $\omega = 30 \text{ rad/s}$ and $k = 0.5 \text{ cm}^{-1}$

$$\therefore \text{Frequency, } f = \frac{\omega}{2\pi} = \frac{15}{\pi} \text{ Hz}$$

Amplitude, $A = 0.2 \text{ cm}$

$$\text{and wave speed, } v = \frac{\omega}{k} = \frac{30}{0.5} = 60 \text{ cm s}^{-1}$$

(ii) Particle velocity,

$$v_P(x, t) = \frac{dy}{dt} = -12 \sin(0.5x) \sin(30t)$$

$$\begin{aligned} \therefore v_P(x = 2.4 \text{ cm}, t = 0.8 \text{ s}) \\ = -12 \sin(1.2) \sin(24) \\ = 10.12 \text{ cm s}^{-1} \end{aligned}$$

Example 17.25 The equation of a sound wave in air is given by

$$p = (0.02 \text{ Nm}^{-2}) [\sin(500 \text{ s}^{-1})t - (3 \text{ m}^{-1})x]$$

(i) Find the frequency, wavelength and the speed of sound wave in air.

(ii) If the equilibrium pressure of air is $1.01 \times 10^5 \text{ Nm}^{-2}$, what are the maximum and minimum pressure at a point as the wave passes through that point?

Sol. The given equation is

$$p = 0.02 \sin(500t - 3x) \quad \dots(i)$$

The standard equation is $p = p_0 \sin(\omega t - kx)$ $\dots(ii)$

Comparing Eqs. (i) and (ii), we get

$$p_0 = 0.02 \text{ Nm}^{-2}, \omega = 500 \text{ rad s}^{-1}, k = 3 \text{ m}^{-1}$$

$$(i) \text{ Therefore, } f = \frac{\omega}{2\pi} = \frac{500}{2\pi} = \frac{250}{\pi} \text{ Hz}$$

$$k = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{2\pi}{k} = \frac{2\pi}{3} \text{ m}$$

$$\text{and } v = f\lambda = \frac{250}{\pi} \times \frac{2\pi}{3} = \frac{500}{3} \text{ ms}^{-1}$$

(ii) Here, pressure, $p_{\max} = p'_0 + p_0$

$$\begin{aligned} &= (1.01 \times 10^5 + 0.02) \text{ Nm}^{-2} \\ &= 101000.02 \text{ Nm}^{-2} \end{aligned}$$

$$\begin{aligned} \text{and } p_{\min} &= p'_0 - p_0 = (1.01 \times 10^5 - 0.02) \text{ Nm}^{-2} \\ &= 100999.98 \text{ Nm}^{-2} \end{aligned}$$

Here, p'_0 = atmospheric pressure

Example 17.26 Consider a wave propagating in the negative x -direction whose frequency is 100 Hz at $t = 5 \text{ s}$, the displacement associated with the wave is given by

$$y = 0.5 \cos(0.1x)$$

Here, x and y are measured in centimetres and t in seconds. Obtain the displacement (as a function of x) at $t = 10 \text{ s}$. What is the wavelength and velocity associated with the wave?

Sol. A wave travelling in negative x -direction can be represented as, $y(x, t) = A \cos(kx + \omega t + \phi)$

$$\text{At } t = 5 \text{ s, } y(x, t = 5) = A \cos(kx + 5\omega + \phi)$$

Comparing this with the given equation, we get

$$A = 0.5 \text{ cm}, k = 0.1 \text{ cm}^{-1} \text{ and } 5\omega + \phi = 0 \quad \dots(i)$$

$$\text{Now, } \lambda = \frac{2\pi}{k} = \frac{2\pi}{0.1} = 20\pi \text{ cm}$$

$$\omega = 2\pi f = 200\pi \text{ rad s}^{-1} \quad (\because f = 100 \text{ Hz})$$

$$\therefore v = \frac{\omega}{k} = \frac{200\pi}{0.1} = 2000\pi \text{ cm s}^{-1}$$

From Eq. (i), we get

$$\phi = -5\omega$$

$$\text{At } t = 10 \text{ s,}$$

$$y(x, t = 10) = 0.5 \cos(0.1x + 10\omega - 5\omega) \\ = 0.5 \cos(0.1x + 5\omega)$$

Substituting $\omega = 200\pi \text{ rad s}^{-1}$, we get

$$y(x, t = 10) = 0.5 \cos(0.1x + 1000\pi) \text{ cm}$$

Example 17.27 For a wave described by $y = A \sin(\omega t - kx)$, consider the following points

$$(i) x = 0 \quad (ii) x = \frac{\pi}{4k} \quad (iii) x = \frac{\pi}{2k} \quad (iv) x = \frac{3\pi}{4k}$$

For a particle at each of these points at $t = 0$, describe whether the particle is moving or not and in what direction and describe whether the particle is speeding up, slowing down or instantaneously not accelerating.

Sol. Given, $y = A \sin(\omega t - kx)$

Particle velocity, $v_p(x, t) = \frac{dy}{dt} = \omega A \cos(\omega t - kx)$ and particle acceleration,

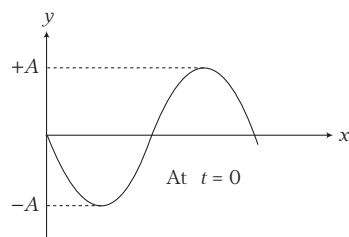
$$a_p(x, t) = \frac{d^2y}{dt^2} = -\omega^2 A \sin(\omega t - kx)$$

$$(i) t = 0 \text{ and } x = 0, v_p = +\omega A \text{ and } a_p = 0$$

i.e. Particle is moving upwards but its acceleration is zero.

Note Direction of velocity can be obtained in a different manner as under,

$$\text{At } t = 0, y = A \sin(-kx) = -A \sin kx$$



i.e. y - x graph is as shown in figure. At $x = 0$, slope is negative. Therefore, particle velocity is positive ($v_p = -v \times \text{slope}$) as the wave is travelling along positive x -direction.

$$(ii) t = 0, x = \frac{\pi}{4k} \Rightarrow kx = \frac{\pi}{4}$$

$$v_p = \omega A \cos\left(-\frac{\pi}{4}\right) = +\frac{\omega A}{\sqrt{2}}$$

$$\text{and } a_p = -\omega^2 A \sin\left(-\frac{\pi}{4}\right) = +\frac{\omega^2 A}{\sqrt{2}}$$

Velocity of particle is positive, i.e. the particle is moving upwards (along positive y -direction). Further, v_p and a_p are in the same direction (both are positive).

Hence, the particle is speeding up.

$$(iii) t = 0, x = \frac{\pi}{2k} \Rightarrow kx = \frac{\pi}{2}$$

$$v_p = \omega A \cos\left(-\frac{\pi}{2}\right) = 0$$

$$\text{and } a_p = -\omega^2 A \sin\left(-\frac{\pi}{2}\right) = \omega^2 A$$

i.e. particle is stationary or at its extreme position (i.e. $y = -A$). So, it is speeding up at this instant.

$$(iv) t = 0, x = \frac{3\pi}{4k} \Rightarrow kx = \frac{3\pi}{4}$$

$$\therefore v_p = \omega A \cos\left(-\frac{3\pi}{4}\right) = -\frac{\omega A}{\sqrt{2}}$$

$$a_p = -\omega^2 A \sin\left(-\frac{3\pi}{4}\right) = +\frac{\omega^2 A}{\sqrt{2}}$$

Velocity of particle is negative, i.e. the particle is moving downwards. Further, v_p and a_p are in opposite directions, i.e. the particle is slowing down.

Relation between the phase difference and path difference of a plane progressive wave

At any instant t , if ϕ_1 and ϕ_2 are the phases of two particles whose distances from the origin are x_1 and x_2 respectively, then

$$\phi_1 = (\omega t - kx_1) \text{ and } \phi_2 = (\omega t - kx_2) \\ \Rightarrow \phi_1 - \phi_2 = k(x_2 - x_1)$$

$$\text{Phase difference, } \Delta\phi = \frac{2\pi}{\lambda} (\text{Path difference, } \Delta x)$$

Example 17.28 If the phase difference between two waves is 60° , then find the value of path difference between them.

$$\text{Sol. Given, } \Delta\phi = 60^\circ = \frac{\pi}{3}$$

$$\text{As, } \Delta\phi = \frac{2\pi}{\lambda} \times \Delta x$$

$$\Rightarrow \Delta x = \frac{\Delta\phi}{2\pi} \times \lambda = \frac{\frac{\pi}{3} \times \pi}{3 \times 2\pi} \Rightarrow \Delta x = \frac{\lambda}{6}$$

Relation between phase difference and time difference of a plane progressive wave

If the phase of a particle distance x from the origin is ϕ_1 at time t_1 and ϕ_2 at time t_2 , then $\phi_1 = (\omega t_1 - kx)$ and

$$\phi_2 = (\omega t_2 - kx) \Rightarrow \phi_1 - \phi_2 = \omega(t_1 - t_2)$$

$$\text{Phase difference, } \Delta\phi = \frac{2\pi}{T} (\text{time difference, } \Delta t)$$

Example 17.29 The frequency of a progressive wave travelling in a medium is 40 Hz. Calculate the change in phase at a given place at 0.02 s.

Sol. Given, $\Delta t = 0.02$ s, $v = 40$ Hz and $\Delta t = 0.02$ s

Time period, $T = \frac{1}{v} = \frac{1}{40}$ s

Phase difference, $\Delta\phi = \frac{2\pi}{T} \times \Delta t = \left(\frac{1}{\frac{1}{40}}\right) \times 0.02$
 $= 2\pi \times 40 \times \frac{2}{100} = \frac{8\pi}{5}$ rad

Energy in wave

The energy is associated with every wave motion. Regarding the energy in wave motion, we come across three terms namely, energy density (u), power (P) and intensity (I). Let us consider them one-by-one.

1. Energy density (u)

The energy density is defined as the total mechanical energy (kinetic + potential) per unit volume of the medium through which the wave is passing.

Let the wave is $y = A \sin(\omega t - kx)$

So, kinetic energy per unit volume,

$$\Delta K = \frac{1}{2} \rho \omega^2 A^2 \cos^2(\omega t - kx)$$

Potential energy per unit volume,

$$\Delta U = \frac{1}{2} \rho \omega^2 A^2 \sin^2(\omega t - kx)$$

Total energy per unit volume,

$$\Delta E = \Delta K + \Delta U = \frac{1}{2} \rho \omega^2 A^2$$

Thus, energy density,

$$u = \langle \Delta E \rangle = \langle (\Delta K + \Delta U) \rangle$$

$$u = \frac{1}{2} \rho \omega^2 A^2$$

The following figure shows the graph between the motion of travelling wave in x -direction and displacement of medium particle at time t .

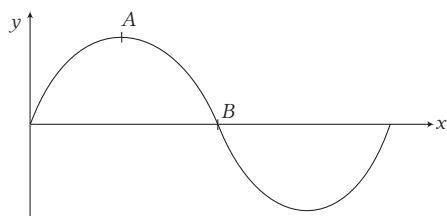


Fig. 17.11 Progressive wave

At A, $\frac{dy}{dx} = 0$, particle velocity, $\frac{dy}{dt} = -v \frac{dy}{dx} = 0$

$\therefore$ Kinetic energy, $K = 0$ and potential energy, $U = U_{\max}$

At B, $\frac{dy}{dx} = \text{maximum}$

$\therefore$ Kinetic energy, $K = K_{\max}$

and potential energy, $U = 0$

2. Power (P)

The instantaneous rate at which energy is transferred along the string, if we consider a transverse wave on a string, is called power.

In unit time, the wave will travel a distance v . If S be the area of cross-section of the string, then volume of this length would be SV . Thus,

Power, $P = \text{Energy density} \times \text{Volume} = \frac{1}{2} \rho \omega^2 A^2 \times SV$

$$P = \frac{1}{2} \rho \omega^2 A^2 SV$$

3. Intensity (I)

Flow of energy per unit area of cross-section of the string in unit time is known as intensity of the wave. Thus,

Intensity, $I = \frac{\text{Power}}{\text{Area of cross-section}} = \frac{P}{S}$

or $I = \frac{1}{2} \rho \omega^2 A^2 v$

Note The intensity of sound waves is given by $I = \frac{p_{\max}^2}{2\rho v}$

where, $p_{\max}$ is the maximum change of pressure in the medium.

The intensity of waves emitting in all directions due to a point source varies inversely as the square of the distance (r).

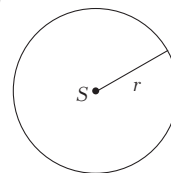


Fig. 17.12 Wave emitted by a sources

i.e. $I = \frac{P}{4\pi r^2}$ or $I \propto \frac{1}{r^2}$

The intensity of waves from a linear source varies inversely as the distance (r) perpendicular to the source, i.e. $I \propto \frac{1}{r}$.

Example 17.30 The faintest sound that the human ear can detect at frequency 1 kHz corresponds to an intensity of about 10^{-12} Wm^{-2} . Determine the pressure amplitude and the maximum displacement associated with this sound, assuming the density of the air = 1.3 kg m^{-3} and velocity of sound in air = 332 ms^{-1} .

Sol. Intensity of sound wave, $I = \frac{p^2}{2\rho v}$

$$\Rightarrow p = \sqrt{I \times 2\rho v} = \sqrt{10^{-12} \times 2 \times 1.3 \times 332}$$

$$= 2.94 \times 10^{-5} \text{ Nm}^{-2}$$

Now, $p = \rho v \omega A$

$$\Rightarrow A = \frac{p}{\rho v \omega} = \frac{2.94 \times 10^{-5}}{1.3 \times 332 \times 2\pi \times 10^3}$$

$$= 1.1 \times 10^{-11} \text{ m}$$

Example 17.31 For a person with normal hearing, the faintest sound that can be heard at a frequency of 400 Hz has a pressure amplitude of about $6.0 \times 10^{-5} \text{ Pa}$. Calculate the corresponding intensity (in Wm^{-2}). (Take, speed of sound in air is 344 m/s and density of air is 1.2 kg m^{-3})

Sol. Intensity, $I = \frac{p_m^2}{2\rho v}$

where, p_m is the pressure amplitude.

$$I = \frac{(6.0 \times 10^{-5})^2}{2 \times 1.2 \times 344} = 4.4 \times 10^{-12} \text{ Wm}^{-2}$$

Example 17.32 A stretched string is forced to transmit transverse waves by means of an oscillator coupled to one end. The string has a diameter of 4 mm. The amplitude of the oscillation is 10^{-4} m and the frequency is 10 Hz. Tension in the string is 100 N and mass density of wire is $4.2 \times 10^3 \text{ kg/m}^3$. Find

- the equation of the waves along the string,
- the energy per unit volume of the wave,
- the average energy flow per unit time across any section of the string, and

(iv) power required to drive the oscillator.

Sol. (i) Speed of transverse waves in the string, $v = \sqrt{\frac{T}{\rho S}}$

(As, $\mu = \rho S$)

Substituting the given values in above equation, we get

$$v = \sqrt{\frac{100}{(4.2 \times 10^3) (\pi/4) (4.0 \times 10^{-3})^2}} = 43.53 \text{ m/s}$$

$$\omega = 2\pi f = 20\pi \text{ rad/s} = 62.83 \text{ rad/s} \quad (\because f = 10 \text{ Hz})$$

$$k = \omega/v = 1.44 \text{ m}^{-1}$$

$\therefore$ Equation of the waves along the string,

$$y(x, t) = A \sin(kx - \omega t) = 10^{-4} \sin(1.44x - 62.83t) \text{ m}$$

(ii) Energy per unit volume of the string,

$$u = \text{energy density} = \frac{1}{2} \rho \omega^2 A^2$$

Substituting all the values in above equation, we get

$$u = \left(\frac{1}{2}\right) (4.2 \times 10^3) (62.83)^2 (10^{-4})^2$$

$$= 8.29 \times 10^{-2} \text{ Jm}^{-3}$$

(iii) Average energy flow per unit time,

$$P = \text{Power} = \left(\frac{1}{2} \rho \omega^2 A^2\right) (Sv) = (u) (Sv)$$

Substituting the values, we get

$$P = (8.29 \times 10^{-2}) \left(\frac{\pi}{4}\right) (4.0 \times 10^{-3})^2 (43.53)$$

$$= 4.53 \times 10^{-5} \text{ Js}^{-1}$$

(iv) Power required to drive the oscillator is obviously $4.53 \times 10^{-5} \text{ W}$.

CHECK POINT 17.2

1. The equation of wave is given by $y = A \sin \omega \left[\frac{x}{v} - k \right]$, where

ω is the angular velocity and v is the linear velocity. The dimensional formula of ω is

- [LT]
- [T]
- [T⁻¹]
- [T²]

2. A wave equation which gives the displacement along y -direction is given by $y = 10^{-4} \sin(60t + x)$, where x and y are in metre and t is time in second. This represents a wave

- travelling with a velocity of 300 ms^{-1} in the -ve x -direction
- of wavelength π metre
- of frequency $\frac{30}{\pi} \text{ Hz}$
- of amplitude 10^4 m travelling along the positive x -direction

3. If the equation of transverse wave is

$$y = 5 \sin 2\pi \left(\frac{t}{0.04} - \frac{x}{40} \right)$$

where, distance is in cm and time in second, then the wavelength of wave will be

- 20 cm
- 40 cm
- 60 cm
- None of these

4. A wave of amplitude $A = 0.2 \text{ m}$, velocity $v = 360 \text{ ms}^{-1}$ and wavelength 60 m is travelling along positive X -axis, then the correct expression for the wave is

- $y = 0.2 \sin 2\pi \left(6t + \frac{x}{60} \right)$
- $y = 0.2 \sin \pi \left(6t + \frac{x}{60} \right)$
- $y = 0.2 \sin 2\pi \left(6t - \frac{x}{60} \right)$
- $y = 0.2 \sin \pi \left(6t - \frac{x}{60} \right)$

5. The equation of a progressive wave is

$$y = 8 \sin \left[\pi \left(\frac{t}{10} - \frac{x}{4} \right) + \frac{\pi}{3} \right]$$

- 8 m
- 4 m
- 2 m
- 10 m

6. In a wave motion $y = a \sin(kx - \omega t)$, y can represent

- velocity
- acceleration
- Displacement
- All of these

7. In the equation $y = A \sin(kx - \omega t + \phi_0)$, the term phase is defined as

(a) ϕ_0 (b) $\phi_0 - \omega t$
(c) $kx + \phi_0$ (d) $kx - \omega t - \phi_0$

8. The equations of displacements of two waves are given as

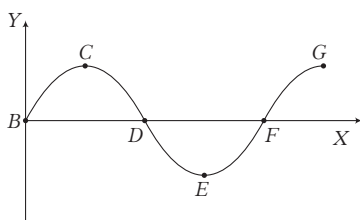
$$y_1 = 10 \sin \left(3\pi t + \frac{\pi}{3} \right)$$

$$y_2 = 5 (\sin 5\pi t + \sqrt{3} \cos 5\pi t)$$

Then, what is the ratio of their amplitudes?

(a) 1 : 2 (b) 2 : 1
(c) 1 : 1 (d) None of these

9. The diagram shows the propagation of a wave. Which points are in same phase?



(a) F and G (b) C and E
(c) B and G (d) B and F

10. A transverse wave is described by the equation

$$Y = Y_0 \sin 2\pi \left(ft - \frac{X}{\lambda} \right).$$

The maximum particle velocity is equal to four times the wave velocity, if

(a) $\lambda = \frac{\pi Y_0}{4}$ (b) $\lambda = \frac{\pi Y_0}{2}$
(c) $\lambda = \pi Y_0$ (d) $\lambda = 2\pi Y_0$

11. The equation of a transverse wave on a stretched string is

$$\text{given by } y = 0.05 \sin 2\pi \left(\frac{t}{0.002} - \frac{x}{0.1} \right),$$

where x and y are expressed in metre and t in second. The speed of the wave is

(a) 100 ms^{-1} (b) 50 ms^{-1}
(c) 200 ms^{-1} (d) 400 ms^{-1}

12. The frequency of a tuning fork with amplitude $A = 1 \text{ cm}$ is 250 Hz . The maximum velocity of any particle in air is equal to

(a) $10\pi \text{ ms}^{-1}$ (b) $5\pi \text{ ms}^{-1}$
(c) 2π (d) None of these

13. A progressive wave in a medium is represented by the equation

$$y = 0.1 \sin \left(10\pi t - \frac{5}{11} \pi x \right)$$

where, y and x are in cm and t is in second. The maximum speed of a particle of the medium due to the wave is

(a) 1 cms^{-1} (b) 10 cms^{-1}
(c) $\pi \text{ cms}^{-1}$ (d) $10 \pi \text{ cms}^{-1}$

14. A wave is represented by the equation

$$y = A \sin \left(10\pi x + 15\pi t + \frac{\pi}{3} \right)$$

where, x and y are in metre and t is in second. The expression represents

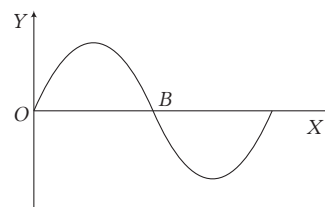
(a) a wave travelling in positive x -direction with a velocity 1.5 ms^{-1}
(b) a wave travelling in negative x -direction with a velocity 1.5 ms^{-1}
(c) a wave travelling in the negative x -direction having a wavelength 0.4 m
(d) a wave travelling in positive x -direction of wavelength 0.1 m

15. If the wave equation $y = 0.08 \sin \frac{2\pi}{\lambda} (200t - x)$, then the

velocity of the wave (ms^{-1}) will be

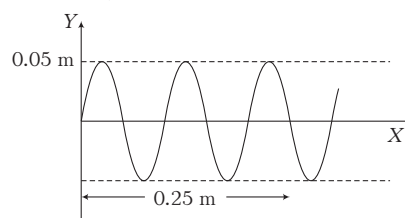
(a) $400\sqrt{2}$ (b) $200\sqrt{2}$ (c) 400 (d) 200

16. Figure shows the wave $y = A \sin(\omega t - kx)$. What is the magnitude of slope of the curve at B?



(a) $\frac{\omega}{A}$ (b) $\frac{k}{A}$
(c) kA (d) ωA

17. If the speed of the wave shown in the figure is 330 ms^{-1} in the given medium, then the equation of the wave propagating in the positive x -direction will be (all quantities are in MKS units)



(a) $y = 0.05 \sin 2\pi (4000t - 125x)$
(b) $y = 0.05 \sin 2\pi (4000t - 1225x)$
(c) $y = 0.05 \sin 2\pi (3300t - 10x)$
(d) $y = 0.05 \sin 2\pi (3300x - 10t)$

18. The phase difference between two points separated by 0.8 m in a wave of frequency 120 Hz is 0.5π . The velocity of wave will be

(a) 720 ms^{-1} (b) 384 ms^{-1}
(c) 256 ms^{-1} (d) 144 ms^{-1}

19. The distance between two points differing in phase by 60° on a wave having wave velocity 360 ms^{-1} and frequency 500 Hz is

(a) 0.36 m (b) 0.18 m
(c) 0.48 m (d) 0.12 m

20. Equation of a progressive wave is given by

$$y = 0.2 \cos \pi \left(0.04 t + 0.02x - \frac{\pi}{6} \right)$$

The distance is expressed in cm and time in second. What will be the minimum distance between two particles having the phase difference of $\frac{\pi}{2}$?

- (a) 4 cm (b) 8 cm
(c) 25 cm (d) 12.5 cm

21. A transverse wave $y = 0.05 \sin(20\pi x - 50\pi t)$ in metre, is propagating along +ve X -axis on a string. A light insect starts crawling on the string with the velocity of 5 cms^{-1} at $t = 0$ along the +ve X -axis from a point, where $x = 5 \text{ cm}$. After 5 s, the difference in the phase of its position is equal to

- (a) 150π (b) 250π (c) 10π (d) 5π

22. Two sources of sound A and B produce the wave of 350 Hz, in the same phase. The particle P is vibrating under the

influence of these two waves, if the amplitudes at the point P produced by the two waves are 0.3 mm and 0.4 mm, then the resultant amplitude of the point P , when $AP - BP = 25 \text{ cm}$ and the velocity of sound is 350 ms^{-1} , will be

- (a) 0.7 mm (b) 0.1 mm
(c) 0.2 mm (d) 0.5 mm

23. Two waves represented by the following equations are travelling in the same medium

$$y_1 = 5 \sin 2\pi (75t - 0.25x),$$

$$y_2 = 10 \sin 2\pi (150t - 0.50x).$$

The intensity ratio $\frac{I_1}{I_2}$ of the two waves is

- (a) 1 : 2 (b) 1 : 4
(c) 1 : 8 (d) 1 : 16

PRINCIPLE OF SUPERPOSITION OF WAVES

Two or more waves can travel simultaneously in a medium without affecting the motion of one another. The resultant displacement of each particle of the medium at any instant is equal to the vector sum of the individual displacements produced by the two waves. This principle is called principle of superposition.

If $y_1, y_2, y_3, \dots$ are the displacements of particle at a particular time due to individual waves, then the resultant displacement is given by

$$y = y_1 + y_2 + \dots$$

Consider the superposition of two sinusoidal waves of same frequency at a point. Let us assume that the two waves are travelling in the same direction with same velocity. The equation of the two waves reaching at a point can be written as

$$y_1 = A_1 \sin(kx - \omega t)$$

and $y_2 = A_2 \sin(kx - \omega t + \phi)$

The resultant displacement of the point, where the waves meet is

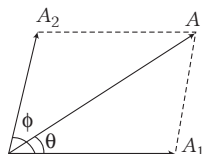


Fig. 17.13 Resultant displacement

$$y = y_1 + y_2 = A_1 \sin(kx - \omega t) + A_2 \sin(kx - \omega t + \phi)$$

$$= A_1 \sin(kx - \omega t) + A_2 \sin(kx - \omega t) \cos \phi$$

$$+ A_2 \cos(kx - \omega t) \sin \phi$$

$$= (A_1 + A_2 \cos \phi) \sin(kx - \omega t) + A_2 \sin \phi \cos(kx - \omega t)$$

Substituting $A_1 + A_2 \cos \phi = A \cos \theta$

and $A_2 \sin \phi = A \sin \theta$ in above equation, we get

$$\therefore y = A \cos \theta \sin(kx - \omega t) + A \sin \theta \cos(kx - \omega t)$$

or $y = A \sin(kx - \omega t + \theta)$

or $A^2 = (A_1 + A_2 \cos \phi)^2 + (A_2 \sin \phi)^2$

or $A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \phi}$

Example 17.33 Two harmonic waves are represented in SI units by

$$y_1(x, t) = 0.2 \sin(x - 3.0t)$$

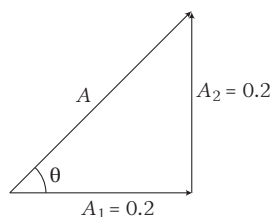
and $y_2(x, t) = 0.2 \sin(x - 3.0t + \phi)$

(i) Write an expression for the sum $y = y_1 + y_2$ for $\phi = \pi/2 \text{ rad}$.

(ii) Suppose the phase difference ϕ between the waves is unknown and the amplitude of their sum is 0.32 m, what is ϕ ?

Sol. (i) $y = y_1 + y_2 = 0.2 \sin(x - 3.0t) + 0.2 \sin\left(x - 3.0t + \frac{\pi}{2}\right)$

$$= A \sin(x - 3.0t + \theta)$$



Here, $A = \sqrt{(0.2)^2 + (0.2)^2} = 0.28 \text{ m}$ and $\theta = \frac{\pi}{4}$

$$\therefore y = 0.28 \sin\left(x - 3.0t + \frac{\pi}{4}\right)$$

(ii) Since, the amplitude of the resulting wave is 0.32 m and $A = 0.2 \text{ m}$, we get

$$0.32 = \sqrt{(0.2)^2 + (0.2)^2 + (2)(0.2)(0.2) \cos \phi}$$

Solving this, we get $\phi = \pm 1.29 \text{ rad}$

Interference of waves

When two coherent waves of same frequency propagate in same direction and superimpose on each other, then the intensity of resultant wave becomes maximum at some points and minimum at some other points, this phenomenon of intensity variation is called **interference**.

If two sinusoidal waves S_1 and S_2 of same angular frequency ω meet at a point B , where phase difference between them is ϕ and path difference is Δx as shown in figure, then according to the principle of superposition, the resulting amplitude is given by

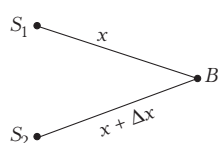


Fig. 17.14 Path difference

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2 \cos \phi \quad \dots(i)$$

As, intensity of a wave is given by

$$I = \frac{1}{2} \rho A^2 \omega^2 v, \text{ i.e. } I \propto A^2$$

So, if ρ , ω and v are same for both interfering waves, Eq. (i) can also be written as

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi \quad \dots(ii)$$

From Eqs. (i) and (ii), we see that the resulting amplitude A and intensity I depends on the phase difference ϕ between the interfering waves. If the resultant wave has maximum amplitude, then the interference is called **constructive interference**.

For maximum amplitude, $\cos \phi = +1$

or $\phi = 0, 2\pi, \dots, 2n\pi$

Resultant amplitude, $A = A_{\max} = A_1 + A_2$

$\therefore$ Resultant intensity, $I = I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$

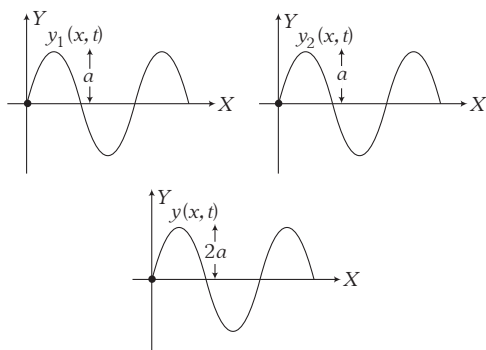


Fig. 17.15 Graphs for constructive interference

If the resultant wave has minimum amplitude, then the interference is called **destructive interference**.

For minimum amplitude, $\cos \phi = -1$

or

$$\phi = \pi, 3\pi, \dots, 2(n+1)\pi$$

Resultant amplitude, $A = A_{\min} = A_1 - A_2$

$\therefore$ Resultant intensity, $I = I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$

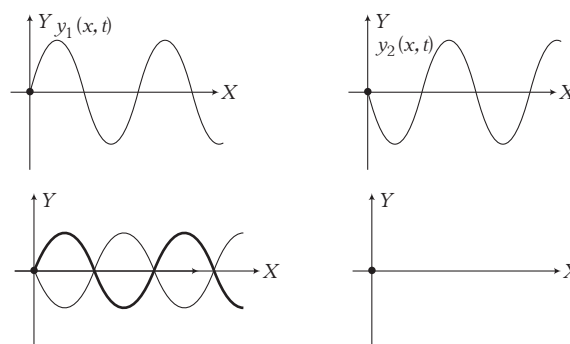


Fig. 17.16 Graph of destructive interference

The graph shown below represents the variation of resultant intensity, I with $(I_1 + I_2)$, i.e. sum of individual intensities.

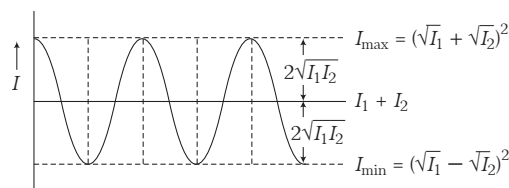


Fig. 17.17 Graph of resultant intensity

From this graph, it is clear that $I_{\max}$ exceeds the sum of individual intensities $(I_1 + I_2)$ by $2\sqrt{I_1 I_2}$ while $I_{\min}$ lacks $(I_1 + I_2)$ by $2\sqrt{I_1 I_2}$, i.e. in interference, energy is neither created nor be destroyed but is redistributed.

Note Detailed discussion on interference will be done in the chapter of wave optics.

Example 17.34 Two waves of equal frequencies have their amplitudes in the ratio of 3 : 5. They are superimposed on each other. Calculate the ratio of maximum and minimum intensities of the resultant wave.

Sol. Given, $\frac{A_1}{A_2} = \frac{3}{5}$

$$\therefore \frac{\sqrt{I_1}}{\sqrt{I_2}} = \frac{3}{5} \quad (\text{As, } I \propto A^2)$$

Maximum intensity is obtained, where

$$\cos \phi = 1 \text{ and } I_{\max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

Minimum intensity is obtained, where

$$\cos \phi = -1 \quad \text{and} \quad I_{\min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\begin{aligned} \text{Hence,} \quad \frac{I_{\max}}{I_{\min}} &= \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2 = \left(\frac{\sqrt{I_1/I_2} + 1}{\sqrt{I_1/I_2} - 1} \right)^2 \\ &= \left(\frac{3/5 + 1}{3/5 - 1} \right)^2 = \frac{64}{4} = \frac{16}{1} \end{aligned}$$

Example 17.35 Two coherent sound sources are at distances $x_1 = 0.2 \text{ m}$ and $x_2 = 0.48 \text{ m}$ from a point. Calculate the intensity of the resultant wave at that point, if the frequency of each wave is $f = 400 \text{ Hz}$ and velocity of wave in the medium is $v = 448 \text{ ms}^{-1}$. The intensity of each wave is $I_0 = 60 \text{ Wm}^{-2}$.

Sol. Path difference, $\Delta x = x_2 - x_1 = 0.48 - 0.2 = 0.28 \text{ m}$

$$\begin{aligned} \text{Now, phase difference, } \phi &= \frac{2\pi}{\lambda} \Delta x = \left(\frac{2\pi f}{v} \right) \Delta x \\ &= \frac{2\pi (400) (0.28)}{448} = \frac{\pi}{2} \end{aligned}$$

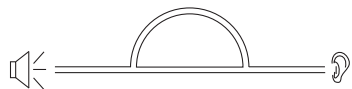
Now, intensity of resultant wave,

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

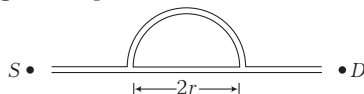
$$\begin{aligned} \text{or} \quad I &= I_0 + I_0 + 2I_0 \cos \left(\frac{\pi}{2} \right) \\ &= 2I_0 = 2(60) = 120 \text{ Wm}^{-2} \end{aligned}$$

Example 17.36 Figure shows a tube structure in which a sound signal is sent from one end and received at the other end. The semicircular part has a radius of 20.0 cm . The frequency of the sound source can be varied electronically between 1000 and 4000 Hz .

Find the frequencies at which maxima of intensity are detected. The speed of sound in air $= 340 \text{ m/s}$.



Sol. According to the question,



$$\text{Path difference, } \Delta x = \pi r - 2r = (\pi - 2)r$$

For maximum intensity,

$$\Delta x = n\lambda$$

$$\Rightarrow (\pi - 2)r = n\lambda = \frac{nv}{f}$$

$$\begin{aligned} \Rightarrow f &= \frac{nv}{(\pi - 2)r} = \frac{n \times 340}{(3.14 - 2) \times 0.2} \\ &= 1491n \text{ Hz} \end{aligned}$$

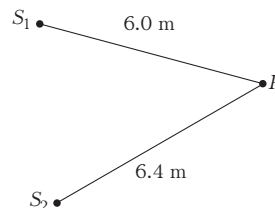
If $n = 1$, then $f = 1491 \text{ Hz}$

If $n = 2$, then $f = 2982 \text{ Hz}$

If $n = 3$, then $f = 4473 \text{ Hz}$

The frequencies in the given range are 1491 Hz and 2982 Hz .

Example 17.37 Two sources are placed from a person P as shown in figure. The speed of sound in air is 320 m/s .



If sound signal is continuously varied from 500 Hz to 2500 Hz , for which frequency listener will hear minimum sound intensity?

Sol. Path difference, $\Delta x = S_2P - S_1P = 6.4 - 6.0 = 0.4 \text{ m}$

For minimum sound intensity,

$$\Delta x = (2n + 1) \frac{\lambda}{2} = \frac{(2n + 1) v}{2 f}$$

$$\Rightarrow f = \frac{(2n + 1)v}{2\Delta x} = \frac{(2n + 1)320}{2(0.4)} = 400(2n + 1) \text{ Hz}$$

where, $n = 0, 1, 2, \dots$

If $n = 0$, then $f = 400 \text{ Hz}$

If $n = 1$, then $f = 1200 \text{ Hz}$

If $n = 2$, then $f = 2000 \text{ Hz}$

If $n = 3$, then $f = 2800 \text{ Hz}$

The frequencies in the given range are 1200 Hz and 2000 Hz .

Beats

When two sound waves of same amplitude travelling in same direction with slightly different frequencies superimpose, the intensity varies periodically with time, this phenomenon is called beats.

A complete cycle of maximum intensity and minimum intensity is called one beat and number of beats heard per second is called **beat frequency**. Beat frequency is equal to the difference of frequencies of two interfering waves.

Let us consider two harmonic waves of nearly equal angular frequencies ω_1 and ω_2 . Further, let the waves have equal amplitude and zero phase difference. Let the initial phase angle for each waves be $\phi = \frac{\pi}{2}$.

Then, at $x = 0$, the two waves can be given by

$$S_1 = a \cos \omega_1 t \quad \text{and} \quad S_2 = a \cos \omega_2 t$$

Symbol S refers to longitudinal displacement and not the transverse one.

Now, according to principle of superposition, the resultant displacement is $S = S_1 + S_2$.

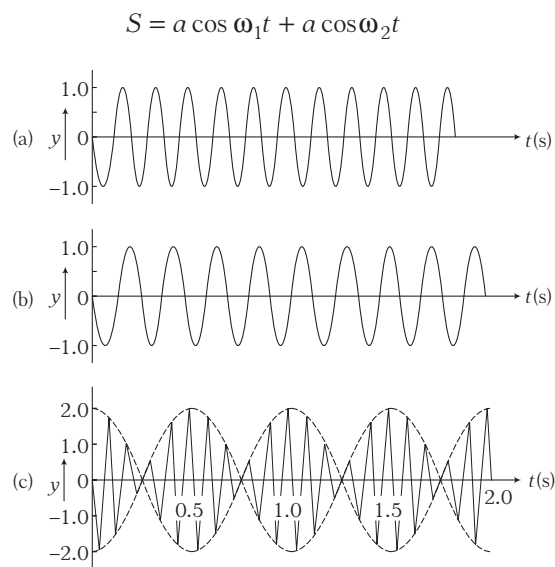


Fig. 17.18 Resultant wave

Using the identity

$$\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right),$$

$$S = 2a \cos \left(\frac{\omega_1 - \omega_2}{2} t \right) \cos \left(\frac{\omega_1 + \omega_2}{2} t \right)$$

If we write $\omega_b = \left(\frac{\omega_1 + \omega_2}{2} \right)$ and $\omega_a = \left(\frac{\omega_1 - \omega_2}{2} \right)$

$$\therefore S = [2a \cos \omega_a t] \cos \omega_b t \quad \dots(i)$$

Here, the resultant wave is a cosine function whose amplitude is $(2a \cos \omega_a t)$. The amplitude is maximum when $\cos \omega_a t$ has the value +1 or -1, which happens twice in each repetition of the cosine function.

Therefore, angular frequency of beat,

$$\omega_{\text{beat}} = 2\omega_a = 2 \left(\frac{\omega_1 - \omega_2}{2} \right)$$

$$\Rightarrow \omega_{\text{beat}} = \omega_1 - \omega_2$$

Also, $\omega = 2\pi f$

$$\therefore \boxed{v_{\text{beat}} = f_1 - f_2}$$

The reciprocal of beat frequency is beat period given by

$$\boxed{T = \frac{1}{f} = \frac{1}{f_1 - f_2}}$$

Essential condition for hearing beats

For beats to be audible, the difference in the frequency of the two sound waves should not exceed 10. If the difference is more than 10, we shall hear more than 10 beats per second. But due to persistence of hearing, our ear is not able to distinguish between two sounds as

separate, if the time interval between them is less than $(1/10)$ th of a second.

Hence, beats heard will not be distinct, if the number of beats produced per second is more than 10.

Example 17.38 The string of a violin emits a note of 400 Hz at its correct tension. The string is bit taut and produces 5 beats per second with a tuning fork of frequency 400 Hz. Find frequency of the note emitted by this taut string.

Sol. The frequency of vibration of a string increases with increase in the tension. Thus, the note emitted by the string will be a little more than 400 Hz. As it produces 5 beats per second with the 400 Hz tuning fork, the frequency will be 405 Hz.

Example 17.39 Consider the two identical piano strings, each tuned exactly to the 420 Hz. The tension in any one of the strings is increased by 2.0%. If they are now struck, what is the beat frequency between the fundamentals of the two strings? (Take, length of the strings = 65 cm)

Sol. If v_1, v_1, T_1 and v_2, v_2 and T_2 are the frequencies, velocities and tensions in the first and second strings respectively, then

$$\frac{v_2}{v_1} = \frac{v_2 / 2L}{v_1 / 2L} = \frac{v_2}{v_1}$$

$$\Rightarrow \frac{v_2}{v_1} = \frac{\frac{1}{2L} \sqrt{T_2 / \mu}}{\frac{1}{2L} \sqrt{T_1 / \mu}} = \sqrt{\frac{T_2}{T_1}}$$

Since, it is given that the tension in one string is 2% larger than the other.

$$\therefore T_2 = T_1 + \frac{2T_1}{100} = 1.02 T_1$$

$$\therefore \frac{v_2}{v_1} = \sqrt{\frac{1.02 T_1}{T_1}} = 1.01$$

Now, the frequency of the tightened string,

$$v_2 = v_1(1.01) = 1.01 \times 420 = 424.2 \text{ Hz}$$

$\therefore$ Beat frequency,

$$v_{\text{beat}} = v_2 - v_1 = 424.2 - 420 = 4.2 \text{ Hz}$$

Practical applications of beats

Practical applications of beats are given below

(i) Determination of frequency of tuning fork

Suppose v_1 is the known frequency of tuning fork A and v_2 is the unknown frequency of tuning fork B. When the two tuning forks are sounded together, suppose they produce b beats per second.

Then,

$$\boxed{v_2 = v_1 + b \text{ or } v_1 - b}$$

The exact frequency may be determined by any of the following two methods

Loading method Attach a little wax to the prong of the tuning fork *B*, i.e. decrease in its frequency. Again, find the number of beats produced per second.

If the beat frequency decreases on loading the prong of the tuning fork of unknown frequency, then the unknown frequency is greater than the known frequency,

$$\text{i.e.} \quad \nu_2 = \nu_1 + b$$

On the other hand, if the beat frequency increases on loading the prong of the tuning fork of unknown frequency, then the unknown frequency is less than the known frequency,

$$\text{i.e.} \quad \nu_2 = \nu_1 - b$$

Filing method If a prong of the tuning fork *B* is filled, its frequency increases. Again, note the **number** of beats produced per second.

If on filing the prong of *B*, the beat frequency decreases, then

$$\nu_2 = \nu_1 - b$$

(ii) Utilisation in electronics

Since, it is difficult to make low frequency oscillators. In practice, two high frequency oscillators with a small difference in their frequencies are used. Their low beat frequency ($\nu_1 - \nu_2$) serves the purpose of low frequency oscillator.

Example 17.40 A tuning fork produces 3 beats per second when sounded together with a fork of frequency 364 Hz. When the first fork is loaded with a little wax, then the number of beats becomes two per second. What is the frequency of the first fork?

Sol. Number of beats, $\nu = \pm (\nu_1 - \nu_2)$

$$\begin{aligned} \Rightarrow \quad \nu_1 &= \nu_2 + \nu \text{ or } \nu_2 - \nu \\ \Rightarrow \quad \nu_1 &= 364 + 3 = 367 \text{ Hz} \\ \text{or} \quad \nu_1 &= 364 - 3 = 361 \text{ Hz} \end{aligned}$$

Loading a fork with wax decreases its frequency.

On loading the first fork, the number of beats produced per second decreases, therefore $\nu_1 = 367 \text{ Hz}$.

Example 17.41 Two tuning forks *P* and *Q* when set vibrating, give 4 beats/s. If a prong of the fork *P* is filled, the beats are reduced to 2 s^{-1} . Determine the frequency of *P*, if that of *Q* is 250 Hz.

Sol. There are four beats between *P* and *Q*, therefore the possible frequencies of *P* are 246 Hz or 254 Hz (i.e. 250 ± 4) Hz.

When the prong of *P* is filled, its frequency becomes greater than the original frequency.

If we assume that the original frequency of *P* is 254 Hz, then on filing, its frequency will be greater than 254 Hz. The beats between *P* and *Q* will be more than 4. But it is given that the beats are reduced to 2, therefore 254 Hz is not possible.

Therefore, the required frequency must be 246 Hz.

(This is true because on filing, the frequency may increase to 248 Hz, giving 2 beats with *Q* of frequency 250 Hz).

Example 17.42 A column of air and a tuning fork produces 4 beats/s. When sounding together, the tuning fork gives the lower note. The temperature of air is 15°C . When the temperature falls to 10°C , the two produce 3 beats/s. Find the frequency of the tuning fork.

Sol. The frequency of the air column is given by $f = v/\lambda$, where v is the velocity of sound in air and λ is the wavelength.

But $v = \sqrt{\frac{\gamma RT}{M}}$, thus it is dependent of temperature.

$$\Rightarrow \quad f \propto \sqrt{T} \quad (\because v = \lambda f)$$

Let the frequency of the tuning fork be f .

Thus, the frequency of the air column at $15^\circ \text{C} = f + 4$

and the frequency of the air column at $10^\circ \text{C} = f + 3$

As, the frequency decreases with decrease in temperature,

$$\Rightarrow \quad \frac{f+4}{f+3} = \sqrt{\frac{288}{283}} \Rightarrow \frac{f+4}{f+3} = 1.00879$$

$$\Rightarrow \quad f + 4 = 1.00879f + 3.02638$$

$$\Rightarrow \quad 8.79 \times 10^{-3} f = 0.97362 \Rightarrow f = 110.76 \text{ Hz}$$

Example 17.43 Wavelengths of two notes in air are $\left(\frac{90}{175}\right) \text{ m}$ and $\left(\frac{90}{173}\right) \text{ m}$. Each note produces 4 beats/s with a third note of a fixed frequency. Calculate the velocity of sound in air.

Sol. Given, $\lambda_1 = \frac{90}{175} \text{ m}$ and $\lambda_2 = \frac{90}{173} \text{ m}$

Let f_1 and f_2 be the corresponding frequencies and v be the velocity of sound in air.

$$v = \lambda_1 f_1 \text{ and } v = \lambda_2 f_2$$

$$\therefore \quad f_1 = \frac{v}{\lambda_1} \text{ and } f_2 = \frac{v}{\lambda_2}$$

Since, $\lambda_2 > \lambda_1$

$$\therefore \quad f_1 > f_2$$

Let f be the frequency of the third note.

$$\therefore \quad f_1 - f = 4$$

$$\text{and} \quad f - f_2 = 4$$

$$\therefore \quad f_1 - f_2 = 8$$

$$\therefore \quad \frac{v}{\lambda_1} - \frac{v}{\lambda_2} = 8$$

$$\Rightarrow \quad v \left(\frac{175}{90} - \frac{173}{90} \right) = 8$$

$$\Rightarrow \quad v \left(\frac{2}{90} \right) = 8$$

Therefore, $v = 360 \text{ ms}^{-1}$

CHECK POINT 17.3

- Law of superposition is applicable to
 - light waves only
 - sound waves only
 - transverse waves only
 - All kinds of waves
- If two waves of same frequency and same amplitude, on superimposition produced a resultant disturbance of the same amplitude, the waves differ in phase by
 - π
 - $\frac{2\pi}{3}$
 - $\frac{\pi}{2}$
 - zero
- If two waves having amplitudes $2A$ and A and same frequencies and velocities, propagate in the same direction in the same phase, the resulting amplitude will be
 - $3A$
 - $\sqrt{5}A$
 - $\sqrt{2}A$
 - A
- The displacement of the interfering sound waves are $y_1 = 4\sin\omega t$ and $y_2 = 3\sin\left(\omega t + \frac{\pi}{2}\right)$. What is the amplitude of the resultant wave?
 - 5
 - 7
 - 1
 - 0
- If the phase difference between the two waves is 2π during superposition, then the resultant amplitude is
 - maximum
 - minimum
 - maximum or minimum
 - None of these
- The periodic waves of amplitudes A_1 and A_2 pass through a region. If $A_1 > A_2$, the difference in the maximum and minimum resultant amplitudes possible is
 - $2A_1$
 - $2A_2$
 - $A_1 + A_2$
 - $A_1 - A_2$
- Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction, the sum of the maximum and minimum intensities is
 - $(\sqrt{I_1} - \sqrt{I_2})^2$
 - $2(I_1 + I_2)$
 - $I_1 + I_2$
 - $(\sqrt{I_1} + \sqrt{I_2})^2$
- The superposing waves are represented by the following equations, $y_1 = 5\sin 2\pi(10t - 0.1x)$
 $y_2 = 10\sin 2\pi(20t - 0.2x)$
 Ratio of maximum and minimum intensities will be
 - 1
 - 9
 - 4
 - 3
- Beats are the result of
 - diffraction
 - destructive interference
 - constructive interference
 - superposition of two waves of nearly equal frequencies
- Maximum number of beats frequency heard by a human being is
 - 10
 - 4
 - 20
 - 6
- The minimum intensity of sound is zero at a point due to two sources of nearly equal frequencies when
 - two sources are vibrating in opposite phase
 - the amplitude of two sources are equal
 - at the point of observation, the amplitudes of two SHM produced by two sources are equal and both the SHM are along the same straight line
 - Both the source are in the same phase
- If $\lambda_1 = 100$ cm, $\lambda_2 = 90$ cm and velocity of sound = 396 ms^{-1} , the number of beats (in Hz) are
 - 4
 - 2
 - 3
 - 44
- Following two wave trains are approaching each other.
 $y_1 = a \sin 2000 \pi t$, $y_2 = a \sin 2008 \pi t$
 The number of beats heard per second is
 - 8
 - 4
 - 1
 - zero
- Two sound waves of lengths 1 m and 1.01 m in a gas produce 10 beats in 3 s. The velocity of sound in gas is
 - 360 ms^{-1}
 - 300 ms^{-1}
 - 337 ms^{-1}
 - 330 ms^{-1}
- An accurate and reliable audio oscillator is used to standardise a tuning fork. When the oscillator reading is 514 Hz, two beats are heard per second. When the oscillator reading is 510 Hz, the beat frequency is 6 Hz. The frequency of the tuning fork (in Hz) is
 - 506
 - 510
 - 516
 - 512
- The frequency of tuning forks A and B are respectively 3% more and 2% less than the frequency of tuning fork C . When A and B are simultaneously excited 5 beats per second are produced. Then, the frequency of the tuning fork A (in Hz) is
 - 98
 - 100
 - 103
 - 105
- A tuning fork of frequency 100 Hz when sound together with another tuning fork of unknown frequency produces 2 beats per second. On loading the tuning fork whose frequency is not known and sounded together with the same tuning fork produces one beat, then the frequency of the unknown tuning fork (in Hz) is
 - 102
 - 98
 - 99
 - 101
- An unknown tuning fork sounded together with a tuning fork of frequency 256 Hz emits two beats. On loading tuning fork of frequency 256 Hz, the number of beats heard is 1 per second. The frequency of unknown tuning fork (in Hz) is
 - 257
 - 258
 - 256
 - 254
- 25 tuning forks are arranged in series in the order of decreasing frequency. Any two successive forks produce 3 beats/s. If the frequency of the first tuning fork is the octave of the last fork, the frequency of the 21st fork is
 - 72 Hz
 - 288 Hz
 - 84 Hz
 - 87 Hz

REFLECTION OF WAVES

When a progressive wave travelling through a medium reaches a rigid boundary, it gets reflected and this phenomenon is called reflection of waves.

If the wave is reflected from a free end, then the reflected wave has the same phase as that of the incident wave and in case of a hard boundary, the reflected wave is out of phase with respect to incident wave.

If equation of incident travelling wave is

$$y_i(x, t) = a \sin(kx - \omega t)$$

At a soft boundary, equation of reflected wave will be

$$y_r(x, t) = a \sin(kx + \omega t)$$

The reflection of waves travelling along a stretched string and reflected by a rigid boundary is shown below.

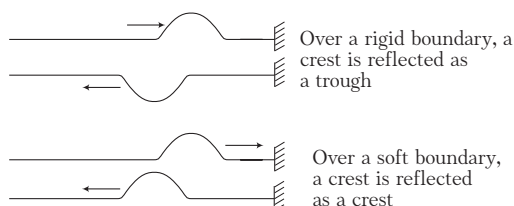


Fig. 17.19 Reflection of waves

If equation of **incident travelling wave** is

$$y_i(x, t) = a \sin(kx - \omega t)$$

At a rigid boundary, the reflected wave will be given by

$$y_r(x, t) = a \sin(kx + \omega t + \pi) = -a \sin(kx + \omega t)$$

Different results related to reflection of waves are given below in tabular form

Wave property	Reflection	Transmission (Refraction)
v	does not change	change
f, T, ω	do not change	do not change
λ, k	do not change	change
A, I	change	change
ϕ	$\Delta\phi = 0$, from a free end $\Delta\phi = \pi$, from a rigid boundary	does not change

Echo

Multiple reflection of sound is called an echo.
e.g. Reflection of sound from hill or cliff.

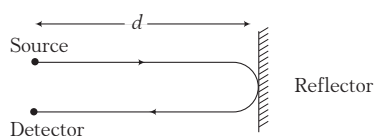


Fig. 17.20 Reflection of sound

If the distance of reflector from the source is d , then

$$2d = vt$$

Hence, v = speed of sound and t = the time of echo.

$$\therefore d = \frac{vt}{2}$$

Since, the effect of ordinary sound remains on our ear for $1/10$ s, therefore if the sound returns to the starting point before $1/10$ s, then it will not be distinguished from the original sound and no echo will be heard. Therefore, the minimum distance of the reflector is

$$d = \frac{v \times t}{2} = \left(\frac{330}{2}\right)\left(\frac{1}{10}\right) = 16.5 \text{ m}$$

Example 17.44 The echo of a gunshot is heard 5 s after it is fired. Calculate the distance of the surface which reflects the sound. (Take, velocity of sound is 332 ms^{-1})

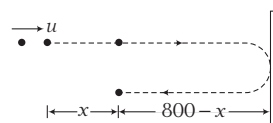
Sol. Distance of the surface which reflects the sound,

$$d = \frac{v \times t}{2} = \frac{332 \times 5}{2} = 166 \times 5 = 830 \text{ m}$$

Example 17.45 (i) An engine approaches a hill with a constant speed. When it is at a distance of 0.8 km, it blows a whistle whose echo is heard by the driver after 4 s.

If speed of engine in air is 330 m/s . Calculate the speed of engine. (ii) A person standing between two parallel hills fires a gun. He hears the first echo after 1.5 s and second after 2.5 s. If speed of sound is 332 m/s , calculate the distance between the hills. When will he hear the third echo?

Sol. (i) The given situation is shown in the following figure.



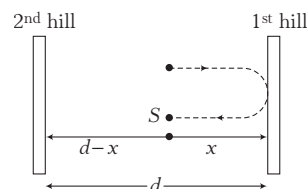
Distance travelled by sound when it again meets the person

$$= 800 + (800 - x) \\ = 1600 - ut = 1600 - u \times 4$$

$$\text{Now, } \frac{1600 - 4u}{v} = 4$$

$$\Rightarrow 1600 - 4u = 4v \Rightarrow 1600 - 4u = 4 \times 330 \\ 4u = 1600 - 1320 \\ 4u = 280 \Rightarrow u = 70 \text{ ms}^{-1}$$

(ii) The given situation is shown in the following figure.



where, d is the distance between two parallel hills and x is the distance between the man and the hill in front of it (i.e. hill 1).

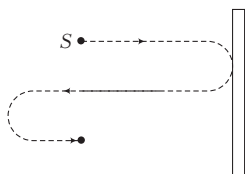
∴ The person hears the first echo after 1.5 s,

$$1.5 = \frac{2x}{v} \quad \dots(i)$$

$$\text{For second echo,} \quad 2.5 = \frac{2(d-x)}{v} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$4 = \frac{2d}{v} \Rightarrow d = 2v = 2 \times 332 = 664 \text{ m}$$



For third echo, sound will be reflected by one hill and then by another hill and then to person

$$= \frac{2d}{v} = \frac{2 \times 664}{332} = 4 \text{ s}$$

Standing/stationary waves

When two harmonic waves of equal frequency and amplitude travelling through a medium (say, string) in opposite directions superimpose each other, we get stationary waves.

Suppose the two waves are

$$y_1 = A \sin(kx - \omega t) \quad \text{and} \quad y_2 = A \sin(kx + \omega t)$$

By the principle of superposition, the resultant wave will be

$$y = y_1 + y_2$$

$$\text{or} \quad y = A [\sin(kx - \omega t) + \sin(kx + \omega t)]$$

Now, using the identity,

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right),$$

$$\text{we obtain} \quad y = 2A \sin(kx) \cos(\omega t) \quad \dots(i)$$

$$\text{or} \quad y = 2A \sin \left(\frac{2\pi x}{\lambda} \right) \cos \left(\frac{2\pi t}{T} \right) \quad \dots(ii)$$

This expression is different from wave representations that we have encountered up till now. It does not have the form $f(x \pm vt)$ or $f(ax \pm bt)$ and therefore, does not describe a travelling wave.

From Eqs. (i) and (ii), it is clear that amplitude of resultant wave depends on the position x (resultant amplitude = $2A \sin kx$). The points at which the amplitude is zero are called **nodes**. At nodes, there is no motion. The points at which the amplitude is maximum are called **antinodes**. Also, the distance between successive nodes or antinodes

is $\frac{\lambda}{2}$, whereas distance between adjacent node and

antinode is $\frac{\lambda}{4}$.

Position of nodes Nodes are the points on the string, where the amplitude of oscillation of constituents is zero.

$$\text{i.e.} \quad \sin kx = 0$$

$$kx = n\pi \quad (\text{where, } n = 0, 1, \dots)$$

$$\frac{2\pi}{\lambda} x = n\pi \quad \boxed{x = \frac{n\lambda}{2}}$$

Position of antinodes Antinodes are the points on the string, where the amplitude of oscillation of the constituents is maximum.

For maximum amplitude,

$$\sin kx = \pm 1 \Rightarrow kx = (2n+1) \frac{\pi}{2}$$

where, $n = 0, 1, 2, \dots$

$$\frac{2\pi}{\lambda} x = (2n+1) \frac{\pi}{2}$$

or

$$\boxed{x = (2n+1) \frac{\lambda}{4}}$$

Note

(i) Following equations also represent stationary waves

- $y = 2a \sin(kx) \sin(\omega t)$
- $y = 2a \cos(kx) \sin(\omega t)$
- $y = 2a \cos(kx) \cos(\omega t)$

(ii) In stationary waves, the system cannot oscillate with any arbitrary frequency but frequencies are characterised by a set of natural frequencies called **normal modes of oscillation**.

Comparison between travelling waves and stationary waves

Travelling waves	Stationary waves
These waves advance in a medium with definite velocity.	These waves remain stationary between two boundaries in the medium.
In these waves, all particles of the medium oscillate with same frequency and amplitude.	In these waves, all particles except nodes oscillate with same frequency but different amplitude. It is zero at nodes and maximum at antinodes.
At any instant, phase of vibration varies continuously from one particle to the other, i.e. phase difference between two particles can have any value between 0 and 2π .	At any instant, the phase of all particles between two successive nodes is the same, but phase of particles on one side of a node is opposite to the phase of particles on the other side of the node, i.e. phase difference between any two particles can be either zero or π .
In these waves, all the particles of the medium do not pass through their mean positions simultaneously at any instant.	In these waves, all particles of the medium pass through their mean positions simultaneously twice in each time period.

Some important points related with stationary waves

Some important points related with stationary waves are given below

- (i) Standing wave is an example of interference. Node means destructive interference and antinode means constructive interference.
- (ii) Two identical waves moving in opposite directions along the string will still produce standing waves even, if their amplitudes are unequal. This is the case when an incident travelling wave is only partially reflected from a boundary. The resulting superposition of two waves having different amplitudes and travelling in opposite directions gives a standing wave pattern of waves whose envelope is shown below in figure.

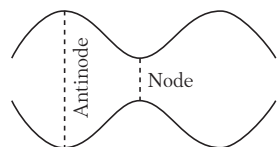


Fig. 17.21 Envelope of resultant superposition of two waves

The standing wave ratio (SWR) is defined as

$$\frac{A_{\max}}{A_{\min}} = \frac{A_i + A_r}{A_i - A_r}$$

where, A_i and A_r are the amplitudes of incident and reflected rays, respectively.

For 100% reflection, $\text{SWR} = \infty$ and for no reflection, $\text{SWR} = 1$.

- (iii) In this pattern, at antinode position, displacement is maximum and hence, velocity is maximum but strain is minimum.

Strain = Slope of stationary wave pattern, *i.e.* $\frac{dy}{dx}$.

At nodal position, displacement and velocity are minimum but strain is maximum.

- (iv) When a wave is reflected off a real surface, part of energy is absorbed by the surface. As a result, energy, intensity and amplitude of reflected wave is always less than that of incident wave. Two waves differ in their amplitudes having same frequency and wavelength and propagate in reverse or opposite direction always give stationary wave pattern by their superposition.

- (v) The intensity of a travelling wave is given by

$$I = \frac{1}{2} \rho A^2 \omega^3 v$$

i.e. $I \propto A^2$

So, we can write, $\frac{I_1}{I_2} = \left(\frac{A_1}{A_2} \right)^2$, if ρ , ω and v are same

for two waves.

e.g. When an incident travelling wave is partly reflected and partly transmitted from a boundary, we can write

$$\frac{I_i}{I_r} = \left(\frac{A_i}{A_r} \right)^2$$

The incident and reflected waves are in the same medium, hence they have same values of ρ and v .

But we cannot write

$$\frac{I_i}{I_t} = \left(\frac{A_i}{A_t} \right)^2$$

where, A_t is the amplitude of transmitted wave.

The incident and transmitted waves have different value of ρ and v .

- (vi) In standing wave, nodes are permanently at rest, so no energy can be transmitted across them, *i.e.* energy of one region is confined in that region. However, this energy oscillates between elastic potential energy and kinetic energy of the particles of the medium.

When particles are at their mean positions, kinetic energy is maximum while elastic potential energy is minimum. When particles are at their extreme positions, kinetic energy is minimum while elastic potential energy is maximum.

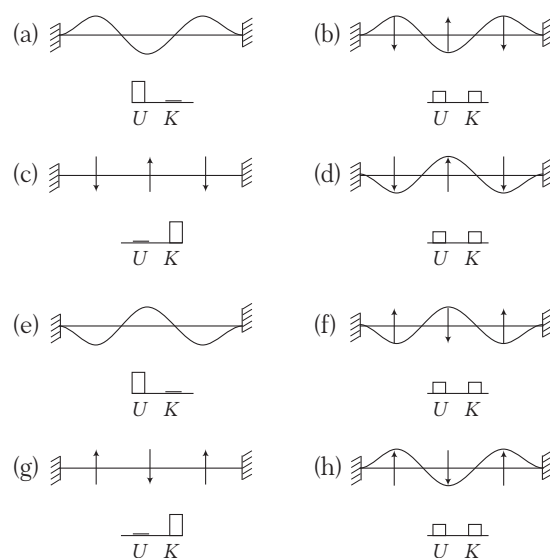


Fig. 17.22 Energy oscillation

In the figure, we have shown how the energy associated with the oscillating string shifts back and forth between kinetic energy K and potential energy U during one cycle.

Example 17.46 The equation given below represents a stationary wave set up in a medium
 $y = 12 \sin(4\pi x) \sin(40\pi t)$

where, y and x are in cm and t is in second. Calculate the amplitude, wavelength and velocity of the component waves.

Sol. On compare the given equation with stationary wave equation, $y = 2A \sin kx \sin \omega t$, we get

$$2A = 12 \text{ cm}, k = 4\pi \text{ cm}^{-1}$$

$$\text{and } \omega = 40\pi \text{ rad s}^{-1}$$

Therefore, amplitude of the component wave = 6 cm

$$\text{As, } k = 4\pi \Rightarrow \frac{2\pi}{\lambda} = 4\pi$$

$$\Rightarrow \lambda = \frac{1}{2} = 0.5 \text{ cm}$$

Further, velocity of the wave,

$$v = \frac{\omega}{k} = \frac{40\pi}{4\pi} = 10 \text{ cm s}^{-1}$$

Example 17.47 The vibrations of a string fixed at both ends are described by the equation

$$y = (5.00 \text{ mm}) \sin [(1.57 \text{ cm}^{-1})x] \sin [(314 \text{ s}^{-1})t]$$

If the length of the string is 10.0 cm, locate the nodes and the antinodes. How many loops are formed in the vibration?

Sol. The given equation can be written as

$$y = 5 \sin(1.57x) \sin(314t)$$

$$= 5 \sin\left(\frac{\pi x}{2}\right) \sin(100\pi t) \quad \dots(i)$$

General equation of standing wave,

$$y = 2A \sin(kx) \sin(\omega t) \quad \dots(ii)$$

Comparing both the equations, we get

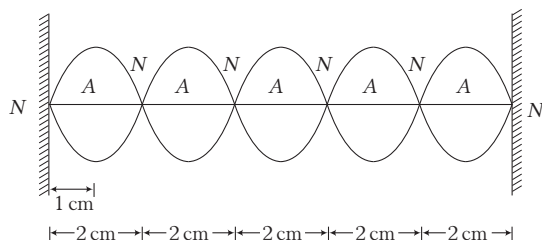
$$k = \frac{\pi}{2} \text{ cm}^{-1}, \omega = 100\pi \text{ rad s}^{-1}$$

$$\therefore k = \frac{2\pi}{\lambda} = \frac{\pi}{2} \Rightarrow \lambda = 4 \text{ cm}$$

$$v = \frac{\omega}{k} = \frac{100\pi}{\pi/2} = 200 \text{ cm s}^{-1} = 2 \text{ m s}^{-1}$$

$$\therefore \text{Length, } L = n \frac{\lambda}{2} \Rightarrow 10 = n \times \frac{4}{2} \Rightarrow n = 5$$

Hence, string is vibrating in 5 loops.



Nodes are formed at $L = \frac{n\lambda}{2}$, whereas antinodes are formed at $(2n+1)\frac{\lambda}{2}$.

$\therefore$ Nodes, $x = 0, 2, 4, 6, 8, 10 \text{ cm}$

and antinodes, $x = 1, 3, 5, 7, 9 \text{ cm}$

Example 17.48 A standing wave is formed by two harmonic waves $y_1 = A \sin(kx - \omega t)$ and $y_2 = A \sin(kx + \omega t)$ travelling on a string in opposite directions. Mass density of the string is ρ and area of cross-section is S . Find the total mechanical energy between two adjacent nodes on the string.

Sol. The distance between two adjacent nodes is $\lambda/2$ or π/k .

$$\left(\because k = \frac{2\pi}{\lambda} \right)$$

$\therefore$ Volume of string between two nodes,

$$V = (\text{Area of cross-section}) (\text{Distance between two nodes})$$

$$= (S) \left(\frac{\pi}{k} \right)$$

Energy density (energy per unit volume) of a travelling wave,

$$u = \frac{1}{2} \rho A^2 \omega^2$$

A standing wave is formed by two identical waves travelling in opposite directions. Therefore, the energy stored between two nodes in a standing wave,

$$E = 2 (\text{energy stored in a distance of } \pi/k \text{ of a travelling wave})$$

$$= 2 (\text{energy density}) (\text{volume}) = 2 \left(\frac{1}{2} \rho A^2 \omega^2 \right) \left(\frac{\pi S}{k} \right)$$

$$\text{or } E = \frac{\rho A^2 \omega^2 \pi S}{k}$$

Terms related to standing waves

Few important terms related to standing waves are given below

(i) Fundamental tones, overtones and harmonics of a sound source

When a sound source produces sound waves, it consists of mixture of many frequencies. This mixture of sound is called **note**. If produced sound contains only one frequency, then it is called **tone**.

Tone of minimum frequency is called **fundamental tone**.

Overtones The tones which are having frequencies greater than fundamental tones are called overtones.

Harmonics Sound produced from a source consists of frequencies which contain fundamental frequency as well as frequencies which are multiples of fundamental frequency, called harmonics.

Odd multiples of fundamental frequency are called **odd harmonics**. Even multiples of fundamental frequency are called **even harmonics**.

Octave The tone which has frequency twice as that of fundamental frequency is called octave. Frequencies of higher multiples are called **higher tones** and frequencies of lower multiples are called **lower tones**.

(ii) Musical scale

The arrangement of notes having definite ratio in respective fundamental frequencies is called a musical scale.

Musical scales are of two types

(a) **Diatonic scale** It is known as 'Sargam' in Indian system. It contains eight notes with definite ratios in their frequencies. The note of lowest frequency is called key note and that of highest frequency (which is double of first) is called an octave. Harmonium, piano, etc., are based on this scale.

(b) **Tempered scale** It contains 13 notes. The ratio of frequencies of successive notes is $2^{1/12}$.

(iii) Musical interval

The ratio of the frequencies of two notes is called the musical interval.

Following are the names of some musical intervals

- Unison, $\frac{n_2}{n_1} = 1$
- Octave, $\frac{n_2}{n_1} = 2$
- Major tone, $\frac{n_2}{n_1} = \frac{9}{8}$
- Minor tone, $\frac{n_2}{n_1} = \frac{10}{9}$
- Semi tone, $\frac{n_2}{n_1} = \frac{16}{15}$
- Fifth tone, $\frac{n_2}{n_1} = \frac{3}{2}$

(iv) Unison

If the frequencies of two notes are equal, then vibrating bodies are said to be in unison.

VIBRATIONS OF AIR COLUMNS IN ORGAN PIPES

Organ pipes are musical instruments which are used to produce musical sound by blowing air into the pipe. This pipe is a cylindrical tube which may be open at one end called **closed organ pipe** [Fig. 17.23(a)] or open at both ends called **open organ pipe** [Fig. 17.23(b)].

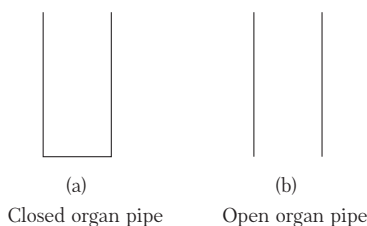


Fig. 17.23 Organ pipes

If the air in pipe at its open end is made to vibrate, longitudinal wave is produced. This wave travels along the pipe towards its far end and is reflected back.

Thus, due to superposition of incident and reflected waves, stationary waves are formed in pipe.

Standing waves in a closed organ pipe

Closed organ pipes are those which are closed at one end and open at the other. A glass tube partially filled with water is an example of such a system. A node is formed at closed end and an antinode is formed at open end. If the length of air column is L , then the closed end can be denoted by $x = 0$, while the open end can be taken as $x = L$.

Position of nodes

Nodes are the points of zero oscillation.

$$\begin{aligned} \therefore \quad \sin kx &= 0 \\ \Rightarrow \quad kx &= n\pi \\ \frac{2\pi}{\lambda} x &= n\pi \end{aligned}$$

$$\text{or} \quad x = \frac{n\lambda}{2} \quad \dots(i)$$

where, $n = 0, 1, 2, \dots$

Position of antinodes

Antinodes are the points of maximum displacement.

$$\therefore \quad \sin kx = \pm 1 \Rightarrow kx = (2n - 1) \frac{\pi}{2}$$

$$\text{or} \quad x = (2n - 1) \frac{\lambda}{4} \quad \left(\because k = 2 \frac{\pi}{\lambda} \right) \dots(ii)$$

where, $n = 1, 2, \dots$

Modes of vibration of air columns in closed organ pipe

At the open end $x = L$, an antinode is formed

$$\therefore \quad L = (2n - 1) \frac{\lambda}{4} \Rightarrow \lambda = \frac{4L}{(2n - 1)}$$

where, $n = 1, 2, \dots$

The corresponding frequencies are given by

$$f = \frac{v}{\lambda} \Rightarrow f = \frac{(2n - 1)v}{4L}$$

(i) First mode

For $n = 1, \lambda = 4L$

$$\text{or} \quad L = \frac{\lambda}{4} \text{ and } f_1 = \frac{v}{4L}$$

where, f_1 = fundamental frequency,

v = speed of sound in air

and L = length of pipe.

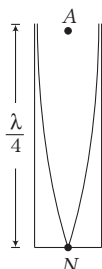


Fig. 17.24 First mode

This mode is called **fundamental mode** or **1st harmonic**.

(ii) **Second mode**

$$\text{For } n = 2, L = \frac{3\lambda}{4} \Rightarrow f_2 = \frac{3v}{4L} \text{ or } f_2 = 3f_1$$

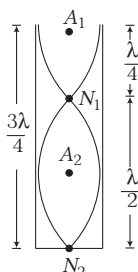


Fig. 17.25 Second mode

This mode is called **3rd harmonic** or **1st overtone**.

(iii) **Third mode**

$$\text{For } n = 3, L = \frac{5\lambda}{4}, f_3 = \frac{5v}{4L} \Rightarrow f_3 = 5f_1$$

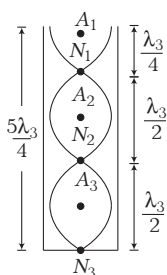


Fig. 17.26 Third mode

This mode is called **5th harmonic** or **2nd overtone**.

From the above discussion, we can say that in a closed organ pipe, only odd harmonics are present. Ratio of harmonics is $f_1 : f_2 : f_3 \dots = 1 : 3 : 5 \dots$

Ratio of overtones = $3 : 5 : 7 \dots$

Also, $f_n = (2n - 1)f_1$ and frequency of n th overtone,

$$f_n = (2n + 1)f_1$$

Example 17.49 Calculate the fundamental frequency of a closed organ pipe of length 66.4 cm at 0°C , if the velocity of sound at 0°C is 332 ms^{-1} .

Sol. Fundamental frequency of a closed organ pipe,

$$f_1 = \frac{v}{4L} = \frac{332}{4 \times 66.4} \times 100 = 125 \text{ Hz}$$

Example 17.50 A 1.0 m long tube is closed at one end. A stretched wire is placed near the open end. The wire is 0.3 m long and has a mass of 0.01 kg. It is held fixed at both ends and vibrates in its fundamental mode. It sets the air column in the tube into vibration at its fundamental frequency by resonance. Find

- the frequency of oscillation of the air column and
- the tension in the wire.

(Take, speed of sound in air = 330 ms^{-1})

Sol. (i) Fundamental frequency of closed pipe = $\frac{v}{4l}$

$$= \frac{330}{4 \times 1} = 82.5 \text{ Hz}$$

(ii) At resonance,

fundamental frequency of stretched wire (at both ends)
= fundamental frequency of air column

$$\therefore \frac{v}{2l} = 82.5 \text{ Hz}$$

$$\therefore \frac{\sqrt{T/\mu}}{2l} = 82.5 \text{ or } T = \mu (2 \times 0.3 \times 82.5)^2$$

$$= \left(\frac{0.01}{0.3} \right) (2 \times 0.3 \times 82.5)^2 = 81.675 \text{ N}$$

Example 17.51 For a certain organ pipe, three successive resonance frequencies are observed at 425, 595 and 765 Hz, respectively. Taking the speed of sound in air to be 340 ms^{-1} .

- Explain whether the pipe is closed at one end or open at both ends.
- Determine the fundamental frequency and length of the pipe.

Sol. (i) The given frequencies are in the ratio $5 : 7 : 9$. As the frequencies are odd multiples of 85 Hz, the pipe must be closed at one end.

(ii) Now, the fundamental frequency is the lowest one, i.e. 85 Hz.

$$\therefore 85 = \frac{v}{4l} \Rightarrow l = \frac{340}{4 \times 85} = 1 \text{ m}$$

Example 17.52 Two tuning forks A and B sounded together give 8 beats/s. With an air resonance tube closed at one end, the two forks give resonances when the two air columns are 32 cm and 33 cm, respectively. Calculate the frequencies of forks.

Sol. Let the frequency of the first fork be f_1 and that of second be f_2 .

$$\text{Then, } f_1 = \frac{v}{4 \times 32} \text{ and } f_2 = \frac{v}{4 \times 33}$$

$$\Rightarrow \quad f_1 > f_2$$

$$\therefore \quad f_1 - f_2 = 8 \quad \dots(i)$$

$$\text{and} \quad \frac{f_1}{f_2} = \frac{33}{32} \quad \dots(ii)$$

Solving Eqs. (i) and (ii), we get

$$\text{and} \quad \begin{aligned} f_1 &= 264 \text{ Hz} \\ f_2 &= 256 \text{ Hz} \end{aligned}$$

Standing waves in an open organ pipe

Open organ pipes are those which are open at both ends. Therefore, a displacement antinode or pressure node is formed at each end. The standing wave in an open organ pipe is given by

$$y(x, t) = 2A \cos kx \cos \omega t$$

If the length of air column is L , then y = maximum at $x = 0$ and at $x = L$.

Position of antinodes

For position of antinodes,

$$\cos kx = \pm 1 \Rightarrow kx = n\pi, \text{ where } n = 0, 1, 2, 3, \dots$$

$$\frac{2\pi}{\lambda} x = n\pi \Rightarrow x = \frac{n\lambda}{2}$$

Further at $x = L$, an antinode is formed

$$\Rightarrow \quad L = \frac{n\lambda}{2} \Rightarrow \lambda = \frac{2L}{n}$$

Modes of vibrations of air column in open organ pipe

The corresponding frequencies are given by

$$f = \frac{v}{\lambda} \Rightarrow f = \frac{nv}{2L}$$

where, $n = 1, 2, 3,$

(i) First mode

$$\text{For } n = 1, \lambda_1 = 2L \Rightarrow L = \frac{\lambda_1}{2} \Rightarrow f_1 = \frac{v}{2L}$$

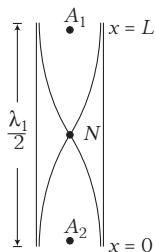


Fig. 17.27 First mode

This mode is called **fundamental mode** or **1st harmonic**. Variation of excess pressure and displacement of an open organ pipe is shown in above figure.

(ii) Second mode

$$\text{For } n = 2, \lambda_2 = \frac{2L}{2} \Rightarrow L = \frac{2\lambda_2}{2}$$

$$\Rightarrow f_2 = \frac{2v}{2L} \quad \text{or} \quad f_2 = 2f_1$$

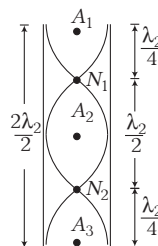


Fig. 17.28 Second mode

This is called **2nd harmonic** or **1st overtone**.

(iii) Third mode

$$\text{For } n = 3, \lambda_3 = \frac{2L}{3} \Rightarrow L = \frac{3\lambda_3}{2}$$

$$\Rightarrow f_3 = \frac{3v}{2L} \quad \text{or} \quad f_3 = 3f_1$$

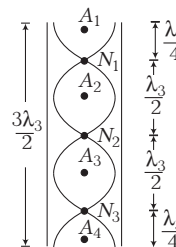


Fig. 17.29 Third mode

This is called **3rd harmonic** or **2nd overtone**.

On generalising, $f_n = nf_1$

From the above discussion, we can say that in open organ pipe, all (even and odd) harmonics are present. Ratio of harmonics is $f_1 : f_2 : f_3 \dots = 1 : 2 : 3 \dots$ and ratio of overtones = $2 : 3 : 4 : 5 \dots$

Frequency of n th over tone is $f_n = (n + 1)f_1$.

Important points regarding vibration of air columns in organ pipe

- (i) If for an organ pipe, velocity of stationary waves remains constant, i.e. $v = \text{constant}$, then with decrease in length of vibrating air column, wavelength ($\lambda \propto L$) will decrease and frequency or pitch will increase and *vice-versa*. This is the reason why the pitch increases gradually as empty vessel fills slowly.
- (ii) For an organ pipe, if $f = \text{constant}$, then $v \propto \lambda$
But $\lambda \propto L$, therefore $v \propto L$

Frequency, $f = \frac{v}{\lambda} = \text{constant}$, i.e. the frequency of

an organ pipe will remain unchanged, if the ratio of speed of sound to its wavelength remains constant.

(iii) For a given organ pipe, if $L = \text{constant}$, so λ will also be constant as $\lambda \propto L$. Therefore, $f \propto v$, so

(a) with rise in temperature, velocity will increase (as $v \propto \sqrt{T}$), so pitch or frequency will increase.

(b) with change in gas in the pipe, v will change (as $v \propto \sqrt{\frac{1}{M}}$), so pitch or frequency will change.

(c) with increase in moisture, v will increase and so the pitch will increase.

Example 17.53 Calculate the frequency of 2nd harmonic in an open organ pipe of length 34 cm, if the velocity of sound is 340 ms^{-1} .

Sol. In an open organ pipe, the frequency of the second harmonic, $v_2 = 2 \times \frac{v}{2L} = \frac{2 \times 340}{2 \times 0.34} = 1000 \text{ Hz}$

Example 17.54 Third overtone of a closed organ pipe is in unison with fourth harmonic of an open organ pipe. Find the ratio of the lengths of the pipes.

Sol. Third overtone of closed organ pipe means seventh harmonic.

Given, $(f_7)_{\text{closed}} = (f_4)_{\text{open}}$

$$7 \left(\frac{v}{4l_c} \right) = 4 \left(\frac{v}{2l_o} \right) \quad \therefore \quad \frac{l_c}{l_o} = \frac{7}{8}$$

Example 17.55 An open organ pipe has a fundamental frequency of 300 Hz. The first overtone of a closed organ pipe has the same frequency as the first overtone of the open pipe. How long is each pipe? (Take, speed of sound in air = 330 ms^{-1})

Sol. Fundamental frequency of an open pipe,

$$f_1 = \frac{v}{2l_o}$$

$$\therefore \quad l_o = \frac{v}{2f_1} = \frac{330}{2 \times 300} = 0.55 \text{ m} = 55 \text{ cm}$$

Given, first overtone of closed pipe = first overtone of open pipe

$$\therefore \quad 3 \left(\frac{v}{4l_c} \right) = 2 \left(\frac{v}{2l_o} \right)$$

$$\therefore \quad l_c = \frac{3}{4} l_o = \left(\frac{3}{4} \right) (0.55) = 0.4125 \text{ m} = 41.25 \text{ cm}$$

Example 17.56 A flute which we treat as a pipe open at both ends is 60 cm long.

(i) What is the fundamental frequency when all the holes are covered?

(ii) How far from the mouthpiece should a hole be uncovered for the fundamental frequency to be 330 Hz? (Take, speed of sound in air = 340 ms^{-1}).

Sol. (i) Fundamental frequency when the pipe is open at both

$$\text{ends, } f_1 = \frac{v}{2l} = \frac{340}{2 \times 0.6} = 283.33 \text{ Hz}$$

(ii) Suppose the hole is uncovered at a length l from the mouthpiece, the fundamental frequency will be

$$f'_1 = \frac{v}{2l}$$

$$\therefore \quad l = \frac{v}{2f'_1} = \frac{340}{2 \times 330} = 0.515 \text{ m} = 51.5 \text{ cm}$$

Note Opening holes in the side effectively shortens the length of the resonance column, thus increasing the frequency.

Example 17.57 (i) A cylindrical metal tube has a length of 60 cm and is open at both ends. Find the frequencies between 1000 Hz and 2000 Hz at which the air column in the tube can resonate. (Take, speed of sound in air = 340 m/s)

(ii) Find the greatest length of an organ pipe open at both ends that will have its fundamental frequencies in the normal hearing range (30-2000 Hz). (Take, speed of sound in air = 340 ms^{-1})

(iii) Two successive resonance frequencies in an open organ pipe are 1944 Hz and 2592 Hz, respectively. Find the length of tube. (Take, speed of sound in air is 324 ms^{-1})

Sol. (i) For open pipe, fundamental frequency,

$$f = \frac{v}{2l} = \frac{340}{2 \times 0.6} = 283.33 \text{ Hz}$$

Possible frequencies of open pipe = nf

where, $n = 1, 2, 3, \dots$

$\therefore$ Possible frequencies of open pipe =

(284, 568, 852, 1136, 1420, 1704, 1988) Hz

Frequencies between 1000 and 2000 Hz

= 1136 Hz, 1420 Hz, 1704 Hz, 1988 Hz

(ii) Fundamental frequency, $f = \frac{v}{2l} \Rightarrow l = \frac{v}{2f}$

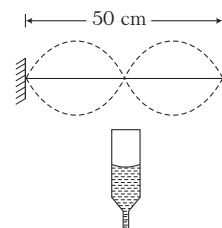
Hence, l will be maximum, if f is minimum

$$\therefore \quad l_{\text{max}} = \frac{340}{2 \times 30} = 5.6 \text{ m}$$

(iii) The difference in successive frequencies of a pipe (open or closed) = $\frac{v}{2l}$

$$\Rightarrow 2592 - 1944 = \frac{324}{2l} \Rightarrow l = \frac{1}{4} \text{ m} = 0.25 \text{ m}$$

Example 17.58 Consider the situation shown in figure. The wire which has a mass of 5 g oscillates in its second harmonic and sets the air column in the tube into vibrations in its fundamental mode. Assuming that the speed of sound in air is 340 ms^{-1} , find the tension in the wire.



Sol. Frequency of wire, $f_2 = \frac{2}{2l} \sqrt{\frac{T}{\mu}} = \frac{1}{l} \sqrt{\frac{T}{M/l}}$

$$= \frac{1}{0.5} \sqrt{\frac{T}{5 \times 10^{-3}/0.5}}$$

$$= 2\sqrt{100T} = 20\sqrt{T}$$

Fundamental frequency of closed pipe,

$$f_1 = \frac{v}{4L} = \frac{340}{4 \times 1} = 85$$

$$\therefore f_2 = f_1 \Rightarrow 20\sqrt{T} = 85$$

Therefore, $T = (4.25)^2 = 18.06 \text{ N}$

Example 17.59 The first overtone of an open organ pipe beats with the first overtone of a closed organ pipe with a beat frequency of 2.2 Hz. The fundamental frequency of the closed organ pipe is 110 Hz. Find the lengths of the pipes. (Take, speed of sound in air is $v = 330 \text{ ms}^{-1}$)

Sol. Let l_1 and l_2 be the lengths of closed and open pipes, respectively.

Fundamental frequency of closed organ pipe is given by

$$f_1 = \frac{v}{4l_1}$$

Here, $v = \text{speed of sound in air} = 330 \text{ ms}^{-1}$

But $f_1 = 110 \text{ Hz}$ (Given)

Therefore, $\frac{v}{4l_1} = 110 \text{ Hz}$

$$\Rightarrow l_1 = \frac{v}{4 \times 110} = \frac{330}{4 \times 110} \text{ m} = 0.75 \text{ m}$$

Frequency of first overtone of closed organ pipe,

$$f_3 = 3f_1 = 3(110) \text{ Hz} = 330 \text{ Hz}$$

This produces a beat frequency of 2.2 Hz with first overtone of open organ pipe.

Therefore, first overtone frequency of open organ pipe is either

$$(330 + 2.2) \text{ Hz} = 332.2 \text{ Hz} \quad \text{or} \quad (330 - 2.2) \text{ Hz} = 327.8 \text{ Hz}$$

If first overtone frequency is 332.2 Hz, then

$$2 \left(\frac{v}{2l_2} \right) = 332.2 \text{ Hz} \quad \text{or} \quad l_2 = \frac{v}{332.2} = \frac{330}{332.2} \text{ m} = 0.99 \text{ m}$$

and if it is 327.8 Hz, then

$$2 \left(\frac{v}{2l_2} \right) = 327.8 \text{ Hz} \quad \text{or} \quad l_2 = \frac{v}{327.8} \text{ m} = \frac{330}{327.8} \text{ m}$$

$$= 1.0067 \text{ m}$$

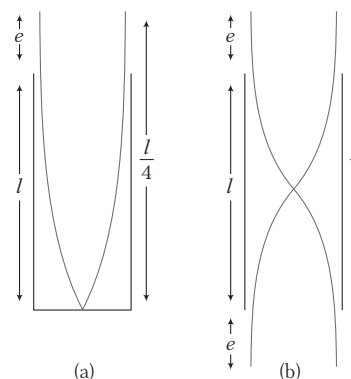
Therefore, length of the closed organ pipe is $l_1 = 0.75 \text{ m}$ while length of open pipe is either $l_2 = 0.99 \text{ m}$ or 1.0067 m .

End correction

It was found that the antinode is not formed exactly at the open end of the **organ pipe** but due to finite momentum

of the particles, the reflection takes place a little above the open end, that is why the antinode is formed a little above the open end. For obtaining true frequency, an end correction must be applied which takes account of this additional length.

This is denoted by e . If length of organ pipe is l and end correction is e , then length of air column in closed pipe will be $(l + e)$ and in open pipe, $(l + 2e)$.



For closed organ pipe For open organ pipe

Fig. 17.30 End correction in organ pipe

Thus, for a closed organ pipe replace L by $L + e$

$$\therefore f_1 = \frac{v}{4(L + e)}$$

and for an open organ pipe replace L by $L + 2e$

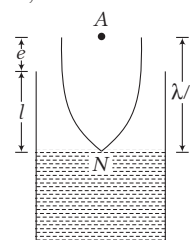
$$\therefore f_2 = \frac{v}{2(L + 2e)}$$

Note The value of end correction e is $0.6r$ for closed organ pipe and $1.2r$ for an open organ pipe, where r is the radius of the pipe.

Example 17.60

- In a resonance tube experiment to determine the speed of sound in air, a pipe of diameter 5 cm is used. The air column in pipe resonates with a tuning fork of frequency 480 Hz, when minimum length of air column is 16 cm. Find the speed of sound in air at room temperature.
- A tube of certain diameter and of length 48 cm is open at both ends. Its fundamental frequency of resonance is found to be 320 Hz. The velocity of sound in air is 320 m/s. Estimate the diameter of the tube. One end of the tube is now closed. Calculate the lowest frequency of resonance for the tube.

Sol. (i) Given, diameter, $D = 5 \text{ cm}$



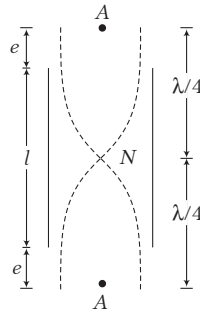
$$\frac{\lambda}{4} = l + e = l + 0.6r = l + 0.3D$$

$$(\because r = \frac{D}{2})$$

$$\lambda = 4(l + 0.3D) = 4(16 + 0.3 \times 5) \\ = 70 \text{ cm} = 0.7 \text{ m}$$

$$\text{Now, speed of sound, } v = f\lambda \\ = 480 \times 0.7 = 336 \text{ ms}^{-1}$$

$$(ii) \text{ Given, } l = 48 \text{ cm, } f = 320 \text{ Hz}$$



$$\text{Speed, } v = f\lambda \Rightarrow \lambda = \frac{v}{f} = \frac{320}{320} = 1 \text{ m} = 100 \text{ cm}$$

$$\frac{\lambda}{2} = l + 2e \Rightarrow \lambda = 2(l + 2e)$$

$$100 = 2(48 + 2e) \Rightarrow e = 1 \text{ cm}$$

$$\Rightarrow 0.6r = 0.3D = 1$$

$$\Rightarrow D = 10/3 \text{ cm} = 3.33 \text{ cm}$$

$$\text{For closed pipe, } \frac{\lambda}{4} = l + e = 48 + 1 = 49$$

$$\lambda = 196 \text{ cm} = 1.96 \text{ m}$$

$$\therefore \text{Speed, } v = f\lambda$$

$$\Rightarrow f = \frac{v}{\lambda} = \frac{320}{1.96} = 163.3 \text{ Hz}$$

many different values. The distance between adjacent nodes is $\lambda/2$, so that in a string of length l there must be exactly an integral number n of half wavelengths, $\lambda/2$. That is,

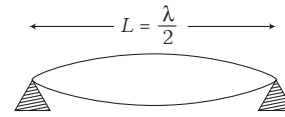


Fig. 17.31 Vibration string

$$\frac{n\lambda}{2} = l \text{ or } \lambda = \frac{2l}{n} \quad (\text{where, } n = 1, 2, 3, \dots)$$

Laws of transverse vibration of a stretched string

- (i) **Law of length** For a given string under a tension (T), the frequency of the vibrating string is inversely proportional to the vibrating length (L) of the string.

$$i.e. \quad f \propto \frac{1}{L} \Rightarrow fL = \text{constant} \Rightarrow f_1 L_1 = f_2 L_2$$

- (ii) **Law of tension** The frequency of the uniform string of given length is proportional to square root of the tension (T) in the string.

$$i.e. \quad f \propto \sqrt{T} \Rightarrow \frac{f}{\sqrt{T}} = \text{constant} \Rightarrow \frac{f_1}{f_2} = \sqrt{\frac{T_1}{T_2}}$$

- (iii) **Law of mass** For a given vibrating length and tension of the string, the frequency of the vibrating string is inversely proportional to the square root of the mass per unit length of the string.

$$i.e. \quad f \propto \frac{1}{\sqrt{\mu}}$$

Therefore, with increase in mass of the string frequency decreases.

As,

Mass per unit length of the string = Volume of unit length $\times$ Density

$$\text{or} \quad \mu = \pi r^2 \rho$$

where, ρ = density of the wire, r = radius of the wire

$$\therefore \quad f \propto \frac{1}{\sqrt{\pi r^2 \rho}}$$

Now, law of mass can be categorised in two parts as follows

- (a) **Law of radius** If length (L), density (ρ) and tension (T) of a vibrating wire are fixed, then frequency of vibration of the wire is inversely proportional to its radius.

$$i.e. \quad f \propto \frac{1}{r}$$

VIBRATIONS OF STRETCHED STRING

In an unbounded continuous medium, there is no restriction on the frequencies or wavelengths of the standing waves. However, if the waves are confined in space, (e.g. when a string is tied at both ends) standing waves can be set up for a discrete set of frequencies or wavelengths.

Consider a string of definite length l , rigidly held at both ends. When we set up a sinusoidal wave on such a string, it gets reflected from the fixed ends. By the superposition of two identical waves travelling in opposite directions, standing waves are established on the string. Here, the only requirement we have to satisfy is that the end points should be nodes, as these points cannot oscillate.

The nodes are permanently at rest. There may be any number of nodes in between or none at all, so that the wavelength associated with the standing waves can take

- (b) **Law of density** If length (L), radius (r) and tension (T) of a vibrating wire are fixed, then frequency of vibration of the wire is inversely proportional to square root of its density.

$$\text{i.e.} \quad f \propto \frac{1}{\sqrt{\rho}}$$

Example 17.61 A wire stretched between two rigid supports vibrates in its fundamental mode with a frequency of 45 Hz. The mass of the wire is 3.5×10^{-2} kg and its linear mass density is 4×10^{-2} kgm⁻¹. What is the speed of transverse wave on string and tension in the wire?

Sol. Length of wire = $\frac{\text{Mass of wire}}{\text{Linear density}} = \frac{3.5 \times 10^{-2}}{4 \times 10^{-2}} = 0.875 \text{ m}$

$$\text{Now, } f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$\therefore \text{ Tension in the wire, } T = 4f^2 l^2 \mu$$

$$= 4(45)^2 (0.875)^2 (4 \times 10^{-2}) = 248.1 \text{ N}$$

$$\text{Speed of transverse wave, } v = f\lambda = f(2l)$$

$$= 45 \times 2 \times 0.875 = 78.75 \text{ ms}^{-1}$$

Example 17.62 A string on a musical instrument is 50 cm long and its fundamental frequency is 270 Hz. Calculate the length of the string, if the desired frequency of 1000 Hz is to be produced.

Sol. As, $f \propto \frac{1}{l} \Rightarrow \frac{l_2}{l_1} = \frac{f_1}{f_2}$

$$\Rightarrow l_2 = l_1 \left(\frac{f_1}{f_2} \right)$$

$$= 50 \times \left(\frac{270}{1000} \right) = 13.5 \text{ cm}$$

Example 17.63 The tension in a piano wire is 10 N. What should be the tension in the wire to produce a note of double the frequency.

Sol. As, $f \propto \sqrt{T}$

$$\therefore \frac{f_1}{f_2} = \frac{\sqrt{T_1}}{\sqrt{T_2}}$$

$$\Rightarrow \frac{f}{2f} = \frac{\sqrt{10}}{\sqrt{T_2}}$$

$$\Rightarrow T_2 = 40 \text{ N}$$

Standing waves in a string fixed at both ends

Consider a stretched string having length L fixed at both ends. Let the one end be fixed at $x = 0$ while the other one at $x = L$, and at $x = 0$ and $x = L$, nodes will be obtained as these ends are fixed.

As, $L = \frac{n\lambda}{2}, n = 1, 2, 3, \dots$

$\Rightarrow$ Possible wavelengths of stationary waves,

$$\lambda = \frac{2L}{n}, n = 1, 2, 3, \dots$$

and corresponding frequencies are

$$f = \frac{nv}{2L}, n = 1, 2, 3, \dots$$

Here, v is the speed of the wave in the given medium.

This relation gives the expression for the natural frequency of vibration.

Modes of vibration

The manner in which the string vibrates and gives rise to a standing wave is known as mode of vibration of the string.

First mode of vibration

The mode of vibration in which string will vibrate in one segment or the lowest possible natural frequency [as shown in Fig. 17.32(a)] is called **fundamental mode** or **first harmonic**.

As, $L = \frac{n\lambda}{2}$

For first harmonic, put $n = 1$

$$\therefore L = \frac{\lambda_1}{2} \text{ and } f_1 = \frac{v}{2L}$$

Second mode of vibration

The mode of vibration in which string will vibrate in two segments [as shown in Fig. 17.32(b)] is called **second harmonic** or **first overtone**.

For the second harmonic, put

$$n = 2 \text{ in } L = \frac{n\lambda}{2}$$

$$\therefore L = \lambda_2$$

and

$$f_2 = \frac{2v}{2L}$$

or

$$f_2 = 2f_1$$

As, $f = \frac{nv}{2L}$

If $n = 3$, the frequency, $f = \frac{3v}{2L}$ or $f = 3f_1$ is called **third harmonic** and so on.

Figure below shows the first four harmonics of a stretched string fixed at both ends.

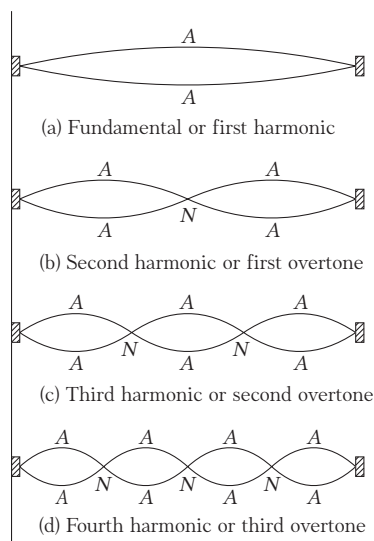


Fig. 17.32 Modes of vibration of a stretched string fixed at both ends

The vibration of a string will be a superposition of different modes. In this case, $f_1 : f_2 : f_3 = 1 : 2 : 3 \dots$, i.e. all harmonics (odd as well as even) can be produced.

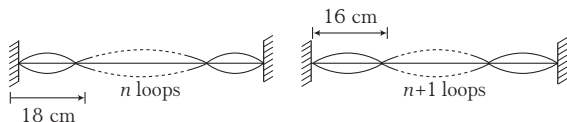
Note Musical instruments like sitar or violin are based on this principle.

Example 17.64 A string fixed at both ends has consecutive standing wave modes for which the distances between adjacent nodes are 18 cm and 16 cm respectively.

(i) What is the minimum possible length of the string?

(ii) If the tension is 10 N and the linear mass density is 4 gm^{-1} , what is the fundamental frequency?

Sol. (i)



Let l be the length of the string. Then,

$$18n = l \quad \dots(i)$$

$$16(n+1) = l \quad \dots(ii)$$

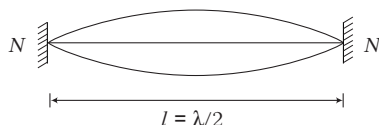
From Eqs. (i) and (ii), we get

$$n = 8 \text{ and } l = 144 \text{ cm}$$

Therefore, the minimum possible length of the string can be 144 cm.

(ii) For fundamental frequency, $l = \lambda/2$

$$\text{or } \lambda = 2l = 288 \text{ cm} = 2.88 \text{ m}$$



Speed of wave on the string,

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{10}{4 \times 10^{-3}}} = 50 \text{ ms}^{-1}$$

$$\therefore \text{Fundamental frequency, } f = \frac{v}{\lambda} = \frac{50}{2.88} = 17.36 \text{ Hz}$$

Example 17.65 The vibrations of a string of length 60 cm fixed at both ends are represented by the equation

$$y = 4 \sin \left(\frac{\pi x}{15} \right) \cos (96 \pi t), \text{ where } x \text{ and } y \text{ are in cm and } t \text{ in}$$

second.

(i) What is the maximum displacement of a point at $x = 5 \text{ cm}$ and $t = 0$?

(ii) Where are the nodes located along the string?

(iii) What is the velocity of the particle at $x = 7.5 \text{ cm}$ and at $t = 0.25 \text{ s}$?

(iv) Write down the equations of the component waves whose superposition gives the above wave.

Sol. Given, $y = 4 \sin (\pi x/15) \cos (96 \pi t)$. It can be broken up into

$$y = 2 \left[\sin \left(\frac{\pi x}{15} + 96 \pi t \right) + \sin \left(\frac{\pi x}{15} - 96 \pi t \right) \right]$$

Thus, the waves are of the same amplitude and frequency but travelling in opposite directions which thus, superimpose to give a standing wave.

(i) At $x = 5 \text{ cm}$, the standing wave equation gives

$$y = 4 \sin (5\pi/15) \cos (96 \pi t) = 4 \times \sqrt{3}/2 \cos (96 \pi t)$$

$$\therefore \text{Maximum displacement} = 2\sqrt{3} \text{ cm}$$

(ii) The nodes are the points which are permanently at rest.

Thus, they are those points for which

$$\sin (\pi x/15) = 0$$

$$\text{i.e. } \pi x/15 = n\pi$$

$$\Rightarrow n = 0, 1, 2, 3, 4, \dots$$

$$x = 15n, \text{ i.e. at } x = 0, 15, 30, 45 \text{ and } 60 \text{ cm.}$$

(iii) The particle velocity is equal to

$$\left(\frac{\partial y}{\partial t} \right) = 4 (96 \pi) \sin \left(\frac{\pi x}{15} \right) (-\sin 96 \pi t)$$

$$= -384 \pi \sin \left(\frac{\pi x}{15} \right) \sin (96 \pi t)$$

At $x = 7.5$ and $t = 0.25$, we get

$$\left(\frac{\partial y}{\partial t} \right) = -384 \pi \sin \left(\frac{\pi 7.5}{15} \right) \sin (96 \pi 0.25)$$

$$= -384 \pi \sin \left(\frac{\pi}{2} \right) \sin (24 \pi) = 0$$

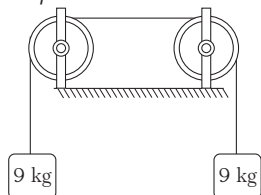
(iv) The equations of the component waves are

$$y_1 = 2 \sin \left(\frac{\pi x}{15} + 96 \pi t \right)$$

$$\text{and } y_2 = 2 \sin \left(\frac{\pi x}{15} - 96 \pi t \right)$$

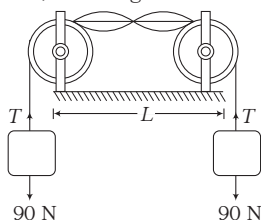
Example 17.66

- (i) The length of the wire shown in figure between the pulleys is 1.5 m and its mass is 12.0 g. Find the frequency of vibration with which the wire vibrates in two loops leaving the middle point of the wire between the pulleys at rest.



- (ii) A string fixed at both ends, vibrates in a resonant mode with a separation of 2.0 cm between the consecutive nodes. For the next higher resonant frequency, this separation is reduced to 1.6 cm. Find the length of the string.

Sol. (i) Given, $L = 1.5$ m, $M = 12$ g



$$T = 90 \text{ N} \quad (\text{Acting downward})$$

$$\therefore \mu = \frac{M}{L} = \frac{12 \times 10^{-3}}{1.5} = 8 \times 10^{-3} \text{ kg m}^{-1}$$

$$\text{As, } f = \frac{2v}{2L} \quad (\because n = 2)$$

$$f = \frac{2}{2L} \sqrt{\frac{T}{\mu}} = \frac{1}{1.5} \sqrt{\frac{90}{8 \times 10^{-3}}} = 70.71 \text{ Hz}$$

$$(ii) \text{ For } k\text{th harmonic, } k \cdot \frac{\lambda}{2} = L \Rightarrow k(2) = L \quad \dots(i)$$

For $(k+1)$ th harmonic,

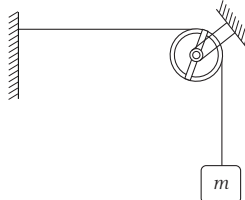
$$(k+1) \frac{\lambda'}{2} = L \Rightarrow (k+1)(1.6) = L \quad \dots(ii)$$

From Eqs. (i) and (ii), $2k = 1.6(k+1)$

$$0.4k = 1.6 \Rightarrow k = 4$$

$$\text{Hence, } L = 2k = 2 \times 4 = 8 \text{ cm}$$

Example 17.67 Figure shows a string stretched by a block going over a pulley. The string vibrates in its tenth harmonic in unison with a particular tuning fork.

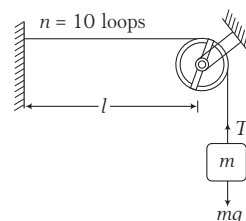


When a beaker containing water is brought under the block, so that the block is completely dipped into the beaker, the string vibrates in its eleventh harmonic. Find the density of the material of the block.

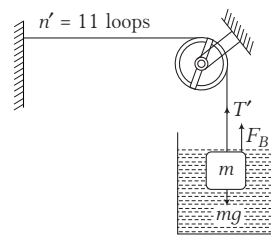
Sol. According to the question,

$$\text{Frequency, } f = \frac{10}{2l} \sqrt{\frac{T}{\mu}} = \frac{10}{2l} \sqrt{\frac{mg}{\mu}} \quad \dots(i)$$

The free body diagram of the given situation is shown below.



$$\begin{aligned} \therefore T' + F_B &= mg \\ T' &= mg - F_B = mg - \rho_w \frac{m}{\sigma} g \\ &= mg \left(1 - \frac{\rho_w}{\sigma} \right) \end{aligned}$$



where, σ is density of material of block and ρ_w is density of water.

$$f = \frac{11}{2l'} \sqrt{\frac{T'}{\mu}} = \frac{11}{2l'} \sqrt{\frac{mg \left(1 - \frac{\rho_w}{\sigma} \right)}{\mu}} \quad \dots(ii)$$

Equating Eqs. (i) and (ii), we get

$$\begin{aligned} \frac{10}{2l} \sqrt{\frac{mg}{\mu}} &= \frac{11}{2l'} \sqrt{\frac{mg \left(1 - \frac{\rho_w}{\sigma} \right)}{\mu}} \\ \Rightarrow 10 &= 11 \sqrt{1 - \frac{\rho_w}{\sigma}} \\ \Rightarrow 1 - \frac{\rho_w}{\sigma} &= \left(\frac{10}{11} \right)^2 = \frac{100}{121} \\ \Rightarrow \frac{\rho_w}{\sigma} &= 1 - \frac{100}{121} = \frac{21}{121} \\ \Rightarrow \sigma &= \frac{121}{21} \rho_w = \frac{121}{21} \text{ g/cm}^3 = 5.8 \text{ g/cm}^3 \end{aligned}$$

Vibration of a string fixed at one end

Standing waves can also be produced on a string or rod which is fixed at one end and whose other end is free to move in a transverse direction. Here, in this case, we will have antinode at the free end and node at the fixed end.

Now, consider the equation of standing wave,

$$y(x, t) = 2a \sin kx \cos \omega t$$

As, we are having antinode at the end $x = L$

$$\sin kL = \pm 1 \Rightarrow kL = (2n + 1) (\pi/2)$$

where, $n = 0, 1, 2, \dots$

$$\frac{2\pi}{\lambda} \times L = (2n + 1) \frac{\pi}{2}$$

$$\Rightarrow \frac{2L}{\lambda} = (2n + 1) \times \frac{1}{2}$$

$$\text{or } \frac{2Lf}{v} = (2n + 1) \frac{1}{2}$$

$\therefore$ Frequency of the vibration,

$$f = \frac{(2n + 1)}{4L} v$$

where, f is frequency and v is the speed of the wave.

The above frequencies are the normal frequencies of vibration. The fundamental frequency is obtained when $n = 0$.

$$\therefore f_0 = \frac{v}{4L} \quad \text{(first harmonic)}$$

The overtone frequencies are

$$f_1 = \frac{3v}{4L} = 3f_0 \quad \text{(third harmonic)}$$

$$f_2 = \frac{5v}{4L} = 5f_0 \quad \text{(fifth harmonic)}$$

$$f_3 = \frac{7v}{4L} = 7f_0 \quad \text{(seventh harmonic)}$$

$$f_0 : f_1 : f_2 : f_3 \dots = 1 : 3 : 5 : 7 \dots$$

Here, we see that only odd harmonics are present (*i.e.* contains odd multiples of the fundamental frequency). The figure below shows shapes of the string.

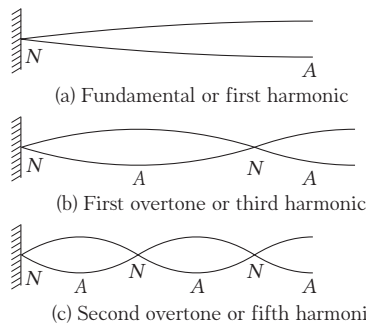


Fig. 17.33 Vibration of a string fixed at one end

Vibration of composite strings

Suppose two strings of different materials and lengths are joined end-to-end and tied between clamps as shown in the figure.

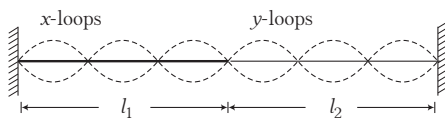


Fig. 17.34 Vibration of composite strings

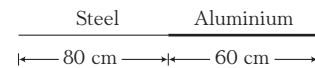
Now, after plucking, stationary waves are established only at those frequencies which matches with any one harmonic of both the independent strings S_1 and S_2 .

As the frequency of the wave in both strings must be same, so

$$\frac{x}{2l_1} \sqrt{\frac{T}{\mu_1}} = \frac{y}{2l_2} \sqrt{\frac{T}{\mu_2}}$$

$$\text{or } \frac{x}{y} = \frac{l_1}{l_2} \sqrt{\frac{\mu_1}{\mu_2}} = \frac{l_1}{l_2} \sqrt{\frac{\rho_1 A_1}{\rho_2 A_2}}$$

Example 17.68 Figure shows an aluminium wire of length 60 cm joined to a steel wire of length 80 cm and stretched between two fixed supports. The tension produced is 40 N. The cross-sectional area of the steel wire is 1.0 mm^2 and that of the aluminium wire is 3.0 mm^2 . What could be the minimum frequency of a tuning fork which can produce standing waves in the system with the joint as a node? The density of aluminium is 2.6 g/cm^3 and that of steel is 7.8 g/cm^3 .



Sol. According to the question,

Steel	Aluminium
$A_s = 1 \text{ mm}^2$	$A_a = 3 \text{ mm}^2$
$\rho_s = 7.8 \text{ g/cm}^3$	$\rho_a = 2.6 \text{ g/cm}^3$
$L_s = 80 \text{ cm}$	$L_a = 60 \text{ cm}$

Let p th harmonic of steel wire coincides with q th harmonic of aluminium wire.

$$n = \frac{p}{2L_s} \sqrt{\frac{T}{\rho_s A_s}} = \frac{q}{2L_a} \sqrt{\frac{T}{\rho_a A_a}}$$

$$\frac{p}{q} = \frac{L_s}{L_a} \sqrt{\frac{\rho_s A_s}{\rho_a A_a}} = \frac{80}{60} \sqrt{\frac{7.8 \times 1}{2.6 \times 3}} = \frac{4}{3}$$

4th harmonic of steel wire coincides with 3rd harmonic of aluminium wire, the minimum frequency of tuning fork can be determined by taking $p = 4$ or $q = 3$

$$n_{\min} = \frac{p}{2L_s} \sqrt{\frac{T}{\rho_s A_s}}$$

$$= \frac{4}{2 \times 0.8} \sqrt{\frac{40}{7.8 \times 10^3 \times 1 \times 10^{-6}}}$$

$$= 2.5 \times 71.6 = 179 \text{ Hz}$$

Speed of a transverse wave in a stretched string

Speed of transverse wave in a stretched string is given by

$$v = \sqrt{\frac{T}{\mu}}$$

where, T is tension and μ is linear mass density of the string. If radius of the string is r having volume V and density of the material of the string is ρ , then

$$\mu = \frac{V}{L} \times \rho = \pi r^2 \rho$$

$\therefore$

$$v = \sqrt{\frac{T}{\pi r^2 \rho}}$$

When a block of mass M is suspended from the string, then

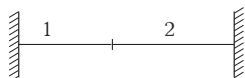
$$T = Mg$$

The speed of the wave along a stretched ideal string depends only on the tension and the linear mass density of the string and does not depend on the frequency of the wave.

The frequency of the wave is determined by the source that generates the wave. The wavelength is given by

$$\lambda = \frac{v}{f}$$

Example 17.69 Two strings 1 and 2 are taut between two fixed supports (as shown in figure) such that the tension in both strings is same. Mass per unit length of 2 is more than that of 1. Explain which string is denser for a transverse travelling wave.



Sol. Speed of a transverse wave on a string, $v = \sqrt{\frac{T}{\mu}}$

$$\text{or } v \propto \frac{1}{\sqrt{\mu}}$$

Now, $\mu_2 > \mu_1$ (Given)

$$\therefore v_2 < v_1$$

i.e. Medium 2 is denser and medium 1 is rarer.

Example 17.70 A steel wire has a length of 12 m and a mass of 2.10 kg. What should be the tension in the wire, so that the speed of a transverse wave on the wire equals to the speed of sound in dry air at 20°C ($= 343 \text{ ms}^{-1}$)?

Sol. Given, speed of sound in air, $v = 343 \text{ ms}^{-1}$

Length of the wire, $l = 12.0 \text{ m}$

Total mass of the wire, $M = 2.10 \text{ kg}$

Therefore, mass per unit length of the wire,

$$\mu = \frac{M}{l} = \frac{2.10}{12.0} = 0.175 \text{ kg m}^{-1}$$

$$\text{Now, } v = \sqrt{\frac{T}{\mu}}$$

$$\text{or } T = v^2 \mu = (343)^2 \times 0.175 = 20588.6 \text{ N} = 2.06 \times 10^4 \text{ N}$$

Example 17.71 One end of 12 m long rubber tube with a total mass of 0.9 kg is fastened to a fixed support. A cord attached to the other end passes over a pulley and supports

an object with a mass of 5 kg. The tube is struck to a transverse blow at one end. Find the time required for the pulse to reach the other end. (Take, $g = 9.8 \text{ ms}^{-2}$)

Sol. Tension in the rubber tube AB,

$$T = mg \text{ or } T = (5.0)(9.8) = 49 \text{ N}$$

Mass per unit length of rubber tube,

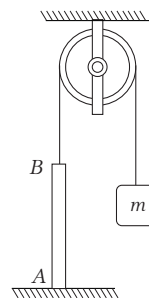
$$\mu = \frac{0.9}{12} = 0.075 \text{ kg m}^{-1}$$

$\therefore$ Speed of wave on the tube,

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{49}{0.075}} = 25.56 \text{ ms}^{-1}$$

$\therefore$ The required time,

$$t = \frac{AB}{v} = \frac{12}{25.56} = 0.47 \text{ s}$$



Example 17.72 A wave moves with speed 300 ms^{-1} on a wire, which is under a tension of 500 N. Find how much the tension must be changed to increase the speed to 312 ms^{-1} .

Sol. Speed of a transverse wave on the wire, $v = \sqrt{\frac{T}{\mu}}$... (i)

Differentiating with respect to tension, we get

$$\frac{dv}{dT} = \frac{1}{2\sqrt{\mu T}} \quad \dots (ii)$$

On dividing Eq. (ii) by Eq. (i), we get

$$\frac{dv}{v} = \frac{1}{2} \frac{dT}{T} \text{ or } dT = (2T) \frac{dv}{v}$$

Substituting the values in above equation, we get

$$dT = \frac{(2)(500)(312 - 300)}{300} = 40 \text{ N}$$

i.e. Tension should be increased by 40 N.

Example 17.73 A wire of uniform cross-section is stretched between two points 100 cm apart. The wire is fixed at one end and a weight is hung over a pulley at the other end. A weight of 9 kg produces a fundamental frequency of 750 Hz.

(i) What is the velocity of the wave in wire ?

(ii) If the weight is reduced to 4 kg, what is the velocity of wave ? What is the wavelength and frequency ?

Sol. (i) Here, $L = 100 \text{ cm}$ and $f_1 = 750 \text{ Hz}$

$$\therefore v_1 = 2Lf_1 = 2 \times 100 \times 750 = 150000 \text{ cms}^{-1} = 1500 \text{ ms}^{-1}$$

$$(ii) \therefore v_1 = \sqrt{\frac{T_1}{\mu}} \text{ and } v_2 = \sqrt{\frac{T_2}{\mu}}$$

$$\therefore \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} \Rightarrow \frac{v_2}{1500} = \sqrt{\frac{4}{9}}$$

$$\therefore v_2 = 1000 \text{ ms}^{-1}$$

Wavelength, $\lambda_2 = 2L = 200 \text{ cm} = 2 \text{ m}$

$$\text{Frequency, } f_2 = \frac{v_2}{\lambda_2} = \frac{1000}{2} = 500 \text{ Hz}$$

Example 17.74

- (i) A wire having a linear mass density $6 \times 10^{-3} \text{ kg m}^{-1}$ is stretched between two rigid supports with a tension of 540 N. The wire resonates at a frequency of 410 Hz. The next higher frequency at which the same wire resonates is 480 Hz. Find the length of the wire.
- (ii) A 40 cm wire having a mass of 3.2 g is stretched between two fixed supports 40.05 cm apart. In its fundamental mode, the wire vibrates at 220 Hz. If the area of cross-section of the wire is 10^{-6} m^2 , find its Young's modulus.

Sol. (i) Given, $\mu = 6 \times 10^{-3} \text{ kg m}^{-1}$ and $T = 540 \text{ N}$

Let frequency of n th harmonic be 410 Hz, then

frequency of $(n + 1)$ th harmonic be 480 Hz.

Fundamental frequency = $480 - 410 = 70 \text{ Hz}$

$$f_1 = \frac{1}{2l} \sqrt{\frac{T}{\mu}} \Rightarrow 70 = \frac{1}{2l} \sqrt{\frac{540}{6 \times 10^{-3}}} = \frac{1}{2l} \times 300$$

$$\therefore \text{Length of the wire, } l = \frac{150}{70} = 2.1 \text{ m}$$

(ii) Given, $L = 40 \text{ cm}$, $M = 3.2 \text{ g}$,

$$\Delta L = 40.05 - 40 = 0.05 \text{ cm},$$

$$f = 220 \text{ Hz and } A = 10^{-6} \text{ m}^2$$

$$\text{Young modulus, } Y = \frac{T/A}{\Delta L/L}$$

$$\text{Tension, } T = AY \frac{\Delta L}{L} = \frac{AY(0.05)}{40} = \frac{AY}{800}$$

$$\Rightarrow T = \frac{AY}{800} = \frac{10^{-6} Y}{800}$$

$$\mu = \frac{M}{L} = \frac{3.2 \times 10^{-3}}{40 \times 10^{-2}} = 8 \times 10^{-3} \text{ kg m}^{-1}$$

$$\text{Frequency, } f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

$$\text{or } 220 = \frac{1}{2 \times 0.4} \sqrt{\frac{10^{-6} Y}{800 \times 8 \times 10^{-3}}}$$

$$176 = \sqrt{\frac{Y}{6.4 \times 10^6}}$$

$$\text{Young's modulus, } Y = 6.4 \times 10^6 \times (176)^2 = 1.98 \times 10^{11} \text{ Nm}^{-2}$$

CHECK POINT 17.4

- Echo is based on which amongst the given phenomenon that sound wave follows?
 - Refraction
 - Reflection
 - Interference
 - Resonance
- Stationary waves are formed when
 - two waves of equal amplitude and equal frequency travel along the same path in opposite directions
 - two waves of equal wavelength and equal amplitude travel along the same path with equal speeds in opposite directions
 - two waves of equal wavelength and equal phase travel along the same path with equal speed
 - two waves of equal amplitude and equal speed travel along the same path in opposite directions
- In a stationary wave, all the particles
 - on either side of a node vibrate in same phase
 - in the region between two nodes vibrate in same phase
 - in the region between two antinodes vibrate in same phase
 - of the medium vibrate in same phase
- Stationary waves
 - transmit energy
 - do not transmit energy
 - have nodes and antinodes
 - Both (b) and (c)
- Stationary waves are so called because in them,
 - the particles of the medium are not disturbed at all
 - the particles of the medium do not execute SHM
 - there occurs no flow of energy along the wave
 - the interference effect cannot be observed
- When a stationary wave is formed, then its frequency is
 - same as that of the individual waves
 - twice that of the individual waves
 - half that of the individual waves
 - None of the above
- A wave represented by the equation $y = a \cos(kx - \omega t)$ is superposed with another wave to form a stationary wave such that the point $x = 0$ is a node. The equation of the other wave is
 - $a \sin(kx + \omega t)$
 - $-a \cos(kx + \omega t)$
 - $-a \cos(kx - \omega t)$
 - $-a \sin(kx - \omega t)$
- A standing wave is represented by

$$y = a \sin(100t) \cos(0.01x)$$
 where, t is in second and x is in metre. Then, the velocity of the constituent wave is
 - 10^2 ms^{-1}
 - 10 ms^{-1}
 - 1 ms^{-1}
 - 10^4 ms^{-1}
- Two transverse sinusoidal waves travel in opposite directions along a string. The speed of transverse waves in the string is 0.5 cm s^{-1} . Each has an amplitude of 3.0 cm and wavelength of 6.0 cm. The equation for the resultant wave is
 - $y = 6 \sin \frac{\pi t}{6} \cos \frac{\pi x}{3}$
 - $y = 6 \sin \frac{\pi x}{3} \cos \frac{\pi t}{6}$
 - Both may be correct
 - Both are wrong
- In stationary waves,
 - change in pressure and density are maximum at nodes
 - change in pressure and density are minimum at nodes
 - strain is zero at nodes
 - energy is maximum at nodes

11. In a resonance column experiment, the first resonance is obtained when the level of the water in tube is 20 cm from the open end. Resonance will also be obtained when the water level is at a distance of
 (a) 40 cm from the open end
 (b) 60 cm from the open end
 (c) 80 cm from the open end
 (d) 90 cm from the open end
12. Velocity of sound in air is 320 ms^{-1} . A pipe closed at one end has length of 1 m. Neglecting end corrections, the air column in the pipe can resonate for sound of frequency
 (a) 80 Hz
 (b) 240 Hz
 (c) 400 Hz
 (d) All of these
13. A resonating column has resonant frequencies as 100 Hz, 300 Hz, 500 Hz. Then, it may be
 (a) an open pipe
 (b) a pipe closed at both ends
 (c) pipe closed at one end
 (d) Data insufficient
14. A tuning fork of frequency 340 Hz is sounded above a cylindrical tube 1 m long. Water is slowly poured into the tube. If the speed of sound is 340 ms^{-1} , at what levels of water in the tube will the sound of the tuning fork be appreciably intensified?
 (a) 25 cm, 75 cm
 (b) 20 cm, 80 cm
 (c) 23 cm, 67 cm
 (d) 40 cm, 80 cm
15. For a certain organ pipe, three successive resonance frequencies are observed at 425, 595 and 765 Hz. The speed of sound in air is 340 ms^{-1} . The pipe is a
 (a) closed pipe of length 1 m
 (b) closed pipe of length 2 m
 (c) open pipe of length 1 m
 (d) open pipe of length 2 m
16. Two closed pipes produce 10 beats per second when emitting their fundamental notes. If their lengths are in ratio of 25 : 26. Then, their fundamental frequency (in Hz) are
 (a) 270, 280
 (b) 260, 270
 (c) 260, 250
 (d) 260, 280
17. The fundamental frequency of a vibrating organ pipe is 10 Hz. Select the wrong alternative.
 (a) The first overtone is 200 Hz
 (b) The first overtone may be 400 Hz
 (c) The first overtone may be 300 Hz
 (d) None of the above
18. A closed organ pipe and an open organ pipe of same length produce 2 beats when they are set into vibrations simultaneously in their fundamental mode. The length of open organ pipe is now halved and of closed organ pipe is doubled. The number of beats produced will be
 (a) 8
 (b) 7
 (c) 4
 (d) 2
19. In a resonance pipe, the first and second resonances are obtained at depths 22.7 cm and 70.2 cm, respectively. What will be the end correction?
 (a) 1.05 cm
 (b) 1115.5 cm
 (c) 92.5 cm
 (d) 113.5 cm
20. In the experiment for the determination of the speed of sound in air using the resonance column method, the length of the air column that resonates in the fundamental mode, with a tuning fork is 0.1 m. When this length is changed to 0.35 m, the same tuning fork resonates with the first overtone. Calculate the end correction.
 (a) 0.012 m
 (b) 0.025 m
 (c) 0.05 m
 (d) 0.024 m
21. When the string of a sonometer of length L between the bridges vibrates in the second overtone, the amplitude of vibration is maximum at
 (a) $\left(\frac{L}{2}\right)$
 (b) $\left(\frac{L}{4}\right)$ and $\left(\frac{3L}{4}\right)$
 (c) $\frac{L}{6}$ and $\frac{3L}{6}$
 (d) $\frac{L}{8}$, $\frac{3L}{8}$, $\frac{5L}{8}$
22. A 1 cm long string vibrates with fundamental frequency of 256 Hz. If the length is reduced to $\frac{1}{4}$ cm keeping the tension unaltered, the new fundamental frequency will be (in Hz)
 (a) 64
 (b) 256
 (c) 512
 (d) 1024
23. In order to double the frequency of the fundamental note emitted by a stretched string, the length is reduced to $\frac{3}{4}$ th of the original length and the tension is changed. The factor by which the tension is to be changed, is
 (a) $\frac{3}{8}$
 (b) $\frac{2}{3}$
 (c) $\frac{8}{9}$
 (d) $\frac{9}{4}$
24. To increase the frequency from 100 Hz to 400 Hz, the tension in the string has to be changed by
 (a) 4 times
 (b) 16 times
 (c) 2 times
 (d) None of these
25. Fundamental frequency of a sonometer wire is n . If the tension is made 3 times and; length and diameter are also increased 3 times, what is the new frequency?
 (a) $\frac{n}{3\sqrt{3}}$
 (b) $3n$
 (c) $\sqrt{3}n$
 (d) $\frac{n}{\sqrt{3}}$
26. The correct graph between the frequency n and square root of density (ρ) of a wire, keeping its length, radius and tension constant, is
 (a)
 (b)
 (c)
 (d)
27. For a string clamped at both ends, which of the following wave equation is valid for a stationary wave set up in it? (Origin is at one end of string)
 (a) $y = A \sin kx \sin \omega t$
 (b) $y = A \cos kx \sin \omega t$
 (c) $y = A \cos kx \cos \omega t$
 (d) All of the above

28. The vibrations of a string of length 60 cm fixed at both ends are represented by the equation $y = 4 \sin\left(\frac{\pi x}{15}\right) \cos(96\pi t)$,

where x and y are in cm and t in second. The maximum displacement at $x = 5$ cm is

- (a) $2\sqrt{3}$ cm (b) $3\sqrt{2}$ cm
(c) $\sqrt{2}$ cm (d) $\sqrt{3}$ cm

29. Resonance is a special case of

- (a) undamped vibration (b) damped vibration
(c) forced vibration (d) natural vibration

30. A string of length l fixed at both the ends is vibrating in two segments. The wavelength of the corresponding wave is

- (a) $\frac{l}{4}$ (b) $\frac{l}{2}$
(c) l (d) $2l$

31. A string of length 1 m has the mass per unit length 0.1 g cm^{-1} . What would be the fundamental frequency of vibration of this string under tension of 400 N?

- (a) 400 Hz (b) 100 Hz
(c) 50 Hz (d) 200 Hz

DOPPLER'S EFFECT

If a wave source and a receiver are moving relative to each other, the frequency observed by the receiver (f') is different from the actual source frequency (f). This phenomenon is called as Doppler's effect, named after the Austrian physicist Christian Johann Doppler (1803–1853), who discovered it in light waves.

It is a wave phenomenon which is associated with sound as well as other electromagnetic waves like light.

Doppler's effect in sound

Perhaps you might have noticed how the sound of a vehicle's horn changes as the vehicle moves past you. The frequency (pitch) of the sound you hear as the vehicle approaches you is higher than the frequency you hear as it moves away from you. This happens because of Doppler's effect. Generally, the Doppler's effect applies to waves.

Let us apply it to sound waves. We have consider the special case in which the source and observer move along the line joining them.

We will use the following symbols,

v = speed of sound, v_s = speed of source

and v_o = speed of observer.

Firstly, we will consider two special cases. Then, the results obtained can be generalised. To observe Doppler's effect, relative separation between source and observer must be changing.

Now, consider the following situations.

1. When sound source and observer both are stationary : separation between source and observer remains same

A stationary point source of sound can produce sound waves in a medium. When this point source is at rest, then sound waves propagate in all directions in the form of spherical wavefronts with speed of sound (v) and wavelength (λ) in the medium. These wavefronts are equally spaced as shown in figure. These sound waves are observed by the stationary observer.

Now, due to fluctuations in the pressure of the medium, this observer hears sound.

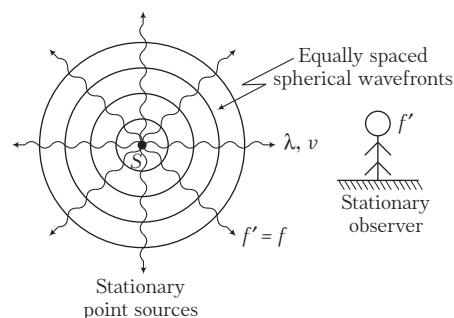


Fig. 17.35

If both source and observer are stationary, then due to equal spacing of wavefronts, reaching at the observer, the observer observes actual frequency (f) of the source, i.e. $f' = f$.

e.g. When horn of a stationary car shown in the

Fig. 17.36 is blowing with frequency f , the frequency of sound observed by a stationary observer standing on the road is also f .

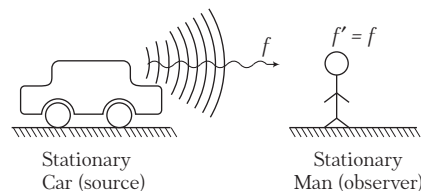


Fig. 17.36

Example 17.75 A train is moving with a speed of 100 km/h. Now, suddenly it blows horn with a frequency of 100 Hz. If speed of sound in air is 330 ms^{-1} , then find the apparent frequency heard by a passenger sitting in the train.

Sol. Here, speed of the passenger (observer) is same as the source (train). Therefore, no Doppler's effect will be observed. Thus, apparent frequency heard by the passenger,

$$f' = f = \text{actual frequency of horn} = 100 \text{ Hz}$$

2. When sound source and observer are moving towards each other : separation between source and observer decreases

In this situation, due to decrease in separation between source and observer, the wavefronts are unequally spaced.

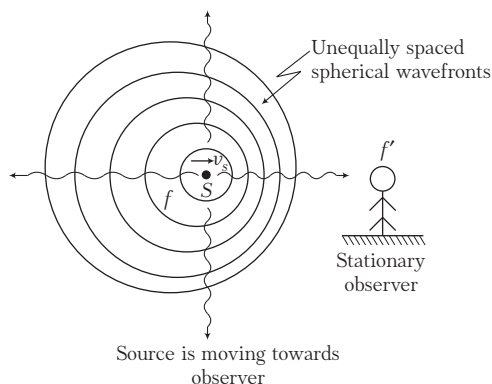


Fig. 17.37

In the given figure, when a sound source of frequency f , moves towards a stationary observer, the separation between the wavefronts towards the observer decreases and hence, the wavelength of sound in the medium decreases.

Therefore, frequency (f') of sound observed by the observer (apparent frequency) increases, i.e. $f' > f$.

e.g. A horn blowing car is moving towards a stationary observer as shown in Fig. 17.38. The frequency (f') of sound observed by the observer is more than the actual frequency of horn (f), i.e. $f' > f$

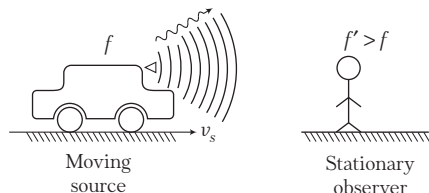


Fig. 17.38

3. When sound source and observer are moving away from each other : separation between source and observer increases

In this situation, due to increase in separation between source and observer, the wavefronts are again unequally spaced. In the following figure, a sound source of frequency f , moves away from a stationary observer.

The separation between the wavefronts towards the observer increases and hence, the wavelength of sound in the medium increases. Therefore, frequency (f') of sound observed by the observer (apparent frequency) decreases, i.e. $f' < f$.

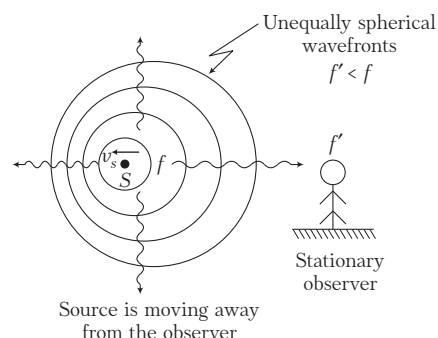


Fig. 17.39

e.g. A horn blowing car is moving away from a stationary man (observer) standing on the road as shown in the Fig. 17.40. The frequency (f') of sound observed by the observer is less than the actual frequency of horn (f), i.e. $f' < f$.

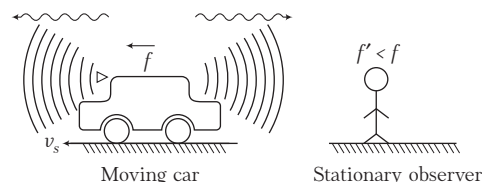


Fig. 17.40

Mathematical expression for apparent frequency due to Doppler's effect in sound

Fundamental cause behind the Doppler's effect is the relative motion between the sound source and observer. Therefore, following three cases are possible

- I. When sound source is moving and the observer is stationary.
- II. When sound source is stationary and observer is moving.
- III. When both sound source and the observer are moving.

Case I When sound source is moving and the observer is stationary.

In this case, following two situations are possible

1. Sound source is moving towards a stationary observer

Let a sound source (S) produces sound waves of frequency f_o and speed v in air.

When both the source and observer are stationary, waves are equally spaced and apparent wavelength of sound observed by the observer,

$$\lambda_a = \frac{v}{f_o} = \lambda_o = \text{actual wavelength of the waves.}$$

Now, the source is moving towards the observer with speed v_s , therefore waves towards the observer are compressed in comparison with actual wavelength λ_o as shown in figure.

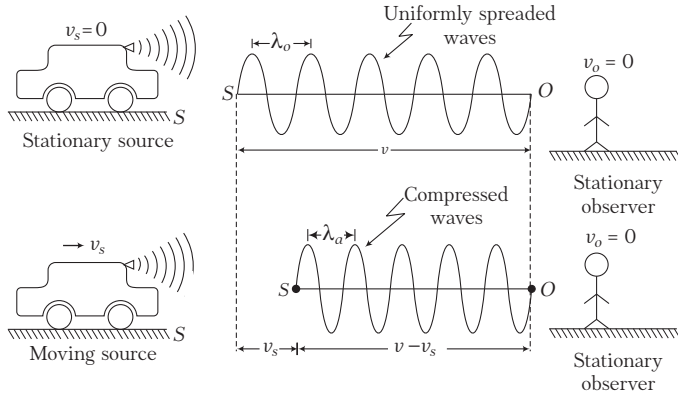


Fig. 17.41

Initially, sound waves were spreaded in the length of v in one second. Now, due to motion of the source, the waves are compressed and spreaded in the length $(v - v_s)$ in one second.

Therefore, apparent wavelength of sound received by the observer,

$$\lambda_a = \frac{v - v_s}{f_o}$$

$\therefore$

$$\frac{v - v_s}{f_o} < \frac{v}{f_o}$$

$\therefore \lambda_a < \lambda_o$, i.e. apparent wavelength decreases.

Also, apparent frequency,

$$f_a = \frac{v}{\lambda_a} = \frac{v}{\frac{v - v_s}{f_o}}$$

$\Rightarrow$

$$f_a = \left(\frac{v}{v - v_s} \right) f_o$$

where, f_a = apparent frequency,

v = speed of sound in the medium,

v_s = speed of source

and f_o = original frequency of sound source.

$\therefore$

$$\frac{v}{v - v_s} > 1$$

$\Rightarrow$

$$f_a > f_o$$

Thus, apparent frequency of sound observed by the observer is greater than the actual frequency of sound source.

Example 17.76 An ambulance is moving with blowing horn towards a person standing on the road with a speed of 54 kmh^{-1} . What frequency does the person hear, if the velocity of sound in air is 330 ms^{-1} and frequency of horn is 450 Hz ?

Sol. Given, frequency of horn, $f = 450 \text{ Hz}$

Speed of ambulance, $v_s = 54 \text{ kmh}^{-1} = 15 \text{ ms}^{-1}$

$v_o = 0$ and speed of sound in air, $v = 330 \text{ ms}^{-1}$

As source is moving towards a stationary observer, therefore frequency heard by the observer,

$$\begin{aligned} f' &= \left(\frac{v}{v - v_s} \right) f = \left(\frac{330}{330 - 15} \right) 450 \text{ Hz} \\ &= \frac{330}{315} \times 450 \text{ Hz} = 471.43 \text{ Hz} \end{aligned}$$

2. Sound source is moving away from a stationary observer

Let the source of sound now moves away from the stationary observer. As, the wavelength of sound observed when both the source and observer are stationary is

$$\lambda_a = \frac{v}{f_o} = \lambda_o = \text{actual wavelength.}$$

Now, when the source moves away from the observer, the waves are expanded as shown in figure.

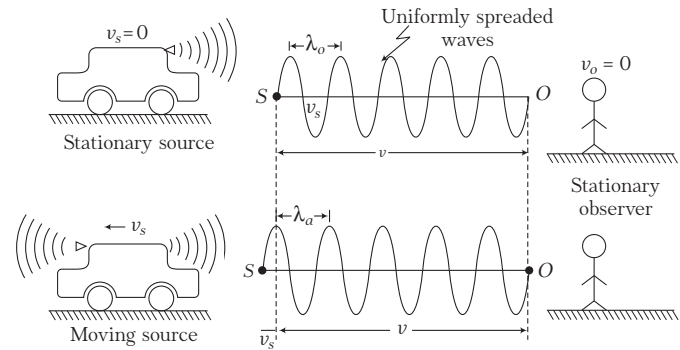


Fig. 17.42

Initially, sound waves were spreaded in the length of v in one second. When the source (car) moves away from the source now the waves are expanded and spreaded in the length of $(v + v_s)$ in one second. Therefore, apparent wavelength of sound received by the observer,

$$\lambda_a = \frac{v + v_s}{f_o}$$

$\therefore$

$$\frac{v + v_s}{f_o} > \frac{v}{f_o}$$

$\therefore \lambda_a > \lambda_o$, i.e. apparent wavelength increases.

Also, apparent frequency,

$$f_a = \frac{v}{\lambda_a} = \frac{v}{\frac{v + v_s}{f_o}} \Rightarrow f_a = \left(\frac{v}{v + v_s} \right) f_o$$

where, f_a = apparent frequency,
 v = speed of sound in the medium,
 v_s = speed of source
 and f_o = original frequency of sound source.

$$\therefore \frac{v}{v + v_s} < 1 \Rightarrow f_a < f_o$$

Thus, apparent frequency of sound observed by the observer is less than the actual frequency of sound source.

Example 17.77 A listener is at rest and a police siren is moving away from the listener at 60 ms^{-1} . What frequency does the listener hear, given that the velocity of sound in air is 340 ms^{-1} and frequency of siren is 500 Hz ?

Sol. In this case, observer is at rest and source is moving away from observer.

Therefore, observed frequency,

$$\begin{aligned} f' &= \left(\frac{v}{v + v_s} \right) f = \left(\frac{340}{340 + 60} \right) 500 \\ &= \frac{340}{400} \times 500 = \frac{1700}{4} = 425 \text{ Hz} \end{aligned}$$

Case II When sound source is stationary and observer is moving.

In this case, following two situations are possible

1. Observer is moving towards a stationary sound source

Let the observer now moves towards a stationary source of sound. As the frequency of sound observed when both the source and observer are stationary is actual frequency f_o .

Now, when the observer is moving towards a stationary sound source, the number of waves received by the observer in one second increases as explained in the figure.

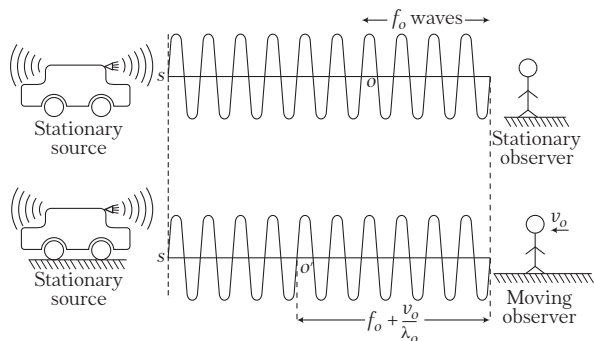


Fig. 17.43

Initially, number of waves received by the observer in one second = f_o .

Now, when the observer moves through a distance v_o in one second, number of waves received = $f_o + \frac{v_o}{\lambda_o}$.

$$\therefore \text{Apparent frequency, } f_a = f_o + \frac{v_o}{\lambda_o}$$

$$\Rightarrow f_a = f_o + \frac{v_o}{v/f_o} \quad \left(\because \lambda_o = \frac{v}{f_o} \right)$$

$$\Rightarrow f_a = \left(\frac{v + v_o}{v} \right) f_o$$

$$\therefore \frac{v + v_o}{v} > 1$$

$\therefore f_a > f_o$, i.e. apparent frequency observed by the observer is greater than the actual frequency.

Example 17.78 A man is travelling in a train towards the station with a speed of 50 ms^{-1} . Calculate the apparent frequency heard by him of a whistle which is blown at the station with a frequency 200 Hz . Velocity of sound in air is 350 ms^{-1} .

Sol. In this case, observer is moving and source is at rest.

Therefore, apparent frequency heard by the observer,

$$\begin{aligned} f' &= \left(\frac{v + v_o}{v} \right) f = \left(\frac{350 + 50}{350} \right) \times 200 \\ &= \frac{400}{350} \times 200 = 228.57 \text{ Hz} \end{aligned}$$

2. Observer is moving away from a stationary sound source

Let the observer now moves away from a stationary source of sound. Initially, when both source and observer were stationary, number of waves received by the observer in one second was f_o . Now, when the observer is moving away from the stationary source, the number of waves received by the observer in one second decreases as explained in the figure.

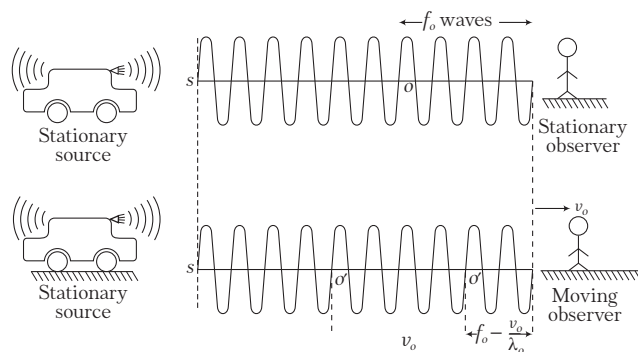


Fig. 17.44

Initially, number of waves received by the observer in one second = f_o .

Now, when the observer moves away through a distance v_o in one second, number of waves received is $f_o - \frac{v_o}{\lambda_o}$.

$$\therefore \text{Apparent frequency, } f_a = f_o - \frac{v_o}{\lambda_o}$$

$$\Rightarrow f_a = f_o - \frac{v_o}{v/f_o} \quad \left(\because \lambda_o = \frac{v}{f_o} \right)$$

$$\Rightarrow f_a = \left(\frac{v - v_o}{v} \right) f_o$$

$$\therefore \frac{v - v_o}{v} < 1$$

$\therefore f_a < f_o$, i.e. apparent frequency observed by the observer is less than the actual frequency.

Example 17.79 An observer moves away from a stationary source of sound with a velocity one-fifth of the velocity of sound. Find the percentage decrease in the apparent frequency. (Take, sound in air = 320 ms^{-1})

Sol. Given, $v_o = \frac{v}{5}$

$$\Rightarrow v_o = \frac{320}{5} = 64 \text{ ms}^{-1}$$

When observer moves away from the stationary source, then

$$f' = \left(\frac{v - v_o}{v} \right) f$$

$$\Rightarrow f' = \left(\frac{320 - 64}{320} \right) f$$

$$\Rightarrow f' = \left(\frac{256}{320} \right) f$$

$$\Rightarrow \frac{f'}{f} = \frac{256}{320}$$

Hence, percentage decrease in frequency,

$$\left(\frac{f' - f}{f} \right) = \left(\frac{320 - 256}{320} \times 100 \right) \%$$

$$f = \left(\frac{64}{320} \times 100 \right) \% = 20\%$$

Case III When both the sound source and the observer are moving

Consider the situation as shown in figure in which the sound source and the observer both are moving.

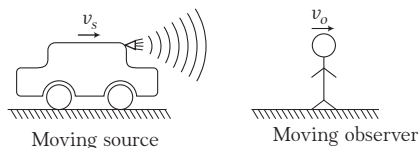


Fig. 17.45

We know that, when observer is stationary and sound source is moving towards the observer, then apparent frequency,

$$f_{a1} = \left(\frac{v}{v - v_s} \right) f \quad \dots (i)$$

Similarly, when source is stationary and the observer is moving away from the source, then apparent frequency,

$$f_{a2} = \left(\frac{v - v_o}{v} \right) f_{a1} \quad \dots (ii)$$

When both the source and observer are moving as shown in the Fig. 17.45, then from Eqs. (i) and (ii), we get

$$f_a = f_o \left(\frac{v - v_o}{v - v_s} \right)$$

When both or any one of the source and observer is moving in opposite direction of velocity of sound, then we have f_o put $-v_o$ and $-v_s$ in the place of v_o and v_s , respectively.

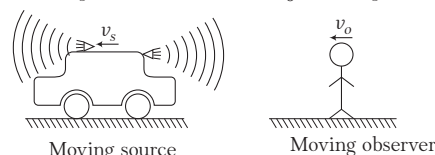


Fig. 17.46

Therefore, for the situation as shown in Fig. 17.46, in which both the source and observer moves leftward and sound waves moves rightward.

Apparent frequency observed by the observer,

$$f_a = f_o \left(\frac{v + v_o}{v + v_s} \right)$$

Note The equation $f_a = f_o \left(\frac{v - v_o}{v - v_s} \right)$ can be used as general equation for Doppler's effect. When any one of the source or observer is at rest, then put $v_o = 0$ or $v_s = 0$.

Example 17.80 A policeman is chasing a thief from behind, while riding on a bike which is moving with a speed of 54 kmh^{-1} . The thief is also riding on a bike moving with a speed of 72 kmh^{-1} . When the policeman blows horn of the bike having frequency of 200 Hz . Find the apparent frequency heard by the thief. (Take, speed of sound = 330 ms^{-1}).

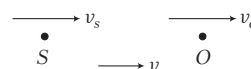
Sol. Given, $v_s = 54 \text{ kmh}^{-1} = 15 \text{ ms}^{-1}$, $v_o = 72 \text{ kmh}^{-1} = 20 \text{ ms}^{-1}$, $v = 330 \text{ ms}^{-1}$ and $f = 200 \text{ Hz}$

$\therefore$ Apparent frequency observed by the thief,

$$\begin{aligned} f' &= \left(\frac{v - v_o}{v - v_s} \right) f = \left(\frac{330 - 20}{330 - 15} \right) \times 200 \\ &= \frac{310}{315} \times 200 = 196.82 \text{ Hz} \end{aligned}$$

Example 17.81 A car moving at 72 km/h finds another car in front of it going in the same direction at 36 km/h . The first car sounds a horn that has a dominant frequency of 900 Hz . What will be the apparent frequency heard by the driver in the front car? (Take, speed of sound in air = 330 ms^{-1})

Sol. The given situation is shown below.



Velocity of source, $v_s = 72 \text{ km/h}$

$$= 72 \times \frac{5}{18}$$

$$= 20 \text{ m/s}$$

Velocity of observer, $v_o = 36 \text{ km/h}$

$$= 36 \times \frac{5}{18}$$

$$= 10 \text{ m/s}$$

$$v = 330 \text{ ms}^{-1} \text{ and } f = 900 \text{ Hz}$$

$$\text{Apparent frequency, } f' = f \left(\frac{v - v_o}{v - v_s} \right)$$

$$= 900 \left(\frac{330 - 10}{330 - 20} \right)$$

$$= 929 \text{ Hz}$$

For different situations of the source and the observer, the apparent frequencies observed due to Doppler's effect are tabulated as follows

Observer	Source)	Apparent frequency
$\rightarrow v_o$	Stationary ($v_s = 0$)	$f_a = \left(\frac{v + v_o}{v} \right) f_o$
$v_o \leftarrow$	Stationary ($v_s = 0$)	$f_a = \left(\frac{v - v_o}{v} \right) f_o$
Stationary ($v_o = 0$)	$\rightarrow v_s$	$f_a = \left(\frac{v}{v + v_s} \right) f_o$
Stationary ($v_o = 0$)	$v_s \leftarrow$	$f_a = \left(\frac{v}{v - v_s} \right) f_o$
$\rightarrow v_o$	$\rightarrow v_s$	$f_a = \left(\frac{v + v_o}{v + v_s} \right) f_o$
$v_o \leftarrow$	$\rightarrow v_s$	$f_a = \left(\frac{v - v_o}{v + v_s} \right) f_o$
$\rightarrow v_o$	$v_s \leftarrow$	$f_a = \left(\frac{v + v_o}{v - v_s} \right) f_o$
$v_o \leftarrow$	$v_s \leftarrow$	$f_a = \left(\frac{v - v_o}{v - v_s} \right) f_o$

It is obvious from the table that the apparent frequency has maximum value of $\left(\frac{v + v_o}{v - v_s} \right) f_o$, when source and observer are moving towards each other. Similarly, apparent frequency has minimum value of $\left(\frac{v - v_o}{v + v_s} \right) f_o$, when source and observer are moving away from each other. Because of this reason, when two vehicles cross each other, then frequency decreases suddenly.

Similarly, when the source and the observer are moving in the same direction such that $v_o = v_s$, then no Doppler's effect is observed. This is because there is no relative motion between the source and observer and actual frequency is observed by the observer.

Example 17.82 A source of sound and an observer are moving away from each other with a speed of 20 ms^{-1} . Find the apparent frequency heard by the observer, if original frequency of the source is 100 Hz . (Take, speed of sound in air $= 330 \text{ ms}^{-1}$)

Sol. Given, $v_o = v_s = 20 \text{ ms}^{-1}$

Apparent frequency heard by the observer,

$$f' = \left(\frac{v - v_o}{v + v_s} \right) f = \left(\frac{330 - 20}{330 + 20} \right) 100 \text{ Hz}$$

$$= \left(\frac{310}{350} \right) 100 \text{ Hz} = 88.57 \text{ Hz}$$

Example 17.83 A source of sound is approaching an observer with speed of 30 ms^{-1} and the observer is approaching the source with a speed of 50 ms^{-1} . Find the fractional change in the frequency of sound (speed of sound in air $= 330 \text{ ms}^{-1}$).

Sol. Given, $v_s = 30 \text{ ms}^{-1} \Rightarrow v_o = 50 \text{ ms}^{-1}$

$$\text{Apparent frequency, } f' = f \left[\frac{v + v_o}{v - v_s} \right]$$

$$\Rightarrow f' = f \left[\frac{330 + 50}{330 - 30} \right]$$

$$\Rightarrow f' = f \left[\frac{380}{300} \right]$$

$$\Rightarrow f' = f \left[\frac{38}{30} \right]$$

$\therefore$ Fractional change in frequency,

$$\left(\frac{f' - f}{f} \right) = \frac{38 - 30}{30} = \frac{8}{30} = \frac{4}{15}$$

Special cases of Doppler's effect in sound

In this section, we are going to deal with various typical cases of Doppler's effect in sound.

Case I Doppler's effect in reflected sound

When the sound wave is reflected from a reflector (wall/cliff), the observer receives two notes: one directly from the source and other from the reflector. If the two frequencies are different, then the observer may observe beats. The following situation explains this fact.

1. A source is moving towards a wall

(Doppler's effect in reflected sound)

Consider a source of sound moving towards a stationary cliff/wall with speed v_s emitting a sound of frequency, f .

There is a stationary observer (O).

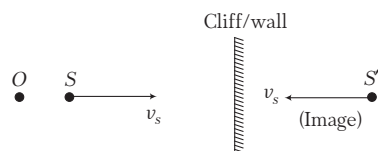


Fig. 17.47

Apparent frequency of sound heard by the observer directly from the source in this case is

$$f' = \left(\frac{v}{v + v_s} \right) f \quad (\text{source is moving away})$$

where, f = frequency of source, v_s = speed of source and v = speed of sound

To find frequency of reflected sound heard by the observer, assume image of the source S' in the wall.

Now, this image will act as a sound source moving towards the observer with speed v_s . Apparent frequency heard by the observer is $f'' = \left(\frac{v}{v - v_s} \right) f$

Beat frequency heard by the observer,

$$f_b = f'' - f' = \left(\frac{v}{v - v_s} - \frac{v}{v + v_s} \right) f$$

$$f_b = \left(\frac{2v_s v}{v^2 - v_s^2} \right) f$$

Note If car is approaching a wall, then frequency of reflected sound heard by the observer in the car,

$$f = \left(\frac{v + u}{v - u} \right) f$$

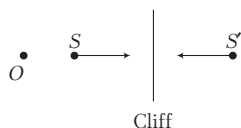
where, u is the velocity of car.

$$\therefore \text{Beat frequency, } f_b = f \left(\frac{v + u}{v - u} \right) - f = \left(\frac{2u}{v - u} \right) f$$

Example 17.84 Imagine siren emitting a sound of frequency 1000 Hz moves away from you toward a cliff at a speed of 10 ms^{-1} .

- What is the frequency of the sound you hear coming directly from the siren?
- What is the frequency of the sound you hear reflected off from the cliff?
- What beat frequency would you hear? Take the speed of sound in air as 330 ms^{-1}

Sol. The given situation is shown below.



- Frequency of sound reaching directly to us (by S),

$$f_1 = \left(\frac{v}{v + v_s} \right) f = \left(\frac{330}{330 + 10} \right) (1000) = 970.6 \text{ Hz}$$

- Frequency of sound which is reflected off from the cliff (from S'),

$$f_2 = \left(\frac{v}{v - v_s} \right) f = \left(\frac{330}{330 - 10} \right) (1000) = 1031.3 \text{ Hz}$$

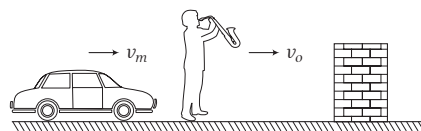
- Beat frequency = $f_2 - f_1 = 60.7 \text{ Hz}$

Example 17.85 A band playing music at a frequency f is moving towards a wall at a speed v_b . A car is following the band with a speed v_m . If v is the speed of sound, obtain an expression for the beat frequency heard by the driver sitting in the car.

Sol. Here $\rightarrow v_o = v_m \quad v_s = v_b \rightarrow$

Frequency received by driver from band,

$$f_1 = f \left(\frac{v - v_o}{v + v_s} \right) = f \left(\frac{v - v_m}{v + v_b} \right) \quad \dots(i)$$



Here, $\rightarrow v_s = v_b \quad v_o = 0 \rightarrow$

$$f' = f \left(\frac{v - v_o}{v + v_s} \right) = f \left(\frac{v}{v + v_b} \right)$$

Now, wall will act as source of frequency f' and driver as observer.

Again, $\rightarrow v_o = v_m \quad v_s = v_b \rightarrow$

Frequency received by driver after reflection from wall,

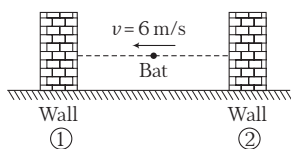
$$f_2 = f' \left(\frac{v - v_o}{v + v_s} \right) = f \left(\frac{v}{v + v_b} \right) \left(\frac{v + v_m}{v} \right) = f \left(\frac{v + v_m}{v - v_b} \right)$$

Beat frequency heard by driver,

$$\begin{aligned} f_2 - f_1 &= f \left(\frac{v + v_m}{v - v_b} \right) - f \left(\frac{v - v_m}{v + v_b} \right) \\ &= \frac{f}{(v^2 - v_b^2)} [(v + v_m)(v + v_b) - (v - v_m)(v - v_b)] \\ &= \frac{f}{(v^2 - v_b^2)} \times [v^2 + vv_b + v_mv + v_mv_b] - [v^2 - vv_b - v_mv + v_mv_b] \\ &= \frac{f \cdot 2v(v_b + v_m)}{(v^2 - v_b^2)} = \frac{2fv(v_b + v_m)}{(v^2 - v_b^2)} \end{aligned}$$

Example 17.86 A bat emitting an ultrasonic wave of frequency 2×10^4 flies at a speed of 6 m/s between two parallel walls. Find the two frequencies heard by the bat and the beat frequency between the two. The speed of sound is 330 m/s

Sol. Frequency received by bat after reflection from wall (1),



$$f_1 = f \left(\frac{v+u}{v-u} \right) = (2 \times 10^4) \left(\frac{330+6}{330-6} \right)$$

$$= 2.03 \times 10^4 \text{ Hz}$$

Frequency received by bat after reflection from wall (2),

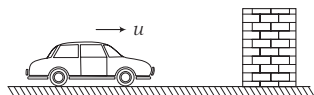
$$f_2 = f \left(\frac{v-u}{v+u} \right) = (2 \times 10^4) \left(\frac{330-6}{330+6} \right)$$

$$= 1.9 \times 10^4 \text{ Hz}$$

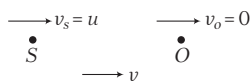
$$\text{Beat frequency} = f_1 - f_2 = 2.03 \times 10^4 - 1.9 \times 10^4 = 1300 \text{ Hz}$$

Example 17.87 The driver of a car approaching a vertical wall notices that the frequency of horn of his car changes from 400 Hz to 480 Hz, when it gets reflected from the wall. Find the speed of car if that of sound is 330 m/s.

Sol. Let u be the speed of car towards the wall.

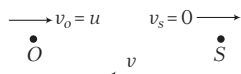


(a) **Frequency received at wall** The car is acting as source emitting frequency f and wall as stationary source.



$$f' = f \left(\frac{v-v_o}{v-v_s} \right) = f \left(\frac{v}{v+u} \right)$$

(b) **Frequency received by driver** Now, wall will act as a source, emitting frequency f' and driver as observer.



$$f'' = f' \left(\frac{v+v_o}{v-v_s} \right) = f \left(\frac{v}{v+u} \right) \left(\frac{v+u}{v} \right)$$

$$f'' = f \left(\frac{v+u}{v-u} \right)$$

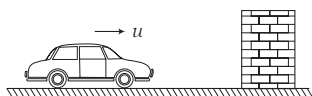
$$480 = 400 \left(\frac{330+u}{330-u} \right)$$

$$1.2(330-u) = 330+u$$

$$\Rightarrow 2.2u = 66 \Rightarrow u = 30 \text{ m/s}$$

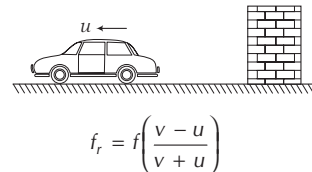
Note Frequency of reflected sound

(a) When car is moving towards wall,



$$f_r = f \left(\frac{v+u}{v-u} \right)$$

(b) When car is moving away from wall,



$$f_r = f \left(\frac{v-u}{v+u} \right)$$

Case II Sound source and observer are moving in a direction making an angle with the line joining them

Till now, we were assuming that v_o and v_s are along the line joining source and observer.

If the motion is along some other direction, join S and O by a straight line, then the components of velocities along the line joining source and observer are considered.

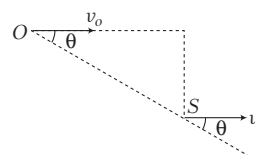


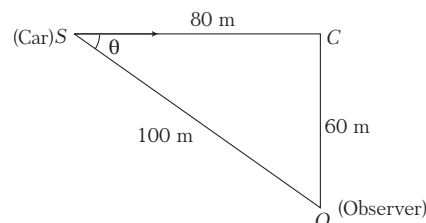
Fig. 17.48 Source is moving at an angle with the observer

For example, in the figure shown,

$$f' = \left(\frac{v + v_o \cos \theta}{v + v_s \cos \theta} \right) f$$

Example 17.88 A car approaching a crossing at a speed of 20 ms^{-1} sounds a horn of frequency 500 Hz when it is at a distance of 80 m from the crossing. Speed of sound in air is 330 ms^{-1} . What frequency is heard by an observer 60 m from the crossing on the straight road which crosses car road at right angles?

Sol. The given situation is as shown in figure.



$$\cos \theta = \frac{80}{100} = \frac{4}{5}$$

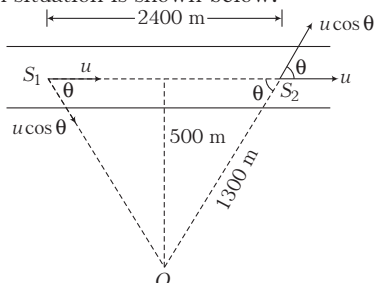
$\therefore$ Apparent frequency,

$$f_a = \left(\frac{v}{v - v_s \cos \theta} \right) f = \left(\frac{330}{330 - 20 \times \frac{4}{5}} \right) 500$$

$$= 525.5 \text{ Hz}$$

Example 17.89 Two electric trains run at the same speed of 72 km/h along the same track and in the same direction with a separation of 2.4 km between them. The two trains simultaneously sound brief whistles. A person is situated at a perpendicular distance of 500 m from the track and is equidistant from the two trains at the instant of the whistling. If both the whistles were at 500 Hz and the speed of sound in air is 340 m/s, find the frequencies heard by the person.

Sol. The given situation is shown below.



Speed of trains, $u = 72 \text{ km/h} = 20 \text{ m/s}$

From the figure, $\cos \theta = 12/13$

Frequency received at O, due to S_1 ,

$$f_1 = f \left(\frac{v}{v - u \cos \theta} \right) = 500 \left(\frac{340}{340 + \frac{20 \times 12}{13}} \right) = 529 \text{ Hz}$$

Frequency received at O, due to S_2 ,

$$f_2 = f \left(\frac{v}{v + u \cos \theta} \right) = 500 \left(\frac{340}{340 + \frac{20 \times 12}{13}} \right) = 474 \text{ Hz}$$

Case III Rotating Source/Observer

Let a source of sound/observer is rotating in a circle of radius r with angular velocity ω .

Now, following situations are possible

1. When source is rotating

- (i) When source moves towards the observer, then the frequency heard will be maximum.

$$f_{\max} = \left(\frac{v}{v - v_s} \right) f$$

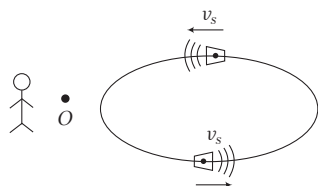


Fig. 17.49 Rotation of source towards and away from the observer

where, v = speed of sound, v_s = speed of source and f = frequency of sound from the source.

- (ii) When source moves away from the observer, then the frequency heard will be minimum.

$$f_{\min} = \left(\frac{v}{v + v_s} \right) f$$

- (iii) If observer is situated at centre of the circle, then there will be no change in frequency of sound heard.

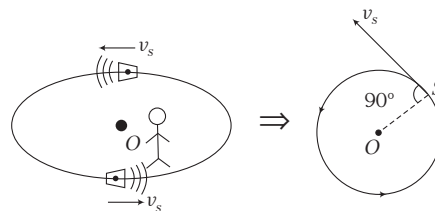


Fig. 17.50 Rotation of source when observer is at centre of the circle

2. When observer is rotating

- (i) When the observer moves towards the source, then the frequency heard will be maximum and is given by

$$f_{\max} = \left(\frac{v + v_o}{v} \right) f$$

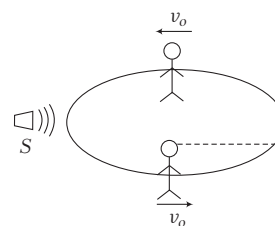


Fig. 17.51 Rotation of observer towards and away from source

- (ii) When the observer moves away from the source, then the heard frequency will be minimum and is given by

$$f_{\min} = \left(\frac{v - v_o}{v} \right) f$$

- (iii) When source is situated at the centre of the circle, there will be no change in frequency heard.

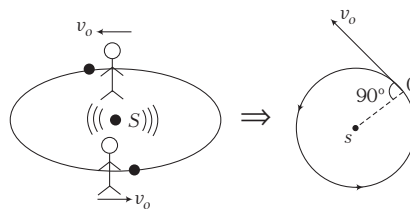
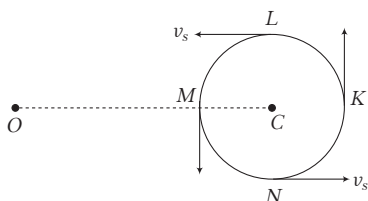


Fig. 17.52 When source is at centre of circle while observer is rotated

Example 17.90 A whistle of frequency 540 Hz rotates in a circle of radius 2 m at a linear speed of 30 ms^{-1} . What are the lowest and highest frequencies heard by an observer a long distance away at rest with respect to the centre of circle. Take speed of sound in air as 330 ms^{-1} . Can the apparent frequency be ever equal to actual frequency?

Sol. Apparent frequency will be minimum when the source is at N and is moving away from the observer



$$f_{\min} = \left(\frac{v}{v + v_s} \right) f = \left(\frac{330}{330 + 30} \right) (540) = 495 \text{ Hz}$$

Frequency will be maximum when source is at L and is approaching the observer.

$$f_{\max} = \left(\frac{v}{v - v_s} \right) f = \left(\frac{330}{330 - 30} \right) (540) = 594 \text{ Hz}$$

Further, when source is at M and K, angle between velocity of source and line joining source and observer is 90° or $v_s \cos \theta = v_s \cos 90^\circ = 0$. So, there will be no change in the apparent frequency.

Effect of motion of medium (air) on apparent frequency

Till now while studying Doppler's Effect, we have assumed medium at rest. If medium is in motion, then following cases are possible.

- (i) **When medium (air) moves from source to observer that is in the direction of velocity of sound** As the medium (air) itself starts moving, appropriate changes must be made in general equation of Doppler's effect. If wind blows at a speed v_w from the source to the observer, as shown in fig., take $v \rightarrow v + v_w$ (both in numerator and denominator) i.e.,

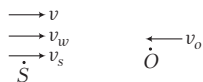


Fig. 17.53

$$f' = \left(\frac{v + v_w \pm v_o}{v + v_w \mp v_s} \right) f$$

- (ii) **When medium (air) moves from observer to source that is in the opposite direction of velocity of sound** As the wind blows in opposite direction of velocity of sound, as shown in figure,

take $v \rightarrow v - v_w$ (both in numerator and denominator) i.e.,

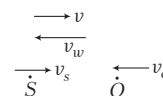


Fig. 17.54

$$f' = \left(\frac{v - v_w \pm v_o}{v - v_w \mp v_s} \right) f$$

Note Assuming medium, source and observer in motion, general equation for Doppler's effect can be written as

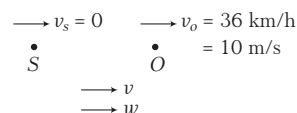
$$f' = \left(\frac{v \pm v_w \pm v_o}{v \pm v_w \mp v_s} \right) f$$

According to situation and given conditions, we will put signs of velocities in above equation.

There will be no doppler effect i.e., frequency will not change when both source and observer are at rest and wind blows.

Example 17.91 A person going away from a factory on his scooter at a speed of 36 km/h listens to the siren of the factory. If the main frequency of the siren is 600 Hz and a wind is blowing along the direction of the scooter at 36 km/h, find the main frequency as heard by the person. Velocity of sound in air = 340 m/s)

Sol. The given situation is shown below



Wind speed, $w = 36 \text{ km/h} = 10 \text{ m/s}$

$$f' = f \left(\frac{v + w - v_o}{v + w - v_s} \right) = 600 \left(\frac{340 + 10 - 10}{340 + 10 - 0} \right) = 582.86 \text{ Hz}$$

Applications of Doppler's effect in sound

Applications of Doppler's effect in sound are given below

- The change in frequency due to Doppler effect is used to measure velocities in diverse areas such as military, medical science, astrophysics, etc.
- In Doppler's effect, a sound wave of known frequency is sent towards a moving object. Some part of that wave is reflected from the object whose frequency can be measured by monitoring the stations. This change in frequency (i.e., due to reflection of the sound wave) is called doppler shift.
- It is used at airports to guide aircraft and in the military to detect enemy aircraft.
- Doctors use doppler echocardiography to examine the heart. It uses high frequency sound waves to create an image of the heart while the use of doppler technology allows to find the speed and direction of blood flow by using the concept of doppler effect.

Doppler's effect in light

Light waves also show Doppler's effect. If a light source is moving away from a stationary observer, then the frequency of light waves appears to be decreased and wavelength appears to be increased and *vice-versa*. If the light source or the observer is moving with a velocity v , such that the distance between them is decreasing, then the apparent frequency of the source will be given by

$$f' = f \sqrt{\frac{1 + (v/c)}{1 - (v/c)}}$$

where, f = frequency emitted by source, v is relative velocity of approach and c is speed of light. If the distance between the light source and the observer is increasing, then the apparent frequency of the source is given by

$$f' = f \sqrt{\frac{1 - (v/c)}{1 + (v/c)}}$$

where, v = relative velocity of recession.
The change in wavelength can be determined by

$$\Delta\lambda = \frac{v}{c} \cdot \lambda$$

Note Doppler's effect in light depends only on the relative motion between the source and the observer similarly the Doppler's effect in sound also depends upon whether the source is moving or the observer is moving.

Red shift and violet shift

If the light source is moving away from the observer, the shift in the spectrum is towards red and it is called Red shift. If it is moving towards the observer, the shift is towards the violet and it is called violet shift.

Red shift $\Delta\lambda = \lambda' - \lambda = \left(\frac{v}{c}\right) \lambda$

or
$$\Delta\lambda = \frac{v}{c} \lambda$$

Violet shift (or blue shift)

$$\Delta\lambda = \lambda' - \lambda = -\left(\frac{v}{c}\right) \lambda$$

or
$$\Delta\lambda = -\frac{v}{c} \lambda$$

In case of approach, frequency increases while wavelength decreases *i.e.*, shift $\Delta\lambda$ is towards blue end of the spectrum while in case of recession, frequency decreases and wavelength increases *i.e.*, shift is towards red end.

Example 17.92 How fast one must move to see a red light signal as a green one?

(Take, $f_R = 4.8 \times 10^{14}$ Hz and $f_G = 5.6 \times 10^{14}$ Hz)

Sol. As, we know, $f' = f \left(\frac{1 + (v/c)}{1 - (v/c)} \right)^{1/2}$

$$\therefore 5.6 \times 10^{14} = 4.8 \times 10^{14} \left(\frac{1 + (v/c)}{1 - (v/c)} \right)^{1/2}$$

$$\Rightarrow \frac{7}{6} = \left(\frac{1 + (v/c)}{1 - (v/c)} \right)^{1/2} \Rightarrow \frac{49}{36} = \left(\frac{1 + (v/c)}{1 - (v/c)} \right)$$

$$\Rightarrow 49 - 49 \left(\frac{v}{c} \right) = 36 + 36 \left(\frac{v}{c} \right) \Rightarrow 85 \left(\frac{v}{c} \right) = 13$$

$$\Rightarrow v = \frac{13}{85} \times 3 \times 10^8 = \frac{39}{85} \times 10^8 = 4.59 \times 10^7 \text{ ms}^{-1}$$

Example 17.93 A star which is emitting radiation at a wavelength of $5000 \text{ \AA}$ is approaching the earth with a velocity of $1.50 \times 10^6 \text{ ms}^{-1}$. Calculate the change in wavelength of the radiation as received on the earth.

Sol. The phenomenon of apparent change in frequency (or wavelength) of the light due to the relative motion between the source of light and the observer is called Doppler's effect in light.

So,
$$\Delta\lambda = \lambda \times \frac{v}{c}$$

Given, wavelength, $\lambda = 5000 \text{ \AA}$

Velocity of source, $v = 1.5 \times 10^6 \text{ ms}^{-1}$ and $c = 3 \times 10^8 \text{ ms}^{-1}$

$$\therefore \Delta\lambda = 5000 \times \frac{1.5 \times 10^6}{3 \times 10^8} = 25 \text{ \AA}$$

Applications of Doppler's effect in light

Applications of Doppler's effect in light are given below

(i) Estimation of velocity of stars and galaxies

Spectral lines of elements like sodium, hydrogen and helium are found in spectrum coming with stars and galaxies. We compare these spectral lines with standard spectral lines of an element in the laboratory. If spectral lines of the star shifts towards red wavelength (red shift), then the star must be moving away from the earth.

Similarly, if the spectral lines are shifting towards violet wavelength (violet shift), then the star must be moving towards the earth.

If $\Delta\lambda$ is doppler shift in spectral lines, then

$$\Delta\lambda = \frac{\lambda_0 v}{c}$$

Using this formula, we can find speed of stars and galaxies (v).

- (ii) **RADAR** Radio Detection and Ranging abbreviated as RADAR is a method used to find position and speed of a plane or ship. Radio waves are sent from RADAR centre towards ship and returns to it after reflection from the plane or ship.

Example 17.94 An astronaut is approaching the moon. He sends out a radio signal of frequency 5000 MHz and the frequency of echo is different from that of the original

frequency by 100 kHz. Find his velocity of approach with respect to the moon.

Sol. Let the shift in frequency be Δn .

$$\Delta n = 100 \times 10^3 \text{ Hz}, v = \text{relative speed}$$

$$\begin{aligned} \text{Then, } \frac{\Delta n}{n} &= \frac{2v}{c} \Rightarrow v = \frac{\Delta n}{2n} c = \frac{100 \times 10^3 \times 3 \times 10^8}{2 \times 5000 \times 10^6} \\ &= 3000 \text{ ms}^{-1} = 3 \text{ km s}^{-1} \end{aligned}$$

CHECK POINT 17.5

- The Doppler's effect is applicable for
 - light waves
 - sound waves
 - matter waves
 - Both (a) and (b)
- Doppler's effect in sound is due to
 - motion of source
 - motion of observer
 - relative motion of source and observer
 - None of the above
- An observer is moving towards the stationary source of sound, then
 - apparent frequency will be less than the real frequency
 - apparent frequency will be greater than the real frequency
 - apparent frequency will be equal to real frequency
 - only the quality of sound will change
- A siren placed at a railway platform is emitting sound of frequency 5 kHz. A passenger sitting in a moving train A records a frequency of 5.5 kHz while train approaches the siren. During his return journey in a different train B, he records a frequency of 6 kHz while approaching the same siren. The ratio of the velocity of train B to that of train A is
 - 4/3
 - 2
 - 5/3
 - 8/5
- An observer moves towards a stationary source of sound, with velocity one-fifth of the velocity of sound. What is the percentage increase in the apparent frequency?
 - 5%
 - 20%
 - zero
 - 0.5%
- A source emits a sound of frequency 400 Hz, but the listener hears it to be 390 Hz. Then,
 - the listener is moving towards the source
 - the source is moving towards the listener
 - the listener is moving away from the source
 - the listener has a defective ear
- A source of sound emits waves with frequency f (in Hz) and has speed v ms⁻¹. Two observers move away from this source in opposite directions each with a speed $0.2v$ relative to the source. The ratio of frequencies heard by the two observers will be
 - 3 : 2
 - 2 : 3
 - 1 : 1
 - 4 : 10
- Doppler shift in frequency does not depend upon
 - the frequency of the wave produced
 - the velocity of the source
 - the velocity of the observer
 - distance from the source to the listener
- A source of sound of frequency 600 Hz is placed inside water. The speed of sound in water is 1500 ms⁻¹ and in air is 300 ms⁻¹. The frequency of sound recorded by an observer who is standing in air is
 - 200 Hz
 - 3000 Hz
 - 120 Hz
 - 600 Hz
- When an observer moves towards a stationary source with certain velocity, he observes an apparent frequency f . When it moves away from the source with same velocity, the observed frequency is $0.8f$. If the velocity of sound is v , then the speed of the observer is
 - $\frac{v}{4}$
 - $\frac{v}{8}$
 - $\frac{v}{9}$
 - None of these
- Two whistles A and B have frequencies 660 Hz and 590 Hz respectively. An observer is standing in the middle of line joining the two sources. Source B and observer are moving towards right with velocity 30 ms⁻¹ and A is standing to the left side. If the velocity of sound in air is 300 ms⁻¹. The number of beats per second listened by the observer is
 - 8
 - 6
 - 4
 - 2
- If a particle is travelling with a speed of 0.9 of the speed of sound and is emitting radiations of frequency 1 kHz and moving towards the observer, what is its apparent frequency (in kHz)?
 - 1.1
 - 2.0
 - 0.1
 - 10
- A train moves towards a stationary observer with speed 34 ms⁻¹. The train sounds a whistle and its frequency registered by the observer is f_1 . If the train's speed is reduced to 17 ms⁻¹, the frequency registered is f_2 . If the speed of sound is 340 ms⁻¹, then ratio $\frac{f_1}{f_2}$ is
 - $\frac{18}{19}$
 - $\frac{17}{18}$
 - $\frac{18}{17}$
 - $\frac{19}{18}$
- The frequency of a whistle is 300 Hz. It is approaching towards an observer with a speed one-third of the speed of sound. The frequency of sound as heard by the observer will be
 - 450 Hz
 - 300 Hz
 - 400 Hz
 - 425 Hz
- The wavelength is 120 cm when the source is stationary. If the source is moving with relative velocity of 60 ms⁻¹ towards the observer, then the wavelength of the sound wave reaching to the observer will be (velocity of sound = 330 ms⁻¹)
 - 98 cm
 - 140 cm
 - 120 cm
 - 1440 cm

16. A source of sound is moving with constant velocity of 20 ms^{-1} emitting a note of frequency 1000 Hz . The ratio of frequencies observed by a stationary observer while the source is approaching him and after it crosses him will be (speed of sound, $v = 340 \text{ ms}^{-1}$)
- (a) $9 : 8$ (b) $8 : 9$
(c) $1 : 1$ (d) $9 : 10$

17. A car sounding a horn of frequency 1000 Hz passes an observer. The ratio of frequencies of the horn noted by the observer before and after passing of the car is $11 : 9$. If the speed of sound is v , the speed of the car is
- (a) $\frac{1}{10} v$ (b) $\frac{1}{2} v$
(c) $\frac{1}{5} v$ (d) v

18. A car sounding its horn at 480 Hz moves towards a high wall at a speed of 20 ms^{-1} . The frequency of the reflected sound heard by the man sitting in the car will be nearest to (speed of sound = 330 ms^{-1})
- (a) 480 Hz (b) 510 Hz (c) 540 Hz (d) 570 Hz

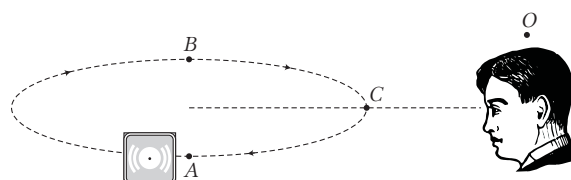
19. A source of sound with frequency 256 Hz is moving with velocity v towards a wall. When the observer is between source and the wall, he finds that the frequency of two waves received directly from the source is x and the frequency of the waves received after reflection from the wall is y , then
- (a) $x > y$ (b) $x < y$
(c) $x = y$ (d) Nothing can be said

20. A source of sound S of frequency 500 Hz situated between a stationary observer O and moves towards the wall with a speed of 2 ms^{-1} . If the velocity of sound is 332 ms^{-1} , then the number of beats per second heard by the observer is (approximately)
- (a) 8 (b) 6 (c) 4 (d) 2

21. Two cars are moving on two perpendicular roads towards a crossing with uniform speeds of 72 kmh^{-1} and 36 kmh^{-1} . If first car blows horn of frequency 280 Hz , then the frequency of horn heard by the driver of second car when line joining the cars make 45° angle with the roads, will be
- (a) 321 Hz (b) 298 Hz
(c) 289 Hz (d) 280 Hz

22. A table is revolving on its axis at 5 revolutions per second. A sound source of frequency 1000 Hz is fixed on the table at 70 cm from the axis. The minimum frequency heard by a listener standing at a distance very far from the table will be (speed of sound = 352 ms^{-1})
- (a) 1000 Hz (b) 1066 Hz
(c) 941 Hz (d) 352 Hz

23. A small source of sound moves on a circle as shown in the figure and an observer is standing at O . Let n_1 , n_2 and n_3 be the frequencies heard when the source is at A , B and C respectively. Then



- (a) $n_1 > n_2 > n_3$ (b) $n_2 > n_3 > n_1$
(c) $n_1 = n_2 > n_3$ (d) $n_2 > n_1 > n_3$
24. A whistle emitting a sound of frequency 440 Hz is tied to a string of 1.5 m length and rotated with an angular velocity of 20 rad s^{-1} in the horizontal plane. Then the range of frequencies heard by an observer stationed at a large distance from the whistle will be ($v = 330 \text{ ms}^{-1}$)
- (a) 400.0 Hz to 484.0 Hz
(b) 403.3 Hz to 480.0 Hz
(c) 400.0 Hz to 480.0 Hz
(d) 403.3 Hz to 484.0 Hz

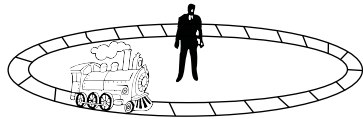
25. The speed of sound in air is v . Both the source and observer are moving towards each other with equal speed u . The speed of wind is w from source to observer. Then, the ratio $\left(\frac{f}{f_o}\right)$ of the apparent frequency to the actual frequency is given by

- (a) $\frac{v + u}{v - u}$ (b) $\frac{v + w + u}{v + w - u}$
(c) $\frac{v + w + u}{v - w - u}$ (d) $\frac{v - w + u}{v - w - u}$

Chapter Exercises

(A) Taking it together

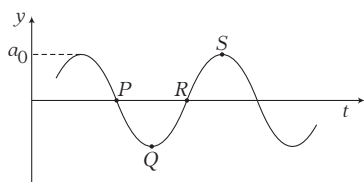
Associated questions of the chapter for advanced level practice

- 1 The frequency of a sound wave is n and its velocity is v . If the frequency is increased to $4n$, the velocity of the wave will be
(a) v (b) $2v$ (c) $4v$ (d) $v/4$
- 2 What is the phase difference at a given instant of time between two particles 25 m apart, when the wave $y(x, t) = 0.03 \sin \pi(2t - 0.01x)$ travels in a medium?
(a) $\frac{\pi}{8}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) π
- 3 The temperature at which the speed of sound in air becomes double of its value at 27°C is
(a) 327°C (b) 127°C (c) 927°C (d) -123°C
4. With propagation of longitudinal waves through a medium, the quantity transmitted is [NCERT Exemplar]
(a) matter
(b) energy
(c) energy and matter
(d) energy, matter and momentum
- 5 The ratio of the maximum velocity of a particle to the velocity of wave is
(a) 1 (b) ωA
(c) kA (d) No unique value exists
- 6 Sound velocity is maximum in
(a) H_2 (b) N_2 (c) He (d) O_2
- 7 If the temperature increases, then what happens to the frequency of the sound produced by the organ pipe?
(a) Increases (b) Decreases
(c) Unchanged (d) Not definite
- 8 In a sinusoidal wave, the minimum time required for a particular point to move from maximum displacement to zero displacement is 0.17 s. The frequency of the wave is
(a) 1.47 Hz (b) 0.36 Hz (c) 0.73 Hz (d) 2.49 Hz
- 9 A sound wave is passing through air column in the form of compressions and rarefactions. In consecutive compressions and rarefactions, [NCERT Exemplar]
(a) density remains constant
(b) Boyle's law is obeyed
(c) bulk modulus of air oscillates
(d) there is no transfer of heat
- 10 An engine is moving on a circular track with a constant speed. It is blowing a whistle of frequency 500 Hz. The frequency received by an observer standing stationary at the centre of the track is

(a) 500 Hz
(b) more than 500 Hz
(c) less than 500 Hz
(d) more or less than 500 Hz depending on the actual speed of the engine
- 11 The extension in a string obeying Hooke's law is x . The speed of transverse waves in the stretched string is v . If the extension in the string is increased to $1.5x$, the speed of transverse waves in it will be
(a) $1.22v$ (b) $2v$ (c) $1.5v$ (d) v
- 12 If the frequency of sound produced by a siren increases from 400 Hz to 1200 Hz while the wave amplitude remains constant, the ratio of the intensity of the 1200 Hz to that of the 400 Hz wave will be
(a) 1 : 1 (b) 1 : 3 (c) 3 : 1 (d) 9 : 1
- 13 A man standing on a cliff claps his hand and hears its echo after 1 s. If sound is reflected from mountain and velocity of sound in air is 340 ms^{-1} . Then, the distance between the man and reflection point is
(a) 680 m (b) 340 m
(c) 85 m (d) 170 m
- 14 A wave of frequency 400 Hz has a wave velocity of 300 ms^{-1} . The phase difference between two displacements at a certain point at time $t = 10^{-3} \text{ s}$ apart is
(a) 72° (b) 102°
(c) 180° (d) 144°
- 15 In a travelling wave
$$y = 0.1 \sin \pi \left(x - 330t + \frac{2}{3} \right) \text{ (SI units)}$$
The phase difference between $x_1 = 3 \text{ m}$ and $x_2 = 3.5 \text{ m}$ is
(a) $\frac{\pi}{2}$ (b) π (c) $\frac{3\pi}{2}$ (d) 2π

- 16** The apparent wavelength of the light from a star, moving away from the earth is 0.01% more than its real wavelength. The speed of the star with respect to earth is
(a) 10 km/s (b) 15 km/s (c) 30 km/s (d) 60 km/s
- 17** A star is moving away from the earth with a velocity of 100 km/s. If the velocity of light is 3×10^8 m/s, then the shift of its spectral line of wavelength $5700 \text{ \AA}$ due to Doppler effect is
(a) $0.63 \text{ \AA}$ (b) $1.90 \text{ \AA}$
(c) $3.80 \text{ \AA}$ (d) $5.70 \text{ \AA}$
- 18** Two waves represented by $y_1 = 10 \sin 200\pi t$ and $y_2 = 20 \sin \left[200\pi t + \frac{\pi}{2} \right]$ at a point are superimposed at a particular instant. The amplitude of the resultant wave is
(a) 30 (b) $10\sqrt{5}$
(c) 50 (d) 10
- 19** Two strings of copper are stretched to the same tension. If their cross-sectional areas are in the ratio 1 : 4, then the respective wave velocities will be in the ratio
(a) 4 : 1 (b) 2 : 1
(c) 1 : 2 (d) 1 : 4
- 20** Under the same conditions of pressure and temperature, the velocity of sound in oxygen and hydrogen gases are v_O and v_H , then
(a) $v_H = 2v_O$ (b) $v_H = 4v_O$
(c) $v_O = 4v_H$ (d) $v_H = v_O$
- 21** Equation of a progressive wave is given by $y = a \sin \pi \left[\frac{t}{2} - \frac{x}{4} \right]$, where t is in second and x in metre. The distance through which the wave travels in 8 s (in metre) is
(a) 8 (b) 16 (c) 2 (d) 4
- 22** If you set up the seventh harmonic on a string fixed at both ends, how many nodes and antinodes are set up in it?
(a) 8, 7 (b) 7, 7 (c) 8, 9 (d) 9, 8
- 23** Two sinusoidal waves with same wavelengths and amplitudes travel in opposite directions along a string with a speed 10 ms^{-1} . If the minimum time interval between two instants when the string is flat is 0.5 s, the wavelength of the waves is
(a) 25 m (b) 20 m (c) 15 m (d) 10 m
- 24** In the given progressive wave
$$y = 5 \sin (100\pi t - 0.4\pi x)$$
What is the wave velocity?
(a) 350 (b) 250 (c) 200 (d) 180
- 25** An organ pipe P_1 is closed at one end and vibrating in its first overtone and another pipe P_2 opened at both ends vibrating in its third overtone are in resonance with a given tuning fork. Then, the ratio of lengths of P_1 and P_2 is
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{8}{3}$ (d) $\frac{3}{8}$
- 26** Four wires of identical lengths, diameters and of the same material are stretched on a sonometer wire. If the ratio of their tensions is 1 : 4 : 9 : 16, then the ratio of their fundamental frequencies is
(a) 16 : 9 : 4 : 1 (b) 4 : 3 : 2 : 1
(c) 1 : 4 : 2 : 16 (d) 1 : 2 : 3 : 4
- 27** A tube closed at one end containing air produces when excited the fundamental note of frequency 512 kHz. If the tube is open at both ends, the fundamental frequency that can be excited is (in Hz)
(a) 1024 (b) 512 (c) 256 (d) 128
- 28** The linear density of a vibrating string is $10^{-4} \text{ kg m}^{-1}$. A transverse wave is propagating on the string, which is described by the equation $y = 0.02 \sin (x + 30t)$, where x and y are in metre and the time t is in second. The tension in the string is
(a) 0.09 N (b) 0.36 N (c) 0.9 N (d) 3.6 N
- 29** Two identical strings A and B are stretched at tensions T_A and T_B . A tuning fork is used to set them in vibration. A vibrates in its fundamental mode and B in its second harmonic mode, then
(a) $T_A = 2T_B$ (b) $T_A = 4T_B$
(c) $2T_A = T_B$ (d) $4T_A = 4T_B$
- 30** A train of sound waves is propagated along a wide pipe and it is reflected from an open end. If the amplitude of the waves is 0.002 cm, the frequency is 1000 Hz and the wavelength is 40 cm, the amplitude of vibration at a point 10 cm from open end inside the pipe will be
(a) zero (b) 0.002 cm (c) 0.001 (d) 0.005 cm
- 31** In a resonance column, two successive resonances are obtained at depths 30 cm and 50 cm, respectively. The next resonance will be obtained at a depth
(a) 60 cm (b) 80 cm (c) 70 cm (d) 90 cm
- 32** The displacement y of a particle on a straight line is given by $y = f(x, t)$, as a function of time. Which of the following functions does not represent wave motion?
(a) $y = A \sin (kx - \omega t)$ (b) $y = A \sin^2 (kx - \omega t)$
(c) $y = A \sin (k^2 x^2 - \omega^2 t^2)$ (d) $y = A \sin \left(kx + \omega t + \frac{\pi}{10} \right)$

- 33** The intensity of sound gets reduced by 20% on passing through a slab. The reduction in intensity on passing through two consecutive slabs is
(a) 40% (b) 36% (c) 30% (d) 50%

- 34** A wave motion has the function $y = a_0 \sin(\omega t - kx)$. The graph in figure shows how the displacement y at a fixed point varies with time t . Which one of the labelled points shows a displacement equal to that at the position $x = \frac{\pi}{2k}$ at time $t = 0$?



- (a) P (b) Q (c) R (d) S
- 35** In a medium in which a transverse progressive wave is travelling, the phase difference between two points with a separation of 1.25 cm is $\left(\frac{\pi}{3}\right)$. If the frequency of wave is 1000 Hz, its velocity will be
(a) 75 ms⁻¹ (b) 125 ms⁻¹ (c) 100 ms⁻¹ (d) 50 ms⁻¹
- 36** Equation of a plane progressive wave is given by $y = 0.6 \sin 2\pi \left(t - \frac{x}{2}\right)$. On reflection from a denser medium, its amplitude becomes (2/3) of the amplitude of the incident wave. The equation of the reflected wave is [NCERT Exemplar]
(a) $y = 0.6 \sin 2\pi \left(t + \frac{x}{2}\right)$ (b) $y = -0.4 \sin 2\pi \left(t + \frac{x}{2}\right)$
(c) $y = 0.4 \sin 2\pi \left(t + \frac{x}{2}\right)$ (d) $y = -0.4 \sin 2\pi \left(t - \frac{x}{2}\right)$

- 37** The equation of a stationary wave is $y = 0.8 \cos \left(\frac{\pi x}{20}\right) \sin 200\pi t$, where x is in cm and t is in second. The separation between consecutive nodes will be
(a) 20 cm (b) 10 cm (c) 40 cm (d) 30 cm
- 38** In the equation for a stationary wave given by $y = 5 \cos \frac{\pi x}{25} \sin 100 \pi t$ (x is in cm and t in second), a node will not occur at distance x . The value of x is
(a) 25 cm (b) 62.5 cm
(c) 12.5 cm (d) 37.5 cm

- 39** A wave of frequency 100 Hz is sent along a string towards a fixed end. When this wave travels back after reflection, a node is formed at a distance of

10 cm from the fixed end of the string. The speed of incident wave is

- (a) 48 ms⁻¹ (b) 20 ms⁻¹ (c) 10 ms⁻¹ (d) 15 ms⁻¹
- 40** Two strings A and B have lengths l_A and l_B and carry masses M_A and M_B at their lower ends, the upper ends being supported by rigid supports. If n_A and n_B are their fundamental frequencies of their vibrations, the ratio of n_A and n_B is
(a) $\frac{M_A l_A}{M_B l_B}$ (b) $\frac{l_A}{l_B} \sqrt{\frac{M_A}{M_B}}$
(c) $\frac{M_A}{M_B} \sqrt{\frac{l_A}{l_B}}$ (d) None of these

- 41** A sound is produced between two vertical parallel walls. The echo from one wall is heard after 2 s while from the other 2 s after the first echo. The speed of sound in air is 340 ms⁻¹. Then,
(a) the distance between two walls is 680 m
(b) the distance between two walls is 1020 m
(c) the next echo will be heard after 8 s from the instant original sound was produced
(d) None of the above

- 42** A standing wave is produced on a string fixed at one end and free at the other. The length of the string
(a) must be an odd integral multiple of $\frac{\lambda}{4}$
(b) must be an integral multiple of $\frac{\lambda}{2}$
(c) must be an integral multiple of λ
(d) may be an integral multiple of $\frac{\lambda}{2}$

- 43** Stationary waves of frequency 300 Hz are formed in a medium in which the velocity of sound is 1200 ms⁻¹. The distance between a node and the neighbouring antinode is
(a) 1 m (b) 2 m (c) 3 m (d) 4 m

- 44** Two waves are passing through a region in the same direction at the same time. If the equations of these waves are $y_1 = a \sin \frac{2\pi}{\lambda} (vt - x)$

$$\text{and } y_2 = b \sin \frac{2\pi}{\lambda} [(vt - x) + x_0],$$

then the amplitude of the resultant wave for $x_0 = \lambda/2$, is

- (a) $|a - b|$ (b) $(a + b)$ (c) zero (d) $\sqrt{a^2 + b^2}$
- 45** Two periodic waves of amplitudes a and b pass through a region at the same time in the same direction. If $a > b$, the difference in the maximum and minimum possible amplitudes is
(a) $a + b$ (b) $a - b$ (c) $2a$ (d) $2b$

- 46** A tube of length 1.05 m is closed at one end. If the velocity of sound in air is 336 ms^{-1} , then the fundamental and the next higher overtone (in Hz) are respectively
 (a) 80, 160 (b) 80, 240 (c) 160, 320 (d) 160, 480
- 47** Two travelling waves $y_1 = A \sin[k(x - ct)]$ and $y_2 = A \sin[k(x + ct)]$ are superimposed on string. The distance between adjacent nodes is
 (a) $\frac{c}{\pi}$ (b) $\frac{c}{2\pi}$ (c) $\frac{\pi}{2k}$ (d) $\frac{\pi}{k}$
- 48** An engine approaches a hill with a constant speed. When it is at a distance of 0.9 km, it blows a whistle whose echo is heard by the driver after 5 s. If speed of sound in air is 330 m/s, the speed of engine is



- (a) 10 m/s (b) 20 m/s (c) 30 m/s (d) 40 m/s
- 49** Velocity of sound in open-ended tube is 330 ms^{-1} , the frequency of tuning fork is 1.1 kHz and the length of tube = 30 cm. In which harmonic will it oscillate?
 (a) 2nd (b) 3rd (c) 4th (d) 5th
- 50** Consider the three waves z_1 , z_2 and z_3 as $z_1 = A \sin(kx - \omega t)$, $z_2 = A \sin(kx + \omega t)$ and $z_3 = A \sin(ky - \omega t)$. Which of the following represents a standing wave?
 (a) $z_1 + z_2$ (b) $z_2 + z_3$ (c) $z_3 + z_1$ (d) $z_1 + z_2 + z_3$
- 51** Standing waves are produced in 10 m long stretched string. If the string vibrates in 5 segments and wave velocity is 20 ms^{-1} , then its frequency will be
 (a) 5 Hz (b) 2 Hz (c) 10 Hz (d) 15 Hz
- 52** A wire of length l having tension T and radius r vibrates with natural frequency f . Another wire of same metal with length $2l$ having tension $2T$ and radius $2r$ will vibrate with natural frequency
 (a) f (b) $2f$ (c) $2\sqrt{2} f$ (d) $\frac{f}{2\sqrt{2}}$
- 53** A source of sound is travelling towards a stationary observer. The frequency of sound heard by the observer is three times the original frequency. The velocity of sound is $v \text{ ms}^{-1}$. The speed of source will be
 (a) $\frac{2}{3} v$ (b) v (c) $\frac{3}{2} v$ (d) $3v$
- 54** A string is hanging from a rigid support. A transverse wave pulse is set up at the free end. The velocity v of the pulse related to the distance x covered by it is given as

- (a) $v \propto \sqrt{x}$ (b) $v \propto x$
 (c) $v \propto \frac{1}{x}$ (d) None of these

- 55** Two sounding bodies producing progressive waves given by $y_1 = 4 \sin(400\pi t)$ and $y_2 = 3 \sin(404\pi t)$, where t is in second, are situated near the ears of a person. The person will hear
 (a) 2 beats per second with intensity ratio 4/3 between maxima and minima
 (b) 2 beats per second with intensity ratio 49 between maxima and minima
 (c) 4 beats per second with intensity ratio 49 between maxima and minima
 (d) 4 beats per second with intensity ratio 4/3 between maxima and minima
- 56** A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 , respectively. The compressibility $\left(= \frac{1}{\text{bulk modulus}} \right)$ of gases are equal in both the pipes. Both the pipes are vibrating in their first overtones with same frequency. The length of the open organ pipe is
 (a) $\frac{L}{3}$ (b) $\frac{4L}{3}$ (c) $\frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$ (d) $\frac{4L}{3} \sqrt{\frac{\rho_2}{\rho_1}}$
- 57** A stretched rope having linear mass density $5 \times 10^{-2} \text{ kgm}^{-1}$ is under a tension of 80 N. The power that has to be supplied to the rope to generate harmonic waves at a frequency of 60 Hz and an amplitude of 6 cm is
 (a) 362 W (b) 251 W (c) 511 W (d) 416 W
- 58** Two vibrating strings of same material but lengths l and $2l$ have radii $2r$ and r , respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes, the string of length l with frequency n_1 and the other with frequency n_2 . The ratio of $\frac{n_1}{n_2}$ is given by
 (a) 2 (b) 4 (c) 8 (d) 1
- 59** An organ pipe open from both ends produces 5 beats per second when vibrated with a source of frequency 200 Hz in its fundamental mode. The second harmonic of the same pipe produces 10 beats per second with a source of frequency 420 Hz. The fundamental frequency of pipe is
 (a) 195 Hz (b) 205 Hz
 (c) 190 Hz (d) 210 Hz
- 60** A transverse sinusoidal wave of amplitude A , wavelength λ and frequency n is travelling on a stretched string. The maximum speed of particle is $(1/10)\text{th}$ the speed of propagation of the wave.

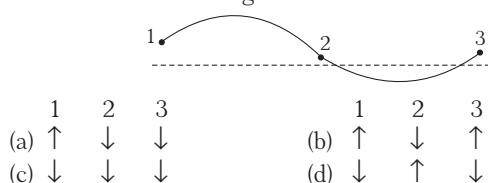
- If $A = 10^{-3}$ m and $v = 10 \text{ ms}^{-1}$, then λ and n are given by
- (a) $\lambda = 2\pi \times 10^{-2}$ m (b) $\lambda = 10^{-3}$ m
 (c) $n = \frac{10^4}{2\pi}$ Hz (d) $n = 10^4$ Hz
- 61** Three waves, $y_1 = A \sin(kx - \omega t)$
 $y_2 = A \sin(kx - \omega t + \phi)$
 and $y_3 = A \sin(kx - \omega t + 2\phi)$
 are superimposed, so that $y_1 + y_2 + y_3 = 0$ at all positions. Then, the value of ϕ is
 (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{4\pi}{3}$ (d) $\frac{\pi}{2}$
- 62** A cylindrical resonance tube, open at both ends, has a fundamental frequency f in air. If half of the length is dipped vertically in water, the fundamental of the air column will be
 (a) $\frac{3f}{2}$ (b) $2f$ (c) f (d) $\frac{f}{2}$
- 63** In a resonance tube, closed at one end by a smooth moving piston and the other end open, exhibits the first three resonance lengths of air column, L_1 , L_2 and L_3 for the same tuning fork. Then, they are related as
 (a) $L_3 = 2L_2 = 4L_1$ (b) $L_3 = \frac{5}{3}L_2 = 5L_1$
 (c) $(L_3 - L_2) = \left(\frac{L_2 - L_1}{2}\right)$ (d) $(L_3 - L_2) = 2(L_2 - L_1)$
- 64** An open pipe is in resonance in its 2nd harmonic with tuning fork of frequency f_1 . Now, it is closed at one end. If the frequency of the tuning fork is increased slowly from f_1 , then again a resonance is obtained with a frequency f_2 . If in this case, the pipe vibrates n th harmonic, then
 (a) $n = 3, f_1 = \frac{3}{4}f_2$ (b) $n = 3, f_2 = \frac{5}{4}f_1$
 (c) $n = 5, f_2 = \frac{5}{4}f_1$ (d) $n = 5, f_2 = \frac{3}{4}f_1$
- 65** If the speed of longitudinal waves is equal to 10 times the speed of the transverse waves in a stretched wire of material which has modulus of elasticity E , then the stress in the wire is
 (a) $10E$ (b) $100E$ (c) $\frac{E}{10}$ (d) $\frac{E}{100}$
- 66** Three waves of equal frequencies having amplitudes $10 \mu\text{m}$, $4 \mu\text{m}$ and $7 \mu\text{m}$ arrive at a given point with successive phase difference of $\frac{\pi}{2}$. The amplitude of the resulting wave (in μm) is given by
 (a) 7 (b) 6 (c) 5 (d) 4
- 67** First overtone frequency of a closed organ pipe is equal to the first overtone frequency of an open organ pipe. Further, n th harmonic of closed organ pipe is also equal to the m th harmonic of open pipe, where the respective values of n and m are
 (a) 5, 4 (b) 7, 5 (c) 9, 6 (d) 7, 3
- 68** A pipe closed at one end produces a fundamental note of 412 Hz. It is cut into two pieces of equal length, the fundamental notes produced by the two piece are
 (a) 824 Hz, 1648 Hz (b) 412 Hz, 824 Hz
 (c) 206 Hz, 412 Hz (d) 206 Hz, 824 Hz
- 69** Two waves are propagating to the point P by two sources A and B of equal frequency. The amplitude of every wave at P is a and the phase of A is ahead by $\frac{\pi}{3}$ than that of B and the distance AP is greater than BP by 50 cm. If the wavelength is 1m, then the resultant amplitude at the point P will be
 (a) $2a$ (b) $a\sqrt{3}$
 (c) $a\sqrt{2}$ (d) a
- 70** A person speaking normally produces a sound intensity of 40 dB at a distance of 1 m. If the threshold intensity for reasonable audibility is 20 dB, the maximum distance at which person can be heard clearly is
 (a) 4 m (b) 5 m (c) 10 m (d) 20 m
- 71** A stationary wave set up on a string have the equation $y = (2 \text{ mm}) \sin[(6.25 \text{ m}^{-1}) x \cos(\omega t)]$. This stationary wave is created by two identical waves each of amplitude A moving in opposite directions along the string. Then,
 (a) $A = 2 \text{ mm}$
 (b) $A = 4 \text{ mm}$
 (c) the smallest length of the string is 50 cm
 (d) the smallest length of the string is 2 m
- 72** When a vibrating tuning fork of frequency 512 Hz is held above the mouth of a resonance tube of adjustable length, the first two successive positions of resonance occur when the length of the air columns are 15.4 cm and 48.6 cm, respectively. Then, the velocity of sound is
 (a) $512(48.6 - 15.4) \text{ cm s}^{-1}$
 (b) $1024(48.6 - 15.4) \text{ cm s}^{-1}$
 (c) $256(48.6 - 15.4) \text{ cm s}^{-1}$
 (d) $2 \times 512(48.6 + 15.4) \text{ cm s}^{-1}$
- 73** A tuning fork of frequency 480 Hz produces 10 beats per second when sounded with a vibrating sonometer string. What must be the frequency of the

string, if a slight increase in tension produces more beats per second than before?

- (a) 460 Hz (b) 470 Hz
(c) 480 Hz (d) 490 Hz

- 74** In a resonance tube, the first resonance with tuning fork occurs at 16 cm and second at 49 cm. If the velocity of sound is 330 m/s, the frequency of tuning fork is
(a) 500 Hz (b) 300 Hz (c) 330 Hz (d) 165 Hz

- 75** The diagram shows an instantaneous position of a string as a transverse progressive wave travels along it from left to right. Which one of the following correctly shows the direction of velocity of points 1, 2 and 3 on the string?



- 76** Two sources A and B are sounding notes of frequency 680 Hz. A listener moves from A to B with a constant velocity u . If the speed of sound is 340 ms^{-1} , what must be the value of u , so that he hears 10 beats per second?
(a) 2.0 ms^{-1} (b) 2.5 ms^{-1} (c) 3.0 ms^{-1} (d) 3.5 ms^{-1}

- 77** A sonometer wire has a length 114 cm between two fixed ends. Where should two bridges be placed, so as to divide the wire into three segments whose fundamental frequencies are in the ratio 1 : 3 : 4?
(a) $l_1 = 72 \text{ cm}$, $l_2 = 24 \text{ cm}$, $l_3 = 18 \text{ cm}$
(b) $l_1 = 60 \text{ cm}$, $l_2 = 40 \text{ cm}$, $l_3 = 14 \text{ cm}$
(c) $l_1 = 52 \text{ cm}$, $l_2 = 30 \text{ cm}$, $l_3 = 32 \text{ cm}$
(d) $l_1 = 65 \text{ cm}$, $l_2 = 30 \text{ cm}$, $l_3 = 19 \text{ cm}$

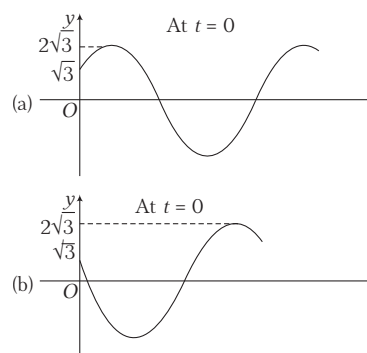
- 78** The amplitude of a wave represented by displacement equation $y = \frac{1}{\sqrt{a}} \sin \omega t \pm \frac{1}{\sqrt{b}} \cos \omega t$ will be
(a) $\frac{a+b}{ab}$ (b) $\frac{\sqrt{a} + \sqrt{b}}{ab}$ (c) $\frac{\sqrt{a} \pm \sqrt{b}}{ab}$ (d) $\sqrt{\frac{a+b}{ab}}$

- 79** Two ends of a stretched wire of length L are fixed at $x = 0$ and $x = L$. In one experiment, the displacement is $y_1 = A \sin\left(\frac{\pi x}{L}\right) \sin \omega t$ and energy is E_1 and in another experiment, its displacement is $y_2 = A \sin\left(\frac{2\pi x}{L}\right) \sin 2\omega t$ and energy is E_2 , then
(a) $E_2 = E_1$ (b) $E_2 = 2E_1$ (c) $E_2 = 4E_1$ (d) $E_2 = 16E_1$

- 80** Two closed organ pipes A and B have the same length. A is wider than B . They resonate in the fundamental mode at frequencies n_A and n_B respectively, then

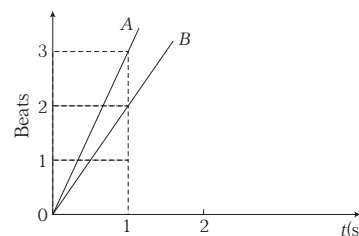
- (a) $n_A = n_B$
(b) $n_A > n_B$
(c) $n_A < n_B$
(d) Either (b) or (c) depending on the ratio of their diameter

- 81** Identify the figure, which correctly represents the given wave function $y = 2\sqrt{3} \sin \pi \left(x - 2t + \frac{1}{6} \right)$



- (c) Both graphs are possible
(d) None of the two graphs are possible

- 82** Two tuning forks P and Q are vibrated together. The numbers of beats produced are represented by the straight line OA in the following graph.



After loading Q with wax, again these are vibrated together and the beats produced are represented by the line OB . If the frequency of P is 341 Hz, the frequency of Q will be

- (a) 341 Hz (b) 338 Hz
(c) 344 Hz (d) None of these

- 83** A string of length 0.4 m and mass 10^{-2} kg is tightly clamped at its ends. The tension in the string is 1.6 N. Identical wave pulses are produced at one end at equal intervals of time Δt . The minimum value of Δt which allows constructive interference between successive pulses is
(a) 0.05 s (b) 0.10 s
(c) 0.20 s (d) 0.40 s

- 84** The displacement due to a wave moving in the positive x -direction is given by $y = \frac{1}{(1+x^2)}$ at time

$t = 0$ and by $y = \frac{1}{[1+(x-1)^2]}$ at $t = 2$ s, where x

and y are in metre. The velocity of the wave (in ms^{-1}) is

- (a) 0.5 (b) 1 (c) 2 (d) 4

- 85** The maximum pressure variation that the human ear can tolerate in loud sound is about 30 Nm^{-2} . The corresponding maximum displacement for a sound wave in air having a frequency of 10^3 Hz is (Take, velocity of sound in air is 300 ms^{-1} and density of air is 1.5 kg m^{-3})

- (a) $\frac{2\pi}{3} \times 10^{-2} \text{ m}$ (b) $\frac{2 \times 10^{-4}}{\pi} \text{ m}$
(c) $\frac{\pi}{3} \times 10^{-2} \text{ m}$ (d) $\frac{10^{-4}}{3\pi} \text{ m}$

- 86** A standing wave is maintained in a homogeneous string of cross-sectional area s and density ρ . It is formed by the superposition of two waves travelling in opposite directions given by the equation $y_1 = a \sin(\omega t - kx)$ and $y_2 = 2a \sin(\omega t + kx)$. The total mechanical energy confined between the sections corresponding to the adjacent antinodes is

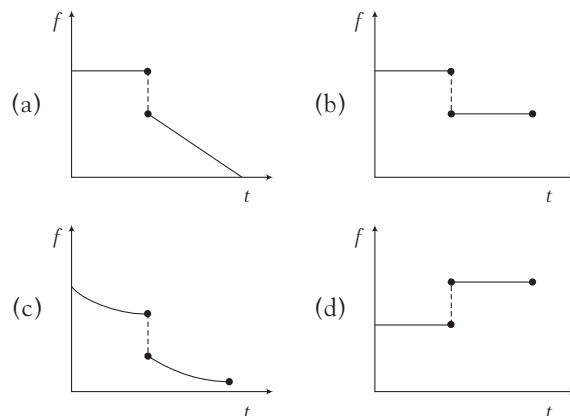
- (a) $\frac{3\pi\rho\omega^2 a^2}{2k}$ (b) $\frac{\pi\rho\omega^2 a^2}{2k}$
(c) $\frac{5\pi\rho\omega^2 a^2}{2k}$ (d) $\frac{2\pi\rho\omega^2 a^2}{k}$

- 87** A glass tube of 1.0 m length is filled with water. The water can be drained out slowly at the bottom of the tube. If a vibrating tuning fork of frequency 500 Hz is brought at the upper end of the tube and the velocity of sound is 330 ms^{-1} , then the total number of resonance obtained will be
(a) 4 (b) 3 (c) 2 (d) 1

- 88** A locomotive engine approaches a railway station and whistles at a frequency of 400 Hz. A stationary observer on the platform observes a change of 40 Hz as the engine passes across him. If the velocity of sound is 340 ms^{-1} , the speed of the engine is
(a) 34 ms^{-1} (b) 40 ms^{-1}
(c) 17 ms^{-1} (d) 20 ms^{-1}

- 89** A sound source emitting sound at constant frequency is moving towards a stationary observer.

Which of the following curve best represents the frequency received by observer f as a function of time?



- 90** A string of mass 2.5 kg is under tension of 200 N. The length of the stretched string is 20.0 m. If the transverse jerk is struck at one end of the string, the disturbance will reach the other end in [NCERT Exemplar]

- (a) 1 s (b) 0.5 s
(c) 2 s (d) data given is insufficient

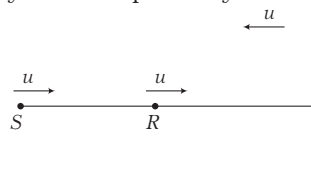
- 91** A train approaching a hill at a speed of 40 km/h sounds a whistle of frequency 580 Hz when it is at a distance of 1 km from a hill. A wind with a speed of 40 km/h is blowing in the direction of motion of the train. Find the frequency of the whistle as heard by an observer on the hill. (Take, speed of sound in air = 1200 km/h)
(a) 400 Hz (b) 500 Hz
(c) 600 Hz (d) 350 Hz

- 92** A whistle of frequency 500 Hz tied to the end of a string of length 1.2 m revolves at 400 rev/min. A listener standing some distance away in the plane of rotation of whistle hears frequencies in the range (Take, speed of sound = 340 m/s)
(a) 436 Hz to 586 Hz (b) 426 Hz to 574 Hz
(c) 426 Hz to 584 Hz (d) 436 Hz to 674 Hz

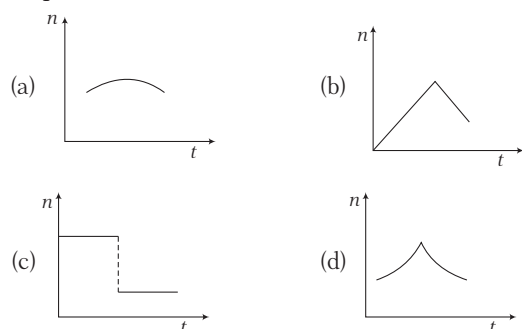
- 93** A source of sound emitting a 1200 Hz note travels along a straight line at a speed of 170 m/s. A detector is placed at a distance of 200 m from the line of motion of the source. Find the frequency of sound received by the detector at the instant when the source gets closest to it. (Take, speed of sound in air = 340 ms^{-1})
(a) 1600 Hz (b) 1000 Hz
(c) 1700 Hz (d) 1200 Hz

- 94** A point source of sound S of natural frequency 256 Hz and a receiver R are moving along same line with speed $u = 20 \text{ ms}^{-1}$ towards a reflecting surface which is approaching them with speed u as shown in figure.

If speed of sound in air is 330 ms^{-1} , then wavelength and frequency of reflected wave received by R are respectively

- 
- (a) $\frac{330}{289} \text{ m}$ and 289 Hz (b) $\frac{340}{289} \text{ m}$ and 289 Hz
(c) $\frac{330}{272} \text{ m}$ and 289 Hz (d) $\frac{340}{272} \text{ m}$ and 289 Hz

- 95** A train whistling at constant frequency is moving towards a station at a constant speed v . The train goes past a stationary observer on the station. The frequency n of the sound as heard by the observer is plotted as a function of time t (figure). Identify the expected curve. [NCERT Exemplar]

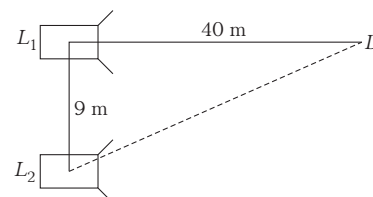


- 96** A stone is hung in air from a wire which is stretched over a sonometer. The bridges of the sonometer are $L \text{ cm}$ apart when the wire is in unison with a tuning fork of frequency N . When the stone is completely immersed in water, the length between the bridges is $l \text{ cm}$. For re-establishing unison, the specific gravity of the material of the stone is

- (a) $\frac{L^2}{L^2 + l^2}$ (b) $\frac{L^2 - l^2}{L^2}$
(c) $\frac{L^2}{L^2 - l^2}$ (d) $\frac{L^2 + l^2}{L^2}$

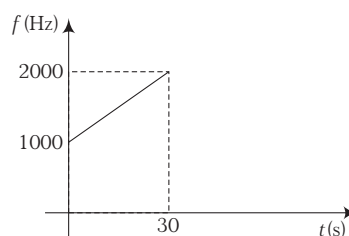
- 97** The loudspeakers L_1 and L_2 driven by a common oscillator and amplifier are set up as shown in the figure. As the frequency of the oscillator increases from zero, the detector at D recorded a series of maximum and minimum signals.

What is the frequency at which the first maximum is observed? (Take, speed of sound = 330 ms^{-1})



- (a) 165 Hz (b) 330 Hz
(c) 495 Hz (d) 660 Hz

- 98** A detector is released from rest over a source of sound of frequency $f_o = 10^3 \text{ Hz}$. The frequency observed by the detector at time t is plotted in the graph. The speed of sound in air is (Take, $g = 10 \text{ ms}^{-2}$)



- (a) 330 ms^{-1} (b) 350 ms^{-1}
(c) 300 ms^{-1} (d) 310 ms^{-1}

- 99** The string vibrates in 4 loops when a 50 g weight is placed in the pan of weight 15 g. To make the string to vibrate in 6 loops the weight that has to be removed from the pan is

- (a) 0.0007 kg-wt (b) 0.0021 kg-wt
(c) 0.036 kg-wt (d) 0.0029 kg-wt

- 100** The frequency of a sonometer wire is f . The frequency becomes $f/2$ when the mass producing the tension is completely immersed in water and on immersing the mass in a certain liquid, frequency becomes $f/3$. The relative density of the liquid is

- (a) 1.32 (b) 1.03
(c) 1.41 (d) 1.18

- 101** Two speakers connected to the same source of fixed frequency are placed 2 m apart in a box. A sensitive microphone placed at a distance of 4 m from the mid-point along the perpendicular bisector shows maximum response. The box is slowly rotated till the speakers are in line with the microphone. The distance between the mid-point of the speakers and the microphone remains unchanged. Exactly 5 maximum responses (including the initial and last one) are observed in the microphone in doing this. The wavelength of the sound wave is

- (a) 0.8 m (b) 0.5 m
(c) 0.2 m (d) 1.6 m

- 102** A stone hangs from the free end of a sonometer wire whose vibrating length, when tuned to a tuning fork is 40 cm. When the stone hangs wholly immersed in water, the resonant length is reduced to 30 cm in same mode of oscillation. The relative density of the stone is

- (a) $\frac{16}{9}$ (b) $\frac{16}{7}$
(c) $\frac{16}{5}$ (d) $\frac{16}{3}$

- 103** In a large room, a person receives direct sound waves from a source 120 metres away from him. He also receives waves from the same source which reaches him after being reflected from the 25 metres high ceiling at a point midway between them. The two waves interfere constructively for wavelength (in m) of

- (a) 20, 20/3, 20/5, etc (b) 10, 5, 2.5, etc
(c) 10, 20, 30, etc (d) 15, 25, 35, etc

(B) Medical entrance special format questions

Assertion and reason

Directions (Q. Nos. 1-5) These questions consists of two statements each printed as Assertion and Reason. While answering these questions, you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) If Assertion is correct but Reason is incorrect.
(d) If Assertion is incorrect but Reason is correct.

- 1 Assertion** If temperature is increased, fundamental frequency of an close pipe will increase.

Reason By increasing the temperature, speed of sound wave will increase.

- 2 Assertion** In Doppler's effect, if the source is moving, change in frequency is observed due to change in wavelength.

Reason By the motion of source, wavelength always decreases.

- 3 Assertion** Velocity of particles while crossing mean position varies from maximum at antinodes to zero at nodes.

Reason Amplitude of vibration at antinodes is maximum and at nodes, the amplitude is zero and all particles between two successive nodes cross the mean position together.

- 4 Assertion** Sound produced by an open organ pipe is richer than the sound produced by a closed organ pipe.

Reason Outside air can enter the pipe from both ends, in case of open organ pipe.

- 5 Assertion** Speed of sound in O_2 gas at temperature T is v . If temperature is increased to $2T$, the molecular oxygen breaks into atomic oxygen. Then, speed of sound under these conditions will become $2v$.

Reason With increase in humidity in air, speed of sound increases.

Statement based questions

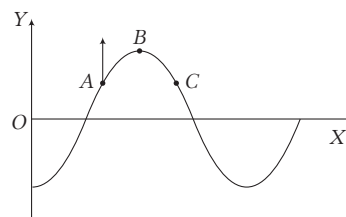
- 1** Choose the correct statement regarding wave motion. [NCERT Exemplar]

- (a) Mechanical transverse waves can propagate through all mediums
(b) Longitudinal waves can propagate through solids only
(c) Mechanical transverse waves can propagate through solids only
(d) Longitudinal waves can propagate through vacuum

- 2** A wave disturbance in a medium is described by $y(x, t) = 0.02 \cos\left(50\pi t + \frac{\pi}{2}\right) \cos(10\pi x)$, where x and y are in metre and t in second. Choose the correct statement.

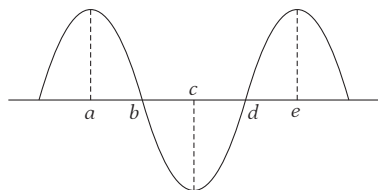
- (a) A node occurs at $x = 0.15$ m
(b) An antinode occurs at $x = 0.3$ m
(c) The wave velocity is 5 ms^{-1}
(d) All of the above

- 3** A wave is travelling along a string. At an instant, shape of the string is as shown in figure. At this instant, point A is moving upwards. Which of the following statement is correct?



- (a) The wave is travelling to the right
(b) Displacement amplitude of the wave is equal to displacement of B at this instant
(c) At this instant, velocity of C is also directed upwards
(d) Phase difference between A and C may be equal to $\pi/2$

- 4 The rope shown at an instant is carrying a wave travelling towards right, created by a source vibrating at a frequency n .



Choose the correct statement(s).

- (a) The speed of the wave is $4n \times ab$
 (b) The phase difference between b and e is $\frac{3\pi}{2}$
 (c) Both (a) and (b) are correct
 (d) Neither (a) nor (b) are incorrect
- 5 I. In a standing wave, energy is transmitted from one region of space to other.
 II. In a standing wave, all the particles cross their mean position together.
- Which of the above statement(s) is/are correct?
- (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 6 I. In a closed organ pipe, standing waves can be formed.
 II. In a closed organ pipe (closed at one end), only odd harmonics are present.
 III. The harmonics which are present in a pipe, open at both ends are odd harmonics only.

Which of the following statement(s) related to organ pipe is/are correct?

- (a) Only I (b) Both II and III
 (c) Both I and II (d) I, II and III

Match the columns

- 1 A wave is transmitted from a denser to rarer medium. Then, match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Frequency of wave	(p) will increase
(B) Speed of wave	(q) will decrease
(C) Wavelength of wave	(r) will remain unchanged
(D) Amplitude of wave	(s) may increase or decrease

Codes

- | | | | | | | | |
|-------|---|---|---|-------|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) q | r | p | s | (b) p | q | r | p |
| (c) r | p | p | p | (d) p | r | q | s |

- 2 Source has frequency f . Source and observer both have same speed. For the apparent frequency observed by observer, match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Observer is approaching the source but source is receding from the observer	(p) more than f
(B) Observer and source both are approaching towards each other	(q) less than f
(C) Observer and source both are receding from each other	(r) equal to f
(D) Source is approaching but observer is receding	(s) None

Codes

- | | | | | | | | |
|-------|---|---|---|-------|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) p | q | r | s | (b) r | p | q | r |
| (c) r | q | p | s | (d) r | p | s | q |

- 3 The equation of a stationary wave is (all in SI units)

$$y = (0.06) \sin(2\pi x) \cos(5\pi t)$$

Match the following columns and mark the correct option from the codes given below.

Column I	Column II
(A) Amplitude of constituent wave	(p) 0.06
(B) Position of node at $x = \dots$ m	(q) 0.5
(C) Position of antinode at $x = \dots$ m	(r) 0.25
(D) Amplitude at $x = \frac{3}{4}$ m	(s) 0.03

Codes

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) s | q | r | p |
| (b) p | r | r | q |
| (c) p | q | r | s |
| (d) p | r | s | q |

- 4 Fundamental frequency of closed pipe is 100 Hz and that of an open pipe is 200 Hz. Match the following columns and mark the correct option from the codes given below ($v_s = 330$ m/s).

Column I	Column II
(A) Length of closed pipe	(p) 0.825 m
(B) Length of open pipe	(q) 1.65 m
(C) Lowest harmonic of closed pipe which is equal to any of the harmonic of open pipe	(r) 5 m
	(s) None

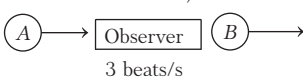
Codes

- | | | | | | |
|-------|---|---|-------|---|---|
| A | B | C | A | B | C |
| (a) p | q | s | (b) r | p | s |
| (c) p | p | s | (d) p | q | r |

(C) Medical entrances' gallery

Collection of questions asked in NEET & various medical entrance exams

- 1 In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be [NEET 2020]
(a) 524 Hz (b) 536 Hz (c) 537 Hz (d) 523 Hz
- 2 The length of the string of a musical instrument is 90 cm and has a fundamental frequency of 120 Hz. Where should it be pressed to produce fundamental frequency of 180 Hz? [NEET 2020]
(a) 75 cm (b) 60 cm (c) 45 cm (d) 80 cm
- 3 A tuning fork with frequency 800 Hz produces resonance in a resonance column tube with upper end open and lower end closed by water surface. Successive resonance are observed at length 9.75 cm, 31.25 cm and 52.75 cm. The speed of sound in air is [NEET 2019]
(a) 500 ms^{-1} (b) 156 ms^{-1} (c) 344 ms^{-1} (d) 172 ms^{-1}
- 4 If the speed of sound in air is 330 m/s, then find the number of tones present in an open organ pipe of length 1 m whose frequency is ≤ 1000 Hz. [AIIMS 2019]
(a) 2 (b) 4 (c) 8 (d) 6
- 5 A string wave equation is given by $y = 0.002 \sin(300t - 15x)$ and mass density is ($\mu = 0.1 \text{ kg/m}$). Then, find the tension force in the string. [AIIMS 2019]
(a) 30 N (b) 20 N (c) 40 N (d) 45 N
- 6 Two sources of sound S_1 and S_2 are moving towards and away from a stationary observer with the same speed, respectively. Observer detects 3 beats per second. Find the value of speed of sources (approximately). (Take, $f_1 = f_2 = 500$ Hz and speed of sound in air = 330 ms^{-1}) [AIIMS 2019]



3 beats/s

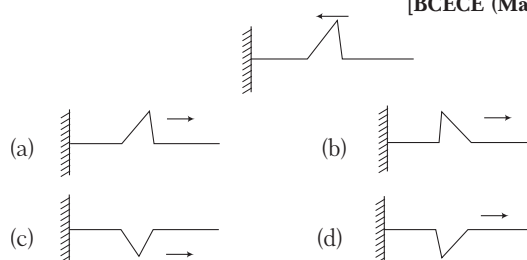
(a) 1 ms^{-1} (b) 2 ms^{-1}
(c) 3 ms^{-1} (d) 4 ms^{-1}
- 7 The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is [NEET 2018]
(a) 12.5 cm (b) 8 cm (c) 13.3 cm (d) 16 cm
- 8 A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C, two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is [NEET 2018]
(a) 350 ms^{-1} (b) 339 ms^{-1}
(c) 330 ms^{-1} (d) 300 ms^{-1}
- 9 **Assertion** To hear different beats, difference of the frequencies of two sources should be less than 10.
Reason More the number of beats more is the confusion. [AIIMS 2018]
(a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Assertion is incorrect but Reason is correct.
- 10 A steel rod 100 cm long is clamped at its middle. The fundamental frequency of longitudinal vibrations of the rod are given to be 2.53 kHz. What is the speed of sound in steel? [AIIMS 2018]
(a) 6.2 kms^{-1} (b) 5.06 kms^{-1} (c) 7.23 kms^{-1} (d) 7.45 kms^{-1}
- 11 If speed of sound in air is 340 m/s and in water is 1480 m/s. If frequency of sound is 1000 kHz, then find the value of wavelength in water. [JIPMER 2018]
(a) 2.96 mm (b) 1.48 mm (c) 0.74 mm (d) 1 mm
- 12 When two tuning forks A and B are sounded together, x beat/s are heard. Frequency of A is n . Now, when one prong of fork B is loaded with a little wax, the number of beats decreases. The frequency of fork B is [JIPMER 2018]
(a) $n + x$ (b) $n - x$ (c) $n + 2x$ (d) $n - 2x$
- 13 An organ pipe open on both ends in the n th harmonic is in resonance with a source of 1000 Hz. The length of pipe is 16.6 cm and speed of sound in air is 332 m/s. Find the value of n . [JIPMER 2018]
(a) 3 (b) 2 (c) 1 (d) 4
- 14 The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system? [NEET 2017]
(a) 10 Hz (b) 20 Hz (c) 30 Hz (d) 40 Hz
- 15 The driver of a car travelling with speed 30 ms^{-1} towards a hill sounds a horn of frequency 600 Hz. If the velocity of sound in air is 330 ms^{-1} , the frequency of reflected sound as heard by driver is [AIIMS 2017]
(a) 550 Hz (b) 555.5 Hz
(c) 720 Hz (d) 500 Hz

- 16** Two cars moving in opposite directions approach each other with speeds of 22 ms^{-1} and 16.5 ms^{-1} , respectively. The driver of the first car blows a horn having a frequency 400 Hz . The frequency heard by the driver of the second car is (Take, velocity of sound = 340 ms^{-1}) [NEET 2017]
 (a) 350 Hz (b) 361 Hz
 (c) 411 Hz (d) 448 Hz
- 17** A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 ms^{-1} . Then, the frequency of sound that the observer hears in the echo reflected from the cliff is (Take, velocity of sound in air = 330 ms^{-1}) [NEET 2016]
 (a) 800 Hz (b) 838 Hz
 (c) 885 Hz (d) 765 Hz
- 18** An air column closed at one end and open at the other resonates with a tuning fork when the smallest length of the column is 50 cm . The next larger length of the column resonating with the same tuning fork is [NEET 2016]
 (a) 100 cm (b) 150 cm
 (c) 200 cm (d) 66.7 cm
- 19** A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2 . The ratio λ_2/λ_1 is [NEET 2016]
 (a) $\sqrt{\frac{m_1 + m_2}{m_2}}$ (b) $\sqrt{\frac{m_2}{m_1}}$
 (c) $\sqrt{\frac{m_1 + m_2}{m_1}}$ (d) $\sqrt{\frac{m_1}{m_2}}$
- 20** When two displacements represented by $y_1 = a \sin(\omega t)$ and $y_2 = b \cos(\omega t)$ are superimposed, the motion is [CBSE AIPMT 2015]
 (a) not a simple harmonic
 (b) simple harmonic with amplitude $\frac{a}{b}$
 (c) simple harmonic with amplitude $\sqrt{a^2 + b^2}$
 (d) simple harmonic with amplitude $\frac{(a+b)}{2}$
- 21** The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is [CBSE AIPMT 2015]
 (a) 80 cm (b) 100 cm (c) 120 cm (d) 140 cm
- 22** If the intensity is increased by a factor of 20, then how many decibels in the sound level will increase? [AIIMS 2015]
 (a) 18 (b) 13 (c) 9 (d) 7
- 23** On the same path, the source and observer are moving in such a way that the distance between these two increases with the time. The speeds of source and observer are same and equal to 10 ms^{-1} with respect to the ground while no wind is blowing. The apparent frequency received by observer is 1950 Hz , then the original frequency must be (Take, the speed of sound in present medium is 340 ms^{-1}) [AIIMS 2015]
 (a) 2068 Hz (b) 2100 Hz (c) 1903 Hz (d) 602 Hz
- 24** The displacement of a particle of a string carrying a travelling wave is given by $y = (3 \text{ cm}) \sin 6.28(0.50x - 50t)$ ($x \rightarrow$ in cm, $t \rightarrow$ in second). What would be the speed of the wave? [UK PMT 2015]
 (a) 200 cms^{-1} (b) 150 cms^{-1}
 (c) 100 cms^{-1} (d) 50 cms^{-1}
- 25** In a medium in which a transverse progressive wave travelling, the phase difference between the points with a separation of 1.25 cm is $\frac{\pi}{4}$. If the frequency of wave is 1000 Hz , the velocity in the medium is [EAMCET 2015]
 (a) 10^4 ms^{-1} (b) 125 ms^{-1}
 (c) 100 ms^{-1} (d) 10 ms^{-1}
- 26** Two wires, of the same diameter and same material have lengths 1.2 m and 2 m , vibrate with the same fundamental frequency. If the shorter wire is stretched by a force of 36 kg-wt , then the tension in the longer wire is [EAMCET 2015]
 (a) 50 kg-wt (b) 200 kg-wt
 (c) 100 kg-wt (d) 400 kg-wt
- 27** If a closed organ pipe has the same third harmonic frequency as that of an open organ pipe, then their respective lengths are in the ratio [Kerala CEE 2015]
 (a) $1 : 1$ (b) $1 : 2$ (c) $1 : 4$ (d) $3 : 4$
 (e) $4 : 5$
- 28** Given that $y = A \sin\left(\frac{2\pi}{\lambda}(ct - x)\right)$, where x and y are measured in metres. Which of the following statement is true? [CG PMT 2015]
 (a) Unit of λ is same as that of x and A
 (b) Unit of λ is same as that of x and not of A
 (c) Unit of c is same as that of $\frac{2\pi}{\lambda}$
 (d) Unit of $(ct - x)$ is same as that of $\frac{2\pi}{\lambda}$

- 29** A radar sends radio waves of frequency ν towards an aeroplane moving with velocity v_A . A change $\Delta\nu$ is observed in the frequency of reflected wave which is higher than the original frequency. The velocity of aeroplane ($v_A \ll c$) is [CG PMT 2015]
 (a) $c \frac{\Delta\nu}{\nu}$ (b) $2 \frac{c\nu}{\Delta\nu}$ (c) $c \frac{\Delta\nu}{2\nu}$ (d) $\frac{\Delta\nu}{2c\nu}$
- 30** A train is moving with a uniform speed of 33 m/s and an observer is approaching the train with the same speed. If the train blows a whistle of frequency 1000 Hz and the velocity of sound is 333 m/s, then the apparent frequency of the sound that the observer hears is [WB JEE 2015]
 (a) 1220 Hz (b) 1099 Hz (c) 1110 Hz (d) 1200 Hz
- 31** A sound source is approaching the listener. The frequency of sound is 100 cps while passing through the listener, it is 50 cps. The frequency when the observer moves with the source is [UP CPMT 2015]
 (a) 77.3 Hz (b) 66.67 Hz (c) 63.33 Hz (d) 133.3 Hz
- 32** When sound wave is refracted from air to water, which of the following will remain unchanged? [UP CPMT 2015]
 (a) Wave number (b) Wavelength
 (c) Frequency (d) Wave velocity
- 33** The path difference between two waves
 $y_1 = a_1 \sin\left(\omega t - \frac{2\pi x}{\lambda}\right)$ and
 $y_2 = a_2 \cos\left(\omega t - \frac{2\pi x}{\lambda} + \phi\right)$ [Manipal 2015]
 (a) $\frac{\lambda}{2\pi}(\phi + \pi/2)$ (b) $\frac{\lambda}{2\pi}(\phi)$ (c) $\frac{2\pi}{\lambda}(\phi - \pi/2)$ (d) $\frac{2\pi}{\lambda}(\phi)$
- 34** A stretched string is vibrating in the second overtone, then the number of nodes and antinodes between the ends of the string are respectively [KCET 2015]
 (a) 3 and 4 (b) 4 and 3 (c) 2 and 3 (d) 3 and 2
- 35** When two tuning forks A and B are sounded together, 4 beats per second are heard. The frequency of the fork B is 384 Hz. When one of the prongs of the fork A is filled and sounded with B, the beat frequency increases, then the frequency of the fork A is [KCET 2015]
 (a) 379 Hz (b) 380 Hz (c) 389 Hz (d) 388 Hz
- 36** A wire having linear mass density $9 \times 10^{-3} \text{ kg/m}^3$ is stretched between two clamps 1m apart and is subjected to an extension of $4.9 \times 10^{-4} \text{ m}$. The lowest frequency of wave produced in the wire is (Take, $Y = 9 \times 10^{10} \text{ N/m}^2$) [UP CPMT 2015]
 (a) 47 Hz (b) 42 Hz (c) 35 Hz (d) 37 Hz
- 37** Identify the correct statement. [Kerala CEE 2014]
 (a) Transverse wave can propagate in gases.
 (b) Transverse wave consists of compressions and rarefactions.
 (c) Longitudinal wave can propagate in solids, liquids and gases.
 (d) In a longitudinal wave, particles of the medium vibrate perpendicular to the direction of propagation.
 (e) In longitudinal wave, the higher density corresponds of rarefactions.
- 38** When a wave travels in a medium, displacement of a particle is given by $y = a \sin 2\pi(bt - cx)$, where a , b and c are constants. The maximum particle velocity will be twice the wave velocity, if [MHT CET 2014]
 (a) $b = ac$ (b) $b = \frac{1}{ac}$ (c) $c = \pi a$ (d) $c = \frac{1}{\pi a}$
- 39** The displacement y of a particle in a medium can be expressed as $y = 10^{-6} \sin\left(100t + 20x + \frac{x}{4}\right)$, where t is in second and x in metre. The speed of the wave is [UK PMT 2014]
 (a) 2000 ms^{-1} (b) 5 ms^{-1} (c) 20 ms^{-1} (d) $5\pi \text{ ms}^{-1}$
- 40** If n_1 , n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by [CBSE AIPMT 2014, 2012]
 (a) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$ (b) $\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n_1}} + \frac{1}{\sqrt{n_2}} + \frac{1}{\sqrt{n_3}}$
 (c) $\sqrt{n} = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3}$ (d) $n = n_1 + n_2 + n_3$
- 41** The number of possible natural oscillations of air column in a pipe closed at one end of length 85 cm whose frequencies lie below 1250 Hz are (Take, velocity of sound = 340 ms^{-1}) [CBSE AIPMT 2014]
 (a) 4 (b) 5 (c) 7 (d) 6
- 42** A whistle whose air column is open at both ends has a fundamental frequency of 5100 Hz. If the speed of sound in air is 340 ms^{-1} , the length of the whistle (in cm) is [WB JEE 2014]
 (a) 5/3 (b) 10/3 (c) 5 (d) 20/3
- 43** A closed pipe and an open pipe of same length produce 2 beats, when they are set into vibration simultaneously in their fundamental mode. If the length of the open pipe is halved and that of closed pipe is doubled, and if they are vibrating in the fundamental mode, then the number of beats produced is [EAMCET 2014]
 (a) 4 (b) 7 (c) 2 (d) 8

- 44** A pipe of 30 cm long open at both the ends produces harmonics. Which harmonic mode of pipe resonates at 1.1 kHz source? [KCET 2014]
(Take, speed of sound in air = 330 ms^{-1})
(a) Fifth harmonic (b) Fourth harmonic
(c) Third harmonic (d) Second harmonic
- 45** A metallic wire of 1 m length has a mass of $10 \times 10^{-3} \text{ kg}$. If a tension of 100 N is applied to a wire, what is the speed of transverse wave? [KCET 2014]
(a) 100 ms^{-1} (b) 10 ms^{-1} (c) 200 ms^{-1} (d) 0.1 ms^{-1}
- 46** A speeding motorcyclist sees traffic jam ahead of him. He slows down to 36 km h^{-1} . He finds that traffic has eased and a car moving ahead of him at 18 km h^{-1} is honking at a frequency of 1392 Hz. If the speed of sound is 343 ms^{-1} , then the frequency of the honk as heard by him will be [CBSE AIPMT 2014]
(a) 1332 Hz (b) 1372 Hz (c) 1412 Hz (d) 1454 Hz
- 47** A wave travelling in the positive x -direction having displacement along y -direction as 1 m, wavelength $2\pi \text{ m}$ and frequency of $1/\pi \text{ Hz}$ is represented by [NEET 2013]
(a) $y = \sin(x - 2t)$ (b) $y = \sin(2\pi x - 2\pi t)$
(c) $y = \sin(10\pi x - 20\pi t)$ (d) $y = \sin(2\pi x + 2\pi t)$
- 48** If we study the vibration of a pipe open at both ends, then which of the following statements is not true? [NEET 2013]
(a) Open end will be antinode
(b) Odd harmonics of the fundamental frequency will be generated
(c) All harmonics of the fundamental frequency will be generated
(d) Pressure change will be maximum at both ends
- 49** A source of unknown frequency gives 4 beats/s when sounded with a source of known frequency 250 Hz. The second harmonic of the source of unknown frequency gives five beats per second when sounded with a source of frequency 513 Hz. The unknown frequency is [NEET 2013]
(a) 254 Hz (b) 246 Hz (c) 240 Hz (d) 260 Hz
- 50** The fundamental frequency of an open pipe of length 1m, if the speed of sound in air is 340 ms^{-1} is [J&K CET 2013]
(a) 340 Hz (b) 170 Hz (c) 680 Hz (d) 85 Hz
- 51** A whistle with frequency 1020 Hz is blown at a station. A man travelling in train moving towards the station at 30 ms^{-1} hears the sound of the whistle. If the speed of sound is 340 ms^{-1} , the apparent frequency heard by him is [J&K CET 2013]
(a) 1020 Hz (b) 1110 Hz (c) 2040 Hz (d) 610 Hz
- 52** The beats are the examples of [J&K CET 2013]
(a) simple harmonic motion
(b) interference of two or more waves having same amplitude but slightly different frequencies in the same direction
(c) interference of two or more waves having different amplitude but same frequencies in the same direction
(d) interference of two or more waves having same amplitude but slightly different frequencies in the perpendicular direction
- 53** A wire of length L and linear density m is stretched by a force T and the frequency is n_1 . Another wire of same material of length $2L$ and same linear density is stretched by a force $9T$ and its frequency is n_2 . Then, the value of (n_2/n_1) is [EAMCET 2013]
(a) 4 : 1 (b) 1 : 3
(c) 3 : 2 (d) 1 : 2
- 54** A pipe closed at one end and open at the other end resonates with a sound of frequency 135 Hz and also with 165 Hz, but not at any other frequency intermediate between these two. Then, the frequency of the fundamental note of the pipe is [EAMCET 2013]
(a) 15 Hz (b) 60 Hz
(c) 7.5 Hz (d) 30 Hz
- 55** The equation of a transverse wave is given by $y = 0.05 \sin \pi (2t - 0.02x)$, where x, y are in metre and t is in second. The minimum distance of separation between two particles which are in phase and the wave velocity are respectively [KCET 2013]
(a) 50 m, 50 ms^{-1} (b) 100 m, 100 ms^{-1}
(c) 50 m, 100 ms^{-1} (d) 100 m, 50 ms^{-1}
- 56** The frequency of the second overtone of the open pipe is equal to the frequency of the first overtone of the closed pipe. The ratio of the lengths of the open pipe and the closed pipe is [KCET 2013]
(a) 2 : 1 (b) 1 : 2
(c) 1 : 3 (d) 3 : 1
- 57** A person with vibrating tuning fork of frequency 338 Hz is moving towards a vertical wall with a speed of 2 ms^{-1} . Velocity of sound in air is 340 ms^{-1} . The number of beats heard by that person per second is [KCET 2013]
(a) 2 (b) 4
(c) 6 (d) 8
- 58** A travelling wave reflected at an open boundary undergoes a phase change of [Kerala CEE 2013]
(a) π (b) 0 (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{3}$
(e) $\pi/4$

- 59** A tube open at both the ends has a fundamental frequency f in air. If the half tube is dipped vertically in water, then the fundamental frequency of the air column is [Kerala CEE 2013]
 (a) $\frac{f}{2}$ (b) $\frac{3f}{2}$ (c) $\frac{3f}{4}$ (d) f
 (e) $2f$
- 60** Two waves each of amplitude A_0 produce a resultant wave of amplitude A_0 on superposition. The phase difference between the two waves is [MP PMT 2013]
 (a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$
- 61** Frequencies of two sound waves are 334 Hz and 326 Hz. If the velocity of the sound is 330 ms^{-1} , then the beat period when the two waves superimpose, is [MP PMT 2013]
 (a) 8 s (b) 1 s (c) 0.125 s (d) 0.25 s
- 62** A stationary source emits a whistle at a frequency of 200 Hz. If the velocity of propagation of sound is 340 ms^{-1} , then the observed frequency, if the observer is moving away from the source at 25 ms^{-1} , will be [MP PMT 2013]
 (a) 185 Hz (b) 215 Hz
 (c) 175 Hz (d) 225 Hz
- 63** The wavelength of a wave in a medium is 0.5m. The phase difference between the oscillations at two points in the medium due to this wave is $\frac{\pi}{5}$. What is the minimum distance between these points? [MP PMT 2013]
 (a) 0.05 m (b) 0.1m
 (c) 0.25 m (d) 0.15 m
- 64 Assertion** It is not possible to have interference between the waves produced by two violins.
Reason For interference of two waves, the phase difference between the waves must remain constant. [AIIMS 2012]
 (a) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (c) Assertion is correct but Reason is incorrect.
 (d) Both Assertion and Reason are incorrect.
- 65** An open pipe is suddenly closed at one end. As a result, the frequency of third harmonic of the closed pipe is found to be higher by 100 Hz than that of the fundamental frequency of the open pipe. The fundamental frequency of the open pipe will be [BHU 2012]
 (a) 200 Hz (b) 240 Hz
 (c) 300 Hz (d) 480 Hz
- 66** A man standing in front of a hill beats a drum at regular intervals. The rate of drumming is generally increased and he finds that the echo is not heard distinctly when the rate becomes 40 per minute. Then, he moves closer to hill by 90 m and he again finds that echo is again not heard when the drumming rate becomes 60 per minute. The distance between the mountain and initial position of the man is [BHU 2012]
 (a) 180 m (b) 205 m
 (c) 270 m (d) 300 m
- 67** The equations of displacement of two waves are
 $y_1 = 10 \sin(3\pi t + \pi/3)$
 and $y_2 = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t)$
 What is the ratio of their amplitude? [UP CPMT 2012]
 (a) 1:2 (b) 2:1
 (c) 1:1 (d) None of these
- 68** Among the following equations, which one represents the spherical progressive wave? [UP CPMT 2012]
 (a) $y = r \sin \omega t$
 (b) $y = \frac{a}{r} \sin(\omega t - kx)$
 (c) $y = \frac{a}{\sqrt{r}} \sin(\omega t - kx)$
 (d) $y = \sqrt{\frac{a}{r}} \sin(\omega t - kx)$
- 69** If the shift in a star light is towards red end, then [UP CPMT 2012]
 (a) the star is approaching the earth
 (b) the star receding from the earth
 (c) the apparent frequency is lesser than actual
 (d) Both (b) and (c)
- 70** Two open organ pipes, sounded simultaneously, are given 5 beats per second between the fundamentals. If the shorter pipe is 1.10 m long, the length of the longer pipe is (Take, velocity of sound in air = 340 m s^{-1}) [AMU 2012]
 (a) 1.22 m (b) 1.18 m (c) 1.14 m (d) 1.12 m
- 71** If an incident pulse, moving towards a rigid support is as shown, then among the following pulse which one represents the reflected pulse correctly? [BCECE (Mains) 2012]



- 72** The fundamental frequency of vibration of 1 cm long string is 256 Hz. If the length of the string is reduced to 1/4 cm keeping the tension unaltered, the new fundamental frequency will be

[BCECE (Mains) 2012]

- (a) 256 Hz (b) 512 Hz
(c) 1024 Hz (d) 2048 Hz
- 73** A wave travelling along positive X-axis is given by $y = A \sin(\omega t - kx)$. If it is reflected from a rigid boundary such that 80% amplitude is reflected, then equation of reflected wave is

[JCECE 2012]

- (a) $y = A \sin(\omega t + 0.8 kx)$ (b) $y = -0.8A \sin(\omega t + kx)$
(c) $y = A \sin(\omega t + kx)$ (d) $y = 0.8A \sin(\omega t + kx)$
- 74** Two waves are represented by the equations
 $y_1 = A \sin(\omega t + kx + 0.57)$ m and
 $y_2 = A \cos(\omega t + kx)$ m, where x is in metre and t is in second. The phase difference between them is

[CBSE AIPMT 2011]

- (a) 1.25 rad (b) 1.57 rad
(c) 0.57 rad (d) 1.0 rad
- 75 Assertion** The fundamental frequency of an open organ pipe increases as the temperature is increased.
Reason As the temperature increases, the velocity of sound increases more rapidly than length of the pipe.

[AIIMS 2011]

- (a) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
(b) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(c) Assertion is correct but Reason is incorrect.
(d) Both Assertion and Reason are incorrect.
- 76** A source of sound S is moving with a velocity of 50 ms^{-1} towards a stationary observer. The observer measures the frequency of the source as 1000 Hz. What will be the apparent frequency of the source when it is moving away from the observer after crossing him? (The velocity of the sound in the medium is 350 ms^{-1} .)

[AIIMS 2011]

- (a) 750 Hz (b) 857 Hz
(c) 1143 Hz (d) 1333 Hz
- 77** A sound absorber attenuates the sound level by 20 dB. The intensity decreases by a factor of

[JIPMER 2011]

- (a) 1000 (b) 10000
(c) 10 (d) 100
- 78** Each of the properties of sound in Column I primarily depends on one of the quantities in Column II. Select the correct answer (matching Column I with Column II) as per codes given below the columns.

[Haryana PMT, CG PMT 2011]

Column I		Column II	
(A)	Loudness	(p)	Waveform
(B)	Pitch	(q)	Frequency
(C)	Quality	(r)	Intensity

Codes

A	B	C	A	B	C
(a) p	q	r	(b) r	q	p
(c) q	r	p	(d) q	p	r

- 79** A theatre of volume $100 \times 40 \times 10 \text{ m}^3$ can accommodate 1000 visitors. The reverberation time of the theatre when empty is 8.5 s. If the theatre is now filled with 500 visitors, occupying the front-half seats, the reverberation time changes to 6.2 s. The average absorption coefficient of each visitor is nearly

[EAMCET 2011]

- (a) 0.6 (b) 0.5
(c) 0.45 (d) 0.7
- 80** Two tuning forks A and B produce notes of frequencies 258 Hz and 262 Hz. An unknown note sounded with A produces certain beats. When the same note is sounded with B , the beat frequency gets doubled. The unknown frequency is

[KCET 2011]

- (a) 250 Hz (b) 252 Hz
(c) 254 Hz (d) 245 Hz
- 81** In a stationary wave, all the particles

[OJEE 2011]

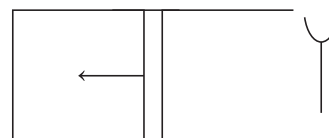
- (a) on either side of a node vibrate in same phase
(b) in the region between two nodes vibrate in same phase
(c) in the region between two antinodes vibrate in same phase
(d) of the medium vibrate in same phase
- 82** The fundamental frequency of a closed pipe is 220 Hz. If $\left(\frac{1}{4}\right)$ th of the pipe is filled with water,

then the frequency of the first overtone of the pipe now is

[Haryana PMT 2011]

- (a) 220 Hz (b) 440 Hz
(c) 880 Hz (d) 1760 Hz
- 83** A piston fitted in cylindrical pipe is pulled as shown in the figure. A tuning fork is sounded at open end and loudest sound is heard at open length 13 cm, 41 cm and 69 cm. The frequency of tuning fork, if velocity of sound is 350 ms^{-1} , is

[DUMET 2011]



- (a) 1250 Hz (b) 625 Hz
(c) 417 Hz (d) 715 Hz

- 84** Two sources are at a finite distance apart. They emit sound of wavelength λ . An observer situated between them on line joining the sources, approaches towards one source with speed u , then number of beats heard per second by observer will be [UP CPMT 2011]

(a) $\frac{2u}{\lambda}$ (b) $\frac{u}{\lambda}$
 (c) $\frac{u}{2\lambda}$ (d) $\frac{\lambda}{u}$

- 85** A source of sound producing wavelength of 50 cm is moving away from stationary observer with $(1/5)$ th speed of sound. Then, what is the wavelength of sound heard by observer? [EAMCET 2011]
 (a) 70 cm (b) 55 cm (c) 40 cm (d) 60 cm
- 86** A man stands in a narrow, steep-sided valley. When he shouts, he hears two echoes, one after 1 s and other after 2 s. If the velocity of sound in air is 330 m/s, the width of the valley is [MGIMS 2011]
 (a) 330 m (b) 495 m (c) 660 m (d) 990 m

ANSWERS

CHECK POINT 17.1

1. (b)	2. (a)	3. (a)	4. (c)	5. (c)	6. (b)	7. (c)	8. (b)	9. (b)	10. (d)
11. (d)	12. (b)	13. (c)	14. (c)	15. (c)	16. (c)	17. (b)	18. (a)	19. (c)	20. (d)
21. (c)	22. (c)	23. (d)	24. (c)	25. (b)	26. (c)	27. (c)	28. (d)	29. (c)	30. (a)

CHECK POINT 17.2

1. (c)	2. (c)	3. (b)	4. (c)	5. (a)	6. (d)	7. (d)	8. (c)	9. (d)	10. (b)
11. (b)	12. (b)	13. (c)	14. (b)	15. (d)	16. (c)	17. (c)	18. (b)	19. (d)	20. (c)
21. (d)	22. (d)	23. (d)							

CHECK POINT 17.3

1. (d)	2. (b)	3. (a)	4. (a)	5. (a)	6. (b)	7. (b)	8. (b)	9. (d)	10. (a)
11. (c)	12. (d)	13. (b)	14. (c)	15. (c)	16. (c)	17. (a)	18. (d)	19. (c)	

CHECK POINT 17.4

1. (b)	2. (a)	3. (b)	4. (d)	5. (c)	6. (a)	7. (b)	8. (d)	9. (c)	10. (b)
11. (b)	12. (d)	13. (c)	14. (a)	15. (a)	16. (c)	17. (c)	18. (b)	19. (a)	20. (b)
21. (c)	22. (d)	23. (d)	24. (b)	25. (a)	26. (c)	27. (a)	28. (a)	29. (c)	30. (c)
31. (b)									

CHECK POINT 17.5

1. (d)	2. (c)	3. (b)	4. (b)	5. (b)	6. (c)	7. (c)	8. (d)	9. (d)	10. (c)
11. (c)	12. (d)	13. (d)	14. (a)	15. (a)	16. (a)	17. (a)	18. (c)	19. (c)	20. (b)
21. (b)	22. (c)	23. (b)	24. (d)	25. (b)					

(A) Taking it together

1. (a)	2. (b)	3. (c)	4. (b)	5. (c)	6. (a)	7. (a)	8. (a)	9. (d)	10. (a)
11. (a)	12. (d)	13. (d)	14. (d)	15. (a)	16. (c)	17. (b)	18. (b)	19. (b)	20. (b)
21. (b)	22. (a)	23. (d)	24. (b)	25. (d)	26. (d)	27. (a)	28. (a)	29. (b)	30. (a)
31. (c)	32. (c)	33. (b)	34. (b)	35. (a)	36. (b)	37. (a)	38. (a)	39. (b)	40. (d)
41. (b)	42. (a)	43. (a)	44. (a)	45. (d)	46. (b)	47. (d)	48. (c)	49. (a)	50. (a)
51. (a)	52. (d)	53. (a)	54. (a)	55. (b)	56. (c)	57. (c)	58. (d)	59. (b)	60. (a)
61. (a)	62. (c)	63. (b)	64. (c)	65. (d)	66. (c)	67. (c)	68. (a)	69. (d)	70. (c)
71. (c)	72. (b)	73. (d)	74. (a)	75. (d)	76. (b)	77. (a)	78. (d)	79. (c)	80. (c)
81. (a)	82. (c)	83. (b)	84. (a)	85. (d)	86. (c)	87. (b)	88. (c)	89. (b)	90. (b)
91. (c)	92. (a)	93. (a)	94. (b)	95. (c)	96. (c)	97. (b)	98. (c)	99. (c)	100. (d)
101. (b)	102. (b)	103. (a)							

(B) Medical entrance special format questions

• Assertion and reason

1. (a) 2. (c) 3. (a) 4. (b) 5. (b)

• Statement based questions

1. (c) 2. (d) 3. (b) 4. (c) 5. (b) 6. (c)

• Match the columns

1. (c) 2. (b) 3. (a) 4. (c)

(C) Medical entrances' gallery

1. (a)	2. (b)	3. (c)	4. (d)	5. (c)	6. (a)	7. (c)	8. (b)	9. (b)	10. (b)
11. (b)	12. (a)	13. (c)	14. (b)	15. (c)	16. (d)	17. (b)	18. (b)	19. (a)	20. (c)
21. (c)	22. (b)	23. (a)	24. (c)	25. (c)	26. (c)	27. (b)	28. (a)	29. (c)	30. (a)
31. (b)	32. (c)	33. (a)	34. (c)	35. (d)	36. (c)	37. (c)	38. (d)	39. (b)	40. (a)
41. (d)	42. (b)	43. (b)	44. (d)	45. (a)	46. (c)	47. (a)	48. (d)	49. (a)	50. (b)
51. (b)	52. (b)	53. (c)	54. (a)	55. (b)	56. (a)	57. (b)	58. (b)	59. (d)	60. (b)
61. (c)	62. (b)	63. (d)	64. (a)	65. (a)	66. (c)	67. (c)	68. (b)	69. (d)	70. (c)
71. (c)	72. (c)	73. (b)	74. (d)	75. (a)	76. (a)	77. (d)	78. (b)	79. (d)	80. (c)
81. (b)	82. (c)	83. (b)	84. (a)	85. (d)	86. (b)				

Hints & Explanations

• CHECK POINT 17.1

- 1 (b) The number of waves present in unit length of the medium is known as wave number. The SI unit of wave number is m^{-1} .
- 2 (a) A wavefront is a line or surface on which the disturbance has the same phase at all points.
- 3 (a) The waves which are associated with the motion of microscopic particles such as electrons, protons, neutrons atoms, molecules, etc., are called matter waves.
- 4 (c) Velocity of wave is given by

$$v = n\lambda \Rightarrow n = \frac{v}{\lambda}$$
- 5 (c) Speed the wave, $v = 60 \text{ ms}^{-1}$
 Frequency of the wave, $\nu = 1200 \text{ min}^{-1} = \frac{1200}{60} = 20 \text{ s}^{-1}$
 Now, wavelength, $\lambda = \frac{v}{\nu} = \frac{60}{20} \Rightarrow \lambda = 3 \text{ m}$
- 6 (b) Movement of string of a sitar or violin is an example of transverse waves.
- 7 (c) Vibrations of air column above the surface of water in the tube of a resonance apparatus are longitudinal.
- 8 (b) Distance between a compression and adjoining rarefaction in a pressure wave is $\frac{\lambda}{2}$.
- 9 (b) Frequency range of sound in audible region is 20 Hz-20 kHz.
- 10 (d) (i) Ultrasonic waves can be used to detect submarines icebergs.
 (ii) Ultrasonic waves can be used to clean clothes, fine machinery parts, etc.
 (iii) These waves can be used to kill smaller animals like rats, fish, frogs, etc.
- 11 (d) Sound wave transfer both energy and momentum.
- 12 (b) Water waves produced by a motorboat sailing in water are both longitudinal and transverse because the waves produce transverse as well as lateral vibrations in the particles of the medium.

- 13 (c) Velocity of sound through gaseous medium, $v = \sqrt{\frac{\gamma RT}{M}}$.

For diatomic gas, $\gamma = 1.4$ and $T = 273 \text{ K}$

$$\therefore \gamma T \approx 382$$

$\therefore$ The velocity of sound through a diatomic gaseous medium,

$$v = \sqrt{\frac{382R}{M}}$$

- 14 (c) Ratio of velocities, $\frac{v_{\text{O}_2}}{v_{\text{He}}} = \frac{\sqrt{\frac{7}{5RT}}}{\sqrt{\frac{32}{5 \times 3RT}}} = 0.32$

or

$$v_{\text{O}_2} = 0.32 v_{\text{He}}$$

$$\text{Time taken} = \frac{1}{0.32} T = 3T$$

- 15 (c) Velocity of sound in gas, $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{\frac{\gamma}{M}}$

$$\Rightarrow \frac{v_{\text{N}_2}}{v_{\text{He}}} = \sqrt{\frac{\gamma_{\text{N}_2}}{\gamma_{\text{He}}} \times \frac{M_{\text{He}}}{M_{\text{N}_2}}} = \sqrt{\frac{\frac{7}{5} \times 4}{\frac{5}{3} \times 28}} = \frac{\sqrt{3}}{5}$$

- 16 (c) Velocity of sound in gaseous medium,

$$v = \sqrt{\frac{\gamma RT}{M}} \text{ or } v \propto T^{1/2}$$

For small percentage change,

$$\% \text{ decrease in } v = \frac{1}{2} (\% \text{ decrease in } T) = \frac{1}{2} (1\%) = 0.5\%$$

- 17 (b) Velocity of sound in gas, $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{T}$

$$\therefore \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{T + 600}{T}} = \sqrt{3}$$

The initial temperature of the gas, $T = 300 \text{ K} = 27^\circ \text{C}$

- 18 (a) Given, $\nu = 220 \text{ Hz}$, $\lambda_1 = 1.5 \text{ m}$,

$$T_1 = 0^\circ \text{C} = 273 \text{ K}$$

$$v_1 = \nu \lambda_1 = 220 \times 1.5 = 330 \text{ ms}^{-1}$$

$$T_2 = 27^\circ \text{C} = 27 + 273 = 300 \text{ K}$$

$$\text{Velocity, } v_2 = v_1 \sqrt{\frac{T_2}{T_1}} = 330 \sqrt{\frac{300}{273}} = 345.9 \text{ ms}^{-1}$$

$$\text{Wavelength, } \lambda_2 = \frac{v_2}{\nu} = \frac{345.9}{220} = 1.57 \text{ m}$$

$$\text{Increase in wavelength} = \lambda_2 - \lambda_1 = 1.57 - 1.5 = 0.07 \text{ m}$$

- 19 (c) Speed of sound wave in a medium, $v \propto \sqrt{T}$

(where, T is temperature of the medium)

Clearly, when temperature changes, speed also changes.

Now,

$$v = \nu \lambda$$

where, ν is frequency and λ is wavelength.

$\therefore$ Frequency (ν) remains fixed,

$$\therefore v \propto \lambda \text{ or } \lambda \propto v \Rightarrow \lambda \propto \sqrt{T}$$

Hence, with change in temperature of the medium, wavelength (λ) of sound waves changes.

- 20 (d) Speed of sound, $v \propto \sqrt{T}$ and it is independent of pressure.

- 21 (c) When a sound wave goes from one medium to another, its frequency does not change but wavelength and speed change.

- 22 (c) Due to presence of moisture, density of air decreases.

As, speed of sound in air is given by

$$v = \sqrt{\frac{\gamma p}{\rho}}$$

For air, γ and p are constants.

$$\therefore v \propto \frac{1}{\sqrt{\rho}}, \text{ where } \rho \text{ is density of air.}$$

$$\frac{v_2}{v_1} = \sqrt{\frac{\rho_1}{\rho_2}}$$

where, ρ_1 is density of dry air and ρ_2 is density of moist air.

$$\text{As, } \rho_2 < \rho_1 \Rightarrow \frac{v_2}{v_1} > 1 \Rightarrow v_2 > v_1$$

Hence, speed of sound wave in air increases with increase in humidity.

23 (d) Velocity, $v = n\lambda$

As frequency n remains constant,

$$v \propto \lambda \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} = \frac{2/3}{3/10} = \frac{20}{9}$$

24 (c) The rms speed of the molecule of gas, $v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = c \dots(i)$

$$\text{As we know that, } \gamma = 1 + \frac{2}{f} = \frac{4}{3}$$

Velocity of sound in air,

$$v_{\text{sound}} = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{4RT}{3M}} \dots(ii)$$

From Eqs. (i) and (ii), we get

$$v_{\text{sound}} = \frac{2}{3} c$$

25 (b) Mach number is the ratio of speed of source to the speed of sound.

$$\text{i.e. Mach number} = \frac{\text{Speed of source}}{\text{Speed of sound}}$$

26 (c) Intensity level $= 10 \log \left(\frac{I}{I_0} \right)$

$$30 = 10 \log \left(\frac{I}{I_0} \right) \text{ or } 3 = \log \left(\frac{I}{I_0} \right)$$

$$\therefore \frac{I}{I_0} = 10^3 = 100$$

27 (c) Intensity level (in dB) $= 10 \log_{10} \left(\frac{I}{I_0} \right)$

$$\text{Here, } I_0 = 10^{-12} \text{ Wm}^{-2}$$

$$\begin{aligned} \therefore \text{Intensity level} &= 10 \log_{10} \frac{2 \times 10^{-8}}{10^{-12}} \\ &= 10(4.3) = 43 \text{ dB} \end{aligned}$$

28 (d) Here, $90 - 40 = 10 \log \frac{I_1}{I_0} - 10 \log \frac{I_2}{I_0}$

$$\text{or } 50 = 10 \log \left(\frac{I_1}{I_2} \right)$$

$$\therefore \frac{I_1}{I_2} = 10^5$$

29 (c) As, $L_2 - L_1 = 10 \log \frac{I_2}{I_0} - 10 \log \frac{I_1}{I_0} = 10 \log \left(\frac{I_2}{I_1} \right)$

$$L_2 - 4 = 10 \log (2) \quad \left(\because \frac{I_2}{I_1} = 2 \right)$$

$$\therefore L_2 = 7 \text{ dB}$$

30 (a) The pitch is the highness or lowness of a tone, related to wave frequency.

• CHECK POINT 17.2

1 (c) Here, $\omega \cdot \frac{x}{v}$ is dimensionless

$$\Rightarrow \omega \left[\frac{L}{LT^{-1}} \right] = [M^0 L^0 T^0] \Rightarrow [\omega T] = [M^0 L^0 T^0]$$

$$\text{or } [\omega] = \frac{[M^0 L^0 T^0]}{[T]} \Rightarrow [\omega] = [T^{-1}]$$

2 (c) Angular velocity, $\omega = 2\pi f$ or $60 = 2\pi f$

$$\therefore \text{Frequency, } f = \frac{30}{\pi} \text{ Hz}$$

3 (b) Here, $k = \frac{2\pi}{40} = \frac{2\pi}{\lambda}$

$$\therefore \text{Wavelength, } \lambda = 40 \text{ cm}$$

4 (c) $k = \frac{2\pi}{\lambda} = \frac{2\pi}{60}$ and $\omega = vk = (360) \left(\frac{2\pi}{60} \right) = 12\pi$

$$\therefore y = 0.2 \sin 2\pi \left(6t - \frac{x}{60} \right)$$

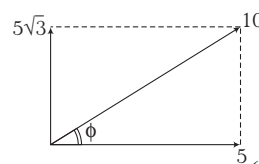
5 (a) From the given equation, $k = \frac{2\pi}{\lambda} = \frac{\pi}{4} \Rightarrow \lambda = 8 \text{ m}$

6 (d) In case of sound wave, y can represent pressure and displacement, while in case of an electromagnetic wave, it represents electric and magnetic fields.

In general, y is any general physical quantity which is made to oscillate at one place and these oscillations are propagated to other places also.

7 (d) Here, the phase is $(kx - \omega t + \phi_0)$.

8 (c) Given, $y_2 = 5 (\sin 5\pi t + \sqrt{3} \cos 5\pi t)$



This can also be written as $y_2 = 10 \sin \left(5\pi t + \frac{\pi}{3} \right)$

Now, $A_1 = 10$ and $A_2 = 10$

$$\therefore \frac{A_1}{A_2} = \frac{1}{1}$$

9 (d) Points B and F are in same phase as they are λ distance apart.

10 (b) We have, $(v_p)_{\text{max}} = 4v$ or $Y_0 \omega = 4(f \lambda)$ or $Y_0 (2\pi f) = 4f \lambda$

$$\therefore \text{Wavelength, } \lambda = \frac{\pi Y_0}{2}$$

$$11 \text{ (b) Speed of wave} = \frac{\text{Coefficient of } t}{\text{Coefficient of } x} = \frac{0.002}{\frac{2\pi}{0.1}} = 50 \text{ ms}^{-1}$$

12 (b) Maximum particle velocity,

$$(v_p)_{\max} = \omega A = 2\pi f A \\ = 2\pi (250) (10^{-2}) = 5\pi \text{ ms}^{-1}$$

13 (c) Maximum particle speed,

$$(v_p)_{\max} = \omega A = (10\pi) (0.1) = \pi \text{ cm s}^{-1}$$

14 (b) Here, $\omega = 15\pi \text{ rad s}^{-1}$, $k = 10\pi \text{ rad m}^{-1}$

$$v = \frac{\omega}{k} = 1.5 \text{ ms}^{-1} \Rightarrow \lambda = \frac{2\pi}{k} = \frac{2\pi}{10\pi} = 0.2 \text{ m}$$

Positive sign between kx and ωt means wave is travelling in negative x -direction.

15 (d) Comparing the given equation with standard wave equation,

$$y = a \sin \frac{2\pi}{\lambda} (vt - x)$$

$\therefore$ We get $v = 200 \text{ m/s}$

16 (c) The particle velocity is maximum at B and is given by

$$\frac{dy}{dt} = (v_p)_{\max} = \omega A$$

Also, wave velocity, $\frac{dx}{dt} = v = \frac{\omega}{k}$

So, slope, $\frac{dy}{dx} = \frac{(v_p)_{\max}}{v} = kA$

17 (c) Here, $A = 0.05 \text{ m}$, $\frac{5\lambda}{2} = 0.25 \Rightarrow \lambda = 0.1 \text{ m}$

Now, standard equation of wave, $y = A \sin \frac{2\pi}{\lambda} (vt - x)$

Putting the given values, we get

$$y = 0.05 \sin 2\pi (3300t - 10x)$$

18 (b) Phase difference, $\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x \Rightarrow \lambda = \left(\frac{2\pi}{0.5\pi} \right) (0.8) = 3.2 \text{ m}$

$\therefore$ Velocity of wave, $v = f \lambda = 120 \times 3.2 = 384 \text{ ms}^{-1}$

19 (d) Phase difference, $\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x$

The distance between two points,

$$\Delta x = \frac{(\Delta\phi)(\lambda)}{2\pi} = \frac{(\Delta\phi)(v/f)}{2\pi} = \frac{(\pi/3)(360/500)}{2\pi} \\ = 0.12 \text{ m}$$

20 (c) Comparing the given equation with $y = a \cos (\omega t + kx - \phi)$, we get

$$k = \frac{2\pi}{\lambda} = 0.02 \Rightarrow \lambda = 100 \text{ cm}$$

$$\Delta\phi = \frac{\pi}{2}$$

Hence, path difference between them,

$$\Delta x = \frac{\lambda}{2\pi} \times \Delta\phi = \frac{\lambda}{2\pi} \times \frac{\pi}{2} = \frac{\lambda}{4} = \frac{100}{4} = 25 \text{ cm}$$

21 (d) Phase difference, $\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x = k (\Delta x) = k (vt)$

Here, vt = distance travelled by insect in given time interval.

$$\text{or } \Delta\phi = (20\pi) (5 \times 10^{-2} \times 5) = 5\pi$$

22 (d) Wavelength, $\lambda = \frac{v}{n} = \frac{350}{350} = 1 \text{ m} = 100 \text{ cm}$

Also, path difference (Δx) between the waves at the point of observation is $AP - BP = 25 \text{ cm}$. Hence,

$$\text{Phase difference, } \Delta\phi = \frac{2\pi}{\lambda} (\Delta x) = \frac{2\pi}{1} \times \left(\frac{25}{100} \right) = \frac{\pi}{2}$$

$$\Rightarrow \text{Resultant amplitude, } A = \sqrt{(a_1)^2 + (a_2)^2} \\ = \sqrt{(0.3)^2 + (0.4)^2} = 0.5 \text{ mm}$$

23 (d) Ratio of intensities, $\frac{I_1}{I_2} = \frac{A_1^2 \omega_1^2}{A_2^2 \omega_2^2} = \left(\frac{5}{10} \right)^2 \left(\frac{2\pi \times 75}{2\pi \times 150} \right)^2 = \frac{1}{16}$

• CHECK POINT 17.3

1 (d) Law of superposition is applicable to all kinds of waves.

2 (b) Given, resultant disturbance, $a^2 = a^2 + a^2 + 2a^2 \cos \theta$

$$\Rightarrow \cos \theta = -\frac{1}{2} \Rightarrow \theta = \frac{2\pi}{3}$$

3 (a) In the same phase, $\phi = 0$. So, resultant amplitude $= a_1 + a_2 = 2A + A = 3A$

4 (a) The amplitude of the resultant wave, $A = \sqrt{a_1^2 + a_2^2} = \sqrt{(4)^2 + (3)^2} = \sqrt{25} = 5$

5 (a) Phase difference of 2π means constructive interference. So, resultant amplitude will be maximum.

6 (b) Maximum amplitude, $A_{\max} = A_1 + A_2$

Minimum amplitude, $A_{\min} = A_1 - A_2$

Difference, $A_{\max} - A_{\min} = A_1 + A_2 - (A_1 - A_2) = 2A_2$

7 (b) $I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$ and $I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$

$\therefore$ Sum of maximum and minimum intensities $= 2(I_1 + I_2)$

8 (b) $\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \frac{(5 + 10)^2}{(5 - 10)^2} = \frac{9}{1}$

9 (d) Beats are formed due to superposition of two waves of nearly equal frequencies.

11 (c) If two waves of nearly equal frequencies superpose, they give $I_{\min} = 0$, if they both travel in straight line, i.e. $\phi = 0$ and they have equal amplitudes.

12 (d) Number of beats $= f_2 - f_1 = \frac{396}{0.9} - \frac{396}{1.0} = 44 \text{ Hz}$

13 (b) Beat frequency $= f_2 - f_1 = \frac{\omega_2 - \omega_1}{2\pi} = \frac{2008\pi - 2000\pi}{2\pi} = 4 \text{ Hz}$

14 (c) Given, $f_1 - f_2 = \frac{10}{3} \Rightarrow \frac{v}{1} - \frac{v}{1.01} = \frac{10}{3}$

$$\Rightarrow v = \frac{10 \times 1.01}{3 \times 0.01} = 337 \text{ ms}^{-1}$$

- 15** (c) With frequency 514 Hz, two beats are heard. Therefore, frequency of tuning fork can be either 512 Hz or 516 Hz. With 510 Hz, it gives beats of frequency 6 Hz. Hence, frequency of tuning fork is 516 Hz.

- 16** (c) Let f be the frequency of fork C , then

$$f_A = f + \frac{3f}{100} = \frac{103f}{100} \text{ and } f_B = f - \frac{2f}{100} = \frac{98f}{100}$$

$$\text{But } f_A - f_B = 5 \Rightarrow \frac{103f}{100} - \frac{98f}{100} = 5 \Rightarrow 5f = 500$$

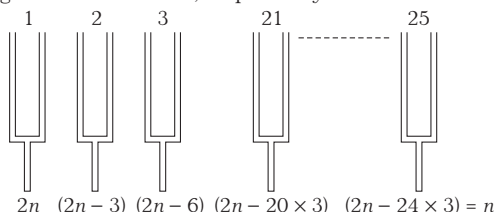
$$\Rightarrow f = 100 \text{ Hz}$$

$$\therefore f_A = \frac{(103)(100)}{100} = 103 \text{ Hz}$$

- 17** (a) Unknown frequency may be 102 Hz or 98 Hz. On loading this tuning fork, its frequency will decrease. Suppose they become 101 Hz or 97 Hz. With 100 Hz, beat frequency will be 1 Hz or 3 Hz. Given frequency is 1 Hz. Hence, unknown frequency should be 102 Hz.

- 18** (d) Frequency of unknown tuning fork can be 258 Hz or 254 Hz. On loading the given tuning fork, its frequency will decrease. Let it becomes 255 Hz. With 254 Hz, beat frequency is 1 Hz which is given. Hence, frequency of unknown tuning fork is 254 Hz.

- 19** (c) According to the question, frequencies of first and last tuning forks are $2n$ and n , respectively.



Hence, frequency in given arrangement are as follows

$$\Rightarrow 2n - 24 \times 3 = n \Rightarrow n = 72 \text{ Hz}$$

So, frequency of 21st tuning fork,

$$n_{21} = (2 \times 72 - 20 \times 3) = 84 \text{ Hz}$$

• CHECK POINT 17.4

- (b) Echo is based on the phenomenon of reflection of sound waves.
- (a) When two waves of same frequency and amplitude moving along the same path but in opposite directions superimpose, we get stationary waves.
- (b) In a stationary wave, all the particles in one particular segment (i.e. between two nodes) vibrate in the same phase.
- (d) Stationary waves do not transmit energy in the medium.
- (c) In stationary waves, all particles of the medium pass through their mean position simultaneously twice in each time period. These wave do not transmit energy in the medium.
- (a) If $y_{\text{incident}} = a \sin(\omega t - kx)$ and $y_{\text{stationary}} = a \sin(\omega t) \cos kx$
Frequencies of both incident and stationary waves are same, i.e. ω .
- (b) The given wave is travelling in positive x -direction. The other wave should travel in negative x -direction. Further, at

$x = 0$, net displacement by both the waves should always be zero for any value of t , as it is node. Only option (b) satisfies both these conditions.

- 8** (d) Velocity of wave, $v = \frac{\omega}{k} = \frac{100}{0.01} = 10^4 \text{ ms}^{-1}$

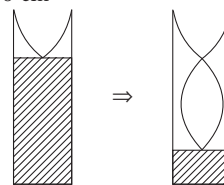
- 9** (c) Standing wave will be formed by the superposition of two waves.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{6} = \frac{\pi}{3} \text{ cm}^{-1} \Rightarrow \omega = vk = 0.5 \times \frac{\pi}{3} = \frac{\pi}{6} \text{ s}^{-1}$$

$$A_{\text{max}} = 2A = 6 \text{ cm}$$

- 10** (b) In stationary waves, all particles except nodes oscillate with same frequency but amplitude is zero at nodes and maximum at antinodes.

- 11** (b) Here, $\frac{\lambda}{4} = 20 \text{ cm}$



Next resonance will be obtained at $\frac{3\lambda}{4}$ or 60 cm.

So, resonance will be obtained when the water level is at a distance of 60 cm from the open end.

- 12** (d) Fundamental frequency of closed pipe, $f_1 = \frac{v}{4l} = 80 \text{ Hz}$.

Next is $f_2 = 3f_1 = 240 \text{ Hz} \rightarrow$ first overtone

and $f_3 = 5f_1 = 400 \text{ Hz} \rightarrow$ second overtone.

- 13** (c) Here, $f_1 : f_2 : f_3 = 1 : 3 : 5$. Odd harmonics are obtained in case of a pipe closed at one end.

- 14** (a) We have, $\lambda = \frac{v}{f} = 1 \text{ m}$

$$\frac{\lambda}{4} = 0.25 \text{ m} = 25 \text{ cm} \Rightarrow \frac{3\lambda}{4} = 0.75 \text{ m} = 75 \text{ cm}$$

$$\frac{5\lambda}{4} = 1.25 \text{ m} = 125 \text{ cm}$$

These are the lengths of air column. Hence, lengths of water column will be 75 cm and 25 cm.

- 15** (a) Ratio of three successive resonance frequencies,

$$f_1 : f_2 : f_3 = 425 : 595 : 765 = 5 : 7 : 9$$

Since, these are odd harmonics, so pipe will be closed.

$$\text{Further, } 5 \left(\frac{v}{4l} \right) = 425$$

$$\therefore \text{The length of closed pipe, } l = \frac{425 \times 4}{5v} = \frac{425 \times 4}{5 \times 340} = 1 \text{ m}$$

- 16** (c) Here, $n_1 - n_2 = 10$... (i)

$$\text{Using } n_1 = \frac{v}{4l_1} \text{ and } n_2 = \frac{v}{4l_2}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{l_2}{l_1} = \frac{26}{25} \text{ ... (ii)}$$

On solving Eqs. (i) and (ii), we get

$$n_1 = 260 \text{ Hz, } n_2 = 250 \text{ Hz}$$

- 17** (c) If it is closed pipe, first overtone will be 300 Hz and if it is open pipe, first overtone will be 200 Hz.

- 18** (b) We have, $f - \frac{f}{2} = 2 \Rightarrow f = 4 \text{ Hz}$

Here, $f = \frac{v}{2l}$ = fundamental frequency of open pipe

and $\frac{f}{2} = \frac{v}{4l}$ = fundamental frequency of closed pipe

Now, new beat frequency

$$= 2f - \frac{f}{2} = \frac{7f}{4} = \frac{7}{4} \times 4 = 7 \text{ Hz}$$

- 19** (a) For end correction x ,

$$\frac{l_2 + x}{l_1 + x} = \frac{3\lambda/4}{\lambda/4} = 3$$

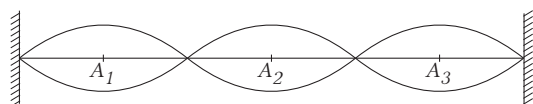
$$\Rightarrow x = \frac{l_2 - 3l_1}{2} = \frac{70.2 - 3 \times 22.7}{2} = 1.05 \text{ cm}$$

- 20** (b) Let x be the end correction, then according to question,

$$\frac{v}{4(l_1 + x)} = \frac{3v}{4(l_2 + x)}$$

$$\Rightarrow x = 2.5 \text{ cm} = 0.025 \text{ m}$$

- 21** (c) Second overtone means three loops.



$$\therefore 3 \frac{\lambda}{2} = L \text{ or } \lambda = \frac{2L}{3}$$

Antinodes will be obtained at $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$ or $\frac{L}{6}, \frac{3L}{6}$ and $\frac{5L}{6}$

- 22** (d) We have, $n \propto \frac{1}{l} \Rightarrow \frac{l_2}{l_1} = \frac{n_1}{n_2}$

$$\therefore \text{Fundamental frequency, } n_2 = \frac{l_1}{l_2} n_1 = \frac{1 \times 256}{1/4} = 1024 \text{ Hz}$$

- 23** (d) Frequency, $n = \frac{1}{2l} \sqrt{\frac{T}{\mu}} \propto \frac{\sqrt{T}}{l}$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{n_2}{n_1}\right)^2 \left(\frac{l_2}{l_1}\right)^2 = (2)^2 \left(\frac{3}{4}\right)^2 = \frac{9}{4}$$

- 24** (b) As, $n \propto \sqrt{T}$

$$\therefore T \propto n^2 \text{ and } T' \propto (4n)^2$$

Tension in the string, $T' = 16T$

- 25** (a) Fundamental frequency,

$$n = \frac{v}{2l} = \frac{\sqrt{T/\mu}}{2l} = \frac{\sqrt{T/\rho s}}{2l} = \frac{\sqrt{T/\rho \pi r^2}}{2l}$$

$$\therefore n \propto \frac{\sqrt{T}}{2l} \times \frac{1}{\sqrt{\rho \pi r^2}}$$

Given that tension, diameter and length all are made 3 times.

$\therefore n$ will become $\frac{1}{3\sqrt{3}}$ times.

- 26** (c) As, frequency, $n = \frac{p}{2l} \sqrt{\frac{T}{\pi r^2 \rho}} \Rightarrow n \propto \frac{1}{\sqrt{\rho}}$

i.e. Graph between n and $\sqrt{\rho}$ will be hyperbola.

- 27** (a) Node should be at $x = 0$.

This is possible when $A_x = A \sin kx \sin \omega t$.

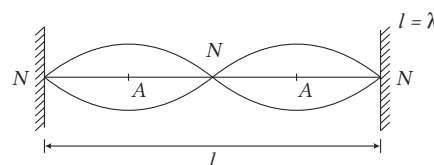
- 28** (a) $A_x = 4 \sin\left(\frac{\pi x}{15}\right)$

At $x = 5 \text{ cm}$,

$$A_x = 4 \sin \frac{\pi}{3} = 2\sqrt{3} \text{ cm}$$

- 29** (c) The state of resonance arises in case of forced vibrations.

- 30** (c) The following figure shows string vibrating in two segments.



The wavelength of the corresponding wave is l .

- 31** (b) Mass per unit length,

$$\mu = 0.1 \text{ g cm}^{-1} = \frac{0.1 \times 10^{-3}}{10^{-2}} = 0.01 \text{ kg m}^{-1}$$

$$\text{Velocity, } v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{400}{0.01}} = 200 \text{ ms}^{-1}$$

$$\text{Fundamental frequency of this string} = \frac{v}{2l} = \frac{200}{2 \times 1} = 100 \text{ Hz}$$

• CHECK POINT 17.5

- 1** (d) The Doppler's effect is applicable for both light and sound waves.

- 2** (c) When source and observer are moving relative to each other, the frequency observed by the receiver is different from the actual source frequency. This effect is called the Doppler's effect.

- 3** (b) The observer is moving towards the stationary source of sound, then apparent frequency will be greater than the real frequency.

- 4** (b) When passenger is sitting in moving train A,

$$5.5 = 5 \left(\frac{v + v_A}{v} \right)$$

$$\text{or } \frac{v_A}{v} = \frac{1}{10} \quad \dots(i)$$

When passenger is sitting in moving train B,

$$6 = 5 \left(\frac{v + v_B}{v} \right) \text{ or } \frac{v_B}{v} = \frac{1}{5} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{v_B}{v_A} = 2$$

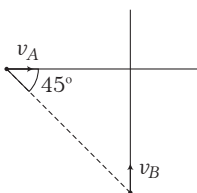
- 21 (b) Velocity, $v_A = 72 \text{ km h}^{-1} = 20 \text{ ms}^{-1}$

Velocity, $v_B = 36 \text{ km h}^{-1} = 10 \text{ ms}^{-1}$

Frequency of horn heard by the driver,

$$f' = f \left(\frac{v + v_B \cos 45^\circ}{v - v_A \cos 45^\circ} \right)$$

$$= 280 \left(\frac{340 + 10/\sqrt{2}}{340 - 20/\sqrt{2}} \right) = 298 \text{ Hz}$$



- 22 (c) For source, $v_s = r \omega = 0.70 \times 2\pi \times 5 = 22 \text{ ms}^{-1}$

Minimum frequency is heard when the source is receding the

man. It is given by $f_{\min} = f \left(\frac{v}{v + v_s} \right) = 1000 \times \left(\frac{352}{352 + 22} \right)$

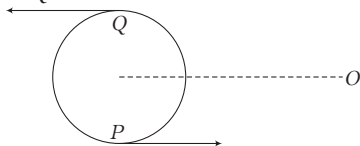
$$= 941 \text{ Hz}$$

- 23 (b) At point A, source is moving away from observer, so apparent frequency $n_1 < n$ (actual frequency). At point B, source is coming towards observer, so apparent frequency $n_2 > n$ and point C, source is moving perpendicular to observer, so $n_3 = n$.

Hence, $n_2 > n_3 > n_1$

- 24 (d) Velocity of source (or whistle), $v_s = R \omega = 30 \text{ ms}^{-1}$

Maximum frequency will be heard when whistle is at P and minimum at Q.



At P: $f_{\max} = f \left(\frac{v}{v - v_s} \right) = 440 \left(\frac{330}{330 - 30} \right) = 484 \text{ Hz}$ and

At Q: $f_{\min} = f \left(\frac{v}{v + v_s} \right) = 440 \left(\frac{330}{330 + 30} \right) = 403.3 \text{ Hz}$

- 25 (b) When wind blows at a speed v_w from the source to the observer, take $v \rightarrow v + v_w$ in equation

$$f' = \left(\frac{v + v_o}{v - v_s} \right) f_o$$

$$f' = \left(\frac{v + w + u}{v + w - u} \right) f_o \quad \left[\begin{array}{l} \because v_o = v_s = u \\ \text{and } v_w = w \end{array} \right]$$

$$\therefore \frac{f}{f_o} = \frac{v + w + u}{v + w - u}$$

(A) Taking it together

- 1 (a) Velocity of sound is independent of frequency. Therefore, it is same for frequencies n and $4n$, i.e. v .

- 2 (b) Phase difference,

$$\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x = k \cdot \Delta x = \pi \times 0.01 \times 25 = \frac{\pi}{4}$$

- 3 (c) Speed of sound, $v \propto \sqrt{T}$

$$\therefore \frac{v}{2v} = \frac{\sqrt{273 + 27}}{\sqrt{T}} \text{ or } T = 1200 \text{ K} = 927^\circ \text{ C}$$

- 4 (b) Propagation of longitudinal waves through a medium leads to transmission of energy and momentum through the medium without transmission of matter.

- 5 (c) $(v_p)_{\max} / v = (\omega A) / (\omega / k) = kA$

- 6 (a) Velocity of sound in gases, $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \frac{1}{\sqrt{M}}$. Since, M is minimum for H_2 , so sound velocity is maximum in H_2 .

- 7 (a) Due to rise in temperature, the speed of sound increases. Since $f \propto v$, hence frequency increases.

- 8 (a) Here, $\frac{T}{4} = 0.17 \text{ s}$

$$\therefore f = \frac{1}{T} = \frac{1}{4 \times 0.17} \text{ Hz}$$

The frequency of the wave, $f = 1.47 \text{ Hz}$

- 9 (d) As here,

- (i) Due to compressions and rarefactions, density of the medium (air) changes. At compressed regions, density is maximum and at rarefactions, density is minimum.
- (ii) As density is changing, so Boyle's law is not obeyed.
- (iii) Bulk modulus remains same.
- (iv) The time of compression and rarefaction is too small, i.e. we can assume adiabatic process and hence, no transfer of heat.

- 10 (a) Since, there is no relative motion between the source and listener, so apparent frequency is equal to original frequency.

- 11 (a) As, $v \propto \sqrt{T}$, so $T \propto x$

$$\therefore \frac{v'}{v} = \sqrt{\frac{1.5x}{x}}$$

$$\Rightarrow v' = \sqrt{1.5} \cdot v = 1.22v$$

- 12 (d) Intensity, $I = \frac{1}{2} \rho \omega^2 A^2 v$ or $I \propto \omega^2$ or $I \propto f^2$

$$\therefore \frac{I_2}{I_1} = \left(\frac{f_2}{f_1} \right)^2 = \left(\frac{1200}{400} \right)^2 = 9$$

- 13 (d) If d is the distance between man and reflecting surface of sound, then for hearing echo,

$$2d = v \times t \Rightarrow d = \frac{340 \times 1}{2} = 170 \text{ m}$$

- 14 (d) Phase difference,

$$\Delta\phi = \left(\frac{2\pi}{T} \right) (\Delta t) = \left(\frac{2\pi}{1/400} \right) (10^{-3}) = 0.8\pi = 144^\circ$$

15. (a) Phase difference, $\Delta\phi = k \cdot \Delta x$

$$\Rightarrow \Delta\phi = \frac{2\pi}{\lambda} \times \Delta x = \pi \times (3.5 - 3) = \frac{\pi}{2} \quad \left(\because k = \frac{2\pi}{\lambda} = \pi \right)$$

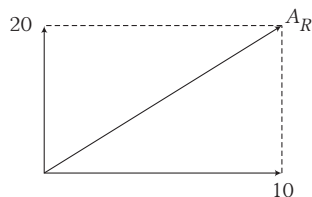
- 16 (c) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow \frac{0.01}{100} = \frac{v}{3 \times 10^8}$

$$\Rightarrow v = 3 \times 10^4 \text{ m/s} = 30 \text{ km/s}$$

- 17 (b) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow \Delta\lambda = \frac{v}{c} \lambda = \frac{100 \times 10^3}{3 \times 10^8} \times 5700 = 1.90 \text{ \AA}$

- 18 (b) Both waves are 90° out of phase.

Hence, $A_R = \sqrt{(10)^2 + (20)^2} = 10\sqrt{5}$



19 (b) $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\rho s}}$ or $v \propto \frac{1}{\sqrt{s}}$

$\therefore \frac{v_1}{v_2} = \sqrt{\frac{s_2}{s_1}} = \sqrt{\frac{4}{1}} \Rightarrow \frac{v_1}{v_2} = \frac{2}{1}$

- 20 (b) Both are diatomic. Hence, $\gamma = \frac{C_p}{C_v}$ for both is same.

$v = \sqrt{\frac{\gamma RT}{M}}$

$\therefore \frac{v_H}{v_O} = \sqrt{\frac{M_O}{M_H}} = \sqrt{\frac{32}{2}} = 4$

- 21 (b) Velocity, $v = \frac{\text{Coefficient of } t}{\text{Coefficient of } x} = \frac{1/2}{1/4} = 2 \text{ ms}^{-1}$

The distance through which the wave travels in 8 s,

$d = vt = 2 \times 8 = 16 \text{ m}$

- 22 (a) 7th harmonic means 7 loops.

In one loop, there are 2 nodes and 1 antinode. In 7 loops, there will be 8 nodes and 7 antinodes.

- 23 (d) Minimum time interval between two instants when the string is flat = $\frac{T}{2} = 0.5 \text{ s} \Rightarrow T = 1 \text{ s}$

Hence, $\lambda = v \times T = 10 \times 1 = 10 \text{ m}$

- 24 (b) Here, $\omega = 100\pi$, $k = 0.4\pi$

Wave velocity, $v = \frac{\omega}{k} = \frac{100\pi}{0.4\pi} = \frac{1000}{4} = 250 \text{ ms}^{-1}$

- 25 (d) First overtone of closed (P_1) = Third overtone of open (P_2)

$\therefore 3\left(\frac{v}{4l_1}\right) = 4\left(\frac{v}{2l_2}\right) \Rightarrow \frac{l_1}{l_2} = \frac{3}{8}$

- 26 (d) Fundamental frequency = $\frac{v}{2l} = \frac{\sqrt{T/\mu}}{2l} \Rightarrow f \propto \sqrt{T}$

So, the ratio of fundamental frequencies = 1 : 2 : 3 : 4.

- 27 (a) Fundamental frequency of closed pipe is $\frac{v}{4l}$ and that of

open pipe is $\frac{v}{2l}$ which is two times that of the closed pipe.

i.e. 1024 Hz.

28 (a) $v = \frac{\omega}{k} = \sqrt{\frac{T}{\mu}}$

$\therefore T = \mu \left(\frac{\omega}{k}\right)^2 = 10^{-4} \left(\frac{30}{1}\right)^2 = 0.09 \text{ N}$

- 29 (b) Given, $\frac{v_A}{2l} = 2\left(\frac{v_B}{2l}\right)$ or $v_A = 2v_B$ but $v \propto \sqrt{T}$

$\therefore T_A = 4T_B$

- 30 (a) At 10 cm distance from open end or at distance $\frac{\lambda}{4}$ from the open end, there should be a node.

- 31 (c) Here, $\frac{\lambda}{2} = (50 - 30) \text{ cm} = 20 \text{ cm}$

Next resonance will be obtained at $50 + \frac{\lambda}{2} = 70 \text{ cm}$

- 32 (c) All functions of x and t of type $(ax \pm bt)$ represent a wave.

So, $y = A \sin(k^2x^2 - \omega^2t^2)$ does not represent wave motion.

- 33 (b) $100 \xrightarrow{\text{First slab}} 80 \xrightarrow{\text{Second slab}} 80\% \text{ of } 80 = 64$

$\therefore$ Reduction in intensity on passing through two slabs = 36%

- 34 (b) At $t = 0$ and $x = \frac{\pi}{2k}$,

displacement,

$y = a_0 \sin\left(-k \times \frac{\pi}{2x}\right)$

$= -a_0 \sin \frac{\pi}{2} = -a_0$

Point of maximum displacement (a_0) in negative direction is Q.

- 35 (a) Phase difference, $\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x$

or $\lambda = \frac{2\pi}{\Delta\phi} \cdot \Delta x = \frac{2\pi}{(\pi/3)} \times 1.25 \times 10^{-2} \text{ m} = 7.5 \times 10^{-2} \text{ m}$

Velocity, $v = f \lambda = 1000 \times 7.5 \times 10^{-2} = 75 \text{ ms}^{-1}$

- 36 (b) Amplitude of reflected wave,

$A_r = \frac{2}{3} \times A_i = \frac{2}{3} \times 0.6 = 0.4 \text{ units}$

Given equation of incident wave,

$y_i = 0.6 \sin 2\pi\left(t - \frac{x}{2}\right)$

Equation of reflected wave, $y_r = A_r \sin 2\pi\left(t + \frac{x}{2} + \pi\right)$

($\therefore$ At denser medium, phase changes by π)

The positive sign is due to reversal of direction of propagation.

So, $y_r = -0.4 \sin 2\pi\left(t + \frac{x}{2}\right)$ [$\because \sin(\pi + \theta) = -\sin\theta$]

- 37 (a) Given, $k = \frac{2\pi}{\lambda} = \frac{\pi}{20} \Rightarrow \lambda = 40 \text{ cm}$

Separation between two consecutive nodes

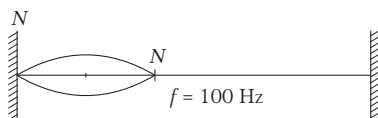
$= \frac{\lambda}{2} = \frac{40}{2} = 20 \text{ cm}$

- 38 (a) Node will occur, when

$kx = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \text{ etc.} \quad \left(\because k = \frac{\pi}{25}\right)$

$\Rightarrow x = 12.5 \text{ cm}, 37.5 \text{ cm}, 62.5 \text{ cm}, \text{ etc.}$

- 39 (b) Here, $\frac{\lambda}{2} = 10 \text{ cm} = 0.1 \text{ m} \Rightarrow \lambda = 0.2 \text{ m}$

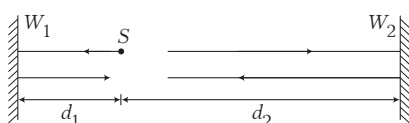


Now, speed of incident wave, $v = f \lambda = 20 \text{ ms}^{-1}$

- 40 (d) Frequency, $n \propto \frac{\sqrt{T}}{l}$ or $n \propto \frac{\sqrt{M}}{l}$

$$\therefore \frac{n_A}{n_B} = \frac{l_B}{l_A} \sqrt{\frac{M_A}{M_B}}$$

- 41 (b) According to question, $2d_1 = 340 \times t_1$



$$\therefore d_1 = 340 \text{ m} \quad (t_1 = 2 \text{ s})$$

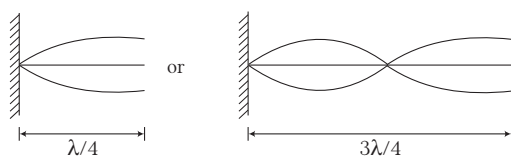
$$2d_2 = 340 \times t_2$$

$$\therefore d_2 = 680 \text{ m} \quad (t_2 = t_1 + 2 = 4 \text{ s})$$

$$\therefore \text{Distance between walls} = d_1 + d_2 = 1020 \text{ m}$$

Next echo will be heard at 6 s not at 8 s. Because sound wave reflected from wall W_2 will be reflected again by wall W_1 in next 2 s.

- 42 (a) The standing wave can be shown as



$\therefore$ The length of the string must be an odd multiple of $\frac{\lambda}{4}$.

- 43 (a) Required distance = $\frac{\lambda}{4} = \frac{v/n}{4} = \frac{1200}{4 \times 300} = 1 \text{ m}$

- 44 (a) Path difference of $x_0 = \frac{\lambda}{2}$ corresponds to a phase difference of π . They will interfere destructively and resultant amplitude is $|a - b|$.

- 45 (d) $A_{\max} = a + b$ and $A_{\min} = a - b$

$$\therefore A_{\max} - A_{\min} = 2b$$

- 46 (b) Fundamental frequency, $f_1 = \frac{v}{4l} = \frac{336}{4 \times 1.05} = 80 \text{ Hz}$

Next higher overtone will be at frequency,

$$f_3 = 3f_1 = 240 \text{ Hz}$$

- 47 (d) The distance between adjacent nodes, $x = \frac{\lambda}{2} \Rightarrow \lambda = 2x$

$$\text{Also, } k = \frac{2\pi}{\lambda}$$

$$\text{Hence, } x = \frac{\pi}{k}$$

- 48 (c) If the speed of engine is v , the distance travelled by engine in 5 s will be $5v$ and hence, the distance travelled by sound in reaching the hill and coming back to the moving driver = $900 + (900 - 5v) = 1800 - 5v$. So, the time interval between original sound and its echo,

$$t = \frac{(1800 - 5v)}{330} = 5 \Rightarrow v = 30 \text{ m/s}$$

- 49 (a) Let the tube will oscillate in n th harmonic. Then,

$$n \left(\frac{v}{2l} \right) = 1.1 \times 10^3$$

$$\therefore n = \frac{2.2 \times 10^3 \times l}{v} = \frac{2.2 \times 10^3 \times 0.3}{330} = 2$$

- 50 (a) Wave, $z_1 = A \sin(kx - \omega t)$ is travelling towards positive x -direction.

Wave, $z_2 = A \sin(kx + \omega t)$ is travelling towards negative x -direction.

Wave, $z_3 = A \sin(ky - \omega t)$ is travelling towards positive y -direction.

Since, waves z_1 and z_2 are travelling along the same line in opposite directions, so they will produce a standing wave.

- 51 (a) Five segments means five loops. Length of one loop is $\frac{\lambda}{2}$.

$$\text{Hence, } 5 \frac{\lambda}{2} = 10 \text{ m}$$

$$\therefore \lambda = 4 \text{ m}$$

$$\text{Now, frequency, } f = \frac{v}{\lambda} = 5 \text{ Hz}$$

- 52 (d) $f = \frac{v}{2l} = \frac{\sqrt{T/\mu}}{2l} = \frac{\sqrt{T/\rho s}}{2l} = \frac{\sqrt{T/\pi r^2 \rho}}{2l}$ ($\because s = \pi r^2$)

$$\therefore f \propto \frac{\sqrt{T}}{rl}$$

Now, tension, length and radius all are doubled.

$$\text{Hence, } f' = \frac{f}{2\sqrt{2}}$$

- 53 (a) $f' = f \left(\frac{v}{v - v_s} \right) \Rightarrow \frac{f'}{f} = \frac{v}{v - v_s}$

$$\Rightarrow \frac{v}{v - v_s} = 3 \Rightarrow v_s = \frac{2v}{3}$$

- 54 (a) Tension at distance x from free end,

$$T_x = w_x = m_x g = (\mu x g)$$

$$\sqrt{\frac{T_x}{\mu}} = \sqrt{x g} \quad \text{or} \quad v = \sqrt{x g}$$

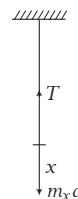
$$\therefore v \propto \sqrt{x}$$

- 55 (b) Beat frequency, $f_b = f_2 - f_1 = \frac{\omega_2 - \omega_1}{2\pi}$

$$= \frac{404\pi - 400\pi}{2\pi} = 2 \text{ Hz}$$

Further, $A_1 = 4$ and $A_2 = 3$.

$$\text{Hence, } \frac{I_{\max}}{I_{\min}} = \left(\frac{A_1 + A_2}{A_1 - A_2} \right)^2 = 49$$



- 56** (c) Frequency of first overtone of closed pipe = Frequency of first overtone of open pipe

$$\Rightarrow \frac{3v_1}{4L_1} = \frac{v_2}{L_2}$$

$$\Rightarrow \frac{3}{4L_1} \sqrt{\frac{B}{\rho_1}} = \frac{1}{L_2} \sqrt{\frac{B}{\rho_2}}$$

$$\Rightarrow \text{Length of the open pipe, } L_2 = \frac{4L_1}{3} \sqrt{\frac{\rho_1}{\rho_2}} = \frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$$

- 57** (c) Speed of the wave, $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{80}{5 \times 10^{-2}}} = 40 \text{ ms}^{-1}$

$$\begin{aligned} \therefore \text{Power, } P &= \frac{1}{2} \rho \omega^2 A^2 s v = \frac{1}{2} \mu (2\pi f)^2 v A^2 \quad (\text{As, } \rho s = \mu) \\ &= \frac{1}{2} \times 5 \times 10^{-2} (2\pi \times 60)^2 \times 40 \times (0.06)^2 \\ &= 511 \text{ W} \end{aligned}$$

- 58** (d) Fundamental frequency, $f = \frac{v}{2l} = \frac{\sqrt{T/\mu}}{2l}$

where, μ = mass per unit length = ρs

$$\text{or } f = \frac{1}{2l} \sqrt{\frac{T}{\rho s}} \quad \text{but } s = \pi r^2$$

$$\therefore f \propto \frac{1}{rl} \quad \text{or } \frac{f_1}{f_2} = \frac{r_2 l_2}{r_1 l_1} = \frac{r \times 2l}{2r \times l} = 1$$

- 59** (b) Initially, number of beats per second = 5

$\therefore$ Frequency of pipe = $200 \pm 5 = 195 \text{ Hz}$ or 205 Hz

Frequency of second harmonic of the pipe = $2n$ and number of beats in this case = 10

$$\therefore f_1 = 2 \times 195 = 390 \text{ Hz and } f_2 = 2 \times 205 = 410 \text{ Hz}$$

$\therefore$ For 10 beats with 420 Hz, it must be $f_2 = 410$

$$\therefore f = 205 \text{ Hz}$$

- 60** (a) Maximum velocity,

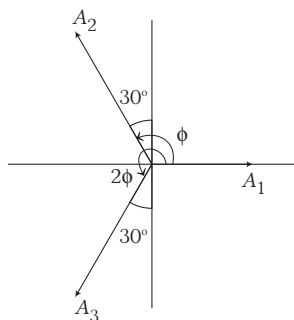
$$v_{\max} = A\omega = \frac{v}{10} = \frac{10}{10} = 1 \text{ ms}^{-1}$$

$$\Rightarrow A\omega = A \times 2\pi n = 1 \Rightarrow n = \frac{10^3}{2\pi} \quad (\because A = 10^{-3} \text{ m})$$

$$\text{Since, } v = n\lambda \Rightarrow \lambda = \frac{v}{n} = \frac{10}{10^3 / 2\pi} = 2\pi \times 10^{-2} \text{ m}$$

- 61** (a) Phase, $\phi = \frac{2\pi}{3} = 120^\circ$

$$\Rightarrow 2\phi = 240^\circ \quad (\because y_1 + y_2 + y_3 = 0)$$



The amplitudes of all the three are equal to A acting at angles as shown in figure. We can see that resultant of these three come out to be zero.

- 62** (c) Frequency, $f = \frac{v}{2l}$ and $f' = \frac{v}{4(l/2)} = \frac{v}{2l} = f$

when half of the length is dipped in water, it will become closed pipe.

- 63** (b) $L_1 = \frac{\lambda}{4}$, $L_2 = \frac{3\lambda}{4}$ and $L_3 = \frac{5\lambda}{4}$ $\left(\because L = (2n-1)\frac{\lambda}{4} \right)$

- 64** (c) For open pipe, $f_1 = 2 \left(\frac{v}{2l} \right) = \frac{v}{l}$

For closed pipe, $f_2 = n \left(\frac{v}{4l} \right)$, $n = 1, 3, 5, 7, \dots$

$$f_2 > f_1$$

$$\text{So, } f_2 = \frac{n}{4} \cdot f_1 \text{ and } n > 4$$

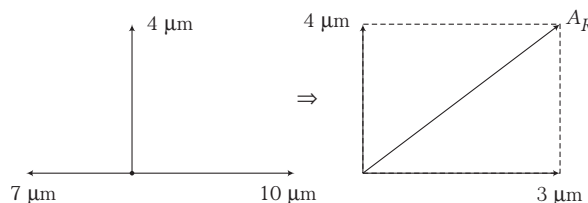
- 65** (d) According to the question, $\sqrt{\frac{E}{\rho}} = 10 \sqrt{\frac{T}{\mu}}$

$$\Rightarrow 100 \left(\frac{T}{\mu} \right) = \frac{E}{\rho} \Rightarrow \frac{T}{s} = \frac{E\mu}{100\rho s}$$

Since, $\rho s = \mu$

$$\therefore \text{Stress} = \frac{T}{s} = \frac{E}{100}$$

- 66** (c) Using vector algebra,



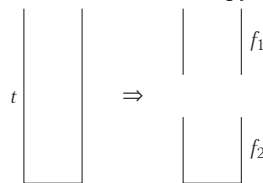
$$\text{Resulting wave, } A_R = \sqrt{(4)^2 + (3)^2} = 5 \mu\text{m}$$

- 67** (c) Given, $3 \left(\frac{v}{4l_c} \right) = 2 \left(\frac{v}{2l_o} \right)$ or $\frac{l_c}{l_o} = \frac{3}{4}$

$$\text{Now, } n \left(\frac{v}{4l_c} \right) = m \left(\frac{v}{2l_o} \right) \text{ or } \frac{n}{m} = 2 \frac{l_c}{l_o} = \frac{3}{2} = \frac{9}{6}$$

Thus, ratio n/m should be $3/2$ but n is only odd, while m may be even or odd.

- 68** (a) Fundamental frequency, $f = \frac{v}{4l} = 412 \text{ Hz}$



After cutting, frequencies of fundamental notes produced,

$$f_1 = \frac{v}{2(l/2)} = \frac{v}{l} = 4 \times 412 = 1648 \text{ Hz}$$

$$\text{and } f_2 = \frac{v}{4(l/2)} = \frac{v}{2l} = 2 \times 412 = 824 \text{ Hz}$$

- 69 (d) Path difference, $\Delta x = 50 \text{ cm} = \frac{1}{2} \text{ m}$

$$\therefore \text{Phase difference, } \Delta\phi = \frac{2\pi}{\lambda} \times \Delta x \Rightarrow \phi = \frac{2\pi}{1} \times \frac{1}{2} = \pi$$

$$\text{Total phase difference} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\Rightarrow A = \sqrt{a^2 + a^2 + 2a^2 \cos\left(\frac{2\pi}{3}\right)} = a$$

- 70 (c) Sound level, $L = 10 \log_{10} \left(\frac{I}{I_0} \right)$, where $I_0 = 10^{-12} \text{ Wm}^{-2}$

$$\Rightarrow 40 = 10 \log_{10} \left(\frac{I_1}{I_0} \right) \Rightarrow \frac{I_1}{I_0} = 10^4 \quad \dots(i)$$

$$\text{Also, } 20 = 10 \log_{10} \left(\frac{I_2}{I_0} \right) \Rightarrow \frac{I_2}{I_0} = 10^2 \quad \dots(ii)$$

$$\Rightarrow \frac{I_2}{I_1} = 10^{-2} = \frac{r_1^2}{r_2^2}$$

$$\Rightarrow r_2^2 = 100 r_1^2 \Rightarrow r_2 = 10 \text{ m} \quad (\because r_1 = 1 \text{ m})$$

- 71 (c) Amplitude, $A = \frac{2}{2} = 1 \text{ mm}$

From the given equation, we can see that $k = 6.25 \text{ m}^{-1}$.

The smallest length of string is equivalent to a single loop or length of string equal to $\frac{\lambda}{2}$ or $\frac{\pi}{k}$.

$$\therefore L_{\min} = \frac{\pi}{k} = \frac{\pi}{6.25} = 0.5 \text{ m} = 50 \text{ cm}$$

- 72 (b) Difference in lengths will be equal to one loop or $\frac{\lambda}{2}$.

$$\therefore \lambda = 2(48.6 - 15.4) \text{ cm}$$

$$\text{Now, } v = f \lambda = 512 \times 2(48.6 - 15.4)$$

$$= 1024(48.6 - 15.4) \text{ cm s}^{-1}$$

- 73 (d) Frequency of string may be 470 Hz or 490 Hz.

With increase in tension, frequency of string will increase. If initially, it is 470 Hz, beat frequency with given tuning fork will decrease. If it is 490 Hz, beat frequency will increase.

- 74 (a) For closed pipe, $l_1 = \frac{v}{4f}$, $l_2 = \frac{3v}{4f}$

$$\text{Now, } v = 2f(l_2 - l_1)$$

$$\Rightarrow f = \frac{v}{2(l_2 - l_1)} = \frac{330}{2(0.49 - 0.16)} = 500 \text{ Hz}$$

- 75 (d) Particle velocity (v_p) = $-v \times \text{Slope of the graph at that point}$.

At point 1 Slope of the curve is positive, hence particle velocity is negative or downward ($\downarrow$).

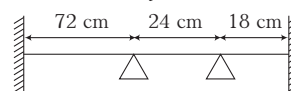
At point 2 Slope is negative, hence particle velocity is positive or upwards ($\uparrow$).

At point 3 Again, slope of the curve is positive, hence particle velocity is negative or downward ($\downarrow$).

- 76 (b) As, per question, $f_1 - f_2 = 10$

$$\therefore (680) \left(\frac{340 + u}{340} \right) - (680) \left(\frac{340 - u}{340} \right) = 10 \Rightarrow u = 2.5 \text{ ms}^{-1}$$

- 77 (a) Frequency of wire, $f \propto \frac{1}{l}$



$$\therefore l_1 : l_2 : l_3 = \frac{1}{f_1} : \frac{1}{f_2} : \frac{1}{f_3} = \frac{1}{1} : \frac{1}{3} : \frac{1}{4} \text{ or } l_1 : l_2 : l_3 = 12 : 4 : 3$$

$$\therefore l_1 = \frac{12}{19} \times 114 = 72 \text{ cm}$$

$$l_2 = \frac{4}{19} \times 114 = 24 \text{ cm and } l_3 = \frac{3}{19} \times 114 = 18 \text{ cm}$$

- 78 (d) Displacement, $y = \frac{1}{\sqrt{a}} \sin \omega t \pm \frac{1}{\sqrt{b}} \sin \left(\omega t + \frac{\pi}{2} \right)$

$$\text{Here, phase difference} = \frac{\pi}{2}$$

$\therefore$ The resultant amplitude

$$= \sqrt{\left(\frac{1}{\sqrt{a}} \right)^2 + \left(\frac{1}{\sqrt{b}} \right)^2} = \sqrt{\frac{1}{a} + \frac{1}{b}} = \sqrt{\frac{a+b}{ab}}$$

- 79 (c) $E = \text{Energy density} \times \text{Volume} = \frac{1}{2} \rho \omega^2 A^2 sL$

or $E \propto \omega^2$, rest all quantities are common for both the waves.

$$\omega_2 = 2\omega \text{ and } \omega_1 = \omega$$

$$\therefore E_2 = 4E_1$$

- 80 (c) In closed organ pipe, first resonance occurs at $\lambda/4$. So, in fundamental mode of vibration of organ pipe $\lambda/4 = (1 + 0.3d)$, where $0.3d$ is necessary end correction, then frequency of vibration, $n = \frac{v}{\lambda} = \frac{v}{4(l + 0.3d)}$.

As l is same, wide pipe A will resonate at a lower frequency.

So, $n_A < n_B$.

- 81 (a) At $x = 0$, $t = 0$, $y = \sqrt{3}$

Further, if x is slightly greater than 0, y is slightly greater than $\sqrt{3}$. Therefore, the correct graph is (a).

- 82 (c) $n_Q = 341 \pm 3 = 344 \text{ Hz or } 338 \text{ Hz}$

On waxing Q , the number of beats decreases, hence $n_Q = 344 \text{ Hz}$.

- 83 (b) For string, $\frac{\text{Mass}}{\text{Length}} = \mu = \frac{10^{-2}}{0.4} = 2.5 \times 10^{-2} \text{ kgm}^{-1}$

$$\therefore \text{Velocity, } v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{1.6}{2.5 \times 10^{-2}}} = 8 \text{ ms}^{-1}$$

For constructive interference between successive pulses,

$$\Delta t_{\min} = \frac{2l}{v} = \frac{2(0.4)}{8} = 0.10 \text{ s}$$

- 84 (a) In a wave equation, x and t must be related in the form $(x \pm vt)$.

$$\text{We rewrite the given equations } y = \frac{1}{1 + (x - vt)^2}$$

$$\text{For } t = 0, y = \frac{1}{(1 + x^2)} \text{ and}$$

$$\text{For } t = 2 \text{ s, } y = \frac{1}{[1 + (x - 2v)^2]} = \frac{1}{[1 + (x - 1)^2]}$$

$$\Rightarrow 2v = 1 \text{ or } v = 0.5 \text{ ms}^{-1}$$

$$85 \text{ (d)} \quad (\Delta p)_{\max} = B A k$$

$$\therefore A = \frac{(\Delta p)_{\max}}{B k} \quad \dots(i)$$

$$\text{Here, } v = \sqrt{\frac{B}{\rho}} = \frac{\omega}{k} \Rightarrow k = \omega \sqrt{\frac{\rho}{B}} = 2\pi f \sqrt{\frac{\rho}{B}}$$

$$\text{Further, } B = \rho v^2$$

Substituting in Eq. (i), we get

$$A = \frac{(\Delta p)_{\max}}{2\pi f \sqrt{B\rho}} = \frac{(\Delta p)_{\max}}{2\pi f \rho v} \quad (\because B = \rho v^2)$$

Substituting all the values, we get

$$A = \frac{30}{2\pi \times 10^3 \times 1.5 \times 300} = \frac{10^{-4}}{3\pi} \text{ m}$$

$$86 \text{ (c)} \text{ Distance between two successive antinodes is } \frac{\lambda}{2} \text{ or } \frac{\pi}{k}.$$

$$\therefore \text{ Volume between two antinodes will be } \frac{\pi}{k} \cdot s.$$

Let u_1 and u_2 be the energy densities due to two waves, then

$$E = (u_1 + u_2) \times \text{volume}$$

$$= \left[\frac{1}{2} \rho \omega^2 a^2 + \frac{1}{2} \rho \omega^2 (2a)^2 \right] \frac{\pi}{k} \cdot s = \frac{5}{2} \frac{\rho \omega^2 a^2 \pi s}{k}$$

$$87 \text{ (b)} \text{ Wavelength, } \lambda = \frac{v}{f} = \frac{330}{500} = 0.66 \text{ m}$$

Length of air column at resonance can be $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$, etc.

or 0.165 m, 0.495 m, 0.825 m, 1.155 m, etc.

But length of tube is only 1 m. Hence, first three resonances will be obtained.

$$88 \text{ (c)} \text{ Given, } f_1 - f_2 = 40$$

$$\text{or } f \left(\frac{340}{340 - v_s} \right) - f \left(\frac{340}{340 + v_s} \right) = 40$$

$$\text{or } 400 \left(\frac{340}{340 - v_s} \right) - 400 \left(\frac{340}{340 + v_s} \right) = 40$$

Applying Binomial theorem and then solving, we get

$$v_s \approx 17 \text{ ms}^{-1}$$

$$89 \text{ (b)} \text{ While approaching, } f > f_o \text{ but } f = \text{constant.}$$

While receding, $f < f_o$ but $f = \text{constant.}$

$$90 \text{ (b)} \text{ Mass, } m = 2.5 \text{ kg}$$

μ = mass per unit length

$$= \frac{m}{l} = \frac{2.5 \text{ kg}}{20} = \frac{1.25}{10} = 0.125 \text{ kg m}^{-1}$$

$$\text{Speed, } v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{200}{0.125}}$$

(Speed of transverse waves in any string)

$$l = v \times t \Rightarrow 20 = \sqrt{\frac{200}{0.125}} \times t$$

$\Rightarrow$ Required time,

$$t = 20 \times \sqrt{\frac{125}{2 \times 10^5}} = 20 \times \sqrt{\frac{25 \times 5}{2 \times 10^5}}$$

$$= 20 \times \sqrt{25 \times \frac{1}{0.4 \times 10^5}}$$

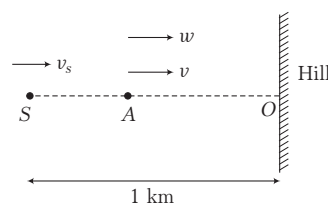
$$= 20 \times 5 \sqrt{\frac{1}{4 \times 10^4}} = \frac{20 \times 5}{2 \times 10^2} = \frac{1}{2} = 0.5 \text{ s}$$

$$91 \text{ (c)} \text{ Speed of sound in air, } v = 1200 \text{ km/h}$$

Speed of source, $v_s = 40 \text{ km/h}$

Speed of wind, $w = 40 \text{ km/h}$

Frequency emitted by source, $f = 580 \text{ Hz}$



Frequency received at hill,

$$f' = f \left(\frac{v + w - v_o}{v + w - v_s} \right) = 580 \left(\frac{1200 + 40 - 0}{1200 + 40 - 40} \right)$$

$$= 599.33 \text{ Hz} \approx 600 \text{ Hz}$$

$$92 \text{ (a)} \text{ The linear velocity of whistle,}$$

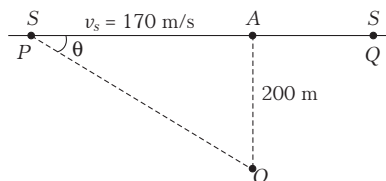
$$v_s = r\omega = 1.2 \times 2\pi \frac{400}{60} = 50 \text{ m/s}$$

When whistle approaches the listener, heard frequency will be maximum and when whistle recedes away, heard frequency will be minimum.

$$\text{So, } f_{\max} = f \left(\frac{v}{v - v_s} \right) = 500 \left(\frac{340}{290} \right) = 586.2 \text{ Hz} \approx 586 \text{ Hz}$$

$$\text{and } f_{\min} = f \left(\frac{v}{v + v_s} \right) = 500 \left(\frac{340}{390} \right) = 435.89 \text{ Hz} \approx 436 \text{ Hz}$$

$$93 \text{ (a)} \text{ When source is nearest to } O, \text{ i.e. at } A, \text{ detector will receive that frequency which was emitted by source sometime before, assume when source is at } P. \text{ Let sound takes time } t \text{ to reach from } P \text{ to } O.$$



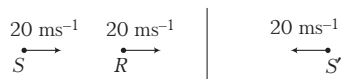
$$PO = vt = 340t, \quad PA = v_s t = 170t$$

$$\cos \theta = \frac{PA}{PO} = \frac{170t}{340t} = \frac{1}{2}$$

Frequency received at O,

$$f' = f \left(\frac{v}{v - v_s \cos \theta} \right) = 1200 \left(\frac{340}{340 - \frac{170}{2}} \right) = 1600 \text{ Hz}$$

- 94 (b) Let the reflecting surface be at rest.



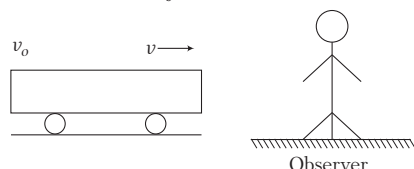
$$f' = 256 \left(\frac{330 + 20}{330 - 20} \right) \text{ Hz} = 289 \text{ Hz}$$

$$\text{and } \lambda' = \frac{v'}{f'} = \frac{(330 + 10)}{289} = \frac{340}{289} \text{ m}$$

- 95 (c) Let the original frequency of the source is n_o .

Let the speed of sound wave in the medium is v .

As observer is stationary,



When train is approaching,

$$\text{Apparent frequency, } n_a = \left(\frac{v}{v - v_s} \right) n_o \Rightarrow n_a > n_o$$

When the train is going away from the observer,

$$\text{Apparent frequency, } n_a = \left(\frac{v}{v + v_s} \right) n_o \Rightarrow n_a < n_o$$

Hence, the expected curve is (c).

- 96 (c) Frequency of vibration stretched string, $n = \frac{L}{2} \sqrt{\frac{T}{m}}$

When the stone is completely immersed in water, length changes but frequency does not change.

$$l \propto \sqrt{T} \Rightarrow \frac{L}{l} = \sqrt{\frac{V\rho g}{V(\rho - 1)g}}$$

$$\Rightarrow \frac{L}{l} = \sqrt{\frac{\rho}{\rho - 1}} \Rightarrow \rho = \frac{L^2}{L^2 - l^2}$$

- 97 (b) We have, $L_2 D = \sqrt{(40)^2 + (9)^2} = 41 \text{ m}$

Path difference, $\Delta x = L_2 D - L_1 D = 1 \text{ m}$

$$\text{For maximum, } \Delta x = 2n \frac{\lambda}{2}$$

$$\text{For } n = 1 \Rightarrow 2(1) \frac{\lambda}{2} = 1 \Rightarrow \lambda = 1 \Rightarrow f = \frac{v}{\lambda} = 330 \text{ Hz}$$

- 98 (c) Frequency, $f = f_o \left(\frac{v + v_o}{v} \right) = 10^3 \left(1 + \frac{v_o}{v} \right)$

$$= 10^3 + \frac{10^3}{v} (gt)$$

$$= 10^3 + \left(\frac{10^4}{v} \right) t$$

$$(\because v_o = gt)$$

Slope of f - t line should be equal to $\frac{10^4}{v}$.

$$\therefore \frac{1000}{30} = \frac{10^4}{v} \text{ or } v = 300 \text{ ms}^{-1}$$

- 99 (c) Frequency of vibration of string is given by

$$n = \frac{1}{2l} \sqrt{\frac{T}{m}} \Rightarrow P\sqrt{T} = \text{constant}$$

where, P is number of loops formed.

$$\frac{P_1}{P_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\text{Hence, } \frac{4}{6} = \sqrt{\frac{T_2}{(50 + 15)}} \Rightarrow T_2 = 28.8 \text{ gf}$$

Hence, weight to be removed from the pan

$$= T_1 - T_2 = 65 - 28.8 = 36.2 \text{ g force}$$

$$= 0.036 \text{ kg-wt}$$

- 100 (d) Frequency $\propto \sqrt{T}$

$$\frac{f}{f/2} = \sqrt{\frac{T}{T - F_w}} \quad (F_w = \text{upthrust in water})$$

$$\therefore 1 - \frac{F_w}{T} = \frac{1}{4} \text{ or } \frac{F_w}{T} = \frac{3}{4} \quad \dots(i)$$

$$\text{In second case, } \frac{f}{f/3} = \sqrt{\frac{T}{T - F_l}} \quad (F_l = \text{upthrust in liquid})$$

$$\Rightarrow 1 - \frac{F_l}{T} = \frac{1}{9}$$

$$\therefore \frac{F_l}{T} = \frac{8}{9} \quad \dots(ii)$$

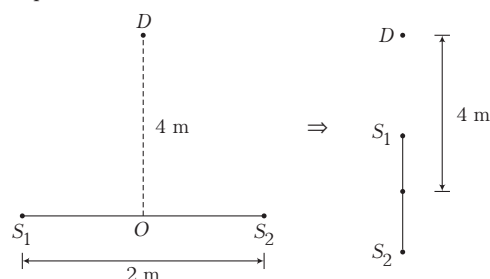
From Eqs. (i) and (ii), we get

$$\frac{F_l}{F_w} = \frac{8}{9} \times \frac{4}{3} = 1.18$$

Now, upthrust is in the ratio of relative density.

- 101 (b) In the starting, $\Delta x = 0$ and maximum intensity is observed.

Next maxima will correspond to $\Delta x = \lambda$, i.e. 5th maxima will correspond to



$$\Delta x = 4\lambda. \text{ But } \Delta x = S_1 S_2 = 2 \text{ m}$$

$$\therefore 4\lambda = 2 \text{ or } \lambda = 0.5 \text{ m}$$

[Here S_1 and S_2 are two speakers placed 2 m apart and D is microphone at 4 m from point O (mid-point of $S_1 S_2$)]

- 102 (b) We have, $f \propto \frac{\sqrt{T}}{l} \Rightarrow f$ remains same.

$$\text{Hence, } \frac{\sqrt{T_1}}{l_1} = \frac{\sqrt{T_2}}{l_2} \text{ or } \sqrt{\frac{T_1}{T_2}} = \frac{l_1}{l_2} = \frac{40}{30} = \frac{4}{3}$$

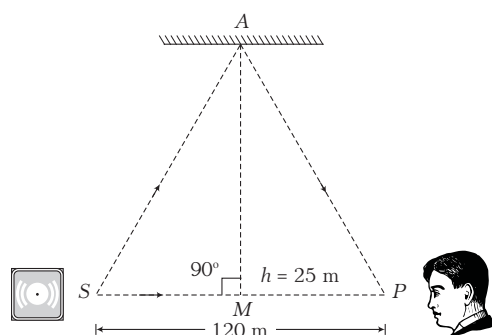
$$\text{or } \sqrt{\frac{w}{w - F}} = \frac{4}{3}$$

$$\therefore \sqrt{\frac{Vdg}{Vdg - V \times 1 \times g}} = \frac{4}{3} \quad \text{or} \quad \sqrt{\frac{d}{d-1}} = \frac{4}{3}$$

$$\therefore \frac{d}{d-1} = \frac{16}{9} \quad \text{or} \quad d = \frac{16}{7}$$

- 103** (a) Let S be the source of sound and P the person or listener. The waves from S reach point P directly following the path SMP and being reflected from the ceiling at point A following the path SAP . M is mid-point of SP (i.e. $SM = MP$) and $\angle SMA = 90^\circ$

Path difference between waves, $\Delta x = SAP - SMP$



$$\text{Now, } SAP = SA + AP = 2(SA)$$

$$= 2\sqrt{(SM)^2 + (MA)^2} = 2\sqrt{(60)^2 + (25)^2} = 130 \text{ m}$$

$$\therefore \text{Path difference, } \Delta x = SAP - SMP = 130 - 120 = 10 \text{ m}$$

Path difference due to reflection from ceiling = $\frac{\lambda}{2}$

$$\therefore \text{If } \Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \frac{5\lambda}{2}, \text{ etc., interference will be constructive}$$

$$\therefore \lambda = 20 \text{ m}, \frac{20}{3} \text{ m}, \frac{20}{5} \text{ m, etc.}$$

(B) Medical entrance special format questions

• Assertion and reason

- 1** (a) Frequency, $f = \frac{v}{\lambda}$

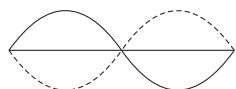
$$\text{Now, velocity, } v = \sqrt{\frac{\gamma RT}{M}}$$

$$\therefore f \propto \sqrt{T} \text{ as } v \propto \sqrt{T}$$

- 2** (c) When source moves towards observer, wavelength is modified as a result frequency changes. The effective wavelength is given by

$$\lambda' = \left(\frac{v - v_s}{f} \right)$$

- 3** (a) Stationary wave can be represented as



A node is a place of zero amplitude and an antinode is a place of maximum amplitude.

- 4** (b) Sound produced by an open organ pipe is richer because it contains all harmonics and frequency of fundamental note in an open organ pipe is twice the fundamental frequency in a closed pipe of same length.

- 5** (b) Velocity, $v \propto \sqrt{\frac{T}{M}}$

T has become double and M has become half. Therefore, v will become two times. Further, with increase in humidity in air, density of air decreases. Therefore, speed of sound increases.

• Statement based questions

- 1** (c) When mechanical transverse wave propagates through a medium, the constituents of the medium oscillate perpendicular to wave motion causing change in shape, i.e. each element of the medium is subjected to shearing stress. Solids and strings have shear modulus, hence they can sustain shearing stress.

Fluids have no shape of their own, they yield to shearing stress. This is why transverse waves are possible in solids and strings but not in fluids.

- 2** (d) Given, $y = 0.02 \cos(10\pi x) \cos\left(50\pi t + \frac{\pi}{2}\right)$

At node, amplitude = 0

$$\Rightarrow \cos(10\pi x) = 0 \Rightarrow 10\pi x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\Rightarrow x = \frac{1}{20} = 0.05 \text{ m}, 0.15 \text{ m}$$

At antinode, amplitude is maximum,

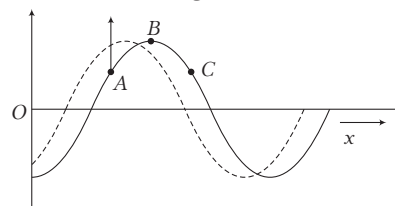
$$\Rightarrow \cos(10\pi x) = \pm 1 \Rightarrow 10\pi x = 0, \pi, 2\pi \dots$$

$$\Rightarrow x = 0, 0.1 \text{ m}, 0.2 \text{ m}, 0.3 \text{ m} \dots$$

$$v = \frac{50\pi}{10\pi} = 5 \text{ ms}^{-1}$$

- 3** (b) Since, A is moving upwards, therefore after an elemental time interval, the wave will be as shown dotted in adjacent figure. It means, the wave is travelling leftward. Therefore, (a) is wrong. Displacement amplitude of the wave means maximum possible displacement of medium particles due to propagation of the wave, which is equal to the displacement at B at the instant shown in figure. Hence (b) is correct.

From figure, it is clear that C is moving downward at this instant. Hence, (c) is wrong.



- 4** (c) Speed = $n\lambda = n(4ab) = 4n \times ab$ $\left(\text{As, } ab = \frac{\lambda}{4} \right)$

$$\text{Path difference between } b \text{ and } e \text{ is } \frac{3\lambda}{4}.$$

$$\text{So, the phase difference, } \Delta\phi = \frac{2\pi}{\lambda} \times \text{path difference}$$

$$= \frac{2\pi}{\lambda} \times \frac{3\lambda}{4} = \frac{3\pi}{2}$$

- 5 (b) Statement II is correct but Statement I is incorrect. It can be corrected as:

In a standing wave, the disturbance produced and energy of one region remains confined to that region.

- 6 (c) In a closed organ pipe, two waves travelling in opposite direction superimpose with each other to develop a wave pattern which is standing or stationary.

Possible wavelengths of stationary waves in this type of organ pipe is given as

$$\lambda_n = \frac{4L}{(2n+1)} \text{ or } \lambda_n = 4L, \frac{4L}{3}, \frac{4L}{5}, \dots$$

As frequency, $v = \frac{v}{\lambda}$ (On putting, $n = 0, 1, 2, 3, \dots$)

So, frequencies of n th harmonics can be given as

$$v_n = \frac{v}{4L}, \frac{3v}{4L}, \frac{5v}{4L}, \dots$$

$$\therefore v_1 : v_2 : v_3 : \dots = 1 : 3 : 5$$

So, only odd harmonics are present in case of closed organ pipe.

However, in case of open organ pipe,

frequencies of n th harmonics, $v_n = \frac{nv}{2L}$

Thus, even and odd or all the harmonics are present.

So, Statements I and II are correct but III is incorrect.

• Match the columns

- 1 (c) Frequency is the property of source. It remains unchanged. In rarer medium, speed of wave is more.

$$\lambda = \frac{v}{f} \text{ or } \lambda \propto v$$

$$A_2 = \left(\frac{2v_2}{v_1 + v_2} \right) A_1$$

As, $v_2 > v_1$

$$\therefore A_2 > A_1$$

Hence, $A \rightarrow r$, $B \rightarrow p$, $C \rightarrow p$, $D \rightarrow r$.

- 2 (b) A. $f' = f \left(\frac{v + v_o}{v + v_s} \right) = f$ B. $f' = f \left(\frac{v + v_o}{v - v_s} \right) > f$

$$C. f' = f \left(\frac{v - v_o}{v + v_s} \right) < f \quad D. f' = f \left(\frac{v - v_o}{v - v_s} \right) = f$$

Hence, $A \rightarrow r$, $B \rightarrow p$, $C \rightarrow q$, $D \rightarrow r$.

- 3 (a) General equation for stationary wave is

$$y = 2A \sin(kx) \cos(\omega t)$$

Comparing it with given equation, we get

$$\therefore 2A = 0.06 \Rightarrow A = 0.03$$

Position of nodes is given by 0.5m, whereas antinodes are formed at 0.25m.

At $x = (3/4)$ m, amplitude will remain same i.e., 0.06.

Hence, $A \rightarrow s$, $B \rightarrow q$, $C \rightarrow r$, $D \rightarrow p$.

- 4 (c) For closed pipe, $100 = \frac{v}{4l_c}$

$$\therefore \text{Length of closed pipe, } l_c = \frac{v}{400} = \frac{330}{400} = 0.825 \text{ m}$$

$$\text{For open pipe, } 200 = \frac{v}{2l_o}$$

$$\therefore \text{Length of open pipe, } l_o = \frac{v}{400} = \frac{330}{400} = 0.825 \text{ m}$$

Different frequencies of closed pipes will be 100 Hz, 300 Hz, 500 Hz, etc., and different frequencies of open pipe are 200 Hz, 400 Hz, 600 Hz, etc., i.e. none of them matches with each other.

Hence, $A \rightarrow p$, $B \rightarrow p$, $C \rightarrow s$.

(C) Medical entrance gallery

- 1 (a) In case of beats formation, unknown frequency,

$$v_B = v_A \pm \text{beats}$$

where, $v_A = 530$ Hz and beats = 6 Hz

$$\therefore v_B = 530 \pm 6 = 536 \text{ Hz or } 524 \text{ Hz}$$

When tension in B is slightly decreased, then beat frequency increases to 7 Hz. This is possible, if we take original frequency of B as 524 Hz. Hence, correct option is (a).

- 2 (b) Given, length of string of musical instrument,

$$l_1 = 90 \text{ cm} = 0.9 \text{ m}$$

Fundamental frequency, $f_1 = 120$ Hz

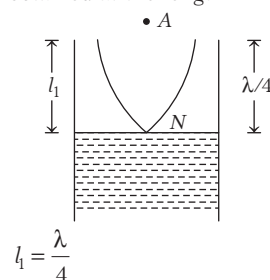
$$f_2 = 180 \text{ Hz}$$

$$\text{As, } f \propto \frac{1}{l} \Rightarrow \frac{f_1}{f_2} = \frac{l_2}{l_1}$$

$$\Rightarrow l_2 = \frac{f_1 l_1}{f_2} = \frac{120 \times 0.9}{180}$$

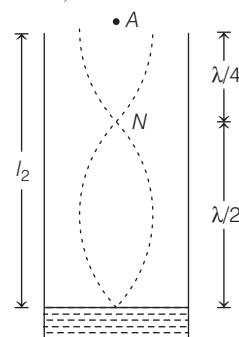
$$\Rightarrow l_2 = \frac{2}{3} \times 0.9 = 0.6 \text{ m} = 60 \text{ cm}$$

- 3 (c) For vibrating tuning fork over a resonance tube, the first resonance is obtained at the length



...(i)

For second resonance,



$$l_2 = \frac{\lambda}{4} + \frac{\lambda}{2} = \frac{3\lambda}{4} \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$l_2 - l_1 = \frac{3\lambda}{4} - \frac{\lambda}{4} = \frac{\lambda}{2}$$

$$\Rightarrow \lambda = 2(l_2 - l_1)$$

$$\Rightarrow v = 2f(l_2 - l_1) \quad \dots(iii) \quad \left(\because \lambda = \frac{v}{f} \right)$$

Here, $f = 800$ Hz, $l_1 = 9.75$ cm and $l_2 = 31.25$ cm

Substituting the given values in Eq. (iii), we get

$$v = 2 \times 800(31.25 - 9.75)$$

$$= 34400 \text{ cm/s} = 344 \text{ ms}^{-1}$$

- 4** (d) Given, speed of sound, $v = 330$ m/s

Length of open organ pipe, $l = 1$ m

Frequency, $f = 1000$ Hz

For open organ pipe, fundamental frequency,

$$f_o = \frac{v}{2l} = \frac{330}{2 \times 1} = 165 \text{ Hz}$$

$\therefore$ Number of tones present in the open organ pipe

$$= \frac{f}{f_o} = \frac{1000}{165} = 6.06 \approx 6$$

- 5** (c) Given, wave equation, $y = 0.002 \sin(300t - 15x)$

Mass density, $\mu = 0.1$ kg/m

On comparing the given wave equation with

$$y = A \sin(\omega t - kx), \text{ we get}$$

$$\omega = 300, k = 15$$

$$\therefore \text{Velocity of wave, } v = \frac{\omega}{k} = \frac{300}{15} = 20 \text{ m/s}$$

Wave speed in string is also given by

$$v = \sqrt{\frac{T}{\mu}}$$

where, T = tension in the string

and μ = mass per unit length of string.

$$\therefore 20 = \sqrt{\frac{T}{0.1}}$$

$$\Rightarrow 400 = \frac{T}{0.1}$$

$$\Rightarrow T = 40 \text{ N}$$

- 6** (a) Given, frequency of sound produced sources S_1 and S_2 , i.e.

$$f_1 = f_2 = 500 \text{ Hz}$$

Speed of sound in air, $v = 330 \text{ ms}^{-1}$

Beat frequency = 3 Hz

Let v_s be the speed of source. When first source is moving towards the observer and f'_1 be apparent frequency heard by the observer, then

$$f'_1 = \frac{v}{v - v_s} \times f_1 = \frac{330}{330 - v_s} \times 500$$

When second source is moving away from the observer and f'_2 be apparent frequency heard by the observer, then

$$f'_2 = \frac{v}{v + v_s} \times f_2 = \frac{330}{330 + v_s} \times 500$$

Now, beat frequency = $f'_1 - f'_2 = 3$

$$\Rightarrow \frac{330}{330 - v_s} \times 500 - \frac{330}{330 + v_s} \times 500 = 3$$

$$\Rightarrow \frac{500 \times 2v_s \times 330}{(330)^2 - v_s^2} = 3$$

Since, $v_s^2 \ll (330)^2$

$$\Rightarrow \frac{500 \times 2v_s \times 330}{(330)^2} = 3$$

$$\Rightarrow v_s = 0.99 \approx 1 \text{ ms}^{-1}$$

- 7** (c) Fundamental frequency for an open organ pipe is given as

$$f = \frac{v}{2L}$$

where, L is the length of the open organ pipe.

Third harmonic for a closed organ pipe is given as $f' = \frac{3v}{4L'}$

where, L' is the length of closed organ pipe.

According to the question, $f = f'$

$$\frac{v}{2L} = \frac{3v}{4L'}$$

$$\Rightarrow L = \frac{2}{3}L'$$

Given, $L' = 20$ cm

$$\Rightarrow L = \frac{2}{3} \times 20 \text{ cm} = \frac{40}{3} \text{ cm} = 13.3 \text{ cm}$$

- 8** (b) For first resonance, $l_1 = \frac{\lambda}{4}$

For second resonance, $l_2 = \frac{3\lambda}{4}$

$$\therefore (l_2 - l_1) = \frac{3\lambda}{4} - \frac{\lambda}{4}$$

$$\text{or } \lambda = 2(l_2 - l_1) \quad \dots(i)$$

As, velocity of sound wave is given as

$$v = \nu \lambda$$

where, ν is the frequency.

$$\Rightarrow v = \nu [2(l_2 - l_1)] \quad \dots(ii)$$

[From Eq. (i)]

Here, $\nu = 320$ Hz,

$$l_2 = 0.73 \text{ m}, l_1 = 0.20 \text{ m} \quad (\text{Given})$$

Putting the values in Eq. (ii), we get

$$v = 320 [2(0.73 - 0.20)] = 2 \times 320 \times 0.53$$

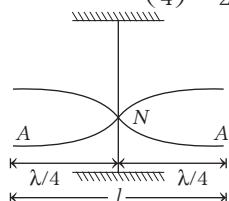
$$= 339.2 \text{ ms}^{-1} \approx 339 \text{ ms}^{-1}$$

- 9** (b) According to the property of persistence of hearing, the impression of a sound heard persists on our mind for $\frac{1}{10}$ s. To hear distinct beats, time interval between two successive

beats should be greater than $\frac{1}{10}$ s. Hence, difference in two frequencies must be less than 10 Hz.

Frequency of 10 Hz means 10 beats per second, i.e. if the number of beats is more than 10, then they will not be audible or more is the confusion.

- 10 (b) In fundamental mode, $l = 2 \left(\frac{\lambda}{4} \right) = \frac{\lambda}{2}$



$$\Rightarrow \lambda = 2l \quad \dots(i)$$

Given, $l = 100 \text{ cm} = 100 \times 10^{-2} \text{ m}$, $v = 2.53 \text{ kHz}$

$$= 2.53 \times 10^3 \text{ Hz}$$

As, $v = v\lambda = v \times 2l$ [From Eq. (i)]

$$= 2.53 \times 10^3 \times 2 \times 100 \times 10^{-2}$$

$$= 5.06 \times 10^3 \text{ ms}^{-1} = 5.06 \text{ kms}^{-1}$$

- 11 (b) Given, $v_w = 1480 \text{ m/s}$, $f = 1000 \text{ kHz}$

$$\text{Speed, } v_w = f\lambda \Rightarrow \lambda = \frac{v_w}{f} = \frac{1480}{1000 \times 10^3} = 1.48 \text{ mm}$$

- 12 (a) Frequency of B can be $n + x$ or $n - x$.

When B is loaded with wax, its frequency will decrease. So, if its original frequency is $(n + x)$, then beat frequency will decrease.

- 13 (c) Given, $f = 1000 \text{ Hz}$, $v = 332 \text{ m/s}$,

$$l = 16.6 \text{ cm} = 16.6 \times 10^{-2} \text{ m}$$

Frequency in an organ pipe open on both ends,

$$f = \frac{nv}{2l} \Rightarrow n = \frac{2fl}{v}$$

$$\text{or } n = \frac{2 \times 1000 \times 16.6 \times 10^{-2}}{332} = 1$$

- 14 (b) Frequency of n th harmonic in a closed end tube,

$$f = \frac{(2n-1)v}{4l}$$

where, $n = 1, 2, 3, \dots$

Also, only odd harmonics exists in a closed end tube.

Now, given two nearest harmonics are of frequency 220 Hz and 260 Hz.

$$\therefore \frac{(2n-1)v}{4l} = 220 \text{ Hz} \quad \dots(i)$$

Next harmonic occurs at

$$\frac{(2n+1)v}{4l} = 260 \text{ Hz} \quad \dots(ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$\frac{[(2n+1) - (2n-1)]v}{4l} = 260 - 220$$

$$2 \left(\frac{v}{4l} \right) = 40 \Rightarrow \frac{v}{4l} = 20 \text{ Hz}$$

$\therefore$ Fundamental frequency of the system

$$= \frac{v}{4l} = 20 \text{ Hz}$$

- 15 (c) When the sound is reflected from the cliff, it approaches the driver of the car. Therefore, the driver acts as an observer and both source (car) and observer are moving.

In this case, the apparent frequency heard by the driver (observer),

$$f' = f \left(\frac{v + v_o}{v - v_s} \right)$$

Given, $f = 600 \text{ Hz}$, $v = 330 \text{ ms}^{-1}$ and $v_o = 30 \text{ ms}^{-1}$

$$f' = 600 \left(\frac{330 + 30}{330 - 30} \right) = 720 \text{ Hz}$$

- 16 (d) When both source and observer are moving towards each other, apparent frequency is given by

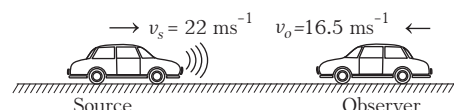
$$f_a = f_o \left(\frac{v + v_o}{v - v_s} \right)$$

where, f_o = original frequency of source = 400 Hz,

v_s = speed of source = 22 ms^{-1} ,

v_o = speed of observer = 16.5 ms^{-1}

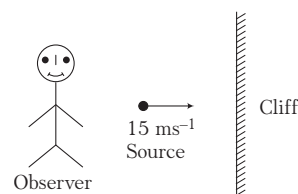
and v = speed of sound = 340 ms^{-1} .



Frequency heard by the driver in the second car,

$$f_a = f_o \left(\frac{v + v_o}{v - v_s} \right) = 400 \left(\frac{340 + 16.5}{340 - 22} \right) = 448.4 \text{ Hz} \approx 448 \text{ Hz}$$

- 17 (b) The given situation can be shown as below.



Frequency of sound that the observer hear in the echo reflected from the cliff is given by

$$f' = \left(\frac{v}{v - v_s} \right) f$$

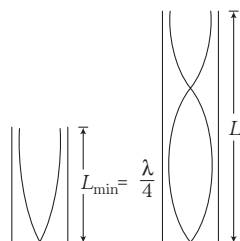
where, f = original frequency of source,

v = velocity of sound

and v_s = velocity of source.

$$\text{So, } f' = \left(\frac{330}{330 - 15} \right) 800 = 838.09 \text{ Hz} \approx 838 \text{ Hz}$$

- 18** (b) The smallest length of the air column is associated with fundamental mode of vibration of the air column as shown in the figure.



$$\therefore L_{\min} = \frac{\lambda}{4} \Rightarrow 50 \text{ cm} = \frac{\lambda}{4} \Rightarrow \lambda = 200 \text{ cm}$$

The next higher length of the air column,

$$L = \frac{\lambda}{4} + \frac{\lambda}{2} = \frac{\lambda + 2\lambda}{4} = \frac{3\lambda}{4} = \frac{3}{4} \times 200 = 150 \text{ cm}$$

- 19** (a) Wavelength of transverse pulse, $\lambda = \frac{v}{f}$... (i)

(where, v = velocity of the wave, f = frequency of the wave)

$$\text{As, } v = \sqrt{\frac{T}{\mu}} \quad \dots \text{(ii)}$$

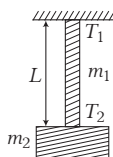
(where, T = tension in the rope, μ = mass per unit length of the rope)

From Eqs. (i) and (ii), we get

$$\lambda = \frac{1}{f} \sqrt{\frac{T}{\mu}} \Rightarrow \lambda \propto \sqrt{T}$$

So, for two different cases, we get

$$\frac{\lambda_2}{\lambda_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{m_1 + m_2}{m_2}}$$



- 20** (c) Given, $y_1 = a \sin \omega t$ and $y_2 = b \cos \omega t = b \sin \left(\omega t + \frac{\pi}{2} \right)$

The resultant displacement is given by

$$y = y_1 + y_2 = \sqrt{a^2 + b^2} \sin(\omega t + \phi)$$

Hence, the motion of superimposed wave is simple harmonic with amplitude $\sqrt{a^2 + b^2}$.

- 21** (c) The fundamental frequencies of closed and open organ pipe are given as $v_c = \frac{v}{4l}$ and $v_o = \frac{v}{2l}$, respectively.

Given, the second overtone (i.e. third harmonic) of open pipe is equal to the fundamental frequency of closed pipe,

$$\text{i.e. } 3v_o = v_c \Rightarrow 3 \frac{v}{2l} = \frac{v}{4l}$$

$$\Rightarrow l = 6l = 6 \times 20 = 120 \text{ cm}$$

- 22** (b) Let the initial intensity be I and corresponding decibels level be β_1 . When intensity is increased to $20I$, let the corresponding increase in decibels level be β_2 . Thus,

$$\beta_1 = 10 \log (I / I_0) \text{ and } \beta_2 = 10 \log \frac{20I}{I_0}$$

$$\Rightarrow \beta_2 - \beta_1 = 10 \log \left(\frac{20I}{I} \right) = 10 \log 20 = 13 \text{ dB}$$

- 23** (a) Let the original frequency be f , then apparent frequency,

$$f' = \frac{v - u_o}{v + u_s} \times f$$

where, u_o = speed of observer,

u_s = speed of source

and v = speed of sound wave.

$$\Rightarrow 1950 = \frac{340 - 10}{340 + 10} f$$

$$\Rightarrow \text{Frequency, } f = \frac{35}{33} \times 1950 = 2068.18 \text{ Hz} \approx 2068 \text{ Hz}$$

- 24** (c) The displacement of the particle is given by

$$y = 3 \sin 6.28 (0.50x - 50t)$$

$$\text{Speed of wave, } v = \frac{\text{Coefficient of } t}{\text{Coefficient of } x} = \frac{50}{0.50} = 100 \text{ cm s}^{-1}$$

- 25** (c) Given, path difference = $1.25 \text{ cm} = 1.25 \times 10^{-2} \text{ m}$

$$\text{Phase difference} = \frac{\pi}{4}$$

Frequency, $n = 1000 \text{ Hz}$

As, phase difference = $\frac{2\pi}{\lambda}$ (path difference)

$$\Rightarrow \frac{\pi}{4} = \frac{2\pi}{\lambda} \times 1.25 \times 10^{-2}$$

$$\Rightarrow \lambda = 2 \times 4 \times 1.25 \times 10^{-2}$$

$$\therefore \text{Wavelength, } \lambda = 10 \times 10^{-2} = \frac{1}{10} \text{ m}$$

Velocity of wave, $v = f\lambda$

$$\Rightarrow v = 1000 \times \frac{1}{10} = 100 \text{ ms}^{-1}$$

- 26** (c) Given, $l_1 = 1.2 \text{ m}$, $\mu_1 = \mu$,

$$F_1 = 36 \text{ kg-wt and } l_2 = 2 \text{ m}$$

Let, $\mu_2 = \mu \Rightarrow F_2 = F$

$$v_2 = v_1$$

As fundamental frequency,

$$v_1 = \frac{1}{2l_1} \sqrt{\frac{F_1}{\mu_1}} = \frac{1}{2 \times 1.2} \sqrt{\frac{36}{\mu}} \quad \dots \text{(i)}$$

$$v_2 = \frac{1}{2 \times 2} \sqrt{\frac{F}{\mu}} \quad \dots \text{(ii)}$$

On dividing Eq. (i) by Eq. (ii), we get

$$\frac{v_1}{v_2} = \frac{\frac{1}{2.4} \sqrt{\frac{36}{\mu}}}{\frac{1}{4} \sqrt{\frac{F}{\mu}}} = 1 \quad \text{(Given)}$$

$$\Rightarrow \frac{1}{2.4} \sqrt{\frac{36}{\mu}} = \frac{1}{4} \sqrt{\frac{F}{\mu}} \Rightarrow \frac{24}{2.4} = \sqrt{F}$$

$$\therefore F = 100 \text{ kg-wt}$$

- 27** (b) According to the question, $\frac{3v}{4l_c} = \frac{3v}{2l_o} \Rightarrow \frac{4l_c}{2l_o} = 1$

$$\Rightarrow \frac{l_c}{l_o} = \frac{1}{2}$$

28 (a) Here, $y = A \sin\left(\frac{2\pi}{\lambda}(ct - x)\right)$

where, x and y are measured in metres.

A is amplitude of wave, whose unit is metre and x is also in metre (given).

$\therefore$ Unit of λ is same as that of x and A , i.e. metre.

However, unit of c (speed of light wave) is ms^{-1} , which is not same as the unit of $2\pi/\lambda$, i.e. metre^{-1} . Similarly, unit of $(ct - x)$ is metre which is not same as the unit of $2\pi/\lambda$.

- 29** (c) Since, the Doppler shift affects the wave incident upon the aeroplane as well as the wave reflected back to the radar, the change in frequency observed by a radar due to an aeroplane moving at relative velocity v_A (given) is twice than that from the aeroplane emitting the wave, i.e.

$$\Delta v = 2 \cdot \frac{v_A}{c} v$$

Therefore, the velocity of aeroplane, $v_A = c \frac{\Delta v}{2v}$

- 30** (a) Apparent frequency,

$$v = \left(\frac{v + u_o}{v - u_s} \right) v_o = \left(\frac{333 + 33}{333 - 33} \right) \times 1000$$

$$= \frac{366}{300} \times 1000 = 1220 \text{ Hz}$$

- 31** (b) From formula of Doppler's effect,

$$f' = f \left(\frac{v}{v - v_s} \right) \Rightarrow 100 = \frac{fv}{v - v_s}$$

$$\Rightarrow v - v_s = \frac{fv}{100} \quad \dots (i)$$

In case of recession of source, $f' = f \left(\frac{v}{v + v_s} \right) \Rightarrow 50 = \frac{fv}{v + v_s}$

$$\Rightarrow v + v_s = \frac{fv}{50} \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$2v = \frac{3fv}{100}$$

$$\therefore f = \frac{200}{3} = 66.67 \text{ Hz}$$

- 32** (c) While refraction of sound in a medium, the frequency of sound wave will remain unchanged.

33 (a) Given, $y_2 = a_2 \cos\left(\omega t - \frac{2\pi x}{\lambda} + \phi\right)$

or $y_2 = a_2 \sin\left[\frac{\pi}{2} + \left(\omega t - \frac{2\pi x}{\lambda} + \phi\right)\right]$

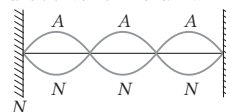
and $y_1 = a_1 \sin\left(\omega t - \frac{2\pi x}{\lambda}\right)$

Phase difference between two waves $= \left(\frac{\pi}{2} + \phi\right)$

Path difference $= \frac{\lambda}{2\pi} \times \text{Phase difference}$

$$\therefore \text{Path difference} = \frac{\lambda}{2\pi} \left(\frac{\pi}{2} + \phi\right)$$

- 34** (c) A stretched string is vibrating such that
2nd overtone = 3rd harmonic



There are 2 nodes and 3 antinodes present.

- 35** (d) Given, frequency of fork B, $v_B = 384 \text{ Hz}$
Beat frequency, $v_A - v_B = 4$, so $v_A = 388$ or 380
 $\therefore$ When one of prong of A is filled, then beat frequency increase.
 $\therefore$ Frequency of fork A, $v_A = 388 \text{ Hz}$

- 36** (c) The fundamental frequency of vibrations,

$$f = \frac{1}{2L} \sqrt{\frac{F}{\mu}} = \frac{1}{2L} \sqrt{\frac{T}{\rho A}}$$

(where, T = tension and $\mu = \rho A$)

Tension due to elasticity, $T = YA \left(\frac{\Delta L}{L}\right)$ $\left(\because Y = \frac{\text{Stress}}{\text{Strain}}\right)$

$$\Rightarrow f = \frac{1}{2L} \sqrt{\frac{YA}{\rho A} \times \frac{\Delta L}{L}} = \frac{1}{2L} \sqrt{\frac{Y\Delta L}{\rho L}}$$

Substituting the given values in above equation, we get

$$f = \frac{1}{2 \times 1} \sqrt{\frac{9 \times 10^{10} \times 4.9 \times 10^{-4}}{9 \times 10^3 \times 1}} = 35 \text{ Hz}$$

- 37** (c) The correct statement is longitudinal wave can propagate in solids, liquids and gases.

- 38** (d) Given, $y = a \sin 2\pi(bt - cx)$

On comparing this equation with general equation

$$y = r \sin\left(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda}\right), \text{ we get}$$

$$\frac{2\pi}{T} = \omega = 2\pi b, r = a$$

$$\lambda = \frac{1}{c} \text{ and } T = \frac{1}{b}$$

Maximum particle velocity, $\omega r = 2\pi ba$

Wave velocity, $v = \frac{\lambda}{T} = \frac{b}{c}$

Given, maximum particle velocity $= 2 \times$ wave velocity

$$2\pi ba = 2 \times \frac{b}{c} \Rightarrow c = \frac{1}{\pi a}$$

- 39** (b) The displacement is given by

$$y = 10^{-6} \sin\left(100t + 20x + \frac{\pi}{4}\right)$$

Comparing with standard equation $y = A \sin(\omega t + kx + \phi)$, we get

$$\omega = 100, k = 20$$

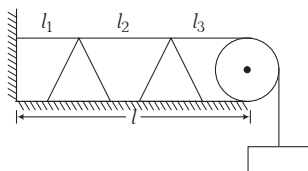
where, ω is angular frequency and k is angular wave number.

Now, $k = \frac{2\pi}{\lambda} = \frac{2\pi v}{\lambda v} = \frac{\omega}{v} = 20$

$$\Rightarrow v = \frac{\omega}{20} = \frac{100}{20} = 5 \text{ ms}^{-1}$$

So, the speed of the wave is 5 ms^{-1} .

- 40 (a)** If fundamental frequency of each part is calculated, then the fundamental frequency of complete wire can be calculated.



For 1st part, $n_1 = \frac{v}{2l_1} \Rightarrow l_1 = \frac{v}{2n_1}$

For 2nd part, $n_2 = \frac{v}{2l_2} \Rightarrow l_2 = \frac{v}{2n_2}$

For 3rd part, $n_3 = \frac{v}{2l_3} \Rightarrow l_3 = \frac{v}{2n_3}$

For the complete wire, $n = \frac{v}{2l} \Rightarrow l = \frac{v}{2n}$

Now, $l = l_1 + l_2 + l_3$
 $\frac{v}{2n} = \frac{v}{2n_1} + \frac{v}{2n_2} + \frac{v}{2n_3}$

$\Rightarrow \frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$

- 41 (d)** For a pipe closed at one end,

$f_n = n \left(\frac{v}{4l} \right)$, here n is an odd number

$= n \left(\frac{340}{4 \times 85 \times 10^{-2}} \right) = n (100)$

Here, n is an odd number, so for the given condition, n can go upto $n = 11$ because for $n = 13$, condition will not be valid.

$n = 1, 3, 5, 7, 9, 11$

So, number of possible natural oscillations could be 6.

- 42 (b)** For an open pipe, the frequency,

$f = \frac{v}{2l} \Rightarrow 5100 = \frac{340}{2 \times l}$

or $l = \frac{340}{5100 \times 2} \Rightarrow l = \frac{2}{30 \times 2} = \frac{1}{30} \text{ m}$

Therefore, $l = \frac{100}{30} = \frac{10}{3} \text{ cm}$

- 43 (b)** For closed organ pipe, fundamental mode, $n_1 = \frac{v}{4l}$... (i)

and for open organ pipe, $n_2 = \frac{v}{2l}$... (ii)

According to the question, $n_2 - n_1 = 2$ (beats)

$\Rightarrow \frac{v}{2l} - \frac{v}{4l} = 2 \Rightarrow \frac{2v - v}{4l} = 2 \Rightarrow v = 8l$... (iii)

When open organ pipe is halved and closed organ pipe is doubled, let the two produce n beats.

$\frac{v}{l} - \frac{v}{8l} = n \Rightarrow \frac{8v - v}{8l} = n$

$\Rightarrow \frac{7v}{8l} = n \Rightarrow n = \frac{7(8l)}{8l} = 7$

- 44 (d)** According to the mode of vibration of air column in an open organ pipe,

First mode of vibration, $v_1 = \frac{v}{2l}$... (i)

where, v_1 = frequency of vibration,

v = speed of sound and l = length of organ pipe.

The frequency of n th mode of vibration,

$v_n = n \times v_1 = n \times \left(\frac{v}{2l} \right)$... (ii)

Given, $v_n = 1.1 \text{ kHz} = 1100 \text{ Hz}$, $v = 330 \text{ ms}^{-1}$ and

$l = 30 \text{ cm} = 0.30 \text{ m}$

Substituting these values in Eq. (ii), we get

$1100 = n \times \left(\frac{330}{2 \times 0.30} \right)$

$\Rightarrow n = \frac{1100 \times 2 \times 0.30}{330} \Rightarrow n = \frac{660}{330} = 2$

So, second harmonic mode of pipe resonates.

- 45 (a)** The linear mass density is defined as a measure of mass per unit of length.

$\therefore \text{Linear mass density, } \mu = \frac{\text{Mass}}{\text{Length}} = \frac{10 \times 10^{-3}}{1}$
 $= 10 \times 10^{-3} \text{ kg m}^{-1}$

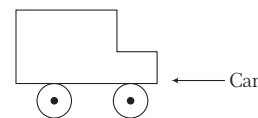
$\therefore$ The speed of transverse wave,

$v = \sqrt{\frac{T}{\mu}}$ (where, μ = volume per unit $\times$ density)

$= \sqrt{\frac{100}{10 \times 10^{-3}}} = \sqrt{10 \times 10^3} = 1 \times 10^2 = 100 \text{ ms}^{-1}$

- 46 (c)** As, both observer and source are moving, we can use the formula of apparent frequency as

$\xrightarrow{36 \text{ kmh}^{-1}} \quad \xrightarrow{18 \text{ kmh}^{-1}}$
 $f = 1392 \text{ Hz}$



$f = f_o \left(\frac{v + v_o}{v + v_s} \right) = 1392 \left(\frac{343 + 10}{343 + 5} \right)$

$= 1392 \left(\frac{353}{348} \right) = 1412 \text{ Hz}$

- 47 (a)** Given, amplitude, $A = 1 \text{ m}$

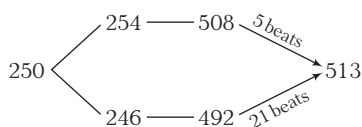
The wave is represented by the equation,

$y = A \sin(kx - \omega t)$

$= \sin \left(\frac{2\pi}{2\pi} x - 2\pi \times \frac{1}{\pi} t \right) = \sin(x - 2t)$

- 48 (d)** Statement (d) is not true because at the open ends, pressure change will be zero.

49 (a) Here,



Hence, unknown frequency is 254 Hz.

50 (b) For open pipe, $v = \frac{v}{2l}$

Given, $v = 340 \text{ ms}^{-1}$, $l = 1 \text{ m}$

$$\therefore v = \frac{340}{2 \times 1} = 170 \text{ Hz}$$

51 (b) Apparent frequency $n' = n \left(\frac{v + v_o}{v} \right)$

$$\Rightarrow n' = 1020 \left(\frac{340 + 30}{340} \right)$$

$$\Rightarrow n' = 1020 \times 1.088 \Rightarrow n' = 1110 \text{ Hz}$$

52 (b) When two sound waves of nearly equal or slightly different frequencies and equal or nearly equal amplitudes travelling along the same direction superimpose at a given point, the resultant sound intensity alternately rises and falls. This periodic waxing of sound at a given position is called beats.

53 (c) We know that, the frequency, $n = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$

Here, μ is a constant, so $n \propto \frac{\sqrt{T}}{l}$

Now, we can calculate the value of n_2 / n_1 as

$$\frac{n_2}{n_1} = \frac{l_1}{l_2} \times \sqrt{\frac{T_2}{T_1}} = \frac{1}{2} \times \sqrt{\frac{9}{1}} = \frac{\sqrt{9}}{2} = \frac{3}{2}$$

Hence, the ratio of $\frac{n_2}{n_1}$ is $\frac{3}{2}$.

54 (a) The frequency of the fundamental note of the pipe is

$$n_2 - n_1 = 2 \left(\frac{v}{4l} \right) \Rightarrow \frac{v}{4l} = \frac{n_2 - n_1}{2}$$

Given, $n_1 = 135 \text{ Hz}$ and $n_2 = 165 \text{ Hz}$

$$\text{Now, } \frac{v}{4l} = \frac{165 - 135}{2} = \frac{30}{2} = 15 \text{ Hz}$$

55 (b) Given, $y = 0.05 \sin \pi (2t - 0.02x)$

$$\text{or } y = 0.05 \sin (2\pi t - 0.02\pi x)$$

On comparing this equation with standard equation,

$$y = A \sin (\omega t - kx)$$

where, $\omega = \frac{2\pi}{T}$ and $k = \frac{2\pi}{\lambda}$, we have

$$\frac{2\pi}{\lambda} = 0.02\pi \Rightarrow \lambda = 100 \text{ m}$$

$$\text{Also, } \omega = \frac{2\pi}{T} = 2\pi \text{ or } \frac{1}{T} = 1 \Rightarrow v = 1 \text{ Hz}$$

$$\text{So, } v = v\lambda = 1 \times 100 = 100 \text{ ms}^{-1}$$

56 (a) The frequency of first overtone for closed pipe and second overtone for open pipe are respectively

$$f_c = \frac{3v}{4l_c} \text{ and } f_o = \frac{3v}{2l_o}$$

$$\text{Given, } f_c = f_o$$

$$\text{i.e. } \frac{3v}{4l_c} = \frac{3v}{2l_o} \Rightarrow \frac{l_o}{l_c} = \frac{2}{1}$$

57 (b) As the person having tuning fork is moving towards a wall, therefore $v_s = v_o = 2 \text{ ms}^{-1}$ (Given)

$$\text{So, } \frac{f'}{f} = \frac{v + v_o}{v - v_o}$$

$$\Rightarrow \frac{f' - f}{f} = \frac{v + v_o - v + v_o}{v - v_o}$$

$$\Rightarrow \frac{\Delta f}{f} = \frac{2v_o}{v - v_o}$$

$$\Delta f = \frac{2v_o}{v - v_o} \times f = \frac{2 \times 2 \times 338}{(340 - 2)} = 4 \text{ Hz}$$

$\therefore$ Number of beats heard = 4

58 (b) When a wave travels from denser medium, it gets reflected at the boundary of a rarer medium; it undergoes no phase change.

59 (d) In given case, pipe will behave as closed pipe. We know the frequency for a open pipe is $f = \frac{v}{2l}$

But frequency after filling with water, $f' = \frac{v}{4l'}$

$$\text{Here, } l' = \frac{l}{2}, \text{ so that } f' = \frac{l}{4l'/2}$$

$$f' = \frac{2v}{4l} = \frac{v}{2l}, \quad f' = f$$

60 (b) Phase difference between two waves is given by

$$A_0 = (A_0^2 + A_0^2 + 2A_0 \cdot A_0 \cos \phi)^{1/2}$$

$$\Rightarrow \cos \phi = -\frac{1}{2} \Rightarrow \phi = \frac{2\pi}{3}$$

61 (c) Given, $f_1 = 334 \text{ Hz}$, $f_2 = 326 \text{ Hz}$, $v = 330 \text{ ms}^{-1}$

Combined frequency, $f = f_1 - f_2 = 334 - 326 = 8 \text{ Hz}$

$$\text{Time period, } T = \frac{1}{f} = \frac{1}{8} = 0.125 \text{ s}$$

62 (b) As we know, $f' = f \left(\frac{v}{v - v_s} \right)$

Given, $f = 200 \text{ Hz}$, $v = 340 \text{ ms}^{-1}$, $v_s = 25 \text{ ms}^{-1}$

From Doppler's effect,

$$f' = 200 \left(\frac{340}{340 - 25} \right) = \frac{200 \times 340}{315} \Rightarrow f' = 215 \text{ Hz}$$

63 (d) Phase difference,

$$\Delta \phi = \frac{2\pi \Delta x}{\lambda} \Rightarrow \Delta x = \frac{\Delta \phi \lambda}{2\pi} = \frac{\pi}{5} \times \frac{0.5}{2\pi} = 0.15 \text{ m}$$

64 (a) Since, the initial phase difference between the two waves coming from different violins changes, therefore the waves produced by two different violins does not interfere. Two waves interfere only when the phase difference between them remain constant throughout.

65 (a) Let l be the length of the pipe and v be the velocity of sound in the air column, then the fundamental frequency of an open pipe = $\frac{v}{2l}$

Fundamental frequency of closed pipe = $\frac{v}{4l}$

Frequency of third harmonic of closed pipe = $\frac{3v}{4l}$

According to given condition,

$$\frac{3v}{4l} = \frac{v}{2l} + 100$$

$$\Rightarrow \frac{v}{2l} = 200 \text{ Hz}$$

66 (c) For not hearing the echo, the time interval between the beats of drum must be equal to that of time of echo.

$$\Rightarrow t_1 = \frac{2d}{v} = \frac{60}{40} = \frac{3}{2} \Rightarrow 2d = \frac{3}{2}v \quad \dots(i)$$

$$\text{Also, } t_2 = \frac{2(d-90)}{v} = \frac{60}{60} = 1$$

$$\Rightarrow 2d - 180 = v \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$\frac{3}{2}v - 180 = v \Rightarrow v = 360 \text{ ms}^{-1}$$

$$\therefore \frac{2d}{360} = \frac{3}{2} \Rightarrow d = \frac{3}{2} \times \frac{360}{2} = 270 \text{ m}$$

67 (c) Here, $A_1 = 10, A_2 = \sqrt{25 + 75} = 10$

$$\therefore A_1 : A_2 = 1 : 1$$

68 (b) Intensity of spherical wave, $I \propto \frac{1}{r^2}$

$$\text{Also, } I \propto a^2$$

$$\therefore a \propto \frac{1}{r}$$

$\therefore$ The spherical progressive wave can be represented as

$$y = \frac{1}{r} \sin(\omega t - kx)$$

Among the given options, $y = \frac{a}{r} \sin(\omega t - kx)$ is correct.

69 (d) When the shift in star light is towards red end, wavelength increases and the apparent frequency is less than the actual. The star must be receding away from the earth.

70 (c) Here, $n_1 - n_2 = 5$

$$\frac{v}{2l_1} - \frac{v}{2l_2} = 5 \quad \left(\because n = \frac{v}{2l} \right)$$

$$\Rightarrow \frac{1}{l_1} - \frac{1}{l_2} = \frac{5 \times 2}{v}$$

$$\Rightarrow \frac{1}{1.10} - \frac{1}{l_2} = \frac{10}{340}$$

$$\Rightarrow -\frac{1}{l_2} = -0.87$$

$$\text{or } l_2 = \frac{1}{0.87}$$

$$\text{Therefore, } l_2 = 1.14 \text{ m}$$

71 (c) When pulse is reflected from a rigid support, the pulse is inverted both lengthwise and sideways.

72 (c) We know that, $n \propto \frac{1}{l}$

$$\begin{aligned} \therefore \frac{n_2}{n_1} &= \frac{l_1}{l_2} \Rightarrow n_2 = \frac{l_1}{l_2} \times n_1 \\ &= \frac{1 \times 256}{1/4} = 1024 \text{ Hz} \end{aligned}$$

73 (b) On getting reflected from a rigid boundary, the wave suffers reflection.

$$\text{Hence, if } y_{\text{incident}} = A \sin(\omega t - kx)$$

$$\begin{aligned} \text{Then, } y_{\text{reflected}} &= (0.8A) \sin\{\omega t - k(-x) + \pi\} \\ &= -0.8A \sin(\omega t + kx) \end{aligned}$$

an additional phase change of π .

74 (d) Given, $y_1 = A \sin(\omega t + kx + 0.57) \text{ m}$

$$\text{and } y_2 = A \cos(\omega t + kx) \text{ m}$$

$$\Rightarrow y_2 = A \sin\left(\frac{\pi}{2} + \omega t + kx\right) \text{ m}$$

$$\begin{aligned} \text{Phase difference, } \Delta\phi &= \phi_2 - \phi_1 = \frac{\pi}{2} - 0.57 \\ &= 1.57 - 0.57 = 1 \text{ rad} \end{aligned}$$

75 (a) The fundamental frequency of an open organ pipe is $n = \frac{v}{2l}$. As temperature increases, both v and l increase but v increases more rapidly than l . Hence, the fundamental frequency increases as the temperature increases.

76 (a) When the source is coming to the stationary observer,

$$n' = \left(\frac{v}{v - v_s} \right) n$$

$$\Rightarrow 1000 = \left(\frac{350}{350 - 50} \right) n$$

$$\text{or } n = \frac{1000 \times 300}{350} \text{ Hz}$$

When the source is moving away from the stationary observer,

$$n'' = \left(\frac{v}{v + v_s} \right) n$$

$$= \left(\frac{350}{350 + 50} \right) \left(\frac{1000 \times 300}{350} \right) = 750 \text{ Hz}$$

77 (d) We know that, $\beta = 10 \log \left(\frac{I}{I_0} \right)$

$$\Rightarrow \beta_1 = 10 \log \left(\frac{I_1}{I_0} \right)$$

and $\beta_2 = 10 \log \left(\frac{I_2}{I_0} \right)$

Now, $\beta_2 - \beta_1 = 20$

$$\Rightarrow 10 \log \left(\frac{I_2}{I_0} \right) - 10 \log \left(\frac{I_1}{I_0} \right) = 20$$

$$\Rightarrow 10 \log \left(\frac{I_2}{I_1} \right) = 20$$

$$\Rightarrow I_2 = 100 I_1$$

79 (d) Number of visitors = 1000

Volume of theatre = $100 \times 40 \times 10 \text{ m}^3$

Case I Volume acquired by one visitor

$$= \frac{100 \times 40 \times 10}{1000} = 40 \text{ m}^3$$

Reverberation time = 8.5 s

Case II Volume acquired by one visitor

$$= \frac{100 \times 40 \times 10}{500} = 80 \text{ m}^3$$

Reverberation time = 6.2 s

The average absorption coefficient,

$$\eta = \frac{40 \times 8.5}{80 \times 6.2} = \frac{85}{124} \approx 0.7$$

80 (c) Given, $n_A = 258 \text{ Hz}$ and $n_B = 262 \text{ Hz}$

If it produces x beats with 258 and $2x$ with 262, then

$$262 - (258 - x) = 2x$$

$$\Rightarrow 262 - 258 + x = 2x \Rightarrow x = 4$$

Therefore, unknown frequency, $n = 258 - x = 254 \text{ Hz}$

81 (b) Phase is expressed as relative displacement between two corresponding peaks of a waveform. In stationary wave, at any instant the phase of all particles between two successive nodes is the same, but phase of particles on one side of node is opposite to the phase of particles on the other side of node, i.e. phase difference between any two particles can be either 0 or π .

82 (c) Fundamental frequency of closed pipe,

$$n = \frac{v}{4l} = 220 \text{ Hz}$$

$$\Rightarrow v = 220 \times 4l \quad \dots(i)$$

If $\left(\frac{1}{4}\right)$ th of the pipe is filled with water, then remaining

length of air column is $\frac{3l}{4}$.

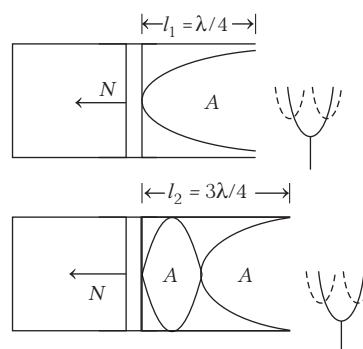
$$\text{Now, fundamental frequency} = \frac{v}{4\left(\frac{3l}{4}\right)} = \frac{v}{3l}$$

First overtone = $3 \times$ Fundamental frequency

$$= 3 \times \frac{v}{3l} = \frac{3 \times 220 \times 4l}{3l} \quad [\text{From Eq. (i)}]$$

$$= 880 \text{ Hz}$$

83 (b) In a closed organ pipe in which length of air-column can be increased or decreased. The first resonance occurs at $\lambda / 4$ and second resonance occurs at $3\lambda / 4$.



Thus, at first resonance, $\frac{\lambda}{4} = 13 \text{ cm} \quad \dots(i)$

and at second resonance, $\frac{3\lambda}{4} = 41 \text{ cm} \quad \dots(ii)$

Subtracting Eq. (i) from Eq. (ii), we get

$$\frac{3\lambda}{4} - \frac{\lambda}{4} = 41 - 13$$

$$\Rightarrow \frac{\lambda}{2} = 28$$

$$\Rightarrow \lambda = 56 \text{ cm}$$

Hence, frequency of tuning fork,

$$n = \frac{v}{\lambda} = \frac{350}{56 \times 10^{-2}} = 625 \text{ Hz}$$

84 (a) As, $n_1 = \left(\frac{v+u}{v} \right) n$ and $n_2 = \left(\frac{v-u}{v} \right) n$

Number of beats per second,

$$x = n_1 - n_2 = \left(\frac{v+u}{v} \right) n - \left(\frac{v-u}{v} \right) n$$

$$= \left(\frac{v+u-v+u}{v} \right) n = 2u \left(\frac{n}{v} \right) = \frac{2u}{\lambda}$$

85 (d) Wavelength, $\lambda' = \lambda \left(\frac{v+v_s}{v} \right)$

$$\lambda' = 50 \left(\frac{330 + (330/5)}{330} \right)$$

$$\Rightarrow \lambda' = 50 \left(\frac{330 + 66}{330} \right)$$

$$= 50 \left(\frac{396}{330} \right) = 50 \times \frac{6}{5}$$

$$= 60 \text{ cm}$$

86 (b) Let s_1 and s_2 be the distances of man from either side of valley, then $2s_1 = vt_1$ and $2s_2 = vt_2$

$$\Rightarrow 2(s_1 + s_2) = v(t_1 + t_2)$$

$$\Rightarrow s_1 + s_2 = \frac{v(t_1 + t_2)}{2}$$

$$\therefore d = s_1 + s_2 = \frac{330 \times (1+2)}{2}$$

$$\Rightarrow d = \frac{330 \times 3}{2} = 495 \text{ m}$$